UNIVERSITY MATHEMATICS COMPETITIONS

IMC

Full Preparation Guide

Theory, methods, and problem solving

Study.
Attempt.
Reconstruct.

An indexed atlas of mathematical ideas and proofs



$$\zeta_k = e^{2\pi i k/17}$$

Leonidas Askianakis

334 problems | 401 solution records | 1994–2024

Revised edition | 16 September 2026

REVISED EDITION

IMC Full Preparation Guide

Theory, methods, and problem solving

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Edition	16 September 2026
Coverage	334 problems, 1994–2024
Solution corpus	401 recorded solution records
Organization	Theory, methods, problem types, recognition cues, proof templates, problems, and solutions

Reading and navigation. Begin with the contents or the reading routes. Links in the chapters lead to locations within this volume. Sources and provenance at the end contains the explicitly identified website links.

Readers are invited to contact the author with corrections, alternative proofs, and source clarifications. Please identify the edition and the problem or theorem concerned.

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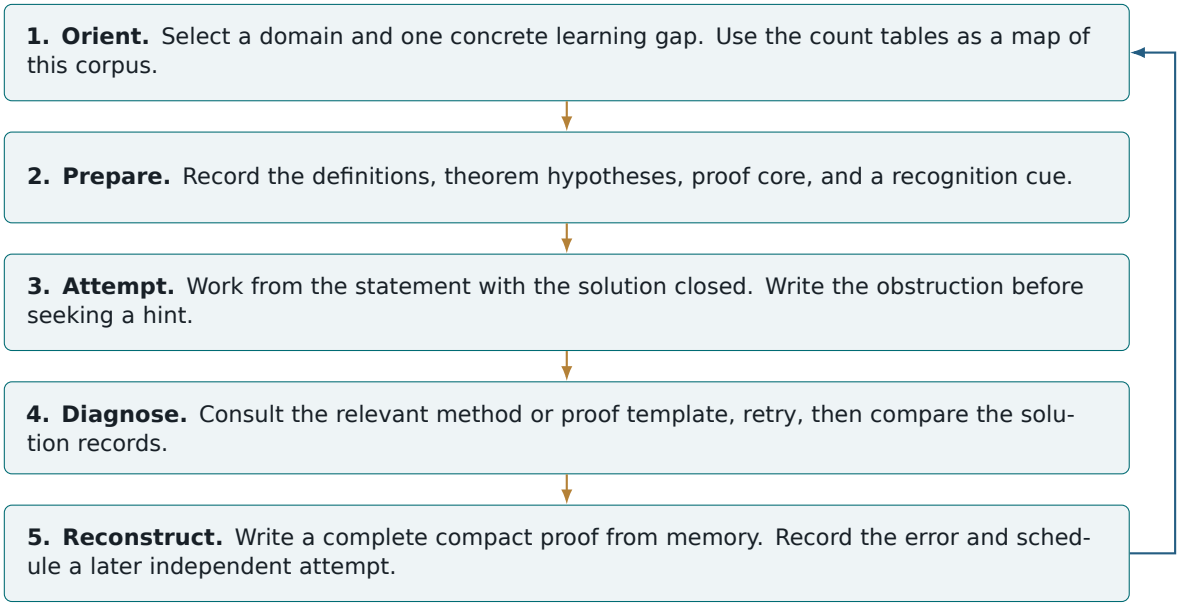
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Chapter 1

Reading Routes and Corpus Overview

1.1 How this book works

The guide is an indexed training atlas. Its layers connect a problem to the theory needed to read it, the methods that can solve it, the cues that suggest those methods, and the proof structure needed to present a complete argument. Use the contents for a broad route and the colored identifiers for focused work.



Before opening a solution: state the goal, list unused hypotheses, record the approaches tried, and formulate one possible intermediate claim. If blocked, consult one method cue, then attempt the problem again.

Three entry routes. A new topic begins with the theory atlas and one accessible example. An unfamiliar problem begins with an independent attempt and a diagnosis of missing theory. A timed session mixes familiar and unfamiliar domains, followed by a written error analysis.

1.2 Navigation and complete proofs

Element	Use
Problem ID	1994-D1-P1: catalog metadata and full statement.
Statement	Open statement: read the problem with solutions closed.
Solution	Open solution: compare recorded variants and their continuation pages.
Theory ID	T-LA-RANK: hypotheses, proof core and related exercises.
Method ID	Method reference: reusable moves linked to their theory.
Strategy / template	Proof strategies, recognition cues and templates.
Correction ID	MATH-001: the controlling replacement for an identified legacy passage.
Source note	An internal provenance entry; website links appear only in the closing source register.

Open the PDF bookmarks panel to move between sections. Every exercise has a linked learning route; each recorded solution identifies its own theory and methods. The reference entries contain a Back to previous view button and explicit return links to all mapped exercises. If your reader does not support the history button, use the exercise ID or [R] link in the entry. Long statements and solutions continue in labeled, breakable boxes; the continuation belongs to the same problem ID.

Compact does not mean incomplete. Define the objects; state the decisive claim; justify the nontrivial implication; check exceptional cases, hypotheses and equality conditions; then state the conclusion. A remembered method is useful only when it produces a valid written proof.

Record after each attempt: time spent, time to the first useful idea, help used, unused hypothesis, failure mode, and a date for another attempt. A possible starting schedule is 1, 3, 7 and 21 days; adapt it to demonstrated recall and your calendar. These intervals are a practical schedule, not a claimed optimum.

Minimal revision card. Keep the trigger, decisive lemma, proof skeleton, and principal trap. Then reconstruct the argument without the card. Compare alternatives only after you can justify one complete solution.

1.3 Scope and interpretation of annotations

This historical corpus contains 334 problems from 1994 through 2024. It is not a survey of later competitions. The catalog retains fingerprints, primary domains, structural types, theory cues, methods, triggers, pace labels, preparation buckets and trap notes. **Primary domain and type** refer to the archived solution-led classification ledger. **Method counts** allow several occurrences per problem. **Linked theory and methods** are checked against the recorded solutions in this revision; the count tables retain the historical annotation scheme and do not count the new navigation edges. **Trigger cues** retain the historical annotations. **Pace** is an editorial label, not a measured solving-time distribution. The tables below describe this coded corpus. They neither predict future contest topics nor establish a uniquely optimal allocation of study time. The correction register distinguishes verified arithmetic from unverified classification judgments.

1.4 Annotation-count overview

These counts are generated from the 334 catalog records bundled with this edition. The mutually exclusive domain, primary-type and pace columns each sum to 334. Essential-method annotations may overlap.

Primary domains

Annotation	Problems
Real analysis / calculus	91
Linear algebra / matrices	71
Algebra / number theory	53
Complex analysis / polynomials	35
Combinatorics	35
Geometry / convexity / metric	24
Functional equations / dynamics	20
Probability / measure	5

Primary structural types

Annotation	Problems
inequality/sharp bound	85
structural proof	68
existence/construction	43
classification/find-all	40
exact computation	22
limit/asymptotic evaluation	20
counting/enumeration	20
nonexistence/counterexample	19
recurrence/convergence	15
game/strategy	2

Essential-method annotations

Annotation	Problems
Explicit construction / witness / counterexample	94
Direct structural / algebraic / invariant argument	92
Polynomial identity / roots / factorization / Vieta	82
Counting / pigeonhole / averaging / incidence	72
Induction / recurrence propagation	59
Limit / Riemann-sum / asymptotic comparison	50
Auxiliary function / monotonicity	45
Integral estimate / IBP / Fubini / representation	44
Modular arithmetic / divisibility / finite field	41
Compactness / extremal point / topology	39
Rank / dimension / linear dependence	33
Rolle / MVT / Taylor / interlacing	30
Spectral / minimal-polynomial argument	30
Group / ring structure	28
Cauchy / norm / inner-product inequality	27
Determinant / block / trace identities	27
Roots of unity / complex-modulus method	25
Convexity / Jensen / tangent method	17
Graph encoding	17

Editorial pace labels

Annotation	Problems
Long/deep	141
Medium	82
Quick/standard	79
Medium/trap	32

Annual denominators

Period	Years	Problems/year	Total
1994–2008	15	12	180
2009–2019	11	10	110
2020–2022	3	8	24
2023–2024	2	10	20

The labels describe the recorded preparation corpus. Frequency is not difficulty or pedagogical priority. Select a topic using your diagnosed gaps, then use linked examples to test whether you can recognize and apply the method. See DATA-001 and CLASS-002 for the limitations of the retained annotation layers.

Chapter 2

Complete Problem-by-Problem Catalog

This catalog retains all 334 historical problem records and their pedagogical annotations. Every full statement is native, searchable LaTeX and can continue across a page boundary. The problem IDs remain stable even when the official printed question number differs from the guide's day-local numbering.

How to read a card. The primary domain and problem type follow the recovered classification ledger. Linked theory, methods and strategies are reviewed against the recorded solution routes and open their canonical reference entries. The remaining secondary cues, triggers, preparation buckets and pace estimates retain the historical annotations, with the explicit corrections in the correction register applied. These annotations are not independently measured frequencies or forecasts. The complete original values and every changed field remain in the accompanying data files.

Read the statement and attempt a proof before opening the linked solution variants. Use the theory and method layers to resolve a specific obstacle, then return to your attempt. A pace label is a legacy planning cue, not a validated difficulty score.

Theory reference | Method reference

2.1 1994

Day 1

1994-D1-P1 Quick/standard

Fingerprint count zero entries in inverse of positive symmetric matrix
Primary domain Linear algebra / matrices
Problem type inequality/sharp bound
Linked methods Read the entries before choosing a theorem [M-LA-ENTRY]
Secondary cues Combinatorics / discrete structures; Real analysis / calculus
Linked theory Determinants, block identities and inverses [T-LA-DETERMINANT]; Rank, dimension and exact linear constraints [T-LA-RANK]
Recognition triggers matrix word; counting/permutation trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]
Correction note The explicit CLASS-001 corrections are incorporated in this card; the accompanying change ledger records the withdrawn labels.
Navigation Full statement All solution variants Source note Learning route

1994-D1-P1 continued

Full statement: 1994-D1-P1

Day 1; official label P1; 13 points; legacy source-page label 2

a) Let A be a $n \times n$, $n \geq 2$, symmetric, invertible matrix with real positive elements. Show that $z_n \leq n^2 - 2n$, where z_n is the number of zero elements in A^{-1}.

b) How many zero elements are there in the inverse of the $n \times n$ matrix

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & \ldots & 1 \\ 1 & 2 & 2 & 2 & \ldots & 2 \\ 1 & 2 & 1 & 1 & \ldots & 1 \\ 1 & 2 & 1 & 2 & \ldots & 2 \\ \ldots & \ldots & \ldots & \ldots & \ldots & \ldots \\ 1 & 2 & 1 & 2 & \ldots & \ldots \end{pmatrix} ?$$

Open all solution variants | Source note

1994-D1-P2 Quick/standard

Fingerprint differential inequality f'+f^2>=-1 forces interval length >= pi
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Design an auxiliary derivative [M-AN-AUXILIARY]
Secondary cues Functional equations / dynamics
Linked theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Differential inequalities and integrating factors [T-AN-ODE]
Recognition triggers continuous/differentiable trigger; integral/limit trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Control the limiting operation before taking the limit [S-ANALYTIC]; Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1994-D1-P2

Day 1; official label P2; 13 points; legacy source-page label 2

Let $f \in C^1(a,b)$, $\lim_{x \to a+} f(x) = +\infty$, $\lim_{x \to b-} f(x) = -\infty$ and $f'(x) + f^2(x) \geq -1$ for $x \in (a,b)$. Prove that $b - a \geq \pi$ and give an example where $b - a = \pi$.

Open all solution variants | Source note

1994-D1-P3 Medium

Fingerprint n irrationals with all positive rational combinations irrational
Primary domain Algebra / number theory
Problem type existence/construction
Linked methods Eliminate rational relations, then exploit discreteness [M-AL-RATIONAL]
Secondary cues Geometry / convexity / metric; Functional equations / dynamics
Linked theory Rational linear relations and discrete-value rigidity [T-AL-RATIONALITY]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Recognition triggers wording-specific
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Reduce size while preserving the defining conditions [S-INDUCTION]; Make the contradiction quantitative or structural [S-CONTRADICTION]
Correction note The explicit CLASS-001 corrections are incorporated in this card; the accompanying

1994-D1-P3 continued

change ledger records the withdrawn labels.

Navigation Full statement All solution variants Source note Learning route

Full statement: 1994-D1-P3

Day 1; official label P3; 13 points; legacy source-page label 2

Given a set S of $2n-1$, $n \in \mathbb{N}$, different irrational numbers. Prove that there are n different elements $x_1, x_2, \ldots, x_n \in S$ such that for all non-negative rational numbers $a_1, a_2, \ldots, a_n$ with $a_1+a_2+\cdots+a_n > 0$ we have that $a_1x_1 + a_2x_2 + \cdots + a_nx_n$ is an irrational number.

Open all solution variants | Source note

1994-D1-P4 Medium

Fingerprint commutator relation FG-GF=aF forces F nilpotent
Primary domain Linear algebra / matrices
Problem type structural proof
Linked methods Propagate a commutator by telescoping [M-LA-DERIVATION]; Treat the unknown matrix as a vector [M-LA-MATRIXOP]
Secondary cues Real analysis / calculus
Linked theory Trace pairing and commutator calculus [T-LA-TRACE]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Nilpotents, projections and involutions [T-LA-NILPOTENT]
Recognition triggers wording-specific
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Reduce size while preserving the defining conditions [S-INDUCTION]; Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1994-D1-P4

Day 1; official label P4; 18 points; legacy source-page label 3

Let $\alpha \in \mathbb{R} \setminus \{0\}$ and suppose that F and G are linear maps (operators) from $\mathbb{R}^n$ into $\mathbb{R}^n$ satisfying $F \circ G - G \circ F = \alpha F$.
a) Show that for all $k \in \mathbb{N}$ one has $F^k \circ G - G \circ F^k = \alpha k F^k$.

b) Show that there exists $k \geq 1$ such that $F^k = 0$.

Open all solution variants | Source note

1994-D1-P5 Long/deep

Fingerprint limit of integral of f(x)g(nx) for periodic g
Primary domain Real analysis / calculus
Problem type limit/asymptotic evaluation
Linked methods Expose the mesh and control its tails [M-AN-RIEMANN]; Pair an integral through symmetry [M-AN-PAIR]
Secondary cues Real analysis / calculus
Linked theory Continuity, compactness, and order [T-AN-CONTINUITY]; Riemann sums, periodic averaging, and tails [T-AN-RIEMANN]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Trigonometric identities and safe branches [T-AN-TRIG]
Recognition triggers integral/limit trigger; functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Control the limiting operation before taking the limit [S-ANALYTIC]; Pass between local constraints and a global conclusion [S-LOCALGLOBAL]
Navigation Full statement All solution variants Source note Learning route

1994-D1-P5 continued

Full statement: 1994-D1-P5

Day 1; official label P5; 18 points; legacy source-page label 4

a) Let $f \in C[0,b]$, $g \in C(\mathbb{R})$ and let g be periodic with period b. Prove that $\int_0^b f(x)g(nx)\,dx$ has a limit as $n \to \infty$ and

$$\lim_{n\to\infty} \int_0^b f(x)g(nx)\,dx = \frac{1}{b}\int_0^b f(x)\,dx \cdot \int_0^b g(x)\,dx.$$

b) Find

$$\lim_{n\to\infty} \int_0^\pi \frac{\sin x}{1+3\cos^2 nx}\,dx.$$

Open all solution variants | Source note

1994-D1-P6 Long/deep

Fingerprint bound integer points on convex curve via derivative constraints
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Localize an extremum or an accumulation point [M-AN-COMPACT]; Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]
Secondary cues Real analysis / calculus; Functional equations / dynamics; Probability / measure / expectation
Linked theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers continuous/differentiable trigger; integer/prime/mod trigger; geometry/metric trigger; functional-equation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1994-D1-P6

Day 1; official label P6; 25 points; legacy source-page label 5

Let $f \in C^2[0,N]$ and $|f'(x)| < 1$, $f''(x) > 0$ for every $x \in [0,N]$. Let $0 \le m_0 < m_1 < \cdots < m_k \le N$ be integers such that $n_i = f(m_i)$ are also integers for $i = 0,1,\ldots,k$. Denote $b_i = n_i - n_{i-1}$ and $a_i = m_i - m_{i-1}$ for $i = 1,2,\ldots,k$.
a) Prove that

$$-1 < \frac{b_1}{a_1} < \frac{b_2}{a_2} < \cdots < \frac{b_k}{a_k} < 1.$$

b) Prove that for every choice of $A > 1$ there are no more than N/A indices j such that $a_j > A$.

c) Prove that $k \le 3N^{2/3}$ (i.e. there are no more than $3N^{2/3}$ integer points on the curve $y = f(x)$, $x \in [0,N]$).

Open all solution variants | Source note

Day 2

1994-D2-P1 Quick/standard

Fingerprint Gronwall-type: |f'|<=lambda|f| with f(a)=0 forces f=0
Primary domain Real analysis / calculus
Problem type structural proof
Linked methods Design an auxiliary derivative [M-AN-AUXILIARY]; Use a first crossing or a forced-growth contradiction [M-AN-BARRIER]
Secondary cues Real analysis / calculus
Linked theory Continuity, compactness, and order [T-AN-CONTINUITY]; Differential inequalities and integrating factors [T-AN-ODE]
Recognition triggers continuous/differentiable trigger; functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Make the contradiction quantitative or structural [S-CONTRADICTION]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1994-D2-P1

Day 2; official label P7; 14 points; legacy source-page label 6

Let $f \in C^1[a,b]$, $f(a) = 0$ and suppose that $\lambda \in \mathbb{R}$, $\lambda > 0$, is such that

$$|f'(x)| \leq \lambda |f(x)|$$

for all $x \in [a,b]$. Is it true that $f(x) = 0$ for all $x \in [a,b]$?

Open all solution variants | Source note

1994-D2-P2 Quick/standard

Fingerprint global extrema and critical points of two-variable function
Primary domain Real analysis / calculus
Problem type classification/find-all
Linked methods Localize an extremum or an accumulation point [M-AN-COMPACT]
Secondary cues Real analysis / calculus; Linear algebra / matrix theory
Linked theory Continuity, compactness, and order [T-AN-CONTINUITY]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Recognition triggers det/rank/eigen trigger; geometry/metric trigger; functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Choose an extremal object and force rigidity [S-EXTREMAL]; Prove necessity and sufficiency separately [S-CLASSIFICATION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1994-D2-P2

Day 2; official label P8; 14 points; legacy source-page label 6

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be given by $f(x,y) = (x^2 - y^2)e^{-x^2-y^2}$.
a) Prove that f attains its minimum and its maximum.

b) Determine all points (x,y) such that $\frac{\partial f}{\partial x}(x,y) = \frac{\partial f}{\partial y}(x,y) = 0$ and determine for which of them f has global or local minimum or maximum.

Open all solution variants | Source note

1994-D2-P3 Medium

Fingerprint mean-value argument yields point where f^(n+1)=f
Primary domain Real analysis / calculus
Problem type existence/construction
Linked methods Count zeros and subtract an interpolant [M-AN-ROLLE]; Design an auxiliary derivative [M-AN-AUXILIARY]
Secondary cues Real analysis / calculus
Linked theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Differential inequalities and integrating factors [T-AN-ODE]
Recognition triggers continuous/differentiable trigger; functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1994-D2-P3

Day 2; official label P9; 14 points; legacy source-page label 7

Let f be a real-valued function with $n+1$ derivatives at each point of $\mathbb{R}$. Show that for each pair of real numbers a, b, $a < b$, such that

$$\ln\left(\frac{f(b)+f'(b)+\cdots+f^{(n)}(b)}{f(a)+f'(a)+\cdots+f^{(n)}(a)}\right)=b-a$$

there is a number c in the open interval (a,b) for which

$$f^{(n+1)}(c)=f(c).$$

Note that ln denotes the natural logarithm.

Open all solution variants | Source note

1994-D2-P4 Medium

Fingerprint dimension of commutant of diagonal matrix
Primary domain Linear algebra / matrices
Problem type exact computation
Linked methods Read the entries before choosing a theorem [M-LA-ENTRY]; Choose a basis adapted to the structure [M-LA-BASIS]
Secondary cues Real analysis / calculus; Complex analysis / polynomials; Algebra / number theory
Linked theory Rank, dimension and exact linear constraints [T-LA-RANK]; Adapted bases, centralizers and common eigenspaces [T-LA-INVARIANT]
Recognition triggers matrix word; continuous/differentiable trigger; polynomial/root trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Choose the objects and count them without ambiguity [S-COUNTING]
Navigation Full statement All solution variants Source note Learning route

1994-D2-P4 continued

Full statement: 1994-D2-P4

Day 2; official label P10; 18 points; legacy source-page label 7

Let A be a $n \times n$ diagonal matrix with characteristic polynomial

$$(x-c_1)^{d_1}(x-c_2)^{d_2}\ldots(x-c_k)^{d_k},$$

where $c_1, c_2, \ldots, c_k$ are distinct (which means that c_1 appears d_1 times on the diagonal, c_2 appears d_2 times on the diagonal, etc. and $d_1+d_2+\cdots+d_k=n$). Let V be the space of all $n \times n$ matrices B such that $AB=BA$. Prove that the dimension of V is

$$d_1^2+d_2^2+\cdots+d_k^2.$$

Open all solution variants | Source note

1994-D2-P5 Long/deep

Fingerprint permute zero-sum vectors keeping partial sums bounded
Primary domain Geometry / convexity / metric
Problem type existence/construction
Linked methods Strengthen the statement before the induction step [M-DS-INDUCT]
Secondary cues Linear algebra / matrix theory; Real analysis / calculus; Geometry / convexity / metric
Linked theory Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Inner products, orthogonality and polarization [T-LA-INNER]
Recognition triggers integer/prime/mod trigger; counting/permutation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Build a witness and verify every constraint [S-CONSTRUCTION]; Reduce size while preserving the defining conditions [S-INDUCTION]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1994-D2-P5

Day 2; official label P11; 18 points; legacy source-page label 8

Let $x_1, x_2, \ldots, x_k$ be vectors of m-dimensional Euclidian

space, such that $x_1+x_2+\cdots+x_k=0$. Show that there exists a permutation π of the integers $\{1,2,\ldots,k\}$ such that

$$\left\|\sum_{i=1}^{n} x_{\pi(i)}\right\| \leq \left(\sum_{i=1}^{k}\|x_i\|^2\right)^{1/2} \quad \text{for each } n=1,2,\ldots,k.$$

Note that $\|\cdot\|$ denotes the Euclidian norm.

Open all solution variants | Source note

1994-D2-P6 Long/deep

Fingerprint asymptotic limit of sum of 1/(ln k ln(N-k))
Primary domain Real analysis / calculus
Problem type limit/asymptotic evaluation
Linked methods Separate a main range from exceptional scales [M-AN-SPLIT]
Secondary cues Combinatorics / discrete structures
Linked theory Scale separation and limit comparisons [T-AN-ASYMPTOTICS]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers integral/limit trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.

1994-D2-P6 continued

Proof strategies Build a chain of bounds and track equality [S-BOUND]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1994-D2-P6

Day 2; official label P12; 22 points; legacy source-page label 9

Find

$$\lim_{N\to\infty} \frac{\ln^2 N}{N} \sum_{k=2}^{N-2} \frac{1}{\ln k \cdot \ln(N-k)}.$$

Note that ln denotes the natural logarithm.

Open all solution variants | Source note

2.2 1995

Day 1

1995-D1-P1 Quick/standard

Fingerprint shifted-column matrices have rank n-1 and zero eigenvalues
Primary domain Linear algebra / matrices
Problem type structural proof
Linked methods Choose a basis adapted to the structure [M-LA-BASIS]
Secondary cues None recorded
Linked theory Rank, dimension and exact linear constraints [T-LA-RANK]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Nilpotents, projections and involutions [T-LA-NILPOTENT]
Recognition triggers matrix word; det/rank/eigen trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1995-D1-P1

Day 1; official label P1; 10 points; legacy source-page label 11

Let X be a nonsingular matrix with columns $X_1, X_2, \ldots, X_n$. Let Y be a matrix with columns $X_2, X_3, \ldots, X_n, 0$. Show that the matrices $A = YX^{-1}$ and $B = X^{-1}Y$ have rank $n-1$ and have only 0's for eigenvalues.

Open all solution variants | Source note

1995-D1-P2 Medium/trap

Fingerprint lower bound integral of f^2 from tail integral condition
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]; Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]
Secondary cues Functional equations / dynamics
Linked theory Nonnegative sums, Fubini, and integral kernels [T-AN-FUBINI]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]
Recognition triggers continuous/differentiable trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.

1995-D1-P2 continued

Proof strategies Expose the structure before calculating [S-STRUCTURE]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1995-D1-P2

Day 1; official label P2; 15 points; legacy source-page label 11

Let f be a continuous function on $[0,1]$ such that for every $x \in [0,1]$ we have $\int_x^1 f(t)\,dt \geq \frac{1-x^2}{2}$. Show that $\int_0^1 f^2(t)\,dt \geq \frac{1}{3}$.

Open all solution variants | Source note

1995-D1-P3 Medium

Fingerprint limits of f′,f″ at 0 force f/f′ to 0
Primary domain Real analysis / calculus
Problem type limit/asymptotic evaluation
Linked methods Expand near equality, bound away from it [M-AN-LOCAL]; Compare growth through derivatives [M-AN-RATIO]
Secondary cues Functional equations / dynamics; Combinatorics / discrete structures
Linked theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers continuous/differentiable trigger; integral/limit trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Control the limiting operation before taking the limit [S-ANALYTIC]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1995-D1-P3

Day 1; official label P3; 15 points; legacy source-page label 11

Let f be twice continuously differentiable on $(0,+\infty)$ such that $\lim_{x\to 0+} f'(x) = -\infty$ and $\lim_{x\to 0+} f''(x) = +\infty$. Show that

$$\lim_{x\to 0+} \frac{f(x)}{f'(x)} = 0.$$

Open all solution variants | Source note

1995-D1-P4 Medium

Fingerprint injectivity and range of integral of dt/ln t
Primary domain Real analysis / calculus
Problem type classification/find-all
Linked methods Design an auxiliary derivative [M-AN-AUXILIARY]; Pair an integral through symmetry [M-AN-PAIR]
Secondary cues Functional equations / dynamics; Linear algebra / matrix theory
Linked theory Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Continuity, compactness, and order [T-AN-CONTINUITY]
Recognition triggers functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Control the limiting operation before taking the limit [S-ANALYTIC]

1995-D1-P4 continued

Navigation Full statement All solution variants Source note Learning route

Full statement: 1995-D1-P4

Day 1; official label P4; 15 points; legacy source-page label 12

Let $F : (1, \infty) \to \mathbb{R}$ be the function defined by

$$F(x) := \int_x^{x^2} \frac{dt}{\ln t}.$$

Show that F is one-to-one (i.e. injective) and find the range (i.e. set of values) of F.

Open all solution variants | Source note

1995-D1-P5 Long/deep

Fingerprint A+tB nilpotent for n+1 values forces A,B nilpotent
Primary domain Linear algebra / matrices
Problem type structural proof
Linked methods Compare coefficients in a matrix polynomial [M-LA-PARAMETER]
Secondary cues Complex analysis / polynomials
Linked theory Nilpotents, projections and involutions [T-LA-NILPOTENT]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers matrix word; det/rank/eigen trigger; integral/limit trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1995-D1-P5

Day 1; official label P5; 20 points; legacy source-page label 13

Let A and B be real $n \times n$ matrices. Assume that there exist $n+1$ different real numbers $t_1, t_2, \ldots, t_{n+1}$ such that the matrices

$$C_i = A + t_i B, \quad i = 1, 2, \ldots, n+1,$$

are nilpotent (i.e. $C_i^n = 0$).

Show that both A and B are nilpotent.

Open all solution variants | Source note

1995-D1-P6 Long/deep

Fingerprint uniform constant in inequality on curve |x|^p+|y|^p=2
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Expand near equality, bound away from it [M-AN-LOCAL]; Choose the correct convexity comparison [M-AN-CONVEX]
Secondary cues Functional equations / dynamics; Combinatorics / discrete structures
Linked theory Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Continuity, compactness, and order [T-AN-CONTINUITY]; Taylor remainders and local scale [T-AN-TAYLOR]
Recognition triggers wording-specific
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Pass between local constraints and a global conclusion [S-LOCALGLOBAL]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

1995-D1-P6 continued

Full statement: 1995-D1-P6

Day 1; official label P6; 25 points; legacy source-page label 13

Let $p > 1$. Show that there exists a constant $K_p > 0$ such that for every $x, y \in \mathbb{R}$ satisfying $|x|^p + |y|^p = 2$, we have

$$(x - y)^2 \leq K_p \left(4 - (x + y)^2\right).$$

Open all solution variants | Source note

Day 2

1995-D2-P1 Quick/standard

Fingerprint antisymmetry of A and representation as cross product
Primary domain Linear algebra / matrices
Problem type structural proof
Linked methods Read the entries before choosing a theorem [M-LA-ENTRY]
Secondary cues Geometry / convexity / metric; Real analysis / calculus
Linked theory Inner products, orthogonality and polarization [T-LA-INNER]
Recognition triggers matrix word
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Translate geometry into a controlled representation [S-GEOMETRIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1995-D2-P1

Day 2; official label P7; 10 points; legacy source-page label 15

Let A be 3×3 real matrix such that the vectors Au and u are orthogonal for each column vector $u \in \mathbb{R}^3$. Prove that:
a) $A^\top = -A$, where $A^\top$ denotes the transpose of the matrix A;

b) there exists a vector $v \in \mathbb{R}^3$ such that $Au = v \times u$ for every $u \in \mathbb{R}^3$, where $v \times u$ denotes the vector product in $\mathbb{R}^3$.

Open all solution variants | Source note

1995-D2-P2 Quick/standard

Fingerprint evaluate sum of b_n 2^n for nested-radical recurrence
Primary domain Real analysis / calculus
Problem type exact computation
Linked methods Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]; Find a finite telescoping identity [M-AN-TELESCOPE]
Secondary cues Geometry / convexity / metric
Linked theory Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Series, finite cancellation, and summation by parts [T-AN-SERIES]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Recognition triggers wording-specific
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

1995-D2-P2 continued

Full statement: 1995-D2-P2

Day 2; official label P8; 15 points; legacy source-page label 15

Let $\{b_n\}_{n=0}^{\infty}$ be a sequence of positive real numbers such that $b_0 = 1$, $b_n = 2 + \sqrt{b_{n-1}} - 2\sqrt{1 + \sqrt{b_{n-1}}}$. Calculate

$$\sum_{n=1}^{\infty} b_n 2^n.$$

Open all solution variants | Source note

1995-D2-P3 Medium

Fingerprint roots of 2zP'-nP stay on unit circle
Primary domain Complex analysis / polynomials
Problem type structural proof
Linked methods Use a logarithmic derivative to expose root geometry [M-AL-LOGDERIVATIVE]
Secondary cues Geometry / convexity / metric
Linked theory Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]; Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]
Recognition triggers polynomial/root trigger; geometry/metric trigger
Preparation route CP-Polynomials/Complex
Trap watch Separate polynomial identity from pointwise testing; check multiplicities, zero polynomial cases, and real-vs-complex roots.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Translate geometry into a controlled representation [S-GEOMETRIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1995-D2-P3

Day 2; official label P9; 15 points; legacy source-page label 16

Let all roots of an n-th degree polynomial $P(z)$ with complex coefficients lie on the unit circle in the complex plane. Prove that all roots of the polynomial

$$2zP'(z) - nP(z)$$

lie on the same circle.

Open all solution variants | Source note

1995-D2-P4 Medium/trap

Fingerprint approximate odd continuous functions by odd polynomials
Primary domain Real analysis / calculus
Problem type existence/construction
Linked methods Preserve structure while approximating [M-AN-APPROXIMATE]; Separate a main range from exceptional scales [M-AN-SPLIT]
Secondary cues Functional equations / dynamics; Real analysis / calculus; Geometry / convexity / metric
Linked theory Uniform approximation and positive kernels [T-AN-APPROXIMATION]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers continuous/differentiable trigger; integer/prime/mod trigger; functional-equation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Build a witness and verify every constraint [S-CONSTRUCTION]; Pass between local constraints and a global conclusion [S-LOCALGLOBAL]
Navigation Full statement All solution variants Source note Learning route

1995-D2-P4 continued

Full statement: 1995-D2-P4

Day 2; official label P10; 15 points; legacy source-page label 17

a) Prove that for every $\varepsilon > 0$ there is a positive integer n and real numbers $\lambda_1, \dots, \lambda_n$ such that

$$\max_{x \in [-1,1]} \left| x - \sum_{k=1}^{n} \lambda_k x^{2k+1} \right| < \varepsilon.$$

b) Prove that for every odd continuous function f on $[-1,1]$ and for every $\varepsilon > 0$ there is a positive integer n and real numbers $\mu_1, \dots, \mu_n$ such that

$$\max_{x \in [-1,1]} \left| f(x) - \sum_{k=1}^{n} \mu_k x^{2k+1} \right| < \varepsilon.$$

Recall that f is odd means that $f(x) = -f(-x)$ for all $x \in [-1,1]$.

Open all solution variants | Source note

1995-D2-P5 Long/deep

Fingerprint sign changes and zeros of cosine sums
Primary domain Real analysis / calculus
Problem type structural proof
Linked methods Exploit identities before estimating oscillations [M-AN-TRIG]; Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]
Secondary cues Functional equations / dynamics; Linear algebra / matrix theory; Algebra / number theory
Linked theory Trigonometric identities and safe branches [T-AN-TRIG]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Continuity, compactness, and order [T-AN-CONTINUITY]; Series, finite cancellation, and summation by parts [T-AN-SERIES]
Recognition triggers functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Make the contradiction quantitative or structural [S-CONTRADICTION]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1995-D2-P5

Day 2; official label P11; 10+15 points; legacy source-page label 18

a) Prove that every function of the form

$$f(x) = \frac{a_0}{2} + \cos x + \sum_{n=2}^{N} a_n \cos(nx)$$

with $|a_0| < 1$, has positive as well as negative values in the period $[0, 2\pi)$.

b) Prove that the function

$$F(x) = \sum_{n=1}^{100} \cos(n^{\frac{3}{2}} x)$$

has at least 40 zeros in the interval $(0, 1000)$.

Open all solution variants | Source note

1995-D2-P6 Long/deep

Fingerprint bounded orthonormal system has no pointwise convergent subsequence
Primary domain Real analysis / calculus
Problem type nonexistence/counterexample
Linked methods Pass a limit through an integral safely [M-AN-DOMINATE]
Secondary cues Probability / measure / expectation; Functional equations / dynamics
Linked theory Dominated convergence and orthonormal separation [T-AN-MEASURE]; Inner products, orthogonality and polarization [T-LA-INNER]
Recognition triggers continuous/differentiable trigger; integral/limit trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Make the contradiction quantitative or structural [S-CONTRADICTION]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1995-D2-P6

Day 2; official label P12; 20 points; legacy source-page label 20

Suppose that $\{f_n\}_{n=1}^{\infty}$ is a sequence of continuous functions on the interval $[0,1]$ such that

$$\int_0^1 f_m(x) f_n(x)\,dx = \begin{cases} 1 & \text{if } n = m \\ 0 & \text{if } n \neq m \end{cases}$$

and

$$\sup\{|f_n(x)| : x \in [0,1] \text{ and } n = 1, 2, \dots\} < +\infty.$$

Show that there exists no subsequence $\{f_{n_k}\}$ of $\{f_n\}$ such that $\lim_{k\to\infty} f_{n_k}(x)$ exists for all $x \in [0,1]$.

Open all solution variants | Source note

2.3 1996

Day 1

1996-D1-P1 Quick/standard

Fingerprint determinant of symmetric Toeplitz matrix of arithmetic progression
Primary domain Linear algebra / matrices
Problem type exact computation
Linked methods Eliminate systematically and track the invariant [M-LA-ELIM]
Secondary cues None recorded
Linked theory Determinants, block identities and inverses [T-LA-DETERMINANT]
Recognition triggers det/rank/eigen trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

1996-D1-P1 continued

Full statement: 1996-D1-P1

Day 1; official label P1; 10 points; legacy source-page label 22

Let for $j = 0, \dots, n$, $a_j = a_0 + jd$, where a_0, d are fixed real numbers. Put

$$A = \begin{pmatrix} a_0 & a_1 & a_2 & \dots & a_n \\ a_1 & a_0 & a_1 & \dots & a_{n-1} \\ a_2 & a_1 & a_0 & \dots & a_{n-2} \\ \dots & \dots & \dots & \dots & \dots \\ a_n & a_{n-1} & a_{n-2} & \dots & a_0 \end{pmatrix}.$$

Calculate $\det(A)$, where $\det(A)$ denotes the determinant of A.

Open all solution variants | Source note

1996-D1-P2 Quick/standard

Fingerprint evaluate definite integral of sin nx/((1+2^x)sin x)
Primary domain Real analysis / calculus
Problem type exact computation
Linked methods Pair an integral through symmetry [M-AN-PAIR]; Exploit identities before estimating oscillations [M-AN-TRIG]; Find a finite telescoping identity [M-AN-TELESCOPE]
Secondary cues Geometry / convexity / metric
Linked theory Trigonometric identities and safe branches [T-AN-TRIG]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Recognition triggers integral/limit trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Reduce size while preserving the defining conditions [S-INDUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1996-D1-P2

Day 1; official label P2; 10 points; legacy source-page label 23

Evaluate the definite integral

$$\int_{-\pi}^{\pi} \frac{\sin nx}{(1+2^x)\sin x}\, dx,$$

where n is a natural number.

Open all solution variants | Source note

1996-D1-P3 Medium/trap

Fingerprint involutions diagonalize; max number of commuting involutions
Primary domain Linear algebra / matrices
Problem type inequality/sharp bound
Linked methods Choose a basis adapted to the structure [M-LA-BASIS]; Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]
Secondary cues Combinatorics / discrete structures; Geometry / convexity / metric; Functional equations / dynamics
Linked theory Nilpotents, projections and involutions [T-LA-NILPOTENT]; Adapted bases, centralizers and common eigenspaces [T-LA-INVARIANT]
Recognition triggers det/rank/eigen trigger; counting/permutation trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.

1996-D1-P3 continued

Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Reduce size while preserving the defining conditions [S-INDUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1996-D1-P3

Day 1; official label P3; 15 points; legacy source-page label 24

The linear operator A on the vector space V is called an involution if $A^2 = E$ where E is the identity operator on V. Let $\dim V = n < \infty$.
(i) Prove that for every involution A on V there exists a basis of V consisting of eigenvectors of A.

(ii) Find the maximal number of distinct pairwise commuting involutions on V.

Open all solution variants | Source note

1996-D1-P4 Medium

Fingerprint growth rate limsup |a_n|^{1/n} for convolution recurrence
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]
Secondary cues Geometry / convexity / metric
Linked theory Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Recognition triggers integral/limit trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Reduce size while preserving the defining conditions [S-INDUCTION]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1996-D1-P4

Day 1; official label P4; 15 points; legacy source-page label 24

Let $a_1 = 1$, $a_n = \dfrac{1}{n}\sum_{k=1}^{n-1} a_k a_{n-k}$ for $n \geq 2$. Show that
(i) $\limsup_{n\to\infty} |a_n|^{1/n} < 2^{-1/2}$;

(ii) $\limsup_{n\to\infty} |a_n|^{1/n} \geq 2/3$.

Open all solution variants | Source note

1996-D1-P5 Long/deep

Fingerprint asymptotics of n times integral of f^n
Primary domain Real analysis / calculus
Problem type limit/asymptotic evaluation
Linked methods Separate a main range from exceptional scales [M-AN-SPLIT]; Expand near equality, bound away from it [M-AN-LOCAL]; Choose the correct convexity comparison [M-AN-CONVEX]
Secondary cues Functional equations / dynamics
Linked theory Scale separation and limit comparisons [T-AN-ASYMPTOTICS]; Taylor remainders and local scale [T-AN-TAYLOR]; Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Recognition triggers continuous/differentiable trigger; integral/limit trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Control the limiting operation before taking the limit [S-ANALYTIC]; Make the contradiction

1996-D1-P5 continued

quantitative or structural [S-CONTRADICTION]

Navigation Full statement All solution variants Source note Learning route

Full statement: 1996-D1-P5

Day 1; official label P5; 25 points; legacy source-page label 25

(i) Let a, b be real numbers such that $b \leq 0$ and $1 + ax + bx^2 \geq 0$ for every x in $[0, 1]$. Prove that

$$\lim_{n\to+\infty} n \int_0^1 (1 + ax + bx^2)^n dx = \begin{cases} -\dfrac{1}{a} & \text{if } a < 0, \\ +\infty & \text{if } a \geq 0. \end{cases}$$

(ii) Let $f : [0, 1] \to [0, \infty)$ be a function with a continuous second derivative and let $f''(x) \leq 0$ for every x in $[0, 1]$. Suppose that $L = \lim_{n\to\infty} n \int_0^1 (f(x))^n dx$ exists and $0 < L < +\infty$. Prove that f' has a constant sign and $\min_{x\in[0,1]} |f'(x)| = L^{-1}$.

Open all solution variants | Source note

1996-D1-P6 Long/deep

Fingerprint compare upper and lower content of plane sets
Primary domain Geometry / convexity / metric
Problem type inequality/sharp bound
Linked methods Push covers forward or move toward a nearest point [M-DS-METRIC]; Strengthen the statement before the induction step [M-DS-INDUCT]; Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Secondary cues Combinatorics / discrete structures; Functional equations / dynamics; Real analysis / calculus
Linked theory Metric contractions, covers, and nearest points [T-DS-METRIC]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Continuity, compactness, and order [T-AN-CONTINUITY]
Recognition triggers graph/color/set trigger; geometry/metric trigger; functional-equation trigger
Preparation route GEO-Geometry/Metricif
Trap watch Control degenerate configurations and make clear which coordinates or affine transformations preserve the quantity.
Proof strategies Translate geometry into a controlled representation [S-GEOMETRIC]; Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1996-D1-P6

Day 1; official label P6; 25 points; legacy source-page label 27

Upper content of a subset E of the plane $\mathbb{R}^2$ is defined as

$$\mathcal{C}(E) = \inf\left\{\sum_{i=1}^{n} \operatorname{diam}(E_i)\right\}$$

where inf is taken over all finite families of sets $E_1, \ldots, E_n$, $n \in \mathbb{N}$, in $\mathbb{R}^2$ such that $E \subset \bigcup_{i=1}^{n} E_i$.

Lower content of E is defined as

$$\mathcal{K}(E) = \sup\{\text{lenght}(L) : L \text{ is a closed line segment onto which } E \text{ can be contracted}\}.$$

Show that
(a) $\mathcal{C}(L) = \text{lenght}(L)$ if L is a closed line segment;
(b) $\mathcal{C}(E) \geq \mathcal{K}(E)$;
(c) the equality in (b) needs not hold even if E is compact.

1996-D1-P6 continued

Hint. If $E = T \cup T'$ where T is the triangle with vertices $(-2, 2)$, $(2, 2)$ and $(0, 4)$, and T' is its reflexion about the x-axis, then $\mathcal{C}(E) = 8 > \mathcal{K}(E)$.

Remarks: All distances used in this problem are Euclidian. Diameter of a set E is $\operatorname{diam}(E) = \sup\{\operatorname{dist}(x, y) : x, y \in E\}$. Contraction of a set E to a set F is a mapping $f : E \mapsto F$ such that $\operatorname{dist}(f(x), f(y)) \leq \operatorname{dist}(x, y)$ for all $x, y \in E$. A set E can be contracted onto a set F if there is a contraction f of E to F which is onto, i.e., such that $f(E) = F$. Triangle is defined as the union of the three segments joining its vertices, i.e., it does not contain the interior.

Open all solution variants | Source note

Day 2

1996-D2-P1 Quick/standard

Fingerprint iterates converge iff consecutive differences vanish
Primary domain Real analysis / calculus
Problem type recurrence/convergence
Linked methods Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]; Use a first crossing or a forced-growth contradiction [M-AN-BARRIER]
Secondary cues Real analysis / calculus; Geometry / convexity / metric
Linked theory Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Continuity, compactness, and order [T-AN-CONTINUITY]
Recognition triggers continuous/differentiable trigger; integral/limit trigger; functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Make the contradiction quantitative or structural [S-CONTRADICTION]; Pass between local constraints and a global conclusion [S-LOCALGLOBAL]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1996-D2-P1

Day 2; official label P7; 10 points; legacy source-page label 29

Prove that if $f : [0, 1] \to [0, 1]$ is a continuous function, then the sequence of iterates $x_{n+1} = f(x_n)$ converges if and only if

$$\lim_{n \to \infty} (x_{n+1} - x_n) = 0.$$

Open all solution variants | Source note

1996-D2-P2 Quick/standard

Fingerprint rationality of cosh k-theta propagates to cosh theta
Primary domain Algebra / number theory
Problem type structural proof
Linked methods Find a recurrence that reveals the invariant [M-AL-RECURRENCE]; Eliminate rational relations, then exploit discreteness [M-AL-RATIONAL]
Secondary cues Real analysis / calculus
Linked theory Linear recurrences and generating functions [T-AL-RECURRENCE]; Rational linear relations and discrete-value rigidity [T-AL-RATIONALITY]
Recognition triggers wording-specific
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

1996-D2-P2 continued

Full statement: 1996-D2-P2

Day 2; official label P8; 10 points; legacy source-page label 30

Let θ be a positive real number and let $\cosh t = \dfrac{e^t + e^{-t}}{2}$ denote the hyperbolic cosine. Show that if $k \in \mathbb{N}$ and both $\cosh k\theta$ and $\cosh(k+1)\theta$ are rational, then so is $\cosh \theta$.

Open all solution variants | Source note

1996-D2-P3 Medium

Fingerprint unipotent subgroup of matrix group not finitely generated
Primary domain Algebra / number theory
Problem type structural proof
Linked methods Prove generation or obstruct it by an image [M-AL-GENERATORS]
Secondary cues Linear algebra / matrix theory; Combinatorics / discrete structures; Geometry / convexity / metric
Linked theory Groups, orders, generators, and quotients [T-AL-GROUP]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Recognition triggers matrix word; group/ring/field trigger; counting/permutation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1996-D2-P3

Day 2; official label P9; 15 points; legacy source-page label 31

Let G be the subgroup of $GL_2(\mathbb{R})$, generated by A and B, where

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Let H consist of those matrices $\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ in G for which $a_{11} = a_{22} = 1$.
(a) Show that H is an abelian subgroup of G.
(b) Show that H is not finitely generated.

Remarks. $GL_2(\mathbb{R})$ denotes, as usual, the group (under matrix multiplication) of all 2×2 invertible matrices with real entries (elements). Abelian means commutative. A group is finitely generated if there are a finite number of elements of the group such that every other element of the group can be obtained from these elements using the group operation.

Open all solution variants | Source note

1996-D2-P4 Long/deep

Fingerprint maximal inscribed disc touches every quarter-circle arc
Primary domain Geometry / convexity / metric
Problem type structural proof
Linked methods Contradict an extremum by a controlled perturbation [M-DS-PERTURB]; Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Secondary cues Real analysis / calculus; Combinatorics / discrete structures
Linked theory Convex bodies, supporting functionals, and packing [T-DS-CONVEX]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Continuity, compactness, and order [T-AN-CONTINUITY]
Recognition triggers geometry/metric trigger
Preparation route GEO-Geometry/Metricif
Trap watch Control degenerate configurations and make clear which coordinates or affine transformations

1996-D2-P4 continued

preserve the quantity.
Proof strategies Choose an extremal object and force rigidity [S-EXTREMAL]; Make the contradiction quantitative or structural [S-CONTRADICTION]; Translate geometry into a controlled representation [S-GEOMETRIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1996-D2-P4

Day 2; official label P10; 20 points; legacy source-page label 32

Let B be a bounded closed convex symmetric (with respect to the origin) set in $\mathbb{R}^2$ with boundary the curve Γ. Let B have the property that the ellipse of maximal area contained in B is the disc D of radius 1 centered at the origin with boundary the circle C. Prove that $A \cap \Gamma \neq \emptyset$ for any arc A of C of length $l(A) \geq \frac{\pi}{2}$.

Open all solution variants | Source note

1996-D2-P5 Long/deep

Fingerprint limit and error bound for series nx/(n^2+x)^2
Primary domain Real analysis / calculus
Problem type limit/asymptotic evaluation
Linked methods Expose the mesh and control its tails [M-AN-RIEMANN]
Secondary cues Real analysis / calculus; Probability / measure / expectation
Linked theory Riemann sums, periodic averaging, and tails [T-AN-RIEMANN]; Taylor remainders and local scale [T-AN-TAYLOR]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Recognition triggers integral/limit trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Control the limiting operation before taking the limit [S-ANALYTIC]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1996-D2-P5

Day 2; official label P11; 20 points; legacy source-page label 32

(i) Prove that

$$\lim_{x \to +\infty} \sum_{n=1}^{\infty} \frac{nx}{(n^2+x)^2} = \frac{1}{2}.$$

(ii) Prove that there is a positive constant c such that for every $x \in [1, \infty)$ we have

$$\left| \sum_{n=1}^{\infty} \frac{nx}{(n^2+x)^2} - \frac{1}{2} \right| \leq \frac{c}{x}.$$

Open all solution variants | Source note

1996-D2-P6 Long/deep

Fingerprint Carleman's inequality and sharpness of constant e
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]; Find a finite telescoping identity [M-AN-TELESCOPE]
Secondary cues Geometry / convexity / metric
Linked theory Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Nonnegative sums, Fubini, and integral kernels [T-AN-FUBINI]; Series, finite cancellation, and summation by parts [T-AN-SERIES]
Recognition triggers wording-specific

1996-D2-P6 continued

Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1996-D2-P6

Day 2; official label P12; 25 points; legacy source-page label 34

(Carleman's inequality)

(i) Prove that for every sequence $\{a_n\}_{n=1}^{\infty}$, such that $a_n > 0$, $n = 1, 2, \ldots$ and $\sum_{n=1}^{\infty} a_n < \infty$, we have

$$\sum_{n=1}^{\infty} (a_1 a_2 \cdots a_n)^{1/n} < e \sum_{n=1}^{\infty} a_n,$$

where e is the natural log base.

(ii) Prove that for every $\varepsilon > 0$ there exists a sequence $\{a_n\}_{n=1}^{\infty}$, such that $a_n > 0$, $n = 1, 2, \ldots$, $\sum_{n=1}^{\infty} a_n < \infty$ and

$$\sum_{n=1}^{\infty} (a_1 a_2 \cdots a_n)^{1/n} > (e - \varepsilon) \sum_{n=1}^{\infty} a_n.$$

Open all solution variants | Source note

2.4 1997

Day 1

1997-D1-P1 Quick/standard

Fingerprint limit of Riemann-type sum of ln(k/n+eps)
Primary domain Real analysis / calculus
Problem type limit/asymptotic evaluation
Linked methods Expose the mesh and control its tails [M-AN-RIEMANN]; Separate a main range from exceptional scales [M-AN-SPLIT]
Secondary cues None recorded
Linked theory Riemann sums, periodic averaging, and tails [T-AN-RIEMANN]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Recognition triggers integral/limit trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

1997-D1-P1 continued

Full statement: 1997-D1-P1

Day 1; official label P1; legacy source-page label 36

Let $\{\varepsilon_n\}_{n=1}^{\infty}$ be a sequence of positive real numbers, such that $\lim_{n\to\infty} \varepsilon_n = 0$. Find

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \ln\left(\frac{k}{n} + \varepsilon_n\right),$$

where ln denotes the natural logarithm.

Open all solution variants | Source note

1997-D1-P2 Quick/standard

Fingerprint do specific rearrangements preserve series convergence
Primary domain Real analysis / calculus
Problem type recurrence/convergence
Linked methods Test convergence through blocks and differences [M-AN-SERIES]
Secondary cues Geometry / convexity / metric
Linked theory Series, finite cancellation, and summation by parts [T-AN-SERIES]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers integral/limit trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1997-D1-P2

Day 1; official label P2; legacy source-page label 37

Suppose $\sum_{n=1}^{\infty} a_n$ converges. Do the following sums have to converge as well?

a) $a_1 + a_2 + a_4 + a_3 + a_8 + a_7 + a_6 + a_5 + a_{16} + a_{15} + \cdots + a_9 + a_{32} + \cdots$

b) $a_1 + a_2 + a_3 + a_4 + a_5 + a_7 + a_6 + a_8 + a_9 + a_{11} + a_{13} + a_{15} + a_{10} + a_{12} + a_{14} + a_{16} + a_{17} + a_{19} + \cdots$

Justify your answers.

Open all solution variants | Source note

1997-D1-P3 Medium

Fingerprint A^2+B^2=AB with invertible commutator forces 3|n
Primary domain Linear algebra / matrices
Problem type structural proof
Linked methods Factor while preserving matrix order [M-LA-FACTOR]
Secondary cues Linear algebra / matrix theory; Real analysis / calculus
Linked theory Determinants, block identities and inverses [T-LA-DETERMINANT]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers matrix word; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

1997-D1-P3 continued

Full statement: 1997-D1-P3

Day 1; official label P3; legacy source-page label 37

Let A and B be real $n \times n$ matrices such that $A^2 + B^2 = AB$. Prove that if $BA - AB$ is an invertible matrix then n is divisible by 3.

Open all solution variants | Source note

1997-D1-P4 Medium/trap

Fingerprint unique infinite product expansion and rationality criterion
Primary domain Algebra / number theory
Problem type classification/find-all
Linked methods Choose the next factor and control the entire tail [M-AL-GREEDY]
Secondary cues Real analysis / calculus
Linked theory Greedy products, tail bounds, and integer descent [T-AL-GREEDY]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Build a witness and verify every constraint [S-CONSTRUCTION]; Choose an extremal object and force rigidity [S-EXTREMAL]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1997-D1-P4

Day 1; official label P4; legacy source-page label 38

Let α be a real number, $1 < \alpha < 2$.
a) Show that α has a unique representation as an infinite product

$$\alpha = \left(1 + \frac{1}{n_1}\right)\left(1 + \frac{1}{n_2}\right)\ldots$$

where each n_i is a positive integer satisfying

$$n_i^2 \leq n_{i+1}.$$

b) Show that α is rational if and only if its infinite product has the following property:

For some m and all $k \geq m$,

$$n_{k+1} = n_k^2.$$

Open all solution variants | Source note

1997-D1-P5 Long/deep

Fingerprint p-norm minimality of x against lattice translates
Primary domain Geometry / convexity / metric
Problem type inequality/sharp bound
Linked methods Split a family by traces, pieces, or labels [M-DS-PARTITION]; Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Secondary cues Real analysis / calculus; Geometry / convexity / metric; Linear algebra / matrix theory
Linked theory Metric contractions, covers, and nearest points [T-DS-METRIC]; Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.

1997-D1-P5 continued

Proof strategies Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]; Prove necessity and sufficiency separately [S-CLASSIFICATION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1997-D1-P5

Day 1; official label P5; legacy source-page label 40

For a natural n consider the hyperplane

$$\mathbb{R}_0^n = \left\{ x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n : \sum_{i=1}^{n} x_i = 0 \right\}$$

and the lattice $\mathbb{Z}_0^n = \{y \in \mathbb{R}_0^n : \text{all } y_i \text{ are integers}\}$. Define the (quasi–)norm in $\mathbb{R}^n$ by $\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}$ if $0 < p < \infty$, and $\|x\|_\infty = \max_i |x_i|$.
a) Let $x \in \mathbb{R}_0^n$ be such that

$$\max_i x_i - \min_i x_i \leq 1.$$

For every $p \in [1, \infty]$ and for every $y \in \mathbb{Z}_0^n$ prove that

$$\|x\|_p \leq \|x + y\|_p.$$

b) For every $p \in (0, 1)$, show that there is an n and an $x \in \mathbb{R}_0^n$ with $\max_i x_i - \min_i x_i \leq 1$ and an $y \in \mathbb{Z}_0^n$ such that

$$\|x\|_p > \|x + y\|_p.$$

Open all solution variants | Source note

1997-D1-P6 Long/deep

Fingerprint intersecting family of finite sets and finite core
Primary domain Combinatorics
Problem type nonexistence/counterexample
Linked methods Split a family by traces, pieces, or labels [M-DS-PARTITION]; Strengthen the statement before the induction step [M-DS-INDUCT]
Secondary cues None recorded
Linked theory Finite set systems and antichains [T-DS-SETS]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers graph/color/set trigger
Preparation route CMB-Combinatoricsi
Trap watch Avoid overcounting; define the encoded object precisely and check extremal/minimal cases.
Proof strategies Reduce size while preserving the defining conditions [S-INDUCTION]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1997-D1-P6

Day 1; official label P6; legacy source-page label 42

Suppose that F is a family of finite subsets of $\mathbb{N}$ and for any two sets $A, B \in F$ we have $A \cap B \neq \emptyset$.
a) Is it true that there is a finite subset Y of $\mathbb{N}$ such that for any $A, B \in F$ we have $A \cap B \cap Y \neq \emptyset$?
b) Is the statement a) true if we suppose in addition that all of the members of F have the same size?

Justify your answers.

Open all solution variants | Source note

Day 2

1997-D2-P1 Quick/standard

Fingerprint boundedness of derivative of sqrt(f)/f' near 0
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Expand near equality, bound away from it [M-AN-LOCAL]
Secondary cues Functional equations / dynamics
Linked theory Taylor remainders and local scale [T-AN-TAYLOR]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Recognition triggers continuous/differentiable trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Pass between local constraints and a global conclusion [S-LOCALGLOBAL]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1997-D2-P1

Day 2; official label P7; legacy source-page label 44

Let f be a $C^3(\mathbb{R})$ non-negative function, $f(0) = f'(0) = 0$, $0 < f''(0)$. Let

$$g(x) = \left(\frac{\sqrt{f(x)}}{f'(x)} \right)'$$

for $x \neq 0$ and $g(0) = 0$. Show that g is bounded in some neighbourhood of 0. Does the theorem hold for $f \in C^2(\mathbb{R})$?

Open all solution variants | Source note

1997-D2-P2 Quick/standard

Fingerprint block matrix identity det M det H = det A
Primary domain Linear algebra / matrices
Problem type structural proof
Linked methods Exploit a compatible block decomposition [M-LA-BLOCK]
Secondary cues None recorded
Linked theory Determinants, block identities and inverses [T-LA-DETERMINANT]
Recognition triggers matrix word; det/rank/eigen trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1997-D2-P2

Day 2; official label P8; legacy source-page label 45

Let M be an invertible matrix of dimension $2n \times 2n$, represented in block form as

$$M = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \quad \text{and} \quad M^{-1} = \begin{bmatrix} E & F \\ G & H \end{bmatrix}.$$

Show that $\det M \cdot \det H = \det A$.

Open all solution variants | Source note

1997-D2-P3 Medium/trap

Fingerprint convergence of alternating series with sin(log n)/n^alpha
Primary domain Real analysis / calculus
Problem type recurrence/convergence
Linked methods Test convergence through blocks and differences [M-AN-SERIES]
Secondary cues Functional equations / dynamics; Geometry / convexity / metric
Linked theory Series, finite cancellation, and summation by parts [T-AN-SERIES]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Trigonometric identities and safe branches [T-AN-TRIG]
Recognition triggers wording-specific
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1997-D2-P3

Day 2; official label P9; legacy source-page label 45

Show that $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} \sin(\log n)}{n^{\alpha}}$ converges if and only if $\alpha > 0$.

Open all solution variants | Source note

1997-D2-P4 Medium

Fingerprint linear trace-like functionals on matrix space are tr(AC)
Primary domain Linear algebra / matrices
Problem type classification/find-all
Linked methods Use trace to extract a scalar invariant [M-LA-TRACE]; Prove a dimension obstruction and attain it [M-LA-DIMENSION]
Secondary cues Functional equations / dynamics
Linked theory Dual bases, kernels and separating functionals [T-LA-DUALITY]; Trace pairing and commutator calculus [T-LA-TRACE]; Rank, dimension and exact linear constraints [T-LA-RANK]
Recognition triggers matrix word; functional-equation trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1997-D2-P4

Day 2; official label P10; legacy source-page label 46

a) Let the mapping $f : M_n \to \mathbb{R}$ from the space $M_n = \mathbb{R}^{n^2}$ of $n \times n$ matrices with real entries to reals be linear, i.e.:

$$f(A+B) = f(A) + f(B), \quad f(cA) = cf(A) \tag{1}$$

for any $A, B \in M_n$, $c \in \mathbb{R}$. Prove that there exists a unique matrix $C \in M_n$ such that $f(A) = \mathrm{tr}(AC)$ for any $A \in M_n$. (If $A = \{a_{ij}\}_{i,j=1}^{n}$ then $\mathrm{tr}(A) = \sum_{i=1}^{n} a_{ii}$).

b) Suppose in addition to (1) that

$$f(A \cdot B) = f(B \cdot A) \tag{2}$$

for any $A, B \in M_n$. Prove that there exists $\lambda \in \mathbb{R}$ such that $f(A) = \lambda \cdot \mathrm{tr}(A)$.

Open all solution variants | Source note

1997-D2-P5 Long/deep

Fingerprint every bijection is product of two involutions
Primary domain Algebra / number theory
Problem type existence/construction
Linked methods Work orbit by orbit and track parity [M-AL-ORBIT]
Secondary cues Real analysis / calculus; Combinatorics / discrete structures; Functional equations / dynamics
Linked theory Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]; Groups, orders, generators, and quotients [T-AL-GROUP]
Recognition triggers wording-specific
Preparation route GEO-Geometry/Metricif
Trap watch Control degenerate configurations and make clear which coordinates or affine transformations preserve the quantity.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1997-D2-P5

Day 2; official label P11; legacy source-page label 47

Let X be an arbitrary set, let f be an one-to-one function mapping X onto itself. Prove that there exist mappings $g_1, g_2 : X \to X$ such that $f = g_1 \circ g_2$ and $g_1 \circ g_1 = id = g_2 \circ g_2$, where id denotes the identity mapping on X.

Open all solution variants | Source note

1997-D2-P6 Long/deep

Fingerprint can continuous function cross axis uncountably often
Primary domain Real analysis / calculus
Problem type nonexistence/counterexample
Linked methods Use a dense set, an enumeration, or a category argument [M-AN-TOPOLOGY]
Secondary cues Functional equations / dynamics; Combinatorics / discrete structures; Algebra / number theory
Linked theory Countability, accumulation, and Cantor constructions [T-AN-SETS]; Continuity, compactness, and order [T-AN-CONTINUITY]; Uniform approximation and positive kernels [T-AN-APPROXIMATION]
Recognition triggers continuous/differentiable trigger; counting/permutation trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a witness and verify every constraint [S-CONSTRUCTION]; Pass between local constraints and a global conclusion [S-LOCALGLOBAL]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1997-D2-P6

Day 2; official label P12; legacy source-page label 47

Let $f : [0, 1] \to \mathbb{R}$ be a continuous function. Say that f "crosses the axis" at x if $f(x) = 0$ but in any neighbourhood of x there are y, z with $f(y) < 0$ and $f(z) > 0$.
a) Give an example of a continuous function that "crosses the axis" infiniteley often.
b) Can a continuous function "cross the axis" uncountably often?

Justify your answer.

Open all solution variants | Source note

2.5 1998

Day 1

1998-D1-P1 Long/deep

Fingerprint dimension of maps preserving nested invariant subspaces
Primary domain Linear algebra / matrices
Problem type exact computation
Linked methods Choose a basis adapted to the structure [M-LA-BASIS]
Secondary cues Combinatorics / discrete structures
Linked theory Adapted bases, centralizers and common eigenspaces [T-LA-INVARIANT]; Rank, dimension and exact linear constraints [T-LA-RANK]
Recognition triggers wording-specific
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Choose the objects and count them without ambiguity [S-COUNTING]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1998-D1-P1

Day 1; official label P1; 20 points; legacy source-page label 49

Let V be a 10-dimensional real vector space and U_1 and U_2 two linear subspaces such that $U_1 \subseteq U_2$, $\dim_{\mathbb{R}} U_1 = 3$ and $\dim_{\mathbb{R}} U_2 = 6$. Let $\mathcal{E}$ be the set of all linear maps $T : V \longrightarrow V$ which have U_1 and U_2 as invariant subspaces (i.e., $T(U_1) \subseteq U_1$ and $T(U_2) \subseteq U_2$). Calculate the dimension of $\mathcal{E}$ as a real vector space.

Open all solution variants | Source note

1998-D1-P2 Quick/standard

Fingerprint two permutations generating all of S_n
Primary domain Algebra / number theory
Problem type structural proof
Linked methods Prove generation or obstruct it by an image [M-AL-GENERATORS]
Secondary cues Algebra / number theory; Geometry / convexity / metric
Linked theory Groups, orders, generators, and quotients [T-AL-GROUP]; Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]
Recognition triggers counting/permutation trigger
Preparation route CMB-Combinatoricsi
Trap watch Avoid overcounting; define the encoded object precisely and check extremal/minimal cases.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1998-D1-P2

Day 1; official label P2; legacy source-page label 49

Prove that the following proposition holds for $n = 3$ (5 points) and $n = 5$ (7 points), and does not hold for $n = 4$ (8 points).

"For any permutation π_1 of $\{1, 2, \ldots, n\}$ different from the identity there is a permutation π_2 such that any permutation π can be obtained from π_1 and π_2 using only compositions (for example, $\pi = \pi_1 \circ \pi_1 \circ \pi_2 \circ \pi_1$)."

Open all solution variants | Source note

1998-D1-P3 Medium

Fingerprint limit and value of integral of logistic iterates
Primary domain Functional equations / dynamics
Problem type exact computation
Linked methods Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]; Strengthen the statement before the induction step [M-DS-INDUCT]
Secondary cues Geometry / convexity / metric; Functional equations / dynamics
Linked theory Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Dominated convergence and orthonormal separation [T-AN-MEASURE]
Recognition triggers integral/limit trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Control the limiting operation before taking the limit [S-ANALYTIC]; Reduce size while preserving the defining conditions [S-INDUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1998-D1-P3

Day 1; official label P3; legacy source-page label 50

Let $f(x) = 2x(1-x)$, $x \in \mathbb{R}$. Define

$$f_n = \underbrace{f \circ \ldots \circ f}_{n}.$$

a) (10 points) Find $\lim_{n\to\infty} \int_0^1 f_n(x)\,dx$.

b) (10 points) Compute $\int_0^1 f_n(x)\,dx$ for $n = 1, 2, \ldots$.

Open all solution variants | Source note

1998-D1-P4 Long/deep

Fingerprint exists xi with f f′ + f″ = 0
Primary domain Real analysis / calculus
Problem type existence/construction
Linked methods Design an auxiliary derivative [M-AN-AUXILIARY]; Count zeros and subtract an interpolant [M-AN-ROLLE]
Secondary cues Functional equations / dynamics
Linked theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Continuity, compactness, and order [T-AN-CONTINUITY]; Differential inequalities and integrating factors [T-AN-ODE]
Recognition triggers continuous/differentiable trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Control the limiting operation before taking the limit [S-ANALYTIC]; Prove necessity and sufficiency separately [S-CLASSIFICATION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1998-D1-P4

Day 1; official label P4; 20 points; legacy source-page label 50

The function $f : \mathbb{R} \to \mathbb{R}$ is twice differentiable and satisfies $f(0) = 2$, $f'(0) = -2$ and $f(1) = 1$. Prove that there exists a real number $\xi \in (0, 1)$ for which

$$f(\xi) \cdot f'(\xi) + f''(\xi) = 0.$$

Open all solution variants | Source note

1998-D1-P5

Long/deep

Fingerprint (n-1)P'^2 >= nPP" for real-rooted polynomials
Primary domain Complex analysis / polynomials
Problem type inequality/sharp bound
Linked methods Use a logarithmic derivative to expose root geometry [M-AL-LOGDERIVATIVE]
Secondary cues Complex analysis / polynomials
Linked theory Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]
Recognition triggers polynomial/root trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1998-D1-P5

Day 1; official label P5; legacy source-page label 51

Let P be an algebraic polynomial of degree n having only real zeros and real coefficients.
a) (15 points) Prove that for every real x the following inequality holds:

$$(n-1)(P'(x))^2 \geq nP(x)P''(x). \tag{2}$$

b) (5 points) Examine the cases of equality.

Open all solution variants | Source note

1998-D1-P6

Long/deep

Fingerprint bound integral by pi/4 from symmetric constraint
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Pair an integral through symmetry [M-AN-PAIR]
Secondary cues Functional equations / dynamics; Combinatorics / discrete structures
Linked theory Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Trigonometric identities and safe branches [T-AN-TRIG]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]
Recognition triggers continuous/differentiable trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1998-D1-P6

Day 1; official label P6; legacy source-page label 51

Let $f : [0,1] \to \mathbb{R}$ be a continuous function with the property that for any x and y in the interval,

$$xf(y) + yf(x) \leq 1.$$

a) (15 points) Show that

$$\int_0^1 f(x)\,dx \leq \frac{\pi}{4}.$$

b) (5 points) Find a function, satisfying the condition, for which there is equality.

Open all solution variants | Source note

Day 2

1998-D2-P1 Long/deep

Fingerprint functional vanishing on common kernel is linear combination
Primary domain Linear algebra / matrices
Problem type structural proof
Linked methods Prove a dimension obstruction and attain it [M-LA-DIMENSION]
Secondary cues Functional equations / dynamics; Real analysis / calculus; Combinatorics / discrete structures
Linked theory Dual bases, kernels and separating functionals [T-LA-DUALITY]; Rank, dimension and exact linear constraints [T-LA-RANK]
Recognition triggers functional-equation trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Reduce size while preserving the defining conditions [S-INDUCTION]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1998-D2-P1

Day 2; official label P7; 20 points; legacy source-page label 52

Let V be a real vector space, and let $f, f_1, f_2, \ldots, f_k$ be linear maps from V to $\mathbb{R}$. Suppose that $f(x) = 0$ whenever $f_1(x) = f_2(x) = \cdots = f_k(x) = 0$. Prove that f is a linear combination of $f_1, f_2, \ldots, f_k$.

Open all solution variants | Source note

1998-D2-P2 Long/deep

Fingerprint extremal second derivative of bounded cubic polynomials
Primary domain Complex analysis / polynomials
Problem type inequality/sharp bound
Linked methods Turn nodal bounds into a sharp linear functional [M-AL-INTERPOLATE]
Secondary cues Complex analysis / polynomials; Functional equations / dynamics
Linked theory Interpolation, divided differences, and extremal signs [T-AL-INTERPOLATION]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Recognition triggers polynomial/root trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Choose an extremal object and force rigidity [S-EXTREMAL]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1998-D2-P2

Day 2; official label P8; 20 points; legacy source-page label 52

Let

$$P = \left\{ f : f(x) = \sum_{k=0}^{3} a_k x^k,\ a_k \in \mathbb{R},\ |f(\pm 1)| \le 1,\ \left|f\left(\pm \tfrac{1}{2}\right)\right| \le 1 \right\}.$$

Evaluate

$$\sup_{f \in P} \max_{-1 \le x \le 1} |f''(x)|$$

and find all polynomials $f \in P$ for which the above "sup" is attained.

Open all solution variants | Source note

1998-D2-P3 Long/deep

Fingerprint periodic points of tent map nonempty and finite
Primary domain Functional equations / dynamics
Problem type existence/construction
Linked methods Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]; Strengthen the statement before the induction step [M-DS-INDUCT]
Secondary cues Real analysis / calculus; Combinatorics / discrete structures
Linked theory Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]; Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Recognition triggers geometry/metric trigger; functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Reduce size while preserving the defining conditions [S-INDUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1998-D2-P3

Day 2; official label P9; 20 points; legacy source-page label 53

Let $0 < c < 1$ and

$$f(x) = \begin{cases} \dfrac{x}{c} & \text{for } x \in [0, c], \\ \dfrac{1-x}{1-c} & \text{for } x \in [c, 1]. \end{cases}$$

We say that p is an n-periodic point if

$$\underbrace{f(f(\ldots f}_{n}(p))) = p$$

and n is the smallest number with this property. Prove that for every $n \geq 1$ the set of n-periodic points is non-empty and finite.

Open all solution variants | Source note

1998-D2-P4 Long/deep

Fingerprint count monotone functions with f(k)=f(f(k+1))
Primary domain Functional equations / dynamics
Problem type counting/enumeration
Linked methods Force structure by admissible substitutions [M-DS-FUNCTIONAL]; Choose information that makes an encoding injective [M-DS-ENCODE]
Secondary cues Real analysis / calculus; Combinatorics / discrete structures; Linear algebra / matrix theory
Linked theory Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]; Finite counting and incidence identities [T-DS-COUNTING]; Extremal choice, exchange arguments, and invariants [T-DS-EXTREMAL]
Recognition triggers counting/permutation trigger; graph/color/set trigger; functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Choose the objects and count them without ambiguity [S-COUNTING]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1998-D2-P4

Day 2; official label P10; 20 points; legacy source-page label 53

Let $A_n = \{1, 2, \ldots, n\}$, where $n \geq 3$. Let F be the family of all non-constant functions $f : A_n \to A_n$ satisfying the following conditions:
(1) $f(k) \leq f(k+1)$ for $k = 1, 2, \ldots, n-1$,

1998-D2-P4 continued

(2) $f(k) = f(f(k+1))$ for $k = 1, 2, \ldots, n-1$.

Find the number of functions in F.

Open all solution variants | Source note

1998-D2-P5 Long/deep

Fingerprint pairwise tangent spheres yield countable intersection set
Primary domain Geometry / convexity / metric
Problem type counting/enumeration
Linked methods Choose information that makes an encoding injective [M-DS-ENCODE]; Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Secondary cues Combinatorics / discrete structures; Real analysis / calculus; Algebra / number theory
Linked theory Metric contractions, covers, and nearest points [T-DS-METRIC]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Countability, accumulation, and Cantor constructions [T-AN-SETS]; Continuity, compactness, and order [T-AN-CONTINUITY]
Recognition triggers counting/permutation trigger; graph/color/set trigger; geometry/metric trigger
Preparation route GEO-Geometry/Metricif
Trap watch Control degenerate configurations and make clear which coordinates or affine transformations preserve the quantity.
Proof strategies Translate geometry into a controlled representation [S-GEOMETRIC]; Choose the objects and count them without ambiguity [S-COUNTING]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1998-D2-P5

Day 2; official label P11; 20 points; legacy source-page label 54

Suppose that S is a family of spheres (i.e., surfaces of balls of positive radius) in $\mathbb{R}^n$, $n \geq 2$, such that the intersection of any two contains at most one point. Prove that the set M of those points that belong to at least two different spheres from S is countable.

Open all solution variants | Source note

1998-D2-P6 Long/deep

Fingerprint differentiability of function supported on countable set
Primary domain Real analysis / calculus
Problem type existence/construction
Linked methods Turn local bad intervals into a global measure bound [M-AN-COVER]; Use a dense set, an enumeration, or a category argument [M-AN-TOPOLOGY]
Secondary cues Functional equations / dynamics; Algebra / number theory
Linked theory Interval coverings and level-set integration [T-AN-COVERING]; Countability, accumulation, and Cantor constructions [T-AN-SETS]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Series, finite cancellation, and summation by parts [T-AN-SERIES]
Recognition triggers continuous/differentiable trigger; geometry/metric trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a witness and verify every constraint [S-CONSTRUCTION]; Pass between local constraints and a global conclusion [S-LOCALGLOBAL]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1998-D2-P6

Day 2; official label P12; 20 points; legacy source-page label 54

Let $f : (0,1) \to [0,\infty)$ be a function that is zero except at the distinct points $a_1, a_2, \ldots$. Let $b_n = f(a_n)$.

1998-D2-P6 continued

(a) Prove that if $\sum_{n=1}^{\infty} b_n < \infty$, then f is differentiable at at least one point $x \in (0,1)$.

(b) Prove that for any sequence of non-negative real numbers $(b_n)_{n=1}^{\infty}$, with $\sum_{n=1}^{\infty} b_n = \infty$, there exists a sequence $(a_n)_{n=1}^{\infty}$ such that the function f defined as above is nowhere differentiable.

Open all solution variants | Source note

2.6 1999

Day 1

1999-D1-P1 Quick/standard

Fingerprint matrices with A^3=A+I exist and have positive determinant
Primary domain Linear algebra / matrices
Problem type existence/construction
Linked methods Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]
Secondary cues Complex analysis / polynomials; Algebra / number theory; Real analysis / calculus
Linked theory Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Determinants, block identities and inverses [T-LA-DETERMINANT]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Continuity, compactness, and order [T-AN-CONTINUITY]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Recognition triggers matrix word; det/rank/eigen trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1999-D1-P1

Day 1; official label P1; 6+14 points; legacy source-page label 56

a) Show that for any $m \in \mathbb{N}$ there exists a real $m \times m$ matrix A such that $A^3 = A + I$, where I is the $m \times m$ identity matrix. (6 points)

b) Show that $\det A > 0$ for every real $m \times m$ matrix satisfying $A^3 = A + I$. (14 points)

Open all solution variants | Source note

1999-D1-P2 Long/deep

Fingerprint bijection of N making sum pi(n)/n^2 converge
Primary domain Real analysis / calculus
Problem type nonexistence/counterexample
Linked methods Test convergence through blocks and differences [M-AN-SERIES]; Find a finite telescoping identity [M-AN-TELESCOPE]; Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]
Secondary cues Combinatorics / discrete structures; Algebra / number theory
Linked theory Series, finite cancellation, and summation by parts [T-AN-SERIES]; Finite counting and incidence identities [T-DS-COUNTING]; Nonnegative sums, Fubini, and integral kernels [T-AN-FUBINI]
Recognition triggers wording-specific
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Make the contradiction quantitative or structural [S-CONTRADICTION]

1999-D1-P2 continued

Navigation Full statement All solution variants Source note Learning route

Full statement: 1999-D1-P2

Day 1; official label P2; 20 points; legacy source-page label 56

Does there exist a bijective map $\pi : \mathbb{N} \to \mathbb{N}$ such that

$$\sum_{n=1}^{\infty} \frac{\pi(n)}{n^2} < \infty?$$

Open all solution variants | Source note

1999-D1-P3 Long/deep

Fingerprint bounded weighted sums of differences force f constant
Primary domain Functional equations / dynamics
Problem type structural proof
Linked methods Force structure by admissible substitutions [M-DS-FUNCTIONAL]
Secondary cues Functional equations / dynamics; Real analysis / calculus; Geometry / convexity / metric
Linked theory Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers integer/prime/mod trigger; functional-equation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1999-D1-P3

Day 1; official label P3; 20 points; legacy source-page label 57

Suppose that a function $f : \mathbb{R} \to \mathbb{R}$ satisfies the inequality

$$\left| \sum_{k=1}^{n} 3^k \big(f(x + ky) - f(x - ky) \big) \right| \leq 1 \tag{1}$$

for every positive integer n and for all $x, y \in \mathbb{R}$. Prove that f is a constant function.

Open all solution variants | Source note

1999-D1-P4 Long/deep

Fingerprint find monotone f with f(x^2/f(x))=x
Primary domain Functional equations / dynamics
Problem type classification/find-all
Linked methods Force structure by admissible substitutions [M-DS-FUNCTIONAL]; Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]
Secondary cues Real analysis / calculus
Linked theory Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]; Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Make the contradiction quantitative or structural [S-CONTRADICTION]

1999-D1-P4 continued

Navigation Full statement All solution variants Source note Learning route

Full statement: 1999-D1-P4

Day 1; official label P4; 20 points; legacy source-page label 57

Find all strictly monotonic functions $f : (0, +\infty) \to (0, +\infty)$ such that $f\left(\frac{x^2}{f(x)}\right) \equiv x$.

Open all solution variants | Source note

1999-D1-P5 Long/deep

Fingerprint 2n marked grid points contain row-column closed cycle
Primary domain Combinatorics
Problem type existence/construction
Linked methods Encode a relation as a graph and extract a cycle [M-DS-GRAPH]; Strengthen the statement before the induction step [M-DS-INDUCT]
Secondary cues Real analysis / calculus; Algebra / number theory; Geometry / convexity / metric
Linked theory Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]; Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Recognition triggers geometry/metric trigger
Preparation route CMB-Combinatoricsi
Trap watch Avoid overcounting; define the encoded object precisely and check extremal/minimal cases.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Reduce size while preserving the defining conditions [S-INDUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1999-D1-P5

Day 1; official label P5; 20 points; legacy source-page label 57

Suppose that $2n$ points of an $n \times n$ grid are marked. Show that for some $k > 1$ one can select $2k$ distinct marked points, say $a_1, \ldots, a_{2k}$, such that a_1 and a_2 are in the same row, a_2 and a_3 are in the same column, ..., a_{2k-1} and a_{2k} are in the same row, and a_{2k} and a_1 are in the same column.

Open all solution variants | Source note

1999-D1-P6 Long/deep

Fingerprint uniform constant c_p in pointwise derivative bound
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Turn local bad intervals into a global measure bound [M-AN-COVER]; Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]
Secondary cues Real analysis / calculus; Probability / measure / expectation
Linked theory Interval coverings and level-set integration [T-AN-COVERING]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Recognition triggers continuous/differentiable trigger; functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Make the contradiction quantitative or structural [S-CONTRADICTION]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1999-D1-P6

Day 1; official label P6; 10+10 points; legacy source-page label 58

a) For each $1 < p < \infty$ find a constant $c_p < \infty$ for which the following statement holds: If $f : [-1, 1] \to \mathbb{R}$ is a continuously differentiable function satisfying $f(1) > f(-1)$ and $|f'(y)| \leq 1$ for all

1999-D1-P6 continued

$y \in [-1,1]$, then there is an $x \in [-1,1]$ such that $f'(x) > 0$ and $|f(y) - f(x)| \leq c_p (f'(x))^{1/p} |y - x|$ for all $y \in [-1,1]$. (10 points)

b) Does such a constant also exist for $p = 1$? (10 points)

Open all solution variants | Source note

Day 2

1999-D2-P1 Long/deep

Fingerprint ring with all squares zero satisfies 2abc=0
Primary domain Algebra / number theory
Problem type structural proof
Linked methods Polarize identities and isolate commutators [M-AL-COMMUTATOR]
Secondary cues None recorded
Linked theory Noncommutative rings, polarization, and torsion [T-AL-RING]
Recognition triggers group/ring/field trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1999-D2-P1

Day 2; official label P7; 20 points; legacy source-page label 59

Suppose that in a not necessarily commutative ring R the square of any element is 0. Prove that $abc + abc = 0$ for any three elements a, b, c.

Open all solution variants | Source note

1999-D2-P2 Long/deep

Fingerprint probability dice-roll sum divisible by 5
Primary domain Probability / measure
Problem type counting/enumeration
Linked methods Count outcomes or condition on a useful structure [M-AN-PROBABILITY]; Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]; Encode convolution or denominators by a power series [M-AN-GENERATING]
Secondary cues Probability / measure / expectation; Combinatorics / discrete structures; Real analysis / calculus
Linked theory Finite probability, conditioning, and expectations [T-AN-PROBABILITY]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Finite counting and incidence identities [T-DS-COUNTING]; Roots-of-unity filters and positive trigonometric kernels [T-AL-FOURIER]
Recognition triggers integral/limit trigger; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

1999-D2-P2 continued

Full statement: 1999-D2-P2

Day 2; official label P8; 20 points; legacy source-page label 59

We throw a dice (which selects one of the numbers $1, 2, \dots, 6$ with equal probability) n times. What is the probability that the sum of the values is divisible by 5?

Open all solution variants | Source note

1999-D2-P3 Long/deep

Fingerprint x_i>=-1 with zero cube sum bounds linear sum
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]
Secondary cues None recorded
Linked theory Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers integral/limit trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1999-D2-P3

Day 2; official label P9; 20 points; legacy source-page label 60

Assume that $x_1, \dots, x_n \geq -1$ and $\sum_{i=1}^{n} x_i^3 = 0$. Prove that $\sum_{i=1}^{n} x_i \leq \frac{n}{3}$.

Open all solution variants | Source note

1999-D2-P4 Long/deep

Fingerprint no positive f with f^2(x)>=f(x+y)(f(x)+y)
Primary domain Functional equations / dynamics
Problem type nonexistence/counterexample
Linked methods Force structure by admissible substitutions [M-DS-FUNCTIONAL]; Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]
Secondary cues Real analysis / calculus
Linked theory Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Recognition triggers functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Make the contradiction quantitative or structural [S-CONTRADICTION]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1999-D2-P4

Day 2; official label P10; 20 points; legacy source-page label 60

Prove that there exists no function $f : (0, +\infty) \to (0, +\infty)$ such that $f^2(x) \geq f(x + y)(f(x) + y)$ for any $x, y > 0$.

Open all solution variants | Source note

1999-D2-P5

Long/deep

Fingerprint idempotent word relation reduces words to length 8
Primary domain Combinatorics
Problem type structural proof
Linked methods Reduce equivalent words with a congruence [M-DS-REWRITE]; Strengthen the statement before the induction step [M-DS-INDUCT]
Secondary cues Geometry / convexity / metric
Linked theory Extremal choice, exchange arguments, and invariants [T-DS-EXTREMAL]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Recognition triggers wording-specific
Preparation route CMB-Combinatoricsi
Trap watch Avoid overcounting; define the encoded object precisely and check extremal/minimal cases.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Reduce size while preserving the defining conditions [S-INDUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1999-D2-P5

Day 2; official label P11; 20 points; legacy source-page label 60

Let S be the set of all words consisting of the letters x, y, z, and consider an equivalence relation $\sim$ on S satisfying the following conditions: for arbitrary words $u, v, w \in S$
(i) $uu \sim u$;
(ii) if $v \sim w$, then $uv \sim uw$ and $vu \sim wu$.

Show that every word in S is equivalent to a word of length at most 8.

Open all solution variants | Source note

1999-D2-P6

Long/deep

Fingerprint large Fourier coefficient exists for small subset of Z_n
Primary domain Algebra / number theory
Problem type inequality/sharp bound
Linked methods Evaluate at roots of unity or average over them [M-AL-UNITY]
Secondary cues Functional equations / dynamics; Complex analysis / polynomials; Geometry / convexity / metric
Linked theory Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers polynomial/root trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Translate geometry into a controlled representation [S-GEOMETRIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 1999-D2-P6

Day 2; official label P12; 20 points; legacy source-page label 61

Let A be a subset of $\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z}$ containing at most $\frac{1}{100}\ln n$ elements. Define the rth Fourier coefficient of A for $r \in \mathbb{Z}_n$ by

$$f(r) = \sum_{s \in A} \exp\left(\frac{2\pi i}{n} sr\right).$$

Prove that there exists an $r \neq 0$, such that $\left|f(r)\right| \geq \frac{|A|}{2}$.

Open all solution variants | Source note

2.7 2000

Day 1

2000-D1-P1 Quick/standard

Fingerprint fixed point of monotone function on unit interval
Primary domain Functional equations / dynamics
Problem type nonexistence/counterexample
Linked methods Force structure by admissible substitutions [M-DS-FUNCTIONAL]; Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]
Secondary cues Functional equations / dynamics; Combinatorics / discrete structures
Linked theory Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]; Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]
Recognition triggers functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2000-D1-P1

Day 1; official label P1; legacy source-page label 62

Is it true that if $f : [0,1] \to [0,1]$ is
a) monotone increasing
b) monotone decreasing

then there exists an $x \in [0,1]$ for which $f(x) = x$?

Open all solution variants | Source note

2000-D1-P2 Medium/trap

Fingerprint complex pairs where two quintic polynomials agree
Primary domain Complex analysis / polynomials
Problem type classification/find-all
Linked methods Compare coefficients after a structural substitution [M-AL-COEFFICIENT]; Translate root restrictions into symmetric coefficients [M-AL-VIETA]
Secondary cues Complex analysis / polynomials
Linked theory Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]; Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]
Recognition triggers wording-specific
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2000-D1-P2

Day 1; official label P2; legacy source-page label 62

Let $p(x) = x^5 + x$ and $q(x) = x^5 + x^2$. Find all pairs (w, z) of complex numbers with $w \neq z$ for which $p(w) = p(z)$ and $q(w) = q(z)$.

Open all solution variants | Source note

2000-D1-P3 Medium

Fingerprint rank-one commutator AB-BA squares to zero
Primary domain Linear algebra / matrices
Problem type structural proof
Linked methods Use trace to extract a scalar invariant [M-LA-TRACE]; Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]
Secondary cues Complex analysis / polynomials; Real analysis / calculus
Linked theory Rank, dimension and exact linear constraints [T-LA-RANK]; Trace pairing and commutator calculus [T-LA-TRACE]; Nilpotents, projections and involutions [T-LA-NILPOTENT]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]
Recognition triggers matrix word; det/rank/eigen trigger; continuous/differentiable trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2000-D1-P3

Day 1; official label P3; legacy source-page label 63

A and B are square complex matrices of the same size and

$$\operatorname{rank}(AB - BA) = 1.$$

Show that $(AB - BA)^2 = 0$.

Open all solution variants | Source note

2000-D1-P4 Medium

Fingerprint inequalities for sums of decreasing positive sequences
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]; Find a finite telescoping identity [M-AN-TELESCOPE]
Secondary cues None recorded
Linked theory Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Nonnegative sums, Fubini, and integral kernels [T-AN-FUBINI]; Series, finite cancellation, and summation by parts [T-AN-SERIES]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Recognition triggers wording-specific
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2000-D1-P4

Day 1; official label P4; legacy source-page label 63

a) Show that if (x_i) is a decreasing sequence of positive numbers then

$$\left(\sum_{i=1}^{n} x_i^2\right)^{1/2} \leq \sum_{i=1}^{n} \frac{x_i}{\sqrt{i}}.$$

b) Show that there is a constant C so that if (x_i) is a decreasing sequence of positive numbers then

$$\sum_{m=1}^{\infty} \frac{1}{\sqrt{m}} \left(\sum_{i=m}^{\infty} x_i^2\right)^{1/2} \leq C \sum_{i=1}^{\infty} x_i.$$

Open all solution variants | Source note

2000-D1-P5 Long/deep

Fingerprint three idempotents summing to zero in characteristic-zero ring
Primary domain Algebra / number theory
Problem type structural proof
Linked methods Polarize identities and isolate commutators [M-AL-COMMUTATOR]
Secondary cues Linear algebra / matrix theory
Linked theory Noncommutative rings, polarization, and torsion [T-AL-RING]
Recognition triggers integer/prime/mod trigger; group/ring/field trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2000-D1-P5

Day 1; official label P5; legacy source-page label 64

Let R be a ring of characteristic zero (not necessarily commutative). Let e, f and g be idempotent elements of R satisfying $e+f+g=0$. Show that $e=f=g=0$.

(R is of characteristic zero means that, if $a \in R$ and n is a positive integer, then $na \neq 0$ unless $a=0$. An idempotent x is an element satisfying $x=x^2$.)

Open all solution variants | Source note

2000-D1-P6 Long/deep

Fingerprint recursive sequence approaches inverse of integral function
Primary domain Real analysis / calculus
Problem type limit/asymptotic evaluation
Linked methods Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]; Compare growth through derivatives [M-AN-RATIO]; Separate a main range from exceptional scales [M-AN-SPLIT]
Secondary cues Real analysis / calculus; Algebra / number theory
Linked theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers continuous/differentiable trigger; integral/limit trigger; functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Control the limiting operation before taking the limit [S-ANALYTIC]; Pass between local constraints and a global conclusion [S-LOCALGLOBAL]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2000-D1-P6

Day 1; official label P6; legacy source-page label 64

Let $f : \mathbb{R} \to (0,\infty)$ be an increasing differentiable function for which $\lim_{x\to\infty} f(x) = \infty$ and f' is bounded.

Let $F(x) = \int_0^x f$. Define the sequence (a_n) inductively by

$$a_0 = 1, \qquad a_{n+1} = a_n + \frac{1}{f(a_n)}, \tag{1}$$

and the sequence (b_n) simply by $b_n = F^{-1}(n)$. Prove that $\lim_{n\to\infty}(a_n - b_n) = 0$.

Open all solution variants | Source note

Day 2

2000-D2-P1 Quick/standard

Fingerprint partition unit square and cube into n smaller cubes
Primary domain Geometry / convexity / metric
Problem type existence/construction
Linked methods Split a family by traces, pieces, or labels [M-DS-PARTITION]; Strengthen the statement before the induction step [M-DS-INDUCT]
Secondary cues Algebra / number theory; Real analysis / calculus; Combinatorics / discrete structures
Linked theory Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]
Recognition triggers geometry/metric trigger
Preparation route GEO-Geometry/Metricif
Trap watch Control degenerate configurations and make clear which coordinates or affine transformations preserve the quantity.
Proof strategies Build a witness and verify every constraint [S-CONSTRUCTION]; Reduce size while preserving the defining conditions [S-INDUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2000-D2-P1

Day 2; official label P7; legacy source-page label 66

a) Show that the unit square can be partitioned into n smaller squares if n is large enough.

b) Let $d \geq 2$. Show that there is a constant $N(d)$ such that, whenever $n \geq N(d)$, a d-dimensional unit cube can be partitioned into n smaller cubes.

Open all solution variants | Source note

2000-D2-P2 Quick/standard

Fingerprint local minima dense for nowhere monotone continuous function
Primary domain Real analysis / calculus
Problem type structural proof
Linked methods Localize an extremum or an accumulation point [M-AN-COMPACT]
Secondary cues Functional equations / dynamics; Combinatorics / discrete structures
Linked theory Continuity, compactness, and order [T-AN-CONTINUITY]
Recognition triggers continuous/differentiable trigger; geometry/metric trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Choose an extremal object and force rigidity [S-EXTREMAL]; Pass between local constraints and a global conclusion [S-LOCALGLOBAL]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2000-D2-P2

Day 2; official label P8; legacy source-page label 66

Let f be continuous and nowhere monotone on $[0,1]$. Show that the set of points on which f attains local minima is dense in $[0,1]$.

(A function is nowhere monotone if there exists no interval where the function is monotone. A set is dense if each non-empty open interval contains at least one element of the set.)

Open all solution variants | Source note

2000-D2-P3 Medium

Fingerprint polynomial takes values 0 or 1 at n+1 points
Primary domain Complex analysis / polynomials
Problem type counting/enumeration
Linked methods Replace many equal values by too many roots [M-AL-ROOTCOUNT]
Secondary cues Algebra / number theory; Combinatorics / discrete structures; Real analysis / calculus
Linked theory Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Recognition triggers polynomial/root trigger
Preparation route CP-Polynomials/Complex
Trap watch Separate polynomial identity from pointwise testing; check multiplicities, zero polynomial cases, and real-vs-complex roots.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2000-D2-P3

Day 2; official label P9; legacy source-page label 66

Let $p(z)$ be a polynomial of degree n with complex coefficients. Prove that there exist at least $n+1$ complex numbers z for which $p(z)$ is 0 or 1.

Open all solution variants | Source note

2000-D2-P4 Medium

Fingerprint areas between sextic polynomial and tangent line
Primary domain Complex analysis / polynomials
Problem type inequality/sharp bound
Linked methods Factor first and compare irreducible exponents [M-AL-FACTOR]; Compare coefficients after a structural substitution [M-AL-COEFFICIENT]
Secondary cues Combinatorics / discrete structures; Geometry / convexity / metric; Real analysis / calculus
Linked theory Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Recognition triggers polynomial/root trigger; graph/color/set trigger; geometry/metric trigger
Preparation route CP-Polynomials/Complex
Trap watch Separate polynomial identity from pointwise testing; check multiplicities, zero polynomial cases, and real-vs-complex roots.
Proof strategies Translate geometry into a controlled representation [S-GEOMETRIC]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2000-D2-P4

Day 2; official label P10; legacy source-page label 67

Suppose the graph of a polynomial of degree 6 is tangent to a straight line at 3 points A_1, A_2, A_3, where A_2 lies between A_1 and A_3.
a) Prove that if the lengths of the segments A_1A_2 and A_2A_3 are equal, then the areas of the figures bounded by these segments and the graph of the polynomial are equal as well.

b) Let $k = \dfrac{A_2A_3}{A_1A_2}$, and let K be the ratio of the areas of the appropriate figures. Prove that

$$\frac{2}{7}k^5 < K < \frac{7}{2}k^5.$$

Open all solution variants | Source note

2000-D2-P5

Long/deep

Fingerprint find positive functions with f(x)f(yf(x))=f(x+y)
Primary domain Functional equations / dynamics
Problem type classification/find-all
Linked methods Force structure by admissible substitutions [M-DS-FUNCTIONAL]
Secondary cues Real analysis / calculus
Linked theory Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Dominated convergence and orthonormal separation [T-AN-MEASURE]
Recognition triggers functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2000-D2-P5

Day 2; official label P11; legacy source-page label 68

Let $\mathbb{R}^+$ be the set of positive real numbers. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that for all $x, y \in \mathbb{R}^+$

$$f(x)f(yf(x)) = f(x + y).$$

Open all solution variants | Source note

2000-D2-P6

Long/deep

Fingerprint p(e^AB) nilpotent iff p(e^BA) nilpotent
Primary domain Linear algebra / matrices
Problem type structural proof
Linked methods Transfer spectral information between products [M-LA-TRANSFER]
Secondary cues Complex analysis / polynomials; Algebra / number theory; Real analysis / calculus
Linked theory Matrix power series and transfer between AB and BA [T-LA-SERIES]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Nilpotents, projections and involutions [T-LA-NILPOTENT]
Recognition triggers matrix word; det/rank/eigen trigger; integral/limit trigger; polynomial/root trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2000-D2-P6

Day 2; official label P12; legacy source-page label 69

For an $m \times m$ real matrix A, e^A is defined as $\sum_{n=0}^{\infty} \frac{1}{n!} A^n$. (The sum is convergent for all matrices.)

Prove or disprove, that for all real polynomials p and $m \times m$ real matrices A and B, $p(e^{AB})$ is nilpotent if and only if $p(e^{BA})$ is nilpotent. (A matrix A is nilpotent if $A^k = 0$ for some positive integer k.)

Open all solution variants | Source note

2.8 2001

Day 1

2001-D1-P1 Quick/standard

Fingerprint sum of transversal entries in numbered square matrix
Primary domain Combinatorics
Problem type exact computation
Linked methods Choose information that makes an encoding injective [M-DS-ENCODE]
Secondary cues Linear algebra / matrix theory; Real analysis / calculus; Combinatorics / discrete structures
Linked theory Finite counting and incidence identities [T-DS-COUNTING]; Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]
Recognition triggers matrix word; integral/limit trigger; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Choose the objects and count them without ambiguity [S-COUNTING]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2001-D1-P1

Day 1; official label P1; legacy source-page label 70

Let n be a positive integer. Consider an $n \times n$ matrix with entries $1, 2, \ldots, n^2$ written in order starting top left and moving along each row in turn left–to–right. We choose n entries of the matrix such that exactly one entry is chosen in each row and each column. What are the possible values of the sum of the selected entries?

Open all solution variants | Source note

2001-D1-P2 Quick/standard

Fingerprint coprime-order elements with (ab)^t=e are trivial
Primary domain Algebra / number theory
Problem type nonexistence/counterexample
Linked methods Prove generation or obstruct it by an image [M-AL-GENERATORS]
Secondary cues Combinatorics / discrete structures; Linear algebra / matrix theory
Linked theory Groups, orders, generators, and quotients [T-AL-GROUP]; Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Recognition triggers integer/prime/mod trigger; group/ring/field trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2001-D1-P2

Day 1; official label P2; legacy source-page label 70

Let r, s, t be positive integers which are pairwise relatively prime. If a and b are elements of a commutative multiplicative group with unity element e, and $a^r = b^s = (ab)^t = e$, prove that $a = b = e$.

Does the same conclusion hold if a and b are elements of an arbitrary non-commutative group?

Open all solution variants | Source note

2001-D1-P3 Medium

Fingerprint limit of (1-t) times series as t approaches 1
Primary domain Real analysis / calculus
Problem type limit/asymptotic evaluation
Linked methods Expose the mesh and control its tails [M-AN-RIEMANN]
Secondary cues None recorded
Linked theory Riemann sums, periodic averaging, and tails [T-AN-RIEMANN]; Scale separation and limit comparisons [T-AN-ASYMPTOTICS]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Recognition triggers integral/limit trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2001-D1-P3

Day 1; official label P3; legacy source-page label 70

Find $\lim_{t\nearrow 1}(1-t)\sum_{n=1}^{\infty}\frac{t^n}{1+t^n}$, where $t\nearrow 1$ means that t approaches 1 from below.

Open all solution variants | Source note

2001-D1-P4 Medium

Fingerprint qth roots of unity are roots of constrained polynomial
Primary domain Complex analysis / polynomials
Problem type structural proof
Linked methods Evaluate at roots of unity or average over them [M-AL-UNITY]
Secondary cues Algebra / number theory; Real analysis / calculus; Geometry / convexity / metric
Linked theory Roots of unity, cyclotomic polynomials, and algebraic norms [T-AL-CYCLOTOMIC]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]
Recognition triggers polynomial/root trigger; integer/prime/mod trigger
Preparation route CP-Polynomials/Complex
Trap watch Separate polynomial identity from pointwise testing; check multiplicities, zero polynomial cases, and real-vs-complex roots.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2001-D1-P4

Day 1; official label P4; legacy source-page label 71

Let k be a positive integer. Let $p(x)$ be a polynomial of degree n each of whose coefficients is -1, 1 or 0, and which is divisible by $(x-1)^k$. Let q be a prime such that $\frac{q}{\ln q}<\frac{k}{\ln(n+1)}$. Prove that the complex qth roots of unity are roots of the polynomial $p(x)$.

Open all solution variants | Source note

2001-D1-P5 Long/deep

Fingerprint nonscalar matrix similar to one nonzero diagonal entry
Primary domain Linear algebra / matrices
Problem type existence/construction
Linked methods Choose a basis adapted to the structure [M-LA-BASIS]
Secondary cues Complex analysis / polynomials; Geometry / convexity / metric
Linked theory Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Adapted bases, centralizers and common eigenspaces [T-LA-INVARIANT]; Trace pairing and commutator calculus [T-LA-TRACE]

2001-D1-P5 continued

Recognition triggers matrix word
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Reduce size while preserving the defining conditions [S-INDUCTION]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2001-D1-P5

Day 1; official label P5; legacy source-page label 71

Let A be an $n \times n$ complex matrix such that $A \neq \lambda I$ for all $\lambda \in \mathbb{C}$. Prove that A is similar to a matrix having at most one non-zero entry on the main diagonal.

Open all solution variants | Source note

2001-D1-P6 Long/deep

Fingerprint limit of f/g from differential relation constraint
Primary domain Real analysis / calculus
Problem type limit/asymptotic evaluation
Linked methods Design an auxiliary derivative [M-AN-AUXILIARY]; Compare growth through derivatives [M-AN-RATIO]
Secondary cues Real analysis / calculus; Linear algebra / matrix theory
Linked theory Differential inequalities and integrating factors [T-AN-ODE]; Scale separation and limit comparisons [T-AN-ASYMPTOTICS]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Recognition triggers continuous/differentiable trigger; integral/limit trigger; functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2001-D1-P6

Day 1; official label P6; legacy source-page label 72

Suppose that the differentiable functions $a, b, f, g : \mathbb{R} \to \mathbb{R}$ satisfy

$$f(x) \geq 0,\ f'(x) \geq 0,\ g(x) > 0,\ g'(x) > 0 \quad \text{for all } x \in \mathbb{R},$$

$$\lim_{x\to\infty} a(x) = A > 0, \quad \lim_{x\to\infty} b(x) = B > 0, \quad \lim_{x\to\infty} f(x) = \lim_{x\to\infty} g(x) = \infty,$$

and

$$\frac{f'(x)}{g'(x)} + a(x)\frac{f(x)}{g(x)} = b(x).$$

Prove that

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \frac{B}{A+1}.$$

Open all solution variants | Source note

Day 2

2001-D2-P1 Quick/standard

Fingerprint factor of 1+x+...+x^(r+s) has 0/1 coefficients
Primary domain Complex analysis / polynomials
Problem type structural proof
Linked methods Compare coefficients after a structural substitution [M-AL-COEFFICIENT]
Secondary cues Algebra / number theory; Complex analysis / polynomials; Combinatorics / discrete structures
Linked theory Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]
Recognition triggers integer/prime/mod trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Reduce size while preserving the defining conditions [S-INDUCTION]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2001-D2-P1

Day 2; official label P7; legacy source-page label 74

Let $r, s \geq 1$ be integers and $a_0, a_1, \ldots, a_{r-1}, b_0, b_1, \ldots, b_{s-1}$ be real non-negative numbers such that

$$(a_0 + a_1 x + a_2 x^2 + \cdots + a_{r-1} x^{r-1} + x^r)(b_0 + b_1 x + b_2 x^2 + \cdots + b_{s-1} x^{s-1} + x^s) = \\ = 1 + x + x^2 + \cdots + x^{r+s-1} + x^{r+s}.$$

Prove that each a_i and each b_j equals either 0 or 1.

Open all solution variants | Source note

2001-D2-P2 Quick/standard

Fingerprint nested radical sequences converge with rate 8^-n
Primary domain Real analysis / calculus
Problem type recurrence/convergence
Linked methods Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]; Expand near equality, bound away from it [M-AN-LOCAL]
Secondary cues Functional equations / dynamics
Linked theory Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Taylor remainders and local scale [T-AN-TAYLOR]; Trigonometric identities and safe branches [T-AN-TRIG]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Recognition triggers integral/limit trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Reduce size while preserving the defining conditions [S-INDUCTION]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2001-D2-P2

Day 2; official label P8; legacy source-page label 74

Let $a_0 = \sqrt{2}$, $b_0 = 2$, $a_{n+1} = \sqrt{2 - \sqrt{4 - a_n^2}}$, $b_{n+1} = \dfrac{2b_n}{2 + \sqrt{4 + b_n^2}}$.

a) Prove that the sequences (a_n), (b_n) are decreasing and converge to 0.

2001-D2-P2 continued

b) Prove that the sequence $(2^n a_n)$ is increasing, the sequence $(2^n b_n)$ is decreasing and that these two sequences converge to the same limit.

c) Prove that there is a positive constant C such that for all n the following inequality holds: $0 < b_n - a_n < \frac{C}{8^n}$.

Open all solution variants | Source note

2001-D2-P3 Medium

Fingerprint max points on sphere pairwise distance exceeding sqrt(2)
Primary domain Geometry / convexity / metric
Problem type inequality/sharp bound
Linked methods Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]; Strengthen the statement before the induction step [M-DS-INDUCT]
Secondary cues Real analysis / calculus; Combinatorics / discrete structures; Functional equations / dynamics
Linked theory Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Inner products, orthogonality and polarization [T-LA-INNER]
Recognition triggers counting/permutation trigger; geometry/metric trigger
Preparation route GEO-Geometry/Metricif
Trap watch Control degenerate configurations and make clear which coordinates or affine transformations preserve the quantity.
Proof strategies Translate geometry into a controlled representation [S-GEOMETRIC]; Reduce size while preserving the defining conditions [S-INDUCTION]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2001-D2-P3

Day 2; official label P9; legacy source-page label 75

Find the maximum number of points on a sphere of radius 1 in $\mathbb{R}^n$ such that the distance between any two of these points is strictly greater than $\sqrt{2}$.

Open all solution variants | Source note

2001-D2-P4 Medium

Fingerprint vanishing principal minors force nilpotency and triangular form
Primary domain Linear algebra / matrices
Problem type structural proof
Linked methods Exploit a compatible block decomposition [M-LA-BLOCK]; Interpret matrix multiplication as a count [M-LA-WALKS]
Secondary cues Combinatorics / discrete structures; Real analysis / calculus; Complex analysis / polynomials
Linked theory Determinants, block identities and inverses [T-LA-DETERMINANT]; Nilpotents, projections and involutions [T-LA-NILPOTENT]; Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]
Recognition triggers matrix word; det/rank/eigen trigger; counting/permutation trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Choose an extremal object and force rigidity [S-EXTREMAL]; Make the contradiction quantitative or structural [S-CONTRADICTION]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

2001-D2-P4 continued

Full statement: 2001-D2-P4

Day 2; official label P10; legacy source-page label 76

Let $A = (a_{k,\ell})_{k,\ell=1,\dots,n}$ be an $n \times n$ complex matrix such that for each $m \in \{1, \dots, n\}$ and $1 \le j_1 < \cdots < j_m \le n$ the determinant of the matrix $(a_{j_k,j_\ell})_{k,\ell=1,\dots,m}$ is zero. Prove that $A^n = 0$ and that there exists a permutation $\sigma \in S_n$ such that the matrix

$$(a_{\sigma(k),\sigma(\ell)})_{k,\ell=1,\dots,n}$$

has all of its nonzero elements above the diagonal.

Open all solution variants | Source note

2001-D2-P5 Long/deep

Fingerprint no function satisfies f(x+y) >= f(x)+yf(f(x))
Primary domain Functional equations / dynamics
Problem type nonexistence/counterexample
Linked methods Force structure by admissible substitutions [M-DS-FUNCTIONAL]; Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]
Secondary cues Real analysis / calculus
Linked theory Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Make the contradiction quantitative or structural [S-CONTRADICTION]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2001-D2-P5

Day 2; official label P11; legacy source-page label 77

Let $\mathbb{R}$ be the set of real numbers. Prove that there is no function $f : \mathbb{R} \to \mathbb{R}$ with $f(0) > 0$, and such that

$$f(x + y) \ge f(x) + yf(f(x)) \quad \text{for all } x, y \in \mathbb{R}.$$

Open all solution variants | Source note

2001-D2-P6 Long/deep

Fingerprint bound product of sines by value at pi/3
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]; Exploit identities before estimating oscillations [M-AN-TRIG]
Secondary cues Real analysis / calculus; Functional equations / dynamics; Geometry / convexity / metric
Linked theory Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Trigonometric identities and safe branches [T-AN-TRIG]
Recognition triggers integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

2001-D2-P6 continued

Full statement: 2001-D2-P6

Day 2; official label P12; legacy source-page label 78

For each positive integer n, let $f_n(\vartheta) = \sin\vartheta \cdot \sin(2\vartheta) \cdot \sin(4\vartheta) \cdots \sin(2^n\vartheta)$. For all real ϑ and all n, prove that

$$|f_n(\vartheta)| \leq \frac{2}{\sqrt{3}}|f_n(\pi/3)|.$$

Open all solution variants | Source note

2.9 2002

Day 1

2002-D1-P1 Quick/standard

Fingerprint reflection swaps vertices and intersections of standard parabolas
Primary domain Geometry / convexity / metric
Problem type structural proof
Linked methods Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Secondary cues Combinatorics / discrete structures; Complex analysis / polynomials; Linear algebra / matrix theory
Linked theory Conics, reflections, and inversion [T-DS-TRANSFORMS]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]
Recognition triggers polynomial/root trigger; graph/color/set trigger; geometry/metric trigger
Preparation route GEO-Geometry/Metricif
Trap watch Control degenerate configurations and make clear which coordinates or affine transformations preserve the quantity.
Proof strategies Translate geometry into a controlled representation [S-GEOMETRIC]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2002-D1-P1

Day 1; official label P1; legacy source-page label 79

A standard parabola is the graph of a quadratic polynomial $y = x^2 + ax + b$ with leading coefficient 1. Three standard parabolas with vertices V_1, V_2, V_3 intersect pairwise at points A_1, A_2, A_3. Let $A \mapsto s(A)$ be the reflection of the plane with respect to the x axis.

Prove that standard parabolas with vertices $s(A_1)$, $s(A_2)$, $s(A_3)$ intersect pairwise at the points $s(V_1)$, $s(V_2)$, $s(V_3)$.

Open all solution variants | Source note

2002-D1-P2 Medium/trap

Fingerprint existence of positive f with f'(x)=f(f(x))
Primary domain Functional equations / dynamics
Problem type nonexistence/counterexample
Linked methods Force structure by admissible substitutions [M-DS-FUNCTIONAL]
Secondary cues Real analysis / calculus
Linked theory Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Recognition triggers continuous/differentiable trigger; functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.

2002-D1-P2 continued

Proof strategies Control the limiting operation before taking the limit [S-ANALYTIC]; Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2002-D1-P2

Day 1; official label P2; legacy source-page label 79

Does there exist a continuously differentiable function $f : \mathbb{R} \to \mathbb{R}$ such that for every $x \in \mathbb{R}$ we have $f(x) > 0$ and $f'(x) = f(f(x))$?

Open all solution variants | Source note

2002-D1-P3 Medium

Fingerprint identity relating reciprocal binomials and powers of two
Primary domain Algebra / number theory
Problem type structural proof
Linked methods Find a recurrence that reveals the invariant [M-AL-RECURRENCE]
Secondary cues Geometry / convexity / metric; Real analysis / calculus; Functional equations / dynamics
Linked theory Linear recurrences and generating functions [T-AL-RECURRENCE]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Reduce size while preserving the defining conditions [S-INDUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2002-D1-P3

Day 1; official label P3; legacy source-page label 80

Let n be a positive integer and let

$$a_k = \frac{1}{\binom{n}{k}}, \qquad b_k = 2^{k-n}, \qquad \text{for } k = 1, 2, \ldots, n.$$

Show that

$$\frac{a_1 - b_1}{1} + \frac{a_2 - b_2}{2} + \cdots + \frac{a_n - b_n}{n} = 0. \tag{1}$$

Open all solution variants | Source note

2002-D1-P4 Medium

Fingerprint closed orbit of iterated continuous map is finite
Primary domain Functional equations / dynamics
Problem type structural proof
Linked methods Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]
Secondary cues Functional equations / dynamics; Geometry / convexity / metric
Linked theory Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]; Continuity, compactness, and order [T-AN-CONTINUITY]; Countability, accumulation, and Cantor constructions [T-AN-SETS]; Baire class one and controlled discontinuities [T-AN-BAIRE]
Recognition triggers continuous/differentiable trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Control the limiting operation before taking the limit [S-ANALYTIC]; Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

2002-D1-P4 continued

Full statement: 2002-D1-P4

Day 1; official label P4; legacy source-page label 80

Let $f : [a,b] \to [a,b]$ be a continuous function and let $p \in [a,b]$. Define $p_0 = p$ and $p_{n+1} = f(p_n)$ for $n = 0, 1, 2, \dots$. Suppose that the set $T_p = \{p_n : n = 0, 1, 2, \dots\}$ is closed, i.e., if $x \notin T_p$ then there is a $\delta > 0$ such that for all $x' \in T_p$ we have $|x' - x| \geq \delta$. Show that T_p has finitely many elements.

Open all solution variants | Source note

2002-D1-P5 Long/deep

Fingerprint functions attaining every value uncountably many times
Primary domain Real analysis / calculus
Problem type nonexistence/counterexample
Linked methods Use a dense set, an enumeration, or a category argument [M-AN-TOPOLOGY]; Turn local bad intervals into a global measure bound [M-AN-COVER]
Secondary cues Combinatorics / discrete structures; Functional equations / dynamics
Linked theory Countability, accumulation, and Cantor constructions [T-AN-SETS]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Interval coverings and level-set integration [T-AN-COVERING]
Recognition triggers continuous/differentiable trigger; counting/permutation trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Make the contradiction quantitative or structural [S-CONTRADICTION]; Choose the objects and count them without ambiguity [S-COUNTING]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2002-D1-P5

Day 1; official label P5; legacy source-page label 81

Prove or disprove the following statements:
(a) There exists a monotone function $f : [0,1] \to [0,1]$ such that for each $y \in [0,1]$ the equation $f(x) = y$ has uncountably many solutions x.

(b) There exists a continuously differentiable function $f : [0,1] \to [0,1]$ such that for each $y \in [0,1]$ the equation $f(x) = y$ has uncountably many solutions x.

Open all solution variants | Source note

2002-D1-P6 Long/deep

Fingerprint bound matrix power norms from consecutive-power difference decay
Primary domain Linear algebra / matrices
Problem type inequality/sharp bound
Linked methods Combine norm bounds with algebraic decay [M-LA-NORM]
Secondary cues Real analysis / calculus; Linear algebra / matrix theory; Geometry / convexity / metric
Linked theory Operator norms and convergence of matrix powers [T-LA-NORM]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers matrix word; integral/limit trigger; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Control the limiting operation before taking the limit [S-ANALYTIC]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

2002-D1-P6 continued

Full statement: 2002-D1-P6

Day 1; official label P6; legacy source-page label 82

For an $n \times n$ matrix M with real entries let $\|M\| = \sup_{x \in \mathbb{R}^n \setminus \{0\}} \frac{\|Mx\|_2}{\|x\|_2}$, where $\|\cdot\|_2$ denotes the Euclidean norm on $\mathbb{R}^n$. Assume that an $n \times n$ matrix A with real entries satisfies $\|A^k - A^{k-1}\| \leq \frac{1}{2002k}$ for all positive integers k. Prove that $\|A^k\| \leq 2002$ for all positive integers k.

Open all solution variants | Source note

Day 2

2002-D2-P1 Quick/standard

Fingerprint determinant of matrix with 2 diagonal, (-1)^|i-j| off-diagonal
Primary domain Linear algebra / matrices
Problem type exact computation
Linked methods Eliminate systematically and track the invariant [M-LA-ELIM]
Secondary cues None recorded
Linked theory Determinants, block identities and inverses [T-LA-DETERMINANT]
Recognition triggers matrix word; det/rank/eigen trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2002-D2-P1

Day 2; official label P7; legacy source-page label 84

Compute the determinant of the $n \times n$ matrix $A = [a_{ij}]$,

$$a_{ij} = \begin{cases} (-1)^{|i-j|}, & \text{if } i \neq j, \\ 2, & \text{if } i = j. \end{cases}$$

Open all solution variants | Source note

2002-D2-P2 Quick/standard

Fingerprint two contestants jointly solve all six problems
Primary domain Combinatorics
Problem type existence/construction
Linked methods Count incidences with their exact multiplicity [M-DS-DOUBLECOUNT]
Secondary cues None recorded
Linked theory Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers wording-specific
Preparation route CMB-Combinatoricsi
Trap watch Avoid overcounting; define the encoded object precisely and check extremal/minimal cases.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

2002-D2-P2 continued

Full statement: 2002-D2-P2

Day 2; official label P8; legacy source-page label 84

Two hundred students participated in a mathematical contest. They had 6 problems to solve. It is known that each problem was correctly solved by at least 120 participants. Prove that there must be two participants such that every problem was solved by at least one of these two students.

Open all solution variants | Source note

2002-D2-P3 Medium

Fingerprint product of two exponential-type series is integer
Primary domain Algebra / number theory
Problem type structural proof
Linked methods Find a recurrence that reveals the invariant [M-AL-RECURRENCE]
Secondary cues Real analysis / calculus; Geometry / convexity / metric; Linear algebra / matrix theory
Linked theory Linear recurrences and generating functions [T-AL-RECURRENCE]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Recognition triggers integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Reduce size while preserving the defining conditions [S-INDUCTION]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2002-D2-P3

Day 2; official label P9; legacy source-page label 85

For each $n \geq 1$ let

$$a_n = \sum_{k=0}^{\infty} \frac{k^n}{k!}, \qquad b_n = \sum_{k=0}^{\infty} (-1)^k \frac{k^n}{k!}.$$

Show that $a_n \cdot b_n$ is an integer.

Open all solution variants | Source note

2002-D2-P4 Medium

Fingerprint tetrahedron face-angle inequality with dihedral cosines
Primary domain Geometry / convexity / metric
Problem type inequality/sharp bound
Linked methods Turn a spherical comparison into an endpoint integral [M-DS-SPHERICAL]; Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Secondary cues Real analysis / calculus; Combinatorics / discrete structures
Linked theory Spherical distance and signed dihedral derivatives [T-DS-SPHERICAL]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Inner products, orthogonality and polarization [T-LA-INNER]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Recognition triggers wording-specific
Preparation route GEO-Geometry/Metricif
Trap watch Control degenerate configurations and make clear which coordinates or affine transformations preserve the quantity.
Proof strategies Translate geometry into a controlled representation [S-GEOMETRIC]; Control the limiting operation before taking the limit [S-ANALYTIC]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

2002-D2-P4 continued

Full statement: 2002-D2-P4

Day 2; official label P10; legacy source-page label 85

In the tetrahedron $OABC$, let $\angle BOC = \alpha$, $\angle COA = \beta$ and $\angle AOB = \gamma$. Let σ be the angle between the faces OAB and OAC, and let τ be the angle between the faces OBA and OBC. Prove that

$$\gamma > \beta \cdot \cos\sigma + \alpha \cdot \cos\tau.$$

Open all solution variants | Source note

2002-D2-P5 Long/deep

Fingerprint characterize matrices with A·conj(A)=I as S·conj(S)^-1
Primary domain Linear algebra / matrices
Problem type classification/find-all
Linked methods Compare coefficients in a matrix polynomial [M-LA-PARAMETER]; Complete the equation to a product equal to identity [M-LA-INVERSE]
Secondary cues Complex analysis / polynomials; Real analysis / calculus; Geometry / convexity / metric
Linked theory Determinants, block identities and inverses [T-LA-DETERMINANT]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]
Recognition triggers matrix word
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Build a witness and verify every constraint [S-CONSTRUCTION]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2002-D2-P5

Day 2; official label P11; legacy source-page label 86

Let A be an $n \times n$ matrix with complex entries and suppose that $n > 1$. Prove that

$$A\overline{A} = I_n \iff \exists S \in GL_n(\mathbb{C}) \text{ such that } A = S\overline{S}^{-1}.$$

(If $A = [a_{ij}]$ then $\overline{A} = [\overline{a_{ij}}]$, where $\overline{a_{ij}}$ is the complex conjugate of a_{ij}; $GL_n(\mathbb{C})$ denotes the set of all $n \times n$ invertible matrices with complex entries, and I_n is the identity matrix.)

Open all solution variants | Source note

2002-D2-P6 Long/deep

Fingerprint co-coercivity inequality for Lipschitz gradient of convex function
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Choose the correct convexity comparison [M-AN-CONVEX]
Secondary cues Real analysis / calculus
Linked theory Lipschitz derivatives and convex smoothness [T-AN-SMOOTH]; Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Inner products, orthogonality and polarization [T-LA-INNER]
Recognition triggers functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

2002-D2-P6 continued

Full statement: 2002-D2-P6

Day 2; official label P12; legacy source-page label 87

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function whose gradient $\nabla f = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}\right)$ exists at every point of $\mathbb{R}^n$ and satisfies the condition

$$\exists L > 0 \ \forall x_1, x_2 \in \mathbb{R}^n \quad \|\nabla f(x_1) - \nabla f(x_2)\| \le L\|x_1 - x_2\|.$$

Prove that

$$\forall x_1, x_2 \in \mathbb{R}^n \quad \|\nabla f(x_1) - \nabla f(x_2)\|^2 \le L\langle \nabla f(x_1) - \nabla f(x_2),\ x_1 - x_2\rangle. \tag{1}$$

In this formula $\langle a, b\rangle$ denotes the scalar product of the vectors a and b.

Open all solution variants | Source note

2.10 2003

Day 1

2003-D1-P1 Quick/standard

Fingerprint growth limit of sequence with a(n+1)>1.5a(n)
Primary domain Real analysis / calculus
Problem type recurrence/convergence
Linked methods Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]
Secondary cues None recorded
Linked theory Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers integral/limit trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a witness and verify every constraint [S-CONSTRUCTION]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2003-D1-P1

Day 1; official label P1; 10+10 points; legacy source-page label 88

(a) Let $a_1, a_2, \dots$ be a sequence of real numbers such that $a_1 = 1$ and $a_{n+1} > \frac{3}{2}a_n$ for all n. Prove that the sequence

$$\frac{a_n}{\left(\frac{3}{2}\right)^{n-1}}$$

has a finite limit or tends to infinity. (10 points)

(b) Prove that for all $\alpha > 1$ there exists a sequence $a_1, a_2, \dots$ with the same properties such that

$$\lim \frac{a_n}{\left(\frac{3}{2}\right)^{n-1}} = \alpha.$$

(10 points)

Open all solution variants | Source note

2003-D1-P2

Long/deep

Fingerprint field characteristic when sum-replacement permutes 51 elements
Primary domain Algebra / number theory
Problem type classification/find-all
Linked methods Compare coefficients after a structural substitution [M-AL-COEFFICIENT]; Work orbit by orbit and track parity [M-AL-ORBIT]
Secondary cues Real analysis / calculus; Combinatorics / discrete structures
Linked theory Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]
Recognition triggers integral/limit trigger; integer/prime/mod trigger; group/ring/field trigger; counting/permutation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2003-D1-P2

Day 1; official label P2; 20 points; legacy source-page label 88

Let $a_1, a_2 \dots, a_{51}$ be non-zero elements of a field. We simultaneously replace each element with the sum of the 50 remaining ones. In this way we get a sequence $b_1 \dots, b_{51}$. If this new sequence is a permutation of the original one, what can be the characteristic of the field? (The characteristic of a field is p, if p is the smallest positive integer such that $\underbrace{x + x + \cdots + x}_{p} = 0$ for any element x of the field. If there exists no such p, the characteristic is 0.)

Open all solution variants | Source note

2003-D1-P3

Long/deep

Fingerprint powers of matrix with 3A^3=A^2+A+I converge
Primary domain Linear algebra / matrices
Problem type recurrence/convergence
Linked methods Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]
Secondary cues Complex analysis / polynomials; Real analysis / calculus
Linked theory Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Operator norms and convergence of matrix powers [T-LA-NORM]; Nilpotents, projections and involutions [T-LA-NILPOTENT]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers matrix word
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Control the limiting operation before taking the limit [S-ANALYTIC]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2003-D1-P3

Day 1; official label P3; 20 points; legacy source-page label 88

Let A be an $n \times n$ real matrix such that $3A^3 = A^2 + A + I$ (I is the identity matrix). Show that the sequence A^k converges to an idempotent matrix. (A matrix B is called idempotent if $B^2 = B$.)

Open all solution variants | Source note

2003-D1-P4

Long/deep

Fingerprint partition positive integers so that aA=bB
Primary domain Algebra / number theory
Problem type classification/find-all
Linked methods Track one prime or one extreme divisor [M-AL-VALUATION]
Secondary cues Functional equations / dynamics; Linear algebra / matrix theory; Real analysis / calculus
Linked theory Prime valuations, divisors, and factorials [T-AL-VALUATION]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Recognition triggers det/rank/eigen trigger; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2003-D1-P4

Day 1; official label P4; 20 points; legacy source-page label 88

Determine the set of all pairs (a, b) of positive integers for which the set of positive integers can be decomposed into two sets A and B such that $a \cdot A = b \cdot B$.

Open all solution variants | Source note

2003-D1-P5

Long/deep

Fingerprint limit of iterated integral averages of continuous function
Primary domain Real analysis / calculus
Problem type limit/asymptotic evaluation
Linked methods Preserve structure while approximating [M-AN-APPROXIMATE]; Separate a main range from exceptional scales [M-AN-SPLIT]
Secondary cues Functional equations / dynamics; Geometry / convexity / metric; Combinatorics / discrete structures
Linked theory Uniform approximation and positive kernels [T-AN-APPROXIMATION]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Nonnegative sums, Fubini, and integral kernels [T-AN-FUBINI]
Recognition triggers det/rank/eigen trigger; continuous/differentiable trigger; integral/limit trigger; counting/permutation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Pass between local constraints and a global conclusion [S-LOCALGLOBAL]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2003-D1-P5

Day 1; official label P5; 20 points; legacy source-page label 89

Let $g : [0,1] \to \mathbb{R}$ be a continuous function and let $f_n : [0,1] \to \mathbb{R}$ be a sequence of functions defined by $f_0(x) = g(x)$ and

$$f_{n+1}(x) = \frac{1}{x} \int_0^x f_n(t)\, dt \quad (x \in (0,1],\ n = 0, 1, 2, \dots).$$

Determine $\lim_{n\to\infty} f_n(x)$ for every $x \in (0,1]$.

Open all solution variants | Source note

2003-D1-P6 Long/deep

Fingerprint coefficient inequalities for left-half-plane root polynomials
Primary domain Complex analysis / polynomials
Problem type inequality/sharp bound
Linked methods Compare coefficients after a structural substitution [M-AL-COEFFICIENT]
Secondary cues Real analysis / calculus; Geometry / convexity / metric; Functional equations / dynamics
Linked theory Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Recognition triggers polynomial/root trigger; functional-equation trigger
Preparation route CP-Polynomials/Complex
Trap watch Separate polynomial identity from pointwise testing; check multiplicities, zero polynomial cases, and real-vs-complex roots.
Proof strategies Reduce size while preserving the defining conditions [S-INDUCTION]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2003-D1-P6

Day 1; official label P6; 20 points; legacy source-page label 91

Let $f(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$ be a polynomial with real coefficients. Prove that if all roots of f lie in the left half-plane $\{z \in \mathbb{C} : \operatorname{Re} z < 0\}$ then

$$a_k a_{k+3} < a_{k+1} a_{k+2}$$

holds for every $k = 0, 1, \ldots, n-3$.

Open all solution variants | Source note

Day 2

2003-D2-P1 Quick/standard

Fingerprint AB+A+B=0 implies matrices commute
Primary domain Linear algebra / matrices
Problem type structural proof
Linked methods Complete the equation to a product equal to identity [M-LA-INVERSE]
Secondary cues None recorded
Linked theory Determinants, block identities and inverses [T-LA-DETERMINANT]
Recognition triggers matrix word
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2003-D2-P1

Day 2; official label P7; legacy source-page label 92

Let A and B be $n \times n$ real matrices such that $AB + A + B = 0$. Prove that $AB = BA$.

Open all solution variants | Source note

2003-D2-P2 Quick/standard

Fingerprint limit of integral of sin^m(t)/t^n over [x,2x]
Primary domain Real analysis / calculus
Problem type limit/asymptotic evaluation
Linked methods Separate a main range from exceptional scales [M-AN-SPLIT]; Pair an integral through symmetry [M-AN-PAIR]
Secondary cues None recorded
Linked theory Trigonometric identities and safe branches [T-AN-TRIG]; Scale separation and limit comparisons [T-AN-ASYMPTOTICS]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Recognition triggers integral/limit trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2003-D2-P2

Day 2; official label P8; legacy source-page label 92

Evaluate the limit

$$\lim_{x\to 0+}\int_x^{2x}\frac{\sin^m t}{t^n}\,dt \quad (m,n\in\mathbb{N}).$$

Open all solution variants | Source note

2003-D2-P3 Medium

Fingerprint points with unique nearest point in closed set dense
Primary domain Geometry / convexity / metric
Problem type structural proof
Linked methods Push covers forward or move toward a nearest point [M-DS-METRIC]
Secondary cues Geometry / convexity / metric; Real analysis / calculus
Linked theory Metric contractions, covers, and nearest points [T-DS-METRIC]; Continuity, compactness, and order [T-AN-CONTINUITY]; Inner products, orthogonality and polarization [T-LA-INNER]
Recognition triggers geometry/metric trigger
Preparation route CMB-Combinatoricsi
Trap watch Avoid overcounting; define the encoded object precisely and check extremal/minimal cases.
Proof strategies Translate geometry into a controlled representation [S-GEOMETRIC]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2003-D2-P3

Day 2; official label P9; legacy source-page label 92

Let A be a closed subset of $\mathbb{R}^n$ and let B be the set of all those points $b\in\mathbb{R}^n$ for which there exists exactly one point $a_0\in A$ such that

$$|a_0-b|=\inf_{a\in A}|a-b|.$$

Prove that B is dense in $\mathbb{R}^n$; that is, the closure of B is $\mathbb{R}^n$.

Open all solution variants | Source note

2003-D2-P4 Medium/trap

Fingerprint triple systems covering pairs with closure condition
Primary domain Combinatorics
Problem type classification/find-all
Linked methods Choose information that makes an encoding injective [M-DS-ENCODE]
Secondary cues Combinatorics / discrete structures

2003-D2-P4 continued

Linked theory Finite set systems and antichains [T-DS-SETS]; Groups, orders, generators, and quotients [T-AL-GROUP]; Rank, dimension and exact linear constraints [T-LA-RANK]
Recognition triggers integer/prime/mod trigger; graph/color/set trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2003-D2-P4

Day 2; official label P10; legacy source-page label 93

Find all positive integers n for which there exists a family F of three-element subsets of $S = \{1, 2, \dots, n\}$ satisfying the following two conditions:
(i) for any two different elements $a, b \in S$, there exists exactly one $A \in F$ containing both a, b;
(ii) if a, b, c, x, y, z are elements of S such that if

$\{a, b, x\}, \{a, c, y\}, \{b, c, z\} \in F$, then $\{x, y, z\} \in F$.

Open all solution variants | Source note

2003-D2-P5 Long/deep

Fingerprint dominating two-variable function by g(x)+g(y), Q versus R
Primary domain Real analysis / calculus
Problem type existence/construction
Linked methods Use a dense set, an enumeration, or a category argument [M-AN-TOPOLOGY]
Secondary cues Functional equations / dynamics; Combinatorics / discrete structures
Linked theory Countability, accumulation, and Cantor constructions [T-AN-SETS]; Continuity, compactness, and order [T-AN-CONTINUITY]
Recognition triggers functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a witness and verify every constraint [S-CONSTRUCTION]; Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2003-D2-P5

Day 2; official label P11; legacy source-page label 94

(a) Show that for each function $f : \mathbb{Q} \times \mathbb{Q} \to \mathbb{R}$ there exists a function $g : \mathbb{Q} \to \mathbb{R}$ such that $f(x, y) \leq g(x) + g(y)$ for all $x, y \in \mathbb{Q}$.

(b) Find a function $f : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ for which there is no function $g : \mathbb{R} \to \mathbb{R}$ such that $f(x, y) \leq g(x) + g(y)$ for all $x, y \in \mathbb{R}$.

Open all solution variants | Source note

2003-D2-P6 Long/deep

Fingerprint limit of weighted sum of convolution-recursive sequence
Primary domain Real analysis / calculus
Problem type limit/asymptotic evaluation
Linked methods Encode convolution or denominators by a power series [M-AN-GENERATING]
Secondary cues Functional equations / dynamics; Combinatorics / discrete structures
Linked theory Power series and generating functions [T-AN-POWER]; Differential inequalities and integrating factors [T-AN-ODE]; Series, finite cancellation, and summation by parts [T-AN-SERIES]
Recognition triggers integral/limit trigger

2003-D2-P6 continued

Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2003-D2-P6

Day 2; official label P12; legacy source-page label 94

Let $(a_n)_{n\in\mathbb{N}}$ be the sequence defined by

$$a_0 = 1, \qquad a_{n+1} = \frac{1}{n+1}\sum_{k=0}^{n}\frac{a_k}{n-k+2}.$$

Find the limit

$$\lim_{n\to\infty}\sum_{k=0}^{n}\frac{a_k}{2^k},$$

if it exists.

Open all solution variants | Source note

2.11 2004

Day 1

2004-D1-P1 Long/deep

Fingerprint bounded finite-subset sums force set countability
Primary domain Combinatorics
Problem type counting/enumeration
Linked methods Split a family by traces, pieces, or labels [M-DS-PARTITION]
Secondary cues Linear algebra / matrix theory; Algebra / number theory; Geometry / convexity / metric
Linked theory Countability, accumulation, and Cantor constructions [T-AN-SETS]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers counting/permutation trigger
Preparation route CMB-Combinatoricsi
Trap watch Avoid overcounting; define the encoded object precisely and check extremal/minimal cases.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2004-D1-P1

Day 1; official label P1; 20 points; legacy source-page label 96

Let S be an infinite set of real numbers such that $|s_1 + s_2 + \cdots + s_k| < 1$ for every finite subset $\{s_1, s_2, \ldots, s_k\} \subset S$. Show that S is countable.

Open all solution variants | Source note

2004-D1-P2 Long/deep

Fingerprint count real roots of 2004-fold iterate of x^2-1
Primary domain Functional equations / dynamics
Problem type counting/enumeration
Linked methods Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]; Strengthen the statement before the induction step [M-DS-INDUCT]

2004-D1-P2 continued

Secondary cues Combinatorics / discrete structures; Geometry / convexity / metric
Linked theory Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Finite counting and incidence identities [T-DS-COUNTING]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers counting/permutation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Reduce size while preserving the defining conditions [S-INDUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2004-D1-P2

Day 1; official label P2; 20 points; legacy source-page label 96

Let $P(x) = x^2 - 1$. How many distinct real solutions does the following equation have:

$$\underbrace{P(P(\dots(P}_{2004}(x))\dots)) = 0\ ?$$

Open all solution variants | Source note

2004-D1-P3 Medium

Fingerprint range of angle sums given sine sum equals one
Primary domain Real analysis / calculus
Problem type limit/asymptotic evaluation
Linked methods Choose the correct convexity comparison [M-AN-CONVEX]; Localize an extremum or an accumulation point [M-AN-COMPACT]
Secondary cues Combinatorics / discrete structures; Linear algebra / matrix theory; Functional equations / dynamics
Linked theory Continuity, compactness, and order [T-AN-CONTINUITY]; Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Trigonometric identities and safe branches [T-AN-TRIG]; Scale separation and limit comparisons [T-AN-ASYMPTOTICS]
Recognition triggers integral/limit trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Choose an extremal object and force rigidity [S-EXTREMAL]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2004-D1-P3

Day 1; official label P3; 10+10 points; legacy source-page label 96

Let S_n be the set of all sums $\sum_{k=1}^{n} x_k$, where $n \geq 2$, $0 \leq x_1, x_2, \dots, x_n \leq \frac{\pi}{2}$ and

$$\sum_{k=1}^{n} \sin x_k = 1.$$

a) Show that S_n is an interval. [10 points]

b) Let l_n be the length of S_n. Find $\lim_{n\to\infty} l_n$. [10 points]

Open all solution variants | Source note

2004-D1-P4

Long/deep

Fingerprint half-white sphere intersections force points on one sphere
Primary domain Geometry / convexity / metric
Problem type structural proof
Linked methods Count incidences with their exact multiplicity [M-DS-DOUBLECOUNT]; Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Secondary cues Combinatorics / discrete structures; Real analysis / calculus
Linked theory Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Finite counting and incidence identities [T-DS-COUNTING]; Rank, dimension and exact linear constraints [T-LA-RANK]
Recognition triggers integral/limit trigger; graph/color/set trigger; geometry/metric trigger
Preparation route GEO-Geometry/Metricif
Trap watch Control degenerate configurations and make clear which coordinates or affine transformations preserve the quantity.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Make the contradiction quantitative or structural [S-CONTRADICTION]; Translate geometry into a controlled representation [S-GEOMETRIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2004-D1-P4

Day 1; official label P4; 20 points; legacy source-page label 97

Suppose $n \geq 4$ and let M be a finite set of n points in $\mathbb{R}^3$, no four of which lie in a plane. Assume that the points can be coloured black or white so that any sphere which intersects M in at least four points has the property that exactly half of the points in the intersection of M and the sphere are white. Prove that all of the points in M lie on one sphere.

Open all solution variants | Source note

2004-D1-P5

Long/deep

Fingerprint Ramsey-type: monotone sequence with doubling gaps exists
Primary domain Combinatorics
Problem type existence/construction
Linked methods Split a family by traces, pieces, or labels [M-DS-PARTITION]; Strengthen the statement before the induction step [M-DS-INDUCT]
Secondary cues Combinatorics / discrete structures
Linked theory Order, colouring, and binomial threshold induction [T-DS-RAMSEY]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers wording-specific
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Reduce size while preserving the defining conditions [S-INDUCTION]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2004-D1-P5

Day 1; official label P5; 20 points; legacy source-page label 97

Let X be a set of $\binom{2k-4}{k-2}+1$ real numbers, $k \geq 2$. Prove that there exists a monotone sequence $\{x_i\}_{i=1}^{k} \subseteq X$ such that

$$|x_{i+1}-x_1| \geq 2|x_i-x_1|$$

for all $i = 2, \ldots, k-1$.

Open all solution variants | Source note

2004-D1-P6 Long/deep

Fingerprint sum of (log z)^-4 over branches is rational
Primary domain Complex analysis / polynomials
Problem type exact computation
Linked methods Build a holomorphic function with a useful boundary [M-AL-ANALYTIC]
Secondary cues Complex analysis / polynomials; Functional equations / dynamics; Geometry / convexity / metric
Linked theory Holomorphic bounds, poles, and logarithmic-branch sums [T-AL-COMPLEXANALYSIS]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers integral/limit trigger; polynomial/root trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Control the limiting operation before taking the limit [S-ANALYTIC]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2004-D1-P6

Day 1; official label P6; 10+10 points; legacy source-page label 98

For every complex number $z \notin \{0,1\}$ define

$$f(z) := \sum (\log z)^{-4},$$

where the sum is over all branches of the complex logarithm.
a) Show that there are two polynomials P and Q such that $f(z) = P(z)/Q(z)$ for all $z \in \mathbb{C} \setminus \{0,1\}$. [10 points]

b) Show that for all $z \in \mathbb{C} \setminus \{0,1\}$

$$f(z) = z\,\frac{z^2+4z+1}{6(z-1)^4}.$$

[10 points]

Open all solution variants | Source note

Day 2

2004-D2-P1 Long/deep

Fingerprint recover BA from given product AB
Primary domain Linear algebra / matrices
Problem type exact computation
Linked methods Exploit a compatible block decomposition [M-LA-BLOCK]; Complete the equation to a product equal to identity [M-LA-INVERSE]
Secondary cues None recorded
Linked theory Determinants, block identities and inverses [T-LA-DETERMINANT]; Rank, dimension and exact linear constraints [T-LA-RANK]
Recognition triggers matrix word
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

2004-D2-P1 continued

Full statement: 2004-D2-P1

Day 2; official label P7; 20 points; legacy source-page label 100

Let A be a real 4×2 matrix and B be a real 2×4 matrix such that

$$AB = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}.$$

Find BA.

Open all solution variants | Source note

2004-D2-P2 Long/deep

Fingerprint compare integrals of sqrt(1+f) given sqrt(f) constraints
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Choose the correct convexity comparison [M-AN-CONVEX]; Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]
Secondary cues Real analysis / calculus; Combinatorics / discrete structures; Geometry / convexity / metric
Linked theory Convex graph length and ordered primitives [T-AN-ARCLENGTH]; Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]
Recognition triggers continuous/differentiable trigger; functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Translate geometry into a controlled representation [S-GEOMETRIC]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2004-D2-P2

Day 2; official label P8; 20 points; legacy source-page label 100

Let $f, g : [a, b] \to [0, \infty)$ be continuous and non-decreasing functions such that for each $x \in [a, b]$ we have

$$\int_a^x \sqrt{f(t)}\, dt \leq \int_a^x \sqrt{g(t)}\, dt$$

and $\int_a^b \sqrt{f(t)}\, dt = \int_a^b \sqrt{g(t)}\, dt$.

Prove that $\int_a^b \sqrt{1 + f(t)}\, dt \geq \int_a^b \sqrt{1 + g(t)}\, dt$.

Open all solution variants | Source note

2004-D2-P3 Long/deep

Fingerprint point in disk with large distance sum to n points
Primary domain Geometry / convexity / metric
Problem type existence/construction
Linked methods Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Secondary cues Real analysis / calculus; Algebra / number theory
Linked theory Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Inner products, orthogonality and polarization [T-LA-INNER]
Recognition triggers integral/limit trigger; geometry/metric trigger
Preparation route GEO-Geometry/Metricif
Trap watch Control degenerate configurations and make clear which coordinates or affine transformations preserve the quantity.

2004-D2-P3 continued

Proof strategies Translate geometry into a controlled representation [S-GEOMETRIC]; Build a witness and verify every constraint [S-CONSTRUCTION]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2004-D2-P3

Day 2; official label P9; 20 points; legacy source-page label 100

Let D be the closed unit disk in the plane, and let $p_1, p_2, \ldots, p_n$ be fixed points in D. Show that there exists a point p in D such that the sum of the distances of p to each of $p_1, p_2, \ldots, p_n$ is greater than or equal to 1.

Open all solution variants | Source note

2004-D2-P4 Long/deep

Fingerprint eigenvalues of operator X -> MX+XM^T
Primary domain Linear algebra / matrices
Problem type exact computation
Linked methods Treat the unknown matrix as a vector [M-LA-MATRIXOP]; Lift products of inner products to new vectors [M-LA-TENSOR]
Secondary cues Complex analysis / polynomials; Geometry / convexity / metric; Real analysis / calculus
Linked theory Tensor products and operators on matrix spaces [T-LA-TENSOR]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Continuity, compactness, and order [T-AN-CONTINUITY]
Recognition triggers matrix word; det/rank/eigen trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2004-D2-P4

Day 2; official label P10; 20 points; legacy source-page label 101

For $n \geq 1$ let M be an $n \times n$ complex matrix with distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_k$, with multiplicities $m_1, m_2, \ldots, m_k$, respectively. Consider the linear operator L_M defined by $L_M(X) = MX + XM^T$, for any complex $n \times n$ matrix X. Find its eigenvalues and their multiplicities. (M^T denotes the transpose of M; that is, if $M = (m_{k,l})$, then $M^T = (m_{l,k})$.)

Open all solution variants | Source note

2004-D2-P5 Long/deep

Fingerprint bound double integral of 1/(1/x+|ln y|-1)
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]; Pair an integral through symmetry [M-AN-PAIR]; Choose the correct convexity comparison [M-AN-CONVEX]
Secondary cues None recorded
Linked theory Nonnegative sums, Fubini, and integral kernels [T-AN-FUBINI]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]
Recognition triggers wording-specific
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Control the limiting operation before taking the limit [S-ANALYTIC]

2004-D2-P5 continued

Navigation Full statement All solution variants Source note Learning route

Full statement: 2004-D2-P5

Day 2; official label P11; 20 points; legacy source-page label 101

Prove that

$$\int_0^1 \int_0^1 \frac{dx\,dy}{x^{-1} + |\ln y| - 1} \leq 1.$$

Open all solution variants | Source note

2004-D2-P6 Long/deep

Fingerprint element sums of recursive block matrix powers coincide
Primary domain Linear algebra / matrices
Problem type counting/enumeration
Linked methods Interpret matrix multiplication as a count [M-LA-WALKS]; Exploit a compatible block decomposition [M-LA-BLOCK]
Secondary cues Combinatorics / discrete structures; Real analysis / calculus; Algebra / number theory
Linked theory Finite counting and incidence identities [T-DS-COUNTING]; Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]; Rank, dimension and exact linear constraints [T-LA-RANK]
Recognition triggers matrix word; integral/limit trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2004-D2-P6

Day 2; official label P12; 20 points; legacy source-page label 102

For $n \geq 0$ define matrices A_n and B_n as follows: $A_0 = B_0 = (1)$ and for every $n > 0$

$$A_n = \begin{pmatrix} A_{n-1} & A_{n-1} \\ A_{n-1} & B_{n-1} \end{pmatrix} \quad \text{and} \quad B_n = \begin{pmatrix} A_{n-1} & A_{n-1} \\ A_{n-1} & 0 \end{pmatrix}.$$

Denote the sum of all elements of a matrix M by $S(M)$. Prove that $S(A_n^{k-1}) = S(A_k^{n-1})$ for every $n, k \geq 1$.

Open all solution variants | Source note

2.12 2005

Day 1

2005-D1-P1 Quick/standard

Fingerprint rank of matrix with entries i+j
Primary domain Linear algebra / matrices
Problem type exact computation
Linked methods Prove a dimension obstruction and attain it [M-LA-DIMENSION]; Eliminate systematically and track the invariant [M-LA-ELIM]
Secondary cues Geometry / convexity / metric; Real analysis / calculus
Linked theory Rank, dimension and exact linear constraints [T-LA-RANK]; Determinants, block identities and inverses [T-LA-DETERMINANT]
Recognition triggers matrix word; det/rank/eigen trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2005-D1-P1

Day 1; official label P1; legacy source-page label 103

Let A be the $n \times n$ matrix, whose (i, j)th entry is $i + j$ for all $i, j = 1, 2, \ldots, n$. What is the rank of A?

Open all solution variants | Source note

2005-D1-P2 Quick/standard

Fingerprint count constrained ternary strings, prove cardinality identity
Primary domain Combinatorics
Problem type counting/enumeration
Linked methods Partition into states and propagate a recurrence [M-DS-STATE]; Encode, decode, and verify a bijection [M-DS-BIJECTION]; Strengthen the statement before the induction step [M-DS-INDUCT]
Secondary cues Real analysis / calculus; Geometry / convexity / metric; Combinatorics / discrete structures
Linked theory Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Recognition triggers integer/prime/mod trigger; counting/permutation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

2005-D1-P2 continued

Full statement: 2005-D1-P2

Day 1; official label P2; legacy source-page label 103

For an integer $n \geq 3$ consider the sets

$$S_n = \{(x_1, x_2, \dots, x_n) : \forall i \; x_i \in \{0, 1, 2\}\}$$

$$A_n = \{(x_1, x_2, \dots, x_n) \in S_n : \forall i \leq n-2 \; |\{x_i, x_{i+1}, x_{i+2}\}| \neq 1\}$$

and

$$B_n = \{(x_1, x_2, \dots, x_n) \in S_n : \forall i \leq n-1 \; (x_i = x_{i+1} \Rightarrow x_i \neq 0)\}.$$

Prove that $|A_{n+1}| = 3 \cdot |B_n|$.

($|A|$ denotes the number of elements of the set A.)

Open all solution variants | Source note

2005-D1-P3 Medium

Fingerprint bound integral expression via maximum of derivative
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]
Secondary cues Functional equations / dynamics
Linked theory Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Recognition triggers continuous/differentiable trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2005-D1-P3

Day 1; official label P3; legacy source-page label 104

Let $f : \mathbb{R} \to [0, \infty)$ be a continuously differentiable function. Prove that

$$\left| \int_0^1 f^3(x)\,dx - f^2(0) \int_0^1 f(x)\,dx \right| \leq \max_{0 \leq x \leq 1} |f'(x)| \left(\int_0^1 f(x)\,dx \right)^2.$$

Open all solution variants | Source note

2005-D1-P4 Long/deep

Fingerprint polynomials with permuted coefficients and all rational roots
Primary domain Complex analysis / polynomials
Problem type classification/find-all
Linked methods Translate root restrictions into symmetric coefficients [M-AL-VIETA]; Factor first and compare irreducible exponents [M-AL-FACTOR]
Secondary cues Real analysis / calculus; Algebra / number theory; Combinatorics / discrete structures
Linked theory Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]; Polynomial factorization and coprime divisibility [T-AL-FACTORIZATION]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Prime valuations, divisors, and factorials [T-AL-VALUATION]
Recognition triggers polynomial/root trigger; counting/permutation trigger
Preparation route CP-Polynomials/Complex

2005-D1-P4 continued

Trap watch Separate polynomial identity from pointwise testing; check multiplicities, zero polynomial cases, and real-vs-complex roots.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2005-D1-P4

Day 1; official label P4; legacy source-page label 105

Find all polynomials $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ ($a_n \neq 0$) satisfying the following two conditions:
(i) $(a_0, a_1, \ldots, a_n)$ is a permutation of the numbers $(0, 1, \ldots, n)$
and

(ii) all roots of $P(x)$ are rational numbers.

Open all solution variants | Source note

2005-D1-P5 Long/deep

Fingerprint differential inequality forces f to vanish at infinity
Primary domain Real analysis / calculus
Problem type limit/asymptotic evaluation
Linked methods Design an auxiliary derivative [M-AN-AUXILIARY]; Compare growth through derivatives [M-AN-RATIO]
Secondary cues Functional equations / dynamics
Linked theory Differential inequalities and integrating factors [T-AN-ODE]; Scale separation and limit comparisons [T-AN-ASYMPTOTICS]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Recognition triggers continuous/differentiable trigger; integral/limit trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Control the limiting operation before taking the limit [S-ANALYTIC]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2005-D1-P5

Day 1; official label P5; legacy source-page label 106

Let $f : (0, \infty) \to \mathbb{R}$ be a twice continuously differentiable function such that

$$|f''(x) + 2xf'(x) + (x^2 + 1)f(x)| \leq 1$$

for all x. Prove that $\lim_{x\to\infty} f(x) = 0$.

Open all solution variants | Source note

2005-D1-P6 Long/deep

Fingerprint subgroups generated by mth powers, commutativity under gcd
Primary domain Algebra / number theory
Problem type structural proof
Linked methods Polarize identities and isolate commutators [M-AL-COMMUTATOR]; Prove generation or obstruct it by an image [M-AL-GENERATORS]
Secondary cues Functional equations / dynamics; Linear algebra / matrix theory; Combinatorics / discrete structures
Linked theory Groups, orders, generators, and quotients [T-AL-GROUP]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Recognition triggers group/ring/field trigger

2005-D1-P6 continued

Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2005-D1-P6

Day 1; official label P6; legacy source-page label 106

Given a group G, denote by $G(m)$ the subgroup generated by the mth powers of elements of G. If $G(m)$ and $G(n)$ are commutative, prove that $G(\gcd(m,n))$ is also commutative. ($\gcd(m,n)$ denotes the greatest common divisor of m and n.)

Open all solution variants | Source note

Day 2

2005-D2-P1 Quick/standard

Fingerprint total length where quadratic has modulus below one
Primary domain Complex analysis / polynomials
Problem type inequality/sharp bound
Linked methods Compare coefficients after a structural substitution [M-AL-COEFFICIENT]
Secondary cues Functional equations / dynamics
Linked theory Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]
Recognition triggers integral/limit trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Prove necessity and sufficiency separately [S-CLASSIFICATION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2005-D2-P1

Day 2; official label P7; legacy source-page label 107

Let $f(x) = x^2 + bx + c$, where b and c are real numbers, and let

$$M = \{x \in \mathbb{R} : |f(x)| < 1\}.$$

Clearly the set M is either empty or consists of disjoint open intervals. Denote the sum of their lengths by $|M|$. Prove that

$$|M| \leq 2\sqrt{2}.$$

Open all solution variants | Source note

2005-D2-P2 Medium/trap

Fingerprint must f be polynomial if all powers are
Primary domain Complex analysis / polynomials
Problem type structural proof
Linked methods Factor first and compare irreducible exponents [M-AL-FACTOR]
Secondary cues Algebra / number theory; Real analysis / calculus; Functional equations / dynamics
Linked theory Polynomial factorization and coprime divisibility [T-AL-FACTORIZATION]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers polynomial/root trigger; functional-equation trigger
Preparation route CP-Polynomials/Complex

2005-D2-P2 continued

Trap watch Separate polynomial identity from pointwise testing; check multiplicities, zero polynomial cases, and real-vs-complex roots.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2005-D2-P2

Day 2; official label P8; legacy source-page label 107

Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $(f(x))^n$ is a polynomial for every $n = 2, 3, \ldots$. Does it follow that f is a polynomial?

Open all solution variants | Source note

2005-D2-P3 Medium

Fingerprint max dimension of subspace with trace(XY)=0
Primary domain Linear algebra / matrices
Problem type inequality/sharp bound
Linked methods Prove a dimension obstruction and attain it [M-LA-DIMENSION]; Use trace to extract a scalar invariant [M-LA-TRACE]
Secondary cues Real analysis / calculus
Linked theory Trace pairing and commutator calculus [T-LA-TRACE]; Rank, dimension and exact linear constraints [T-LA-RANK]; Inner products, orthogonality and polarization [T-LA-INNER]
Recognition triggers matrix word; det/rank/eigen trigger; integral/limit trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2005-D2-P3

Day 2; official label P9; legacy source-page label 108

In the linear space of all real $n \times n$ matrices, find the maximum possible dimension of a linear subspace V such that

$$\forall X, Y \in V \quad \operatorname{trace}(XY) = 0.$$

(The trace of a matrix is the sum of the diagonal entries.)

Open all solution variants | Source note

2005-D2-P4 Medium

Fingerprint intermediate point identity for third derivative
Primary domain Real analysis / calculus
Problem type existence/construction
Linked methods Count zeros and subtract an interpolant [M-AN-ROLLE]
Secondary cues Real analysis / calculus; Combinatorics / discrete structures
Linked theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers continuous/differentiable trigger; functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

2005-D2-P4 continued

Full statement: 2005-D2-P4

Day 2; official label P10; legacy source-page label 108

Prove that if $f : \mathbb{R} \to \mathbb{R}$ is three times differentiable, then there exists a real number $\xi \in (-1, 1)$ such that

$$\frac{f'''(\xi)}{6} = \frac{f(1) - f(-1)}{2} - f'(0).$$

Open all solution variants | Source note

2005-D2-P5 Long/deep

Fingerprint radii where Lipschitz-gradient function attains unique maximum
Primary domain Geometry / convexity / metric
Problem type classification/find-all
Linked methods Contradict an extremum by a controlled perturbation [M-DS-PERTURB]; Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Secondary cues Real analysis / calculus; Geometry / convexity / metric
Linked theory Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Metric contractions, covers, and nearest points [T-DS-METRIC]; Lipschitz derivatives and convex smoothness [T-AN-SMOOTH]; Continuity, compactness, and order [T-AN-CONTINUITY]
Recognition triggers continuous/differentiable trigger; functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Translate geometry into a controlled representation [S-GEOMETRIC]; Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2005-D2-P5

Day 2; official label P11; legacy source-page label 109

Find all $r > 0$ such that whenever $f : \mathbb{R}^2 \to \mathbb{R}$ is a differentiable function such that $|\operatorname{grad} f(0,0)| = 1$ and $|\operatorname{grad} f(u) - \operatorname{grad} f(v)| \le |u - v|$ for all $u, v \in \mathbb{R}^2$, then the maximum of f on the disk $\{u \in \mathbb{R}^2 : |u| \le r\}$ is attained at exactly one point. ($\operatorname{grad} f(u) = (\partial_1 f(u), \partial_2 f(u))$ is the gradient vector of f at the point u. For a vector $u = (a, b)$, $|u| = \sqrt{a^2 + b^2}$.)

Open all solution variants | Source note

2005-D2-P6 Long/deep

Fingerprint integer Mobius transformation fixing quadratic irrational
Primary domain Algebra / number theory
Problem type existence/construction
Linked methods Construct roots or units before the coefficients [M-AL-PARAMETRIZE]; Find a recurrence that reveals the invariant [M-AL-RECURRENCE]
Secondary cues Linear algebra / matrix theory; Complex analysis / polynomials; Real analysis / calculus
Linked theory Quadratic norms and Pell-type constructions [T-AL-PELL]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Determinants, block identities and inverses [T-LA-DETERMINANT]
Recognition triggers matrix word; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

2005-D2-P6 continued

Full statement: 2005-D2-P6

Day 2; official label P12; legacy source-page label 109

Prove that if p and q are rational numbers and $r = p + q\sqrt{7}$, then there exists a matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \neq \pm \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ with integer entries and with $ad - bc = 1$ such that

$$\frac{ar + b}{cr + d} = r.$$

Open all solution variants | Source note

2.13 2006

Day 1

2006-D1-P1 Long/deep

Fingerprint continuity, monotonicity, surjectivity implications for real functions
Primary domain Real analysis / calculus
Problem type nonexistence/counterexample
Linked methods Use a dense set, an enumeration, or a category argument [M-AN-TOPOLOGY]
Secondary cues Functional equations / dynamics; Combinatorics / discrete structures; Geometry / convexity / metric
Linked theory Continuity, compactness, and order [T-AN-CONTINUITY]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Recognition triggers continuous/differentiable trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2006-D1-P1

Day 1; official label P1; 20 points; legacy source-page label 111

Let $f : \mathbb{R} \to \mathbb{R}$ be a real function. Prove or disprove each of the following statements.
(a) If f is continuous and range$(f) = \mathbb{R}$ then f is monotonic.
(b) If f is monotonic and range$(f) = \mathbb{R}$ then f is continuous.

(c) If f is monotonic and f is continuous then range$(f) = \mathbb{R}$.

Open all solution variants | Source note

2006-D1-P2 Long/deep

Fingerprint count x below 10^2006 with x^2-x divisible
Primary domain Algebra / number theory
Problem type counting/enumeration
Linked methods Choose a modulus that exposes the obstruction [M-AL-MODULAR]; Construct and count compatible local residues [M-AL-CRT]
Secondary cues Real analysis / calculus; Combinatorics / discrete structures
Linked theory Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Prime valuations, divisors, and factorials [T-AL-VALUATION]
Recognition triggers integer/prime/mod trigger; counting/permutation trigger

2006-D1-P2 continued

Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Pass between local constraints and a global conclusion [S-LOCALGLOBAL]; Choose the objects and count them without ambiguity [S-COUNTING]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2006-D1-P2

Day 1; official label P2; 20 points; legacy source-page label 111

Find the number of positive integers x satisfying the following two conditions:
1. $x < 10^{2006}$;

2. $x^2 - x$ is divisible by 10^{2006}.

Open all solution variants | Source note

2006-D1-P3 Long/deep

Fingerprint factor integer matrix matching determinant factorization
Primary domain Algebra / number theory
Problem type existence/construction
Linked methods Track one prime or one extreme divisor [M-AL-VALUATION]
Secondary cues Linear algebra / matrix theory; Real analysis / calculus; Geometry / convexity / metric
Linked theory Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Prime valuations, divisors, and factorials [T-AL-VALUATION]; Determinants, block identities and inverses [T-LA-DETERMINANT]
Recognition triggers matrix word; det/rank/eigen trigger; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Pass between local constraints and a global conclusion [S-LOCALGLOBAL]; Reduce size while preserving the defining conditions [S-INDUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2006-D1-P3

Day 1; official label P3; 20 points; legacy source-page label 111

Let A be an $n \times n$-matrix with integer entries and $b_1, \ldots, b_k$ be integers satisfying $\det A = b_1 \cdot \cdots \cdot b_k$. Prove that there exist $n \times n$-matrices $B_1, \ldots, B_k$ with integer entries such that $A = B_1 \cdot \cdots \cdot B_k$ and $\det B_i = b_i$ for all $i = 1, \ldots, k$.

Open all solution variants | Source note

2006-D1-P4 Long/deep

Fingerprint integer-valued rational function must be polynomial
Primary domain Complex analysis / polynomials
Problem type structural proof
Linked methods Eliminate rational relations, then exploit discreteness [M-AL-RATIONAL]
Secondary cues Algebra / number theory; Functional equations / dynamics; Real analysis / calculus
Linked theory Rational linear relations and discrete-value rigidity [T-AL-RATIONALITY]; Polynomial factorization and coprime divisibility [T-AL-FACTORIZATION]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Rank, dimension and exact linear constraints [T-LA-RANK]
Recognition triggers polynomial/root trigger; integer/prime/mod trigger; functional-equation trigger
Preparation route CP-Polynomials/Complex
Trap watch Separate polynomial identity from pointwise testing; check multiplicities, zero polynomial cases, and real-vs-complex roots.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

2006-D1-P4 continued

Full statement: 2006-D1-P4

Day 1; official label P4; 20 points; legacy source-page label 112

Let f be a rational function (i.e. the quotient of two real polynomials) and suppose that $f(n)$ is an integer for infinitely many integers n. Prove that f is a polynomial.

Open all solution variants | Source note

2006-D1-P5 Long/deep

Fingerprint compare cubic power sums given quadratic, quartic equalities
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Count zeros and subtract an interpolant [M-AN-ROLLE]; Compare growth through derivatives [M-AN-RATIO]
Secondary cues Functional equations / dynamics
Linked theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers continuous/differentiable trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Make the contradiction quantitative or structural [S-CONTRADICTION]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2006-D1-P5

Day 1; official label P5; 20 points; legacy source-page label 112

Let $a, b, c, d, e > 0$ be real numbers such that $a^2 + b^2 + c^2 = d^2 + e^2$ and $a^4 + b^4 + c^4 = d^4 + e^4$. Compare the numbers $a^3 + b^3 + c^3$ and $d^3 + e^3$.

Open all solution variants | Source note

2006-D1-P6 Long/deep

Fingerprint coefficient sequences making generalized Rolle theorem hold
Primary domain Real analysis / calculus
Problem type classification/find-all
Linked methods Count zeros and subtract an interpolant [M-AN-ROLLE]; Use a first crossing or a forced-growth contradiction [M-AN-BARRIER]
Secondary cues Functional equations / dynamics; Complex analysis / polynomials; Linear algebra / matrix theory
Linked theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Differential inequalities and integrating factors [T-AN-ODE]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers continuous/differentiable trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

2006-D1-P6 continued

Full statement: 2006-D1-P6

Day 1; official label P6; 20 points; legacy source-page label 113

Find all sequences $a_0, a_1, \ldots, a_n$ of real numbers where $n \geq 1$ and $a_n \neq 0$, for which the following statement is true:

If $f : \mathbb{R} \to \mathbb{R}$ is an n times differentiable function and $x_0 < x_1 < \cdots < x_n$ are real numbers such that $f(x_0) = f(x_1) = \cdots = f(x_n) = 0$ then there exists an $h \in (x_0, x_n)$ for which

$$a_0 f(h) + a_1 f'(h) + \cdots + a_n f^{(n)}(h) = 0.$$

Open all solution variants | Source note

Day 2

2006-D2-P1 Long/deep

Fingerprint polygon triangulations with vertex-degree parity conditions
Primary domain Geometry / convexity / metric
Problem type existence/construction
Linked methods Strengthen the statement before the induction step [M-DS-INDUCT]; Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Secondary cues Combinatorics / discrete structures; Geometry / convexity / metric; Real analysis / calculus
Linked theory Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Recognition triggers integer/prime/mod trigger; counting/permutation trigger; graph/color/set trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Reduce size while preserving the defining conditions [S-INDUCTION]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2006-D2-P1

Day 2; official label P7; 20 points; legacy source-page label 114

Let V be a convex polygon with n vertices.
(a) Prove that if n is divisible by 3 then V can be triangulated (i.e. dissected into non-overlapping triangles whose vertices are vertices of V) so that each vertex of V is the vertex of an odd number of triangles.

(b) Prove that if n is not divisible by 3 then V can be triangulated so that there are exactly two vertices that are the vertices of an even number of the triangles.

Open all solution variants | Source note

2006-D2-P2 Long/deep

Fingerprint functions mapping intervals to equal-length closed intervals
Primary domain Functional equations / dynamics
Problem type classification/find-all
Linked methods Force structure by admissible substitutions [M-DS-FUNCTIONAL]; Push covers forward or move toward a nearest point [M-DS-METRIC]
Secondary cues Real analysis / calculus; Linear algebra / matrix theory
Linked theory Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]; Metric contractions, covers, and nearest points [T-DS-METRIC]; Continuity, compactness, and order [T-AN-CONTINUITY]
Recognition triggers functional-equation trigger

2006-D2-P2 continued

Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2006-D2-P2

Day 2; official label P8; 20 points; legacy source-page label 114

Find all functions $f : \mathbb{R} \longrightarrow \mathbb{R}$ such that for any real numbers $a < b$, the image $f([a, b])$ is a closed interval of length $b - a$.

Open all solution variants | Source note

2006-D2-P3 Long/deep

Fingerprint compare tan(sin x) and sin(tan x)
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Exploit identities before estimating oscillations [M-AN-TRIG]; Choose the correct convexity comparison [M-AN-CONVEX]
Secondary cues Functional equations / dynamics
Linked theory Trigonometric identities and safe branches [T-AN-TRIG]; Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Recognition triggers wording-specific
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Prove necessity and sufficiency separately [S-CLASSIFICATION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2006-D2-P3

Day 2; official label P9; 20 points; legacy source-page label 115

Compare $\tan(\sin x)$ and $\sin(\tan x)$ for all $x \in (0, \frac{\pi}{2})$.

Open all solution variants | Source note

2006-D2-P4 Long/deep

Fingerprint rational pairwise distances force rational linear dependence
Primary domain Linear algebra / matrices
Problem type structural proof
Linked methods Encode distances and orthogonality by a Gram matrix [M-LA-GRAM]
Secondary cues Linear algebra / matrix theory; Real analysis / calculus; Geometry / convexity / metric
Linked theory Gram matrices and positive semidefinite forms [T-LA-GRAM]; Inner products, orthogonality and polarization [T-LA-INNER]; Determinants, block identities and inverses [T-LA-DETERMINANT]; Rank, dimension and exact linear constraints [T-LA-RANK]
Recognition triggers wording-specific
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

2006-D2-P4 continued

Full statement: 2006-D2-P4

Day 2; official label P10; 20 points; legacy source-page label 115

Let v_0 be the zero vector in $\mathbb{R}^n$ and let $v_1, v_2, \ldots, v_{n+1} \in \mathbb{R}^n$ be such that the Euclidean norm $|v_i - v_j|$ is rational for every $0 \leq i, j \leq n+1$. Prove that $v_1, \ldots, v_{n+1}$ are linearly dependent over the rationals.

Open all solution variants | Source note

2006-D2-P5 Long/deep

Fingerprint infinitely many coprime pairs giving three integer roots
Primary domain Algebra / number theory
Problem type existence/construction
Linked methods Construct roots or units before the coefficients [M-AL-PARAMETRIZE]; Translate root restrictions into symmetric coefficients [M-AL-VIETA]
Secondary cues Complex analysis / polynomials; Combinatorics / discrete structures; Real analysis / calculus
Linked theory Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]; Prime valuations, divisors, and factorials [T-AL-VALUATION]
Recognition triggers polynomial/root trigger; integer/prime/mod trigger; counting/permutation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2006-D2-P5

Day 2; official label P11; 20 points; legacy source-page label 116

Prove that there exists an infinite number of relatively prime pairs (m, n) of positive integers such that the equation

$$(x+m)^3 = nx$$

has three distinct integer roots.

Open all solution variants | Source note

2006-D2-P6 Long/deep

Fingerprint simultaneous similarity of three 2x2 matrix triples
Primary domain Linear algebra / matrices
Problem type existence/construction
Linked methods Choose a basis adapted to the structure [M-LA-BASIS]; Exploit a compatible block decomposition [M-LA-BLOCK]
Secondary cues None recorded
Linked theory Adapted bases, centralizers and common eigenspaces [T-LA-INVARIANT]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Trace pairing and commutator calculus [T-LA-TRACE]; Determinants, block identities and inverses [T-LA-DETERMINANT]
Recognition triggers matrix word; det/rank/eigen trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

2006-D2-P6 continued

Full statement: 2006-D2-P6

Day 2; official label P12; 20 points; legacy source-page label 116

Let A_i, B_i, S_i $(i = 1, 2, 3)$ be invertible real 2×2 matrices such that
(1) not all A_i have a common real eigenvector;
(2) $A_i = S_i^{-1} B_i S_i$ for all $i = 1, 2, 3$;
(3) $A_1 A_2 A_3 = B_1 B_2 B_3 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

Prove that there is an invertible real 2×2 matrix S such that $A_i = S^{-1} B_i S$ for all $i = 1, 2, 3$.

Open all solution variants | Source note

2.14 2007

Day 1

2007-D1-P1 Medium/trap

Fingerprint quadratic with all values divisible by five
Primary domain Algebra / number theory
Problem type structural proof
Linked methods Choose a modulus that exposes the obstruction [M-AL-MODULAR]; Replace many equal values by too many roots [M-AL-ROOTCOUNT]
Secondary cues Complex analysis / polynomials; Functional equations / dynamics
Linked theory Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]
Recognition triggers polynomial/root trigger; integer/prime/mod trigger; functional-equation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Pass between local constraints and a global conclusion [S-LOCALGLOBAL]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2007-D1-P1

Day 1; official label P1; legacy source-page label 118

Let f be a polynomial of degree 2 with integer coefficients. Suppose that $f(k)$ is divisible by 5 for every integer k. Prove that all coefficients of f are divisible by 5.

Open all solution variants | Source note

2007-D1-P2 Quick/standard

Fingerprint extremal ranks of matrix containing 1..n^2
Primary domain Linear algebra / matrices
Problem type inequality/sharp bound
Linked methods Prove a dimension obstruction and attain it [M-LA-DIMENSION]; Reduce integer matrices modulo an integer [M-LA-RESIDUE]
Secondary cues Linear algebra / matrix theory; Real analysis / calculus
Linked theory Rank, dimension and exact linear constraints [T-LA-RANK]; Determinants, block identities and inverses [T-LA-DETERMINANT]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Recognition triggers matrix word; det/rank/eigen trigger; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.

2007-D1-P2 continued

Proof strategies Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2007-D1-P2

Day 1; official label P2; legacy source-page label 118

Let $n \geq 2$ be an integer. What is the minimal and maximal possible rank of an $n \times n$ matrix whose n^2 entries are precisely the numbers $1, 2, \ldots, n^2$?

Open all solution variants | Source note

2007-D1-P3 Medium/trap

Fingerprint quadratic forms expressible as 2x2 determinant combinations
Primary domain Linear algebra / matrices
Problem type classification/find-all
Linked methods Prove a dimension obstruction and attain it [M-LA-DIMENSION]; Exploit a compatible block decomposition [M-LA-BLOCK]
Secondary cues Complex analysis / polynomials; Real analysis / calculus
Linked theory Rank, dimension and exact linear constraints [T-LA-RANK]; Determinants, block identities and inverses [T-LA-DETERMINANT]
Recognition triggers matrix word; det/rank/eigen trigger; polynomial/root trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Build a witness and verify every constraint [S-CONSTRUCTION]; Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2007-D1-P3

Day 1; official label P3; legacy source-page label 118

Call a polynomial $P(x_1, \ldots, x_k)$ good if there exist 2×2 real matrices $A_1, \ldots, A_k$ such that

$$P(x_1, \ldots, x_k) = \det\left(\sum_{i=1}^{k} x_i A_i\right).$$

Find all values of k for which all homogeneous polynomials with k variables of degree 2 are good. (A polynomial is homogeneous if each term has the same total degree.)

Open all solution variants | Source note

2007-D1-P4 Medium

Fingerprint count triples multiplying to identity across partition
Primary domain Algebra / number theory
Problem type counting/enumeration
Linked methods Convert group data into orbits or cosets [M-AL-ACTION]
Secondary cues Combinatorics / discrete structures
Linked theory Groups, orders, generators, and quotients [T-AL-GROUP]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers group/ring/field trigger; counting/permutation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]
Navigation Full statement All solution variants Source note Learning route

2007-D1-P4 continued

Full statement: 2007-D1-P4

Day 1; official label P4; legacy source-page label 119

Let G be a finite group. For arbitrary sets $U, V, W \subset G$, denote by N_{UVW} the number of triples $(x, y, z) \in U \times V \times W$ for which xyz is the unity.

Suppose that G is partitioned into three sets A, B and C (i.e. sets A, B, C are pairwise disjoint and $G = A \cup B \cup C$). Prove that $N_{ABC} = N_{CBA}$.

Open all solution variants | Source note

2007-D1-P5 Long/deep

Fingerprint vanishing arithmetic-progression sums force zero function
Primary domain Complex analysis / polynomials
Problem type structural proof
Linked methods Take the gcd of annihilating shift polynomials [M-AL-SHIFT]
Secondary cues Algebra / number theory; Real analysis / calculus; Linear algebra / matrix theory
Linked theory Finite differences and shift-operator ideals [T-AL-DIFFERENCE]; Polynomial factorization and coprime divisibility [T-AL-FACTORIZATION]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers integer/prime/mod trigger; functional-equation trigger
Preparation route CP-Polynomials/Complex
Trap watch Separate polynomial identity from pointwise testing; check multiplicities, zero polynomial cases, and real-vs-complex roots.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2007-D1-P5

Day 1; official label P5; legacy source-page label 119

Let n be a positive integer and $a_1, \ldots, a_n$ be arbitrary integers. Suppose that a function $f : \mathbb{Z} \to \mathbb{R}$ satisfies $\sum_{i=1}^{n} f(k + a_i \ell) = 0$ whenever k and ℓ are integers and $\ell \neq 0$. Prove that $f = 0$.

Open all solution variants | Source note

2007-D1-P6 Long/deep

Fingerprint max nonzero coefficients given bound on unit circle
Primary domain Complex analysis / polynomials
Problem type classification/find-all
Linked methods Evaluate at roots of unity or average over them [M-AL-UNITY]; Replace many equal values by too many roots [M-AL-ROOTCOUNT]
Secondary cues Combinatorics / discrete structures; Algebra / number theory; Real analysis / calculus
Linked theory Roots-of-unity filters and positive trigonometric kernels [T-AL-FOURIER]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]
Recognition triggers polynomial/root trigger; integer/prime/mod trigger; counting/permutation trigger
Preparation route CP-Polynomials/Complex
Trap watch Separate polynomial identity from pointwise testing; check multiplicities, zero polynomial cases, and real-vs-complex roots.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Choose an extremal object and force rigidity [S-EXTREMAL]
Navigation Full statement All solution variants Source note Learning route

2007-D1-P6 continued

Full statement: 2007-D1-P6

Day 1; official label P6; legacy source-page label 120

How many nonzero coefficients can a polynomial $P(z)$ have if its coefficients are integers and $|P(z)| \leq 2$ for any complex number z of unit length?

Open all solution variants | Source note

Day 2

2007-D2-P1 Quick/standard

Fingerprint must graphs movable onto scaled copies be lines
Primary domain Real analysis / calculus
Problem type nonexistence/counterexample
Linked methods Design an auxiliary derivative [M-AN-AUXILIARY]
Secondary cues Functional equations / dynamics
Linked theory Continuity, compactness, and order [T-AN-CONTINUITY]; Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]
Recognition triggers continuous/differentiable trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a witness and verify every constraint [S-CONSTRUCTION]
Correction note The explicit CLASS-001 corrections are incorporated in this card; the accompanying change ledger records the withdrawn labels.
Navigation Full statement All solution variants Source note Learning route

Full statement: 2007-D2-P1

Day 2; official label P7; legacy source-page label 121

Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function. Suppose that for any $c > 0$, the graph of f can be moved to the graph of cf using only a translation or a rotation. Does this imply that $f(x) = ax + b$ for some real numbers a and b?

Open all solution variants | Source note

2007-D2-P2 Quick/standard

Fingerprint fourth-power sum divisible by 29 forces 29^4
Primary domain Algebra / number theory
Problem type structural proof
Linked methods Choose a modulus that exposes the obstruction [M-AL-MODULAR]
Secondary cues None recorded
Linked theory Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Prime valuations, divisors, and factorials [T-AL-VALUATION]
Recognition triggers integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Pass between local constraints and a global conclusion [S-LOCALGLOBAL]
Correction note The explicit CLASS-001 corrections are incorporated in this card; the accompanying change ledger records the withdrawn labels.
Navigation Full statement All solution variants Source note Learning route

2007-D2-P2 continued

Full statement: 2007-D2-P2

Day 2; official label P8; legacy source-page label 121

Let x, y, and z be integers such that $S = x^4 + y^4 + z^4$ is divisible by 29. Show that S is divisible by 29^4.

Open all solution variants | Source note

2007-D2-P3 Medium

Fingerprint fixed point of monotone map on closed set
Primary domain Real analysis / calculus
Problem type existence/construction
Linked methods Localize an extremum or an accumulation point [M-AN-COMPACT]; Use a first crossing or a forced-growth contradiction [M-AN-BARRIER]
Secondary cues Functional equations / dynamics; Combinatorics / discrete structures
Linked theory Continuity, compactness, and order [T-AN-CONTINUITY]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers continuous/differentiable trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Choose an extremal object and force rigidity [S-EXTREMAL]; Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2007-D2-P3

Day 2; official label P9; legacy source-page label 121

Let C be a nonempty closed bounded subset of the real line and $f : C \to C$ be a nondecreasing continuous function. Show that there exists a point $p \in C$ such that $f(p) = p$.

(A set is closed if its complement is a union of open intervals. A function g is nondecreasing if $g(x) \leq g(y)$ for all $x \leq y$.)

Open all solution variants | Source note

2007-D2-P4 Medium

Fingerprint determinant of circulant-type 0-1-2 matrix
Primary domain Linear algebra / matrices
Problem type exact computation
Linked methods Eliminate systematically and track the invariant [M-LA-ELIM]; Exploit a compatible block decomposition [M-LA-BLOCK]
Secondary cues Linear algebra / matrix theory
Linked theory Determinants, block identities and inverses [T-LA-DETERMINANT]; Rank, dimension and exact linear constraints [T-LA-RANK]
Recognition triggers matrix word; det/rank/eigen trigger; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

2007-D2-P4 continued

Full statement: 2007-D2-P4

Day 2; official label P10; legacy source-page label 121

Let $n > 1$ be an odd positive integer and $A = (a_{ij})_{i,j=1\ldots n}$ be the $n \times n$ matrix with

$$a_{ij} = \begin{cases} 2 & \text{if } i = j \\ 1 & \text{if } i - j \equiv \pm 2 \pmod{n} \\ 0 & \text{otherwise.} \end{cases}$$

Find $\det A$.

Open all solution variants | Source note

2007-D2-P5 Long/deep

Fingerprint minimal size for commuting nilpotents with nonzero product
Primary domain Linear algebra / matrices
Problem type inequality/sharp bound
Linked methods Prove a dimension obstruction and attain it [M-LA-DIMENSION]; Choose a basis adapted to the structure [M-LA-BASIS]
Secondary cues Linear algebra / matrix theory; Real analysis / calculus; Combinatorics / discrete structures
Linked theory Nilpotents, projections and involutions [T-LA-NILPOTENT]; Rank, dimension and exact linear constraints [T-LA-RANK]; Dual bases, kernels and separating functionals [T-LA-DUALITY]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers matrix word; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2007-D2-P5

Day 2; official label P11; legacy source-page label 122

For each positive integer k, find the smallest number n_k for which there exist real $n_k \times n_k$ matrices $A_1, A_2, \ldots, A_k$ such that all of the following conditions hold:
(1) $A_1^2 = A_2^2 = \cdots = A_k^2 = 0$,
(2) $A_i A_j = A_j A_i$ for all $1 \leq i, j \leq k$, and

(3) $A_1 A_2 \ldots A_k \neq 0$.

Open all solution variants | Source note

2007-D2-P6 Long/deep

Fingerprint iterating f+f' eventually yields all-real roots
Primary domain Complex analysis / polynomials
Problem type recurrence/convergence
Linked methods Use a logarithmic derivative to expose root geometry [M-AL-LOGDERIVATIVE]
Secondary cues Real analysis / calculus; Geometry / convexity / metric; Functional equations / dynamics
Linked theory Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Recognition triggers polynomial/root trigger
Preparation route CP-Polynomials/Complex
Trap watch Separate polynomial identity from pointwise testing; check multiplicities, zero polynomial cases, and real-vs-complex roots.
Proof strategies Reduce size while preserving the defining conditions [S-INDUCTION]; Control the limiting

2007-D2-P6 continued

operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2007-D2-P6

Day 2; official label P12; legacy source-page label 123

Let $f \neq 0$ be a polynomial with real coefficients. Define the sequence $f_0, f_1, f_2, \ldots$ of polynomials by $f_0 = f$ and $f_{n+1} = f_n + f_n'$ for every $n \geq 0$. Prove that there exists a number N such that for every $n \geq N$, all roots of f_n are real.

Open all solution variants | Source note

2.15 2008

Day 1

2008-D1-P1 Medium/trap

Fingerprint continuous functions preserving rationality of differences
Primary domain Functional equations / dynamics
Problem type classification/find-all
Linked methods Force structure by admissible substitutions [M-DS-FUNCTIONAL]
Secondary cues Algebra / number theory; Real analysis / calculus
Linked theory Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]; Continuity, compactness, and order [T-AN-CONTINUITY]; Countability, accumulation, and Cantor constructions [T-AN-SETS]
Recognition triggers continuous/differentiable trigger; functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2008-D1-P1

Day 1; official label P1; legacy source-page label 125

Find all continuous functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) - f(y)$ is rational for all reals x and y such that $x - y$ is rational.

Open all solution variants | Source note

2008-D1-P2 Quick/standard

Fingerprint multiplicativity-like condition forces evaluation functional
Primary domain Linear algebra / matrices
Problem type structural proof
Linked methods Test a linear map on a basis and normalize its values [M-LA-DUAL]
Secondary cues Functional equations / dynamics; Complex analysis / polynomials; Real analysis / calculus
Linked theory Dual bases, kernels and separating functionals [T-LA-DUALITY]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers polynomial/root trigger; functional-equation trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Reduce size while preserving the defining conditions [S-INDUCTION]; Prove necessity and sufficiency separately [S-CLASSIFICATION]
Navigation Full statement All solution variants Source note Learning route

2008-D1-P2 continued

Full statement: 2008-D1-P2

Day 1; official label P2; legacy source-page label 125

Denote by V the real vector space of all real polynomials in one variable, and let $P : V \to \mathbb{R}$ be a linear map. Suppose that for all $f, g \in V$ with $P(fg) = 0$ we have $P(f) = 0$ or $P(g) = 0$. Prove that there exist real numbers x_0, c such that $P(f) = cf(x_0)$ for all $f \in V$.

Open all solution variants | Source note

2008-D1-P3 Medium/trap

Fingerprint integer a making all p(a_i) divide p(a)
Primary domain Algebra / number theory
Problem type existence/construction
Linked methods Construct and count compatible local residues [M-AL-CRT]
Secondary cues Complex analysis / polynomials; Linear algebra / matrix theory; Real analysis / calculus
Linked theory Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Prime valuations, divisors, and factorials [T-AL-VALUATION]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers polynomial/root trigger; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Build a witness and verify every constraint [S-CONSTRUCTION]; Pass between local constraints and a global conclusion [S-LOCALGLOBAL]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2008-D1-P3

Day 1; official label P3; legacy source-page label 126

Let p be a polynomial with integer coefficients and let $a_1 < a_2 < \cdots < a_k$ be integers.
a) Prove that there exists $a \in \mathbb{Z}$ such that $p(a_i)$ divides $p(a)$ for all $i = 1, 2, \ldots, k$.

b) Does there exist an $a \in \mathbb{Z}$ such that the product $p(a_1) \cdot p(a_2) \cdot \cdots \cdot p(a_k)$ divides $p(a)$?

Open all solution variants | Source note

2008-D1-P4 Long/deep

Fingerprint smallest set of triples dominating all special triples
Primary domain Combinatorics
Problem type classification/find-all
Linked methods Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]; Repair an extremal configuration by a local exchange [M-DS-EXTREMAL]; Split a family by traces, pieces, or labels [M-DS-PARTITION]
Secondary cues Real analysis / calculus
Linked theory Convex bodies, supporting functionals, and packing [T-DS-CONVEX]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Extremal choice, exchange arguments, and invariants [T-DS-EXTREMAL]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers wording-specific
Preparation route CMB-Combinatoricsi
Trap watch Avoid overcounting; define the encoded object precisely and check extremal/minimal cases.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2008-D1-P4

Day 1; official label P4; legacy source-page label 126

We say a triple (a_1, a_2, a_3) of nonnegative reals is better than another triple (b_1, b_2, b_3) if two out of the three following inequalities $a_1 > b_1$, $a_2 > b_2$, $a_3 > b_3$ are satisfied. We call a triple (x, y, z) special

2008-D1-P4 continued

if x, y, z are nonnegative and $x + y + z = 1$. Find all natural numbers n for which there is a set S of n special triples such that for any given special triple we can find at least one better triple in S.

Open all solution variants | Source note

2008-D1-P5 Long/deep

Fingerprint normal subgroup with more automorphisms than group
Primary domain Algebra / number theory
Problem type existence/construction
Linked methods Convert group data into orbits or cosets [M-AL-ACTION]
Secondary cues Linear algebra / matrix theory; Geometry / convexity / metric; Combinatorics / discrete structures
Linked theory Groups, orders, generators, and quotients [T-AL-GROUP]; Group actions, stabilizers, and semidirect products [T-AL-ACTION]; Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Rank, dimension and exact linear constraints [T-LA-RANK]
Recognition triggers group/ring/field trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Build a witness and verify every constraint [S-CONSTRUCTION]; Choose the objects and count them without ambiguity [S-COUNTING]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2008-D1-P5

Day 1; official label P5; legacy source-page label 126

Does there exist a finite group G with a normal subgroup H such that $|\operatorname{Aut} H| > |\operatorname{Aut} G|$?

Open all solution variants | Source note

2008-D1-P6 Long/deep

Fingerprint parity of permutation counts by total displacement
Primary domain Combinatorics
Problem type counting/enumeration
Linked methods Encode the statistic and extract the required coefficient [M-DS-GENERATING]
Secondary cues Linear algebra / matrix theory; Real analysis / calculus; Complex analysis / polynomials
Linked theory Generating functions, coefficient filters, and finite differences [T-DS-GENERATING]; Finite counting and incidence identities [T-DS-COUNTING]; Determinants, block identities and inverses [T-LA-DETERMINANT]; Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Recognition triggers counting/permutation trigger
Preparation route CMB-Combinatoricsi
Trap watch Avoid overcounting; define the encoded object precisely and check extremal/minimal cases.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2008-D1-P6

Day 1; official label P6; legacy source-page label 125

For a permutation $\sigma = (i_1, i_2, \dots, i_n)$ of $(1, 2, \dots, n)$ define $D(\sigma) = \sum_{k=1}^{n} |i_k - k|$. Let $Q(n, d)$ be the number of permutations σ of $(1, 2, \dots, n)$ with $d = D(\sigma)$. Prove that $Q(n, d)$ is even for $d \geq 2n$.

Open all solution variants | Source note

Day 2

2008-D2-P1 Quick/standard

Fingerprint divisibility transfer between trinomials via roots of unity
Primary domain Complex analysis / polynomials
Problem type structural proof
Linked methods Evaluate at roots of unity or average over them [M-AL-UNITY]; Replace many equal values by too many roots [M-AL-ROOTCOUNT]
Secondary cues Complex analysis / polynomials; Algebra / number theory; Functional equations / dynamics
Linked theory Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]; Roots of unity, cyclotomic polynomials, and algebraic norms [T-AL-CYCLOTOMIC]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers polynomial/root trigger; integer/prime/mod trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2008-D2-P1

Day 2; official label P7; legacy source-page label 127

Let n, k be positive integers and suppose that the polynomial $x^{2k}-x^k+1$ divides $x^{2n}+x^n+1$. Prove that $x^{2k}+x^k+1$ divides $x^{2n}+x^n+1$.

Open all solution variants | Source note

2008-D2-P2 Quick/standard

Fingerprint ellipses sharing focus meet in at most two points
Primary domain Geometry / convexity / metric
Problem type structural proof
Linked methods Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Secondary cues Linear algebra / matrix theory; Complex analysis / polynomials; Functional equations / dynamics
Linked theory Conics, reflections, and inversion [T-DS-TRANSFORMS]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers geometry/metric trigger
Preparation route GEO-Geometry/Metricif
Trap watch Control degenerate configurations and make clear which coordinates or affine transformations preserve the quantity.
Proof strategies Translate geometry into a controlled representation [S-GEOMETRIC]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2008-D2-P2

Day 2; official label P8; legacy source-page label 127

Two different ellipses are given. One focus of the first ellipse coincides with one focus of the second ellipse. Prove that the ellipses have at most two points in common.

Open all solution variants | Source note

2008-D2-P3 Medium

Fingerprint power of two divides binomial sum with 5^k
Primary domain Algebra / number theory
Problem type structural proof
Linked methods Find a recurrence that reveals the invariant [M-AL-RECURRENCE]
Secondary cues Real analysis / calculus

2008-D2-P3 continued

Linked theory Linear recurrences and generating functions [T-AL-RECURRENCE]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2008-D2-P3

Day 2; official label P9; legacy source-page label 127

Let n be a positive integer. Prove that 2^{n-1} divides

$$\sum_{0\le k<n/2} \binom{n}{2k+1} 5^k.$$

Open all solution variants | Source note

2008-D2-P4 Medium

Fingerprint degree bound on divisor from many integer roots
Primary domain Algebra / number theory
Problem type inequality/sharp bound
Linked methods Replace many equal values by too many roots [M-AL-ROOTCOUNT]
Secondary cues Functional equations / dynamics; Complex analysis / polynomials; Combinatorics / discrete structures
Linked theory Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Prime valuations, divisors, and factorials [T-AL-VALUATION]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers polynomial/root trigger; integer/prime/mod trigger; group/ring/field trigger; functional-equation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2008-D2-P4

Day 2; official label P10; legacy source-page label 127

Let $\mathbb{Z}[x]$ be the ring of polynomials with integer coefficients, and let $f(x), g(x) \in \mathbb{Z}[x]$ be nonconstant polynomials such that $g(x)$ divides $f(x)$ in $\mathbb{Z}[x]$. Prove that if the polynomial $f(x) - 2008$ has at least 81 distinct integer roots, then the degree of $g(x)$ is greater than 5.

Open all solution variants | Source note

2008-D2-P5 Long/deep

Fingerprint determinant of prime-sum indicator matrix is perfect square
Primary domain Linear algebra / matrices
Problem type structural proof
Linked methods Exploit a compatible block decomposition [M-LA-BLOCK]
Secondary cues Linear algebra / matrix theory; Real analysis / calculus; Complex analysis / polynomials
Linked theory Determinants, block identities and inverses [T-LA-DETERMINANT]; Rank, dimension and exact linear constraints [T-LA-RANK]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Recognition triggers matrix word; det/rank/eigen trigger; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in

2008-D2-P5 continued

Z, Q, a field, a ring, or a group.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2008-D2-P5

Day 2; official label P11; legacy source-page label 128

Let n be a positive integer, and consider the matrix $A = (a_{ij})_{1 \le i,j \le n}$, where

$$a_{ij} = \begin{cases} 1 & \text{if } i+j \text{ is a prime number,} \\ 0 & \text{otherwise.} \end{cases}$$

Prove that $|\det A| = k^2$ for some integer k.

Open all solution variants | Source note

2008-D2-P6 Long/deep

Fingerprint equidistant set in Hilbert space yields orthonormal system
Primary domain Linear algebra / matrices
Problem type existence/construction
Linked methods Encode distances and orthogonality by a Gram matrix [M-LA-GRAM]
Secondary cues Geometry / convexity / metric; Real analysis / calculus; Combinatorics / discrete structures
Linked theory Completeness, barycentres and equidistant sets [T-LA-HILBERT]; Inner products, orthogonality and polarization [T-LA-INNER]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers counting/permutation trigger; geometry/metric trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Translate geometry into a controlled representation [S-GEOMETRIC]; Control the limiting operation before taking the limit [S-ANALYTIC]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2008-D2-P6

Day 2; official label P12; legacy source-page label 128

Let $\mathcal{H}$ be an infinite-dimensional real Hilbert space, let $d > 0$, and suppose that S is a set of points (not necessarily countable) in $\mathcal{H}$ such that the distance between any two distinct points in S is equal to d. Show that there is a point $y \in \mathcal{H}$ such that

$$\left\{ \frac{\sqrt{2}}{d}(x-y) : x \in S \right\}$$

is an orthonormal system of vectors in $\mathcal{H}$.

Open all solution variants | Source note

2.16 2009

Day 1

2009-D1-P1 Medium/trap

Fingerprint does f<=g on rationals extend to reals
Primary domain Real analysis / calculus
Problem type nonexistence/counterexample
Linked methods Use a dense set, an enumeration, or a category argument [M-AN-TOPOLOGY]
Secondary cues Functional equations / dynamics; Algebra / number theory; Combinatorics / discrete structures
Linked theory Continuity, compactness, and order [T-AN-CONTINUITY]; Countability, accumulation, and Cantor constructions [T-AN-SETS]
Recognition triggers continuous/differentiable trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2009-D1-P1

Day 1; official label P1; legacy source-page label 129

Suppose that f and g are real-valued functions on the real line and $f(r) \leq g(r)$ for every rational r. Does this imply that $f(x) \leq g(x)$ for every real x if
a) f and g are non-decreasing?

b) f and g are continuous?

Open all solution variants | Source note

2009-D1-P2 Quick/standard

Fingerprint matrix identity (A-B)C=BA^{-1} implies C(A-B)=A^{-1}B
Primary domain Linear algebra / matrices
Problem type structural proof
Linked methods Complete the equation to a product equal to identity [M-LA-INVERSE]
Secondary cues None recorded
Linked theory Determinants, block identities and inverses [T-LA-DETERMINANT]
Recognition triggers matrix word
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2009-D1-P2

Day 1; official label P2; legacy source-page label 129

Let A, B and C be real square matrices of the same size, and suppose that A is invertible. Prove that if $(A-B)C = BA^{-1}$, then $C(A-B) = A^{-1}B$.

Open all solution variants | Source note

2009-D1-P3 Medium

Fingerprint shortest friendship cycle in graph with degree condition
Primary domain Combinatorics
Problem type classification/find-all
Linked methods Count incidences with their exact multiplicity [M-DS-DOUBLECOUNT]; Encode a relation as a graph and extract a cycle [M-DS-GRAPH]
Secondary cues Real analysis / calculus
Linked theory Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers det/rank/eigen trigger; counting/permutation trigger
Preparation route CMB-Combinatoricsi
Trap watch Avoid overcounting; define the encoded object precisely and check extremal/minimal cases.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2009-D1-P3

Day 1; official label P3; legacy source-page label 129

In a town every two residents who are not friends have a friend in common, and no one is a friend of everyone else. Let us number the residents from 1 to n and let a_i be the number of friends of the i-th resident. Suppose that $\sum_{i=1}^{n} a_i^2 = n^2 - n$. Let k be the smallest number of residents (at least three) who can be seated at a round table in such a way that any two neighbors are friends. Determine all possible values of k.

Open all solution variants | Source note

2009-D1-P4 Medium/trap

Fingerprint convex coefficient damping preserves max modulus bound
Primary domain Complex analysis / polynomials
Problem type inequality/sharp bound
Linked methods Evaluate at roots of unity or average over them [M-AL-UNITY]; Build a holomorphic function with a useful boundary [M-AL-ANALYTIC]
Secondary cues Complex analysis / polynomials; Linear algebra / matrix theory; Geometry / convexity / metric
Linked theory Roots-of-unity filters and positive trigonometric kernels [T-AL-FOURIER]; Holomorphic bounds, poles, and logarithmic-branch sums [T-AL-COMPLEXANALYSIS]; Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]
Recognition triggers continuous/differentiable trigger; polynomial/root trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2009-D1-P4

Day 1; official label P4; legacy source-page label 130

Let $p(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_n z^n$ be a complex polynomial. Suppose that $1 = c_0 \geq c_1 \geq \ldots \geq c_n \geq 0$ is a sequence of real numbers which is convex (i.e. $2c_k \leq c_{k-1} + c_{k+1}$ for every $k = 1, 2, \ldots, n-1$), and consider the polynomial

$$q(z) = c_0 a_0 + c_1 a_1 z + c_2 a_2 z^2 + \cdots + c_n a_n z^n.$$

Prove that

$$\max_{|z| \leq 1} |q(z)| \leq \max_{|z| \leq 1} |p(z)|.$$

Open all solution variants | Source note

2009-D1-P5 Long/deep

Fingerprint weighted circumcenter identity for subdivided n-simplex
Primary domain Geometry / convexity / metric
Problem type structural proof
Linked methods Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]; Strengthen the statement before the induction step [M-DS-INDUCT]
Secondary cues Geometry / convexity / metric; Combinatorics / discrete structures
Linked theory Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Inner products, orthogonality and polarization [T-LA-INNER]; Determinants, block identities and inverses [T-LA-DETERMINANT]; Rank, dimension and exact linear constraints [T-LA-RANK]
Recognition triggers integer/prime/mod trigger; graph/color/set trigger; geometry/metric trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Translate geometry into a controlled representation [S-GEOMETRIC]; Reduce size while preserving the defining conditions [S-INDUCTION]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2009-D1-P5

Day 1; official label P5; legacy source-page label 131

Let n be a positive integer. An n-simplex in $\mathbb{R}^n$ is given by $n+1$ points $P_0, P_1, \ldots, P_n$, called its vertices, which do not all belong to the same hyperplane. For every n-simplex S we denote by $v(S)$ the volume of S, and we write $C(S)$ for the center of the unique sphere containing all the vertices of S.

Suppose that P is a point inside an n-simplex S. Let S_i be the n-simplex obtained from S by replacing its i-th vertex by P. Prove that

$$v(S_0)C(S_0) + v(S_1)C(S_1) + \cdots + v(S_n)C(S_n) = v(S)C(S).$$

Open all solution variants | Source note

Day 2

2009-D2-P1 Quick/standard

Fingerprint volume of region closer to point than line
Primary domain Geometry / convexity / metric
Problem type exact computation
Linked methods Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Secondary cues Real analysis / calculus; Complex analysis / polynomials
Linked theory Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Conics, reflections, and inversion [T-DS-TRANSFORMS]; Determinants, block identities and inverses [T-LA-DETERMINANT]
Recognition triggers geometry/metric trigger
Preparation route GEO-Geometry/Metricif
Trap watch Control degenerate configurations and make clear which coordinates or affine transformations preserve the quantity.
Proof strategies Translate geometry into a controlled representation [S-GEOMETRIC]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

2009-D2-P1 continued

Full statement: 2009-D2-P1

Day 2; official label P6; legacy source-page label 133

Let ℓ be a line and P a point in $\mathbb{R}^3$. Let S be the set of points X such that the distance from X to ℓ is greater than or equal to two times the distance between X and P. If the distance from P to ℓ is $d > 0$, find the volume of S.

Open all solution variants | Source note

2009-D2-P2 Quick/standard

Fingerprint differential inequality forces lower bound 3e^{2x}-2e^{3x}
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Design an auxiliary derivative [M-AN-AUXILIARY]
Secondary cues Real analysis / calculus
Linked theory Differential inequalities and integrating factors [T-AN-ODE]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Recognition triggers continuous/differentiable trigger; functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2009-D2-P2

Day 2; official label P7; legacy source-page label 133

Suppose $f : \mathbb{R} \to \mathbb{R}$ is a two times differentiable function satisfying $f(0) = 1$, $f'(0) = 0$, and for all $x \in [0, \infty)$,

$$f''(x) - 5f'(x) + 6f(x) \geq 0.$$

Prove that for all $x \in [0, \infty)$,

$$f(x) \geq 3e^{2x} - 2e^{3x}.$$

Open all solution variants | Source note

2009-D2-P3 Medium

Fingerprint A^2B+BA^2=2ABA implies commutator nilpotent
Primary domain Linear algebra / matrices
Problem type structural proof
Linked methods Propagate a commutator by telescoping [M-LA-DERIVATION]; Use trace to extract a scalar invariant [M-LA-TRACE]
Secondary cues Linear algebra / matrix theory
Linked theory Trace pairing and commutator calculus [T-LA-TRACE]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Nilpotents, projections and involutions [T-LA-NILPOTENT]
Recognition triggers matrix word; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Reduce size while preserving the defining conditions [S-INDUCTION]
Navigation Full statement All solution variants Source note Learning route

2009-D2-P3 continued

Full statement: 2009-D2-P3

Day 2; official label P8; legacy source-page label 134

Let $A, B \in M_n(\mathbb{C})$ be two $n \times n$ matrices such that

$$A^2B + BA^2 = 2ABA.$$

Prove that there exists a positive integer k such that $(AB - BA)^k = 0$.

Open all solution variants | Source note

2009-D2-P4 Medium

Fingerprint count polynomials in composition-closed set over F_p
Primary domain Algebra / number theory
Problem type exact computation
Linked methods Choose a modulus that exposes the obstruction [M-AL-MODULAR]; Prove generation or obstruct it by an image [M-AL-GENERATORS]
Secondary cues Algebra / number theory; Combinatorics / discrete structures; Real analysis / calculus
Linked theory Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers polynomial/root trigger; integer/prime/mod trigger; group/ring/field trigger; counting/permutation trigger
Preparation route CP-Polynomials/Complex
Trap watch Separate polynomial identity from pointwise testing; check multiplicities, zero polynomial cases, and real-vs-complex roots.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Choose the objects and count them without ambiguity [S-COUNTING]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2009-D2-P4

Day 2; official label P9; legacy source-page label 134

Let p be a prime number and $\mathbb{F}_p$ be the field of residues modulo p. Let W be the smallest set of polynomials with coefficients in $\mathbb{F}_p$ such that

- the polynomials $x + 1$ and $x^{p-2} + x^{p-3} + \cdots + x^2 + 2x + 1$ are in W, and
- for any polynomials $h_1(x)$ and $h_2(x)$ in W the polynomial $r(x)$, which is the remainder of $h_1(h_2(x))$ modulo $x^p - x$, is also in W.

How many polynomials are there in W?

Open all solution variants | Source note

2009-D2-P5 Long/deep

Fingerprint column-space dimension bound for minimal covering matrix spaces
Primary domain Linear algebra / matrices
Problem type inequality/sharp bound
Linked methods Use a generic parameter to force an image inclusion [M-LA-PERTURB]; Prove a dimension obstruction and attain it [M-LA-DIMENSION]
Secondary cues Linear algebra / matrix theory; Combinatorics / discrete structures; Real analysis / calculus
Linked theory Rank, dimension and exact linear constraints [T-LA-RANK]; Evaluation maps and rank under perturbation [T-LA-PERTURBATION]; Dual bases, kernels and separating functionals [T-LA-DUALITY]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Recognition triggers matrix word; integer/prime/mod trigger; group/ring/field trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in

2009-D2-P5 continued

Z, Q, a field, a ring, or a group.
Proof strategies Choose an extremal object and force rigidity [S-EXTREMAL]; Reduce size while preserving the defining conditions [S-INDUCTION]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2009-D2-P5

Day 2; official label P10; 8+2 points; legacy source-page label 135

Let M be the vector space of $m \times p$ real matrices. For a vector subspace $S \subseteq M$, denote by $\delta(S)$ the dimension of the vector space generated by all columns of all matrices in S.
Say that a vector subspace $T \subseteq M$ is a covering matrix space if

$$\bigcup_{A \in T, A \neq 0} \ker A = \mathbb{R}^p.$$

Such a T is minimal if it does not contain a proper vector subspace $S \subset T$ which is also a covering matrix space.
(a) (8 points) Let T be a minimal covering matrix space and let $n = \dim T$. Prove that

$$\delta(T) \leq \binom{n}{2}.$$

(b) (2 points) Prove that for every positive integer n we can find m and p, and a minimal covering matrix space T as above such that $\dim T = n$ and $\delta(T) = \binom{n}{2}$.

Open all solution variants | Source note

2.17 2010

Day 1

2010-D1-P1 Quick/standard

Fingerprint lower bound for integral of (x^2+1)e^{-x^2}
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Count zeros and subtract an interpolant [M-AN-ROLLE]; Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]; Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]
Secondary cues Real analysis / calculus
Linked theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]
Recognition triggers wording-specific
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2010-D1-P1

Day 1; official label P1; 10 points; legacy source-page label 137

Let $0 < a < b$. Prove that

$$\int_a^b (x^2+1)e^{-x^2}dx \geq e^{-a^2} - e^{-b^2}.$$

Open all solution variants | Source note

2010-D1-P2

Quick/standard

Fingerprint sum series 1/((4k+1)(4k+2)(4k+3)(4k+4))
Primary domain Real analysis / calculus
Problem type exact computation
Linked methods Encode convolution or denominators by a power series [M-AN-GENERATING]; Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]; Find a finite telescoping identity [M-AN-TELESCOPE]; Test convergence through blocks and differences [M-AN-SERIES]
Secondary cues Functional equations / dynamics
Linked theory Power series and generating functions [T-AN-POWER]; Nonnegative sums, Fubini, and integral kernels [T-AN-FUBINI]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Series, finite cancellation, and summation by parts [T-AN-SERIES]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers integral/limit trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2010-D1-P2

Day 1; official label P2; 10 points; legacy source-page label 137

Compute the sum of the series

$$\sum_{k=0}^{\infty} \frac{1}{(4k+1)(4k+2)(4k+3)(4k+4)} = \frac{1}{1\cdot 2\cdot 3\cdot 4} + \frac{1}{5\cdot 6\cdot 7\cdot 8} + \cdots.$$

Open all solution variants | Source note

2010-D1-P3

Medium

Fingerprint limit of product over x_{n+1} for quadratic recurrence
Primary domain Functional equations / dynamics
Problem type limit/asymptotic evaluation
Linked methods Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]; Strengthen the statement before the induction step [M-DS-INDUCT]
Secondary cues Real analysis / calculus
Linked theory Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers integral/limit trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Control the limiting operation before taking the limit [S-ANALYTIC]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2010-D1-P3

Day 1; official label P3; 10 points; legacy source-page label 139

Define the sequence $x_1, x_2, \ldots$ inductively by $x_1 = \sqrt{5}$ and $x_{n+1} = x_n^2 - 2$ for each $n \geq 1$. Compute

$$\lim_{n\to\infty} \frac{x_1 \cdot x_2 \cdot x_3 \cdots x_n}{x_{n+1}}.$$

Open all solution variants | Source note

2010-D1-P4

Medium

Fingerprint ax^n+by^n misses finitely many integers forces n=1
Primary domain Algebra / number theory
Problem type structural proof
Linked methods Choose a modulus that exposes the obstruction [M-AL-MODULAR]
Secondary cues Combinatorics / discrete structures; Real analysis / calculus
Linked theory Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Prime valuations, divisors, and factorials [T-AL-VALUATION]
Recognition triggers integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Pass between local constraints and a global conclusion [S-LOCALGLOBAL]; Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2010-D1-P4

Day 1; official label P4; 10 points; legacy source-page label 139

Let a, b be two integers and suppose that n is a positive integer for which the set

$$\mathbb{Z} \setminus \{ax^n + by^n \mid x, y \in \mathbb{Z}\}$$

is finite. Prove that $n = 1$.

Open all solution variants | Source note

2010-D1-P5

Long/deep

Fingerprint propagate 1+2abc>=a^2+b^2+c^2 to all powers n
Primary domain Linear algebra / matrices
Problem type inequality/sharp bound
Linked methods Encode distances and orthogonality by a Gram matrix [M-LA-GRAM]; Lift products of inner products to new vectors [M-LA-TENSOR]; Read the entries before choosing a theorem [M-LA-ENTRY]
Secondary cues Linear algebra / matrix theory; Real analysis / calculus; Geometry / convexity / metric
Linked theory Gram matrices and positive semidefinite forms [T-LA-GRAM]; Tensor products and operators on matrix spaces [T-LA-TENSOR]; Determinants, block identities and inverses [T-LA-DETERMINANT]; Inner products, orthogonality and polarization [T-LA-INNER]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers continuous/differentiable trigger; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2010-D1-P5

Day 1; official label P5; 10 points; legacy source-page label 139

Suppose that a, b, c are real numbers in the interval $[-1, 1]$ such that

$$1 + 2abc \geq a^2 + b^2 + c^2.$$

Prove that

$$1 + 2(abc)^n \geq a^{2n} + b^{2n} + c^{2n}$$

for all positive integers n.

Open all solution variants | Source note

Day 2

2010-D2-P1 Quick/standard

Fingerprint convergence of x_{n+1}=x_n cos x_n and sin variant
Primary domain Real analysis / calculus
Problem type recurrence/convergence
Linked methods Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]
Secondary cues Functional equations / dynamics; Algebra / number theory
Linked theory Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Trigonometric identities and safe branches [T-AN-TRIG]; Continuity, compactness, and order [T-AN-CONTINUITY]
Recognition triggers wording-specific
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2010-D2-P1

Day 2; official label P6; 5+5 points; legacy source-page label 141

(a) A sequence $x_1, x_2, \ldots$ of real numbers satisfies

$$x_{n+1} = x_n \cos x_n \quad \text{for all } n \geq 1.$$

Does it follow that this sequence converges for all initial values x_1? (5 points)

(b) A sequence $y_1, y_2, \ldots$ of real numbers satisfies

$$y_{n+1} = y_n \sin y_n \quad \text{for all } n \geq 1.$$

Does it follow that this sequence converges for all initial values y_1? (5 points)

Open all solution variants | Source note

2010-D2-P2 Quick/standard

Fingerprint product inequality for sequence with gaps at least one
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]; Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]
Secondary cues Geometry / convexity / metric; Algebra / number theory
Linked theory Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Recognition triggers wording-specific
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Reduce size while preserving the defining conditions [S-INDUCTION]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2010-D2-P2

Day 2; official label P7; 10 points; legacy source-page label 141

Let $a_0, a_1, \ldots, a_n$ be positive real numbers such that $a_{k+1} - a_k \geq 1$ for all $k = 0, 1, \ldots, n-1$. Prove that

$$1 + \frac{1}{a_0}\left(1 + \frac{1}{a_1 - a_0}\right) \cdots \left(1 + \frac{1}{a_n - a_0}\right) \leq \left(1 + \frac{1}{a_0}\right)\left(1 + \frac{1}{a_1}\right) \cdots \left(1 + \frac{1}{a_n}\right).$$

Open all solution variants | Source note

2010-D2-P3 Medium/trap

Fingerprint permutation subgroup with unique fixed point per element
Primary domain Algebra / number theory
Problem type structural proof
Linked methods Convert group data into orbits or cosets [M-AL-ACTION]
Secondary cues Combinatorics / discrete structures; Real analysis / calculus; Functional equations / dynamics
Linked theory Group actions, stabilizers, and semidirect products [T-AL-ACTION]; Groups, orders, generators, and quotients [T-AL-GROUP]
Recognition triggers group/ring/field trigger; counting/permutation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2010-D2-P3

Day 2; official label P8; 10 points; legacy source-page label 142

Denote by S_n the group of permutations of the sequence $(1, 2, \dots, n)$. Suppose that G is a subgroup of S_n, such that for every $\pi \in G \setminus \{e\}$ there exists a unique $k \in \{1, 2, \dots, n\}$ for which $\pi(k) = k$. (Here e is the unit element in the group S_n.) Show that this k is the same for all $\pi \in G \setminus \{e\}$.

Open all solution variants | Source note

2010-D2-P4 Medium/trap

Fingerprint columns of powers of symmetric F_2 matrix have zero
Primary domain Linear algebra / matrices
Problem type structural proof
Linked methods Exploit parity with linear algebra over F2 [M-LA-FINITEFIELD]
Secondary cues Linear algebra / matrix theory; Real analysis / calculus; Geometry / convexity / metric
Linked theory Linear algebra in characteristic two [T-LA-FINITEFIELD]
Recognition triggers matrix word; integer/prime/mod trigger; group/ring/field trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Reduce size while preserving the defining conditions [S-INDUCTION]; Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2010-D2-P4

Day 2; official label P9; 10 points; legacy source-page label 143

Let A be a symmetric $m \times m$ matrix over the two-element field all of whose diagonal entries are zero. Prove that for every positive integer n each column of the matrix A^n has a zero entry.

Open all solution variants | Source note

2010-D2-P5 Long/deep

Fingerprint prime averaging condition forces f identically zero
Primary domain Complex analysis / polynomials
Problem type structural proof
Linked methods Take the gcd of annihilating shift polynomials [M-AL-SHIFT]
Secondary cues Algebra / number theory; Functional equations / dynamics; Real analysis / calculus
Linked theory Finite differences and shift-operator ideals [T-AL-DIFFERENCE]; Roots of unity, cyclotomic polynomials, and algebraic norms [T-AL-CYCLOTOMIC]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]

2010-D2-P5 continued

Recognition triggers integer/prime/mod trigger; functional-equation trigger
Preparation route CP-Polynomials/Complex
Trap watch Separate polynomial identity from pointwise testing; check multiplicities, zero polynomial cases, and real-vs-complex roots.
Proof strategies Pass between local constraints and a global conclusion [S-LOCALGLOBAL]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2010-D2-P5

Day 2; official label P10; 10 points; legacy source-page label 144

Suppose that for a function $f : \mathbb{R} \to \mathbb{R}$ and real numbers $a < b$ one has $f(x) = 0$ for all $x \in (a, b)$. Prove that $f(x) = 0$ for all $x \in \mathbb{R}$ if

$$\sum_{k=0}^{p-1} f\left(y + \frac{k}{p}\right) = 0$$

for every prime number p and every real number y.

Open all solution variants | Source note

2.18 2011

Day 1

2011-D1-P1 Quick/standard

Fingerprint shadow points force f(a)=f(b) on interval
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Localize an extremum or an accumulation point [M-AN-COMPACT]
Secondary cues Real analysis / calculus; Combinatorics / discrete structures
Linked theory Continuity, compactness, and order [T-AN-CONTINUITY]
Recognition triggers continuous/differentiable trigger; geometry/metric trigger; functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Choose an extremal object and force rigidity [S-EXTREMAL]; Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2011-D1-P1

Day 1; official label P1; 10 points; legacy source-page label 145

Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function. A point x is called a shadow point if there exists a point $y \in \mathbb{R}$ with $y > x$ such that $f(y) > f(x)$. Let $a < b$ be real numbers and suppose that

- all the points of the open interval $I = (a, b)$ are shadow points;
- a and b are not shadow points.

Prove that
a) $f(x) \leq f(b)$ for all $a < x < b$;
b) $f(a) = f(b)$.

(José Luis Díaz-Barrero, Barcelona)

Open all solution variants | Source note

2011-D1-P2

Medium/trap

Fingerprint existence of 3x3 matrix with tr=0, A^2+A^t=I
Primary domain Linear algebra / matrices
Problem type nonexistence/counterexample
Linked methods Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]; Use trace to extract a scalar invariant [M-LA-TRACE]
Secondary cues Real analysis / calculus; Complex analysis / polynomials
Linked theory Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Trace pairing and commutator calculus [T-LA-TRACE]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]
Recognition triggers matrix word; det/rank/eigen trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2011-D1-P2

Day 1; official label P2; 10 points; legacy source-page label 145

Does there exist a real 3×3 matrix A such that $\mathrm{tr}(A) = 0$ and $A^2 + A^t = I$? ($\mathrm{tr}(A)$ denotes the trace of A, A^t is the transpose of A, and I is the identity matrix.)

(Moubinool Omarjee, Paris)

Open all solution variants | Source note

2011-D1-P3

Medium

Fingerprint minimal n with x^n-1 in ideal mod p
Primary domain Algebra / number theory
Problem type classification/find-all
Linked methods Choose a modulus that exposes the obstruction [M-AL-MODULAR]
Secondary cues Complex analysis / polynomials; Functional equations / dynamics; Geometry / convexity / metric
Linked theory Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Groups, orders, generators, and quotients [T-AL-GROUP]
Recognition triggers polynomial/root trigger; integer/prime/mod trigger; functional-equation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Prove necessity and sufficiency separately [S-CLASSIFICATION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2011-D1-P3

Day 1; official label P3; 10 points; legacy source-page label 145

Let p be a prime number. Call a positive integer n interesting if

$$x^n - 1 = (x^p - x + 1)f(x) + pg(x)$$

for some polynomials f and g with integer coefficients.
a) Prove that the number $p^p - 1$ is interesting.
b) For which p is $p^p - 1$ the minimal interesting number?

(Eugene Goryachko and Fedor Petrov, St. Petersburg)

Open all solution variants | Source note

2011-D1-P4 Medium

Fingerprint inclusion-exclusion polynomial of union sizes is nondecreasing
Primary domain Combinatorics
Problem type structural proof
Linked methods Realize all inclusion probabilities on one probability space [M-DS-COUPLING]; Count incidences with their exact multiplicity [M-DS-DOUBLECOUNT]
Secondary cues Real analysis / calculus; Probability / measure / expectation; Functional equations / dynamics
Linked theory Finite counting and incidence identities [T-DS-COUNTING]; Finite probability, conditioning, and expectations [T-AN-PROBABILITY]
Recognition triggers counting/permutation trigger; functional-equation trigger
Preparation route CMB-Combinatoricsi
Trap watch Avoid overcounting; define the encoded object precisely and check extremal/minimal cases.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2011-D1-P4

Day 1; official label P4; 10 points; legacy source-page label 146

Let $A_1, A_2, \ldots, A_n$ be finite, nonempty sets. Define the function

$$f(t) = \sum_{k=1}^{n} \sum_{1 \le i_1 < i_2 < \cdots < i_k \le n} (-1)^{k-1} t^{|A_{i_1} \cup A_{i_2} \cup \cdots \cup A_{i_k}|}.$$

Prove that f is nondecreasing on $[0, 1]$.
($|A|$ denotes the number of elements in A.)

(Levon Nurbekyan and Vardan Voskanyan, Yerevan)

Open all solution variants | Source note

2011-D1-P5 Long/deep

Fingerprint zero-sum 2n-subset among 4n-1 vectors over F_2
Primary domain Linear algebra / matrices
Problem type existence/construction
Linked methods Exploit parity with linear algebra over F2 [M-LA-FINITEFIELD]; Prove a dimension obstruction and attain it [M-LA-DIMENSION]
Secondary cues Linear algebra / matrix theory; Real analysis / calculus; Probability / measure / expectation
Linked theory Linear algebra in characteristic two [T-LA-FINITEFIELD]; Rank, dimension and exact linear constraints [T-LA-RANK]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers integer/prime/mod trigger; group/ring/field trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Reduce size while preserving the defining conditions [S-INDUCTION]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2011-D1-P5

Day 1; official label P5; 10 points; legacy source-page label 146

Let n be a positive integer and let V be a $(2n-1)$-dimensional vector space over the two-element field. Prove that for arbitrary vectors $v_1, \ldots, v_{4n-1} \in V$, there exists a sequence $1 \le i_1 < \cdots < i_{2n} \le 4n-1$ of indices such that $v_{i_1} + \cdots + v_{i_{2n}} = 0$.

(Ilya Bogdanov, Moscow and Géza Kós, Budapest)

Open all solution variants | Source note

Day 2

2011-D2-P1 Quick/standard

Fingerprint limit of tanh-like addition recurrence
Primary domain Real analysis / calculus
Problem type recurrence/convergence
Linked methods Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]; Exploit identities before estimating oscillations [M-AN-TRIG]
Secondary cues Functional equations / dynamics; Geometry / convexity / metric
Linked theory Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Trigonometric identities and safe branches [T-AN-TRIG]
Recognition triggers integral/limit trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Control the limiting operation before taking the limit [S-ANALYTIC]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2011-D2-P1

Day 2; official label P6; 10 points; legacy source-page label 147

Let $(a_n)_{n=0}^{\infty}$ be a sequence with $\frac{1}{2} < a_n < 1$ for all $n \geq 0$. Define the sequence $(x_n)_{n=0}^{\infty}$ by

$$x_0 = a_0, \qquad x_{n+1} = \frac{a_{n+1} + x_n}{1 + a_{n+1}x_n} \quad (n \geq 0).$$

What are the possible values of $\lim_{n\to\infty} x_n$? Can such a sequence diverge?

(Johnson Olaleru, Lagos)

Open all solution variants | Source note

2011-D2-P2 Long/deep

Fingerprint three-gender matching guaranteed when k>=3n/4
Primary domain Combinatorics
Problem type existence/construction
Linked methods Repair an extremal configuration by a local exchange [M-DS-EXTREMAL]; Match classes in stages and verify Hall each time [M-DS-MATCHING]; Split a family by traces, pieces, or labels [M-DS-PARTITION]
Secondary cues Combinatorics / discrete structures; Geometry / convexity / metric; Real analysis / calculus
Linked theory Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]; Finite counting and incidence identities [T-DS-COUNTING]; Extremal choice, exchange arguments, and invariants [T-DS-EXTREMAL]; Hall matching and augmenting paths [T-DS-MATCHING]
Recognition triggers group/ring/field trigger; counting/permutation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Build a witness and verify every constraint [S-CONSTRUCTION]; Choose an extremal object and force rigidity [S-EXTREMAL]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2011-D2-P2

Day 2; official label P7; 10 points; legacy source-page label 147

An alien race has three genders: male, female, and emale. A married triple consists of three persons, one from each gender, who all like each other. Any person is allowed to belong to at most one married triple. A special feature of this race is that feelings are always mutual — if x likes y, then y likes x.

2011-D2-P2 continued

The race is sending an expedition to colonize a planet. The expedition has n males, n females, and n emales. It is known that every expedition member likes at least k persons of each of the two other genders. The problem is to create as many married triples as possible to produce healthy offspring so the colony could grow and prosper.
a) Show that if n is even and $k = \frac{n}{2}$, then it might be impossible to create even one married triple.
b) Show that if $k \geq \frac{3n}{4}$, then it is always possible to create n disjoint married triples, thus marrying all of the expedition members.

(Fedor Duzhin and Nick Gravin, Singapore)

Open all solution variants | Source note

2011-D2-P3 Medium

Fingerprint evaluate telescoping sum of products of logarithms
Primary domain Real analysis / calculus
Problem type exact computation
Linked methods Find a finite telescoping identity [M-AN-TELESCOPE]
Secondary cues Real analysis / calculus; Geometry / convexity / metric
Linked theory Series, finite cancellation, and summation by parts [T-AN-SERIES]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers det/rank/eigen trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2011-D2-P3

Day 2; official label P8; 10 points; legacy source-page label 148

Determine the value of

$$\sum_{n=1}^{\infty} \ln\left(1+\frac{1}{n}\right) \cdot \ln\left(1+\frac{1}{2n}\right) \cdot \ln\left(1+\frac{1}{2n+1}\right).$$

(Gerhard Woeginger, Utrecht)

Open all solution variants | Source note

2011-D2-P4 Medium/trap

Fingerprint divided differences integer implies a-b divides f(a)-f(b)
Primary domain Algebra / number theory
Problem type structural proof
Linked methods Turn nodal bounds into a sharp linear functional [M-AL-INTERPOLATE]; Compare coefficients after a structural substitution [M-AL-COEFFICIENT]
Secondary cues Functional equations / dynamics; Complex analysis / polynomials; Real analysis / calculus
Linked theory Interpolation, divided differences, and extremal signs [T-AL-INTERPOLATION]; Finite differences and shift-operator ideals [T-AL-DIFFERENCE]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Recognition triggers polynomial/root trigger; integer/prime/mod trigger; functional-equation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Reduce size while preserving the defining conditions [S-INDUCTION]; Expose the structure before calculating [S-STRUCTURE]

2011-D2-P4 continued

Navigation Full statement All solution variants Source note Learning route

Full statement: 2011-D2-P4

Day 2; official label P9; 10 points; legacy source-page label 149

Let $f(x)$ be a polynomial with real coefficients of degree n. Suppose that $\dfrac{f(k)-f(m)}{k-m}$ is an integer for all integers $0 \le k < m \le n$. Prove that $a-b$ divides $f(a)-f(b)$ for all pairs of distinct integers a and b.

(Fedor Petrov, St. Petersburg)

Open all solution variants | Source note

2011-D2-P5 Long/deep

Fingerprint iterated vertex reflections of polygon return identity
Primary domain Geometry / convexity / metric
Problem type structural proof
Linked methods Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]; Choose information that makes an encoding injective [M-DS-ENCODE]
Secondary cues Real analysis / calculus; Combinatorics / discrete structures; Linear algebra / matrix theory
Linked theory Conics, reflections, and inversion [T-DS-TRANSFORMS]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Adjacent-transposition relations and finite presentations [T-AL-COXETER]; Groups, orders, generators, and quotients [T-AL-GROUP]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers graph/color/set trigger
Preparation route GEO-Geometry/Metricif
Trap watch Control degenerate configurations and make clear which coordinates or affine transformations preserve the quantity.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Translate geometry into a controlled representation [S-GEOMETRIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2011-D2-P5

Day 2; official label P10; 10 points; legacy source-page label 150

Let $F = A_0A_1 \ldots A_n$ be a convex polygon in the plane. Define for all $1 \le k \le n-1$ the operation f_k which replaces F with a new polygon

$$f_k(F) = A_0 \ldots A_{k-1}A'_kA_{k+1} \ldots A_n,$$

where A'_k is the point symmetric to A_k with respect to the perpendicular bisector of $A_{k-1}A_{k+1}$. Prove that $(f_1 \circ f_2 \circ \ldots \circ f_{n-1})^n(F) = F$. We suppose that all operations are well-defined on the polygons, to which they are applied, i.e. results are convex polygons again. (A_0, $A_1, \ldots, A_n$ are the vertices of F in consecutive order.)

(Mikhail Khristoforov, St. Petersburg)

Open all solution variants | Source note

2.19 2012

Day 1

2012-D1-P1 Quick/standard

Fingerprint p(n)-p(n-1) counts partitions into parts >1
Primary domain Combinatorics
Problem type counting/enumeration
Linked methods Encode, decode, and verify a bijection [M-DS-BIJECTION]; Split a family by traces, pieces, or labels [M-DS-PARTITION]; Encode the statistic and extract the required coefficient [M-DS-GENERATING]
Secondary cues Combinatorics / discrete structures; Real analysis / calculus; Geometry / convexity / metric
Linked theory Finite counting and incidence identities [T-DS-COUNTING]; Generating functions, coefficient filters, and finite differences [T-DS-GENERATING]
Recognition triggers integral/limit trigger; integer/prime/mod trigger; counting/permutation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2012-D1-P1

Day 1; official label P1; 10 points; legacy source-page label 151

For every positive integer n, let $p(n)$ denote the number of ways to express n as a sum of positive integers. For instance, $p(4) = 5$ because

$$4 = 3 + 1 = 2 + 2 = 2 + 1 + 1 = 1 + 1 + 1 + 1.$$

Also define $p(0) = 1$.
Prove that $p(n) - p(n-1)$ is the number of ways to express n as a sum of integers each of which is strictly greater than 1.

(Proposed by Fedor Duzhin, Nanyang Technological University)

Open all solution variants | Source note

2012-D1-P2 Quick/standard

Fingerprint minimal rank of hollow matrix with positive off-diagonal
Primary domain Linear algebra / matrices
Problem type inequality/sharp bound
Linked methods Read the entries before choosing a theorem [M-LA-ENTRY]; Prove a dimension obstruction and attain it [M-LA-DIMENSION]
Secondary cues Linear algebra / matrix theory; Real analysis / calculus
Linked theory Rank, dimension and exact linear constraints [T-LA-RANK]
Recognition triggers matrix word; det/rank/eigen trigger; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

2012-D1-P2 continued

Full statement: 2012-D1-P2

Day 1; official label P2; 10 points; legacy source-page label 151

Let n be a fixed positive integer. Determine the smallest possible rank of an $n \times n$ matrix that has zeros along the main diagonal and strictly positive real numbers off the main diagonal.

(Proposed by Ilya Bogdanov and Grigoriy Chelnokov, MIPT, Moscow)

Open all solution variants | Source note

2012-D1-P3 Medium

Fingerprint last-move-loses game generating symmetric group
Primary domain Algebra / number theory
Problem type game/strategy
Linked methods Prove generation or obstruct it by an image [M-AL-GENERATORS]; Work orbit by orbit and track parity [M-AL-ORBIT]
Secondary cues Combinatorics / discrete structures; Geometry / convexity / metric
Linked theory Groups, orders, generators, and quotients [T-AL-GROUP]; Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]
Recognition triggers integer/prime/mod trigger; group/ring/field trigger; counting/permutation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Choose an extremal object and force rigidity [S-EXTREMAL]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2012-D1-P3

Day 1; official label P3; 10 points; legacy source-page label 151

Given an integer $n > 1$, let S_n be the group of permutations of the numbers $1, 2, \ldots, n$. Two players, A and B, play the following game. Taking turns, they select elements (one element at a time) from the group S_n. It is forbidden to select an element that has already been selected. The game ends when the selected elements generate the whole group S_n. The player who made the last move loses the game. The first move is made by A. Which player has a winning strategy?

(Proposed by Fedor Petrov, St. Petersburg State University)

Open all solution variants | Source note

2012-D1-P4 Medium

Fingerprint f'(t)>f(f(t)) implies f(f(f(t)))<=0
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Use a first crossing or a forced-growth contradiction [M-AN-BARRIER]
Secondary cues Functional equations / dynamics
Linked theory Differential inequalities and integrating factors [T-AN-ODE]; Continuity, compactness, and order [T-AN-CONTINUITY]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Recognition triggers continuous/differentiable trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Make the contradiction quantitative or structural [S-CONTRADICTION]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

2012-D1-P4 continued

Full statement: 2012-D1-P4

Day 1; official label P4; 10 points; legacy source-page label 151

Let $f : \mathbb{R} \to \mathbb{R}$ be a continuously differentiable function that satisfies $f'(t) > f(f(t))$ for all $t \in \mathbb{R}$. Prove that $f(f(f(t))) \leq 0$ for all $t \geq 0$.

(Proposed by Tomáš Bárta, Charles University, Prague)

Open all solution variants | Source note

2012-D1-P5 Long/deep

Fingerprint irreducibility of X^{2^n}(X+a)^{2^n}+1 over Q
Primary domain Complex analysis / polynomials
Problem type structural proof
Linked methods Factor first and compare irreducible exponents [M-AL-FACTOR]; Replace a minimal algebraic counterexample by a smaller one [M-AL-DESCENT]
Secondary cues Algebra / number theory
Linked theory Polynomial factorization and coprime divisibility [T-AL-FACTORIZATION]; Roots of unity, cyclotomic polynomials, and algebraic norms [T-AL-CYCLOTOMIC]; Prime valuations, divisors, and factorials [T-AL-VALUATION]
Recognition triggers polynomial/root trigger; integer/prime/mod trigger; group/ring/field trigger
Preparation route CP-Polynomials/Complex
Trap watch Separate polynomial identity from pointwise testing; check multiplicities, zero polynomial cases, and real-vs-complex roots.
Proof strategies Make the contradiction quantitative or structural [S-CONTRADICTION]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2012-D1-P5

Day 1; official label P5; 10 points; legacy source-page label 151

Let a be a rational number and let n be a positive integer. Prove that the polynomial $X^{2^n}(X+a)^{2^n}+1$ is irreducible in the ring $\mathbb{Q}[X]$ of polynomials with rational coefficients.

(Proposed by Vincent Jugé, École Polytechnique, Paris)

Open all solution variants | Source note

Day 2

2012-D2-P1 Quick/standard

Fingerprint coefficient assignment game for divisibility by m(x)
Primary domain Complex analysis / polynomials
Problem type game/strategy
Linked methods Compare coefficients after a structural substitution [M-AL-COEFFICIENT]
Secondary cues Functional equations / dynamics; Algebra / number theory
Linked theory Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Rank, dimension and exact linear constraints [T-LA-RANK]
Recognition triggers polynomial/root trigger; integer/prime/mod trigger; functional-equation trigger
Preparation route CP-Polynomials/Complex
Trap watch Separate polynomial identity from pointwise testing; check multiplicities, zero polynomial cases, and real-vs-complex roots.
Proof strategies Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

2012-D2-P1 continued

Full statement: 2012-D2-P1

Day 2; official label P6; 10 points; legacy source-page label 152

Consider a polynomial

$$f(x) = x^{2012} + a_{2011}x^{2011} + \cdots + a_1 x + a_0.$$

Albert Einstein and Homer Simpson are playing the following game. In turn, they choose one of the coefficients $a_0, \ldots, a_{2011}$ and assign a real value to it. Albert has the first move. Once a value is assigned to a coefficient, it cannot be changed any more. The game ends after all the coefficients have been assigned values.
Homer's goal is to make $f(x)$ divisible by a fixed polynomial $m(x)$ and Albert's goal is to prevent this.
(a) Which of the players has a winning strategy if $m(x) = x - 2012$?
(b) Which of the players has a winning strategy if $m(x) = x^2 + 1$?

(Proposed by Fedor Duzhin, Nanyang Technological University)

Open all solution variants | Source note

2012-D2-P2 Quick/standard

Fingerprint convergence and value of sum a_{k+1}/a_k
Primary domain Real analysis / calculus
Problem type recurrence/convergence
Linked methods Find a finite telescoping identity [M-AN-TELESCOPE]; Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]
Secondary cues Geometry / convexity / metric
Linked theory Series, finite cancellation, and summation by parts [T-AN-SERIES]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers det/rank/eigen trigger; integral/limit trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2012-D2-P2

Day 2; official label P7; 10 points; legacy source-page label 152

Define the sequence $a_0, a_1, \ldots$ inductively by $a_0 = 1$, $a_1 = \frac{1}{2}$ and

$$a_{n+1} = \frac{n a_n^2}{1 + (n+1)a_n} \quad \text{for } n \geq 1.$$

Show that the series $\sum_{k=0}^{\infty} \frac{a_{k+1}}{a_k}$ converges and determine its value.

(Proposed by Christophe Debry, KU Leuven, Belgium)

Open all solution variants | Source note

2012-D2-P3 Medium

Fingerprint finiteness of n with n!+1 dividing (2012n)!
Primary domain Algebra / number theory
Problem type structural proof
Linked methods Track one prime or one extreme divisor [M-AL-VALUATION]
Secondary cues None recorded
Linked theory Prime valuations, divisors, and factorials [T-AL-VALUATION]; Factorial growth and the prime-

2012-D2-P3 continued

number input [T-AL-PRIMEGROWTH]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2012-D2-P3

Day 2; official label P8; 10 points; legacy source-page label 152

Is the set of positive integers n such that $n! + 1$ divides $(2012n)!$ finite or infinite?

(Proposed by Fedor Petrov, St. Petersburg State University)

Open all solution variants | Source note

2012-D2-P4 Medium/trap

Fingerprint all a admitting cyclic solution x_i(1-x_{i+1})=a
Primary domain Linear algebra / matrices
Problem type classification/find-all
Linked methods Linearize a fractional recurrence in dimension two [M-LA-PROJECTIVE]; Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]
Secondary cues Real analysis / calculus
Linked theory Fractional-linear recurrences and matrix powers [T-LA-PROJECTIVE]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2012-D2-P4

Day 2; official label P9; 10 points; legacy source-page label 152

Let $n \geq 2$ be an integer. Find all real numbers a such that there exist real numbers x_1, ..., x_n satisfying

$$x_1(1-x_2) = x_2(1-x_3) = \cdots = x_{n-1}(1-x_n) = x_n(1-x_1) = a. \tag{1}$$

(Proposed by Walther Janous and Gerhard Kirchner, Innsbruck)

Open all solution variants | Source note

2012-D2-P5 Long/deep

Fingerprint Plunnecke inequality for iterated sumsets
Primary domain Combinatorics
Problem type inequality/sharp bound
Linked methods Repair an extremal configuration by a local exchange [M-DS-EXTREMAL]; Strengthen the statement before the induction step [M-DS-INDUCT]; Count incidences with their exact multiplicity [M-DS-DOUBLECOUNT]
Secondary cues Real analysis / calculus; Combinatorics / discrete structures
Linked theory Sumsets and the minimal-growth lemma [T-DS-ADDITIVE]; Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Groups, orders, generators, and quotients [T-AL-GROUP]
Recognition triggers integer/prime/mod trigger; group/ring/field trigger

2012-D2-P5 continued

Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Choose an extremal object and force rigidity [S-EXTREMAL]; Reduce size while preserving the defining conditions [S-INDUCTION]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2012-D2-P5

Day 2; official label P10; 10 points; legacy source-page label 152

Let $c \geq 1$ be a real number. Let G be an abelian group and let $A \subset G$ be a finite set satisfying $|A + A| \leq c|A|$, where $X + Y := \{x + y \mid x \in X,\ y \in Y\}$ and $|Z|$ denotes the cardinality of Z. Prove that

$$|\underbrace{A + A + \cdots + A}_{k \text{ times}}| \leq c^k |A|$$

for every positive integer k. (Plünnecke's inequality)

(Proposed by Przemyslaw Mazur, Jagiellonian University)

Open all solution variants | Source note

2.20 2013

Day 1

2013-D1-P1 Quick/standard

Fingerprint eigenvalue of AB exceeds 1 for symmetric matrices
Primary domain Linear algebra / matrices
Problem type inequality/sharp bound
Linked methods Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]; Combine norm bounds with algebraic decay [M-LA-NORM]
Secondary cues None recorded
Linked theory Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Operator norms and convergence of matrix powers [T-LA-NORM]
Recognition triggers matrix word; det/rank/eigen trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Translate geometry into a controlled representation [S-GEOMETRIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2013-D1-P1

Day 1; official label P1; 10 points; legacy source-page label 153

Let A and B be real symmetric matrices with all eigenvalues strictly greater than 1. Let λ be a real eigenvalue of matrix AB. Prove that $|\lambda| > 1$.

(Proposed by Pavel Kozhevnikov, MIPT, Moscow)

Open all solution variants | Source note

2013-D1-P2 Quick/standard

Fingerprint exists xi with f''=f(1+2tan^2)
Primary domain Real analysis / calculus
Problem type existence/construction
Linked methods Count zeros and subtract an interpolant [M-AN-ROLLE]; Design an auxiliary derivative [M-AN-AUXILIARY]
Secondary cues Functional equations / dynamics
Linked theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Trigonometric identities and safe branches [T-AN-TRIG]
Recognition triggers continuous/differentiable trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2013-D1-P2

Day 1; official label P2; 10 points; legacy source-page label 153

Let $f : \mathbb{R} \to \mathbb{R}$ be a twice differentiable function. Suppose $f(0) = 0$. Prove that there exists $\xi \in (-\pi/2, \pi/2)$ such that

$$f''(\xi) = f(\xi)(1 + 2\tan^2 \xi).$$

(Proposed by Karen Keryan, Yerevan State University, Yerevan, Armenia)

Open all solution variants | Source note

2013-D1-P3 Medium

Fingerprint minimum trips so every two students meet
Primary domain Combinatorics
Problem type inequality/sharp bound
Linked methods Count incidences with their exact multiplicity [M-DS-DOUBLECOUNT]; Split a family by traces, pieces, or labels [M-DS-PARTITION]
Secondary cues Real analysis / calculus
Linked theory Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers counting/permutation trigger
Preparation route CMB-Combinatoricsi
Trap watch Avoid overcounting; define the encoded object precisely and check extremal/minimal cases.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Build a witness and verify every constraint [S-CONSTRUCTION]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2013-D1-P3

Day 1; official label P3; 10 points; legacy source-page label 153

There are $2n$ students in a school ($n \in \mathbb{N}$, $n \geq 2$). Each week n students go on a trip. After several trips the following condition was fulfilled: every two students were together on at least one trip. What is the minimum number of trips needed for this to happen?

(Proposed by Oleksandr Rybak, Kiev, Ukraine)

Open all solution variants | Source note

2013-D1-P4 Medium

Fingerprint symmetric inequality in power sums A,B,C
Primary domain Complex analysis / polynomials
Problem type inequality/sharp bound
Linked methods Translate root restrictions into symmetric coefficients [M-AL-VIETA]
Secondary cues None recorded
Linked theory Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]
Recognition triggers wording-specific
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2013-D1-P4

Day 1; official label P4; 10 points; legacy source-page label 153

Let $n \geq 3$ and let $x_1, \ldots, x_n$ be nonnegative real numbers. Define $A = \sum_{i=1}^{n} x_i$, $B = \sum_{i=1}^{n} x_i^2$ and $C = \sum_{i=1}^{n} x_i^3$. Prove that

$$(n+1)A^2B + (n-2)B^2 \geq A^4 + (2n-2)AC.$$

(Proposed by Géza Kós, Eötvös University, Budapest)

Open all solution variants | Source note

2013-D1-P5 Long/deep

Fingerprint complex sequence with sum a_n^p converging iff p nonprime
Primary domain Complex analysis / polynomials
Problem type existence/construction
Linked methods Translate root restrictions into symmetric coefficients [M-AL-VIETA]
Secondary cues Real analysis / calculus; Complex analysis / polynomials
Linked theory Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Build a witness and verify every constraint [S-CONSTRUCTION]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2013-D1-P5

Day 1; official label P5; 10 points; legacy source-page label 153

Does there exist a sequence (a_n) of complex numbers such that for every positive integer p we have that $\sum_{n=1}^{\infty} a_n^p$ converges if and only if p is not a prime?

(Proposed by Tomáš Bárta, Charles University, Prague)

Open all solution variants | Source note

Day 2

2013-D2-P1 Quick/standard

Fingerprint bound |z^3+1| given |z+1|>2
Primary domain Complex analysis / polynomials
Problem type inequality/sharp bound
Linked methods Compare coefficients after a structural substitution [M-AL-COEFFICIENT]
Secondary cues None recorded
Linked theory Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]
Recognition triggers wording-specific
Preparation route CP-Polynomials/Complex
Trap watch Separate polynomial identity from pointwise testing; check multiplicities, zero polynomial cases, and real-vs-complex roots.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Translate geometry into a controlled representation [S-GEOMETRIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2013-D2-P1

Day 2; official label P6; 10 points; legacy source-page label 154

Let z be a complex number with $|z+1|>2$. Prove that $|z^3+1|>1$.

(Proposed by Walther Janous and Gerhard Kirchner, Innsbruck)

Open all solution variants | Source note

2013-D2-P2 Quick/standard

Fingerprint alternating sum of floor functions for coprime p,q
Primary domain Algebra / number theory
Problem type exact computation
Linked methods Construct and count compatible local residues [M-AL-CRT]
Secondary cues None recorded
Linked theory Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2013-D2-P2

Day 2; official label P7; 10 points; legacy source-page label 154

Let p and q be relatively prime positive integers. Prove that

$$\sum_{k=0}^{pq-1}(-1)^{\lfloor \frac{k}{p}\rfloor+\lfloor \frac{k}{q}\rfloor}=\begin{cases}0 & \text{if } pq \text{ is even,}\\ 1 & \text{if } pq \text{ is odd.}\end{cases} \qquad (*)$$

(Here $\lfloor x\rfloor$ denotes the integer part of x.)

(Proposed by Alexander Bolbot, State University, Novosibirsk)

Open all solution variants | Source note

2013-D2-P3 Medium

Fingerprint unit vector nearly orthogonal to d unit vectors
Primary domain Linear algebra / matrices
Problem type existence/construction
Linked methods Prove a dimension obstruction and attain it [M-LA-DIMENSION]; Test a linear map on a basis and normalize its values [M-LA-DUAL]
Secondary cues Geometry / convexity / metric
Linked theory Dual bases, kernels and separating functionals [T-LA-DUALITY]; Inner products, orthogonality and polarization [T-LA-INNER]; Rank, dimension and exact linear constraints [T-LA-RANK]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers wording-specific
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2013-D2-P3

Day 2; official label P8; 10 points; legacy source-page label 154

Suppose that $v_1, \dots, v_d$ are unit vectors in $\mathbb{R}^d$. Prove that there exists a unit vector u such that

$$|u \cdot v_i| \leq 1/\sqrt{d}$$

for $i = 1, 2, \dots, d$.
(Here $\cdot$ denotes the usual scalar product on $\mathbb{R}^d$.)

(Proposed by Tomasz Tkocz, University of Warwick)

Open all solution variants | Source note

2013-D2-P4 Medium/trap

Fingerprint infinite set with all pairwise sums square-free
Primary domain Algebra / number theory
Problem type existence/construction
Linked methods Construct by excluding a sparse set of bad choices [M-AL-SIEVE]
Secondary cues Real analysis / calculus
Linked theory Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Prime valuations, divisors, and factorials [T-AL-VALUATION]; Finite counting and incidence identities [T-DS-COUNTING]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers integral/limit trigger; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Build a witness and verify every constraint [S-CONSTRUCTION]; Reduce size while preserving the defining conditions [S-INDUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2013-D2-P4

Day 2; official label P9; 10 points; legacy source-page label 154

Does there exist an infinite set M consisting of positive integers such that for any $a, b \in M$, with $a < b$, the sum $a + b$ is square-free?
(A positive integer is called square-free if no perfect square greater than 1 divides it.)

(Proposed by Fedor Petrov, St. Petersburg State University)

Open all solution variants | Source note

2013-D2-P5 Long/deep

Fingerprint parity of good necklace colorings count
Primary domain Combinatorics
Problem type counting/enumeration
Linked methods Strengthen the statement before the induction step [M-DS-INDUCT]; Split a family by traces, pieces, or labels [M-DS-PARTITION]; Pair objects canonically to prove parity [M-DS-INVOLUTION]; Partition into states and propagate a recurrence [M-DS-STATE]
Secondary cues Geometry / convexity / metric
Linked theory Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Finite counting and incidence identities [T-DS-COUNTING]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Recognition triggers counting/permutation trigger
Preparation route CMB-Combinatoricsi
Trap watch Avoid overcounting; define the encoded object precisely and check extremal/minimal cases.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2013-D2-P5

Day 2; official label P10; 10 points; legacy source-page label 154

Consider a circular necklace with 2013 beads. Each bead can be painted either white or green. A painting of the necklace is called good, if among any 21 successive beads there is at least one green bead. Prove that the number of good paintings of the necklace is odd.
(Two paintings that differ on some beads, but can be obtained from each other by rotating or flipping the necklace, are counted as different paintings.)

(Proposed by Vsevolod Bykov and Oleksandr Rybak, Kiev)

Open all solution variants | Source note

2.21 2014

Day 1

2014-D1-P1 Quick/standard

Fingerprint pairs (trace, det) giving unique symmetric 2x2 matrix
Primary domain Linear algebra / matrices
Problem type classification/find-all
Linked methods Read the entries before choosing a theorem [M-LA-ENTRY]; Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]
Secondary cues None recorded
Linked theory Determinants, block identities and inverses [T-LA-DETERMINANT]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]
Recognition triggers matrix word; det/rank/eigen trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]
Navigation Full statement All solution variants Source note Learning route

2014-D1-P1 continued

Full statement: 2014-D1-P1

Day 1; official label P1; 10 points; legacy source-page label 155

Determine all pairs (a, b) of real numbers for which there exists a unique symmetric 2×2 matrix M with real entries satisfying $\operatorname{trace}(M) = a$ and $\det(M) = b$.

(Proposed by Stephan Wagner, Stellenbosch University)

Open all solution variants | Source note

2014-D1-P2 Medium/trap

Fingerprint asymptotics of partial sums of staircase sequence
Primary domain Real analysis / calculus
Problem type classification/find-all
Linked methods Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]; Separate a main range from exceptional scales [M-AN-SPLIT]
Secondary cues None recorded
Linked theory Scale separation and limit comparisons [T-AN-ASYMPTOTICS]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers integral/limit trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2014-D1-P2

Day 1; official label P2; 10 points; legacy source-page label 155

Consider the following sequence

$$(a_n)_{n=1}^{\infty} = (1, 1, 2, 1, 2, 3, 1, 2, 3, 4, 1, 2, 3, 4, 5, 1, \dots).$$

Find all pairs (α, β) of positive real numbers such that $\lim_{n\to\infty} \dfrac{\sum_{k=1}^{n} a_k}{n^{\alpha}} = \beta$.

(Proposed by Tomas Barta, Charles University, Prague)

Open all solution variants | Source note

2014-D1-P3 Medium

Fingerprint coefficients giving n real roots under any signs
Primary domain Complex analysis / polynomials
Problem type existence/construction
Linked methods Compare coefficients after a structural substitution [M-AL-COEFFICIENT]
Secondary cues None recorded
Linked theory Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Continuity, compactness, and order [T-AN-CONTINUITY]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Recognition triggers wording-specific
Preparation route CP-Polynomials/Complex
Trap watch Separate polynomial identity from pointwise testing; check multiplicities, zero polynomial cases, and real-vs-complex roots.
Proof strategies Build a witness and verify every constraint [S-CONSTRUCTION]; Reduce size while preserving the defining conditions [S-INDUCTION]
Navigation Full statement All solution variants Source note Learning route

2014-D1-P3 continued

Full statement: 2014-D1-P3

Day 1; official label P3; 10 points; legacy source-page label 155

Let n be a positive integer. Show that there are positive real numbers $a_0, a_1, \ldots, a_n$ such that for each choice of signs the polynomial

$$\pm a_n x^n \pm a_{n-1} x^{n-1} \pm \cdots \pm a_1 x \pm a_0$$

has n distinct real roots.

(Proposed by Stephan Neupert, TUM, München)

Open all solution variants | Source note

2014-D1-P4 Medium

Fingerprint first exponent in perfect number factorization is even
Primary domain Algebra / number theory
Problem type structural proof
Linked methods Track one prime or one extreme divisor [M-AL-VALUATION]
Secondary cues Geometry / convexity / metric; Real analysis / calculus
Linked theory Prime valuations, divisors, and factorials [T-AL-VALUATION]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Recognition triggers integral/limit trigger; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Choose an extremal object and force rigidity [S-EXTREMAL]; Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2014-D1-P4

Day 1; official label P4; 10 points; legacy source-page label 155

Let $n > 6$ be a perfect number, and let $n = p_1^{e_1} \cdots p_k^{e_k}$ be its prime factorisation with $1 < p_1 < \cdots < p_k$. Prove that e_1 is an even number.
A number n is perfect if $s(n) = 2n$, where $s(n)$ is the sum of the divisors of n.

(Proposed by Javier Rodrigo, Universidad Pontificia Comillas)

Open all solution variants | Source note

2014-D1-P5 Long/deep

Fingerprint bound self-intersections of equiangular closed broken line
Primary domain Geometry / convexity / metric
Problem type inequality/sharp bound
Linked methods Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]; Choose information that makes an encoding injective [M-DS-ENCODE]; Repair an extremal configuration by a local exchange [M-DS-EXTREMAL]
Secondary cues None recorded
Linked theory Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Finite counting and incidence identities [T-DS-COUNTING]; Extremal choice, exchange arguments, and invariants [T-DS-EXTREMAL]
Recognition triggers wording-specific
Preparation route GEO-Geometry/Metricif
Trap watch Control degenerate configurations and make clear which coordinates or affine transformations preserve the quantity.
Proof strategies Translate geometry into a controlled representation [S-GEOMETRIC]; Choose an extremal object and force rigidity [S-EXTREMAL]; Choose the objects and count them without ambiguity [S-COUNTING]; Make the contradiction quantitative or structural [S-CONTRADICTION]

2014-D1-P5 continued

Navigation Full statement All solution variants Source note Learning route

Full statement: 2014-D1-P5

Day 1; official label P5; 10 points; legacy source-page label 155

Let $A_1A_2 \ldots A_{3n}$ be a closed broken line consisting of $3n$ line segments in the Euclidean plane. Suppose that no three of its vertices are collinear, and for each index $i = 1, 2, \ldots, 3n$, the triangle $A_iA_{i+1}A_{i+2}$ has counterclockwise orientation and $\angle A_iA_{i+1}A_{i+2} = 60^\circ$, using the notation $A_{3n+1} = A_1$ and $A_{3n+2} = A_2$. Prove that the number of self-intersections of the broken line is at most $\frac{3}{2}n^2 - 2n + 1$.

(Proposed by Martin Langer)

Open all solution variants | Source note

Day 2

2014-D2-P1 Quick/standard

Fingerprint digit condition forces infinitely many missing integers
Primary domain Combinatorics
Problem type existence/construction
Linked methods Choose information that makes an encoding injective [M-DS-ENCODE]; Encode, decode, and verify a bijection [M-DS-BIJECTION]
Secondary cues None recorded
Linked theory Finite counting and incidence identities [T-DS-COUNTING]; Countability, accumulation, and Cantor constructions [T-AN-SETS]
Recognition triggers wording-specific
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Build a witness and verify every constraint [S-CONSTRUCTION]; Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2014-D2-P1

Day 2; official label P6; 10 points; legacy source-page label 156

For a positive integer x, denote its nth decimal digit by $d_n(x)$, i.e. $d_n(x) \in \{0, 1, \ldots, 9\}$ and $x = \sum_{n=1}^{\infty} d_n(x)10^{n-1}$. Suppose that for some sequence $(a_n)_{n=1}^{\infty}$, there are only finitely many zeros in the sequence $(d_n(a_n))_{n=1}^{\infty}$. Prove that there are infinitely many positive integers that do not occur in the sequence $(a_n)_{n=1}^{\infty}$.

(Proposed by Alexander Bolbot, State University, Novosibirsk)

Open all solution variants | Source note

2014-D2-P2 Quick/standard

Fingerprint diagonal products dominate eigenvalue products for symmetric matrix
Primary domain Linear algebra / matrices
Problem type inequality/sharp bound
Linked methods Use trace to extract a scalar invariant [M-LA-TRACE]
Secondary cues Linear algebra / matrix theory
Linked theory Trace pairing and commutator calculus [T-LA-TRACE]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Inner products, orthogonality and polarization [T-LA-INNER]
Recognition triggers matrix word; det/rank/eigen trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equali-

2014-D2-P2 continued

ty/sharpness cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2014-D2-P2

Day 2; official label P7; 10 points; legacy source-page label 156

Let $A = (a_{ij})_{i,j=1}^{n}$ be a symmetric $n \times n$ matrix with real entries, and let $\lambda_1, \lambda_2, \ldots, \lambda_n$ denote its eigenvalues. Show that

$$\sum_{1 \le i < j \le n} a_{ii} a_{jj} \ge \sum_{1 \le i < j \le n} \lambda_i \lambda_j,$$

and determine all matrices for which equality holds.

(Proposed by Martin Niepel, Comenius University, Bratislava)

Open all solution variants | Source note

2014-D2-P3 Medium

Fingerprint bound nth derivative of sin(x)/x
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Design an auxiliary derivative [M-AN-AUXILIARY]; Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]
Secondary cues Functional equations / dynamics
Linked theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Trigonometric identities and safe branches [T-AN-TRIG]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Recognition triggers functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2014-D2-P3

Day 2; official label P8; 10 points; legacy source-page label 156

Let $f(x) = \frac{\sin x}{x}$, for $x > 0$, and let n be a positive integer. Prove that $\left|f^{(n)}(x)\right| < \frac{1}{n+1}$, where $f^{(n)}$ denotes the nth derivative of f.

(Proposed by Alexander Bolbot, State University, Novosibirsk)

Open all solution variants | Source note

2014-D2-P4 Medium

Fingerprint minimal size of k-generic subset in R^n
Primary domain Geometry / convexity / metric
Problem type inequality/sharp bound
Linked methods Repair an extremal configuration by a local exchange [M-DS-EXTREMAL]; Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Secondary cues Combinatorics / discrete structures
Linked theory Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Extremal choice, exchange arguments, and invariants [T-DS-EXTREMAL]; Rank, dimension and exact linear constraints [T-LA-RANK]
Recognition triggers wording-specific
Preparation route GEO-Geometry/Metricif
Trap watch Control degenerate configurations and make clear which coordinates or affine transformations preserve the quantity.
Proof strategies Choose an extremal object and force rigidity [S-EXTREMAL]; Translate geometry into a con-

2014-D2-P4 continued

trolled representation [S-GEOMETRIC]; Build a witness and verify every constraint [S-CONSTRUCTION]; Build a chain of bounds and track equality [S-BOUND]

Navigation Full statement All solution variants Source note Learning route

Full statement: 2014-D2-P4

Day 2; official label P9; 10 points; legacy source-page label 156

We say that a subset of $\mathbb{R}^n$ is k-almost contained by a hyperplane if there are less than k points in that set which do not belong to the hyperplane. We call a finite set of points k-generic if there is no hyperplane that k-almost contains the set. For each pair of positive integers k and n, find the minimal number $d(k,n)$ such that every finite k-generic set in $\mathbb{R}^n$ contains a k-generic subset with at most $d(k,n)$ elements.

(Proposed by Shachar Carmeli, Weizmann Inst. and Lev Radzivilovsky, Tel Aviv Univ.)

Open all solution variants | Source note

2014-D2-P5 Long/deep

Fingerprint derangement-type identity for restricted permutations
Primary domain Combinatorics
Problem type counting/enumeration
Linked methods Encode, decode, and verify a bijection [M-DS-BIJECTION]; Strengthen the statement before the induction step [M-DS-INDUCT]
Secondary cues Geometry / convexity / metric
Linked theory Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]
Recognition triggers wording-specific
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Reduce size while preserving the defining conditions [S-INDUCTION]; Expose the structure before calculating [S-STRUCTURE]

Navigation Full statement All solution variants Source note Learning route

Full statement: 2014-D2-P5

Day 2; official label P10; 10 points; legacy source-page label 156

For every positive integer n, denote by D_n the number of permutations $(x_1,\dots,x_n)$ of $(1,2,\dots,n)$ such that $x_j \neq j$ for every $1 \leq j \leq n$. For $1 \leq k \leq \frac{n}{2}$, denote by $\Delta(n,k)$ the number of permutations of $(1,2,\dots,n)$ such that $x_i = k+i$ for every $1 \leq i \leq k$ and $x_j \neq j$ for every $1 \leq j \leq n$. Prove that

$$\Delta(n,k) = \sum_{i=0}^{k-1} \binom{k-1}{i} \frac{D_{(n+1)-(k+i)}}{n-(k+i)}.$$

(Proposed by Combinatorics; Ferdowsi University of Mashhad, Iran; Mirzavaziri)

Open all solution variants | Source note

2.22 2015

Day 1

2015-D1-P1 Quick/standard

Fingerprint inverse sum identity forces equal determinants
Primary domain Linear algebra / matrices
Problem type structural proof
Linked methods Complete the equation to a product equal to identity [M-LA-INVERSE]; Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]
Secondary cues Complex analysis / polynomials; Algebra / number theory; Geometry / convexity / metric
Linked theory Determinants, block identities and inverses [T-LA-DETERMINANT]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers matrix word; det/rank/eigen trigger; integer/prime/mod trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2015-D1-P1

Day 1; official label P1; 10 points; legacy source-page label 157

For any integer $n \geq 2$ and two $n \times n$ matrices with real entries A, B that satisfy the equation

$$A^{-1} + B^{-1} = (A + B)^{-1}$$

prove that $\det(A) = \det(B)$.
Does the same conclusion follow for matrices with complex entries?

(Proposed by Zbigniew Skoczylas, Wrocław University of Technology)

Open all solution variants | Source note

2015-D1-P2 Quick/standard

Fingerprint sum of binary-complement function bounded by n^2/4
Primary domain Algebra / number theory
Problem type inequality/sharp bound
Linked methods Compare coefficients after a structural substitution [M-AL-COEFFICIENT]
Secondary cues Functional equations / dynamics; Geometry / convexity / metric; Real analysis / calculus
Linked theory Linear recurrences and generating functions [T-AL-RECURRENCE]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers integer/prime/mod trigger; functional-equation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Prove necessity and sufficiency separately [S-CLASSIFICATION]
Navigation Full statement All solution variants Source note Learning route

2015-D1-P2 continued

Full statement: 2015-D1-P2

Day 1; official label P2; 10 points; legacy source-page label 157

For a positive integer n, let $f(n)$ be the number obtained by writing n in binary and replacing every 0 with 1 and vice versa. For example, $n = 23$ is 10111 in binary, so $f(n)$ is 1000 in binary, therefore $f(23) = 8$. Prove that

$$\sum_{k=1}^{n} f(k) \leq \frac{n^2}{4}.$$

When does equality hold?

(Proposed by Stephan Wagner, Stellenbosch University)

Open all solution variants | Source note

2015-D1-P3 Medium

Fingerprint rationality of sum 1/F(2^n) for linear recurrence
Primary domain Algebra / number theory
Problem type exact computation
Linked methods Find a recurrence that reveals the invariant [M-AL-RECURRENCE]
Secondary cues Geometry / convexity / metric; Algebra / number theory; Real analysis / calculus
Linked theory Linear recurrences and generating functions [T-AL-RECURRENCE]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Prime valuations, divisors, and factorials [T-AL-VALUATION]
Recognition triggers det/rank/eigen trigger; functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Control the limiting operation before taking the limit [S-ANALYTIC]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2015-D1-P3

Day 1; official label P3; 10 points; legacy source-page label 157

Let $F(0) = 0$, $F(1) = \frac{3}{2}$, and $F(n) = \frac{5}{2}F(n-1) - F(n-2)$ for $n \geq 2$.

Determine whether or not $\displaystyle\sum_{n=0}^{\infty} \frac{1}{F(2^n)}$ is a rational number.

(Proposed by Gerhard Woeginger, Eindhoven University of Technology)

Open all solution variants | Source note

2015-D1-P4 Medium

Fingerprint integer combination of arctangents equal to arctan(16)
Primary domain Algebra / number theory
Problem type nonexistence/counterexample
Linked methods Track one prime or one extreme divisor [M-AL-VALUATION]
Secondary cues Real analysis / calculus; Geometry / convexity / metric
Linked theory Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]; Prime valuations, divisors, and factorials [T-AL-VALUATION]
Recognition triggers det/rank/eigen trigger; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

2015-D1-P4 continued

Full statement: 2015-D1-P4

Day 1; official label P4; 10 points; legacy source-page label 157

Determine whether or not there exist 15 integers $m_1, \dots, m_{15}$ such that

$$\sum_{k=1}^{15} m_k \cdot \arctan(k) = \arctan(16). \tag{1}$$

(Proposed by Gerhard Woeginger, Eindhoven University of Technology)

Open all solution variants | Source note

2015-D1-P5 Long/deep

Fingerprint many obtuse angles at interior point of simplex
Primary domain Geometry / convexity / metric
Problem type inequality/sharp bound
Linked methods Encode a relation as a graph and extract a cycle [M-DS-GRAPH]; Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Secondary cues Geometry / convexity / metric; Linear algebra / matrix theory
Linked theory Convex bodies, supporting functionals, and packing [T-DS-CONVEX]; Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]; Inner products, orthogonality and polarization [T-LA-INNER]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]
Recognition triggers geometry/metric trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Translate geometry into a controlled representation [S-GEOMETRIC]; Expose the structure before calculating [S-STRUCTURE]; Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2015-D1-P5

Day 1; official label P5; 10 points; legacy source-page label 157

Let $n \geq 2$, let $A_1, A_2, \dots, A_{n+1}$ be $n+1$ points in the n-dimensional Euclidean space, not lying on the same hyperplane, and let B be a point strictly inside the convex hull of $A_1, A_2, \dots, A_{n+1}$. Prove that $\angle A_i B A_j > 90^\circ$ holds for at least n pairs (i, j) with $1 \leq i < j \leq n+1$.

(Proposed by Géza Kós, Eötvös University, Budapest)

Open all solution variants | Source note

Day 2

2015-D2-P6 Long/deep

Fingerprint sum of 1/sqrt(n(n+1)) less than 2
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]; Find a finite telescoping identity [M-AN-TELESCOPE]
Secondary cues None recorded
Linked theory Series, finite cancellation, and summation by parts [T-AN-SERIES]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]
Recognition triggers wording-specific
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.

2015-D2-P6 continued

Proof strategies Build a chain of bounds and track equality [S-BOUND]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2015-D2-P6

Day 2; official label P6; 10 points; legacy source-page label 158

Prove that

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}(n+1)} < 2.$$

(Proposed by Ivan Krijan, University of Zagreb)

Open all solution variants | Source note

2015-D2-P7 Medium

Fingerprint limit of averaged integral of A^(1/x)
Primary domain Real analysis / calculus
Problem type limit/asymptotic evaluation
Linked methods Separate a main range from exceptional scales [M-AN-SPLIT]; Compare growth through derivatives [M-AN-RATIO]; Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]
Secondary cues None recorded
Linked theory Scale separation and limit comparisons [T-AN-ASYMPTOTICS]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Recognition triggers integral/limit trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Control the limiting operation before taking the limit [S-ANALYTIC]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2015-D2-P7

Day 2; official label P7; 10 points; legacy source-page label 158

Compute

$$\lim_{A\to+\infty} \frac{1}{A} \int_1^A A^{\frac{1}{x}}\, dx.$$

(Proposed by Jan Šustek, University of Ostrava)

Open all solution variants | Source note

2015-D2-P8 Medium

Fingerprint weighted count of 26-letter words equals 3^75
Primary domain Combinatorics
Problem type exact computation
Linked methods Encode the statistic and extract the required coefficient [M-DS-GENERATING]; Count incidences with their exact multiplicity [M-DS-DOUBLECOUNT]
Secondary cues Real analysis / calculus
Linked theory Generating functions, coefficient filters, and finite differences [T-DS-GENERATING]; Finite counting and incidence identities [T-DS-COUNTING]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers integral/limit trigger; counting/permutation trigger
Preparation route CMB-Combinatoricsi
Trap watch Avoid overcounting; define the encoded object precisely and check extremal/minimal cases.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

2015-D2-P8 continued

Full statement: 2015-D2-P8

Day 2; official label P8; 10 points; legacy source-page label 158

Consider all 26^{26} words of length 26 in the Latin alphabet. Define the weight of a word as $1/(k+1)$, where k is the number of letters not used in this word. Prove that the sum of the weights of all words is 3^{75}.

(Proposed by Fedor Petrov, St. Petersburg State University)

Open all solution variants | Source note

2015-D2-P9 Long/deep

Fingerprint max dimension of linear space of t-normal matrices
Primary domain Linear algebra / matrices
Problem type inequality/sharp bound
Linked methods Prove a dimension obstruction and attain it [M-LA-DIMENSION]; Use trace to extract a scalar invariant [M-LA-TRACE]
Secondary cues Complex analysis / polynomials; Geometry / convexity / metric
Linked theory Rank, dimension and exact linear constraints [T-LA-RANK]; Trace pairing and commutator calculus [T-LA-TRACE]; Dual bases, kernels and separating functionals [T-LA-DUALITY]; Adapted bases, centralizers and common eigenspaces [T-LA-INVARIANT]
Recognition triggers matrix word; det/rank/eigen trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2015-D2-P9

Day 2; official label P9; 10 points; legacy source-page label 158

An $n \times n$ complex matrix A is called t-normal if $AA^t = A^tA$ where A^t is the transpose of A. For each n, determine the maximum dimension of a linear space of complex $n \times n$ matrices consisting of t-normal matrices.

(Proposed by Shachar Carmeli, Weizmann Institute of Science)

Open all solution variants | Source note

2015-D2-P10 Long/deep

Fingerprint integer polynomial exceeds e^(-n) somewhere on [0,1]
Primary domain Complex analysis / polynomials
Problem type inequality/sharp bound
Linked methods Build a holomorphic function with a useful boundary [M-AL-ANALYTIC]
Secondary cues Real analysis / calculus; Complex analysis / polynomials
Linked theory Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Factorial growth and the prime-number input [T-AL-PRIMEGROWTH]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers polynomial/root trigger; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

2015-D2-P10 continued

Full statement: 2015-D2-P10

Day 2; official label P10; 10 points; legacy source-page label 158

Let n be a positive integer, and let $p(x)$ be a polynomial of degree n with integer coefficients. Prove that

$$\max_{0\le x\le 1}|p(x)| > \frac{1}{e^n}.$$

(Proposed by Géza Kós, Eötvös University, Budapest)

Open all solution variants | Source note

2.23 2016

Day 1

2016-D1-P1 Quick/standard

Fingerprint function with infinitely many zeros vanishes at endpoint
Primary domain Real analysis / calculus
Problem type structural proof
Linked methods Localize an extremum or an accumulation point [M-AN-COMPACT]; Use a dense set, an enumeration, or a category argument [M-AN-TOPOLOGY]
Secondary cues Real analysis / calculus
Linked theory Continuity, compactness, and order [T-AN-CONTINUITY]; Countability, accumulation, and Cantor constructions [T-AN-SETS]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Trigonometric identities and safe branches [T-AN-TRIG]
Recognition triggers continuous/differentiable trigger; functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Make the contradiction quantitative or structural [S-CONTRADICTION]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2016-D1-P1

Day 1; official label P1; 10 points; legacy source-page label 159

Let $f:[a,b]\to\mathbb{R}$ be continuous on $[a,b]$ and differentiable on (a,b). Suppose that f has infinitely many zeros, but there is no $x\in(a,b)$ with $f(x)=f'(x)=0$.
(a) Prove that $f(a)f(b)=0$.
(b) Give an example of such a function on $[0,1]$.

(Proposed by Alexandr Bolbot, Novosibirsk State University)

Open all solution variants | Source note

2016-D1-P2 Quick/standard

Fingerprint at most n matrices with pairwise zero products
Primary domain Linear algebra / matrices
Problem type inequality/sharp bound
Linked methods Prove a dimension obstruction and attain it [M-LA-DIMENSION]
Secondary cues Real analysis / calculus; Linear algebra / matrix theory
Linked theory Rank, dimension and exact linear constraints [T-LA-RANK]; Dual bases, kernels and separating functionals [T-LA-DUALITY]
Recognition triggers matrix word; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory

2016-D1-P2 continued

Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2016-D1-P2

Day 1; official label P2; 10 points; legacy source-page label 159

Let k and n be positive integers. A sequence $(A_1, \dots, A_k)$ of $n \times n$ real matrices is preferred by Ivan the Confessor if $A_i^2 \neq 0$ for $1 \leq i \leq k$, but $A_i A_j = 0$ for $1 \leq i, j \leq k$ with $i \neq j$. Show that $k \leq n$ in all preferred sequences, and give an example of a preferred sequence with $k = n$ for each n.

(Proposed by Fedor Petrov, St. Petersburg State University)

Open all solution variants | Source note

2016-D1-P3 Medium

Fingerprint Cauchy-Schwarz-type bound for sums (ab-b^2)/(a+b)
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]
Secondary cues Real analysis / calculus
Linked theory Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Inner products, orthogonality and polarization [T-LA-INNER]
Recognition triggers integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2016-D1-P3

Day 1; official label P3; 10 points; legacy source-page label 159

Let n be a positive integer. Also let $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ be real numbers such that $a_i + b_i > 0$ for $i = 1, 2, \dots, n$. Prove that

$$\sum_{i=1}^{n} \frac{a_i b_i - b_i^2}{a_i + b_i} \leq \frac{\sum_{i=1}^{n} a_i \cdot \sum_{i=1}^{n} b_i - \left(\sum_{i=1}^{n} b_i\right)^2}{\sum_{i=1}^{n} (a_i + b_i)}.$$

(Proposed by Daniel Strzelecki, Nicolaus Copernicus University in Toruń, Poland)

Open all solution variants | Source note

2016-D1-P4 Medium

Fingerprint union-closed family contains three large sets
Primary domain Combinatorics
Problem type inequality/sharp bound
Linked methods Count incidences with their exact multiplicity [M-DS-DOUBLECOUNT]; Split a family by traces, pieces, or labels [M-DS-PARTITION]; Repair an extremal configuration by a local exchange [M-DS-EXTREMAL]; Strengthen the statement before the induction step [M-DS-INDUCT]
Secondary cues Combinatorics / discrete structures; Real analysis / calculus
Linked theory Finite set systems and antichains [T-DS-SETS]; Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Extremal choice,

2016-D1-P4 continued

exchange arguments, and invariants [T-DS-EXTREMAL]
Recognition triggers integer/prime/mod trigger; graph/color/set trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Reduce size while preserving the defining conditions [S-INDUCTION]; Choose an extremal object and force rigidity [S-EXTREMAL]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2016-D1-P4

Day 1; official label P4; 10 points; legacy source-page label 159

Let $n \geq k$ be positive integers, and let $\mathcal{F}$ be a family of finite sets with the following properties:
(i) $\mathcal{F}$ contains at least $\binom{n}{k} + 1$ distinct sets containing exactly k elements;
(ii) for any two sets $A, B \in \mathcal{F}$, their union $A \cup B$ also belongs to $\mathcal{F}$.
Prove that $\mathcal{F}$ contains at least three sets with at least n elements.

(Proposed by Fedor Petrov, St. Petersburg State University)

Open all solution variants | Source note

2016-D1-P5 Long/deep

Fingerprint permutation inversions divisible by n+1, prime comparisons
Primary domain Combinatorics
Problem type existence/construction
Linked methods Encode the statistic and extract the required coefficient [M-DS-GENERATING]; Choose information that makes an encoding injective [M-DS-ENCODE]
Secondary cues Combinatorics / discrete structures; Real analysis / calculus; Functional equations / dynamics
Linked theory Generating functions, coefficient filters, and finite differences [T-DS-GENERATING]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]; Roots-of-unity filters and positive trigonometric kernels [T-AL-FOURIER]
Recognition triggers integer/prime/mod trigger; counting/permutation trigger; functional-equation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Control the limiting operation before taking the limit [S-ANALYTIC]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2016-D1-P5

Day 1; official label P5; 10 points; legacy source-page label 159

Let S_n denote the set of permutations of the sequence $(1, 2, \ldots, n)$. For every permutation $\pi = (\pi_1, \ldots, \pi_n) \in S_n$, let $\mathrm{inv}(\pi)$ be the number of pairs $1 \leq i < j \leq n$ with $\pi_i > \pi_j$; i.e. the number of inversions in π. Denote by $f(n)$ the number of permutations $\pi \in S_n$ for which $\mathrm{inv}(\pi)$ is divisible by $n+1$.
Prove that there exist infinitely many primes p such that $f(p-1) > \frac{(p-1)!}{p}$, and infinitely many primes p such that $f(p-1) < \frac{(p-1)!}{p}$.

(Proposed by Fedor Petrov, St. Petersburg State University)

Open all solution variants | Source note

Day 2

2016-D2-P6 Long/deep

Fingerprint double sum bound under weighted normalization condition
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Find a finite telescoping identity [M-AN-TELESCOPE]; Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]
Secondary cues None recorded
Linked theory Nonnegative sums, Fubini, and integral kernels [T-AN-FUBINI]; Series, finite cancellation, and summation by parts [T-AN-SERIES]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]
Recognition triggers wording-specific
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2016-D2-P6

Day 2; official label P6; 10 points; legacy source-page label 160

Let $(x_1, x_2, \dots)$ be a sequence of positive real numbers satisfying $\sum_{n=1}^{\infty} \frac{x_n}{2n-1} = 1$. Prove that

$$\sum_{k=1}^{\infty} \sum_{n=1}^{k} \frac{x_n}{k^2} \leq 2.$$

(Proposed by Gerhard J. Woeginger, The Netherlands)

Open all solution variants | Source note

2016-D2-P7 Medium

Fingerprint minimize integral of f with f(x)+f(y)>=|x-y|
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Pair an integral through symmetry [M-AN-PAIR]
Secondary cues Functional equations / dynamics
Linked theory Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]
Recognition triggers continuous/differentiable trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2016-D2-P7

Day 2; official label P7; 10 points; legacy source-page label 160

Today, Ivan the Confessor prefers continuous functions $f : [0, 1] \to \mathbb{R}$ satisfying $f(x) + f(y) \geq |x - y|$ for all pairs $x, y \in [0, 1]$. Find the minimum of $\int_0^1 f$ over all preferred functions.

(Proposed by Fedor Petrov, St. Petersburg State University)

Open all solution variants | Source note

2016-D2-P8 Medium

Fingerprint involution-like function on Z_n forces n=2 mod 4
Primary domain Algebra / number theory
Problem type structural proof
Linked methods Work orbit by orbit and track parity [M-AL-ORBIT]
Secondary cues Functional equations / dynamics; Real analysis / calculus
Linked theory Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]; Groups, orders, generators, and quotients [T-AL-GROUP]
Recognition triggers integer/prime/mod trigger; group/ring/field trigger; functional-equation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2016-D2-P8

Day 2; official label P8; 10 points; legacy source-page label 160

Let n be a positive integer, and denote by $\mathbb{Z}_n$ the ring of integers modulo n. Suppose that there exists a function $f : \mathbb{Z}_n \to \mathbb{Z}_n$ satisfying the following three properties:
(i) $f(x) \neq x$,
(ii) $f(f(x)) = x$,
(iii) $f(f(f(x+1)+1)+1) = x$ for all $x \in \mathbb{Z}_n$.
Prove that $n \equiv 2 \pmod 4$.

(Proposed by Ander Lamaison Vidarte, Berlin Mathematical School, Germany)

Open all solution variants | Source note

2016-D2-P9 Long/deep

Fingerprint log-concavity of lattice point counts in l1-balls
Primary domain Combinatorics
Problem type inequality/sharp bound
Linked methods Strengthen the statement before the induction step [M-DS-INDUCT]; Split a family by traces, pieces, or labels [M-DS-PARTITION]; Encode the statistic and extract the required coefficient [M-DS-GENERATING]
Secondary cues Real analysis / calculus; Functional equations / dynamics; Combinatorics / discrete structures
Linked theory Log-concave sequences and coefficient smoothing [T-DS-LOGCONCAVITY]; Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Generating functions, coefficient filters, and finite differences [T-DS-GENERATING]
Recognition triggers integer/prime/mod trigger; counting/permutation trigger; functional-equation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Reduce size while preserving the defining conditions [S-INDUCTION]; Build a chain of bounds and track equality [S-BOUND]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2016-D2-P9

Day 2; official label P9; 10 points; legacy source-page label 160

Let k be a positive integer. For each nonnegative integer n, let $f(n)$ be the number of solutions $(x_1, \ldots, x_k) \in \mathbb{Z}^k$ of the inequality $|x_1| + \cdots + |x_k| \leq n$. Prove that for every $n \geq 1$, we have $f(n-1)f(n+1) \leq f(n)^2$.

(Proposed by Esteban Arreaga, Renan Finder and José Madrid, IMPA, Rio de Janeiro)

Open all solution variants | Source note

2016-D2-P10 Long/deep

Fingerprint norm bound for powers of matrix with small eigenvalues
Primary domain Linear algebra / matrices
Problem type inequality/sharp bound
Linked methods Combine norm bounds with algebraic decay [M-LA-NORM]; Exploit a compatible block decomposition [M-LA-BLOCK]
Secondary cues Real analysis / calculus; Complex analysis / polynomials; Geometry / convexity / metric
Linked theory Operator norms and convergence of matrix powers [T-LA-NORM]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]
Recognition triggers matrix word; det/rank/eigen trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2016-D2-P10

Day 2; official label P10; 10 points; legacy source-page label 160

Let A be a $n \times n$ complex matrix whose eigenvalues have absolute value at most 1. Prove that

$$\|A^n\| \leq \frac{n}{\ln 2}\|A\|^{n-1}.$$

(Here $\|B\| = \sup_{\|x\| \leq 1} \|Bx\|$ for every $n \times n$ matrix B and $\|x\| = \sqrt{\sum_{i=1}^{n} |x_i|^2}$ for every complex vector $x \in \mathbb{C}^n$.)

(Proposed by Ian Morris and Fedor Petrov, St. Petersburg State University)

Open all solution variants | Source note

2.24 2017

Day 1

2017-D1-P1 Quick/standard

Fingerprint eigenvalues of real matrix with A^2=A^T
Primary domain Linear algebra / matrices
Problem type classification/find-all
Linked methods Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]
Secondary cues Complex analysis / polynomials; Algebra / number theory; Geometry / convexity / metric
Linked theory Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]
Recognition triggers matrix word; det/rank/eigen trigger; integer/prime/mod trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

2017-D1-P1 continued

Full statement: 2017-D1-P1

Day 1; official label P1; 10 points; legacy source-page label 161

Determine all complex numbers λ for which there exist a positive integer n and a real $n \times n$ matrix A such that $A^2 = A^T$ and λ is an eigenvalue of A.

(Proposed by Alexandr Bolbot, Novosibirsk State University)

Open all solution variants | Source note

2017-D1-P2 Quick/standard

Fingerprint Lipschitz derivative implies (f')^2<2Lf
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]; Choose the correct convexity comparison [M-AN-CONVEX]
Secondary cues Functional equations / dynamics
Linked theory Lipschitz derivatives and convex smoothness [T-AN-SMOOTH]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Recognition triggers functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2017-D1-P2

Day 1; official label P2; 10 points; legacy source-page label 161

Let $f : \mathbb{R} \to (0, \infty)$ be a differentiable function, and suppose that there exists a constant $L > 0$ such that

$$\left|f'(x) - f'(y)\right| \leq L\left|x - y\right|$$

for all x, y. Prove that

$$\left(f'(x)\right)^2 < 2Lf(x)$$

holds for all x.

(Proposed by Jan Šustek, University of Ostrava)

Open all solution variants | Source note

2017-D1-P3 Medium/trap

Fingerprint iterated divisor-product squares pattern prescribed by any set S
Primary domain Algebra / number theory
Problem type existence/construction
Linked methods Choose a modulus that exposes the obstruction [M-AL-MODULAR]; Find a recurrence that reveals the invariant [M-AL-RECURRENCE]
Secondary cues Real analysis / calculus; Geometry / convexity / metric; Combinatorics / discrete structures
Linked theory Prime valuations, divisors, and factorials [T-AL-VALUATION]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Recognition triggers det/rank/eigen trigger; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Build a witness and verify every constraint [S-CONSTRUCTION]; Reduce size while preserving the defining conditions [S-INDUCTION]

2017-D1-P3 continued

Navigation Full statement All solution variants Source note Learning route

Full statement: 2017-D1-P3

Day 1; official label P3; 10 points; legacy source-page label 161

For any positive integer m, denote by $P(m)$ the product of positive divisors of m (e.g. $P(6) = 36$). For every positive integer n define the sequence

$$a_1(n) = n, \qquad a_{k+1}(n) = P(a_k(n)) \quad (k = 1, 2, \dots, 2016).$$

Determine whether for every set $S \subseteq \{1, 2, \dots, 2017\}$, there exists a positive integer n such that the following condition is satisfied:
For every k with $1 \leq k \leq 2017$, the number $a_k(n)$ is a perfect square if and only if $k \in S$.

(Proposed by Matko Ljulj, University of Zagreb)

Open all solution variants | Source note

2017-D1-P4 Medium

Fingerprint select group where many have exactly two friends
Primary domain Combinatorics
Problem type existence/construction
Linked methods Random selection and the expected number of successes [M-DS-EXPECTATION]
Secondary cues Combinatorics / discrete structures
Linked theory Finite probability, conditioning, and expectations [T-AN-PROBABILITY]; Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers group/ring/field trigger; counting/permutation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Build a witness and verify every constraint [S-CONSTRUCTION]; Choose the objects and count them without ambiguity [S-COUNTING]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2017-D1-P4

Day 1; official label P4; 10 points; legacy source-page label 161

There are n people in a city, and each of them has exactly 1000 friends (friendship is always symmetric). Prove that it is possible to select a group S of people such that at least $n/2017$ persons in S have exactly two friends in S.

(Proposed by Rooholah Majdodin and Fedor Petrov, St. Petersburg State University)

Open all solution variants | Source note

2017-D1-P5 Long/deep

Fingerprint bound common roots of f and z^n-1
Primary domain Complex analysis / polynomials
Problem type inequality/sharp bound
Linked methods Evaluate at roots of unity or average over them [M-AL-UNITY]
Secondary cues Real analysis / calculus; Functional equations / dynamics; Algebra / number theory
Linked theory Roots-of-unity filters and positive trigonometric kernels [T-AL-FOURIER]; Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers continuous/differentiable trigger; polynomial/root trigger; integer/prime/mod trigger; functional-equation trigger
Preparation route CP-Polynomials/Complex
Trap watch Separate polynomial identity from pointwise testing; check multiplicities, zero polynomial cases, and real-vs-complex roots.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure

2017-D1-P5 continued

before calculating [S-STRUCTURE]

Navigation Full statement All solution variants Source note Learning route

Full statement: 2017-D1-P5

Day 1; official label P5; 10 points; legacy source-page label 161

Let k and n be positive integers with $n \geq k^2 - 3k + 4$, and let

$$f(z) = z^{n-1} + c_{n-2}z^{n-2} + \cdots + c_0$$

be a polynomial with complex coefficients such that

$$c_0 c_{n-2} = c_1 c_{n-3} = \cdots = c_{n-2} c_0 = 0.$$

Prove that $f(z)$ and $z^n - 1$ have at most $n - k$ common roots.

(Proposed by Vsevolod Lev and Fedor Petrov, St. Petersburg State University)

Open all solution variants | Source note

Day 2

2017-D2-P6 Long/deep

Fingerprint limit of integral of f(nx) equals L
Primary domain Real analysis / calculus
Problem type limit/asymptotic evaluation
Linked methods Separate a main range from exceptional scales [M-AN-SPLIT]; Compare growth through derivatives [M-AN-RATIO]
Secondary cues Functional equations / dynamics
Linked theory Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Scale separation and limit comparisons [T-AN-ASYMPTOTICS]
Recognition triggers continuous/differentiable trigger; integral/limit trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Control the limiting operation before taking the limit [S-ANALYTIC]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2017-D2-P6

Day 2; official label P6; 10 points; legacy source-page label 162

Let $f : [0; +\infty) \to \mathbb{R}$ be a continuous function such that $\lim\limits_{x \to +\infty} f(x) = L$ exists (it may be finite or infinite). Prove that

$$\lim_{n \to \infty} \int_0^1 f(nx)\, dx = L.$$

(Proposed by Alexandr Bolbot, Novosibirsk State University)

Open all solution variants | Source note

2017-D2-P7 Medium/trap

Fingerprint finitely many n make q_n all-real-rooted
Primary domain Complex analysis / polynomials
Problem type structural proof
Linked methods Translate root restrictions into symmetric coefficients [M-AL-VIETA]
Secondary cues Algebra / number theory; Geometry / convexity / metric

2017-D2-P7 continued

Linked theory Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers polynomial/root trigger; integer/prime/mod trigger
Preparation route CP-Polynomials/Complex
Trap watch Separate polynomial identity from pointwise testing; check multiplicities, zero polynomial cases, and real-vs-complex roots.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2017-D2-P7

Day 2; official label P7; 10 points; legacy source-page label 162

Let $p(x)$ be a nonconstant polynomial with real coefficients. For every positive integer n, let

$$q_n(x) = (x+1)^n p(x) + x^n p(x+1).$$

Prove that there are only finitely many numbers n such that all roots of $q_n(x)$ are real.

(Proposed by Alexandr Bolbot, Novosibirsk State University)

Open all solution variants | Source note

2017-D2-P8 Medium

Fingerprint integer eigenvalues with binomial multiplicities for recursive block matrices
Primary domain Linear algebra / matrices
Problem type exact computation
Linked methods Lift products of inner products to new vectors [M-LA-TENSOR]
Secondary cues Linear algebra / matrix theory; Real analysis / calculus; Functional equations / dynamics
Linked theory Tensor products and operators on matrix spaces [T-LA-TENSOR]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers matrix word; det/rank/eigen trigger; integer/prime/mod trigger; functional-equation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Reduce size while preserving the defining conditions [S-INDUCTION]; Choose the objects and count them without ambiguity [S-COUNTING]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2017-D2-P8

Day 2; official label P8; 10 points; legacy source-page label 162

Define the sequence $A_1, A_2, \dots$ of matrices by the following recurrence:

$$A_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad A_{n+1} = \begin{pmatrix} A_n & I_{2^n} \\ I_{2^n} & A_n \end{pmatrix} \quad (n = 1, 2, \dots)$$

where I_m is the $m \times m$ identity matrix.
Prove that A_n has $n+1$ distinct integer eigenvalues $\lambda_0 < \lambda_1 < \cdots < \lambda_n$ with multiplicities $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$, respectively.

(Proposed by Snježana Majstorović, University of J. J. Strossmayer in Osijek, Croatia)

Open all solution variants | Source note

2017-D2-P9 Long/deep

Fingerprint limit of recursively defined function sequence
Primary domain Functional equations / dynamics
Problem type recurrence/convergence
Linked methods Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]; Pass a monotone functional iteration through its integral operator [M-DS-VOLTERRA]
Secondary cues Functional equations / dynamics
Linked theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]; Uniform approximation and positive kernels [T-AN-APPROXIMATION]
Recognition triggers det/rank/eigen trigger; continuous/differentiable trigger; integral/limit trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Control the limiting operation before taking the limit [S-ANALYTIC]; Reduce size while preserving the defining conditions [S-INDUCTION]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2017-D2-P9

Day 2; official label P9; 10 points; legacy source-page label 162

Define the sequence $f_1, f_2, \cdots : [0,1) \to \mathbb{R}$ of continuously differentiable functions by the following recurrence:

$$f_1 = 1; \qquad f'_{n+1} = f_n f_{n+1} \text{ on } (0,1), \text{ and } f_{n+1}(0) = 1.$$

Show that $\lim_{n\to\infty} f_n(x)$ exists for every $x \in [0,1)$ and determine the limit function.

(Proposed by Tomáš Bárta, Charles University, Prague)

Open all solution variants | Source note

2017-D2-P10 Long/deep

Fingerprint perimeters of inverted homothetic triangles blow up
Primary domain Geometry / convexity / metric
Problem type inequality/sharp bound
Linked methods Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]; Split a family by traces, pieces, or labels [M-DS-PARTITION]
Secondary cues Geometry / convexity / metric
Linked theory Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers integer/prime/mod trigger; geometry/metric trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Translate geometry into a controlled representation [S-GEOMETRIC]; Control the limiting operation before taking the limit [S-ANALYTIC]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

2017-D2-P10 continued

Full statement: 2017-D2-P10

Day 2; official label P10; 10 points; legacy source-page label 162

Let K be an equilateral triangle in the plane. Prove that for every $p > 0$ there exists an $\varepsilon > 0$ with the following property: If n is a positive integer, and $T_1, \dots, T_n$ are non-overlapping triangles inside K such that each of them is homothetic to K with a negative ratio, and

$$\sum_{\ell=1}^{n} \text{area}(T_\ell) > \text{area}(K) - \varepsilon,$$

then

$$\sum_{\ell=1}^{n} \text{perimeter}(T_\ell) > p.$$

(Proposed by Fedor Malyshev, Steklov Math. Inst. and Ilya Bogdanov, MIPT, Moscow)

Open all solution variants | Source note

2.25 2018

Day 1

2018-D1-P1 Quick/standard

Fingerprint equivalence of convergence via intermediate sequence c_n
Primary domain Real analysis / calculus
Problem type recurrence/convergence
Linked methods Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]
Secondary cues None recorded
Linked theory Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Series, finite cancellation, and summation by parts [T-AN-SERIES]
Recognition triggers wording-specific
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2018-D1-P1

Day 1; official label P1; 10 points; legacy source-page label 163

Let $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ be two sequences of positive numbers. Show that the following statements are equivalent:

(1) There is a sequence $(c_n)_{n=1}^{\infty}$ of positive numbers such that $\sum_{n=1}^{\infty} \frac{a_n}{c_n}$ and $\sum_{n=1}^{\infty} \frac{c_n}{b_n}$ both converge;

(2) $\sum_{n=1}^{\infty} \sqrt{\frac{a_n}{b_n}}$ converges.

(Proposed by Tomáš Bárta, Charles University, Prague)

Open all solution variants | Source note

2018-D1-P2 Quick/standard

Fingerprint field with multiplicative group isomorphic to additive group
Primary domain Algebra / number theory
Problem type nonexistence/counterexample
Linked methods Prove generation or obstruct it by an image [M-AL-GENERATORS]; Replace many equal values by too many roots [M-AL-ROOTCOUNT]
Secondary cues None recorded
Linked theory Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Groups, orders, generators, and quotients [T-AL-GROUP]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers group/ring/field trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2018-D1-P2

Day 1; official label P2; 10 points; legacy source-page label 163

Does there exist a field such that its multiplicative group is isomorphic to its additive group?

(Proposed by Alexandre Chapovalov, New York University, Abu Dhabi)

Open all solution variants | Source note

2018-D1-P3 Medium

Fingerprint rational a making 4x4 matrix a rational square
Primary domain Linear algebra / matrices
Problem type classification/find-all
Linked methods Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]
Secondary cues Algebra / number theoryi
Linked theory Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Determinants, block identities and inverses [T-LA-DETERMINANT]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers matrix word; det/rank/eigen trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Make the contradiction quantitative or structural [S-CONTRADICTION]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2018-D1-P3

Day 1; official label P3; 10 points; legacy source-page label 163

Determine all rational numbers a for which the matrix

$$\begin{pmatrix} a & -a & -1 & 0 \\ a & -a & 0 & -1 \\ 1 & 0 & a & -a \\ 0 & 1 & a & -a \end{pmatrix}$$

is the square of a matrix with all rational entries.

(Proposed by Daniël Kroes, University of California, San Diego)

Open all solution variants | Source note

2018-D1-P4 Medium/trap

Fingerprint mean value at geometric mean characterizes functions
Primary domain Functional equations / dynamics
Problem type classification/find-all
Linked methods Force structure by admissible substitutions [M-DS-FUNCTIONAL]
Secondary cues Functional equations / dynamics
Linked theory Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Recognition triggers functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2018-D1-P4

Day 1; official label P4; 10 points; legacy source-page label 163

Find all differentiable functions $f : (0, \infty) \to \mathbb{R}$ such that

$$f(b) - f(a) = (b - a)f'(\sqrt{ab}) \quad \text{for all } a, b > 0. \tag{2}$$

(Proposed by Orif Ibrogimov, National University of Uzbekistan)

Open all solution variants | Source note

2018-D1-P5 Long/deep

Fingerprint side-sum bound in equiangular pq-gon with integer sides
Primary domain Complex analysis / polynomials
Problem type inequality/sharp bound
Linked methods Evaluate at roots of unity or average over them [M-AL-UNITY]; Construct and count compatible local residues [M-AL-CRT]
Secondary cues Real analysis / calculus; Geometry / convexity / metric
Linked theory Roots of unity, cyclotomic polynomials, and algebraic norms [T-AL-CYCLOTOMIC]; Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Translate geometry into a controlled representation [S-GEOMETRIC]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2018-D1-P5

Day 1; official label P5; 10 points; legacy source-page label 163

Let p and q be prime numbers with $p < q$. Suppose that in a convex polygon $P_1P_2 \ldots P_{pq}$ all angles are equal and the side lengths are distinct positive integers. Prove that

$$P_1P_2 + P_2P_3 + \cdots + P_kP_{k+1} \geq \frac{k^3 + k}{2}$$

holds for every integer k with $1 \leq k \leq p$.

(Proposed by Ander Lamaison Vidarte, Berlin Mathematical School, Berlin)

Open all solution variants | Source note

Day 2

2018-D2-P6 Long/deep

Fingerprint smallest dimension for chain-orthogonal nonzero vectors
Primary domain Linear algebra / matrices
Problem type inequality/sharp bound
Linked methods Prove a dimension obstruction and attain it [M-LA-DIMENSION]
Secondary cues Geometry / convexity / metric; Probability / measure / expectation; Combinatorics / discrete structures
Linked theory Inner products, orthogonality and polarization [T-LA-INNER]; Rank, dimension and exact linear constraints [T-LA-RANK]
Recognition triggers integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2018-D2-P6

Day 2; official label P6; 10 points; legacy source-page label 164

Let k be a positive integer. Find the smallest positive integer n for which there exist k nonzero vectors $v_1, \ldots, v_k$ in $\mathbb{R}^n$ such that for every pair i, j of indices with $|i-j|>1$ the vectors v_i and v_j are orthogonal.

(Proposed by Alexey Balitskiy, Moscow Institute of Physics and Technology and M.I.T.)

Open all solution variants | Source note

2018-D2-P7 Medium

Fingerprint convergence of total variation for cubic recurrence
Primary domain Functional equations / dynamics
Problem type recurrence/convergence
Linked methods Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]; Strengthen the statement before the induction step [M-DS-INDUCT]
Secondary cues Geometry / convexity / metric
Linked theory Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers integral/limit trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Reduce size while preserving the defining conditions [S-INDUCTION]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2018-D2-P7

Day 2; official label P7; 10 points; legacy source-page label 164

Let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers such that $a_0 = 0$ and

$$a_{n+1}^3 = a_n^2 - 8 \quad \text{for } n = 0, 1, 2, \ldots$$

Prove that the following series is convergent:

$$\sum_{n=0}^{\infty} |a_{n+1} - a_n|. \tag{1}$$

(Proposed by Orif Ibrogimov, National University of Uzbekistan)

Open all solution variants | Source note

2018-D2-P8 Medium/trap

Fingerprint count frog lattice paths to (n,n,n)
Primary domain Combinatorics
Problem type counting/enumeration
Linked methods Encode, decode, and verify a bijection [M-DS-BIJECTION]; Strengthen the statement before the induction step [M-DS-INDUCT]
Secondary cues Real analysis / calculus; Combinatorics / discrete structures; Geometry / convexity / metric
Linked theory Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Recognition triggers det/rank/eigen trigger; integer/prime/mod trigger; counting/permutation trigger; geometry/metric trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]; Reduce size while preserving the defining conditions [S-INDUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2018-D2-P8

Day 2; official label P8; 10 points; legacy source-page label 164

Let $\Omega = \{(x, y, z) \in \mathbb{Z}^3 : y + 1 \geq x \geq y \geq z \geq 0\}$. A frog moves along the points of Ω by jumps of length 1. For every positive integer n, determine the number of paths the frog can take to reach (n, n, n) starting from $(0, 0, 0)$ in exactly $3n$ jumps.

(Proposed by Fedor Petrov and Anatoly Vershik, St. Petersburg State University)

Open all solution variants | Source note

2018-D2-P9 Long/deep

Fingerprint monic pairs with P|Q^2+1 and Q|P^2+1
Primary domain Complex analysis / polynomials
Problem type classification/find-all
Linked methods Replace a minimal algebraic counterexample by a smaller one [M-AL-DESCENT]
Secondary cues Algebra / number theoryi
Linked theory Polynomial factorization and coprime divisibility [T-AL-FACTORIZATION]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]
Recognition triggers det/rank/eigen trigger; polynomial/root trigger
Preparation route CP-Polynomials/Complex
Trap watch Separate polynomial identity from pointwise testing; check multiplicities, zero polynomial cases, and real-vs-complex roots.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Choose an extremal object and force rigidity [S-EXTREMAL]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2018-D2-P9

Day 2; official label P9; 10 points; legacy source-page label 164

Determine all pairs $P(x)$, $Q(x)$ of complex polynomials with leading coefficient 1 such that $P(x)$ divides $Q(x)^2 + 1$ and $Q(x)$ divides $P(x)^2 + 1$.

(Proposed by Rodrigo Angelo, Princeton University and Matheus Secco, PUC, Rio de Janeiro)

Open all solution variants | Source note

2018-D2-P10 Long/deep

Fingerprint limit of alternating lattice sum (-1)^(a+b)/(a^2+b^2)
Primary domain Real analysis / calculus
Problem type limit/asymptotic evaluation
Linked methods Separate a main range from exceptional scales [M-AN-SPLIT]; Expose the mesh and control its tails [M-AN-RIEMANN]
Secondary cues None recorded
Linked theory Riemann sums, periodic averaging, and tails [T-AN-RIEMANN]; Scale separation and limit comparisons [T-AN-ASYMPTOTICS]; Series, finite cancellation, and summation by parts [T-AN-SERIES]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]
Recognition triggers integral/limit trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Translate geometry into a controlled representation [S-GEOMETRIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2018-D2-P10

Day 2; official label P10; 10 points; legacy source-page label 164

For $R > 1$ let $D_R = \{(a, b) \in \mathbb{Z}^2 : 0 < a^2 + b^2 < R\}$. Compute

$$\lim_{R\to\infty} \sum_{(a,b)\in D_R} \frac{(-1)^{a+b}}{a^2 + b^2}.$$

(Proposed by Rodrigo Angelo, Princeton University and Matheus Secco, PUC, Rio de Janeiro)

Open all solution variants | Source note

2.26 2019

Day 1

2019-D1-P1 Quick/standard

Fingerprint evaluate infinite telescoping product
Primary domain Real analysis / calculus
Problem type exact computation
Linked methods Find a finite telescoping identity [M-AN-TELESCOPE]
Secondary cues None recorded
Linked theory Series, finite cancellation, and summation by parts [T-AN-SERIES]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers wording-specific
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

2019-D1-P1 continued

Full statement: 2019-D1-P1

Day 1; official label P1; 10 points; legacy source-page label 165

Evaluate the product

$$\prod_{n=3}^{\infty} \frac{(n^3+3n)^2}{n^6-64}.$$

Proposed by Orif Ibrogimov, ETH Zurich and National University of Uzbekistan and Karen Keryan, Yerevan State University and American University of Armenia, Yerevan

Open all solution variants | Source note

2019-D1-P2 Quick/standard

Fingerprint find YEARs making circulant linear system degenerate
Primary domain Linear algebra / matrices
Problem type classification/find-all
Linked methods Eliminate systematically and track the invariant [M-LA-ELIM]
Secondary cues None recorded
Linked theory Rank, dimension and exact linear constraints [T-LA-RANK]; Determinants, block identities and inverses [T-LA-DETERMINANT]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Inner products, orthogonality and polarization [T-LA-INNER]
Recognition triggers wording-specific
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2019-D1-P2

Day 1; official label P2; 10 points; legacy source-page label 165

A four-digit number $YEAR$ is called very good if the system

$$\begin{aligned} Yx + Ey + Az + Rw &= Y \\ Rx + Yy + Ez + Aw &= E \\ Ax + Ry + Yz + Ew &= A \\ Ex + Ay + Rz + Yw &= R \end{aligned}$$

of linear equations in the variables x, y, z and w has at least two solutions. Find all very good $YEAR$s in the 21st century.
(The 21st century starts in 2001 and ends in 2100.)

Proposed by Tomáš Bárta, Charles University, Prague

Open all solution variants | Source note

2019-D1-P3 Medium

Fingerprint bound integral of xf(x) given differential inequality
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Design an auxiliary derivative [M-AN-AUXILIARY]; Choose the correct convexity comparison [M-AN-CONVEX]
Secondary cues Functional equations / dynamics
Linked theory Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Recognition triggers functional-equation trigger
Preparation route RA-Analysis Core

2019-D1-P3 continued

Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2019-D1-P3

Day 1; official label P3; 10 points; legacy source-page label 165

Let $f : (-1, 1) \to \mathbb{R}$ be a twice differentiable function such that

$$2f'(x) + xf''(x) \geq 1 \quad \text{for } x \in (-1, 1).$$

Prove that

$$\int_{-1}^{1} xf(x)\,dx \geq \frac{1}{3}.$$

Proposed by Orif Ibrogimov, ETH Zurich and National University of Uzbekistan and Karim Rakhimov, Scuola Normale Superiore and National University of Uzbekistan

Open all solution variants | Source note

2019-D1-P4 Medium

Fingerprint prove recurrence terms are all integers
Primary domain Algebra / number theory
Problem type structural proof
Linked methods Find a recurrence that reveals the invariant [M-AL-RECURRENCE]
Secondary cues Real analysis / calculus; Geometry / convexity / metric; Functional equations / dynamics
Linked theory Linear recurrences and generating functions [T-AL-RECURRENCE]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers integer/prime/mod trigger; functional-equation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Reduce size while preserving the defining conditions [S-INDUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2019-D1-P4

Day 1; official label P4; 10 points; legacy source-page label 165

Define the sequence $a_0, a_1, \ldots$ of numbers by the following recurrence:

$$a_0 = 1, \quad a_1 = 2, \quad (n+3)a_{n+2} = (6n+9)a_{n+1} - na_n \quad \text{for } n \geq 0.$$

Prove that all terms of this sequence are integers.

Proposed by Khakimboy Egamberganov, ICTP, Italy

Open all solution variants | Source note

2019-D1-P5 Long/deep

Fingerprint integer matrix equation equal to 2019 times identity
Primary domain Linear algebra / matrices
Problem type nonexistence/counterexample
Linked methods Factor while preserving matrix order [M-LA-FACTOR]; Reduce integer matrices modulo an integer [M-LA-RESIDUE]
Secondary cues Linear algebra / matrix theory; Geometry / convexity / metric
Linked theory Determinants, block identities and inverses [T-LA-DETERMINANT]; Congruences, Bézout, and

2019-D1-P5 continued

the Chinese remainder theorem [T-AL-CONGRUENCE]
Recognition triggers matrix word; det/rank/eigen trigger; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2019-D1-P5

Day 1; official label P5; 10 points; legacy source-page label 165

Determine whether there exist an odd positive integer n and $n \times n$ matrices A and B with integer entries, that satisfy the following conditions:
(1) $\det(B) = 1$;
(2) $AB = BA$;
(3) $A^4 + 4A^2B^2 + 16B^4 = 2019I$.
(Here I denotes the $n \times n$ identity matrix.)

Proposed by Orif Ibrogimov, ETH Zurich and National University of Uzbekistan

Open all solution variants | Source note

Day 2

2019-D2-P6 Long/deep

Fingerprint existence of point where f equals g-prime
Primary domain Real analysis / calculus
Problem type existence/construction
Linked methods Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]; Count zeros and subtract an interpolant [M-AN-ROLLE]
Secondary cues Real analysis / calculus
Linked theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Recognition triggers continuous/differentiable trigger; integral/limit trigger; functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Control the limiting operation before taking the limit [S-ANALYTIC]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2019-D2-P6

Day 2; official label P6; 10 points; legacy source-page label 166

Let $f, g : \mathbb{R} \longrightarrow \mathbb{R}$ be continuous functions such that g is differentiable. Assume that $(f(0) - g'(0))(g'(1) - f(1)) > 0$. Show that there exists a point $c \in (0, 1)$ such that $f(c) = g'(c)$.

Proposed by Fereshteh Malek, K. N. Toosi University of Technology

Open all solution variants | Source note

2019-D2-P7 Medium

Fingerprint convergence of series with smallest-factorial-divisor terms
Primary domain Algebra / number theory
Problem type recurrence/convergence
Linked methods Track one prime or one extreme divisor [M-AL-VALUATION]
Secondary cues Real analysis / calculus

2019-D2-P7 continued

Linked theory Prime valuations, divisors, and factorials [T-AL-VALUATION]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers det/rank/eigen trigger; integral/limit trigger; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2019-D2-P7

Day 2; official label P7; 10 points; legacy source-page label 166

Let $C = \{4, 6, 8, 9, 10, \dots\}$ be the set of composite positive integers. For each $n \in C$ let a_n be the smallest positive integer k such that $k!$ is divisible by n. Determine whether the following series converges:

$$\sum_{n \in C} \left(\frac{a_n}{n}\right)^n. \tag{1}$$

Proposed by Orif Ibrogimov, ETH Zurich and National University of Uzbekistan

Open all solution variants | Source note

2019-D2-P8 Medium

Fingerprint bound number of subsets with fixed sum
Primary domain Combinatorics
Problem type inequality/sharp bound
Linked methods Choose information that makes an encoding injective [M-DS-ENCODE]
Secondary cues Real analysis / calculus
Linked theory Finite counting and incidence identities [T-DS-COUNTING]; Inner products, orthogonality and polarization [T-LA-INNER]
Recognition triggers integral/limit trigger; counting/permutation trigger; graph/color/set trigger
Preparation route CMB-Combinatoricsi
Trap watch Avoid overcounting; define the encoded object precisely and check extremal/minimal cases.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2019-D2-P8

Day 2; official label P8; 10 points; legacy source-page label 166

Let $x_1, \dots, x_n$ be real numbers. For any set $I \subset \{1, 2, \dots, n\}$ let $s(I) = \sum_{i \in I} x_i$. Assume that the function $I \mapsto s(I)$ takes on at least 1.8^n values where I runs over all 2^n subsets of $\{1, 2, \dots, n\}$. Prove that the number of sets $I \subset \{1, 2, \dots, n\}$ for which $s(I) = 2019$ does not exceed 1.7^n.

Proposed by Fedor Part and Fedor Petrov, St. Petersburg State University

Open all solution variants | Source note

2019-D2-P9 Long/deep

Fingerprint invertible matrices satisfying commutator identity, find n
Primary domain Linear algebra / matrices
Problem type classification/find-all
Linked methods Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]; Exploit a compatible block decomposition [M-LA-BLOCK]
Secondary cues Linear algebra / matrix theory
Linked theory Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Determinants, block identities and inverses [T-LA-DETERMINANT]

2019-D2-P9 continued

Recognition triggers matrix word; det/rank/eigen trigger; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2019-D2-P9

Day 2; official label P9; 10 points; legacy source-page label 166

Determine all positive integers n for which there exist $n \times n$ real invertible matrices A and B that satisfy $AB - BA = B^2A$.

Proposed by Karen Keryan, Yerevan State University & American University of Armenia, Yerevan

Open all solution variants | Source note

2019-D2-P10 Long/deep

Fingerprint compare convex hull triangle vs quadrilateral probabilities
Primary domain Probability / measure
Problem type inequality/sharp bound
Linked methods Condition and average a convex-hull event [M-AN-GEOMETRIC]; Count outcomes or condition on a useful structure [M-AN-PROBABILITY]
Secondary cues Combinatorics / discrete structures; Geometry / convexity / metric; Real analysis / calculus
Linked theory Uniform random points and convex-hull events [T-AN-GEOMETRICPROB]; Finite probability, conditioning, and expectations [T-AN-PROBABILITY]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]
Recognition triggers graph/color/set trigger; geometry/metric trigger
Preparation route PRB-Probability/Measure
Trap watch Conditioning, independence, and zero-measure/endpoint cases need explicit verification.
Proof strategies Translate geometry into a controlled representation [S-GEOMETRIC]; Choose the objects and count them without ambiguity [S-COUNTING]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2019-D2-P10

Day 2; official label P10; 10 points; legacy source-page label 166

2019 points are chosen at random, independently, and distributed uniformly in the unit disc $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$. Let C be the convex hull of the chosen points. Which probability is larger: that C is a polygon with three vertices, or a polygon with four vertices?

Proposed by Fedor Petrov, St. Petersburg State University

Open all solution variants | Source note

2.27 2020

Day 1

2020-D1-P1 Quick/standard

Fingerprint count words with even counts of two letters
Primary domain Combinatorics
Problem type counting/enumeration
Linked methods Split a family by traces, pieces, or labels [M-DS-PARTITION]; Encode, decode, and verify a bijection [M-DS-BIJECTION]; Strengthen the statement before the induction step [M-DS-INDUCT]; Encode the statistic and extract the required coefficient [M-DS-GENERATING]; Partition into states and propagate a recurrence [M-DS-STATE]
Secondary cues Combinatorics / discrete structures; Geometry / convexity / metric; Real analysis / calculus
Linked theory Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Generating functions, coefficient filters, and finite differences [T-DS-GENERATING]
Recognition triggers integer/prime/mod trigger; counting/permutation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2020-D1-P1

Day 1; official label P1; legacy source-page label 167

Let n be a positive integer. Compute the number of words w (finite sequences of letters) that satisfy all the following three properties:
(1) w consists of n letters, all of them are from the alphabet $\{a, b, c, d\}$;
(2) w contains an even number of letters a;
(3) w contains an even number of letters b.
(For example, for $n = 2$ there are 6 such words: aa, bb, cc, dd, cd and dc.)

Armend Sh. Shabani, University of Prishtina

Open all solution variants | Source note

2020-D1-P2 Quick/standard

Fingerprint trace identity from rank-one commutator condition
Primary domain Linear algebra / matrices
Problem type structural proof
Linked methods Use trace to extract a scalar invariant [M-LA-TRACE]
Secondary cues Combinatorics / discrete structures; Real analysis / calculus
Linked theory Trace pairing and commutator calculus [T-LA-TRACE]; Rank, dimension and exact linear constraints [T-LA-RANK]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]
Recognition triggers matrix word; det/rank/eigen trigger; integral/limit trigger; counting/permutation trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

2020-D1-P2 continued

Full statement: 2020-D1-P2

Day 1; official label P2; legacy source-page label 167

Let A and B be $n \times n$ real matrices such that

$$\operatorname{rk}(AB - BA + I) = 1$$

where I is the $n \times n$ identity matrix.
Prove that

$$\operatorname{trace}(ABAB) - \operatorname{trace}(A^2B^2) = \frac{1}{2}n(n-1).$$

($\operatorname{rk}(M)$ denotes the rank of matrix M, i.e., the maximum number of linearly independent columns in M. $\operatorname{trace}(M)$ denotes the trace of M, that is the sum of diagonal elements in M.)

Rustam Turdibaev, V. I. Romanovskiy Institute of Mathematics

Open all solution variants | Source note

2020-D1-P3 Medium

Fingerprint approximate symmetric convex body by few-vertex polytope
Primary domain Geometry / convexity / metric
Problem type existence/construction
Linked methods Repair an extremal configuration by a local exchange [M-DS-EXTREMAL]; Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Secondary cues Combinatorics / discrete structures; Real analysis / calculus
Linked theory Convex bodies, supporting functionals, and packing [T-DS-CONVEX]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Extremal choice, exchange arguments, and invariants [T-DS-EXTREMAL]; Metric contractions, covers, and nearest points [T-DS-METRIC]
Recognition triggers graph/color/set trigger; geometry/metric trigger
Preparation route GEO-Geometry/Convexity
Trap watch Track convexity, symmetry, dimension, and the required containment or approximation factor.
Proof strategies Choose an extremal object and force rigidity [S-EXTREMAL]; Translate geometry into a controlled representation [S-GEOMETRIC]; Build a chain of bounds and track equality [S-BOUND]; Make the contradiction quantitative or structural [S-CONTRADICTION]
Correction note The explicit CLASS-001 corrections are incorporated in this card; the accompanying change ledger records the withdrawn labels.
Navigation Full statement All solution variants Source note Learning route

Full statement: 2020-D1-P3

Day 1; official label P3; legacy source-page label 167

Let $d \geq 2$ be an integer. Prove that there exists a constant $C(d)$ such that the following holds: For any convex polytope $K \subset \mathbb{R}^d$, which is symmetric about the origin, and any $\varepsilon \in (0,1)$, there exists a convex polytope $L \subset \mathbb{R}^d$ with at most $C(d)\varepsilon^{1-d}$ vertices such that

$$(1-\varepsilon)K \subseteq L \subseteq K.$$

(For a real α, a set $T \subset \mathbb{R}^d$ with nonempty interior is a convex polytope with at most α vertices, if T is a convex hull of a set $X \subset \mathbb{R}^d$ of at most α points, i.e., $T = \{\sum_{x \in X} t_x x \mid t_x \geq 0, \sum_{x \in X} t_x = 1\}$. For a real λ, put $\lambda K = \{\lambda x \mid x \in K\}$. A set $T \subset \mathbb{R}^d$ is symmetric about the origin if $(-1)T = T$.)

Fedor Petrov, St. Petersburg State University

Open all solution variants | Source note

2020-D1-P4 Medium

Fingerprint inequality for polynomial with prescribed difference equation
Primary domain Complex analysis / polynomials
Problem type inequality/sharp bound
Linked methods Build a holomorphic function with a useful boundary [M-AL-ANALYTIC]; Compare coefficients after a structural substitution [M-AL-COEFFICIENT]
Secondary cues Complex analysis / polynomials
Linked theory Finite differences and shift-operator ideals [T-AL-DIFFERENCE]; Holomorphic bounds, poles, and logarithmic-branch sums [T-AL-COMPLEXANALYSIS]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Recognition triggers polynomial/root trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Control the limiting operation before taking the limit [S-ANALYTIC]; Reduce size while preserving the defining conditions [S-INDUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2020-D1-P4

Day 1; official label P4; legacy source-page label 167

A polynomial p with real coefficients satisfies the equation $p(x+1)-p(x)=x^{100}$ for all $x\in\mathbb{R}$. Prove that $p(1-t)\geqslant p(t)$ for $0\leqslant t\leqslant 1/2$.

Daniil Klyuev, St. Petersburg State University

Open all solution variants | Source note

Day 2

2020-D2-P5 Long/deep

Fingerprint classify positive functions with differential inequality
Primary domain Real analysis / calculus
Problem type classification/find-all
Linked methods Design an auxiliary derivative [M-AN-AUXILIARY]; Choose the correct convexity comparison [M-AN-CONVEX]
Secondary cues Functional equations / dynamics
Linked theory Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Recognition triggers continuous/differentiable trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2020-D2-P5

Day 2; official label P5; legacy source-page label 168

Find all twice continuously differentiable functions $f:\mathbb{R}\to(0,+\infty)$ satisfying

$$f''(x)f(x)\geq 2(f'(x))^2$$

for all $x\in\mathbb{R}$.

Karen Keryan, Yerevan State University & American University of Armenia, Yerevan

Open all solution variants | Source note

2020-D2-P6

Long/deep

Fingerprint primes where cubic has unique root mod p
Primary domain Algebra / number theory
Problem type classification/find-all
Linked methods Choose a modulus that exposes the obstruction [M-AL-MODULAR]; Factor first and compare irreducible exponents [M-AL-FACTOR]; Translate root restrictions into symmetric coefficients [M-AL-VIETA]
Secondary cues None recorded
Linked theory Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]; Polynomial factorization and coprime divisibility [T-AL-FACTORIZATION]
Recognition triggers integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2020-D2-P6

Day 2; official label P6; legacy source-page label 168

Find all prime numbers p for which there exists a unique $a \in \{1, 2, \ldots, p\}$ such that $a^3 - 3a + 1$ is divisible by p.

Géza Kós, Loránd Eötvös University, Budapest

Open all solution variants | Source note

2020-D2-P7

Medium

Fingerprint subgroups with given indices are conjugate
Primary domain Algebra / number theory
Problem type structural proof
Linked methods Convert group data into orbits or cosets [M-AL-ACTION]
Secondary cues Real analysis / calculus; Combinatorics / discrete structures
Linked theory Groups, orders, generators, and quotients [T-AL-GROUP]; Group actions, stabilizers, and semidirect products [T-AL-ACTION]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers integer/prime/mod trigger; group/ring/field trigger; counting/permutation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2020-D2-P7

Day 2; official label P7; legacy source-page label 168

Let G be a group and $n \geq 2$ be an integer. Let H_1 and H_2 be two subgroups of G that satisfy

$$[G : H_1] = [G : H_2] = n \quad \text{and} \quad [G : (H_1 \cap H_2)] = n(n-1).$$

Prove that H_1 and H_2 are conjugate in G.
(Here $[G : H]$ denotes the index of the subgroup H, i.e. the number of distinct left cosets xH of H in G. The subgroups H_1 and H_2 are conjugate if there exists an element $g \in G$ such that $g^{-1}H_1g = H_2$.)

Ilya Bogdanov and Alexander Matushkin, Moscow Institute of Physics and Technology

Open all solution variants | Source note

2020-D2-P8 Medium

Fingerprint asymptotics of alternating binomial sum of logarithms
Primary domain Real analysis / calculus
Problem type limit/asymptotic evaluation
Linked methods Encode convolution or denominators by a power series [M-AN-GENERATING]; Separate a main range from exceptional scales [M-AN-SPLIT]
Secondary cues None recorded
Linked theory Nonnegative sums, Fubini, and integral kernels [T-AN-FUBINI]; Scale separation and limit comparisons [T-AN-ASYMPTOTICS]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Finite counting and incidence identities [T-DS-COUNTING]; Series, finite cancellation, and summation by parts [T-AN-SERIES]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers integral/limit trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2020-D2-P8

Day 2; official label P8; legacy source-page label 168

Compute

$$\lim_{n\to\infty} \frac{1}{\log\log n} \sum_{k=1}^{n} (-1)^k \binom{n}{k} \log k.$$

(Here log denotes the natural logarithm.)

Fedor Petrov, St. Petersburg State University

Open all solution variants | Source note

2.28 2021

Day 1

2021-D1-P1 Quick/standard

Fingerprint unique solution of matrix equation with nilpotent A
Primary domain Linear algebra / matrices
Problem type structural proof
Linked methods Treat the unknown matrix as a vector [M-LA-MATRIXOP]; Complete the equation to a product equal to identity [M-LA-INVERSE]
Secondary cues None recorded
Linked theory Nilpotents, projections and involutions [T-LA-NILPOTENT]; Determinants, block identities and inverses [T-LA-DETERMINANT]
Recognition triggers matrix word
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2021-D1-P1

Day 1; official label P1; 10 points; legacy source-page label 169

Let A be a real $n \times n$ matrix such that $A^3 = 0$.

2021-D1-P1 continued

(a) Prove that there is a unique real $n \times n$ matrix X that satisfies the equation

$$X + AX + XA^2 = A.$$

(b) Express X in terms of A.

(proposed by Bekhzod Kurbonboev, Institute of Mathematics, Tashkent)

Open all solution variants | Source note

2021-D1-P2 Quick/standard

Fingerprint subset probability independent of parameter
Primary domain Probability / measure
Problem type structural proof
Linked methods Count outcomes or condition on a useful structure [M-AN-PROBABILITY]
Secondary cues Probability / measure / expectation; Combinatorics / discrete structures
Linked theory Finite probability, conditioning, and expectations [T-AN-PROBABILITY]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers integer/prime/mod trigger; graph/color/set trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2021-D1-P2

Day 1; official label P2; 10 points; legacy source-page label 169

Let n and k be fixed positive integers, and let a be an arbitrary non-negative integer. Choose a random k-element subset X of $\{1, 2, \dots, k+a\}$ uniformly (i.e., all k-element subsets are chosen with the same probability) and, independently of X, choose a random n-element subset Y of $\{1, \dots, k+n+a\}$ uniformly.
Prove that the probability

$$\mathsf{P}(\min(Y) > \max(X))$$

does not depend on a.

(proposed by Fedor Petrov, St. Petersburg State University)

Open all solution variants | Source note

2021-D1-P3 Medium

Fingerprint supremum of interval length allowing fine partition sequence
Primary domain Combinatorics
Problem type inequality/sharp bound
Linked methods Split a family by traces, pieces, or labels [M-DS-PARTITION]; Strengthen the statement before the induction step [M-DS-INDUCT]
Secondary cues Combinatorics / discrete structures
Linked theory Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers geometry/metric trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

2021-D1-P3 continued

Full statement: 2021-D1-P3

Day 1; official label P3; 10 points; legacy source-page label 169

We say that a positive real number d is good if there exists an infinite sequence $a_1, a_2, a_3, \cdots \in (0, d)$ such that for each n, the points $a_1, \ldots, a_n$ partition the interval $[0, d]$ into segments of length at most $1/n$ each. Find

$$\sup\{d \mid d \text{ is good}\}.$$

(proposed by Josef Tkadlec)

Open all solution variants | Source note

2021-D1-P4 Medium/trap

Fingerprint show f is pointwise limit of continuous functions
Primary domain Real analysis / calculus
Problem type existence/construction
Linked methods Use a dense set, an enumeration, or a category argument [M-AN-TOPOLOGY]; Preserve structure while approximating [M-AN-APPROXIMATE]
Secondary cues Functional equations / dynamics
Linked theory Baire class one and controlled discontinuities [T-AN-BAIRE]; Continuity, compactness, and order [T-AN-CONTINUITY]; Uniform approximation and positive kernels [T-AN-APPROXIMATION]; Countability, accumulation, and Cantor constructions [T-AN-SETS]
Recognition triggers continuous/differentiable trigger; integral/limit trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Build a witness and verify every constraint [S-CONSTRUCTION]; Pass between local constraints and a global conclusion [S-LOCALGLOBAL]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2021-D1-P4

Day 1; official label P4; 10 points; legacy source-page label 169

Let $f : \mathbb{R} \to \mathbb{R}$ be a function. Suppose that for every $\varepsilon > 0$, there exists a function $g : \mathbb{R} \to (0, \infty)$ such that for every pair (x, y) of real numbers,

$$\text{if } |x - y| < \min\{g(x), g(y)\}, \text{ then } |f(x) - f(y)| < \varepsilon.$$

Prove that f is the pointwise limit of a sequence of continuous $\mathbb{R} \to \mathbb{R}$ functions, i.e., there is a sequence $h_1, h_2, \ldots$ of continuous $\mathbb{R} \to \mathbb{R}$ functions such that $\lim_{n\to\infty} h_n(x) = f(x)$ for every $x \in \mathbb{R}$.

(proposed by Camille Mau, Nanyang Technological University, Singapore)

Open all solution variants | Source note

Day 2

2021-D2-P5 Long/deep

Fingerprint determinant bound from symmetric matrix equation family
Primary domain Linear algebra / matrices
Problem type inequality/sharp bound
Linked methods Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]
Secondary cues Linear algebra / matrix theory; Real analysis / calculus
Linked theory Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Determinants, block identities and inverses [T-LA-DETERMINANT]
Recognition triggers matrix word; det/rank/eigen trigger; integer/prime/mod trigger

2021-D2-P5 continued

Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2021-D2-P5

Day 2; official label P5; 10 points; legacy source-page label 170

Let A be a real $n \times n$ matrix and suppose that for every positive integer m there exists a real symmetric matrix B such that

$$2021B = A^m + B^2.$$

Prove that $|\det A| \leq 1$.

(proposed by Rafael Filipe dos Santos, Instituto Militar de Engenharia, Rio de Janeiro)

Open all solution variants | Source note

2021-D2-P6 Long/deep

Fingerprint no embedding of GL2 mod p into Sp
Primary domain Algebra / number theory
Problem type nonexistence/counterexample
Linked methods Prove generation or obstruct it by an image [M-AL-GENERATORS]
Secondary cues Linear algebra / matrix theory; Combinatorics / discrete structures
Linked theory Groups, orders, generators, and quotients [T-AL-GROUP]; Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]; Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]
Recognition triggers matrix word; integer/prime/mod trigger; group/ring/field trigger; counting/permutation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2021-D2-P6

Day 2; official label P6; 10 points; legacy source-page label 170

For a prime number p, let $\mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z})$ be the group of invertible 2×2 matrices of residues modulo p, and let S_p be the symmetric group (the group of all permutations) on p elements. Show that there is no injective group homomorphism $\varphi : \mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z}) \to S_p$.

(proposed by Thiago Landim, Sorbonne University, Paris)

Open all solution variants | Source note

2021-D2-P7 Medium

Fingerprint bound f(0) by maximum of f times monic polynomial
Primary domain Complex analysis / polynomials
Problem type inequality/sharp bound
Linked methods Build a holomorphic function with a useful boundary [M-AL-ANALYTIC]
Secondary cues Real analysis / calculus; Functional equations / dynamics; Combinatorics / discrete structures
Linked theory Holomorphic bounds, poles, and logarithmic-branch sums [T-AL-COMPLEXANALYSIS]; Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]
Recognition triggers polynomial/root trigger; functional-equation trigger
Preparation route CP-Polynomials/Complex
Trap watch Separate polynomial identity from pointwise testing; check multiplicities, zero polynomial cases,

2021-D2-P7 continued

and real-vs-complex roots.
Proof strategies Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2021-D2-P7

Day 2; official label P7; 10 points; legacy source-page label 170

Let $D \subseteq \mathbb{C}$ be an open set containing the closed unit disk $\{z : |z| \leq 1\}$. Let $f : D \to \mathbb{C}$ be a holomorphic function, and let $p(z)$ be a monic polynomial. Prove that

$$\big|f(0)\big| \leq \max_{|z|=1} \big|f(z)p(z)\big|.$$

(proposed by Lars Hörmander)

Open all solution variants | Source note

2021-D2-P8 Medium

Fingerprint max unit vectors with orthogonal pair in any triple
Primary domain Linear algebra / matrices
Problem type inequality/sharp bound
Linked methods Encode distances and orthogonality by a Gram matrix [M-LA-GRAM]; Use trace to extract a scalar invariant [M-LA-TRACE]
Secondary cues Geometry / convexity / metric; Combinatorics / discrete structures
Linked theory Gram matrices and positive semidefinite forms [T-LA-GRAM]; Trace pairing and commutator calculus [T-LA-TRACE]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]
Recognition triggers integer/prime/mod trigger; counting/permutation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2021-D2-P8

Day 2; official label P8; 10 points; legacy source-page label 170

Let n be a positive integer. At most how many distinct unit vectors can be selected in $\mathbb{R}^n$ such that from any three of them, at least two are orthogonal?

(proposed by Alexander Polyanskii, Moscow Institute of Physics and Technology; based on results of Paul Erdős and Moshe Rosenfeld)

Open all solution variants | Source note

2.29 2022

Day 1

2022-D1-P1 Quick/standard

Fingerprint integral lower bound from multiplicative symmetry f(x)f(1-x)=1
Primary domain Real analysis / calculus
Problem type inequality/sharp bound
Linked methods Pair an integral through symmetry [M-AN-PAIR]; Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]
Secondary cues Functional equations / dynamics; Real analysis / calculus
Linked theory Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Inner products, orthogonality and polarization [T-LA-INNER]
Recognition triggers functional-equation trigger
Preparation route PRB-Probability/Measure
Trap watch Conditioning, independence, and zero-measure/endpoint cases need explicit verification.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2022-D1-P1

Day 1; official label P1; 10 points; legacy source-page label 171

Let $f : [0,1] \to (0,\infty)$ be an integrable function such that $f(x) \cdot f(1-x) = 1$ for all $x \in [0,1]$. Prove that

$$\int_0^1 f(x)\,dx \geq 1.$$

(proposed by Mike Daas, Universiteit Leiden)

Open all solution variants | Source note

2022-D1-P2 Quick/standard

Fingerprint matrices where A plus power equals transpose
Primary domain Linear algebra / matrices
Problem type classification/find-all
Linked methods Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]; Use trace to extract a scalar invariant [M-LA-TRACE]
Secondary cues Linear algebra / matrix theory; Real analysis / calculus
Linked theory Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Nilpotents, projections and involutions [T-LA-NILPOTENT]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Trace pairing and commutator calculus [T-LA-TRACE]
Recognition triggers matrix word; det/rank/eigen trigger; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2022-D1-P2

Day 1; official label P2; 10 points; legacy source-page label 171

Let n be a positive integer. Find all $n \times n$ real matrices A with only real eigenvalues satisfying

$$A + A^k = A^T$$

for some integer $k \geq n$.

2022-D1-P2 continued

(A^T denotes the transpose of A.)

(proposed by Camille Mau, Nanyang Technological University)

Open all solution variants | Source note

2022-D1-P3 Medium

Fingerprint count flea walk strategies modulo p
Primary domain Algebra / number theory
Problem type counting/enumeration
Linked methods Choose a modulus that exposes the obstruction [M-AL-MODULAR]; Find a recurrence that reveals the invariant [M-AL-RECURRENCE]
Secondary cues Combinatorics / discrete structures; Functional equations / dynamics
Linked theory Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]; Linear recurrences and generating functions [T-AL-RECURRENCE]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers integer/prime/mod trigger; counting/permutation trigger; functional-equation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2022-D1-P3

Day 1; official label P3; 10 points; legacy source-page label 171

Let p be a prime number. A flea is staying at point 0 of the real line. At each minute, the flea has three possibilities: to stay at its position, or to move by 1 to the left or to the right. After $p-1$ minutes, it wants to be at 0 again. Denote by $f(p)$ the number of its strategies to do this (for example, $f(3)=3$: it may either stay at 0 for the entire time, or go to the left and then to the right, or go to the right and then to the left). Find $f(p)$ modulo p.

(proposed by Fedor Petrov, St. Petersburg)

Open all solution variants | Source note

2022-D1-P4 Medium

Fingerprint loglog bounds for colouring triples of integers
Primary domain Combinatorics
Problem type inequality/sharp bound
Linked methods Encode a relation as a graph and extract a cycle [M-DS-GRAPH]; Choose information that makes an encoding injective [M-DS-ENCODE]
Secondary cues Real analysis / calculus; Combinatorics / discrete structures
Linked theory Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers integer/prime/mod trigger; counting/permutation trigger; graph/color/set trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Choose the objects and count them without ambiguity [S-COUNTING]; Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

2022-D1-P4 continued

Full statement: 2022-D1-P4

Day 1; official label P4; 10 points; legacy source-page label 171

Let $n > 3$ be an integer. Let Ω be the set of all triples of distinct elements of $\{1, 2, \dots, n\}$. Let m denote the minimal number of colours which suffice to colour Ω so that whenever $1 \leq a < b < c < d \leq n$, the triples $\{a, b, c\}$ and $\{b, c, d\}$ have different colours. Prove that

$$\frac{1}{100} \log \log n \leqslant m \leqslant 100 \log \log n.$$

(proposed by Danila Cherkashin, St. Petersburg)

Open all solution variants | Source note

Day 2

2022-D2-P5 Long/deep

Fingerprint count red monochromatic triangles given blue count
Primary domain Combinatorics
Problem type counting/enumeration
Linked methods Count incidences with their exact multiplicity [M-DS-DOUBLECOUNT]; Encode a relation as a graph and extract a cycle [M-DS-GRAPH]
Secondary cues Geometry / convexity / metric
Linked theory Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers counting/permutation trigger; graph/color/set trigger
Preparation route CMB-Combinatoricsi
Trap watch Avoid overcounting; define the encoded object precisely and check extremal/minimal cases.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Pass between local constraints and a global conclusion [S-LOCALGLOBAL]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2022-D2-P5

Day 2; official label P5; 10 points; legacy source-page label 172

We colour all the sides and diagonals of a regular polygon P with 43 vertices either red or blue in such a way that every vertex is an endpoint of 20 red segments and 22 blue segments. A triangle formed by vertices of P is called monochromatic if all of its sides have the same colour. Suppose that there are 2022 blue monochromatic triangles. How many red monochromatic triangles are there?

(proposed by Mike Daas, Universiteit Leiden)

Open all solution variants | Source note

2022-D2-P6 Long/deep

Fingerprint permutation with cyclic product sum two mod p
Primary domain Algebra / number theory
Problem type existence/construction
Linked methods Choose a modulus that exposes the obstruction [M-AL-MODULAR]; Construct roots or units before the coefficients [M-AL-PARAMETRIZE]; Compare coefficients after a structural substitution [M-AL-COEFFICIENT]
Secondary cues Combinatorics / discrete structures
Linked theory Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers integer/prime/mod trigger; counting/permutation trigger

2022-D2-P6 continued

Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2022-D2-P6

Day 2; official label P6; 10 points; legacy source-page label 172

Let $p > 2$ be a prime number. Prove that there is a permutation $(x_1, x_2, \dots, x_{p-1})$ of the numbers $(1, 2, \dots, p-1)$ such that

$$x_1x_2 + x_2x_3 + \cdots + x_{p-2}x_{p-1} \equiv 2 \pmod{p}.$$

(proposed by Giorgi Arabidze, Tbilisi Free University, Georgia)

Open all solution variants | Source note

2022-D2-P7 Medium

Fingerprint rank bound for anticommuting idempotent matrices
Primary domain Linear algebra / matrices
Problem type inequality/sharp bound
Linked methods Use trace to extract a scalar invariant [M-LA-TRACE]; Prove a dimension obstruction and attain it [M-LA-DIMENSION]
Secondary cues Complex analysis / polynomials; Real analysis / calculus
Linked theory Nilpotents, projections and involutions [T-LA-NILPOTENT]; Trace pairing and commutator calculus [T-LA-TRACE]; Rank, dimension and exact linear constraints [T-LA-RANK]; Dual bases, kernels and separating functionals [T-LA-DUALITY]
Recognition triggers matrix word; det/rank/eigen trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Build a chain of bounds and track equality [S-BOUND]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2022-D2-P7

Day 2; official label P7; 10 points; legacy source-page label 172

Let $A_1, A_2, \dots, A_k$ be $n \times n$ idempotent complex matrices such that

$$A_iA_j = -A_jA_i \quad \text{for all } i \neq j.$$

Prove that at least one of the given matrices has rank $\leq \frac{n}{k}$.

(A matrix A is called idempotent if $A^2 = A$.)

(proposed by Danila Belousov, Novosibirsk)

Open all solution variants | Source note

2022-D2-P8 Medium

Fingerprint expected vertices of intersection of random convex hulls
Primary domain Probability / measure
Problem type counting/enumeration
Linked methods Condition and average a convex-hull event [M-AN-GEOMETRIC]; Count outcomes or condition on a useful structure [M-AN-PROBABILITY]; Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]
Secondary cues Combinatorics / discrete structures; Probability / measure / expectation; Geometry / convexity / metric
Linked theory Uniform random points and convex-hull events [T-AN-GEOMETRICPROB]; Finite probability, conditioning, and expectations [T-AN-PROBABILITY]; Finite counting and incidence identities [T-DS-

2022-D2-P8 continued

COUNTING]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Recognition triggers integer/prime/mod trigger; counting/permutation trigger; graph/color/set trigger; geometry/metric trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Choose the objects and count them without ambiguity [S-COUNTING]; Translate geometry into a controlled representation [S-GEOMETRIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2022-D2-P8

Day 2; official label P8; 10 points; legacy source-page label 172

Let $n, k \geq 3$ be integers, and let S be a circle. Let n blue points and k red points be chosen uniformly and independently at random on the circle S. Denote by F the intersection of the convex hull of the red points and the convex hull of the blue points. Let m be the number of vertices of the convex polygon F (in particular, $m = 0$ when F is empty). Find the expected value of m.

(proposed by Fedor Petrov, St. Petersburg)

Open all solution variants | Source note

2.30 2023

Day 1

2023-D1-P1 Quick/standard

Fingerprint smooth functions satisfying scaling functional equation
Primary domain Functional equations / dynamics
Problem type classification/find-all
Linked methods Force structure by admissible substitutions [M-DS-FUNCTIONAL]; Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]
Secondary cues Functional equations / dynamics
Linked theory Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]; Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Continuity, compactness, and order [T-AN-CONTINUITY]
Recognition triggers continuous/differentiable trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2023-D1-P1

Day 1; official label P1; 10 points; legacy source-page label 173

Find all functions $f : \mathbb{R} \to \mathbb{R}$ that have a continuous second derivative and for which the equality $f(7x + 1) = 49f(x)$ holds for all $x \in \mathbb{R}$.

(proposed by Alex Avdiushenko, Neapolis University Paphos, Cyprus)

Open all solution variants | Source note

2023-D1-P2 Quick/standard

Fingerprint prove sixth power of matrix is identity
Primary domain Linear algebra / matrices
Problem type structural proof
Linked methods Complete the equation to a product equal to identity [M-LA-INVERSE]
Secondary cues Complex analysis / polynomials; Real analysis / calculus
Linked theory Determinants, block identities and inverses [T-LA-DETERMINANT]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]
Recognition triggers matrix word; continuous/differentiable trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2023-D1-P2

Day 1; official label P2; 10 points; legacy source-page label 173

Let A, B and C be $n \times n$ matrices with complex entries satisfying

$$A^2 = B^2 = C^2 \quad \text{and} \quad B^3 = ABC + 2I.$$

Prove that $A^6 = I$.

(proposed by Mike Daas, Universiteit Leiden)

Open all solution variants | Source note

2023-D1-P3 Medium/trap

Fingerprint two-variable polynomials multiplicative under complex multiplication
Primary domain Complex analysis / polynomials
Problem type classification/find-all
Linked methods Compare coefficients after a structural substitution [M-AL-COEFFICIENT]; Factor first and compare irreducible exponents [M-AL-FACTOR]
Secondary cues None recorded
Linked theory Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Polynomial factorization and coprime divisibility [T-AL-FACTORIZATION]; Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]
Recognition triggers polynomial/root trigger
Preparation route CP-Polynomials/Complex
Trap watch Separate polynomial identity from pointwise testing; check multiplicities, zero polynomial cases, and real-vs-complex roots.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2023-D1-P3

Day 1; official label P3; 10 points; legacy source-page label 173

Find all polynomials P in two variables with real coefficients satisfying the identity

$$P(x, y)P(z, t) = P(xz - yt, xt + yz).$$

(proposed by Giorgi Arabidze, Free University of Tbilisi, Georgia)

Open all solution variants | Source note

2023-D1-P4 Medium

Fingerprint possible residues given complete residue system condition
Primary domain Algebra / number theory
Problem type classification/find-all
Linked methods Choose a modulus that exposes the obstruction [M-AL-MODULAR]; Evaluate at roots of unity or average over them [M-AL-UNITY]
Secondary cues None recorded
Linked theory Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Groups, orders, generators, and quotients [T-AL-GROUP]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2023-D1-P4

Day 1; official label P4; 10 points; legacy source-page label 173

Let p be a prime number and let k be a positive integer. Suppose that the numbers $a_i = i^k + i$ for $i = 0, 1, \ldots, p-1$ form a complete residue system modulo p. What is the set of possible remainders of a_2 upon division by p?

(proposed by Tigran Hakobyan, Yerevan State University, Armenia)

Open all solution variants | Source note

2023-D1-P5 Long/deep

Fingerprint lower bound k! on preferred permutations via swaps
Primary domain Combinatorics
Problem type inequality/sharp bound
Linked methods Repair an extremal configuration by a local exchange [M-DS-EXTREMAL]; Count incidences with their exact multiplicity [M-DS-DOUBLECOUNT]
Secondary cues Combinatorics / discrete structures; Real analysis / calculus; Probability / measure / expectation
Linked theory Extremal choice, exchange arguments, and invariants [T-DS-EXTREMAL]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers integer/prime/mod trigger; counting/permutation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Choose an extremal object and force rigidity [S-EXTREMAL]; Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2023-D1-P5

Day 1; official label P5; 10 points; legacy source-page label 173

Fix positive integers n and k such that $2 \leq k \leq n$ and a set M consisting of n fruits. A permutation is a sequence $x = (x_1, x_2, \ldots, x_n)$ such that $\{x_1, \ldots, x_n\} = M$. Ivan prefers some (at least one) of these permutations. He realized that for every preferred permutation x, there exist k indices $i_1 < i_2 < \cdots < i_k$ with the following property: for every $1 \leq j < k$, if he swaps x_{i_j} and $x_{i_{j+1}}$, he obtains another preferred permutation.
Prove that he prefers at least $k!$ permutations.

(proposed by Ivan Mitrofanov, École Normale Superieur Paris)

Open all solution variants | Source note

Day 2

2023-D2-P6 Long/deep

Fingerprint matrix reachability under row-column multiply operations
Primary domain Linear algebra / matrices
Problem type nonexistence/counterexample
Linked methods Eliminate systematically and track the invariant [M-LA-ELIM]
Secondary cues Algebra / number theory
Linked theory Determinants, block identities and inverses [T-LA-DETERMINANT]
Recognition triggers matrix word
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Make the contradiction quantitative or structural [S-CONTRADICTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2023-D2-P6

Day 2; official label P6; 10 points; legacy source-page label 174

Ivan writes the matrix $\begin{pmatrix} 2 & 3 \\ 2 & 4 \end{pmatrix}$ on the board. Then he performs the following operation on the matrix several times:

- he chooses a row or a column of the matrix, and
- he multiplies or divides the chosen row or column entry-wise by the other row or column, respectively.

Can Ivan end up with the matrix $\begin{pmatrix} 2 & 4 \\ 2 & 3 \end{pmatrix}$ after finitely many steps?

(proposed by Alex Avdiushenko, Neapolis University Paphos, Cyprus)

Open all solution variants | Source note

2023-D2-P7 Medium/trap

Fingerprint alpha values forcing mean-value-type equation f+alpha=f'
Primary domain Real analysis / calculus
Problem type classification/find-all
Linked methods Count zeros and subtract an interpolant [M-AN-ROLLE]; Design an auxiliary derivative [M-AN-AUXILIARY]; Compare growth through derivatives [M-AN-RATIO]
Secondary cues Functional equations / dynamics
Linked theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Differential inequalities and integrating factors [T-AN-ODE]
Recognition triggers det/rank/eigen trigger; continuous/differentiable trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

2023-D2-P7 continued

Full statement: 2023-D2-P7

Day 2; official label P7; 10 points; legacy source-page label 174

Let V be the set of all continuous functions $f : [0,1] \to \mathbb{R}$, differentiable on $(0,1)$, with the property that $f(0) = 0$ and $f(1) = 1$. Determine all $\alpha \in \mathbb{R}$ such that for every $f \in V$, there exists some $\xi \in (0,1)$ such that

$$f(\xi) + \alpha = f'(\xi).$$

(proposed by Mike Daas, Leiden University)

Open all solution variants | Source note

2023-D2-P8 Medium/trap

Fingerprint inequality for sums of tree distances and reciprocals
Primary domain Combinatorics
Problem type inequality/sharp bound
Linked methods Split a family by traces, pieces, or labels [M-DS-PARTITION]
Secondary cues Geometry / convexity / metric; Real analysis / calculus
Linked theory Trees, unique paths, and distance sums [T-DS-TREES]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]
Recognition triggers integral/limit trigger; counting/permutation trigger; graph/color/set trigger; geometry/metric trigger
Preparation route CMB-Combinatoricsi
Trap watch Avoid overcounting; define the encoded object precisely and check extremal/minimal cases.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Translate geometry into a controlled representation [S-GEOMETRIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2023-D2-P8

Day 2; official label P8; 10 points; legacy source-page label 174

Let T be a tree with n vertices; that is, a connected simple graph on n vertices that contains no cycle. For every pair u, v of vertices, let $d(u,v)$ denote the distance between u and v, that is, the number of edges in the shortest path in T that connects u with v.

Consider the sums

$$W(T) = \sum_{\substack{\{u,v\} \subseteq V(T) \\ u \neq v}} d(u,v) \quad \text{and} \quad H(T) = \sum_{\substack{\{u,v\} \subseteq V(T) \\ u \neq v}} \frac{1}{d(u,v)}.$$

Prove that

$$W(T) \cdot H(T) \geq \frac{(n-1)^3(n+2)}{4}.$$

(proposed by Slobodan Filipovski, University of Primorska, Koper)

Open all solution variants | Source note

2023-D2-P9 Long/deep

Fingerprint max volume convex bodies with disjoint coordinate projections
Primary domain Geometry / convexity / metric
Problem type inequality/sharp bound
Linked methods Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]; Split a family by traces, pieces, or labels [M-DS-PARTITION]
Secondary cues Linear algebra / matrix theory; Combinatorics / discrete structures; Real analysis / calculus
Linked theory Convex bodies, supporting functionals, and packing [T-DS-CONVEX]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Finite counting and incidence identities [T-DS-COUNTING]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]

2023-D2-P9 continued

Recognition triggers graph/color/set trigger; geometry/metric trigger
Preparation route GEO-Geometry/Metricif
Trap watch Control degenerate configurations and make clear which coordinates or affine transformations preserve the quantity.
Proof strategies Translate geometry into a controlled representation [S-GEOMETRIC]; Choose the objects and count them without ambiguity [S-COUNTING]; Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2023-D2-P9

Day 2; official label P9; 10 points; legacy source-page label 174

We say that a real number V is good if there exist two closed convex subsets X, Y of the unit cube in $\mathbb{R}^3$, with volume V each, such that for each of the three coordinate planes (that is, the planes spanned by any two of the three coordinate axes), the projections of X and Y onto that plane are disjoint.
Find $\sup\{V \mid V \text{ is good}\}$.

(proposed by Josef Tkadlec and Arseniy Akopyan)

Open all solution variants | Source note

2023-D2-P10 Long/deep

Fingerprint denominator growth of factorial partial sums of e
Primary domain Algebra / number theory
Problem type existence/construction
Linked methods Track one prime or one extreme divisor [M-AL-VALUATION]; Replace many equal values by too many roots [M-AL-ROOTCOUNT]
Secondary cues Functional equations / dynamics; Geometry / convexity / metric
Linked theory Prime valuations, divisors, and factorials [T-AL-VALUATION]; Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Factorial growth and the prime-number input [T-AL-PRIMEGROWTH]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers det/rank/eigen trigger; integer/prime/mod trigger; functional-equation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Build a chain of bounds and track equality [S-BOUND]; Choose the objects and count them without ambiguity [S-COUNTING]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2023-D2-P10

Day 2; official label P10; 10 points; legacy source-page label 174

For every positive integer n, let $f(n)$, $g(n)$ be the minimal positive integers such that

$$1+\frac{1}{1!}+\frac{1}{2!}+\cdots+\frac{1}{n!}=\frac{f(n)}{g(n)}.$$

Determine whether there exists a positive integer n for which $g(n)>n^{0.999n}$.

(proposed by Fedor Petrov, St. Petersburg State University)

Open all solution variants | Source note

2.31 2024

Day 1

2024-D1-P1 Quick/standard

Fingerprint unit complex numbers making a+b+ab real
Primary domain Complex analysis / polynomials
Problem type classification/find-all
Linked methods Compare coefficients after a structural substitution [M-AL-COEFFICIENT]; Factor first and compare irreducible exponents [M-AL-FACTOR]
Secondary cues None recorded
Linked theory Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers det/rank/eigen trigger
Preparation route CP-Polynomials/Complex
Trap watch Separate polynomial identity from pointwise testing; check multiplicities, zero polynomial cases, and real-vs-complex roots.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Translate geometry into a controlled representation [S-GEOMETRIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2024-D1-P1

Day 1; official label P1; 10 points; legacy source-page label 175

Determine all pairs $(a, b) \in \mathbb{C} \times \mathbb{C}$ satisfying

$$|a| = |b| = 1 \quad \text{and} \quad a + b + a\bar{b} \in \mathbb{R}.$$

(proposed by Mike Daas, Universiteit Leiden)

Open all solution variants | Source note

2024-D1-P2 Quick/standard

Fingerprint limit of normalized hyperfactorial logarithm
Primary domain Real analysis / calculus
Problem type limit/asymptotic evaluation
Linked methods Expose the mesh and control its tails [M-AN-RIEMANN]
Secondary cues Geometry / convexity / metric; Functional equations / dynamics
Linked theory Riemann sums, periodic averaging, and tails [T-AN-RIEMANN]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Scale separation and limit comparisons [T-AN-ASYMPTOTICS]
Recognition triggers integral/limit trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2024-D1-P2

Day 1; official label P2; 10 points; legacy source-page label 175

For $n = 1, 2, \ldots$ let

$$S_n = \log\left(\sqrt[n^2]{1^1 \cdot 2^2 \cdot \cdots \cdot n^n} \right) - \log(\sqrt{n}),$$

where log denotes the natural logarithm. Find $\lim_{n\to\infty} S_n$.

(proposed by Sergey Chernov, Belarusian State University, Minsk)

Open all solution variants | Source note

2024-D1-P3 Medium/trap

Fingerprint 0-1 matrix whose square is all-ones
Primary domain Linear algebra / matrices
Problem type classification/find-all
Linked methods Read the entries before choosing a theorem [M-LA-ENTRY]; Exploit a compatible block decomposition [M-LA-BLOCK]
Secondary cues Linear algebra / matrix theory
Linked theory Rank, dimension and exact linear constraints [T-LA-RANK]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers matrix word; integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2024-D1-P3

Day 1; official label P3; 10 points; legacy source-page label 175

For which positive integers n does there exist an $n \times n$ matrix A whose entries are all in $\{0, 1\}$, such that A^2 is the matrix of all ones?

(proposed by Alex Avdiushenko, Neapolis University Paphos, Cyprus)

Open all solution variants | Source note

2024-D1-P4 Medium

Fingerprint subgroup from sliding word windows independent of sequence
Primary domain Algebra / number theory
Problem type structural proof
Linked methods Prove generation or obstruct it by an image [M-AL-GENERATORS]
Secondary cues Real analysis / calculus
Linked theory Groups, orders, generators, and quotients [T-AL-GROUP]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers integer/prime/mod trigger; group/ring/field trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Reduce size while preserving the defining conditions [S-INDUCTION]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2024-D1-P4

Day 1; official label P4; 10 points; legacy source-page label 175

Let g and h be two distinct elements of a group G, and let n be a positive integer. Consider a sequence $w = (w_1, w_2, \dots)$ which is not eventually periodic and where each w_i is either g or h. Denote by H the subgroup of G generated by all elements of the form $w_k w_{k+1} \dots w_{k+n-1}$ with $k \geq 1$. Prove that H does not depend on the choice of the sequence w (but may depend on n).

(proposed by Ivan Mitrofanov, Saarland University)

Open all solution variants | Source note

2024-D1-P5 Long/deep

Fingerprint hull containment probability monotone in distance from origin
Primary domain Probability / measure
Problem type inequality/sharp bound
Linked methods Condition and average a convex-hull event [M-AN-GEOMETRIC]
Secondary cues Probability / measure / expectation; Combinatorics / discrete structures; Geometry / convexity / metric
Linked theory Uniform random points and convex-hull events [T-AN-GEOMETRICPROB]; Finite probability, conditioning, and expectations [T-AN-PROBABILITY]; Facets and signed simplex decompositions [T-DS-POLYTOPE]
Recognition triggers integral/limit trigger; integer/prime/mod trigger; geometry/metric trigger; functional-equation trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Translate geometry into a controlled representation [S-GEOMETRIC]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2024-D1-P5

Day 1; official label P5; 10 points; legacy source-page label 175

Let $n > d$ be positive integers. Choose n independent, uniformly distributed random points $x_1, \dots, x_n$ in the unit ball $B \subset \mathbb{R}^d$ centered at the origin. For a point $p \in B$ denote by $f(p)$ the probability that the convex hull of $x_1, \dots, x_n$ contains p. Prove that if $p, q \in B$ and the distance of p from the origin is smaller than the distance of q from the origin, then $f(p) \geq f(q)$.

(proposed by Fedor Petrov, St Petersburg State University)

Open all solution variants | Source note

Day 2

2024-D2-P6 Long/deep

Fingerprint local maximum pattern for rational-to-integer functions
Primary domain Combinatorics
Problem type existence/construction
Linked methods Split a family by traces, pieces, or labels [M-DS-PARTITION]; Repair an extremal configuration by a local exchange [M-DS-EXTREMAL]
Secondary cues Real analysis / calculus
Linked theory Countability, accumulation, and Cantor constructions [T-AN-SETS]; Finite counting and incidence identities [T-DS-COUNTING]; Extremal choice, exchange arguments, and invariants [T-DS-EXTREMAL]
Recognition triggers functional-equation trigger
Preparation route FEQ-Functional Equations/Dynamics
Trap watch Do not use continuity, monotonicity, or injectivity unless it is given or proved; check fixed/periodic edge cases.
Proof strategies Make the contradiction quantitative or structural [S-CONTRADICTION]; Expose the structure before calculating [S-STRUCTURE]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2024-D2-P6

Day 2; official label P6; 10 points; legacy source-page label 176

Prove that for any function $f : \mathbb{Q} \to \mathbb{Z}$, there exist $a, b, c \in \mathbb{Q}$ such that $a < b < c$, $f(b) \geq f(a)$, and $f(b) \geq f(c)$.

(proposed by Mehdi Golafshan & Markus A. Whiteland, University of Liège, Liège)

Open all solution variants | Source note

2024-D2-P7

Medium/trap

Fingerprint possible det(AB) given polynomial matrix identity
Primary domain Linear algebra / matrices
Problem type classification/find-all
Linked methods Factor while preserving matrix order [M-LA-FACTOR]; Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]
Secondary cues Complex analysis / polynomials; Algebra / number theoryi
Linked theory Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Determinants, block identities and inverses [T-LA-DETERMINANT]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Recognition triggers matrix word; det/rank/eigen trigger; integer/prime/mod trigger
Preparation route LA-Matrix Core
Trap watch Do not assume commutativity; check real/complex eigenvalues, rank defects, and whether similarity or transpose preserves the needed invariant.
Proof strategies Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]
Navigation Full statement All solution variants Source note Learning route

Full statement: 2024-D2-P7

Day 2; official label P7; 10 points; legacy source-page label 176

Let n be a positive integer. Suppose that A and B are invertible $n \times n$ matrices with complex entries such that $A + B = I$ (where I is the identity matrix) and

$$(A^2 + B^2)(A^4 + B^4) = A^5 + B^5.$$

Find all possible values of $\det(AB)$ for the given n.

(proposed by Sergey Bondarev, Sergey Chernov, Belarusian State University, Minsk)

Open all solution variants | Source note

2024-D2-P8

Medium

Fingerprint limit bounds for nonlinear recurrence over 2^n
Primary domain Functional equations / dynamics
Problem type recurrence/convergence
Linked methods Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]; Strengthen the statement before the induction step [M-DS-INDUCT]
Secondary cues Functional equations / dynamics; Geometry / convexity / metric
Linked theory Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Recognition triggers integral/limit trigger; functional-equation trigger
Preparation route RA-Analysis Core
Trap watch Justify endpoints, limit interchanges, monotonicity, differentiability hypotheses, and equality/sharpness cases.
Proof strategies Reduce size while preserving the defining conditions [S-INDUCTION]; Build a chain of bounds and track equality [S-BOUND]; Control the limiting operation before taking the limit [S-ANALYTIC]
Navigation Full statement All solution variants Source note Learning route

2024-D2-P8 continued

Full statement: 2024-D2-P8

Day 2; official label P8; 10 points; legacy source-page label 176

Define the sequence $x_1, x_2, \ldots$ by the initial terms $x_1 = 2$, $x_2 = 4$, and the recurrence relation

$$x_{n+2} = 3x_{n+1} - 2x_n + \frac{2^n}{x_n} \quad \text{for } n \geq 1.$$

Prove that $\lim_{n\to\infty} \frac{x_n}{2^n}$ exists and satisfies

$$\frac{1+\sqrt{3}}{2} \leq \lim_{n\to\infty} \frac{x_n}{2^n} \leq \frac{3}{2}.$$

(proposed by Karen Keryan, Yerevan State University & American University of Armenia, Armenia)

Open all solution variants | Source note

2024-D2-P9 Long/deep

Fingerprint parity of number of nice monotone matrices
Primary domain Combinatorics
Problem type counting/enumeration
Linked methods Encode a relation as a graph and extract a cycle [M-DS-GRAPH]; Encode, decode, and verify a bijection [M-DS-BIJECTION]
Secondary cues Linear algebra / matrix theory; Real analysis / calculus; Combinatorics / discrete structures
Linked theory Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]; Finite counting and incidence identities [T-DS-COUNTING]
Recognition triggers matrix word; counting/permutation trigger
Preparation route COMB-Counting/Enumeration
Trap watch Verify that any pairing or counting map preserves every defining matrix condition; account for fixed points explicitly.
Proof strategies Expose the structure before calculating [S-STRUCTURE]; Choose the objects and count them without ambiguity [S-COUNTING]
Correction note The explicit CLASS-001 corrections are incorporated in this card; the accompanying change ledger records the withdrawn labels.
Navigation Full statement All solution variants Source note Learning route

Full statement: 2024-D2-P9

Day 2; official label P9; 10 points; legacy source-page label 176

A matrix $A = (a_{ij})$ is called nice, if it has the following properties:
(i) the set of all entries of A is $\{1, 2, \ldots, 2t\}$ for some integer t;
(ii) the entries are non-decreasing in every row and in every column: $a_{i,j} \leq a_{i,j+1}$ and $a_{i,j} \leq a_{i+1,j}$;
(iii) equal entries can appear only in the same row or the same column: if $a_{i,j} = a_{k,\ell}$, then either $i = k$ or $j = \ell$;
(iv) for each $s = 1, 2, \ldots, 2t-1$, there exist $i \neq k$ and $j \neq \ell$ such that $a_{i,j} = s$ and $a_{k,\ell} = s+1$.
Prove that for any positive integers m and n, the number of nice $m \times n$ matrices is even.
For example, the only two nice 2×3 matrices are $\begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \end{pmatrix}$ and $\begin{pmatrix} 1 & 1 & 3 \\ 2 & 4 & 4 \end{pmatrix}$.

(proposed by Fedor Petrov, St Petersburg State University)

Open all solution variants | Source note

2024-D2-P10 Long/deep

Fingerprint divisor congruences for almost prime integers
Primary domain Algebra / number theory
Problem type structural proof
Linked methods Choose a modulus that exposes the obstruction [M-AL-MODULAR]
Secondary cues None retained after correction
Linked theory Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Groups, orders, generators, and quotients [T-AL-GROUP]; Prime valuations, divisors, and factorials [T-AL-VALUATION]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Recognition triggers integer/prime/mod trigger
Preparation route ANT-Algebra/Number Theory
Trap watch Watch zero cases, invertibility modulo primes, gcd assumptions, and whether statements are in Z, Q, a field, a ring, or a group.
Proof strategies Pass between local constraints and a global conclusion [S-LOCALGLOBAL]; Expose the structure before calculating [S-STRUCTURE]
Correction note The explicit CLASS-001 corrections are incorporated in this card; the accompanying change ledger records the withdrawn labels.
Navigation Full statement All solution variants Source note Learning route

Full statement: 2024-D2-P10

Day 2; official label P10; 10 points; legacy source-page label 176

We say that a square-free positive integer n is almost prime if

$$n \mid x^{d_1} + x^{d_2} + \cdots + x^{d_k} - kx$$

for all integers x, where $1 = d_1 < d_2 < \cdots < d_k = n$ are all the positive divisors of n. Suppose that r is a Fermat prime (i.e. it is a prime of the form $2^{2^m} + 1$ for an integer $m \geq 0$), p is a prime divisor of an almost prime integer n, and $p \equiv 1 \pmod{r}$. Show that, with the above notation, $d_i \equiv 1 \pmod{r}$ for all $1 \leq i \leq k$.
(An integer n is called square-free if it is not divisible by d^2 for any integer $d > 1$.)

(proposed by Tigran Hakobyan, Yerevan State University, Vanadzor, Armenia)

Open all solution variants | Source note

Turn the catalog into a working practice system

Each exercise now has a single learning route linking its theory, methods and strategy. The reference entries collect the reverse links, so you can study a topic and choose a related exercise. Keep four connected records:

1. **Theory:** definitions, hypotheses, counterexamples, equality cases, and a short proof core.
2. **Methods:** recognition cues, reusable moves, a proof skeleton, and representative problems.
3. **Triggers:** the precise words or structures that suggested a useful approach.
4. **Performance:** attempt time, help used, failure mode, and independent reattempt date.

Next layer	Purpose
Theory reference	Map a knowledge gap to prerequisites and linked examples.
Method reference	Practice a reusable move across distinct problems.
Types and triggers	Recognize the structural task before selecting a method.
Proof strategies	Convert an idea into a complete written argument.
Solution corpus	Compare recorded solutions after an independent attempt.

The retained atlas contains historical annotation tables and identified passages with correction links. Where a page carries a correction notice, follow it before using the statement. A topic label is a search aid; verify that its hypothesis and conclusion fit the problem in hand.

Theory, Methods, Triggers and Proof Templates

The following atlas retains the original preparation architecture and stable identifiers. The native catalog and solution chapters are linked throughout. Identified historical defects carry controlling correction links; annotation counts are navigation data, not forecasts of future competitions.

Solution-Linked Theory Reference

These compact entries state the hypotheses, identities and proof ideas used by the recorded problem solutions. A theory entry describes what is true; a method describes how to use it; a strategy organizes the proof. Each entry has one canonical location. Exercise tags lead here, and the return lists lead back to the exercises.

Read selectively. Start from an exercise's learning route. Open the relevant entry, check its hypotheses, and return to your attempt before reading the solution. The links cover the recorded proof routes; they do not assert that every listed technique is necessary for every possible proof. **Return navigation.** Back to previous view uses the PDF reader's history command. If that reader does not support it, use the explicit exercise ID or [R] links printed in every entry.

Linear algebra and matrices

Rank, dimension and exact linear constraints [T-LA-RANK]

Determinants, block identities and inverses [T-LA-DETERMINANT]

Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]

Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]

Trace pairing and commutator calculus [T-LA-TRACE]

Nilpotents, projections and involutions [T-LA-NILPOTENT]

Dual bases, kernels and separating functionals [T-LA-DUALITY]

Inner products, orthogonality and polarization [T-LA-INNER]

Gram matrices and positive semidefinite forms [T-LA-GRAM]

Adapted bases, centralizers and common eigenspaces [T-LA-INVARIANT]

Operator norms and convergence of matrix powers [T-LA-NORM]

Tensor products and operators on matrix spaces [T-LA-TENSOR]

Linear algebra in characteristic two [T-LA-FINITEFIELD]

Completeness, barycentres and equidistant sets [T-LA-HILBERT]

Fractional-linear recurrences and matrix powers [T-LA-PROJECTIVE]

Matrix power series and transfer between AB and BA [T-LA-SERIES]

Evaluation maps and rank under perturbation [T-LA-PERTURBATION]

Analysis, limits and probability

Continuity, compactness, and order [T-AN-CONTINUITY]

Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]

Taylor remainders and local scale [T-AN-TAYLOR]

Primitives, substitution, and integration by parts [T-AN-INTEGRAL]

Nonnegative sums, Fubini, and integral kernels [T-AN-FUBINI]

Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]

Series, finite cancellation, and summation by parts [T-AN-SERIES]

Power series and generating functions [T-AN-POWER]

Riemann sums, periodic averaging, and tails [T-AN-RIEMANN]

Scale separation and limit comparisons [T-AN-ASYMPTOTICS]

Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]

Lipschitz derivatives and convex smoothness [T-AN-SMOOTH]

Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]

Differential inequalities and integrating factors [T-AN-ODE]
Uniform approximation and positive kernels [T-AN-APPROXIMATION]
Dominated convergence and orthonormal separation [T-AN-MEASURE]
Countability, accumulation, and Cantor constructions [T-AN-SETS]
Baire class one and controlled discontinuities [T-AN-BAIRE]
Interval coverings and level-set integration [T-AN-COVERING]
Trigonometric identities and safe branches [T-AN-TRIG]
Finite probability, conditioning, and expectations [T-AN-PROBABILITY]
Uniform random points and convex-hull events [T-AN-GEOMETRICPROB]
Convex graph length and ordered primitives [T-AN-ARCLENGTH]

Algebra, number theory and complex polynomials

Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Polynomial factorization and coprime divisibility [T-AL-FACTORIZATION]
Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]
Interpolation, divided differences, and extremal signs [T-AL-INTERPOLATION]
Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]
Prime valuations, divisors, and factorials [T-AL-VALUATION]
Groups, orders, generators, and quotients [T-AL-GROUP]
Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]
Group actions, stabilizers, and semidirect products [T-AL-ACTION]
Noncommutative rings, polarization, and torsion [T-AL-RING]
Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]
Roots of unity, cyclotomic polynomials, and algebraic norms [T-AL-CYCLOTOMIC]
Quadratic norms and Pell-type constructions [T-AL-PELL]
Linear recurrences and generating functions [T-AL-RECURRENCE]
Finite differences and shift-operator ideals [T-AL-DIFFERENCE]
Rational linear relations and discrete-value rigidity [T-AL-RATIONALITY]
Greedy products, tail bounds, and integer descent [T-AL-GREEDY]
Holomorphic bounds, poles, and logarithmic-branch sums [T-AL-COMPLEXANALYSIS]
Roots-of-unity filters and positive trigonometric kernels [T-AL-FOURIER]
Factorial growth and the prime-number input [T-AL-PRIMEGROWTH]
Adjacent-transposition relations and finite presentations [T-AL-COXETER]

Counting, geometry and functional equations

Finite counting and incidence identities [T-DS-COUNTING]
Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]
Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]
Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]
Metric contractions, covers, and nearest points [T-DS-METRIC]
Convex bodies, supporting functionals, and packing [T-DS-CONVEX]
Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]
Extremal choice, exchange arguments, and invariants [T-DS-EXTREMAL]
Finite set systems and antichains [T-DS-SETS]
Hall matching and augmenting paths [T-DS-MATCHING]
Conics, reflections, and inversion [T-DS-TRANSFORMS]

Order, colouring, and binomial threshold induction [T-DS-RAMSEY]

Spherical distance and signed dihedral derivatives [T-DS-SPHERICAL]

Facets and signed simplex decompositions [T-DS-POLYTOPE]

Generating functions, coefficient filters, and finite differences [T-DS-GENERATING]

Log-concave sequences and coefficient smoothing [T-DS-LOGCONCAVITY]

Sumsets and the minimal-growth lemma [T-DS-ADDITIVE]

Trees, unique paths, and distance sums [T-DS-TREES]

Linear algebra and matrices

T-LA-RANK Rank, dimension and exact linear constraints

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Setting. Let $T : V \to W$ be linear over a field K, with $\dim V < \infty$. Its kernel is $\{v : Tv = 0\}$ and its image is $T(V)$. Extending a basis of the kernel to a basis of V proves

$$\dim V = \dim \ker T + \dim \operatorname{im} T.$$

For subspaces $U, W \subseteq V$, extending a basis of their intersection gives $\dim(U+W) = \dim U + \dim W - \dim(U \cap W)$. Consequently, $U \cap W = \{0\}$ forces $\dim U + \dim W \le \dim V$.

Matrix toolkit. Rank equals column rank, row rank, and the largest order of a nonzero minor. Compatible matrices satisfy

$$\operatorname{rank}(AB) \le \min(\operatorname{rank} A, \operatorname{rank} B), \qquad \operatorname{rank}(A+B) \le \operatorname{rank} A + \operatorname{rank} B.$$

For A of size $m \times r$ and B of size $r \times p$, also $\operatorname{rank}(AB) \ge \operatorname{rank} A + \operatorname{rank} B - r$: restrict A to $\operatorname{im} B$ and bound its kernel. A rank-one matrix is uv^T with $u, v \neq 0$; hence $(uv^T)^2 = (v^T u)uv^T$.

Example. A matrix with entries $a_i + b_j$ is $a\mathbf{1}^T + \mathbf{1}b^T$, so its rank is at most two. A nonzero 2×2 minor certifies equality. This separates the upper bound from the matching lower bound.

Checks. A square system $Ax = b$ has multiple solutions exactly when it is consistent and $\ker A \neq \{0\}$; singularity alone does not ensure consistency. Rank of a real matrix is unchanged on extending scalars to $\mathbb{C}$, because its nonzero minors are unchanged. Subspace dimension, however, must always name its scalar field.

Extended atlas LA-04

Reference notes 4

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P1 [R]; 1994-D2-P4 [R]; 1995-D1-P1 [R]; 1997-D2-P4 [R]; 1998-D1-P1 [R]; 1998-D2-P1 [R]; 2000-D1-P3 [R]; 2003-D2-P4 [R]; 2004-D1-P4 [R]; 2004-D2-P1 [R]; 2004-D2-P6 [R]; 2005-D1-P1 [R]; 2005-D2-P3 [R]; 2006-D1-P4 [R]; 2006-D2-P4 [R]; 2007-D1-P2 [R]; 2007-D1-P3 [R]; 2007-D2-P4 [R]; 2007-D2-P5 [R]; 2008-D1-P5 [R]; 2008-D2-P5 [R]; 2009-D1-P5 [R]; 2009-D2-P5 [R]; 2011-D1-P5 [R]; 2012-D1-P2 [R]; 2012-D2-P1 [R]; 2013-D2-P3 [R]; 2014-D2-P4 [R]; 2015-D2-P9 [R]; 2016-D1-P2 [R]; 2018-D2-P6 [R]; 2019-D1-P2 [R]; 2020-D1-P2 [R]; 2022-D2-P7 [R]; 2024-D1-P3 [R]

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T-LA-DETERMINANT Determinants, block identities and inverses

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Definition and rules. For a square matrix over a commutative field, $\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_i a_{i,\sigma(i)}$. Adding a multiple of one row to another preserves the determinant; interchanging two rows changes its sign; scaling one row scales it. The same rules hold for columns. Triangular matrices have determinant equal to the product of their diagonal entries, and $\det(AB) = \det A \det B$.

Inverses. A square matrix is invertible exactly when its determinant is nonzero. If square matrices satisfy $XY = I$, then Y is injective and therefore invertible; thus $YX = I$ as well. Rectangular matrices do not satisfy this conclusion. An inverse of a matrix with rational entries again has rational entries, by the adjugate formula.

Blocks. If P is invertible, block elimination gives

$$\det \begin{pmatrix} P & Q \\ R & S \end{pmatrix} = \det P \, \det(S - RP^{-1}Q).$$

The order of factors is essential. Without an invertible diagonal block, use direct multiplication instead. For example, if $M = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$ and $M^{-1} = \left(\begin{smallmatrix} E & F \\ G & H \end{smallmatrix}\right)$ with equal square blocks, then

$$M \begin{pmatrix} I & F \\ 0 & H \end{pmatrix} = \begin{pmatrix} A & 0 \\ C & I \end{pmatrix}, \qquad \det M \det H = \det A.$$

Here A and H need not be invertible.

T-LA-DETERMINANT Determinants, block identities and inverses (continued)

Pattern checks. For $r \times r$ matrices C, D, $\det \left(\begin{smallmatrix} 0 & C \\ D & 0 \end{smallmatrix} \right) = (-1)^r \det C \det D$. In particular, $D = C^T$ yields a signed square. An integer determinant may be reduced modulo a prime or composite modulus because its defining expression uses only sums and products; division modulo that modulus requires an additional invertibility check.

Integer operations. An integer square matrix is unimodular when its determinant is ± 1; its inverse then has integer entries. Integer row additions, row swaps and row sign changes are unimodular operations. Repeated additions and swaps implement the Euclidean algorithm on a column, replacing its entries by their greatest common divisor and zeros. Thus elimination over the integers must use these reversible operations rather than division by an arbitrary nonunit pivot.

Extended atlas LA-01; LA-02; LA-03

Reference notes 4

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P1 [R]; 1996-D1-P1 [R]; 1997-D1-P3 [R]; 1997-D2-P2 [R]; 1999-D1-P1 [R]; 2001-D2-P4 [R]; 2002-D2-P1 [R]; 2002-D2-P5 [R]; 2003-D2-P1 [R]; 2004-D2-P1 [R]; 2005-D1-P1 [R]; 2005-D2-P6 [R]; 2006-D1-P3 [R]; 2006-D2-P4 [R]; 2006-D2-P6 [R]; 2007-D1-P2 [R]; 2007-D1-P3 [R]; 2007-D2-P4 [R]; 2008-D1-P6 [R]; 2008-D2-P5 [R]; 2009-D1-P2 [R]; 2009-D1-P5 [R]; 2009-D2-P1 [R]; 2010-D1-P5 [R]; 2014-D1-P1 [R]; 2015-D1-P1 [R]; 2018-D1-P3 [R]; 2019-D1-P2 [R]; 2019-D1-P5 [R]; 2019-D2-P9 [R]; 2021-D1-P1 [R]; 2021-D2-P5 [R]; 2023-D1-P2 [R]; 2023-D2-P6 [R]; 2024-D2-P7 [R]

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T-LA-SPECTRAL Eigenvalues, similarity and orthogonal diagonalization

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Rank, dimension and exact linear constraints [T-LA-RANK]

Eigenvalues. For $A \in M_n(\mathbb{C})$, an eigenvalue λ satisfies $Av = \lambda v$ for some $v \neq 0$; equivalently, $\det(\lambda I - A) = 0$. Algebraic multiplicity is multiplicity as a root; geometric multiplicity is $\dim \ker(A - \lambda I)$ and cannot exceed it. Eigenvectors for distinct eigenvalues are linearly independent, so there are at most n distinct eigenvalues.

Similarity. If $B = P^{-1}AP$, then A and B represent the same operator in different bases. Rank, trace, characteristic polynomial and all polynomial identities are preserved. Entrywise positivity and the Euclidean operator norm need not be preserved. Over $\mathbb{C}$, every matrix is unitarily similar to an upper-triangular matrix: choose a unit eigenvector and apply induction to its orthogonal complement block. Its diagonal lists all eigenvalues, with algebraic multiplicity.

Spectral theorem. A real symmetric matrix is orthogonally diagonalizable; a complex Hermitian matrix ($A^* = A$) is unitarily diagonalizable with real eigenvalues. More generally a complex normal matrix ($AA^* = A^*A$) is unitarily diagonalizable. In a triangular normal matrix, comparing the first diagonal entries of TT^* and T^*T forces the rest of its first row to vanish; induction proves the claim.

Example. If real symmetric A has all eigenvalues at least $c > 0$, then $\|Ax\|_2 \geq c\|x\|_2$. If $c > 1$, it strictly expands every nonzero vector.

Traps. Repeated eigenvalues do not by themselves imply a scalar matrix: $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix} \right)$ is a counterexample. A real matrix may have nonreal conjugate eigenvalues. Complex transpose-normality $AA^T = A^TA$ is not Hermitian normality and does not give the usual spectral theorem.

Extended atlas LA-07; LA-09

Reference notes 4

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P4 [R]; 1995-D1-P1 [R]; 1999-D1-P1 [R]; 2000-D1-P3 [R]; 2001-D1-P5 [R]; 2002-D2-P5 [R]; 2004-D2-P4 [R]; 2006-D2-P6 [R]; 2011-D1-P2 [R]; 2013-D1-P1 [R]; 2014-D1-P1 [R]; 2014-D2-P2 [R]; 2016-D2-P10 [R]; 2017-D1-P1 [R]; 2017-D2-P8 [R]; 2019-D1-P2 [R]; 2019-D2-P9 [R]; 2021-D2-P5 [R]; 2021-D2-P8 [R]; 2022-D1-P2 [R]

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T-LA-POLYNOMIAL Annihilating polynomials and spectral mapping

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]

Definitions. For $p(t) = \sum_j c_j t^j$, define $p(A) = \sum_j c_j A^j$. Polynomials in one fixed matrix commute. The monic polynomial of least degree with $m_A(A) = 0$ is the minimal polynomial. Division with remainder shows that m_A divides every annihilating polynomial. Cayley–Hamilton states that $\chi_A(A) = 0$, so $\deg m_A \le n$.

Consequences. If $Av = \lambda v$, then $p(A)v = p(\lambda)v$. Triangularizing A proves the stronger statement: the eigenvalues of $p(A)$ are $p(\lambda_1), \dots, p(\lambda_n)$, with multiplicities combined when values coincide. A matrix is diagonalizable over K precisely when its minimal polynomial splits over K into distinct linear factors. Indeed, for distinct roots λ_i, the interpolation polynomials

$$e_i(t) = \prod_{j \neq i} \frac{t - \lambda_j}{\lambda_i - \lambda_j}$$

give projections $e_i(A)$ onto complementary eigenspaces.

Trace and determinant. Triangularization yields $\operatorname{tr} p(A) = \sum_i p(\lambda_i)$ and $\det p(A) = \prod_i p(\lambda_i)$. For real A, nonreal eigenvalues occur in conjugate pairs; each nonzero pair contributes $|\lambda|^2 > 0$ to its determinant.

Examples. If $A^2 = A$, the polynomial $t(t-1)$ proves diagonalizability over any field. If $A^4 = A$, its spectrum is among 0 and the cube roots of unity; one must still construct admissible real matrices to show that all proposed values occur.

Checks. The implication $p(A) = 0 \Rightarrow p(\lambda) = 0$ is necessary, not sufficient for a matrix identity when Jordan blocks are present. Over $\mathbb{Q}$, irreducible factors constrain degrees of m_A; factoring only over $\mathbb{C}$ can lose this arithmetic information.

Extended atlas LA-07

Reference notes 4

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1999-D1-P1 [R]; 2000-D2-P6 [R]; 2003-D1-P3 [R]; 2004-D2-P4 [R]; 2009-D2-P3 [R]; 2011-D1-P2 [R]; 2012-D2-P4 [R]; 2014-D1-P1 [R]; 2015-D1-P1 [R]; 2016-D2-P10 [R]; 2017-D1-P1 [R]; 2018-D1-P3 [R]; 2019-D2-P9 [R]; 2020-D1-P2 [R]; 2021-D2-P5 [R]; 2021-D2-P8 [R]; 2022-D1-P2 [R]; 2023-D1-P2 [R]; 2024-D2-P7 [R]

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T-LA-TRACE Trace pairing and commutator calculus

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Rank, dimension and exact linear constraints [T-LA-RANK]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]

Cyclicity. For compatible matrices, $\operatorname{tr}(XY) = \operatorname{tr}(YX)$ by summing $x_{ij}y_{ji}$. Consequently trace is invariant under cyclic rotation of a product, and every commutator $[A, B] = AB - BA$ has trace zero. This permits cyclic rotations, not arbitrary permutations.

Pairing. On $M_n(K)$, the bilinear form $(X, Y) \mapsto \operatorname{tr}(XY)$ is nondegenerate because $\operatorname{tr}(XE_{ji}) = x_{ij}$. Thus every linear functional is uniquely $X \mapsto \operatorname{tr}(XC)$. Over $\mathbb{R}$, $\operatorname{tr}(S^2) = \sum_{ij} s_{ij}^2 > 0$ for every nonzero symmetric S. Therefore a subspace on which $\operatorname{tr}(XY)$ vanishes identically has trivial intersection with the symmetric matrices. The dimension formula then gives an upper bound $n(n-1)/2$, attained by the strictly upper-triangular matrices.

Derivations. The map $\Delta(X) = [A, X]$ satisfies

$$\Delta(XY) = \Delta(X)Y + X\Delta(Y), \qquad [F^k, G] = \sum_{j=0}^{k-1} F^{k-1-j}[F, G]F^j.$$

Factor order must be retained. If $\Delta^2(B) = 0$, repeated Leibniz expansion gives $\Delta^k(B^k) = k!\Delta(B)^k$; every surviving term differentiates each factor exactly once.

T-LA-TRACE Trace pairing and commutator calculus (continued)

Example and trap. If a rank-one commutator is $C = uv^T$, then $C^2 = (v^T u)C = \operatorname{tr}(C)C = 0$. Trace zero alone is insufficient: $\operatorname{diag}(1, -1)$ has trace zero and is invertible. Over characteristic zero, vanishing of $\operatorname{tr}(A^j)$ for $1 \le j \le n$ forces $\chi_A(t) = t^n$ by Newton's identities, hence nilpotency; this conclusion requires all those power sums, not just the first. Writing $\chi_A(t) = t^n + c_1 t^{n-1} + \cdots + c_n$ and $s_j = \operatorname{tr}(A^j)$, Newton's recurrence is

$$kc_k + \sum_{j=1}^{k} c_{k-j} s_j = 0, \qquad c_0 = 1, \quad 1 \le k \le n.$$

When all s_j vanish, induction gives $c_k = 0$; division by k explains the characteristic-zero assumption.

Extended atlas LA-05; LA-06; LA-12

Reference notes 4

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P4 [R]; 1997-D2-P4 [R]; 2000-D1-P3 [R]; 2001-D1-P5 [R]; 2005-D2-P3 [R]; 2006-D2-P6 [R]; 2009-D2-P3 [R]; 2011-D1-P2 [R]; 2014-D2-P2 [R]; 2015-D2-P9 [R]; 2020-D1-P2 [R]; 2021-D2-P8 [R]; 2022-D1-P2 [R]; 2022-D2-P7 [R]

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T-LA-NILPOTENT Nilpotents, projections and involutions

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Prerequisites Rank, dimension and exact linear constraints [T-LA-RANK]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]

Nilpotency. An operator N is nilpotent if $N^r = 0$ for some positive integer r. On an n-dimensional space, nilpotency implies $N^n = 0$: the kernel chain strictly increases until it reaches the whole space. Over a splitting field, the converse spectral test is that every eigenvalue is zero, by Cayley–Hamilton. Nilpotents are generally not diagonalizable; a diagonalizable nilpotent is zero.

Finite inverse. If $N^r = 0$, direct multiplication gives

$$(I + N)^{-1} = I - N + N^2 - \cdots + (-1)^{r-1} N^{r-1}.$$

This identity applies equally to operators acting on matrix spaces. Left multiplication $L_A(X) = AX$ and right multiplication $R_B(X) = XB$ commute, even when A and B do not. If $A^r = B^s = 0$, the binomial expansion gives $(L_A + R_B)^{r+s-1} = 0$.

Projections. For $P^2 = P$, $V = \operatorname{im} P \oplus \ker P$, since $v = Pv + (v - Pv)$. In an adapted basis $P = \operatorname{diag}(I_r, 0)$, so $\operatorname{tr} P = r = \operatorname{rank} P$ in characteristic zero. This need not be an orthogonal projection unless $P = P^*$.

Involutions. Over characteristic different from two, $A^2 = I$ gives complementary projections $(I + A)/2$ and $(I - A)/2$ onto the $+1$ and -1 eigenspaces. A commuting family of such operators can be simultaneously diagonalized and has at most 2^n distinct members.

Example. The shift $Ne_j = e_{j+1}$ for $j < n$, $Ne_n = 0$, has rank $n - 1$ but only zero eigenvalues. Thus counting nonzero eigenvalues does not compute rank without diagonalizability. In characteristic two, a nonidentity square-zero perturbation $I + N$ with $N^2 = 0$ is an involution that need not diagonalize.

$$e_1 \xrightarrow{N} e_2 \xrightarrow{N} e_3 \xrightarrow{N} 0$$
$$N^3 = 0, \quad \operatorname{rank} N = 2$$

Extended atlas LA-08

Reference notes 4

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P4 [R]; 1995-D1-P1 [R]; 1995-D1-P5 [R]; 1996-D1-P3 [R]; 2000-D1-P3 [R]; 2000-D2-P6 [R]; 2001-D2-P4 [R]; 2003-D1-P3 [R]; 2007-D2-P5 [R]; 2009-D2-P3 [R]; 2021-D1-P1 [R]; 2022-D1-P2 [R]; 2022-D2-P7 [R]

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T-LA-DUALITY Dual bases, kernels and separating functionals

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Prerequisites Rank, dimension and exact linear constraints [T-LA-RANK]

Dual coordinates. For a basis $v_1, \dots, v_n$ of a finite-dimensional space, the dual basis $f_1, \dots, f_n$ consists of the linear functionals $f_i(v_j) = \delta_{ij}$. Thus $v = \sum_i f_i(v) v_i$. On a real inner-product space, each f_i is represented by a unique vector w_i with $f_i(v) = \langle w_i, v \rangle$. If the columns of an invertible matrix V are v_i, the columns of V^{-T} are w_i.

Kernel criterion. Let $f, f_1, \dots, f_k : V \to K$ be linear, with V possibly infinite-dimensional. If f vanishes on $\bigcap_i \ker f_i$, then $f \in \operatorname{span}(f_1, \dots, f_k)$. Define $F(v) = (f_1(v), \dots, f_k(v))$. The rule $\ell(F(v)) = f(v)$ is well-defined because $\ker F \subseteq \ker f$. Extend ℓ from $\operatorname{im} F$ to K^k, and write it as a linear combination of coordinate functionals. This proves the criterion without assuming the f_i independent.

Separating a dependence. To prove vectors v_i independent, find linear maps L_i such that $L_i(v_j) = 0$ for $j \neq i$ and $L_i(v_i) \neq 0$. Applying L_i to a putative relation isolates its ith coefficient. The maps may be matrix operators rather than scalar functionals.

Example. If $A_i A_j = 0$ for $i \neq j$ and $A_i^2 \neq 0$, choose $v_i \in \operatorname{im} A_i$ with $A_i v_i \neq 0$. Then the maps A_i separate the v_i, proving that their number cannot exceed the ambient dimension.

Check. Linear independence of functionals is a property of their values on all vectors; checking a single vector cannot establish independence of a family with at least two functionals.

Extended atlas LA-11

Reference notes 4

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1997-D2-P4 [R]; 1998-D2-P1 [R]; 2007-D2-P5 [R]; 2008-D1-P2 [R]; 2009-D2-P5 [R]; 2013-D2-P3 [R]; 2015-D2-P9 [R]; 2016-D1-P2 [R]; 2022-D2-P7 [R]

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T-LA-INNER Inner products, orthogonality and polarization

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Prerequisites Rank, dimension and exact linear constraints [T-LA-RANK]

Setting. A real inner product is a symmetric bilinear form with $\langle x, x \rangle > 0$ for $x \neq 0$. Over $\mathbb{C}$ use a positive-definite Hermitian form and conjugate transpose. Its norm satisfies $\|x\|^2 = \langle x, x \rangle$ and the Cauchy–Schwarz inequality $|\langle x, y \rangle| \leq \|x\| \|y\|$, with equality precisely when x, y are linearly dependent. The proof minimizes $\|x - ty\|^2$ over the appropriate scalar t.

Orthogonal decomposition. For a finite-dimensional subspace U, Gram–Schmidt gives an orthonormal basis e_i, and $P_U x = \sum_i \langle x, e_i \rangle e_i$ in the real case. Then $x - P_U x \perp U$ and $\|x\|^2 = \|P_U x\|^2 + \|x - P_U x\|^2$. Nonzero pairwise orthogonal vectors are linearly independent; their number cannot exceed dimension.

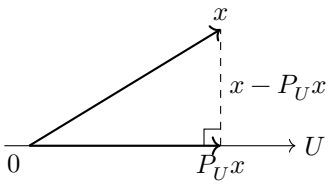
Polarization. In a real inner-product space,

$$2\langle u, v \rangle = \|u\|^2 + \|v\|^2 - \|u - v\|^2.$$

For a real bilinear form B, the identity $B(x, x) = 0$ for all x implies $B(u, v) + B(v, u) = 0$, by expanding $B(u + v, u + v)$. Thus $x^T A x = 0$ for every real x forces $A^T = -A$.

Example. If v_i are unit vectors and w_i their dual basis, then $1 = \langle w_i, v_i \rangle \leq \|w_i\|$. Averaging over signs gives $2^{-n} \sum_{\varepsilon_i = \pm 1} \| \sum_i \varepsilon_i w_i \|^2 = \sum_i \|w_i\|^2$; all mixed terms cancel. This provides a long vector with prescribed small normalized inner products.

Trap. The bilinear form $x^T y$ on complex vectors, or on a finite field, is not positive definite. Arguments using positivity of squared norms must not be transferred to those settings.



Extended atlas LA-15

T-LA-INNER Inner products, orthogonality and polarization (continued)

Reference notes 4

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D2-P5 [R]; 1995-D2-P1 [R]; 1995-D2-P6 [R]; 2001-D2-P3 [R]; 2002-D2-P4 [R]; 2002-D2-P6 [R]; 2003-D2-P3 [R]; 2004-D2-P3 [R]; 2005-D2-P3 [R]; 2006-D2-P4 [R]; 2008-D2-P6 [R]; 2009-D1-P5 [R]; 2010-D1-P5 [R]; 2013-D2-P3 [R]; 2014-D2-P2 [R]; 2015-D1-P5 [R]; 2016-D1-P3 [R]; 2018-D2-P6 [R]; 2019-D1-P2 [R]; 2019-D2-P8 [R]; 2022-D1-P1 [R]

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T-LA-GRAM Gram matrices and positive semidefinite forms

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Inner products, orthogonality and polarization [T-LA-INNER]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Determinants, block identities and inverses [T-LA-DETERMINANT]

Dictionary. Given vectors $v_1, \dots, v_m$ in a real Euclidean space, their Gram matrix is $G = (\langle v_i, v_j \rangle)$. For a coefficient column c,

$$c^T G c = \left\| \sum_i c_i v_i \right\|^2 \geq 0.$$

Hence G is positive semidefinite (PSD), and its rank equals the dimension of the vectors' span. It is positive definite exactly when the vectors are independent. Conversely, every real symmetric PSD matrix has a factorization $G = V^T V$, obtained by orthogonally diagonalizing it and taking the nonnegative square roots of its eigenvalues.

Tests. A real symmetric matrix is PSD exactly when all eigenvalues are nonnegative, equivalently when all principal minors are nonnegative. For sufficiency of the minor test, the coefficients of $\det(tI + G)$ are sums of principal minors; if all are nonnegative, this polynomial is positive for $t > 0$, excluding negative eigenvalues. Checking only leading principal minors is not sufficient for semidefiniteness. Strict positivity of all leading principal minors does characterize positive definiteness.

Distance example. If distances from the origin and between vectors are rational, polarization makes the Gram entries rational. Choose a real basis from the vectors. Its nonsingular Gram matrix G gives coordinates $c = G^{-1} b \in \mathbb{Q}^r$ for any remaining vector, because its inner products b are rational.

Three-vector example. The matrix $\left(\begin{smallmatrix} 1 & a & b \\ a & 1 & c \\ b & c & 1 \end{smallmatrix}\right)$ is PSD exactly when $|a|, |b|, |c| \leq 1$ and $1 + 2abc \geq a^2 + b^2 + c^2$. Its entries then have a unit-vector realization. Tensoring those vectors proves that every positive integer entrywise power remains PSD.

Checks. Entrywise powers are different from matrix powers. A PSD matrix may be singular; never invert its Gram matrix before verifying independence.

Extended atlas LA-15

Reference notes 4

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2006-D2-P4 [R]; 2010-D1-P5 [R]; 2021-D2-P8 [R]

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T-LA-INVARIANT Adapted bases, centralizers and common eigenspaces

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Rank, dimension and exact linear constraints [T-LA-RANK]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]

Invariant subspaces. A subspace $U \subseteq V$ is invariant under A if $A(U) \subseteq U$. A basis beginning with a basis of U gives the block form $\left(\begin{smallmatrix} B & C \\ 0 & D \end{smallmatrix}\right)$, using columns for images of basis vectors. For a flag $0 = U_0 \subset U_1 \subset \cdots \subset U_r = V$, extend bases successively. If $d_i = \dim U_i - \dim U_{i-1}$, the space of

T-LA-INVARIANT Adapted bases, centralizers and common eigenspaces (continued)

operators preserving the flag has dimension $\sum_{i\le j} d_i d_j$: exactly the blocks below the block diagonal vanish.

Commutants. If $AB = BA$, then every eigenspace of A is B-invariant, since $(A-\lambda I)Bv = B(A-\lambda I)v$. If $A = \operatorname{diag}(\lambda_i)$, then $AB = BA$ is equivalent to $(\lambda_i - \lambda_j)b_{ij} = 0$. Equal-eigenvalue groups of sizes $d_1, \dots, d_s$ therefore give commutant dimension $\sum_i d_i^2$.

Common diagonalization. A pairwise commuting family of operators each diagonalizable over the same field has a common eigenbasis. Split into eigenspaces of a nonscalar member; the others preserve those spaces, and their restrictions remain diagonalizable because their minimal polynomials still divide square-free polynomials. Induct on dimension. Scalar members impose no extra restriction.

Worked count. A flag of dimensions $3, 6, 10$ has block sizes $3, 3, 4$, so its preserving endomorphisms have dimension $3 \cdot 3 + 3 \cdot 6 + 4 \cdot 10 = 67$.

Trap. Simultaneous similarity means using one basis for the entire family. Separate similarities for each member do not automatically combine. Commutativity alone supplies invariant subspaces but not diagonalizability; the identity commutes with a nondiagonalizable Jordan block.

Extended atlas LA-09; LA-10

Reference notes 4

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D2-P4 [R]; 1996-D1-P3 [R]; 1998-D1-P1 [R]; 2001-D1-P5 [R]; 2006-D2-P6 [R]; 2015-D2-P9 [R]

Back to previous view | Theory index | Method index | Strategy index

T-LA-NORM Operator norms and convergence of matrix powers

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]

Induced norm. For a real or complex matrix between Euclidean spaces, $\|A\| = \sup_{\|x\|_2=1} \|Ax\|_2$. The triangle inequality and $\|AB\| \le \|A\|\|B\|$ follow directly from the definition. Unitary changes of basis preserve this norm, while arbitrary similarities need not. Restricting the input or projecting the output orthogonally cannot increase the norm. Every eigenvalue obeys $|\lambda| \le \|A\|$.

Power behavior. For diagonalizable $A = PDP^{-1}$, $\|A^k\| \le \|P\|\|P^{-1}\| \max_i |\lambda_i|^k$. For a Jordan block $J = \lambda I + N$ of size r,

$$J^k = \sum_{j=0}^{\min(k,r-1)} \binom{k}{j} \lambda^{k-j} N^j.$$

Thus eigenvalues of modulus less than one contribute decaying terms, even with Jordan blocks. In finite dimension, A^k converges precisely when all eigenvalues lie inside the open unit disk except possibly 1, whose Jordan blocks must have size one. The limit is the projection onto the 1-eigenspace along the other generalized eigenspaces.

Telescoping. If $\sum_{k\ge 0} \|A^{k+1} - A^k\| < \infty$, the powers form a Cauchy sequence and converge. A bound C/k alone is not summable; extra algebraic structure is needed to improve it. For any convergent power sequence, continuity of multiplication gives $L^2 = L$, since $A^{2k} \to L$ and $(A^k)^2 \to L^2$.

Trap and example. Eigenvalues of modulus at most one do not imply bounded powers: $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)^k = \left(\begin{smallmatrix} 1 & k \\ 0 & 1 \end{smallmatrix}\right)$. For a symmetric matrix the norm is the largest absolute eigenvalue; for general matrices only the inequality in that direction is automatic.

Extended atlas LA-14

Reference notes 4

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2002-D1-P6 [R]; 2003-D1-P3 [R]; 2013-D1-P1 [R]; 2016-D2-P10 [R]

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T-LA-TENSOR Tensor products and operators on matrix spaces

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Gram matrices and positive semidefinite forms [T-LA-GRAM]

Coordinates. For columns $u \in \mathbb{R}^r$ and $v \in \mathbb{R}^s$, the tensor $u \otimes v$ has coordinates $u_i v_j$ indexed by pairs (i, j). A Kronecker product $A \otimes B$ is the block matrix with block $a_{ij}B$. Directly from coordinates,

$$(A \otimes B)(u \otimes v) = Au \otimes Bv, \qquad \langle u \otimes v, x \otimes y \rangle = \langle u, x \rangle \langle v, y \rangle.$$

The same inner-product identity holds for complex Hermitian products. Tensors of basis vectors form a basis of the product space, so diagonalizable A, B give a full eigenbasis for $A \otimes B$ with eigenvalues $\lambda_i \mu_j$.

Sums and multiplicities. The Kronecker sum $A \otimes I + I \otimes B$ has eigenvalues $\lambda_i + \mu_j$. Triangularization proves this without assuming diagonalizability. Coincident sums contribute their multiplicities together. The matrix-space operator $X \mapsto MX + XM^T$ is represented, after choosing an ordering of matrix entries, by such a Kronecker sum. Its dimension is n^2, not n.

PSD product. If $g_{ij} = \langle u_i, u_j \rangle$ and $h_{ij} = \langle v_i, v_j \rangle$, then $g_{ij} h_{ij}$ is the Gram matrix of $u_i \otimes v_i$. Hence the entrywise product of two PSD matrices is PSD. Iteration realizes g_{ij}^k using $u_i^{\otimes k}$, for positive integers k.

Example. Repeatedly adding a two-state operator with eigenvalues $-1, +1$ produces sums $-n+2j$, with multiplicity $\binom{n}{j}$: choose the j positive contributions. Count multiplicities to check the total dimension $\sum_j \binom{n}{j} = 2^n$.

Trap. A tensor product multiplies dimensions and eigenvalues; a direct sum adds dimensions and merely joins the two spectra.

Reference notes 4

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2004-D2-P4 [R]; 2010-D1-P5 [R]; 2017-D2-P8 [R]

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T-LA-FINITEFIELD Linear algebra in characteristic two

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Rank, dimension and exact linear constraints [T-LA-RANK]

Field and rank. The field $\mathbb{F}_2 = \{0, 1\}$ has addition modulo two, so $1 + 1 = 0$ and $-x = x$. Gaussian elimination, rank-nullity, bases and quotient spaces work as over any field. Positivity, length and ordering do not. For columns $x, y \in \mathbb{F}_2^m$, the form $B(x, y) = x^T y$ is bilinear and nondegenerate, but $B(x, x) = 0$ need not imply $x = 0$; for example $(1, 1)^T$ is isotropic.

Alternating forms. A form is alternating if $B(x, x) = 0$ for every x. A matrix represents an alternating form over $\mathbb{F}_2$ exactly when it is symmetric with zero diagonal. The cross terms in $x^T Ax$ cancel in pairs. Let $\mathbf{1}$ be the all-ones vector. Because $x_i^2 = x_i$,

$$x^T x = x^T \mathbf{1}.$$

If $A = A^T$, then $x^T A^{2k} x = (A^k x)^T (A^k x)$; if its diagonal is zero, $x^T A^{2k+1} x = (A^k x)^T A (A^k x) = 0$. These identities permit induction by halving an exponent.

Affine and quotient viewpoint. Translating an even-sized collection leaves its sum unchanged, since an even multiple of any vector is zero. The affine span becomes a linear space after translation by one of its points. Quotienting by a nonzero vector w identifies x with $x + w$ and reduces dimension by one; a basis of the quotient lifts to a basis after adjoining w.

Example. To choose one vector from each pair (a_i, b_i) with a prescribed sum, encode the choice by $\varepsilon_i \in \mathbb{F}_2$: $\sum_i a_i + \sum_i \varepsilon_i (a_i + b_i)$. If the pair sums form a basis, every target sum has exactly one such choice.

Reference notes 4

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2010-D2-P4 [R]; 2011-D1-P5 [R]

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T-LA-HILBERT Completeness, barycentres and equidistant sets

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Inner products, orthogonality and polarization [T-LA-INNER]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]

Setting. A real Hilbert space is a complete real inner-product space: every norm-Cauchy sequence converges in the space. Its closed finite-dimensional subspaces have orthogonal projections. In an infinite-dimensional Hilbert space, every finite-dimensional subspace has a nonzero orthogonal complement.

Equidistant identity. Suppose distinct points $x_1, \dots, x_r$ have common pairwise distance $d > 0$. For real coefficients with $\sum_i a_i = 0$, expansion of squared norms gives

$$\left\| \sum_i a_i x_i \right\|^2 = -\frac{1}{2} \sum_{i,j} a_i a_j \|x_i - x_j\|^2 = \frac{d^2}{2} \sum_i a_i^2.$$

The zero-sum condition removes the dependence on the chosen origin. Polarization yields all corresponding scalar products.

Finite centre. For $\bar{x} = r^{-1} \sum_i x_i$, one gets $\|x_i - \bar{x}\|^2 = d^2(r-1)/(2r)$ and $\langle x_i - \bar{x}, x_j - \bar{x} \rangle = -d^2/(2r)$ for $i \neq j$. A perpendicular displacement z with $\|z\|^2 = d^2/(2r)$ therefore makes the vectors $x_i - (\bar{x} + z)$ pairwise orthogonal, each of norm $d/\sqrt{2}$.

Infinite centre. For a sequence of distinct equidistant points, the averages $y_r = r^{-1} \sum_{i=1}^{r} x_i$ satisfy $\|y_s - y_r\|^2 = (d^2/2)(1/r - 1/s)$ when $s > r$. They are Cauchy, hence converge. Adjoining finitely many further points changes none of these limits. Testing one or two added points extends the norm and pairwise-orthogonality conclusions even to an uncountable equidistant set.

Checks. The empty and singleton cases need separate harmless choices. The assumptions $d > 0$, completeness, and available perpendicular directions are essential to the corresponding steps; finite-dimensional ambient space may not have room for the required displacement.

Reference notes 16

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2008-D2-P6 [R]

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T-LA-PROJECTIVE Fractional-linear recurrences and matrix powers

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]

Homogeneous coordinates. The real projective line consists of one-dimensional subspaces of $\mathbb{R}^2$. Write its points as $[u : v]$, with $[u : v] = [cu : cv]$ for $c \neq 0$; finite x corresponds to $[x : 1]$ and infinity to $[1 : 0]$. An invertible matrix $M = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ acts by $[u : v] \mapsto [au + bv : cu + dv]$, which is $x \mapsto (ax + b)/(cx + d)$ whenever the denominator is nonzero.

Iteration. Composition corresponds to matrix multiplication, and a nonzero scalar multiple of M gives the same projective map. Thus an n-periodic projective point is exactly a real eigenline of M^n. This converts a rational cyclic recurrence into a two-dimensional spectral question.

Example. For $\varphi(x) = 1 - a/x$ with $a > 1/4$, take $M = \left(\begin{smallmatrix} 1 & -a \\ 1 & 0 \end{smallmatrix}\right)$. Its two distinct eigenvalues are $\sqrt{a}\, e^{\pm i\theta}$, where $0 < \theta < \pi/2$ and $\cos\theta = 1/(2\sqrt{a})$. Their nth powers are real precisely when $n\theta \in \pi\mathbb{Z}$. They are then equal, so diagonalizability of M gives $M^n = \lambda^n I$. Otherwise M^n has no real eigenline and hence no real periodic projective point of period dividing n.

Domain check. A periodic orbit in the projective line is not automatically an admissible real solution: discard starting points whose iterates hit zero, infinity, or any forbidden denominator. When the iterate is the identity, there are only finitely many such excluded preimages, leaving permissible choices. Conversely, any finite real cyclic solution supplies a legitimate eigenline, proving necessity.

Reference notes 18

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2012-D2-P4 [R]

T-LA-PROJECTIVE Fractional-linear recurrences and matrix powers (continued)

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T-LA-SERIES Matrix power series and transfer between AB and BA

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Operator norms and convergence of matrix powers [T-LA-NORM]; Determinants, block identities and inverses [T-LA-DETERMINANT]; Nilpotents, projections and involutions [T-LA-NILPOTENT]

Convergence. If the scalar power series $f(z)=\sum_{j\ge 0} c_j z^j$ converges absolutely for every complex z, define $f(A)=\sum_j c_j A^j$. Submultiplicativity gives $\sum_j \|c_j A^j\| \le \sum_j |c_j|\|A\|^j < \infty$. Thus rearrangement of absolutely convergent series and multiplication of two such series are valid. In particular, $e^A = \sum_{j\ge 0} A^j/j!$ exists for every square matrix.

Product transfer. If A is $m\times n$ and B is $n\times m$, block elimination in $\left(\begin{smallmatrix} I_m & A \\ -B & I_n \end{smallmatrix}\right)$ gives

$$\det(I_m + AB) = \det(I_n + BA).$$

Substituting a scalar multiple and clearing powers of that scalar compares characteristic polynomials. For square matrices of the same size, AB and BA have identical characteristic polynomials even if neither factor is invertible. With different sizes, zero multiplicities differ by the dimension difference.

Series transfer. For equal-size square A, B, set $C=\sum_{j\ge 1} c_j (BA)^{j-1}B$. The identities $(AB)^j = A(BA)^{j-1}B$ and $B(AB)^j=(BA)^jB$ yield $f(AB)=c_0 I + AC$ and $f(BA)=c_0 I + CA$. Consequently these matrices have the same characteristic polynomial. Their nilpotency is equivalent because a square matrix is nilpotent exactly when its characteristic polynomial is a pure power of t.

Example and trap. For a polynomial p, $f(z)=p(e^z)$ is entire, so this comparison applies to $p(e^{AB})$ and $p(e^{BA})$. It does not assert equality or similarity of those matrices. Also $e^{A+B}=e^Ae^B$ requires $AB=BA$ as a sufficient hypothesis; the formula is false for arbitrary matrices.

Extended atlas LA-13; LA-03

Reference notes 4

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2000-D2-P6 [R]

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T-LA-PERTURBATION Evaluation maps and rank under perturbation

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Rank, dimension and exact linear constraints [T-LA-RANK]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]

Polynomial minors. If the entries of a matrix $F(t)$ are polynomials in one real variable and a particular $r\times r$ minor is nonzero at $t=0$, that minor is a nonzero polynomial. It vanishes at only finitely many real values. Therefore $\operatorname{rank} F(t)\ge r$ outside a finite exceptional set. Several such requirements can be imposed simultaneously by avoiding the finite union of their exceptional sets.

Evaluation. Let $\mathcal{T}$ be an n-dimensional space of linear maps $V\to W$. For $x\in V$, define $E_x:\mathcal{T}\to W$ by $E_x(T)=Tx$. The assertion that some nonzero $T\in\mathcal{T}$ annihilates every chosen x separately is exactly $\ker E_x \ne \{0\}$ for every x. The annihilating operator may depend on x; this is different from a common kernel vector for all T.

Local rank argument. Suppose $\operatorname{rank} E_{x_0} = n-1$ and all evaluations have rank at most $n-1$. A nonzero $(n-1)$-minor stays nonzero for small t, so E_{x_0+tx} has constant rank $n-1$ there. Normalize one coordinate of a nonzero kernel vector $T(t)$ to be constant. Solving the remaining linear equations by that invertible minor makes $T(t)$ rational, hence differentiable near zero. Differentiating $T(t)(x_0+tx)=0$ gives

$$T(0)x + T'(0)x_0 = 0.$$

Thus $T(0)x$ lies in the evaluation image $\mathcal{T}x_0$.

T-LA-PERTURBATION Evaluation maps and rank under perturbation (continued)

Checks. The normalization requires a nonzero coordinate at $t = 0$. Rank can drop at exceptional parameters, so arbitrary substitution is unsafe. The derivative identity differentiates both the operator and its argument; interchanging these two terms gives an incorrect proof.
Reference notes 17

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2009-D2-P5 [R]

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Analysis, limits and probability

T-AN-CONTINUITY Continuity, compactness, and order

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Recall. A function is continuous at x if $x_n \to x$ implies $f(x_n) \to f(x)$. On a nonempty compact subset $K \subset \mathbb{R}^d$, a continuous real function is bounded, attains both extrema, and is uniformly continuous. In Euclidean space, compact means closed and bounded. Every sequence in K has a convergent subsequence with limit in K.

Existence tools. A continuous image of an interval is an interval (the intermediate value theorem). More generally, a continuous image of a connected set is connected. A convex set is path connected: join two points by their line segment and apply the intermediate value theorem to the restriction along that segment. Thus a continuous real function on a nonempty compact convex constraint set takes every value between its attained minimum and maximum.

Proof core. To transfer $f \le g$ from a dense set to its closure, approximate each point by a sequence in the dense set and pass to the limit; both functions must be continuous there. A monotone function instead has one-sided limits, but may jump. A monotone surjection $\mathbb{R} \to \mathbb{R}$ cannot jump because the skipped interval would be missing from its range.

Example and trap. For a continuous function on an unbounded domain, first exclude the exterior of a large ball from competing with a known value, then use compactness inside the ball. A stationary point alone need not be an extremum: $x^2 - y^2$ has a saddle at the origin. Closedness also matters: x has no maximum on $(0, 1)$.

Extended atlas RA-01; RA-02; RA-03

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P5 [R]; 1994-D2-P1 [R]; 1994-D2-P2 [R]; 1995-D1-P4 [R]; 1995-D1-P6 [R]; 1995-D2-P5 [R]; 1996-D1-P6 [R]; 1996-D2-P1 [R]; 1996-D2-P4 [R]; 1997-D2-P6 [R]; 1998-D1-P4 [R]; 1998-D2-P5 [R]; 1999-D1-P1 [R]; 2000-D2-P2 [R]; 2002-D1-P4 [R]; 2003-D2-P3 [R]; 2003-D2-P5 [R]; 2004-D1-P3 [R]; 2004-D2-P4 [R]; 2005-D2-P5 [R]; 2006-D1-P1 [R]; 2006-D2-P2 [R]; 2007-D2-P1 [R]; 2007-D2-P3 [R]; 2008-D1-P1 [R]; 2009-D1-P1 [R]; 2010-D2-P1 [R]; 2011-D1-P1 [R]; 2012-D1-P4 [R]; 2014-D1-P3 [R]; 2016-D1-P1 [R]; 2021-D1-P4 [R]; 2023-D1-P1 [R]

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T-AN-DERIVATIVE Rolle, mean values, and the Darboux property

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Prerequisites Continuity, compactness, and order [T-AN-CONTINUITY]

Hypotheses. If f is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then $f'(c) = 0$ for some $c \in (a, b)$. Subtracting the chord gives the mean value theorem,

$$f(b) - f(a) = f'(c)(b - a).$$

For two such functions f, g, Cauchy's form is

$$[f(b) - f(a)]g'(c) = [g(b) - g(a)]f'(c).$$

Divide only when the denominator is nonzero; $g' \neq 0$ on the interval is a convenient sufficient condition. The derivative sign controls monotonicity by the mean value theorem.

Zeros and interpolation. Repeated Rolle applications give at least $r - 1$ distinct derivative zeros between r distinct zeros. If $f \in C^n$ on an interval containing the nodes and an interpolating polynomial p of degree at most n matches f at $n+1$ distinct nodes, its leading coefficient is $f^{(n)}(\xi)/n!$. Hermite interpolation also counts prescribed derivative matches: write out the intervening Rolle points to avoid counting an endpoint twice.

Derivatives need not be continuous. They still have the intermediate value property. To prove it, subtract tx from a differentiable function whose endpoint derivatives lie on opposite sides of t; an interior extremum gives derivative t. This is Darboux's theorem. For continuous u and differentiable v, write $u - v' = (\int_a^x u - v)'$ before applying it.

T-AN-DERIVATIVE Rolle, mean values, and the Darboux property (continued)

Example and trap. To force $f' - f = 0$, apply Rolle to $e^{-x}f(x)$, rather than to f itself. An infinite set of zeros accumulating at an interior point forces both f and f' to vanish there, by the difference quotient. Accumulation at an endpoint does not give an interior derivative zero.
Extended atlas RA-04

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P2 [R]; 1994-D1-P6 [R]; 1994-D2-P2 [R]; 1994-D2-P3 [R]; 1995-D1-P3 [R]; 1995-D1-P4 [R]; 1995-D2-P2 [R]; 1997-D2-P1 [R]; 1997-D2-P3 [R]; 1998-D1-P4 [R]; 1998-D2-P2 [R]; 1998-D2-P6 [R]; 1999-D1-P1 [R]; 1999-D1-P6 [R]; 2000-D1-P6 [R]; 2000-D2-P3 [R]; 2000-D2-P4 [R]; 2000-D2-P5 [R]; 2001-D1-P6 [R]; 2002-D1-P2 [R]; 2002-D1-P5 [R]; 2002-D2-P4 [R]; 2004-D2-P5 [R]; 2005-D1-P3 [R]; 2005-D1-P5 [R]; 2005-D2-P4 [R]; 2006-D1-P1 [R]; 2006-D1-P5 [R]; 2006-D1-P6 [R]; 2006-D2-P3 [R]; 2007-D2-P6 [R]; 2009-D2-P2 [R]; 2009-D2-P5 [R]; 2010-D1-P1 [R]; 2012-D1-P4 [R]; 2013-D1-P2 [R]; 2013-D1-P4 [R]; 2014-D1-P3 [R]; 2014-D2-P3 [R]; 2016-D1-P1 [R]; 2017-D1-P2 [R]; 2017-D2-P9 [R]; 2017-D2-P10 [R]; 2018-D1-P4 [R]; 2019-D1-P3 [R]; 2019-D1-P4 [R]; 2019-D2-P6 [R]; 2020-D1-P4 [R]; 2020-D2-P5 [R]; 2023-D1-P1 [R]; 2023-D2-P7 [R]

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T-AN-TAYLOR Taylor remainders and local scale

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Prerequisites Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Recall. If $f \in C^m$ near a, then

$$f(a+h) = \sum_{j=0}^{m} \frac{f^{(j)}(a)}{j!} h^j + o(|h|^m).$$

If $f^{(m+1)}$ is continuous, the remainder after degree m equals $f^{(m+1)}(\xi)h^{m+1}/(m+1)!$ for an intermediate ξ. In particular a bounded next derivative gives an $O(|h|^{m+1})$ remainder. Here $r(h) = o(|h|^m)$ means $r(h)/|h|^m \to 0$; $O(|h|^m)$ means this quotient is bounded near zero.
Proof use. Expand only to the first order at which numerator and denominator cease cancelling. If $A(h) = ah^r + o(h^r)$ and $B(h) = bh^r + o(h^r)$ with $b \neq 0$, then $A/B \to a/b$. For a bounded quotient it is enough that the numerator is $O(|h|^r)$ and the denominator has absolute value at least $c|h|^r$ locally.
Example. If $f(0) = f'(0) = 0$, $f''(0) = 2c > 0$, and $f \in C^3$, then $f = cx^2 + O(|x|^3)$, $f' = 2cx + O(x^2)$, $f'' = 2c + O(|x|)$. Hence $(f')^2 - 2ff'' = O(|x|^3)$, while $2(f')^2\sqrt{f}$ has size $|x|^3$. The absolute value in $\sqrt{x^2} = |x|$ is essential.
Trap. Do not differentiate an unspecified little-o remainder. Expand each derivative using its own smoothness. A C^2 assumption gives only $o(x^2)$ for f, not the cubic bound above. Local estimates near every equality point must be combined with compactness away from those points to obtain a global constant.
Extended atlas RA-05

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1995-D1-P6 [R]; 1996-D1-P5 [R]; 1996-D2-P5 [R]; 1997-D2-P1 [R]; 2001-D2-P2 [R]

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T-AN-INTEGRAL Primitives, substitution, and integration by parts

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Prerequisites Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Recall. For continuous f on $[a,b]$, $F(x) = \int_a^x f(t)\,dt$ satisfies $F' = f$. If $u, v \in C^1[a,b]$, then

$$\int_a^b uv' = [uv]_a^b - \int_a^b u'v.$$

T-AN-INTEGRAL Primitives, substitution, and integration by parts (continued)

For $\phi \in C^1[\alpha, \beta]$ and continuous f on an interval containing its image,

$$\int_\alpha^\beta f(\phi(t))\phi'(t)\,dt = F(\phi(\beta)) - F(\phi(\alpha)).$$

This form tracks orientation automatically. A symmetric substitution such as $x \mapsto 1 - x$ can pair two unequal-looking integrands without changing their integrals.

Weighted identities. If $f \geq 0$ and $F(x) = \int_a^x f$, then $F \geq 0$ and $\int_a^b fF = F(b)^2/2$. A tail primitive $G(x) = -\int_x^b f$ instead satisfies $G' = f$ and $G \leq 0$. Choose the primitive whose boundary term matches the prescribed endpoint data.

Parameter rule. If H and $\partial_s H$ are continuous near the relevant compact region and a, b are differentiable, then

$$\frac{d}{ds}\int_{a(s)}^{b(s)} H(s,t)\,dt = H(s,b(s))b'(s) - H(s,a(s))a'(s) + \int_{a(s)}^{b(s)} \partial_s H(s,t)\,dt.$$

For improper integrals, additional local domination is needed before this interchange.

Example and trap. Since $\sin x/x = \int_0^1 \cos(tx)\,dt$, its nth derivative has magnitude at most $\int_0^1 t^n\,dt = 1/(n+1)$; for $x > 0$ the bound is strict because the trigonometric factor has magnitude below one on a set of positive length. Always include boundary terms, and never replace an improper integral by an undefined endpoint value.

Extended atlas RA-08

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P5 [R]; 1995-D1-P2 [R]; 1995-D1-P4 [R]; 1995-D2-P5 [R]; 1996-D1-P2 [R]; 1996-D1-P5 [R]; 1996-D2-P5 [R]; 1997-D1-P1 [R]; 1998-D1-P3 [R]; 1998-D1-P6 [R]; 1999-D1-P6 [R]; 2000-D1-P4 [R]; 2000-D1-P6 [R]; 2000-D2-P4 [R]; 2001-D1-P3 [R]; 2002-D2-P4 [R]; 2002-D2-P6 [R]; 2003-D1-P5 [R]; 2003-D2-P2 [R]; 2004-D2-P2 [R]; 2004-D2-P5 [R]; 2005-D1-P3 [R]; 2005-D1-P5 [R]; 2007-D2-P6 [R]; 2009-D2-P2 [R]; 2010-D1-P1 [R]; 2010-D1-P2 [R]; 2014-D2-P3 [R]; 2015-D2-P7 [R]; 2015-D2-P10 [R]; 2016-D2-P7 [R]; 2017-D1-P2 [R]; 2017-D2-P6 [R]; 2017-D2-P9 [R]; 2017-D2-P10 [R]; 2019-D1-P3 [R]; 2019-D2-P6 [R]; 2020-D2-P8 [R]; 2021-D1-P3 [R]; 2022-D1-P1 [R]; 2022-D2-P8 [R]; 2023-D2-P9 [R]; 2024-D1-P2 [R]

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T-AN-FUBINI Nonnegative sums, Fubini, and integral kernels

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Prerequisites Primitives, substitution, and integration by parts [T-AN-INTEGRAL]

Recall. For a nonnegative measurable function on a product of sigma-finite measure spaces, Tonelli's theorem permits exchanging integrals; their common value may be $+\infty$. For an absolutely integrable function, Fubini's theorem permits the exchange and gives finite iterated integrals. Counting measure yields the same statements for double series. With signed terms, convergence of just one iterated sum does not suffice.

Triangle identity. For continuous f on $[0,1]$, integrating over $0 \leq x \leq t \leq 1$ gives

$$\int_0^1 \int_x^1 f(t)\,dt\,dx = \int_0^1 tf(t)\,dt.$$

More generally, r repeated integrations of f from zero give the kernel

$$\int_{0<t_r<\cdots<t_1<x} f(t_r)\,dt_r\cdots dt_1 = \frac{1}{(r-1)!}\int_0^x (x-t)^{r-1}f(t)\,dt.$$

The coefficient is the volume of an ordered simplex; forgetting its factorial changes the answer.

Series example. If $x_n \geq 0$, then

$$\sum_{k\geq 1}\frac{1}{k^2}\sum_{n\leq k} x_n = \sum_{n\geq 1} x_n \sum_{k\geq n}\frac{1}{k^2}.$$

T-AN-FUBINI Nonnegative sums, Fubini, and integral kernels (continued)

One may establish a finite upper bound only after this exchange: nonnegativity already justifies it. For a two-variable change of variables, rewrite the integration region as well as the integrand and include the absolute Jacobian. Prove statements first on truncated regions if an integrand is singular at a boundary, then pass monotonically to the limit.

Extended atlas RA-09

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1995-D1-P2 [R]; 1996-D2-P6 [R]; 1999-D1-P2 [R]; 2000-D1-P4 [R]; 2003-D1-P5 [R]; 2004-D2-P5 [R]; 2010-D1-P2 [R]; 2016-D2-P6 [R]; 2020-D2-P8 [R]

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T-AN-CONVERGENCE Sequences, subsequences, and recurrence limits

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Prerequisites Continuity, compactness, and order [T-AN-CONTINUITY]

Recall. A real sequence converges exactly when it is Cauchy. A nondecreasing sequence has limit equal to its supremum, possibly $+\infty$; bounded monotone sequences therefore converge. For a bounded sequence, the least and greatest subsequential limits are $\liminf a_n$ and $\limsup a_n$. Equality of these two numbers proves convergence.

Recurrences. For $x_{n+1} = F(x_n)$, first prove that the starting value and all iterates belong to an invariant interval where the formula is defined. Then establish monotonicity, a contraction estimate, or control of all cluster points. Only after convergence is known may continuity yield $\ell = F(\ell)$. A fixed-point equation identifies candidates, not convergence.

Error estimate. If $|x_{n+1} - \ell| \le q|x_n - \ell|$ with $0 \le q < 1$, induction gives $|x_n - \ell| \le q^n|x_0 - \ell|$. For growth recurrences with $r > 0$, divide by the expected scale: $a_{n+1} > ra_n$ makes a_n/r^{n-1} increasing. Conversely, a prescribed increasing sequence approaching α constructs examples with that normalized limit.

Trap. The condition $x_{n+1} - x_n \to 0$ alone does not imply convergence: $\sin(\log n)$ is a bounded counterexample. In continuous interval iteration, a separate crossing argument can turn vanishing increments into convergence: an orbit cannot repeatedly cross an interval that both its step size and its nonzero displacement forbid it to enter. Also, convergence of $|x_n|$ does not imply convergence of x_n; show eventual sign stability or express the next iterate continuously in terms of $|x_n|$.

Extended atlas RA-03; RA-17

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D2-P6 [R]; 1995-D1-P3 [R]; 1995-D2-P2 [R]; 1996-D1-P4 [R]; 1996-D2-P1 [R]; 1997-D1-P1 [R]; 1997-D1-P2 [R]; 1997-D1-P4 [R]; 1997-D1-P5 [R]; 1998-D1-P3 [R]; 1999-D1-P3 [R]; 1999-D1-P4 [R]; 2000-D1-P6 [R]; 2001-D2-P2 [R]; 2001-D2-P5 [R]; 2002-D1-P6 [R]; 2002-D2-P3 [R]; 2003-D1-P1 [R]; 2003-D1-P3 [R]; 2003-D1-P5 [R]; 2004-D1-P6 [R]; 2006-D1-P5 [R]; 2007-D2-P3 [R]; 2008-D2-P6 [R]; 2010-D1-P3 [R]; 2010-D2-P1 [R]; 2011-D2-P1 [R]; 2012-D2-P2 [R]; 2013-D1-P5 [R]; 2013-D2-P4 [R]; 2014-D1-P2 [R]; 2015-D1-P3 [R]; 2015-D2-P10 [R]; 2016-D1-P5 [R]; 2017-D2-P6 [R]; 2017-D2-P9 [R]; 2017-D2-P10 [R]; 2018-D2-P7 [R]; 2019-D1-P4 [R]; 2019-D2-P7 [R]; 2021-D1-P3 [R]; 2024-D2-P8 [R]

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T-AN-SERIES Series, finite cancellation, and summation by parts

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Prerequisites Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]

Recall. The series $\sum a_n$ means the limit of its partial sums. Convergence implies $a_n \to 0$, but the converse fails. For $a_n \ge 0$, bounded partial sums are equivalent to convergence. Absolute convergence permits arbitrary rearrangement; conditional convergence does not. Grouping consecutive terms of a convergent series preserves its sum, whereas proving convergence of the grouped series also requires control of incomplete groups.

T-AN-SERIES Series, finite cancellation, and summation by parts (continued)

Abel summation. With $A_k = \sum_{j=1}^{k} a_j$, the finite identity is

$$\sum_{k=1}^{N} a_k b_k = A_N b_N + \sum_{k=1}^{N-1} A_k (b_k - b_{k+1}).$$

Bounded A_k and monotone $b_k \to 0$ imply convergence of $\sum a_k b_k$ (Dirichlet). More generally, finite total variation $\sum |b_{k+1} - b_k| < \infty$ together with $b_k \to 0$ suffices. Pairing an alternating series with nonmonotone amplitudes is valid when the paired differences are absolutely summable and the omitted last term tends to zero.
Telescoping. Work at a finite cutoff: $\sum_{k=1}^{N} (u_k - u_{k+1}) = u_1 - u_{N+1}$. For positive factors, likewise cancel a finite product before passing to its limit. A finite nonzero infinite product can alternatively be studied through convergence of the sum of logarithms.
Example and trap. $\sum_{k \geq n} 1/[k(k+1)] = 1/n$ is a useful tail bound after factorization. Never split a convergent combination into divergent component series; keep matching cutoffs until cancellation is complete. A rearrangement must be checked at every partial sum, not merely at block endpoints.
Extended atlas RA-11

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1995-D2-P2 [R]; 1995-D2-P5 [R]; 1996-D2-P6 [R]; 1997-D1-P2 [R]; 1997-D2-P3 [R]; 1998-D2-P6 [R]; 1999-D1-P2 [R]; 2000-D1-P4 [R]; 2003-D2-P6 [R]; 2010-D1-P2 [R]; 2011-D2-P3 [R]; 2012-D2-P2 [R]; 2015-D2-P6 [R]; 2016-D2-P6 [R]; 2018-D1-P1 [R]; 2018-D2-P10 [R]; 2019-D1-P1 [R]; 2020-D2-P8 [R]

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T-AN-POWER Power series and generating functions

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Prerequisites Series, finite cancellation, and summation by parts [T-AN-SERIES]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Recall. For $F(z) = \sum_{n \geq 0} a_n z^n$, its convergence radius satisfies $R^{-1} = \limsup |a_n|^{1/n}$, with the usual conventions for zero and infinity. Inside $|z| < R$, termwise differentiation and integration are valid, uniformly on every smaller closed disk. Products are Cauchy convolutions whenever the series are absolutely convergent.
Recurrence translation. Multiplication by z shifts indices, differentiation multiplies a coefficient by its index, and a convolution becomes a product. For example,

$$(n+1)a_{n+1} = \sum_{k=0}^{n} a_k b_{n-k} \quad \Longrightarrow \quad F' = FB,$$

where $B(z) = \sum b_n z^n$. Prove a coefficient bound first, or explicitly work formally and later justify the analytic evaluation. The initial coefficient supplies the integration constant when solving the differential equation.
Endpoints. If $\sum a_n$ converges, Abel's theorem gives $\lim_{r \uparrow 1} \sum a_n r^n = \sum a_n$. Absolute summability also gives uniform convergence on $[0, 1]$ by direct tail bounds. These are separate justifications: a formula proved for $r < 1$ is not automatically valid at $r = 1$.
Example and trap. A denominator $(4k+1)(4k+2)(4k+3)(4k+4)$ suggests four integrations of $\sum x^{4k} = 1/(1-x^4)$. The derivative series has an endpoint singularity even when the integrated series converges there. Initial values and endpoint limits must therefore be taken in the correct order. Coefficientwise formal identities do not license substituting a point outside the convergence disk.
Extended atlas RA-17

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2003-D2-P6 [R]; 2010-D1-P2 [R]

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T-AN-RIEMANN Riemann sums, periodic averaging, and tails

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Prerequisites Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]

Recall. For continuous f on $[a,b]$, a partition with mesh tending to zero satisfies $\sum f(\xi_j)\Delta x_j \to \int_a^b f$. An estimate using the modulus of continuity $\omega_f(\delta) = \sup_{|x-y|\le\delta} |f(x)-f(y)|$ bounds the error by $(b-a)\omega_f(\text{mesh})$.

Infinite and singular ranges. A sum $h\sum_{k\ge1} f(kh)$ is not justified by finite-interval Riemann integration alone. Prove convergence on a fixed compact interval and control the remainder uniformly in h. If $f \ge 0$ is eventually decreasing and integrable, the tail rectangles are trapped by neighboring tail integrals. Near a singular endpoint, first cut off $(0,\delta)$ and bound its sum and integral separately. For instance $x\log x$ extends continuously by zero, whereas $\log x$ needs an improper-integral comparison.

Periodic averaging. For continuous f on $[0,b]$ and continuous b-periodic g,

$$\int_0^b f(x)g(nx)\,dx \longrightarrow \frac{1}{b}\left(\int_0^b f\right)\left(\int_0^b g\right).$$

Partition into n periods, freeze f at one point in each cell, and bound the error by $\omega_f(b/n)\int_0^b |g|$. The factor $1/b$ is necessary.

Example. Setting $h = x^{-1/2}$ turns $\sum_{n\ge1} nx/(n^2+x)^2$ into $h\sum_{n\ge1} f(nh)$ with $f(t) = t/(1+t^2)^2$. The limiting integral is $1/2$; its infinite tail must still be justified.

Extended atlas RA-10; RA-12

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P5 [R]; 1996-D2-P5 [R]; 1997-D1-P1 [R]; 2001-D1-P3 [R]; 2018-D2-P10 [R]; 2024-D1-P2 [R]

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T-AN-ASYMPTOTICS Scale separation and limit comparisons

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Prerequisites Taylor remainders and local scale [T-AN-TAYLOR]; Riemann sums, periodic averaging, and tails [T-AN-RIEMANN]

Recall. Write $a_n \sim b_n$ when $a_n/b_n \to 1$. It is legitimate only where $b_n \neq 0$. Squeezing and $\liminf$ / $\limsup$ bounds often avoid assumptions that a desired limit already exists. Logarithms grow more slowly than every positive power: $(\log x)^m/x^\delta \to 0$ for fixed $m,\delta>0$.

Derivative comparison. A useful l'Hospital form is: if differentiable F,G satisfy $G'>0$, $G(x)\to+\infty$, and $F'/G' \to L$ as $x\to+\infty$, then $F/G\to L$ (also for extended real L). No numerator divergence is needed in this form. Prove it by bounding F' between $(L\pm\varepsilon)G'$ eventually and applying the mean value theorem; fixed initial constants vanish after division by G. A $0/0$ endpoint form follows from Cauchy's mean value theorem.

Concentration scale. For f continuous on $[0,1]$, continuously differentiable near zero, with $0\le f\le 1$, unique maximum $f(0)=1$, and $f'(0)=-c<0$, one has

$$n\int_0^1 f(x)^n\,dx \longrightarrow 1/c.$$

Indeed, on a small interval compare f with $1-(c\pm\varepsilon)x$; away from zero, $f\le q<1$ so $nq^n\to0$. A quadratic rather than linear drop has scale $n^{-1/2}$, so multiplying the integral by n generally diverges.

Trap. A pointwise approximation on a moving domain is insufficient. Name the central range and tails, prove uniform error bounds there, and take auxiliary limits in the stated order. Likewise, a convergent subsequence determines a candidate limit but not convergence of the full sequence.

Extended atlas RA-13

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.

T-AN-ASYMPTOTICS Scale separation and limit comparisons (continued)

1994-D2-P6 [R]; 1996-D1-P5 [R]; 2001-D1-P3 [R]; 2001-D1-P6 [R]; 2003-D2-P2 [R]; 2004-D1-P3 [R]; 2005-D1-P5 [R]; 2014-D1-P2 [R]; 2015-D2-P7 [R]; 2017-D2-P6 [R]; 2018-D2-P10 [R]; 2020-D2-P8 [R]; 2024-D1-P2 [R]

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T-AN-CONVEXITY Convexity, Jensen, and supporting lines

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Prerequisites Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]

Definition. On a convex set C, f is convex when

$$f((1-t)x+ty) \le (1-t)f(x)+tf(y) \quad (0 \le t \le 1).$$

On an interval, a differentiable f is convex exactly when f' is nondecreasing; if $f \in C^2$, this is equivalent to $f'' \ge 0$. Concavity reverses the inequalities. For $x < y < z$, convexity orders the secant slopes:

$$\frac{f(y)-f(x)}{y-x} \le \frac{f(z)-f(x)}{z-x} \le \frac{f(z)-f(y)}{z-y}.$$

Three forms. A convex graph lies below each chord and above each supporting tangent. For differentiable f on C,

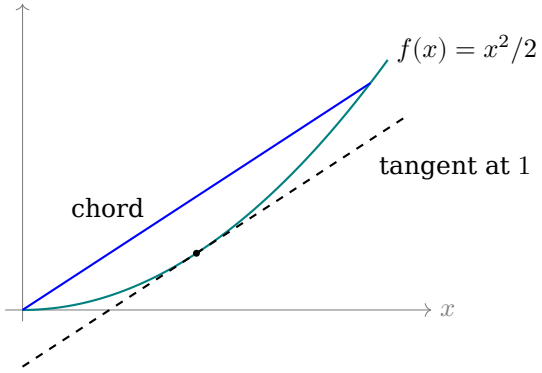
$$f(y) \ge f(x) + \langle \nabla f(x), y-x \rangle.$$

Jensen's inequality is $f(\sum \lambda_i x_i) \le \sum \lambda_i f(x_i)$ for $\lambda_i \ge 0$, $\sum \lambda_i = 1$. It follows from the two-point definition by induction; for strictly convex f, equality with positive weights requires all x_i equal.

Example. Concavity of sine on $[0,\pi/2]$ gives $2x/\pi \le \sin x$ from its chord and $\sum \sin x_i/n \le \sin(\sum x_i/n)$ from Jensen. These different forms bound opposite sides of a constrained angle sum.

Global rigidity and traps. A positive concave function on all of $\mathbb{R}$ is constant: a nonzero supporting slope would force negative values in one direction. The conclusion fails on a bounded interval. Check the entire interval of convexity before applying Jensen, and retain the inequality sign in the tangent estimate; touching at one point does not make the function equal to its tangent.

Continuity. Every finite convex function on an open interval is continuous there. On a compact subinterval of its domain, choose one point on either side; the ordered secant slopes bound every intermediate secant slope in absolute value. This gives a local Lipschitz bound. Endpoint continuity is not automatic from convexity alone.



The chord is above the convex graph on its endpoint interval; the tangent is below it.

Extended atlas RA-07

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.

1994-D1-P6 [R]; 1995-D1-P3 [R]; 1995-D1-P6 [R]; 1996-D1-P5 [R]; 1997-D1-P5 [R]; 2002-D2-P6 [R]; 2004-D1-P3 [R]; 2004-D2-P2 [R]; 2004-D2-P5 [R]; 2006-D2-P3 [R]; 2009-D1-P4 [R]; 2017-D2-P9 [R]; 2019-D1-P3 [R]; 2020-D1-P4 [R]; 2020-D2-P5 [R]

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T-AN-SMOOTH Lipschitz derivatives and convex smoothness

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Prerequisites Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Inner products, orthogonality and polarization [T-LA-INNER]

Hypothesis. Let $f:\mathbb{R}^d\to\mathbb{R}$ be differentiable with L-Lipschitz gradient, $L>0$: $\|\nabla f(x)-\nabla f(y)\|\le L\|x-y\|$. Integrating along the segment from x to y and applying Cauchy–Schwarz gives the descent bound

$$\left|f(y)-f(x)-\langle\nabla f(x),y-x\rangle\right|\le\tfrac{L}{2}\|y-x\|^2.$$

This uses gradient continuity, not existence of a Hessian. Taking $y=x-\nabla f(x)/L$ shows that for $f\ge 0$, $\|\nabla f(x)\|^2\le 2Lf(x)$. If $f>0$ everywhere, the inequality is strict because $f(y)>0$.

Convex strengthening. If f is also convex, subtract the tangent at x: $g(z)=f(z)-f(x)-\langle\nabla f(x),z-x\rangle\ge 0$. The descent bound at y gives

$$f(y)-f(x)-\langle\nabla f(x),y-x\rangle\ge\frac{\|\nabla f(y)-\nabla f(x)\|^2}{2L}.$$

Add the same inequality with x,y exchanged to obtain

$$\|\nabla f(y)-\nabla f(x)\|^2\le L\langle\nabla f(y)-\nabla f(x),y-x\rangle.$$

Proof obligations. The descent step subtracts the vector $\nabla g(y)/L$, not its scalar norm. The chosen point must belong to the domain; the global hypothesis above ensures this. The constant L is positive before division. Convexity is essential for the last inequality, but is unnecessary for the positive-function gradient bound. Equality in the nonnegative case is possible, for example for a suitable quadratic; strict positivity rules that case out at the descent point.

Extended atlas RA-19

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2002-D2-P6 [R]; 2005-D2-P5 [R]; 2017-D1-P2 [R]

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T-AN-INEQUALITIES Cauchy–Schwarz, AM–GM, and sharpness

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Prerequisites Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Inner products, orthogonality and polarization [T-LA-INNER]

Core forms. For real u_i,v_i,

$$(\sum u_iv_i)^2\le(\sum u_i^2)(\sum v_i^2).$$

Equality means the two vectors are proportional (including the zero case). Taking $u_i=x_i/\sqrt{w_i}$ and $v_i=\sqrt{w_i}$, where $w_i>0$, gives

$$\sum\frac{x_i^2}{w_i}\ge\frac{(\sum x_i)^2}{\sum w_i}.$$

No sign condition on the x_i is needed. The integral version follows from $0\le\int(u-tv)^2$; use it only when the squares are integrable.

For positive a_i and weights $\lambda_i\ge 0$ with total one, weighted AM–GM says $\prod a_i^{\lambda_i}\le\sum\lambda_ia_i$. It follows from concavity of log; with positive weights, equality requires equal a_i. Useful scalar consequences include $\log(1+t)\le t$ for $t>-1$ and $e^t\ge 1+t$.

Certificate viewpoint. To exploit $\sum x_i^3=0$ with $x_i\ge -1$, sum the pointwise nonnegative identity

$$x^3-\tfrac{3}{4}x+\tfrac{1}{4}=(x+1)(x-\tfrac{1}{2})^2.$$

This shows why equality candidates guide the factorization. For integral constraints, a square such as $\int(f-x)^2\ge 0$ can expose exactly the controlled moment.

Sharpness and traps. Check every equality condition against all original constraints. A best constant need not be attained: use an admissible sequence approaching equality. To pass a positive-term inequality to an infinite series, prove it for finite partial sums and control the remainder, or

T-AN-INEQUALITIES Cauchy–Schwarz, AM–GM, and sharpness (continued)

invoke nonnegative monotone limits. Denominator positivity must be established before cross-multiplication.

Extended atlas RA-07; RA-08

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1995-D1-P2 [R]; 1996-D2-P6 [R]; 1998-D1-P5 [R]; 1998-D1-P6 [R]; 1999-D2-P3 [R]; 2000-D1-P4 [R]; 2001-D2-P6 [R]; 2005-D1-P4 [R]; 2005-D2-P1 [R]; 2006-D2-P3 [R]; 2010-D1-P1 [R]; 2010-D1-P5 [R]; 2010-D2-P2 [R]; 2011-D2-P3 [R]; 2013-D1-P4 [R]; 2013-D2-P1 [R]; 2015-D2-P6 [R]; 2016-D1-P3 [R]; 2016-D2-P6 [R]; 2016-D2-P7 [R]; 2016-D2-P10 [R]; 2018-D1-P1 [R]; 2021-D2-P8 [R]; 2022-D1-P1 [R]; 2023-D2-P8 [R]

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T-AN-ODE Differential inequalities and integrating factors

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]

Recall. For continuous a, b on an interval, set $\mu(x) = \exp(\int_{x_0}^{x} a(t)\,dt) > 0$. Then

$$(\mu f)' = \mu(f' + af).$$

Thus $f' + af \geq b$ implies, for $x \geq x_0$,

$$\mu(x)f(x) \geq f(x_0) + \int_{x_0}^{x} \mu(t)b(t)\,dt.$$

The integration direction reverses for $x < x_0$. A nonnegative derivative implies monotonicity even without continuous second derivatives.

Nonlinear transform. If $f' \geq -\Phi(f)$ with $\Phi > 0$, let $H'(u) = 1/\Phi(u)$. Then $(H(f(x)) + x)' \geq 0$. For $\Phi(u) = 1 + u^2$, $H(u) = \arctan u$; infinite endpoint values become finite angles. Check the sign and domain before dividing, taking reciprocals, or taking logarithms.

Factorization. Operators such as $D^2 - 5D + 6 = (D-3)(D-2)$ suggest two successive first-order comparisons. With a variable coefficient, expand carefully: $(D + x)^2 f = f'' + 2xf' + (x^2 + 1)f$. The extra 1 comes from differentiating x.

Barrier and uniqueness. If $|f'| \leq \lambda|f|$ and $f(a) = 0$, a positive component would make $\log f - \lambda x$ nonincreasing and force a positive value at its zero endpoint, a contradiction. First-crossing arguments similarly forbid a solution from crossing a level when its derivative there points strictly in the opposite direction.

Trap. For functions defined only on $(0, \infty)$, begin integration at any fixed $x_0 > 0$, not at an undefined value $f(0)$. Endpoint limits must be justified separately.

Extended atlas RA-06

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P2 [R]; 1994-D2-P1 [R]; 1994-D2-P3 [R]; 1998-D1-P4 [R]; 2001-D1-P6 [R]; 2003-D2-P6 [R]; 2005-D1-P5 [R]; 2006-D1-P6 [R]; 2009-D2-P2 [R]; 2012-D1-P4 [R]; 2023-D2-P7 [R]

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T-AN-APPROXIMATION Uniform approximation and positive kernels

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Continuity, compactness, and order [T-AN-CONTINUITY]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]

Recall. Uniform convergence means $\|f_n - f\|_\infty \to 0$. It preserves continuity on a common domain and permits passage of integrals over a bounded interval. Pointwise convergence lacks these guarantees. Every continuous function on a compact interval is uniformly approximable by polynomials (Weierstrass).

Constructive proof core. On $[0,1]$, the Bernstein polynomial is

$$B_n f(x) = \sum_{k=0}^{n} f(k/n) \binom{n}{k} x^k (1-x)^{n-k}.$$

The nonnegative weights total one. Their mean is x and their variance is $x(1-x)/n \le 1/(4n)$. Split at $|k/n - x| < \delta$: the near error is at most $\omega_f(\delta)$ and the far error is at most $2\|f\|_\infty/(4n\delta^2)$. Choose δ first, then n, to prove uniform convergence.

Preserve constraints. For odd f on $[-1,1]$, replace an approximating p by $(p(x) - p(-x))/2$ without increasing its error. If the allowed basis excludes x, replace that term by $x[1-(1-x^2)^N]$. Its error $|x|(1-x^2)^N$ is at most δ when $|x| \le \delta$ and at most $(1-\delta^2)^N$ elsewhere.

Positive kernels. More generally, averaging against $K_n \ge 0$ with integral one approximates a target value when the kernel mass outside each neighborhood of that point tends to zero. Estimate the local error by continuity and the remote error by its mass.

Trap. Small uniform function error does not imply small derivative error. Preserve parity, normalization, and endpoint constraints explicitly; they are not automatic consequences of approximation.

Dini's theorem. If continuous f_n on a compact set decrease pointwise to a continuous f, convergence is uniform; the increasing case follows by negation. For fixed $\varepsilon > 0$, the open sets $\{f_n - f < \varepsilon\}$ increase and cover the compact set, so one already covers it. Compactness and continuity of the limit are essential.

Extended atlas RA-14

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1995-D2-P4 [R]; 1997-D2-P6 [R]; 2003-D1-P5 [R]; 2017-D2-P9 [R]; 2021-D1-P4 [R]

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T-AN-MEASURE Dominated convergence and orthonormal separation

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Inner products, orthogonality and polarization [T-LA-INNER]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]

Recall. A property holds almost everywhere if its exceptional set has measure zero. For measurable $f_n \to f$ almost everywhere, a single integrable function G satisfying $|f_n| \le G$ implies $\int f_n \to \int f$ (dominated convergence). A constant bound is integrable on a finite-measure space, not on all of $\mathbb{R}$. For $0 \le f_n \uparrow f$, monotone convergence permits $\int f_n \uparrow \int f$, including an infinite limit.

L^2 recall. The norm is $\|u\|_2 = (\int |u|^2)^{1/2}$ and the inner product is $\langle u, v\rangle = \int u\overline{v}$. An orthonormal family satisfies $\langle u_j, u_k\rangle = \delta_{jk}$, so distinct members have squared distance 2.

Useful proof. On a finite-measure space, suppose an orthonormal subsequence converges pointwise and all its members satisfy $|u_k| \le M$. Its limit satisfies $|u| \le M$ and

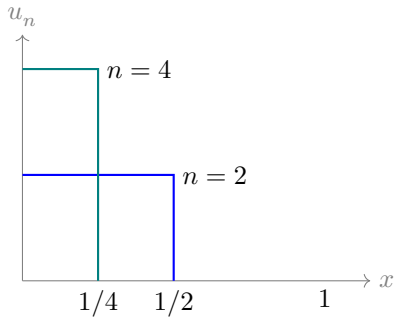
$$|u_k - u|^2 \le 4M^2.$$

Dominated convergence gives $\|u_k - u\|_2 \to 0$. The triangle inequality then forces $\|u_k - u_j\|_2 \to 0$ as $j, k \to \infty$, contradicting their distance $\sqrt{2}$. Completeness of the orthonormal family is unnecessary; extending it to a bounded complete family is not a justified step.

Trap and counterexample. Pointwise convergence alone does not control mass concentrated in narrow spikes. On $[0,1]$, $u_n(x) = n\mathbf{1}_{(0,1/n)}(x)$ tends pointwise to zero but has integral one. Check the domination of the actual quantity being integrated: a bound for u_n must be squared when passing an L^2 norm.

T-AN-MEASURE Dominated convergence and orthonormal separation (continued)

Monotone differentiation. A monotone real function on an interval is differentiable with a finite derivative almost everywhere (Lebesgue's theorem). Thus any nondegenerate subinterval contains a differentiability point. This is stronger than the elementary claim that its discontinuities are countable, and it is a separate theorem: continuity alone does not imply almost-everywhere differentiability.



For $u_n = n\mathbf{1}_{(0,1/n)}$, each rectangle has area one even as its width tends to zero; endpoint values are irrelevant to its integral.

Extended atlas RA-15

Reference notes 24; 10

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1995-D2-P6 [R]; 1998-D1-P3 [R]; 2000-D2-P5 [R]

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T-AN-SETS Countability, accumulation, and Cantor constructions

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Continuity, compactness, and order [T-AN-CONTINUITY]

Recall. A countable union of countable sets is countable. Pairwise disjoint nonempty open intervals form a countable family: assign a different rational number to each interval. This does not make every closed or nowhere dense set countable. The Cantor set is compact, has no isolated points, and is uncountable: its points admit ternary expansions using only digits zero and two, encoding binary sequences.

Accumulation. Every infinite bounded subset of $\mathbb{R}$ has an accumulation point. A differentiable function constant on a sequence of distinct points converging to an interior point c has derivative zero at c. A monotone function's level sets are intervals or singletons; consequently only countably many levels can have more than one point.

Construction core. Define a function to be zero on a closed set and place triangular tents on complementary intervals. If a tent on an interval of length ℓ has height at most ℓ, continuity holds at every boundary and accumulation point. On successive generations of the Cantor construction choose alternating signs and heights tending to zero. Every neighborhood of each Cantor point then sees both signs, giving uncountably many zero crossings.

Enumeration versus a continuum. On a countable domain, enumerate points and let the nth bound dominate the finite first n by n table. On an uncountable interval, decomposing into bounded-value sets forces at least one infinite set and hence an accumulation point; a function cannot tend to infinity at every finite point while remaining bounded along that subsequence.

Trap. Density and cardinality are different: the rationals are countable and dense. Digit expansions can be nonunique at endpoints; use a fixed binary convention or an injection from sequences rather than casually asserting uniqueness of every expansion.

Coding sequences. The set of real sequences has the same cardinality as $\mathbb{R}$: represent reals by binary sequences with a fixed convention, and enumerate the pairs of natural-number digit indices to encode a sequence of sequences by one sequence. The reverse injection is given by constant sequences; the two injections give a bijection by the Cantor–Bernstein theorem.

Extended atlas RA-16

T-AN-SETS Countability, accumulation, and Cantor constructions (continued)

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1997-D2-P6 [R]; 1998-D2-P5 [R]; 1998-D2-P6 [R]; 2002-D1-P4 [R]; 2002-D1-P5 [R]; 2003-D2-P5 [R]; 2004-D1-P1 [R]; 2008-D1-P1 [R]; 2009-D1-P1 [R]; 2014-D2-P1 [R]; 2016-D1-P1 [R]; 2021-D1-P4 [R]; 2024-D2-P6 [R]

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T-AN-BAIRE Baire class one and controlled discontinuities

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Countability, accumulation, and Cantor constructions [T-AN-SETS]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]

Definitions. A real function is of Baire class one if it is a pointwise limit of continuous functions. An F_σ set is a countable union of closed sets; a G_δ set is a countable intersection of open sets. A set is nowhere dense when its closure has empty interior. Baire's category theorem says that a nonempty complete metric space is not a countable union of nowhere dense sets; relative topology is used inside closed subsets.

Characterizations on $\mathbb{R}$. A real-valued f is of Baire class one if and only if each restriction $f|_F$, for nonempty closed $F \subset \mathbb{R}$, has a point of continuity. Equivalently, the inverse image of every open subset of $\mathbb{R}$ is F_σ. It suffices to verify that both $\{f \geq c\}$ and $\{f \leq c\}$ are G_δ for every real c.

Gauge mechanism. Suppose each $\varepsilon > 0$ admits $g_\varepsilon(x) > 0$ such that $|x-y| < \min(g_\varepsilon(x), g_\varepsilon(y))$ forces $|f(x) - f(y)| < \varepsilon$. Set

$$U_{n,k} = \bigcup_{f(y) \geq c} \left(y - \min(1/k, g_{1/n}(y)),\ y + \min(1/k, g_{1/n}(y))\right).$$

Then $\{f \geq c\} = \bigcap_{n,k \geq 1} U_{n,k}$. One inclusion uses $y = x$. For the other, if $f(x) < c$, choose $1/n < c - f(x)$ and $1/k < g_{1/n}(x)$; membership would produce a forbidden nearby y. Apply the same argument to $-f$.

Trap. The gauge may approach zero arbitrarily fast and need not be continuous. Ordinary uniform continuity cannot be inferred. The displayed characterization is a theorem about real-valued functions on $\mathbb{R}$, not a license to transfer it to arbitrary domains or codomains.

Extended atlas RA-16

Reference notes 21

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2002-D1-P4 [R]; 2021-D1-P4 [R]

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T-AN-COVERING Interval coverings and level-set integration

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Dominated convergence and orthonormal separation [T-AN-MEASURE]; Nonnegative sums, Fubini, and integral kernels [T-AN-FUBINI]

Measure tools. The measure of a countable union is at most the sum of the measures. Thus intervals of total length less than one cannot cover $(0,1)$. A countable exceptional set has measure zero. For a measurable $v \geq 0$, the layer-cake identity is

$$\int v(x)\,dx = \int_0^\infty |\{x : v(x) > t\}|\,dt,$$

obtained by integrating $\mathbf{1}_{0<t<v(x)}$ in the two orders.

Covering estimate. For integrable $v \geq 0$ on a bounded interval and $\lambda > 0$, let E_λ consist of points contained in some subinterval I with $\int_I v > \lambda |I|$. Then

$$|E_\lambda| \leq \frac{3}{\lambda} \int v.$$

T-AN-COVERING Interval coverings and level-set integration (continued)

For a finite collection of witnessing intervals, greedily choose a longest remaining interval and discard intervals meeting it. Chosen intervals are disjoint; each discarded interval lies in the concentric triple of a chosen one. Their total length is at most $\lambda^{-1}\int v$. Approximate the full open union by compact subsets to pass to the general case.

Application. A bound $|\{g > t\}| \le Ct^{-1/p}$ is integrable over $0 < t < 1$ when $p > 1$, with integral $Cp/(p-1)$. The failure at $p = 1$ explains why a borderline counterexample may be needed.

Differentiability construction. Let $f(a_n) = b_n \ge 0$ at distinct centers and $f = 0$ elsewhere. If $\sum b_n$ is finite, equivalently the heights satisfy $\sum b_n < \infty$, choose $c_n \to \infty$ with $\sum c_n b_n$ small. Outside intervals of radius $c_n b_n$, difference quotients at the spike centers are at most $1/c_n$. Exclude the countable centers separately and handle any finitely many early spikes before taking a local limit. Zero-height spikes require no division by b_n.

Extended atlas RA-20

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1998-D2-P6 [R]; 1999-D1-P6 [R]; 2002-D1-P5 [R]

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T-AN-TRIG Trigonometric identities and safe branches

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]

Recall. Addition and product identities include

$$\sin(u+v) = \sin u\cos v + \cos u\sin v, \qquad 2\cos u\cos v = \cos(u-v) + \cos(u+v),$$

$$\sin u - \sin v = 2\cos\tfrac{u+v}{2}\sin\tfrac{u-v}{2}.$$

On $(0,\pi)$, $\sin x/x$ decreases: its derivative numerator is $x\cos x - \sin x$, whose derivative is $-x\sin x < 0$ and whose limit at zero is zero. On $[0,\pi/2]$, sine and cosine are concave; tangent is increasing on each interval between consecutive poles, since its derivative there is $\sec^2 x > 0$.

Orthogonality. On $[0,2\pi]$, distinct positive integer cosine frequencies have zero product integral, and $\int \cos^2(nx) = \pi$. The product-to-sum formula proves this. Multiplication by the nonnegative weight $1 \pm \cos x$ can isolate one coefficient while preserving the sign of an integrand.

Recurrence coordinates. Half-angle identities turn nested radicals into sine or tangent values, but the initial angle and nonnegative square-root branch must be verified. Likewise,

$$\tanh(u+v) = \frac{\tanh u + \tanh v}{1 + \tanh u\tanh v}, \qquad \operatorname{artanh} x = \tfrac{1}{2}\log\frac{1+x}{1-x}$$

for $|x| < 1$. This transforms a rational recurrence into addition.

Trap. Periodicity creates two-cycles even when absolute values decrease. A tangent argument may cross a pole or leave the interval on which concavity was used. Check these intervals explicitly, and do not substitute the amplitude of a trigonometric factor for its signed value when an exact identity is needed.

Extended atlas RA-18

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P5 [R]; 1995-D2-P5 [R]; 1996-D1-P2 [R]; 1997-D2-P3 [R]; 1998-D1-P6 [R]; 2001-D2-P2 [R]; 2001-D2-P6 [R]; 2003-D2-P2 [R]; 2004-D1-P3 [R]; 2006-D2-P3 [R]; 2010-D2-P1 [R]; 2011-D2-P1 [R]; 2013-D1-P2 [R]; 2014-D2-P3 [R]; 2016-D1-P1 [R]

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T-AN-PROBABILITY Finite probability, conditioning, and expectations

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Prerequisites Finite counting and incidence identities [T-DS-COUNTING]

Recall. In a finite uniform sample space, probability is the number of favorable outcomes divided by the total number. Independence means joint probabilities of specified events factor. Conditional probability satisfies $\mathsf{P}(A \cap B) = \mathsf{P}(A \mid B)\mathsf{P}(B)$ when $\mathsf{P}(B) > 0$, and summing over a partition gives the law of total probability.

Indicator method. For an event A, $\mathbf{1}_A$ equals one on A and zero otherwise, with expectation $\mathsf{E}\mathbf{1}_A = \mathsf{P}(A)$. For a finite sum,

$$\mathsf{E}\sum_i X_i = \sum_i \mathsf{E}X_i,$$

without any independence requirement. This converts a difficult count into the probabilities that individual candidates contribute.

Convolution. For independent nonnegative integer-valued X, Y, their probability generating functions multiply: $\mathsf{E}z^{X+Y} = (\mathsf{E}z^X)(\mathsf{E}z^Y)$ wherever these expectations converge. Residue probabilities also obey a finite recurrence: $p_{n+1}(r) = \sum_s q(s)p_n(r-s)$, with all indices taken modulo the fixed modulus. Specify the initial point mass.

Roots-of-unity filter. For a finite polynomial $F(z) = \sum a_j z^j$ and $\omega = e^{2\pi i/m}$,

$$\sum_{j \equiv r \pmod m} a_j = \frac{1}{m}\sum_{\ell=0}^{m-1} \omega^{-r\ell} F(\omega^\ell).$$

The inner geometric sum equals one for the selected residue and zero otherwise.

Trap. A bijection proves probabilities only when it preserves the sampling weights. Random subsets of fixed size have dependent memberships. Exchangeability permits equal marginal probabilities, not multiplication of those probabilities. After conditioning, recompute the conditional law before invoking independence.

Extended atlas PRB-01; PRB-02; PRB-03

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1999-D2-P2 [R]; 2011-D1-P4 [R]; 2017-D1-P4 [R]; 2019-D2-P10 [R]; 2021-D1-P2 [R]; 2022-D2-P8 [R]; 2024-D1-P5 [R]

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T-AN-GEOMETRICPROB Uniform random points and convex-hull events

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Finite probability, conditioning, and expectations [T-AN-PROBABILITY]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]

Uniform law. If X is uniform in a measurable region D of finite positive volume, then $\mathsf{P}(X \in A) = |A \cap D|/|D|$. For a circle use normalized arc length, not area. For independent points, conditioning on a fixed geometric configuration often makes the remaining event a product of area or arc-length ratios.

Null degeneracies. A random point with a density lies in a fixed hyperplane with probability zero. Conditioning on earlier points and taking a finite union therefore removes coincidences and unwanted affine dependencies when the relevant hyperplane is genuinely lower dimensional. Distinguish this almost-sure reduction from a deterministic assertion about every configuration.

Hull counts. In a nondegenerate convex polygon, vertices and edges have the same count. Write an intersection-edge count as a sum of indicators over candidate chords. A chord intersects the other polygon only when the other color occurs on both sides. On a circle, the order of independently sampled labels is uniformly distributed; count color changes and separately subtract configurations where changes exist but the hull intersection is empty.

Symmetry averaging. If the law of each X_i is centrally symmetric and the points are independent, then all sign changes $(\pm X_1, \dots, \pm X_n)$ have the same joint law. Average a deterministic inequality

T-AN-GEOMETRICPROB Uniform random points and convex-hull events (continued)

over all 2^n sign choices and then take expectations. Radial symmetry can reduce two query points to the same radius line, but does not itself prove monotonicity along it.

Integral example. Conditioning on a uniform arc length $x \in [0,1]$ leads to beta integrals. For integers $a, b \geq 1$, repeated integration by parts gives $\int_0^1 x^{a-1}(1-x)^{b-1}\,dx = (a-1)!(b-1)!/(a+b-1)!$.

Extended atlas PRB-04; RA-20

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2019-D2-P10 [R]; 2022-D2-P8 [R]; 2024-D1-P5 [R]

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T-AN-ARCLENGTH Convex graph length and ordered primitives

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]

Recall. For $F \in C^1[a,b]$, the graph length is $\int_a^b \sqrt{1+F'(x)^2}\,dx$. This follows by applying the mean value theorem on a partition to the lengths of the secant segments and taking a Riemann-sum limit. A convex graph can also be approximated by its inscribed polygonal graphs.

Comparison theorem. Let u, v be continuous nondecreasing functions, and let $F(x) = \int_a^x u$, $G(x) = \int_a^x v$. If $F \leq G$ and $F(b) = G(b)$, then for $\Phi(s) = \sqrt{1+s^2}$,

$$\int_a^b \Phi(u)\,dx \geq \int_a^b \Phi(v)\,dx.$$

Indeed, convexity gives $\Phi(u) - \Phi(v) \geq \Phi'(v)(u-v)$. Set $H = F - G \leq 0$, so $H(a) = H(b) = 0$ and $H' = u - v$. Since $w = \Phi'(v) = v/\sqrt{1+v^2}$ is continuous nondecreasing,

$$\int_a^b wH' = [wH]_a^b - \int_a^b H\,dw \geq 0.$$

Here $\int H\,dw$ is the Riemann–Stieltjes integral, defined by limits of sums $\sum H(\xi_i)[w(x_i) - w(x_{i-1})]$. It exists for continuous H and monotone w; all increments of w are nonnegative. Its integration-by-parts identity follows by telescoping finite sums.

Example and trap. Apply this with $u = \sqrt{f}$, $v = \sqrt{g}$ when cumulative integrals are ordered. The resulting length inequality runs opposite to the pointwise ordering of the primitives. Do not infer an ordering of F' and G' from $F \leq G$; derivatives may cross repeatedly. Shared endpoint values and monotonicity of the derivatives are the decisive hypotheses.

Extended atlas RA-07; RA-08

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2004-D2-P2 [R]

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Algebra, number theory and complex polynomials

T-AL-POLYNOMIAL Polynomial identities, degree, and multiplicity

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A polynomial over a field K is a finite expression $P(X)=\sum_{j=0}^{d} a_j X^j$ with $a_d \neq 0$. Division by $X-c$ gives $P(X)=(X-c)Q(X)+P(c)$, so $P(c)=0$ exactly when $X-c$ divides P. Repeated division proves that a nonzero degree-d polynomial has at most d distinct roots in any extension field. Consequently two polynomials of degree at most d agreeing at $d+1$ points are identical. Over $\mathbb{C}$, the fundamental theorem of algebra supplies exactly d roots counted with multiplicity.

The multiplicity of c is the largest m for which $(X-c)^m \mid P$. In characteristic zero, writing $P=(X-c)^m R$ with $R(c)\neq 0$ shows that P' has multiplicity exactly $m-1$ there. Thus a tangency to a line L gives a double root of $P-L$; three distinct tangencies of a sextic force $P-L=A\prod_{j=1}^{3}(X-c_j)^2$.

For integer coefficients, $a-b \mid P(a)-P(b)$ follows termwise from $a^j-b^j=(a-b)\sum_{r=0}^{j-1} a^{j-1-r}b^r$.

Proof core. Turn each specified value into a factor, count degrees, and treat any zero polynomial separately. For several variables, regard an identity as a polynomial in one variable whose coefficients are polynomials in the others; over an infinite field, repeated univariate root counting justifies coefficient comparison.

Example and trap. A cubic taking the same value at four distinct real points must be constant, a contradiction. Over $\mathbb{F}_p$, however, X^p-X is nonzero and vanishes at every field element: equality as functions is not equality as polynomials. In characteristic p, differentiation can erase multiplicity information, since $(X^p)'=0$.

Extended atlas ANT-04; CP-13

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1995-D1-P5 [R]; 1995-D2-P3 [R]; 1995-D2-P4 [R]; 1997-D1-P3 [R]; 1999-D1-P1 [R]; 1999-D2-P3 [R]; 2000-D1-P2 [R]; 2000-D2-P3 [R]; 2000-D2-P4 [R]; 2001-D1-P4 [R]; 2001-D2-P1 [R]; 2003-D1-P6 [R]; 2004-D1-P2 [R]; 2004-D1-P6 [R]; 2005-D1-P4 [R]; 2005-D2-P1 [R]; 2005-D2-P2 [R]; 2005-D2-P4 [R]; 2005-D2-P6 [R]; 2006-D1-P4 [R]; 2006-D1-P6 [R]; 2007-D1-P1 [R]; 2007-D1-P5 [R]; 2007-D1-P6 [R]; 2008-D1-P2 [R]; 2008-D1-P3 [R]; 2008-D2-P1 [R]; 2008-D2-P2 [R]; 2008-D2-P3 [R]; 2008-D2-P4 [R]; 2009-D2-P4 [R]; 2009-D2-P5 [R]; 2010-D1-P2 [R]; 2010-D1-P5 [R]; 2011-D1-P3 [R]; 2011-D2-P3 [R]; 2011-D2-P4 [R]; 2011-D2-P5 [R]; 2012-D2-P1 [R]; 2012-D2-P4 [R]; 2013-D1-P5 [R]; 2013-D2-P1 [R]; 2014-D1-P3 [R]; 2015-D1-P1 [R]; 2015-D1-P2 [R]; 2015-D2-P8 [R]; 2015-D2-P10 [R]; 2016-D1-P5 [R]; 2016-D2-P10 [R]; 2017-D2-P7 [R]; 2018-D1-P2 [R]; 2018-D1-P3 [R]; 2018-D2-P7 [R]; 2018-D2-P9 [R]; 2019-D1-P1 [R]; 2020-D1-P4 [R]; 2020-D2-P6 [R]; 2020-D2-P8 [R]; 2022-D2-P6 [R]; 2023-D1-P3 [R]; 2023-D1-P4 [R]; 2023-D2-P10 [R]; 2024-D1-P1 [R]; 2024-D2-P7 [R]

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T-AL-FACTORIZATION Polynomial factorization and coprime divisibility

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Prerequisites Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]

For a field K, the ring $K[X]$ has Euclidean division by degree. The Euclidean algorithm gives a greatest common divisor D and polynomials A, B with $AP+BQ=D$. A nonconstant polynomial is irreducible if it is not a product of two polynomials of positive degree. Factoring into monic irreducibles is unique, apart from an overall nonzero constant. In particular, if $\gcd(P,Q)=1$ and $P \mid QR$, then $P \mid R$: multiply a Bézout identity by R.

An integer polynomial is primitive when the gcd of its coefficients is 1. Gauss's lemma says that a product of primitive integer polynomials is primitive, and a primitive integer polynomial factors over $\mathbb{Q}$ exactly when it factors over $\mathbb{Z}$ into factors of positive degree. One proof of the first assertion reduces a proposed common prime divisor modulo that prime: the product of two nonzero polynomials over a field cannot vanish. Clearing denominators then yields the second assertion. A rational root u/v in lowest terms of $a_d X^d+\cdots+a_0$ satisfies $u \mid a_0$ and $v \mid a_d$.

Exponent bookkeeping. If nonzero polynomials satisfy $P^3=Q^2$, unique factorization gives $3v_R(P)=2v_R(Q)$ for every monic irreducible R. Thus each exponent in P is even and each exponent in Q is divisible by 3. This explains why two coprime powers can recover a polynomial function.

T-AL-FACTORIZATION Polynomial factorization and coprime divisibility (continued)

Example and trap. If a reduced rational function P/Q has polynomial square, then $Q^2 \mid P^2$ forces Q constant. This argument concerns rational functions; if it was obtained by dividing pointwise by another function, separately verify the points at which that original divisor vanishes. Irreducibility is always relative to the coefficient field: X^2+1 is irreducible over $\mathbb{R}$ but splits over $\mathbb{C}$.
Extended atlas ANT-04

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2005-D1-P4 [R]; 2005-D2-P2 [R]; 2006-D1-P4 [R]; 2007-D1-P5 [R]; 2012-D1-P5 [R]; 2018-D2-P9 [R]; 2020-D2-P6 [R]; 2023-D1-P3 [R]

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T-AL-ROOTS Root identities, symmetric sums, and logarithmic derivatives

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Prerequisites Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]

Let $P(X) = a\prod_{j=1}^{n}(X-r_j)$ over $\mathbb{C}$, with roots listed with multiplicity. Viète's formulas identify the coefficient of X^{n-k} as $a(-1)^k e_k$, where e_k is the kth elementary symmetric sum of the roots. If $s_m = \sum_j r_j^m$ and $e_0 = 1$, Newton's identities read

$$ke_k = \sum_{j=1}^{k}(-1)^{j-1}e_{k-j}s_j \qquad (1 \le k \le n).$$

They follow by comparing coefficients in the logarithmic derivative of $\prod_j(1+r_jt)$. Over characteristic zero, prescribed first n power sums therefore determine $e_1,\dots,e_n$, and a monic polynomial realizes those sums through its complex roots.

Away from zeros,

$$\frac{P'}{P} = \sum_j \frac{1}{X-r_j}, \qquad \left(\frac{P'}{P}\right)' = -\sum_j \frac{1}{(X-r_j)^2}.$$

For real roots and real x outside the zero set, this derivative is strictly negative. Cauchy–Schwarz gives $n\sum_j(x-r_j)^{-2} \ge (\sum_j(x-r_j)^{-1})^2$ and hence $(n-1)P'(x)^2 \ge nP(x)P''(x)$. At a root the inequality is checked directly; for $n>1$, equality there means a multiple root. At a nonroot, equality holds precisely when all roots coincide.

Example. If all roots are real, $s_2 \ge 0$ implies $a_{n-1}^2 - 2a_na_{n-2} \ge 0$. Computing only three leading coefficients can therefore rule out real-rootedness of a large family. Rolle's theorem gives another route: derivatives retain real roots, allowing reduction to a cubic.

Traps. Never divide by P at a root. Nonnegative coefficients alone do not imply real-rootedness. For a real polynomial with all roots of negative real part and positive leading coefficient, real linear factors and conjugate-pair quadratic factors have positive coefficients; this explains positivity, but any stronger coefficient inequality still requires its own argument.
Extended atlas ANT-05; CP-11

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1995-D2-P3 [R]; 1998-D1-P5 [R]; 2000-D1-P2 [R]; 2001-D2-P1 [R]; 2003-D1-P6 [R]; 2005-D1-P4 [R]; 2006-D2-P5 [R]; 2007-D2-P6 [R]; 2013-D1-P4 [R]; 2013-D1-P5 [R]; 2017-D2-P7 [R]; 2018-D2-P9 [R]; 2020-D2-P6 [R]

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T-AL-INTERPOLATION Interpolation, divided differences, and extremal signs

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Prerequisites Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]

Let $x_0, \dots, x_n$ be distinct elements of a field and define

$$L_j(X) = \prod_{i \neq j} \frac{X - x_i}{x_j - x_i}.$$

Every polynomial P of degree at most n satisfies $P(X) = \sum_j P(x_j) L_j(X)$. Indeed the difference has degree at most n and vanishes at all $n+1$ nodes. The distinctness hypothesis makes each denominator invertible. Differentiating gives $P^{(r)}(t) = \sum_j P(x_j) L_j^{(r)}(t)$ over characteristic zero.

If x_j, t are real and $|P(x_j)| \le 1$, then $|P^{(r)}(t)| \le \sum_j |L_j^{(r)}(t)|$. This bound is attained at that fixed t by choosing the nodal values with signs matching $L_j^{(r)}(t)$, or with all signs reversed. Terms having zero weight impose no sign restriction. Thus an extremal polynomial can be constructed, rather than merely bounded. If a remaining derivative is affine on an interval, the maximum of its absolute value occurs at an endpoint, reducing the optimization to finitely many such sign choices.

For consecutive integer nodes, the Newton basis $\binom{X}{k} = X(X-1)\cdots(X-k+1)/k!$ is often preferable. Writing $\Delta P(X) = P(X+1) - P(X)$ gives

$$P(X) = \sum_{k=0}^{n} \Delta^k P(0) \binom{X}{k}.$$

This follows because $\Delta\binom{X}{k} = \binom{X}{k-1}$ and because the values at $0, \dots, n$ determine P.

Example and trap. $\binom{X}{2}$ takes integer values at every integer but has noninteger monomial coefficients. Integer-valued is therefore weaker than integer-coefficient. For extremal interpolation, a guessed alternating pattern must be checked against the actual signs of the differentiated basis; node order alone does not determine it. In finite characteristic, denominators such as $k!$ need not be invertible.

Extended atlas CP-10

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1998-D2-P2 [R]; 2011-D2-P4 [R]

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T-AL-CONGRUENCE Congruences, Bézout, and the Chinese remainder theorem

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For integers a, b and $m \ge 1$, $a \equiv b \pmod m$ means $m \mid a-b$. Addition, multiplication, and evaluation of integer polynomials preserve congruence. Cancellation by u modulo m is valid when $\gcd(u, m) = 1$; the Euclidean algorithm then produces v with $uv \equiv 1 \pmod m$. More generally, $ux \equiv b \pmod m$ is solvable exactly when $\gcd(u, m) \mid b$.

For pairwise coprime positive moduli $m_1, \dots, m_s$, the residue map

$$\mathbb{Z}/(m_1 \cdots m_s)\mathbb{Z} \longrightarrow \prod_{j=1}^{s} \mathbb{Z}/m_j\mathbb{Z}$$

is a bijection. To construct its inverse, put $M = \prod_j m_j$, $M_j = M/m_j$, choose $u_j M_j \equiv 1 \pmod{m_j}$, and take $x = \sum_j a_j u_j M_j$. Uniqueness follows because a difference divisible by every coprime m_j is divisible by M. With two noncoprime moduli, compatibility requires $a_1 \equiv a_2 \pmod{\gcd(m_1, m_2)}$, and uniqueness is modulo their lcm.

Example. Since consecutive integers are coprime, $x(x-1) \equiv 0 \pmod{p^a}$ forces $x \equiv 0$ or $1 \pmod{p^a}$. An integer with s distinct prime factors consequently has 2^s idempotent residue classes modulo it. Excluding the representative 0 changes the final count.

Local obstruction. If an integer-valued expression misses even one residue modulo m, it misses infinitely many integers. This turns a global representation claim into a finite calculation.

T-AL-CONGRUENCE Congruences, Bézout, and the Chinese remainder theorem (continued)

Traps. Coprimality belongs to the moduli, not merely the prescribed residues. From $m \mid xy$ one cannot conclude $m \mid x$ or $m \mid y$ for composite m: $6 \mid 2 \cdot 3$ is the simplest counterexample. Split the modulus into prime powers before assigning factors.
Two Euclidean prime constructions. If all primes congruent to 3 modulo 4 were $p_1, \dots, p_r$, the number $4p_1 \cdots p_r - 1$ would have a prime divisor congruent to 3 modulo 4 outside that list. For primes congruent to 1 modulo 4, use $(2p_1 \cdots p_r)^2 + 1$. Each prime divisor q is odd, and a square root of -1 modulo q has order 4, so $4 \mid q - 1$; again none of the listed primes divides the constructed number.
Extended atlas ANT-01; ANT-02

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1996-D2-P3 [R]; 1999-D2-P2 [R]; 2000-D2-P1 [R]; 2001-D1-P2 [R]; 2003-D1-P4 [R]; 2005-D1-P2 [R]; 2005-D1-P6 [R]; 2005-D2-P6 [R]; 2006-D1-P2 [R]; 2006-D1-P3 [R]; 2006-D2-P1 [R]; 2007-D1-P1 [R]; 2007-D1-P2 [R]; 2008-D1-P3 [R]; 2008-D1-P6 [R]; 2008-D2-P5 [R]; 2010-D1-P4 [R]; 2010-D2-P5 [R]; 2011-D2-P4 [R]; 2013-D2-P2 [R]; 2013-D2-P4 [R]; 2013-D2-P5 [R]; 2014-D1-P4 [R]; 2015-D1-P2 [R]; 2016-D1-P5 [R]; 2017-D1-P3 [R]; 2018-D1-P5 [R]; 2019-D1-P5 [R]; 2022-D2-P6 [R]; 2024-D2-P10 [R]

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T-AL-FINITEFIELD Finite fields, Frobenius, and polynomial functions

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Prerequisites Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
A field has characteristic 0 or a prime p: if the least positive integer annihilating 1 were composite, it would create zero divisors. In characteristic p, the binomial coefficients between the endpoints of row p are divisible by p, so $(x+y)^p = x^p + y^p$. The map $x \mapsto x^p$ is the Frobenius homomorphism. On $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$, Fermat's theorem gives $x^p = x$ for every x.
Every function $\mathbb{F}_p \to \mathbb{F}_p$ has a unique representing polynomial of degree at most $p-1$. Existence follows by interpolation at all p elements, and uniqueness by the root bound. Division by $X^p - X$ thus converts polynomial functions to their canonical representatives. On $\mathbb{F}_p^*$, positive exponents may instead be reduced modulo $p-1$; at 0, treat the constant term separately.
The multiplicative group of a finite field is cyclic. A short proof lets m be the lcm of all element orders: in an abelian group one can combine elements attaining each maximal prime-power part to obtain an element of order m. All elements solve $X^m = 1$, so the root bound gives $|\mathbb{F}_p^*| \le m$, while Lagrange gives $m \le p-1$. Hence $m = p-1$. It follows that $x \mapsto x^k$ has image size $(p-1)/\gcd(k, p-1)$ on nonzero elements, with equal fiber sizes.
Example and trap. A quadratic vanishing at all five residues modulo 5 has every coefficient divisible by 5. This fails for degree 5, since $X^5 - X$ vanishes everywhere. A quadratic discriminant test requires a separate characteristic-2 case. In quotient rings $\mathbb{F}_p[X]/(F)$, cancellation of X needs $\gcd(X, F) = 1$; the quotient is a field only when F is irreducible.
Extended atlas ANT-03
Reference notes 5

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2003-D1-P2 [R]; 2007-D1-P1 [R]; 2007-D2-P2 [R]; 2008-D1-P5 [R]; 2009-D2-P4 [R]; 2010-D1-P4 [R]; 2011-D1-P3 [R]; 2018-D1-P2 [R]; 2020-D2-P6 [R]; 2021-D2-P6 [R]; 2022-D1-P3 [R]; 2022-D2-P6 [R]; 2023-D1-P4 [R]; 2023-D2-P10 [R]; 2024-D2-P10 [R]

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T-AL-VALUATION Prime valuations, divisors, and factorials

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For a prime p and a nonzero rational number x, write $x = p^r u/v$ with $p \nmid uv$ and define $v_p(x) = r \in \mathbb{Z}$; put $v_p(0) = +\infty$. Then $v_p(xy) = v_p(x) + v_p(y)$ and $v_p(x+y) \geq \min(v_p(x), v_p(y))$, with equality when the two valuations differ. The latter assertion follows by factoring out the smaller power of p; the remaining sum is not divisible by p. Thus cancellation can increase valuation only when the lowest orders tie.

For positive integers a, b, $a \mid b$ exactly when $v_p(a) \leq v_p(b)$ for every prime. The valuation of a gcd is the minimum and that of an lcm the maximum. A positive integer is a square precisely when all its prime valuations are even. Coprime factors of a square are therefore squares separately.

If $n = \prod p^{e_p}$, independent choices of exponents give

$$\tau(n) = \prod_p (e_p + 1), \qquad \sigma(n) = \prod_p (1 + p + \cdots + p^{e_p}), \qquad \prod_{d \mid n} d = n^{\tau(n)/2}.$$

The last identity pairs d with n/d; it remains valid for squares, whose middle divisor is $\sqrt{n}$. Counting multiples of $p, p^2, \ldots$ yields Legendre's formula

$$v_p(N!) = \sum_{j \geq 1} \lfloor N/p^j \rfloor .$$

Example. In a rational identity $257R^2 = \prod_{k=1}^{15} (1 + k^2)^{m_k}$ with integer m_k and $R \neq 0$, the left valuation at 257 is odd, while the right valuation is zero. Negative exponents do not invalidate this obstruction because valuations are defined on rationals.

Traps. A composite-base exponent need not be a prime valuation. Divisibility of a sum cannot be obtained by adding the summands' valuations. Always separate 0, signs, and cancellation before making a parity argument.

Extended atlas ANT-06

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2003-D1-P4 [R]; 2005-D1-P4 [R]; 2006-D1-P2 [R]; 2006-D1-P3 [R]; 2006-D2-P5 [R]; 2007-D2-P2 [R]; 2008-D1-P3 [R]; 2008-D2-P4 [R]; 2010-D1-P4 [R]; 2012-D1-P5 [R]; 2012-D2-P3 [R]; 2013-D2-P4 [R]; 2014-D1-P4 [R]; 2015-D1-P3 [R]; 2015-D1-P4 [R]; 2017-D1-P3 [R]; 2019-D2-P7 [R]; 2023-D2-P10 [R]; 2024-D2-P10 [R]

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T-AL-GROUP Groups, orders, generators, and quotients

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A group has associative multiplication, an identity e, and inverses. The subgroup $\langle S \rangle$ generated by S consists of finite words in elements of S and their inverses. A nonempty set H is a subgroup if $xy^{-1} \in H$ whenever $x, y \in H$. The order of g is the least positive r satisfying $g^r = e$, if such an integer exists; then $g^m = e$ exactly when $r \mid m$. If $g^a = g^b = e$, Bézout gives $g^{\gcd(a,b)} = e$.

For finite groups, left cosets of H partition G into equally sized sets, proving $|G| = [G : H]|H|$ and Lagrange's divisibility theorem. If $K \leq H \leq G$ have finite indices, $[G : K] = [G : H][H : K]$. A subgroup N is normal if $gNg^{-1} = N$ for all g; only then does coset multiplication define the quotient group G/N. Images of a generating set generate every homomorphic image, giving useful obstructions.

For commuting elements a, b of coprime finite orders r, s, the product has order rs. Indeed $(ab)^m = e$ implies $a^m = b^{-m}$ lies in both cyclic subgroups, whose intersection has order dividing both r and s, hence is trivial. Thus $r, s \mid m$.

Example. The upper triangular matrices $U(x) = \left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right)$ obey $U(x)U(y) = U(x+y)$. Dyadic parameters form a subgroup, but any finite collection has bounded powers of 2 in its denominators, so it cannot generate all $U(2^{-n})$.

Traps. $(ab)^m = a^m b^m$ needs commutativity. Equal subgroup indices do not imply equality, conjugacy, or normality. An injective homomorphism preserves element orders exactly; a general homomorphism can decrease them.

T-AL-GROUP Groups, orders, generators, and quotients (continued)

Extended atlas ANT-07; ANT-09

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1996-D2-P3 [R]; 1997-D2-P5 [R]; 1998-D1-P2 [R]; 2001-D1-P2 [R]; 2003-D2-P4 [R]; 2005-D1-P6 [R]; 2007-D1-P4 [R]; 2008-D1-P5 [R]; 2010-D2-P3 [R]; 2011-D1-P3 [R]; 2011-D2-P5 [R]; 2012-D1-P3 [R]; 2012-D2-P5 [R]; 2016-D2-P8 [R]; 2018-D1-P2 [R]; 2020-D2-P7 [R]; 2021-D2-P6 [R]; 2023-D1-P4 [R]; 2024-D1-P4 [R]; 2024-D2-P10 [R]

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T-AL-PERMUTATION Permutation cycles, parity, and orbit decomposition

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Prerequisites Groups, orders, generators, and quotients [T-AL-GROUP]

A permutation is a bijection of a set to itself. On a finite set, its disjoint cycles partition the set and its order is the lcm of the cycle lengths. A cycle of length r is a product of $r-1$ transpositions, so its sign is $(-1)^{r-1}$. Sign is a homomorphism $S_n \to \{1,-1\}$; its kernel is the alternating group A_n. In particular, a fixed-point-free involution on n points consists of $n/2$ disjoint transpositions and has sign $(-1)^{n/2}$.

Adjacent transpositions generate S_n: successive swaps move each required symbol to its position. Conjugation transports a transposition, $\sigma(a\ b)\sigma^{-1} = (\sigma(a)\ \sigma(b))$. A full cycle and a transposition joining consecutive elements on that cycle therefore generate S_n. The adjacency condition matters: an arbitrary transposition paired with a composite-length cycle may generate a smaller subgroup.

For an arbitrary bijection $f : X \to X$, the orbits $\{f^k(x) : k \in \mathbb{Z}\}$ are either finite cycles or copies of $\mathbb{Z}$. On the latter, f is the shift $j \mapsto j+1$, and the reflections $j \mapsto -j$ and $j \mapsto 1-j$ compose to that shift. The same formulas modulo r handle a finite r-cycle. Choosing one base point per orbit and combining these definitions expresses f as two involutions; for arbitrary families of orbits this choice is understood in the usual setting with the axiom of choice.

Example and trap. A permutation of p points with order divisible by a prime p must contain a p-cycle, hence has order exactly p. It cannot have order $2p$. Keep composition order fixed, normally applying the rightmost permutation first; confusing this convention changes explicit generator calculations.

Extended atlas ANT-08

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1997-D2-P5 [R]; 1998-D1-P2 [R]; 2001-D1-P1 [R]; 2001-D1-P2 [R]; 2003-D1-P2 [R]; 2008-D1-P6 [R]; 2009-D2-P4 [R]; 2012-D1-P3 [R]; 2014-D2-P5 [R]; 2016-D1-P5 [R]; 2016-D2-P8 [R]; 2021-D2-P6 [R]; 2022-D1-P3 [R]

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T-AL-ACTION Group actions, stabilizers, and semidirect products

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Prerequisites Groups, orders, generators, and quotients [T-AL-GROUP]

An action of G on X assigns bijections $x \mapsto gx$ with $ex = x$ and $(gh)x = g(hx)$. The orbit Gx is the set of reachable points, and the stabilizer $G_x = \{g : gx = x\}$ is a subgroup. The map $gG_x \mapsto gx$ is a bijection, proving $|Gx| = [G : G_x]$ even when the group is infinite and the index finite. Stabilizers in one orbit are conjugate: $G_{gx} = gG_xg^{-1}$.

For an action on a finite set, count pairs (g,x) with $gx = x$ in either order. This gives $\sum_g |\operatorname{Fix}(g)| = \sum_x |G_x|$ and, for finite G, Burnside's formula $|X/G| = |G|^{-1}\sum_g |\operatorname{Fix}(g)|$. On G/H, the stabilizer of the base coset H is H; on a pair of base cosets (H_1, H_2) it is $H_1 \cap H_2$. Stabilizers of arbitrary pairs are intersections of the corresponding conjugates, not necessarily that same intersection.

To build a group with prescribed action, take a homomorphism $\rho : T \to \operatorname{Aut}(H)$ and define

$$(h,t)(h',t') = (h\rho(t)(h'), tt').$$

T-AL-ACTION Group actions, stabilizers, and semidirect products (continued)

This semidirect product contains a normal copy of H. Associativity follows from $\rho(tt') = \rho(t)\rho(t')$. Automorphisms are determined by images of generators and preserve element orders, so the number of candidate images gives an upper bound; an exact count requires checking the relations.
Example. Automorphisms of $(\mathbb{F}_2)^3$ are invertible linear maps, numbering $(8-1)(8-2)(8-4)$ by choosing independent columns. A cyclic coordinate shift gives a concrete action for a semidirect product.
Trap. A product H_1H_2 of subgroups need not be a subgroup. Coset counting and group multiplication must be kept distinct.
Extended atlas ANT-09; ANT-10

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2008-D1-P5 [R]; 2010-D2-P3 [R]; 2020-D2-P7 [R]

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T-AL-RING Noncommutative rings, polarization, and torsion

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In an associative ring, addition is commutative but multiplication need not be. Expand products in order: $(a+b)^2 = a^2 + ab + ba + b^2$. An idempotent satisfies $e^2 = e$; a nilpotent has some positive power zero. If every element has square zero, substituting $a+b$ gives $ab = -ba$ for every pair. Applying this to products as well as individual factors and using associativity yields $abc = -abc$, hence $2abc = 0$. One may not infer $abc = 0$ without an appropriate additive-torsion assumption.
For the additive commutator $[a,b] = ab - ba$, direct expansion gives $[a,bc] = [a,b]c + b[a,c]$. When e, f, g are idempotent and $e+f+g = 0$, squaring $g = -e - f$ gives $ef + fe = 2g$. Commuting with e then yields $3[e,f] = 0$. If the additive group has no nonzero element killed by a positive integer, $[e,f] = 0$, and the remaining identities reduce the problem to commuting algebra.
Essential distinction. For a unital ring, characteristic zero means $n1 \neq 0$ for every positive integer n. It does not by itself imply $na = 0 \Rightarrow a = 0$ for arbitrary a. The ring $\mathbb{Z}\times\mathbb{F}_3$ has characteristic zero but contains $a = (0,1) \neq 0$ with $3a = 0$. Its three equal idempotents $a+a+a = 0$ show why the stronger, explicitly stated torsion-free hypothesis is needed in such a problem.
Proof routine and trap. Polarize a universal identity by replacing one variable with a sum; then compare the resulting mixed terms. Substitute products to exploit associativity, and only cancel an integer multiple when the ring's additive group permits it. Group commutators $a^{-1}b^{-1}ab$ and ring commutators $ab - ba$ are different operations and must not be interchanged.
Extended atlas ANT-11

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1999-D2-P1 [R]; 2000-D1-P5 [R]

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T-AL-COMPLEX Complex coordinates, conjugation, and unit-circle geometry

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Write $z = x + iy$, $\bar{z} = x - iy$, and $|z| = \sqrt{z\bar{z}}$. Then z is real exactly when $z = \bar{z}$, and $|z| = 1$ implies $\bar{z} = 1/z$. This converts many geometric or reality conditions on the unit circle into polynomial equations. Euler's formula $e^{it} = \cos t + i\sin t$ gives an alternative conversion into trigonometric products; arguments are taken modulo 2π, not as a globally single-valued additive function.
Multiplication by $re^{i\theta}$ scales lengths by r and rotates by θ. A closed polygon with side lengths a_j and successive directions $e^{i\theta_j}$ therefore satisfies $\sum_j a_j e^{i\theta_j} = 0$. For an equiangular convex N-gon the exterior turn is $2\pi/N$, producing a polynomial relation at a primitive Nth root of unity.

T-AL-COMPLEX Complex coordinates, conjugation, and unit-circle geometry (continued)

The identities $|u+v| \le |u|+|v|$ and $\operatorname{Re} u \le |u|$ are often strongest when equality is tracked: nonzero summands attain equality in the triangle inequality only when they lie on a common ray. Another useful calculation, for $z \neq \alpha$, is

$$\operatorname{Re}\frac{z+\alpha}{z-\alpha} = \frac{|z|^2-|\alpha|^2}{|z-\alpha|^2}.$$

It transforms unit-circle root questions into a common sign argument.

Example. If a, b are unit complex numbers, multiply any identity containing $\bar{a}, \bar{b}$ by the nonzero product ab to obtain a polynomial factorization. Alternatively write $a = e^{ix}, b = e^{iy}$ and use sum-to-product formulas to classify the vanishing imaginary part.

Traps. $|P(\bar{z})|$ need not equal $|P(z)|$ for complex coefficients. The correct reflected polynomial is $P^*(z) = z^n\overline{P(1/\bar{z})}$, whose boundary modulus equals $|P(z)|$. Extracting square roots or dividing by a factor can lose branches or zero cases; record them explicitly.

Extended atlas CP-01

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1995-D2-P3 [R]; 1999-D2-P6 [R]; 2000-D1-P2 [R]; 2001-D1-P4 [R]; 2007-D1-P6 [R]; 2008-D2-P1 [R]; 2013-D2-P1 [R]; 2015-D1-P4 [R]; 2017-D1-P5 [R]; 2018-D1-P5 [R]; 2021-D2-P7 [R]; 2023-D1-P3 [R]; 2024-D1-P1 [R]

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T-AL-CYCLOTOMIC Roots of unity, cyclotomic polynomials, and algebraic norms

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Prerequisites Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Polynomial factorization and coprime divisibility [T-AL-FACTORIZATION]

The Nth roots of unity are $e^{2\pi ij/N}$, $0 \le j < N$, and a primitive root has multiplicative order exactly N. Their minimal polynomial over $\mathbb{Q}$ is the cyclotomic polynomial

$$\Phi_N(X) = \prod_{\substack{1\le j\le N\\ \gcd(j,N)=1}} (X - e^{2\pi ij/N}), \qquad X^N - 1 = \prod_{d\mid N} \Phi_d(X).$$

The polynomials Φ_N are monic with integer coefficients and irreducible over $\mathbb{Q}$, of degree $\varphi(N)$. Thus an integer polynomial vanishing at one primitive Nth root is divisible by Φ_N and vanishes at every primitive root. For a prime q, $\Phi_q = 1 + X + \cdots + X^{q-1}$; irreducibility follows from Eisenstein after substituting $X+1$.

If $R \in \mathbb{Z}[X]$ and ζ is primitive, the product $\prod_{\gcd(j,N)=1} R(\zeta^j)$ is an integer. One direct justification uses the determinant of multiplication by R on the free integer module $\mathbb{Z}[X]/(\Phi_N)$; over $\mathbb{C}$ its eigenvalues are exactly those values. If none vanish, the integer has absolute value at least 1. In the prime case, $\prod_{j=1}^{q-1}(1-\zeta^j) = \Phi_q(1) = q$, a useful lower bound against small coefficient estimates.

Proof core. Translate angle or periodicity information into a root of unity, identify its exact order, and use the minimal polynomial before comparing coefficients or norms.

Example and trap. A polynomial of degree below $q-1$ with rational coefficients cannot vanish at a primitive qth root unless it is zero. Irreducibility of $F(X)$ does not automatically imply irreducibility of $F(G(X))$: X^2+1 is irreducible over $\mathbb{Q}$, but substitutions need a separate argument. In positive characteristic dividing N, X^N-1 has repeated roots, so the characteristic-zero distinct-root arguments cannot be transferred unchanged.

Extended atlas CP-06

Reference notes 23

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2001-D1-P4 [R]; 2008-D2-P1 [R]; 2010-D2-P5 [R]; 2012-D1-P5 [R]; 2018-D1-P5 [R]

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T-AL-PELL Quadratic norms and Pell-type constructions

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Prerequisites Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]

For a positive nonsquare integer D, the conjugation $a+b\sqrt{D}\mapsto a-b\sqrt{D}$ preserves addition and multiplication. The norm is

$$N(a+b\sqrt{D})=a^2-Db^2,$$

and is multiplicative. Consequently, if integers u,v satisfy $u^2-Dv^2=1$ and $u+v\sqrt{D}>1$ with $v>0$, then $(u+v\sqrt{D})^k=u_k+v_k\sqrt{D}$ yields integer solutions of the same norm equation for every $k\geq 1$. This construction requires only one explicitly verified nontrivial unit; it does not require a general existence theorem.

Multiplication supplies a reversible recurrence

$$\begin{pmatrix}u_{k+1}\\ v_{k+1}\end{pmatrix}=\begin{pmatrix}u & Dv\\ v & u\end{pmatrix}\begin{pmatrix}u_k\\ v_k\end{pmatrix},$$

whose determinant is 1. Modulo any positive integer m the state space is finite and the transition invertible, so the orbit of $(1,0)$ is periodic from its first step. Some positive power therefore returns to $(1,0)$ modulo m, ensuring $m\mid v_k$.

A fractional linear map $(ar+b)/(cr+d)$ fixes a quadratic irrational r exactly when $cr^2+(d-a)r-b=0$, provided the denominator is nonzero. Comparing this polynomial with the minimal quadratic of r translates the fixed-point condition into coefficient ratios. Combining it with $ad-bc=1$ gives a Pell-type norm equation. Reversible recurrence modulo a denominator then restores integral coefficients.

Example. $8^2-7\cdot 3^2=1$, so powers of $8+3\sqrt{7}$ generate the needed integer pairs for $D=7$.

Traps. Finite-state recurrence gives eventual periodicity only in general; immediate return to the initial state uses invertibility. Check parity when reconstructing a,d from $a+d$ and $a-d$, and verify both nontriviality and $cr+d\neq 0$ at the end.

Extended atlas ANT-12

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.

2005-D2-P6 [R]

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T-AL-RECURRENCE Linear recurrences and generating functions

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Prerequisites Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]

For a constant-coefficient recurrence $u_{n+2}=Au_{n+1}+Bu_n$, a trial sequence r^n leads to $r^2-Ar-B=0$. Distinct characteristic roots r,s give $u_n=\alpha r^n+\beta s^n$, determined by the initial values; a repeated nonzero root gives $(\alpha+\beta n)r^n$. These formulas are verified by substitution and uniqueness of a recurrence with specified initial data. Degenerate cases, especially zero roots, are better handled directly.

The ordinary generating function is the formal series $U(X)=\sum_{n\geq 0}u_nX^n$. Formal multiplication uses finite convolution in each coefficient:

$$[X^n]U(X)V(X)=\sum_{j=0}^{n}u_jv_{n-j}.$$

Index shifts become multiplication by X, and coefficients weighted by n are encoded by $XU'(X)$. Hence a recurrence with polynomial coefficients often becomes a differential equation for U. An algebraic identity may then yield a second recurrence that makes integrality transparent. For example, $U=1+XU+XU^2$ and $u_0=1$ imply $u_{n+1}=u_n+\sum_{j=0}^{n}u_ju_{n-j}$, proving all u_n are integers.

Analytic use. A formal identity requires no convergence. To evaluate a generating series at a nonzero number, prove convergence there; termwise derivatives are justified inside its radius of convergence. Telescoping should first be written for a finite partial sum and then followed by a tail estimate.

T-AL-RECURRENCE Linear recurrences and generating functions (continued)

Example and trap. The recurrence $u_{n+2} = \frac{5}{2}u_{n+1} - u_n$, $u_0 = 0, u_1 = \frac{3}{2}$, gives $u_n = 2^n - 2^{-n}$. When converting its reciprocal into a difference along doubled indices, verify the algebra before claiming telescoping. A plausible pattern in the first few values does not prove either the formula or integrality.

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1996-D2-P2 [R]; 2002-D1-P3 [R]; 2002-D2-P3 [R]; 2008-D2-P3 [R]; 2015-D1-P2 [R]; 2015-D1-P3 [R]; 2019-D1-P4 [R]; 2022-D1-P3 [R]

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T-AL-DIFFERENCE Finite differences and shift-operator ideals

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Polynomial factorization and coprime divisibility [T-AL-FACTORIZATION]

The forward difference $\Delta P(X) = P(X+1) - P(X)$ lowers the degree of a nonconstant polynomial by one in characteristic zero: if P has leading term aX^d, then ΔP has leading term daX^{d-1}. Thus solving $\Delta P = Q$ among polynomials determines P uniquely up to a constant and can be done coefficient by coefficient. Newton's expansion in the basis $\binom{X}{k}$ makes this discrete integration especially simple.

For functions on $\mathbb{Z}$ or $\mathbb{R}$, let $T_h f(x) = f(x+h)$. Polynomial expressions $A(T_h)$ are finite linear combinations of shifts. If $A(T_h)f = B(T_h)f = 0$, then every polynomial combination $(UA + VB)(T_h)$ also annihilates f. The annihilating polynomials therefore form an ideal in $K[X]$, generated by their gcd. Bézout's identity can replace several long recurrences by one short recurrence. Negative shifts require Laurent polynomials, or an initial common translation that makes every exponent nonnegative.

If $a_0 f(x) + \cdots + a_d f(x+dh) = 0$ with $a_0 a_d \neq 0$, a zero interval longer than $d|h|$ propagates step by step in both directions. At each step all but one value are known, and the nonzero endpoint coefficient determines the remaining one.

Example. The annihilators $1+X$ and $1+X+X^2$ have gcd 1, so their simultaneous shift recurrences force $f = 0$.

Traps. The ideal of annihilators is defined for a fixed step unless the hypothesis supplies all steps. A root 0 of its generator needs separate exclusion before concluding that repeated powers of a root give a root of unity. Degree lowering can fail in characteristic p, since $\Delta X^p = 1$ there. Distinguish ordinary derivatives from finite differences throughout an induction.

Extended atlas ANT-04

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2007-D1-P5 [R]; 2010-D2-P5 [R]; 2011-D2-P4 [R]; 2020-D1-P4 [R]

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T-AL-RATIONALITY Rational linear relations and discrete-value rigidity

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Prerequisites Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]

The real numbers form a vector space over $\mathbb{Q}$, and the rational numbers are a subspace. Passing to the quotient $\mathbb{R}/\mathbb{Q}$ translates "this real number is rational" into "this vector is zero". Thus a statement that every nontrivial nonnegative rational combination of selected numbers is irrational is a statement excluding zero from their positive rational cone. This is weaker than linear independence, which excludes relations with arbitrary rational signs.

Elimination principle. Suppose no nontrivial nonnegative combination of $x_1, \ldots, x_r$ is rational. If an additional irrational y creates such a relation, its coefficient must be positive, so it can be normalized and solved for: $y = q - \sum_i b_i x_i$, $q \in \mathbb{Q}$, $b_i \geq 0$, with at least one $b_i > 0$. Substituting

T-AL-RATIONALITY Rational linear relations and discrete-value rigidity (continued)

several such relations into another nonnegative relation produces a contradiction. Positivity is what prevents complete cancellation of the remaining coefficients.

A second rigidity principle combines rational linear algebra with discreteness. A finite homogeneous linear system with rational coefficients has a basis of rational solutions after Gaussian elimination. Infinitely many homogeneous rational-coefficient constraints on a fixed finite-dimensional coefficient vector are equivalent to a finite subsystem, because their row span has finite dimension. After recovering rational coefficients, clearing a common denominator often creates an integer-valued error $E(n)$ tending to 0 along an unbounded sequence of integers. Eventually $|E(n)| < 1$, so $E(n) = 0$; a polynomial numerator with infinitely many zeros vanishes identically.

Example and trap. If $R \in \mathbb{Q}[X]$, $Q \in \mathbb{Q}[X] \setminus \{0\}$, $\deg R < \deg Q$, and $R(n)/Q(n)$ is an integer at infinitely many unbounded integer arguments, the rational function must vanish. But a sum of irrational numbers may be rational, as $\sqrt{2} + (2 - \sqrt{2}) = 2$ shows. Rationality of a function at finitely many points does not imply rational coefficients or a polynomial identity.

Extended atlas ANT-05

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P3 [R]; 1996-D2-P2 [R]; 2006-D1-P4 [R]

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T-AL-GREEDY Greedy products, tail bounds, and integer descent

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Prerequisites Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]

For $M > 1$, the finite identity

$$\prod_{j=0}^{r-1}(1 + M^{-2^j}) = \frac{1 - M^{-2^r}}{1 - M^{-1}}$$

follows by repeatedly factoring $1 - t^2 = (1 - t)(1 + t)$. Letting $r \to \infty$ gives a product equal to $M/(M-1)$. This is an exact tail bound for multiplicative expansions whose successive denominators at least square.

For $1 < \alpha < 2$, define a remaining ratio $\theta_0 = \alpha$ and choose n_k as the least positive integer with $1 + 1/n_k < \theta_{k-1}$. Set $\theta_k = \theta_{k-1}/(1 + 1/n_k)$. Minimality gives

$$1 + \frac{1}{n_k} < \theta_{k-1} \le 1 + \frac{1}{n_k - 1}.$$

Consequently $1 < \theta_k \le 1 + 1/(n_k^2 - 1)$, and the next choice satisfies $n_{k+1} \ge n_k^2$. Since $n_1 \ge 2$, the remaining ratio tends to 1, proving convergence of the constructed product to α.

Uniqueness core. A larger early factor would already exhaust the required ratio, whereas a smaller early factor cannot be compensated by the largest permitted tail; use the exact infinite product above to justify this second statement. Both inequalities are needed.

Rationality core. When $\theta_k = p_k/q_k$ with integer numerators and denominators propagated without cancellation, strict nonterminal choices make the positive integer $p_k - q_k$ strictly decrease. This cannot continue indefinitely, so a rational input eventually reaches the extremal tail. Conversely that explicit tail is rational.

Example and trap. Starting with $M = 2$ gives $(1 + 1/2)(1 + 1/4)(1 + 1/16)\cdots = 2$. The strict sign in the greedy choice determines endpoint behavior. Convergence of the factors to 1 alone never proves convergence of an infinite product; here the tail estimate supplies the missing control.

Extended atlas ANT-14

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1997-D1-P4 [R]

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T-AL-COMPLEXANALYSIS Holomorphic bounds, poles, and logarithmic-branch sums

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]

If f is holomorphic on a neighborhood of a closed disk, Cauchy's formula gives $f(0) = (2\pi)^{-1}\int_0^{2\pi} f(e^{it})\,dt$ for the unit disk, so $|f(0)| \le \max_{|z|=1}|f(z)|$. The maximum-modulus principle extends this boundary bound throughout the disk. A real harmonic function continuous on a closed bounded domain attains its minimum and maximum on the boundary unless constant; real parts of holomorphic functions, and sums with antiholomorphic conjugates, are harmonic.

For a monic polynomial P of degree n, define $P^*(z) = z^n\overline{P(1/\bar z)}$ by expanding coefficients. It is a polynomial with $P^*(0) = 1$, and $|P^*(z)| = |P(z)|$ on the unit circle. Applying the boundary bound to fP^* proves $|f(0)| \le \max_{|z|=1}|f(z)P(z)|$.

Branch sums. For $z \neq 0, 1$, choose a local logarithm w and write $S_m(z) = \sum_{k\in\mathbb{Z}}(w + 2\pi ik)^{-m}$, $m \ge 2$. The series and its derivative converge uniformly on compact subsets away from the poles, by comparison with $\sum |k|^{-m}$. Changing the logarithm merely reindexes the sum. The cotangent partial-fraction expansion gives $S_2 = z/(z-1)^2$, and termwise differentiation yields

$$S_{m+1} = -\frac{z}{m}S_m', \qquad S_4 = \frac{z(z^2+4z+1)}{6(z-1)^4}.$$

A meromorphic function on the sphere is rational: subtract all principal parts, leaving an entire function bounded at infinity, then apply Liouville's theorem.

Traps. Pointwise convergence does not justify termwise complex differentiation. The first-power branch sum requires a specified symmetric summation convention. Reflected coefficients must be conjugated. For an unbounded strip, apply the harmonic principle first to bounded rectangles and control their horizontal boundaries uniformly.

Extended atlas CP-12

Reference notes 3; 6

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2004-D1-P6 [R]; 2009-D1-P4 [R]; 2020-D1-P4 [R]; 2021-D2-P7 [R]

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T-AL-FOURIER Roots-of-unity filters and positive trigonometric kernels

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]

Let $\zeta = e^{2\pi i/N}$. The geometric-series formula yields

$$\sum_{j=0}^{N-1}\zeta^{jm} = \begin{cases} N, & N \mid m, \\ 0, & N \nmid m. \end{cases}$$

For $P(z) = \sum_m a_m z^m$, averaging $P(\zeta^j)$ therefore selects the coefficients whose indices are multiples of N. Multiplying by ζ^{-jr} selects a different residue class. Continuous orthogonality similarly gives $a_m = (2\pi)^{-1}\int_0^{2\pi} P(e^{it})e^{-imt}\,dt$ and Parseval's identity $(2\pi)^{-1}\int |P(e^{it})|^2 dt = \sum_m |a_m|^2$.

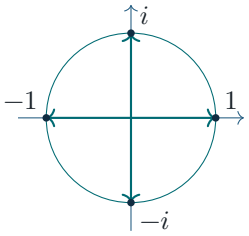
A nonnegative normalized kernel contracts the supremum norm: if $K \ge 0$ and $(2\pi)^{-1}\int K = 1$, then the absolute value of its circular average against a function bounded by M is at most M. The elementary Fejér kernel is

$$F_k(e^{it}) = \left|1 + e^{it} + \cdots + e^{i(k-1)t}\right|^2 = \sum_{|j|<k}(k - |j|)e^{ijt}.$$

Its integral divided by 2π is k, not 1. For a decreasing convex coefficient sequence $c_0 = 1$, $c_n = 0$, set $c_{n+1} = 0$ and $d_k = c_{k-1} - 2c_k + c_{k+1} \ge 0$. Two finite telescopings show $\sum_{j=-n}^{n} c_{|j|}e^{ijt} = \sum_{k=1}^{n} d_k F_k(e^{it}) \ge 0$, with normalized integral 1. A nonzero endpoint c_n is separated first as a convex multiple of the original polynomial.

T-AL-FOURIER Roots-of-unity filters and positive trigonometric kernels (continued)

Example and trap. Averaging a polynomial at rotated Nth roots can isolate its constant and leading terms and force equality in a modulus bound. Positivity of a kernel is a proof obligation, not a consequence of real-valuedness. Check normalization before applying contraction. In sparse Fourier arguments, placing phases in the same short arc can replace Fourier inversion with a pigeonhole estimate on real parts.
The four arrows below are the fourth roots of unity; opposite vectors cancel exactly. This is the filter at exponent $m = 1$ for $N = 4$.



$$1 + i + (-1) + (-i) = 0$$

Extended atlas CP-06

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1999-D2-P2 [R]; 2007-D1-P6 [R]; 2009-D1-P4 [R]; 2016-D1-P5 [R]; 2017-D1-P5 [R]

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T-AL-PRIMEGROWTH Factorial growth and the prime-number input

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Prime valuations, divisors, and factorials [T-AL-VALUATION]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
The integral comparison for the increasing function $\log t$ gives

$$n \log n - n + 1 \le \log(n!) \le (n+1)\log(n+1) - n.$$

Thus $\log(n!) = n \log n - n + O(\log n)$, enough to show that $n!$ eventually exceeds C^n for each fixed $C > 0$. This elementary estimate often proves finiteness after a divisibility condition produces an upper bound of exponential size. Multinomial coefficients also give $(a_1 + \cdots + a_r)!/(a_1! \cdots a_r!) \in \mathbb{Z}$ and, by summing all multinomial terms, $(rn)!/(n!)^r \le r^{rn}$.
Let $L(N) = \operatorname{lcm}(1, \dots, N)$. Prime factorization gives the exact identity

$$\log L(N) = \sum_{p^j \le N} \log p = \psi(N).$$

The prime number theorem implies $\psi(N)/N \to 1$. This is a substantial theorem, not an elementary consequence of Legendre's formula. In a proof using it, for each fixed $\varepsilon > 0$ one obtains $L(N) < \exp((1+\varepsilon)N)$ only after N is sufficiently large. A proof relying on this input should identify it explicitly.
Application pattern. If a positive rational integral $J_k = \int_0^1 P(x)^{2k}\,dx$ has denominator dividing $L(2kd+1)$, then $J_k \ge 1/L(2kd+1)$. Comparing with $\|P\|_\infty^{2k}$ and taking $2k$th roots converts denominator growth into an analytic lower bound.
Traps. A limit of strict inequalities need only yield a weak inequality. For an integer polynomial of positive degree d, to exclude equality at e^{-d}, separately note that its maximum on $[0,1]$ is algebraic: endpoints and real critical points are algebraic, and polynomial values there are algebraic. Then use the transcendence of e. Bounds proved for sufficiently large integers do not justify claims about every small integer.
Reference notes 7

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2012-D2-P3 [R]; 2015-D2-P10 [R]; 2023-D2-P10 [R]

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T-AL-COXETER Adjacent-transposition relations and finite presentations

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Prerequisites Groups, orders, generators, and quotients [T-AL-GROUP]; Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]

The adjacent transpositions $s_i = (i\ i+1)$ generate S_n and satisfy

$$s_i^2 = e, \qquad s_i s_j = s_j s_i\ (|i-j|>1), \qquad s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}.$$

These relations are also a presentation of S_n: any group generated by elements with these relations is a quotient of S_n. The generation assertion follows by moving the desired entry into its place by neighboring swaps. The relations follow by checking the two, three, or four moved positions.

Why the relations suffice. In the abstract group defined by them, repeated use of commutation and the braid relation moves an occurrence of s_{n-1} through a word in lower-index generators. Together with $s_{n-1}^2 = e$, this gives a normal form uv, where u uses only $s_1, \dots, s_{n-2}$ and v is one of $1, s_{n-1}, s_{n-1}s_{n-2}, \dots, s_{n-1}\cdots s_1$. This reduction can be checked when appending each generator and then iterated over a word. Inductively there are at most $(n-1)!\,n = n!$ normal forms. The natural surjection to S_n has at least $n!$ distinct images, hence is an isomorphism.

Consequence. For any elements a_i satisfying those same relations, the substitution $s_i \mapsto a_i$ defines a homomorphism. Since $s_1 \cdots s_{n-1}$ is an n-cycle, $(a_1 \cdots a_{n-1})^n = e$. The order may be a proper divisor of n in a quotient.

Trap. Proving that chosen permutations satisfy the relations only produces a homomorphism from the presented group; it does not prove injectivity. The normal-form count is what closes that gap. Distant commutation alone is insufficient: the adjacent braid relation is essential.

Reference notes 1

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2011-D2-P5 [R]

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Counting, geometry and functional equations

T-DS-COUNTING Finite counting and incidence identities

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A finite counting argument begins with a specified set of objects. Disjoint classes add; independent successive choices multiply. The number of ordered selections of k distinct elements from an n-element set is $n!/(n-k)!$; forgetting their order gives $\binom{n}{k}$. In particular,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}, \qquad \sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

The first identity separates subsets according to whether a distinguished element is present. Use $\binom{n}{k} = 0$ when $k < 0$ or $k > n$.

For a relation $R \subseteq A \times B$ between finite sets,

$$|R| = \sum_{a \in A} |\{b : (a,b) \in R\}| = \sum_{b \in B} |\{a : (a,b) \in R\}|.$$

The same identity holds for weighted incidences: sum the same weight on both sides. If every bad pair of contestants misses a common problem, then the number of bad pairs is at most the sum, over problems, of the number of pairs missing that problem. This is an upper bound because a pair may be counted repeatedly.

For finite sets $E_1, \dots, E_m$, inclusion–exclusion is

$$\left| \bigcup_i E_i \right| = \sum_{\emptyset \neq I \subseteq \{1,\dots,m\}} (-1)^{|I|+1} \left| \bigcap_{i \in I} E_i \right|.$$

Proof core: an object occurring in exactly $r \geq 1$ sets contributes $\sum_{j=1}^{r} (-1)^{j+1} \binom{r}{j} = 1$. A union bound retains only the singleton terms. The pigeonhole principle says that a map from a larger finite set to a smaller one cannot be injective; always identify the map and the capacity of each fibre. Distinguish ordered pairs from unordered pairs, distinct elements from repeated choices, and a count from an upper estimate. These distinctions determine equality cases.

Among permutations of n labelled objects, a prescribed relative order of k distinguished objects occurs exactly $n!/k!$ times: choose their k positions, insert them in that order, and arrange the remaining objects. For a nonempty finite set, toggling one fixed element pairs even-sized and odd-sized subsets.

Extended atlas CMB-02; CMB-07

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P6 [R]; 1997-D1-P6 [R]; 1998-D2-P3 [R]; 1998-D2-P4 [R]; 1999-D1-P2 [R]; 1999-D1-P5 [R]; 1999-D2-P2 [R]; 1999-D2-P6 [R]; 2001-D1-P1 [R]; 2002-D1-P3 [R]; 2002-D2-P2 [R]; 2004-D1-P1 [R]; 2004-D1-P2 [R]; 2004-D1-P4 [R]; 2004-D1-P5 [R]; 2004-D2-P6 [R]; 2005-D1-P2 [R]; 2007-D1-P4 [R]; 2007-D2-P5 [R]; 2008-D1-P4 [R]; 2008-D1-P6 [R]; 2008-D2-P4 [R]; 2009-D1-P3 [R]; 2011-D1-P4 [R]; 2011-D1-P5 [R]; 2011-D2-P2 [R]; 2012-D1-P1 [R]; 2012-D2-P3 [R]; 2012-D2-P5 [R]; 2013-D1-P3 [R]; 2013-D2-P2 [R]; 2013-D2-P3 [R]; 2013-D2-P4 [R]; 2013-D2-P5 [R]; 2014-D1-P2 [R]; 2014-D1-P5 [R]; 2014-D2-P1 [R]; 2014-D2-P5 [R]; 2015-D2-P8 [R]; 2016-D1-P4 [R]; 2016-D2-P9 [R]; 2017-D1-P4 [R]; 2017-D1-P5 [R]; 2017-D2-P8 [R]; 2018-D1-P5 [R]; 2018-D2-P8 [R]; 2019-D2-P8 [R]; 2020-D1-P1 [R]; 2020-D2-P7 [R]; 2020-D2-P8 [R]; 2021-D1-P2 [R]; 2021-D1-P3 [R]; 2022-D1-P3 [R]; 2022-D1-P4 [R]; 2022-D2-P5 [R]; 2022-D2-P8 [R]; 2023-D1-P5 [R]; 2023-D2-P9 [R]; 2023-D2-P10 [R]; 2024-D1-P3 [R]; 2024-D1-P4 [R]; 2024-D2-P6 [R]; 2024-D2-P9 [R]

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T-DS-INDUCTION Strong induction, state recurrences, and constructive invariants

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Prerequisites Finite counting and incidence identities [T-DS-COUNTING]

To prove $P(n)$ for integers $n \geq n_0$, establish the base cases and show that the earlier admissible cases imply $P(n)$. Strong induction permits all smaller parameters. A step $n \mapsto n+3$ needs three residue-class bases; one base does not suffice. For pairs of parameters, use a stated well-founded order, such as their sum or lexicographic order. A strengthened statement is often necessary because deletion changes more than one parameter.

For a recursive construction, the induction hypothesis must record the invariant needed for the next choice. If vectors v_i sum to zero and s is the sum already selected, then

$$\sum_{i \text{ unused}} \langle s, v_i \rangle = -\|s\|^2 \leq 0.$$

Some unused vector has nonpositive inner product with s, so choosing it gives $\|s+v_i\|^2 \leq \|s\|^2 + \|v_i\|^2$. Induction propagates this stronger bound, not merely the final desired estimate.

For constrained words, a state must retain exactly the information that controls the allowed next letters. If a_n counts words ending in a repeated pair and b_n counts the other admissible endings, derive each transition by an explicit extension or deletion. Writing the count vector $u_{n+1} = Au_n$ is valid only when the states are disjoint, exhaustive, and every extension is counted once. Two sequences with the same order-r recurrence agree after r matching initial values; their closed forms need not be found.

An induction can also prove a finite reduction: each legal simplification must decrease a nonnegative integer, such as length. Prove that the reduced object remains in the permitted class and that its solution lifts back. A decreasing quantity guarantees termination, but says nothing by itself about uniqueness of the resulting normal form.

Extended atlas CMB-05; CMB-14

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P3 [R]; 1994-D2-P5 [R]; 1996-D1-P4 [R]; 1996-D1-P6 [R]; 1997-D1-P6 [R]; 1998-D1-P3 [R]; 1998-D2-P3 [R]; 1999-D1-P4 [R]; 1999-D1-P5 [R]; 1999-D2-P2 [R]; 1999-D2-P4 [R]; 1999-D2-P5 [R]; 2000-D2-P1 [R]; 2001-D2-P2 [R]; 2001-D2-P3 [R]; 2002-D1-P3 [R]; 2002-D2-P3 [R]; 2003-D1-P6 [R]; 2004-D1-P2 [R]; 2004-D1-P5 [R]; 2005-D1-P2 [R]; 2006-D2-P1 [R]; 2007-D2-P6 [R]; 2009-D1-P5 [R]; 2010-D1-P3 [R]; 2010-D2-P2 [R]; 2011-D2-P1 [R]; 2011-D2-P4 [R]; 2012-D2-P5 [R]; 2013-D2-P5 [R]; 2014-D1-P3 [R]; 2014-D2-P5 [R]; 2016-D1-P4 [R]; 2016-D2-P9 [R]; 2017-D1-P3 [R]; 2018-D2-P8 [R]; 2020-D1-P1 [R]; 2020-D1-P4 [R]; 2021-D1-P3 [R]; 2024-D1-P4 [R]; 2024-D2-P8 [R]

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T-DS-GRAPH Graphs, walks, cycles, and local degree counts

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Prerequisites Finite counting and incidence identities [T-DS-COUNTING]

A finite simple undirected graph has a vertex set V and an edge set consisting of unordered pairs of distinct vertices. Its degree $d(v)$ counts the neighbours of v. Counting edge endpoints gives

$$\sum_{v \in V} d(v) = 2|E|.$$

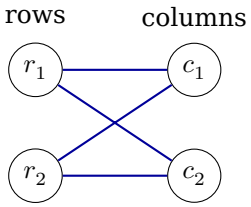
In particular, the number of vertices of odd degree is even.

An ordered walk of length two is (u, v, w) with $uv, vw \in E$; $u = w$ is permitted. There are $\sum_v d(v)^2$ such walks. Walks with distinct endpoints through v number $d(v)(d(v)-1)$, while unordered pairs of neighbours number $\binom{d(v)}{2}$.

A path has no repeated vertices; a cycle closes a path of at least three vertices. Every forest with N vertices and c components has $N - c$ edges: remove a leaf from each nontrivial tree inductively. A connected graph contains a spanning tree: delete one edge from each cycle until none remains, preserving connectedness. Thus a connected graph on N vertices has at least $N-1$ edges. Therefore a graph with at least N edges contains a cycle. A graph of minimum degree at least two also contains a cycle: a longest path has an endpoint whose extra neighbour lies earlier on the path.

T-DS-GRAPH Graphs, walks, cycles, and local degree counts (continued)

A bipartite graph splits into two vertex classes with every edge crossing between them; all its cycles have even length. Encode a marked matrix cell (i, j) as the edge between row r_i and column c_j. A cycle then alternates rows and columns, exactly the desired sequence of marked cells. The encoding, rather than the geometry of a drawn grid, is the relevant structure.
In a two-colouring of the edges of a complete graph, an unordered triple is either monochromatic or has two mixed-colour corners. Consequently the number of nonmonochromatic triangles is $\frac{1}{2}\sum_v d_R(v)d_B(v)$. This identity is a useful example of counting the same object with its multiplicity.
Never confuse a closed walk with a simple cycle, or an average-degree bound with a minimum-degree bound.
A directed graph uses ordered edges. A proper vertex colouring assigns different colours at the endpoints of each edge, regardless of its orientation. If its edges are covered by r classes, none containing a directed walk of length two (including a return), encode each vertex by the classes of its outgoing edges. The two endpoints of any edge have different encodings: its class occurs at its source and cannot occur at its target. Hence this gives a proper colouring with at most 2^r colours. Overlapping edge classes are allowed, but each must satisfy the stated path restriction.



A four-cycle encodes four distinct marked cells; crossing edges in this drawing do not create additional vertices.
Extended atlas CMB-03; CMB-11

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1999-D1-P5 [R]; 2001-D2-P4 [R]; 2004-D2-P6 [R]; 2009-D1-P3 [R]; 2011-D2-P2 [R]; 2015-D1-P5 [R]; 2017-D1-P4 [R]; 2022-D1-P4 [R]; 2022-D2-P5 [R]; 2024-D2-P9 [R]

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T-DS-GEOMETRY Euclidean geometry, simplices, and barycentric weights

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Prerequisites Inner products, orthogonality and polarization [T-LA-INNER]; Determinants, block identities and inverses [T-LA-DETERMINANT]
For an integer $d \geq 1$, in $\mathbb{R}^d$ fix the Euclidean inner product and norm. Squared distances satisfy

$$\|x - y\|^2 = \|x\|^2 + \|y\|^2 - 2\langle x, y\rangle.$$

Thus two unit vectors have distance greater than $\sqrt{2}$ exactly when their inner product is negative.
A regular simplex of $d+1$ unit vectors has pairwise inner product $-1/d$: start with $e_i - \frac{1}{d+1}\mathbf{1}$ in the sum-zero hyperplane of $\mathbb{R}^{d+1}$ and normalize. Its pairwise distance is $\sqrt{2 + 2/d}$.
An affine simplex $S = \operatorname{conv}(p_0, \dots, p_d)$ is nondegenerate when $p_1 - p_0, \dots, p_d - p_0$ are linearly independent. Its volume is

$$\operatorname{vol}(S) = \frac{1}{d!}\left|\det(p_1 - p_0, \dots, p_d - p_0)\right|.$$

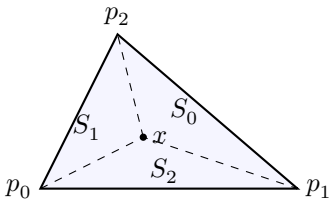
Every point of its affine span has unique barycentric coordinates $x = \sum_i \lambda_i p_i$, $\sum_i \lambda_i = 1$. It lies in S iff all $\lambda_i \geq 0$, and in its interior iff all are positive. For an interior point, replacing p_i by x forms a simplex S_i with $\lambda_i = \operatorname{vol}(S_i)/\operatorname{vol}(S)$, by multilinearity of the determinant. Hence these volumes sum to $\operatorname{vol}(S)$.
The circumcentre c is the unique solution of the linear equations

$$2\langle c, p_i - p_0\rangle = \|p_i\|^2 - \|p_0\|^2 \quad (1 \leq i \leq d).$$

They come from cancelling $\|c\|^2$ in equal squared distances. This linearization is safer than manipulating separate square roots. Orthogonal projections do not increase distance, while a general

T-DS-GEOMETRY Euclidean geometry, simplices, and barycentric weights (continued)

invertible affine map need not preserve angles or circles. In volume arguments, track its absolute determinant. Outside the simplex, barycentric coefficients may be negative; unsigned sub-simplex volumes no longer give those coefficients.



Extended atlas GEO-02; GEO-03; GEO-06; GEO-11

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1996-D2-P4 [R]; 1998-D2-P5 [R]; 2000-D2-P1 [R]; 2001-D2-P3 [R]; 2002-D1-P1 [R]; 2002-D2-P4 [R]; 2004-D1-P4 [R]; 2004-D2-P2 [R]; 2004-D2-P3 [R]; 2005-D2-P5 [R]; 2006-D2-P1 [R]; 2008-D1-P4 [R]; 2008-D2-P2 [R]; 2009-D1-P5 [R]; 2009-D2-P1 [R]; 2011-D2-P5 [R]; 2014-D1-P5 [R]; 2014-D2-P4 [R]; 2015-D1-P5 [R]; 2017-D2-P10 [R]; 2018-D2-P10 [R]; 2019-D2-P10 [R]; 2020-D1-P3 [R]; 2022-D2-P8 [R]; 2023-D2-P9 [R]

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T-DS-FUNCTIONAL Functional equations: domain, injectivity, and regularity

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Prerequisites Continuity, compactness, and order [T-AN-CONTINUITY]

Treat a functional equation as a family of identities, one for each admissible tuple of arguments. Before substituting 0, a negative value, a quotient, or an image $f(x)$, check that every argument remains in the domain. A proposed formula is a solution only after substitution verifies the original equation and its range requirements.

Injectivity permits $f(u) = f(v) \Rightarrow u = v$; strict monotonicity implies injectivity, but weak monotonicity does not. Surjectivity permits replacing an arbitrary range value by $f(x)$; it must first be proved. Equations that compare $f(x + y)$ with $f(x)$ can force monotonicity. A positive periodic monotone function is constant: for a positive period p and $x < y$, choose n with $x + np \geq y$ and trap $f(y)$ between equal endpoint values.

For the additive equation $g(x + y) = g(x) + g(y)$ on $\mathbb{R}$, algebra gives $g(q) = qg(1)$ for rational q. Continuity at one point then yields $g(x) = cx$ on $\mathbb{R}$ by rational approximation. Monotonicity is another sufficient regularity assumption. Without regularity, one cannot simply assert linearity. Likewise a continuous rational-valued function on an interval is constant, since a nontrivial interval between two values contains an irrational number and the intermediate value theorem applies.

A useful rigidity device is a uniform estimate $|f(u) - f(v)| \leq \varepsilon_n$ for every n, with $\varepsilon_n \to 0$. Fix u, v before choosing auxiliary substitutions depending on n; then $f(u) = f(v)$. For a contradiction, a bound such as $f(x + h) \leq f(x) - c$ with fixed $h, c > 0$ can be iterated until positivity fails. Do not differentiate an arbitrary functional equation until differentiability has been established or a justified bootstrap supplies it.

Extended atlas M-FE-02; M-FE-05; M-FE-06

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1998-D2-P4 [R]; 1999-D1-P3 [R]; 1999-D1-P4 [R]; 1999-D2-P4 [R]; 2000-D1-P1 [R]; 2000-D2-P5 [R]; 2001-D2-P5 [R]; 2002-D1-P2 [R]; 2006-D2-P2 [R]; 2007-D2-P1 [R]; 2008-D1-P1 [R]; 2018-D1-P4 [R]; 2023-D1-P1 [R]

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T-DS-METRIC Metric contractions, covers, and nearest points

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Prerequisites Continuity, compactness, and order [T-AN-CONTINUITY]; Inner products, orthogonality and polarization [T-LA-INNER]

For a nonempty bounded subset E of Euclidean space, its diameter is $\operatorname{diam} E = \sup_{x,y\in E} \|x-y\|$. A map $f : E \to F$ is L-Lipschitz if $\|f(x)-f(y)\| \le L\|x-y\|$ for every pair. Here a contraction in a covering problem may mean $L = 1$; the strict constant $L < 1$ used in the contraction mapping theorem is a stronger hypothesis. Lipschitz maps are continuous and satisfy $\operatorname{diam} f(A) \le L \operatorname{diam} A$ for $A \subseteq E$.

Define the finite-cover content

$$C(E) = \inf\left\{ \sum_{i=1}^{m} \operatorname{diam} E_i : E \subseteq \bigcup_{i=1}^{m} E_i \right\}.$$

Then $C(f(E)) \le LC(E)$: first intersect each covering set with the domain E, map these intersections, and then take the infimum. This intersection is essential when f is not defined outside E. For a line segment I, $C(I) = \text{length}(I)$: project every covering set onto its line and enclose the projection in an interval of length at most its diameter; finite interval covers have total length at least that of I. No countable-cover or small-diameter condition is imposed here, so this is not automatically Hausdorff measure.

A nonempty closed set $A \subseteq \mathbb{R}^d$ has a nearest point to every b: a minimizing sequence can be confined to a closed bounded ball, hence has a convergent subsequence whose limit is in A. Uniqueness can fail, for example at the centre of a circle. If a is one nearest point and z is strictly between a and b, then a is the unique nearest point to z. The triangle inequality and its Euclidean equality case prove this: another minimizer would also be nearest to b and collinear with b, z in the same direction, forcing it to equal a. Thus such unique-nearest-point locations approach every b. The nonempty assumption cannot be omitted.

Extended atlas GEO-08; GEO-10

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1996-D1-P6 [R]; 1997-D1-P5 [R]; 1998-D2-P5 [R]; 2003-D2-P3 [R]; 2005-D2-P5 [R]; 2006-D2-P2 [R]; 2020-D1-P3 [R]

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T-DS-CONVEX Convex bodies, supporting functionals, and packing

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Prerequisites Continuity, compactness, and order [T-AN-CONTINUITY]; Inner products, orthogonality and polarization [T-LA-INNER]

A set $K \subseteq \mathbb{R}^d$ is convex if it contains $(1-t)x + ty$ whenever $x, y \in K$ and $0 \le t \le 1$. A convex body is compact, convex, and has nonempty interior. The convex hull of a finite set consists of its convex combinations. A point p outside a compact convex set C can be strictly separated by a linear functional: choose a nearest point $q \in C$, put $u = p - q$, and differentiate $\|p - q - t(x-q)\|^2$ at $t = 0$ to get

$$\langle u, x\rangle \le \langle u, q\rangle < \langle u, p\rangle \qquad (x \in C).$$

This supplies the needed finite-dimensional separation result directly.

If $K = -K$ is a convex body containing 0 in its interior, its gauge $\|x\|_K = \inf\{t > 0 : x \in tK\}$ is a norm. Positive homogeneity follows by scaling, definiteness by boundedness, symmetry by $K = -K$, and the triangle inequality by convexity. Without symmetry the gauge need not be a norm. For nonnegative scalars a, b, convexity gives $aK + bK = (a+b)K$; volume scales as $\operatorname{vol}(tK) = t^d \operatorname{vol}(K)$. Choose boundary points separated by at least ε in this norm. The translates $p_i + (\varepsilon/2)\operatorname{int} K$ are disjoint. They lie in the shell between $(1-\varepsilon/2)K$ and $(1+\varepsilon/2)K$, for $0 < \varepsilon < 1$, by the reverse triangle inequality. Hence

$$m(\varepsilon/2)^d \le (1+\varepsilon/2)^d - (1-\varepsilon/2)^d.$$

This finite bound permits a maximal separated set. Maximality makes it a boundary net; separation then tests whether its convex hull contains a prescribed smaller copy of K. Always distinguish max-

T-DS-CONVEX Convex bodies, supporting functionals, and packing (continued)

imal by inclusion from maximum cardinality. Local perturbations of an extremal inscribed ellipse additionally need a uniform positive clearance from the body's boundary, obtained by compactness before selecting the perturbation size.

Extended atlas GEO-04; GEO-13

Reference notes 20

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1996-D2-P4 [R]; 2008-D1-P4 [R]; 2015-D1-P5 [R]; 2020-D1-P3 [R]; 2023-D2-P9 [R]

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T-DS-DYNAMICS Iteration, fixed points, and periodic orbits

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Prerequisites Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Continuity, compactness, and order [T-AN-CONTINUITY]

For a self-map $f : X \to X$, write $f^0 = \mathrm{id}$ and $f^{n+1} = f \circ f^n$. An orbit is $x_n = f^n(x_0)$. Repetition $x_i = x_j$ with $i < j$ makes the orbit eventually periodic; injectivity of f is not needed. A point has least period r when $f^r(x) = x$ and no smaller positive iterate fixes it. If $f^n(x) = x$, then $r \mid n$, by Euclidean division of n by r. Thus solving $f^n(x) = x$ does not by itself count points of least period n.

For real recurrences, first find an invariant interval, then monotonicity or another trapping estimate. A bounded monotone sequence converges; continuity at its limit justifies $L = f(L)$. A fixed-point equation only lists candidates: for $f(x) = x^2$ on $[0, 1]$, both 0 and 1 are fixed, but every starting point in $(0, 1)$ converges to 0. A conjugacy $h \circ f = g \circ h$, with bijective h, transfers iterates exactly: $h \circ f^n = g^n \circ h$. It must be valid on the whole orbit.

An increasing self-map of a nonempty compact real interval $[u, v]$ has a fixed point even without continuity. For $A = \{x : f(x) \geq x\}$, let $a = \sup A$. Monotonicity gives $f(a) \geq a$, so $f(f(a)) \geq f(a)$; hence $f(a) \in A$ and $f(a) \leq a$. A decreasing discontinuous map need not have a fixed point.

Finally, a nonempty countable compact metric space has an isolated point. Otherwise enumerate its points and choose nested nonempty closed balls, radii tending to zero, successively avoiding the next point. Compactness supplies a point in their intersection, contradicting the enumeration. This is useful for a continuous self-map of a compact metric space with a closed orbit: if an infinite orbit has no repetition and x_m is its earliest accumulation point, continuity makes every later orbit point an accumulation point. Removing the finitely many earlier isolated points leaves a countable compact space without isolated points, a contradiction.

Extended atlas M-FE-01; M-FE-03

Reference notes 11

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1998-D1-P3 [R]; 1998-D2-P3 [R]; 1999-D1-P4 [R]; 2000-D1-P1 [R]; 2002-D1-P4 [R]; 2004-D1-P2 [R]; 2010-D1-P3 [R]; 2017-D2-P9 [R]; 2018-D2-P7 [R]; 2023-D1-P1 [R]; 2024-D2-P8 [R]

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T-DS-EXTREMAL Extremal choice, exchange arguments, and invariants

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Prerequisites Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]

An extremal argument chooses an admissible object minimizing or maximizing a specified quantity, then proves that any forbidden feature would allow improvement. Existence is automatic for a nonempty finite set of objects. For an infinite family, justify attainment by compactness and continuity, a well-order on nonnegative integers, or another explicit argument. A supremum need not be attained: $(0, 1)$ has supremum 1 and no largest element.

T-DS-EXTREMAL Extremal choice, exchange arguments, and invariants (continued)

A potential Φ taking nonnegative integer values forces a repair process to terminate if every legal repair decreases Φ. For example, partition three equal-sized classes into triples and count incompatible pairs within triples. A swap of two members of the same class changes only their incidences inside the two affected triples. If both moved members like all their new partners and one previously disliked a partner, the potential strictly decreases. One must prove that an admissible improving swap exists whenever the potential is positive; termination alone does not produce that fact.

An invariant remains unchanged by every move, whereas a monovariant changes in one direction. To count objects modulo two, an involution ι with $\iota^2 = \mathrm{id}$ partitions a finite set into fixed points and pairs, giving

$$|X| \equiv |\{x : \iota(x) = x\}| \pmod 2.$$

A fixed-point-free involution proves even cardinality. Define the distinguished feature being changed canonically and verify that it is still selected after the first move; otherwise a plausible pairing may fail to be an involution.

For rewriting words modulo relations, the equivalence must be a congruence: replacing a subword inside either left or right context preserves equivalence. A proof may temporarily increase length before reaching a shorter equivalent word. Consequently failure of a greedy local shortening rule does not establish minimality. State the actual decreasing quantity between completed reduction stages, not at every intermediate rewrite.

Extended atlas CMB-01; CMB-09; CMB-12; CMB-13

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1998-D2-P4 [R]; 1999-D2-P5 [R]; 2008-D1-P4 [R]; 2011-D2-P2 [R]; 2014-D1-P5 [R]; 2014-D2-P4 [R]; 2016-D1-P4 [R]; 2020-D1-P3 [R]; 2023-D1-P5 [R]; 2024-D2-P6 [R]

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T-DS-SETS Finite set systems and antichains

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Prerequisites Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]

A family $\mathcal{F}$ of subsets of an n-element set is an antichain when no distinct members contain one another. A maximal chain has one set of every size and is obtained by ordering the n elements. An r-element set belongs to $r!(n-r)!$ such chains. Since an antichain meets each chain at most once, double counting yields the Lubell inequality

$$\sum_{A \in \mathcal{F}} \frac{1}{\binom{n}{|A|}} \le 1, \qquad |\mathcal{F}| \le \binom{n}{\lfloor n/2 \rfloor}.$$

All sets of a fixed middle size attain the latter bound. If positive weights are assigned to the underlying elements, subsets with the same weight sum form an antichain: strict containment strictly increases that sum. Positivity is essential; zero weights destroy this implication.

A family is intersecting if any two of its members meet, and union-closed if the union of any two members belongs to the family. Pairwise intersection need not imply a common element: $\{1,2\},\{2,3\},\{1,3\}$ are a minimal counterexample. For intersecting families of bounded finite sizes, it is often useful to fix a finite reference set B and partition the family by its traces $A \cap B$. There are only finitely many traces. For two disjoint nonempty traces, remove them; any required intersection of the original sets must occur outside B. This reduces cardinality parameters and makes strong induction applicable.

A set family can instead be described by 0–1 incidence vectors. Coordinate sums count element frequencies, while unions correspond to coordinatewise maximum. Counting incidences requires care when identical sets arise from different constructions: either retain labels as part of the objects or prove the construction injective. For a disjoint partition of the ground set, write each member as the union of its traces and check which choices remain independent. Do not infer that every choice of traces occurs merely because each trace occurs separately.

Extended atlas CMB-04; CMB-15

T-DS-SETS Finite set systems and antichains (continued)

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1997-D1-P6 [R]; 2003-D2-P4 [R]; 2016-D1-P4 [R]

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T-DS-MATCHING Hall matching and augmenting paths

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Prerequisites Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]; Finite counting and incidence identities [T-DS-COUNTING]

In a finite bipartite graph with vertex classes A, B, a matching is a set of edges with disjoint endpoints. It saturates A if every vertex of A is incident to a matching edge. For $S \subseteq A$, let $N(S)$ be the set of its neighbours in B. Hall's theorem states that a matching saturating A exists if and only if

$$|N(S)| \geq |S| \qquad \text{for every } S \subseteq A.$$

Necessity follows because the partners of vertices of S are distinct elements of $N(S)$. For sufficiency, choose a maximum matching. If $a \in A$ is unmatched, follow alternating paths: an unmatched edge from A to B, then a matched edge back. Reaching an unmatched vertex of B would permit flipping the path and enlarging the matching. Otherwise let S and T be the reached vertices in A and B. Every neighbour of S is in T, and matching edges biject T with $S \setminus \{a\}$, contradicting Hall's condition.

For $|A| = |B| = n$, minimum degree at least $n/2$ on both sides implies a perfect matching. If $0 < |S| \leq n/2$, the neighbours of one vertex already give $|N(S)| \geq n/2 \geq |S|$. If $|S| > n/2$ and $N(S) \neq B$, a vertex outside $N(S)$ has fewer than $n/2$ possible neighbours, impossible.

Multi-class matching is not an automatic instance of Hall's theorem. A useful reduction first matches two classes, then treats each matched pair as a new vertex and joins it to members of the third class compatible with both partners. The common-neighbour lower bound $|U \cap V| \geq |U| + |V| - n$ controls this second graph. Prove Hall's condition again; merely checking individual degrees on one side is insufficient.

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2011-D2-P2 [R]

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T-DS-TRANSFORMS Conics, reflections, and inversion

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Prerequisites Inner products, orthogonality and polarization [T-LA-INNER]; Determinants, block identities and inverses [T-LA-DETERMINANT]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]

A Euclidean isometry preserves distances: translations and orthogonal linear maps are the standard examples. Uniform scaling by $r > 0$ multiplies distances by r and d-dimensional volume by r^d. A general invertible linear map instead scales volume by $|\det A|$ and usually changes angles. Completing squares reduces a positive definite quadratic inequality to an ellipsoid; its volume is the unit-ball volume times the product of its semiaxes.

A parabola with vertical axis and leading coefficient 1 has equation $y = (x - v)^2 + w$, where (v, w) is its vertex. Incidence can therefore be checked by one scalar identity. For a noncircular ellipse, a focus F, directrix ℓ, and eccentricity $0 < e < 1$ give $|X - F| = e\, d(X, \ell)$. All its points lie on the same side of the directrix as F, so the distance can be replaced by one consistently signed affine expression. Squaring or discarding this sign can introduce a second, spurious line. A circle is the separate $e = 0$ limiting case, not a finite-directrix ellipse.

Inversion of radius $\rho > 0$ about the origin is

$$I(x) = \frac{\rho^2 x}{\|x\|^2} \quad (x \neq 0), \qquad \|I(x) - I(y)\| = \frac{\rho^2 \|x - y\|}{\|x\|\,\|y\|}.$$

T-DS-TRANSFORMS Conics, reflections, and inversion (continued)

The distance formula follows by expanding the squared norm and is valid in every Euclidean dimension. Inversion is its own inverse. It sends spheres or planes not passing through the origin to spheres or planes; substitute its formula into their quadratic equations to determine which. It does not preserve lengths, straight segments, or convexity. A triangle formed by inverted vertices need not be the image of the original filled triangle. Track the excluded centre and denominator bounds before using inversion in a limit.

Extended atlas GEO-01; GEO-05; GEO-07

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2002-D1-P1 [R]; 2008-D2-P2 [R]; 2009-D2-P1 [R]; 2011-D2-P5 [R]

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T-DS-RAMSEY Order, colouring, and binomial threshold induction

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Prerequisites Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]

In extremal colouring, labels often encode the largest forbidden structure ending at an object. For distinct real numbers $a_1, \dots, a_N$, let u_i and v_i be the longest increasing and decreasing subsequence lengths ending at a_i. If $i < j$, then $a_i < a_j$ gives $u_j > u_i$, and $a_i > a_j$ gives $v_j > v_i$. Thus all pairs (u_i, v_i) are distinct. If no increasing subsequence has length r and no decreasing one has length s, then $N \le (r-1)(s-1)$. Distinctness matters; repeated entries require a chosen weak/strict convention.

The two-colour Ramsey numbers satisfy

$$R(r,s) \le R(r-1,s) + R(r,s-1), \qquad R(2,s) = s.$$

At one vertex, partition its neighbours by edge colour and apply the appropriate smaller case. Pascal's identity gives $R(r,s) \le \binom{r+s-2}{r-1}$. For real-number sets with a doubling-gap condition, the same recurrence can arise by splitting at the midpoint of the extreme elements: a long sequence in the lower half can be extended by the maximum; use the symmetric extension for the upper half. Prove the geometric extension separately from the numerical threshold.

For triples of ordered integers, colours can encode first differing binary digits. If $i < j < k$, define the highest binary position where i, j differ and similarly for j, k. Comparing these positions reduces a triple condition to an order condition on integer labels. An upper bound then needs an injectivity or distinct-label argument; a lower bound requires an explicit colouring and verification for every ordered quadruple or other forbidden pattern. Keep these tasks separate. The existence of a colouring attaining a lower bound does not prove that every colouring needs that many colours, and reversing the binary-digit order can reverse the desired inequalities.

Extended atlas CMB-10

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2004-D1-P5 [R]

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T-DS-SPHERICAL Spherical distance and signed dihedral derivatives

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Prerequisites Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]

On the unit sphere, the shorter spherical distance between nonantipodal unit vectors u, v is $\arccos\langle u, v\rangle \in [0, \pi)$. A unit-speed great circle can be written $X(t) = a\cos t + b\sin t$ for orthonormal

T-DS-SPHERICAL Spherical distance and signed dihedral derivatives (continued)

a, b; thus $X'' = -X$, $\|X'\| = 1$, and $X \perp X'$. Fix a unit vector c outside the plane spanned by a, b. Then

$$f(t) = \arccos\langle c, X(t)\rangle$$

is smooth and satisfies $0 < f(t) < \pi$ and $|f'(t)| < 1$. Indeed, the component of c in the great-circle plane has norm strictly less than 1, and differentiating $\cos f = \langle c, X\rangle$ yields the strict derivative bound. Twice differentiating gives

$$f'' = \cot f\,(1 - f'^2).$$

For $0 < u < \pi$, $u \cot u < 1$: the function $\sin u - u\cos u$ starts at zero and has derivative $u \sin u > 0$. Consequently

$$(ff')' = f'^2 + f \cot f\,(1 - f'^2) < 1.$$

Integrating relates endpoint distances and their signed directional derivatives to the arc length. The derivative of distance at an endpoint is minus the scalar product of the travel tangent with the unit tangent pointing toward c. At the final endpoint, reversing the travel tangent changes the sign. This explains why the two endpoint dihedral terms enter with different derivative signs before subtraction.

The noncoplanarity hypothesis is essential for strictness. If c lies on the great circle, distance may have derivative 1 or -1, or fail to be differentiable where the arc passes through c or its antipode. Spherical distance is not the Euclidean chord length; the latter is $2\sin(f/2)$ and cannot be substituted in the differential identity.

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2002-D2-P4 [R]

Back to previous view | Theory index | Method index | Strategy index

T-DS-POLYTOPE Facets and signed simplex decompositions

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Convex bodies, supporting functionals, and packing [T-DS-CONVEX]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]

A full-dimensional polytope is the convex hull of finitely many points spanning $\mathbb{R}^d$. A supporting hyperplane bounds a closed half-space containing the polytope and meets it; a facet is such an intersection of dimension $d-1$. A simplicial polytope has exactly d affinely independent vertices on each facet. General position rules out extra coplanar vertices.

Let P be a full-dimensional polytope and o a point on none of its facet hyperplanes. For each facet F, set $Q_F = \mathrm{conv}(\{o\} \cup F)$. Away from boundaries,

$$\mathbf{1}_P(x) = \sum_F \epsilon_F \mathbf{1}_{Q_F}(x), \qquad \epsilon_F = \begin{cases} +1, & o \text{ lies on the interior side of } F, \\ -1, & o \text{ lies on the exterior side of } F. \end{cases}$$

Follow a generic ray from o: if o is outside, its entry and exit facets contribute -1 and $+1$. Before entry they cancel; inside only the exit contribution remains; beyond exit neither contributes. If o is inside, there is only an exit. Rays through lower-dimensional faces are excluded from this argument. They form a measure-zero exceptional set in integral applications.

When P is simplicial, each Q_F is a simplex. This permits averaging over sign choices of vertices while retaining the sign of every contribution. Unsigned volumes or indicators are insufficient when o lies outside P.

Extended atlas GEO-13
Reference notes 22

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2024-D1-P5 [R]

Back to previous view | Theory index | Method index | Strategy index

T-DS-GENERATING Generating functions, coefficient filters, and finite differences

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Finite counting and incidence identities [T-DS-COUNTING]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]

Encoding. For a sequence $(a_n)_{n\ge 0}$, its ordinary generating function is the formal series $A(z) = \sum_{n\ge 0} a_n z^n$. Equality means equality of every coefficient; no analytic convergence is needed. Multiplication records independent choices because

$$[z^n]A(z)B(z) = \sum_{j=0}^{n} a_j b_{n-j}.$$

An infinite product is legitimate formally when each coefficient receives only finitely many contributions. For example,

$$\prod_{j\ge 1}(1-z^j)^{-1} = \sum_{n\ge 0} p(n) z^n$$

counts integer partitions: the exponent selected from the jth factor is the multiplicity of part j. Omitting that factor forbids that part. Each partition is unordered; an ordered composition requires a different encoding.

Residue filter. If $P(z) = \sum_{r=0}^{N} c_r z^r$, $m \ge 1$, and $\zeta = e^{2\pi i/m}$, then for any integer a,

$$\sum_{r\equiv a \pmod m} c_r = \frac{1}{m}\sum_{j=0}^{m-1} \zeta^{-aj} P(\zeta^j).$$

Indeed, the inner geometric sum is m when $m \mid r-a$ and zero otherwise. Use all m roots and retain the factor $1/m$. For parity the filter reduces to $(P(1)+P(-1))/2$. In several variables, average independently over each sign; this selects words with prescribed even letter counts.

Finite differences. Let $\Delta f(x) = f(x+1) - f(x)$. For a polynomial over a field of characteristic zero, of degree at most d,

$$\Delta^{d+1} f(x) = \sum_{j=0}^{d+1} (-1)^{d+1-j} \binom{d+1}{j} f(x+j) = 0.$$

Each difference lowers positive degree by one; expanding the iterated operator proves the displayed formula. This turns polynomial identities into alternating weighted counts. For $f(x) = x^2$, the third difference is zero. Before interpreting a sum, specify exactly which objects and which multiplicities a coefficient counts. A formal identity does not license substituting a numerical value outside convergence; the root filter above is stated for a polynomial.

Extended atlas CMB-06; M-CMB-06; M-CMB-12

Reference notes 18; 13; 14

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2008-D1-P6 [R]; 2012-D1-P1 [R]; 2015-D2-P8 [R]; 2016-D1-P5 [R]; 2016-D2-P9 [R]; 2020-D1-P1 [R]

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T-DS-LOGCONCAVITY Log-concave sequences and coefficient smoothing

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Generating functions, coefficient filters, and finite differences [T-DS-GENERATING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]

Definition and test. A strictly positive sequence $(a_n)_{n\ge 0}$ is log-concave when $a_n^2 \ge a_{n-1}a_{n+1}$ for every $n \ge 1$. Equivalently,

$$q_n := \frac{a_{n-1}}{a_n} \quad (n \ge 1) \quad \text{is nondecreasing.}$$

T-DS-LOGCONCAVITY Log-concave sequences and coefficient smoothing (continued)

This follows by dividing the defining inequality by $a_n a_{n+1} > 0$. Thus a multiplicative inequality becomes a monotonicity assertion. Positivity is essential for this ratio formulation; zeros require separate support conditions.

Adjacent-sum preservation. Put $a_{-1} = 0$ and $b_n = a_n + \lambda a_{n-1}$, where $\lambda \geq 0$. With $q_0 = 0$,

$$\frac{b_{n-1}}{b_n} = \frac{q_n(1+\lambda q_{n-1})}{1+\lambda q_n}.$$

The right side is nondecreasing in each of $q_{n-1}, q_n \geq 0$, since $q/(1+\lambda q)$ is increasing. Therefore (b_n) is log-concave. The boundary $n = 1$ is included by $q_0 = 0$; do not discard it.

Partial sums. Let $h_n = \sum_{j=0}^{n} a_j$. Then

$$\frac{h_{n-1}}{h_n} = \frac{\sum_{j=0}^{n} a_j q_j}{\sum_{j=0}^{n} a_j} \leq q_n \leq q_{n+1}.$$

Hence $h_n a_n \geq h_{n-1} a_{n+1}$, which is precisely $h_n^2 \geq h_{n-1} h_{n+1}$. Partial sums therefore preserve log-concavity too. Combining the two rules proves preservation for $a_n + 2\sum_{j<n} a_j = h_n + h_{n-1}$.

Coefficient example. For a fixed integer $k \geq 0$, the coefficients of $(1-z)^{-k-1}$ are $a_n = \binom{k+n}{k}$ and $q_n = n/(k+n)$, a nondecreasing sequence. Multiplication of the generating function by $(1+z)$ performs the adjacent-sum operation. Repeating it k times proves log-concavity of the coefficients of $(1+z)^k/(1-z)^{k+1}$. This isolates the reusable argument behind the lattice-ball count; no general convolution theorem is needed.

Trap. Increasing terms are not automatically log-concave. For instance, $1, 2, 5$ increase but $2^2 < 1 \cdot 5$. Verify adjacent ratios, and keep the direction of the required inequality consistent with the chosen ratio.

Extended atlas M-CMB-06; M-CMB-05

Reference notes 15

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2016-D2-P9 [R]

Back to previous view | Theory index | Method index | Strategy index

T-DS-ADDITIVE Sumsets and the minimal-growth lemma

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Groups, orders, generators, and quotients [T-AL-GROUP]

Setting. In an abelian group, put $X + Y = \{x + y : x \in X, y \in Y\}$ and let kA denote a k-fold sumset. These are sets, so repeated representations count once. Translation preserves cardinality. Suppose A is finite and nonempty. Choose a nonempty $B \subseteq A$ minimizing

$$\alpha = \frac{|A+B|}{|B|}.$$

Then $|A+U| \geq \alpha|U|$ for every $U \subseteq B$, including the empty set.

Minimal-growth lemma. For every finite D,

$$|A+B+D| \leq \alpha|B+D|.$$

Induct on $|D|$; the empty and singleton cases are immediate. Upon adjoining $x \notin D$, set $U = \{b \in B : b+x \in B+D\}$. Exactly $|B|-|U|$ new elements enter $B+D$. Moreover, $A+U+x$ already belongs to $A+B+D$, so at most

$$|A+B| - |A+U| \leq \alpha(|B| - |U|)$$

new elements enter $A+B+D$. Adding this inequality to the induction hypothesis proves the lemma. This is the reason to minimize a growth ratio, rather than just the size of a set.

T-DS-ADDITIVE Sumsets and the minimal-growth lemma (continued)

Iteration. Substituting $D = (k-1)A$ gives $|kA + B| \le \alpha^k |B|$ by induction. Any fixed $b \in B$ gives the inclusion $kA + b \subseteq kA + B$. Thus if $|A + A| \le c|A|$, then $\alpha \le c$ and

$$|kA| \le c^k |A| \qquad (k \ge 1).$$

Handle $A = \emptyset$ separately, without forming a ratio.
Example and trap. For a finite subgroup H, $H + H = H$ and $kH = H$, so $c = 1$ is exact. The inequality $|X + Y| \le |X||Y|$ counts representations very inefficiently when translates overlap; the lemma measures that overlap. Commutativity is needed for the stated sumset iteration; do not transfer it unchanged to arbitrary ordered products in a nonabelian group.
Extended atlas M-CMB-08; M-CMB-02
Reference notes 2; 18

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2012-D2-P5 [R]

Back to previous view | Theory index | Method index | Strategy index

T-DS-TREES Trees, unique paths, and distance sums

Back to previous view | Theory index | Method index | Strategy index
Prerequisites Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]; Finite counting and incidence identities [T-DS-COUNTING]
Definitions. A finite simple graph is a tree if it is connected and contains no cycle. Between two vertices of a tree there is exactly one simple path: connectedness gives one, while two distinct paths would form a cycle. The distance $d(u, v)$ is the number of its edges. In any graph,

$$\sum_{v \in V} \deg(v) = 2|E|,$$

since each edge contributes at both endpoints. In particular the number of odd-degree vertices is even; no assumption of connectedness is needed for this parity consequence.
Tree edge count. A tree on $n \ge 1$ vertices has $n - 1$ edges. For $n \ge 2$, both endpoints of a longest simple path have degree one: an extra neighbour either extends the path or creates a cycle. Remove such a leaf and its edge. The remaining graph stays connected and acyclic, so induction proves the count. The one-vertex case has zero edges. Conversely, a connected graph with $n - 1$ edges is a tree; having $n - 1$ edges alone is insufficient.
Distance bookkeeping. There are $\binom{n}{2}$ unordered distinct vertex pairs. Exactly $n - 1$ pairs in a tree have distance one, and the others have distance at least two. More generally, deleting edge e splits a tree into components of sizes $s_e, n - s_e$. A path uses e exactly when its endpoints are in different components, so

$$\sum_{\{u,v\}} d(u, v) = \sum_{e \in E} s_e(n - s_e).$$

This is a double count of pairs consisting of an edge and an endpoint pair whose path uses it.
Example. In the star on n vertices, the $n-1$ centre-leaf distances equal one and the $\binom{n-1}{2}$ leaf-leaf distances equal two. Thus its distance sum is $(n-1)^2$. This identifies a natural equality candidate before applying inequalities. Keep ordered and unordered pairs distinct: switching conventions multiplies each distance sum by two.
Extended atlas M-CMB-03; M-CMB-02
Reference notes 9; 8; 19

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2023-D2-P8 [R]

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DATA NOTICE — fixed-quota counts are exploratory metadata, not empirical frequencies; click DATA-001.

6 Enhanced Theory Breakdown Atlas

This section is the first expansion layer after the problem-by-problem tagging pass. The aim is not to repeat the problem statements, but to extract the mathematical theory that repeatedly appears in the IMC corpus and to record exactly where each item is needed.

How to use this atlas. A theory node is a reusable prerequisite: it gives the theorem or structural fact to know, the way it normally appears in IMC solutions, and the problem IDs where the node was assigned. The assignment is intentionally preparation-oriented: a problem may have several nodes because a clean solution often combines one primary theorem with two or three execution tools.

Coverage audit. Every detected problem in the previous catalog has at least one detailed theory node assigned. The complete machine-readable mapping is included in the ZIP as `data/imc_problem_theory_catalog.csv` and `data/imc_problem_theory_catalog.json`.

6.1 Theory-node frequency by domain

Domain	Theory node	Problems
Linear algebra / matrix theory	LA-01 - Matrix products, inverses, and one-sided inverse identities	14
Linear algebra / matrix theory	LA-02 - Determinant evaluation by row/column operations and triangular forms	51
Linear algebra / matrix theory	LA-03 - Block matrices, Schur complements, and determinant identities	114
Linear algebra / matrix theory	LA-04 - Rank-nullity, image/kernel, and rank of factorizations	87
Linear algebra / matrix theory	LA-05 - Trace cyclicity, trace pairing, and zero-trace subspaces	33
Linear algebra / matrix theory	LA-06 - Commutators and spectral obstructions	20
Linear algebra / matrix theory	LA-07 - Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton	51
Linear algebra / matrix theory	LA-08 - Nilpotent, idempotent, involution, and square-zero structures	14
Linear algebra / matrix theory	LA-09 - Simultaneous diagonalization and commuting families	3
Linear algebra / matrix theory	LA-10 - Invariant subspaces and adapted bases	30
Linear algebra / matrix theory	LA-11 - Linear functionals, kernels, and dual span criteria	6
Linear algebra / matrix theory	LA-12 - Matrix subspaces constrained by trace bilinear forms	1
Linear algebra / matrix theory	LA-13 - Matrix exponentials and polynomial functional calculus	1
Linear algebra / matrix theory	LA-14 - Operator norm, power boundedness, and spectral-radius style estimates	1
Linear algebra / matrix theory	LA-15 - Gram matrices and rational linear dependence	7
Linear algebra / matrix theory	LA-16 - Anticommuting and rank constraints for special matrices	0
Real analysis / calculus	RA-01 - Compactness, extreme values, and closed-bounded reduction	37
Real analysis / calculus	RA-02 - Intermediate value, connectedness, density, and continuity transfer	44
Real analysis / calculus	RA-03 - Monotone functions, fixed points, and iteration on intervals	15
Real analysis / calculus	RA-04 - Rolle, mean-value theorem, and divided differences	33
Real analysis / calculus	RA-05 - Taylor expansion, local asymptotics, and little-o bookkeeping	50
Real analysis / calculus	RA-06 - Auxiliary functions, integrating factors, and differential inequalities	62
Real analysis / calculus	RA-07 - Convexity, concavity, Jensen, and tangent/chord inequalities	179

DATA NOTICE — fixed-quota counts are exploratory metadata, not empirical frequencies; click DATA-001.

Domain	Theory node	Problems
Real analysis / calculus	RA-08 - Integral inequalities, integration by parts, and weighted estimates	199
Real analysis / calculus	RA-09 - Fubini/Tonelli and change of variables	39
Real analysis / calculus	RA-10 - Riemann sums and sum-integral asymptotics	73
Real analysis / calculus	RA-11 - Series convergence, rearrangements, and alternating/Dirichlet tools	67
Real analysis / calculus	RA-12 - Uniform continuity and periodic averaging	1
Real analysis / calculus	RA-13 - L'Hospital, ratio limits, and growth comparisons	33
Real analysis / calculus	RA-14 - Weierstrass approximation and polynomial approximation	1
Real analysis / calculus	RA-15 - L2 spaces, orthonormal systems, and dominated convergence	1
Real analysis / calculus	RA-16 - Baire/category, dense sets, and pathological continuous functions	6
Real analysis / calculus	RA-17 - Sequence recurrences, monotone bounds, and generating functions	22
Real analysis / calculus	RA-18 - Trigonometric and hyperbolic identities with monotonicity	14
Real analysis / calculus	RA-19 - Gradient/Lipschitz-gradient optimization and convex smoothness	3
Real analysis / calculus	RA-20 - Measure and probability-integral estimates	2
Complex analysis / polynomials	CP-01 - Polynomial identity and more-roots-than-degree principle	68
Complex analysis / polynomials	CP-02 - Factorization, irreducibility, and Gauss-style integer polynomial arguments	24
Complex analysis / polynomials	CP-03 - Vieta, logarithmic derivatives, and real-root inequalities	28
Complex analysis / polynomials	CP-04 - Unit-circle roots and conjugation identities	13
Complex analysis / polynomials	CP-05 - Roots of unity, cyclotomic filters, and discrete Fourier tricks	24
Complex analysis / polynomials	CP-06 - Cauchy formula and coefficient extraction on the unit circle	7
Complex analysis / polynomials	CP-07 - Maximum modulus principle	0
Complex analysis / polynomials	CP-08 - Laurent expansions, isolated singularities, and rationality of meromorphic functions	1
Complex analysis / polynomials	CP-09 - Fejer kernels and positive trigonometric kernels	1
Complex analysis / polynomials	CP-10 - Interpolation, Lagrange basis, and Chebyshev extremality	2
Complex analysis / polynomials	CP-11 - Stable polynomials and coefficient inequalities	3
Complex analysis / polynomials	CP-12 - Partial fractions and cotangent-type expansions	1
Complex analysis / polynomials	CP-13 - Polynomial functional identities and algebraic composition	8
Algebra / number theory	ANT-01 - Modular arithmetic and divisibility	92
Algebra / number theory	ANT-02 - Chinese remainder theorem and complete residue systems	30
Algebra / number theory	ANT-03 - Finite fields, field characteristic, and Fermat/Frobenius tools	49
Algebra / number theory	ANT-04 - Integer polynomials, finite differences, and value divisibility	22
Algebra / number theory	ANT-05 - Rational roots, Vieta, and rationality constraints	5
Algebra / number theory	ANT-06 - GCD, valuations, prime powers, and coprimality	42
Algebra / number theory	ANT-07 - Group basics: orders, cyclic subgroups, and generators	52
Algebra / number theory	ANT-08 - Permutation groups, cycles, parity, and generation	2

DATA NOTICE — fixed-quota counts are exploratory metadata, not empirical frequencies; click DATA-001.

Domain	Theory node	Problems
Algebra / number theory	ANT-09 - Normal subgroups, quotient groups, and group actions	12
Algebra / number theory	ANT-10 - Automorphisms and semidirect products	27
Algebra / number theory	ANT-11 - Rings, idempotents, nil rings, and noncommutativity	33
Algebra / number theory	ANT-12 - Diophantine equations, Pell-type recurrences, and quadratic irrationals	2
Algebra / number theory	ANT-13 - Lattices and norms/quasinorms	2
Algebra / number theory	ANT-14 - Greedy infinite products and rationality tails	1
Algebra / number theory	ANT-15 - Finite abelian groups and vector spaces over finite fields	1
Combinatorics / discrete structures	CMB-01 - Pigeonhole principle and extremal element selection	68
Combinatorics / discrete structures	CMB-02 - Double counting and degree-sum methods	56
Combinatorics / discrete structures	CMB-03 - Graph encoding, cycles, and bipartite representations	68
Combinatorics / discrete structures	CMB-04 - Set systems, intersecting families, and induction on sizes	23
Combinatorics / discrete structures	CMB-05 - Recurrences and state partitions	4
Combinatorics / discrete structures	CMB-06 - Generating functions for counting	14
Combinatorics / discrete structures	CMB-07 - Inclusion-exclusion and root-of-unity filters	1
Combinatorics / discrete structures	CMB-08 - Permutations, inversions, and permutation statistics	5
Combinatorics / discrete structures	CMB-09 - Explicit construction and counterexample design	36
Combinatorics / discrete structures	CMB-10 - Erdos-Szekeres style monotone subsequences and binomial thresholds	1
Combinatorics / discrete structures	CMB-11 - Grid, lattice, and alternating-cycle arguments	2
Combinatorics / discrete structures	CMB-12 - Coloring, partitions, and parity	48
Combinatorics / discrete structures	CMB-13 - Words, equivalence relations, and rewriting systems	3
Combinatorics / discrete structures	CMB-14 - Transfer matrices and matrix powers for counting	1
Combinatorics / discrete structures	CMB-15 - Finite designs and pair-determines-third structures	1
Geometry / convexity / metric	GEO-01 - Coordinate and affine normalization	2
Geometry / convexity / metric	GEO-02 - Euclidean distances and inner-product identities	19
Geometry / convexity / metric	GEO-03 - Projections and orthogonal decompositions	25
Geometry / convexity / metric	GEO-04 - Convex bodies, supporting geometry, and John-ellipse ideas	7
Geometry / convexity / metric	GEO-05 - Conics: parabolas, ellipses, foci, and directrices	4
Geometry / convexity / metric	GEO-06 - Simplex, barycentric coordinates, and circumcenters	6
Geometry / convexity / metric	GEO-07 - Area, volume, projection, and ellipsoid computation	28
Geometry / convexity / metric	GEO-08 - Metric content, diameter covers, and contractions	1
Geometry / convexity / metric	GEO-09 - Hilbert-space equidistant sets and orthonormal translations	6
Geometry / convexity / metric	GEO-10 - Closest points to closed sets and density of uniqueness	1

DATA NOTICE — fixed-quota counts are exploratory metadata, not empirical frequencies; click DATA-001.

Domain	Theory node	Problems
Geometry / convexity / metric	GEO-11 - Unit-sphere configurations and regular simplex bounds	1
Geometry / convexity / metric	GEO-12 - Polygon triangulation and parity of incidences	1
Geometry / convexity / metric	GEO-13 - Convex hulls, random polytopes, and geometric probability	1
Geometry / convexity / metric	GEO-14 - Cube and square dissection by number-theoretic semigroups	1
Functional equations / dynamics	FEQ-01 - Substitution of special values and normalization	7
Functional equations / dynamics	FEQ-02 - Injectivity and monotonicity forced by a functional equation	8
Functional equations / dynamics	FEQ-03 - Fixed points, iterates, and orbit closure	6
Functional equations / dynamics	FEQ-04 - Recurrence transformations and conjugacy	3
Functional equations / dynamics	FEQ-05 - Differentiable functional equations	0
Functional equations / dynamics	FEQ-06 - Piecewise-linear maps and periodic points	1
Functional equations / dynamics	FEQ-07 - Bijections as products of involutions and symmetry decompositions	1
Functional equations / dynamics	FEQ-08 - No-regularity functions and order patterns on Q or R	6
Functional equations / dynamics	FEQ-09 - Image-length preserving functions	1
Probability / measure / expectation	PRB-01 - Discrete probability via residues and recurrences	1
Probability / measure / expectation	PRB-02 - Generating functions for probability distributions	1
Probability / measure / expectation	PRB-03 - Expectation, linearity, and probabilistic method	1
Probability / measure / expectation	PRB-04 - Continuous uniform distribution and geometric probability	0
Probability / measure / expectation	PRB-05 - Measure, variation, and differentiability almost everywhere	2

6.2 Domain-level theory requirements

6.2.1 Linear algebra / matrix theory

The linear algebra layer is not just determinant computation. It contains rank-nullity, invariant subspaces, centralizers, trace forms, commutators, nilpotent/idempotent structure, spectral arguments, and matrix-space dimension counts.

- **LA-01.** Matrix products, inverses, and one-sided inverse identities (assigned to 14 problems).
- **LA-02.** Determinant evaluation by row/column operations and triangular forms (assigned to 51 problems).
- **LA-03.** Block matrices, Schur complements, and determinant identities (assigned to 114 problems).
- **LA-04.** Rank-nullity, image/kernel, and rank of factorizations (assigned to 87 problems).
- **LA-05.** Trace cyclicity, trace pairing, and zero-trace subspaces (assigned to 33 problems).
- **LA-06.** Commutators and spectral obstructions (assigned to 20 problems).
- **LA-07.** Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton (assigned to 51 problems).
- **LA-08.** Nilpotent, idempotent, involution, and square-zero structures (assigned to 14 problems).
- **LA-09.** Simultaneous diagonalization and commuting families (assigned to 3 problems).
- **LA-10.** Invariant subspaces and adapted bases (assigned to 30 problems).
- **LA-11.** Linear functionals, kernels, and dual span criteria (assigned to 6 problems).
- **LA-12.** Matrix subspaces constrained by trace bilinear forms (assigned to 1 problems).
- **LA-13.** Matrix exponentials and polynomial functional calculus (assigned to 1 problems).
- **LA-14.** Operator norm, power boundedness, and spectral-radius style estimates (assigned to 1 problems).

- **LA-15.** Gram matrices and rational linear dependence (assigned to 7 problems).
- **LA-16.** Anticommuting and rank constraints for special matrices (assigned to 0 problems).

6.2.2 Real analysis / calculus

The analysis layer is driven by compactness, derivative theorems, auxiliary functions, convexity, integral inequalities, Riemann-sum asymptotics, and series/sequence control.

- **RA-01.** Compactness, extreme values, and closed-bounded reduction (assigned to 37 problems).
- **RA-02.** Intermediate value, connectedness, density, and continuity transfer (assigned to 44 problems).
- **RA-03.** Monotone functions, fixed points, and iteration on intervals (assigned to 15 problems).
- **RA-04.** Rolle, mean-value theorem, and divided differences (assigned to 33 problems).
- **RA-05.** Taylor expansion, local asymptotics, and little-o bookkeeping (assigned to 50 problems).
- **RA-06.** Auxiliary functions, integrating factors, and differential inequalities (assigned to 62 problems).
- **RA-07.** Convexity, concavity, Jensen, and tangent/chord inequalities (assigned to 179 problems).
- **RA-08.** Integral inequalities, integration by parts, and weighted estimates (assigned to 199 problems).
- **RA-09.** Fubini/Tonelli and change of variables (assigned to 39 problems).
- **RA-10.** Riemann sums and sum-integral asymptotics (assigned to 73 problems).
- **RA-11.** Series convergence, rearrangements, and alternating/Dirichlet tools (assigned to 67 problems).
- **RA-12.** Uniform continuity and periodic averaging (assigned to 1 problems).
- **RA-13.** L'Hospital, ratio limits, and growth comparisons (assigned to 33 problems).
- **RA-14.** Weierstrass approximation and polynomial approximation (assigned to 1 problems).
- **RA-15.** L2 spaces, orthonormal systems, and dominated convergence (assigned to 1 problems).
- **RA-16.** Baire/category, dense sets, and pathological continuous functions (assigned to 6 problems).
- **RA-17.** Sequence recurrences, monotone bounds, and generating functions (assigned to 22 problems).
- **RA-18.** Trigonometric and hyperbolic identities with monotonicity (assigned to 14 problems).
- **RA-19.** Gradient/Lipschitz-gradient optimization and convex smoothness (assigned to 3 problems).
- **RA-20.** Measure and probability-integral estimates (assigned to 2 problems).

6.2.3 Complex analysis / polynomials

The polynomial/complex layer includes root-counting, factorization, roots of unity, unit-circle methods, holomorphic estimates, Laurent expansions, Fejer kernels, and stable polynomial coefficient inequalities.

- **CP-01.** Polynomial identity and more-roots-than-degree principle (assigned to 68 problems).
- **CP-02.** Factorization, irreducibility, and Gauss-style integer polynomial arguments (assigned to 24 problems).
- **CP-03.** Vieta, logarithmic derivatives, and real-root inequalities (assigned to 28 problems).
- **CP-04.** Unit-circle roots and conjugation identities (assigned to 13 problems).
- **CP-05.** Roots of unity, cyclotomic filters, and discrete Fourier tricks (assigned to 24 problems).
- **CP-06.** Cauchy formula and coefficient extraction on the unit circle (assigned to 7 problems).
- **CP-07.** Maximum modulus principle (assigned to 0 problems).
- **CP-08.** Laurent expansions, isolated singularities, and rationality of meromorphic functions (assigned to 1 problems).
- **CP-09.** Fejer kernels and positive trigonometric kernels (assigned to 1 problems).
- **CP-10.** Interpolation, Lagrange basis, and Chebyshev extremality (assigned to 2 problems).
- **CP-11.** Stable polynomials and coefficient inequalities (assigned to 3 problems).
- **CP-12.** Partial fractions and cotangent-type expansions (assigned to 1 problems).
- **CP-13.** Polynomial functional identities and algebraic composition (assigned to 8 problems).

6.2.4 Algebra / number theory

The algebra layer includes modular arithmetic, CRT, finite fields, integer polynomial divisibility, rationality, group/ring structure, permutation groups, automorphisms, and Diophantine constructions.

- **ANT-01.** Modular arithmetic and divisibility (assigned to 92 problems).
- **ANT-02.** Chinese remainder theorem and complete residue systems (assigned to 30 problems).
- **ANT-03.** Finite fields, field characteristic, and Fermat/Frobenius tools (assigned to 49 problems).
- **ANT-04.** Integer polynomials, finite differences, and value divisibility (assigned to 22 problems).
- **ANT-05.** Rational roots, Vieta, and rationality constraints (assigned to 5 problems).
- **ANT-06.** GCD, valuations, prime powers, and coprimality (assigned to 42 problems).
- **ANT-07.** Group basics: orders, cyclic subgroups, and generators (assigned to 52 problems).
- **ANT-08.** Permutation groups, cycles, parity, and generation (assigned to 2 problems).
- **ANT-09.** Normal subgroups, quotient groups, and group actions (assigned to 12 problems).
- **ANT-10.** Automorphisms and semidirect products (assigned to 27 problems).
- **ANT-11.** Rings, idempotents, nil rings, and noncommutativity (assigned to 33 problems).

- `ANT-12`. Diophantine equations, Pell-type recurrences, and quadratic irrationals (assigned to 2 problems).
- `ANT-13`. Lattices and norms/quasinorms (assigned to 2 problems).
- `ANT-14`. Greedy infinite products and rationality tails (assigned to 1 problems).
- `ANT-15`. Finite abelian groups and vector spaces over finite fields (assigned to 1 problems).

6.2.5 Combinatorics / discrete structures

The combinatorics layer is a method layer as much as a content layer: double counting, pigeonhole, graph encoding, recurrences, set systems, coloring parity, and constructions recur across many nominally algebraic or geometric problems.

- `CMB-01`. Pigeonhole principle and extremal element selection (assigned to 68 problems).
- `CMB-02`. Double counting and degree-sum methods (assigned to 56 problems).
- `CMB-03`. Graph encoding, cycles, and bipartite representations (assigned to 68 problems).
- `CMB-04`. Set systems, intersecting families, and induction on sizes (assigned to 23 problems).
- `CMB-05`. Recurrences and state partitions (assigned to 4 problems).
- `CMB-06`. Generating functions for counting (assigned to 14 problems).
- `CMB-07`. Inclusion-exclusion and root-of-unity filters (assigned to 1 problems).
- `CMB-08`. Permutations, inversions, and permutation statistics (assigned to 5 problems).
- `CMB-09`. Explicit construction and counterexample design (assigned to 36 problems).
- `CMB-10`. Erdos-Szekeres style monotone subsequences and binomial thresholds (assigned to 1 problems).
- `CMB-11`. Grid, lattice, and alternating-cycle arguments (assigned to 2 problems).
- `CMB-12`. Coloring, partitions, and parity (assigned to 48 problems).
- `CMB-13`. Words, equivalence relations, and rewriting systems (assigned to 3 problems).
- `CMB-14`. Transfer matrices and matrix powers for counting (assigned to 1 problems).
- `CMB-15`. Finite designs and pair-determines-third structures (assigned to 1 problems).

6.2.6 Geometry / convexity / metric

The geometry layer is mostly analytic and linear-algebraic: coordinates, inner products, projections, conics, convex bodies, simplex/circumcenter identities, metric covers, and Hilbert-space geometry.

- `GEO-01`. Coordinate and affine normalization (assigned to 2 problems).
- `GEO-02`. Euclidean distances and inner-product identities (assigned to 19 problems).
- `GEO-03`. Projections and orthogonal decompositions (assigned to 25 problems).
- `GEO-04`. Convex bodies, supporting geometry, and John-ellipse ideas (assigned to 7 problems).
- `GEO-05`. Conics: parabolas, ellipses, foci, and directrices (assigned to 4 problems).
- `GEO-06`. Simplex, barycentric coordinates, and circumcenters (assigned to 6 problems).
- `GEO-07`. Area, volume, projection, and ellipsoid computation (assigned to 28 problems).
- `GEO-08`. Metric content, diameter covers, and contractions (assigned to 1 problems).
- `GEO-09`. Hilbert-space equidistant sets and orthonormal translations (assigned to 6 problems).
- `GEO-10`. Closest points to closed sets and density of uniqueness (assigned to 1 problems).
- `GEO-11`. Unit-sphere configurations and regular simplex bounds (assigned to 1 problems).
- `GEO-12`. Polygon triangulation and parity of incidences (assigned to 1 problems).
- `GEO-13`. Convex hulls, random polytopes, and geometric probability (assigned to 1 problems).
- `GEO-14`. Cube and square dissection by number-theoretic semigroups (assigned to 1 problems).

6.2.7 Functional equations / dynamics

The functional-equation layer focuses on substitution, monotonicity/injectivity extraction, fixed points, iterates, recurrence conjugacy, and no-regularity order arguments.

- **FEQ-01.** Substitution of special values and normalization (assigned to 7 problems).
- **FEQ-02.** Injectivity and monotonicity forced by a functional equation (assigned to 8 problems).
- **FEQ-03.** Fixed points, iterates, and orbit closure (assigned to 6 problems).
- **FEQ-04.** Recurrence transformations and conjugacy (assigned to 3 problems).
- **FEQ-05.** Differentiable functional equations (assigned to 0 problems).
- **FEQ-06.** Piecewise-linear maps and periodic points (assigned to 1 problems).
- **FEQ-07.** Bijections as products of involutions and symmetry decompositions (assigned to 1 problems).
- **FEQ-08.** No-regularity functions and order patterns on Q or R (assigned to 6 problems).
- **FEQ-09.** Image-length preserving functions (assigned to 1 problems).

6.2.8 Probability / measure / expectation

The probability layer uses residue recurrences, generating functions, linearity of expectation, continuous geometric probability, and measure/covering arguments.

- **PRB-01.** Discrete probability via residues and recurrences (assigned to 1 problems).
- **PRB-02.** Generating functions for probability distributions (assigned to 1 problems).
- **PRB-03.** Expectation, linearity, and probabilistic method (assigned to 1 problems).
- **PRB-04.** Continuous uniform distribution and geometric probability (assigned to 0 problems).
- **PRB-05.** Measure, variation, and differentiability almost everywhere (assigned to 2 problems).

CORRECTION MATH-008 — legacy statement superseded; click for canonical replacement.

7 Detailed Theory Nodes with Problem Occurrences

7.1 Linear algebra / matrix theory

LA-01 - Matrix products, inverses, and one-sided inverse identities

Bucket	LA matrix identities
Core theory	For square matrices, a right inverse is also a left inverse: if $XY = I$, then X and Y are invertible and $YX = I$. Algebraic identities such as $(A + I)(B + I) = I$ or $(A - B)(C + A^{-1}) = I$ often turn a noncommutative-looking equation into an inverse statement.
IMC use	Use this whenever the statement contains a matrix equation that can be completed to a product equal to I. After factoring, switch the order only because the factors have been proved inverse, not because matrices commute.
Trap watch	Never cancel or reorder matrix factors before proving invertibility; record which factor is square.
Appears in	1994-D1-P1, 1996-D1-P1, 1996-D2-P3, 2002-D2-P1, 2002-D2-P5, 2003-D2-P1, 2004-D2-P1, 2009-D1-P2 2013-D1-P1, 2014-D1-P1, 2018-D1-P3, 2019-D2-P9, 2021-D1-P1, 2023-D2-P6

LA-02 - Determinant evaluation by row/column operations and triangular forms

Bucket	LA determinants
Core theory	Determinants are preserved by adding a multiple of one row/column to another, change sign under swaps, and scale under row/column scaling. A common IMC target is to transform a structured matrix into triangular or block triangular form.
IMC use	Use for explicit determinant computations, sign/parity determinant arguments, and matrices with entries depending on $i + j$, $\lvert i - j \rvert$, parity, primes, or a recurrence.
Trap watch	Track row/column operations exactly; if operations depend on parity or cyclic order, verify small n separately.
Appears in	1994-D1-P1, 1995-D1-P1, 1995-D2-P1, 1996-D1-P1, 1997-D1-P3, 1997-D2-P2, 1997-D2-P4, 1999-D1-P1 2000-D1-P3, 2001-D1-P1, 2001-D1-P5, 2001-D2-P4, 2002-D2-P1, 2003-D2-P1, 2004-D2-P1, 2004-D2-P6 2005-D1-P1, 2005-D2-P3, 2005-D2-P6, 2006-D1-P3, 2006-D2-P6, 2007-D1-P2, 2007-D1-P3, 2007-D2-P4 2008-D1-P6, 2008-D2-P5, 2009-D1-P2, 2009-D1-P5, 2009-D2-P5, 2010-D1-P5, 2010-D2-P4, 2011-D1-P2 2012-D1-P2, 2013-D1-P1, 2014-D1-P1, 2015-D1-P1, 2016-D2-P10, 2017-D1-P1, 2018-D1-P3, 2019-D1-P5 2019-D2-P9, 2020-D1-P2, 2021-D1-P1, 2021-D2-P5, 2022-D2-P7, 2023-D1-P2, 2023-D2-P6, 2023-D2-P10 2024-D1-P1, 2024-D2-P7, 2024-D2-P9

LA-03 - Block matrices, Schur complements, and determinant identities

Bucket	LA block methods
Core theory	For compatible square blocks, block multiplication and determinant identities often reduce a large matrix equation to a smaller one. Typical facts include $\det(M)\det(H) = \det(A)$ for inverse block pairs in suitable block decompositions, and AC and CA having the same nonzero eigenvalues.
IMC use	Use when a problem explicitly gives a block matrix, a product AB versus BA, or a matrix of a linear operator on a matrix space.
Trap watch	Schur complement formulas require invertibility of the chosen block; if it is not given, use block multiplication identities instead.
Appears in	1994-D1-P1, 1994-D2-P2, 1995-D1-P1, 1995-D1-P2, 1995-D2-P1, 1995-D2-P2, 1996-D1-P1, 1996-D1-P2 1997-D1-P3, 1997-D2-P2, 1997-D2-P4, 1998-D1-P1, 1998-D1-P5, 1999-D1-P1, 1999-D1-P5, 1999-D2-P2 1999-D2-P3, 2000-D1-P2, 2000-D1-P3, 2000-D1-P4, 2000-D2-P6, 2001-D1-P1, 2001-D1-P3, 2001-D1-P5 2001-D2-P4, 2002-D2-P1, 2002-D2-P2, 2003-D2-P1, 2004-D2-P1, 2004-D2-P4, 2004-D2-P6, 2005-D1-P1 2005-D2-P3, 2005-D2-P6, 2006-D1-P3, 2006-D2-P2, 2006-D2-P3, 2006-D2-P6, 2007-D1-P2, 2007-D1-P3 2007-D2-P1, 2007-D2-P4, 2008-D1-P6, 2008-D1-P1, 2008-D2-P1, 2008-D2-P5, 2009-D1-P2, 2009-D1-P5 2009-D2-P3, 2009-D2-P5, 2010-D1-P2, 2010-D1-P3, 2010-D1-P5, 2010-D2-P4, 2011-D1-P2, 2011-D1-P5 2011-D2-P2, 2011-D2-P3, 2012-D1-P2, 2012-D2-P4, 2013-D1-P1, 2013-D1-P3, 2013-D1-P4, 2013-D2-P3 2013-D2-P5, 2014-D1-P1, 2014-D1-P2, 2014-D2-P2, 2014-D2-P3, 2014-D2-P5, 2015-D1-P1, 2015-D1-P2 2015-D1-P3, 2015-D2-P6, 2015-D2-P7, 2015-D2-P9, 2016-D1-P1, 2016-D1-P2, 2016-D2-P6, 2016-D2-P7 2017-D1-P1, 2017-D1-P2, 2017-D2-P6, 2017-D2-P8, 2018-D1-P1, 2018-D1-P3, 2018-D1-P4, 2018-D2-P8 2018-D2-P10, 2019-D1-P1, 2019-D1-P2, 2019-D1-P3, 2019-D1-P5, 2019-D2-P9, 2020-D1-P2, 2020-D1-P4 2020-D2-P5, 2020-D2-P6, 2020-D2-P8, 2021-D1-P1, 2021-D1-P3, 2021-D1-P4, 2021-D2-P5, 2022-D1-P1 2022-D1-P2, 2022-D2-P7, 2023-D1-P1, 2023-D2-P6, 2024-D1-P1, 2024-D1-P2, 2024-D2-P6, 2024-D2-P7 2024-D2-P8, 2024-D2-P9

CORRECTION MATH-009 — legacy statement superseded; click for canonical replacement.

LA-04 - Rank-nullity, image/kernel, and rank of factorizations

Bucket	LA rank/dimension
Core theory	Rank-nullity gives $\dim V = \dim \ker T + \dim \operatorname{im} T$. For products, $\operatorname{rank}(AB) \leq \min(\operatorname{rank} A, \operatorname{rank} B)$ and column spaces/images often reveal the exact rank.
IMC use	Use for rank minimization/maximization, rank-one perturbations, factorizations AB, and problems asking for the dimension of a matrix subspace.
Trap watch	Rank statements over $\mathbb{R}$ and $\mathbb{C}$ can differ only through spectral issues; dimension counts must specify the ground field.
Appears in	1994-D1-P1, 1994-D1-P4, 1994-D2-P4, 1994-D2-P5, 1995-D1-P1, 1995-D1-P5, 1996-D1-P1, 1996-D1-P3 1996-D1-P6, 1997-D1-P3, 1997-D2-P2, 1997-D2-P4, 1998-D1-P1, 1998-D2-P1, 1999-D1-P1, 2000-D1-P3 2000-D2-P1, 2001-D1-P5, 2001-D2-P3, 2001-D2-P4, 2002-D1-P5, 2002-D1-P6, 2002-D2-P1, 2003-D2-P1 2004-D1-P3, 2004-D2-P1, 2004-D2-P4, 2005-D1-P1, 2005-D1-P3, 2005-D2-P3, 2006-D1-P3, 2006-D1-P6 2006-D2-P2, 2006-D2-P4, 2006-D2-P6, 2007-D1-P2, 2007-D1-P3, 2007-D1-P5, 2007-D2-P4, 2007-D2-P5 2008-D1-P2, 2008-D2-P5, 2008-D2-P6, 2009-D1-P2, 2009-D1-P3, 2009-D1-P4, 2009-D1-P5, 2009-D2-P3 2009-D2-P5, 2010-D1-P5, 2010-D2-P4, 2011-D1-P2, 2011-D1-P5, 2011-D2-P4, 2011-D2-P5, 2012-D1-P2 2012-D1-P4, 2013-D1-P1, 2013-D1-P2, 2014-D1-P1, 2014-D2-P2, 2015-D1-P1, 2015-D1-P5, 2015-D2-P9 2016-D1-P1, 2016-D2-P10, 2017-D1-P1, 2017-D1-P2, 2017-D2-P9, 2018-D1-P3, 2018-D1-P4, 2019-D1-P3 2019-D1-P5, 2019-D2-P9, 2019-D2-P10, 2020-D1-P2, 2020-D2-P5, 2021-D1-P1, 2022-D1-P1, 2022-D2-P7 2023-D1-P2, 2023-D2-P6, 2023-D2-P7, 2023-D2-P9, 2024-D1-P1, 2024-D1-P5, 2024-D2-P7

LA-05 - Trace cyclicity, trace pairing, and zero-trace subspaces

Bucket	LA trace
Core theory	The trace is cyclic: $\operatorname{tr}(XY) = \operatorname{tr}(YX)$ and hence $\operatorname{tr}([X,Y]) = 0$. The bilinear form $(X,Y) \mapsto \operatorname{tr}(XY)$ identifies the dual of the matrix space and detects symmetric or triangular subspaces.
IMC use	Use for commutator problems, linear functionals on matrix spaces, bilinear orthogonality conditions, and trace constraints.
Trap watch	Cyclicity is safe for products under trace, not outside trace. Do not infer $XY = YX$ from equal traces.
Appears in	1995-D1-P1, 1995-D2-P1, 1996-D1-P1, 1997-D2-P2, 1997-D2-P4, 1999-D1-P1, 2000-D1-P3, 2001-D1-P5 2001-D2-P4, 2002-D2-P1, 2003-D2-P1, 2004-D2-P1, 2005-D1-P1, 2005-D2-P3, 2005-D2-P6, 2006-D2-P6 2007-D1-P2, 2007-D1-P3, 2007-D2-P4, 2009-D1-P2, 2009-D2-P3, 2010-D1-P5, 2011-D1-P2, 2012-D1-P2 2013-D1-P1, 2014-D1-P1, 2015-D1-P1, 2019-D1-P5, 2020-D1-P2, 2021-D1-P1, 2022-D2-P7, 2023-D2-P6 2024-D2-P7

LA-06 - Commutators and spectral obstructions

Bucket	LA commutators
Core theory	A commutator $AB - BA$ has trace zero. If it has very small rank or satisfies a polynomial identity, trace and Jordan-form restrictions can force nilpotency or impossibility.
IMC use	Use when a problem contains $AB - BA$, $FG - GF$, or an operator $X \mapsto AX - XA$.
Trap watch	Trace zero alone is weak; combine it with rank, eigenvalue multiplicity, or finite-dimensionality.
Appears in	1994-D1-P4, 1995-D1-P5, 1996-D1-P1, 1997-D1-P3, 1997-D2-P4, 2000-D1-P3, 2000-D1-P5, 2000-D2-P6 2001-D1-P5, 2002-D2-P1, 2003-D1-P3, 2003-D2-P1, 2004-D2-P1, 2005-D1-P6, 2009-D2-P3, 2013-D1-P1 2014-D1-P1, 2020-D1-P2, 2021-D1-P1, 2022-D2-P7

LA-07 - Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton

Bucket	LA spectral theory
Core theory	If $p(A) = 0$, every eigenvalue of A is a root of p. Cayley-Hamilton gives $\chi_A(A) = 0$. Distinct roots of the minimal polynomial imply diagonalizability over a splitting field.
IMC use	Use for convergence of matrix powers, nilpotency, idempotency, diagonalization, determinant/eigenvalue constraints, and polynomial equations in matrices.
Trap watch	Real matrices may have complex eigenvalues; diagonalization over $\mathbb{C}$ may be needed even when entries are real.
Appears in	1994-D1-P4, 1994-D2-P2, 1995-D1-P1, 1995-D1-P4, 1995-D1-P5, 1995-D2-P5, 1996-D1-P1, 1996-D1-P3 1996-D1-P5, 1997-D1-P5, 1997-D2-P2, 1998-D2-P1, 1998-D2-P4, 1999-D1-P1, 2000-D1-P3, 2000-D1-P5 2000-D2-P6, 2001-D1-P5, 2001-D1-P6, 2002-D1-P1, 2002-D2-P1, 2002-D2-P3, 2002-D2-P5, 2003-D1-P3 2003-D1-P4, 2003-D2-P1, 2004-D1-P1, 2004-D1-P5, 2004-D2-P1, 2004-D2-P4, 2006-D2-P2, 2006-D2-P6 2008-D1-P3, 2009-D1-P2, 2009-D2-P3, 2010-D2-P1, 2011-D1-P2, 2013-D1-P1, 2014-D1-P1, 2014-D2-P2 2015-D1-P1, 2015-D2-P9, 2016-D2-P10, 2017-D1-P1, 2017-D2-P8, 2018-D1-P3, 2021-D1-P1, 2022-D1-P2 2023-D1-P2, 2023-D2-P6, 2024-D2-P7

LA-08 - Nilpotent, idempotent, involution, and square-zero structures

Bucket	LA special matrices
Core theory	Nilpotent matrices have all eigenvalues 0; idempotents have eigenvalues in $\{0, 1\}$; involutions have eigenvalues in $\{\pm 1\}$ when the characteristic is not 2. These facts often convert algebraic equations into diagonal or block forms.
IMC use	Use for matrix equations involving powers, commuting involutions, square-zero products, and rank bounds for idempotents.
Trap watch	Do not assume diagonalizability for nilpotents. Idempotents and involutions are diagonalizable only when the minimal polynomial has distinct roots.
Appears in	1994-D1-P4, 1995-D1-P1, 1995-D1-P5, 1996-D1-P3, 2000-D1-P3, 2000-D1-P5, 2000-D2-P6, 2001-D2-P4 2003-D1-P3, 2005-D1-P6, 2007-D2-P5, 2009-D1-P2, 2018-D1-P3, 2022-D2-P7

LA-09 - Simultaneous diagonalization and commuting families

Bucket	LA simultaneous structure
Core theory	A commuting family of diagonalizable operators over an algebraically closed field is simultaneously diagonalizable under standard finite-dimensional hypotheses. For diagonal matrices, the commutant is block diagonal according to equal eigenvalues.
IMC use	Use when several operators commute or preserve common eigenspaces; typical conclusions are dimension counts and maximum cardinality of families.
Trap watch	Commuting alone does not imply diagonalizable. First prove each operator is diagonalizable or use invariant subspaces carefully.
Appears in	1994-D2-P4, 1996-D1-P3, 2006-D2-P6

LA-10 - Invariant subspaces and adapted bases

Bucket	LA adapted basis
Core theory	If a chain of subspaces is invariant, choose a basis adapted to the chain; the matrix becomes block upper triangular, and free entries can be counted.
IMC use	Use in dimension-of-endomorphism-space problems, simultaneous preservation of subspaces, and centralizer computations.
Trap watch	Check the order of basis blocks: $T(U) \subset U$ imposes zero blocks below, not above, depending on column convention.
Appears in	1994-D2-P4, 1994-D2-P5, 1995-D1-P1, 1996-D1-P3, 1996-D1-P6, 1997-D2-P2, 1997-D2-P4, 1998-D1-P1 1998-D2-P1, 2001-D1-P5, 2001-D2-P3, 2002-D1-P5, 2002-D1-P6, 2004-D2-P4, 2005-D2-P3, 2006-D2-P2 2006-D2-P4, 2007-D1-P2, 2007-D2-P5, 2008-D1-P2, 2009-D1-P3, 2009-D1-P4, 2009-D1-P5, 2010-D2-P4 2011-D1-P5, 2011-D2-P5, 2012-D1-P2, 2015-D1-P5, 2015-D2-P9, 2023-D2-P9

CORRECTION MATH-010, MATH-011, MATH-012 — legacy statement superseded; click for canonical replacement.

LA-11 - Linear functionals, kernels, and dual span criteria

Bucket	LA duality
Core theory	A linear functional that vanishes on $\bigcap_i \ker f_i$ belongs to the span of $f_1, \dots, f_k$. Equivalently, the annihilator of an intersection of kernels is the span of the functionals.
IMC use	Use for abstract vector-space problems where one functional is forced by the vanishing of several others.
Trap watch	Independence of the given functionals may need to be arranged by discarding redundant ones.
Appears in	1997-D2-P2, 1997-D2-P4, 1998-D1-P1, 1998-D2-P1, 2008-D1-P2, 2015-D2-P9

LA-12 - Matrix subspaces constrained by trace bilinear forms

Bucket	LA trace geometry
Core theory	The trace form separates symmetric and skew/triangular subspaces. If $\operatorname{tr}(XY) = 0$ for all X, Y in a subspace, then the subspace cannot contain nonzero symmetric matrices, giving a dimension bound.
IMC use	Use in maximum-dimension matrix subspace problems with quadratic trace constraints.
Trap watch	The form $\operatorname{tr}(XY)$ is not positive definite on all matrices; positivity appears on symmetric matrices via $\operatorname{tr}(A^2)$.
Appears in	2005-D2-P3

LA-13 - Matrix exponentials and polynomial functional calculus

Bucket	LA functional calculus
Core theory	For a matrix X, $q(e^X)$ can be expanded as a convergent power series in X. Identities comparing AB and BA often reduce to the fact that AC and CA have the same characteristic polynomial.
IMC use	Use when exponentials of products AB and BA appear, especially with nilpotency of polynomial expressions.
Trap watch	Do not use $e^{AB} = e^{BA}$; compare characteristic polynomials or nonzero spectra instead.
Appears in	2000-D2-P6

LA-14 - Operator norm, power boundedness, and spectral-radius style estimates

Bucket	LA norms
Core theory	A sequence of operator powers can be controlled by estimates on differences $T^{n+1} - T^n$. Exponential decay of these differences implies norm convergence and power boundedness.
IMC use	Use for norm inequalities involving all positive powers of a matrix or linear operator.
Trap watch	Norm convergence is stronger than pointwise convergence. Watch constants and finite-dimensional vs Banach-space arguments.
Appears in	2002-D1-P6

LA-15 - Gram matrices and rational linear dependence

Bucket	LA Gram matrices
Core theory	Distances determine inner products by $2\langle u, v\rangle = \|u\|^2 + \|v\|^2 - \|u - v\|^2$. If the Gram matrix has rational entries and is invertible, coordinates of another vector obtained by inner products are rational.
IMC use	Use for rational-distance vector problems and rational linear-dependence conclusions.
Trap watch	First reduce to a real basis; then solve the linear system over $\mathbb{Q}$.
Appears in	1995-D2-P1, 2002-D2-P6, 2006-D2-P4, 2008-D2-P6, 2013-D2-P3, 2018-D2-P6, 2021-D2-P8

LA-16 - Anticommuting and rank constraints for special matrices

Bucket	LA special rank bounds
Core theory	Anticommuting idempotents or square-zero maps impose rank and dimension constraints. A common proof decomposes space into images/kernels or uses trace/rank averaging.
IMC use	Use when many idempotents, nilpotents, or anticommuting matrices are given and a rank bound is requested.
Trap watch	Pairwise anticommutation is not pairwise orthogonality; translate the algebraic condition into image/kernel constraints.
Appears in	No direct automatic assignment; keep as supporting background.

7.2 Real analysis / calculus

RA-01 - Compactness, extreme values, and closed-bounded reduction

Bucket	RA compactness
Core theory	A continuous function on a compact set attains its minimum and maximum. For unbounded domains, first show extrema cannot occur outside a compact sublevel region.
IMC use	Use in multivariable extrema, closed-set nearest-point arguments, convex-body problems, and existence of maximizers/minimizers.
Trap watch	Compactness requires closed and bounded in finite-dimensional Euclidean spaces; infinite-dimensional Hilbert spaces behave differently.
Appears in	1994-D2-P2, 1995-D1-P6, 1995-D2-P4, 1995-D2-P6, 1996-D1-P6, 1996-D2-P4, 1998-D2-P2, 2000-D1-P1 2000-D2-P2, 2001-D1-P3, 2001-D2-P6, 2002-D1-P4, 2002-D2-P6, 2003-D2-P3, 2004-D1-P3, 2005-D2-P1 2005-D2-P5, 2006-D2-P3, 2007-D2-P3, 2008-D1-P1, 2011-D1-P1, 2013-D1-P3, 2013-D1-P4, 2014-D1-P2 2014-D2-P3, 2014-D2-P5, 2015-D2-P6, 2015-D2-P7, 2015-D2-P9, 2016-D2-P6, 2016-D2-P7, 2018-D1-P1 2018-D1-P4, 2018-D2-P10, 2019-D1-P1, 2019-D1-P3, 2020-D2-P8

RA-02 - Intermediate value, connectedness, density, and continuity transfer

Bucket	RA continuity
Core theory	Continuous images of intervals are intervals; continuous functions preserve limits. Dense subsets such as $\mathbb{Q}$ determine continuous functions on $\mathbb{R}$ when the relevant equality or inequality is closed under limits.
IMC use	Use for interval image problems, monotone/continuous comparisons on rationals, and connectedness of constraint sets.
Trap watch	A monotone function need not be continuous unless surjectivity or no jumps is proved.
Appears in	1994-D2-P2, 1995-D1-P2, 1995-D1-P6, 1995-D2-P4, 1995-D2-P6, 1996-D1-P6, 1996-D2-P1, 1996-D2-P4 1998-D1-P4, 1998-D1-P6, 1998-D2-P5, 2000-D2-P2, 2002-D1-P4, 2003-D2-P3, 2004-D1-P3, 2005-D1-P3 2006-D1-P1, 2006-D1-P5, 2006-D2-P2, 2007-D2-P1, 2007-D2-P3, 2008-D1-P1, 2009-D1-P1, 2009-D1-P3 2011-D1-P1, 2012-D1-P4, 2013-D1-P2, 2013-D1-P4, 2014-D2-P3, 2015-D2-P6, 2016-D1-P1, 2016-D2-P6 2016-D2-P7, 2017-D2-P6, 2017-D2-P9, 2018-D1-P4, 2019-D1-P1, 2019-D2-P6, 2020-D2-P5, 2021-D1-P3 2021-D1-P4, 2023-D1-P1, 2023-D2-P7, 2023-D2-P8

RA-03 - Monotone functions, fixed points, and iteration on intervals

Bucket	RA monotonicity/fixed points
Core theory	A nondecreasing self-map of a compact interval has a fixed point. For continuous interval maps, convergence of iterates can often be characterized by the vanishing of increments plus cluster-point analysis.
IMC use	Use for monotone maps, closed invariant sets, fixed points, and iteration sequences.
Trap watch	A decreasing map may fail to have a fixed point if discontinuous. Iteration convergence needs cluster-point control.
Appears in	1994-D2-P6, 1995-D1-P4, 1998-D2-P3, 1999-D2-P4, 2000-D1-P1, 2000-D1-P4, 2000-D2-P2, 2001-D2-P2 2004-D1-P5, 2004-D2-P2, 2007-D2-P3, 2010-D1-P1, 2011-D1-P4, 2015-D2-P6, 2019-D1-P1

RA-04 - Rolle, mean-value theorem, and divided differences

Bucket	RA Rolle/MVT
Core theory	Rolle's theorem and the mean-value theorem convert repeated zeros or equal endpoint values into derivative zeros. Divided differences satisfy $f[x_0, \dots, x_n] = f^{(n)}(\xi)/n!$ under differentiability.
IMC use	Use whenever many zeros are given, a derivative value must exist, or endpoint data must imply a higher-derivative identity.
Trap watch	Check differentiability order and that all points lie in the open interval where the conclusion is asserted.
Appears in	1994-D1-P2, 1994-D1-P6, 1994-D2-P1, 1994-D2-P3, 1995-D1-P3, 1996-D1-P5, 1996-D2-P5, 1996-D2-P6 1997-D2-P1, 1997-D2-P3, 1998-D1-P4, 1999-D1-P6, 2000-D1-P2, 2000-D1-P6, 2000-D2-P5, 2001-D1-P6 2001-D2-P6, 2002-D1-P2, 2002-D1-P5, 2005-D1-P3, 2005-D1-P5, 2005-D2-P4, 2006-D1-P5, 2006-D1-P6 2007-D2-P6, 2009-D2-P2, 2010-D1-P1, 2012-D1-P4, 2013-D1-P2, 2017-D1-P5, 2023-D1-P1, 2023-D1-P2 2023-D2-P7

RA-05 - Taylor expansion, local asymptotics, and little-o bookkeeping

Bucket	RA Taylor
Core theory	Near a point, C^k functions can be expanded with a controlled remainder. Competition solutions often need only the first nonzero term and the sign/size of the remainder.
IMC use	Use for local boundedness, sharp constants near equality, ratio limits, and endpoint asymptotics.
Trap watch	Do not expand beyond available smoothness; distinguish $O(x^k)$ from $o(x^k)$.
Appears in	1994-D1-P2, 1994-D1-P5, 1994-D1-P6, 1994-D2-P1, 1994-D2-P3, 1994-D2-P4, 1994-D2-P6, 1995-D1-P3 1995-D1-P6, 1995-D2-P2, 1996-D1-P5, 1996-D2-P5, 1996-D2-P6, 1997-D1-P1, 1997-D2-P1, 1997-D2-P3 1998-D1-P4, 1999-D1-P6, 2000-D1-P2, 2000-D2-P5, 2001-D1-P3, 2001-D1-P6, 2001-D2-P6, 2002-D1-P2 2004-D1-P6, 2005-D1-P3, 2005-D1-P5, 2005-D2-P4, 2006-D1-P5, 2006-D1-P6, 2007-D2-P6, 2009-D2-P2 2010-D1-P1, 2010-D1-P3, 2010-D1-P5, 2010-D2-P5, 2012-D1-P4, 2013-D1-P2, 2014-D1-P2, 2015-D2-P7 2017-D1-P5, 2017-D2-P6, 2017-D2-P9, 2018-D2-P10, 2020-D2-P8, 2021-D1-P4, 2023-D1-P1, 2023-D1-P2 2023-D2-P7, 2024-D1-P2

RA-06 - Auxiliary functions, integrating factors, and differential inequalities

Bucket	RA differential inequalities
Core theory	A differential inequality is often solved by choosing an auxiliary expression whose derivative has a fixed sign, such as $\arctan f + x$, $e^{-\lambda x} f(x)$, or $f' + xf$.
IMC use	Use for uniqueness, lower/upper bounds, exponential comparison, and ODE-like inequalities.
Trap watch	If taking logarithms, first prove the function has the required sign on the interval.
Appears in	1994-D1-P2, 1994-D2-P1, 1994-D2-P3, 1994-D2-P6, 1995-D1-P2, 1995-D1-P4, 1996-D1-P2, 1998-D1-P3 1998-D1-P4, 1998-D1-P6, 1999-D1-P2, 1999-D1-P4, 1999-D2-P3, 1999-D2-P4, 2000-D1-P2, 2000-D1-P4 2000-D1-P6, 2000-D2-P2, 2001-D1-P3, 2001-D1-P6, 2003-D1-P1, 2003-D2-P2, 2004-D1-P5, 2004-D2-P5 2005-D1-P3, 2005-D1-P5, 2006-D1-P1, 2006-D1-P5, 2006-D2-P3, 2007-D2-P3, 2009-D2-P2, 2010-D1-P1 2010-D2-P1, 2011-D2-P1, 2012-D1-P4, 2012-D2-P2, 2013-D1-P2, 2013-D1-P4, 2013-D2-P3, 2014-D1-P2 2014-D2-P2, 2014-D2-P3, 2014-D2-P5, 2015-D1-P4, 2015-D2-P6, 2015-D2-P7, 2016-D2-P6, 2016-D2-P7 2017-D1-P2, 2017-D2-P6, 2018-D1-P1, 2018-D1-P4, 2018-D2-P7, 2018-D2-P10, 2019-D1-P1, 2019-D1-P3 2020-D2-P5, 2020-D2-P8, 2021-D1-P3, 2021-D1-P4, 2023-D1-P1, 2024-D1-P2

RA-07 - Convexity, concavity, Jensen, and tangent/chord inequalities

Bucket	RA convexity
Core theory	For convex f, tangent lines lie below the graph and Jensen gives $f(\sum \lambda_i x_i) \le \sum \lambda_i f(x_i)$. For concave functions the inequalities reverse. Equality cases are usually decisive.
IMC use	Use for inequalities, sharp constants, integral estimates, chord comparisons, and geometric convexity.
Trap watch	Check the interval of convexity and strictness conditions for equality.
Appears in	1994-D1-P1, 1994-D1-P2, 1994-D1-P4, 1994-D1-P5, 1994-D1-P6, 1994-D2-P1, 1994-D2-P2, 1994-D2-P5 1994-D2-P6, 1995-D1-P2, 1995-D1-P3, 1995-D1-P4, 1995-D1-P6, 1995-D2-P2, 1995-D2-P4, 1995-D2-P5 1996-D1-P2, 1996-D1-P4, 1996-D1-P5, 1996-D2-P2, 1996-D2-P4, 1996-D2-P5, 1996-D2-P6, 1997-D1-P1 1997-D1-P2, 1997-D1-P4, 1997-D1-P5, 1997-D2-P1, 1997-D2-P3, 1998-D1-P2, 1998-D1-P3, 1998-D1-P4 1998-D1-P5, 1998-D1-P6, 1998-D2-P1, 1998-D2-P2, 1998-D2-P3, 1998-D2-P4, 1998-D2-P5, 1998-D2-P6 1999-D1-P2, 1999-D1-P3, 1999-D1-P4, 1999-D1-P6, 1999-D2-P3, 1999-D2-P4, 1999-D2-P6, 2000-D1-P1 2000-D1-P4, 2000-D2-P1, 2000-D2-P2, 2000-D2-P3, 2000-D2-P4, 2000-D2-P5, 2001-D1-P4, 2001-D1-P6 2001-D2-P1, 2001-D2-P2, 2001-D2-P3, 2001-D2-P5, 2001-D2-P6, 2002-D1-P2, 2002-D1-P3, 2002-D1-P6 2002-D2-P3, 2002-D2-P6, 2003-D1-P4, 2003-D1-P5, 2003-D1-P6, 2003-D2-P2, 2003-D2-P3, 2003-D2-P5 2003-D2-P6, 2004-D1-P1, 2004-D1-P2, 2004-D1-P3, 2004-D1-P5, 2004-D1-P6, 2004-D2-P2, 2004-D2-P3 2004-D2-P4, 2004-D2-P5, 2004-D2-P6, 2005-D1-P1, 2005-D1-P2, 2005-D1-P3, 2005-D1-P5, 2005-D1-P6 2005-D2-P1, 2005-D2-P5, 2006-D1-P1, 2006-D1-P5, 2006-D1-P6, 2006-D2-P1, 2006-D2-P2, 2006-D2-P3 2006-D2-P4, 2007-D1-P2, 2007-D1-P3, 2007-D1-P5, 2007-D1-P6, 2007-D2-P3, 2007-D2-P6, 2008-D1-P1 2008-D1-P2, 2008-D1-P4, 2008-D1-P5, 2008-D2-P3, 2008-D2-P6, 2009-D1-P1, 2009-D1-P4, 2009-D2-P1 2009-D2-P2, 2010-D1-P1, 2010-D1-P2, 2010-D1-P3, 2010-D1-P4, 2010-D1-P5, 2010-D2-P1, 2010-D2-P2 2010-D2-P3, 2010-D2-P4, 2011-D1-P1, 2011-D1-P4, 2011-D2-P1, 2011-D2-P3, 2011-D2-P5, 2012-D1-P4 2012-D2-P2, 2012-D2-P4, 2012-D2-P5, 2013-D1-P3, 2013-D1-P4, 2013-D2-P3, 2014-D2-P2, 2014-D2-P5 2015-D1-P1, 2015-D1-P2, 2015-D1-P3, 2015-D1-P5, 2015-D2-P10, 2016-D1-P2, 2016-D1-P3, 2016-D1-P4 2016-D1-P5, 2016-D2-P6, 2016-D2-P7, 2016-D2-P9, 2016-D2-P10, 2017-D1-P2, 2017-D1-P3, 2017-D1-P5 2018-D1-P5, 2018-D2-P7, 2018-D2-P8, 2019-D1-P3, 2019-D1-P4, 2019-D2-P10, 2020-D1-P3, 2020-D1-P4 2020-D2-P5, 2020-D2-P7, 2021-D2-P5, 2021-D2-P7, 2022-D1-P1, 2022-D1-P2, 2022-D1-P4, 2022-D2-P7 2022-D2-P8, 2023-D1-P5, 2023-D2-P8, 2023-D2-P9, 2024-D1-P2, 2024-D1-P4, 2024-D1-P5, 2024-D2-P6 2024-D2-P8, 2024-D2-P9, 2024-D2-P10

RA-08 - Integral inequalities, integration by parts, and weighted estimates

Bucket	RA integral inequalities
Core theory	Integration by parts transfers derivatives between factors; nonnegative integrands and pointwise bounds integrate to global inequalities. Introducing primitives often converts a cubic or square integral into a boundary term plus an error.
IMC use	Use in L^p inequalities, integral constraints, monotone integrals, and estimates of functions by their derivatives.
Trap watch	Boundary terms are often the whole point; compute their signs before discarding them.
Appears in	1994-D1-P2, 1994-D1-P3, 1994-D1-P4, 1994-D1-P5, 1994-D1-P6, 1994-D2-P1, 1994-D2-P2, 1994-D2-P3 1994-D2-P5, 1994-D2-P6, 1995-D1-P2, 1995-D1-P4, 1995-D1-P5, 1995-D1-P6, 1995-D2-P2, 1995-D2-P5 1995-D2-P6, 1996-D1-P2, 1996-D1-P4, 1996-D1-P5, 1996-D2-P1, 1996-D2-P2, 1996-D2-P4, 1996-D2-P5 1996-D2-P6, 1997-D1-P1, 1997-D1-P2, 1997-D1-P4, 1997-D1-P5, 1997-D1-P6, 1997-D2-P1, 1997-D2-P3 1998-D1-P2, 1998-D1-P3, 1998-D1-P4, 1998-D1-P5, 1998-D1-P6, 1998-D2-P1, 1998-D2-P2, 1998-D2-P6 1999-D1-P2, 1999-D1-P3, 1999-D1-P4, 1999-D1-P6, 1999-D2-P3, 1999-D2-P4, 1999-D2-P5, 1999-D2-P6 2000-D1-P1, 2000-D1-P4, 2000-D1-P6, 2000-D2-P2, 2000-D2-P4, 2000-D2-P5, 2000-D2-P6, 2001-D1-P1 2001-D1-P3, 2001-D2-P1, 2001-D2-P2, 2001-D2-P4, 2001-D2-P5, 2001-D2-P6, 2002-D1-P2, 2002-D1-P3 2002-D1-P4, 2002-D1-P5, 2002-D1-P6, 2002-D2-P3, 2002-D2-P4, 2002-D2-P6, 2003-D1-P1, 2003-D1-P2 2003-D1-P3, 2003-D1-P4, 2003-D1-P5, 2003-D1-P6, 2003-D2-P2, 2003-D2-P5, 2003-D2-P6, 2004-D1-P2 2004-D1-P3, 2004-D1-P4, 2004-D1-P5, 2004-D1-P6, 2004-D2-P2, 2004-D2-P3, 2004-D2-P4, 2004-D2-P5 2004-D2-P6, 2005-D1-P1, 2005-D1-P3, 2005-D1-P4, 2005-D1-P5, 2005-D2-P1, 2005-D2-P2, 2005-D2-P3 2005-D2-P5, 2006-D1-P1, 2006-D1-P3, 2006-D1-P4, 2006-D1-P5, 2006-D1-P6, 2006-D2-P1, 2006-D2-P3 2006-D2-P4, 2007-D1-P2, 2007-D1-P5, 2007-D1-P6, 2007-D2-P2, 2007-D2-P3, 2007-D2-P6, 2008-D1-P1 2008-D1-P2, 2008-D1-P4, 2008-D2-P3, 2008-D2-P6, 2009-D1-P1, 2009-D1-P4, 2009-D1-P5, 2009-D2-P1 2009-D2-P2, 2009-D2-P3, 2009-D2-P5, 2010-D1-P1, 2010-D1-P2, 2010-D1-P3, 2010-D1-P4, 2010-D2-P1 2010-D2-P2, 2010-D2-P3, 2011-D1-P1, 2011-D1-P5, 2011-D2-P1, 2011-D2-P2, 2011-D2-P3, 2011-D2-P4 2011-D2-P5, 2012-D1-P1, 2012-D1-P4, 2012-D2-P2, 2012-D2-P4, 2013-D1-P2, 2013-D1-P3, 2013-D1-P4 2013-D1-P5, 2013-D2-P3, 2013-D2-P4, 2014-D1-P2, 2014-D1-P4, 2014-D2-P2, 2014-D2-P3, 2014-D2-P5 2015-D1-P2, 2015-D1-P3, 2015-D1-P5, 2015-D2-P6, 2015-D2-P7, 2015-D2-P8, 2015-D2-P10, 2016-D1-P2 2016-D1-P3, 2016-D1-P4, 2016-D2-P6, 2016-D2-P7, 2016-D2-P9, 2016-D2-P10, 2017-D1-P2, 2017-D1-P3 2017-D2-P6, 2018-D1-P1, 2018-D1-P4, 2018-D1-P5, 2018-D2-P7, 2018-D2-P8, 2018-D2-P10, 2019-D1-P1 2019-D1-P3, 2019-D1-P4, 2019-D2-P6, 2019-D2-P7, 2019-D2-P8, 2020-D1-P3, 2020-D1-P4, 2020-D2-P5 2020-D2-P8, 2021-D1-P3, 2021-D1-P4, 2021-D2-P5, 2022-D1-P1, 2022-D1-P2, 2022-D1-P4, 2023-D1-P1 2023-D1-P5, 2023-D2-P8, 2024-D1-P2, 2024-D1-P4, 2024-D2-P6, 2024-D2-P8, 2024-D2-P10

RA-09 - Fubini/Tonelli and change of variables

Bucket	RA Fubini
Core theory	For nonnegative or absolutely integrable functions, interchange the order of integration and use substitutions to convert regions or kernels into simpler forms.
IMC use	Use for nested integrals, averaging operators, double integrals over the unit square, and kernel formulas.
Trap watch	Justify interchanging integrals or sums when signs are not nonnegative.
Appears in	1994-D2-P1, 1994-D2-P3, 1995-D1-P2, 1995-D2-P5, 1996-D1-P2, 1996-D1-P4, 1996-D2-P1, 1997-D1-P1 1997-D1-P2, 1997-D1-P4, 1998-D1-P3, 1998-D1-P6, 1999-D1-P4, 1999-D2-P3, 1999-D2-P4, 2001-D1-P1 2002-D1-P3, 2003-D1-P1, 2003-D2-P6, 2004-D1-P2, 2004-D2-P5, 2005-D2-P1, 2007-D2-P2, 2008-D1-P4 2010-D1-P2, 2010-D2-P1, 2010-D2-P2, 2011-D2-P1, 2011-D2-P3, 2012-D1-P1, 2012-D2-P2, 2013-D1-P5 2013-D2-P4, 2014-D1-P4, 2015-D2-P8, 2018-D1-P1, 2018-D2-P7, 2019-D2-P6, 2019-D2-P7

RA-10 - Riemann sums and sum-integral asymptotics

Bucket	RA Riemann sums
Core theory	If $h \to 0$, sums of the form $h \sum f(nh)$ approximate $\int f$. For infinite ranges, split into a compact main part and a tail controlled uniformly.
IMC use	Use for limits of sums, logarithmic sums, periodic averaging, and asymptotic counting.
Trap watch	Tail control is the common missing step; do not use finite-interval Riemann sums on an infinite domain without a tail estimate.
Appears in	1994-D1-P2, 1994-D1-P5, 1994-D2-P6, 1995-D1-P2, 1995-D1-P3, 1995-D2-P2, 1995-D2-P5, 1995-D2-P6 1996-D1-P4, 1996-D1-P5, 1996-D2-P1, 1996-D2-P5, 1997-D1-P1, 1997-D1-P2, 1997-D2-P1, 1997-D2-P3 1998-D1-P3, 1999-D1-P2, 1999-D2-P3, 2000-D1-P4, 2000-D1-P6, 2000-D2-P5, 2001-D1-P3, 2001-D1-P6 2001-D2-P2, 2001-D2-P5, 2002-D1-P4, 2002-D1-P6, 2003-D1-P1, 2003-D1-P3, 2003-D1-P5, 2003-D2-P2 2003-D2-P5, 2003-D2-P6, 2004-D2-P4, 2004-D2-P5, 2005-D1-P5, 2006-D1-P1, 2006-D1-P4, 2006-D2-P3 2007-D2-P6, 2009-D1-P1, 2010-D1-P2, 2010-D1-P3, 2010-D2-P1, 2010-D2-P5, 2011-D1-P1, 2011-D2-P1 2013-D1-P2, 2013-D1-P4, 2014-D1-P2, 2014-D1-P4, 2014-D2-P2, 2014-D2-P3, 2014-D2-P5, 2015-D2-P6 2015-D2-P7, 2016-D2-P6, 2017-D1-P2, 2017-D2-P6, 2017-D2-P9, 2018-D1-P1, 2018-D1-P4, 2018-D2-P10 2019-D1-P1, 2019-D1-P3, 2020-D2-P5, 2020-D2-P8, 2021-D1-P3, 2021-D1-P4, 2023-D1-P1, 2024-D1-P2 2024-D2-P8

RA-11 - Series convergence, rearrangements, and alternating/Dirichlet tools

Bucket	RA series
Core theory	Convergence can be altered by nonabsolute rearrangements. Alternating and Dirichlet-type arguments reduce convergence to decay plus bounded variation or controlled differences.
IMC use	Use for rearranged series, alternating trigonometric series, Carleman-type inequalities, and positive series estimates.
Trap watch	Conditional convergence is not preserved by arbitrary permutations.
Appears in	1994-D1-P3, 1994-D2-P1, 1994-D2-P3, 1994-D2-P5, 1995-D1-P5, 1995-D1-P6, 1995-D2-P5, 1995-D2-P6 1996-D1-P2, 1996-D1-P4, 1996-D1-P5, 1996-D2-P1, 1996-D2-P5, 1996-D2-P6, 1997-D1-P1, 1997-D1-P2 1997-D1-P4, 1997-D1-P5, 1997-D1-P6, 1997-D2-P3, 1998-D1-P3, 1998-D1-P6, 1998-D2-P1, 1999-D1-P2 1999-D1-P4, 1999-D2-P3, 1999-D2-P4, 1999-D2-P5, 2000-D2-P5, 2001-D1-P1, 2001-D2-P2, 2002-D1-P2 2002-D1-P3, 2002-D2-P4, 2003-D1-P1, 2003-D1-P3, 2003-D2-P6, 2004-D1-P2, 2004-D1-P5, 2005-D1-P1 2005-D2-P1, 2005-D2-P2, 2006-D1-P1, 2006-D1-P4, 2006-D1-P6, 2006-D2-P4, 2007-D2-P2, 2008-D1-P2 2008-D1-P4, 2010-D1-P2, 2010-D2-P1, 2010-D2-P2, 2010-D2-P3, 2011-D1-P1, 2011-D2-P1, 2011-D2-P3 2012-D1-P1, 2012-D2-P2, 2013-D1-P5, 2013-D2-P4, 2014-D1-P4, 2015-D2-P8, 2018-D1-P1, 2018-D2-P7 2019-D2-P6, 2019-D2-P7, 2019-D2-P8

RA-12 - Uniform continuity and periodic averaging

Bucket	RA oscillatory averaging
Core theory	Continuous functions on compact intervals are uniformly continuous. Products with rapidly oscillating periodic factors average to the product of means after partitioning into small intervals.
IMC use	Use for limits of $\int f(x)g(nx)\,dx$ and periodic integral averaging.
Trap watch	Keep the period and interval length consistent; normalize by the period correctly.
Appears in	1994-D1-P5

CORRECTION MATH-013, MATH-014 — legacy statement superseded; click for canonical replacement.

RA-13 - L'Hospital, ratio limits, and growth comparisons

Bucket	RA ratio limits
Core theory	When numerator and denominator have compatible divergent or vanishing behavior, derivative ratios can determine function ratios. Growth comparisons often reduce to exponentials, logarithms, or powers.
IMC use	Use for endpoint ratio limits, exponential weights, and asymptotic estimates.
Trap watch	LH́ospital requires the appropriate hypotheses; for inequalities, use comparison rather than equality of limits when needed.
Appears in	1994-D1-P2, 1994-D1-P5, 1994-D2-P6, 1995-D1-P3, 1995-D1-P4, 1995-D2-P2, 1996-D1-P4, 1996-D2-P1 1996-D2-P5, 1997-D1-P1, 1997-D1-P2, 1997-D2-P1, 2000-D1-P6, 2001-D1-P3, 2001-D1-P6, 2001-D2-P5 2003-D1-P1, 2003-D2-P2, 2004-D1-P6, 2004-D2-P5, 2005-D1-P5, 2010-D1-P2, 2010-D1-P3, 2011-D2-P3 2014-D1-P2, 2015-D2-P7, 2017-D2-P6, 2017-D2-P9, 2018-D2-P10, 2020-D2-P8, 2021-D1-P4, 2024-D1-P2 2024-D2-P8

RA-14 - Weierstrass approximation and polynomial approximation

Bucket	RA approximation
Core theory	Every continuous function on a compact interval can be uniformly approximated by polynomials. Symmetry can be imposed by averaging $p(x)$ with $p(-x)$ to obtain even or odd approximants.
IMC use	Use for uniform approximation by restricted polynomial families and Chebyshev-style extremal examples.
Trap watch	Approximate the symmetric part, then handle missing low-degree terms separately if the basis omits them.
Appears in	1995-D2-P4

RA-15 - L2 spaces, orthonormal systems, and dominated convergence

Bucket	RA functional analysis
Core theory	In L^2, orthonormality gives coefficient constraints and no pointwise-convergent bounded subsequence can contradict norm preservation when dominated convergence applies.
IMC use	Use for orthonormal continuous functions, Hilbert-space arguments, and bounded convergence of integrals.
Trap watch	Pointwise convergence plus uniform boundedness allows dominated convergence, but only on finite measure domains.
Appears in	1995-D2-P6

RA-16 - Baire/category, dense sets, and pathological continuous functions

Bucket	RA topology/pathology
Core theory	Closed intervals cannot be countable unions of nowhere dense sets. Cantor-type constructions create uncountable closed sets with prescribed local behavior.
IMC use	Use for dense local minima, uncountably many zero crossings, and existence/nonexistence of functions with many level sets.
Trap watch	Countability and uncountability arguments must distinguish isolated intervals from point sets.
Appears in	1997-D2-P6, 1998-D1-P4, 1998-D2-P6, 2000-D2-P2, 2002-D1-P5, 2003-D2-P5

CORRECTION MATH-015 — legacy statement superseded; click for canonical replacement.

RA-17 - Sequence recurrences, monotone bounds, and generating functions

Bucket	RA recurrences
Core theory	Recursive sequences are attacked by monotone normalization, induction estimates, telescoping transformations, or generating functions $F(x) = \sum a_n x^n$.
IMC use	Use for recurrence limits, coefficient sums, and nonlinear-looking sequences that become telescoping after substitution.
Trap watch	Check radius of convergence before manipulating generating functions; induction bounds need verified base cases.
Appears in	1995-D2-P2, 1996-D1-P4, 1998-D2-P6, 2001-D2-P2, 2002-D2-P3, 2003-D1-P1, 2003-D1-P5, 2003-D1-P6 2003-D2-P6, 2005-D1-P2, 2007-D1-P4, 2008-D2-P3, 2010-D2-P2, 2011-D2-P1, 2012-D2-P2, 2015-D1-P5 2017-D2-P8, 2017-D2-P9, 2018-D2-P7, 2019-D1-P4, 2021-D1-P3, 2024-D2-P8

RA-18 - Trigonometric and hyperbolic identities with monotonicity

Bucket	RA trig
Core theory	Use standard identities, product-to-sum formulas, monotonicity of $\sin x/x$, and hyperbolic recurrences such as $\cosh((m+1)t) = 2\cosh t \cosh(mt) - \cosh((m-1)t)$.
IMC use	Use for trig integrals, zero-counting of trig sums, hyperbolic rationality, and trigonometric inequalities.
Trap watch	Track the interval: many monotonicity claims only hold on $(0, \pi/2)$ or $(0, \pi)$.
Appears in	1995-D2-P5, 1996-D1-P2, 1996-D2-P2, 1997-D2-P3, 2000-D1-P4, 2001-D2-P2, 2001-D2-P6, 2003-D2-P2 2004-D1-P3, 2004-D2-P5, 2006-D2-P3, 2014-D2-P3, 2015-D1-P4, 2017-D1-P2

RA-19 - Gradient/Lipschitz-gradient optimization and convex smoothness

Bucket	RA smooth convexity
Core theory	For convex differentiable f with L-Lipschitz gradient, descent estimates give $\|\nabla f(x) - \nabla f(y)\|^2 \le L\langle \nabla f(x) - \nabla f(y), x - y\rangle$. Lipschitz gradients also control uniqueness of maximizers under geometric constraints.
IMC use	Use in convex analysis, gradient inequalities, and maximum-location problems on disks or balls.
Trap watch	Gradient Lipschitzness is a global condition; local Hessian intuition must be converted into an inequality.
Appears in	1994-D2-P2, 2002-D2-P6, 2005-D2-P5

RA-20 - Measure and probability-integral estimates

Bucket	RA/probability bridge
Core theory	Integral estimates can be interpreted probabilistically, and probability problems can often be turned into expectation or measure integrals. Symmetry and independence are the main reductions.
IMC use	Use for random points, expected geometric quantities, and inequalities for integrable functions.
Trap watch	Distinguish deterministic inequalities from almost-sure statements; justify independence and symmetry.
Appears in	2019-D2-P10, 2022-D1-P1

7.3 Complex analysis / polynomials

CORRECTION MATH-016 — legacy statement superseded; click for canonical replacement.

CP-01 - Polynomial identity and more-roots-than-degree principle

Bucket	CP polynomial identity
Core theory	A nonzero polynomial of degree d over a field has at most d roots. Therefore, if it has more than d roots, it is identically zero.
IMC use	Use for polynomial divisibility, coefficient vanishing, finite-field polynomial functions, and interpolation-style arguments.
Trap watch	Over finite fields, different polynomials can define the same function; reduce modulo $x^p - x$ when working on $\mathbb{F}_p$.
Appears in	1994-D2-P4, 1995-D1-P5, 1995-D2-P3, 1995-D2-P4, 1996-D2-P2, 1998-D1-P2, 1998-D1-P5, 1998-D2-P2 1999-D1-P1, 1999-D2-P2, 1999-D2-P6, 2000-D1-P6, 2000-D2-P3, 2000-D2-P4, 2000-D2-P6, 2001-D1-P4 2001-D2-P1, 2002-D1-P1, 2002-D2-P3, 2003-D1-P3, 2003-D1-P6, 2003-D2-P2, 2004-D1-P6, 2005-D1-P4 2005-D2-P2, 2005-D2-P6, 2006-D1-P4, 2006-D1-P6, 2006-D2-P5, 2006-D2-P6, 2007-D1-P1, 2007-D1-P3 2007-D1-P5, 2007-D1-P6, 2007-D2-P6, 2008-D1-P6, 2008-D1-P2, 2008-D1-P3, 2008-D2-P1, 2008-D2-P2 2008-D2-P4, 2008-D2-P5, 2008-D2-P6, 2009-D1-P3, 2009-D1-P4, 2009-D2-P1, 2009-D2-P3, 2009-D2-P4 2009-D2-P5, 2010-D2-P5, 2011-D1-P2, 2011-D1-P3, 2011-D1-P5, 2011-D2-P2, 2011-D2-P4, 2012-D1-P5 2012-D2-P1, 2013-D2-P1, 2014-D1-P3, 2015-D2-P10, 2017-D1-P1, 2017-D1-P5, 2017-D2-P7, 2018-D2-P9 2020-D1-P4, 2021-D2-P7, 2023-D1-P3, 2024-D1-P1

CP-02 - Factorization, irreducibility, and Gauss-style integer polynomial arguments

Bucket	CP factorization
Core theory	Integer polynomial divisibility often uses Gauss lemma, primitive factorization, and reduction modulo primes. Cyclotomic and simple irreducibility facts turn one root into all conjugate roots.
IMC use	Use for $\mathbb{Z}[x]$ divisibility, rational functions integer-valued infinitely often, and roots of unity.
Trap watch	Divisibility in values is not the same as divisibility in $\mathbb{Z}[x]$.
Appears in	1995-D2-P3, 1998-D1-P5, 1998-D2-P2, 2000-D2-P3, 2001-D1-P4, 2001-D2-P1, 2005-D2-P2, 2006-D1-P4 2007-D1-P1, 2008-D1-P3, 2008-D2-P1, 2008-D2-P2, 2008-D2-P4, 2009-D2-P1, 2012-D1-P5, 2012-D2-P1 2013-D2-P1, 2015-D2-P10, 2017-D1-P5, 2017-D2-P7, 2018-D2-P9, 2020-D1-P4, 2021-D2-P7, 2023-D1-P3

CP-03 - Vieta, logarithmic derivatives, and real-root inequalities

Bucket	CP log derivative
Core theory	For $P(x) = \prod_i (x - r_i)$, $P'(x)/P(x) = \sum_i 1/(x - r_i)$. This converts root-location assumptions into sums of squares or coefficient inequalities.
IMC use	Use for polynomials with real roots, roots on the unit circle, and inequalities involving P, P', P''.
Trap watch	Handle points that are roots of P separately before dividing by $P(x)$.
Appears in	1995-D2-P3, 1998-D1-P5, 1998-D2-P2, 2000-D2-P3, 2000-D2-P4, 2001-D1-P4, 2003-D1-P6, 2005-D1-P4 2005-D2-P2, 2006-D1-P4, 2007-D1-P1, 2007-D1-P5, 2007-D2-P6, 2008-D2-P1, 2009-D2-P4, 2010-D2-P5 2012-D1-P5, 2012-D2-P1, 2013-D2-P1, 2014-D1-P3, 2015-D2-P10, 2017-D1-P5, 2017-D2-P7, 2018-D2-P9 2020-D1-P4, 2021-D2-P7, 2023-D1-P3, 2024-D1-P1

CP-04 - Unit-circle roots and conjugation identities

Bucket	CP unit circle
Core theory	When $\|\alpha\| = 1$, identities involving $z, \bar{z}$, and $\frac{z+\alpha}{z-alpha}$ reveal signs of real parts and force roots onto the unit circle.
IMC use	Use for root-location preservation under derivative-like transformations and conditions $\|a\| = \|b\| = 1$.
Trap watch	Do not divide by zero at existing roots; remove common factors or argue by continuity.
Appears in	1995-D2-P3, 2000-D1-P2, 2000-D2-P3, 2002-D2-P5, 2008-D2-P1, 2013-D2-P1, 2015-D1-P1, 2016-D2-P10 2017-D1-P1, 2018-D2-P9, 2021-D2-P7, 2023-D1-P2, 2024-D2-P7

CORRECTION MATH-017, MATH-018, MATH-019 — legacy statement superseded; click for canonical replacement.

CP-05 - Roots of unity, cyclotomic filters, and discrete Fourier tricks

Bucket	CP roots of unity
Core theory	The root-of-unity filter extracts residue classes: $\frac{1}{m}\sum_{j=0}^{m-1}\omega^{jN}$ equals 1 if $m \mid N$ and 0 otherwise. Cyclotomic irreducibility can force all primitive roots to appear.
IMC use	Use for probability modulo m, coefficient sums by congruence class, divisibility by cyclotomic polynomials, and Fourier coefficients of finite sets.
Trap watch	Check whether all roots, primitive roots, or all nontrivial roots are needed.
Appears in	1995-D2-P3, 1997-D1-P3, 1999-D2-P2, 1999-D2-P6, 2000-D1-P2, 2000-D2-P3, 2001-D1-P4, 2001-D2-P3 2002-D2-P5, 2007-D1-P5, 2007-D1-P6, 2008-D2-P1, 2010-D2-P5, 2013-D2-P1, 2014-D1-P3, 2016-D2-P10 2017-D1-P1, 2017-D1-P5, 2017-D2-P7, 2018-D2-P9, 2021-D2-P7, 2023-D1-P2, 2024-D1-P1, 2024-D2-P7

CP-06 - Cauchy formula and coefficient extraction on the unit circle

Bucket	CP Cauchy
Core theory	For holomorphic p, coefficients are integrals over the unit circle. Cauchy formulas let coefficient multipliers be represented by convolution kernels.
IMC use	Use for polynomial coefficient transformations bounded by sup norms and holomorphic estimates on the unit disk.
Trap watch	The maximum on the closed disk is controlled by the boundary only for holomorphic functions.
Appears in	2007-D1-P5, 2007-D1-P6, 2009-D1-P4, 2013-D2-P1, 2014-D1-P3, 2017-D2-P7, 2023-D1-P3

CP-07 - Maximum modulus principle

Bucket	CP maximum modulus
Core theory	A nonconstant holomorphic function reaches its maximum modulus on the boundary of a compact domain. This underlies many unit-disk inequalities.
IMC use	Use for bounding values inside the disk from boundary data, often after multiplying by a polynomial.
Trap watch	The function must be holomorphic on a neighborhood of the closed domain, not merely inside it with boundary singularities.
Appears in	No direct automatic assignment; keep as supporting background.

CP-08 - Laurent expansions, isolated singularities, and rationality of meromorphic functions

Bucket	CP meromorphic
Core theory	A meromorphic function on the Riemann sphere is rational. Local Laurent expansions determine principal parts and hence the rational function when poles are known.
IMC use	Use for sums over logarithm branches, pole-order computations, and rational expressions from analytic data.
Trap watch	Uniform convergence on compact subsets away from singularities must be checked before termwise analytic operations.
Appears in	2004-D1-P6

CP-09 - Fejer kernels and positive trigonometric kernels

Bucket	CP Fourier positivity
Core theory	The Fejer kernel $F_k(e^{it}) = \lvert 1 + e^{it} + \cdots + e^{i(k-1)t}\rvert^2$ is nonnegative. Convex coefficient sequences decompose into nonnegative combinations of Fejer kernels.
IMC use	Use for sup-norm contraction of polynomial coefficient multipliers and positivity of trigonometric sums.
Trap watch	Convexity of the coefficient sequence is the exact hypothesis that makes the kernel nonnegative.
Appears in	2009-D1-P4

CP-10 - Interpolation, Lagrange basis, and Chebyshev extremality

Bucket	CP interpolation
Core theory	A polynomial of degree at most n is determined by $n+1$ values; derivatives can be written using Lagrange basis derivatives. Chebyshev polynomials often give extremal oscillation.
IMC use	Use for bounded polynomial values at selected points and maximizing derivatives.
Trap watch	The extremal sign pattern must match the signs of the derivative of the Lagrange basis at the target point.
Appears in	1995-D2-P4, 1998-D2-P2

CP-11 - Stable polynomials and coefficient inequalities

Bucket	CP stable polynomials
Core theory	Polynomials with all roots in the left half-plane have positive coefficients after normalization and satisfy log-concavity-like coefficient inequalities through factorization into positive linear and quadratic factors.
IMC use	Use for inequalities among coefficients under root-location assumptions.
Trap watch	Keep track of the order of coefficients and signs after normalizing the leading coefficient.
Appears in	2003-D1-P6, 2014-D1-P3, 2023-D1-P3

CP-12 - Partial fractions and cotangent-type expansions

Bucket	CP partial fractions
Core theory	Classical partial fractions such as cotangent expansions evaluate sums over arithmetic progressions of poles. Differentiating produces higher-power denominator sums.
IMC use	Use for explicit sums over all logarithm branches or equally spaced complex poles.
Trap watch	Termwise differentiation requires locally uniform convergence away from poles.
Appears in	2004-D1-P6

CP-13 - Polynomial functional identities and algebraic composition

Bucket	CP identities
Core theory	Polynomial identities in several variables are solved by substituting special values, comparing degrees, and using the identity theorem for polynomials.
IMC use	Use for classifying polynomials satisfying multiplicative or composition identities and for systems $p(w)=p(z), q(w)=q(z)$.
Trap watch	Special-value substitutions must not lose solutions; after guessing a form, verify the full identity.
Appears in	2000-D1-P2, 2007-D1-P3, 2013-D2-P1, 2014-D1-P3, 2017-D2-P7, 2018-D2-P9, 2023-D1-P3, 2024-D1-P1

7.4 Algebra / number theory

ANT-01 - Modular arithmetic and divisibility

Bucket	ANT modular
Core theory	Divisibility questions are often local modulo prime powers. Reduce congruences modulo primes first, then lift or combine information.
IMC use	Use for integer polynomials, prime divisibility, residue classes, and modular nonexistence.
Trap watch	Check whether the modulus is prime, prime power, or composite; cancellation requires invertibility.
Appears in	1994-D1-P3, 1994-D1-P6, 1997-D1-P3, 1997-D1-P4, 1999-D1-P3, 1999-D2-P1, 1999-D2-P2, 2000-D1-P5 2000-D2-P1, 2001-D1-P1, 2001-D1-P2, 2001-D1-P4, 2001-D2-P6, 2002-D1-P3, 2003-D1-P2, 2003-D1-P4 2005-D1-P2, 2005-D1-P4, 2005-D2-P2, 2005-D2-P6, 2006-D1-P2, 2006-D1-P3, 2006-D2-P1, 2006-D2-P5 2007-D1-P1, 2007-D2-P2, 2007-D2-P4, 2008-D1-P3, 2008-D1-P5, 2008-D2-P3, 2008-D2-P4, 2008-D2-P5 2009-D2-P4, 2010-D1-P4, 2010-D2-P4, 2010-D2-P5, 2011-D1-P3, 2011-D1-P5, 2011-D2-P4, 2012-D1-P1 2012-D1-P2, 2012-D1-P3, 2012-D2-P1, 2012-D2-P3, 2012-D2-P4, 2012-D2-P5, 2013-D1-P5, 2013-D2-P2 2013-D2-P4, 2014-D1-P4, 2014-D2-P1, 2015-D1-P2, 2015-D1-P4, 2015-D2-P10, 2016-D1-P2, 2016-D1-P3 2016-D1-P4, 2016-D1-P5, 2016-D2-P8, 2016-D2-P9, 2017-D1-P3, 2017-D1-P4, 2017-D2-P8, 2017-D2-P10 2018-D1-P2, 2018-D1-P5, 2018-D2-P6, 2018-D2-P8, 2019-D1-P2, 2019-D1-P4, 2019-D1-P5, 2019-D2-P7 2019-D2-P9, 2020-D1-P1, 2020-D1-P3, 2020-D2-P6, 2021-D1-P2, 2021-D2-P5, 2021-D2-P6, 2021-D2-P8 2022-D1-P2, 2022-D1-P3, 2022-D1-P4, 2022-D2-P6, 2023-D1-P4, 2023-D1-P5, 2023-D2-P10, 2024-D1-P3 2024-D1-P4, 2024-D1-P5, 2024-D2-P9, 2024-D2-P10

ANT-02 - Chinese remainder theorem and complete residue systems

Bucket	ANT CRT
Core theory	If moduli are coprime, congruences modulo each modulus combine uniquely modulo the product. Complete residue systems remain complete under bijective polynomial maps over finite fields.
IMC use	Use for constructing integers with simultaneous congruence constraints and counting idempotents modulo 10^k.
Trap watch	Uniqueness is modulo the product; remember to exclude disallowed representatives such as zero if the problem says positive.
Appears in	1999-D1-P3, 2006-D1-P2, 2006-D2-P5, 2007-D1-P1, 2007-D2-P2, 2007-D2-P4, 2010-D1-P4, 2012-D2-P1 2012-D2-P4, 2013-D1-P5, 2013-D2-P2, 2014-D1-P4, 2014-D2-P1, 2015-D1-P4, 2016-D1-P3, 2016-D1-P5 2016-D2-P8, 2017-D2-P10, 2018-D1-P5, 2018-D2-P6, 2019-D1-P2, 2019-D2-P7, 2020-D2-P6, 2021-D2-P6 2022-D1-P3, 2022-D2-P6, 2023-D1-P4, 2023-D2-P10, 2024-D1-P3, 2024-D2-P10

ANT-03 - Finite fields, field characteristic, and Fermat/Frobenius tools

Bucket	ANT finite fields
Core theory	Over $\mathbb{F}_p$, Fermat gives $a^{p-1} = 1$ for nonzero a, and polynomial functions are naturally considered modulo $x^p - x$. Field characteristic controls additive cancellation.
IMC use	Use for finite-field polynomial permutations, characteristic restrictions, and group actions over $\mathbb{F}_p$.
Trap watch	A polynomial of high degree may define the same function as a lower-degree remainder modulo $x^p - x$.
Appears in	1997-D1-P3, 1999-D2-P1, 1999-D2-P2, 2000-D2-P1, 2000-D2-P3, 2001-D1-P2, 2001-D1-P4, 2003-D1-P2 2005-D1-P2, 2005-D1-P4, 2005-D2-P2, 2005-D2-P6, 2006-D1-P2, 2006-D1-P3, 2006-D2-P5, 2007-D1-P1 2007-D2-P2, 2007-D2-P4, 2008-D1-P3, 2008-D2-P4, 2008-D2-P5, 2009-D2-P4, 2010-D1-P4, 2010-D2-P4 2010-D2-P5, 2011-D1-P3, 2011-D1-P5, 2011-D2-P4, 2012-D1-P3, 2012-D2-P1, 2013-D1-P5, 2013-D2-P2 2014-D1-P4, 2014-D2-P1, 2016-D1-P3, 2016-D1-P5, 2016-D2-P8, 2018-D1-P2, 2018-D1-P5, 2019-D1-P2 2019-D2-P7, 2020-D2-P6, 2021-D2-P6, 2022-D1-P3, 2022-D2-P6, 2023-D1-P4, 2023-D2-P10, 2024-D1-P3 2024-D2-P10

ANT-04 - Integer polynomials, finite differences, and value divisibility

Bucket	ANT integer polynomials
Core theory	For integer polynomials, $p(a+mq) \equiv p(a) \pmod{q}$. A degree-$d$ polynomial over a field with more than d roots is zero.
IMC use	Use for proving coefficient divisibility from value divisibility and constructing values divisible by prescribed polynomial values.
Trap watch	Testing finitely many residues works only when the degree is less than the prime modulus or when using finite differences.
Appears in	1995-D2-P4, 1996-D2-P2, 1998-D1-P5, 2001-D2-P1, 2002-D2-P3, 2005-D1-P4, 2006-D1-P3, 2006-D1-P4 2007-D1-P1, 2008-D1-P3, 2008-D2-P1, 2008-D2-P4, 2011-D1-P3, 2012-D1-P5, 2012-D2-P1, 2014-D1-P3 2014-D2-P1, 2015-D2-P10, 2017-D2-P7, 2018-D2-P9, 2020-D1-P4, 2023-D1-P3

ANT-05 - Rational roots, Vieta, and rationality constraints

Bucket	ANT rationality
Core theory	Rational-root and Vieta arguments translate root rationality into coefficient and product/sum constraints. For irrationality questions, positive rational linear combinations are controlled by induction and contradiction.
IMC use	Use for polynomials with rational roots, hyperbolic rationality, irrational subset selection, and rational functions.
Trap watch	Do not divide by a coefficient that may be zero; isolate zero roots first.
Appears in	1994-D1-P3, 1996-D2-P2, 2005-D1-P4, 2006-D1-P5, 2018-D1-P3

ANT-06 - GCD, valuations, prime powers, and coprimality

Bucket	ANT valuations
Core theory	Coprime factors can be assigned independently. Valuations measure powers of a prime dividing an integer and turn multiplicative decompositions into additive comparisons.
IMC use	Use for decompositions of positive integers, products equaling powers, and divisibility by prime powers.
Trap watch	When xy is divisible by p^k and x, y are not coprime, valuation splitting is necessary.
Appears in	1999-D1-P3, 1999-D2-P1, 2000-D2-P1, 2002-D1-P3, 2003-D1-P4, 2007-D2-P2, 2008-D2-P3, 2012-D1-P1 2012-D1-P2, 2012-D1-P3, 2012-D2-P3, 2012-D2-P4, 2013-D2-P2, 2014-D2-P1, 2015-D1-P2, 2015-D1-P4 2016-D1-P2, 2016-D1-P3, 2016-D1-P4, 2016-D2-P9, 2017-D1-P3, 2017-D2-P8, 2017-D2-P10, 2018-D2-P6 2018-D2-P8, 2019-D1-P2, 2019-D1-P4, 2019-D1-P5, 2019-D2-P9, 2020-D1-P1, 2020-D1-P3, 2020-D2-P6 2021-D2-P5, 2021-D2-P8, 2022-D1-P2, 2022-D1-P3, 2022-D2-P6, 2023-D1-P4, 2023-D1-P5, 2023-D2-P10 2024-D1-P3, 2024-D2-P10

ANT-07 - Group basics: orders, cyclic subgroups, and generators

Bucket	ANT groups
Core theory	Element orders, generated subgroups, and relations determine many finite-group arguments. In abelian groups, Bezout identities in exponents are especially powerful.
IMC use	Use for group equations, pairwise coprime exponents, and subgroup generation.
Trap watch	Exponent arithmetic using $(ab)^n = a^n b^n$ requires commutativity unless justified.
Appears in	1996-D2-P3, 1999-D1-P3, 1999-D2-P1, 2000-D1-P5, 2001-D1-P2, 2003-D1-P2, 2003-D2-P4, 2005-D1-P6 2006-D1-P2, 2007-D1-P4, 2007-D2-P5, 2008-D1-P5, 2008-D2-P3, 2009-D2-P5, 2010-D2-P2, 2010-D2-P3 2011-D1-P3, 2011-D2-P2, 2011-D2-P4, 2012-D1-P3, 2012-D1-P5, 2012-D2-P3, 2012-D2-P4, 2012-D2-P5 2013-D2-P2, 2014-D2-P1, 2015-D1-P2, 2015-D1-P4, 2016-D1-P2, 2016-D1-P3, 2016-D2-P8, 2016-D2-P9 2017-D1-P4, 2017-D2-P8, 2017-D2-P10, 2018-D1-P2, 2018-D2-P6, 2019-D1-P2, 2019-D1-P4, 2019-D1-P5 2019-D2-P9, 2020-D1-P1, 2020-D2-P6, 2020-D2-P7, 2021-D2-P5, 2021-D2-P6, 2022-D1-P2, 2022-D2-P6 2023-D2-P10, 2024-D1-P3, 2024-D1-P4, 2024-D2-P10

ANT-08 - Permutation groups, cycles, parity, and generation

Bucket	ANT permutation groups
Core theory	Cycles, transpositions, and parity describe symmetric and alternating groups. Adjacent transpositions generate S_n; selected 3-cycles generate A_n.
IMC use	Use for proving generation/non-generation of permutation groups and for constructing permutations with modular properties.
Trap watch	Cycle notation composition order must be consistent throughout the proof.
Appears in	1998-D1-P2, 2001-D1-P2

ANT-09 - Normal subgroups, quotient groups, and group actions

Bucket	ANT quotients/actions
Core theory	Normal subgroups allow quotient groups. Group actions turn algebraic relations into orbit-stabilizer or index constraints.
IMC use	Use for subgroup-index problems, conjugacy of subgroups, and generation obstructions through quotients.
Trap watch	A subgroup of the same index need not be normal; prove normality before passing to a quotient.
Appears in	1996-D2-P3, 1999-D2-P1, 2005-D1-P6, 2010-D2-P3, 2012-D1-P3, 2012-D2-P5, 2016-D2-P8, 2017-D1-P4 2018-D1-P2, 2020-D2-P7, 2021-D2-P6, 2024-D1-P4

ANT-10 - Automorphisms and semidirect products

Bucket	ANT automorphisms
Core theory	A semidirect product $H \rtimes T$ is built from an action of T on H. Automorphisms preserve element orders, normal subgroups, and structural decompositions.
IMC use	Use for constructing finite groups with prescribed automorphism behavior.
Trap watch	Counting automorphisms by images of generators requires proving those images satisfy the same defining relations.
Appears in	1996-D2-P3, 1998-D1-P2, 1999-D1-P1, 1999-D1-P5, 1999-D2-P1, 2000-D1-P5, 2001-D1-P2, 2003-D1-P2 2003-D2-P4, 2004-D2-P3, 2005-D1-P6, 2007-D1-P4, 2007-D2-P5, 2008-D1-P5, 2010-D2-P2, 2010-D2-P3 2011-D1-P3, 2011-D2-P4, 2012-D1-P3, 2012-D1-P5, 2012-D2-P5, 2016-D2-P8, 2017-D1-P4, 2018-D1-P2 2020-D2-P7, 2021-D2-P6, 2024-D1-P4

ANT-11 - Rings, idempotents, nil rings, and noncommutativity

Bucket	ANT rings
Core theory	In noncommutative rings, expansion of $(a+b)^2$ still gives $ab+ba$ when squares vanish. Idempotent equations often become commutator relations under characteristic assumptions.
IMC use	Use for ring problems with idempotents, nilpotents, characteristic zero, and noncommutative algebra.
Trap watch	Do not commute factors; use associativity and given identities only.
Appears in	1996-D2-P3, 1999-D1-P5, 1999-D2-P1, 2000-D1-P5, 2001-D1-P2, 2003-D1-P2, 2003-D2-P4, 2005-D1-P6 2006-D1-P2, 2007-D1-P4, 2007-D2-P5, 2008-D1-P5, 2009-D2-P4, 2010-D2-P3, 2011-D1-P3, 2011-D2-P2 2012-D1-P3, 2012-D1-P5, 2012-D2-P3, 2012-D2-P5, 2013-D2-P2, 2014-D2-P1, 2016-D2-P8, 2017-D1-P4 2018-D1-P2, 2019-D1-P2, 2020-D2-P6, 2020-D2-P7, 2021-D2-P6, 2022-D2-P6, 2023-D2-P10, 2024-D1-P3 2024-D1-P4

ANT-12 - Diophantine equations, Pell-type recurrences, and quadratic irrationals

Bucket	ANT Diophantine
Core theory	Quadratic irrational fixed points of fractional linear transformations lead to Pell-type equations. Recurrences from fundamental units provide infinitely many solutions.
IMC use	Use for constructing integer matrices fixing quadratic irrationals and cubic equations with integer roots.
Trap watch	Verify determinant-one and nontriviality conditions after solving the Pell equation.
Appears in	2005-D2-P6, 2006-D2-P5

CORRECTION MATH-020 — legacy statement superseded; click for canonical replacement.

ANT-13 - Lattices and norms/quasinorms

Bucket	ANT lattices
Core theory	Integer lattice shifts interact with convex norms for $p \geq 1$ and can fail for quasi-norms $0 < p < 1$. Coordinate splitting by signs is often required.
IMC use	Use for lattice minimization and norm comparison problems.
Trap watch	Convexity fails for $p < 1$; do not use triangle inequality in the quasi-norm case.
Appears in	1994-D1-P6, 1997-D1-P5

ANT-14 - Greedy infinite products and rationality tails

Bucket	ANT greedy products
Core theory	Greedy multiplicative expansions are controlled by a tail identity such as $\prod_{j\geq 0}(1+M^{-2^j}) = M/(M-1)$. Rationality can force eventual equality in the greedy recurrence.
IMC use	Use for unique infinite product representations and rationality criteria.
Trap watch	A greedy inequality must be sharp in both directions to prove uniqueness.
Appears in	1997-D1-P4

ANT-15 - Finite abelian groups and vector spaces over finite fields

Bucket	ANT finite abelian
Core theory	Finite abelian groups of exponent 2 are vector spaces over $\mathbb{F}_2$ and have order a power of two. This structure often classifies triple systems.
IMC use	Use for finite designs where each pair determines a third element and associativity-like conditions appear.
Trap watch	The added identity element must be included before claiming group structure.
Appears in	2003-D2-P4

7.5 Combinatorics / discrete structures

CMB-01 - Pigeonhole principle and extremal element selection

Bucket	CMB pigeonhole
Core theory	If more objects than boxes are distributed, one box contains many objects. Extremal choices (minimal counterexample, largest element, shortest cycle) often force structure.
IMC use	Use for divisibility value collisions, selecting common rows/columns, and minimal counterexample arguments.
Trap watch	Define the boxes explicitly; hidden boxes often are residue classes, divisors, or graph neighborhoods.
Appears in	1994-D1-P1, 1994-D1-P3, 1994-D1-P6, 1994-D2-P4, 1995-D1-P3, 1996-D1-P3, 1996-D2-P3, 1996-D2-P6 1997-D1-P6, 1997-D2-P6, 1998-D2-P3, 1998-D2-P4, 1998-D2-P5, 1998-D2-P6, 1999-D1-P2, 1999-D1-P5 1999-D2-P6, 2000-D1-P1, 2001-D2-P1, 2001-D2-P3, 2002-D1-P4, 2002-D1-P5, 2002-D2-P2, 2002-D2-P6 2003-D1-P5, 2003-D2-P3, 2003-D2-P5, 2004-D1-P1, 2004-D1-P2, 2004-D1-P4, 2004-D2-P6, 2005-D1-P2 2006-D1-P2, 2006-D2-P1, 2006-D2-P5, 2007-D1-P4, 2007-D1-P6, 2008-D1-P6, 2008-D2-P4, 2008-D2-P6 2009-D1-P1, 2009-D1-P3, 2009-D2-P5, 2010-D1-P4, 2011-D1-P4, 2011-D2-P2, 2012-D1-P1, 2013-D1-P3 2013-D2-P5, 2015-D2-P8, 2016-D1-P5, 2016-D2-P7, 2016-D2-P9, 2017-D1-P4, 2018-D2-P8, 2019-D2-P8 2020-D1-P1, 2020-D1-P2, 2020-D2-P7, 2021-D1-P2, 2021-D2-P8, 2022-D1-P3, 2022-D1-P4, 2022-D2-P5 2022-D2-P8, 2023-D2-P8, 2024-D1-P5, 2024-D2-P9

CMB-02 - Double counting and degree-sum methods

Bucket	CMB double counting
Core theory	Count the same set of objects in two ways, often by vertices versus edges, pairs versus problems, or walks via middle vertex.
IMC use	Use for contest scores, graph degree constraints, set intersections, and probability counts.
Trap watch	Specify ordered vs unordered objects; many IMC counts hinge on this distinction.
Appears in	1994-D1-P1, 1994-D2-P4, 1995-D1-P3, 1996-D1-P3, 1996-D2-P3, 1996-D2-P6, 1997-D2-P6, 1998-D2-P3 1998-D2-P4, 1998-D2-P5, 1998-D2-P6, 1999-D1-P5, 2000-D1-P1, 2001-D2-P1, 2001-D2-P3, 2002-D1-P4 2002-D1-P5, 2002-D2-P2, 2003-D1-P5, 2003-D2-P5, 2004-D1-P1, 2004-D1-P2, 2004-D1-P4, 2004-D2-P6 2005-D1-P2, 2006-D1-P2, 2006-D2-P5, 2007-D1-P4, 2007-D1-P6, 2008-D1-P6, 2008-D2-P4, 2009-D1-P1 2009-D1-P3, 2010-D1-P4, 2011-D1-P4, 2011-D2-P2, 2012-D1-P1, 2013-D1-P3, 2013-D2-P5, 2015-D2-P8 2016-D1-P5, 2016-D2-P9, 2017-D1-P4, 2018-D2-P8, 2019-D2-P8, 2020-D1-P1, 2020-D1-P2, 2020-D2-P7 2021-D1-P2, 2021-D2-P8, 2022-D1-P3, 2022-D1-P4, 2022-D2-P5, 2022-D2-P8, 2023-D2-P8, 2024-D2-P9

CMB-03 - Graph encoding, cycles, and bipartite representations

Bucket	CMB graph encoding
Core theory	Encode relations as graphs. Cycles, degree sums, connectedness, and bipartite representations convert combinatorial statements into graph-theoretic ones.
IMC use	Use for grids, friendship conditions, colored diagonals, trees, and matrix zero/nonzero patterns.
Trap watch	When encoding a grid, rows and columns often become vertices and marked cells become edges.
Appears in	1994-D1-P3, 1994-D1-P5, 1994-D2-P3, 1995-D1-P4, 1995-D2-P1, 1996-D1-P6, 1996-D2-P2, 1997-D1-P6 1997-D2-P5, 1997-D2-P6, 1998-D1-P1, 1998-D1-P2, 1998-D2-P3, 1998-D2-P4, 1998-D2-P5, 1999-D1-P4 1999-D1-P5, 1999-D1-P6, 1999-D2-P5, 2000-D2-P4, 2001-D1-P2, 2001-D2-P4, 2001-D2-P5, 2002-D1-P1 2002-D2-P2, 2002-D2-P4, 2002-D2-P5, 2003-D1-P4, 2003-D2-P3, 2003-D2-P4, 2004-D1-P1, 2004-D1-P3 2004-D2-P2, 2005-D2-P1, 2005-D2-P4, 2006-D2-P1, 2007-D2-P1, 2008-D1-P6, 2008-D1-P1, 2008-D1-P4 2008-D2-P5, 2009-D1-P3, 2009-D1-P5, 2009-D2-P4, 2011-D1-P1, 2011-D1-P4, 2011-D2-P5, 2012-D2-P3 2012-D2-P5, 2013-D1-P3, 2013-D2-P4, 2013-D2-P5, 2014-D2-P4, 2016-D1-P4, 2017-D1-P3, 2019-D2-P7 2019-D2-P8, 2019-D2-P10, 2020-D1-P3, 2021-D1-P2, 2022-D1-P4, 2022-D2-P5, 2022-D2-P8, 2023-D1-P4 2023-D1-P5, 2023-D2-P7, 2023-D2-P8, 2023-D2-P9

CMB-04 - Set systems, intersecting families, and induction on sizes

Bucket	CMB set systems
Core theory	Intersecting families are often handled by induction on set size, splitting by intersection with a fixed set, or constructing a finite transversal.
IMC use	Use for finite set families with pairwise intersections and uniform-size assumptions.
Trap watch	Uniformity assumptions are often essential; construct counterexamples without them.
Appears in	1997-D1-P6, 1997-D2-P5, 1997-D2-P6, 1998-D1-P1, 1999-D2-P5, 2002-D2-P2, 2004-D1-P1, 2005-D2-P4 2007-D2-P1, 2008-D1-P4, 2011-D1-P4, 2012-D2-P3, 2013-D2-P5, 2014-D2-P4, 2016-D1-P4, 2017-D1-P3 2019-D2-P8, 2020-D1-P3, 2021-D1-P2, 2022-D2-P5, 2023-D1-P4, 2023-D1-P5, 2023-D2-P7

CMB-05 - Recurrences and state partitions

Bucket	CMB recurrences
Core theory	Partition objects by a finite state, derive a recurrence, verify initial conditions, and solve or compare recurrences.
IMC use	Use for words, sequences under local constraints, matrix-defined recurrences, and counting functions.
Trap watch	The state must contain enough memory to extend objects without ambiguity.
Appears in	1998-D2-P4, 2002-D2-P2, 2005-D1-P2, 2013-D2-P5

CMB-06 - Generating functions for counting

Bucket	CMB generating functions
Core theory	Generating functions translate recurrences and weighted counts into algebraic identities. Coefficient extraction and radius/convergence considerations then solve the count.
IMC use	Use for sequences, expected sums, probability distributions, and recurrence-defined counts.
Trap watch	Formal power series do not require convergence, but analytic evaluations do.
Appears in	2000-D1-P6, 2002-D1-P3, 2002-D2-P3, 2003-D1-P6, 2003-D2-P6, 2008-D2-P3, 2012-D2-P2, 2015-D2-P8 2017-D2-P8, 2017-D2-P9, 2018-D2-P7, 2019-D1-P4, 2021-D1-P3, 2024-D2-P8

CMB-07 - Inclusion-exclusion and root-of-unity filters

Bucket	CMB filters
Core theory	Counting by congruence class can be done with roots of unity; exclusion of forbidden configurations is handled by complements or inclusion-exclusion.
IMC use	Use for dice sums, modulo counts, and divisibility by residues in combinatorial settings.
Trap watch	Root-of-unity filters count residues of exponents/sums; ensure the generating function encodes the correct statistic.
Appears in	1999-D2-P2

CMB-08 - Permutations, inversions, and permutation statistics

Bucket	CMB permutations
Core theory	Permutations encode selections with one item per row/column. Statistics such as inversions or displacement can be represented by determinants or generating functions.
IMC use	Use for permutation enumeration, determinant parity of permutation statistics, and row-column selection problems.
Trap watch	Distinguish permutation as a function from its one-line notation.
Appears in	2001-D1-P1, 2003-D1-P2, 2009-D2-P4, 2013-D2-P5, 2022-D2-P6

CMB-09 - Explicit construction and counterexample design

Bucket	CMB constructions
Core theory	Construction problems require preserving all hypotheses while forcing or breaking the conclusion. Small examples, periodic patterns, and extremal configurations are the standard sources.
IMC use	Use for yes/no questions, nonexistence disproofs, and sharpness examples.
Trap watch	A counterexample must satisfy every hypothesis exactly, including continuity, monotonicity, or finiteness.
Appears in	1994-D1-P4, 1994-D2-P5, 1995-D1-P6, 1995-D2-P6, 1997-D1-P2, 1997-D1-P4, 1997-D1-P6, 1997-D2-P1 1997-D2-P6, 1998-D1-P2, 1999-D1-P2, 1999-D1-P6, 1999-D2-P5, 2000-D1-P1, 2002-D1-P2, 2003-D2-P4 2004-D2-P3, 2006-D1-P1, 2007-D2-P1, 2007-D2-P3, 2007-D2-P5, 2008-D1-P3, 2008-D1-P5, 2009-D1-P1 2011-D1-P2, 2013-D1-P5, 2013-D2-P4, 2015-D1-P2, 2016-D1-P1, 2016-D1-P2, 2018-D1-P2, 2019-D2-P6 2020-D1-P1, 2022-D1-P3, 2024-D1-P3, 2024-D2-P9

CMB-10 - Erdos-Szekeres style monotone subsequences and binomial thresholds

Bucket	CMB monotone subsequences
Core theory	Many monotone-subsequence thresholds are proved by induction using binomial identities and splitting by a median or extremal element.
IMC use	Use for finding long monotone sequences with extra growth constraints.
Trap watch	The induction parameter may be two-dimensional: length of increasing and decreasing alternatives.
Appears in	2004-D1-P5

CMB-11 - Grid, lattice, and alternating-cycle arguments

Bucket	CMB grid cycles
Core theory	Represent marked grid cells as edges in a bipartite graph between row-vertices and column-vertices. A cycle in the graph becomes an alternating row/column sequence.
IMC use	Use for grid problems asking for alternating selections or cycles.
Trap watch	Deleting rows/columns with too few points preserves inequalities only if the count is adjusted correctly.
Appears in	1999-D1-P5, 2023-D2-P6

CMB-12 - Coloring, partitions, and parity

Bucket	CMB coloring/parity
Core theory	Coloring and partition problems often require parity counts, degree constraints, or induction preserving parity.
IMC use	Use for colored graphs, triangulations with odd/even vertex incidences, and partitions into sets with algebraic equality.
Trap watch	Parity can change at boundary vertices; track local contributions, not just total counts.
Appears in	1994-D1-P3, 1997-D1-P6, 1997-D2-P5, 1997-D2-P6, 1998-D1-P1, 1998-D2-P3, 1998-D2-P4, 1999-D1-P6 1999-D2-P5, 2000-D2-P4, 2001-D2-P5, 2002-D1-P1, 2002-D2-P2, 2002-D2-P4, 2003-D1-P4, 2003-D2-P3 2004-D1-P1, 2004-D1-P4, 2004-D2-P2, 2005-D2-P1, 2005-D2-P4, 2006-D2-P1, 2007-D1-P4, 2007-D2-P1 2008-D1-P6, 2008-D1-P1, 2008-D1-P4, 2011-D1-P4, 2012-D2-P3, 2013-D2-P2, 2013-D2-P4, 2014-D2-P4 2016-D1-P4, 2016-D1-P5, 2017-D1-P3, 2019-D2-P8, 2019-D2-P10, 2020-D1-P3, 2021-D1-P2, 2022-D1-P4 2022-D2-P5, 2022-D2-P8, 2023-D1-P4, 2023-D1-P5, 2023-D2-P7, 2023-D2-P8, 2023-D2-P9, 2024-D1-P5

CMB-13 - Words, equivalence relations, and rewriting systems

Bucket	CMB words
Core theory	In word problems, local rewriting rules reduce length if a repeated block appears. Lemmas about absorbing words with all letters create global bounded-length representatives.
IMC use	Use for equivalence relations on words and finite reduction algorithms.
Trap watch	A rewrite rule must be compatible with left and right concatenation before applying it inside a word.
Appears in	1999-D2-P5, 2015-D2-P8, 2020-D1-P1

CMB-14 - Transfer matrices and matrix powers for counting

Bucket	CMB transfer matrix
Core theory	If local compatibility between states is encoded by a matrix A, then entries or sums of A^{k-1} count sequences or table fillings of length k.
IMC use	Use for counting binary tables, local constraints, and symmetry $F_{n,k} = F_{k,n}$.
Trap watch	Make sure the matrix state represents a whole column/row, not a single cell.
Appears in	2004-D2-P6

CMB-15 - Finite designs and pair-determines-third structures

Bucket	CMB designs
Core theory	A system where every pair belongs to exactly one triple can often be encoded by a binary operation $a * b = c$. Additional associativity-like axioms may turn it into a vector space over $\mathbb{F}_2$.
IMC use	Use for Steiner triple-system-like problems and classification of allowed sizes.
Trap watch	The operation is initially undefined on equal elements; adding an identity often completes the algebraic structure.
Appears in	2003-D2-P4

7.6 Geometry / convexity / metric

CORRECTION MATH-021 — legacy statement superseded; click for canonical replacement.

GEO-01 - Coordinate and affine normalization

Bucket	GEO coordinates
Core theory	Translate, rotate, scale, or choose coordinates to simplify distances, areas, conics, and vectors while preserving the quantity of interest.
IMC use	Use at the start of most Euclidean geometry, conic, and volume problems.
Trap watch	Only affine-invariant quantities survive affine transformations; Euclidean distances require isometries or controlled scalings.
Appears in	2014-D1-P5, 2014-D2-P4

GEO-02 - Euclidean distances and inner-product identities

Bucket	GEO inner products
Core theory	Use $\|u-v\|^2 = \|u\|^2 + \|v\|^2 - 2\langle u, v\rangle$. Spherical distance constraints often become inner-product inequalities.
IMC use	Use for rational distances, vectors with prescribed sums, tetrahedron angle estimates, and unit-sphere configurations.
Trap watch	Square distances only after confirming all quantities are nonnegative.
Appears in	1994-D2-P5, 1995-D2-P1, 1996-D1-P6, 1998-D1-P6, 2002-D1-P6, 2004-D1-P4, 2004-D2-P3, 2005-D2-P5 2006-D2-P4, 2008-D1-P5, 2008-D2-P2, 2010-D1-P1, 2010-D1-P5, 2013-D2-P3, 2014-D1-P5, 2014-D2-P4 2015-D2-P9, 2018-D2-P6, 2021-D2-P8

GEO-03 - Projections and orthogonal decompositions

Bucket	GEO projections
Core theory	Orthogonal projection shortens distances and decomposes vectors into parallel and perpendicular components. Projecting geometric regions can compare volumes or areas.
IMC use	Use for sphere/conic intersections, convex geometry, tetrahedra, and nearest-point arguments.
Trap watch	A projection can collapse dimensions; track signed area/volume if orientation matters.
Appears in	1995-D2-P3, 1996-D2-P4, 1997-D1-P5, 1997-D2-P5, 1998-D2-P5, 1999-D2-P6, 2002-D1-P1, 2002-D2-P4 2002-D2-P5, 2004-D1-P4, 2004-D2-P2, 2004-D2-P3, 2005-D2-P5, 2008-D2-P2, 2009-D2-P1, 2011-D2-P5 2014-D1-P5, 2015-D1-P5, 2017-D2-P10, 2018-D1-P5, 2019-D2-P10, 2022-D2-P5, 2022-D2-P8, 2023-D2-P9 2024-D1-P5

GEO-04 - Convex bodies, supporting geometry, and John-ellipse ideas

Bucket	GEO convex bodies
Core theory	For convex bodies, support and containment arguments compare maximal ellipses or balls. If a larger ellipse fits after a perturbation, maximality is contradicted.
IMC use	Use for convex symmetric planar bodies and maximal inscribed ellipse problems.
Trap watch	The proof often needs a compactness gap between an arc and the boundary.
Appears in	1996-D2-P4, 2011-D2-P3, 2014-D1-P5, 2014-D2-P4, 2014-D2-P5, 2015-D1-P3, 2015-D1-P4

GEO-05 - Conics: parabolas, ellipses, foci, and directrices

Bucket	GEO conics
Core theory	A standard parabola with vertex (v, w) has equation $y = (x-v)^2 + w$. Ellipses can be described by focus-directrix ratios or quadratic equations.
IMC use	Use for conic incidence/tangency, common-focus ellipse intersections, and reflected parabolas.
Trap watch	Tangency to a line imposes double roots after subtracting the line equation.
Appears in	2000-D2-P4, 2002-D1-P1, 2008-D2-P2, 2014-D1-P5

CORRECTION MATH-007, MATH-022 — legacy statement superseded; click for canonical replacement.

GEO-06 - Simplex, barycentric coordinates, and circumcenters

Bucket	GEO simplex
Core theory	Barycentric weights are proportional to opposite simplex volumes. Circumcenters are characterized by equal distances to all vertices.
IMC use	Use for volume-weighted circumcenter identities and regular-simplex configurations.
Trap watch	Volume weights must use consistent orientation or absolute volumes depending on the statement.
Appears in	1997-D2-P5, 2008-D2-P2, 2009-D1-P5, 2009-D2-P1, 2014-D1-P5, 2017-D2-P10

GEO-07 - Area, volume, projection, and ellipsoid computation

Bucket	GEO volume
Core theory	Volumes transform by the absolute determinant of a linear map. Ellipsoid volumes follow from scaling the unit ball.
IMC use	Use for solid regions defined by distance inequalities, cube dissections, and projected areas.
Trap watch	Complete the square before identifying the ellipsoid axes.
Appears in	1995-D2-P3, 1996-D2-P4, 1997-D1-P5, 1997-D2-P5, 1998-D2-P5, 1999-D2-P6, 2000-D2-P1, 2002-D1-P1 2002-D2-P4, 2002-D2-P5, 2004-D1-P4, 2004-D2-P2, 2004-D2-P3, 2005-D2-P5, 2008-D2-P2, 2009-D2-P1 2011-D2-P5, 2014-D1-P5, 2014-D2-P4, 2015-D1-P5, 2017-D2-P10, 2018-D1-P5, 2019-D2-P10, 2021-D2-P7 2022-D2-P5, 2022-D2-P8, 2023-D2-P9, 2024-D1-P5

GEO-08 - Metric content, diameter covers, and contractions

Bucket	GEO metric content
Core theory	A contraction cannot increase diameters. Covering a segment by finitely many sets forces the sum of diameters to be at least the segment length.
IMC use	Use for metric content versus lower content and compact counterexamples.
Trap watch	The covering sets need not be intervals; project or contract them to a segment.
Appears in	1996-D1-P6

GEO-09 - Hilbert-space equidistant sets and orthonormal translations

Bucket	GEO Hilbert
Core theory	A set of mutually equidistant points in a Hilbert space is a translate of a scaled orthonormal system. The center is obtained as a limit of barycenters for countable subsets.
IMC use	Use for infinite-dimensional equidistant sets and orthonormal-system construction.
Trap watch	Completeness of the Hilbert space is used to take the barycenter limit.
Appears in	1995-D2-P1, 2002-D2-P6, 2008-D2-P6, 2013-D2-P3, 2018-D2-P6, 2021-D2-P8

GEO-10 - Closest points to closed sets and density of uniqueness

Bucket	GEO nearest point
Core theory	A closed set intersects a compact ball around an exterior point, so nearest points exist. If the nearest point is not unique, moving toward one nearest point produces nearby points with a unique nearest point.
IMC use	Use for density of points with unique nearest projection onto a closed set.
Trap watch	Closedness is needed for existence; uniqueness is obtained by geometry of balls, not by convexity.
Appears in	2003-D2-P3

GEO-11 - Unit-sphere configurations and regular simplex bounds

Bucket	GEO sphere configurations
Core theory	On the unit sphere, $\|x-y\| > \sqrt{2}$ is equivalent to $\langle x, y\rangle < 0$. At most $n+1$ such points exist in $\mathbb{R}^n$, with equality given by a regular simplex.
IMC use	Use for maximum number of points on a sphere with obtuse pairwise angles.
Trap watch	Induction via projection must preserve strict negativity of inner products.
Appears in	2001-D2-P3

GEO-12 - Polygon triangulation and parity of incidences

Bucket	GEO triangulations
Core theory	Triangulations can be built inductively by adding three vertices and preserving parity at old vertices while controlling new vertices.
IMC use	Use for parity-constrained triangulations of convex polygons.
Trap watch	Base cases modulo 3 are essential.
Appears in	2006-D2-P1

GEO-13 - Convex hulls, random polytopes, and geometric probability

Bucket	GEO random geometry
Core theory	Random geometric problems usually reduce by symmetry, conditioning on extremal points, or comparing volumes/areas of regions.
IMC use	Use for probability involving random points in disks/balls and convex hull events.
Trap watch	Independence of points and conditioning on positions must be explicit.
Appears in	2002-D2-P4

GEO-14 - Cube and square dissection by number-theoretic semigroups

Bucket	GEO dissections
Core theory	Refining one cube into a^d smaller cubes increases the count by $a^d - 1$. If two such increments are coprime, the Frobenius coin lemma gives all sufficiently large counts.
IMC use	Use for partitioning cubes/squares into a prescribed number of smaller cubes.
Trap watch	Find two coprime increments $a^d - 1$; not every pair works in every dimension.
Appears in	2000-D2-P1

7.7 Functional equations / dynamics

FEQ-01 - Substitution of special values and normalization

Bucket	FEQ substitutions
Core theory	Start by substituting $0, 1, x, y = f(x)$, or symmetric choices; normalize by translating/scaling when possible. Then prove any guessed family satisfies the equation.
IMC use	Use for most functional equations over $\mathbb{R}$, $\mathbb{Q}$, or positive reals.
Trap watch	Do not assume continuity or monotonicity unless the equation implies it or it is given.
Appears in	2005-D2-P4, 2007-D2-P1, 2009-D2-P2, 2015-D1-P3, 2016-D1-P1, 2019-D2-P6, 2024-D2-P6

FEQ-02 - Injectivity and monotonicity forced by a functional equation

Bucket	FEQ monotonicity
Core theory	Equations of the form $f(x)f(yf(x)) = f(x+y)$ or $f(x^2/f(x)) = x$ often force monotonicity, injectivity, or a reciprocal linear form.
IMC use	Use for classification of functions on positive reals or intervals.
Trap watch	If injectivity is used, prove it before canceling arguments inside f.
Appears in	1999-D1-P4, 2000-D2-P5, 2005-D2-P4, 2009-D2-P2, 2015-D1-P3, 2016-D1-P1, 2019-D2-P6, 2024-D2-P6

CORRECTION MATH-023, MATH-024 — legacy statement superseded; click for canonical replacement.

FEQ-03 - Fixed points, iterates, and orbit closure

Bucket	FEQ dynamics
Core theory	For interval maps, cluster points and continuity control iterates. Closed orbits of continuous maps on compact intervals are often finite unless accumulation behavior contradicts closedness.
IMC use	Use for iteration convergence, finite orbit proofs, and periodic point counts.
Trap watch	A pointwise closed orbit is not automatically finite without compactness and continuity arguments.
Appears in	1996-D2-P1, 2002-D1-P4, 2010-D1-P3, 2015-D1-P3, 2016-D1-P1, 2024-D2-P6

FEQ-04 - Recurrence transformations and conjugacy

Bucket	FEQ recurrences
Core theory	Nonlinear recurrences often simplify after substituting a new variable, conjugating the map, or identifying an exact iterate formula.
IMC use	Use for iterated quadratic maps, nested radicals, and recurrence limits.
Trap watch	Verify the transformed recurrence preserves the original domain.
Appears in	1998-D1-P3, 2003-D1-P5, 2004-D1-P2

FEQ-05 - Differentiable functional equations

Bucket	FEQ smooth
Core theory	When a functional equation includes differentiability, differentiate the equation, analyze scaling factors, and use fixed points or growth to force polynomial/exponential forms.
IMC use	Use for smooth equations such as $f(ax+b)=a^2f(x)$ and derivative constraints.
Trap watch	Find the fixed point of the affine map before iterating or differentiating repeatedly.
Appears in	No direct automatic assignment; keep as supporting background.

FEQ-06 - Piecewise-linear maps and periodic points

Bucket	FEQ piecewise dynamics
Core theory	For piecewise monotone maps, the graph of f^n has many monotone branches; intersections with the diagonal count fixed points of iterates.
IMC use	Use for proving existence and finiteness of exact n-periodic points.
Trap watch	Exact period requires excluding fixed points of smaller iterates.
Appears in	1998-D2-P3

FEQ-07 - Bijections as products of involutions and symmetry decompositions

Bucket	FEQ involutions
Core theory	A bijection decomposes into disjoint orbits; each finite or infinite orbit can be modeled geometrically so the map is a product of two involutions.
IMC use	Use for abstract set bijections and decompositions into symmetries.
Trap watch	Handle finite cycles and infinite orbits separately.
Appears in	1997-D2-P5

FEQ-08 - No-regularity functions and order patterns on Q or R

Bucket	FEQ no regularity
Core theory	For arbitrary functions without regularity, use order/combinatorial arguments rather than limits. On $\mathbb{Q}$, density and finite pigeonhole patterns can force local maxima-like triples.
IMC use	Use for functions from $\mathbb{Q}$ or arbitrary sets where no continuity is available.
Trap watch	Do not invoke IVT, compactness, or monotonicity unless it is part of the statement.
Appears in	1999-D1-P3, 1999-D2-P4, 2001-D2-P5, 2002-D1-P2, 2003-D2-P5, 2024-D2-P6

CORRECTION MATH-025 — legacy statement superseded; click for canonical replacement.

FEQ-09 - Image-length preserving functions

Bucket	FEQ interval images
Core theory	If every interval image is a closed interval of the same length, the function is 1-Lipschitz and endpoint extrema force monotonicity; hence $f(x) = x + c$ or $f(x) = -x + c$.
IMC use	Use for functions preserving interval lengths or ranges.
Trap watch	The maximum/minimum on an interval must occur at endpoints only after equality in the Lipschitz chain is analyzed.
Appears in	2006-D2-P2

7.8 Probability / measure / expectation

PRB-01 - Discrete probability via residues and recurrences

Bucket	PRB discrete
Core theory	A Markov recurrence on residues or a generating function can compute probabilities of sums modulo m.
IMC use	Use for dice sums, random walks modulo primes, and residue probabilities.
Trap watch	State transition probabilities must sum to one and residues are modulo the correct integer.
Appears in	1999-D2-P2

PRB-02 - Generating functions for probability distributions

Bucket	PRB generating functions
Core theory	The probability generating function encodes sums of independent variables as products. Coefficients or root-of-unity filters extract desired events.
IMC use	Use for sums of independent discrete variables and random subsets.
Trap watch	Normalize coefficients by the total number of outcomes.
Appears in	2022-D1-P1

PRB-03 - Expectation, linearity, and probabilistic method

Bucket	PRB expectation
Core theory	Linearity of expectation requires no independence. A positive average proves existence of an object at least as good as the average.
IMC use	Use for random choices, expected counts, and probabilistic existence.
Trap watch	Independence is only needed for products or distribution factorization, not for linearity.
Appears in	2021-D1-P2

PRB-04 - Continuous uniform distribution and geometric probability

Bucket	PRB geometry
Core theory	Continuous geometric probability is volume/area normalized by the ambient domain. Symmetry, conditioning, and convex hull geometry are typical reductions.
IMC use	Use for random points in disks, balls, circles, and convex hulls.
Trap watch	Events involving exact coincidences on continuous distributions often have probability zero.
Appears in	No direct automatic assignment; keep as supporting background.

CORRECTION MATH-026 — legacy statement superseded; click for canonical replacement.

PRB-05 - Measure, variation, and differentiability almost everywhere

Bucket	PRB measure
Core theory	Finite total variation implies differentiability almost everywhere; covering arguments with small total length can force at least one point with small difference quotients.
IMC use	Use for sparse functions supported on countable sets and differentiability existence/nonexistence.
Trap watch	A countable exceptional support can still be dense; use interval covers, not isolation.
Appears in	1998-D2-P6, 2022-D1-P1

CLASSIFICATION NOTICE — cited legacy assignments are withdrawn pending quota-free recoding; click CLASS-001.

8 Problem-to-Theory Matrix

This table is the second pass through the corpus: every problem is assigned a primary detailed theory node and supporting nodes. The full titles are intentionally included so the table can be used directly as the first draft of the later theory database.

8.1 1994

Problem	Main theory	Supporting detailed theory nodes
1994-D1-P1	CMB-01 Pigeonhole principle and extremal element selection	LA-01 Matrix products, inverses, and one-sided inverse identities; LA-04 Rank-nullity, image/kernel, and rank of factorizations; CMB-02 Double counting and degree-sum methods; LA-02 Determinant evaluation by row/column operations and triangular forms; LA-03 Block matrices, Schur complements, and determinant identities; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities
1994-D1-P2	RA-06 Auxiliary functions, integrating factors, and differential inequalities	RA-13 L'Hospital, ratio limits, and growth comparisons; RA-10 Riemann sums and sum-integral asymptotics; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-07 Convexity, concavity, Jensen, and tangent/-chord inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-04 Rolle, mean-value theorem, and divided differences
1994-D1-P3	ANT-05 Rational roots, Vieta, and rationality constraints	CMB-01 Pigeonhole principle and extremal element selection; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; CMB-03 Graph encoding, cycles, and bipartite representations; CMB-12 Coloring, partitions, and parity; ANT-01 Modular arithmetic and divisibility
1994-D1-P4	LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton	LA-06 Commutators and spectral obstructions; LA-08 Nilpotent, idempotent, involution, and square-zero structures; RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-04 Rank-nullity, image/kernel, and rank of factorizations; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; CMB-09 Explicit construction and counterexample design
1994-D1-P5	RA-10 Riemann sums and sum-integral asymptotics	RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-12 Uniform continuity and periodic averaging; RA-13 L'Hospital, ratio limits, and growth comparisons; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; CMB-03 Graph encoding, cycles, and bipartite representations
1994-D1-P6	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	ANT-13 Lattices and norms/quasinorms; CMB-01 Pigeonhole principle and extremal element selection; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-04 Rolle, mean-value theorem, and divided differences; ANT-01 Modular arithmetic and divisibility; RA-08 Integral inequalities, integration by parts, and weighted estimates
1994-D2-P1	RA-06 Auxiliary functions, integrating factors, and differential inequalities	RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-04 Rolle, mean-value theorem, and divided differences; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-09 Fubini/Tonelli and change of variables

Problem	Main theory	Supporting detailed theory nodes
1994-D2-P2	RA-01 Compactness, extreme values, and closed-bounded reduction	RA-19 Gradient/Lipschitz-gradient optimization and convex smoothness; RA-02 Intermediate value, connectedness, density, and continuity transfer; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; LA-03 Block matrices, Schur complements, and determinant identities; RA-08 Integral inequalities, integration by parts, and weighted estimates
1994-D2-P3	RA-04 Rolle, mean-value theorem, and divided differences	RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-09 Fubini/Tonelli and change of variables; CMB-03 Graph encoding, cycles, and bipartite representations
1994-D2-P4	LA-10 Invariant subspaces and adapted bases	LA-09 Simultaneous diagonalization and commuting families; CP-01 Polynomial identity and more-roots-than-degree principle; LA-04 Rank-nullity, image/kernel, and rank of factorizations; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping
1994-D2-P5	CMB-09 Explicit construction and counterexample design	GEO-02 Euclidean distances and inner-product identities; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-10 Invariant subspaces and adapted bases; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools
1994-D2-P6	RA-10 Riemann sums and sum-integral asymptotics	RA-13 L'Hospital, ratio limits, and growth comparisons; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-03 Monotone functions, fixed points, and iteration on intervals; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping

8.2 1995

Problem	Main theory	Supporting detailed theory nodes
1995-D1-P1	LA-04 Rank-nullity, image/kernel, and rank of factorizations	LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; LA-08 Nilpotent, idempotent, involution, and square-zero structures; LA-02 Determinant evaluation by row/column operations and triangular forms; LA-03 Block matrices, Schur complements, and determinant identities; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; LA-10 Invariant subspaces and adapted bases
1995-D1-P2	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-09 Fubini/Tonelli and change of variables; RA-02 Intermediate value, connectedness, density, and continuity transfer; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-10 Riemann sums and sum-integral asymptotics; LA-03 Block matrices, Schur complements, and determinant identities

Problem	Main theory	Supporting detailed theory nodes
1995-D1-P3	RA-04 Rolle, mean-value theorem, and divided differences	RA-13 L'Hospital, ratio limits, and growth comparisons; RA-10 Riemann sums and sum-integral asymptotics; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-07 Convexity, concavity, Jensen, and tangent/-chord inequalities
1995-D1-P4	RA-03 Monotone functions, fixed points, and iteration on intervals	RA-13 L'Hospital, ratio limits, and growth comparisons; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-03 Graph encoding, cycles, and bipartite representations
1995-D1-P5	CP-01 Polynomial identity and more-roots-than-degree principle	LA-08 Nilpotent, idempotent, involution, and square-zero structures; LA-06 Commutators and spectral obstructions; RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools
1995-D1-P6	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	RA-01 Compactness, extreme values, and closed-bounded reduction; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-02 Intermediate value, connectedness, density, and continuity transfer; CMB-09 Explicit construction and counterexample design; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools
1995-D2-P1	GEO-02 Euclidean distances and inner-product identities	LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; LA-02 Determinant evaluation by row/column operations and triangular forms; LA-03 Block matrices, Schur complements, and determinant identities; LA-15 Gram matrices and rational linear dependence; GEO-09 Hilbert-space equidistant sets and orthonormal translations; CMB-03 Graph encoding, cycles, and bipartite representations
1995-D2-P2	RA-17 Sequence recurrences, monotone bounds, and generating functions	RA-10 Riemann sums and sum-integral asymptotics; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-13 L'Hospital, ratio limits, and growth comparisons; LA-03 Block matrices, Schur complements, and determinant identities; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping
1995-D2-P3	CP-01 Polynomial identity and more-roots-than-degree principle	CP-04 Unit-circle roots and conjugation identities; CP-03 Vieta, logarithmic derivatives, and real-root inequalities; CP-05 Roots of unity, cyclotomic filters, and discrete Fourier tricks; GEO-03 Projections and orthogonal decompositions; GEO-07 Area, volume, projection, and ellipsoid computation; CP-02 Factorization, irreducibility, and Gauss-style integer polynomial arguments
1995-D2-P4	RA-14 Weierstrass approximation and polynomial approximation	CP-10 Interpolation, Lagrange basis, and Chebyshev extremality; CP-01 Polynomial identity and more-roots-than-degree principle; RA-01 Compactness, extreme values, and closed-bounded reduction; RA-02 Intermediate value, connectedness, density, and continuity transfer; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; ANT-04 Integer polynomials, finite differences, and value divisibility

Problem	Main theory	Supporting detailed theory nodes
1995-D2-P5	RA-08 Integral inequalities, integration by parts, and weighted estimates	RA-18 Trigonometric and hyperbolic identities with monotonicity; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-10 Riemann sums and sum-integral asymptotics; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-09 Fubini/Tonelli and change of variables
1995-D2-P6	RA-15 L2 spaces, orthonormal systems, and dominated convergence	RA-02 Intermediate value, connectedness, density, and continuity transfer; RA-10 Riemann sums and sum-integral asymptotics; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-01 Compactness, extreme values, and closed-bounded reduction; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; CMB-09 Explicit construction and counterexample design

8.3 1996

Problem	Main theory	Supporting detailed theory nodes
1996-D1-P1	LA-02 Determinant evaluation by row/column operations and triangular forms	LA-03 Block matrices, Schur complements, and determinant identities; LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; LA-01 Matrix products, inverses, and one-sided inverse identities; LA-06 Commutators and spectral obstructions
1996-D1-P2	RA-18 Trigonometric and hyperbolic identities with monotonicity	RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-09 Fubini/Tonelli and change of variables; RA-06 Auxiliary functions, integrating factors, and differential inequalities; LA-03 Block matrices, Schur complements, and determinant identities
1996-D1-P3	LA-08 Nilpotent, idempotent, involution, and square-zero structures	LA-09 Simultaneous diagonalization and commuting families; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; LA-04 Rank-nullity, image/kernel, and rank of factorizations; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; LA-10 Invariant subspaces and adapted bases
1996-D1-P4	RA-17 Sequence recurrences, monotone bounds, and generating functions	RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-10 Riemann sums and sum-integral asymptotics; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-09 Fubini/Tonelli and change of variables; RA-13 L'Hospital, ratio limits, and growth comparisons
1996-D1-P5	RA-10 Riemann sums and sum-integral asymptotics	RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-04 Rolle, mean-value theorem, and divided differences; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton

Problem	Main theory	Supporting detailed theory nodes
1996-D1-P6	GEO-08 Metric content, diameter covers, and contractions	RA-01 Compactness, extreme values, and closed-bounded reduction; RA-02 Intermediate value, connectedness, density, and continuity transfer; LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-10 Invariant subspaces and adapted bases; CMB-03 Graph encoding, cycles, and bipartite representations; GEO-02 Euclidean distances and inner-product identities
1996-D2-P1	FEQ-03 Fixed points, iterates, and orbit closure	RA-02 Intermediate value, connectedness, density, and continuity transfer; RA-10 Riemann sums and sum-integral asymptotics; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-09 Fubini/Tonelli and change of variables; RA-13 L'Hospital, ratio limits, and growth comparisons
1996-D2-P2	RA-18 Trigonometric and hyperbolic identities with monotonicity	ANT-05 Rational roots, Vieta, and rationality constraints; CP-01 Polynomial identity and more-roots-than-degree principle; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; ANT-04 Integer polynomials, finite differences, and value divisibility; RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-03 Graph encoding, cycles, and bipartite representations
1996-D2-P3	ANT-07 Group basics: orders, cyclic subgroups, and generators	LA-01 Matrix products, inverses, and one-sided inverse identities; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; ANT-11 Rings, idempotents, nil rings, and noncommutativity; ANT-10 Automorphisms and semidirect products; ANT-09 Normal subgroups, quotient groups, and group actions
1996-D2-P4	GEO-04 Convex bodies, supporting geometry, and John-ellipse ideas	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-01 Compactness, extreme values, and closed-bounded reduction; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-02 Intermediate value, connectedness, density, and continuity transfer; GEO-07 Area, volume, projection, and ellipsoid computation; GEO-03 Projections and orthogonal decompositions
1996-D2-P5	RA-10 Riemann sums and sum-integral asymptotics	RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-13 L'Hospital, ratio limits, and growth comparisons; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-04 Rolle, mean-value theorem, and divided differences; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools
1996-D2-P6	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; RA-04 Rolle, mean-value theorem, and divided differences; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping

8.4 1997

Problem	Main theory	Supporting detailed theory nodes
1997-D1-P1	RA-10 Riemann sums and sum-integral asymptotics	RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-13 L'Hospital, ratio limits, and growth comparisons; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-09 Fubini/Tonelli and change of variables; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping
1997-D1-P2	RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools	RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-10 Riemann sums and sum-integral asymptotics; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-09 Fubini/Tonelli and change of variables; RA-13 L'Hospital, ratio limits, and growth comparisons; CMB-09 Explicit construction and counterexample design
1997-D1-P3	LA-06 Commutators and spectral obstructions	CP-05 Roots of unity, cyclotomic filters, and discrete Fourier tricks; LA-02 Determinant evaluation by row/column operations and triangular forms; LA-03 Block matrices, Schur complements, and determinant identities; ANT-01 Modular arithmetic and divisibility; LA-04 Rank-nullity, image/kernel, and rank of factorizations; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools
1997-D1-P4	ANT-14 Greedy infinite products and rationality tails	RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; ANT-01 Modular arithmetic and divisibility; CMB-09 Explicit construction and counterexample design; RA-09 Fubini/Tonelli and change of variables
1997-D1-P5	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	ANT-13 Lattices and norms/quasinorms; RA-08 Integral inequalities, integration by parts, and weighted estimates; GEO-03 Projections and orthogonal decompositions; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; GEO-07 Area, volume, projection, and ellipsoid computation; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools
1997-D1-P6	CMB-04 Set systems, intersecting families, and induction on sizes	CMB-09 Explicit construction and counterexample design; CMB-03 Graph encoding, cycles, and bipartite representations; CMB-12 Coloring, partitions, and parity; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; CMB-01 Pigeonhole principle and extremal element selection
1997-D2-P1	RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping	CMB-09 Explicit construction and counterexample design; RA-13 L'Hospital, ratio limits, and growth comparisons; RA-10 Riemann sums and sum-integral asymptotics; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-04 Rolle, mean-value theorem, and divided differences; RA-08 Integral inequalities, integration by parts, and weighted estimates
1997-D2-P2	LA-03 Block matrices, Schur complements, and determinant identities	LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-02 Determinant evaluation by row/column operations and triangular forms; LA-10 Invariant subspaces and adapted bases; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; LA-11 Linear functionals, kernels, and dual span criteria; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton

Problem	Main theory	Supporting detailed theory nodes
1997-D2-P3	RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools	RA-18 Trigonometric and hyperbolic identities with monotonicity; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-04 Rolle, mean-value theorem, and divided differences; RA-10 Riemann sums and sum-integral asymptotics; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping
1997-D2-P4	LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces	LA-11 Linear functionals, kernels, and dual span criteria; LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-03 Block matrices, Schur complements, and determinant identities; LA-10 Invariant subspaces and adapted bases; LA-02 Determinant evaluation by row/column operations and triangular forms; LA-06 Commutators and spectral obstructions
1997-D2-P5	FEQ-07 Bijections as products of involutions and symmetry decompositions	CMB-03 Graph encoding, cycles, and bipartite representations; GEO-07 Area, volume, projection, and ellipsoid computation; GEO-03 Projections and orthogonal decompositions; CMB-12 Coloring, partitions, and parity; GEO-06 Simplex, barycentric coordinates, and circumcenters; CMB-04 Set systems, intersecting families, and induction on sizes
1997-D2-P6	RA-16 Baire/category, dense sets, and pathological continuous functions	CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; CMB-09 Explicit construction and counterexample design; CMB-03 Graph encoding, cycles, and bipartite representations; CMB-12 Coloring, partitions, and parity; CMB-04 Set systems, intersecting families, and induction on sizes

8.5 1998

Problem	Main theory	Supporting detailed theory nodes
1998-D1-P1	LA-10 Invariant subspaces and adapted bases	LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-11 Linear functionals, kernels, and dual span criteria; CMB-03 Graph encoding, cycles, and bipartite representations; CMB-12 Coloring, partitions, and parity; LA-03 Block matrices, Schur complements, and determinant identities; CMB-04 Set systems, intersecting families, and induction on sizes
1998-D1-P2	ANT-08 Permutation groups, cycles, parity, and generation	CP-01 Polynomial identity and more-roots-than-degree principle; RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-03 Graph encoding, cycles, and bipartite representations; CMB-09 Explicit construction and counterexample design; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; ANT-10 Automorphisms and semidirect products
1998-D1-P3	FEQ-04 Recurrence transformations and conjugacy	RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-10 Riemann sums and sum-integral asymptotics; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-09 Fubini/Tonelli and change of variables
1998-D1-P4	RA-04 Rolle, mean-value theorem, and divided differences	RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-02 Intermediate value, connectedness, density, and continuity transfer; RA-16 Baire/category, dense sets, and pathological continuous functions

Problem	Main theory	Supporting detailed theory nodes
1998-D1-P5	CP-03 Vieta, logarithmic derivatives, and real-root inequalities	CP-01 Polynomial identity and more-roots-than-degree principle; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; CP-02 Factorization, irreducibility, and Gauss-style integer polynomial arguments; ANT-04 Integer polynomials, finite differences, and value divisibility; LA-03 Block matrices, Schur complements, and determinant identities
1998-D1-P6	RA-08 Integral inequalities, integration by parts, and weighted estimates	GEO-02 Euclidean distances and inner-product identities; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-09 Fubini/Tonelli and change of variables; RA-02 Intermediate value, connectedness, density, and continuity transfer; RA-06 Auxiliary functions, integrating factors, and differential inequalities
1998-D2-P1	LA-11 Linear functionals, kernels, and dual span criteria	LA-04 Rank-nullity, image/kernel, and rank of factorizations; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; LA-10 Invariant subspaces and adapted bases; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools
1998-D2-P2	CP-10 Interpolation, Lagrange basis, and Chebyshev extremality	CP-01 Polynomial identity and more-roots-than-degree principle; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-01 Compactness, extreme values, and closed-bounded reduction; CP-02 Factorization, irreducibility, and Gauss-style integer polynomial arguments; CP-03 Vieta, logarithmic derivatives, and real-root inequalities
1998-D2-P3	FEQ-06 Piecewise-linear maps and periodic points	CMB-03 Graph encoding, cycles, and bipartite representations; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; CMB-12 Coloring, partitions, and parity; RA-03 Monotone functions, fixed points, and iteration on intervals
1998-D2-P4	CMB-05 Recurrences and state partitions	CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; CMB-03 Graph encoding, cycles, and bipartite representations; CMB-12 Coloring, partitions, and parity
1998-D2-P5	CMB-01 Pigeonhole principle and extremal element selection	GEO-03 Projections and orthogonal decompositions; CMB-02 Double counting and degree-sum methods; RA-02 Intermediate value, connectedness, density, and continuity transfer; GEO-07 Area, volume, projection, and ellipsoid computation; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; CMB-03 Graph encoding, cycles, and bipartite representations
1998-D2-P6	RA-16 Baire/category, dense sets, and pathological continuous functions	PRB-05 Measure, variation, and differentiability almost everywhere; RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-17 Sequence recurrences, monotone bounds, and generating functions

8.6 1999

Problem	Main theory	Supporting detailed theory nodes
1999-D1-P1	LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton	LA-02 Determinant evaluation by row/column operations and triangular forms; CP-01 Polynomial identity and more-roots-than-degree principle; LA-03 Block matrices, Schur complements, and determinant identities; LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; ANT-10 Automorphisms and semidirect products
1999-D1-P2	RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools	CMB-01 Pigeonhole principle and extremal element selection; CMB-09 Explicit construction and counterexample design; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-10 Riemann sums and sum-integral asymptotics
1999-D1-P3	FEQ-08 No-regularity functions and order patterns on Q or R	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; ANT-01 Modular arithmetic and divisibility; RA-08 Integral inequalities, integration by parts, and weighted estimates; ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators; ANT-02 Chinese remainder theorem and complete residue systems
1999-D1-P4	FEQ-02 Injectivity and monotonicity forced by a functional equation	RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-09 Fubini/Tonelli and change of variables; CMB-03 Graph encoding, cycles, and bipartite representations
1999-D1-P5	CMB-03 Graph encoding, cycles, and bipartite representations	CMB-11 Grid, lattice, and alternating-cycle arguments; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; LA-03 Block matrices, Schur complements, and determinant identities; ANT-10 Automorphisms and semidirect products; ANT-11 Rings, idempotents, nil rings, and noncommutativity
1999-D1-P6	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-04 Rolle, mean-value theorem, and divided differences; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; CMB-09 Explicit construction and counterexample design; CMB-03 Graph encoding, cycles, and bipartite representations; CMB-12 Coloring, partitions, and parity
1999-D2-P1	ANT-11 Rings, idempotents, nil rings, and noncommutativity	ANT-07 Group basics: orders, cyclic subgroups, and generators; ANT-10 Automorphisms and semidirect products; ANT-09 Normal subgroups, quotient groups, and group actions; ANT-01 Modular arithmetic and divisibility; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; ANT-06 GCD, valuations, prime powers, and coprimality
1999-D2-P2	CP-05 Roots of unity, cyclotomic filters, and discrete Fourier tricks	PRB-01 Discrete probability via residues and recurrences; ANT-01 Modular arithmetic and divisibility; CP-01 Polynomial identity and more-roots-than-degree principle; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; CMB-07 Inclusion-exclusion and root-of-unity filters; LA-03 Block matrices, Schur complements, and determinant identities

Problem	Main theory	Supporting detailed theory nodes
1999-D2-P3	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-09 Fubini/-Tonelli and change of variables; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-10 Riemann sums and sum-integral asymptotics; LA-03 Block matrices, Schur complements, and determinant identities
1999-D2-P4	FEQ-08 No-regularity functions and order patterns on Q or R	RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-09 Fubini/Tonelli and change of variables; RA-03 Monotone functions, fixed points, and iteration on intervals; RA-06 Auxiliary functions, integrating factors, and differential inequalities
1999-D2-P5	CMB-13 Words, equivalence relations, and rewriting systems	CMB-03 Graph encoding, cycles, and bipartite representations; CMB-09 Explicit construction and counterexample design; CMB-12 Coloring, partitions, and parity; CMB-04 Set systems, intersecting families, and induction on sizes; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools
1999-D2-P6	CP-05 Roots of unity, cyclotomic filters, and discrete Fourier tricks	CMB-01 Pigeonhole principle and extremal element selection; CP-01 Polynomial identity and more-roots-than-degree principle; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; GEO-03 Projections and orthogonal decompositions; GEO-07 Area, volume, projection, and ellipsoid computation

8.7 2000

Problem	Main theory	Supporting detailed theory nodes
2000-D1-P1	RA-03 Monotone functions, fixed points, and iteration on intervals	CMB-09 Explicit construction and counterexample design; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-01 Compactness, extreme values, and closed-bounded reduction
2000-D1-P2	CP-13 Polynomial functional identities and algebraic composition	RA-04 Rolle, mean-value theorem, and divided differences; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; CP-04 Unit-circle roots and conjugation identities; CP-05 Roots of unity, cyclotomic filters, and discrete Fourier tricks; LA-03 Block matrices, Schur complements, and determinant identities; RA-06 Auxiliary functions, integrating factors, and differential inequalities
2000-D1-P3	LA-06 Commutators and spectral obstructions	LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-08 Nilpotent, idempotent, involution, and square-zero structures; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; LA-02 Determinant evaluation by row/column operations and triangular forms; LA-03 Block matrices, Schur complements, and determinant identities

Problem	Main theory	Supporting detailed theory nodes
2000-D1-P4	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-03 Monotone functions, fixed points, and iteration on intervals; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-18 Trigonometric and hyperbolic identities with monotonicity; RA-10 Riemann sums and sum-integral asymptotics; LA-03 Block matrices, Schur complements, and determinant identities
2000-D1-P5	ANT-11 Rings, idempotents, nil rings, and noncommutativity	LA-06 Commutators and spectral obstructions; ANT-07 Group basics: orders, cyclic subgroups, and generators; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; LA-08 Nilpotent, idempotent, involution, and square-zero structures; ANT-10 Automorphisms and semidirect products; ANT-01 Modular arithmetic and divisibility
2000-D1-P6	RA-06 Auxiliary functions, integrating factors, and differential inequalities	RA-13 L'Hospital, ratio limits, and growth comparisons; RA-08 Integral inequalities, integration by parts, and weighted estimates; CP-01 Polynomial identity and more-roots-than-degree principle; RA-04 Rolle, mean-value theorem, and divided differences; CMB-06 Generating functions for counting; RA-10 Riemann sums and sum-integral asymptotics
2000-D2-P1	GEO-14 Cube and square dissection by number-theoretic semigroups	ANT-06 GCD, valuations, prime powers, and coprimality; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; ANT-01 Modular arithmetic and divisibility; GEO-07 Area, volume, projection, and ellipsoid computation; LA-04 Rank-nullity, image/kernel, and rank of factorizations; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities
2000-D2-P2	RA-01 Compactness, extreme values, and closed-bounded reduction	RA-16 Baire/category, dense sets, and pathological continuous functions; RA-02 Intermediate value, connectedness, density, and continuity transfer; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-03 Monotone functions, fixed points, and iteration on intervals; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-06 Auxiliary functions, integrating factors, and differential inequalities
2000-D2-P3	CP-01 Polynomial identity and more-roots-than-degree principle	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; CP-03 Vieta, logarithmic derivatives, and real-root inequalities; CP-05 Roots of unity, cyclotomic filters, and discrete Fourier tricks; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; CP-02 Factorization, irreducibility, and Gauss-style integer polynomial arguments; CP-04 Unit-circle roots and conjugation identities
2000-D2-P4	CP-03 Vieta, logarithmic derivatives, and real-root inequalities	CP-01 Polynomial identity and more-roots-than-degree principle; GEO-05 Conics: parabolas, ellipses, foci, and directrices; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; CMB-03 Graph encoding, cycles, and bipartite representations; RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-12 Coloring, partitions, and parity
2000-D2-P5	FEQ-02 Injectivity and monotonicity forced by a functional equation	RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-10 Riemann sums and sum-integral asymptotics; RA-04 Rolle, mean-value theorem, and divided differences; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools

Problem	Main theory	Supporting detailed theory nodes
2000-D2-P6	LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton	LA-03 Block matrices, Schur complements, and determinant identities; LA-13 Matrix exponentials and polynomial functional calculus; CP-01 Polynomial identity and more-roots-than-degree principle; LA-08 Nilpotent, idempotent, involution, and square-zero structures; LA-06 Commutators and spectral obstructions; RA-08 Integral inequalities, integration by parts, and weighted estimates

8.8 2001

Problem	Main theory	Supporting detailed theory nodes
2001-D1-P1	CMB-08 Permutations, inversions, and permutation statistics	LA-03 Block matrices, Schur complements, and determinant identities; LA-02 Determinant evaluation by row/column operations and triangular forms; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; ANT-01 Modular arithmetic and divisibility; RA-09 Fubini/Tonelli and change of variables
2001-D1-P2	ANT-07 Group basics: orders, cyclic subgroups, and generators	ANT-08 Permutation groups, cycles, parity, and generation; ANT-01 Modular arithmetic and divisibility; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; CMB-03 Graph encoding, cycles, and bipartite representations; ANT-11 Rings, idempotents, nil rings, and noncommutativity; ANT-10 Automorphisms and semidirect products
2001-D1-P3	RA-10 Riemann sums and sum-integral asymptotics	RA-13 L'Hospital, ratio limits, and growth comparisons; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-03 Block matrices, Schur complements, and determinant identities; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-01 Compactness, extreme values, and closed-bounded reduction
2001-D1-P4	CP-05 Roots of unity, cyclotomic filters, and discrete Fourier tricks	CP-02 Factorization, irreducibility, and Gauss-style integer polynomial arguments; CP-01 Polynomial identity and more-roots-than-degree principle; ANT-01 Modular arithmetic and divisibility; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; CP-03 Vieta, logarithmic derivatives, and real-root inequalities
2001-D1-P5	LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton	LA-10 Invariant subspaces and adapted bases; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; LA-02 Determinant evaluation by row/column operations and triangular forms; LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-03 Block matrices, Schur complements, and determinant identities; LA-06 Commutators and spectral obstructions
2001-D1-P6	RA-13 L'Hospital, ratio limits, and growth comparisons	RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-10 Riemann sums and sum-integral asymptotics; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-04 Rolle, mean-value theorem, and divided differences

Problem	Main theory	Supporting detailed theory nodes
2001-D2-P1	ANT-04 Integer polynomials, finite differences, and value divisibility	CP-02 Factorization, irreducibility, and Gauss-style integer polynomial arguments; CP-01 Polynomial identity and more-roots-than-degree principle; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates
2001-D2-P2	RA-17 Sequence recurrences, monotone bounds, and generating functions	RA-18 Trigonometric and hyperbolic identities with monotonicity; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-10 Riemann sums and sum-integral asymptotics; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-03 Monotone functions, fixed points, and iteration on intervals; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools
2001-D2-P3	GEO-11 Unit-sphere configurations and regular simplex bounds	CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; LA-04 Rank-nullity, image/kernel, and rank of factorizations; CP-05 Roots of unity, cyclotomic filters, and discrete Fourier tricks; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; LA-10 Invariant subspaces and adapted bases
2001-D2-P4	CMB-03 Graph encoding, cycles, and bipartite representations	LA-08 Nilpotent, idempotent, involution, and square-zero structures; LA-02 Determinant evaluation by row/column operations and triangular forms; LA-03 Block matrices, Schur complements, and determinant identities; RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces
2001-D2-P5	FEQ-08 No-regularity functions and order patterns on Q or R	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-10 Riemann sums and sum-integral asymptotics; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-13 L'Hospital, ratio limits, and growth comparisons; CMB-03 Graph encoding, cycles, and bipartite representations; CMB-12 Coloring, partitions, and parity
2001-D2-P6	RA-18 Trigonometric and hyperbolic identities with monotonicity	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-04 Rolle, mean-value theorem, and divided differences; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; ANT-01 Modular arithmetic and divisibility; RA-01 Compactness, extreme values, and closed-bounded reduction; RA-08 Integral inequalities, integration by parts, and weighted estimates

8.9 2002

Problem	Main theory	Supporting detailed theory nodes
2002-D1-P1	GEO-05 Conics: parabolas, ellipses, foci, and directrices	CP-01 Polynomial identity and more-roots-than-degree principle; CMB-03 Graph encoding, cycles, and bipartite representations; GEO-07 Area, volume, projection, and ellipsoid computation; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; GEO-03 Projections and orthogonal decompositions; CMB-12 Coloring, partitions, and parity

Problem	Main theory	Supporting detailed theory nodes
2002-D1-P2	FEQ-08 No-regularity functions and order patterns on Q or R	RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-09 Explicit construction and counterexample design; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-04 Rolle, mean-value theorem, and divided differences; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools
2002-D1-P3	ANT-06 GCD, valuations, prime powers, and coprimality	CMB-06 Generating functions for counting; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; ANT-01 Modular arithmetic and divisibility; RA-09 Fubini/Tonelli and change of variables
2002-D1-P4	RA-01 Compactness, extreme values, and closed-bounded reduction	FEQ-03 Fixed points, iterates, and orbit closure; RA-02 Intermediate value, connectedness, density, and continuity transfer; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-10 Riemann sums and sum-integral asymptotics; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods
2002-D1-P5	RA-16 Baire/category, dense sets, and pathological continuous functions	LA-04 Rank-nullity, image/kernel, and rank of factorizations; RA-04 Rolle, mean-value theorem, and divided differences; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-10 Invariant subspaces and adapted bases
2002-D1-P6	LA-14 Operator norm, power boundedness, and spectral-radius style estimates	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; GEO-02 Euclidean distances and inner-product identities; LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-10 Invariant subspaces and adapted bases; RA-10 Riemann sums and sum-integral asymptotics
2002-D2-P1	LA-02 Determinant evaluation by row/column operations and triangular forms	LA-03 Block matrices, Schur complements, and determinant identities; LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; LA-06 Commutators and spectral obstructions; LA-01 Matrix products, inverses, and one-sided inverse identities
2002-D2-P2	CMB-02 Double counting and degree-sum methods	CMB-03 Graph encoding, cycles, and bipartite representations; CMB-12 Coloring, partitions, and parity; CMB-04 Set systems, intersecting families, and induction on sizes; CMB-01 Pigeonhole principle and extremal element selection; LA-03 Block matrices, Schur complements, and determinant identities; CMB-05 Recurrences and state partitions
2002-D2-P3	RA-17 Sequence recurrences, monotone bounds, and generating functions	CMB-06 Generating functions for counting; CP-01 Polynomial identity and more-roots-than-degree principle; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; ANT-04 Integer polynomials, finite differences, and value divisibility

Problem	Main theory	Supporting detailed theory nodes
2002-D2-P4	GEO-03 Projections and orthogonal decompositions	GEO-13 Convex hulls, random polytopes, and geometric probability; CMB-03 Graph encoding, cycles, and bipartite representations; GEO-07 Area, volume, projection, and ellipsoid computation; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; CMB-12 Coloring, partitions, and parity
2002-D2-P5	LA-01 Matrix products, inverses, and one-sided inverse identities	LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; CP-05 Roots of unity, cyclotomic filters, and discrete Fourier tricks; GEO-03 Projections and orthogonal decompositions; GEO-07 Area, volume, projection, and ellipsoid computation; CP-04 Unit-circle roots and conjugation identities; CMB-03 Graph encoding, cycles, and bipartite representations
2002-D2-P6	RA-19 Gradient/Lipschitz-gradient optimization and convex smoothness	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; LA-15 Gram matrices and rational linear dependence; RA-01 Compactness, extreme values, and closed-bounded reduction; RA-08 Integral inequalities, integration by parts, and weighted estimates; GEO-09 Hilbert-space equidistant sets and orthonormal translations; CMB-01 Pigeonhole principle and extremal element selection

8.10 2003

Problem	Main theory	Supporting detailed theory nodes
2003-D1-P1	RA-17 Sequence recurrences, monotone bounds, and generating functions	RA-10 Riemann sums and sum-integral asymptotics; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-09 Fubini/Tonelli and change of variables; RA-13 L'Hospital, ratio limits, and growth comparisons
2003-D1-P2	ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools	CMB-08 Permutations, inversions, and permutation statistics; ANT-01 Modular arithmetic and divisibility; ANT-07 Group basics: orders, cyclic subgroups, and generators; ANT-11 Rings, idempotents, nil rings, and noncommutativity; ANT-10 Automorphisms and semidirect products; RA-08 Integral inequalities, integration by parts, and weighted estimates
2003-D1-P3	LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton	CP-01 Polynomial identity and more-roots-than-degree principle; LA-08 Nilpotent, idempotent, involution, and square-zero structures; LA-06 Commutators and spectral obstructions; RA-10 Riemann sums and sum-integral asymptotics; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-08 Integral inequalities, integration by parts, and weighted estimates
2003-D1-P4	ANT-06 GCD, valuations, prime powers, and coprimality	LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; ANT-01 Modular arithmetic and divisibility; RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-03 Graph encoding, cycles, and bipartite representations; CMB-12 Coloring, partitions, and parity

Problem	Main theory	Supporting detailed theory nodes
2003-D1-P5	RA-08 Integral inequalities, integration by parts, and weighted estimates	FEQ-04 Recurrence transformations and conjugacy; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; RA-10 Riemann sums and sum-integral asymptotics; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-17 Sequence recurrences, monotone bounds, and generating functions
2003-D1-P6	CP-11 Stable polynomials and coefficient inequalities	CP-01 Polynomial identity and more-roots-than-degree principle; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-17 Sequence recurrences, monotone bounds, and generating functions; CP-03 Vieta, logarithmic derivatives, and real-root inequalities; CMB-06 Generating functions for counting
2003-D2-P1	LA-01 Matrix products, inverses, and one-sided inverse identities	LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; LA-02 Determinant evaluation by row/column operations and triangular forms; LA-03 Block matrices, Schur complements, and determinant identities; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; LA-06 Commutators and spectral obstructions
2003-D2-P2	RA-13 L'Hospital, ratio limits, and growth comparisons	RA-18 Trigonometric and hyperbolic identities with monotonicity; CP-01 Polynomial identity and more-roots-than-degree principle; RA-10 Riemann sums and sum-integral asymptotics; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-06 Auxiliary functions, integrating factors, and differential inequalities
2003-D2-P3	GEO-10 Closest points to closed sets and density of uniqueness	RA-01 Compactness, extreme values, and closed-bounded reduction; RA-02 Intermediate value, connectedness, density, and continuity transfer; CMB-01 Pigeonhole principle and extremal element selection; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; CMB-03 Graph encoding, cycles, and bipartite representations; CMB-12 Coloring, partitions, and parity
2003-D2-P4	CMB-15 Finite designs and pair-determines-third structures	ANT-15 Finite abelian groups and vector spaces over finite fields; ANT-07 Group basics: orders, cyclic subgroups, and generators; ANT-11 Rings, idempotents, nil rings, and noncommutativity; ANT-10 Automorphisms and semidirect products; CMB-03 Graph encoding, cycles, and bipartite representations; CMB-09 Explicit construction and counterexample design
2003-D2-P5	RA-16 Baire/category, dense sets, and pathological continuous functions	FEQ-08 No-regularity functions and order patterns on Q or R; RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; RA-10 Riemann sums and sum-integral asymptotics; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities
2003-D2-P6	RA-17 Sequence recurrences, monotone bounds, and generating functions	CMB-06 Generating functions for counting; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-10 Riemann sums and sum-integral asymptotics; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-09 Fubini/Tonelli and change of variables

8.11 2004

Problem	Main theory	Supporting detailed theory nodes
2004-D1-P1	CMB-01 Pigeonhole principle and extremal element selection	CMB-02 Double counting and degree-sum methods; CMB-03 Graph encoding, cycles, and bipartite representations; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; CMB-12 Coloring, partitions, and parity; CMB-04 Set systems, intersecting families, and induction on sizes
2004-D1-P2	FEQ-04 Recurrence transformations and conjugacy	RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-09 Fubini/Tonelli and change of variables
2004-D1-P3	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	RA-18 Trigonometric and hyperbolic identities with monotonicity; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-01 Compactness, extreme values, and closed-bounded reduction; RA-02 Intermediate value, connectedness, density, and continuity transfer; CMB-03 Graph encoding, cycles, and bipartite representations; LA-04 Rank-nullity, image/kernel, and rank of factorizations
2004-D1-P4	GEO-03 Projections and orthogonal decompositions	GEO-02 Euclidean distances and inner-product identities; CMB-01 Pigeonhole principle and extremal element selection; RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-02 Double counting and degree-sum methods; CMB-12 Coloring, partitions, and parity; GEO-07 Area, volume, projection, and ellipsoid computation
2004-D1-P5	CMB-10 Erdos-Szekeres style monotone subsequences and binomial thresholds	RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; RA-03 Monotone functions, fixed points, and iteration on intervals; RA-06 Auxiliary functions, integrating factors, and differential inequalities
2004-D1-P6	CP-12 Partial fractions and cotangent-type expansions	CP-08 Laurent expansions, isolated singularities, and rationality of meromorphic functions; CP-01 Polynomial identity and more-roots-than-degree principle; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-13 L'Hospital, ratio limits, and growth comparisons
2004-D2-P1	LA-03 Block matrices, Schur complements, and determinant identities	LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; LA-01 Matrix products, inverses, and one-sided inverse identities; LA-02 Determinant evaluation by row/column operations and triangular forms; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; LA-06 Commutators and spectral obstructions
2004-D2-P2	GEO-07 Area, volume, projection, and ellipsoid computation	RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; CMB-03 Graph encoding, cycles, and bipartite representations; GEO-03 Projections and orthogonal decompositions; CMB-12 Coloring, partitions, and parity; RA-03 Monotone functions, fixed points, and iteration on intervals

Problem	Main theory	Supporting detailed theory nodes
2004-D2-P3	GEO-02 Euclidean distances and inner-product identities	RA-08 Integral inequalities, integration by parts, and weighted estimates; GEO-07 Area, volume, projection, and ellipsoid computation; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; GEO-03 Projections and orthogonal decompositions; ANT-10 Automorphisms and semidirect products; CMB-09 Explicit construction and counterexample design
2004-D2-P4	LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton	LA-03 Block matrices, Schur complements, and determinant identities; LA-04 Rank-nullity, image/kernel, and rank of factorizations; RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-10 Invariant subspaces and adapted bases; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-10 Riemann sums and sum-integral asymptotics
2004-D2-P5	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	RA-09 Fubini/Tonelli and change of variables; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-10 Riemann sums and sum-integral asymptotics; RA-13 L'Hospital, ratio limits, and growth comparisons; RA-18 Trigonometric and hyperbolic identities with monotonicity
2004-D2-P6	CMB-14 Transfer matrices and matrix powers for counting	RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; LA-02 Determinant evaluation by row/column operations and triangular forms; LA-03 Block matrices, Schur complements, and determinant identities; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities

8.12 2005

Problem	Main theory	Supporting detailed theory nodes
2005-D1-P1	LA-04 Rank-nullity, image/kernel, and rank of factorizations	LA-02 Determinant evaluation by row/column operations and triangular forms; LA-03 Block matrices, Schur complements, and determinant identities; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools
2005-D1-P2	CMB-05 Recurrences and state partitions	ANT-01 Modular arithmetic and divisibility; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-17 Sequence recurrences, monotone bounds, and generating functions
2005-D1-P3	RA-08 Integral inequalities, integration by parts, and weighted estimates	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-04 Rolle, mean-value theorem, and divided differences; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-02 Intermediate value, connectedness, density, and continuity transfer; LA-04 Rank-nullity, image/kernel, and rank of factorizations; RA-06 Auxiliary functions, integrating factors, and differential inequalities

Problem	Main theory	Supporting detailed theory nodes
2005-D1-P4	ANT-04 Integer polynomials, finite differences, and value divisibility	CP-01 Polynomial identity and more-roots-than-degree principle; ANT-05 Rational roots, Vieta, and rationality constraints; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; RA-08 Integral inequalities, integration by parts, and weighted estimates; ANT-01 Modular arithmetic and divisibility; CP-03 Vieta, logarithmic derivatives, and real-root inequalities
2005-D1-P5	RA-13 L'Hospital, ratio limits, and growth comparisons	RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-10 Riemann sums and sum-integral asymptotics; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-04 Rolle, mean-value theorem, and divided differences
2005-D1-P6	ANT-07 Group basics: orders, cyclic subgroups, and generators	ANT-09 Normal subgroups, quotient groups, and group actions; LA-06 Commutators and spectral obstructions; ANT-11 Rings, idempotents, nil rings, and noncommutativity; ANT-10 Automorphisms and semidirect products; LA-08 Nilpotent, idempotent, involution, and square-zero structures; RA-07 Convexity, concavity, Jensen, and tangent/-chord inequalities
2005-D2-P1	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-01 Compactness, extreme values, and closed-bounded reduction; RA-09 Fubini/Tonelli and change of variables; CMB-03 Graph encoding, cycles, and bipartite representations; CMB-12 Coloring, partitions, and parity
2005-D2-P2	CP-01 Polynomial identity and more-roots-than-degree principle	CP-03 Vieta, logarithmic derivatives, and real-root inequalities; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; CP-02 Factorization, irreducibility, and Gauss-style integer polynomial arguments; ANT-01 Modular arithmetic and divisibility
2005-D2-P3	LA-12 Matrix subspaces constrained by trace bilinear forms	LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-03 Block matrices, Schur complements, and determinant identities; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-10 Invariant subspaces and adapted bases; LA-02 Determinant evaluation by row/column operations and triangular forms
2005-D2-P4	RA-04 Rolle, mean-value theorem, and divided differences	RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; CMB-03 Graph encoding, cycles, and bipartite representations; CMB-12 Coloring, partitions, and parity; CMB-04 Set systems, intersecting families, and induction on sizes; FEQ-01 Substitution of special values and normalization; FEQ-02 Injectivity and monotonicity forced by a functional equation
2005-D2-P5	RA-19 Gradient/Lipschitz-gradient optimization and convex smoothness	GEO-02 Euclidean distances and inner-product identities; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; GEO-07 Area, volume, projection, and ellipsoid computation; GEO-03 Projections and orthogonal decompositions; RA-01 Compactness, extreme values, and closed-bounded reduction

Problem	Main theory	Supporting detailed theory nodes
2005-D2-P6	ANT-12 Diophantine equations, Pell-type recurrences, and quadratic irrationals	ANT-01 Modular arithmetic and divisibility; LA-03 Block matrices, Schur complements, and determinant identities; CP-01 Polynomial identity and more-roots-than-degree principle; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; LA-02 Determinant evaluation by row/column operations and triangular forms; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces

8.13 2006

Problem	Main theory	Supporting detailed theory nodes
2006-D1-P1	CMB-09 Explicit construction and counterexample design	RA-02 Intermediate value, connectedness, density, and continuity transfer; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-10 Riemann sums and sum-integral asymptotics; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-06 Auxiliary functions, integrating factors, and differential inequalities
2006-D1-P2	ANT-01 Modular arithmetic and divisibility	ANT-02 Chinese remainder theorem and complete residue systems; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; ANT-07 Group basics: orders, cyclic subgroups, and generators; ANT-11 Rings, idempotents, nil rings, and noncommutativity
2006-D1-P3	ANT-04 Integer polynomials, finite differences, and value divisibility	ANT-01 Modular arithmetic and divisibility; LA-02 Determinant evaluation by row/column operations and triangular forms; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; LA-03 Block matrices, Schur complements, and determinant identities; LA-04 Rank-nullity, image/kernel, and rank of factorizations; RA-08 Integral inequalities, integration by parts, and weighted estimates
2006-D1-P4	CP-02 Factorization, irreducibility, and Gauss-style integer polynomial arguments	ANT-04 Integer polynomials, finite differences, and value divisibility; CP-01 Polynomial identity and more-roots-than-degree principle; RA-10 Riemann sums and sum-integral asymptotics; CP-03 Vieta, logarithmic derivatives, and real-root inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools
2006-D1-P5	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	ANT-05 Rational roots, Vieta, and rationality constraints; RA-04 Rolle, mean-value theorem, and divided differences; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-02 Intermediate value, connectedness, density, and continuity transfer; RA-06 Auxiliary functions, integrating factors, and differential inequalities
2006-D1-P6	CP-01 Polynomial identity and more-roots-than-degree principle	RA-04 Rolle, mean-value theorem, and divided differences; RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-04 Rank-nullity, image/kernel, and rank of factorizations; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools

Problem	Main theory	Supporting detailed theory nodes
2006-D2-P1	CMB-12 Coloring, partitions, and parity	GEO-12 Polygon triangulation and parity of incidences; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; CMB-03 Graph encoding, cycles, and bipartite representations; ANT-01 Modular arithmetic and divisibility; RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-01 Pigeonhole principle and extremal element selection
2006-D2-P2	FEQ-09 Image-length preserving functions	LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-10 Invariant subspaces and adapted bases; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; RA-02 Intermediate value, connectedness, density, and continuity transfer; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; LA-03 Block matrices, Schur complements, and determinant identities
2006-D2-P3	RA-18 Trigonometric and hyperbolic identities with monotonicity	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-10 Riemann sums and sum-integral asymptotics; LA-03 Block matrices, Schur complements, and determinant identities; RA-01 Compactness, extreme values, and closed-bounded reduction
2006-D2-P4	GEO-02 Euclidean distances and inner-product identities	LA-15 Gram matrices and rational linear dependence; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-10 Invariant subspaces and adapted bases; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools
2006-D2-P5	ANT-12 Diophantine equations, Pell-type recurrences, and quadratic irrationals	ANT-01 Modular arithmetic and divisibility; CP-01 Polynomial identity and more-roots-than-degree principle; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; CMB-02 Double counting and degree-sum methods; CMB-01 Pigeonhole principle and extremal element selection; ANT-02 Chinese remainder theorem and complete residue systems
2006-D2-P6	LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton	LA-09 Simultaneous diagonalization and commuting families; LA-02 Determinant evaluation by row/column operations and triangular forms; LA-03 Block matrices, Schur complements, and determinant identities; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; CP-01 Polynomial identity and more-roots-than-degree principle; LA-04 Rank-nullity, image/kernel, and rank of factorizations

8.14 2007

Problem	Main theory	Supporting detailed theory nodes
2007-D1-P1	CP-01 Polynomial identity and more-roots-than-degree principle	ANT-04 Integer polynomials, finite differences, and value divisibility; ANT-01 Modular arithmetic and divisibility; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; ANT-02 Chinese remainder theorem and complete residue systems; CP-02 Factorization, irreducibility, and Gauss-style integer polynomial arguments; CP-03 Vieta, logarithmic derivatives, and real-root inequalities

Problem	Main theory	Supporting detailed theory nodes
2007-D1-P2	LA-04 Rank-nullity, image/kernel, and rank of factorizations	LA-02 Determinant evaluation by row/column operations and triangular forms; LA-03 Block matrices, Schur complements, and determinant identities; RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-10 Invariant subspaces and adapted bases; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces
2007-D1-P3	LA-04 Rank-nullity, image/kernel, and rank of factorizations	CP-13 Polynomial functional identities and algebraic composition; CP-01 Polynomial identity and more-roots-than-degree principle; LA-02 Determinant evaluation by row/-column operations and triangular forms; LA-03 Block matrices, Schur complements, and determinant identities; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities
2007-D1-P4	CMB-02 Double counting and degree-sum methods	ANT-07 Group basics: orders, cyclic subgroups, and generators; CMB-01 Pigeonhole principle and extremal element selection; ANT-11 Rings, idempotents, nil rings, and noncommutativity; ANT-10 Automorphisms and semidirect products; CMB-12 Coloring, partitions, and parity; RA-17 Sequence recurrences, monotone bounds, and generating functions
2007-D1-P5	CP-05 Roots of unity, cyclotomic filters, and discrete Fourier tricks	CP-06 Cauchy formula and coefficient extraction on the unit circle; CP-01 Polynomial identity and more-roots-than-degree principle; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; CP-03 Vieta, logarithmic derivatives, and real-root inequalities; LA-04 Rank-nullity, image/kernel, and rank of factorizations
2007-D1-P6	CP-05 Roots of unity, cyclotomic filters, and discrete Fourier tricks	CP-01 Polynomial identity and more-roots-than-degree principle; CP-06 Cauchy formula and coefficient extraction on the unit circle; RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities
2007-D2-P1	CMB-09 Explicit construction and counterexample design	FEQ-01 Substitution of special values and normalization; CMB-03 Graph encoding, cycles, and bipartite representations; CMB-12 Coloring, partitions, and parity; CMB-04 Set systems, intersecting families, and induction on sizes; RA-02 Intermediate value, connectedness, density, and continuity transfer; LA-03 Block matrices, Schur complements, and determinant identities
2007-D2-P2	ANT-01 Modular arithmetic and divisibility	ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; ANT-02 Chinese remainder theorem and complete residue systems; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-09 Fubini/Tonelli and change of variables; ANT-06 GCD, valuations, prime powers, and coprimality
2007-D2-P3	RA-03 Monotone functions, fixed points, and iteration on intervals	RA-02 Intermediate value, connectedness, density, and continuity transfer; RA-01 Compactness, extreme values, and closed-bounded reduction; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; CMB-09 Explicit construction and counterexample design; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-06 Auxiliary functions, integrating factors, and differential inequalities

Problem	Main theory	Supporting detailed theory nodes
2007-D2-P4	LA-02 Determinant evaluation by row/column operations and triangular forms	ANT-01 Modular arithmetic and divisibility; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; LA-03 Block matrices, Schur complements, and determinant identities; LA-04 Rank-nullity, image/kernel, and rank of factorizations; ANT-02 Chinese remainder theorem and complete residue systems; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces
2007-D2-P5	CMB-09 Explicit construction and counterexample design	LA-08 Nilpotent, idempotent, involution, and square-zero structures; LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-10 Invariant subspaces and adapted bases; ANT-07 Group basics: orders, cyclic subgroups, and generators; ANT-11 Rings, idempotents, nil rings, and noncommutativity; ANT-10 Automorphisms and semidirect products
2007-D2-P6	CP-01 Polynomial identity and more-roots-than-degree principle	RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-04 Rolle, mean-value theorem, and divided differences; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-10 Riemann sums and sum-integral asymptotics; CP-03 Vieta, logarithmic derivatives, and real-root inequalities

8.15 2008

Problem	Main theory	Supporting detailed theory nodes
2008-D1-P6	CMB-01 Pigeonhole principle and extremal element selection	LA-03 Block matrices, Schur complements, and determinant identities; CP-01 Polynomial identity and more-roots-than-degree principle; LA-02 Determinant evaluation by row/column operations and triangular forms; CMB-02 Double counting and degree-sum methods; CMB-03 Graph encoding, cycles, and bipartite representations; CMB-12 Coloring, partitions, and parity
2008-D1-P1	RA-02 Intermediate value, connectedness, density, and continuity transfer	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-01 Compactness, extreme values, and closed-bounded reduction; RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-03 Graph encoding, cycles, and bipartite representations; CMB-12 Coloring, partitions, and parity; LA-03 Block matrices, Schur complements, and determinant identities
2008-D1-P2	CP-01 Polynomial identity and more-roots-than-degree principle	RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-04 Rank-nullity, image/kernel, and rank of factorizations; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; LA-10 Invariant subspaces and adapted bases; LA-11 Linear functionals, kernels, and dual span criteria; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools
2008-D1-P3	ANT-01 Modular arithmetic and divisibility	CP-01 Polynomial identity and more-roots-than-degree principle; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; ANT-04 Integer polynomials, finite differences, and value divisibility; CMB-09 Explicit construction and counterexample design; CP-02 Factorization, irreducibility, and Gauss-style integer polynomial arguments

Problem	Main theory	Supporting detailed theory nodes
2008-D1-P4	RA-08 Integral inequalities, integration by parts, and weighted estimates	CMB-03 Graph encoding, cycles, and bipartite representations; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; CMB-12 Coloring, partitions, and parity; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; CMB-04 Set systems, intersecting families, and induction on sizes; RA-09 Fubini/Tonelli and change of variables
2008-D1-P5	ANT-07 Group basics: orders, cyclic subgroups, and generators	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; ANT-10 Automorphisms and semidirect products; GEO-02 Euclidean distances and inner-product identities; CMB-09 Explicit construction and counterexample design; ANT-01 Modular arithmetic and divisibility; ANT-11 Rings, idempotents, nil rings, and noncommutativity
2008-D2-P1	CP-02 Factorization, irreducibility, and Gauss-style integer polynomial arguments	CP-05 Roots of unity, cyclotomic filters, and discrete Fourier tricks; CP-01 Polynomial identity and more-roots-than-degree principle; ANT-04 Integer polynomials, finite differences, and value divisibility; CP-03 Vieta, logarithmic derivatives, and real-root inequalities; CP-04 Unit-circle roots and conjugation identities; LA-03 Block matrices, Schur complements, and determinant identities
2008-D2-P2	GEO-05 Conics: parabolas, ellipses, foci, and directrices	CP-01 Polynomial identity and more-roots-than-degree principle; GEO-07 Area, volume, projection, and ellipsoid computation; GEO-03 Projections and orthogonal decompositions; GEO-02 Euclidean distances and inner-product identities; GEO-06 Simplex, barycentric coordinates, and circumcenters; CP-02 Factorization, irreducibility, and Gauss-style integer polynomial arguments
2008-D2-P3	ANT-01 Modular arithmetic and divisibility	CMB-06 Generating functions for counting; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-17 Sequence recurrences, monotone bounds, and generating functions; RA-08 Integral inequalities, integration by parts, and weighted estimates; ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators
2008-D2-P4	CMB-01 Pigeonhole principle and extremal element selection	ANT-04 Integer polynomials, finite differences, and value divisibility; CP-01 Polynomial identity and more-roots-than-degree principle; ANT-01 Modular arithmetic and divisibility; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; CMB-02 Double counting and degree-sum methods; CP-02 Factorization, irreducibility, and Gauss-style integer polynomial arguments
2008-D2-P5	LA-02 Determinant evaluation by row/column operations and triangular forms	CMB-03 Graph encoding, cycles, and bipartite representations; LA-04 Rank-nullity, image/kernel, and rank of factorizations; ANT-01 Modular arithmetic and divisibility; LA-03 Block matrices, Schur complements, and determinant identities; CP-01 Polynomial identity and more-roots-than-degree principle; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools
2008-D2-P6	GEO-09 Hilbert-space equidistant sets and orthonormal translations	LA-15 Gram matrices and rational linear dependence; LA-04 Rank-nullity, image/kernel, and rank of factorizations; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; CP-01 Polynomial identity and more-roots-than-degree principle; RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-01 Pigeonhole principle and extremal element selection

8.16 2009

Problem	Main theory	Supporting detailed theory nodes
2009-D1-P1	CMB-09 Explicit construction and counterexample design	RA-02 Intermediate value, connectedness, density, and continuity transfer; RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; RA-10 Riemann sums and sum-integral asymptotics; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities
2009-D1-P2	LA-01 Matrix products, inverses, and one-sided inverse identities	LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; LA-02 Determinant evaluation by row/column operations and triangular forms; LA-08 Nilpotent, idempotent, involution, and square-zero structures; LA-03 Block matrices, Schur complements, and determinant identities; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces
2009-D1-P3	CMB-02 Double counting and degree-sum methods	CMB-03 Graph encoding, cycles, and bipartite representations; CMB-01 Pigeonhole principle and extremal element selection; CP-01 Polynomial identity and more-roots-than-degree principle; LA-04 Rank-nullity, image/kernel, and rank of factorizations; RA-02 Intermediate value, connectedness, density, and continuity transfer; LA-10 Invariant subspaces and adapted bases
2009-D1-P4	CP-06 Cauchy formula and coefficient extraction on the unit circle	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; CP-09 Fejer kernels and positive trigonometric kernels; CP-01 Polynomial identity and more-roots-than-degree principle; LA-04 Rank-nullity, image/kernel, and rank of factorizations; RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-10 Invariant subspaces and adapted bases
2009-D1-P5	GEO-06 Simplex, barycentric coordinates, and circumcenters	LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-03 Block matrices, Schur complements, and determinant identities; LA-02 Determinant evaluation by row/column operations and triangular forms; RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-03 Graph encoding, cycles, and bipartite representations; LA-10 Invariant subspaces and adapted bases
2009-D2-P1	GEO-07 Area, volume, projection, and ellipsoid computation	CP-01 Polynomial identity and more-roots-than-degree principle; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; GEO-03 Projections and orthogonal decompositions; GEO-06 Simplex, barycentric coordinates, and circumcenters; RA-08 Integral inequalities, integration by parts, and weighted estimates; CP-02 Factorization, irreducibility, and Gauss-style integer polynomial arguments
2009-D2-P2	RA-06 Auxiliary functions, integrating factors, and differential inequalities	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-04 Rolle, mean-value theorem, and divided differences; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-08 Integral inequalities, integration by parts, and weighted estimates; FEQ-01 Substitution of special values and normalization; FEQ-02 Injectivity and monotonicity forced by a functional equation
2009-D2-P3	LA-06 Commutators and spectral obstructions	LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; LA-04 Rank-nullity, image/kernel, and rank of factorizations; CP-01 Polynomial identity and more-roots-than-degree principle; RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-03 Block matrices, Schur complements, and determinant identities

Problem	Main theory	Supporting detailed theory nodes
2009-D2-P4	ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools	CP-01 Polynomial identity and more-roots-than-degree principle; CMB-08 Permutations, inversions, and permutation statistics; ANT-01 Modular arithmetic and divisibility; CMB-03 Graph encoding, cycles, and bipartite representations; ANT-11 Rings, idempotents, nil rings, and noncommutativity; CP-03 Vieta, logarithmic derivatives, and real-root inequalities
2009-D2-P5	LA-04 Rank-nullity, image/kernel, and rank of factorizations	CMB-01 Pigeonhole principle and extremal element selection; LA-03 Block matrices, Schur complements, and determinant identities; LA-02 Determinant evaluation by row/-column operations and triangular forms; CP-01 Polynomial identity and more-roots-than-degree principle; RA-08 Integral inequalities, integration by parts, and weighted estimates; ANT-07 Group basics: orders, cyclic subgroups, and generators

8.17 2010

Problem	Main theory	Supporting detailed theory nodes
2010-D1-P1	RA-04 Rolle, mean-value theorem, and divided differences	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-03 Monotone functions, fixed points, and iteration on intervals; GEO-02 Euclidean distances and inner-product identities
2010-D1-P2	RA-08 Integral inequalities, integration by parts, and weighted estimates	RA-10 Riemann sums and sum-integral asymptotics; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-09 Fubini/Tonelli and change of variables; RA-13 L'Hospital, ratio limits, and growth comparisons; LA-03 Block matrices, Schur complements, and determinant identities
2010-D1-P3	RA-10 Riemann sums and sum-integral asymptotics	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-13 L'Hospital, ratio limits, and growth comparisons; FEQ-03 Fixed points, iterates, and orbit closure; LA-03 Block matrices, Schur complements, and determinant identities; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping
2010-D1-P4	ANT-01 Modular arithmetic and divisibility	ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; ANT-02 Chinese remainder theorem and complete residue systems
2010-D1-P5	LA-03 Block matrices, Schur complements, and determinant identities	LA-02 Determinant evaluation by row/column operations and triangular forms; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; GEO-02 Euclidean distances and inner-product identities; LA-04 Rank-nullity, image/kernel, and rank of factorizations; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces

Problem	Main theory	Supporting detailed theory nodes
2010-D2-P1	RA-08 Integral inequalities, integration by parts, and weighted estimates	RA-10 Riemann sums and sum-integral asymptotics; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-09 Fubini/Tonelli and change of variables
2010-D2-P2	RA-08 Integral inequalities, integration by parts, and weighted estimates	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-17 Sequence recurrences, monotone bounds, and generating functions; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; ANT-10 Automorphisms and semidirect products; RA-09 Fubini/Tonelli and change of variables; ANT-07 Group basics: orders, cyclic subgroups, and generators
2010-D2-P3	ANT-07 Group basics: orders, cyclic subgroups, and generators	RA-08 Integral inequalities, integration by parts, and weighted estimates; ANT-11 Rings, idempotents, nil rings, and noncommutativity; ANT-10 Automorphisms and semidirect products; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; ANT-09 Normal subgroups, quotient groups, and group actions
2010-D2-P4	LA-04 Rank-nullity, image/kernel, and rank of factorizations	ANT-01 Modular arithmetic and divisibility; LA-03 Block matrices, Schur complements, and determinant identities; LA-10 Invariant subspaces and adapted bases; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; RA-07 Convexity, concavity, Jensen, and tangent/-chord inequalities; LA-02 Determinant evaluation by row/-column operations and triangular forms
2010-D2-P5	CP-01 Polynomial identity and more-roots-than-degree principle	ANT-01 Modular arithmetic and divisibility; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-10 Riemann sums and sum-integral asymptotics; CP-03 Vieta, logarithmic derivatives, and real-root inequalities; CP-05 Roots of unity, cyclotomic filters, and discrete Fourier tricks

8.18 2011

Problem	Main theory	Supporting detailed theory nodes
2011-D1-P1	RA-01 Compactness, extreme values, and closed-bounded reduction	RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-02 Intermediate value, connectedness, density, and continuity transfer; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-10 Riemann sums and sum-integral asymptotics; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; CMB-03 Graph encoding, cycles, and bipartite representations
2011-D1-P2	LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton	LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; CP-01 Polynomial identity and more-roots-than-degree principle; LA-03 Block matrices, Schur complements, and determinant identities; LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-02 Determinant evaluation by row/column operations and triangular forms; CMB-09 Explicit construction and counterexample design

Problem	Main theory	Supporting detailed theory nodes
2011-D1-P3	ANT-01 Modular arithmetic and divisibility	CP-01 Polynomial identity and more-roots-than-degree principle; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; ANT-07 Group basics: orders, cyclic subgroups, and generators; ANT-04 Integer polynomials, finite differences, and value divisibility; ANT-10 Automorphisms and semidirect products; ANT-11 Rings, idempotents, nil rings, and noncommutativity
2011-D1-P4	CMB-01 Pigeonhole principle and extremal element selection	CMB-02 Double counting and degree-sum methods; CMB-03 Graph encoding, cycles, and bipartite representations; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; CMB-12 Coloring, partitions, and parity; RA-03 Monotone functions, fixed points, and iteration on intervals; CMB-04 Set systems, intersecting families, and induction on sizes
2011-D1-P5	LA-04 Rank-nullity, image/kernel, and rank of factorizations	CP-01 Polynomial identity and more-roots-than-degree principle; RA-08 Integral inequalities, integration by parts, and weighted estimates; ANT-01 Modular arithmetic and divisibility; LA-10 Invariant subspaces and adapted bases; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; LA-03 Block matrices, Schur complements, and determinant identities
2011-D2-P1	RA-08 Integral inequalities, integration by parts, and weighted estimates	RA-10 Riemann sums and sum-integral asymptotics; RA-17 Sequence recurrences, monotone bounds, and generating functions; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-09 Fubini/Tonelli and change of variables
2011-D2-P2	CP-01 Polynomial identity and more-roots-than-degree principle	RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; LA-03 Block matrices, Schur complements, and determinant identities; ANT-07 Group basics: orders, cyclic subgroups, and generators; ANT-11 Rings, idempotents, nil rings, and noncommutativity
2011-D2-P3	RA-08 Integral inequalities, integration by parts, and weighted estimates	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-09 Fubini/Tonelli and change of variables; LA-03 Block matrices, Schur complements, and determinant identities; RA-13 L'Hospital, ratio limits, and growth comparisons; GEO-04 Convex bodies, supporting geometry, and John-ellipse ideas
2011-D2-P4	CP-01 Polynomial identity and more-roots-than-degree principle	ANT-01 Modular arithmetic and divisibility; RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-04 Rank-nullity, image/kernel, and rank of factorizations; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; ANT-07 Group basics: orders, cyclic subgroups, and generators; ANT-10 Automorphisms and semidirect products
2011-D2-P5	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	CMB-03 Graph encoding, cycles, and bipartite representations; RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-10 Invariant subspaces and adapted bases; GEO-07 Area, volume, projection, and ellipsoid computation; GEO-03 Projections and orthogonal decompositions

8.19 2012

Problem	Main theory	Supporting detailed theory nodes
2012-D1-P1	CMB-01 Pigeonhole principle and extremal element selection	CMB-02 Double counting and degree-sum methods; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; ANT-01 Modular arithmetic and divisibility; RA-09 Fubini/Tonelli and change of variables; ANT-06 GCD, valuations, prime powers, and coprimality
2012-D1-P2	LA-04 Rank-nullity, image/kernel, and rank of factorizations	LA-02 Determinant evaluation by row/column operations and triangular forms; LA-03 Block matrices, Schur complements, and determinant identities; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; ANT-01 Modular arithmetic and divisibility; LA-10 Invariant subspaces and adapted bases; ANT-06 GCD, valuations, prime powers, and coprimality
2012-D1-P3	ANT-07 Group basics: orders, cyclic subgroups, and generators	ANT-11 Rings, idempotents, nil rings, and noncommutativity; ANT-10 Automorphisms and semidirect products; ANT-01 Modular arithmetic and divisibility; ANT-09 Normal subgroups, quotient groups, and group actions; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; ANT-06 GCD, valuations, prime powers, and coprimality
2012-D1-P4	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	RA-04 Rolle, mean-value theorem, and divided differences; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-02 Intermediate value, connectedness, density, and continuity transfer; LA-04 Rank-nullity, image/kernel, and rank of factorizations; RA-06 Auxiliary functions, integrating factors, and differential inequalities
2012-D1-P5	CP-01 Polynomial identity and more-roots-than-degree principle	ANT-11 Rings, idempotents, nil rings, and noncommutativity; CP-02 Factorization, irreducibility, and Gauss-style integer polynomial arguments; CP-03 Vieta, logarithmic derivatives, and real-root inequalities; ANT-07 Group basics: orders, cyclic subgroups, and generators; ANT-10 Automorphisms and semidirect products; ANT-04 Integer polynomials, finite differences, and value divisibility
2012-D2-P1	CP-01 Polynomial identity and more-roots-than-degree principle	ANT-01 Modular arithmetic and divisibility; CP-03 Vieta, logarithmic derivatives, and real-root inequalities; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; CP-02 Factorization, irreducibility, and Gauss-style integer polynomial arguments; ANT-04 Integer polynomials, finite differences, and value divisibility; ANT-02 Chinese remainder theorem and complete residue systems
2012-D2-P2	RA-08 Integral inequalities, integration by parts, and weighted estimates	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-17 Sequence recurrences, monotone bounds, and generating functions; RA-09 Fubini/Tonelli and change of variables; CMB-06 Generating functions for counting; RA-06 Auxiliary functions, integrating factors, and differential inequalities
2012-D2-P3	ANT-01 Modular arithmetic and divisibility	CMB-03 Graph encoding, cycles, and bipartite representations; CMB-12 Coloring, partitions, and parity; CMB-04 Set systems, intersecting families, and induction on sizes; ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators; ANT-11 Rings, idempotents, nil rings, and noncommutativity

Problem	Main theory	Supporting detailed theory nodes
2012-D2-P4	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	ANT-01 Modular arithmetic and divisibility; RA-08 Integral inequalities, integration by parts, and weighted estimates; ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators; LA-03 Block matrices, Schur complements, and determinant identities; ANT-02 Chinese remainder theorem and complete residue systems
2012-D2-P5	ANT-07 Group basics: orders, cyclic subgroups, and generators	ANT-11 Rings, idempotents, nil rings, and noncommutativity; ANT-10 Automorphisms and semidirect products; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; ANT-01 Modular arithmetic and divisibility; ANT-09 Normal subgroups, quotient groups, and group actions; CMB-03 Graph encoding, cycles, and bipartite representations

8.20 2013

Problem	Main theory	Supporting detailed theory nodes
2013-D1-P1	LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton	LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-02 Determinant evaluation by row/column operations and triangular forms; LA-01 Matrix products, inverses, and one-sided inverse identities; LA-03 Block matrices, Schur complements, and determinant identities; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; LA-06 Commutators and spectral obstructions
2013-D1-P2	RA-04 Rolle, mean-value theorem, and divided differences	RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; LA-04 Rank-nullity, image/kernel, and rank of factorizations; RA-02 Intermediate value, connectedness, density, and continuity transfer; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-10 Riemann sums and sum-integral asymptotics
2013-D1-P3	CMB-01 Pigeonhole principle and extremal element selection	CMB-02 Double counting and degree-sum methods; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-01 Compactness, extreme values, and closed-bounded reduction; RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-03 Block matrices, Schur complements, and determinant identities; CMB-03 Graph encoding, cycles, and bipartite representations
2013-D1-P4	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-03 Block matrices, Schur complements, and determinant identities; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-10 Riemann sums and sum-integral asymptotics; RA-01 Compactness, extreme values, and closed-bounded reduction; RA-02 Intermediate value, connectedness, density, and continuity transfer
2013-D1-P5	ANT-01 Modular arithmetic and divisibility	ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; CMB-09 Explicit construction and counterexample design; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; ANT-02 Chinese remainder theorem and complete residue systems; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-09 Fubini/Tonelli and change of variables

Problem	Main theory	Supporting detailed theory nodes
2013-D2-P1	CP-01 Polynomial identity and more-roots-than-degree principle	CP-05 Roots of unity, cyclotomic filters, and discrete Fourier tricks; CP-04 Unit-circle roots and conjugation identities; CP-03 Vieta, logarithmic derivatives, and real-root inequalities; CP-13 Polynomial functional identities and algebraic composition; CP-02 Factorization, irreducibility, and Gauss-style integer polynomial arguments; CP-06 Cauchy formula and coefficient extraction on the unit circle
2013-D2-P2	ANT-01 Modular arithmetic and divisibility	ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; ANT-02 Chinese remainder theorem and complete residue systems; ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators; ANT-11 Rings, idempotents, nil rings, and noncommutativity; CMB-12 Coloring, partitions, and parity
2013-D2-P3	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	RA-08 Integral inequalities, integration by parts, and weighted estimates; GEO-02 Euclidean distances and inner-product identities; GEO-09 Hilbert-space equidistant sets and orthonormal translations; LA-15 Gram matrices and rational linear dependence; LA-03 Block matrices, Schur complements, and determinant identities; RA-06 Auxiliary functions, integrating factors, and differential inequalities
2013-D2-P4	CMB-09 Explicit construction and counterexample design	RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; ANT-01 Modular arithmetic and divisibility; RA-09 Fubini/Tonelli and change of variables; CMB-03 Graph encoding, cycles, and bipartite representations; CMB-12 Coloring, partitions, and parity
2013-D2-P5	CMB-01 Pigeonhole principle and extremal element selection	CMB-02 Double counting and degree-sum methods; LA-03 Block matrices, Schur complements, and determinant identities; CMB-03 Graph encoding, cycles, and bipartite representations; CMB-08 Permutations, inversions, and permutation statistics; CMB-04 Set systems, intersecting families, and induction on sizes; CMB-05 Recurrences and state partitions

8.21 2014

Problem	Main theory	Supporting detailed theory nodes
2014-D1-P1	LA-02 Determinant evaluation by row/column operations and triangular forms	LA-03 Block matrices, Schur complements, and determinant identities; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-06 Commutators and spectral obstructions; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; LA-01 Matrix products, inverses, and one-sided inverse identities
2014-D1-P2	RA-10 Riemann sums and sum-integral asymptotics	RA-13 L'Hospital, ratio limits, and growth comparisons; RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-03 Block matrices, Schur complements, and determinant identities; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-01 Compactness, extreme values, and closed-bounded reduction

Problem	Main theory	Supporting detailed theory nodes
2014-D1-P3	CP-01 Polynomial identity and more-roots-than-degree principle	CP-03 Vieta, logarithmic derivatives, and real-root inequalities; ANT-04 Integer polynomials, finite differences, and value divisibility; CP-05 Roots of unity, cyclotomic filters, and discrete Fourier tricks; CP-06 Cauchy formula and coefficient extraction on the unit circle; CP-11 Stable polynomials and coefficient inequalities; CP-13 Polynomial functional identities and algebraic composition
2014-D1-P4	ANT-01 Modular arithmetic and divisibility	ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; RA-08 Integral inequalities, integration by parts, and weighted estimates; ANT-02 Chinese remainder theorem and complete residue systems; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-09 Fubini/Tonelli and change of variables; RA-10 Riemann sums and sum-integral asymptotics
2014-D1-P5	GEO-07 Area, volume, projection, and ellipsoid computation	GEO-03 Projections and orthogonal decompositions; GEO-06 Simplex, barycentric coordinates, and circumcenters; GEO-01 Coordinate and affine normalization; GEO-02 Euclidean distances and inner-product identities; GEO-04 Convex bodies, supporting geometry, and John-ellipse ideas; GEO-05 Conics: parabolas, ellipses, foci, and directrices
2014-D2-P1	ANT-01 Modular arithmetic and divisibility	ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators; ANT-02 Chinese remainder theorem and complete residue systems; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; ANT-11 Rings, idempotents, nil rings, and noncommutativity; ANT-04 Integer polynomials, finite differences, and value divisibility
2014-D2-P2	LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-03 Block matrices, Schur complements, and determinant identities; LA-04 Rank-nullity, image/kernel, and rank of factorizations; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-10 Riemann sums and sum-integral asymptotics
2014-D2-P3	LA-03 Block matrices, Schur complements, and determinant identities	RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-10 Riemann sums and sum-integral asymptotics; RA-18 Trigonometric and hyperbolic identities with monotonicity; RA-01 Compactness, extreme values, and closed-bounded reduction; RA-02 Intermediate value, connectedness, density, and continuity transfer
2014-D2-P4	CMB-03 Graph encoding, cycles, and bipartite representations	CMB-12 Coloring, partitions, and parity; CMB-04 Set systems, intersecting families, and induction on sizes; GEO-01 Coordinate and affine normalization; GEO-02 Euclidean distances and inner-product identities; GEO-04 Convex bodies, supporting geometry, and John-ellipse ideas; GEO-07 Area, volume, projection, and ellipsoid computation
2014-D2-P5	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-10 Riemann sums and sum-integral asymptotics; RA-06 Auxiliary functions, integrating factors, and differential inequalities; GEO-04 Convex bodies, supporting geometry, and John-ellipse ideas; LA-03 Block matrices, Schur complements, and determinant identities; RA-01 Compactness, extreme values, and closed-bounded reduction

8.22 2015

Problem	Main theory	Supporting detailed theory nodes
2015-D1-P1	LA-02 Determinant evaluation by row/column operations and triangular forms	LA-03 Block matrices, Schur complements, and determinant identities; LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; CP-04 Unit-circle roots and conjugation identities
2015-D1-P2	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	CMB-09 Explicit construction and counterexample design; ANT-01 Modular arithmetic and divisibility; RA-08 Integral inequalities, integration by parts, and weighted estimates; ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators; LA-03 Block matrices, Schur complements, and determinant identities
2015-D1-P3	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-03 Block matrices, Schur complements, and determinant identities; GEO-04 Convex bodies, supporting geometry, and John-ellipse ideas; FEQ-01 Substitution of special values and normalization; FEQ-02 Injectivity and monotonicity forced by a functional equation; FEQ-03 Fixed points, iterates, and orbit closure
2015-D1-P4	ANT-01 Modular arithmetic and divisibility	ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators; GEO-04 Convex bodies, supporting geometry, and John-ellipse ideas; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-18 Trigonometric and hyperbolic identities with monotonicity; ANT-02 Chinese remainder theorem and complete residue systems
2015-D1-P5	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	LA-04 Rank-nullity, image/kernel, and rank of factorizations; RA-17 Sequence recurrences, monotone bounds, and generating functions; LA-10 Invariant subspaces and adapted bases; GEO-03 Projections and orthogonal decompositions; RA-08 Integral inequalities, integration by parts, and weighted estimates; GEO-07 Area, volume, projection, and ellipsoid computation
2015-D2-P6	LA-03 Block matrices, Schur complements, and determinant identities	RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-10 Riemann sums and sum-integral asymptotics; RA-01 Compactness, extreme values, and closed-bounded reduction; RA-02 Intermediate value, connectedness, density, and continuity transfer; RA-03 Monotone functions, fixed points, and iteration on intervals
2015-D2-P7	RA-10 Riemann sums and sum-integral asymptotics	RA-13 L'Hospital, ratio limits, and growth comparisons; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-06 Auxiliary functions, integrating factors, and differential inequalities; LA-03 Block matrices, Schur complements, and determinant identities; RA-01 Compactness, extreme values, and closed-bounded reduction
2015-D2-P8	CMB-01 Pigeonhole principle and extremal element selection	CMB-02 Double counting and degree-sum methods; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-09 Fubini/Tonelli and change of variables; CMB-13 Words, equivalence relations, and rewriting systems; CMB-06 Generating functions for counting

Problem	Main theory	Supporting detailed theory nodes
2015-D2-P9	LA-04 Rank-nullity, image/kernel, and rank of factorizations	LA-10 Invariant subspaces and adapted bases; LA-03 Block matrices, Schur complements, and determinant identities; GEO-02 Euclidean distances and inner-product identities; LA-11 Linear functionals, kernels, and dual span criteria; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; RA-01 Compactness, extreme values, and closed-bounded reduction
2015-D2-P10	CP-01 Polynomial identity and more-roots-than-degree principle	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; ANT-04 Integer polynomials, finite differences, and value divisibility; ANT-01 Modular arithmetic and divisibility; RA-08 Integral inequalities, integration by parts, and weighted estimates; CP-02 Factorization, irreducibility, and Gauss-style integer polynomial arguments; CP-03 Vieta, logarithmic derivatives, and real-root inequalities

8.23 2016

Problem	Main theory	Supporting detailed theory nodes
2016-D1-P1	CMB-09 Explicit construction and counterexample design	RA-02 Intermediate value, connectedness, density, and continuity transfer; FEQ-01 Substitution of special values and normalization; FEQ-02 Injectivity and monotonicity forced by a functional equation; FEQ-03 Fixed points, iterates, and orbit closure; LA-03 Block matrices, Schur complements, and determinant identities; LA-04 Rank-nullity, image/kernel, and rank of factorizations
2016-D1-P2	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	ANT-01 Modular arithmetic and divisibility; CMB-09 Explicit construction and counterexample design; RA-08 Integral inequalities, integration by parts, and weighted estimates; ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators; LA-03 Block matrices, Schur complements, and determinant identities
2016-D1-P3	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	ANT-01 Modular arithmetic and divisibility; RA-08 Integral inequalities, integration by parts, and weighted estimates; ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators; ANT-02 Chinese remainder theorem and complete residue systems; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools
2016-D1-P4	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	CMB-03 Graph encoding, cycles, and bipartite representations; CMB-12 Coloring, partitions, and parity; ANT-01 Modular arithmetic and divisibility; CMB-04 Set systems, intersecting families, and induction on sizes; RA-08 Integral inequalities, integration by parts, and weighted estimates; ANT-06 GCD, valuations, prime powers, and coprimality
2016-D1-P5	ANT-01 Modular arithmetic and divisibility	ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; CMB-02 Double counting and degree-sum methods; CMB-01 Pigeonhole principle and extremal element selection; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; ANT-02 Chinese remainder theorem and complete residue systems; CMB-12 Coloring, partitions, and parity

Problem	Main theory	Supporting detailed theory nodes
2016-D2-P6	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-03 Block matrices, Schur complements, and determinant identities; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-10 Riemann sums and sum-integral asymptotics; RA-01 Compactness, extreme values, and closed-bounded reduction; RA-02 Intermediate value, connectedness, density, and continuity transfer
2016-D2-P7	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	RA-01 Compactness, extreme values, and closed-bounded reduction; RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-01 Pigeonhole principle and extremal element selection; RA-02 Intermediate value, connectedness, density, and continuity transfer; LA-03 Block matrices, Schur complements, and determinant identities; RA-06 Auxiliary functions, integrating factors, and differential inequalities
2016-D2-P8	ANT-01 Modular arithmetic and divisibility	ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; ANT-07 Group basics: orders, cyclic subgroups, and generators; ANT-11 Rings, idempotents, nil rings, and noncommutativity; ANT-10 Automorphisms and semidirect products; ANT-02 Chinese remainder theorem and complete residue systems; ANT-09 Normal subgroups, quotient groups, and group actions
2016-D2-P9	CMB-01 Pigeonhole principle and extremal element selection	CMB-02 Double counting and degree-sum methods; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; ANT-01 Modular arithmetic and divisibility; RA-08 Integral inequalities, integration by parts, and weighted estimates; ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators
2016-D2-P10	LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; LA-04 Rank-nullity, image/kernel, and rank of factorizations; RA-08 Integral inequalities, integration by parts, and weighted estimates; CP-04 Unit-circle roots and conjugation identities; CP-05 Roots of unity, cyclotomic filters, and discrete Fourier tricks; LA-02 Determinant evaluation by row/column operations and triangular forms

8.24 2017

Problem	Main theory	Supporting detailed theory nodes
2017-D1-P1	LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton	CP-04 Unit-circle roots and conjugation identities; CP-05 Roots of unity, cyclotomic filters, and discrete Fourier tricks; LA-02 Determinant evaluation by row/column operations and triangular forms; LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-03 Block matrices, Schur complements, and determinant identities; CP-01 Polynomial identity and more-roots-than-degree principle
2017-D1-P2	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-10 Riemann sums and sum-integral asymptotics; RA-18 Trigonometric and hyperbolic identities with monotonicity; LA-03 Block matrices, Schur complements, and determinant identities; LA-04 Rank-nullity, image/kernel, and rank of factorizations

Problem	Main theory	Supporting detailed theory nodes
2017-D1-P3	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	ANT-01 Modular arithmetic and divisibility; CMB-03 Graph encoding, cycles, and bipartite representations; CMB-12 Coloring, partitions, and parity; RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-04 Set systems, intersecting families, and induction on sizes; ANT-06 GCD, valuations, prime powers, and coprimality
2017-D1-P4	CMB-01 Pigeonhole principle and extremal element selection	ANT-07 Group basics: orders, cyclic subgroups, and generators; CMB-02 Double counting and degree-sum methods; ANT-11 Rings, idempotents, nil rings, and noncommutativity; ANT-10 Automorphisms and semidirect products; ANT-09 Normal subgroups, quotient groups, and group actions; ANT-01 Modular arithmetic and divisibility
2017-D1-P5	CP-01 Polynomial identity and more-roots-than-degree principle	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; CP-03 Vieta, logarithmic derivatives, and real-root inequalities; CP-05 Roots of unity, cyclotomic filters, and discrete Fourier tricks; RA-04 Rolle, mean-value theorem, and divided differences; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; CP-02 Factorization, irreducibility, and Gauss-style integer polynomial arguments
2017-D2-P6	RA-10 Riemann sums and sum-integral asymptotics	RA-13 L'Hospital, ratio limits, and growth comparisons; RA-02 Intermediate value, connectedness, density, and continuity transfer; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-06 Auxiliary functions, integrating factors, and differential inequalities; LA-03 Block matrices, Schur complements, and determinant identities
2017-D2-P7	CP-01 Polynomial identity and more-roots-than-degree principle	CP-03 Vieta, logarithmic derivatives, and real-root inequalities; CP-02 Factorization, irreducibility, and Gauss-style integer polynomial arguments; ANT-04 Integer polynomials, finite differences, and value divisibility; CP-13 Polynomial functional identities and algebraic composition; CP-05 Roots of unity, cyclotomic filters, and discrete Fourier tricks; CP-06 Cauchy formula and coefficient extraction on the unit circle
2017-D2-P8	LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton	RA-17 Sequence recurrences, monotone bounds, and generating functions; ANT-01 Modular arithmetic and divisibility; CMB-06 Generating functions for counting; LA-03 Block matrices, Schur complements, and determinant identities; ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators
2017-D2-P9	RA-10 Riemann sums and sum-integral asymptotics	RA-17 Sequence recurrences, monotone bounds, and generating functions; RA-13 L'Hospital, ratio limits, and growth comparisons; CMB-06 Generating functions for counting; LA-04 Rank-nullity, image/kernel, and rank of factorizations; RA-02 Intermediate value, connectedness, density, and continuity transfer; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping
2017-D2-P10	GEO-03 Projections and orthogonal decompositions	GEO-07 Area, volume, projection, and ellipsoid computation; ANT-01 Modular arithmetic and divisibility; GEO-06 Simplex, barycentric coordinates, and circumcenters; ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators; ANT-02 Chinese remainder theorem and complete residue systems

8.25 2018

Problem	Main theory	Supporting detailed theory nodes
2018-D1-P1	RA-08 Integral inequalities, integration by parts, and weighted estimates	RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-09 Fubini/Tonelli and change of variables; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-10 Riemann sums and sum-integral asymptotics; LA-03 Block matrices, Schur complements, and determinant identities; RA-01 Compactness, extreme values, and closed-bounded reduction
2018-D1-P2	ANT-07 Group basics: orders, cyclic subgroups, and generators	ANT-11 Rings, idempotents, nil rings, and noncommutativity; CMB-09 Explicit construction and counterexample design; ANT-10 Automorphisms and semidirect products; ANT-09 Normal subgroups, quotient groups, and group actions; ANT-01 Modular arithmetic and divisibility; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools
2018-D1-P3	LA-02 Determinant evaluation by row/column operations and triangular forms	LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; LA-03 Block matrices, Schur complements, and determinant identities; LA-08 Nilpotent, idempotent, involution, and square-zero structures; ANT-05 Rational roots, Vieta, and rationality constraints; LA-01 Matrix products, inverses, and one-sided inverse identities
2018-D1-P4	LA-03 Block matrices, Schur complements, and determinant identities	LA-04 Rank-nullity, image/kernel, and rank of factorizations; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-10 Riemann sums and sum-integral asymptotics; RA-01 Compactness, extreme values, and closed-bounded reduction; RA-02 Intermediate value, connectedness, density, and continuity transfer
2018-D1-P5	ANT-01 Modular arithmetic and divisibility	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; ANT-02 Chinese remainder theorem and complete residue systems; GEO-03 Projections and orthogonal decompositions; GEO-07 Area, volume, projection, and ellipsoid computation; RA-08 Integral inequalities, integration by parts, and weighted estimates
2018-D2-P6	ANT-01 Modular arithmetic and divisibility	GEO-09 Hilbert-space equidistant sets and orthonormal translations; LA-15 Gram matrices and rational linear dependence; GEO-02 Euclidean distances and inner-product identities; ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators; ANT-02 Chinese remainder theorem and complete residue systems
2018-D2-P7	RA-08 Integral inequalities, integration by parts, and weighted estimates	RA-17 Sequence recurrences, monotone bounds, and generating functions; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-09 Fubini/-Tonelli and change of variables; CMB-06 Generating functions for counting; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities
2018-D2-P8	CMB-01 Pigeonhole principle and extremal element selection	CMB-02 Double counting and degree-sum methods; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; ANT-01 Modular arithmetic and divisibility; RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-03 Block matrices, Schur complements, and determinant identities; ANT-06 GCD, valuations, prime powers, and coprimality

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Problem	Main theory	Supporting detailed theory nodes
2018-D2-P9	CP-01 Polynomial identity and more-roots-than-degree principle	CP-02 Factorization, irreducibility, and Gauss-style integer polynomial arguments; CP-03 Vieta, logarithmic derivatives, and real-root inequalities; CP-05 Roots of unity, cyclotomic filters, and discrete Fourier tricks; CP-04 Unit-circle roots and conjugation identities; ANT-04 Integer polynomials, finite differences, and value divisibility; CP-13 Polynomial functional identities and algebraic composition
2018-D2-P10	RA-10 Riemann sums and sum-integral asymptotics	RA-13 L'Hospital, ratio limits, and growth comparisons; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-06 Auxiliary functions, integrating factors, and differential inequalities; LA-03 Block matrices, Schur complements, and determinant identities; RA-01 Compactness, extreme values, and closed-bounded reduction

8.26 2019

Problem	Main theory	Supporting detailed theory nodes
2019-D1-P1	LA-03 Block matrices, Schur complements, and determinant identities	RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-10 Riemann sums and sum-integral asymptotics; RA-01 Compactness, extreme values, and closed-bounded reduction; RA-02 Intermediate value, connectedness, density, and continuity transfer; RA-03 Monotone functions, fixed points, and iteration on intervals
2019-D1-P2	ANT-01 Modular arithmetic and divisibility	ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators; LA-03 Block matrices, Schur complements, and determinant identities; ANT-02 Chinese remainder theorem and complete residue systems; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; ANT-11 Rings, idempotents, nil rings, and noncommutativity
2019-D1-P3	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-10 Riemann sums and sum-integral asymptotics; LA-03 Block matrices, Schur complements, and determinant identities; LA-04 Rank-nullity, image/kernel, and rank of factorizations; RA-01 Compactness, extreme values, and closed-bounded reduction
2019-D1-P4	RA-17 Sequence recurrences, monotone bounds, and generating functions	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; ANT-01 Modular arithmetic and divisibility; RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-06 Generating functions for counting; ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators
2019-D1-P5	LA-02 Determinant evaluation by row/column operations and triangular forms	LA-03 Block matrices, Schur complements, and determinant identities; LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; ANT-01 Modular arithmetic and divisibility; ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators

Problem	Main theory	Supporting detailed theory nodes
2019-D2-P6	RA-08 Integral inequalities, integration by parts, and weighted estimates	CMB-09 Explicit construction and counterexample design; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; RA-09 Fubini/Tonelli and change of variables; RA-02 Intermediate value, connectedness, density, and continuity transfer; FEQ-01 Substitution of special values and normalization; FEQ-02 Injectivity and monotonicity forced by a functional equation
2019-D2-P7	ANT-01 Modular arithmetic and divisibility	ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools; ANT-02 Chinese remainder theorem and complete residue systems; RA-09 Fubini/Tonelli and change of variables; CMB-03 Graph encoding, cycles, and bipartite representations
2019-D2-P8	CMB-01 Pigeonhole principle and extremal element selection	CMB-02 Double counting and degree-sum methods; CMB-03 Graph encoding, cycles, and bipartite representations; CMB-12 Coloring, partitions, and parity; RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-04 Set systems, intersecting families, and induction on sizes; RA-11 Series convergence, rearrangements, and alternating/Dirichlet tools
2019-D2-P9	ANT-01 Modular arithmetic and divisibility	LA-04 Rank-nullity, image/kernel, and rank of factorizations; ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators; LA-01 Matrix products, inverses, and one-sided inverse identities; LA-02 Determinant evaluation by row/column operations and triangular forms; LA-03 Block matrices, Schur complements, and determinant identities
2019-D2-P10	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	CMB-03 Graph encoding, cycles, and bipartite representations; RA-20 Measure and probability-integral estimates; LA-04 Rank-nullity, image/kernel, and rank of factorizations; CMB-12 Coloring, partitions, and parity; GEO-03 Projections and orthogonal decompositions; GEO-07 Area, volume, projection, and ellipsoid computation

8.27 2020

Problem	Main theory	Supporting detailed theory nodes
2020-D1-P1	CMB-01 Pigeonhole principle and extremal element selection	CMB-02 Double counting and degree-sum methods; CMB-09 Explicit construction and counterexample design; ANT-01 Modular arithmetic and divisibility; ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators; CMB-13 Words, equivalence relations, and rewriting systems
2020-D1-P2	LA-04 Rank-nullity, image/kernel, and rank of factorizations	LA-03 Block matrices, Schur complements, and determinant identities; LA-02 Determinant evaluation by row/column operations and triangular forms; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; LA-06 Commutators and spectral obstructions
2020-D1-P3	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	CMB-03 Graph encoding, cycles, and bipartite representations; CMB-12 Coloring, partitions, and parity; ANT-01 Modular arithmetic and divisibility; RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-04 Set systems, intersecting families, and induction on sizes; ANT-06 GCD, valuations, prime powers, and coprimality

Problem	Main theory	Supporting detailed theory nodes
2020-D1-P4	CP-01 Polynomial identity and more-roots-than-degree principle	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; CP-02 Factorization, irreducibility, and Gauss-style integer polynomial arguments; CP-03 Vieta, logarithmic derivatives, and real-root inequalities; ANT-04 Integer polynomials, finite differences, and value divisibility; LA-03 Block matrices, Schur complements, and determinant identities
2020-D2-P5	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-03 Block matrices, Schur complements, and determinant identities; RA-02 Intermediate value, connectedness, density, and continuity transfer; LA-04 Rank-nullity, image/kernel, and rank of factorizations; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-10 Riemann sums and sum-integral asymptotics
2020-D2-P6	ANT-01 Modular arithmetic and divisibility	ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; ANT-02 Chinese remainder theorem and complete residue systems; ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators; ANT-11 Rings, idempotents, nil rings, and noncommutativity; LA-03 Block matrices, Schur complements, and determinant identities
2020-D2-P7	ANT-07 Group basics: orders, cyclic subgroups, and generators	CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; ANT-11 Rings, idempotents, nil rings, and noncommutativity; ANT-10 Automorphisms and semidirect products; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; ANT-09 Normal subgroups, quotient groups, and group actions
2020-D2-P8	RA-10 Riemann sums and sum-integral asymptotics	RA-13 L'Hospital, ratio limits, and growth comparisons; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-06 Auxiliary functions, integrating factors, and differential inequalities; LA-03 Block matrices, Schur complements, and determinant identities; RA-01 Compactness, extreme values, and closed-bounded reduction

8.28 2021

Problem	Main theory	Supporting detailed theory nodes
2021-D1-P1	LA-04 Rank-nullity, image/kernel, and rank of factorizations	LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; LA-02 Determinant evaluation by row/column operations and triangular forms; LA-01 Matrix products, inverses, and one-sided inverse identities; LA-03 Block matrices, Schur complements, and determinant identities; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; LA-06 Commutators and spectral obstructions
2021-D1-P2	CMB-03 Graph encoding, cycles, and bipartite representations	CMB-12 Coloring, partitions, and parity; ANT-01 Modular arithmetic and divisibility; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; CMB-04 Set systems, intersecting families, and induction on sizes; PRB-03 Expectation, linearity, and probabilistic method

Problem	Main theory	Supporting detailed theory nodes
2021-D1-P3	RA-17 Sequence recurrences, monotone bounds, and generating functions	CMB-06 Generating functions for counting; LA-03 Block matrices, Schur complements, and determinant identities; RA-02 Intermediate value, connectedness, density, and continuity transfer; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-10 Riemann sums and sum-integral asymptotics
2021-D1-P4	RA-10 Riemann sums and sum-integral asymptotics	RA-13 L'Hospital, ratio limits, and growth comparisons; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-02 Intermediate value, connectedness, density, and continuity transfer; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-06 Auxiliary functions, integrating factors, and differential inequalities; LA-03 Block matrices, Schur complements, and determinant identities
2021-D2-P5	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	ANT-01 Modular arithmetic and divisibility; RA-08 Integral inequalities, integration by parts, and weighted estimates; ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators; LA-02 Determinant evaluation by row/column operations and triangular forms; LA-03 Block matrices, Schur complements, and determinant identities
2021-D2-P6	ANT-01 Modular arithmetic and divisibility	ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; ANT-07 Group basics: orders, cyclic subgroups, and generators; ANT-11 Rings, idempotents, nil rings, and noncommutativity; ANT-02 Chinese remainder theorem and complete residue systems; ANT-10 Automorphisms and semidirect products; ANT-09 Normal subgroups, quotient groups, and group actions
2021-D2-P7	CP-01 Polynomial identity and more-roots-than-degree principle	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; CP-03 Vieta, logarithmic derivatives, and real-root inequalities; CP-05 Roots of unity, cyclotomic filters, and discrete Fourier tricks; GEO-07 Area, volume, projection, and ellipsoid computation; CP-02 Factorization, irreducibility, and Gauss-style integer polynomial arguments; CP-04 Unit-circle roots and conjugation identities
2021-D2-P8	CMB-01 Pigeonhole principle and extremal element selection	ANT-01 Modular arithmetic and divisibility; CMB-02 Double counting and degree-sum methods; GEO-02 Euclidean distances and inner-product identities; GEO-09 Hilbert-space equidistant sets and orthonormal translations; LA-15 Gram matrices and rational linear dependence; ANT-06 GCD, valuations, prime powers, and coprimality

8.29 2022

Problem	Main theory	Supporting detailed theory nodes
2022-D1-P1	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	RA-20 Measure and probability-integral estimates; RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-04 Rank-nullity, image/kernel, and rank of factorizations; PRB-05 Measure, variation, and differentiability almost everywhere; PRB-02 Generating functions for probability distributions; LA-03 Block matrices, Schur complements, and determinant identities

Problem	Main theory	Supporting detailed theory nodes
2022-D1-P2	LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; ANT-01 Modular arithmetic and divisibility; RA-08 Integral inequalities, integration by parts, and weighted estimates; LA-03 Block matrices, Schur complements, and determinant identities; ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators
2022-D1-P3	ANT-01 Modular arithmetic and divisibility	ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; ANT-02 Chinese remainder theorem and complete residue systems; CMB-09 Explicit construction and counterexample design; ANT-06 GCD, valuations, prime powers, and coprimality
2022-D1-P4	CMB-01 Pigeonhole principle and extremal element selection	CMB-02 Double counting and degree-sum methods; CMB-12 Coloring, partitions, and parity; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; CMB-03 Graph encoding, cycles, and bipartite representations; ANT-01 Modular arithmetic and divisibility; RA-08 Integral inequalities, integration by parts, and weighted estimates
2022-D2-P5	CMB-03 Graph encoding, cycles, and bipartite representations	CMB-12 Coloring, partitions, and parity; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; GEO-03 Projections and orthogonal decompositions; GEO-07 Area, volume, projection, and ellipsoid computation; CMB-04 Set systems, intersecting families, and induction on sizes
2022-D2-P6	ANT-01 Modular arithmetic and divisibility	ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; ANT-02 Chinese remainder theorem and complete residue systems; ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators; ANT-11 Rings, idempotents, nil rings, and noncommutativity; CMB-08 Permutations, inversions, and permutation statistics
2022-D2-P7	LA-04 Rank-nullity, image/kernel, and rank of factorizations	LA-02 Determinant evaluation by row/column operations and triangular forms; LA-08 Nilpotent, idempotent, involution, and square-zero structures; LA-03 Block matrices, Schur complements, and determinant identities; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; LA-06 Commutators and spectral obstructions; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities
2022-D2-P8	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	CMB-03 Graph encoding, cycles, and bipartite representations; CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; CMB-12 Coloring, partitions, and parity; GEO-03 Projections and orthogonal decompositions; GEO-07 Area, volume, projection, and ellipsoid computation

8.30 2023

Problem	Main theory	Supporting detailed theory nodes
2023-D1-P1	RA-04 Rolle, mean-value theorem, and divided differences	RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-02 Intermediate value, connectedness, density, and continuity transfer; LA-03 Block matrices, Schur complements, and determinant identities; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-10 Riemann sums and sum-integral asymptotics

Problem	Main theory	Supporting detailed theory nodes
2023-D1-P2	RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping	RA-04 Rolle, mean-value theorem, and divided differences; LA-04 Rank-nullity, image/kernel, and rank of factorizations; CP-04 Unit-circle roots and conjugation identities; CP-05 Roots of unity, cyclotomic filters, and discrete Fourier tricks; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; LA-02 Determinant evaluation by row/column operations and triangular forms
2023-D1-P3	CP-01 Polynomial identity and more-roots-than-degree principle	CP-03 Vieta, logarithmic derivatives, and real-root inequalities; CP-02 Factorization, irreducibility, and Gauss-style integer polynomial arguments; CP-13 Polynomial functional identities and algebraic composition; ANT-04 Integer polynomials, finite differences, and value divisibility; CP-06 Cauchy formula and coefficient extraction on the unit circle; CP-11 Stable polynomials and coefficient inequalities
2023-D1-P4	ANT-01 Modular arithmetic and divisibility	ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; ANT-02 Chinese remainder theorem and complete residue systems; CMB-12 Coloring, partitions, and parity; CMB-03 Graph encoding, cycles, and bipartite representations; CMB-04 Set systems, intersecting families, and induction on sizes; ANT-06 GCD, valuations, prime powers, and coprimality
2023-D1-P5	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	ANT-01 Modular arithmetic and divisibility; CMB-03 Graph encoding, cycles, and bipartite representations; CMB-12 Coloring, partitions, and parity; RA-08 Integral inequalities, integration by parts, and weighted estimates; CMB-04 Set systems, intersecting families, and induction on sizes; ANT-06 GCD, valuations, prime powers, and coprimality
2023-D2-P6	LA-02 Determinant evaluation by row/column operations and triangular forms	LA-03 Block matrices, Schur complements, and determinant identities; LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; CMB-11 Grid, lattice, and alternating-cycle arguments; LA-01 Matrix products, inverses, and one-sided inverse identities; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces
2023-D2-P7	RA-04 Rolle, mean-value theorem, and divided differences	RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; CMB-03 Graph encoding, cycles, and bipartite representations; CMB-12 Coloring, partitions, and parity; CMB-04 Set systems, intersecting families, and induction on sizes; LA-04 Rank-nullity, image/kernel, and rank of factorizations; RA-02 Intermediate value, connectedness, density, and continuity transfer
2023-D2-P8	CMB-03 Graph encoding, cycles, and bipartite representations	CMB-01 Pigeonhole principle and extremal element selection; CMB-02 Double counting and degree-sum methods; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-02 Intermediate value, connectedness, density, and continuity transfer; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; CMB-12 Coloring, partitions, and parity
2023-D2-P9	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	GEO-07 Area, volume, projection, and ellipsoid computation; LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-10 Invariant subspaces and adapted bases; GEO-03 Projections and orthogonal decompositions; CMB-03 Graph encoding, cycles, and bipartite representations; CMB-12 Coloring, partitions, and parity

Problem	Main theory	Supporting detailed theory nodes
2023-D2-P10	ANT-01 Modular arithmetic and divisibility	ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators; ANT-02 Chinese remainder theorem and complete residue systems; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; ANT-11 Rings, idempotents, nil rings, and noncommutativity; LA-02 Determinant evaluation by row/column operations and triangular forms

8.31 2024

Problem	Main theory	Supporting detailed theory nodes
2024-D1-P1	CP-01 Polynomial identity and more-roots-than-degree principle	CP-03 Vieta, logarithmic derivatives, and real-root inequalities; CP-05 Roots of unity, cyclotomic filters, and discrete Fourier tricks; CP-13 Polynomial functional identities and algebraic composition; LA-02 Determinant evaluation by row/column operations and triangular forms; LA-03 Block matrices, Schur complements, and determinant identities; LA-04 Rank-nullity, image/kernel, and rank of factorizations
2024-D1-P2	RA-10 Riemann sums and sum-integral asymptotics	RA-13 L'Hospital, ratio limits, and growth comparisons; RA-05 Taylor expansion, local asymptotics, and little-o bookkeeping; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-06 Auxiliary functions, integrating factors, and differential inequalities; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; LA-03 Block matrices, Schur complements, and determinant identities
2024-D1-P3	CMB-09 Explicit construction and counterexample design	ANT-01 Modular arithmetic and divisibility; ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators; ANT-02 Chinese remainder theorem and complete residue systems; ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; ANT-11 Rings, idempotents, nil rings, and noncommutativity
2024-D1-P4	ANT-07 Group basics: orders, cyclic subgroups, and generators	ANT-11 Rings, idempotents, nil rings, and noncommutativity; ANT-10 Automorphisms and semidirect products; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; ANT-01 Modular arithmetic and divisibility; ANT-09 Normal subgroups, quotient groups, and group actions; RA-08 Integral inequalities, integration by parts, and weighted estimates
2024-D1-P5	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	GEO-07 Area, volume, projection, and ellipsoid computation; CMB-12 Coloring, partitions, and parity; GEO-03 Projections and orthogonal decompositions; LA-04 Rank-nullity, image/kernel, and rank of factorizations; ANT-01 Modular arithmetic and divisibility; CMB-01 Pigeonhole principle and extremal element selection
2024-D2-P6	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities	RA-08 Integral inequalities, integration by parts, and weighted estimates; FEQ-08 No-regularity functions and order patterns on Q or R; FEQ-01 Substitution of special values and normalization; FEQ-02 Injectivity and monotonicity forced by a functional equation; FEQ-03 Fixed points, iterates, and orbit closure; LA-03 Block matrices, Schur complements, and determinant identities

Problem	Main theory	Supporting detailed theory nodes
2024-D2-P7	LA-02 Determinant evaluation by row/column operations and triangular forms	LA-03 Block matrices, Schur complements, and determinant identities; LA-04 Rank-nullity, image/kernel, and rank of factorizations; LA-05 Trace cyclicity, trace pairing, and zero-trace subspaces; LA-07 Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; CP-04 Unit-circle roots and conjugation identities; CP-05 Roots of unity, cyclotomic filters, and discrete Fourier tricks
2024-D2-P8	RA-10 Riemann sums and sum-integral asymptotics	RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; RA-17 Sequence recurrences, monotone bounds, and generating functions; RA-08 Integral inequalities, integration by parts, and weighted estimates; RA-13 L'Hospital, ratio limits, and growth comparisons; CMB-06 Generating functions for counting; LA-03 Block matrices, Schur complements, and determinant identities
2024-D2-P9	CMB-01 Pigeonhole principle and extremal element selection	CMB-02 Double counting and degree-sum methods; LA-03 Block matrices, Schur complements, and determinant identities; LA-02 Determinant evaluation by row/column operations and triangular forms; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; CMB-09 Explicit construction and counterexample design; ANT-01 Modular arithmetic and divisibility
2024-D2-P10	ANT-01 Modular arithmetic and divisibility	ANT-03 Finite fields, field characteristic, and Fermat/Frobenius tools; RA-07 Convexity, concavity, Jensen, and tangent/chord inequalities; ANT-02 Chinese remainder theorem and complete residue systems; RA-08 Integral inequalities, integration by parts, and weighted estimates; ANT-06 GCD, valuations, prime powers, and coprimality; ANT-07 Group basics: orders, cyclic subgroups, and generators

9 Full IMC Theory Manual: Statements, Properties, and Compact Proofs

This section expands the retained theory index into a compact, self-contained theory manual. Each card corresponds to one theory node already used in the problem-by-problem catalog and in the problem-to-theory matrix. The aim is not to reproduce complete textbook chapters; it is to state exactly the reusable facts that appear in the IMC corpus, give the proof mechanism when it is short enough to be useful in contest conditions, and connect the card back to the occurrence ledger already retained above. **Coverage audit.** The manual contains one theory card for every detailed theory node in the enhanced index: **107 cards**. The retained catalog contains **334 problem entries**; every problem is linked to at least one main theory node and to supporting theory nodes in the matrix above. **Reading rule.** For a first pass, read only the *must know* and *core facts*. For a proof-writing pass, read the compact proof sketches and reproduce them without looking. For problem review, use the occurrence count and examples to jump back to the problem catalog.

9.1 Literature source map

The cards are based on standard references used throughout undergraduate olympiad-level and early graduate mathematical training. The citations below are anchors for the underlying theory; the problem occurrence lists are derived from the uploaded IMC corpus.

Domain	Main literature anchors
Linear algebra / matrix theory	[6], [19], [18]
Real analysis / calculus	[28], [4], [17]
Complex analysis / polynomials	[1], [11], [24]
Algebra / number theory	[15], [5], [20]
Combinatorics / discrete structures	[2], [30], [29], [31]
Geometry / convexity / metric	[7], [26], [9]
Functional equations / dynamics	[22], [14], [28]
Probability / measure / expectation	[16], [8], [27]

9.2 Linear algebra / matrix theory

Domain anchors. [6], [19], [18].

9.2.1 LA-01: Matrix products, inverses, and one-sided inverse identities

Must know	Matrix algebra as algebra of linear maps; inverse rules; one-sided inverses for square matrices.
Occurrence	14 tagged problem(s). Selected anchors: 1994-D1-P1; 1996-D1-P1; 1996-D2-P3; 2002-D2-P1; 2002-D2-P5; 2003-D2-P1; 2004-D2-P1; 2009-D1-P2; 2013-D1-P1; 2014-D1-P1; 2018-D1-P3; 2019-D2-P9; 2021-D1-P1; 2023-D2-P6
Original trap note	Never cancel or reorder matrix factors before proving invertibility; record which factor is square.
Literature	[6], [18]

Core facts and properties.

1. If $A, B \in M_n(F)$ and $AB = I$, then A and B are invertible and $BA = I$.
2. If A is invertible, then $(AB)^{-1} = B^{-1}A^{-1}$ whenever B is invertible, and $(A^{-1})^T = (A^T)^{-1}$.
3. Equations like $AB + A + B = 0$ should be completed to $(A+I)(B+I) = I$, which is an inverse statement rather than a commutativity statement.

Compact proof mechanisms.

1. For $AB = I$, the linear map B is injective and A is surjective; in finite dimension injective/surjective are equivalent, hence both are bijective and $B = A^{-1}$.
2. The inverse of AB reverses order because $(AB)(B^{-1}A^{-1}) = AIA^{-1} = I$ and similarly on the other side.

3. In IMC solutions, first rewrite to a product equal to I; only after that may the order be reversed.

9.2.2 LA-02: Determinant evaluation by row/column operations and triangular forms

Must know	Determinants, elementary row/column operations, triangular matrices, parity of permutations, and expansion.
Occurrence	51 tagged problem(s). Selected anchors: 1994-D1-P1; 1995-D1-P1; 1995-D2-P1; 1996-D1-P1; 1997-D1-P3; 1997-D2-P2; 1997-D2-P4; 1999-D1-P1; 2000-D1-P3; 2001-D1-P1; 2001-D1-P5; 2001-D2-P4; 2002-D2-P1; 2003-D2-P1; 2004-D2-P1; 2004-D2-P6; 2005-D1-P1; 2005-D2-P3; ... (full list retained in the theory-node index above; 51 total).
Original trap note	Track row/column operations exactly; if operations depend on parity or cyclic order, verify small n separately.
Literature	[19], [18]

Core facts and properties.

1. Adding a multiple of one row to another preserves determinant; swapping two rows changes sign; multiplying one row by c multiplies determinant by c.
2. A triangular matrix has determinant equal to the product of diagonal entries.
3. For a block triangular matrix $\begin{pmatrix} A & * \\ 0 & D \end{pmatrix}$, $\det = \det A \det D$.

Compact proof mechanisms.

1. The determinant is the alternating multilinear form normalized by $\det I = 1$; the elementary rules follow directly from multilinearity and alternation.
2. Laplace expansion is obtained by grouping determinant terms by the chosen row/column index.
3. When a structured determinant appears, try to manufacture zeros or repeated rows/columns; this is usually faster than expanding.

9.2.3 LA-03: Block matrices, Schur complements, and determinant identities

Must know	Block multiplication, Schur complement logic, determinants of block triangular matrices, and rank identities for rectangular factorizations.
Occurrence	114 tagged problem(s). Selected anchors: 1994-D1-P1; 1994-D2-P2; 1995-D1-P1; 1995-D1-P2; 1995-D2-P1; 1995-D2-P2; 1996-D1-P1; 1996-D1-P2; 1997-D1-P3; 1997-D2-P2; 1997-D2-P4; 1998-D1-P1; 1998-D1-P5; 1999-D1-P1; 1999-D1-P5; 1999-D2-P2; 1999-D2-P3; 2000-D1-P2; ... (full list retained in the theory-node index above; 114 total).
Original trap note	Schur complement formulas require invertibility of the chosen block; if it is not given, use block multiplication identities instead.
Literature	[19]

Core facts and properties.

1. If A is invertible, $\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det(A) \det(D - CA^{-1}B)$.
2. For rectangular A and B, the nonzero eigenvalues of AB and BA agree with algebraic multiplicity after padding by zeros.
3. Block equations often become ordinary equations by comparing blocks of $MN = I$ or M^{-1}.

Compact proof mechanisms.

1. The Schur formula follows by multiplying on the left by $\begin{pmatrix} I & 0 \\ -CA^{-1} & I \end{pmatrix}$, which has determinant 1.
2. For AB and BA, use Sylvester's determinant identity $\det(I + tAB) = \det(I + tBA)$; compare characteristic polynomials up to powers of λ.
3. Block inverse identities come from writing $MM^{-1} = I$ and equating the four blocks.

9.2.4 LA-04: Rank-nullity, image/kernel, and rank of factorizations

Must know	Rank-nullity, dimension counting, images and kernels, ranks of products, and factorization rank bounds.
Occurrence	87 tagged problem(s). Selected anchors: 1994-D1-P1; 1994-D1-P4; 1994-D2-P4; 1994-D2-P5; 1995-D1-P1; 1995-D1-P5; 1996-D1-P1; 1996-D1-P3; 1996-D1-P6; 1997-D1-P3; 1997-D2-P2; 1997-D2-P4; 1998-D1-P1; 1998-D2-P1; 1999-D1-P1; 2000-D1-P3; 2000-D2-P1; 2001-D1-P5; ... (full list retained in the theory-node index above; 87 total).
Original trap note	Rank statements over $\mathbb{R}$ and $\mathbb{C}$ can differ only through spectral issues; dimension counts must specify the ground field.
Literature	[6]

Core facts and properties.

1. Rank-nullity: $\dim V = \dim \ker T + \dim \operatorname{im} T$.
2. $\operatorname{rank}(AB) \le \min(\operatorname{rank} A, \operatorname{rank} B)$ and $\operatorname{rank}(A+B) \le \operatorname{rank} A + \operatorname{rank} B$.
3. A subspace of $M_{m\times n}$ whose columns all lie in a fixed r-dimensional subspace has dimension at most rn.

Compact proof mechanisms.

1. Rank-nullity is the basis theorem: extend a basis of the kernel to a basis of the domain; the remaining vectors map to a basis of the image.
2. For products, AB maps first through B and then A, so its image is contained in $\operatorname{im} A$ and is the image under A of a subspace of $\operatorname{im} B$.
3. Dimension bounds in IMC usually require choosing an adapted basis, not writing all equations.

9.2.5 LA-05: Trace cyclicity, trace pairing, and zero-trace subspaces

Must know	Trace, cyclicity, trace pairing $\langle A, B\rangle = \operatorname{tr}(AB)$, and zero-trace subspaces.
Occurrence	33 tagged problem(s). Selected anchors: 1995-D1-P1; 1995-D2-P1; 1996-D1-P1; 1997-D2-P2; 1997-D2-P4; 1999-D1-P1; 2000-D1-P3; 2001-D1-P5; 2001-D2-P4; 2002-D2-P1; 2003-D2-P1; 2004-D2-P1; 2005-D1-P1; 2005-D2-P3; 2005-D2-P6; 2006-D2-P6; 2007-D1-P2; 2007-D1-P3; ... (full list retained in the theory-node index above; 33 total).
Original trap note	Cyclicity is safe for products under trace, not outside trace. Do not infer $XY = YX$ from equal traces.
Literature	[19], [18]

Core facts and properties.

1. $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ for compatible square products; hence $\operatorname{tr}[A, B] = 0$.
2. The linear functionals on $M_n(F)$ are exactly $A \mapsto \operatorname{tr}(AC)$ for a unique C.
3. The subspace of trace-zero matrices has codimension 1 and is spanned by off-diagonal E_{ij} and $E_{ii} - E_{nn}$.

Compact proof mechanisms.

1. Cyclicity follows by expanding: $\operatorname{tr}(AB) = \sum_{i,j} a_{ij} b_{ji} = \operatorname{tr}(BA)$.
2. For the trace pairing, take the matrix units E_{ij}; the coefficient of A_{ij} is recovered by $\operatorname{tr}(AE_{ji})$.
3. Many commutator impossibility problems are solved by taking trace or trace of powers.

9.2.6 LA-06: Commutators and spectral obstructions

Must know	Commutators $AB - BA$, derivations, spectral obstruction, trace-zero consequences, and nilpotency tests.
Occurrence	20 tagged problem(s). Selected anchors: 1994-D1-P4; 1995-D1-P5; 1996-D1-P1; 1997-D1-P3; 1997-D2-P4; 2000-D1-P3; 2000-D1-P5; 2000-D2-P6; 2001-D1-P5; 2002-D2-P1; 2003-D1-P3; 2003-D2-P1; 2004-D2-P1; 2005-D1-P6; 2009-D2-P3; 2013-D1-P1; 2014-D1-P1; 2020-D1-P2; ... (full list retained in the theory-node index above; 20 total).
Original trap note	Trace zero alone is weak; combine it with rank, eigenvalue multiplicity, or finite-dimensionality.
Literature	[19]

Core facts and properties.

1. Every commutator has trace zero; if a rank-one matrix has trace zero, all its eigenvalues are zero, so it is nilpotent of index at most 2.

2. For $\Delta_A(X) = AX - XA$, Δ_A is a derivation: $\Delta_A(XY) = \Delta_A(X)Y + X\Delta_A(Y)$.
3. If $[F, G] = \alpha F$ with $\alpha \neq 0$, powers often satisfy $[F^k, G] = k\alpha F^k$; infinitely many eigenvalues of a finite-dimensional operator are impossible.

Compact proof mechanisms.

1. Trace zero of a commutator is immediate from cyclicity. Rank one leaves at most one nonzero eigenvalue, whose sum is the trace; hence all eigenvalues vanish.
2. The derivation rule is $A(XY) - (XY)A = (AX - XA)Y + X(AY - YA)$.
3. The power identity telescopes: $F^kG - GF^k = \sum_i F^{k-i}(FG - GF)F^{i-1}$.

9.2.7 LA-07: Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton

Must know	Eigenvalues, characteristic and minimal polynomials, Cayley-Hamilton, diagonalizability, Jordan intuition, and spectral mapping.
Occurrence	51 tagged problem(s). Selected anchors: 1994-D1-P4; 1994-D2-P2; 1995-D1-P1; 1995-D1-P4; 1995-D1-P5; 1995-D2-P5; 1996-D1-P1; 1996-D1-P3; 1996-D1-P5; 1997-D1-P5; 1997-D2-P2; 1998-D2-P1; 1998-D2-P4; 1999-D1-P1; 2000-D1-P3; 2000-D1-P5; 2000-D2-P6; 2001-D1-P5; ... (full list retained in the theory-node index above; 51 total).
Original trap note	Real matrices may have complex eigenvalues; diagonalization over $\mathbb{C}$ may be needed even when entries are real.
Literature	[6], [19]

Core facts and properties.

1. Cayley-Hamilton: every square matrix satisfies its characteristic polynomial.
2. The minimal polynomial divides every annihilating polynomial and has the same irreducible factors as needed to determine diagonalizability.
3. If $p(A) = 0$, then every eigenvalue λ of A satisfies $p(\lambda) = 0$; if A is diagonalizable, $p(A)$ acts by $p(\lambda)$ on eigenspaces.

Compact proof mechanisms.

1. Cayley-Hamilton can be proved by triangularization over $\mathbb{C}$ or by adjugate identity $\operatorname{adj}(xI - A)(xI - A) = \det(xI - A)I$ and polynomial comparison.
2. Diagonalizability follows if the minimal polynomial splits into distinct linear factors; decompose by kernels of relatively prime factors.
3. Matrix-power convergence is reduced to powers of eigenvalues and Jordan blocks.

9.2.8 LA-08: Nilpotent, idempotent, involution, and square-zero structures

Must know	Nilpotents, idempotents, involutions, projections, square-zero maps, and their canonical consequences.
Occurrence	14 tagged problem(s). Selected anchors: 1994-D1-P4; 1995-D1-P1; 1995-D1-P5; 1996-D1-P3; 2000-D1-P3; 2000-D1-P5; 2000-D2-P6; 2001-D2-P4; 2003-D1-P3; 2005-D1-P6; 2007-D2-P5; 2009-D1-P2; 2018-D1-P3; 2022-D2-P7
Original trap note	Do not assume diagonalizability for nilpotents. Idempotents and involutions are diagonalizable only when the minimal polynomial has distinct roots.
Literature	[6]

Core facts and properties.

1. Nilpotent means $N^k = 0$; all eigenvalues are 0. Conversely over $\mathbb{C}$, if all eigenvalues are 0, then some power is 0.
2. Idempotent $P^2 = P$ has eigenvalues in $\{0, 1\}$ and is a projection onto $\operatorname{im} P$ along $\ker P$.
3. Involution $A^2 = I$ has eigenvalues ± 1 and is diagonalizable if the field characteristic is not 2.

Compact proof mechanisms.

1. For idempotents, every v decomposes as $v = Pv + (v - Pv)$ with $Pv \in \operatorname{im} P$ and $v - Pv \in \ker P$.
2. For involutions, $P = (A + I)/2$ is idempotent, so a basis adapted to P diagonalizes $A = 2P - I$.
3. Nilpotent matrices are triangularizable with zero diagonal over $\mathbb{C}$, so high powers vanish.

9.2.9 LA-09: Simultaneous diagonalization and commuting families

Must know	Commuting diagonalizable families, simultaneous invariant subspaces, and common eigenvectors.
Occurrence	3 tagged problem(s). Selected anchors: 1994-D2-P4; 1996-D1-P3; 2006-D2-P6
Original trap note	Commuting alone does not imply diagonalizable. First prove each operator is diagonalizable or use invariant subspaces carefully.
Literature	[6], [18]

Core facts and properties.

1. A commuting family of diagonalizable operators over an algebraically closed field is simultaneously diagonalizable if each operator is diagonalizable and the family is finite.
2. If $AB = BA$, every eigenspace of A is B-invariant.
3. Pairwise commuting involutions can be simultaneously diagonalized and hence are represented by sign choices on a basis.

Compact proof mechanisms.

1. If $Av = \lambda v$, then $A(Bv) = B(Av) = \lambda Bv$, so Bv stays in the eigenspace.
2. Induct on dimension: restrict all operators to an eigenspace of one operator and diagonalize there.
3. For n-dimensional involutions, simultaneous diagonal forms have only ± 1 entries, giving at most 2^n distinct sign patterns.

9.2.10 LA-10: Invariant subspaces and adapted bases

Must know	Invariant subspaces, adapted bases, block triangular forms, and subspace flags.
Occurrence	30 tagged problem(s). Selected anchors: 1994-D2-P4; 1994-D2-P5; 1995-D1-P1; 1996-D1-P3; 1996-D1-P6; 1997-D2-P2; 1997-D2-P4; 1998-D1-P1; 1998-D2-P1; 2001-D1-P5; 2001-D2-P3; 2002-D1-P5; 2002-D1-P6; 2004-D2-P4; 2005-D2-P3; 2006-D2-P2; 2006-D2-P4; 2007-D1-P2; ... (full list retained in the theory-node index above; 30 total).
Original trap note	Check the order of basis blocks: $T(U) \subset U$ imposes zero blocks below, not above, depending on column convention.
Literature	[6]

Core facts and properties.

1. If $U \subset V$ is invariant under T, a basis of U extended to a basis of V gives a block upper triangular matrix for T.
2. For a flag $U_1 \subset U_2 \subset \cdots \subset V$, an adapted basis gives block upper triangular form with block sizes equal to dimension jumps.
3. Counting parameters in such block forms gives dimensions of spaces of endomorphisms preserving the flag.

Compact proof mechanisms.

1. Write basis vectors in order: first a basis of U, then extra vectors. Images of the first block stay in U, causing zeros below that block.
2. For multiple subspaces, extend the basis successively along the flag.
3. The free entries in the resulting block pattern are exactly the dimension count.

9.2.11 LA-11: Linear functionals, kernels, and dual span criteria

Must know	Linear functionals, kernels, annihilators, and the criterion that one functional lies in the span of others.
Occurrence	6 tagged problem(s). Selected anchors: 1997-D2-P2; 1997-D2-P4; 1998-D1-P1; 1998-D2-P1; 2008-D1-P2; 2015-D2-P9
Original trap note	Independence of the given functionals may need to be arranged by discarding redundant ones.
Literature	[6]

Core facts and properties.

1. If $f_i : V \to F$ are linear and $\bigcap_i \ker f_i \subseteq \ker f$, then f is a linear combination of the f_i.
2. The annihilator of a subspace $U \subset V$ is $U^0 = \{\phi \in V^* : \phi(U) = 0\}$ and has dimension $\dim V - \dim U$ in finite dimension.

3. Choosing dual test vectors a_j with $f_i(a_j) = \delta_{ij}$ is the constructive proof.

Compact proof mechanisms.

1. Define $T : V \to F^k$ by $T(v) = (f_1(v), \dots, f_k(v))$. The hypothesis says f is constant on fibers of T, so it factors through $\operatorname{im} T$; every functional on $\operatorname{im} T$ is linear combination of coordinates.
2. If the f_i are independent, construct a_j inductively outside previous kernels; otherwise discard dependent ones.
3. This is the right tool when a problem says: whenever several linear equations vanish, another one vanishes.

9.2.12 LA-12: Matrix subspaces constrained by trace bilinear forms

Must know	Subspaces of matrices constrained by bilinear forms such as $\operatorname{tr}(XY) = 0$.
Occurrence	1 tagged problem(s). Selected anchors: `2005-D2-P3`
Original trap note	The form $\operatorname{tr}(XY)$ is not positive definite on all matrices; positivity appears on symmetric matrices via $\operatorname{tr}(A^2)$.
Literature	[19]

Core facts and properties.

1. The symmetric matrices form a subspace of dimension $n(n+1)/2$; strictly upper triangular matrices have dimension $n(n-1)/2$.
2. If A is real symmetric and nonzero, $\operatorname{tr}(A^2) = \sum_{i,j} a_{ij}^2 > 0$.
3. A subspace V with $\operatorname{tr}(XY) = 0$ for all $X, Y \in V$ intersects the symmetric subspace trivially.

Compact proof mechanisms.

1. If a nonzero symmetric $A \in V$, then taking $X = Y = A$ gives $\operatorname{tr}(A^2) > 0$, contradiction.
2. Thus $\dim V + \dim S \le n^2$, yielding $\dim V \le n(n-1)/2$; the strictly upper triangular subspace attains equality because products are strictly upper triangular and trace zero.
3. This is a model for maximal isotropic subspace dimension arguments.

9.2.13 LA-13: Matrix exponentials and polynomial functional calculus

Must know	Matrix exponentials, power series in matrices, and polynomial functional calculus.
Occurrence	1 tagged problem(s). Selected anchors: `2000-D2-P6`
Original trap note	Do not use $e^{AB} = e^{BA}$; compare characteristic polynomials or nonzero spectra instead.
Literature	[19]

Core facts and properties.

1. $e^A = \sum_{n \ge 0} A^n/n!$ converges for every matrix and commutes with A.
2. If q is a polynomial or convergent power series, then $q(AB)$ and $q(BA)$ often have related nonzero spectra.
3. Cayley-Hamilton can reduce an infinite power series in A to a polynomial in A of degree $< n$.

Compact proof mechanisms.

1. Convergence follows from any submultiplicative matrix norm: $\sum \|A\|^n/n! = e^{\|A\|}$.
2. For $q(e^{AB}) = c_0 I + AC$ and $q(e^{BA}) = c_0 I + CA$, use equality of characteristic polynomials for AC and CA.
3. Nilpotency of $p(e^{AB})$ is characterized by characteristic polynomial x^n and then Cayley-Hamilton.

9.2.14 LA-14: Operator norm, power boundedness, and spectral-radius style estimates

Must know	Operator norm, power boundedness, spectral-radius style estimates, and convergence of matrix powers.
Occurrence	1 tagged problem(s). Selected anchors: `2002-D1-P6`
Original trap note	Norm convergence is stronger than pointwise convergence. Watch constants and finite-dimensional vs Banach-space arguments.
Literature	[19], [28]

Core facts and properties.

1. The Euclidean operator norm is $\|A\| = \sup_{x \neq 0} \|Ax\|_2 / \|x\|_2$ and is submultiplicative.
2. If $\sum_k \|A^{k+1} - A^k\|$ converges, then A^k is Cauchy in operator norm.
3. Exponential decay of differences implies convergence and power boundedness.

Compact proof mechanisms.

1. Submultiplicativity follows from $\|ABx\| \leq \|A\|\|Bx\| \leq \|A\|\|B\|\|x\|$.
2. To prove power boundedness from difference estimates, write $A^n = A^0 + \sum_{k<n}(A^{k+1} - A^k)$ when the series is summable.
3. IMC operator-norm problems often hide a scalar recurrence for $a_n = \|A^{n+1} - A^n\|$.

9.2.15 LA-15: Gram matrices and rational linear dependence

Must know	Gram matrices, inner products from distances, positive definiteness, and rational linear dependence.
Occurrence	7 tagged problem(s). Selected anchors: 1995-D2-P1; 2002-D2-P6; 2006-D2-P4; 2008-D2-P6; 2013-D2-P3; 2018-D2-P6; 2021-D2-P8
Original trap note	First reduce to a real basis; then solve the linear system over $\mathbb{Q}$.
Literature	[6], [7]

Core facts and properties.

1. The Gram matrix $G = (\langle v_i, v_j \rangle)$ is positive semidefinite; it is invertible iff the vectors are linearly independent.
2. Distances determine inner products by $2\langle u, v \rangle = \|u\|^2 + \|v\|^2 - \|u - v\|^2$.
3. If all relevant squared distances are rational, then the Gram matrix has rational entries; solving $G\lambda = w$ gives rational coordinates whenever G is invertible.

Compact proof mechanisms.

1. For any coefficients c_i, $c^T G c = \|\sum c_i v_i\|^2 \geq 0$.
2. Invertibility of G follows because $c^T G c = 0$ means $\sum c_i v_i = 0$; linear independence forces $c = 0$.
3. Rational dependence arguments reduce geometry to solving a rational linear system.

9.2.16 LA-16: Anticommuting and rank constraints for special matrices

Must know	Special anticommuting or low-rank constraints; use trace, determinant, rank, and minimal polynomial obstructions.
Occurrence	0 tagged problem(s). Selected anchors: No direct past-problem occurrence detected in the current parsed corpus; keep as prerequisite/background node.
Original trap note	Pairwise anticommutation is not pairwise orthogonality; translate the algebraic condition into image/kernel constraints.
Literature	[19]

Core facts and properties.

1. If $AB = -BA$, then spectra are symmetric under multiplication by -1 on invariant pieces.
2. Rank constraints combine with trace constraints: a low-rank trace-zero matrix has very restricted Jordan form.
3. For square-zero maps $N^2 = 0$, $\operatorname{im} N \subseteq \ker N$, so $\operatorname{rank} N \leq n/2$.

Compact proof mechanisms.

1. $N^2 = 0$ gives $N(Nv) = 0$, hence every image vector lies in the kernel; rank-nullity gives $\operatorname{rank} N \leq \dim \ker N = n - \operatorname{rank} N$.
2. Trace and determinant are similarity invariants, so special identities should be pushed to canonical form when possible.
3. This is currently a low-frequency node but covers many hidden special-matrix cases.

9.3 Real analysis / calculus

Domain anchors. [28], [4], [17].

9.3.1 RA-01: Compactness, extreme values, and closed-bounded reduction

Must know	Compactness on intervals or closed bounded sets, existence of maxima/minima, and reduction of noncompact problems to compact sublevel regions.
Occurrence	37 tagged problem(s). Selected anchors: 1994-D2-P2; 1995-D1-P6; 1995-D2-P4; 1995-D2-P6; 1996-D1-P6; 1996-D2-P4; 1998-D2-P2; 2000-D1-P1; 2000-D2-P2; 2001-D1-P3; 2001-D2-P6; 2002-D1-P4; 2002-D2-P6; 2003-D2-P3; 2004-D1-P3; 2005-D2-P1; 2005-D2-P5; 2006-D2-P3; ... (full list retained in the theory-node index above; 37 total).
Original trap note	Compactness requires closed and bounded in finite-dimensional Euclidean spaces; infinite-dimensional Hilbert spaces behave differently.
Literature	[28], [4]

Core facts and properties.

1. Extreme Value Theorem: a continuous real function on a compact set attains its maximum and minimum.
2. Heine-Borel in $\mathbb{R}^n$: compact iff closed and bounded.
3. If a function tends to a smaller bound at infinity, global extrema occur inside a sufficiently large compact set.

Compact proof mechanisms.

1. EVT follows from sequential compactness: take a maximizing sequence, pass to a convergent subsequence, and use continuity.
2. Closed and bounded subsets of $\mathbb{R}^n$ are sequentially compact by Bolzano-Weierstrass plus closedness.
3. In contest solutions, first show the extremum cannot lie outside a compact set; then apply EVT.

9.3.2 RA-02: Intermediate value, connectedness, density, and continuity transfer

Must know	Continuity transfer from dense sets, intermediate value theorem, connected images, and interval arguments.
Occurrence	44 tagged problem(s). Selected anchors: 1994-D2-P2; 1995-D1-P2; 1995-D1-P6; 1995-D2-P4; 1995-D2-P6; 1996-D1-P6; 1996-D2-P1; 1996-D2-P4; 1998-D1-P4; 1998-D1-P6; 1998-D2-P5; 2000-D2-P2; 2002-D1-P4; 2003-D2-P3; 2004-D1-P3; 2005-D1-P3; 2006-D1-P1; 2006-D1-P5; ... (full list retained in the theory-node index above; 44 total).
Original trap note	A monotone function need not be continuous unless surjectivity or no jumps is proved.
Literature	[28]

Core facts and properties.

1. Intermediate Value Theorem: continuous images of intervals are intervals.
2. If two continuous functions agree or satisfy an inequality on a dense set, the equality/inequality extends to the closure.
3. A continuous function from a connected set to $\mathbb{R}$ has connected image, hence an interval.

Compact proof mechanisms.

1. IVT follows because if a value between $f(a)$ and $f(b)$ is missed, the preimages of two open half-lines separate the interval.
2. For dense-set transfer, take rationals $r_n \to x$ and pass to limits using continuity.
3. Use this for rational restrictions, images of compact/connected sets, and zero-crossing arguments.

9.3.3 RA-03: Monotone functions, fixed points, and iteration on intervals

Must know	Monotone maps, fixed points, iterates on compact intervals, cluster points, and convergence criteria.
Occurrence	15 tagged problem(s). Selected anchors: 1994-D2-P6; 1995-D1-P4; 1998-D2-P3; 1999-D2-P4; 2000-D1-P1; 2000-D1-P4; 2000-D2-P2; 2001-D2-P2; 2004-D1-P5; 2004-D2-P2; 2007-D2-P3; 2010-D1-P1; 2011-D1-P4; 2015-D2-P6; 2019-D1-P1
Original trap note	A decreasing map may fail to have a fixed point if discontinuous. Iteration convergence needs cluster-point control.
Literature	[28], [14]

Core facts and properties.

1. If $f : [a, b] \to [a, b]$ is continuous, then f has a fixed point because $f(x) - x$ changes sign or vanishes at endpoints.

2. If f is nondecreasing on a compact interval, the set $\{x : f(x) \geq x\}$ has a useful extremal point.
3. For iterates $x_{n+1} = f(x_n)$, cluster points and the condition $x_{n+1} - x_n \to 0$ constrain possible nonconvergence.

Compact proof mechanisms.

1. For fixed points, $g(a) = f(a) - a \geq 0$ and $g(b) = f(b) - b \leq 0$; IVT gives $g(p) = 0$.
2. Monotone fixed-point proofs use $p = \sup\{x : f(x) > x\}$ and compare $f(p)$ with p.
3. If an orbit has two cluster points and jumps vanish, it cannot cross an interval where $|f(x) - x|$ is bounded below.

9.3.4 RA-04: Rolle, mean-value theorem, and divided differences

Must know	Rolle's theorem, Mean Value Theorem, Cauchy MVT, divided differences, and repeated zero arguments.
Occurrence	33 tagged problem(s). Selected anchors: 1994-D1-P2; 1994-D1-P6; 1994-D2-P1; 1994-D2-P3; 1995-D1-P3; 1996-D1-P5; 1996-D2-P5; 1996-D2-P6; 1997-D2-P1; 1997-D2-P3; 1998-D1-P4; 1999-D1-P6; 2000-D1-P2; 2000-D1-P6; 2000-D2-P5; 2001-D1-P6; 2001-D2-P6; 2002-D1-P2; ... (full list retained in the theory-node index above; 33 total).
Original trap note	Check differentiability order and that all points lie in the open interval where the conclusion is asserted.
Literature	[28], [4]

Core facts and properties.

1. Rolle: if f is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then $f'(c) = 0$.
2. MVT: $f(b) - f(a) = f'(c)(b - a)$ for some $c \in (a, b)$.
3. Repeated Rolle: $n + 1$ zeros force a zero of the nth derivative; Hermite divided differences handle repeated nodes.

Compact proof mechanisms.

1. Rolle follows by EVT: an interior extremum has derivative zero unless the function is constant.
2. MVT applies Rolle to $f(x) - \ell(x)$ where ℓ is the secant line.
3. Divided-difference identities are compact ways to encode interpolating polynomial remainders.

9.3.5 RA-05: Taylor expansion, local asymptotics, and little-o bookkeeping

Must know	Taylor expansions, local asymptotics, little-o estimates, comparison near singular or equality points.
Occurrence	50 tagged problem(s). Selected anchors: 1994-D1-P2; 1994-D1-P5; 1994-D1-P6; 1994-D2-P1; 1994-D2-P3; 1994-D2-P4; 1994-D2-P6; 1995-D1-P3; 1995-D1-P6; 1995-D2-P2; 1996-D1-P5; 1996-D2-P5; 1996-D2-P6; 1997-D1-P1; 1997-D2-P1; 1997-D2-P3; 1998-D1-P4; 1999-D1-P6; ... (full list retained in the theory-node index above; 50 total).
Original trap note	Do not expand beyond available smoothness; distinguish $O(x^k)$ from $o(x^k)$.
Literature	[28], [4]

Core facts and properties.

1. Taylor: if f has k derivatives, $f(x + h) = \sum_{j=0}^{k} f^{(j)}(x)h^j/j! + o(h^k)$ under appropriate regularity.
2. For C^{k+1} functions, the Lagrange remainder is $f^{(k+1)}(\xi)h^{k+1}/(k + 1)!$.
3. Asymptotic quotient: if numerator and denominator have first nonzero terms ah^m and bh^m, the quotient tends to a/b.

Compact proof mechanisms.

1. Taylor with remainder follows by repeated MVT or integral remainder.
2. Little-o bookkeeping is algebraic: $o(h^r) + O(h^s) = o(h^t)$ only when the orders justify it; always track the minimum order.
3. Use Taylor at equality/extremal points to identify sharp constants and local obstructions.

9.3.6 RA-06: Auxiliary functions, integrating factors, and differential inequalities

Must know	Auxiliary functions, integrating factors, differential inequalities, and one-dimensional comparison arguments.
Occurrence	62 tagged problem(s). Selected anchors: 1994-D1-P2; 1994-D2-P1; 1994-D2-P3; 1994-D2-P6; 1995-D1-P2; 1995-D1-P4; 1996-D1-P2; 1998-D1-P3; 1998-D1-P4; 1998-D1-P6; 1999-D1-P2; 1999-D1-P4; 1999-D2-P3; 1999-D2-P4; 2000-D1-P2; 2000-D1-P4; 2000-D1-P6; 2000-D2-P2; ... (full list retained in the theory-node index above; 62 total).
Original trap note	If taking logarithms, first prove the function has the required sign on the interval.
Literature	[28], [4]

Core facts and properties.

1. A differential inequality often becomes monotonicity after multiplying by an integrating factor.
2. If $g'(x) + a(x)g(x) \geq 0$, then $(g(x)e^{\int a})' \geq 0$.
3. Logarithmic differentiation: if $f > 0$, $(\log f)' = f'/f$ turns multiplicative bounds into additive ones.

Compact proof mechanisms.

1. Compute $(ge^{A})' = e^{A}(g' + A'g)$ and choose $A' = a$.
2. For $|f'| \leq \lambda|f|$ and $f(a) = 0$, argue on an interval where f has one sign and apply $\log f - \lambda x$.
3. The art is choosing the auxiliary expression whose derivative is exactly the given inequality.

9.3.7 RA-07: Convexity, concavity, Jensen, and tangent/chord inequalities

Must know	Convexity/concavity, Jensen, tangent/chord inequalities, strict convexity, equality cases, and AM-GM/HM consequences.
Occurrence	179 tagged problem(s). Selected anchors: 1994-D1-P1; 1994-D1-P2; 1994-D1-P4; 1994-D1-P5; 1994-D1-P6; 1994-D2-P1; 1994-D2-P2; 1994-D2-P5; 1994-D2-P6; 1995-D1-P2; 1995-D1-P3; 1995-D1-P4; 1995-D1-P6; 1995-D2-P2; 1995-D2-P4; 1995-D2-P5; 1996-D1-P2; 1996-D1-P4; ... (full list retained in the theory-node index above; 179 total).
Original trap note	Check the interval of convexity and strictness conditions for equality.
Literature	[17], [26], [9]

Core facts and properties.

1. For convex ϕ, $\phi(tx + (1-t)y) \leq t\phi(x) + (1-t)\phi(y)$; for concave, reverse.
2. Jensen: $\phi(\sum w_i x_i) \leq \sum w_i \phi(x_i)$ for convex ϕ and $w_i \geq 0$, $\sum w_i = 1$.
3. Tangent line: differentiable convex ϕ satisfies $\phi(y) \geq \phi(x) + \phi'(x)(y - x)$; chords lie above the graph.

Compact proof mechanisms.

1. Jensen follows by induction from the two-point convexity definition.
2. The tangent inequality follows because slopes of secants of a convex function are monotone.
3. Strict convexity gives equality only when all relevant variables coincide, except for zero-weight cases.

9.3.8 RA-08: Integral inequalities, integration by parts, and weighted estimates

Must know	Integral inequalities, integration by parts, weighted estimates, comparison, Cauchy-Schwarz/Hölder, and energy-type estimates.
Occurrence	199 tagged problem(s). Selected anchors: 1994-D1-P2; 1994-D1-P3; 1994-D1-P4; 1994-D1-P5; 1994-D1-P6; 1994-D2-P1; 1994-D2-P2; 1994-D2-P3; 1994-D2-P5; 1994-D2-P6; 1995-D1-P2; 1995-D1-P4; 1995-D1-P5; 1995-D1-P6; 1995-D2-P2; 1995-D2-P5; 1995-D2-P6; 1996-D1-P2; ... (full list retained in the theory-node index above; 199 total).
Original trap note	Boundary terms are often the whole point; compute their signs before discarding them.
Literature	[4], [17]

Core facts and properties.

1. Integration by parts: $\int_a^b uv' = uv|_a^b - \int_a^b u'v$.
2. Cauchy-Schwarz: $(\int fg)^2 \leq (\int f^2)(\int g^2)$; discrete and continuous versions share the same proof.
3. If $F(x) = \int_a^x f$, then constraints on all partial integrals can be exploited by Fubini or integration by parts.

Compact proof mechanisms.

1. Integration by parts follows from integrating the product rule $(uv)' = u'v + uv'$.
2. Cauchy-Schwarz follows from nonnegativity of $\int (f - tg)^2$ and discriminant ≤ 0.
3. When an integral condition is given for all x, integrate it over x to swap order and create weighted moments.

9.3.9 RA-09: Fubini/Tonelli and change of variables

Must know	Fubini/Tonelli, changing order of integration/summation, substitutions, and geometric interpretation of repeated integrals.
Occurrence	39 tagged problem(s). Selected anchors: 1994-D2-P1; 1994-D2-P3; 1995-D1-P2; 1995-D2-P5; 1996-D1-P2; 1996-D1-P4; 1996-D2-P1; 1997-D1-P1; 1997-D1-P2; 1997-D1-P4; 1998-D1-P3; 1998-D1-P6; 1999-D1-P4; 1999-D2-P3; 1999-D2-P4; 2001-D1-P1; 2002-D1-P3; 2003-D1-P1; ... (full list retained in the theory-node index above; 39 total).
Original trap note	Justify interchanging integrals or sums when signs are not nonnegative.
Literature	[28], [4]

Core facts and properties.

1. Tonelli permits swapping order for nonnegative functions; Fubini permits it for absolutely integrable functions.
2. Change of variables in one dimension: $\int f(\phi(u))\phi'(u)du$ with orientation handled by limits.
3. Simplex integrals often collapse by recognizing symmetry over ordered variables.

Compact proof mechanisms.

1. For continuous functions on compact rectangles, Fubini follows from Riemann integration and uniform approximation by step functions.
2. Ordered-region formula: $n! \int_{a \leq x_1 \leq \cdots \leq x_n \leq b} \prod h(x_i) = \left(\int_a^b h \right)^n$ because the cube is partitioned into $n!$ orderings up to measure-zero diagonals.
3. Always check nonnegativity or absolute convergence before swapping infinite sums/integrals.

9.3.10 RA-10: Riemann sums and sum-integral asymptotics

Must know	Riemann sums, sum-integral asymptotics, endpoint splitting, and main-term/error estimates.
Occurrence	73 tagged problem(s). Selected anchors: 1994-D1-P2; 1994-D1-P5; 1994-D2-P6; 1995-D1-P2; 1995-D1-P3; 1995-D2-P2; 1995-D2-P5; 1995-D2-P6; 1996-D1-P4; 1996-D1-P5; 1996-D2-P1; 1996-D2-P5; 1997-D1-P1; 1997-D1-P2; 1997-D2-P1; 1997-D2-P3; 1998-D1-P3; 1999-D1-P2; ... (full list retained in the theory-node index above; 73 total).
Original trap note	Tail control is the common missing step; do not use finite-interval Riemann sums on an infinite domain without a tail estimate.
Literature	[28], [4]

Core facts and properties.

1. If f is Riemann integrable on $[a, b]$, then $\frac{b-a}{n} \sum_{k=1}^{n} f(a + k(b-a)/n) \to \int_a^b f$.
2. For infinite-domain sums, truncate tails uniformly and then apply compact Riemann sums.
3. Endpoint singularities require splitting into small endpoint ranges plus a regular middle.

Compact proof mechanisms.

1. Riemann-sum convergence is the definition/integrability criterion via mesh size tending to zero.
2. Tail control: choose M so $\int_M^\infty |f|$ is small and dominate the sum tail by monotonicity or comparison.
3. In logarithmic sums, isolate the first/last M terms and choose M depending slowly on N.

9.3.11 RA-11: Series convergence, rearrangements, and alternating/Dirichlet tools

Must know	Series convergence, rearrangements, alternating tests, Dirichlet/Abel summation, absolute versus conditional convergence.
Occurrence	67 tagged problem(s). Selected anchors: 1994-D1-P3; 1994-D2-P1; 1994-D2-P3; 1994-D2-P5; 1995-D1-P5; 1995-D1-P6; 1995-D2-P5; 1995-D2-P6; 1996-D1-P2; 1996-D1-P4; 1996-D1-P5; 1996-D2-P1; 1996-D2-P5; 1996-D2-P6; 1997-D1-P1; 1997-D1-P2; 1997-D1-P4; 1997-D1-P5; ... (full list retained in the theory-node index above; 67 total).
Original trap note	Conditional convergence is not preserved by arbitrary permutations.
Literature	[28], [4]

Core facts and properties.

1. Absolute convergence implies convergence and any rearrangement has the same sum.
2. Leibniz alternating test: if $a_n \downarrow 0$, then $\sum(-1)^n a_n$ converges.
3. Dirichlet test: bounded partial sums of a_n and monotone $b_n \to 0$ imply $\sum a_n b_n$ converges.

Compact proof mechanisms.

1. Absolute convergence makes tails small uniformly over finite rearrangements.
2. Alternating convergence follows because even and odd partial sums are monotone and squeeze each other.
3. Dirichlet is summation by parts: $\sum a_n b_n = A_N b_N + \sum A_n(b_n - b_{n+1})$.

9.3.12 RA-12: Uniform continuity and periodic averaging

Must know	Uniform continuity and periodic averaging/oscillatory cancellation.
Occurrence	1 tagged problem(s). Selected anchors: 1994-D1-P5
Original trap note	Keep the period and interval length consistent; normalize by the period correctly.
Literature	[28]

Core facts and properties.

1. A continuous function on a compact interval is uniformly continuous.
2. For periodic g of period b, $\int_0^b f(x)g(nx)dx \to b^{-1}(\int_0^b f)(\int_0^b g)$ when f is continuous.
3. If g has mean zero, high-frequency periodic oscillation averages out against continuous f.

Compact proof mechanisms.

1. Uniform continuity follows from compactness: otherwise two sequences with distance tending to zero but function gap bounded below have convergent subsequences with the same limit.
2. Partition $[0, b]$ into n equal subintervals; approximate f by a constant on each subinterval; periodicity makes each integral of $g(nx)$ equal to $n^{-1}\int_0^b g$.
3. The error is controlled by the modulus of continuity of f.

9.3.13 RA-13: L'Hospital, ratio limits, and growth comparisons

Must know	L'Hospital's rule, ratio limits, exponential/logarithmic growth comparisons, and dominant terms.
Occurrence	33 tagged problem(s). Selected anchors: 1994-D1-P2; 1994-D1-P5; 1994-D2-P6; 1995-D1-P3; 1995-D1-P4; 1995-D2-P2; 1996-D1-P4; 1996-D2-P1; 1996-D2-P5; 1997-D1-P1; 1997-D1-P2; 1997-D2-P1; 2000-D1-P6; 2001-D1-P3; 2001-D1-P6; 2001-D2-P5; 2003-D1-P1; 2003-D2-P2; ... (full list retained in the theory-node index above; 33 total).
Original trap note	L\{}'Hospital requires the appropriate hypotheses; for inequalities, use comparison rather than equality of limits when needed.
Literature	[28], [4]

Core facts and properties.

1. L'Hospital: for $0/0$ or ∞/∞ forms under hypotheses, $\lim f/g = \lim f'/g'$ if the latter exists and $g' \neq 0$ near the point.
2. Exponentials dominate powers; powers dominate logarithms as $x \to \infty$.
3. Ratio comparison: if $a_n/b_n \to L$ with b_n positive and summable/divergent appropriately, transfer convergence or asymptotics.

Compact proof mechanisms.

1. L'Hospital follows from Cauchy's Mean Value Theorem applied on intervals approaching the limit point.
2. Growth hierarchy follows by repeated L'Hospital, e.g. $\log x/x^{\alpha} \to 0$ and $x^m/e^{cx} \to 0$.
3. In IMC, L'Hospital often appears after an integrating factor or asymptotic quotient has been isolated.

9.3.14 RA-14: Weierstrass approximation and polynomial approximation

Must know	Weierstrass approximation, odd/even polynomial approximation, and reduction to polynomial subspaces.
Occurrence	1 tagged problem(s). Selected anchors: 1995-D2-P4
Original trap note	Approximate the symmetric part, then handle missing low-degree terms separately if the basis omits them.
Literature	[28], [4]

Core facts and properties.

1. Weierstrass: every continuous function on $[a, b]$ is uniformly approximable by polynomials.
2. If f is odd on $[-1, 1]$, approximate by p and replace by $(p(x) - p(-x))/2$ to get an odd polynomial.
3. Specific missing monomials can sometimes be approximated by identities such as $x(1 - x^2)^n \to 0$ uniformly.

Compact proof mechanisms.

1. The odd projection works because $f(-x) = -f(x)$ and uniform error is preserved by symmetrization.
2. For $x(1 - x^2)^n$, split $|x| \le 1 - \delta$ and near endpoints, or maximize directly; it tends uniformly to zero.
3. Use this when a function approximation problem excludes the linear term or forces parity.

9.3.15 RA-15: L2 spaces, orthonormal systems, and dominated convergence

Must know	L^2 inner products, orthonormal systems, completeness, bounded convergence/dominated convergence logic.
Occurrence	1 tagged problem(s). Selected anchors: 1995-D2-P6
Original trap note	Pointwise convergence plus uniform boundedness allows dominated convergence, but only on finite measure domains.
Literature	[28], [27]

Core facts and properties.

1. In $L^2[0, 1]$, $\langle f, g\rangle = \int_0^1 fg$ and orthonormality means $\langle f_m, f_n\rangle = \delta_{mn}$.
2. A complete orthonormal system has the property that if a function is orthogonal to every basis vector, it is zero in L^2.
3. Dominated convergence: if $f_n \to f$ pointwise and $|f_n| \le g \in L^1$, then $\int f_n \to \int f$.

Compact proof mechanisms.

1. Completeness means the closed span is all of L^2; orthogonal complement of the closed span is $\{0\}$.
2. Dominated convergence follows by controlling large tails of g and using uniform smallness on finite-measure sets; in contest use it when bounded functions on $[0, 1]$ converge pointwise.
3. A pointwise convergent bounded subsequence of an orthonormal system would converge weakly to zero and in norm to one, contradiction via dominated convergence.

9.3.16 RA-16: Baire/category, dense sets, and pathological continuous functions

Must know	Baire/category, dense sets, Cantor-type constructions, and pathological continuous functions.
Occurrence	6 tagged problem(s). Selected anchors: 1997-D2-P6; 1998-D1-P4; 1998-D2-P6; 2000-D2-P2; 2002-D1-P5; 2003-D2-P5
Original trap note	Countability and uncountability arguments must distinguish isolated intervals from point sets.
Literature	[28], [27]

Core facts and properties.

1. A dense set meets every nonempty open interval; a nowhere dense set has closure with empty interior.

2. Baire theorem: a complete metric space cannot be a countable union of nowhere dense sets.
3. The Cantor set is closed, uncountable, nowhere dense, and often supports uncountably many exceptional points.

Compact proof mechanisms.

1. Baire is usually proved by nested closed balls with radii tending to zero avoiding each nowhere dense set.
2. The Cantor set is uncountable by binary-to-ternary expansion using digits $0, 2$.
3. Continuous functions with infinitely/uncountably many zero crossings are made by summing uniformly convergent localized tent functions.

9.3.17 RA-17: Sequence recurrences, monotone bounds, and generating functions

Must know	Sequence recurrences, monotone bounds, telescoping, and generating functions as analytic objects.
Occurrence	22 tagged problem(s). Selected anchors: 1995-D2-P2; 1996-D1-P4; 1998-D2-P6; 2001-D2-P2; 2002-D2-P3; 2003-D1-P1; 2003-D1-P5; 2003-D1-P6; 2003-D2-P6; 2005-D1-P2; 2007-D1-P4; 2008-D2-P3; 2010-D2-P2; 2011-D2-P1; 2012-D2-P2; 2015-D1-P5; 2017-D2-P8; 2017-D2-P9; ... (full list retained in the theory-node index above; 22 total).
Original trap note	Check radius of convergence before manipulating generating functions; induction bounds need verified base cases.
Literature	[4], [29]

Core facts and properties.

1. Monotone bounded sequences converge; increasing unbounded sequences tend to infinity.
2. A recurrence can often be transformed by a substitution into a telescoping difference.
3. Generating function $F(x) = \sum a_n x^n$ converts convolution recurrences into product or differential equations.

Compact proof mechanisms.

1. Monotone convergence of real sequences follows from completeness: the supremum is the limit.
2. For convolution $a_{n+1} = \sum_k a_k a_{n-k}$, multiplying by x^n and summing turns recurrence into algebraic/differential equations for F.
3. Check radius/convergence or work formally, depending on the problem.

9.3.18 RA-18: Trigonometric and hyperbolic identities with monotonicity

Must know	Trigonometric and hyperbolic identities with monotonicity, products, and inequalities.
Occurrence	14 tagged problem(s). Selected anchors: 1995-D2-P5; 1996-D1-P2; 1996-D2-P2; 1997-D2-P3; 2000-D1-P4; 2001-D2-P2; 2001-D2-P6; 2003-D2-P2; 2004-D1-P3; 2004-D2-P5; 2006-D2-P3; 2014-D2-P3; 2015-D1-P4; 2017-D1-P2
Original trap note	Track the interval: many monotonicity claims only hold on $(0, \pi/2)$ or $(0, \pi)$.
Literature	[4]

Core facts and properties.

1. $\sin u \cos v + \cos u \sin v = \sin(u + v)$, $\cosh(u + v) = \cosh u \cosh v + \sinh u \sinh v$.
2. $\sin x / x$ decreases on $(0, \pi)$, a standard tool for small-interval integral estimates.
3. Roots on the unit circle convert trigonometric identities into algebraic identities in e^{ix}.

Compact proof mechanisms.

1. For $\sin x / x$, differentiate: $(x \cos x - \sin x)/x^2 < 0$ because $\tan x > x$ on $(0, \pi/2)$ and symmetry handles the rest needed in IMC estimates.
2. Hyperbolic recurrence $\cosh((m+1)t) = 2 \cosh t \cosh(mt) - \cosh((m-1)t)$ follows from exponential definitions.
3. Always check the interval; sine/cosine monotonicity changes across quadrants.

9.3.19 RA-19: Gradient/Lipschitz-gradient optimization and convex smoothness

Must know	Convex differentiable functions with Lipschitz gradient, smoothness inequalities, co-coercivity, and optimization geometry.
Occurrence	3 tagged problem(s). Selected anchors: 1994-D2-P2; 2002-D2-P6; 2005-D2-P5
Original trap note	Gradient Lipschitzness is a global condition; local Hessian intuition must be converted into an inequality.
Literature	[26], [9]

Core facts and properties.

1. If ∇f is L-Lipschitz, then $f(y) \le f(x) + \langle \nabla f(x), y-x\rangle + \frac{L}{2}\|y-x\|^2$.
2. For convex differentiable f, $f(y) \ge f(x) + \langle \nabla f(x), y-x\rangle$.
3. Baillon-Haddad/co-coercivity: for convex L-smooth f, $\|\nabla f(x) - \nabla f(y)\|^2 \le L\langle \nabla f(x) - \nabla f(y), x-y\rangle$.

Compact proof mechanisms.

1. Smoothness inequality follows by integrating $\frac{d}{dt} f(x + t(y-x))$ and bounding gradient variation by $Lt\|y-x\|$.
2. Convex tangent inequality follows from one-dimensional convexity along the segment.
3. Co-coercivity is obtained by applying the lower bound $g(y) \ge \|\nabla g(y)\|^2/(2L)$ to $g(z) = f(z) - f(x) - \langle \nabla f(x), z-x\rangle$ and symmetrizing.

9.3.20 RA-20: Measure and probability-integral estimates

Must know	Measure/probability-integral estimates: zero-measure exceptional sets, length/area normalization, and integrability bounds.
Occurrence	2 tagged problem(s). Selected anchors: 2019-D2-P10; 2022-D1-P1
Original trap note	Distinguish deterministic inequalities from almost-sure statements; justify independence and symmetry.
Literature	[8], [27]

Core facts and properties.

1. A countable set has Lebesgue measure zero, but can be dense.
2. On a finite-measure domain, bounded pointwise convergence permits dominated convergence with constant dominator.
3. Geometric probability is normalized measure when the distribution is uniform.

Compact proof mechanisms.

1. Countable measure zero follows by covering the nth point by an interval of length $\varepsilon/2^n$.
2. Dominated convergence handles limits of integrals once uniform boundedness is known.
3. Distinguish cardinality (countable/uncountable) from measure; IMC problems often exploit the gap.

9.4 Complex analysis / polynomials

Domain anchors. [1], [11], [24].

9.4.1 CP-01: Polynomial identity and more-roots-than-degree principle

Must know	Polynomial identities, more-roots-than-degree, coefficient comparison, and finite interpolation.
Occurrence	68 tagged problem(s). Selected anchors: 1994-D2-P4; 1995-D1-P5; 1995-D2-P3; 1995-D2-P4; 1996-D2-P2; 1998-D1-P2; 1998-D1-P5; 1998-D2-P2; 1999-D1-P1; 1999-D2-P2; 1999-D2-P6; 2000-D1-P6; 2000-D2-P3; 2000-D2-P4; 2000-D2-P6; 2001-D1-P4; 2001-D2-P1; 2002-D1-P1; ... (full list retained in the theory-node index above; 68 total).
Original trap note	Over finite fields, different polynomials can define the same function; reduce modulo $x^p - x$ when working on $\mathbb{F}_p$.
Literature	[15], [24]

Core facts and properties.

1. A nonzero polynomial over a field of degree d has at most d roots.

2. If two polynomials of degree at most d agree at $d+1$ distinct points, they are identical.
3. A polynomial identity in one or more variables can be proved by fixing all but one variable and applying the one-variable result iteratively.

Compact proof mechanisms.

1. If $p(a)=0$, then $x-a$ divides $p(x)$ by division with remainder; induction gives the root bound.
2. For equality of polynomials, apply the root bound to their difference.
3. Over finite fields, remember the root bound holds for degree $< |F|$ but functions may be represented by many polynomials modulo $x^q - x$.

9.4.2 CP-02: Factorization, irreducibility, and Gauss-style integer polynomial arguments

Must know	Factorization, irreducibility, Gauss lemma, integer polynomial divisibility, and rational functions.
Occurrence	24 tagged problem(s). Selected anchors: 1995-D2-P3; 1998-D1-P5; 1998-D2-P2; 2000-D2-P3; 2001-D1-P4; 2001-D2-P1; 2005-D2-P2; 2006-D1-P4; 2007-D1-P1; 2008-D1-P3; 2008-D2-P1; 2008-D2-P2; 2008-D2-P4; 2009-D2-P1; 2012-D1-P5; 2012-D2-P1; 2013-D2-P1; 2015-D2-P10; ... (full list retained in the theory-node index above; 24 total).
Original trap note	Divisibility in values is not the same as divisibility in $\mathbb{Z}[x]$.
Literature	[15], [20]

Core facts and properties.

1. Gauss lemma: a primitive product in $\mathbb{Z}[x]$ is primitive; divisibility over $\mathbb{Q}[x]$ of primitive integer polynomials descends to $\mathbb{Z}[x]$ up to units.
2. Rational root theorem: if p/q in lowest terms is a root of $a_n x^n + \cdots + a_0$, then $p \mid a_0$ and $q \mid a_n$.
3. Unique factorization in $\mathbb{Z}[x]$ controls polynomial products with integer coefficients.

Compact proof mechanisms.

1. Gauss lemma follows by reducing modulo a prime that divides all coefficients of a product and reaching contradiction with primitivity.
2. Rational root theorem follows by multiplying by q^n and comparing divisibility of the first and last terms.
3. When integer-valued divisibility is given at infinitely many integers, reduce to a rational-polynomial quotient and compare growth.

9.4.3 CP-03: Vieta, logarithmic derivatives, and real-root inequalities

Must know	Vieta relations, logarithmic derivatives, real-root inequalities, and interlacing-type identities.
Occurrence	28 tagged problem(s). Selected anchors: 1995-D2-P3; 1998-D1-P5; 1998-D2-P2; 2000-D2-P3; 2000-D2-P4; 2001-D1-P4; 2003-D1-P6; 2005-D1-P4; 2005-D2-P2; 2006-D1-P4; 2007-D1-P1; 2007-D1-P5; 2007-D2-P6; 2008-D2-P1; 2009-D2-P4; 2010-D2-P5; 2012-D1-P5; 2012-D2-P1; ... (full list retained in the theory-node index above; 28 total).
Original trap note	Handle points that are roots of P separately before dividing by $P(x)$.
Literature	[24], [1]

Core facts and properties.

1. If $P(x) = a\prod_i (x-r_i)$, then coefficients are elementary symmetric functions of the roots.
2. For x not a root, $P'(x)/P(x) = \sum_i 1/(x-r_i)$.
3. If all roots are real, sums of squares from logarithmic derivatives produce inequalities such as Laguerre-type bounds.

Compact proof mechanisms.

1. Logarithmic derivative follows by differentiating $\log P = \log a + \sum \log(x-r_i)$.
2. Second derivative identity follows by differentiating or expanding pairs: $P''/P = 2\sum_{i<j} 1/((x-r_i)(x-r_j))$.
3. Equality cases typically require all roots coinciding or a multiple root at the evaluation point.

9.4.4 CP-04: Unit-circle roots and conjugation identities

Must know	Unit-circle roots, conjugation, reciprocal polynomials, and modulus identities.
Occurrence	13 tagged problem(s). Selected anchors: 1995-D2-P3; 2000-D1-P2; 2000-D2-P3; 2002-D2-P5; 2008-D2-P1; 2013-D2-P1; 2015-D1-P1; 2016-D2-P10; 2017-D1-P1; 2018-D2-P9; 2021-D2-P7; 2023-D1-P2; 2024-D2-P7
Original trap note	Do not divide by zero at existing roots; remove common factors or argue by continuity.
Literature	[1], [24]

Core facts and properties.

1. If $|\alpha| = 1$, then $\bar{\alpha} = 1/\alpha$.
2. For $|\alpha| = 1$, $\operatorname{Re} \frac{z+\alpha}{z-\alpha} = \frac{|z|^2-1}{|z-\alpha|^2}$ for $z \neq \alpha$.
3. Self-inversive/reciprocal transformations convert roots inside/outside the unit circle.

Compact proof mechanisms.

1. The displayed real-part identity is obtained by multiplying numerator and denominator by $\bar{z} - \bar{\alpha}$ and using $\bar{\alpha} = 1/\alpha$.
2. If a sum of terms has real part strictly nonzero off the unit circle, its zeros must lie on the unit circle.
3. Always treat multiple roots separately; logarithmic derivatives may have poles.

9.4.5 CP-05: Roots of unity, cyclotomic filters, and discrete Fourier tricks

Must know	Roots of unity, cyclotomic filters, Fourier tricks on finite cyclic groups, and divisibility by cyclotomic polynomials.
Occurrence	24 tagged problem(s). Selected anchors: 1995-D2-P3; 1997-D1-P3; 1999-D2-P2; 1999-D2-P6; 2000-D1-P2; 2000-D2-P3; 2001-D1-P4; 2001-D2-P3; 2002-D2-P5; 2007-D1-P5; 2007-D1-P6; 2008-D2-P1; 2010-D2-P5; 2013-D2-P1; 2014-D1-P3; 2016-D2-P10; 2017-D1-P1; 2017-D1-P5; … (full list retained in the theory-node index above; 24 total).
Original trap note	Check whether all roots, primitive roots, or all nontrivial roots are needed.
Literature	[20], [15]

Core facts and properties.

1. If $\omega = e^{2\pi i/q}$, then $\frac{1}{q}\sum_{j=0}^{q-1} \omega^{jm} = 1$ if $q \mid m$ and 0 otherwise.
2. $x^{q-1} + \cdots + x + 1$ is the qth cyclotomic polynomial for prime q and is irreducible over $\mathbb{Q}$.
3. Root-of-unity filters count coefficients in selected congruence classes.

Compact proof mechanisms.

1. The filter is a finite geometric series.
2. Prime cyclotomic irreducibility follows by substituting $x + 1$ and applying Eisenstein at q.
3. In polynomial-divisibility problems, evaluate at a primitive root and exploit Galois conjugacy/irreducibility.

9.4.6 CP-06: Cauchy formula and coefficient extraction on the unit circle

Must know	Cauchy coefficient formula, maximum on the unit circle, Parseval-style coefficient extraction.
Occurrence	7 tagged problem(s). Selected anchors: 2007-D1-P5; 2007-D1-P6; 2009-D1-P4; 2013-D2-P1; 2014-D1-P3; 2017-D2-P7; 2023-D1-P3
Original trap note	The maximum on the closed disk is controlled by the boundary only for holomorphic functions.
Literature	[1], [11]

Core facts and properties.

1. For a polynomial $p(z) = \sum a_j z^j$, $a_j = \frac{1}{2\pi i}\int_{|z|=1} p(z) z^{-j-1}\, dz$.
2. Equivalently, $a_j = \frac{1}{2\pi}\int_0^{2\pi} p(e^{it}) e^{-ijt} dt$.
3. Averaging p over roots of unity extracts coefficients in residue classes.

Compact proof mechanisms.

1. Cauchy's formula follows from parametrizing the circle and the orthogonality $\int_0^{2\pi} e^{i(k-j)t} dt = 0$ unless $k = j$.

2. On $|z| = 1$, bounds on $|p(z)|$ give bounds on integrals and coefficients.
3. Use with positive kernels when coefficient multipliers are convex or monotone.

9.4.7 CP-07: Maximum modulus principle

Must know	Maximum modulus principle and polynomial/holomorphic maximum reduction to boundary.
Occurrence	0 tagged problem(s). Selected anchors: No direct past-problem occurrence detected in the current parsed corpus; keep as prerequisite/background node.
Original trap note	The function must be holomorphic on a neighborhood of the closed domain, not merely inside it with boundary singularities.
Literature	[1], [11]

Core facts and properties.

1. If f is holomorphic and nonconstant on a connected domain, $|f|$ has no strict interior maximum.
2. For a polynomial, $\max_{|z|\le 1} |p(z)| = \max_{|z|=1} |p(z)|$.
3. If equality/maximum occurs inside for nonconstant holomorphic f, the function must be constant on the component.

Compact proof mechanisms.

1. Maximum modulus follows from Cauchy's integral formula: $f(a)$ is an average of nearby boundary values, so if $|f(a)|$ is maximal then equality forces constant argument and local constancy.
2. The polynomial disk version follows by applying the principle to the unit disk and compactness on the boundary.
3. Use when a problem asks for a uniform bound on $|p|$ over a disk.

9.4.8 CP-08: Laurent expansions, isolated singularities, and rationality of meromorphic functions

Must know	Laurent expansions, isolated singularities, residues when needed, and rationality of meromorphic functions with finite poles.
Occurrence	1 tagged problem(s). Selected anchors: 2004-D1-P6
Original trap note	Uniform convergence on compact subsets away from singularities must be checked before termwise analytic operations.
Literature	[1], [11]

Core facts and properties.

1. An isolated singularity is removable, a pole, or essential; a meromorphic function on the Riemann sphere is rational.
2. If a function is holomorphic except finitely many poles and has a pole/removable behavior at infinity, it is rational.
3. Laurent expansions identify pole order and principal part.

Compact proof mechanisms.

1. Subtract the principal parts at each pole; the remainder is entire and bounded by polynomial growth, hence a polynomial by Liouville/generalized Liouville.
2. Pole order at a zero of $\log z$ can be found by Taylor expanding $\log(1+w)$.
3. Functional symmetries such as $f(z) = f(1/z)$ reduce unknown coefficients in rational forms.

9.4.9 CP-09: Fejer kernels and positive trigonometric kernels

Must know	Fejer kernels, positive trigonometric kernels, convex coefficient multipliers, and boundary norm contractions.
Occurrence	1 tagged problem(s). Selected anchors: 2009-D1-P4
Original trap note	Convexity of the coefficient sequence is the exact hypothesis that makes the kernel nonnegative.
Literature	[21], [1]

Core facts and properties.

1. The Fejer kernel $F_k(e^{it}) = \sum_{j=-k+1}^{k-1}(k-|j|)e^{ijt} = |1+e^{it}+\cdots+e^{i(k-1)t}|^2 \geq 0$.
2. Convex decreasing coefficient sequences can be decomposed into nonnegative sums of Fejer kernels via second differences.
3. Positive kernels of integral 2π define averaging operators that do not increase the sup norm on the unit circle.

Compact proof mechanisms.

1. The square formula follows by multiplying the geometric sum by its conjugate.
2. If $d_k = c_{k-1} - 2c_k + c_{k+1} \geq 0$, then the graph of c_j is a nonnegative sum of triangular tents, which are Fejer coefficients.
3. The contraction proof is $|\int Kp| \leq \|p\|_\infty \int K$ when $K \geq 0$.

9.4.10 CP-10: Interpolation, Lagrange basis, and Chebyshev extremality

Must know	Interpolation, Lagrange basis, finite differences, and Chebyshev extremality in polynomial inequalities.
Occurrence	2 tagged problem(s). Selected anchors: 1995-D2-P4; 1998-D2-P2
Original trap note	The extremal sign pattern must match the signs of the derivative of the Lagrange basis at the target point.
Literature	[25], [4]

Core facts and properties.

1. Lagrange interpolation: for distinct nodes x_i, every polynomial of degree $\leq n$ is $p(x) = \sum_i p(x_i)L_i(x)$.
2. Linear functionals on finite-dimensional polynomial spaces attain suprema over boxes at vertices.
3. Chebyshev polynomial T_n minimizes the sup norm among monic/leading-coefficient constrained polynomials and equioscillates.

Compact proof mechanisms.

1. Lagrange basis $L_i(x) = \prod_{j\neq i}(x-x_j)/(x_i-x_j)$ satisfies $L_i(x_j) = \delta_{ij}$, proving the formula.
2. For a linear objective with constraints $|p(x_i)| \leq 1$, the maximum occurs by choosing signs matching coefficients.
3. Equioscillation prevents improvement because alternating extreme errors force many zeros in a difference polynomial.

9.4.11 CP-11: Stable polynomials and coefficient inequalities

Must know	Stable polynomials, roots in half-planes, coefficient log-concavity/Newton-type inequalities.
Occurrence	3 tagged problem(s). Selected anchors: 2003-D1-P6; 2014-D1-P3; 2023-D1-P3
Original trap note	Keep track of the order of coefficients and signs after normalizing the leading coefficient.
Literature	[24], [17]

Core facts and properties.

1. If real polynomial roots lie in the left half-plane, real linear and positive quadratic factors have coefficients of one sign.
2. Newton-type inequalities and log-concavity compare adjacent coefficients for real-rooted or stable-related polynomials.
3. Products preserve many coefficient inequalities if each factor has positive coefficients and the inequality is convolution-stable.

Compact proof mechanisms.

1. Linear factors from negative real roots and quadratic factors from conjugate roots with negative real part have positive coefficients after global sign normalization.
2. Coefficient inequalities often follow by induction on multiplying by a positive quadratic factor and checking algebraic nonnegative terms.
3. Always first normalize the sign so all coefficients are positive.

9.4.12 CP-12: Partial fractions and cotangent-type expansions

Must know	Partial fractions and cotangent-type expansions for sums over logarithm branches or lattice shifts.
Occurrence	1 tagged problem(s). Selected anchors: 2004-D1-P6
Original trap note	Termwise differentiation requires locally uniform convergence away from poles.
Literature	[11], [1]

Core facts and properties.

1. The classical expansion $\sum_{k\in\mathbb{Z}} \frac{1}{w+2\pi ik} = \frac{i}{2}\cot(iw/2)$ holds in symmetric principal value sense.
2. Differentiating termwise gives sums of higher negative powers.
3. Substituting $w = \log z$ converts branch sums into rational functions of z.

Compact proof mechanisms.

1. The cotangent expansion is derived from residues of $\pi\cot\pi z$ or Mittag-Leffler theorem.
2. Higher-power sums converge absolutely, so termwise differentiation is justified locally uniformly away from poles.
3. After the first identity, repeated application of $z\frac{d}{dz}$ gives higher powers of $(\log z)^{-1}$.

9.4.13 CP-13: Polynomial functional identities and algebraic composition

Must know	Polynomial functional identities, composition, divisibility of $p(q(x))$, and algebraic transformations.
Occurrence	8 tagged problem(s). Selected anchors: 2000-D1-P2; 2007-D1-P3; 2013-D2-P1; 2014-D1-P3; 2017-D2-P7; 2018-D2-P9; 2023-D1-P3; 2024-D1-P1
Original trap note	Special-value substitutions must not lose solutions; after guessing a form, verify the full identity.
Literature	[15], [20]

Core facts and properties.

1. If $p(x)-p(y)$ vanishes when $x=y$, it is divisible by $x-y$; the quotient encodes equal-value pairs.
2. Composition respects degree: $\deg p\circ q = (\deg p)(\deg q)$ for nonconstant polynomials over a field.
3. Polynomial remainders modulo $x^q - x$ represent functions on finite fields $\mathbb{F}_q$.

Compact proof mechanisms.

1. Divisibility by $x-y$ follows by treating $p(x)-p(y)$ as a polynomial in x and evaluating at $x=y$.
2. Degree of composition follows from leading terms.
3. For finite fields, every element satisfies $a^q - a = 0$, so reducing modulo $x^q - x$ preserves all values on $\mathbb{F}_q$.

9.5 Algebra / number theory

Domain anchors. [15], [5], [20].

9.5.1 ANT-01: Modular arithmetic and divisibility

Must know	Congruences, modular arithmetic, residues, divisibility, and reducing equations modulo primes or prime powers.
Occurrence	92 tagged problem(s). Selected anchors: 1994-D1-P3; 1994-D1-P6; 1997-D1-P3; 1997-D1-P4; 1999-D1-P3; 1999-D2-P1; 1999-D2-P2; 2000-D1-P5; 2000-D2-P1; 2001-D1-P1; 2001-D1-P2; 2001-D1-P4; 2001-D2-P6; 2002-D1-P3; 2003-D1-P2; 2003-D1-P4; 2005-D1-P2; 2005-D1-P4; ... (full list retained in the theory-node index above; 92 total).
Original trap note	Check whether the modulus is prime, prime power, or composite; cancellation requires invertibility.
Literature	[20]

Core facts and properties.

1. Congruence $a \equiv b \pmod m$ means $m \mid a-b$ and is compatible with addition and multiplication.
2. If a congruence has no solution modulo m, the original integer equation has no solution.

3. For a prime p, nonzero residues form a multiplicative group modulo p.

Compact proof mechanisms.

1. Compatibility follows because divisibility is preserved by sums and by multiplying by integers.
2. Reduction modulo m is a ring homomorphism $\mathbb{Z} \to \mathbb{Z}/m\mathbb{Z}$.
3. Use small residue tables for powers when the modulus is fixed, e.g. fourth powers mod 29.

9.5.2 ANT-02: Chinese remainder theorem and complete residue systems

Must know	Chinese Remainder Theorem, complete residue systems, and lifting/residue splitting.
Occurrence	30 tagged problem(s). Selected anchors: 1999-D1-P3; 2006-D1-P2; 2006-D2-P5; 2007-D1-P1; 2007-D2-P2; 2007-D2-P4; 2010-D1-P4; 2012-D2-P1; 2012-D2-P4; 2013-D1-P5; 2013-D2-P2; 2014-D1-P4; 2014-D2-P1; 2015-D1-P4; 2016-D1-P3; 2016-D1-P5; 2016-D2-P8; 2017-D2-P10; ... (full list retained in the theory-node index above; 30 total).
Original trap note	Uniqueness is modulo the product; remember to exclude disallowed representatives such as zero if the problem says positive.
Literature	[20]

Core facts and properties.

1. If m, n are coprime, then $x \mapsto (x \bmod m, x \bmod n)$ is a bijection $\mathbb{Z}/mn\mathbb{Z} \to \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$.
2. Linear congruence $ax \equiv b \pmod m$ is solvable iff $\gcd(a, m) \mid b$.
3. A complete residue system remains complete after multiplication by a unit.

Compact proof mechanisms.

1. CRT follows by Bezout: choose $um + vn = 1$ and combine residues as $avn + bum$.
2. Linear congruence criterion follows after dividing by $d = \gcd(a, m)$ and using invertibility of a/d modulo m/d.
3. Use CRT when divisibility splits across coprime prime powers.

9.5.3 ANT-03: Finite fields, field characteristic, and Fermat/Frobenius tools

Must know	Finite fields, field characteristic, Fermat's theorem, Frobenius, and polynomial functions over finite fields.
Occurrence	49 tagged problem(s). Selected anchors: 1997-D1-P3; 1999-D2-P1; 1999-D2-P2; 2000-D2-P1; 2000-D2-P3; 2001-D1-P2; 2001-D1-P4; 2003-D1-P2; 2005-D1-P2; 2005-D1-P4; 2005-D2-P2; 2005-D2-P6; 2006-D1-P2; 2006-D1-P3; 2006-D2-P5; 2007-D1-P1; 2007-D2-P2; 2007-D2-P4; ... (full list retained in the theory-node index above; 49 total).
Original trap note	A polynomial of high degree may define the same function as a lower-degree remainder modulo $x^p - x$.
Literature	[15], [20]

Core facts and properties.

1. In a field of characteristic p, $p \cdot 1 = 0$ and $(x + y)^p = x^p + y^p$.
2. Fermat: for prime p, $a^p \equiv a \pmod p$ for all integers a.
3. A polynomial function on $\mathbb{F}_p$ has a unique representative of degree $< p$ modulo $x^p - x$.

Compact proof mechanisms.

1. Frobenius follows because intermediate binomial coefficients $\binom{p}{k}$ are divisible by p.
2. Fermat follows since nonzero residues form a group of order $p - 1$; $a^{p-1} = 1$ for $a \neq 0$.
3. The quotient by $x^p - x$ works because $x^p - x$ vanishes at all field elements.

CORRECTION MATH-002 — legacy statement superseded; click for canonical replacement.

9.5.4 ANT-04: Integer polynomials, finite differences, and value divisibility

Must know	Integer polynomials, finite differences, integer-valued divisibility, and polynomial values modulo primes.
Occurrence	22 tagged problem(s). Selected anchors: 1995-D2-P4; 1996-D2-P2; 1998-D1-P5; 2001-D2-P1; 2002-D2-P3; 2005-D1-P4; 2006-D1-P3; 2006-D1-P4; 2007-D1-P1; 2008-D1-P3; 2008-D2-P1; 2008-D2-P4; 2011-D1-P3; 2012-D1-P5; 2012-D2-P1; 2014-D1-P3; 2014-D2-P1; 2015-D2-P10; ... (full list retained in the theory-node index above; 22 total).
Original trap note	Testing finitely many residues works only when the degree is less than the prime modulus or when using finite differences.
Literature	[20]

Core facts and properties.

1. If $p \in \mathbb{Z}[x]$, then $a - b \mid p(a) - p(b)$ for integers a, b.
2. Finite differences lower degree: $\Delta p(x) = p(x+1) - p(x)$ has degree one less if $\deg p > 0$.
3. A degree d polynomial over a field that vanishes on more than d residues is the zero polynomial.

Compact proof mechanisms.

1. The divisibility $a - b \mid p(a) - p(b)$ follows from $x^k - y^k = (x-y)(x^{k-1} + \cdots + y^{k-1})$.
2. Finite difference leading term: $(x+1)^d - x^d = dx^{d-1} + \cdots$.
3. Quadratic or low-degree divisibility for every integer is often solved by checking enough residues modulo a prime.

9.5.5 ANT-05: Rational roots, Vieta, and rationality constraints

Must know	Rational roots, Vieta, and rationality constraints in number-theoretic settings.
Occurrence	5 tagged problem(s). Selected anchors: 1994-D1-P3; 1996-D2-P2; 2005-D1-P4; 2006-D1-P5; 2018-D1-P3
Original trap note	Do not divide by a coefficient that may be zero; isolate zero roots first.
Literature	[15], [20]

Core facts and properties.

1. Rational root theorem and Vieta's formulas connect rational/integer roots to coefficient divisibility.
2. If a monic integer polynomial has rational roots, then all roots are integers.
3. Symmetric expressions in algebraic conjugates are rational/integer when coefficients are rational/integer.

Compact proof mechanisms.

1. Monic rational-root integrality follows from rational root theorem with leading coefficient 1.
2. Vieta is derived by expanding $a\prod(x - r_i)$ and comparing coefficients.
3. For hyperbolic/trigonometric rationality problems, use recurrences to express higher multiples as polynomials in the base value.

9.5.6 ANT-06: GCD, valuations, prime powers, and coprimality

Must know	GCD, Bezout, valuations, prime powers, and coprimality.
Occurrence	42 tagged problem(s). Selected anchors: 1999-D1-P3; 1999-D2-P1; 2000-D2-P1; 2002-D1-P3; 2003-D1-P4; 2007-D2-P2; 2008-D2-P3; 2012-D1-P1; 2012-D1-P2; 2012-D1-P3; 2012-D2-P3; 2012-D2-P4; 2013-D2-P2; 2014-D2-P1; 2015-D1-P2; 2015-D1-P4; 2016-D1-P2; 2016-D1-P3; ... (full list retained in the theory-node index above; 42 total).
Original trap note	When xy is divisible by p^k and x, y are not coprime, valuation splitting is necessary.
Literature	[20]

Core facts and properties.

1. Bezout: $\gcd(a, b) = d$ iff there exist integers u, v with $ua + vb = d$.
2. Euclid's lemma: if p prime divides ab, then $p \mid a$ or $p \mid b$.
3. Valuation $v_p(n)$ tracks exponent of p in n and satisfies $v_p(ab) = v_p(a) + v_p(b)$.

Compact proof mechanisms.

1. Bezout follows from the Euclidean algorithm and the smallest positive integer combination.
2. Euclid's lemma follows from Bezout applied to p and a.
3. Prime-power arguments should be done prime by prime; never infer divisibility by a product without coprimality.

9.5.7 ANT-07: Group basics: orders, cyclic subgroups, and generators

Must know	Group basics: orders, cyclic subgroups, generators, Lagrange theorem, and element order arguments.
Occurrence	52 tagged problem(s). Selected anchors: 1996-D2-P3; 1999-D1-P3; 1999-D2-P1; 2000-D1-P5; 2001-D1-P2; 2003-D1-P2; 2003-D2-P4; 2005-D1-P6; 2006-D1-P2; 2007-D1-P4; 2007-D2-P5; 2008-D1-P5; 2008-D2-P3; 2009-D2-P5; 2010-D2-P2; 2010-D2-P3; 2011-D1-P3; 2011-D2-P2; ... (full list retained in the theory-node index above; 52 total).
Original trap note	Exponent arithmetic using $(ab)^n = a^n b^n$ requires commutativity unless justified.
Literature	[15], [5]

Core facts and properties.

1. Lagrange: the order of a subgroup divides the order of a finite group.
2. The order of an element divides the group order and is the size of the cyclic subgroup it generates.
3. If $a^m = e$ and $\gcd(m, n) = 1$, Bezout exponents can extract powers such as $a = a^{um+vn}$.

Compact proof mechanisms.

1. Cosets of a subgroup partition the group and all have equal size, proving Lagrange.
2. The cyclic subgroup $\langle a \rangle$ consists of powers until the first return to e.
3. In commutative groups, combine exponent equations; in noncommutative groups, avoid rearranging products.

9.5.8 ANT-08: Permutation groups, cycles, parity, and generation

Must know	Permutation groups, cycles, parity, transpositions, and generation.
Occurrence	2 tagged problem(s). Selected anchors: 1998-D1-P2; 2001-D1-P2
Original trap note	Cycle notation composition order must be consistent throughout the proof.
Literature	[15], [5]

Core facts and properties.

1. Every permutation decomposes into disjoint cycles, uniquely up to order and cyclic rotation.
2. Transpositions generate S_n; adjacent transpositions also generate S_n.
3. The sign homomorphism records parity of the number of transpositions in a decomposition.

Compact proof mechanisms.

1. Cycle decomposition follows by following orbits of the permutation action on $\{1, \ldots, n\}$.
2. Any transposition (ij) is a product of adjacent transpositions; any cycle is product of transpositions.
3. Generation problems are solved by producing enough transpositions or a known generating set.

9.5.9 ANT-09: Normal subgroups, quotient groups, and group actions

Must know	Normal subgroups, quotients, group actions, orbit-stabilizer, and invariants under conjugation.
Occurrence	12 tagged problem(s). Selected anchors: 1996-D2-P3; 1999-D2-P1; 2005-D1-P6; 2010-D2-P3; 2012-D1-P3; 2012-D2-P5; 2016-D2-P8; 2017-D1-P4; 2018-D1-P2; 2020-D2-P7; 2021-D2-P6; 2024-D1-P4
Original trap note	A subgroup of the same index need not be normal; prove normality before passing to a quotient.
Literature	[15], [5]

Core facts and properties.

1. A subgroup N is normal iff left cosets equal right cosets iff $gNg^{-1} = N$ for all g.

CORRECTION MATH-003 — legacy statement superseded; click for canonical replacement.

2. Quotient group G/N exists exactly when N is normal.
3. Orbit-stabilizer: $|Gx| = |G|/|G_x|$ for finite group actions.

Compact proof mechanisms.

1. Normality ensures coset multiplication $(gN)(hN) = ghN$ is well-defined.
2. Orbit-stabilizer follows from the bijection $G/G_x \to Gx$.
3. Use quotients to prove non-generation by showing the generated image would be too small/cyclic/abelian.

9.5.10 ANT-10: Automorphisms and semidirect products

Must know	Automorphisms, semidirect products, and groups built from actions.
Occurrence	27 tagged problem(s). Selected anchors: 1996-D2-P3; 1998-D1-P2; 1999-D1-P1; 1999-D1-P5; 1999-D2-P1; 2000-D1-P5; 2001-D1-P2; 2003-D1-P2; 2003-D2-P4; 2004-D2-P3; 2005-D1-P6; 2007-D1-P4; 2007-D2-P5; 2008-D1-P5; 2010-D2-P2; 2010-D2-P3; 2011-D1-P3; 2011-D2-P4; ... (full list retained in the theory-node index above; 27 total).
Original trap note	Counting automorphisms by images of generators requires proving those images satisfy the same defining relations.
Literature	[15], [5]

Core facts and properties.

1. $\mathrm{Aut}(G)$ is the group of isomorphisms $G \to G$.
2. A semidirect product $H \rtimes_\phi K$ has multiplication $(h,k)(h',k') = (h\phi_k(h'), kk')$.
3. Automorphisms preserve element orders, centers, commutator subgroups, and other characteristic structures.

Compact proof mechanisms.

1. Associativity in a semidirect product follows from $\phi_{kk'} = \phi_k\phi_{k'}$.
2. To bound $|\mathrm{Aut}\, G|$, determine images of a generating set and use order/invariant constraints.
3. For vector spaces over $\mathbb{F}_q$, automorphisms are invertible matrices: $|GL_n(q)| = (q^n-1)(q^n-q)\cdots(q^n-q^{n-1})$.

9.5.11 ANT-11: Rings, idempotents, nil rings, and noncommutativity

Must know	Rings, idempotents, nil rings, characteristic, noncommutativity, and commutator caution.
Occurrence	33 tagged problem(s). Selected anchors: 1996-D2-P3; 1999-D1-P5; 1999-D2-P1; 2000-D1-P5; 2001-D1-P2; 2003-D1-P2; 2003-D2-P4; 2005-D1-P6; 2006-D1-P2; 2007-D1-P4; 2007-D2-P5; 2008-D1-P5; 2009-D2-P4; 2010-D2-P3; 2011-D1-P3; 2011-D2-P2; 2012-D1-P3; 2012-D1-P5; ... (full list retained in the theory-node index above; 33 total).
Original trap note	Do not commute factors; use associativity and given identities only.
Literature	[15], [5]

Core facts and properties.

1. Ring characteristic is the additive order of 1 if it exists; characteristic zero means no nonzero element is killed by a positive integer.
2. Idempotent $e^2 = e$; nilpotent $x^n = 0$; square-zero identities often imply anticommutation via $(a+b)^2 = 0$.
3. In noncommutative rings, expand in order and never reorder factors.

Compact proof mechanisms.

1. If every square is zero, $(a+b)^2 = 0$ gives $ab + ba = 0$.
2. Idempotent equations like $e + f + g = 0$ can be multiplied on left/right by idempotents to force commutativity or vanishing under characteristic assumptions.
3. Characteristic zero allows cancellation of integer multiples: $na = 0 \Rightarrow a = 0$.

9.5.12 ANT-12: Diophantine equations, Pell-type recurrences, and quadratic irrationals

Must know	Diophantine equations, Pell-type recurrences, and quadratic irrationals.
Occurrence	2 tagged problem(s). Selected anchors: 2005-D2-P6; 2006-D2-P5
Original trap note	Verify determinant-one and nontriviality conditions after solving the Pell equation.
Literature	[20]

Core facts and properties.

1. Pell equation $x^2 - Dy^2 = 1$ has infinitely many solutions for nonsquare $D > 0$.
2. Quadratic irrationals have conjugates; rational/integer expressions are often obtained by symmetric functions of conjugates.
3. Matrices in $SL_2(\mathbb{Z})$ acting by fractional linear transformations fix roots of quadratic equations.

Compact proof mechanisms.

1. Pell solutions arise from powers of a fundamental unit in $\mathbb{Z}[\sqrt{D}]$.
2. If r satisfies $ur^2 + vr + w = 0$, the condition $(ar + b)/(cr + d) = r$ is equivalent to a scalar multiple of this polynomial.
3. Use congruence/recurrence of Pell solutions to meet divisibility side conditions.

9.5.13 ANT-13: Lattices and norms/quasinorms

Must know	Lattices, integer hyperplanes, norms/quasinorms, and nearest lattice point inequalities.
Occurrence	2 tagged problem(s). Selected anchors: 1994-D1-P6; 1997-D1-P5
Original trap note	Convexity fails for $p < 1$; do not use triangle inequality in the quasi-norm case.
Literature	[26], [17]

Core facts and properties.

1. A lattice is a discrete additive subgroup generated by a basis over $\mathbb{Z}$.
2. For $p \geq 1$, $\|x\|_p = (\sum |x_i|^p)^{1/p}$ is a norm; for $0 < p < 1$ it is not convex and triangle inequality fails.
3. Integer shifts in a zero-sum hyperplane are controlled by sign partitions and convexity of $|t|^p$ for $p \geq 1$.

Compact proof mechanisms.

1. Convexity of $|t|^p$ for $p \geq 1$ gives comparison inequalities coordinatewise.
2. For $0 < p < 1$, concavity on $\mathbb{R}_+$ lets mass concentration decrease the quasi-norm, producing counterexamples.
3. Always use $\sum y_i = 0$ when shifting by a lattice vector in $\mathbb{Z}_0^n$.

9.5.14 ANT-14: Greedy infinite products and rationality tails

Must know	Greedy infinite products and rationality tails.
Occurrence	1 tagged problem(s). Selected anchors: 1997-D1-P4
Original trap note	A greedy inequality must be sharp in both directions to prove uniqueness.
Literature	[2], [20]

Core facts and properties.

1. Greedy algorithms choose the least integer satisfying a local inequality, producing uniqueness by forced first choice.
2. Products $\prod_{j\geq 0}(1 + M^{-2^j}) = M/(M-1)$ telescope via binary expansion.
3. For rationality, track numerator-denominator differences as positive integers and show strict descent unless the tail becomes telescoping.

Compact proof mechanisms.

1. The product identity follows because each subset of powers $1, 2, 4, \ldots$ represents a unique exponent, so the expansion is $1 + M^{-1} + M^{-2} + \cdots$.
2. Uniqueness follows: a wrong greedy choice makes the remaining ratio either below 1 or too large to be corrected by the maximal possible tail.
3. Descent on positive integers is the standard contradiction for rational nonterminating greedy processes.

9.5.15 ANT-15: Finite abelian groups and vector spaces over finite fields

Must know	Finite abelian groups and vector spaces over finite fields, especially elementary abelian 2-groups.
Occurrence	1 tagged problem(s). Selected anchors: 2003-D2-P4
Original trap note	The added identity element must be included before claiming group structure.
Literature	[15], [5]

Core facts and properties.

1. A finite abelian group in which every nonidentity element has order 2 is a vector space over $\mathbb{F}_2$.
2. Such a group has order 2^m for some m.
3. Operations satisfying $a * a = 0$, commutativity, associativity, and identity 0 define an elementary abelian 2-group.

Compact proof mechanisms.

1. Define scalar multiplication by $0v = 0$, $1v = v$; group addition is the vector addition. The order-2 condition gives $v + v = 0$.
2. A finite vector space of dimension m over $\mathbb{F}_2$ has 2^m elements.
3. Design problems where any pair determines a third often hide this group law.

9.6 Combinatorics / discrete structures

Domain anchors. [2], [30], [29], [31].

9.6.1 CMB-01: Pigeonhole principle and extremal element selection

Must know	Pigeonhole principle, extremal element selection, minimal counterexamples, and finite collision arguments.
Occurrence	68 tagged problem(s). Selected anchors: 1994-D1-P1; 1994-D1-P3; 1994-D1-P6; 1994-D2-P4; 1995-D1-P3; 1996-D1-P3; 1996-D2-P3; 1996-D2-P6; 1997-D1-P6; 1997-D2-P6; 1998-D2-P3; 1998-D2-P4; 1998-D2-P5; 1998-D2-P6; 1999-D1-P2; 1999-D1-P5; 1999-D2-P6; 2000-D1-P1; ... (full list retained in the theory-node index above; 68 total).
Original trap note	Define the boxes explicitly; hidden boxes often are residue classes, divisors, or graph neighborhoods.
Literature	[2]

Core facts and properties.

1. Pigeonhole: more than kn objects in n boxes forces some box to contain at least $k + 1$ objects.
2. Extremal principle: in a finite nonempty set, choose an object maximizing/minimizing a useful parameter.
3. Minimal counterexample arguments work by proving any minimal bad object can be reduced or improved.

Compact proof mechanisms.

1. Pigeonhole is counting: if every box had at most k objects, total would be at most kn.
2. Extremal arguments are proof by contradiction plus finite existence of max/min.
3. In IMC, use this for residue collisions, repeated values, largest/smallest elements, and local-to-global contradictions.

9.6.2 CMB-02: Double counting and degree-sum methods

Must know	Double counting, degree sums, walk counts, and incidence matrices.
Occurrence	56 tagged problem(s). Selected anchors: 1994-D1-P1; 1994-D2-P4; 1995-D1-P3; 1996-D1-P3; 1996-D2-P3; 1996-D2-P6; 1997-D2-P6; 1998-D2-P3; 1998-D2-P4; 1998-D2-P5; 1998-D2-P6; 1999-D1-P5; 2000-D1-P1; 2001-D2-P1; 2001-D2-P3; 2002-D1-P4; 2002-D1-P5; 2002-D2-P2; ... (full list retained in the theory-node index above; 56 total).
Original trap note	Specify ordered vs unordered objects; many IMC counts hinge on this distinction.
Literature	[30], [31]

Core facts and properties.

1. Count the same finite set in two ways, usually by summing over rows versus columns of an incidence relation.

2. Graph handshaking: $\sum_v \deg(v) = 2|E|$.
3. Counting walks of length 2 from degrees gives $\sum_v d(v)^2$.

Compact proof mechanisms.

1. The handshaking lemma counts incidences (v, e) with $v \in e$; each edge contributes two.
2. Double counting often proves existence by average: if total exceeds a threshold, one object exceeds the average.
3. When equality holds in a double-counting inequality, inspect where injectivity/surjectivity became equality.

9.6.3 CMB-03: Graph encoding, cycles, and bipartite representations

Must know	Graph encoding, cycles, bipartite representations, directed acyclic graphs, and topological order.
Occurrence	68 tagged problem(s). Selected anchors: 1994-D1-P3; 1994-D1-P5; 1994-D2-P3; 1995-D1-P4; 1995-D2-P1; 1996-D1-P6; 1996-D2-P2; 1997-D1-P6; 1997-D2-P5; 1997-D2-P6; 1998-D1-P1; 1998-D1-P2; 1998-D2-P3; 1998-D2-P4; 1998-D2-P5; 1999-D1-P4; 1999-D1-P5; 1999-D1-P6; ... (full list retained in the theory-node index above; 68 total).
Original trap note	When encoding a grid, rows and columns often become vertices and marked cells become edges.
Literature	[31], [2]

Core facts and properties.

1. A graph with N vertices and at least N edges contains a cycle.
2. Bipartite graphs encode grid row/column incidence and alternating row-column cycles.
3. A finite directed graph is acyclic iff it has a topological ordering.

Compact proof mechanisms.

1. A forest on N vertices has at most $N - 1$ edges; prove by induction deleting leaves.
2. Grid marked points become edges between row vertices and column vertices; cycles translate to alternating marked-point sequences.
3. Every finite DAG has a source; remove it inductively to topologically order vertices.

9.6.4 CMB-04: Set systems, intersecting families, and induction on sizes

Must know	Set systems, intersecting families, induction on sizes, and finite hitting sets.
Occurrence	23 tagged problem(s). Selected anchors: 1997-D1-P6; 1997-D2-P5; 1997-D2-P6; 1998-D1-P1; 1999-D2-P5; 2002-D2-P2; 2004-D1-P1; 2005-D2-P4; 2007-D2-P1; 2008-D1-P4; 2011-D1-P4; 2012-D2-P3; 2013-D2-P5; 2014-D2-P4; 2016-D1-P4; 2017-D1-P3; 2019-D2-P8; 2020-D1-P3; ... (full list retained in the theory-node index above; 23 total).
Original trap note	Uniformity assumptions are often essential; construct counterexamples without them.
Literature	[30]

Core facts and properties.

1. For finite set families, partition by intersection with a fixed set to reduce cardinalities.
2. Intersecting-family arguments often require a finite transversal/hitting set under bounded set-size hypotheses.
3. Induction on pair (r, s) is natural when two uniform families interact.

Compact proof mechanisms.

1. Fix A_0, B_0 and classify members by $C = A \cap (A_0 \cup B_0)$; if two classes are disjoint, remove fixed parts and reduce sizes.
2. Finite union of finite hitting sets remains finite, closing induction.
3. Without uniform bounded sizes, counterexamples can push intersections farther and farther out.

9.6.5 CMB-05: Recurrences and state partitions

Must know	Recurrences and state partitions for finite strings, functions, and constrained configurations.
Occurrence	4 tagged problem(s). Selected anchors: 1998-D2-P4; 2002-D2-P2; 2005-D1-P2; 2013-D2-P5
Original trap note	The state must contain enough memory to extend objects without ambiguity.
Literature	[29]

Core facts and properties.

1. Partition configurations by the last state or last few symbols to get a finite linear recurrence.
2. If two sequences satisfy the same recurrence and enough initial values agree, they are identical.
3. State refinement should remember exactly the information needed to extend by one step.

Compact proof mechanisms.

1. For a recurrence of order r, induction proves equality after r initial conditions.
2. In constrained words, states are usually suffixes of length equal to the forbidden pattern length minus one.
3. Avoid overcounting by making the state classes disjoint and exhaustive.

9.6.6 CMB-06: Generating functions for counting

Must know	Generating functions for counting and coefficient extraction.
Occurrence	14 tagged problem(s). Selected anchors: 2000-D1-P6; 2002-D1-P3; 2002-D2-P3; 2003-D1-P6; 2003-D2-P6; 2008-D2-P3; 2012-D2-P2; 2015-D2-P8; 2017-D2-P8; 2017-D2-P9; 2018-D2-P7; 2019-D1-P4; 2021-D1-P3; 2024-D2-P8
Original trap note	Formal power series do not require convergence, but analytic evaluations do.
Literature	[29], [32]

Core facts and properties.

1. Ordinary generating function $A(x) = \sum a_n x^n$ turns recurrences into algebraic or differential equations.
2. Convolution corresponds to multiplication: coefficient of x^n in $A(x)B(x)$ is $\sum_k a_k b_{n-k}$.
3. Probability generating functions encode sums of independent discrete variables by products.

Compact proof mechanisms.

1. Multiplication proof is coefficient collection in the Cauchy product.
2. Differentiation marks indices: $xA'(x) = \sum n a_n x^n$.
3. After solving a generating-function equation, extract the value needed, often at a special x inside the convergence radius.

9.6.7 CMB-07: Inclusion-exclusion and root-of-unity filters

Must know	Inclusion-exclusion and root-of-unity filters for counting by congruence or avoiding properties.
Occurrence	1 tagged problem(s). Selected anchors: 1999-D2-P2
Original trap note	Root-of-unity filters count residues of exponents/sums; ensure the generating function encodes the correct statistic.
Literature	[29], [2]

Core facts and properties.

1. Inclusion-exclusion: count objects with none of the bad properties by alternating sums over intersections of bad property sets.
2. Root-of-unity filters select terms whose exponent lies in a residue class.
3. These methods combine when forbidden conditions are periodic or modular.

Compact proof mechanisms.

1. Inclusion-exclusion follows by checking each object with exactly r bad properties contributes $\sum_{j=0}^{r} (-1)^j \binom{r}{j} = 0$ unless $r = 0$.
2. Root-of-unity filters follow from finite geometric sums.
3. Use filters to count dice sums, word sums, and polynomial coefficient classes.

9.6.8 CMB-08: Permutations, inversions, and permutation statistics

Must know	Permutations, inversions, displacement statistics, determinant sign sums, and parity.
Occurrence	5 tagged problem(s). Selected anchors: 2001-D1-P1; 2003-D1-P2; 2009-D2-P4; 2013-D2-P5; 2022-D2-P6
Original trap note	Distinguish permutation as a function from its one-line notation.
Literature	[29]

Core facts and properties.

1. The determinant expansion sums over permutations with sign $(-1)^{\operatorname{inv}\sigma}$.
2. Permutation statistics can be encoded as coefficients of determinants or generating functions.
3. Parity of a count may be obtained by reducing a signed sum modulo 2, where signs disappear.

Compact proof mechanisms.

1. Determinant expansion is the alternating multilinear formula in columns.
2. If $Q(n,d)$ counts permutations with statistic d, its parity equals the coefficient modulo 2 in the signed determinant generating function.
3. Evaluate structured determinants to show high-degree coefficients vanish.

9.6.9 CMB-09: Explicit construction and counterexample design

Must know	Explicit construction and counterexample design.
Occurrence	36 tagged problem(s). Selected anchors: 1994-D1-P4; 1994-D2-P5; 1995-D1-P6; 1995-D2-P6; 1997-D1-P2; 1997-D1-P4; 1997-D1-P6; 1997-D2-P1; 1997-D2-P6; 1998-D1-P2; 1999-D1-P2; 1999-D1-P6; 1999-D2-P5; 2000-D1-P1; 2002-D1-P2; 2003-D2-P4; 2004-D2-P3; 2006-D1-P1; ... (full list retained in the theory-node index above; 36 total).
Original trap note	A counterexample must satisfy every hypothesis exactly, including continuity, monotonicity, or finiteness.
Literature	[2]

Core facts and properties.

1. To prove existence, construct objects with parameters and verify all constraints explicitly.
2. To disprove a statement, violate one necessary hypothesis while preserving the others.
3. Sharpness examples usually come from equality cases of inequalities or extremal small configurations.

Compact proof mechanisms.

1. A construction proof is complete only after checking membership, all constraints, and the target conclusion.
2. Counterexamples should be minimal: small dimension, step functions, special matrices, or modular residues.
3. When a problem asks 'does there exist', test boundary cases before searching for a theorem.

9.6.10 CMB-10: Erdos-Szekeres style monotone subsequences and binomial thresholds

Must know	Erdos-Szekeres style monotone subsequences, binomial thresholds, and induction on two target lengths.
Occurrence	1 tagged problem(s). Selected anchors: 2004-D1-P5
Original trap note	The induction parameter may be two-dimensional: length of increasing and decreasing alternatives.
Literature	[2]

Core facts and properties.

1. The classical Erdős-Szekeres theorem uses labels: sequence length $(r-1)(s-1)+1$ forces increasing length r or decreasing length s.
2. Binomial thresholds often satisfy Pascal recurrences for induction with a split into lower/upper halves.
3. Strengthened monotonicity conditions can be preserved by appending an extremal endpoint.

Compact proof mechanisms.

1. For binomial threshold $\binom{k+l-4}{k-2}+1$, split at midpoint; Pascal identity gives one side large enough for induction.

2. If induction gives a shorter increasing sequence in the lower half, append the maximum to satisfy doubling distance.
3. The proof is a template: split, use Pascal, append extremal element.

9.6.11 CMB-11: Grid, lattice, and alternating-cycle arguments

Must know	Grid, lattice, and alternating-cycle arguments.
Occurrence	2 tagged problem(s). Selected anchors: 1999-D1-P5; 2023-D2-P6
Original trap note	Deleting rows/columns with too few points preserves inequalities only if the count is adjusted correctly.
Literature	[31]

Core facts and properties.

1. Marked cells in an $n \times k$ grid correspond to edges of a bipartite graph between row and column vertices.
2. A cycle in this graph gives an alternating sequence of marked points sharing rows and columns.
3. Deleting rows/columns with at most one mark preserves surplus-edge conditions.

Compact proof mechanisms.

1. Graph encoding: cell (r, c) becomes edge $r - c$; an even cycle alternates row and column vertices.
2. In deletion proofs, each deletion reduces vertices and marks by at most one, preserving marks at least vertices.
3. When no sparse row/column remains, follow alternating choices until repetition forces a cycle.

9.6.12 CMB-12: Coloring, partitions, and parity

Must know	Coloring, partitions, parity, and sign assignments.
Occurrence	48 tagged problem(s). Selected anchors: 1994-D1-P3; 1997-D1-P6; 1997-D2-P5; 1997-D2-P6; 1998-D1-P1; 1998-D2-P3; 1998-D2-P4; 1999-D1-P6; 1999-D2-P5; 2000-D2-P4; 2001-D2-P5; 2002-D1-P1; 2002-D2-P2; 2002-D2-P4; 2003-D1-P4; 2003-D2-P3; 2004-D1-P1; 2004-D1-P4; ... (full list retained in the theory-node index above; 48 total).
Original trap note	Parity can change at boundary vertices; track local contributions, not just total counts.
Literature	[2]

Core facts and properties.

1. Parity is invariant modulo 2; many incidence sums are controlled by whether counts are odd/even.
2. Colorings often encode a function $f : M \to \{-1, 1\}$ so balanced conditions become sums equal to zero.
3. Partition arguments split a set into disjoint classes and count contributions classwise.

Compact proof mechanisms.

1. A balance condition 'exactly half white' becomes $\sum f = 0$ on each tested subset.
2. Parity of triangulation incidences changes predictably when adding ears or triangles.
3. Always specify whether partitions are ordered or unordered and whether empty parts are allowed.

9.6.13 CMB-13: Words, equivalence relations, and rewriting systems

Must know	Words, equivalence relations, rewriting systems, and finite reduction.
Occurrence	3 tagged problem(s). Selected anchors: 1999-D2-P5; 2015-D2-P8; 2020-D1-P1
Original trap note	A rewrite rule must be compatible with left and right concatenation before applying it inside a word.
Literature	[23]

Core facts and properties.

1. A rewriting rule is valid in a congruence if it can be used inside any context: $v \sim w \Rightarrow uv \sim uw$ and $vu \sim wu$.
2. Idempotent word rule $uu \sim u$ deletes repeated adjacent blocks.
3. To prove bounded normal length, show every long word is equivalent to a shorter one.

Compact proof mechanisms.

1. If a word contains all letters, append a suitable suffix so that a prescribed following word can be absorbed using $uu \sim u$.
2. For long words, split into prefix-middle-suffix and use absorption from both sides to remove the middle.
3. This is semigroup congruence reasoning; every step must preserve context.

9.6.14 CMB-14: Transfer matrices and matrix powers for counting

Must know	Transfer matrices and matrix powers for counting constrained arrays.
Occurrence	1 tagged problem(s). Selected anchors: `2004-D2-P6`
Original trap note	Make sure the matrix state represents a whole column/row, not a single cell.
Literature	[29]

Core facts and properties.

1. If states are columns/rows and compatibility is encoded by a 0-1 matrix A, then the number of length-k sequences is a sum of entries of A^{k-1}.
2. Symmetry of a two-dimensional counting problem may imply $S(A_n^{k-1}) = S(A_k^{n-1})$ after reinterpreting both as the same array count.
3. Recursive block construction of transfer matrices mirrors adding one row/column.

Compact proof mechanisms.

1. Matrix multiplication counts paths: $(A^{k-1})_{ij}$ counts compatible state sequences from i to j.
2. Summing all entries removes fixed endpoints.
3. Combinatorial interpretation is often simpler than algebraic manipulation of the matrices.

9.6.15 CMB-15: Finite designs and pair-determines-third structures

Must know	Finite designs, pair-determines-third structures, and Steiner triple systems with algebraic closure.
Occurrence	1 tagged problem(s). Selected anchors: `2003-D2-P4`
Original trap note	The operation is initially undefined on equal elements; adding an identity often completes the algebraic structure.
Literature	[30], [15]

Core facts and properties.

1. A family of triples in which every pair lies in exactly one triple defines a partial operation $a * b = c$.
2. Additional associativity-like closure can force an elementary abelian 2-group structure after adding an identity.
3. Such systems exist on $2^m - 1$ nonzero vectors of $\mathbb{F}_2^m$ with triples $\{a, b, a + b\}$.

Compact proof mechanisms.

1. Define $a * a = 0$ and $a * 0 = a$; pair uniqueness gives cancellation and order two.
2. The closure condition gives associativity for distinct elements; case checking extends it to all elements with 0.
3. Finite elementary abelian group size is 2^m, so the original set has size $2^m - 1$.

9.7 Geometry / convexity / metric

Domain anchors. [7], [26], [9].

9.7.1 GEO-01: Coordinate and affine normalization

Must know	Coordinate and affine normalization: translating, rotating, scaling, and choosing convenient axes.
Occurrence	2 tagged problem(s). Selected anchors: 2014-D1-P5; 2014-D2-P4
Original trap note	Only affine-invariant quantities survive affine transformations; Euclidean distances require isometries or controlled scalings.
Literature	[7]

Core facts and properties.

1. Euclidean distances are preserved by translations and orthogonal transformations; ratios/affine incidence are preserved by affine maps.
2. A problem with a line and point in $\mathbb{R}^3$ can often be normalized to a coordinate axis and one coordinate point.
3. Volume scales by absolute determinant under a linear map.

Compact proof mechanisms.

1. Orthogonal transformations preserve dot products, hence distances and angles.
2. Affine maps preserve barycentric coordinates and ratios along lines but not necessarily angles.
3. After solving in normalized coordinates, transform back and multiply volume/area by the determinant factor.

9.7.2 GEO-02: Euclidean distances and inner-product identities

Must know	Euclidean distances, dot-product identities, and distance-to-inner-product conversion.
Occurrence	19 tagged problem(s). Selected anchors: 1994-D2-P5; 1995-D2-P1; 1996-D1-P6; 1998-D1-P6; 2002-D1-P6; 2004-D1-P4; 2004-D2-P3; 2005-D2-P5; 2006-D2-P4; 2008-D1-P5; 2008-D2-P2; 2010-D1-P1; 2010-D1-P5; 2013-D2-P3; 2014-D1-P5; 2014-D2-P4; 2015-D2-P9; 2018-D2-P6; ... (full list retained in the theory-node index above; 19 total).
Original trap note	Square distances only after confirming all quantities are nonnegative.
Literature	[7], [6]

Core facts and properties.

1. $\|u-v\|^2 = \|u\|^2 + \|v\|^2 - 2\langle u, v\rangle$.
2. Orthogonality is $\langle u, v\rangle = 0$; angle cosine is $\langle u, v\rangle/(\|u\|\|v\|)$.
3. Pairwise equal-distance configurations are studied through their Gram matrices.

Compact proof mechanisms.

1. The distance identity follows by expanding $\langle u-v, u-v\rangle$.
2. For rational distances from the origin and pairwise rational distances, inner products are rational after squaring.
3. Use squared distances to avoid square roots whenever possible.

9.7.3 GEO-03: Projections and orthogonal decompositions

Must know	Projections, orthogonal decompositions, nearest points, and projecting geometric regions.
Occurrence	25 tagged problem(s). Selected anchors: 1995-D2-P3; 1996-D2-P4; 1997-D1-P5; 1997-D2-P5; 1998-D2-P5; 1999-D2-P6; 2002-D1-P1; 2002-D2-P4; 2002-D2-P5; 2004-D1-P4; 2004-D2-P2; 2004-D2-P3; 2005-D2-P5; 2008-D2-P2; 2009-D2-P1; 2011-D2-P5; 2014-D1-P5; 2015-D1-P5; ... (full list retained in the theory-node index above; 25 total).
Original trap note	A projection can collapse dimensions; track signed area/volume if orientation matters.
Literature	[7], [12]

Core facts and properties.

1. Orthogonal projection onto a closed subspace minimizes distance uniquely in Hilbert/Euclidean space.
2. Projection onto a line/plane is linear and reduces distances in perpendicular directions.
3. The signed area/volume of a projection is multiplied by the cosine of the angle between normals.

Compact proof mechanisms.

1. For a subspace U, decompose $x = u + u^{\perp}$; then $\|x-v\|^2 = \|u-v\|^2 + \|u^{\perp}\|^2$ for $v \in U$.

2. Projection length/area formulas follow from decomposing an orthonormal basis and taking determinants.
3. Use projection to turn 3D angle/area statements into planar inequalities.

9.7.4 GEO-04: Convex bodies, supporting geometry, and John-ellipse ideas

Must know	Convex bodies, supporting geometry, symmetric convex sets, and John-ellipse ideas.
Occurrence	7 tagged problem(s). Selected anchors: 1996-D2-P4; 2011-D2-P3; 2014-D1-P5; 2014-D2-P4; 2014-D2-P5; 2015-D1-P3; 2015-D1-P4
Original trap note	The proof often needs a compactness gap between an arc and the boundary.
Literature	[26], [7]

Core facts and properties.

1. A convex set contains every segment between its points; a supporting line/hyperplane touches the set while keeping it on one side.
2. John's ellipsoid is the maximum-volume ellipsoid contained in a convex body; in symmetric planar problems the unit disk may be the maximal ellipse.
3. If a boundary arc is missing, a slight ellipse deformation can create a larger contained ellipse, contradiction.

Compact proof mechanisms.

1. Compactness gives existence of distance between a compact arc and a closed boundary when disjoint.
2. Ellipse deformation is checked by explicit semiaxes and area comparison.
3. Convexity ensures that small neighborhoods inside a missing boundary arc remain inside the body.

9.7.5 GEO-05: Conics: parabolas, ellipses, foci, and directrices

Must know	Conics: parabolas, ellipses, foci/directrices, reflections, and algebraic equations.
Occurrence	4 tagged problem(s). Selected anchors: 2000-D2-P4; 2002-D1-P1; 2008-D2-P2; 2014-D1-P5
Original trap note	Tangency to a line imposes double roots after subtracting the line equation.
Literature	[7], [13]

Core facts and properties.

1. Standard parabola with vertex (v, w) has equation $y = (x - v)^2 + w$.
2. An ellipse can be described by focus-directrix data with eccentricity $e < 1$.
3. Two conics generally intersect in at most four points, but extra shared focus/directrix relations may reduce intersections to a line and hence at most two points.

Compact proof mechanisms.

1. The parabola reflection identity follows by substituting coordinates: $b = (a - v)^2 + w$ iff $-w = (v - a)^2 - b$.
2. Focus-directrix condition $FX = e\, d(X, \ell)$ lets equality for two ellipses eliminate FX and gives a linear relation between distances to directrices.
3. A line intersects a nondegenerate ellipse in at most two points by substituting a linear parameter into a quadratic equation.

9.7.6 GEO-06: Simplex, barycentric coordinates, and circumcenters

Must know	Simplex, barycentric coordinates, volumes, and circumcenters.
Occurrence	6 tagged problem(s). Selected anchors: 1997-D2-P5; 2008-D2-P2; 2009-D1-P5; 2009-D2-P1; 2014-D1-P5; 2017-D2-P10
Original trap note	Volume weights must use consistent orientation or absolute volumes depending on the statement.
Literature	[7]

Core facts and properties.

1. A point inside a simplex has barycentric coordinates $P = \sum \lambda_i P_i$, $\lambda_i > 0$, $\sum \lambda_i = 1$.
2. The volume of the sub-simplex replacing vertex P_i by P is λ_i times the original volume.
3. A circumcenter is the unique point equidistant from all simplex vertices when they are affinely independent.

Compact proof mechanisms.

1. Barycentric-volume relation follows because both volumes use the same opposite face and heights scale linearly with distance to that face.
2. The circumcenter solves linear equations obtained by subtracting squared-distance equations.
3. Vector identities with volume weights are often barycentric identities in disguise.

9.7.7 GEO-07: Area, volume, projection, and ellipsoid computation

Must know	Area, volume, projection, ellipsoid computation, and determinant scaling.
Occurrence	28 tagged problem(s). Selected anchors: 1995-D2-P3; 1996-D2-P4; 1997-D1-P5; 1997-D2-P5; 1998-D2-P5; 1999-D2-P6; 2000-D2-P1; 2002-D1-P1; 2002-D2-P4; 2002-D2-P5; 2004-D1-P4; 2004-D2-P2; 2004-D2-P3; 2005-D2-P5; 2008-D2-P2; 2009-D2-P1; 2011-D2-P5; 2014-D1-P5; ... (full list retained in the theory-node index above; 28 total).
Original trap note	Complete the square before identifying the ellipsoid axes.
Literature	[4], [7]

Core facts and properties.

1. Volume of the unit ball in $\mathbb{R}^3$ is $4\pi/3$; an ellipsoid with semiaxes a, b, c has volume $(4\pi/3)abc$.
2. A linear transformation scales n-dimensional volume by $|\det T|$.
3. Area under polynomial curves is computed by integrating the vertical difference.

Compact proof mechanisms.

1. Ellipsoid volume follows by applying the diagonal map $(u, v, w) \mapsto (au, bv, cw)$ to the unit ball.
2. Determinant scaling follows first for boxes/parallelepipeds and then by approximation.
3. When inequalities define a quadratic region, complete squares and diagonalize/scale to a unit ball.

9.7.8 GEO-08: Metric content, diameter covers, and contractions

Must know	Metric content, diameter covers, contractions, and length lower bounds.
Occurrence	1 tagged problem(s). Selected anchors: 1996-D1-P6
Original trap note	The covering sets need not be intervals; project or contract them to a segment.
Literature	[10]

Core facts and properties.

1. Diameter of a set is $\sup d(x, y)$; a contraction is a 1-Lipschitz map.
2. A cover of a segment by finitely many sets has sum of diameters at least the segment length.
3. If a contraction maps E onto a segment L, then any cover of E maps to a cover of L with no larger diameters.

Compact proof mechanisms.

1. Segment-cover lower bound follows by induction on number of cover sets: remove a set covering an endpoint and shorten the segment by at most its diameter.
2. Contractions satisfy $\operatorname{diam} f(E_i) \le \operatorname{diam} E_i$.
3. Projection onto an axis is a contraction, giving lower bounds for upper content.

9.7.9 GEO-09: Hilbert-space equidistant sets and orthonormal translations

Must know	Hilbert-space equidistant sets and orthonormal translations.
Occurrence	6 tagged problem(s). Selected anchors: 1995-D2-P1; 2002-D2-P6; 2008-D2-P6; 2013-D2-P3; 2018-D2-P6; 2021-D2-P8
Original trap note	Completeness of the Hilbert space is used to take the barycenter limit.
Literature	[12]

Core facts and properties.

1. If vectors e_i are orthonormal, then points $(d/\sqrt{2})e_i$ are pairwise distance d.
2. Conversely, a set of pairwise equidistant points in a Hilbert space is a translate of such a scaled orthonormal set.

3. Centers of finite equidistant sets are averages adjusted in an orthogonal direction; infinite centers arise as limits of averages.

Compact proof mechanisms.

1. For equidistant points x_i, compute inner products of differences using the polarization identity; the values match the standard simplex model.
2. Averages $y_n = (x_1 + \cdots + x_n)/n$ form a Cauchy sequence because $\|y_m - y_n\|^2$ can be computed universally.
3. The limit y satisfies $\langle x_i - y, x_j - y\rangle = 0$ and $\|x_i - y\| = d/\sqrt{2}$.

9.7.10 GEO-10: Closest points to closed sets and density of uniqueness

Must know	Closest points to closed sets and density of uniqueness.
Occurrence	1 tagged problem(s). Selected anchors: `2003-D2-P3`
Original trap note	Closedness is needed for existence; uniqueness is obtained by geometry of balls, not by convexity.
Literature	[7], [10]

Core facts and properties.

1. A closed set intersected with a sufficiently large closed ball is compact, so nearest points exist.
2. If a point has two nearest points, moving slightly toward one of them often creates a unique nearest point.
3. The set of points with unique nearest point to a closed set is dense in Euclidean space.

Compact proof mechanisms.

1. Existence: choose minimizing sequence; it lies in a bounded ball and has a convergent subsequence whose limit lies in the closed set.
2. If a_0 is a nearest point to b_0, then for any point on the open segment (a_0, b_0) the ball centered there with radius to a_0 lies inside the original nearest ball and touches only at a_0.
3. This proves points with unique nearest point arbitrarily close to b_0.

9.7.11 GEO-11: Unit-sphere configurations and regular simplex bounds

Must know	Unit-sphere configurations and regular simplex bounds.
Occurrence	1 tagged problem(s). Selected anchors: `2001-D2-P3`
Original trap note	Induction via projection must preserve strict negativity of inner products.
Literature	[7], [13]

Core facts and properties.

1. On the unit sphere, distance $> \sqrt{2}$ is equivalent to negative inner product.
2. At most $n + 1$ points in $\mathbb{R}^n$ can have pairwise negative inner products after normalization in the relevant induction.
3. A regular simplex centered at the origin has $n + 1$ vertices on the unit sphere with pairwise inner product $-1/n$.

Compact proof mechanisms.

1. Distance equivalence: $\|x - y\|^2 = 2 - 2\langle x, y\rangle$ for unit vectors.
2. The upper bound follows by fixing one point and projecting the remaining points to its orthogonal complement with normalized coordinates, reducing dimension by one.
3. The simplex construction gives equality and verifies distances $\sqrt{2 + 2/n} > \sqrt{2}$.

9.7.12 GEO-12: Polygon triangulation and parity of incidences

Must know	Polygon triangulation and parity of incidences.
Occurrence	1 tagged problem(s). Selected anchors: `2006-D2-P1`
Original trap note	Base cases modulo 3 are essential.
Literature	[7], [2]

Core facts and properties.

1. Every convex n-gon can be triangulated into $n-2$ triangles.
2. Adding three vertices by a fixed local triangulation changes vertex triangle-incidence parity in a controlled way.
3. Induction modulo 3 controls whether all vertices can have odd incidence or exactly two even incidences.

Compact proof mechanisms.

1. Triangulate a smaller polygon and add three triangles connecting the old boundary edge to the three new vertices; old endpoint incidences increase by two, preserving parity.
2. New vertices have odd incidences in the construction.
3. Initial cases $n = 3, 4, 5$ establish the three congruence classes.

9.7.13 GEO-13: Convex hulls, random polytopes, and geometric probability

Must know	Convex hulls, random polytopes, and geometric probability.
Occurrence	1 tagged problem(s). Selected anchors: `2002-D2-P4`
Original trap note	Independence of points and conditioning on positions must be explicit.
Literature	[26], [16]

Core facts and properties.

1. Convex hull is the set of all convex combinations of a point set.
2. Uniform geometric probability is area/volume ratio in the domain.
3. Linearity of expectation can count expected vertices/faces/areas without independence for the summands.

Compact proof mechanisms.

1. A point lies in a convex hull iff it is a convex combination; Caratheodory reduces to at most $n+1$ points in $\mathbb{R}^n$.
2. Expected value of a sum of indicators is the sum of probabilities of each indicator.
3. Geometric probability problems should first identify the correct configuration space and measure.

9.7.14 GEO-14: Cube and square dissection by number-theoretic semigroups

Must know	Cube and square dissection by number-theoretic semigroups.
Occurrence	1 tagged problem(s). Selected anchors: `2000-D2-P1`
Original trap note	Find two coprime increments $a^d - 1$; not every pair works in every dimension.
Literature	[2]

Core facts and properties.

1. If a cube is dissected into n cubes, refining one small cube into a^d equal cubes changes the count by $a^d - 1$.
2. If u, v are coprime positive integers, every sufficiently large integer is $ux + vy$ with $x, y \geq 0$.
3. It suffices to find two coprime numbers of the form $a^d - 1$.

Compact proof mechanisms.

1. The semigroup lemma follows because multiples $0, u, 2u, \ldots, (v-1)u$ cover all residues mod v; choose the representative not exceeding $(v-1)u$.
2. Refinement count adds $a^d - 1$ because one cube is replaced by a^d cubes.
3. For dimension d, $2^d - 1$ and $(2^d - 1)^d - 1$ are coprime by a simple modular check.

9.8 Functional equations / dynamics

Domain anchors. [22], [14], [28].

CORRECTION MATH-023 — legacy statement superseded; click for canonical replacement.

9.8.1 FEQ-01: Substitution of special values and normalization

Must know	Substitution of special values, normalization, and reducing functional equations to simpler parameters.
Occurrence	7 tagged problem(s). Selected anchors: 2005-D2-P4; 2007-D2-P1; 2009-D2-P2; 2015-D1-P3; 2016-D1-P1; 2019-D2-P6; 2024-D2-P6
Original trap note	Do not assume continuity or monotonicity unless the equation implies it or it is given.
Literature	[22]

Core facts and properties.

1. Set variables to identities, zeros, equal values, or expressions that cancel a term.
2. Normalize by defining auxiliary ratios such as $g(x) = f(x)/x$ when the equation is homogeneous.
3. After proving monotonicity/injectivity, substitute transformed variables to get rigidity.

Compact proof mechanisms.

1. Special values are justified only when they lie in the domain; for $\mathbb{R}_+$ avoid substituting zero.
2. If an equation is scale-invariant, ratios often become constants by iteration.
3. The clean solution usually proves a qualitative property first, then solves an algebraic equation.

9.8.2 FEQ-02: Injectivity and monotonicity forced by a functional equation

Must know	Injectivity and monotonicity forced by a functional equation.
Occurrence	8 tagged problem(s). Selected anchors: 1999-D1-P4; 2000-D2-P5; 2005-D2-P4; 2009-D2-P2; 2015-D1-P3; 2016-D1-P1; 2019-D2-P6; 2024-D2-P6
Original trap note	If injectivity is used, prove it before canceling arguments inside f.
Literature	[22]

Core facts and properties.

1. If f is strictly monotone, it is injective; if an equation implies $f(u) = f(v)$, then $u = v$.
2. Positive-valued equations can force $f \le 1$, decreasing behavior, or sign restrictions.
3. Surjectivity/range constraints plus monotonicity often force continuity or rule out jumps.

Compact proof mechanisms.

1. Monotonicity may follow by comparing $f(x + y)$ with $f(x)$ or writing the equation as a difference quotient.
2. Injectivity lets one equate arguments after producing equal function values through two substitutions.
3. Never assume monotonicity from positivity; derive it explicitly from the equation.

9.8.3 FEQ-03: Fixed points, iterates, and orbit closure

Must know	Fixed points, iterates, orbit closure, and dynamical constraints.
Occurrence	6 tagged problem(s). Selected anchors: 1996-D2-P1; 2002-D1-P4; 2010-D1-P3; 2015-D1-P3; 2016-D1-P1; 2024-D2-P6
Original trap note	A pointwise closed orbit is not automatically finite without compactness and continuity arguments.
Literature	[14], [22]

Core facts and properties.

1. A periodic point of period n satisfies $f^n(p) = p$ and no smaller positive iterate returns p.
2. A closed forward orbit in a compact interval must be finite under certain continuity constraints.
3. Piecewise monotone maps allow counting fixed points of iterates by graph intersections.

Compact proof mechanisms.

1. If an infinite orbit has a convergent subsequence and the orbit set is closed, the limit is an orbit point; continuity shifts accumulation to later iterates.
2. For tent-like maps, each branch of f^n crosses the identity exactly once, giving finitely many periodic candidates.
3. Subtract lower-period fixed sets to prove existence of exact period.

9.8.4 FEQ-04: Recurrence transformations and conjugacy

Must know	Recurrence transformations, conjugacy, and closed-form iteration.
Occurrence	3 tagged problem(s). Selected anchors: 1998-D1-P3; 2003-D1-P5; 2004-D1-P2
Original trap note	Verify the transformed recurrence preserves the original domain.
Literature	[14], [4]

Core facts and properties.

1. A recurrence may simplify under a change of variables, e.g. nested radicals to powers or trigonometric half-angles.
2. Conjugacy $h \circ f \circ h^{-1}$ turns iteration of f into iteration of a simpler map.
3. Monotone bounded recurrences converge to fixed points of the limiting map.

Compact proof mechanisms.

1. For $a_{n+1} = \sqrt{a_n}$ after substitution, solve by repeated exponents.
2. Trigonometric identities $2\sin(\theta/2) = \sqrt{2 - 2\cos\theta}$ and tangent half-angle often linearize radicals.
3. Always verify the chosen branch/sign of square roots.

9.8.5 FEQ-05: Differentiable functional equations

Must know	Differentiable functional equations and derivative constraints.
Occurrence	0 tagged problem(s). Selected anchors: No direct past-problem occurrence detected in the current parsed corpus; keep as prerequisite/background node.
Original trap note	Find the fixed point of the affine map before iterating or differentiating repeatedly.
Literature	[22], [28]

Core facts and properties.

1. If a functional equation holds for all small increments, differentiate with respect to the free variable only after regularity is given.
2. Differentiable solutions often satisfy an ODE obtained from the functional equation at a special value.
3. If f^2 and f^3 are polynomials, then $f = f^3/f^2$ away from zeros and extends polynomially.

Compact proof mechanisms.

1. Differentiate identities on open domains and then use continuity to extend to boundary points if needed.
2. Polynomial quotient proof uses unique factorization or cancellation in rational functions.
3. Be careful: equations stated only on discrete sets or without continuity cannot be differentiated.

9.8.6 FEQ-06: Piecewise-linear maps and periodic points

Must know	Piecewise-linear maps and periodic points.
Occurrence	1 tagged problem(s). Selected anchors: 1998-D2-P3
Original trap note	Exact period requires excluding fixed points of smaller iterates.
Literature	[14]

Core facts and properties.

1. For a full tent map with two monotone branches, f^n has 2^n monotone linear branches.
2. If each branch connects opposite sides of the unit square, it intersects the diagonal exactly once.
3. Exact-period points exist because lower-period fixed points account for fewer than all fixed points of f^n.

Compact proof mechanisms.

1. Branch count doubles at each iterate because each branch maps onto the full interval and then splits under the next application.
2. A nonconstant linear branch from 0 to 1 or 1 to 0 crosses $y = x$ exactly once.
3. The union of fixed-point sets for periods $< n$ has at most $2 + 4 + \cdots + 2^{n-1} < 2^n$.

9.8.7 FEQ-07: Bijections as products of involutions and symmetry decompositions

Must know	Bijections as products of involutions and symmetry decompositions.
Occurrence	1 tagged problem(s). Selected anchors: 1997-D2-P5
Original trap note	Handle finite cycles and infinite orbits separately.
Literature	[22]

Core facts and properties.

1. Every bijection decomposes into disjoint orbits under iteration: finite cycles or bi-infinite chains.
2. A rotation of a finite regular polygon is a product of two reflections.
3. The shift on $\mathbb{Z}$ is a product of two involutions, e.g. reflections about 0 and $1/2$.

Compact proof mechanisms.

1. Orbits are equivalence classes under $x \sim f^n(x)$, and f preserves each orbit.
2. In a finite cycle, choose two reflection axes whose angle composes to the required rotation.
3. On an infinite orbit indexed by $\mathbb{Z}$, define involutions $g_2(m) = -m$ and $g_1(m) = 1 - m$; then $g_1 g_2(m) = m + 1$.

9.8.8 FEQ-08: No-regularity functions and order patterns on Q or R

Must know	No-regularity functions and order patterns on $\mathbb{Q}$ or $\mathbb{R}$.
Occurrence	6 tagged problem(s). Selected anchors: 1999-D1-P3; 1999-D2-P4; 2001-D2-P5; 2002-D1-P2; 2003-D2-P5; 2024-D2-P6
Original trap note	Do not invoke IVT, compactness, or monotonicity unless it is part of the statement.
Literature	[22], [28]

Core facts and properties.

1. Without continuity/measurability, functions on countable sets can be defined recursively or by enumeration.
2. For $\mathbb{Q}$, a bijection to $\mathbb{N}$ lets one define bounds depending on finite initial segments.
3. On $\mathbb{R}$, cardinality arguments can construct functions defeating every candidate bound.

Compact proof mechanisms.

1. For a function $f : \mathbb{Q}^2 \to \mathbb{R}$, enumerate rationals and set $g(x)$ as a maximum over finitely many previously enumerated pairs.
2. For impossibility on $\mathbb{R}$, encode every real sequence as one real using $|\mathbb{R}^{\mathbb{N}}| = |\mathbb{R}|$.
3. State clearly where choice/enumeration is used and where continuity would make the result false.

9.8.9 FEQ-09: Image-length preserving functions

Must know	Image-length preserving functions and isometries of intervals.
Occurrence	1 tagged problem(s). Selected anchors: 2006-D2-P2
Original trap note	The maximum/minimum on an interval must occur at endpoints only after equality in the Lipschitz chain is analyzed.
Literature	[22]

Core facts and properties.

1. If $f([a,b])$ is an interval of length $b-a$, then $|f(x) - f(y)| \le |x - y|$ for all x, y.
2. A continuous 1-Lipschitz map that preserves interval diameter on every interval must be monotone.
3. The only solutions on $\mathbb{R}$ are $f(x) = x + c$ and $f(x) = -x + c$.

Compact proof mechanisms.

1. The Lipschitz bound follows because $f(x), f(y) \in f([x,y])$, whose length is $y - x$.
2. If $f([x,y])$ has full length, its extrema must occur at endpoints; otherwise the endpoint distance would be smaller.
3. Monotonicity plus exact endpoint differences gives $f(x) - x$ constant or $f(x) + x$ constant.

9.9 Probability / measure / expectation

Domain anchors. [16], [8], [27].

9.9.1 PRB-01: Discrete probability via residues and recurrences

Must know	Discrete probability via residues and recurrences.
Occurrence	1 tagged problem(s). Selected anchors: 1999-D2-P2
Original trap note	State transition probabilities must sum to one and residues are modulo the correct integer.
Literature	[16]

Core facts and properties.

1. A probability distribution on residues modulo m evolves by convolution with the step distribution.
2. Symmetry or recurrence can reveal a nearly uniform distribution plus a decaying correction.
3. A combinatorial partition of outcomes may count favorable residues exactly.

Compact proof mechanisms.

1. For dice sums, write $p_n^{(r)} = \frac{1}{6}\sum_{i=1}^{6} p_{n-1}^{(r-i)}$.
2. Guess the formula from small n and prove by induction through the recurrence.
3. Alternatively group sequences into blocks whose sums form complete residue systems, with one exceptional sequence if present.

9.9.2 PRB-02: Generating functions for probability distributions

Must know	Generating functions for probability distributions.
Occurrence	1 tagged problem(s). Selected anchors: 2022-D1-P1
Original trap note	Normalize coefficients by the total number of outcomes.
Literature	[16], [29]

Core facts and properties.

1. The probability generating function of a sum of independent integer variables is the product of their generating functions.
2. The probability of a residue class modulo m is extracted by a root-of-unity filter.
3. For equal dice, $((x + x^2 + \cdots + x^6)/6)^n$ encodes the sum distribution.

Compact proof mechanisms.

1. Independence gives product because coefficients multiply and convolve probabilities.
2. Root-of-unity extraction is $\frac{1}{m}\sum_{j=0}^{m-1} \omega^{-jr} F(\omega^j)$.
3. Use this when direct recurrence is possible but slower.

9.9.3 PRB-03: Expectation, linearity, and probabilistic method

Must know	Expectation, linearity, and probabilistic method.
Occurrence	1 tagged problem(s). Selected anchors: 2021-D1-P2
Original trap note	Independence is only needed for products or distribution factorization, not for linearity.
Literature	[3], [16]

Core facts and properties.

1. Linearity: $\mathbb{E}[\sum X_i] = \sum \mathbb{E}[X_i]$, no independence required.
2. If the average value of a random variable is μ, some outcome has value at least μ and some has value at most μ.
3. Indicators turn counts into sums of 0-1 variables.

Compact proof mechanisms.

1. Linearity follows directly from finite sums or integrals defining expectation.
2. The existence claim is contradiction: if all values were below μ, the average would be below μ.
3. Use indicators for expected number of covered elements, collisions, or good pairs.

9.9.4 PRB-04: Continuous uniform distribution and geometric probability

Must know	Continuous uniform distribution and geometric probability.
Occurrence	0 tagged problem(s). Selected anchors: No direct past-problem occurrence detected in the current parsed corpus; keep as prerequisite/background node.
Original trap note	Events involving exact coincidences on continuous distributions often have probability zero.
Literature	[8], [16]

Core facts and properties.

1. For uniform choice in a region D, $\mathbb{P}(E) = \mathrm{vol}(E)/\mathrm{vol}(D)$ for measurable $E \subseteq D$.
2. Independently chosen points give product measure on D^n.
3. Boundary events of lower dimension usually have probability zero.

Compact proof mechanisms.

1. Uniform probability is normalized Lebesgue measure.
2. Product measure corresponds to independent coordinates/points.
3. For geometric probabilities, draw the configuration space first; the random object may live in a higher-dimensional product region.

9.9.5 PRB-05: Measure, variation, and differentiability almost everywhere

Must know	Measure, variation, and differentiability almost everywhere.
Occurrence	2 tagged problem(s). Selected anchors: `1998-D2-P6`; `2022-D1-P1`
Original trap note	A countable exceptional support can still be dense; use interval covers, not isolation.
Literature	[27], [8]

Core facts and properties.

1. A function of bounded variation on an interval is differentiable almost everywhere.
2. A sparse spike function can be differentiable at points outside carefully chosen small intervals.
3. If intervals with total length less than the interval length cover all bad spike neighborhoods, some good point remains.

Compact proof mechanisms.

1. Cover the point a_n with radius proportional to spike height times a growing factor; finite total weighted height makes total cover length small.
2. Outside all intervals, every difference quotient is bounded by the reciprocal of the growing factor, hence tends to zero.
3. For nowhere differentiability construction, choose intervals around spikes that cover every point infinitely often.

10 Complete Theory Properties Manual: Exact Statements and Compact Proof Cores

This section is an additive expansion of the previous booklet. Nothing in the original categorization, occurrence lists, compact theory-node atlas, or problem-to-theory matrix is removed. The aim is to spell out the exact mathematical properties, formulas, theorem statements, and compact proof cores that were previously only named or summarized. Each theory node keeps its occurrence anchors, so the reader can see both the statement of the theory and where it appeared in the IMC corpus.

Scope convention. The statements below are not a replacement for a full textbook, but they are the competition-level theory core needed by the detected past problems. If a node has zero detected occurrence, it is kept as a background node because it is adjacent to detected material and prevents gaps in nearby solutions.

10.1 Linear algebra / matrix theory: exact theorem bank

Literature anchor: [6, 18, 19]. **Global theorem/property bank.**

1. **One-sided inverse theorem.** If $A, B \in M_n(F)$ and $AB = I$, then A is surjective, hence invertible, and $B = A^{-1}$; therefore $BA = I$. The proof is rank-nullity in finite dimension, not commutativity.
2. **Inverse product rules.** For invertible matrices, $(AB)^{-1} = B^{-1}A^{-1}$, $(A^T)^{-1} = (A^{-1})^T$, and $(A^*)^{-1} = (A^{-1})^*$. Verify by multiplying on both sides.
3. **Determinant laws.** $\det(AB) = \det A \det B$, $\det A^T = \det A$, $\det A^{-1} = 1/\det A$, and triangular/block-triangular matrices have determinant equal to the product of diagonal/block determinants.
4. **Elementary determinant operations.** Adding a multiple of one row/column to another preserves determinant; row/column swaps change sign; scaling one row/column by c scales determinant by c.
5. **Rank-nullity and dimension.** For $T : V \to W$, $\dim V = \dim \ker T + \operatorname{rank} T$. Also $\operatorname{rank}(AB) \leq \min(\operatorname{rank} A, \operatorname{rank} B)$ and $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.
6. **Trace cyclicity.** $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ and $\operatorname{tr}(A_1 \cdots A_k) = \operatorname{tr}(A_2 \cdots A_k A_1)$. Therefore every commutator $AB - BA$ has trace zero.
7. **Cayley-Hamilton and spectrum.** Every matrix satisfies its characteristic polynomial. If $p(A) = 0$, then every eigenvalue of A is a root of p. If the minimal polynomial splits into distinct linear factors, then A is diagonalizable.
8. **Special operators.** If $N^m = 0$, then $I - N$ is invertible with inverse $I + N + \cdots + N^{m-1}$. If $P^2 = P$, then $V = \ker P \oplus \operatorname{im} P$. If $A^2 = I$ in characteristic not 2, then $(I \pm A)/2$ are complementary projections.

Node-by-node properties, formulas, proof cores, and occurrence anchors.

10.1.1 LA-01: Matrix products, inverses, and one-sided inverse identities

Occurrence anchors	1994-D1-P1; 1996-D1-P1; 1996-D2-P3; 2002-D2-P1; 2002-D2-P5; 2003-D2-P1; 2004-D2-P1; 2009-D1-P2; 2013-D1-P1; 2014-D1-P1; 2018-D1-P3; 2019-D2-P9; 2021-D1-P1; 2023-D2-P6
Original node statement	For square matrices, a right inverse is also a left inverse: if $XY = I$, then X and Y are invertible and $YX = I$. Algebraic identities such as $(A + I)(B + I) = I$ or $(A - B)(C + A\text{\^}\{-1\}) = I$ often turn a noncommutative-looking equation into an inverse statement.
Literature	[6, 18, 19]

Exact properties and formulas to know.

1. If $AB = I$ for square matrices, then $BA = I$; proof: A is surjective, hence invertible.
2. For invertible A, B, $(AB)^{-1} = B^{-1}A^{-1}$.
3. Equations like $(A + I)(B + I) = I$ allow reversing the product only after inverse status is proved.
4. Block equations from $MM^{-1} = I$ and $M^{-1}M = I$ are often enough; do not assume block commutation.

Compact proof cores for this node.

1. Choose a basis adapted to kernels, images, invariant subspaces, or eigenspaces.
2. Use trace, determinant, rank, and characteristic polynomial as similarity-invariant obstructions.
3. Preserve multiplication order unless cyclicity of trace or inverse-factor reversal has been justified.

Use reminder. Use this whenever the statement contains a matrix equation that can be completed to a product equal to I. After factoring, switch the order only because the factors have been proved inverse, not because matrices commute.

Trap reminder. Never cancel or reorder matrix factors before proving invertibility; record which factor is square.

10.1.2 LA-02: Determinant evaluation by row/column operations and triangular forms

Occurrence anchors	1994-D1-P1; 1995-D1-P1; 1995-D2-P1; 1996-D1-P1; 1997-D1-P3; 1997-D2-P2; 1997-D2-P4; 1999-D1-P1; 2000-D1-P3; 2001-D1-P1; 2001-D1-P5; 2001-D2-P4; 2002-D2-P1; 2003-D2-P1; 2004-D2-P1; 2004-D2-P6; 2005-D1-P1; 2005-D2-P3; 2005-D2-P6; 2006-D1-P3; 2006-D2-P6; 2007-D1-P2; 2007-D1-P3; 2007-D2-P4; 2008-D1-P6; 2008-D2-P5; ... (+25 more in data/imc_theory_node_index.csv)
Original node statement	Determinants are preserved by adding a multiple of one row/column to another, change sign under swaps, and scale under row/column scaling. A common IMC target is to transform a structured matrix into triangular or block triangular form.
Literature	[6, 18, 19]

Exact properties and formulas to know.

1. Leibniz formula: $\det A = \sum_{\sigma \in S_n} \operatorname{sgn}\sigma \prod_i a_{i\sigma(i)}$.
2. Laplace expansion along a sparse row/column follows from multilinearity.
3. Elementary row/column additions preserve determinant.
4. Triangular determinant is the product of diagonal entries.

Compact proof cores for this node.

1. Choose a basis adapted to kernels, images, invariant subspaces, or eigenspaces.
2. Use trace, determinant, rank, and characteristic polynomial as similarity-invariant obstructions.
3. Preserve multiplication order unless cyclicity of trace or inverse-factor reversal has been justified.

Use reminder. Use for explicit determinant computations, sign/parity determinant arguments, and matrices with entries depending on $i+j$, $|i-j|$, parity, primes, or a recurrence.

Trap reminder. Track row/column operations exactly; if operations depend on parity or cyclic order, verify small n separately.

10.1.3 LA-03: Block matrices, Schur complements, and determinant identities

Occurrence anchors	1994-D1-P1; 1994-D2-P2; 1995-D1-P1; 1995-D1-P2; 1995-D2-P1; 1995-D2-P2; 1996-D1-P1; 1996-D1-P2; 1997-D1-P3; 1997-D2-P2; 1997-D2-P4; 1998-D1-P1; 1998-D1-P5; 1999-D1-P1; 1999-D1-P5; 1999-D2-P2; 1999-D2-P3; 2000-D1-P2; 2000-D1-P3; 2000-D1-P4; 2000-D2-P6; 2001-D1-P1; 2001-D1-P3; 2001-D1-P5; 2001-D2-P4; 2002-D2-P1; ... (+88 more in data/imc_theory_node_index.csv)
Original node statement	For compatible square blocks, block multiplication and determinant identities often reduce a large matrix equation to a smaller one. Typical facts include $\{\}det(M)\{\}det(H) = \{\}det(A)$ for inverse block pairs in suitable block decompositions, and AC and CA having the same nonzero eigenvalues.
Literature	[6, 18, 19]

Exact properties and formulas to know.

1. Leibniz formula: $\det A = \sum_{\sigma \in S_n} \operatorname{sgn}\sigma \prod_i a_{i\sigma(i)}$.
2. Laplace expansion along a sparse row/column follows from multilinearity.
3. Elementary row/column additions preserve determinant.
4. Triangular determinant is the product of diagonal entries.

Compact proof cores for this node.

1. Choose a basis adapted to kernels, images, invariant subspaces, or eigenspaces.
2. Use trace, determinant, rank, and characteristic polynomial as similarity-invariant obstructions.
3. Preserve multiplication order unless cyclicity of trace or inverse-factor reversal has been justified.

Use reminder. Use when a problem explicitly gives a block matrix, a product AB versus BA, or a matrix of a linear operator on a matrix space.

Trap reminder. Schur complement formulas require invertibility of the chosen block; if it is not given, use block multiplication identities instead.

10.1.4 LA-04: Rank-nullity, image/kernel, and rank of factorizations

Occurrence anchors	1994-D1-P1; 1994-D1-P4; 1994-D2-P4; 1994-D2-P5; 1995-D1-P1; 1995-D1-P5; 1996-D1-P1; 1996-D1-P3; 1996-D1-P6; 1997-D1-P3; 1997-D2-P2; 1997-D2-P4; 1998-D1-P1; 1998-D2-P1; 1999-D1-P1; 2000-D1-P3; 2000-D2-P1; 2001-D1-P5; 2001-D2-P3; 2001-D2-P4; 2002-D1-P5; 2002-D1-P6; 2002-D2-P1; 2003-D2-P1; 2004-D1-P3; 2004-D2-P1; ... (+61 more in data/imc_theory_node_index.csv)
Original node statement	Rank-nullity gives \{}$dimV = \{}$dim\{}$kerT + \{}$dim\{}$operatorname{im}T$. For products, \{}$operatorname{rank}(AB)\{}$le\{}$min(\{}$operatorname{rank}A, \{}$operatorname{rank}B)$ and column spaces/images often reveal the exact rank.
Literature	[6, 18, 19]

Exact properties and formulas to know.

1. Rank-nullity: $\dim V = \dim \ker T + \dim \operatorname{im} T$.
2. $\operatorname{rank}(AB) \le \min(\operatorname{rank} A, \operatorname{rank} B)$.
3. $\operatorname{rank}(A+B) \le \operatorname{rank} A + \operatorname{rank} B$.
4. $\dim(U+W) = \dim U + \dim W - \dim(U \cap W)$.

Compact proof cores for this node.

1. Choose a basis adapted to kernels, images, invariant subspaces, or eigenspaces.
2. Use trace, determinant, rank, and characteristic polynomial as similarity-invariant obstructions.
3. Preserve multiplication order unless cyclicity of trace or inverse-factor reversal has been justified.

Use reminder. Use for rank minimization/maximization, rank-one perturbations, factorizations AB, and problems asking for the dimension of a matrix subspace.

Trap reminder. Rank statements over \{}$mathbbR$ and \{}$mathbbC$ can differ only through spectral issues; dimension counts must specify the ground field.

10.1.5 LA-05: Trace cyclicity, trace pairing, and zero-trace subspaces

Occurrence anchors	1995-D1-P1; 1995-D2-P1; 1996-D1-P1; 1997-D2-P2; 1997-D2-P4; 1999-D1-P1; 2000-D1-P3; 2001-D1-P5; 2001-D2-P4; 2002-D2-P1; 2003-D2-P1; 2004-D2-P1; 2005-D1-P1; 2005-D2-P3; 2005-D2-P6; 2006-D2-P6; 2007-D1-P2; 2007-D1-P3; 2007-D2-P4; 2009-D1-P2; 2009-D2-P3; 2010-D1-P5; 2011-D1-P2; 2012-D1-P2; 2013-D1-P1; 2014-D1-P1; ... (+7 more in data/imc_theory_node_index.csv)
Original node statement	The trace is cyclic: \{}$operatorname{tr}(XY) = \{}$operatorname{tr}(YX)$ and hence \{}$operatorname{tr}([X,Y]) = 0$. The bilinear form $(X,Y)\{}$mapsto\{}$operatorname{tr}(XY)$ identifies the dual of the matrix space and detects symmetric or triangular subspaces.
Literature	[6, 18, 19]

Exact properties and formulas to know.

1. $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ by expanding entries.
2. $\operatorname{tr}(A_1 \cdots A_k)$ is invariant under cyclic shifts.
3. Every commutator has trace zero.
4. The trace pairing with elementary matrices identifies every linear functional on M_n as $A \mapsto \operatorname{tr}(AC)$.

Compact proof cores for this node.

1. Choose a basis adapted to kernels, images, invariant subspaces, or eigenspaces.
2. Use trace, determinant, rank, and characteristic polynomial as similarity-invariant obstructions.
3. Preserve multiplication order unless cyclicity of trace or inverse-factor reversal has been justified.

Use reminder. Use for commutator problems, linear functionals on matrix spaces, bilinear orthogonality conditions, and trace constraints.

Trap reminder. Cyclicity is safe for products under trace, not outside trace. Do not infer $XY = YX$ from equal traces.

10.1.6 LA-06: Commutators and spectral obstructions

Occurrence anchors	1994-D1-P4; 1995-D1-P5; 1996-D1-P1; 1997-D1-P3; 1997-D2-P4; 2000-D1-P3; 2000-D1-P5; 2000-D2-P6; 2001-D1-P5; 2002-D2-P1; 2003-D1-P3; 2003-D2-P1; 2004-D2-P1; 2005-D1-P6; 2009-D2-P3; 2013-D1-P1; 2014-D1-P1; 2020-D1-P2; 2021-D1-P1; 2022-D2-P7
Original node statement	A commutator $AB - BA$ has trace zero. If it has very small rank or satisfies a polynomial identity, trace and Jordan-form restrictions can force nilpotency or impossibility.
Literature	[6, 18, 19]

Exact properties and formulas to know.

1. $[A, B] = AB - BA$ and $\mathrm{tr}[A, B] = 0$.
2. The map $\Delta_A(X) = AX - XA$ is a derivation: $\Delta_A(XY) = \Delta_A(X)Y + X\Delta_A(Y)$.
3. Rank-one trace-zero matrices have all eigenvalues zero.
4. A finite-dimensional operator cannot have infinitely many distinct eigenvalues.

Compact proof cores for this node.

1. Choose a basis adapted to kernels, images, invariant subspaces, or eigenspaces.
2. Use trace, determinant, rank, and characteristic polynomial as similarity-invariant obstructions.
3. Preserve multiplication order unless cyclicity of trace or inverse-factor reversal has been justified.

Use reminder. Use when a problem contains $AB - BA$, $FG - GF$, or an operator $X\{\}mapstoAX - XA$.

Trap reminder. Trace zero alone is weak; combine it with rank, eigenvalue multiplicity, or finite-dimensionality.

10.1.7 LA-07: Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton

Occurrence anchors	1994-D1-P4; 1994-D2-P2; 1995-D1-P1; 1995-D1-P4; 1995-D1-P5; 1995-D2-P5; 1996-D1-P1; 1996-D1-P3; 1996-D1-P5; 1997-D1-P5; 1997-D2-P2; 1998-D2-P1; 1998-D2-P4; 1999-D1-P1; 2000-D1-P3; 2000-D1-P5; 2000-D2-P6; 2001-D1-P5; 2001-D1-P6; 2002-D1-P1; 2002-D2-P1; 2002-D2-P3; 2002-D2-P5; 2003-D1-P3; 2003-D1-P4; 2003-D2-P1; ... (+25 more in data/imc_theory_node_index.csv)
Original node statement	If $p(A) = 0$, every eigenvalue of A is a root of p. Cayley-Hamilton gives $\{\}chi_A(A) = 0$. Distinct roots of the minimal polynomial imply diagonalizability over a splitting field.
Literature	[6, 18, 19]

Exact properties and formulas to know.

1. Similar matrices have the same characteristic polynomial.
2. If $p(A) = 0$, every eigenvalue of A is a root of p.
3. Cayley-Hamilton: $\chi_A(A) = 0$.
4. Minimal polynomial with distinct linear factors implies diagonalizability over the splitting field.

Compact proof cores for this node.

1. Choose a basis adapted to kernels, images, invariant subspaces, or eigenspaces.
2. Use trace, determinant, rank, and characteristic polynomial as similarity-invariant obstructions.
3. Preserve multiplication order unless cyclicity of trace or inverse-factor reversal has been justified.

Use reminder. Use for convergence of matrix powers, nilpotency, idempotency, diagonalization, determinant/eigenvalue constraints, and polynomial equations in matrices.

Trap reminder. Real matrices may have complex eigenvalues; diagonalization over $\{\}mathbbC$ may be needed even when entries are real.

10.1.8 LA-08: Nilpotent, idempotent, involution, and square-zero structures

Occurrence anchors	1994-D1-P4; 1995-D1-P1; 1995-D1-P5; 1996-D1-P3; 2000-D1-P3; 2000-D1-P5; 2000-D2-P6; 2001-D2-P4; 2003-D1-P3; 2005-D1-P6; 2007-D2-P5; 2009-D1-P2; 2018-D1-P3; 2022-D2-P7
Original node statement	Nilpotent matrices have all eigenvalues 0; idempotents have eigenvalues in $\{\}\{0, 1\{\}\}$; involutions have eigenvalues in $\{\}\{\{\}pm1\{\}\}$ when the characteristic is not 2. These facts often convert algebraic equations into diagonal or block forms.
Literature	[6, 18, 19]

Exact properties and formulas to know.

1. Nilpotent means $N^m = 0$; then all eigenvalues are 0.
2. Idempotent $P^2 = P$ gives $V = \ker P \oplus \operatorname{im} P$.
3. Involution $A^2 = I$ has eigenvalues only ± 1 in characteristic not 2.
4. Square-zero $S^2 = 0$ implies $\operatorname{im} S \subseteq \ker S$.

Compact proof cores for this node.

1. Choose a basis adapted to kernels, images, invariant subspaces, or eigenspaces.
2. Use trace, determinant, rank, and characteristic polynomial as similarity-invariant obstructions.
3. Preserve multiplication order unless cyclicity of trace or inverse-factor reversal has been justified.

Use reminder. Use for matrix equations involving powers, commuting involutions, square-zero products, and rank bounds for idempotents.

Trap reminder. Do not assume diagonalizability for nilpotents. Idempotents and involutions are diagonalizable only when the minimal polynomial has distinct roots.

10.1.9 LA-09: Simultaneous diagonalization and commuting families

Occurrence anchors	1994-D2-P4; 1996-D1-P3; 2006-D2-P6
Original node statement	A commuting family of diagonalizable operators over an algebraically closed field is simultaneously diagonalizable under standard finite-dimensional hypotheses. For diagonal matrices, the commutant is block diagonal according to equal eigenvalues.
Literature	[6, 18, 19]

Exact properties and formulas to know.

1. Commuting operators preserve eigenspaces of one another.
2. Pairwise commuting diagonalizable operators are simultaneously diagonalizable over a splitting field.
3. Commuting involutions have a common eigenbasis with entries ± 1.
4. There are at most 2^n distinct diagonal involution patterns on an n-dimensional space.

Compact proof cores for this node.

1. Choose a basis adapted to kernels, images, invariant subspaces, or eigenspaces.
2. Use trace, determinant, rank, and characteristic polynomial as similarity-invariant obstructions.
3. Preserve multiplication order unless cyclicity of trace or inverse-factor reversal has been justified.

Use reminder. Use when several operators commute or preserve common eigenspaces; typical conclusions are dimension counts and maximum cardinality of families.

Trap reminder. Commuting alone does not imply diagonalizable. First prove each operator is diagonalizable or use invariant subspaces carefully.

10.1.10 LA-10: Invariant subspaces and adapted bases

Occurrence anchors	1994-D2-P4; 1994-D2-P5; 1995-D1-P1; 1996-D1-P3; 1996-D1-P6; 1997-D2-P2; 1997-D2-P4; 1998-D1-P1; 1998-D2-P1; 2001-D1-P5; 2001-D2-P3; 2002-D1-P5; 2002-D1-P6; 2004-D2-P4; 2005-D2-P3; 2006-D2-P2; 2006-D2-P4; 2007-D1-P2; 2007-D2-P5; 2008-D1-P2; 2009-D1-P3; 2009-D1-P4; 2009-D1-P5; 2010-D2-P4; 2011-D1-P5; 2011-D2-P5; ... (+4 more in data/imc_theory_node_index.csv)
Original node statement	If a chain of subspaces is invariant, choose a basis adapted to the chain; the matrix becomes block upper triangular, and free entries can be counted.
Literature	[6, 18, 19]

Exact properties and formulas to know.

1. If U is invariant under T, an adapted basis makes the matrix block upper triangular.
2. A flag of invariant subspaces creates a block upper-triangular zero pattern.
3. Dimension of a matrix space is counted by free entries after basis adaptation.
4. Invariant subspace constraints are linear equations on matrix entries.

Compact proof cores for this node.

1. Choose a basis adapted to kernels, images, invariant subspaces, or eigenspaces.
2. Use trace, determinant, rank, and characteristic polynomial as similarity-invariant obstructions.

3. Preserve multiplication order unless cyclicity of trace or inverse-factor reversal has been justified.

Use reminder. Use in dimension-of-endomorphism-space problems, simultaneous preservation of subspaces, and centralizer computations.

Trap reminder. Check the order of basis blocks: $T(U)\backslash\{\}subsetU$ imposes zero blocks below, not above, depending on column convention.

10.1.11 LA-11: Linear functionals, kernels, and dual span criteria

Occurrence anchors	1997-D2-P2; 1997-D2-P4; 1998-D1-P1; 1998-D2-P1; 2008-D1-P2; 2015-D2-P9
Original node statement	A linear functional that vanishes on $\backslash\{\}bigcap_i\backslash\{\}kerf_i$ belongs to the span of $f_1, \backslash\{\}dots, f_k$. Equivalently, the annihilator of an intersection of kernels is the span of the functionals.
Literature	[6, 18, 19]

Exact properties and formulas to know.

1. Rank-nullity: $\dim V = \dim\ker T + \dim\operatorname{im} T$.
2. $\operatorname{rank}(AB) \le \min(\operatorname{rank} A, \operatorname{rank} B)$.
3. $\operatorname{rank}(A+B) \le \operatorname{rank} A + \operatorname{rank} B$.
4. $\dim(U+W) = \dim U + \dim W - \dim(U \cap W)$.

Compact proof cores for this node.

1. Choose a basis adapted to kernels, images, invariant subspaces, or eigenspaces.
2. Use trace, determinant, rank, and characteristic polynomial as similarity-invariant obstructions.
3. Preserve multiplication order unless cyclicity of trace or inverse-factor reversal has been justified.

Use reminder. Use for abstract vector-space problems where one functional is forced by the vanishing of several others.

Trap reminder. Independence of the given functionals may need to be arranged by discarding redundant ones.

10.1.12 LA-12: Matrix subspaces constrained by trace bilinear forms

Occurrence anchors	2005-D2-P3
Original node statement	The trace form separates symmetric and skew/triangular subspaces. If $\backslash\{\}operatorname{tr}(XY) = 0$ for all X, Y in a subspace, then the subspace cannot contain nonzero symmetric matrices, giving a dimension bound.
Literature	[6, 18, 19]

Exact properties and formulas to know.

1. $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ by expanding entries.
2. $\operatorname{tr}(A_1 \cdots A_k)$ is invariant under cyclic shifts.
3. Every commutator has trace zero.
4. The trace pairing with elementary matrices identifies every linear functional on M_n as $A \mapsto \operatorname{tr}(AC)$.

Compact proof cores for this node.

1. Choose a basis adapted to kernels, images, invariant subspaces, or eigenspaces.
2. Use trace, determinant, rank, and characteristic polynomial as similarity-invariant obstructions.
3. Preserve multiplication order unless cyclicity of trace or inverse-factor reversal has been justified.

Use reminder. Use in maximum-dimension matrix subspace problems with quadratic trace constraints.

Trap reminder. The form $\backslash\{\}operatorname{tr}(XY)$ is not positive definite on all matrices; positivity appears on symmetric matrices via $\backslash\{\}operatorname{tr}(A\text{\^{}}2)$.

10.1.13 LA-13: Matrix exponentials and polynomial functional calculus

Occurrence anchors	2000-D2-P6
Original node statement	For a matrix X, $q(e\^X)$ can be expanded as a convergent power series in X. Identities comparing AB and BA often reduce to the fact that AC and CA have the same characteristic polynomial.
Literature	[6, 18, 19]

Exact properties and formulas to know.

1. If $\cap_i \ker f_i \subseteq \ker f$, then $f \in \operatorname{span}(f_i)$.
2. Independent functionals admit a dual family of vectors a_j with $f_i(a_j) = \delta_{ij}$.
3. The quotient by $\cap_i \ker f_i$ reduces the statement to finite-dimensional duality.
4. Contrapositive: if f is outside the span, some vector is killed by all f_i but not by f.

Compact proof cores for this node.

1. Choose a basis adapted to kernels, images, invariant subspaces, or eigenspaces.
2. Use trace, determinant, rank, and characteristic polynomial as similarity-invariant obstructions.
3. Preserve multiplication order unless cyclicity of trace or inverse-factor reversal has been justified.

Use reminder. Use when exponentials of products AB and BA appear, especially with nilpotency of polynomial expressions.

Trap reminder. Do not use $e\^\{AB\} = e\^\{BA\}$; compare characteristic polynomials or nonzero spectra instead.

10.1.14 LA-14: Operator norm, power boundedness, and spectral-radius style estimates

Occurrence anchors	2002-D1-P6
Original node statement	A sequence of operator powers can be controlled by estimates on differences $T\^\{n+1\} - T\^n$. Exponential decay of these differences implies norm convergence and power boundedness.
Literature	[6, 18, 19]

Exact properties and formulas to know.

1. Operator norm is $\|A\| = \sup_{x \neq 0} \|Ax\|/\|x\|$.
2. Submultiplicativity: $\|AB\| \leq \|A\|\|B\|$.
3. Power convergence follows if tails $\sum \|A^{k+1} - A^k\|$ are controlled.
4. Norm bounds are stronger than spectral-radius heuristics and require explicit estimates.

Compact proof cores for this node.

1. Choose a basis adapted to kernels, images, invariant subspaces, or eigenspaces.
2. Use trace, determinant, rank, and characteristic polynomial as similarity-invariant obstructions.
3. Preserve multiplication order unless cyclicity of trace or inverse-factor reversal has been justified.

Use reminder. Use for norm inequalities involving all positive powers of a matrix or linear operator.

Trap reminder. Norm convergence is stronger than pointwise convergence. Watch constants and finite-dimensional vs Banach-space arguments.

10.1.15 LA-15: Gram matrices and rational linear dependence

Occurrence anchors	1995-D2-P1; 2002-D2-P6; 2006-D2-P4; 2008-D2-P6; 2013-D2-P3; 2018-D2-P6; 2021-D2-P8
Original node statement	Distances determine inner products by $2\{\}langleu, v\{\}rangle = \{\}\|u\{\}\|\^2 + \{\}\|v\{\}\|\^2 - \{\}\|u - v\{\}\|\^2$. If the Gram matrix has rational entries and is invertible, coordinates of another vector obtained by inner products are rational.
Literature	[6, 18, 19]

Exact properties and formulas to know.

1. Gram matrix $G = (\langle v_i, v_j \rangle)$ is positive semidefinite.
2. G is invertible iff v_i are linearly independent.
3. Polarization gives inner products from squared distances.
4. Rational Gram systems produce rational coordinates by solving $G\lambda = w$.

Compact proof cores for this node.

1. Choose a basis adapted to kernels, images, invariant subspaces, or eigenspaces.
2. Use trace, determinant, rank, and characteristic polynomial as similarity-invariant obstructions.
3. Preserve multiplication order unless cyclicity of trace or inverse-factor reversal has been justified.

Use reminder. Use for rational-distance vector problems and rational linear-dependence conclusions.

Trap reminder. First reduce to a real basis; then solve the linear system over $\{\}mathbbQ$.

10.1.16 LA-16: Anticommuting and rank constraints for special matrices

Occurrence anchors	nan
Original node statement	Anticommuting idempotents or square-zero maps impose rank and dimension constraints. A common proof decomposes space into images/kernels or uses trace/rank averaging.
Literature	[6, 18, 19]

Exact properties and formulas to know.

1. Rank-nullity: $\dim V = \dim\ker T + \dim\operatorname{im} T$.
2. $\operatorname{rank}(AB) \le \min(\operatorname{rank} A, \operatorname{rank} B)$.
3. $\operatorname{rank}(A+B) \le \operatorname{rank} A + \operatorname{rank} B$.
4. $\dim(U+W) = \dim U + \dim W - \dim(U \cap W)$.

Compact proof cores for this node.

1. Choose a basis adapted to kernels, images, invariant subspaces, or eigenspaces.
2. Use trace, determinant, rank, and characteristic polynomial as similarity-invariant obstructions.
3. Preserve multiplication order unless cyclicity of trace or inverse-factor reversal has been justified.

Use reminder. Use when many idempotents, nilpotents, or anticommuting matrices are given and a rank bound is requested.

Trap reminder. Pairwise anticommutation is not pairwise orthogonality; translate the algebraic condition into image/kernel constraints.

10.2 Real analysis / calculus: exact theorem bank

Literature anchor: [28, 4, 17, 27]. **Global theorem/property bank.**

1. **Heine-Borel and EVT.** In $\mathbb{R}^n$, compact means closed and bounded. A continuous function on a compact set attains maximum and minimum; on a noncompact domain first prove the extremum cannot escape.
2. **Density and continuity transfer.** If D is dense and f, g are continuous with $f \leq g$ on D, then $f \leq g$ everywhere. Equalities extend in the same way.
3. **Connectedness/IVT.** Continuous images of connected sets are connected. Connected subsets of $\mathbb{R}$ are intervals; this gives the intermediate value theorem.
4. **Rolle and mean value.** If $f(a) = f(b)$, some $c \in (a, b)$ satisfies $f'(c) = 0$. The MVT gives $f(b) - f(a) = f'(c)(b-a)$.
5. **Taylor and local order.** If f is smooth enough, $f(a+h) = \sum_{j=0}^{k} f^{(j)}(a)h^j/j! + o(h^k)$. The first nonzero term controls sign, sharp constants, and local asymptotics.
6. **Convexity/Jensen.** Convexity is $f(tx + (1-t)y) \leq tf(x) + (1-t)f(y)$. Jensen is the finite weighted version. If differentiable and convex, $f(y) \geq f(x) + f'(x)(y-x)$.
7. **Integrating factors.** For $h' + ah = r$, with $A' = a$, $(he^A)' = re^A$. Inequalities of this form give monotonicity and exponential comparison estimates.
8. **Integral tools.** Integration by parts is $\int uv' = uv|_a^b - \int u'v$. Cauchy-Schwarz follows from $\int (f - tg)^2 \geq 0$. Fubini/Tonelli requires absolute integrability or nonnegativity.
9. **Sums and series.** Riemann sums follow from uniform continuity on compact intervals. Improper domains require main/tail splitting. Absolute convergence allows rearrangement; conditional series need block or summation-by-parts control.

Node-by-node properties, formulas, proof cores, and occurrence anchors.

10.2.1 RA-01: Compactness, extreme values, and closed-bounded reduction

Occurrence anchors	1994-D2-P2; 1995-D1-P6; 1995-D2-P4; 1995-D2-P6; 1996-D1-P6; 1996-D2-P4; 1998-D2-P2; 2000-D1-P1; 2000-D2-P2; 2001-D1-P3; 2001-D2-P6; 2002-D1-P4; 2002-D2-P6; 2003-D2-P3; 2004-D1-P3; 2005-D2-P1; 2005-D2-P5; 2006-D2-P3; 2007-D2-P3; 2008-D1-P1; 2011-D1-P1; 2013-D1-P3; 2013-D1-P4; 2014-D1-P2; 2014-D2-P3; 2014-D2-P5; ... (+11 more in data/imc_theory_node_index.csv)
Original node statement	A continuous function on a compact set attains its minimum and maximum. For unbounded domains, first show extrema cannot occur outside a compact sublevel region.
Literature	[28, 4, 17, 27]

Exact properties and formulas to know.

1. Closed bounded subsets of $\mathbb{R}^n$ are compact.
2. Continuous functions on compact sets attain extrema.
3. Closed subsets of compact sets are compact.
4. Noncompact problems require a preliminary bound confining extrema to a compact set.

Compact proof cores for this node.

1. Create an auxiliary function whose derivative has a sign or whose endpoint values match.
2. Reduce existence to EVT/IVT/Rolle after verifying compactness and differentiability.
3. For limits, isolate the main term and bound the error or tail explicitly.

Use reminder. Use in multivariable extrema, closed-set nearest-point arguments, convex-body problems, and existence of maximizers/minimizers.

Trap reminder. Compactness requires closed and bounded in finite-dimensional Euclidean spaces; infinite-dimensional Hilbert spaces behave differently.

10.2.2 RA-02: Intermediate value, connectedness, density, and continuity transfer

Occurrence anchors	1994-D2-P2; 1995-D1-P2; 1995-D1-P6; 1995-D2-P4; 1995-D2-P6; 1996-D1-P6; 1996-D2-P1; 1996-D2-P4; 1998-D1-P4; 1998-D1-P6; 1998-D2-P5; 2000-D2-P2; 2002-D1-P4; 2003-D2-P3; 2004-D1-P3; 2005-D1-P3; 2006-D1-P1; 2006-D1-P5; 2006-D2-P2; 2007-D2-P1; 2007-D2-P3; 2008-D1-P1; 2009-D1-P1; 2009-D1-P3; 2011-D1-P1; 2012-D1-P4; ... (+18 more in data/imc_theory_node_index.csv)
Original node statement	Continuous images of intervals are intervals; continuous functions preserve limits. Dense subsets such as \{}$mathbbQ$ determine continuous functions on \{}$mathbbR$ when the relevant equality or inequality is closed under limits.
Literature	[28, 4, 17, 27]

Exact properties and formulas to know.

1. Continuous images of intervals are intervals.
2. Dense-set inequalities extend by taking limits.
3. A monotone function has one-sided limits at every point.
4. Jumps in monotone functions prevent interval-surjectivity across the jump.

Compact proof cores for this node.

1. Create an auxiliary function whose derivative has a sign or whose endpoint values match.
2. Reduce existence to EVT/IVT/Rolle after verifying compactness and differentiability.
3. For limits, isolate the main term and bound the error or tail explicitly.

Use reminder. Use for interval image problems, monotone/continuous comparisons on rationals, and connectedness of constraint sets.

Trap reminder. A monotone function need not be continuous unless surjectivity or no jumps is proved.

10.2.3 RA-03: Monotone functions, fixed points, and iteration on intervals

Occurrence anchors	1994-D2-P6; 1995-D1-P4; 1998-D2-P3; 1999-D2-P4; 2000-D1-P1; 2000-D1-P4; 2000-D2-P2; 2001-D2-P2; 2004-D1-P5; 2004-D2-P2; 2007-D2-P3; 2010-D1-P1; 2011-D1-P4; 2015-D2-P6; 2019-D1-P1
Original node statement	A nondecreasing self-map of a compact interval has a fixed point. For continuous interval maps, convergence of iterates can often be characterized by the vanishing of increments plus cluster-point analysis.
Literature	[28, 4, 17, 27]

Exact properties and formulas to know.

1. A nondecreasing self-map of a compact interval has a fixed point.
2. A bounded monotone sequence converges.
3. For continuous iteration, cluster points pass through f.
4. If increments tend to zero, an orbit cannot jump across a neighborhood where $|f(x)-x|$ is bounded below.

Compact proof cores for this node.

1. Create an auxiliary function whose derivative has a sign or whose endpoint values match.
2. Reduce existence to EVT/IVT/Rolle after verifying compactness and differentiability.
3. For limits, isolate the main term and bound the error or tail explicitly.

Use reminder. Use for monotone maps, closed invariant sets, fixed points, and iteration sequences.

Trap reminder. A decreasing map may fail to have a fixed point if discontinuous. Iteration convergence needs cluster-point control.

10.2.4 RA-04: Rolle, mean-value theorem, and divided differences

Occurrence anchors	1994-D1-P2; 1994-D1-P6; 1994-D2-P1; 1994-D2-P3; 1995-D1-P3; 1996-D1-P5; 1996-D2-P5; 1996-D2-P6; 1997-D2-P1; 1997-D2-P3; 1998-D1-P4; 1999-D1-P6; 2000-D1-P2; 2000-D1-P6; 2000-D2-P5; 2001-D1-P6; 2001-D2-P6; 2002-D1-P2; 2002-D1-P5; 2005-D1-P3; 2005-D1-P5; 2005-D2-P4; 2006-D1-P5; 2006-D1-P6; 2007-D2-P6; 2009-D2-P2; ... (+7 more in data/imc_theory_node_index.csv)
Original node statement	Rolle's theorem and the mean-value theorem convert repeated zeros or equal endpoint values into derivative zeros. Divided differences satisfy $f[x_0, \{\}dots, x_n] = f\^{}\{(n)\}(\{\}xi)/n!$ under differentiability.
Literature	[28, 4, 17, 27]

Exact properties and formulas to know.

1. Rolle: equal endpoint values force an interior derivative zero.
2. MVT: $f(b) - f(a) = f'(c)(b - a)$.
3. Repeated Rolle: $m + 1$ zeros force a zero of $f^{(m)}$.
4. Divided difference: $f[x_0, \ldots, x_n] = f^{(n)}(\xi)/n!$.

Compact proof cores for this node.

1. Create an auxiliary function whose derivative has a sign or whose endpoint values match.
2. Reduce existence to EVT/IVT/Rolle after verifying compactness and differentiability.
3. For limits, isolate the main term and bound the error or tail explicitly.

Use reminder. Use whenever many zeros are given, a derivative value must exist, or endpoint data must imply a higher-derivative identity.

Trap reminder. Check differentiability order and that all points lie in the open interval where the conclusion is asserted.

10.2.5 RA-05: Taylor expansion, local asymptotics, and little-o bookkeeping

Occurrence anchors	1994-D1-P2; 1994-D1-P5; 1994-D1-P6; 1994-D2-P1; 1994-D2-P3; 1994-D2-P4; 1994-D2-P6; 1995-D1-P3; 1995-D1-P6; 1995-D2-P2; 1996-D1-P5; 1996-D2-P5; 1996-D2-P6; 1997-D1-P1; 1997-D2-P1; 1997-D2-P3; 1998-D1-P4; 1999-D1-P6; 2000-D1-P2; 2000-D2-P5; 2001-D1-P3; 2001-D1-P6; 2001-D2-P6; 2002-D1-P2; 2004-D1-P6; 2005-D1-P3; ... (+24 more in data/imc_theory_node_index.csv)
Original node statement	Near a point, $C\^{}k$ functions can be expanded with a controlled remainder. Competition solutions often need only the first nonzero term and the sign/size of the remainder.
Literature	[28, 4, 17, 27]

Exact properties and formulas to know.

1. Taylor: $f(a + h) = \sum_{j=0}^{k} f^{(j)}(a)h^j/j! + o(h^k)$.
2. If the first nonzero term is even/odd, it controls local extremum/sign behavior.
3. $O(h^k)$ means bounded by $C|h|^k$; $o(h^k)$ means ratio to h^k tends to 0.
4. For quotient limits, compare the first nonzero terms.

Compact proof cores for this node.

1. Create an auxiliary function whose derivative has a sign or whose endpoint values match.
2. Reduce existence to EVT/IVT/Rolle after verifying compactness and differentiability.
3. For limits, isolate the main term and bound the error or tail explicitly.

Use reminder. Use for local boundedness, sharp constants near equality, ratio limits, and endpoint asymptotics.

Trap reminder. Do not expand beyond available smoothness; distinguish $O(x\^{}k)$ from $o(x\^{}k)$.

10.2.6 RA-06: Auxiliary functions, integrating factors, and differential inequalities

Occurrence anchors	1994-D1-P2; 1994-D2-P1; 1994-D2-P3; 1994-D2-P6; 1995-D1-P2; 1995-D1-P4; 1996-D1-P2; 1998-D1-P3; 1998-D1-P4; 1998-D1-P6; 1999-D1-P2; 1999-D1-P4; 1999-D2-P3; 1999-D2-P4; 2000-D1-P2; 2000-D1-P4; 2000-D1-P6; 2000-D2-P2; 2001-D1-P3; 2001-D1-P6; 2003-D1-P1; 2003-D2-P2; 2004-D1-P5; 2004-D2-P5; 2005-D1-P3; 2005-D1-P5; ... (+36 more in data/imc_theory_node_index.csv)
Original node statement	A differential inequality is often solved by choosing an auxiliary expression whose derivative has a fixed sign, such as \{}$arctanf + x$, e^{−\{}$lambdax$}$f(x)$, or $f' + xf$.
Literature	[28, 4, 17, 27]

Exact properties and formulas to know.

1. If $h' + ah = r$ and $A' = a$, then $(he^A)' = re^A$.
2. If $(\Phi(f, x))' \geq 0$, then Φ is monotone and endpoint limits give inequalities.
3. Completing derivatives such as $(f^2/2 + f')'$ often creates the target expression.
4. Differential inequalities are integrated only after sign and regularity hypotheses are verified.

Compact proof cores for this node.

1. Create an auxiliary function whose derivative has a sign or whose endpoint values match.
2. Reduce existence to EVT/IVT/Rolle after verifying compactness and differentiability.
3. For limits, isolate the main term and bound the error or tail explicitly.

Use reminder. Use for uniqueness, lower/upper bounds, exponential comparison, and ODE-like inequalities.

Trap reminder. If taking logarithms, first prove the function has the required sign on the interval.

10.2.7 RA-07: Convexity, concavity, Jensen, and tangent/chord inequalities

Occurrence anchors	1994-D1-P1; 1994-D1-P2; 1994-D1-P4; 1994-D1-P5; 1994-D1-P6; 1994-D2-P1; 1994-D2-P2; 1994-D2-P5; 1994-D2-P6; 1995-D1-P2; 1995-D1-P3; 1995-D1-P4; 1995-D1-P6; 1995-D2-P2; 1995-D2-P4; 1995-D2-P5; 1996-D1-P2; 1996-D1-P4; 1996-D1-P5; 1996-D2-P2; 1996-D2-P4; 1996-D2-P5; 1996-D2-P6; 1997-D1-P1; 1997-D1-P2; 1997-D1-P4; ... (+153 more in data/imc_theory_node_index.csv)
Original node statement	For convex f, tangent lines lie below the graph and Jensen gives $f(\{}sum\{}lambda_ix_i)\{}le\{}sum\{}lambda_if(x_i)$. For concave functions the inequalities reverse. Equality cases are usually decisive.
Literature	[28, 4, 17, 27]

Exact properties and formulas to know.

1. Convex: $f(tx + (1-t)y) \leq tf(x) + (1-t)f(y)$.
2. Jensen: $f(\sum \lambda_i x_i) \leq \sum \lambda_i f(x_i)$.
3. Differentiable convex functions satisfy tangent support $f(y) \geq f(x) + f'(x)(y - x)$.
4. Twice differentiable f with $f'' \geq 0$ is convex.

Compact proof cores for this node.

1. Create an auxiliary function whose derivative has a sign or whose endpoint values match.
2. Reduce existence to EVT/IVT/Rolle after verifying compactness and differentiability.
3. For limits, isolate the main term and bound the error or tail explicitly.

Use reminder. Use for inequalities, sharp constants, integral estimates, chord comparisons, and geometric convexity.

Trap reminder. Check the interval of convexity and strictness conditions for equality.

10.2.8 RA-08: Integral inequalities, integration by parts, and weighted estimates

Occurrence anchors	1994-D1-P2; 1994-D1-P3; 1994-D1-P4; 1994-D1-P5; 1994-D1-P6; 1994-D2-P1; 1994-D2-P2; 1994-D2-P3; 1994-D2-P5; 1994-D2-P6; 1995-D1-P2; 1995-D1-P4; 1995-D1-P5; 1995-D1-P6; 1995-D2-P2; 1995-D2-P5; 1995-D2-P6; 1996-D1-P2; 1996-D1-P4; 1996-D1-P5; 1996-D2-P1; 1996-D2-P2; 1996-D2-P4; 1996-D2-P5; 1996-D2-P6; 1997-D1-P1; ... (+173 more in data/imc_theory_node_index.csv)
Original node statement	Integration by parts transfers derivatives between factors; nonnegative integrands and pointwise bounds integrate to global inequalities. Introducing primitives often converts a cubic or square integral into a boundary term plus an error.
Literature	[28, 4, 17, 27]

Exact properties and formulas to know.

1. Integration by parts: $\int uv' = uv|_a^b - \int u'v$.
2. Cauchy-Schwarz: $(\int fg)^2 \le (\int f^2)(\int g^2)$.
3. If $0 \le f \le g$, then $\int f \le \int g$.
4. Equality cases often require proportionality or support constraints.

Compact proof cores for this node.

1. Create an auxiliary function whose derivative has a sign or whose endpoint values match.
2. Reduce existence to EVT/IVT/Rolle after verifying compactness and differentiability.
3. For limits, isolate the main term and bound the error or tail explicitly.

Use reminder. Use in $L\hat{}p$ inequalities, integral constraints, monotone integrals, and estimates of functions by their derivatives.

Trap reminder. Boundary terms are often the whole point; compute their signs before discarding them.

10.2.9 RA-09: Fubini/Tonelli and change of variables

Occurrence anchors	1994-D2-P1; 1994-D2-P3; 1995-D1-P2; 1995-D2-P5; 1996-D1-P2; 1996-D1-P4; 1996-D2-P1; 1997-D1-P1; 1997-D1-P2; 1997-D1-P4; 1998-D1-P3; 1998-D1-P6; 1999-D1-P4; 1999-D2-P3; 1999-D2-P4; 2001-D1-P1; 2002-D1-P3; 2003-D1-P1; 2003-D2-P6; 2004-D1-P2; 2004-D2-P5; 2005-D2-P1; 2007-D2-P2; 2008-D1-P4; 2010-D1-P2; 2010-D2-P1; ... (+13 more in data/imc_theory_node_index.csv)
Original node statement	For nonnegative or absolutely integrable functions, interchange the order of integration and use substitutions to convert regions or kernels into simpler forms.
Literature	[28, 4, 17, 27]

Exact properties and formulas to know.

1. Tonelli permits swapping nonnegative sums/integrals.
2. Fubini permits swapping absolutely integrable integrals.
3. Change of variables multiplies by the absolute Jacobian.
4. Draw or write the transformed domain explicitly before integrating.

Compact proof cores for this node.

1. Create an auxiliary function whose derivative has a sign or whose endpoint values match.
2. Reduce existence to EVT/IVT/Rolle after verifying compactness and differentiability.
3. For limits, isolate the main term and bound the error or tail explicitly.

Use reminder. Use for nested integrals, averaging operators, double integrals over the unit square, and kernel formulas.

Trap reminder. Justify interchanging integrals or sums when signs are not nonnegative.

10.2.10 RA-10: Riemann sums and sum-integral asymptotics

Occurrence anchors	1994-D1-P2; 1994-D1-P5; 1994-D2-P6; 1995-D1-P2; 1995-D1-P3; 1995-D2-P2; 1995-D2-P5; 1995-D2-P6; 1996-D1-P4; 1996-D1-P5; 1996-D2-P1; 1996-D2-P5; 1997-D1-P1; 1997-D1-P2; 1997-D2-P1; 1997-D2-P3; 1998-D1-P3; 1999-D1-P2; 1999-D2-P3; 2000-D1-P4; 2000-D1-P6; 2000-D2-P5; 2001-D1-P3; 2001-D1-P6; 2001-D2-P2; 2001-D2-P5; ... (+47 more in data/imc_theory_node_index.csv)
Original node statement	If $h\backslash\{\}to0$, sums of the form $h\backslash\{\}sumf(nh)$ approximate $\backslash\{\}intf$. For infinite ranges, split into a compact main part and a tail controlled uniformly.
Literature	[28, 4, 17, 27]

Exact properties and formulas to know.

1. Taylor: $f(a+h)=\sum_{j=0}^{k} f^{(j)}(a)h^j/j!+o(h^k)$.
2. If the first nonzero term is even/odd, it controls local extremum/sign behavior.
3. $O(h^k)$ means bounded by $C|h|^k$; $o(h^k)$ means ratio to h^k tends to 0.
4. For quotient limits, compare the first nonzero terms.

Compact proof cores for this node.

1. Create an auxiliary function whose derivative has a sign or whose endpoint values match.
2. Reduce existence to EVT/IVT/Rolle after verifying compactness and differentiability.
3. For limits, isolate the main term and bound the error or tail explicitly.

Use reminder. Use for limits of sums, logarithmic sums, periodic averaging, and asymptotic counting.

Trap reminder. Tail control is the common missing step; do not use finite-interval Riemann sums on an infinite domain without a tail estimate.

10.2.11 RA-11: Series convergence, rearrangements, and alternating/Dirichlet tools

Occurrence anchors	1994-D1-P3; 1994-D2-P1; 1994-D2-P3; 1994-D2-P5; 1995-D1-P5; 1995-D1-P6; 1995-D2-P5; 1995-D2-P6; 1996-D1-P2; 1996-D1-P4; 1996-D1-P5; 1996-D2-P1; 1996-D2-P5; 1996-D2-P6; 1997-D1-P1; 1997-D1-P2; 1997-D1-P4; 1997-D1-P5; 1997-D1-P6; 1997-D2-P3; 1998-D1-P3; 1998-D1-P6; 1998-D2-P1; 1999-D1-P2; 1999-D1-P4; 1999-D2-P3; ... (+41 more in data/imc_theory_node_index.csv)
Original node statement	Convergence can be altered by nonabsolute rearrangements. Alternating and Dirichlet-type arguments reduce convergence to decay plus bounded variation or controlled differences.
Literature	[28, 4, 17, 27]

Exact properties and formulas to know.

1. Absolute convergence implies convergence and permits rearrangement.
2. Alternating test requires decreasing terms tending to zero.
3. Dirichlet follows from summation by parts and bounded partial sums.
4. Conditional rearrangements need explicit block/tail control.

Compact proof cores for this node.

1. Create an auxiliary function whose derivative has a sign or whose endpoint values match.
2. Reduce existence to EVT/IVT/Rolle after verifying compactness and differentiability.
3. For limits, isolate the main term and bound the error or tail explicitly.

Use reminder. Use for rearranged series, alternating trigonometric series, Carleman-type inequalities, and positive series estimates.

Trap reminder. Conditional convergence is not preserved by arbitrary permutations.

10.2.12 RA-12: Uniform continuity and periodic averaging

Occurrence anchors	1994-D1-P5
Original node statement	Continuous functions on compact intervals are uniformly continuous. Products with rapidly oscillating periodic factors average to the product of means after partitioning into small intervals.
Literature	[28, 4, 17, 27]

Exact properties and formulas to know.

1. Every continuous function on a compact interval is uniformly continuous.
2. The modulus $\omega_f(\delta) \to 0$ controls replacement by endpoint values.
3. Periodic functions have equal integrals over translated full periods.
4. Fast oscillation averages to the mean when the slow factor is uniformly continuous.

Compact proof cores for this node.

1. Create an auxiliary function whose derivative has a sign or whose endpoint values match.
2. Reduce existence to EVT/IVT/Rolle after verifying compactness and differentiability.
3. For limits, isolate the main term and bound the error or tail explicitly.

Use reminder. Use for limits of $\backslash\{\}intf(x)g(nx)\backslash\{\},dx$ and periodic integral averaging.

Trap reminder. Keep the period and interval length consistent; normalize by the period correctly.

10.2.13 RA-13: L'Hospital, ratio limits, and growth comparisons

Occurrence anchors	1994-D1-P2; 1994-D1-P5; 1994-D2-P6; 1995-D1-P3; 1995-D1-P4; 1995-D2-P2; 1996-D1-P4; 1996-D2-P1; 1996-D2-P5; 1997-D1-P1; 1997-D1-P2; 1997-D2-P1; 2000-D1-P6; 2001-D1-P3; 2001-D1-P6; 2001-D2-P5; 2003-D1-P1; 2003-D2-P2; 2004-D1-P6; 2004-D2-P5; 2005-D1-P5; 2010-D1-P2; 2010-D1-P3; 2011-D2-P3; 2014-D1-P2; 2015-D2-P7; ... (+7 more in data/imc_theory_node_index.csv)
Original node statement	When numerator and denominator have compatible divergent or vanishing behavior, derivative ratios can determine function ratios. Growth comparisons often reduce to exponentials, logarithms, or powers.
Literature	[28, 4, 17, 27]

Exact properties and formulas to know.

1. LHospital applies to $0/0$ or ∞/∞ forms under the usual derivative-ratio hypotheses.
2. Exponential weights can dominate polynomial or subexponential factors.
3. Growth comparisons require verifying denominator tends to infinity or zero.
4. Ratio limits should be supported by monotonicity, LHospital, or squeeze estimates.

Compact proof cores for this node.

1. Create an auxiliary function whose derivative has a sign or whose endpoint values match.
2. Reduce existence to EVT/IVT/Rolle after verifying compactness and differentiability.
3. For limits, isolate the main term and bound the error or tail explicitly.

Use reminder. Use for endpoint ratio limits, exponential weights, and asymptotic estimates.

Trap reminder. L\{}'Hospital requires the appropriate hypotheses; for inequalities, use comparison rather than equality of limits when needed.

10.2.14 RA-14: Weierstrass approximation and polynomial approximation

Occurrence anchors	1995-D2-P4
Original node statement	Every continuous function on a compact interval can be uniformly approximated by polynomials. Symmetry can be imposed by averaging $p(x)$ with $p(-x)$ to obtain even or odd approximants.
Literature	[28, 4, 17, 27]

Exact properties and formulas to know.

1. Weierstrass approximation: continuous functions on $[a,b]$ are uniformly approximated by polynomials.
2. Odd/even symmetrization preserves uniform approximation and parity.
3. Polynomial bases may omit low-degree terms only after those terms are approximated separately.
4. Uniform norm is $\|f\|_\infty = \max|f|$.

Compact proof cores for this node.

1. Create an auxiliary function whose derivative has a sign or whose endpoint values match.
2. Reduce existence to EVT/IVT/Rolle after verifying compactness and differentiability.
3. For limits, isolate the main term and bound the error or tail explicitly.

Use reminder. Use for uniform approximation by restricted polynomial families and Chebyshev-style extremal examples.

Trap reminder. Approximate the symmetric part, then handle missing low-degree terms separately if the basis omits them.

10.2.15 RA-15: L2 spaces, orthonormal systems, and dominated convergence

Occurrence anchors	1995-D2-P6
Original node statement	In L^2, orthonormality gives coefficient constraints and no pointwise-convergent bounded subsequence can contradict norm preservation when dominated convergence applies.
Literature	[28, 4, 17, 27]

Exact properties and formulas to know.

1. Orthonormality in L^2 means $\int f_m f_n = \delta_{mn}$.
2. Dominated convergence passes limits through integrals under a common integrable bound.
3. A complete orthonormal system has only zero vector orthogonal to all basis elements.
4. Pointwise convergence plus uniform boundedness can contradict preservation of L^2 norm.

Compact proof cores for this node.

1. Create an auxiliary function whose derivative has a sign or whose endpoint values match.
2. Reduce existence to EVT/IVT/Rolle after verifying compactness and differentiability.
3. For limits, isolate the main term and bound the error or tail explicitly.

Use reminder. Use for orthonormal continuous functions, Hilbert-space arguments, and bounded convergence of integrals.

Trap reminder. Pointwise convergence plus uniform boundedness allows dominated convergence, but only on finite measure domains.

10.2.16 RA-16: Baire/category, dense sets, and pathological continuous functions

Occurrence anchors	1997-D2-P6; 1998-D1-P4; 1998-D2-P6; 2000-D2-P2; 2002-D1-P5; 2003-D2-P5
Original node statement	Closed intervals cannot be countable unions of nowhere dense sets. Cantor-type constructions create uncountable closed sets with prescribed local behavior.
Literature	[28, 4, 17, 27]

Exact properties and formulas to know.

1. Baire: a complete metric space is not a countable union of nowhere dense sets.
2. A set is dense if every nonempty open interval meets it.
3. Cantor-type constructions give uncountable closed nowhere dense sets.
4. Local extrema can be forced densely by nonmonotonicity on every interval.

Compact proof cores for this node.

1. Create an auxiliary function whose derivative has a sign or whose endpoint values match.
2. Reduce existence to EVT/IVT/Rolle after verifying compactness and differentiability.
3. For limits, isolate the main term and bound the error or tail explicitly.

Use reminder. Use for dense local minima, uncountably many zero crossings, and existence/nonexistence of functions with many level sets.

Trap reminder. Countability and uncountability arguments must distinguish isolated intervals from point sets.

10.2.17 RA-17: Sequence recurrences, monotone bounds, and generating functions

Occurrence anchors	1995-D2-P2; 1996-D1-P4; 1998-D2-P6; 2001-D2-P2; 2002-D2-P3; 2003-D1-P1; 2003-D1-P5; 2003-D1-P6; 2003-D2-P6; 2005-D1-P2; 2007-D1-P4; 2008-D2-P3; 2010-D2-P2; 2011-D2-P1; 2012-D2-P2; 2015-D1-P5; 2017-D2-P8; 2017-D2-P9; 2018-D2-P7; 2019-D1-P4; 2021-D1-P3; 2024-D2-P8
Original node statement	Recursive sequences are attacked by monotone normalization, induction estimates, telescoping transformations, or generating functions $F(x) = \{\}suma_nx\^n$.
Literature	[28, 4, 17, 27]

Exact properties and formulas to know.

1. A nondecreasing self-map of a compact interval has a fixed point.
2. A bounded monotone sequence converges.
3. For continuous iteration, cluster points pass through f.
4. If increments tend to zero, an orbit cannot jump across a neighborhood where $|f(x) - x|$ is bounded below.

Compact proof cores for this node.

1. Create an auxiliary function whose derivative has a sign or whose endpoint values match.
2. Reduce existence to EVT/IVT/Rolle after verifying compactness and differentiability.
3. For limits, isolate the main term and bound the error or tail explicitly.

Use reminder. Use for recurrence limits, coefficient sums, and nonlinear-looking sequences that become telescoping after substitution.

Trap reminder. Check radius of convergence before manipulating generating functions; induction bounds need verified base cases.

10.2.18 RA-18: Trigonometric and hyperbolic identities with monotonicity

Occurrence anchors	1995-D2-P5; 1996-D1-P2; 1996-D2-P2; 1997-D2-P3; 2000-D1-P4; 2001-D2-P2; 2001-D2-P6; 2003-D2-P2; 2004-D1-P3; 2004-D2-P5; 2006-D2-P3; 2014-D2-P3; 2015-D1-P4; 2017-D1-P2
Original node statement	Use standard identities, product-to-sum formulas, monotonicity of $\{\}sinx/x$, and hyperbolic recurrences such as $\{\}cosh((m+1)t) = 2\{\}cosht\{\}cosh(mt) - \{\}cosh((m-1)t)$.
Literature	[28, 4, 17, 27]

Exact properties and formulas to know.

1. $\sin 2x = 2\sin x\cos x$ and $\cos(u \pm v) = \cos u\cos v \mp \sin u\sin v$.
2. $\cosh(u+v) = \cosh u\cosh v + \sinh u\sinh v$.
3. $\cosh^2 u - \sinh^2 u = 1$.
4. Monotonicity of $\sin x/x$ near zero is a standard comparison tool.

Compact proof cores for this node.

1. Create an auxiliary function whose derivative has a sign or whose endpoint values match.
2. Reduce existence to EVT/IVT/Rolle after verifying compactness and differentiability.
3. For limits, isolate the main term and bound the error or tail explicitly.

Use reminder. Use for trig integrals, zero-counting of trig sums, hyperbolic rationality, and trigonometric inequalities.

Trap reminder. Track the interval: many monotonicity claims only hold on $(0, \{\}pi/2)$ or $(0, \{\}pi)$.

10.2.19 RA-19: Gradient/Lipschitz-gradient optimization and convex smoothness

Occurrence anchors	1994-D2-P2; 2002-D2-P6; 2005-D2-P5
Original node statement	For convex differentiable f with L-Lipschitz gradient, descent estimates give $\{\}\|\{\}nablaf(x) - \{\}nablaf(y)\{\}\|\^2\{\}leL\{\}langle\{\}nablaf(x) - \{\}nablaf(y), x - y\{\}rangle$. Lipschitz gradients also control uniqueness of maximizers under geometric constraints.
Literature	[28, 4, 17, 27]

Exact properties and formulas to know.

1. L-Lipschitz gradient: $\|\nabla f(x) - \nabla f(y)\| \le L\|x - y\|$.

2. Convexity gives monotonicity of the gradient: $\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq 0$.
3. Cocoercivity: $\|\nabla f(x) - \nabla f(y)\|^2 \leq L\langle \nabla f(x) - \nabla f(y), x - y \rangle$.
4. Descent lemma: $f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + L\|y - x\|^2/2$.

Compact proof cores for this node.

1. Create an auxiliary function whose derivative has a sign or whose endpoint values match.
2. Reduce existence to EVT/IVT/Rolle after verifying compactness and differentiability.
3. For limits, isolate the main term and bound the error or tail explicitly.

Use reminder. Use in convex analysis, gradient inequalities, and maximum-location problems on disks or balls.

Trap reminder. Gradient Lipschitzness is a global condition; local Hessian intuition must be converted into an inequality.

10.2.20 RA-20: Measure and probability-integral estimates

Occurrence anchors	2019-D2-P10; 2022-D1-P1
Original node statement	Integral estimates can be interpreted probabilistically, and probability problems can often be turned into expectation or measure integrals. Symmetry and independence are the main reductions.
Literature	[28, 4, 17, 27]

Exact properties and formulas to know.

1. Integration by parts: $\int uv' = uv|_a^b - \int u'v$.
2. Cauchy-Schwarz: $(\int fg)^2 \leq (\int f^2)(\int g^2)$.
3. If $0 \leq f \leq g$, then $\int f \leq \int g$.
4. Equality cases often require proportionality or support constraints.

Compact proof cores for this node.

1. Create an auxiliary function whose derivative has a sign or whose endpoint values match.
2. Reduce existence to EVT/IVT/Rolle after verifying compactness and differentiability.
3. For limits, isolate the main term and bound the error or tail explicitly.

Use reminder. Use for random points, expected geometric quantities, and inequalities for integrable functions.

Trap reminder. Distinguish deterministic inequalities from almost-sure statements; justify independence and symmetry.

10.3 Complex analysis / polynomials: exact theorem bank

Literature anchor: [1, 11, 24, 21]. **Global theorem/property bank.**

1. **More roots than degree.** A nonzero polynomial of degree d over a field has at most d roots. Hence agreement at $d+1$ points implies polynomial identity.
2. **Factor theorem and multiplicity.** $P(a)=0$ iff $(x-a)$ divides $P(x)$. Multiplicity m means $(x-a)^m \mid P$ and $(x-a)^{m+1} \nmid P$.
3. **Vieta and logarithmic derivative.** For $P(x)=c\prod_i (x-r_i)$, $P'/P=\sum_i 1/(x-r_i)$ away from roots. Vieta identifies coefficients with elementary symmetric sums of roots.
4. **Unit-circle identities.** If $|\alpha|=1$, then $\bar{\alpha}=1/\alpha$. Also $\mathrm{Re}\,\frac{z+\alpha}{z-\alpha}=\frac{|z|^2-1}{|z-\alpha|^2}$ for $z\neq\alpha$.
5. **Roots of unity filter.** For $\omega=e^{2\pi i/m}$, $m^{-1}\sum_{j=0}^{m-1}\omega^{j(k-r)}$ is 1 if $k\equiv r \pmod m$ and 0 otherwise.
6. **Coefficient extraction.** $a_k=\frac{1}{2\pi i}\int_{|z|=1}P(z)z^{-k-1}\,dz$. Equivalently, Fourier orthogonality extracts exponents.
7. **Maximum modulus.** A nonconstant holomorphic function cannot have a strict interior maximum of modulus; on a disk the maximum is attained on the boundary.
8. **Meromorphic rationality.** A meromorphic function on the Riemann sphere is rational. In practice, if all finite singularities are poles and infinity is not essential, the function is rational.

Node-by-node properties, formulas, proof cores, and occurrence anchors.

10.3.1 CP-01: Polynomial identity and more-roots-than-degree principle

Occurrence anchors	1994-D2-P4; 1995-D1-P5; 1995-D2-P3; 1995-D2-P4; 1996-D2-P2; 1998-D1-P2; 1998-D1-P5; 1998-D2-P2; 1999-D1-P1; 1999-D2-P2; 1999-D2-P6; 2000-D1-P6; 2000-D2-P3; 2000-D2-P4; 2000-D2-P6; 2001-D1-P4; 2001-D2-P1; 2002-D1-P1; 2002-D2-P3; 2003-D1-P3; 2003-D1-P6; 2003-D2-P2; 2004-D1-P6; 2005-D1-P4; 2005-D2-P2; 2005-D2-P6; ... (+42 more in data/imc_theory_node_index.csv)
Original node statement	A nonzero polynomial of degree d over a field has at most d roots. Therefore, if it has more than d roots, it is identically zero.
Literature	[1, 11, 24, 21]

Exact properties and formulas to know.

1. A nonzero polynomial of degree d has at most d roots.
2. Two degree-d polynomials agreeing at $d+1$ points are equal.
3. Polynomial identities over infinite fields can be checked on infinitely many points.
4. Multivariable identities reduce by induction on variables.

Compact proof cores for this node.

1. Use factorization/root counting before analytic machinery.
2. Translate geometric root constraints into conjugation, real-part, or modulus identities.
3. For coefficient questions, use comparison, interpolation, or Cauchy/Fourier extraction.

Use reminder. Use for polynomial divisibility, coefficient vanishing, finite-field polynomial functions, and interpolation-style arguments.

Trap reminder. Over finite fields, different polynomials can define the same function; reduce modulo x^p-x when working on \{}$mathbbF_p$.

10.3.2 CP-02: Factorization, irreducibility, and Gauss-style integer polynomial arguments

Occurrence anchors	1995-D2-P3; 1998-D1-P5; 1998-D2-P2; 2000-D2-P3; 2001-D1-P4; 2001-D2-P1; 2005-D2-P2; 2006-D1-P4; 2007-D1-P1; 2008-D1-P3; 2008-D2-P1; 2008-D2-P2; 2008-D2-P4; 2009-D2-P1; 2012-D1-P5; 2012-D2-P1; 2013-D2-P1; 2015-D2-P10; 2017-D1-P5; 2017-D2-P7; 2018-D2-P9; 2020-D1-P4; 2021-D2-P7; 2023-D1-P3
Original node statement	Integer polynomial divisibility often uses Gauss lemma, primitive factorization, and reduction modulo primes. Cyclotomic and simple irreducibility facts turn one root into all conjugate roots.
Literature	[1, 11, 24, 21]

Exact properties and formulas to know.

1. Gauss lemma relates factorization in $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$.

2. Irreducibility over $\mathbb{Q}$ forces conjugate algebraic roots to appear together.
3. Content times primitive part separates coefficient gcd from structural factors.
4. Eisenstein applies after suitable shifts in many integer-polynomial problems.

Compact proof cores for this node.

1. Use factorization/root counting before analytic machinery.
2. Translate geometric root constraints into conjugation, real-part, or modulus identities.
3. For coefficient questions, use comparison, interpolation, or Cauchy/Fourier extraction.

Use reminder. Use for \{}$mathbbZ[x]$ divisibility, rational functions integer-valued infinitely often, and roots of unity.

Trap reminder. Divisibility in values is not the same as divisibility in \{}$mathbbZ[x]$.

10.3.3 CP-03: Vieta, logarithmic derivatives, and real-root inequalities

Occurrence anchors	1995-D2-P3; 1998-D1-P5; 1998-D2-P2; 2000-D2-P3; 2000-D2-P4; 2001-D1-P4; 2003-D1-P6; 2005-D1-P4; 2005-D2-P2; 2006-D1-P4; 2007-D1-P1; 2007-D1-P5; 2007-D2-P6; 2008-D2-P1; 2009-D2-P4; 2010-D2-P5; 2012-D1-P5; 2012-D2-P1; 2013-D2-P1; 2014-D1-P3; 2015-D2-P10; 2017-D1-P5; 2017-D2-P7; 2018-D2-P9; 2020-D1-P4; 2021-D2-P7; ... (+2 more in data/imc_theory_node_index.csv)
Original node statement	For $P(x) = \{}prod_i(x - r_i)$, $P'(x)/P(x) = \{}sum_i1/(x - r_i)$. This converts root-location assumptions into sums of squares or coefficient inequalities.
Literature	[1, 11, 24, 21]

Exact properties and formulas to know.

1. Vieta formulas express coefficients as elementary symmetric sums.
2. $P'/P = \sum 1/(x - r_i)$ away from roots.
3. Multiple roots must be handled separately before dividing by P.
4. Sums of squares from logarithmic derivatives prove real-root inequalities.

Compact proof cores for this node.

1. Use factorization/root counting before analytic machinery.
2. Translate geometric root constraints into conjugation, real-part, or modulus identities.
3. For coefficient questions, use comparison, interpolation, or Cauchy/Fourier extraction.

Use reminder. Use for polynomials with real roots, roots on the unit circle, and inequalities involving P, P', P''.

Trap reminder. Handle points that are roots of P separately before dividing by $P(x)$.

10.3.4 CP-04: Unit-circle roots and conjugation identities

Occurrence anchors	1995-D2-P3; 2000-D1-P2; 2000-D2-P3; 2002-D2-P5; 2008-D2-P1; 2013-D2-P1; 2015-D1-P1; 2016-D2-P10; 2017-D1-P1; 2018-D2-P9; 2021-D2-P7; 2023-D1-P2; 2024-D2-P7
Original node statement	When $\lvert\alpha\rvert = 1$, identities involving $z, \bar{z}$, and $\frac{z+\alpha}{z-\alpha}$ reveal signs of real parts and force roots onto the unit circle.
Literature	[1, 11, 24, 21]

Exact properties and formulas to know.

1. $\lvert\alpha\rvert = 1$ implies $\bar{\alpha} = 1/\alpha$.
2. $\mathrm{Re}((z + \alpha)/(z - \alpha)) = (\lvert z\rvert^2 - 1)/\lvert z - \alpha\rvert^2$.
3. Unit-circle root arguments often prove no zero off the circle.
4. Real-coefficient polynomials have conjugate-pair roots.

Compact proof cores for this node.

1. Use factorization/root counting before analytic machinery.
2. Translate geometric root constraints into conjugation, real-part, or modulus identities.
3. For coefficient questions, use comparison, interpolation, or Cauchy/Fourier extraction.

Use reminder. Use for root-location preservation under derivative-like transformations and conditions $\lvert a\rvert = \lvert b\rvert = 1$.

Trap reminder. Do not divide by zero at existing roots; remove common factors or argue by continuity.

10.3.5 CP-05: Roots of unity, cyclotomic filters, and discrete Fourier tricks

Occurrence anchors	1995-D2-P3; 1997-D1-P3; 1999-D2-P2; 1999-D2-P6; 2000-D1-P2; 2000-D2-P3; 2001-D1-P4; 2001-D2-P3; 2002-D2-P5; 2007-D1-P5; 2007-D1-P6; 2008-D2-P1; 2010-D2-P5; 2013-D2-P1; 2014-D1-P3; 2016-D2-P10; 2017-D1-P1; 2017-D1-P5; 2017-D2-P7; 2018-D2-P9; 2021-D2-P7; 2023-D1-P2; 2024-D1-P1; 2024-D2-P7
Original node statement	The root-of-unity filter extracts residue classes: \{}$frac1m\{}sum_{j = 0}^{m - 1}\{}omega^{jN}$ equals 1 if $m\{}midN$ and 0 otherwise. Cyclotomic irreducibility can force all primitive roots to appear.
Literature	[1, 11, 24, 21]

Exact properties and formulas to know.

1. $\sum_{j=0}^{m-1} \omega^{jk} = 0$ unless $m \mid k$.
2. $x^m - 1 = \prod_{j=0}^{m-1}(x - \omega^j)$.
3. Cyclotomic factors collect primitive roots.
4. Fourier filtering extracts residue classes of exponents or sums.

Compact proof cores for this node.

1. Use factorization/root counting before analytic machinery.
2. Translate geometric root constraints into conjugation, real-part, or modulus identities.
3. For coefficient questions, use comparison, interpolation, or Cauchy/Fourier extraction.

Use reminder. Use for probability modulo m, coefficient sums by congruence class, divisibility by cyclotomic polynomials, and Fourier coefficients of finite sets.

Trap reminder. Check whether all roots, primitive roots, or all nontrivial roots are needed.

10.3.6 CP-06: Cauchy formula and coefficient extraction on the unit circle

Occurrence anchors	2007-D1-P5; 2007-D1-P6; 2009-D1-P4; 2013-D2-P1; 2014-D1-P3; 2017-D2-P7; 2023-D1-P3
Original node statement	For holomorphic p, coefficients are integrals over the unit circle. Cauchy formulas let coefficient multipliers be represented by convolution kernels.
Literature	[1, 11, 24, 21]

Exact properties and formulas to know.

1. Cauchy coefficient formula extracts a_k from a contour integral.
2. On $|z| = 1$, $\int z^j \bar{z}^k$ gives orthogonality.
3. Parseval-type coefficient bounds follow by integrating $|P|^2$.
4. Kernels built from coefficients must be checked for nonnegativity.

Compact proof cores for this node.

1. Use factorization/root counting before analytic machinery.
2. Translate geometric root constraints into conjugation, real-part, or modulus identities.
3. For coefficient questions, use comparison, interpolation, or Cauchy/Fourier extraction.

Use reminder. Use for polynomial coefficient transformations bounded by sup norms and holomorphic estimates on the unit disk.

Trap reminder. The maximum on the closed disk is controlled by the boundary only for holomorphic functions.

10.3.7 CP-07: Maximum modulus principle

Occurrence anchors	nan
Original node statement	A nonconstant holomorphic function reaches its maximum modulus on the boundary of a compact domain. This underlies many unit-disk inequalities.
Literature	[1, 11, 24, 21]

Exact properties and formulas to know.

1. Maximum of $|f|$ on a closed disk occurs on the boundary for holomorphic f.
2. If a maximum occurs in the interior for nonconstant holomorphic f, contradiction.

3. Remove poles/singularities before applying maximum modulus.
4. For polynomials on the unit disk, only boundary values need be checked.

Compact proof cores for this node.

1. Use factorization/root counting before analytic machinery.
2. Translate geometric root constraints into conjugation, real-part, or modulus identities.
3. For coefficient questions, use comparison, interpolation, or Cauchy/Fourier extraction.

Use reminder. Use for bounding values inside the disk from boundary data, often after multiplying by a polynomial.

Trap reminder. The function must be holomorphic on a neighborhood of the closed domain, not merely inside it with boundary singularities.

10.3.8 CP-08: Laurent expansions, isolated singularities, and rationality of meromorphic functions

Occurrence anchors	2004-D1-P6
Original node statement	A meromorphic function on the Riemann sphere is rational. Local Laurent expansions determine principal parts and hence the rational function when poles are known.
Literature	[1, 11, 24, 21]

Exact properties and formulas to know.

1. Laurent series separates principal and analytic parts.
2. Pole order is the highest negative exponent.
3. A meromorphic function on the sphere is rational.
4. Behavior at infinity is studied by substituting $w = 1/z$.

Compact proof cores for this node.

1. Use factorization/root counting before analytic machinery.
2. Translate geometric root constraints into conjugation, real-part, or modulus identities.
3. For coefficient questions, use comparison, interpolation, or Cauchy/Fourier extraction.

Use reminder. Use for sums over logarithm branches, pole-order computations, and rational expressions from analytic data.

Trap reminder. Uniform convergence on compact subsets away from singularities must be checked before termwise analytic operations.

10.3.9 CP-09: Fejer kernels and positive trigonometric kernels

Occurrence anchors	2009-D1-P4
Original node statement	The Fejer kernel $F_k(e\^\{it\}) = \|1 + e\^\{it\} + \\\{\}cdots + e\^\{i(k-1)t\}\|\^2$ is nonnegative. Convex coefficient sequences decompose into nonnegative combinations of Fejer kernels.
Literature	[1, 11, 24, 21]

Exact properties and formulas to know.

1. Fejer kernel is nonnegative: $F_k(e^{it}) = |1 + e^{it} + \cdots + e^{i(k-1)t}|^2$.
2. Convex sequences decompose via second differences.
3. Nonnegative kernels preserve sup norm under convolution/averaging.
4. Boundary maximum plus positive kernel gives polynomial coefficient inequalities.

Compact proof cores for this node.

1. Use factorization/root counting before analytic machinery.
2. Translate geometric root constraints into conjugation, real-part, or modulus identities.
3. For coefficient questions, use comparison, interpolation, or Cauchy/Fourier extraction.

Use reminder. Use for sup-norm contraction of polynomial coefficient multipliers and positivity of trigonometric sums.

Trap reminder. Convexity of the coefficient sequence is the exact hypothesis that makes the kernel nonnegative.

10.3.10 CP-10: Interpolation, Lagrange basis, and Chebyshev extremality

Occurrence anchors	1995-D2-P4; 1998-D2-P2
Original node statement	A polynomial of degree at most n is determined by $n+1$ values; derivatives can be written using Lagrange basis derivatives. Chebyshev polynomials often give extremal oscillation.
Literature	[1, 11, 24, 21]

Exact properties and formulas to know.

1. Lagrange interpolation reconstructs degree-$\leq n$ polynomials from $n+1$ values.
2. $\ell_j(x_i) = \delta_{ij}$ for Lagrange basis.
3. Chebyshev $T_n(\cos\theta) = \cos n\theta$.
4. Alternation characterizes sharp sup-norm extremals.

Compact proof cores for this node.

1. Use factorization/root counting before analytic machinery.
2. Translate geometric root constraints into conjugation, real-part, or modulus identities.
3. For coefficient questions, use comparison, interpolation, or Cauchy/Fourier extraction.

Use reminder. Use for bounded polynomial values at selected points and maximizing derivatives.

Trap reminder. The extremal sign pattern must match the signs of the derivative of the Lagrange basis at the target point.

10.3.11 CP-11: Stable polynomials and coefficient inequalities

Occurrence anchors	2003-D1-P6; 2014-D1-P3; 2023-D1-P3
Original node statement	Polynomials with all roots in the left half-plane have positive coefficients after normalization and satisfy log-concavity-like coefficient inequalities through factorization into positive linear and quadratic factors.
Literature	[1, 11, 24, 21]

Exact properties and formulas to know.

1. Use root counting before heavy analysis.
2. Factor at known roots.
3. Extract coefficients by comparing powers or by Cauchy/Fourier tools.
4. Check multiplicities and exceptional roots separately.

Compact proof cores for this node.

1. Use factorization/root counting before analytic machinery.
2. Translate geometric root constraints into conjugation, real-part, or modulus identities.
3. For coefficient questions, use comparison, interpolation, or Cauchy/Fourier extraction.

Use reminder. Use for inequalities among coefficients under root-location assumptions.

Trap reminder. Keep track of the order of coefficients and signs after normalizing the leading coefficient.

10.3.12 CP-12: Partial fractions and cotangent-type expansions

Occurrence anchors	2004-D1-P6
Original node statement	Classical partial fractions such as cotangent expansions evaluate sums over arithmetic progressions of poles. Differentiating produces higher-power denominator sums.
Literature	[1, 11, 24, 21]

Exact properties and formulas to know.

1. Use root counting before heavy analysis.
2. Factor at known roots.
3. Extract coefficients by comparing powers or by Cauchy/Fourier tools.
4. Check multiplicities and exceptional roots separately.

Compact proof cores for this node.

1. Use factorization/root counting before analytic machinery.
2. Translate geometric root constraints into conjugation, real-part, or modulus identities.
3. For coefficient questions, use comparison, interpolation, or Cauchy/Fourier extraction.

Use reminder. Use for explicit sums over all logarithm branches or equally spaced complex poles.

Trap reminder. Termwise differentiation requires locally uniform convergence away from poles.

10.3.13 CP-13: Polynomial functional identities and algebraic composition

Occurrence anchors	2000-D1-P2; 2007-D1-P3; 2013-D2-P1; 2014-D1-P3; 2017-D2-P7; 2018-D2-P9; 2023-D1-P3; 2024-D1-P1
Original node statement	Polynomial identities in several variables are solved by substituting special values, comparing degrees, and using the identity theorem for polynomials.
Literature	[1, 11, 24, 21]

Exact properties and formulas to know.

1. Use root counting before heavy analysis.
2. Factor at known roots.
3. Extract coefficients by comparing powers or by Cauchy/Fourier tools.
4. Check multiplicities and exceptional roots separately.

Compact proof cores for this node.

1. Use factorization/root counting before analytic machinery.
2. Translate geometric root constraints into conjugation, real-part, or modulus identities.
3. For coefficient questions, use comparison, interpolation, or Cauchy/Fourier extraction.

Use reminder. Use for classifying polynomials satisfying multiplicative or composition identities and for systems $p(w) = p(z), q(w) = q(z)$.

Trap reminder. Special-value substitutions must not lose solutions; after guessing a form, verify the full identity.

CORRECTION MATH-002 — legacy statement superseded; click for canonical replacement.

10.4 Algebra / number theory: exact theorem bank

Literature anchor: [5, 15, 20]. **Global theorem/property bank.**

1. **Bezout and Euclid.** $\gcd(a,b) = d$ iff $ua + vb = d$ for some integers u, v. If $\gcd(a,b) = 1$ and $a \mid bc$, then $a \mid c$.
2. **CRT.** If m, n are coprime, every pair of residues modulo m, n has a unique solution modulo mn; Bezout coefficients give it explicitly.
3. **Prime fields.** In $\mathbb{F}_p$, nonzero elements form a group of order $p-1$, so $a^{p-1} = 1$ for $a \neq 0$ and $a^p = a$ for all a.
4. **Polynomial root bound over fields.** A degree-d nonzero polynomial has at most d roots. A degree $< p$ polynomial over $\mathbb{F}_p$ with all p roots is zero.
5. **Integer polynomial difference.** If $P \in \mathbb{Z}[x]$, then $h \mid P(x+h) - P(x)$ for integer x, h, termwise from the binomial theorem.
6. **Rational root theorem.** If $P(x) = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$ and s/t in lowest terms is a root, then $s \mid a_0$ and $t \mid a_n$.
7. **Valuations.** $v_p(ab) = v_p(a) + v_p(b)$ and $v_p(a+b) \geq \min(v_p(a), v_p(b))$. They turn divisibility into additive inequalities.
8. **Groups and rings.** Element orders divide finite group order. Quotients require normal subgroups. In noncommutative rings, preserve order: $(a+b)^2 = a^2 + ab + ba + b^2$.

Node-by-node properties, formulas, proof cores, and occurrence anchors.

10.4.1 ANT-01: Modular arithmetic and divisibility

Occurrence anchors	1994-D1-P3; 1994-D1-P6; 1997-D1-P3; 1997-D1-P4; 1999-D1-P3; 1999-D2-P1; 1999-D2-P2; 2000-D1-P5; 2000-D2-P1; 2001-D1-P1; 2001-D1-P2; 2001-D1-P4; 2001-D2-P6; 2002-D1-P3; 2003-D1-P2; 2003-D1-P4; 2005-D1-P2; 2005-D1-P4; 2005-D2-P2; 2005-D2-P6; 2006-D1-P2; 2006-D1-P3; 2006-D2-P1; 2006-D2-P5; 2007-D1-P1; 2007-D2-P2; ... (+66 more in data/imc_theory_node_index.csv)
Original node statement	Divisibility questions are often local modulo prime powers. Reduce congruences modulo primes first, then lift or combine information.
Literature	[5, 15, 20]

Exact properties and formulas to know.

1. Congruence is compatible with addition, multiplication, and integer polynomials.
2. Power residues modulo primes are computed in the cyclic group $\mathbb{F}_p^\times$.
3. Divisibility by a prime power is tracked by valuation.
4. Reduce an integer statement modulo a strategic prime to force contradiction.

Compact proof cores for this node.

1. Reduce to fields, prime powers, or coprime CRT components.
2. Use Bezout, valuations, and root bounds to turn divisibility into structure.
3. In group/ring problems, state whether commutativity, normality, or characteristic assumptions are actually available.

Use reminder. Use for integer polynomials, prime divisibility, residue classes, and modular nonexistence.

Trap reminder. Check whether the modulus is prime, prime power, or composite; cancellation requires invertibility.

10.4.2 ANT-02: Chinese remainder theorem and complete residue systems

Occurrence anchors	1999-D1-P3; 2006-D1-P2; 2006-D2-P5; 2007-D1-P1; 2007-D2-P2; 2007-D2-P4; 2010-D1-P4; 2012-D2-P1; 2012-D2-P4; 2013-D1-P5; 2013-D2-P2; 2014-D1-P4; 2014-D2-P1; 2015-D1-P4; 2016-D1-P3; 2016-D1-P5; 2016-D2-P8; 2017-D2-P10; 2018-D1-P5; 2018-D2-P6; 2019-D1-P2; 2019-D2-P7; 2020-D2-P6; 2021-D2-P6; 2022-D1-P3; 2022-D2-P6; ... (+4 more in data/imc_theory_node_index.csv)
Original node statement	If moduli are coprime, congruences modulo each modulus combine uniquely modulo the product. Complete residue systems remain complete under bijective polynomial maps over finite fields.
Literature	[5, 15, 20]

Exact properties and formulas to know.

1. CRT gives unique solutions modulo a product of coprime moduli.

2. Complete residue systems stay complete under multiplication by a unit.
3. Bezout coefficients construct the CRT solution explicitly.
4. Composite moduli are handled prime-power componentwise.

Compact proof cores for this node.

1. Reduce to fields, prime powers, or coprime CRT components.
2. Use Bezout, valuations, and root bounds to turn divisibility into structure.
3. In group/ring problems, state whether commutativity, normality, or characteristic assumptions are actually available.

Use reminder. Use for constructing integers with simultaneous congruence constraints and counting idempotents modulo $10\hat{}k$.

Trap reminder. Uniqueness is modulo the product; remember to exclude disallowed representatives such as zero if the problem says positive.

10.4.3 ANT-03: Finite fields, field characteristic, and Fermat/Frobenius tools

Occurrence anchors	1997-D1-P3; 1999-D2-P1; 1999-D2-P2; 2000-D2-P1; 2000-D2-P3; 2001-D1-P2; 2001-D1-P4; 2003-D1-P2; 2005-D1-P2; 2005-D1-P4; 2005-D2-P2; 2005-D2-P6; 2006-D1-P2; 2006-D1-P3; 2006-D2-P5; 2007-D1-P1; 2007-D2-P2; 2007-D2-P4; 2008-D1-P3; 2008-D2-P4; 2008-D2-P5; 2009-D2-P4; 2010-D1-P4; 2010-D2-P4; 2010-D2-P5; 2011-D1-P3; ... (+23 more in data/imc_theory_node_index.csv)
Original node statement	Over \{}$mathbbF_p$, Fermat gives $a\hat{}\{p-1\} = 1$ for nonzero a, and polynomial functions are naturally considered modulo $x\hat{}p - x$. Field characteristic controls additive cancellation.
Literature	[5, 15, 20]

Exact properties and formulas to know.

1. $a^p = a$ in $\mathbb{F}_p$.
2. $(x+y)^p = x^p + y^p$ in characteristic p.
3. Nonzero elements have order dividing $p-1$.
4. Degree-$< p$ polynomial functions on $\mathbb{F}_p$ are represented uniquely.

Compact proof cores for this node.

1. Reduce to fields, prime powers, or coprime CRT components.
2. Use Bezout, valuations, and root bounds to turn divisibility into structure.
3. In group/ring problems, state whether commutativity, normality, or characteristic assumptions are actually available.

Use reminder. Use for finite-field polynomial permutations, characteristic restrictions, and group actions over \{}$mathbbF_p$.

Trap reminder. A polynomial of high degree may define the same function as a lower-degree remainder modulo $x\hat{}p - x$.

10.4.4 ANT-04: Integer polynomials, finite differences, and value divisibility

Occurrence anchors	1995-D2-P4; 1996-D2-P2; 1998-D1-P5; 2001-D2-P1; 2002-D2-P3; 2005-D1-P4; 2006-D1-P3; 2006-D1-P4; 2007-D1-P1; 2008-D1-P3; 2008-D2-P1; 2008-D2-P4; 2011-D1-P3; 2012-D1-P5; 2012-D2-P1; 2014-D1-P3; 2014-D2-P1; 2015-D2-P10; 2017-D2-P7; 2018-D2-P9; 2020-D1-P4; 2023-D1-P3
Original node statement	For integer polynomials, $p(a+mq)\backslash\{\}equivp(a)\backslash\{\}pmodq$. A degree-$d$ polynomial over a field with more than d roots is zero.
Literature	[5, 15, 20]

Exact properties and formulas to know.

1. Congruence is compatible with addition, multiplication, and integer polynomials.
2. Power residues modulo primes are computed in the cyclic group $\mathbb{F}_p^\times$.
3. Divisibility by a prime power is tracked by valuation.
4. Reduce an integer statement modulo a strategic prime to force contradiction.

Compact proof cores for this node.

1. Reduce to fields, prime powers, or coprime CRT components.
2. Use Bezout, valuations, and root bounds to turn divisibility into structure.
3. In group/ring problems, state whether commutativity, normality, or characteristic assumptions are actually available.

Use reminder. Use for proving coefficient divisibility from value divisibility and constructing values divisible by prescribed polynomial values.

Trap reminder. Testing finitely many residues works only when the degree is less than the prime modulus or when using finite differences.

10.4.5 ANT-05: Rational roots, Vieta, and rationality constraints

Occurrence anchors	1994-D1-P3; 1996-D2-P2; 2005-D1-P4; 2006-D1-P5; 2018-D1-P3
Original node statement	Rational-root and Vieta arguments translate root rationality into coefficient and product/sum constraints. For irrationality questions, positive rational linear combinations are controlled by induction and contradiction.
Literature	[5, 15, 20]

Exact properties and formulas to know.

1. Rational root theorem imposes numerator/denominator divisibility.
2. Vieta converts rational roots into coefficient equations.
3. Nonnegative coefficients exclude positive roots except the zero polynomial.
4. Clearing denominators turns rational linear factors into integer factors.

Compact proof cores for this node.

1. Reduce to fields, prime powers, or coprime CRT components.
2. Use Bezout, valuations, and root bounds to turn divisibility into structure.
3. In group/ring problems, state whether commutativity, normality, or characteristic assumptions are actually available.

Use reminder. Use for polynomials with rational roots, hyperbolic rationality, irrational subset selection, and rational functions.

Trap reminder. Do not divide by a coefficient that may be zero; isolate zero roots first.

10.4.6 ANT-06: GCD, valuations, prime powers, and coprimality

Occurrence anchors	1999-D1-P3; 1999-D2-P1; 2000-D2-P1; 2002-D1-P3; 2003-D1-P4; 2007-D2-P2; 2008-D2-P3; 2012-D1-P1; 2012-D1-P2; 2012-D1-P3; 2012-D2-P3; 2012-D2-P4; 2013-D2-P2; 2014-D2-P1; 2015-D1-P2; 2015-D1-P4; 2016-D1-P2; 2016-D1-P3; 2016-D1-P4; 2016-D2-P9; 2017-D1-P3; 2017-D2-P8; 2017-D2-P10; 2018-D2-P6; 2018-D2-P8; 2019-D1-P2; ... (+16 more in data/imc_theory_node_index.csv)
Original node statement	Coprime factors can be assigned independently. Valuations measure powers of a prime dividing an integer and turn multiplicative decompositions into additive comparisons.
Literature	[5, 15, 20]

Exact properties and formulas to know.

1. Bezout gives integer combinations of coprime numbers.
2. Euclid lemma: if $a \mid bc$ and $(a, b) = 1$, then $a \mid c$.
3. Valuations add over products.
4. Pairwise coprime factors can be assigned independently in a divisibility condition.

Compact proof cores for this node.

1. Reduce to fields, prime powers, or coprime CRT components.
2. Use Bezout, valuations, and root bounds to turn divisibility into structure.
3. In group/ring problems, state whether commutativity, normality, or characteristic assumptions are actually available.

Use reminder. Use for decompositions of positive integers, products equaling powers, and divisibility by prime powers.

Trap reminder. When xy is divisible by $p\hat{}k$ and x, y are not coprime, valuation splitting is necessary.

10.4.7 ANT-07: Group basics: orders, cyclic subgroups, and generators

Occurrence anchors	1996-D2-P3; 1999-D1-P3; 1999-D2-P1; 2000-D1-P5; 2001-D1-P2; 2003-D1-P2; 2003-D2-P4; 2005-D1-P6; 2006-D1-P2; 2007-D1-P4; 2007-D2-P5; 2008-D1-P5; 2008-D2-P3; 2009-D2-P5; 2010-D2-P2; 2010-D2-P3; 2011-D1-P3; 2011-D2-P2; 2011-D2-P4; 2012-D1-P3; 2012-D1-P5; 2012-D2-P3; 2012-D2-P4; 2012-D2-P5; 2013-D2-P2; 2014-D2-P1; ... (+26 more in data/imc_theory_node_index.csv)
Original node statement	Element orders, generated subgroups, and relations determine many finite-group arguments. In abelian groups, Bezout identities in exponents are especially powerful.
Literature	[5, 15, 20]

Exact properties and formulas to know.

1. Lagrange: subgroup order divides group order.
2. Element order divides group order in finite groups.
3. Normal subgroups define quotients.
4. Automorphisms are determined by generator images satisfying relations.

Compact proof cores for this node.

1. Reduce to fields, prime powers, or coprime CRT components.
2. Use Bezout, valuations, and root bounds to turn divisibility into structure.
3. In group/ring problems, state whether commutativity, normality, or characteristic assumptions are actually available.

Use reminder. Use for group equations, pairwise coprime exponents, and subgroup generation.

Trap reminder. Exponent arithmetic using $(ab)\hat{}n = a\hat{}nb\hat{}n$ requires commutativity unless justified.

10.4.8 ANT-08: Permutation groups, cycles, parity, and generation

Occurrence anchors	1998-D1-P2; 2001-D1-P2
Original node statement	Cycles, transpositions, and parity describe symmetric and alternating groups. Adjacent transpositions generate S_n; selected 3-cycles generate A_n.
Literature	[5, 15, 20]

Exact properties and formulas to know.

1. Lagrange: subgroup order divides group order.
2. Element order divides group order in finite groups.
3. Normal subgroups define quotients.
4. Automorphisms are determined by generator images satisfying relations.

Compact proof cores for this node.

1. Reduce to fields, prime powers, or coprime CRT components.
2. Use Bezout, valuations, and root bounds to turn divisibility into structure.
3. In group/ring problems, state whether commutativity, normality, or characteristic assumptions are actually available.

Use reminder. Use for proving generation/non-generation of permutation groups and for constructing permutations with modular properties.

Trap reminder. Cycle notation composition order must be consistent throughout the proof.

10.4.9 ANT-09: Normal subgroups, quotient groups, and group actions

Occurrence anchors	1996-D2-P3; 1999-D2-P1; 2005-D1-P6; 2010-D2-P3; 2012-D1-P3; 2012-D2-P5; 2016-D2-P8; 2017-D1-P4; 2018-D1-P2; 2020-D2-P7; 2021-D2-P6; 2024-D1-P4
Original node statement	Normal subgroups allow quotient groups. Group actions turn algebraic relations into orbit-stabilizer or index constraints.
Literature	[5, 15, 20]

Exact properties and formulas to know.

1. Lagrange: subgroup order divides group order.
2. Element order divides group order in finite groups.
3. Normal subgroups define quotients.
4. Automorphisms are determined by generator images satisfying relations.

Compact proof cores for this node.

1. Reduce to fields, prime powers, or coprime CRT components.
2. Use Bezout, valuations, and root bounds to turn divisibility into structure.
3. In group/ring problems, state whether commutativity, normality, or characteristic assumptions are actually available.

Use reminder. Use for subgroup-index problems, conjugacy of subgroups, and generation obstructions through quotients.

Trap reminder. A subgroup of the same index need not be normal; prove normality before passing to a quotient.

10.4.10 ANT-10: Automorphisms and semidirect products

Occurrence anchors	1996-D2-P3; 1998-D1-P2; 1999-D1-P1; 1999-D1-P5; 1999-D2-P1; 2000-D1-P5; 2001-D1-P2; 2003-D1-P2; 2003-D2-P4; 2004-D2-P3; 2005-D1-P6; 2007-D1-P4; 2007-D2-P5; 2008-D1-P5; 2010-D2-P2; 2010-D2-P3; 2011-D1-P3; 2011-D2-P4; 2012-D1-P3; 2012-D1-P5; 2012-D2-P5; 2016-D2-P8; 2017-D1-P4; 2018-D1-P2; 2020-D2-P7; 2021-D2-P6; ... (+1 more in data/imc_theory_node_index.csv)
Original node statement	A semidirect product $H\{\}rtimesT$ is built from an action of T on H. Automorphisms preserve element orders, normal subgroups, and structural decompositions.
Literature	[5, 15, 20]

Exact properties and formulas to know.

1. Lagrange: subgroup order divides group order.
2. Element order divides group order in finite groups.
3. Normal subgroups define quotients.
4. Automorphisms are determined by generator images satisfying relations.

Compact proof cores for this node.

1. Reduce to fields, prime powers, or coprime CRT components.
2. Use Bezout, valuations, and root bounds to turn divisibility into structure.
3. In group/ring problems, state whether commutativity, normality, or characteristic assumptions are actually available.

Use reminder. Use for constructing finite groups with prescribed automorphism behavior.

Trap reminder. Counting automorphisms by images of generators requires proving those images satisfy the same defining relations.

10.4.11 ANT-11: Rings, idempotents, nil rings, and noncommutativity

Occurrence anchors	1996-D2-P3; 1999-D1-P5; 1999-D2-P1; 2000-D1-P5; 2001-D1-P2; 2003-D1-P2; 2003-D2-P4; 2005-D1-P6; 2006-D1-P2; 2007-D1-P4; 2007-D2-P5; 2008-D1-P5; 2009-D2-P4; 2010-D2-P3; 2011-D1-P3; 2011-D2-P2; 2012-D1-P3; 2012-D1-P5; 2012-D2-P3; 2012-D2-P5; 2013-D2-P2; 2014-D2-P1; 2016-D2-P8; 2017-D1-P4; 2018-D1-P2; 2019-D1-P2; ... (+7 more in data/imc_theory_node_index.csv)
Original node statement	In noncommutative rings, expansion of $(a+b)\^2$ still gives $ab+ba$ when squares vanish. Idempotent equations often become commutator relations under characteristic assumptions.
Literature	[5, 15, 20]

Exact properties and formulas to know.

1. Noncommutative expansions preserve order.
2. If $x^2=0$ for all x, then $ab=-ba$.
3. Idempotent means $e^2=e$.
4. Characteristic zero allows cancellation of $nx=0 \Rightarrow x=0$.

Compact proof cores for this node.

1. Reduce to fields, prime powers, or coprime CRT components.
2. Use Bezout, valuations, and root bounds to turn divisibility into structure.
3. In group/ring problems, state whether commutativity, normality, or characteristic assumptions are actually available.

Use reminder. Use for ring problems with idempotents, nilpotents, characteristic zero, and noncommutative algebra.

Trap reminder. Do not commute factors; use associativity and given identities only.

10.4.12 ANT-12: Diophantine equations, Pell-type recurrences, and quadratic irrationals

Occurrence anchors	2005-D2-P6; 2006-D2-P5
Original node statement	Quadratic irrational fixed points of fractional linear transformations lead to Pell-type equations. Recurrences from fundamental units provide infinitely many solutions.
Literature	[5, 15, 20]

Exact properties and formulas to know.

1. Pell equations arise from norms $x^2 - Dy^2$.
2. Powers of norm-one units generate recurrences.
3. Quadratic irrational fixed points satisfy fractional-linear equations.
4. Integer solutions must be checked for coprimality when requested.

Compact proof cores for this node.

1. Reduce to fields, prime powers, or coprime CRT components.
2. Use Bezout, valuations, and root bounds to turn divisibility into structure.
3. In group/ring problems, state whether commutativity, normality, or characteristic assumptions are actually available.

Use reminder. Use for constructing integer matrices fixing quadratic irrationals and cubic equations with integer roots.

Trap reminder. Verify determinant-one and nontriviality conditions after solving the Pell equation.

10.4.13 ANT-13: Lattices and norms/quasinorms

Occurrence anchors	1994-D1-P6; 1997-D1-P5
Original node statement	Integer lattice shifts interact with convex norms for $p\backslash\{\}ge1$ and can fail for quasi-norms $0 < p < 1$. Coordinate splitting by signs is often required.
Literature	[5, 15, 20]

Exact properties and formulas to know.

1. $\mathbb{Z}_0^n$ is generated by $e_i - e_j$.
2. For $p \geq 1$, ℓ_p is a norm; for $0 < p < 1$, triangle inequality fails.
3. Convexity of $|x|^p$ holds for $p \geq 1$.
4. Lattice translations preserve congruence classes and integer sums.

Compact proof cores for this node.

1. Reduce to fields, prime powers, or coprime CRT components.
2. Use Bezout, valuations, and root bounds to turn divisibility into structure.
3. In group/ring problems, state whether commutativity, normality, or characteristic assumptions are actually available.

Use reminder. Use for lattice minimization and norm comparison problems.

Trap reminder. Convexity fails for $p < 1$; do not use triangle inequality in the quasi-norm case.

10.4.14 ANT-14: Greedy infinite products and rationality tails

Occurrence anchors	1997-D1-P4
Original node statement	Greedy multiplicative expansions are controlled by a tail identity such as \{}$prod_{j\{}ge0}(1+M\^{}\{-2\^{}j\}) = M/(M-1)$. Rationality can force eventual equality in the greedy recurrence.
Literature	[5, 15, 20]

Exact properties and formulas to know.

1. Rational root theorem imposes numerator/denominator divisibility.
2. Vieta converts rational roots into coefficient equations.
3. Nonnegative coefficients exclude positive roots except the zero polynomial.
4. Clearing denominators turns rational linear factors into integer factors.

Compact proof cores for this node.

1. Reduce to fields, prime powers, or coprime CRT components.
2. Use Bezout, valuations, and root bounds to turn divisibility into structure.
3. In group/ring problems, state whether commutativity, normality, or characteristic assumptions are actually available.

Use reminder. Use for unique infinite product representations and rationality criteria.

Trap reminder. A greedy inequality must be sharp in both directions to prove uniqueness.

10.4.15 ANT-15: Finite abelian groups and vector spaces over finite fields

Occurrence anchors	2003-D2-P4
Original node statement	Finite abelian groups of exponent 2 are vector spaces over \{}$mathbbF_2$ and have order a power of two. This structure often classifies triple systems.
Literature	[5, 15, 20]

Exact properties and formulas to know.

1. $a^p = a$ in $\mathbb{F}_p$.
2. $(x+y)^p = x^p + y^p$ in characteristic p.
3. Nonzero elements have order dividing $p-1$.
4. Degree-$< p$ polynomial functions on $\mathbb{F}_p$ are represented uniquely.

Compact proof cores for this node.

1. Reduce to fields, prime powers, or coprime CRT components.
2. Use Bezout, valuations, and root bounds to turn divisibility into structure.
3. In group/ring problems, state whether commutativity, normality, or characteristic assumptions are actually available.

Use reminder. Use for finite designs where each pair determines a third element and associativity-like conditions appear.

Trap reminder. The added identity element must be included before claiming group structure.

10.5 Combinatorics / discrete structures: exact theorem bank

Literature anchor: [29, 30, 2, 31]. **Global theorem/property bank.**

1. **Pigeonhole.** N objects in m boxes force a box with at least $\lceil N/m \rceil$ objects. Contrapositive forms are often sharper.
2. **Double counting.** For incidences $I \subseteq A \times B$, $|I| = \sum_a |I_a| = \sum_b |I^b|$. Degree-sum and walk counts are graph versions.
3. **Graph cycles.** A forest on v vertices and c components has $v - c$ edges. Thus a graph with at least v edges has a cycle.
4. **Bipartite encoding.** Rows/columns, sets/elements, or people/problems can be two vertex classes; incidences become edges and alternating cycles become combinatorial configurations.
5. **Recurrences and states.** Partition objects into states so the next object depends only on the state. This yields scalar recurrences or matrix recurrences $v_{n+1} = Mv_n$.
6. **Generating functions.** Ordinary generating functions multiply by convolution: $[x^n]A(x)B(x) = \sum_{k=0}^{n} a_k b_{n-k}$.
7. **Inclusion-exclusion.** $|\cup A_i| = \sum |A_i| - \sum |A_i \cap A_j| + \cdots$. Use only when intersections are controlled.
8. **Permutation statistics.** $\operatorname{sgn} \sigma = (-1)^{\operatorname{inv} \sigma}$. Determinants encode signed sums over permutations; reducing modulo 2 ignores signs.

Node-by-node properties, formulas, proof cores, and occurrence anchors.

10.5.1 CMB-01: Pigeonhole principle and extremal element selection

Occurrence anchors	1994-D1-P1; 1994-D1-P3; 1994-D1-P6; 1994-D2-P4; 1995-D1-P3; 1996-D1-P3; 1996-D2-P3; 1996-D2-P6; 1997-D1-P6; 1997-D2-P6; 1998-D2-P3; 1998-D2-P4; 1998-D2-P5; 1998-D2-P6; 1999-D1-P2; 1999-D1-P5; 1999-D2-P6; 2000-D1-P1; 2001-D2-P1; 2001-D2-P3; 2002-D1-P4; 2002-D1-P5; 2002-D2-P2; 2002-D2-P6; 2003-D1-P5; 2003-D2-P3; ... (+42 more in data/imc_theory_node_index.csv)
Original node statement	If more objects than boxes are distributed, one box contains many objects. Extremal choices (minimal counterexample, largest element, shortest cycle) often force structure.
Literature	[29, 30, 2, 31]

Exact properties and formulas to know.

1. N objects in m boxes force a box with $\lceil N/m \rceil$ objects.
2. Contrapositive: if every box has at most r objects, total is at most mr.
3. Pigeonhole with divisors requires counting all possible divisors/values.
4. Extremal selection often chooses a maximal/minimal counterexample.

Compact proof cores for this node.

1. Define the incidence set, graph, state space, or generating function explicitly.
2. Choose an extremal object or induction split that strictly reduces the parameter.
3. After construction, verify every original hypothesis, not just the intended pattern.

Use reminder. Use for divisibility value collisions, selecting common rows/columns, and minimal counterexample arguments.

Trap reminder. Define the boxes explicitly; hidden boxes often are residue classes, divisors, or graph neighborhoods.

10.5.2 CMB-02: Double counting and degree-sum methods

Occurrence anchors	1994-D1-P1; 1994-D2-P4; 1995-D1-P3; 1996-D1-P3; 1996-D2-P3; 1996-D2-P6; 1997-D2-P6; 1998-D2-P3; 1998-D2-P4; 1998-D2-P5; 1998-D2-P6; 1999-D1-P5; 2000-D1-P1; 2001-D2-P1; 2001-D2-P3; 2002-D1-P4; 2002-D1-P5; 2002-D2-P2; 2003-D1-P5; 2003-D2-P5; 2004-D1-P1; 2004-D1-P2; 2004-D1-P4; 2004-D2-P6; 2005-D1-P2; 2006-D1-P2; ... (+30 more in data/imc_theory_node_index.csv)
Original node statement	Count the same set of objects in two ways, often by vertices versus edges, pairs versus problems, or walks via middle vertex.
Literature	[29, 30, 2, 31]

Exact properties and formulas to know.

1. Count incidences by left and right projections.
2. $\sum_v d(v) = 2|E|$.
3. Number of length-two walks in a graph is $\sum_v d(v)^2$.
4. Equality cases in double counts often imply uniqueness of witnesses.

Compact proof cores for this node.

1. Define the incidence set, graph, state space, or generating function explicitly.
2. Choose an extremal object or induction split that strictly reduces the parameter.
3. After construction, verify every original hypothesis, not just the intended pattern.

Use reminder. Use for contest scores, graph degree constraints, set intersections, and probability counts.

Trap reminder. Specify ordered vs unordered objects; many IMC counts hinge on this distinction.

10.5.3 CMB-03: Graph encoding, cycles, and bipartite representations

Occurrence anchors	1994-D1-P3; 1994-D1-P5; 1994-D2-P3; 1995-D1-P4; 1995-D2-P1; 1996-D1-P6; 1996-D2-P2; 1997-D1-P6; 1997-D2-P5; 1997-D2-P6; 1998-D1-P1; 1998-D1-P2; 1998-D2-P3; 1998-D2-P4; 1998-D2-P5; 1999-D1-P4; 1999-D1-P5; 1999-D1-P6; 1999-D2-P5; 2000-D2-P4; 2001-D1-P2; 2001-D2-P4; 2001-D2-P5; 2002-D1-P1; 2002-D2-P2; 2002-D2-P4; ... (+42 more in data/imc_theory_node_index.csv)
Original node statement	Encode relations as graphs. Cycles, degree sums, connectedness, and bipartite representations convert combinatorial statements into graph-theoretic ones.
Literature	[29, 30, 2, 31]

Exact properties and formulas to know.

1. A graph with minimum degree at least 2 contains a cycle.
2. Bipartite graph cycles have even length.
3. DAGs admit topological orderings.
4. Grid marked points become edges between row and column vertices.

Compact proof cores for this node.

1. Define the incidence set, graph, state space, or generating function explicitly.
2. Choose an extremal object or induction split that strictly reduces the parameter.
3. After construction, verify every original hypothesis, not just the intended pattern.

Use reminder. Use for grids, friendship conditions, colored diagonals, trees, and matrix zero/nonzero patterns.

Trap reminder. When encoding a grid, rows and columns often become vertices and marked cells become edges.

10.5.4 CMB-04: Set systems, intersecting families, and induction on sizes

Occurrence anchors	1997-D1-P6; 1997-D2-P5; 1997-D2-P6; 1998-D1-P1; 1999-D2-P5; 2002-D2-P2; 2004-D1-P1; 2005-D2-P4; 2007-D2-P1; 2008-D1-P4; 2011-D1-P4; 2012-D2-P3; 2013-D2-P5; 2014-D2-P4; 2016-D1-P4; 2017-D1-P3; 2019-D2-P8; 2020-D1-P3; 2021-D1-P2; 2022-D2-P5; 2023-D1-P4; 2023-D1-P5; 2023-D2-P7
Original node statement	Intersecting families are often handled by induction on set size, splitting by intersection with a fixed set, or constructing a finite transversal.
Literature	[29, 30, 2, 31]

Exact properties and formulas to know.

1. Uniform set systems allow induction on set sizes.
2. A transversal/hitting set intersects every member.
3. Intersecting families have pairwise nonempty intersections.
4. Partition by intersection with a fixed finite core to reduce parameters.

Compact proof cores for this node.

1. Define the incidence set, graph, state space, or generating function explicitly.
2. Choose an extremal object or induction split that strictly reduces the parameter.
3. After construction, verify every original hypothesis, not just the intended pattern.

Use reminder. Use for finite set families with pairwise intersections and uniform-size assumptions.

Trap reminder. Uniformity assumptions are often essential; construct counterexamples without them.

10.5.5 CMB-05: Recurrences and state partitions

Occurrence anchors	1998-D2-P4; 2002-D2-P2; 2005-D1-P2; 2013-D2-P5
Original node statement	Partition objects by a finite state, derive a recurrence, verify initial conditions, and solve or compare recurrences.
Literature	[29, 30, 2, 31]

Exact properties and formulas to know.

1. State recurrences partition by last symbol, last row, or boundary state.
2. Linear recurrence plus initial values determines all terms.
3. Matrix recurrence $v_{n+1} = Mv_n$ counts walks/words.
4. Generating functions solve convolution recurrences.

Compact proof cores for this node.

1. Define the incidence set, graph, state space, or generating function explicitly.
2. Choose an extremal object or induction split that strictly reduces the parameter.
3. After construction, verify every original hypothesis, not just the intended pattern.

Use reminder. Use for words, sequences under local constraints, matrix-defined recurrences, and counting functions.

Trap reminder. The state must contain enough memory to extend objects without ambiguity.

10.5.6 CMB-06: Generating functions for counting

Occurrence anchors	2000-D1-P6; 2002-D1-P3; 2002-D2-P3; 2003-D1-P6; 2003-D2-P6; 2008-D2-P3; 2012-D2-P2; 2015-D2-P8; 2017-D2-P8; 2017-D2-P9; 2018-D2-P7; 2019-D1-P4; 2021-D1-P3; 2024-D2-P8
Original node statement	Generating functions translate recurrences and weighted counts into algebraic identities. Coefficient extraction and radius/convergence considerations then solve the count.
Literature	[29, 30, 2, 31]

Exact properties and formulas to know.

1. Coefficient of product is convolution.
2. Differentiation multiplies coefficients by n.
3. Composition of generating functions encodes substitution of structures.
4. Radius/convergence is irrelevant for formal power series unless analytic limits are used.

Compact proof cores for this node.

1. Define the incidence set, graph, state space, or generating function explicitly.
2. Choose an extremal object or induction split that strictly reduces the parameter.
3. After construction, verify every original hypothesis, not just the intended pattern.

Use reminder. Use for sequences, expected sums, probability distributions, and recurrence-defined counts.

Trap reminder. Formal power series do not require convergence, but analytic evaluations do.

10.5.7 CMB-07: Inclusion-exclusion and root-of-unity filters

Occurrence anchors	1999-D2-P2
Original node statement	Counting by congruence class can be done with roots of unity; exclusion of forbidden configurations is handled by complements or inclusion-exclusion.
Literature	[29, 30, 2, 31]

Exact properties and formulas to know.

1. Inclusion-exclusion alternates intersections.
2. Root-of-unity filters isolate congruence classes.

3. Complement counting is often shorter than direct counting.
4. Check overlaps before subtracting.

Compact proof cores for this node.

1. Define the incidence set, graph, state space, or generating function explicitly.
2. Choose an extremal object or induction split that strictly reduces the parameter.
3. After construction, verify every original hypothesis, not just the intended pattern.

Use reminder. Use for dice sums, modulo counts, and divisibility by residues in combinatorial settings.

Trap reminder. Root-of-unity filters count residues of exponents/sums; ensure the generating function encodes the correct statistic.

10.5.8 CMB-08: Permutations, inversions, and permutation statistics

Occurrence anchors	2001-D1-P1; 2003-D1-P2; 2009-D2-P4; 2013-D2-P5; 2022-D2-P6
Original node statement	Permutations encode selections with one item per row/column. Statistics such as inversions or displacement can be represented by determinants or generating functions.
Literature	[29, 30, 2, 31]

Exact properties and formulas to know.

1. Inversions determine permutation parity.
2. Determinants sum over signed permutations.
3. Permutation statistics can be encoded by generating polynomials.
4. Modulo 2, signs disappear.

Compact proof cores for this node.

1. Define the incidence set, graph, state space, or generating function explicitly.
2. Choose an extremal object or induction split that strictly reduces the parameter.
3. After construction, verify every original hypothesis, not just the intended pattern.

Use reminder. Use for permutation enumeration, determinant parity of permutation statistics, and row-column selection problems.

Trap reminder. Distinguish permutation as a function from its one-line notation.

10.5.9 CMB-09: Explicit construction and counterexample design

Occurrence anchors	1994-D1-P4; 1994-D2-P5; 1995-D1-P6; 1995-D2-P6; 1997-D1-P2; 1997-D1-P4; 1997-D1-P6; 1997-D2-P1; 1997-D2-P6; 1998-D1-P2; 1999-D1-P2; 1999-D1-P6; 1999-D2-P5; 2000-D1-P1; 2002-D1-P2; 2003-D2-P4; 2004-D2-P3; 2006-D1-P1; 2007-D2-P1; 2007-D2-P3; 2007-D2-P5; 2008-D1-P3; 2008-D1-P5; 2009-D1-P1; 2011-D1-P2; 2013-D1-P5; ... (+10 more in data/imc_theory_node_index.csv)
Original node statement	Construction problems require preserving all hypotheses while forcing or breaking the conclusion. Small examples, periodic patterns, and extremal configurations are the standard sources.
Literature	[29, 30, 2, 31]

Exact properties and formulas to know.

1. A counterexample must satisfy every hypothesis exactly.
2. Sharp constructions should realize equality in the upper bound.
3. Random or algebraic constructions require explicit verification.
4. Boundary cases often invalidate otherwise good constructions.

Compact proof cores for this node.

1. Define the incidence set, graph, state space, or generating function explicitly.
2. Choose an extremal object or induction split that strictly reduces the parameter.
3. After construction, verify every original hypothesis, not just the intended pattern.

Use reminder. Use for yes/no questions, nonexistence disproofs, and sharpness examples.

Trap reminder. A counterexample must satisfy every hypothesis exactly, including continuity, monotonicity, or finiteness.

10.5.10 CMB-10: Erdos-Szekeres style monotone subsequences and binomial thresholds

Occurrence anchors	2004-D1-P5
Original node statement	Many monotone-subsequence thresholds are proved by induction using binomial identities and splitting by a median or extremal element.
Literature	[29, 30, 2, 31]

Exact properties and formulas to know.

1. Erdos-Szekeres arguments use two-parameter induction.
2. Binomial recurrences match the inductive split.
3. Increasing/decreasing alternatives are dual.
4. Extending a shorter sequence needs a distance/growth check.

Compact proof cores for this node.

1. Define the incidence set, graph, state space, or generating function explicitly.
2. Choose an extremal object or induction split that strictly reduces the parameter.
3. After construction, verify every original hypothesis, not just the intended pattern.

Use reminder. Use for finding long monotone sequences with extra growth constraints.

Trap reminder. The induction parameter may be two-dimensional: length of increasing and decreasing alternatives.

10.5.11 CMB-11: Grid, lattice, and alternating-cycle arguments

Occurrence anchors	1999-D1-P5; 2023-D2-P6
Original node statement	Represent marked grid cells as edges in a bipartite graph between row-vertices and column-vertices. A cycle in the graph becomes an alternating row/column sequence.
Literature	[29, 30, 2, 31]

Exact properties and formulas to know.

1. Encode the structure explicitly.
2. Count or construct with all edge cases checked.
3. Use induction only with a decreasing parameter.
4. Translate back from graph/word/generating-function language to the original problem.

Compact proof cores for this node.

1. Define the incidence set, graph, state space, or generating function explicitly.
2. Choose an extremal object or induction split that strictly reduces the parameter.
3. After construction, verify every original hypothesis, not just the intended pattern.

Use reminder. Use for grid problems asking for alternating selections or cycles.

Trap reminder. Deleting rows/columns with too few points preserves inequalities only if the count is adjusted correctly.

10.5.12 CMB-12: Coloring, partitions, and parity

Occurrence anchors	1994-D1-P3; 1997-D1-P6; 1997-D2-P5; 1997-D2-P6; 1998-D1-P1; 1998-D2-P3; 1998-D2-P4; 1999-D1-P6; 1999-D2-P5; 2000-D2-P4; 2001-D2-P5; 2002-D1-P1; 2002-D2-P2; 2002-D2-P4; 2003-D1-P4; 2003-D2-P3; 2004-D1-P1; 2004-D1-P4; 2004-D2-P2; 2005-D2-P1; 2005-D2-P4; 2006-D2-P1; 2007-D1-P4; 2007-D2-P1; 2008-D1-P6; 2008-D1-P1; ... (+22 more in data/imc_theory_node_index.csv)
Original node statement	Coloring and partition problems often require parity counts, degree constraints, or induction preserving parity.
Literature	[29, 30, 2, 31]

Exact properties and formulas to know.

1. Encode the structure explicitly.
2. Count or construct with all edge cases checked.
3. Use induction only with a decreasing parameter.
4. Translate back from graph/word/generating-function language to the original problem.

Compact proof cores for this node.

1. Define the incidence set, graph, state space, or generating function explicitly.
2. Choose an extremal object or induction split that strictly reduces the parameter.
3. After construction, verify every original hypothesis, not just the intended pattern.

Use reminder. Use for colored graphs, triangulations with odd/even vertex incidences, and partitions into sets with algebraic equality.

Trap reminder. Parity can change at boundary vertices; track local contributions, not just total counts.

10.5.13 CMB-13: Words, equivalence relations, and rewriting systems

Occurrence anchors	1999-D2-P5; 2015-D2-P8; 2020-D1-P1
Original node statement	In word problems, local rewriting rules reduce length if a repeated block appears. Lemmas about absorbing words with all letters create global bounded-length representatives.
Literature	[29, 30, 2, 31]

Exact properties and formulas to know.

1. Encode the structure explicitly.
2. Count or construct with all edge cases checked.
3. Use induction only with a decreasing parameter.
4. Translate back from graph/word/generating-function language to the original problem.

Compact proof cores for this node.

1. Define the incidence set, graph, state space, or generating function explicitly.
2. Choose an extremal object or induction split that strictly reduces the parameter.
3. After construction, verify every original hypothesis, not just the intended pattern.

Use reminder. Use for equivalence relations on words and finite reduction algorithms.

Trap reminder. A rewrite rule must be compatible with left and right concatenation before applying it inside a word.

10.5.14 CMB-14: Transfer matrices and matrix powers for counting

Occurrence anchors	2004-D2-P6
Original node statement	If local compatibility between states is encoded by a matrix A, then entries or sums of A^{k-1} count sequences or table fillings of length k.
Literature	[29, 30, 2, 31]

Exact properties and formulas to know.

1. Encode the structure explicitly.
2. Count or construct with all edge cases checked.
3. Use induction only with a decreasing parameter.
4. Translate back from graph/word/generating-function language to the original problem.

Compact proof cores for this node.

1. Define the incidence set, graph, state space, or generating function explicitly.
2. Choose an extremal object or induction split that strictly reduces the parameter.
3. After construction, verify every original hypothesis, not just the intended pattern.

Use reminder. Use for counting binary tables, local constraints, and symmetry $F_{n,k} = F_{k,n}$.

Trap reminder. Make sure the matrix state represents a whole column/row, not a single cell.

10.5.15 CMB-15: Finite designs and pair-determines-third structures

Occurrence anchors	2003-D2-P4
Original node statement	A system where every pair belongs to exactly one triple can often be encoded by a binary operation $a * b = c$. Additional associativity-like axioms may turn it into a vector space over $\{\}mathbbF_2$.
Literature	[29, 30, 2, 31]

Exact properties and formulas to know.

1. Encode the structure explicitly.
2. Count or construct with all edge cases checked.
3. Use induction only with a decreasing parameter.
4. Translate back from graph/word/generating-function language to the original problem.

Compact proof cores for this node.

1. Define the incidence set, graph, state space, or generating function explicitly.
2. Choose an extremal object or induction split that strictly reduces the parameter.
3. After construction, verify every original hypothesis, not just the intended pattern.

Use reminder. Use for Steiner triple-system-like problems and classification of allowed sizes.

Trap reminder. The operation is initially undefined on equal elements; adding an identity often completes the algebraic structure.

10.6 Geometry / convexity / metric: exact theorem bank

Literature anchor: [7, 13, 26, 10]. **Global theorem/property bank.**

1. **Squared distance.** $\|x-y\|^2 = \|x\|^2 + \|y\|^2 - 2\langle x, y\rangle$. This turns distance data into inner-product equations.
2. **Projection theorem.** In Euclidean space, $v = p + q$ uniquely with $p \in U$ and $q \perp U$; p is the closest point of U to v.
3. **Affine normalization.** Invertible affine maps preserve lines, convexity, ratios on a line, and barycentric coordinates. Volumes scale by $|\det T|$.
4. **Barycentric coordinates.** In a simplex, $P = \sum \lambda_i P_i$ with $\sum \lambda_i = 1$; P lies inside iff all $\lambda_i \geq 0$.
5. **Gram matrices.** $G = (\langle v_i, v_j\rangle)$ is positive semidefinite; it is invertible iff the vectors are independent. Rational Gram entries lead to rational coefficient systems.
6. **Ellipsoid volume.** $\sum (x_i/a_i)^2 \leq 1$ has volume $\omega_n \prod_i a_i$, by scaling the unit ball.
7. **Conics.** A standard parabola with vertex (v, w) is $y = (x-v)^2 + w$. Ellipse focus-directrix form is $\operatorname{dist}(X, F) = e \operatorname{dist}(X, \ell)$ with $0 < e < 1$.
8. **Contractions.** If f is 1-Lipschitz, then $\operatorname{diam} f(E) \leq \operatorname{diam} E$; this is the basic property behind metric content comparisons.

Node-by-node properties, formulas, proof cores, and occurrence anchors.

10.6.1 GEO-01: Coordinate and affine normalization

Occurrence anchors	2014-D1-P5; 2014-D2-P4
Original node statement	Translate, rotate, scale, or choose coordinates to simplify distances, areas, conics, and vectors while preserving the quantity of interest.
Literature	[7, 13, 26, 10]

Exact properties and formulas to know.

1. Translations and rotations preserve all Euclidean distances and angles.
2. Affine maps preserve lines, ratios, and convexity.
3. Volume scales by the determinant of the linear part.
4. Normalize coordinates to place centers/vertices at convenient positions.

Compact proof cores for this node.

1. Normalize coordinates, then replace distances by squared-distance/inner-product identities.
2. Use projections, Gram matrices, barycentric coordinates, or determinants to algebraize the configuration.
3. Check whether the chosen transformation preserves lengths or only affine/convex data.

Use reminder. Use at the start of most Euclidean geometry, conic, and volume problems.

Trap reminder. Only affine-invariant quantities survive affine transformations; Euclidean distances require isometries or controlled scalings.

10.6.2 GEO-02: Euclidean distances and inner-product identities

Occurrence anchors	1994-D2-P5; 1995-D2-P1; 1996-D1-P6; 1998-D1-P6; 2002-D1-P6; 2004-D1-P4; 2004-D2-P3; 2005-D2-P5; 2006-D2-P4; 2008-D1-P5; 2008-D2-P2; 2010-D1-P1; 2010-D1-P5; 2013-D2-P3; 2014-D1-P5; 2014-D2-P4; 2015-D2-P9; 2018-D2-P6; 2021-D2-P8
Original node statement	Use \{}|u - v\{}|^2 = \{}|u\{}|^2 + \{}|v\{}|^2 - 2\{}langleu, v\{}rangle. Spherical distance constraints often become inner-product inequalities.
Literature	[7, 13, 26, 10]

Exact properties and formulas to know.

1. $\|x-y\|^2 = \|x\|^2 + \|y\|^2 - 2\langle x, y\rangle$.
2. Polarization gives inner products from norms.
3. On the unit sphere, distance $> \sqrt{2}$ means negative inner product.
4. Rational distances produce rational squared distances.

Compact proof cores for this node.

1. Normalize coordinates, then replace distances by squared-distance/inner-product identities.
2. Use projections, Gram matrices, barycentric coordinates, or determinants to algebraize the configuration.

3. Check whether the chosen transformation preserves lengths or only affine/convex data.

Use reminder. Use for rational distances, vectors with prescribed sums, tetrahedron angle estimates, and unit-sphere configurations.

Trap reminder. Square distances only after confirming all quantities are nonnegative.

10.6.3 GEO-03: Projections and orthogonal decompositions

Occurrence anchors	1995-D2-P3; 1996-D2-P4; 1997-D1-P5; 1997-D2-P5; 1998-D2-P5; 1999-D2-P6; 2002-D1-P1; 2002-D2-P4; 2002-D2-P5; 2004-D1-P4; 2004-D2-P2; 2004-D2-P3; 2005-D2-P5; 2008-D2-P2; 2009-D2-P1; 2011-D2-P5; 2014-D1-P5; 2015-D1-P5; 2017-D2-P10; 2018-D1-P5; 2019-D2-P10; 2022-D2-P5; 2022-D2-P8; 2023-D2-P9; 2024-D1-P5
Original node statement	Orthogonal projection shortens distances and decomposes vectors into parallel and perpendicular components. Projecting geometric regions can compare volumes or areas.
Literature	[7, 13, 26, 10]

Exact properties and formulas to know.

1. Orthogonal projection is 1-Lipschitz.
2. Projection minimizes distance to a subspace.
3. Projected area is multiplied by cosine of the angle to the projection plane.
4. Orthogonal decomposition gives Pythagorean identities.

Compact proof cores for this node.

1. Normalize coordinates, then replace distances by squared-distance/inner-product identities.
2. Use projections, Gram matrices, barycentric coordinates, or determinants to algebraize the configuration.
3. Check whether the chosen transformation preserves lengths or only affine/convex data.

Use reminder. Use for sphere/conic intersections, convex geometry, tetrahedra, and nearest-point arguments.

Trap reminder. A projection can collapse dimensions; track signed area/volume if orientation matters.

10.6.4 GEO-04: Convex bodies, supporting geometry, and John-ellipse ideas

Occurrence anchors	1996-D2-P4; 2011-D2-P3; 2014-D1-P5; 2014-D2-P4; 2014-D2-P5; 2015-D1-P3; 2015-D1-P4
Original node statement	For convex bodies, support and containment arguments compare maximal ellipses or balls. If a larger ellipse fits after a perturbation, maximality is contradicted.
Literature	[7, 13, 26, 10]

Exact properties and formulas to know.

1. Convexity contains line segments between points.
2. A supporting line/hyperplane separates a convex body from an exterior point.
3. Maximal ellipses are contradicted by constructing a larger ellipse inside.
4. Origin symmetry often reduces arcs or directions by rotation/reflection.

Compact proof cores for this node.

1. Normalize coordinates, then replace distances by squared-distance/inner-product identities.
2. Use projections, Gram matrices, barycentric coordinates, or determinants to algebraize the configuration.
3. Check whether the chosen transformation preserves lengths or only affine/convex data.

Use reminder. Use for convex symmetric planar bodies and maximal inscribed ellipse problems.

Trap reminder. The proof often needs a compactness gap between an arc and the boundary.

10.6.5 GEO-05: Conics: parabolas, ellipses, foci, and directrices

Occurrence anchors	2000-D2-P4; 2002-D1-P1; 2008-D2-P2; 2014-D1-P5
Original node statement	A standard parabola with vertex (v, w) has equation $y = (x - v)\text{^}2 + w$. Ellipses can be described by focus-directrix ratios or quadratic equations.
Literature	[7, 13, 26, 10]

Exact properties and formulas to know.

1. Standard parabola: $y = (x - v)^2 + w$.
2. Ellipse focus-directrix form uses eccentricity $e < 1$.
3. Two conics usually intersect in at most four complex points, but extra structure can reduce common points.
4. Reflection across axes changes signs of coordinates in equations.

Compact proof cores for this node.

1. Normalize coordinates, then replace distances by squared-distance/inner-product identities.
2. Use projections, Gram matrices, barycentric coordinates, or determinants to algebraize the configuration.
3. Check whether the chosen transformation preserves lengths or only affine/convex data.

Use reminder. Use for conic incidence/tangency, common-focus ellipse intersections, and reflected parabolas.

Trap reminder. Tangency to a line imposes double roots after subtracting the line equation.

10.6.6 GEO-06: Simplex, barycentric coordinates, and circumcenters

Occurrence anchors	1997-D2-P5; 2008-D2-P2; 2009-D1-P5; 2009-D2-P1; 2014-D1-P5; 2017-D2-P10
Original node statement	Barycentric weights are proportional to opposite simplex volumes. Circumcenters are characterized by equal distances to all vertices.
Literature	[7, 13, 26, 10]

Exact properties and formulas to know.

1. Translations and rotations preserve all Euclidean distances and angles.
2. Affine maps preserve lines, ratios, and convexity.
3. Volume scales by the determinant of the linear part.
4. Normalize coordinates to place centers/vertices at convenient positions.

Compact proof cores for this node.

1. Normalize coordinates, then replace distances by squared-distance/inner-product identities.
2. Use projections, Gram matrices, barycentric coordinates, or determinants to algebraize the configuration.
3. Check whether the chosen transformation preserves lengths or only affine/convex data.

Use reminder. Use for volume-weighted circumcenter identities and regular-simplex configurations.

Trap reminder. Volume weights must use consistent orientation or absolute volumes depending on the statement.

10.6.7 GEO-07: Area, volume, projection, and ellipsoid computation

Occurrence anchors	1995-D2-P3; 1996-D2-P4; 1997-D1-P5; 1997-D2-P5; 1998-D2-P5; 1999-D2-P6; 2000-D2-P1; 2002-D1-P1; 2002-D2-P4; 2002-D2-P5; 2004-D1-P4; 2004-D2-P2; 2004-D2-P3; 2005-D2-P5; 2008-D2-P2; 2009-D2-P1; 2011-D2-P5; 2014-D1-P5; 2014-D2-P4; 2015-D1-P5; 2017-D2-P10; 2018-D1-P5; 2019-D2-P10; 2021-D2-P7; 2022-D2-P5; 2022-D2-P8; ... (+2 more in data/imc_theory_node_index.csv)
Original node statement	Volumes transform by the absolute determinant of a linear map. Ellipsoid volumes follow from scaling the unit ball.
Literature	[7, 13, 26, 10]

Exact properties and formulas to know.

1. Orthogonal projection is 1-Lipschitz.
2. Projection minimizes distance to a subspace.
3. Projected area is multiplied by cosine of the angle to the projection plane.
4. Orthogonal decomposition gives Pythagorean identities.

Compact proof cores for this node.

1. Normalize coordinates, then replace distances by squared-distance/inner-product identities.
2. Use projections, Gram matrices, barycentric coordinates, or determinants to algebraize the configuration.
3. Check whether the chosen transformation preserves lengths or only affine/convex data.

Use reminder. Use for solid regions defined by distance inequalities, cube dissections, and projected areas.

Trap reminder. Complete the square before identifying the ellipsoid axes.

10.6.8 GEO-08: Metric content, diameter covers, and contractions

Occurrence anchors	1996-D1-P6
Original node statement	A contraction cannot increase diameters. Covering a segment by finitely many sets forces the sum of diameters to be at least the segment length.
Literature	[7, 13, 26, 10]

Exact properties and formulas to know.

1. Convert geometry to coordinates or inner products.
2. Use projections and barycentric coordinates.
3. Check whether transformations preserve metric or only affine data.
4. Handle degeneracies explicitly.

Compact proof cores for this node.

1. Normalize coordinates, then replace distances by squared-distance/inner-product identities.
2. Use projections, Gram matrices, barycentric coordinates, or determinants to algebraize the configuration.
3. Check whether the chosen transformation preserves lengths or only affine/convex data.

Use reminder. Use for metric content versus lower content and compact counterexamples.

Trap reminder. The covering sets need not be intervals; project or contract them to a segment.

10.6.9 GEO-09: Hilbert-space equidistant sets and orthonormal translations

Occurrence anchors	1995-D2-P1; 2002-D2-P6; 2008-D2-P6; 2013-D2-P3; 2018-D2-P6; 2021-D2-P8
Original node statement	A set of mutually equidistant points in a Hilbert space is a translate of a scaled orthonormal system. The center is obtained as a limit of barycenters for countable subsets.
Literature	[7, 13, 26, 10]

Exact properties and formulas to know.

1. Hilbert spaces are complete inner-product spaces.
2. Equidistant sets have constant pairwise squared distances.
3. Barycenters of finite subsets form Cauchy sequences in the equidistant construction.
4. Orthonormality means inner products are Kronecker deltas.

Compact proof cores for this node.

1. Normalize coordinates, then replace distances by squared-distance/inner-product identities.
2. Use projections, Gram matrices, barycentric coordinates, or determinants to algebraize the configuration.
3. Check whether the chosen transformation preserves lengths or only affine/convex data.

Use reminder. Use for infinite-dimensional equidistant sets and orthonormal-system construction.

Trap reminder. Completeness of the Hilbert space is used to take the barycenter limit.

10.6.10 GEO-10: Closest points to closed sets and density of uniqueness

Occurrence anchors	2003-D2-P3
Original node statement	A closed set intersects a compact ball around an exterior point, so nearest points exist. If the nearest point is not unique, moving toward one nearest point produces nearby points with a unique nearest point.
Literature	[7, 13, 26, 10]

Exact properties and formulas to know.

1. Closed subsets of $\mathbb{R}^n$ have nearest points after compact-ball reduction.
2. Uniqueness is dense by moving toward one nearest point.
3. Balls tangent at one point separate competing nearest points.
4. Closedness is needed to attain the infimum.

Compact proof cores for this node.

1. Normalize coordinates, then replace distances by squared-distance/inner-product identities.
2. Use projections, Gram matrices, barycentric coordinates, or determinants to algebraize the configuration.
3. Check whether the chosen transformation preserves lengths or only affine/convex data.

Use reminder. Use for density of points with unique nearest projection onto a closed set.

Trap reminder. Closedness is needed for existence; uniqueness is obtained by geometry of balls, not by convexity.

10.6.11 GEO-11: Unit-sphere configurations and regular simplex bounds

Occurrence anchors	2001-D2-P3
Original node statement	On the unit sphere, \{}$\|x-y$\{}$\| >$ \{}$sqrt2$ is equivalent to \{}$langlex, y$\{}$rangle < 0$. At most $n+1$ such points exist in \{}$mathbbR$^n, with equality given by a regular simplex.
Literature	[7, 13, 26, 10]

Exact properties and formulas to know.

1. Simplex volume is determinant divided by $n!$.
2. Barycentric weights equal opposite sub-simplex volume ratios.
3. Circumcenter solves equal-distance linear equations.
4. Affine maps preserve barycentric identities and scale all relevant simplex volumes uniformly.

Compact proof cores for this node.

1. Normalize coordinates, then replace distances by squared-distance/inner-product identities.
2. Use projections, Gram matrices, barycentric coordinates, or determinants to algebraize the configuration.
3. Check whether the chosen transformation preserves lengths or only affine/convex data.

Use reminder. Use for maximum number of points on a sphere with obtuse pairwise angles.

Trap reminder. Induction via projection must preserve strict negativity of inner products.

10.6.12 GEO-12: Polygon triangulation and parity of incidences

Occurrence anchors	2006-D2-P1
Original node statement	Triangulations can be built inductively by adding three vertices and preserving parity at old vertices while controlling new vertices.
Literature	[7, 13, 26, 10]

Exact properties and formulas to know.

1. Convert geometry to coordinates or inner products.
2. Use projections and barycentric coordinates.
3. Check whether transformations preserve metric or only affine data.
4. Handle degeneracies explicitly.

Compact proof cores for this node.

1. Normalize coordinates, then replace distances by squared-distance/inner-product identities.
2. Use projections, Gram matrices, barycentric coordinates, or determinants to algebraize the configuration.
3. Check whether the chosen transformation preserves lengths or only affine/convex data.

Use reminder. Use for parity-constrained triangulations of convex polygons.

Trap reminder. Base cases modulo 3 are essential.

10.6.13 GEO-13: Convex hulls, random polytopes, and geometric probability

Occurrence anchors	2002-D2-P4
Original node statement	Random geometric problems usually reduce by symmetry, conditioning on extremal points, or comparing volumes/areas of regions.
Literature	[7, 13, 26, 10]

Exact properties and formulas to know.

1. Convert geometry to coordinates or inner products.
2. Use projections and barycentric coordinates.
3. Check whether transformations preserve metric or only affine data.
4. Handle degeneracies explicitly.

Compact proof cores for this node.

1. Normalize coordinates, then replace distances by squared-distance/inner-product identities.
2. Use projections, Gram matrices, barycentric coordinates, or determinants to algebraize the configuration.
3. Check whether the chosen transformation preserves lengths or only affine/convex data.

Use reminder. Use for probability involving random points in disks/balls and convex hull events.

Trap reminder. Independence of points and conditioning on positions must be explicit.

10.6.14 GEO-14: Cube and square dissection by number-theoretic semigroups

Occurrence anchors	`2000-D2-P1`
Original node statement	Refining one cube into $a^{\wedge}d$ smaller cubes increases the count by $a^{\wedge}d-1$. If two such increments are coprime, the Frobenius coin lemma gives all sufficiently large counts.
Literature	[7, 13, 26, 10]

Exact properties and formulas to know.

1. Convert geometry to coordinates or inner products.
2. Use projections and barycentric coordinates.
3. Check whether transformations preserve metric or only affine data.
4. Handle degeneracies explicitly.

Compact proof cores for this node.

1. Normalize coordinates, then replace distances by squared-distance/inner-product identities.
2. Use projections, Gram matrices, barycentric coordinates, or determinants to algebraize the configuration.
3. Check whether the chosen transformation preserves lengths or only affine/convex data.

Use reminder. Use for partitioning cubes/squares into a prescribed number of smaller cubes.

Trap reminder. Find two coprime increments $a^{\wedge}d-1$; not every pair works in every dimension.

10.7 Functional equations / dynamics: exact theorem bank

Literature anchor: [22, 14]. **Global theorem/property bank.**

1. **Special-value substitution.** First test neutral values, equal variables, variables giving a known argument, or choices that make a term vanish.
2. **Injectivity.** If $f(u) = f(v)$ can be inserted into the equation to force $u = v$, then f is injective and output equalities may be canceled.
3. **Monotonicity from inequalities.** Equations/inequalities involving $f(x + y)$ or $f(x) - f(x + y)$ can force monotonicity before continuity or differentiability is known.
4. **Iteration.** Exact period n means $f^n(x) = x$ and $f^k(x) \neq x$ for $1 \leq k < n$. Fixed points of f^n must be filtered by smaller periods.
5. **Conjugacy.** If $hfh^{-1} = g$, then f and g have equivalent orbits after the change of variables h.
6. **Orbit decomposition.** A bijection decomposes a set into finite cycles and bi-infinite orbits under integer powers.
7. **No-regularity warning.** Without continuity or measurability, functions may be built by enumeration or cardinal diagonalization; do not apply IVT or compactness.

Node-by-node properties, formulas, proof cores, and occurrence anchors.

10.7.1 FEQ-01: Substitution of special values and normalization

Occurrence anchors	2005-D2-P4; 2007-D2-P1; 2009-D2-P2; 2015-D1-P3; 2016-D1-P1; 2019-D2-P6; 2024-D2-P6
Original node statement	Start by substituting $0, 1, x, y = f(x)$, or symmetric choices; normalize by translating/scaling when possible. Then prove any guessed family satisfies the equation.
Literature	[22, 14]

Exact properties and formulas to know.

1. Test special values first.
2. Normalize constants by translation/scaling when invariant.
3. Substitute arguments that make two sides share a term.
4. Check constant solutions before nonconstant ones.

Compact proof cores for this node.

1. Substitute special values and prove injectivity/monotonicity before cancellation.
2. Use iteration, conjugacy, or orbit decomposition once the map structure is known.
3. Plug every candidate solution back into the original equation, including constants and boundary cases.

Use reminder. Use for most functional equations over $\{\}mathbbR$, $\{\}mathbbQ$, or positive reals.

Trap reminder. Do not assume continuity or monotonicity unless the equation implies it or it is given.

10.7.2 FEQ-02: Injectivity and monotonicity forced by a functional equation

Occurrence anchors	1999-D1-P4; 2000-D2-P5; 2005-D2-P4; 2009-D2-P2; 2015-D1-P3; 2016-D1-P1; 2019-D2-P6; 2024-D2-P6
Original node statement	Equations of the form $f(x)f(yf(x)) = f(x + y)$ or $f(x\text{^}2/f(x)) = x$ often force monotonicity, injectivity, or a reciprocal linear form.
Literature	[22, 14]

Exact properties and formulas to know.

1. Injectivity follows if equal outputs force equal inputs.
2. Monotonicity may follow from sign of $f(x + y) - f(x)$.
3. Once injective, cancel f from both sides.
4. Do not assume continuity unless given or proved.

Compact proof cores for this node.

1. Substitute special values and prove injectivity/monotonicity before cancellation.
2. Use iteration, conjugacy, or orbit decomposition once the map structure is known.
3. Plug every candidate solution back into the original equation, including constants and boundary cases.

Use reminder. Use for classification of functions on positive reals or intervals.

Trap reminder. If injectivity is used, prove it before canceling arguments inside f.

10.7.3 FEQ-03: Fixed points, iterates, and orbit closure

Occurrence anchors	1996-D2-P1; 2002-D1-P4; 2010-D1-P3; 2015-D1-P3; 2016-D1-P1; 2024-D2-P6
Original node statement	For interval maps, cluster points and continuity control iterates. Closed orbits of continuous maps on compact intervals are often finite unless accumulation behavior contradicts closedness.
Literature	[22, 14]

Exact properties and formulas to know.

1. Fixed point: $f(x) = x$.
2. Periodic point: $f^n(x) = x$.
3. Exact period excludes smaller periods.
4. Orbit closure under continuous maps sends cluster points to cluster points.

Compact proof cores for this node.

1. Substitute special values and prove injectivity/monotonicity before cancellation.
2. Use iteration, conjugacy, or orbit decomposition once the map structure is known.
3. Plug every candidate solution back into the original equation, including constants and boundary cases.

Use reminder. Use for iteration convergence, finite orbit proofs, and periodic point counts.

Trap reminder. A pointwise closed orbit is not automatically finite without compactness and continuity arguments.

10.7.4 FEQ-04: Recurrence transformations and conjugacy

Occurrence anchors	1998-D1-P3; 2003-D1-P5; 2004-D1-P2
Original node statement	Nonlinear recurrences often simplify after substituting a new variable, conjugating the map, or identifying an exact iterate formula.
Literature	[22, 14]

Exact properties and formulas to know.

1. Change variables to linearize recurrence.
2. Conjugacy preserves orbit structure.
3. Monotone bounded recurrence converges to fixed point.
4. Trigonometric substitutions reveal half-angle recurrences.

Compact proof cores for this node.

1. Substitute special values and prove injectivity/monotonicity before cancellation.
2. Use iteration, conjugacy, or orbit decomposition once the map structure is known.
3. Plug every candidate solution back into the original equation, including constants and boundary cases.

Use reminder. Use for iterated quadratic maps, nested radicals, and recurrence limits.

Trap reminder. Verify the transformed recurrence preserves the original domain.

10.7.5 FEQ-05: Differentiable functional equations

Occurrence anchors	nan
Original node statement	When a functional equation includes differentiability, differentiate the equation, analyze scaling factors, and use fixed points or growth to force polynomial/exponential forms.
Literature	[22, 14]

Exact properties and formulas to know.

1. Extract special values.
2. Prove regularity/injectivity/monotonicity before using it.

3. Use iteration or substitution systematically.
4. Verify final candidate functions directly.

Compact proof cores for this node.

1. Substitute special values and prove injectivity/monotonicity before cancellation.
2. Use iteration, conjugacy, or orbit decomposition once the map structure is known.
3. Plug every candidate solution back into the original equation, including constants and boundary cases.

Use reminder. Use for smooth equations such as $f(ax+b)=a\hat{}2f(x)$ and derivative constraints.

Trap reminder. Find the fixed point of the affine map before iterating or differentiating repeatedly.

10.7.6 FEQ-06: Piecewise-linear maps and periodic points

Occurrence anchors	1998-D2-P3
Original node statement	For piecewise monotone maps, the graph of $f\hat{}n$ has many monotone branches; intersections with the diagonal count fixed points of iterates.
Literature	[22, 14]

Exact properties and formulas to know.

1. Extract special values.
2. Prove regularity/injectivity/monotonicity before using it.
3. Use iteration or substitution systematically.
4. Verify final candidate functions directly.

Compact proof cores for this node.

1. Substitute special values and prove injectivity/monotonicity before cancellation.
2. Use iteration, conjugacy, or orbit decomposition once the map structure is known.
3. Plug every candidate solution back into the original equation, including constants and boundary cases.

Use reminder. Use for proving existence and finiteness of exact n-periodic points.

Trap reminder. Exact period requires excluding fixed points of smaller iterates.

10.7.7 FEQ-07: Bijections as products of involutions and symmetry decompositions

Occurrence anchors	1997-D2-P5
Original node statement	A bijection decomposes into disjoint orbits; each finite or infinite orbit can be modeled geometrically so the map is a product of two involutions.
Literature	[22, 14]

Exact properties and formulas to know.

1. Involution means $g^2=\mathrm{id}$.
2. Finite cycles are rotations of polygons.
3. Rotations are products of two reflections.
4. The shift on $\mathbb{Z}$ is product of two reflections.

Compact proof cores for this node.

1. Substitute special values and prove injectivity/monotonicity before cancellation.
2. Use iteration, conjugacy, or orbit decomposition once the map structure is known.
3. Plug every candidate solution back into the original equation, including constants and boundary cases.

Use reminder. Use for abstract set bijections and decompositions into symmetries.

Trap reminder. Handle finite cycles and infinite orbits separately.

10.7.8 FEQ-08: No-regularity functions and order patterns on Q or R

Occurrence anchors	1999-D1-P3; 1999-D2-P4; 2001-D2-P5; 2002-D1-P2; 2003-D2-P5; 2024-D2-P6
Original node statement	For arbitrary functions without regularity, use order/combinatorial arguments rather than limits. On $\{\}mathbbQ$, density and finite pigeonhole patterns can force local maxima-like triples.
Literature	[22, 14]

Exact properties and formulas to know.

1. Countable domains permit enumeration.
2. Uncountable domains permit cardinal diagonalization.
3. Without regularity, wild functions are legal.
4. State explicitly where choice/enumeration enters.

Compact proof cores for this node.

1. Substitute special values and prove injectivity/monotonicity before cancellation.
2. Use iteration, conjugacy, or orbit decomposition once the map structure is known.
3. Plug every candidate solution back into the original equation, including constants and boundary cases.

Use reminder. Use for functions from $\{\}mathbbQ$ or arbitrary sets where no continuity is available.

Trap reminder. Do not invoke IVT, compactness, or monotonicity unless it is part of the statement.

10.7.9 FEQ-09: Image-length preserving functions

Occurrence anchors	2006-D2-P2
Original node statement	If every interval image is a closed interval of the same length, the function is 1-Lipschitz and endpoint extrema force monotonicity; hence $f(x) = x + c$ or $f(x) = -x + c$.
Literature	[22, 14]

Exact properties and formulas to know.

1. Image of every interval is an interval of the same length.
2. This implies 1-Lipschitz.
3. Extrema on every interval must occur at endpoints.
4. Monotonicity then gives $f(x) = x + c$ or $f(x) = -x + c$.

Compact proof cores for this node.

1. Substitute special values and prove injectivity/monotonicity before cancellation.
2. Use iteration, conjugacy, or orbit decomposition once the map structure is known.
3. Plug every candidate solution back into the original equation, including constants and boundary cases.

Use reminder. Use for functions preserving interval lengths or ranges.

Trap reminder. The maximum/minimum on an interval must occur at endpoints only after equality in the Lipschitz chain is analyzed.

10.8 Probability / measure / expectation: exact theorem bank

Literature anchor: [16, 8, 3, 27]. **Global theorem/property bank.**

1. **Linearity.** $\mathbb{E}\sum X_i = \sum \mathbb{E}X_i$ without independence. Independence is needed for product distributions, not for linearity.
2. **Indicators.** A count is $N = \sum_i 1_{E_i}$, so $\mathbb{E}N = \sum_i \mathbb{P}(E_i)$.
3. **Residue recurrences.** Distributions modulo m evolve by convolution: $p_{n+1}^{(r)} = \sum_s p_n^{(r-s)}\mu(s)$.
4. **Probability generating functions.** If independent integer variables have generating functions F_i, their sum has generating function $\prod_i F_i$.
5. **Root-of-unity probability filter.** $\mathbb{P}(S \equiv r \pmod m) = m^{-1}\sum_{j=0}^{m-1} \omega^{-jr} F(\omega^j)$.
6. **Uniform geometric probability.** Uniform probability on a region is normalized volume; independent points use product measure.
7. **Interval-cover estimates.** If intervals in $[0,1]$ have total length < 1, they cannot cover the whole interval. This gives existence points outside bad covers.

Node-by-node properties, formulas, proof cores, and occurrence anchors.

10.8.1 PRB-01: Discrete probability via residues and recurrences

Occurrence anchors	1999-D2-P2
Original node statement	A Markov recurrence on residues or a generating function can compute probabilities of sums modulo m.
Literature	[16, 8, 3, 27]

Exact properties and formulas to know.

1. Residue distributions evolve by convolution.
2. Transition probabilities in each row sum to 1.
3. Uniform distribution is often stationary.
4. Exceptional eigenmodes decay by powers of the step bias.

Compact proof cores for this node.

1. Specify the probability space and encode counts by indicators or generating functions.
2. Use linearity/convolution/root filters only after normalization is clear.
3. Finish with an existence-from-average or exact coefficient/probability computation.

Use reminder. Use for dice sums, random walks modulo primes, and residue probabilities.

Trap reminder. State transition probabilities must sum to one and residues are modulo the correct integer.

10.8.2 PRB-02: Generating functions for probability distributions

Occurrence anchors	2022-D1-P1
Original node statement	The probability generating function encodes sums of independent variables as products. Coefficients or root-of-unity filters extract desired events.
Literature	[16, 8, 3, 27]

Exact properties and formulas to know.

1. Probability generating functions multiply for independent sums.
2. Coefficients are probabilities.
3. Root-of-unity filters extract congruence classes.
4. Normalize by total number of outcomes or probability weights.

Compact proof cores for this node.

1. Specify the probability space and encode counts by indicators or generating functions.
2. Use linearity/convolution/root filters only after normalization is clear.
3. Finish with an existence-from-average or exact coefficient/probability computation.

Use reminder. Use for sums of independent discrete variables and random subsets.

Trap reminder. Normalize coefficients by the total number of outcomes.

10.8.3 PRB-03: Expectation, linearity, and probabilistic method

Occurrence anchors	2021-D1-P2
Original node statement	Linearity of expectation requires no independence. A positive average proves existence of an object at least as good as the average.
Literature	[16, 8, 3, 27]

Exact properties and formulas to know.

1. Linearity does not require independence.
2. Indicators turn counts into sums.
3. Average value implies existence of an outcome at least the average.
4. Probabilistic method proves existence by positive expectation.

Compact proof cores for this node.

1. Specify the probability space and encode counts by indicators or generating functions.
2. Use linearity/convolution/root filters only after normalization is clear.
3. Finish with an existence-from-average or exact coefficient/probability computation.

Use reminder. Use for random choices, expected counts, and probabilistic existence.

Trap reminder. Independence is only needed for products or distribution factorization, not for linearity.

10.8.4 PRB-04: Continuous uniform distribution and geometric probability

Occurrence anchors	nan
Original node statement	Continuous geometric probability is volume/area normalized by the ambient domain. Symmetry, conditioning, and convex hull geometry are typical reductions.
Literature	[16, 8, 3, 27]

Exact properties and formulas to know.

1. Probability is normalized measure or count.
2. Independence corresponds to product measure.
3. Finite total interval length gives covering estimates.
4. Almost-everywhere statements require measure hypotheses.

Compact proof cores for this node.

1. Specify the probability space and encode counts by indicators or generating functions.
2. Use linearity/convolution/root filters only after normalization is clear.
3. Finish with an existence-from-average or exact coefficient/probability computation.

Use reminder. Use for random points in disks, balls, circles, and convex hulls.

Trap reminder. Events involving exact coincidences on continuous distributions often have probability zero.

10.8.5 PRB-05: Measure, variation, and differentiability almost everywhere

Occurrence anchors	1998-D2-P6; 2022-D1-P1
Original node statement	Finite total variation implies differentiability almost everywhere; covering arguments with small total length can force at least one point with small difference quotients.
Literature	[16, 8, 3, 27]

Exact properties and formulas to know.

1. Probability is normalized measure or count.
2. Independence corresponds to product measure.
3. Finite total interval length gives covering estimates.
4. Almost-everywhere statements require measure hypotheses.

Compact proof cores for this node.

1. Specify the probability space and encode counts by indicators or generating functions.
2. Use linearity/convolution/root filters only after normalization is clear.
3. Finish with an existence-from-average or exact coefficient/probability computation.

Use reminder. Use for sparse functions supported on countable sets and differentiability existence/nonexistence.

Trap reminder. A countable exceptional support can still be dense; use interval covers, not isolation.

11 Exhaustive Theory Properties and Theorem Reference

This section is an additive expansion of the previous theory manual. Nothing in the earlier catalog, occurrence atlas, problem-to-theory matrix, or compact theorem manual is removed. The goal here is to state the properties themselves in a form that can be used directly while solving past IMC problems. For each theory node, the entry below records exact statements, compact proof cores, and exam-use checks. Literature anchors point to standard references already listed in the bibliography.

11.1 How to read this section

For each theory point, use the following hierarchy.

1. **Exact facts** are the statements you should be able to quote or rederive quickly.
2. **Proof cores** are compact derivations that prevent black-box use of a theorem when the IMC solution requires justification.
3. **Exam checks** are the conditions that usually cause errors: invertibility, commutativity, endpoint behavior, compactness, field characteristic, equality cases, and domain restrictions.

11.2 Global examples of the required level of explicitness

1. **Inverse-matrix property.** If $A, B \in M_n(F)$ and $AB = I$, then A and B are invertible and $BA = I$. Proof: $1 = \det(AB) = \det A \det B$, so A is invertible; left-multiply by A^{-1} to get $B = A^{-1}$, hence $BA = I$. This is the exact property behind all one-sided inverse completions in matrix problems.
2. **Convexity property.** If f is differentiable and convex on an interval, then $f(y) \geq f(x) + f'(x)(y - x)$ for all x, y. Proof: slopes of secants of a convex function are monotone; equivalently, apply the definition of convexity to points on the segment and pass to a derivative limit. This is the exact property behind tangent-line inequalities and Jensen-type arguments.
3. **Polynomial root-count property.** A nonzero polynomial of degree d over a field has at most d roots. Proof: if a is a root then $(x - a)$ divides the polynomial; repeat by induction. This is the exact property behind all "more roots than degree" arguments, including matrix-polynomial entries.

11.3 Linear algebra / matrix theory

Foundation checklist. All matrix arguments in this corpus reduce to linear maps on finite-dimensional spaces. The non-negotiable toolkit is: bases and change of basis, matrix multiplication as composition, row and column operations, rank-nullity, determinants, trace, eigenvalues, characteristic and minimal polynomials, canonical-form intuition, and bilinear forms. A matrix equality must be treated as an equality of linear maps; factors may be cancelled only after the relevant map is known to be invertible. The most common IMC failure is to commute matrices accidentally.

11.3.1 LA-01. Matrix products, inverses, and one-sided inverse identities

Domain and bucket. Linear algebra / matrix theory; LA matrix identities. **Literature anchors.** [19, 18, 6].

Prior compact node statement. For square matrices, a right inverse is also a left inverse: if $XY = I$, then X and Y are invertible and $YX = I$. Algebraic identities such as $(A + I)(B + I) = I$ or $(A - B)(C + A^{-1}) = I$ often turn a noncommutative-looking equation into an inverse statement.

Exact facts to know.

1. If $A, B \in M_n(F)$ and $AB = I$, then A and B are invertible and $BA = I$. Equivalently, a one-sided inverse of a square matrix is automatically the two-sided inverse.
2. If A and B are invertible, then $(AB)^{-1} = B^{-1}A^{-1}$, $(A^T)^{-1} = (A^{-1})^T$, and over $\mathbb{C}$, $(A^*)^{-1} = (A^{-1})^*$.
3. Cancellation is valid only with invertibility: $AX = AY$ implies $X = Y$ if A is invertible; $XA = YA$ implies $X = Y$ if A is invertible.
4. Completion identities are exact: $AB + A + B = 0$ is equivalent to $(A + I)(B + I) = I$; $(A - B)C = BA^{-1}$ is equivalent to $(A - B)(C + A^{-1}) = I$.
5. For square matrices, $\det(AB) = \det A \det B$; hence $AB = I$ implies $\det A \det B = 1$ and both factors are nonsingular.

Compact proof cores.

1. From $AB = I$ obtain $\det A \det B = 1$, so A is invertible. Multiplying $AB = I$ on the left by A^{-1} gives $B = A^{-1}$ and hence $BA = I$.
2. For inverse-of-product, compute $(AB)(B^{-1}A^{-1}) = I = (B^{-1}A^{-1})(AB)$.
3. For completion identities, expand both sides and group terms without changing order; after a product equals I, reverse the order only through the one-sided-inverse theorem.

How it appears in IMC work. **Use.** Use this whenever the statement contains a matrix equation that can be completed to a product equal to I. After factoring, switch the order only because the factors have been proved inverse, not because matrices commute.

Trap watch. Never cancel or reorder matrix factors before proving invertibility; record which factor is square.

Exam checks.

- Do not infer $AB = BA$ from an equation until an inverse identity has forced it.
- When a problem gives A invertible, try to add A^{-1} or I to complete a product.

Occurrence count. 14 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D1-P1;1996-D1-P1;1996-D2-P3;2002-D2-P1;2002-D2-P5;2003-D2-P1;2004-D2-P1;2009-D1-P2;2013-D1-P1;2014-D1-P1;2018-D1-P3;2019-D2-P9;2021-D1-P1;2023-D2-P6

11.3.2 LA-02. Determinant evaluation by row/column operations and triangular forms

Domain and bucket. Linear algebra / matrix theory; LA determinants. **Literature anchors.** [19, 18, 6].

Prior compact node statement. Determinants are preserved by adding a multiple of one row/column to another, change sign under swaps, and scale under row/column scaling. A common IMC target is to transform a structured matrix into triangular or block triangular form.

Exact facts to know.

1. Determinant row operations: swapping two rows changes sign; multiplying one row by c multiplies the determinant by c; adding a multiple of one row to another preserves the determinant.

2. The same statements hold for column operations. A sequence of row and column additions can be mixed, but every scaling and swap must be tracked.
3. If a matrix is upper or lower triangular, its determinant is the product of diagonal entries.
4. Laplace expansion along a row or column with many zeros reduces determinant size; it is often best after first creating zeros by additions.
5. For block triangular matrices $\begin{pmatrix} A & B \\ 0 & D \end{pmatrix}$ or $\begin{pmatrix} A & 0 \\ C & D \end{pmatrix}$, $\det = \det A \det D$.

Compact proof cores.

1. The row-operation rules follow from multilinearity and alternation of the determinant in its rows.
2. Triangular determinant follows by induction using expansion along the first row or by observing that only the identity permutation contributes.
3. Block triangular determinant follows because any contributing permutation must preserve the two diagonal blocks.

How it appears in IMC work. **Use.** Use for explicit determinant computations, sign/parity determinant arguments, and matrices with entries depending on $i+j$, $|i-j|$, parity, primes, or a recurrence.

Trap watch. Track row/column operations exactly; if operations depend on parity or cyclic order, verify small n separately.

Exam checks.

- Before computing, ask whether the pattern becomes triangular after first differences, cyclic differences, or parity reordering.

Occurrence count. 51 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D1-P1;1995-D1-P1;1995-D2-P1;1996-D1-P1;1997-D1-P3;1997-D2-P2;1997-D2-P4;1999-D1-P1;2000-D1-P3;2001-D1-P1;2001-D1-P5;2001-D2-P4;2002-D2-P1;2003-D2-P1;2004-D2-P1;2004-D2-P6;2005-D1-P1;2005-D2-P3;2005-D2-P6;2006-D1-P3;2006-D2-P6;2007-D1-P2;2007-D1-P3;2007-D2-P4;2008-D1-P6;2008-D2-P5;2009-D1-P2;2009-D1-P5;2009-D2-P5;2010-D1-P5;2010-D2-P4;2011-D1-P2;2012-D1-P2;2013-D1-P1;2014-D1-P1;2015-D1-P1;2016-D2-P10;2017-D1-P1;2018-D1-P3;2019-D1-P5;2019-D2-P9;2020-D1-P2;2021-D1-P1;2021-D2-P5;2022-D2-P7;2023-D1-P2;2023-D2-P6;2023-D2-P10;2024-D1-P1;2024-D2-P7;2024-D2-P9

11.3.3 LA-03. Block matrices, Schur complements, and determinant identities

Domain and bucket. Linear algebra / matrix theory; LA block methods. **Literature anchors.** [19, 18, 6].

Prior compact node statement. For compatible square blocks, block multiplication and determinant identities often reduce a large matrix equation to a smaller one. Typical facts include $\det(M)\det(H) = \det(A)$ for inverse block pairs in suitable block decompositions, and AC and CA having the same nonzero eigenvalues.

Exact facts to know.

1. Block multiplication is ordinary multiplication with compatible blocks: $\begin{pmatrix} A & B \\ C & D \end{pmatrix}\begin{pmatrix} E & F \\ G & H \end{pmatrix} = \begin{pmatrix} AE+BG & AF+BH \\ CE+DG & CF+DH \end{pmatrix}$.
2. Schur determinant: if A is invertible, $\det\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det(A)\det(D-CA^{-1}B)$. If D is invertible, it also equals $\det(D)\det(A-BD^{-1}C)$.
3. If $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ is invertible with inverse $M^{-1} = \begin{pmatrix} E & F \\ G & H \end{pmatrix}$ and the block sizes match, then block equations such as $AF+BH=0$ and $CF+DH=I$ are legitimate identities.
4. For rectangular-compatible A, B, AB and BA have the same nonzero eigenvalues with the same algebraic multiplicities; in particular $\det(I+AB) = \det(I+BA)$ when both sides are square.
5. Block diagonal matrices satisfy $\chi_{A\oplus D}(t) = \chi_A(t)\chi_D(t)$ and functions act blockwise: $p(A\oplus D) = p(A)\oplus p(D)$.

Compact proof cores.

1. Schur complement is proved by multiplying $\begin{pmatrix} I & 0 \\ -CA^{-1} & I \end{pmatrix}\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} A & B \\ 0 & D-CA^{-1}B \end{pmatrix}$ and taking determinants.
2. The identity $\det(I+AB) = \det(I+BA)$ follows by comparing characteristic polynomials or multiplying the block matrices $\begin{pmatrix} I & A \\ -B & I \end{pmatrix}$ in two ways.
3. Inverse block identities come directly from $MM^{-1} = I$ and $M^{-1}M = I$.

How it appears in IMC work. **Use.** Use when a problem explicitly gives a block matrix, a product AB versus BA, or a matrix of a linear operator on a matrix space.

Trap watch. Schur complement formulas require invertibility of the chosen block; if it is not given, use block multiplication identities instead.

Exam checks.

- Use Schur complements only when the chosen diagonal block is invertible; otherwise use elementary block equations.

Occurrence count. 114 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D1-P1;1994-D2-P2;1995-D1-P1;1995-D1-P2;1995-D2-P1;1995-D2-P2;1996-D1-P1;1996-D1-P2;1997-D1-P3;1997-D2-P2;1997-D2-P4;1998-D1-P1;1998-D1-P5;1999-D1-P1;1999-D1-P5;1999-D2-P2;1999-D2-P3;2000-D1-P2;2000-D1-P3;2000-D1-P4;2000-D2-P6;2001-D1-P1;2001-D1-P3;2001-D1-P5;2001-D2-P4;2002-D2-P1;2002-D2-P2;2003-D2-P1;2004-D2-P1;2004-D2-P4;2004-D2-P6;2005-D1-P1;2005-D2-P3;2005-D2-P6;2006-D1-P3;2006-D2-P2;2006-D2-P3;2006-D2-P6;2007-D1-P2;2007-D1-P3;2007-D2-P1;2007-D2-P4;2008-D1-P6;2008-D1-P1;2008-D2-P1;2008-D2-P5;2009-D1-P2;2009-D1-P5;2009-D2-P3;2009-D2-P5;2010-D1-P2;2010-D1-P3;2010-D1-P5;2010-D2-P4;2011-D1-P2;2011-D1-P5;2011-D2-P2;2011-D2-P3;2012-D1-P2;2012-D2-P4;2013-D1-P1;2013-D1-P3;2013-D1-P4;2013-D2-P3;2013-D2-P5;2014-D1-P1;2014-D1-P2;2014-D2-P2;2014-D2-P3;2014-D2-P5;2015-D1-P1;2015-D1-P2;2015-D1-P3;2015-D2-P6;2015-D2-P7;2015-D2-P9;2016-D1-P1;2016-D1-P2;2016-D2-P6;2016-D2-P7;2017-D1-P1;2017-D1-P2;2017-D2-P6;2017-D2-P8;2018-D1-P1;2018-D1-P3;2018-D1-P4;2018-D2-P8;2018-D2-P10;2019-D1-P1;2019-D1-P2;2019-D1-P3;2019-D1-P5;2019-D2-P9;2020-D1-P2;2020-D1-P4;2020-D2-P5;2020-D2-P6;2020-D2-P8;2021-D1-P1;2021-D1-P3;2021-D1-P4;2021-D2-P5;2022-D1-P1;2022-D1-P2;2022-D2-P7;2023-D1-P1;2023-D2-P6;2024-D1-P1;2024-D1-P2;2024-D2-P6;2024-D2-P7;2024-D2-P8;2024-D2-P9

11.3.4 LA-04. Rank-nullity, image/kernel, and rank of factorizations

Domain and bucket. Linear algebra / matrix theory; LA rank/dimension. **Literature anchors.** [19, 18, 6].

Prior compact node statement. Rank-nullity gives $\dim V = \dim\ker T + \dim\operatorname{im} T$. For products, $\operatorname{rank}(AB) \le \min(\operatorname{rank} A, \operatorname{rank} B)$ and column spaces/images often reveal the exact rank.

Exact facts to know.

1. Rank-nullity: for $T : V \to W$, $\dim V = \dim\ker T + \operatorname{rank} T$.
2. Rank inequalities: $\operatorname{rank}(AB) \le \min(\operatorname{rank} A, \operatorname{rank} B)$ and $\operatorname{rank}(A+B) \le \operatorname{rank} A + \operatorname{rank} B$.
3. Sylvester inequality: if A is $m \times n$ and B is $n \times p$, then $\operatorname{rank}(AB) \ge \operatorname{rank} A + \operatorname{rank} B - n$.
4. Rank is invariant under multiplication on the left or right by invertible matrices.
5. A matrix has rank one exactly when it can be written as uv^T with nonzero column vectors u, v; then its image is the line spanned by u and its kernel is $v^\perp$ under the standard pairing.

Compact proof cores.

1. Rank-nullity follows by extending a basis of the kernel to a basis of the domain and observing that the images of the added basis vectors form a basis of the image.
2. Rank invariance follows because invertible maps do not change dimension of images.
3. Sylvester follows from $\dim B(\ker A \cap \operatorname{im} B)$ or from dimension inequalities for images and kernels.

How it appears in IMC work. **Use.** Use for rank minimization/maximization, rank-one perturbations, factorizations AB, and problems asking for the dimension of a matrix subspace.

Trap watch. Rank statements over $\mathbb{R}$ and $\mathbb{C}$ can differ only through spectral issues; dimension counts must specify the ground field.

Exam checks.

- Dimension-count problems usually need both an upper bound and a construction attaining it.

Occurrence count. 87 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D1-P1;1994-D1-P4;1994-D2-P4;1994-D2-P5;1995-D1-P1;1995-D1-P5;1996-D1-P1;1996-D1-P3;1996-D1-P6;1997-D1-P3;1997-D2-P2;1997-D2-P4;1998-D1-P1;1998-D2-P1;1999-D1-P1;2000-D1-P3;2000-D2-P1;2001-D1-P5;2001-D2-P3;2001-D2-P4;2002-D1-P5;2002-D1-P6;2002-D2-P1;2003-D2-P1;2004-D1-P3;2004-D2-P1;2004-D2-P4;2005-D1-P1;2005-D1-P3;2005-D2-P3;2006-D1-P3;2006-D1-P6;2006-D2-P2;2006-D2-P4;2006-D2-P6;2007-D1-P2;2007-D1-P3;2007-D1-P5;2007-D2-P4;2007-D2-P5;2008-D1-P2;2008-D2-P5;2008-D2-P6;2009-D1-P2;2009-D1-P3;2009-D1-P4;2009-D1-P5;2009-D2-P3;2009-D2-P5;2010-D1-P5;2010-D2-P4;2011-D1-P2;2011-D1-P5;2011-D2-P4;2011-D2-P5;2012-D1-P2;2012-D1-P4;2013-D1-P1;2013-D1-P2;2014-D1-P1;2014-D2-P2;2015-D1-P1;2015-D1-P5;2015-D2-P9;2016-D1-P1;2016-D2-P10;2017-D1-P1;2017-D1-P2;2017-D2-P9;2018-D1-P3;2018-D1-P4;2019-D1-P3;2019-D1-P5;2019-D2-P9;2019-D2-P10;2020-D1-P2;2020-D2-P5;2021-D1-P1;2022-D1-P1;2022-D2-P7;2023-D1-P2;2023-D2-P6;2023-D2-P7;2023-D2-P9;2024-D1-P1;2024-D1-P5;2024-D2-P7

11.3.5 LA-05. Trace cyclicity, trace pairing, and zero-trace subspaces

Domain and bucket. Linear algebra / matrix theory; LA trace. **Literature anchors.** [19, 18, 6].

Prior compact node statement. The trace is cyclic: $\operatorname{tr}(XY) = \operatorname{tr}(YX)$ and hence $\operatorname{tr}([X,Y]) = 0$. The bilinear form $(X,Y) \mapsto \operatorname{tr}(XY)$ identifies the dual of the matrix space and detects symmetric or triangular subspaces.

Exact facts to know.

1. Trace is linear and $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ for square-compatible A, B. More generally $\operatorname{tr}(A_1 \cdots A_m) = \operatorname{tr}(A_m A_1 \cdots A_{m-1})$.
2. Every commutator has trace zero: $\operatorname{tr}(AB - BA) = 0$.
3. The trace pairing $\langle A, B \rangle = \operatorname{tr}(AB)$ is a nondegenerate bilinear form on $M_n(F)$; with $\operatorname{tr}(A^T B)$ it becomes the Frobenius inner product over $\mathbb{R}$.
4. The subspace of trace-zero matrices has codimension one in $M_n(F)$.
5. If a linear functional L on $M_n(F)$ is given, there is a unique C such that $L(A) = \operatorname{tr}(AC)$ for all A.

Compact proof cores.

1. For cyclicity, write entries: $\operatorname{tr}(AB) = \sum_i \sum_j a_{ij} b_{ji} = \sum_j \sum_i b_{ji} a_{ij} = \operatorname{tr}(BA)$.
2. Nondegeneracy follows by testing against elementary matrices E_{ji}, since $\operatorname{tr}(AE_{ji}) = a_{ij}$.
3. The representation of linear functionals follows because elementary matrices form a basis and the previous formula selects each coefficient.

How it appears in IMC work. **Use.** Use for commutator problems, linear functionals on matrix spaces, bilinear orthogonality conditions, and trace constraints.

Trap watch. Cyclicity is safe for products under trace, not outside trace. Do not infer $XY = YX$ from equal traces.

Exam checks.

- Use trace inside a trace only; cyclicity does not let you commute factors outside trace.

Occurrence count. 33 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1995-D1-P1;1995-D2-P1;1996-D1-P1;1997-D2-P2;1997-D2-P4;1999-D1-P1;2000-D1-P3;2001-D1-P5;2001-D2-P4;2002-D2-P1;2003-D2-P1;2004-D2-P1;2005-D1-P1;2005-D2-P3;2005-D2-P6;2006-D2-P6;2007-D1-P2;2007-D1-P3;2007-D2-P4;2009-D1-P2;2009-D2-P3;2010-D1-P5;2011-D1-P2;2012-D1-P2;2013-D1-P1;2014-D1-P1;2015-D1-P1;2019-D1-P5;2020-D1-P2;2021-D1-P1;2022-D2-P7;2023-D2-P6;2024-D2-P7

11.3.6 LA-06. Commutators and spectral obstructions

Domain and bucket. Linear algebra / matrix theory; LA commutators. **Literature anchors.** [19, 18, 6].

Prior compact node statement. A commutator $AB - BA$ has trace zero. If it has very small rank or satisfies a polynomial identity, trace and Jordan-form restrictions can force nilpotency or impossibility.

Exact facts to know.

1. For $\mathrm{ad}_A(X) = AX - XA$, ad_A is a linear operator on $M_n(F)$ and a derivation: $\mathrm{ad}_A(XY) = \mathrm{ad}_A(X)Y + X\,\mathrm{ad}_A(Y)$.
2. If $C = AB - BA$ has rank 1, then $\operatorname{tr} C = 0$ and hence its only possible nonzero eigenvalue is zero; therefore all eigenvalues are zero.
3. A rank-one nilpotent matrix satisfies $C^2 = 0$: write $C = uv^T$; $\operatorname{tr} C = v^T u = 0$, so $C^2 = u(v^T u)v^T = 0$.
4. If $FG - GF = \alpha F$ with $\alpha \neq 0$, then repeated commutator identities give $F^k G - GF^k = \alpha k F^k$; finite-dimensionality forces nilpotency of F.
5. Trace powers often detect eigenvalue constraints: $\operatorname{tr}(C^m)$ equals the sum of mth powers of eigenvalues over an algebraic closure.

Compact proof cores.

1. The derivation identity is obtained by adding and subtracting $AXY - XAY$.
2. The rank-one nilpotent claim follows from $C = uv^T$ and the equality $\operatorname{tr} C = v^T u$.
3. The identity for F^k is a telescoping sum $F^k G - GF^k = \sum_{i=0}^{k-1} F^{k-1-i}(FG - GF)F^i$.

How it appears in IMC work. **Use.** Use when a problem contains $AB - BA$, $FG - GF$, or an operator $X \mapsto AX - XA$.

Trap watch. Trace zero alone is weak; combine it with rank, eigenvalue multiplicity, or finite-dimensionality.

Exam checks.

- Combine trace-zero with rank or with a polynomial identity; trace-zero alone rarely solves the problem.

Occurrence count. 20 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D1-P4;1995-D1-P5;1996-D1-P1;1997-D1-P3;1997-D2-P4;2000-D1-P3;2000-D1-P5;2000-D2-P6;2001-D1-P5;2002-D2-P1;2003-D1-P3;2003-D2-P1;2004-D2-P1;2005-D1-P6;2009-D2-P3;2013-D1-P1;2014-D1-P1;2020-D1-P2;2021-D1-P1;2022-D2-P7

11.3.7 LA-07. Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton

Domain and bucket. Linear algebra / matrix theory; LA spectral theory. **Literature anchors.** [19, 18, 6].

Prior compact node statement. If $p(A) = 0$, every eigenvalue of A is a root of p. Cayley-Hamilton gives $\chi_A(A) = 0$. Distinct roots of the minimal polynomial imply diagonalizability over a splitting field.

Exact facts to know.

1. If $p(A) = 0$ and $Av = \lambda v$ with $v \neq 0$, then $p(\lambda) = 0$. Thus eigenvalues of A lie among roots of any annihilating polynomial.
2. Cayley-Hamilton: every square matrix satisfies its characteristic polynomial $\chi_A(A) = 0$.
3. The minimal polynomial m_A divides every annihilating polynomial and has the same roots as the possible eigenvalues, with Jordan-block sizes encoded by powers.
4. If m_A splits into distinct linear factors, then A is diagonalizable over the splitting field.
5. Spectral mapping for polynomials: eigenvalues of $p(A)$ are $p(\lambda)$ for eigenvalues λ of A, with multiplicities controlled by Jordan structure.

Compact proof cores.

1. For eigenvalue substitution, apply $p(A)$ to v: $p(A)v = p(\lambda)v$.
2. Cayley-Hamilton may be proved by triangularization over an algebraic closure or by adjugate identity for $tI - A$.
3. Distinct-root diagonalization follows from decomposing $V = \bigoplus \ker(A - \lambda_i I)$ using Bezout polynomials for the coprime factors.

How it appears in IMC work. **Use.** Use for convergence of matrix powers, nilpotency, idempotency, diagonalization, determinant/eigenvalue constraints, and polynomial equations in matrices.

Trap watch. Real matrices may have complex eigenvalues; diagonalization over $\mathbb{C}$ may be needed even when entries are real.

Exam checks.

CORRECTION MATH-004 — legacy statement superseded; click for canonical replacement.

- For real matrices, do spectral arguments over $\mathbb{C}$ when needed, then translate real conclusions back.

Occurrence count. 51 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D1-P4;1994-D2-P2;1995-D1-P1;1995-D1-P4;1995-D1-P5;1995-D2-P5;1996-D1-P1;1996-D1-P3;1996-D1-P5;1997-D1-P5;1997-D2-P2;1998-D2-P1;1998-D2-P4;1999-D1-P1;2000-D1-P3;2000-D1-P5;2000-D2-P6;2001-D1-P5;2001-D1-P6;2002-D1-P1;2002-D2-P1;2002-D2-P3;2002-D2-P5;2003-D1-P3;2003-D1-P4;2003-D2-P1;2004-D1-P1;2004-D1-P5;2004-D2-P1;2004-D2-P4;2006-D2-P2;2006-D2-P6;2008-D1-P3;2009-D1-P2;2009-D2-P3;2010-D2-P1;2011-D1-P2;2013-D1-P1;2014-D1-P1;2014-D2-P2;2015-D1-P1;2015-D2-P9;2016-D2-P10;2017-D1-P1;2017-D2-P8;2018-D1-P3;2021-D1-P1;2022-D1-P2;2023-D1-P2;2023-D2-P6;2024-D2-P7

11.3.8 LA-08. Nilpotent, idempotent, involution, and square-zero structures

Domain and bucket. Linear algebra / matrix theory; LA special matrices. **Literature anchors.** [19, 18, 6].

Prior compact node statement. Nilpotent matrices have all eigenvalues 0; idempotents have eigenvalues in $\{0,1\}$; involutions have eigenvalues in $\{\pm 1\}$ when the characteristic is not 2. These facts often convert algebraic equations into diagonal or block forms.

Exact facts to know.

1. Nilpotent means $N^k = 0$ for some k; over an algebraically closed field this is equivalent to all eigenvalues being 0. For $n \times n$ matrices, nilpotent implies $N^n = 0$.
2. Idempotent means $P^2 = P$; the minimal polynomial divides $x(x-1)$, so P is diagonalizable with eigenvalues in $\{0,1\}$ and $\operatorname{rank} P = \operatorname{tr} P$.
3. Involution means $A^2 = I$; over characteristic not 2, A is diagonalizable with eigenvalues ± 1, and $\frac{1}{2}(I \pm A)$ are complementary projections.
4. Square-zero means $N^2 = 0$; then $\operatorname{im} N \subseteq \ker N$ and $\operatorname{rank} N \le n/2$.
5. If commuting square-zero operators A_i have a nonzero product $A_1 \cdots A_k$, then the generated vectors $A_S v$ for subsets S are a natural candidate for 2^k independent vectors.

Compact proof cores.

1. Idempotent diagonalizability follows from the distinct-root polynomial $x(x-1)$.
2. For square-zero, applying N to an image vector gives zero, proving $\operatorname{im} N \subseteq \ker N$ and the rank bound by rank-nullity.
3. Independence of subset vectors is often proved by annihilating all vectors after a given one with the complementary product.

How it appears in IMC work. **Use.** Use for matrix equations involving powers, commuting involutions, square-zero products, and rank bounds for idempotents.

Trap watch. Do not assume diagonalizability for nilpotents. Idempotents and involutions are diagonalizable only when the minimal polynomial has distinct roots.

Exam checks.

- When a problem says nilpotent, check whether it asks for nilpotency index, zero eigenvalues, trace powers, or triangularization.

Occurrence count. 14 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D1-P4;1995-D1-P1;1995-D1-P5;1996-D1-P3;2000-D1-P3;2000-D1-P5;2000-D2-P6;2001-D2-P4;2003-D1-P3;2005-D1-P6;2007-D2-P5;2009-D1-P2;2018-D1-P3;2022-D2-P7

11.3.9 LA-09. Simultaneous diagonalization and commuting families

Domain and bucket. Linear algebra / matrix theory; LA simultaneous structure. **Literature anchors.** [19, 18, 6].

Prior compact node statement. A commuting family of diagonalizable operators over an algebraically closed field is simultaneously diagonalizable under standard finite-dimensional hypotheses. For diagonal matrices, the commutant is block diagonal according to equal eigenvalues.

Exact facts to know.

1. If a family of diagonalizable linear operators over an algebraically closed field commutes pairwise, then it is simultaneously diagonalizable.
2. If A and B commute, every eigenspace of A is invariant under B.
3. Commuting involutions over characteristic not 2 are simultaneously diagonalizable with diagonal entries ± 1.
4. A family of simultaneously diagonalizable $n \times n$ involutions has at most 2^n distinct elements.
5. If matrices are normal and commute over $\mathbb{C}$, unitary simultaneous diagonalization is available, but IMC problems usually need only the algebraic diagonalizable version.

Compact proof cores.

1. Use induction on dimension: choose an eigenspace of one operator; commuting makes it invariant under all others; diagonalize restrictions.
2. The 2^n bound follows because each common eigenbasis vector receives a sign, so an involution is determined by an n-tuple in $\{\pm 1\}^n$.
3. The invariance statement is $A(Bv) = B(Av) = \lambda Bv$ for $Av = \lambda v$.

How it appears in IMC work. **Use.** Use when several operators commute or preserve common eigenspaces; typical conclusions are dimension counts and maximum cardinality of families.

Trap watch. Commuting alone does not imply diagonalizable. First prove each operator is diagonalizable or use invariant subspaces carefully.

Exam checks.

- Check that each operator is diagonalizable; commuting alone is not enough.

Occurrence count. 3 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D2-P4;1996-D1-P3;2006-D2-P6

11.3.10 LA-10. Invariant subspaces and adapted bases

Domain and bucket. Linear algebra / matrix theory; LA adapted basis. **Literature anchors.** [19, 18, 6].

Prior compact node statement. If a chain of subspaces is invariant, choose a basis adapted to the chain; the matrix becomes block upper triangular, and free entries can be counted.

Exact facts to know.

1. A subspace U is T-invariant iff $T(U) \subseteq U$. In a basis beginning with a basis of U, the matrix of T is block upper triangular.
2. If $U_1 \subseteq U_2 \subseteq V$ are invariant, a basis adapted to the flag gives a block upper triangular matrix with diagonal block sizes $\dim U_1$, $\dim U_2 - \dim U_1$, and $\dim V - \dim U_2$.
3. The dimension of a matrix space preserving a flag is the number of freely choosable entries in the adapted block upper triangular pattern.
4. Common invariant subspaces often arise as kernels, images, eigenspaces, or spans generated by cyclic vectors.
5. Triangularization over $\mathbb{C}$ is possible for a single operator; simultaneous triangularization needs stronger hypotheses such as solvable Lie algebra conditions, but IMC usually uses explicit flags.

Compact proof cores.

1. Adapted-basis form follows because the image of a basis vector from U has no component outside U.
2. Dimension counting is counting free matrix entries after zeros forced by invariance.
3. Kernels and images are invariant under commuting operators: if $ST = TS$, then $T(\ker S) \subseteq \ker S$ and $T(\operatorname{im} S) \subseteq \operatorname{im} S$.

How it appears in IMC work. **Use.** Use in dimension-of-endomorphism-space problems, simultaneous preservation of subspaces, and centralizer computations.

Trap watch. Check the order of basis blocks: $T(U) \subset U$ imposes zero blocks below, not above, depending on column convention.

Exam checks.

- When asked for dimension, draw the zero pattern first and then count free entries.

Occurrence count. 30 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D2-P4;1994-D2-P5;1995-D1-P1;1996-D1-P3;1996-D1-P6;1997-D2-P2;1997-D2-P4;1998-D1-P1;1998-D2-P1;2001-D1-P5;2001-D2-P3;2002-D1-P5;2002-D1-P6;2004-D2-P4;2005-D2-P3;2006-D2-P2;2006-D2-P4;2007-D1-P2;2007-D2-P5;2008-D1-P2;2009-D1-P3;2009-D1-P4;2009-D1-P5;2010-D2-P4;2011-D1-P5;2011-D2-P5;2012-D1-P2;2015-D1-P5;2015-D2-P9;2023-D2-P9

11.3.11 LA-11. Linear functionals, kernels, and dual span criteria

Domain and bucket. Linear algebra / matrix theory; LA duality. **Literature anchors.** [19, 18, 6].

Prior compact node statement. A linear functional that vanishes on $\bigcap_i \ker f_i$ belongs to the span of $f_1, \dots, f_k$. Equivalently, the annihilator of an intersection of kernels is the span of the functionals.

Exact facts to know.

1. For linear functionals $f, f_1, \dots, f_k$ on V, if $\bigcap_i \ker f_i \subseteq \ker f$, then $f \in \operatorname{span}\{f_1, \dots, f_k\}$.
2. Equivalently, the annihilator of $\bigcap_i \ker f_i$ is the span of the f_i.
3. If $f_1, \dots, f_k$ are independent, there exist vectors $a_1, \dots, a_k$ with $f_i(a_j) = \delta_{ij}$.
4. Linear maps to $\mathbb{R}^k$ encode simultaneous constraints: $F(x) = (f_1(x), \dots, f_k(x))$. Then the previous criterion says f factors through F.
5. Finite-dimensional duality is not required for the main criterion when only finitely many functionals are involved.

Compact proof cores.

1. Construct a_j inductively or use surjectivity of F when the f_i are independent.
2. For any x, subtract $\sum f_i(x)a_i$ to land in $\bigcap \ker f_i$; applying f gives the linear combination formula.
3. The annihilator equality follows by the same construction.

How it appears in IMC work. **Use.** Use for abstract vector-space problems where one functional is forced by the vanishing of several others.

Trap watch. Independence of the given functionals may need to be arranged by discarding redundant ones.

Exam checks.

- Use this when a problem says one linear condition vanishes whenever several others vanish.

Occurrence count. 6 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1997-D2-P2;1997-D2-P4;1998-D1-P1;1998-D2-P1;2008-D1-P2;2015-D2-P9

11.3.12 LA-12. Matrix subspaces constrained by trace bilinear forms

Domain and bucket. Linear algebra / matrix theory; LA trace geometry. **Literature anchors.** [19, 18, 6].

Prior compact node statement. The trace form separates symmetric and skew/triangular subspaces. If $\operatorname{tr}(XY) = 0$ for all X, Y in a subspace, then the subspace cannot contain nonzero symmetric matrices, giving a dimension bound.

Exact facts to know.

1. With $\langle X, Y \rangle = \operatorname{tr}(XY)$, a subspace V satisfying $\operatorname{tr}(XY) = 0$ for all $X, Y \in V$ is totally isotropic for this bilinear form.

2. The symmetric matrices form a subspace of dimension $n(n+1)/2$; the strictly upper triangular matrices form a subspace of dimension $n(n-1)/2$.
3. If A is real symmetric and nonzero, then $\operatorname{tr}(A^2) = \sum_{i,j} a_{ij}^2 > 0$.
4. Therefore a totally isotropic subspace for $\operatorname{tr}(XY)$ contains no nonzero symmetric matrix.
5. Using $M_n(\mathbb{R}) =$ symmetric $\oplus$ skew-symmetric as vector spaces gives sharp dimension bounds.

Compact proof cores.

1. The positivity formula is $\operatorname{tr}(A^2) = \operatorname{tr}(A^T A)$ for symmetric A.
2. If $V \cap S = \{0\}$, then $\dim V + \dim S \le n^2$; hence $\dim V \le n(n-1)/2$.
3. Strictly upper triangular matrices attain the bound because a product of two strictly upper triangular matrices has zero diagonal and hence trace zero.

How it appears in IMC work. **Use.** Use in maximum-dimension matrix subspace problems with quadratic trace constraints.

Trap watch. The form $\operatorname{tr}(XY)$ is not positive definite on all matrices; positivity appears on symmetric matrices via $\operatorname{tr}(A^2)$.

Exam checks.

- Look for a positive definite subspace that intersects the required subspace trivially.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2005-D2-P3

11.3.13 LA-13. Matrix exponentials and polynomial functional calculus

Domain and bucket. Linear algebra / matrix theory; LA functional calculus. **Literature anchors.** [19, 12].

Prior compact node statement. For a matrix X, $q(e^X)$ can be expanded as a convergent power series in X. Identities comparing AB and BA often reduce to the fact that AC and CA have the same characteristic polynomial.

Exact facts to know.

1. The matrix exponential is $e^A = \sum_{m \ge 0} A^m/m!$ and converges absolutely in every matrix norm.
2. If A and B commute, then $e^{A+B} = e^A e^B$; without commutativity this identity can fail.
3. For any polynomial or convergent power series q, the matrix $q(A)$ is defined by substituting A and is a polynomial in A after Cayley-Hamilton reduction.
4. If $q(z) = \sum c_m z^m$, then $q(AB) = c_0 I + A\left(\sum_{m \ge 1} c_m (BA)^{m-1} B\right)$ and $q(BA) = c_0 I + \left(\sum_{m \ge 1} c_m (BA)^{m-1} B\right) A$ when sizes match.
5. Matrices AC and CA have the same nonzero eigenvalues; this allows nilpotency of $p(e^{AB})$ and $p(e^{BA})$ to be compared.

Compact proof cores.

1. Absolute convergence follows from submultiplicativity: $\sum \|A\|^m/m! = e^{\|A\|}$.
2. The formulas for $q(AB)$ and $q(BA)$ use $(AB)^m = A(BA)^{m-1}B$ for $m \ge 1$.
3. Nilpotency follows from characteristic polynomial equality and Cayley-Hamilton.

How it appears in IMC work. **Use.** Use when exponentials of products AB and BA appear, especially with nilpotency of polynomial expressions.

Trap watch. Do not use $e^{AB} = e^{BA}$; compare characteristic polynomials or nonzero spectra instead.

Exam checks.

- Never use $e^{A+B} = e^A e^B$ unless commutativity is known.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2000-D2-P6

11.3.14 LA-14. Operator norm, power boundedness, and spectral-radius style estimates

Domain and bucket. Linear algebra / matrix theory; LA norms. **Literature anchors.** [19, 12].

Prior compact node statement. A sequence of operator powers can be controlled by estimates on differences $T^{n+1} - T^n$. Exponential decay of these differences implies norm convergence and power boundedness.

Exact facts to know.

1. For the Euclidean operator norm, $\|A\| = \sup_{x \neq 0} \|Ax\|_2 / \|x\|_2$ and $\|AB\| \leq \|A\|\|B\|$.
2. Power bounded means $\sup_k \|A^k\| < \infty$. Convergence of A^k in norm implies power boundedness.
3. The spectral radius satisfies $\rho(A) \leq \|A\|$ and $\rho(A) = \lim_{k\to\infty} \|A^k\|^{1/k}$ in finite dimensions.
4. Difference estimates such as $\|A^{k+1} - A^k\|$ can be turned into Cauchy estimates by telescoping if they decay fast enough.
5. If A^k converges to P, then $AP = P$ and $P^2 = P$ when multiplication is continuous and the limit commutes with A through powers.

Compact proof cores.

1. Submultiplicativity follows by taking suprema after $\|ABx\| \leq \|A\|\|Bx\|$.
2. Telescoping: $A^m - A^n = \sum_{j=n}^{m-1} (A^{j+1} - A^j)$.
3. If $A^k \to P$, then $A^{k+1} \to AP$ and also $A^{k+1} \to P$, so $AP = P$; similarly $PA = P$ and $P^2 = P$.

How it appears in IMC work. **Use.** Use for norm inequalities involving all positive powers of a matrix or linear operator.

Trap watch. Norm convergence is stronger than pointwise convergence. Watch constants and finite-dimensional vs Banach-space arguments.

Exam checks.

- Check whether a problem gives a norm estimate on powers or on differences of powers; the strategy differs.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2002-D1-P6

11.3.15 LA-15. Gram matrices and rational linear dependence

Domain and bucket. Linear algebra / matrix theory; LA Gram matrices. **Literature anchors.** [19, 7].

Prior compact node statement. Distances determine inner products by $2\langle u, v\rangle = \|u\|^2 + \|v\|^2 - \|u - v\|^2$. If the Gram matrix has rational entries and is invertible, coordinates of another vector obtained by inner products are rational.

Exact facts to know.

1. The Gram matrix of vectors $v_1, \ldots, v_m$ is $G = (\langle v_i, v_j\rangle)$. It is symmetric positive semidefinite.
2. The vectors are linearly independent iff G is positive definite iff $\det G \neq 0$.
3. Distance and inner products are related by $\|u - v\|^2 = \|u\|^2 + \|v\|^2 - 2\langle u, v\rangle$.
4. If all squared distances among vectors and to the origin are rational, then all Gram entries are rational.
5. If $v_{n+1} = \sum_i \lambda_i v_i$ and the Gram matrix of $v_1, \ldots, v_n$ is rational invertible while the right-hand inner-product vector is rational, then $\lambda_i \in \mathbb{Q}$.

Compact proof cores.

1. Positive semidefiniteness follows from $c^T Gc = \|\sum c_i v_i\|^2 \geq 0$.
2. The distance formula is obtained by expanding the square.
3. Rational coefficients follow by solving $G\lambda = w$ and noting G^{-1} has rational entries when G is rational with nonzero determinant.

How it appears in IMC work. **Use.** Use for rational-distance vector problems and rational linear-dependence conclusions.

Trap watch. First reduce to a real basis; then solve the linear system over $\mathbb{Q}$.

Exam checks.

- Convert geometry with rational distances to linear algebra through the Gram matrix.

Occurrence count. 7 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1995-D2-P1;2002-D2-P6;2006-D2-P4;2008-D2-P6;2013-D2-P3;2018-D2-P6;2021-D2-P8

11.3.16 LA-16. Anticommuting and rank constraints for special matrices

Domain and bucket. Linear algebra / matrix theory; LA special rank bounds. **Literature anchors.** [19, 18, 6].

Prior compact node statement. Anticommuting idempotents or square-zero maps impose rank and dimension constraints. A common proof decomposes space into images/kernels or uses trace/rank averaging.

Exact facts to know.

1. If $A_i^2 = 0$, then $\operatorname{im} A_i \subseteq \ker A_i$ and $\operatorname{rank} A_i \leq n/2$.
2. If square-zero maps commute and $A_1 \cdots A_k v \neq 0$, then every A_i acts like adding the label i in the subset-vector construction.
3. If $AB = -BA$ and the field has characteristic not 2, traces of odd products often vanish and eigenvalues may occur in sign pairs under extra invertibility assumptions.
4. Rank constraints usually come from nested image-kernel inclusions or from constructing many independent vectors by applying products of the A_i.
5. For nilpotent or square-zero products, order matters unless commutativity is explicitly given.

Compact proof cores.

1. The rank bound follows immediately from image contained in kernel.
2. Independence of subset-product vectors is proved by applying complementary products to kill all later vectors in a size-compatible ordering.
3. Anticommutation spectral pairing follows from $AB = -BA$: if $Av = \lambda v$ and $Bv \neq 0$, then $A(Bv) = -\lambda Bv$.

How it appears in IMC work. **Use.** Use when many idempotents, nilpotents, or anticommuting matrices are given and a rank bound is requested.

Trap watch. Pairwise anticommutation is not pairwise orthogonality; translate the algebraic condition into image/kernel constraints.

Exam checks.

- This is a low-frequency but high-value node for problems asking for minimal matrix size.

Occurrence count. 0 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances.

11.4 Real analysis / calculus

Foundation checklist. The real-analysis layer uses compactness, continuity, connectedness, differentiability, Taylor expansion, convexity, integration, series, and asymptotics. A valid solution must state where compactness is used, justify endpoint limits, specify differentiability hypotheses before differentiating, and control tails when passing from finite sums to infinite sums or integrals.

11.4.1 RA-01. Compactness, extreme values, and closed-bounded reduction

Domain and bucket. Real analysis / calculus; RA compactness. **Literature anchors.** [28, 27, 17].

Prior compact node statement. A continuous function on a compact set attains its minimum and maximum. For unbounded domains, first show extrema cannot occur outside a compact sublevel region.

Exact facts to know.

1. Heine-Borel: a subset of $\mathbb{R}^n$ is compact iff it is closed and bounded.
2. Extreme value theorem: a continuous real-valued function on a compact set attains its maximum and minimum.
3. Bolzano-Weierstrass: every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.
4. Closed subsets contain limits of convergent sequences from the set; compactness plus continuity often turns an infimum into a minimum.
5. To prove an extremum is not outside a compact region, first obtain a uniform bound outside the region that is worse than a known interior value.

Compact proof cores.

1. Extreme values follow because continuous images of compact sets are compact subsets of $\mathbb{R}$, hence closed and bounded.
2. Heine-Borel is usually used as a black-box theorem in IMC solutions.
3. Compactness reduction is proved by bounding the tail region and then applying the extreme value theorem on the remaining compact set.

How it appears in IMC work. **Use.** Use in multivariable extrema, closed-set nearest-point arguments, convex-body problems, and existence of maximizers/minimizers.

Trap watch. Compactness requires closed and bounded in finite-dimensional Euclidean spaces; infinite-dimensional Hilbert spaces behave differently.

Exam checks.

- When a function is defined on a noncompact set, first isolate a compact sublevel or superlevel region.

Occurrence count. 37 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D2-P2;1995-D1-P6;1995-D2-P4;1995-D2-P6;1996-D1-P6;1996-D2-P4;1998-D2-P2;2000-D1-P1;2000-D2-P2;2001-D1-P3;2001-D2-P6;2002-D1-P4;2002-D2-P6;2003-D2-P3;2004-D1-P3;2005-D2-P1;2005-D2-P5;2006-D2-P3;2007-D2-P3;2008-D1-P1;2011-D1-P1;2013-D1-P3;2013-D1-P4;2014-D1-P2;2014-D2-P3;2014-D2-P5;2015-D2-P6;2015-D2-P7;2015-D2-P9;2016-D2-P6;2016-D2-P7;2018-D1-P1;2018-D1-P4;2018-D2-P10;2019-D1-P1;2019-D1-P3;2020-D2-P8

11.4.2 RA-02. Intermediate value, connectedness, density, and continuity transfer

Domain and bucket. Real analysis / calculus; RA continuity. **Literature anchors.** [28, 27, 17].

Prior compact node statement. Continuous images of intervals are intervals; continuous functions preserve limits. Dense subsets such as $\mathbb{Q}$ determine continuous functions on $\mathbb{R}$ when the relevant equality or inequality is closed under limits.

Exact facts to know.

1. Intermediate value theorem: if f is continuous on $[a, b]$, then every value between $f(a)$ and $f(b)$ is attained.
2. Continuous images of connected sets are connected; the connected subsets of $\mathbb{R}$ are intervals.
3. If two continuous functions agree on a dense subset of a domain, they agree everywhere.
4. If h is continuous and $h(q) \geq 0$ for all rational q, then $h(x) \geq 0$ for all real x.
5. A continuous real-valued function with only rational values on an interval is constant, because its image is connected and contained in the totally disconnected set $\mathbb{Q}$.

Compact proof cores.

1. IVT is exactly connectedness of $[a, b]$ plus continuity.
2. Dense transfer: choose rational $q_n \to x$ and pass to the limit by continuity.
3. The rational-valued continuous-image claim follows because the only connected subsets of $\mathbb{Q}$ are singletons.

How it appears in IMC work. **Use.** Use for interval image problems, monotone/continuous comparisons on rationals, and connectedness of constraint sets.

Trap watch. A monotone function need not be continuous unless surjectivity or no jumps is proved.

Exam checks.

- Density arguments require continuity of the function being passed to the limit, not merely monotonicity.

Occurrence count. 44 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D2-P2;1995-D1-P2;1995-D1-P6;1995-D2-P4;1995-D2-P6;1996-D1-P6;1996-D2-P1;1996-D2-P4;1998-D1-P4;1998-D1-P6;1998-D2-P5;2000-D2-P2;2002-D1-P4;2003-D2-P3;2004-D1-P3;2005-D1-P3;2006-D1-P1;2006-D1-P5;2006-D2-P2;2007-D2-P1;2007-D2-P3;2008-D1-P1;2009-D1-P1;2009-D1-P3;2011-D1-P1;2012-D1-P4;2013-D1-P2;2013-D1-P4;2014-D2-P3;2015-D2-P6;2016-D1-P1;2016-D2-P6;2016-D2-P7;2017-D2-P6;2017-D2-P9;2018-D1-P4;2019-D1-P1;2019-D2-P6;2020-D2-P5;2021-D1-P3;2021-D1-P4;2023-D1-P1;2023-D2-P7;2023-D2-P8

11.4.3 RA-03. Monotone functions, fixed points, and iteration on intervals

Domain and bucket. Real analysis / calculus; RA monotonicity/fixed points. **Literature anchors.** [28, 27, 17].

Prior compact node statement. A nondecreasing self-map of a compact interval has a fixed point. For continuous interval maps, convergence of iterates can often be characterized by the vanishing of increments plus cluster-point analysis.

Exact facts to know.

1. A monotone function on an interval has one-sided limits at every point.
2. If $f : [0, 1] \to [0, 1]$ is continuous, then it has a fixed point: apply IVT to $f(x) - x$.
3. If $f : [0, 1] \to [0, 1]$ is nondecreasing, a fixed point exists even without continuity, by a supremum argument.
4. Monotone bounded sequences converge.
5. For iterates $x_{n+1} = f(x_n)$, convergence implies $x_{n+1} - x_n \to 0$; conversely, with continuity and interval constraints, cluster-point separation can force convergence in several IMC problems.

Compact proof cores.

1. For the monotone fixed point, let $A = \{x : f(x) > x\}$ and study $\sup A$.
2. Monotone sequence convergence follows from completeness of $\mathbb{R}$ by taking the supremum or infimum.
3. Orbit closure arguments use compactness to obtain cluster points and continuity to map cluster points forward.

How it appears in IMC work. **Use.** Use for monotone maps, closed invariant sets, fixed points, and iteration sequences.

Trap watch. A decreasing map may fail to have a fixed point if discontinuous. Iteration convergence needs cluster-point control.

Exam checks.

- Decreasing maps may fail the fixed-point statement without continuity; always check direction.

Occurrence count. 15 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D2-P6;1995-D1-P4;1998-D2-P3;1999-D2-P4;2000-D1-P1;2000-D1-P4;2000-D2-P2;2001-D2-P2;2004-D1-P5;2004-D2-P2;2007-D2-P3;2010-D1-P1;2011-D1-P4;2015-D2-P6;2019-D1-P1

CORRECTION MATH-027 — legacy statement superseded; click for canonical replacement.

11.4.4 RA-04. Rolle, mean-value theorem, and divided differences

Domain and bucket. Real analysis / calculus; RA Rolle/MVT. **Literature anchors.** [28, 27, 17].

Prior compact node statement. Rolle's theorem and the mean-value theorem convert repeated zeros or equal endpoint values into derivative zeros. Divided differences satisfy $f[x_0, \dots, x_n] = f^{(n)}(\xi)/n!$ under differentiability.

Exact facts to know.

1. Rolle: if f is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then $f'(c) = 0$ for some $c \in (a, b)$.
2. Mean value theorem: $f(b) - f(a) = f'(c)(b - a)$ for some $c \in (a, b)$.
3. Cauchy mean value theorem: for differentiable f, g with $g' \neq 0$, $\frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(c)}{g'(c)}$.
4. Divided differences with repeated nodes satisfy $f[x_0, \dots, x_n] = f^{(n)}(\xi)/n!$ under the usual differentiability assumptions.
5. Repeated Rolle is the default way to turn many zeros or interpolation conditions into a high-derivative conclusion.

Compact proof cores.

1. Rolle follows from the extreme value theorem and Fermatś theorem at an interior extremum.
2. MVT follows by applying Rolle to $f(x) - \ell(x)$ where ℓ is the secant line.
3. Divided-difference form follows by applying Rolle repeatedly to the interpolation error.

How it appears in IMC work. **Use.** Use whenever many zeros are given, a derivative value must exist, or endpoint data must imply a higher-derivative identity.

Trap watch. Check differentiability order and that all points lie in the open interval where the conclusion is asserted.

Exam checks.

- When a problem has $n + 1$ zeros or endpoint data, immediately try repeated Rolle or interpolation.

Occurrence count. 33 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D1-P2;1994-D1-P6;1994-D2-P1;1994-D2-P3;1995-D1-P3;1996-D1-P5;1996-D2-P5;1996-D2-P6;1997-D2-P1;1997-D2-P3;1998-D1-P4;1999-D1-P6;2000-D1-P2;2000-D1-P6;2000-D2-P5;2001-D1-P6;2001-D2-P6;2002-D1-P2;2002-D1-P5;2005-D1-P3;2005-D1-P5;2005-D2-P4;2006-D1-P5;2006-D1-P6;2007-D2-P6;2009-D2-P2;2010-D1-P1;2012-D1-P4;2013-D1-P2;2017-D1-P5;2023-D1-P1;2023-D1-P2;2023-D2-P7

11.4.5 RA-05. Taylor expansion, local asymptotics, and little-o bookkeeping

Domain and bucket. Real analysis / calculus; RA Taylor. **Literature anchors.** [28, 27, 17].

Prior compact node statement. Near a point, C^k functions can be expanded with a controlled remainder. Competition solutions often need only the first nonzero term and the sign/size of the remainder.

Exact facts to know.

1. Taylor with Peano remainder: if f is k times differentiable at a, then $f(a+h) = \sum_{j=0}^{k} f^{(j)}(a)h^j/j! + o(h^k)$ under the standard hypotheses.
2. Taylor with Lagrange remainder: if $f \in C^{k+1}$ near a, then the remainder is $f^{(k+1)}(\xi)h^{k+1}/(k+1)!$.
3. If $f(a) = f'(a) = 0$ and $f''(a) > 0$, then locally $f(a + h) = \frac{1}{2}f''(a)h^2 + o(h^2)$ and f is nonnegative near a if the statement assumes global nonnegativity.
4. Little-o algebra: $o(h^r) + o(h^r) = o(h^r)$ and $h^s o(h^r) = o(h^{r+s})$.
5. Asymptotic comparison after a substitution must track the sign and domain of the small parameter.

Compact proof cores.

1. Taylor formula follows by repeated integration or by applying Cauchy MVT to the remainder.
2. Quadratic local behavior follows by substituting the Taylor expansion with zero constant and linear terms.
3. Little-o rules follow directly from dividing by the target power and taking limits.

How it appears in IMC work. **Use.** Use for local boundedness, sharp constants near equality, ratio limits, and endpoint asymptotics.

Trap watch. Do not expand beyond available smoothness; distinguish $O(x^k)$ from $o(x^k)$.

Exam checks.

- For local blow-up or boundedness, identify the first nonzero term in numerator and denominator.

Occurrence count. 50 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D1-P2;1994-D1-P5;1994-D1-P6;1994-D2-P1;1994-D2-P3;1994-D2-P4;1994-D2-P6;1995-D1-P3;1995-D1-P6;1995-D2-P2;1996-D1-P5;1996-D2-P5;1996-D2-P6;1997-D1-P1;1997-D2-P1;1997-D2-P3;1998-D1-P4;1999-D1-P6;2000-D1-P2;2000-D2-P5;2001-D1-P3;2001-D1-P6;2001-D2-P6;2002-D1-P2;2004-D1-P6;2005-D1-P3;2005-D1-P5;2005-D2-P4;2006-D1-P5;2006-D1-P6;2007-D2-P6;2009-D2-P2;2010-D1-P1;2010-D1-P3;2010-D1-P5;2010-D2-P5;2012-D1-P4;2013-D1-P2;2014-D1-P2;2015-D2-P7;2017-D1-P5;2017-D2-P6;2017-D2-P9;2018-D2-P10;2020-D2-P8;2021-D1-P4;2023-D1-P1;2023-D1-P2;2023-D2-P7;2024-D1-P2

11.4.6 RA-06. Auxiliary functions, integrating factors, and differential inequalities

Domain and bucket. Real analysis / calculus; RA differential inequalities. **Literature anchors.** [28, 27, 17].

Prior compact node statement. A differential inequality is often solved by choosing an auxiliary expression whose derivative has a fixed sign, such as $\arctan f + x$, $e^{-\lambda x} f(x)$, or $f' + xf$.

Exact facts to know.

1. For a first-order linear differential inequality $h'(x) + a(x)h(x) \geq b(x)$, multiply by the integrating factor $e^{\int a}$ to obtain $(he^{\int a})' \geq be^{\int a}$.
2. If $|f'| \leq \lambda |f|$ and $f(a) = 0$, then $f \equiv 0$ on the interval; this is the uniqueness mechanism behind Gronwall-type arguments.
3. Auxiliary functions are chosen so that the desired expression is the derivative: e.g. $g = \frac{1}{2}f^2 + f'$ makes $g' = ff' + f''$.
4. For inequalities involving $f' + f^2$, use transformations like $\arctan f$ because $(\arctan f)' = f'/(1 + f^2)$.
5. For higher-order linear inequalities, factor the differential operator when possible, e.g. $D^2 - 5D + 6 = (D - 2)(D - 3)$.

Compact proof cores.

1. Integrating-factor proof is one line: differentiate he^A where $A' = a$.
2. For $|f'| \leq \lambda |f|$, on any subinterval where $f > 0$ consider $\log f - \lambda x$; endpoint contradiction gives zero.
3. Operator factorization reduces the inequality to two successive first-order inequalities.

How it appears in IMC work. **Use.** Use for uniqueness, lower/upper bounds, exponential comparison, and ODE-like inequalities.

Trap watch. If taking logarithms, first prove the function has the required sign on the interval.

Exam checks.

- When a derivative inequality looks artificial, search for the exact derivative that generated it.

Occurrence count. 62 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D1-P2;1994-D2-P1;1994-D2-P3;1994-D2-P6;1995-D1-P2;1995-D1-P4;1996-D1-P2;1998-D1-P3;1998-D1-P4;1998-D1-P6;1999-D1-P2;1999-D1-P4;1999-D2-P3;1999-D2-P4;2000-D1-P2;2000-D1-P4;2000-D1-P6;2000-D2-P2;2001-D1-P3;2001-D1-P6;2003-D1-P1;2003-D2-P2;2004-D1-P5;2004-D2-P5;2005-D1-P3;2005-D1-P5;2006-D1-P1;2006-D1-P5;2006-D2-P3;2007-D2-P3;2009-D2-P2;2010-D1-P1;2010-D2-P1;2011-D2-P1;2012-D1-P4;2012-D2-P2;2013-D1-P2;2013-D1-P4;2013-D2-P3;2014-D1-P2;2014-D2-P2;2014-D2-P3;2014-D2-P5;2015-D1-P4;2015-D2-P6;2015-D2-P7;2016-D2-P6;2016-D2-P7;2017-D1-P2;2017-D2-P6;2018-D1-P1;2018-D1-P4;2018-D2-P7;2018-D2-P10;2019-D1-P1;2019-D1-P3;2020-D2-P5;2020-D2-P8;2021-D1-P3;2021-D1-P4;2023-D1-P1;2024-D1-P2

11.4.7 RA-07. Convexity, concavity, Jensen, and tangent/chord inequalities

Domain and bucket. Real analysis / calculus; RA convexity. **Literature anchors.** [17, 26, 9].

Prior compact node statement. For convex f, tangent lines lie below the graph and Jensen gives $f(\sum \lambda_i x_i) \leq \sum \lambda_i f(x_i)$. For concave functions the inequalities reverse. Equality cases are usually decisive.

Exact facts to know.

1. Convexity: $f(tx+(1-t)y) \le tf(x)+(1-t)f(y)$ for $t \in [0,1]$. Concavity reverses the inequality.
2. For differentiable convex f, tangent lines support the graph: $f(y) \ge f(x)+f'(x)(y-x)$.
3. Jensen: for convex f, $f(\sum \lambda_i x_i) \le \sum \lambda_i f(x_i)$ when $\lambda_i \ge 0$ and $\sum \lambda_i = 1$. Reverse for concave f.
4. For twice differentiable functions on an interval, $f'' \ge 0$ implies convexity and $f'' \le 0$ implies concavity.
5. Strict convexity gives equality in Jensen only when all relevant points are equal or when weights vanish.

Compact proof cores.

1. Jensen follows by induction from the two-point definition of convexity.
2. The tangent-line inequality follows from monotonicity of slopes of a convex function.
3. The f'' criterion follows because f' is monotone exactly when secant slopes increase.

How it appears in IMC work. **Use.** Use for inequalities, sharp constants, integral estimates, chord comparisons, and geometric convexity.

Trap watch. Check the interval of convexity and strictness conditions for equality.

Exam checks.

- Always state whether you need convexity of f, $\log f$, $|x|^p$, or a transformed function.

Occurrence count. 179 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D1-P1;1994-D1-P2;1994-D1-P4;1994-D1-P5;1994-D1-P6;1994-D2-P1;1994-D2-P2;1994-D2-P5;1994-D2-P6;1995-D1-P2;1995-D1-P3;1995-D1-P4;1995-D1-P6;1995-D2-P2;1995-D2-P4;1995-D2-P5;1996-D1-P2;1996-D1-P4;1996-D1-P5;1996-D2-P2;1996-D2-P4;1996-D2-P5;1996-D2-P6;1997-D1-P1;1997-D1-P2;1997-D1-P4;1997-D1-P5;1997-D2-P1;1997-D2-P3;1998-D1-P2;1998-D1-P3;1998-D1-P4;1998-D1-P5;1998-D1-P6;1998-D2-P1;1998-D2-P2;1998-D2-P3;1998-D2-P4;1998-D2-P5;1998-D2-P6;1999-D1-P2;1999-D1-P3;1999-D1-P4;1999-D1-P6;1999-D2-P3;1999-D2-P4;1999-D2-P6;2000-D1-P1;2000-D1-P4;2000-D2-P1;2000-D2-P2;2000-D2-P3;2000-D2-P4;2000-D2-P5;2001-D1-P4;2001-D1-P6;2001-D2-P1;2001-D2-P2;2001-D2-P3;2001-D2-P5;2001-D2-P6;2002-D1-P2;2002-D1-P3;2002-D1-P6;2002-D2-P3;2002-D2-P6;2003-D1-P4;2003-D1-P5;2003-D1-P6;2003-D2-P2;2003-D2-P3;2003-D2-P5;2003-D2-P6;2004-D1-P1;2004-D1-P2;2004-D1-P3;2004-D1-P5;2004-D1-P6;2004-D2-P2;2004-D2-P3;2004-D2-P4;2004-D2-P5;2004-D2-P6;2005-D1-P1;2005-D1-P2;2005-D1-P3;2005-D1-P5;2005-D1-P6;2005-D2-P1;2005-D2-P5;2006-D1-P1;2006-D1-P5;2006-D1-P6;2006-D2-P1;2006-D2-P2;2006-D2-P3;2006-D2-P4;2007-D1-P2;2007-D1-P3;2007-D1-P5;2007-D1-P6;2007-D2-P3;2007-D2-P6;2008-D1-P1;2008-D1-P2;2008-D1-P4;2008-D1-P5;2008-D2-P3;2008-D2-P6;2009-D1-P1;2009-D1-P4;2009-D2-P1;2009-D2-P2;2010-D1-P1;2010-D1-P2;2010-D1-P3;2010-D1-P4;2010-D1-P5;2010-D2-P1;2010-D2-P2;2010-D2-P3;2010-D2-P4;2011-D1-P1;2011-D1-P4;2011-D2-P1;2011-D2-P3;2011-D2-P5;2012-D1-P4;2012-D2-P2;2012-D2-P4;2012-D2-P5;2013-D1-P3;2013-D1-P4;2013-D2-P3;2014-D2-P2;2014-D2-P5;2015-D1-P1;2015-D1-P2;2015-D1-P3;2015-D1-P5;2015-D2-P10;2016-D1-P2;2016-D1-P3;2016-D1-P4;2016-D1-P5;2016-D2-P6;2016-D2-P7;2016-D2-P9;2016-D2-P10;2017-D1-P2;2017-D1-P3;2017-D1-P5;2018-D1-P5;2018-D2-P7;2018-D2-P8;2019-D1-P3;2019-D1-P4;2019-D2-P10;2020-D1-P3;2020-D1-P4;2020-D2-P5;2020-D2-P7;2021-D2-P5;2021-D2-P7;2022-D1-P1;2022-D1-P2;2022-D1-P4;2022-D2-P7;2022-D2-P8;2023-D1-P5;2023-D2-P8;2023-D2-P9;2024-D1-P2;2024-D1-P4;2024-D1-P5;2024-D2-P6;2024-D2-P8;2024-D2-P9;2024-D2-P10

11.4.8 RA-08. Integral inequalities, integration by parts, and weighted estimates

Domain and bucket. Real analysis / calculus; RA integral inequalities. **Literature anchors.** [28, 27, 17].

Prior compact node statement. Integration by parts transfers derivatives between factors; nonnegative integrands and pointwise bounds integrate to global inequalities. Introducing primitives often converts a cubic or square integral into a boundary term plus an error.

Exact facts to know.

1. Integration by parts: $\int_a^b uv' = uv|_a^b - \int_a^b u'v$.
2. If $0 \le h \le k$, then $\int h \le \int k$ on the same domain.
3. Cauchy-Schwarz integral form: $\left(\int fg\right)^2 \le \int f^2 \int g^2$.
4. Holder: $\int |fg| \le \|f\|_p \|g\|_q$ for $1/p + 1/q = 1$.
5. Weighted estimates often introduce primitives $F(x) = \int_a^x f$ or tails $F(x) = \int_x^b f$ to convert a product into a boundary term plus an error.

Compact proof cores.

1. Integration by parts follows by integrating $(uv)' = u'v + uv'$.

2. Cauchy follows by nonnegativity of $\int (f - tg)^2$ as a quadratic in t.
3. Weighted primitive tricks are verified by differentiating the chosen primitive and checking boundary values.

How it appears in IMC work. **Use.** Use in L^p inequalities, integral constraints, monotone integrals, and estimates of functions by their derivatives.

Trap watch. Boundary terms are often the whole point; compute their signs before discarding them.

Exam checks.

- Boundary signs are decisive; do not discard boundary terms automatically.

Occurrence count. 199 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D1-P2;1994-D1-P3;1994-D1-P4;1994-D1-P5;1994-D1-P6;1994-D2-P1;1994-D2-P2;1994-D2-P3;1994-D2-P5;1994-D2-P6;1995-D1-P2;1995-D1-P4;1995-D1-P5;1995-D1-P6;1995-D2-P2;1995-D2-P5;1995-D2-P6;1996-D1-P2;1996-D1-P4;1996-D1-P5;1996-D2-P1;1996-D2-P2;1996-D2-P4;1996-D2-P5;1996-D2-P6;1997-D1-P1;1997-D1-P2;1997-D1-P4;1997-D1-P5;1997-D1-P6;1997-D2-P1;1997-D2-P3;1998-D1-P2;1998-D1-P3;1998-D1-P4;1998-D1-P5;1998-D1-P6;1998-D2-P1;1998-D2-P2;1998-D2-P6;1999-D1-P2;1999-D1-P3;1999-D1-P4;1999-D1-P6;1999-D2-P3;1999-D2-P4;1999-D2-P5;1999-D2-P6;2000-D1-P1;2000-D1-P4;2000-D1-P6;2000-D2-P2;2000-D2-P4;2000-D2-P5;2000-D2-P6;2001-D1-P1;2001-D1-P3;2001-D2-P1;2001-D2-P2;2001-D2-P4;2001-D2-P5;2001-D2-P6;2002-D1-P2;2002-D1-P3;2002-D1-P4;2002-D1-P5;2002-D1-P6;2002-D2-P3;2002-D2-P4;2002-D2-P6;2003-D1-P1;2003-D1-P2;2003-D1-P3;2003-D1-P4;2003-D1-P5;2003-D1-P6;2003-D2-P2;2003-D2-P5;2003-D2-P6;2004-D1-P2;2004-D1-P3;2004-D1-P4;2004-D1-P5;2004-D1-P6;2004-D2-P2;2004-D2-P3;2004-D2-P4;2004-D2-P5;2004-D2-P6;2005-D1-P1;2005-D1-P3;2005-D1-P4;2005-D1-P5;2005-D2-P1;2005-D2-P2;2005-D2-P3;2005-D2-P5;2006-D1-P1;2006-D1-P3;2006-D1-P4;2006-D1-P5;2006-D1-P6;2006-D2-P1;2006-D2-P3;2006-D2-P4;2007-D1-P2;2007-D1-P5;2007-D1-P6;2007-D2-P2;2007-D2-P3;2007-D2-P6;2008-D1-P1;2008-D1-P2;2008-D1-P4;2008-D2-P3;2008-D2-P6;2009-D1-P1;2009-D1-P4;2009-D1-P5;2009-D2-P1;2009-D2-P2;2009-D2-P3;2009-D2-P5;2010-D1-P1;2010-D1-P2;2010-D1-P3;2010-D1-P4;2010-D2-P1;2010-D2-P2;2010-D2-P3;2011-D1-P1;2011-D1-P5;2011-D2-P1;2011-D2-P2;2011-D2-P3;2011-D2-P4;2011-D2-P5;2012-D1-P1;2012-D1-P4;2012-D2-P2;2012-D2-P4;2013-D1-P2;2013-D1-P3;2013-D1-P4;2013-D1-P5;2013-D2-P3;2013-D2-P4;2014-D1-P2;2014-D1-P4;2014-D2-P2;2014-D2-P3;2014-D2-P5;2015-D1-P2;2015-D1-P3;2015-D1-P5;2015-D2-P6;2015-D2-P7;2015-D2-P8;2015-D2-P10;2016-D1-P2;2016-D1-P3;2016-D1-P4;2016-D2-P6;2016-D2-P7;2016-D2-P9;2016-D2-P10;2017-D1-P2;2017-D1-P3;2017-D2-P6;2018-D1-P1;2018-D1-P4;2018-D1-P5;2018-D2-P7;2018-D2-P8;2018-D2-P10;2019-D1-P1;2019-D1-P3;2019-D1-P4;2019-D2-P6;2019-D2-P7;2019-D2-P8;2020-D1-P3;2020-D1-P4;2020-D2-P5;2020-D2-P8;2021-D1-P3;2021-D1-P4;2021-D2-P5;2022-D1-P1;2022-D1-P2;2022-D1-P4;2023-D1-P1;2023-D1-P5;2023-D2-P8;2024-D1-P2;2024-D1-P4;2024-D2-P6;2024-D2-P8;2024-D2-P10

11.4.9 RA-09. Fubini/Tonelli and change of variables

Domain and bucket. Real analysis / calculus; RA Fubini. **Literature anchors.** [28, 27, 17].

Prior compact node statement. For nonnegative or absolutely integrable functions, interchange the order of integration and use substitutions to convert regions or kernels into simpler forms.

Exact facts to know.

1. Tonelli: nonnegative measurable functions may be integrated in either order, including infinite values.
2. Fubini: absolutely integrable functions allow interchange of integration order and give finite equal iterated integrals.
3. For continuous functions on compact rectangles, Fubini is directly valid.
4. Change of variables in one dimension: $\int_{\phi(a)}^{\phi(b)} f(u)du = \int_a^b f(\phi(t))\phi'(t)dt$ for monotone C^1 substitutions, with orientation handled by limits.
5. For triangular regions, switch order by rewriting inequalities carefully; many IMC nested averages become kernels after this step.

Compact proof cores.

1. The compact continuous case follows from uniform approximation by step functions or standard Riemann integration theorems.
2. Change-of-variable formula is the chain rule integrated.
3. Tonelli/Fubini justifications are automatic for nonnegative integrands or bounded continuous integrands on finite domains.

How it appears in IMC work. **Use.** Use for nested integrals, averaging operators, double integrals over the unit square, and kernel formulas.

Trap watch. Justify interchanging integrals or sums when signs are not nonnegative.

Exam checks.

- If signs change on an infinite domain, show absolute integrability or split into controlled pieces.

Occurrence count. 39 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D2-P1;1994-D2-P3;1995-D1-P2;1995-D2-P5;1996-D1-P2;1996-D1-P4;1996-D2-P1;1997-D1-P1;1997-D1-P2;1997-D1-P4;1998-D1-P3;1998-D1-P6;1999-D1-P4;1999-D2-P3;1999-D2-P4;2001-D1-P1;2002-D1-P3;2003-D1-P1;2003-D2-P6;2004-D1-P2;2004-D2-P5;2005-D2-P1;2007-D2-P2;2008-D1-P4;2010-D1-P2;2010-D2-P1;2010-D2-P2;2011-D2-P1;2011-D2-P3;2012-D1-P1;2012-D2-P2;2013-D1-P5;2013-D2-P4;2014-D1-P4;2015-D2-P8;2018-D1-P1;2018-D2-P7;2019-D2-P6;2019-D2-P7

11.4.10 RA-10. Riemann sums and sum-integral asymptotics

Domain and bucket. Real analysis / calculus; RA Riemann sums. **Literature anchors.** [28, 27, 17].

Prior compact node statement. If $h \to 0$, sums of the form $h\sum f(nh)$ approximate $\int f$. For infinite ranges, split into a compact main part and a tail controlled uniformly.

Exact facts to know.

1. Finite Riemann sums: if f is Riemann integrable on $[a,b]$, then $\frac{b-a}{n}\sum_{k=1}^{n} f(a+k(b-a)/n) \to \int_a^b f$.
2. For $h \to 0^+$, sums $h\sum_{n\geq 1} f(nh)$ approximate $\int_0^\infty f(t)dt$ when f is integrable and tails are uniformly controlled.
3. Logarithmic sums often use k/n substitutions or $h = -\log t$ substitutions.
4. Euler-Maclaurin is rarely necessary; monotonicity tail bounds usually suffice.
5. Tail splitting principle: choose M large, control the tail uniformly beyond M, then pass to the limit on $[0,M]$.

Compact proof cores.

1. Finite Riemann sums are the definition/characterization of Riemann integrability.
2. Infinite-range result follows by first truncating to $[0,M]$, applying finite Riemann sums, then bounding $\sum_{nh>M}$ by an integral tail.
3. Monotone tail comparison uses rectangles under or over the graph.

How it appears in IMC work. **Use.** Use for limits of sums, logarithmic sums, periodic averaging, and asymptotic counting.

Trap watch. Tail control is the common missing step; do not use finite-interval Riemann sums on an infinite domain without a tail estimate.

Exam checks.

- Never ignore singular endpoints; isolate them or show they are integrable.

Occurrence count. 73 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D1-P2;1994-D1-P5;1994-D2-P6;1995-D1-P2;1995-D1-P3;1995-D2-P2;1995-D2-P5;1995-D2-P6;1996-D1-P4;1996-D1-P5;1996-D2-P1;1996-D2-P5;1997-D1-P1;1997-D1-P2;1997-D2-P1;1997-D2-P3;1998-D1-P3;1999-D1-P2;1999-D2-P3;2000-D1-P4;2000-D1-P6;2000-D2-P5;2001-D1-P3;2001-D1-P6;2001-D2-P2;2001-D2-P5;2002-D1-P4;2002-D1-P6;2003-D1-P1;2003-D1-P3;2003-D1-P5;2003-D2-P2;2003-D2-P5;2003-D2-P6;2004-D2-P4;2004-D2-P5;2005-D1-P5;2006-D1-P1;2006-D1-P4;2006-D2-P3;2007-D2-P6;2009-D1-P1;2010-D1-P2;2010-D1-P3;2010-D2-P1;2010-D2-P5;2011-D1-P1;2011-D2-P1;2013-D1-P2;2013-D1-P4;2014-D1-P2;2014-D1-P4;2014-D2-P2;2014-D2-P3;2014-D2-P5;2015-D2-P6;2015-D2-P7;2016-D2-P6;2017-D1-P2;2017-D2-P6;2017-D2-P9;2018-D1-P1;2018-D1-P4;2018-D2-P10;2019-D1-P1;2019-D1-P3;2020-D2-P5;2020-D2-P8;2021-D1-P3;2021-D1-P4;2023-D1-P1;2024-D1-P2;2024-D2-P8

11.4.11 RA-11. Series convergence, rearrangements, and alternating/Dirichlet tools

Domain and bucket. Real analysis / calculus; RA series. **Literature anchors.** [28, 27, 17].

Prior compact node statement. Convergence can be altered by nonabsolute rearrangements. Alternating and Dirichlet-type arguments reduce convergence to decay plus bounded variation or controlled differences.

Exact facts to know.

1. Absolute convergence implies convergence and is invariant under rearrangement.
2. Conditional convergence may be destroyed or changed by rearrangement.
3. Alternating test: if $a_n \downarrow 0$, then $\sum(-1)^n a_n$ converges.
4. Dirichlet test: if partial sums of a_n are bounded and $b_n \downarrow 0$, then $\sum a_n b_n$ converges.
5. If $\sum |a_{n+1} - a_n| < \infty$ and $a_n \to 0$, then the alternating pair grouping $\sum(a_{2n-1} - a_{2n})$ converges absolutely.

Compact proof cores.

1. Alternating test follows because even and odd partial sums are monotone and squeeze each other.
2. Dirichlet follows by summation by parts.
3. Rearrangement sensitivity is demonstrated by block permutations of a conditionally convergent series.

How it appears in IMC work. **Use.** Use for rearranged series, alternating trigonometric series, Carleman-type inequalities, and positive series estimates.

Trap watch. Conditional convergence is not preserved by arbitrary permutations.

Exam checks.

- For a permuted series, write explicit partial sums at block endpoints.

Occurrence count. 67 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D1-P3;1994-D2-P1;1994-D2-P3;1994-D2-P5;1995-D1-P5;1995-D1-P6;1995-D2-P5;1995-D2-P6;1996-D1-P2;1996-D1-P4;1996-D1-P5;1996-D2-P1;1996-D2-P5;1996-D2-P6;1997-D1-P1;1997-D1-P2;1997-D1-P4;1997-D1-P5;1997-D1-P6;1997-D2-P3;1998-D1-P3;1998-D1-P6;1998-D2-P1;1999-D1-P2;1999-D1-P4;1999-D2-P3;1999-D2-P4;1999-D2-P5;2000-D2-P5;2001-D1-P1;2001-D2-P2;2002-D1-P2;2002-D1-P3;2002-D2-P4;2003-D1-P1;2003-D1-P3;2003-D2-P6;2004-D1-P2;2004-D1-P5;2005-D1-P1;2005-D2-P1;2005-D2-P2;2006-D1-P1;2006-D1-P4;2006-D1-P6;2006-D2-P4;2007-D2-P2;2008-D1-P2;2008-D1-P4;2010-D1-P2;2010-D2-P1;2010-D2-P2;2010-D2-P3;2011-D1-P1;2011-D2-P1;2011-D2-P3;2012-D1-P1;2012-D2-P2;2013-D1-P5;2013-D2-P4;2014-D1-P4;2015-D2-P8;2018-D1-P1;2018-D2-P7;2019-D2-P6;2019-D2-P7;2019-D2-P8

11.4.12 RA-12. Uniform continuity and periodic averaging

Domain and bucket. Real analysis / calculus; RA oscillatory averaging. **Literature anchors.** [28, 27, 17].

Prior compact node statement. Continuous functions on compact intervals are uniformly continuous. Products with rapidly oscillating periodic factors average to the product of means after partitioning into small intervals.

Exact facts to know.

1. Continuous functions on compact intervals are uniformly continuous.
2. If g has period b, then $\int_{(k-1)b/n}^{kb/n} g(nx)dx = \frac{1}{n}\int_0^b g(u)du$ after $u = nx - b(k-1)$.
3. Periodic averaging: $\int_0^b f(x)g(nx)dx \to \frac{1}{b}\int_0^b f \int_0^b g$ for continuous f and periodic continuous g.
4. The error is controlled by the modulus of continuity $\omega(f, b/n)\int_0^b |g|$.
5. If the mean of g is zero, the limit is zero.

Compact proof cores.

1. Partition $[0, b]$ into n intervals, replace f by its endpoint value on each interval, and use periodicity to compute each oscillatory integral.
2. Uniform continuity makes the replacement error tend to zero.
3. The endpoint sum then becomes a Riemann sum for $\int f$.

How it appears in IMC work. **Use.** Use for limits of $\int f(x)g(nx)\,dx$ and periodic integral averaging.

Trap watch. Keep the period and interval length consistent; normalize by the period correctly.

Exam checks.

- Normalize by the period b; missing the factor $1/b$ is the common error.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D1-P5

11.4.13 RA-13. L'Hospital, ratio limits, and growth comparisons

Domain and bucket. Real analysis / calculus; RA ratio limits. **Literature anchors.** [28, 27, 17].

Prior compact node statement. When numerator and denominator have compatible divergent or vanishing behavior, derivative ratios can determine function ratios. Growth comparisons often reduce to exponentials, logarithms, or powers.

Exact facts to know.

1. L'Hospital: for $0/0$ or ∞/∞ forms, if $g' \neq 0$ and $f'/g' \to L$ under the standard hypotheses, then $f/g \to L$.
2. Growth hierarchy: powers of x dominate powers of $\log x$; exponentials dominate powers.
3. For endpoint limits, Cauchy MVT can prove ratio estimates without directly invoking L'Hospital.
4. If numerator is bounded and denominator tends to infinity, the ratio tends to zero.
5. Asymptotic equivalence $f \sim g$ means $f/g \to 1$ and can be multiplied/divided when denominators stay nonzero eventually.

Compact proof cores.

1. L'Hospital follows from Cauchy MVT applied on intervals approaching the endpoint.
2. Growth hierarchy follows by repeated L'Hospital or standard exponential/log substitutions.
3. Equivalent functions preserve limits by writing $f = h(1 + o(1))$.

How it appears in IMC work. **Use.** Use for endpoint ratio limits, exponential weights, and asymptotic estimates.

Trap watch. L'Hospital requires the appropriate hypotheses; for inequalities, use comparison rather than equality of limits when needed.

Exam checks.

- Check the indeterminate form first; L'Hospital is not a symbolic simplifier.

Occurrence count. 33 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D1-P2;1994-D1-P5;1994-D2-P6;1995-D1-P3;1995-D1-P4;1995-D2-P2;1996-D1-P4;1996-D2-P1;1996-D2-P5;1997-D1-P1;1997-D1-P2;1997-D2-P1;2000-D1-P6;2001-D1-P3;2001-D1-P6;2001-D2-P5;2003-D1-P1;2003-D2-P2;2004-D1-P6;2004-D2-P5;2005-D1-P5;2010-D1-P2;2010-D1-P3;2011-D2-P3;2014-D1-P2;2015-D2-P7;2017-D2-P6;2017-D2-P9;2018-D2-P10;2020-D2-P8;2021-D1-P4;2024-D1-P2;2024-D2-P8

11.4.14 RA-14. Weierstrass approximation and polynomial approximation

Domain and bucket. Real analysis / calculus; RA approximation. **Literature anchors.** [28, 25].

Prior compact node statement. Every continuous function on a compact interval can be uniformly approximated by polynomials. Symmetry can be imposed by averaging $p(x)$ with $p(-x)$ to obtain even or odd approximants.

Exact facts to know.

1. Weierstrass approximation: every continuous function on $[a, b]$ is uniformly approximable by polynomials.
2. If f is odd on $[-1, 1]$ and p approximates f, then $q(x) = \frac{1}{2}(p(x) - p(-x))$ is an odd polynomial approximating f at least as well up to the same error bound.
3. Odd polynomials have only odd powers; even polynomials have only even powers.
4. Missing basis terms can sometimes be approximated by identities such as $x(1 - x^2)^n \to 0$ uniformly.
5. Chebyshev polynomials often realize extremal uniform norms under interpolation constraints.

CORRECTION MATH-028 — legacy statement superseded; click for canonical replacement.

Compact proof cores.

1. Symmetrization proof is direct from $f(-x) = -f(x)$ and the triangle inequality.
2. Uniform convergence of $x(1-x^2)^n$ to 0 is obtained by maximizing on $[0,1]$ or splitting near 1.
3. Chebyshev extremality follows from alternation: a lower-degree perturbation cannot reduce all alternating extrema.

How it appears in IMC work. **Use.** Use for uniform approximation by restricted polynomial families and Chebyshev-style extremal examples.

Trap watch. Approximate the symmetric part, then handle missing low-degree terms separately if the basis omits them.

Exam checks.

- Approximate the symmetry class, not an arbitrary polynomial, when the target has parity.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1995-D2-P4

11.4.15 RA-15. L2 spaces, orthonormal systems, and dominated convergence

Domain and bucket. Real analysis / calculus; RA functional analysis. **Literature anchors.** [27, 12].

Prior compact node statement. In L^2, orthonormality gives coefficient constraints and no pointwise-convergent bounded subsequence can contradict norm preservation when dominated convergence applies.

Exact facts to know.

1. In $L^2[a,b]$, $\langle f,g\rangle = \int_a^b fg$ and $\|f\|_2^2 = \int f^2$.
2. An orthonormal system satisfies $\langle f_m, f_n\rangle = \delta_{mn}$.
3. Bessel inequality: for orthonormal e_n, $\sum_{n=1}^{N} |\langle f, e_n\rangle|^2 \le \|f\|^2$.
4. Dominated convergence: if $f_n \to f$ pointwise and $|f_n| \le g$ with g integrable, then $\int f_n \to \int f$.
5. Uniform boundedness of continuous functions on a finite interval gives domination by a constant.

Compact proof cores.

1. Bessel follows by expanding $\|f - \sum_{n=1}^{N}\langle f, e_n\rangle e_n\|^2 \ge 0$.
2. Dominated convergence is a standard measure theorem; in IMC use it when a uniform bound and finite measure are explicit.
3. Orthogonality to a complete orthonormal system forces a function to be zero in L^2.

How it appears in IMC work. **Use.** Use for orthonormal continuous functions, Hilbert-space arguments, and bounded convergence of integrals.

Trap watch. Pointwise convergence plus uniform boundedness allows dominated convergence, but only on finite measure domains.

Exam checks.

- Do not conclude pointwise zero from L^2 zero except almost everywhere, unless continuity upgrades it.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1995-D2-P6

11.4.16 RA-16. Baire/category, dense sets, and pathological continuous functions

Domain and bucket. Real analysis / calculus; RA topology/pathology. **Literature anchors.** [28, 27, 17].

Prior compact node statement. Closed intervals cannot be countable unions of nowhere dense sets. Cantor-type constructions create uncountable closed sets with prescribed local behavior.

Exact facts to know.

1. A set is dense in an interval if every nonempty open subinterval intersects it.
2. Baire category theorem: a complete metric space cannot be a countable union of nowhere dense closed sets.
3. A continuous nowhere-monotone function has local extrema densely, by applying nonmonotonicity on both sides and the extreme value theorem.
4. Cantor-type constructions can provide uncountable closed nowhere-dense sets and continuous functions with prescribed oscillation patterns.
5. Closed countable orbit sets in compact intervals often contradict accumulation behavior unless finite.

Compact proof cores.

1. Density of local minima for nowhere-monotone functions follows by finding $p < q \le x \le r < s$ with $f(p) > f(q)$ and $f(r) < f(s)$, then minimizing on $[p, s]$.
2. Baire contradiction: a closed countable set with all points isolated cannot be a complete interval-like compact set.
3. Cantor constructions use uniform convergence of piecewise-linear functions.

How it appears in IMC work. **Use.** Use for dense local minima, uncountably many zero crossings, and existence/nonexistence of functions with many level sets.

Trap watch. Countability and uncountability arguments must distinguish isolated intervals from point sets.

Exam checks.

- When a set is claimed closed and infinite in a compact interval, look for accumulation points.

Occurrence count. 6 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1997-D2-P6;1998-D1-P4;1998-D2-P6;2000-D2-P2;2002-D1-P5;2003-D2-P5

11.4.17 RA-17. Sequence recurrences, monotone bounds, and generating functions

Domain and bucket. Real analysis / calculus; RA recurrences. **Literature anchors.** [28, 27, 17].

Prior compact node statement. Recursive sequences are attacked by monotone normalization, induction estimates, telescoping transformations, or generating functions $F(x) = \sum a_n x^n$.

Exact facts to know.

1. Monotone recursive sequences are controlled by fixed points and invariant intervals.
2. Generating function principle: if a_n satisfies a convolution recurrence, $A(x) = \sum a_n x^n$ often satisfies an algebraic or differential equation.
3. If $a_{n+1} = \frac{1}{n+1} \sum_{k=0}^{n} a_k/(n-k+2)$, differentiating the generating function transforms the recurrence into $A'(x) = A(x) \sum_{m \ge 0} x^m/(m+2)$.
4. Inductive upper/lower exponential bounds imply estimates for $\limsup |a_n|^{1/n}$.
5. Telescoping recurrences should be rewritten so consecutive terms cancel visibly.

Compact proof cores.

1. Generating-function derivation is justified inside the radius of convergence by absolute convergence or formal-power-series algebra.
2. Limsup bounds follow from $|a_n| \le Cq^n$ eventually and from subsequential lower estimates.
3. Monotone recursion convergence follows by bounding the sequence and identifying possible limits.

How it appears in IMC work. **Use.** Use for recurrence limits, coefficient sums, and nonlinear-looking sequences that become telescoping after substitution.

Trap watch. Check radius of convergence before manipulating generating functions; induction bounds need verified base cases.

Exam checks.

- Before solving a recurrence, compute several first values and look for a substitution.

Occurrence count. 22 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1995-D2-P2;1996-D1-P4;1998-D2-P6;2001-D2-P2;2002-D2-P3;2003-D1-P1;2003-D1-P5;2003-D1-P6;2003-D2-P6;2005-D1-P2;2007-D1-P4;2008-D2-P3;2010-D2-P2;2011-D2-P1;2012-D2-P2;2015-D1-P5;2017-D2-P8;2017-D2-P9;2018-D2-P7;2019-D1-P4;2021-D1-P3;2024-D2-P8

11.4.18 RA-18. Trigonometric and hyperbolic identities with monotonicity

Domain and bucket. Real analysis / calculus; RA trig. **Literature anchors.** [28, 27, 17].

Prior compact node statement. Use standard identities, product-to-sum formulas, monotonicity of $\sin x/x$, and hyperbolic recurrences such as $\cosh((m+1)t) = 2\cosh t\cosh(mt) - \cosh((m-1)t)$.

Exact facts to know.

1. Basic identities: $\sin 2x = 2\sin x\cos x$, $\cosh(a+b) = \cosh a\cosh b + \sinh a\sinh b$, and $\cosh((m+1)t) = 2\cosh t\cosh(mt) - \cosh((m-1)t)$.
2. $\sin x/x \to 1$ as $x \to 0$ and is decreasing on $(0,\pi)$.
3. Products of sines often benefit from pairing $\sin\theta$ and $\sin 2\theta$ or using AM-GM on trigonometric factors.
4. Hyperbolic identities convert rationality of $\cosh kt$ into algebraic recurrences.
5. Complex exponentials convert trigonometric sums/products into polynomial or roots-of-unity problems.

Compact proof cores.

1. The hyperbolic recurrence follows by adding formulas for $\cosh((m+1)t)$ and $\cosh((m-1)t)$.
2. Monotonicity of $\sin x/x$ follows from derivative sign $x\cos x - \sin x < 0$ on $(0,\pi)$.
3. Complex exponential identities follow from $e^{ix} = \cos x + i\sin x$.

How it appears in IMC work. **Use.** Use for trig integrals, zero-counting of trig sums, hyperbolic rationality, and trigonometric inequalities.

Trap watch. Track the interval: many monotonicity claims only hold on $(0,\pi/2)$ or $(0,\pi)$.

Exam checks.

- Check angle ranges before using concavity/monotonicity of sine or cosine.

Occurrence count. 14 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1995-D2-P5;1996-D1-P2;1996-D2-P2;1997-D2-P3;2000-D1-P4;2001-D2-P2;2001-D2-P6;2003-D2-P2;2004-D1-P3;2004-D2-P5;2006-D2-P3;2014-D2-P3;2015-D1-P4;2017-D1-P2

11.4.19 RA-19. Gradient/Lipschitz-gradient optimization and convex smoothness

Domain and bucket. Real analysis / calculus; RA smooth convexity. **Literature anchors.** [26, 9].

Prior compact node statement. For convex differentiable f with L-Lipschitz gradient, descent estimates give $\|\nabla f(x) - \nabla f(y)\|^2 \le L\langle \nabla f(x) - \nabla f(y), x - y\rangle$. Lipschitz gradients also control uniqueness of maximizers under geometric constraints.

Exact facts to know.

1. If f is convex and differentiable, then $\langle \nabla f(x) - \nabla f(y), x - y\rangle \ge 0$ (monotonicity of the gradient).
2. If ∇f is L-Lipschitz, then $f(y) \le f(x) + \langle \nabla f(x), y - x\rangle + \frac{L}{2}\|y - x\|^2$.
3. For convex f with L-Lipschitz gradient, the Baillon-Haddad/co-coercivity inequality holds: $\|\nabla f(x) - \nabla f(y)\|^2 \le L\langle \nabla f(x) - \nabla f(y), x - y\rangle$.
4. A differentiable function on a compact disk attains its maximum; if the gradient is nonzero in the interior, the maximum lies on the boundary.
5. Gradient Lipschitz bounds allow comparison of gradients inside balls: $\|\nabla f(u) - \nabla f(0)\| \le L\|u\|$.

Compact proof cores.

1. Gradient monotonicity follows by applying the first-order convex inequality in both directions and adding.
2. The descent lemma follows by integrating $\frac{d}{dt} f(x + t(y - x))$ and using Lipschitzness of ∇f.
3. Co-coercivity follows by applying the descent lemma to f after subtracting a tangent plane and minimizing along the negative gradient direction, then symmetrizing.

How it appears in IMC work. **Use.** Use in convex analysis, gradient inequalities, and maximum-location problems on disks or balls.

Trap watch. Gradient Lipschitzness is a global condition; local Hessian intuition must be converted into an inequality.

Exam checks.

- For maximum-uniqueness problems, compare boundary normal directions with the Lipschitz gradient ball.

Occurrence count. 3 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D2-P2;2002-D2-P6;2005-D2-P5

11.4.20 RA-20. Measure and probability-integral estimates

Domain and bucket. Real analysis / calculus; RA/probability bridge. **Literature anchors.** [28, 27, 17].

Prior compact node statement. Integral estimates can be interpreted probabilistically, and probability problems can often be turned into expectation or measure integrals. Symmetry and independence are the main reductions.

Exact facts to know.

1. Markov/Chebyshev-style integral estimates: if $h \geq 0$, then measure of $\{h > t\}$ is bounded by $t^{-1} \int h$.
2. Layer-cake formula: for nonnegative h, $\int h = \int_0^\infty \mu\{h > t\}\, dt$ under standard measure hypotheses.
3. Finite variation of a function implies differentiability almost everywhere, but IMC uses this mainly as a remark; direct covering arguments are preferred.
4. Covering intervals with total length less than the ambient interval cannot cover the whole interval.
5. If intervals have total length infinite, one can often place them to cover each point infinitely often by a constructive dense scheduling.

Compact proof cores.

1. Markov follows because $\int h \geq \int_{h>t} h \geq t\mu\{h > t\}$.
2. Layer-cake follows from Fubini applied to $h(x) = \int_0^{h(x)} dt$.
3. Finite-cover length arguments use subadditivity of length.

How it appears in IMC work. **Use.** Use for random points, expected geometric quantities, and inequalities for integrable functions.

Trap watch. Distinguish deterministic inequalities from almost-sure statements; justify independence and symmetry.

Exam checks.

- For nowhere differentiability constructions, show every point is hit by arbitrarily close spikes.

Occurrence count. 2 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2019-D2-P10;2022-D1-P1

11.5 Complex analysis / polynomials

Foundation checklist. Polynomial problems rely on the root-count principle, factorization, Vieta formulae, logarithmic derivatives, roots of unity, Cauchy coefficient extraction, and elementary meromorphic-function ideas. The safest rule is: convert a global statement about values into a polynomial identity only after checking that there are more roots than the degree or that the identity holds on a set with an accumulation point in a domain of holomorphy.

11.5.1 CP-01. Polynomial identity and more-roots-than-degree principle

Domain and bucket. Complex analysis / polynomials; CP polynomial identity. **Literature anchors.** [11, 24, 21].

Prior compact node statement. A nonzero polynomial of degree d over a field has at most d roots. Therefore, if it has more than d roots, it is identically zero.

Exact facts to know.

1. A nonzero polynomial of degree n over a field has at most n roots.
2. If two polynomials of degree at most n agree at $n+1$ distinct points, they are identical.
3. If a polynomial in parameter t with matrix entries vanishes at more points than its degree, every coefficient matrix is zero.
4. For multivariate polynomials, fixing all variables but one can reduce to the one-variable identity principle.
5. Holomorphic identity theorem: if holomorphic functions agree on a set with an accumulation point inside the domain, they agree on the connected component.

Compact proof cores.

1. The root bound follows by induction using the factor theorem.
2. The equality criterion applies the root bound to the difference.
3. Matrix polynomial vanishing is entrywise polynomial vanishing.

How it appears in IMC work. **Use.** Use for polynomial divisibility, coefficient vanishing, finite-field polynomial functions, and interpolation-style arguments.

Trap watch. Over finite fields, different polynomials can define the same function; reduce modulo $x^p - x$ when working on $\mathbb{F}_p$.

Exam checks.

- Whenever $n+1$ values are given for a degree-n object, try coefficient annihilation.

Occurrence count. 68 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D2-P4;1995-D1-P5;1995-D2-P3;1995-D2-P4;1996-D2-P2;1998-D1-P2;1998-D1-P5;1998-D2-P2;1999-D1-P1;1999-D2-P2;1999-D2-P6;2000-D1-P6;2000-D2-P3;2000-D2-P4;2000-D2-P6;2001-D1-P4;2001-D2-P1;2002-D1-P1;2002-D2-P3;2003-D1-P3;2003-D1-P6;2003-D2-P2;2004-D1-P6;2005-D1-P4;2005-D2-P2;2005-D2-P6;2006-D1-P4;2006-D1-P6;2006-D2-P5;2006-D2-P6;2007-D1-P1;2007-D1-P3;2007-D1-P5;2007-D1-P6;2007-D2-P6;2008-D1-P6;2008-D1-P2;2008-D1-P3;2008-D2-P1;2008-D2-P2;2008-D2-P4;2008-D2-P5;2008-D2-P6;2009-D1-P3;2009-D1-P4;2009-D2-P1;2009-D2-P3;2009-D2-P4;2009-D2-P5;2010-D2-P5;2011-D1-P2;2011-D1-P3;2011-D1-P5;2011-D2-P2;2011-D2-P4;2012-D1-P5;2012-D2-P1;2013-D2-P1;2014-D1-P3;2015-D2-P10;2017-D1-P1;2017-D1-P5;2017-D2-P7;2018-D2-P9;2020-D1-P4;2021-D2-P7;2023-D1-P3;2024-D1-P1

11.5.2 CP-02. Factorization, irreducibility, and Gauss-style integer polynomial arguments

Domain and bucket. Complex analysis / polynomials; CP factorization. **Literature anchors.** [11, 24, 21].

Prior compact node statement. Integer polynomial divisibility often uses Gauss lemma, primitive factorization, and reduction modulo primes. Cyclotomic and simple irreducibility facts turn one root into all conjugate roots.

Exact facts to know.

1. Gauss lemma: a primitive polynomial in $\mathbb{Z}[x]$ that factors over $\mathbb{Q}[x]$ factors over $\mathbb{Z}[x]$ up to units.
2. Rational root theorem: if p/q in lowest terms is a root of $a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$, then $p \mid a_0$ and $q \mid a_n$.

3. If an integer polynomial is divisible by a monic integer polynomial over $\mathbb{Q}[x]$, the quotient lies in $\mathbb{Z}[x]$.
4. Irreducibility of cyclotomic-like polynomials often implies that if one primitive root is a root, all conjugate primitive roots are roots.
5. Polynomial division over a field gives $f = qg + r$ with $\deg r < \deg g$.

Compact proof cores.

1. Gauss lemma follows from content multiplicativity.
2. Rational root theorem follows by substituting p/q and using coprimality.
3. Conjugate-root closure follows because the minimal polynomial of the algebraic number divides the polynomial.

How it appears in IMC work. **Use.** Use for $\mathbb{Z}[x]$ divisibility, rational functions integer-valued infinitely often, and roots of unity.

Trap watch. Divisibility in values is not the same as divisibility in $\mathbb{Z}[x]$.

Exam checks.

- Check whether factorization is in $\mathbb{Z}[x]$, $\mathbb{Q}[x]$, $\mathbb{R}[x]$, or $\mathbb{C}[x]$.

Occurrence count. 24 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1995-D2-P3;1998-D1-P5;1998-D2-P2;2000-D2-P3;2001-D1-P4;2001-D2-P1;2005-D2-P2;2006-D1-P4;2007-D1-P1;2008-D1-P3;2008-D2-P1;2008-D2-P2;2008-D2-P4;2009-D2-P1;2012-D1-P5;2012-D2-P1;2013-D2-P1;2015-D2-P10;2017-D1-P5;2017-D2-P7;2018-D2-P9;2020-D1-P4;2021-D2-P7;2023-D1-P3

11.5.3 CP-03. Vieta, logarithmic derivatives, and real-root inequalities

Domain and bucket. Complex analysis / polynomials; CP log derivative. **Literature anchors.** [11, 24, 21].

Prior compact node statement. For $P(x) = \prod_i (x - r_i)$, $P'(x)/P(x) = \sum_i 1/(x - r_i)$. This converts root-location assumptions into sums of squares or coefficient inequalities.

Exact facts to know.

1. Vieta: if $P(x) = a_n \prod_{i=1}^{n} (x - r_i)$, then coefficients are elementary symmetric polynomials in the roots.
2. Logarithmic derivative: $P'(x)/P(x) = \sum_i 1/(x - r_i)$ for x not a root.
3. For real-rooted P, inequalities such as $(n-1)(P')^2 \geq nPP''$ follow from sums of squares in the logarithmic derivative.
4. Multiple roots must be handled separately because P'/P is undefined at roots.
5. If roots lie in a half-plane or on a circle, pair conjugate roots to infer signs or modulus identities for coefficients.

Compact proof cores.

1. Logarithmic derivative follows by differentiating $\log \prod (x - r_i)$ formally or by product rule.
2. The Laguerre-type inequality reduces to $\sum_{i<j} (1/(x - r_i) - 1/(x - r_j))^2 \geq 0$.
3. Vieta follows from expanding the product.

How it appears in IMC work. **Use.** Use for polynomials with real roots, roots on the unit circle, and inequalities involving P, P', P''.

Trap watch. Handle points that are roots of P separately before dividing by $P(x)$.

Exam checks.

- At a root, prove the inequality separately before dividing by $P(x)$.

Occurrence count. 28 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1995-D2-P3;1998-D1-P5;1998-D2-P2;2000-D2-P3;2000-D2-P4;2001-D1-P4;2003-D1-P6;2005-D1-P4;2005-D2-P2;2006-D1-P4;2007-D1-P1;2007-D1-P5;2007-D2-P6;2008-D2-P1;2009-D2-P4;2010-D2-P5;2012-D1-P5;2012-D2-P1;2013-D2-P1;2014-D1-P3;2015-D2-P10;2017-D1-P5;2017-D2-P7;2018-D2-P9;2020-D1-P4;2021-D2-P7;2023-D1-P3;2024-D1-P1

11.5.4 CP-04. Unit-circle roots and conjugation identities

Domain and bucket. Complex analysis / polynomials; CP unit circle. **Literature anchors.** [11, 24, 21].

Prior compact node statement. When $|\alpha| = 1$, identities involving $z, \bar{z}$, and $\frac{z+\alpha}{z-\alpha}$ reveal signs of real parts and force roots onto the unit circle.

Exact facts to know.

1. If $|\alpha| = 1$, then $\bar{\alpha} = 1/\alpha$.
2. For $|\alpha| = 1$, $\operatorname{Re} \frac{z+\alpha}{z-\alpha} = \frac{|z|^2-1}{|z-\alpha|^2}$ for $z \neq \alpha$.
3. If a polynomial has real coefficients, nonreal roots occur in conjugate pairs.
4. Unit-circle root problems often use reciprocal polynomials $z^n \overline{P(1/\bar{z})}$.
5. On $|z| = 1$, conjugation converts powers by $\overline{z^k} = z^{-k}$.

Compact proof cores.

1. The real-part identity is obtained by multiplying numerator and denominator by $\bar{z} - \bar{\alpha}$ and using $\bar{\alpha} = 1/\alpha$.
2. Conjugate-pair property follows by conjugating the equation $P(z) = 0$ for real coefficients.
3. Reciprocal-polynomial identities follow by substituting $1/\bar{z}$ and clearing denominators.

How it appears in IMC work. **Use.** Use for root-location preservation under derivative-like transformations and conditions $|a| = |b| = 1$.

Trap watch. Do not divide by zero at existing roots; remove common factors or argue by continuity.

Exam checks.

- Check whether roots are on the unit circle or merely inside it; the identities differ.

Occurrence count. 13 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1995-D2-P3;2000-D1-P2;2000-D2-P3;2002-D2-P5;2008-D2-P1;2013-D2-P1;2015-D1-P1;2016-D2-P10;2017-D1-P1;2018-D2-P9;2021-D2-P7;2023-D1-P2;2024-D2-P7

11.5.5 CP-05. Roots of unity, cyclotomic filters, and discrete Fourier tricks

Domain and bucket. Complex analysis / polynomials; CP roots of unity. **Literature anchors.** [11, 24, 21].

Prior compact node statement. The root-of-unity filter extracts residue classes: $\frac{1}{m}\sum_{j=0}^{m-1} \omega^{jN}$ equals 1 if $m \mid N$ and 0 otherwise. Cyclotomic irreducibility can force all primitive roots to appear.

Exact facts to know.

1. If $\omega = e^{2\pi i/m}$, then $1 + \omega^j + \cdots + \omega^{(m-1)j} = 0$ unless $m \mid j$, in which case the sum is m.
2. Root-of-unity filter: $\frac{1}{m}\sum_{r=0}^{m-1} \omega^{-ar} F(\omega^r)$ extracts coefficients whose index is congruent to $a \pmod m$.
3. $x^m - 1 = \prod_{r=0}^{m-1}(x - \omega^r)$ and $\Phi_m(x)$ is the product over primitive mth roots.
4. If p is prime, $1 + x + \cdots + x^{p-1}$ is irreducible over $\mathbb{Q}$.
5. DFT arguments on finite cyclic groups translate convolution into multiplication and residue counting into coefficient extraction.

Compact proof cores.

1. The geometric-series identity proves the filter immediately.
2. Prime cyclotomic irreducibility follows by applying Eisenstein to $(x^p - 1)/(x - 1)$ after shifting $x \mapsto x + 1$.
3. DFT coefficient extraction follows by expanding F and using orthogonality of roots of unity.

How it appears in IMC work. **Use.** Use for probability modulo m, coefficient sums by congruence class, divisibility by cyclotomic polynomials, and Fourier coefficients of finite sets.

Trap watch. Check whether all roots, primitive roots, or all nontrivial roots are needed.

Exam checks.

- Use roots of unity when congruence classes of exponents or sums modulo m appear.

Occurrence count. 24 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1995-D2-P3;1997-D1-P3;1999-D2-P2;1999-D2-P6;2000-D1-P2;2000-D2-P3;2001-D1-P4;2001-D2-P3;2002-D2-P5;2007-D1-P5;2007-D1-P6;2008-D2-P1;2010-D2-P5;2013-D2-P1;2014-D1-P3;2016-D2-P10;2017-D1-P1;2017-D1-P5;2017-D2-P7;2018-D2-P9;2021-D2-P7;2023-D1-P2;2024-D1-P1;2024-D2-P7

11.5.6 CP-06. Cauchy formula and coefficient extraction on the unit circle

Domain and bucket. Complex analysis / polynomials; CP Cauchy. **Literature anchors.** [11, 24, 21].

Prior compact node statement. For holomorphic p, coefficients are integrals over the unit circle. Cauchy formulas let coefficient multipliers be represented by convolution kernels.

Exact facts to know.

1. Cauchy coefficient formula: if $P(z) = \sum a_k z^k$, then $a_k = \frac{1}{2\pi i} \int_{|z|=1} P(z) z^{-k-1}\, dz$.
2. Equivalently, on $|z| = 1$, $a_k = \frac{1}{2\pi} \int_0^{2\pi} P(e^{it}) e^{-ikt} dt$.
3. Maximum modulus on a closed disk for polynomials occurs on the boundary.
4. Parseval on the unit circle: $\sum |a_k|^2 = \frac{1}{2\pi} \int_0^{2\pi} |P(e^{it})|^2 dt$.
5. Coefficient multipliers can be represented by convolution kernels on the unit circle.

Compact proof cores.

1. Cauchy formula follows from $\int_{|z|=1} z^m dz = 0$ unless $m = -1$.
2. Maximum boundary property follows from the maximum modulus principle.
3. Parseval follows by multiplying Fourier series and using orthogonality.

How it appears in IMC work. **Use.** Use for polynomial coefficient transformations bounded by sup norms and holomorphic estimates on the unit disk.

Trap watch. The maximum on the closed disk is controlled by the boundary only for holomorphic functions.

Exam checks.

- When bounding transformed coefficients, look for a positive kernel with integral 1.

Occurrence count. 7 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2007-D1-P5;2007-D1-P6;2009-D1-P4;2013-D2-P1;2014-D1-P3;2017-D2-P7;2023-D1-P3

11.5.7 CP-07. Maximum modulus principle

Domain and bucket. Complex analysis / polynomials; CP maximum modulus. **Literature anchors.** [11, 24, 21].

Prior compact node statement. A nonconstant holomorphic function reaches its maximum modulus on the boundary of a compact domain. This underlies many unit-disk inequalities.

Exact facts to know.

1. Maximum modulus principle: a nonconstant holomorphic function on a domain cannot attain a strict maximum of modulus in the interior.
2. For a polynomial, $\max_{|z|\le 1} |P(z)| = \max_{|z|=1} |P(z)|$.
3. If equality or extremality is achieved in an integral bound, the equality case of triangle inequality may force constant argument.

4. Minimum modulus can be applied to $1/f$ only when f has no zeros in the domain.
5. For rational functions, check poles before applying maximum modulus to a region.

Compact proof cores.

1. Maximum modulus follows from Cauchy integral formula or the open mapping theorem.
2. Polynomial disk statement follows because polynomials are entire and continuous on the closed disk.
3. Minimum modulus statement follows by applying maximum modulus to $1/f$.

How it appears in IMC work. **Use.** Use for bounding values inside the disk from boundary data, often after multiplying by a polynomial.

Trap watch. The function must be holomorphic on a neighborhood of the closed domain, not merely inside it with boundary singularities.

Exam checks.

- Use maximum modulus for boundary reduction, not for arbitrary real-variable inequalities.

Occurrence count. 0 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances.

11.5.8 CP-08. Laurent expansions, isolated singularities, and rationality of meromorphic functions

Domain and bucket. Complex analysis / polynomials; CP meromorphic. **Literature anchors.** [11, 24, 21].

Prior compact node statement. A meromorphic function on the Riemann sphere is rational. Local Laurent expansions determine principal parts and hence the rational function when poles are known.

Exact facts to know.

1. An isolated singularity is removable, a pole, or essential according to the Laurent expansion.
2. A meromorphic function on the Riemann sphere is rational.
3. If a meromorphic function has only one pole at a of order at most m and vanishes at infinity sufficiently, then it has form $P(z)/(z-a)^m$ with bounded degree.
4. Laurent principal parts determine a meromorphic function on the sphere up to a polynomial; behavior at infinity fixes the polynomial.
5. Uniform convergence on compact sets permits termwise holomorphy and differentiation of series.

Compact proof cores.

1. Rationality on the sphere follows because subtracting principal parts at all poles leaves an entire function bounded at infinity, hence constant.
2. Pole order is read from the lowest negative power in the Laurent series.
3. Behavior at infinity is analyzed by substituting $w = 1/z$.

How it appears in IMC work. **Use.** Use for sums over logarithm branches, pole-order computations, and rational expressions from analytic data.

Trap watch. Uniform convergence on compact subsets away from singularities must be checked before termwise analytic operations.

Exam checks.

- When summing over logarithm branches, first establish local uniform convergence away from singularities.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2004-D1-P6

CORRECTION MATH-001, MATH-006 — legacy statement superseded; click for canonical replacement.

11.5.9 CP-09. Fejer kernels and positive trigonometric kernels

Domain and bucket. Complex analysis / polynomials; CP Fourier positivity. **Literature anchors.** [21, 17].

Prior compact node statement. The Fejer kernel $F_k(e^{it}) = |1 + e^{it} + \cdots + e^{i(k-1)t}|^2$ is nonnegative. Convex coefficient sequences decompose into nonnegative combinations of Fejer kernels.

Exact facts to know.

1. Fejer kernel: $F_k(e^{it}) = \sum_{j=-(k-1)}^{k-1}(k - |j|)e^{ijt} = |1 + e^{it} + \cdots + e^{i(k-1)t}|^2 \geq 0$.
2. Convex coefficient sequences decompose into nonnegative linear combinations of Fejer kernels by second differences.
3. A positive kernel of integral 2π defines an averaging operator that does not increase the sup norm on the unit circle.
4. Coefficient multipliers generated by convex decreasing sequences are contractions in the disk algebra sup norm.
5. The boundary maximum principle transfers unit-circle bounds to disk bounds.

Compact proof cores.

1. The square formula is direct multiplication of a geometric sum with its conjugate.
2. Second-difference decomposition is discrete summation by parts.
3. The sup-norm contraction follows from $|\int Kp| \leq \int K|p| \leq \|p\|_\infty \int K$ for $K \geq 0$.

How it appears in IMC work. **Use.** Use for sup-norm contraction of polynomial coefficient multipliers and positivity of trigonometric sums.

Trap watch. Convexity of the coefficient sequence is the exact hypothesis that makes the kernel nonnegative.

Exam checks.

- Look for convexity of coefficient weights; it is usually a Fejer-kernel signal.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2009-D1-P4

11.5.10 CP-10. Interpolation, Lagrange basis, and Chebyshev extremality

Domain and bucket. Complex analysis / polynomials; CP interpolation. **Literature anchors.** [25, 24].

Prior compact node statement. A polynomial of degree at most n is determined by $n + 1$ values; derivatives can be written using Lagrange basis derivatives. Chebyshev polynomials often give extremal oscillation.

Exact facts to know.

1. Lagrange basis: for distinct nodes $x_0, \ldots, x_n$, $\ell_k(x) = \prod_{j\neq k}(x - x_j)/(x_k - x_j)$ and $p(x) = \sum p(x_k)\ell_k(x_k)$ for degree at most n.
2. Derivatives of the interpolating polynomial can be written as linear combinations of node values with coefficients $\ell_k^{(r)}(x)$.
3. Linear functionals on a finite-dimensional polynomial space attain their maximum over a box of node values at a vertex of the box.
4. Chebyshev polynomial T_n satisfies $T_n(\cos\theta) = \cos n\theta$ and has alternating extrema ± 1 on $[-1, 1]$.
5. Alternation characterizes minimax extremal polynomials in the standard Chebyshev setting.

Compact proof cores.

1. Lagrange formula is verified by checking equality at all $n + 1$ nodes.
2. Box maximum of a linear functional follows because a linear function on a polytope attains maxima at vertices.
3. Chebyshev identity follows from de Moivre formula.

How it appears in IMC work. **Use.** Use for bounded polynomial values at selected points and maximizing derivatives.

Trap watch. The extremal sign pattern must match the signs of the derivative of the Lagrange basis at the target point.

Exam checks.

- Interpolation problems often reduce to choosing signs of node values.

Occurrence count. 2 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1995-D2-P4;1998-D2-P2

11.5.11 CP-11. Stable polynomials and coefficient inequalities

Domain and bucket. Complex analysis / polynomials; CP stable polynomials. **Literature anchors.** [11, 24, 21].

Prior compact node statement. Polynomials with all roots in the left half-plane have positive coefficients after normalization and satisfy log-concavity-like coefficient inequalities through factorization into positive linear and quadratic factors.

Exact facts to know.

1. A real polynomial with all roots in the open left half-plane factors into positive linear and positive quadratic real factors after an overall sign adjustment.
2. Such polynomials have coefficients of one sign, usually taken positive.
3. Log-concavity-type inequalities for coefficients can be proved by induction through multiplication by positive quadratic factors.
4. Hurwitz stability often implies strict coefficient inequalities when all roots have negative real part.
5. Conjugate pairs give quadratic factors $x^2 + px + q$ with $p, q > 0$.

Compact proof cores.

1. For a root $-a$ with $a > 0$, the linear factor is $x + a$. For a conjugate pair with negative real part, the quadratic factor has positive coefficients.
2. Inductive coefficient inequality follows by expanding coefficients after multiplying by x^2+px+q and grouping terms nonnegatively.
3. Strictness comes from positivity of at least one decisive term.

How it appears in IMC work. **Use.** Use for inequalities among coefficients under root-location assumptions.

Trap watch. Keep track of the order of coefficients and signs after normalizing the leading coefficient.

Exam checks.

- When proving coefficient inequalities, extend coefficients by zeros outside the range.

Occurrence count. 3 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2003-D1-P6;2014-D1-P3;2023-D1-P3

11.5.12 CP-12. Partial fractions and cotangent-type expansions

Domain and bucket. Complex analysis / polynomials; CP partial fractions. **Literature anchors.** [11, 21].

Prior compact node statement. Classical partial fractions such as cotangent expansions evaluate sums over arithmetic progressions of poles. Differentiating produces higher-power denominator sums.

Exact facts to know.

1. The classical partial fraction identity $\sum_{k\in\mathbb{Z}} \frac{1}{w+2\pi ik} = \frac{i}{2}\cot(iw/2)$ is a standard cotangent expansion under symmetric summation.
2. Differentiating termwise gives sums of higher powers $(w+2\pi ik)^{-m}$ when convergence is uniform on compact subsets away from poles.
3. With $w = \log z$, branches of logarithm are $\log z + 2\pi ik$.
4. Differentiation with respect to z uses $d/dz = (1/z)d/dw$ when $w = \log z$.
5. Higher reciprocal-power sums over logarithm branches can become rational functions of z.

Compact proof cores.

1. Cotangent expansion follows from the Mittag-Leffler theorem or residue comparison.
2. Termwise differentiation is justified by locally uniform convergence of differentiated series away from poles.
3. Rational forms emerge after substituting $e^w = z$ into cotangent expressions.

How it appears in IMC work. **Use.** Use for explicit sums over all logarithm branches or equally spaced complex poles.

Trap watch. Termwise differentiation requires locally uniform convergence away from poles.

Exam checks.

- Use symmetric limits for conditionally convergent cotangent sums.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2004-D1-P6

11.5.13 CP-13. Polynomial functional identities and algebraic composition

Domain and bucket. Complex analysis / polynomials; CP identities. **Literature anchors.** [11, 24, 21].

Prior compact node statement. Polynomial identities in several variables are solved by substituting special values, comparing degrees, and using the identity theorem for polynomials.

Exact facts to know.

1. If $P(Q(x))$ or $P(x) - P(y)$ appears, factor differences: $P(x) - P(y) = (x - y)R(x, y)$.
2. Polynomial composition respects degree: $\deg(P \circ Q) = \deg P \deg Q$ for nonconstant polynomials over a field.
3. Iterated polynomial equations can be analyzed by preimage trees and forbidden ranges.
4. Functional identities between polynomials are often solved by comparing leading terms, roots, or special values.
5. For $P(x) = x^2 - 1$, $P_n(x) = 0$ can be counted recursively through equations $P_{n-1}(x) = \pm 1$ or related values.

Compact proof cores.

1. Difference factorization follows from $x^m - y^m = (x - y)(x^{m-1} + \cdots + y^{m-1})$.
2. Degree composition follows from leading terms.
3. Preimage counting uses induction and checks which target values lie in the real range.

How it appears in IMC work. **Use.** Use for classifying polynomials satisfying multiplicative or composition identities and for systems $p(w) = p(z), q(w) = q(z)$.

Trap watch. Special-value substitutions must not lose solutions; after guessing a form, verify the full identity.

Exam checks.

- For pairs $P(w) = P(z)$ and $Q(w) = Q(z)$, divide by $w - z$ only after excluding $w = z$.

Occurrence count. 8 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2000-D1-P2;2007-D1-P3;2013-D2-P1;2014-D1-P3;2017-D2-P7;2018-D2-P9;2023-D1-P3;2024-D1-P1

11.6 Algebra / number theory

Foundation checklist. The algebra-number-theory layer uses congruences, divisibility, finite fields, integer polynomials, groups, rings, quotient structures, and valuations. Keep the ambient structure explicit: a statement true in a field may fail in a ring, and a statement true in an abelian group may fail in a noncommutative group.

11.6.1 ANT-01. Modular arithmetic and divisibility

Domain and bucket. Algebra / number theory; ANT modular. **Literature anchors.** [15, 20].

Prior compact node statement. Divisibility questions are often local modulo prime powers. Reduce congruences modulo primes first, then lift or combine information.

Exact facts to know.

1. Congruence $a \equiv b \pmod{m}$ means $m \mid a-b$ and is compatible with addition, subtraction, and multiplication.
2. If p is prime, $\mathbb{Z}/p\mathbb{Z}$ is a field; every nonzero residue has an inverse.
3. Polynomial congruences can be studied by reducing coefficients modulo m.
4. If $a \equiv b \pmod{m}$, then $P(a) \equiv P(b) \pmod{m}$ for every integer polynomial P.
5. To prove divisibility by a prime power, first force each variable to be divisible by the prime when possible, then lift the exponent.

Compact proof cores.

1. Compatibility follows because divisibility is preserved under integer linear combinations and products.
2. Prime field inverse follows from Bezout: $\gcd(a,p)=1$ gives $ua+vp=1$.
3. Polynomial evaluation congruence follows by repeated compatibility.

How it appears in IMC work. **Use.** Use for integer polynomials, prime divisibility, residue classes, and modular nonexistence.

Trap watch. Check whether the modulus is prime, prime power, or composite; cancellation requires invertibility.

Exam checks.

- When a sum of powers is divisible by a prime, list or characterize the power residues.

Occurrence count. 92 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D1-P3;1994-D1-P6;1997-D1-P3;1997-D1-P4;1999-D1-P3;1999-D2-P1;1999-D2-P2;2000-D1-P5;2000-D2-P1;2001-D1-P1;2001-D1-P2;2001-D1-P4;2001-D2-P6;2002-D1-P3;2003-D1-P2;2003-D1-P4;2005-D1-P2;2005-D1-P4;2005-D2-P2;2005-D2-P6;2006-D1-P2;2006-D1-P3;2006-D2-P1;2006-D2-P5;2007-D1-P1;2007-D2-P2;2007-D2-P4;2008-D1-P3;2008-D1-P5;2008-D2-P3;2008-D2-P4;2008-D2-P5;2009-D2-P4;2010-D1-P4;2010-D2-P4;2010-D2-P5;2011-D1-P3;2011-D1-P5;2011-D2-P4;2012-D1-P1;2012-D1-P2;2012-D1-P3;2012-D2-P1;2012-D2-P3;2012-D2-P4;2012-D2-P5;2013-D1-P5;2013-D2-P2;2013-D2-P4;2014-D1-P4;2014-D2-P1;2015-D1-P2;2015-D1-P4;2015-D2-P10;2016-D1-P2;2016-D1-P3;2016-D1-P4;2016-D1-P5;2016-D2-P8;2016-D2-P9;2017-D1-P3;2017-D1-P4;2017-D2-P8;2017-D2-P10;2018-D1-P2;2018-D1-P5;2018-D2-P6;2018-D2-P8;2019-D1-P2;2019-D1-P4;2019-D1-P5;2019-D2-P7;2019-D2-P9;2020-D1-P1;2020-D1-P3;2020-D2-P6;2021-D1-P2;2021-D2-P5;2021-D2-P6;2021-D2-P8;2022-D1-P2;2022-D1-P3;2022-D1-P4;2022-D2-P6;2023-D1-P4;2023-D1-P5;2023-D2-P10;2024-D1-P3;2024-D1-P4;2024-D1-P5;2024-D2-P9;2024-D2-P10

11.6.2 ANT-02. Chinese remainder theorem and complete residue systems

Domain and bucket. Algebra / number theory; ANT CRT. **Literature anchors.** [15, 20].

Prior compact node statement. If moduli are coprime, congruences modulo each modulus combine uniquely modulo the product. Complete residue systems remain complete under bijective polynomial maps over finite fields.

Exact facts to know.

1. Chinese remainder theorem: if $m_1,\dots,m_r$ are pairwise coprime, then the map $\mathbb{Z}/(m_1\cdots m_r) \to \prod \mathbb{Z}/m_i$ is an isomorphism.
2. Every system $x \equiv a_i \pmod{m_i}$ with pairwise coprime moduli has a unique solution modulo $M=\prod m_i$.

3. A complete residue system modulo m is any set containing exactly one representative of each class.
4. If $\gcd(a, m) = 1$, then $0, a, 2a, \ldots, (m-1)a$ is a complete residue system modulo m.
5. Frobenius coin lemma: if a, b are coprime positive integers, all sufficiently large integers are $ax + by$ with $x, y \geq 0$.

Compact proof cores.

1. CRT follows by constructing idempotents $e_i \equiv 1 \pmod{m_i}$ and $e_i \equiv 0 \pmod{m_j}$ for $j \neq i$.
2. Complete residue multiplication by a is injective modulo m and hence bijective.
3. Coin lemma follows by choosing x among $0, \ldots, b-1$ so $ax \equiv N \pmod{b}$ and taking N large enough for $y \geq 0$.

How it appears in IMC work. **Use.** Use for constructing integers with simultaneous congruence constraints and counting idempotents modulo 10^k.

Trap watch. Uniqueness is modulo the product; remember to exclude disallowed representatives such as zero if the problem says positive.

Exam checks.

- Use CRT when a condition splits over coprime prime powers.

Occurrence count. 30 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1999-D1-P3;2006-D1-P2;2006-D2-P5;2007-D1-P1;2007-D2-P2;2007-D2-P4;2010-D1-P4;2012-D2-P1;2012-D2-P4;2013-D1-P5;2013-D2-P2;2014-D1-P4;2014-D2-P1;2015-D1-P4;2016-D1-P3;2016-D1-P5;2016-D2-P8;2017-D2-P10;2018-D1-P5;2018-D2-P6;2019-D1-P2;2019-D2-P7;2020-D2-P6;2021-D2-P6;2022-D1-P3;2022-D2-P6;2023-D1-P4;2023-D2-P10;2024-D1-P3;2024-D2-P10

11.6.3 ANT-03. Finite fields, field characteristic, and Fermat/Frobenius tools

Domain and bucket. Algebra / number theory; ANT finite fields. **Literature anchors.** [15, 20].

Prior compact node statement. Over $\mathbb{F}_p$, Fermat gives $a^{p-1} = 1$ for nonzero a, and polynomial functions are naturally considered modulo $x^p - x$. Field characteristic controls additive cancellation.

Exact facts to know.

1. A field of characteristic p contains a prime subfield isomorphic to $\mathbb{F}_p$; characteristic 0 contains a copy of $\mathbb{Q}$.
2. In characteristic p, Frobenius holds: $(x + y)^p = x^p + y^p$.
3. Fermat: in $\mathbb{F}_p$, $a^p = a$ for all a, and $a^{p-1} = 1$ for $a \neq 0$.
4. A polynomial over a field of degree d has at most d roots unless it is zero.
5. Every function $\mathbb{F}_p \to \mathbb{F}_p$ has a unique representing polynomial of degree at most $p - 1$ modulo $x^p - x$.

Compact proof cores.

1. Frobenius follows because binomial coefficients $\binom{p}{k}$ are divisible by p for $0 < k < p$.
2. Fermat follows because the nonzero elements form a finite multiplicative group of order $p - 1$.
3. Polynomial representation follows by reducing modulo $x^p - x$ and using interpolation on p points.

How it appears in IMC work. **Use.** Use for finite-field polynomial permutations, characteristic restrictions, and group actions over $\mathbb{F}_p$.

Trap watch. A polynomial of high degree may define the same function as a lower-degree remainder modulo $x^p - x$.

Exam checks.

- Always identify whether the problem is in an arbitrary field or specifically a finite field.

Occurrence count. 49 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1997-D1-P3;1999-D2-P1;1999-D2-P2;2000-D2-P1;2000-D2-P3;2001-D1-P2;2001-D1-P4;2003-D1-P2;2005-D1-P2;2005-D1-P4;2005-D2-P2;2005-D2-P6;2006-D1-P2;2006-D1-P3;2006-D2-P5;2007-D1-P1;2007-D2-P2;2007-D2-P4;2008-D1-P3;2008-D2-P4;2008-D2-P5;2009-D2-P4;2010-D1-P4;2010-D2-P4;2010-D2-P5;2011-D1-P3;2011-D1-P5;2011-D2-P4;2012-D1-P3;2012-D2-P1;2013-D1-P5;

2013-D2-P2;2014-D1-P4;2014-D2-P1;2016-D1-P3;2016-D1-P5;2016-D2-P8;2018-D1-P2;2018-D1-P5;2019-D1-P2;2019-D2-P7;2020-D2-P6;
2021-D2-P6;2022-D1-P3;2022-D2-P6;2023-D1-P4;2023-D2-P10;2024-D1-P3;2024-D2-P10

11.6.4 ANT-04. Integer polynomials, finite differences, and value divisibility

Domain and bucket. Algebra / number theory; ANT integer polynomials. **Literature anchors.** [15, 20].

Prior compact node statement. For integer polynomials, $p(a+mq) \equiv p(a) \pmod{q}$. A degree-$d$ polynomial over a field with more than d roots is zero.

Exact facts to know.

1. For $P \in \mathbb{Z}[x]$, $a-b \mid P(a)-P(b)$ for integers a, b.
2. Finite differences $\Delta P(x) = P(x+1)-P(x)$ lower degree by one and preserve integer-valuedness properties.
3. If a degree-d polynomial over $\mathbb{F}_p$ vanishes at more than d residues, it is the zero polynomial modulo p.
4. Integer-valued divisibility constraints are often tested at $0, 1, -1, \ldots, d$ or reduced modulo primes.
5. If $P(k)$ is divisible by p for all integers k and $\deg P < p$, then all coefficients of P are divisible by p.

Compact proof cores.

1. The divisibility $a-b \mid P(a)-P(b)$ follows from $a^m - b^m$ factorization.
2. Degree lowering of Δ follows by cancellation of leading terms.
3. The coefficient conclusion follows because the reduction modulo p has all p roots and degree below p.

How it appears in IMC work. **Use.** Use for proving coefficient divisibility from value divisibility and constructing values divisible by prescribed polynomial values.

Trap watch. Testing finitely many residues works only when the degree is less than the prime modulus or when using finite differences.

Exam checks.

- Check degree relative to modulus prime before using root-count over $\mathbb{F}_p$.

Occurrence count. 22 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1995-D2-P4;1996-D2-P2;1998-D1-P5;2001-D2-P1;2002-D2-P3;2005-D1-P4;2006-D1-P3;2006-D1-P4;2007-D1-P1
;2008-D1-P3;2008-D2-P1;2008-D2-P4;2011-D1-P3;2012-D1-P5;2012-D2-P1;2014-D1-P3;2014-D2-P1;2015-D2-P10;2017-D2-P7;2018-D2-P9;
2020-D1-P4;2023-D1-P3

11.6.5 ANT-05. Rational roots, Vieta, and rationality constraints

Domain and bucket. Algebra / number theory; ANT rationality. **Literature anchors.** [15, 20].

Prior compact node statement. Rational-root and Vieta arguments translate root rationality into coefficient and product/sum constraints. For irrationality questions, positive rational linear combinations are controlled by induction and contradiction.

Exact facts to know.

1. Rational root theorem gives divisibility constraints on numerator and denominator of rational roots of integer polynomials.
2. If a monic polynomial has integer coefficients, all rational roots are integers.
3. Vieta translates rational roots into rational elementary symmetric sums.
4. If all roots are rational and coefficients are integers, factorization into linear rational factors can often be made integral after clearing contents.
5. Positive coefficients exclude positive roots.

Compact proof cores.

1. Rational-root proof uses $q \mid a_n p^n$ and $p \mid a_0 q^n$ with $\gcd(p, q) = 1$.
2. Monic rational roots are integers because $q \mid 1$.
3. No positive roots with positive coefficients follows because $P(x) > 0$ for $x > 0$.

CORRECTION MATH-002 — legacy statement superseded; click for canonical replacement.

How it appears in IMC work. **Use.** Use for polynomials with rational roots, hyperbolic rationality, irrational subset selection, and rational functions.

Trap watch. Do not divide by a coefficient that may be zero; isolate zero roots first.

Exam checks.

- Combine rational roots with coefficient permutations or positivity to make strong degree bounds.

Occurrence count. 5 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D1-P3;1996-D2-P2;2005-D1-P4;2006-D1-P5;2018-D1-P3

11.6.6 ANT-06. GCD, valuations, prime powers, and coprimality

Domain and bucket. Algebra / number theory; ANT valuations. **Literature anchors.** [15, 20].

Prior compact node statement. Coprime factors can be assigned independently. Valuations measure powers of a prime dividing an integer and turn multiplicative decompositions into additive comparisons.

Exact facts to know.

1. Bezout: $\gcd(a,b)=d$ iff there exist integers u,v with $ua+vb=d$.
2. Euclid lemma: if p prime divides ab, then $p \mid a$ or $p \mid b$.
3. Valuation $v_p(n)$ is the exponent of p in n; $v_p(ab)=v_p(a)+v_p(b)$ and $v_p(a+b) \geq \min(v_p(a), v_p(b))$.
4. If $\gcd(a,b)=1$ and $a \mid bc$, then $a \mid c$.
5. When a product is a perfect power and factors are coprime, each factor is the corresponding perfect power up to sign constraints.

Compact proof cores.

1. Bezout follows from the Euclidean algorithm.
2. Euclid lemma follows from Bezout for p and a.
3. Valuation rules follow from unique prime factorization.

How it appears in IMC work. **Use.** Use for decompositions of positive integers, products equaling powers, and divisibility by prime powers.

Trap watch. When xy is divisible by p^k and x, y are not coprime, valuation splitting is necessary.

Exam checks.

- For determinant or integer-matrix factorization problems, isolate gcds on diagonal entries.

Occurrence count. 42 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1999-D1-P3;1999-D2-P1;2000-D2-P1;2002-D1-P3;2003-D1-P4;2007-D2-P2;2008-D2-P3;2012-D1-P1;2012-D1-P2;2012-D1-P3;2012-D2-P3;2012-D2-P4;2013-D2-P2;2014-D2-P1;2015-D1-P2;2015-D1-P4;2016-D1-P2;2016-D1-P3;2016-D1-P4;2016-D2-P9;2017-D1-P3;2017-D2-P8;2017-D2-P10;2018-D2-P6;2018-D2-P8;2019-D1-P2;2019-D1-P4;2019-D1-P5;2019-D2-P9;2020-D1-P1;2020-D1-P3;2020-D2-P6;2021-D2-P5;2021-D2-P8;2022-D1-P2;2022-D1-P3;2022-D2-P6;2023-D1-P4;2023-D1-P5;2023-D2-P10;2024-D1-P3;2024-D2-P10

11.6.7 ANT-07. Group basics: orders, cyclic subgroups, and generators

Domain and bucket. Algebra / number theory; ANT groups. **Literature anchors.** [15, 20].

Prior compact node statement. Element orders, generated subgroups, and relations determine many finite-group arguments. In abelian groups, Bezout identities in exponents are especially powerful.

Exact facts to know.

1. The order of an element g in a finite group divides the order of the group.
2. If $g^m=e$, then the order of g divides m.
3. A cyclic group generated by g consists of all powers g^k and has one subgroup for each divisor of its order.
4. In an abelian group, powers distribute: $(ab)^n=a^n b^n$. This can fail in noncommutative groups.
5. If $\gcd(r,s)=1$ and $g^r=g^s=e$, then $g=e$.

Compact proof cores.

1. Lagrange theorem proves order divisibility by cosets.
2. The gcd-power statement follows from Bezout: $g = g^{ur+vs}$.
3. Abelian power distribution follows by commuting factors to group all aś and bś.

How it appears in IMC work. **Use.** Use for group equations, pairwise coprime exponents, and subgroup generation.

Trap watch. Exponent arithmetic using $(ab)^n = a^n b^n$ requires commutativity unless justified.

Exam checks.

- When using powers of products, verify commutativity or replace with conjugation identities.

Occurrence count. 52 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1996-D2-P3;1999-D1-P3;1999-D2-P1;2000-D1-P5;2001-D1-P2;2003-D1-P2;2003-D2-P4;2005-D1-P6;2006-D1-P2;2007-D1-P4;2007-D2-P5;2008-D1-P5;2008-D2-P3;2009-D2-P5;2010-D2-P2;2010-D2-P3;2011-D1-P3;2011-D2-P2;2011-D2-P4;2012-D1-P3;2012-D1-P5;2012-D2-P3;2012-D2-P4;2012-D2-P5;2013-D2-P2;2014-D2-P1;2015-D1-P2;2015-D1-P4;2016-D1-P2;2016-D1-P3;2016-D2-P8;2016-D2-P9;2017-D1-P4;2017-D2-P8;2017-D2-P10;2018-D1-P2;2018-D2-P6;2019-D1-P2;2019-D1-P4;2019-D1-P5;2019-D2-P9;2020-D1-P1;2020-D2-P6;2020-D2-P7;2021-D2-P5;2021-D2-P6;2022-D1-P2;2022-D2-P6;2023-D2-P10;2024-D1-P3;2024-D1-P4;2024-D2-P10

11.6.8 ANT-08. Permutation groups, cycles, parity, and generation

Domain and bucket. Algebra / number theory; ANT permutation groups. **Literature anchors.** [15, 20].

Prior compact node statement. Cycles, transpositions, and parity describe symmetric and alternating groups. Adjacent transpositions generate S_n; selected 3-cycles generate A_n.

Exact facts to know.

1. Every permutation decomposes uniquely into disjoint cycles up to order of cycles and cyclic rotation.
2. The sign of a k-cycle is $(-1)^{k-1}$; the alternating group A_n consists of even permutations.
3. Transpositions generate S_n; adjacent transpositions also generate S_n.
4. A cycle and a transposition with suitable connectivity often generate S_n.
5. Conjugation relabels cycles: $\sigma(i_1 \cdots i_k)\sigma^{-1} = (\sigma(i_1) \cdots \sigma(i_k))$.

Compact proof cores.

1. Cycle decomposition is obtained by following orbits of the permutation action.
2. Sign of a cycle follows by writing it as $k-1$ transpositions.
3. Adjacent transpositions sort any permutation, proving generation.

How it appears in IMC work. **Use.** Use for proving generation/non-generation of permutation groups and for constructing permutations with modular properties.

Trap watch. Cycle notation composition order must be consistent throughout the proof.

Exam checks.

- For generation problems, reduce to producing enough transpositions or enough 3-cycles.

Occurrence count. 2 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1998-D1-P2;2001-D1-P2

11.6.9 ANT-09. Normal subgroups, quotient groups, and group actions

Domain and bucket. Algebra / number theory; ANT quotients/actions. **Literature anchors.** [15, 20].

Prior compact node statement. Normal subgroups allow quotient groups. Group actions turn algebraic relations into orbit-stabilizer or index constraints.

Exact facts to know.

1. A subgroup $N \triangleleft G$ is normal iff $gNg^{-1} = N$ for all $g \in G$.
2. If N is normal, the quotient group G/N has multiplication $(gN)(hN) = ghN$.
3. If images of generators commute or generate in a quotient, the quotient can obstruct generation in G.
4. Group actions convert algebraic structure into permutations; orbit-stabilizer: $|Gx| = [G : G_x]$.
5. Conjugacy classes are orbits under conjugation.

Compact proof cores.

1. Quotient multiplication is well-defined exactly because of normality.
2. Orbit-stabilizer follows because $gG_x \mapsto gx$ is a bijection.
3. Generation obstruction: if generated subgroup maps to a proper subgroup of the quotient, it cannot be all of G.

How it appears in IMC work. **Use.** Use for subgroup-index problems, conjugacy of subgroups, and generation obstructions through quotients.

Trap watch. A subgroup of the same index need not be normal; prove normality before passing to a quotient.

Exam checks.

- When a normal subgroup is given, inspect the quotient before trying explicit generation.

Occurrence count. 12 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1996-D2-P3;1999-D2-P1;2005-D1-P6;2010-D2-P3;2012-D1-P3;2012-D2-P5;2016-D2-P8;2017-D1-P4;2018-D1-P2;2020-D2-P7;2021-D2-P6;2024-D1-P4

11.6.10 ANT-10. Automorphisms and semidirect products

Domain and bucket. Algebra / number theory; ANT automorphisms. **Literature anchors.** [15].

Prior compact node statement. A semidirect product $H \rtimes T$ is built from an action of T on H. Automorphisms preserve element orders, normal subgroups, and structural decompositions.

Exact facts to know.

1. An automorphism is a bijective homomorphism from a group to itself.
2. For a finite-dimensional vector space over $\mathbb{F}_q$, $|GL_n(\mathbb{F}_q)| = (q^n - 1)(q^n - q) \cdots (q^n - q^{n-1})$.
3. A semidirect product $H \rtimes_\varphi K$ has multiplication $(h, k)(h_1, k_1) = (h\varphi_k(h_1), kk_1)$.
4. Element orders in semidirect products can be computed by repeated multiplication and the action.
5. An automorphism of a generated group is determined by images of a generating set, but those images must satisfy the defining relations.

Compact proof cores.

1. The GL_n count chooses columns one at a time, each outside the span of previous columns.
2. Semidirect multiplication is associative because φ is a homomorphism $K \to \mathrm{Aut}(H)$.
3. Generator-image bound follows because homomorphisms are determined on words in the generators.

How it appears in IMC work. **Use.** Use for constructing finite groups with prescribed automorphism behavior.

Trap watch. Counting automorphisms by images of generators requires proving those images satisfy the same defining relations.

Exam checks.

- To show $|\mathrm{Aut}\, H| > |\mathrm{Aut}\, G|$, count H exactly and bound G by generator images.

Occurrence count. 27 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

CORRECTION MATH-003, MATH-029 — legacy statement superseded; click for canonical replacement.

Tagged appearances. 1996-D2-P3;1998-D1-P2;1999-D1-P1;1999-D1-P5;1999-D2-P1;2000-D1-P5;2001-D1-P2;2003-D1-P2;2003-D2-P4;2004-D2-P3;2005-D1-P6;2007-D1-P4;2007-D2-P5;2008-D1-P5;2010-D2-P2;2010-D2-P3;2011-D1-P3;2011-D2-P4;2012-D1-P3;2012-D1-P5;2012-D2-P5;2016-D2-P8;2017-D1-P4;2018-D1-P2;2020-D2-P7;2021-D2-P6;2024-D1-P4

11.6.11 ANT-11. Rings, idempotents, nil rings, and noncommutativity

Domain and bucket. Algebra / number theory; ANT rings. **Literature anchors.** [15, 20].

Prior compact node statement. In noncommutative rings, expansion of $(a+b)^2$ still gives $ab+ba$ when squares vanish. Idempotent equations often become commutator relations under characteristic assumptions.

Exact facts to know.

1. A ring may be noncommutative; multiplication order must be preserved.
2. Idempotent means $e^2 = e$; nilpotent means $x^k = 0$ for some k; square-zero means $x^2 = 0$.
3. If every element of a ring has square zero, then $(a+b)^2 = 0$ gives $ab + ba = 0$.
4. In characteristic zero, $na = 0$ implies $a = 0$ for every positive integer n.
5. Commutators $[a, b] = ab - ba$ measure noncommutativity and often become central or zero under idempotent relations.

Compact proof cores.

1. The square-zero anticommutation identity is obtained by expanding $(a + b)^2$ and using $a^2 = b^2 = 0$.
2. Characteristic-zero cancellation follows from the definition.
3. Idempotent equations are manipulated by multiplying on the left or right, preserving order.

How it appears in IMC work. **Use.** Use for ring problems with idempotents, nilpotents, characteristic zero, and noncommutative algebra.

Trap watch. Do not commute factors; use associativity and given identities only.

Exam checks.

- Never rearrange ring factors unless commutativity has been proved.

Occurrence count. 33 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1996-D2-P3;1999-D1-P5;1999-D2-P1;2000-D1-P5;2001-D1-P2;2003-D1-P2;2003-D2-P4;2005-D1-P6;2006-D1-P2;2007-D1-P4;2007-D2-P5;2008-D1-P5;2009-D2-P4;2010-D2-P3;2011-D1-P3;2011-D2-P2;2012-D1-P3;2012-D1-P5;2012-D2-P3;2012-D2-P5;2013-D2-P2;2014-D2-P1;2016-D2-P8;2017-D1-P4;2018-D1-P2;2019-D1-P2;2020-D2-P6;2020-D2-P7;2021-D2-P6;2022-D2-P6;2023-D2-P10;2024-D1-P3;2024-D1-P4

11.6.12 ANT-12. Diophantine equations, Pell-type recurrences, and quadratic irrationals

Domain and bucket. Algebra / number theory; ANT Diophantine. **Literature anchors.** [20, 15].

Prior compact node statement. Quadratic irrational fixed points of fractional linear transformations lead to Pell-type equations. Recurrences from fundamental units provide infinitely many solutions.

Exact facts to know.

1. Pell-type equations $x^2 - Dy^2 = N$ are solved or supplied by units in $\mathbb{Z}[\sqrt{D}]$ when D is nonsquare positive.
2. Quadratic irrational $r = p + q\sqrt{D}$ has conjugate $\bar{r} = p - q\sqrt{D}$ and a quadratic minimal polynomial over $\mathbb{Q}$.
3. Mobius fixed equation $(ar + b)/(cr + d) = r$ is equivalent to $cr^2 + (d - a)r - b = 0$.
4. Matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with determinant 1 act on real quadratic irrationals by fractional linear transformations.
5. Linear recurrences from units, e.g. $(u + v\sqrt{D})^n = k_n + \ell_n\sqrt{D}$, provide infinitely many congruence-controlled solutions.

Compact proof cores.

1. The fixed equation follows by cross multiplying.
2. Norm identity $N(k+\ell\sqrt{D}) = k^2 - Dl^2$ gives Pell equations.
3. Periodicity modulo M follows because pairs in a linear recurrence have finitely many residues modulo M.

How it appears in IMC work. **Use.** Use for constructing integer matrices fixing quadratic irrationals and cubic equations with integer roots.

Trap watch. Verify determinant-one and nontriviality conditions after solving the Pell equation.

Exam checks.

- For quadratic irrationals, use conjugates and norms to turn the problem into integer equations.

Occurrence count. 2 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2005-D2-P6;2006-D2-P5

11.6.13 ANT-13. Lattices and norms/quasinorms

Domain and bucket. Algebra / number theory; ANT lattices. **Literature anchors.** [15, 20].

Prior compact node statement. Integer lattice shifts interact with convex norms for $p \geq 1$ and can fail for quasi-norms $0 < p < 1$. Coordinate splitting by signs is often required.

Exact facts to know.

1. A lattice in a real vector space is a discrete additive subgroup generated by a basis of a subspace.
2. For $p \geq 1$, $\|x\|_p = (\sum |x_i|^p)^{1/p}$ is a norm; for $0 < p < 1$ it fails the triangle inequality and is only a quasinorm-like expression.
3. Convexity of $|t|^p$ for $p \geq 1$ underlies many ℓ_p inequalities.
4. For $0 < p < 1$, concavity on positive reals can reverse expected minimization behavior.
5. Integer lattice shifts in a hyperplane can be analyzed by signs and coordinate partitions.

Compact proof cores.

1. Minkowski inequality proves the triangle inequality for $p \geq 1$.
2. Failure for $p < 1$ can be seen from $(a+b)^p < a^p + b^p$ for positive a, b.
3. Lattice-shift inequalities often reduce to checking how integer changes alter positive and negative coordinates.

How it appears in IMC work. **Use.** Use for lattice minimization and norm comparison problems.

Trap watch. Convexity fails for $p < 1$; do not use triangle inequality in the quasi-norm case.

Exam checks.

- Check whether p is below or above 1 before using convexity.

Occurrence count. 2 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D1-P6;1997-D1-P5

11.6.14 ANT-14. Greedy infinite products and rationality tails

Domain and bucket. Algebra / number theory; ANT greedy products. **Literature anchors.** [15, 20].

Prior compact node statement. Greedy multiplicative expansions are controlled by a tail identity such as $\prod_{j\geq 0}(1+M^{-2^j}) = M/(M-1)$. Rationality can force eventual equality in the greedy recurrence.

Exact facts to know.

1. Greedy algorithms choose the next integer by a sharp inequality, e.g. the least n such that $1 + 1/n < \theta$.
2. Infinite products converge when logarithms of factors form a convergent series.
3. The product $\prod_{j\geq 0}(1 + M^{-2^j}) = \frac{M}{M-1}$ follows from telescoping $\prod(1 + x^{2^j}) = 1/(1 - x)$ for $|x| < 1$.
4. Rationality tails can be proved by showing a positive integer sequence would strictly decrease unless the tail terminates periodically.
5. Uniqueness follows when each greedy choice is forced by a two-sided inequality.

Compact proof cores.

1. The telescoping product follows from $(1 - x)\prod_{j=0}^{N}(1 + x^{2^j}) = 1 - x^{2^{N+1}}$ and $N \to \infty$.
2. Greedy uniqueness is proved by showing any smaller or larger choice prevents the residual ratio from tending to 1.
3. The decreasing-integer contradiction uses rational numerator-denominator differences.

How it appears in IMC work. **Use.** Use for unique infinite product representations and rationality criteria.

Trap watch. A greedy inequality must be sharp in both directions to prove uniqueness.

Exam checks.

- For infinite products, work with residual ratios after the first k factors.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1997-D1-P4

11.6.15 ANT-15. Finite abelian groups and vector spaces over finite fields

Domain and bucket. Algebra / number theory; ANT finite abelian. **Literature anchors.** [15, 20].

Prior compact node statement. Finite abelian groups of exponent 2 are vector spaces over $\mathbb{F}_2$ and have order a power of two. This structure often classifies triple systems.

Exact facts to know.

1. Every finite abelian group of exponent 2 is a vector space over $\mathbb{F}_2$.
2. If every nonzero element has order 2, the group operation can be written additively with $x + x = 0$.
3. Such a group has order 2^m for some m.
4. Pair-determines-third systems often define $a * b = c$ and become vector spaces over $\mathbb{F}_2$ after adjoining a zero element.
5. Binary-coordinate construction realizes the group $(\mathbb{F}_2)^m$ explicitly.

Compact proof cores.

1. Associativity and commutativity turn the operation into an abelian group; $x * x = 0$ gives exponent 2.
2. A finite vector space over $\mathbb{F}_2$ of dimension m has 2^m elements.
3. The binary-coordinate model uses componentwise addition modulo 2.

How it appears in IMC work. **Use.** Use for finite designs where each pair determines a third element and associativity-like conditions appear.

Trap watch. The added identity element must be included before claiming group structure.

Exam checks.

- When every pair determines a third element, try defining an algebraic operation.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2003-D2-P4

11.7 Combinatorics / discrete structures

Foundation checklist. The discrete layer uses encodings, counting twice, extremal choices, graph models, recurrences, generating functions, parity, and construction. A solution should identify the objects being counted, specify whether order matters, and check that the encoding is injective, surjective, or bijective as claimed.

11.7.1 CMB-01. Pigeonhole principle and extremal element selection

Domain and bucket. Combinatorics / discrete structures; CMB pigeonhole. **Literature anchors.** [29, 30, 31].

Prior compact node statement. If more objects than boxes are distributed, one box contains many objects. Extremal choices (minimal counterexample, largest element, shortest cycle) often force structure.

Exact facts to know.

1. Pigeonhole principle: if N objects are placed in m boxes, some box has at least $\lceil N/m \rceil$ objects.
2. Extremal principle: choose a minimal or maximal object satisfying a condition and show it has forced structure.
3. Finite descent: a positive integer parameter cannot decrease indefinitely.
4. Greedy selection is valid when each step preserves a stated invariant.
5. In covering/counting problems, compare total possible incidences with required incidences to force an empty or nonempty box.

Compact proof cores.

1. Pigeonhole proof is counting: if every box had at most $r-1$, total would be at most $m(r-1)$.
2. Extremal arguments prove contradiction by modifying the extremal object to a better one.
3. Greedy invariants are proved inductively.

How it appears in IMC work. **Use.** Use for divisibility value collisions, selecting common rows/columns, and minimal counterexample arguments.

Trap watch. Define the boxes explicitly; hidden boxes often are residue classes, divisors, or graph neighborhoods.

Exam checks.

- Identify the boxes explicitly; vague pigeonhole arguments lose rigor.

Occurrence count. 68 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D1-P1;1994-D1-P3;1994-D1-P6;1994-D2-P4;1995-D1-P3;1996-D1-P3;1996-D2-P3;1996-D2-P6;1997-D1-P6;1997-D2-P6;1998-D2-P3;1998-D2-P4;1998-D2-P5;1998-D2-P6;1999-D1-P2;1999-D1-P5;1999-D2-P6;2000-D1-P1;2001-D2-P1;2001-D2-P3;2002-D1-P4;2002-D1-P5;2002-D2-P2;2002-D2-P6;2003-D1-P5;2003-D2-P3;2003-D2-P5;2004-D1-P1;2004-D1-P2;2004-D1-P4;2004-D2-P6;2005-D1-P2;2006-D1-P2;2006-D2-P1;2006-D2-P5;2007-D1-P4;2007-D1-P6;2008-D1-P6;2008-D2-P4;2008-D2-P6;2009-D1-P1;2009-D1-P3;2009-D2-P5;2010-D1-P4;2011-D1-P4;2011-D2-P2;2012-D1-P1;2013-D1-P3;2013-D2-P5;2015-D2-P8;2016-D1-P5;2016-D2-P7;2016-D2-P9;2017-D1-P4;2018-D2-P8;2019-D2-P8;2020-D1-P1;2020-D1-P2;2020-D2-P7;2021-D1-P2;2021-D2-P8;2022-D1-P3;2022-D1-P4;2022-D2-P5;2022-D2-P8;2023-D2-P8;2024-D1-P5;2024-D2-P9

11.7.2 CMB-02. Double counting and degree-sum methods

Domain and bucket. Combinatorics / discrete structures; CMB double counting. **Literature anchors.** [29, 30, 31].

Prior compact node statement. Count the same set of objects in two ways, often by vertices versus edges, pairs versus problems, or walks via middle vertex.

Exact facts to know.

1. Double counting counts the same finite set of incidences in two ways.
2. Degree-sum formula for graphs: $\sum_v \deg v = 2|E|$.
3. Counting walks of length two gives $\sum_v \deg(v)^2$ in a simple graph when walks are ordered by middle vertex.
4. If an injection maps a set of size N into another set of size N, it is bijective.
5. Equality cases in counting inequalities usually force uniqueness conditions.

Compact proof cores.

1. Degree-sum counts endpoint-edge incidences.
2. Length-two walk count fixes the middle vertex and chooses first and last neighbors independently.
3. Equality in injection-count comparison follows because finite injective map between equal-size sets is surjective.

How it appears in IMC work. **Use.** Use for contest scores, graph degree constraints, set intersections, and probability counts.

Trap watch. Specify ordered vs unordered objects; many IMC counts hinge on this distinction.

Exam checks.

- After proving an inequality by counting, analyze equality; IMC often asks for structure from equality.

Occurrence count. 56 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D1-P1;1994-D2-P4;1995-D1-P3;1996-D1-P3;1996-D2-P3;1996-D2-P6;1997-D2-P6;1998-D2-P3;1998-D2-P4;1998-D2-P5;1998-D2-P6;1999-D1-P5;2000-D1-P1;2001-D2-P1;2001-D2-P3;2002-D1-P4;2002-D1-P5;2002-D2-P2;2003-D1-P5;2003-D2-P5;2004-D1-P1;2004-D1-P2;2004-D1-P4;2004-D2-P6;2005-D1-P2;2006-D1-P2;2006-D2-P5;2007-D1-P4;2007-D1-P6;2008-D1-P6;2008-D2-P4;2009-D1-P1;2009-D1-P3;2010-D1-P4;2011-D1-P4;2011-D2-P2;2012-D1-P1;2013-D1-P3;2013-D2-P5;2015-D2-P8;2016-D1-P5;2016-D2-P9;2017-D1-P4;2018-D2-P8;2019-D2-P8;2020-D1-P1;2020-D1-P2;2020-D2-P7;2021-D1-P2;2021-D2-P8;2022-D1-P3;2022-D1-P4;2022-D2-P5;2022-D2-P8;2023-D2-P8;2024-D2-P9

11.7.3 CMB-03. Graph encoding, cycles, and bipartite representations

Domain and bucket. Combinatorics / discrete structures; CMB graph encoding. **Literature anchors.** [29, 30, 31].

Prior compact node statement. Encode relations as graphs. Cycles, degree sums, connectedness, and bipartite representations convert combinatorial statements into graph-theoretic ones.

Exact facts to know.

1. A relation can often be encoded as a graph with vertices as objects and edges as incidences or compatibility.
2. A graph with N vertices and at least N edges contains a cycle.
3. Bipartite graph cycles alternate between the two vertex classes.
4. Marked grid points correspond to edges between row-vertices and column-vertices.
5. Shortest-cycle arguments forbid chords or shortcuts because they would create smaller cycles.

Compact proof cores.

1. A forest on N vertices has at most $N-1$ edges; prove by induction or components formula $|E| \le |V| - c$.
2. Grid-to-bipartite encoding sends a marked point (r, c) to edge rc.
3. Shortcut contradiction replaces part of a cycle by a shorter path.

How it appears in IMC work. **Use.** Use for grids, friendship conditions, colored diagonals, trees, and matrix zero/nonzero patterns.

Trap watch. When encoding a grid, rows and columns often become vertices and marked cells become edges.

Exam checks.

- When rows and columns alternate in a statement, build a bipartite graph immediately.

Occurrence count. 68 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D1-P3;1994-D1-P5;1994-D2-P3;1995-D1-P4;1995-D2-P1;1996-D1-P6;1996-D2-P2;1997-D1-P6;1997-D2-P5;1997-D2-P6;1998-D1-P1;1998-D1-P2;1998-D2-P3;1998-D2-P4;1998-D2-P5;1999-D1-P4;1999-D1-P5;1999-D1-P6;1999-D2-P5;2000-D2-P4;2001-D1-P2;2001-D2-P4;2001-D2-P5;2002-D1-P1;2002-D2-P2;2002-D2-P4;2002-D2-P5;2003-D1-P4;2003-D2-P3;2003-D2-P4;2004-D1-P1;2004-D1-P3;2004-D2-P2;2005-D2-P1;2005-D2-P4;2006-D2-P1;2007-D2-P1;2008-D1-P6;2008-D1-P1;2008-D1-P4;2008-D2-P5;2009-D1-P3;2009-D1-P5;2009-D2-P4;2011-D1-P1;2011-D1-P4;2011-D2-P5;2012-D2-P3;2012-D2-P5;2013-D1-P3;2013-D2-P4;2013-D2-P5;2014-D2-P4;2016-D1-P4;2017-D1-P3;2019-D2-P7;2019-D2-P8;2019-D2-P10;2020-D1-P3;2021-D1-P2;2022-D1-P4;2022-D2-P5;2022-D2-P8;2023-D1-P4;2023-D1-P5;2023-D2-P7;2023-D2-P8;2023-D2-P9

11.7.4 CMB-04. Set systems, intersecting families, and induction on sizes

Domain and bucket. Combinatorics / discrete structures; CMB set systems. **Literature anchors.** [29, 30, 31].

Prior compact node statement. Intersecting families are often handled by induction on set size, splitting by intersection with a fixed set, or constructing a finite transversal.

Exact facts to know.

1. Intersecting family: every two sets in the family meet.
2. Uniform family: all sets have the same size; induction can use traces on a fixed set and reduce sizes.
3. If a family is partitioned by intersection with a fixed finite set, each subfamily can be analyzed separately.
4. Finite hitting-set conclusions often require uniformity; without it counterexamples can drift to infinity.
5. Shadow/restriction arguments remove fixed intersections and reduce parameters.

Compact proof cores.

1. Partition by $A \cap U = C$ for fixed finite U; there are finitely many C.
2. If two trace classes have disjoint traces, their original sets must intersect outside U, enabling induction on smaller set sizes.
3. Counterexamples without uniformity use pairs of sets with intersections escaping every finite Y.

How it appears in IMC work. **Use.** Use for finite set families with pairwise intersections and uniform-size assumptions.

Trap watch. Uniformity assumptions are often essential; construct counterexamples without them.

Exam checks.

- Check whether all members have the same size before using finite hitting arguments.

Occurrence count. 23 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1997-D1-P6;1997-D2-P5;1997-D2-P6;1998-D1-P1;1999-D2-P5;2002-D2-P2;2004-D1-P1;2005-D2-P4;2007-D2-P1;2008-D1-P4;2011-D1-P4;2012-D2-P3;2013-D2-P5;2014-D2-P4;2016-D1-P4;2017-D1-P3;2019-D2-P8;2020-D1-P3;2021-D1-P2;2022-D2-P5;2023-D1-P4;2023-D1-P5;2023-D2-P7

11.7.5 CMB-05. Recurrences and state partitions

Domain and bucket. Combinatorics / discrete structures; CMB recurrences. **Literature anchors.** [29, 30, 31].

Prior compact node statement. Partition objects by a finite state, derive a recurrence, verify initial conditions, and solve or compare recurrences.

Exact facts to know.

1. State partition method: split objects by the last symbol/state and derive recurrences among state counts.
2. Linear recurrences are solved by initial values plus transition equations.
3. If two sequences satisfy the same recurrence and matching enough initial values, they are equal.
4. Automaton states should encode exactly the memory needed to test the next extension.
5. For ternary strings with local constraints, states often depend on equality/inequality of the last two symbols.

Compact proof cores.

1. Recurrences follow by appending one symbol and checking which previous states permit it.
2. Equality of recurrence sequences follows by induction.
3. Minimal state correctness requires proving every valid extension is counted once and only once.

How it appears in IMC work. **Use.** Use for words, sequences under local constraints, matrix-defined recurrences, and counting functions.

Trap watch. The state must contain enough memory to extend objects without ambiguity.

Exam checks.

- Define states before counting; do not count sequences only by total length if the rule is local.

Occurrence count. 4 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1998-D2-P4;2002-D2-P2;2005-D1-P2;2013-D2-P5

11.7.6 CMB-06. Generating functions for counting

Domain and bucket. Combinatorics / discrete structures; CMB generating functions. **Literature anchors.** [32, 29].

Prior compact node statement. Generating functions translate recurrences and weighted counts into algebraic identities. Coefficient extraction and radius/convergence considerations then solve the count.

Exact facts to know.

1. Ordinary generating function: $A(x) = \sum a_n x^n$ turns convolution into multiplication.
2. Exponential generating function: $E(x) = \sum a_n x^n / n!$ is natural for labeled structures and sums involving $1/k!$.
3. Coefficient extraction notation: $[x^n]A(x) = a_n$.
4. Recurrences with polynomial coefficients often become differential equations for generating functions.
5. Probability generating functions encode sums of independent discrete variables by products.

Compact proof cores.

1. Convolution property follows from Cauchy product of series.
2. Differential relation follows because $A'(x) = \sum (n+1)a_{n+1}x^n$.
3. Independent-sum generating function is product because coefficients count all decompositions.

How it appears in IMC work. **Use.** Use for sequences, expected sums, probability distributions, and recurrence-defined counts.

Trap watch. Formal power series do not require convergence, but analytic evaluations do.

Exam checks.

- Check convergence if treating generating functions analytically; formal power series need no convergence.

Occurrence count. 14 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2000-D1-P6;2002-D1-P3;2002-D2-P3;2003-D1-P6;2003-D2-P6;2008-D2-P3;2012-D2-P2;2015-D2-P8;2017-D2-P8;2017-D2-P9;2018-D2-P7;2019-D1-P4;2021-D1-P3;2024-D2-P8

11.7.7 CMB-07. Inclusion-exclusion and root-of-unity filters

Domain and bucket. Combinatorics / discrete structures; CMB filters. **Literature anchors.** [29, 30, 31].

Prior compact node statement. Counting by congruence class can be done with roots of unity; exclusion of forbidden configurations is handled by complements or inclusion-exclusion.

Exact facts to know.

1. Inclusion-exclusion: $|\cup A_i| = \sum |A_i| - \sum |A_i \cap A_j| + \cdots$.
2. Root-of-unity filters count objects with a statistic congruent to a fixed residue.
3. For residues modulo m, $\frac{1}{m}\sum_{r=0}^{m-1} \omega^{r(s-a)}$ is 1 if $s \equiv a \pmod m$ and 0 otherwise.
4. Generating functions plus filters give exact probabilities for dice and digit sums modulo m.
5. Complement counting is often simpler than direct inclusion-exclusion when few bad events are possible.

Compact proof cores.

1. Inclusion-exclusion follows by checking each element contained in exactly r sets contributes $\sum_{j=1}^{r}(-1)^{j+1}\binom{r}{j}=1$.
2. Root-filter identity follows from finite geometric sums.
3. Generating function for sums follows from multiplying single-trial polynomials.

How it appears in IMC work. **Use.** Use for dice sums, modulo counts, and divisibility by residues in combinatorial settings.

Trap watch. Root-of-unity filters count residues of exponents/sums; ensure the generating function encodes the correct statistic.

Exam checks.

- Use filters when the phrase ”divisible by” or ”modulo” appears with counting.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1999-D2-P2

11.7.8 CMB-08. Permutations, inversions, and permutation statistics

Domain and bucket. Combinatorics / discrete structures; CMB permutations. **Literature anchors.** [29, 30, 31].

Prior compact node statement. Permutations encode selections with one item per row/column. Statistics such as inversions or displacement can be represented by determinants or generating functions.

Exact facts to know.

1. Inversion number $\operatorname{inv}(\sigma)$ counts pairs $i<j$ with $\sigma(i)>\sigma(j)$.
2. The sign of a permutation is $(-1)^{\operatorname{inv}(\sigma)}$.
3. Determinant expansion is $\det(a_{ij})=\sum_{\sigma\in S_n}\operatorname{sgn}(\sigma)\prod_i a_{i,\sigma(i)}$.
4. Permutation statistics can be encoded as exponents in determinant or generating-function expansions.
5. Parity of counts can be obtained by evaluating a signed enumeration modulo 2.

Compact proof cores.

1. Determinant expansion is the Leibniz formula.
2. Sign-inversion formula follows because adjacent swaps change inversion parity and generate S_n.
3. If signed and unsigned counts are congruent modulo 2, determinant coefficients give parity of statistics.

How it appears in IMC work. **Use.** Use for permutation enumeration, determinant parity of permutation statistics, and row-column selection problems.

Trap watch. Distinguish permutation as a function from its one-line notation.

Exam checks.

- For parity of permutation counts, build a determinant whose monomial exponent is the statistic.

Occurrence count. 5 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2001-D1-P1;2003-D1-P2;2009-D2-P4;2013-D2-P5;2022-D2-P6

11.7.9 CMB-09. Explicit construction and counterexample design

Domain and bucket. Combinatorics / discrete structures; CMB constructions. **Literature anchors.** [29, 30, 31].

Prior compact node statement. Construction problems require preserving all hypotheses while forcing or breaking the conclusion. Small examples, periodic patterns, and extremal configurations are the standard sources.

Exact facts to know.

1. To disprove a universal statement, identify the weakest hypothesis and construct an object violating the conclusion while preserving all assumptions.
2. Extremal constructions should be checked for all parameter ranges and edge cases.
3. Counterexamples often use discontinuities, noncompactness, noncommutativity, conditional convergence, or small dimensions.
4. For existence, explicit construction plus verification is sufficient; uniqueness is separate.
5. When a lower bound is proved, a matching construction is required for optimality.

Compact proof cores.

1. Counterexample validation is a direct checklist against hypotheses and conclusion.
2. Optimality proof is two-sided: upper/lower bound plus construction.
3. Parameter construction often begins with small cases and then uses induction or scaling.

How it appears in IMC work. **Use.** Use for yes/no questions, nonexistence disproofs, and sharpness examples.

Trap watch. A counterexample must satisfy every hypothesis exactly, including continuity, monotonicity, or finiteness.

Exam checks.

- Every construction should end with a line verifying the original variables, not just transformed variables.

Occurrence count. 36 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D1-P4;1994-D2-P5;1995-D1-P6;1995-D2-P6;1997-D1-P2;1997-D1-P4;1997-D1-P6;1997-D2-P1;1997-D2-P6;1998-D1-P2;1999-D1-P2;1999-D1-P6;1999-D2-P5;2000-D1-P1;2002-D1-P2;2003-D2-P4;2004-D2-P3;2006-D1-P1;2007-D2-P1;2007-D2-P3;2007-D2-P5;2008-D1-P3;2008-D1-P5;2009-D1-P1;2011-D1-P2;2013-D1-P5;2013-D2-P4;2015-D1-P2;2016-D1-P1;2016-D1-P2;2018-D1-P2;2019-D2-P6;2020-D1-P1;2022-D1-P3;2024-D1-P3;2024-D2-P9

11.7.10 CMB-10. Erdos-Szekeres style monotone subsequences and binomial thresholds

Domain and bucket. Combinatorics / discrete structures; CMB monotone subsequences. **Literature anchors.** [29, 30, 31].

Prior compact node statement. Many monotone-subsequence thresholds are proved by induction using binomial identities and splitting by a median or extremal element.

Exact facts to know.

1. Erdos-Szekeres monotone subsequence principle: long enough sequences force increasing or decreasing subsequences of prescribed length.
2. Binomial-threshold induction: $\binom{k+l-4}{k-2}+1$ splits according to two halves using Pascal identity.
3. Extending a monotone sequence by a far endpoint can preserve growth conditions such as distance doubling.
4. Minimum and maximum elements are natural splitters for ordered real sets.
5. Sharpness usually comes from recursive constructions matching Pascal recurrence.

Compact proof cores.

1. Induction uses Pascal identity to force one half large enough for a smaller parameter.
2. Distance doubling is preserved because elements in the opposite half are at least as far from the first point as needed.
3. Sharpness constructions reverse the induction by joining extremal examples in two separated intervals.

How it appears in IMC work. **Use.** Use for finding long monotone sequences with extra growth constraints.

Trap watch. The induction parameter may be two-dimensional: length of increasing and decreasing alternatives.

Exam checks.

- When a binomial coefficient appears in a sequence problem, look for Pascal induction.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2004-D1-P5

11.7.11 CMB-11. Grid, lattice, and alternating-cycle arguments

Domain and bucket. Combinatorics / discrete structures; CMB grid cycles. **Literature anchors.** [29, 30, 31].

Prior compact node statement. Represent marked grid cells as edges in a bipartite graph between row-vertices and column-vertices. A cycle in the graph becomes an alternating row/column sequence.

Exact facts to know.

1. An $n \times k$ grid with marked points is naturally a bipartite graph between rows and columns.
2. If every remaining row and column has degree at least two, repeatedly following alternate row/column choices must eventually create a cycle.
3. Deleting rows or columns of degree at most one preserves inequalities of the form marked points at least rows plus columns.
4. Lattice points on curves with convexity can be controlled by slopes and denominator counts.
5. Alternating cycles in grids translate back to point sequences sharing rows and columns alternately.

Compact proof cores.

1. Deletion proof preserves $E \geq R + C$ because both sides drop by at most one.
2. Cycle creation follows by finiteness of the graph and degree at least two.
3. Slope denominator count bounds rational slopes with denominator at most A.

How it appears in IMC work.

Use. Use for grid problems asking for alternating selections or cycles.

Trap watch. Deleting rows/columns with too few points preserves inequalities only if the count is adjusted correctly.

Exam checks.

- Grid alternating statements are graph-cycle statements in disguise.

Occurrence count. 2 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1999-D1-P5;2023-D2-P6

11.7.12 CMB-12. Coloring, partitions, and parity

Domain and bucket. Combinatorics / discrete structures; CMB coloring/parity. **Literature anchors.** [29, 30, 31].

Prior compact node statement. Coloring and partition problems often require parity counts, degree constraints, or induction preserving parity.

Exact facts to know.

1. Partition problems often become coloring problems where each color class has a forbidden property.
2. Parity invariant: if a quantity changes by an even number at each step, its parity is fixed.
3. Triangulation parity can be tracked by counting incidences of vertices with triangles.
4. If n is divisible by 3, polygon triangulation parity can be built inductively by adding three vertices.
5. Coloring constraints on spheres or finite sets often translate to sums of ± 1 over special subsets.

Compact proof cores.

1. Parity invariance follows by reducing each operation modulo 2.
2. Incidence sums count each triangle three times, so congruences modulo 2 or 3 can appear.
3. Coloring-to-sum translation uses white/black as $-1/+1$ and rewrites balance as zero sum.

How it appears in IMC work. **Use.** Use for colored graphs, triangulations with odd/even vertex incidences, and partitions into sets with algebraic equality.

Trap watch. Parity can change at boundary vertices; track local contributions, not just total counts.

Exam checks.

- Encode colors numerically when ”exactly half” or parity appears.

Occurrence count. 48 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D1-P3;1997-D1-P6;1997-D2-P5;1997-D2-P6;1998-D1-P1;1998-D2-P3;1998-D2-P4;1999-D1-P6;1999-D2-P5;2000-D2-P4;2001-D2-P5;2002-D1-P1;2002-D2-P2;2002-D2-P4;2003-D1-P4;2003-D2-P3;2004-D1-P1;2004-D1-P4;2004-D2-P2;2005-D2-P1;2005-D2-P4;2006-D2-P1;2007-D1-P4;2007-D2-P1;2008-D1-P6;2008-D1-P1;2008-D1-P4;2011-D1-P4;2012-D2-P3;2013-D2-P2;2013-D2-P4;2014-D2-P4;2016-D1-P4;2016-D1-P5;2017-D1-P3;2019-D2-P8;2019-D2-P10;2020-D1-P3;2021-D1-P2;2022-D1-P4;2022-D2-P5;2022-D2-P8;2023-D1-P4;2023-D1-P5;2023-D2-P7;2023-D2-P8;2023-D2-P9;2024-D1-P5

11.7.13 CMB-13. Words, equivalence relations, and rewriting systems

Domain and bucket. Combinatorics / discrete structures; CMB words. **Literature anchors.** [23, 29].

Prior compact node statement. In word problems, local rewriting rules reduce length if a repeated block appears. Lemmas about absorbing words with all letters create global bounded-length representatives.

Exact facts to know.

1. A rewriting system is generated by local equivalence rules; congruence means rules remain valid inside larger words.
2. Idempotent word relation $uu \sim u$ allows deletion of doubled blocks.
3. If a word contains all letters, appending suitable words can return it to an equivalent shorter anchor.
4. Long words are reduced by finding repeated blocks or splitting into prefix-middle-suffix with universal letters.
5. Normal-form goals require proving every word is equivalent to one below a fixed length.

Compact proof cores.

1. Congruence closure follows by applying allowed substitutions inside a context.
2. The deletion $uvvw \sim uvw$ follows by applying $vv \sim v$ inside a larger word.
3. Universal-letter lemma is proved one appended letter at a time.

How it appears in IMC work. **Use.** Use for equivalence relations on words and finite reduction algorithms.

Trap watch. A rewrite rule must be compatible with left and right concatenation before applying it inside a word.

Exam checks.

- For word problems, mark blocks explicitly; vague ”reduce repeatedly” is not enough.

Occurrence count. 3 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1999-D2-P5;2015-D2-P8;2020-D1-P1

11.7.14 CMB-14. Transfer matrices and matrix powers for counting

Domain and bucket. Combinatorics / discrete structures; CMB transfer matrix. **Literature anchors.** [29, 32].

Prior compact node statement. If local compatibility between states is encoded by a matrix A, then entries or sums of A^{k-1} count sequences or table fillings of length k.

Exact facts to know.

1. A transfer matrix A encodes allowed transitions between states; the number of length-k paths is a sum of entries of A^{k-1}.
2. If states are binary columns, entries of A can represent whether adjacent columns avoid a forbidden pattern.
3. Symmetry of rectangular grids can show $F_{n,k} = F_{k,n}$ by transposing fillings.
4. Recursive block definitions of matrices can encode local compatibility by adding one coordinate.
5. Matrix powers count walks because $(A^m)_{ij}$ counts paths of length m from i to j when A is a $0/1$ adjacency matrix.

Compact proof cores.

1. Path-count proof is induction on m using matrix multiplication.
2. Grid transposition is a bijection between $n \times k$ and $k \times n$ fillings preserving forbidden 2×2 all-one blocks.
3. Block recursion is verified by splitting states according to first or last bit.

How it appears in IMC work. **Use.** Use for counting binary tables, local constraints, and symmetry $F_{n,k} = F_{k,n}$.

Trap watch. Make sure the matrix state represents a whole column/row, not a single cell.

Exam checks.

- When a matrix is recursively defined with $0/1$ blocks, search for a counting interpretation.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2004-D2-P6

11.7.15 CMB-15. Finite designs and pair-determines-third structures

Domain and bucket. Combinatorics / discrete structures; CMB designs. **Literature anchors.** [29, 30, 31].

Prior compact node statement. A system where every pair belongs to exactly one triple can often be encoded by a binary operation $a * b = c$. Additional associativity-like axioms may turn it into a vector space over $\mathbb{F}_2$.

Exact facts to know.

1. A Steiner triple-like system has the property that every pair lies in exactly one triple.
2. Pair-determines-third lets one define $a * b = c$ when $\{a, b, c\}$ is a block.
3. Additional closure axioms may imply associativity of the induced operation after adjoining a zero.
4. Finite elementary abelian 2-groups produce examples with $n = 2^m - 1$ nonzero elements.
5. The blocks are triples $\{a, b, a + b\}$ in $(\mathbb{F}_2)^m \setminus \{0\}$.

Compact proof cores.

1. Uniqueness of a block through a pair makes $*$ well-defined.
2. Associativity is checked from the closure axiom on triples.
3. The finite vector-space construction verifies every pair of nonzero distinct elements has nonzero sum distinct from both.

How it appears in IMC work. **Use.** Use for Steiner triple-system-like problems and classification of allowed sizes.

Trap watch. The operation is initially undefined on equal elements; adding an identity often completes the algebraic structure.

Exam checks.

- When triples contain each pair once, try finite vector spaces over $\mathbb{F}_2$.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2003-D2-P4

11.8 Geometry / convexity / metric

Foundation checklist. The geometry layer is mainly analytic: translate, rotate, scale, square distances, project onto subspaces, use convexity, compute volumes by affine change of variables, and convert simplex or sphere statements into Gram-matrix identities. Always state which transformations preserve the relevant quantity.

11.8.1 GEO-01. Coordinate and affine normalization

Domain and bucket. Geometry / convexity / metric; GEO coordinates. **Literature anchors.** [7, 13, 10].

Prior compact node statement. Translate, rotate, scale, or choose coordinates to simplify distances, areas, conics, and vectors while preserving the quantity of interest.

Exact facts to know.

1. Translations, rotations, and reflections preserve Euclidean distances, angles, areas, and volumes.
2. Invertible affine maps preserve collinearity, ratios along a line, convexity, and incidence, but scale volumes by $|\det A|$.
3. Coordinate normalization may place a point at the origin, a line on an axis, or a simplex in a convenient basis.
4. For conics and quadrics, completing the square gives canonical coordinates.
5. WLOG transformations must preserve the exact statement being proved.

Compact proof cores.

1. Distance preservation under orthogonal maps follows from $\|Qx - Qy\|^2 = (x-y)^T Q^T Q (x-y)$.
2. Volume scaling under a linear map follows first for parallelepipeds and then by decomposition/measure theory.
3. Completing the square is algebraic rearrangement of quadratic forms.

How it appears in IMC work. **Use.** Use at the start of most Euclidean geometry, conic, and volume problems.

Trap watch. Only affine-invariant quantities survive affine transformations; Euclidean distances require isometries or controlled scalings.

Exam checks.

- After normalizing, state how to transform back.

Occurrence count. 2 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2014-D1-P5;2014-D2-P4

11.8.2 GEO-02. Euclidean distances and inner-product identities

Domain and bucket. Geometry / convexity / metric; GEO inner products. **Literature anchors.** [7, 13, 10].

Prior compact node statement. Use $\|u-v\|^2 = \|u\|^2 + \|v\|^2 - 2\langle u, v\rangle$. Spherical distance constraints often become inner-product inequalities.

Exact facts to know.

1. Polarization: $2\langle u, v\rangle = \|u\|^2 + \|v\|^2 - \|u-v\|^2$.
2. Parallelogram identity: $\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2$.
3. Cosine rule: $\|u-v\|^2 = \|u\|^2 + \|v\|^2 - 2\|u\|\|v\|\cos\theta$.
4. Distance constraints on a sphere become inner-product constraints.
5. Rational distances squared imply rational inner products when distances to the origin are rational.

Compact proof cores.

1. Polarization is obtained by expanding $\|u-v\|^2$.
2. Parallelogram identity follows by expanding both squares.
3. Sphere distance relation: for unit u, v, $\|u-v\|^2 = 2 - 2\langle u, v\rangle$.

How it appears in IMC work. **Use.** Use for rational distances, vectors with prescribed sums, tetrahedron angle estimates, and unit-sphere configurations.

Trap watch. Square distances only after confirming all quantities are nonnegative.

Exam checks.

- Square distances before comparing; this usually linearizes the condition.

Occurrence count. 19 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1994-D2-P5;1995-D2-P1;1996-D1-P6;1998-D1-P6;2002-D1-P6;2004-D1-P4;2004-D2-P3;2005-D2-P5;2006-D2-P4;2008-D1-P5;2008-D2-P2;2010-D1-P1;2010-D1-P5;2013-D2-P3;2014-D1-P5;2014-D2-P4;2015-D2-P9;2018-D2-P6;2021-D2-P8

11.8.3 GEO-03. Projections and orthogonal decompositions

Domain and bucket. Geometry / convexity / metric; GEO projections. **Literature anchors.** [7, 13, 10].

Prior compact node statement. Orthogonal projection shortens distances and decomposes vectors into parallel and perpendicular components. Projecting geometric regions can compare volumes or areas.

Exact facts to know.

1. Orthogonal projection $P_U x$ is the unique point in a closed subspace U minimizing distance to x.
2. $x - P_U x$ is orthogonal to U.
3. Projection onto a hyperplane or line converts distances and volumes into lower-dimensional data.
4. A contraction is a map f with $d(f(x), f(y)) \leq d(x, y)$; projections are contractions.
5. For vectors, Pythagoras: if $u \perp v$, then $\|u + v\|^2 = \|u\|^2 + \|v\|^2$.

Compact proof cores.

1. Projection theorem in finite dimensions follows by minimizing a quadratic function or solving normal equations.
2. Contraction of orthogonal projection follows from $\|P(x - y)\| \leq \|x - y\|$.
3. Pythagoras follows from expanding the square and using zero inner product.

How it appears in IMC work. **Use.** Use for sphere/conic intersections, convex geometry, tetrahedra, and nearest-point arguments.

Trap watch. A projection can collapse dimensions; track signed area/volume if orientation matters.

Exam checks.

- When a problem mentions distance to a set/line/subspace, project.

Occurrence count. 25 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1995-D2-P3;1996-D2-P4;1997-D1-P5;1997-D2-P5;1998-D2-P5;1999-D2-P6;2002-D1-P1;2002-D2-P4;2002-D2-P5;2004-D1-P4;2004-D2-P2;2004-D2-P3;2005-D2-P5;2008-D2-P2;2009-D2-P1;2011-D2-P5;2014-D1-P5;2015-D1-P5;2017-D2-P10;2018-D1-P5;2019-D2-P10;2022-D2-P5;2022-D2-P8;2023-D2-P9;2024-D1-P5

11.8.4 GEO-04. Convex bodies, supporting geometry, and John-ellipse ideas

Domain and bucket. Geometry / convexity / metric; GEO convex bodies. **Literature anchors.** [26, 7].

Prior compact node statement. For convex bodies, support and containment arguments compare maximal ellipses or balls. If a larger ellipse fits after a perturbation, maximality is contradicted.

Exact facts to know.

1. A convex set contains the segment between any two of its points.
2. A centrally symmetric convex body satisfies $B = -B$ after centering at the origin.
3. A supporting hyperplane touches a convex body while the body lies in one closed half-space.

CORRECTION MATH-030 — legacy statement superseded; click for canonical replacement.

4. John ellipsoid theorem: every convex body has a unique maximal-volume inscribed ellipsoid; in symmetric cases it provides strong contact-point constraints.
5. If an arc of the unit circle lies strictly inside a convex body, a slight anisotropic stretch can create a larger inscribed ellipse unless a contact condition blocks it.

Compact proof cores.

1. Existence of maximal ellipsoid follows from compactness of admissible ellipsoids under suitable parametrization.
2. Uniqueness and contact conditions are standard convex-geometry facts; IMC applications often need only a local contradiction by perturbing an ellipse.
3. Supporting hyperplanes follow from separation theorems for convex sets.

How it appears in IMC work. **Use.** Use for convex symmetric planar bodies and maximal inscribed ellipse problems.

Trap watch. The proof often needs a compactness gap between an arc and the boundary.

Exam checks.

- For John-ellipse problems, look for a perturbation increasing area while staying inside the body.

Occurrence count. 7 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1996-D2-P4;2011-D2-P3;2014-D1-P5;2014-D2-P4;2014-D2-P5;2015-D1-P3;2015-D1-P4

11.8.5 GEO-05. Conics: parabolas, ellipses, foci, and directrices

Domain and bucket. Geometry / convexity / metric; GEO conics. **Literature anchors.** [7, 13, 10].

Prior compact node statement. A standard parabola with vertex (v, w) has equation $y = (x-v)^2 + w$. Ellipses can be described by focus-directrix ratios or quadratic equations.

Exact facts to know.

1. Standard parabola with vertex (v, w) and leading coefficient 1 has equation $y = (x-v)^2 + w$.
2. An ellipse can be described by focus-directrix data: distance to focus equals eccentricity times distance to directrix, with $0 < e < 1$.
3. Two conics usually meet in at most four points, but common-focus geometry can reduce possible intersections to a line.
4. Reflection across the x-axis sends (x, y) to $(x, -y)$ and preserves horizontal distances.
5. Completing squares transforms quadratic equations into canonical conic equations.

Compact proof cores.

1. Parabola vertex formula follows by completing square in $x^2 + ax + b$.
2. Focus-directrix equivalence is a standard conic theorem.
3. Common-focus ellipse intersection: equal focus distance gives a linear relation between distances to directrices, hence a line condition.

How it appears in IMC work. **Use.** Use for conic incidence/tangency, common-focus ellipse intersections, and reflected parabolas.

Trap watch. Tangency to a line imposes double roots after subtracting the line equation.

Exam checks.

- For conic incidence, write equations in canonical coordinates rather than drawing only.

Occurrence count. 4 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2000-D2-P4;2002-D1-P1;2008-D2-P2;2014-D1-P5

11.8.6 GEO-06. Simplex, barycentric coordinates, and circumcenters

Domain and bucket. Geometry / convexity / metric; GEO simplex. **Literature anchors.** [7, 13].

Prior compact node statement. Barycentric weights are proportional to opposite simplex volumes. Circumcenters are characterized by equal distances to all vertices.

Exact facts to know.

1. Barycentric coordinates of P in simplex $P_0 \dots P_n$ are weights $\lambda_i \geq 0$, $\sum \lambda_i = 1$, $P = \sum \lambda_i P_i$.
2. Volume ratio: replacing vertex P_i by P gives $v(S_i) = \lambda_i v(S)$ for an interior point.
3. The circumcenter of a simplex is the unique point equidistant from all vertices when the vertices are affinely independent.
4. Affine independence of $n+1$ points in $\mathbb{R}^n$ is equivalent to nonzero simplex volume.
5. Orthogonal bisector hyperplanes determine circumcenters by a linear system.

Compact proof cores.

1. Volume ratio follows because the altitude to the opposite face scales by the barycentric coordinate.
2. Circumcenter linear system comes from subtracting equations $\|X - P_i\|^2 = \|X - P_0\|^2$.
3. Uniqueness follows because the resulting linear equations have independent normals $P_i - P_0$.

How it appears in IMC work. **Use.** Use for volume-weighted circumcenter identities and regular-simplex configurations.

Trap watch. Volume weights must use consistent orientation or absolute volumes depending on the statement.

Exam checks.

- For simplex weighted identities, replace volumes by barycentric coordinates.

Occurrence count. 6 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1997-D2-P5;2008-D2-P2;2009-D1-P5;2009-D2-P1;2014-D1-P5;2017-D2-P10

11.8.7 GEO-07. Area, volume, projection, and ellipsoid computation

Domain and bucket. Geometry / convexity / metric; GEO volume. **Literature anchors.** [7, 13, 10].

Prior compact node statement. Volumes transform by the absolute determinant of a linear map. Ellipsoid volumes follow from scaling the unit ball.

Exact facts to know.

1. Area/volume of a linear image scales by the absolute determinant of the linear map.
2. The unit ball in $\mathbb{R}^3$ has volume $4\pi/3$; an ellipsoid with semiaxes a, b, c has volume $4\pi abc/3$.
3. Projection of a region can only decrease lengths in the projected direction; signed projected area scales by cosine of the angle between planes.
4. For a quadric inequality, completing squares and rescaling coordinates gives volume.
5. For polynomial graph areas, subtract the tangent line and integrate the resulting nonnegative polynomial.

Compact proof cores.

1. Ellipsoid volume follows from the diagonal linear map sending the unit ball to the ellipsoid.
2. Cosine area scaling follows from projecting a parallelogram and using cross products.
3. Graph area between curves is an integral of vertical difference after sign is known.

How it appears in IMC work. **Use.** Use for solid regions defined by distance inequalities, cube dissections, and projected areas.

Trap watch. Complete the square before identifying the ellipsoid axes.

Exam checks.

CORRECTION MATH-007 — legacy statement superseded; click for canonical replacement.

- Always identify the Jacobian or determinant of the scaling map.

Occurrence count. 28 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1995-D2-P3;1996-D2-P4;1997-D1-P5;1997-D2-P5;1998-D2-P5;1999-D2-P6;2000-D2-P1;2002-D1-P1;2002-D2-P4;2002-D2-P5;2004-D1-P4;2004-D2-P2;2004-D2-P3;2005-D2-P5;2008-D2-P2;2009-D2-P1;2011-D2-P5;2014-D1-P5;2014-D2-P4;2015-D1-P5;2017-D2-P10;2018-D1-P5;2019-D2-P10;2021-D2-P7;2022-D2-P5;2022-D2-P8;2023-D2-P9;2024-D1-P5

11.8.8 GEO-08. Metric content, diameter covers, and contractions

Domain and bucket. Geometry / convexity / metric; GEO metric content. **Literature anchors.** [7, 13, 10].

Prior compact node statement. A contraction cannot increase diameters. Covering a segment by finitely many sets forces the sum of diameters to be at least the segment length.

Exact facts to know.

1. Diameter: $\operatorname{diam}(E) = \sup\{d(x,y) : x, y \in E\}$.
2. Upper content by finite diameter covers is subadditive by definition.
3. A contraction cannot increase diameters: $\operatorname{diam}(f(E)) \le \operatorname{diam}(E)$.
4. If a set contracts onto a segment L, every finite cover of the set induces a finite cover of L with no larger total diameters.
5. The upper content of a segment equals its length.

Compact proof cores.

1. Contraction diameter inequality follows directly from pairwise distance inequality.
2. Segment content lower bound follows by induction on number of covering sets or by projecting onto the line.
3. Content inequality $C(E) \ge K(E)$ follows by applying the segment result to images of a cover under a contraction.

How it appears in IMC work. **Use.** Use for metric content versus lower content and compact counterexamples.

Trap watch. The covering sets need not be intervals; project or contract them to a segment.

Exam checks.

- For metric content, projections and contractions are the primary tools.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1996-D1-P6

11.8.9 GEO-09. Hilbert-space equidistant sets and orthonormal translations

Domain and bucket. Geometry / convexity / metric; GEO Hilbert. **Literature anchors.** [12, 10].

Prior compact node statement. A set of mutually equidistant points in a Hilbert space is a translate of a scaled orthonormal system. The center is obtained as a limit of barycenters for countable subsets.

Exact facts to know.

1. An orthonormal system satisfies $\langle e_i, e_j \rangle = \delta_{ij}$.
2. If points x_i are pairwise at distance d, their centered finite average $\bar{x}$ gives inner products $\langle x_i - \bar{x}, x_j - \bar{x} \rangle$ depending only on d and the number of points.
3. In infinite-dimensional Hilbert space, Cauchy sequences of finite averages can converge to a center y.
4. After the correct translation, scaled vectors $\sqrt{2}(x-y)/d$ become orthonormal.
5. Completeness of Hilbert space is essential for the limit center.

Compact proof cores.

1. Finite centered inner products are computed by expanding averages and using equal distances.
2. Cauchy property of averages follows from explicit norm computations.
3. Orthonormality in the limit follows by continuity of the inner product.

How it appears in IMC work. **Use.** Use for infinite-dimensional equidistant sets and orthonormal-system construction.

Trap watch. Completeness of the Hilbert space is used to take the barycenter limit.

Exam checks.

- For equidistant sets, center the configuration before checking orthogonality.

Occurrence count. 6 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1995-D2-P1;2002-D2-P6;2008-D2-P6;2013-D2-P3;2018-D2-P6;2021-D2-P8

11.8.10 GEO-10. Closest points to closed sets and density of uniqueness

Domain and bucket. Geometry / convexity / metric; GEO nearest point. **Literature anchors.** [7, 13, 10].

Prior compact node statement. A closed set intersects a compact ball around an exterior point, so nearest points exist. If the nearest point is not unique, moving toward one nearest point produces nearby points with a unique nearest point.

Exact facts to know.

1. For a nonempty closed set $A \subset \mathbb{R}^n$, the distance $d(b, A) = \inf_{a \in A} \|b - a\|$ is attained locally by compactness after intersecting with a large ball.
2. If a nearest point is not unique, points on an open segment from one nearest point to b may have unique nearest point.
3. The set of points with a unique nearest point to a closed set is dense in $\mathbb{R}^n$.
4. Closedness of A is used to ensure existence of nearest points, not uniqueness.
5. Ball tangency geometry can certify uniqueness of nearest points.

Compact proof cores.

1. Existence: choose a minimizing sequence and restrict it to a compact ball around b.
2. Uniqueness along a segment follows because the smaller ball around the segment point lies inside the original nearest ball and touches at only one point.
3. Density follows by perturbing arbitrary b toward one chosen nearest point.

How it appears in IMC work. **Use.** Use for density of points with unique nearest projection onto a closed set.

Trap watch. Closedness is needed for existence; uniqueness is obtained by geometry of balls, not by convexity.

Exam checks.

- For nearest-point problems, draw nested balls and their boundary intersections.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2003-D2-P3

11.8.11 GEO-11. Unit-sphere configurations and regular simplex bounds

Domain and bucket. Geometry / convexity / metric; GEO sphere configurations. **Literature anchors.** [7, 13, 10].

Prior compact node statement. On the unit sphere, $\|x - y\| > \sqrt{2}$ is equivalent to $\langle x, y \rangle < 0$. At most $n + 1$ such points exist in $\mathbb{R}^n$, with equality given by a regular simplex.

Exact facts to know.

1. On the unit sphere, $\|x-y\| > \sqrt{2}$ is equivalent to $\langle x, y\rangle < 0$.
2. At most $n+1$ unit vectors in $\mathbb{R}^n$ can have pairwise negative inner products.
3. Regular simplex vertices on the sphere satisfy $\langle v_i, v_j\rangle = -1/n$ for $i \neq j$.
4. A Gram matrix with diagonal 1 and off-diagonal $-1/n$ is positive semidefinite of rank n.
5. Inductive projection can reduce the dimension after fixing one point.

Compact proof cores.

1. Distance-inner-product equivalence follows from $\|x-y\|^2 = 2 - 2\langle x, y\rangle$.
2. The upper bound follows by induction: fix a point and project the rest to the orthogonal complement after sign normalization.
3. Regular simplex construction is verified by computing the Gram matrix.

How it appears in IMC work. **Use.** Use for maximum number of points on a sphere with obtuse pairwise angles.

Trap watch. Induction via projection must preserve strict negativity of inner products.

Exam checks.

- Unit-sphere distance thresholds almost always become inner-product sign conditions.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2001-D2-P3

11.8.12 GEO-12. Polygon triangulation and parity of incidences

Domain and bucket. Geometry / convexity / metric; GEO triangulations. **Literature anchors.** [7, 13, 10].

Prior compact node statement. Triangulations can be built inductively by adding three vertices and preserving parity at old vertices while controlling new vertices.

Exact facts to know.

1. A triangulation of a convex n-gon uses $n-2$ triangles.
2. Adding a block of three vertices can preserve parity of old vertex incidences while setting new incidences odd.
3. Vertex-triangle incidence parity is tracked modulo 2.
4. Base cases $n = 3, 4, 5$ often determine induction by steps of 3.
5. Convexity ensures diagonals used in triangulations lie inside the polygon.

Compact proof cores.

1. $n-2$ triangle count follows by induction after drawing one diagonal.
2. Parity preservation is verified by counting how many new triangles touch each old endpoint.
3. Induction by $n \mapsto n+3$ covers congruence classes modulo 3.

How it appears in IMC work. **Use.** Use for parity-constrained triangulations of convex polygons.

Trap watch. Base cases modulo 3 are essential.

Exam checks.

- For polygon parity, draw the base cases and the three-vertex extension.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2006-D2-P1

11.8.13 GEO-13. Convex hulls, random polytopes, and geometric probability

Domain and bucket. Geometry / convexity / metric; GEO random geometry. **Literature anchors.** [7, 13, 10].

Prior compact node statement. Random geometric problems usually reduce by symmetry, conditioning on extremal points, or comparing volumes/areas of regions.

Exact facts to know.

1. The convex hull of a set is the smallest convex set containing it, equivalently all finite convex combinations.
2. Random polytopes or geometric probability problems often reduce to integrating over ordered coordinates or using symmetry.
3. Affine invariance can normalize a simplex or convex body before computing probabilities.
4. Expectation of geometric quantities may be computed by indicator integration.
5. Boundary cases usually have probability zero under continuous distributions.

Compact proof cores.

1. Convex-combination characterization follows by intersection of all convex sets containing the points.
2. Indicator integration proof: expected measure of a random set is integral of the probability a fixed point belongs.
3. Affine invariance follows by applying the same invertible affine map to all sampled points.

How it appears in IMC work. **Use.** Use for probability involving random points in disks/balls and convex hull events.

Trap watch. Independence of points and conditioning on positions must be explicit.

Exam checks.

- Use symmetry and affine normalization before attempting coordinates.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2002-D2-P4

11.8.14 GEO-14. Cube and square dissection by number-theoretic semigroups

Domain and bucket. Geometry / convexity / metric; GEO dissections. **Literature anchors.** [7, 13, 10].

Prior compact node statement. Refining one cube into a^d smaller cubes increases the count by $a^d - 1$. If two such increments are coprime, the Frobenius coin lemma gives all sufficiently large counts.

Exact facts to know.

1. Refining one cube in a dissection into a^d congruent smaller cubes increases the number of cubes by $a^d - 1$.
2. If two increments u, v are coprime, all sufficiently large integers are nonnegative combinations of u and v.
3. For square dissections, increments $2^2 - 1 = 3$ and $3^2 - 1 = 8$ are coprime.
4. For dimension d, $2^d - 1$ and $(2^d - 1)^d - 1$ are coprime.
5. Number-theoretic semigroups turn one initial dissection into all sufficiently large counts.

Compact proof cores.

1. Refinement count is $n \mapsto n - 1 + a^d = n + (a^d - 1)$.
2. Coprime semigroup lemma is the Frobenius coin argument via residues.
3. The gcd claim follows because if $r = 2^d - 1$, then $(r)^d - 1 \equiv (-1)^d - 1$ modulo factors of r with the stated construction checked by Euclidean reduction.

How it appears in IMC work. **Use.** Use for partitioning cubes/squares into a prescribed number of smaller cubes.

Trap watch. Find two coprime increments $a^d - 1$; not every pair works in every dimension.

Exam checks.

- For dissection counts, look for two coprime refinement increments.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2000-D2-P1

11.9 Functional equations / dynamics

Foundation checklist. Functional-equation problems require special substitutions, normalization, monotonicity or injectivity extraction, iteration/orbit analysis, and counterexample discipline. First decide whether regularity is present; without continuity or monotonicity, use algebraic or order-based arguments rather than limits.

11.9.1 FEQ-01. Substitution of special values and normalization

Domain and bucket. Functional equations / dynamics; FEQ substitutions. **Literature anchors.** [22, 14].

Prior compact node statement. Start by substituting $0, 1, x, y = f(x)$, or symmetric choices; normalize by translating/scaling when possible. Then prove any guessed family satisfies the equation.

Exact facts to know.

1. Substitute special values such as 0, 1, x, $f(x)$, or values making an argument equal to a known constant.
2. Normalize by defining ratios like $g(x) = f(x)/x$ or shifts like $h(x) = f(x) - x$ when the equation is homogeneous or affine.
3. If the equation is symmetric in variables, swap variables and compare.
4. If an equation holds for all $y > 0$, choose y depending on x to force a contradiction or fixed value.
5. Always verify the substituted values lie in the stated domain.

Compact proof cores.

1. Special substitutions are justified because the identity is universal over the domain.
2. Normalization is reversible when denominators are nonzero and domain restrictions are respected.
3. Comparing swapped equations eliminates common terms.

How it appears in IMC work. **Use.** Use for most functional equations over $\mathbb{R}$, $\mathbb{Q}$, or positive reals.

Trap watch. Do not assume continuity or monotonicity unless the equation implies it or it is given.

Exam checks.

- Write the domain beside the equation before choosing substitutions.

Occurrence count. 7 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2005-D2-P4;2007-D2-P1;2009-D2-P2;2015-D1-P3;2016-D1-P1;2019-D2-P6;2024-D2-P6

11.9.2 FEQ-02. Injectivity and monotonicity forced by a functional equation

Domain and bucket. Functional equations / dynamics; FEQ monotonicity. **Literature anchors.** [22, 14].

Prior compact node statement. Equations of the form $f(x)f(yf(x)) = f(x+y)$ or $f(x^2/f(x)) = x$ often force monotonicity, injectivity, or a reciprocal linear form.

Exact facts to know.

1. If a functional equation implies $f(u) = f(v) \Rightarrow u = v$, then f is injective and arguments inside f may be equated.
2. If $f(x+y)$ is expressed monotonically in y through a positive factor, the equation may force monotonicity.
3. Positive-valued equations often imply bounds like $f \leq 1$ or f decreasing by choosing a special y.
4. Strict monotonicity plus dense/interval domain can upgrade algebraic identities to global form.
5. If f is monotone and has full range $\mathbb{R}$, jumps are impossible, hence continuity follows.

Compact proof cores.

1. Injectivity extraction follows by applying the equation to both sides and using equality of outputs.
2. Monotonicity follows when $x < y$ can be represented as $x + t$ and the equation gives an ordered comparison.
3. Jump removal for monotone full-range functions follows because a jump misses an interval of values.

How it appears in IMC work. **Use.** Use for classification of functions on positive reals or intervals.

Trap watch. If injectivity is used, prove it before canceling arguments inside f.

Exam checks.

- Once injectivity is proved, revisit the original equation and cancel f safely.

Occurrence count. 8 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1999-D1-P4;2000-D2-P5;2005-D2-P4;2009-D2-P2;2015-D1-P3;2016-D1-P1;2019-D2-P6;2024-D2-P6

11.9.3 FEQ-03. Fixed points, iterates, and orbit closure

Domain and bucket. Functional equations / dynamics; FEQ dynamics. **Literature anchors.** [22, 14].

Prior compact node statement. For interval maps, cluster points and continuity control iterates. Closed orbits of continuous maps on compact intervals are often finite unless accumulation behavior contradicts closedness.

Exact facts to know.

1. For continuous $f : [a, b] \to [a, b]$, fixed points exist by IVT.
2. For iterates $p_{n+1} = f(p_n)$, if the orbit repeats then the orbit set is finite and eventually periodic.
3. A closed infinite orbit in a compact interval has an accumulation point in the orbit.
4. Continuity maps cluster points of p_n to cluster points of p_{n+1}.
5. If the set of orbit points is closed and all points are distinct, accumulation behavior creates contradiction in several compact interval settings.

Compact proof cores.

1. Fixed point theorem applies to $f(x) - x$ at endpoints.
2. Repetition $p_m = p_n$ implies periodicity from deterministic iteration.
3. Cluster-point statements follow from compactness and continuity.

How it appears in IMC work. **Use.** Use for iteration convergence, finite orbit proofs, and periodic point counts.

Trap watch. A pointwise closed orbit is not automatically finite without compactness and continuity arguments.

Exam checks.

- For orbit closure problems, first split into repeated-orbit and all-distinct cases.

Occurrence count. 6 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1996-D2-P1;2002-D1-P4;2010-D1-P3;2015-D1-P3;2016-D1-P1;2024-D2-P6

11.9.4 FEQ-04. Recurrence transformations and conjugacy

Domain and bucket. Functional equations / dynamics; FEQ recurrences. **Literature anchors.** [22, 14].

Prior compact node statement. Nonlinear recurrences often simplify after substituting a new variable, conjugating the map, or identifying an exact iterate formula.

Exact facts to know.

1. Conjugacy transforms iteration: if $h \circ f \circ h^{-1} = g$, then $f^n = h^{-1} \circ g^n \circ h$.
2. Quadratic maps like $2x(1 - x)$ can be centered at $1/2$ to reveal powers.
3. Recurrences involving radicals often simplify after setting a new variable such as $a_n = 1 + \sqrt{b_n}$.
4. Monotone iteration can identify pointwise limits by solving $L = f(L)$.
5. Exact formulas should be proved by induction after guessing from first cases.

Compact proof cores.

1. Conjugacy identity follows by induction on n.
2. Centered formula for logistic-like map follows by expanding around the fixed point.
3. Radical recurrence substitution is verified by direct algebra and induction.

How it appears in IMC work. **Use.** Use for iterated quadratic maps, nested radicals, and recurrence limits.

Trap watch. Verify the transformed recurrence preserves the original domain.

Exam checks.

- For iterates, seek a coordinate where the map becomes squaring, doubling, or shifting.

Occurrence count. 3 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1998-D1-P3;2003-D1-P5;2004-D1-P2

11.9.5 FEQ-05. Differentiable functional equations

Domain and bucket. Functional equations / dynamics; FEQ smooth. **Literature anchors.** [22, 14].

Prior compact node statement. When a functional equation includes differentiability, differentiate the equation, analyze scaling factors, and use fixed points or growth to force polynomial/exponential forms.

Exact facts to know.

1. Differentiable functional equations can be differentiated only after verifying differentiability of all composed terms.
2. If differentiability holds almost everywhere because of monotonicity, an equation may first imply monotonicity and then differential form.
3. Equations of form $f(x)f(yf(x)) = f(x+y)$ can lead to an ODE after dividing by y and taking $y \to 0^+$.
4. ODE solutions must be substituted back into the original equation to eliminate extraneous constants.
5. Initial or limiting value at 0^+ often emerges from the equation even if 0 is not in the domain.

Compact proof cores.

1. Difference quotient limits convert the functional equation to $f' = -af^2$ under the needed regularity.
2. Solving $f' = -af^2$ gives $1/f = ax + b$.
3. Substitution back fixes constants such as $b = 1$.

How it appears in IMC work. **Use.** Use for smooth equations such as $f(ax+b) = a^2 f(x)$ and derivative constraints.

Trap watch. Find the fixed point of the affine map before iterating or differentiating repeatedly.

Exam checks.

- Differentiating too early is dangerous; derive monotonicity or continuity first.

Occurrence count. 0 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances.

11.9.6 FEQ-06. Piecewise-linear maps and periodic points

Domain and bucket. Functional equations / dynamics; FEQ piecewise dynamics. **Literature anchors.** [14, 22].

Prior compact node statement. For piecewise monotone maps, the graph of f^n has many monotone branches; intersections with the diagonal count fixed points of iterates.

Exact facts to know.

1. For piecewise-linear maps, iterates are piecewise linear with branches determined by itineraries.
2. Periodic points of period dividing n are fixed points of f^n.
3. If every branch of f^n crosses the diagonal once, then $f^n(x) = x$ has one solution per branch.
4. Exact period n points are fixed points of f^n not fixed by any f^k for $k < n$.
5. Counting exact periodic points can be done by subtracting lower-period fixed points or bounding their total.

Compact proof cores.

1. Piecewise-linear branch count follows by induction on iteration.
2. Diagonal crossing uniqueness follows from monotonicity and endpoint values of each branch.
3. Exact-period nonemptiness follows when fixed points of f^n outnumber all lower-period fixed points.

How it appears in IMC work. **Use.** Use for proving existence and finiteness of exact n-periodic points.

Trap watch. Exact period requires excluding fixed points of smaller iterates.

Exam checks.

- Distinguish ”period dividing n” from ”exact period n”.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1998-D2-P3

11.9.7 FEQ-07. Bijections as products of involutions and symmetry decompositions

Domain and bucket. Functional equations / dynamics; FEQ involutions. **Literature anchors.** [22, 14].

Prior compact node statement. A bijection decomposes into disjoint orbits; each finite or infinite orbit can be modeled geometrically so the map is a product of two involutions.

Exact facts to know.

1. Every bijection decomposes into disjoint orbits under the cyclic action of its iterates.
2. On a finite cycle, a rotation is a product of two reflections/involutions.
3. On an infinite bi-infinite orbit indexed by $\mathbb{Z}$, the shift $m \mapsto m + 1$ is a product of two reflections.
4. An involution is a map g with $g^2 = \mathrm{id}$.
5. Constructing maps orbit-by-orbit gives global maps because orbits are disjoint.

Compact proof cores.

1. Orbit decomposition follows from equivalence relation $x \sim y$ iff $y = f^n(x)$ for some $n \in \mathbb{Z}$.
2. Finite-cycle reflection product can be checked geometrically or by modular arithmetic.
3. Infinite case: reflections about 0 and $1/2$ compose to a unit shift.

How it appears in IMC work. **Use.** Use for abstract set bijections and decompositions into symmetries.

Trap watch. Handle finite cycles and infinite orbits separately.

Exam checks.

- For arbitrary sets, avoid ordering the whole set; work on each orbit.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1997-D2-P5

11.9.8 FEQ-08. No-regularity functions and order patterns on Q or R

Domain and bucket. Functional equations / dynamics; FEQ no regularity. **Literature anchors.** [22, 14].

Prior compact node statement. For arbitrary functions without regularity, use order/combinatorial arguments rather than limits. On $\mathbb{Q}$, density and finite pigeonhole patterns can force local maxima-like triples.

Exact facts to know.

1. Without regularity, functions on $\mathbb{Q}$ or $\mathbb{R}$ can be defined using enumerations, Hamel-basis style ideas, or arbitrary choices.
2. For countable domains like $\mathbb{Q}$, one can dominate any two-variable function by constructing g along an enumeration.
3. For uncountable domains, diagonalization over all sequences/functions can build counterexamples.
4. Order patterns on dense sets can force linearity only when continuity or monotonicity is also present.
5. No-regularity counterexamples should be explicit about choice or enumeration.

Compact proof cores.

1. Enumeration domination: define $g(x)$ as a maximum over finitely many earlier indexed values.
2. Diagonal counterexample chooses a row of a two-variable function to exceed a proposed g at some indexed coordinate.
3. Density plus continuity extends rational identities to real identities.

How it appears in IMC work. **Use.** Use for functions from $\mathbb{Q}$ or arbitrary sets where no continuity is available.

Trap watch. Do not invoke IVT, compactness, or monotonicity unless it is part of the statement.

Exam checks.

- Ask first: is the domain countable and is any regularity assumed?

Occurrence count. 6 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1999-D1-P3;1999-D2-P4;2001-D2-P5;2002-D1-P2;2003-D2-P5;2024-D2-P6

11.9.9 FEQ-09. Image-length preserving functions

Domain and bucket. Functional equations / dynamics; FEQ interval images. **Literature anchors.** [22, 14].

Prior compact node statement. If every interval image is a closed interval of the same length, the function is 1-Lipschitz and endpoint extrema force monotonicity; hence $f(x) = x + c$ or $f(x) = -x + c$.

Exact facts to know.

1. If $f([a, b])$ is a closed interval of length $b - a$ for every $a < b$, then $|f(x) - f(y)| \leq |x - y|$ for all x, y.
2. The previous inequality implies f is 1-Lipschitz and hence continuous.
3. If the image interval has full length $b-a$, the maximum and minimum over $[a, b]$ must be attained at endpoints.
4. Therefore f is monotone.
5. A monotone 1-Lipschitz function whose every interval image has length equal to the interval length is $f(x) = x + c$ or $f(x) = -x + c$.

Compact proof cores.

1. Lipschitz follows because $f(x), f(y)$ lie in an image interval of length $|x - y|$.
2. Endpoint-extremum argument: if max/min occur at interior points, their distance is shorter than $b - a$, contradiction.
3. Monotone plus equality of interval lengths gives $f(y) - f(x) = \pm(y - x)$ with constant sign, hence affine.

How it appears in IMC work. **Use.** Use for functions preserving interval lengths or ranges.

Trap watch. The maximum/minimum on an interval must occur at endpoints only after equality in the Lipschitz chain is analyzed.

Exam checks.

- Image-length preservation is a rigidity condition; convert it to Lipschitz plus equality cases.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2006-D2-P2

11.10 Probability / measure / expectation

Foundation checklist. Probability problems in the corpus are elementary but structurally useful: encode events by residues, use generating functions, apply linearity of expectation, or translate a probabilistic statement into counting. Always distinguish exact probability from asymptotic probability.

11.10.1 PRB-01. Discrete probability via residues and recurrences

Domain and bucket. Probability / measure / expectation; PRB discrete. **Literature anchors.** [16, 8].

Prior compact node statement. A Markov recurrence on residues or a generating function can compute probabilities of sums modulo m.

Exact facts to know.

1. For sums modulo m, define state probabilities $p_n^{(r)}$ and write a transition recurrence.
2. If increments are equally likely, the transition matrix is circulant and often diagonalized by roots of unity.
3. Exceptional sequences can be grouped into collections representing all residues equally.
4. Exact probability equals favorable sequences divided by total sequences.
5. Initial condition at $n = 0$ is concentrated at residue 0.

Compact proof cores.

1. Transition recurrence sums over possible die results or increments.
2. Roots-of-unity diagonalization follows from character orthogonality.
3. Grouping proof partitions all sequences except a special one into residue-balanced classes.

How it appears in IMC work. **Use.** Use for dice sums, random walks modulo primes, and residue probabilities.

Trap watch. State transition probabilities must sum to one and residues are modulo the correct integer.

Exam checks.

- For modulo probability, decide between recurrence, grouping, or generating-function filter.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1999-D2-P2

11.10.2 PRB-02. Generating functions for probability distributions

Domain and bucket. Probability / measure / expectation; PRB generating functions. **Literature anchors.** [16, 8].

Prior compact node statement. The probability generating function encodes sums of independent variables as products. Coefficients or root-of-unity filters extract desired events.

Exact facts to know.

1. Probability generating function of a nonnegative integer-valued random variable X is $G_X(t) = E[t^X]$.
2. For independent sums, generating functions multiply.
3. Coefficient $[t^k]G(t)$ gives $P(X = k)$ when G is a probability generating polynomial/series.
4. Root-of-unity filters extract probabilities of congruence classes.
5. For a fair die, one-step generating function is $(t + t^2 + \cdots + t^6)/6$.

Compact proof cores.

1. Multiplication follows from $E[t^{X+Y}] = E[t^X]E[t^Y]$ for independent X, Y.
2. Coefficient extraction is the definition of the generating polynomial.
3. Congruence extraction uses $\frac{1}{m}\sum \omega^{r(X-a)}$ as an indicator.

How it appears in IMC work. **Use.** Use for sums of independent discrete variables and random subsets.

Trap watch. Normalize coefficients by the total number of outcomes.

Exam checks.

- Use PGFs when independent repeated trials appear.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2022-D1-P1

11.10.3 PRB-03. Expectation, linearity, and probabilistic method

Domain and bucket. Probability / measure / expectation; PRB expectation. **Literature anchors.** [16, 8].

Prior compact node statement. Linearity of expectation requires no independence. A positive average proves existence of an object at least as good as the average.

Exact facts to know.

1. Linearity of expectation: $E[\sum X_i] = \sum E[X_i]$ without independence.
2. Indicator variables convert counts into sums of $0/1$ variables.
3. Probabilistic method: if expected number of bad events is less than 1, some outcome has no bad event.
4. Symmetry can compute probabilities without enumerating all outcomes.
5. Variance or second-moment methods are rarely needed in the IMC corpus but expectation is common in disguised counting.

Compact proof cores.

1. Linearity follows from finite sums or from integral linearity.
2. Indicator expectation is $P(A)$.
3. Probabilistic existence follows because a nonnegative integer-valued random variable with expectation < 1 must be zero for some outcome.

How it appears in IMC work. **Use.** Use for random choices, expected counts, and probabilistic existence.

Trap watch. Independence is only needed for products or distribution factorization, not for linearity.

Exam checks.

- When a problem asks for existence of a configuration, consider counting bad events by expectation.

Occurrence count. 1 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 2021-D1-P2

11.10.4 PRB-04. Continuous uniform distribution and geometric probability

Domain and bucket. Probability / measure / expectation; PRB geometry. **Literature anchors.** [16, 8].

Prior compact node statement. Continuous geometric probability is volume/area normalized by the ambient domain. Symmetry, conditioning, and convex hull geometry are typical reductions.

Exact facts to know.

1. Uniform distribution on a region assigns probability proportional to volume.
2. Geometric probability often reduces to computing the measure of a subset defined by inequalities.
3. Affine transformations scale numerator and denominator volumes by the same factor when applied to the whole sample space.
4. Boundary sets of lower dimension have probability zero under continuous uniform distributions.
5. Independent continuous points give product measure on the Cartesian product of regions.

Compact proof cores.

1. Probability proportional to volume is the definition of uniform distribution on finite-volume regions.
2. Boundary-zero claims follow because hyperplanes and smooth surfaces have Lebesgue measure zero.
3. Affine invariance follows from determinant scaling in both favorable and total volumes.

How it appears in IMC work. **Use.** Use for random points in disks, balls, circles, and convex hulls.

Trap watch. Events involving exact coincidences on continuous distributions often have probability zero.

Exam checks.

- Normalize geometry first, then integrate.

Occurrence count. 0 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances.

11.10.5 PRB-05. Measure, variation, and differentiability almost everywhere

Domain and bucket. Probability / measure / expectation; PRB measure. **Literature anchors.** [16, 8].

Prior compact node statement. Finite total variation implies differentiability almost everywhere; covering arguments with small total length can force at least one point with small difference quotients.

Exact facts to know.

1. A function of bounded variation on an interval is differentiable almost everywhere, but many IMC solutions avoid this theorem by direct interval arguments.
2. If $\sum b_n < \infty$, intervals centered at a_n with lengths proportional to $c_n b_n$ can have total length less than one after choosing $c_n \to \infty$ slowly.
3. Outside all such intervals, spike heights divided by distances can tend to zero, giving differentiability with derivative zero.
4. If $\sum b_n = \infty$, intervals with radii comparable to b_n can be arranged to hit every point infinitely often.
5. Infinite hitting by spikes prevents differentiability with derivative zero at every point.

Compact proof cores.

1. Summable case: choose $c_n \to \infty$ with $\sum c_n b_n < \infty$ by grouping tails.
2. Then total interval length can be made < 1, leaving a point outside all intervals.
3. Divergent case uses a covering scheduling so arbitrarily small spikes occur near each point.

How it appears in IMC work. **Use.** Use for sparse functions supported on countable sets and differentiability existence/nonexistence.

Trap watch. A countable exceptional support can still be dense; use interval covers, not isolation.

Exam checks.

- Differentiability of spike functions is controlled by height-to-distance ratios.

Occurrence count. 2 tagged problem entries. The full problem-to-theory matrix earlier in this booklet remains the authoritative cross-reference.

Tagged appearances. 1998-D2-P6;2022-D1-P1

11.11 Coverage audit for this expansion

The table below records that every theory node from the previous theory-node index has a corresponding detailed property entry in this section. Nodes with zero tagged occurrences are retained because they were part of the method/theory scaffold and are useful for nearby variants.

Node	Theory point	Tags	Status
LA-01	Matrix products, inverses, and one-sided inverse identities	14	expanded
LA-02	Determinant evaluation by row/column operations and triangular forms	51	expanded
LA-03	Block matrices, Schur complements, and determinant identities	114	expanded
LA-04	Rank-nullity, image/kernel, and rank of factorizations	87	expanded
LA-05	Trace cyclicity, trace pairing, and zero-trace subspaces	33	expanded
LA-06	Commutators and spectral obstructions	20	expanded
LA-07	Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton	51	expanded
LA-08	Nilpotent, idempotent, involution, and square-zero structures	14	expanded
LA-09	Simultaneous diagonalization and commuting families	3	expanded
LA-10	Invariant subspaces and adapted bases	30	expanded
LA-11	Linear functionals, kernels, and dual span criteria	6	expanded
LA-12	Matrix subspaces constrained by trace bilinear forms	1	expanded
LA-13	Matrix exponentials and polynomial functional calculus	1	expanded
LA-14	Operator norm, power boundedness, and spectral-radius style estimates	1	expanded
LA-15	Gram matrices and rational linear dependence	7	expanded
LA-16	Anticommuting and rank constraints for special matrices	0	expanded
RA-01	Compactness, extreme values, and closed-bounded reduction	37	expanded
RA-02	Intermediate value, connectedness, density, and continuity transfer	44	expanded
RA-03	Monotone functions, fixed points, and iteration on intervals	15	expanded
RA-04	Rolle, mean-value theorem, and divided differences	33	expanded
RA-05	Taylor expansion, local asymptotics, and little-o bookkeeping	50	expanded
RA-06	Auxiliary functions, integrating factors, and differential inequalities	62	expanded
RA-07	Convexity, concavity, Jensen, and tangent/chord inequalities	179	expanded
RA-08	Integral inequalities, integration by parts, and weighted estimates	199	expanded
RA-09	Fubini/Tonelli and change of variables	39	expanded
RA-10	Riemann sums and sum-integral asymptotics	73	expanded
RA-11	Series convergence, rearrangements, and alternating/Dirichlet tools	67	expanded
RA-12	Uniform continuity and periodic averaging	1	expanded
RA-13	L'Hospital, ratio limits, and growth comparisons	33	expanded
RA-14	Weierstrass approximation and polynomial approximation	1	expanded
RA-15	L2 spaces, orthonormal systems, and dominated convergence	1	expanded
RA-16	Baire/category, dense sets, and pathological continuous functions	6	expanded
RA-17	Sequence recurrences, monotone bounds, and generating functions	22	expanded
RA-18	Trigonometric and hyperbolic identities with monotonicity	14	expanded
RA-19	Gradient/Lipschitz-gradient optimization and convex smoothness	3	expanded
RA-20	Measure and probability-integral estimates	2	expanded
CP-01	Polynomial identity and more-roots-than-degree principle	68	expanded
CP-02	Factorization, irreducibility, and Gauss-style integer polynomial arguments	24	expanded
CP-03	Vieta, logarithmic derivatives, and real-root inequalities	28	expanded

Node	Theory point	Tags	Status
CP-04	Unit-circle roots and conjugation identities	13	expanded
CP-05	Roots of unity, cyclotomic filters, and discrete Fourier tricks	24	expanded
CP-06	Cauchy formula and coefficient extraction on the unit circle	7	expanded
CP-07	Maximum modulus principle	0	expanded
CP-08	Laurent expansions, isolated singularities, and rationality of meromorphic functions	1	expanded
CP-09	Fejer kernels and positive trigonometric kernels	1	expanded
CP-10	Interpolation, Lagrange basis, and Chebyshev extremality	2	expanded
CP-11	Stable polynomials and coefficient inequalities	3	expanded
CP-12	Partial fractions and cotangent-type expansions	1	expanded
CP-13	Polynomial functional identities and algebraic composition	8	expanded
ANT-01	Modular arithmetic and divisibility	92	expanded
ANT-02	Chinese remainder theorem and complete residue systems	30	expanded
ANT-03	Finite fields, field characteristic, and Fermat/Frobenius tools	49	expanded
ANT-04	Integer polynomials, finite differences, and value divisibility	22	expanded
ANT-05	Rational roots, Vieta, and rationality constraints	5	expanded
ANT-06	GCD, valuations, prime powers, and coprimality	42	expanded
ANT-07	Group basics: orders, cyclic subgroups, and generators	52	expanded
ANT-08	Permutation groups, cycles, parity, and generation	2	expanded
ANT-09	Normal subgroups, quotient groups, and group actions	12	expanded
ANT-10	Automorphisms and semidirect products	27	expanded
ANT-11	Rings, idempotents, nil rings, and noncommutativity	33	expanded
ANT-12	Diophantine equations, Pell-type recurrences, and quadratic irrationals	2	expanded
ANT-13	Lattices and norms/quasinorms	2	expanded
ANT-14	Greedy infinite products and rationality tails	1	expanded
ANT-15	Finite abelian groups and vector spaces over finite fields	1	expanded
CMB-01	Pigeonhole principle and extremal element selection	68	expanded
CMB-02	Double counting and degree-sum methods	56	expanded
CMB-03	Graph encoding, cycles, and bipartite representations	68	expanded
CMB-04	Set systems, intersecting families, and induction on sizes	23	expanded
CMB-05	Recurrences and state partitions	4	expanded
CMB-06	Generating functions for counting	14	expanded
CMB-07	Inclusion-exclusion and root-of-unity filters	1	expanded
CMB-08	Permutations, inversions, and permutation statistics	5	expanded
CMB-09	Explicit construction and counterexample design	36	expanded
CMB-10	Erdos-Szekeres style monotone subsequences and binomial thresholds	1	expanded
CMB-11	Grid, lattice, and alternating-cycle arguments	2	expanded
CMB-12	Coloring, partitions, and parity	48	expanded
CMB-13	Words, equivalence relations, and rewriting systems	3	expanded
CMB-14	Transfer matrices and matrix powers for counting	1	expanded
CMB-15	Finite designs and pair-determines-third structures	1	expanded
GEO-01	Coordinate and affine normalization	2	expanded
GEO-02	Euclidean distances and inner-product identities	19	expanded
GEO-03	Projections and orthogonal decompositions	25	expanded
GEO-04	Convex bodies, supporting geometry, and John-ellipse ideas	7	expanded
GEO-05	Conics: parabolas, ellipses, foci, and directrices	4	expanded
GEO-06	Simplex, barycentric coordinates, and circumcenters	6	expanded
GEO-07	Area, volume, projection, and ellipsoid computation	28	expanded
GEO-08	Metric content, diameter covers, and contractions	1	expanded
GEO-09	Hilbert-space equidistant sets and orthonormal translations	6	expanded

Node	Theory point	Tags	Status
GEO-10	Closest points to closed sets and density of uniqueness	1	expanded
GEO-11	Unit-sphere configurations and regular simplex bounds	1	expanded
GEO-12	Polygon triangulation and parity of incidences	1	expanded
GEO-13	Convex hulls, random polytopes, and geometric probability	1	expanded
GEO-14	Cube and square dissection by number-theoretic semigroups	1	expanded
FEQ-01	Substitution of special values and normalization	7	expanded
FEQ-02	Injectivity and monotonicity forced by a functional equation	8	expanded
FEQ-03	Fixed points, iterates, and orbit closure	6	expanded
FEQ-04	Recurrence transformations and conjugacy	3	expanded
FEQ-05	Differentiable functional equations	0	expanded
FEQ-06	Piecewise-linear maps and periodic points	1	expanded
FEQ-07	Bijections as products of involutions and symmetry decompositions	1	expanded
FEQ-08	No-regularity functions and order patterns on Q or R	6	expanded
FEQ-09	Image-length preserving functions	1	expanded
PRB-01	Discrete probability via residues and recurrences	1	expanded
PRB-02	Generating functions for probability distributions	1	expanded
PRB-03	Expectation, linearity, and probabilistic method	1	expanded
PRB-04	Continuous uniform distribution and geometric probability	0	expanded
PRB-05	Measure, variation, and differentiability almost everywhere	2	expanded

12 Theory-to-Database Extraction Plan

The next stage should split this enhanced booklet into separate working databases. The present ZIP already contains the two core tables needed for that step.

Database	What to extract from this booklet
Theory database	One row per theory node: statement, proof prerequisites, examples, and occurrence list.
Method database	For each theory node, extract the execution move: e.g. trace cyclicity, adapted basis, root-of-unity filter, Jensen tangent line, Riemann-sum split.
Trigger database	From the problem-to-theory matrix, record the wording that activates each node: e.g. “commutator”, “all roots real”, “unit circle”, “closed bounded”, “exactly one triple”.
Drill database	For each problem, use the main node as the drill category and the supporting nodes as prerequisites.
Mistake database	Use the trap-watch lines in the detailed nodes as the first list of recurring failure modes.

AUDIT CLOSURE REPLACED — residual errors may remain; independent mathematical review is required.

13 Coverage Audit and Missing-Theory Closure Addendum

13.1 Audit protocol

This section was added after re-checking the expanded theory booklet against the merged IMC problem corpus and the existing problem-to-theory matrix. The goal of the audit was not to duplicate previous material, but to close theory gaps where the earlier booklet used a broad tag such as “linear algebra”, “convexity”, or “polynomial roots” while an actual IMC solution needs a more precise theorem, property, or compact proof pattern.

The audit used three checks:

1. **Problem-level sufficiency.** For every detected problem entry, ask whether the previous theory manual plus its property cards contains the specific theorem needed to start and finish the official-style solution.
2. **Property-level specificity.** When a solution uses a named or hidden fact - for example trace cyclicity, Fejer kernel positivity, the squarefree-minimal-polynomial criterion, or a finite-field root count - the exact statement and proof core must appear explicitly.
3. **Occurrence retention.** Every added theory closure below records where it appears, using the same IMC problem IDs as the previous catalog. Full problem-level mappings are also included in the CSV/JSON audit files in the source package.

Audit outcome. The previous booklet already contained the broad theory map. The following addendum strengthens it with exact theory closures. After these closures, the remaining gaps are not theory gaps but method-execution gaps: choosing the right lemma, arranging the algebra, and writing a complete proof under time pressure.

13.2 High-impact correction and strengthening notes

The following entries do not remove earlier tags; they refine them. They mark places where a broad previous tag was mathematically true but insufficiently precise for a self-contained theory booklet.

Problem	Added or corrected theory requirement
1994-D1-P1	Add positive weighted sign-cancellation lemma for inverse columns; previous convexity tag was too broad.
1994-D1-P3	Add rational cone/induction obstruction; not an integral or graph problem.
1995-D1-P1	Add shift/Jordan nilpotent matrix model $Y = XJ$.
1995-D2-P1	Add polarization step from $(Au, u) = 0$ to skew-symmetry and cross-product parametrization.
1995-D2-P3	Add unit-circle logarithmic-derivative real-part identity.
1996-D1-P3	Add simultaneous diagonalization of commuting involutions.
1996-D1-P5	Add endpoint Laplace principle for $n \int f^n$.
1996-D2-P6	Add weighted AM-GM/Carleman constant mechanism and near-extremal sequence.
1997-D2-P4	Add trace pairing and commutator subspace equals trace-zero matrices.
1998-D1-P5	Add Laguerre inequality for real-rooted polynomials.
1999-D2-P5	Add idempotent word-reduction lemma for semigroups.
2000-D2-P6	Add functional calculus for $q(e^{AB})$ and AC/CA spectra.
2001-D1-P4	Add cyclotomic irreducibility and product $\prod(1 - \zeta^j) = q$.
2002-D1-P6	Add operator-norm sequence lemma separate from finite-dimensional spectral theory.
2002-D2-P6	Add Lipschitz-gradient co-coercivity proof.
2004-D1-P6	Add cotangent partial-fraction/log-branch meromorphic argument.
2005-D2-P6	Add Pell-equation stabilizer of a quadratic irrational.
2007-D2-P5	Add 2^k lower bound for commuting square-zero matrices.
2008-D2-P6	Add Hilbert equidistant-set barycenter/Cauchy construction.
2009-D1-P4	Add Fejer kernel positivity for convex coefficient multipliers.
2009-D2-P3	Add derivation binomial formula for iterated commutators.
2013-D2-P4	Add square-free pair construction via probabilistic/CRT selection, separate from generic number theory.
2016-D2-P10	Add Jordan/norm growth estimate for matrices with spectral radius at most one.
2020-D1-P2	Add rank-one perturbation plus trace identity for commutator expressions.
2022-D2-P7	Add anticommuting idempotents/rank averaging theory.

Problem	Added or corrected theory requirement
2024-D2-P10	Add square-free almost-prime divisor condition and finite-field check for all divisors.

13.3 New closure-node index

ID	Domain	Closure theorem/property	Occ.
AUD-LA-01	Linear algebra	Positive inverse sign lemma and zero-count columns	1
AUD-LA-02	Linear algebra	Shift matrices, nilpotent Jordan blocks, and similarity	8
AUD-LA-03	Linear algebra	Commutators, trace cyclicity, and nilpotent commutators	25
AUD-LA-04	Linear algebra	Simultaneous diagonalization of commuting diagonalizable operators	1
AUD-LA-05	Linear algebra	Skew-symmetric matrices and the cross product in three dimensions	1
AUD-LA-06	Linear algebra	Trace pairing and the dual of the full matrix space	1
AUD-LA-07	Linear algebra	Squarefree minimal polynomials and convergence of matrix powers	11
AUD-LA-08	Linear algebra	Sylvester identities and nonzero spectra of AB and BA	5
AUD-LA-09	Linear algebra	Functional calculus for polynomials and matrix exponentials	1
AUD-LA-10	Linear algebra	Real matrices, conjugate eigenpairs, and determinant sign	3
AUD-LA-11	Linear algebra	Symmetric/skew decomposition and trace-isotropic subspaces	2
AUD-LA-12	Linear algebra	Commuting square-zero matrices and the exterior-algebra model	5
AUD-LA-13	Linear algebra	Derivation binomial formula for iterated commutators	2
AUD-LA-14	Linear algebra	Covering matrix spaces and low-rank elements	0
AUD-LA-15	Linear algebra	Kronecker-sum/Lyapunov operator spectrum	1
AUD-LA-16	Linear algebra	Complex conjugation cocycles: solving $A\overline{A} = I$ as $A = S\overline{S}^{-1}$	1
AUD-RA-01	Real analysis	First-order differential inequalities and integrating factors	31
AUD-RA-02	Real analysis	Gronwall-type zero uniqueness	2
AUD-RA-03	Real analysis	Periodic averaging and equidistributed Riemann blocks	55
AUD-RA-04	Real analysis	Endpoint Laplace principle for $n\int f^n$	32
AUD-RA-05	Real analysis	Riemann sums on infinite intervals and tail control	52
AUD-RA-06	Real analysis	Rolle chains and divided differences with repeated nodes	60
AUD-RA-07	Real analysis	Compactness, extrema, and localization on unbounded domains	32
AUD-RA-08	Real analysis	Convexity, strict convexity, and equality cases	228
AUD-RA-09	Real analysis	Dominated convergence in bounded orthonormal systems	1
AUD-RA-10	Real analysis	Monotone functions, discontinuities, and range gaps	7
AUD-RA-11	Real analysis	Baire category and closed orbits of interval maps	2
AUD-RA-12	Real analysis	Cantor sets and continuous functions crossing zero uncountably often	1
AUD-RA-13	Real analysis	Nowhere monotone functions and dense local extrema	2
AUD-RA-14	Real analysis	Sparse spike functions and differentiability by interval covers	1

ID	Domain	Closure theorem/property	Occ.
AUD-RA-15	Real analysis	Lipschitz gradients and co-coercivity of convex functions	2
AUD-RA-16	Real analysis	Conditionally convergent series under structured rearrangements	1
AUD-PC-01	Polynomials/complex	Polynomial identity, roots versus degree, and finite-field vanishing	107
AUD-PC-02	Polynomials/complex	Vieta, logarithmic derivatives, and real-rooted Laguerre inequality	1
AUD-PC-03	Polynomials/complex	Unit-circle root localization via logarithmic derivative	1
AUD-PC-04	Polynomials/complex	Hurwitz stability and coefficient log-concavity type inequalities	1
AUD-PC-05	Polynomials/complex	Cyclotomic irreducibility and roots-of-unity divisibility	28
AUD-PC-06	Polynomials/complex	Root-of-unity filters for congruence classes	10
AUD-PC-07	Polynomials/complex	Fejer kernels and positive trigonometric multipliers	1
AUD-PC-08	Polynomials/complex	Chebyshev interpolation extremality on $[-1,1]$	54
AUD-PC-09	Polynomials/complex	Weierstrass approximation with symmetry projection	1
AUD-PC-10	Polynomials/complex	Cotangent partial fractions and logarithm-branch sums	1
AUD-PC-11	Polynomials/complex	Integer-valued congruence of polynomial values	4
AUD-PC-12	Polynomials/complex	Rational functions with many integer values	0
AUD-PC-13	Polynomials/complex	Ordinary generating functions for recurrences and weighted sums	69
AUD-PC-14	Polynomials/complex	Pell equations and quadratic irrational stabilizers in $\mathrm{SL}_2(\mathbb{Z})$	2
AUD-ANT-01	Algebra/number theory	Chinese remainder theorem and idempotents modulo prime powers	3
AUD-ANT-02	Algebra/number theory	Finite fields, Fermat, and polynomial permutations	95
AUD-ANT-03	Algebra/number theory	Quartic residues and modular obstruction by tables	1
AUD-ANT-04	Algebra/number theory	Field characteristic and parity of additive inverse pairing	2
AUD-ANT-05	Algebra/number theory	Gauss lemma and factorization of integer polynomials	1
AUD-ANT-06	Algebra/number theory	Power subgroups, commutators, and central intersections	7
AUD-ANT-07	Algebra/number theory	Noncommutative rings: square-zero and idempotent identities	2
AUD-ANT-08	Algebra/number theory	Permutation-group generators and parity/normal-subgroup obstructions	3
AUD-ANT-09	Algebra/number theory	Semidirect products and automorphism-count counterexamples	0
AUD-ANT-10	Algebra/number theory	Elementary abelian 2-groups and Steiner triple systems	2
AUD-ANT-11	Algebra/number theory	Dyadic-rational subgroups inside upper triangular matrices	2
AUD-ANT-12	Algebra/number theory	Words modulo idempotent relations	2
AUD-CMB-01	Combinatorics	Double counting walks, degree squares, and graph girth	1
AUD-CMB-02	Combinatorics	Bipartite graph encoding of grids and alternating cycles	1
AUD-CMB-03	Combinatorics	Pigeonhole with divisor values and polynomial fibers	52
AUD-CMB-04	Combinatorics	Intersecting finite set families and inductive hitting sets	2

ID	Domain	Closure theorem/property	Occ.
AUD-CMB-05	Combinatorics	Erdos-Szekeres style monotone subsequences with growth gaps	11
AUD-CMB-06	Combinatorics	Transfer matrices for binary tables and symmetry of dimensions	1
AUD-CMB-07	Combinatorics	State recurrences and last-symbol classification	47
AUD-CMB-08	Combinatorics	Permutation statistics via determinants	12
AUD-CMB-09	Combinatorics	Residue-class grouping of words and dice sequences	1
AUD-GEO-01	Geometry	Gram matrices and rational dependence from rational distances	1
AUD-GEO-02	Geometry	Projection, signed area, and volume scaling	26
AUD-GEO-03	Geometry	Simplex volume, barycentric coordinates, and circumcenter identity	2
AUD-GEO-04	Geometry	Regular simplex bound on spherical codes with negative inner products	2
AUD-GEO-05	Geometry	Ellipse focus-directrix form and conic intersections	1
AUD-GEO-06	Geometry	John ellipse contact-arc obstruction	1
AUD-GEO-07	Geometry	Metric content, diameter covers, and Lipschitz contractions	1
AUD-GEO-08	Geometry	Closest-point uniqueness is dense for closed sets	2
AUD-GEO-09	Geometry	Countability by rational encoding of sphere intersections	4
AUD-GEO-10	Geometry	Polygon triangulation parity	2
AUD-GEO-11	Geometry	Cube dissection and the Frobenius coin lemma	2
AUD-PRB-01	Probability/measure	Expectation, symmetry, and geometric probability in convex hulls	5
AUD-PRB-02	Probability/measure	Measure-preserving symmetry and integrable reciprocal inequalities	1

14 Added Exact Theory Closures

Each card below is intended to be directly usable while solving past IMC problems. The statements are compact but explicit: they include hypotheses, proof core, use pattern, and occurrence references.

AUD-LA-01 - Positive inverse sign lemma and zero-count columns

Domain. Linear algebra. **References.** Horn-Johnson; Hoffman-Kunze.

Gap closed. The previous inverse-matrix entry stated algebraic identities but did not state the sign argument needed when a positive row is orthogonal to a column of the inverse.

Statement/properties. If $a_1, \dots, a_n > 0$ and $\sum_i a_i b_i = 0$ with not all b_i zero, then the vector b has at least one positive and at least one negative component. Consequently a nonzero column orthogonal to a strictly positive row has at least two nonzero entries. For an invertible matrix $A = (a_{ij})$ with $A^{-1} = B = (b_{ij})$, the identity $AB = I$ gives $\sum_i a_{ki} b_{im} = \delta_{km}$. For $k \neq m$, each column m of B is orthogonal to every non-matching row of A.

Compact proof core. If all nonzero b_i had the same sign, then $\sum a_i b_i$ would have that sign because all weights a_i are positive. Hence cancellation to zero requires both signs. Applied column-by-column to $AB = I$, this proves the required nonzero-count lower bounds.

Use pattern. Use whenever a matrix has positive entries and an inverse column is forced to have zero scalar product with a positive vector. The invariant is not determinant size but sign cancellation.

Occurrence references. 1994-D1-P1.

AUD-LA-02 - Shift matrices, nilpotent Jordan blocks, and similarity

Domain. Linear algebra. **References.** Horn-Johnson; Axler.

Gap closed. Nilpotent shift matrices appeared in the corpus but the previous theory section did not explicitly isolate their rank, eigenvalues, and similarity use.

Statement/properties. Let J be the $n \times n$ shift matrix with $Je_j = e_{j+1}$ for $1 \leq j < n$ and $Je_n = 0$. Then $\operatorname{rank} J = n - 1$, $J^n = 0$, and all eigenvalues of J are 0. If X is invertible, then XJX^{-1} has the same rank, characteristic polynomial, minimal polynomial, nilpotency index, and Jordan block structure as J.

Compact proof core. The image of J is $\operatorname{span}(e_2, \dots, e_n)$ and its kernel is $\operatorname{span}(e_n)$, giving rank $n - 1$. The equation $J^n = 0$ implies every eigenvalue λ satisfies $\lambda^n = 0$. Similarity preserves all linear-map invariants because it is only a change of basis.

Use pattern. When columns are shifted, write the new matrix as $Y = XJ$; then $YX^{-1} = XJX^{-1}$ and $X^{-1}Y = J$.

Occurrence references. 1995-D1-P1; 1995-D1-P5; 2000-D1-P5; 2000-D2-P6; 2003-D1-P3; 2005-D1-P6; 2009-D2-P3; 2022-D2-P7.

AUD-LA-03 - Commutators, trace cyclicity, and nilpotent commutators

Domain. Linear algebra. **References.** Horn-Johnson; Hoffman-Kunze.

Gap closed. Several problems rely on commutators not merely as notation but on cyclic trace, derivations, and spectral consequences.

Statement/properties. For square matrices over a field of characteristic 0, $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ and therefore $\operatorname{tr}(AB - BA) = 0$. More generally, $\operatorname{tr}(XYZ) = \operatorname{tr}(ZXY) = \operatorname{tr}(YZX)$. If C has rank 1 and trace 0, then all eigenvalues of C are 0, so C is nilpotent; over $\mathbb{C}$, rank 1 then implies $C^2 = 0$. If $\Delta(X) = AX - XA$, then Δ is a derivation: $\Delta(XY) = \Delta(X)Y + X\Delta(Y)$.

Compact proof core. Trace cyclicity follows by summing entries: $\sum_i (AB)_{ii} = \sum_{i,j} a_{ij} b_{ji} = \sum_j (BA)_{jj}$. A rank-one matrix has at most one nonzero eigenvalue; the sum of eigenvalues equals the trace, hence this possible nonzero eigenvalue is also zero. The derivation identity follows by adding and subtracting XAY.

Use pattern. Use for equations involving $AB - BA$, $A^2B + BA^2 - 2ABA$, or rank-one commutators. First prove a trace/eigenvalue obstruction, then apply Cayley-Hamilton or Jordan form.

Occurrence references. 1995-D1-P5; 1997-D2-P2; 1997-D2-P4; 1998-D1-P1; 1999-D2-P1; 2000-D1-P3; 2000-D1-P5; 2000-D2-P6; 2001-D1-P2; 2001-D1-P5; 2003-D1-P3; 2003-D2-P1; 2004-D2-P1; 2005-D1-P6; 2005-D2-P3; 2005-D2-P6; 2006-D2-P6; 2009-D2-P3; 2011-D1-P2; 2014-D1-P1; 2018-D1-P3; 2019-D2-P9; (+3 more in CSV).

AUD-LA-04 - Simultaneous diagonalization of commuting diagonalizable operators

Domain. Linear algebra. **References.** Axler; Hoffman-Kunze.

Gap closed. The involution problems require the exact common-eigenbasis theorem, not just diagonalization of one operator.

Statement/properties. If $T_1, \dots, T_m$ are pairwise commuting diagonalizable operators on a finite-dimensional vector space over an algebraically closed field, then they are simultaneously diagonalizable. An involution $A^2 = I$ over $\mathbb{R}$ is diagonalizable with eigenvalues in $\{1, -1\}$ because its minimal polynomial divides $(x-1)(x+1)$, which has distinct roots.

Compact proof core. For one operator T_1, decompose the space into eigenspaces. Since $T_i T_1 = T_1 T_i$, each eigenspace of T_1 is invariant under every T_i. Apply induction on dimension inside each eigenspace. The involution claim follows from the squarefree-minimal-polynomial criterion.

Use pattern. Use to count commuting involutions: in a common eigenbasis each operator is a diagonal sign matrix, so at most 2^n distinct involutions occur.

Occurrence references. 1996-D1-P3.

AUD-LA-05 - Skew-symmetric matrices and the cross product in three dimensions

Domain. Linear algebra. **References.** Horn-Johnson; Coxeter.

Gap closed. The previous matrix theory did not state the polarization step from $(Au, u) = 0$ to skew-symmetry, nor the 3-dimensional cross-product representation.

Statement/properties. A real matrix A satisfies $\langle Au, u \rangle = 0$ for every u iff $A^T = -A$. In $\mathbb{R}^3$, every skew-symmetric matrix has the form

$$A = \begin{pmatrix} 0 & -v_3 & v_2 \\ v_3 & 0 & -v_1 \\ -v_2 & v_1 & 0 \end{pmatrix},$$

and then $Au = v \times u$ for all $u \in \mathbb{R}^3$.

Compact proof core. Taking $u = e_i$ gives diagonal entries zero. Taking $u = e_i + e_j$ gives $a_{ij} + a_{ji} = 0$. The displayed matrix is obtained by reading off $v_1 = -a_{23}$, $v_2 = a_{13}$, $v_3 = -a_{12}$.

Use pattern. Use when all vectors are orthogonal to their images, especially in 3D problems with the vector product.

Occurrence references. 1995-D2-P1.

AUD-LA-06 - Trace pairing and the dual of the full matrix space

Domain. Linear algebra. **References.** Horn-Johnson; Hoffman-Kunze.

Gap closed. The trace pairing was used for linear functionals on matrix spaces and for maximal trace-isotropic subspaces, but it needed an explicit nondegeneracy statement.

Statement/properties. The bilinear form $\langle A, B \rangle = \operatorname{tr}(AB)$ on $M_n(F)$ is nondegenerate. Therefore every linear functional $\varphi : M_n(F) \to F$ has a unique representation $\varphi(A) = \operatorname{tr}(AC)$ for some $C \in M_n(F)$. The commutator subspace $[M_n, M_n]$ is exactly the trace-zero subspace.

Compact proof core. For elementary matrices, $\operatorname{tr}(E_{ij}C) = c_{ji}$, so C is determined by values of φ on E_{ij}. The trace-zero basis consists of E_{ij} for $i \neq j$ and $E_{ii} - E_{nn}$; each is an explicit commutator.

Use pattern. Use to turn an abstract linear functional into a matrix, and to prove that any functional invariant under cyclic products is a scalar multiple of trace.

Occurrence references. 1997-D2-P4.

AUD-LA-07 - Squarefree minimal polynomials and convergence of matrix powers

Domain. Linear algebra. **References.** Axler; Horn-Johnson.

Gap closed. Matrix-power convergence problems require the exact squarefree-minimal-polynomial and spectral-radius criteria.

Statement/properties. If the minimal polynomial of A splits into distinct linear factors, then A is diagonalizable. If $A = SDS^{-1}$ and every eigenvalue of A has modulus < 1 or equals 1 with no nontrivial Jordan block, then A^k converges to the projection onto the 1-eigenspace along the remaining eigenspaces. The limit is idempotent.

Compact proof core. The squarefree criterion follows from primary decomposition. For diagonal D, powers act entrywise: $\lambda^k \to 0$ when $|\lambda| < 1$ and $1^k = 1$. Similarity transports the limit; an entrywise $0/1$ diagonal limit squares to itself.

Use pattern. Use for polynomial equations in matrices such as $p(A) = 0$: first locate roots of p, then infer diagonalization and asymptotic powers.

Occurrence references. 1997-D2-P2; 1998-D1-P1; 1999-D1-P1; 2003-D1-P3; 2003-D2-P1; 2004-D2-P1; 2006-D2-P6; 2016-D2-P10; 2018-D1-P3; 2021-D1-P1; 2023-D2-P6.

AUD-LA-08 - Sylvester identities and nonzero spectra of AB and BA

Domain. Linear algebra. **References.** Horn-Johnson.

Gap closed. The matrix-exponential and block determinant problems need the precise relation between AB and BA beyond the square case.

Statement/properties. For compatible matrices A and B, the nonzero eigenvalues of AB and BA coincide with algebraic multiplicities. If both products are square, then $\det(I + AB) = \det(I + BA)$. More generally $\chi_{AB}(t) = t^{m-n}\chi_{BA}(t)$ when A is $m \times n$, B is $n \times m$, and $m \geq n$.

Compact proof core. Sylvester's determinant identity follows from block determinant multiplication. The characteristic-polynomial relation follows by applying it to $I - tAB$ and $I - tBA$ and comparing coefficients.

Use pattern. Use whenever a problem asks to compare $p(AB)$ and $p(BA)$, especially for nilpotency or matrix exponentials.

Occurrence references. 2000-D1-P3; 2000-D2-P6; 2001-D1-P2; 2009-D2-P3; 2020-D1-P2.

AUD-LA-09 - Functional calculus for polynomials and matrix exponentials

Domain. Linear algebra. **References.** Horn-Johnson.

Gap closed. The previous theory covered polynomials in matrices but did not explicitly connect $q(e^X)$ with a power series in X.

Statement/properties. For every polynomial q and square matrix X, $q(e^X)$ is a convergent power series in X: $q(e^X) = \sum_{r \geq 0} c_r X^r$. If $X = AB$, then terms with $r \geq 1$ can often be rewritten as $A(BA)^{r-1}B$, giving $q(e^{AB}) = c_0 I + AC$ and $q(e^{BA}) = c_0 I + CA$ for a suitable C.

Compact proof core. The exponential series converges absolutely in any matrix norm. Substituting it into the polynomial and multiplying series gives a new absolutely convergent series. The identity $(AB)^r = A(BA)^{r-1}B$ is immediate by associativity.

Use pattern. Use to reduce nilpotency of $p(e^{AB})$ versus $p(e^{BA})$ to the common characteristic polynomial of AC and CA.

Occurrence references. 2000-D2-P6.

AUD-LA-10 - Real matrices, conjugate eigenpairs, and determinant sign

Domain. Linear algebra. **References.** Horn-Johnson; Axler.

Gap closed. Several real-matrix determinant problems depend on complex eigenvalues occurring in conjugate pairs.

Statement/properties. If A is real, its characteristic polynomial has real coefficients; hence nonreal eigenvalues occur in conjugate pairs with the same algebraic multiplicity. The determinant equals the product of all complex eigenvalues counted with algebraic multiplicity. Thus a conjugate pair contributes $\lambda\overline{\lambda} = |\lambda|^2 > 0$ to a real determinant.

Compact proof core. Real coefficients imply $p(\overline{z}) = \overline{p(z)}$, so zeros are conjugate. The determinant is the constant term of the characteristic polynomial up to sign, equivalently the product of eigenvalues over $\mathbb{C}$.

Use pattern. Use when a real matrix satisfies a scalar polynomial equation whose roots include one real root and a complex conjugate pair.

Occurrence references. 1999-D1-P1; 2013-D1-P1; 2022-D1-P2.

AUD-LA-11 - Symmetric/skew decomposition and trace-isotropic subspaces

Domain. Linear algebra. **References.** Horn-Johnson; Hoffman-Kunze.

Gap closed. Maximal-dimension matrix-space problems need the exact symmetric-subspace obstruction.

Statement/properties. Every real matrix decomposes uniquely as $X = S + K$ where $S^T = S$ and $K^T = -K$. The symmetric subspace has dimension $n(n+1)/2$ and the strictly upper triangular subspace has dimension $n(n-1)/2$. If a linear subspace $V \subset M_n(\mathbb{R})$ satisfies $\operatorname{tr}(XY) = 0$ for all $X, Y \in V$, then V contains no nonzero symmetric matrix, because $\operatorname{tr}(S^2) = \sum_{i,j} s_{ij}^2 > 0$ for nonzero symmetric S.

Compact proof core. Use $S = (X + X^T)/2$ and $K = (X - X^T)/2$. The trace-square formula for symmetric S follows from $\operatorname{tr}(S^2) = \operatorname{tr}(S^T S)$. Dimension then follows from $V \cap \operatorname{Sym}_n = 0$.

Use pattern. Use for maximal isotropic subspaces under the trace form and for spaces of triangular or normal-like matrices.

Occurrence references. 2015-D2-P9; 2021-D2-P5.

AUD-LA-12 - Commuting square-zero matrices and the exterior-algebra model

Domain. Linear algebra. **References.** Axler; Hoffman-Kunze.

Gap closed. The lower bound 2^k for commuting square-zero matrices needs an explicit independence argument.

Statement/properties. If $A_1, \dots, A_k$ commute, $A_i^2 = 0$, and $A_1 \cdots A_k v \neq 0$, then the vectors $A_X v = \prod_{i \in X} A_i v$, indexed by all subsets $X \subset \{1, \dots, k\}$, are linearly independent under a suitable subset order. Hence the ambient dimension is at least 2^k. The model attaining this bound is the vector space with basis $[S]$ for subsets S and $A_i[S] = [S \cup \{i\}]$ if $i \notin S$, otherwise 0.

Compact proof core. For each X, applying A_{X^c} kills $A_Y v$ whenever Y is not contained in X in the required order, but sends $A_X v$ to $A_{\{1,\dots,k\}} v \neq 0$. This triangular annihilation proves independence.

Use pattern. Use when a product of commuting nilpotents is nonzero and minimal dimension is requested.

Occurrence references. 1999-D2-P1; 2000-D1-P3; 2007-D2-P5; 2016-D1-P2; 2024-D1-P3.

AUD-LA-13 - Derivation binomial formula for iterated commutators

Domain. Linear algebra. **References.** Horn-Johnson.

Gap closed. The nilpotency proof for $AB - BA$ from second commutator vanishing uses a full iterated derivation formula.

Statement/properties. Let $\Delta(X) = AX - XA$. Then Δ is a derivation and

$$\Delta^r(X_1 \cdots X_m) = \sum_{s_1 + \cdots + s_m = r} \binom{r}{s_1, \dots, s_m} \Delta^{s_1}(X_1) \cdots \Delta^{s_m}(X_m).$$

If $\Delta^2(B) = 0$, then $\Delta^k(B^k) = k!(\Delta B)^k$ and $\Delta^{k+1}(B^k) = 0$.

Compact proof core. The formula follows by induction from the Leibniz rule. If $\Delta^2(B) = 0$, all terms vanish except those differentiating each factor exactly once.

Use pattern. Use for equations equivalent to $A^2B + BA^2 - 2ABA = 0$, because that condition is $\Delta^2(B) = 0$.

Occurrence references. 2009-D2-P3; 2009-D2-P5.

AUD-LA-14 - Covering matrix spaces and low-rank elements

Domain. Linear algebra. **References.** Horn-Johnson; Axler.

Gap closed. The advanced matrix-space problem requires a specialized lemma absent from ordinary rank-nullity notes.

Statement/properties. A subspace T of $m \times p$ matrices is covering if $\bigcup_{0 \neq A \in T} \ker A = \mathbb{R}^p$. If T is a minimal covering matrix space with $\dim T = n$, then one can find $A \in T$ with $\operatorname{rank} A \leq n-1$. Iterating quotient projections yields $\delta(T) \leq (n-1) + (n-2) + \cdots + 1 = \binom{n}{2}$.

Compact proof core. Choose a codimension-one noncovering subspace and a vector avoiding all its kernels. Minimality supplies an additional matrix killing that vector. If its image were too large, perturbing the vector would avoid all kernels, contradiction. Project out its image and iterate.

Use pattern. Use only for minimal covering matrix-space questions; it is a rank-nullity method with an extremal noncovering hyperplane.

Occurrence references. No direct occurrence assigned by the audit; retained as prerequisite/background closure..

AUD-LA-15 - Kronecker-sum/Lyapunov operator spectrum

Domain. Linear algebra. **References.** Horn-Johnson.

Gap closed. The operator $X \mapsto MX + XM^T$ needs an explicit eigenmatrix construction and multiplicity rule.

Statement/properties. If M has eigenvalues λ_i with algebraic multiplicities m_i, then the linear operator $L(X) = MX + XM^T$ on $M_n(\mathbb{C})$ has eigenvalues $\lambda_i + \lambda_j$ with multiplicities $m_i m_j$ for ordered pairs (i, j); equal sums must be combined. For diagonalizable M, eigenmatrices are $v_i v_j^T$.

Compact proof core. If $Mv_i = \lambda_i v_i$ and $Mv_j = \lambda_j v_j$, then $L(v_i v_j^T) = \lambda_i v_i v_j^T + \lambda_j v_i v_j^T$. General multiplicities follow by perturbation or Jordan/Kronecker form.

Use pattern. Use for linear operators on matrix spaces defined by left and right multiplication.

Occurrence references. 2004-D2-P4.

AUD-LA-16 - Complex conjugation cocycles: solving $A\overline{A} = I$ as $A = S\overline{S}^{-1}$

Domain. Linear algebra. **References.** Horn-Johnson.

Gap closed. The conjugation equation is a finite-dimensional Hilbert-90 style statement; the constructive matrix $S = wA + \overline{w}I$ should be explicit.

Statement/properties. If $A\overline{A} = I$, then there exists $S \in GL_n(\mathbb{C})$ such that $A = S\overline{S}^{-1}$, equivalently $A\overline{S} = S$. A constructive choice is $S = wA + \overline{w}I$ for a complex number $w \neq 0$ with $\overline{w}/w$ not among the finitely many eigenvalues of $-A$ on the unit circle.

Compact proof core. Compute $A\overline{S} = A(\overline{w}\overline{A} + wI) = \overline{w}I + wA = S$. Singular choices of S correspond to finitely many values of $\overline{w}/w$, so choose w avoiding them.

Use pattern. Use for matrix equations involving entrywise complex conjugation, not transpose or adjoint.

Occurrence references. 2002-D2-P5.

AUD-RA-01 - First-order differential inequalities and integrating factors

Domain. Real analysis. **References.** Rudin; Apostol.

Gap closed. Many official solutions create an auxiliary function whose derivative has a sign; the exact integrating-factor template needs to be explicit.

Statement/properties. For $u'(x) + a(x)u(x) \geq h(x)$, multiply by $\mu(x) = \exp \int a(x)dx$ to get $(\mu u)' \geq \mu h$. If $u'(x) \leq \lambda u(x)$ on an interval and $u > 0$, then $(\log u - \lambda x)' \leq 0$. If $u'(x) + xu(x)$ is bounded on $[0, \infty)$, then $u(x) \to 0$ after multiplying by $e^{x^2/2}$ and using L'Hospital or comparison.

Compact proof core. The product rule gives $(\mu u)' = \mu(u' + au)$. The logarithmic form follows by dividing by a positive solution. The Gaussian integrating factor grows faster than its integral: $\int_0^x e^{t^2/2}dt / e^{x^2/2} \to 0$.

CORRECTION MATH-005 — legacy statement superseded; click for canonical replacement.

Use pattern. Use for inequalities involving f', f'', endpoint data, or expressions like $f'' + 2xf' + (x^2+1)f$.

Occurrence references. 1994-D1-P2; 1994-D1-P6; 1994-D2-P1; 1994-D2-P3; 1995-D1-P3; 1997-D2-P1; 1998-D1-P4; 1998-D2-P2; 1999-D1-P6; 2000-D1-P6; 2001-D1-P6; 2002-D1-P2; 2005-D1-P3; 2005-D1-P5; 2005-D2-P4; 2007-D2-P1; 2007-D2-P6; 2009-D2-P2; 2012-D1-P4; 2013-D1-P2; 2014-D2-P3; 2015-D2-P6; (+9 more in CSV).

AUD-RA-02 - Gronwall-type zero uniqueness

Domain. Real analysis. **References.** Rudin; Apostol.

Gap closed. The zero-uniqueness problem uses a local logarithmic contradiction that was only broadly tagged as differential inequality.

Statement/properties. If $f \in C^1([a,b])$, $f(a) = 0$, and $|f'| \le \lambda |f|$, then $f \equiv 0$. A common variant: if a nonnegative continuous function satisfies $u(t) \le C \int_a^t u(s)ds$, then $u \equiv 0$.

Compact proof core. If f first becomes nonzero, restrict to an interval where f has fixed sign. Then $(\log|f| - \lambda x)' \le 0$ or ≥ 0 gives a positive lower bound as one approaches the previous zero, contradiction. The integral version follows by applying the same argument to $U(t) = \int_a^t u$.

Use pattern. Use when the derivative is bounded by the function itself and an initial zero is given.

Occurrence references. 1994-D2-P1; 2010-D2-P5.

AUD-RA-03 - Periodic averaging and equidistributed Riemann blocks

Domain. Real analysis. **References.** Rudin; Katznelson.

Gap closed. The periodic integral limit needs the exact block decomposition and uniform-continuity error.

Statement/properties. If $f \in C[0,b]$ and g is continuous and b-periodic, then

$$\lim_{n\to\infty} \int_0^b f(x)g(nx)dx = \frac{1}{b}\int_0^b f(x)dx \int_0^b g(x)dx.$$

A uniform version holds when f is uniformly continuous and g is integrable over one period.

Compact proof core. Split $[0,b]$ into n intervals of length b/n. On each interval replace f by a sample value. The periodicity of $g(nx)$ makes each block integral equal to $n^{-1}\int_0^b g$. The error is bounded by the modulus of continuity of f times $\int_0^b |g|$.

Use pattern. Use for integrals with $g(nx)$ or fast oscillations under a continuous slow factor.

Occurrence references. 1994-D1-P2; 1994-D1-P5; 1994-D2-P6; 1995-D1-P3; 1995-D2-P2; 1995-D2-P6; 1996-D1-P4; 1996-D1-P5; 1996-D2-P1; 1996-D2-P5; 1997-D1-P1; 1997-D1-P2; 1997-D2-P1; 1997-D2-P3; 1998-D1-P3; 1998-D2-P3; 2000-D1-P6; 2000-D2-P5; 2001-D1-P3; 2001-D1-P6; 2001-D2-P2; 2001-D2-P5; (+33 more in CSV).

AUD-RA-04 - Endpoint Laplace principle for $n\int f^n$

Domain. Real analysis. **References.** Rudin; Apostol.

Gap closed. The endpoint maximum asymptotic was not previously stated as a reusable theorem.

Statement/properties. If $f(0) = 1$, $f(x) < 1$ for $x > 0$, and $f(x) = 1 + ax + O(x^2)$ with $a < 0$, then

$$n\int_0^A f(x)^n dx \to -1/a.$$

If the first nonzero departure from 1 at an interior maximum is quadratic and negative, then the scale is $\sqrt{n}$, so $n\int f^n$ diverges like $\sqrt{n}$.

Compact proof core. Compare $f(x)^n$ with e^{anx} near 0 and bound the tail by a geometric/exponential estimate. At an interior stationary maximum, Taylor's theorem gives $f(x) \ge 1 - C(x-x_0)^2$ locally, whose nth power has width $n^{-1/2}$, causing n times the integral to diverge.

Use pattern. Use in concavity problems where the limit $0 < L < \infty$ forces the maximum to occur at an endpoint with nonzero derivative.

Occurrence references. 1995-D2-P6; 1996-D1-P4; 1996-D1-P5; 1996-D2-P1; 1996-D2-P5; 1997-D1-P1; 1997-D1-P2; 1997-D2-P3; 1998-D1-P3; 2000-D1-P6; 2000-D2-P5; 2001-D2-P2; 2002-D1-P4; 2002-D1-P6; 2003-D1-P1; 2003-D1-P3; 2003-D1-P5; 2003-D2-P5; 2003-D2-P6; 2004-D1-P3; 2004-D1-P6; 2004-D2-P4; (+10 more in CSV).

AUD-RA-05 - Riemann sums on infinite intervals and tail control

Domain. Real analysis. **References.** Rudin; Apostol.

Gap closed. Infinite Riemann sums require a separate tail estimate, not just compact-interval convergence.

Statement/properties. If f is Riemann integrable on $[0,\infty)$ and has monotone or dominated tails, then $h\sum_{n\geq 1} f(nh) \to \int_0^\infty f(t)dt$ as $h \downarrow 0$. A standard sufficient condition is that f is continuous, nonnegative, eventually decreasing, and integrable.

Compact proof core. First prove convergence on $[0,R]$. For the tail use monotonicity to compare $h\sum_{nh\geq R} f(nh)$ with an adjacent integral tail. Let $R \to \infty$ after taking $h \to 0$.

Use pattern. Use for sums of the form $\sum nx/(n^2+x)^2$ or scaled logarithmic sums.

Occurrence references. 1994-D1-P2; 1994-D1-P5; 1994-D2-P6; 1995-D1-P3; 1995-D2-P2; 1995-D2-P6; 1996-D1-P4; 1996-D1-P5; 1996-D2-P1; 1996-D2-P5; 1997-D1-P1; 1997-D1-P2; 1997-D2-P1; 1997-D2-P3; 1998-D1-P3; 2000-D1-P6; 2000-D2-P5; 2001-D1-P3; 2001-D1-P6; 2001-D2-P2; 2001-D2-P5; 2002-D1-P4; (+30 more in CSV).

AUD-RA-06 - Rolle chains and divided differences with repeated nodes

Domain. Real analysis. **References.** Rudin; Apostol.

Gap closed. Several problems use repeated applications of Rolle; one uses the repeated-node divided-difference theorem explicitly.

Statement/properties. If a differentiable function has $m+1$ distinct zeros, then its derivative has at least m zeros between them; iterating gives zeros of higher derivatives. The generalized mean-value theorem for divided differences says that if f is n times differentiable, then $f[x_0,\ldots,x_n] = f^{(n)}(\xi)/n!$ for some ξ in the convex hull, with repeated nodes interpreted by derivatives.

Compact proof core. Ordinary Rolle gives one derivative zero between each consecutive pair. Iterate. Divided differences follow by subtracting the Hermite interpolation polynomial and applying the Rolle chain to the error.

Use pattern. Use for problems with many zeros, endpoint values, or exact linear combinations of derivatives at an intermediate point.

Occurrence references. 1994-D1-P2; 1994-D1-P6; 1994-D2-P1; 1994-D2-P3; 1994-D2-P4; 1995-D1-P3; 1996-D1-P5; 1996-D1-P6; 1996-D2-P5; 1996-D2-P6; 1997-D2-P1; 1997-D2-P3; 1998-D1-P4; 1998-D2-P6; 1999-D1-P6; 2000-D1-P2; 2000-D1-P3; 2000-D1-P6; 2000-D2-P4; 2000-D2-P5; 2000-D2-P6; 2001-D1-P6; (+38 more in CSV).

AUD-RA-07 - Compactness, extrema, and localization on unbounded domains

Domain. Real analysis. **References.** Rudin; Apostol.

Gap closed. The previous compactness node needed the common IMC localization step: prove extrema cannot occur outside a compact set.

Statement/properties. A continuous function on a compact set attains its minimum and maximum. On an unbounded domain, first prove that outside a compact set all values are strictly below a known candidate maximum or above a known candidate minimum; then apply Weierstrass on the compact part. In finite-dimensional Euclidean space, closed and bounded equals compact.

Compact proof core. Weierstrass follows from sequential compactness. Localization reduces the problem to a compact set by comparing tails to explicit test values.

Use pattern. Use for functions like polynomial times e^{-r^2} and for global-extremum classification.

Occurrence references. 1994-D2-P2; 1995-D1-P6; 1995-D2-P4; 1995-D2-P6; 1996-D1-P6; 1996-D2-P4; 1998-D2-P2; 1998-D2-P5; 1999-D1-P1; 2000-D1-P1; 2000-D2-P2; 2001-D2-P3; 2001-D2-P4; 2001-D2-P6; 2002-D1-P4; 2002-D2-P6; 2003-D2-P3; 2004-D1-P3; 2005-D2-P1; 2005-D2-P3; 2005-D2-P5; 2006-D2-P2; (+10 more in CSV).

AUD-RA-08 - Convexity, strict convexity, and equality cases

Domain. Real analysis. **References.** Hardy-Polya-Littlewood; Rockafellar; Boyd-Vandenberghe.

Gap closed. The broad convexity entry needed explicit equality conditions and the exact forms used in IMC inequalities.

Statement/properties. For convex ϕ, $\phi(tx+(1-t)y) \leq t\phi(x)+(1-t)\phi(y)$; for strict convexity equality with $0<t<1$ forces $x=y$. Jensen: $\phi(\sum w_i x_i) \leq \sum w_i \phi(x_i)$. A differentiable convex function lies above its tangent:

$\phi(y) \geq \phi(x) + \phi'(x)(y-x)$; a concave function lies below its tangent and above its chords. AM-HM gives $\frac{1}{n}\sum a_i \geq n/\sum 1/a_i$ for positive a_i.

Compact proof core. Jensen follows by induction from the definition. The tangent inequality follows by monotonicity of slopes. Equality cases are inherited from strict convexity or from linearity on the relevant interval.

Use pattern. Use for $|x|^p$, sine/arcsine on convexity intervals, coefficient inequalities, and optimization with equality cases.

Occurrence references. 1994-D1-P1; 1994-D1-P2; 1994-D1-P3; 1994-D1-P4; 1994-D1-P5; 1994-D1-P6; 1994-D2-P1; 1994-D2-P2; 1994-D2-P5; 1994-D2-P6; 1995-D1-P2; 1995-D1-P3; 1995-D1-P4; 1995-D1-P6; 1995-D2-P1; 1995-D2-P2; 1995-D2-P3; 1995-D2-P4; 1995-D2-P5; 1996-D1-P2; 1996-D1-P3; 1996-D1-P4; (+206 more in CSV).

AUD-RA-09 - Dominated convergence in bounded orthonormal systems

Domain. Real analysis. **References.** Royden-Fitzpatrick; Conway1990.

Gap closed. The no-pointwise-subsequence problem uses dominated convergence plus completeness/totality in L^2.

Statement/properties. If $|f_n| \leq M$ and $f_{n_k}(x) \to f(x)$ pointwise on a finite-measure space, then $\int h f_{n_k} \to \int hf$ for every bounded measurable h, and also $\int f_{n_k}^2 \to \int f^2$. In a complete orthonormal system, if f is orthogonal to every basis element, then $f = 0$ in L^2.

Compact proof core. Dominated convergence applies to hf_{n_k} and $f_{n_k}^2$. Orthogonality to a dense span implies zero inner product against every L^2 vector, hence norm zero.

Use pattern. Use to contradict $\|f_{n_k}\|_2 = 1$ after a pointwise limit is forced to be zero.

Occurrence references. 1995-D2-P6.

AUD-RA-10 - Monotone functions, discontinuities, and range gaps

Domain. Real analysis. **References.** Rudin; Royden-Fitzpatrick.

Gap closed. Several problems rely on exact structure of monotone functions, level sets, and jumps.

Statement/properties. A monotone real function has left and right limits at every point. If it has a jump at a, its range omits the open interval between the one-sided limits, except possibly the single value $f(a)$. Level sets of a monotone function are intervals or points, and there are at most countably many disjoint nondegenerate intervals in $\mathbb{R}$.

Compact proof core. One-sided limits exist by monotone boundedness on each side. Disjoint open intervals each contain a rational number, giving countability.

Use pattern. Use for monotone-continuity equivalences, fixed points on closed bounded subsets, and level-set cardinality questions.

Occurrence references. 2000-D1-P1; 2000-D2-P2; 2002-D1-P5; 2004-D1-P5; 2006-D1-P1; 2007-D2-P3; 2011-D1-P4.

AUD-RA-11 - Baire category and closed orbits of interval maps

Domain. Real analysis. **References.** Rudin; Devaney.

Gap closed. The orbit-closure problem has a short Baire interpretation and a sequential proof; both need to be available.

Statement/properties. A nonempty complete metric space cannot be a countable union of nowhere dense closed sets. For a continuous map on a compact interval, an infinite orbit has cluster points. If the orbit set itself is closed and countable, Baire forces an isolated point; dynamics then force eventual periodicity, hence finiteness.

Compact proof core. Compactness gives a convergent subsequence. If an infinite closed orbit had every point an accumulation point, it would be a countable closed space without isolated points, contradicting Baire.

Use pattern. Use when an orbit $T_p = \{p, f(p), f^2(p), \ldots\}$ is assumed closed.

Occurrence references. 1996-D2-P1; 2002-D1-P4.

CORRECTION MATH-031 — legacy statement superseded; click for canonical replacement.

AUD-RA-12 - Cantor sets and continuous functions crossing zero uncountably often

Domain. Real analysis. **References.** Rudin; Royden-Fitzpatrick.

Gap closed. The uncountable-crossing construction requires Cantor-set encoding and uniform convergence of tent functions.

Statement/properties. The middle-thirds Cantor set is uncountable, closed, has empty interior, and its complement is a countable union of disjoint open intervals. A uniformly convergent sum of continuous functions is continuous. By placing alternating-sign triangular functions on the removed intervals with amplitudes tending to zero, one can build a continuous function that changes sign arbitrarily near every Cantor point.

Compact proof core. Uncountability follows by mapping binary expansions to ternary expansions using digits 0 and 2. Uniform convergence follows from the Weierstrass M-test. Density of removed intervals of alternating levels gives sign changes near each Cantor point.

Use pattern. Use for examples of continuous functions with large zero/crossing sets.

Occurrence references. 1997-D2-P6.

AUD-RA-13 - Nowhere monotone functions and dense local extrema

Domain. Real analysis. **References.** Rudin.

Gap closed. The proof that local minima are dense is a compactness argument with two nonmonotone subintervals; it needed its own card.

Statement/properties. If f is continuous and nowhere monotone on an interval, then every nonempty open subinterval contains a local minimum and a local maximum. In particular, the set of local minima is dense.

Compact proof core. Inside $(x-\alpha, x+\alpha)$ choose $p < q \le x$ with $f(p) > f(q)$ and $x \le r < s$ with $f(r) < f(s)$. A global minimum on $[p, s]$ cannot occur at p or s, so it occurs at an interior point and is local.

Use pattern. Use whenever continuity plus failure of monotonicity on every interval appears.

Occurrence references. 2000-D2-P2; 2003-D2-P3.

AUD-RA-14 - Sparse spike functions and differentiability by interval covers

Domain. Real analysis. **References.** Royden-Fitzpatrick.

Gap closed. The sparse function problem uses a covering argument not captured by generic differentiability theory.

Statement/properties. If f is zero except at points a_n with values $b_n \ge 0$ and $\sum b_n < \infty$, choose $c_n \to \infty$ with $\sum c_n b_n < 1/2$. The intervals $(a_n - c_n b_n, a_n + c_n b_n)$ do not cover $(0,1)$, so a point outside all intervals is a differentiability point with derivative 0. Conversely, if $\sum b_n = \infty$, choose a_n so intervals $(a_n - \beta_n, a_n + \beta_n)$ cover every point infinitely often to prevent differentiability.

Compact proof core. Outside all intervals, $|f(a_n)|/|a_n - x_0| \le 1/c_n \to 0$, and all other points have quotient zero. For divergence, infinite recurrence of spikes keeps difference quotients bounded away from zero arbitrarily close to each point.

Use pattern. Use for functions supported on countable sets with prescribed spike heights.

Occurrence references. 1998-D2-P6.

AUD-RA-15 - Lipschitz gradients and co-coercivity of convex functions

Domain. Real analysis. **References.** Boyd-Vandenberghe; Rockafellar.

Gap closed. The convex-gradient problem requires a theorem-like inequality stronger than ordinary convexity.

Statement/properties. If f is convex and differentiable on $\mathbb{R}^n$ with L-Lipschitz gradient, then

$$\|\nabla f(x) - \nabla f(y)\|^2 \le L\langle \nabla f(x) - \nabla f(y), x - y\rangle.$$

This is the Baillon-Haddad co-coercivity inequality.

Compact proof core. Apply the descent lemma to $g(z) = f(z) - f(x) - \langle \nabla f(x), z - x\rangle$, whose minimum is 0 at x. Evaluate g at $y - (1/L)\nabla g(y)$ to get $g(y) \ge \|\nabla g(y)\|^2/(2L)$; repeat with x, y swapped and add.

Use pattern. Use for inequalities combining gradient differences and inner products under convexity plus Lipschitz gradient.

Occurrence references. 2002-D2-P6; 2017-D1-P2.

AUD-RA-16 - Conditionally convergent series under structured rearrangements

Domain. Real analysis. **References.** Rudin; Apostol.

Gap closed. Block rearrangement problems require a direct partial-sum criterion, not only the Riemann rearrangement theorem.

Statement/properties. If $\sum a_n$ converges with partial sums $S_n \to S$, then any rearrangement whose partial sums can be expressed using only finitely many tails $S_m - S$ with $m \to \infty$ converges to S. A structured rearrangement can diverge when it systematically groups enough positive terms before compensating negative terms; alternating $1/\sqrt{n}$ is a standard test source.

Compact proof core. For convergence, write rearranged block partial sums explicitly in terms of original partial sums and use the Cauchy criterion. For divergence, estimate the excess positive block below by length times its smallest term.

Use pattern. Use for block permutations of a conditionally convergent series.

Occurrence references. 1997-D1-P2.

AUD-PC-01 - Polynomial identity, roots versus degree, and finite-field vanishing

Domain. Polynomials/complex. **References.** Ahlfors; Dummit-Foote.

Gap closed. This most-used fact should be stated both over infinite fields and over finite fields with the degree bound.

Statement/properties. A nonzero polynomial of degree d over a field has at most d roots. Thus if it has more than d roots, it is the zero polynomial. Over $\mathbb{F}_p$, every polynomial function $\mathbb{F}_p \to \mathbb{F}_p$ has a unique representative of degree at most $p-1$, obtained modulo $x^p - x$.

Compact proof core. If a is a root, divide by $x-a$ and induct on degree. In $\mathbb{F}_p$, Fermat gives $x^p - x = 0$ for all field elements; division by $x^p - x$ reduces functions without changing values.

Use pattern. Use for congruence polynomials, finite-field functions, and equality of polynomial identities from many roots.

Occurrence references. 1994-D2-P4; 1995-D1-P5; 1995-D2-P3; 1995-D2-P4; 1996-D2-P2; 1996-D2-P6; 1997-D1-P3; 1998-D1-P2; 1998-D1-P5; 1998-D2-P2; 1999-D1-P1; 1999-D2-P2; 1999-D2-P6; 2000-D1-P2; 2000-D1-P3; 2000-D2-P1; 2000-D2-P3; 2000-D2-P4; 2001-D1-P1; 2001-D1-P2; 2001-D1-P4; 2001-D1-P5; (+85 more in CSV).

AUD-PC-02 - Vieta, logarithmic derivatives, and real-rooted Laguerre inequality

Domain. Polynomials/complex. **References.** Marden; Rivlin.

Gap closed. Real-rooted polynomial inequalities need the exact logarithmic-derivative identities.

Statement/properties. If $P(x) = c\prod_{i=1}^{n}(x-r_i)$ with real roots, then for x not a root,

$$P'(x)/P(x) = \sum_i \frac{1}{x-r_i}, \qquad P''(x)/P(x) = 2\sum_{i<j} \frac{1}{(x-r_i)(x-r_j)}.$$

Therefore $(n-1)(P')^2 \geq nPP''$, with equality for all x exactly when all roots coincide.

Compact proof core. Differentiate $\log P$. Substituting the formulas reduces the difference to $\sum_{i<j}(1/(x-r_i) - 1/(x-r_j))^2 \geq 0$, after clearing by $P(x)^2$; roots are handled by continuity or multiplicity.

Use pattern. Use for polynomial inequalities under the hypothesis "all roots real.

Occurrence references. 1998-D1-P5.

AUD-PC-03 - Unit-circle root localization via logarithmic derivative

Domain. Polynomials/complex. **References.** Marden; Ahlfors.

Gap closed. The unit-circle problem uses a specialized real-part identity.

Statement/properties. If $|\alpha| = 1$, then

$$\operatorname{Re} \frac{z+\alpha}{z-\alpha} = \frac{|z|^2 - 1}{|z-\alpha|^2}.$$

For $P(z) = \prod(z - \alpha_i)$ with all $|\alpha_i| = 1$, the zeros of $2zP'(z) - nP(z)$ lie on $|z| = 1$.

Compact proof core. Compute the real part by multiplying numerator and denominator by $\overline{z} - \overline{\alpha}$. Then divide $2zP' - nP$ by P and sum the displayed real parts. Off the unit circle, the real part cannot vanish.

Use pattern. Use for root-location problems involving $zP'(z)$ and roots on a circle.

Occurrence references. 1995-D2-P3.

AUD-PC-04 - Hurwitz stability and coefficient log-concavity type inequalities

Domain. Polynomials/complex. **References.** Marden; Hardy-Polya-Littlewood.

Gap closed. The left-half-plane coefficient problem requires factorization into positive linear/quadratic factors and an induction inequality.

Statement/properties. A real polynomial with all roots in $\operatorname{Re} z < 0$ factors over $\mathbb{R}$ into linear factors $az + b$ and quadratic factors $az^2 + bz + c$ with all coefficients of each factor having the same sign. After multiplying by -1 if needed, all coefficients are positive. For such stable polynomials, the IMC inequality $a_k a_{k+3} < a_{k+1} a_{k+2}$ follows by induction after removing a positive quadratic factor.

Compact proof core. Real negative roots yield positive linear factors; conjugate pairs in the left half-plane yield quadratics with positive coefficients. The induction step expands coefficients of $(z^2 + pz + q)g$ and reduces the target inequality to a sum of nonnegative inequalities for the coefficients of g, with one strict term.

Use pattern. Use for coefficient inequalities under left-half-plane root hypotheses.

Occurrence references. 2003-D1-P6.

AUD-PC-05 - Cyclotomic irreducibility and roots-of-unity divisibility

Domain. Polynomials/complex. **References.** Ireland-Rosen; Dummit-Foote.

Gap closed. Several root-of-unity divisor problems need irreducibility of cyclotomic polynomials and product estimates.

Statement/properties. For prime q, $\Phi_q(x) = 1 + x + \cdots + x^{q-1}$ is irreducible over $\mathbb{Q}$. If an integer polynomial has one primitive qth root as a root, it is divisible by Φ_q and all primitive qth roots are roots. Also $\prod_{j=1}^{q-1}(1 - \zeta_q^j) = q$.

Compact proof core. Irreducibility follows from Eisenstein after substituting $x+1$ into Φ_q. The product identity is $\Phi_q(1) = q$.

Use pattern. Use when divisibility by a root-of-unity factor must force all roots in a conjugacy class.

Occurrence references. 1995-D2-P3; 1999-D1-P1; 1999-D2-P2; 1999-D2-P6; 2000-D1-P2; 2000-D2-P3; 2001-D1-P4; 2001-D1-P5; 2001-D2-P3; 2001-D2-P4; 2002-D2-P5; 2006-D1-P6; 2007-D1-P5; 2007-D1-P6; 2008-D2-P1; 2008-D2-P3; 2013-D1-P5; 2013-D2-P1; 2015-D1-P1; 2015-D2-P9; 2016-D2-P10; 2017-D1-P1; (+6 more in CSV).

AUD-PC-06 - Root-of-unity filters for congruence classes

Domain. Polynomials/complex. **References.** Katznelson; Stanley.

Gap closed. Probability and counting modulo m problems need the exact filter formula.

Statement/properties. Let $\omega = e^{2\pi i/m}$. For a polynomial or generating function $F(x) = \sum a_k x^k$, the sum of coefficients with $k \equiv r \pmod m$ is

$$\frac{1}{m} \sum_{j=0}^{m-1} \omega^{-rj} F(\omega^j).$$

Compact proof core. Use $m^{-1} \sum_{j=0}^{m-1} \omega^{j(k-r)}$, which equals 1 if $k \equiv r \pmod m$ and 0 otherwise.

Use pattern. Use for dice sums modulo 5, Fourier coefficients of subsets of $\mathbb{Z}_n$, and binomial congruence sums.

Occurrence references. 1995-D2-P3; 1999-D2-P2; 1999-D2-P6; 2002-D2-P5; 2007-D1-P1; 2007-D1-P6; 2009-D2-P4; 2018-D2-P6; 2021-D2-P6; 2023-D1-P4.

AUD-PC-07 - Fejer kernels and positive trigonometric multipliers

Domain. Polynomials/complex. **References.** Katznelson; Ahlfors.

Gap closed. The convex coefficient multiplier problem depends on positivity of Fejer kernels.

Statement/properties. The Fejer kernel

$$F_k(e^{it}) = \sum_{j=-k+1}^{k-1} (k-|j|)e^{ijt} = |1 + e^{it} + \cdots + e^{i(k-1)t}|^2 \geq 0.$$

If $1 = c_0 \geq c_1 \geq \cdots \geq c_n = 0$ is convex, then $K(e^{it}) = \sum_{j=-n}^{n} c_{|j|} e^{ijt}$ is a nonnegative linear combination of Fejer kernels, hence $K \geq 0$ and has integral 2π.

Compact proof core. Set $d_k = c_{k-1} - 2c_k + c_{k+1} \geq 0$ with $c_{n+1} = 0$. Summing $d_k F_k$ reconstructs the coefficient profile. Positivity and normalization make convolution with K a contraction in sup norm.

Use pattern. Use for coefficient-damping inequalities on the unit disk.

Occurrence references. 2009-D1-P4.

AUD-PC-08 - Chebyshev interpolation extremality on $[-1, 1]$

Domain. Polynomials/complex. **References.** Rivlin.

Gap closed. The cubic second-derivative extremum problem needs Lagrange interpolation plus Chebyshev alternation.

Statement/properties. For nodes $x_0, \ldots, x_m$, the Lagrange basis ℓ_k satisfies $p(x) = \sum p(x_k)\ell_k(x)$ for $\deg p \leq m$. If a linear functional $L(p) = \sum c_k p(x_k)$ is optimized under $|p(x_k)| \leq 1$, its maximum is $\sum |c_k|$, attained by $p(x_k) = \operatorname{sign} c_k$. On $[-1, 1]$, $T_3(x) = 4x^3 - 3x$ alternates at $-1, -1/2, 1/2, 1$ and has $T_3''(1) = 24$.

Compact proof core. Interpolation gives L as a finite signed sum of node values; the box constraint is optimized at signs. The Chebyshev polynomial is verified directly and has the required alternation.

Use pattern. Use for extremal polynomial problems with finitely many pointwise constraints.

Occurrence references. 1994-D2-P4; 1995-D1-P5; 1998-D1-P5; 1998-D2-P2; 1999-D2-P6; 2000-D1-P2; 2000-D2-P3; 2000-D2-P4; 2001-D1-P4; 2001-D1-P5; 2001-D2-P1; 2001-D2-P4; 2002-D1-P1; 2003-D1-P6; 2005-D1-P4; 2005-D2-P2; 2005-D2-P6; 2006-D1-P4; 2006-D1-P6; 2006-D2-P5; 2007-D1-P1; 2007-D1-P3; (+32 more in CSV).

AUD-PC-09 - Weierstrass approximation with symmetry projection

Domain. Polynomials/complex. **References.** Rudin; Rivlin.

Gap closed. The approximation problem requires the exact odd-symmetrization and replacement of the linear term.

Statement/properties. Every continuous function on $[a, b]$ is uniformly approximable by polynomials. If f is odd on $[-1, 1]$, then after approximating by p, the polynomial $q(x) = (p(x) - p(-x))/2$ is odd and approximates f with no larger error. The term x can be uniformly approximated by polynomials in $x^3, x^5, \ldots$ using $x(1-x^2)^N \to 0$ uniformly.

Compact proof core. Weierstrass gives p. Odd projection preserves approximation because f is odd. The identity $x - \sum_{k=1}^{N} (-1)^{k+1} \binom{N}{k} x^{2k+1} = x(1-x^2)^N$ tends uniformly to zero.

Use pattern. Use for constrained approximation by odd powers excluding the linear term.

Occurrence references. 1995-D2-P4.

AUD-PC-10 - Cotangent partial fractions and logarithm-branch sums

Domain. Polynomials/complex. **References.** Ahlfors; Conway1978.

Gap closed. The logarithm-branch rational-function problem needs the standard cotangent partial fraction and differentiation.

Statement/properties. The partial fraction expansion

$$\sum_{k\in\mathbb{Z}} \frac{1}{w+2\pi ik} = \frac{i}{2}\cot\left(\frac{iw}{2}\right)$$

holds in the symmetric principal-value sense. Differentiating gives sums of $(w+2\pi ik)^{-m}$. With $w=\log z$, the branch sum over all logarithms becomes a rational function of z; for $m=4$, it equals $z(z^2+4z+1)/(6(z-1)^4)$.

Compact proof core. Use the classical Mittag-Leffler expansion of cotangent. Differentiate termwise on compact subsets avoiding poles. Substitute $e^w=z$ and simplify using $\cot(iw/2)$ in exponentials.

Use pattern. Use only for complex logarithm branch sums and meromorphic partial fractions.

Occurrence references. 2004-D1-P6.

AUD-PC-11 - Integer-valued congruence of polynomial values

Domain. Polynomials/complex. **References.** Ireland-Rosen; Dummit-Foote.

Gap closed. The divisibility-by-values problems need the congruence identity $p(a+mq)\equiv p(a) \pmod q$.

Statement/properties. If $p\in\mathbb{Z}[x]$ and $u\equiv v \pmod m$, then $p(u)\equiv p(v) \pmod m$. Therefore, by CRT one can choose a simultaneously satisfying congruences modulo factors of $p(a_i)$, and then $p(a_i)\mid p(a)$ for all i when the moduli have been made coprime by splitting their lcm.

Compact proof core. Each monomial satisfies $u^r-v^r=(u-v)(u^{r-1}+\cdots+v^{r-1})$, so $m\mid u-v$ implies $m\mid p(u)-p(v)$. CRT solves the simultaneous congruences.

Use pattern. Use for polynomial divisibility values at finitely many integer arguments.

Occurrence references. 2007-D1-P1; 2008-D1-P3; 2008-D2-P4; 2011-D1-P3.

AUD-PC-12 - Rational functions with many integer values

Domain. Polynomials/complex. **References.** Dummit-Foote; Apostol.

Gap closed. The rational-function problem needs a proof that the denominator must be constant.

Statement/properties. If $f=p/q\in\mathbb{Q}(x)$ takes integer values at infinitely many integers and $p,q\in\mathbb{Z}[x]$ are coprime, then after Euclidean division $p=qs+r$ with $\deg r<\deg q$. Since $r(n)/q(n)\to 0$ along large integers but is rational with bounded denominator after clearing coefficients and also integral for infinitely many n, it must vanish infinitely often; hence $r=0$ and $f=s$ is a polynomial.

Compact proof core. Clear denominators first. For large n in the infinite set, the integer $N(f(n)-s(n))=Nr(n)/q(n)$ has absolute value <1, hence is 0. Thus r has infinitely many roots.

Use pattern. Use when a rational function is integer-valued on an infinite integer subset.

Occurrence references. No direct occurrence assigned by the audit; retained as prerequisite/background closure..

AUD-PC-13 - Ordinary generating functions for recurrences and weighted sums

Domain. Polynomials/complex. **References.** Wilf; Stanley.

Gap closed. Several recurrence/sum problems are shortest with generating functions, which needed a standalone theory card.

Statement/properties. For a sequence (a_n), its ordinary generating function $A(x)=\sum_{n\ge 0}a_nx^n$ converts convolution into multiplication: if $c_n=\sum_{k=0}^{n}a_kb_{n-k}$, then $C(x)=A(x)B(x)$. Recurrences with polynomial coefficients often become differential equations for $A(x)$ because $xA'(x)=\sum na_nx^n$.

Compact proof core. Multiply the series and compare coefficients. Differentiation is termwise within the radius of convergence or formal in the ring of formal power series.

Use pattern. Use for recursively defined sequences, partition functions, and sums with binomial or factorial weights.

Occurrence references. 1994-D2-P4; 1995-D1-P5; 1995-D2-P4; 1996-D2-P2; 1996-D2-P6; 1998-D1-P2; 1998-D1-P5; 1998-D2-P6; 1999-D2-P2; 2000-D2-P1; 2000-D2-P3; 2000-D2-P4; 2001-D1-P4; 2001-D2-P1; 2001-D2-P2; 2002-D1-P1; 2002-D2-P3; 2003-D1-P1; 2003-D1-P2; 2003-D1-P5; 2003-D1-P6; 2003-D2-P2; (+47 more in CSV).

AUD-PC-14 - Pell equations and quadratic irrational stabilizers in $\mathrm{SL}_2(\mathbb{Z})$

Domain. Polynomials/complex. **References.** Ireland-Rosen; Artin.

Gap closed. The quadratic irrational fixed-point problem needs the link between Mobius stabilizers and Pell equations.

Statement/properties. A Mobius transformation $z \mapsto (az+b)/(cz+d)$ with $ad-bc=1$ fixes a quadratic irrational r iff $cr^2+(d-a)r-b=0$ is a multiple of the minimal polynomial of r. The determinant condition then becomes a Pell-type equation $s^2-Dt^2=4$. Such equations have infinitely many nontrivial solutions when D is a nonsquare positive integer.

Compact proof core. Cross-multiply the fixed-point equation. Matching coefficients with the minimal polynomial gives $b, c, d-a$ in terms of one parameter. The determinant is equivalent to a norm equation in $\mathbb{Z}[\sqrt{D}]$. Units give infinitely many solutions.

Use pattern. Use for $SL_2(\mathbb{Z})$ matrices fixing rational or quadratic irrational numbers.

Occurrence references. 2002-D1-P1; 2005-D2-P6.

AUD-ANT-01 - Chinese remainder theorem and idempotents modulo prime powers

Domain. Algebra/number theory. **References.** Ireland-Rosen; Dummit-Foote.

Gap closed. Counting $x^2 \equiv x$ modulo 10^n requires CRT idempotents explicitly.

Statement/properties. If m and n are coprime, then $\mathbb{Z}/(mn) \cong \mathbb{Z}/m \times \mathbb{Z}/n$. Solutions of $x^2 \equiv x \pmod N$ correspond to choices $x \equiv 0$ or 1 modulo each prime-power divisor of N. Thus modulo 2^a5^b there are four residue classes, one of which is 0.

Compact proof core. From $x(x-1) \equiv 0 \pmod{p^r}$ and $\gcd(x, x-1) = 1$, one factor is divisible by p^r. CRT combines the independent choices.

Use pattern. Use for divisibility of x^2-x by powers of 10 or other squarefree-prime decompositions.

Occurrence references. 2000-D1-P5; 2005-D1-P6; 2006-D1-P2.

AUD-ANT-02 - Finite fields, Fermat, and polynomial permutations

Domain. Algebra/number theory. **References.** Dummit-Foote; Ireland-Rosen.

Gap closed. Finite-field permutation problems require exact facts about x^{p-1} and polynomial-function representatives.

Statement/properties. In $\mathbb{F}_p$, $x^p = x$ for all x and $x^{p-1} = 1$ for $x \neq 0$. A polynomial $f \in \mathbb{F}_p[x]$ is a permutation polynomial iff the function $x \mapsto f(x)$ is bijective on the p field elements. The polynomial $x^p - x$ is the product of all monic linear factors over $\mathbb{F}_p$.

Compact proof core. The nonzero elements form a multiplicative group of order $p-1$, giving Fermat. The factorization follows because all p elements are roots and the degree is p.

Use pattern. Use for residue systems, polynomial functions on $\mathbb{Z}/p\mathbb{Z}$, and GL actions over finite fields.

Occurrence references. 1994-D1-P6; 1994-D2-P5; 1997-D1-P3; 1997-D1-P4; 1997-D1-P5; 1999-D1-P3; 1999-D2-P2; 2000-D2-P1; 2001-D1-P1; 2001-D1-P2; 2001-D2-P6; 2002-D1-P3; 2002-D2-P3; 2003-D1-P2; 2003-D1-P4; 2003-D2-P4; 2005-D1-P2; 2005-D1-P4; 2005-D2-P2; 2005-D2-P6; 2006-D1-P2; 2006-D1-P3; (+73 more in CSV).

AUD-ANT-03 - Quartic residues and modular obstruction by tables

Domain. Algebra/number theory. **References.** Ireland-Rosen.

Gap closed. The 29-divisibility problem is a quartic-residue obstruction; the table should be seen as a standard finite-field residue check.

Statement/properties. Modulo an odd prime p, the set of fourth powers in $\mathbb{F}_p$ is the image of the homomorphism $x \mapsto x^4$ on $\mathbb{F}_p^\times$, together with 0. Its size is $1 + (p-1)/\gcd(4, p-1)$. For $p = 29$, this gives 8 nonzero fourth-power classes plus 0; direct checking shows no solution to $1 + y^4 + z^4 \equiv 0 \pmod{29}$.

Compact proof core. The multiplicative group is cyclic. The image size of the fourth-power map is $(p-1)/\gcd(4, p-1)$. Once the small image is listed, the obstruction is finite checking.

Use pattern. Use when a sum of fourth powers is divisible by a prime and one variable can be normalized to 1.

Occurrence references. 2007-D2-P2.

AUD-ANT-04 - Field characteristic and parity of additive inverse pairing

Domain. Algebra/number theory. **References.** Artin; Dummit-Foote.

Gap closed. The characteristic problem needs the exact separation between characteristic 2, characteristic 7, and pairing by additive inverses.

Statement/properties. In a field of characteristic p, $mx = 0$ for all x iff $p \mid m$; in characteristic 0, only $m = 0$ annihilates all elements. If a finite odd-length list is permuted into its additive negatives in characteristic not 2, then one element would have to be self-negative, hence zero, unless another obstruction applies.

Compact proof core. The prime subfield is either $\mathbb{Q}$ or $\mathbb{F}_p$. In characteristic not 2, $x = -x$ implies $2x = 0$, hence $x = 0$. Pairing nonzero elements with their distinct negatives requires an even number of entries.

Use pattern. Use for problems where a global sum equation forces each transformed entry to be the negative of an original entry.

Occurrence references. 1994-D1-P3; 2003-D1-P2.

AUD-ANT-05 - Gauss lemma and factorization of integer polynomials

Domain. Algebra/number theory. **References.** Dummit-Foote; Artin.

Gap closed. Several integer-polynomial problems require moving between $\mathbb{Q}[x]$ and $\mathbb{Z}[x]$ without losing integrality.

Statement/properties. Gauss lemma: a primitive polynomial in $\mathbb{Z}[x]$ that factors over $\mathbb{Q}[x]$ factors over $\mathbb{Z}[x]$ into primitive factors. If an integer polynomial has all rational roots, then it factors over $\mathbb{Z}[x]$ as a product of integer linear factors after extracting the leading coefficient among the factors. The rational-root theorem: a rational root u/v in lowest terms satisfies $u \mid a_0$ and $v \mid a_n$.

Compact proof core. Primitive content is multiplicative. Clearing denominators and comparing contents transfers factorizations to $\mathbb{Z}[x]$. The rational-root theorem follows by substituting u/v and using coprimality.

Use pattern. Use for coefficient-permutation polynomials and irreducibility in $\mathbb{Q}[x]$.

Occurrence references. 2005-D1-P4.

AUD-ANT-06 - Power subgroups, commutators, and central intersections

Domain. Algebra/number theory. **References.** Dummit-Foote; Artin.

Gap closed. The $G(m)$ theorem needs exact commutator manipulation.

Statement/properties. Let $G(m)$ be the subgroup generated by mth powers. If $d = \gcd(m, n)$, then $\langle G(m), G(n) \rangle = G(d)$ by Bezout. If $G(m)$ and $G(n)$ are abelian, then commutators between generators a^m and b^n lie in the central intersection $G(m) \cap G(n)$; exponent relations can then force them to be trivial.

Compact proof core. Bezout expresses a dth power as a product of an mth power and an nth power. For $z = [a^m, b^n]$, rewrite it both as an mth-power expression and an nth-power expression to get centrality. Then use coprime exponent squares to conclude $z = e$.

Use pattern. Use when commutativity of power-generated subgroups must descend to a gcd-power subgroup.

Occurrence references. 1997-D2-P4; 2000-D1-P5; 2005-D1-P6; 2005-D2-P3; 2011-D1-P2; 2014-D1-P1; 2020-D1-P2.

AUD-ANT-07 - Noncommutative rings: square-zero and idempotent identities

Domain. Algebra/number theory. **References.** Artin; Dummit-Foote.

Gap closed. The ring problems require safe expansion without commutativity and attention to characteristic.

Statement/properties. In any ring where $x^2 = 0$ for every x, expansion of $(a+b)^2 = 0$ gives $ab + ba = 0$. Repeated anticommutation yields $abc = -abc$, hence $2abc = 0$. In a characteristic-zero ring, if idempotents e, f, g satisfy $e + f + g = 0$, then $e = f = g = 0$.

Compact proof core. For square-zero rings, expand only preserving order. For idempotents, write $g = -(e+f)$ and compare $g^2 = g$ to get $ef + fe = 2g$; commutator algebra shows $ef = fe$, then $e + f + ef = 0$ forces $e = f = g$ and $3e = 0$.

Use pattern. Use whenever a noncommutative ring problem gives identities for every element or idempotents.

Occurrence references. 1999-D2-P1; 2000-D1-P5.

AUD-ANT-08 - Permutation-group generators and parity/normal-subgroup obstructions

Domain. Algebra/number theory. **References.** Dummit-Foote; Artin.

Gap closed. Several problems rely on explicit generators of S_n and on quotient obstructions.

Statement/properties. The adjacent transpositions generate S_n, and 3-cycles generate A_n for $n \geq 3$. A transposition plus an n-cycle generates S_n. In S_4, the Klein four subgroup $V_4 = \{e, (12)(34), (13)(24), (14)(23)\}$ is normal and $S_4/V_4 \cong S_3$, which is not cyclic.

Compact proof core. Adjacent transpositions bubble-sort any permutation. Every even permutation is a product of 3-cycles. Normality of V_4 follows by conjugation preserving cycle type and the three double transpositions.

Use pattern. Use for generator existence/nonexistence in small symmetric groups and for games selecting permutations.

Occurrence references. 1996-D2-P3; 1998-D1-P2; 2009-D2-P5.

AUD-ANT-09 - Semidirect products and automorphism-count counterexamples

Domain. Algebra/number theory. **References.** Dummit-Foote; Artin.

Gap closed. The automorphism problem needs a concrete semidirect product construction and element-order count.

Statement/properties. If a group T acts on H by automorphisms $\tau : T \to \operatorname{Aut} H$, the semidirect product $H \rtimes T$ has multiplication $(h,t)(h',t') = (h\tau_t(h'), tt')$. For $H = \mathbb{F}_2^3$, $|\operatorname{Aut} H| = |GL_3(\mathbb{F}_2)| = (8-1)(8-2)(8-4) = 168$. A cyclic order-3 action by coordinate shift gives a group G of order 24 with far fewer automorphisms, showing $|\operatorname{Aut} H| > |\operatorname{Aut} G|$ with $H \triangleleft G$.

Compact proof core. The order of $GL_3(\mathbb{F}_2)$ counts choices of independent columns. In the semidirect product, element orders constrain automorphisms because automorphisms preserve order and are determined by images of generators.

Use pattern. Use for constructing finite-group counterexamples involving normal subgroups and automorphism counts.

Occurrence references. No direct occurrence assigned by the audit; retained as prerequisite/background closure..

AUD-ANT-10 - Elementary abelian 2-groups and Steiner triple systems

Domain. Algebra/number theory. **References.** Dummit-Foote; Van-Lint-Wilson.

Gap closed. The triple-system problem needs the equivalence with vector spaces over $\mathbb{F}_2$.

Statement/properties. A finite abelian group in which every nonidentity element has order 2 is a vector space over $\mathbb{F}_2$, hence has order 2^m. A Steiner triple operation on S can be completed by adding 0 and defining $a * a = 0$, $a * 0 = a$; associativity then turns $S \cup \{0\}$ into such a group. Thus $|S| = 2^m - 1$.

Compact proof core. Commutativity, identity, and $a*a=0$ give elementary abelian structure once associativity is checked. Conversely, in $\mathbb{F}_2^n \setminus \{0\}$, triples $\{a,b,a+b\}$ satisfy the unique-pair and associativity conditions.

Use pattern. Use for set systems of triples with exactly one block through each pair and algebraic closure conditions.

Occurrence references. 2003-D2-P4; 2012-D2-P5.

AUD-ANT-11 - Dyadic-rational subgroups inside upper triangular matrices

Domain. Algebra/number theory. **References.** Artin; Dummit-Foote.

Gap closed. The non-finitely-generated subgroup problem needs the conjugation formula that creates dyadic translations.

Statement/properties. In $GL_2(\mathbb{R})$, let $A=\operatorname{diag}(2,1)$ and $B=M(1)=\begin{pmatrix}1 & 1\\ 0 & 1\end{pmatrix}$. Then $A^{-n}BA^n=M(2^{-n})$. The subgroup $\{M(x): x\in\mathbb{Z}[1/2]\}$ is isomorphic to the additive group of dyadic rationals and is not finitely generated.

Compact proof core. Compute $A^{-n}M(1)A^n=M(2^{-n})$. A finitely generated subgroup of $\mathbb{Z}[1/2]$ has denominators bounded by the largest power of 2 appearing among generators, but 2^{-n} requires unbounded denominators.

Use pattern. Use for upper-triangular matrix groups generated by dilation and shear.

Occurrence references. 1995-D2-P6; 1996-D2-P3.

AUD-ANT-12 - Words modulo idempotent relations

Domain. Algebra/number theory. **References.** Lothaire.

Gap closed. The word-reduction problem uses a lemma about words containing all letters; this was absent from general combinatorics notes.

Statement/properties. Under relations $uu\sim u$ and compatibility with concatenation, any adjacent repeated block can be deleted: $uvvw\sim uvw$. If a word u contains all letters and v is arbitrary, one can append a word w so that $uvw\sim u$. Consequently any word on three letters of length >8 is equivalent to a shorter word, and all words reduce to length at most 8.

Compact proof core. For one added letter, find its occurrence inside u and use $aa\sim a$ after grouping. For a general word, process letters one by one. Then split a long word into first four, middle, last four letters; the first and last blocks must contain all letters unless reducible already.

Use pattern. Use for semigroup/word equivalence relations generated by idempotence.

Occurrence references. 1999-D2-P5; 2021-D1-P3.

AUD-CMB-01 - Double counting walks, degree squares, and graph girth

Domain. Combinatorics. **References.** West; Van-Lint-Wilson.

Gap closed. The friendship graph problem uses a precise injection from ordered vertex pairs to length-two walks.

Statement/properties. In a simple graph, the number of ordered walks of length 2 is $\sum_v d(v)^2$. If every nonadjacent pair has a common neighbor, one can map each ordered distinct pair to a length-two walk: adjacent pairs use (u,v,u), nonadjacent pairs use a common neighbor. Equality with n^2-n forces uniqueness and excludes triangles and 4-cycles.

Compact proof core. Count walks by their middle vertex. The map is injective because a walk with equal endpoints comes from an edge, while a walk with unequal endpoints identifies the nonadjacent ordered pair. Equality forces surjectivity and the stated exclusions.

Use pattern. Use for graph problems with degree-square sums and common-neighbor hypotheses.

Occurrence references. 2017-D1-P4.

AUD-CMB-02 - Bipartite graph encoding of grids and alternating cycles

Domain. Combinatorics. **References.** West.

Gap closed. The grid-cycle problem needs the row-column graph model explicitly.

Statement/properties. Represent an $r \times c$ grid by a bipartite graph with row vertices and column vertices; a marked cell is an edge. An alternating row/column cycle in the graph is exactly a sequence of marked grid points alternating same row and same column. A graph with N vertices and at least N edges contains a cycle.

Compact proof core. A forest on N vertices has at most $N-1$ edges. Any cycle in a bipartite row-column graph alternates row and column vertices; the corresponding edges are the marked points.

Use pattern. Use for grid problems asking for alternating horizontal/vertical marked-point cycles.

Occurrence references. 1999-D1-P5.

AUD-CMB-03 - Pigeonhole with divisor values and polynomial fibers

Domain. Combinatorics. **References.** Aigner-Ziegler; Van-Lint-Wilson.

Gap closed. The divisor-fiber polynomial problem combines finite divisor count with degree-root obstruction.

Statement/properties. If a polynomial $g \in \mathbb{Z}[x]$ takes values in a finite set D on N distinct inputs, then some value is assumed at least $\lceil N/|D| \rceil$ times. If $g(x) - c$ has more roots than its degree, then $g \equiv c$; otherwise $\deg g$ is at least the number of distinct roots of that fiber.

Compact proof core. This is the pigeonhole principle plus the polynomial root-degree lemma.

Use pattern. Use when $g(a_i)$ must divide a fixed integer such as 2008.

Occurrence references. 1994-D1-P1; 1994-D2-P4; 1996-D1-P3; 1996-D2-P3; 1996-D2-P6; 1997-D2-P6; 1998-D2-P3; 1998-D2-P4; 1998-D2-P5; 1998-D2-P6; 1999-D1-P5; 2000-D1-P1; 2001-D2-P1; 2002-D1-P5; 2003-D2-P5; 2004-D1-P1; 2004-D1-P2; 2004-D1-P4; 2004-D2-P6; 2005-D1-P2; 2006-D1-P2; 2006-D2-P1; (+30 more in CSV).

AUD-CMB-04 - Intersecting finite set families and inductive hitting sets

Domain. Combinatorics. **References.** Van-Lint-Wilson; Aigner-Ziegler.

Gap closed. The intersecting-family problem uses a two-family induction with fixed set sizes.

Statement/properties. If $\mathcal{F}$ and $\mathcal{G}$ are cross-intersecting families of finite subsets of $\mathbb{N}$, all members of sizes r and s, then there is a finite set Y meeting every intersection $A \cap B$ with $A \in \mathcal{F}$, $B \in \mathcal{G}$. The proof is by induction on (r, s) after partitioning by intersections with fixed $A_0 \cup B_0$.

Compact proof core. Fix A_0, B_0. For each nonempty trace $C \subset A_0 \cup B_0$, restrict to sets with trace C. If two traces intersect, the finite union itself works. If traces are disjoint, delete them and apply induction to smaller set sizes outside $A_0 \cup B_0$.

Use pattern. Use for same-size intersecting families over an infinite ground set.

Occurrence references. 1997-D1-P6; 2009-D1-P2.

AUD-CMB-05 - Erdos-Szekeres style monotone subsequences with growth gaps

Domain. Combinatorics. **References.** Aigner-Ziegler; Van-Lint-Wilson.

Gap closed. The exponential-gap monotone sequence problem is a strengthened induction on two target lengths.

Statement/properties. For integers $k, l \geq 2$, any set of $\binom{k+l-4}{k-2} + 1$ real numbers contains either an increasing length-k sequence or a decreasing length-l sequence whose distances from the first term at least double at each step. Splitting at the midpoint between minimum and maximum gives the Pascal recurrence.

Compact proof core. Let m, M be min and max, split X into lower and upper halves. If the lower half is large, induction gives either a decreasing sequence or an increasing sequence extendable by M; the midpoint split ensures the final gap doubles. The upper-half case is symmetric.

Use pattern. Use for monotone sequence selection with a binomial threshold and multiplicative gap condition.

Occurrence references. 1996-D1-P1; 1997-D1-P1; 2004-D1-P1; 2004-D1-P4; 2004-D1-P5; 2006-D2-P2; 2010-D1-P5; 2010-D2-P2; 2012-D1-P2; 2016-D2-P6; 2019-D2-P8.

AUD-CMB-06 - Transfer matrices for binary tables and symmetry of dimensions

Domain. Combinatorics. **References.** Stanley; Wilf.

Gap closed. The binary-table matrix recurrence problem needs the transfer-matrix interpretation.

Statement/properties. For constrained $n \times k$ binary arrays, encode each column or row by a binary state. A transfer matrix T_n has entry 1 when two adjacent states are compatible. Then the number of length-k arrays is the sum of entries of T_n^{k-1}. If the constraint is symmetric under transposition, the count also equals the analogous expression with n and k swapped.

Compact proof core. A valid table is exactly a walk of length $k-1$ in the state graph. Matrix powers count walks. Transposing tables gives a bijection between $n \times k$ and $k \times n$ valid arrays.

Use pattern. Use for array-filling problems with local adjacency constraints.

Occurrence references. 2004-D2-P6.

AUD-CMB-07 - State recurrences and last-symbol classification

Domain. Combinatorics. **References.** Stanley; Wilf.

Gap closed. Many enumeration problems are shortest by splitting objects by the final local state.

Statement/properties. When a constraint is local, split the set of length-n objects according to the last one or two symbols. Derive a finite linear recurrence for the vector of state counts. If two objects satisfy the same recurrence and have matching initial values, they have the same count for all n.

Compact proof core. Appending a new symbol depends only on the stored state, so counts update by a fixed transition matrix. Equality follows by induction from the recurrence and initial values.

Use pattern. Use for word/function counts, local sequence constraints, and finite automaton enumerations.

Occurrence references. 1994-D2-P1; 1996-D2-P3; 1997-D1-P6; 1997-D2-P6; 1998-D2-P4; 1999-D1-P2; 2000-D2-P1; 2001-D2-P5; 2002-D1-P2; 2002-D2-P2; 2002-D2-P3; 2003-D1-P1; 2003-D1-P2; 2004-D1-P1; 2004-D1-P2; 2006-D1-P2; 2007-D1-P4; 2008-D2-P3; 2010-D2-P2; 2011-D1-P4; 2012-D1-P1; 2012-D2-P2; (+25 more in CSV).

AUD-CMB-08 - Permutation statistics via determinants

Domain. Combinatorics. **References.** Stanley.

Gap closed. The parity of permutation counts with displacement statistic uses a determinant generating function.

Statement/properties. For a matrix $M(x) = (x^{|i-j|})_{i,j=1}^{n}$, the determinant expands as

$$\det M(x) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) x^{\sum_i |\sigma(i)-i|}.$$

Modulo 2, signs disappear, so the coefficient parity equals the parity of the number of permutations with that displacement statistic.

Compact proof core. This is the Leibniz determinant formula. Row/column operations can then compute the determinant explicitly, e.g. $(1-x^2)^{n-1}$ in the IMC problem.

Use pattern. Use when a parity question over permutations has a natural determinant expansion.

Occurrence references. 2002-D2-P1; 2005-D1-P1; 2005-D2-P3; 2007-D1-P2; 2007-D1-P3; 2008-D1-P6; 2009-D1-P5; 2010-D1-P5; 2011-D2-P2; 2015-D1-P1; 2024-D2-P7; 2024-D2-P9.

AUD-CMB-09 - Residue-class grouping of words and dice sequences

Domain. Combinatorics. **References.** Feller; Stanley.

Gap closed. The dice modulo problem has both a probabilistic recurrence and a direct grouping proof.

Statement/properties. Sequences over digits $1,\dots,6$ can be partitioned, except for the all-six sequence, into blocks of five whose digit sums cover all residue classes modulo 5. Equivalently, the Markov chain on residues has transition matrix whose nontrivial eigenvalues have modulus $1/6$.

Compact proof core. Group each non-all-six sequence by the position of its first non-six digit and vary that digit through $1,\dots,5$. The sums differ by a complete residue system modulo 5.

Use pattern. Use for exact probabilities of digit sums modulo a fixed modulus.

Occurrence references. 1999-D2-P2.

AUD-GEO-01 - Gram matrices and rational dependence from rational distances

Domain. Geometry. **References.** Berger; Coxeter.

Gap closed. The rational-distance problem is linear algebra through Gram matrices; this needed an explicit statement.

Statement/properties. If $v_0 = 0$ and all distances $\|v_i - v_j\|$ are rational, then all inner products $\langle v_i, v_j\rangle$ are rational because

$$2\langle v_i, v_j\rangle = \|v_i\|^2 + \|v_j\|^2 - \|v_i - v_j\|^2.$$

If $v_1,\dots,v_n$ are real-linearly independent, their Gram matrix is invertible; solving $G\lambda = w$ with rational G, w gives rational coefficients.

Compact proof core. The polarization identity gives rational entries. A Gram matrix of independent vectors is positive definite, hence invertible. The inverse of an invertible rational matrix has rational entries by the adjugate formula.

Use pattern. Use for rational-distance configurations in $\mathbb{R}^n$ with $n+2$ points.

Occurrence references. 2006-D2-P4.

AUD-GEO-02 - Projection, signed area, and volume scaling

Domain. Geometry. **References.** Berger; Coxeter.

Gap closed. Several geometry solutions compare areas or volumes after orthogonal or linear projection.

Statement/properties. Orthogonal projection of a planar region onto another plane multiplies signed area by the cosine of the angle between the planes. A linear map with singular scale factors $s_1,\dots,s_n$ multiplies n-dimensional volume by $|\det T| = \prod s_i$. The volume of the unit ball in $\mathbb{R}^3$ is $4\pi/3$; an ellipsoid with semiaxes a, b, c has volume $(4\pi/3)abc$.

Compact proof core. Area scaling follows by applying projection to an orthonormal basis of the plane. Volume scaling is the determinant definition of volume. The ellipsoid is the linear image of the unit ball.

Use pattern. Use for tetrahedron angle projections, ellipsoid volumes, and geometric probability.

Occurrence references. 1996-D1-P3; 1996-D2-P4; 1997-D1-P5; 1997-D2-P5; 1998-D2-P5; 1999-D1-P5; 2002-D2-P4; 2003-D2-P3; 2004-D1-P4; 2004-D2-P2; 2004-D2-P3; 2005-D2-P5; 2006-D2-P1; 2008-D2-P2; 2009-D2-P1; 2011-D2-P5; 2014-D1-P5; 2015-D1-P5; 2017-D2-P10; 2018-D1-P5; 2019-D2-P10; 2022-D2-P5; (+4 more in CSV).

AUD-GEO-03 - Simplex volume, barycentric coordinates, and circumcenter identity

Domain. Geometry. **References.** Berger; Coxeter.

Gap closed. The simplex-circumcenter identity requires barycentric volume weights and affine dependence of centers.

Statement/properties. If P lies inside an n-simplex S with vertices $P_0,\dots,P_n$, then the volumes $v(S_i)$ of the simplices obtained by replacing P_i by P are barycentric weights: $\sum_i v(S_i) = v(S)$ and $P = \sum_i (v(S_i)/v(S))P_i$. For circumcenters, the IMC identity

$$\sum_i v(S_i)C(S_i) = v(S)C(S)$$

holds.

Compact proof core. Volume weights are barycentric coordinates by decomposing S into sub-simplices. The circumcenter identity is proved by showing both sides have equal squared-distance equations against all vertices, or inducting using projection onto facets.

Use pattern. Use for affine geometry problems involving replacing simplex vertices by an interior point.

Occurrence references. 2009-D1-P5; 2015-D1-P5.

AUD-GEO-04 - Regular simplex bound on spherical codes with negative inner products

Domain. Geometry. **References.** Coxeter; Berger.

Gap closed. The maximum number of sphere points at pairwise distance $> \sqrt{2}$ is an inner-product simplex bound.

Statement/properties. For unit vectors u_i, the condition $\|u_i - u_j\| > \sqrt{2}$ is equivalent to $\langle u_i, u_j \rangle < 0$. In $\mathbb{R}^n$, at most $n+1$ unit vectors can have pairwise negative inner products; equality is realized by the vertices of a regular simplex centered at the origin, with pairwise inner product $-1/n$.

Compact proof core. Induct by fixing one vector and projecting the rest onto its orthogonal complement after noting they lie in the opposite open hemisphere. The construction is the standard regular simplex from vectors $e_i - \frac{1}{n+1}\mathbf{1}$ in the hyperplane $\sum x_i = 0$.

Use pattern. Use for spherical distance configurations and simplex extremal examples.

Occurrence references. 2001-D2-P3; 2021-D2-P8.

AUD-GEO-05 - Ellipse focus-directrix form and conic intersections

Domain. Geometry. **References.** Coxeter; Berger.

Gap closed. The ellipse intersection problem uses the focus-directrix definition, not general quartic intersection.

Statement/properties. An ellipse is the locus of points whose distance to a focus divided by distance to a directrix is a fixed eccentricity $0 < e < 1$. If two ellipses share a focus, a common point must satisfy $e_1 d(X, l_1) = e_2 d(X, l_2)$ with the correct signed side choices; this is a line. A line meets a nondegenerate ellipse in at most two points.

Compact proof core. Use signed distances to directrices. The equality of scaled signed distances is linear. Substitution of a line into a quadratic equation gives a quadratic polynomial, hence at most two intersections unless the line is contained, impossible for an ellipse.

Use pattern. Use for conics sharing a focus or directrix constraints.

Occurrence references. 2008-D2-P2.

AUD-GEO-06 - John ellipse contact-arc obstruction

Domain. Geometry. **References.** Rockafellar; Boyd-Vandenberghe.

Gap closed. The John-ellipse problem requires the perturbation idea: if a long arc has no contact, enlarge the disk slightly into an ellipse.

Statement/properties. If the unit disk is the maximal-area ellipse contained in a centrally symmetric convex body $B \subset \mathbb{R}^2$, then every arc of the unit circle of length at least $\pi/2$ meets ∂B. Otherwise an ellipse stretched slightly along the arc direction and compressed just enough through the endpoints has area larger than π and remains inside B, contradiction.

Compact proof core. A compact arc disjoint from the boundary has positive distance to the boundary, allowing a small perturbation. The chosen ellipse has semiaxes producing area $> \pi$ while still lying inside the local slack and inside the disk elsewhere.

Use pattern. Use for maximal inscribed ellipses and contact-point distribution.

Occurrence references. 1996-D2-P4.

AUD-GEO-07 - Metric content, diameter covers, and Lipschitz contractions

Domain. Geometry. **References.** Burago-Burago-Ivanov.

Gap closed. The metric-content problem needs exact one-dimensional cover length and Lipschitz monotonicity.

Statement/properties. For a line segment L, the infimum of $\sum_i \operatorname{diam} E_i$ over finite covers equals length L. If f is 1-Lipschitz, then $\operatorname{diam} f(E_i) \le \operatorname{diam} E_i$, so any contraction of E onto L gives $C(E) \ge \text{length } L$.

Compact proof core. For a segment, any finite cover projected to the line covers the interval, and interval cover lengths dominate the segment length. Lipschitz maps do not increase diameters, so covers push forward to covers.

Use pattern. Use for geometric measure-like problems with diameter covers and contractions.

Occurrence references. 1996-D1-P6.

AUD-GEO-08 - Closest-point uniqueness is dense for closed sets

Domain. Geometry. **References.** Burago-Burago-Ivanov; Rudin.

Gap closed. Nearest-point density requires moving along a segment to a nearest point and comparing balls.

Statement/properties. For a nonempty closed set $A \subset \mathbb{R}^n$, the set of points having a unique nearest point in A is dense. If $b \notin A$ and a_0 is a nearest point, then every point on the open segment (a_0, b) has a_0 as its unique nearest point.

Compact proof core. Closedness plus compact intersection with a large ball gives existence of a nearest point. For a point c between a_0 and b, the ball centered at c through a_0 lies inside the ball centered at b through a_0 and touches its boundary only at a_0, forcing uniqueness.

Use pattern. Use for metric projection problems onto arbitrary closed sets.

Occurrence references. 2000-D2-P2; 2003-D2-P3.

AUD-GEO-09 - Countability by rational encoding of sphere intersections

Domain. Geometry. **References.** Burago-Burago-Ivanov; Aigner-Ziegler.

Gap closed. The sphere-intersection countability proof uses separability and unique reconstruction from rational witnesses.

Statement/properties. In $\mathbb{R}^n$, every nonempty open set contains a rational point. If each pair of spheres intersects in at most one point, then each point belonging to at least two spheres can be encoded by finitely many rational points chosen in components of the complement of the two spheres. Since rational tuples are countable and the encoding is injective, the set of such points is countable.

Compact proof core. Choose rational witnesses in the three relevant components. If another intersection point gave the same witnesses, nesting relations among the corresponding components force the two intersection points to coincide.

Use pattern. Use for countability of special points in separable metric spaces via rational certificates.

Occurrence references. 1998-D1-P6; 1998-D2-P5; 2004-D1-P1; 2008-D2-P6.

AUD-GEO-10 - Polygon triangulation parity

Domain. Geometry. **References.** Coxeter; West.

Gap closed. The triangulation problem is a parity-preserving induction by adding three vertices.

Statement/properties. A triangulation of an n-gon has $n-2$ triangles. The parity of the number of incident triangles at vertices can be controlled by induction: adding three consecutive vertices and three triangles changes selected old incidences by 2 and gives the new vertices odd incidence.

Compact proof core. Verify base cases $n = 3, 4, 5$. For $n+3$, triangulate the polygon with the last three vertices removed, then add three triangles along the final chain; old parities are preserved because two triangles are added at the endpoints.

Use pattern. Use for parity constraints in triangulations of convex polygons.

Occurrence references. 2011-D2-P5; 2018-D1-P5.

AUD-GEO-11 - Cube dissection and the Frobenius coin lemma

Domain. Geometry. **References.** Aigner-Ziegler; Van-Lint-Wilson.

Gap closed. The cube dissection problem needs a number-theoretic semigroup lemma.

Statement/properties. If a, b are coprime positive integers, every sufficiently large integer is $ax + by$ with $x, y \geq 0$. A cube dissection into N smaller cubes can be refined into $N + (r^d - 1)$ cubes by subdividing one cube into r^d equal cubes. Therefore two coprime numbers of the form $r^d - 1$ imply all sufficiently large counts are possible.

Compact proof core. Residues $0, a, 2a, \ldots, (b-1)a$ cover all classes modulo b; choose the representative not exceeding $(b-1)a$, then the remaining quotient is nonnegative. Refinement increases the number of pieces by $r^d - 1$.

Use pattern. Use for dissection counts in squares/cubes and eventual semigroup arguments.

Occurrence references. 2000-D2-P1; 2023-D2-P9.

AUD-PRB-01 - Expectation, symmetry, and geometric probability in convex hulls

Domain. Probability/measure. **References.** Feller; Billingsley; Alon-Spencer.

Gap closed. The geometric probability entries require symmetry, conditioning, and monotonicity of random convex hulls beyond elementary counting.

Statement/properties. For independent uniform points in a symmetric region, distributional symmetries can compare events by applying measure-preserving transformations. Conditioning on all but one point often reduces a convex-hull event to an area/angle fraction. Linearity of expectation applies without independence; indicator variables convert counts or probabilities into integrals.

Compact proof core. Use invariance of Lebesgue measure under rotations/reflections and Fubini/Tonelli to condition on subsets of random variables. Indicators satisfy $\mathbb{E}\sum I_i = \sum \mathbb{E} I_i$.

Use pattern. Use for random points in disks, balls, circles, and random polytope comparisons.

Occurrence references. 2015-D1-P5; 2019-D2-P10; 2021-D1-P2; 2022-D2-P8; 2024-D1-P5.

AUD-PRB-02 - Measure-preserving symmetry and integrable reciprocal inequalities

Domain. Probability/measure. **References.** Billingsley; Hardy-Polya-Littlewood.

Gap closed. The integral reciprocal problem is a symmetry plus AM-GM argument.

Statement/properties. If $f > 0$ is integrable on $[0, 1]$ and $f(x)f(1-x) = 1$, then

$$\int_0^1 f(x)dx = \frac{1}{2}\int_0^1 (f(x) + f(1-x))dx \geq \int_0^1 1\,dx = 1.$$

Compact proof core. The substitution $u = 1 - x$ preserves Lebesgue measure. AM-GM gives $f(x) + f(1-x) \geq 2\sqrt{f(x)f(1-x)} = 2$.

Use pattern. Use for symmetric reciprocal functional integral inequalities.

Occurrence references. 2022-D1-P1.

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15 Problem-by-Problem Theory Sufficiency Audit

This table records the additional closure nodes assigned during the audit. It should be read together with the earlier problem catalog and theory-node occurrence lists. A row marked “closed” means that the problem has an explicit theory route in the previous manual plus the addendum above; it does not mean that the method execution is automatic.

Problem	Primary domain	Problem type	Added closure nodes	PDF p.
1994-D1-P1	Linear algebra / matrix theory	inequality / sharp bound	AUD-LA-01; AUD-RA-08; AUD-CMB-03	2
1994-D1-P2	Real analysis / calculus	inequality / sharp bound	AUD-RA-01; AUD-RA-03; AUD-RA-05; AUD-RA-06; AUD-RA-08	2
1994-D1-P3	Algebra / number theory	proof of structural property	AUD-ANT-04; AUD-RA-08	2
1994-D1-P4	Linear algebra / matrix theory	existence / construction	AUD-RA-08	3
1994-D1-P5	Functional equations / dynamics	limit / asymptotic evaluation	AUD-RA-03; AUD-RA-05; AUD-RA-08	4
1994-D1-P6	Algebra / number theory	inequality / sharp bound	AUD-RA-01; AUD-RA-06; AUD-RA-08; AUD-ANT-02	5
1994-D2-P1	Functional equations / dynamics	nonexistence / counterexample	AUD-RA-01; AUD-RA-02; AUD-RA-06; AUD-RA-08; AUD-CMB-07	6
1994-D2-P2	Functional equations / dynamics	classification / find all	AUD-RA-07; AUD-RA-08	6
1994-D2-P3	Functional equations / dynamics	proof of structural property	AUD-RA-01; AUD-RA-06	7
1994-D2-P4	Linear algebra / matrix theory	proof of structural property	AUD-RA-06; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-CMB-03	7
1994-D2-P5	Algebra / number theory	existence / construction	AUD-RA-08; AUD-ANT-02	8
1994-D2-P6	Real analysis / calculus	limit / asymptotic evaluation	AUD-RA-03; AUD-RA-05; AUD-RA-08	9
1995-D1-P1	Linear algebra / matrix theory	proof of structural property	AUD-LA-02	11
1995-D1-P2	Real analysis / calculus	inequality / sharp bound	AUD-RA-08	11
1995-D1-P3	Real analysis / calculus	limit / asymptotic evaluation	AUD-RA-01; AUD-RA-03; AUD-RA-05; AUD-RA-06; AUD-RA-08	11
1995-D1-P4	Real analysis / calculus	proof of structural property	AUD-RA-08	12
1995-D1-P5	Linear algebra / matrix theory	proof of structural property	AUD-LA-02; AUD-LA-03; AUD-PC-01; AUD-PC-08; AUD-PC-13	13
1995-D1-P6	Real analysis / calculus	existence / construction	AUD-RA-07; AUD-RA-08	13
1995-D2-P1	Linear algebra / matrix theory	existence / construction	AUD-LA-05; AUD-RA-08	15
1995-D2-P2	Real analysis / calculus	exact computation	AUD-RA-03; AUD-RA-05; AUD-RA-08	15
1995-D2-P3	Complex analysis / polynomials	proof of structural property	AUD-RA-08; AUD-PC-01; AUD-PC-03; AUD-PC-05; AUD-PC-06	16
1995-D2-P4	Algebra / number theory	proof of structural property	AUD-RA-07; AUD-RA-08; AUD-PC-01; AUD-PC-09; AUD-PC-13	17
1995-D2-P5	Real analysis / calculus	proof of structural property	AUD-RA-08	18
1995-D2-P6	Real analysis / calculus	existence / construction	AUD-RA-09; AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-07; AUD-ANT-11	20
1996-D1-P1	Linear algebra / matrix theory	exact computation	AUD-CMB-05	22

Problem	Primary domain	Problem type	Added closure nodes	PDF p.
1996-D1-P2	Real analysis / calculus	exact computation	AUD-RA-08	23
1996-D1-P3	Linear algebra / matrix theory	existence / construction	AUD-LA-04; AUD-RA-08; AUD-CMB-03; AUD-GEO-02	24
1996-D1-P4	Real analysis / calculus	inequality / sharp bound	AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08	24
1996-D1-P5	Real analysis / calculus	inequality / sharp bound	AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-06; AUD-RA-08	25
1996-D1-P6	Geometry / convexity / metric	inequality / sharp bound	AUD-RA-06; AUD-RA-07; AUD-RA-08; AUD-GEO-07	27
1996-D2-P1	Functional equations / dynamics	limit / asymptotic evaluation	AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08; AUD-RA-11	29
1996-D2-P2	Algebra / number theory	proof of structural property	AUD-RA-08; AUD-PC-01; AUD-PC-13	30
1996-D2-P3	Algebra / number theory	counting / enumeration	AUD-RA-08; AUD-ANT-08; AUD-ANT-11; AUD-CMB-03; AUD-CMB-07	31
1996-D2-P4	Geometry / convexity / metric	inequality / sharp bound	AUD-RA-07; AUD-RA-08; AUD-GEO-02; AUD-GEO-06	32
1996-D2-P5	Functional equations / dynamics	inequality / sharp bound	AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-06; AUD-RA-08	32
1996-D2-P6	Real analysis / calculus	existence / construction	AUD-RA-06; AUD-RA-08; AUD-PC-01; AUD-PC-13; AUD-CMB-03	34
1997-D1-P1	Real analysis / calculus	limit / asymptotic evaluation	AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08; AUD-CMB-05	36
1997-D1-P2	Real analysis / calculus	proof / reasoning	AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08; AUD-RA-16	37
1997-D1-P3	Algebra / number theory	proof of structural property	AUD-PC-01; AUD-ANT-02	37
1997-D1-P4	Algebra / number theory	inequality / sharp bound	AUD-RA-08; AUD-ANT-02	38
1997-D1-P5	Algebra / number theory	inequality / sharp bound	AUD-RA-08; AUD-GEO-02; AUD-ANT-02	40
1997-D1-P6	Combinatorics / discrete structures	nonexistence / counterexample	AUD-CMB-04; AUD-CMB-07	42
1997-D2-P1	Real analysis / calculus	inequality / sharp bound	AUD-RA-01; AUD-RA-03; AUD-RA-05; AUD-RA-06; AUD-RA-08	44
1997-D2-P2	Linear algebra / matrix theory	proof of structural property	AUD-LA-03; AUD-LA-07	45
1997-D2-P3	Real analysis / calculus	proof of structural property	AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-06; AUD-RA-08	45
1997-D2-P4	Linear algebra / matrix theory	existence / construction	AUD-LA-06; AUD-LA-03; AUD-ANT-06	46
1997-D2-P5	Geometry / convexity / metric	proof of structural property	AUD-RA-08; AUD-GEO-02	47
1997-D2-P6	Real analysis / calculus	counting / enumeration	AUD-RA-12; AUD-CMB-03; AUD-CMB-07	47
1998-D1-P1	Linear algebra / matrix theory	exact computation	AUD-LA-03; AUD-LA-07	49
1998-D1-P2	Combinatorics / discrete structures	proof of structural property	AUD-ANT-08; AUD-RA-08; AUD-PC-01; AUD-PC-13	49
1998-D1-P3	Real analysis / calculus	exact computation	AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08	50
1998-D1-P4	Real analysis / calculus	existence / construction	AUD-RA-01; AUD-RA-06; AUD-RA-08	50
1998-D1-P5	Real analysis / calculus	inequality / sharp bound	AUD-PC-02; AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13	51

Problem	Primary domain	Problem type	Added closure nodes	PDF p.
1998-D1-P6	Real analysis / calculus	inequality / sharp bound	AUD-GEO-09; AUD-RA-08	51
1998-D2-P1	Linear algebra / matrix theory	proof of structural property	AUD-RA-08	52
1998-D2-P2	Real analysis / calculus	classification / find all	AUD-RA-01; AUD-RA-07; AUD-RA-08; AUD-PC-01; AUD-PC-08	52
1998-D2-P3	Functional equations / dynamics	inequality / sharp bound	AUD-RA-03; AUD-RA-08; AUD-CMB-03	53
1998-D2-P4	Functional equations / dynamics	counting / enumeration	AUD-RA-08; AUD-CMB-03; AUD-CMB-07	53
1998-D2-P5	Geometry / convexity / metric	counting / enumeration	AUD-RA-07; AUD-RA-08; AUD-CMB-03; AUD-GEO-02; AUD-GEO-09	54
1998-D2-P6	Real analysis / calculus	existence / construction	AUD-RA-06; AUD-RA-08; AUD-RA-14; AUD-PC-13; AUD-CMB-03	54
1999-D1-P1	Linear algebra / matrix theory	existence / construction	AUD-LA-07; AUD-LA-10; AUD-RA-07; AUD-PC-01; AUD-PC-05	56
1999-D1-P2	Real analysis / calculus	nonexistence / counterexample	AUD-RA-08; AUD-CMB-07	56
1999-D1-P3	Algebra / number theory	inequality / sharp bound	AUD-RA-08; AUD-ANT-02	57
1999-D1-P4	Functional equations / dynamics	classification / find all	AUD-RA-08	57
1999-D1-P5	Combinatorics / discrete structures	proof of structural property	AUD-RA-08; AUD-CMB-02; AUD-CMB-03; AUD-GEO-02	57
1999-D1-P6	Functional equations / dynamics	proof / reasoning	AUD-RA-01; AUD-RA-06; AUD-RA-08	58
1999-D2-P1	Algebra / number theory	proof of structural property	AUD-LA-03; AUD-LA-12; AUD-ANT-07	59
1999-D2-P2	Algebra / number theory	counting / enumeration	AUD-CMB-09; AUD-PC-01; AUD-PC-05; AUD-PC-06; AUD-PC-13; AUD-ANT-02	59
1999-D2-P3	Real analysis / calculus	inequality / sharp bound	AUD-RA-08	60
1999-D2-P4	Functional equations / dynamics	existence / construction	AUD-RA-08	60
1999-D2-P5	Combinatorics / discrete structures	proof of structural property	AUD-RA-08; AUD-ANT-12	60
1999-D2-P6	Real analysis / calculus	existence / construction	AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-06; AUD-PC-08	61
2000-D1-P1	Real analysis / calculus	existence / construction	AUD-RA-07; AUD-RA-08; AUD-RA-10; AUD-CMB-03	62
2000-D1-P2	Real analysis / calculus	classification / find all	AUD-RA-06; AUD-PC-01; AUD-PC-05; AUD-PC-08	62
2000-D1-P3	Linear algebra / matrix theory	proof of structural property	AUD-LA-03; AUD-LA-08; AUD-LA-12; AUD-RA-06; AUD-PC-01	63
2000-D1-P4	Real analysis / calculus	inequality / sharp bound	AUD-RA-08	63
2000-D1-P5	Algebra / number theory	proof of structural property	AUD-LA-02; AUD-LA-03; AUD-ANT-01; AUD-ANT-06; AUD-ANT-07	64
2000-D1-P6	Functional equations / dynamics	inequality / sharp bound	AUD-RA-01; AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-06	64
2000-D2-P1	Geometry / convexity / metric	inequality / sharp bound	AUD-GEO-11; AUD-RA-08; AUD-PC-01; AUD-PC-13; AUD-ANT-02; AUD-CMB-07	66
2000-D2-P2	Real analysis / calculus	existence / construction	AUD-RA-07; AUD-RA-08; AUD-RA-10; AUD-RA-13; AUD-GEO-08	66
2000-D2-P3	Complex analysis / polynomials	proof of structural property	AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-08; AUD-PC-13	66

Problem	Primary domain	Problem type	Added closure nodes	PDF p.
2000-D2-P4	Complex analysis / polynomials	inequality / sharp bound	AUD-RA-06; AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13	67
2000-D2-P5	Functional equations / dynamics	classification / find all	AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-06; AUD-RA-08	68
2000-D2-P6	Linear algebra / matrix theory	nonexistence / counterexample	AUD-LA-02; AUD-LA-03; AUD-LA-08; AUD-LA-09; AUD-RA-06	69
2001-D1-P1	Algebra / number theory	proof / reasoning	AUD-ANT-02; AUD-PC-01	70
2001-D1-P2	Algebra / number theory	proof of structural property	AUD-LA-03; AUD-LA-08; AUD-PC-01; AUD-ANT-02	70
2001-D1-P3	Real analysis / calculus	limit / asymptotic evaluation	AUD-RA-03; AUD-RA-05	70
2001-D1-P4	Complex analysis / polynomials	proof of structural property	AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-08; AUD-PC-13	71
2001-D1-P5	Linear algebra / matrix theory	proof of structural property	AUD-LA-03; AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-08	71
2001-D1-P6	Functional equations / dynamics	limit / asymptotic evaluation	AUD-RA-01; AUD-RA-03; AUD-RA-05; AUD-RA-06; AUD-RA-08	72
2001-D2-P1	Real analysis / calculus	proof of structural property	AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-CMB-03	74
2001-D2-P2	Real analysis / calculus	inequality / sharp bound	AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08; AUD-PC-13	74
2001-D2-P3	Geometry / convexity / metric	inequality / sharp bound	AUD-GEO-04; AUD-RA-06; AUD-RA-07; AUD-RA-08; AUD-PC-01; AUD-PC-05	75
2001-D2-P4	Linear algebra / matrix theory	existence / construction	AUD-RA-07; AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-08	76
2001-D2-P5	Functional equations / dynamics	nonexistence / counterexample	AUD-RA-03; AUD-RA-05; AUD-RA-08; AUD-CMB-07	77
2001-D2-P6	Algebra / number theory	inequality / sharp bound	AUD-RA-06; AUD-RA-07; AUD-RA-08; AUD-ANT-02	78
2002-D1-P1	Geometry / convexity / metric	proof of structural property	AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-PC-14	79
2002-D1-P2	Functional equations / dynamics	nonexistence / counterexample	AUD-RA-01; AUD-RA-06; AUD-RA-08; AUD-CMB-07	79
2002-D1-P3	Algebra / number theory	proof of structural property	AUD-RA-08; AUD-ANT-02	80
2002-D1-P4	Real analysis / calculus	proof of structural property	AUD-RA-11; AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-07; AUD-RA-08	80
2002-D1-P5	Real analysis / calculus	existence / construction	AUD-RA-06; AUD-RA-10; AUD-CMB-03	81
2002-D1-P6	Algebra / number theory	inequality / sharp bound	AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-06; AUD-RA-08	82
2002-D2-P1	Linear algebra / matrix theory	exact computation	AUD-CMB-08	84
2002-D2-P2	Combinatorics / discrete structures	proof of structural property	AUD-CMB-07	84
2002-D2-P3	Algebra / number theory	proof of structural property	AUD-RA-08; AUD-PC-01; AUD-PC-13; AUD-CMB-07; AUD-ANT-02	85
2002-D2-P4	Geometry / convexity / metric	proof of structural property	AUD-RA-08; AUD-GEO-02	85
2002-D2-P5	Linear algebra / matrix theory	proof of structural property	AUD-LA-16; AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-06	86
2002-D2-P6	Functional equations / dynamics	inequality / sharp bound	AUD-RA-07; AUD-RA-08; AUD-RA-15	87
2003-D1-P1	Real analysis / calculus	existence / construction	AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-PC-13; AUD-CMB-07	88

Problem	Primary domain	Problem type	Added closure nodes	PDF p.
2003-D1-P2	Algebra / number theory	existence / construction	AUD-PC-01; AUD-PC-13; AUD-ANT-04; AUD-CMB-07; AUD-ANT-02	88
2003-D1-P3	Linear algebra / matrix theory	proof of structural property	AUD-LA-02; AUD-LA-03; AUD-LA-07; AUD-RA-03; AUD-RA-04	88
2003-D1-P4	Algebra / number theory	proof / reasoning	AUD-RA-08; AUD-ANT-02	88
2003-D1-P5	Real analysis / calculus	existence / construction	AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08; AUD-PC-13	89
2003-D1-P6	Complex analysis / polynomials	proof of structural property	AUD-RA-08; AUD-PC-01; AUD-PC-04; AUD-PC-08; AUD-PC-13	91
2003-D2-P1	Linear algebra / matrix theory	proof of structural property	AUD-LA-03; AUD-LA-07	92
2003-D2-P2	Real analysis / calculus	limit / asymptotic evaluation	AUD-RA-03; AUD-RA-05; AUD-RA-08; AUD-PC-01; AUD-PC-13	92
2003-D2-P3	Combinatorics / discrete structures	existence / construction	AUD-RA-07; AUD-RA-08; AUD-RA-13; AUD-GEO-02; AUD-GEO-08	92
2003-D2-P4	Algebra / number theory	classification / find all	AUD-RA-08; AUD-ANT-10; AUD-ANT-02	93
2003-D2-P5	Real analysis / calculus	existence / construction	AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08; AUD-CMB-03	94
2003-D2-P6	Real analysis / calculus	limit / asymptotic evaluation	AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08; AUD-PC-13	94
2004-D1-P1	Combinatorics / discrete structures	counting / enumeration	AUD-RA-08; AUD-CMB-03; AUD-CMB-05; AUD-GEO-09; AUD-CMB-07	96
2004-D1-P2	Real analysis / calculus	counting / enumeration	AUD-RA-08; AUD-CMB-03; AUD-CMB-07	96
2004-D1-P3	Real analysis / calculus	limit / asymptotic evaluation	AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-07; AUD-RA-08	96
2004-D1-P4	Geometry / convexity / metric	proof of structural property	AUD-CMB-05; AUD-RA-08; AUD-CMB-03; AUD-GEO-02	97
2004-D1-P5	Real analysis / calculus	existence / construction	AUD-RA-08; AUD-RA-10; AUD-CMB-05	97
2004-D1-P6	Real analysis / calculus	proof of structural property	AUD-PC-10; AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-06; AUD-RA-08	98
2004-D2-P1	Linear algebra / matrix theory	exact computation	AUD-LA-03; AUD-LA-07	100
2004-D2-P2	Functional equations / dynamics	inequality / sharp bound	AUD-RA-08; AUD-GEO-02	100
2004-D2-P3	Geometry / convexity / metric	existence / construction	AUD-RA-08; AUD-GEO-02	100
2004-D2-P4	Linear algebra / matrix theory	proof / reasoning	AUD-LA-15; AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08	101
2004-D2-P5	Real analysis / calculus	inequality / sharp bound	AUD-RA-08	101
2004-D2-P6	Linear algebra / matrix theory	inequality / sharp bound	AUD-RA-08; AUD-CMB-03; AUD-CMB-06	102
2005-D1-P1	Linear algebra / matrix theory	proof / reasoning	AUD-RA-08; AUD-CMB-08	103
2005-D1-P2	Algebra / number theory	counting / enumeration	AUD-RA-08; AUD-PC-01; AUD-PC-13; AUD-ANT-02; AUD-CMB-03	103
2005-D1-P3	Real analysis / calculus	inequality / sharp bound	AUD-RA-01; AUD-RA-06; AUD-RA-08	104
2005-D1-P4	Complex analysis / polynomials	classification / find all	AUD-ANT-05; AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-ANT-02	105
2005-D1-P5	Real analysis / calculus	limit / asymptotic evaluation	AUD-RA-01; AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-06	106

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2005-D1-P6	Algebra / number theory	proof of structural property	AUD-LA-02; AUD-LA-03; AUD-RA-08; AUD-ANT-01; AUD-ANT-06	106
2005-D2-P1	Real analysis / calculus	inequality / sharp bound	AUD-RA-07; AUD-RA-08	107
2005-D2-P2	Complex analysis / polynomials	proof / reasoning	AUD-RA-04; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-ANT-02	107
2005-D2-P3	Linear algebra / matrix theory	inequality / sharp bound	AUD-LA-03; AUD-RA-07; AUD-RA-08; AUD-ANT-06; AUD-CMB-08	108
2005-D2-P4	Functional equations / dynamics	existence / construction	AUD-RA-01; AUD-RA-06	108
2005-D2-P5	Functional equations / dynamics	classification / find all	AUD-RA-06; AUD-RA-07; AUD-RA-08; AUD-GEO-02	109
2005-D2-P6	Algebra / number theory	existence / construction	AUD-PC-14; AUD-LA-03; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-ANT-02	109
2006-D1-P1	Real analysis / calculus	nonexistence / counterexample	AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08; AUD-RA-10	111
2006-D1-P2	Algebra / number theory	counting / enumeration	AUD-ANT-01; AUD-RA-08; AUD-PC-01; AUD-ANT-02; AUD-CMB-03; AUD-CMB-07	111
2006-D1-P3	Algebra / number theory	proof of structural property	AUD-RA-08; AUD-PC-01; AUD-ANT-02	111
2006-D1-P4	Complex analysis / polynomials	proof of structural property	AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-PC-01; AUD-PC-08	112
2006-D1-P5	Real analysis / calculus	proof / reasoning	AUD-RA-06; AUD-RA-08	112
2006-D1-P6	Real analysis / calculus	classification / find all	AUD-RA-06; AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-08	113
2006-D2-P1	Algebra / number theory	counting / enumeration	AUD-RA-08; AUD-PC-01; AUD-ANT-02; AUD-CMB-03; AUD-GEO-02	114
2006-D2-P2	Functional equations / dynamics	classification / find all	AUD-RA-07; AUD-RA-08; AUD-CMB-05	114
2006-D2-P3	Real analysis / calculus	proof / reasoning	AUD-RA-08	115
2006-D2-P4	Algebra / number theory	proof of structural property	AUD-GEO-01; AUD-RA-08	115
2006-D2-P5	Algebra / number theory	existence / construction	AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-ANT-02; AUD-CMB-03	116
2006-D2-P6	Linear algebra / matrix theory	proof of structural property	AUD-LA-03; AUD-LA-07; AUD-RA-06; AUD-PC-01; AUD-PC-13	116
2007-D1-P1	Algebra / number theory	proof of structural property	AUD-PC-01; AUD-PC-06; AUD-PC-08; AUD-PC-11; AUD-PC-13	118
2007-D1-P2	Algebra / number theory	inequality / sharp bound	AUD-RA-08; AUD-CMB-08; AUD-ANT-02	118
2007-D1-P3	Linear algebra / matrix theory	classification / find all	AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-CMB-08	118
2007-D1-P4	Algebra / number theory	counting / enumeration	AUD-PC-13; AUD-CMB-03; AUD-CMB-07	119
2007-D1-P5	Complex analysis / polynomials	proof of structural property	AUD-RA-06; AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-08	119
2007-D1-P6	Complex analysis / polynomials	counting / enumeration	AUD-RA-06; AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-06	120
2007-D2-P1	Real analysis / calculus	proof / reasoning	AUD-RA-01; AUD-RA-06	121
2007-D2-P2	Algebra / number theory	proof of structural property	AUD-PC-01; AUD-ANT-02; AUD-ANT-03	121
2007-D2-P3	Real analysis / calculus	existence / construction	AUD-RA-07; AUD-RA-08; AUD-RA-10	121

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2007-D2-P4	Algebra / number theory	exact computation	AUD-PC-01; AUD-ANT-02	121
2007-D2-P5	Algebra / number theory	proof / reasoning	AUD-LA-12; AUD-RA-08; AUD-ANT-02	122
2007-D2-P6	Complex analysis / polynomials	existence / construction	AUD-RA-01; AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-06	123
2008-D1-P6	Combinatorics / discrete structures	inequality / sharp bound	AUD-CMB-08; AUD-RA-06; AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13	125
2008-D1-P1	Functional equations / dynamics	classification / find all	AUD-RA-07; AUD-RA-08	125
2008-D1-P2	Linear algebra / matrix theory	proof of structural property	AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13	125
2008-D1-P3	Algebra / number theory	existence / construction	AUD-PC-01; AUD-PC-08; AUD-PC-11; AUD-PC-13; AUD-ANT-02	126
2008-D1-P4	Combinatorics / discrete structures	classification / find all	AUD-RA-08	126
2008-D1-P5	Algebra / number theory	nonexistence / counterexample	AUD-RA-03; AUD-RA-05; AUD-RA-06; AUD-RA-08; AUD-PC-01	126
2008-D2-P1	Real analysis / calculus	proof of structural property	AUD-PC-01; AUD-PC-05; AUD-PC-08; AUD-PC-13; AUD-ANT-02	127
2008-D2-P2	Geometry / convexity / metric	proof of structural property	AUD-GEO-05; AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-GEO-02	127
2008-D2-P3	Algebra / number theory	inequality / sharp bound	AUD-RA-08; AUD-PC-05; AUD-PC-13; AUD-CMB-07; AUD-ANT-02	127
2008-D2-P4	Algebra / number theory	proof of structural property	AUD-CMB-03; AUD-PC-01; AUD-PC-08; AUD-PC-11; AUD-PC-13; AUD-ANT-02	127
2008-D2-P5	Algebra / number theory	proof of structural property	AUD-RA-06; AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13	128
2008-D2-P6	Linear algebra / matrix theory	counting / enumeration	AUD-RA-08; AUD-PC-01; AUD-PC-13; AUD-CMB-03; AUD-GEO-09	128
2009-D1-P1	Real analysis / calculus	proof / reasoning	AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08; AUD-CMB-03	129
2009-D1-P2	Linear algebra / matrix theory	proof of structural property	AUD-CMB-04	129
2009-D1-P3	Combinatorics / discrete structures	classification / find all	AUD-RA-06; AUD-RA-07; AUD-RA-08; AUD-PC-01; AUD-PC-13	129
2009-D1-P4	Real analysis / calculus	inequality / sharp bound	AUD-PC-07; AUD-RA-06; AUD-RA-07; AUD-RA-08; AUD-PC-01; AUD-PC-08	130
2009-D1-P5	Algebra / number theory	proof of structural property	AUD-RA-08; AUD-CMB-08; AUD-GEO-03; AUD-ANT-02	131
2009-D2-P1	Geometry / convexity / metric	proof / reasoning	AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-GEO-02	133
2009-D2-P2	Functional equations / dynamics	inequality / sharp bound	AUD-RA-01; AUD-RA-06; AUD-RA-08	133
2009-D2-P3	Algebra / number theory	existence / construction	AUD-LA-02; AUD-LA-03; AUD-LA-08; AUD-LA-13; AUD-RA-08	134
2009-D2-P4	Complex analysis / polynomials	counting / enumeration	AUD-PC-01; AUD-PC-06; AUD-PC-08; AUD-PC-13; AUD-ANT-02	134
2009-D2-P5	Algebra / number theory	inequality / sharp bound	AUD-LA-13; AUD-RA-08; AUD-PC-01; AUD-PC-13; AUD-ANT-08	135
2010-D1-P1	Functional equations / dynamics	inequality / sharp bound	AUD-RA-06; AUD-RA-08	137
2010-D1-P2	Real analysis / calculus	exact computation	AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08	137
2010-D1-P3	Functional equations / dynamics	limit / asymptotic evaluation	AUD-RA-03; AUD-RA-05; AUD-RA-08	139

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2010-D1-P4	Algebra / number theory	proof of structural property	AUD-RA-08; AUD-PC-01; AUD-ANT-02; AUD-CMB-03	139
2010-D1-P5	Algebra / number theory	inequality / sharp bound	AUD-RA-06; AUD-RA-08; AUD-CMB-05; AUD-CMB-08; AUD-ANT-02	139
2010-D2-P1	Real analysis / calculus	proof / reasoning	AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08	141
2010-D2-P2	Real analysis / calculus	inequality / sharp bound	AUD-RA-08; AUD-PC-13; AUD-CMB-05; AUD-CMB-07	141
2010-D2-P3	Algebra / number theory	existence / construction	AUD-RA-08	142
2010-D2-P4	Algebra / number theory	proof of structural property	AUD-RA-08; AUD-PC-01; AUD-ANT-02	143
2010-D2-P5	Complex analysis / polynomials	proof of structural property	AUD-RA-02; AUD-RA-03; AUD-RA-05; AUD-RA-06; AUD-PC-01	144
2011-D1-P1	Functional equations / dynamics	existence / construction	AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-07; AUD-RA-08	145
2011-D1-P2	Linear algebra / matrix theory	nonexistence / counterexample	AUD-LA-03; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-ANT-06	145
2011-D1-P3	Algebra / number theory	inequality / sharp bound	AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-11; AUD-PC-13	145
2011-D1-P4	Combinatorics / discrete structures	counting / enumeration	AUD-RA-08; AUD-RA-10; AUD-CMB-03; AUD-CMB-07	146
2011-D1-P5	Algebra / number theory	existence / construction	AUD-RA-06; AUD-RA-08; AUD-PC-01; AUD-PC-13; AUD-ANT-02	146
2011-D2-P1	Real analysis / calculus	limit / asymptotic evaluation	AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08; AUD-PC-13	147
2011-D2-P2	Algebra / number theory	inequality / sharp bound	AUD-RA-08; AUD-PC-01; AUD-PC-13; AUD-CMB-03; AUD-CMB-08	147
2011-D2-P3	Functional equations / dynamics	exact computation	AUD-RA-08	148
2011-D2-P4	Algebra / number theory	proof of structural property	AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-ANT-02	149
2011-D2-P5	Geometry / convexity / metric	proof of structural property	AUD-RA-08; AUD-GEO-02; AUD-GEO-10	150
2012-D1-P1	Algebra / number theory	counting / enumeration	AUD-RA-08; AUD-CMB-03; AUD-CMB-07; AUD-ANT-02	151
2012-D1-P2	Algebra / number theory	proof / reasoning	AUD-CMB-05; AUD-ANT-02	151
2012-D1-P3	Algebra / number theory	proof / reasoning	AUD-RA-08; AUD-ANT-02	151
2012-D1-P4	Real analysis / calculus	inequality / sharp bound	AUD-RA-01; AUD-RA-06; AUD-RA-08	151
2012-D1-P5	Complex analysis / polynomials	proof of structural property	AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-ANT-02	151
2012-D2-P1	Complex analysis / polynomials	proof / reasoning	AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-ANT-02	152
2012-D2-P2	Real analysis / calculus	proof of structural property	AUD-RA-08; AUD-PC-13; AUD-CMB-07	152
2012-D2-P3	Algebra / number theory	proof / reasoning	AUD-ANT-02; AUD-PC-01	152
2012-D2-P4	Algebra / number theory	classification / find all	AUD-RA-08; AUD-ANT-02	152
2012-D2-P5	Algebra / number theory	inequality / sharp bound	AUD-RA-08; AUD-ANT-10	152
2013-D1-P1	Linear algebra / matrix theory	proof of structural property	AUD-LA-10	153

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2013-D1-P2	Real analysis / calculus	existence / construction	AUD-RA-01; AUD-RA-06	153
2013-D1-P3	Combinatorics / discrete structures	counting / enumeration	AUD-RA-07; AUD-RA-08; AUD-CMB-03; AUD-CMB-07	153
2013-D1-P4	Real analysis / calculus	inequality / sharp bound	AUD-RA-08	153
2013-D1-P5	Algebra / number theory	nonexistence / counterexample	AUD-PC-01; AUD-PC-05; AUD-PC-08; AUD-ANT-02; AUD-CMB-07	153
2013-D2-P1	Complex analysis / polynomials	proof of structural property	AUD-PC-01; AUD-PC-05; AUD-PC-08	154
2013-D2-P2	Algebra / number theory	proof of structural property	AUD-PC-01; AUD-ANT-02	154
2013-D2-P3	Real analysis / calculus	existence / construction	AUD-RA-08	154
2013-D2-P4	Algebra / number theory	nonexistence / counterexample	AUD-ANT-02; AUD-PC-01; AUD-CMB-07	154
2013-D2-P5	Combinatorics / discrete structures	counting / enumeration	AUD-RA-08; AUD-CMB-03; AUD-CMB-07	154
2014-D1-P1	Linear algebra / matrix theory	classification / find all	AUD-LA-03; AUD-ANT-06	155
2014-D1-P2	Real analysis / calculus	classification / find all	AUD-RA-03; AUD-RA-05	155
2014-D1-P3	Complex analysis / polynomials	proof / reasoning	AUD-PC-01; AUD-PC-08; AUD-PC-13	155
2014-D1-P4	Algebra / number theory	proof / reasoning	AUD-RA-08; AUD-PC-01; AUD-ANT-02	155
2014-D1-P5	Geometry / convexity / metric	proof / reasoning	AUD-RA-08; AUD-GEO-02	155
2014-D2-P1	Algebra / number theory	proof / reasoning	AUD-ANT-02; AUD-PC-01	156
2014-D2-P2	Real analysis / calculus	classification / find all	AUD-RA-08	156
2014-D2-P3	Real analysis / calculus	proof / reasoning	AUD-RA-01; AUD-RA-06	156
2014-D2-P4	Geometry / convexity / metric	inequality / sharp bound	AUD-RA-08	156
2014-D2-P5	Real analysis / calculus	proof / reasoning	AUD-RA-08	156
2015-D1-P1	Linear algebra / matrix theory	proof of structural property	AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-08; AUD-CMB-08	157
2015-D1-P2	Algebra / number theory	inequality / sharp bound	AUD-RA-08; AUD-ANT-02	157
2015-D1-P3	Functional equations / dynamics	proof / reasoning	AUD-RA-08	157
2015-D1-P4	Algebra / number theory	proof / reasoning	AUD-RA-08; AUD-ANT-02	157
2015-D1-P5	Real analysis / calculus	inequality / sharp bound	AUD-GEO-03; AUD-RA-08; AUD-PC-13; AUD-CMB-07; AUD-GEO-02; AUD-PRB-01	157
2015-D2-P6	Real analysis / calculus	proof of structural property	AUD-RA-01; AUD-RA-06	158
2015-D2-P7	Real analysis / calculus	limit / asymptotic evaluation	AUD-RA-03; AUD-RA-05	158
2015-D2-P8	Combinatorics / discrete structures	counting / enumeration	AUD-CMB-03; AUD-CMB-07	158
2015-D2-P9	Linear algebra / matrix theory	inequality / sharp bound	AUD-LA-11; AUD-RA-07; AUD-RA-08; AUD-PC-01; AUD-PC-05	158

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2015-D2-P10	Algebra / number theory	inequality / sharp bound	AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-ANT-02	158
2016-D1-P1	Functional equations / dynamics	proof of structural property	AUD-RA-01	159
2016-D1-P2	Algebra / number theory	inequality / sharp bound	AUD-LA-12; AUD-RA-08; AUD-ANT-02	159
2016-D1-P3	Algebra / number theory	inequality / sharp bound	AUD-RA-08; AUD-ANT-02	159
2016-D1-P4	Algebra / number theory	proof of structural property	AUD-RA-08; AUD-ANT-02	159
2016-D1-P5	Algebra / number theory	counting / enumeration	AUD-RA-08; AUD-PC-01; AUD-ANT-02; AUD-CMB-03; AUD-CMB-07	159
2016-D2-P6	Real analysis / calculus	inequality / sharp bound	AUD-RA-08; AUD-CMB-05	160
2016-D2-P7	Real analysis / calculus	proof / reasoning	AUD-RA-07; AUD-RA-08	160
2016-D2-P8	Algebra / number theory	existence / construction	AUD-PC-01; AUD-ANT-02	160
2016-D2-P9	Algebra / number theory	inequality / sharp bound	AUD-RA-08; AUD-CMB-03; AUD-ANT-02	160
2016-D2-P10	Linear algebra / matrix theory	inequality / sharp bound	AUD-LA-07; AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-08	160
2017-D1-P1	Linear algebra / matrix theory	classification / find all	AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-08; AUD-ANT-02	161
2017-D1-P2	Real analysis / calculus	existence / construction	AUD-RA-15; AUD-RA-01; AUD-RA-08	161
2017-D1-P3	Algebra / number theory	existence / construction	AUD-RA-08; AUD-ANT-02	161
2017-D1-P4	Algebra / number theory	counting / enumeration	AUD-CMB-01; AUD-CMB-03; AUD-CMB-07	161
2017-D1-P5	Complex analysis / polynomials	proof of structural property	AUD-RA-06; AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-08	161
2017-D2-P6	Real analysis / calculus	limit / asymptotic evaluation	AUD-RA-03; AUD-RA-05	162
2017-D2-P7	Complex analysis / polynomials	proof of structural property	AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-ANT-02	162
2017-D2-P8	Algebra / number theory	proof of structural property	AUD-PC-13; AUD-CMB-07; AUD-ANT-02	162
2017-D2-P9	Real analysis / calculus	limit / asymptotic evaluation	AUD-RA-01; AUD-RA-03; AUD-RA-05; AUD-PC-13; AUD-CMB-07	162
2017-D2-P10	Algebra / number theory	existence / construction	AUD-RA-08; AUD-GEO-02; AUD-ANT-02	162
2018-D1-P1	Real analysis / calculus	proof of structural property	AUD-RA-01; AUD-RA-06	163
2018-D1-P2	Algebra / number theory	nonexistence / counterexample	AUD-ANT-02; AUD-PC-01; AUD-CMB-07	163
2018-D1-P3	Linear algebra / matrix theory	classification / find all	AUD-LA-03; AUD-LA-07	163
2018-D1-P4	Real analysis / calculus	classification / find all	AUD-RA-01	163
2018-D1-P5	Algebra / number theory	inequality / sharp bound	AUD-RA-08; AUD-PC-01; AUD-ANT-02; AUD-GEO-02; AUD-GEO-10	163
2018-D2-P6	Algebra / number theory	proof / reasoning	AUD-RA-08; AUD-PC-06; AUD-ANT-02	164
2018-D2-P7	Real analysis / calculus	proof of structural property	AUD-RA-08; AUD-PC-13; AUD-CMB-07	164

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2018-D2-P8	Algebra / number theory	counting / enumeration	AUD-RA-08; AUD-CMB-03; AUD-CMB-07; AUD-ANT-02	164
2018-D2-P9	Complex analysis / polynomials	classification / find all	AUD-PC-01; AUD-PC-05; AUD-PC-08; AUD-PC-13	164
2018-D2-P10	Real analysis / calculus	limit / asymptotic evaluation	AUD-RA-03; AUD-RA-05	164
2019-D1-P1	Real analysis / calculus	exact computation	AUD-RA-01; AUD-RA-06	165
2019-D1-P2	Algebra / number theory	classification / find all	AUD-ANT-02; AUD-PC-01	165
2019-D1-P3	Real analysis / calculus	inequality / sharp bound	AUD-RA-01; AUD-RA-08	165
2019-D1-P4	Algebra / number theory	proof of structural property	AUD-RA-08; AUD-PC-13; AUD-CMB-07; AUD-ANT-02	165
2019-D1-P5	Algebra / number theory	proof / reasoning	AUD-RA-08; AUD-ANT-02	165
2019-D2-P6	Functional equations / dynamics	existence / construction	AUD-RA-01; AUD-RA-06	166
2019-D2-P7	Algebra / number theory	proof / reasoning	AUD-PC-01; AUD-ANT-02	166
2019-D2-P8	Combinatorics / discrete structures	counting / enumeration	AUD-CMB-03; AUD-CMB-05; AUD-CMB-07	166
2019-D2-P9	Algebra / number theory	classification / find all	AUD-LA-03; AUD-ANT-02	166
2019-D2-P10	Probability / measure / expectation	counting / enumeration	AUD-RA-08; AUD-CMB-03; AUD-GEO-02; AUD-PRB-01; AUD-CMB-07	166
2020-D1-P1	Algebra / number theory	exact computation	AUD-RA-08; AUD-CMB-03; AUD-ANT-02	167
2020-D1-P2	Linear algebra / matrix theory	inequality / sharp bound	AUD-LA-03; AUD-LA-08; AUD-RA-07; AUD-ANT-06; AUD-CMB-03	167
2020-D1-P3	Algebra / number theory	existence / construction	AUD-RA-08; AUD-ANT-02	167
2020-D1-P4	Real analysis / calculus	inequality / sharp bound	AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13	167
2020-D2-P5	Real analysis / calculus	classification / find all	AUD-RA-01; AUD-RA-08	168
2020-D2-P6	Algebra / number theory	classification / find all	AUD-PC-01; AUD-ANT-02	168
2020-D2-P7	Algebra / number theory	existence / construction	AUD-RA-08; AUD-CMB-03; AUD-ANT-02	168
2020-D2-P8	Real analysis / calculus	limit / asymptotic evaluation	AUD-RA-03; AUD-RA-05	168
2021-D1-P1	Linear algebra / matrix theory	proof of structural property	AUD-LA-07	169
2021-D1-P2	Algebra / number theory	counting / enumeration	AUD-CMB-03; AUD-PRB-01; AUD-CMB-07; AUD-ANT-02	169
2021-D1-P3	Real analysis / calculus	existence / construction	AUD-PC-13; AUD-ANT-12; AUD-CMB-07	169
2021-D1-P4	Real analysis / calculus	existence / construction	AUD-RA-03; AUD-RA-05	169
2021-D2-P5	Algebra / number theory	existence / construction	AUD-LA-11; AUD-RA-08; AUD-ANT-02	170
2021-D2-P6	Algebra / number theory	proof of structural property	AUD-PC-01; AUD-PC-06; AUD-ANT-02	170
2021-D2-P7	Complex analysis / polynomials	inequality / sharp bound	AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-08; AUD-PC-13	170

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2021-D2-P8	Algebra / number theory	counting / enumeration	AUD-RA-08; AUD-CMB-03; AUD-GEO-04; AUD-CMB-07; AUD-ANT-02	170
2022-D1-P1	Probability / measure / expectation	inequality / sharp bound	AUD-RA-08; AUD-PRB-02	171
2022-D1-P2	Algebra / number theory	classification / find all	AUD-LA-10; AUD-RA-08; AUD-ANT-02	171
2022-D1-P3	Algebra / number theory	counting / enumeration	AUD-PC-01; AUD-ANT-02; AUD-CMB-03; AUD-CMB-07	171
2022-D1-P4	Algebra / number theory	inequality / sharp bound	AUD-RA-08; AUD-CMB-03; AUD-ANT-02	171
2022-D2-P5	Combinatorics / discrete structures	counting / enumeration	AUD-RA-08; AUD-CMB-03; AUD-GEO-02; AUD-CMB-07	172
2022-D2-P6	Algebra / number theory	proof of structural property	AUD-PC-01; AUD-ANT-02	172
2022-D2-P7	Linear algebra / matrix theory	inequality / sharp bound	AUD-LA-02; AUD-LA-03; AUD-RA-08; AUD-PC-01; AUD-PC-05	172
2022-D2-P8	Algebra / number theory	counting / enumeration	AUD-RA-08; AUD-CMB-03; AUD-GEO-02; AUD-PRB-01; AUD-CMB-07	172
2023-D1-P1	Real analysis / calculus	classification / find all	AUD-RA-06	173
2023-D1-P2	Linear algebra / matrix theory	proof of structural property	AUD-RA-06; AUD-PC-01; AUD-PC-05; AUD-PC-08	173
2023-D1-P3	Complex analysis / polynomials	classification / find all	AUD-PC-01; AUD-PC-08; AUD-PC-13	173
2023-D1-P4	Algebra / number theory	proof / reasoning	AUD-PC-01; AUD-PC-06; AUD-ANT-02	173
2023-D1-P5	Algebra / number theory	proof of structural property	AUD-RA-08; AUD-ANT-02	173
2023-D2-P6	Linear algebra / matrix theory	proof / reasoning	AUD-LA-03; AUD-LA-07	174
2023-D2-P7	Real analysis / calculus	classification / find all	AUD-RA-06	174
2023-D2-P8	Combinatorics / discrete structures	inequality / sharp bound	AUD-RA-07; AUD-RA-08; AUD-CMB-03; AUD-GEO-02	174
2023-D2-P9	Geometry / convexity / metric	proof / reasoning	AUD-RA-08; AUD-GEO-02; AUD-GEO-11	174
2023-D2-P10	Algebra / number theory	existence / construction	AUD-RA-08; AUD-ANT-02	174
2024-D1-P1	Complex analysis / polynomials	classification / find all	AUD-PC-08; AUD-PC-01	175
2024-D1-P2	Real analysis / calculus	limit / asymptotic evaluation	AUD-RA-03; AUD-RA-05; AUD-RA-08	175
2024-D1-P3	Algebra / number theory	nonexistence / counterexample	AUD-LA-12; AUD-CMB-07; AUD-ANT-02	175
2024-D1-P4	Algebra / number theory	proof of structural property	AUD-RA-03; AUD-RA-08; AUD-ANT-02	175
2024-D1-P5	Algebra / number theory	inequality / sharp bound	AUD-RA-08; AUD-CMB-03; AUD-GEO-02; AUD-PRB-01; AUD-ANT-02	175
2024-D2-P6	Functional equations / dynamics	inequality / sharp bound	AUD-RA-08	176
2024-D2-P7	Linear algebra / matrix theory	classification / find all	AUD-PC-01; AUD-PC-05; AUD-PC-08; AUD-CMB-08; AUD-ANT-02	176
2024-D2-P8	Real analysis / calculus	inequality / sharp bound	AUD-RA-03; AUD-RA-05; AUD-RA-08; AUD-PC-13; AUD-CMB-07	176
2024-D2-P9	Algebra / number theory	counting / enumeration	AUD-RA-08; AUD-CMB-03; AUD-CMB-08; AUD-CMB-07; AUD-ANT-02	176

Problem	Primary domain	Problem type	Added closure nodes	PDF p.
2024-D2-P10	Algebra / number theory	proof of structural property	AUD-RA-08; AUD-PC-01; AUD-ANT-02	176

16 Audit Conclusion

The added closure nodes cover the specific missing theory families found in the cross-check: sign-cancellation in positive matrix inverses, nilpotent shift models, commutator trace/nilpotency, simultaneous diagonalization, trace pairing, matrix exponential functional calculus, endpoint Laplace asymptotics, Fejer kernels, cyclotomic irreducibility, finite-field polynomial representatives, Pell-type stabilizers, sparse-spike differentiability, metric projection uniqueness, and the major combinatorial encoding lemmas.

For the next database stage, the recommended split is now:

1. keep the previous 107 theory nodes as the broad syllabus map;
2. use the audit nodes as exact theorem/property cards;
3. build a method database from the proof cores above;
4. build a drill database by problem ID using `imc_problem_theory_sufficiency_audit.csv`;
5. build a trap database from the correction table and from each node’s use pattern.

Solution-Linked Method Reference

Choose a move after identifying its hypotheses. The theory links state the mathematical facts behind it; the proof program explains the operations and checks. Routes from individual solution records distinguish methods used by different proofs.

Linear algebra and matrices

Read the entries before choosing a theorem [M-LA-ENTRY]

Eliminate systematically and track the invariant [M-LA-ELIM]

Exploit a compatible block decomposition [M-LA-BLOCK]

Complete the equation to a product equal to identity [M-LA-INVERSE]

Prove a dimension obstruction and attain it [M-LA-DIMENSION]

Choose a basis adapted to the structure [M-LA-BASIS]

Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]

Use trace to extract a scalar invariant [M-LA-TRACE]

Propagate a commutator by telescoping [M-LA-DERIVATION]

Treat the unknown matrix as a vector [M-LA-MATRIXOP]

Encode distances and orthogonality by a Gram matrix [M-LA-GRAM]

Combine norm bounds with algebraic decay [M-LA-NORM]

Exploit parity with linear algebra over F2 [M-LA-FINITEFIELD]

Transfer spectral information between products [M-LA-TRANSFER]

Lift products of inner products to new vectors [M-LA-TENSOR]

Linearize a fractional recurrence in dimension two [M-LA-PROJECTIVE]

Use a generic parameter to force an image inclusion [M-LA-PERTURB]

Interpret matrix multiplication as a count [M-LA-WALKS]

Compare coefficients in a matrix polynomial [M-LA-PARAMETER]

Factor while preserving matrix order [M-LA-FACTOR]

Test a linear map on a basis and normalize its values [M-LA-DUAL]

Reduce integer matrices modulo an integer [M-LA-RESIDUE]

Analysis, limits and probability

Design an auxiliary derivative [M-AN-AUXILIARY]

Count zeros and subtract an interpolant [M-AN-ROLLE]

Expand near equality, bound away from it [M-AN-LOCAL]

Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]

Pair an integral through symmetry [M-AN-PAIR]

Expose the mesh and control its tails [M-AN-RIEMANN]

Separate a main range from exceptional scales [M-AN-SPLIT]

Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]

Find a finite telescoping identity [M-AN-TELESCOPE]

Test convergence through blocks and differences [M-AN-SERIES]

Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]

Choose the correct convexity comparison [M-AN-CONVEX]

Localize an extremum or an accumulation point [M-AN-COMPACT]

Preserve structure while approximating [M-AN-APPROXIMATE]

Pass a limit through an integral safely [M-AN-DOMINATE]

Use a dense set, an enumeration, or a category argument [M-AN-TOPOLOGY]

Turn local bad intervals into a global measure bound [M-AN-COVER]

Encode convolution or denominators by a power series [M-AN-GENERATING]

Count outcomes or condition on a useful structure [M-AN-PROBABILITY]

Condition and average a convex-hull event [M-AN-GEOMETRIC]

Compare growth through derivatives [M-AN-RATIO]

Exploit identities before estimating oscillations [M-AN-TRIG]

Use a first crossing or a forced-growth contradiction [M-AN-BARRIER]

Algebra, number theory and complex polynomials

Replace many equal values by too many roots [M-AL-ROOTCOUNT]

Factor first and compare irreducible exponents [M-AL-FACTOR]

Translate root restrictions into symmetric coefficients [M-AL-VIETA]

Use a logarithmic derivative to expose root geometry [M-AL-LOGDERIVATIVE]

Turn nodal bounds into a sharp linear functional [M-AL-INTERPOLATE]

Choose a modulus that exposes the obstruction [M-AL-MODULAR]

Construct and count compatible local residues [M-AL-CRT]

Track one prime or one extreme divisor [M-AL-VALUATION]

Compare coefficients after a structural substitution [M-AL-COEFFICIENT]

Prove generation or obstruct it by an image [M-AL-GENERATORS]

Convert group data into orbits or cosets [M-AL-ACTION]

Polarize identities and isolate commutators [M-AL-COMMUTATOR]

Work orbit by orbit and track parity [M-AL-ORBIT]

Find a recurrence that reveals the invariant [M-AL-RECURRENCE]

Take the gcd of annihilating shift polynomials [M-AL-SHIFT]

Evaluate at roots of unity or average over them [M-AL-UNITY]

Build a holomorphic function with a useful boundary [M-AL-ANALYTIC]

Eliminate rational relations, then exploit discreteness [M-AL-RATIONAL]

Replace a minimal algebraic counterexample by a smaller one [M-AL-DESCENT]

Choose the next factor and control the entire tail [M-AL-GREEDY]

Construct by excluding a sparse set of bad choices [M-AL-SIEVE]

Construct roots or units before the coefficients [M-AL-PARAMETRIZE]

Counting, geometry and functional equations

Count incidences with their exact multiplicity [M-DS-DOUBLECOUNT]

Encode, decode, and verify a bijection [M-DS-BIJECTION]

Strengthen the statement before the induction step [M-DS-INDUCT]

Partition into states and propagate a recurrence [M-DS-STATE]

Encode a relation as a graph and extract a cycle [M-DS-GRAPH]

Force structure by admissible substitutions [M-DS-FUNCTIONAL]

Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]

Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]

Contradict an extremum by a controlled perturbation [M-DS-PERTURB]

Push covers forward or move toward a nearest point [M-DS-METRIC]

Repair an extremal configuration by a local exchange [M-DS-EXTREMAL]

Pair objects canonically to prove parity [M-DS-INVOLUTION]

Split a family by traces, pieces, or labels [M-DS-PARTITION]

Match classes in stages and verify Hall each time [M-DS-MATCHING]

Reduce equivalent words with a congruence [M-DS-REWRITE]

Random selection and the expected number of successes [M-DS-EXPECTATION]

Encode the statistic and extract the required coefficient [M-DS-GENERATING]

Choose information that makes an encoding injective [M-DS-ENCODE]

Realize all inclusion probabilities on one probability space [M-DS-COUPLING]

Pass a monotone functional iteration through its integral operator [M-DS-VOLTERRA]

Turn a spherical comparison into an endpoint integral [M-DS-SPHERICAL]

Linear algebra and matrices

M-LA-ENTRY Read the entries before choosing a theorem

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Rank, dimension and exact linear constraints [T-LA-RANK]; Inner products, orthogonality and polarization [T-LA-INNER]

Trigger. Positivity, zero patterns, prescribed columns, or a bilinear identity holding for every vector.

1. Write the relevant product entry exactly: $(AB)_{ij} = \sum_k a_{ik}b_{kj}$. Separate diagonal from off-diagonal constraints.
2. If a positive-weighted sum vanishes, a nonzero coefficient list must contain both signs. First justify that the list is nonzero; an invertible matrix has no zero column.
3. For a quadratic identity in all real x, test e_i and $e_i + e_j$, or polarize. This extracts diagonal and symmetric off-diagonal coefficients.
4. Convert the local constraints into a global count or canonical matrix shape, and verify the proposed extremal example by multiplication.

Example. For $n \geq 2$, an off-diagonal entry of $AA^{-1} = I$ is a strictly positive weighted sum of one inverse column, equal to zero. Every column therefore has at least two nonzero entries. Positivity here means entrywise strict positivity, not positive definiteness. These are unrelated hypotheses: a positive definite matrix can have zero or negative off-diagonal entries.

Extended atlas M-LA-17

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P1 [R]; 1994-D2-P4 [R]; 1995-D2-P1 [R]; 2010-D1-P5 [R]; 2012-D1-P2 [R]; 2014-D1-P1 [R]; 2024-D1-P3 [R]

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M-LA-ELIM Eliminate systematically and track the invariant

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Prerequisites Rank, dimension and exact linear constraints [T-LA-RANK]; Determinants, block identities and inverses [T-LA-DETERMINANT]

Trigger. Entries depend on $i + j$, $|i - j|$, parity, or simple recurrences.

1. Decide whether rank, determinant, or the solution set is to be preserved. Arbitrary invertible row operations preserve rank; determinant additionally requires sign and scale tracking.
2. Take consecutive differences to lower the degree of an index-dependent pattern. Work bottom to top when each subtraction must use the old preceding row.
3. Use column operations to clear the remaining pattern, then read a triangular determinant or exhibit independent rows.
4. For a linear system, apply each row operation to its right-hand side as well. For a rank bound, pair a spanning decomposition with a nonzero minor.

Example. The rows $(i+j)_j$ lie in the span of $(1, \dots, n)$ and $(1, \dots, 1)$. The first 2×2 minor is $2\cdot 4 - 3\cdot 3 = -1$, proving rank two when $n \geq 2$. **Check.** Similarity is more restrictive than arbitrary row and column elimination. An elimination that computes rank correctly may destroy the spectrum.

Extended atlas M-LA-01

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1996-D1-P1 [R]; 2002-D2-P1 [R]; 2005-D1-P1 [R]; 2007-D2-P4 [R]; 2019-D1-P2 [R]; 2023-D2-P6 [R]

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M-LA-BLOCK Exploit a compatible block decomposition

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Prerequisites Determinants, block identities and inverses [T-LA-DETERMINANT]; Rank, dimension and exact linear constraints [T-LA-RANK]

Trigger. Rectangular factors, nested subspaces, parity partitions, or a repeated block recurrence.

1. Write dimensions beside the blocks; verify that every product is defined.
2. Choose a basis or reorder indices so that the given relation has forced zero or identity blocks. A simultaneous row and column permutation preserves determinant; a single permutation multiplies it by the sign of that permutation.
3. Multiply the blocks explicitly. An identity block equation $PQ = I$ with square blocks of the same size permits reversing that particular product.
4. Only use a Schur complement after proving the pivot block invertible. Otherwise multiply by a suitable block triangular matrix and take determinants.

Example. A symmetric bipartite pattern becomes $\left(\begin{smallmatrix} 0 & C \\ C^T & 0 \end{smallmatrix}\right)$. Equal block sizes give determinant $(-1)^r(\det C)^2$; unequal sizes force a kernel by dimension. **Checks.** Inverse of the whole block matrix does not imply that either diagonal block is invertible. Algebraic identities for scalar 2×2 determinants cannot be applied to noncommuting blocks without justification.

Extended atlas M-LA-03; M-LA-14

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1997-D2-P2 [R]; 2001-D2-P4 [R]; 2004-D2-P1 [R]; 2004-D2-P6 [R]; 2006-D2-P6 [R]; 2007-D1-P3 [R]; 2007-D2-P4 [R]; 2008-D2-P5 [R]; 2016-D2-P10 [R]; 2019-D2-P9 [R]; 2024-D1-P3 [R]

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M-LA-INVERSE Complete the equation to a product equal to identity

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Prerequisites Determinants, block identities and inverses [T-LA-DETERMINANT]; Nilpotents, projections and involutions [T-LA-NILPOTENT]

Trigger. Matrix equations involving $AB + A + B$, inverses, or a factor multiplied on only one side.

1. Add the missing constant or product term so that the expression factors in its existing order.
2. Obtain $XY = I$ for square matrices of the same size. Infer invertibility from finite-dimensional rank-nullity or determinants.
3. Reverse the product only now: $YX = I$. Expand this new equality to obtain the desired identity.
4. If an unknown matrix is determined by this procedure, substitute the candidate back; uniqueness alone is not existence.

Example. From $AB+A+B = 0$ obtain $(A+I)(B+I) = I$, then $(B+I)(A+I) = I$, and subtract to get $AB = BA$. If $N^r = 0$, the finite geometric inverse for $I + N$ often completes the same route without a determinant. **Check.** Left and right inverse coincide for square finite-dimensional operators. They need not coincide for rectangular matrices or operators on infinite-dimensional spaces.

Extended atlas M-LA-02

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2002-D2-P5 [R]; 2003-D2-P1 [R]; 2004-D2-P1 [R]; 2009-D1-P2 [R]; 2015-D1-P1 [R]; 2021-D1-P1 [R]; 2023-D1-P2 [R]

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M-LA-DIMENSION Prove a dimension obstruction and attain it

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Prerequisites Rank, dimension and exact linear constraints [T-LA-RANK]; Dual bases, kernels and separating functionals [T-LA-DUALITY]

Trigger. Maximum size of a matrix family, minimum rank, or a subspace subject to vanishing conditions.

1. Name the ambient field and space. Translate each condition to a kernel, image, direct sum or evaluation map.
2. Bound dimension by rank-nullity, an intersection formula, or separating a proposed linear dependence with a carefully chosen operator.
3. For a family with pairwise annihilating products, choose vectors on which their own operator is nonzero; applying each operator isolates one coefficient. Alternatively, build a strictly descending chain of kernel intersections.
4. Supply a construction attaining the bound and check every condition, including nonzero requirements and edge cases.

Example. A real matrix subspace orthogonal to itself under $\operatorname{tr}(XY)$ cannot contain a nonzero symmetric matrix. Comparing dimensions with the symmetric subspace bounds it by $n(n-1)/2$; strictly upper-triangular matrices attain this. **Check.** A chain of k strict dimension drops in an n-dimensional space yields $k \leq n$, not the reverse. A parameter count must also prove that its parameters are independent.

Extended atlas M-LA-04; M-LA-16

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1997-D2-P4 [R]; 1998-D2-P1 [R]; 2005-D1-P1 [R]; 2005-D2-P3 [R]; 2007-D1-P2 [R]; 2007-D1-P3 [R]; 2007-D2-P5 [R]; 2009-D2-P5 [R]; 2011-D1-P5 [R]; 2012-D1-P2 [R]; 2013-D2-P3 [R]; 2015-D2-P9 [R]; 2016-D1-P2 [R]; 2018-D2-P6 [R]; 2022-D2-P7 [R]

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M-LA-BASIS Choose a basis adapted to the structure

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Adapted bases, centralizers and common eigenspaces [T-LA-INVARIANT]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]

Trigger. An invariant flag, commuting family, shift action, or simultaneous similarity problem.

1. Start with a basis of the smallest prescribed subspace and extend it through each larger subspace.
2. Compute images of basis vectors, using the column convention. Record forced zero blocks before counting free entries.
3. For commuting diagonalizable operators, split into eigenspaces of a nonscalar member and recurse on each invariant part.
4. In simultaneous-conjugacy problems, retain the freedom of basis changes commuting with the first normalized matrix; use it to normalize the next one.

Example. For a diagonal matrix, commuting matrices have zero entries between unequal-eigenvalue blocks. A block of multiplicity d contributes d^2 free entries. **Checks.** Explicitly distinguish a common eigenbasis from several unrelated eigenbases. Scalar operators create no restriction. When a chosen block unexpectedly becomes scalar, an induction that assumes non-scalarity needs a separate adjustment; simply reapplying the inductive hypothesis leaves a gap.

Extended atlas M-LA-05; M-LA-08; M-LA-09

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D2-P4 [R]; 1995-D1-P1 [R]; 1996-D1-P3 [R]; 1998-D1-P1 [R]; 2001-D1-P5 [R]; 2006-D2-P6 [R]; 2007-D2-P5 [R]

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M-LA-SPECTRAL Reduce an operator identity to its scalar roots

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Nilpotents, projections and involutions [T-LA-NILPOTENT]

Trigger. A polynomial in a matrix, transposes expressible in terms of that matrix, or desired determinant and power information.

1. Derive a genuine one-variable identity $p(A) = 0$, preserving order until all terms are polynomials in the same matrix.
2. Factor p over the relevant field and restrict the eigenvalues. Use the minimal polynomial to determine whether repeated-root Jordan blocks remain possible.
3. Convert the allowed spectrum to trace, determinant or convergence information. For real matrices, match nonreal conjugate multiplicities.
4. For a classification, construct matrices realizing every claimed possibility. Scalar blocks and real 2×2 rotation blocks often suffice.

Example. If $A^2 = A^T$, transposing and squaring gives $A^4 = A$. This restricts eigenvalues to zero and cube roots of unity. Real rotation blocks supply the nonreal pair. **Checks.** A scalar equation for eigenvalues does not alone prove the corresponding matrix equation. Convergence at eigenvalue one requires absence of nontrivial Jordan blocks; the bound $|\lambda| \leq 1$ alone is insufficient.

Extended atlas M-LA-06; M-LA-12; M-LA-13

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1996-D1-P3 [R]; 1999-D1-P1 [R]; 2000-D1-P3 [R]; 2003-D1-P3 [R]; 2011-D1-P2 [R]; 2012-D2-P4 [R]; 2013-D1-P1 [R]; 2014-D1-P1 [R]; 2015-D1-P1 [R]; 2017-D1-P1 [R]; 2018-D1-P3 [R]; 2019-D2-P9 [R]; 2021-D2-P5 [R]; 2022-D1-P2 [R]; 2024-D2-P7 [R]

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M-LA-TRACE Use trace to extract a scalar invariant

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Trace pairing and commutator calculus [T-LA-TRACE]; Rank, dimension and exact linear constraints [T-LA-RANK]

Trigger. A commutator, a rank-one perturbation of identity, trace-pairing subspaces, or comparisons between diagonal entries and eigenvalues.

1. Expand the target under trace and group cyclically equivalent terms. Do not commute factors outside trace.
2. Apply the specific extra structure. Rank one gives $C^2 = \operatorname{tr}(C)C$; symmetry gives $\operatorname{tr}(A^2) = \sum_{ij} a_{ij}^2$.
3. If nilpotency is sought, show all first n power traces vanish, or derive a stronger low-rank polynomial identity. One zero trace is inadequate.
4. Track the equality case: a sum of nonnegative entry squares vanishes exactly when each indicated entry is zero.

Example. For $X = AB - BA$, cyclicity gives $\operatorname{tr}(X^2) = 2\operatorname{tr}(ABAB) - 2\operatorname{tr}(A^2B^2)$. If $X + I$ has rank one, its trace is n and $(X+I)^2 = n(X+I)$, so $\operatorname{tr}(X^2) = n(n-1)$. **Check.** The bilinear pairing $\operatorname{tr}(XY)$ is not a positive inner product on all matrices, especially over $\mathbb{C}$.

Extended atlas M-LA-07; M-LA-16

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1997-D2-P4 [R]; 2000-D1-P3 [R]; 2005-D2-P3 [R]; 2009-D2-P3 [R]; 2011-D1-P2 [R]; 2014-D2-P2 [R]; 2015-D2-P9 [R]; 2020-D1-P2 [R]; 2021-D2-P8 [R]; 2022-D1-P2 [R]; 2022-D2-P7 [R]

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M-LA-DERIVATION Propagate a commutator by telescoping

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Trace pairing and commutator calculus [T-LA-TRACE]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Nilpotents, projections and involutions [T-LA-NILPOTENT]

Trigger. A relation $[F, G] = \alpha F$ with $\alpha \neq 0$, or a vanishing iterated commutator.

1. Expand $[F^k, G]$ as a telescoping sum, preserving the factor order in each term.
2. Substitute the original relation; for $[F, G] = \alpha F$ the result is $[F^k, G] = k\alpha F^k$.
3. Regard $X \mapsto [X, G]$ as an operator on the n^2-dimensional matrix space. If every F^k were nonzero, the infinitely many distinct values $k\alpha$ would be eigenvalues, impossible in characteristic zero.
4. For $\Delta^2 B = 0$, use the repeated Leibniz rule and Cayley–Hamilton on B: $\Delta^n(B^n) = n!(\Delta B)^n = 0$.

Check. Characteristic zero is needed both for distinct $k\alpha$ and for cancelling factorials. The argument concerns the spectrum of a matrix-space operator, not the spectrum of F itself. Once nilpotency is established, finite dimension improves an unspecified vanishing power to $F^n = 0$.

Extended atlas M-LA-18; M-LA-10

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P4 [R]; 2009-D2-P3 [R]

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M-LA-MATRIXOP Treat the unknown matrix as a vector

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Nilpotents, projections and involutions [T-LA-NILPOTENT]; Tensor products and operators on matrix spaces [T-LA-TENSOR]

Trigger. A linear equation in an unknown matrix, or the spectrum of $X \mapsto AX + XB$.

1. Define a linear map on $M_n(K)$, a space of dimension n^2. Matrix units provide a concrete basis if needed.
2. Use left and right multiplication operators: they commute with each other, so polynomial identities can simplify their sum even if the original matrices do not commute.
3. For nilpotent left and right factors, invert identity plus their sum by a finite geometric series. This proves both existence and uniqueness.
4. For spectral problems, tensor bases or triangularization give pairwise sums of eigenvalues; add multiplicities when sums coincide.

Example. If $A^3 = 0$, the equation $X + AX + XA^2 = A$ has linear part $I + L_A + R_{A^2}$. Since $L_A^3 = R_{A^2}^2 = 0$ and these operators commute, their sum is nilpotent. Direct substitution then verifies $X = A - A^2$.

Check. Solving formally for a candidate is insufficient if reversibility has not been proved. Always establish invertibility of the linear operator or check both the candidate and the homogeneous equation.

Extended atlas M-LA-10

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P4 [R]; 2004-D2-P4 [R]; 2021-D1-P1 [R]

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M-LA-GRAM Encode distances and orthogonality by a Gram matrix

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Gram matrices and positive semidefinite forms [T-LA-GRAM]; Inner products, orthogonality and polarization [T-LA-INNER]

Trigger. Rational distances, many orthogonal pairs, unit-vector restrictions, or an inequality containing a correlation determinant.

1. Translate if useful and compute inner products from squared distances using polarization.
2. Form the Gram matrix, recording its rank bound, PSD property and diagonal entries.
3. Convert the geometric condition into algebra: zero inner products are zero entries; triple restrictions can make diagonal entries of a matrix cube vanish.
4. Use rank, traces, eigenvalue bounds or a rational linear system, then translate back and verify any equality construction.

Example. Independent vectors with rational mutual distances and rational distances from the origin have an invertible rational Gram matrix. Any additional vector in their span with the same distance property has rational coordinates in this basis. **For equidistant sets.** A finite barycentre or a norm-limit of barycentres supplies a centre; the zero-sum coefficient identity makes its norm calculations exact. **Checks.** The rank bound comes from the actual ambient dimension. For complex vectors use conjugate transpose. A singular Gram matrix requires choosing an independent subfamily before attempting inversion.

Extended atlas M-LA-11

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2006-D2-P4 [R]; 2008-D2-P6 [R]; 2010-D1-P5 [R]; 2021-D2-P8 [R]

Back to previous view | Theory index | Method index | Strategy index

M-LA-NORM Combine norm bounds with algebraic decay

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Operator norms and convergence of matrix powers [T-LA-NORM]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]

Trigger. Norms of powers, polynomial restrictions on a matrix, or bounds on consecutive-power differences.

1. Record submultiplicativity, unitary invariance and any contraction induced by a block restriction.
2. Use Cayley–Hamilton to obtain a second power bound. If $|\lambda_i| \le 1$ and $\chi_A(t) = t^n + \sum_{j=1}^{n} c_j t^{n-j}$, then $|c_j| \le \binom{n}{j}$, and $\|A^n\| \le \sum_{j=1}^{n} \binom{n}{j} \|A\|^{n-j}$.
3. Combine this with the trivial $\|A^n\| \le \|A\|^n$; split at the intersection of the two bounds instead of using either everywhere.
4. For difference estimates, set $a_k = \|A^{k+1} - A^k\|$. A summable majorant permits telescoping; a harmonic majorant requires an additional improvement argument.

Dyadic refinement. When $a_{2k} - a_{2k+1} \le a_k^2$ and $a_{2k+1} - a_{2k+2} \le a_k a_{k+1}$, define $c_l = \sup_{k \ge 2^l} (k+1) a_k$. Telescoping the difference inequalities yields $c_{l+1} \le 4c_l^2$. If $4c_0 < 1$, this decreases doubly exponentially and makes $\sum a_k$ finite. **Check.** Retain the correct power $n-j$ in the Cayley–Hamilton estimate; a reversed index can change the bound.

Extended atlas M-LA-06

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2002-D1-P6 [R]; 2013-D1-P1 [R]; 2016-D2-P10 [R]

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M-LA-FINITEFIELD Exploit parity with linear algebra over F2

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Prerequisites Linear algebra in characteristic two [T-LA-FINITEFIELD]; Rank, dimension and exact linear constraints [T-LA-RANK]

Trigger. Zero-one matrices used modulo two, parity of vector sums, or a symmetric matrix with zero diagonal over $\mathbb{F}_2$.

1. Move completely to $\mathbb{F}_2$ and distinguish its bilinear form from a real inner product.
2. For an alternating matrix, use $x^T Ax = 0$. Split powers into odd and even cases; even powers let you halve the exponent.
3. For prescribed-cardinality zero sums, pair vectors and encode a choice from each pair by coefficients of their pair sums.
4. Build a suitable basis inductively by quotienting out one nonzero pair sum, then lift the quotient basis and restore that pair.

Example. If pair sums $a_i + b_i$ form a basis, choosing one member of each pair realizes any target difference by solving a linear system over $\mathbb{F}_2$. The number of selected vectors remains fixed throughout the switching process. **Checks.** Arbitrary linear dependence gives some zero sum, not necessarily one of the prescribed cardinality. Keep the combinatorial size constraint alongside the rank argument. Positivity and nonzero squared norms cannot be assumed in this field.

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2010-D2-P4 [R]; 2011-D1-P5 [R]

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M-LA-TRANSFER Transfer spectral information between products

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Prerequisites Matrix power series and transfer between AB and BA [T-LA-SERIES]; Determinants, block identities and inverses [T-LA-DETERMINANT]

Trigger. Both AB and BA occur, perhaps inside an exponential or a polynomial.

1. Check dimensions. Equal-size square products have the same characteristic polynomial; rectangular products agree only on the nonzero spectrum with algebraic multiplicities.
2. If one factor is invertible, exhibit the similarity directly. Otherwise use $\det(I + AB) = \det(I + BA)$; singular factors do not invalidate this identity.
3. For an entire series f, isolate its constant term and write $f(AB) = cI + AC$, $f(BA) = cI + CA$ using one convergent series C.
4. Transfer exactly the characteristic-polynomial information needed: determinant, eigenvalue multiplicities or nilpotency. Avoid claiming the matrices are equal.

Example. Nilpotency of $p(e^{AB})$ is equivalent to nilpotency of $p(e^{BA})$, since both have the same characteristic polynomial and nilpotency is equivalent to that polynomial being t^n. **Check.** Similarity of AB and BA can fail when both factors are singular. Prove only the spectral statement actually required.

Extended atlas M-LA-15

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2000-D2-P6 [R]

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M-LA-TENSOR Lift products of inner products to new vectors

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Prerequisites Tensor products and operators on matrix spaces [T-LA-TENSOR]; Gram matrices and positive semidefinite forms [T-LA-GRAM]

Trigger. Entrywise powers of a Gram matrix, products of correlations, or recursive block spectra.

1. First exhibit a Gram representation or independent eigenbases of the factors.
2. Replace v_i by $v_i^{\otimes k}$ to turn $\langle v_i, v_j \rangle$ into its kth power. Norms transform by the same rule.
3. For a Kronecker sum, tensor eigenvectors of the factors and add their eigenvalues.
4. Count all multiplicities and verify that their sum equals the dimension of the lifted space.

Example. A PSD correlation matrix with diagonal one remains PSD after every positive integer entrywise power. Its nonnegative determinant then gives the corresponding powered three-variable inequality. **Checks.** A general scalar entrywise function need not preserve PSD. Here the construction proves preservation specifically for positive integer powers. Repeated eigenvalue sums must be combined, and eigenvector counting requires full bases or a separate triangularization argument.

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2004-D2-P4 [R]; 2010-D1-P5 [R]; 2017-D2-P8 [R]

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M-LA-PROJECTIVE Linearize a fractional recurrence in dimension two

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Prerequisites Fractional-linear recurrences and matrix powers [T-LA-PROJECTIVE]

Trigger. A cyclic recurrence $x_{i+1} = (ax_i + b)/(cx_i + d)$.

1. Treat constant solutions and zero denominators separately before introducing projective coordinates.
2. Represent the recurrence by an invertible 2×2 matrix and replace n iterations by its nth power.
3. Translate a periodic point into a real eigenline. Analyze whether conjugate eigenvalues can acquire real nth powers.
4. When the power is scalar, discard the finite set of starting points whose orbit hits a forbidden denominator, then choose any remaining real starting point.

Example. The matrix for $x \mapsto 1 - a/x$ is $\left(\begin{smallmatrix} 1 & -a \\ 1 & 0 \end{smallmatrix}\right)$. For $a > 1/4$, write its eigenvalues as $\sqrt{a}e^{\pm i\theta}$. Periodicity forces $n\theta \in \pi\mathbb{Z}$. **Check.** A projective solution involving infinity does not solve the original real equations. Establish admissibility explicitly after the spectral calculation.

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2012-D2-P4 [R]

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M-LA-PERTURB Use a generic parameter to force an image inclusion

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Prerequisites Evaluation maps and rank under perturbation [T-LA-PERTURBATION]; Rank, dimension and exact linear constraints [T-LA-RANK]

Trigger. A minimal space of maps covers vectors by its kernels, or a rank statement must hold away from exceptional parameters.

1. Express the condition through evaluation maps $E_x(T) = Tx$ and choose an evaluation of maximal rank.

M-LA-PERTURB Use a generic parameter to force an image inclusion (continued)

2. Perturb the evaluation point to $x + ty$. A minor nonzero at $t = 0$ is a nonzero polynomial, hence remains nonzero except at finitely many parameters.
3. Combine this generic injectivity with the assumed covering property to contradict any proposed image outside the required subspace.
4. Alternatively, normalize a one-dimensional kernel locally and differentiate its rational parametrization; use $T(0)y + T'(0)x = 0$ to obtain the image inclusion.

Check. The derivative route needs locally constant rank and a fixed nonzero normalization coordinate. Minimality is used to find a point not killed by any nonzero map in a proper subspace; it does not imply that every evaluation has the same rank. **Afterward.** Quotient the image by the forced subspace and apply dimension induction, recording every loss in dimension.

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2009-D2-P5 [R]

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M-LA-WALKS Interpret matrix multiplication as a count

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Rank, dimension and exact linear constraints [T-LA-RANK]; Finite counting and incidence identities [T-DS-COUNTING]; Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]

Trigger. A zero-one matrix encodes allowed transitions, or a matrix recursion resembles a combinatorial state recursion.

1. Specify precisely what each row and column state represents and when one transition is allowed.
2. Expand $(A^k)_{ij} = \sum a_{ii_1}a_{i_1i_2}\cdots a_{i_{k-1}j}$. Each nonzero summand corresponds to one admissible sequence with k transitions.
3. Sum over endpoints if the target is the sum of all entries. Explain why every desired configuration is represented once.
4. Search for a bijection, such as transposing a rectangular configuration, that proves equality of two counts.

Example. If states are binary columns of height n and transitions forbid adjacent all-one 2×2 squares, the sum of entries of A^{k-1} counts allowable $n \times k$ arrays. Transposition exchanges n and k.
Check. Powers count walks, which may repeat states. They do not count simple paths unless the state system enforces that additional restriction. Distinguish adjacent forbidden subarrays from arbitrary submatrices.

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2001-D2-P4 [R]; 2004-D2-P6 [R]

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M-LA-PARAMETER Compare coefficients in a matrix polynomial

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Prerequisites Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]

Trigger. A relation holds at many scalar parameters or a matrix is built as $A + tB$.

1. Expand entrywise as a polynomial in the scalar parameter, retaining the order of matrix factors. Noncommutativity changes coefficients, not the degree bound.
2. Bound the degree of each entry. A degree at most r polynomial vanishing at $r + 1$ distinct field elements is identically zero.

M-LA-PARAMETER Compare coefficients in a matrix polynomial (continued)

3. Compare the constant and highest-degree coefficients, which are often the desired pure powers of A and B.
4. For an existence argument, use the converse viewpoint: a nonzero determinant polynomial has only finitely many roots, so choose a parameter outside them.

Example. In $(A+tB)^n$, the constant coefficient is A^n and the coefficient of t^n is B^n. If the expression vanishes at $n+1$ distinct parameters, both matrices are nilpotent. **Check.** Do not replace mixed coefficients by scalar binomial expressions unless A and B commute. Over a finite field, there may be too few available parameters for the argument.

Extended atlas M-LA-06

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1995-D1-P5 [R]; 2002-D2-P5 [R]

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M-LA-FACTOR Factor while preserving matrix order

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Prerequisites Determinants, block identities and inverses [T-LA-DETERMINANT]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]

Trigger. A matrix polynomial in two variables, an invertible commutator, or a determinant obstruction.

1. First determine which factors commute. An identity such as $B = I - A$ makes all relevant expressions polynomials in one matrix; absent such a relation, preserve every product order.
2. Test a factorization by full expansion before cancelling any factor. Scalar roots of unity can organize cancellations without requiring commutativity.
3. Take determinants after the matrix identity has been established. Over the reals, conjugate complex factors have product of determinants $|\det S|^2$.
4. Separate necessary scalar conditions from existence: a determinant equation alone does not guarantee a matrix solution. Supply and verify admissible blocks for the converse.

Example. If $A^2 + B^2 = AB$ and $\omega^2 + \omega + 1 = 0$, then $(A + \omega B)(A + \overline{\omega} B) = \omega(BA - AB)$ for real A, B. Nonzero determinant forces ω^n real. **Check.** Cancellation by a matrix requires the appropriate one-sided inverse. An invertible product of square matrices does make both factors invertible, but this fact must be stated before cancellation.

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1997-D1-P3 [R]; 2019-D1-P5 [R]; 2024-D2-P7 [R]

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M-LA-DUAL Test a linear map on a basis and normalize its values

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Prerequisites Dual bases, kernels and separating functionals [T-LA-DUALITY]; Inner products, orthogonality and polarization [T-LA-INNER]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]

Trigger. A linear functional is constrained by its kernel or multiplication, or vectors with prescribed scalar products are needed.

1. Choose a basis or an independent subfamily of the given functionals. Identify which basis values determine the whole map by linearity.
2. Normalize a nonzero constant or a dual coordinate, proving first that it is nonzero. If polynomial inputs occur, translating the variable can normalize the value of the degree-one monomial.

M-LA-DUAL Test a linear map on a basis and normalize its values (continued)

3. Use the extra hypothesis to force the remaining basis values, often by induction or by factoring a polynomial whose value is zero.
4. For geometric dual vectors, form signed sums; cancellation of mixed terms in the average squared norm finds a sufficiently long sum, then normalize it.

Example. A dual family $\langle w_i, v_j\rangle = \delta_{ij}$ gives $|\langle \sum_i \varepsilon_i w_i, v_j\rangle| = 1$ for every choice of signs. One signed sum has squared norm at least $\sum_i \|w_i\|^2$, by averaging. **Check.** A multiplicativity implication is not equality $P(fg) = P(f)P(g)$. Use precisely the given zero-product condition at each step; do not replace it by an unproved stronger algebra homomorphism property.

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2008-D1-P2 [R]; 2013-D2-P3 [R]

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M-LA-RESIDUE Reduce integer matrices modulo an integer

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Prerequisites Determinants, block identities and inverses [T-LA-DETERMINANT]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]

Trigger. Integer entries, an odd/even size constraint, or a determinant required to lie in an impossible residue class.

1. Choose a modulus that simplifies the coefficients or distinguishes the proposed determinant values. Entrywise reduction commutes with addition and multiplication.
2. Reduce the matrix identity before taking determinants. The determinant polynomial also commutes with reduction, even modulo a composite integer.
3. Compare the scalar residue classes: a fourth power is zero or one modulo four, whereas an odd power of a number congruent to three remains three.
4. For constructions, place odd entries on a triangular diagonal and even entries above it; the determinant is then odd and hence nonzero over the integers.

Checks. A nonzero determinant modulo a prime proves nonsingularity over the rationals. Zero determinant modulo that prime does not prove rational singularity. Division is safe only by a unit; working modulo four does not give a field, but determinant multiplication remains valid. **Example.** An integer identity $A^4 \equiv 3I \pmod 4$ in odd dimension would force $(\det A)^4 \equiv 3 \pmod 4$, an immediate contradiction.

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2007-D1-P2 [R]; 2019-D1-P5 [R]

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Analysis, limits and probability

M-AN-AUXILIARY Design an auxiliary derivative

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Prerequisites Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Differential inequalities and integrating factors [T-AN-ODE]

Trigger. The target contains a combination of f, f', f'', an endpoint relation, or a desired monotonicity statement. Work backwards from the expression: product and chain rules suggest $f^2/2+f'$, $e^{-x}f$, $xf-x^2/2$, or arctan $f+x$.

Procedure. Define the auxiliary function on an explicit interval. Differentiate it exactly before estimating. If its derivative has one sign, compare endpoint values; if equal endpoint values are known, apply Rolle. If the expression is a derivative only after division, first exclude zeros or divide the domain into nonzero components.

Obligations. For open endpoints, take justified one-sided limits rather than writing undefined values. A monotonicity inequality gives a necessary bound; supply an admissible example to prove sharpness. For $f' \geq -(1+f^2)$, the exact identity $(\arctan f + x)' = f'/(1+f^2) + 1$ makes infinite values finite and exposes the interval-length obstruction. The substitution is derived from the denominator, not guessed from an unrelated function family.

Extended atlas M-RA-01; M-RA-02

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P2 [R]; 1994-D2-P1 [R]; 1994-D2-P3 [R]; 1995-D1-P4 [R]; 1998-D1-P4 [R]; 2001-D1-P6 [R]; 2005-D1-P5 [R]; 2007-D2-P1 [R]; 2009-D2-P2 [R]; 2013-D1-P2 [R]; 2014-D2-P3 [R]; 2019-D1-P3 [R]; 2020-D2-P5 [R]; 2023-D2-P7 [R]

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M-AN-ROLLE Count zeros and subtract an interpolant

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Prerequisites Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Differential inequalities and integrating factors [T-AN-ODE]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]

Trigger. Several zeros, endpoint values, or derivative values must force an interior identity. Choose a low-degree polynomial or explicit comparison function that matches every prescribed datum. Subtract it from f so the hypothesis becomes a list of zeros.

Procedure. Draw or list the ordered zeros. Apply Rolle between consecutive distinct zeros. Add any derivative zeros supplied directly by Hermite data, then repeat until the desired derivative order is reached. For an operator $D-c$, apply ordinary Rolle to $e^{-cx}f$; factor a real-rooted polynomial in D into such first-order factors and repeat.

Obligations. Count multiplicities through actual derivative conditions, not by informal repeated labels. The nth derivative of a leading term ax^n is $n!a$, so retain the factorial. A comparison function with constant target expression can show that a proposed universal parameter is necessary as well as sufficient. For nonreal characteristic roots, oscillatory exponential solutions may instead yield counterexamples.

Extended atlas M-RA-03

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D2-P3 [R]; 1998-D1-P4 [R]; 2005-D2-P4 [R]; 2006-D1-P5 [R]; 2006-D1-P6 [R]; 2010-D1-P1 [R]; 2013-D1-P2 [R]; 2019-D2-P6 [R]; 2023-D2-P7 [R]

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M-AN-LOCAL Expand near equality, bound away from it

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Prerequisites Taylor remainders and local scale [T-AN-TAYLOR]; Continuity, compactness, and order [T-AN-CONTINUITY]

Trigger. A quotient has a vanishing denominator, or a uniform constant is required near suspected equality points. Identify all zeros of the denominator on the constraint set before expanding.

Procedure. Parameterize a neighborhood of each exceptional point and Taylor-expand the numerator and denominator to the first surviving order. Prove that the denominator has a fixed positive leading coefficient and bound the quotient locally. On the remaining compact part, continuity and nonvanishing of the denominator give a finite maximum. Take the maximum of the finitely many bounds.

Obligations. Record available differentiability and bound each derivative's remainder separately. For a square root of a quadratic zero use $|h|$, not h. If a smoothness assumption is weakened, choose a fractional-power perturbation whose lower derivatives remain continuous but whose target quotient diverges. This tests necessity of the hypothesis without changing the main argument.

Extended atlas M-RA-04

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1995-D1-P3 [R]; 1995-D1-P6 [R]; 1996-D1-P5 [R]; 1997-D2-P1 [R]; 2001-D2-P2 [R]

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M-AN-PRIMITIVE Turn a cumulative constraint into a weighted identity

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Prerequisites Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Nonnegative sums, Fubini, and integral kernels [T-AN-FUBINI]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]

Trigger. The hypothesis controls $\int_x^b f$, while the target contains a moment, a square, or a cubic. Introduce a primitive whose endpoint value is zero at the convenient end.

Procedure. Integrate the given inequality over its free endpoint, or integrate by parts to transfer a derivative onto a controlled factor. Sketch the triangular integration region and exchange order when continuity, nonnegativity, or absolute integrability justifies it. Then use a nonnegative square or a derivative bound.

Example. A lower bound on $\int_x^1 f$ yields a lower bound on $\int_0^1 tf(t)\,dt$. Expanding $\int_0^1 (f-t)^2 \geq 0$ turns that information into a bound on $\int f^2$.

Obligations. Compute the boundary term and the sign of the primitive before discarding anything. If $G = -\int_x^b f$, then $G' = f$ but $G \leq 0$ for $f \geq 0$; this sign is exactly what turns $\int -2Gf$ into a positive endpoint square.

Extended atlas M-RA-11

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1995-D1-P2 [R]; 1999-D1-P6 [R]; 2004-D2-P2 [R]; 2004-D2-P5 [R]; 2005-D1-P3 [R]; 2010-D1-P1 [R]; 2010-D1-P2 [R]; 2014-D2-P3 [R]; 2015-D2-P7 [R]; 2017-D1-P2 [R]; 2019-D2-P6 [R]; 2022-D2-P8 [R]

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M-AN-PAIR Pair an integral through symmetry

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Prerequisites Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]

Trigger. The constraint links two arguments, such as x and $1-x$, or a weight simplifies when x is reflected. Write the target integral twice using substitutions adapted to that relation.

Procedure. Add the two representations, apply the pointwise hypothesis to the paired integrands, and integrate the resulting elementary bound. Sometimes the right substitution is trigonometric: $x = \sin t$ and $x = \cos t$ pair the weights $\cos t$ and $\sin t$ on $[0, \pi/2]$.

M-AN-PAIR Pair an integral through symmetry (continued)

Example. If $f(x)f(1-x) = 1$ and $f > 0$, then $f(x) + f(1-x) \geq 2$; pairing gives $\int_0^1 f \geq 1$. With $f(x)+f(y) \geq |x-y|$, choose a pairing with constant separation and test an absolute-value extremizer.
Obligations. Include Jacobians and transformed endpoints. Check the proposed extremizer against the original two-variable condition for every pair, not merely along the pairing curve. Equality of two integrals does not imply equality of their integrands.
Extended atlas M-RA-11; M-RA-18

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P5 [R]; 1995-D1-P4 [R]; 1996-D1-P2 [R]; 1998-D1-P6 [R]; 2003-D2-P2 [R]; 2004-D2-P5 [R]; 2016-D2-P7 [R]; 2022-D1-P1 [R]

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M-AN-RIEMANN Expose the mesh and control its tails

Back to previous view | Theory index | Method index | Strategy index
Prerequisites Riemann sums, periodic averaging, and tails [T-AN-RIEMANN]; Taylor remainders and local scale [T-AN-TAYLOR]
Trigger. A large-parameter sum contains k/n, $k/\sqrt{x}$, or nh after a logarithmic substitution. Rewrite it exactly as a mesh width times a sampled function.
Procedure. Identify the limiting integral on a fixed compact interval. Prove that the integrand is Riemann integrable there, filling removable endpoint values explicitly. If the range is infinite or singular, establish a uniform sum–integral tail bound before taking the mesh limit. For periodic products, partition into full periods and freeze the slow factor using its modulus of continuity.
Refinement. If an error rate matters, use symmetric cells. Taylor's linear term integrates to zero on a centered cell, leaving a second-derivative remainder. Sum its absolute integral and account separately for the initial half-cell.
Obligations. Integrability by itself does not guarantee arbitrary infinite Riemann sums converge; monotone-tail or comparable uniform control is needed. A parameter-dependent small offset must be squeezed uniformly or held fixed while the mesh limit is taken.
Extended atlas M-RA-06; M-RA-13

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P5 [R]; 1996-D2-P5 [R]; 1997-D1-P1 [R]; 2001-D1-P3 [R]; 2018-D2-P10 [R]; 2024-D1-P2 [R]

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M-AN-SPLIT Separate a main range from exceptional scales

Back to previous view | Theory index | Method index | Strategy index
Prerequisites Scale separation and limit comparisons [T-AN-ASYMPTOTICS]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Trigger. Pointwise limits fail to control a moving domain, or only a narrow neighborhood contributes. Divide the sum or integral into regions where different estimates are effective.
Procedure. Use a sharp uniform approximation on the main region and coarse bounds on small or remote regions. Choose a cutoff that makes every error small after the requested normalization. Alternatively keep cutoffs fixed, take the main limit, and only then send the cutoffs to their extreme values.
Example. For a logarithmic convolution sum, discard two endpoint blocks of length M and use $\log k \sim \log N$ uniformly in the middle. A choice such as $M = N/(\log N)^2$ balances a small endpoint mass with logarithmic equivalence. For f^n, first locate the maximum of f, then compare its endpoint drop to linear or quadratic models.
Obligations. State exactly which constants depend on auxiliary parameters. Show both lower and upper bounds, and never choose a cutoff using an unproved value of the desired limit.
Extended atlas M-RA-07; M-RA-14

M-AN-SPLIT Separate a main range from exceptional scales (continued)

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D2-P6 [R]; 1995-D2-P4 [R]; 1996-D1-P5 [R]; 1997-D1-P1 [R]; 2000-D1-P6 [R]; 2003-D1-P5 [R]; 2003-D2-P2 [R]; 2014-D1-P2 [R]; 2015-D2-P7 [R]; 2017-D2-P6 [R]; 2018-D2-P10 [R]; 2020-D2-P8 [R]

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M-AN-RECURRENCE Normalize, trap, and identify a recurrence

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]

Trigger. Repeated radicals, rational updates, or convolution recurrences obscure monotonicity. Seek an error from a candidate limit, a scale-normalized term, or an invertible substitution.

Procedure. Prove domain invariance first. Derive the transformed recurrence exactly; establish monotonicity and bounds, a geometric contraction, or induction barriers cq^n. Verify all initial cases, especially the short convolution ranges. Identify the limit only after convergence has been established.

Example. If an update gives $1 - x_{n+1} = r_n(1 - x_n)$ with $0 < r_n < 1/2$, induction gives a uniform geometric error. A half-angle identity may instead reveal an explicit formula; confirm its initial index before using asymptotics.

Obligations. A preserved algebraic relation between two sequences can prove their normalized limits agree, but it does not itself guarantee either limit exists. For a bounded continuous iteration with vanishing increments, a cluster-point crossing argument may replace monotonicity. Search for fixed points and cycles when a universal convergence assertion is asked.

Extended atlas M-RA-15

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1995-D2-P2 [R]; 1996-D1-P4 [R]; 1996-D2-P1 [R]; 1999-D2-P2 [R]; 2000-D1-P6 [R]; 2001-D2-P2 [R]; 2003-D1-P1 [R]; 2010-D2-P1 [R]; 2010-D2-P2 [R]; 2011-D2-P1 [R]; 2012-D2-P2 [R]; 2014-D1-P2 [R]

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M-AN-TELESCOPE Find a finite telescoping identity

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Prerequisites Series, finite cancellation, and summation by parts [T-AN-SERIES]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]

Trigger. The target is an infinite series or product tied to a recurrence, or adjacent indices differ by a simple factor. Look for $u_n - u_{n+1}$, ratios u_n/u_{n+1}, or cancellation between dyadic blocks.

Procedure. Prove the identity for a finite upper limit N. Cancel interior terms and list all lower and upper boundary terms explicitly. Estimate the surviving remainder to prove it converges to the asserted value. Rational functions often telescope after factorization or partial fractions; logarithms can expose multiplicative structure.

Example. A recurrence may give $a_{n+1}/a_n = na_n - (n+1)a_{n+1}$. The finite sum is then a constant minus $(N+1)a_{N+1}$. First show convergence of the positive series, then obtain a geometric bound on a_n to eliminate the remainder.

Obligations. Do not split divergent components into separate infinite sums. If a strict bound is required, exhibit a strict loss in at least one fixed term and bound the remaining tail non-strictly; a limit of strict finite inequalities alone need not stay strict.

Extended atlas M-RA-15

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1995-D2-P2 [R]; 1996-D1-P2 [R]; 1996-D2-P6 [R]; 1999-D1-P2 [R]; 2000-D1-P4 [R]; 2010-D1-P2 [R]; 2011-D2-P3 [R]; 2012-D2-P2 [R]; 2015-D2-P6 [R]; 2016-D2-P6 [R]; 2019-D1-P1 [R]

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M-AN-SERIES Test convergence through blocks and differences

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Series, finite cancellation, and summation by parts [T-AN-SERIES]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]

Trigger. A rearrangement, alternating amplitude, or permutation changes the visible cancellation. Translate the question into a condition on partial sums, not individual terms alone.

Procedure. For a rearrangement within blocks, express each intermediate partial sum through a bounded number of original partial sums. For an alternating series with nonmonotone amplitudes b_n, pair consecutive terms and bound $|b_{n+1} - b_n|$ using the mean value theorem. For a divergence claim, find arbitrarily late blocks with mass bounded below by a fixed positive constant, violating the Cauchy criterion.

Example. Distinct positive integers assigned to $2N$ positions include at least N values exceeding N; a denominator bounded by $(3N)^2$ then gives a fixed positive block mass.

Obligations. Check the last unpaired term tends to zero. For parameter thresholds, prove necessity by the term test using an explicit subsequence or integer-phase argument. Absolute convergence, conditional convergence, and mere boundedness of partial sums are different conclusions.

Extended atlas M-RA-12

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1997-D1-P2 [R]; 1997-D2-P3 [R]; 1999-D1-P2 [R]; 2010-D1-P2 [R]

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M-AN-WEIGHTS Choose a square, factor, or AM–GM weight

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Prerequisites Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Nonnegative sums, Fubini, and integral kernels [T-AN-FUBINI]

Trigger. The target mixes squares, denominators, moments, or products. Rewrite it until a standard inequality uses exactly the supplied constraint.

Procedure. For a rational sum, separate a linear term from a weighted square and apply $\sum x_i^2/w_i \geq (\sum x_i)^2/\sum w_i$. For moment constraints, search for a pointwise factorization vanishing at anticipated equality values. For products, choose weights whose product telescopes; AM–GM then produces a summable triangular kernel.

Sharpness. Trace equality in every step. If equality is incompatible with summability, truncate a near-extremal sequence and append a small positive summable tail; prove the ratio approaches the claimed best constant.

Obligations. Establish positive denominators before multiplying. A sum-of-squares certificate must be nonnegative on the full allowed domain, including negative variables. When infinite sums are involved, nonnegative terms justify exchanging the order, but do not forget to prove the final comparison series finite.

Extended atlas M-RA-10

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P6 [R]; 1995-D1-P2 [R]; 1995-D2-P5 [R]; 1996-D2-P6 [R]; 1999-D1-P2 [R]; 1999-D2-P3 [R]; 2000-D1-P4 [R]; 2001-D2-P6 [R]; 2010-D1-P1 [R]; 2010-D2-P2 [R]; 2015-D2-P6 [R]; 2016-D1-P3 [R]; 2016-D2-P6 [R]; 2018-D1-P1 [R]; 2022-D1-P1 [R]

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M-AN-CONVEX Choose the correct convexity comparison

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Prerequisites Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Lipschitz derivatives and convex smoothness [T-AN-SMOOTH]

Trigger. Equal-weight constraints, increasing derivatives, concave profiles, or gradient Lipschitz bounds suggest convexity.

Procedure. Prove convexity on the actual domain by a derivative or secant-slope test. Use a tangent for a lower bound, a chord for an upper bound, or Jensen for a fixed average. To compare constrained ranges, separately prove the constraint set is connected and that both endpoint bounds are attained.

Smooth case. Subtract the tangent at a base point so the resulting convex function has minimum zero. Take one step in the direction $-\nabla g/L$ and apply the descent estimate. Interchanging the base and evaluation points cancels function values and exposes a gradient inequality.

Obligations. A positive concave function on the entire line is constant, but the same is false on a finite interval. Equality depends on strictness and weights. Deriving $g'' \geq 0$ gives g above its tangent; it never makes g identically equal to that tangent.

Extended atlas M-RA-09

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1995-D1-P6 [R]; 1996-D1-P5 [R]; 2002-D2-P6 [R]; 2004-D1-P3 [R]; 2004-D2-P2 [R]; 2004-D2-P5 [R]; 2006-D2-P3 [R]; 2017-D1-P2 [R]; 2019-D1-P3 [R]; 2020-D2-P5 [R]

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M-AN-COMPACT Localize an extremum or an accumulation point

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Prerequisites Continuity, compactness, and order [T-AN-CONTINUITY]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]

Trigger. A global maximum is asserted on a noncompact domain, a function is nowhere monotone, or infinitely many zeros are present.

Procedure. For extrema, exhibit a value that the function cannot reach outside a sufficiently large compact set, then apply the extreme value theorem inside it. Examine interior critical points and boundary points separately. To force a local extremum, choose endpoints whose values are both worse than an interior value and minimize or maximize on that closed interval.

Accumulation form. Select a convergent subsequence from an infinite bounded set. Transfer values by continuity; if differentiability is available at the limit point, apply the difference quotient there. This can force a zero to accumulate only at an endpoint.

Obligations. Closedness ensures the limiting point belongs to the domain. A stationary point is only a candidate; a change of sign along two directions detects a saddle. When using a supremum-defined point, prove it lies in the closed set and that the required inequality survives passage to the limit.

Extended atlas M-RA-08

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P6 [R]; 1994-D2-P2 [R]; 2000-D2-P2 [R]; 2004-D1-P3 [R]; 2007-D2-P3 [R]; 2011-D1-P1 [R]; 2016-D1-P1 [R]

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M-AN-APPROXIMATE Preserve structure while approximating

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Prerequisites Uniform approximation and positive kernels [T-AN-APPROXIMATION]; Nonnegative sums, Fubini, and integral kernels [T-AN-FUBINI]

Trigger. A continuous function must be approximated by a restricted polynomial family or by iterated averaging operators.

Procedure. Obtain a uniform approximation, symmetrize to preserve parity, then replace any forbidden basis terms by explicit allowed approximants. Allocate an error budget to each replacement, scaled by its coefficient. For positive kernels, verify nonnegativity and total mass one before splitting the integral into a small neighborhood and its complement.

Alternative. A positive averaging operator preserves inequalities. Bound the initial function above and below by monotone continuous envelopes, prove the iterated limits for those envelopes, then squeeze the original iterates.

Obligations. Uniform continuity provides one scale valid at all points; pointwise continuity gives a scale depending on the point. Normalize the kernel exactly, including any factorial from an ordered integration region. Do not infer convergence of derivatives from uniform convergence of functions, and do not treat a positive but unnormalized kernel as an average.

Extended atlas M-RA-17

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1995-D2-P4 [R]; 2003-D1-P5 [R]; 2021-D1-P4 [R]

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M-AN-DOMINATE Pass a limit through an integral safely

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Dominated convergence and orthonormal separation [T-AN-MEASURE]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]

Trigger. A pointwise limit is known, but an integral or norm must also converge.

Procedure. State the pointwise or almost-everywhere convergence. Identify one integrable majorant for the exact integrand and invoke dominated convergence, or prove the sequence is nonnegative increasing and use monotone convergence. If neither fits, use uniform convergence on compact pieces plus uniform tail estimates.

Norm contradiction. For an allegedly convergent uniformly bounded orthonormal subsequence, apply dominated convergence to its squared difference from the pointwise limit. This makes it Cauchy in L^2, contradicting the fixed distance $\sqrt{2}$ between distinct orthonormal members.

Obligations. A uniform numerical bound gives an integrable majorant only on a finite-measure domain. In an L^2 argument, square the bound and account for the limit as well. Do not assume an orthonormal family is complete, and do not augment it by additional functions with unproved boundedness properties.

Extended atlas M-RA-12

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1995-D2-P6 [R]

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M-AN-TOPOLOGY Use a dense set, an enumeration, or a category argument

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Countability, accumulation, and Cantor constructions [T-AN-SETS]; Baire class one and controlled discontinuities [T-AN-BAIRE]

Trigger. The problem compares rational and real domains, uncountable fibers, dense extrema, or pointwise limits of continuous functions.

Procedure. Identify which of density, countability, compactness, and completeness is actually used. An enumeration supports finite-stage constructions; a dense sequence supports continuity transfer. Infinite bounded sets supply accumulation points. For a Cantor construction, specify the closed residual set, the values on every complementary interval, and a uniform vanishing-height bound proving continuity.

Baire route. For a gauge condition, prove closed sublevel and superlevel inverse images are G_δ using unions of explicitly sized intervals, then apply the real Baire-one characterization. This gives a precise route without assuming the gauge has continuity.

Obligations. A countable set may be dense. An uncountable set need not contain an interval. Category and Lebesgue measure are different notions of smallness. Where a proof uses a relative closed set, all balls, neighborhoods, and continuity statements must be interpreted in its relative topology.

Extended atlas M-RA-16

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1997-D2-P6 [R]; 1998-D2-P6 [R]; 2002-D1-P5 [R]; 2003-D2-P5 [R]; 2006-D1-P1 [R]; 2009-D1-P1 [R]; 2016-D1-P1 [R]; 2021-D1-P4 [R]

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M-AN-COVER Turn local bad intervals into a global measure bound

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Interval coverings and level-set integration [T-AN-COVERING]; Dominated convergence and orthonormal separation [T-AN-MEASURE]

Trigger. Every candidate point is assumed to violate a desired slope bound, or a spike construction must leave at least one differentiability point.

Procedure. Assign a witnessing interval to each bad point. For average-value violations, choose a disjoint subfamily whose triples cover the original finite family, and sum the integrals on the disjoint intervals. Pass from compact subsets to the full bad set. Then integrate the resulting level-set estimate using the layer-cake formula.

Spike route. For a summable height sequence, make slowly increasing weights whose weighted sum remains small. Intervals with these weighted radii cannot cover the whole unit interval. Outside them, late spike difference quotients tend to zero; remove finitely many early spikes by shrinking the neighborhood.

Obligations. Total length greater than one does not by itself prove coverage; a divergent-length construction requires an explicit placement. Exclude countable centers separately and treat zero heights without division. Track whether the endpoint exponent makes the level-set integral finite or logarithmically divergent.

Extended atlas M-RA-16

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1998-D2-P6 [R]; 1999-D1-P6 [R]; 2002-D1-P5 [R]

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M-AN-GENERATING Encode convolution or denominators by a power series

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Power series and generating functions [T-AN-POWER]; Nonnegative sums, Fubini, and integral kernels [T-AN-FUBINI]; Series, finite cancellation, and summation by parts [T-AN-SERIES]

Trigger. A recurrence contains a convolution, or a summand has consecutive factors in its denominator.

Procedure. Define the generating function with its initial coefficient. Prove a radius bound by induction or absolute comparison. Translate the recurrence into a product or differential equation inside that radius. Solve with the correct initial condition and evaluate only at a justified point. Consecutive denominator factors often arise by repeated integration of a geometric series.

Alternative encoding. Express a logarithm or reciprocal as an integral and sum a finite binomial expansion inside it. For example $\log k = \int_0^\infty (e^{-t} - e^{-kt})\,dt/t$ for $k > 0$. Keep the cancellation inside the integral; the two terms are separately singular at zero.

Obligations. Separate formal coefficient algebra from analytic convergence. An endpoint evaluation requires Abel's theorem, uniform convergence, or a direct integral bound. Finite sums can be interchanged freely with convergent integrals; infinite sums require an additional justification.

Extended atlas M-RA-15

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1999-D2-P2 [R]; 2003-D2-P6 [R]; 2010-D1-P2 [R]; 2020-D2-P8 [R]

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M-AN-PROBABILITY Count outcomes or condition on a useful structure

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Prerequisites Finite probability, conditioning, and expectations [T-AN-PROBABILITY]; Finite counting and incidence identities [T-DS-COUNTING]

Trigger. A random discrete event has a hidden bijection, residue state, or count of contributing objects.

Procedure. Specify the equiprobable sample space. Count favorable outcomes by a bijection that retains all constraints; alternatively condition on a partial configuration and sum the conditional probabilities. For repeated independent choices, use a residue recurrence or a probability generating function. For an expected count, introduce one indicator per candidate and sum expectations.

Example. If all elements of one chosen subset must precede all elements of another, their ordered union determines the pair. Check that the earliest chosen elements automatically lie in the smaller allowed ambient set before accepting the bijection.

Obligations. Equal marginal distributions do not imply independence. Independence is needed for a product formula, but not for linearity of expectation. A roots-of-unity filter applies to a generating polynomial with normalized probabilities; omitting the factor $1/m$ changes the answer.

Extended atlas PRB-01; PRB-02; PRB-03

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1999-D2-P2 [R]; 2019-D2-P10 [R]; 2021-D1-P2 [R]; 2022-D2-P8 [R]

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M-AN-GEOMETRIC Condition and average a convex-hull event

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Prerequisites Uniform random points and convex-hull events [T-AN-GEOMETRICPROB]; Facets and signed simplex decompositions [T-DS-POLYTOPE]

Trigger. Random hull vertices, chord intersections, or containment probabilities are requested. Translate the geometry into an indicator condition before integrating.

Procedure. Condition on a small fixed configuration; express the remaining probability by area or arc-length ratios. Count candidate edges with indicators, or exploit the uniform cyclic order of

M-AN-GEOMETRIC Condition and average a convex-hull event (continued)

labels. Treat configurations with empty intersections separately even when their color sequence changes twice.

Symmetry route. For centrally symmetric independent points, average over all sign choices. A convex polytope can be decomposed along generic rays from the origin into signed pyramids over its facets: entering and leaving crossings contribute opposite signs. After summing sign choices, nonnegative pyramid coefficients yield monotonicity along each ray because a pyramid containing the origin contains the intervening segment.

Obligations. Justify general position almost surely and handle query points on exceptional boundaries by a limiting or null-event argument. A symmetry statement alone proves equality at equal radii, not the direction of an inequality at different radii. Verify signs and facet membership before counting the pyramid coefficients.

Extended atlas PRB-04

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2019-D2-P10 [R]; 2022-D2-P8 [R]; 2024-D1-P5 [R]

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M-AN-RATIO Compare growth through derivatives

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Scale separation and limit comparisons [T-AN-ASYMPTOTICS]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]

Trigger. A quotient of an integral and its scale, or a differential relation between two increasing functions, suggests growth comparison.

Procedure. Transform the quotient so its derivative ratio contains the given expression. Check the denominator derivative is nonzero and the appropriate endpoint behavior holds. Prove that the derivative ratio converges before invoking l'Hospital. For differential inequalities, bound the derivative ratio between fixed constants with an arbitrary epsilon and integrate those bounds.

Example. Multiplying by an exponential weight can turn a second-order expression into two successive derivative ratios. Start from the final bounded-over-divergent expression and reason backwards; writing a chain of limits before establishing their existence conceals the logic.

Obligations. A numerator need not diverge in the monotone unbounded-denominator version, but the derivative ratio limit is still essential. Keep a fixed positive base point if zero is outside the domain. If no legitimate l'Hospital form is available, use the mean value theorem and a squeeze instead.

Extended atlas M-RA-05

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1995-D1-P3 [R]; 2000-D1-P6 [R]; 2001-D1-P6 [R]; 2005-D1-P5 [R]; 2006-D1-P5 [R]; 2015-D2-P7 [R]; 2017-D2-P6 [R]; 2023-D2-P7 [R]

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M-AN-TRIG Exploit identities before estimating oscillations

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Prerequisites Trigonometric identities and safe branches [T-AN-TRIG]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]

Trigger. A sine quotient, nested half-angle recurrence, weighted trigonometric product, or zero-counting integral is present.

Procedure. Use addition or product-to-sum formulas to separate frequencies or convert two successive indices into a simple difference. Orthogonality may isolate one low frequency. For zero counting, compare the unweighted integral with an integral against a test factor of absolute value at most one; a function of constant sign would force the weighted integral's magnitude to be no larger.

M-AN-TRIG Exploit identities before estimating oscillations (continued)

Recurrence route. Introduce an angle or inverse hyperbolic coordinate only after checking its range. Verify the initial value and update formula, then use monotonicity or small-angle asymptotics.
Obligations. State the interval on which sine or cosine is concave and tangent is finite. Avoid using a sign-changing multiplier in an inequality without splitting cases. A global trigonometric comparison often needs one interval handled by derivative signs and its complement handled by a simple amplitude bound.
Extended atlas M-RA-18

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1995-D2-P5 [R]; 1996-D1-P2 [R]; 2001-D2-P6 [R]; 2006-D2-P3 [R]; 2011-D2-P1 [R]

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M-AN-BARRIER Use a first crossing or a forced-growth contradiction

Back to previous view | Theory index | Method index | Strategy index
Prerequisites Differential inequalities and integrating factors [T-AN-ODE]; Continuity, compactness, and order [T-AN-CONTINUITY]
Trigger. A differential inequality controls the derivative whenever the function reaches a chosen level, or a composition $f(f(t))$ forces escalating growth.
Procedure. Suppose the function crosses a barrier and take the first crossing time as an infimum. Prove by continuity that equality holds there. The derivative sign dictated by the hypothesis must be consistent with the direction of crossing; if it is not, the crossing is impossible. Iterate the resulting order information only on ranges where it is already established.
Growth form. A hypothetical limit may force $f' > c > 0$, then $f' > f$, and finally $f' > ce^f$. Multiplying the last inequality by e^{-f} yields a bounded quantity with derivative bounded away from zero, contradicting existence on an infinite interval.
Obligations. Distinguish a first crossing from an attained infimum at the initial point and examine the correct one-sided neighborhood in each case. For compositions, verify every intermediate argument lies in the range where the previous bound holds. Close endpoint cases by continuity only after proving the interior statement.
Extended atlas M-RA-02

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D2-P1 [R]; 1996-D2-P1 [R]; 2006-D1-P6 [R]; 2007-D2-P3 [R]; 2012-D1-P4 [R]

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Algebra, number theory and complex polynomials

M-AL-ROOTCOUNT Replace many equal values by too many roots

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Prerequisites Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]

Subtract the specified value from the polynomial, or subtract the two proposed formulas. Count distinct zeros and compare with the degree. If zeros come from different factors or different levels, record disjointness; if tangency is given, count multiplicity instead. Over a finite field reduce the degree before invoking the bound. When many integer values lie in a small divisor set, use the pigeonhole principle to obtain repeated values first. Finish by checking that the polynomial claimed to vanish identically was not assumed nonconstant. A root count establishes an identity, not merely an estimate.

Extended atlas M-ANT-06

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2000-D2-P3 [R]; 2007-D1-P1 [R]; 2007-D1-P6 [R]; 2008-D2-P1 [R]; 2008-D2-P4 [R]; 2018-D1-P2 [R]; 2023-D2-P10 [R]

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M-AL-FACTOR Factor first and compare irreducible exponents

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Prerequisites Polynomial factorization and coprime divisibility [T-AL-FACTORIZATION]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]

Separate zero polynomials, zero roots, and nonzero scalar factors. Factor over the field or ring actually specified, using Gauss's lemma when moving between integer and rational coefficients. In an identity of powers, compare each irreducible factor's exponent. In a divisibility problem, remove gcds before cancelling. Translate the resulting degree and coefficient restrictions back into candidate formulas, and substitute every candidate into the original conditions. Do not infer that composition preserves irreducibility. For noncommuting rings this method is usually unavailable; expansion in order is required instead.

Extended atlas M-ANT-05

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2000-D2-P4 [R]; 2005-D1-P4 [R]; 2005-D2-P2 [R]; 2012-D1-P5 [R]; 2020-D2-P6 [R]; 2023-D1-P3 [R]; 2024-D1-P1 [R]

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M-AL-VIETA Translate root restrictions into symmetric coefficients

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Prerequisites Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]

Normalize the leading coefficient, label roots with multiplicity, and compute only the symmetric sums needed. Use Viète or Newton identities to replace the first few coefficients by sums of roots, squares, or products. A nonnegative sum, such as the sum of squares of real roots, yields a necessary coefficient inequality. For a constructive problem, prescribe roots first and read off coefficients; then verify integrality, coprimality, and distinctness separately. Isolate zero roots before dividing by their product. When reducing to a low-degree derivative, justify that Rolle's theorem retains real roots.

Extended atlas M-ANT-07

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2000-D1-P2 [R]; 2005-D1-P4 [R]; 2006-D2-P5 [R]; 2013-D1-P4 [R]; 2013-D1-P5 [R]; 2017-D2-P7 [R]; 2020-D2-P6 [R]

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M-AL-LOGDERIVATIVE Use a logarithmic derivative to expose root geometry

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Prerequisites Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]; Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]

Factor the polynomial and write its logarithmic derivative as a sum of reciprocals away from its roots. For real-rooted problems, differentiate this sum or complete it to a sum of squares. For unit-circle roots, combine terms into $(z+\alpha)/(z-\alpha)$ and inspect real parts. A common sign rules out cancellation. Revisit all original roots separately because the quotient is undefined there. If seeking equality, track equality in each square or inequality; a conclusion about equality at nonroots need not describe equality at multiple roots.

Extended atlas M-ANT-07

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1995-D2-P3 [R]; 1998-D1-P5 [R]; 2007-D2-P6 [R]

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M-AL-INTERPOLATE Turn nodal bounds into a sharp linear functional

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Prerequisites Interpolation, divided differences, and extremal signs [T-AL-INTERPOLATION]

Choose the distinct interpolation nodes supplied by the problem and build their Lagrange basis. Express the desired derivative or evaluation as a linear combination of the nodal values. Bound by the sum of the absolute weights, then choose signs matching those weights to demonstrate attainability. If the target point is variable, first reduce its optimization to endpoints or another justified finite set. Finally identify all equality cases, including weights that vanish. For integer-value questions, switch to the binomial basis rather than assuming integer monomial coefficients.

Extended atlas M-ANT-04

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1998-D2-P2 [R]; 2011-D2-P4 [R]

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M-AL-MODULAR Choose a modulus that exposes the obstruction

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]

Inspect the target prime powers or the parity and choose a small informative modulus. Reduce every condition explicitly, normalizing a nonzero variable by its inverse only when that inverse exists. Enumerate power residues, use a finite-field root count, or use Frobenius to simplify large exponents. If a global representation misses a residue, the whole arithmetic progression is excluded. For a lifting argument, expand $P(x+tp^k)$ to the precision required and check that the linear coefficient is invertible before claiming a unique lift. Treat exceptional primes separately.

Extended atlas M-ANT-01; M-ANT-09

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2006-D1-P2 [R]; 2007-D1-P1 [R]; 2007-D2-P2 [R]; 2009-D2-P4 [R]; 2010-D1-P4 [R]; 2011-D1-P3 [R]; 2017-D1-P3 [R]; 2020-D2-P6 [R]; 2022-D1-P3 [R]; 2022-D2-P6 [R]; 2023-D1-P4 [R]; 2024-D2-P10 [R]

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M-AL-CRT Construct and count compatible local residues

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Prerequisites Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Prime valuations, divisors, and factorials [T-AL-VALUATION]

Factor the modulus into prime powers and solve the condition locally. Verify independence by coprimality, then invoke the Chinese remainder theorem to combine exactly one residue for each local choice. Count residue classes first and impose positivity or interval restrictions afterward. For several polynomial values, assign each prime power to an input where its required exponent is largest, prescribe that input modulo that prime power, and combine the prescriptions. This avoids imposing incompatible congruences modulo overlapping polynomial values. Zero values must be isolated before taking lcms or products.

Extended atlas M-ANT-02

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2006-D1-P2 [R]; 2008-D1-P3 [R]; 2013-D2-P2 [R]; 2018-D1-P5 [R]

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M-AL-VALUATION Track one prime or one extreme divisor

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Prerequisites Prime valuations, divisors, and factorials [T-AL-VALUATION]

Select a prime forced by the expression, often the smallest prime factor or a prime absent from every other factor. Apply its valuation to products and quotients, preserving the sign of exponents. For sums, first check whether the least valuation occurs uniquely; only then can it be read off without cancellation. Use parity of valuations for squares and Legendre's formula for factorials. A bound on the smallest prime may also force a specific neighboring prime or an impossible divisor sum. Complete the argument for every required prime, and restore coprimality and zero cases.

Extended atlas M-ANT-03

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2003-D1-P4 [R]; 2006-D1-P3 [R]; 2012-D2-P3 [R]; 2014-D1-P4 [R]; 2015-D1-P4 [R]; 2019-D2-P7 [R]; 2023-D2-P10 [R]

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M-AL-COEFFICIENT Compare coefficients after a structural substitution

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Prerequisites Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Polynomial factorization and coprime divisibility [T-AL-FACTORIZATION]

Choose a substitution that simplifies the operation in the problem: center a quadratic, replace complex multiplication by variables $x + iy$ and $x - iy$, or remove a known line of tangency. Compare the highest and lowest coefficients first, then only the coefficients that constrain the remaining parameters. For a polynomial identity on an infinite field, justify coefficient comparison by the root bound. In inequalities, preserve nonnegativity under each factor multiplication. A finite check of candidate coefficients is the final verification, not a substitute for excluding higher-degree possibilities.

Extended atlas M-ANT-16

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2000-D1-P2 [R]; 2000-D2-P4 [R]; 2001-D2-P1 [R]; 2003-D1-P2 [R]; 2003-D1-P6 [R]; 2005-D2-P1 [R]; 2011-D2-P4 [R]; 2012-D2-P1 [R]; 2013-D2-P1 [R]; 2014-D1-P3 [R]; 2015-D1-P2 [R]; 2020-D1-P4 [R]; 2022-D2-P6 [R]; 2023-D1-P3 [R]; 2024-D1-P1 [R]

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M-AL-GENERATORS Prove generation or obstruct it by an image

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Prerequisites Groups, orders, generators, and quotients [T-AL-GROUP]; Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]

Describe generated elements as words, then produce useful short words: conjugates, powers, transpositions, or parameter sums. To prove equality of subgroups, establish both inclusions. To disprove finite generation, find a quantity bounded for words in any finite proposed generating set, such as allowed denominators. To disprove generation of a finite group, pass to a quotient where the images cannot generate, or compare preserved element orders under a proposed embedding. Prove normality before forming a quotient and keep a consistent composition order in all explicit permutation calculations.

Extended atlas M-ANT-10; M-ANT-12

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1996-D2-P3 [R]; 1998-D1-P2 [R]; 2001-D1-P2 [R]; 2005-D1-P6 [R]; 2009-D2-P4 [R]; 2012-D1-P3 [R]; 2018-D1-P2 [R]; 2021-D2-P6 [R]; 2024-D1-P4 [R]

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M-AL-ACTION Convert group data into orbits or cosets

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Prerequisites Group actions, stabilizers, and semidirect products [T-AL-ACTION]

Specify the acted-on set and the action, often cosets or a product of two coset sets. Compute the stabilizer at an explicit base point and use orbit–stabilizer. Count fixed pairs in two orders, or show that a complement of a large orbit has exactly one point in each row and column and therefore induces an equivariant bijection. Conclude conjugacy from stabilizers in the same transitive action. For examples, prescribe a semidirect product action and bound automorphisms through images of proven generators; arbitrary candidate images give only an upper bound.

Extended atlas M-ANT-10; M-ANT-14

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2007-D1-P4 [R]; 2008-D1-P5 [R]; 2010-D2-P3 [R]; 2020-D2-P7 [R]

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M-AL-COMMUTATOR Polarize identities and isolate commutators

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Prerequisites Noncommutative rings, polarization, and torsion [T-AL-RING]; Groups, orders, generators, and quotients [T-AL-GROUP]

Expand a universal identity after replacing an element by a sum or product. Keep multiplication in its original order. Identify the mixed term as a commutator or anticommutator and use the same identity again on products. In groups, place a commutator in two known abelian normal subgroups to make it central in the subgroup they generate; centrality there permits controlled power formulas. Apply Bézout to resulting coprime exponent restrictions. In rings, do not cancel a factor such as 2 or 3 until the additive torsion hypothesis has been checked explicitly.

Extended atlas M-ANT-11

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1999-D2-P1 [R]; 2000-D1-P5 [R]; 2005-D1-P6 [R]

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M-AL-ORBIT Work orbit by orbit and track parity

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Prerequisites Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]; Groups, orders, generators, and quotients [T-AL-GROUP]

Decompose a permutation or bijection into its invariant orbits. Replace a finite orbit by residues modulo its length and an infinite orbit by the integer shift. Build the desired maps in those coordinates and join them across disjoint orbits. For a finite permutation equation, apply the sign homomorphism or compare cycle lengths and orders before attempting a pointwise construction. A fixed-point-free involution pairs every element, so its number of transpositions is known exactly. Explicitly justify any choice of orbit representatives when the underlying set is arbitrary.

Extended atlas M-ANT-10

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1997-D2-P5 [R]; 2003-D1-P2 [R]; 2012-D1-P3 [R]; 2016-D2-P8 [R]

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M-AL-RECURRENCE Find a recurrence that reveals the invariant

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Prerequisites Linear recurrences and generating functions [T-AL-RECURRENCE]

Write the first few terms only to identify a plausible recurrence or generating function. Then derive it algebraically by shifting indices, comparing coefficients, or using the characteristic equation. To prove integrality, replace a recurrence with denominators by an equivalent convolution recurrence with integer coefficients. To evaluate a series, seek a finite telescoping identity and show its remainder tends to zero; nonnegative double sums may instead be reordered with justification. When controlling residues, consider the entire recurrence state and establish invertibility if return to the initial state is required.

Extended atlas M-ANT-16

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1996-D2-P2 [R]; 2002-D1-P3 [R]; 2002-D2-P3 [R]; 2005-D2-P6 [R]; 2008-D2-P3 [R]; 2015-D1-P3 [R]; 2017-D1-P3 [R]; 2019-D1-P4 [R]; 2022-D1-P3 [R]

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M-AL-SHIFT Take the gcd of annihilating shift polynomials

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Prerequisites Finite differences and shift-operator ideals [T-AL-DIFFERENCE]; Roots of unity, cyclotomic polynomials, and algebraic norms [T-AL-CYCLOTOMIC]

Encode a finite linear relation among translates as $A(T_h)f = 0$. Fix the step and collect all such polynomials into an ideal. Use the Euclidean algorithm to replace them by their gcd, exploiting the special structure of their roots, often roots of unity. Establish that both endpoint coefficients of the resulting recurrence are nonzero. A short recurrence can propagate known zero values from an interval; a constant gcd forces every value to vanish immediately. If original exponents are negative, first translate all indices or work explicitly with Laurent polynomials.

Extended atlas M-ANT-05; M-ANT-08

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2007-D1-P5 [R]; 2010-D2-P5 [R]

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M-AL-UNITY Evaluate at roots of unity or average over them

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Prerequisites Roots of unity, cyclotomic polynomials, and algebraic norms [T-AL-CYCLOTOMIC]; Roots-of-unity filters and positive trigonometric kernels [T-AL-FOURIER]

Decide whether the useful object is a single primitive root or the complete set of roots. For a single root, determine its order and invoke its minimal polynomial; take an algebraic norm when integer coefficients make a nonzero product at least one. For a complete set, use the root-of-unity filter to kill all unwanted exponents. Track equality in any modulus bound term by term. In geometry, encode closure as a polynomial relation first. Distinguish the original polynomial from its function on a finite cyclic set and check repeated roots in positive characteristic.

Extended atlas M-ANT-08

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1999-D2-P6 [R]; 2001-D1-P4 [R]; 2007-D1-P6 [R]; 2008-D2-P1 [R]; 2009-D1-P4 [R]; 2017-D1-P5 [R]; 2018-D1-P5 [R]; 2023-D1-P4 [R]

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M-AL-ANALYTIC Build a holomorphic function with a useful boundary

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Prerequisites Holomorphic bounds, poles, and logarithmic-branch sums [T-AL-COMPLEXANALYSIS]

Construct a function whose boundary modulus equals the expression to be bounded and whose interior value encodes the target. For a monic polynomial use its conjugate-reciprocal polynomial, preserving modulus on the unit circle and setting its value at zero to one. Apply Cauchy's formula or the maximum principle on a compact domain. For a series, first establish locally uniform convergence away from singularities, then differentiate or identify poles. A strip argument needs uniform bounds on the extra sides of a large rectangle. Record every normalization constant when differentiating branch sums.

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2004-D1-P6 [R]; 2009-D1-P4 [R]; 2015-D2-P10 [R]; 2020-D1-P4 [R]; 2021-D2-P7 [R]

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M-AL-RATIONAL Eliminate rational relations, then exploit discreteness

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Prerequisites Rational linear relations and discrete-value rigidity [T-AL-RATIONALITY]

Normalize one nonzero coefficient in each rational relation and substitute to eliminate the newly introduced elements. Preserve signs if the argument concerns nonnegative combinations; at least one surviving coefficient must remain positive. For rational functions taking integer values, use rational linear equations to recover rational coefficients, divide polynomially, and clear a common denominator. The proper-fraction remainder tends to zero along an unbounded integer set, yet is integer-valued there, so it eventually vanishes. Use infinitely many roots to finish. Do not replace an unbounded set by a one-sided limit without checking its direction.

Extended atlas M-ANT-15

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P3 [R]; 1996-D2-P2 [R]; 2006-D1-P4 [R]

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M-AL-DESCENT Replace a minimal algebraic counterexample by a smaller one

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Prerequisites Polynomial factorization and coprime divisibility [T-AL-FACTORIZATION]; Prime valuations, divisors, and factorials [T-AL-VALUATION]

Assume a counterexample and minimize a positive integer measure, such as a sum of degrees, a denominator, or the largest coordinate of a primitive triple. Use a factorization or the other root of a quadratic equation to construct another admissible object. Verify all conditions before comparing sizes: integrality, positivity, coprimality, nontriviality, and monicity as appropriate. Then prove strict decrease of the selected measure. In polynomial Vieta jumping, $(Q^2+1)/P$ is a polynomial only because the original divisibility holds; constant cases and leading coefficients must be checked independently.

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2012-D1-P5 [R]; 2018-D2-P9 [R]

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M-AL-GREEDY Choose the next factor and control the entire tail

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Prerequisites Greedy products, tail bounds, and integer descent [T-AL-GREEDY]

Define the remaining ratio explicitly. Choose the smallest allowed denominator making the next factor strictly smaller than that ratio; convert minimality into two-sided inequalities. Derive the growth condition on successive denominators and prove the remaining ratio tends to one. For uniqueness, compare with the largest possible tail after the first discrepancy. For rationality, propagate integer numerator and denominator pairs and exhibit a positive integer that strictly decreases unless the extremal tail has been reached. A local greedy choice is insufficient without a global tail bound.

Extended atlas M-ANT-17

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1997-D1-P4 [R]

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M-AL-SIEVE Construct by excluding a sparse set of bad choices

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Prerequisites Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Prime valuations, divisors, and factorials [T-AL-VALUATION]

Extend a finite admissible configuration by choosing a new element in an arithmetic progression. Arrange the progression to preserve all small-prime conditions. For each remaining forbidden prime square and each old element, count the bad parameters among $1,\dots,N$: one residue class contributes at most N/p^2+1. The monitored linear expressions are positive and $O(N)$, so a prime square dividing one has $p=O(\sqrt{N})$. Sum these upper bounds even though bad sets may overlap. Use a convergent reciprocal-square tail and an $O(\sqrt{N})$ error to show fewer than N choices are bad. Choose a sufficiently large surviving parameter so the sequence increases, then repeat the induction.

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2013-D2-P4 [R]

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M-AL-PARAMETRIZE Construct roots or units before the coefficients

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Prerequisites Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]; Quadratic norms and Pell-type constructions [T-AL-PELL]; Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]

Express the desired object through variables with a built-in relation: roots summing to zero for a cubic, powers of a quadratic unit for a Pell equation, or inverses of all nonzero finite-field elements for a permutation. Compute the induced coefficients and constraints exactly. Introduce a scale only after determining which denominators it must clear. To prove infinitely many admissible objects, retain a free parameter and show distinctness and coprimality for the chosen parameter family. Verify the original equation and all exceptional cases, rather than assuming the parameterization is automatically sufficient.

Extended atlas M-ANT-13

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2005-D2-P6 [R]; 2006-D2-P5 [R]; 2022-D2-P6 [R]

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Counting, geometry and functional equations

M-DS-DOUBLECOUNT Count incidences with their exact multiplicity

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Prerequisites Finite counting and incidence identities [T-DS-COUNTING]; Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]

Specify a finite incidence set, such as $(\text{object}, \text{marked feature})$, before summing. Count it by fixing the object and then by fixing the feature. For a weighted identity, assign the same weight to each incidence in both counts. For an inequality, state whether one count is exact or merely an upper bound.

To force a good object, count the bad ones: if each bad object has at least one witness and the total number of witness incidences is smaller than the number of all objects, some object has no witness. For graph problems, decide whether two-step walks may return to their starting vertex. The formulas $\sum d(v)^2$, $\sum d(v)(d(v)-1)$, and $\sum \binom{d(v)}{2}$ count different objects.

Finish by checking equality: every overcount or inequality must be sharp simultaneously. This can force unique common neighbours, forbidden short cycles, or an exact structural classification. Do not discard this information when the hypothesis itself gives equality.

Extended atlas M-CMB-02

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2002-D2-P2 [R]; 2004-D1-P4 [R]; 2009-D1-P3 [R]; 2011-D1-P4 [R]; 2012-D2-P5 [R]; 2013-D1-P3 [R]; 2015-D2-P8 [R]; 2016-D1-P4 [R]; 2022-D2-P5 [R]; 2023-D1-P5 [R]

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M-DS-BIJECTION Encode, decode, and verify a bijection

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Prerequisites Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]

Define a map from the counted objects to a simpler family, including all parameters needed to reconstruct the original. Verify that every image is admissible. Then write an inverse and check both compositions. Surjectivity without uniqueness proves only an upper or lower bound, not equality.

For a word over $\mathbb{Z}/3\mathbb{Z}$ with initial letter fixed as $x_0 = 0$, the difference word $y_i = x_i - x_{i-1}$ is inverted by $x_i = \sum_{j=1}^{i} y_j$. A local equality of three consecutive letters becomes two consecutive zero differences. This translates a word restriction into a different, simpler restriction.

For subsets, paths, partitions, or permutations, explicitly handle empty objects and endpoints. When forgetting labels or order, prove every fibre has the same size before dividing a count. If some objects have extra symmetry, uniform division generally fails; count their stabilizers or use an orbit argument.

Extended atlas M-CMB-10

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2005-D1-P2 [R]; 2012-D1-P1 [R]; 2014-D2-P1 [R]; 2014-D2-P5 [R]; 2018-D2-P8 [R]; 2020-D1-P1 [R]; 2024-D2-P9 [R]

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M-DS-INDUCT Strengthen the statement before the induction step

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Prerequisites Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]

Choose the decreasing parameter and write precisely what survives deletion or refinement. If a square grid becomes rectangular after removing a row, formulate the inductive statement for arbitrary rectangles. If two set families lose different-sized traces, induct on the sum of their uniform set sizes. Record all necessary base cases.

For constructive induction, make the induction hypothesis include the quantitative invariant used to select the next piece. For example, a zero-sum vector ordering can maintain $\|s_j\|^2 \le \sum_{i\le j}\|v_{\pi(i)}\|^2$; the next vector is chosen with $\langle s_j, v\rangle \le 0$.

After applying the smaller case, verify the lifting operation one condition at a time. In a triangulation step adding three vertices, enumerate the new triangles incident at every old attachment vertex; adding two preserves parity. A schematic drawing helps check incidences, but does not replace that count. Distinguish induction by one from induction by a fixed residue-class step.

Extended atlas M-CMB-08; M-CMB-11

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D2-P5 [R]; 1996-D1-P6 [R]; 1997-D1-P6 [R]; 1998-D1-P3 [R]; 1998-D2-P3 [R]; 1999-D1-P5 [R]; 1999-D2-P5 [R]; 2000-D2-P1 [R]; 2001-D2-P3 [R]; 2004-D1-P2 [R]; 2004-D1-P5 [R]; 2005-D1-P2 [R]; 2006-D2-P1 [R]; 2009-D1-P5 [R]; 2010-D1-P3 [R]; 2012-D2-P5 [R]; 2013-D2-P5 [R]; 2014-D2-P5 [R]; 2016-D1-P4 [R]; 2016-D2-P9 [R]; 2018-D2-P7 [R]; 2018-D2-P8 [R]; 2020-D1-P1 [R]; 2021-D1-P3 [R]; 2024-D2-P8 [R]

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M-DS-STATE Partition into states and propagate a recurrence

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Prerequisites Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Finite counting and incidence identities [T-DS-COUNTING]

Read the forbidden local pattern and retain the shortest suffix needed to decide which next symbols are legal. Partition admissible objects into disjoint states and tabulate all transitions, with multiplicities. If u_n is the state-count column vector, establish $u_{n+1} = Au_n$ from explicit extension/deletion maps.

To prove a counting identity, derive recurrences for both sides and match enough initial values; solving the recurrence is optional. For parity-constrained words, states may be the parity vector of selected letter counts. Appending a letter either flips one coordinate or leaves the state unchanged. Check the empty word separately when beginning at $n = 0$.

The state must contain enough information: total length alone does not determine whether the next symbol creates a repeated block. Conversely, unnecessary memory makes the recurrence harder to solve. If transitions depend on position, use A_n rather than silently assuming a constant matrix.

Extended atlas M-CMB-05

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2005-D1-P2 [R]; 2013-D2-P5 [R]; 2020-D1-P1 [R]

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M-DS-GRAPH Encode a relation as a graph and extract a cycle

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Prerequisites Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]

Choose vertices for the entities and edges for the relation. In a marked matrix use one class for rows and another for columns; the marked cells themselves are edges, not vertices. Translate the required configuration into an ordinary graph property.

For a cycle, either compare edges with vertices using the forest formula or repeatedly delete vertices of degree at most one. Show the numerical invariant prevents the entire graph from disappearing. The remaining nonempty graph has minimum degree at least two, so a longest path produces a cycle. In a bipartite graph the cycle is automatically even.

Translate the cycle back and verify distinctness of the requested objects. An arbitrary closed walk may repeat vertices and edges; trim repetitions or use a simple cycle from the start. For girth arguments, take a shortest cycle before introducing a shortcut, and check that the shortcut really produces a shorter simple cycle rather than merely a shorter closed walk.

Extended atlas M-CMB-03; M-CMB-04

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1999-D1-P5 [R]; 2009-D1-P3 [R]; 2015-D1-P5 [R]; 2022-D1-P4 [R]; 2022-D2-P5 [R]; 2024-D2-P9 [R]

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M-DS-FUNCTIONAL Force structure by admissible substitutions

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Prerequisites Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]; Dominated convergence and orthonormal separation [T-AN-MEASURE]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]

Start with the domain and range, then substitute equal variables, fixed values, or arguments chosen to make two inner expressions coincide. Use positivity to justify division. Derive monotonicity, injectivity, or a translation identity before relying on it.

To compare $f(u)$ and $f(v)$, solve for auxiliary variables making them the two arguments in a known estimate. For instance $x - ny = u$ and $x + ny = v$, for $n \geq 1$, are achieved by $x = (u+v)/2$ and $y = (v-u)/(2n)$. A uniform bound tending to zero then proves equality for the fixed pair u, v.

For classification, separate exceptional constant or zero cases before cancelling factors. Establish a candidate on a dense subset only when continuity allows extension. Finally substitute every remaining parameterized candidate into the original equation and range conditions. Do not differentiate merely because a closed-form answer appears smooth; prove regularity or cite the applicable monotone differentiability theorem and justify the subsequent bootstrap.

A regularity checkpoint for the positive equation $f(x)f(yf(x)) = f(x+y)$ is useful. Once decrease is known, take $y \downarrow 0$ at one continuity point to get $f(0+) = 1$. For $t \uparrow x$, the same identity gives $f(x) = f(t)f((x-t)f(t))$, proving left continuity; right continuity follows directly. At one differentiability point, obtain the finite limit $\lim_{z \downarrow 0}(f(z)-1)/z = -a$. The positive-increment quotient is then $f(x)^2$ times this quotient at $z = hf(x)$. For negative increments use the displayed identity with $t = x + h$; continuity gives the same limit $-af(x)^2$. This supplies an ordinary two-sided derivative everywhere before solving the reciprocal-linear differential equation.

Extended atlas M-FE-02; M-FE-05

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1998-D2-P4 [R]; 1999-D1-P3 [R]; 1999-D1-P4 [R]; 1999-D2-P4 [R]; 2000-D1-P1 [R]; 2000-D2-P5 [R]; 2001-D2-P5 [R]; 2002-D1-P2 [R]; 2006-D2-P2 [R]; 2008-D1-P1 [R]; 2018-D1-P4 [R]; 2023-D1-P1 [R]

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M-DS-ITERATE Trap an orbit and telescope its transformed recurrence

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Prerequisites Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]

Find an invariant domain for the iteration and check that all denominators remain nonzero. Factor $F(x) - x$ to identify monotone regions and possible limiting fixed points. Prove convergence or divergence before substituting a limiting value.

Search for a transform h turning the recurrence into $h(x_{n+1}) = a_n h(x_n)$ or $h(x_{n+1}) - h(x_n) = b_n$. Products or sums then telescope. For $x_{n+1} = x_n^2 - 2$, setting $y_n = x_n^2$ gives $y_{n+1} - 4 = y_n(y_n - 4)$; this controls the desired product. Check the initial factor and the sign before taking square roots.

When counting iterated preimages, partition target values into ranges on which each inverse branch exists. Prove the branches are disjoint and count distinct roots, not multiplicity. For periodic points, subtract or bound smaller-period points only after distinguishing least period from a solution of $F^n(x) = x$.

Extended atlas M-FE-01; M-FE-03; M-FE-06

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1998-D1-P3 [R]; 1998-D2-P3 [R]; 1999-D1-P4 [R]; 1999-D2-P4 [R]; 2000-D1-P1 [R]; 2001-D2-P5 [R]; 2002-D1-P4 [R]; 2004-D1-P2 [R]; 2010-D1-P3 [R]; 2017-D2-P9 [R]; 2018-D2-P7 [R]; 2023-D1-P1 [R]; 2024-D2-P8 [R]

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M-DS-GEOMETRIC Normalize coordinates and linearize squared distances

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Prerequisites Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Conics, reflections, and inversion [T-DS-TRANSFORMS]; Inner products, orthogonality and polarization [T-LA-INNER]

Use translations and orthogonal coordinates to put a distinguished point at the origin or a distinguished line on an axis. State which quantities the normalization preserves; an invertible affine map preserves incidence and ratios on a line, but not angles or Euclidean distance.

Write distances as norms or inner products. Equal-distance equations become linear after subtracting their squares. In a simplex, express an interior point using barycentric coordinates and replace the coefficients by opposite sub-simplex volume ratios. For a sphere configuration, normalize radius and use $\|u - v\|^2 = 2 - 2\langle u, v\rangle$.

For a quadratic region, complete squares, identify each semiaxis, and multiply the unit-ball volume by the absolute determinant of the scaling. Squaring an inequality is equivalent only after both sides are known nonnegative. Keep degenerate cases separate: a zero-dimensional simplex, a coincident point, or a zero denominator can invalidate an otherwise correct calculation.

Extended atlas M-GEO-01; M-GEO-03; M-GEO-04; M-GEO-11

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1996-D1-P6 [R]; 1996-D2-P4 [R]; 1997-D1-P5 [R]; 1998-D2-P5 [R]; 2001-D2-P3 [R]; 2002-D1-P1 [R]; 2002-D2-P4 [R]; 2004-D1-P4 [R]; 2004-D2-P3 [R]; 2005-D2-P5 [R]; 2006-D2-P1 [R]; 2008-D1-P4 [R]; 2008-D2-P2 [R]; 2009-D1-P5 [R]; 2009-D2-P1 [R]; 2011-D2-P5 [R]; 2014-D1-P5 [R]; 2014-D2-P4 [R]; 2015-D1-P5 [R]; 2017-D2-P10 [R]; 2020-D1-P3 [R]; 2023-D2-P9 [R]

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M-DS-PERTURB Contradict an extremum by a controlled perturbation

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Prerequisites Convex bodies, supporting functionals, and packing [T-DS-CONVEX]; Continuity, compactness, and order [T-AN-CONTINUITY]; Lipschitz derivatives and convex smoothness [T-AN-SMOOTH]

Assume an extremizer lacks the required contact or structure. Isolate a compact portion lying strictly inside the feasible region and obtain a positive uniform clearance from the boundary. Define an explicit one-parameter deformation that improves the objective while leaving the other constraints unchanged or easier to satisfy.

For an inscribed ellipse, choose its semiaxes so designated contact points remain on the ellipse. Compare the area product exactly; a first-order calculation may miss an improvement that begins at second order. Split the new ellipse into the part already inside the original disk and the outward part controlled by the clearance. Only then select a small enough parameter.

At a constrained smooth maximum, first exclude an interior critical point using a nonzero gradient. On a sphere, the gradient is a scalar multiple of the outward normal; its scalar has the appropriate nonnegative sign for a maximum over the closed ball. In the case $\|\nabla f(0)\| = 1$, a 1-Lipschitz gradient and radius $0 < r \le 1/2$, compare the outward radial gradients to contradict the Lipschitz estimate if two distinct maximizers existed. A Lipschitz gradient by itself does not guarantee uniqueness. Existence of the maximum comes from compactness.

Extended atlas M-GEO-06

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1996-D2-P4 [R]; 2005-D2-P5 [R]

Back to previous view | Theory index | Method index | Strategy index

M-DS-METRIC Push covers forward or move toward a nearest point

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Metric contractions, covers, and nearest points [T-DS-METRIC]

For a lower bound on a covering cost, map the set onto a simpler object by a nonexpansive map, commonly an orthogonal projection onto a segment. Intersect covering sets with the map's domain, push them forward, and compare each image diameter with the original one. Taking an infimum preserves the direction of the inequality; an arbitrary cover should be used until the last step.

For nearest-point uniqueness, choose a distance minimizer using compactness of a bounded closed slice of the set. Move the observation point a positive fraction toward that minimizer. The smaller closed ball is internally tangent to the original distance ball at only the chosen point, so a second minimizer is impossible. Let the fraction tend to zero to obtain density.

Record nonemptiness of the closed set and nondegeneracy of the displacement. In finite-cover content, neither the cover members nor their images need be open. Do not import measure additivity without proving the exact interval-cover or projection estimate needed.

Extended atlas M-GEO-07; M-GEO-10

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1996-D1-P6 [R]; 2003-D2-P3 [R]; 2006-D2-P2 [R]

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M-DS-EXTREMAL Repair an extremal configuration by a local exchange

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Extremal choice, exchange arguments, and invariants [T-DS-EXTREMAL]; Finite counting and incidence identities [T-DS-COUNTING]

Choose a configuration minimizing a nonnegative integer defect count, such as incompatible pairs, inversions, or uncovered requirements. Compute the exact change under a local exchange. Unchanged terms should be identified explicitly; controlling only one newly improved term can miss several newly created defects.

M-DS-EXTREMAL Repair an extremal configuration by a local exchange (continued)

If every defective configuration admits a strict improvement, a minimum has defect zero. Alternatively, repeated repairs terminate because the defect count cannot decrease indefinitely. To prove an exchange exists, count the forbidden choices and show their union does not exhaust all candidates. Overlap between forbidden sets can be decisive at the threshold.
For a sharp lower bound, add a construction and verify all its constraints. Minimal by inclusion, minimum size, and minimum defect are different notions. Select the one that yields the local property actually used later. For an infinite family, prove attainment or use a terminating discrete potential instead of assuming an optimizer exists.
Extended atlas M-CMB-08

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2008-D1-P4 [R]; 2011-D2-P2 [R]; 2012-D2-P5 [R]; 2014-D1-P5 [R]; 2014-D2-P4 [R]; 2016-D1-P4 [R]; 2020-D1-P3 [R]; 2023-D1-P5 [R]; 2024-D2-P6 [R]

Back to previous view | Theory index | Method index | Strategy index

M-DS-INVOLUTION Pair objects canonically to prove parity

Back to previous view | Theory index | Method index | Strategy index
Prerequisites Extremal choice, exchange arguments, and invariants [T-DS-EXTREMAL]; Finite counting and incidence identities [T-DS-COUNTING]
Choose a reversible local change and a rule that selects where to apply it. Prove four facts: the changed object remains admissible; the selected feature is unchanged by the operation; applying the operation twice returns the original; and the fixed points are exactly those claimed.
A fixed-point-free involution proves the number of objects even. More generally, all two-element orbits contribute zero modulo two, so the parity is that of the fixed-point set. If a second involution acts on that smaller set, repeat the argument there. For sign-reversing sums, show the two objects have opposite weights of equal magnitude.
In an ordered array, changing the first nontrivial block can preserve the choice of block only if all earlier blocks remain identical and the chosen one stays nontrivial. Monotonicity across its boundary must be checked, not only inside the block. Swapping entries without a canonical selection rule is not yet a pairing.
Extended atlas M-CMB-07

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2013-D2-P5 [R]

Back to previous view | Theory index | Method index | Strategy index

M-DS-PARTITION Split a family by traces, pieces, or labels

Back to previous view | Theory index | Method index | Strategy index
Prerequisites Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Finite counting and incidence identities [T-DS-COUNTING]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Choose a finite partition that turns a global condition into smaller conditions: traces on a fixed set, half-intervals between extreme elements, or pieces of a geometric dissection. Verify the classes are disjoint and exhaustive before adding their sizes.
In an intersecting-family induction, fix a finite reference set and group members by their intersections with it. If two traces already meet, that intersection supplies a witness. If they are disjoint, remove both nonempty traces and invoke the smaller-size hypothesis on the residual families. A finite union of the finitely many resulting witnesses remains finite.
For a cube dissection, refining one cube into a^d cubes increases the piece count by a^d-1. To reach every sufficiently large count, find two coprime increments u, v. Every $m \geq (u-1)v$ has the form $ru+sv$ with $r, s \geq 0$: choose $0 \leq s < u$ in the required residue class modulo u. Account for the initial piece count separately.

M-DS-PARTITION Split a family by traces, pieces, or labels (continued)

Extended atlas M-CMB-09

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1997-D1-P5 [R]; 1997-D1-P6 [R]; 2000-D2-P1 [R]; 2004-D1-P1 [R]; 2004-D1-P5 [R]; 2008-D1-P4 [R]; 2011-D2-P2 [R]; 2012-D1-P1 [R]; 2013-D1-P3 [R]; 2013-D2-P5 [R]; 2016-D1-P4 [R]; 2016-D2-P9 [R]; 2017-D2-P10 [R]; 2020-D1-P1 [R]; 2021-D1-P3 [R]; 2023-D2-P8 [R]; 2023-D2-P9 [R]; 2024-D2-P6 [R]

Back to previous view | Theory index | Method index | Strategy index

M-DS-MATCHING Match classes in stages and verify Hall each time

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Prerequisites Hall matching and augmenting paths [T-DS-MATCHING]; Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]

Translate permitted pairings into a finite bipartite graph. For each subset S on the side to be matched, estimate its whole neighbourhood $N(S)$; individual vertex degrees alone are not Hall's condition. Frequently split into small and large S, using a vertex outside $N(S)$ to contradict a degree lower bound in the large case.

For three classes, first construct a matching between two of them. Form an auxiliary graph whose vertices on one side are these matched pairs. Connect a pair to a member of the third class only when both compatibility requirements hold. Bound common neighbours by inclusion–exclusion and verify Hall again.

The first matching need not be unique. Prove the second-stage estimates hold for every possible first matching or explicitly select a matching with a needed extremal property. Finish by translating the two matchings into disjoint triples; this verifies both compatibility and that no original object is used twice.

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2011-D2-P2 [R]

Back to previous view | Theory index | Method index | Strategy index

M-DS-REWRITE Reduce equivalent words with a congruence

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Extremal choice, exchange arguments, and invariants [T-DS-EXTREMAL]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]

List the permitted relations and prove they may be used inside arbitrary left and right contexts. Establish a short absorption or extension lemma before attempting a long word. Split the word into controlled prefix, middle, and suffix, and use the lemma to remove the middle modulo the relations.

To prove every word has a representative of bounded length, it suffices to show every longer word has a strictly shorter equivalent representative. Intermediate expansions are allowed; the final length is the decreasing integer used for induction. An idempotence relation $uu \sim u$ does not imply commutativity, so letters cannot be reordered without proof.

When checking examples, retain the parentheses showing which repeated subword is replaced. Local inability to shorten does not prove a word minimal: a temporary expansion can create a different repeated block. This distinction is crucial when a naive greedy reduction becomes stuck.

Extended atlas M-CMB-15

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1999-D2-P5 [R]

Back to previous view | Theory index | Method index | Strategy index

M-DS-EXPECTATION Random selection and the expected number of successes

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Finite probability, conditioning, and expectations [T-AN-PROBABILITY]; Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]

Trigger. An existence problem asks for a subfamily or vertex set with many local successes, while direct greedy choices interfere with one another. Introduce independent Bernoulli choices with a tunable probability p.

Operation. Specify the successful event E_i at each object and let $X = \sum_i I_{E_i}$ count successes. Calculate each $\mathbb{P}(E_i)$, sum by linearity, optimize in p, and conclude that some outcome has $X \geq \mathbb{E}X$. For a d-regular simple graph on n vertices, if success means a selected vertex with exactly r selected neighbours, then

$$\mathbb{E}X = n\binom{d}{r}p^{r+1}(1-p)^{d-r}.$$

For $0 \leq r < d$, logarithmic differentiation gives the maximizing value $p = (r+1)/(d+1)$. With $d = 1000, r = 2$, this is the parameter used in the IMC construction.

Proof checkpoint. The success indicators overlap and need not be independent. Independence is required only for multiplying the selection probabilities within one local event. Name X explicitly: it is generally different from the selected-set size. If an exact-degree event is involved, it is not monotone under adding vertices; do not assume that a coupling which enlarges the set can only improve the count. The final inference is existence from the average, not a probability-of-success estimate.

Extended atlas M-CMB-13

Reference notes 12

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2017-D1-P4 [R]

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M-DS-GENERATING Encode the statistic and extract the required coefficient

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Generating functions, coefficient filters, and finite differences [T-DS-GENERATING]; Finite counting and incidence identities [T-DS-COUNTING]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]

Assign each object a monomial whose exponents record the relevant statistics. Multiply factors only for independent choices, and sum over mutually exclusive cases. State whether the series is formal or analytically convergent. For partitions, a factor $(1-z^j)^{-1}$ permits any number of parts of size j; deleting it forbids that part.

To select exponents in a residue class modulo m, average over all m roots of unity with the correct phase factor and the normalization $1/m$. For parity, the evaluations at 1 and -1 suffice; independent parity constraints require independent sign averages. A determinant expansion can record signed permutations; modulo two its signs disappear, so a determinant coefficient becomes a parity count. If an alternating binomial sum contains a polynomial of degree d, compare it with the $(d+1)$st finite difference. Match the sign convention and the shift before concluding it vanishes. Finally translate the extracted coefficient back to the original labelled objects, including boundary and empty cases.

For asymptotic sign comparisons after a root filter, combine conjugate terms before estimating. If the leading pair has real contribution L_n, prove $\sum |\text{remaining terms}| < |L_n|$. A largest complex modulus alone does not determine the real sign, since the leading pair may nearly cancel. Bound its trigonometric phase away from zero on the actual integer subsequence used.

Extended atlas M-CMB-06; M-CMB-12

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2008-D1-P6 [R]; 2012-D1-P1 [R]; 2015-D2-P8 [R]; 2016-D1-P5 [R]; 2016-D2-P9 [R]; 2020-D1-P1 [R]

Back to previous view | Theory index | Method index | Strategy index

M-DS-ENCODE Choose information that makes an encoding injective

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Finite counting and incidence identities [T-DS-COUNTING]; Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]

Replace an object by a word, vector, subset, or graph label that captures the obstruction. To prove a count bound, exhibit the map and recover the original object uniquely from its image. It is insufficient to count possible labels without proving that different objects receive different labels. For fixed-sum subset families, select one representative for each distinct subset-sum value. An equality of ternary difference vectors can be dotted with the weight vector to recover that value; uniqueness of the selected representative then recovers both subsets. Selecting distinct subsets with equal sums would invalidate this argument.

In a directed graph, an edge cover with no directed walk of length two (including a return) in any one class yields a binary vertex label: the set of classes of outgoing edges. The class of a given edge belongs to its source label and not its target label. For digit-diagonal arguments, choose a digit differing from the corresponding sequence digit, then check separately indices inside and beyond a finite prefix. Infinite decimal or binary expansions require a uniqueness convention; finite integer digits avoid that ambiguity.

Extended atlas M-CMB-01; M-CMB-10

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1998-D2-P4 [R]; 1998-D2-P5 [R]; 2001-D1-P1 [R]; 2003-D2-P4 [R]; 2011-D2-P5 [R]; 2014-D1-P5 [R]; 2014-D2-P1 [R]; 2016-D1-P5 [R]; 2019-D2-P8 [R]; 2022-D1-P4 [R]

Back to previous view | Theory index | Method index | Strategy index

M-DS-COUPLING Realize all inclusion probabilities on one probability space

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Finite probability, conditioning, and expectations [T-AN-PROBABILITY]; Finite counting and incidence identities [T-DS-COUNTING]

For a finite ground set Ω, choose independent uniform random variables U_x on $[0,1]$ and set $X_t = \{x : U_x \le t\}$. If $s \le t$, then $X_s \subseteq X_t$ for every outcome, not just on average. Therefore any increasing event E, meaning $A \in E$ and $A \subseteq B$ imply $B \in E$, has a nondecreasing probability $\mathbb{P}(X_t \in E)$.

The event that X_t contains at least one of fixed subsets $A_1, \dots, A_n$ is increasing. Independence gives $\mathbb{P}(C \subseteq X_t) = t^{|C|}$; applying inclusion–exclusion identifies its probability with the alternating union-size polynomial in the problem. The coupling proves its monotonicity without differentiating a complicated sum.

Check the event itself for monotonicity. The event that a chosen vertex has exactly two chosen neighbours is not increasing: adding a third neighbour destroys it. Such an event can still be treated by expectation, but this coupling does not establish monotonicity for it.

Extended atlas M-CMB-13

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2011-D1-P4 [R]

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M-DS-VOLTERRA Pass a monotone functional iteration through its integral operator

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]; Uniform approximation and positive kernels [T-AN-APPROXIMATION]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]

On a fixed compact interval $[0,b]$, define $Tg(x) = \exp(\int_0^x g(t)\,dt)$. For continuous g, h with sup norms at most M, the mean value theorem for the exponential gives

$$\|Tg - Th\|_\infty \le be^{bM}\|g - h\|_\infty.$$

M-DS-VOLTERRA Pass a monotone functional iteration through its integral operator (continued)

Indeed, each exponent has absolute value at most bM, while their difference is at most $b\|g-h\|_\infty$. Thus uniform convergence passes through T; the bound need not make T a contraction.
First prove pointwise monotonicity and an upper bound for the iterates. Establish continuity of the limit, for example from finite convexity on the open interval and a separate endpoint squeeze. Dini's theorem then upgrades monotone convergence of continuous functions to uniform convergence on each compact subinterval. Pass to the limit in $g_{n+1} = Tg_n$. Differentiating the resulting integral identity gives the limiting differential equation and its initial condition. Solve that equation only on its valid interval, and prove the claimed convergence or divergence at excluded endpoints separately.

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2017-D2-P9 [R]

Back to previous view | Theory index | Method index | Strategy index

M-DS-SPHERICAL Turn a spherical comparison into an endpoint integral

Back to previous view | Theory index | Method index | Strategy index

Prerequisites Spherical distance and signed dihedral derivatives [T-DS-SPHERICAL]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]

Normalize the three rays at a tetrahedron vertex to unit length. Parametrize the minor great-circle arc between two of them at unit speed and let $f(t)$ be its spherical distance to the third. Verify that noncoplanarity excludes the singular values $0, \pi$ and gives the strict estimate $|f'| < 1$.
Express the endpoint derivatives as signed cosines of the dihedral angles. Differentiate $\cos f = \langle c, X\rangle$ twice, using $X'' = -X$, to obtain $f'' = \cot f(1 - f'^2)$. The scalar inequality $u\cot u < 1$ for $0 < u < \pi$ then gives $(ff')' < 1$. Integrating yields the desired endpoint expression.
Keep track of the tangent orientation at the final endpoint: the forward travel tangent is the negative of the tangent pointing back along the arc. Confusing these signs reverses a dihedral term. Strictness requires the nondegenerate tetrahedron hypothesis, not merely distinct unit vectors.

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
2002-D2-P4 [R]

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17 Method Sheets and Solution-Move Atlas

This section is appended to the existing categorization, theory breakdown, theorem-property manual, and coverage audit. It does not replace any prior section. Its purpose is different from the theory chapter: a theory node says *what mathematical fact is available*; a method node says *what move converts the problem into a proof*. The same theorem may appear under several methods, and the same method may be used in linear algebra, analysis, geometry, and combinatorics.

Method granularity. A method is recorded when it is repeatable across problems: row elimination, trace-of-commutator, auxiliary derivative, Jensen/tangent, roots-of-unity filtering, graph encoding, compactness/extremal selection, etc. A problem may have several method nodes because a typical IMC solution is a sequence of moves rather than one theorem.

How to read the method catalog. For every method sheet below, the occurrence list says where the method is used or naturally activated by the official problem/solution structure. The problem-by-problem matrix at the end gives the recommended core method sequence for each problem.

17.1 Method-domain frequency overview

Method domain	Assigned problem-links
Real analysis	749
Algebra, number theory, and polynomials	572
Universal proof infrastructure	546
Linear algebra and matrices	319
Combinatorics and discrete structures	297
Functional equations and dynamics	85
Geometry, convexity, and metric methods	72
Complex and Fourier methods	32

17.2 Highest-frequency method nodes

ID	Method	Problems
M-RA-09	Convexity, Jensen, tangent, and chord inequalities	161
M-UNI-07	Special substitution and parameter freezing	141
M-RA-01	Auxiliary function with derivative sign	135
M-ANT-01	Reduction modulo a prime or finite field	116
M-UNI-09	Invariant extraction and invariant comparison	115
M-UNI-03	Extremal construction and equality-case engineering	103
M-ANT-03	GCD, valuation, and prime-power bookkeeping	102
M-RA-08	Compactness plus extreme value theorem	86
M-ANT-06	Degree versus number of roots	84
M-ANT-16	Polynomial identity by coefficient comparison	83
M-CMB-01	Pigeonhole threshold and collision forcing	72
M-UNI-05	Induction with strengthened hypothesis	72
M-LA-04	Rank-nullity and dimension-count obstruction	69
M-RA-06	Riemann sum and sum-integral conversion	66
M-CMB-02	Double counting ordered or weighted objects	66
M-LA-03	Block decomposition and Schur-style bookkeeping	65
M-RA-07	Split into main range and tail/error range	63
M-RA-03	Rolle chain and divided-difference localization	60
M-ANT-05	Factorization and unique factorization in Z[x]	59
M-UNI-08	Sign, zero, and boundary case splitting	58
M-FE-02	Monotonicity forced by functional equation	55
M-CMB-03	Graph encoding of a relation	47
M-LA-06	Spectral/minimal-polynomial reduction	47

ID	Method	Problems
M-ANT-02	Chinese remainder and residue construction	45
M-UNI-06	Compactness or extremal-point selection	43
M-RA-11	Integration by parts, Fubini, and weighted identities	39
M-LA-01	Row/column elimination to triangular or sparse form	34
M-RA-02	Integrating factor and ODE comparison	33
M-CMB-04	Cycle extraction and alternating-cycle method	33
M-RA-05	L-Hospital and ratio/growth comparison	30

17.3 Complete method sheets

Each method sheet contains the repeatable operation, when it should be triggered, the proof skeleton to train, the trap watch, and where it appeared in the corpus.

17.3.1 Algebra, number theory, and polynomials

M-ANT-01: Reduction modulo a prime or finite field *116 problem links*

Core move	Move an integer or polynomial problem to F_p, where finite field root counts, Fermat, or residue enumeration applies.
When to use	Use for divisibility, complete residue systems, polynomial values modulo p, and impossibility statements.
Proof skeleton	1. Reduce all hypotheses modulo p. 2. Use field properties. 3. Derive contradiction or classification. 4. Lift conclusion to integers if needed.
Trap watch	Modulo composite rings have zero divisors; use prime moduli when inversion is needed.
Appears in	1994-D1-P3; 1994-D1-P6; 1994-D2-P5; 1995-D2-P4; 1996-D2-P2; 1997-D1-P3; 1997-D1-P4; 1997-D1-P5; 1999-D1-P3; 1999-D2-P1; 1999-D2-P2; 2000-D1-P5; 2000-D2-P1; 2001-D1-P1; 2001-D1-P2; 2001-D1-P4; 2001-D2-P1; 2001-D2-P6; 2002-D1-P3; 2002-D2-P3; 2003-D1-P2; 2003-D1-P4; 2003-D2-P4; 2004-D1-P6; 2005-D1-P2; 2005-D1-P4; 2005-D1-P6; 2005-D2-P2; 2005-D2-P6; 2006-D1-P2; 2006-D1-P3; 2006-D1-P4; 2006-D2-P1; 2006-D2-P4; 2006-D2-P5; 2007-D1-P1; 2007-D1-P2; 2007-D1-P5; 2007-D2-P2; 2007-D2-P4; 2007-D2-P5; 2008-D1-P3; 2008-D1-P5; 2008-D2-P1; 2008-D2-P3; 2008-D2-P4; 2008-D2-P5; 2009-D2-P3; 2009-D2-P4; 2010-D1-P4; 2010-D1-P5; 2010-D2-P3; 2010-D2-P4; 2010-D2-P5; 2011-D1-P3; 2011-D1-P5; 2012-D1-P1; 2012-D1-P2; 2012-D1-P3; 2012-D1-P5; ... (+56 more)

M-ANT-02: Chinese remainder and residue construction *45 problem links*

Core move	Satisfy several congruences simultaneously by CRT or construct a value with prescribed residues.
When to use	Use for divisibility by multiple values, idempotent congruences, and integer-polynomial value control.
Proof skeleton	1. Factor the modulus or required divisors. 2. Set congruences. 3. Apply CRT. 4. Verify divisibility target.
Trap watch	CRT requires coprime moduli; split common factors before applying.
Appears in	1999-D1-P3; 1999-D2-P2; 2000-D2-P1; 2001-D1-P2; 2001-D1-P4; 2003-D1-P2; 2005-D1-P2; 2005-D1-P4; 2005-D2-P6; 2006-D1-P2; 2006-D1-P3; 2006-D2-P5; 2007-D1-P1; 2007-D2-P2; 2007-D2-P4; 2008-D1-P3; 2008-D1-P5; 2008-D2-P4; 2008-D2-P5; 2009-D2-P4; 2010-D1-P4; 2010-D2-P4; 2010-D2-P5; 2011-D1-P3; 2011-D1-P5; 2012-D2-P4; 2013-D1-P5; 2013-D2-P2; 2014-D1-P4; 2014-D2-P1; 2015-D1-P4; 2016-D1-P3; 2016-D1-P5; 2016-D2-P8; 2018-D1-P5; 2018-D2-P6; 2019-D1-P2; 2020-D2-P6; 2021-D2-P6; 2022-D1-P3; 2022-D2-P6; 2023-D1-P4; 2023-D2-P10; 2024-D1-P3; 2024-D2-P10

M-ANT-03: GCD, valuation, and prime-power bookkeeping *102 problem links*

Core move	Track divisibility through gcds, prime exponents, and coprimality constraints.
When to use	Use for determinant factorization, roots with integer coefficients, divisibility by powers, and group order arguments.
Proof skeleton	1. Factor relevant integers. 2. Compare valuations. 3. Use coprimality to assign factors. 4. Reassemble.
Trap watch	Sign and unit factors must be handled in Z and polynomial rings.
Appears in	1994-D1-P3; 1994-D1-P6; 1994-D2-P5; 1995-D2-P4; 1996-D2-P2; 1997-D1-P3; 1997-D1-P4; 1997-D1-P5; 1999-D1-P3; 1999-D2-P1; 2000-D1-P5; 2000-D2-P1; 2001-D1-P1; 2001-D1-P2; 2001-D1-P4; 2001-D2-P1; 2001-D2-P6; 2002-D1-P3; 2002-D2-P3; 2003-D1-P2; 2003-D1-P4; 2003-D2-P4; 2005-D1-P2; 2005-D1-P6; 2006-D1-P2; 2006-D1-P3; 2006-D1-P4; 2006-D2-P1; 2006-D2-P4; 2007-D1-P2; 2007-D1-P5; 2007-D2-P2; 2007-D2-P4; 2007-D2-P5; 2008-D2-P1; 2008-D2-P3; 2009-D2-P4; 2010-D1-P4; 2010-D1-P5; 2010-D2-P4; 2010-D2-P5; 2011-D1-P3; 2011-D1-P5; 2012-D1-P1; 2012-D1-P2; 2012-D1-P3; 2012-D1-P5; 2012-D2-P3; 2012-D2-P4; 2012-D2-P5; 2013-D1-P5; 2013-D2-P2; 2013-D2-P4; 2014-D1-P4; 2014-D2-P1; 2015-D1-P1; 2015-D1-P2; 2015-D1-P4; 2015-D2-P10; 2016-D1-P2; ... (+42 more)

M-ANT-04: Finite differences and integer polynomial values

1 problem links

Core move	Use values at consecutive integers or finite differences to recover divisibility of coefficients.
When to use	Use for integer polynomials whose values are divisible by a modulus.
Proof skeleton	1. Evaluate at enough integer points. 2. Take linear combinations/finite differences. 3. Infer coefficient divisibility.
Trap watch	Degree determines how many values are needed; more roots than degree only works over a field.
Appears in	2014-D2-P1

M-ANT-05: Factorization and unique factorization in Z[x]

59 problem links

Core move	Use factorization into irreducibles/linear factors and uniqueness to control rational roots, coefficient signs, and divisibility.
When to use	Use for polynomials with rational/integer roots, all roots rational, and factor divisibility.
Proof skeleton	1. Factor over Q or Z. 2. Normalize units/content. 3. Compare irreducible factors. 4. Extract coefficient/root information.
Trap watch	Use Gauss lemma when moving between Q[x] and Z[x].
Appears in	1994-D2-P4; 1995-D1-P5; 1995-D2-P3; 1995-D2-P4; 1996-D2-P2; 1998-D1-P2; 1998-D1-P5; 1998-D2-P2; 1999-D1-P1; 1999-D2-P2; 1999-D2-P6; 2000-D1-P6; 2000-D2-P3; 2000-D2-P4; 2000-D2-P6; 2001-D1-P4; 2001-D2-P1; 2002-D1-P1; 2002-D2-P3; 2003-D1-P3; 2003-D1-P6; 2003-D2-P2; 2005-D1-P4; 2005-D2-P2; 2006-D1-P4; 2006-D1-P6; 2006-D2-P5; 2006-D2-P6; 2007-D1-P1; 2007-D1-P3; 2007-D1-P5; 2007-D1-P6; 2007-D2-P6; 2008-D1-P6; 2008-D1-P2; 2008-D1-P3; 2008-D2-P1; 2008-D2-P2; 2008-D2-P4; 2008-D2-P6; 2009-D1-P4; 2009-D2-P1; 2009-D2-P3; 2009-D2-P4; 2009-D2-P5; 2010-D2-P5; 2011-D1-P2; 2011-D1-P3; 2011-D1-P5; 2011-D2-P4; 2012-D1-P5; 2012-D2-P1; 2015-D2-P10; 2017-D1-P5; 2017-D2-P7; 2018-D2-P9; 2020-D1-P4; 2021-D2-P7; 2023-D1-P3

M-ANT-06: Degree versus number of roots

84 problem links

Core move	A nonzero polynomial of degree d over a field has at most d roots; a polynomial identity follows from enough pointwise agreement.
When to use	Use for polynomial identities, finite fields, rational functions, and zero-count contradictions.
Proof skeleton	1. Convert the statement to zeros of a polynomial. 2. Count roots. 3. Compare with degree. 4. Conclude zero polynomial or contradiction.
Trap watch	In finite fields, functions may agree on all field elements without polynomials being identical unless degree is controlled.
Appears in	1994-D2-P4; 1995-D1-P5; 1995-D2-P3; 1995-D2-P4; 1996-D2-P2; 1998-D1-P2; 1998-D1-P5; 1998-D2-P2; 1999-D1-P1; 1999-D2-P2; 1999-D2-P6; 2000-D1-P6; 2000-D2-P3; 2000-D2-P4; 2000-D2-P6; 2001-D1-P1; 2001-D1-P4; 2001-D2-P1; 2002-D1-P1; 2002-D2-P3; 2003-D1-P2; 2003-D1-P3; 2003-D1-P6; 2003-D2-P2; 2004-D1-P6; 2005-D1-P4; 2005-D2-P2; 2005-D2-P6; 2006-D1-P2; 2006-D1-P4; 2006-D1-P6; 2006-D2-P5; 2006-D2-P6; 2007-D1-P1; 2007-D1-P3; 2007-D1-P5; 2007-D1-P6; 2007-D2-P2; 2007-D2-P6; 2008-D1-P6; 2008-D1-P2; 2008-D1-P3; 2008-D2-P1; 2008-D2-P2; 2008-D2-P4; 2008-D2-P5; 2008-D2-P6; 2009-D1-P3; 2009-D1-P4; 2009-D2-P1; 2009-D2-P3; 2009-D2-P4; 2009-D2-P5; 2010-D2-P5; 2011-D1-P2; 2011-D1-P3; 2011-D1-P5; 2011-D2-P2; 2011-D2-P4; 2012-D1-P5; ... (+24 more)

M-ANT-07: Vieta, symmetric sums, and logarithmic derivative

6 problem links

Core move	Translate root information into coefficient inequalities or expressions through Vieta or P'/P=sum 1/(x-r_i).
When to use	Use for real-root inequalities, roots on the unit circle, and coefficient comparisons.
Proof skeleton	1. Factor by roots. 2. Write coefficient or logarithmic-derivative identities. 3. Apply algebraic inequality/sign. 4. Analyze equality.
Trap watch	Multiple roots require multiplicities in sums and products.
Appears in	1994-D1-P3; 2003-D1-P6; 2006-D1-P5; 2014-D1-P3; 2023-D1-P3; 2024-D1-P1

M-ANT-08: Roots of unity and cyclotomic irreducibility

18 problem links

Core move	Evaluate polynomials at roots of unity, use cyclotomic irreducibility, or filter coefficients by residue classes.
When to use	Use for divisibility of lacunary polynomials, Fourier filters, and residue-class sums.
Proof skeleton	1. Pick a primitive root. 2. Evaluate or factor. 3. Use irreducibility/conjugates or averaging. 4. Translate to divisibility/counting.
Trap watch	Primitive and nonprimitive roots have different minimal polynomials.
Appears in	1995-D2-P3; 1999-D2-P2; 1999-D2-P6; 2000-D1-P2; 2000-D2-P3; 2001-D1-P4; 2001-D2-P3; 2002-D2-P5; 2007-D1-P5; 2007-D1-P6; 2008-D2-P1; 2013-D2-P1; 2016-D2-P10; 2017-D1-P1; 2017-D1-P5; 2018-D2-P9; 2021-D2-P7; 2023-D1-P2

M-ANT-09: Finite-field function representative and permutation argument

7 problem links

Core move	Every function F_p -> F_p has a unique representative polynomial of degree at most p-1 modulo x^p-x; bijections correspond to permutations.
When to use	Use for composition-closed polynomial sets over finite fields.
Proof skeleton	1. Reduce modulo x^p-x. 2. Interpret polynomials as functions. 3. Identify generated permutations. 4. Count distinct representatives.
Trap watch	Polynomial equality as functions is weaker than formal equality unless reduced degree is fixed.
Appears in	2003-D2-P4; 2012-D2-P3; 2013-D2-P2; 2016-D1-P3; 2018-D2-P6; 2020-D2-P6; 2023-D1-P4

M-ANT-10: Group action, orbit, and generator method

29 problem links

Core move	Translate algebraic structure into group actions, generators, cycles, or orbits.
When to use	Use for permutation groups, automorphisms, finite group partitions, and generation questions.
Proof skeleton	1. Identify generators or action. 2. Compute orbits/cycles. 3. Use known generation or normality. 4. Conclude group equality or obstruction.
Trap watch	Showing elements generate a subgroup is not enough; prove it contains standard generators or has full size.
Appears in	1994-D2-P5; 1996-D2-P3; 1998-D1-P2; 1999-D1-P3; 1999-D2-P1; 2000-D1-P5; 2001-D1-P2; 2003-D2-P4; 2005-D1-P6; 2007-D1-P4; 2010-D2-P3; 2012-D1-P3; 2012-D1-P5; 2012-D2-P3; 2012-D2-P4; 2012-D2-P5; 2015-D1-P2; 2015-D1-P4; 2016-D1-P2; 2016-D1-P5; 2016-D2-P8; 2016-D2-P9; 2017-D1-P4; 2018-D1-P2; 2020-D2-P7; 2021-D2-P6; 2022-D2-P6; 2023-D1-P5; 2024-D1-P4

M-ANT-11: Commutator and center manipulation in groups/rings

15 problem links

Core move	Rewrite noncommutative expressions using commutators, centrality, idempotency, or nil identities.
When to use	Use for rings with x^2=0, idempotents, commutative subgroups, and power-generated groups.
Proof skeleton	1. Linearize identities by substituting sums. 2. Move terms with commutators. 3. Prove centrality or cancellation. 4. Finish with torsion/characteristic.
Trap watch	Characteristic assumptions matter; integer cancellation may fail in finite characteristic.
Appears in	1996-D2-P3; 1999-D2-P1; 2000-D1-P5; 2001-D1-P2; 2005-D1-P6; 2007-D1-P4; 2010-D2-P3; 2012-D1-P3; 2012-D1-P5; 2012-D2-P5; 2016-D2-P8; 2017-D1-P4; 2018-D1-P2; 2021-D2-P6; 2024-D1-P4

M-ANT-12: Normal subgroup and quotient obstruction

1 problem links

Core move	Pass to a quotient group to show a proposed generating set cannot generate the whole group.
When to use	Use for finite group generation, automorphisms, and normal subgroup counterexamples.
Proof skeleton	1. Identify a normal subgroup. 2. Map generators to the quotient. 3. Show quotient generated image is too small. 4. Conclude impossibility.
Trap watch	The subgroup must be normal to form a quotient.
Appears in	2005-D1-P6

M-ANT-13: Pell-type recurrence or quadratic irrational stabilizer

1 problem links

Core move	Use units of quadratic fields or periodic recurrences to create integer matrices/stabilizers.
When to use	Use for quadratic irrationals fixed by fractional linear transformations.
Proof skeleton	1. Derive a Pell-like equation. 2. Produce infinitely many solutions using unit recurrence. 3. Choose a congruent solution. 4. Build the matrix/object.
Trap watch	Need nontrivial solution not equal to identity and determinant/sign verification.
Appears in	2005-D2-P6

M-ANT-14: Semidirect product or explicit algebraic construction

2 problem links

Core move	Build a group/ring/vector-space example by prescribing an action or multiplication rule.
When to use	Use for automorphism counterexamples, finite group constructions, and noncommutative examples.
Proof skeleton	1. Choose normal part and acting part. 2. Define action. 3. Verify group law. 4. Count orders/automorphisms or required property.
Trap watch	Construction must be associative and the action must be a homomorphism.
Appears in	1999-D2-P1; 2000-D1-P5

M-ANT-15: Rationality and linear dependence over Q *2 problem links*

Core move	Convert rational/irrational constraints into linear dependence over Q or into rational coefficient equations.
When to use	Use for irrational subset selection, rational distances, and rational roots.
Proof skeleton	1. Identify the Q-vector space structure. 2. Suppose rational dependence. 3. Combine relations with nonnegative/rational coefficients. 4. Contradict independence/irrationality.
Trap watch	Nonnegative coefficients restrict allowed linear combinations; preserve positivity when combining relations.
Appears in	1994-D1-P3; 2006-D1-P5

M-ANT-16: Polynomial identity by coefficient comparison *83 problem links*

Core move	Turn a functional/algebraic equation into equality of coefficients or homogeneous-degree components.
When to use	Use for multivariable polynomial identities and classification of polynomial forms.
Proof skeleton	1. Expand both sides. 2. Compare degrees or coefficients. 3. Solve the resulting algebraic system. 4. Check all candidates.
Trap watch	Do not ignore the zero polynomial unless it is excluded.
Appears in	1994-D2-P4; 1995-D1-P5; 1995-D2-P3; 1995-D2-P4; 1996-D2-P2; 1998-D1-P2; 1998-D1-P5; 1998-D2-P2; 1999-D1-P1; 1999-D2-P2; 1999-D2-P6; 2000-D1-P2; 2000-D1-P6; 2000-D2-P3; 2000-D2-P4; 2000-D2-P6; 2001-D1-P4; 2001-D1-P5; 2001-D2-P1; 2002-D1-P1; 2002-D2-P3; 2003-D1-P3; 2003-D1-P6; 2003-D2-P2; 2004-D1-P6; 2005-D1-P4; 2005-D2-P2; 2005-D2-P6; 2006-D1-P4; 2006-D1-P6; 2006-D2-P5; 2006-D2-P6; 2007-D1-P1; 2007-D1-P3; 2007-D1-P5; 2007-D1-P6; 2007-D2-P2; 2007-D2-P6; 2008-D1-P6; 2008-D1-P2; 2008-D1-P3; 2008-D2-P1; 2008-D2-P2; 2008-D2-P4; 2008-D2-P5; 2008-D2-P6; 2009-D1-P3; 2009-D1-P4; 2009-D2-P1; 2009-D2-P3; 2009-D2-P4; 2009-D2-P5; 2010-D2-P5; 2011-D1-P2; 2011-D1-P3; 2011-D1-P5; 2011-D2-P2; 2011-D2-P4; 2012-D1-P5; 2012-D2-P1; ... (+23 more)

M-ANT-17: Greedy expansion or tail-rationality method *1 problem links*

Core move	Define terms greedily so inequalities force uniqueness; characterize rationality by eventual tail behavior.
When to use	Use for infinite products, Egyptian-fraction-like representations, and greedy algorithms.
Proof skeleton	1. Choose the next term minimally. 2. Prove growth/termination or convergence. 3. Bound the tail. 4. Analyze rationality by descent.
Trap watch	Strict inequalities are needed to prevent nonunique boundary choices.
Appears in	1997-D1-P4

17.3.2 Combinatorics and discrete structures

M-CMB-01: Pigeonhole threshold and collision forcing *72 problem links*

Core move	Use finite capacity to force two objects into the same class or to force enough objects with a property.
When to use	Use for residue classes, marked grids, divisors, roots, set families, and finite coverings.
Proof skeleton	1. Define classes. 2. Bound number/size of classes. 3. Compare with total objects. 4. Extract collision or surplus.
Trap watch	The classes must cover all objects and the capacity bound must be sharp enough.
Appears in	1994-D1-P1; 1994-D1-P3; 1994-D1-P6; 1994-D2-P4; 1995-D1-P3; 1996-D1-P3; 1996-D2-P3; 1996-D2-P6; 1997-D1-P6; 1997-D2-P6; 1998-D1-P5; 1998-D2-P3; 1998-D2-P4; 1998-D2-P5; 1998-D2-P6; 1999-D1-P5; 1999-D2-P5; 1999-D2-P6; 2000-D1-P1; 2001-D2-P1; 2001-D2-P3; 2002-D1-P4; 2002-D1-P5; 2002-D2-P2; 2003-D1-P5; 2003-D2-P3; 2003-D2-P4; 2003-D2-P5; 2004-D1-P1; 2004-D1-P2; 2004-D1-P4; 2004-D2-P6; 2005-D1-P2; 2006-D1-P2; 2006-D1-P6; 2006-D2-P1; 2006-D2-P5; 2007-D1-P4; 2007-D1-P6; 2008-D1-P6; 2008-D1-P4; 2008-D1-P5; 2008-D2-P4; 2008-D2-P6; 2009-D1-P1; 2009-D1-P3; 2010-D1-P4; 2011-D1-P4; 2011-D2-P2; 2012-D1-P1; 2013-D1-P3; 2013-D2-P5; 2015-D2-P8; 2016-D1-P5; 2016-D2-P7; 2016-D2-P9; 2017-D1-P4; 2018-D2-P8; 2019-D2-P8; 2020-D1-P1; ... (+12 more)

M-CMB-02: Double counting ordered or weighted objects

66 problem links

Core move	Count the same triples, walks, pairs, incidences, or weights in two ways.
When to use	Use for graphs, contest-solvers, set systems, probabilities, and degree-sum arguments.
Proof skeleton	1. Define the counted object precisely. 2. Count by first parameter. 3. Count by second parameter. 4. Compare/equate.
Trap watch	Ordered versus unordered counts are a common source of factor errors.
Appears in	1994-D1-P1; 1994-D2-P4; 1995-D1-P3; 1996-D1-P3; 1996-D2-P3; 1996-D2-P6; 1997-D1-P6; 1997-D2-P6; 1998-D2-P3; 1998-D2-P4; 1998-D2-P5; 1998-D2-P6; 1999-D1-P5; 1999-D2-P5; 2000-D1-P1; 2001-D2-P1; 2001-D2-P3; 2002-D1-P4; 2002-D1-P5; 2002-D2-P2; 2003-D1-P5; 2003-D2-P3; 2003-D2-P4; 2003-D2-P5; 2004-D1-P1; 2004-D1-P2; 2004-D1-P4; 2004-D2-P6; 2005-D1-P2; 2006-D1-P2; 2006-D1-P6; 2006-D2-P1; 2006-D2-P5; 2007-D1-P4; 2007-D1-P6; 2008-D1-P6; 2008-D1-P4; 2008-D1-P5; 2008-D2-P4; 2008-D2-P6; 2009-D1-P1; 2009-D1-P3; 2010-D1-P4; 2011-D1-P4; 2011-D2-P2; 2012-D1-P1; 2013-D1-P3; 2013-D2-P5; 2015-D2-P8; 2016-D1-P5; 2016-D2-P9; 2017-D1-P4; 2018-D2-P8; 2019-D2-P8; 2020-D1-P1; 2020-D1-P2; 2020-D2-P7; 2021-D1-P2; 2021-D2-P8; 2022-D1-P3; ... (+6 more)

M-CMB-03: Graph encoding of a relation

47 problem links

Core move	Encode rows/columns, friendships, intersections, or marked positions as vertices and edges.
When to use	Use when the statement contains pairwise relations, grids, trees, cycles, or incidence structures.
Proof skeleton	1. Define vertices and edges. 2. Translate hypotheses. 3. Apply graph theorem or cycle/degree count. 4. Decode the result.
Trap watch	The encoded graph may be directed, bipartite, simple, or multigraph; choose correctly.
Appears in	1996-D1-P6; 1997-D2-P5; 1997-D2-P6; 1998-D1-P1; 1998-D2-P3; 1998-D2-P5; 1999-D1-P5; 1999-D2-P5; 2000-D2-P4; 2001-D1-P2; 2001-D2-P4; 2002-D1-P1; 2002-D2-P2; 2002-D2-P4; 2003-D1-P4; 2004-D1-P1; 2004-D1-P3; 2004-D1-P4; 2004-D2-P2; 2005-D2-P4; 2006-D2-P1; 2007-D2-P1; 2008-D1-P4; 2009-D1-P3; 2009-D1-P5; 2009-D2-P4; 2011-D1-P4; 2011-D2-P2; 2011-D2-P5; 2013-D1-P3; 2013-D2-P5; 2014-D2-P4; 2015-D2-P8; 2016-D1-P4; 2016-D2-P9; 2017-D1-P3; 2018-D2-P8; 2019-D2-P8; 2019-D2-P10; 2020-D1-P3; 2021-D1-P2; 2022-D1-P4; 2022-D2-P5; 2022-D2-P8; 2023-D2-P7; 2023-D2-P8; 2024-D1-P5

M-CMB-04: Cycle extraction and alternating-cycle method

33 problem links

Core move	From minimum degree or edge count, extract a cycle and translate it back into an alternating sequence.
When to use	Use for grid marked-point problems and bipartite graph cycles.
Proof skeleton	1. Delete degree 0/1 vertices if needed. 2. Walk until repetition. 3. Extract a simple cycle. 4. Decode alternating objects.
Trap watch	Odd repetitions in a constructed walk may require shortening before decoding.
Appears in	1996-D1-P6; 1997-D1-P6; 1997-D2-P5; 1998-D1-P1; 1998-D2-P3; 1999-D1-P5; 2000-D2-P4; 2001-D1-P2; 2001-D2-P4; 2002-D1-P1; 2002-D2-P2; 2002-D2-P4; 2003-D1-P4; 2004-D1-P3; 2004-D2-P2; 2006-D2-P1; 2007-D2-P1; 2009-D1-P2; 2009-D1-P3; 2009-D1-P5; 2009-D2-P4; 2011-D2-P5; 2013-D2-P5; 2014-D2-P4; 2016-D1-P4; 2019-D2-P8; 2019-D2-P10; 2020-D1-P3; 2022-D1-P4; 2022-D2-P5; 2022-D2-P8; 2023-D2-P7; 2023-D2-P8

M-CMB-05: State recurrence or automaton partition

12 problem links

Core move	Partition objects by last state, boundary condition, or finite memory to derive a recurrence.
When to use	Use for words, strings, nonconstant functions on finite sets, and recursive matrices.
Proof skeleton	1. Choose states with enough memory. 2. Derive transitions. 3. Compute initial conditions. 4. Solve or compare recurrences.
Trap watch	State definitions must be mutually exclusive and exhaustive.
Appears in	1998-D2-P4; 2004-D1-P1; 2004-D2-P6; 2005-D1-P2; 2006-D1-P2; 2013-D1-P3; 2013-D2-P5; 2015-D2-P8; 2017-D1-P4; 2018-D2-P8; 2019-D2-P8; 2021-D1-P2

M-CMB-06: Generating function encoding

11 problem links

Core move	Encode a sequence or probability distribution as a power series and extract coefficients or values.
When to use	Use for recurrences, sums of powers, probability distributions, and counting by statistics.
Proof skeleton	1. Define the generating function. 2. Translate recurrence/constraint. 3. Solve functional equation. 4. Extract coefficient/value.
Trap watch	Convergence or formal-power-series context must be clear.
Appears in	2002-D1-P3; 2003-D1-P6; 2003-D2-P6; 2008-D2-P3; 2012-D2-P2; 2017-D2-P8; 2018-D2-P7; 2019-D1-P4; 2021-D1-P3; 2022-D1-P1; 2024-D2-P8

M-CMB-07: Involution, parity, and sign-reversing cancellation

12 problem links

Core move	Pair objects to show counts are even, coefficients vanish, or determinants have parity constraints.
When to use	Use for permutation statistics, determinant coefficient parity, and cancellation sums.
Proof skeleton	1. Define the involution or pairing. 2. Show it preserves the set and flips sign/parity. 3. Identify fixed points. 4. Conclude.
Trap watch	An involution must be reversible and fixed-point analysis must be complete.
Appears in	1998-D1-P2; 2002-D2-P4; 2003-D1-P4; 2005-D1-P1; 2007-D1-P2; 2010-D2-P3; 2012-D1-P3; 2013-D2-P5; 2016-D1-P5; 2019-D2-P8; 2022-D2-P6; 2023-D1-P5

M-CMB-08: Extremal set or minimal counterexample induction

6 problem links

Core move	Choose a smallest/largest set, element, interval, or family and exploit its extremality.
When to use	Use for intersecting families, finite subsets of N, coverings, and set-system structure.
Proof skeleton	1. Choose an extremal object. 2. Split by intersection with it. 3. Apply induction or contradiction. 4. Combine finitely many witnesses.
Trap watch	The extremal object must exist; infinite families need a finite-size parameter.
Appears in	2004-D1-P1; 2004-D1-P5; 2010-D2-P2; 2012-D1-P2; 2016-D2-P6; 2019-D2-P8

M-CMB-09: Partition, refinement, and cover compression

0 problem links

Core move	Split a set/family into structured parts or compress a cover without increasing cost.
When to use	Use for covering content, finite-family problems, and triangulations.
Proof skeleton	1. Partition by a local pattern. 2. Solve each part. 3. Combine witnesses/covers. 4. Bound total cost or parity.
Trap watch	Refinement must not destroy the property being counted or covered.
Appears in	

M-CMB-10: Explicit bijection and encoding-decoding proof

24 problem links

Core move	Construct a reversible map between two sets so that counting equality becomes obvious.
When to use	Use for word conditions, residue strings, binary/ternary encodings, and matrix-power counting.
Proof skeleton	1. Define the encoding. 2. Prove it maps into the target. 3. Define inverse. 4. Verify compositions are identity.
Trap watch	Every component of the inverse must be well-defined.
Appears in	1996-D2-P3; 1997-D1-P6; 1997-D2-P6; 1998-D2-P4; 2002-D2-P2; 2004-D1-P1; 2004-D1-P2; 2004-D2-P6; 2006-D1-P2; 2007-D1-P4; 2011-D1-P4; 2012-D1-P1; 2013-D1-P3; 2013-D2-P5; 2015-D2-P8; 2016-D1-P5; 2017-D1-P4; 2018-D2-P8; 2019-D2-P8; 2019-D2-P10; 2021-D1-P2; 2022-D1-P3; 2022-D2-P5; 2024-D1-P3

M-CMB-11: Erdos-Szekeres/binomial threshold induction

0 problem links

Core move	Use a binomial recurrence threshold to force increasing or decreasing structured subsequences.
When to use	Use for monotone sequence extraction with growth constraints.
Proof skeleton	1. Prove a two-parameter stronger lemma. 2. Split by a median/extreme. 3. Apply induction to one side. 4. Append an endpoint.
Trap watch	The threshold recurrence must match the split sizes exactly.
Appears in	

M-CMB-12: Root-of-unity filter for residues

6 problem links

Core move	Average over q-th roots of unity to count objects whose statistic is congruent to a residue.
When to use	Use for dice/residue probabilities, sums divisible by m, and coefficient filtering.
Proof skeleton	1. Build generating function. 2. Evaluate at roots of unity. 3. Average. 4. Simplify residue cases.
Trap watch	The filter counts modulo q; check whether zero term or empty sequence is included.
Appears in	1995-D2-P3; 1999-D2-P2; 1999-D2-P6; 2001-D2-P3; 2002-D2-P5; 2007-D1-P6

M-CMB-13: Indicator and expectation method

6 problem links

Core move	Express a random quantity as a sum of indicator variables and use linearity or symmetry.
When to use	Use for probability, geometric random points, and existence by averaging.
Proof skeleton	1. Define indicators. 2. Compute expectations using symmetry. 3. Sum. 4. Convert expectation to probability/existence.
Trap watch	Independence is not needed for linearity but is needed for product probabilities.
Appears in	2011-D1-P4; 2019-D2-P10; 2021-D1-P2; 2022-D1-P1; 2022-D2-P8; 2024-D1-P5

M-CMB-14: Topological ordering of an acyclic directed graph

2 problem links

Core move	Show a directed graph has no cycle, then order vertices so all edges point forward.
When to use	Use for matrix zero patterns and principal-minor conditions.
Proof skeleton	1. Build directed graph from nonzero entries. 2. Rule out cycles by minimal-cycle determinant. 3. Topologically sort. 4. Obtain triangular form.
Trap watch	The minimal cycle argument must exclude all alternative cycle permutations.
Appears in	2001-D2-P4; 2015-D2-P7

M-CMB-15: Rewriting-system reduction *0 problem links*

Core move	Use equivalence rules to shorten words or reduce algebraic expressions to bounded length.
When to use	Use for word problems with identities like uu~u and context closure.
Proof skeleton	1. Prove a lemma for absorbing appended words. 2. Find repeated subword or complete alphabet block. 3. Apply reductions. 4. Iterate to bounded length.
Trap watch	The rewriting rule must be used only in allowed contexts.
Appears in	

17.3.3 Complex and Fourier methods

M-CX-01: Cauchy coefficient formula and boundary maximum *15 problem links*

Core move	Represent coefficients by unit-circle integrals and use maximum modulus or boundary control.
When to use	Use for polynomial coefficient multipliers, bounded polynomials on the disk, and trigonometric kernels.
Proof skeleton	1. Move maximum to boundary. 2. Express coefficients/integral kernel. 3. Bound by sup norm. 4. Use kernel positivity if needed.
Trap watch	The function must be holomorphic on the relevant domain for maximum modulus/Cauchy.
Appears in	1995-D2-P3; 1999-D2-P6; 2000-D2-P3; 2005-D2-P2; 2006-D1-P4; 2007-D1-P5; 2012-D1-P5; 2012-D2-P1; 2013-D2-P1; 2014-D1-P3; 2017-D2-P7; 2018-D2-P9; 2021-D2-P7; 2023-D1-P3; 2024-D1-P1

M-CX-02: Laurent expansion and isolated singularity rationality *1 problem links*

Core move	Analyze poles/zeros and behavior at infinity to prove a meromorphic function is rational and determine it.
When to use	Use for sums over logarithm branches, partial fractions, and rationality of meromorphic expressions.
Proof skeleton	1. Prove local uniform convergence away from singularities. 2. Locate poles/zeros. 3. Bound at infinity. 4. Determine rational form from principal parts.
Trap watch	Branch choices and singularities at 0, 1, infinity must be handled explicitly.
Appears in	2004-D1-P6

M-CX-03: Unit-circle root real-part/log-derivative test *1 problem links*

Core move	Use expressions like Re((z+a)/(z-a)) to show zeros must lie on |z|=1.
When to use	Use for polynomials whose roots are all on the unit circle.
Proof skeleton	1. Factor the polynomial. 2. Express transformed polynomial divided by original. 3. Take real parts. 4. Exclude |z| not equal to 1.
Trap watch	Avoid evaluating at roots of the denominator; handle multiplicities by continuity if needed.
Appears in	1995-D2-P3

M-CX-04: Fejer kernel positivity *1 problem links*

Core move	Decompose a convex coefficient multiplier into nonnegative Fejer kernels to preserve disk sup norm.
When to use	Use for convex sequences of multipliers and trigonometric polynomial inequalities.
Proof skeleton	1. Write second differences. 2. Decompose into Fejer kernels. 3. Use positivity and integral one. 4. Apply sup norm bound.
Trap watch	Convexity direction and endpoint normalization determine nonnegative coefficients.
Appears in	2009-D1-P4

M-CX-05: Trigonometric-to-complex conversion *13 problem links*

Core move	Replace sine/cosine products or sums by complex exponentials or roots of unity.
When to use	Use for trigonometric products, zero counts, residue class probabilities, and Fourier coefficient filters.
Proof skeleton	1. Express trig terms as exponentials. 2. Identify cancellation/orthogonality. 3. Sum or bound. 4. Convert back to real statement.
Trap watch	Track real and imaginary parts; not every complex equality follows from real-part equality.
Appears in	1995-D2-P3; 1995-D2-P5; 1999-D2-P6; 2000-D1-P4; 2001-D2-P3; 2002-D2-P5; 2004-D2-P5; 2006-D2-P3; 2007-D1-P6; 2014-D2-P3; 2015-D1-P4; 2017-D1-P2; 2021-D2-P7

M-CX-06: Cotangent partial-fraction differentiation *1 problem links*

Core move	Start from the cotangent partial-fraction expansion and differentiate to obtain sums of inverse powers.
When to use	Use for sums over logarithm branches and rational functions built from periodic poles.
Proof skeleton	1. Write base cotangent expansion. 2. Substitute logarithm variable. 3. Differentiate with chain rule. 4. Simplify to rational form.
Trap watch	Symmetric summation must be justified; ordinary one-sided sums may diverge.
Appears in	2004-D1-P6

M-CX-07: Multiplicity count for values 0 and 1 *0 problem links*

Core move	Compare zeros of p and p-1 with zeros of p’ to prove p takes values 0 or 1 at many distinct points.
When to use	Use for polynomial covering/value problems over C.
Proof skeleton	1. Count multiplicities of zeros of p and p-1. 2. Subtract critical multiplicities from p’. 3. Use degree of derivative. 4. Bound distinct points.
Trap watch	Critical points shared by p and p-1 cannot overlap because the value differs.
Appears in	

17.3.4 Functional equations and dynamics

M-FE-01: Iterates, fixed points, and cluster-point trapping *16 problem links*

Core move	Study orbits x_{n+1}=f(x_n), fixed points, periodic points, and closed orbit sets.
When to use	Use for interval maps, continuous dynamics, and convergence of iterates.
Proof skeleton	1. Identify cluster points. 2. Use continuity to transfer behavior. 3. Show jumps cannot cross a forbidden interval. 4. Conclude convergence/periodicity/finite orbit.
Trap watch	Continuity matters; without it, orbit-cluster arguments can fail.
Appears in	1994-D2-P6; 1995-D2-P6; 1996-D2-P1; 1998-D1-P3; 1998-D2-P3; 1999-D2-P4; 2000-D1-P4; 2001-D2-P6; 2004-D2-P3; 2010-D1-P3; 2015-D1-P3; 2015-D2-P6; 2017-D2-P9; 2019-D1-P1; 2024-D1-P4; 2024-D2-P6

M-FE-02: Monotonicity forced by functional equation

55 problem links

Core move	Derive monotonicity, injectivity, boundedness, or strictness from the functional equation before solving.
When to use	Use for positive-valued equations, image-length preserving maps, and order constraints.
Proof skeleton	1. Substitute to get inequality/equality. 2. Prove monotonicity/injectivity. 3. Use it to cancel arguments. 4. Solve algebraically.
Trap watch	Monotonicity may be nonstrict; injectivity requires strictness or separate proof.
Appears in	1994-D1-P2; 1994-D1-P6; 1994-D2-P1; 1994-D2-P2; 1994-D2-P3; 1994-D2-P6; 1995-D1-P3; 1995-D1-P4; 1996-D1-P5; 1996-D2-P5; 1997-D1-P4; 1998-D1-P3; 1998-D2-P3; 1998-D2-P4; 1999-D1-P4; 1999-D1-P6; 1999-D2-P4; 2000-D1-P1; 2000-D1-P4; 2000-D1-P6; 2000-D2-P2; 2000-D2-P5; 2001-D1-P6; 2001-D2-P2; 2001-D2-P5; 2002-D1-P2; 2002-D1-P5; 2002-D1-P6; 2003-D1-P1; 2003-D1-P5; 2003-D2-P2; 2004-D1-P5; 2004-D2-P2; 2005-D2-P4; 2006-D1-P1; 2006-D2-P2; 2007-D2-P3; 2007-D2-P6; 2008-D1-P1; 2009-D1-P1; 2009-D2-P2; 2010-D1-P1; 2010-D1-P3; 2010-D2-P1; 2011-D1-P1; 2011-D1-P4; 2011-D2-P1; 2011-D2-P3; 2015-D1-P3; 2015-D2-P6; 2016-D1-P1; 2019-D1-P1; 2019-D2-P6; 2024-D2-P6; 2024-D2-P9

M-FE-03: Composition, inversion, and conjugacy

12 problem links

Core move	Compose the equation with f, f^{-1}, shifts, or transformations to expose invariant ratios or conjugate maps.
When to use	Use for f(x^2/f(x))=x, f(x)f(yf(x))=f(x+y), and polynomial composition identities.
Proof skeleton	1. Define a new variable or ratio. 2. Compose/substitute repeatedly. 3. Derive invariant orbit. 4. Use monotonicity/growth to force constancy.
Trap watch	Compositions require domain/range compatibility and invertibility when used.
Appears in	2000-D1-P2; 2005-D2-P2; 2007-D1-P3; 2012-D2-P1; 2013-D2-P1; 2014-D1-P3; 2015-D2-P10; 2017-D2-P7; 2018-D2-P9; 2020-D1-P4; 2023-D1-P3; 2024-D1-P1

M-FE-04: Orbit decomposition for bijections

1 problem links

Core move	Decompose a bijection into finite or infinite orbits and solve the problem on each orbit.
When to use	Use for representing a bijection as product of involutions or symmetries.
Proof skeleton	1. Define orbit under f. 2. Treat finite cycles and infinite chains separately. 3. Construct local involutions. 4. Combine across disjoint orbits.
Trap watch	Infinite orbits may be bi-infinite if the map is bijective; use f^n for n in Z.
Appears in	1997-D2-P5

M-FE-05: Translation/rational-shift invariance

0 problem links

Core move	Show f(x+q)-f(x) is constant for rational q or derive affine form from dense subgroup behavior.
When to use	Use for continuous functions with rational-difference constraints.
Proof skeleton	1. Fix q. 2. Show the difference is continuous with rational values, hence constant. 3. Compute on Q. 4. Extend by density.
Trap watch	Continuity is needed to turn rational-valued continuous functions into constants.
Appears in	

M-FE-06: Nonexistence by growth iteration

1 problem links

Core move	Iterate a functional inequality to force values to become negative, infinite, or contradict boundedness/positivity.
When to use	Use for functional inequalities on positive domains and impossible differential/functional systems.
Proof skeleton	1. Derive monotone/growth inequality. 2. Iterate over a sequence. 3. Accumulate contradiction with positivity or boundedness. 4. Close all cases.
Trap watch	Check the chosen iterates remain in the domain.
Appears in	2024-D2-P6

17.3.5 Geometry, convexity, and metric methods

M-GEO-01: Coordinate normalization and algebraization *18 problem links*

Core move	Choose coordinates so a line, focus, center, vertex, or simplex has simple form; translate geometry into equations.
When to use	Use for conics, distance inequalities, ellipsoids, parabolas, spheres, and volume computations.
Proof skeleton	1. Place origin/axes conveniently. 2. Write equations. 3. Simplify algebraically. 4. Interpret result geometrically.
Trap watch	Coordinate choices may hide degeneracies; check special cases.
Appears in	1996-D2-P4; 1998-D2-P5; 2000-D2-P1; 2002-D1-P1; 2002-D2-P4; 2002-D2-P6; 2004-D1-P4; 2008-D2-P2; 2009-D2-P1; 2011-D2-P5; 2014-D1-P5; 2014-D2-P4; 2015-D1-P5; 2018-D1-P5; 2019-D2-P10; 2020-D1-P3; 2022-D2-P5; 2023-D2-P9

M-GEO-02: Projection to a lower-dimensional object *3 problem links*

Core move	Project a geometric configuration onto a line/plane/subspace to compare lengths, areas, volumes, or distances.
When to use	Use for slices, ellipses, contractions, closest points, and sphere configurations.
Proof skeleton	1. Choose projection. 2. Track how distances/areas change. 3. Apply lower-dimensional inequality. 4. Lift conclusion.
Trap watch	Projection may collapse information; ensure the projected inequality is strong enough.
Appears in	1997-D1-P5; 2014-D1-P5; 2015-D1-P5

M-GEO-03: Squared-distance and inner-product identities *26 problem links*

Core move	Replace geometric distance constraints by algebraic inner-product equalities/inequalities.
When to use	Use for rational distances, simplex bounds, Hilbert equidistant sets, and unit sphere packings.
Proof skeleton	1. Square distances. 2. Use norm identities. 3. Convert to Gram matrix or angle condition. 4. Apply linear algebra.
Trap watch	Squaring inequalities is safe only for nonnegative quantities.
Appears in	1995-D2-P1; 1996-D1-P6; 1996-D2-P4; 1997-D1-P5; 1997-D2-P5; 1998-D2-P5; 2002-D1-P1; 2002-D1-P6; 2002-D2-P4; 2003-D2-P3; 2004-D1-P4; 2004-D2-P3; 2005-D2-P5; 2006-D2-P4; 2009-D2-P1; 2013-D2-P3; 2014-D1-P5; 2014-D2-P4; 2015-D1-P5; 2015-D2-P9; 2017-D2-P10; 2018-D2-P6; 2021-D2-P8; 2023-D2-P8; 2023-D2-P9; 2024-D1-P5

M-GEO-04: Barycentric coordinates and volume weights *2 problem links*

Core move	Express an interior point or circumcenter identity using signed/positive simplex volumes as weights.
When to use	Use for simplex subdivisions, centers, and affine identities.
Proof skeleton	1. Express point in barycentric coordinates. 2. Relate coordinates to sub-simplex volumes. 3. Prove vector identity. 4. Check normalization.
Trap watch	Orientation and sign must be handled if points leave the simplex.
Appears in	2009-D1-P5; 2014-D1-P5

M-GEO-05: Affine invariance and change of metric scale

3 problem links

Core move	Apply an affine map or linear transformation that preserves incidence/convexity while rescaling volumes or distances predictably.
When to use	Use for ellipsoid volume, simplex normalization, and cube/square dissections.
Proof skeleton	1. Choose map. 2. Track determinant/scale. 3. Solve normalized geometry. 4. Undo scaling.
Trap watch	Affine maps do not preserve Euclidean distances or angles except under isometries/scalings.
Appears in	1997-D1-P5; 2000-D2-P1; 2014-D1-P5

M-GEO-06: Convex support and John-ellipse perturbation

4 problem links

Core move	Use maximality of an inscribed ellipse/ball and perturb it to contradict missing contact points.
When to use	Use for convex bodies with maximal-area ellipse or supporting geometry.
Proof skeleton	1. Assume a contact arc/region is missing. 2. Perturb ellipse to increase area while staying inside. 3. Contradict maximality.
Trap watch	The perturbation must remain inside the convex body; compact distance from boundary usually supplies margin.
Appears in	1996-D2-P4; 2011-D2-P3; 2014-D2-P5; 2015-D1-P3

M-GEO-07: Metric contraction and diameter-cover comparison

0 problem links

Core move	Compare upper content or cover diameter with length of a contracted image.
When to use	Use for metric content, contractions, and examples where inequalities are strict.
Proof skeleton	1. Cover the set. 2. Apply contraction to each piece. 3. Use length lower bound on image cover. 4. Optimize or construct strict example.
Trap watch	Contractions are 1-Lipschitz but need not be linear or injective.
Appears in	

M-GEO-08: Conic, sphere, and directrix algebra

1 problem links

Core move	Use standard equations or focus-directrix definitions of conics/spheres to reduce intersections to algebraic systems.
When to use	Use for ellipses with common focus, parabolas, and sphere intersection problems.
Proof skeleton	1. Write conic/sphere equation. 2. Compare equations. 3. Reduce to line/quadratic intersection. 4. Count points.
Trap watch	Degenerate conics and tangent cases may reduce the number of intersections.
Appears in	1998-D1-P5

M-GEO-09: Regular simplex and equidistant orthonormal translation

1 problem links

Core move	Translate an equidistant set by its barycenter/limit center to obtain orthogonal vectors.
When to use	Use for Hilbert-space equidistant sets and maximum size on spheres.
Proof skeleton	1. Compute inner products from distances. 2. Choose finite barycenter or Cauchy-limit center. 3. Translate. 4. Normalize to orthonormal system.
Trap watch	Infinite sets require completeness to pass to the limiting center.
Appears in	2008-D1-P6

M-GEO-10: Closest-point uniqueness by moving toward a minimizer *1 problem links*

Core move	If a closed set has multiple nearest points from b, move slightly toward one nearest point to obtain uniqueness nearby.
When to use	Use for density of points with unique nearest projection.
Proof skeleton	1. Pick a nearest point by compact reduction. 2. Move along segment. 3. Use ball-intersection geometry to prove uniqueness. 4. Conclude density.
Trap watch	Closedness is needed for existence of nearest point on bounded truncation.
Appears in	2004-D1-P1

M-GEO-11: Volume, projection, and ellipsoid computation *13 problem links*

Core move	Convert a geometric region to a standard ellipsoid or use projected areas/volumes to compare quantities.
When to use	Use for 3D distance regions, tetrahedron angle inequalities, and projected convex figures.
Proof skeleton	1. Write inequality/equation. 2. Complete squares or project. 3. Transform to standard body. 4. Multiply by scale determinant.
Trap watch	Volume scale is the absolute determinant of the linear transformation.
Appears in	1996-D2-P4; 1998-D2-P5; 2000-D2-P1; 2002-D1-P1; 2002-D2-P6; 2004-D1-P4; 2011-D2-P5; 2015-D1-P5; 2018-D1-P5; 2019-D2-P10; 2020-D1-P3; 2022-D2-P5; 2023-D2-P9

17.3.6 Linear algebra and matrices

M-LA-01: Row/column elimination to triangular or sparse form *34 problem links*

Core move	Use determinant-preserving row/column additions, swaps, and factorizations to expose triangular, diagonal, Toeplitz, circulant, or block structure.
When to use	Use for explicit determinant/rank computations and matrices with entries depending on indices, parity, primes, or recurrences.
Proof skeleton	1. Perform operations that preserve or track determinant. 2. Create zeros systematically. 3. Reduce to triangular/block form. 4. Compute determinant/rank.
Trap watch	Track determinant sign and scaling; row and column operations are not interchangeable in noncommutative-looking identities.
Appears in	1994-D1-P1; 1995-D1-P1; 1995-D2-P1; 1996-D1-P1; 1997-D1-P3; 1997-D2-P2; 1999-D1-P1; 1999-D1-P5; 2001-D1-P1; 2001-D2-P4; 2002-D2-P1; 2004-D2-P6; 2005-D1-P1; 2005-D2-P6; 2006-D1-P3; 2006-D2-P6; 2007-D1-P2; 2007-D1-P3; 2007-D2-P4; 2008-D1-P6; 2008-D2-P5; 2009-D1-P5; 2009-D2-P5; 2010-D1-P5; 2010-D2-P4; 2012-D1-P2; 2014-D1-P1; 2015-D1-P1; 2019-D1-P5; 2020-D1-P2; 2022-D2-P7; 2023-D2-P6; 2024-D2-P7; 2024-D2-P9

M-LA-02: Complete a matrix equation to an inverse identity *8 problem links*

Core move	Rewrite equations such as AB+A+B=0 or (A-B)C=BA^{-1} as a product XY=I, then use that for square matrices YX=I.
When to use	Use when a matrix equation can be rearranged into (A+I)(B+I)=I or similar.
Proof skeleton	1. Add/subtract I or a missing term. 2. Factor. 3. Infer invertibility and reverse product. 4. Expand the reversed identity.
Trap watch	Do not reverse products until the inverse relation has been proved.
Appears in	1994-D1-P1; 1995-D1-P1; 1997-D1-P3; 1997-D2-P2; 2002-D2-P5; 2009-D1-P2; 2019-D2-P9; 2024-D2-P7

M-LA-03: Block decomposition and Schur-style bookkeeping

65 problem links

Core move	Partition matrices into compatible blocks, multiply blocks explicitly, or use Schur-complement-like identities when invertibility is available.
When to use	Use when a problem gives block matrices, rectangular factorizations AB/BA, or a subspace decomposition.
Proof skeleton	1. Choose blocks adapted to the hypothesis. 2. Write product equations blockwise. 3. Solve forced block identities. 4. Recombine determinant/rank/eigenvalue data.
Trap watch	Schur complements require an invertible block; otherwise use direct block multiplication.
Appears in	1994-D1-P1; 1995-D1-P1; 1995-D2-P1; 1996-D1-P1; 1997-D1-P3; 1997-D2-P2; 1997-D2-P4; 1998-D1-P1; 1999-D1-P1; 1999-D1-P5; 1999-D2-P2; 1999-D2-P3; 2000-D1-P3; 2000-D1-P4; 2001-D1-P1; 2001-D1-P3; 2001-D1-P5; 2001-D2-P4; 2002-D2-P1; 2003-D2-P1; 2004-D2-P1; 2004-D2-P6; 2005-D1-P1; 2005-D2-P6; 2006-D1-P3; 2006-D2-P3; 2006-D2-P6; 2007-D1-P2; 2007-D1-P3; 2007-D2-P4; 2008-D1-P6; 2008-D2-P5; 2009-D1-P5; 2009-D2-P5; 2010-D1-P5; 2010-D2-P4; 2011-D2-P2; 2011-D2-P3; 2013-D1-P4; 2014-D1-P1; 2014-D2-P2; 2014-D2-P3; 2014-D2-P5; 2015-D1-P1; 2015-D1-P2; 2015-D1-P3; 2016-D1-P2; 2016-D2-P6; 2018-D1-P1; 2018-D1-P3; 2018-D1-P4; 2018-D2-P10; 2019-D1-P3; 2019-D1-P5; 2019-D2-P9; 2020-D1-P2; 2020-D2-P5; 2020-D2-P8; 2022-D1-P1; 2022-D2-P7; ... (+5 more)

M-LA-04: Rank-nullity and dimension-count obstruction

69 problem links

Core move	Turn existence/nonexistence into dimensions of image, kernel, span, intersection, or matrix subspace.
When to use	Use for rank bounds, spaces of matrices, linear functionals, invariant subspaces, and construction of common vectors.
Proof skeleton	1. Define the linear map/subspace. 2. Bound kernel or image. 3. Apply rank-nullity or dimension formula. 4. Translate back.
Trap watch	Dimension counts prove existence but not uniqueness; check whether the selected vector is nonzero.
Appears in	1994-D1-P1; 1994-D1-P4; 1994-D2-P4; 1994-D2-P5; 1995-D1-P1; 1995-D1-P5; 1995-D2-P1; 1996-D1-P1; 1996-D1-P3; 1996-D1-P6; 1997-D2-P2; 1997-D2-P4; 1998-D1-P1; 1998-D2-P1; 2000-D1-P3; 2000-D2-P1; 2001-D1-P5; 2001-D2-P3; 2002-D1-P5; 2002-D1-P6; 2002-D2-P1; 2002-D2-P5; 2003-D1-P3; 2003-D2-P1; 2004-D1-P3; 2004-D2-P1; 2004-D2-P4; 2005-D1-P1; 2005-D1-P3; 2005-D2-P3; 2006-D2-P2; 2006-D2-P4; 2007-D1-P2; 2007-D1-P3; 2007-D2-P5; 2008-D1-P2; 2008-D2-P5; 2008-D2-P6; 2009-D1-P2; 2009-D1-P3; 2009-D1-P4; 2009-D1-P5; 2009-D2-P5; 2010-D2-P4; 2011-D1-P5; 2011-D2-P4; 2011-D2-P5; 2012-D1-P2; 2012-D1-P4; 2013-D1-P1; 2013-D1-P2; 2014-D1-P1; 2015-D1-P1; 2015-D1-P5; 2015-D2-P9; 2016-D1-P2; 2016-D2-P10; 2017-D1-P1; 2018-D1-P3; 2018-D1-P4; ... (+9 more)

M-LA-05: Similarity, adapted basis, and canonical model

5 problem links

Core move	Replace a matrix/operator by a simpler similar form: shift matrix, diagonal/Jordan form, invariant-subspace block form, or basis adapted to a flag.
When to use	Use for nilpotent shifts, commuting matrices, invariant subspaces, and matrices defined by their action on a basis.
Proof skeleton	1. Build or choose the basis. 2. Express the operator in canonical form. 3. Read rank/eigenvalues/nilpotency. 4. Transfer back by similarity.
Trap watch	Similarity preserves determinant, trace, rank, eigenvalues, nilpotency, but not entrywise positivity or sparsity.
Appears in	1995-D1-P1; 1998-D2-P1; 2007-D2-P5; 2015-D2-P9; 2021-D2-P5

M-LA-06: Spectral/minimal-polynomial reduction

47 problem links

Core move	Use the polynomial satisfied by a matrix to restrict eigenvalues, diagonalizability, convergence of powers, or determinant sign.
When to use	Use when the problem contains p(A)=0, powers of A, idempotent/nilpotent/involution, or convergence of A^k.
Proof skeleton	1. Factor the polynomial. 2. Locate possible eigenvalues. 3. Use minimal polynomial/Cayley-Hamilton. 4. Infer diagonalizable/nilpotent/convergent behavior.
Trap watch	Repeated polynomial roots and real-vs-complex eigenvalues require separate checks.
Appears in	1994-D1-P4; 1994-D2-P4; 1995-D1-P1; 1995-D1-P5; 1995-D2-P1; 1996-D1-P1; 1996-D1-P3; 1997-D2-P2; 1997-D2-P4; 1998-D1-P1; 1998-D2-P1; 1999-D1-P1; 2000-D1-P3; 2000-D1-P5; 2000-D2-P6; 2001-D1-P5; 2002-D2-P1; 2002-D2-P5; 2003-D1-P3; 2003-D2-P1; 2004-D2-P1; 2004-D2-P4; 2005-D1-P1; 2005-D2-P3; 2006-D2-P6; 2008-D1-P2; 2009-D1-P2; 2009-D2-P3; 2010-D2-P4; 2011-D1-P2; 2012-D1-P2; 2013-D1-P1; 2014-D1-P1; 2014-D2-P2; 2015-D1-P1; 2015-D2-P9; 2016-D2-P10; 2017-D1-P1; 2017-D2-P8; 2018-D1-P3; 2021-D1-P1; 2022-D1-P2; 2022-D2-P5; 2022-D2-P7; 2023-D1-P2; 2023-D2-P6; 2024-D2-P7

M-LA-07: Trace cyclicity and commutator annihilation

21 problem links

Core move	Use tr(XY)=tr(YX), tr(AB-BA)=0, and trace powers to force zero spectra, scalar identities, or vanishing on commutator subspaces.
When to use	Use for commutators, rank-one commutators, trace-pairing subspaces, and functionals satisfying f(AB)=f(BA).
Proof skeleton	1. Express the object as a commutator or trace pairing. 2. Apply cyclicity. 3. Convert trace constraints to eigenvalue/nilpotency or subspace vanishing.
Trap watch	Trace zero alone does not imply nilpotent unless combined with rank/spectral constraints.
Appears in	1996-D1-P1; 1997-D1-P3; 1997-D2-P4; 2000-D1-P3; 2000-D1-P5; 2001-D1-P5; 2002-D2-P1; 2003-D2-P1; 2004-D2-P1; 2005-D1-P6; 2005-D2-P3; 2005-D2-P6; 2006-D2-P6; 2009-D2-P3; 2011-D1-P2; 2013-D1-P1; 2014-D1-P1; 2019-D2-P9; 2020-D1-P2; 2021-D1-P1; 2022-D2-P7

M-LA-08: Simultaneous diagonalization or common eigenvector induction

0 problem links

Core move	For commuting diagonalizable operators, build a common eigenbasis by restricting to eigenspaces.
When to use	Use for families of involutions, commuting matrices, and finite sets of diagonalizable operators.
Proof skeleton	1. Pick one operator. 2. Decompose into eigenspaces. 3. Show other operators preserve each eigenspace. 4. Induct.
Trap watch	Commuting alone is not enough without diagonalizability or additional hypotheses.
Appears in	

M-LA-09: Invariant subspace flag and matrix-shape count

1 problem links

Core move	Choose a basis compatible with U1 subset U2 subset V or with an invariant flag, then count free entries in the resulting block upper-triangular form.
When to use	Use when the statement asks for dimension of endomorphisms preserving subspaces.
Proof skeleton	1. Extend a basis along the flag. 2. Write the forced zero pattern. 3. Count free entries. 4. Verify all such matrices work.
Trap watch	Free-entry counts depend on row/column convention; check whether columns encode images of basis vectors.
Appears in	1998-D2-P1

M-LA-10: Linear operator on a matrix space

2 problem links

Core move	View a matrix expression X -> AX-XA, X -> MX+XM^T, or X -> AXB as a linear operator on an n^2-dimensional space.
When to use	Use when matrices themselves are variables or when infinitely many eigenvalues would contradict finite dimension.
Proof skeleton	1. Define the operator on M_n. 2. Identify eigenvectors/eigenvalues or kernel. 3. Use finite-dimensional constraints. 4. Translate to the original matrix statement.
Trap watch	Do not confuse eigenvalues of the matrix-space operator with eigenvalues of the original matrix.
Appears in	1994-D1-P4; 2022-D1-P2

M-LA-11: Gram matrix and distance-to-inner-product conversion

1 problem links

Core move	Convert squared distances into inner products via 2<u,v>=||u||^2+||v||^2-||u-v||^2, then use invertibility of Gram matrices.
When to use	Use for rational distances, equidistant sets, Hilbert geometry, simplex bounds, and rational linear dependence.
Proof skeleton	1. Choose a basis of vectors. 2. Form the Gram matrix. 3. Use distance data to prove entries lie in the desired field. 4. Solve linear coordinates.
Trap watch	Gram invertibility requires linear independence over the relevant field.
Appears in	2006-D2-P4

M-LA-12: Nilpotency from zero spectrum plus finite dimension

25 problem links

Core move	Show all eigenvalues are zero and then use Jordan form/Cayley-Hamilton to conclude some power is zero.
When to use	Use for commutators, rank-one matrices, matrix equations, and polynomial constraints.
Proof skeleton	1. Prove spectral values vanish. 2. Invoke Jordan form or Cayley-Hamilton. 3. Bound or assert a nilpotent power.
Trap watch	Zero trace is insufficient; all power sums or a rank/spectrum argument is needed.
Appears in	1994-D1-P4; 1995-D1-P1; 1995-D1-P5; 1996-D1-P3; 1997-D1-P3; 2000-D1-P3; 2000-D1-P5; 2000-D2-P6; 2001-D2-P4; 2002-D2-P5; 2003-D1-P3; 2004-D2-P4; 2005-D2-P3; 2006-D2-P6; 2009-D2-P3; 2011-D1-P2; 2013-D1-P1; 2014-D1-P1; 2014-D2-P2; 2016-D2-P10; 2017-D1-P1; 2017-D2-P8; 2019-D2-P9; 2022-D1-P2; 2022-D2-P7

M-LA-13: Determinant through eigenvalue or root product

20 problem links

Core move	Compute determinant as product of eigenvalues, often after deriving possible roots of a polynomial or diagonalizing a circulant/shift matrix.
When to use	Use when a matrix satisfies p(A)=0, is circulant-like, or has a special adjacency structure.
Proof skeleton	1. Identify eigenvalues or their allowed set. 2. Control multiplicities. 3. Multiply eigenvalues. 4. Handle real conjugate pairs.
Trap watch	For real matrices, complex nonreal eigenvalues occur in conjugate pairs.
Appears in	1995-D1-P1; 1996-D1-P1; 1997-D2-P2; 2000-D1-P3; 2001-D1-P5; 2002-D2-P1; 2003-D2-P1; 2004-D2-P1; 2007-D1-P3; 2007-D2-P4; 2009-D1-P2; 2013-D1-P1; 2014-D1-P1; 2015-D1-P1; 2016-D2-P10; 2018-D1-P3; 2019-D1-P5; 2021-D1-P1; 2023-D2-P6; 2024-D2-P7

M-LA-14: Parity or zero-pattern block determinant split

1 problem links

Core move	Permute rows/columns to reveal bipartite block structure, square determinant, or forced zero determinant.
When to use	Use for matrices whose nonzero entries depend on parity, prime sums, odd/even positions, or bipartite adjacency.
Proof skeleton	1. Partition indices. 2. Permute rows/columns. 3. Read block form. 4. Compute determinant as zero or square.
Trap watch	Permutation changes determinant sign; absolute value often avoids the sign issue.
Appears in	2008-D2-P5

M-LA-15: AB versus BA transfer

11 problem links

Core move	Use that AB and BA have the same nonzero eigenvalues, or that characteristic polynomials of I+AB and I+BA match in compatible dimensions.
When to use	Use for rectangular factorizations, products AB/BA, and matrix exponentials with AB and BA.
Proof skeleton	1. Express target as cI+AB and cI+BA. 2. Compare characteristic polynomials/nonzero spectra. 3. Transfer nilpotency or determinant data.
Trap watch	Zero eigenvalue multiplicities may differ if matrices are rectangular; check dimensions.
Appears in	2000-D1-P3; 2000-D2-P6; 2006-D2-P3; 2013-D1-P1; 2013-D1-P4; 2014-D2-P3; 2014-D2-P5; 2015-D1-P2; 2015-D1-P3; 2018-D1-P4; 2023-D1-P1

M-LA-16: Trace pairing and orthogonal matrix subspaces

7 problem links

Core move	Use the bilinear form (X,Y)=tr(XY) or tr(X^T Y) to bound dimensions or identify functionals.
When to use	Use for matrix subspaces with tr(XY)=0, linear functionals on matrices, and symmetric/skew-symmetric decompositions.
Proof skeleton	1. Identify a nondegenerate pairing. 2. Compute an orthogonal complement. 3. Bound dimensions. 4. Exhibit an extremal subspace.
Trap watch	tr(XY) is not positive definite on all matrices; use symmetric matrices when positivity is needed.
Appears in	1997-D2-P2; 1998-D1-P1; 2003-D2-P1; 2004-D2-P1; 2018-D1-P3; 2021-D1-P1; 2023-D2-P6

M-LA-17: Sign forcing from positive matrices or inner products

1 problem links

Core move	Use positivity of entries or norms to force each column/row/vector combination to contain positive and negative contributions.
When to use	Use in inverse-matrix zero-counting, norm inequalities, and orthogonal vector selection.
Proof skeleton	1. Write the zero equation as a positive-weighted sum. 2. Infer sign diversity among coefficients. 3. Count nonzero entries or choose a vector with nonpositive inner product.
Trap watch	Strict positivity is essential; nonnegative entries may allow all-zero terms.
Appears in	1994-D1-P1

M-LA-18: Commutator telescoping through powers

1 problem links

Core move	Expand F^kG-GF^k as a telescoping sum of F^{k-i}(FG-GF)F^{i-1}.
When to use	Use when a commutator relation is given and powers of one operator appear.
Proof skeleton	1. Write the telescoping sum. 2. Insert the base commutator. 3. Sum identical terms. 4. Use the result spectrally or dimensionally.
Trap watch	Order of factors matters; telescoping is the safe way to move G past powers of F.
Appears in	1994-D1-P4

17.3.7 Real analysis

M-RA-01: Auxiliary function with derivative sign *135 problem links*

Core move	Create a function whose derivative equals the desired expression or has a controlled sign.
When to use	Use for inequalities, uniqueness, endpoint constraints, and differential identities.
Proof skeleton	1. Guess the antiderivative/integrating expression. 2. Differentiate. 3. Use monotonicity. 4. Evaluate endpoints or limits.
Trap watch	Endpoint limits must be justified if the interval is open.
Appears in	1994-D1-P2; 1994-D1-P5; 1994-D1-P6; 1994-D2-P1; 1994-D2-P3; 1994-D2-P6; 1995-D1-P2; 1995-D1-P3; 1995-D1-P4; 1995-D2-P2; 1995-D2-P4; 1995-D2-P5; 1995-D2-P6; 1996-D1-P2; 1996-D1-P4; 1996-D1-P5; 1996-D2-P1; 1996-D2-P3; 1996-D2-P6; 1997-D1-P2; 1997-D1-P4; 1997-D2-P1; 1997-D2-P3; 1997-D2-P5; 1997-D2-P6; 1998-D1-P3; 1998-D1-P4; 1998-D1-P5; 1998-D1-P6; 1998-D2-P2; 1998-D2-P3; 1998-D2-P4; 1998-D2-P6; 1999-D1-P2; 1999-D1-P4; 1999-D1-P6; 1999-D2-P3; 1999-D2-P4; 2000-D1-P1; 2000-D1-P2; 2000-D1-P4; 2000-D1-P6; 2000-D2-P2; 2000-D2-P5; 2001-D1-P3; 2001-D1-P6; 2001-D2-P2; 2001-D2-P5; 2002-D1-P2; 2002-D1-P4; 2002-D1-P5; 2002-D1-P6; 2002-D2-P6; 2003-D1-P1; 2003-D1-P5; 2003-D2-P2; 2003-D2-P5; 2003-D2-P6; 2004-D1-P2; 2004-D1-P5; ... (+75 more)

M-RA-02: Integrating factor and ODE comparison *33 problem links*

Core move	Transform h’(x)+a(x)h(x) into a derivative of h(x) times an exponential or weight.
When to use	Use for first-order differential inequalities and growth/decay estimates.
Proof skeleton	1. Identify the homogeneous equation. 2. Multiply by integrating factor. 3. Integrate inequality. 4. Compare with explicit barrier.
Trap watch	Sign of the integrating factor is positive, but inequality direction depends on the multiplier and interval direction.
Appears in	1994-D1-P5; 1994-D2-P1; 1994-D2-P3; 1995-D1-P4; 1996-D1-P2; 1996-D2-P3; 1997-D2-P1; 1997-D2-P5; 1998-D1-P4; 1999-D1-P4; 1999-D1-P6; 1999-D2-P3; 2000-D1-P6; 2000-D2-P5; 2002-D2-P6; 2003-D1-P5; 2003-D2-P5; 2004-D2-P5; 2005-D1-P4; 2005-D1-P5; 2005-D2-P4; 2007-D1-P1; 2007-D1-P4; 2008-D1-P5; 2008-D2-P2; 2009-D2-P2; 2010-D1-P1; 2010-D2-P3; 2011-D2-P4; 2012-D2-P5; 2020-D2-P7; 2021-D2-P6; 2024-D1-P4

M-RA-03: Rolle chain and divided-difference localization *60 problem links*

Core move	Use repeated Rolle/MVT or divided differences to force a derivative of high order to take a prescribed value.
When to use	Use when a function has many zeros, equal endpoint values, interpolation data, or derivative constraints.
Proof skeleton	1. Build an interpolation polynomial if needed. 2. Subtract it. 3. Apply Rolle repeatedly. 4. Read the derivative value.
Trap watch	Need enough differentiability and distinct/interior points for each Rolle step.
Appears in	1994-D1-P2; 1994-D1-P6; 1994-D2-P1; 1994-D2-P3; 1994-D2-P4; 1995-D1-P2; 1995-D1-P3; 1995-D2-P6; 1996-D1-P5; 1996-D2-P1; 1996-D2-P5; 1996-D2-P6; 1997-D2-P1; 1997-D2-P3; 1997-D2-P6; 1998-D1-P4; 1998-D1-P6; 1998-D2-P6; 1999-D1-P6; 2000-D1-P2; 2000-D1-P6; 2000-D2-P2; 2000-D2-P5; 2001-D1-P6; 2001-D2-P6; 2002-D1-P2; 2002-D1-P4; 2002-D1-P5; 2004-D2-P2; 2005-D1-P3; 2005-D1-P5; 2005-D2-P4; 2005-D2-P5; 2006-D1-P1; 2006-D1-P5; 2006-D1-P6; 2007-D2-P1; 2007-D2-P3; 2007-D2-P6; 2008-D1-P1; 2009-D2-P2; 2010-D1-P1; 2010-D1-P5; 2011-D1-P1; 2012-D1-P4; 2013-D1-P2; 2014-D2-P3; 2015-D2-P6; 2016-D1-P1; 2016-D2-P7; 2017-D1-P5; 2017-D2-P6; 2017-D2-P9; 2018-D1-P1; 2019-D1-P1; 2019-D2-P6; 2020-D2-P5; 2023-D1-P1; 2023-D1-P2; 2023-D2-P7

M-RA-04: Taylor expansion and local asymptotic bookkeeping *12 problem links*

Core move	Expand near an endpoint or critical point to compare leading terms and estimate errors.
When to use	Use for sharp constants, differentiability counterexamples, local extrema, and limits.
Proof skeleton	1. Choose expansion center. 2. Compute leading nonzero terms. 3. Control remainder. 4. Compare orders.
Trap watch	Little-o terms must be uniform enough over the domain used.
Appears in	1994-D1-P5; 1995-D1-P6; 1996-D1-P5; 1996-D2-P5; 1997-D1-P1; 1997-D2-P1; 1997-D2-P3; 2000-D2-P5; 2001-D1-P6; 2004-D1-P6; 2005-D1-P5; 2009-D2-P2

M-RA-05: L-Hospital and ratio/growth comparison

30 problem links

Core move	Convert a ratio limit into a derivative ratio, or compare numerator growth with denominator growth.
When to use	Use for limits involving exponentials, integrals, inverse functions, and asymptotic normalizations.
Proof skeleton	1. Check indeterminate form and denominator behavior. 2. Differentiate numerator/denominator. 3. Evaluate limit. 4. Return to original ratio.
Trap watch	Do not apply l-Hospital without verifying hypotheses or monotone unbounded denominator conditions.
Appears in	1994-D1-P2; 1994-D1-P5; 1994-D2-P6; 1995-D1-P3; 1995-D1-P6; 1995-D2-P2; 1996-D1-P4; 1996-D2-P1; 1996-D2-P5; 1997-D1-P1; 1997-D1-P2; 1997-D2-P1; 2001-D1-P3; 2001-D1-P6; 2001-D2-P5; 2003-D1-P1; 2003-D2-P2; 2004-D1-P6; 2005-D1-P5; 2010-D1-P2; 2010-D1-P3; 2014-D1-P2; 2015-D2-P7; 2017-D2-P6; 2017-D2-P9; 2018-D2-P10; 2020-D2-P8; 2021-D1-P4; 2024-D1-P2; 2024-D2-P8

M-RA-06: Riemann sum and sum-integral conversion

66 problem links

Core move	Recognize normalized sums as Riemann sums, often after scaling variables.
When to use	Use for limits of sums, logarithmic products, and series depending on a large parameter.
Proof skeleton	1. Choose mesh size. 2. Rewrite summand as h f(kh). 3. Check integrability/tail. 4. Evaluate integral.
Trap watch	Infinite domains require tail control, not only local Riemann convergence.
Appears in	1994-D1-P2; 1994-D1-P5; 1994-D2-P6; 1995-D1-P3; 1995-D2-P2; 1995-D2-P5; 1996-D1-P2; 1996-D1-P4; 1996-D1-P5; 1996-D2-P1; 1996-D2-P5; 1997-D1-P1; 1997-D1-P2; 1997-D2-P3; 1997-D2-P4; 1998-D1-P3; 1998-D2-P4; 1999-D1-P2; 2000-D2-P4; 2000-D2-P5; 2000-D2-P6; 2001-D1-P3; 2001-D1-P6; 2001-D2-P2; 2001-D2-P5; 2002-D1-P4; 2002-D1-P6; 2003-D1-P1; 2003-D1-P2; 2003-D1-P3; 2003-D2-P2; 2003-D2-P5; 2003-D2-P6; 2004-D1-P3; 2004-D2-P3; 2004-D2-P4; 2004-D2-P6; 2005-D1-P5; 2005-D2-P1; 2005-D2-P3; 2006-D1-P1; 2006-D1-P3; 2006-D1-P4; 2009-D1-P1; 2010-D1-P2; 2010-D1-P3; 2010-D2-P1; 2010-D2-P5; 2011-D1-P1; 2011-D2-P1; 2012-D1-P1; 2012-D2-P2; 2013-D2-P4; 2014-D1-P2; 2015-D2-P7; 2015-D2-P8; 2016-D2-P8; 2017-D2-P6; 2017-D2-P9; 2018-D2-P7; ... (+6 more)

M-RA-07: Split into main range and tail/error range

63 problem links

Core move	Divide sums or integrals into small, central, and tail regions to isolate the main contribution.
When to use	Use for asymptotics, logarithmic sums, Laplace-type estimates, and convergence bounds.
Proof skeleton	1. Pick cutoff depending on parameter. 2. Bound tails crudely. 3. Estimate main range sharply. 4. Optimize cutoff.
Trap watch	The cutoff must make all error terms vanish at the needed rate.
Appears in	1994-D1-P5; 1994-D2-P6; 1995-D1-P3; 1995-D2-P2; 1995-D2-P5; 1996-D1-P2; 1996-D1-P4; 1996-D1-P5; 1996-D2-P1; 1996-D2-P5; 1997-D1-P1; 1997-D1-P2; 1997-D2-P3; 1997-D2-P4; 1998-D1-P3; 1998-D2-P4; 1999-D1-P2; 2000-D2-P4; 2000-D2-P5; 2000-D2-P6; 2001-D1-P3; 2001-D1-P6; 2001-D2-P2; 2001-D2-P5; 2002-D1-P6; 2003-D1-P1; 2003-D1-P2; 2003-D1-P3; 2003-D2-P2; 2003-D2-P5; 2003-D2-P6; 2004-D2-P3; 2004-D2-P4; 2004-D2-P6; 2005-D1-P5; 2005-D2-P1; 2005-D2-P3; 2006-D1-P1; 2006-D1-P3; 2006-D1-P4; 2009-D1-P1; 2010-D1-P2; 2010-D1-P3; 2010-D2-P1; 2010-D2-P5; 2011-D1-P1; 2011-D2-P1; 2012-D1-P1; 2012-D2-P2; 2013-D2-P4; 2014-D1-P2; 2015-D2-P7; 2015-D2-P8; 2016-D2-P8; 2017-D2-P6; 2017-D2-P9; 2018-D2-P7; 2018-D2-P10; 2019-D2-P7; 2020-D2-P8; ... (+3 more)

M-RA-08: Compactness plus extreme value theorem

86 problem links

Core move	Reduce an unbounded problem to a compact set and use continuity to attain maxima/minima.
When to use	Use for global extrema, maximum/minimum classification, and bounded closed geometry.
Proof skeleton	1. Show values outside a compact set cannot be extremal. 2. Apply EVT. 3. Analyze critical/boundary points.
Trap watch	The compact set must contain all possible extrema; do not assume it.
Appears in	1994-D2-P2; 1994-D2-P6; 1995-D1-P2; 1995-D1-P4; 1995-D1-P6; 1995-D2-P2; 1995-D2-P4; 1995-D2-P5; 1995-D2-P6; 1996-D1-P2; 1996-D1-P4; 1996-D1-P5; 1996-D1-P6; 1996-D2-P4; 1996-D2-P6; 1997-D2-P3; 1998-D1-P4; 1998-D1-P5; 1998-D1-P6; 1998-D2-P2; 1999-D1-P1; 1999-D1-P2; 1999-D2-P3; 2000-D1-P1; 2000-D1-P2; 2000-D1-P4; 2000-D2-P2; 2001-D1-P3; 2001-D2-P2; 2001-D2-P3; 2001-D2-P4; 2001-D2-P6; 2002-D1-P4; 2002-D1-P5; 2002-D2-P6; 2003-D1-P1; 2003-D2-P3; 2004-D1-P2; 2004-D1-P3; 2004-D1-P5; 2004-D2-P5; 2005-D1-P3; 2005-D2-P1; 2005-D2-P3; 2005-D2-P5; 2006-D1-P5; 2006-D2-P2; 2006-D2-P3; 2007-D2-P1; 2007-D2-P3; 2007-D2-P6; 2008-D1-P1; 2008-D2-P1; 2009-D1-P4; 2010-D2-P2; 2011-D1-P1; 2012-D1-P4; 2013-D1-P2; 2013-D1-P3; 2013-D1-P4; ... (+26 more)

M-RA-09: Convexity, Jensen, tangent, and chord inequalities

161 problem links

Core move	Use convex/concave structure to compare averages, chords, tangents, or equality cases.
When to use	Use for inequalities, optimization, norm estimates, and concave/convex functions.
Proof skeleton	1. Identify convexity sign. 2. Choose Jensen/tangent/chord form. 3. Apply to the correct weights. 4. Analyze equality.
Trap watch	Convexity direction reverses for concave functions; equality cases depend on strictness.
Appears in	1994-D1-P1; 1994-D1-P2; 1994-D1-P3; 1994-D1-P4; 1994-D1-P6; 1994-D2-P1; 1994-D2-P2; 1994-D2-P6; 1995-D1-P2; 1995-D1-P4; 1995-D1-P6; 1995-D2-P2; 1995-D2-P5; 1996-D1-P2; 1996-D1-P4; 1996-D1-P5; 1996-D2-P2; 1996-D2-P4; 1996-D2-P5; 1996-D2-P6; 1997-D1-P1; 1997-D1-P2; 1997-D1-P5; 1997-D2-P1; 1997-D2-P3; 1998-D1-P2; 1998-D1-P3; 1998-D1-P4; 1998-D1-P5; 1998-D1-P6; 1998-D2-P1; 1998-D2-P2; 1998-D2-P3; 1998-D2-P5; 1998-D2-P6; 1999-D1-P2; 1999-D1-P3; 1999-D1-P4; 1999-D1-P6; 1999-D2-P3; 1999-D2-P4; 1999-D2-P5; 2000-D1-P4; 2000-D2-P1; 2000-D2-P2; 2000-D2-P3; 2000-D2-P4; 2001-D1-P4; 2001-D1-P6; 2001-D2-P2; 2001-D2-P5; 2001-D2-P6; 2002-D1-P2; 2002-D1-P3; 2002-D1-P6; 2002-D2-P3; 2002-D2-P4; 2002-D2-P6; 2003-D1-P4; 2003-D1-P6; ... (+101 more)

M-RA-10: Cauchy-Schwarz, Holder, and norm comparison

23 problem links

Core move	Convert sums/integrals/inner products into norm bounds and equality conditions.
When to use	Use for L2 inequalities, Euclidean vector problems, and bounding integrals by energy.
Proof skeleton	1. Identify product or inner product. 2. Apply the norm inequality. 3. Simplify with known constraints. 4. Check equality.
Trap watch	Equality requires proportionality; do not ignore when sharpness is asked.
Appears in	1994-D1-P6; 1994-D2-P2; 1994-D2-P5; 1995-D2-P1; 1996-D2-P6; 1997-D1-P5; 1998-D2-P1; 2002-D1-P3; 2002-D1-P6; 2004-D2-P3; 2005-D2-P5; 2006-D2-P4; 2008-D1-P5; 2008-D2-P2; 2008-D2-P6; 2009-D1-P4; 2009-D2-P1; 2010-D1-P1; 2010-D1-P5; 2015-D1-P5; 2015-D2-P9; 2021-D2-P8; 2023-D2-P8

M-RA-11: Integration by parts, Fubini, and weighted identities

39 problem links

Core move	Move derivatives or weights between functions, reverse order of integration, or express a constraint as an integral identity.
When to use	Use for integral inequalities, moment constraints, and estimates with cumulative integrals.
Proof skeleton	1. Define cumulative function/weight. 2. Apply integration by parts or Fubini. 3. Use sign/monotonicity. 4. Bound the result.
Trap watch	Boundary terms often carry the main contribution; compute them exactly.
Appears in	1994-D2-P1; 1994-D2-P3; 1995-D1-P2; 1995-D2-P5; 1996-D1-P2; 1996-D1-P4; 1996-D2-P6; 1997-D1-P1; 1997-D1-P2; 1998-D1-P3; 1998-D1-P6; 1999-D1-P4; 1999-D2-P3; 1999-D2-P4; 2001-D1-P1; 2002-D1-P3; 2003-D1-P1; 2003-D2-P6; 2004-D1-P2; 2004-D2-P5; 2005-D2-P1; 2007-D2-P2; 2008-D1-P4; 2010-D1-P2; 2010-D2-P1; 2010-D2-P2; 2011-D2-P1; 2011-D2-P3; 2012-D1-P1; 2012-D2-P2; 2013-D1-P5; 2013-D2-P4; 2014-D1-P4; 2015-D2-P8; 2018-D1-P1; 2019-D2-P6; 2019-D2-P7; 2019-D2-P10; 2022-D1-P1

M-RA-12: Squeeze, dominated convergence, and monotone convergence

1 problem links

Core move	Pass limits through functions or integrals by trapping between known limits or using uniform bounds.
When to use	Use for iterated functions, orthonormal systems, pointwise subsequences, and integral limits.
Proof skeleton	1. Establish pointwise or monotone convergence. 2. Find a dominating bound. 3. Pass to the limit. 4. Compare with invariant norm/mass.
Trap watch	Pointwise convergence alone is not enough for integrals without domination or monotonicity.
Appears in	2003-D1-P5

M-RA-13: Periodic averaging and oscillation cancellation

1 problem links

Core move	Partition an interval into periods or small cells so a rapidly oscillating factor averages to its mean.
When to use	Use for integrals of f(x)g(nx), trigonometric sums, and periodic kernels.
Proof skeleton	1. Partition into n subintervals. 2. Freeze the slow factor. 3. Use periodicity of the fast factor. 4. Control the error by uniform continuity.
Trap watch	The slow factor must be uniformly continuous or otherwise controlled.
Appears in	1994-D1-P5

M-RA-14: Endpoint Laplace-type asymptotic comparison

0 problem links

Core move	For n\int f(x)^n dx, locate the maximum of f and expand near it.
When to use	Use for powers of functions, concave functions, and sharp endpoint constants.
Proof skeleton	1. Show the maximum equals the relevant scale. 2. Prove it occurs at an endpoint or handle interior contradiction. 3. Compare with linear/quadratic model. 4. Pass to limit.
Trap watch	Interior flat maxima may change the scale; rule them out before using endpoint asymptotics.
Appears in	

M-RA-15: Recurrence transformed to monotone quotient or telescoping form

17 problem links

Core move	Normalize a recurrence so the transformed sequence is monotone, telescoping, or bounded by barriers.
When to use	Use for nonlinear sequences, products, and iteration limits.
Proof skeleton	1. Guess normalization. 2. Derive recurrence for normalized variables. 3. Prove monotonicity/bounds. 4. Identify the limit.
Trap watch	The normalization must match the dominant growth rate.
Appears in	1998-D2-P6; 2001-D2-P2; 2002-D2-P3; 2003-D1-P1; 2003-D1-P6; 2003-D2-P6; 2008-D2-P3; 2010-D2-P2; 2011-D2-P1; 2012-D2-P2; 2014-D1-P3; 2015-D1-P5; 2017-D2-P8; 2018-D2-P7; 2019-D1-P4; 2021-D1-P3; 2024-D2-P8

M-RA-16: Density, Baire/category, and pathological construction *20 problem links*

Core move	Use dense sets, countability, accumulation points, or category-style reasoning to force or forbid pathological behavior.
When to use	Use for continuous functions with many zeros, dense extrema, closed orbit sets, and pointwise convergence counterexamples.
Proof skeleton	1. Identify dense/countable/closed structure. 2. Use accumulation/interval/category property. 3. Derive continuity or differentiability contradiction.
Trap watch	Avoid assuming countable sets cannot be dense; distinguish density from uncountability.
Appears in	1994-D2-P2; 1995-D1-P6; 1995-D2-P6; 1996-D1-P6; 1996-D2-P4; 1998-D1-P4; 1998-D2-P5; 2000-D2-P2; 2002-D1-P4; 2003-D2-P3; 2003-D2-P5; 2004-D1-P3; 2004-D2-P3; 2006-D2-P2; 2007-D2-P3; 2008-D1-P1; 2009-D1-P3; 2021-D2-P7; 2023-D2-P8; 2023-D2-P9

M-RA-17: Approximation by polynomials or kernels *1 problem links*

Core move	Approximate continuous functions by polynomials or use positive kernels to control coefficient multipliers.
When to use	Use for Weierstrass approximation, Chebyshev extremality, Fejer kernels, and polynomial interpolation estimates.
Proof skeleton	1. Approximate the target in sup norm. 2. Symmetrize if needed. 3. Replace missing low-degree terms by an approximation. 4. Bound the error.
Trap watch	Approximation must preserve parity or endpoint constraints if these are part of the target.
Appears in	2011-D2-P2

M-RA-18: Trigonometric/hyperbolic identity plus monotonicity *1 problem links*

Core move	Use identities such as sin addition, cosh recurrences, product-to-sum, or tangent monotonicity to transform inequalities.
When to use	Use for trigonometric limits, roots on the unit circle, and hyperbolic rationality problems.
Proof skeleton	1. Rewrite with standard identity. 2. Reduce to monotonicity/sign. 3. Compare endpoints or recurrences.
Trap watch	Check branch/period restrictions; inverse trigonometric functions introduce domain conditions.
Appears in	1995-D2-P5

17.3.8 Universal proof infrastructure

M-UNI-01: Normalization, WLOG reduction, and scaling *0 problem links*

Core move	Replace the given object by an equivalent normalized one: shift the origin, scale a variable, divide by a leading coefficient, assume an order, or conjugate/change basis.
When to use	Use when the statement is invariant under translation, scaling, relabeling, conjugation, multiplying by a nonzero constant, or choosing coordinates.
Proof skeleton	1. State the invariant transformation. 2. Normalize the parameter or object. 3. Solve the normalized problem. 4. Translate the conclusion back.
Trap watch	Check that the transformation preserves all hypotheses, especially integer, positivity, interval-endpoint, and determinant/sign conditions.
Appears in	

M-UNI-02: Small cases, equality cases, and example probing *7 problem links*

Core move	Before proving, test boundary values, n=1, n=2, zero/equality cases, and expected extremizers.
When to use	Use for classification, sharp constants, recurrence guesses, determinant patterns, and counterexample design.
Proof skeleton	1. Compute minimal cases. 2. Identify the repeated invariant. 3. Guess the statement. 4. Prove by a general argument.
Trap watch	Do not present numerical pattern recognition as proof; use it only to find the invariant.
Appears in	1996-D1-P1; 1998-D1-P1; 2002-D2-P1; 2007-D2-P4; 2011-D2-P3; 2019-D1-P1; 2020-D1-P1

M-UNI-03: Extremal construction and equality-case engineering

103 problem links

Core move	Build an example by forcing equality in the inequalities or structural constraints used in the proof.
When to use	Use when the problem asks for best constant, maximum/minimum, existence, sharpness, or a counterexample.
Proof skeleton	1. Identify the equality conditions of the upper/lower bound. 2. Solve the equality equations. 3. Verify all original hypotheses. 4. Record uniqueness if needed.
Trap watch	Construction must be checked against every hypothesis, not only against the target equality.
Appears in	1994-D1-P2; 1994-D1-P4; 1994-D2-P1; 1994-D2-P5; 1995-D1-P2; 1995-D1-P6; 1995-D2-P1; 1995-D2-P6; 1996-D1-P3; 1997-D1-P2; 1997-D1-P4; 1997-D1-P5; 1997-D1-P6; 1997-D2-P1; 1997-D2-P4; 1997-D2-P6; 1998-D1-P2; 1998-D1-P4; 1998-D1-P6; 1998-D2-P6; 1999-D1-P2; 1999-D1-P3; 1999-D1-P6; 1999-D2-P3; 1999-D2-P4; 1999-D2-P5; 2000-D1-P1; 2000-D1-P4; 2000-D2-P2; 2001-D2-P3; 2001-D2-P5; 2001-D2-P6; 2002-D1-P2; 2002-D1-P5; 2003-D1-P2; 2003-D2-P3; 2003-D2-P4; 2004-D1-P5; 2004-D2-P2; 2004-D2-P3; 2004-D2-P5; 2005-D1-P3; 2005-D2-P1; 2005-D2-P4; 2006-D1-P1; 2007-D1-P2; 2007-D2-P1; 2007-D2-P3; 2007-D2-P5; 2008-D1-P3; 2008-D1-P5; 2008-D2-P3; 2008-D2-P4; 2008-D2-P6; 2009-D1-P1; 2009-D2-P2; 2009-D2-P5; 2010-D1-P1; 2010-D1-P4; 2010-D2-P1; ... (+43 more)

M-UNI-04: Contradiction with minimal or maximal counterexample

7 problem links

Core move	Assume failure and choose a counterexample minimal in size, degree, dimension, number of terms, or extremal value.
When to use	Use in graph cycles, set systems, words, finite families, divisibility obstructions, and compactness problems.
Proof skeleton	1. Define a well-ordered complexity. 2. Choose a minimal counterexample. 3. Remove, contract, split, or shorten. 4. Contradict minimality.
Trap watch	The chosen minimal parameter must actually decrease under the operation.
Appears in	1997-D1-P6; 1999-D1-P2; 2002-D1-P2; 2013-D1-P5; 2013-D2-P4; 2018-D1-P2; 2024-D1-P3

M-UNI-05: Induction with strengthened hypothesis

72 problem links

Core move	Prove a stronger statement so that the induction step has enough room to preserve the invariant.
When to use	Use for sequences, set systems, triangulations, polynomial recurrences, finite families, and recursive maps.
Proof skeleton	1. Strengthen the target. 2. Prove base cases. 3. Remove or add one structured piece. 4. Reassemble without losing hypotheses.
Trap watch	Weak induction often fails; strengthen the statement before forcing the step.
Appears in	1994-D1-P3; 1994-D2-P3; 1994-D2-P5; 1995-D1-P4; 1995-D1-P5; 1996-D1-P3; 1996-D1-P4; 1996-D1-P6; 1996-D2-P2; 1997-D1-P3; 1997-D1-P4; 1997-D1-P6; 1997-D2-P2; 1997-D2-P3; 1997-D2-P5; 1998-D1-P3; 1998-D2-P1; 1999-D1-P4; 1999-D1-P5; 1999-D2-P1; 1999-D2-P5; 2000-D1-P3; 2000-D2-P3; 2001-D1-P2; 2001-D1-P5; 2001-D2-P1; 2001-D2-P2; 2002-D1-P3; 2002-D1-P4; 2002-D2-P2; 2002-D2-P3; 2002-D2-P4; 2002-D2-P5; 2003-D1-P5; 2003-D1-P6; 2003-D2-P6; 2004-D1-P2; 2004-D1-P4; 2004-D1-P5; 2005-D1-P2; 2005-D1-P6; 2006-D1-P3; 2006-D1-P6; 2006-D2-P1; 2006-D2-P4; 2007-D2-P2; 2008-D1-P2; 2008-D1-P3; 2009-D1-P2; 2009-D1-P5; 2009-D2-P3; 2010-D1-P3; 2010-D2-P2; 2010-D2-P4; 2011-D1-P5; 2011-D2-P1; 2011-D2-P4; 2012-D2-P2; 2013-D1-P1; 2013-D2-P1; ... (+12 more)

M-UNI-06: Compactness or extremal-point selection

43 problem links

Core move	Use closed-boundedness, finite choice, or an extremal element to obtain an object where inequalities can be tested sharply.
When to use	Use for existence of extrema, convex bodies, closest points, continuous functions, finite graph/set arguments, and density proofs.
Proof skeleton	1. Reduce to a compact or finite set. 2. Select an extremal point. 3. Use local perturbation or comparison. 4. Conclude existence/contradiction.
Trap watch	Verify compactness; closedness or boundedness may be missing in open intervals or noncompact spaces.
Appears in	1994-D2-P2; 1995-D1-P6; 1995-D2-P1; 1995-D2-P4; 1995-D2-P6; 1996-D1-P3; 1996-D1-P6; 1996-D2-P4; 1997-D2-P4; 1998-D1-P4; 1998-D2-P2; 1999-D1-P1; 1999-D2-P4; 2000-D1-P1; 2000-D2-P2; 2001-D2-P4; 2001-D2-P6; 2002-D2-P6; 2003-D1-P2; 2003-D2-P3; 2004-D1-P3; 2004-D1-P5; 2005-D2-P1; 2005-D2-P3; 2005-D2-P4; 2005-D2-P5; 2006-D2-P2; 2007-D2-P3; 2007-D2-P6; 2011-D1-P1; 2013-D1-P2; 2013-D1-P3; 2013-D2-P3; 2015-D2-P9; 2016-D2-P7; 2017-D1-P2; 2017-D1-P3; 2017-D2-P10; 2019-D2-P6; 2020-D1-P2; 2021-D1-P3; 2021-D2-P5; 2023-D2-P10

M-UNI-07: Special substitution and parameter freezing

141 problem links

Core move	Substitute special values, set one variable to zero, let parameters coincide, freeze all but one variable, or use a strategically chosen test object.
When to use	Use in functional equations, polynomial identities, matrix equations, inequalities, and group/ring identities.
Proof skeleton	1. Identify a variable that can be controlled. 2. Substitute a value that kills terms or reveals symmetry. 3. Iterate the substitution if needed.
Trap watch	Special values may be outside the domain; check positivity, nonzero, interval, and invertibility restrictions.
Appears in	1994-D1-P5; 1994-D2-P1; 1994-D2-P2; 1994-D2-P3; 1995-D1-P2; 1995-D1-P4; 1995-D2-P2; 1996-D1-P2; 1996-D2-P1; 1996-D2-P3; 1996-D2-P5; 1997-D1-P1; 1997-D2-P1; 1997-D2-P5; 1998-D1-P6; 1998-D2-P2; 1999-D1-P3; 1999-D1-P4; 1999-D1-P6; 1999-D2-P1; 1999-D2-P3; 1999-D2-P4; 2000-D1-P2; 2000-D1-P6; 2000-D2-P5; 2001-D1-P1; 2001-D1-P3; 2001-D2-P5; 2002-D1-P2; 2002-D2-P2; 2002-D2-P6; 2003-D1-P4; 2003-D1-P5; 2003-D2-P1; 2003-D2-P5; 2004-D2-P1; 2004-D2-P2; 2004-D2-P4; 2004-D2-P5; 2005-D1-P1; 2005-D1-P3; 2005-D1-P4; 2005-D1-P5; 2005-D2-P2; 2005-D2-P4; 2005-D2-P5; 2006-D1-P5; 2006-D2-P2; 2006-D2-P3; 2007-D1-P1; 2007-D1-P4; 2007-D2-P1; 2007-D2-P5; 2008-D1-P1; 2008-D1-P4; 2008-D2-P2; 2008-D2-P3; 2009-D1-P2; 2009-D2-P1; 2009-D2-P2; ... (+81 more)

M-UNI-08: Sign, zero, and boundary case splitting

58 problem links

Core move	Split the proof according to zero/nonzero, positive/negative, equality/strict inequality, boundary/interior, or singular/nonsingular cases.
When to use	Use when a formula requires division, a maximum may occur at the boundary, roots have multiplicities, or matrices may be singular.
Proof skeleton	1. Isolate degenerate cases. 2. Solve the generic case using stronger tools. 3. Return to edge cases explicitly.
Trap watch	Do not divide by an expression before proving it is nonzero.
Appears in	1994-D1-P2; 1994-D1-P4; 1994-D2-P2; 1994-D2-P5; 1995-D1-P6; 1995-D2-P6; 1997-D1-P2; 1997-D1-P4; 1997-D1-P6; 1997-D2-P6; 1998-D1-P2; 1998-D2-P6; 1999-D1-P2; 1999-D1-P4; 1999-D1-P6; 1999-D2-P5; 2000-D1-P1; 2002-D1-P2; 2003-D2-P4; 2004-D2-P3; 2006-D1-P1; 2007-D2-P1; 2007-D2-P3; 2007-D2-P5; 2008-D1-P1; 2008-D1-P3; 2008-D1-P4; 2009-D1-P1; 2009-D2-P5; 2010-D1-P4; 2010-D2-P1; 2011-D1-P2; 2011-D2-P5; 2012-D2-P4; 2013-D1-P5; 2013-D2-P4; 2014-D1-P2; 2014-D2-P2; 2015-D1-P2; 2016-D1-P1; 2016-D1-P2; 2018-D1-P2; 2018-D1-P3; 2018-D1-P4; 2018-D2-P9; 2019-D1-P2; 2019-D2-P6; 2020-D1-P1; 2020-D2-P5; 2020-D2-P6; 2022-D1-P2; 2022-D1-P3; 2023-D1-P1; 2023-D1-P3; 2023-D2-P7; 2024-D1-P1; 2024-D1-P3; 2024-D2-P9

M-UNI-09: Invariant extraction and invariant comparison

115 problem links

Core move	Replace the object by an invariant: degree, trace, determinant, rank, parity, residue, dimension, sign pattern, total mass, or graph cycle count.
When to use	Use whenever the statement is preserved by allowed operations or when direct construction is too large.
Proof skeleton	1. Identify the invariant. 2. Show it is preserved or bounded. 3. Evaluate it in a terminal/known case. 4. Compare with the target.
Trap watch	The invariant must distinguish the required alternatives; a weak invariant may not close the proof.
Appears in	1994-D1-P3; 1994-D2-P3; 1995-D1-P2; 1995-D1-P4; 1995-D1-P5; 1995-D2-P2; 1996-D1-P1; 1996-D2-P1; 1996-D2-P2; 1997-D1-P1; 1998-D1-P1; 1998-D1-P6; 1998-D2-P1; 1999-D1-P3; 1999-D1-P5; 1999-D2-P1; 1999-D2-P5; 2000-D1-P2; 2000-D2-P3; 2001-D1-P1; 2001-D1-P3; 2001-D1-P5; 2002-D1-P3; 2002-D2-P1; 2002-D2-P2; 2002-D2-P4; 2003-D1-P4; 2003-D2-P1; 2004-D1-P4; 2004-D2-P1; 2004-D2-P4; 2005-D1-P1; 2005-D1-P3; 2005-D2-P2; 2006-D1-P5; 2006-D2-P3; 2007-D2-P2; 2007-D2-P4; 2007-D2-P5; 2008-D1-P2; 2008-D1-P4; 2008-D2-P3; 2009-D1-P2; 2010-D1-P2; 2011-D2-P3; 2012-D1-P2; 2012-D1-P3; 2012-D1-P4; 2012-D2-P1; 2012-D2-P3; 2012-D2-P4; 2013-D1-P1; 2013-D1-P2; 2013-D1-P4; 2013-D2-P1; 2013-D2-P2; 2013-D2-P3; 2014-D1-P2; 2014-D1-P3; 2014-D1-P4; ... (+55 more)

M-UNI-10: Peeling, deletion, and reduction to a core

0 problem links

Core move	Delete rows, vertices, intervals, low-degree objects, or redundant pieces until a dense/minimal core remains.
When to use	Use in graph cycle arguments, grid problems, coverings, set systems, and determinant recurrences.
Proof skeleton	1. Define the deletion rule. 2. Show the hypothesis survives. 3. Stop at a nonempty core. 4. Extract the needed structure.
Trap watch	Check that the procedure cannot delete everything unless that already contradicts the hypothesis.
Appears in	

17.4 Method occurrence index

This compact index is intended for database export: it maps each method to all tagged problem IDs.

ID	Method	Problem IDs
M-RA-09	Convexity, Jensen, tangent, and chord inequalities	1994-D1-P1; 1994-D1-P2; 1994-D1-P3; 1994-D1-P4; 1994-D1-P6; 1994-D2-P1; 1994-D2-P2; 1994-D2-P6; 1995-D1-P2; 1995-D1-P4; 1995-D1-P6; 1995-D2-P2; 1995-D2-P5; 1996-D1-P2; 1996-D1-P4; 1996-D1-P5; 1996-D2-P2; 1996-D2-P4; 1996-D2-P5; 1996-D2-P6; 1997-D1-P1; 1997-D1-P2; 1997-D1-P5; 1997-D2-P1; 1997-D2-P3; 1998-D1-P2; 1998-D1-P3; 1998-D1-P4; 1998-D1-P5; 1998-D1-P6; 1998-D2-P1; 1998-D2-P2; 1998-D2-P3; 1998-D2-P5; 1998-D2-P6; 1999-D1-P2; 1999-D1-P3; 1999-D1-P4; 1999-D1-P6; 1999-D2-P3; 1999-D2-P4; 1999-D2-P5; 2000-D1-P4; 2000-D2-P1; 2000-D2-P2; 2000-D2-P3; 2000-D2-P4; 2001-D1-P4; 2001-D1-P6; 2001-D2-P2; 2001-D2-P5; 2001-D2-P6; 2002-D1-P2; 2002-D1-P3; 2002-D1-P6; 2002-D2-P3; 2002-D2-P4; 2002-D2-P6; 2003-D1-P4; 2003-D1-P6; 2003-D2-P3; 2003-D2-P6; 2004-D1-P1; 2004-D1-P2; 2004-D1-P3; 2004-D1-P5; 2004-D1-P6; 2004-D2-P2; 2004-D2-P4; 2004-D2-P5; 2005-D1-P1; 2005-D1-P2; 2005-D1-P3; 2005-D1-P6; 2005-D2-P1; 2005-D2-P5; 2006-D1-P1; 2006-D1-P5; 2006-D2-P1; 2006-D2-P2; 2006-D2-P3; 2006-D2-P4; 2007-D1-P2; 2007-D1-P5; 2008-D1-P1; 2008-D1-P2; 2008-D1-P4; 2008-D2-P3; 2009-D1-P4; 2009-D2-P1; 2009-D2-P2; 2010-D1-P1; 2010-D1-P2; 2010-D1-P3; 2010-D1-P4; 2010-D1-P5; 2010-D2-P1; 2010-D2-P2; 2010-D2-P3; 2011-D1-P1; 2011-D1-P3; 2011-D1-P4; 2011-D2-P1; 2011-D2-P3; 2011-D2-P5; 2012-D1-P3; 2012-D1-P4; 2012-D2-P2; 2012-D2-P4; 2012-D2-P5; 2013-D1-P3; 2013-D1-P4; 2013-D2-P3; 2013-D2-P5; 2014-D1-P5; 2014-D2-P2; 2014-D2-P4; 2014-D2-P5; 2015-D1-P2; 2015-D1-P3; 2015-D1-P4; 2015-D1-P5; 2015-D2-P9; 2015-D2-P10; 2016-D1-P2; 2016-D1-P3; 2016-D1-P4; 2016-D2-P6; 2016-D2-P7; 2016-D2-P9; 2016-D2-P10; 2017-D1-P2; 2017-D1-P3; 2017-D2-P10; 2018-D1-P5; 2018-D2-P6; 2018-D2-P7; 2018-D2-P8; 2019-D1-P3; 2019-D1-P4; 2019-D1-P5; 2019-D2-P10; 2020-D1-P3; 2020-D1-P4; 2020-D2-P5; 2021-D2-P5; 2021-D2-P7; 2022-D1-P1; 2022-D1-P2; 2022-D1-P4; 2022-D2-P7; 2022-D2-P8; 2023-D1-P5; 2023-D2-P8; 2023-D2-P9; 2023-D2-P10; 2024-D1-P2; 2024-D1-P5; 2024-D2-P6; 2024-D2-P8; 2024-D2-P10

ID	Method	Problem IDs
M-UNI-07	Special substitution and parameter freezing	1994-D1-P5; 1994-D2-P1; 1994-D2-P2; 1994-D2-P3; 1995-D1-P2; 1995-D1-P4; 1995-D2-P2; 1996-D1-P2; 1996-D2-P1; 1996-D2-P3; 1996-D2-P5; 1997-D1-P1; 1997-D2-P1; 1997-D2-P5; 1998-D1-P6; 1998-D2-P2; 1999-D1-P3; 1999-D1-P4; 1999-D1-P6; 1999-D2-P1; 1999-D2-P3; 1999-D2-P4; 2000-D1-P2; 2000-D1-P6; 2000-D2-P5; 2001-D1-P1; 2001-D1-P3; 2001-D2-P5; 2002-D1-P2; 2002-D2-P2; 2002-D2-P6; 2003-D1-P4; 2003-D1-P5; 2003-D2-P1; 2003-D2-P5; 2004-D2-P1; 2004-D2-P2; 2004-D2-P4; 2004-D2-P5; 2005-D1-P1; 2005-D1-P3; 2005-D1-P4; 2005-D1-P5; 2005-D2-P2; 2005-D2-P4; 2005-D2-P5; 2006-D1-P5; 2006-D2-P2; 2006-D2-P3; 2007-D1-P1; 2007-D1-P4; 2007-D2-P1; 2007-D2-P5; 2008-D1-P1; 2008-D1-P4; 2008-D2-P2; 2008-D2-P3; 2009-D1-P2; 2009-D2-P1; 2009-D2-P2; 2010-D1-P1; 2010-D1-P2; 2010-D1-P3; 2010-D2-P3; 2011-D2-P3; 2011-D2-P4; 2012-D1-P2; 2012-D1-P3; 2012-D1-P4; 2012-D2-P1; 2012-D2-P3; 2012-D2-P4; 2012-D2-P5; 2013-D1-P2; 2013-D1-P4; 2013-D2-P1; 2013-D2-P3; 2014-D1-P1; 2014-D1-P2; 2014-D1-P3; 2014-D1-P4; 2014-D1-P5; 2014-D2-P1; 2014-D2-P2; 2014-D2-P3; 2014-D2-P4; 2014-D2-P5; 2015-D1-P3; 2015-D1-P4; 2015-D2-P6; 2015-D2-P7; 2016-D1-P1; 2016-D1-P3; 2016-D1-P4; 2016-D2-P6; 2016-D2-P7; 2017-D1-P2; 2017-D1-P3; 2017-D2-P6; 2017-D2-P9; 2017-D2-P10; 2018-D1-P1; 2018-D1-P3; 2018-D1-P4; 2018-D2-P6; 2018-D2-P7; 2018-D2-P9; 2018-D2-P10; 2019-D1-P1; 2019-D1-P2; 2019-D1-P3; 2019-D1-P4; 2019-D1-P5; 2019-D2-P6; 2019-D2-P7; 2019-D2-P9; 2020-D2-P5; 2020-D2-P6; 2020-D2-P7; 2020-D2-P8; 2021-D1-P1; 2021-D1-P3; 2021-D1-P4; 2021-D2-P5; 2021-D2-P6; 2021-D2-P8; 2022-D1-P1; 2022-D1-P2; 2023-D1-P1; 2023-D1-P2; 2023-D1-P3; 2023-D1-P4; 2023-D1-P5; 2023-D2-P6; 2023-D2-P7; 2023-D2-P10; 2024-D1-P1; 2024-D1-P2; 2024-D1-P4; 2024-D2-P6; 2024-D2-P8

ID	Method	Problem IDs
M-RA-01	Auxiliary function with derivative sign	1994-D1-P2; 1994-D1-P5; 1994-D1-P6; 1994-D2-P1; 1994-D2-P3; 1994-D2-P6; 1995-D1-P2; 1995-D1-P3; 1995-D1-P4; 1995-D2-P2; 1995-D2-P4; 1995-D2-P5; 1995-D2-P6; 1996-D1-P2; 1996-D1-P4; 1996-D1-P5; 1996-D2-P1; 1996-D2-P3; 1996-D2-P6; 1997-D1-P2; 1997-D1-P4; 1997-D2-P1; 1997-D2-P3; 1997-D2-P5; 1997-D2-P6; 1998-D1-P3; 1998-D1-P4; 1998-D1-P5; 1998-D1-P6; 1998-D2-P2; 1998-D2-P3; 1998-D2-P4; 1998-D2-P6; 1999-D1-P2; 1999-D1-P4; 1999-D1-P6; 1999-D2-P3; 1999-D2-P4; 2000-D1-P1; 2000-D1-P2; 2000-D1-P4; 2000-D1-P6; 2000-D2-P2; 2000-D2-P5; 2001-D1-P3; 2001-D1-P6; 2001-D2-P2; 2001-D2-P5; 2002-D1-P2; 2002-D1-P4; 2002-D1-P5; 2002-D1-P6; 2002-D2-P6; 2003-D1-P1; 2003-D1-P5; 2003-D2-P2; 2003-D2-P5; 2003-D2-P6; 2004-D1-P2; 2004-D1-P5; 2004-D2-P2; 2004-D2-P5; 2005-D1-P3; 2005-D1-P4; 2005-D1-P5; 2005-D2-P1; 2005-D2-P4; 2005-D2-P5; 2006-D1-P1; 2006-D1-P5; 2006-D1-P6; 2006-D2-P2; 2006-D2-P3; 2006-D2-P5; 2007-D1-P1; 2007-D1-P4; 2007-D2-P1; 2007-D2-P3; 2007-D2-P6; 2008-D1-P1; 2008-D1-P5; 2008-D2-P1; 2008-D2-P2; 2009-D1-P1; 2009-D1-P5; 2009-D2-P2; 2010-D1-P1; 2010-D1-P2; 2010-D1-P5; 2010-D2-P1; 2010-D2-P2; 2010-D2-P3; 2011-D1-P1; 2011-D1-P4; 2011-D2-P1; 2011-D2-P4; 2012-D1-P4; 2012-D2-P2; 2012-D2-P5; 2013-D1-P2; 2013-D1-P4; 2013-D2-P3; 2014-D1-P2; 2014-D2-P2; 2014-D2-P3; 2014-D2-P5; 2015-D2-P6; 2015-D2-P7; 2016-D1-P1; 2016-D2-P6; 2016-D2-P7; 2017-D1-P2; 2017-D1-P5; 2017-D2-P6; 2017-D2-P9; 2018-D1-P1; 2018-D1-P4; 2018-D2-P7; 2018-D2-P10; 2019-D1-P1; 2019-D1-P3; 2019-D2-P6; 2020-D1-P4; 2020-D2-P5; 2020-D2-P7; 2020-D2-P8; 2021-D1-P3; 2021-D1-P4; 2021-D2-P6; 2023-D1-P1; 2023-D1-P2; 2023-D2-P7; 2024-D1-P2; 2024-D1-P4; 2024-D2-P9
M-ANT-01	Reduction modulo a prime or finite field	1994-D1-P3; 1994-D1-P6; 1994-D2-P5; 1995-D2-P4; 1996-D2-P2; 1997-D1-P3; 1997-D1-P4; 1997-D1-P5; 1999-D1-P3; 1999-D2-P1; 1999-D2-P2; 2000-D1-P5; 2000-D2-P1; 2001-D1-P1; 2001-D1-P2; 2001-D1-P4; 2001-D2-P1; 2001-D2-P6; 2002-D1-P3; 2002-D2-P3; 2003-D1-P2; 2003-D1-P4; 2003-D2-P4; 2004-D1-P6; 2005-D1-P2; 2005-D1-P4; 2005-D1-P6; 2005-D2-P2; 2005-D2-P6; 2006-D1-P2; 2006-D1-P3; 2006-D1-P4; 2006-D2-P1; 2006-D2-P4; 2006-D2-P5; 2007-D1-P1; 2007-D1-P2; 2007-D1-P5; 2007-D2-P2; 2007-D2-P4; 2007-D2-P5; 2008-D1-P3; 2008-D1-P5; 2008-D2-P1; 2008-D2-P3; 2008-D2-P4; 2008-D2-P5; 2009-D2-P3; 2009-D2-P4; 2010-D1-P4; 2010-D1-P5; 2010-D2-P3; 2010-D2-P4; 2010-D2-P5; 2011-D1-P3; 2011-D1-P5; 2012-D1-P1; 2012-D1-P2; 2012-D1-P3; 2012-D1-P5; 2012-D2-P1; 2012-D2-P3; 2012-D2-P4; 2012-D2-P5; 2013-D1-P5; 2013-D2-P2; 2013-D2-P4; 2014-D1-P4; 2014-D2-P1; 2015-D1-P1; 2015-D1-P2; 2015-D1-P4; 2015-D2-P10; 2016-D1-P2; 2016-D1-P3; 2016-D1-P4; 2016-D1-P5; 2016-D2-P8; 2016-D2-P9; 2017-D1-P1; 2017-D1-P3; 2017-D1-P4; 2017-D1-P5; 2017-D2-P7; 2017-D2-P8; 2017-D2-P10; 2018-D1-P2; 2018-D1-P5; 2018-D2-P6; 2018-D2-P8; 2019-D1-P2; 2019-D1-P4; 2019-D1-P5; 2019-D2-P7; 2019-D2-P9; 2020-D1-P1; 2020-D1-P3; 2020-D2-P6; 2020-D2-P7; 2021-D1-P2; 2021-D2-P5; 2021-D2-P6; 2021-D2-P8; 2022-D1-P2; 2022-D1-P3; 2022-D1-P4; 2022-D2-P6; 2022-D2-P8; 2023-D1-P4; 2023-D1-P5; 2023-D2-P10; 2024-D1-P3; 2024-D1-P4; 2024-D1-P5; 2024-D2-P7; 2024-D2-P10

ID	Method	Problem IDs
M-UNI-09	Invariant extraction and invariant comparison	1994-D1-P3; 1994-D2-P3; 1995-D1-P2; 1995-D1-P4; 1995-D1-P5; 1995-D2-P2; 1996-D1-P1; 1996-D2-P1; 1996-D2-P2; 1997-D1-P1; 1998-D1-P1; 1998-D1-P6; 1998-D2-P1; 1999-D1-P3; 1999-D1-P5; 1999-D2-P1; 1999-D2-P5; 2000-D1-P2; 2000-D2-P3; 2001-D1-P1; 2001-D1-P3; 2001-D1-P5; 2002-D1-P3; 2002-D2-P1; 2002-D2-P2; 2002-D2-P4; 2003-D1-P4; 2003-D2-P1; 2004-D1-P4; 2004-D2-P1; 2004-D2-P4; 2005-D1-P1; 2005-D1-P3; 2005-D2-P2; 2006-D1-P5; 2006-D2-P3; 2007-D2-P2; 2007-D2-P4; 2007-D2-P5; 2008-D1-P2; 2008-D1-P4; 2008-D2-P3; 2009-D1-P2; 2010-D1-P2; 2011-D2-P3; 2012-D1-P2; 2012-D1-P3; 2012-D1-P4; 2012-D2-P1; 2012-D2-P3; 2012-D2-P4; 2013-D1-P1; 2013-D1-P2; 2013-D1-P4; 2013-D2-P1; 2013-D2-P2; 2013-D2-P3; 2014-D1-P2; 2014-D1-P3; 2014-D1-P4; 2014-D1-P5; 2014-D2-P1; 2014-D2-P3; 2014-D2-P4; 2014-D2-P5; 2015-D1-P3; 2015-D1-P4; 2015-D2-P6; 2015-D2-P7; 2016-D1-P1; 2016-D1-P3; 2016-D1-P4; 2016-D2-P6; 2016-D2-P7; 2017-D1-P2; 2017-D1-P3; 2017-D2-P6; 2017-D2-P8; 2017-D2-P9; 2017-D2-P10; 2018-D1-P1; 2018-D1-P3; 2018-D1-P4; 2018-D2-P6; 2018-D2-P7; 2018-D2-P10; 2019-D1-P1; 2019-D1-P2; 2019-D1-P3; 2019-D1-P4; 2019-D1-P5; 2019-D2-P7; 2019-D2-P9; 2020-D1-P1; 2020-D2-P5; 2020-D2-P8; 2021-D1-P1; 2021-D1-P3; 2021-D1-P4; 2021-D2-P5; 2021-D2-P8; 2022-D1-P1; 2022-D2-P6; 2023-D1-P1; 2023-D1-P2; 2023-D1-P4; 2023-D1-P5; 2023-D2-P6; 2023-D2-P7; 2023-D2-P10; 2024-D1-P1; 2024-D1-P2; 2024-D2-P6; 2024-D2-P8; 2024-D2-P10
M-UNI-03	Extremal construction and equality-case engineering	1994-D1-P2; 1994-D1-P4; 1994-D2-P1; 1994-D2-P5; 1995-D1-P2; 1995-D1-P6; 1995-D2-P1; 1995-D2-P6; 1996-D1-P3; 1997-D1-P2; 1997-D1-P4; 1997-D1-P5; 1997-D1-P6; 1997-D2-P1; 1997-D2-P4; 1997-D2-P6; 1998-D1-P2; 1998-D1-P4; 1998-D1-P6; 1998-D2-P6; 1999-D1-P2; 1999-D1-P3; 1999-D1-P6; 1999-D2-P3; 1999-D2-P4; 1999-D2-P5; 2000-D1-P1; 2000-D1-P4; 2000-D2-P2; 2001-D2-P3; 2001-D2-P5; 2001-D2-P6; 2002-D1-P2; 2002-D1-P5; 2003-D1-P2; 2003-D2-P3; 2003-D2-P4; 2004-D1-P5; 2004-D2-P2; 2004-D2-P3; 2004-D2-P5; 2005-D1-P3; 2005-D2-P1; 2005-D2-P4; 2006-D1-P1; 2007-D1-P2; 2007-D2-P1; 2007-D2-P3; 2007-D2-P5; 2008-D1-P3; 2008-D1-P5; 2008-D2-P3; 2008-D2-P4; 2008-D2-P6; 2009-D1-P1; 2009-D2-P2; 2009-D2-P5; 2010-D1-P1; 2010-D1-P4; 2010-D2-P1; 2010-D2-P2; 2011-D1-P2; 2011-D1-P3; 2011-D2-P2; 2011-D2-P5; 2012-D1-P4; 2013-D1-P2; 2013-D1-P4; 2013-D1-P5; 2013-D2-P3; 2013-D2-P4; 2014-D2-P4; 2015-D1-P2; 2015-D2-P10; 2016-D1-P1; 2016-D1-P2; 2016-D1-P3; 2016-D2-P6; 2016-D2-P8; 2016-D2-P9; 2016-D2-P10; 2017-D1-P2; 2017-D1-P3; 2017-D2-P10; 2018-D1-P2; 2018-D1-P5; 2019-D1-P3; 2019-D2-P6; 2020-D1-P1; 2020-D1-P3; 2020-D1-P4; 2021-D1-P3; 2021-D1-P4; 2021-D2-P5; 2022-D1-P1; 2022-D1-P3; 2022-D1-P4; 2022-D2-P7; 2023-D2-P10; 2024-D1-P3; 2024-D1-P5; 2024-D2-P6; 2024-D2-P9

ID	Method	Problem IDs
M-ANT-03	GCD, valuation, and prime-power bookkeeping	1994-D1-P3; 1994-D1-P6; 1994-D2-P5; 1995-D2-P4; 1996-D2-P2; 1997-D1-P3; 1997-D1-P4; 1997-D1-P5; 1999-D1-P3; 1999-D2-P1; 2000-D1-P5; 2000-D2-P1; 2001-D1-P1; 2001-D1-P2; 2001-D1-P4; 2001-D2-P1; 2001-D2-P6; 2002-D1-P3; 2002-D2-P3; 2003-D1-P2; 2003-D1-P4; 2003-D2-P4; 2005-D1-P2; 2005-D1-P6; 2006-D1-P2; 2006-D1-P3; 2006-D1-P4; 2006-D2-P1; 2006-D2-P4; 2007-D1-P2; 2007-D1-P5; 2007-D2-P2; 2007-D2-P4; 2007-D2-P5; 2008-D2-P1; 2008-D2-P3; 2009-D2-P4; 2010-D1-P4; 2010-D1-P5; 2010-D2-P4; 2010-D2-P5; 2011-D1-P3; 2011-D1-P5; 2012-D1-P1; 2012-D1-P2; 2012-D1-P3; 2012-D1-P5; 2012-D2-P3; 2012-D2-P4; 2012-D2-P5; 2013-D1-P5; 2013-D2-P2; 2013-D2-P4; 2014-D1-P4; 2014-D2-P1; 2015-D1-P1; 2015-D1-P2; 2015-D1-P4; 2015-D2-P10; 2016-D1-P2; 2016-D1-P3; 2016-D1-P4; 2016-D1-P5; 2016-D2-P8; 2016-D2-P9; 2017-D1-P1; 2017-D1-P3; 2017-D1-P4; 2017-D1-P5; 2017-D2-P7; 2017-D2-P8; 2017-D2-P10; 2018-D1-P2; 2018-D1-P5; 2018-D2-P6; 2018-D2-P8; 2019-D1-P2; 2019-D1-P4; 2019-D1-P5; 2019-D2-P7; 2019-D2-P9; 2020-D1-P1; 2020-D1-P3; 2020-D2-P6; 2020-D2-P7; 2021-D1-P2; 2021-D2-P5; 2021-D2-P6; 2021-D2-P8; 2022-D1-P2; 2022-D1-P3; 2022-D1-P4; 2022-D2-P6; 2022-D2-P8; 2023-D1-P4; 2023-D1-P5; 2023-D2-P10; 2024-D1-P3; 2024-D1-P4; 2024-D1-P5; 2024-D2-P7; 2024-D2-P10
M-RA-08	Compactness plus extreme value theorem	1994-D2-P2; 1994-D2-P6; 1995-D1-P2; 1995-D1-P4; 1995-D1-P6; 1995-D2-P2; 1995-D2-P4; 1995-D2-P5; 1995-D2-P6; 1996-D1-P2; 1996-D1-P4; 1996-D1-P5; 1996-D1-P6; 1996-D2-P4; 1996-D2-P6; 1997-D2-P3; 1998-D1-P4; 1998-D1-P5; 1998-D1-P6; 1998-D2-P2; 1999-D1-P1; 1999-D1-P2; 1999-D2-P3; 2000-D1-P1; 2000-D1-P2; 2000-D1-P4; 2000-D2-P2; 2001-D1-P3; 2001-D2-P2; 2001-D2-P3; 2001-D2-P4; 2001-D2-P6; 2002-D1-P4; 2002-D1-P5; 2002-D2-P6; 2003-D1-P1; 2003-D2-P3; 2004-D1-P2; 2004-D1-P3; 2004-D1-P5; 2004-D2-P5; 2005-D1-P3; 2005-D2-P1; 2005-D2-P3; 2005-D2-P5; 2006-D1-P5; 2006-D2-P2; 2006-D2-P3; 2007-D2-P1; 2007-D2-P3; 2007-D2-P6; 2008-D1-P1; 2008-D2-P1; 2009-D1-P4; 2010-D2-P2; 2011-D1-P1; 2012-D1-P4; 2013-D1-P2; 2013-D1-P3; 2013-D1-P4; 2013-D2-P3; 2014-D1-P2; 2014-D2-P2; 2014-D2-P3; 2014-D2-P5; 2015-D2-P6; 2015-D2-P7; 2015-D2-P9; 2016-D2-P6; 2016-D2-P7; 2017-D1-P2; 2017-D2-P6; 2018-D1-P1; 2018-D1-P4; 2018-D2-P10; 2019-D1-P1; 2019-D1-P3; 2020-D1-P2; 2020-D1-P4; 2020-D2-P5; 2020-D2-P8; 2021-D1-P3; 2021-D1-P4; 2023-D1-P1; 2023-D2-P7; 2024-D1-P2

ID	Method	Problem IDs
M-ANT-06	Degree versus number of roots	1994-D2-P4; 1995-D1-P5; 1995-D2-P3; 1995-D2-P4; 1996-D2-P2; 1998-D1-P2; 1998-D1-P5; 1998-D2-P2; 1999-D1-P1; 1999-D2-P2; 1999-D2-P6; 2000-D1-P6; 2000-D2-P3; 2000-D2-P4; 2000-D2-P6; 2001-D1-P1; 2001-D1-P4; 2001-D2-P1; 2002-D1-P1; 2002-D2-P3; 2003-D1-P2; 2003-D1-P3; 2003-D1-P6; 2003-D2-P2; 2004-D1-P6; 2005-D1-P4; 2005-D2-P2; 2005-D2-P6; 2006-D1-P2; 2006-D1-P4; 2006-D1-P6; 2006-D2-P5; 2006-D2-P6; 2007-D1-P1; 2007-D1-P3; 2007-D1-P5; 2007-D1-P6; 2007-D2-P2; 2007-D2-P6; 2008-D1-P6; 2008-D1-P2; 2008-D1-P3; 2008-D2-P1; 2008-D2-P2; 2008-D2-P4; 2008-D2-P5; 2008-D2-P6; 2009-D1-P3; 2009-D1-P4; 2009-D2-P1; 2009-D2-P3; 2009-D2-P4; 2009-D2-P5; 2010-D2-P5; 2011-D1-P2; 2011-D1-P3; 2011-D1-P5; 2011-D2-P2; 2011-D2-P4; 2012-D1-P5; 2012-D2-P1; 2012-D2-P3; 2013-D2-P1; 2013-D2-P2; 2014-D1-P3; 2014-D1-P4; 2014-D2-P1; 2015-D2-P10; 2017-D1-P1; 2017-D1-P5; 2017-D2-P7; 2018-D1-P2; 2018-D1-P5; 2018-D2-P9; 2019-D1-P2; 2019-D2-P7; 2020-D1-P4; 2020-D2-P6; 2021-D2-P7; 2022-D2-P6; 2023-D1-P3; 2023-D1-P4; 2024-D1-P1; 2024-D2-P10
M-ANT-16	Polynomial identity by coefficient comparison	1994-D2-P4; 1995-D1-P5; 1995-D2-P3; 1995-D2-P4; 1996-D2-P2; 1998-D1-P2; 1998-D1-P5; 1998-D2-P2; 1999-D1-P1; 1999-D2-P2; 1999-D2-P6; 2000-D1-P2; 2000-D1-P6; 2000-D2-P3; 2000-D2-P4; 2000-D2-P6; 2001-D1-P4; 2001-D1-P5; 2001-D2-P1; 2002-D1-P1; 2002-D2-P3; 2003-D1-P3; 2003-D1-P6; 2003-D2-P2; 2004-D1-P6; 2005-D1-P4; 2005-D2-P2; 2005-D2-P6; 2006-D1-P4; 2006-D1-P6; 2006-D2-P5; 2006-D2-P6; 2007-D1-P1; 2007-D1-P3; 2007-D1-P5; 2007-D1-P6; 2007-D2-P2; 2007-D2-P6; 2008-D1-P6; 2008-D1-P2; 2008-D1-P3; 2008-D2-P1; 2008-D2-P2; 2008-D2-P4; 2008-D2-P5; 2008-D2-P6; 2009-D1-P3; 2009-D1-P4; 2009-D2-P1; 2009-D2-P3; 2009-D2-P4; 2009-D2-P5; 2010-D2-P5; 2011-D1-P2; 2011-D1-P3; 2011-D1-P5; 2011-D2-P2; 2011-D2-P4; 2012-D1-P5; 2012-D2-P1; 2012-D2-P3; 2013-D1-P5; 2013-D2-P1; 2013-D2-P2; 2014-D1-P3; 2014-D1-P4; 2014-D2-P1; 2015-D1-P1; 2015-D2-P10; 2016-D2-P10; 2017-D1-P1; 2017-D1-P5; 2017-D2-P7; 2018-D2-P9; 2019-D1-P2; 2020-D1-P4; 2020-D2-P6; 2021-D2-P7; 2023-D1-P2; 2023-D1-P3; 2023-D1-P4; 2024-D1-P1; 2024-D2-P10
M-CMB-01	Pigeonhole threshold and collision forcing	1994-D1-P1; 1994-D1-P3; 1994-D1-P6; 1994-D2-P4; 1995-D1-P3; 1996-D1-P3; 1996-D2-P3; 1996-D2-P6; 1997-D1-P6; 1997-D2-P6; 1998-D1-P5; 1998-D2-P3; 1998-D2-P4; 1998-D2-P5; 1998-D2-P6; 1999-D1-P5; 1999-D2-P5; 1999-D2-P6; 2000-D1-P1; 2001-D2-P1; 2001-D2-P3; 2002-D1-P4; 2002-D1-P5; 2002-D2-P2; 2003-D1-P5; 2003-D2-P3; 2003-D2-P4; 2003-D2-P5; 2004-D1-P1; 2004-D1-P2; 2004-D1-P4; 2004-D2-P6; 2005-D1-P2; 2006-D1-P2; 2006-D1-P6; 2006-D2-P1; 2006-D2-P5; 2007-D1-P4; 2007-D1-P6; 2008-D1-P6; 2008-D1-P4; 2008-D1-P5; 2008-D2-P4; 2008-D2-P6; 2009-D1-P1; 2009-D1-P3; 2010-D1-P4; 2011-D1-P4; 2011-D2-P2; 2012-D1-P1; 2013-D1-P3; 2013-D2-P5; 2015-D2-P8; 2016-D1-P5; 2016-D2-P7; 2016-D2-P9; 2017-D1-P4; 2018-D2-P8; 2019-D2-P8; 2020-D1-P1; 2020-D1-P2; 2020-D2-P7; 2021-D1-P2; 2021-D2-P8; 2022-D1-P3; 2022-D1-P4; 2022-D2-P5; 2022-D2-P8; 2023-D2-P8; 2023-D2-P9; 2024-D1-P5; 2024-D2-P9

ID	Method	Problem IDs
M-UNI-05	Induction with strengthened hypothesis	1994-D1-P3; 1994-D2-P3; 1994-D2-P5; 1995-D1-P4; 1995-D1-P5; 1996-D1-P3; 1996-D1-P4; 1996-D1-P6; 1996-D2-P2; 1997-D1-P3; 1997-D1-P4; 1997-D1-P6; 1997-D2-P2; 1997-D2-P3; 1997-D2-P5; 1998-D1-P3; 1998-D2-P1; 1999-D1-P4; 1999-D1-P5; 1999-D2-P1; 1999-D2-P5; 2000-D1-P3; 2000-D2-P3; 2001-D1-P2; 2001-D1-P5; 2001-D2-P1; 2001-D2-P2; 2002-D1-P3; 2002-D1-P4; 2002-D2-P2; 2002-D2-P3; 2002-D2-P4; 2002-D2-P5; 2003-D1-P5; 2003-D1-P6; 2003-D2-P6; 2004-D1-P2; 2004-D1-P4; 2004-D1-P5; 2005-D1-P2; 2005-D1-P6; 2006-D1-P3; 2006-D1-P6; 2006-D2-P1; 2006-D2-P4; 2007-D2-P2; 2008-D1-P2; 2008-D1-P3; 2009-D1-P2; 2009-D1-P5; 2009-D2-P3; 2010-D1-P3; 2010-D2-P2; 2010-D2-P4; 2011-D1-P5; 2011-D2-P1; 2011-D2-P4; 2012-D2-P2; 2013-D1-P1; 2013-D2-P1; 2013-D2-P2; 2015-D2-P6; 2016-D1-P1; 2016-D1-P4; 2017-D2-P7; 2017-D2-P8; 2018-D1-P1; 2019-D1-P4; 2021-D1-P1; 2022-D2-P6; 2023-D1-P5; 2024-D2-P10
M-LA-04	Rank-nullity and dimension-count obstruction	1994-D1-P1; 1994-D1-P4; 1994-D2-P4; 1994-D2-P5; 1995-D1-P1; 1995-D1-P5; 1995-D2-P1; 1996-D1-P1; 1996-D1-P3; 1996-D1-P6; 1997-D2-P2; 1997-D2-P4; 1998-D1-P1; 1998-D2-P1; 2000-D1-P3; 2000-D2-P1; 2001-D1-P5; 2001-D2-P3; 2002-D1-P5; 2002-D1-P6; 2002-D2-P1; 2002-D2-P5; 2003-D1-P3; 2003-D2-P1; 2004-D1-P3; 2004-D2-P1; 2004-D2-P4; 2005-D1-P1; 2005-D1-P3; 2005-D2-P3; 2006-D2-P2; 2006-D2-P4; 2007-D1-P2; 2007-D1-P3; 2007-D2-P5; 2008-D1-P2; 2008-D2-P5; 2008-D2-P6; 2009-D1-P2; 2009-D1-P3; 2009-D1-P4; 2009-D1-P5; 2009-D2-P5; 2010-D2-P4; 2011-D1-P5; 2011-D2-P4; 2011-D2-P5; 2012-D1-P2; 2012-D1-P4; 2013-D1-P1; 2013-D1-P2; 2014-D1-P1; 2015-D1-P1; 2015-D1-P5; 2015-D2-P9; 2016-D1-P2; 2016-D2-P10; 2017-D1-P1; 2018-D1-P3; 2018-D1-P4; 2019-D1-P3; 2020-D1-P2; 2021-D1-P1; 2022-D2-P7; 2023-D1-P2; 2023-D2-P6; 2023-D2-P9; 2024-D1-P3; 2024-D2-P7
M-RA-06	Riemann sum and sum-integral conversion	1994-D1-P2; 1994-D1-P5; 1994-D2-P6; 1995-D1-P3; 1995-D2-P2; 1995-D2-P5; 1996-D1-P2; 1996-D1-P4; 1996-D1-P5; 1996-D2-P1; 1996-D2-P5; 1997-D1-P1; 1997-D1-P2; 1997-D2-P3; 1997-D2-P4; 1998-D1-P3; 1998-D2-P4; 1999-D1-P2; 2000-D2-P4; 2000-D2-P5; 2000-D2-P6; 2001-D1-P3; 2001-D1-P6; 2001-D2-P2; 2001-D2-P5; 2002-D1-P4; 2002-D1-P6; 2003-D1-P1; 2003-D1-P2; 2003-D1-P3; 2003-D2-P2; 2003-D2-P5; 2003-D2-P6; 2004-D1-P3; 2004-D2-P3; 2004-D2-P4; 2004-D2-P6; 2005-D1-P5; 2005-D2-P1; 2005-D2-P3; 2006-D1-P1; 2006-D1-P3; 2006-D1-P4; 2009-D1-P1; 2010-D1-P2; 2010-D1-P3; 2010-D2-P1; 2010-D2-P5; 2011-D1-P1; 2011-D2-P1; 2012-D1-P1; 2012-D2-P2; 2013-D2-P4; 2014-D1-P2; 2015-D2-P7; 2015-D2-P8; 2016-D2-P8; 2017-D2-P6; 2017-D2-P9; 2018-D2-P7; 2018-D2-P10; 2019-D2-P7; 2020-D2-P8; 2021-D1-P4; 2024-D1-P2; 2024-D2-P8

ID	Method	Problem IDs
M-CMB-02	Double counting ordered or weighted objects	1994-D1-P1; 1994-D2-P4; 1995-D1-P3; 1996-D1-P3; 1996-D2-P3; 1996-D2-P6; 1997-D1-P6; 1997-D2-P6; 1998-D2-P3; 1998-D2-P4; 1998-D2-P5; 1998-D2-P6; 1999-D1-P5; 1999-D2-P5; 2000-D1-P1; 2001-D2-P1; 2001-D2-P3; 2002-D1-P4; 2002-D1-P5; 2002-D2-P2; 2003-D1-P5; 2003-D2-P3; 2003-D2-P4; 2003-D2-P5; 2004-D1-P1; 2004-D1-P2; 2004-D1-P4; 2004-D2-P6; 2005-D1-P2; 2006-D1-P2; 2006-D1-P6; 2006-D2-P1; 2006-D2-P5; 2007-D1-P4; 2007-D1-P6; 2008-D1-P6; 2008-D1-P4; 2008-D1-P5; 2008-D2-P4; 2008-D2-P6; 2009-D1-P1; 2009-D1-P3; 2010-D1-P4; 2011-D1-P4; 2011-D2-P2; 2012-D1-P1; 2013-D1-P3; 2013-D2-P5; 2015-D2-P8; 2016-D1-P5; 2016-D2-P9; 2017-D1-P4; 2018-D2-P8; 2019-D2-P8; 2020-D1-P1; 2020-D1-P2; 2020-D2-P7; 2021-D1-P2; 2021-D2-P8; 2022-D1-P3; 2022-D1-P4; 2022-D2-P5; 2022-D2-P8; 2023-D2-P8; 2023-D2-P9; 2024-D2-P9
M-LA-03	Block decomposition and Schur-style bookkeeping	1994-D1-P1; 1995-D1-P1; 1995-D2-P1; 1996-D1-P1; 1997-D1-P3; 1997-D2-P2; 1997-D2-P4; 1998-D1-P1; 1999-D1-P1; 1999-D1-P5; 1999-D2-P2; 1999-D2-P3; 2000-D1-P3; 2000-D1-P4; 2001-D1-P1; 2001-D1-P3; 2001-D1-P5; 2001-D2-P4; 2002-D2-P1; 2003-D2-P1; 2004-D2-P1; 2004-D2-P6; 2005-D1-P1; 2005-D2-P6; 2006-D1-P3; 2006-D2-P3; 2006-D2-P6; 2007-D1-P2; 2007-D1-P3; 2007-D2-P4; 2008-D1-P6; 2008-D2-P5; 2009-D1-P5; 2009-D2-P5; 2010-D1-P5; 2010-D2-P4; 2011-D2-P2; 2011-D2-P3; 2013-D1-P4; 2014-D1-P1; 2014-D2-P2; 2014-D2-P3; 2014-D2-P5; 2015-D1-P1; 2015-D1-P2; 2015-D1-P3; 2016-D1-P2; 2016-D2-P6; 2018-D1-P1; 2018-D1-P3; 2018-D1-P4; 2018-D2-P10; 2019-D1-P3; 2019-D1-P5; 2019-D2-P9; 2020-D1-P2; 2020-D2-P5; 2020-D2-P8; 2022-D1-P1; 2022-D2-P7; 2023-D1-P1; 2023-D2-P6; 2024-D2-P6; 2024-D2-P7; 2024-D2-P9
M-RA-07	Split into main range and tail/error range	1994-D1-P5; 1994-D2-P6; 1995-D1-P3; 1995-D2-P2; 1995-D2-P5; 1996-D1-P2; 1996-D1-P4; 1996-D1-P5; 1996-D2-P1; 1996-D2-P5; 1997-D1-P1; 1997-D1-P2; 1997-D2-P3; 1997-D2-P4; 1998-D1-P3; 1998-D2-P4; 1999-D1-P2; 2000-D2-P4; 2000-D2-P5; 2000-D2-P6; 2001-D1-P3; 2001-D1-P6; 2001-D2-P2; 2001-D2-P5; 2002-D1-P6; 2003-D1-P1; 2003-D1-P2; 2003-D1-P3; 2003-D2-P2; 2003-D2-P5; 2003-D2-P6; 2004-D2-P3; 2004-D2-P4; 2004-D2-P6; 2005-D1-P5; 2005-D2-P1; 2005-D2-P3; 2006-D1-P1; 2006-D1-P3; 2006-D1-P4; 2009-D1-P1; 2010-D1-P2; 2010-D1-P3; 2010-D2-P1; 2010-D2-P5; 2011-D1-P1; 2011-D2-P1; 2012-D1-P1; 2012-D2-P2; 2013-D2-P4; 2014-D1-P2; 2015-D2-P7; 2015-D2-P8; 2016-D2-P8; 2017-D2-P6; 2017-D2-P9; 2018-D2-P7; 2018-D2-P10; 2019-D2-P7; 2020-D2-P8; 2021-D1-P4; 2024-D1-P2; 2024-D2-P8

ID	Method	Problem IDs
M-RA-03	Rolle chain and divided-difference localization	1994-D1-P2; 1994-D1-P6; 1994-D2-P1; 1994-D2-P3; 1994-D2-P4; 1995-D1-P2; 1995-D1-P3; 1995-D2-P6; 1996-D1-P5; 1996-D2-P1; 1996-D2-P5; 1996-D2-P6; 1997-D2-P1; 1997-D2-P3; 1997-D2-P6; 1998-D1-P4; 1998-D1-P6; 1998-D2-P6; 1999-D1-P6; 2000-D1-P2; 2000-D1-P6; 2000-D2-P2; 2000-D2-P5; 2001-D1-P6; 2001-D2-P6; 2002-D1-P2; 2002-D1-P4; 2002-D1-P5; 2004-D2-P2; 2005-D1-P3; 2005-D1-P5; 2005-D2-P4; 2005-D2-P5; 2006-D1-P1; 2006-D1-P5; 2006-D1-P6; 2007-D2-P1; 2007-D2-P3; 2007-D2-P6; 2008-D1-P1; 2009-D2-P2; 2010-D1-P1; 2010-D1-P5; 2011-D1-P1; 2012-D1-P4; 2013-D1-P2; 2014-D2-P3; 2015-D2-P6; 2016-D1-P1; 2016-D2-P7; 2017-D1-P5; 2017-D2-P6; 2017-D2-P9; 2018-D1-P1; 2019-D1-P1; 2019-D2-P6; 2020-D2-P5; 2023-D1-P1; 2023-D1-P2; 2023-D2-P7
M-ANT-05	Factorization and unique factorization in Z[x]	1994-D2-P4; 1995-D1-P5; 1995-D2-P3; 1995-D2-P4; 1996-D2-P2; 1998-D1-P2; 1998-D1-P5; 1998-D2-P2; 1999-D1-P1; 1999-D2-P2; 1999-D2-P6; 2000-D1-P6; 2000-D2-P3; 2000-D2-P4; 2000-D2-P6; 2001-D1-P4; 2001-D2-P1; 2002-D1-P1; 2002-D2-P3; 2003-D1-P3; 2003-D1-P6; 2003-D2-P2; 2005-D1-P4; 2005-D2-P2; 2006-D1-P4; 2006-D1-P6; 2006-D2-P5; 2006-D2-P6; 2007-D1-P1; 2007-D1-P3; 2007-D1-P5; 2007-D1-P6; 2007-D2-P6; 2008-D1-P6; 2008-D1-P2; 2008-D1-P3; 2008-D2-P1; 2008-D2-P2; 2008-D2-P4; 2008-D2-P6; 2009-D1-P4; 2009-D2-P1; 2009-D2-P3; 2009-D2-P4; 2009-D2-P5; 2010-D2-P5; 2011-D1-P2; 2011-D1-P3; 2011-D1-P5; 2011-D2-P4; 2012-D1-P5; 2012-D2-P1; 2015-D2-P10; 2017-D1-P5; 2017-D2-P7; 2018-D2-P9; 2020-D1-P4; 2021-D2-P7; 2023-D1-P3
M-UNI-08	Sign, zero, and boundary case splitting	1994-D1-P2; 1994-D1-P4; 1994-D2-P2; 1994-D2-P5; 1995-D1-P6; 1995-D2-P6; 1997-D1-P2; 1997-D1-P4; 1997-D1-P6; 1997-D2-P6; 1998-D1-P2; 1998-D2-P6; 1999-D1-P2; 1999-D1-P4; 1999-D1-P6; 1999-D2-P5; 2000-D1-P1; 2002-D1-P2; 2003-D2-P4; 2004-D2-P3; 2006-D1-P1; 2007-D2-P1; 2007-D2-P3; 2007-D2-P5; 2008-D1-P1; 2008-D1-P3; 2008-D1-P4; 2009-D1-P1; 2009-D2-P5; 2010-D1-P4; 2010-D2-P1; 2011-D1-P2; 2011-D2-P5; 2012-D2-P4; 2013-D1-P5; 2013-D2-P4; 2014-D1-P2; 2014-D2-P2; 2015-D1-P2; 2016-D1-P1; 2016-D1-P2; 2018-D1-P2; 2018-D1-P3; 2018-D1-P4; 2018-D2-P9; 2019-D1-P2; 2019-D2-P6; 2020-D1-P1; 2020-D2-P5; 2020-D2-P6; 2022-D1-P2; 2022-D1-P3; 2023-D1-P1; 2023-D1-P3; 2023-D2-P7; 2024-D1-P1; 2024-D1-P3; 2024-D2-P9
M-FE-02	Monotonicity forced by functional equation	1994-D1-P2; 1994-D1-P6; 1994-D2-P1; 1994-D2-P2; 1994-D2-P3; 1994-D2-P6; 1995-D1-P3; 1995-D1-P4; 1996-D1-P5; 1996-D2-P5; 1997-D1-P4; 1998-D1-P3; 1998-D2-P3; 1998-D2-P4; 1999-D1-P4; 1999-D1-P6; 1999-D2-P4; 2000-D1-P1; 2000-D1-P4; 2000-D1-P6; 2000-D2-P2; 2000-D2-P5; 2001-D1-P6; 2001-D2-P2; 2001-D2-P5; 2002-D1-P2; 2002-D1-P5; 2002-D1-P6; 2003-D1-P1; 2003-D1-P5; 2003-D2-P2; 2004-D1-P5; 2004-D2-P2; 2005-D2-P4; 2006-D1-P1; 2006-D2-P2; 2007-D2-P3; 2007-D2-P6; 2008-D1-P1; 2009-D1-P1; 2009-D2-P2; 2010-D1-P1; 2010-D1-P3; 2010-D2-P1; 2011-D1-P1; 2011-D1-P4; 2011-D2-P1; 2011-D2-P3; 2015-D1-P3; 2015-D2-P6; 2016-D1-P1; 2019-D1-P1; 2019-D2-P6; 2024-D2-P6; 2024-D2-P9

ID	Method	Problem IDs
M-CMB-03	Graph encoding of a relation	1996-D1-P6; 1997-D2-P5; 1997-D2-P6; 1998-D1-P1; 1998-D2-P3; 1998-D2-P5; 1999-D1-P5; 1999-D2-P5; 2000-D2-P4; 2001-D1-P2; 2001-D2-P4; 2002-D1-P1; 2002-D2-P2; 2002-D2-P4; 2003-D1-P4; 2004-D1-P1; 2004-D1-P3; 2004-D1-P4; 2004-D2-P2; 2005-D2-P4; 2006-D2-P1; 2007-D2-P1; 2008-D1-P4; 2009-D1-P3; 2009-D1-P5; 2009-D2-P4; 2011-D1-P4; 2011-D2-P2; 2011-D2-P5; 2013-D1-P3; 2013-D2-P5; 2014-D2-P4; 2015-D2-P8; 2016-D1-P4; 2016-D2-P9; 2017-D1-P3; 2018-D2-P8; 2019-D2-P8; 2019-D2-P10; 2020-D1-P3; 2021-D1-P2; 2022-D1-P4; 2022-D2-P5; 2022-D2-P8; 2023-D2-P7; 2023-D2-P8; 2024-D1-P5
M-LA-06	Spectral/minimal-polynomial reduction	1994-D1-P4; 1994-D2-P4; 1995-D1-P1; 1995-D1-P5; 1995-D2-P1; 1996-D1-P1; 1996-D1-P3; 1997-D2-P2; 1997-D2-P4; 1998-D1-P1; 1998-D2-P1; 1999-D1-P1; 2000-D1-P3; 2000-D1-P5; 2000-D2-P6; 2001-D1-P5; 2002-D2-P1; 2002-D2-P5; 2003-D1-P3; 2003-D2-P1; 2004-D2-P1; 2004-D2-P4; 2005-D1-P1; 2005-D2-P3; 2006-D2-P6; 2008-D1-P2; 2009-D1-P2; 2009-D2-P3; 2010-D2-P4; 2011-D1-P2; 2012-D1-P2; 2013-D1-P1; 2014-D1-P1; 2014-D2-P2; 2015-D1-P1; 2015-D2-P9; 2016-D2-P10; 2017-D1-P1; 2017-D2-P8; 2018-D1-P3; 2021-D1-P1; 2022-D1-P2; 2022-D2-P5; 2022-D2-P7; 2023-D1-P2; 2023-D2-P6; 2024-D2-P7
M-ANT-02	Chinese remainder and residue construction	1999-D1-P3; 1999-D2-P2; 2000-D2-P1; 2001-D1-P2; 2001-D1-P4; 2003-D1-P2; 2005-D1-P2; 2005-D1-P4; 2005-D2-P6; 2006-D1-P2; 2006-D1-P3; 2006-D2-P5; 2007-D1-P1; 2007-D2-P2; 2007-D2-P4; 2008-D1-P3; 2008-D1-P5; 2008-D2-P4; 2008-D2-P5; 2009-D2-P4; 2010-D1-P4; 2010-D2-P4; 2010-D2-P5; 2011-D1-P3; 2011-D1-P5; 2012-D2-P4; 2013-D1-P5; 2013-D2-P2; 2014-D1-P4; 2014-D2-P1; 2015-D1-P4; 2016-D1-P3; 2016-D1-P5; 2016-D2-P8; 2018-D1-P5; 2018-D2-P6; 2019-D1-P2; 2020-D2-P6; 2021-D2-P6; 2022-D1-P3; 2022-D2-P6; 2023-D1-P4; 2023-D2-P10; 2024-D1-P3; 2024-D2-P10
M-UNI-06	Compactness or extremal-point selection	1994-D2-P2; 1995-D1-P6; 1995-D2-P1; 1995-D2-P4; 1995-D2-P6; 1996-D1-P3; 1996-D1-P6; 1996-D2-P4; 1997-D2-P4; 1998-D1-P4; 1998-D2-P2; 1999-D1-P1; 1999-D2-P4; 2000-D1-P1; 2000-D2-P2; 2001-D2-P4; 2001-D2-P6; 2002-D2-P6; 2003-D1-P2; 2003-D2-P3; 2004-D1-P3; 2004-D1-P5; 2005-D2-P1; 2005-D2-P3; 2005-D2-P4; 2005-D2-P5; 2006-D2-P2; 2007-D2-P3; 2007-D2-P6; 2011-D1-P1; 2013-D1-P2; 2013-D1-P3; 2013-D2-P3; 2015-D2-P9; 2016-D2-P7; 2017-D1-P2; 2017-D1-P3; 2017-D2-P10; 2019-D2-P6; 2020-D1-P2; 2021-D1-P3; 2021-D2-P5; 2023-D2-P10
M-RA-11	Integration by parts, Fubini, and weighted identities	1994-D2-P1; 1994-D2-P3; 1995-D1-P2; 1995-D2-P5; 1996-D1-P2; 1996-D1-P4; 1996-D2-P6; 1997-D1-P1; 1997-D1-P2; 1998-D1-P3; 1998-D1-P6; 1999-D1-P4; 1999-D2-P3; 1999-D2-P4; 2001-D1-P1; 2002-D1-P3; 2003-D1-P1; 2003-D2-P6; 2004-D1-P2; 2004-D2-P5; 2005-D2-P1; 2007-D2-P2; 2008-D1-P4; 2010-D1-P2; 2010-D2-P1; 2010-D2-P2; 2011-D2-P1; 2011-D2-P3; 2012-D1-P1; 2012-D2-P2; 2013-D1-P5; 2013-D2-P4; 2014-D1-P4; 2015-D2-P8; 2018-D1-P1; 2019-D2-P6; 2019-D2-P7; 2019-D2-P10; 2022-D1-P1

ID	Method	Problem IDs
M-LA-01	Row/column elimination to triangular or sparse form	1994-D1-P1; 1995-D1-P1; 1995-D2-P1; 1996-D1-P1; 1997-D1-P3; 1997-D2-P2; 1999-D1-P1; 1999-D1-P5; 2001-D1-P1; 2001-D2-P4; 2002-D2-P1; 2004-D2-P6; 2005-D1-P1; 2005-D2-P6; 2006-D1-P3; 2006-D2-P6; 2007-D1-P2; 2007-D1-P3; 2007-D2-P4; 2008-D1-P6; 2008-D2-P5; 2009-D1-P5; 2009-D2-P5; 2010-D1-P5; 2010-D2-P4; 2012-D1-P2; 2014-D1-P1; 2015-D1-P1; 2019-D1-P5; 2020-D1-P2; 2022-D2-P7; 2023-D2-P6; 2024-D2-P7; 2024-D2-P9
M-RA-02	Integrating factor and ODE comparison	1994-D1-P5; 1994-D2-P1; 1994-D2-P3; 1995-D1-P4; 1996-D1-P2; 1996-D2-P3; 1997-D2-P1; 1997-D2-P5; 1998-D1-P4; 1999-D1-P4; 1999-D1-P6; 1999-D2-P3; 2000-D1-P6; 2000-D2-P5; 2002-D2-P6; 2003-D1-P5; 2003-D2-P5; 2004-D2-P5; 2005-D1-P4; 2005-D1-P5; 2005-D2-P4; 2007-D1-P1; 2007-D1-P4; 2008-D1-P5; 2008-D2-P2; 2009-D2-P2; 2010-D1-P1; 2010-D2-P3; 2011-D2-P4; 2012-D2-P5; 2020-D2-P7; 2021-D2-P6; 2024-D1-P4
M-CMB-04	Cycle extraction and alternating-cycle method	1996-D1-P6; 1997-D1-P6; 1997-D2-P5; 1998-D1-P1; 1998-D2-P3; 1999-D1-P5; 2000-D2-P4; 2001-D1-P2; 2001-D2-P4; 2002-D1-P1; 2002-D2-P2; 2002-D2-P4; 2003-D1-P4; 2004-D1-P3; 2004-D2-P2; 2006-D2-P1; 2007-D2-P1; 2009-D1-P2; 2009-D1-P3; 2009-D1-P5; 2009-D2-P4; 2011-D2-P5; 2013-D2-P5; 2014-D2-P4; 2016-D1-P4; 2019-D2-P8; 2019-D2-P10; 2020-D1-P3; 2022-D1-P4; 2022-D2-P5; 2022-D2-P8; 2023-D2-P7; 2023-D2-P8
M-RA-05	L-Hospital and ratio/growth comparison	1994-D1-P2; 1994-D1-P5; 1994-D2-P6; 1995-D1-P3; 1995-D1-P6; 1995-D2-P2; 1996-D1-P4; 1996-D2-P1; 1996-D2-P5; 1997-D1-P1; 1997-D1-P2; 1997-D2-P1; 2001-D1-P3; 2001-D1-P6; 2001-D2-P5; 2003-D1-P1; 2003-D2-P2; 2004-D1-P6; 2005-D1-P5; 2010-D1-P2; 2010-D1-P3; 2014-D1-P2; 2015-D2-P7; 2017-D2-P6; 2017-D2-P9; 2018-D2-P10; 2020-D2-P8; 2021-D1-P4; 2024-D1-P2; 2024-D2-P8
M-ANT-10	Group action, orbit, and generator method	1994-D2-P5; 1996-D2-P3; 1998-D1-P2; 1999-D1-P3; 1999-D2-P1; 2000-D1-P5; 2001-D1-P2; 2003-D2-P4; 2005-D1-P6; 2007-D1-P4; 2010-D2-P3; 2012-D1-P3; 2012-D1-P5; 2012-D2-P3; 2012-D2-P4; 2012-D2-P5; 2015-D1-P2; 2015-D1-P4; 2016-D1-P2; 2016-D1-P5; 2016-D2-P8; 2016-D2-P9; 2017-D1-P4; 2018-D1-P2; 2020-D2-P7; 2021-D2-P6; 2022-D2-P6; 2023-D1-P5; 2024-D1-P4
M-GEO-03	Squared-distance and inner-product identities	1995-D2-P1; 1996-D1-P6; 1996-D2-P4; 1997-D1-P5; 1997-D2-P5; 1998-D2-P5; 2002-D1-P1; 2002-D1-P6; 2002-D2-P4; 2003-D2-P3; 2004-D1-P4; 2004-D2-P3; 2005-D2-P5; 2006-D2-P4; 2009-D2-P1; 2013-D2-P3; 2014-D1-P5; 2014-D2-P4; 2015-D1-P5; 2015-D2-P9; 2017-D2-P10; 2018-D2-P6; 2021-D2-P8; 2023-D2-P8; 2023-D2-P9; 2024-D1-P5
M-LA-12	Nilpotency from zero spectrum plus finite dimension	1994-D1-P4; 1995-D1-P1; 1995-D1-P5; 1996-D1-P3; 1997-D1-P3; 2000-D1-P3; 2000-D1-P5; 2000-D2-P6; 2001-D2-P4; 2002-D2-P5; 2003-D1-P3; 2004-D2-P4; 2005-D2-P3; 2006-D2-P6; 2009-D2-P3; 2011-D1-P2; 2013-D1-P1; 2014-D1-P1; 2014-D2-P2; 2016-D2-P10; 2017-D1-P1; 2017-D2-P8; 2019-D2-P9; 2022-D1-P2; 2022-D2-P7

ID	Method	Problem IDs
M-CMB-10	Explicit bijection and encoding-decoding proof	1996-D2-P3; 1997-D1-P6; 1997-D2-P6; 1998-D2-P4; 2002-D2-P2; 2004-D1-P1; 2004-D1-P2; 2004-D2-P6; 2006-D1-P2; 2007-D1-P4; 2011-D1-P4; 2012-D1-P1; 2013-D1-P3; 2013-D2-P5; 2015-D2-P8; 2016-D1-P5; 2017-D1-P4; 2018-D2-P8; 2019-D2-P8; 2019-D2-P10; 2021-D1-P2; 2022-D1-P3; 2022-D2-P5; 2024-D1-P3
M-RA-10	Cauchy-Schwarz, Holder, and norm comparison	1994-D1-P6; 1994-D2-P2; 1994-D2-P5; 1995-D2-P1; 1996-D2-P6; 1997-D1-P5; 1998-D2-P1; 2002-D1-P3; 2002-D1-P6; 2004-D2-P3; 2005-D2-P5; 2006-D2-P4; 2008-D1-P5; 2008-D2-P2; 2008-D2-P6; 2009-D1-P4; 2009-D2-P1; 2010-D1-P1; 2010-D1-P5; 2015-D1-P5; 2015-D2-P9; 2021-D2-P8; 2023-D2-P8
M-LA-07	Trace cyclicity and commutator annihilation	1996-D1-P1; 1997-D1-P3; 1997-D2-P4; 2000-D1-P3; 2000-D1-P5; 2001-D1-P5; 2002-D2-P1; 2003-D2-P1; 2004-D2-P1; 2005-D1-P6; 2005-D2-P3; 2005-D2-P6; 2006-D2-P6; 2009-D2-P3; 2011-D1-P2; 2013-D1-P1; 2014-D1-P1; 2019-D2-P9; 2020-D1-P2; 2021-D1-P1; 2022-D2-P7
M-RA-16	Density, Baire/category, and pathological construction	1994-D2-P2; 1995-D1-P6; 1995-D2-P6; 1996-D1-P6; 1996-D2-P4; 1998-D1-P4; 1998-D2-P5; 2000-D2-P2; 2002-D1-P4; 2003-D2-P3; 2003-D2-P5; 2004-D1-P3; 2004-D2-P3; 2006-D2-P2; 2007-D2-P3; 2008-D1-P1; 2009-D1-P3; 2021-D2-P7; 2023-D2-P8; 2023-D2-P9
M-LA-13	Determinant through eigenvalue or root product	1995-D1-P1; 1996-D1-P1; 1997-D2-P2; 2000-D1-P3; 2001-D1-P5; 2002-D2-P1; 2003-D2-P1; 2004-D2-P1; 2007-D1-P3; 2007-D2-P4; 2009-D1-P2; 2013-D1-P1; 2014-D1-P1; 2015-D1-P1; 2016-D2-P10; 2018-D1-P3; 2019-D1-P5; 2021-D1-P1; 2023-D2-P6; 2024-D2-P7
M-GEO-01	Coordinate normalization and algebraization	1996-D2-P4; 1998-D2-P5; 2000-D2-P1; 2002-D1-P1; 2002-D2-P4; 2002-D2-P6; 2004-D1-P4; 2008-D2-P2; 2009-D2-P1; 2011-D2-P5; 2014-D1-P5; 2014-D2-P4; 2015-D1-P5; 2018-D1-P5; 2019-D2-P10; 2020-D1-P3; 2022-D2-P5; 2023-D2-P9
M-ANT-08	Roots of unity and cyclotomic irreducibility	1995-D2-P3; 1999-D2-P2; 1999-D2-P6; 2000-D1-P2; 2000-D2-P3; 2001-D1-P4; 2001-D2-P3; 2002-D2-P5; 2007-D1-P5; 2007-D1-P6; 2008-D2-P1; 2013-D2-P1; 2016-D2-P10; 2017-D1-P1; 2017-D1-P5; 2018-D2-P9; 2021-D2-P7; 2023-D1-P2
M-RA-15	Recurrence transformed to monotone quotient or telescoping form	1998-D2-P6; 2001-D2-P2; 2002-D2-P3; 2003-D1-P1; 2003-D1-P6; 2003-D2-P6; 2008-D2-P3; 2010-D2-P2; 2011-D2-P1; 2012-D2-P2; 2014-D1-P3; 2015-D1-P5; 2017-D2-P8; 2018-D2-P7; 2019-D1-P4; 2021-D1-P3; 2024-D2-P8
M-FE-01	Iterates, fixed points, and cluster-point trapping	1994-D2-P6; 1995-D2-P6; 1996-D2-P1; 1998-D1-P3; 1998-D2-P3; 1999-D2-P4; 2000-D1-P4; 2001-D2-P6; 2004-D2-P3; 2010-D1-P3; 2015-D1-P3; 2015-D2-P6; 2017-D2-P9; 2019-D1-P1; 2024-D1-P4; 2024-D2-P6
M-CX-01	Cauchy coefficient formula and boundary maximum	1995-D2-P3; 1999-D2-P6; 2000-D2-P3; 2005-D2-P2; 2006-D1-P4; 2007-D1-P5; 2012-D1-P5; 2012-D2-P1; 2013-D2-P1; 2014-D1-P3; 2017-D2-P7; 2018-D2-P9; 2021-D2-P7; 2023-D1-P3; 2024-D1-P1
M-ANT-11	Commutator and center manipulation in groups/rings	1996-D2-P3; 1999-D2-P1; 2000-D1-P5; 2001-D1-P2; 2005-D1-P6; 2007-D1-P4; 2010-D2-P3; 2012-D1-P3; 2012-D1-P5; 2012-D2-P5; 2016-D2-P8; 2017-D1-P4; 2018-D1-P2; 2021-D2-P6; 2024-D1-P4
M-CX-05	Trigonometric-to-complex conversion	1995-D2-P3; 1995-D2-P5; 1999-D2-P6; 2000-D1-P4; 2001-D2-P3; 2002-D2-P5; 2004-D2-P5; 2006-D2-P3; 2007-D1-P6; 2014-D2-P3; 2015-D1-P4; 2017-D1-P2; 2021-D2-P7

ID	Method	Problem IDs
M-GEO-11	Volume, projection, and ellipsoid computation	1996-D2-P4; 1998-D2-P5; 2000-D2-P1; 2002-D1-P1; 2002-D2-P6; 2004-D1-P4; 2011-D2-P5; 2015-D1-P5; 2018-D1-P5; 2019-D2-P10; 2020-D1-P3; 2022-D2-P5; 2023-D2-P9
M-CMB-05	State recurrence or automaton partition	1998-D2-P4; 2004-D1-P1; 2004-D2-P6; 2005-D1-P2; 2006-D1-P2; 2013-D1-P3; 2013-D2-P5; 2015-D2-P8; 2017-D1-P4; 2018-D2-P8; 2019-D2-P8; 2021-D1-P2
M-CMB-07	Involution, parity, and sign-reversing cancellation	1998-D1-P2; 2002-D2-P4; 2003-D1-P4; 2005-D1-P1; 2007-D1-P2; 2010-D2-P3; 2012-D1-P3; 2013-D2-P5; 2016-D1-P5; 2019-D2-P8; 2022-D2-P6; 2023-D1-P5
M-FE-03	Composition, inversion, and conjugacy	2000-D1-P2; 2005-D2-P2; 2007-D1-P3; 2012-D2-P1; 2013-D2-P1; 2014-D1-P3; 2015-D2-P10; 2017-D2-P7; 2018-D2-P9; 2020-D1-P4; 2023-D1-P3; 2024-D1-P1
M-RA-04	Taylor expansion and local asymptotic bookkeeping	1994-D1-P5; 1995-D1-P6; 1996-D1-P5; 1996-D2-P5; 1997-D1-P1; 1997-D2-P1; 1997-D2-P3; 2000-D2-P5; 2001-D1-P6; 2004-D1-P6; 2005-D1-P5; 2009-D2-P2
M-LA-15	AB versus BA transfer	2000-D1-P3; 2000-D2-P6; 2006-D2-P3; 2013-D1-P1; 2013-D1-P4; 2014-D2-P3; 2014-D2-P5; 2015-D1-P2; 2015-D1-P3; 2018-D1-P4; 2023-D1-P1
M-CMB-06	Generating function encoding	2002-D1-P3; 2003-D1-P6; 2003-D2-P6; 2008-D2-P3; 2012-D2-P2; 2017-D2-P8; 2018-D2-P7; 2019-D1-P4; 2021-D1-P3; 2022-D1-P1; 2024-D2-P8
M-LA-02	Complete a matrix equation to an inverse identity	1994-D1-P1; 1995-D1-P1; 1997-D1-P3; 1997-D2-P2; 2002-D2-P5; 2009-D1-P2; 2019-D2-P9; 2024-D2-P7
M-UNI-04	Contradiction with minimal or maximal counterexample	1997-D1-P6; 1999-D1-P2; 2002-D1-P2; 2013-D1-P5; 2013-D2-P4; 2018-D1-P2; 2024-D1-P3
M-LA-16	Trace pairing and orthogonal matrix subspaces	1997-D2-P2; 1998-D1-P1; 2003-D2-P1; 2004-D2-P1; 2018-D1-P3; 2021-D1-P1; 2023-D2-P6
M-UNI-02	Small cases, equality cases, and example probing	1996-D1-P1; 1998-D1-P1; 2002-D2-P1; 2007-D2-P4; 2011-D2-P3; 2019-D1-P1; 2020-D1-P1
M-ANT-09	Finite-field function representative and permutation argument	2003-D2-P4; 2012-D2-P3; 2013-D2-P2; 2016-D1-P3; 2018-D2-P6; 2020-D2-P6; 2023-D1-P4
M-CMB-13	Indicator and expectation method	2011-D1-P4; 2019-D2-P10; 2021-D1-P2; 2022-D1-P1; 2022-D2-P8; 2024-D1-P5
M-ANT-07	Vieta, symmetric sums, and logarithmic derivative	1994-D1-P3; 2003-D1-P6; 2006-D1-P5; 2014-D1-P3; 2023-D1-P3; 2024-D1-P1
M-CMB-08	Extremal set or minimal counterexample induction	2004-D1-P1; 2004-D1-P5; 2010-D2-P2; 2012-D1-P2; 2016-D2-P6; 2019-D2-P8
M-CMB-12	Root-of-unity filter for residues	1995-D2-P3; 1999-D2-P2; 1999-D2-P6; 2001-D2-P3; 2002-D2-P5; 2007-D1-P6
M-LA-05	Similarity, adapted basis, and canonical model	1995-D1-P1; 1998-D2-P1; 2007-D2-P5; 2015-D2-P9; 2021-D2-P5
M-GEO-06	Convex support and John-ellipse perturbation	1996-D2-P4; 2011-D2-P3; 2014-D2-P5; 2015-D1-P3
M-GEO-05	Affine invariance and change of metric scale	1997-D1-P5; 2000-D2-P1; 2014-D1-P5
M-GEO-02	Projection to a lower-dimensional object	1997-D1-P5; 2014-D1-P5; 2015-D1-P5
M-LA-10	Linear operator on a matrix space	1994-D1-P4; 2022-D1-P2
M-CMB-14	Topological ordering of an acyclic directed graph	2001-D2-P4; 2015-D2-P7
M-ANT-14	Semidirect product or explicit algebraic construction	1999-D2-P1; 2000-D1-P5
M-GEO-04	Barycentric coordinates and volume weights	2009-D1-P5; 2014-D1-P5
M-ANT-15	Rationality and linear dependence over Q	1994-D1-P3; 2006-D1-P5

ID	Method	Problem IDs
M-RA-12	Squeeze, dominated convergence, and monotone convergence	2003-D1-P5
M-LA-17	Sign forcing from positive matrices or inner products	1994-D1-P1
M-RA-13	Periodic averaging and oscillation cancellation	1994-D1-P5
M-CX-06	Cotangent partial-fraction differentiation	2004-D1-P6
M-CX-02	Laurent expansion and isolated singularity rationality	2004-D1-P6
M-CX-04	Fejer kernel positivity	2009-D1-P4
M-CX-03	Unit-circle root real-part/log-derivative test	1995-D2-P3
M-ANT-17	Greedy expansion or tail-rationality method	1997-D1-P4
M-ANT-13	Pell-type recurrence or quadratic irrational stabilizer	2005-D2-P6
M-ANT-12	Normal subgroup and quotient obstruction	2005-D1-P6
M-ANT-04	Finite differences and integer polynomial values	2014-D2-P1
M-FE-06	Nonexistence by growth iteration	2024-D2-P6
M-GEO-09	Regular simplex and equidistant orthonormal translation	2008-D1-P6
M-GEO-08	Conic, sphere, and directrix algebra	1998-D1-P5
M-FE-04	Orbit decomposition for bijections	1997-D2-P5
M-LA-11	Gram matrix and distance-to-inner-product conversion	2006-D2-P4
M-RA-17	Approximation by polynomials or kernels	2011-D2-P2
M-LA-14	Parity or zero-pattern block determinant split	2008-D2-P5
M-LA-09	Invariant subspace flag and matrix-shape count	1998-D2-P1
M-RA-18	Trigonometric/hyperbolic identity plus monotonicity	1995-D2-P5
M-GEO-10	Closest-point uniqueness by moving toward a minimizer	2004-D1-P1
M-LA-18	Commutator telescoping through powers	1994-D1-P4
M-CMB-09	Partition, refinement, and cover compression	
M-CMB-11	Erdos-Szekeres/binomial threshold induction	
M-CMB-15	Rewriting-system reduction	
M-CX-07	Multiplicity count for values 0 and 1	
M-FE-05	Translation/rational-shift invariance	
M-GEO-07	Metric contraction and diameter-cover comparison	
M-LA-08	Simultaneous diagonalization or common eigenvector induction	
M-RA-14	Endpoint Laplace-type asymptotic comparison	
M-UNI-01	Normalization, WLOG reduction, and scaling	

ID	Method	Problem IDs
M-UNI-10	Peeling, deletion, and reduction to a core	

18 Problem-by-Problem Method Audit

This table is the method-level analog of the previous theory sufficiency audit. It is the sweep through every detected problem entry. The *core method sequence* gives the recommended solution-move order to train before looking at the official solution. The *method IDs* can be joined with the method-node database in the ZIP package.

18.1 1994

18.1.1 Day 1

1994-D1-P1 (PDF p. 2) *inequality / sharp bound*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Sign forcing from positive matrices or inner products; 3. Double counting ordered or weighted objects
Method IDs	M-CMB-01; M-LA-17; M-CMB-02; M-LA-01; M-LA-03; M-LA-02; M-RA-09; M-LA-04
Method titles	Pigeonhole threshold and collision forcing; Sign forcing from positive matrices or inner products; Double counting ordered or weighted objects; Row/column elimination to triangular or sparse form; Block decomposition and Schur-style bookkeeping; Complete a matrix equation to an inverse identity; Convexity, Jensen, tangent, and chord inequalities; Rank-nullity and dimension-count obstruction
Theory dependencies	Pigeonhole principle and extremal element selection; Matrix products, inverses, and one-sided inverse identities; Rank-nullity, image/kernel, and rank of factorizations
Cue extraction	prior broad method tags: determinant / block manipulation, double counting / pigeonhole | statement triggers: matrix word; counting/permutation trigger | audit closures: AUD-LA-01; AUD-RA-08; AUD-CMB-03
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1994-D1-P2 (PDF p. 2) *inequality / sharp bound*

Primary domain	Real analysis / calculus
Core method sequence	1. Auxiliary function with derivative sign; 2. Extremal construction and equality-case engineering; 3. Monotonicity forced by functional equation
Method IDs	M-RA-01; M-UNI-03; M-FE-02; M-UNI-08; M-RA-03; M-RA-09; M-RA-05; M-RA-06
Method titles	Auxiliary function with derivative sign; Extremal construction and equality-case engineering; Monotonicity forced by functional equation; Sign, zero, and boundary case splitting; Rolle chain and divided-difference localization; Convexity, Jensen, tangent, and chord inequalities; L-Hospital and ratio/growth comparison; Riemann sum and sum-integral conversion
Theory dependencies	Auxiliary functions, integrating factors, and differential inequalities; L'Hospital, ratio limits, and growth comparisons; Riemann sums and sum-integral asymptotics
Cue extraction	prior broad method tags: monotonicity argument, construction / counterexample | statement triggers: continuous/differentiable trigger; integral/limit trigger; functional-equation trigger | audit closures: AUD-RA-01; AUD-RA-03; AUD-RA-05; AUD-RA-06; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1994-D1-P3 (PDF p. 2) *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Induction with strengthened hypothesis; 2. Reduction modulo a prime or finite field; 3. GCD, valuation, and prime-power bookkeeping
Method IDs	M-UNI-05; M-ANT-01; M-ANT-03; M-ANT-15; M-UNI-09; M-RA-09; M-ANT-07; M-CMB-01
Method titles	Induction with strengthened hypothesis; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Rationality and linear dependence over Q; Invariant extraction and invariant comparison; Convexity, Jensen, tangent, and chord inequalities; Vieta, symmetric sums, and logarithmic derivative; Pigeonhole threshold and collision forcing
Theory dependencies	Rational roots, Vieta, and rationality constraints; Pigeonhole principle and extremal element selection; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: induction | statement triggers: wording-specific | audit closures: AUD-ANT-04; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1994-D1-P4 (PDF p. 3) *existence / construction*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Nilpotency from zero spectrum plus finite dimension; 2. Commutator telescoping through powers; 3. Linear operator on a matrix space
Method IDs	M-LA-12; M-LA-18; M-LA-10; M-LA-06; M-UNI-03; M-UNI-08; M-LA-04; M-RA-09
Method titles	Nilpotency from zero spectrum plus finite dimension; Commutator telescoping through powers; Linear operator on a matrix space; Spectral/minimal-polynomial reduction; Extremal construction and equality-case engineering; Sign, zero, and boundary case splitting; Rank-nullity and dimension-count obstruction; Convexity, Jensen, tangent, and chord inequalities
Theory dependencies	Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; Commutators and spectral obstructions; Nilpotent, idempotent, involution, and square-zero structures
Cue extraction	prior broad method tags: spectral decomposition / eigenvalues, construction / counterexample | statement triggers: wording-specific | audit closures: AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1994-D1-P5 (PDF p. 4) *limit / asymptotic evaluation*

Primary domain	Functional equations / dynamics
Core method sequence	1. Periodic averaging and oscillation cancellation; 2. L-Hospital and ratio/growth comparison; 3. Special substitution and parameter freezing
Method IDs	M-RA-13; M-RA-05; M-UNI-07; M-RA-04; M-RA-01; M-RA-02; M-RA-06; M-RA-07
Method titles	Periodic averaging and oscillation cancellation; L-Hospital and ratio/growth comparison; Special substitution and parameter freezing; Taylor expansion and local asymptotic bookkeeping; Auxiliary function with derivative sign; Integrating factor and ODE comparison; Riemann sum and sum-integral conversion; Split into main range and tail/error range
Theory dependencies	Riemann sums and sum-integral asymptotics; Integral inequalities, integration by parts, and weighted estimates; Uniform continuity and periodic averaging
Cue extraction	prior broad method tags: auxiliary function / substitution, l'Hospital / asymptotic expansion | statement triggers: integral/limit trigger; functional-equation trigger | audit closures: AUD-RA-03; AUD-RA-05; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1994-D1-P6 (PDF p. 5) *inequality / sharp bound*

Primary domain	Algebra / number theory
Core method sequence	1. Convexity, Jensen, tangent, and chord inequalities; 2. Pigeonhole threshold and collision forcing; 3. Auxiliary function with derivative sign
Method IDs	M-RA-09; M-CMB-01; M-RA-01; M-FE-02; M-ANT-01; M-ANT-03; M-RA-03; M-RA-10
Method titles	Convexity, Jensen, tangent, and chord inequalities; Pigeonhole threshold and collision forcing; Auxiliary function with derivative sign; Monotonicity forced by functional equation; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Rolle chain and divided-difference localization; Cauchy-Schwarz, Holder, and norm comparison
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Lattices and norms/quasinorms; Pigeonhole principle and extremal element selection
Cue extraction	prior broad method tags: monotonicity argument, convexity / Jensen / tangent | statement triggers: continuous/differentiable trigger; integer/prime/mod trigger; geometry/metric trigger; functional-equation trigger | audit closures: AUD-RA-01; AUD-RA-06; AUD-RA-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.1.2 Day 2

1994-D2-P1 (PDF p. 6) *nonexistence / counterexample*

Primary domain	Functional equations / dynamics
Core method sequence	1. Auxiliary function with derivative sign; 2. Monotonicity forced by functional equation; 3. Rolle chain and divided-difference localization
Method IDs	M-RA-01; M-FE-02; M-RA-03; M-RA-02; M-RA-09; M-RA-11; M-UNI-07; M-UNI-03
Method titles	Auxiliary function with derivative sign; Monotonicity forced by functional equation; Rolle chain and divided-difference localization; Integrating factor and ODE comparison; Convexity, Jensen, tangent, and chord inequalities; Integration by parts, Fubini, and weighted identities; Special substitution and parameter freezing; Extremal construction and equality-case engineering
Theory dependencies	Auxiliary functions, integrating factors, and differential inequalities; Integral inequalities, integration by parts, and weighted estimates; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: monotonicity argument | statement triggers: continuous/differentiable trigger; functional-equation trigger | audit closures: AUD-RA-01; AUD-RA-02; AUD-RA-06; AUD-RA-08; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1994-D2-P2 (PDF p. 6) *classification / find all*

Primary domain	Functional equations / dynamics
Core method sequence	1. Compactness plus extreme value theorem; 2. Density, Baire/category, and pathological construction; 3. Compactness or extremal-point selection
Method IDs	M-RA-08; M-RA-16; M-UNI-06; M-UNI-07; M-RA-09; M-RA-10; M-FE-02; M-UNI-08
Method titles	Compactness plus extreme value theorem; Density, Baire/category, and pathological construction; Compactness or extremal-point selection; Special substitution and parameter freezing; Convexity, Jensen, tangent, and chord inequalities; Cauchy-Schwarz, Holder, and norm comparison; Monotonicity forced by functional equation; Sign, zero, and boundary case splitting
Theory dependencies	Compactness, extreme values, and closed-bounded reduction; Gradient/Lipschitz-gradient optimization and convex smoothness; Intermediate value, connectedness, density, and continuity transfer
Cue extraction	prior broad method tags: compactness / extremal point, topological / density argument | statement triggers: det/rank/eigen trigger; geometry/metric trigger; functional-equation trigger | audit closures: AUD-RA-07; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1994-D2-P3 (PDF p. 7) *proof of structural property*

Primary domain	Functional equations / dynamics
Core method sequence	1. Auxiliary function with derivative sign; 2. Special substitution and parameter freezing; 3. Integrating factor and ODE comparison
Method IDs	M-RA-01; M-UNI-07; M-RA-02; M-RA-03; M-RA-11; M-FE-02; M-UNI-05; M-UNI-09
Method titles	Auxiliary function with derivative sign; Special substitution and parameter freezing; Integrating factor and ODE comparison; Rolle chain and divided-difference localization; Integration by parts, Fubini, and weighted identities; Monotonicity forced by functional equation; Induction with strengthened hypothesis; Invariant extraction and invariant comparison
Theory dependencies	Rolle, mean-value theorem, and divided differences; Auxiliary functions, integrating factors, and differential inequalities; Taylor expansion, local asymptotics, and little-o bookkeeping
Cue extraction	prior broad method tags: auxiliary function / substitution | statement triggers: continuous/differentiable trigger; functional-equation trigger | audit closures: AUD-RA-01; AUD-RA-06
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1994-D2-P4 (PDF p. 7) *proof of structural property*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Rank-nullity and dimension-count obstruction
Method IDs	M-ANT-16; M-ANT-06; M-LA-04; M-CMB-01; M-CMB-02; M-ANT-05; M-LA-06; M-RA-03
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Rank-nullity and dimension-count obstruction; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Factorization and unique factorization in Z[x]; Spectral/minimal-polynomial reduction; Rolle chain and divided-difference localization
Theory dependencies	Invariant subspaces and adapted bases; Simultaneous diagonalization and commuting families; Polynomial identity and more-roots-than-degree principle
Cue extraction	prior broad method tags: rank-nullity / dimension count, polynomial identity / factorization, double counting / pigeonhole | statement triggers: matrix word; continuous/differentiable trigger; polynomial/root trigger | audit closures: AUD-RA-06; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-CMB-03
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1994-D2-P5 (PDF p. 8) *existence / construction*

Primary domain	Algebra / number theory
Core method sequence	1. Extremal construction and equality-case engineering; 2. Rank-nullity and dimension-count obstruction; 3. Cauchy-Schwarz, Holder, and norm comparison
Method IDs	M-UNI-03; M-LA-04; M-RA-10; M-UNI-05; M-UNI-08; M-ANT-01; M-ANT-03; M-ANT-10
Method titles	Extremal construction and equality-case engineering; Rank-nullity and dimension-count obstruction; Cauchy-Schwarz, Holder, and norm comparison; Induction with strengthened hypothesis; Sign, zero, and boundary case splitting; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Group action, orbit, and generator method
Theory dependencies	Explicit construction and counterexample design; Euclidean distances and inner-product identities; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: induction, Cauchy-Schwarz / norm inequality, rank-nullity / dimension count | statement triggers: integer/prime/mod trigger; counting/permutation trigger | audit closures: AUD-RA-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1994-D2-P6 (PDF p. 9) *limit / asymptotic evaluation*

Primary domain	Real analysis / calculus
Core method sequence	1. Auxiliary function with derivative sign; 2. Monotonicity forced by functional equation; 3. Riemann sum and sum-integral conversion
Method IDs	M-RA-01; M-FE-02; M-RA-06; M-RA-07; M-RA-05; M-RA-09; M-RA-08; M-FE-01
Method titles	Auxiliary function with derivative sign; Monotonicity forced by functional equation; Riemann sum and sum-integral conversion; Split into main range and tail/error range; L-Hospital and ratio/growth comparison; Convexity, Jensen, tangent, and chord inequalities; Compactness plus extreme value theorem; Iterates, fixed points, and cluster-point trapping
Theory dependencies	Riemann sums and sum-integral asymptotics; L'Hospital, ratio limits, and growth comparisons; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: monotonicity argument | statement triggers: integral/limit trigger | audit closures: AUD-RA-03; AUD-RA-05; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.2 1995

18.2.1 Day 1

1995-D1-P1 (PDF p. 11) *proof of structural property*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Nilpotency from zero spectrum plus finite dimension; 2. Similarity, adapted basis, and canonical model; 3. Rank-nullity and dimension-count obstruction
Method IDs	M-LA-12; M-LA-05; M-LA-04; M-LA-06; M-LA-01; M-LA-03; M-LA-02; M-LA-13
Method titles	Nilpotency from zero spectrum plus finite dimension; Similarity, adapted basis, and canonical model; Rank-nullity and dimension-count obstruction; Spectral/minimal-polynomial reduction; Row/column elimination to triangular or sparse form; Block decomposition and Schur-style bookkeeping; Complete a matrix equation to an inverse identity; Determinant through eigenvalue or root product
Theory dependencies	Rank-nullity, image/kernel, and rank of factorizations; Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; Nilpotent, idempotent, involution, and square-zero structures
Cue extraction	prior broad method tags: spectral decomposition / eigenvalues, determinant / block manipulation, rank-nullity / dimension count | statement triggers: matrix word; det/rank/eigen trigger | audit closures: AUD-LA-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1995-D1-P2 (PDF p. 11) *inequality / sharp bound*

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Auxiliary function with derivative sign
Method IDs	M-UNI-07; M-UNI-09; M-RA-01; M-RA-03; M-RA-09; M-RA-11; M-RA-08; M-UNI-03
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Auxiliary function with derivative sign; Rolle chain and divided-difference localization; Convexity, Jensen, tangent, and chord inequalities; Integration by parts, Fubini, and weighted identities; Compactness plus extreme value theorem; Extremal construction and equality-case engineering
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Integral inequalities, integration by parts, and weighted estimates; Fubini/Tonelli and change of variables
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: continuous/differentiable trigger; functional-equation trigger | audit closures: AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1995-D1-P3 (PDF p. 11) *limit / asymptotic evaluation*

Primary domain	Real analysis / calculus
Core method sequence	1. Auxiliary function with derivative sign; 2. Rolle chain and divided-difference localization; 3. Pigeonhole threshold and collision forcing
Method IDs	M-RA-01; M-RA-03; M-CMB-01; M-CMB-02; M-FE-02; M-RA-06; M-RA-07; M-RA-05
Method titles	Auxiliary function with derivative sign; Rolle chain and divided-difference localization; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Monotonicity forced by functional equation; Riemann sum and sum-integral conversion; Split into main range and tail/error range; L-Hospital and ratio/growth comparison
Theory dependencies	Rolle, mean-value theorem, and divided differences; L'Hospital, ratio limits, and growth comparisons; Riemann sums and sum-integral asymptotics
Cue extraction	prior broad method tags: Rolle / mean-value theorem, monotonicity argument, double counting / pigeonhole | statement triggers: continuous/differentiable trigger; integral/limit trigger; functional-equation trigger | audit closures: AUD-RA-01; AUD-RA-03; AUD-RA-05; AUD-RA-06; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1995-D1-P4 (PDF p. 12) *proof of structural property*

Primary domain	Real analysis / calculus
Core method sequence	1. Auxiliary function with derivative sign; 2. Monotonicity forced by functional equation; 3. Integrating factor and ODE comparison
Method IDs	M-RA-01; M-FE-02; M-RA-02; M-UNI-07; M-RA-09; M-RA-08; M-UNI-05; M-UNI-09
Method titles	Auxiliary function with derivative sign; Monotonicity forced by functional equation; Integrating factor and ODE comparison; Special substitution and parameter freezing; Convexity, Jensen, tangent, and chord inequalities; Compactness plus extreme value theorem; Induction with strengthened hypothesis; Invariant extraction and invariant comparison
Theory dependencies	Monotone functions, fixed points, and iteration on intervals; L'Hospital, ratio limits, and growth comparisons; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: auxiliary function / substitution, monotonicity argument | statement triggers: functional-equation trigger | audit closures: AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1995-D1-P5 (PDF p. 13) *proof of structural property*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Degree versus number of roots; 2. Nilpotency from zero spectrum plus finite dimension; 3. Polynomial identity by coefficient comparison
Method IDs	M-ANT-06; M-LA-12; M-ANT-16; M-ANT-05; M-LA-06; M-LA-04; M-UNI-05; M-UNI-09
Method titles	Degree versus number of roots; Nilpotency from zero spectrum plus finite dimension; Polynomial identity by coefficient comparison; Factorization and unique factorization in Z[x]; Spectral/minimal-polynomial reduction; Rank-nullity and dimension-count obstruction; Induction with strengthened hypothesis; Invariant extraction and invariant comparison
Theory dependencies	Polynomial identity and more-roots-than-degree principle; Nilpotent, idempotent, involution, and square-zero structures; Commutators and spectral obstructions
Cue extraction	prior broad method tags: polynomial identity / factorization | statement triggers: matrix word; det/rank/eigen trigger; integral/limit trigger | audit closures: AUD-LA-02; AUD-LA-03; AUD-PC-01; AUD-PC-08; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1995-D1-P6 (PDF p. 13) *existence / construction*

Primary domain	Real analysis / calculus
Core method sequence	1. Compactness plus extreme value theorem; 2. Extremal construction and equality-case engineering; 3. Compactness or extremal-point selection
Method IDs	M-RA-08; M-UNI-03; M-UNI-06; M-RA-09; M-RA-04; M-RA-16; M-RA-05; M-UNI-08
Method titles	Compactness plus extreme value theorem; Extremal construction and equality-case engineering; Compactness or extremal-point selection; Convexity, Jensen, tangent, and chord inequalities; Taylor expansion and local asymptotic bookkeeping; Density, Baire/category, and pathological construction; L-Hospital and ratio/growth comparison; Sign, zero, and boundary case splitting
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Compactness, extreme values, and closed-bounded reduction; Taylor expansion, local asymptotics, and little-o bookkeeping
Cue extraction	prior broad method tags: compactness / extremal point, convexity / Jensen / tangent, construction / counterexample | statement triggers: wording-specific | audit closures: AUD-RA-07; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.2.2 Day 2

1995-D2-P1 (PDF p. 15) *existence / construction*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Row/column elimination to triangular or sparse form; 2. Block decomposition and Schur-style bookkeeping; 3. Squared-distance and inner-product identities
Method IDs	M-LA-01; M-LA-03; M-GEO-03; M-RA-10; M-LA-04; M-LA-06; M-UNI-03; M-UNI-06
Method titles	Row/column elimination to triangular or sparse form; Block decomposition and Schur-style bookkeeping; Squared-distance and inner-product identities; Cauchy-Schwarz, Holder, and norm comparison; Rank-nullity and dimension-count obstruction; Spectral/minimal-polynomial reduction; Extremal construction and equality-case engineering; Compactness or extremal-point selection
Theory dependencies	Euclidean distances and inner-product identities; Trace cyclicity, trace pairing, and zero-trace subspaces; Determinant evaluation by row/column operations and triangular forms
Cue extraction	prior broad method tags: determinant / block manipulation | statement triggers: matrix word | audit closures: AUD-LA-05; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1995-D2-P2 (PDF p. 15) *exact computation*

Primary domain	Real analysis / calculus
Core method sequence	1. Invariant extraction and invariant comparison; 2. Special substitution and parameter freezing; 3. L-Hospital and ratio/growth comparison
Method IDs	M-UNI-09; M-UNI-07; M-RA-05; M-RA-06; M-RA-07; M-RA-09; M-RA-01; M-RA-08
Method titles	Invariant extraction and invariant comparison; Special substitution and parameter freezing; L-Hospital and ratio/growth comparison; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Convexity, Jensen, tangent, and chord inequalities; Auxiliary function with derivative sign; Compactness plus extreme value theorem
Theory dependencies	Sequence recurrences, monotone bounds, and generating functions; Riemann sums and sum-integral asymptotics; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: wording-specific | audit closures: AUD-RA-03; AUD-RA-05; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1995-D2-P3 (PDF p. 16) *proof of structural property*

Primary domain	Complex analysis / polynomials
Core method sequence	1. Unit-circle root real-part/log-derivative test; 2. Polynomial identity by coefficient comparison; 3. Degree versus number of roots
Method IDs	M-CX-03; M-ANT-16; M-ANT-06; M-CX-05; M-ANT-08; M-ANT-05; M-CMB-12; M-CX-01
Method titles	Unit-circle root real-part/log-derivative test; Polynomial identity by coefficient comparison; Degree versus number of roots; Trigonometric-to-complex conversion; Roots of unity and cyclotomic irreducibility; Factorization and unique factorization in Z[x]; Root-of-unity filter for residues; Cauchy coefficient formula and boundary maximum
Theory dependencies	Polynomial identity and more-roots-than-degree principle; Unit-circle roots and conjugation identities; Vieta, logarithmic derivatives, and real-root inequalities
Cue extraction	prior broad method tags: polynomial identity / factorization, roots of unity / Fourier filter | statement triggers: polynomial/root trigger; geometry/metric trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-PC-03; AUD-PC-05; AUD-PC-06
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1995-D2-P4 (PDF p. 17) *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Compactness plus extreme value theorem
Method IDs	M-ANT-16; M-ANT-06; M-RA-08; M-UNI-06; M-ANT-05; M-ANT-01; M-ANT-03; M-RA-01
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Compactness plus extreme value theorem; Compactness or extremal-point selection; Factorization and unique factorization in Z[x]; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Auxiliary function with derivative sign
Theory dependencies	Weierstrass approximation and polynomial approximation; Interpolation, Lagrange basis, and Chebyshev extremality; Polynomial identity and more-roots-than-degree principle
Cue extraction	prior broad method tags: compactness / extremal point, polynomial identity / factorization | statement triggers: continuous/differentiable trigger; integer/prime/mod trigger; functional-equation trigger | audit closures: AUD-RA-07; AUD-RA-08; AUD-PC-01; AUD-PC-09; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1995-D2-P5 (PDF p. 18) *proof of structural property*

Primary domain	Real analysis / calculus
Core method sequence	1. Trigonometric-to-complex conversion; 2. Trigonometric/hyperbolic identity plus monotonicity; 3. Riemann sum and sum-integral conversion
Method IDs	M-CX-05; M-RA-18; M-RA-06; M-RA-07; M-RA-09; M-RA-11; M-RA-01; M-RA-08
Method titles	Trigonometric-to-complex conversion; Trigonometric/hyperbolic identity plus monotonicity; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Convexity, Jensen, tangent, and chord inequalities; Integration by parts, Fubini, and weighted identities; Auxiliary function with derivative sign; Compactness plus extreme value theorem
Theory dependencies	Integral inequalities, integration by parts, and weighted estimates; Trigonometric and hyperbolic identities with monotonicity; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: Riemann-sum / integral comparison | statement triggers: functional-equation trigger | audit closures: AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1995-D2-P6 (PDF p. 20) *existence / construction*

Primary domain	Real analysis / calculus
Core method sequence	1. Extremal construction and equality-case engineering; 2. Density, Baire/category, and pathological construction; 3. Sign, zero, and boundary case splitting
Method IDs	M-UNI-03; M-RA-16; M-UNI-08; M-RA-01; M-FE-01; M-RA-03; M-RA-08; M-UNI-06
Method titles	Extremal construction and equality-case engineering; Density, Baire/category, and pathological construction; Sign, zero, and boundary case splitting; Auxiliary function with derivative sign; Iterates, fixed points, and cluster-point trapping; Rolle chain and divided-difference localization; Compactness plus extreme value theorem; Compactness or extremal-point selection
Theory dependencies	L2 spaces, orthonormal systems, and dominated convergence; Intermediate value, connectedness, density, and continuity transfer; Riemann sums and sum-integral asymptotics
Cue extraction	prior broad method tags: construction / counterexample, topological / density argument | statement triggers: continuous/differentiable trigger; integral/limit trigger | audit closures: AUD-RA-09; AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-07; AUD-ANT-11
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.3 1996

18.3.1 Day 1

1996-D1-P1 **(PDF p. 22)** *exact computation*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Row/column elimination to triangular or sparse form; 2. Block decomposition and Schur-style bookkeeping; 3. Rank-nullity and dimension-count obstruction
Method IDs	M-LA-01; M-LA-03; M-LA-04; M-LA-06; M-LA-07; M-LA-13; M-UNI-02; M-UNI-09
Method titles	Row/column elimination to triangular or sparse form; Block decomposition and Schur-style bookkeeping; Rank-nullity and dimension-count obstruction; Spectral/minimal-polynomial reduction; Trace cyclicity and commutator annihilation; Determinant through eigenvalue or root product; Small cases, equality cases, and example probing; Invariant extraction and invariant comparison
Theory dependencies	Determinant evaluation by row/column operations and triangular forms; Block matrices, Schur complements, and determinant identities; Rank-nullity, image/kernel, and rank of factorizations
Cue extraction	prior broad method tags: determinant / block manipulation | statement triggers: det/rank/eigen trigger | audit closures: AUD-CMB-05
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1996-D1-P2 **(PDF p. 23)** *exact computation*

Primary domain	Real analysis / calculus
Core method sequence	1. Auxiliary function with derivative sign; 2. Integrating factor and ODE comparison; 3. Special substitution and parameter freezing
Method IDs	M-RA-01; M-RA-02; M-UNI-07; M-RA-06; M-RA-07; M-RA-09; M-RA-11; M-RA-08
Method titles	Auxiliary function with derivative sign; Integrating factor and ODE comparison; Special substitution and parameter freezing; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Convexity, Jensen, tangent, and chord inequalities; Integration by parts, Fubini, and weighted identities; Compactness plus extreme value theorem
Theory dependencies	Trigonometric and hyperbolic identities with monotonicity; Integral inequalities, integration by parts, and weighted estimates; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: auxiliary function / substitution | statement triggers: integral/limit trigger | audit closures: AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1996-D1-P3 **(PDF p. 24)** *existence / construction*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Spectral/minimal-polynomial reduction; 2. Nilpotency from zero spectrum plus finite dimension; 3. Rank-nullity and dimension-count obstruction
Method IDs	M-LA-06; M-LA-12; M-LA-04; M-CMB-01; M-CMB-02; M-UNI-05; M-UNI-03; M-UNI-06
Method titles	Spectral/minimal-polynomial reduction; Nilpotency from zero spectrum plus finite dimension; Rank-nullity and dimension-count obstruction; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Induction with strengthened hypothesis; Extremal construction and equality-case engineering; Compactness or extremal-point selection
Theory dependencies	Nilpotent, idempotent, involution, and square-zero structures; Simultaneous diagonalization and commuting families; Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton
Cue extraction	prior broad method tags: induction, spectral decomposition / eigenvalues, rank-nullity / dimension count | statement triggers: det/rank/eigen trigger; counting/permutation trigger | audit closures: AUD-LA-04; AUD-RA-08; AUD-CMB-03; AUD-GEO-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1996-D1-P4 (PDF p. 24)

inequality / sharp bound

Primary domain	Real analysis / calculus
Core method sequence	1. Induction with strengthened hypothesis; 2. Riemann sum and sum-integral conversion; 3. Split into main range and tail/error range
Method IDs	M-UNI-05; M-RA-06; M-RA-07; M-RA-09; M-RA-05; M-RA-11; M-RA-01; M-RA-08
Method titles	Induction with strengthened hypothesis; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Convexity, Jensen, tangent, and chord inequalities; L-Hospital and ratio/growth comparison; Integration by parts, Fubini, and weighted identities; Auxiliary function with derivative sign; Compactness plus extreme value theorem
Theory dependencies	Sequence recurrences, monotone bounds, and generating functions; Integral inequalities, integration by parts, and weighted estimates; Riemann sums and sum-integral asymptotics
Cue extraction	prior broad method tags: induction | statement triggers: integral/limit trigger | audit closures: AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1996-D1-P5 (PDF p. 25)

inequality / sharp bound

Primary domain	Real analysis / calculus
Core method sequence	1. Riemann sum and sum-integral conversion; 2. Split into main range and tail/error range; 3. Rolle chain and divided-difference localization
Method IDs	M-RA-06; M-RA-07; M-RA-03; M-RA-01; M-FE-02; M-RA-09; M-RA-04; M-RA-08
Method titles	Riemann sum and sum-integral conversion; Split into main range and tail/error range; Rolle chain and divided-difference localization; Auxiliary function with derivative sign; Monotonicity forced by functional equation; Convexity, Jensen, tangent, and chord inequalities; Taylor expansion and local asymptotic bookkeeping; Compactness plus extreme value theorem
Theory dependencies	Riemann sums and sum-integral asymptotics; Taylor expansion, local asymptotics, and little-o bookkeeping; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: Rolle / mean-value theorem, monotonicity argument, Riemann-sum / integral comparison | statement triggers: continuous/differentiable trigger; integral/limit trigger; functional-equation trigger | audit closures: AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-06; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1996-D1-P6 (PDF p. 27)

inequality / sharp bound

Primary domain	Geometry / convexity / metric
Core method sequence	1. Compactness plus extreme value theorem; 2. Graph encoding of a relation; 3. Cycle extraction and alternating-cycle method
Method IDs	M-RA-08; M-CMB-03; M-CMB-04; M-LA-04; M-RA-16; M-UNI-06; M-UNI-05; M-GEO-03
Method titles	Compactness plus extreme value theorem; Graph encoding of a relation; Cycle extraction and alternating-cycle method; Rank-nullity and dimension-count obstruction; Density, Baire/category, and pathological construction; Compactness or extremal-point selection; Induction with strengthened hypothesis; Squared-distance and inner-product identities
Theory dependencies	Metric content, diameter covers, and contractions; Compactness, extreme values, and closed-bounded reduction; Intermediate value, connectedness, density, and continuity transfer
Cue extraction	prior broad method tags: induction, compactness / extremal point, rank-nullity / dimension count | statement triggers: graph/color/set trigger; geometry/metric trigger; functional-equation trigger | audit closures: AUD-RA-06; AUD-RA-07; AUD-RA-08; AUD-GEO-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.3.2 Day 2

1996-D2-P1 (PDF p. 29) *limit / asymptotic evaluation*

Primary domain	Functional equations / dynamics
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Riemann sum and sum-integral conversion
Method IDs	M-UNI-07; M-UNI-09; M-RA-06; M-RA-07; M-FE-01; M-RA-01; M-RA-03; M-RA-05
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Iterates, fixed points, and cluster-point trapping; Auxiliary function with derivative sign; Rolle chain and divided-difference localization; L-Hospital and ratio/growth comparison
Theory dependencies	Fixed points, iterates, and orbit closure; Intermediate value, connectedness, density, and continuity transfer; Riemann sums and sum-integral asymptotics
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: continuous/differentiable trigger; integral/limit trigger; functional-equation trigger | audit closures: AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08; AUD-RA-11
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1996-D2-P2 (PDF p. 30) *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Degree versus number of roots; 2. Polynomial identity by coefficient comparison; 3. Induction with strengthened hypothesis
Method IDs	M-ANT-06; M-ANT-16; M-UNI-05; M-ANT-05; M-RA-09; M-ANT-01; M-ANT-03; M-UNI-09
Method titles	Degree versus number of roots; Polynomial identity by coefficient comparison; Induction with strengthened hypothesis; Factorization and unique factorization in Z[x]; Convexity, Jensen, tangent, and chord inequalities; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Invariant extraction and invariant comparison
Theory dependencies	Trigonometric and hyperbolic identities with monotonicity; Rational roots, Vieta, and rationality constraints; Polynomial identity and more-roots-than-degree principle
Cue extraction	prior broad method tags: induction, polynomial identity / factorization | statement triggers: wording-specific | audit closures: AUD-RA-08; AUD-PC-01; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1996-D2-P3 (PDF p. 31) *counting / enumeration*

Primary domain	Algebra / number theory
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Double counting ordered or weighted objects; 3. Auxiliary function with derivative sign
Method IDs	M-CMB-01; M-CMB-02; M-RA-01; M-RA-02; M-UNI-07; M-ANT-10; M-ANT-11; M-CMB-10
Method titles	Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Auxiliary function with derivative sign; Integrating factor and ODE comparison; Special substitution and parameter freezing; Group action, orbit, and generator method; Commutator and center manipulation in groups/rings; Explicit bijection and encoding-decoding proof
Theory dependencies	Group basics: orders, cyclic subgroups, and generators; Matrix products, inverses, and one-sided inverse identities; Pigeonhole principle and extremal element selection
Cue extraction	prior broad method tags: auxiliary function / substitution, double counting / pigeonhole | statement triggers: matrix word; group/ring/field trigger; counting/permutation trigger | audit closures: AUD-RA-08; AUD-ANT-08; AUD-ANT-11; AUD-CMB-03; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1996-D2-P4 (PDF p. 32) *inequality / sharp bound*

Primary domain	Geometry / convexity / metric
Core method sequence	1. Convex support and John-ellipse perturbation; 2. Convexity, Jensen, tangent, and chord inequalities; 3. Compactness plus extreme value theorem
Method IDs	M-GEO-06; M-RA-09; M-RA-08; M-RA-16; M-UNI-06; M-GEO-01; M-GEO-11; M-GEO-03
Method titles	Convex support and John-ellipse perturbation; Convexity, Jensen, tangent, and chord inequalities; Compactness plus extreme value theorem; Density, Baire/category, and pathological construction; Compactness or extremal-point selection; Coordinate normalization and algebraization; Volume, projection, and ellipsoid computation; Squared-distance and inner-product identities
Theory dependencies	Convex bodies, supporting geometry, and John-ellipse ideas; Convexity, concavity, Jensen, and tangent/chord inequalities; Compactness, extreme values, and closed-bounded reduction
Cue extraction	prior broad method tags: compactness / extremal point, convexity / Jensen / tangent, topological / density argument | statement triggers: geometry/metric trigger | audit closures: AUD-RA-07; AUD-RA-08; AUD-GEO-02; AUD-GEO-06
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1996-D2-P5 (PDF p. 32) *inequality / sharp bound*

Primary domain	Functional equations / dynamics
Core method sequence	1. Taylor expansion and local asymptotic bookkeeping; 2. L-Hospital and ratio/growth comparison; 3. Riemann sum and sum-integral conversion
Method IDs	M-RA-04; M-RA-05; M-RA-06; M-RA-07; M-RA-09; M-RA-03; M-FE-02; M-UNI-07
Method titles	Taylor expansion and local asymptotic bookkeeping; L-Hospital and ratio/growth comparison; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Convexity, Jensen, tangent, and chord inequalities; Rolle chain and divided-difference localization; Monotonicity forced by functional equation; Special substitution and parameter freezing
Theory dependencies	Riemann sums and sum-integral asymptotics; Taylor expansion, local asymptotics, and little-o bookkeeping; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: Riemann-sum / integral comparison, l'Hospital / asymptotic expansion | statement triggers: integral/limit trigger | audit closures: AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-06; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1996-D2-P6 (PDF p. 34) *existence / construction*

Primary domain	Real analysis / calculus
Core method sequence	1. Cauchy-Schwarz, Holder, and norm comparison; 2. Integration by parts, Fubini, and weighted identities; 3. Pigeonhole threshold and collision forcing
Method IDs	M-RA-10; M-RA-11; M-CMB-01; M-CMB-02; M-RA-03; M-RA-09; M-RA-01; M-RA-08
Method titles	Cauchy-Schwarz, Holder, and norm comparison; Integration by parts, Fubini, and weighted identities; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Rolle chain and divided-difference localization; Convexity, Jensen, tangent, and chord inequalities; Auxiliary function with derivative sign; Compactness plus extreme value theorem
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Series convergence, rearrangements, and alternating/Dirichlet tools; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: double counting / pigeonhole | statement triggers: wording-specific | audit closures: AUD-RA-06; AUD-RA-08; AUD-PC-01; AUD-PC-13; AUD-CMB-03
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.4 1997

18.4.1 Day 1

1997-D1-P1 (PDF p. 36) *limit / asymptotic evaluation*

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Riemann sum and sum-integral conversion
Method IDs	M-UNI-07; M-UNI-09; M-RA-06; M-RA-07; M-RA-04; M-RA-05; M-RA-09; M-RA-11
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Taylor expansion and local asymptotic bookkeeping; L-Hospital and ratio/growth comparison; Convexity, Jensen, tangent, and chord inequalities; Integration by parts, Fubini, and weighted identities
Theory dependencies	Riemann sums and sum-integral asymptotics; Integral inequalities, integration by parts, and weighted estimates; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: integral/limit trigger | audit closures: AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08; AUD-CMB-05
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1997-D1-P2 (PDF p. 37) *proof / reasoning*

Primary domain	Real analysis / calculus
Core method sequence	1. Extremal construction and equality-case engineering; 2. Sign, zero, and boundary case splitting; 3. L-Hospital and ratio/growth comparison
Method IDs	M-UNI-03; M-UNI-08; M-RA-05; M-RA-06; M-RA-07; M-RA-09; M-RA-11; M-RA-01
Method titles	Extremal construction and equality-case engineering; Sign, zero, and boundary case splitting; L-Hospital and ratio/growth comparison; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Convexity, Jensen, tangent, and chord inequalities; Integration by parts, Fubini, and weighted identities; Auxiliary function with derivative sign
Theory dependencies	Series convergence, rearrangements, and alternating/Dirichlet tools; Integral inequalities, integration by parts, and weighted estimates; Riemann sums and sum-integral asymptotics
Cue extraction	prior broad method tags: construction / counterexample | statement triggers: integral/limit trigger | audit closures: AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08; AUD-RA-16
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1997-D1-P3 (PDF p. 37) *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Row/column elimination to triangular or sparse form; 2. Block decomposition and Schur-style bookkeeping; 3. Reduction modulo a prime or finite field
Method IDs	M-LA-01; M-LA-03; M-ANT-01; M-ANT-03; M-LA-07; M-LA-02; M-LA-12; M-UNI-05
Method titles	Row/column elimination to triangular or sparse form; Block decomposition and Schur-style bookkeeping; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Trace cyclicity and commutator annihilation; Complete a matrix equation to an inverse identity; Nilpotency from zero spectrum plus finite dimension; Induction with strengthened hypothesis
Theory dependencies	Commutators and spectral obstructions; Roots of unity, cyclotomic filters, and discrete Fourier tricks; Determinant evaluation by row/column operations and triangular forms
Cue extraction	prior broad method tags: determinant / block manipulation | statement triggers: matrix word; integer/prime/mod trigger | audit closures: AUD-PC-01; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1997-D1-P4 (PDF p. 38) *inequality / sharp bound*

Primary domain	Algebra / number theory
Core method sequence	1. Greedy expansion or tail-rationality method; 2. Extremal construction and equality-case engineering; 3. Monotonicity forced by functional equation
Method IDs	M-ANT-17; M-UNI-03; M-FE-02; M-RA-01; M-UNI-05; M-UNI-08; M-ANT-01; M-ANT-03
Method titles	Greedy expansion or tail-rationality method; Extremal construction and equality-case engineering; Monotonicity forced by functional equation; Auxiliary function with derivative sign; Induction with strengthened hypothesis; Sign, zero, and boundary case splitting; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping
Theory dependencies	Greedy infinite products and rationality tails; Integral inequalities, integration by parts, and weighted estimates; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: induction, monotonicity argument, construction / counterexample | statement triggers: integer/prime/mod trigger | audit closures: AUD-RA-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1997-D1-P5 (PDF p. 40) *inequality / sharp bound*

Primary domain	Algebra / number theory
Core method sequence	1. Cauchy-Schwarz, Holder, and norm comparison; 2. Reduction modulo a prime or finite field; 3. Squared-distance and inner-product identities
Method IDs	M-RA-10; M-ANT-01; M-GEO-03; M-ANT-03; M-RA-09; M-UNI-03; M-GEO-02; M-GEO-05
Method titles	Cauchy-Schwarz, Holder, and norm comparison; Reduction modulo a prime or finite field; Squared-distance and inner-product identities; GCD, valuation, and prime-power bookkeeping; Convexity, Jensen, tangent, and chord inequalities; Extremal construction and equality-case engineering; Projection to a lower-dimensional object; Affine invariance and change of metric scale
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Lattices and norms/quasinorms; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: Cauchy-Schwarz / norm inequality | statement triggers: integer/prime/mod trigger | audit closures: AUD-RA-08; AUD-GEO-02; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1997-D1-P6 (PDF p. 42) *nonexistence / counterexample*

Primary domain	Combinatorics / discrete structures
Core method sequence	1. Extremal construction and equality-case engineering; 2. Induction with strengthened hypothesis; 3. Sign, zero, and boundary case splitting
Method IDs	M-UNI-03; M-UNI-05; M-UNI-08; M-CMB-01; M-CMB-02; M-CMB-04; M-UNI-04; M-CMB-10
Method titles	Extremal construction and equality-case engineering; Induction with strengthened hypothesis; Sign, zero, and boundary case splitting; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Cycle extraction and alternating-cycle method; Contradiction with minimal or maximal counterexample; Explicit bijection and encoding-decoding proof
Theory dependencies	Set systems, intersecting families, and induction on sizes; Explicit construction and counterexample design; Graph encoding, cycles, and bipartite representations
Cue extraction	prior broad method tags: induction, construction / counterexample | statement triggers: graph/color/set trigger | audit closures: AUD-CMB-04; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.4.2 Day 2

1997-D2-P1 (PDF p. 44)

inequality / sharp bound

Primary domain	Real analysis / calculus
Core method sequence	1. Auxiliary function with derivative sign; 2. L-Hospital and ratio/growth comparison; 3. Taylor expansion and local asymptotic bookkeeping
Method IDs	M-RA-01; M-RA-05; M-RA-04; M-RA-02; M-UNI-07; M-RA-03; M-RA-09; M-UNI-03
Method titles	Auxiliary function with derivative sign; L-Hospital and ratio/growth comparison; Taylor expansion and local asymptotic bookkeeping; Integrating factor and ODE comparison; Special substitution and parameter freezing; Rolle chain and divided-difference localization; Convexity, Jensen, tangent, and chord inequalities; Extremal construction and equality-case engineering
Theory dependencies	Taylor expansion, local asymptotics, and little-o bookkeeping; Explicit construction and counterexample design; L'Hospital, ratio limits, and growth comparisons
Cue extraction	prior broad method tags: auxiliary function / substitution, l'Hospital / asymptotic expansion | statement triggers: continuous/differentiable trigger; functional-equation trigger | audit closures: AUD-RA-01; AUD-RA-03; AUD-RA-05; AUD-RA-06; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1997-D2-P2 (PDF p. 45)

proof of structural property

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Block decomposition and Schur-style bookkeeping; 2. Row/column elimination to triangular or sparse form; 3. Rank-nullity and dimension-count obstruction
Method IDs	M-LA-03; M-LA-01; M-LA-04; M-LA-13; M-LA-02; M-LA-06; M-LA-16; M-UNI-05
Method titles	Block decomposition and Schur-style bookkeeping; Row/column elimination to triangular or sparse form; Rank-nullity and dimension-count obstruction; Determinant through eigenvalue or root product; Complete a matrix equation to an inverse identity; Spectral/minimal-polynomial reduction; Trace pairing and orthogonal matrix subspaces; Induction with strengthened hypothesis
Theory dependencies	Block matrices, Schur complements, and determinant identities; Rank-nullity, image/kernel, and rank of factorizations; Determinant evaluation by row/column operations and triangular forms
Cue extraction	prior broad method tags: determinant / block manipulation, rank-nullity / dimension count | statement triggers: matrix word; det/rank/eigen trigger | audit closures: AUD-LA-03; AUD-LA-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1997-D2-P3 (PDF p. 45)

proof of structural property

Primary domain	Real analysis / calculus
Core method sequence	1. Rolle chain and divided-difference localization; 2. Taylor expansion and local asymptotic bookkeeping; 3. Riemann sum and sum-integral conversion
Method IDs	M-RA-03; M-RA-04; M-RA-06; M-RA-07; M-RA-09; M-RA-01; M-RA-08; M-UNI-05
Method titles	Rolle chain and divided-difference localization; Taylor expansion and local asymptotic bookkeeping; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Convexity, Jensen, tangent, and chord inequalities; Auxiliary function with derivative sign; Compactness plus extreme value theorem; Induction with strengthened hypothesis
Theory dependencies	Series convergence, rearrangements, and alternating/Dirichlet tools; Trigonometric and hyperbolic identities with monotonicity; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: Rolle / mean-value theorem | statement triggers: wording-specific | audit closures: AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-06; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1997-D2-P4 (PDF p. 46) *existence / construction*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Rank-nullity and dimension-count obstruction; 2. Trace cyclicity and commutator annihilation; 3. Riemann sum and sum-integral conversion
Method IDs	M-LA-04; M-LA-07; M-RA-06; M-RA-07; M-LA-03; M-LA-06; M-UNI-03; M-UNI-06
Method titles	Rank-nullity and dimension-count obstruction; Trace cyclicity and commutator annihilation; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Block decomposition and Schur-style bookkeeping; Spectral/minimal-polynomial reduction; Extremal construction and equality-case engineering; Compactness or extremal-point selection
Theory dependencies	Trace cyclicity, trace pairing, and zero-trace subspaces; Linear functionals, kernels, and dual span criteria; Rank-nullity, image/kernel, and rank of factorizations
Cue extraction	prior broad method tags: trace trick / commutator, rank-nullity / dimension count | statement triggers: matrix word; functional-equation trigger | audit closures: AUD-LA-06; AUD-LA-03; AUD-ANT-06
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1997-D2-P5 (PDF p. 47) *proof of structural property*

Primary domain	Geometry / convexity / metric
Core method sequence	1. Orbit decomposition for bijections; 2. Graph encoding of a relation; 3. Cycle extraction and alternating-cycle method
Method IDs	M-FE-04; M-CMB-03; M-CMB-04; M-RA-01; M-RA-02; M-UNI-07; M-GEO-03; M-UNI-05
Method titles	Orbit decomposition for bijections; Graph encoding of a relation; Cycle extraction and alternating-cycle method; Auxiliary function with derivative sign; Integrating factor and ODE comparison; Special substitution and parameter freezing; Squared-distance and inner-product identities; Induction with strengthened hypothesis
Theory dependencies	Bijections as products of involutions and symmetry decompositions; Graph encoding, cycles, and bipartite representations; Area, volume, projection, and ellipsoid computation
Cue extraction	prior broad method tags: auxiliary function / substitution, graph encoding / cycles | statement triggers: wording-specific | audit closures: AUD-RA-08; AUD-GEO-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1997-D2-P6 (PDF p. 47) *counting / enumeration*

Primary domain	Real analysis / calculus
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Double counting ordered or weighted objects; 3. Extremal construction and equality-case engineering
Method IDs	M-CMB-01; M-CMB-02; M-UNI-03; M-UNI-08; M-RA-01; M-RA-03; M-CMB-10; M-CMB-03
Method titles	Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Extremal construction and equality-case engineering; Sign, zero, and boundary case splitting; Auxiliary function with derivative sign; Rolle chain and divided-difference localization; Explicit bijection and encoding-decoding proof; Graph encoding of a relation
Theory dependencies	Baire/category, dense sets, and pathological continuous functions; Pigeonhole principle and extremal element selection; Double counting and degree-sum methods
Cue extraction	prior broad method tags: double counting / pigeonhole, construction / counterexample | statement triggers: continuous/differentiable trigger; counting/permutation trigger; functional-equation trigger | audit closures: AUD-RA-12; AUD-CMB-03; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.5 1998

18.5.1 Day 1

1998-D1-P1 (PDF p. 49) *exact computation*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Rank-nullity and dimension-count obstruction; 2. Block decomposition and Schur-style bookkeeping; 3. Spectral/minimal-polynomial reduction
Method IDs	M-LA-04; M-LA-03; M-LA-06; M-UNI-02; M-UNI-09; M-LA-16; M-CMB-03; M-CMB-04
Method titles	Rank-nullity and dimension-count obstruction; Block decomposition and Schur-style bookkeeping; Spectral/minimal-polynomial reduction; Small cases, equality cases, and example probing; Invariant extraction and invariant comparison; Trace pairing and orthogonal matrix subspaces; Graph encoding of a relation; Cycle extraction and alternating-cycle method
Theory dependencies	Invariant subspaces and adapted bases; Rank-nullity, image/kernel, and rank of factorizations; Linear functionals, kernels, and dual span criteria
Cue extraction	prior broad method tags: rank-nullity / dimension count | statement triggers: wording-specific | audit closures: AUD-LA-03; AUD-LA-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1998-D1-P2 (PDF p. 49) *proof of structural property*

Primary domain	Combinatorics / discrete structures
Core method sequence	1. Degree versus number of roots; 2. Polynomial identity by coefficient comparison; 3. Extremal construction and equality-case engineering
Method IDs	M-ANT-06; M-ANT-16; M-UNI-03; M-ANT-05; M-UNI-08; M-ANT-10; M-CMB-07; M-RA-09
Method titles	Degree versus number of roots; Polynomial identity by coefficient comparison; Extremal construction and equality-case engineering; Factorization and unique factorization in Z[x]; Sign, zero, and boundary case splitting; Group action, orbit, and generator method; Involution, parity, and sign-reversing cancellation; Convexity, Jensen, tangent, and chord inequalities
Theory dependencies	Permutation groups, cycles, parity, and generation; Polynomial identity and more-roots-than-degree principle; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: polynomial identity / factorization, construction / counterexample | statement triggers: counting/permutation trigger | audit closures: AUD-ANT-08; AUD-RA-08; AUD-PC-01; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1998-D1-P3 (PDF p. 50) *exact computation*

Primary domain	Real analysis / calculus
Core method sequence	1. Auxiliary function with derivative sign; 2. Monotonicity forced by functional equation; 3. Induction with strengthened hypothesis
Method IDs	M-RA-01; M-FE-02; M-UNI-05; M-RA-06; M-RA-07; M-FE-01; M-RA-09; M-RA-11
Method titles	Auxiliary function with derivative sign; Monotonicity forced by functional equation; Induction with strengthened hypothesis; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Iterates, fixed points, and cluster-point trapping; Convexity, Jensen, tangent, and chord inequalities; Integration by parts, Fubini, and weighted identities
Theory dependencies	Recurrence transformations and conjugacy; Integral inequalities, integration by parts, and weighted estimates; Riemann sums and sum-integral asymptotics
Cue extraction	prior broad method tags: induction, monotonicity argument | statement triggers: integral/limit trigger; functional-equation trigger | audit closures: AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1998-D1-P4 (PDF p. 50) *existence / construction*

Primary domain	Real analysis / calculus
Core method sequence	1. Density, Baire/category, and pathological construction; 2. Auxiliary function with derivative sign; 3. Rolle chain and divided-difference localization
Method IDs	M-RA-16; M-RA-01; M-RA-03; M-RA-09; M-RA-08; M-UNI-03; M-UNI-06; M-RA-02
Method titles	Density, Baire/category, and pathological construction; Auxiliary function with derivative sign; Rolle chain and divided-difference localization; Convexity, Jensen, tangent, and chord inequalities; Compactness plus extreme value theorem; Extremal construction and equality-case engineering; Compactness or extremal-point selection; Integrating factor and ODE comparison
Theory dependencies	Rolle, mean-value theorem, and divided differences; Auxiliary functions, integrating factors, and differential inequalities; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: topological / density argument | statement triggers: continuous/differentiable trigger; functional-equation trigger | audit closures: AUD-RA-01; AUD-RA-06; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1998-D1-P5 (PDF p. 51) *inequality / sharp bound*

Primary domain	Real analysis / calculus
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Conic, sphere, and directrix algebra; 3. Polynomial identity by coefficient comparison
Method IDs	M-CMB-01; M-GEO-08; M-ANT-16; M-ANT-06; M-ANT-05; M-RA-09; M-RA-01; M-RA-08
Method titles	Pigeonhole threshold and collision forcing; Conic, sphere, and directrix algebra; Polynomial identity by coefficient comparison; Degree versus number of roots; Factorization and unique factorization in Z[x]; Convexity, Jensen, tangent, and chord inequalities; Auxiliary function with derivative sign; Compactness plus extreme value theorem
Theory dependencies	Vieta, logarithmic derivatives, and real-root inequalities; Polynomial identity and more-roots-than-degree principle; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: polynomial identity / factorization | statement triggers: polynomial/root trigger | audit closures: AUD-PC-02; AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1998-D1-P6 (PDF p. 51) *inequality / sharp bound*

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Auxiliary function with derivative sign
Method IDs	M-UNI-07; M-UNI-09; M-RA-01; M-RA-03; M-RA-09; M-RA-11; M-RA-08; M-UNI-03
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Auxiliary function with derivative sign; Rolle chain and divided-difference localization; Convexity, Jensen, tangent, and chord inequalities; Integration by parts, Fubini, and weighted identities; Compactness plus extreme value theorem; Extremal construction and equality-case engineering
Theory dependencies	Integral inequalities, integration by parts, and weighted estimates; Euclidean distances and inner-product identities; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: continuous/differentiable trigger; functional-equation trigger | audit closures: AUD-GEO-09; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.5.2 Day 2

1998-D2-P1 (PDF p. 52) *proof of structural property*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Rank-nullity and dimension-count obstruction; 2. Induction with strengthened hypothesis; 3. Cauchy-Schwarz, Holder, and norm comparison
Method IDs	M-LA-04; M-UNI-05; M-RA-10; M-LA-06; M-RA-09; M-UNI-09; M-LA-05; M-LA-09
Method titles	Rank-nullity and dimension-count obstruction; Induction with strengthened hypothesis; Cauchy-Schwarz, Holder, and norm comparison; Spectral/minimal-polynomial reduction; Convexity, Jensen, tangent, and chord inequalities; Invariant extraction and invariant comparison; Similarity, adapted basis, and canonical model; Invariant subspace flag and matrix-shape count
Theory dependencies	Linear functionals, kernels, and dual span criteria; Rank-nullity, image/kernel, and rank of factorizations; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: induction, Cauchy-Schwarz / norm inequality, rank-nullity / dimension count | statement triggers: functional-equation trigger | audit closures: AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1998-D2-P2 (PDF p. 52) *classification / find all*

Primary domain	Real analysis / calculus
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Compactness plus extreme value theorem
Method IDs	M-ANT-16; M-ANT-06; M-RA-08; M-ANT-05; M-UNI-06; M-RA-01; M-RA-09; M-UNI-07
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Compactness plus extreme value theorem; Factorization and unique factorization in Z[x]; Compactness or extremal-point selection; Auxiliary function with derivative sign; Convexity, Jensen, tangent, and chord inequalities; Special substitution and parameter freezing
Theory dependencies	Interpolation, Lagrange basis, and Chebyshev extremality; Polynomial identity and more-roots-than-degree principle; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: compactness / extremal point, polynomial identity / factorization | statement triggers: polynomial/root trigger; functional-equation trigger | audit closures: AUD-RA-01; AUD-RA-07; AUD-RA-08; AUD-PC-01; AUD-PC-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1998-D2-P3 (PDF p. 53) *inequality / sharp bound*

Primary domain	Functional equations / dynamics
Core method sequence	1. Monotonicity forced by functional equation; 2. Graph encoding of a relation; 3. Pigeonhole threshold and collision forcing
Method IDs	M-FE-02; M-CMB-03; M-CMB-01; M-CMB-02; M-CMB-04; M-RA-01; M-FE-01; M-RA-09
Method titles	Monotonicity forced by functional equation; Graph encoding of a relation; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Cycle extraction and alternating-cycle method; Auxiliary function with derivative sign; Iterates, fixed points, and cluster-point trapping; Convexity, Jensen, tangent, and chord inequalities
Theory dependencies	Piecewise-linear maps and periodic points; Graph encoding, cycles, and bipartite representations; Pigeonhole principle and extremal element selection
Cue extraction	prior broad method tags: monotonicity argument, double counting / pigeonhole, graph encoding / cycles | statement triggers: geometry/metric trigger; functional-equation trigger | audit closures: AUD-RA-03; AUD-RA-08; AUD-CMB-03
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1998-D2-P4 (PDF p. 53) *counting / enumeration*

Primary domain	Functional equations / dynamics
Core method sequence	1. Monotonicity forced by functional equation; 2. Pigeonhole threshold and collision forcing; 3. Double counting ordered or weighted objects
Method IDs	M-FE-02; M-CMB-01; M-CMB-02; M-RA-01; M-RA-06; M-RA-07; M-CMB-10; M-CMB-05
Method titles	Monotonicity forced by functional equation; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Auxiliary function with derivative sign; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Explicit bijection and encoding-decoding proof; State recurrence or automaton partition
Theory dependencies	Recurrences and state partitions; Pigeonhole principle and extremal element selection; Double counting and degree-sum methods
Cue extraction	prior broad method tags: monotonicity argument, double counting / pigeonhole | statement triggers: counting/permutation trigger; graph/color/set trigger; functional-equation trigger | audit closures: AUD-RA-08; AUD-CMB-03; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1998-D2-P5 (PDF p. 54) *counting / enumeration*

Primary domain	Geometry / convexity / metric
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Double counting ordered or weighted objects; 3. Density, Baire/category, and pathological construction
Method IDs	M-CMB-01; M-CMB-02; M-RA-16; M-GEO-01; M-GEO-11; M-GEO-03; M-CMB-03; M-RA-09
Method titles	Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Density, Baire/category, and pathological construction; Coordinate normalization and algebraization; Volume, projection, and ellipsoid computation; Squared-distance and inner-product identities; Graph encoding of a relation; Convexity, Jensen, tangent, and chord inequalities
Theory dependencies	Pigeonhole principle and extremal element selection; Projections and orthogonal decompositions; Double counting and degree-sum methods
Cue extraction	prior broad method tags: double counting / pigeonhole, topological / density argument | statement triggers: counting/permutation trigger; graph/color/set trigger; geometry/metric trigger | audit closures: AUD-RA-07; AUD-RA-08; AUD-CMB-03; AUD-GEO-02; AUD-GEO-09
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1998-D2-P6 (PDF p. 54) *existence / construction*

Primary domain	Real analysis / calculus
Core method sequence	1. Extremal construction and equality-case engineering; 2. Pigeonhole threshold and collision forcing; 3. Double counting ordered or weighted objects
Method IDs	M-UNI-03; M-CMB-01; M-CMB-02; M-UNI-08; M-RA-01; M-RA-03; M-RA-09; M-RA-15
Method titles	Extremal construction and equality-case engineering; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Sign, zero, and boundary case splitting; Auxiliary function with derivative sign; Rolle chain and divided-difference localization; Convexity, Jensen, tangent, and chord inequalities; Recurrence transformed to monotone quotient or telescoping form
Theory dependencies	Baire/category, dense sets, and pathological continuous functions; Measure, variation, and differentiability almost everywhere; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: double counting / pigeonhole, construction / counterexample | statement triggers: continuous/differentiable trigger; geometry/metric trigger; functional-equation trigger | audit closures: AUD-RA-06; AUD-RA-08; AUD-RA-14; AUD-PC-13; AUD-CMB-03
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.6 1999

18.6.1 Day 1

1999-D1-P1 (PDF p. 56) *existence / construction*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Row/column elimination to triangular or sparse form; 2. Spectral/minimal-polynomial reduction; 3. Degree versus number of roots
Method IDs	M-LA-01; M-LA-06; M-ANT-06; M-ANT-16; M-UNI-06; M-RA-08; M-LA-03; M-ANT-05
Method titles	Row/column elimination to triangular or sparse form; Spectral/minimal-polynomial reduction; Degree versus number of roots; Polynomial identity by coefficient comparison; Compactness or extremal-point selection; Compactness plus extreme value theorem; Block decomposition and Schur-style bookkeeping; Factorization and unique factorization in Z[x]
Theory dependencies	Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; Determinant evaluation by row/column operations and triangular forms; Polynomial identity and more-roots-than-degree principle
Cue extraction	prior broad method tags: compactness / extremal point, spectral decomposition / eigenvalues, determinant / block manipulation | statement triggers: matrix word; det/rank/eigen trigger | audit closures: AUD-LA-07; AUD-LA-10; AUD-RA-07; AUD-PC-01; AUD-PC-05
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1999-D1-P2 (PDF p. 56) *nonexistence / counterexample*

Primary domain	Real analysis / calculus
Core method sequence	1. Extremal construction and equality-case engineering; 2. Sign, zero, and boundary case splitting; 3. Riemann sum and sum-integral conversion
Method IDs	M-UNI-03; M-UNI-08; M-RA-06; M-RA-07; M-RA-01; M-RA-09; M-RA-08; M-UNI-04
Method titles	Extremal construction and equality-case engineering; Sign, zero, and boundary case splitting; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Auxiliary function with derivative sign; Convexity, Jensen, tangent, and chord inequalities; Compactness plus extreme value theorem; Contradiction with minimal or maximal counterexample
Theory dependencies	Series convergence, rearrangements, and alternating/Dirichlet tools; Pigeonhole principle and extremal element selection; Explicit construction and counterexample design
Cue extraction	prior broad method tags: construction / counterexample | statement triggers: wording-specific | audit closures: AUD-RA-08; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1999-D1-P3 (PDF p. 57) *inequality / sharp bound*

Primary domain	Algebra / number theory
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Reduction modulo a prime or finite field
Method IDs	M-UNI-07; M-UNI-09; M-ANT-01; M-ANT-03; M-RA-09; M-UNI-03; M-ANT-02; M-ANT-10
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Convexity, Jensen, tangent, and chord inequalities; Extremal construction and equality-case engineering; Chinese remainder and residue construction; Group action, orbit, and generator method
Theory dependencies	No-regularity functions and order patterns on Q or R; Convexity, concavity, Jensen, and tangent/chord inequalities; Modular arithmetic and divisibility
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: integer/prime/mod trigger; functional-equation trigger | audit closures: AUD-RA-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1999-D1-P4 (PDF p. 57) *classification / find all*

Primary domain	Functional equations / dynamics
Core method sequence	1. Auxiliary function with derivative sign; 2. Special substitution and parameter freezing; 3. Monotonicity forced by functional equation
Method IDs	M-RA-01; M-UNI-07; M-FE-02; M-RA-02; M-UNI-05; M-RA-09; M-RA-11; M-UNI-08
Method titles	Auxiliary function with derivative sign; Special substitution and parameter freezing; Monotonicity forced by functional equation; Integrating factor and ODE comparison; Induction with strengthened hypothesis; Convexity, Jensen, tangent, and chord inequalities; Integration by parts, Fubini, and weighted identities; Sign, zero, and boundary case splitting
Theory dependencies	Injectivity and monotonicity forced by a functional equation; Integral inequalities, integration by parts, and weighted estimates; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: induction, auxiliary function / substitution, monotonicity argument | statement triggers: functional-equation trigger | audit closures: AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1999-D1-P5 (PDF p. 57)

proof of structural property

Primary domain	Combinatorics / discrete structures
Core method sequence	1. Graph encoding of a relation; 2. Cycle extraction and alternating-cycle method; 3. Double counting ordered or weighted objects
Method IDs	M-CMB-03; M-CMB-04; M-CMB-02; M-CMB-01; M-LA-03; M-LA-01; M-UNI-05; M-UNI-09
Method titles	Graph encoding of a relation; Cycle extraction and alternating-cycle method; Double counting ordered or weighted objects; Pigeonhole threshold and collision forcing; Block decomposition and Schur-style bookkeeping; Row/column elimination to triangular or sparse form; Induction with strengthened hypothesis; Invariant extraction and invariant comparison
Theory dependencies	Graph encoding, cycles, and bipartite representations; Grid, lattice, and alternating-cycle arguments; Pigeonhole principle and extremal element selection
Cue extraction	prior broad method tags: determinant / block manipulation, double counting / pigeonhole, graph encoding / cycles | statement triggers: geometry/metric trigger | audit closures: AUD-RA-08; AUD-CMB-02; AUD-CMB-03; AUD-GEO-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1999-D1-P6 (PDF p. 58)

proof / reasoning

Primary domain	Functional equations / dynamics
Core method sequence	1. Auxiliary function with derivative sign; 2. Special substitution and parameter freezing; 3. Extremal construction and equality-case engineering
Method IDs	M-RA-01; M-UNI-07; M-UNI-03; M-RA-02; M-UNI-08; M-RA-03; M-RA-09; M-FE-02
Method titles	Auxiliary function with derivative sign; Special substitution and parameter freezing; Extremal construction and equality-case engineering; Integrating factor and ODE comparison; Sign, zero, and boundary case splitting; Rolle chain and divided-difference localization; Convexity, Jensen, tangent, and chord inequalities; Monotonicity forced by functional equation
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Integral inequalities, integration by parts, and weighted estimates; Rolle, mean-value theorem, and divided differences
Cue extraction	prior broad method tags: auxiliary function / substitution, construction / counterexample | statement triggers: continuous/differentiable trigger; functional-equation trigger | audit closures: AUD-RA-01; AUD-RA-06; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.6.2 Day 2

1999-D2-P1 (PDF p. 59) *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Invariant extraction and invariant comparison; 2. Special substitution and parameter freezing; 3. Group action, orbit, and generator method
Method IDs	M-UNI-09; M-UNI-07; M-ANT-10; M-ANT-11; M-ANT-01; M-ANT-03; M-ANT-14; M-UNI-05
Method titles	Invariant extraction and invariant comparison; Special substitution and parameter freezing; Group action, orbit, and generator method; Commutator and center manipulation in groups/rings; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Semidirect product or explicit algebraic construction; Induction with strengthened hypothesis
Theory dependencies	Rings, idempotents, nil rings, and noncommutativity; Group basics: orders, cyclic subgroups, and generators; Automorphisms and semidirect products
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: group/ring/field trigger | audit closures: AUD-LA-03; AUD-LA-12; AUD-ANT-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1999-D2-P2 (PDF p. 59) *counting / enumeration*

Primary domain	Algebra / number theory
Core method sequence	1. Root-of-unity filter for residues; 2. Reduction modulo a prime or finite field; 3. Degree versus number of roots
Method IDs	M-CMB-12; M-ANT-01; M-ANT-06; M-ANT-08; M-ANT-16; M-LA-03; M-ANT-02; M-ANT-05
Method titles	Root-of-unity filter for residues; Reduction modulo a prime or finite field; Degree versus number of roots; Roots of unity and cyclotomic irreducibility; Polynomial identity by coefficient comparison; Block decomposition and Schur-style bookkeeping; Chinese remainder and residue construction; Factorization and unique factorization in Z[x]
Theory dependencies	Roots of unity, cyclotomic filters, and discrete Fourier tricks; Discrete probability via residues and recurrences; Modular arithmetic and divisibility
Cue extraction	prior broad method tags: induction, determinant / block manipulation, polynomial identity / factorization | statement triggers: integral/limit trigger; integer/prime/mod trigger | audit closures: AUD-CMB-09; AUD-PC-01; AUD-PC-05; AUD-PC-06; AUD-PC-13; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1999-D2-P3 (PDF p. 60) *inequality / sharp bound*

Primary domain	Real analysis / calculus
Core method sequence	1. Auxiliary function with derivative sign; 2. Integrating factor and ODE comparison; 3. Special substitution and parameter freezing
Method IDs	M-RA-01; M-RA-02; M-UNI-07; M-RA-09; M-RA-11; M-RA-08; M-UNI-03; M-LA-03
Method titles	Auxiliary function with derivative sign; Integrating factor and ODE comparison; Special substitution and parameter freezing; Convexity, Jensen, tangent, and chord inequalities; Integration by parts, Fubini, and weighted identities; Compactness plus extreme value theorem; Extremal construction and equality-case engineering; Block decomposition and Schur-style bookkeeping
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Integral inequalities, integration by parts, and weighted estimates; Series convergence, rearrangements, and alternating/Dirichlet tools
Cue extraction	prior broad method tags: auxiliary function / substitution | statement triggers: integral/limit trigger | audit closures: AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1999-D2-P4 (PDF p. 60) *existence / construction*

Primary domain	Functional equations / dynamics
Core method sequence	1. Monotonicity forced by functional equation; 2. Auxiliary function with derivative sign; 3. Convexity, Jensen, tangent, and chord inequalities
Method IDs	M-FE-02; M-RA-01; M-RA-09; M-RA-11; M-UNI-07; M-UNI-03; M-UNI-06; M-FE-01
Method titles	Monotonicity forced by functional equation; Auxiliary function with derivative sign; Convexity, Jensen, tangent, and chord inequalities; Integration by parts, Fubini, and weighted identities; Special substitution and parameter freezing; Extremal construction and equality-case engineering; Compactness or extremal-point selection; Iterates, fixed points, and cluster-point trapping
Theory dependencies	No-regularity functions and order patterns on Q or R; Integral inequalities, integration by parts, and weighted estimates; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: monotonicity argument | statement triggers: functional-equation trigger | audit closures: AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1999-D2-P5 (PDF p. 60) *proof of structural property*

Primary domain	Combinatorics / discrete structures
Core method sequence	1. Extremal construction and equality-case engineering; 2. Sign, zero, and boundary case splitting; 3. Induction with strengthened hypothesis
Method IDs	M-UNI-03; M-UNI-08; M-UNI-05; M-CMB-01; M-CMB-02; M-UNI-09; M-RA-09; M-CMB-03
Method titles	Extremal construction and equality-case engineering; Sign, zero, and boundary case splitting; Induction with strengthened hypothesis; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Invariant extraction and invariant comparison; Convexity, Jensen, tangent, and chord inequalities; Graph encoding of a relation
Theory dependencies	Words, equivalence relations, and rewriting systems; Graph encoding, cycles, and bipartite representations; Explicit construction and counterexample design
Cue extraction	prior broad method tags: construction / counterexample | statement triggers: wording-specific | audit closures: AUD-RA-08; AUD-ANT-12
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

1999-D2-P6 (PDF p. 61) *existence / construction*

Primary domain	Real analysis / calculus
Core method sequence	1. Roots of unity and cyclotomic irreducibility; 2. Pigeonhole threshold and collision forcing; 3. Trigonometric-to-complex conversion
Method IDs	M-ANT-08; M-CMB-01; M-CX-05; M-ANT-16; M-ANT-06; M-CMB-12; M-ANT-05; M-CX-01
Method titles	Roots of unity and cyclotomic irreducibility; Pigeonhole threshold and collision forcing; Trigonometric-to-complex conversion; Polynomial identity by coefficient comparison; Degree versus number of roots; Root-of-unity filter for residues; Factorization and unique factorization in Z[x]; Cauchy coefficient formula and boundary maximum
Theory dependencies	Roots of unity, cyclotomic filters, and discrete Fourier tricks; Pigeonhole principle and extremal element selection; Polynomial identity and more-roots-than-degree principle
Cue extraction	prior broad method tags: polynomial identity / factorization, roots of unity / Fourier filter | statement triggers: polynomial/root trigger; functional-equation trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-06; AUD-PC-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.7 2000

18.7.1 Day 1

2000-D1-P1 (PDF p. 62) *existence / construction*

Primary domain	Real analysis / calculus
Core method sequence	1. Compactness plus extreme value theorem; 2. Extremal construction and equality-case engineering; 3. Compactness or extremal-point selection
Method IDs	M-RA-08; M-UNI-03; M-UNI-06; M-RA-01; M-CMB-01; M-CMB-02; M-FE-02; M-UNI-08
Method titles	Compactness plus extreme value theorem; Extremal construction and equality-case engineering; Compactness or extremal-point selection; Auxiliary function with derivative sign; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Monotonicity forced by functional equation; Sign, zero, and boundary case splitting
Theory dependencies	Monotone functions, fixed points, and iteration on intervals; Explicit construction and counterexample design; Pigeonhole principle and extremal element selection
Cue extraction	prior broad method tags: compactness / extremal point, monotonicity argument, double counting / pigeonhole | statement triggers: functional-equation trigger | audit closures: AUD-RA-07; AUD-RA-08; AUD-RA-10; AUD-CMB-03
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2000-D1-P2 (PDF p. 62) *classification / find all*

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Polynomial identity by coefficient comparison
Method IDs	M-UNI-07; M-UNI-09; M-ANT-16; M-RA-01; M-ANT-08; M-FE-03; M-RA-03; M-RA-08
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Polynomial identity by coefficient comparison; Auxiliary function with derivative sign; Roots of unity and cyclotomic irreducibility; Composition, inversion, and conjugacy; Rolle chain and divided-difference localization; Compactness plus extreme value theorem
Theory dependencies	Polynomial functional identities and algebraic composition; Rolle, mean-value theorem, and divided differences; Taylor expansion, local asymptotics, and little-o bookkeeping
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: wording-specific | audit closures: AUD-RA-06; AUD-PC-01; AUD-PC-05; AUD-PC-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2000-D1-P3 (PDF p. 63)

proof of structural property

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Rank-nullity and dimension-count obstruction; 2. Spectral/minimal-polynomial reduction; 3. Trace cyclicity and commutator annihilation
Method IDs	M-LA-04; M-LA-06; M-LA-07; M-LA-12; M-LA-03; M-LA-15; M-LA-13; M-UNI-05
Method titles	Rank-nullity and dimension-count obstruction; Spectral/minimal-polynomial reduction; Trace cyclicity and commutator annihilation; Nilpotency from zero spectrum plus finite dimension; Block decomposition and Schur-style bookkeeping; AB versus BA transfer; Determinant through eigenvalue or root product; Induction with strengthened hypothesis
Theory dependencies	Commutators and spectral obstructions; Rank-nullity, image/kernel, and rank of factorizations; Nilpotent, idempotent, involution, and square-zero structures
Cue extraction	prior broad method tags: spectral decomposition / eigenvalues, trace trick / commutator, rank-nullity / dimension count | statement triggers: matrix word; det/rank/eigen trigger; continuous/differentiable trigger | audit closures: AUD-LA-03; AUD-LA-08; AUD-LA-12; AUD-RA-06; AUD-PC-01
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2000-D1-P4 (PDF p. 63)

inequality / sharp bound

Primary domain	Real analysis / calculus
Core method sequence	1. Auxiliary function with derivative sign; 2. Monotonicity forced by functional equation; 3. Convexity, Jensen, tangent, and chord inequalities
Method IDs	M-RA-01; M-FE-02; M-RA-09; M-RA-08; M-UNI-03; M-CX-05; M-FE-01; M-LA-03
Method titles	Auxiliary function with derivative sign; Monotonicity forced by functional equation; Convexity, Jensen, tangent, and chord inequalities; Compactness plus extreme value theorem; Extremal construction and equality-case engineering; Trigonometric-to-complex conversion; Iterates, fixed points, and cluster-point trapping; Block decomposition and Schur-style bookkeeping
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Integral inequalities, integration by parts, and weighted estimates; Monotone functions, fixed points, and iteration on intervals
Cue extraction	prior broad method tags: monotonicity argument | statement triggers: wording-specific | audit closures: AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2000-D1-P5 (PDF p. 64)

proof of structural property

Primary domain	Algebra / number theory
Core method sequence	1. Trace cyclicity and commutator annihilation; 2. Spectral/minimal-polynomial reduction; 3. Group action, orbit, and generator method
Method IDs	M-LA-07; M-LA-06; M-ANT-10; M-ANT-11; M-LA-12; M-ANT-03; M-ANT-01; M-ANT-14
Method titles	Trace cyclicity and commutator annihilation; Spectral/minimal-polynomial reduction; Group action, orbit, and generator method; Commutator and center manipulation in groups/rings; Nilpotency from zero spectrum plus finite dimension; GCD, valuation, and prime-power bookkeeping; Reduction modulo a prime or finite field; Semidirect product or explicit algebraic construction
Theory dependencies	Rings, idempotents, nil rings, and noncommutativity; Commutators and spectral obstructions; Group basics: orders, cyclic subgroups, and generators
Cue extraction	prior broad method tags: trace trick / commutator | statement triggers: integer/prime/mod trigger; group/ring/field trigger | audit closures: AUD-LA-02; AUD-LA-03; AUD-ANT-01; AUD-ANT-06; AUD-ANT-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2000-D1-P6 (PDF p. 64) *inequality / sharp bound*

Primary domain	Functional equations / dynamics
Core method sequence	1. Auxiliary function with derivative sign; 2. Rolle chain and divided-difference localization; 3. Monotonicity forced by functional equation
Method IDs	M-RA-01; M-RA-03; M-FE-02; M-UNI-07; M-ANT-06; M-ANT-16; M-RA-02; M-ANT-05
Method titles	Auxiliary function with derivative sign; Rolle chain and divided-difference localization; Monotonicity forced by functional equation; Special substitution and parameter freezing; Degree versus number of roots; Polynomial identity by coefficient comparison; Integrating factor and ODE comparison; Factorization and unique factorization in Z[x]
Theory dependencies	Auxiliary functions, integrating factors, and differential inequalities; L'Hospital, ratio limits, and growth comparisons; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: induction, auxiliary function / substitution, Rolle / mean-value theorem | statement triggers: continuous/differentiable trigger; integral/limit trigger; functional-equation trigger | audit closures: AUD-RA-01; AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-06
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.7.2 Day 2

2000-D2-P1 (PDF p. 66) *inequality / sharp bound*

Primary domain	Geometry / convexity / metric
Core method sequence	1. Reduction modulo a prime or finite field; 2. Chinese remainder and residue construction; 3. Rank-nullity and dimension-count obstruction
Method IDs	M-ANT-01; M-ANT-02; M-LA-04; M-GEO-01; M-GEO-11; M-RA-09; M-GEO-05; M-ANT-03
Method titles	Reduction modulo a prime or finite field; Chinese remainder and residue construction; Rank-nullity and dimension-count obstruction; Coordinate normalization and algebraization; Volume, projection, and ellipsoid computation; Convexity, Jensen, tangent, and chord inequalities; Affine invariance and change of metric scale; GCD, valuation, and prime-power bookkeeping
Theory dependencies	Cube and square dissection by number-theoretic semigroups; GCD, valuations, prime powers, and coprimality; Finite fields, field characteristic, and Fermat/Frobenius tools
Cue extraction	prior broad method tags: rank-nullity / dimension count, modular / finite-field reduction | statement triggers: geometry/metric trigger | audit closures: AUD-GEO-11; AUD-RA-08; AUD-PC-01; AUD-PC-13; AUD-ANT-02; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2000-D2-P2 (PDF p. 66) *existence / construction*

Primary domain	Real analysis / calculus
Core method sequence	1. Auxiliary function with derivative sign; 2. Compactness plus extreme value theorem; 3. Compactness or extremal-point selection
Method IDs	M-RA-01; M-RA-08; M-UNI-06; M-RA-16; M-FE-02; M-RA-03; M-RA-09; M-UNI-03
Method titles	Auxiliary function with derivative sign; Compactness plus extreme value theorem; Compactness or extremal-point selection; Density, Baire/category, and pathological construction; Monotonicity forced by functional equation; Rolle chain and divided-difference localization; Convexity, Jensen, tangent, and chord inequalities; Extremal construction and equality-case engineering
Theory dependencies	Compactness, extreme values, and closed-bounded reduction; Baire/category, dense sets, and pathological continuous functions; Intermediate value, connectedness, density, and continuity transfer
Cue extraction	prior broad method tags: compactness / extremal point, monotonicity argument, topological / density argument | statement triggers: continuous/differentiable trigger; geometry/metric trigger | audit closures: AUD-RA-07; AUD-RA-08; AUD-RA-10; AUD-RA-13; AUD-GEO-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2000-D2-P3 (PDF p. 66) *proof of structural property*

Primary domain	Complex analysis / polynomials
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Factorization and unique factorization in Z[x]
Method IDs	M-ANT-16; M-ANT-06; M-ANT-05; M-ANT-08; M-RA-09; M-CX-01; M-UNI-05; M-UNI-09
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Factorization and unique factorization in Z[x]; Roots of unity and cyclotomic irreducibility; Convexity, Jensen, tangent, and chord inequalities; Cauchy coefficient formula and boundary maximum; Induction with strengthened hypothesis; Invariant extraction and invariant comparison
Theory dependencies	Polynomial identity and more-roots-than-degree principle; Convexity, concavity, Jensen, and tangent/chord inequalities; Vieta, logarithmic derivatives, and real-root inequalities
Cue extraction	prior broad method tags: polynomial identity / factorization | statement triggers: polynomial/root trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-08; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2000-D2-P4 (PDF p. 67) *inequality / sharp bound*

Primary domain	Complex analysis / polynomials
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Graph encoding of a relation
Method IDs	M-ANT-16; M-ANT-06; M-CMB-03; M-CMB-04; M-RA-09; M-ANT-05; M-RA-06; M-RA-07
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Graph encoding of a relation; Cycle extraction and alternating-cycle method; Convexity, Jensen, tangent, and chord inequalities; Factorization and unique factorization in Z[x]; Riemann sum and sum-integral conversion; Split into main range and tail/error range
Theory dependencies	Vieta, logarithmic derivatives, and real-root inequalities; Polynomial identity and more-roots-than-degree principle; Conics: parabolas, ellipses, foci, and directrices
Cue extraction	prior broad method tags: convexity / Jensen / tangent, polynomial identity / factorization, graph encoding / cycles | statement triggers: polynomial/root trigger; graph/color/set trigger; geometry/metric trigger | audit closures: AUD-RA-06; AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2000-D2-P5 (PDF p. 68) *classification / find all*

Primary domain	Functional equations / dynamics
Core method sequence	1. Auxiliary function with derivative sign; 2. Special substitution and parameter freezing; 3. Monotonicity forced by functional equation
Method IDs	M-RA-01; M-UNI-07; M-FE-02; M-RA-02; M-RA-03; M-RA-04; M-RA-06; M-RA-07
Method titles	Auxiliary function with derivative sign; Special substitution and parameter freezing; Monotonicity forced by functional equation; Integrating factor and ODE comparison; Rolle chain and divided-difference localization; Taylor expansion and local asymptotic bookkeeping; Riemann sum and sum-integral conversion; Split into main range and tail/error range
Theory dependencies	Injectivity and monotonicity forced by a functional equation; Integral inequalities, integration by parts, and weighted estimates; Taylor expansion, local asymptotics, and little-o bookkeeping
Cue extraction	prior broad method tags: auxiliary function / substitution, monotonicity argument | statement triggers: functional-equation trigger | audit closures: AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-06; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2000-D2-P6 (PDF p. 69) *nonexistence / counterexample*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. AB versus BA transfer; 2. Spectral/minimal-polynomial reduction; 3. Degree versus number of roots
Method IDs	M-LA-15; M-LA-06; M-ANT-06; M-ANT-16; M-LA-12; M-ANT-05; M-RA-06; M-RA-07
Method titles	AB versus BA transfer; Spectral/minimal-polynomial reduction; Degree versus number of roots; Polynomial identity by coefficient comparison; Nilpotency from zero spectrum plus finite dimension; Factorization and unique factorization in Z[x]; Riemann sum and sum-integral conversion; Split into main range and tail/error range
Theory dependencies	Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; Block matrices, Schur complements, and determinant identities; Matrix exponentials and polynomial functional calculus
Cue extraction	prior broad method tags: spectral decomposition / eigenvalues, polynomial identity / factorization | statement triggers: matrix word; det/rank/eigen trigger; integral/limit trigger; polynomial/root trigger | audit closures: AUD-LA-02; AUD-LA-03; AUD-LA-08; AUD-LA-09; AUD-RA-06
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.8 2001

18.8.1 Day 1

2001-D1-P1 (PDF p. 70) *proof / reasoning*

Primary domain	Algebra / number theory
Core method sequence	1. Row/column elimination to triangular or sparse form; 2. Block decomposition and Schur-style bookkeeping; 3. Reduction modulo a prime or finite field
Method IDs	M-LA-01; M-LA-03; M-ANT-01; M-ANT-03; M-RA-11; M-UNI-07; M-UNI-09; M-ANT-06
Method titles	Row/column elimination to triangular or sparse form; Block decomposition and Schur-style bookkeeping; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Integration by parts, Fubini, and weighted identities; Special substitution and parameter freezing; Invariant extraction and invariant comparison; Degree versus number of roots
Theory dependencies	Permutations, inversions, and permutation statistics; Block matrices, Schur complements, and determinant identities; Determinant evaluation by row/column operations and triangular forms
Cue extraction	prior broad method tags: determinant / block manipulation | statement triggers: matrix word; integral/limit trigger; integer/prime/mod trigger | audit closures: AUD-ANT-02; AUD-PC-01
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2001-D1-P2 (PDF p. 70) *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Graph encoding of a relation; 3. Cycle extraction and alternating-cycle method
Method IDs	M-ANT-01; M-CMB-03; M-CMB-04; M-ANT-02; M-ANT-03; M-ANT-10; M-ANT-11; M-UNI-05
Method titles	Reduction modulo a prime or finite field; Graph encoding of a relation; Cycle extraction and alternating-cycle method; Chinese remainder and residue construction; GCD, valuation, and prime-power bookkeeping; Group action, orbit, and generator method; Commutator and center manipulation in groups/rings; Induction with strengthened hypothesis
Theory dependencies	Group basics: orders, cyclic subgroups, and generators; Permutation groups, cycles, parity, and generation; Modular arithmetic and divisibility
Cue extraction	prior broad method tags: modular / finite-field reduction, graph encoding / cycles | statement triggers: integer/prime/mod trigger; group/ring/field trigger | audit closures: AUD-LA-03; AUD-LA-08; AUD-PC-01; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2001-D1-P3 (PDF p. 70) *limit / asymptotic evaluation*

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Riemann sum and sum-integral conversion
Method IDs	M-UNI-07; M-UNI-09; M-RA-06; M-RA-07; M-RA-01; M-RA-08; M-RA-05; M-LA-03
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Auxiliary function with derivative sign; Compactness plus extreme value theorem; L-Hospital and ratio/growth comparison; Block decomposition and Schur-style bookkeeping
Theory dependencies	Riemann sums and sum-integral asymptotics; L'Hospital, ratio limits, and growth comparisons; Taylor expansion, local asymptotics, and little-o bookkeeping
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: integral/limit trigger | audit closures: AUD-RA-03; AUD-RA-05
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2001-D1-P4 (PDF p. 71) *proof of structural property*

Primary domain	Complex analysis / polynomials
Core method sequence	1. Roots of unity and cyclotomic irreducibility; 2. Polynomial identity by coefficient comparison; 3. Degree versus number of roots
Method IDs	M-ANT-08; M-ANT-16; M-ANT-06; M-ANT-01; M-ANT-05; M-ANT-02; M-ANT-03; M-RA-09
Method titles	Roots of unity and cyclotomic irreducibility; Polynomial identity by coefficient comparison; Degree versus number of roots; Reduction modulo a prime or finite field; Factorization and unique factorization in Z[x]; Chinese remainder and residue construction; GCD, valuation, and prime-power bookkeeping; Convexity, Jensen, tangent, and chord inequalities
Theory dependencies	Roots of unity, cyclotomic filters, and discrete Fourier tricks; Factorization, irreducibility, and Gauss-style integer polynomial arguments; Polynomial identity and more-roots-than-degree principle
Cue extraction	prior broad method tags: polynomial identity / factorization, modular / finite-field reduction | statement triggers: polynomial/root trigger; integer/prime/mod trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-08; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2001-D1-P5 (PDF p. 71) *proof of structural property*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Trace cyclicity and commutator annihilation; 2. Induction with strengthened hypothesis; 3. Spectral/minimal-polynomial reduction
Method IDs	M-LA-07; M-UNI-05; M-LA-06; M-LA-04; M-ANT-16; M-LA-03; M-LA-13; M-UNI-09
Method titles	Trace cyclicity and commutator annihilation; Induction with strengthened hypothesis; Spectral/minimal-polynomial reduction; Rank-nullity and dimension-count obstruction; Polynomial identity by coefficient comparison; Block decomposition and Schur-style bookkeeping; Determinant through eigenvalue or root product; Invariant extraction and invariant comparison
Theory dependencies	Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; Invariant subspaces and adapted bases; Trace cyclicity, trace pairing, and zero-trace subspaces
Cue extraction	prior broad method tags: induction, trace trick / commutator | statement triggers: matrix word | audit closures: AUD-LA-03; AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2001-D1-P6 (PDF p. 72) *limit / asymptotic evaluation*

Primary domain	Functional equations / dynamics
Core method sequence	1. L-Hospital and ratio/growth comparison; 2. Taylor expansion and local asymptotic bookkeeping; 3. Auxiliary function with derivative sign
Method IDs	M-RA-05; M-RA-04; M-RA-01; M-RA-03; M-RA-06; M-RA-07; M-RA-09; M-FE-02
Method titles	L-Hospital and ratio/growth comparison; Taylor expansion and local asymptotic bookkeeping; Auxiliary function with derivative sign; Rolle chain and divided-difference localization; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Convexity, Jensen, tangent, and chord inequalities; Monotonicity forced by functional equation
Theory dependencies	L'Hospital, ratio limits, and growth comparisons; Auxiliary functions, integrating factors, and differential inequalities; Taylor expansion, local asymptotics, and little-o bookkeeping
Cue extraction	prior broad method tags: l'Hospital / asymptotic expansion | statement triggers: continuous/differentiable trigger; integral/limit trigger; functional-equation trigger | audit closures: AUD-RA-01; AUD-RA-03; AUD-RA-05; AUD-RA-06; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.8.2 Day 2

2001-D2-P1 (PDF p. 74)

proof of structural property

Primary domain	Real analysis / calculus
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Induction with strengthened hypothesis
Method IDs	M-ANT-16; M-ANT-06; M-UNI-05; M-ANT-05; M-CMB-01; M-CMB-02; M-ANT-01; M-ANT-03
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Induction with strengthened hypothesis; Factorization and unique factorization in Z[x]; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping
Theory dependencies	Integer polynomials, finite differences, and value divisibility; Factorization, irreducibility, and Gauss-style integer polynomial arguments; Polynomial identity and more-roots-than-degree principle
Cue extraction	prior broad method tags: induction, polynomial identity / factorization, double counting / pigeonhole | statement triggers: integer/prime/mod trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-CMB-03
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2001-D2-P2 (PDF p. 74)

inequality / sharp bound

Primary domain	Real analysis / calculus
Core method sequence	1. Auxiliary function with derivative sign; 2. Monotonicity forced by functional equation; 3. Induction with strengthened hypothesis
Method IDs	M-RA-01; M-FE-02; M-UNI-05; M-RA-09; M-RA-06; M-RA-07; M-RA-15; M-RA-08
Method titles	Auxiliary function with derivative sign; Monotonicity forced by functional equation; Induction with strengthened hypothesis; Convexity, Jensen, tangent, and chord inequalities; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Recurrence transformed to monotone quotient or telescoping form; Compactness plus extreme value theorem
Theory dependencies	Sequence recurrences, monotone bounds, and generating functions; Trigonometric and hyperbolic identities with monotonicity; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: induction, monotonicity argument | statement triggers: integral/limit trigger | audit closures: AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2001-D2-P3 (PDF p. 75)

inequality / sharp bound

Primary domain	Geometry / convexity / metric
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Double counting ordered or weighted objects; 3. Roots of unity and cyclotomic irreducibility
Method IDs	M-CMB-01; M-CMB-02; M-ANT-08; M-UNI-03; M-RA-08; M-CMB-12; M-LA-04; M-CX-05
Method titles	Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Roots of unity and cyclotomic irreducibility; Extremal construction and equality-case engineering; Compactness plus extreme value theorem; Root-of-unity filter for residues; Rank-nullity and dimension-count obstruction; Trigonometric-to-complex conversion
Theory dependencies	Unit-sphere configurations and regular simplex bounds; Pigeonhole principle and extremal element selection; Double counting and degree-sum methods
Cue extraction	prior broad method tags: compactness / extremal point, rank-nullity / dimension count, roots of unity / Fourier filter | statement triggers: counting/permutation trigger; geometry/metric trigger | audit closures: AUD-GEO-04; AUD-RA-06; AUD-RA-07; AUD-RA-08; AUD-PC-01; AUD-PC-05
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2001-D2-P4 (PDF p. 76) *existence / construction*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Nilpotency from zero spectrum plus finite dimension; 2. Topological ordering of an acyclic directed graph; 3. Row/column elimination to triangular or sparse form
Method IDs	M-LA-12; M-CMB-14; M-LA-01; M-UNI-06; M-RA-08; M-CMB-03; M-CMB-04; M-LA-03
Method titles	Nilpotency from zero spectrum plus finite dimension; Topological ordering of an acyclic directed graph; Row/column elimination to triangular or sparse form; Compactness or extremal-point selection; Compactness plus extreme value theorem; Graph encoding of a relation; Cycle extraction and alternating-cycle method; Block decomposition and Schur-style bookkeeping
Theory dependencies	Graph encoding, cycles, and bipartite representations; Nilpotent, idempotent, involution, and square-zero structures; Determinant evaluation by row/column operations and triangular forms
Cue extraction	prior broad method tags: compactness / extremal point, determinant / block manipulation, graph encoding / cycles | statement triggers: matrix word; det/rank/eigen trigger; counting/permutation trigger | audit closures: AUD-RA-07; AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2001-D2-P5 (PDF p. 77) *nonexistence / counterexample*

Primary domain	Functional equations / dynamics
Core method sequence	1. Monotonicity forced by functional equation; 2. Auxiliary function with derivative sign; 3. L-Hospital and ratio/growth comparison
Method IDs	M-FE-02; M-RA-01; M-RA-05; M-RA-06; M-RA-07; M-RA-09; M-UNI-07; M-UNI-03
Method titles	Monotonicity forced by functional equation; Auxiliary function with derivative sign; L-Hospital and ratio/growth comparison; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Convexity, Jensen, tangent, and chord inequalities; Special substitution and parameter freezing; Extremal construction and equality-case engineering
Theory dependencies	No-regularity functions and order patterns on Q or R; Convexity, concavity, Jensen, and tangent/chord inequalities; Riemann sums and sum-integral asymptotics
Cue extraction	prior broad method tags: monotonicity argument | statement triggers: functional-equation trigger | audit closures: AUD-RA-03; AUD-RA-05; AUD-RA-08; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2001-D2-P6 (PDF p. 78) *inequality / sharp bound*

Primary domain	Algebra / number theory
Core method sequence	1. Compactness plus extreme value theorem; 2. Compactness or extremal-point selection; 3. Reduction modulo a prime or finite field
Method IDs	M-RA-08; M-UNI-06; M-ANT-01; M-ANT-03; M-FE-01; M-RA-09; M-RA-03; M-UNI-03
Method titles	Compactness plus extreme value theorem; Compactness or extremal-point selection; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Iterates, fixed points, and cluster-point trapping; Convexity, Jensen, tangent, and chord inequalities; Rolle chain and divided-difference localization; Extremal construction and equality-case engineering
Theory dependencies	Trigonometric and hyperbolic identities with monotonicity; Convexity, concavity, Jensen, and tangent/chord inequalities; Rolle, mean-value theorem, and divided differences
Cue extraction	prior broad method tags: compactness / extremal point | statement triggers: integer/prime/mod trigger | audit closures: AUD-RA-06; AUD-RA-07; AUD-RA-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.9 2002

18.9.1 Day 1

2002-D1-P1 (PDF p. 79) *proof of structural property*

Primary domain	Geometry / convexity / metric
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Graph encoding of a relation
Method IDs	M-ANT-16; M-ANT-06; M-CMB-03; M-CMB-04; M-ANT-05; M-GEO-01; M-GEO-11; M-GEO-03
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Graph encoding of a relation; Cycle extraction and alternating-cycle method; Factorization and unique factorization in Z[x]; Coordinate normalization and algebraization; Volume, projection, and ellipsoid computation; Squared-distance and inner-product identities
Theory dependencies	Conics: parabolas, ellipses, foci, and directrices; Polynomial identity and more-roots-than-degree principle; Graph encoding, cycles, and bipartite representations
Cue extraction	prior broad method tags: polynomial identity / factorization, graph encoding / cycles | statement triggers: polynomial/root trigger; graph/color/set trigger; geometry/metric trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-PC-14
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2002-D1-P2 (PDF p. 79) *nonexistence / counterexample*

Primary domain	Functional equations / dynamics
Core method sequence	1. Auxiliary function with derivative sign; 2. Extremal construction and equality-case engineering; 3. Monotonicity forced by functional equation
Method IDs	M-RA-01; M-UNI-03; M-FE-02; M-UNI-08; M-RA-03; M-RA-09; M-UNI-07; M-UNI-04
Method titles	Auxiliary function with derivative sign; Extremal construction and equality-case engineering; Monotonicity forced by functional equation; Sign, zero, and boundary case splitting; Rolle chain and divided-difference localization; Convexity, Jensen, tangent, and chord inequalities; Special substitution and parameter freezing; Contradiction with minimal or maximal counterexample
Theory dependencies	No-regularity functions and order patterns on Q or R; Integral inequalities, integration by parts, and weighted estimates; Explicit construction and counterexample design
Cue extraction	prior broad method tags: monotonicity argument, construction / counterexample | statement triggers: continuous/differentiable trigger; functional-equation trigger | audit closures: AUD-RA-01; AUD-RA-06; AUD-RA-08; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2002-D1-P3 (PDF p. 80) *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Induction with strengthened hypothesis; 2. Reduction modulo a prime or finite field; 3. GCD, valuation, and prime-power bookkeeping
Method IDs	M-UNI-05; M-ANT-01; M-ANT-03; M-RA-09; M-RA-11; M-UNI-09; M-CMB-06; M-RA-10
Method titles	Induction with strengthened hypothesis; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Convexity, Jensen, tangent, and chord inequalities; Integration by parts, Fubini, and weighted identities; Invariant extraction and invariant comparison; Generating function encoding; Cauchy-Schwarz, Holder, and norm comparison
Theory dependencies	GCD, valuations, prime powers, and coprimality; Generating functions for counting; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: induction | statement triggers: integer/prime/mod trigger | audit closures: AUD-RA-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2002-D1-P4 (PDF p. 80) *proof of structural property*

Primary domain	Real analysis / calculus
Core method sequence	1. Induction with strengthened hypothesis; 2. Pigeonhole threshold and collision forcing; 3. Double counting ordered or weighted objects
Method IDs	M-UNI-05; M-CMB-01; M-CMB-02; M-RA-16; M-RA-01; M-RA-03; M-RA-08; M-RA-06
Method titles	Induction with strengthened hypothesis; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Density, Baire/category, and pathological construction; Auxiliary function with derivative sign; Rolle chain and divided-difference localization; Compactness plus extreme value theorem; Riemann sum and sum-integral conversion
Theory dependencies	Compactness, extreme values, and closed-bounded reduction; Fixed points, iterates, and orbit closure; Intermediate value, connectedness, density, and continuity transfer
Cue extraction	prior broad method tags: induction, double counting / pigeonhole, topological / density argument | statement triggers: continuous/differentiable trigger; functional-equation trigger | audit closures: AUD-RA-11; AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-07; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2002-D1-P5 (PDF p. 81) *existence / construction*

Primary domain	Real analysis / calculus
Core method sequence	1. Rolle chain and divided-difference localization; 2. Auxiliary function with derivative sign; 3. Pigeonhole threshold and collision forcing
Method IDs	M-RA-03; M-RA-01; M-CMB-01; M-CMB-02; M-LA-04; M-FE-02; M-RA-08; M-UNI-03
Method titles	Rolle chain and divided-difference localization; Auxiliary function with derivative sign; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Rank-nullity and dimension-count obstruction; Monotonicity forced by functional equation; Compactness plus extreme value theorem; Extremal construction and equality-case engineering
Theory dependencies	Baire/category, dense sets, and pathological continuous functions; Rank-nullity, image/kernel, and rank of factorizations; Rolle, mean-value theorem, and divided differences
Cue extraction	prior broad method tags: Rolle / mean-value theorem, monotonicity argument, rank-nullity / dimension count | statement triggers: continuous/differentiable trigger; counting/permutation trigger; functional-equation trigger | audit closures: AUD-RA-06; AUD-RA-10; AUD-CMB-03
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2002-D1-P6 (PDF p. 82) *inequality / sharp bound*

Primary domain	Algebra / number theory
Core method sequence	1. Cauchy-Schwarz, Holder, and norm comparison; 2. Rank-nullity and dimension-count obstruction; 3. Monotonicity forced by functional equation
Method IDs	M-RA-10; M-LA-04; M-FE-02; M-RA-01; M-GEO-03; M-RA-09; M-RA-06; M-RA-07
Method titles	Cauchy-Schwarz, Holder, and norm comparison; Rank-nullity and dimension-count obstruction; Monotonicity forced by functional equation; Auxiliary function with derivative sign; Squared-distance and inner-product identities; Convexity, Jensen, tangent, and chord inequalities; Riemann sum and sum-integral conversion; Split into main range and tail/error range
Theory dependencies	Operator norm, power boundedness, and spectral-radius style estimates; Convexity, concavity, Jensen, and tangent/chord inequalities; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: monotonicity argument, Cauchy-Schwarz / norm inequality, rank-nullity / dimension count | statement triggers: matrix word; integral/limit trigger; integer/prime/mod trigger | audit closures: AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-06; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.9.2 Day 2

2002-D2-P1 (PDF p. 84) *exact computation*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Row/column elimination to triangular or sparse form; 2. Block decomposition and Schur-style bookkeeping; 3. Determinant through eigenvalue or root product
Method IDs	M-LA-01; M-LA-03; M-LA-13; M-LA-04; M-LA-06; M-LA-07; M-UNI-02; M-UNI-09
Method titles	Row/column elimination to triangular or sparse form; Block decomposition and Schur-style bookkeeping; Determinant through eigenvalue or root product; Rank-nullity and dimension-count obstruction; Spectral/minimal-polynomial reduction; Trace cyclicity and commutator annihilation; Small cases, equality cases, and example probing; Invariant extraction and invariant comparison
Theory dependencies	Determinant evaluation by row/column operations and triangular forms; Block matrices, Schur complements, and determinant identities; Rank-nullity, image/kernel, and rank of factorizations
Cue extraction	prior broad method tags: determinant / block manipulation | statement triggers: matrix word; det/rank/eigen trigger | audit closures: AUD-CMB-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2002-D2-P2 (PDF p. 84) *proof of structural property*

Primary domain	Combinatorics / discrete structures
Core method sequence	1. Invariant extraction and invariant comparison; 2. Special substitution and parameter freezing; 3. Pigeonhole threshold and collision forcing
Method IDs	M-UNI-09; M-UNI-07; M-CMB-01; M-CMB-02; M-UNI-05; M-CMB-10; M-CMB-03; M-CMB-04
Method titles	Invariant extraction and invariant comparison; Special substitution and parameter freezing; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Induction with strengthened hypothesis; Explicit bijection and encoding-decoding proof; Graph encoding of a relation; Cycle extraction and alternating-cycle method
Theory dependencies	Double counting and degree-sum methods; Graph encoding, cycles, and bipartite representations; Coloring, partitions, and parity
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: wording-specific | audit closures: AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2002-D2-P3 (PDF p. 85) *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Degree versus number of roots; 2. Polynomial identity by coefficient comparison; 3. Induction with strengthened hypothesis
Method IDs	M-ANT-06; M-ANT-16; M-UNI-05; M-ANT-05; M-ANT-01; M-ANT-03; M-RA-09; M-RA-15
Method titles	Degree versus number of roots; Polynomial identity by coefficient comparison; Induction with strengthened hypothesis; Factorization and unique factorization in Z[x]; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Convexity, Jensen, tangent, and chord inequalities; Recurrence transformed to monotone quotient or telescoping form
Theory dependencies	Sequence recurrences, monotone bounds, and generating functions; Generating functions for counting; Polynomial identity and more-roots-than-degree principle
Cue extraction	prior broad method tags: induction, polynomial identity / factorization | statement triggers: integer/prime/mod trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-PC-13; AUD-CMB-07; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2002-D2-P4 (PDF p. 85) *proof of structural property*

Primary domain	Geometry / convexity / metric
Core method sequence	1. Graph encoding of a relation; 2. Cycle extraction and alternating-cycle method; 3. Squared-distance and inner-product identities
Method IDs	M-CMB-03; M-CMB-04; M-GEO-03; M-GEO-01; M-UNI-05; M-UNI-09; M-RA-09; M-CMB-07
Method titles	Graph encoding of a relation; Cycle extraction and alternating-cycle method; Squared-distance and inner-product identities; Coordinate normalization and algebraization; Induction with strengthened hypothesis; Invariant extraction and invariant comparison; Convexity, Jensen, tangent, and chord inequalities; Involution, parity, and sign-reversing cancellation
Theory dependencies	Projections and orthogonal decompositions; Convex hulls, random polytopes, and geometric probability; Graph encoding, cycles, and bipartite representations
Cue extraction	prior broad method tags: graph encoding / cycles | statement triggers: wording-specific | audit closures: AUD-RA-08; AUD-GEO-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2002-D2-P5 (PDF p. 86) *proof of structural property*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Complete a matrix equation to an inverse identity; 2. Spectral/minimal-polynomial reduction; 3. Roots of unity and cyclotomic irreducibility
Method IDs	M-LA-02; M-LA-06; M-ANT-08; M-CMB-12; M-CX-05; M-LA-12; M-LA-04; M-UNI-05
Method titles	Complete a matrix equation to an inverse identity; Spectral/minimal-polynomial reduction; Roots of unity and cyclotomic irreducibility; Root-of-unity filter for residues; Trigonometric-to-complex conversion; Nilpotency from zero spectrum plus finite dimension; Rank-nullity and dimension-count obstruction; Induction with strengthened hypothesis
Theory dependencies	Matrix products, inverses, and one-sided inverse identities; Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; Roots of unity, cyclotomic filters, and discrete Fourier tricks
Cue extraction	prior broad method tags: spectral decomposition / eigenvalues, roots of unity / Fourier filter | statement triggers: matrix word | audit closures: AUD-LA-16; AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-06
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2002-D2-P6 (PDF p. 87) *inequality / sharp bound*

Primary domain	Functional equations / dynamics
Core method sequence	1. Convexity, Jensen, tangent, and chord inequalities; 2. Compactness plus extreme value theorem; 3. Special substitution and parameter freezing
Method IDs	M-RA-09; M-RA-08; M-UNI-07; M-UNI-06; M-RA-01; M-RA-02; M-GEO-01; M-GEO-11
Method titles	Convexity, Jensen, tangent, and chord inequalities; Compactness plus extreme value theorem; Special substitution and parameter freezing; Compactness or extremal-point selection; Auxiliary function with derivative sign; Integrating factor and ODE comparison; Coordinate normalization and algebraization; Volume, projection, and ellipsoid computation
Theory dependencies	Gradient/Lipschitz-gradient optimization and convex smoothness; Convexity, concavity, Jensen, and tangent/chord inequalities; Gram matrices and rational linear dependence
Cue extraction	prior broad method tags: auxiliary function / substitution, compactness / extremal point, convexity / Jensen / tangent | statement triggers: functional-equation trigger | audit closures: AUD-RA-07; AUD-RA-08; AUD-RA-15
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.10 2003

18.10.1 Day 1

2003-D1-P1 (PDF p. 88) *existence / construction*

Primary domain	Real analysis / calculus
Core method sequence	1. Auxiliary function with derivative sign; 2. Monotonicity forced by functional equation; 3. L-Hospital and ratio/growth comparison
Method IDs	M-RA-01; M-FE-02; M-RA-05; M-RA-06; M-RA-07; M-RA-15; M-RA-11; M-RA-08
Method titles	Auxiliary function with derivative sign; Monotonicity forced by functional equation; L-Hospital and ratio/growth comparison; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Recurrence transformed to monotone quotient or telescoping form; Integration by parts, Fubini, and weighted identities; Compactness plus extreme value theorem
Theory dependencies	Sequence recurrences, monotone bounds, and generating functions; Riemann sums and sum-integral asymptotics; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: monotonicity argument | statement triggers: integral/limit trigger | audit closures: AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-PC-13; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2003-D1-P2 (PDF p. 88) *existence / construction*

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Chinese remainder and residue construction; 3. GCD, valuation, and prime-power bookkeeping
Method IDs	M-ANT-01; M-ANT-02; M-ANT-03; M-RA-06; M-RA-07; M-UNI-03; M-UNI-06; M-ANT-06
Method titles	Reduction modulo a prime or finite field; Chinese remainder and residue construction; GCD, valuation, and prime-power bookkeeping; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Extremal construction and equality-case engineering; Compactness or extremal-point selection; Degree versus number of roots
Theory dependencies	Finite fields, field characteristic, and Fermat/Frobenius tools; Permutations, inversions, and permutation statistics; Modular arithmetic and divisibility
Cue extraction	prior broad method tags: modular / finite-field reduction | statement triggers: integral/limit trigger; integer/prime/mod trigger; group/ring/field trigger; counting/permutation trigger | audit closures: AUD-PC-01; AUD-PC-13; AUD-ANT-04; AUD-CMB-07; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2003-D1-P3 (PDF p. 88) *proof of structural property*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Spectral/minimal-polynomial reduction; 2. Nilpotency from zero spectrum plus finite dimension; 3. Degree versus number of roots
Method IDs	M-LA-06; M-LA-12; M-ANT-06; M-ANT-16; M-ANT-05; M-RA-06; M-RA-07; M-LA-04
Method titles	Spectral/minimal-polynomial reduction; Nilpotency from zero spectrum plus finite dimension; Degree versus number of roots; Polynomial identity by coefficient comparison; Factorization and unique factorization in Z[x]; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Rank-nullity and dimension-count obstruction
Theory dependencies	Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; Polynomial identity and more-roots-than-degree principle; Nilpotent, idempotent, involution, and square-zero structures
Cue extraction	prior broad method tags: spectral decomposition / eigenvalues, polynomial identity / factorization | statement triggers: matrix word | audit closures: AUD-LA-02; AUD-LA-03; AUD-LA-07; AUD-RA-03; AUD-RA-04
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2003-D1-P4 (PDF p. 88) *proof / reasoning*

Primary domain	Algebra / number theory
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Reduction modulo a prime or finite field
Method IDs	M-UNI-07; M-UNI-09; M-ANT-01; M-ANT-03; M-RA-09; M-CMB-03; M-CMB-04; M-CMB-07
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Convexity, Jensen, tangent, and chord inequalities; Graph encoding of a relation; Cycle extraction and alternating-cycle method; Involution, parity, and sign-reversing cancellation
Theory dependencies	GCD, valuations, prime powers, and coprimality; Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: det/rank/eigen trigger; integer/prime/mod trigger | audit closures: AUD-RA-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2003-D1-P5 (PDF p. 89) *existence / construction*

Primary domain	Real analysis / calculus
Core method sequence	1. Squeeze, dominated convergence, and monotone convergence; 2. Auxiliary function with derivative sign; 3. Pigeonhole threshold and collision forcing
Method IDs	M-RA-12; M-RA-01; M-CMB-01; M-CMB-02; M-FE-02; M-RA-02; M-UNI-05; M-UNI-07
Method titles	Squeeze, dominated convergence, and monotone convergence; Auxiliary function with derivative sign; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Monotonicity forced by functional equation; Integrating factor and ODE comparison; Induction with strengthened hypothesis; Special substitution and parameter freezing
Theory dependencies	Integral inequalities, integration by parts, and weighted estimates; Recurrence transformations and conjugacy; Pigeonhole principle and extremal element selection
Cue extraction	prior broad method tags: induction, auxiliary function / substitution, monotonicity argument | statement triggers: det/rank/eigen trigger; continuous/differentiable trigger; integral/limit trigger; counting/permutation trigger | audit closures: AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2003-D1-P6 (PDF p. 91) *proof of structural property*

Primary domain	Complex analysis / polynomials
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Induction with strengthened hypothesis
Method IDs	M-ANT-16; M-ANT-06; M-UNI-05; M-ANT-05; M-RA-09; M-RA-15; M-ANT-07; M-CMB-06
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Induction with strengthened hypothesis; Factorization and unique factorization in Z[x]; Convexity, Jensen, tangent, and chord inequalities; Recurrence transformed to monotone quotient or telescoping form; Vieta, symmetric sums, and logarithmic derivative; Generating function encoding
Theory dependencies	Stable polynomials and coefficient inequalities; Polynomial identity and more-roots-than-degree principle; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: induction, polynomial identity / factorization | statement triggers: polynomial/root trigger; functional-equation trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-PC-04; AUD-PC-08; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.10.2 Day 2

2003-D2-P1 (PDF p. 92) *proof of structural property*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Invariant extraction and invariant comparison; 2. Special substitution and parameter freezing; 3. Rank-nullity and dimension-count obstruction
Method IDs	M-UNI-09; M-UNI-07; M-LA-04; M-LA-06; M-LA-03; M-LA-16; M-LA-07; M-LA-13
Method titles	Invariant extraction and invariant comparison; Special substitution and parameter freezing; Rank-nullity and dimension-count obstruction; Spectral/minimal-polynomial reduction; Block decomposition and Schur-style bookkeeping; Trace pairing and orthogonal matrix subspaces; Trace cyclicity and commutator annihilation; Determinant through eigenvalue or root product
Theory dependencies	Matrix products, inverses, and one-sided inverse identities; Rank-nullity, image/kernel, and rank of factorizations; Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: matrix word | audit closures: AUD-LA-03; AUD-LA-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2003-D2-P2 (PDF p. 92) *limit / asymptotic evaluation*

Primary domain	Real analysis / calculus
Core method sequence	1. Auxiliary function with derivative sign; 2. Degree versus number of roots; 3. Polynomial identity by coefficient comparison
Method IDs	M-RA-01; M-ANT-06; M-ANT-16; M-ANT-05; M-FE-02; M-RA-06; M-RA-07; M-RA-05
Method titles	Auxiliary function with derivative sign; Degree versus number of roots; Polynomial identity by coefficient comparison; Factorization and unique factorization in Z[x]; Monotonicity forced by functional equation; Riemann sum and sum-integral conversion; Split into main range and tail/error range; L-Hospital and ratio/growth comparison
Theory dependencies	L'Hospital, ratio limits, and growth comparisons; Trigonometric and hyperbolic identities with monotonicity; Polynomial identity and more-roots-than-degree principle
Cue extraction	prior broad method tags: monotonicity argument, polynomial identity / factorization | statement triggers: integral/limit trigger | audit closures: AUD-RA-03; AUD-RA-05; AUD-RA-08; AUD-PC-01; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2003-D2-P3 (PDF p. 92) *existence / construction*

Primary domain	Combinatorics / discrete structures
Core method sequence	1. Compactness or extremal-point selection; 2. Compactness plus extreme value theorem; 3. Density, Baire/category, and pathological construction
Method IDs	M-UNI-06; M-RA-08; M-RA-16; M-CMB-01; M-RA-09; M-CMB-02; M-UNI-03; M-GEO-03
Method titles	Compactness or extremal-point selection; Compactness plus extreme value theorem; Density, Baire/category, and pathological construction; Pigeonhole threshold and collision forcing; Convexity, Jensen, tangent, and chord inequalities; Double counting ordered or weighted objects; Extremal construction and equality-case engineering; Squared-distance and inner-product identities
Theory dependencies	Closest points to closed sets and density of uniqueness; Compactness, extreme values, and closed-bounded reduction; Intermediate value, connectedness, density, and continuity transfer
Cue extraction	prior broad method tags: compactness / extremal point, topological / density argument | statement triggers: geometry/metric trigger | audit closures: AUD-RA-07; AUD-RA-08; AUD-RA-13; AUD-GEO-02; AUD-GEO-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2003-D2-P4 (PDF p. 93) *classification / find all*

Primary domain	Algebra / number theory
Core method sequence	1. Sign, zero, and boundary case splitting; 2. Extremal construction and equality-case engineering; 3. Reduction modulo a prime or finite field
Method IDs	M-UNI-08; M-UNI-03; M-ANT-01; M-ANT-03; M-CMB-01; M-CMB-02; M-ANT-09; M-ANT-10
Method titles	Sign, zero, and boundary case splitting; Extremal construction and equality-case engineering; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Finite-field function representative and permutation argument; Group action, orbit, and generator method
Theory dependencies	Finite designs and pair-determines-third structures; Finite abelian groups and vector spaces over finite fields; Group basics: orders, cyclic subgroups, and generators
Cue extraction	prior broad method tags: construction / counterexample | statement triggers: integer/prime/mod trigger; graph/color/set trigger | audit closures: AUD-RA-08; AUD-ANT-10; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2003-D2-P5 (PDF p. 94) *existence / construction*

Primary domain	Real analysis / calculus
Core method sequence	1. Density, Baire/category, and pathological construction; 2. Auxiliary function with derivative sign; 3. Pigeonhole threshold and collision forcing
Method IDs	M-RA-16; M-RA-01; M-CMB-01; M-CMB-02; M-RA-02; M-UNI-07; M-RA-06; M-RA-07
Method titles	Density, Baire/category, and pathological construction; Auxiliary function with derivative sign; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Integrating factor and ODE comparison; Special substitution and parameter freezing; Riemann sum and sum-integral conversion; Split into main range and tail/error range
Theory dependencies	Baire/category, dense sets, and pathological continuous functions; No-regularity functions and order patterns on Q or R; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: auxiliary function / substitution, double counting / pigeonhole, topological / density argument | statement triggers: functional-equation trigger | audit closures: AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08; AUD-CMB-03
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2003-D2-P6 (PDF p. 94) *limit / asymptotic evaluation*

Primary domain	Real analysis / calculus
Core method sequence	1. Induction with strengthened hypothesis; 2. Riemann sum and sum-integral conversion; 3. Split into main range and tail/error range
Method IDs	M-UNI-05; M-RA-06; M-RA-07; M-RA-09; M-RA-15; M-CMB-06; M-RA-11; M-RA-01
Method titles	Induction with strengthened hypothesis; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Convexity, Jensen, tangent, and chord inequalities; Recurrence transformed to monotone quotient or telescoping form; Generating function encoding; Integration by parts, Fubini, and weighted identities; Auxiliary function with derivative sign
Theory dependencies	Sequence recurrences, monotone bounds, and generating functions; Generating functions for counting; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: induction, generating functions | statement triggers: integral/limit trigger | audit closures: AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.11 2004

18.11.1 Day 1

2004-D1-P1 (PDF p. 96) *counting / enumeration*

Primary domain	Combinatorics / discrete structures
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Double counting ordered or weighted objects; 3. Explicit bijection and encoding-decoding proof
Method IDs	M-CMB-01; M-CMB-02; M-CMB-10; M-CMB-03; M-CMB-08; M-RA-09; M-CMB-05; M-GEO-10
Method titles	Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Explicit bijection and encoding-decoding proof; Graph encoding of a relation; Extremal set or minimal counterexample induction; Convexity, Jensen, tangent, and chord inequalities; State recurrence or automaton partition; Closest-point uniqueness by moving toward a minimizer
Theory dependencies	Pigeonhole principle and extremal element selection; Double counting and degree-sum methods; Graph encoding, cycles, and bipartite representations
Cue extraction	prior broad method tags: double counting / pigeonhole | statement triggers: counting/permutation trigger | audit closures: AUD-RA-08; AUD-CMB-03; AUD-CMB-05; AUD-GEO-09; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2004-D1-P2 (PDF p. 96) *counting / enumeration*

Primary domain	Real analysis / calculus
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Double counting ordered or weighted objects; 3. Induction with strengthened hypothesis
Method IDs	M-CMB-01; M-CMB-02; M-UNI-05; M-CMB-10; M-RA-09; M-RA-11; M-RA-01; M-RA-08
Method titles	Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Induction with strengthened hypothesis; Explicit bijection and encoding-decoding proof; Convexity, Jensen, tangent, and chord inequalities; Integration by parts, Fubini, and weighted identities; Auxiliary function with derivative sign; Compactness plus extreme value theorem
Theory dependencies	Recurrence transformations and conjugacy; Integral inequalities, integration by parts, and weighted estimates; Pigeonhole principle and extremal element selection
Cue extraction	prior broad method tags: induction, double counting / pigeonhole | statement triggers: counting/permutation trigger | audit closures: AUD-RA-08; AUD-CMB-03; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2004-D1-P3 (PDF p. 96) *limit / asymptotic evaluation*

Primary domain	Real analysis / calculus
Core method sequence	1. Compactness plus extreme value theorem; 2. Convexity, Jensen, tangent, and chord inequalities; 3. Graph encoding of a relation
Method IDs	M-RA-08; M-RA-09; M-CMB-03; M-CMB-04; M-LA-04; M-RA-16; M-UNI-06; M-RA-06
Method titles	Compactness plus extreme value theorem; Convexity, Jensen, tangent, and chord inequalities; Graph encoding of a relation; Cycle extraction and alternating-cycle method; Rank-nullity and dimension-count obstruction; Density, Baire/category, and pathological construction; Compactness or extremal-point selection; Riemann sum and sum-integral conversion
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Trigonometric and hyperbolic identities with monotonicity; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: compactness / extremal point, convexity / Jensen / tangent, rank-nullity / dimension count | statement triggers: integral/limit trigger | audit closures: AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-07; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2004-D1-P4 (PDF p. 97) *proof of structural property*

Primary domain	Geometry / convexity / metric
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Double counting ordered or weighted objects; 3. Coordinate normalization and algebraization
Method IDs	M-CMB-01; M-CMB-02; M-GEO-01; M-GEO-11; M-GEO-03; M-UNI-05; M-UNI-09; M-CMB-03
Method titles	Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Coordinate normalization and algebraization; Volume, projection, and ellipsoid computation; Squared-distance and inner-product identities; Induction with strengthened hypothesis; Invariant extraction and invariant comparison; Graph encoding of a relation
Theory dependencies	Projections and orthogonal decompositions; Euclidean distances and inner-product identities; Pigeonhole principle and extremal element selection
Cue extraction	prior broad method tags: double counting / pigeonhole | statement triggers: integral/limit trigger; graph/color/set trigger; geometry/metric trigger | audit closures: AUD-CMB-05; AUD-RA-08; AUD-CMB-03; AUD-GEO-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2004-D1-P5 (PDF p. 97)

existence / construction

Primary domain	Real analysis / calculus
Core method sequence	1. Auxiliary function with derivative sign; 2. Monotonicity forced by functional equation; 3. Induction with strengthened hypothesis
Method IDs	M-RA-01; M-FE-02; M-UNI-05; M-RA-09; M-RA-08; M-UNI-03; M-UNI-06; M-CMB-08
Method titles	Auxiliary function with derivative sign; Monotonicity forced by functional equation; Induction with strengthened hypothesis; Convexity, Jensen, tangent, and chord inequalities; Compactness plus extreme value theorem; Extremal construction and equality-case engineering; Compactness or extremal-point selection; Extremal set or minimal counterexample induction
Theory dependencies	Erdos-Szekeres style monotone subsequences and binomial thresholds; Integral inequalities, integration by parts, and weighted estimates; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: induction, monotonicity argument | statement triggers: wording-specific | audit closures: AUD-RA-08; AUD-RA-10; AUD-CMB-05
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2004-D1-P6 (PDF p. 98)

proof of structural property

Primary domain	Real analysis / calculus
Core method sequence	1. Laurent expansion and isolated singularity rationality; 2. Cotangent partial-fraction differentiation; 3. Degree versus number of roots
Method IDs	M-CX-02; M-CX-06; M-ANT-06; M-ANT-16; M-RA-04; M-RA-05; M-RA-09; M-ANT-01
Method titles	Laurent expansion and isolated singularity rationality; Cotangent partial-fraction differentiation; Degree versus number of roots; Polynomial identity by coefficient comparison; Taylor expansion and local asymptotic bookkeeping; L-Hospital and ratio/growth comparison; Convexity, Jensen, tangent, and chord inequalities; Reduction modulo a prime or finite field
Theory dependencies	Partial fractions and cotangent-type expansions; Laurent expansions, isolated singularities, and rationality of meromorphic functions; Polynomial identity and more-roots-than-degree principle
Cue extraction	prior broad method tags: convexity / Jensen / tangent, polynomial identity / factorization, modular / finite-field reduction | statement triggers: integral/limit trigger; polynomial/root trigger; functional-equation trigger | audit closures: AUD-PC-10; AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-06; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.11.2 Day 2

2004-D2-P1 (PDF p. 100) *exact computation*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Invariant extraction and invariant comparison; 2. Special substitution and parameter freezing; 3. Rank-nullity and dimension-count obstruction
Method IDs	M-UNI-09; M-UNI-07; M-LA-04; M-LA-06; M-LA-03; M-LA-16; M-LA-07; M-LA-13
Method titles	Invariant extraction and invariant comparison; Special substitution and parameter freezing; Rank-nullity and dimension-count obstruction; Spectral/minimal-polynomial reduction; Block decomposition and Schur-style bookkeeping; Trace pairing and orthogonal matrix subspaces; Trace cyclicity and commutator annihilation; Determinant through eigenvalue or root product
Theory dependencies	Block matrices, Schur complements, and determinant identities; Rank-nullity, image/kernel, and rank of factorizations; Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: matrix word | audit closures: AUD-LA-03; AUD-LA-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2004-D2-P2 (PDF p. 100) *inequality / sharp bound*

Primary domain	Functional equations / dynamics
Core method sequence	1. Auxiliary function with derivative sign; 2. Convexity, Jensen, tangent, and chord inequalities; 3. Monotonicity forced by functional equation
Method IDs	M-RA-01; M-RA-09; M-FE-02; M-CMB-03; M-CMB-04; M-RA-03; M-UNI-07; M-UNI-03
Method titles	Auxiliary function with derivative sign; Convexity, Jensen, tangent, and chord inequalities; Monotonicity forced by functional equation; Graph encoding of a relation; Cycle extraction and alternating-cycle method; Rolle chain and divided-difference localization; Special substitution and parameter freezing; Extremal construction and equality-case engineering
Theory dependencies	Area, volume, projection, and ellipsoid computation; Integral inequalities, integration by parts, and weighted estimates; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: monotonicity argument, convexity / Jensen / tangent, graph encoding / cycles | statement triggers: continuous/differentiable trigger; functional-equation trigger | audit closures: AUD-RA-08; AUD-GEO-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2004-D2-P3 (PDF p. 100) *existence / construction*

Primary domain	Geometry / convexity / metric
Core method sequence	1. Extremal construction and equality-case engineering; 2. Density, Baire/category, and pathological construction; 3. Sign, zero, and boundary case splitting
Method IDs	M-UNI-03; M-RA-16; M-UNI-08; M-GEO-03; M-RA-10; M-FE-01; M-RA-06; M-RA-07
Method titles	Extremal construction and equality-case engineering; Density, Baire/category, and pathological construction; Sign, zero, and boundary case splitting; Squared-distance and inner-product identities; Cauchy-Schwarz, Holder, and norm comparison; Iterates, fixed points, and cluster-point trapping; Riemann sum and sum-integral conversion; Split into main range and tail/error range
Theory dependencies	Euclidean distances and inner-product identities; Integral inequalities, integration by parts, and weighted estimates; Area, volume, projection, and ellipsoid computation
Cue extraction	prior broad method tags: construction / counterexample, topological / density argument | statement triggers: integral/limit trigger; geometry/metric trigger | audit closures: AUD-RA-08; AUD-GEO-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2004-D2-P4 (PDF p. 101) *proof / reasoning*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Spectral/minimal-polynomial reduction; 2. Rank-nullity and dimension-count obstruction; 3. Nilpotency from zero spectrum plus finite dimension
Method IDs	M-LA-06; M-LA-04; M-LA-12; M-RA-06; M-RA-07; M-RA-09; M-UNI-07; M-UNI-09
Method titles	Spectral/minimal-polynomial reduction; Rank-nullity and dimension-count obstruction; Nilpotency from zero spectrum plus finite dimension; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Convexity, Jensen, tangent, and chord inequalities; Special substitution and parameter freezing; Invariant extraction and invariant comparison
Theory dependencies	Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; Block matrices, Schur complements, and determinant identities; Rank-nullity, image/kernel, and rank of factorizations
Cue extraction	prior broad method tags: spectral decomposition / eigenvalues, rank-nullity / dimension count | statement triggers: matrix word; det/rank/eigen trigger | audit closures: AUD-LA-15; AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2004-D2-P5 (PDF p. 101) *inequality / sharp bound*

Primary domain	Real analysis / calculus
Core method sequence	1. Convexity, Jensen, tangent, and chord inequalities; 2. Auxiliary function with derivative sign; 3. Integrating factor and ODE comparison
Method IDs	M-RA-09; M-RA-01; M-RA-02; M-UNI-07; M-RA-11; M-RA-08; M-UNI-03; M-CX-05
Method titles	Convexity, Jensen, tangent, and chord inequalities; Auxiliary function with derivative sign; Integrating factor and ODE comparison; Special substitution and parameter freezing; Integration by parts, Fubini, and weighted identities; Compactness plus extreme value theorem; Extremal construction and equality-case engineering; Trigonometric-to-complex conversion
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Fubini/Tonelli and change of variables; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: auxiliary function / substitution, convexity / Jensen / tangent | statement triggers: wording-specific | audit closures: AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2004-D2-P6 (PDF p. 102) *inequality / sharp bound*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. State recurrence or automaton partition; 2. Explicit bijection and encoding-decoding proof; 3. Pigeonhole threshold and collision forcing
Method IDs	M-CMB-05; M-CMB-10; M-CMB-01; M-CMB-02; M-LA-01; M-LA-03; M-RA-06; M-RA-07
Method titles	State recurrence or automaton partition; Explicit bijection and encoding-decoding proof; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Row/column elimination to triangular or sparse form; Block decomposition and Schur-style bookkeeping; Riemann sum and sum-integral conversion; Split into main range and tail/error range
Theory dependencies	Transfer matrices and matrix powers for counting; Integral inequalities, integration by parts, and weighted estimates; Pigeonhole principle and extremal element selection
Cue extraction	prior broad method tags: determinant / block manipulation, double counting / pigeonhole | statement triggers: matrix word; integral/limit trigger | audit closures: AUD-RA-08; AUD-CMB-03; AUD-CMB-06
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.12 2005

18.12.1 Day 1

2005-D1-P1 (PDF p. 103) *proof / reasoning*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Rank-nullity and dimension-count obstruction; 2. Row/column elimination to triangular or sparse form; 3. Block decomposition and Schur-style bookkeeping
Method IDs	M-LA-04; M-LA-01; M-LA-03; M-RA-09; M-LA-06; M-UNI-07; M-UNI-09; M-CMB-07
Method titles	Rank-nullity and dimension-count obstruction; Row/column elimination to triangular or sparse form; Block decomposition and Schur-style bookkeeping; Convexity, Jensen, tangent, and chord inequalities; Spectral/minimal-polynomial reduction; Special substitution and parameter freezing; Invariant extraction and invariant comparison; Involution, parity, and sign-reversing cancellation
Theory dependencies	Rank-nullity, image/kernel, and rank of factorizations; Determinant evaluation by row/column operations and triangular forms; Block matrices, Schur complements, and determinant identities
Cue extraction	prior broad method tags: determinant / block manipulation, rank-nullity / dimension count | statement triggers: matrix word; det/rank/eigen trigger | audit closures: AUD-RA-08; AUD-CMB-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2005-D1-P2 (PDF p. 103) *counting / enumeration*

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Pigeonhole threshold and collision forcing; 3. Double counting ordered or weighted objects
Method IDs	M-ANT-01; M-CMB-01; M-CMB-02; M-ANT-02; M-UNI-05; M-ANT-03; M-CMB-05; M-RA-09
Method titles	Reduction modulo a prime or finite field; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Chinese remainder and residue construction; Induction with strengthened hypothesis; GCD, valuation, and prime-power bookkeeping; State recurrence or automaton partition; Convexity, Jensen, tangent, and chord inequalities
Theory dependencies	Recurrences and state partitions; Modular arithmetic and divisibility; Finite fields, field characteristic, and Fermat/Frobenius tools
Cue extraction	prior broad method tags: induction, modular / finite-field reduction, double counting / pigeonhole | statement triggers: integer/prime/mod trigger; counting/permutation trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-PC-13; AUD-ANT-02; AUD-CMB-03
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2005-D1-P3 (PDF p. 104)

inequality / sharp bound

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Auxiliary function with derivative sign
Method IDs	M-UNI-07; M-UNI-09; M-RA-01; M-RA-03; M-RA-09; M-RA-08; M-UNI-03; M-LA-04
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Auxiliary function with derivative sign; Rolle chain and divided-difference localization; Convexity, Jensen, tangent, and chord inequalities; Compactness plus extreme value theorem; Extremal construction and equality-case engineering; Rank-nullity and dimension-count obstruction
Theory dependencies	Integral inequalities, integration by parts, and weighted estimates; Convexity, concavity, Jensen, and tangent/chord inequalities; Rolle, mean-value theorem, and divided differences
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: continuous/differentiable trigger | audit closures: AUD-RA-01; AUD-RA-06; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2005-D1-P4 (PDF p. 105)

classification / find all

Primary domain	Complex analysis / polynomials
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Reduction modulo a prime or finite field
Method IDs	M-ANT-16; M-ANT-06; M-ANT-01; M-UNI-07; M-ANT-02; M-ANT-05; M-RA-01; M-RA-02
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Reduction modulo a prime or finite field; Special substitution and parameter freezing; Chinese remainder and residue construction; Factorization and unique factorization in Z[x]; Auxiliary function with derivative sign; Integrating factor and ODE comparison
Theory dependencies	Integer polynomials, finite differences, and value divisibility; Polynomial identity and more-roots-than-degree principle; Rational roots, Vieta, and rationality constraints
Cue extraction	prior broad method tags: auxiliary function / substitution, polynomial identity / factorization, modular / finite-field reduction | statement triggers: polynomial/root trigger; counting/permutation trigger | audit closures: AUD-ANT-05; AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2005-D1-P5 (PDF p. 106)

limit / asymptotic evaluation

Primary domain	Real analysis / calculus
Core method sequence	1. Auxiliary function with derivative sign; 2. Taylor expansion and local asymptotic bookkeeping; 3. L-Hospital and ratio/growth comparison
Method IDs	M-RA-01; M-RA-04; M-RA-05; M-RA-02; M-UNI-07; M-RA-03; M-RA-06; M-RA-07
Method titles	Auxiliary function with derivative sign; Taylor expansion and local asymptotic bookkeeping; L-Hospital and ratio/growth comparison; Integrating factor and ODE comparison; Special substitution and parameter freezing; Rolle chain and divided-difference localization; Riemann sum and sum-integral conversion; Split into main range and tail/error range
Theory dependencies	L'Hospital, ratio limits, and growth comparisons; Auxiliary functions, integrating factors, and differential inequalities; Taylor expansion, local asymptotics, and little-o bookkeeping
Cue extraction	prior broad method tags: auxiliary function / substitution, l'Hospital / asymptotic expansion | statement triggers: continuous/differentiable trigger; integral/limit trigger; functional-equation trigger | audit closures: AUD-RA-01; AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-06
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2005-D1-P6 (PDF p. 106) *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Induction with strengthened hypothesis; 2. Trace cyclicity and commutator annihilation; 3. Group action, orbit, and generator method
Method IDs	M-UNI-05; M-LA-07; M-ANT-10; M-ANT-11; M-ANT-03; M-ANT-12; M-RA-09; M-ANT-01
Method titles	Induction with strengthened hypothesis; Trace cyclicity and commutator annihilation; Group action, orbit, and generator method; Commutator and center manipulation in groups/rings; GCD, valuation, and prime-power bookkeeping; Normal subgroup and quotient obstruction; Convexity, Jensen, tangent, and chord inequalities; Reduction modulo a prime or finite field
Theory dependencies	Group basics: orders, cyclic subgroups, and generators; Normal subgroups, quotient groups, and group actions; Commutators and spectral obstructions
Cue extraction	prior broad method tags: induction, trace trick / commutator | statement triggers: group/ring/field trigger | audit closures: AUD-LA-02; AUD-LA-03; AUD-RA-08; AUD-ANT-01; AUD-ANT-06
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.12.2 Day 2

2005-D2-P1 (PDF p. 107) *inequality / sharp bound*

Primary domain	Real analysis / calculus
Core method sequence	1. Compactness plus extreme value theorem; 2. Compactness or extremal-point selection; 3. Riemann sum and sum-integral conversion
Method IDs	M-RA-08; M-UNI-06; M-RA-06; M-RA-07; M-RA-09; M-RA-11; M-RA-01; M-UNI-03
Method titles	Compactness plus extreme value theorem; Compactness or extremal-point selection; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Convexity, Jensen, tangent, and chord inequalities; Integration by parts, Fubini, and weighted identities; Auxiliary function with derivative sign; Extremal construction and equality-case engineering
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Integral inequalities, integration by parts, and weighted estimates; Series convergence, rearrangements, and alternating/Dirichlet tools
Cue extraction	prior broad method tags: compactness / extremal point | statement triggers: integral/limit trigger; functional-equation trigger | audit closures: AUD-RA-07; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2005-D2-P2 (PDF p. 107) *proof / reasoning*

Primary domain	Complex analysis / polynomials
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Factorization and unique factorization in Z[x]
Method IDs	M-ANT-16; M-ANT-06; M-ANT-05; M-ANT-01; M-CX-01; M-UNI-07; M-UNI-09; M-FE-03
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Factorization and unique factorization in Z[x]; Reduction modulo a prime or finite field; Cauchy coefficient formula and boundary maximum; Special substitution and parameter freezing; Invariant extraction and invariant comparison; Composition, inversion, and conjugacy
Theory dependencies	Polynomial identity and more-roots-than-degree principle; Vieta, logarithmic derivatives, and real-root inequalities; Finite fields, field characteristic, and Fermat/Frobenius tools
Cue extraction	prior broad method tags: polynomial identity / factorization | statement triggers: polynomial/root trigger; functional-equation trigger | audit closures: AUD-RA-04; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2005-D2-P3 (PDF p. 108) *inequality / sharp bound*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Trace cyclicity and commutator annihilation; 2. Rank-nullity and dimension-count obstruction; 3. Compactness plus extreme value theorem
Method IDs	M-LA-07; M-LA-04; M-RA-08; M-UNI-06; M-LA-06; M-LA-12; M-RA-06; M-RA-07
Method titles	Trace cyclicity and commutator annihilation; Rank-nullity and dimension-count obstruction; Compactness plus extreme value theorem; Compactness or extremal-point selection; Spectral/minimal-polynomial reduction; Nilpotency from zero spectrum plus finite dimension; Riemann sum and sum-integral conversion; Split into main range and tail/error range
Theory dependencies	Matrix subspaces constrained by trace bilinear forms; Rank-nullity, image/kernel, and rank of factorizations; Block matrices, Schur complements, and determinant identities
Cue extraction	prior broad method tags: compactness / extremal point, trace trick / commutator, rank-nullity / dimension count | statement triggers: matrix word; det/rank/eigen trigger; integral/limit trigger | audit closures: AUD-LA-03; AUD-RA-07; AUD-RA-08; AUD-ANT-06; AUD-CMB-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2005-D2-P4 (PDF p. 108) *existence / construction*

Primary domain	Functional equations / dynamics
Core method sequence	1. Rolle chain and divided-difference localization; 2. Auxiliary function with derivative sign; 3. Special substitution and parameter freezing
Method IDs	M-RA-03; M-RA-01; M-UNI-07; M-RA-02; M-FE-02; M-UNI-03; M-UNI-06; M-CMB-03
Method titles	Rolle chain and divided-difference localization; Auxiliary function with derivative sign; Special substitution and parameter freezing; Integrating factor and ODE comparison; Monotonicity forced by functional equation; Extremal construction and equality-case engineering; Compactness or extremal-point selection; Graph encoding of a relation
Theory dependencies	Rolle, mean-value theorem, and divided differences; Taylor expansion, local asymptotics, and little-o bookkeeping; Graph encoding, cycles, and bipartite representations
Cue extraction	prior broad method tags: auxiliary function / substitution, Rolle / mean-value theorem | statement triggers: continuous/differentiable trigger; functional-equation trigger | audit closures: AUD-RA-01; AUD-RA-06
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2005-D2-P5 (PDF p. 109)

classification / find all

Primary domain	Functional equations / dynamics
Core method sequence	1. Compactness plus extreme value theorem; 2. Compactness or extremal-point selection; 3. Rolle chain and divided-difference localization
Method IDs	M-RA-08; M-UNI-06; M-RA-03; M-RA-01; M-UNI-07; M-RA-09; M-GEO-03; M-RA-10
Method titles	Compactness plus extreme value theorem; Compactness or extremal-point selection; Rolle chain and divided-difference localization; Auxiliary function with derivative sign; Special substitution and parameter freezing; Convexity, Jensen, tangent, and chord inequalities; Squared-distance and inner-product identities; Cauchy-Schwarz, Holder, and norm comparison
Theory dependencies	Gradient/Lipschitz-gradient optimization and convex smoothness; Euclidean distances and inner-product identities; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: compactness / extremal point | statement triggers: continuous/differentiable trigger; functional-equation trigger | audit closures: AUD-RA-06; AUD-RA-07; AUD-RA-08; AUD-GEO-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2005-D2-P6 (PDF p. 109)

existence / construction

Primary domain	Algebra / number theory
Core method sequence	1. Pell-type recurrence or quadratic irrational stabilizer; 2. Reduction modulo a prime or finite field; 3. Polynomial identity by coefficient comparison
Method IDs	M-ANT-13; M-ANT-01; M-ANT-16; M-ANT-06; M-LA-03; M-LA-01; M-LA-07; M-ANT-02
Method titles	Pell-type recurrence or quadratic irrational stabilizer; Reduction modulo a prime or finite field; Polynomial identity by coefficient comparison; Degree versus number of roots; Block decomposition and Schur-style bookkeeping; Row/column elimination to triangular or sparse form; Trace cyclicity and commutator annihilation; Chinese remainder and residue construction
Theory dependencies	Diophantine equations, Pell-type recurrences, and quadratic irrationals; Modular arithmetic and divisibility; Block matrices, Schur complements, and determinant identities
Cue extraction	prior broad method tags: trace trick / commutator, determinant / block manipulation, polynomial identity / factorization | statement triggers: matrix word; integer/prime/mod trigger | audit closures: AUD-PC-14; AUD-LA-03; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.13 2006

18.13.1 Day 1

2006-D1-P1 **(PDF p. 111)** *nonexistence / counterexample*

Primary domain	Real analysis / calculus
Core method sequence	1. Auxiliary function with derivative sign; 2. Extremal construction and equality-case engineering; 3. Monotonicity forced by functional equation
Method IDs	M-RA-01; M-UNI-03; M-FE-02; M-UNI-08; M-RA-03; M-RA-06; M-RA-07; M-RA-09
Method titles	Auxiliary function with derivative sign; Extremal construction and equality-case engineering; Monotonicity forced by functional equation; Sign, zero, and boundary case splitting; Rolle chain and divided-difference localization; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Convexity, Jensen, tangent, and chord inequalities
Theory dependencies	Explicit construction and counterexample design; Intermediate value, connectedness, density, and continuity transfer; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: monotonicity argument, construction / counterexample | statement triggers: continuous/differentiable trigger | audit closures: AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08; AUD-RA-10
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2006-D1-P2 **(PDF p. 111)** *counting / enumeration*

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Pigeonhole threshold and collision forcing; 3. Double counting ordered or weighted objects
Method IDs	M-ANT-01; M-CMB-01; M-CMB-02; M-ANT-02; M-ANT-03; M-CMB-10; M-CMB-05; M-ANT-06
Method titles	Reduction modulo a prime or finite field; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Chinese remainder and residue construction; GCD, valuation, and prime-power bookkeeping; Explicit bijection and encoding-decoding proof; State recurrence or automaton partition; Degree versus number of roots
Theory dependencies	Modular arithmetic and divisibility; Chinese remainder theorem and complete residue systems; Finite fields, field characteristic, and Fermat/Frobenius tools
Cue extraction	prior broad method tags: modular / finite-field reduction, double counting / pigeonhole | statement triggers: integer/prime/mod trigger; counting/permutation trigger | audit closures: AUD-ANT-01; AUD-RA-08; AUD-PC-01; AUD-ANT-02; AUD-CMB-03; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2006-D1-P3 **(PDF p. 111)** *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Row/column elimination to triangular or sparse form; 3. Induction with strengthened hypothesis
Method IDs	M-ANT-01; M-LA-01; M-UNI-05; M-LA-03; M-ANT-02; M-RA-06; M-RA-07; M-ANT-03
Method titles	Reduction modulo a prime or finite field; Row/column elimination to triangular or sparse form; Induction with strengthened hypothesis; Block decomposition and Schur-style bookkeeping; Chinese remainder and residue construction; Riemann sum and sum-integral conversion; Split into main range and tail/error range; GCD, valuation, and prime-power bookkeeping
Theory dependencies	Integer polynomials, finite differences, and value divisibility; Modular arithmetic and divisibility; Determinant evaluation by row/column operations and triangular forms
Cue extraction	prior broad method tags: induction, determinant / block manipulation, modular / finite-field reduction | statement triggers: matrix word; det/rank/eigen trigger; integer/prime/mod trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2006-D1-P4 (PDF p. 112) *proof of structural property*

Primary domain	Complex analysis / polynomials
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Factorization and unique factorization in Z[x]
Method IDs	M-ANT-16; M-ANT-06; M-ANT-05; M-ANT-01; M-ANT-03; M-RA-06; M-RA-07; M-CX-01
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Factorization and unique factorization in Z[x]; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Cauchy coefficient formula and boundary maximum
Theory dependencies	Factorization, irreducibility, and Gauss-style integer polynomial arguments; Integer polynomials, finite differences, and value divisibility; Polynomial identity and more-roots-than-degree principle
Cue extraction	prior broad method tags: polynomial identity / factorization | statement triggers: polynomial/root trigger; integer/prime/mod trigger; functional-equation trigger | audit closures: AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-PC-01; AUD-PC-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2006-D1-P5 (PDF p. 112) *proof / reasoning*

Primary domain	Real analysis / calculus
Core method sequence	1. Rolle chain and divided-difference localization; 2. Auxiliary function with derivative sign; 3. Convexity, Jensen, tangent, and chord inequalities
Method IDs	M-RA-03; M-RA-01; M-RA-09; M-RA-08; M-UNI-07; M-UNI-09; M-ANT-07; M-ANT-15
Method titles	Rolle chain and divided-difference localization; Auxiliary function with derivative sign; Convexity, Jensen, tangent, and chord inequalities; Compactness plus extreme value theorem; Special substitution and parameter freezing; Invariant extraction and invariant comparison; Vieta, symmetric sums, and logarithmic derivative; Rationality and linear dependence over Q
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Rational roots, Vieta, and rationality constraints; Rolle, mean-value theorem, and divided differences
Cue extraction	prior broad method tags: Rolle / mean-value theorem | statement triggers: continuous/differentiable trigger | audit closures: AUD-RA-06; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2006-D1-P6 (PDF p. 113) *classification / find all*

Primary domain	Real analysis / calculus
Core method sequence	1. Rolle chain and divided-difference localization; 2. Polynomial identity by coefficient comparison; 3. Degree versus number of roots
Method IDs	M-RA-03; M-ANT-16; M-ANT-06; M-ANT-05; M-CMB-01; M-CMB-02; M-UNI-05; M-RA-01
Method titles	Rolle chain and divided-difference localization; Polynomial identity by coefficient comparison; Degree versus number of roots; Factorization and unique factorization in Z[x]; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Induction with strengthened hypothesis; Auxiliary function with derivative sign
Theory dependencies	Polynomial identity and more-roots-than-degree principle; Rolle, mean-value theorem, and divided differences; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: induction, Rolle / mean-value theorem, polynomial identity / factorization | statement triggers: continuous/differentiable trigger; functional-equation trigger | audit closures: AUD-RA-06; AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.13.2 Day 2

2006-D2-P1 (PDF p. 114) *counting / enumeration*

Primary domain	Algebra / number theory
Core method sequence	1. Graph encoding of a relation; 2. Convexity, Jensen, tangent, and chord inequalities; 3. Cycle extraction and alternating-cycle method
Method IDs	M-CMB-03; M-RA-09; M-CMB-04; M-CMB-01; M-CMB-02; M-UNI-05; M-ANT-01; M-ANT-03
Method titles	Graph encoding of a relation; Convexity, Jensen, tangent, and chord inequalities; Cycle extraction and alternating-cycle method; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Induction with strengthened hypothesis; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping
Theory dependencies	Coloring, partitions, and parity; Polygon triangulation and parity of incidences; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: induction, convexity / Jensen / tangent, double counting / pigeonhole | statement triggers: integer/prime/mod trigger; counting/permutation trigger; graph/color/set trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-ANT-02; AUD-CMB-03; AUD-GEO-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2006-D2-P2 (PDF p. 114) *classification / find all*

Primary domain	Functional equations / dynamics
Core method sequence	1. Rank-nullity and dimension-count obstruction; 2. Monotonicity forced by functional equation; 3. Compactness plus extreme value theorem
Method IDs	M-LA-04; M-FE-02; M-RA-08; M-RA-16; M-RA-01; M-UNI-06; M-UNI-07; M-RA-09
Method titles	Rank-nullity and dimension-count obstruction; Monotonicity forced by functional equation; Compactness plus extreme value theorem; Density, Baire/category, and pathological construction; Auxiliary function with derivative sign; Compactness or extremal-point selection; Special substitution and parameter freezing; Convexity, Jensen, tangent, and chord inequalities
Theory dependencies	Image-length preserving functions; Rank-nullity, image/kernel, and rank of factorizations; Invariant subspaces and adapted bases
Cue extraction	prior broad method tags: compactness / extremal point, monotonicity argument, rank-nullity / dimension count | statement triggers: functional-equation trigger | audit closures: AUD-RA-07; AUD-RA-08; AUD-CMB-05
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2006-D2-P3 (PDF p. 115)

proof / reasoning

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Auxiliary function with derivative sign
Method IDs	M-UNI-07; M-UNI-09; M-RA-01; M-RA-08; M-RA-09; M-CX-05; M-LA-03; M-LA-15
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Auxiliary function with derivative sign; Compactness plus extreme value theorem; Convexity, Jensen, tangent, and chord inequalities; Trigonometric-to-complex conversion; Block decomposition and Schur-style bookkeeping; AB versus BA transfer
Theory dependencies	Trigonometric and hyperbolic identities with monotonicity; Convexity, concavity, Jensen, and tangent/chord inequalities; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: wording-specific | audit closures: AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2006-D2-P4 (PDF p. 115)

proof of structural property

Primary domain	Algebra / number theory
Core method sequence	1. Squared-distance and inner-product identities; 2. Gram matrix and distance-to-inner-product conversion; 3. Cauchy-Schwarz, Holder, and norm comparison
Method IDs	M-GEO-03; M-LA-11; M-RA-10; M-LA-04; M-RA-09; M-ANT-01; M-ANT-03; M-UNI-05
Method titles	Squared-distance and inner-product identities; Gram matrix and distance-to-inner-product conversion; Cauchy-Schwarz, Holder, and norm comparison; Rank-nullity and dimension-count obstruction; Convexity, Jensen, tangent, and chord inequalities; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Induction with strengthened hypothesis
Theory dependencies	Euclidean distances and inner-product identities; Gram matrices and rational linear dependence; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: Cauchy-Schwarz / norm inequality, rank-nullity / dimension count | statement triggers: wording-specific | audit closures: AUD-GEO-01; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2006-D2-P5 (PDF p. 116)

existence / construction

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Polynomial identity by coefficient comparison; 3. Degree versus number of roots
Method IDs	M-ANT-01; M-ANT-16; M-ANT-06; M-CMB-01; M-CMB-02; M-ANT-02; M-ANT-05; M-RA-01
Method titles	Reduction modulo a prime or finite field; Polynomial identity by coefficient comparison; Degree versus number of roots; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Chinese remainder and residue construction; Factorization and unique factorization in Z[x]; Auxiliary function with derivative sign
Theory dependencies	Diophantine equations, Pell-type recurrences, and quadratic irrationals; Modular arithmetic and divisibility; Polynomial identity and more-roots-than-degree principle
Cue extraction	prior broad method tags: auxiliary function / substitution, polynomial identity / factorization, modular / finite-field reduction | statement triggers: polynomial/root trigger; integer/prime/mod trigger; counting/permutation trigger | audit closures: AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-ANT-02; AUD-CMB-03
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2006-D2-P6 (PDF p. 116) *proof of structural property*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Spectral/minimal-polynomial reduction; 2. Degree versus number of roots; 3. Polynomial identity by coefficient comparison
Method IDs	M-LA-06; M-ANT-06; M-ANT-16; M-LA-03; M-LA-01; M-LA-07; M-ANT-05; M-LA-12
Method titles	Spectral/minimal-polynomial reduction; Degree versus number of roots; Polynomial identity by coefficient comparison; Block decomposition and Schur-style bookkeeping; Row/column elimination to triangular or sparse form; Trace cyclicity and commutator annihilation; Factorization and unique factorization in Z[x]; Nilpotency from zero spectrum plus finite dimension
Theory dependencies	Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; Simultaneous diagonalization and commuting families; Determinant evaluation by row/column operations and triangular forms
Cue extraction	prior broad method tags: spectral decomposition / eigenvalues, trace trick / commutator, determinant / block manipulation | statement triggers: matrix word; det/rank/eigen trigger | audit closures: AUD-LA-03; AUD-LA-07; AUD-RA-06; AUD-PC-01; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.14 2007

18.14.1 Day 1

2007-D1-P1 (PDF p. 118) *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Polynomial identity by coefficient comparison; 3. Degree versus number of roots
Method IDs	M-ANT-01; M-ANT-16; M-ANT-06; M-ANT-02; M-ANT-05; M-RA-01; M-RA-02; M-UNI-07
Method titles	Reduction modulo a prime or finite field; Polynomial identity by coefficient comparison; Degree versus number of roots; Chinese remainder and residue construction; Factorization and unique factorization in Z[x]; Auxiliary function with derivative sign; Integrating factor and ODE comparison; Special substitution and parameter freezing
Theory dependencies	Polynomial identity and more-roots-than-degree principle; Integer polynomials, finite differences, and value divisibility; Modular arithmetic and divisibility
Cue extraction	prior broad method tags: auxiliary function / substitution, polynomial identity / factorization, modular / finite-field reduction | statement triggers: polynomial/root trigger; integer/prime/mod trigger; functional-equation trigger | audit closures: AUD-PC-01; AUD-PC-06; AUD-PC-08; AUD-PC-11; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2007-D1-P2 (PDF p. 118) *inequality / sharp bound*

Primary domain	Algebra / number theory
Core method sequence	1. Rank-nullity and dimension-count obstruction; 2. Row/column elimination to triangular or sparse form; 3. Block decomposition and Schur-style bookkeeping
Method IDs	M-LA-04; M-LA-01; M-LA-03; M-ANT-01; M-ANT-03; M-RA-09; M-UNI-03; M-CMB-07
Method titles	Rank-nullity and dimension-count obstruction; Row/column elimination to triangular or sparse form; Block decomposition and Schur-style bookkeeping; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Convexity, Jensen, tangent, and chord inequalities; Extremal construction and equality-case engineering; Involution, parity, and sign-reversing cancellation
Theory dependencies	Rank-nullity, image/kernel, and rank of factorizations; Determinant evaluation by row/column operations and triangular forms; Block matrices, Schur complements, and determinant identities
Cue extraction	prior broad method tags: determinant / block manipulation, rank-nullity / dimension count | statement triggers: matrix word; det/rank/eigen trigger; integer/prime/mod trigger | audit closures: AUD-RA-08; AUD-CMB-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2007-D1-P3 (PDF p. 118) *classification / find all*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Row/column elimination to triangular or sparse form
Method IDs	M-ANT-16; M-ANT-06; M-LA-01; M-LA-03; M-ANT-05; M-LA-13; M-LA-04; M-FE-03
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Row/column elimination to triangular or sparse form; Block decomposition and Schur-style bookkeeping; Factorization and unique factorization in Z[x]; Determinant through eigenvalue or root product; Rank-nullity and dimension-count obstruction; Composition, inversion, and conjugacy
Theory dependencies	Rank-nullity, image/kernel, and rank of factorizations; Polynomial functional identities and algebraic composition; Polynomial identity and more-roots-than-degree principle
Cue extraction	prior broad method tags: determinant / block manipulation, polynomial identity / factorization | statement triggers: matrix word; det/rank/eigen trigger; polynomial/root trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-CMB-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2007-D1-P4 (PDF p. 119) *counting / enumeration*

Primary domain	Algebra / number theory
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Double counting ordered or weighted objects; 3. Auxiliary function with derivative sign
Method IDs	M-CMB-01; M-CMB-02; M-RA-01; M-RA-02; M-UNI-07; M-ANT-10; M-ANT-11; M-CMB-10
Method titles	Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Auxiliary function with derivative sign; Integrating factor and ODE comparison; Special substitution and parameter freezing; Group action, orbit, and generator method; Commutator and center manipulation in groups/rings; Explicit bijection and encoding-decoding proof
Theory dependencies	Double counting and degree-sum methods; Group basics: orders, cyclic subgroups, and generators; Pigeonhole principle and extremal element selection
Cue extraction	prior broad method tags: auxiliary function / substitution, double counting / pigeonhole | statement triggers: group/ring/field trigger; counting/permutation trigger | audit closures: AUD-PC-13; AUD-CMB-03; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2007-D1-P5 (PDF p. 119) *proof of structural property*

Primary domain	Complex analysis / polynomials
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Factorization and unique factorization in Z[x]
Method IDs	M-ANT-16; M-ANT-06; M-ANT-05; M-ANT-01; M-ANT-03; M-CX-01; M-ANT-08; M-RA-09
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Factorization and unique factorization in Z[x]; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Cauchy coefficient formula and boundary maximum; Roots of unity and cyclotomic irreducibility; Convexity, Jensen, tangent, and chord inequalities
Theory dependencies	Roots of unity, cyclotomic filters, and discrete Fourier tricks; Cauchy formula and coefficient extraction on the unit circle; Polynomial identity and more-roots-than-degree principle
Cue extraction	prior broad method tags: polynomial identity / factorization | statement triggers: integer/prime/mod trigger; functional-equation trigger | audit closures: AUD-RA-06; AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2007-D1-P6 (PDF p. 120) *counting / enumeration*

Primary domain	Complex analysis / polynomials
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Pigeonhole threshold and collision forcing
Method IDs	M-ANT-16; M-ANT-06; M-CMB-01; M-CMB-02; M-ANT-08; M-CMB-12; M-ANT-05; M-CX-05
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Roots of unity and cyclotomic irreducibility; Root-of-unity filter for residues; Factorization and unique factorization in Z[x]; Trigonometric-to-complex conversion
Theory dependencies	Roots of unity, cyclotomic filters, and discrete Fourier tricks; Polynomial identity and more-roots-than-degree principle; Cauchy formula and coefficient extraction on the unit circle
Cue extraction	prior broad method tags: polynomial identity / factorization, roots of unity / Fourier filter, double counting / pigeonhole | statement triggers: polynomial/root trigger; integer/prime/mod trigger; counting/permutation trigger | audit closures: AUD-RA-06; AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-06
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.14.2 Day 2

2007-D2-P1 (PDF p. 121) *proof / reasoning*

Primary domain	Real analysis / calculus
Core method sequence	1. Graph encoding of a relation; 2. Cycle extraction and alternating-cycle method; 3. Extremal construction and equality-case engineering
Method IDs	M-CMB-03; M-CMB-04; M-UNI-03; M-UNI-08; M-RA-01; M-RA-03; M-UNI-07; M-RA-08
Method titles	Graph encoding of a relation; Cycle extraction and alternating-cycle method; Extremal construction and equality-case engineering; Sign, zero, and boundary case splitting; Auxiliary function with derivative sign; Rolle chain and divided-difference localization; Special substitution and parameter freezing; Compactness plus extreme value theorem
Theory dependencies	Explicit construction and counterexample design; Substitution of special values and normalization; Graph encoding, cycles, and bipartite representations
Cue extraction	prior broad method tags: graph encoding / cycles, construction / counterexample | statement triggers: continuous/differentiable trigger; graph/color/set trigger; functional-equation trigger | audit closures: AUD-RA-01; AUD-RA-06
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2007-D2-P2 (PDF p. 121) *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Chinese remainder and residue construction; 3. GCD, valuation, and prime-power bookkeeping
Method IDs	M-ANT-01; M-ANT-02; M-ANT-03; M-RA-11; M-UNI-05; M-UNI-09; M-ANT-06; M-ANT-16
Method titles	Reduction modulo a prime or finite field; Chinese remainder and residue construction; GCD, valuation, and prime-power bookkeeping; Integration by parts, Fubini, and weighted identities; Induction with strengthened hypothesis; Invariant extraction and invariant comparison; Degree versus number of roots; Polynomial identity by coefficient comparison
Theory dependencies	Modular arithmetic and divisibility; Finite fields, field characteristic, and Fermat/Frobenius tools; Chinese remainder theorem and complete residue systems
Cue extraction	prior broad method tags: modular / finite-field reduction | statement triggers: integer/prime/mod trigger | audit closures: AUD-PC-01; AUD-ANT-02; AUD-ANT-03
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2007-D2-P3 (PDF p. 121) *existence / construction*

Primary domain	Real analysis / calculus
Core method sequence	1. Auxiliary function with derivative sign; 2. Extremal construction and equality-case engineering; 3. Monotonicity forced by functional equation
Method IDs	M-RA-01; M-UNI-03; M-FE-02; M-RA-16; M-UNI-08; M-RA-03; M-RA-08; M-UNI-06
Method titles	Auxiliary function with derivative sign; Extremal construction and equality-case engineering; Monotonicity forced by functional equation; Density, Baire/category, and pathological construction; Sign, zero, and boundary case splitting; Rolle chain and divided-difference localization; Compactness plus extreme value theorem; Compactness or extremal-point selection
Theory dependencies	Monotone functions, fixed points, and iteration on intervals; Intermediate value, connectedness, density, and continuity transfer; Compactness, extreme values, and closed-bounded reduction
Cue extraction	prior broad method tags: monotonicity argument, construction / counterexample, topological / density argument | statement triggers: continuous/differentiable trigger; functional-equation trigger | audit closures: AUD-RA-07; AUD-RA-08; AUD-RA-10
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2007-D2-P4 (PDF p. 121) *exact computation*

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Row/column elimination to triangular or sparse form; 3. Chinese remainder and residue construction
Method IDs	M-ANT-01; M-LA-01; M-ANT-02; M-LA-03; M-ANT-03; M-LA-13; M-UNI-02; M-UNI-09
Method titles	Reduction modulo a prime or finite field; Row/column elimination to triangular or sparse form; Chinese remainder and residue construction; Block decomposition and Schur-style bookkeeping; GCD, valuation, and prime-power bookkeeping; Determinant through eigenvalue or root product; Small cases, equality cases, and example probing; Invariant extraction and invariant comparison
Theory dependencies	Determinant evaluation by row/column operations and triangular forms; Modular arithmetic and divisibility; Finite fields, field characteristic, and Fermat/Frobenius tools
Cue extraction	prior broad method tags: determinant / block manipulation, modular / finite-field reduction | statement triggers: matrix word; det/rank/eigen trigger; integer/prime/mod trigger | audit closures: AUD-PC-01; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2007-D2-P5 (PDF p. 122) *proof / reasoning*

Primary domain	Algebra / number theory
Core method sequence	1. Rank-nullity and dimension-count obstruction; 2. Similarity, adapted basis, and canonical model; 3. Extremal construction and equality-case engineering
Method IDs	M-LA-04; M-LA-05; M-UNI-03; M-UNI-08; M-ANT-01; M-ANT-03; M-UNI-07; M-UNI-09
Method titles	Rank-nullity and dimension-count obstruction; Similarity, adapted basis, and canonical model; Extremal construction and equality-case engineering; Sign, zero, and boundary case splitting; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Special substitution and parameter freezing; Invariant extraction and invariant comparison
Theory dependencies	Explicit construction and counterexample design; Nilpotent, idempotent, involution, and square-zero structures; Rank-nullity, image/kernel, and rank of factorizations
Cue extraction	prior broad method tags: rank-nullity / dimension count, construction / counterexample | statement triggers: matrix word; integer/prime/mod trigger | audit closures: AUD-LA-12; AUD-RA-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2007-D2-P6 (PDF p. 123) *existence / construction*

Primary domain	Complex analysis / polynomials
Core method sequence	1. Rolle chain and divided-difference localization; 2. Polynomial identity by coefficient comparison; 3. Degree versus number of roots
Method IDs	M-RA-03; M-ANT-16; M-ANT-06; M-UNI-06; M-RA-01; M-ANT-05; M-FE-02; M-RA-08
Method titles	Rolle chain and divided-difference localization; Polynomial identity by coefficient comparison; Degree versus number of roots; Compactness or extremal-point selection; Auxiliary function with derivative sign; Factorization and unique factorization in Z[x]; Monotonicity forced by functional equation; Compactness plus extreme value theorem
Theory dependencies	Polynomial identity and more-roots-than-degree principle; Taylor expansion, local asymptotics, and little-o bookkeeping; Rolle, mean-value theorem, and divided differences
Cue extraction	prior broad method tags: induction, Rolle / mean-value theorem, compactness / extremal point | statement triggers: polynomial/root trigger | audit closures: AUD-RA-01; AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-06
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.15 2008

18.15.1 Day 1

2008-D1-P6 (PDF p. 125) *inequality / sharp bound*

Primary domain	Combinatorics / discrete structures
Core method sequence	1. Regular simplex and equidistant orthonormal translation; 2. Pigeonhole threshold and collision forcing; 3. Double counting ordered or weighted objects
Method IDs	M-GEO-09; M-CMB-01; M-CMB-02; M-ANT-16; M-ANT-06; M-LA-01; M-LA-03; M-ANT-05
Method titles	Regular simplex and equidistant orthonormal translation; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Polynomial identity by coefficient comparison; Degree versus number of roots; Row/column elimination to triangular or sparse form; Block decomposition and Schur-style bookkeeping; Factorization and unique factorization in Z[x]
Theory dependencies	Pigeonhole principle and extremal element selection; Block matrices, Schur complements, and determinant identities; Polynomial identity and more-roots-than-degree principle
Cue extraction	prior broad method tags: determinant / block manipulation, polynomial identity / factorization, double counting / pigeonhole | statement triggers: counting/permutation trigger | audit closures: AUD-CMB-08; AUD-RA-06; AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2008-D1-P1 (PDF p. 125) *classification / find all*

Primary domain	Functional equations / dynamics
Core method sequence	1. Density, Baire/category, and pathological construction; 2. Auxiliary function with derivative sign; 3. Rolle chain and divided-difference localization
Method IDs	M-RA-16; M-RA-01; M-RA-03; M-UNI-07; M-RA-08; M-RA-09; M-FE-02; M-UNI-08
Method titles	Density, Baire/category, and pathological construction; Auxiliary function with derivative sign; Rolle chain and divided-difference localization; Special substitution and parameter freezing; Compactness plus extreme value theorem; Convexity, Jensen, tangent, and chord inequalities; Monotonicity forced by functional equation; Sign, zero, and boundary case splitting
Theory dependencies	Intermediate value, connectedness, density, and continuity transfer; Convexity, concavity, Jensen, and tangent/chord inequalities; Compactness, extreme values, and closed-bounded reduction
Cue extraction	prior broad method tags: topological / density argument | statement triggers: continuous/differentiable trigger; functional-equation trigger | audit closures: AUD-RA-07; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2008-D1-P2 (PDF p. 125) *proof of structural property*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Induction with strengthened hypothesis
Method IDs	M-ANT-16; M-ANT-06; M-UNI-05; M-ANT-05; M-LA-04; M-RA-09; M-LA-06; M-UNI-09
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Induction with strengthened hypothesis; Factorization and unique factorization in Z[x]; Rank-nullity and dimension-count obstruction; Convexity, Jensen, tangent, and chord inequalities; Spectral/minimal-polynomial reduction; Invariant extraction and invariant comparison
Theory dependencies	Polynomial identity and more-roots-than-degree principle; Integral inequalities, integration by parts, and weighted estimates; Rank-nullity, image/kernel, and rank of factorizations
Cue extraction	prior broad method tags: induction, polynomial identity / factorization | statement triggers: polynomial/root trigger; functional-equation trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2008-D1-P3 (PDF p. 126) *existence / construction*

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Polynomial identity by coefficient comparison; 3. Degree versus number of roots
Method IDs	M-ANT-01; M-ANT-16; M-ANT-06; M-UNI-03; M-ANT-05; M-ANT-02; M-UNI-05; M-UNI-08
Method titles	Reduction modulo a prime or finite field; Polynomial identity by coefficient comparison; Degree versus number of roots; Extremal construction and equality-case engineering; Factorization and unique factorization in Z[x]; Chinese remainder and residue construction; Induction with strengthened hypothesis; Sign, zero, and boundary case splitting
Theory dependencies	Modular arithmetic and divisibility; Polynomial identity and more-roots-than-degree principle; Finite fields, field characteristic, and Fermat/Frobenius tools
Cue extraction	prior broad method tags: induction, polynomial identity / factorization, modular / finite-field reduction | statement triggers: polynomial/root trigger; integer/prime/mod trigger | audit closures: AUD-PC-01; AUD-PC-08; AUD-PC-11; AUD-PC-13; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2008-D1-P4 (PDF p. 126) *classification / find all*

Primary domain	Combinatorics / discrete structures
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Convexity, Jensen, tangent, and chord inequalities
Method IDs	M-UNI-07; M-UNI-09; M-RA-09; M-RA-11; M-CMB-01; M-CMB-02; M-UNI-08; M-CMB-03
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Convexity, Jensen, tangent, and chord inequalities; Integration by parts, Fubini, and weighted identities; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Sign, zero, and boundary case splitting; Graph encoding of a relation
Theory dependencies	Integral inequalities, integration by parts, and weighted estimates; Graph encoding, cycles, and bipartite representations; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: wording-specific | audit closures: AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2008-D1-P5 (PDF p. 126) *nonexistence / counterexample*

Primary domain	Algebra / number theory
Core method sequence	1. Cauchy-Schwarz, Holder, and norm comparison; 2. Reduction modulo a prime or finite field; 3. Extremal construction and equality-case engineering
Method IDs	M-RA-10; M-ANT-01; M-UNI-03; M-ANT-02; M-CMB-01; M-CMB-02; M-RA-01; M-RA-02
Method titles	Cauchy-Schwarz, Holder, and norm comparison; Reduction modulo a prime or finite field; Extremal construction and equality-case engineering; Chinese remainder and residue construction; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Auxiliary function with derivative sign; Integrating factor and ODE comparison
Theory dependencies	Group basics: orders, cyclic subgroups, and generators; Convexity, concavity, Jensen, and tangent/chord inequalities; Automorphisms and semidirect products
Cue extraction	prior broad method tags: auxiliary function / substitution, Cauchy-Schwarz / norm inequality, modular / finite-field reduction | statement triggers: group/ring/field trigger | audit closures: AUD-RA-03; AUD-RA-05; AUD-RA-06; AUD-RA-08; AUD-PC-01
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.15.2 Day 2

2008-D2-P1 (PDF p. 127) *proof of structural property*

Primary domain	Real analysis / calculus
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Factorization and unique factorization in Z[x]
Method IDs	M-ANT-16; M-ANT-06; M-ANT-05; M-ANT-01; M-ANT-03; M-ANT-08; M-RA-01; M-RA-08
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Factorization and unique factorization in Z[x]; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Roots of unity and cyclotomic irreducibility; Auxiliary function with derivative sign; Compactness plus extreme value theorem
Theory dependencies	Factorization, irreducibility, and Gauss-style integer polynomial arguments; Roots of unity, cyclotomic filters, and discrete Fourier tricks; Polynomial identity and more-roots-than-degree principle
Cue extraction	prior broad method tags: polynomial identity / factorization | statement triggers: polynomial/root trigger; integer/prime/mod trigger | audit closures: AUD-PC-01; AUD-PC-05; AUD-PC-08; AUD-PC-13; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2008-D2-P2 (PDF p. 127) *proof of structural property*

Primary domain	Geometry / convexity / metric
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Factorization and unique factorization in Z[x]
Method IDs	M-ANT-16; M-ANT-06; M-ANT-05; M-RA-01; M-RA-02; M-RA-10; M-UNI-07; M-GEO-01
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Factorization and unique factorization in Z[x]; Auxiliary function with derivative sign; Integrating factor and ODE comparison; Cauchy-Schwarz, Holder, and norm comparison; Special substitution and parameter freezing; Coordinate normalization and algebraization
Theory dependencies	Conics: parabolas, ellipses, foci, and directrices; Polynomial identity and more-roots-than-degree principle; Area, volume, projection, and ellipsoid computation
Cue extraction	prior broad method tags: auxiliary function / substitution, Cauchy-Schwarz / norm inequality, polynomial identity / factorization | statement triggers: geometry/metric trigger | audit closures: AUD-GEO-05; AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-GEO-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2008-D2-P3 (PDF p. 127) *inequality / sharp bound*

Primary domain	Algebra / number theory
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Reduction modulo a prime or finite field
Method IDs	M-UNI-07; M-UNI-09; M-ANT-01; M-ANT-03; M-RA-09; M-RA-15; M-CMB-06; M-UNI-03
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Convexity, Jensen, tangent, and chord inequalities; Recurrence transformed to monotone quotient or telescoping form; Generating function encoding; Extremal construction and equality-case engineering
Theory dependencies	Modular arithmetic and divisibility; Generating functions for counting; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: integer/prime/mod trigger | audit closures: AUD-RA-08; AUD-PC-05; AUD-PC-13; AUD-CMB-07; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2008-D2-P4 (PDF p. 127) *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Polynomial identity by coefficient comparison; 3. Degree versus number of roots
Method IDs	M-ANT-01; M-ANT-16; M-ANT-06; M-ANT-05; M-CMB-01; M-CMB-02; M-ANT-02; M-UNI-03
Method titles	Reduction modulo a prime or finite field; Polynomial identity by coefficient comparison; Degree versus number of roots; Factorization and unique factorization in Z[x]; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Chinese remainder and residue construction; Extremal construction and equality-case engineering
Theory dependencies	Pigeonhole principle and extremal element selection; Integer polynomials, finite differences, and value divisibility; Polynomial identity and more-roots-than-degree principle
Cue extraction	prior broad method tags: polynomial identity / factorization, modular / finite-field reduction, double counting / pigeonhole | statement triggers: polynomial/root trigger; integer/prime/mod trigger; group/ring/field trigger; functional-equation trigger | audit closures: AUD-CMB-03; AUD-PC-01; AUD-PC-08; AUD-PC-11; AUD-PC-13; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2008-D2-P5 (PDF p. 128) *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Parity or zero-pattern block determinant split; 2. Reduction modulo a prime or finite field; 3. Row/column elimination to triangular or sparse form
Method IDs	M-LA-14; M-ANT-01; M-LA-01; M-ANT-16; M-ANT-06; M-LA-03; M-LA-04; M-ANT-02
Method titles	Parity or zero-pattern block determinant split; Reduction modulo a prime or finite field; Row/column elimination to triangular or sparse form; Polynomial identity by coefficient comparison; Degree versus number of roots; Block decomposition and Schur-style bookkeeping; Rank-nullity and dimension-count obstruction; Chinese remainder and residue construction
Theory dependencies	Determinant evaluation by row/column operations and triangular forms; Graph encoding, cycles, and bipartite representations; Rank-nullity, image/kernel, and rank of factorizations
Cue extraction	prior broad method tags: determinant / block manipulation, rank-nullity / dimension count, polynomial identity / factorization | statement triggers: matrix word; det/rank/eigen trigger; integer/prime/mod trigger | audit closures: AUD-RA-06; AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2008-D2-P6 (PDF p. 128) *counting / enumeration*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Cauchy-Schwarz, Holder, and norm comparison; 3. Double counting ordered or weighted objects
Method IDs	M-CMB-01; M-RA-10; M-CMB-02; M-LA-04; M-ANT-06; M-ANT-16; M-ANT-05; M-UNI-03
Method titles	Pigeonhole threshold and collision forcing; Cauchy-Schwarz, Holder, and norm comparison; Double counting ordered or weighted objects; Rank-nullity and dimension-count obstruction; Degree versus number of roots; Polynomial identity by coefficient comparison; Factorization and unique factorization in Z[x]; Extremal construction and equality-case engineering
Theory dependencies	Hilbert-space equidistant sets and orthonormal translations; Gram matrices and rational linear dependence; Rank-nullity, image/kernel, and rank of factorizations
Cue extraction	prior broad method tags: Cauchy-Schwarz / norm inequality, rank-nullity / dimension count, polynomial identity / factorization | statement triggers: counting/permutation trigger; geometry/metric trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-PC-13; AUD-CMB-03; AUD-GEO-09
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.16 2009

18.16.1 Day 1

2009-D1-P1 (PDF p. 129) *proof / reasoning*

Primary domain	Real analysis / calculus
Core method sequence	1. Auxiliary function with derivative sign; 2. Pigeonhole threshold and collision forcing; 3. Double counting ordered or weighted objects
Method IDs	M-RA-01; M-CMB-01; M-CMB-02; M-UNI-03; M-FE-02; M-UNI-08; M-RA-06; M-RA-07
Method titles	Auxiliary function with derivative sign; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Extremal construction and equality-case engineering; Monotonicity forced by functional equation; Sign, zero, and boundary case splitting; Riemann sum and sum-integral conversion; Split into main range and tail/error range
Theory dependencies	Explicit construction and counterexample design; Intermediate value, connectedness, density, and continuity transfer; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: monotonicity argument, double counting / pigeonhole, construction / counterexample | statement triggers: continuous/differentiable trigger; functional-equation trigger | audit closures: AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08; AUD-CMB-03
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2009-D1-P2 (PDF p. 129)

proof of structural property

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Invariant extraction and invariant comparison; 2. Special substitution and parameter freezing; 3. Complete a matrix equation to an inverse identity
Method IDs	M-UNI-09; M-UNI-07; M-LA-02; M-LA-06; M-LA-04; M-LA-13; M-UNI-05; M-CMB-04
Method titles	Invariant extraction and invariant comparison; Special substitution and parameter freezing; Complete a matrix equation to an inverse identity; Spectral/minimal-polynomial reduction; Rank-nullity and dimension-count obstruction; Determinant through eigenvalue or root product; Induction with strengthened hypothesis; Cycle extraction and alternating-cycle method
Theory dependencies	Matrix products, inverses, and one-sided inverse identities; Rank-nullity, image/kernel, and rank of factorizations; Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: matrix word | audit closures: AUD-CMB-04
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2009-D1-P3 (PDF p. 129)

classification / find all

Primary domain	Combinatorics / discrete structures
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Double counting ordered or weighted objects; 3. Degree versus number of roots
Method IDs	M-CMB-01; M-CMB-02; M-ANT-06; M-ANT-16; M-CMB-03; M-CMB-04; M-LA-04; M-RA-16
Method titles	Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Degree versus number of roots; Polynomial identity by coefficient comparison; Graph encoding of a relation; Cycle extraction and alternating-cycle method; Rank-nullity and dimension-count obstruction; Density, Baire/category, and pathological construction
Theory dependencies	Double counting and degree-sum methods; Graph encoding, cycles, and bipartite representations; Pigeonhole principle and extremal element selection
Cue extraction	prior broad method tags: rank-nullity / dimension count, polynomial identity / factorization, double counting / pigeonhole | statement triggers: det/rank/eigen trigger; counting/permutation trigger | audit closures: AUD-RA-06; AUD-RA-07; AUD-RA-08; AUD-PC-01; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2009-D1-P4 (PDF p. 130) *inequality / sharp bound*

Primary domain	Real analysis / calculus
Core method sequence	1. Fejer kernel positivity; 2. Convexity, Jensen, tangent, and chord inequalities; 3. Polynomial identity by coefficient comparison
Method IDs	M-CX-04; M-RA-09; M-ANT-16; M-ANT-06; M-RA-08; M-LA-04; M-RA-10; M-ANT-05
Method titles	Fejer kernel positivity; Convexity, Jensen, tangent, and chord inequalities; Polynomial identity by coefficient comparison; Degree versus number of roots; Compactness plus extreme value theorem; Rank-nullity and dimension-count obstruction; Cauchy-Schwarz, Holder, and norm comparison; Factorization and unique factorization in Z[x]
Theory dependencies	Cauchy formula and coefficient extraction on the unit circle; Convexity, concavity, Jensen, and tangent/chord inequalities; Fejer kernels and positive trigonometric kernels
Cue extraction	prior broad method tags: induction, compactness / extremal point, convexity / Jensen / tangent | statement triggers: continuous/differentiable trigger; polynomial/root trigger | audit closures: AUD-PC-07; AUD-RA-06; AUD-RA-07; AUD-RA-08; AUD-PC-01; AUD-PC-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2009-D1-P5 (PDF p. 131) *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Barycentric coordinates and volume weights; 2. Graph encoding of a relation; 3. Cycle extraction and alternating-cycle method
Method IDs	M-GEO-04; M-CMB-03; M-CMB-04; M-UNI-05; M-LA-01; M-LA-03; M-LA-04; M-RA-01
Method titles	Barycentric coordinates and volume weights; Graph encoding of a relation; Cycle extraction and alternating-cycle method; Induction with strengthened hypothesis; Row/column elimination to triangular or sparse form; Block decomposition and Schur-style bookkeeping; Rank-nullity and dimension-count obstruction; Auxiliary function with derivative sign
Theory dependencies	Simplex, barycentric coordinates, and circumcenters; Rank-nullity, image/kernel, and rank of factorizations; Block matrices, Schur complements, and determinant identities
Cue extraction	prior broad method tags: induction, auxiliary function / substitution, determinant / block manipulation | statement triggers: integer/prime/mod trigger; graph/color/set trigger; geometry/metric trigger | audit closures: AUD-RA-08; AUD-CMB-08; AUD-GEO-03; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.16.2 Day 2

2009-D2-P1 (PDF p. 133) *proof / reasoning*

Primary domain	Geometry / convexity / metric
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Factorization and unique factorization in Z[x]
Method IDs	M-ANT-16; M-ANT-06; M-ANT-05; M-GEO-03; M-RA-10; M-RA-09; M-GEO-01; M-UNI-07
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Factorization and unique factorization in Z[x]; Squared-distance and inner-product identities; Cauchy-Schwarz, Holder, and norm comparison; Convexity, Jensen, tangent, and chord inequalities; Coordinate normalization and algebraization; Special substitution and parameter freezing
Theory dependencies	Area, volume, projection, and ellipsoid computation; Polynomial identity and more-roots-than-degree principle; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: polynomial identity / factorization | statement triggers: geometry/metric trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-GEO-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2009-D2-P2 (PDF p. 133) *inequality / sharp bound*

Primary domain	Functional equations / dynamics
Core method sequence	1. Auxiliary function with derivative sign; 2. Special substitution and parameter freezing; 3. Integrating factor and ODE comparison
Method IDs	M-RA-01; M-UNI-07; M-RA-02; M-RA-03; M-RA-09; M-FE-02; M-UNI-03; M-RA-04
Method titles	Auxiliary function with derivative sign; Special substitution and parameter freezing; Integrating factor and ODE comparison; Rolle chain and divided-difference localization; Convexity, Jensen, tangent, and chord inequalities; Monotonicity forced by functional equation; Extremal construction and equality-case engineering; Taylor expansion and local asymptotic bookkeeping
Theory dependencies	Auxiliary functions, integrating factors, and differential inequalities; Convexity, concavity, Jensen, and tangent/chord inequalities; Rolle, mean-value theorem, and divided differences
Cue extraction	prior broad method tags: auxiliary function / substitution | statement triggers: continuous/differentiable trigger; functional-equation trigger | audit closures: AUD-RA-01; AUD-RA-06; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2009-D2-P3 (PDF p. 134) *existence / construction*

Primary domain	Algebra / number theory
Core method sequence	1. Trace cyclicity and commutator annihilation; 2. Nilpotency from zero spectrum plus finite dimension; 3. Degree versus number of roots
Method IDs	M-LA-07; M-LA-12; M-ANT-06; M-ANT-16; M-LA-06; M-ANT-05; M-UNI-05; M-ANT-01
Method titles	Trace cyclicity and commutator annihilation; Nilpotency from zero spectrum plus finite dimension; Degree versus number of roots; Polynomial identity by coefficient comparison; Spectral/minimal-polynomial reduction; Factorization and unique factorization in Z[x]; Induction with strengthened hypothesis; Reduction modulo a prime or finite field
Theory dependencies	Commutators and spectral obstructions; Trace cyclicity, trace pairing, and zero-trace subspaces; Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton
Cue extraction	prior broad method tags: induction, spectral decomposition / eigenvalues, trace trick / commutator | statement triggers: matrix word; integer/prime/mod trigger | audit closures: AUD-LA-02; AUD-LA-03; AUD-LA-08; AUD-LA-13; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2009-D2-P4 (PDF p. 134) *counting / enumeration*

Primary domain	Complex analysis / polynomials
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Reduction modulo a prime or finite field; 3. Degree versus number of roots
Method IDs	M-ANT-16; M-ANT-01; M-ANT-06; M-CMB-03; M-CMB-04; M-ANT-02; M-ANT-05; M-ANT-03
Method titles	Polynomial identity by coefficient comparison; Reduction modulo a prime or finite field; Degree versus number of roots; Graph encoding of a relation; Cycle extraction and alternating-cycle method; Chinese remainder and residue construction; Factorization and unique factorization in Z[x]; GCD, valuation, and prime-power bookkeeping
Theory dependencies	Finite fields, field characteristic, and Fermat/Frobenius tools; Polynomial identity and more-roots-than-degree principle; Permutations, inversions, and permutation statistics
Cue extraction	prior broad method tags: polynomial identity / factorization, modular / finite-field reduction, graph encoding / cycles | statement triggers: polynomial/root trigger; integer/prime/mod trigger; group/ring/field trigger; counting/permutation trigger | audit closures: AUD-PC-01; AUD-PC-06; AUD-PC-08; AUD-PC-13; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2009-D2-P5 (PDF p. 135) *inequality / sharp bound*

Primary domain	Algebra / number theory
Core method sequence	1. Degree versus number of roots; 2. Polynomial identity by coefficient comparison; 3. Extremal construction and equality-case engineering
Method IDs	M-ANT-06; M-ANT-16; M-UNI-03; M-LA-01; M-LA-03; M-LA-04; M-ANT-05; M-UNI-08
Method titles	Degree versus number of roots; Polynomial identity by coefficient comparison; Extremal construction and equality-case engineering; Row/column elimination to triangular or sparse form; Block decomposition and Schur-style bookkeeping; Rank-nullity and dimension-count obstruction; Factorization and unique factorization in Z[x]; Sign, zero, and boundary case splitting
Theory dependencies	Rank-nullity, image/kernel, and rank of factorizations; Pigeonhole principle and extremal element selection; Block matrices, Schur complements, and determinant identities
Cue extraction	prior broad method tags: determinant / block manipulation, rank-nullity / dimension count, polynomial identity / factorization | statement triggers: matrix word; integer/prime/mod trigger; group/ring/field trigger | audit closures: AUD-LA-13; AUD-RA-08; AUD-PC-01; AUD-PC-13; AUD-ANT-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.17 2010

18.17.1 Day 1

2010-D1-P1 (PDF p. 137) *inequality / sharp bound*

Primary domain	Functional equations / dynamics
Core method sequence	1. Auxiliary function with derivative sign; 2. Monotonicity forced by functional equation; 3. Rolle chain and divided-difference localization
Method IDs	M-RA-01; M-FE-02; M-RA-03; M-UNI-07; M-RA-02; M-RA-10; M-RA-09; M-UNI-03
Method titles	Auxiliary function with derivative sign; Monotonicity forced by functional equation; Rolle chain and divided-difference localization; Special substitution and parameter freezing; Integrating factor and ODE comparison; Cauchy-Schwarz, Holder, and norm comparison; Convexity, Jensen, tangent, and chord inequalities; Extremal construction and equality-case engineering
Theory dependencies	Rolle, mean-value theorem, and divided differences; Convexity, concavity, Jensen, and tangent/chord inequalities; Taylor expansion, local asymptotics, and little-o bookkeeping
Cue extraction	prior broad method tags: auxiliary function / substitution, Rolle / mean-value theorem, monotonicity argument | statement triggers: wording-specific | audit closures: AUD-RA-06; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2010-D1-P2 (PDF p. 137) *exact computation*

Primary domain	Real analysis / calculus
Core method sequence	1. Invariant extraction and invariant comparison; 2. Special substitution and parameter freezing; 3. Riemann sum and sum-integral conversion
Method IDs	M-UNI-09; M-UNI-07; M-RA-06; M-RA-07; M-RA-05; M-RA-09; M-RA-11; M-RA-01
Method titles	Invariant extraction and invariant comparison; Special substitution and parameter freezing; Riemann sum and sum-integral conversion; Split into main range and tail/error range; L-Hospital and ratio/growth comparison; Convexity, Jensen, tangent, and chord inequalities; Integration by parts, Fubini, and weighted identities; Auxiliary function with derivative sign
Theory dependencies	Integral inequalities, integration by parts, and weighted estimates; Riemann sums and sum-integral asymptotics; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: integral/limit trigger | audit closures: AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2010-D1-P3 (PDF p. 139) *limit / asymptotic evaluation*

Primary domain	Functional equations / dynamics
Core method sequence	1. Induction with strengthened hypothesis; 2. Riemann sum and sum-integral conversion; 3. Split into main range and tail/error range
Method IDs	M-UNI-05; M-RA-06; M-RA-07; M-RA-05; M-RA-09; M-FE-02; M-UNI-07; M-FE-01
Method titles	Induction with strengthened hypothesis; Riemann sum and sum-integral conversion; Split into main range and tail/error range; L-Hospital and ratio/growth comparison; Convexity, Jensen, tangent, and chord inequalities; Monotonicity forced by functional equation; Special substitution and parameter freezing; Iterates, fixed points, and cluster-point trapping
Theory dependencies	Riemann sums and sum-integral asymptotics; Convexity, concavity, Jensen, and tangent/chord inequalities; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: induction | statement triggers: integral/limit trigger | audit closures: AUD-RA-03; AUD-RA-05; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2010-D1-P4 (PDF p. 139) *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Chinese remainder and residue construction; 3. Pigeonhole threshold and collision forcing
Method IDs	M-ANT-01; M-ANT-02; M-CMB-01; M-CMB-02; M-UNI-03; M-UNI-08; M-ANT-03; M-RA-09
Method titles	Reduction modulo a prime or finite field; Chinese remainder and residue construction; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Extremal construction and equality-case engineering; Sign, zero, and boundary case splitting; GCD, valuation, and prime-power bookkeeping; Convexity, Jensen, tangent, and chord inequalities
Theory dependencies	Modular arithmetic and divisibility; Finite fields, field characteristic, and Fermat/Frobenius tools; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: modular / finite-field reduction, double counting / pigeonhole, construction / counterexample | statement triggers: integer/prime/mod trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-ANT-02; AUD-CMB-03
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2010-D1-P5 (PDF p. 139) *inequality / sharp bound*

Primary domain	Algebra / number theory
Core method sequence	1. Row/column elimination to triangular or sparse form; 2. Block decomposition and Schur-style bookkeeping; 3. Cauchy-Schwarz, Holder, and norm comparison
Method IDs	M-LA-01; M-LA-03; M-RA-10; M-ANT-01; M-ANT-03; M-RA-03; M-RA-01; M-RA-09
Method titles	Row/column elimination to triangular or sparse form; Block decomposition and Schur-style bookkeeping; Cauchy-Schwarz, Holder, and norm comparison; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Rolle chain and divided-difference localization; Auxiliary function with derivative sign; Convexity, Jensen, tangent, and chord inequalities
Theory dependencies	Block matrices, Schur complements, and determinant identities; Determinant evaluation by row/column operations and triangular forms; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: Cauchy-Schwarz / norm inequality, determinant / block manipulation | statement triggers: continuous/differentiable trigger; integer/prime/mod trigger | audit closures: AUD-RA-06; AUD-RA-08; AUD-CMB-05; AUD-CMB-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.17.2 Day 2

2010-D2-P1 (PDF p. 141) *proof / reasoning*

Primary domain	Real analysis / calculus
Core method sequence	1. Auxiliary function with derivative sign; 2. Monotonicity forced by functional equation; 3. Extremal construction and equality-case engineering
Method IDs	M-RA-01; M-FE-02; M-UNI-03; M-UNI-08; M-RA-06; M-RA-07; M-RA-09; M-RA-11
Method titles	Auxiliary function with derivative sign; Monotonicity forced by functional equation; Extremal construction and equality-case engineering; Sign, zero, and boundary case splitting; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Convexity, Jensen, tangent, and chord inequalities; Integration by parts, Fubini, and weighted identities
Theory dependencies	Integral inequalities, integration by parts, and weighted estimates; Riemann sums and sum-integral asymptotics; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: monotonicity argument, construction / counterexample | statement triggers: wording-specific | audit closures: AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2010-D2-P2 (PDF p. 141) *inequality / sharp bound*

Primary domain	Real analysis / calculus
Core method sequence	1. Induction with strengthened hypothesis; 2. Convexity, Jensen, tangent, and chord inequalities; 3. Recurrence transformed to monotone quotient or telescoping form
Method IDs	M-UNI-05; M-RA-09; M-RA-15; M-RA-11; M-RA-01; M-RA-08; M-UNI-03; M-CMB-08
Method titles	Induction with strengthened hypothesis; Convexity, Jensen, tangent, and chord inequalities; Recurrence transformed to monotone quotient or telescoping form; Integration by parts, Fubini, and weighted identities; Auxiliary function with derivative sign; Compactness plus extreme value theorem; Extremal construction and equality-case engineering; Extremal set or minimal counterexample induction
Theory dependencies	Integral inequalities, integration by parts, and weighted estimates; Convexity, concavity, Jensen, and tangent/chord inequalities; Sequence recurrences, monotone bounds, and generating functions
Cue extraction	prior broad method tags: induction | statement triggers: wording-specific | audit closures: AUD-RA-08; AUD-PC-13; AUD-CMB-05; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2010-D2-P3 (PDF p. 142) *existence / construction*

Primary domain	Algebra / number theory
Core method sequence	1. Auxiliary function with derivative sign; 2. Integrating factor and ODE comparison; 3. Special substitution and parameter freezing
Method IDs	M-RA-01; M-RA-02; M-UNI-07; M-ANT-10; M-ANT-11; M-CMB-07; M-RA-09; M-ANT-01
Method titles	Auxiliary function with derivative sign; Integrating factor and ODE comparison; Special substitution and parameter freezing; Group action, orbit, and generator method; Commutator and center manipulation in groups/rings; Involution, parity, and sign-reversing cancellation; Convexity, Jensen, tangent, and chord inequalities; Reduction modulo a prime or finite field
Theory dependencies	Group basics: orders, cyclic subgroups, and generators; Integral inequalities, integration by parts, and weighted estimates; Rings, idempotents, nil rings, and noncommutativity
Cue extraction	prior broad method tags: auxiliary function / substitution | statement triggers: group/ring/field trigger; counting/permutation trigger | audit closures: AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2010-D2-P4 (PDF p. 143) *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Induction with strengthened hypothesis; 3. Row/column elimination to triangular or sparse form
Method IDs	M-ANT-01; M-UNI-05; M-LA-01; M-LA-03; M-LA-04; M-ANT-02; M-ANT-03; M-LA-06
Method titles	Reduction modulo a prime or finite field; Induction with strengthened hypothesis; Row/column elimination to triangular or sparse form; Block decomposition and Schur-style bookkeeping; Rank-nullity and dimension-count obstruction; Chinese remainder and residue construction; GCD, valuation, and prime-power bookkeeping; Spectral/minimal-polynomial reduction
Theory dependencies	Rank-nullity, image/kernel, and rank of factorizations; Modular arithmetic and divisibility; Block matrices, Schur complements, and determinant identities
Cue extraction	prior broad method tags: induction, determinant / block manipulation, rank-nullity / dimension count | statement triggers: matrix word; integer/prime/mod trigger; group/ring/field trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2010-D2-P5 (PDF p. 144) *proof of structural property*

Primary domain	Complex analysis / polynomials
Core method sequence	1. Reduction modulo a prime or finite field; 2. Polynomial identity by coefficient comparison; 3. Degree versus number of roots
Method IDs	M-ANT-01; M-ANT-16; M-ANT-06; M-ANT-02; M-ANT-05; M-ANT-03; M-RA-06; M-RA-07
Method titles	Reduction modulo a prime or finite field; Polynomial identity by coefficient comparison; Degree versus number of roots; Chinese remainder and residue construction; Factorization and unique factorization in Z[x]; GCD, valuation, and prime-power bookkeeping; Riemann sum and sum-integral conversion; Split into main range and tail/error range
Theory dependencies	Polynomial identity and more-roots-than-degree principle; Modular arithmetic and divisibility; Finite fields, field characteristic, and Fermat/Frobenius tools
Cue extraction	prior broad method tags: polynomial identity / factorization, modular / finite-field reduction | statement triggers: integer/prime/mod trigger; functional-equation trigger | audit closures: AUD-RA-02; AUD-RA-03; AUD-RA-05; AUD-RA-06; AUD-PC-01
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.18 2011

18.18.1 Day 1

2011-D1-P1 (PDF p. 145) *existence / construction*

Primary domain	Functional equations / dynamics
Core method sequence	1. Compactness or extremal-point selection; 2. Compactness plus extreme value theorem; 3. Auxiliary function with derivative sign
Method IDs	M-UNI-06; M-RA-08; M-RA-01; M-RA-03; M-RA-06; M-RA-07; M-RA-09; M-FE-02
Method titles	Compactness or extremal-point selection; Compactness plus extreme value theorem; Auxiliary function with derivative sign; Rolle chain and divided-difference localization; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Convexity, Jensen, tangent, and chord inequalities; Monotonicity forced by functional equation
Theory dependencies	Compactness, extreme values, and closed-bounded reduction; Integral inequalities, integration by parts, and weighted estimates; Intermediate value, connectedness, density, and continuity transfer
Cue extraction	prior broad method tags: compactness / extremal point | statement triggers: continuous/differentiable trigger; geometry/metric trigger; functional-equation trigger | audit closures: AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-07; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2011-D1-P2 (PDF p. 145) *nonexistence / counterexample*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Trace cyclicity and commutator annihilation; 2. Nilpotency from zero spectrum plus finite dimension; 3. Polynomial identity by coefficient comparison
Method IDs	M-LA-07; M-LA-12; M-ANT-16; M-LA-06; M-UNI-03; M-ANT-06; M-ANT-05; M-UNI-08
Method titles	Trace cyclicity and commutator annihilation; Nilpotency from zero spectrum plus finite dimension; Polynomial identity by coefficient comparison; Spectral/minimal-polynomial reduction; Extremal construction and equality-case engineering; Degree versus number of roots; Factorization and unique factorization in Z[x]; Sign, zero, and boundary case splitting
Theory dependencies	Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; Trace cyclicity, trace pairing, and zero-trace subspaces; Polynomial identity and more-roots-than-degree principle
Cue extraction	prior broad method tags: spectral decomposition / eigenvalues, trace trick / commutator, polynomial identity / factorization | statement triggers: matrix word; det/rank/eigen trigger | audit closures: AUD-LA-03; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-ANT-06
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2011-D1-P3 (PDF p. 145) *inequality / sharp bound*

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Polynomial identity by coefficient comparison; 3. Degree versus number of roots
Method IDs	M-ANT-01; M-ANT-16; M-ANT-06; M-ANT-02; M-ANT-05; M-ANT-03; M-RA-09; M-UNI-03
Method titles	Reduction modulo a prime or finite field; Polynomial identity by coefficient comparison; Degree versus number of roots; Chinese remainder and residue construction; Factorization and unique factorization in Z[x]; GCD, valuation, and prime-power bookkeeping; Convexity, Jensen, tangent, and chord inequalities; Extremal construction and equality-case engineering
Theory dependencies	Modular arithmetic and divisibility; Polynomial identity and more-roots-than-degree principle; Finite fields, field characteristic, and Fermat/Frobenius tools
Cue extraction	prior broad method tags: polynomial identity / factorization, modular / finite-field reduction | statement triggers: polynomial/root trigger; integer/prime/mod trigger; functional-equation trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-11; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2011-D1-P4 (PDF p. 146) *counting / enumeration*

Primary domain	Combinatorics / discrete structures
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Double counting ordered or weighted objects; 3. Monotonicity forced by functional equation
Method IDs	M-CMB-01; M-CMB-02; M-FE-02; M-CMB-13; M-RA-01; M-CMB-10; M-CMB-03; M-RA-09
Method titles	Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Monotonicity forced by functional equation; Indicator and expectation method; Auxiliary function with derivative sign; Explicit bijection and encoding-decoding proof; Graph encoding of a relation; Convexity, Jensen, tangent, and chord inequalities
Theory dependencies	Pigeonhole principle and extremal element selection; Double counting and degree-sum methods; Graph encoding, cycles, and bipartite representations
Cue extraction	prior broad method tags: monotonicity argument, double counting / pigeonhole, probabilistic / expectation | statement triggers: counting/permutation trigger; functional-equation trigger | audit closures: AUD-RA-08; AUD-RA-10; AUD-CMB-03; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2011-D1-P5 (PDF p. 146) *existence / construction*

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Degree versus number of roots; 3. Polynomial identity by coefficient comparison
Method IDs	M-ANT-01; M-ANT-06; M-ANT-16; M-LA-04; M-ANT-02; M-ANT-05; M-UNI-05; M-ANT-03
Method titles	Reduction modulo a prime or finite field; Degree versus number of roots; Polynomial identity by coefficient comparison; Rank-nullity and dimension-count obstruction; Chinese remainder and residue construction; Factorization and unique factorization in Z[x]; Induction with strengthened hypothesis; GCD, valuation, and prime-power bookkeeping
Theory dependencies	Rank-nullity, image/kernel, and rank of factorizations; Polynomial identity and more-roots-than-degree principle; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: induction, rank-nullity / dimension count, polynomial identity / factorization | statement triggers: integer/prime/mod trigger; group/ring/field trigger | audit closures: AUD-RA-06; AUD-RA-08; AUD-PC-01; AUD-PC-13; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.18.2 Day 2

2011-D2-P1 (PDF p. 147) *limit / asymptotic evaluation*

Primary domain	Real analysis / calculus
Core method sequence	1. Auxiliary function with derivative sign; 2. Monotonicity forced by functional equation; 3. Induction with strengthened hypothesis
Method IDs	M-RA-01; M-FE-02; M-UNI-05; M-RA-06; M-RA-07; M-RA-09; M-RA-15; M-RA-11
Method titles	Auxiliary function with derivative sign; Monotonicity forced by functional equation; Induction with strengthened hypothesis; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Convexity, Jensen, tangent, and chord inequalities; Recurrence transformed to monotone quotient or telescoping form; Integration by parts, Fubini, and weighted identities
Theory dependencies	Integral inequalities, integration by parts, and weighted estimates; Riemann sums and sum-integral asymptotics; Sequence recurrences, monotone bounds, and generating functions
Cue extraction	prior broad method tags: induction, monotonicity argument | statement triggers: integral/limit trigger | audit closures: AUD-RA-03; AUD-RA-04; AUD-RA-05; AUD-RA-08; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2011-D2-P2 (PDF p. 147) *inequality / sharp bound*

Primary domain	Algebra / number theory
Core method sequence	1. Approximation by polynomials or kernels; 2. Degree versus number of roots; 3. Polynomial identity by coefficient comparison
Method IDs	M-RA-17; M-ANT-06; M-ANT-16; M-UNI-03; M-CMB-03; M-CMB-01; M-CMB-02; M-LA-03
Method titles	Approximation by polynomials or kernels; Degree versus number of roots; Polynomial identity by coefficient comparison; Extremal construction and equality-case engineering; Graph encoding of a relation; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Block decomposition and Schur-style bookkeeping
Theory dependencies	Polynomial identity and more-roots-than-degree principle; Integral inequalities, integration by parts, and weighted estimates; Pigeonhole principle and extremal element selection
Cue extraction	prior broad method tags: determinant / block manipulation, polynomial identity / factorization, double counting / pigeonhole | statement triggers: group/ring/field trigger; counting/permutation trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-PC-13; AUD-CMB-03; AUD-CMB-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2011-D2-P3 (PDF p. 148) *exact computation*

Primary domain	Functional equations / dynamics
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Convexity, Jensen, tangent, and chord inequalities
Method IDs	M-UNI-07; M-UNI-09; M-RA-09; M-RA-11; M-FE-02; M-UNI-02; M-GEO-06; M-LA-03
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Convexity, Jensen, tangent, and chord inequalities; Integration by parts, Fubini, and weighted identities; Monotonicity forced by functional equation; Small cases, equality cases, and example probing; Convex support and John-ellipse perturbation; Block decomposition and Schur-style bookkeeping
Theory dependencies	Integral inequalities, integration by parts, and weighted estimates; Convexity, concavity, Jensen, and tangent/chord inequalities; Series convergence, rearrangements, and alternating/Dirichlet tools
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: det/rank/eigen trigger | audit closures: AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2011-D2-P4 (PDF p. 149) *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Induction with strengthened hypothesis
Method IDs	M-ANT-16; M-ANT-06; M-UNI-05; M-LA-04; M-ANT-05; M-RA-01; M-RA-02; M-UNI-07
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Induction with strengthened hypothesis; Rank-nullity and dimension-count obstruction; Factorization and unique factorization in Z[x]; Auxiliary function with derivative sign; Integrating factor and ODE comparison; Special substitution and parameter freezing
Theory dependencies	Polynomial identity and more-roots-than-degree principle; Modular arithmetic and divisibility; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: induction, auxiliary function / substitution, rank-nullity / dimension count | statement triggers: polynomial/root trigger; integer/prime/mod trigger; functional-equation trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2011-D2-P5 (PDF p. 150) *proof of structural property*

Primary domain	Geometry / convexity / metric
Core method sequence	1. Convexity, Jensen, tangent, and chord inequalities; 2. Graph encoding of a relation; 3. Cycle extraction and alternating-cycle method
Method IDs	M-RA-09; M-CMB-03; M-CMB-04; M-LA-04; M-UNI-03; M-UNI-08; M-GEO-11; M-GEO-01
Method titles	Convexity, Jensen, tangent, and chord inequalities; Graph encoding of a relation; Cycle extraction and alternating-cycle method; Rank-nullity and dimension-count obstruction; Extremal construction and equality-case engineering; Sign, zero, and boundary case splitting; Volume, projection, and ellipsoid computation; Coordinate normalization and algebraization
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Graph encoding, cycles, and bipartite representations; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: convexity / Jensen / tangent, rank-nullity / dimension count, graph encoding / cycles | statement triggers: graph/color/set trigger | audit closures: AUD-RA-08; AUD-GEO-02; AUD-GEO-10
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.19 2012

18.19.1 Day 1

2012-D1-P1 (PDF p. 151) *counting / enumeration*

Primary domain	Algebra / number theory
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Double counting ordered or weighted objects; 3. Reduction modulo a prime or finite field
Method IDs	M-CMB-01; M-CMB-02; M-ANT-01; M-ANT-03; M-RA-06; M-RA-07; M-CMB-10; M-RA-11
Method titles	Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Explicit bijection and encoding-decoding proof; Integration by parts, Fubini, and weighted identities
Theory dependencies	Pigeonhole principle and extremal element selection; Double counting and degree-sum methods; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: double counting / pigeonhole | statement triggers: integral/limit trigger; integer/prime/mod trigger; counting/permutation trigger | audit closures: AUD-RA-08; AUD-CMB-03; AUD-CMB-07; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2012-D1-P2 (PDF p. 151)

proof / reasoning

Primary domain	Algebra / number theory
Core method sequence	1. Rank-nullity and dimension-count obstruction; 2. Reduction modulo a prime or finite field; 3. GCD, valuation, and prime-power bookkeeping
Method IDs	M-LA-04; M-ANT-01; M-ANT-03; M-LA-06; M-UNI-07; M-UNI-09; M-CMB-08; M-LA-01
Method titles	Rank-nullity and dimension-count obstruction; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Spectral/minimal-polynomial reduction; Special substitution and parameter freezing; Invariant extraction and invariant comparison; Extremal set or minimal counterexample induction; Row/column elimination to triangular or sparse form
Theory dependencies	Rank-nullity, image/kernel, and rank of factorizations; Determinant evaluation by row/column operations and triangular forms; Block matrices, Schur complements, and determinant identities
Cue extraction	prior broad method tags: rank-nullity / dimension count | statement triggers: matrix word; det/rank/eigen trigger; integer/prime/mod trigger | audit closures: AUD-CMB-05; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2012-D1-P3 (PDF p. 151)

proof / reasoning

Primary domain	Algebra / number theory
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Group action, orbit, and generator method
Method IDs	M-UNI-07; M-UNI-09; M-ANT-10; M-ANT-01; M-ANT-03; M-ANT-11; M-CMB-07; M-RA-09
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Group action, orbit, and generator method; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Commutator and center manipulation in groups/rings; Involution, parity, and sign-reversing cancellation; Convexity, Jensen, tangent, and chord inequalities
Theory dependencies	Group basics: orders, cyclic subgroups, and generators; Rings, idempotents, nil rings, and noncommutativity; Automorphisms and semidirect products
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: integer/prime/mod trigger; group/ring/field trigger; counting/permutation trigger | audit closures: AUD-RA-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2012-D1-P4 (PDF p. 151)

inequality / sharp bound

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Auxiliary function with derivative sign
Method IDs	M-UNI-07; M-UNI-09; M-RA-01; M-RA-03; M-RA-09; M-RA-08; M-UNI-03; M-LA-04
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Auxiliary function with derivative sign; Rolle chain and divided-difference localization; Convexity, Jensen, tangent, and chord inequalities; Compactness plus extreme value theorem; Extremal construction and equality-case engineering; Rank-nullity and dimension-count obstruction
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Rolle, mean-value theorem, and divided differences; Taylor expansion, local asymptotics, and little-o bookkeeping
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: continuous/differentiable trigger; functional-equation trigger | audit closures: AUD-RA-01; AUD-RA-06; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2012-D1-P5 (PDF p. 151) *proof of structural property*

Primary domain	Complex analysis / polynomials
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Factorization and unique factorization in Z[x]
Method IDs	M-ANT-16; M-ANT-06; M-ANT-05; M-ANT-01; M-ANT-10; M-ANT-11; M-ANT-03; M-CX-01
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Factorization and unique factorization in Z[x]; Reduction modulo a prime or finite field; Group action, orbit, and generator method; Commutator and center manipulation in groups/rings; GCD, valuation, and prime-power bookkeeping; Cauchy coefficient formula and boundary maximum
Theory dependencies	Polynomial identity and more-roots-than-degree principle; Rings, idempotents, nil rings, and noncommutativity; Factorization, irreducibility, and Gauss-style integer polynomial arguments
Cue extraction	prior broad method tags: polynomial identity / factorization | statement triggers: polynomial/root trigger; integer/prime/mod trigger; group/ring/field trigger | audit closures: AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.19.2 Day 2

2012-D2-P1 (PDF p. 152) *proof / reasoning*

Primary domain	Complex analysis / polynomials
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Factorization and unique factorization in Z[x]
Method IDs	M-ANT-16; M-ANT-06; M-ANT-05; M-ANT-01; M-CX-01; M-UNI-07; M-UNI-09; M-FE-03
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Factorization and unique factorization in Z[x]; Reduction modulo a prime or finite field; Cauchy coefficient formula and boundary maximum; Special substitution and parameter freezing; Invariant extraction and invariant comparison; Composition, inversion, and conjugacy
Theory dependencies	Polynomial identity and more-roots-than-degree principle; Modular arithmetic and divisibility; Vieta, logarithmic derivatives, and real-root inequalities
Cue extraction	prior broad method tags: polynomial identity / factorization | statement triggers: polynomial/root trigger; integer/prime/mod trigger; functional-equation trigger | audit closures: AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2012-D2-P2 (PDF p. 152) *proof of structural property*

Primary domain	Real analysis / calculus
Core method sequence	1. Induction with strengthened hypothesis; 2. Riemann sum and sum-integral conversion; 3. Split into main range and tail/error range
Method IDs	M-UNI-05; M-RA-06; M-RA-07; M-RA-01; M-RA-09; M-RA-15; M-CMB-06; M-RA-11
Method titles	Induction with strengthened hypothesis; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Auxiliary function with derivative sign; Convexity, Jensen, tangent, and chord inequalities; Recurrence transformed to monotone quotient or telescoping form; Generating function encoding; Integration by parts, Fubini, and weighted identities
Theory dependencies	Integral inequalities, integration by parts, and weighted estimates; Convexity, concavity, Jensen, and tangent/chord inequalities; Series convergence, rearrangements, and alternating/Dirichlet tools
Cue extraction	prior broad method tags: induction | statement triggers: det/rank/eigen trigger; integral/limit trigger | audit closures: AUD-RA-08; AUD-PC-13; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2012-D2-P3 (PDF p. 152) *proof / reasoning*

Primary domain	Algebra / number theory
Core method sequence	1. Finite-field function representative and permutation argument; 2. Special substitution and parameter freezing; 3. Invariant extraction and invariant comparison
Method IDs	M-ANT-09; M-UNI-07; M-UNI-09; M-ANT-01; M-ANT-03; M-ANT-06; M-ANT-16; M-ANT-10
Method titles	Finite-field function representative and permutation argument; Special substitution and parameter freezing; Invariant extraction and invariant comparison; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Degree versus number of roots; Polynomial identity by coefficient comparison; Group action, orbit, and generator method
Theory dependencies	Modular arithmetic and divisibility; Graph encoding, cycles, and bipartite representations; Coloring, partitions, and parity
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: integer/prime/mod trigger | audit closures: AUD-ANT-02; AUD-PC-01
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2012-D2-P4 (PDF p. 152) *classification / find all*

Primary domain	Algebra / number theory
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Reduction modulo a prime or finite field
Method IDs	M-UNI-07; M-UNI-09; M-ANT-01; M-ANT-03; M-RA-09; M-UNI-08; M-ANT-02; M-ANT-10
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Convexity, Jensen, tangent, and chord inequalities; Sign, zero, and boundary case splitting; Chinese remainder and residue construction; Group action, orbit, and generator method
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Modular arithmetic and divisibility; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: integer/prime/mod trigger | audit closures: AUD-RA-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2012-D2-P5 (PDF p. 152) *inequality / sharp bound*

Primary domain	Algebra / number theory
Core method sequence	1. Auxiliary function with derivative sign; 2. Integrating factor and ODE comparison; 3. Special substitution and parameter freezing
Method IDs	M-RA-01; M-RA-02; M-UNI-07; M-ANT-10; M-ANT-11; M-RA-09; M-ANT-01; M-ANT-03
Method titles	Auxiliary function with derivative sign; Integrating factor and ODE comparison; Special substitution and parameter freezing; Group action, orbit, and generator method; Commutator and center manipulation in groups/rings; Convexity, Jensen, tangent, and chord inequalities; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping
Theory dependencies	Group basics: orders, cyclic subgroups, and generators; Rings, idempotents, nil rings, and noncommutativity; Automorphisms and semidirect products
Cue extraction	prior broad method tags: auxiliary function / substitution | statement triggers: integer/prime/mod trigger; group/ring/field trigger | audit closures: AUD-RA-08; AUD-ANT-10
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.20 2013

18.20.1 Day 1

2013-D1-P1 (PDF p. 153) *proof of structural property*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. AB versus BA transfer; 2. Spectral/minimal-polynomial reduction; 3. Nilpotency from zero spectrum plus finite dimension
Method IDs	M-LA-15; M-LA-06; M-LA-12; M-LA-04; M-LA-07; M-LA-13; M-UNI-05; M-UNI-09
Method titles	AB versus BA transfer; Spectral/minimal-polynomial reduction; Nilpotency from zero spectrum plus finite dimension; Rank-nullity and dimension-count obstruction; Trace cyclicity and commutator annihilation; Determinant through eigenvalue or root product; Induction with strengthened hypothesis; Invariant extraction and invariant comparison
Theory dependencies	Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; Rank-nullity, image/kernel, and rank of factorizations; Determinant evaluation by row/column operations and triangular forms
Cue extraction	prior broad method tags: spectral decomposition / eigenvalues | statement triggers: matrix word; det/rank/eigen trigger | audit closures: AUD-LA-10
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2013-D1-P2 (PDF p. 153) *existence / construction*

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Auxiliary function with derivative sign
Method IDs	M-UNI-07; M-UNI-09; M-RA-01; M-RA-03; M-RA-08; M-UNI-03; M-UNI-06; M-LA-04
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Auxiliary function with derivative sign; Rolle chain and divided-difference localization; Compactness plus extreme value theorem; Extremal construction and equality-case engineering; Compactness or extremal-point selection; Rank-nullity and dimension-count obstruction
Theory dependencies	Rolle, mean-value theorem, and divided differences; Taylor expansion, local asymptotics, and little-o bookkeeping; Rank-nullity, image/kernel, and rank of factorizations
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: continuous/differentiable trigger; functional-equation trigger | audit closures: AUD-RA-01; AUD-RA-06
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2013-D1-P3 (PDF p. 153) *counting / enumeration*

Primary domain	Combinatorics / discrete structures
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Double counting ordered or weighted objects; 3. Compactness plus extreme value theorem
Method IDs	M-CMB-01; M-CMB-02; M-RA-08; M-UNI-06; M-CMB-10; M-CMB-03; M-RA-09; M-CMB-05
Method titles	Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Compactness plus extreme value theorem; Compactness or extremal-point selection; Explicit bijection and encoding-decoding proof; Graph encoding of a relation; Convexity, Jensen, tangent, and chord inequalities; State recurrence or automaton partition
Theory dependencies	Pigeonhole principle and extremal element selection; Double counting and degree-sum methods; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: compactness / extremal point, double counting / pigeonhole | statement triggers: counting/permutation trigger | audit closures: AUD-RA-07; AUD-RA-08; AUD-CMB-03; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2013-D1-P4 (PDF p. 153) *inequality / sharp bound*

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Convexity, Jensen, tangent, and chord inequalities
Method IDs	M-UNI-07; M-UNI-09; M-RA-09; M-RA-01; M-RA-08; M-UNI-03; M-LA-03; M-LA-15
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Convexity, Jensen, tangent, and chord inequalities; Auxiliary function with derivative sign; Compactness plus extreme value theorem; Extremal construction and equality-case engineering; Block decomposition and Schur-style bookkeeping; AB versus BA transfer
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Integral inequalities, integration by parts, and weighted estimates; Block matrices, Schur complements, and determinant identities
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: wording-specific | audit closures: AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2013-D1-P5 (PDF p. 153) *nonexistence / counterexample*

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Extremal construction and equality-case engineering; 3. Chinese remainder and residue construction
Method IDs	M-ANT-01; M-UNI-03; M-ANT-02; M-UNI-08; M-ANT-03; M-ANT-16; M-RA-11; M-UNI-04
Method titles	Reduction modulo a prime or finite field; Extremal construction and equality-case engineering; Chinese remainder and residue construction; Sign, zero, and boundary case splitting; GCD, valuation, and prime-power bookkeeping; Polynomial identity by coefficient comparison; Integration by parts, Fubini, and weighted identities; Contradiction with minimal or maximal counterexample
Theory dependencies	Modular arithmetic and divisibility; Finite fields, field characteristic, and Fermat/Frobenius tools; Explicit construction and counterexample design
Cue extraction	prior broad method tags: modular / finite-field reduction, construction / counterexample | statement triggers: integer/prime/mod trigger | audit closures: AUD-PC-01; AUD-PC-05; AUD-PC-08; AUD-ANT-02; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.20.2 Day 2

2013-D2-P1 (PDF p. 154) *proof of structural property*

Primary domain	Complex analysis / polynomials
Core method sequence	1. Invariant extraction and invariant comparison; 2. Special substitution and parameter freezing; 3. Polynomial identity by coefficient comparison
Method IDs	M-UNI-09; M-UNI-07; M-ANT-16; M-CX-01; M-ANT-06; M-ANT-08; M-FE-03; M-UNI-05
Method titles	Invariant extraction and invariant comparison; Special substitution and parameter freezing; Polynomial identity by coefficient comparison; Cauchy coefficient formula and boundary maximum; Degree versus number of roots; Roots of unity and cyclotomic irreducibility; Composition, inversion, and conjugacy; Induction with strengthened hypothesis
Theory dependencies	Polynomial identity and more-roots-than-degree principle; Roots of unity, cyclotomic filters, and discrete Fourier tricks; Unit-circle roots and conjugation identities
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: wording-specific | audit closures: AUD-PC-01; AUD-PC-05; AUD-PC-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2013-D2-P2 (PDF p. 154) *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Chinese remainder and residue construction; 3. GCD, valuation, and prime-power bookkeeping
Method IDs	M-ANT-01; M-ANT-02; M-ANT-03; M-UNI-05; M-UNI-09; M-ANT-06; M-ANT-16; M-ANT-09
Method titles	Reduction modulo a prime or finite field; Chinese remainder and residue construction; GCD, valuation, and prime-power bookkeeping; Induction with strengthened hypothesis; Invariant extraction and invariant comparison; Degree versus number of roots; Polynomial identity by coefficient comparison; Finite-field function representative and permutation argument
Theory dependencies	Modular arithmetic and divisibility; Finite fields, field characteristic, and Fermat/Frobenius tools; Chinese remainder theorem and complete residue systems
Cue extraction	prior broad method tags: modular / finite-field reduction | statement triggers: integer/prime/mod trigger | audit closures: AUD-PC-01; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2013-D2-P3 (PDF p. 154) *existence / construction*

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Auxiliary function with derivative sign
Method IDs	M-UNI-07; M-UNI-09; M-RA-01; M-RA-09; M-GEO-03; M-RA-08; M-UNI-03; M-UNI-06
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Auxiliary function with derivative sign; Convexity, Jensen, tangent, and chord inequalities; Squared-distance and inner-product identities; Compactness plus extreme value theorem; Extremal construction and equality-case engineering; Compactness or extremal-point selection
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Integral inequalities, integration by parts, and weighted estimates; Euclidean distances and inner-product identities
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: wording-specific | audit closures: AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2013-D2-P4 (PDF p. 154) *nonexistence / counterexample*

Primary domain	Algebra / number theory
Core method sequence	1. Extremal construction and equality-case engineering; 2. Sign, zero, and boundary case splitting; 3. Reduction modulo a prime or finite field
Method IDs	M-UNI-03; M-UNI-08; M-ANT-01; M-ANT-03; M-RA-06; M-RA-07; M-RA-11; M-UNI-04
Method titles	Extremal construction and equality-case engineering; Sign, zero, and boundary case splitting; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Integration by parts, Fubini, and weighted identities; Contradiction with minimal or maximal counterexample
Theory dependencies	Explicit construction and counterexample design; Integral inequalities, integration by parts, and weighted estimates; Series convergence, rearrangements, and alternating/Dirichlet tools
Cue extraction	prior broad method tags: construction / counterexample | statement triggers: integral/limit trigger; integer/prime/mod trigger | audit closures: AUD-ANT-02; AUD-PC-01; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2013-D2-P5 (PDF p. 154) *counting / enumeration*

Primary domain	Combinatorics / discrete structures
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Double counting ordered or weighted objects; 3. Explicit bijection and encoding-decoding proof
Method IDs	M-CMB-01; M-CMB-02; M-CMB-10; M-CMB-05; M-CMB-03; M-RA-09; M-CMB-04; M-CMB-07
Method titles	Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Explicit bijection and encoding-decoding proof; State recurrence or automaton partition; Graph encoding of a relation; Convexity, Jensen, tangent, and chord inequalities; Cycle extraction and alternating-cycle method; Involution, parity, and sign-reversing cancellation
Theory dependencies	Pigeonhole principle and extremal element selection; Double counting and degree-sum methods; Block matrices, Schur complements, and determinant identities
Cue extraction	prior broad method tags: double counting / pigeonhole | statement triggers: counting/permutation trigger | audit closures: AUD-RA-08; AUD-CMB-03; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.21 2014

18.21.1 Day 1

2014-D1-P1 (PDF p. 155) *classification / find all*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Trace cyclicity and commutator annihilation; 2. Row/column elimination to triangular or sparse form; 3. Block decomposition and Schur-style bookkeeping
Method IDs	M-LA-07; M-LA-01; M-LA-03; M-LA-13; M-LA-12; M-LA-04; M-LA-06; M-UNI-07
Method titles	Trace cyclicity and commutator annihilation; Row/column elimination to triangular or sparse form; Block decomposition and Schur-style bookkeeping; Determinant through eigenvalue or root product; Nilpotency from zero spectrum plus finite dimension; Rank-nullity and dimension-count obstruction; Spectral/minimal-polynomial reduction; Special substitution and parameter freezing
Theory dependencies	Determinant evaluation by row/column operations and triangular forms; Block matrices, Schur complements, and determinant identities; Trace cyclicity, trace pairing, and zero-trace subspaces
Cue extraction	prior broad method tags: trace trick / commutator, determinant / block manipulation | statement triggers: matrix word; det/rank/eigen trigger | audit closures: AUD-LA-03; AUD-ANT-06
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2014-D1-P2 (PDF p. 155) *classification / find all*

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Riemann sum and sum-integral conversion
Method IDs	M-UNI-07; M-UNI-09; M-RA-06; M-RA-07; M-RA-01; M-RA-08; M-RA-05; M-UNI-08
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Auxiliary function with derivative sign; Compactness plus extreme value theorem; L-Hospital and ratio/growth comparison; Sign, zero, and boundary case splitting
Theory dependencies	Riemann sums and sum-integral asymptotics; L'Hospital, ratio limits, and growth comparisons; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: integral/limit trigger | audit closures: AUD-RA-03; AUD-RA-05
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2014-D1-P3 (PDF p. 155) *proof / reasoning*

Primary domain	Complex analysis / polynomials
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Polynomial identity by coefficient comparison
Method IDs	M-UNI-07; M-UNI-09; M-ANT-16; M-CX-01; M-ANT-06; M-FE-03; M-ANT-07; M-RA-15
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Polynomial identity by coefficient comparison; Cauchy coefficient formula and boundary maximum; Degree versus number of roots; Composition, inversion, and conjugacy; Vieta, symmetric sums, and logarithmic derivative; Recurrence transformed to monotone quotient or telescoping form
Theory dependencies	Polynomial identity and more-roots-than-degree principle; Vieta, logarithmic derivatives, and real-root inequalities; Integer polynomials, finite differences, and value divisibility
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: wording-specific | audit closures: AUD-PC-01; AUD-PC-08; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2014-D1-P4 (PDF p. 155) *proof / reasoning*

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Chinese remainder and residue construction; 3. GCD, valuation, and prime-power bookkeeping
Method IDs	M-ANT-01; M-ANT-02; M-ANT-03; M-RA-11; M-UNI-07; M-UNI-09; M-ANT-06; M-ANT-16
Method titles	Reduction modulo a prime or finite field; Chinese remainder and residue construction; GCD, valuation, and prime-power bookkeeping; Integration by parts, Fubini, and weighted identities; Special substitution and parameter freezing; Invariant extraction and invariant comparison; Degree versus number of roots; Polynomial identity by coefficient comparison
Theory dependencies	Modular arithmetic and divisibility; Finite fields, field characteristic, and Fermat/Frobenius tools; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: modular / finite-field reduction | statement triggers: integral/limit trigger; integer/prime/mod trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2014-D1-P5 (PDF p. 155) *proof / reasoning*

Primary domain	Geometry / convexity / metric
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Squared-distance and inner-product identities
Method IDs	M-UNI-07; M-UNI-09; M-GEO-03; M-GEO-01; M-RA-09; M-GEO-02; M-GEO-04; M-GEO-05
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Squared-distance and inner-product identities; Coordinate normalization and algebraization; Convexity, Jensen, tangent, and chord inequalities; Projection to a lower-dimensional object; Barycentric coordinates and volume weights; Affine invariance and change of metric scale
Theory dependencies	Area, volume, projection, and ellipsoid computation; Projections and orthogonal decompositions; Simplex, barycentric coordinates, and circumcenters
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: wording-specific | audit closures: AUD-RA-08; AUD-GEO-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.21.2 Day 2

2014-D2-P1 (PDF p. 156) *proof / reasoning*

Primary domain	Algebra / number theory
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Reduction modulo a prime or finite field
Method IDs	M-UNI-07; M-UNI-09; M-ANT-01; M-ANT-03; M-ANT-06; M-ANT-16; M-ANT-02; M-ANT-04
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Degree versus number of roots; Polynomial identity by coefficient comparison; Chinese remainder and residue construction; Finite differences and integer polynomial values
Theory dependencies	Modular arithmetic and divisibility; GCD, valuations, prime powers, and coprimality; Group basics: orders, cyclic subgroups, and generators
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: wording-specific | audit closures: AUD-ANT-02; AUD-PC-01
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2014-D2-P2 (PDF p. 156) *classification / find all*

Primary domain	Real analysis / calculus
Core method sequence	1. Spectral/minimal-polynomial reduction; 2. Nilpotency from zero spectrum plus finite dimension; 3. Auxiliary function with derivative sign
Method IDs	M-LA-06; M-LA-12; M-RA-01; M-RA-09; M-RA-08; M-UNI-07; M-UNI-08; M-LA-03
Method titles	Spectral/minimal-polynomial reduction; Nilpotency from zero spectrum plus finite dimension; Auxiliary function with derivative sign; Convexity, Jensen, tangent, and chord inequalities; Compactness plus extreme value theorem; Special substitution and parameter freezing; Sign, zero, and boundary case splitting; Block decomposition and Schur-style bookkeeping
Theory dependencies	Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; Convexity, concavity, Jensen, and tangent/chord inequalities; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: spectral decomposition / eigenvalues | statement triggers: matrix word; det/rank/eigen trigger | audit closures: AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2014-D2-P3 (PDF p. 156) *proof / reasoning*

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Auxiliary function with derivative sign
Method IDs	M-UNI-07; M-UNI-09; M-RA-01; M-RA-08; M-RA-03; M-CX-05; M-LA-03; M-LA-15
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Auxiliary function with derivative sign; Compactness plus extreme value theorem; Rolle chain and divided-difference localization; Trigonometric-to-complex conversion; Block decomposition and Schur-style bookkeeping; AB versus BA transfer
Theory dependencies	Block matrices, Schur complements, and determinant identities; Auxiliary functions, integrating factors, and differential inequalities; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: functional-equation trigger | audit closures: AUD-RA-01; AUD-RA-06
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2014-D2-P4 (PDF p. 156) *inequality / sharp bound*

Primary domain	Geometry / convexity / metric
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Coordinate normalization and algebraization
Method IDs	M-UNI-07; M-UNI-09; M-GEO-01; M-GEO-03; M-RA-09; M-UNI-03; M-CMB-03; M-CMB-04
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Coordinate normalization and algebraization; Squared-distance and inner-product identities; Convexity, Jensen, tangent, and chord inequalities; Extremal construction and equality-case engineering; Graph encoding of a relation; Cycle extraction and alternating-cycle method
Theory dependencies	Graph encoding, cycles, and bipartite representations; Coloring, partitions, and parity; Set systems, intersecting families, and induction on sizes
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: wording-specific | audit closures: AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2014-D2-P5 (PDF p. 156) *proof / reasoning*

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Auxiliary function with derivative sign
Method IDs	M-UNI-07; M-UNI-09; M-RA-01; M-RA-08; M-RA-09; M-GEO-06; M-LA-03; M-LA-15
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Auxiliary function with derivative sign; Compactness plus extreme value theorem; Convexity, Jensen, tangent, and chord inequalities; Convex support and John-ellipse perturbation; Block decomposition and Schur-style bookkeeping; AB versus BA transfer
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Integral inequalities, integration by parts, and weighted estimates; Riemann sums and sum-integral asymptotics
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: wording-specific | audit closures: AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.22 2015

18.22.1 Day 1

2015-D1-P1 (PDF p. 157) *proof of structural property*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Row/column elimination to triangular or sparse form; 2. Block decomposition and Schur-style bookkeeping; 3. Determinant through eigenvalue or root product
Method IDs	M-LA-01; M-LA-03; M-LA-13; M-ANT-01; M-ANT-03; M-LA-04; M-LA-06; M-ANT-16
Method titles	Row/column elimination to triangular or sparse form; Block decomposition and Schur-style bookkeeping; Determinant through eigenvalue or root product; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Rank-nullity and dimension-count obstruction; Spectral/minimal-polynomial reduction; Polynomial identity by coefficient comparison
Theory dependencies	Determinant evaluation by row/column operations and triangular forms; Block matrices, Schur complements, and determinant identities; Rank-nullity, image/kernel, and rank of factorizations
Cue extraction	prior broad method tags: determinant / block manipulation | statement triggers: matrix word; det/rank/eigen trigger; integer/prime/mod trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-08; AUD-CMB-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2015-D1-P2 (PDF p. 157) *inequality / sharp bound*

Primary domain	Algebra / number theory
Core method sequence	1. Extremal construction and equality-case engineering; 2. Sign, zero, and boundary case splitting; 3. Reduction modulo a prime or finite field
Method IDs	M-UNI-03; M-UNI-08; M-ANT-01; M-ANT-03; M-RA-09; M-ANT-10; M-LA-03; M-LA-15
Method titles	Extremal construction and equality-case engineering; Sign, zero, and boundary case splitting; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Convexity, Jensen, tangent, and chord inequalities; Group action, orbit, and generator method; Block decomposition and Schur-style bookkeeping; AB versus BA transfer
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Explicit construction and counterexample design; Modular arithmetic and divisibility
Cue extraction	prior broad method tags: construction / counterexample | statement triggers: integer/prime/mod trigger; functional-equation trigger | audit closures: AUD-RA-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2015-D1-P3 (PDF p. 157) *proof / reasoning*

Primary domain	Functional equations / dynamics
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Monotonicity forced by functional equation
Method IDs	M-UNI-07; M-UNI-09; M-FE-02; M-RA-09; M-FE-01; M-GEO-06; M-LA-03; M-LA-15
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Monotonicity forced by functional equation; Convexity, Jensen, tangent, and chord inequalities; Iterates, fixed points, and cluster-point trapping; Convex support and John-ellipse perturbation; Block decomposition and Schur-style bookkeeping; AB versus BA transfer
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Integral inequalities, integration by parts, and weighted estimates; Block matrices, Schur complements, and determinant identities
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: det/rank/eigen trigger; functional-equation trigger | audit closures: AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2015-D1-P4 (PDF p. 157) *proof / reasoning*

Primary domain	Algebra / number theory
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Reduction modulo a prime or finite field
Method IDs	M-UNI-07; M-UNI-09; M-ANT-01; M-ANT-03; M-RA-09; M-ANT-02; M-ANT-10; M-CX-05
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Convexity, Jensen, tangent, and chord inequalities; Chinese remainder and residue construction; Group action, orbit, and generator method; Trigonometric-to-complex conversion
Theory dependencies	Modular arithmetic and divisibility; GCD, valuations, prime powers, and coprimality; Group basics: orders, cyclic subgroups, and generators
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: det/rank/eigen trigger; integer/prime/mod trigger | audit closures: AUD-RA-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2015-D1-P5 (PDF p. 157) *inequality / sharp bound*

Primary domain	Real analysis / calculus
Core method sequence	1. Convexity, Jensen, tangent, and chord inequalities; 2. Rank-nullity and dimension-count obstruction; 3. Squared-distance and inner-product identities
Method IDs	M-RA-09; M-LA-04; M-GEO-03; M-GEO-11; M-RA-10; M-GEO-01; M-GEO-02; M-RA-15
Method titles	Convexity, Jensen, tangent, and chord inequalities; Rank-nullity and dimension-count obstruction; Squared-distance and inner-product identities; Volume, projection, and ellipsoid computation; Cauchy-Schwarz, Holder, and norm comparison; Coordinate normalization and algebraization; Projection to a lower-dimensional object; Recurrence transformed to monotone quotient or telescoping form
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Rank-nullity, image/kernel, and rank of factorizations; Sequence recurrences, monotone bounds, and generating functions
Cue extraction	prior broad method tags: convexity / Jensen / tangent, rank-nullity / dimension count | statement triggers: geometry/metric trigger | audit closures: AUD-GEO-03; AUD-RA-08; AUD-PC-13; AUD-CMB-07; AUD-GEO-02; AUD-PRB-01
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.22.2 Day 2

2015-D2-P6 (PDF p. 158) *proof of structural property*

Primary domain	Real analysis / calculus
Core method sequence	1. Invariant extraction and invariant comparison; 2. Special substitution and parameter freezing; 3. Auxiliary function with derivative sign
Method IDs	M-UNI-09; M-UNI-07; M-RA-01; M-RA-08; M-UNI-05; M-RA-03; M-FE-01; M-FE-02
Method titles	Invariant extraction and invariant comparison; Special substitution and parameter freezing; Auxiliary function with derivative sign; Compactness plus extreme value theorem; Induction with strengthened hypothesis; Rolle chain and divided-difference localization; Iterates, fixed points, and cluster-point trapping; Monotonicity forced by functional equation
Theory dependencies	Block matrices, Schur complements, and determinant identities; Auxiliary functions, integrating factors, and differential inequalities; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: wording-specific | audit closures: AUD-RA-01; AUD-RA-06
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2015-D2-P7 (PDF p. 158) *limit / asymptotic evaluation*

Primary domain	Real analysis / calculus
Core method sequence	1. Topological ordering of an acyclic directed graph; 2. Special substitution and parameter freezing; 3. Invariant extraction and invariant comparison
Method IDs	M-CMB-14; M-UNI-07; M-UNI-09; M-RA-06; M-RA-07; M-RA-01; M-RA-08; M-RA-05
Method titles	Topological ordering of an acyclic directed graph; Special substitution and parameter freezing; Invariant extraction and invariant comparison; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Auxiliary function with derivative sign; Compactness plus extreme value theorem; L-Hospital and ratio/growth comparison
Theory dependencies	Riemann sums and sum-integral asymptotics; L'Hospital, ratio limits, and growth comparisons; Taylor expansion, local asymptotics, and little-o bookkeeping
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: integral/limit trigger | audit closures: AUD-RA-03; AUD-RA-05
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2015-D2-P8 (PDF p. 158) *counting / enumeration*

Primary domain	Combinatorics / discrete structures
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Double counting ordered or weighted objects; 3. Riemann sum and sum-integral conversion
Method IDs	M-CMB-01; M-CMB-02; M-RA-06; M-RA-07; M-CMB-10; M-RA-11; M-CMB-05; M-CMB-03
Method titles	Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Explicit bijection and encoding-decoding proof; Integration by parts, Fubini, and weighted identities; State recurrence or automaton partition; Graph encoding of a relation
Theory dependencies	Pigeonhole principle and extremal element selection; Double counting and degree-sum methods; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: double counting / pigeonhole | statement triggers: integral/limit trigger; counting/permutation trigger | audit closures: AUD-CMB-03; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2015-D2-P9 (PDF p. 158) *inequality / sharp bound*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Cauchy-Schwarz, Holder, and norm comparison; 2. Rank-nullity and dimension-count obstruction; 3. Compactness plus extreme value theorem
Method IDs	M-RA-10; M-LA-04; M-RA-08; M-UNI-06; M-GEO-03; M-LA-06; M-RA-09; M-LA-05
Method titles	Cauchy-Schwarz, Holder, and norm comparison; Rank-nullity and dimension-count obstruction; Compactness plus extreme value theorem; Compactness or extremal-point selection; Squared-distance and inner-product identities; Spectral/minimal-polynomial reduction; Convexity, Jensen, tangent, and chord inequalities; Similarity, adapted basis, and canonical model
Theory dependencies	Rank-nullity, image/kernel, and rank of factorizations; Invariant subspaces and adapted bases; Block matrices, Schur complements, and determinant identities
Cue extraction	prior broad method tags: compactness / extremal point, Cauchy-Schwarz / norm inequality, rank-nullity / dimension count | statement triggers: matrix word; det/rank/eigen trigger | audit closures: AUD-LA-11; AUD-RA-07; AUD-RA-08; AUD-PC-01; AUD-PC-05
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2015-D2-P10 (PDF p. 158) *inequality / sharp bound*

Primary domain	Algebra / number theory
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Factorization and unique factorization in Z[x]
Method IDs	M-ANT-16; M-ANT-06; M-ANT-05; M-ANT-01; M-ANT-03; M-RA-09; M-UNI-03; M-FE-03
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Factorization and unique factorization in Z[x]; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Convexity, Jensen, tangent, and chord inequalities; Extremal construction and equality-case engineering; Composition, inversion, and conjugacy
Theory dependencies	Polynomial identity and more-roots-than-degree principle; Convexity, concavity, Jensen, and tangent/chord inequalities; Integer polynomials, finite differences, and value divisibility
Cue extraction	prior broad method tags: polynomial identity / factorization | statement triggers: polynomial/root trigger; integer/prime/mod trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.23 2016

18.23.1 Day 1

2016-D1-P1 (PDF p. 159) *proof of structural property*

Primary domain	Functional equations / dynamics
Core method sequence	1. Extremal construction and equality-case engineering; 2. Sign, zero, and boundary case splitting; 3. Auxiliary function with derivative sign
Method IDs	M-UNI-03; M-UNI-08; M-RA-01; M-RA-03; M-FE-02; M-UNI-07; M-UNI-05; M-UNI-09
Method titles	Extremal construction and equality-case engineering; Sign, zero, and boundary case splitting; Auxiliary function with derivative sign; Rolle chain and divided-difference localization; Monotonicity forced by functional equation; Special substitution and parameter freezing; Induction with strengthened hypothesis; Invariant extraction and invariant comparison
Theory dependencies	Explicit construction and counterexample design; Intermediate value, connectedness, density, and continuity transfer; Substitution of special values and normalization
Cue extraction	prior broad method tags: construction / counterexample | statement triggers: continuous/differentiable trigger; functional-equation trigger | audit closures: AUD-RA-01
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2016-D1-P2 (PDF p. 159) *inequality / sharp bound*

Primary domain	Algebra / number theory
Core method sequence	1. Extremal construction and equality-case engineering; 2. Sign, zero, and boundary case splitting; 3. Reduction modulo a prime or finite field
Method IDs	M-UNI-03; M-UNI-08; M-ANT-01; M-ANT-03; M-RA-09; M-LA-04; M-ANT-10; M-LA-03
Method titles	Extremal construction and equality-case engineering; Sign, zero, and boundary case splitting; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Convexity, Jensen, tangent, and chord inequalities; Rank-nullity and dimension-count obstruction; Group action, orbit, and generator method; Block decomposition and Schur-style bookkeeping
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Modular arithmetic and divisibility; Explicit construction and counterexample design
Cue extraction	prior broad method tags: construction / counterexample | statement triggers: matrix word; integer/prime/mod trigger | audit closures: AUD-LA-12; AUD-RA-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2016-D1-P3 (PDF p. 159) *inequality / sharp bound*

Primary domain	Algebra / number theory
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Reduction modulo a prime or finite field
Method IDs	M-UNI-07; M-UNI-09; M-ANT-01; M-ANT-03; M-RA-09; M-UNI-03; M-ANT-02; M-ANT-09
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Convexity, Jensen, tangent, and chord inequalities; Extremal construction and equality-case engineering; Chinese remainder and residue construction; Finite-field function representative and permutation argument
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Modular arithmetic and divisibility; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: integer/prime/mod trigger | audit closures: AUD-RA-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2016-D1-P4 (PDF p. 159) *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Invariant extraction and invariant comparison; 2. Special substitution and parameter freezing; 3. Reduction modulo a prime or finite field
Method IDs	M-UNI-09; M-UNI-07; M-ANT-01; M-ANT-03; M-UNI-05; M-RA-09; M-CMB-03; M-CMB-04
Method titles	Invariant extraction and invariant comparison; Special substitution and parameter freezing; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Induction with strengthened hypothesis; Convexity, Jensen, tangent, and chord inequalities; Graph encoding of a relation; Cycle extraction and alternating-cycle method
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Graph encoding, cycles, and bipartite representations; Coloring, partitions, and parity
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: integer/prime/mod trigger; graph/color/set trigger | audit closures: AUD-RA-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2016-D1-P5 (PDF p. 159) *counting / enumeration*

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Pigeonhole threshold and collision forcing; 3. Double counting ordered or weighted objects
Method IDs	M-ANT-01; M-CMB-01; M-CMB-02; M-ANT-02; M-CMB-07; M-ANT-10; M-ANT-03; M-CMB-10
Method titles	Reduction modulo a prime or finite field; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Chinese remainder and residue construction; Involution, parity, and sign-reversing cancellation; Group action, orbit, and generator method; GCD, valuation, and prime-power bookkeeping; Explicit bijection and encoding-decoding proof
Theory dependencies	Modular arithmetic and divisibility; Finite fields, field characteristic, and Fermat/Frobenius tools; Double counting and degree-sum methods
Cue extraction	prior broad method tags: modular / finite-field reduction, double counting / pigeonhole | statement triggers: integer/prime/mod trigger; counting/permutation trigger; functional-equation trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-ANT-02; AUD-CMB-03; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.23.2 Day 2

2016-D2-P6 (PDF p. 160) *inequality / sharp bound*

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Convexity, Jensen, tangent, and chord inequalities
Method IDs	M-UNI-07; M-UNI-09; M-RA-09; M-RA-01; M-RA-08; M-UNI-03; M-CMB-08; M-LA-03
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Convexity, Jensen, tangent, and chord inequalities; Auxiliary function with derivative sign; Compactness plus extreme value theorem; Extremal construction and equality-case engineering; Extremal set or minimal counterexample induction; Block decomposition and Schur-style bookkeeping
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Integral inequalities, integration by parts, and weighted estimates; Block matrices, Schur complements, and determinant identities
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: wording-specific | audit closures: AUD-RA-08; AUD-CMB-05
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2016-D2-P7 (PDF p. 160) *proof / reasoning*

Primary domain	Real analysis / calculus
Core method sequence	1. Compactness plus extreme value theorem; 2. Compactness or extremal-point selection; 3. Auxiliary function with derivative sign
Method IDs	M-RA-08; M-UNI-06; M-RA-01; M-RA-03; M-RA-09; M-UNI-07; M-UNI-09; M-CMB-01
Method titles	Compactness plus extreme value theorem; Compactness or extremal-point selection; Auxiliary function with derivative sign; Rolle chain and divided-difference localization; Convexity, Jensen, tangent, and chord inequalities; Special substitution and parameter freezing; Invariant extraction and invariant comparison; Pigeonhole threshold and collision forcing
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Compactness, extreme values, and closed-bounded reduction; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: compactness / extremal point | statement triggers: continuous/differentiable trigger; functional-equation trigger | audit closures: AUD-RA-07; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2016-D2-P8 (PDF p. 160) *existence / construction*

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Chinese remainder and residue construction; 3. GCD, valuation, and prime-power bookkeeping
Method IDs	M-ANT-01; M-ANT-02; M-ANT-03; M-ANT-10; M-ANT-11; M-RA-06; M-RA-07; M-UNI-03
Method titles	Reduction modulo a prime or finite field; Chinese remainder and residue construction; GCD, valuation, and prime-power bookkeeping; Group action, orbit, and generator method; Commutator and center manipulation in groups/rings; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Extremal construction and equality-case engineering
Theory dependencies	Modular arithmetic and divisibility; Finite fields, field characteristic, and Fermat/Frobenius tools; Group basics: orders, cyclic subgroups, and generators
Cue extraction	prior broad method tags: modular / finite-field reduction | statement triggers: integer/prime/mod trigger; group/ring/field trigger; functional-equation trigger | audit closures: AUD-PC-01; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2016-D2-P9 (PDF p. 160) *inequality / sharp bound*

Primary domain	Algebra / number theory
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Double counting ordered or weighted objects; 3. Reduction modulo a prime or finite field
Method IDs	M-CMB-01; M-CMB-02; M-ANT-01; M-ANT-03; M-RA-09; M-UNI-03; M-CMB-03; M-ANT-10
Method titles	Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Convexity, Jensen, tangent, and chord inequalities; Extremal construction and equality-case engineering; Graph encoding of a relation; Group action, orbit, and generator method
Theory dependencies	Pigeonhole principle and extremal element selection; Double counting and degree-sum methods; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: double counting / pigeonhole | statement triggers: integer/prime/mod trigger; counting/permutation trigger; functional-equation trigger | audit closures: AUD-RA-08; AUD-CMB-03; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2016-D2-P10 (PDF p. 160) *inequality / sharp bound*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Spectral/minimal-polynomial reduction; 2. Nilpotency from zero spectrum plus finite dimension; 3. Convexity, Jensen, tangent, and chord inequalities
Method IDs	M-LA-06; M-LA-12; M-RA-09; M-LA-04; M-ANT-16; M-ANT-08; M-LA-13; M-UNI-03
Method titles	Spectral/minimal-polynomial reduction; Nilpotency from zero spectrum plus finite dimension; Convexity, Jensen, tangent, and chord inequalities; Rank-nullity and dimension-count obstruction; Polynomial identity by coefficient comparison; Roots of unity and cyclotomic irreducibility; Determinant through eigenvalue or root product; Extremal construction and equality-case engineering
Theory dependencies	Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; Convexity, concavity, Jensen, and tangent/chord inequalities; Rank-nullity, image/kernel, and rank of factorizations
Cue extraction	prior broad method tags: spectral decomposition / eigenvalues | statement triggers: matrix word; det/rank/eigen trigger | audit closures: AUD-LA-07; AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.24 2017

18.24.1 Day 1

2017-D1-P1 (PDF p. 161) *classification / find all*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Spectral/minimal-polynomial reduction; 2. Nilpotency from zero spectrum plus finite dimension; 3. Reduction modulo a prime or finite field
Method IDs	M-LA-06; M-LA-12; M-ANT-01; M-ANT-03; M-ANT-16; M-LA-04; M-ANT-06; M-ANT-08
Method titles	Spectral/minimal-polynomial reduction; Nilpotency from zero spectrum plus finite dimension; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Polynomial identity by coefficient comparison; Rank-nullity and dimension-count obstruction; Degree versus number of roots; Roots of unity and cyclotomic irreducibility
Theory dependencies	Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; Unit-circle roots and conjugation identities; Roots of unity, cyclotomic filters, and discrete Fourier tricks
Cue extraction	prior broad method tags: spectral decomposition / eigenvalues | statement triggers: matrix word; det/rank/eigen trigger; integer/prime/mod trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2017-D1-P2 (PDF p. 161) *existence / construction*

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Auxiliary function with derivative sign
Method IDs	M-UNI-07; M-UNI-09; M-RA-01; M-RA-09; M-RA-08; M-UNI-03; M-UNI-06; M-CX-05
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Auxiliary function with derivative sign; Convexity, Jensen, tangent, and chord inequalities; Compactness plus extreme value theorem; Extremal construction and equality-case engineering; Compactness or extremal-point selection; Trigonometric-to-complex conversion
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Integral inequalities, integration by parts, and weighted estimates; Auxiliary functions, integrating factors, and differential inequalities
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: functional-equation trigger | audit closures: AUD-RA-15; AUD-RA-01; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2017-D1-P3 (PDF p. 161) *existence / construction*

Primary domain	Algebra / number theory
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Reduction modulo a prime or finite field
Method IDs	M-UNI-07; M-UNI-09; M-ANT-01; M-ANT-03; M-RA-09; M-UNI-03; M-UNI-06; M-CMB-03
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Convexity, Jensen, tangent, and chord inequalities; Extremal construction and equality-case engineering; Compactness or extremal-point selection; Graph encoding of a relation
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Modular arithmetic and divisibility; Graph encoding, cycles, and bipartite representations
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: det/rank/eigen trigger; integer/prime/mod trigger | audit closures: AUD-RA-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2017-D1-P4 (PDF p. 161) *counting / enumeration*

Primary domain	Algebra / number theory
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Double counting ordered or weighted objects; 3. Group action, orbit, and generator method
Method IDs	M-CMB-01; M-CMB-02; M-ANT-10; M-ANT-11; M-ANT-01; M-ANT-03; M-CMB-10; M-CMB-05
Method titles	Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Group action, orbit, and generator method; Commutator and center manipulation in groups/rings; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Explicit bijection and encoding-decoding proof; State recurrence or automaton partition
Theory dependencies	Pigeonhole principle and extremal element selection; Group basics: orders, cyclic subgroups, and generators; Double counting and degree-sum methods
Cue extraction	prior broad method tags: double counting / pigeonhole | statement triggers: group/ring/field trigger; counting/permutation trigger | audit closures: AUD-CMB-01; AUD-CMB-03; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2017-D1-P5 (PDF p. 161) *proof of structural property*

Primary domain	Complex analysis / polynomials
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Factorization and unique factorization in Z[x]
Method IDs	M-ANT-16; M-ANT-06; M-ANT-05; M-RA-03; M-ANT-01; M-ANT-03; M-RA-01; M-ANT-08
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Factorization and unique factorization in Z[x]; Rolle chain and divided-difference localization; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Auxiliary function with derivative sign; Roots of unity and cyclotomic irreducibility
Theory dependencies	Polynomial identity and more-roots-than-degree principle; Convexity, concavity, Jensen, and tangent/chord inequalities; Vieta, logarithmic derivatives, and real-root inequalities
Cue extraction	prior broad method tags: polynomial identity / factorization | statement triggers: continuous/differentiable trigger; polynomial/root trigger; integer/prime/mod trigger; functional-equation trigger | audit closures: AUD-RA-06; AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.24.2 Day 2

2017-D2-P6 (PDF p. 162) *limit / asymptotic evaluation*

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Riemann sum and sum-integral conversion
Method IDs	M-UNI-07; M-UNI-09; M-RA-06; M-RA-07; M-RA-01; M-RA-03; M-RA-05; M-RA-08
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Auxiliary function with derivative sign; Rolle chain and divided-difference localization; L-Hospital and ratio/growth comparison; Compactness plus extreme value theorem
Theory dependencies	Riemann sums and sum-integral asymptotics; L'Hospital, ratio limits, and growth comparisons; Intermediate value, connectedness, density, and continuity transfer
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: continuous/differentiable trigger; integral/limit trigger; functional-equation trigger | audit closures: AUD-RA-03; AUD-RA-05
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2017-D2-P7 (PDF p. 162) *proof of structural property*

Primary domain	Complex analysis / polynomials
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Factorization and unique factorization in Z[x]
Method IDs	M-ANT-16; M-ANT-06; M-ANT-05; M-ANT-01; M-ANT-03; M-CX-01; M-FE-03; M-UNI-05
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Factorization and unique factorization in Z[x]; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Cauchy coefficient formula and boundary maximum; Composition, inversion, and conjugacy; Induction with strengthened hypothesis
Theory dependencies	Polynomial identity and more-roots-than-degree principle; Vieta, logarithmic derivatives, and real-root inequalities; Factorization, irreducibility, and Gauss-style integer polynomial arguments
Cue extraction	prior broad method tags: polynomial identity / factorization | statement triggers: polynomial/root trigger; integer/prime/mod trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2017-D2-P8 (PDF p. 162) *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Spectral/minimal-polynomial reduction; 2. Nilpotency from zero spectrum plus finite dimension; 3. Reduction modulo a prime or finite field
Method IDs	M-LA-06; M-LA-12; M-ANT-01; M-ANT-03; M-RA-15; M-CMB-06; M-UNI-05; M-UNI-09
Method titles	Spectral/minimal-polynomial reduction; Nilpotency from zero spectrum plus finite dimension; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Recurrence transformed to monotone quotient or telescoping form; Generating function encoding; Induction with strengthened hypothesis; Invariant extraction and invariant comparison
Theory dependencies	Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; Sequence recurrences, monotone bounds, and generating functions; Modular arithmetic and divisibility
Cue extraction	prior broad method tags: spectral decomposition / eigenvalues | statement triggers: matrix word; det/rank/eigen trigger; integer/prime/mod trigger; functional-equation trigger | audit closures: AUD-PC-13; AUD-CMB-07; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2017-D2-P9 (PDF p. 162) *limit / asymptotic evaluation*

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Riemann sum and sum-integral conversion
Method IDs	M-UNI-07; M-UNI-09; M-RA-06; M-RA-07; M-RA-01; M-FE-01; M-RA-03; M-RA-05
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Auxiliary function with derivative sign; Iterates, fixed points, and cluster-point trapping; Rolle chain and divided-difference localization; L-Hospital and ratio/growth comparison
Theory dependencies	Riemann sums and sum-integral asymptotics; Sequence recurrences, monotone bounds, and generating functions; L'Hospital, ratio limits, and growth comparisons
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: det/rank/eigen trigger; continuous/differentiable trigger; integral/limit trigger; functional-equation trigger | audit closures: AUD-RA-01; AUD-RA-03; AUD-RA-05; AUD-PC-13; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2017-D2-P10 (PDF p. 162) *existence / construction*

Primary domain	Algebra / number theory
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Reduction modulo a prime or finite field
Method IDs	M-UNI-07; M-UNI-09; M-ANT-01; M-ANT-03; M-UNI-03; M-UNI-06; M-GEO-03; M-RA-09
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Extremal construction and equality-case engineering; Compactness or extremal-point selection; Squared-distance and inner-product identities; Convexity, Jensen, tangent, and chord inequalities
Theory dependencies	Projections and orthogonal decompositions; Area, volume, projection, and ellipsoid computation; Modular arithmetic and divisibility
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: integer/prime/mod trigger; geometry/metric trigger | audit closures: AUD-RA-08; AUD-GEO-02; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.25 2018

18.25.1 Day 1

2018-D1-P1 (PDF p. 163) *proof of structural property*

Primary domain	Real analysis / calculus
Core method sequence	1. Invariant extraction and invariant comparison; 2. Special substitution and parameter freezing; 3. Auxiliary function with derivative sign
Method IDs	M-UNI-09; M-UNI-07; M-RA-01; M-RA-08; M-RA-11; M-UNI-05; M-RA-03; M-LA-03
Method titles	Invariant extraction and invariant comparison; Special substitution and parameter freezing; Auxiliary function with derivative sign; Compactness plus extreme value theorem; Integration by parts, Fubini, and weighted identities; Induction with strengthened hypothesis; Rolle chain and divided-difference localization; Block decomposition and Schur-style bookkeeping
Theory dependencies	Integral inequalities, integration by parts, and weighted estimates; Series convergence, rearrangements, and alternating/Dirichlet tools; Fubini/Tonelli and change of variables
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: wording-specific | audit closures: AUD-RA-01; AUD-RA-06
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2018-D1-P2 (PDF p. 163) *nonexistence / counterexample*

Primary domain	Algebra / number theory
Core method sequence	1. Extremal construction and equality-case engineering; 2. Sign, zero, and boundary case splitting; 3. Group action, orbit, and generator method
Method IDs	M-UNI-03; M-UNI-08; M-ANT-10; M-ANT-01; M-ANT-11; M-ANT-03; M-UNI-04; M-ANT-06
Method titles	Extremal construction and equality-case engineering; Sign, zero, and boundary case splitting; Group action, orbit, and generator method; Reduction modulo a prime or finite field; Commutator and center manipulation in groups/rings; GCD, valuation, and prime-power bookkeeping; Contradiction with minimal or maximal counterexample; Degree versus number of roots
Theory dependencies	Group basics: orders, cyclic subgroups, and generators; Rings, idempotents, nil rings, and noncommutativity; Explicit construction and counterexample design
Cue extraction	prior broad method tags: construction / counterexample | statement triggers: group/ring/field trigger | audit closures: AUD-ANT-02; AUD-PC-01; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2018-D1-P3 (PDF p. 163) *classification / find all*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Spectral/minimal-polynomial reduction
Method IDs	M-UNI-07; M-UNI-09; M-LA-06; M-LA-04; M-LA-03; M-LA-13; M-UNI-08; M-LA-16
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Spectral/minimal-polynomial reduction; Rank-nullity and dimension-count obstruction; Block decomposition and Schur-style bookkeeping; Determinant through eigenvalue or root product; Sign, zero, and boundary case splitting; Trace pairing and orthogonal matrix subspaces
Theory dependencies	Determinant evaluation by row/column operations and triangular forms; Rank-nullity, image/kernel, and rank of factorizations; Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: matrix word; det/rank/eigen trigger | audit closures: AUD-LA-03; AUD-LA-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2018-D1-P4 (PDF p. 163) *classification / find all*

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Auxiliary function with derivative sign
Method IDs	M-UNI-07; M-UNI-09; M-RA-01; M-RA-08; M-UNI-08; M-LA-03; M-LA-04; M-LA-15
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Auxiliary function with derivative sign; Compactness plus extreme value theorem; Sign, zero, and boundary case splitting; Block decomposition and Schur-style bookkeeping; Rank-nullity and dimension-count obstruction; AB versus BA transfer
Theory dependencies	Block matrices, Schur complements, and determinant identities; Rank-nullity, image/kernel, and rank of factorizations; Auxiliary functions, integrating factors, and differential inequalities
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: functional-equation trigger | audit closures: AUD-RA-01
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2018-D1-P5 (PDF p. 163) *inequality / sharp bound*

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Convexity, Jensen, tangent, and chord inequalities; 3. Chinese remainder and residue construction
Method IDs	M-ANT-01; M-RA-09; M-ANT-02; M-ANT-03; M-GEO-11; M-GEO-01; M-UNI-03; M-ANT-06
Method titles	Reduction modulo a prime or finite field; Convexity, Jensen, tangent, and chord inequalities; Chinese remainder and residue construction; GCD, valuation, and prime-power bookkeeping; Volume, projection, and ellipsoid computation; Coordinate normalization and algebraization; Extremal construction and equality-case engineering; Degree versus number of roots
Theory dependencies	Modular arithmetic and divisibility; Convexity, concavity, Jensen, and tangent/chord inequalities; Finite fields, field characteristic, and Fermat/Frobenius tools
Cue extraction	prior broad method tags: convexity / Jensen / tangent, modular / finite-field reduction | statement triggers: integer/prime/mod trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-ANT-02; AUD-GEO-02; AUD-GEO-10
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.25.2 Day 2

2018-D2-P6 (PDF p. 164) *proof / reasoning*

Primary domain	Algebra / number theory
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Reduction modulo a prime or finite field
Method IDs	M-UNI-07; M-UNI-09; M-ANT-01; M-ANT-03; M-GEO-03; M-ANT-09; M-RA-09; M-ANT-02
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Squared-distance and inner-product identities; Finite-field function representative and permutation argument; Convexity, Jensen, tangent, and chord inequalities; Chinese remainder and residue construction
Theory dependencies	Modular arithmetic and divisibility; Hilbert-space equidistant sets and orthonormal translations; Gram matrices and rational linear dependence
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: integer/prime/mod trigger | audit closures: AUD-RA-08; AUD-PC-06; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2018-D2-P7 (PDF p. 164) *proof of structural property*

Primary domain	Real analysis / calculus
Core method sequence	1. Invariant extraction and invariant comparison; 2. Special substitution and parameter freezing; 3. Riemann sum and sum-integral conversion
Method IDs	M-UNI-09; M-UNI-07; M-RA-06; M-RA-07; M-RA-01; M-RA-09; M-RA-15; M-CMB-06
Method titles	Invariant extraction and invariant comparison; Special substitution and parameter freezing; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Auxiliary function with derivative sign; Convexity, Jensen, tangent, and chord inequalities; Recurrence transformed to monotone quotient or telescoping form; Generating function encoding
Theory dependencies	Integral inequalities, integration by parts, and weighted estimates; Sequence recurrences, monotone bounds, and generating functions; Series convergence, rearrangements, and alternating/Dirichlet tools
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: integral/limit trigger | audit closures: AUD-RA-08; AUD-PC-13; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2018-D2-P8 (PDF p. 164) *counting / enumeration*

Primary domain	Algebra / number theory
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Double counting ordered or weighted objects; 3. Reduction modulo a prime or finite field
Method IDs	M-CMB-01; M-CMB-02; M-ANT-01; M-ANT-03; M-CMB-10; M-RA-09; M-CMB-05; M-CMB-03
Method titles	Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Explicit bijection and encoding-decoding proof; Convexity, Jensen, tangent, and chord inequalities; State recurrence or automaton partition; Graph encoding of a relation
Theory dependencies	Pigeonhole principle and extremal element selection; Double counting and degree-sum methods; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: double counting / pigeonhole | statement triggers: det/rank/eigen trigger; integer/prime/mod trigger; counting/permutation trigger; geometry/metric trigger | audit closures: AUD-RA-08; AUD-CMB-03; AUD-CMB-07; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2018-D2-P9 (PDF p. 164) *classification / find all*

Primary domain	Complex analysis / polynomials
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Factorization and unique factorization in Z[x]
Method IDs	M-ANT-16; M-ANT-06; M-ANT-05; M-ANT-08; M-FE-03; M-CX-01; M-UNI-07; M-UNI-08
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Factorization and unique factorization in Z[x]; Roots of unity and cyclotomic irreducibility; Composition, inversion, and conjugacy; Cauchy coefficient formula and boundary maximum; Special substitution and parameter freezing; Sign, zero, and boundary case splitting
Theory dependencies	Polynomial identity and more-roots-than-degree principle; Factorization, irreducibility, and Gauss-style integer polynomial arguments; Vieta, logarithmic derivatives, and real-root inequalities
Cue extraction	prior broad method tags: polynomial identity / factorization | statement triggers: det/rank/eigen trigger; polynomial/root trigger | audit closures: AUD-PC-01; AUD-PC-05; AUD-PC-08; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2018-D2-P10 (PDF p. 164) *limit / asymptotic evaluation*

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Riemann sum and sum-integral conversion
Method IDs	M-UNI-07; M-UNI-09; M-RA-06; M-RA-07; M-RA-01; M-RA-08; M-RA-05; M-LA-03
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Auxiliary function with derivative sign; Compactness plus extreme value theorem; L-Hospital and ratio/growth comparison; Block decomposition and Schur-style bookkeeping
Theory dependencies	Riemann sums and sum-integral asymptotics; L'Hospital, ratio limits, and growth comparisons; Taylor expansion, local asymptotics, and little-o bookkeeping
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: integral/limit trigger | audit closures: AUD-RA-03; AUD-RA-05
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.26 2019

18.26.1 Day 1

2019-D1-P1 (PDF p. 165) *exact computation*

Primary domain	Real analysis / calculus
Core method sequence	1. Invariant extraction and invariant comparison; 2. Special substitution and parameter freezing; 3. Auxiliary function with derivative sign
Method IDs	M-UNI-09; M-UNI-07; M-RA-01; M-RA-08; M-UNI-02; M-RA-03; M-FE-01; M-FE-02
Method titles	Invariant extraction and invariant comparison; Special substitution and parameter freezing; Auxiliary function with derivative sign; Compactness plus extreme value theorem; Small cases, equality cases, and example probing; Rolle chain and divided-difference localization; Iterates, fixed points, and cluster-point trapping; Monotonicity forced by functional equation
Theory dependencies	Block matrices, Schur complements, and determinant identities; Auxiliary functions, integrating factors, and differential inequalities; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: wording-specific | audit closures: AUD-RA-01; AUD-RA-06
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2019-D1-P2 (PDF p. 165) *classification / find all*

Primary domain	Algebra / number theory
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Reduction modulo a prime or finite field
Method IDs	M-UNI-07; M-UNI-09; M-ANT-01; M-ANT-03; M-UNI-08; M-ANT-06; M-ANT-16; M-ANT-02
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Sign, zero, and boundary case splitting; Degree versus number of roots; Polynomial identity by coefficient comparison; Chinese remainder and residue construction
Theory dependencies	Modular arithmetic and divisibility; GCD, valuations, prime powers, and coprimality; Group basics: orders, cyclic subgroups, and generators
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: wording-specific | audit closures: AUD-ANT-02; AUD-PC-01
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2019-D1-P3 (PDF p. 165) *inequality / sharp bound*

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Auxiliary function with derivative sign
Method IDs	M-UNI-07; M-UNI-09; M-RA-01; M-RA-09; M-RA-08; M-UNI-03; M-LA-03; M-LA-04
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Auxiliary function with derivative sign; Convexity, Jensen, tangent, and chord inequalities; Compactness plus extreme value theorem; Extremal construction and equality-case engineering; Block decomposition and Schur-style bookkeeping; Rank-nullity and dimension-count obstruction
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Integral inequalities, integration by parts, and weighted estimates; Auxiliary functions, integrating factors, and differential inequalities
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: functional-equation trigger | audit closures: AUD-RA-01; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2019-D1-P4 (PDF p. 165) *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Invariant extraction and invariant comparison; 2. Special substitution and parameter freezing; 3. Reduction modulo a prime or finite field
Method IDs	M-UNI-09; M-UNI-07; M-ANT-01; M-ANT-03; M-RA-09; M-RA-15; M-CMB-06; M-UNI-05
Method titles	Invariant extraction and invariant comparison; Special substitution and parameter freezing; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Convexity, Jensen, tangent, and chord inequalities; Recurrence transformed to monotone quotient or telescoping form; Generating function encoding; Induction with strengthened hypothesis
Theory dependencies	Sequence recurrences, monotone bounds, and generating functions; Convexity, concavity, Jensen, and tangent/chord inequalities; Modular arithmetic and divisibility
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: integer/prime/mod trigger; functional-equation trigger | audit closures: AUD-RA-08; AUD-PC-13; AUD-CMB-07; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2019-D1-P5 (PDF p. 165) *proof / reasoning*

Primary domain	Algebra / number theory
Core method sequence	1. Row/column elimination to triangular or sparse form; 2. Block decomposition and Schur-style bookkeeping; 3. Reduction modulo a prime or finite field
Method IDs	M-LA-01; M-LA-03; M-ANT-01; M-ANT-03; M-LA-13; M-UNI-07; M-UNI-09; M-RA-09
Method titles	Row/column elimination to triangular or sparse form; Block decomposition and Schur-style bookkeeping; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Determinant through eigenvalue or root product; Special substitution and parameter freezing; Invariant extraction and invariant comparison; Convexity, Jensen, tangent, and chord inequalities
Theory dependencies	Determinant evaluation by row/column operations and triangular forms; Block matrices, Schur complements, and determinant identities; Rank-nullity, image/kernel, and rank of factorizations
Cue extraction	prior broad method tags: determinant / block manipulation | statement triggers: matrix word; det/rank/eigen trigger; integer/prime/mod trigger | audit closures: AUD-RA-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.26.2 Day 2

2019-D2-P6 (PDF p. 166) *existence / construction*

Primary domain	Functional equations / dynamics
Core method sequence	1. Extremal construction and equality-case engineering; 2. Sign, zero, and boundary case splitting; 3. Auxiliary function with derivative sign
Method IDs	M-UNI-03; M-UNI-08; M-RA-01; M-RA-03; M-FE-02; M-UNI-07; M-RA-11; M-UNI-06
Method titles	Extremal construction and equality-case engineering; Sign, zero, and boundary case splitting; Auxiliary function with derivative sign; Rolle chain and divided-difference localization; Monotonicity forced by functional equation; Special substitution and parameter freezing; Integration by parts, Fubini, and weighted identities; Compactness or extremal-point selection
Theory dependencies	Integral inequalities, integration by parts, and weighted estimates; Explicit construction and counterexample design; Series convergence, rearrangements, and alternating/Dirichlet tools
Cue extraction	prior broad method tags: construction / counterexample | statement triggers: continuous/differentiable trigger; integral/limit trigger; functional-equation trigger | audit closures: AUD-RA-01; AUD-RA-06
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2019-D2-P7 (PDF p. 166) *proof / reasoning*

Primary domain	Algebra / number theory
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Reduction modulo a prime or finite field
Method IDs	M-UNI-07; M-UNI-09; M-ANT-01; M-ANT-03; M-RA-06; M-RA-07; M-RA-11; M-ANT-06
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Integration by parts, Fubini, and weighted identities; Degree versus number of roots
Theory dependencies	Modular arithmetic and divisibility; Finite fields, field characteristic, and Fermat/Frobenius tools; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: det/rank/eigen trigger; integral/limit trigger; integer/prime/mod trigger | audit closures: AUD-PC-01; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2019-D2-P8 (PDF p. 166) *counting / enumeration*

Primary domain	Combinatorics / discrete structures
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Double counting ordered or weighted objects; 3. Explicit bijection and encoding-decoding proof
Method IDs	M-CMB-01; M-CMB-02; M-CMB-10; M-CMB-03; M-CMB-08; M-CMB-05; M-CMB-04; M-CMB-07
Method titles	Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Explicit bijection and encoding-decoding proof; Graph encoding of a relation; Extremal set or minimal counterexample induction; State recurrence or automaton partition; Cycle extraction and alternating-cycle method; Involution, parity, and sign-reversing cancellation
Theory dependencies	Pigeonhole principle and extremal element selection; Double counting and degree-sum methods; Graph encoding, cycles, and bipartite representations
Cue extraction	prior broad method tags: double counting / pigeonhole | statement triggers: integral/limit trigger; counting/permutation trigger; graph/color/set trigger | audit closures: AUD-CMB-03; AUD-CMB-05; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2019-D2-P9 (PDF p. 166) *classification / find all*

Primary domain	Algebra / number theory
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Reduction modulo a prime or finite field
Method IDs	M-UNI-07; M-UNI-09; M-ANT-01; M-ANT-03; M-LA-02; M-LA-07; M-LA-12; M-LA-03
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Complete a matrix equation to an inverse identity; Trace cyclicity and commutator annihilation; Nilpotency from zero spectrum plus finite dimension; Block decomposition and Schur-style bookkeeping
Theory dependencies	Modular arithmetic and divisibility; Rank-nullity, image/kernel, and rank of factorizations; GCD, valuations, prime powers, and coprimality
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: matrix word; det/rank/eigen trigger; integer/prime/mod trigger | audit closures: AUD-LA-03; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2019-D2-P10 (PDF p. 166) *counting / enumeration*

Primary domain	Probability / measure / expectation
Core method sequence	1. Indicator and expectation method; 2. Convexity, Jensen, tangent, and chord inequalities; 3. Graph encoding of a relation
Method IDs	M-CMB-13; M-RA-09; M-CMB-03; M-CMB-04; M-GEO-11; M-GEO-01; M-CMB-10; M-RA-11
Method titles	Indicator and expectation method; Convexity, Jensen, tangent, and chord inequalities; Graph encoding of a relation; Cycle extraction and alternating-cycle method; Volume, projection, and ellipsoid computation; Coordinate normalization and algebraization; Explicit bijection and encoding-decoding proof; Integration by parts, Fubini, and weighted identities
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Graph encoding, cycles, and bipartite representations; Measure and probability-integral estimates
Cue extraction	prior broad method tags: convexity / Jensen / tangent, graph encoding / cycles, probabilistic / expectation | statement triggers: graph/color/set trigger; geometry/metric trigger | audit closures: AUD-RA-08; AUD-CMB-03; AUD-GEO-02; AUD-PRB-01; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.27 2020

18.27.1 Day 1

2020-D1-P1 (PDF p. 167) *exact computation*

Primary domain	Algebra / number theory
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Double counting ordered or weighted objects; 3. Extremal construction and equality-case engineering
Method IDs	M-CMB-01; M-CMB-02; M-UNI-03; M-UNI-08; M-ANT-01; M-ANT-03; M-UNI-02; M-UNI-09
Method titles	Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Extremal construction and equality-case engineering; Sign, zero, and boundary case splitting; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Small cases, equality cases, and example probing; Invariant extraction and invariant comparison
Theory dependencies	Pigeonhole principle and extremal element selection; Double counting and degree-sum methods; Explicit construction and counterexample design
Cue extraction	prior broad method tags: double counting / pigeonhole, construction / counterexample | statement triggers: integer/prime/mod trigger; counting/permutation trigger | audit closures: AUD-RA-08; AUD-CMB-03; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2020-D1-P2 (PDF p. 167) *inequality / sharp bound*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Rank-nullity and dimension-count obstruction; 2. Trace cyclicity and commutator annihilation; 3. Block decomposition and Schur-style bookkeeping
Method IDs	M-LA-04; M-LA-07; M-LA-03; M-RA-08; M-CMB-01; M-CMB-02; M-LA-01; M-UNI-06
Method titles	Rank-nullity and dimension-count obstruction; Trace cyclicity and commutator annihilation; Block decomposition and Schur-style bookkeeping; Compactness plus extreme value theorem; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Row/column elimination to triangular or sparse form; Compactness or extremal-point selection
Theory dependencies	Rank-nullity, image/kernel, and rank of factorizations; Block matrices, Schur complements, and determinant identities; Determinant evaluation by row/column operations and triangular forms
Cue extraction	prior broad method tags: compactness / extremal point, trace trick / commutator, determinant / block manipulation | statement triggers: matrix word; det/rank/eigen trigger; integral/limit trigger; counting/permutation trigger | audit closures: AUD-LA-03; AUD-LA-08; AUD-RA-07; AUD-ANT-06; AUD-CMB-03
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2020-D1-P3 (PDF p. 167) *existence / construction*

Primary domain	Algebra / number theory
Core method sequence	1. Convexity, Jensen, tangent, and chord inequalities; 2. Graph encoding of a relation; 3. Cycle extraction and alternating-cycle method
Method IDs	M-RA-09; M-CMB-03; M-CMB-04; M-ANT-01; M-ANT-03; M-GEO-01; M-GEO-11; M-UNI-03
Method titles	Convexity, Jensen, tangent, and chord inequalities; Graph encoding of a relation; Cycle extraction and alternating-cycle method; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Coordinate normalization and algebraization; Volume, projection, and ellipsoid computation; Extremal construction and equality-case engineering
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Graph encoding, cycles, and bipartite representations; Coloring, partitions, and parity
Cue extraction	prior broad method tags: convexity / Jensen / tangent, graph encoding / cycles | statement triggers: integer/prime/mod trigger; graph/color/set trigger; geometry/metric trigger | audit closures: AUD-RA-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2020-D1-P4 (PDF p. 167) *inequality / sharp bound*

Primary domain	Real analysis / calculus
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Factorization and unique factorization in Z[x]
Method IDs	M-ANT-16; M-ANT-06; M-ANT-05; M-RA-09; M-RA-01; M-RA-08; M-UNI-03; M-FE-03
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Factorization and unique factorization in Z[x]; Convexity, Jensen, tangent, and chord inequalities; Auxiliary function with derivative sign; Compactness plus extreme value theorem; Extremal construction and equality-case engineering; Composition, inversion, and conjugacy
Theory dependencies	Polynomial identity and more-roots-than-degree principle; Convexity, concavity, Jensen, and tangent/chord inequalities; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: polynomial identity / factorization | statement triggers: polynomial/root trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-PC-08; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.27.2 Day 2

2020-D2-P5 (PDF p. 168) *classification / find all*

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Auxiliary function with derivative sign
Method IDs	M-UNI-07; M-UNI-09; M-RA-01; M-RA-03; M-RA-09; M-RA-08; M-UNI-08; M-LA-03
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Auxiliary function with derivative sign; Rolle chain and divided-difference localization; Convexity, Jensen, tangent, and chord inequalities; Compactness plus extreme value theorem; Sign, zero, and boundary case splitting; Block decomposition and Schur-style bookkeeping
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Integral inequalities, integration by parts, and weighted estimates; Block matrices, Schur complements, and determinant identities
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: continuous/differentiable trigger; functional-equation trigger | audit closures: AUD-RA-01; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2020-D2-P6 (PDF p. 168) *classification / find all*

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Chinese remainder and residue construction; 3. GCD, valuation, and prime-power bookkeeping
Method IDs	M-ANT-01; M-ANT-02; M-ANT-03; M-UNI-07; M-UNI-08; M-ANT-06; M-ANT-16; M-ANT-09
Method titles	Reduction modulo a prime or finite field; Chinese remainder and residue construction; GCD, valuation, and prime-power bookkeeping; Special substitution and parameter freezing; Sign, zero, and boundary case splitting; Degree versus number of roots; Polynomial identity by coefficient comparison; Finite-field function representative and permutation argument
Theory dependencies	Modular arithmetic and divisibility; Finite fields, field characteristic, and Fermat/Frobenius tools; Chinese remainder theorem and complete residue systems
Cue extraction	prior broad method tags: modular / finite-field reduction | statement triggers: integer/prime/mod trigger | audit closures: AUD-PC-01; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2020-D2-P7 (PDF p. 168) *existence / construction*

Primary domain	Algebra / number theory
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Double counting ordered or weighted objects; 3. Auxiliary function with derivative sign
Method IDs	M-CMB-01; M-CMB-02; M-RA-01; M-RA-02; M-UNI-07; M-ANT-01; M-ANT-10; M-ANT-03
Method titles	Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Auxiliary function with derivative sign; Integrating factor and ODE comparison; Special substitution and parameter freezing; Reduction modulo a prime or finite field; Group action, orbit, and generator method; GCD, valuation, and prime-power bookkeeping
Theory dependencies	Group basics: orders, cyclic subgroups, and generators; Pigeonhole principle and extremal element selection; Double counting and degree-sum methods
Cue extraction	prior broad method tags: auxiliary function / substitution, double counting / pigeonhole | statement triggers: integer/prime/mod trigger; group/ring/field trigger; counting/permutation trigger | audit closures: AUD-RA-08; AUD-CMB-03; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2020-D2-P8 (PDF p. 168) *limit / asymptotic evaluation*

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Riemann sum and sum-integral conversion
Method IDs	M-UNI-07; M-UNI-09; M-RA-06; M-RA-07; M-RA-01; M-RA-08; M-RA-05; M-LA-03
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Auxiliary function with derivative sign; Compactness plus extreme value theorem; L-Hospital and ratio/growth comparison; Block decomposition and Schur-style bookkeeping
Theory dependencies	Riemann sums and sum-integral asymptotics; L'Hospital, ratio limits, and growth comparisons; Taylor expansion, local asymptotics, and little-o bookkeeping
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: integral/limit trigger | audit closures: AUD-RA-03; AUD-RA-05
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.28 2021

18.28.1 Day 1

2021-D1-P1 (PDF p. 169) *proof of structural property*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Invariant extraction and invariant comparison; 2. Special substitution and parameter freezing; 3. Rank-nullity and dimension-count obstruction
Method IDs	M-UNI-09; M-UNI-07; M-LA-04; M-LA-06; M-LA-16; M-LA-07; M-LA-13; M-UNI-05
Method titles	Invariant extraction and invariant comparison; Special substitution and parameter freezing; Rank-nullity and dimension-count obstruction; Spectral/minimal-polynomial reduction; Trace pairing and orthogonal matrix subspaces; Trace cyclicity and commutator annihilation; Determinant through eigenvalue or root product; Induction with strengthened hypothesis
Theory dependencies	Rank-nullity, image/kernel, and rank of factorizations; Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; Determinant evaluation by row/column operations and triangular forms
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: matrix word | audit closures: AUD-LA-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2021-D1-P2 (PDF p. 169) *counting / enumeration*

Primary domain	Algebra / number theory
Core method sequence	1. Indicator and expectation method; 2. Reduction modulo a prime or finite field; 3. GCD, valuation, and prime-power bookkeeping
Method IDs	M-CMB-13; M-ANT-01; M-ANT-03; M-CMB-01; M-CMB-02; M-CMB-10; M-CMB-03; M-CMB-05
Method titles	Indicator and expectation method; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Explicit bijection and encoding-decoding proof; Graph encoding of a relation; State recurrence or automaton partition
Theory dependencies	Graph encoding, cycles, and bipartite representations; Coloring, partitions, and parity; Modular arithmetic and divisibility
Cue extraction	prior broad method tags: probabilistic / expectation | statement triggers: integer/prime/mod trigger; graph/color/set trigger | audit closures: AUD-CMB-03; AUD-PRB-01; AUD-CMB-07; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2021-D1-P3 (PDF p. 169) *existence / construction*

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Auxiliary function with derivative sign
Method IDs	M-UNI-07; M-UNI-09; M-RA-01; M-RA-15; M-CMB-06; M-RA-08; M-UNI-03; M-UNI-06
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Auxiliary function with derivative sign; Recurrence transformed to monotone quotient or telescoping form; Generating function encoding; Compactness plus extreme value theorem; Extremal construction and equality-case engineering; Compactness or extremal-point selection
Theory dependencies	Sequence recurrences, monotone bounds, and generating functions; Generating functions for counting; Block matrices, Schur complements, and determinant identities
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: geometry/metric trigger | audit closures: AUD-PC-13; AUD-ANT-12; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2021-D1-P4 (PDF p. 169) *existence / construction*

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Auxiliary function with derivative sign
Method IDs	M-UNI-07; M-UNI-09; M-RA-01; M-RA-05; M-RA-06; M-RA-07; M-RA-08; M-UNI-03
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Auxiliary function with derivative sign; L-Hospital and ratio/growth comparison; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Compactness plus extreme value theorem; Extremal construction and equality-case engineering
Theory dependencies	Riemann sums and sum-integral asymptotics; L'Hospital, ratio limits, and growth comparisons; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: continuous/differentiable trigger; integral/limit trigger; functional-equation trigger | audit closures: AUD-RA-03; AUD-RA-05
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.28.2 Day 2

2021-D2-P5 (PDF p. 170) *existence / construction*

Primary domain	Algebra / number theory
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Reduction modulo a prime or finite field
Method IDs	M-UNI-07; M-UNI-09; M-ANT-01; M-ANT-03; M-RA-09; M-UNI-03; M-UNI-06; M-LA-05
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Convexity, Jensen, tangent, and chord inequalities; Extremal construction and equality-case engineering; Compactness or extremal-point selection; Similarity, adapted basis, and canonical model
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Modular arithmetic and divisibility; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: matrix word; det/rank/eigen trigger; integer/prime/mod trigger | audit closures: AUD-LA-11; AUD-RA-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2021-D2-P6 (PDF p. 170)

proof of structural property

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Chinese remainder and residue construction; 3. Auxiliary function with derivative sign
Method IDs	M-ANT-01; M-ANT-02; M-RA-01; M-RA-02; M-UNI-07; M-ANT-10; M-ANT-03; M-ANT-11
Method titles	Reduction modulo a prime or finite field; Chinese remainder and residue construction; Auxiliary function with derivative sign; Integrating factor and ODE comparison; Special substitution and parameter freezing; Group action, orbit, and generator method; GCD, valuation, and prime-power bookkeeping; Commutator and center manipulation in groups/rings
Theory dependencies	Modular arithmetic and divisibility; Finite fields, field characteristic, and Fermat/Frobenius tools; Group basics: orders, cyclic subgroups, and generators
Cue extraction	prior broad method tags: auxiliary function / substitution, modular / finite-field reduction | statement triggers: matrix word; integer/prime/mod trigger; group/ring/field trigger; counting/permutation trigger | audit closures: AUD-PC-01; AUD-PC-06; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2021-D2-P7 (PDF p. 170)

inequality / sharp bound

Primary domain	Complex analysis / polynomials
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Factorization and unique factorization in Z[x]
Method IDs	M-ANT-16; M-ANT-06; M-ANT-05; M-RA-16; M-CX-01; M-CX-05; M-RA-09; M-ANT-08
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Factorization and unique factorization in Z[x]; Density, Baire/category, and pathological construction; Cauchy coefficient formula and boundary maximum; Trigonometric-to-complex conversion; Convexity, Jensen, tangent, and chord inequalities; Roots of unity and cyclotomic irreducibility
Theory dependencies	Polynomial identity and more-roots-than-degree principle; Convexity, concavity, Jensen, and tangent/chord inequalities; Vieta, logarithmic derivatives, and real-root inequalities
Cue extraction	prior broad method tags: polynomial identity / factorization, topological / density argument | statement triggers: polynomial/root trigger; functional-equation trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-PC-05; AUD-PC-08; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2021-D2-P8 (PDF p. 170)

counting / enumeration

Primary domain	Algebra / number theory
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Reduction modulo a prime or finite field
Method IDs	M-UNI-07; M-UNI-09; M-ANT-01; M-ANT-03; M-GEO-03; M-CMB-01; M-CMB-02; M-RA-10
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Squared-distance and inner-product identities; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Cauchy-Schwarz, Holder, and norm comparison
Theory dependencies	Pigeonhole principle and extremal element selection; Modular arithmetic and divisibility; Double counting and degree-sum methods
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: integer/prime/mod trigger; counting/permutation trigger | audit closures: AUD-RA-08; AUD-CMB-03; AUD-GEO-04; AUD-CMB-07; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.29 2022

18.29.1 Day 1

2022-D1-P1 **(PDF p. 171)** *inequality / sharp bound*

Primary domain	Probability / measure / expectation
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Convexity, Jensen, tangent, and chord inequalities
Method IDs	M-UNI-07; M-UNI-09; M-RA-09; M-RA-11; M-CMB-13; M-UNI-03; M-CMB-06; M-LA-03
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Convexity, Jensen, tangent, and chord inequalities; Integration by parts, Fubini, and weighted identities; Indicator and expectation method; Extremal construction and equality-case engineering; Generating function encoding; Block decomposition and Schur-style bookkeeping
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Measure and probability-integral estimates; Integral inequalities, integration by parts, and weighted estimates
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: functional-equation trigger | audit closures: AUD-RA-08; AUD-PRB-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2022-D1-P2 **(PDF p. 171)** *classification / find all*

Primary domain	Algebra / number theory
Core method sequence	1. Spectral/minimal-polynomial reduction; 2. Nilpotency from zero spectrum plus finite dimension; 3. Reduction modulo a prime or finite field
Method IDs	M-LA-06; M-LA-12; M-ANT-01; M-ANT-03; M-RA-09; M-UNI-07; M-UNI-08; M-LA-10
Method titles	Spectral/minimal-polynomial reduction; Nilpotency from zero spectrum plus finite dimension; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Convexity, Jensen, tangent, and chord inequalities; Special substitution and parameter freezing; Sign, zero, and boundary case splitting; Linear operator on a matrix space
Theory dependencies	Eigenvalues, characteristic/minimal polynomials, and Cayley-Hamilton; Convexity, concavity, Jensen, and tangent/chord inequalities; Modular arithmetic and divisibility
Cue extraction	prior broad method tags: spectral decomposition / eigenvalues | statement triggers: matrix word; det/rank/eigen trigger; integer/prime/mod trigger | audit closures: AUD-LA-10; AUD-RA-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2022-D1-P3 **(PDF p. 171)** *counting / enumeration*

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Chinese remainder and residue construction; 3. Pigeonhole threshold and collision forcing
Method IDs	M-ANT-01; M-ANT-02; M-CMB-01; M-CMB-02; M-UNI-03; M-UNI-08; M-ANT-03; M-CMB-10
Method titles	Reduction modulo a prime or finite field; Chinese remainder and residue construction; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Extremal construction and equality-case engineering; Sign, zero, and boundary case splitting; GCD, valuation, and prime-power bookkeeping; Explicit bijection and encoding-decoding proof
Theory dependencies	Modular arithmetic and divisibility; Finite fields, field characteristic, and Fermat/Frobenius tools; Pigeonhole principle and extremal element selection
Cue extraction	prior broad method tags: modular / finite-field reduction, double counting / pigeonhole, construction / counterexample | statement triggers: integer/prime/mod trigger; counting/permutation trigger; functional-equation trigger | audit closures: AUD-PC-01; AUD-ANT-02; AUD-CMB-03; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2022-D1-P4 (PDF p. 171) *inequality / sharp bound*

Primary domain	Algebra / number theory
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Double counting ordered or weighted objects; 3. Reduction modulo a prime or finite field
Method IDs	M-CMB-01; M-CMB-02; M-ANT-01; M-ANT-03; M-RA-09; M-CMB-03; M-UNI-03; M-CMB-04
Method titles	Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Convexity, Jensen, tangent, and chord inequalities; Graph encoding of a relation; Extremal construction and equality-case engineering; Cycle extraction and alternating-cycle method
Theory dependencies	Pigeonhole principle and extremal element selection; Double counting and degree-sum methods; Coloring, partitions, and parity
Cue extraction	prior broad method tags: double counting / pigeonhole | statement triggers: integer/prime/mod trigger; counting/permutation trigger; graph/color/set trigger | audit closures: AUD-RA-08; AUD-CMB-03; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.29.2 Day 2

2022-D2-P5 (PDF p. 172) *counting / enumeration*

Primary domain	Combinatorics / discrete structures
Core method sequence	1. Graph encoding of a relation; 2. Cycle extraction and alternating-cycle method; 3. Volume, projection, and ellipsoid computation
Method IDs	M-CMB-03; M-CMB-04; M-GEO-11; M-GEO-01; M-LA-06; M-CMB-01; M-CMB-02; M-CMB-10
Method titles	Graph encoding of a relation; Cycle extraction and alternating-cycle method; Volume, projection, and ellipsoid computation; Coordinate normalization and algebraization; Spectral/minimal-polynomial reduction; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Explicit bijection and encoding-decoding proof
Theory dependencies	Graph encoding, cycles, and bipartite representations; Coloring, partitions, and parity; Pigeonhole principle and extremal element selection
Cue extraction	prior broad method tags: graph encoding / cycles | statement triggers: counting/permutation trigger; graph/color/set trigger | audit closures: AUD-RA-08; AUD-CMB-03; AUD-GEO-02; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2022-D2-P6 (PDF p. 172)

proof of structural property

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Chinese remainder and residue construction; 3. GCD, valuation, and prime-power bookkeeping
Method IDs	M-ANT-01; M-ANT-02; M-ANT-03; M-ANT-10; M-CMB-07; M-UNI-05; M-UNI-09; M-ANT-06
Method titles	Reduction modulo a prime or finite field; Chinese remainder and residue construction; GCD, valuation, and prime-power bookkeeping; Group action, orbit, and generator method; Involution, parity, and sign-reversing cancellation; Induction with strengthened hypothesis; Invariant extraction and invariant comparison; Degree versus number of roots
Theory dependencies	Modular arithmetic and divisibility; Finite fields, field characteristic, and Fermat/Frobenius tools; Chinese remainder theorem and complete residue systems
Cue extraction	prior broad method tags: modular / finite-field reduction | statement triggers: integer/prime/mod trigger; counting/permutation trigger | audit closures: AUD-PC-01; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2022-D2-P7 (PDF p. 172)

inequality / sharp bound

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Rank-nullity and dimension-count obstruction; 2. Spectral/minimal-polynomial reduction; 3. Nilpotency from zero spectrum plus finite dimension
Method IDs	M-LA-04; M-LA-06; M-LA-12; M-RA-09; M-LA-01; M-LA-03; M-LA-07; M-UNI-03
Method titles	Rank-nullity and dimension-count obstruction; Spectral/minimal-polynomial reduction; Nilpotency from zero spectrum plus finite dimension; Convexity, Jensen, tangent, and chord inequalities; Row/column elimination to triangular or sparse form; Block decomposition and Schur-style bookkeeping; Trace cyclicity and commutator annihilation; Extremal construction and equality-case engineering
Theory dependencies	Rank-nullity, image/kernel, and rank of factorizations; Determinant evaluation by row/column operations and triangular forms; Nilpotent, idempotent, involution, and square-zero structures
Cue extraction	prior broad method tags: rank-nullity / dimension count | statement triggers: matrix word; det/rank/eigen trigger | audit closures: AUD-LA-02; AUD-LA-03; AUD-RA-08; AUD-PC-01; AUD-PC-05
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2022-D2-P8 (PDF p. 172)

counting / enumeration

Primary domain	Algebra / number theory
Core method sequence	1. Convexity, Jensen, tangent, and chord inequalities; 2. Indicator and expectation method; 3. Graph encoding of a relation
Method IDs	M-RA-09; M-CMB-13; M-CMB-03; M-CMB-01; M-CMB-02; M-CMB-04; M-ANT-01; M-ANT-03
Method titles	Convexity, Jensen, tangent, and chord inequalities; Indicator and expectation method; Graph encoding of a relation; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Cycle extraction and alternating-cycle method; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Graph encoding, cycles, and bipartite representations; Pigeonhole principle and extremal element selection
Cue extraction	prior broad method tags: convexity / Jensen / tangent, double counting / pigeonhole, graph encoding / cycles | statement triggers: integer/prime/mod trigger; counting/permutation trigger; graph/color/set trigger; geometry/metric trigger | audit closures: AUD-RA-08; AUD-CMB-03; AUD-GEO-02; AUD-PRB-01; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.30 2023

18.30.1 Day 1

2023-D1-P1 (PDF p. 173) *classification / find all*

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Auxiliary function with derivative sign
Method IDs	M-UNI-07; M-UNI-09; M-RA-01; M-RA-03; M-RA-08; M-UNI-08; M-LA-03; M-LA-15
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Auxiliary function with derivative sign; Rolle chain and divided-difference localization; Compactness plus extreme value theorem; Sign, zero, and boundary case splitting; Block decomposition and Schur-style bookkeeping; AB versus BA transfer
Theory dependencies	Rolle, mean-value theorem, and divided differences; Taylor expansion, local asymptotics, and little-o bookkeeping; Intermediate value, connectedness, density, and continuity transfer
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: continuous/differentiable trigger; functional-equation trigger | audit closures: AUD-RA-06
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2023-D1-P2 (PDF p. 173) *proof of structural property*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Invariant extraction and invariant comparison; 2. Special substitution and parameter freezing; 3. Rolle chain and divided-difference localization
Method IDs	M-UNI-09; M-UNI-07; M-RA-03; M-RA-01; M-LA-04; M-LA-06; M-ANT-16; M-ANT-08
Method titles	Invariant extraction and invariant comparison; Special substitution and parameter freezing; Rolle chain and divided-difference localization; Auxiliary function with derivative sign; Rank-nullity and dimension-count obstruction; Spectral/minimal-polynomial reduction; Polynomial identity by coefficient comparison; Roots of unity and cyclotomic irreducibility
Theory dependencies	Taylor expansion, local asymptotics, and little-o bookkeeping; Rolle, mean-value theorem, and divided differences; Rank-nullity, image/kernel, and rank of factorizations
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: matrix word; continuous/differentiable trigger | audit closures: AUD-RA-06; AUD-PC-01; AUD-PC-05; AUD-PC-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2023-D1-P3 (PDF p. 173) *classification / find all*

Primary domain	Complex analysis / polynomials
Core method sequence	1. Polynomial identity by coefficient comparison; 2. Degree versus number of roots; 3. Factorization and unique factorization in Z[x]
Method IDs	M-ANT-16; M-ANT-06; M-ANT-05; M-CX-01; M-FE-03; M-ANT-07; M-UNI-07; M-UNI-08
Method titles	Polynomial identity by coefficient comparison; Degree versus number of roots; Factorization and unique factorization in Z[x]; Cauchy coefficient formula and boundary maximum; Composition, inversion, and conjugacy; Vieta, symmetric sums, and logarithmic derivative; Special substitution and parameter freezing; Sign, zero, and boundary case splitting
Theory dependencies	Polynomial identity and more-roots-than-degree principle; Vieta, logarithmic derivatives, and real-root inequalities; Factorization, irreducibility, and Gauss-style integer polynomial arguments
Cue extraction	prior broad method tags: polynomial identity / factorization | statement triggers: polynomial/root trigger | audit closures: AUD-PC-01; AUD-PC-08; AUD-PC-13
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2023-D1-P4 (PDF p. 173) *proof / reasoning*

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Chinese remainder and residue construction; 3. GCD, valuation, and prime-power bookkeeping
Method IDs	M-ANT-01; M-ANT-02; M-ANT-03; M-ANT-09; M-UNI-07; M-UNI-09; M-ANT-06; M-ANT-16
Method titles	Reduction modulo a prime or finite field; Chinese remainder and residue construction; GCD, valuation, and prime-power bookkeeping; Finite-field function representative and permutation argument; Special substitution and parameter freezing; Invariant extraction and invariant comparison; Degree versus number of roots; Polynomial identity by coefficient comparison
Theory dependencies	Modular arithmetic and divisibility; Finite fields, field characteristic, and Fermat/Frobenius tools; Chinese remainder theorem and complete residue systems
Cue extraction	prior broad method tags: modular / finite-field reduction | statement triggers: integer/prime/mod trigger | audit closures: AUD-PC-01; AUD-PC-06; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2023-D1-P5 (PDF p. 173) *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Invariant extraction and invariant comparison; 2. Special substitution and parameter freezing; 3. Reduction modulo a prime or finite field
Method IDs	M-UNI-09; M-UNI-07; M-ANT-01; M-ANT-03; M-CMB-07; M-ANT-10; M-UNI-05; M-RA-09
Method titles	Invariant extraction and invariant comparison; Special substitution and parameter freezing; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Involution, parity, and sign-reversing cancellation; Group action, orbit, and generator method; Induction with strengthened hypothesis; Convexity, Jensen, tangent, and chord inequalities
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Modular arithmetic and divisibility; Graph encoding, cycles, and bipartite representations
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: integer/prime/mod trigger; counting/permutation trigger | audit closures: AUD-RA-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.30.2 Day 2

2023-D2-P6 (PDF p. 174) *proof / reasoning*

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Block decomposition and Schur-style bookkeeping; 2. Row/column elimination to triangular or sparse form; 3. Rank-nullity and dimension-count obstruction
Method IDs	M-LA-03; M-LA-01; M-LA-04; M-LA-06; M-LA-16; M-LA-13; M-UNI-07; M-UNI-09
Method titles	Block decomposition and Schur-style bookkeeping; Row/column elimination to triangular or sparse form; Rank-nullity and dimension-count obstruction; Spectral/minimal-polynomial reduction; Trace pairing and orthogonal matrix subspaces; Determinant through eigenvalue or root product; Special substitution and parameter freezing; Invariant extraction and invariant comparison
Theory dependencies	Determinant evaluation by row/column operations and triangular forms; Block matrices, Schur complements, and determinant identities; Rank-nullity, image/kernel, and rank of factorizations
Cue extraction	prior broad method tags: determinant / block manipulation | statement triggers: matrix word | audit closures: AUD-LA-03; AUD-LA-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2023-D2-P7 (PDF p. 174)

classification / find all

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Rolle chain and divided-difference localization
Method IDs	M-UNI-07; M-UNI-09; M-RA-03; M-RA-01; M-RA-08; M-UNI-08; M-CMB-03; M-CMB-04
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Rolle chain and divided-difference localization; Auxiliary function with derivative sign; Compactness plus extreme value theorem; Sign, zero, and boundary case splitting; Graph encoding of a relation; Cycle extraction and alternating-cycle method
Theory dependencies	Rolle, mean-value theorem, and divided differences; Taylor expansion, local asymptotics, and little-o bookkeeping; Graph encoding, cycles, and bipartite representations
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: det/rank/eigen trigger; continuous/differentiable trigger; functional-equation trigger | audit closures: AUD-RA-06
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2023-D2-P8 (PDF p. 174)

inequality / sharp bound

Primary domain	Combinatorics / discrete structures
Core method sequence	1. Graph encoding of a relation; 2. Cycle extraction and alternating-cycle method; 3. Pigeonhole threshold and collision forcing
Method IDs	M-CMB-03; M-CMB-04; M-CMB-01; M-CMB-02; M-RA-16; M-GEO-03; M-RA-10; M-RA-09
Method titles	Graph encoding of a relation; Cycle extraction and alternating-cycle method; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Density, Baire/category, and pathological construction; Squared-distance and inner-product identities; Cauchy-Schwarz, Holder, and norm comparison; Convexity, Jensen, tangent, and chord inequalities
Theory dependencies	Graph encoding, cycles, and bipartite representations; Pigeonhole principle and extremal element selection; Double counting and degree-sum methods
Cue extraction	prior broad method tags: double counting / pigeonhole, graph encoding / cycles, topological / density argument | statement triggers: integral/limit trigger; counting/permutation trigger; graph/color/set trigger; geometry/metric trigger | audit closures: AUD-RA-07; AUD-RA-08; AUD-CMB-03; AUD-GEO-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2023-D2-P9 (PDF p. 174)

proof / reasoning

Primary domain	Geometry / convexity / metric
Core method sequence	1. Convexity, Jensen, tangent, and chord inequalities; 2. Rank-nullity and dimension-count obstruction; 3. Density, Baire/category, and pathological construction
Method IDs	M-RA-09; M-LA-04; M-RA-16; M-GEO-01; M-GEO-11; M-CMB-01; M-CMB-02; M-GEO-03
Method titles	Convexity, Jensen, tangent, and chord inequalities; Rank-nullity and dimension-count obstruction; Density, Baire/category, and pathological construction; Coordinate normalization and algebraization; Volume, projection, and ellipsoid computation; Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Squared-distance and inner-product identities
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Area, volume, projection, and ellipsoid computation; Rank-nullity, image/kernel, and rank of factorizations
Cue extraction	prior broad method tags: convexity / Jensen / tangent, rank-nullity / dimension count, topological / density argument | statement triggers: graph/color/set trigger; geometry/metric trigger | audit closures: AUD-RA-08; AUD-GEO-02; AUD-GEO-11
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2023-D2-P10 (PDF p. 174) *existence / construction*

Primary domain	Algebra / number theory
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Reduction modulo a prime or finite field
Method IDs	M-UNI-07; M-UNI-09; M-ANT-01; M-ANT-03; M-UNI-03; M-UNI-06; M-RA-09; M-ANT-02
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Extremal construction and equality-case engineering; Compactness or extremal-point selection; Convexity, Jensen, tangent, and chord inequalities; Chinese remainder and residue construction
Theory dependencies	Modular arithmetic and divisibility; GCD, valuations, prime powers, and coprimality; Group basics: orders, cyclic subgroups, and generators
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: det/rank/eigen trigger; integer/prime/mod trigger; functional-equation trigger | audit closures: AUD-RA-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.31 2024

18.31.1 Day 1

2024-D1-P1 (PDF p. 175) *classification / find all*

Primary domain	Complex analysis / polynomials
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Polynomial identity by coefficient comparison
Method IDs	M-UNI-07; M-UNI-09; M-ANT-16; M-ANT-06; M-FE-03; M-CX-01; M-UNI-08; M-ANT-07
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Polynomial identity by coefficient comparison; Degree versus number of roots; Composition, inversion, and conjugacy; Cauchy coefficient formula and boundary maximum; Sign, zero, and boundary case splitting; Vieta, symmetric sums, and logarithmic derivative
Theory dependencies	Polynomial identity and more-roots-than-degree principle; Vieta, logarithmic derivatives, and real-root inequalities; Roots of unity, cyclotomic filters, and discrete Fourier tricks
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: det/rank/eigen trigger | audit closures: AUD-PC-08; AUD-PC-01
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2024-D1-P2 (PDF p. 175) *limit / asymptotic evaluation*

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Riemann sum and sum-integral conversion
Method IDs	M-UNI-07; M-UNI-09; M-RA-06; M-RA-07; M-RA-01; M-RA-05; M-RA-09; M-RA-08
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Auxiliary function with derivative sign; L-Hospital and ratio/growth comparison; Convexity, Jensen, tangent, and chord inequalities; Compactness plus extreme value theorem
Theory dependencies	Riemann sums and sum-integral asymptotics; L'Hospital, ratio limits, and growth comparisons; Taylor expansion, local asymptotics, and little-o bookkeeping
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: integral/limit trigger | audit closures: AUD-RA-03; AUD-RA-05; AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2024-D1-P3 (PDF p. 175) *nonexistence / counterexample*

Primary domain	Algebra / number theory
Core method sequence	1. Extremal construction and equality-case engineering; 2. Sign, zero, and boundary case splitting; 3. Reduction modulo a prime or finite field
Method IDs	M-UNI-03; M-UNI-08; M-ANT-01; M-ANT-03; M-UNI-04; M-CMB-10; M-LA-04; M-ANT-02
Method titles	Extremal construction and equality-case engineering; Sign, zero, and boundary case splitting; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Contradiction with minimal or maximal counterexample; Explicit bijection and encoding-decoding proof; Rank-nullity and dimension-count obstruction; Chinese remainder and residue construction
Theory dependencies	Explicit construction and counterexample design; Modular arithmetic and divisibility; GCD, valuations, prime powers, and coprimality
Cue extraction	prior broad method tags: construction / counterexample | statement triggers: matrix word; integer/prime/mod trigger | audit closures: AUD-LA-12; AUD-CMB-07; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2024-D1-P4 (PDF p. 175) *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Auxiliary function with derivative sign; 2. Integrating factor and ODE comparison; 3. Special substitution and parameter freezing
Method IDs	M-RA-01; M-RA-02; M-UNI-07; M-ANT-01; M-ANT-03; M-ANT-10; M-ANT-11; M-FE-01
Method titles	Auxiliary function with derivative sign; Integrating factor and ODE comparison; Special substitution and parameter freezing; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Group action, orbit, and generator method; Commutator and center manipulation in groups/rings; Iterates, fixed points, and cluster-point trapping
Theory dependencies	Group basics: orders, cyclic subgroups, and generators; Rings, idempotents, nil rings, and noncommutativity; Automorphisms and semidirect products
Cue extraction	prior broad method tags: auxiliary function / substitution | statement triggers: integer/prime/mod trigger; group/ring/field trigger | audit closures: AUD-RA-03; AUD-RA-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2024-D1-P5 (PDF p. 175) *inequality / sharp bound*

Primary domain	Algebra / number theory
Core method sequence	1. Indicator and expectation method; 2. Convexity, Jensen, tangent, and chord inequalities; 3. Reduction modulo a prime or finite field
Method IDs	M-CMB-13; M-RA-09; M-ANT-01; M-ANT-03; M-UNI-03; M-CMB-03; M-GEO-03; M-CMB-01
Method titles	Indicator and expectation method; Convexity, Jensen, tangent, and chord inequalities; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Extremal construction and equality-case engineering; Graph encoding of a relation; Squared-distance and inner-product identities; Pigeonhole threshold and collision forcing
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Area, volume, projection, and ellipsoid computation; Coloring, partitions, and parity
Cue extraction	prior broad method tags: convexity / Jensen / tangent, probabilistic / expectation | statement triggers: integral/limit trigger; integer/prime/mod trigger; geometry/metric trigger; functional-equation trigger | audit closures: AUD-RA-08; AUD-CMB-03; AUD-GEO-02; AUD-PRB-01; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

18.31.2 Day 2

2024-D2-P6 (PDF p. 176) *inequality / sharp bound*

Primary domain	Functional equations / dynamics
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Convexity, Jensen, tangent, and chord inequalities
Method IDs	M-UNI-07; M-UNI-09; M-RA-09; M-FE-02; M-UNI-03; M-FE-01; M-FE-06; M-LA-03
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Convexity, Jensen, tangent, and chord inequalities; Monotonicity forced by functional equation; Extremal construction and equality-case engineering; Iterates, fixed points, and cluster-point trapping; Nonexistence by growth iteration; Block decomposition and Schur-style bookkeeping
Theory dependencies	Convexity, concavity, Jensen, and tangent/chord inequalities; Integral inequalities, integration by parts, and weighted estimates; No-regularity functions and order patterns on Q or R
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: functional-equation trigger | audit closures: AUD-RA-08
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2024-D2-P7 (PDF p. 176)

classification / find all

Primary domain	Linear algebra / matrix theory
Core method sequence	1. Complete a matrix equation to an inverse identity; 2. Spectral/minimal-polynomial reduction; 3. Determinant through eigenvalue or root product
Method IDs	M-LA-02; M-LA-06; M-LA-13; M-LA-01; M-LA-03; M-ANT-01; M-ANT-03; M-LA-04
Method titles	Complete a matrix equation to an inverse identity; Spectral/minimal-polynomial reduction; Determinant through eigenvalue or root product; Row/column elimination to triangular or sparse form; Block decomposition and Schur-style bookkeeping; Reduction modulo a prime or finite field; GCD, valuation, and prime-power bookkeeping; Rank-nullity and dimension-count obstruction
Theory dependencies	Determinant evaluation by row/column operations and triangular forms; Block matrices, Schur complements, and determinant identities; Rank-nullity, image/kernel, and rank of factorizations
Cue extraction	prior broad method tags: determinant / block manipulation | statement triggers: matrix word; det/rank/eigen trigger; integer/prime/mod trigger | audit closures: AUD-PC-01; AUD-PC-05; AUD-PC-08; AUD-CMB-08; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2024-D2-P8 (PDF p. 176)

inequality / sharp bound

Primary domain	Real analysis / calculus
Core method sequence	1. Special substitution and parameter freezing; 2. Invariant extraction and invariant comparison; 3. Riemann sum and sum-integral conversion
Method IDs	M-UNI-07; M-UNI-09; M-RA-06; M-RA-07; M-RA-09; M-RA-05; M-RA-15; M-CMB-06
Method titles	Special substitution and parameter freezing; Invariant extraction and invariant comparison; Riemann sum and sum-integral conversion; Split into main range and tail/error range; Convexity, Jensen, tangent, and chord inequalities; L-Hospital and ratio/growth comparison; Recurrence transformed to monotone quotient or telescoping form; Generating function encoding
Theory dependencies	Riemann sums and sum-integral asymptotics; Convexity, concavity, Jensen, and tangent/chord inequalities; Sequence recurrences, monotone bounds, and generating functions
Cue extraction	prior broad method tags: direct structural reasoning | statement triggers: integral/limit trigger; functional-equation trigger | audit closures: AUD-RA-03; AUD-RA-05; AUD-RA-08; AUD-PC-13; AUD-CMB-07
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

2024-D2-P9 (PDF p. 176)

counting / enumeration

Primary domain	Algebra / number theory
Core method sequence	1. Pigeonhole threshold and collision forcing; 2. Double counting ordered or weighted objects; 3. Row/column elimination to triangular or sparse form
Method IDs	M-CMB-01; M-CMB-02; M-LA-01; M-LA-03; M-UNI-03; M-FE-02; M-RA-01; M-UNI-08
Method titles	Pigeonhole threshold and collision forcing; Double counting ordered or weighted objects; Row/column elimination to triangular or sparse form; Block decomposition and Schur-style bookkeeping; Extremal construction and equality-case engineering; Monotonicity forced by functional equation; Auxiliary function with derivative sign; Sign, zero, and boundary case splitting
Theory dependencies	Pigeonhole principle and extremal element selection; Double counting and degree-sum methods; Block matrices, Schur complements, and determinant identities
Cue extraction	prior broad method tags: monotonicity argument, determinant / block manipulation, double counting / pigeonhole | statement triggers: matrix word; integer/prime/mod trigger; counting/permutation trigger | audit closures: AUD-RA-08; AUD-CMB-03; AUD-CMB-08; AUD-CMB-07; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

CLASSIFICATION NOTICE — cited legacy assignments are withdrawn pending quota-free recoding; click CLASS-001.

2024-D2-P10 (PDF p. 176) *proof of structural property*

Primary domain	Algebra / number theory
Core method sequence	1. Reduction modulo a prime or finite field; 2. Chinese remainder and residue construction; 3. GCD, valuation, and prime-power bookkeeping
Method IDs	M-ANT-01; M-ANT-02; M-ANT-03; M-RA-09; M-UNI-05; M-UNI-09; M-ANT-06; M-ANT-16
Method titles	Reduction modulo a prime or finite field; Chinese remainder and residue construction; GCD, valuation, and prime-power bookkeeping; Convexity, Jensen, tangent, and chord inequalities; Induction with strengthened hypothesis; Invariant extraction and invariant comparison; Degree versus number of roots; Polynomial identity by coefficient comparison
Theory dependencies	Modular arithmetic and divisibility; Finite fields, field characteristic, and Fermat/Frobenius tools; Convexity, concavity, Jensen, and tangent/chord inequalities
Cue extraction	prior broad method tags: modular / finite-field reduction | statement triggers: integral/limit trigger; integer/prime/mod trigger | audit closures: AUD-RA-08; AUD-PC-01; AUD-ANT-02
Practice action	Re-solve by writing the invariant, the triggering cue, and the three-line method skeleton before reading the official solution.

19 Method-Training Protocol

The previous table is not only an index; it defines the training routine. For every problem, write the method sequence before writing the full proof. If the sequence cannot be written within ten minutes, the problem should be marked as a recognition failure rather than only as a theory failure.

19.1 Per-problem method extraction checklist

1. **Trigger.** Which words in the statement activated the method? Example: trace/commutator, closed-bounded, polynomial roots, residue system, graph/tree, convex, or unit circle.
2. **Invariant.** What quantity remains fixed or is forced: rank, determinant, trace, degree, sign, residue, total count, volume, or norm?
3. **Reduction.** What object is replaced: matrix basis, auxiliary function, graph, polynomial, finite field, coordinate system, or recurrence variable?
4. **Closure.** What theorem closes the proof: finite-dimensionality, more roots than degree, Rolle, Jensen, compactness, CRT, Cayley-Hamilton, or double counting?
5. **Trap.** What is the first illegal step a rushed solution might make: dividing by zero, assuming commutativity, ignoring endpoints, interchanging limits, or missing equality cases?

19.2 How this method layer should be used next

The next database split should create: (i) one drill deck per method node; (ii) one “recognition cue” deck keyed by statement words; (iii) one proof-skeleton deck keyed by method sequence; and (iv) a timed practice dashboard that records whether failure was due to theory, method recognition, method execution, calculation, or proof-writing rigor.

Strategy and Proof Reference

A strategy is a proof organization rather than a theorem. These programs separate recognition, construction, justification and verification. Their examples illustrate the logic; the linked exercises show its use in this corpus. Expose the structure before calculating [S-STRUCTURE]

Choose an extremal object and force rigidity [S-EXTREMAL]

Reduce size while preserving the defining conditions [S-INDUCTION]

Make the contradiction quantitative or structural [S-CONTRADICTION]

Build a witness and verify every constraint [S-CONSTRUCTION]

Pass between local constraints and a global conclusion [S-LOCALGLOBAL]

Control the limiting operation before taking the limit [S-ANALYTIC]

Prove necessity and sufficiency separately [S-CLASSIFICATION]

Choose the objects and count them without ambiguity [S-COUNTING]

Translate geometry into a controlled representation [S-GEOMETRIC]

Build a chain of bounds and track equality [S-BOUND]

S-STRUCTURE Expose the structure before calculating

Back to previous view | Theory index | Method index | Strategy index

Recognize. Several objects satisfy the same identity, or a long expression is built from a small number of operations. Identify the underlying linear map, polynomial, group operation, or invariant before expanding coordinates.

Proof program. (1) State the ambient field, domain and dimensions. (2) Rewrite the condition as an identity between objects of the same type. (3) Choose coordinates or a decomposition that respect it. (4) Derive the claim in that representation. (5) Translate the conclusion back and check singular or zero cases.

Why it works. An invertible change of coordinates does not change a linear map: if $B = P^{-1}AP$, then $B^k = P^{-1}A^kP$. Thus an identity $p(A) = 0$ can be studied in any basis. It does not follow that $P^{-1}A^{\mathsf{T}}P = B^{\mathsf{T}}$: transpose interacts correctly with orthogonal changes of coordinates, not arbitrary similarities.

Micro-example. If $P^2 = P$, then every v has the decomposition $v = Pv + (v - Pv)$ with terms in $\operatorname{im} P$ and $\ker P$. Their intersection is zero, so an adapted basis gives $\operatorname{diag}(I, 0)$ without solving a characteristic polynomial.

Exit test. Every simplification must preserve the hypotheses actually used. In particular, commuting matrices, symmetric matrices, and diagonalizable matrices are different conditions.

Extended atlas TPL-DIR

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.

1994-D2-P3 [R]; 1994-D2-P4 [R]; 1995-D1-P1 [R]; 1995-D1-P2 [R]; 1995-D1-P5 [R]; 1995-D2-P1 [R]; 1995-D2-P2 [R]; 1995-D2-P3 [R]; 1996-D1-P1 [R]; 1996-D1-P2 [R]; 1996-D2-P2 [R]; 1996-D2-P3 [R]; 1997-D1-P3 [R]; 1997-D2-P2 [R]; 1997-D2-P4 [R]; 1997-D2-P5 [R]; 1998-D1-P1 [R]; 1998-D1-P2 [R]; 1998-D2-P1 [R]; 1999-D1-P5 [R]; 1999-D2-P1 [R]; 1999-D2-P2 [R]; 1999-D2-P3 [R]; 1999-D2-P5 [R]; 2000-D1-P3 [R]; 2000-D1-P4 [R]; 2000-D1-P5 [R]; 2000-D2-P6 [R]; 2001-D1-P1 [R]; 2001-D1-P2 [R]; 2001-D1-P5 [R]; 2001-D2-P1 [R]; 2001-D2-P4 [R]; 2001-D2-P6 [R]; 2002-D1-P1 [R]; 2002-D2-P1 [R]; 2002-D2-P5 [R]; 2002-D2-P6 [R]; 2003-D1-P3 [R]; 2003-D2-P1 [R]; 2003-D2-P4 [R]; 2003-D2-P6 [R]; 2004-D1-P6 [R]; 2004-D2-P1 [R]; 2004-D2-P4 [R]; 2004-D2-P6 [R]; 2005-D1-P1 [R]; 2005-D1-P2 [R]; 2005-D1-P3 [R]; 2005-D1-P5 [R]; 2005-D1-P6 [R]; 2005-D2-P2 [R]; 2005-D2-P4 [R]; 2006-D1-P4 [R]; 2006-D2-P4 [R]; 2006-D2-P6 [R]; 2007-D1-P5 [R]; 2007-D2-P4 [R]; 2008-D1-P6 [R]; 2008-D2-P1 [R]; 2008-D2-P3 [R]; 2008-D2-P5 [R]; 2009-D1-P2 [R]; 2009-D1-P3 [R]; 2009-D1-P5 [R]; 2009-D2-P1 [R]; 2009-D2-P2 [R]; 2009-D2-P3 [R]; 2009-D2-P4 [R]; 2010-D1-P2 [R]; 2010-D1-P3 [R]; 2010-D1-P5 [R]; 2010-D2-P3 [R]; 2010-D2-P5 [R]; 2011-D1-P3 [R]; 2011-D1-P4 [R]; 2011-D2-P1 [R]; 2011-D2-P3 [R]; 2011-D2-P4 [R]; 2011-D2-P5 [R]; 2012-D1-P1 [R]; 2012-D1-P3 [R]; 2012-D1-P5 [R]; 2012-D2-P2 [R]; 2012-D2-P4 [R]; 2013-D1-P2 [R]; 2013-D1-P4 [R]; 2013-D2-P2 [R]; 2013-D2-P5 [R]; 2014-D2-P5 [R]; 2015-D1-P1 [R]; 2015-D1-P3 [R]; 2015-D1-P5 [R]; 2015-D2-P6 [R]; 2015-D2-P8 [R]; 2016-D1-P3 [R]; 2016-D1-P5 [R]; 2016-D2-P6 [R]; 2016-D2-P8 [R]; 2016-D2-P9 [R]; 2017-D1-P5 [R]; 2018-D2-P8 [R]; 2018-D2-P10 [R]; 2019-D1-P1 [R]; 2019-D1-P3 [R]; 2019-D1-P4 [R]; 2019-D2-P6 [R]; 2020-D1-P1 [R]; 2020-D1-P2 [R]; 2020-D2-P6 [R]; 2020-D2-P7 [R]; 2020-D2-P8 [R]; 2021-D1-P1 [R]; 2021-D1-P2 [R]; 2022-D1-P1 [R]; 2022-D1-P3 [R]; 2022-D1-P4 [R]; 2023-D1-P2 [R]; 2023-D1-P4 [R]; 2023-D1-P5 [R]; 2023-D2-P6 [R]; 2024-D1-P2 [R]; 2024-D1-P4 [R]; 2024-D1-P5 [R]; 2024-D2-P6 [R]; 2024-D2-P9 [R]; 2024-D2-P10 [R]

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S-EXTREMAL Choose an extremal object and force rigidity

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Recognize. A finite family admits a size, degree, distance, support, or other ordered statistic. Choose an object whose extremality turns a local modification into an inequality.

Proof program. Specify the admissible family and objective. Prove that an extremum exists: finiteness suffices; for a continuous objective on a compact nonempty set, the extreme-value theorem suffices. Describe an admissible local change and compare its value with the extremum. Equality or impossibility then constrains the chosen object.

Micro-example. In a finite simple graph, choose a longest simple path $v_0, \dots, v_k$. Every neighbour of v_k belongs to the path, since a neighbour outside it would extend the path. This converts an extremal choice into information about the neighbourhood; it does not assert that the path contains every vertex.

S-EXTREMAL Choose an extremal object and force rigidity (continued)

When only an infimum is known. Work with an ε-minimizer or a convergent minimizing sequence, and justify the limiting step. A continuous function on an open or unbounded set need not attain its infimum.
Exit test. Check that the proposed improvement remains admissible, including positivity, distinctness, integrality and boundary constraints. Record the strict inequality responsible for the contradiction.
Extended atlas TPL-EXIST

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D2-P2 [R]; 1996-D2-P4 [R]; 1997-D1-P4 [R]; 1998-D2-P2 [R]; 2000-D2-P2 [R]; 2001-D2-P4 [R]; 2004-D1-P3 [R]; 2007-D1-P6 [R]; 2007-D2-P3 [R]; 2009-D2-P5 [R]; 2011-D1-P1 [R]; 2011-D2-P2 [R]; 2012-D1-P3 [R]; 2012-D2-P5 [R]; 2014-D1-P4 [R]; 2014-D1-P5 [R]; 2014-D2-P4 [R]; 2016-D1-P4 [R]; 2018-D2-P9 [R]; 2020-D1-P3 [R]; 2023-D1-P5 [R]

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S-INDUCTION Reduce size while preserving the defining conditions

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Recognize. The objects have a natural integer size and a smaller instance retains their defining structure. The right induction statement may need an invariant or a stronger conclusion.
Proof program. State $P(n)$ with its full quantifiers. Verify all initial sizes used by the recursion. In the induction step, construct a smaller admissible instance, identify precisely which hypothesis permits invoking $P(k)$, and reconstruct the desired object or inequality at size n.
Micro-example. A finite tree on $n \geq 2$ vertices has a leaf: an endpoint of a longest simple path has no unused neighbour and cannot have a second neighbour along the path without forming a cycle. Removing that leaf preserves connectedness and acyclicity. Induction therefore gives $n-1$ edges, with the one-vertex tree as base case.
Strengthen when necessary. If the reconstruction needs a chosen endpoint, a parity condition, or a prescribed boundary value, put that property into $P(n)$ before starting. It cannot be added after the induction hypothesis has been applied.
Exit test. The reduction must strictly decrease a well-founded parameter. Check every exceptional small case and show that reconstruction neither loses a constraint nor double-counts an object.
Extended atlas TPL-IND

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P3 [R]; 1994-D1-P4 [R]; 1994-D2-P5 [R]; 1996-D1-P2 [R]; 1996-D1-P3 [R]; 1996-D1-P4 [R]; 1997-D1-P6 [R]; 1998-D1-P3 [R]; 1998-D2-P1 [R]; 1998-D2-P3 [R]; 1999-D1-P5 [R]; 1999-D2-P5 [R]; 2000-D2-P1 [R]; 2001-D1-P5 [R]; 2001-D2-P1 [R]; 2001-D2-P2 [R]; 2001-D2-P3 [R]; 2002-D1-P3 [R]; 2002-D2-P3 [R]; 2003-D1-P6 [R]; 2004-D1-P2 [R]; 2004-D1-P5 [R]; 2006-D1-P3 [R]; 2006-D2-P1 [R]; 2007-D2-P6 [R]; 2008-D1-P2 [R]; 2009-D1-P5 [R]; 2009-D2-P3 [R]; 2009-D2-P5 [R]; 2010-D2-P2 [R]; 2010-D2-P4 [R]; 2011-D1-P5 [R]; 2011-D2-P4 [R]; 2012-D2-P5 [R]; 2013-D2-P4 [R]; 2014-D1-P3 [R]; 2014-D2-P5 [R]; 2016-D1-P4 [R]; 2016-D2-P9 [R]; 2017-D1-P3 [R]; 2017-D2-P8 [R]; 2017-D2-P9 [R]; 2018-D2-P7 [R]; 2018-D2-P8 [R]; 2019-D1-P4 [R]; 2020-D1-P4 [R]; 2024-D1-P4 [R]; 2024-D2-P8 [R]

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S-CONTRADICTION Make the contradiction quantitative or structural

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Recognize. The negation of the goal supplies an extra witness, a smallest counterexample, or a strict gap. Write that negation explicitly rather than saying only “suppose not.”
Proof program. Introduce the counterexample with its domain and quantifiers. Preserve its hypotheses through each deduction. Aim for a concrete impossibility: a positive number at most zero, a rank simultaneously above and below the same bound, a forbidden cycle, or two incompatible limits.

S-CONTRADICTION Make the contradiction quantitative or structural (continued)

Micro-example. If a sequence has limits $a \neq b$, choose $\varepsilon = |a-b|/3$. For sufficiently large n, the triangle inequality yields

$$|a-b| \leq |a-x_n| + |x_n-b| < 2|a-b|/3,$$

a contradiction. The positive gap determines the needed scale.
Minimal-counterexample form. Choose the least size of a counterexample only after showing the sizes form a nonempty subset of the nonnegative integers. Construct a smaller counterexample, not merely a smaller object.
Exit test. Identify the exact pair of incompatible statements and which assumption created them. Do not negate a universal statement as another universal statement: the negation of "for every" is "there exists."
Extended atlas TPL-CONTR

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P3 [R]; 1994-D1-P4 [R]; 1994-D2-P1 [R]; 1995-D2-P5 [R]; 1995-D2-P6 [R]; 1996-D1-P5 [R]; 1996-D2-P1 [R]; 1996-D2-P3 [R]; 1996-D2-P4 [R]; 1997-D1-P3 [R]; 1999-D1-P2 [R]; 1999-D1-P4 [R]; 1999-D1-P6 [R]; 1999-D2-P4 [R]; 2001-D1-P4 [R]; 2001-D2-P4 [R]; 2001-D2-P5 [R]; 2002-D1-P2 [R]; 2002-D1-P4 [R]; 2002-D1-P5 [R]; 2003-D2-P5 [R]; 2004-D1-P4 [R]; 2006-D1-P5 [R]; 2007-D1-P3 [R]; 2007-D1-P5 [R]; 2007-D2-P3 [R]; 2010-D1-P4 [R]; 2010-D2-P4 [R]; 2011-D1-P1 [R]; 2011-D1-P2 [R]; 2012-D1-P4 [R]; 2012-D1-P5 [R]; 2014-D1-P4 [R]; 2014-D1-P5 [R]; 2014-D2-P1 [R]; 2015-D1-P4 [R]; 2015-D1-P5 [R]; 2016-D1-P1 [R]; 2017-D2-P7 [R]; 2018-D1-P2 [R]; 2018-D1-P3 [R]; 2019-D1-P5 [R]; 2019-D2-P8 [R]; 2020-D1-P3 [R]; 2021-D2-P5 [R]; 2021-D2-P6 [R]; 2023-D2-P6 [R]; 2024-D2-P6 [R]

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S-CONSTRUCTION Build a witness and verify every constraint

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Recognize. Existence, sharpness or a counterexample is requested. A plausible formula is only the start; every required property must be checked.
Proof program. List the constraints and free parameters. Choose a simple family that automatically satisfies as many constraints as possible. Solve the remaining compatibility conditions. Verify the resulting witness explicitly, including exceptional parameter values.
Micro-example. For a nonzero linear functional f on a finite-dimensional vector space, choose u with $f(u) = 1$ and define $P(v) = f(v)u$. Then $P^2 = P$, $\operatorname{im} P = \operatorname{span}\{u\}$ and $\ker P = \ker f$. Each claim follows directly, so the construction simultaneously provides a projection, its rank and its kernel.
Sharp bounds. Proving an upper bound B and exhibiting admissible examples approaching B shows that B is the supremum. It shows a maximum only when an admissible example attains B.
Exit test. Separate necessity from sufficiency. Check signs, integrality, invertibility, distinctness and normalization; then compute the target quantity. An unverified numerical experiment is a clue for a witness, not the final construction.
Extended atlas TPL-CONSTRUCT; TPL-SHARP

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P1 [R]; 1994-D1-P2 [R]; 1994-D2-P5 [R]; 1995-D2-P4 [R]; 1996-D1-P6 [R]; 1996-D2-P6 [R]; 1997-D1-P2 [R]; 1997-D1-P4 [R]; 1997-D1-P5 [R]; 1997-D1-P6 [R]; 1997-D2-P1 [R]; 1997-D2-P5 [R]; 1997-D2-P6 [R]; 1998-D1-P6 [R]; 1998-D2-P6 [R]; 1999-D1-P1 [R]; 1999-D1-P6 [R]; 2000-D1-P1 [R]; 2000-D2-P1 [R]; 2001-D1-P2 [R]; 2001-D2-P3 [R]; 2002-D2-P2 [R]; 2002-D2-P5 [R]; 2003-D1-P1 [R]; 2003-D1-P4 [R]; 2003-D2-P3 [R]; 2003-D2-P4 [R]; 2003-D2-P5 [R]; 2004-D2-P3 [R]; 2005-D2-P3 [R]; 2005-D2-P5 [R]; 2005-D2-P6 [R]; 2006-D1-P1 [R]; 2006-D1-P6 [R]; 2006-D2-P1 [R]; 2006-D2-P5 [R]; 2007-D1-P2 [R]; 2007-D1-P3 [R]; 2007-D2-P1 [R]; 2007-D2-P5 [R]; 2008-D1-P3 [R]; 2008-D1-P4 [R]; 2008-D1-P5 [R]; 2008-D2-P6 [R]; 2009-D1-P1 [R]; 2009-D2-P5 [R]; 2010-D2-P1 [R]; 2011-D1-P5 [R]; 2011-D2-P2 [R]; 2012-D1-P2 [R]; 2012-D2-P1 [R]; 2013-D1-P3 [R]; 2013-D1-P5 [R]; 2013-D2-P3 [R]; 2013-D2-P4 [R]; 2014-D1-P3 [R]; 2014-D2-P1 [R]; 2014-D2-P4 [R]; 2015-D1-P1 [R]; 2015-D2-P9 [R]; 2016-D1-P1 [R]; 2016-D1-P2 [R]; 2016-D2-P7 [R]; 2017-D1-P1 [R]; 2017-D1-P3 [R]; 2017-D1-P4 [R]; 2018-D1-P1 [R]; 2018-D1-P3 [R]; 2018-D2-P6 [R]; 2019-D2-P9 [R]; 2021-D1-P3 [R]; 2021-D1-P4 [R]; 2021-D2-P8 [R]; 2022-D2-P6 [R]; 2023-D2-P9 [R]; 2024-D1-P3 [R]; 2024-D2-P7 [R]

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S-LOCALGLOBAL Pass between local constraints and a global conclusion

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Recognize. Information is available at individual primes, coordinates, small subsets, intervals, or basis vectors. Determine what compatibility is needed to assemble it.

Proof program. Identify the local tests. Prove they are necessary. Show that the tested family is complete for the desired global property, or invoke a precise assembly theorem with its hypotheses. Finally check compatibility on overlaps.

Micro-example. For positive integers a, b, the relation $a \mid b$ is equivalent to $v_p(a) \le v_p(b)$ for every prime p. The converse follows by prime factorization, not by testing one or two convenient primes. Likewise, congruences modulo pairwise coprime moduli assemble by the Chinese remainder theorem; noncoprime moduli require compatible residues modulo their gcds.

Linear version. Two linear maps agreeing on a basis agree everywhere, by linearity. Agreement on a spanning set suffices; agreement on arbitrary selected vectors need not.

Exit test. State why the local tests exhaust the global obstruction. A polynomial vanishing at all points of a finite field need not be the zero polynomial unless its degree is suitably bounded. A local necessary condition is not automatically sufficient.

Extended atlas TPL-MOD; TPL-POLY

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P5 [R]; 1995-D1-P6 [R]; 1995-D2-P4 [R]; 1996-D2-P1 [R]; 1997-D2-P1 [R]; 1997-D2-P6 [R]; 1998-D2-P6 [R]; 2000-D1-P6 [R]; 2000-D2-P2 [R]; 2003-D1-P5 [R]; 2006-D1-P2 [R]; 2006-D1-P3 [R]; 2007-D1-P1 [R]; 2007-D2-P2 [R]; 2008-D1-P3 [R]; 2010-D1-P4 [R]; 2010-D2-P5 [R]; 2021-D1-P4 [R]; 2022-D2-P5 [R]; 2024-D2-P10 [R]

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S-ANALYTIC Control the limiting operation before taking the limit

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Recognize. A sequence, integral, derivative or infinite sum is compared with a simpler object. Separate the estimate from the passage to the limit.

Proof program. State the convergence mode and domain. Find a uniform bound, dominating function, monotonicity condition or summable majorant as required. Apply the relevant limit theorem with its hypotheses visible. Quantify the remainder whenever a leading-order term is used.

Micro-example. On a finite interval $[a, b]$, uniform convergence $f_n \to f$ of integrable functions gives

$$\left| \int_a^b f_n - \int_a^b f \right| \le (b-a)\|f_n - f\|_\infty \to 0.$$

Pointwise convergence alone does not provide this bound. Similarly, $n\mathbf{1}_{(0,1/n)}$ tends pointwise to zero on $(0, 1]$ while all its integrals equal one.

Asymptotic version. In $F(x) = A(x) + R(x)$, write the exact statement such as $R(x) = o(A(x))$ or $|R(x)| \le Cx^2$ on a specified range. Verify that division by $A(x)$ is legitimate.

Exit test. Never interchange limits, derivatives, integrals or sums solely because each separate expression exists. The theorem authorizing the interchange is part of the proof.

Extended atlas TPL-CONV; TPL-LIMIT

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P2 [R]; 1994-D1-P5 [R]; 1994-D2-P1 [R]; 1994-D2-P3 [R]; 1994-D2-P6 [R]; 1995-D1-P3 [R]; 1995-D1-P4 [R]; 1995-D2-P2 [R]; 1995-D2-P6 [R]; 1996-D1-P5 [R]; 1996-D2-P5 [R]; 1997-D1-P1 [R]; 1997-D2-P3 [R]; 1998-D1-P3 [R]; 1998-D1-P4 [R]; 2000-D1-P6 [R]; 2000-D2-P4 [R]; 2000-D2-P5 [R]; 2000-D2-P6 [R]; 2001-D1-P3 [R]; 2001-D1-P6 [R]; 2001-D2-P2 [R]; 2002-D1-P2 [R]; 2002-D1-P4 [R]; 2002-D1-P6 [R]; 2002-D2-P3 [R]; 2002-D2-P4 [R]; 2003-D1-P1 [R]; 2003-D1-P3 [R]; 2003-D2-P2 [R]; 2003-D2-P6 [R]; 2004-D1-P3 [R]; 2004-D1-P6 [R]; 2004-D2-P4 [R]; 2004-D2-P5 [R]; 2005-D1-P5 [R]; 2005-D2-P4 [R]; 2006-D1-P4 [R]; 2006-D1-P5 [R]; 2007-D2-P6 [R]; 2008-D1-P1 [R]; 2008-D2-P6 [R]; 2009-D1-P4 [R]; 2010-D1-P1 [R]; 2010-D1-P2 [R]; 2010-D1-P3 [R]; 2011-D2-P1 [R]; 2011-D2-P3 [R]; 2012-D1-P4 [R]; 2012-D2-P2 [R]; 2012-D2-P3 [R]; 2013-D1-P2 [R]; 2013-D1-P5 [R]; 2014-D1-P2 [R]; 2014-D2-P3 [R]; 2015-D1-P3 [R]; 2015-D2-P7 [R]; 2015-D2-P10 [R]; 2016-D1-P5 [R]; 2016-D2-P10 [R]; 2017-D1-P2 [R]; 2017-D2-P6 [R]; 2017-D2-P9 [R]; 2017-D2-P10 [R]; 2018-D1-P4 [R]; 2018-D2-P7 [R];

S-ANALYTIC Control the limiting operation before taking the limit (continued)

2019-D1-P1 [R]; 2019-D2-P6 [R]; 2020-D1-P4 [R]; 2020-D2-P5 [R]; 2020-D2-P8 [R]; 2021-D1-P3 [R]; 2023-D1-P1 [R]; 2023-D2-P7 [R]; 2024-D1-P2 [R]; 2024-D2-P8 [R]

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S-CLASSIFICATION Prove necessity and sufficiency separately

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Recognize. The question asks for all functions, matrices, parameters, configurations or equality cases. Useful reductions usually constrain the candidates before they identify them.

Proof program. Extract identities valid for every solution. Use special inputs or structural decompositions to obtain a candidate family. Check every exceptional case omitted by division or inversion. Substitute every surviving candidate into the original conditions.

Micro-example. A complex square matrix satisfying $A^2 = A$ has minimal polynomial dividing $x(x-1)$. The roots are distinct, so it is diagonalizable with eigenvalues in $\{0, 1\}$. Conversely, every matrix similar to $\operatorname{diag}(I_r, 0)$ is idempotent. Both directions are needed for the classification.

Functional-equation caution. Additivity $f(x+y) = f(x) + f(y)$ on $\mathbb{R}$ does not imply $f(x) = cx$ without an extra hypothesis such as continuity at one point, measurability, or local boundedness. The domain and regularity assumptions determine the candidate family.

Exit test. Confirm that each transformation was reversible, or restore the cases it excluded. A necessary eigenvalue condition alone may not determine a matrix; a necessary numerical inequality alone may not supply an admissible configuration.

Extended atlas TPL-CLASS; TPL-FUNC

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D2-P2 [R]; 1995-D1-P4 [R]; 1996-D1-P3 [R]; 1997-D1-P2 [R]; 1997-D1-P5 [R]; 1997-D2-P3 [R]; 1998-D1-P2 [R]; 1998-D1-P4 [R]; 1998-D2-P4 [R]; 1999-D1-P1 [R]; 1999-D1-P3 [R]; 1999-D1-P4 [R]; 2000-D1-P1 [R]; 2000-D1-P2 [R]; 2000-D2-P5 [R]; 2003-D1-P2 [R]; 2003-D1-P4 [R]; 2003-D2-P2 [R]; 2003-D2-P4 [R]; 2005-D1-P4 [R]; 2005-D2-P1 [R]; 2006-D1-P1 [R]; 2006-D1-P6 [R]; 2006-D2-P2 [R]; 2006-D2-P3 [R]; 2006-D2-P6 [R]; 2008-D1-P1 [R]; 2008-D1-P2 [R]; 2008-D1-P4 [R]; 2009-D1-P1 [R]; 2010-D2-P1 [R]; 2011-D1-P2 [R]; 2011-D1-P3 [R]; 2012-D2-P4 [R]; 2014-D1-P1 [R]; 2015-D1-P2 [R]; 2017-D1-P1 [R]; 2018-D1-P1 [R]; 2018-D1-P4 [R]; 2018-D2-P9 [R]; 2019-D1-P2 [R]; 2019-D2-P7 [R]; 2019-D2-P9 [R]; 2020-D2-P5 [R]; 2020-D2-P6 [R]; 2022-D1-P2 [R]; 2023-D1-P1 [R]; 2023-D1-P3 [R]; 2023-D2-P7 [R]; 2024-D1-P1 [R]; 2024-D1-P3 [R]; 2024-D2-P7 [R]

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S-COUNTING Choose the objects and count them without ambiguity

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Recognize. An equality, parity statement or bound concerns a finite family. Decide whether to count directly, by a bijection, by incidences, or by an involution.

Proof program. Define the finite set precisely. Specify every map, including its domain and codomain. Prove injectivity/surjectivity for a bijection, or verify that the selected weights count each object with the claimed multiplicity. Check fixed points separately for an involution.

Micro-example. In a finite simple graph, count pairs (v, e) with vertex v incident to edge e. Summing over vertices gives $\sum_v \deg v$; summing over edges gives $2|E|$. Thus the number of odd-degree vertices is even. Both counts refer to the same incidence set.

Parity form. If $\tau : X \to X$ satisfies $\tau^2 = \mathrm{id}$ and has no fixed points, its orbits are disjoint pairs and $|X|$ is even. Defining τ is insufficient: it must preserve every condition defining X.

Exit test. Account for indistinguishable objects, ordering, symmetries and boundary cases. Uniform multiplicity must be proved before dividing a count by a symmetry factor.

Extended atlas TPL-COUNT

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1994-D1-P6 [R]; 1994-D2-P4 [R]; 1998-D1-P1 [R]; 1998-D2-P3 [R]; 1998-D2-P4 [R]; 1998-D2-P5 [R]; 1999-D1-P2 [R]; 1999-D2-P2 [R]; 1999-D2-P6 [R]; 2000-D2-P3 [R]; 2001-D1-P1 [R]; 2002-D1-P5 [R];

S-COUNTING Choose the objects and count them without ambiguity (continued)

2002-D2-P2 [R]; 2004-D1-P1 [R]; 2004-D1-P2 [R]; 2004-D1-P4 [R]; 2004-D2-P6 [R]; 2005-D1-P2 [R]; 2006-D1-P2 [R]; 2007-D1-P4 [R]; 2008-D1-P5 [R]; 2008-D1-P6 [R]; 2008-D2-P4 [R]; 2009-D1-P3 [R]; 2009-D2-P4 [R]; 2010-D2-P3 [R]; 2011-D1-P4 [R]; 2012-D1-P1 [R]; 2013-D1-P3 [R]; 2013-D2-P2 [R]; 2013-D2-P5 [R]; 2014-D1-P2 [R]; 2014-D1-P5 [R]; 2014-D2-P1 [R]; 2014-D2-P5 [R]; 2015-D2-P8 [R]; 2016-D1-P4 [R]; 2016-D1-P5 [R]; 2017-D1-P4 [R]; 2017-D1-P5 [R]; 2017-D2-P8 [R]; 2018-D2-P8 [R]; 2019-D2-P8 [R]; 2019-D2-P10 [R]; 2020-D1-P1 [R]; 2020-D2-P7 [R]; 2021-D1-P2 [R]; 2022-D1-P3 [R]; 2022-D1-P4 [R]; 2022-D2-P5 [R]; 2022-D2-P8 [R]; 2023-D1-P5 [R]; 2023-D2-P9 [R]; 2023-D2-P10 [R]; 2024-D2-P9 [R]

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S-GEOMETRIC Translate geometry into a controlled representation

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Recognize. Distances, angles, containment or convexity can be simplified by coordinates, a projection, a support direction or an extremal point. Choose the representation that makes the required property linear or quadratic.

Proof program. State the dimension and nondegeneracy conditions. Choose coordinates without losing generality and explain the allowed rigid motion or scaling. Translate geometric claims into equations or inequalities. Prove the translated claim and recover its geometric meaning.

Micro-example. For points x, y in Euclidean space,

$$\|x-y\|^2 = \|x\|^2 + \|y\|^2 - 2\langle x, y\rangle.$$

Equal-distance configurations can therefore be encoded by a Gram matrix, whose positive semidefiniteness and rank impose dimension constraints. An arbitrary symmetric matrix is not automatically a realizable Gram matrix: it must be positive semidefinite.

Figures. Use a diagram to identify a possible relation, then prove it. Perspective and scale are not hypotheses; a planar sketch cannot certify a higher-dimensional intersection or containment.

Exit test. Check collinear, coincident or boundary configurations and the direction of every containment. Orthogonal projection preserves neither all lengths nor all angles, though it does not increase distances.

Extended atlas TPL-GEO

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.
1995-D2-P1 [R]; 1995-D2-P3 [R]; 1996-D1-P6 [R]; 1996-D2-P4 [R]; 1998-D2-P5 [R]; 1999-D2-P6 [R]; 2000-D2-P4 [R]; 2001-D2-P3 [R]; 2002-D1-P1 [R]; 2002-D2-P4 [R]; 2003-D2-P3 [R]; 2004-D1-P4 [R]; 2004-D2-P2 [R]; 2004-D2-P3 [R]; 2005-D2-P5 [R]; 2008-D2-P2 [R]; 2008-D2-P6 [R]; 2009-D1-P5 [R]; 2009-D2-P1 [R]; 2011-D2-P5 [R]; 2013-D1-P1 [R]; 2013-D2-P1 [R]; 2014-D1-P5 [R]; 2014-D2-P4 [R]; 2015-D1-P5 [R]; 2017-D2-P10 [R]; 2018-D1-P5 [R]; 2018-D2-P10 [R]; 2019-D2-P10 [R]; 2020-D1-P3 [R]; 2022-D2-P8 [R]; 2023-D2-P8 [R]; 2023-D2-P9 [R]; 2024-D1-P1 [R]; 2024-D1-P5 [R]

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S-BOUND Build a chain of bounds and track equality

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Recognize. A maximum, minimum or universal estimate is requested. Look for a quantity that is preserved by normalization and a chain of standard inequalities adapted to it.

Proof program. State the sign and domain assumptions before applying an inequality. Normalize only by a positive nonzero quantity. Record equality conditions at each step. Intersect those conditions and verify that a candidate attaining them is admissible.

Micro-example. For nonnegative $a_1, \dots, a_n$ with sum s, Cauchy–Schwarz gives

$$s^2 \le n \sum_{i=1}^{n} a_i^2,$$

S-BOUND Build a chain of bounds and track equality (continued)

with equality exactly when all $a_i = s/n$. The inequality proves a lower bound; the equal tuple proves sharpness.

Nonattainment. A sequence approaching a bound proves sharpness of a supremum or infimum, not existence of an optimizer. If an inequality is strict at any mandatory step, equality cannot occur in the final result.

Exit test. Check directions when multiplying by unknown-sign terms or applying a decreasing function. Inspect zero denominators and endpoint cases. A numerically convincing bound is only a conjecture until the complete admissible domain is covered.

Extended atlas TPL-INEQ; TPL-SHARP

Return to an exercise. Each ID opens its statement; the adjacent R opens its learning route.

1994-D1-P1 [R]; 1994-D1-P2 [R]; 1994-D1-P6 [R]; 1994-D2-P5 [R]; 1994-D2-P6 [R]; 1995-D1-P2 [R]; 1995-D1-P3 [R]; 1995-D1-P6 [R]; 1995-D2-P5 [R]; 1996-D1-P4 [R]; 1996-D1-P6 [R]; 1996-D2-P5 [R]; 1996-D2-P6 [R]; 1997-D1-P1 [R]; 1997-D1-P5 [R]; 1998-D1-P5 [R]; 1998-D1-P6 [R]; 1998-D2-P2 [R]; 1999-D1-P3 [R]; 1999-D1-P6 [R]; 1999-D2-P3 [R]; 1999-D2-P4 [R]; 2000-D1-P4 [R]; 2001-D1-P4 [R]; 2001-D1-P6 [R]; 2001-D2-P5 [R]; 2001-D2-P6 [R]; 2002-D1-P6 [R]; 2002-D2-P4 [R]; 2002-D2-P6 [R]; 2003-D1-P5 [R]; 2003-D1-P6 [R]; 2004-D1-P1 [R]; 2004-D1-P5 [R]; 2004-D2-P2 [R]; 2004-D2-P3 [R]; 2004-D2-P5 [R]; 2005-D1-P1 [R]; 2005-D1-P3 [R]; 2005-D1-P4 [R]; 2005-D2-P1 [R]; 2005-D2-P3 [R]; 2005-D2-P5 [R]; 2006-D2-P2 [R]; 2006-D2-P3 [R]; 2007-D1-P2 [R]; 2007-D1-P6 [R]; 2007-D2-P5 [R]; 2008-D1-P4 [R]; 2008-D2-P2 [R]; 2009-D1-P4 [R]; 2009-D2-P2 [R]; 2010-D1-P1 [R]; 2010-D1-P5 [R]; 2010-D2-P2 [R]; 2011-D2-P2 [R]; 2012-D1-P2 [R]; 2012-D2-P3 [R]; 2012-D2-P5 [R]; 2013-D1-P1 [R]; 2013-D1-P3 [R]; 2013-D1-P4 [R]; 2013-D2-P1 [R]; 2013-D2-P3 [R]; 2014-D2-P2 [R]; 2014-D2-P3 [R]; 2014-D2-P4 [R]; 2015-D1-P2 [R]; 2015-D2-P6 [R]; 2015-D2-P7 [R]; 2015-D2-P9 [R]; 2015-D2-P10 [R]; 2016-D1-P2 [R]; 2016-D1-P3 [R]; 2016-D2-P6 [R]; 2016-D2-P7 [R]; 2016-D2-P9 [R]; 2016-D2-P10 [R]; 2017-D1-P2 [R]; 2017-D1-P4 [R]; 2017-D2-P6 [R]; 2017-D2-P7 [R]; 2017-D2-P9 [R]; 2017-D2-P10 [R]; 2018-D1-P5 [R]; 2018-D2-P6 [R]; 2018-D2-P7 [R]; 2019-D1-P3 [R]; 2019-D2-P7 [R]; 2019-D2-P10 [R]; 2020-D1-P3 [R]; 2021-D1-P3 [R]; 2021-D2-P5 [R]; 2021-D2-P7 [R]; 2021-D2-P8 [R]; 2022-D1-P1 [R]; 2022-D1-P4 [R]; 2022-D2-P7 [R]; 2023-D2-P8 [R]; 2023-D2-P9 [R]; 2023-D2-P10 [R]; 2024-D2-P8 [R]

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Part I

Problem-Type Sheets and Trigger Library Expansion

20 Expansion Purpose

This new layer preserves the preceding categorization, theory, coverage audit, and method sheets, and adds two exam-facing databases: (i) problem-type sheets and (ii) a trigger library. The problem-type sheets answer: *what kind of problem is this?* The trigger library answers: *which words in the statement should activate which methods?* The entries are derived from the prior problem catalog and method audit for all detected IMC problems.

20.1 How to read this expansion

For a given problem, first identify its primary problem-type sheet. Then read the trigger cards attached to the statement. Only after that choose the detailed method sequence from the earlier Method Sheets layer. This gives the intended pipeline: statement cue → problem type → method stack → theory card → proof skeleton.

21 Problem-Type Frequency Overview

ID	Problem-type sheet	Primary	Total linked
PT-INEQ	Inequalities and sharp constants	36	295
PT-LIM	Limits, asymptotics, Riemann sums, and growth comparison	23	191
PT-INT	Integral estimates, averaging, oscillation, and integration by parts	8	143
PT-FE	Functional equations, iterates, and dynamics	58	130
PT-LA-DET	Determinants, rank, inverse identities, and matrix factorizations	23	121
PT-FINITE	Finite fields, modular arithmetic, residues, and prime-power bookkeeping	71	119
PT-COMB	Graph, set-system, permutation, word, and counting structures	29	117
PT-LA-SPEC	Nilpotency, spectra, commutators, idempotents, and operator powers	22	99
PT-GRP	Group, ring, automorphism, and noncommutative algebra patterns	17	91
PT-GEO	Geometry by coordinates, metric identities, convexity, and projection	16	70
PT-POLY	Polynomial roots, coefficient constraints, divisibility, and interpolation	31	68
PT-EXT	Extremal configurations, compactness, existence, and counterexamples	0	35
PT-SEQ	Recurrences, generating functions, and discrete dynamical sequences	0	13

22 Problem-Type Sheets

Each sheet is meant to be used as a one-page-or-two-page decision guide during practice review. Counts distinguish *primary* occurrences, where the type is the main solution lens, and *total* occurrences, where it appears as either a primary or secondary lens.

METHOD-DATA NOTICE — trigger/template fields are not validated frequencies; click CLASS-002.

22.1 PT-LA-DET: Determinants, rank, inverse identities, and matrix factorizations

Core domain	Linear algebra / matrix theory
Primary / total occurrences	23 / 121
Scope	Problems where the decisive object is a determinant, rank, inverse, block product, trace form, or matrix factorization. The work is usually not to compute all entries, but to choose a basis, split into blocks, eliminate rows/columns, or recognize a rank/determinant invariant.
Trigger phrases	determinant of a structured matrix; rank of an n by n matrix; inverse of A; AB or BA with rectangular factors; block form; trace(XY)=0; dimension of a matrix subspace
Linked method IDs	M-LA-01; M-LA-02; M-LA-03; M-LA-04; M-LA-05; M-LA-06; M-LA-08; M-LA-11

Reusable method stack.

1. Row and column operations that preserve determinant/rank
2. Block decomposition and Schur-complement style bookkeeping
3. Rank-nullity and image/kernel dimension count
4. Trace pairing nondegeneracy and symmetric/skew decomposition
5. Factorization rank inequalities rk(AB) <= min(rk A,rk B)

Proof skeleton.

1. Normalize the matrix by row/column permutations or basis choice.
2. Identify the invariant: rank, determinant, trace, kernel, image, or diagonal block.
3. Perform operations that preserve the invariant or transform the matrix to sparse/triangular form.
4. For maximal/minimal statements, give both an upper obstruction and an explicit construction.
5. Check singular edge cases, dimensions, and whether the argument uses commutativity illegally.

Trap watch.

- Assuming AB and BA are the same size or have identical entries.
- Using determinant identities requiring invertibility without checking it.
- Confusing row operations that preserve determinant with those that only preserve rank.
- Forgetting that real matrices may have complex eigenvalues.

Most common trigger cards inside this problem type.

Trigger	Count	Phrase
TR-LA-01	99	rank, dimension, span, image, kernel, invariant subspace
TR-RA-07	72	convex, concave, monotone, nondecreasing
TR-LA-05	68	eigenvalue, characteristic polynomial, diagonalizable, similar
TR-LA-02	45	determinant, inverse, block matrix, triangular form
TR-RA-06	44	series convergence, rearrangement, alternating, absolute convergence
TR-ANT-01	41	prime, modulo, residue, divisible by, p-adic power
TR-LA-03	37	AB-BA, commutator, trace zero
TR-ANT-04	37	ring, idempotent, square zero, noncommutative

Primary occurrences.

1994-D1-P1; 1994-D2-P4; 1995-D2-P1; 1996-D1-P1; 1997-D2-P2; 1998-D1-P1; 2001-D2-P4; 2002-D2-P1; 2003-D2-P1; 2004-D2-P1; 2004-D2-P6; 2005-D1-P1; 2007-D1-P3; 2008-D1-P2; 2008-D2-P6; 2009-D1-P2; 2015-D1-P1; 2015-D2-P9; 2018-D1-P3; 2021-D1-P1; 2023-D1-P2; 2023-D2-P6; 2024-D2-P7

All linked occurrences.

1994-D1-P1; 1994-D1-P4; 1994-D2-P2; 1994-D2-P4; 1994-D2-P5; 1995-D1-P1; 1995-D1-P4; 1995-D1-P5; 1995-D2-P1; 1995-D2-P5; 1996-D1-P1; 1996-D1-P3; 1996-D1-P6; 1996-D2-P3; 1997-D1-P3; 1997-D1-P5; 1997-D2-P2; 1997-D2-P4; 1998-D1-P1; 1998-D2-P1; 1998-D2-P4; 1999-D1-P1; 1999-D1-P5; 1999-D2-P2; 2000-D1-P3; 2000-D1-P5; 2000-D2-P1; 2000-D2-P6; 2001-D1-P1; 2001-D1-P2; 2001-D1-P5; 2001-D1-P6; 2001-D2-P3; 2001-D2-P4; 2002-D1-P1; 2002-D1-P5; 2002-D1-P6; 2002-D2-P1; 2002-D2-P3; 2002-D2-P5; 2003-D1-P3; 2003-D1-P4; 2003-D1-P5; 2003-D2-P1; 2004-D1-P1; 2004-D1-P3; 2004-D2-P1; 2004-D2-P4; 2004-D2-P6; 2005-D1-P1; 2005-D1-P6; 2005-D2-P3; 2005-D2-P6; 2006-D1-P3; 2006-D1-P6; 2006-D2-P2; 2006-D2-P4; 2006-D2-P6; 2007-D1-P2; 2007-D1-P3; 2007-D1-P5; 2007-D2-P4; 2007-D2-P5; 2008-D1-P6; 2008-D1-P2; 2008-D1-P3; 2008-D1-P5; 2008-D2-P2; 2008-D2-P5; 2008-D2-P6; 2009-D1-P2; 2009-D1-P3

; 2009-D1-P4; 2009-D1-P5; 2009-D2-P3; 2009-D2-P5; 2010-D1-P5; 2010-D2-P4; 2011-D1-P2; 2011-D1-P5; 2011-D2-P2; 2011-D2-P3; 2011-D2-P4; 2011-D2-P5 ; 2012-D1-P2; 2012-D2-P2; 2013-D1-P1; 2014-D1-P1; 2014-D2-P2; 2015-D1-P1; 2015-D1-P3; 2015-D1-P4; 2015-D1-P5; 2015-D2-P9; 2016-D1-P2; 2016-D2-P10 ; 2017-D1-P1; 2017-D1-P3; 2017-D2-P8; 2017-D2-P9; 2018-D1-P3; 2018-D2-P8; 2018-D2-P9; 2019-D1-P5; 2019-D2-P7; 2019-D2-P9; 2020-D1-P2; 2021-D1-P1; 2021-D2-P5; 2021-D2-P6; 2022-D1-P2; 2022-D2-P7; 2023-D1-P2; 2023-D2-P6; 2023-D2-P7; 2023-D2-P9; 2023-D2-P10; 2024-D1-P1; 2024-D1-P3; 2024-D2-P7; 2024-D2-P9

22.2 PT-LA-SPEC: Nilpotency, spectra, commutators, idempotents, and operator powers

Core domain	Linear algebra / matrix theory
Primary / total occurrences	22 / 99
Scope	Problems where equations in matrices/operators are solved by eigenvalues, minimal polynomial, Cayley-Hamilton, trace of powers, Jordan form, nilpotency, idempotency, involution structure, or commutator identities.
Trigger phrases	nilpotent; idempotent; involution; all eigenvalues are zero; AB-BA; A^k converges; characteristic/minimal polynomial; A is similar to
Linked method IDs	M-LA-05; M-LA-06; M-LA-07; M-LA-09; M-LA-10; M-LA-12; M-LA-13; M-LA-14

Reusable method stack.

1. Apply the polynomial equation to eigenvalues.
2. Use trace of commutators and trace powers to force spectral data.
3. Use Cayley-Hamilton to convert zero spectrum to nilpotency.
4. Use diagonalization when the minimal polynomial has simple roots.
5. Use simultaneous triangularization/diagonalization when commuting operators appear.

Proof skeleton.

1. Translate the matrix equation into a polynomial or commutator relation.
2. Extract eigenvalue constraints or trace identities.
3. Prove all possible eigenvalues have the claimed behavior.
4. Use finite-dimensionality, Cayley-Hamilton, or Jordan form to pass from eigenvalues to a matrix conclusion.
5. Verify that the constructed or limiting matrix has the required idempotent/nilpotent property.

Trap watch.

- Trace zero alone does not imply nilpotency unless rank or all power traces give enough information.
- Diagonalization requires simple-root minimal polynomial or normality/commuting hypotheses.
- Nilpotent means some power is zero, not merely determinant zero.

Most common trigger cards inside this problem type.

Trigger	Count	Phrase
TR-LA-01	78	rank, dimension, span, image, kernel, invariant subspace
TR-LA-05	76	eigenvalue, characteristic polynomial, diagonalizable, similar
TR-RA-07	54	convex, concave, monotone, nondecreasing
TR-RA-06	37	series convergence, rearrangement, alternating, absolute convergence
TR-LA-02	36	determinant, inverse, block matrix, triangular form
TR-LA-03	33	AB-BA, commutator, trace zero
TR-GEO-01	26	distance to point/line/sphere/ellipse/parabola
TR-ANT-04	26	ring, idempotent, square zero, noncommutative

Primary occurrences.

1994-D1-P4; 1995-D1-P1; 1995-D1-P5; 1996-D1-P3; 1997-D2-P4; 1998-D2-P1; 1999-D1-P1; 2000-D1-P3; 2000-D2-P6; 2001-D1-P5; 2002-D2-P5; 2003-D1-P3; 2004-D2-P4; 2005-D2-P3; 2006-D2-P6; 2011-D1-P2; 2013-D1-P1; 2014-D1-P1; 2016-D2-P10; 2017-D1-P1; 2020-D1-P2; 2022-D2-P7

All linked occurrences.

1994-D1-P1; 1994-D1-P4; 1994-D2-P2; 1994-D2-P4; 1995-D1-P1; 1995-D1-P4; 1995-D1-P5; 1995-D2-P1; 1995-D2-P5; 1996-D1-P1; 1996-D1-P3; 1996-D1-P5; 1996-D1-P6; 1997-D1-P5; 1997-D2-P2; 1997-D2-P4; 1998-D1-P1; 1998-D2-P1; 1998-D2-P4; 1998-D2-P5; 1999-D1-P1; 2000-D1-P3; 2000-D1-P5; 2000-D2-P6; 2001-D1-P2; 2001-D1-P5; 2001-D1-P6; 2001-D2-P4; 2002-D1-P1; 2002-D2-P1; 2002-D2-P3; 2002-D2-P5; 2003-D1-P3; 2003-D1-P4; 2003-D1-P5; 2003-D1-P6; 2003-D2-P1; 2004-D1-P1; 2004-D1-P5; 2004-D1-P6; 2004-D2-P1; 2004-D2-P4; 2004-D2-P6; 2005-D1-P1; 2005-D1-P6; 2005-D2-P3; 2005-D2-P6; 2006-D1-P3; 2006-D2-P2; 2006-D2-P6; 2007-D1-P2; 2007-D1-P3; 2007-D2-P4; 2007-D2-P6; 2008-D1-P2; 2008-D1-P3; 2008-D2-P5; 2008-D2-P6; 2009-D1-P2; 2009-D1-P3; 2009-D1-P4; 2009-D1-P5; 2009-D2-P3; 2010-D2-P1; 2010-D2-P5; 2011-D1-P2; 2011-D2-P2; 2011-D2-P3; 2012-D1-P2; 2012-D2-P2; 2013-D1-P1; 2014-D1-P1; 2014-D2-P2; 2015-D1-P1; 2015-D1-P3; 2015-D1-P4; 2015-D2-P9; 2016-D2-P10; 2017-D1-P1; 2017-D1-P3; 2017-D2-P8; 2017-D2-P9; 2018-D1-P3; 2018-D2-P8; 2018-D2-P9; 2019-D1-P5; 2019-D2-P7; 2019-D2-P9; 2020-D1-P2; 2021-D1-P1; 2021-D2-P5; 2022-D1-P2; 2022-D2-P7; 2023-D1-P2; 2023-D2-P6; 2023-D2-P7; 2023-D2-P10; 2024-D1-P1; 2024-D2-P7

22.3 PT-FE: Functional equations, iterates, and dynamics

Core domain	Functional equations / dynamics
Primary / total occurrences	58 / 130
Scope	Problems where an unknown function or map is constrained by identities, inequalities, iterates, image intervals, monotonicity, periodicity, or composition. The core task is to turn the statement into rigidity, monotonicity, injectivity, or a counterexample.
Trigger phrases	find all functions; for all x,y; f(f(x)); iterate f_n; periodic point; image of every interval; monotone map; closed orbit
Linked method IDs	M-FE-01; M-FE-02; M-FE-03; M-UNI-07; M-UNI-08; M-UNI-03; M-UNI-10

Reusable method stack.

1. Substitute special values and freeze parameters.
2. Extract monotonicity or injectivity from the equation.
3. Use orbits and invariant sets for iterates.
4. Use continuity/compactness to turn local behavior into global rigidity.
5. Construct counterexamples by piecewise, exponential, Cantor, or shift models.

Proof skeleton.

1. Test constants, affine functions, and obvious models.
2. Use special substitutions to obtain boundedness, monotonicity, or injectivity.
3. If iterates appear, study fixed points, cluster points, and orbit closure.
4. If classification is requested, prove that the derived structure forces one family.
5. Check endpoints and domain restrictions.

Trap watch.

- Assuming continuity when it is not given.
- Cancelling f or applying inverse before proving injectivity/nonzero.
- Ignoring disconnected domains or endpoint behavior.

Most common trigger cards inside this problem type.

Trigger	Count	Phrase
TR-RA-07	101	convex, concave, monotone, nondecreasing
TR-RA-02	75	endpoint equalities, repeated zeros, differentiable, Rolle
TR-RA-04	72	lim n->infty of a sum, Riemann sum, asymptotic sum
TR-ANT-04	58	ring, idempotent, square zero, noncommutative
TR-RA-06	49	series convergence, rearrangement, alternating, absolute convergence
TR-RA-05	33	periodic, oscillatory integral, sin(nx), cos(nx)
TR-ANT-01	28	prime, modulo, residue, divisible by, p-adic power
TR-CMB-03	23	town, friends, grid, rows/columns, graph, cycle

Primary occurrences.

1994-D1-P5; 1994-D2-P1; 1994-D2-P2; 1994-D2-P3; 1995-D1-P4; 1996-D2-P1; 1996-D2-P5; 1997-D2-P6; 1998-D1-P4; 1998-D1-P6; 1998-D2-P2; 1998-D2-P3; 1998-D2-P4; 1998-D2-P6; 1999-D1-P2; 1999-D1-P4; 1999-D1-P6; 1999-D2-P4; 2000-D1-P1; 2000-D1-P2; 2000-D1-P6; 2000-D2-P5; 2001-D1-P6; 2001-D2-P5; 2002-D1-P2; 2002-D1-P4; 2002-D1-P5; 2002-D2-P6; 2003-D2-P5; 2004-D2-P2; 2005-D1-P3; 2005-D2-P4; 2005-D2-P5; 2006-D1-P1; 2006-D1-P6; 2006-D2-P2; 2007-D2-P1; 2007-D2-P3; 2008-D1-P1; 2009-D2-P2; 2010-D1-P1; 2010-D1-P3; 2011-D1-P1; 2011-D2-P3; 2012-D1-P4; 2013-D1-P2; 2015-D1-P3; 2016-D1-P1; 2016-D2-P7; 2017-D1-P2; 2018-D1-P4; 2019-D1-P3; 2019-D2-P6; 2020-D2-P5; 2021-D1-P4; 2023-D1-P1; 2023-D2-P7; 2024-D2-P6

All linked occurrences.

1994-D1-P2; 1994-D1-P3; 1994-D1-P5; 1994-D1-P6; 1994-D2-P1; 1994-D2-P2; 1994-D2-P3; 1995-D1-P2; 1995-D1-P3; 1995-D1-P4; 1995-D1-P6; 1995-D2-P4; 1995-D2-P5; 1995-D2-P6; 1996-D1-P3; 1996-D1-P5; 1996-D1-P6; 1996-D2-P1; 1996-D2-P5; 1997-D2-P1; 1997-D2-P3; 1997-D2-P4; 1997-D2-P5; 1997-D2-P6; 1998-D1-P3; 1998-D1-P4; 1998-D1-P6; 1998-D2-P1; 1998-D2-P2; 1998-D2-P3; 1998-D2-P4; 1998-D2-P6; 1999-D1-P2; 1999-D1-P3; 1999-D1-P4; 1999-D1-P6; 1999-D2-P4; 1999-D2-P6; 2000-D1-P1; 2000-D1-P2; 2000-D1-P6; 2000-D2-P2; 2000-D2-P5; 2001-D1-P6; 2001-D2-P2; 2001-D2-P3; 2001-D2-P5; 2001-D2-P6; 2002-D1-P2; 2002-D1-P3; 2002-D1-P4; 2002-D1-P5; 2002-D2-P6; 2003-D1-P4; 2003-D1-P5; 2003-D1-P6; 2003-D2-P5; 2003-D2-P6; 2004-D1-P3; 2004-D1-P6; 2004-D2-P2; 2005-D1-P3; 2005-D1-P5; 2005-D1-P6; 2005-D2-P1; 2005-D2-P2; 2005-D2-P4; 2005-D2-P5; 2006-D1-P1; 2006-D1-P4; 2006-D1-P5; 2006-D1-P6; 2006-D2-P2; 2006-D2-P3; 2007-D1-P1; 2007-D1-P5; 2007-D2-P1; 2007-D2-P3; 2007-D2-P6; 2008-D1-P1; 2008-D1-P2; 2008-D2-P1; 2008-D2-P2; 2008-D2-P4

; 2009-D1-P1; 2009-D2-P2; 2010-D1-P1; 2010-D1-P2; 2010-D1-P3; 2010-D2-P1; 2010-D2-P3; 2010-D2-P5; 2011-D1-P1; 2011-D1-P3; 2011-D1-P4; 2011-D2-P1 ; 2011-D2-P3; 2011-D2-P4; 2012-D1-P4; 2012-D2-P1; 2013-D1-P2; 2014-D2-P3; 2015-D1-P2; 2015-D1-P3; 2016-D1-P1; 2016-D1-P5; 2016-D2-P7; 2016-D2-P8 ; 2016-D2-P9; 2017-D1-P2; 2017-D1-P5; 2017-D2-P6; 2017-D2-P8; 2017-D2-P9; 2018-D1-P4; 2019-D1-P3; 2019-D1-P4; 2019-D2-P6; 2020-D2-P5; 2021-D1-P4; 2021-D2-P7; 2022-D1-P1; 2022-D1-P3; 2023-D1-P1; 2023-D2-P7; 2023-D2-P10; 2024-D1-P2; 2024-D1-P5; 2024-D2-P6; 2024-D2-P8

22.4 PT-INEQ: Inequalities and sharp constants

Core domain	Real analysis / inequalities
Primary / total occurrences	36 / 295
Scope	Problems asking to prove an inequality, find a best constant, compare two expressions, or bound a quantity. These are solved by identifying the source of convexity, sign, norm, integral comparison, or equality cases.
Trigger phrases	prove <= or >=; best constant; sharp; maximum/minimum; compare; positive/nonnegative; bounded by
Linked method IDs	M-RA-09; M-RA-10; M-RA-11; M-RA-12; M-RA-13; M-RA-01; M-UNI-03

Reusable method stack.

1. Cauchy-Schwarz, AM-GM, Holder/Minkowski as algebraic certificates.
2. Convexity/concavity, Jensen, tangent-line and chord-line estimates.
3. Auxiliary function plus derivative sign.
4. Integral inequalities via integration by parts or weighted averaging.
5. Equality-case and extremizer construction.

Proof skeleton.

1. Identify homogeneity or normalize variables.
2. Guess equality cases from small or symmetric examples.
3. Choose the inequality family compatible with the expression.
4. Prove the inequality and then verify all equality/sharpness cases.
5. If constant is asked, give a matching construction.

Trap watch.

- Proving a non-sharp inequality when the statement needs the exact constant.
- Losing equality cases by applying a strict inequality.
- Using Jensen with the wrong convexity direction.

Most common trigger cards inside this problem type.

Trigger	Count	Phrase
TR-RA-07	226	convex, concave, monotone, nondecreasing
TR-ANT-04	134	ring, idempotent, square zero, noncommutative
TR-RA-04	126	lim n->infty of a sum, Riemann sum, asymptotic sum
TR-RA-06	109	series convergence, rearrangement, alternating, absolute convergence
TR-RA-02	107	endpoint equalities, repeated zeros, differentiable, Rolle
TR-ANT-01	94	prime, modulo, residue, divisible by, p-adic power
TR-ANT-03	89	group generated by, subgroup, normal, automorphism, quotient
TR-LA-01	87	rank, dimension, span, image, kernel, invariant subspace

Primary occurrences.

1995-D1-P2; 1995-D1-P6; 1995-D2-P2; 1995-D2-P5; 1995-D2-P6; 1996-D2-P6; 1997-D2-P1; 1997-D2-P3; 1998-D1-P5; 1999-D2-P6; 2000-D1-P4; 2000-D2-P2; 2001-D2-P1; 2001-D2-P2; 2004-D1-P2; 2004-D1-P5; 2004-D2-P5; 2006-D1-P5; 2006-D2-P3; 2008-D2-P1; 2009-D1-P1; 2009-D1-P4; 2010-D2-P1; 2010-D2-P2; 2013-D1-P4; 2013-D2-P3; 2014-D2-P2; 2014-D2-P3; 2014-D2-P5; 2015-D1-P5; 2015-D2-P6; 2016-D2-P6; 2018-D1-P1; 2019-D1-P1; 2020-D1-P4; 2021-D1-P3

All linked occurrences.

1994-D1-P1; 1994-D1-P2; 1994-D1-P3; 1994-D1-P4; 1994-D1-P5; 1994-D1-P6; 1994-D2-P1; 1994-D2-P2; 1994-D2-P5; 1994-D2-P6; 1995-D1-P1; 1995-D1-P2; 1995-D1-P3; 1995-D1-P4; 1995-D1-P6; 1995-D2-P1; 1995-D2-P2; 1995-D2-P3; 1995-D2-P4; 1995-D2-P5; 1995-D2-P6; 1996-D1-P1; 1996-D1-P2; 1996-D1-P3; 1996-D1-P4; 1996-D1-P5; 1996-D1-P6; 1996-D2-P1; 1996-D2-P2; 1996-D2-P3; 1996-D2-P4; 1996-D2-P5; 1996-D2-P6; 1997-D1-P1; 1997-D1-P2; 1997-D1-P3; 1997-D1-P4; 1997-D1-P5; 1997-D1-P6; 1997-D2-P1; 1997-D2-P2; 1997-D2-P3; 1997-D2-P4; 1997-D2-P5; 1997-D2-P6; 1998-D1-P2; 1998-D1-P3; 1998-D1-P4; 1998-D1-P5; 1998-D1-P6; 1998-D2-P1; 1998-D2-P2; 1998-D2-P3; 1998-D2-P4; 1998-D2-P5; 1998-D2-P6; 1999-D1-P1; 1999-D1-P2; 1999-D1-P3; 1999-D1-P4; 1999-D1-P5; 1999-D1-P6; 1999-D2-P2; 1999-D2-P3; 1999-D2-P4; 1999-D2-P5; 1999-D2-P6; 2000-D1-P1; 2000-D1-P2; 2000-D1-P3; 2000-D1-P4; 2000-D1-P6; 2000-D2-P1; 2000-D2-P2; 2000-D2-P3; 2000-D2-P4; 2000-D2-P5; 2001-D1-P1; 2001-D1-P2; 2001-D1-P3; 2001-D1-P4; 2001-D1-P5; 2001-D1-P6; 2001-D2-P1; 2001-D2-P2; 2001-D2-P3; 2001-D2-P4; 2001-D2-P5; 2001-D2-P6; 2002-D1-P1; 2002-D1-P2; 2002-D1-P3; 2002-D1-P4; 2002-D1-P5; 2002-D1-P6; 2002-D2-P1; 2002-D2-P2; 2002-D2-P3; 2002-D2-P4; 2002-D2-P5; 2002-D2-P6; 2003-D1-P1; 2003-D1-P4; 2003-D1-P5; 2003-D1-P6; 2003-D2-P2; 2003-D2-P3; 2003-D2-P4; 2003-D2-P5; 2003-D2-P6; 2004-D1-P1; 2004-D1-P2; 2004-D1-P3; 2004-D1-P4; 2004-D1-P5; 2004-D1-P6; 2004-D2-P2; 2004-D2-P3; 2004-D2-P4; 2004-D2-P5

; 2004-D2-P6; 2005-D1-P1; 2005-D1-P2; 2005-D1-P3; 2005-D1-P4; 2005-D1-P5; 2005-D1-P6; 2005-D2-P1; 2005-D2-P3; 2005-D2-P5; 2005-D2-P6; 2006-D1-P1
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; 2008-D2-P1; 2008-D2-P2; 2008-D2-P3; 2008-D2-P5; 2008-D2-P6; 2009-D1-P1; 2009-D1-P3; 2009-D1-P4; 2009-D1-P5; 2009-D2-P1; 2009-D2-P2; 2009-D2-P3
; 2009-D2-P5; 2010-D1-P1; 2010-D1-P2; 2010-D1-P3; 2010-D1-P4; 2010-D1-P5; 2010-D2-P1; 2010-D2-P2; 2010-D2-P3; 2010-D2-P4; 2011-D1-P1; 2011-D1-P2
; 2011-D1-P3; 2011-D1-P4; 2011-D1-P5; 2011-D2-P1; 2011-D2-P2; 2011-D2-P3; 2011-D2-P4; 2011-D2-P5; 2012-D1-P1; 2012-D1-P2; 2012-D1-P3; 2012-D1-P4
; 2012-D1-P5; 2012-D2-P2; 2012-D2-P3; 2012-D2-P4; 2012-D2-P5; 2013-D1-P2; 2013-D1-P3; 2013-D1-P4; 2013-D1-P5; 2013-D2-P2; 2013-D2-P3; 2013-D2-P4
; 2013-D2-P5; 2014-D1-P1; 2014-D1-P2; 2014-D1-P4; 2014-D1-P5; 2014-D2-P2; 2014-D2-P3; 2014-D2-P4; 2014-D2-P5; 2015-D1-P1; 2015-D1-P2; 2015-D1-P3;
2015-D1-P4; 2015-D1-P5; 2015-D2-P6; 2015-D2-P7; 2015-D2-P8; 2015-D2-P9; 2015-D2-P10; 2016-D1-P2; 2016-D1-P3; 2016-D1-P4; 2016-D1-P5; 2016-D2-P6;
2016-D2-P7; 2016-D2-P8; 2016-D2-P9; 2016-D2-P10; 2017-D1-P1; 2017-D1-P2; 2017-D1-P3; 2017-D1-P5; 2017-D2-P6; 2017-D2-P7; 2017-D2-P9; 2017-D2-P10
; 2018-D1-P1; 2018-D1-P3; 2018-D1-P4; 2018-D1-P5; 2018-D2-P6; 2018-D2-P7; 2018-D2-P8; 2018-D2-P9; 2018-D2-P10; 2019-D1-P1; 2019-D1-P3; 2019-D1-P4
; 2019-D1-P5; 2019-D2-P7; 2019-D2-P8; 2019-D2-P9; 2019-D2-P10; 2020-D1-P1; 2020-D1-P2; 2020-D1-P3; 2020-D1-P4; 2020-D2-P5; 2020-D2-P7; 2020-D2-P8
; 2021-D1-P2; 2021-D1-P3; 2021-D1-P4; 2021-D2-P5; 2021-D2-P7; 2021-D2-P8; 2022-D1-P1; 2022-D1-P2; 2022-D1-P3; 2022-D1-P4; 2022-D2-P5; 2022-D2-P7;
2022-D2-P8; 2023-D1-P1; 2023-D1-P4; 2023-D1-P5; 2023-D2-P6; 2023-D2-P7; 2023-D2-P8; 2023-D2-P9; 2023-D2-P10; 2024-D1-P1; 2024-D1-P2; 2024-D1-P3;
2024-D1-P4; 2024-D1-P5; 2024-D2-P6; 2024-D2-P7; 2024-D2-P8; 2024-D2-P9; 2024-D2-P10

22.5 PT-LIM: Limits, asymptotics, Riemann sums, and growth comparison

Core domain	Real analysis / asymptotics
Primary / total occurrences	23 / 191
Scope	Problems where the answer is a limit or asymptotic estimate for sums, products, sequences, functions, powers, or recursively defined quantities.
Trigger phrases	lim; as n tends to infinity; as x tends to 0 or infinity; O(...); asymptotic; Riemann sum; tail estimate
Linked method IDs	M-RA-05; M-RA-06; M-RA-07; M-RA-08; M-RA-14; M-RA-15

Reusable method stack.

1. Scale variables to expose a Riemann sum.
2. Split into main range and tail range.
3. Use monotone comparison or squeeze.
4. Use Taylor expansion with controlled remainder.
5. Use L-Hospital or logarithms for ratios/products.

Proof skeleton.

1. Identify the natural scale.
2. Separate singular endpoints or tails.
3. Replace the expression by an integral/main term plus error.
4. Bound the error uniformly enough to pass to the limit.
5. Check whether the limit is finite, zero, or infinite.

Trap watch.

- Interchanging limits and infinite sums without uniform/tail control.
- Ignoring singular endpoints.
- Using Taylor expansion outside its valid range.

Most common trigger cards inside this problem type.

Trigger	Count	Phrase
TR-RA-07	138	convex, concave, monotone, nondecreasing
TR-RA-04	133	lim n->infty of a sum, Riemann sum, asymptotic sum
TR-RA-06	117	series convergence, rearrangement, alternating, absolute convergence
TR-RA-02	94	endpoint equalities, repeated zeros, differentiable, Rolle
TR-ANT-04	70	ring, idempotent, square zero, noncommutative
TR-LA-01	44	rank, dimension, span, image, kernel, invariant subspace
TR-ANT-01	43	prime, modulo, residue, divisible by, p-adic power
TR-ANT-03	39	group generated by, subgroup, normal, automorphism, quotient

Primary occurrences.

1994-D1-P2; 1994-D2-P6; 1995-D1-P3; 1996-D1-P4; 1996-D1-P5; 1997-D1-P1; 1998-D1-P3; 2001-D1-P3; 2003-D1-P1; 2003-D1-P5; 2003-D2-P2; 2003-D2-P6; 2004-D1-P3; 2005-D1-P5; 2011-D2-P1; 2014-D1-P2; 2015-D2-P7; 2017-D2-P6; 2017-D2-P9; 2018-D2-P10; 2020-D2-P8; 2024-D1-P2; 2024-D2-P8

All linked occurrences.

1994-D1-P2; 1994-D1-P3; 1994-D1-P4; 1994-D1-P5; 1994-D2-P1; 1994-D2-P3; 1994-D2-P5; 1994-D2-P6; 1995-D1-P2; 1995-D1-P3; 1995-D1-P4; 1995-D1-P5; 1995-D1-P6; 1995-D2-P2; 1995-D2-P5; 1995-D2-P6; 1996-D1-P2; 1996-D1-P4; 1996-D1-P5; 1996-D2-P1; 1996-D2-P4; 1996-D2-P5; 1996-D2-P6; 1997-D1-P1; 1997-D1-P2; 1997-D1-P4; 1997-D1-P5; 1997-D1-P6; 1997-D2-P1; 1997-D2-P3; 1997-D2-P6; 1998-D1-P2; 1998-D1-P3; 1998-D1-P4; 1998-D1-P5; 1998-D1-P6; 1998-D2-P1; 1998-D2-P2; 1998-D2-P6; 1999-D1-P2; 1999-D1-P4; 1999-D2-P2; 1999-D2-P3; 1999-D2-P4; 1999-D2-P5; 1999-D2-P6; 2000-D1-P1; 2000-D1-P2; 2000-D1-P4; 2000-D1-P6; 2000-D2-P2; 2000-D2-P4; 2000-D2-P5; 2000-D2-P6; 2001-D1-P1; 2001-D1-P3; 2001-D1-P5; 2001-D1-P6; 2001-D2-P1; 2001-D2-P2; 2001-D2-P4; 2001-D2-P5; 2002-D1-P1; 2002-D1-P2; 2002-D1-P3; 2002-D1-P4; 2002-D1-P5; 2002-D1-P6; 2002-D2-P3; 2002-D2-P4; 2003-D1-P1; 2003-D1-P2; 2003-D1-P3; 2003-D1-P5; 2003-D1-P6; 2003-D2-P2; 2003-D2-P5; 2003-D2-P6; 2004-D1-P2; 2004-D1-P3; 2004-D1-P4; 2004-D1-P5; 2004-D1-P6; 2004-D2-P3; 2004-D2-P4; 2004-D2-P5; 2004-D2-P6; 2005-D1-P1; 2005-D1-P3; 2005-D1-P4; 2005-D1-P5; 2005-D2-P1; 2005-D2-P2; 2005-D2-P3; 2005-D2-P5; 2005-D2-P6; 2006-D1-P1; 2006-D1-P3; 2006-D1-P4; 2006-D1-P5; 2006-D1-P6; 2006-D2-P1; 2006-D2-P3; 2006-D2-P4; 2006-D2-P5; 2006-D2-P6; 2007-D1-P2; 2007-D1-P5; 2007-D1-P6; 2007-D2-P1; 2007-D2-P2; 2007-D2-P3; 2007-D2-P6; 2008-D1-P2; 2008-D1-P3; 2008-D1-P4; 2008-D1-P5; 2008-D2-P1; 2008-D2-P6; 2009-D1-P1; 2009-D1-P4; 2009-D1-P5; 2009-D2-P3; 2009-D2-P5; 2010-D1-P2; 2010-D1-P3; 2010-D1-P4; 2010-D2-P1; 2010-D2-P2; 2010-D2-P3; 2010-D2-P5; 2011-D1-P1; 2011-D1-P2; 2011-D1-P5; 2011-D2-P1; 2011-D2-P2; 2011-D2-P3; 2011-D2-P4; 2011-D2-P5; 2012-D1-P1; 2012-D1-P4; 2012-D2-P2; 2013-D1-P2; 2013-D1-P4

; 2013-D1-P5; 2013-D2-P3; 2013-D2-P4; 2014-D1-P2; 2014-D1-P4; 2014-D2-P2; 2014-D2-P3; 2014-D2-P5; 2015-D1-P5; 2015-D2-P6; 2015-D2-P7; 2015-D2-P8 ; 2016-D1-P2; 2016-D1-P5; 2016-D2-P6; 2016-D2-P7; 2017-D1-P2; 2017-D1-P3; 2017-D2-P6; 2017-D2-P8; 2017-D2-P9; 2018-D1-P1; 2018-D1-P4; 2018-D2-P7; 2018-D2-P10; 2019-D1-P1; 2019-D1-P3; 2019-D1-P4; 2019-D2-P6; 2019-D2-P7; 2019-D2-P8; 2020-D1-P1; 2020-D1-P2; 2020-D1-P4; 2020-D2-P5; 2020-D2-P8; 2021-D1-P3; 2021-D1-P4; 2023-D1-P1; 2023-D1-P5; 2023-D2-P7; 2023-D2-P8; 2024-D1-P2; 2024-D1-P4; 2024-D1-P5; 2024-D2-P8; 2024-D2-P10

22.6 PT-INT: Integral estimates, averaging, oscillation, and integration by parts

Core domain	Real analysis / integration
Primary / total occurrences	8 / 143
Scope	Problems where integrals are not just notation but the core mechanism: weighted inequalities, periodic averaging, double integrals, orthogonality, integral transforms, or integration by parts.
Trigger phrases	integral; double integral; periodic; orthonormal in L2; integration by parts; average value; oscillatory factor
Linked method IDs	M-RA-06; M-RA-13; M-RA-16; M-RA-17; M-RA-18; M-LA-16

Reusable method stack.

1. Convert hypotheses into integral inequalities using Fubini.
2. Use periodic averaging and uniform continuity.
3. Use integration by parts to move derivatives to controlled terms.
4. Use dominated/monotone convergence when limits and integrals interact.
5. Exploit orthogonality and inner products in L2.

Proof skeleton.

1. Rewrite the integral in the measure/coordinate system suggested by the problem.
2. Identify whether the task is averaging, bounding, or asymptotic.
3. Apply the right integral theorem or elementary substitute.
4. Bound remainders and justify limit passages.
5. Check endpoint singularities and positivity.

Trap watch.

- Forgetting absolute convergence or domination.
- Treating oscillatory integrals as zero without a quantitative estimate.
- Losing signs in integration by parts.

Most common trigger cards inside this problem type.

Trigger	Count	Phrase
TR-RA-07	106	convex, concave, monotone, nondecreasing
TR-RA-06	99	series convergence, rearrangement, alternating, absolute convergence
TR-RA-04	81	lim n->infty of a sum, Riemann sum, asymptotic sum
TR-ANT-04	68	ring, idempotent, square zero, noncommutative
TR-RA-02	65	endpoint equalities, repeated zeros, differentiable, Rolle
TR-ANT-03	35	group generated by, subgroup, normal, automorphism, quotient
TR-RA-05	33	periodic, oscillatory integral, sin(nx), cos(nx)
TR-ANT-05	31	gcd, coprime, relatively prime, Bezout

Primary occurrences.

1996-D1-P2; 1997-D1-P2; 1999-D2-P3; 2004-D1-P6; 2005-D2-P1; 2010-D1-P2; 2012-D2-P2; 2018-D2-P7

All linked occurrences.

1994-D1-P2; 1994-D1-P3; 1994-D1-P4; 1994-D1-P5; 1994-D1-P6; 1994-D2-P1; 1994-D2-P2; 1994-D2-P3; 1994-D2-P5; 1994-D2-P6; 1995-D1-P3; 1995-D1-P5; 1995-D1-P6; 1995-D2-P6; 1996-D1-P2; 1996-D1-P4; 1996-D1-P5; 1996-D2-P1; 1996-D2-P3; 1996-D2-P4; 1996-D2-P5; 1996-D2-P6; 1997-D1-P1; 1997-D1-P2; 1997-D1-P4; 1997-D1-P6; 1997-D2-P3; 1998-D1-P2; 1998-D1-P3; 1998-D1-P6; 1998-D2-P3; 1998-D2-P4; 1998-D2-P6; 1999-D1-P4; 1999-D1-P6; 1999-D2-P2; 1999-D2-P3; 1999-D2-P4; 1999-D2-P5; 2000-D1-P6; 2000-D2-P4; 2000-D2-P5; 2000-D2-P6; 2001-D1-P1; 2001-D1-P3; 2001-D1-P5; 2001-D2-P2; 2001-D2-P4; 2001-D2-P5; 2002-D1-P2; 2002-D1-P3; 2002-D1-P4; 2002-D1-P5; 2002-D1-P6; 2002-D2-P4; 2002-D2-P6; 2003-D1-P1; 2003-D1-P2; 2003-D1-P3; 2003-D2-P2; 2003-D2-P5; 2003-D2-P6; 2004-D1-P2; 2004-D1-P3; 2004-D1-P4; 2004-D1-P5; 2004-D1-P6; 2004-D2-P2; 2004-D2-P3; 2004-D2-P4; 2004-D2-P6; 2005-D1-P1; 2005-D1-P4; 2005-D1-P5; 2005-D2-P1; 2005-D2-P2; 2005-D2-P3; 2005-D2-P4; 2005-D2-P5; 2006-D1-P1; 2006-D1-P4; 2006-D1-P6; 2006-D2-P1; 2006-D2-P2; 2006-D2-P4; 2006-D2-P5; 2006-D2-P6; 2007-D1-P6; 2007-D2-P2; 2008-D1-P1; 2008-D1-P3; 2008-D1-P4; 2008-D2-P6; 2009-D1-P1; 2009-D1-P4; 2009-D2-P2; 2009-D2-P5; 2010-D1-P1; 2010-D1-P2; 2010-D1-P3; 2010-D1-P4; 2010-D2-P1; 2010-D2-P2; 2010-D2-P3; 2011-D1-P1; 2011-D1-P2; 2011-D1-P4; 2011-D1-P5; 2011-D2-P1; 2011-D2-P5; 2012-D1-P1; 2012-D2-P2; 2013-D1-P5; 2013-D2-P4; 2014-D1-P2; 2014-D1-P4; 2015-D1-P3; 2015-D2-P7; 2015-D2-P8; 2016-D1-P1; 2017-D2-P6; 2018-D1-P1; 2018-D2-P6; 2018-D2-P7; 2018-D2-P10; 2019-D2-P6; 2019-D2-P8; 2019-D2-P10; 2020-D1-P2; 2020-D2-P8; 2021-D1-P2; 2021-D1-P4; 2021-D2-P6; 2022-D1-P1; 2022-D2-P8; 2023-D1-P5; 2023-D2-P8; 2024-D1-P2; 2024-D1-P4; 2024-D1-P5; 2024-D2-P6; 2024-D2-P8; 2024-D2-P10

22.7 PT-POLY: Polynomial roots, coefficient constraints, divisibility, and interpolation

Core domain	Complex analysis / polynomials
Primary / total occurrences	31 / 68
Scope	Problems where polynomial structure is decisive: roots, multiplicity, divisibility, integer coefficients, rational roots, Vieta relations, interpolation, or polynomial identities.
Trigger phrases	polynomial; all roots; divides; integer coefficients; rational roots; coefficient permutation; degree n; more roots than degree
Linked method IDs	M-ANT-06; M-ANT-07; M-ANT-08; M-ANT-09; M-ANT-10; M-PC-01; M-PC-02

Reusable method stack.

1. Degree versus number of roots.
2. Factorization over R, C, Z[x], or finite fields.
3. Vieta and symmetric sums.
4. Interpolation and Lagrange basis.
5. Logarithmic derivative for root-location inequalities.

Proof skeleton.

1. Choose the coefficient/root representation.
2. Identify the ring or field in which factorization is valid.
3. Convert the condition into roots, coefficients, or divisibility.
4. Use degree/multiplicity or Vieta to force the answer.
5. Check zero polynomial, leading coefficient, and repeated roots.

Trap watch.

- Using unique factorization over the wrong ring.
- Forgetting repeated roots/multiplicities.
- Assuming rational roots when only complex roots are known.

Most common trigger cards inside this problem type.

Trigger	Count	Phrase
TR-RA-07	40	convex, concave, monotone, nondecreasing
TR-PC-04	27	root of unity, Fourier coefficient, unit circle average
TR-ANT-01	27	prime, modulo, residue, divisible by, p-adic power
TR-LA-01	24	rank, dimension, span, image, kernel, invariant subspace
TR-ANT-04	24	ring, idempotent, square zero, noncommutative
TR-LA-05	20	eigenvalue, characteristic polynomial, diagonalizable, similar
TR-ANT-03	18	group generated by, subgroup, normal, automorphism, quotient
TR-RA-06	17	series convergence, rearrangement, alternating, absolute convergence

Primary occurrences.

1995-D2-P3; 2000-D2-P3; 2000-D2-P4; 2001-D1-P4; 2003-D1-P6; 2005-D1-P4; 2005-D2-P2; 2006-D1-P4; 2006-D2-P5; 2007-D1-P1; 2007-D1-P5; 2007-D1-P6; 2007-D2-P6; 2008-D1-P3; 2008-D2-P3; 2009-D2-P4; 2010-D2-P5; 2011-D1-P3; 2011-D2-P4; 2012-D1-P5; 2012-D2-P1; 2012-D2-P3; 2013-D2-P1; 2014-D1-P3; 2015-D2-P10; 2017-D1-P5; 2017-D2-P7; 2018-D2-P9; 2021-D2-P7; 2023-D1-P3; 2024-D1-P1

All linked occurrences.

1994-D2-P4; 1995-D1-P5; 1995-D2-P3; 1995-D2-P4; 1996-D2-P2; 1996-D2-P6; 1998-D1-P2; 1998-D1-P5; 1998-D2-P2; 1999-D1-P1; 1999-D2-P6; 2000-D1-P2; 2000-D1-P3; 2000-D1-P6; 2000-D2-P3; 2000-D2-P4; 2000-D2-P6; 2001-D1-P4; 2001-D2-P1; 2001-D2-P3; 2002-D2-P5; 2003-D1-P3; 2003-D1-P6; 2003-D2-P2; 2005-D1-P4; 2005-D2-P2; 2006-D1-P4; 2006-D2-P5; 2007-D1-P1; 2007-D1-P3; 2007-D1-P5; 2007-D1-P6; 2007-D2-P6; 2008-D1-P6; 2008-D1-P3; 2008-D2-P1; 2008-D2-P2; 2008-D2-P3; 2008-D2-P4; 2008-D2-P5; 2009-D1-P3; 2009-D2-P1; 2009-D2-P4; 2010-D2-P5; 2011-D1-P3; 2011-D2-P4; 2012-D1-P5; 2012-D2-P1; 2012-D2-P3; 2013-D1-P5; 2013-D2-P1; 2014-D1-P3; 2015-D1-P1; 2015-D2-P9; 2015-D2-P10; 2016-D2-P10; 2017-D1-P1; 2017-D1-P5; 2017-D2-P7; 2018-D2-P9; 2020-D1-P4; 2021-D2-P7; 2022-D2-P7; 2023-D1-P2; 2023-D1-P3; 2023-D2-P6; 2024-D1-P1; 2024-D2-P7

22.8 PT-FINITE: Finite fields, modular arithmetic, residues, and prime-power bookkeeping

Core domain	Algebra / number theory
Primary / total occurrences	71 / 119
Scope	Problems where working modulo primes, finite fields, residues, Chinese remainders, field characteristic, prime powers, or divisibility gives the main obstruction or construction.
Trigger phrases	prime; modulo; divisible by; characteristic of a field; finite field; residue class; CRT; gcd
Linked method IDs	M-ANT-01; M-ANT-02; M-ANT-03; M-ANT-04; M-ANT-05; M-ANT-16

Reusable method stack.

1. Reduce polynomial/function statements modulo p.
2. Use degree versus roots in a finite field.
3. Use Fermat/Frobenius and cyclicity of finite multiplicative groups.
4. Use CRT to combine congruences.
5. Track valuations and coprimality to assign factors.

Proof skeleton.

1. Identify the modulus or prime-power structure.
2. Reduce all objects into the relevant ring/field.
3. Use a finite-field theorem or counting obstruction.
4. Lift or reconstruct with CRT/valuations if needed.
5. Verify exceptional primes and zero divisors.

Trap watch.

- Treating Z/nZ as a field when n is composite.
- Ignoring the prime p=2 or p dividing a denominator.
- Forgetting that polynomial functions over finite fields are modulo x^p-x.

Most common trigger cards inside this problem type.

Trigger	Count	Phrase
TR-ANT-01	107	prime, modulo, residue, divisible by, p-adic power
TR-ANT-03	94	group generated by, subgroup, normal, automorphism, quotient
TR-ANT-04	84	ring, idempotent, square zero, noncommutative
TR-RA-07	74	convex, concave, monotone, nondecreasing
TR-ANT-05	62	gcd, coprime, relatively prime, Bezout
TR-ANT-02	36	field characteristic, F_p, polynomial function modulo x^p-x
TR-LA-01	32	rank, dimension, span, image, kernel, invariant subspace
TR-RA-06	30	series convergence, rearrangement, alternating, absolute convergence

Primary occurrences.

1994-D1-P3; 1994-D1-P6; 1995-D2-P4; 1996-D2-P2; 1997-D1-P3; 1997-D1-P4; 1997-D1-P5; 1999-D1-P3; 1999-D2-P2; 2001-D1-P1; 2001-D2-P6; 2002-D1-P3; 2002-D1-P6; 2002-D2-P3; 2003-D1-P2; 2005-D2-P6; 2006-D1-P2; 2006-D1-P3; 2006-D2-P1; 2006-D2-P4; 2007-D1-P2; 2007-D2-P2; 2007-D2-P4; 2007-D2-P5; 2008-D2-P5; 2009-D1-P5; 2009-D2-P3; 2009-D2-P5; 2010-D1-P4; 2010-D1-P5; 2010-D2-P4; 2011-D1-P5; 2011-D2-P2; 2012-D1-P1; 2012-D1-P2; 2012-D2-P4; 2013-D1-P5; 2013-D2-P2; 2013-D2-P4; 2014-D1-P4; 2014-D2-P1; 2015-D1-P2; 2015-D1-P4; 2016-D1-P2; 2016-D1-P3; 2016-D2-P9; 2017-D1-P3; 2017-D2-P8; 2017-D2-P10; 2018-D1-P5; 2018-D2-P6; 2018-D2-P8; 2019-D1-P2; 2019-D1-P4; 2019-D1-P5; 2019-D2-P7; 2019-D2-P9; 2020-D1-P3; 2020-D2-P6; 2021-D2-P5; 2021-D2-P8; 2022-D1-P2; 2022-D1-P3; 2022-D1-P4; 2022-D2-P8; 2023-D1-P4; 2023-D2-P10; 2024-D1-P3; 2024-D1-P5; 2024-D2-P9; 2024-D2-P10

All linked occurrences.

1994-D1-P3; 1994-D1-P6; 1994-D2-P4; 1995-D2-P4; 1996-D2-P2; 1996-D2-P3; 1997-D1-P3; 1997-D1-P4; 1997-D1-P5; 1999-D1-P3; 1999-D2-P1; 1999-D2-P2; 2000-D1-P5; 2000-D2-P1; 2000-D2-P3; 2001-D1-P1; 2001-D1-P2; 2001-D1-P4; 2001-D2-P1; 2001-D2-P6; 2002-D1-P3; 2002-D1-P6; 2002-D2-P3; 2003-D1-P2; 2003-D2-P4; 2005-D1-P2; 2005-D1-P4; 2005-D2-P2; 2005-D2-P6; 2006-D1-P2; 2006-D1-P3; 2006-D1-P4; 2006-D2-P1; 2006-D2-P4; 2006-D2-P5; 2007-D1-P1; 2007-D1-P2; 2007-D1-P4; 2007-D1-P6; 2007-D2-P2; 2007-D2-P4; 2007-D2-P5; 2008-D1-P5; 2008-D2-P1; 2008-D2-P3; 2008-D2-P4; 2008-D2-P5; 2009-D1-P5; 2009-D2-P3; 2009-D2-P4; 2009-D2-P5; 2010-D1-P4; 2010-D1-P5; 2010-D2-P4; 2010-D2-P5; 2011-D1-P3; 2011-D1-P5; 2011-D2-P2; 2012-D1-P1; 2012-D1-P2

; 2012-D1-P3; 2012-D1-P5; 2012-D2-P1; 2012-D2-P3; 2012-D2-P4; 2012-D2-P5; 2013-D1-P5; 2013-D2-P2; 2013-D2-P4; 2014-D1-P4; 2014-D2-P1; 2015-D1-P1 ; 2015-D1-P2; 2015-D1-P4; 2015-D2-P10; 2016-D1-P2; 2016-D1-P3; 2016-D1-P4; 2016-D1-P5; 2016-D2-P8; 2016-D2-P9; 2017-D1-P1; 2017-D1-P3; 2017-D1-P4 ; 2017-D1-P5; 2017-D2-P7; 2017-D2-P8; 2017-D2-P10; 2018-D1-P2; 2018-D1-P5; 2018-D2-P6; 2018-D2-P8; 2019-D1-P2; 2019-D1-P4; 2019-D1-P5; 2019-D2-P7 ; 2019-D2-P9; 2020-D1-P1; 2020-D1-P3; 2020-D2-P6; 2020-D2-P7; 2021-D1-P2; 2021-D2-P5; 2021-D2-P6; 2021-D2-P8; 2022-D1-P2; 2022-D1-P3; 2022-D1-P4; 2022-D2-P6; 2022-D2-P8; 2023-D1-P4; 2023-D1-P5; 2023-D2-P10; 2024-D1-P3; 2024-D1-P4; 2024-D1-P5; 2024-D2-P7; 2024-D2-P9; 2024-D2-P10

22.9 PT-COMB: Graph, set-system, permutation, word, and counting structures

Core domain	Combinatorics / discrete structures
Primary / total occurrences	29 / 117
Scope	Problems where objects must be counted, encoded, paired, ordered, partitioned, or represented as graphs/words/permutations/set systems.
Trigger phrases	family of sets; permutation; grid; words; graph/friends/town; at least/at most number of objects; coloring; cycle
Linked method IDs	M-CMB-01; M-CMB-02; M-CMB-03; M-CMB-04; M-CMB-05; M-CMB-06; M-CMB-07

Reusable method stack.

1. Double count ordered objects.
2. Build a graph from relations and use cycles/degrees/walks.
3. Use pigeonhole thresholds and extremal choices.
4. Build recurrences or generating functions.
5. Use involutions, bijections, and complement counting.

Proof skeleton.

1. Define the combinatorial object cleanly.
2. Translate conditions into graph/set/word/permutation language.
3. Find the invariant: degree, parity, cycle, count, or closure.
4. Prove upper and lower bounds or build a bijection.
5. Check whether labels, order, and repetitions matter.

Trap watch.

- Counting unordered objects as ordered or vice versa.
- Forgetting that graph cycles must have distinct vertices.
- Missing the construction that proves sharpness.

Most common trigger cards inside this problem type.

Trigger	Count	Phrase
TR-ANT-04	98	ring, idempotent, square zero, noncommutative
TR-RA-07	92	convex, concave, monotone, nondecreasing
TR-CMB-03	50	town, friends, grid, rows/columns, graph, cycle
TR-ANT-03	44	group generated by, subgroup, normal, automorphism, quotient
TR-ANT-01	41	prime, modulo, residue, divisible by, p-adic power
TR-RA-04	40	lim n->infty of a sum, Riemann sum, asymptotic sum
TR-RA-06	38	series convergence, rearrangement, alternating, absolute convergence
TR-CMB-01	38	permutation, inversion, cycle, transposition

Primary occurrences.

1994-D2-P5; 1997-D1-P6; 1998-D1-P2; 1999-D1-P5; 1999-D2-P5; 2002-D2-P2; 2003-D1-P4; 2003-D2-P3; 2003-D2-P4; 2004-D1-P1; 2005-D1-P2; 2008-D1-P6; 2008-D1-P4; 2009-D1-P3; 2011-D1-P4; 2013-D1-P3; 2013-D2-P5; 2015-D2-P8; 2016-D1-P4; 2016-D1-P5; 2019-D2-P8; 2019-D2-P10; 2020-D1-P1; 2021-D1-P2; 2022-D1-P1; 2022-D2-P5; 2022-D2-P6; 2023-D1-P5; 2023-D2-P8

All linked occurrences.

1994-D1-P1; 1994-D1-P5; 1994-D1-P6; 1994-D2-P3; 1994-D2-P4; 1994-D2-P5; 1994-D2-P6; 1995-D1-P3; 1995-D1-P6; 1995-D2-P1; 1995-D2-P4; 1995-D2-P6; 1996-D1-P3; 1996-D2-P2; 1996-D2-P4; 1996-D2-P5; 1996-D2-P6; 1997-D1-P3; 1997-D1-P6; 1997-D2-P3; 1997-D2-P5; 1997-D2-P6; 1998-D1-P1; 1998-D1-P2; 1998-D1-P4; 1998-D1-P6; 1998-D2-P3; 1998-D2-P5; 1998-D2-P6; 1999-D1-P2; 1999-D1-P4; 1999-D1-P5; 1999-D1-P6; 1999-D2-P5; 1999-D2-P6; 2000-D1-P1; 2000-D2-P1; 2000-D2-P2; 2000-D2-P3; 2000-D2-P4; 2000-D2-P5; 2001-D2-P1; 2001-D2-P5; 2002-D1-P4; 2002-D2-P2; 2002-D2-P4; 2002-D2-P5; 2003-D1-P2; 2003-D1-P4; 2003-D2-P3; 2003-D2-P4; 2003-D2-P5; 2003-D2-P6; 2004-D1-P1; 2004-D1-P2; 2004-D1-P4; 2004-D1-P5; 2004-D2-P2; 2005-D1-P2; 2005-D2-P1; 2005-D2-P4; 2005-D2-P5; 2006-D1-P1; 2006-D1-P2; 2006-D2-P1; 2007-D1-P4; 2007-D2-P1; 2007-D2-P3; 2007-D2-P5; 2008-D1-P6; 2008-D1-P1; 2008-D1-P4; 2008-D2-P4; 2009-D1-P1; 2009-D1-P3; 2009-D2-P1; 2009-D2-P4; 2010-D1-P4; 2011-D1-P1; 2011-D1-P4; 2011-D2-P1; 2012-D1-P1; 2012-D1-P3; 2012-D2-P3; 2012-D2-P5; 2013-D1-P3; 2013-D2-P4; 2013-D2-P5; 2014-D2-P4; 2015-D2-P8; 2016-D1-P4; 2016-D1-P5; 2016-D2-P9; 2017-D1-P4; 2018-D2-P6; 2018-D2-P8; 2019-D2-P8; 2019-D2-P10; 2020-D1-P1; 2020-D1-P3; 2020-D2-P7; 2021-D1-P2; 2021-D1-P3; 2021-D2-P6; 2021-D2-P7; 2021-D2-P8; 2022-D1-P1; 2022-D1-P3; 2022-D1-P4; 2022-D2-P5; 2022-D2-P6; 2022-D2-P8; 2023-D1-P4; 2023-D1-P5; 2023-D2-P8; 2023-D2-P9; 2024-D2-P9

22.10 PT-GEO: Geometry by coordinates, metric identities, convexity, and projection

Core domain	Geometry / convexity / metric
Primary / total occurrences	16 / 70
Scope	Problems where the geometry becomes tractable after choosing coordinates, using squared distances, projections, affine transformations, volumes, circumcenters, ellipses/spheres, or Gram matrices.
Trigger phrases	sphere/ellipse/parabola; distance; volume; simplex; convex polygon/body; projection; angle between faces; Hilbert space
Linked method IDs	M-GEO-01; M-GEO-02; M-GEO-03; M-GEO-04; M-GEO-05; M-GEO-06

Reusable method stack.

1. Choose coordinates adapted to a point, line, plane, or center.
2. Square distances and use inner products/Gram matrices.
3. Use affine or barycentric coordinates for simplex/volume problems.
4. Use projection and support arguments for convex objects.
5. Normalize by translations/rotations/scaling.

Proof skeleton.

1. Use symmetries to place the configuration in convenient coordinates.
2. Translate geometric conditions into algebraic equations.
3. Use inner products, Gram matrices, or volume determinants.
4. Prove the geometric claim algebraically or by convexity.
5. Translate back and verify nondegeneracy.

Trap watch.

- Forgetting that squaring inequalities can introduce extraneous cases.
- Using a 2D intuition in higher dimensions without proof.
- Ignoring degenerate configurations.

Most common trigger cards inside this problem type.

Trigger	Count	Phrase
TR-RA-07	68	convex, concave, monotone, nondecreasing
TR-ANT-04	39	ring, idempotent, square zero, noncommutative
TR-ANT-03	24	group generated by, subgroup, normal, automorphism, quotient
TR-RA-02	23	endpoint equalities, repeated zeros, differentiable, Rolle
TR-RA-04	21	lim n->infty of a sum, Riemann sum, asymptotic sum
TR-ANT-01	21	prime, modulo, residue, divisible by, p-adic power
TR-RA-06	18	series convergence, rearrangement, alternating, absolute convergence
TR-CMB-03	18	town, friends, grid, rows/columns, graph, cycle

Primary occurrences.

1996-D1-P6; 1996-D2-P4; 1997-D2-P5; 1998-D2-P5; 2000-D2-P1; 2001-D2-P3; 2002-D1-P1; 2002-D2-P4; 2004-D1-P4; 2004-D2-P3; 2008-D2-P2; 2009-D2-P1; 2011-D2-P5; 2014-D1-P5; 2014-D2-P4; 2023-D2-P9

All linked occurrences.

1995-D2-P1; 1995-D2-P2; 1995-D2-P3; 1996-D1-P2; 1996-D1-P4; 1996-D1-P6; 1996-D2-P1; 1996-D2-P4; 1997-D1-P2; 1997-D2-P5; 1998-D1-P3; 1998-D2-P3; 1998-D2-P5; 1999-D1-P3; 1999-D1-P5; 1999-D2-P5; 2000-D2-P1; 2000-D2-P2; 2001-D1-P4; 2001-D2-P3; 2001-D2-P6; 2002-D1-P1; 2002-D2-P4; 2002-D2-P6; 2003-D2-P3; 2004-D1-P1; 2004-D1-P2; 2004-D1-P4; 2004-D2-P2; 2004-D2-P3; 2005-D1-P2; 2008-D2-P2; 2009-D2-P1; 2010-D1-P5; 2010-D2-P2; 2010-D2-P4; 2011-D1-P3; 2011-D2-P5; 2012-D1-P3; 2013-D2-P3; 2013-D2-P5; 2014-D1-P4; 2014-D1-P5; 2014-D2-P4; 2014-D2-P5; 2015-D1-P2; 2015-D1-P4; 2015-D1-P5; 2015-D2-P9; 2016-D2-P10; 2017-D2-P7; 2017-D2-P10; 2018-D1-P5; 2018-D2-P6; 2018-D2-P7; 2019-D1-P4; 2019-D1-P5; 2019-D2-P10; 2020-D1-P1; 2020-D1-P3; 2021-D1-P3; 2021-D2-P7; 2021-D2-P8; 2022-D2-P5; 2022-D2-P8; 2023-D2-P8; 2023-D2-P9; 2024-D1-P2; 2024-D2-P8; 2024-D2-P10

22.11 PT-EXT: Extremal configurations, compactness, existence, and counterexamples

Core domain	Universal proof infrastructure
Primary / total occurrences	0 / 35
Scope	Problems where the main act is to find or rule out an extremal object, prove existence/nonexistence, build a counterexample, or show a maximum/minimum is attained.
Trigger phrases	exists; does there exist; maximum possible; minimum possible; attains; compact; closed bounded; counterexample; minimal
Linked method IDs	M-UNI-01; M-UNI-02; M-UNI-03; M-UNI-04; M-UNI-05; M-UNI-06; M-UNI-10

Reusable method stack.

1. Use compactness/EVT to get an extremal point.
2. Choose a minimal counterexample or extremal configuration.
3. Perturb or replace the object to contradict extremality.
4. Give an explicit construction for lower bound and an obstruction for upper bound.
5. Separate existence from uniqueness.

Proof skeleton.

1. Decide whether the problem is asking for existence, uniqueness, or sharp bound.
2. If compactness applies, prove the relevant set is compact and the functional continuous.
3. If finite, choose a minimal/maximal counterexample.
4. Derive local improvement or structural contradiction.
5. For sharpness, present the exact construction.

Trap watch.

- Using compactness on nonclosed or unbounded sets.
- Proving only one direction of a sharp statement.
- Not verifying the constructed object satisfies every condition.

Most common trigger cards inside this problem type.

Trigger	Count	Phrase
TR-RA-07	20	convex, concave, monotone, nondecreasing
TR-RA-02	17	endpoint equalities, repeated zeros, differentiable, Rolle
TR-RA-04	14	lim n->infty of a sum, Riemann sum, asymptotic sum
TR-ANT-04	12	ring, idempotent, square zero, noncommutative
TR-UNI-02	11	does there exist, prove/disprove, counterexample
TR-ANT-05	9	gcd, coprime, relatively prime, Bezout
TR-ANT-03	8	group generated by, subgroup, normal, automorphism, quotient
TR-ANT-01	8	prime, modulo, residue, divisible by, p-adic power

Primary occurrences.

All linked occurrences.

1994-D1-P2; 1994-D2-P1; 1997-D1-P2; 1997-D2-P4; 1998-D2-P2; 1999-D1-P2; 1999-D2-P4; 2000-D1-P1; 2002-D1-P2; 2002-D2-P2; 2002-D2-P6; 2003-D1-P1; 2003-D2-P3; 2003-D2-P4; 2005-D2-P4; 2013-D1-P2; 2013-D1-P3; 2013-D2-P3; 2013-D2-P5; 2014-D1-P1; 2015-D2-P8; 2016-D1-P1; 2016-D2-P7; 2016-D2-P8; 2017-D1-P2; 2017-D2-P6; 2017-D2-P10; 2018-D1-P2; 2019-D2-P6; 2020-D2-P6; 2020-D2-P7; 2021-D1-P4; 2022-D1-P4; 2022-D2-P5; 2024-D1-P3

22.12 PT-GRP: Group, ring, automorphism, and noncommutative algebra patterns

Core domain	Algebra / abstract algebra
Primary / total occurrences	17 / 91
Scope	Bridge sheet added for corpus completeness. Several IMC problems are not merely modular arithmetic: they use group generators, quotient groups, normal subgroups, automorphisms, rings, idempotents, and noncommutative expansions.
Trigger phrases	group; subgroup; normal; automorphism; ring; idempotent; commutator; generated by
Linked method IDs	M-ANT-11; M-ANT-12; M-ANT-13; M-ANT-14; M-UNI-09

Reusable method stack.

1. Pass to quotient groups and use generators.
2. Represent elements concretely as permutations/matrices where possible.
3. Expand noncommutative identities without reordering.
4. Use commutators and centrality to force abelian behavior.
5. Use group actions or semidirect products for constructions.

Proof skeleton.

1. Identify whether the algebraic object is commutative.
2. Write generators and relations explicitly.
3. Use quotient or subgroup structure to reduce the problem.
4. For constructions, define the operation/action and verify the axioms.
5. Check identity, inverses, and normality conditions.

Trap watch.

- Commuting factors in a noncommutative group/ring.
- Confusing subgroup generated by powers with set of powers.
- Omitting verification that a proposed subset is a subgroup.

Most common trigger cards inside this problem type.

Trigger	Count	Phrase
TR-ANT-04	81	ring, idempotent, square zero, noncommutative
TR-ANT-03	64	group generated by, subgroup, normal, automorphism, quotient
TR-ANT-01	61	prime, modulo, residue, divisible by, p-adic power
TR-RA-07	54	convex, concave, monotone, nondecreasing
TR-ANT-05	46	gcd, coprime, relatively prime, Bezout
TR-CMB-03	32	town, friends, grid, rows/columns, graph, cycle
TR-CMB-01	22	permutation, inversion, cycle, transposition
TR-CMB-04	21	words, strings, functions on finite sets, colorings

Primary occurrences.

1996-D2-P3; 1999-D2-P1; 2000-D1-P5; 2001-D1-P2; 2005-D1-P6; 2007-D1-P4; 2008-D1-P5; 2008-D2-P4; 2010-D2-P3; 2012-D1-P3; 2012-D2-P5; 2016-D2-P8; 2017-D1-P4; 2018-D1-P2; 2020-D2-P7; 2021-D2-P6; 2024-D1-P4

All linked occurrences.

1994-D2-P3; 1996-D2-P2; 1996-D2-P3; 1997-D1-P3; 1997-D1-P4; 1997-D1-P6; 1997-D2-P5; 1997-D2-P6; 1998-D1-P1; 1998-D1-P4; 1998-D2-P5; 1999-D1-P1; 1999-D1-P3; 1999-D1-P5; 1999-D1-P6; 1999-D2-P1; 2000-D1-P5; 2000-D2-P3; 2001-D1-P2; 2001-D2-P6; 2002-D2-P2; 2003-D1-P2; 2003-D2-P3; 2003-D2-P4; 2004-D2-P3; 2005-D1-P2; 2005-D1-P6; 2005-D2-P4; 2006-D1-P2; 2007-D1-P1; 2007-D1-P4; 2007-D2-P1; 2007-D2-P2; 2007-D2-P3; 2007-D2-P4; 2007-D2-P5; 2008-D1-P6; 2008-D1-P1; 2008-D1-P4; 2008-D1-P5; 2008-D2-P3; 2008-D2-P4; 2009-D2-P1; 2009-D2-P4; 2010-D1-P5; 2010-D2-P2; 2010-D2-P3; 2010-D2-P4; 2011-D1-P4; 2012-D1-P2; 2012-D1-P3; 2012-D1-P5; 2012-D2-P3; 2012-D2-P4; 2012-D2-P5; 2013-D2-P2; 2014-D2-P1; 2014-D2-P4; 2015-D1-P2; 2015-D2-P10; 2016-D1-P2; 2016-D1-P3; 2016-D1-P4; 2016-D2-P8; 2016-D2-P9; 2017-D1-P4; 2017-D2-P10; 2018-D1-P2; 2018-D1-P5; 2019-D1-P2; 2019-D2-P8; 2019-D2-P9; 2019-D2-P10; 2020-D1-P3; 2020-D2-P6; 2020-D2-P7; 2021-D1-P2; 2021-D2-P5; 2021-D2-P6; 2021-D2-P8; 2022-D1-P2; 2022-D1-P3; 2022-D1-P4; 2022-D2-P5; 2022-D2-P6; 2022-D2-P7; 2023-D1-P4; 2023-D2-P9; 2024-D1-P3; 2024-D1-P4; 2024-D2-P9

22.13 PT-SEQ: Recurrences, generating functions, and discrete dynamical sequences

Core domain	Analysis / combinatorics bridge
Primary / total occurrences	0 / 13
Scope	Bridge sheet for recurrent sequences, generating functions, telescoping products, recursive matrices, and iterative maps where the main job is to discover the hidden invariant or closed form.
Trigger phrases	sequence defined by; recurrence; generating function; sum over n; matrix sequence; iterate
Linked method IDs	M-RA-07; M-CMB-05; M-CMB-06; M-UNI-05; M-UNI-07

Reusable method stack.

1. Change variables to linearize the recurrence.
2. Generate telescoping differences.
3. Introduce ordinary generating functions.
4. Use induction after guessing a closed form.
5. Interpret matrix powers or recursive matrices combinatorially.

Proof skeleton.

1. Compute initial terms.
2. Search for substitution, invariant, or telescoping form.
3. Prove the closed form/recurrence by induction or generating function.
4. Use convergence/asymptotics if required.
5. Check initial cases.

Trap watch.

- Guessing a closed form without induction.
- Using a generating function outside its convergence radius without justification.
- Losing initial terms in index shifts.

Most common trigger cards inside this problem type.

Trigger	Count	Phrase
TR-RA-04	11	lim n->infty of a sum, Riemann sum, asymptotic sum
TR-UNI-07	10	recursively defined sequence, iterative construction
TR-RA-02	10	endpoint equalities, repeated zeros, differentiable, Rolle
TR-RA-07	9	convex, concave, monotone, nondecreasing
TR-PC-05	7	generating function, coefficient extraction, recurrence
TR-RA-06	5	series convergence, rearrangement, alternating, absolute convergence
TR-ANT-05	4	gcd, coprime, relatively prime, Bezout
TR-ANT-04	2	ring, idempotent, square zero, noncommutative

Primary occurrences.

All linked occurrences.

1995-D2-P2; 1997-D1-P1; 2000-D1-P4; 2001-D2-P2; 2003-D1-P1; 2007-D1-P4; 2008-D2-P3; 2010-D1-P3; 2015-D1-P5; 2016-D2-P6; 2018-D1-P1; 2018-D2-P7; 2021-D1-P3

23 Problem-by-Problem Problem-Type Audit

This table assigns each detected problem a primary problem-type sheet, secondary sheets, and the trigger cards that should be read before attempting the problem. It is intentionally concise because the detailed method sequence already appears in the previous Method Sheets audit.

Problem	Primary type	Secondary types	Trigger cards	Existing method tags
1994-D1-P1	PT-LA-DET	PT-LA-SPEC; PT-INEQ; PT-COMB	TR-LA-01; TR-LA-02; TR-LA-05; TR-RA-07; TR-CMB-01; TR-CMB-06	determinant / block manipulation; double counting / pigeonhole
1994-D1-P2	PT-LIM	PT-FE; PT-INEQ; PT-INT; PT-EXT	TR-RA-02; TR-RA-04; TR-RA-07; TR-UNI-02	monotonicity argument; construction / counterexample
1994-D1-P3	PT-FINITE	PT-FE; PT-INEQ; PT-LIM; PT-INT	TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	induction
1994-D1-P4	PT-LA-SPEC	PT-LA-DET; PT-INEQ; PT-LIM; PT-INT	TR-LA-01; TR-LA-05; TR-RA-06; TR-RA-07; TR-GEO-01; TR-UNI-02	spectral decomposition / eigenvalues; construction / counterexample
1994-D1-P5	PT-FE	PT-INEQ; PT-LIM; PT-INT; PT-COMB	TR-RA-04; TR-RA-05; TR-RA-07; TR-ANT-04; TR-ANT-05; TR-CMB-03	auxiliary function / substitution; l'Hospital / asymptotic expansion
1994-D1-P6	PT-FINITE	PT-FE; PT-INEQ; PT-INT; PT-COMB	TR-RA-02; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	monotonicity argument; convexity / Jensen / tangent
1994-D2-P1	PT-FE	PT-INEQ; PT-LIM; PT-INT; PT-EXT	TR-RA-02; TR-RA-05; TR-RA-06; TR-RA-07; TR-UNI-02	monotonicity argument
1994-D2-P2	PT-FE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-INT	TR-LA-01; TR-LA-05; TR-RA-01; TR-RA-05; TR-RA-07; TR-GEO-01	compactness / extremal point; topological / density argument
1994-D2-P3	PT-FE	PT-LIM; PT-INT; PT-COMB; PT-GRP	TR-RA-02; TR-RA-05; TR-RA-06; TR-ANT-04; TR-CMB-03; TR-CMB-04	auxiliary function / substitution
1994-D2-P4	PT-LA-DET	PT-LA-SPEC; PT-POLY; PT-FINITE; PT-COMB	TR-LA-01; TR-LA-05; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-CMB-03	rank-nullity / dimension count; polynomial identity / factorization; double counting / pigeonhole
1994-D2-P5	PT-COMB	PT-LA-DET; PT-INEQ; PT-LIM; PT-INT	TR-LA-01; TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04	induction; Cauchy-Schwarz / norm inequality; rank-nullity / dimension count; construction / counterexample
1994-D2-P6	PT-LIM	PT-INEQ; PT-INT; PT-COMB	TR-RA-02; TR-RA-04; TR-RA-07; TR-ANT-05	monotonicity argument
1995-D1-P1	PT-LA-SPEC	PT-LA-DET; PT-INEQ	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-04; TR-LA-05; TR-CMB-03	spectral decomposition / eigenvalues; determinant / block manipulation; rank-nullity / dimension count
1995-D1-P2	PT-INEQ	PT-FE; PT-LIM	TR-RA-02; TR-RA-04; TR-RA-07	direct structural reasoning

Problem	Primary type	Secondary types	Trigger cards	Existing method tags
1995-D1-P3	PT-LIM	PT-FE; PT-INEQ; PT-INT; PT-COMB	TR-RA-02; TR-RA-04; TR-RA-07; TR-ANT-05	Rolle / mean-value theorem; monotonicity argument; double counting / pigeonhole
1995-D1-P4	PT-FE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-LIM	TR-LA-05; TR-RA-02; TR-RA-04; TR-RA-07; TR-ANT-04; TR-CMB-03	auxiliary function / substitution; monotonicity argument
1995-D1-P5	PT-LA-SPEC	PT-LA-DET; PT-LIM; PT-INT; PT-POLY	TR-LA-01; TR-LA-03; TR-LA-04; TR-LA-05; TR-LA-06; TR-RA-04	polynomial identity / factorization
1995-D1-P6	PT-INEQ	PT-FE; PT-LIM; PT-INT; PT-COMB	TR-RA-01; TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-04	compactness / extremal point; convexity / Jensen / tangent; construction / counterexample; l'Hospital / asymptotic expansion; topological / density argument
1995-D2-P1	PT-LA-DET	PT-LA-SPEC; PT-INEQ; PT-COMB; PT-GEO	TR-LA-01; TR-LA-02; TR-LA-05; TR-LA-07; TR-RA-07; TR-ANT-04	determinant / block manipulation
1995-D2-P2	PT-INEQ	PT-LIM; PT-GEO; PT-SEQ	TR-RA-02; TR-RA-04; TR-RA-07; TR-UNI-07	direct structural reasoning
1995-D2-P3	PT-POLY	PT-INEQ; PT-GEO	TR-RA-05; TR-RA-07; TR-PC-01; TR-PC-03; TR-PC-04; TR-GEO-01	polynomial identity / factorization; roots of unity / Fourier filter
1995-D2-P4	PT-FINITE	PT-FE; PT-INEQ; PT-POLY; PT-COMB	TR-RA-01; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	compactness / extremal point; polynomial identity / factorization
1995-D2-P5	PT-INEQ	PT-LA-DET; PT-LA-SPEC; PT-FE; PT-LIM	TR-LA-05; TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-GEO-01	Riemann-sum / integral comparison
1995-D2-P6	PT-INEQ	PT-FE; PT-LIM; PT-INT; PT-COMB	TR-RA-01; TR-RA-02; TR-RA-04; TR-RA-06; TR-CMB-05; TR-UNI-02	construction / counterexample; topological / density argument
1996-D1-P1	PT-LA-DET	PT-LA-SPEC; PT-INEQ	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-GEO-01	determinant / block manipulation
1996-D1-P2	PT-INT	PT-INEQ; PT-LIM; PT-GEO	TR-RA-02; TR-RA-04; TR-RA-05; TR-RA-06; TR-RA-07	auxiliary function / substitution
1996-D1-P3	PT-LA-SPEC	PT-LA-DET; PT-FE; PT-INEQ; PT-COMB	TR-LA-01; TR-LA-04; TR-LA-05; TR-RA-07; TR-ANT-04; TR-CMB-01	induction; spectral decomposition / eigenvalues; rank-nullity / dimension count; double counting / pigeonhole
1996-D1-P4	PT-LIM	PT-INEQ; PT-INT; PT-GEO	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07	induction

Problem	Primary type	Secondary types	Trigger cards	Existing method tags
1996-D1-P5	PT-LIM	PT-LA-SPEC; PT-FE; PT-INEQ; PT-INT	TR-LA-05; TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-04	Rolle / mean-value theorem; monotonicity argument; Riemann-sum / integral comparison
1996-D1-P6	PT-GEO	PT-LA-DET; PT-LA-SPEC; PT-FE; PT-INEQ	TR-LA-01; TR-LA-05; TR-LA-07; TR-RA-01; TR-RA-02; TR-RA-06	induction; compactness / extremal point; rank-nullity / dimension count; graph encoding / cycles; topological / density argument
1996-D2-P1	PT-FE	PT-INEQ; PT-LIM; PT-INT; PT-GEO	TR-RA-02; TR-RA-04; TR-RA-05; TR-RA-06; TR-RA-07; TR-ANT-05	direct structural reasoning
1996-D2-P2	PT-FINITE	PT-INEQ; PT-POLY; PT-COMB; PT-GRP	TR-RA-07; TR-PC-04; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	induction; polynomial identity / factorization
1996-D2-P3	PT-GRP	PT-LA-DET; PT-INEQ; PT-INT; PT-FINITE	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	auxiliary function / substitution; double counting / pigeonhole
1996-D2-P4	PT-GEO	PT-INEQ; PT-LIM; PT-INT; PT-COMB	TR-RA-01; TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-03; TR-CMB-04	compactness / extremal point; convexity / Jensen / tangent; topological / density argument
1996-D2-P5	PT-FE	PT-INEQ; PT-LIM; PT-INT; PT-COMB	TR-RA-02; TR-RA-04; TR-RA-05; TR-RA-06; TR-RA-07; TR-ANT-04	Riemann-sum / integral comparison; l'Hospital / asymptotic expansion
1996-D2-P6	PT-INEQ	PT-LIM; PT-INT; PT-POLY; PT-COMB	TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-03; TR-CMB-04	double counting / pigeonhole
1997-D1-P1	PT-LIM	PT-INEQ; PT-INT; PT-SEQ	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-05; TR-UNI-07	direct structural reasoning
1997-D1-P2	PT-INT	PT-INEQ; PT-LIM; PT-GEO; PT-EXT	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-UNI-02	construction / counterexample
1997-D1-P3	PT-FINITE	PT-LA-DET; PT-INEQ; PT-COMB; PT-GRP	TR-LA-01; TR-LA-02; TR-LA-03; TR-ANT-01; TR-ANT-02; TR-ANT-03	determinant / block manipulation
1997-D1-P4	PT-FINITE	PT-INEQ; PT-LIM; PT-INT; PT-GRP	TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	induction; monotonicity argument; construction / counterexample
1997-D1-P5	PT-FINITE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-LIM	TR-LA-05; TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04	Cauchy-Schwarz / norm inequality
1997-D1-P6	PT-COMB	PT-INEQ; PT-LIM; PT-INT; PT-GRP	TR-RA-06; TR-ANT-04; TR-CMB-02; TR-CMB-03; TR-CMB-04; TR-UNI-02	induction; construction / counterexample
1997-D2-P1	PT-INEQ	PT-FE; PT-LIM	TR-RA-02; TR-RA-03; TR-RA-04; TR-RA-07	auxiliary function / substitution; l'Hospital / asymptotic expansion

Problem	Primary type	Secondary types	Trigger cards	Existing method tags
1997-D2-P2	PT-LA-DET	PT-LA-SPEC; PT-INEQ	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-GEO-01	determinant / block manipulation; rank-nullity / dimension count
1997-D2-P3	PT-INEQ	PT-FE; PT-LIM; PT-INT; PT-COMB	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-PC-05; TR-ANT-04	Rolle / mean-value theorem
1997-D2-P4	PT-LA-SPEC	PT-LA-DET; PT-FE; PT-INEQ; PT-EXT	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-GEO-01	trace trick / commutator; rank-nullity / dimension count
1997-D2-P5	PT-GEO	PT-FE; PT-INEQ; PT-COMB; PT-GRP	TR-RA-07; TR-ANT-04; TR-CMB-01; TR-CMB-03; TR-CMB-04; TR-GEO-02	auxiliary function / substitution; graph encoding / cycles
1997-D2-P6	PT-FE	PT-INEQ; PT-LIM; PT-COMB; PT-GRP	TR-RA-02; TR-RA-04; TR-ANT-04; TR-CMB-01; TR-CMB-03; TR-CMB-04	double counting / pigeonhole; construction / counterexample
1998-D1-P1	PT-LA-DET	PT-LA-SPEC; PT-COMB; PT-GRP	TR-LA-01; TR-LA-05; TR-ANT-04; TR-CMB-03; TR-CMB-04; TR-GEO-01	rank-nullity / dimension count
1998-D1-P2	PT-COMB	PT-INEQ; PT-LIM; PT-INT; PT-POLY	TR-RA-06; TR-RA-07; TR-ANT-03; TR-ANT-04; TR-CMB-01; TR-CMB-03	polynomial identity / factorization; construction / counterexample
1998-D1-P3	PT-LIM	PT-FE; PT-INEQ; PT-INT; PT-GEO	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07	induction; monotonicity argument
1998-D1-P4	PT-FE	PT-INEQ; PT-LIM; PT-COMB; PT-GRP	TR-RA-02; TR-RA-04; TR-RA-07; TR-ANT-04; TR-CMB-03; TR-CMB-04	topological / density argument
1998-D1-P5	PT-INEQ	PT-LIM; PT-POLY	TR-RA-04; TR-RA-07; TR-PC-01	polynomial identity / factorization
1998-D1-P6	PT-FE	PT-INEQ; PT-LIM; PT-INT; PT-COMB	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07	direct structural reasoning
1998-D2-P1	PT-LA-SPEC	PT-LA-DET; PT-FE; PT-INEQ; PT-LIM	TR-LA-01; TR-LA-05; TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-03	induction; Cauchy-Schwarz / norm inequality; rank-nullity / dimension count
1998-D2-P2	PT-FE	PT-INEQ; PT-LIM; PT-POLY; PT-EXT	TR-RA-01; TR-RA-04; TR-RA-07; TR-UNI-03; TR-UNI-04	compactness / extremal point; polynomial identity / factorization
1998-D2-P3	PT-FE	PT-INEQ; PT-INT; PT-COMB; PT-GEO	TR-RA-05; TR-RA-07; TR-ANT-04; TR-CMB-01; TR-CMB-03; TR-CMB-04	monotonicity argument; double counting / pigeonhole; graph encoding / cycles
1998-D2-P4	PT-FE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-INT	TR-LA-05; TR-RA-05; TR-RA-07; TR-ANT-04; TR-CMB-01; TR-CMB-03	monotonicity argument; double counting / pigeonhole
1998-D2-P5	PT-GEO	PT-LA-SPEC; PT-INEQ; PT-COMB; PT-GRP	TR-LA-05; TR-RA-01; TR-RA-07; TR-ANT-04; TR-CMB-01; TR-CMB-03	double counting / pigeonhole; topological / density argument
1998-D2-P6	PT-FE	PT-INEQ; PT-LIM; PT-INT; PT-COMB	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-PC-05; TR-ANT-04	double counting / pigeonhole; construction / counterexample

Problem	Primary type	Secondary types	Trigger cards	Existing method tags
1999-D1-P1	PT-LA-SPEC	PT-LA-DET; PT-INEQ; PT-POLY; PT-GRP	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-PC-04; TR-ANT-03	compactness / extremal point; spectral decomposition / eigenvalues; determinant / block manipulation; polynomial identity / factorization
1999-D1-P2	PT-FE	PT-INEQ; PT-LIM; PT-COMB; PT-EXT	TR-RA-02; TR-RA-04; TR-RA-07; TR-UNI-02	construction / counterexample
1999-D1-P3	PT-FINITE	PT-FE; PT-INEQ; PT-GEO; PT-GRP	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-UNI-01	direct structural reasoning
1999-D1-P4	PT-FE	PT-INEQ; PT-LIM; PT-INT; PT-COMB	TR-RA-05; TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-03; TR-CMB-04	induction; auxiliary function / substitution; monotonicity argument
1999-D1-P5	PT-COMB	PT-LA-DET; PT-INEQ; PT-GEO; PT-GRP	TR-LA-02; TR-RA-07; TR-ANT-03; TR-ANT-04; TR-CMB-01; TR-CMB-03	determinant / block manipulation; double counting / pigeonhole; graph encoding / cycles
1999-D1-P6	PT-FE	PT-INEQ; PT-INT; PT-COMB; PT-GRP	TR-RA-02; TR-RA-05; TR-RA-07; TR-ANT-04; TR-CMB-03; TR-CMB-04	auxiliary function / substitution; construction / counterexample
1999-D2-P1	PT-GRP	PT-FINITE	TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	direct structural reasoning
1999-D2-P2	PT-FINITE	PT-LA-DET; PT-INEQ; PT-LIM; PT-INT	TR-LA-02; TR-RA-04; TR-RA-05; TR-RA-06; TR-PC-04; TR-PC-05	induction; determinant / block manipulation; polynomial identity / factorization; modular / finite-field reduction; roots of unity / Fourier filter; generating functions
1999-D2-P3	PT-INT	PT-INEQ; PT-LIM	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07	auxiliary function / substitution
1999-D2-P4	PT-FE	PT-INEQ; PT-LIM; PT-INT; PT-EXT	TR-RA-05; TR-RA-06; TR-RA-07; TR-UNI-01; TR-UNI-02	monotonicity argument
1999-D2-P5	PT-COMB	PT-INEQ; PT-LIM; PT-INT; PT-GEO	TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-03; TR-CMB-04; TR-UNI-02	construction / counterexample
1999-D2-P6	PT-INEQ	PT-FE; PT-LIM; PT-POLY; PT-COMB	TR-RA-04; TR-RA-05; TR-RA-07; TR-PC-04; TR-ANT-04; TR-CMB-03	polynomial identity / factorization; roots of unity / Fourier filter
2000-D1-P1	PT-FE	PT-INEQ; PT-LIM; PT-COMB; PT-EXT	TR-RA-01; TR-RA-02; TR-RA-04; TR-RA-07; TR-CMB-06; TR-UNI-02	compactness / extremal point; monotonicity argument; double counting / pigeonhole; construction / counterexample
2000-D1-P2	PT-FE	PT-INEQ; PT-LIM; PT-POLY	TR-RA-04; TR-PC-04; TR-UNI-03	direct structural reasoning

Problem	Primary type	Secondary types	Trigger cards	Existing method tags
2000-D1-P3	PT-LA-SPEC	PT-LA-DET; PT-INEQ; PT-POLY	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-PC-04; TR-CMB-02	spectral decomposition / eigenvalues; trace trick / commutator; rank-nullity / dimension count
2000-D1-P4	PT-INEQ	PT-LIM; PT-SEQ	TR-RA-02; TR-RA-04; TR-RA-07; TR-UNI-07	monotonicity argument
2000-D1-P5	PT-GRP	PT-LA-DET; PT-LA-SPEC; PT-FINITE	TR-LA-03; TR-LA-04; TR-LA-05; TR-ANT-01; TR-ANT-02; TR-ANT-03	trace trick / commutator
2000-D1-P6	PT-FE	PT-INEQ; PT-LIM; PT-INT; PT-POLY	TR-RA-04; TR-RA-05; TR-RA-06; TR-RA-07; TR-PC-04; TR-PC-05	induction; auxiliary function / substitution; Rolle / mean-value theorem; monotonicity argument; polynomial identity / factorization
2000-D2-P1	PT-GEO	PT-LA-DET; PT-INEQ; PT-FINITE; PT-COMB	TR-LA-01; TR-RA-07; TR-PC-05; TR-ANT-02; TR-GEO-02; TR-GEO-04	rank-nullity / dimension count; modular / finite-field reduction
2000-D2-P2	PT-INEQ	PT-FE; PT-LIM; PT-COMB; PT-GEO	TR-RA-01; TR-RA-02; TR-RA-04; TR-RA-07; TR-RA-09; TR-RA-10	compactness / extremal point; monotonicity argument; topological / density argument
2000-D2-P3	PT-POLY	PT-INEQ; PT-FINITE; PT-COMB; PT-GRP	TR-RA-07; TR-PC-01; TR-PC-04; TR-ANT-02; TR-ANT-04; TR-CMB-03	polynomial identity / factorization
2000-D2-P4	PT-POLY	PT-INEQ; PT-LIM; PT-INT; PT-COMB	TR-RA-04; TR-RA-06; TR-RA-07; TR-PC-01; TR-ANT-04; TR-CMB-01	convexity / Jensen / tangent; polynomial identity / factorization; graph encoding / cycles; Riemann-sum / integral comparison
2000-D2-P5	PT-FE	PT-INEQ; PT-LIM; PT-INT; PT-COMB	TR-RA-02; TR-RA-04; TR-RA-05; TR-RA-06; TR-RA-07; TR-ANT-04	auxiliary function / substitution; monotonicity argument
2000-D2-P6	PT-LA-SPEC	PT-LA-DET; PT-LIM; PT-INT; PT-POLY	TR-LA-01; TR-LA-03; TR-LA-04; TR-LA-05; TR-RA-04; TR-RA-06	spectral decomposition / eigenvalues; polynomial identity / factorization
2001-D1-P1	PT-FINITE	PT-LA-DET; PT-INEQ; PT-LIM; PT-INT	TR-LA-02; TR-RA-04; TR-RA-06; TR-ANT-01; TR-ANT-03; TR-ANT-04	determinant / block manipulation
2001-D1-P2	PT-GRP	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-FINITE	TR-LA-05; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05	modular / finite-field reduction; graph encoding / cycles
2001-D1-P3	PT-LIM	PT-INEQ; PT-INT	TR-RA-02; TR-RA-04; TR-ANT-05	direct structural reasoning
2001-D1-P4	PT-POLY	PT-INEQ; PT-FINITE; PT-GEO	TR-RA-07; TR-PC-01; TR-PC-04; TR-ANT-01; TR-ANT-02	polynomial identity / factorization; modular / finite-field reduction

Problem	Primary type	Secondary types	Trigger cards	Existing method tags
2001-D1-P5	PT-LA-SPEC	PT-LA-DET; PT-INEQ; PT-LIM; PT-INT	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-RA-06; TR-RA-07	induction; trace trick / commutator
2001-D1-P6	PT-FE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-LIM	TR-LA-05; TR-RA-02; TR-RA-03; TR-RA-04; TR-RA-05; TR-RA-07	l’Hospital / asymptotic expansion
2001-D2-P1	PT-INEQ	PT-LIM; PT-POLY; PT-FINITE; PT-COMB	TR-RA-04; TR-RA-07; TR-ANT-01; TR-CMB-06	induction; polynomial identity / factorization; double counting / pigeonhole
2001-D2-P2	PT-INEQ	PT-FE; PT-LIM; PT-INT; PT-SEQ	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-PC-05; TR-UNI-07	induction; monotonicity argument
2001-D2-P3	PT-GEO	PT-LA-DET; PT-FE; PT-INEQ; PT-POLY	TR-LA-01; TR-RA-01; TR-RA-02; TR-RA-05; TR-RA-07; TR-PC-04	compactness / extremal point; rank-nullity / dimension count; roots of unity / Fourier filter; double counting / pigeonhole; construction / counterexample
2001-D2-P4	PT-LA-DET	PT-LA-SPEC; PT-INEQ; PT-LIM; PT-INT	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-RA-01; TR-RA-06	compactness / extremal point; determinant / block manipulation; graph encoding / cycles
2001-D2-P5	PT-FE	PT-INEQ; PT-LIM; PT-INT; PT-COMB	TR-RA-04; TR-RA-05; TR-RA-07; TR-ANT-04; TR-CMB-03; TR-CMB-04	monotonicity argument
2001-D2-P6	PT-FINITE	PT-FE; PT-INEQ; PT-GEO; PT-GRP	TR-RA-01; TR-RA-02; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04	compactness / extremal point
2002-D1-P1	PT-GEO	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-LIM	TR-LA-05; TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-01; TR-CMB-03	polynomial identity / factorization; graph encoding / cycles
2002-D1-P2	PT-FE	PT-INEQ; PT-LIM; PT-INT; PT-EXT	TR-RA-02; TR-RA-05; TR-RA-06; TR-RA-07; TR-UNI-02	monotonicity argument; construction / counterexample
2002-D1-P3	PT-FINITE	PT-FE; PT-INEQ; PT-LIM; PT-INT	TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	induction
2002-D1-P4	PT-FE	PT-INEQ; PT-LIM; PT-INT; PT-COMB	TR-RA-01; TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-04	induction; double counting / pigeonhole; topological / density argument
2002-D1-P5	PT-FE	PT-LA-DET; PT-INEQ; PT-LIM; PT-INT	TR-LA-01; TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-04	Rolle / mean-value theorem; monotonicity argument; rank-nullity / dimension count; double counting / pigeonhole

Problem	Primary type	Secondary types	Trigger cards	Existing method tags
2002-D1-P6	PT-FINITE	PT-LA-DET; PT-INEQ; PT-LIM; PT-INT	TR-LA-01; TR-LA-07; TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07	monotonicity argument; Cauchy-Schwarz / norm inequality; rank-nullity / dimension count
2002-D2-P1	PT-LA-DET	PT-LA-SPEC; PT-INEQ	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-GEO-01	determinant / block manipulation
2002-D2-P2	PT-COMB	PT-INEQ; PT-GRP; PT-EXT	TR-ANT-04; TR-CMB-03; TR-CMB-04; TR-CMB-06; TR-UNI-05	direct structural reasoning
2002-D2-P3	PT-FINITE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-LIM	TR-LA-05; TR-RA-06; TR-RA-07; TR-PC-05; TR-ANT-01; TR-ANT-03	induction; polynomial identity / factorization
2002-D2-P4	PT-GEO	PT-INEQ; PT-LIM; PT-INT; PT-COMB	TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-01; TR-CMB-03; TR-CMB-04	graph encoding / cycles
2002-D2-P5	PT-LA-SPEC	PT-LA-DET; PT-INEQ; PT-POLY; PT-COMB	TR-LA-01; TR-LA-05; TR-RA-05; TR-RA-07; TR-PC-04; TR-ANT-04	spectral decomposition / eigenvalues; roots of unity / Fourier filter
2002-D2-P6	PT-FE	PT-INEQ; PT-INT; PT-GEO; PT-EXT	TR-LA-07; TR-RA-01; TR-RA-05; TR-RA-07; TR-RA-08; TR-GEO-04	auxiliary function / substitution; compactness / extremal point; convexity / Jensen / tangent
2003-D1-P1	PT-LIM	PT-INEQ; PT-INT; PT-SEQ; PT-EXT	TR-RA-02; TR-RA-04; TR-RA-06; TR-PC-05; TR-UNI-07	monotonicity argument
2003-D1-P2	PT-FINITE	PT-LIM; PT-INT; PT-COMB; PT-GRP	TR-RA-04; TR-RA-06; TR-PC-05; TR-ANT-01; TR-ANT-03; TR-ANT-04	modular / finite-field reduction
2003-D1-P3	PT-LA-SPEC	PT-LA-DET; PT-LIM; PT-INT; PT-POLY	TR-LA-01; TR-LA-03; TR-LA-04; TR-LA-05; TR-RA-04; TR-RA-06	spectral decomposition / eigenvalues; polynomial identity / factorization
2003-D1-P4	PT-COMB	PT-LA-DET; PT-LA-SPEC; PT-FE; PT-INEQ	TR-LA-01; TR-LA-05; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04	direct structural reasoning
2003-D1-P5	PT-LIM	PT-LA-DET; PT-LA-SPEC; PT-FE; PT-INEQ	TR-LA-01; TR-LA-05; TR-RA-02; TR-RA-03; TR-RA-04; TR-RA-06	induction; auxiliary function / substitution; monotonicity argument; double counting / pigeonhole
2003-D1-P6	PT-POLY	PT-LA-SPEC; PT-FE; PT-INEQ; PT-LIM	TR-LA-05; TR-RA-06; TR-RA-07; TR-PC-03; TR-PC-05; TR-GEO-01	induction; polynomial identity / factorization
2003-D2-P1	PT-LA-DET	PT-LA-SPEC	TR-LA-01; TR-LA-02; TR-LA-05; TR-GEO-01	direct structural reasoning
2003-D2-P2	PT-LIM	PT-INEQ; PT-INT; PT-POLY	TR-RA-04; TR-RA-07; TR-PC-04; TR-ANT-05	monotonicity argument; polynomial identity / factorization
2003-D2-P3	PT-COMB	PT-INEQ; PT-GEO; PT-GRP; PT-EXT	TR-RA-01; TR-RA-07; TR-RA-10; TR-ANT-04; TR-CMB-03; TR-CMB-04	compactness / extremal point; topological / density argument

Problem	Primary type	Secondary types	Trigger cards	Existing method tags
2003-D2-P4	PT-COMB	PT-INEQ; PT-FINITE; PT-GRP; PT-EXT	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-03	construction / counterexample
2003-D2-P5	PT-FE	PT-INEQ; PT-LIM; PT-INT; PT-COMB	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-03	auxiliary function / substitution; double counting / pigeonhole; topological / density argument
2003-D2-P6	PT-LIM	PT-FE; PT-INEQ; PT-INT; PT-COMB	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-PC-05; TR-ANT-05	induction; generating functions
2004-D1-P1	PT-COMB	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-GEO	TR-LA-05; TR-RA-07; TR-ANT-04; TR-CMB-01; TR-CMB-03; TR-CMB-04	double counting / pigeonhole
2004-D1-P2	PT-INEQ	PT-LIM; PT-INT; PT-COMB; PT-GEO	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-01	induction; double counting / pigeonhole
2004-D1-P3	PT-LIM	PT-LA-DET; PT-FE; PT-INEQ; PT-INT	TR-LA-01; TR-RA-01; TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07	compactness / extremal point; convexity / Jensen / tangent; rank-nullity / dimension count; graph encoding / cycles; topological / density argument
2004-D1-P4	PT-GEO	PT-INEQ; PT-LIM; PT-INT; PT-COMB	TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-03; TR-CMB-04	double counting / pigeonhole
2004-D1-P5	PT-INEQ	PT-LA-SPEC; PT-LIM; PT-INT; PT-COMB	TR-LA-05; TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-04	induction; monotonicity argument
2004-D1-P6	PT-INT	PT-LA-SPEC; PT-FE; PT-INEQ; PT-LIM	TR-LA-05; TR-RA-04; TR-RA-06; TR-RA-07	convexity / Jensen / tangent; polynomial identity / factorization; modular / finite-field reduction; double counting / pigeonhole; l'Hospital / asymptotic expansion
2004-D2-P1	PT-LA-DET	PT-LA-SPEC	TR-LA-01; TR-LA-02; TR-LA-05; TR-GEO-01	direct structural reasoning
2004-D2-P2	PT-FE	PT-INEQ; PT-INT; PT-COMB; PT-GEO	TR-RA-02; TR-RA-05; TR-RA-07; TR-ANT-04; TR-CMB-01; TR-CMB-03	monotonicity argument; convexity / Jensen / tangent; graph encoding / cycles
2004-D2-P3	PT-GEO	PT-INEQ; PT-LIM; PT-INT; PT-GRP	TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-03; TR-ANT-04; TR-GEO-01	construction / counterexample; topological / density argument
2004-D2-P4	PT-LA-SPEC	PT-LA-DET; PT-INEQ; PT-LIM; PT-INT	TR-LA-01; TR-LA-05; TR-RA-04; TR-RA-06; TR-RA-07; TR-PC-04	spectral decomposition / eigenvalues; rank-nullity / dimension count

Problem	Primary type	Secondary types	Trigger cards	Existing method tags
2004-D2-P5	PT-INEQ	PT-LIM	TR-RA-02; TR-RA-04; TR-RA-07	auxiliary function / substitution; convexity / Jensen / tangent
2004-D2-P6	PT-LA-DET	PT-LA-SPEC; PT-INEQ; PT-LIM; PT-INT	TR-LA-01; TR-LA-02; TR-LA-05; TR-RA-04; TR-RA-06; TR-RA-07	determinant / block manipulation; double counting / pigeonhole
2005-D1-P1	PT-LA-DET	PT-LA-SPEC; PT-INEQ; PT-LIM; PT-INT	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-RA-06; TR-RA-07	determinant / block manipulation; rank-nullity / dimension count
2005-D1-P2	PT-COMB	PT-INEQ; PT-FINITE; PT-GEO; PT-GRP	TR-RA-07; TR-PC-05; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04	induction; modular / finite-field reduction; double counting / pigeonhole
2005-D1-P3	PT-FE	PT-INEQ; PT-LIM	TR-RA-02; TR-RA-04; TR-RA-07	direct structural reasoning
2005-D1-P4	PT-POLY	PT-INEQ; PT-LIM; PT-INT; PT-FINITE	TR-RA-06; TR-RA-07; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-CMB-01	auxiliary function / substitution; polynomial identity / factorization; modular / finite-field reduction
2005-D1-P5	PT-LIM	PT-FE; PT-INEQ; PT-INT	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-05	auxiliary function / substitution; l'Hospital / asymptotic expansion
2005-D1-P6	PT-GRP	PT-LA-DET; PT-LA-SPEC; PT-FE; PT-INEQ	TR-LA-03; TR-LA-04; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04	induction; trace trick / commutator
2005-D2-P1	PT-INT	PT-FE; PT-INEQ; PT-LIM; PT-COMB	TR-RA-01; TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-04	compactness / extremal point
2005-D2-P2	PT-POLY	PT-FE; PT-LIM; PT-INT; PT-FINITE	TR-RA-06; TR-ANT-02	polynomial identity / factorization
2005-D2-P3	PT-LA-SPEC	PT-LA-DET; PT-INEQ; PT-LIM; PT-INT	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-RA-04; TR-RA-06	compactness / extremal point; trace trick / commutator; rank-nullity / dimension count
2005-D2-P4	PT-FE	PT-INT; PT-COMB; PT-GRP; PT-EXT	TR-RA-02; TR-RA-05; TR-ANT-04; TR-CMB-03; TR-CMB-04	auxiliary function / substitution; Rolle / mean-value theorem
2005-D2-P5	PT-FE	PT-INEQ; PT-LIM; PT-INT; PT-COMB	TR-RA-01; TR-RA-02; TR-RA-05; TR-RA-06; TR-RA-07; TR-ANT-04	compactness / extremal point
2005-D2-P6	PT-FINITE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-LIM	TR-LA-01; TR-LA-02; TR-LA-03; TR-RA-06; TR-PC-05; TR-ANT-01	trace trick / commutator; determinant / block manipulation; polynomial identity / factorization; modular / finite-field reduction
2006-D1-P1	PT-FE	PT-INEQ; PT-LIM; PT-INT; PT-COMB	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-03	monotonicity argument; construction / counterexample

Problem	Primary type	Secondary types	Trigger cards	Existing method tags
2006-D1-P2	PT-FINITE	PT-INEQ; PT-COMB; PT-GRP	TR-RA-07; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05	modular / finite-field reduction; double counting / pigeonhole
2006-D1-P3	PT-FINITE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-LIM	TR-LA-01; TR-LA-02; TR-LA-03; TR-RA-04; TR-RA-06; TR-RA-07	induction; determinant / block manipulation; modular / finite-field reduction; Riemann-sum / integral comparison
2006-D1-P4	PT-POLY	PT-FE; PT-LIM; PT-INT; PT-FINITE	TR-RA-04; TR-RA-06; TR-PC-02; TR-ANT-01; TR-ANT-03; TR-ANT-04	polynomial identity / factorization
2006-D1-P5	PT-INEQ	PT-FE; PT-LIM	TR-RA-02; TR-RA-04; TR-RA-07	Rolle / mean-value theorem
2006-D1-P6	PT-FE	PT-LA-DET; PT-INEQ; PT-LIM; PT-INT	TR-LA-01; TR-RA-04; TR-RA-06; TR-RA-07; TR-PC-04; TR-GEO-01	induction; Rolle / mean-value theorem; polynomial identity / factorization; double counting / pigeonhole
2006-D2-P1	PT-FINITE	PT-INEQ; PT-LIM; PT-INT; PT-COMB	TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04	induction; convexity / Jensen / tangent; double counting / pigeonhole; graph encoding / cycles
2006-D2-P2	PT-FE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-INT	TR-LA-01; TR-LA-05; TR-RA-01; TR-RA-05; TR-RA-07; TR-GEO-01	compactness / extremal point; monotonicity argument; rank-nullity / dimension count; topological / density argument
2006-D2-P3	PT-INEQ	PT-FE; PT-LIM	TR-RA-02; TR-RA-04; TR-RA-07	direct structural reasoning
2006-D2-P4	PT-FINITE	PT-LA-DET; PT-INEQ; PT-LIM; PT-INT	TR-LA-01; TR-LA-07; TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-03	Cauchy-Schwarz / norm inequality; rank-nullity / dimension count
2006-D2-P5	PT-POLY	PT-INEQ; PT-LIM; PT-INT; PT-FINITE	TR-RA-06; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05	auxiliary function / substitution; polynomial identity / factorization; modular / finite-field reduction; double counting / pigeonhole
2006-D2-P6	PT-LA-SPEC	PT-LA-DET; PT-INEQ; PT-LIM; PT-INT	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-RA-06; TR-ANT-03	spectral decomposition / eigenvalues; trace trick / commutator; determinant / block manipulation; polynomial identity / factorization

Problem	Primary type	Secondary types	Trigger cards	Existing method tags
2007-D1-P1	PT-POLY	PT-FE; PT-FINITE; PT-GRP	TR-PC-01; TR-PC-02; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04	auxiliary function / substitution; polynomial identity / factorization; modular / finite-field reduction
2007-D1-P2	PT-FINITE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-LIM	TR-LA-01; TR-LA-02; TR-LA-03; TR-RA-06; TR-RA-07; TR-ANT-01	determinant / block manipulation; rank-nullity / dimension count
2007-D1-P3	PT-LA-DET	PT-LA-SPEC; PT-INEQ; PT-POLY	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-RA-07; TR-GEO-01	determinant / block manipulation; polynomial identity / factorization
2007-D1-P4	PT-GRP	PT-FINITE; PT-COMB; PT-SEQ	TR-PC-05; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	auxiliary function / substitution; double counting / pigeonhole
2007-D1-P5	PT-POLY	PT-LA-DET; PT-FE; PT-INEQ; PT-LIM	TR-LA-01; TR-RA-06; TR-RA-07; TR-PC-04; TR-ANT-01; TR-GEO-01	polynomial identity / factorization
2007-D1-P6	PT-POLY	PT-INEQ; PT-LIM; PT-INT; PT-FINITE	TR-RA-05; TR-RA-06; TR-RA-07; TR-PC-04; TR-ANT-01; TR-CMB-01	polynomial identity / factorization; roots of unity / Fourier filter; double counting / pigeonhole; construction / counterexample
2007-D2-P1	PT-FE	PT-INEQ; PT-LIM; PT-COMB; PT-GRP	TR-RA-02; TR-RA-04; TR-ANT-04; TR-CMB-01; TR-CMB-03; TR-CMB-04	graph encoding / cycles; construction / counterexample
2007-D2-P2	PT-FINITE	PT-LIM; PT-INT; PT-GRP	TR-RA-06; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05	modular / finite-field reduction
2007-D2-P3	PT-FE	PT-INEQ; PT-LIM; PT-COMB; PT-GRP	TR-RA-01; TR-RA-02; TR-RA-04; TR-RA-07; TR-ANT-04; TR-CMB-03	monotonicity argument; construction / counterexample; topological / density argument
2007-D2-P4	PT-FINITE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-GRP	TR-LA-01; TR-LA-02; TR-LA-03; TR-ANT-01; TR-ANT-02; TR-ANT-03	determinant / block manipulation; modular / finite-field reduction
2007-D2-P5	PT-FINITE	PT-LA-DET; PT-INEQ; PT-COMB; PT-GRP	TR-LA-01; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	rank-nullity / dimension count; construction / counterexample
2007-D2-P6	PT-POLY	PT-LA-SPEC; PT-FE; PT-INEQ; PT-LIM	TR-LA-05; TR-RA-01; TR-RA-03; TR-RA-04; TR-RA-06; TR-RA-07	induction; Rolle / mean-value theorem; compactness / extremal point; monotonicity argument; polynomial identity / factorization

Problem	Primary type	Secondary types	Trigger cards	Existing method tags
2008-D1-P6	PT-COMB	PT-LA-DET; PT-INEQ; PT-POLY; PT-GRP	TR-LA-01; TR-LA-02; TR-LA-03; TR-RA-07; TR-ANT-04; TR-CMB-01	determinant / block manipulation; polynomial identity / factorization; double counting / pigeonhole
2008-D1-P1	PT-FE	PT-INEQ; PT-INT; PT-COMB; PT-GRP	TR-RA-01; TR-RA-02; TR-RA-05; TR-RA-07; TR-ANT-04; TR-CMB-03	topological / density argument
2008-D1-P2	PT-LA-DET	PT-LA-SPEC; PT-FE; PT-INEQ; PT-LIM	TR-LA-01; TR-LA-05; TR-RA-06; TR-RA-07; TR-GEO-01	induction; polynomial identity / factorization
2008-D1-P3	PT-POLY	PT-LA-DET; PT-LA-SPEC; PT-LIM; PT-INT	TR-LA-05; TR-RA-06; TR-PC-01; TR-PC-02; TR-ANT-01; TR-ANT-02	induction; polynomial identity / factorization; modular / finite-field reduction; construction / counterexample
2008-D1-P4	PT-COMB	PT-INEQ; PT-LIM; PT-INT; PT-GRP	TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-03; TR-CMB-04; TR-UNI-03	direct structural reasoning
2008-D1-P5	PT-GRP	PT-LA-DET; PT-INEQ; PT-LIM; PT-FINITE	TR-LA-01; TR-LA-07; TR-RA-02; TR-RA-04; TR-RA-07; TR-ANT-01	auxiliary function / substitution; Cauchy-Schwarz / norm inequality; modular / finite-field reduction; double counting / pigeonhole; construction / counterexample
2008-D2-P1	PT-INEQ	PT-FE; PT-LIM; PT-POLY; PT-FINITE	TR-RA-04; TR-PC-01; TR-PC-04; TR-ANT-01	polynomial identity / factorization
2008-D2-P2	PT-GEO	PT-LA-DET; PT-FE; PT-INEQ; PT-POLY	TR-RA-07; TR-CMB-06; TR-GEO-01; TR-GEO-02; TR-GEO-04	auxiliary function / substitution; Cauchy-Schwarz / norm inequality; polynomial identity / factorization
2008-D2-P3	PT-POLY	PT-INEQ; PT-FINITE; PT-GRP; PT-SEQ	TR-RA-07; TR-PC-05; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	direct structural reasoning
2008-D2-P4	PT-GRP	PT-FE; PT-POLY; PT-FINITE; PT-COMB	TR-PC-01; TR-PC-02; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04	polynomial identity / factorization; modular / finite-field reduction; double counting / pigeonhole; construction / counterexample
2008-D2-P5	PT-FINITE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-POLY	TR-LA-01; TR-LA-02; TR-LA-03; TR-RA-07; TR-ANT-01; TR-ANT-02	determinant / block manipulation; rank-nullity / dimension count; polynomial identity / factorization; modular / finite-field reduction

Problem	Primary type	Secondary types	Trigger cards	Existing method tags
2008-D2-P6	PT-LA-DET	PT-LA-SPEC; PT-INEQ; PT-LIM; PT-INT	TR-LA-01; TR-LA-05; TR-LA-07; TR-RA-06; TR-RA-07; TR-CMB-01	Cauchy-Schwarz / norm inequality; rank-nullity / dimension count; polynomial identity / factorization; double counting / pigeonhole; construction / counterexample
2009-D1-P1	PT-INEQ	PT-FE; PT-LIM; PT-INT; PT-COMB	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-03; TR-ANT-04	monotonicity argument; double counting / pigeonhole; construction / counterexample
2009-D1-P2	PT-LA-DET	PT-LA-SPEC	TR-LA-01; TR-LA-05; TR-CMB-02; TR-GEO-01	direct structural reasoning
2009-D1-P3	PT-COMB	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-POLY	TR-LA-01; TR-RA-07; TR-CMB-01; TR-CMB-03; TR-CMB-06; TR-UNI-03	rank-nullity / dimension count; polynomial identity / factorization; double counting / pigeonhole; graph encoding / cycles; topological / density argument
2009-D1-P4	PT-INEQ	PT-LA-DET; PT-LA-SPEC; PT-LIM; PT-INT	TR-LA-01; TR-LA-05; TR-RA-04; TR-RA-06; TR-RA-07; TR-GEO-01	induction; compactness / extremal point; convexity / Jensen / tangent; Cauchy-Schwarz / norm inequality; rank-nullity / dimension count; polynomial identity / factorization
2009-D1-P5	PT-FINITE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-LIM	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-LA-07; TR-RA-04	induction; auxiliary function / substitution; determinant / block manipulation; rank-nullity / dimension count; graph encoding / cycles; l'Hospital / asymptotic expansion
2009-D2-P1	PT-GEO	PT-INEQ; PT-POLY; PT-COMB; PT-GRP	TR-RA-07; TR-ANT-04; TR-CMB-03; TR-CMB-04; TR-GEO-01; TR-GEO-02	polynomial identity / factorization
2009-D2-P2	PT-FE	PT-INEQ; PT-INT	TR-RA-02; TR-RA-03; TR-RA-05; TR-RA-07	auxiliary function / substitution
2009-D2-P3	PT-FINITE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-LIM	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-04; TR-LA-05; TR-RA-06	induction; spectral decomposition / eigenvalues; trace trick / commutator; polynomial identity / factorization
2009-D2-P4	PT-POLY	PT-FINITE; PT-COMB; PT-GRP	TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-CMB-01; TR-CMB-03	polynomial identity / factorization; modular / finite-field reduction; graph encoding / cycles

Problem	Primary type	Secondary types	Trigger cards	Existing method tags
2009-D2-P5	PT-FINITE	PT-LA-DET; PT-INEQ; PT-LIM; PT-INT	TR-LA-01; TR-LA-02; TR-LA-03; TR-RA-06; TR-RA-07; TR-ANT-01	determinant / block manipulation; rank-nullity / dimension count; polynomial identity / factorization; construction / counterexample
2010-D1-P1	PT-FE	PT-INEQ; PT-INT	TR-RA-02; TR-RA-05; TR-RA-07	auxiliary function / substitution; Rolle / mean-value theorem; monotonicity argument; Cauchy-Schwarz / norm inequality
2010-D1-P2	PT-INT	PT-FE; PT-INEQ; PT-LIM	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07	direct structural reasoning
2010-D1-P3	PT-FE	PT-INEQ; PT-LIM; PT-INT; PT-SEQ	TR-RA-04; TR-RA-05; TR-RA-07; TR-ANT-05; TR-UNI-07	induction
2010-D1-P4	PT-FINITE	PT-INEQ; PT-LIM; PT-INT; PT-COMB	TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04	modular / finite-field reduction; double counting / pigeonhole; construction / counterexample
2010-D1-P5	PT-FINITE	PT-LA-DET; PT-INEQ; PT-GEO; PT-GRP	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-07; TR-RA-02; TR-RA-07	Cauchy-Schwarz / norm inequality; determinant / block manipulation
2010-D2-P1	PT-INEQ	PT-LA-SPEC; PT-FE; PT-LIM; PT-INT	TR-LA-05; TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-UNI-02	monotonicity argument; construction / counterexample
2010-D2-P2	PT-INEQ	PT-LIM; PT-INT; PT-GEO; PT-GRP	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-PC-05; TR-ANT-03	induction
2010-D2-P3	PT-GRP	PT-FE; PT-INEQ; PT-LIM; PT-INT	TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	auxiliary function / substitution
2010-D2-P4	PT-FINITE	PT-LA-DET; PT-INEQ; PT-GEO; PT-GRP	TR-LA-01; TR-LA-02; TR-LA-07; TR-RA-07; TR-ANT-01; TR-ANT-03	induction; determinant / block manipulation; rank-nullity / dimension count; modular / finite-field reduction
2010-D2-P5	PT-POLY	PT-LA-SPEC; PT-FE; PT-LIM; PT-FINITE	TR-LA-05; TR-RA-04; TR-PC-04; TR-ANT-01; TR-ANT-02	polynomial identity / factorization; modular / finite-field reduction
2011-D1-P1	PT-FE	PT-INEQ; PT-LIM; PT-INT; PT-COMB	TR-RA-01; TR-RA-02; TR-RA-04; TR-RA-05; TR-RA-06; TR-RA-07	compactness / extremal point
2011-D1-P2	PT-LA-SPEC	PT-LA-DET; PT-INEQ; PT-LIM; PT-INT	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-RA-06; TR-GEO-01	spectral decomposition / eigenvalues; trace trick / commutator; polynomial identity / factorization; construction / counterexample

Problem	Primary type	Secondary types	Trigger cards	Existing method tags
2011-D1-P3	PT-POLY	PT-FE; PT-INEQ; PT-FINITE; PT-GEO	TR-RA-07; TR-PC-02; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04	polynomial identity / factorization; modular / finite-field reduction
2011-D1-P4	PT-COMB	PT-FE; PT-INEQ; PT-INT; PT-GRP	TR-RA-07; TR-ANT-04; TR-CMB-01; TR-CMB-03; TR-CMB-04; TR-CMB-05	monotonicity argument; double counting / pigeonhole; probabilistic / expectation
2011-D1-P5	PT-FINITE	PT-LA-DET; PT-INEQ; PT-LIM; PT-INT	TR-LA-01; TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04	induction; rank-nullity / dimension count; polynomial identity / factorization; modular / finite-field reduction
2011-D2-P1	PT-LIM	PT-FE; PT-INEQ; PT-INT; PT-COMB	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-PC-05; TR-ANT-04	induction; monotonicity argument
2011-D2-P2	PT-FINITE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-LIM	TR-LA-02; TR-LA-05; TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-03	determinant / block manipulation; polynomial identity / factorization; double counting / pigeonhole; graph encoding / cycles; construction / counterexample
2011-D2-P3	PT-FE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-LIM	TR-LA-01; TR-RA-05; TR-RA-06; TR-RA-07	direct structural reasoning
2011-D2-P4	PT-POLY	PT-LA-DET; PT-FE; PT-INEQ; PT-LIM	TR-LA-01; TR-RA-06; TR-RA-07; TR-PC-01; TR-ANT-01; TR-ANT-02	induction; auxiliary function / substitution; rank-nullity / dimension count; polynomial identity / factorization
2011-D2-P5	PT-GEO	PT-LA-DET; PT-INEQ; PT-LIM; PT-INT	TR-LA-01; TR-RA-06; TR-RA-07; TR-ANT-03; TR-ANT-04; TR-CMB-01	convexity / Jensen / tangent; rank-nullity / dimension count; graph encoding / cycles; construction / counterexample
2012-D1-P1	PT-FINITE	PT-INEQ; PT-LIM; PT-INT; PT-COMB	TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04	double counting / pigeonhole
2012-D1-P2	PT-FINITE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-GRP	TR-LA-01; TR-LA-02; TR-LA-03; TR-RA-02; TR-ANT-01; TR-ANT-03	rank-nullity / dimension count
2012-D1-P3	PT-GRP	PT-INEQ; PT-FINITE; PT-COMB; PT-GEO	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	direct structural reasoning
2012-D1-P4	PT-FE	PT-INEQ; PT-LIM	TR-RA-02; TR-RA-04; TR-RA-07	direct structural reasoning
2012-D1-P5	PT-POLY	PT-INEQ; PT-FINITE; PT-GRP	TR-ANT-01; TR-ANT-03; TR-ANT-04	polynomial identity / factorization
2012-D2-P1	PT-POLY	PT-FE; PT-FINITE	TR-ANT-01; TR-ANT-02	polynomial identity / factorization

Problem	Primary type	Secondary types	Trigger cards	Existing method tags
2012-D2-P2	PT-INT	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-LIM	TR-LA-01; TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-PC-05	induction
2012-D2-P3	PT-POLY	PT-INEQ; PT-FINITE; PT-COMB; PT-GRP	TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-03; TR-CMB-04	direct structural reasoning
2012-D2-P4	PT-FINITE	PT-INEQ; PT-GRP	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-UNI-03	direct structural reasoning
2012-D2-P5	PT-GRP	PT-INEQ; PT-FINITE; PT-COMB	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-03	auxiliary function / substitution
2013-D1-P1	PT-LA-SPEC	PT-LA-DET	TR-LA-01; TR-LA-05; TR-GEO-01	spectral decomposition / eigenvalues
2013-D1-P2	PT-FE	PT-INEQ; PT-LIM; PT-EXT	TR-RA-02; TR-RA-04	direct structural reasoning
2013-D1-P3	PT-COMB	PT-INEQ; PT-EXT	TR-RA-07; TR-CMB-01; TR-CMB-06; TR-UNI-04; TR-UNI-05	compactness / extremal point; double counting / pigeonhole
2013-D1-P4	PT-INEQ	PT-LIM	TR-RA-02; TR-RA-04; TR-RA-07	direct structural reasoning
2013-D1-P5	PT-FINITE	PT-INEQ; PT-LIM; PT-INT; PT-POLY	TR-RA-06; TR-PC-04; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04	modular / finite-field reduction; construction / counterexample
2013-D2-P1	PT-POLY		TR-PC-04	direct structural reasoning
2013-D2-P2	PT-FINITE	PT-INEQ; PT-GRP	TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05	modular / finite-field reduction
2013-D2-P3	PT-INEQ	PT-LIM; PT-GEO; PT-EXT	TR-LA-07; TR-RA-02; TR-RA-04; TR-RA-07; TR-GEO-04	direct structural reasoning
2013-D2-P4	PT-FINITE	PT-INEQ; PT-LIM; PT-INT; PT-COMB	TR-RA-04; TR-RA-06; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	construction / counterexample
2013-D2-P5	PT-COMB	PT-INEQ; PT-GEO; PT-EXT	TR-RA-07; TR-CMB-01; TR-CMB-06; TR-UNI-05	double counting / pigeonhole
2014-D1-P1	PT-LA-SPEC	PT-LA-DET; PT-INEQ; PT-EXT	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-GEO-01; TR-UNI-03	trace trick / commutator; determinant / block manipulation
2014-D1-P2	PT-LIM	PT-INEQ; PT-INT	TR-RA-02; TR-RA-04; TR-UNI-03	direct structural reasoning
2014-D1-P3	PT-POLY		TR-PC-01	direct structural reasoning
2014-D1-P4	PT-FINITE	PT-INEQ; PT-LIM; PT-INT; PT-GEO	TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-02; TR-ANT-03	modular / finite-field reduction
2014-D1-P5	PT-GEO	PT-INEQ	TR-RA-07; TR-GEO-01; TR-GEO-02; TR-GEO-04	direct structural reasoning
2014-D2-P1	PT-FINITE	PT-GRP	TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	direct structural reasoning
2014-D2-P2	PT-INEQ	PT-LA-DET; PT-LA-SPEC; PT-LIM	TR-LA-01; TR-LA-05; TR-RA-02; TR-RA-04; TR-RA-07; TR-GEO-01	spectral decomposition / eigenvalues
2014-D2-P3	PT-INEQ	PT-FE; PT-LIM	TR-RA-02; TR-RA-04	direct structural reasoning

Problem	Primary type	Secondary types	Trigger cards	Existing method tags
2014-D2-P4	PT-GEO	PT-INEQ; PT-COMB; PT-GRP	TR-RA-07; TR-ANT-04; TR-CMB-03; TR-CMB-04; TR-GEO-01	direct structural reasoning
2014-D2-P5	PT-INEQ	PT-LIM; PT-GEO	TR-RA-02; TR-RA-04; TR-RA-07	direct structural reasoning
2015-D1-P1	PT-LA-DET	PT-LA-SPEC; PT-INEQ; PT-POLY; PT-FINITE	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-RA-07; TR-PC-04	determinant / block manipulation
2015-D1-P2	PT-FINITE	PT-FE; PT-INEQ; PT-GEO; PT-GRP	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-UNI-02	construction / counterexample
2015-D1-P3	PT-FE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-INT	TR-LA-01; TR-RA-02; TR-RA-05; TR-RA-07	direct structural reasoning
2015-D1-P4	PT-FINITE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-GEO	TR-LA-01; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	direct structural reasoning
2015-D1-P5	PT-INEQ	PT-LA-DET; PT-LIM; PT-GEO; PT-SEQ	TR-LA-01; TR-RA-02; TR-RA-04; TR-RA-07; TR-PC-05; TR-GEO-01	convexity / Jensen / tangent; rank-nullity / dimension count
2015-D2-P6	PT-INEQ	PT-LIM	TR-RA-02; TR-RA-04	direct structural reasoning
2015-D2-P7	PT-LIM	PT-INEQ; PT-INT	TR-RA-02; TR-RA-04; TR-ANT-05	direct structural reasoning
2015-D2-P8	PT-COMB	PT-INEQ; PT-LIM; PT-INT; PT-EXT	TR-RA-04; TR-RA-06; TR-CMB-01; TR-CMB-04; TR-UNI-05	double counting / pigeonhole
2015-D2-P9	PT-LA-DET	PT-LA-SPEC; PT-INEQ; PT-POLY; PT-GEO	TR-LA-01; TR-LA-05; TR-RA-07; TR-PC-04; TR-ANT-03; TR-GEO-01	compactness / extremal point; Cauchy-Schwarz / norm inequality; rank-nullity / dimension count
2015-D2-P10	PT-POLY	PT-INEQ; PT-FINITE; PT-GRP	TR-RA-07; TR-PC-01; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	polynomial identity / factorization
2016-D1-P1	PT-FE	PT-INT; PT-EXT	TR-RA-02; TR-RA-05; TR-UNI-02	construction / counterexample
2016-D1-P2	PT-FINITE	PT-LA-DET; PT-INEQ; PT-LIM; PT-GRP	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-GEO-01	construction / counterexample
2016-D1-P3	PT-FINITE	PT-INEQ; PT-GRP	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	direct structural reasoning
2016-D1-P4	PT-COMB	PT-INEQ; PT-FINITE; PT-GRP	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-03	direct structural reasoning
2016-D1-P5	PT-COMB	PT-FE; PT-INEQ; PT-LIM; PT-FINITE	TR-RA-07; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05	modular / finite-field reduction; double counting / pigeonhole
2016-D2-P6	PT-INEQ	PT-LIM; PT-SEQ	TR-RA-02; TR-RA-04; TR-RA-07; TR-UNI-07	direct structural reasoning
2016-D2-P7	PT-FE	PT-INEQ; PT-LIM; PT-EXT	TR-RA-01; TR-RA-02; TR-RA-04; TR-RA-07; TR-UNI-04	compactness / extremal point

Problem	Primary type	Secondary types	Trigger cards	Existing method tags
2016-D2-P8	PT-GRP	PT-FE; PT-INEQ; PT-FINITE; PT-EXT	TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05	modular / finite-field reduction
2016-D2-P9	PT-FINITE	PT-FE; PT-INEQ; PT-COMB; PT-GRP	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	double counting / pigeonhole
2016-D2-P10	PT-LA-SPEC	PT-LA-DET; PT-INEQ; PT-POLY; PT-GEO	TR-LA-01; TR-LA-05; TR-RA-07; TR-PC-04; TR-CMB-06; TR-GEO-01	spectral decomposition / eigenvalues
2017-D1-P1	PT-LA-SPEC	PT-LA-DET; PT-INEQ; PT-POLY; PT-FINITE	TR-LA-01; TR-LA-04; TR-LA-05; TR-RA-07; TR-PC-04; TR-ANT-01	spectral decomposition / eigenvalues
2017-D1-P2	PT-FE	PT-INEQ; PT-LIM; PT-EXT	TR-RA-02; TR-RA-04; TR-RA-07; TR-UNI-01	direct structural reasoning
2017-D1-P3	PT-FINITE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-LIM	TR-LA-01; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	direct structural reasoning
2017-D1-P4	PT-GRP	PT-FINITE; PT-COMB	TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01; TR-CMB-03	double counting / pigeonhole
2017-D1-P5	PT-POLY	PT-FE; PT-INEQ; PT-FINITE	TR-RA-07; TR-PC-04; TR-ANT-01	polynomial identity / factorization
2017-D2-P6	PT-LIM	PT-FE; PT-INEQ; PT-INT; PT-EXT	TR-RA-02; TR-RA-04; TR-ANT-05	direct structural reasoning
2017-D2-P7	PT-POLY	PT-INEQ; PT-FINITE; PT-GEO	TR-RA-07; TR-ANT-01	polynomial identity / factorization
2017-D2-P8	PT-FINITE	PT-LA-DET; PT-LA-SPEC; PT-FE; PT-LIM	TR-LA-01; TR-LA-05; TR-PC-05; TR-ANT-01; TR-ANT-03; TR-ANT-04	spectral decomposition / eigenvalues
2017-D2-P9	PT-LIM	PT-LA-DET; PT-LA-SPEC; PT-FE; PT-INEQ	TR-LA-01; TR-RA-02; TR-RA-04; TR-PC-05; TR-ANT-05; TR-UNI-07	direct structural reasoning
2017-D2-P10	PT-FINITE	PT-INEQ; PT-GEO; PT-GRP; PT-EXT	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-GEO-01	direct structural reasoning
2018-D1-P1	PT-INEQ	PT-LIM; PT-INT; PT-SEQ	TR-RA-02; TR-RA-04; TR-RA-06; TR-UNI-06; TR-UNI-07	direct structural reasoning
2018-D1-P2	PT-GRP	PT-FINITE; PT-EXT	TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-UNI-02	construction / counterexample
2018-D1-P3	PT-LA-DET	PT-LA-SPEC; PT-INEQ	TR-LA-01; TR-LA-05; TR-GEO-01; TR-UNI-03	direct structural reasoning
2018-D1-P4	PT-FE	PT-INEQ; PT-LIM	TR-RA-02; TR-RA-04; TR-UNI-03	direct structural reasoning
2018-D1-P5	PT-FINITE	PT-INEQ; PT-GEO; PT-GRP	TR-RA-07; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05	convexity / Jensen / tangent; modular / finite-field reduction
2018-D2-P6	PT-FINITE	PT-INEQ; PT-INT; PT-COMB; PT-GEO	TR-LA-07; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	direct structural reasoning
2018-D2-P7	PT-INT	PT-INEQ; PT-LIM; PT-GEO; PT-SEQ	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-PC-05; TR-UNI-07	direct structural reasoning
2018-D2-P8	PT-FINITE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-COMB	TR-LA-01; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	double counting / pigeonhole

Problem	Primary type	Secondary types	Trigger cards	Existing method tags
2018-D2-P9	PT-POLY	PT-LA-DET; PT-LA-SPEC; PT-INEQ	TR-LA-01; TR-PC-01; TR-PC-04; TR-UNI-03	polynomial identity / factorization
2018-D2-P10	PT-LIM	PT-INEQ; PT-INT	TR-RA-02; TR-RA-04; TR-ANT-05	direct structural reasoning
2019-D1-P1	PT-INEQ	PT-LIM	TR-RA-02; TR-RA-04	direct structural reasoning
2019-D1-P2	PT-FINITE	PT-GRP	TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-GEO-01; TR-UNI-03	direct structural reasoning
2019-D1-P3	PT-FE	PT-INEQ; PT-LIM	TR-RA-02; TR-RA-04; TR-RA-07	direct structural reasoning
2019-D1-P4	PT-FINITE	PT-FE; PT-INEQ; PT-LIM; PT-GEO	TR-RA-07; TR-PC-05; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	direct structural reasoning
2019-D1-P5	PT-FINITE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-GEO	TR-LA-01; TR-LA-02; TR-LA-03; TR-RA-07; TR-ANT-01; TR-ANT-03	determinant / block manipulation
2019-D2-P6	PT-FE	PT-LIM; PT-INT; PT-EXT	TR-RA-02; TR-RA-04; TR-RA-05; TR-RA-06; TR-UNI-02	construction / counterexample
2019-D2-P7	PT-FINITE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-LIM	TR-LA-01; TR-RA-04; TR-RA-06; TR-ANT-01; TR-ANT-02; TR-ANT-03	direct structural reasoning
2019-D2-P8	PT-COMB	PT-INEQ; PT-LIM; PT-INT; PT-GRP	TR-RA-04; TR-RA-06; TR-ANT-04; TR-CMB-01; TR-CMB-03; TR-CMB-04	double counting / pigeonhole
2019-D2-P9	PT-FINITE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-GRP	TR-LA-01; TR-LA-02; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	direct structural reasoning
2019-D2-P10	PT-COMB	PT-INEQ; PT-INT; PT-GEO; PT-GRP	TR-RA-02; TR-RA-07; TR-ANT-04; TR-CMB-01; TR-CMB-03; TR-CMB-04	convexity / Jensen / tangent; graph encoding / cycles; probabilistic / expectation
2020-D1-P1	PT-COMB	PT-INEQ; PT-LIM; PT-FINITE; PT-GEO	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	double counting / pigeonhole; construction / counterexample
2020-D1-P2	PT-LA-SPEC	PT-LA-DET; PT-INEQ; PT-LIM; PT-INT	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-RA-04; TR-RA-06	compactness / extremal point; trace trick / commutator; determinant / block manipulation; rank-nullity / dimension count; double counting / pigeonhole
2020-D1-P3	PT-FINITE	PT-INEQ; PT-COMB; PT-GEO; PT-GRP	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	convexity / Jensen / tangent; graph encoding / cycles
2020-D1-P4	PT-INEQ	PT-LIM; PT-POLY	TR-RA-04; TR-RA-07	polynomial identity / factorization
2020-D2-P5	PT-FE	PT-INEQ; PT-LIM	TR-RA-02; TR-RA-03; TR-RA-04; TR-RA-07; TR-UNI-03	direct structural reasoning
2020-D2-P6	PT-FINITE	PT-GRP; PT-EXT	TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-UNI-03	modular / finite-field reduction

Problem	Primary type	Secondary types	Trigger cards	Existing method tags
2020-D2-P7	PT-GRP	PT-INEQ; PT-FINITE; PT-COMB; PT-EXT	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	auxiliary function / substitution; double counting / pigeonhole
2020-D2-P8	PT-LIM	PT-INEQ; PT-INT	TR-RA-02; TR-RA-04; TR-ANT-05	direct structural reasoning
2021-D1-P1	PT-LA-DET	PT-LA-SPEC	TR-LA-01; TR-LA-04; TR-LA-05; TR-GEO-01	direct structural reasoning
2021-D1-P2	PT-COMB	PT-INEQ; PT-INT; PT-FINITE; PT-GRP	TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-03; TR-CMB-04	probabilistic / expectation
2021-D1-P3	PT-INEQ	PT-LIM; PT-COMB; PT-GEO; PT-SEQ	TR-RA-02; TR-RA-04; TR-PC-05; TR-CMB-06; TR-UNI-07	direct structural reasoning
2021-D1-P4	PT-FE	PT-INEQ; PT-LIM; PT-INT; PT-EXT	TR-RA-02; TR-RA-04	direct structural reasoning
2021-D2-P5	PT-FINITE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-GRP	TR-LA-01; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	direct structural reasoning
2021-D2-P6	PT-GRP	PT-LA-DET; PT-INT; PT-FINITE; PT-COMB	TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	auxiliary function / substitution; modular / finite-field reduction
2021-D2-P7	PT-POLY	PT-FE; PT-INEQ; PT-COMB; PT-GEO	TR-RA-07; TR-PC-04; TR-PC-06; TR-ANT-04; TR-CMB-03; TR-CMB-04	polynomial identity / factorization; topological / density argument
2021-D2-P8	PT-FINITE	PT-INEQ; PT-COMB; PT-GEO; PT-GRP	TR-LA-07; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	direct structural reasoning
2022-D1-P1	PT-COMB	PT-FE; PT-INEQ; PT-INT	TR-RA-02; TR-RA-07; TR-CMB-05	direct structural reasoning
2022-D1-P2	PT-FINITE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-GRP	TR-LA-01; TR-LA-05; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04	spectral decomposition / eigenvalues
2022-D1-P3	PT-FINITE	PT-FE; PT-INEQ; PT-COMB; PT-GRP	TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	modular / finite-field reduction; double counting / pigeonhole; construction / counterexample
2022-D1-P4	PT-FINITE	PT-INEQ; PT-COMB; PT-GRP; PT-EXT	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	double counting / pigeonhole
2022-D2-P5	PT-COMB	PT-INEQ; PT-GEO; PT-GRP; PT-EXT	TR-RA-02; TR-RA-07; TR-ANT-04; TR-CMB-01; TR-CMB-03; TR-CMB-04	graph encoding / cycles
2022-D2-P6	PT-COMB	PT-FINITE; PT-GRP	TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	modular / finite-field reduction
2022-D2-P7	PT-LA-SPEC	PT-LA-DET; PT-INEQ; PT-POLY; PT-GRP	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-04; TR-LA-05; TR-RA-07	rank-nullity / dimension count
2022-D2-P8	PT-FINITE	PT-INEQ; PT-INT; PT-COMB; PT-GEO	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	convexity / Jensen / tangent; double counting / pigeonhole; graph encoding / cycles; probabilistic / expectation

Problem	Primary type	Secondary types	Trigger cards	Existing method tags
2023-D1-P1	PT-FE	PT-INEQ; PT-LIM	TR-RA-02; TR-RA-04; TR-UNI-03	direct structural reasoning
2023-D1-P2	PT-LA-DET	PT-LA-SPEC; PT-POLY	TR-LA-01; TR-LA-04; TR-LA-05; TR-PC-04; TR-GEO-01	direct structural reasoning
2023-D1-P3	PT-POLY		TR-UNI-03	polynomial identity / factorization
2023-D1-P4	PT-FINITE	PT-INEQ; PT-COMB; PT-GRP	TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-03	modular / finite-field reduction
2023-D1-P5	PT-COMB	PT-INEQ; PT-LIM; PT-INT; PT-FINITE	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	direct structural reasoning
2023-D2-P6	PT-LA-DET	PT-LA-SPEC; PT-INEQ; PT-POLY	TR-LA-01; TR-LA-02; TR-LA-05; TR-CMB-03; TR-GEO-01	determinant / block manipulation
2023-D2-P7	PT-FE	PT-LA-DET; PT-LA-SPEC; PT-INEQ; PT-LIM	TR-LA-01; TR-RA-02; TR-RA-04; TR-ANT-04; TR-CMB-03; TR-CMB-04	direct structural reasoning
2023-D2-P8	PT-COMB	PT-INEQ; PT-LIM; PT-INT; PT-GEO	TR-RA-01; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-01	double counting / pigeonhole; graph encoding / cycles; topological / density argument
2023-D2-P9	PT-GEO	PT-LA-DET; PT-INEQ; PT-COMB; PT-GRP	TR-LA-01; TR-RA-07; TR-ANT-04; TR-CMB-03; TR-CMB-04; TR-GEO-01	convexity / Jensen / tangent; rank-nullity / dimension count; topological / density argument
2023-D2-P10	PT-FINITE	PT-LA-DET; PT-LA-SPEC; PT-FE; PT-INEQ	TR-LA-01; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	direct structural reasoning
2024-D1-P1	PT-POLY	PT-LA-DET; PT-LA-SPEC; PT-INEQ	TR-LA-01; TR-UNI-03	direct structural reasoning
2024-D1-P2	PT-LIM	PT-FE; PT-INEQ; PT-INT; PT-GEO	TR-RA-02; TR-RA-04; TR-RA-07; TR-ANT-05	direct structural reasoning
2024-D1-P3	PT-FINITE	PT-LA-DET; PT-INEQ; PT-GRP; PT-EXT	TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-GEO-01; TR-UNI-02	construction / counterexample
2024-D1-P4	PT-GRP	PT-INEQ; PT-LIM; PT-INT; PT-FINITE	TR-RA-05; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	auxiliary function / substitution
2024-D1-P5	PT-FINITE	PT-FE; PT-INEQ; PT-LIM; PT-INT	TR-RA-04; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	convexity / Jensen / tangent; probabilistic / expectation
2024-D2-P6	PT-FE	PT-INEQ; PT-INT	TR-RA-05; TR-RA-07	direct structural reasoning
2024-D2-P7	PT-LA-DET	PT-LA-SPEC; PT-INEQ; PT-POLY; PT-FINITE	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-PC-04; TR-ANT-01	determinant / block manipulation
2024-D2-P8	PT-LIM	PT-FE; PT-INEQ; PT-INT; PT-GEO	TR-RA-02; TR-RA-04; TR-RA-07; TR-PC-05; TR-UNI-07	direct structural reasoning

Problem	Primary type	Secondary types	Trigger cards	Existing method tags
2024-D2-P9	PT-FINITE	PT-LA-DET; PT-INEQ; PT-COMB; PT-GRP	TR-LA-02; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	monotonicity argument; determinant / block manipulation; double counting / pigeonhole; construction / counterexample
2024-D2-P10	PT-FINITE	PT-INEQ; PT-LIM; PT-INT; PT-GEO	TR-RA-04; TR-RA-07; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04	modular / finite-field reduction

METHOD-DATA NOTICE — trigger/template fields are not validated frequencies; click CLASS-002.

24 Trigger Library: Frequency Overview

Trigger cards translate problem-statement language into method selection. A trigger is not a theorem; it is an exam-time cue that says which theorem or method family should be tested first.

ID	Domain	Statement cue	Count
TR-RA-07	Real analysis	convex, concave, monotone, nondecreasing	226
TR-ANT-04	Rings/noncommutati algebra	ring, idempotent, square zero, noncommutative	154
TR-RA-04	Real analysis	lim n->infty of a sum, Riemann sum, asymptotic sum	133
TR-RA-06	Real analysis	series convergence, rearrangement, alternating, absolute convergence	118
TR-ANT-01	Number theory	prime, modulo, residue, divisible by, p-adic power	114
TR-RA-02	Real analysis	endpoint equalities, repeated zeros, differentiable, Rolle	111
TR-ANT-03	Groups	group generated by, subgroup, normal, automorphism, quotient	107
TR-LA-01	Linear algebra	rank, dimension, span, image, kernel, invariant subspace	99
TR-ANT-05	Arithmetic	gcd, coprime, relatively prime, Bezout	79
TR-LA-05	Linear algebra	eigenvalue, characteristic polynomial, diagonalizable, similar	76
TR-CMB-03	Combinatorics/graph theory	town, friends, grid, rows/columns, graph, cycle	57
TR-CMB-01	Combinatorics	permutation, inversion, cycle, transposition	45
TR-LA-02	Linear algebra	determinant, inverse, block matrix, triangular form	45
TR-GEO-01	Geometry	distance to point/line/sphere/ellipse/parabola	40
TR-RA-05	Real analysis	periodic, oscillatory integral, sin(nx), cos(nx)	39
TR-ANT-02	Finite fields	field characteristic, F_p, polynomial function modulo x^p-x	38
TR-LA-03	Linear algebra	AB-BA, commutator, trace zero	37
TR-CMB-04	Combinatorics	words, strings, functions on finite sets, colorings	33
TR-PC-04	Complex/Fourier	root of unity, Fourier coefficient, unit circle average	30
TR-RA-01	Real analysis	continuous on closed bounded set, compact, attains maximum/minimum	26
TR-PC-05	Generating functions	generating function, coefficient extraction, recurrence	25
TR-UNI-02	Universal	does there exist, prove/disprove, counterexample	17
TR-UNI-03	Universal	find all functions/objects/sequences	16
TR-LA-07	Linear algebra	operator norm, Hilbert space, orthonormal system, inner product	13
TR-PC-01	Polynomials	polynomial divides, more roots than degree, identically zero	13
TR-LA-04	Linear algebra	nilpotent, idempotent, involution, A^2=I, A^k=0	12
TR-UNI-07	Universal/sequences	recursively defined sequence, iterative construction	12
TR-CMB-06	Combinatorics	at least, more than, threshold, pigeonhole	10
TR-RA-03	Real analysis	f' + ..., differential inequality, integrating factor	6
TR-GEO-02	Geometry	simplex, volume, circumcenter, barycentric	5
TR-GEO-04	Metric/Hilbert geometry	orthogonal, projection, Euclidean/Hilbert norm	5
TR-PC-02	Polynomials/number theory	integer coefficients, integer values, divisibility for all integers	5
TR-UNI-05	Universal	maximum/minimum possible, smallest/largest number	4
TR-CMB-02	Combinatorics	family of finite sets, intersection, same size, covering set	3
TR-CMB-05	Probability/counting	dice, probability, random, expected, sum divisible	3

METHOD-DATA NOTICE — trigger/template fields are not validated frequencies; click CLASS-002.

ID	Domain	Statement cue	Count
TR-UNI-01	Universal	for every x,y or for arbitrary parameters	3
TR-UNI-04	Universal/topology	closed, dense, compact, countable, uncountable	3
TR-PC-03	Polynomials/complex	all roots real/on unit circle/left half-plane	2
TR-RA-10	Real analysis/topology	dense, closed, uncountable, nowhere monotone	2
TR-LA-06	Linear algebra	A+tB nilpotent for many t, polynomial in a parameter	1
TR-PC-06	Complex analysis	branches of log, Laurent expansion, cotangent series	1
TR-RA-08	Real analysis	Lipschitz gradient, smooth convex function	1
TR-RA-09	Real analysis	critical point, local/global extremum, gradient zero	1
TR-UNI-06	Universal	if and only if, equivalent, characterize	1
TR-GEO-03	Convex geometry	convex polygon/body, maximal ellipse, triangulation	0
TR-GEO-05	Metric geometry	content, diameter cover, contraction, lattice metric	0
TR-UNI-08	Universal/dynamics	iterates, periodic points, orbit, fixed point	0

METHOD-DATA NOTICE — trigger/template fields are not validated frequencies; click CLASS-002.

25 Trigger Cards

25.1 TR-RA-07: convex, concave, monotone, nondecreasing

Domain	Real analysis
Diagnosis	Use Jensen, tangent/chord bounds, monotonicity, or graph geometry.
First move	Determine convexity direction and write the tangent/chord inequality explicitly.
Method links	M-RA-09; M-RA-01
Trap	Jensen direction reverses for concave functions.
Occurrences	226 problems

Occurrence references.

1994-D1-P1; 1994-D1-P2; 1994-D1-P3; 1994-D1-P4; 1994-D1-P5; 1994-D1-P6; 1994-D2-P1; 1994-D2-P2; 1994-D2-P5; 1994-D2-P6; 1995-D1-P2; 1995-D1-P3; 1995-D1-P4; 1995-D1-P6; 1995-D2-P1; 1995-D2-P2; 1995-D2-P3; 1995-D2-P4; 1995-D2-P5; 1996-D1-P2; 1996-D1-P3; 1996-D1-P4; 1996-D1-P5; 1996-D2-P1; 1996-D2-P2; 1996-D2-P3; 1996-D2-P4; 1996-D2-P5; 1996-D2-P6; 1997-D1-P1; 1997-D1-P2; 1997-D1-P4; 1997-D1-P5; 1997-D2-P1; 1997-D2-P3; 1997-D2-P5; 1998-D1-P2; 1998-D1-P3; 1998-D1-P4; 1998-D1-P5; 1998-D1-P6; 1998-D2-P1; 1998-D2-P2; 1998-D2-P3; 1998-D2-P4; 1998-D2-P5; 1998-D2-P6; 1999-D1-P2; 1999-D1-P3; 1999-D1-P4; 1999-D1-P5; 1999-D1-P6; 1999-D2-P3; 1999-D2-P4; 1999-D2-P5; 1999-D2-P6; 2000-D1-P1; 2000-D1-P4; 2000-D1-P6; 2000-D2-P1; 2000-D2-P2; 2000-D2-P3; 2000-D2-P4; 2000-D2-P5; 2001-D1-P4; 2001-D1-P5; 2001-D1-P6; 2001-D2-P1; 2001-D2-P2; 2001-D2-P3; 2001-D2-P5; 2001-D2-P6; 2002-D1-P1; 2002-D1-P2; 2002-D1-P3; 2002-D1-P4; 2002-D1-P5; 2002-D1-P6; 2002-D2-P3; 2002-D2-P4; 2002-D2-P5; 2002-D2-P6; 2003-D1-P4; 2003-D1-P6; 2003-D2-P2; 2003-D2-P3; 2003-D2-P4; 2003-D2-P5; 2003-D2-P6; 2004-D1-P1; 2004-D1-P2; 2004-D1-P3; 2004-D1-P4; 2004-D1-P5; 2004-D1-P6; 2004-D2-P2; 2004-D2-P3; 2004-D2-P4; 2004-D2-P5; 2004-D2-P6; 2005-D1-P1; 2005-D1-P2; 2005-D1-P3; 2005-D1-P4; 2005-D1-P5; 2005-D1-P6; 2005-D2-P1; 2005-D2-P5; 2006-D1-P1; 2006-D1-P2; 2006-D1-P3; 2006-D1-P5; 2006-D1-P6; 2006-D2-P1; 2006-D2-P2; 2006-D2-P3; 2006-D2-P4; 2007-D1-P2; 2007-D1-P3; 2007-D1-P5; 2007-D1-P6; 2007-D2-P3; 2007-D2-P5; 2007-D2-P6; 2008-D1-P6; 2008-D1-P1; 2008-D1-P2; 2008-D1-P4; 2008-D1-P5; 2008-D2-P2; 2008-D2-P3; 2008-D2-P5; 2008-D2-P6; 2009-D1-P1; 2009-D1-P3; 2009-D1-P4; 2009-D2-P1; 2009-D2-P2; 2009-D2-P5; 2010-D1-P1; 2010-D1-P2; 2010-D1-P3; 2010-D1-P4; 2010-D1-P5; 2010-D2-P1; 2010-D2-P2; 2010-D2-P3; 2010-D2-P4; 2011-D1-P1; 2011-D1-P3; 2011-D1-P4; 2011-D1-P5; 2011-D2-P1; 2011-D2-P2; 2011-D2-P3; 2011-D2-P4; 2011-D2-P5; 2012-D1-P1; 2012-D1-P3; 2012-D1-P4; 2012-D2-P2; 2012-D2-P4; 2012-D2-P5; 2013-D1-P3; 2013-D1-P4; 2013-D2-P3; 2013-D2-P5; 2014-D1-P4; 2014-D1-P5; 2014-D2-P2; 2014-D2-P4; 2014-D2-P5; 2015-D1-P1; 2015-D1-P2; 2015-D1-P3; 2015-D1-P4; 2015-D1-P5; 2015-D2-P9; 2015-D2-P10; 2016-D1-P2; 2016-D1-P3; 2016-D1-P4; 2016-D1-P5; 2016-D2-P6; 2016-D2-P7; 2016-D2-P9; 2016-D2-P10; 2017-D1-P1; 2017-D1-P2; 2017-D1-P3; 2017-D1-P5; 2017-D2-P7; 2017-D2-P10; 2018-D1-P5; 2018-D2-P6; 2018-D2-P7; 2018-D2-P8; 2019-D1-P3; 2019-D1-P4; 2019-D1-P5; 2019-D2-P10; 2020-D1-P1; 2020-D1-P3; 2020-D1-P4; 2020-D2-P5; 2020-D2-P7; 2021-D2-P5; 2021-D2-P7; 2021-D2-P8; 2022-D1-P1; 2022-D1-P2; 2022-D1-P4; 2022-D2-P5; 2022-D2-P7; 2022-D2-P8; 2023-D1-P5; 2023-D2-P8; 2023-D2-P9; 2023-D2-P10; 2024-D1-P2; 2024-D1-P4; 2024-D1-P5; 2024-D2-P6; 2024-D2-P8; 2024-D2-P9; 2024-D2-P10

25.2 TR-ANT-04: ring, idempotent, square zero, noncommutative

Domain	Rings/noncommutative algebra
Diagnosis	Expand identities without commuting factors; use additive commutators.
First move	Expand (a+b)^2 or idempotent equations and preserve factor order.
Method links	M-ANT-14; M-LA-06
Trap	Never reorder terms in a noncommutative ring unless proved.
Occurrences	154 problems

Occurrence references.

1994-D1-P3; 1994-D1-P5; 1994-D1-P6; 1994-D2-P3; 1994-D2-P4; 1994-D2-P5; 1995-D1-P4; 1995-D1-P6; 1995-D2-P1; 1995-D2-P4; 1996-D1-P3; 1996-D1-P5; 1996-D2-P2; 1996-D2-P3; 1996-D2-P4; 1996-D2-P5; 1996-D2-P6; 1997-D1-P4; 1997-D1-P5; 1997-D1-P6; 1997-D2-P3; 1997-D2-P5; 1997-D2-P6; 1998-D1-P1; 1998-D1-P2; 1998-D1-P4; 1998-D2-P1; 1998-D2-P3; 1998-D2-P4; 1998-D2-P5; 1998-D2-P6; 1999-D1-P3; 1999-D1-P4; 1999-D1-P5; 1999-D1-P6; 1999-D2-P1; 1999-D2-P5; 1999-D2-P6; 2000-D2-P3; 2000-D2-P4; 2000-D2-P5; 2001-D1-P1; 2001-D1-P2; 2001-D2-P5; 2001-D2-P6; 2002-D1-P1; 2002-D1-P3; 2002-D1-P4; 2002-D1-P5; 2002-D2-P2; 2002-D2-P4; 2002-D2-P5; 2003-D1-P2; 2003-D1-P4; 2003-D2-P3; 2003-D2-P4; 2003-D2-P5; 2004-D1-P1; 2004-D1-P2; 2004-D1-P4; 2004-D1-P5; 2004-D2-P2; 2004-D2-P3; 2005-D1-P2; 2005-D1-P4; 2005-D1-P6; 2005-D2-P1; 2005-D2-P4; 2005-D2-P5; 2006-D1-P1; 2006-D1-P2; 2006-D1-P4; 2006-D2-P1; 2006-D2-P5; 2007-D1-P1; 2007-D1-P4; 2007-D2-P1; 2007-D2-P2; 2007-D2-P3; 2007-D2-P5; 2008-D1-P6; 2008-D1-P1; 2008-D1-P4; 2008-D2-P3; 2008-D2-P4; 2009-D1-P1; 2009-D2-P1; 2009-D2-P4; 2010-D1-P4; 2010-D2-P3; 2011-D1-P3; 2011-D1-P4; 2011-D1-P5; 2011-D2-P1; 2011-D2-P5; 2012-D1-P1; 2012-D1-P3; 2012-D1-P5; 2012-D2-P3; 2012-D2-P4; 2012-D2-P5; 2013-D1-P5; 2013-D2-P2; 2013-D2-P4; 2014-D2-P1; 2014-D2-P4; 2015-D1-P2; 2015-D1-P4; 2015-D2-P10; 2016-D1-P2; 2016-D1-P3; 2016-D1-P4; 2016-D1-P5; 2016-D2-P8; 2016-D2-P9; 2017-D1-P3; 2017-D1-P4; 2017-D2-P8; 2017-D2-P10; 2018-D1-P2; 2018-D1-P5; 2018-D2-P6; 2018-D2-P8; 2019-D1-P2; 2019-D1-P4; 2019-D2-P8; 2019-D2-P9; 2019-D2-P10; 2020-D1-P1; 2020-D1-P3; 2020-D2-P6; 2020-D2-P7; 2021-D1-P2; 2021-D2-P5; 2021-D2-P6; 2021-D2-P7; 2021-D2-P8; 2022-D1-P2; 2022-D1-P3; 2022-D1-P4; 2022-D2-P5; 2022-D2-P6; 2022-D2-P8; 2023-D1-P4; 2023-D1-P5; 2023-D2-P7; 2023-D2-P8; 2023-D2-P9; 2023-D2-P10; 2024-D1-P3; 2024-D1-P4; 2024-D1-P5; 2024-D2-P9; 2024-D2-P10

25.3 TR-RA-04: lim n->infty of a sum, Riemann sum, asymptotic sum

Domain	Real analysis
Diagnosis	Scale variables and compare to an integral; isolate tails.
First move	Find h and rewrite the sum as h sum f(nh) or split small/large ranges.
Method links	M-RA-06; M-RA-08
Trap	Infinite-domain Riemann sums require tail estimates.
Occurrences	133 problems

Occurrence references.

1994-D1-P2; 1994-D1-P5; 1994-D2-P6; 1995-D1-P2; 1995-D1-P3; 1995-D1-P4; 1995-D1-P5; 1995-D1-P6; 1995-D2-P2; 1995-D2-P5; 1995-D2-P6; 1996-D1-P2; 1996-D1-P4; 1996-D1-P5; 1996-D2-P1; 1996-D2-P5; 1996-D2-P6; 1997-D1-P1; 1997-D1-P2; 1997-D2-P1; 1997-D2-P3; 1997-D2-P6; 1998-D1-P3; 1998-D1-P4; 1998-D1-P5; 1998-D1-P6; 1998-D2-P2; 1998-D2-P6; 1999-D1-P2; 1999-D2-P2; 1999-D2-P3; 1999-D2-P6; 2000-D1-P1; 2000-D1-P2; 2000-D1-P4; 2000-D1-P6; 2000-D2-P2; 2000-D2-P4; 2000-D2-P5; 2000-D2-P6; 2001-D1-P1; 2001-D1-P3; 2001-D1-P6; 2001-D2-P1; 2001-D2-P2; 2001-D2-P5; 2002-D1-P4; 2002-D1-P5; 2002-D1-P6; 2003-D1-P1; 2003-D1-P2; 2003-D1-P3; 2003-D1-P5; 2003-D2-P2; 2003-D2-P5; 2003-D2-P6; 2004-D1-P2; 2004-D1-P3; 2004-D1-P4; 2004-D1-P5; 2004-D1-P6; 2004-D2-P3; 2004-D2-P4; 2004-D2-P5; 2004-D2-P6; 2005-D1-P3; 2005-D1-P5; 2005-D2-P1; 2005-D2-P3; 2006-D1-P1; 2006-D1-P3; 2006-D1-P4

METHOD-DATA NOTICE — trigger/template fields are not validated frequencies; click CLASS-002.

; 2006-D1-P5; 2006-D1-P6; 2006-D2-P3; 2007-D2-P1; 2007-D2-P3; 2007-D2-P6; 2008-D1-P5; 2008-D2-P1; 2009-D1-P1; 2009-D1-P4; 2009-D1-P5; 2010-D1-P2 ; 2010-D1-P3; 2010-D2-P1; 2010-D2-P2; 2010-D2-P5; 2011-D1-P1; 2011-D2-P1; 2012-D1-P1; 2012-D1-P4; 2012-D2-P2; 2013-D1-P2; 2013-D1-P4; 2013-D2-P3 ; 2013-D2-P4; 2014-D1-P2; 2014-D1-P4; 2014-D2-P2; 2014-D2-P3; 2014-D2-P5; 2015-D1-P5; 2015-D2-P6; 2015-D2-P7; 2015-D2-P8; 2016-D2-P6; 2016-D2-P7 ; 2017-D1-P2; 2017-D2-P6; 2017-D2-P9; 2018-D1-P1; 2018-D1-P4; 2018-D2-P7; 2018-D2-P10; 2019-D1-P1; 2019-D1-P3; 2019-D2-P6; 2019-D2-P7; 2019-D2-P8 ; 2020-D1-P2; 2020-D1-P4; 2020-D2-P5; 2020-D2-P8; 2021-D1-P3; 2021-D1-P4; 2023-D1-P1; 2023-D2-P7; 2023-D2-P8; 2024-D1-P2; 2024-D1-P5; 2024-D2-P8; 2024-D2-P10

25.4 TR-RA-06: series convergence, rearrangement, alternating, absolute convergence

Domain	Real analysis
Diagnosis	Use Cauchy/tail control, grouping, Dirichlet/alternating ideas, or counterexample rearrangements.
First move	Compute partial sums under the proposed order and isolate a divergent subsequence if disproving.
Method links	M-RA-07; M-UNI-07
Trap	Conditional convergence is not stable under arbitrary rearrangement.
Occurrences	118 problems

Occurrence references.

1994-D1-P3; 1994-D1-P4; 1994-D2-P1; 1994-D2-P3; 1994-D2-P5; 1995-D1-P6; 1995-D2-P5; 1995-D2-P6; 1996-D1-P2; 1996-D1-P4; 1996-D1-P5; 1996-D1-P6 ; 1996-D2-P1; 1996-D2-P4; 1996-D2-P5; 1996-D2-P6; 1997-D1-P1; 1997-D1-P2; 1997-D1-P4; 1997-D1-P5; 1997-D1-P6; 1997-D2-P3; 1998-D1-P2; 1998-D1-P3 ; 1998-D1-P6; 1998-D2-P1; 1998-D2-P6; 1999-D1-P4; 1999-D2-P2; 1999-D2-P3; 1999-D2-P4; 1999-D2-P5; 2000-D1-P6; 2000-D2-P4; 2000-D2-P5; 2000-D2-P6 ; 2001-D1-P1; 2001-D1-P5; 2001-D2-P2; 2001-D2-P4; 2002-D1-P1; 2002-D1-P2; 2002-D1-P3; 2002-D1-P4; 2002-D1-P5; 2002-D1-P6; 2002-D2-P3; 2002-D2-P4 ; 2003-D1-P1; 2003-D1-P2; 2003-D1-P3; 2003-D1-P5; 2003-D1-P6; 2003-D2-P5; 2003-D2-P6; 2004-D1-P2; 2004-D1-P3; 2004-D1-P4; 2004-D1-P5; 2004-D1-P6 ; 2004-D2-P3; 2004-D2-P4; 2004-D2-P6; 2005-D1-P1; 2005-D1-P4; 2005-D1-P5; 2005-D2-P1; 2005-D2-P2; 2005-D2-P3; 2005-D2-P5; 2005-D2-P6; 2006-D1-P1 ; 2006-D1-P3; 2006-D1-P4; 2006-D1-P6; 2006-D2-P1; 2006-D2-P4; 2006-D2-P5; 2006-D2-P6; 2007-D1-P2; 2007-D1-P5; 2007-D1-P6; 2007-D2-P2; 2007-D2-P6 ; 2008-D1-P2; 2008-D1-P3; 2008-D1-P4; 2008-D2-P6; 2009-D1-P1; 2009-D1-P4; 2009-D2-P3; 2009-D2-P5; 2010-D1-P2; 2010-D1-P4; 2010-D2-P1; 2010-D2-P2 ; 2010-D2-P3; 2011-D1-P1; 2011-D1-P2; 2011-D1-P5; 2011-D2-P1; 2011-D2-P2; 2011-D2-P3; 2011-D2-P4; 2011-D2-P5; 2012-D1-P1; 2012-D2-P2; 2013-D1-P5; 2013-D2-P4; 2014-D1-P4; 2015-D2-P8; 2018-D1-P1; 2018-D2-P7; 2019-D2-P6; 2019-D2-P7; 2019-D2-P8; 2020-D1-P2; 2023-D2-P8

25.5 TR-ANT-01: prime, modulo, residue, divisible by, p-adic power

Domain	Number theory
Diagnosis	Reduce modulo primes and track valuations.
First move	Factor the modulus and solve conditions prime-power by prime-power.
Method links	M-ANT-01; M-ANT-03
Trap	Composite moduli can have zero divisors.
Occurrences	114 problems

Occurrence references.

1994-D1-P3; 1994-D1-P6; 1994-D2-P5; 1995-D2-P4; 1996-D2-P2; 1996-D2-P3; 1997-D1-P3; 1997-D1-P4; 1997-D1-P5; 1999-D1-P3; 1999-D2-P1; 2000-D1-P5 ; 2001-D1-P1; 2001-D1-P2; 2001-D1-P4; 2001-D2-P1; 2001-D2-P6; 2002-D1-P3; 2002-D2-P3; 2003-D1-P2; 2003-D1-P4; 2003-D2-P4; 2005-D1-P2; 2005-D1-P6 ; 2005-D2-P6; 2006-D1-P2; 2006-D1-P4; 2006-D2-P1; 2006-D2-P4; 2006-D2-P5; 2007-D1-P1; 2007-D1-P2; 2007-D1-P4; 2007-D1-P5; 2007-D1-P6; 2007-D2-P2 ; 2007-D2-P4; 2007-D2-P5; 2008-D1-P3; 2008-D1-P5; 2008-D2-P1; 2008-D2-P3; 2008-D2-P4; 2008-D2-P5; 2009-D2-P4; 2009-D2-P5; 2010-D1-P4; 2010-D2-P3 ; 2010-D2-P4; 2010-D2-P5; 2011-D1-P3; 2011-D1-P5; 2011-D2-P2; 2011-D2-P4; 2012-D1-P1; 2012-D1-P2; 2012-D1-P3; 2012-D1-P5; 2012-D2-P1; 2012-D2-P3 ; 2012-D2-P4; 2012-D2-P5; 2013-D1-P5; 2013-D2-P2; 2013-D2-P4; 2014-D1-P4; 2014-D2-P1; 2015-D1-P2; 2015-D1-P4; 2015-D2-P10; 2016-D1-P2; 2016-D1-P3 ; 2016-D1-P4; 2016-D1-P5; 2016-D2-P8; 2016-D2-P9; 2017-D1-P1; 2017-D1-P3; 2017-D1-P4; 2017-D1-P5; 2017-D2-P7; 2017-D2-P8; 2017-D2-P10; 2018-D1-P2 ; 2018-D1-P5; 2018-D2-P6; 2018-D2-P8; 2019-D1-P2; 2019-D1-P4; 2019-D1-P5; 2019-D2-P7; 2019-D2-P9; 2020-D1-P1; 2020-D1-P3; 2020-D2-P6; 2020-D2-P7; 2021-D1-P2; 2021-D2-P5; 2021-D2-P6; 2021-D2-P8; 2022-D1-P2; 2022-D1-P3; 2022-D1-P4; 2022-D2-P6; 2022-D2-P8; 2023-D1-P4; 2023-D1-P5; 2023-D2-P10; 2024-D1-P3; 2024-D1-P4; 2024-D1-P5; 2024-D2-P7; 2024-D2-P9; 2024-D2-P10

25.6 TR-RA-02: endpoint equalities, repeated zeros, differentiable, Rolle

Domain	Real analysis
Diagnosis	Use Rolle chains, divided differences, or interpolation remainders.
First move	Build an auxiliary function that has enough zeros, then apply Rolle repeatedly.
Method links	M-RA-03; M-RA-04
Trap	Check differentiability order and open interval location of the final point.
Occurrences	111 problems

Occurrence references.

1994-D1-P2; 1994-D1-P6; 1994-D2-P1; 1994-D2-P3; 1994-D2-P6; 1995-D1-P2; 1995-D1-P3; 1995-D1-P4; 1995-D1-P6; 1995-D2-P2; 1995-D2-P5; 1995-D2-P6 ; 1996-D1-P2; 1996-D1-P4; 1996-D1-P5; 1996-D1-P6; 1996-D2-P1; 1996-D2-P5; 1997-D1-P1; 1997-D1-P2; 1997-D2-P1; 1997-D2-P3; 1997-D2-P6; 1998-D1-P3 ; 1998-D1-P4; 1998-D1-P6; 1998-D2-P6; 1999-D1-P2; 1999-D1-P6; 1999-D2-P3; 2000-D1-P1; 2000-D1-P4; 2000-D2-P2; 2000-D2-P5; 2001-D1-P3; 2001-D1-P6 ; 2001-D2-P2; 2001-D2-P3; 2001-D2-P6; 2002-D1-P2; 2002-D1-P4; 2002-D1-P5; 2002-D1-P6; 2003-D1-P1; 2003-D1-P5; 2003-D2-P5; 2003-D2-P6; 2004-D1-P2 ; 2004-D1-P3; 2004-D1-P5; 2004-D2-P2; 2004-D2-P5; 2005-D1-P3; 2005-D1-P5; 2005-D2-P1; 2005-D2-P4; 2005-D2-P5; 2006-D1-P1; 2006-D1-P5; 2006-D2-P3 ; 2007-D2-P1; 2007-D2-P3; 2008-D1-P1; 2008-D1-P5; 2009-D1-P1; 2009-D2-P2; 2010-D1-P1; 2010-D1-P2; 2010-D1-P5; 2010-D2-P1; 2010-D2-P2; 2011-D1-P1 ; 2011-D2-P1; 2012-D1-P2; 2012-D1-P4; 2012-D2-P2; 2013-D1-P2; 2013-D1-P4; 2013-D2-P3; 2014-D1-P2; 2014-D2-P2; 2014-D2-P3; 2014-D2-P5; 2015-D1-P3 ; 2015-D1-P5; 2015-D2-P6; 2015-D2-P7; 2016-D1-P1; 2016-D2-P6; 2016-D2-P7; 2017-D1-P2; 2017-D2-P6; 2017-D2-P9; 2018-D1-P1; 2018-D1-P4; 2018-D2-P7; 2018-D2-P10; 2019-D1-P1; 2019-D1-P3; 2019-D2-P6; 2019-D2-P10; 2020-D2-P5; 2020-D2-P8; 2021-D1-P3; 2021-D1-P4; 2022-D1-P1; 2022-D2-P5; 2023-D1-P1; 2023-D2-P7; 2024-D1-P2; 2024-D2-P8

METHOD-DATA NOTICE — trigger/template fields are not validated frequencies; click CLASS-002.

25.7 TR-ANT-03: group generated by, subgroup, normal, automorphism, quotient

Domain	Groups
Diagnosis	Use generators, quotients, group actions, or explicit permutation/matrix models.
First move	Write the generators and identify the quotient or action that preserves the relevant property.
Method links	M-ANT-11; M-ANT-12; M-ANT-13
Trap	Subgroup generated by powers is not merely the set of powers.
Occurrences	107 problems

Occurrence references.

1994-D1-P3; 1994-D1-P6; 1994-D2-P4; 1994-D2-P5; 1995-D2-P4; 1996-D2-P2; 1996-D2-P3; 1997-D1-P3; 1997-D1-P4; 1997-D1-P5; 1998-D1-P2; 1999-D1-P1; 1999-D1-P3; 1999-D1-P5; 1999-D2-P1; 2000-D1-P5; 2001-D1-P1; 2001-D1-P2; 2001-D2-P6; 2002-D1-P3; 2002-D2-P3; 2003-D1-P2; 2003-D1-P4; 2003-D2-P4; 2004-D2-P3; 2005-D1-P2; 2005-D1-P4; 2005-D1-P6; 2006-D1-P2; 2006-D1-P4; 2006-D2-P1; 2006-D2-P4; 2006-D2-P5; 2006-D2-P6; 2007-D1-P1; 2007-D1-P4; 2007-D2-P2; 2007-D2-P4; 2007-D2-P5; 2008-D2-P3; 2008-D2-P4; 2009-D1-P1; 2009-D2-P4; 2010-D1-P4; 2010-D2-P2; 2010-D2-P3; 2010-D2-P4; 2011-D1-P3; 2011-D1-P5; 2011-D2-P2; 2011-D2-P5; 2012-D1-P1; 2012-D1-P2; 2012-D1-P3; 2012-D1-P5; 2012-D2-P3; 2012-D2-P4; 2012-D2-P5; 2013-D1-P5; 2013-D2-P2; 2013-D2-P4; 2014-D1-P4; 2014-D2-P1; 2015-D1-P2; 2015-D1-P4; 2015-D2-P9; 2015-D2-P10; 2016-D1-P2; 2016-D1-P3; 2016-D1-P4; 2016-D1-P5; 2016-D2-P8; 2016-D2-P9; 2017-D1-P3; 2017-D1-P4; 2017-D2-P8; 2017-D2-P10; 2018-D1-P2; 2018-D1-P5; 2018-D2-P6; 2018-D2-P8; 2019-D1-P2; 2019-D1-P4; 2019-D1-P5; 2019-D2-P7; 2019-D2-P9; 2020-D1-P1; 2020-D1-P3; 2020-D2-P6; 2020-D2-P7; 2021-D1-P2; 2021-D2-P5; 2021-D2-P6; 2021-D2-P8; 2022-D1-P2; 2022-D1-P3; 2022-D1-P4; 2022-D2-P6; 2022-D2-P8; 2023-D1-P4; 2023-D1-P5; 2023-D2-P10; 2024-D1-P3; 2024-D1-P4; 2024-D1-P5; 2024-D2-P9; 2024-D2-P10

25.8 TR-LA-01: rank, dimension, span, image, kernel, invariant subspace

Domain	Linear algebra
Diagnosis	Look for rank-nullity, a basis adapted to subspaces, and dimension counting.
First move	Write the relevant linear map and its kernel/image; choose a basis adapted to the flags or invariant subspaces.
Method links	M-LA-04; M-LA-08; M-UNI-04
Trap	Do not count free matrix entries before proving all constraints are independent.
Occurrences	99 problems

Occurrence references.

1994-D1-P1; 1994-D1-P4; 1994-D2-P2; 1994-D2-P4; 1994-D2-P5; 1995-D1-P1; 1995-D1-P5; 1995-D2-P1; 1996-D1-P1; 1996-D1-P3; 1996-D1-P6; 1997-D1-P3; 1997-D2-P2; 1997-D2-P4; 1998-D1-P1; 1998-D2-P1; 1999-D1-P1; 2000-D1-P3; 2000-D2-P1; 2000-D2-P6; 2001-D1-P5; 2001-D2-P3; 2001-D2-P4; 2002-D1-P5; 2002-D1-P6; 2002-D2-P1; 2002-D2-P5; 2003-D1-P3; 2003-D1-P4; 2003-D1-P5; 2003-D2-P1; 2004-D1-P3; 2004-D2-P1; 2004-D2-P4; 2004-D2-P6; 2005-D1-P1; 2005-D2-P3; 2005-D2-P6; 2006-D1-P3; 2006-D1-P6; 2006-D2-P2; 2006-D2-P4; 2006-D2-P6; 2007-D1-P2; 2007-D1-P3; 2007-D1-P5; 2007-D2-P4; 2007-D2-P5; 2008-D1-P6; 2008-D1-P2; 2008-D1-P5; 2008-D2-P5; 2008-D2-P6; 2009-D1-P2; 2009-D1-P3; 2009-D1-P4; 2009-D1-P5; 2009-D2-P3; 2009-D2-P5; 2010-D1-P5; 2010-D2-P4; 2011-D1-P2; 2011-D1-P5; 2011-D2-P3; 2011-D2-P4; 2011-D2-P5; 2012-D1-P2; 2012-D2-P2; 2013-D1-P1; 2014-D1-P1; 2014-D2-P2; 2015-D1-P1; 2015-D1-P3; 2015-D1-P4; 2015-D1-P5; 2015-D2-P9; 2016-D2-P10; 2017-D1-P1; 2017-D1-P3; 2017-D2-P8; 2017-D2-P9; 2018-D1-P3; 2018-D2-P8; 2018-D2-P9; 2019-D1-P5; 2019-D2-P7; 2019-D2-P9; 2020-D1-P2; 2021-D1-P1; 2021-D2-P5; 2022-D1-P2; 2022-D2-P7; 2023-D1-P2; 2023-D2-P6; 2023-D2-P7; 2023-D2-P9; 2023-D2-P10; 2024-D1-P1; 2024-D2-P7

25.9 TR-ANT-05: gcd, coprime, relatively prime, Bezout

Domain	Arithmetic
Diagnosis	Use Bezout, valuations, and prime-factor allocation.
First move	Factor all quantities and write gcd/coprimality relations explicitly.
Method links	M-ANT-03; M-ANT-05
Trap	Units and signs matter in Z and Z[x].
Occurrences	79 problems

Occurrence references.

1994-D1-P3; 1994-D1-P5; 1994-D1-P6; 1994-D2-P6; 1995-D1-P3; 1995-D2-P4; 1996-D2-P1; 1996-D2-P2; 1996-D2-P3; 1997-D1-P1; 1997-D1-P4; 1999-D1-P3; 1999-D2-P1; 2001-D1-P2; 2001-D1-P3; 2002-D1-P3; 2003-D2-P2; 2003-D2-P4; 2003-D2-P6; 2005-D1-P5; 2006-D1-P2; 2006-D2-P5; 2007-D1-P4; 2007-D2-P2; 2007-D2-P5; 2008-D2-P3; 2010-D1-P3; 2010-D2-P3; 2012-D1-P3; 2012-D2-P3; 2012-D2-P4; 2012-D2-P5; 2013-D2-P2; 2013-D2-P4; 2014-D2-P1; 2015-D1-P2; 2015-D1-P4; 2015-D2-P7; 2015-D2-P10; 2016-D1-P2; 2016-D1-P3; 2016-D1-P4; 2016-D1-P5; 2016-D2-P8; 2016-D2-P9; 2017-D1-P3; 2017-D1-P4; 2017-D2-P6; 2017-D2-P9; 2017-D2-P10; 2018-D1-P2; 2018-D1-P5; 2018-D2-P6; 2018-D2-P8; 2018-D2-P10; 2019-D1-P2; 2019-D1-P4; 2019-D2-P9; 2020-D1-P1; 2020-D1-P3; 2020-D2-P6; 2020-D2-P7; 2020-D2-P8; 2021-D1-P2; 2021-D2-P5; 2021-D2-P6; 2021-D2-P8; 2022-D1-P3; 2022-D1-P4; 2022-D2-P6; 2022-D2-P8; 2023-D1-P4; 2023-D1-P5; 2023-D2-P10; 2024-D1-P2; 2024-D1-P3; 2024-D1-P4; 2024-D1-P5; 2024-D2-P9

METHOD-DATA NOTICE — trigger/template fields are not validated frequencies; click CLASS-002.

25.10 TR-LA-05: eigenvalue, characteristic polynomial, diagonalizable, similar

Domain	Linear algebra
Diagnosis	Use spectral mapping, Jordan/canonical form intuition, and similarity invariants.
First move	List eigenvalue constraints and which quantities are invariant under similarity.
Method links	M-LA-05; M-LA-11; M-LA-13
Trap	A real matrix can have complex eigenvalues; do not assume an eigenbasis over R.
Occurrences	76 problems

Occurrence references.

1994-D1-P1; 1994-D1-P4; 1994-D2-P2; 1994-D2-P4; 1995-D1-P1; 1995-D1-P4; 1995-D1-P5; 1995-D2-P1; 1995-D2-P5; 1996-D1-P1; 1996-D1-P3; 1996-D1-P5; 1996-D1-P6; 1997-D1-P5; 1997-D2-P2; 1997-D2-P4; 1998-D1-P1; 1998-D2-P1; 1998-D2-P4; 1998-D2-P5; 1999-D1-P1; 2000-D1-P3; 2000-D1-P5; 2000-D2-P6; 2001-D1-P2; 2001-D1-P5; 2001-D1-P6; 2001-D2-P4; 2002-D1-P1; 2002-D2-P1; 2002-D2-P3; 2002-D2-P5; 2003-D1-P3; 2003-D1-P4; 2003-D1-P5; 2003-D1-P6; 2003-D2-P1; 2004-D1-P1; 2004-D1-P5; 2004-D1-P6; 2004-D2-P1; 2004-D2-P4; 2004-D2-P6; 2005-D1-P1; 2005-D2-P3; 2006-D2-P2; 2006-D2-P6; 2007-D1-P3; 2007-D2-P6; 2008-D1-P2; 2008-D1-P3; 2008-D2-P6; 2009-D1-P2; 2009-D1-P4; 2009-D1-P5; 2009-D2-P3; 2010-D2-P1; 2010-D2-P5; 2011-D1-P2; 2011-D2-P2; 2013-D1-P1; 2014-D1-P1; 2014-D2-P2; 2015-D1-P1; 2015-D2-P9; 2016-D2-P10; 2017-D1-P1; 2017-D2-P8; 2018-D1-P3; 2020-D1-P2; 2021-D1-P1; 2022-D1-P2; 2022-D2-P7; 2023-D1-P2; 2023-D2-P6; 2024-D2-P7

25.11 TR-CMB-03: town, friends, grid, rows/columns, graph, cycle

Domain	Combinatorics/graph theory
Diagnosis	Build a graph and translate the claim into degree/cycle/walk language.
First move	Define vertices and edges so that the desired structure becomes a graph cycle or path.
Method links	M-CMB-03; M-CMB-01
Trap	Graphs may be bipartite or directed depending on the encoding.
Occurrences	57 problems

Occurrence references.

1994-D1-P5; 1994-D2-P3; 1994-D2-P4; 1995-D1-P1; 1995-D1-P4; 1996-D2-P4; 1996-D2-P6; 1997-D1-P6; 1997-D2-P5; 1997-D2-P6; 1998-D1-P1; 1998-D1-P2; 1998-D1-P4; 1998-D2-P1; 1998-D2-P3; 1998-D2-P4; 1998-D2-P5; 1999-D1-P4; 1999-D1-P5; 1999-D1-P6; 1999-D2-P5; 1999-D2-P6; 2000-D2-P3; 2001-D2-P5; 2002-D1-P1; 2002-D2-P2; 2002-D2-P4; 2003-D2-P3; 2003-D2-P4; 2003-D2-P5; 2004-D1-P1; 2004-D1-P4; 2004-D2-P2; 2005-D2-P4; 2006-D1-P1; 2007-D2-P1; 2007-D2-P3; 2008-D1-P1; 2008-D1-P4; 2009-D1-P3; 2009-D2-P1; 2009-D2-P4; 2011-D1-P4; 2012-D2-P3; 2012-D2-P5; 2014-D2-P4; 2016-D1-P4; 2017-D1-P4; 2019-D2-P8; 2019-D2-P10; 2021-D1-P2; 2021-D2-P7; 2022-D2-P5; 2023-D1-P4; 2023-D2-P6; 2023-D2-P7; 2023-D2-P9

25.12 TR-CMB-01: permutation, inversion, cycle, transposition

Domain	Combinatorics
Diagnosis	Use cycle structure, parity, generating functions, or group generation.
First move	Write the permutation in cycles or encode the statistic algebraically.
Method links	M-CMB-04; M-ANT-11
Trap	Composition order conventions can change computations.
Occurrences	45 problems

Occurrence references.

1994-D1-P1; 1996-D1-P3; 1996-D2-P3; 1997-D2-P5; 1997-D2-P6; 1998-D1-P2; 1998-D2-P3; 1998-D2-P4; 1998-D2-P5; 1999-D1-P5; 2000-D2-P4; 2002-D1-P1; 2002-D2-P4; 2004-D1-P1; 2004-D1-P2; 2004-D2-P2; 2005-D1-P4; 2007-D1-P4; 2007-D1-P6; 2007-D2-P1; 2008-D1-P6; 2008-D2-P6; 2009-D1-P3; 2009-D2-P4; 2011-D1-P4; 2011-D2-P5; 2012-D1-P3; 2013-D1-P3; 2013-D2-P5; 2015-D2-P8; 2016-D2-P9; 2017-D1-P4; 2019-D2-P8; 2019-D2-P10; 2020-D1-P1; 2020-D1-P3; 2020-D2-P7; 2021-D2-P6; 2022-D1-P3; 2022-D1-P4; 2022-D2-P5; 2022-D2-P6; 2022-D2-P8; 2023-D1-P5; 2023-D2-P8

25.13 TR-LA-02: determinant, inverse, block matrix, triangular form

Domain	Linear algebra
Diagnosis	Use row/column operations, block multiplication, Schur-type identities, or determinant-preserving reductions.
First move	Try to make the matrix triangular/sparse or complete a product to the identity.
Method links	M-LA-01; M-LA-02; M-LA-03
Trap	Check invertibility before multiplying by inverses.
Occurrences	45 problems

Occurrence references.

1994-D1-P1; 1995-D1-P1; 1995-D2-P1; 1996-D1-P1; 1997-D1-P3; 1997-D2-P2; 1997-D2-P4; 1999-D1-P1; 1999-D1-P5; 1999-D2-P2; 2000-D1-P3; 2001-D1-P1; 2001-D1-P5; 2001-D2-P4; 2002-D2-P1; 2003-D2-P1; 2004-D2-P1; 2004-D2-P6; 2005-D1-P1; 2005-D2-P3; 2005-D2-P6; 2006-D1-P3; 2006-D2-P6; 2007-D1-P2; 2007-D1-P3; 2007-D2-P4; 2008-D1-P6; 2008-D2-P5; 2009-D1-P5; 2009-D2-P3; 2009-D2-P5; 2010-D1-P5; 2010-D2-P4; 2011-D1-P2; 2011-D2-P2; 2012-D1-P2; 2014-D1-P1; 2015-D1-P1; 2019-D1-P5; 2019-D2-P9; 2020-D1-P2; 2022-D2-P7; 2023-D2-P6; 2024-D2-P7; 2024-D2-P9

METHOD-DATA NOTICE — trigger/template fields are not validated frequencies; click CLASS-002.

25.14 TR-GEO-01: distance to point/line/sphere/ellipse/parabola

Domain	Geometry
Diagnosis	Choose coordinates and square distances.
First move	Place the line/point/center at coordinate axes and convert to a quadratic equation.
Method links	M-GEO-01; M-GEO-02
Trap	Squaring can lose the sign/side condition.
Occurrences	40 problems

Occurrence references.

1994-D1-P4; 1994-D2-P2; 1995-D2-P3; 1995-D2-P5; 1996-D1-P1; 1997-D2-P2; 1997-D2-P4; 1998-D1-P1; 2002-D2-P1; 2003-D1-P6; 2003-D2-P1; 2004-D2-P1; 2004-D2-P3; 2006-D1-P6; 2006-D2-P2; 2007-D1-P3; 2007-D1-P5; 2008-D1-P2; 2008-D2-P2; 2009-D1-P2; 2009-D1-P4; 2009-D2-P1; 2011-D1-P2; 2013-D1-P1; 2014-D1-P1; 2014-D1-P5; 2014-D2-P2; 2014-D2-P4; 2015-D1-P5; 2015-D2-P9; 2016-D1-P2; 2016-D2-P10; 2017-D2-P10; 2018-D1-P3; 2019-D1-P2; 2021-D1-P1; 2023-D1-P2; 2023-D2-P6; 2023-D2-P9; 2024-D1-P3

25.15 TR-RA-05: periodic, oscillatory integral, sin(nx), cos(nx)

Domain	Real analysis
Diagnosis	Use averaging, orthogonality, or Fourier-type cancellation.
First move	Partition into periods or multiply by a test trigonometric function.
Method links	M-RA-16; M-PC-04
Trap	Oscillation does not automatically imply zero; quantify the averaging.
Occurrences	39 problems

Occurrence references.

1994-D1-P5; 1994-D2-P1; 1994-D2-P2; 1994-D2-P3; 1995-D2-P3; 1996-D1-P2; 1996-D2-P1; 1996-D2-P5; 1998-D2-P3; 1998-D2-P4; 1999-D1-P4; 1999-D1-P6; 1999-D2-P2; 1999-D2-P4; 1999-D2-P6; 2000-D1-P6; 2000-D2-P5; 2001-D1-P6; 2001-D2-P3; 2001-D2-P5; 2002-D1-P2; 2002-D2-P5; 2002-D2-P6; 2004-D2-P2; 2005-D2-P4; 2005-D2-P5; 2006-D2-P2; 2007-D1-P6; 2008-D1-P1; 2009-D2-P2; 2010-D1-P1; 2010-D1-P3; 2011-D1-P1; 2011-D2-P3; 2015-D1-P3; 2016-D1-P1; 2019-D2-P6; 2024-D1-P4; 2024-D2-P6

25.16 TR-ANT-02: field characteristic, F_p, polynomial function modulo x^p-x

Domain	Finite fields
Diagnosis	Use finite-field arithmetic and Fermat/Frobenius.
First move	Separate cases according to characteristic and use degree/root counts in the field.
Method links	M-ANT-02; M-ANT-06
Trap	A polynomial representative of a function on F_p is unique only modulo x^p-x with degree < p.
Occurrences	38 problems

Occurrence references.

1994-D2-P4; 1997-D1-P3; 2000-D1-P5; 2000-D2-P1; 2000-D2-P3; 2001-D1-P2; 2001-D1-P4; 2005-D1-P2; 2005-D1-P4; 2005-D2-P2; 2006-D1-P2; 2006-D2-P1; 2006-D2-P5; 2007-D1-P1; 2007-D2-P2; 2007-D2-P4; 2008-D1-P3; 2008-D2-P4; 2008-D2-P5; 2009-D2-P4; 2010-D1-P4; 2010-D2-P5; 2011-D1-P3; 2011-D2-P4; 2012-D2-P1; 2013-D1-P5; 2013-D2-P2; 2014-D1-P4; 2016-D1-P5; 2016-D2-P8; 2018-D1-P5; 2019-D2-P7; 2020-D2-P6; 2021-D2-P6; 2022-D1-P3; 2022-D2-P6; 2023-D1-P4; 2024-D2-P10

25.17 TR-LA-03: AB-BA, commutator, trace zero

Domain	Linear algebra
Diagnosis	Trace of commutators is zero; commutator equations often force nilpotency or spectral restrictions.
First move	Take traces, trace powers, or view X -> AX-XA as a linear operator.
Method links	M-LA-06; M-LA-07; M-LA-12
Trap	Trace zero alone is weak; combine it with rank/eigenvalue or all-power information.
Occurrences	37 problems

Occurrence references.

1995-D1-P1; 1995-D1-P5; 1996-D1-P1; 1997-D1-P3; 1997-D2-P2; 1997-D2-P4; 1999-D1-P1; 2000-D1-P3; 2000-D1-P5; 2000-D2-P6; 2001-D1-P5; 2001-D2-P4; 2002-D2-P1; 2003-D1-P3; 2005-D1-P1; 2005-D1-P6; 2005-D2-P3; 2005-D2-P6; 2006-D1-P3; 2006-D2-P6; 2007-D1-P2; 2007-D1-P3; 2007-D2-P4; 2008-D1-P6; 2008-D2-P5; 2009-D1-P5; 2009-D2-P3; 2009-D2-P5; 2010-D1-P5; 2011-D1-P2; 2012-D1-P2; 2014-D1-P1; 2015-D1-P1; 2019-D1-P5; 2020-D1-P2; 2022-D2-P7; 2024-D2-P7

METHOD-DATA NOTICE — trigger/template fields are not validated frequencies; click CLASS-002.

25.18 TR-CMB-04: words, strings, functions on finite sets, colorings

Domain	Combinatorics
Diagnosis	Use recurrences, finite automata, equivalence reductions, or bijections.
First move	Classify states by the last symbols or by the invariant preserved by operations.
Method links	M-CMB-05; M-CMB-06
Trap	Equivalence relations require both directions if constructing normal forms.
Occurrences	33 problems

Occurrence references.

1994-D2-P3; 1996-D2-P4; 1996-D2-P6; 1997-D1-P6; 1997-D2-P5; 1997-D2-P6; 1998-D1-P1; 1998-D1-P4; 1998-D2-P3; 1999-D1-P4; 1999-D1-P6; 1999-D2-P5; 2001-D2-P5; 2002-D2-P2; 2002-D2-P4; 2003-D2-P3; 2004-D1-P1; 2004-D1-P4; 2005-D2-P4; 2007-D2-P1; 2008-D1-P4; 2009-D2-P1; 2011-D1-P4; 2012-D2-P3; 2014-D2-P4; 2015-D2-P8; 2019-D2-P8; 2019-D2-P10; 2021-D1-P2; 2021-D2-P7; 2022-D2-P5; 2023-D2-P7; 2023-D2-P9

25.19 TR-PC-04: root of unity, Fourier coefficient, unit circle average

Domain	Complex/Fourier
Diagnosis	Use root-of-unity filters or averaging on the unit circle.
First move	Average f(1), f(omega), ..., f(omega^{m-1}) to isolate multiples of m.
Method links	M-PC-04; M-CMB-05
Trap	Check primitive root and indexing conventions.
Occurrences	30 problems

Occurrence references.

1995-D2-P3; 1996-D2-P2; 1999-D1-P1; 1999-D2-P2; 1999-D2-P6; 2000-D1-P2; 2000-D1-P3; 2000-D1-P6; 2000-D2-P3; 2001-D1-P4; 2001-D2-P3; 2002-D2-P5; 2003-D2-P2; 2004-D2-P4; 2006-D1-P6; 2007-D1-P5; 2007-D1-P6; 2008-D2-P1; 2010-D2-P5; 2013-D1-P5; 2013-D2-P1; 2015-D1-P1; 2015-D2-P9; 2016-D2-P10; 2017-D1-P1; 2017-D1-P5; 2018-D2-P9; 2021-D2-P7; 2023-D1-P2; 2024-D2-P7

25.20 TR-RA-01: continuous on closed bounded set, compact, attains maximum/minimum

Domain	Real analysis
Diagnosis	Use compactness and the extreme value theorem.
First move	Identify the compact set and continuous function whose extremum is needed.
Method links	M-RA-02; M-UNI-02
Trap	The set must actually be closed and bounded or compact in the relevant space.
Occurrences	26 problems

Occurrence references.

1994-D2-P2; 1995-D1-P6; 1995-D2-P4; 1995-D2-P6; 1996-D1-P6; 1996-D2-P4; 1998-D2-P2; 1998-D2-P5; 2000-D1-P1; 2000-D2-P2; 2001-D2-P3; 2001-D2-P4; 2001-D2-P6; 2002-D1-P4; 2002-D2-P6; 2003-D2-P3; 2004-D1-P3; 2005-D2-P1; 2005-D2-P5; 2006-D2-P2; 2007-D2-P3; 2007-D2-P6; 2008-D1-P1; 2011-D1-P1; 2016-D2-P7; 2023-D2-P8

25.21 TR-PC-05: generating function, coefficient extraction, recurrence

Domain	Generating functions
Diagnosis	Encode recurrence/counting into a formal power series.
First move	Define F(x)=sum a_n x^n and translate the recurrence into an equation for F.
Method links	M-CMB-05; M-RA-07
Trap	Index shifts and initial terms are the usual source of errors.
Occurrences	25 problems

Occurrence references.

1997-D2-P3; 1998-D2-P6; 1999-D2-P2; 2000-D1-P6; 2000-D2-P1; 2001-D2-P2; 2002-D2-P3; 2003-D1-P1; 2003-D1-P2; 2003-D1-P6; 2003-D2-P6; 2005-D1-P2; 2005-D2-P6; 2007-D1-P4; 2008-D2-P3; 2010-D2-P2; 2011-D2-P1; 2012-D2-P2; 2015-D1-P5; 2017-D2-P8; 2017-D2-P9; 2018-D2-P7; 2019-D1-P4; 2021-D1-P3; 2024-D2-P8

METHOD-DATA NOTICE — trigger/template fields are not validated frequencies; click CLASS-002.

25.22 TR-UNI-02: does there exist, prove/disprove, counterexample

Domain	Universal
Diagnosis	Split proof and construction; test extreme or pathological examples.
First move	Try canonical examples: affine, exponential, step-like monotone, Cantor, finite model, or shift.
Method links	M-UNI-03; M-UNI-10
Trap	A counterexample must satisfy every hypothesis exactly.
Occurrences	17 problems

Occurrence references.

1994-D1-P2; 1994-D1-P4; 1994-D2-P1; 1995-D2-P6; 1997-D1-P2; 1997-D1-P6; 1999-D1-P2; 1999-D2-P4; 1999-D2-P5; 2000-D1-P1; 2002-D1-P2; 2010-D2-P1; 2015-D1-P2; 2016-D1-P1; 2018-D1-P2; 2019-D2-P6; 2024-D1-P3

25.23 TR-UNI-03: find all functions/objects/sequences

Domain	Universal
Diagnosis	Classification requires both necessity and sufficiency.
First move	Derive necessary structure, then verify the full family.
Method links	M-UNI-06; M-FE-01
Trap	Forgetting to prove all listed examples actually work.
Occurrences	16 problems

Occurrence references.

1998-D2-P2; 2000-D1-P2; 2008-D1-P4; 2009-D1-P3; 2012-D2-P4; 2014-D1-P1; 2014-D1-P2; 2018-D1-P3; 2018-D1-P4; 2018-D2-P9; 2019-D1-P2; 2020-D2-P5; 2020-D2-P6; 2023-D1-P1; 2023-D1-P3; 2024-D1-P1

25.24 TR-LA-07: operator norm, Hilbert space, orthonormal system, inner product

Domain	Linear algebra
Diagnosis	Expand squared norms and use projection/Gram identities.
First move	Write norm differences via inner products; search for orthogonality.
Method links	M-LA-16; M-GEO-04; M-RA-10
Trap	Norm convergence and pointwise convergence are different.
Occurrences	13 problems

Occurrence references.

1995-D2-P1; 1996-D1-P6; 2002-D1-P6; 2002-D2-P6; 2006-D2-P4; 2008-D1-P5; 2008-D2-P6; 2009-D1-P5; 2010-D1-P5; 2010-D2-P4; 2013-D2-P3; 2018-D2-P6; 2021-D2-P8

25.25 TR-PC-01: polynomial divides, more roots than degree, identically zero

Domain	Polynomials
Diagnosis	Use degree versus roots and factor evaluation.
First move	Evaluate at roots of the divisor or count roots of the difference.
Method links	M-ANT-06; M-PC-01
Trap	A polynomial function over a finite field can vanish everywhere without being the zero polynomial unless degree is controlled.
Occurrences	13 problems

Occurrence references.

1995-D2-P3; 1998-D1-P5; 2000-D2-P3; 2000-D2-P4; 2001-D1-P4; 2007-D1-P1; 2008-D1-P3; 2008-D2-P1; 2008-D2-P4; 2011-D2-P4; 2014-D1-P3; 2015-D2-P10; 2018-D2-P9

METHOD-DATA NOTICE — trigger/template fields are not validated frequencies; click CLASS-002.

25.26 TR-LA-04: nilpotent, idempotent, involution, A^2=I, A^k=0

Domain	Linear algebra
Diagnosis	The minimal polynomial and eigenvalues control the structure.
First move	Apply the polynomial identity to eigenvalues and check whether roots are simple.
Method links	M-LA-05; M-LA-09; M-LA-10
Trap	Nilpotent is stronger than singular; idempotents need not be diagonal in arbitrary rings.
Occurrences	12 problems

Occurrence references.

1995-D1-P1; 1995-D1-P5; 1996-D1-P3; 2000-D1-P5; 2000-D2-P6; 2003-D1-P3; 2005-D1-P6; 2009-D2-P3; 2017-D1-P1; 2021-D1-P1; 2022-D2-P7; 2023-D1-P2

25.27 TR-UNI-07: recursively defined sequence, iterative construction

Domain	Universal/sequences
Diagnosis	Search for substitution, monotonic invariant, telescoping, or generating function.
First move	Compute first terms and try to transform the recurrence.
Method links	M-UNI-05; M-RA-07; M-CMB-05
Trap	Index shifts and base cases.
Occurrences	12 problems

Occurrence references.

1995-D2-P2; 1997-D1-P1; 2000-D1-P4; 2001-D2-P2; 2003-D1-P1; 2010-D1-P3; 2016-D2-P6; 2017-D2-P9; 2018-D1-P1; 2018-D2-P7; 2021-D1-P3; 2024-D2-P8

25.28 TR-CMB-06: at least, more than, threshold, pigeonhole

Domain	Combinatorics
Diagnosis	Use pigeonhole or double counting to force collision/structure.
First move	Count how many objects can avoid the desired conclusion.
Method links	M-CMB-01; M-CMB-02
Trap	Sharp pigeonhole arguments often require counting the complement.
Occurrences	10 problems

Occurrence references.

1994-D1-P1; 2000-D1-P1; 2001-D2-P1; 2002-D2-P2; 2008-D2-P2; 2009-D1-P3; 2013-D1-P3; 2013-D2-P5; 2016-D2-P10; 2021-D1-P3

25.29 TR-RA-03: f’ + ..., differential inequality, integrating factor

Domain	Real analysis
Diagnosis	Convert to a monotone expression by multiplying an integrating factor or differentiating an auxiliary expression.
First move	Search for h such that h’ equals the left side times a positive factor.
Method links	M-RA-01; M-RA-05
Trap	Endpoint limits must be handled with one-sided care.
Occurrences	6 problems

Occurrence references.

1997-D2-P1; 2001-D1-P6; 2003-D1-P5; 2007-D2-P6; 2009-D2-P2; 2020-D2-P5

25.30 TR-GEO-02: simplex, volume, circumcenter, barycentric

Domain	Geometry
Diagnosis	Use affine coordinates, determinants, and barycentric volume weights.
First move	Express the point as a barycentric combination and volumes as determinant weights.
Method links	M-GEO-03; M-GEO-06
Trap	Orientation signs and absolute volumes must be handled consistently.
Occurrences	5 problems

Occurrence references.

1997-D2-P5; 2000-D2-P1; 2008-D2-P2; 2009-D2-P1; 2014-D1-P5

METHOD-DATA NOTICE — trigger/template fields are not validated frequencies; click CLASS-002.

25.31 TR-GEO-04: orthogonal, projection, Euclidean/Hilbert norm

Domain	Metric/Hilbert geometry
Diagnosis	Use inner products, Gram matrices, and projections.
First move	Expand |x-y|^2 and write Gram matrix constraints.
Method links	M-GEO-04; M-LA-16
Trap	Infinite-dimensional Hilbert spaces need completeness for limits.
Occurrences	5 problems

Occurrence references.

2000-D2-P1; 2002-D2-P6; 2008-D2-P2; 2013-D2-P3; 2014-D1-P5

25.32 TR-PC-02: integer coefficients, integer values, divisibility for all integers

Domain	Polynomials/number theory
Diagnosis	Use finite differences, modular reduction, or CRT constructions.
First move	Reduce modulo a prime or evaluate at enough residue classes.
Method links	M-ANT-01; M-ANT-04; M-ANT-06
Trap	Do not divide by a nonunit modulo p.
Occurrences	5 problems

Occurrence references.

2006-D1-P4; 2007-D1-P1; 2008-D1-P3; 2008-D2-P4; 2011-D1-P3

25.33 TR-UNI-05: maximum/minimum possible, smallest/largest number

Domain	Universal
Diagnosis	Prove an upper bound and construct a matching example.
First move	Search for an invariant obstruction and a construction reaching it.
Method links	M-UNI-01; M-UNI-02; M-UNI-03
Trap	The construction must match the exact extremal value.
Occurrences	4 problems

Occurrence references.

2002-D2-P2; 2013-D1-P3; 2013-D2-P5; 2015-D2-P8

25.34 TR-CMB-02: family of finite sets, intersection, same size, covering set

Domain	Combinatorics
Diagnosis	Use induction, extremal set choice, or blocking/hitting set arguments.
First move	Fix one set and split the family by its intersections with a finite core.
Method links	M-CMB-02; M-UNI-05
Trap	Pairwise intersection does not imply common intersection.
Occurrences	3 problems

Occurrence references.

1997-D1-P6; 2000-D1-P3; 2009-D1-P2

25.35 TR-CMB-05: dice, probability, random, expected, sum divisible

Domain	Probability/counting
Diagnosis	Use generating functions, roots of unity, or symmetry classes.
First move	Write the probability generating function and apply a residue filter.
Method links	M-CMB-05; M-PC-04
Trap	Uniform distribution may have a small correction term.
Occurrences	3 problems

Occurrence references.

1995-D2-P6; 2011-D1-P4; 2022-D1-P1

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25.36 TR-UNI-01: for every x,y or for arbitrary parameters

Domain	Universal
Diagnosis	Specialize variables to expose constants, monotonicity, or algebraic constraints.
First move	Plug in neutral values, equal values, or parameter values that kill terms.
Method links	M-UNI-07; M-FE-01
Trap	Ensure the special values lie in the domain.
Occurrences	3 problems

Occurrence references.

1999-D1-P3; 1999-D2-P4; 2017-D1-P2

25.37 TR-UNI-04: closed, dense, compact, countable, uncountable

Domain	Universal/topology
Diagnosis	Use compactness, accumulation points, density, or countability decomposition.
First move	Translate the topological condition into intervals, sequences, or covers.
Method links	M-RA-19; M-UNI-10
Trap	Closed in a subset is not the same as closed in the ambient space.
Occurrences	3 problems

Occurrence references.

1998-D2-P2; 2013-D1-P3; 2016-D2-P7

25.38 TR-PC-03: all roots real/on unit circle/left half-plane

Domain	Polynomials/complex
Diagnosis	Use logarithmic derivative, Vieta, interlacing, or root-location factorization.
First move	Write P′/P as a sum over roots or factor into real linear/quadratic pieces.
Method links	M-ANT-07; M-PC-02; M-PC-03
Trap	Separate repeated roots and roots at the evaluation point.
Occurrences	2 problems

Occurrence references.

1995-D2-P3; 2003-D1-P6

25.39 TR-RA-10: dense, closed, uncountable, nowhere monotone

Domain	Real analysis/topology
Diagnosis	Use density, compactness, Baire/Sard-like measure intuition, or Cantor construction.
First move	Translate the condition into open intervals and accumulation points.
Method links	M-RA-19; M-UNI-10
Trap	Uncountable does not mean positive measure; closed countable sets can have isolated/limit points.
Occurrences	2 problems

Occurrence references.

2000-D2-P2; 2003-D2-P3

25.40 TR-LA-06: A+tB nilpotent for many t, polynomial in a parameter

Domain	Linear algebra
Diagnosis	Entries of a matrix polynomial vanish at too many parameter values.
First move	Write each matrix entry as a polynomial in t and use roots > degree.
Method links	M-ANT-06; M-LA-05
Trap	Degree is bounded entrywise, not by determinant degree unless stated.
Occurrences	1 problems

Occurrence references.

1995-D1-P5

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25.41 TR-PC-06: branches of log, Laurent expansion, cotangent series

Domain	Complex analysis
Diagnosis	Use meromorphic continuation, Laurent coefficients, or known partial fraction expansions.
First move	Identify singularities and compute enough Laurent terms or use cotangent expansion.
Method links	M-PC-06; M-PC-07
Trap	Keep branch sums and local uniform convergence distinct.
Occurrences	1 problems

Occurrence references.

2021-D2-P7

25.42 TR-RA-08: Lipschitz gradient, smooth convex function

Domain	Real analysis
Diagnosis	Use cocoercivity/descent inequalities for convex smooth functions.
First move	Subtract the affine tangent at one point and integrate along a segment.
Method links	M-RA-12; M-GEO-04
Trap	Convexity and Lipschitz gradient are both needed for the sharp inner-product inequality.
Occurrences	1 problems

Occurrence references.

2002-D2-P6

25.43 TR-RA-09: critical point, local/global extremum, gradient zero

Domain	Real analysis
Diagnosis	Use compactness for existence and derivative tests or directional signs for classification.
First move	Locate critical points and compare boundary/at-infinity behavior.
Method links	M-RA-02; M-RA-03
Trap	Gradient zero is necessary for an interior extremum, not sufficient.
Occurrences	1 problems

Occurrence references.

2000-D2-P2

25.44 TR-UNI-06: if and only if, equivalent, characterize

Domain	Universal
Diagnosis	Separate the two directions and identify the stronger direction.
First move	Prove necessity first, then sufficiency; look for a hidden theorem each way.
Method links	M-UNI-06
Trap	One direction often uses a counterexample or construction not present in the other.
Occurrences	1 problems

Occurrence references.

2018-D1-P1

25.45 TR-GEO-03: convex polygon/body, maximal ellipse, triangulation

Domain	Convex geometry
Diagnosis	Use compactness, supporting geometry, and induction on polygon vertices.
First move	Reduce to a normalized configuration and exploit convexity/extremality.
Method links	M-GEO-05; M-UNI-05
Trap	Do not assume strict convexity unless stated.
Occurrences	0 problems

Occurrence references.

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25.46 TR-GEO-05: content, diameter cover, contraction, lattice metric

Domain	Metric geometry
Diagnosis	Use covering estimates, projections, or Lipschitz contraction.
First move	Project or contract the set and compare lengths/diameters.
Method links	M-GEO-07; M-RA-10
Trap	Finite covers and infima/suprema require epsilon arguments.
Occurrences	0 problems

Occurrence references.

25.47 TR-UNI-08: iterates, periodic points, orbit, fixed point

Domain	Universal/dynamics
Diagnosis	Study fixed points, monotone orbits, cluster points, and finite periodic structure.
First move	Write x_{n+1}=f(x_n) and analyze cluster points or graph intersections.
Method links	M-FE-03; M-RA-19
Trap	A closed orbit set under a continuous map need not be finite without extra reasoning.
Occurrences	0 problems

Occurrence references.

26 Problem-by-Problem Trigger Audit

This table is the operational version of the trigger library. It records the exact trigger cards to test when opening each problem. The first trigger is the recommended starting cue; later triggers are secondary routes or proof-completion checks.

Problem	Trigger cards	Trigger phrases
1994-D1-P1	TR-LA-01; TR-LA-02; TR-LA-05; TR-RA-07; TR-CMB-01; TR-CMB-06	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; eigenvalue, characteristic polynomial, diagonalizable, similar; convex, concave, monotone, nondecreasing; permutation, inversion, cycle, transposition; at least, more than, threshold, pigeonhole
1994-D1-P2	TR-RA-02; TR-RA-04; TR-RA-07; TR-UNI-02	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; does there exist, prove/disprove, counterexample
1994-D1-P3	TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
1994-D1-P4	TR-LA-01; TR-LA-05; TR-RA-06; TR-RA-07; TR-GEO-01; TR-UNI-02	rank, dimension, span, image, kernel, invariant subspace; eigenvalue, characteristic polynomial, diagonalizable, similar; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; distance to point/line/sphere/ellipse/parabola; does there exist, prove/disprove, counterexample
1994-D1-P5	TR-RA-04; TR-RA-05; TR-RA-07; TR-ANT-04; TR-ANT-05; TR-CMB-03	lim n->infty of a sum, Riemann sum, asymptotic sum; periodic, oscillatory integral, sin(nx), cos(nx); convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; town, friends, grid, rows/columns, graph, cycle
1994-D1-P6	TR-RA-02; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	endpoint equalities, repeated zeros, differentiable, Rolle; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
1994-D2-P1	TR-RA-02; TR-RA-05; TR-RA-06; TR-RA-07; TR-UNI-02	endpoint equalities, repeated zeros, differentiable, Rolle; periodic, oscillatory integral, sin(nx), cos(nx); series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; does there exist, prove/disprove, counterexample
1994-D2-P2	TR-LA-01; TR-LA-05; TR-RA-01; TR-RA-05; TR-RA-07; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; eigenvalue, characteristic polynomial, diagonalizable, similar; continuous on closed bounded set, compact, attains maximum/minimum; periodic, oscillatory integral, sin(nx), cos(nx); convex, concave, monotone, nondecreasing; distance to point/line/sphere/ellipse/parabola
1994-D2-P3	TR-RA-02; TR-RA-05; TR-RA-06; TR-ANT-04; TR-CMB-03; TR-CMB-04	endpoint equalities, repeated zeros, differentiable, Rolle; periodic, oscillatory integral, sin(nx), cos(nx); series convergence, rearrangement, alternating, absolute convergence; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings
1994-D2-P4	TR-LA-01; TR-LA-05; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-CMB-03	rank, dimension, span, image, kernel, invariant subspace; eigenvalue, characteristic polynomial, diagonalizable, similar; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/columns, graph, cycle

Problem	Trigger cards	Trigger phrases
1994-D2-P5	TR-LA-01; TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04	rank, dimension, span, image, kernel, invariant subspace; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative
1994-D2-P6	TR-RA-02; TR-RA-04; TR-RA-07; TR-ANT-05	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; gcd, coprime, relatively prime, Bezout
1995-D1-P1	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-04; TR-LA-05; TR-CMB-03	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; nilpotent, idempotent, involution, A^2=I, A^k=0; eigenvalue, characteristic polynomial, diagonalizable, similar; town, friends, grid, rows/columns, graph, cycle
1995-D1-P2	TR-RA-02; TR-RA-04; TR-RA-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing
1995-D1-P3	TR-RA-02; TR-RA-04; TR-RA-07; TR-ANT-05	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; gcd, coprime, relatively prime, Bezout
1995-D1-P4	TR-LA-05; TR-RA-02; TR-RA-04; TR-RA-07; TR-ANT-04; TR-CMB-03	eigenvalue, characteristic polynomial, diagonalizable, similar; endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/columns, graph, cycle
1995-D1-P5	TR-LA-01; TR-LA-03; TR-LA-04; TR-LA-05; TR-LA-06; TR-RA-04	rank, dimension, span, image, kernel, invariant subspace; AB-BA, commutator, trace zero; nilpotent, idempotent, involution, A^2=I, A^k=0; eigenvalue, characteristic polynomial, diagonalizable, similar; A+tB nilpotent for many t, polynomial in a parameter; lim n->infty of a sum, Riemann sum, asymptotic sum
1995-D1-P6	TR-RA-01; TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-04	continuous on closed bounded set, compact, attains maximum/minimum; endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative
1995-D2-P1	TR-LA-01; TR-LA-02; TR-LA-05; TR-LA-07; TR-RA-07; TR-ANT-04	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; eigenvalue, characteristic polynomial, diagonalizable, similar; operator norm, Hilbert space, orthonormal system, inner product; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative
1995-D2-P2	TR-RA-02; TR-RA-04; TR-RA-07; TR-UNI-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; recursively defined sequence, iterative construction
1995-D2-P3	TR-RA-05; TR-RA-07; TR-PC-01; TR-PC-03; TR-PC-04; TR-GEO-01	periodic, oscillatory integral, sin(nx), cos(nx); convex, concave, monotone, nondecreasing; polynomial divides, more roots than degree, identically zero; all roots real/on unit circle/left half-plane; root of unity, Fourier coefficient, unit circle average; distance to point/line/sphere/ellipse/parabola
1995-D2-P4	TR-RA-01; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	continuous on closed bounded set, compact, attains maximum/minimum; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout

Problem	Trigger cards	Trigger phrases
1995-D2-P5	TR-LA-05; TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-GEO-01	eigenvalue, characteristic polynomial, diagonalizable, similar; endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; distance to point/-line/sphere/ellipse/parabola
1995-D2-P6	TR-RA-01; TR-RA-02; TR-RA-04; TR-RA-06; TR-CMB-05; TR-UNI-02	continuous on closed bounded set, compact, attains maximum/minimum; endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; dice, probability, random, expected, sum divisible; does there exist, prove/disprove, counterexample
1996-D1-P1	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; eigenvalue, characteristic polynomial, diagonalizable, similar; distance to point/line/sphere/ellipse/-parabola
1996-D1-P2	TR-RA-02; TR-RA-04; TR-RA-05; TR-RA-06; TR-RA-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; periodic, oscillatory integral, sin(nx), cos(nx); series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing
1996-D1-P3	TR-LA-01; TR-LA-04; TR-LA-05; TR-RA-07; TR-ANT-04; TR-CMB-01	rank, dimension, span, image, kernel, invariant subspace; nilpotent, idempotent, involution, A^2=I, A^k=0; eigenvalue, characteristic polynomial, diagonalizable, similar; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; permutation, inversion, cycle, transposition
1996-D1-P4	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing
1996-D1-P5	TR-LA-05; TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-04	eigenvalue, characteristic polynomial, diagonalizable, similar; endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative
1996-D1-P6	TR-LA-01; TR-LA-05; TR-LA-07; TR-RA-01; TR-RA-02; TR-RA-06	rank, dimension, span, image, kernel, invariant subspace; eigenvalue, characteristic polynomial, diagonalizable, similar; operator norm, Hilbert space, orthonormal system, inner product; continuous on closed bounded set, compact, attains maximum/minimum; endpoint equalities, repeated zeros, differentiable, Rolle; series convergence, rearrangement, alternating, absolute convergence
1996-D2-P1	TR-RA-02; TR-RA-04; TR-RA-05; TR-RA-06; TR-RA-07; TR-ANT-05	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; periodic, oscillatory integral, sin(nx), cos(nx); series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; gcd, coprime, relatively prime, Bezout
1996-D2-P2	TR-RA-07; TR-PC-04; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	convex, concave, monotone, nondecreasing; root of unity, Fourier coefficient, unit circle average; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
1996-D2-P3	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; permutation, inversion, cycle, transposition

Problem	Trigger cards	Trigger phrases
1996-D2-P4	TR-RA-01; TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-03; TR-CMB-04	continuous on closed bounded set, compact, attains maximum/minimum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings
1996-D2-P5	TR-RA-02; TR-RA-04; TR-RA-05; TR-RA-06; TR-RA-07; TR-ANT-04	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; periodic, oscillatory integral, sin(nx), cos(nx); series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative
1996-D2-P6	TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-03; TR-CMB-04	lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings
1997-D1-P1	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-05; TR-UNI-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; gcd, coprime, relatively prime, Bezout; recursively defined sequence, iterative construction
1997-D1-P2	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-UNI-02	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; does there exist, prove/disprove, counterexample
1997-D1-P3	TR-LA-01; TR-LA-02; TR-LA-03; TR-ANT-01; TR-ANT-02; TR-ANT-03	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient
1997-D1-P4	TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
1997-D1-P5	TR-LA-05; TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04	eigenvalue, characteristic polynomial, diagonalizable, similar; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative
1997-D1-P6	TR-RA-06; TR-ANT-04; TR-CMB-02; TR-CMB-03; TR-CMB-04; TR-UNI-02	series convergence, rearrangement, alternating, absolute convergence; ring, idempotent, square zero, noncommutative; family of finite sets, intersection, same size, covering set; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings; does there exist, prove/disprove, counterexample
1997-D2-P1	TR-RA-02; TR-RA-03; TR-RA-04; TR-RA-07	endpoint equalities, repeated zeros, differentiable, Rolle; f’ + ..., differential inequality, integrating factor; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing

Problem	Trigger cards	Trigger phrases
1997-D2-P2	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; eigenvalue, characteristic polynomial, diagonalizable, similar; distance to point/line/sphere/ellipse/-parabola
1997-D2-P3	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-PC-05; TR-ANT-04	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; generating function, coefficient extraction, recurrence; ring, idempotent, square zero, noncommutative
1997-D2-P4	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; eigenvalue, characteristic polynomial, diagonalizable, similar; distance to point/line/sphere/ellipse/-parabola
1997-D2-P5	TR-RA-07; TR-ANT-04; TR-CMB-01; TR-CMB-03; TR-CMB-04; TR-GEO-02	convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; permutation, inversion, cycle, transposition; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings; simplex, volume, circumcenter, barycentric
1997-D2-P6	TR-RA-02; TR-RA-04; TR-ANT-04; TR-CMB-01; TR-CMB-03; TR-CMB-04	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; ring, idempotent, square zero, noncommutative; permutation, inversion, cycle, transposition; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings
1998-D1-P1	TR-LA-01; TR-LA-05; TR-ANT-04; TR-CMB-03; TR-CMB-04; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; eigenvalue, characteristic polynomial, diagonalizable, similar; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings; distance to point/line/sphere/ellipse/-parabola
1998-D1-P2	TR-RA-06; TR-RA-07; TR-ANT-03; TR-ANT-04; TR-CMB-01; TR-CMB-03	series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; permutation, inversion, cycle, transposition; town, friends, grid, rows/columns, graph, cycle
1998-D1-P3	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing
1998-D1-P4	TR-RA-02; TR-RA-04; TR-RA-07; TR-ANT-04; TR-CMB-03; TR-CMB-04	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings
1998-D1-P5	TR-RA-04; TR-RA-07; TR-PC-01	lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; polynomial divides, more roots than degree, identically zero
1998-D1-P6	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing
1998-D2-P1	TR-LA-01; TR-LA-05; TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-03	rank, dimension, span, image, kernel, invariant subspace; eigenvalue, characteristic polynomial, diagonalizable, similar; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/columns, graph, cycle

Problem	Trigger cards	Trigger phrases
1998-D2-P2	TR-RA-01; TR-RA-04; TR-RA-07; TR-UNI-03; TR-UNI-04	continuous on closed bounded set, compact, attains maximum/minimum; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; find all functions/objects/sequences; closed, dense, compact, countable, uncountable
1998-D2-P3	TR-RA-05; TR-RA-07; TR-ANT-04; TR-CMB-01; TR-CMB-03; TR-CMB-04	periodic, oscillatory integral, sin(nx), cos(nx); convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; permutation, inversion, cycle, transposition; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings
1998-D2-P4	TR-LA-05; TR-RA-05; TR-RA-07; TR-ANT-04; TR-CMB-01; TR-CMB-03	eigenvalue, characteristic polynomial, diagonalizable, similar; periodic, oscillatory integral, sin(nx), cos(nx); convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; permutation, inversion, cycle, transposition; town, friends, grid, rows/columns, graph, cycle
1998-D2-P5	TR-LA-05; TR-RA-01; TR-RA-07; TR-ANT-04; TR-CMB-01; TR-CMB-03	eigenvalue, characteristic polynomial, diagonalizable, similar; continuous on closed bounded set, compact, attains maximum/minimum; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; permutation, inversion, cycle, transposition; town, friends, grid, rows/-columns, graph, cycle
1998-D2-P6	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-PC-05; TR-ANT-04	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; generating function, coefficient extraction, recurrence; ring, idempotent, square zero, noncommutative
1999-D1-P1	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-PC-04; TR-ANT-03	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; eigenvalue, characteristic polynomial, diagonalizable, similar; root of unity, Fourier coefficient, unit circle average; group generated by, subgroup, normal, automorphism, quotient
1999-D1-P2	TR-RA-02; TR-RA-04; TR-RA-07; TR-UNI-02	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; does there exist, prove/disprove, counterexample
1999-D1-P3	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-UNI-01	convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; for every x,y or for arbitrary parameters
1999-D1-P4	TR-RA-05; TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-03; TR-CMB-04	periodic, oscillatory integral, sin(nx), cos(nx); series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/-columns, graph, cycle; words, strings, functions on finite sets, colorings
1999-D1-P5	TR-LA-02; TR-RA-07; TR-ANT-03; TR-ANT-04; TR-CMB-01; TR-CMB-03	determinant, inverse, block matrix, triangular form; convex, concave, monotone, nondecreasing; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; permutation, inversion, cycle, transposition; town, friends, grid, rows/columns, graph, cycle
1999-D1-P6	TR-RA-02; TR-RA-05; TR-RA-07; TR-ANT-04; TR-CMB-03; TR-CMB-04	endpoint equalities, repeated zeros, differentiable, Rolle; periodic, oscillatory integral, sin(nx), cos(nx); convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings

Problem	Trigger cards	Trigger phrases
1999-D2-P1	TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
1999-D2-P2	TR-LA-02; TR-RA-04; TR-RA-05; TR-RA-06; TR-PC-04; TR-PC-05	determinant, inverse, block matrix, triangular form; lim n->infty of a sum, Riemann sum, asymptotic sum; periodic, oscillatory integral, sin(nx), cos(nx); series convergence, rearrangement, alternating, absolute convergence; root of unity, Fourier coefficient, unit circle average; generating function, coefficient extraction, recurrence
1999-D2-P3	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing
1999-D2-P4	TR-RA-05; TR-RA-06; TR-RA-07; TR-UNI-01; TR-UNI-02	periodic, oscillatory integral, sin(nx), cos(nx); series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; for every x,y or for arbitrary parameters; does there exist, prove/disprove, counterexample
1999-D2-P5	TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-03; TR-CMB-04; TR-UNI-02	series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings; does there exist, prove/disprove, counterexample
1999-D2-P6	TR-RA-04; TR-RA-05; TR-RA-07; TR-PC-04; TR-ANT-04; TR-CMB-03	lim n->infty of a sum, Riemann sum, asymptotic sum; periodic, oscillatory integral, sin(nx), cos(nx); convex, concave, monotone, nondecreasing; root of unity, Fourier coefficient, unit circle average; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/columns, graph, cycle
2000-D1-P1	TR-RA-01; TR-RA-02; TR-RA-04; TR-RA-07; TR-CMB-06; TR-UNI-02	continuous on closed bounded set, compact, attains maximum/minimum; endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; at least, more than, threshold, pigeonhole; does there exist, prove/disprove, counterexample
2000-D1-P2	TR-RA-04; TR-PC-04; TR-UNI-03	lim n->infty of a sum, Riemann sum, asymptotic sum; root of unity, Fourier coefficient, unit circle average; find all functions/objects/sequences
2000-D1-P3	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-PC-04; TR-CMB-02	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; eigenvalue, characteristic polynomial, diagonalizable, similar; root of unity, Fourier coefficient, unit circle average; family of finite sets, intersection, same size, covering set
2000-D1-P4	TR-RA-02; TR-RA-04; TR-RA-07; TR-UNI-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; recursively defined sequence, iterative construction
2000-D1-P5	TR-LA-03; TR-LA-04; TR-LA-05; TR-ANT-01; TR-ANT-02; TR-ANT-03	AB-BA, commutator, trace zero; nilpotent, idempotent, involution, A^2=I, A^k=0; eigenvalue, characteristic polynomial, diagonalizable, similar; prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient
2000-D1-P6	TR-RA-04; TR-RA-05; TR-RA-06; TR-RA-07; TR-PC-04; TR-PC-05	lim n->infty of a sum, Riemann sum, asymptotic sum; periodic, oscillatory integral, sin(nx), cos(nx); series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; root of unity, Fourier coefficient, unit circle average; generating function, coefficient extraction, recurrence

Problem	Trigger cards	Trigger phrases
2000-D2-P1	TR-LA-01; TR-RA-07; TR-PC-05; TR-ANT-02; TR-GEO-02; TR-GEO-04	rank, dimension, span, image, kernel, invariant subspace; convex, concave, monotone, nondecreasing; generating function, coefficient extraction, recurrence; field characteristic, F_p, polynomial function modulo x^p-x; simplex, volume, circumcenter, barycentric; orthogonal, projection, Euclidean/Hilbert norm
2000-D2-P2	TR-RA-01; TR-RA-02; TR-RA-04; TR-RA-07; TR-RA-09; TR-RA-10	continuous on closed bounded set, compact, attains maximum/minimum; endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; critical point, local/global extremum, gradient zero; dense, closed, uncountable, nowhere monotone
2000-D2-P3	TR-RA-07; TR-PC-01; TR-PC-04; TR-ANT-02; TR-ANT-04; TR-CMB-03	convex, concave, monotone, nondecreasing; polynomial divides, more roots than degree, identically zero; root of unity, Fourier coefficient, unit circle average; field characteristic, F_p, polynomial function modulo x^p-x; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/columns, graph, cycle
2000-D2-P4	TR-RA-04; TR-RA-06; TR-RA-07; TR-PC-01; TR-ANT-04; TR-CMB-01	lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; polynomial divides, more roots than degree, identically zero; ring, idempotent, square zero, noncommutative; permutation, inversion, cycle, transposition
2000-D2-P5	TR-RA-02; TR-RA-04; TR-RA-05; TR-RA-06; TR-RA-07; TR-ANT-04	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; periodic, oscillatory integral, sin(nx), cos(nx); series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative
2000-D2-P6	TR-LA-01; TR-LA-03; TR-LA-04; TR-LA-05; TR-RA-04; TR-RA-06	rank, dimension, span, image, kernel, invariant subspace; AB-BA, commutator, trace zero; nilpotent, idempotent, involution, A^2=I, A^k=0; eigenvalue, characteristic polynomial, diagonalizable, similar; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence
2001-D1-P1	TR-LA-02; TR-RA-04; TR-RA-06; TR-ANT-01; TR-ANT-03; TR-ANT-04	determinant, inverse, block matrix, triangular form; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative
2001-D1-P2	TR-LA-05; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05	eigenvalue, characteristic polynomial, diagonalizable, similar; prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2001-D1-P3	TR-RA-02; TR-RA-04; TR-ANT-05	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; gcd, coprime, relatively prime, Bezout
2001-D1-P4	TR-RA-07; TR-PC-01; TR-PC-04; TR-ANT-01; TR-ANT-02	convex, concave, monotone, nondecreasing; polynomial divides, more roots than degree, identically zero; root of unity, Fourier coefficient, unit circle average; prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x

Problem	Trigger cards	Trigger phrases
2001-D1-P5	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-RA-06; TR-RA-07	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; eigenvalue, characteristic polynomial, diagonalizable, similar; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing
2001-D1-P6	TR-LA-05; TR-RA-02; TR-RA-03; TR-RA-04; TR-RA-05; TR-RA-07	eigenvalue, characteristic polynomial, diagonalizable, similar; endpoint equalities, repeated zeros, differentiable, Rolle; f' + ..., differential inequality, integrating factor; lim n->infty of a sum, Riemann sum, asymptotic sum; periodic, oscillatory integral, sin(nx), cos(nx); convex, concave, monotone, nondecreasing
2001-D2-P1	TR-RA-04; TR-RA-07; TR-ANT-01; TR-CMB-06	lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; at least, more than, threshold, pigeonhole
2001-D2-P2	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-PC-05; TR-UNI-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; generating function, coefficient extraction, recurrence; recursively defined sequence, iterative construction
2001-D2-P3	TR-LA-01; TR-RA-01; TR-RA-02; TR-RA-05; TR-RA-07; TR-PC-04	rank, dimension, span, image, kernel, invariant subspace; continuous on closed bounded set, compact, attains maximum/minimum; endpoint equalities, repeated zeros, differentiable, Rolle; periodic, oscillatory integral, sin(nx), cos(nx); convex, concave, monotone, nondecreasing; root of unity, Fourier coefficient, unit circle average
2001-D2-P4	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-RA-01; TR-RA-06	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; eigenvalue, characteristic polynomial, diagonalizable, similar; continuous on closed bounded set, compact, attains maximum/minimum; series convergence, rearrangement, alternating, absolute convergence
2001-D2-P5	TR-RA-04; TR-RA-05; TR-RA-07; TR-ANT-04; TR-CMB-03; TR-CMB-04	lim n->infty of a sum, Riemann sum, asymptotic sum; periodic, oscillatory integral, sin(nx), cos(nx); convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings
2001-D2-P6	TR-RA-01; TR-RA-02; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04	continuous on closed bounded set, compact, attains maximum/minimum; endpoint equalities, repeated zeros, differentiable, Rolle; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative
2002-D1-P1	TR-LA-05; TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-01; TR-CMB-03	eigenvalue, characteristic polynomial, diagonalizable, similar; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; permutation, inversion, cycle, transposition; town, friends, grid, rows/columns, graph, cycle
2002-D1-P2	TR-RA-02; TR-RA-05; TR-RA-06; TR-RA-07; TR-UNI-02	endpoint equalities, repeated zeros, differentiable, Rolle; periodic, oscillatory integral, sin(nx), cos(nx); series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; does there exist, prove/disprove, counterexample

Problem	Trigger cards	Trigger phrases
2002-D1-P3	TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2002-D1-P4	TR-RA-01; TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-04	continuous on closed bounded set, compact, attains maximum/minimum; endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative
2002-D1-P5	TR-LA-01; TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-04	rank, dimension, span, image, kernel, invariant subspace; endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative
2002-D1-P6	TR-LA-01; TR-LA-07; TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07	rank, dimension, span, image, kernel, invariant subspace; operator norm, Hilbert space, orthonormal system, inner product; endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing
2002-D2-P1	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; eigenvalue, characteristic polynomial, diagonalizable, similar; distance to point/line/sphere/ellipse/-parabola
2002-D2-P2	TR-ANT-04; TR-CMB-03; TR-CMB-04; TR-CMB-06; TR-UNI-05	ring, idempotent, square zero, noncommutative; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings; at least, more than, threshold, pigeonhole; maximum/minimum possible, smallest/largest number
2002-D2-P3	TR-LA-05; TR-RA-06; TR-RA-07; TR-PC-05; TR-ANT-01; TR-ANT-03	eigenvalue, characteristic polynomial, diagonalizable, similar; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; generating function, coefficient extraction, recurrence; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient
2002-D2-P4	TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-01; TR-CMB-03; TR-CMB-04	series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; permutation, inversion, cycle, transposition; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings
2002-D2-P5	TR-LA-01; TR-LA-05; TR-RA-05; TR-RA-07; TR-PC-04; TR-ANT-04	rank, dimension, span, image, kernel, invariant subspace; eigenvalue, characteristic polynomial, diagonalizable, similar; periodic, oscillatory integral, sin(nx), cos(nx); convex, concave, monotone, nondecreasing; root of unity, Fourier coefficient, unit circle average; ring, idempotent, square zero, noncommutative
2002-D2-P6	TR-LA-07; TR-RA-01; TR-RA-05; TR-RA-07; TR-RA-08; TR-GEO-04	operator norm, Hilbert space, orthonormal system, inner product; continuous on closed bounded set, compact, attains maximum/minimum; periodic, oscillatory integral, sin(nx), cos(nx); convex, concave, monotone, nondecreasing; Lipschitz gradient, smooth convex function; orthogonal, projection, Euclidean/Hilbert norm
2003-D1-P1	TR-RA-02; TR-RA-04; TR-RA-06; TR-PC-05; TR-UNI-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; generating function, coefficient extraction, recurrence; recursively defined sequence, iterative construction

Problem	Trigger cards	Trigger phrases
2003-D1-P2	TR-RA-04; TR-RA-06; TR-PC-05; TR-ANT-01; TR-ANT-03; TR-ANT-04	lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; generating function, coefficient extraction, recurrence; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative
2003-D1-P3	TR-LA-01; TR-LA-03; TR-LA-04; TR-LA-05; TR-RA-04; TR-RA-06	rank, dimension, span, image, kernel, invariant subspace; AB-BA, commutator, trace zero; nilpotent, idempotent, involution, A^2=I, A^k=0; eigenvalue, characteristic polynomial, diagonalizable, similar; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence
2003-D1-P4	TR-LA-01; TR-LA-05; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04	rank, dimension, span, image, kernel, invariant subspace; eigenvalue, characteristic polynomial, diagonalizable, similar; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative
2003-D1-P5	TR-LA-01; TR-LA-05; TR-RA-02; TR-RA-03; TR-RA-04; TR-RA-06	rank, dimension, span, image, kernel, invariant subspace; eigenvalue, characteristic polynomial, diagonalizable, similar; endpoint equalities, repeated zeros, differentiable, Rolle; f' + ..., differential inequality, integrating factor; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence
2003-D1-P6	TR-LA-05; TR-RA-06; TR-RA-07; TR-PC-03; TR-PC-05; TR-GEO-01	eigenvalue, characteristic polynomial, diagonalizable, similar; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; all roots real/on unit circle/left half-plane; generating function, coefficient extraction, recurrence; distance to point/line/sphere/ellipse/parabola
2003-D2-P1	TR-LA-01; TR-LA-02; TR-LA-05; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; eigenvalue, characteristic polynomial, diagonalizable, similar; distance to point/line/sphere/ellipse/parabola
2003-D2-P2	TR-RA-04; TR-RA-07; TR-PC-04; TR-ANT-05	lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; root of unity, Fourier coefficient, unit circle average; gcd, coprime, relatively prime, Bezout
2003-D2-P3	TR-RA-01; TR-RA-07; TR-RA-10; TR-ANT-04; TR-CMB-03; TR-CMB-04	continuous on closed bounded set, compact, attains maximum/minimum; convex, concave, monotone, nondecreasing; dense, closed, uncountable, nowhere monotone; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/-columns, graph, cycle; words, strings, functions on finite sets, colorings
2003-D2-P4	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-03	convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; town, friends, grid, rows/columns, graph, cycle
2003-D2-P5	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-03	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/-columns, graph, cycle
2003-D2-P6	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-PC-05; TR-ANT-05	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; generating function, coefficient extraction, recurrence; gcd, coprime, relatively prime, Bezout

Problem	Trigger cards	Trigger phrases
2004-D1-P1	TR-LA-05; TR-RA-07; TR-ANT-04; TR-CMB-01; TR-CMB-03; TR-CMB-04	eigenvalue, characteristic polynomial, diagonalizable, similar; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; permutation, inversion, cycle, transposition; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings
2004-D1-P2	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-01	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; permutation, inversion, cycle, transposition
2004-D1-P3	TR-LA-01; TR-RA-01; TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07	rank, dimension, span, image, kernel, invariant subspace; continuous on closed bounded set, compact, attains maximum/minimum; endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing
2004-D1-P4	TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-03; TR-CMB-04	lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings
2004-D1-P5	TR-LA-05; TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-04	eigenvalue, characteristic polynomial, diagonalizable, similar; endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative
2004-D1-P6	TR-LA-05; TR-RA-04; TR-RA-06; TR-RA-07	eigenvalue, characteristic polynomial, diagonalizable, similar; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing
2004-D2-P1	TR-LA-01; TR-LA-02; TR-LA-05; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; eigenvalue, characteristic polynomial, diagonalizable, similar; distance to point/line/sphere/ellipse/parabola
2004-D2-P2	TR-RA-02; TR-RA-05; TR-RA-07; TR-ANT-04; TR-CMB-01; TR-CMB-03	endpoint equalities, repeated zeros, differentiable, Rolle; periodic, oscillatory integral, sin(nx), cos(nx); convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; permutation, inversion, cycle, transposition; town, friends, grid, rows/columns, graph, cycle
2004-D2-P3	TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-03; TR-ANT-04; TR-GEO-01	lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; distance to point/line/-sphere/ellipse/parabola
2004-D2-P4	TR-LA-01; TR-LA-05; TR-RA-04; TR-RA-06; TR-RA-07; TR-PC-04	rank, dimension, span, image, kernel, invariant subspace; eigenvalue, characteristic polynomial, diagonalizable, similar; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; root of unity, Fourier coefficient, unit circle average
2004-D2-P5	TR-RA-02; TR-RA-04; TR-RA-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing

Problem	Trigger cards	Trigger phrases
2004-D2-P6	TR-LA-01; TR-LA-02; TR-LA-05; TR-RA-04; TR-RA-06; TR-RA-07	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; eigenvalue, characteristic polynomial, diagonalizable, similar; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing
2005-D1-P1	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-RA-06; TR-RA-07	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; eigenvalue, characteristic polynomial, diagonalizable, similar; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing
2005-D1-P2	TR-RA-07; TR-PC-05; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04	convex, concave, monotone, nondecreasing; generating function, coefficient extraction, recurrence; prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative
2005-D1-P3	TR-RA-02; TR-RA-04; TR-RA-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing
2005-D1-P4	TR-RA-06; TR-RA-07; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-CMB-01	series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; permutation, inversion, cycle, transposition
2005-D1-P5	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-05	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; gcd, coprime, relatively prime, Bezout
2005-D1-P6	TR-LA-03; TR-LA-04; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04	AB-BA, commutator, trace zero; nilpotent, idempotent, involution, A^2=I, A^k=0; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative
2005-D2-P1	TR-RA-01; TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-04	continuous on closed bounded set, compact, attains maximum/minimum; endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative
2005-D2-P2	TR-RA-06; TR-ANT-02	series convergence, rearrangement, alternating, absolute convergence; field characteristic, F_p, polynomial function modulo x^p-x
2005-D2-P3	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-RA-04; TR-RA-06	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; eigenvalue, characteristic polynomial, diagonalizable, similar; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence
2005-D2-P4	TR-RA-02; TR-RA-05; TR-ANT-04; TR-CMB-03; TR-CMB-04	endpoint equalities, repeated zeros, differentiable, Rolle; periodic, oscillatory integral, sin(nx), cos(nx); ring, idempotent, square zero, noncommutative; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings

Problem	Trigger cards	Trigger phrases
2005-D2-P5	TR-RA-01; TR-RA-02; TR-RA-05; TR-RA-06; TR-RA-07; TR-ANT-04	continuous on closed bounded set, compact, attains maximum/minimum; endpoint equalities, repeated zeros, differentiable, Rolle; periodic, oscillatory integral, sin(nx), cos(nx); series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative
2005-D2-P6	TR-LA-01; TR-LA-02; TR-LA-03; TR-RA-06; TR-PC-05; TR-ANT-01	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; series convergence, rearrangement, alternating, absolute convergence; generating function, coefficient extraction, recurrence; prime, modulo, residue, divisible by, p-adic power
2006-D1-P1	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-03	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/-columns, graph, cycle
2006-D1-P2	TR-RA-07; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05	convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2006-D1-P3	TR-LA-01; TR-LA-02; TR-LA-03; TR-RA-04; TR-RA-06; TR-RA-07	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing
2006-D1-P4	TR-RA-04; TR-RA-06; TR-PC-02; TR-ANT-01; TR-ANT-03; TR-ANT-04	lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; integer coefficients, integer values, divisibility for all integers; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative
2006-D1-P5	TR-RA-02; TR-RA-04; TR-RA-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing
2006-D1-P6	TR-LA-01; TR-RA-04; TR-RA-06; TR-RA-07; TR-PC-04; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; root of unity, Fourier coefficient, unit circle average; distance to point/line/sphere/ellipse/parabola
2006-D2-P1	TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04	series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative
2006-D2-P2	TR-LA-01; TR-LA-05; TR-RA-01; TR-RA-05; TR-RA-07; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; eigenvalue, characteristic polynomial, diagonalizable, similar; continuous on closed bounded set, compact, attains maximum/minimum; periodic, oscillatory integral, sin(nx), cos(nx); convex, concave, monotone, nondecreasing; distance to point/-line/sphere/ellipse/parabola
2006-D2-P3	TR-RA-02; TR-RA-04; TR-RA-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing

Problem	Trigger cards	Trigger phrases
2006-D2-P4	TR-LA-01; TR-LA-07; TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-03	rank, dimension, span, image, kernel, invariant subspace; operator norm, Hilbert space, orthonormal system, inner product; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient
2006-D2-P5	TR-RA-06; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05	series convergence, rearrangement, alternating, absolute convergence; prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2006-D2-P6	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-RA-06; TR-ANT-03	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; eigenvalue, characteristic polynomial, diagonalizable, similar; series convergence, rearrangement, alternating, absolute convergence; group generated by, subgroup, normal, automorphism, quotient
2007-D1-P1	TR-PC-01; TR-PC-02; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04	polynomial divides, more roots than degree, identically zero; integer coefficients, integer values, divisibility for all integers; prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative
2007-D1-P2	TR-LA-01; TR-LA-02; TR-LA-03; TR-RA-06; TR-RA-07; TR-ANT-01	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power
2007-D1-P3	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-RA-07; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; eigenvalue, characteristic polynomial, diagonalizable, similar; convex, concave, monotone, nondecreasing; distance to point/line/sphere/ellipse/parabola
2007-D1-P4	TR-PC-05; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	generating function, coefficient extraction, recurrence; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; permutation, inversion, cycle, transposition
2007-D1-P5	TR-LA-01; TR-RA-06; TR-RA-07; TR-PC-04; TR-ANT-01; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; root of unity, Fourier coefficient, unit circle average; prime, modulo, residue, divisible by, p-adic power; distance to point/line/-sphere/ellipse/parabola
2007-D1-P6	TR-RA-05; TR-RA-06; TR-RA-07; TR-PC-04; TR-ANT-01; TR-CMB-01	periodic, oscillatory integral, sin(nx), cos(nx); series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; root of unity, Fourier coefficient, unit circle average; prime, modulo, residue, divisible by, p-adic power; permutation, inversion, cycle, transposition
2007-D2-P1	TR-RA-02; TR-RA-04; TR-ANT-04; TR-CMB-01; TR-CMB-03; TR-CMB-04	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; ring, idempotent, square zero, noncommutative; permutation, inversion, cycle, transposition; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings

Problem	Trigger cards	Trigger phrases
2007-D2-P2	TR-RA-06; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05	series convergence, rearrangement, alternating, absolute convergence; prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2007-D2-P3	TR-RA-01; TR-RA-02; TR-RA-04; TR-RA-07; TR-ANT-04; TR-CMB-03	continuous on closed bounded set, compact, attains maximum/minimum; endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/columns, graph, cycle
2007-D2-P4	TR-LA-01; TR-LA-02; TR-LA-03; TR-ANT-01; TR-ANT-02; TR-ANT-03	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient
2007-D2-P5	TR-LA-01; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	rank, dimension, span, image, kernel, invariant subspace; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2007-D2-P6	TR-LA-05; TR-RA-01; TR-RA-03; TR-RA-04; TR-RA-06; TR-RA-07	eigenvalue, characteristic polynomial, diagonalizable, similar; continuous on closed bounded set, compact, attains maximum/minimum; f' + ..., differential inequality, integrating factor; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing
2008-D1-P6	TR-LA-01; TR-LA-02; TR-LA-03; TR-RA-07; TR-ANT-04; TR-CMB-01	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; permutation, inversion, cycle, transposition
2008-D1-P1	TR-RA-01; TR-RA-02; TR-RA-05; TR-RA-07; TR-ANT-04; TR-CMB-03	continuous on closed bounded set, compact, attains maximum/minimum; endpoint equalities, repeated zeros, differentiable, Rolle; periodic, oscillatory integral, sin(nx), cos(nx); convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/columns, graph, cycle
2008-D1-P2	TR-LA-01; TR-LA-05; TR-RA-06; TR-RA-07; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; eigenvalue, characteristic polynomial, diagonalizable, similar; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; distance to point/line/sphere/ellipse/parabola
2008-D1-P3	TR-LA-05; TR-RA-06; TR-PC-01; TR-PC-02; TR-ANT-01; TR-ANT-02	eigenvalue, characteristic polynomial, diagonalizable, similar; series convergence, rearrangement, alternating, absolute convergence; polynomial divides, more roots than degree, identically zero; integer coefficients, integer values, divisibility for all integers; prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x
2008-D1-P4	TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-03; TR-CMB-04; TR-UNI-03	series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings; find all functions/objects/sequences

Problem	Trigger cards	Trigger phrases
2008-D1-P5	TR-LA-01; TR-LA-07; TR-RA-02; TR-RA-04; TR-RA-07; TR-ANT-01	rank, dimension, span, image, kernel, invariant subspace; operator norm, Hilbert space, orthonormal system, inner product; endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power
2008-D2-P1	TR-RA-04; TR-PC-01; TR-PC-04; TR-ANT-01	lim n->infty of a sum, Riemann sum, asymptotic sum; polynomial divides, more roots than degree, identically zero; root of unity, Fourier coefficient, unit circle average; prime, modulo, residue, divisible by, p-adic power
2008-D2-P2	TR-RA-07; TR-CMB-06; TR-GEO-01; TR-GEO-02; TR-GEO-04	convex, concave, monotone, nondecreasing; at least, more than, threshold, pigeonhole; distance to point/line/sphere/ellipse/parabola; simplex, volume, circumcenter, barycentric; orthogonal, projection, Euclidean/Hilbert norm
2008-D2-P3	TR-RA-07; TR-PC-05; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	convex, concave, monotone, nondecreasing; generating function, coefficient extraction, recurrence; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2008-D2-P4	TR-PC-01; TR-PC-02; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04	polynomial divides, more roots than degree, identically zero; integer coefficients, integer values, divisibility for all integers; prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative
2008-D2-P5	TR-LA-01; TR-LA-02; TR-LA-03; TR-RA-07; TR-ANT-01; TR-ANT-02	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x
2008-D2-P6	TR-LA-01; TR-LA-05; TR-LA-07; TR-RA-06; TR-RA-07; TR-CMB-01	rank, dimension, span, image, kernel, invariant subspace; eigenvalue, characteristic polynomial, diagonalizable, similar; operator norm, Hilbert space, orthonormal system, inner product; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; permutation, inversion, cycle, transposition
2009-D1-P1	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-03; TR-ANT-04	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative
2009-D1-P2	TR-LA-01; TR-LA-05; TR-CMB-02; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; eigenvalue, characteristic polynomial, diagonalizable, similar; family of finite sets, intersection, same size, covering set; distance to point/line/sphere/ellipse/parabola
2009-D1-P3	TR-LA-01; TR-RA-07; TR-CMB-01; TR-CMB-03; TR-CMB-06; TR-UNI-03	rank, dimension, span, image, kernel, invariant subspace; convex, concave, monotone, nondecreasing; permutation, inversion, cycle, transposition; town, friends, grid, rows/columns, graph, cycle; at least, more than, threshold, pigeonhole; find all functions/objects/sequences
2009-D1-P4	TR-LA-01; TR-LA-05; TR-RA-04; TR-RA-06; TR-RA-07; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; eigenvalue, characteristic polynomial, diagonalizable, similar; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; distance to point/line/sphere/ellipse/parabola

Problem	Trigger cards	Trigger phrases
2009-D1-P5	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-LA-07; TR-RA-04	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; eigenvalue, characteristic polynomial, diagonalizable, similar; operator norm, Hilbert space, orthonormal system, inner product; lim n->infty of a sum, Riemann sum, asymptotic sum
2009-D2-P1	TR-RA-07; TR-ANT-04; TR-CMB-03; TR-CMB-04; TR-GEO-01; TR-GEO-02	convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings; distance to point/line/sphere/ellipse/parabola; simplex, volume, circumcenter, barycentric
2009-D2-P2	TR-RA-02; TR-RA-03; TR-RA-05; TR-RA-07	endpoint equalities, repeated zeros, differentiable, Rolle; f' + ..., differential inequality, integrating factor; periodic, oscillatory integral, sin(nx), cos(nx); convex, concave, monotone, nondecreasing
2009-D2-P3	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-04; TR-LA-05; TR-RA-06	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; nilpotent, idempotent, involution, A^2=I, A^k=0; eigenvalue, characteristic polynomial, diagonalizable, similar; series convergence, rearrangement, alternating, absolute convergence
2009-D2-P4	TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-CMB-01; TR-CMB-03	prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; permutation, inversion, cycle, transposition; town, friends, grid, rows/columns, graph, cycle
2009-D2-P5	TR-LA-01; TR-LA-02; TR-LA-03; TR-RA-06; TR-RA-07; TR-ANT-01	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power
2010-D1-P1	TR-RA-02; TR-RA-05; TR-RA-07	endpoint equalities, repeated zeros, differentiable, Rolle; periodic, oscillatory integral, sin(nx), cos(nx); convex, concave, monotone, nondecreasing
2010-D1-P2	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing
2010-D1-P3	TR-RA-04; TR-RA-05; TR-RA-07; TR-ANT-05; TR-UNI-07	lim n->infty of a sum, Riemann sum, asymptotic sum; periodic, oscillatory integral, sin(nx), cos(nx); convex, concave, monotone, nondecreasing; gcd, coprime, relatively prime, Bezout; recursively defined sequence, iterative construction
2010-D1-P4	TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04	series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative
2010-D1-P5	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-07; TR-RA-02; TR-RA-07	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; operator norm, Hilbert space, orthonormal system, inner product; endpoint equalities, repeated zeros, differentiable, Rolle; convex, concave, monotone, nondecreasing
2010-D2-P1	TR-LA-05; TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-UNI-02	eigenvalue, characteristic polynomial, diagonalizable, similar; endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; does there exist, prove/disprove, counterexample

Problem	Trigger cards	Trigger phrases
2010-D2-P2	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-PC-05; TR-ANT-03	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; generating function, coefficient extraction, recurrence; group generated by, subgroup, normal, automorphism, quotient
2010-D2-P3	TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2010-D2-P4	TR-LA-01; TR-LA-02; TR-LA-07; TR-RA-07; TR-ANT-01; TR-ANT-03	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; operator norm, Hilbert space, orthonormal system, inner product; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient
2010-D2-P5	TR-LA-05; TR-RA-04; TR-PC-04; TR-ANT-01; TR-ANT-02	eigenvalue, characteristic polynomial, diagonalizable, similar; lim n->infty of a sum, Riemann sum, asymptotic sum; root of unity, Fourier coefficient, unit circle average; prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x
2011-D1-P1	TR-RA-01; TR-RA-02; TR-RA-04; TR-RA-05; TR-RA-06; TR-RA-07	continuous on closed bounded set, compact, attains maximum/minimum; endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; periodic, oscillatory integral, sin(nx), cos(nx); series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing
2011-D1-P2	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-RA-06; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; eigenvalue, characteristic polynomial, diagonalizable, similar; series convergence, rearrangement, alternating, absolute convergence; distance to point/line/sphere/ellipse/parabola
2011-D1-P3	TR-RA-07; TR-PC-02; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04	convex, concave, monotone, nondecreasing; integer coefficients, integer values, divisibility for all integers; prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative
2011-D1-P4	TR-RA-07; TR-ANT-04; TR-CMB-01; TR-CMB-03; TR-CMB-04; TR-CMB-05	convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; permutation, inversion, cycle, transposition; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings; dice, probability, random, expected, sum divisible
2011-D1-P5	TR-LA-01; TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04	rank, dimension, span, image, kernel, invariant subspace; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative
2011-D2-P1	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-PC-05; TR-ANT-04	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; generating function, coefficient extraction, recurrence; ring, idempotent, square zero, noncommutative

Problem	Trigger cards	Trigger phrases
2011-D2-P2	TR-LA-02; TR-LA-05; TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-03	determinant, inverse, block matrix, triangular form; eigenvalue, characteristic polynomial, diagonalizable, similar; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient
2011-D2-P3	TR-LA-01; TR-RA-05; TR-RA-06; TR-RA-07	rank, dimension, span, image, kernel, invariant subspace; periodic, oscillatory integral, sin(nx), cos(nx); series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing
2011-D2-P4	TR-LA-01; TR-RA-06; TR-RA-07; TR-PC-01; TR-ANT-01; TR-ANT-02	rank, dimension, span, image, kernel, invariant subspace; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; polynomial divides, more roots than degree, identically zero; prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x
2011-D2-P5	TR-LA-01; TR-RA-06; TR-RA-07; TR-ANT-03; TR-ANT-04; TR-CMB-01	rank, dimension, span, image, kernel, invariant subspace; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; permutation, inversion, cycle, transposition
2012-D1-P1	TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04	lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative
2012-D1-P2	TR-LA-01; TR-LA-02; TR-LA-03; TR-RA-02; TR-ANT-01; TR-ANT-03	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; endpoint equalities, repeated zeros, differentiable, Rolle; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient
2012-D1-P3	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; permutation, inversion, cycle, transposition
2012-D1-P4	TR-RA-02; TR-RA-04; TR-RA-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing
2012-D1-P5	TR-ANT-01; TR-ANT-03; TR-ANT-04	prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative
2012-D2-P1	TR-ANT-01; TR-ANT-02	prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x
2012-D2-P2	TR-LA-01; TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-PC-05	rank, dimension, span, image, kernel, invariant subspace; endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; generating function, coefficient extraction, recurrence
2012-D2-P3	TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-03; TR-CMB-04	prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings

Problem	Trigger cards	Trigger phrases
2012-D2-P4	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-UNI-03	convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; find all functions/objects/sequences
2012-D2-P5	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-03	convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; town, friends, grid, rows/columns, graph, cycle
2013-D1-P1	TR-LA-01; TR-LA-05; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; eigenvalue, characteristic polynomial, diagonalizable, similar; distance to point/line/sphere/ellipse/parabola
2013-D1-P2	TR-RA-02; TR-RA-04	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum
2013-D1-P3	TR-RA-07; TR-CMB-01; TR-CMB-06; TR-UNI-04; TR-UNI-05	convex, concave, monotone, nondecreasing; permutation, inversion, cycle, transposition; at least, more than, threshold, pigeonhole; closed, dense, compact, countable, uncountable; maximum/minimum possible, smallest/largest number
2013-D1-P4	TR-RA-02; TR-RA-04; TR-RA-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing
2013-D1-P5	TR-RA-06; TR-PC-04; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04	series convergence, rearrangement, alternating, absolute convergence; root of unity, Fourier coefficient, unit circle average; prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative
2013-D2-P1	TR-PC-04	root of unity, Fourier coefficient, unit circle average
2013-D2-P2	TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05	prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2013-D2-P3	TR-LA-07; TR-RA-02; TR-RA-04; TR-RA-07; TR-GEO-04	operator norm, Hilbert space, orthonormal system, inner product; endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; orthogonal, projection, Euclidean/Hilbert norm
2013-D2-P4	TR-RA-04; TR-RA-06; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2013-D2-P5	TR-RA-07; TR-CMB-01; TR-CMB-06; TR-UNI-05	convex, concave, monotone, nondecreasing; permutation, inversion, cycle, transposition; at least, more than, threshold, pigeonhole; maximum/minimum possible, smallest/largest number
2014-D1-P1	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-GEO-01; TR-UNI-03	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; eigenvalue, characteristic polynomial, diagonalizable, similar; distance to point/line/sphere/ellipse/-parabola; find all functions/objects/sequences
2014-D1-P2	TR-RA-02; TR-RA-04; TR-UNI-03	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; find all functions/objects/sequences
2014-D1-P3	TR-PC-01	polynomial divides, more roots than degree, identically zero

Problem	Trigger cards	Trigger phrases
2014-D1-P4	TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-01; TR-ANT-02; TR-ANT-03	lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient
2014-D1-P5	TR-RA-07; TR-GEO-01; TR-GEO-02; TR-GEO-04	convex, concave, monotone, nondecreasing; distance to point/line/sphere/ellipse/parabola; simplex, volume, circumcenter, barycentric; orthogonal, projection, Euclidean/Hilbert norm
2014-D2-P1	TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2014-D2-P2	TR-LA-01; TR-LA-05; TR-RA-02; TR-RA-04; TR-RA-07; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; eigenvalue, characteristic polynomial, diagonalizable, similar; endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; distance to point/line/sphere/ellipse/parabola
2014-D2-P3	TR-RA-02; TR-RA-04	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum
2014-D2-P4	TR-RA-07; TR-ANT-04; TR-CMB-03; TR-CMB-04; TR-GEO-01	convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings; distance to point/line/sphere/ellipse/parabola
2014-D2-P5	TR-RA-02; TR-RA-04; TR-RA-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing
2015-D1-P1	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-RA-07; TR-PC-04	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; eigenvalue, characteristic polynomial, diagonalizable, similar; convex, concave, monotone, nondecreasing; root of unity, Fourier coefficient, unit circle average
2015-D1-P2	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-UNI-02	convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; does there exist, prove/disprove, counterexample
2015-D1-P3	TR-LA-01; TR-RA-02; TR-RA-05; TR-RA-07	rank, dimension, span, image, kernel, invariant subspace; endpoint equalities, repeated zeros, differentiable, Rolle; periodic, oscillatory integral, sin(nx), cos(nx); convex, concave, monotone, nondecreasing
2015-D1-P4	TR-LA-01; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	rank, dimension, span, image, kernel, invariant subspace; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2015-D1-P5	TR-LA-01; TR-RA-02; TR-RA-04; TR-RA-07; TR-PC-05; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; generating function, coefficient extraction, recurrence; distance to point/line/sphere/ellipse/parabola
2015-D2-P6	TR-RA-02; TR-RA-04	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum
2015-D2-P7	TR-RA-02; TR-RA-04; TR-ANT-05	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; gcd, coprime, relatively prime, Bezout

Problem	Trigger cards	Trigger phrases
2015-D2-P8	TR-RA-04; TR-RA-06; TR-CMB-01; TR-CMB-04; TR-UNI-05	lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; permutation, inversion, cycle, transposition; words, strings, functions on finite sets, colorings; maximum/minimum possible, smallest/largest number
2015-D2-P9	TR-LA-01; TR-LA-05; TR-RA-07; TR-PC-04; TR-ANT-03; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; eigenvalue, characteristic polynomial, diagonalizable, similar; convex, concave, monotone, nondecreasing; root of unity, Fourier coefficient, unit circle average; group generated by, subgroup, normal, automorphism, quotient; distance to point/-line/sphere/ellipse/parabola
2015-D2-P10	TR-RA-07; TR-PC-01; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	convex, concave, monotone, nondecreasing; polynomial divides, more roots than degree, identically zero; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2016-D1-P1	TR-RA-02; TR-RA-05; TR-UNI-02	endpoint equalities, repeated zeros, differentiable, Rolle; periodic, oscillatory integral, sin(nx), cos(nx); does there exist, prove/disprove, counterexample
2016-D1-P2	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-GEO-01	convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; distance to point/line/sphere/ellipse/parabola
2016-D1-P3	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2016-D1-P4	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-03	convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; town, friends, grid, rows/columns, graph, cycle
2016-D1-P5	TR-RA-07; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05	convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2016-D2-P6	TR-RA-02; TR-RA-04; TR-RA-07; TR-UNI-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; recursively defined sequence, iterative construction
2016-D2-P7	TR-RA-01; TR-RA-02; TR-RA-04; TR-RA-07; TR-UNI-04	continuous on closed bounded set, compact, attains maximum/minimum; endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; closed, dense, compact, countable, uncountable
2016-D2-P8	TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05	prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2016-D2-P9	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; permutation, inversion, cycle, transposition

Problem	Trigger cards	Trigger phrases
2016-D2-P10	TR-LA-01; TR-LA-05; TR-RA-07; TR-PC-04; TR-CMB-06; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; eigenvalue, characteristic polynomial, diagonalizable, similar; convex, concave, monotone, nondecreasing; root of unity, Fourier coefficient, unit circle average; at least, more than, threshold, pigeonhole; distance to point/line/sphere/ellipse/-parabola
2017-D1-P1	TR-LA-01; TR-LA-04; TR-LA-05; TR-RA-07; TR-PC-04; TR-ANT-01	rank, dimension, span, image, kernel, invariant subspace; nilpotent, idempotent, involution, A^2=I, A^k=0; eigenvalue, characteristic polynomial, diagonalizable, similar; convex, concave, monotone, nondecreasing; root of unity, Fourier coefficient, unit circle average; prime, modulo, residue, divisible by, p-adic power
2017-D1-P2	TR-RA-02; TR-RA-04; TR-RA-07; TR-UNI-01	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; for every x,y or for arbitrary parameters
2017-D1-P3	TR-LA-01; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	rank, dimension, span, image, kernel, invariant subspace; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2017-D1-P4	TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01; TR-CMB-03	prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; permutation, inversion, cycle, transposition; town, friends, grid, rows/columns, graph, cycle
2017-D1-P5	TR-RA-07; TR-PC-04; TR-ANT-01	convex, concave, monotone, nondecreasing; root of unity, Fourier coefficient, unit circle average; prime, modulo, residue, divisible by, p-adic power
2017-D2-P6	TR-RA-02; TR-RA-04; TR-ANT-05	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; gcd, coprime, relatively prime, Bezout
2017-D2-P7	TR-RA-07; TR-ANT-01	convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power
2017-D2-P8	TR-LA-01; TR-LA-05; TR-PC-05; TR-ANT-01; TR-ANT-03; TR-ANT-04	rank, dimension, span, image, kernel, invariant subspace; eigenvalue, characteristic polynomial, diagonalizable, similar; generating function, coefficient extraction, recurrence; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative
2017-D2-P9	TR-LA-01; TR-RA-02; TR-RA-04; TR-PC-05; TR-ANT-05; TR-UNI-07	rank, dimension, span, image, kernel, invariant subspace; endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; generating function, coefficient extraction, recurrence; gcd, coprime, relatively prime, Bezout; recursively defined sequence, iterative construction
2017-D2-P10	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-GEO-01	convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; distance to point/line/sphere/ellipse/parabola
2018-D1-P1	TR-RA-02; TR-RA-04; TR-RA-06; TR-UNI-06; TR-UNI-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; if and only if, equivalent, characterize; recursively defined sequence, iterative construction

Problem	Trigger cards	Trigger phrases
2018-D1-P2	TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-UNI-02	prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; does there exist, prove/disprove, counterexample
2018-D1-P3	TR-LA-01; TR-LA-05; TR-GEO-01; TR-UNI-03	rank, dimension, span, image, kernel, invariant subspace; eigenvalue, characteristic polynomial, diagonalizable, similar; distance to point/line/sphere/ellipse/parabola; find all functions/objects/sequences
2018-D1-P4	TR-RA-02; TR-RA-04; TR-UNI-03	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; find all functions/objects/sequences
2018-D1-P5	TR-RA-07; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05	convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2018-D2-P6	TR-LA-07; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	operator norm, Hilbert space, orthonormal system, inner product; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2018-D2-P7	TR-RA-02; TR-RA-04; TR-RA-06; TR-RA-07; TR-PC-05; TR-UNI-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; generating function, coefficient extraction, recurrence; recursively defined sequence, iterative construction
2018-D2-P8	TR-LA-01; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	rank, dimension, span, image, kernel, invariant subspace; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2018-D2-P9	TR-LA-01; TR-PC-01; TR-PC-04; TR-UNI-03	rank, dimension, span, image, kernel, invariant subspace; polynomial divides, more roots than degree, identically zero; root of unity, Fourier coefficient, unit circle average; find all functions/objects/sequences
2018-D2-P10	TR-RA-02; TR-RA-04; TR-ANT-05	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; gcd, coprime, relatively prime, Bezout
2019-D1-P1	TR-RA-02; TR-RA-04	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum
2019-D1-P2	TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-GEO-01; TR-UNI-03	prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; distance to point/line/sphere/ellipse/parabola; find all functions/objects/sequences
2019-D1-P3	TR-RA-02; TR-RA-04; TR-RA-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing
2019-D1-P4	TR-RA-07; TR-PC-05; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	convex, concave, monotone, nondecreasing; generating function, coefficient extraction, recurrence; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout

Problem	Trigger cards	Trigger phrases
2019-D1-P5	TR-LA-01; TR-LA-02; TR-LA-03; TR-RA-07; TR-ANT-01; TR-ANT-03	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient
2019-D2-P6	TR-RA-02; TR-RA-04; TR-RA-05; TR-RA-06; TR-UNI-02	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; periodic, oscillatory integral, sin(nx), cos(nx); series convergence, rearrangement, alternating, absolute convergence; does there exist, prove/disprove, counterexample
2019-D2-P7	TR-LA-01; TR-RA-04; TR-RA-06; TR-ANT-01; TR-ANT-02; TR-ANT-03	rank, dimension, span, image, kernel, invariant subspace; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient
2019-D2-P8	TR-RA-04; TR-RA-06; TR-ANT-04; TR-CMB-01; TR-CMB-03; TR-CMB-04	lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; ring, idempotent, square zero, noncommutative; permutation, inversion, cycle, transposition; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings
2019-D2-P9	TR-LA-01; TR-LA-02; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2019-D2-P10	TR-RA-02; TR-RA-07; TR-ANT-04; TR-CMB-01; TR-CMB-03; TR-CMB-04	endpoint equalities, repeated zeros, differentiable, Rolle; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; permutation, inversion, cycle, transposition; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings
2020-D1-P1	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; permutation, inversion, cycle, transposition
2020-D1-P2	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-RA-04; TR-RA-06	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; eigenvalue, characteristic polynomial, diagonalizable, similar; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence
2020-D1-P3	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; permutation, inversion, cycle, transposition
2020-D1-P4	TR-RA-04; TR-RA-07	lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing
2020-D2-P5	TR-RA-02; TR-RA-03; TR-RA-04; TR-RA-07; TR-UNI-03	endpoint equalities, repeated zeros, differentiable, Rolle; f' + ..., differential inequality, integrating factor; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; find all functions/objects/sequences
2020-D2-P6	TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-UNI-03	prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; find all functions/objects/sequences

Problem	Trigger cards	Trigger phrases
2020-D2-P7	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; permutation, inversion, cycle, transposition
2020-D2-P8	TR-RA-02; TR-RA-04; TR-ANT-05	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; gcd, coprime, relatively prime, Bezout
2021-D1-P1	TR-LA-01; TR-LA-04; TR-LA-05; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; nilpotent, idempotent, involution, A^2=I, A^k=0; eigenvalue, characteristic polynomial, diagonalizable, similar; distance to point/line/sphere/ellipse/parabola
2021-D1-P2	TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-03; TR-CMB-04	prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings
2021-D1-P3	TR-RA-02; TR-RA-04; TR-PC-05; TR-CMB-06; TR-UNI-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; generating function, coefficient extraction, recurrence; at least, more than, threshold, pigeonhole; recursively defined sequence, iterative construction
2021-D1-P4	TR-RA-02; TR-RA-04	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum
2021-D2-P5	TR-LA-01; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	rank, dimension, span, image, kernel, invariant subspace; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2021-D2-P6	TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; permutation, inversion, cycle, transposition
2021-D2-P7	TR-RA-07; TR-PC-04; TR-PC-06; TR-ANT-04; TR-CMB-03; TR-CMB-04	convex, concave, monotone, nondecreasing; root of unity, Fourier coefficient, unit circle average; branches of log, Laurent expansion, cotangent series; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings
2021-D2-P8	TR-LA-07; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	operator norm, Hilbert space, orthonormal system, inner product; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2022-D1-P1	TR-RA-02; TR-RA-07; TR-CMB-05	endpoint equalities, repeated zeros, differentiable, Rolle; convex, concave, monotone, nondecreasing; dice, probability, random, expected, sum divisible
2022-D1-P2	TR-LA-01; TR-LA-05; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04	rank, dimension, span, image, kernel, invariant subspace; eigenvalue, characteristic polynomial, diagonalizable, similar; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative
2022-D1-P3	TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; permutation, inversion, cycle, transposition

Problem	Trigger cards	Trigger phrases
2022-D1-P4	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; permutation, inversion, cycle, transposition
2022-D2-P5	TR-RA-02; TR-RA-07; TR-ANT-04; TR-CMB-01; TR-CMB-03; TR-CMB-04	endpoint equalities, repeated zeros, differentiable, Rolle; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; permutation, inversion, cycle, transposition; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings
2022-D2-P6	TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; permutation, inversion, cycle, transposition
2022-D2-P7	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-04; TR-LA-05; TR-RA-07	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; nilpotent, idempotent, involution, A^2=I, A^k=0; eigenvalue, characteristic polynomial, diagonalizable, similar; convex, concave, monotone, nondecreasing
2022-D2-P8	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; permutation, inversion, cycle, transposition
2023-D1-P1	TR-RA-02; TR-RA-04; TR-UNI-03	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; find all functions/objects/sequences
2023-D1-P2	TR-LA-01; TR-LA-04; TR-LA-05; TR-PC-04; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; nilpotent, idempotent, involution, A^2=I, A^k=0; eigenvalue, characteristic polynomial, diagonalizable, similar; root of unity, Fourier coefficient, unit circle average; distance to point/line/-sphere/ellipse/parabola
2023-D1-P3	TR-UNI-03	find all functions/objects/sequences
2023-D1-P4	TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-03	prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; town, friends, grid, rows/columns, graph, cycle
2023-D1-P5	TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-CMB-01	convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; permutation, inversion, cycle, transposition
2023-D2-P6	TR-LA-01; TR-LA-02; TR-LA-05; TR-CMB-03; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; eigenvalue, characteristic polynomial, diagonalizable, similar; town, friends, grid, rows/columns, graph, cycle; distance to point/-line/sphere/ellipse/parabola
2023-D2-P7	TR-LA-01; TR-RA-02; TR-RA-04; TR-ANT-04; TR-CMB-03; TR-CMB-04	rank, dimension, span, image, kernel, invariant subspace; endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/-columns, graph, cycle; words, strings, functions on finite sets, colorings
2023-D2-P8	TR-RA-01; TR-RA-04; TR-RA-06; TR-RA-07; TR-ANT-04; TR-CMB-01	continuous on closed bounded set, compact, attains maximum/minimum; lim n->infty of a sum, Riemann sum, asymptotic sum; series convergence, rearrangement, alternating, absolute convergence; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; permutation, inversion, cycle, transposition

Problem	Trigger cards	Trigger phrases
2023-D2-P9	TR-LA-01; TR-RA-07; TR-ANT-04; TR-CMB-03; TR-CMB-04; TR-GEO-01	rank, dimension, span, image, kernel, invariant subspace; convex, concave, monotone, nondecreasing; ring, idempotent, square zero, noncommutative; town, friends, grid, rows/columns, graph, cycle; words, strings, functions on finite sets, colorings; distance to point/line/sphere/ellipse/parabola
2023-D2-P10	TR-LA-01; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	rank, dimension, span, image, kernel, invariant subspace; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2024-D1-P1	TR-LA-01; TR-UNI-03	rank, dimension, span, image, kernel, invariant subspace; find all functions/objects/sequences
2024-D1-P2	TR-RA-02; TR-RA-04; TR-RA-07; TR-ANT-05	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; gcd, coprime, relatively prime, Bezout
2024-D1-P3	TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05; TR-GEO-01; TR-UNI-02	prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout; distance to point/line/sphere/ellipse/parabola; does there exist, prove/disprove, counterexample
2024-D1-P4	TR-RA-05; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	periodic, oscillatory integral, sin(nx), cos(nx); convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2024-D1-P5	TR-RA-04; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2024-D2-P6	TR-RA-05; TR-RA-07	periodic, oscillatory integral, sin(nx), cos(nx); convex, concave, monotone, nondecreasing
2024-D2-P7	TR-LA-01; TR-LA-02; TR-LA-03; TR-LA-05; TR-PC-04; TR-ANT-01	rank, dimension, span, image, kernel, invariant subspace; determinant, inverse, block matrix, triangular form; AB-BA, commutator, trace zero; eigenvalue, characteristic polynomial, diagonalizable, similar; root of unity, Fourier coefficient, unit circle average; prime, modulo, residue, divisible by, p-adic power
2024-D2-P8	TR-RA-02; TR-RA-04; TR-RA-07; TR-PC-05; TR-UNI-07	endpoint equalities, repeated zeros, differentiable, Rolle; lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; generating function, coefficient extraction, recurrence; recursively defined sequence, iterative construction
2024-D2-P9	TR-LA-02; TR-RA-07; TR-ANT-01; TR-ANT-03; TR-ANT-04; TR-ANT-05	determinant, inverse, block matrix, triangular form; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative; gcd, coprime, relatively prime, Bezout
2024-D2-P10	TR-RA-04; TR-RA-07; TR-ANT-01; TR-ANT-02; TR-ANT-03; TR-ANT-04	lim n->infty of a sum, Riemann sum, asymptotic sum; convex, concave, monotone, nondecreasing; prime, modulo, residue, divisible by, p-adic power; field characteristic, F_p, polynomial function modulo x^p-x; group generated by, subgroup, normal, automorphism, quotient; ring, idempotent, square zero, noncommutative

27 Exam-Time Decision Flow After This Expansion

1. **Read for triggers, not for solution.** Mark all trigger cards visible in the statement: matrix/rank, roots/divisibility, compactness, convexity, graph/cycle, functional equation, etc.
2. **Choose a problem-type sheet.** Decide whether the problem is primarily determinant/rank, spectral/nilpotent, inequality, limit, integral, polynomial, finite-field, combinatorial, geometric, extremal, group/ring, or recurrence-driven.
3. **Open the method stack.** Use the problem-type sheet to pick the first two or three methods from the previous Method Sheets layer.
4. **Check theory dependencies.** Verify that every theorem used is present in the preceding theory/property section; if not, mark it as a theory-gap candidate for the next audit.
5. **Write a proof skeleton before calculations.** One line for normalization, one line for the invariant, one line for the closing theorem, and one line for equality/edge cases.

27.1 New structured files added in this package

- `imc_problem_type_trigger_catalog.csv/json`: problem-by-problem assignment of problem-type and trigger cards.
- `imc_problem_type_frequency.csv/json`: occurrence counts and references for each problem-type sheet.
- `imc_trigger_library.csv/json`: trigger cards with domains, first moves, method links, traps, and occurrence lists.

METHOD-DATA NOTICE — trigger/template fields are not validated frequencies; click CLASS-002.

Part II

Bucket 5: Proof Templates and Reproducible Proof Programs

28 Expansion Purpose

This final layer preserves all previous categorization, theory, coverage-audit, method, problem-type, and trigger-library material, and appends a proof-template layer. The aim is to turn method recognition into reproducible proof execution. A method says what move is useful; a proof template says how the complete solution should be written and checked. The organization follows the classical problem-solving emphasis on understanding the problem, devising a plan, executing it, and reviewing the result, while making the plan domain-specific for the IMC corpus [35, 33, 34, 36].

28.1 How to use this layer

1. Use the earlier proof-template layer after reading the problem-type sheet and trigger card: statement cue → problem type → method node → proof template.
2. Before reading a solution, write the template ID, the first proof move, the invariant or auxiliary object, and the expected closure theorem.
3. After solving, compare the written proof against the template checklist: hypotheses checked, transformations reversible, edge cases handled, equality/sharpness verified, and conclusion returned to the original problem.
4. The problem-by-problem audit at the end of this part gives a primary proof template and secondary templates for every detected problem entry.

29 Proof-Template Frequency Overview

ID	Template	Primary	Linked	Methods
TPL-MOD	Modular arithmetic, finite-field, CRT, and valuation proof	56	119	7
TPL-COUNT	Counting, double counting, graph encoding, and bijection proof	48	98	9
TPL-POLY	Polynomial roots, coefficients, interpolation, and divisibility proof	37	89	6
TPL-LIMIT	Limit, asymptotic, Riemann-sum, and error-splitting proof	35	73	4
TPL-INEQ	Inequality proof and norm comparison	33	146	2
TPL-DET	Determinant, rank, inverse, and matrix-factorization proof	24	100	9
TPL-FUNC	Functional equation, iterate, fixed-point, and dynamics proof	23	100	6
TPL-SPEC	Spectral, nilpotency, idempotent, and commutator proof	22	63	7
TPL-EXIST	Existence by compactness, extremal choice, or closedness	14	69	2
TPL-GEO	Geometry by coordinates, projection, volume, and metric identities	13	39	11
TPL-GROUP	Group, ring, automorphism, and noncommutative-algebra proof	12	34	4
TPL-CLASS	Classification and find-all proof	7	32	0
TPL-INTEGRAL	Integral identity, Fubini, integration by parts, and oscillation averaging proof	5	39	2
TPL-DERIV	Auxiliary function, Rolle/MVT, Taylor, and differential inequality proof	3	99	5

METHOD-DATA NOTICE — trigger/template fields are not validated frequencies; click CLASS-002.

ID	Template	Primary	Linked	Methods
TPL-COMPLEX	Complex, Fourier, roots-of-unity, and unit-circle proof	1	24	7
TPL-CONTR	Contradiction, obstruction, and minimal counterexample proof	1	14	2
TPL-APPROX	Approximation, kernels, and polynomial-density proof	0	1	2
TPL-CASE	Case split by sign, zero, boundary, and degeneracy	0	18	2
TPL-CONSTRUCT	Explicit construction and counterexample template	0	59	0
TPL-CONV	Convergence, Cauchy, series, and compactness-of-orbits proof	0	0	1
TPL-DIR	Direct structural/invariant proof	0	111	2
TPL-EQUIV	Bidirectional equivalence and reformulation proof	0	80	1
TPL-GF	Recurrence, automaton, and generating-function proof	0	26	3
TPL-IND	Induction, recursive peeling, and strengthened hypothesis	0	51	4
TPL-SHARP	Sharpness, equality case, and extremal constant template	0	166	3
TPL-TOPO	Density, countability, Baire/category, and pathological construction proof	0	10	1

30 Decision Map: Problem Type → Proof Template

This table is the exam-time bridge from Bucket 3 to Bucket 5. The links jump to the problem-type sheet and to the proof-template sheet.

Problem type	Sheet title	Primary template	Secondary templates
PT-INEQ	Inequalities and sharp constants	TPL-INEQ	TPL-SHARP; TPL-CASE; TPL-DERIV
PT-LIM	Limits, asymptotics, Riemann sums, and growth comparison	TPL-LIMIT	TPL-INTEGRAL; TPL-CONV; TPL-CASE
PT-INT	Integral estimates, averaging, oscillation, and integration by parts	TPL-INTEGRAL	TPL-LIMIT; TPL-INEQ; TPL-CONV
PT-FE	Functional equations, iterates, and dynamics	TPL-FUNC	TPL-CLASS; TPL-CONTR; TPL-CASE
PT-LA-DET	Determinants, rank, inverse identities, and matrix factorizations	TPL-DET	TPL-SPEC; TPL-DIR; TPL-CASE
PT-FINITE	Finite fields, modular arithmetic, residues, and prime-power bookkeeping	TPL-MOD	TPL-POLY; TPL-CONTR; TPL-CONSTRUCT
PT-COMB	Graph, set-system, permutation, word, and counting structures	TPL-COUNT	TPL-IND; TPL-GF; TPL-CONSTRUCT
PT-LA-SPEC	Nilpotency, spectra, commutators, idempotents, and operator powers	TPL-SPEC	TPL-DET; TPL-CONTR; TPL-CASE
PT-GRP	Group, ring, automorphism, and noncommutative algebra patterns	TPL-GROUP	TPL-MOD; TPL-CONSTRUCT; TPL-CASE
PT-GEO	Geometry by coordinates, metric identities, convexity, and projection	TPL-GEO	TPL-INEQ; TPL-DET; TPL-EXIST
PT-POLY	Polynomial roots, coefficient constraints, divisibility, and interpolation	TPL-POLY	TPL-MOD; TPL-COMPLEX; TPL-CLASS
PT-EXT	Extremal configurations, compactness, existence, and counterexamples	TPL-EXIST	TPL-CONTR; TPL-CONSTRUCT; TPL-SHARP

Problem type	Sheet title	Primary template	Secondary templates
PT-SEQ	Recurrences, generating functions, and discrete dynamical sequences	TPL-GF	TPL-LIMIT; TPL-IND; TPL-CONV

31 Complete Proof-Template Sheets

31.1 TPL-DIR: Direct structural/invariant proof

Family	Universal
Purpose	Use when the statement is solved by naming the correct invariant and pushing definitions without a long auxiliary construction.
Trigger cues	for all objects satisfying; show that a structural quantity is fixed; same sum/rank/trace/sign; direct structural reasoning; invariant comparison
Method links	M-UNI-01; M-UNI-07
Occurrence count	0 primary problem assignments; 111 linked problem assignments.

Reproducible proof program.

1. Rewrite the statement in the native language of the object: matrix map, graph, polynomial, function, group, or metric configuration.
2. Name the invariant explicitly: determinant, rank, trace, degree, sign, image/kernel, sum of coordinates, residue, length, or norm.
3. Apply only transformations that preserve this invariant, and state why preservation holds.
4. Derive the target statement as an identity, inequality, or contradiction from the invariant.
5. Check degenerate cases separately: zero object, dimension one, empty set, boundary endpoint, singular matrix, or constant polynomial.

Proof-output standard. A short proof where every line is definition -> invariant -> implication. The final sentence must explicitly state that the original object, not only the transformed object, has the desired property.

Trap watch. Leaving the invariant implicit; changing the object by a non-invertible transformation; forgetting zero/boundary cases; using a property only true in a commutative setting.

Representative linked problems.

1994-D1-P5; 1994-D2-P3; 1995-D1-P2; 1995-D1-P4; 1995-D2-P2; 1996-D1-P2; 1996-D2-P1; 1996-D2-P3; 1997-D1-P1; 1997-D2-P1; 1997-D2-P5; 1999-D1-P3; 1999-D1-P4; 1999-D1-P6; 1999-D2-P1; 1999-D2-P3; 2000-D1-P2; 2000-D1-P6; 2000-D2-P5; 2001-D1-P1; 2001-D1-P3; 2002-D2-P2; 2003-D1-P4; 2003-D2-P1; 2004-D2-P1; 2004-D2-P4; 2004-D2-P5; 2005-D1-P1; 2005-D1-P3; 2005-D1-P4; 2005-D1-P5; 2005-D2-P2; 2006-D1-P5; 2006-D2-P3; 2007-D1-P1; 2007-D1-P4; 2008-D1-P4; 2008-D2-P2; 2009-D1-P2; 2009-D2-P1; 2009-D2-P2; 2010-D1-P1; 2010-D1-P2; 2010-D1-P3; 2010-D2-P3; 2011-D2-P3; 2011-D2-P4; 2012-D1-P2; 2012-D1-P3; 2012-D1-P4; 2012-D2-P1; 2012-D2-P3; 2012-D2-P4; 2012-D2-P5; 2013-D1-P4; 2013-D2-P1; 2014-D1-P1; 2014-D1-P2; 2014-D1-P3; 2014-D1-P4; 2014-D1-P5; 2014-D2-P1; 2014-D2-P3; 2014-D2-P4; 2014-D2-P5; 2015-D1-P3; 2015-D1-P4; 2015-D2-P6; 2015-D2-P7; 2016-D1-P1; 2016-D1-P3; 2016-D1-P4; 2016-D2-P6; 2016-D2-P7; 2017-D1-P2; 2017-D1-P3; 2017-D2-P6; 2017-D2-P9; 2018-D1-P1; 2018-D1-P3; ... (+31 more)

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.2 TPL-EQUIV: Bidirectional equivalence and reformulation proof

Family	Universal
Purpose	Use when the problem is best solved by replacing the desired assertion with an equivalent equation, inequality, or geometric condition.
Trigger cues	if and only if; equivalent; characterize; can be written as; the condition is equivalent to; transform the equation
Method links	M-UNI-09
Occurrence count	0 primary problem assignments; 80 linked problem assignments.

Reproducible proof program.

1. Write the target as A iff B, or reduce the problem to a simpler target B.
2. Prove A => B without using the converse accidentally.
3. Prove B => A, explicitly reconstructing the original object or condition.
4. Mark all reversible operations: multiplying by nonzero terms, applying monotone functions, squaring, taking logarithms, passing to roots.
5. When a step is one-way only, isolate it as an implication and verify that it is used only in that direction.

Proof-output standard. A two-column or two-paragraph solution with clear labels for both directions and a final reconstruction step.

Trap watch. Squaring without checking signs; dividing by a quantity that may vanish; proving necessity but not sufficiency; losing domain restrictions under substitution.

Representative linked problems.

1994-D1-P3; 1995-D1-P2; 1995-D1-P5; 1995-D2-P2; 1996-D1-P1; 1996-D2-P1; 1996-D2-P2; 1997-D1-P1; 1998-D1-P1; 1999-D1-P5; 1999-D2-P1; 1999-D2-P5; 2000-D2-P3; 2001-D1-P1; 2001-D1-P3; 2001-D1-P5; 2002-D2-P1; 2002-D2-P2; 2002-D2-P4; 2003-D1-P4; 2003-D2-P1; 2004-D1-P4; 2004-D2-P1; 2005-D1-P1; 2006-D1-P5; 2006-D2-P3; 2007-D2-P2; 2007-D2-P4; 2008-D1-P4; 2009-D1-P2; 2010-D1-P2; 2011-D2-P3; 2012-D1-P2; 2012-D1-P3; 2012-D2-P3; 2012-D2-P4; 2013-D1-P1; 2013-D1-P4; 2013-D2-P1; 2013-D2-P2; 2014-D1-P2; 2014-D1-P3; 2014-D1-P4; 2014-D1-P5; 2014-D2-P1; 2014-D2-P3; 2015-D1-P3; 2015-D1-P4; 2015-D2-P6; 2015-D2-P7; 2016-D1-P3; 2016-D1-P4; 2016-D2-P6; 2017-D2-P6; 2017-D2-P8; 2017-D2-P9; 2018-D1-P1; 2018-D1-P3; 2018-D1-P4; 2018-D2-P6; 2018-D2-P7; 2018-D2-P10; 2019-D1-P1; 2019-D1-P2; 2019-D1-P4; 2019-D1-P5; 2019-D2-P7; 2020-D1-P1; 2020-D2-P8; 2021-D1-P1; 2022-D2-P6; 2023-D1-P2; 2023-D1-P4; 2023-D1-P5; 2023-D2-P6; 2024-D1-P1; 2024-D1-P2; 2024-D2-P6; 2024-D2-P8; 2024-D2-P10

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.3 TPL-IND: Induction, recursive peeling, and strengthened hypothesis

Family	Universal / combinatorics
Purpose	Use when a statement has a size parameter and the official move removes one element/block, adds one step, or proves a stronger two-parameter statement.
Trigger cues	for all n; recursively defined; finite family; delete a row/column; extend from n to n+1; stronger lemma
Method links	M-CMB-09; M-CMB-11; M-UNI-05; M-UNI-10
Occurrence count	0 primary problem assignments; 51 linked problem assignments.

Reproducible proof program.

1. Check the smallest nontrivial values, including degenerate n=1 or n=2 cases.
2. Decide whether ordinary induction is too weak; if so, strengthen to a two-parameter, hereditary, or invariant-preserving statement.
3. Identify the peel: remove an endpoint, row/column, vertex, factor, block, interval, or first violation.
4. Apply the induction hypothesis to the reduced object.
5. Reinsert the removed part and verify that all hypotheses and parity/count/rank conditions survive.
6. If construction is required, describe the recursive construction explicitly.

Proof-output standard. A proof with a visible induction hypothesis, a clearly smaller object, and a verification that the return step preserves the desired conclusion.

Trap watch. Inducting on the wrong parameter; using the desired result for the same size; failing to preserve side conditions; not proving the strengthened lemma actually implies the original problem.

Representative linked problems.

1994-D1-P3; 1994-D2-P3; 1995-D1-P4; 1995-D1-P5; 1996-D1-P4; 1996-D2-P2; 1997-D1-P3; 1997-D2-P2; 1997-D2-P3; 1998-D1-P3; 1998-D2-P1; 1999-D1-P5; 1999-D2-P1; 1999-D2-P5; 2000-D1-P3; 2000-D2-P3; 2001-D1-P2; 2001-D1-P5; 2001-D2-P1; 2002-D1-P3; 2002-D1-P4; 2002-D2-P2; 2002-D2-P3; 2002-D2-P4

; 2002-D2-P5; 2003-D1-P6; 2003-D2-P6; 2004-D1-P2; 2004-D1-P4; 2005-D1-P2; 2005-D1-P6; 2006-D1-P3; 2006-D2-P1; 2007-D2-P2; 2008-D1-P2; 2009-D1-P2 ; 2009-D1-P5; 2010-D1-P3; 2010-D2-P2; 2010-D2-P4; 2011-D2-P1; 2011-D2-P4; 2012-D2-P2; 2013-D1-P1; 2013-D2-P2; 2016-D1-P4; 2017-D2-P7; 2017-D2-P8; 2021-D1-P1; 2022-D2-P6; 2024-D2-P10

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.4 TPL-CONTR: Contradiction, obstruction, and minimal counterexample proof

Family	Universal
Purpose	Use when the natural way to prove a statement is to assume failure and force an impossible invariant, smaller counterexample, or forbidden configuration.
Trigger cues	prove there exists no; impossible; assume not; minimal counterexample; contradiction; obstruction; cannot happen
Method links	M-CMB-08; M-UNI-04
Occurrence count	1 primary problem assignments; 14 linked problem assignments.

Reproducible proof program.

1. Assume the negation in its strongest useful form: no object exists, all objects fail, or choose a minimal counterexample.
2. Extract an invariant that the counterexample must preserve.
3. Produce either a smaller counterexample, a forbidden numerical inequality, or a direct violation of a theorem.
4. If minimality is used, state the ordering and prove the new object is strictly smaller.
5. Close by explicitly identifying the contradiction with the assumption or minimality.

Proof-output standard. A rigorous negation-first proof with a named obstruction and no hidden existential assumptions.

Trap watch. Negating quantifiers incorrectly; assuming a minimal counterexample exists without well-ordering/compactness; deriving a weaker contradiction that does not refute the original assumption.

Representative linked problems.

1994-D2-P1; 1997-D1-P6; 1999-D1-P2; 2001-D2-P5; 2002-D1-P2; 2004-D1-P1; 2006-D1-P1; 2008-D1-P5; 2011-D1-P2; 2013-D1-P5; 2013-D2-P4; 2018-D1-P2; 2019-D2-P8; 2024-D1-P3

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.5 TPL-EXIST: Existence by compactness, extremal choice, or closedness

Family	Universal / analysis
Purpose	Use when a problem asks to prove an object exists but gives no formula; select an extremal object, a limit point, or a compact set where a continuous function attains a value.
Trigger cues	there exists; maximum/minimum attained; closed bounded; compact; dense; nearest point; choose an extremal element
Method links	M-GEO-10; M-UNI-06
Occurrence count	14 primary problem assignments; 69 linked problem assignments.

Reproducible proof program.

1. Identify the ambient compact/finite set or create one by truncation/localization.
2. Choose a point/object minimizing or maximizing a relevant quantity.
3. Use the extremal property to derive local inequalities or structural constraints.
4. Show those constraints force the required object or contradiction.
5. If the original set is not compact, first prove the desired extremum cannot escape to infinity or to a boundary.

Proof-output standard. A proof that first justifies the selection of the extremal object, then uses its extremality in a visible inequality or deformation argument.

Trap watch. Using EVT on a noncompact set; assuming a closest point exists without closedness/local compactness; failing to rule out endpoints or infinity.

Representative linked problems.

1994-D1-P4; 1994-D2-P1; 1994-D2-P5; 1995-D1-P6; 1995-D2-P1; 1995-D2-P4; 1995-D2-P6; 1996-D1-P3; 1996-D2-P4; 1996-D2-P6; 1997-D1-P6; 1997-D2-P4; 1998-D1-P4; 1998-D2-P2; 1998-D2-P6; 1999-D1-P1; 1999-D1-P2; 1999-D2-P4; 1999-D2-P6; 2000-D1-P1; 2000-D2-P2; 2000-D2-P6; 2001-D2-P4; 2001-D2-P5; 2001-D2-P6; 2002-D1-P2; 2002-D1-P5; 2002-D2-P6; 2003-D1-P1; 2003-D1-P2; 2003-D1-P5; 2003-D2-P3; 2003-D2-P5; 2004-D1-P1; 2004-D1-P5; 2004-D2-P3; 2005-D2-P1; 2005-D2-P4; 2005-D2-P6; 2006-D1-P1; 2006-D2-P5; 2007-D2-P3; 2007-D2-P6; 2008-D1-P3; 2008-D1-P5; 2009-D2-P3; 2010-D2-P3; 2011-D1-P1; 2011-D1-P2; 2011-D1-P5; 2013-D1-P2; 2013-D1-P3; 2013-D1-P5; 2013-D2-P3; 2013-D2-P4; 2016-D2-P7; 2016-D2-P8; 2017-D1-P2; 2017-D1-P3; 2017-D2-P10; 2018-D1-P2; 2019-D2-P6; 2020-D1-P3; 2020-D2-P7; 2021-D1-P3; 2021-D1-P4; 2021-D2-P5; 2023-D2-P10; 2024-D1-P3

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.6 TPL-CONSTRUCT: Explicit construction and counterexample template

Family	Universal
Purpose	Use for prove/disprove, sharpness examples, extremal configurations, and problems asking for an object with prescribed properties.
Trigger cues	give an example; construct; disprove; does there exist; equality case; find a sequence/set/matrix
Method links	
Occurrence count	0 primary problem assignments; 59 linked problem assignments.

Reproducible proof program.

1. List the required properties and mark which ones are structural and which are numerical.
2. Choose the smallest model where the structure can appear: diagonal matrix, cyclic group, grid graph, monomial polynomial, two-point function, or extremal sequence.
3. Build the object so the hardest constraint is automatic.
4. Verify every condition in the exact order stated in the problem.

5. For counterexamples, explicitly show the hypotheses hold and the conclusion fails.
6. For infinite families, specify the parameter and prove infinitely many distinct valid objects.

Proof-output standard. A construction followed by a checklist verification, not just a displayed example.

Trap watch. Giving an example that satisfies the conclusion but not all hypotheses; forgetting positivity/invertibility; proving existence for one parameter when the problem requires infinitely many.

Representative linked problems.

1994-D1-P4; 1994-D2-P1; 1994-D2-P5; 1995-D1-P6; 1995-D2-P1; 1995-D2-P6; 1996-D1-P3; 1996-D2-P6; 1997-D1-P6; 1997-D2-P4; 1998-D1-P4; 1998-D2-P6; 1999-D1-P1; 1999-D2-P4; 1999-D2-P6; 2000-D1-P1; 2000-D2-P2; 2000-D2-P6; 2001-D2-P4; 2001-D2-P5; 2002-D1-P2; 2002-D1-P5; 2003-D1-P1; 2003-D1-P2; 2003-D1-P5; 2003-D2-P3; 2003-D2-P5; 2004-D1-P5; 2004-D2-P3; 2005-D2-P4; 2005-D2-P6; 2006-D1-P1; 2006-D2-P5; 2007-D2-P3; 2007-D2-P6; 2008-D1-P3; 2008-D1-P5; 2009-D2-P3; 2010-D2-P3; 2011-D1-P1; 2011-D1-P2; 2011-D1-P5; 2013-D1-P2; 2013-D1-P5; 2013-D2-P3; 2013-D2-P4; 2016-D2-P8; 2017-D1-P2; 2017-D1-P3; 2017-D2-P10; 2018-D1-P2; 2019-D2-P6; 2020-D1-P3; 2020-D2-P7; 2021-D1-P3; 2021-D1-P4; 2021-D2-P5; 2023-D2-P10; 2024-D1-P3

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.7 TPL-CLASS: Classification and find-all proof

Family	Universal
Purpose	Use when the answer is a complete list of functions, polynomials, matrices, sequences, or algebraic objects.
Trigger cues	find all; classify; determine all; what can be; all functions satisfying; all possible values
Method links	
Occurrence count	7 primary problem assignments; 32 linked problem assignments.

Reproducible proof program.

1. Start with an arbitrary solution and derive necessary restrictions.
2. Use substitutions, special values, invariants, or canonical forms to reduce degrees of freedom.
3. Show the remaining candidates form a short list or parameter family.
4. Prove sufficiency by substituting each candidate back into the original statement.
5. Separate degenerate cases before division, logarithms, inverses, roots, or monotonicity arguments.

Proof-output standard. A necessity part plus a sufficiency check. The final answer must be boxed as a set/family, not merely described in prose.

Trap watch. Stopping after deriving candidates; missing constant/zero/singular solutions; using injectivity or monotonicity before proving it.

Representative linked problems.

1994-D2-P2; 1998-D2-P2; 1999-D1-P4; 2000-D1-P2; 2000-D2-P5; 2003-D2-P4; 2005-D1-P4; 2005-D2-P5; 2006-D1-P6; 2006-D2-P2; 2007-D1-P3; 2008-D1-P1; 2008-D1-P4; 2009-D1-P3; 2012-D2-P4; 2014-D1-P1; 2014-D1-P2; 2014-D2-P2; 2017-D1-P1; 2018-D1-P3; 2018-D1-P4; 2018-D2-P9; 2019-D1-P2; 2019-D2-P9; 2020-D2-P5; 2020-D2-P6; 2022-D1-P2; 2023-D1-P1; 2023-D1-P3; 2023-D2-P7; 2024-D1-P1; 2024-D2-P7

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.8 TPL-CASE: Case split by sign, zero, boundary, and degeneracy

Family	Universal
Purpose	Use when a proof is blocked by zero denominators, sign changes, endpoint cases, equality cases, or mutually exclusive structural regimes.
Trigger cues	positive/nonnegative; zero; boundary; equality; invertible/singular; real/complex roots; if one of the factors vanishes
Method links	M-UNI-02; M-UNI-08
Occurrence count	0 primary problem assignments; 18 linked problem assignments.

Reproducible proof program.

1. Identify the operation that is unsafe globally: division, sign comparison, logarithm, inverse, square root, or endpoint differentiation.
2. Split cases exactly where the operation changes validity.
3. Solve the generic case first with the strongest method.
4. Return to degenerate cases and prove they satisfy the result directly or are impossible.
5. In equality problems, keep equality cases through every inequality.

Proof-output standard. A proof where every exceptional case is either solved before the general argument or explicitly ruled out.

Trap watch. Burying a zero case in a phrase like “clearly”; losing equality cases after applying a strict inequality; assuming positivity from nonnegativity.

Representative linked problems.

1996-D1-P1; 1997-D1-P2; 1997-D2-P6; 1998-D1-P1; 1998-D1-P2; 1999-D1-P6; 1999-D2-P5; 2002-D2-P1; 2003-D2-P4; 2007-D2-P1; 2007-D2-P4; 2007-D2-P5; 2010-D1-P4; 2011-D2-P5; 2016-D1-P1; 2020-D1-P1; 2020-D2-P6; 2022-D1-P3

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.9 TPL-INEQ: Inequality proof and norm comparison

Family	Analysis / geometry
Purpose	Use when the target is an inequality, especially with convexity, integrals, inner products, norms, or a sharp constant.
Trigger cues	prove <=, >=; maximum/minimum constant; convex/concave; norm; inner product; integral inequality; AM-GM; Jensen; Cauchy-Schwarz
Method links	M-LA-17; M-RA-10
Occurrence count	33 primary problem assignments; 146 linked problem assignments.

Reproducible proof program.

1. Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.
2. Normalize variables so the equality case becomes visible.
3. Apply the inequality in the correct direction and state its hypotheses.
4. Track equality conditions immediately after each inequality.
5. For sharp constants, prove the upper/lower bound and then construct or approximate an equality case.

Proof-output standard. A proof with named inequality source, equality tracking, and a sharpness paragraph when requested.

Trap watch. Using Jensen in the wrong direction; applying Cauchy-Schwarz where the weights are not positive; not proving optimality of a constant.

Representative linked problems.

1994-D1-P1; 1994-D1-P2; 1994-D1-P6; 1994-D2-P2; 1994-D2-P5; 1994-D2-P6; 1995-D1-P1; 1995-D1-P2; 1995-D1-P3; 1995-D1-P6; 1995-D2-P1; 1995-D2-P2; 1995-D2-P3; 1995-D2-P5; 1995-D2-P6; 1996-D1-P1; 1996-D1-P4; 1996-D1-P5; 1996-D1-P6; 1996-D2-P4; 1996-D2-P5; 1996-D2-P6; 1997-D1-P3; 1997-D1-P4; 1997-D1-P5; 1997-D2-P1; 1997-D2-P2; 1997-D2-P3; 1998-D1-P5; 1998-D1-P6; 1998-D2-P1; 1998-D2-P3; 1999-D1-P3; 1999-D1-P5; 1999-D2-P3; 1999-D2-P6; 2000-D1-P3; 2000-D1-P4; 2000-D1-P6; 2000-D2-P1; 2000-D2-P2; 2000-D2-P4; 2001-D1-P4; 2001-D2-P1; 2001-D2-P2; 2001-D2-P3; 2001-D2-P6; 2002-D1-P3; 2002-D1-P6; 2002-D2-P1; 2002-D2-P2; 2002-D2-P5; 2002-D2-P6; 2003-D2-P2; 2004-D1-P2; 2004-D1-P4; 2004-D1-P5; 2004-D2-P2; 2004-D2-P3; 2004-D2-P5; 2004-D2-P6; 2005-D1-P3; 2005-D1-P5; 2005-D2-P1; 2005-D2-P3; 2005-D2-P5; 2006-D1-P2; 2006-D1-P5; 2006-D2-P1; 2006-D2-P3; 2006-D2-P4; 2006-D2-P6; 2007-D1-P2; 2007-D1-P6; 2008-D1-P6; 2008-D2-P1; 2008-D2-P2; 2008-D2-P3; 2008-D2-P5; 2008-D2-P6; ... (+66 more)

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.10 TPL-SHARP: Sharpness, equality case, and extremal constant template

Family	Universal / inequalities
Purpose	Use when the problem asks for best possible constants, maximum/minimum values, equality cases, or near-extremal examples.
Trigger cues	best constant; maximum/minimum possible; equality; attain; sup; inf; sharp; optimal; smallest/largest
Method links	M-RA-08; M-RA-09; M-UNI-03
Occurrence count	0 primary problem assignments; 166 linked problem assignments.

Reproducible proof program.

1. Guess the extremizer from small cases, symmetry, or equality conditions of standard inequalities.
2. Prove the global bound using an inequality or extremal argument that exposes equality conditions.
3. Verify the proposed extremizer satisfies all constraints.

4. If the extremum is not attained, build a limiting sequence and compute its limiting value.
5. State whether the value is a maximum/minimum or only a supremum/infimum.

Proof-output standard. Bound + extremizer/near-extremizer + equality characterization.

Trap watch. Confusing supremum with maximum; equality in intermediate inequalities being incompatible; ignoring boundary extremizers.

Representative linked problems.

1994-D1-P1; 1994-D1-P2; 1994-D1-P3; 1994-D1-P4; 1994-D2-P2; 1994-D2-P6; 1995-D1-P2; 1995-D1-P4; 1995-D1-P6; 1995-D2-P2; 1995-D2-P4; 1995-D2-P5; 1995-D2-P6; 1996-D1-P2; 1996-D1-P4; 1996-D1-P5; 1996-D1-P6; 1996-D2-P2; 1996-D2-P4; 1996-D2-P5; 1997-D1-P2; 1997-D1-P4; 1997-D1-P5; 1997-D1-P6; 1997-D2-P1; 1997-D2-P3; 1997-D2-P6; 1998-D1-P2; 1998-D1-P4; 1998-D1-P5; 1998-D1-P6; 1998-D2-P1; 1998-D2-P2; 1998-D2-P3; 1998-D2-P5; 1998-D2-P6; 1999-D1-P2; 1999-D1-P3; 1999-D1-P6; 1999-D2-P3; 1999-D2-P4; 1999-D2-P5; 2000-D1-P1; 2000-D1-P4; 2000-D2-P1; 2000-D2-P2; 2000-D2-P3; 2000-D2-P4; 2001-D1-P3; 2001-D1-P4; 2001-D1-P6; 2001-D2-P2; 2001-D2-P3; 2001-D2-P6; 2002-D1-P3; 2002-D2-P3; 2002-D2-P4; 2002-D2-P6; 2003-D1-P2; 2003-D1-P4; 2003-D1-P6; 2003-D2-P3; 2003-D2-P6; 2004-D1-P1; 2004-D1-P2; 2004-D1-P3; 2004-D1-P5; 2004-D2-P2; 2004-D2-P4; 2004-D2-P5; 2005-D1-P1; 2005-D1-P2; 2005-D1-P3; 2005-D1-P6; 2005-D2-P1; 2005-D2-P3; 2005-D2-P5; 2006-D1-P5; 2006-D2-P1; 2006-D2-P2; ... (+86 more)

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.11 TPL-DERIV: Auxiliary function, Rolle/MVT, Taylor, and differential inequality proof

Family	Real analysis
Purpose	Use when differentiability, repeated zeros, endpoint values, or differential inequalities are central.
Trigger cues	differentiable; C^k; endpoint equalities; repeated zeros; f prime; f double prime; differential inequality; Taylor; Rolle
Method links	M-RA-01; M-RA-02; M-RA-03; M-RA-04; M-RA-18
Occurrence count	3 primary problem assignments; 99 linked problem assignments.

Reproducible proof program.

1. Construct an auxiliary function whose derivative is exactly the target expression or has a fixed sign.
2. Use endpoint data to force equal values or zeros of the auxiliary function.
3. Apply Rolle/MVT repeatedly, or Taylor with remainder, at the correct order.
4. For differential inequalities, multiply by an integrating factor or compose with a monotone transform.
5. Check differentiability order and whether endpoints are included.

Proof-output standard. A derivative-localization proof where the target point c/xi appears from Rolle, MVT, Taylor, or monotonicity.

Trap watch. Applying Rolle without equal endpoint values; differentiating a function not known to be differentiable; forgetting that Taylor remainders depend on an intermediate point.

Representative linked problems.

1994-D1-P2; 1994-D1-P5; 1994-D1-P6; 1994-D2-P1; 1994-D2-P3; 1994-D2-P4; 1994-D2-P6; 1995-D1-P2; 1995-D1-P3; 1995-D1-P4; 1995-D2-P4; 1995-D2-P5; 1995-D2-P6; 1996-D1-P2; 1996-D1-P5; 1996-D2-P1; 1996-D2-P3; 1996-D2-P5; 1997-D1-P1; 1997-D1-P2; 1997-D1-P4; 1997-D2-P1; 1997-D2-P3; 1997-D2-P5; 1997-D2-P6; 1998-D1-P3; 1998-D1-P4; 1998-D1-P6; 1998-D2-P3; 1998-D2-P4; 1999-D1-P4; 1999-D1-P6; 1999-D2-P3; 2000-D1-P2; 2000-D1-P4; 2000-D1-P6; 2000-D2-P2; 2000-D2-P5; 2001-D1-P3; 2001-D1-P6; 2002-D1-P2; 2002-D1-P4; 2002-D1-P5; 2003-D1-P5; 2003-D2-P2; 2004-D1-P5; 2004-D1-P6; 2004-D2-P2; 2004-D2-P5; 2005-D1-P3; 2005-D1-P4; 2005-D1-P5; 2005-D2-P4; 2005-D2-P5; 2006-D1-P5; 2006-D1-P6; 2007-D1-P1; 2007-D1-P4; 2007-D2-P1; 2007-D2-P3; 2007-D2-P6; 2008-D1-P1; 2008-D2-P1; 2008-D2-P2; 2009-D1-P1; 2009-D2-P2; 2010-D1-P1; 2010-D1-P5; 2010-D2-P3; 2011-D1-P4; 2011-D2-P1; 2011-D2-P4; 2012-D1-P4; 2012-D2-P2; 2012-D2-P5; 2013-D1-P2; 2014-D1-P2; 2014-D2-P3; 2015-D2-P6; 2015-D2-P7; ... (+19 more)

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.12 TPL-LIMIT: Limit, asymptotic, Riemann-sum, and error-splitting proof

Family	Real analysis
Purpose	Use for limits of sums, integrals, sequences, powers, and scaled quantities.
Trigger cues	lim n to infinity; asymptotic; Riemann sum; main term; tail; logarithm; grows; endpoint dominance
Method links	M-RA-05; M-RA-06; M-RA-07; M-RA-14
Occurrence count	35 primary problem assignments; 73 linked problem assignments.

Reproducible proof program.

1. Identify the scale and normalize the expression.
2. Split into main range and error/tail range if a uniform estimate is not immediate.
3. Replace sums by integrals only after checking mesh size and tail control.
4. Use squeeze, monotone convergence, dominated convergence, Taylor, or l-Hospital as appropriate.
5. State the final error order and why it vanishes under the normalization.

Proof-output standard. A limit proof with a visible main term, controlled remainder, and justified convergence theorem.

Trap watch. Unjustified interchange of limit and infinite sum/integral; wrong scale; ignoring endpoints where the integrand is singular.

Representative linked problems.

1994-D1-P2; 1994-D1-P5; 1994-D2-P6; 1995-D1-P3; 1995-D1-P6; 1995-D2-P2; 1995-D2-P5; 1996-D1-P2; 1996-D1-P4; 1996-D1-P5; 1996-D2-P1; 1996-D2-P5; 1997-D1-P1; 1997-D1-P2; 1997-D2-P1; 1997-D2-P3; 1997-D2-P4; 1998-D1-P3; 1998-D2-P4; 1999-D1-P2; 2000-D2-P4; 2000-D2-P5; 2000-D2-P6; 2001-D1-P3; 2001-D1-P6; 2001-D2-P2; 2001-D2-P5; 2002-D1-P4; 2002-D1-P6; 2003-D1-P1; 2003-D1-P2; 2003-D1-P3; 2003-D1-P5; 2003-D2-P2; 2003-D2-P5; 2003-D2-P6; 2004-D1-P3; 2004-D1-P6; 2004-D2-P3; 2004-D2-P4; 2004-D2-P6; 2005-D1-P5; 2005-D2-P1; 2005-D2-P3; 2006-D1-P1; 2006-D1-P3; 2006-D1-P4; 2006-D2-P6; 2007-D1-P6; 2009-D1-P1; 2010-D1-P2; 2010-D1-P3; 2010-D2-P1; 2010-D2-P5; 2011-D1-P1; 2011-D2-P1; 2012-D1-P1; 2012-D2-P2; 2013-D2-P4; 2014-D1-P2; 2015-D2-P7; 2015-D2-P8; 2016-D2-P8; 2017-D2-P6; 2017-D2-P9; 2018-D2-P7; 2018-D2-P10; 2019-D2-P7; 2019-D2-P8; 2020-D2-P8; 2021-D1-P4; 2024-D1-P2; 2024-D2-P8

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.13 TPL-INTEGRAL: Integral identity, Fubini, integration by parts, and oscillation averaging proof

Family	Real analysis
Purpose	Use when the solution is an integral identity, a weighted average, a periodic/oscillatory cancellation, or a Fubini swap.
Trigger cues	integral; double integral; periodic; sin(nx), cos(nx); average; integration by parts; kernel; weighted identity
Method links	M-RA-11; M-RA-13
Occurrence count	5 primary problem assignments; 39 linked problem assignments.

Reproducible proof program.

1. Look for a weight or kernel that converts the hypothesis into the desired integral.
2. Justify Fubini/Tonelli or finite-sum interchange before changing order.
3. For integration by parts, choose the primitive so boundary terms match known data.
4. For oscillation, split into periods or use periodic averaging/Fourier orthogonality.
5. Estimate the residual using uniform continuity, boundedness, or an integrable majorant.

Proof-output standard. An integral proof where each exchange, boundary term, and error estimate is explicit.

Trap watch. Swapping integrals without absolute convergence or continuity; losing boundary terms; treating oscillatory cancellation as pointwise convergence.

Representative linked problems.

1994-D1-P5; 1994-D2-P3; 1996-D1-P2; 1996-D1-P4; 1996-D2-P6; 1997-D1-P1; 1997-D1-P2; 1998-D1-P3; 1998-D1-P6; 1999-D1-P4; 1999-D2-P3; 1999-D2-P4; 2001-D1-P1; 2002-D1-P3; 2003-D1-P1; 2003-D1-P3; 2003-D2-P6; 2004-D1-P2; 2004-D1-P6; 2004-D2-P5; 2005-D2-P1; 2007-D2-P2; 2008-D1-P4; 2010-D1-P2; 2010-D2-P1; 2010-D2-P2; 2011-D2-P1; 2012-D1-P1; 2012-D2-P2; 2014-D1-P4; 2015-D2-P8; 2018-D1-P1; 2018-D2-P7; 2019-D2-P6; 2019-D2-P7; 2019-D2-P10; 2021-D1-P2; 2022-D1-P1; 2022-D2-P8

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.14 TPL-CONV: Convergence, Cauchy, series, and compactness-of-orbits proof

Family	Analysis
Purpose	Use for convergence of sequences/functions/operators and nonexistence of pointwise convergent subsequences.
Trigger cues	converges; Cauchy; subsequence; pointwise; uniform; bounded; compact orbit; series convergence; rearrangement
Method links	M-RA-12
Occurrence count	0 primary problem assignments; 0 linked problem assignments.

Reproducible proof program.

1. Determine the convergence mode required: pointwise, uniform, norm, almost everywhere, or scalar sequence.
2. Find monotonicity, Cauchy estimates, compactness, or an orthogonality obstruction.
3. If using a subsequence, pass to a limit and check that the limiting object remains in the right space.
4. For series, test absolute convergence, alternating cancellation, or rearrangement stability.

5. Close with a contradiction if a norm/orthogonality/identity is incompatible with the limit.

Proof-output standard. A proof that names the convergence mode and supplies the correct theorem or epsilon estimate.

Trap watch. Changing convergence mode silently; applying dominated convergence without a dominator; assuming conditional series are rearrangement-invariant.

Representative linked problems.

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.15 TPL-DET: Determinant, rank, inverse, and matrix-factorization proof

Family	Linear algebra
Purpose	Use for matrix computations and structural linear-algebra proofs driven by rank, determinant, trace pairing, inverse identities, or block form.
Trigger cues	determinant; rank; inverse; block matrix; AB/BA; trace(XY); dimension of matrix subspace; row/column elimination
Method links	M-LA-01; M-LA-02; M-LA-03; M-LA-04; M-LA-09; M-LA-10; M-LA-14; M-LA-15; M-LA-16
Occurrence count	24 primary problem assignments; 100 linked problem assignments.

Reproducible proof program.

1. Choose a basis, row/column order, or block decomposition adapted to the hypotheses.
2. Preserve the target invariant: determinant, rank, trace, kernel, image, or dimension.
3. Simplify by row/column operations, Schur-style multiplication, or completing an inverse identity.
4. For dimension bounds, prove an upper obstruction and provide a matching construction.
5. Check whether matrices are square, rectangular, invertible, symmetric, or over R/C.

Proof-output standard. A matrix proof with a declared invariant and legal transformations at each step.

Trap watch. Using det(AB)=det(A)det(B) for rectangular matrices; assuming one-sided inverses commute; confusing rank-preserving and determinant-preserving operations.

Representative linked problems.

1994-D1-P1; 1994-D1-P4; 1994-D2-P4; 1994-D2-P5; 1995-D1-P1; 1995-D1-P5; 1995-D2-P1; 1996-D1-P1; 1996-D1-P3; 1996-D1-P6; 1996-D2-P3; 1997-D1-P3; 1997-D1-P5; 1997-D2-P2; 1997-D2-P4; 1998-D1-P1; 1998-D2-P1; 1999-D1-P1; 1999-D1-P5; 1999-D2-P2; 2000-D1-P3; 2000-D1-P5; 2000-D2-P1; 2001-D1-P1; 2001-D1-P2; 2001-D1-P5; 2001-D1-P6; 2001-D2-P4; 2002-D1-P1; 2002-D1-P6; 2002-D2-P1; 2002-D2-P5; 2003-D1-P3; 2003-D2-P1; 2004-D1-P3; 2004-D2-P1; 2004-D2-P4; 2004-D2-P6; 2005-D1-P1; 2005-D2-P3; 2005-D2-P6; 2006-D1-P3; 2006-D2-P2; 2006-D2-P3; 2006-D2-P4; 2006-D2-P6; 2007-D1-P2; 2007-D1-P3; 2007-D1-P5; 2007-D2-P4; 2007-D2-P5; 2008-D1-P2; 2008-D1-P6; 2008-D2-P5; 2008-D2-P6; 2009-D1-P2; 2009-D1-P3; 2009-D1-P4; 2009-D1-P5; 2009-D2-P5; 2010-D1-P5; 2010-D2-P4; 2011-D1-P5; 2011-D2-P2; 2011-D2-P3; 2011-D2-P4; 2011-D2-P5; 2012-D1-P2; 2013-D1-P1; 2013-D1-P4; 2014-D1-P1; 2014-D2-P2; 2014-D2-P3; 2014-D2-P5; 2015-D1-P1; 2015-D1-P2; 2015-D1-P3; 2015-D1-P5; 2015-D2-P9; 2016-D1-P2; ... (+20 more)

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.16 TPL-SPEC: Spectral, nilpotency, idempotent, and commutator proof

Family	Linear algebra
Purpose	Use when eigenvalues, minimal polynomials, trace cyclicity, nilpotency, idempotents, involutions, or commutators are central.
Trigger cues	eigenvalue; characteristic polynomial; minimal polynomial; diagonalizable; similar; nilpotent; idempotent; involution; AB-BA; trace zero
Method links	M-LA-05; M-LA-06; M-LA-07; M-LA-08; M-LA-12; M-LA-13; M-LA-18
Occurrence count	22 primary problem assignments; 63 linked problem assignments.

Reproducible proof program.

1. Translate the equation into a polynomial identity in the matrix or a commutator relation.
2. Apply the identity to eigenvalues or to the minimal polynomial.
3. Use trace cyclicity or trace powers to remove commutator terms.
4. Pass from spectral information to matrix structure using Jordan form, Cayley-Hamilton, or diagonalization criteria.
5. For commuting families, search for simultaneous triangularization/diagonalization or common invariant subspaces.

Proof-output standard. A spectral proof that separates eigenvalue constraints from the finite-dimensional theorem that turns them into a matrix conclusion.

Trap watch. Trace zero alone is too weak; diagonalization requires justification; real matrices may need complex eigenvalues.

Representative linked problems.

1994-D1-P1; 1994-D1-P4; 1994-D2-P4; 1995-D1-P1; 1995-D1-P5; 1996-D1-P1; 1996-D1-P3; 1997-D1-P3; 1997-D2-P2; 1997-D2-P4; 1998-D1-P1; 1998-D2-P1; 1998-D2-P5; 1999-D1-P1; 2000-D1-P3; 2000-D1-P5; 2000-D2-P6; 2001-D1-P5; 2001-D2-P4; 2002-D1-P1; 2002-D2-P1; 2002-D2-P5; 2003-D1-P3; 2003-D1-P6; 2003-D2-P1; 2004-D2-P1; 2004-D2-P4; 2005-D1-P1; 2005-D1-P6; 2005-D2-P3; 2006-D1-P3; 2006-D2-P6; 2007-D1-P3; 2007-D2-P4; 2007-D2-P5; 2008-D1-P2; 2008-D2-P5; 2009-D1-P2; 2009-D2-P3; 2010-D2-P4; 2010-D2-P5; 2011-D1-P2; 2012-D1-P2; 2013-D1-P1; 2014-D1-P1; 2014-D2-P2; 2015-D1-P1; 2015-D2-P9; 2016-D2-P10; 2017-D1-P1; 2017-D2-P8; 2018-D1-P3; 2019-D1-P5; 2019-D2-P9; 2020-D1-P2; 2021-D1-P1; 2021-D2-P5; 2022-D1-P2; 2022-D2-P5; 2022-D2-P7; 2023-D1-P2; 2023-D2-P6; 2024-D2-P7

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.17 TPL-POLY: Polynomial roots, coefficients, interpolation, and divisibility proof

Family	Algebra / complex
Purpose	Use when roots, coefficients, polynomial identities, divisibility, or interpolation control the proof.
Trigger cues	polynomial divides; roots; all roots real/unit circle/left half-plane; integer coefficients; coefficient comparison; more roots than degree
Method links	M-ANT-04; M-ANT-05; M-ANT-06; M-ANT-07; M-ANT-16; M-CX-07
Occurrence count	37 primary problem assignments; 89 linked problem assignments.

Reproducible proof program.

1. Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.

2. Use degree versus number of roots to prove identity or impossibility.
3. For divisibility, evaluate at roots or compare irreducible factors.
4. For integer coefficients, control content, finite differences, or unique factorization.
5. For interpolation/extremal polynomial problems, write the basis or alternation conditions explicitly.

Proof-output standard. A proof that states whether it is using root information, coefficient comparison, or divisibility, then checks degree and multiplicity.

Trap watch. Forgetting multiplicities; applying root-count over a non-field; assuming rational roots imply integer linear factors without clearing denominators.

Representative linked problems.

1994-D1-P3; 1994-D2-P4; 1995-D1-P5; 1995-D2-P3; 1995-D2-P4; 1996-D2-P2; 1998-D1-P2; 1998-D1-P5; 1998-D2-P2; 1999-D1-P1; 1999-D2-P2; 1999-D2-P6; 2000-D1-P2; 2000-D1-P3; 2000-D1-P6; 2000-D2-P3; 2000-D2-P4; 2000-D2-P6; 2001-D1-P4; 2001-D1-P5; 2001-D2-P1; 2002-D1-P1; 2002-D2-P3; 2003-D1-P3; 2003-D1-P6; 2003-D2-P2; 2004-D1-P6; 2005-D1-P4; 2005-D2-P2; 2005-D2-P6; 2006-D1-P2; 2006-D1-P4; 2006-D1-P6; 2006-D2-P5; 2006-D2-P6; 2007-D1-P1; 2007-D1-P3; 2007-D1-P5; 2007-D1-P6; 2007-D2-P2; 2007-D2-P6; 2008-D1-P2; 2008-D1-P3; 2008-D1-P6; 2008-D2-P1; 2008-D2-P2; 2008-D2-P3; 2008-D2-P4; 2008-D2-P5; 2008-D2-P6; 2009-D1-P3; 2009-D1-P4; 2009-D2-P1; 2009-D2-P3; 2009-D2-P4; 2009-D2-P5; 2010-D2-P5; 2011-D1-P2; 2011-D1-P3; 2011-D1-P5; 2011-D2-P2; 2011-D2-P4; 2012-D1-P5; 2012-D2-P1; 2012-D2-P3; 2013-D1-P5; 2013-D2-P1; 2013-D2-P2; 2014-D1-P3; 2014-D1-P4; 2014-D2-P1; 2015-D1-P1; 2015-D2-P10; 2016-D2-P10; 2017-D1-P1; 2017-D1-P5; 2017-D2-P7; 2018-D1-P5; 2018-D2-P9; 2019-D1-P2; ... (+9 more)

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.18 TPL-MOD: Modular arithmetic, finite-field, CRT, and valuation proof

Family	Number theory
Purpose	Use when integer, prime, residue, divisibility, characteristic, or finite-field constraints drive the solution.
Trigger cues	prime; modulo; divisible by; residue; field characteristic; F_p; gcd; coprime; CRT; Fermat; p-th roots
Method links	M-ANT-01; M-ANT-02; M-ANT-03; M-ANT-09; M-ANT-13; M-ANT-15; M-ANT-17
Occurrence count	56 primary problem assignments; 119 linked problem assignments.

Reproducible proof program.

1. Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.
2. Reduce the statement modulo that modulus and preserve all relevant hypotheses.
3. Use finite-field facts: nonzero elements invert, degree bounds root count, Fermat/Frobenius, or cyclicity.
4. Lift the modular conclusion to divisibility, impossibility, or a constructed integer.
5. For CRT, prove the moduli are coprime before simultaneous construction.

Proof-output standard. A modular proof where every reduction and lift is stated explicitly.

Trap watch. Reducing modulo a composite ring and dividing by a zero divisor; ignoring signs/units; failing to prove residue classes are exhaustive.

Representative linked problems.

1994-D1-P3; 1994-D1-P6; 1995-D2-P4; 1996-D2-P2; 1997-D1-P3; 1997-D1-P4; 1997-D1-P5; 1999-D1-P3; 1999-D2-P1; 1999-D2-P2; 2000-D1-P5; 2000-D2-P1; 2001-D1-P1; 2001-D1-P2; 2001-D1-P4; 2001-D2-P1; 2001-D2-P6; 2002-D1-P3; 2002-D1-P6; 2002-D2-P3; 2003-D1-P2; 2003-D1-P4; 2003-D2-P4; 2005-D1-P2; 2005-D1-P4; 2005-D1-P6; 2005-D2-P2; 2005-D2-P6; 2006-D1-P2; 2006-D1-P3; 2006-D1-P4; 2006-D2-P1; 2006-D2-P4; 2006-D2-P5; 2007-D1-P1; 2007-D1-P2; 2007-D1-P4; 2007-D1-P5; 2007-D2-P2; 2007-D2-P4; 2007-D2-P5; 2008-D1-P3; 2008-D1-P5; 2008-D2-P1; 2008-D2-P3; 2008-D2-P4; 2008-D2-P5; 2009-D1-P5; 2009-D2-P3; 2009-D2-P4; 2009-D2-P5; 2010-D1-P4; 2010-D1-P5; 2010-D2-P4; 2010-D2-P5; 2011-D1-P3; 2011-D1-P5; 2011-D2-P2; 2012-D1-P1; 2012-D1-P2; 2012-D1-P3; 2012-D1-P5; 2012-D2-P1; 2012-D2-P3; 2012-D2-P4; 2012-D2-P5; 2013-D1-P5; 2013-D2-P2; 2013-D2-P4; 2014-D1-P4; 2014-D2-P1; 2015-D1-P1; 2015-D1-P2; 2015-D1-P4; 2015-D2-P10; 2016-D1-P2; 2016-D1-P3; 2016-D1-P4; 2016-D1-P5; 2016-D2-P8; ... (+39 more)

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.19 TPL-GROUP: Group, ring, automorphism, and noncommutative-algebra proof

Family	Algebra
Purpose	Use when the statement is about groups, rings, automorphisms, commutators, idempotents, quotient obstructions, or explicit algebraic examples.
Trigger cues	group; subgroup; normal; quotient; automorphism; generated by; ring; idempotent; square zero; noncommutative; center; commutator
Method links	M-ANT-10; M-ANT-11; M-ANT-12; M-ANT-14
Occurrence count	12 primary problem assignments; 34 linked problem assignments.

Reproducible proof program.

1. Translate the statement into generators, relations, subgroups, quotients, or commutators.
2. Preserve order of multiplication unless commutativity has been proved.
3. Use quotient/group-action arguments for obstruction or generator reduction.
4. For rings, expand identities in order and use additive commutators or characteristic constraints.
5. For examples, construct a permutation group, semidirect product, or matrix/ring model and verify relations.

Proof-output standard. An algebraic proof with explicitly ordered products and verified generator/quotient claims.

Trap watch. Commuting factors illegally; confusing subgroup generated by powers with set of powers; omitting normality when passing to a quotient.

Representative linked problems.

1996-D2-P3; 1998-D1-P2; 1999-D1-P3; 1999-D2-P1; 2000-D1-P5; 2001-D1-P2; 2003-D2-P4; 2005-D1-P6; 2007-D1-P4; 2008-D1-P5; 2008-D2-P4; 2009-D2-P4; 2010-D2-P3; 2012-D1-P3; 2012-D1-P5; 2012-D2-P3; 2012-D2-P4; 2012-D2-P5; 2014-D2-P1; 2015-D1-P2; 2015-D1-P4; 2016-D1-P2; 2016-D1-P5; 2016-D2-P8; 2016-D2-P9; 2017-D1-P4; 2018-D1-P2; 2020-D2-P7; 2021-D2-P6; 2022-D1-P4; 2022-D2-P5; 2022-D2-P6; 2023-D1-P5; 2024-D1-P4

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.20 TPL-COUNT: Counting, double counting, graph encoding, and bijection proof

Family	Combinatorics
Purpose	Use when a finite structure should be encoded as a graph, counted two ways, injected into another set, or paired by a bijection/involution.
Trigger cues	number of; family of sets; grid; rows/columns; graph; cycle; permutation; words; coloring; at least; pigeonhole; selected objects
Method links	M-CMB-01; M-CMB-02; M-CMB-03; M-CMB-04; M-CMB-07; M-CMB-10; M-CMB-13; M-CMB-14; M-CMB-15
Occurrence count	48 primary problem assignments; 98 linked problem assignments.

Reproducible proof program.

1. Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.
2. Count the same set of objects in two ways, or build an injection/surjection/bijection.
3. Use pigeonhole only after computing the exact number of boxes and objects.
4. For graph problems, translate cycles, degrees, paths, and connectedness back to the original statement.
5. For parity, define the involution or determinant/sign cancellation explicitly.

Proof-output standard. A finite proof with a named counted set and a reversible decoding step when using bijections.

Trap watch. Counting unordered objects as ordered or vice versa; undefined encoding inverse; pigeonhole with weak thresholds; graph model not equivalent to original problem.

Representative linked problems.

1994-D1-P1; 1994-D1-P6; 1994-D2-P4; 1994-D2-P5; 1995-D1-P3; 1996-D1-P3; 1996-D1-P6; 1996-D2-P3; 1996-D2-P6; 1997-D1-P6; 1997-D2-P5; 1997-D2-P6; 1998-D1-P1; 1998-D1-P2; 1998-D1-P5; 1998-D2-P3; 1998-D2-P4; 1998-D2-P5; 1998-D2-P6; 1999-D1-P5; 1999-D2-P2; 1999-D2-P5; 2000-D1-P1; 2000-D2-P4; 2001-D1-P2; 2001-D2-P1; 2001-D2-P3; 2001-D2-P4; 2002-D1-P1; 2002-D1-P4; 2002-D1-P5; 2002-D2-P2; 2002-D2-P4; 2003-D1-P4; 2003-D1-P5; 2003-D2-P3; 2003-D2-P4; 2003-D2-P5; 2004-D1-P1; 2004-D1-P2; 2004-D1-P3; 2004-D1-P4; 2004-D2-P2; 2004-D2-P6; 2005-D1-P2; 2005-D2-P4; 2006-D1-P2; 2006-D1-P6; 2006-D2-P1; 2006-D2-P5; 2007-D1-P2; 2007-D1-P4; 2007-D1-P6; 2007-D2-P1; 2008-D1-P4; 2008-D1-P6; 2008-D2-P4; 2008-D2-P6; 2009-D1-P1; 2009-D1-P3; 2009-D1-P5; 2009-D2-P4; 2010-D1-P4; 2011-D1-P4; 2011-D2-P2; 2011-D2-P5; 2012-D1-P1; 2012-D1-P3; 2013-D1-P3; 2013-D2-P5; 2014-D2-P4; 2015-D2-P7; 2015-D2-P8; 2016-D1-P4; 2016-D1-P5; 2016-D2-P9; 2017-D1-P4; 2018-D2-P8; 2019-D2-P8; 2019-D2-P10; ... (+18 more)

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.21 TPL-GF: Recurrence, automaton, and generating-function proof

Family	Combinatorics / sequences
Purpose	Use for recurrences, state partitions, coefficient extraction, and sequence identities.
Trigger cues	recursively defined; count words/functions; generating function; coefficient; recurrence; finite automaton; matrix powers
Method links	M-CMB-05; M-CMB-06; M-RA-15
Occurrence count	0 primary problem assignments; 26 linked problem assignments.

Reproducible proof program.

1. Choose states that retain exactly the information needed for the next step.
2. Derive the recurrence or transition matrix from the state rules.
3. Check initial conditions and boundary states.
4. If using a generating function, prove convergence/formality and identify the coefficient extraction target.
5. Solve the recurrence or compare it with another recurrence using identical initial data.

Proof-output standard. A recurrence/generating-function proof whose state definition and initial conditions are explicit.

Trap watch. Missing forbidden boundary transitions; using generating functions analytically when only formal operations were justified; off-by-one indexing.

Representative linked problems.

1998-D2-P4; 2001-D2-P2; 2002-D2-P3; 2003-D1-P1; 2003-D1-P6; 2003-D2-P6; 2004-D1-P1; 2004-D2-P6; 2005-D1-P2; 2006-D1-P2; 2008-D2-P3; 2010-D2-P2; 2012-D2-P2; 2013-D1-P3; 2013-D2-P5; 2015-D1-P5; 2015-D2-P8; 2017-D1-P4; 2017-D2-P8; 2018-D2-P7; 2018-D2-P8; 2019-D1-P4; 2019-D2-P8; 2021-D1-P2; 2021-D1-P3; 2024-D2-P8

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.22 TPL-GEO: Geometry by coordinates, projection, volume, and metric identities

Family	Geometry
Purpose	Use when geometric conditions become algebraic after choosing coordinates, inner products, projections, or barycentric/volume weights.
Trigger cues	distance; sphere; ellipse; parabola; simplex; volume; projection; orthogonal; convex body; contraction; Hilbert space; equidistant
Method links	M-GEO-01; M-GEO-02; M-GEO-03; M-GEO-04; M-GEO-05; M-GEO-06; M-GEO-07; M-GEO-08; M-GEO-09; M-GEO-11; M-LA-11
Occurrence count	13 primary problem assignments; 39 linked problem assignments.

Reproducible proof program.

1. Normalize by translation, rotation, scaling, affine transformation, or coordinate choice.
2. Replace distances by squared distances and introduce inner products/Gram matrices when useful.
3. Use projections or lower-dimensional slices to reduce the geometry.
4. For volume/circumcenter/simplex problems, write barycentric or determinant weights.
5. Translate the algebraic result back into the geometric statement.

Proof-output standard. A coordinate-free-looking result derived from a legal coordinate normalization and a checked algebraic identity.

Trap watch. Squaring distances without preserving inequality direction; ignoring affine transformations that do not preserve lengths; assuming a projection is injective.

Representative linked problems.

1995-D2-P1; 1995-D2-P3; 1996-D1-P6; 1996-D2-P4; 1997-D1-P5; 1997-D2-P5; 1998-D1-P5; 1998-D2-P5; 2000-D2-P1; 2001-D2-P3; 2002-D1-P1; 2002-D2-P4; 2002-D2-P6; 2003-D2-P3; 2004-D1-P4; 2004-D2-P3; 2006-D2-P4; 2008-D1-P6; 2008-D2-P2; 2009-D1-P5; 2009-D2-P1; 2011-D2-P5; 2013-D2-P3; 2013-D2-P5; 2014-D1-P5; 2014-D2-P4; 2014-D2-P5; 2015-D1-P5; 2015-D2-P9; 2017-D2-P10; 2018-D1-P5; 2018-D2-P6; 2019-D2-P10; 2020-D1-P3; 2021-D2-P8; 2022-D2-P5; 2023-D2-P8; 2023-D2-P9; 2024-D1-P5

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.23 TPL-COMPLEX: Complex, Fourier, roots-of-unity, and unit-circle proof

Family	Complex / harmonic
Purpose	Use when roots of unity, unit-circle bounds, Fourier coefficients, Laurent expansions, cotangent series, or complexification simplify the problem.
Trigger cues	unit circle; complex roots of unity; Fourier coefficient; maximum modulus; branches of log; Laurent; trigonometric product; Fejer kernel
Method links	M-ANT-08; M-CMB-12; M-CX-01; M-CX-02; M-CX-03; M-CX-05; M-CX-06
Occurrence count	1 primary problem assignments; 24 linked problem assignments.

Reproducible proof program.

1. Translate real/trigonometric data into complex exponentials or a holomorphic/rational function when appropriate.
2. Use roots of unity as filters for congruence classes or coefficient extraction.
3. For unit-circle root problems, compute real parts or logarithmic derivatives.
4. Use maximum modulus/Cauchy/Laurent only after identifying analyticity and singularities.
5. Translate the complex conclusion back to the real polynomial/trigonometric statement.

Proof-output standard. A complex proof with explicit analytic domain, branch/singularity checks, and real interpretation.

Trap watch. Choosing one branch of logarithm when all branches matter; applying maximum modulus to a non-holomorphic expression; ignoring multiplicities of roots of unity.

Representative linked problems.

1995-D2-P3; 1995-D2-P5; 1999-D2-P2; 1999-D2-P6; 2000-D1-P4; 2000-D2-P3; 2001-D1-P4; 2001-D2-P3; 2002-D2-P5; 2004-D1-P6; 2005-D2-P2; 2006-D1-P4; 2007-D1-P5; 2007-D1-P6; 2008-D2-P1; 2012-D1-P5; 2012-D2-P1; 2013-D2-P1; 2014-D1-P3; 2017-D1-P5; 2017-D2-P7; 2018-D2-P9; 2021-D2-P7; 2023-D1-P3

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.24 TPL-FUNC: Functional equation, iterate, fixed-point, and dynamics proof

Family	Functional equations
Purpose	Use for functions constrained by substitutions, compositions, monotonicity, iteration, or orbit behavior.
Trigger cues	f(x+y); f(f(x)); iterates; periodic point; fixed point; maps onto itself; bijection; monotone forced by equation; translation invariance
Method links	M-FE-01; M-FE-02; M-FE-03; M-FE-04; M-FE-05; M-FE-06
Occurrence count	23 primary problem assignments; 100 linked problem assignments.

Reproducible proof program.

1. Substitute special values and freeze parameters to obtain monotonicity, injectivity, or invariance.
2. If iterates appear, analyze orbits, fixed points, cluster points, or periodic points.
3. If bijections appear, decompose the set into orbits/cycles under the map.
4. Use continuity/monotonicity only after proving it or after it is given.
5. Verify all candidate functions by direct substitution.

Proof-output standard. A substitution-driven proof with a clear route from the equation to rigidity or an explicit family.

Trap watch. Assuming continuity or monotonicity before deriving it; using inverse functions without bijectivity; missing constant solutions.

Representative linked problems.

1994-D1-P2; 1994-D1-P5; 1994-D1-P6; 1994-D2-P1; 1994-D2-P2; 1994-D2-P3; 1994-D2-P6; 1995-D1-P3; 1995-D1-P4; 1996-D1-P5; 1996-D2-P1; 1996-D2-P5; 1997-D1-P4; 1997-D2-P5; 1997-D2-P6; 1998-D1-P3; 1998-D1-P4; 1998-D1-P6; 1998-D2-P2; 1998-D2-P3; 1998-D2-P4; 1998-D2-P6; 1999-D1-P2; 1999-D1-P4; 1999-D1-P6; 1999-D2-P4; 2000-D1-P1; 2000-D1-P2; 2000-D1-P4; 2000-D1-P6; 2000-D2-P5; 2001-D1-P6; 2001-D2-P2; 2001-D2-P5; 2001-D2-P6; 2002-D1-P2; 2002-D1-P4; 2002-D1-P5; 2002-D1-P6; 2002-D2-P6; 2003-D2-P2; 2003-D2-P5; 2004-D2-P2; 2005-D1-P3; 2005-D1-P5; 2005-D2-P2; 2005-D2-P4; 2005-D2-P5; 2006-D1-P1; 2006-D1-P4; 2006-D1-P6; 2006-D2-P2; 2007-D1-P1; 2007-D1-P3; 2007-D2-P1; 2007-D2-P3; 2007-D2-P6; 2008-D1-P1; 2009-D1-P1; 2009-D2-P2; 2010-D1-P1; 2010-D1-P3; 2010-D2-P1; 2010-D2-P5; 2011-D1-P1; 2011-D1-P3; 2011-D1-P4; 2011-D2-P1; 2011-D2-P3; 2012-D1-P4; 2012-D2-P1; 2013-D1-P2; 2013-D2-P1; 2014-D1-P3; 2015-D1-P3; 2015-D2-P6; 2015-D2-P10; 2016-D1-P1; 2016-D1-P5; 2016-D2-P7; ... (+20 more)

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.25 TPL-TOPO: Density, countability, Baire/category, and pathological construction proof

Family	Topology / analysis
Purpose	Use when the problem involves dense sets, closed sets, countability/uncountability, nowhere monotone behavior, or pathological examples.
Trigger cues	dense; closed; countable; uncountable; Cantor set; nowhere monotone; local minima dense; every neighborhood
Method links	M-RA-16
Occurrence count	0 primary problem assignments; 10 linked problem assignments.

Reproducible proof program.

1. Translate the topological property into intervals/neighborhoods/cluster points.
2. For density, take an arbitrary nonempty interval and construct a point inside it.
3. For countability, cover by countably many finite or countable sets.

4. For uncountability/pathology, use Cantor-set or binary/ternary encoding when appropriate.
5. For Baire-type arguments, identify closed nowhere-dense pieces and the contradiction with a complete interval.

Proof-output standard. A proof where every topological quantifier is handled through arbitrary neighborhoods or explicit encodings.

Trap watch. Confusing dense with full measure; claiming a countable union of finite sets without proving the index set is countable; using Baire outside a complete space.

Representative linked problems.

1994-D2-P2; 1996-D2-P4; 1998-D2-P5; 2004-D1-P3; 2006-D2-P2; 2008-D1-P1; 2009-D1-P3; 2021-D2-P7; 2023-D2-P8; 2023-D2-P9

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

31.26 TPL-APPROX: Approximation, kernels, and polynomial-density proof

Family	Analysis / complex
Purpose	Use when functions are approximated by polynomials, kernels, trigonometric sums, or smoothing/averaging procedures.
Trigger cues	approximate uniformly; Weierstrass; kernel; Fejer; continuous odd function; polynomial approximation; max norm
Method links	M-CX-04; M-RA-17
Occurrence count	0 primary problem assignments; 1 linked problem assignments.

Reproducible proof program.

1. Identify the target function class and the norm/topology of approximation.
2. Start from a dense class such as polynomials/trigonometric polynomials, or a positive kernel.
3. Preserve required symmetries by symmetrizing the approximant.
4. Approximate missing basis elements separately if the allowed class is restricted.
5. Combine error bounds with a triangle inequality.

Proof-output standard. An approximation proof with a chosen dense family, symmetry preservation, and explicit error budget.

Trap watch. Approximating in the wrong norm; losing odd/even/positivity constraints; forgetting to split the epsilon budget.

Representative linked problems.

2009-D1-P4

Template fill-in form.

Claim target	What exactly must be proved, computed, classified, or constructed?
Trigger and template	Which phrase forced this template, and which method links are likely needed?
Core object	What invariant, auxiliary function, graph, polynomial, basis, or extremal object will carry the proof?
Closure theorem	What theorem closes the solution: Rolle, compactness, root count, rank-nullity, Cayley-Hamilton, CRT, Jensen, double counting, or another?
Edge checks	Which zero, endpoint, equality, singular, noncommutative, or convergence cases must be checked?
Final line	How does the derived statement translate back to the original problem?

32 Method-Node to Proof-Template Coverage Matrix

Every method node from the previous method atlas is assigned to a proof-template family. This is the compression layer that prevents the method database from becoming too granular during proof-writing practice.

Method	Method title	Template	Template title
M-ANT-01	Reduction modulo a prime or finite field	TPL-MOD	Modular arithmetic, finite-field, CRT, and valuation proof
M-ANT-02	Chinese remainder and residue construction	TPL-MOD	Modular arithmetic, finite-field, CRT, and valuation proof
M-ANT-03	GCD, valuation, and prime-power bookkeeping	TPL-MOD	Modular arithmetic, finite-field, CRT, and valuation proof
M-ANT-04	Finite differences and integer polynomial values	TPL-POLY	Polynomial roots, coefficients, interpolation, and divisibility proof
M-ANT-05	Factorization and unique factorization in Z[x]	TPL-POLY	Polynomial roots, coefficients, interpolation, and divisibility proof
M-ANT-06	Degree versus number of roots	TPL-POLY	Polynomial roots, coefficients, interpolation, and divisibility proof
M-ANT-07	Vieta, symmetric sums, and logarithmic derivative	TPL-POLY	Polynomial roots, coefficients, interpolation, and divisibility proof
M-ANT-08	Roots of unity and cyclotomic irreducibility	TPL-COMPLEX	Complex, Fourier, roots-of-unity, and unit-circle proof
M-ANT-09	Finite-field function representative and permutation argument	TPL-MOD	Modular arithmetic, finite-field, CRT, and valuation proof
M-ANT-10	Group action, orbit, and generator method	TPL-GROUP	Group, ring, automorphism, and noncommutative-algebra proof
M-ANT-11	Commutator and center manipulation in groups/rings	TPL-GROUP	Group, ring, automorphism, and noncommutative-algebra proof
M-ANT-12	Normal subgroup and quotient obstruction	TPL-GROUP	Group, ring, automorphism, and noncommutative-algebra proof
M-ANT-13	Pell-type recurrence or quadratic irrational stabilizer	TPL-MOD	Modular arithmetic, finite-field, CRT, and valuation proof
M-ANT-14	Semidirect product or explicit algebraic construction	TPL-GROUP	Group, ring, automorphism, and noncommutative-algebra proof
M-ANT-15	Rationality and linear dependence over Q	TPL-MOD	Modular arithmetic, finite-field, CRT, and valuation proof
M-ANT-16	Polynomial identity by coefficient comparison	TPL-POLY	Polynomial roots, coefficients, interpolation, and divisibility proof
M-ANT-17	Greedy expansion or tail-rationality method	TPL-MOD	Modular arithmetic, finite-field, CRT, and valuation proof
M-CMB-01	Pigeonhole threshold and collision forcing	TPL-COUNT	Counting, double counting, graph encoding, and bijection proof

Method	Method title	Template	Template title
M-CMB-02	Double counting ordered or weighted objects	TPL-COUNT	Counting, double counting, graph encoding, and bijection proof
M-CMB-03	Graph encoding of a relation	TPL-COUNT	Counting, double counting, graph encoding, and bijection proof
M-CMB-04	Cycle extraction and alternating-cycle method	TPL-COUNT	Counting, double counting, graph encoding, and bijection proof
M-CMB-05	State recurrence or automaton partition	TPL-GF	Recurrence, automaton, and generating-function proof
M-CMB-06	Generating function encoding	TPL-GF	Recurrence, automaton, and generating-function proof
M-CMB-07	Involution, parity, and sign-reversing cancellation	TPL-COUNT	Counting, double counting, graph encoding, and bijection proof
M-CMB-08	Extremal set or minimal counterexample induction	TPL-CONTR	Contradiction, obstruction, and minimal counterexample proof
M-CMB-09	Partition, refinement, and cover compression	TPL-IND	Induction, recursive peeling, and strengthened hypothesis
M-CMB-10	Explicit bijection and encoding-decoding proof	TPL-COUNT	Counting, double counting, graph encoding, and bijection proof
M-CMB-11	Erdos-Szekeres/binomial threshold induction	TPL-IND	Induction, recursive peeling, and strengthened hypothesis
M-CMB-12	Root-of-unity filter for residues	TPL-COMPLEX	Complex, Fourier, roots-of-unity, and unit-circle proof
M-CMB-13	Indicator and expectation method	TPL-COUNT	Counting, double counting, graph encoding, and bijection proof
M-CMB-14	Topological ordering of an acyclic directed graph	TPL-COUNT	Counting, double counting, graph encoding, and bijection proof
M-CMB-15	Rewriting-system reduction	TPL-COUNT	Counting, double counting, graph encoding, and bijection proof
M-CX-01	Cauchy coefficient formula and boundary maximum	TPL-COMPLEX	Complex, Fourier, roots-of-unity, and unit-circle proof
M-CX-02	Laurent expansion and isolated singularity rationality	TPL-COMPLEX	Complex, Fourier, roots-of-unity, and unit-circle proof
M-CX-03	Unit-circle root real-part/log-derivative test	TPL-COMPLEX	Complex, Fourier, roots-of-unity, and unit-circle proof
M-CX-04	Fejer kernel positivity	TPL-APPROX	Approximation, kernels, and polynomial-density proof
M-CX-05	Trigonometric-to-complex conversion	TPL-COMPLEX	Complex, Fourier, roots-of-unity, and unit-circle proof
M-CX-06	Cotangent partial-fraction differentiation	TPL-COMPLEX	Complex, Fourier, roots-of-unity, and unit-circle proof
M-CX-07	Multiplicity count for values 0 and 1	TPL-POLY	Polynomial roots, coefficients, interpolation, and divisibility proof
M-FE-01	Iterates, fixed points, and cluster-point trapping	TPL-FUNC	Functional equation, iterate, fixed-point, and dynamics proof
M-FE-02	Monotonicity forced by functional equation	TPL-FUNC	Functional equation, iterate, fixed-point, and dynamics proof

Method	Method title	Template	Template title
M-FE-03	Composition, inversion, and conjugacy	TPL-FUNC	Functional equation, iterate, fixed-point, and dynamics proof
M-FE-04	Orbit decomposition for bijections	TPL-FUNC	Functional equation, iterate, fixed-point, and dynamics proof
M-FE-05	Translation/rational-shift invariance	TPL-FUNC	Functional equation, iterate, fixed-point, and dynamics proof
M-FE-06	Nonexistence by growth iteration	TPL-FUNC	Functional equation, iterate, fixed-point, and dynamics proof
M-GEO-01	Coordinate normalization and algebraization	TPL-GEO	Geometry by coordinates, projection, volume, and metric identities
M-GEO-02	Projection to a lower-dimensional object	TPL-GEO	Geometry by coordinates, projection, volume, and metric identities
M-GEO-03	Squared-distance and inner-product identities	TPL-GEO	Geometry by coordinates, projection, volume, and metric identities
M-GEO-04	Barycentric coordinates and volume weights	TPL-GEO	Geometry by coordinates, projection, volume, and metric identities
M-GEO-05	Affine invariance and change of metric scale	TPL-GEO	Geometry by coordinates, projection, volume, and metric identities
M-GEO-06	Convex support and John-ellipse perturbation	TPL-GEO	Geometry by coordinates, projection, volume, and metric identities
M-GEO-07	Metric contraction and diameter-cover comparison	TPL-GEO	Geometry by coordinates, projection, volume, and metric identities
M-GEO-08	Conic, sphere, and directrix algebra	TPL-GEO	Geometry by coordinates, projection, volume, and metric identities
M-GEO-09	Regular simplex and equidistant orthonormal translation	TPL-GEO	Geometry by coordinates, projection, volume, and metric identities
M-GEO-10	Closest-point uniqueness by moving toward a minimizer	TPL-EXIST	Existence by compactness, extremal choice, or closedness
M-GEO-11	Volume, projection, and ellipsoid computation	TPL-GEO	Geometry by coordinates, projection, volume, and metric identities
M-LA-01	Row/column elimination to triangular or sparse form	TPL-DET	Determinant, rank, inverse, and matrix-factorization proof
M-LA-02	Complete a matrix equation to an inverse identity	TPL-DET	Determinant, rank, inverse, and matrix-factorization proof
M-LA-03	Block decomposition and Schur-style bookkeeping	TPL-DET	Determinant, rank, inverse, and matrix-factorization proof
M-LA-04	Rank-nullity and dimension-count obstruction	TPL-DET	Determinant, rank, inverse, and matrix-factorization proof
M-LA-05	Similarity, adapted basis, and canonical model	TPL-SPEC	Spectral, nilpotency, idempotent, and commutator proof
M-LA-06	Spectral/minimal-polynomial reduction	TPL-SPEC	Spectral, nilpotency, idempotent, and commutator proof
M-LA-07	Trace cyclicity and commutator annihilation	TPL-SPEC	Spectral, nilpotency, idempotent, and commutator proof

Method	Method title	Template	Template title
M-LA-08	Simultaneous diagonalization or common eigenvector induction	TPL-SPEC	Spectral, nilpotency, idempotent, and commutator proof
M-LA-09	Invariant subspace flag and matrix-shape count	TPL-DET	Determinant, rank, inverse, and matrix-factorization proof
M-LA-10	Linear operator on a matrix space	TPL-DET	Determinant, rank, inverse, and matrix-factorization proof
M-LA-11	Gram matrix and distance-to-inner-product conversion	TPL-GEO	Geometry by coordinates, projection, volume, and metric identities
M-LA-12	Nilpotency from zero spectrum plus finite dimension	TPL-SPEC	Spectral, nilpotency, idempotent, and commutator proof
M-LA-13	Determinant through eigenvalue or root product	TPL-SPEC	Spectral, nilpotency, idempotent, and commutator proof
M-LA-14	Parity or zero-pattern block determinant split	TPL-DET	Determinant, rank, inverse, and matrix-factorization proof
M-LA-15	AB versus BA transfer	TPL-DET	Determinant, rank, inverse, and matrix-factorization proof
M-LA-16	Trace pairing and orthogonal matrix subspaces	TPL-DET	Determinant, rank, inverse, and matrix-factorization proof
M-LA-17	Sign forcing from positive matrices or inner products	TPL-INEQ	Inequality proof and norm comparison
M-LA-18	Commutator telescoping through powers	TPL-SPEC	Spectral, nilpotency, idempotent, and commutator proof
M-RA-01	Auxiliary function with derivative sign	TPL-DERIV	Auxiliary function, Rolle/MVT, Taylor, and differential inequality proof
M-RA-02	Integrating factor and ODE comparison	TPL-DERIV	Auxiliary function, Rolle/MVT, Taylor, and differential inequality proof
M-RA-03	Rolle chain and divided-difference localization	TPL-DERIV	Auxiliary function, Rolle/MVT, Taylor, and differential inequality proof
M-RA-04	Taylor expansion and local asymptotic bookkeeping	TPL-DERIV	Auxiliary function, Rolle/MVT, Taylor, and differential inequality proof
M-RA-05	L-Hospital and ratio/growth comparison	TPL-LIMIT	Limit, asymptotic, Riemann-sum, and error-splitting proof
M-RA-06	Riemann sum and sum-integral conversion	TPL-LIMIT	Limit, asymptotic, Riemann-sum, and error-splitting proof
M-RA-07	Split into main range and tail/error range	TPL-LIMIT	Limit, asymptotic, Riemann-sum, and error-splitting proof
M-RA-08	Compactness plus extreme value theorem	TPL-SHARP	Sharpness, equality case, and extremal constant template
M-RA-09	Convexity, Jensen, tangent, and chord inequalities	TPL-SHARP	Sharpness, equality case, and extremal constant template
M-RA-10	Cauchy-Schwarz, Holder, and norm comparison	TPL-INEQ	Inequality proof and norm comparison
M-RA-11	Integration by parts, Fubini, and weighted identities	TPL-INTEGRAL	Integral identity, Fubini, integration by parts, and oscillation averaging proof
M-RA-12	Squeeze, dominated convergence, and monotone convergence	TPL-CONV	Convergence, Cauchy, series, and compactness-of-orbits proof
M-RA-13	Periodic averaging and oscillation cancellation	TPL-INTEGRAL	Integral identity, Fubini, integration by parts, and oscillation averaging proof
M-RA-14	Endpoint Laplace-type asymptotic comparison	TPL-LIMIT	Limit, asymptotic, Riemann-sum, and error-splitting proof

Method	Method title	Template	Template title
M-RA-15	Recurrence transformed to monotone quotient or telescoping form	TPL-GF	Recurrence, automaton, and generating-function proof
M-RA-16	Density, Baire/category, and pathological construction	TPL-TOPO	Density, countability, Baire/-category, and pathological construction proof
M-RA-17	Approximation by polynomials or kernels	TPL-APPROX	Approximation, kernels, and polynomial-density proof
M-RA-18	Trigonometric/hyperbolic identity plus monotonicity	TPL-DERIV	Auxiliary function, Rolle/MVT, Taylor, and differential in-equality proof
M-UNI-01	Normalization, WLOG reduction, and scaling	TPL-DIR	Direct structural/invariant proof
M-UNI-02	Small cases, equality cases, and example probing	TPL-CASE	Case split by sign, zero, boundary, and degeneracy
M-UNI-03	Extremal construction and equality-case engineering	TPL-SHARP	Sharpness, equality case, and extremal constant template
M-UNI-04	Contradiction with minimal or maximal counterexample	TPL-CONTR	Contradiction, obstruction, and minimal counterexample proof
M-UNI-05	Induction with strengthened hypothesis	TPL-IND	Induction, recursive peeling, and strengthened hypothesis
M-UNI-06	Compactness or extremal-point selection	TPL-EXIST	Existence by compactness, ex-tremal choice, or closedness
M-UNI-07	Special substitution and parameter freez-ing	TPL-DIR	Direct structural/invariant proof
M-UNI-08	Sign, zero, and boundary case splitting	TPL-CASE	Case split by sign, zero, boundary, and degeneracy
M-UNI-09	Invariant extraction and invariant com-parison	TPL-EQUIV	Bidirectional equivalence and reformulation proof
M-UNI-10	Peeling, deletion, and reduction to a core	TPL-IND	Induction, recursive peeling, and strengthened hypothesis

33 Problem-by-Problem Proof-Template Audit

This audit assigns a primary proof template and secondary backup templates to every detected problem entry. The problem link jumps back to the earlier problem card; the template links jump to the reproducible proof programs above.

Problem	Primary template	Secondary templates	First proof move	Type / triggers
1994-D1-P1	TPL-DET	TPL-INEQ; TPL-COUNT; TPL-SHARP; TPL-SPEC	Choose a basis, row/column order, or block decomposition adapted to the hypotheses.	PT-LA-DET; TR-LA-01; TR-LA-02; TR-LA-05
1994-D1-P2	TPL-LIMIT	TPL-INEQ; TPL-DERIV; TPL-SHARP; TPL-FUNC	Identify the scale and normalize the expression.	PT-LIM; TR-RA-02; TR-RA-04; TR-RA-07
1994-D1-P3	TPL-MOD	TPL-IND; TPL-EQUIV; TPL-SHARP; TPL-POLY	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-06; TR-RA-07; TR-ANT-01
1994-D1-P4	TPL-SPEC	TPL-DET; TPL-EXIST; TPL-CONSTRUCT; TPL-SHARP	Translate the equation into a polynomial identity in the matrix or a commutator relation.	PT-LA-SPEC; TR-LA-01; TR-LA-05; TR-RA-06
1994-D1-P5	TPL-LIMIT	TPL-DERIV; TPL-FUNC; TPL-INTEGRAL; TPL-DIR	Identify the scale and normalize the expression.	PT-FE; TR-RA-04; TR-RA-05; TR-RA-07
1994-D1-P6	TPL-MOD	TPL-INEQ; TPL-COUNT; TPL-DERIV; TPL-FUNC	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-02; TR-RA-07; TR-ANT-01
1994-D2-P1	TPL-FUNC	TPL-EXIST; TPL-DERIV; TPL-CONSTRUCT; TPL-CONTR	Substitute special values and freeze parameters to obtain monotonicity, injectivity, or invariance.	PT-FE; TR-RA-02; TR-RA-05; TR-RA-06
1994-D2-P2	TPL-FUNC	TPL-CLASS; TPL-SHARP; TPL-INEQ; TPL-TOPO	Substitute special values and freeze parameters to obtain monotonicity, injectivity, or invariance.	PT-FE; TR-LA-01; TR-LA-05; TR-RA-01
1994-D2-P3	TPL-FUNC	TPL-DERIV; TPL-INTEGRAL; TPL-DIR; TPL-IND	Substitute special values and freeze parameters to obtain monotonicity, injectivity, or invariance.	PT-FE; TR-RA-02; TR-RA-05; TR-RA-06
1994-D2-P4	TPL-POLY	TPL-DET; TPL-COUNT; TPL-SPEC; TPL-DERIV	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-LA-DET; TR-LA-01; TR-LA-05; TR-ANT-02
1994-D2-P5	TPL-COUNT	TPL-EXIST; TPL-CONSTRUCT; TPL-DET; TPL-INEQ	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-LA-01; TR-RA-06; TR-RA-07
1994-D2-P6	TPL-LIMIT	TPL-FUNC; TPL-SHARP; TPL-DERIV; TPL-INEQ	Identify the scale and normalize the expression.	PT-LIM; TR-RA-02; TR-RA-04; TR-RA-07
1995-D1-P1	TPL-SPEC	TPL-DET; TPL-INEQ	Translate the equation into a polynomial identity in the matrix or a commutator relation.	PT-LA-SPEC; TR-LA-01; TR-LA-02; TR-LA-03
1995-D1-P2	TPL-INEQ	TPL-SHARP; TPL-DERIV; TPL-DIR; TPL-EQUIV	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-INEQ; TR-RA-02; TR-RA-04; TR-RA-07
1995-D1-P3	TPL-LIMIT	TPL-COUNT; TPL-DERIV; TPL-FUNC; TPL-INEQ	Identify the scale and normalize the expression.	PT-LIM; TR-RA-02; TR-RA-04; TR-RA-07
1995-D1-P4	TPL-FUNC	TPL-DERIV; TPL-SHARP; TPL-DIR; TPL-IND	Substitute special values and freeze parameters to obtain monotonicity, injectivity, or invariance.	PT-FE; TR-LA-05; TR-RA-02; TR-RA-04
1995-D1-P5	TPL-SPEC	TPL-POLY; TPL-DET; TPL-IND; TPL-EQUIV	Translate the equation into a polynomial identity in the matrix or a commutator relation.	PT-LA-SPEC; TR-LA-01; TR-LA-03; TR-LA-04
1995-D1-P6	TPL-EXIST	TPL-SHARP; TPL-INEQ; TPL-CONSTRUCT; TPL-LIMIT	Identify the ambient compact/finite set or create one by truncation/localization.	PT-INEQ; TR-RA-01; TR-RA-02; TR-RA-04
1995-D2-P1	TPL-DET	TPL-EXIST; TPL-CONSTRUCT; TPL-GEO; TPL-INEQ	Choose a basis, row/column order, or block decomposition adapted to the hypotheses.	PT-LA-DET; TR-LA-01; TR-LA-02; TR-LA-05
1995-D2-P2	TPL-LIMIT	TPL-INEQ; TPL-SHARP; TPL-EQUIV; TPL-DIR	Identify the scale and normalize the expression.	PT-INEQ; TR-RA-02; TR-RA-04; TR-RA-07
1995-D2-P3	TPL-POLY	TPL-COMPLEX; TPL-INEQ; TPL-GEO	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-RA-05; TR-RA-07; TR-PC-01
1995-D2-P4	TPL-MOD	TPL-POLY; TPL-SHARP; TPL-EXIST; TPL-DERIV	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-01; TR-RA-07; TR-ANT-01
1995-D2-P5	TPL-LIMIT	TPL-INEQ; TPL-DERIV; TPL-SHARP; TPL-COMPLEX	Identify the scale and normalize the expression.	PT-INEQ; TR-LA-05; TR-RA-02; TR-RA-04
1995-D2-P6	TPL-EXIST	TPL-INEQ; TPL-CONSTRUCT; TPL-SHARP; TPL-DERIV	Identify the ambient compact/finite set or create one by truncation/localization.	PT-INEQ; TR-RA-01; TR-RA-02; TR-RA-04
1996-D1-P1	TPL-DET	TPL-SPEC; TPL-CASE; TPL-EQUIV; TPL-INEQ	Choose a basis, row/column order, or block decomposition adapted to the hypotheses.	PT-LA-DET; TR-LA-01; TR-LA-02; TR-LA-03

Problem	Primary template	Secondary templates	First proof move	Type / triggers
1996-D1-P2	TPL-INTEGRAL	TPL-LIMIT; TPL-DERIV; TPL-SHARP; TPL-DIR	Look for a weight or kernel that converts the hypothesis into the desired integral.	PT-INT; TR-RA-02; TR-RA-04; TR-RA-05
1996-D1-P3	TPL-SPEC	TPL-EXIST; TPL-COUNT; TPL-CONSTRUCT; TPL-DET	Translate the equation into a polynomial identity in the matrix or a commutator relation.	PT-LA-SPEC; TR-LA-01; TR-LA-04; TR-LA-05
1996-D1-P4	TPL-LIMIT	TPL-INEQ; TPL-SHARP; TPL-INTEGRAL; TPL-IND	Identify the scale and normalize the expression.	PT-LIM; TR-RA-02; TR-RA-04; TR-RA-06
1996-D1-P5	TPL-LIMIT	TPL-INEQ; TPL-DERIV; TPL-FUNC; TPL-SHARP	Identify the scale and normalize the expression.	PT-LIM; TR-LA-05; TR-RA-02; TR-RA-04
1996-D1-P6	TPL-GEO	TPL-INEQ; TPL-COUNT; TPL-DET; TPL-SHARP	Normalize by translation, rotation, scaling, affine transformation, or coordinate choice.	PT-GEO; TR-LA-01; TR-LA-05; TR-LA-07
1996-D2-P1	TPL-LIMIT	TPL-FUNC; TPL-DERIV; TPL-DIR; TPL-EQUIV	Identify the scale and normalize the expression.	PT-FE; TR-RA-02; TR-RA-04; TR-RA-05
1996-D2-P2	TPL-MOD	TPL-POLY; TPL-IND; TPL-SHARP; TPL-EQUIV	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-07; TR-PC-04; TR-ANT-01
1996-D2-P3	TPL-COUNT	TPL-GROUP; TPL-DERIV; TPL-DIR; TPL-DET	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-GRP; TR-RA-07; TR-ANT-01; TR-ANT-03
1996-D2-P4	TPL-GEO	TPL-INEQ; TPL-SHARP; TPL-TOPO; TPL-EXIST	Normalize by translation, rotation, scaling, affine transformation, or coordinate choice.	PT-GEO; TR-RA-01; TR-RA-06; TR-RA-07
1996-D2-P5	TPL-LIMIT	TPL-FUNC; TPL-INEQ; TPL-DERIV; TPL-SHARP	Identify the scale and normalize the expression.	PT-FE; TR-RA-02; TR-RA-04; TR-RA-05
1996-D2-P6	TPL-INEQ	TPL-COUNT; TPL-EXIST; TPL-CONSTRUCT; TPL-INTEGRAL	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-INEQ; TR-RA-04; TR-RA-06; TR-RA-07
1997-D1-P1	TPL-LIMIT	TPL-INTEGRAL; TPL-DIR; TPL-EQUIV; TPL-DERIV	Identify the scale and normalize the expression.	PT-LIM; TR-RA-02; TR-RA-04; TR-RA-06
1997-D1-P2	TPL-LIMIT	TPL-INTEGRAL; TPL-SHARP; TPL-CASE; TPL-DERIV	Identify the scale and normalize the expression.	PT-INT; TR-RA-02; TR-RA-04; TR-RA-06
1997-D1-P3	TPL-MOD	TPL-DET; TPL-SPEC; TPL-IND; TPL-INEQ	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-LA-01; TR-LA-02; TR-LA-03
1997-D1-P4	TPL-MOD	TPL-INEQ; TPL-SHARP; TPL-FUNC; TPL-DERIV	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-06; TR-RA-07; TR-ANT-01
1997-D1-P5	TPL-INEQ	TPL-MOD; TPL-GEO; TPL-SHARP; TPL-DET	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-FINITE; TR-LA-05; TR-RA-06; TR-RA-07
1997-D1-P6	TPL-COUNT	TPL-CONTR; TPL-EXIST; TPL-CONSTRUCT; TPL-SHARP	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-RA-06; TR-ANT-04; TR-CMB-02
1997-D2-P1	TPL-INEQ	TPL-DERIV; TPL-LIMIT; TPL-SHARP; TPL-DIR	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-INEQ; TR-RA-02; TR-RA-03; TR-RA-04
1997-D2-P2	TPL-DET	TPL-SPEC; TPL-IND; TPL-INEQ	Choose a basis, row/column order, or block decomposition adapted to the hypotheses.	PT-LA-DET; TR-LA-01; TR-LA-02; TR-LA-03
1997-D2-P3	TPL-DERIV	TPL-LIMIT; TPL-INEQ; TPL-SHARP; TPL-IND	Construct an auxiliary function whose derivative is exactly the target expression or has a fixed sign.	PT-INEQ; TR-RA-02; TR-RA-04; TR-RA-06
1997-D2-P4	TPL-SPEC	TPL-EXIST; TPL-DET; TPL-CONSTRUCT; TPL-LIMIT	Translate the equation into a polynomial identity in the matrix or a commutator relation.	PT-LA-SPEC; TR-LA-01; TR-LA-02; TR-LA-03
1997-D2-P5	TPL-GEO	TPL-COUNT; TPL-FUNC; TPL-DERIV; TPL-DIR	Normalize by translation, rotation, scaling, affine transformation, or coordinate choice.	PT-GEO; TR-RA-07; TR-ANT-04; TR-CMB-01
1997-D2-P6	TPL-COUNT	TPL-FUNC; TPL-DERIV; TPL-SHARP; TPL-CASE	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-FE; TR-RA-02; TR-RA-04; TR-ANT-04
1998-D1-P1	TPL-DET	TPL-COUNT; TPL-SPEC; TPL-CASE; TPL-EQUIV	Choose a basis, row/column order, or block decomposition adapted to the hypotheses.	PT-LA-DET; TR-LA-01; TR-LA-05; TR-ANT-04
1998-D1-P2	TPL-POLY	TPL-COUNT; TPL-SHARP; TPL-CASE; TPL-GROUP	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-COMB; TR-RA-06; TR-RA-07; TR-ANT-03

Problem	Primary template	Secondary templates	First proof move	Type / triggers
1998-D1-P3	TPL-LIMIT	TPL-FUNC; TPL-INTEGRAL; TPL-DERIV; TPL-IND	Identify the scale and normalize the expression.	PT-LIM; TR-RA-02; TR-RA-04; TR-RA-06
1998-D1-P4	TPL-EXIST	TPL-DERIV; TPL-SHARP; TPL-FUNC; TPL-CONSTRUCT	Identify the ambient compact/finite set or create one by truncation/localization.	PT-FE; TR-RA-02; TR-RA-04; TR-RA-07
1998-D1-P5	TPL-INEQ	TPL-POLY; TPL-SHARP; TPL-COUNT; TPL-GEO	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-INEQ; TR-RA-04; TR-RA-07; TR-PC-01
1998-D1-P6	TPL-INEQ	TPL-SHARP; TPL-FUNC; TPL-DERIV; TPL-INTEGRAL	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-FE; TR-RA-02; TR-RA-04; TR-RA-06
1998-D2-P1	TPL-SPEC	TPL-DET; TPL-INEQ; TPL-IND; TPL-SHARP	Translate the equation into a polynomial identity in the matrix or a commutator relation.	PT-LA-SPEC; TR-LA-01; TR-LA-05; TR-RA-06
1998-D2-P2	TPL-POLY	TPL-CLASS; TPL-FUNC; TPL-SHARP; TPL-EXIST	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-FE; TR-RA-01; TR-RA-04; TR-RA-07
1998-D2-P3	TPL-FUNC	TPL-COUNT; TPL-INEQ; TPL-DERIV; TPL-SHARP	Substitute special values and freeze parameters to obtain monotonicity, injectivity, or invariance.	PT-FE; TR-RA-05; TR-RA-07; TR-ANT-04
1998-D2-P4	TPL-COUNT	TPL-FUNC; TPL-LIMIT; TPL-DERIV; TPL-GF	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-FE; TR-LA-05; TR-RA-05; TR-RA-07
1998-D2-P5	TPL-COUNT	TPL-GEO; TPL-TOPO; TPL-SHARP; TPL-SPEC	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-GEO; TR-LA-05; TR-RA-01; TR-RA-07
1998-D2-P6	TPL-COUNT	TPL-FUNC; TPL-EXIST; TPL-CONSTRUCT; TPL-SHARP	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-FE; TR-RA-02; TR-RA-04; TR-RA-06
1999-D1-P1	TPL-SPEC	TPL-POLY; TPL-EXIST; TPL-DET; TPL-CONSTRUCT	Translate the equation into a polynomial identity in the matrix or a commutator relation.	PT-LA-SPEC; TR-LA-01; TR-LA-02; TR-LA-03
1999-D1-P2	TPL-CONTR	TPL-EXIST; TPL-SHARP; TPL-LIMIT; TPL-FUNC	Assume the negation in its strongest useful form: no object exists, all objects fail, or choose a minimal counterexample.	PT-FE; TR-RA-02; TR-RA-04; TR-RA-07
1999-D1-P3	TPL-MOD	TPL-INEQ; TPL-SHARP; TPL-GROUP; TPL-DIR	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-07; TR-ANT-01; TR-ANT-03
1999-D1-P4	TPL-FUNC	TPL-CLASS; TPL-DERIV; TPL-INTEGRAL; TPL-DIR	Substitute special values and freeze parameters to obtain monotonicity, injectivity, or invariance.	PT-FE; TR-RA-05; TR-RA-06; TR-RA-07
1999-D1-P5	TPL-COUNT	TPL-DET; TPL-IND; TPL-EQUIV; TPL-INEQ	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-LA-02; TR-RA-07; TR-ANT-03
1999-D1-P6	TPL-FUNC	TPL-DERIV; TPL-SHARP; TPL-DIR; TPL-CASE	Substitute special values and freeze parameters to obtain monotonicity, injectivity, or invariance.	PT-FE; TR-RA-02; TR-RA-05; TR-RA-07
1999-D2-P1	TPL-GROUP	TPL-MOD; TPL-EQUIV; TPL-DIR; TPL-IND	Translate the statement into generators, relations, subgroups, quotients, or commutators.	PT-GRP; TR-ANT-01; TR-ANT-03; TR-ANT-04
1999-D2-P2	TPL-MOD	TPL-POLY; TPL-COUNT; TPL-COMPLEX; TPL-DET	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-LA-02; TR-RA-04; TR-RA-05
1999-D2-P3	TPL-INTEGRAL	TPL-INEQ; TPL-SHARP; TPL-DERIV; TPL-DIR	Look for a weight or kernel that converts the hypothesis into the desired integral.	PT-INT; TR-RA-02; TR-RA-04; TR-RA-06
1999-D2-P4	TPL-FUNC	TPL-EXIST; TPL-CONSTRUCT; TPL-SHARP; TPL-INTEGRAL	Substitute special values and freeze parameters to obtain monotonicity, injectivity, or invariance.	PT-FE; TR-RA-05; TR-RA-06; TR-RA-07
1999-D2-P5	TPL-COUNT	TPL-SHARP; TPL-CASE; TPL-IND; TPL-EQUIV	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-RA-06; TR-RA-07; TR-ANT-04
1999-D2-P6	TPL-COMPLEX	TPL-POLY; TPL-INEQ; TPL-EXIST; TPL-CONSTRUCT	Translate real/trigonometric data into complex exponentials or a holomorphic/rational function when appropriate.	PT-INEQ; TR-RA-04; TR-RA-05; TR-RA-07
2000-D1-P1	TPL-EXIST	TPL-FUNC; TPL-COUNT; TPL-CONSTRUCT; TPL-SHARP	Identify the ambient compact/finite set or create one by truncation/localization.	PT-FE; TR-RA-01; TR-RA-02; TR-RA-04
2000-D1-P2	TPL-FUNC	TPL-CLASS; TPL-POLY; TPL-DERIV; TPL-DIR	Substitute special values and freeze parameters to obtain monotonicity, injectivity, or invariance.	PT-FE; TR-RA-04; TR-PC-04; TR-UNI-03
2000-D1-P3	TPL-SPEC	TPL-DET; TPL-IND; TPL-INEQ; TPL-POLY	Translate the equation into a polynomial identity in the matrix or a commutator relation.	PT-LA-SPEC; TR-LA-01; TR-LA-02; TR-LA-03

Problem	Primary template	Secondary templates	First proof move	Type / triggers
2000-D1-P4	TPL-INEQ	TPL-SHARP; TPL-FUNC; TPL-DERIV; TPL-COMPLEX	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-INEQ; TR-RA-02; TR-RA-04; TR-RA-07
2000-D1-P5	TPL-GROUP	TPL-SPEC; TPL-MOD; TPL-DET	Translate the statement into generators, relations, subgroups, quotients, or commutators.	PT-GRP; TR-LA-03; TR-LA-04; TR-LA-05
2000-D1-P6	TPL-FUNC	TPL-POLY; TPL-INEQ; TPL-DERIV; TPL-DIR	Substitute special values and freeze parameters to obtain monotonicity, injectivity, or invariance.	PT-FE; TR-RA-04; TR-RA-05; TR-RA-06
2000-D2-P1	TPL-GEO	TPL-MOD; TPL-INEQ; TPL-DET; TPL-SHARP	Normalize by translation, rotation, scaling, affine transformation, or coordinate choice.	PT-GEO; TR-LA-01; TR-RA-07; TR-PC-05
2000-D2-P2	TPL-EXIST	TPL-SHARP; TPL-INEQ; TPL-CONSTRUCT; TPL-DERIV	Identify the ambient compact/finite set or create one by truncation/localization.	PT-INEQ; TR-RA-01; TR-RA-02; TR-RA-04
2000-D2-P3	TPL-POLY	TPL-COMPLEX; TPL-SHARP; TPL-IND; TPL-EQUIV	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-RA-07; TR-PC-01; TR-PC-04
2000-D2-P4	TPL-POLY	TPL-INEQ; TPL-COUNT; TPL-LIMIT; TPL-SHARP	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-RA-04; TR-RA-06; TR-RA-07
2000-D2-P5	TPL-DERIV	TPL-FUNC; TPL-CLASS; TPL-LIMIT; TPL-DIR	Construct an auxiliary function whose derivative is exactly the target expression or has a fixed sign.	PT-FE; TR-RA-02; TR-RA-04; TR-RA-05
2000-D2-P6	TPL-SPEC	TPL-POLY; TPL-LIMIT; TPL-EXIST; TPL-CONSTRUCT	Translate the equation into a polynomial identity in the matrix or a commutator relation.	PT-LA-SPEC; TR-LA-01; TR-LA-03; TR-LA-04
2001-D1-P1	TPL-MOD	TPL-DET; TPL-INTEGRAL; TPL-DIR; TPL-EQUIV	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-LA-02; TR-RA-04; TR-RA-06
2001-D1-P2	TPL-GROUP	TPL-MOD; TPL-COUNT; TPL-IND; TPL-DET	Translate the statement into generators, relations, subgroups, quotients, or commutators.	PT-GRP; TR-LA-05; TR-ANT-01; TR-ANT-02
2001-D1-P3	TPL-LIMIT	TPL-DIR; TPL-EQUIV; TPL-DERIV; TPL-SHARP	Identify the scale and normalize the expression.	PT-LIM; TR-RA-02; TR-RA-04; TR-ANT-05
2001-D1-P4	TPL-POLY	TPL-MOD; TPL-COMPLEX; TPL-SHARP; TPL-INEQ	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-RA-07; TR-PC-01; TR-PC-04
2001-D1-P5	TPL-SPEC	TPL-DET; TPL-IND; TPL-POLY; TPL-EQUIV	Translate the equation into a polynomial identity in the matrix or a commutator relation.	PT-LA-SPEC; TR-LA-01; TR-LA-02; TR-LA-03
2001-D1-P6	TPL-LIMIT	TPL-FUNC; TPL-DERIV; TPL-SHARP; TPL-DET	Identify the scale and normalize the expression.	PT-FE; TR-LA-05; TR-RA-02; TR-RA-03
2001-D2-P1	TPL-POLY	TPL-COUNT; TPL-MOD; TPL-INEQ; TPL-IND	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-INEQ; TR-RA-04; TR-RA-07; TR-ANT-01
2001-D2-P2	TPL-INEQ	TPL-LIMIT; TPL-FUNC; TPL-SHARP; TPL-GF	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-INEQ; TR-RA-02; TR-RA-04; TR-RA-06
2001-D2-P3	TPL-INEQ	TPL-COMPLEX; TPL-GEO; TPL-COUNT; TPL-SHARP	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-GEO; TR-LA-01; TR-RA-01; TR-RA-02
2001-D2-P4	TPL-DET	TPL-EXIST; TPL-COUNT; TPL-CONSTRUCT; TPL-SPEC	Choose a basis, row/column order, or block decomposition adapted to the hypotheses.	PT-LA-DET; TR-LA-01; TR-LA-02; TR-LA-03
2001-D2-P5	TPL-FUNC	TPL-LIMIT; TPL-EXIST; TPL-CONSTRUCT; TPL-CONTR	Substitute special values and freeze parameters to obtain monotonicity, injectivity, or invariance.	PT-FE; TR-RA-04; TR-RA-05; TR-RA-07
2001-D2-P6	TPL-MOD	TPL-INEQ; TPL-SHARP; TPL-FUNC; TPL-EXIST	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-01; TR-RA-02; TR-RA-07
2002-D1-P1	TPL-GEO	TPL-POLY; TPL-COUNT; TPL-DET; TPL-SPEC	Normalize by translation, rotation, scaling, affine transformation, or coordinate choice.	PT-GEO; TR-LA-05; TR-RA-06; TR-RA-07
2002-D1-P2	TPL-FUNC	TPL-CONTR; TPL-EXIST; TPL-CONSTRUCT; TPL-DERIV	Substitute special values and freeze parameters to obtain monotonicity, injectivity, or invariance.	PT-FE; TR-RA-02; TR-RA-05; TR-RA-06
2002-D1-P3	TPL-MOD	TPL-INTEGRAL; TPL-INEQ; TPL-IND; TPL-SHARP	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-06; TR-RA-07; TR-ANT-01
2002-D1-P4	TPL-COUNT	TPL-FUNC; TPL-DERIV; TPL-LIMIT; TPL-IND	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-FE; TR-RA-01; TR-RA-02; TR-RA-04

Problem	Primary template	Secondary templates	First proof move	Type / triggers
2002-D1-P5	TPL-FUNC	TPL-EXIST; TPL-CONSTRUCT; TPL-DERIV; TPL-COUNT	Substitute special values and freeze parameters to obtain monotonicity, injectivity, or invariance.	PT-FE; TR-LA-01; TR-RA-02; TR-RA-04
2002-D1-P6	TPL-INEQ	TPL-LIMIT; TPL-MOD; TPL-DET; TPL-FUNC	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-FINITE; TR-LA-01; TR-LA-07; TR-RA-02
2002-D2-P1	TPL-DET	TPL-SPEC; TPL-CASE; TPL-EQUIV; TPL-INEQ	Choose a basis, row/column order, or block decomposition adapted to the hypotheses.	PT-LA-DET; TR-LA-01; TR-LA-02; TR-LA-03
2002-D2-P2	TPL-COUNT	TPL-EQUIV; TPL-DIR; TPL-IND; TPL-INEQ	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-ANT-04; TR-CMB-03; TR-CMB-04
2002-D2-P3	TPL-MOD	TPL-POLY; TPL-IND; TPL-SHARP; TPL-GF	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-LA-05; TR-RA-06; TR-RA-07
2002-D2-P4	TPL-GEO	TPL-COUNT; TPL-IND; TPL-EQUIV; TPL-SHARP	Normalize by translation, rotation, scaling, affine transformation, or coordinate choice.	PT-GEO; TR-RA-06; TR-RA-07; TR-ANT-04
2002-D2-P5	TPL-SPEC	TPL-DET; TPL-COMPLEX; TPL-IND; TPL-INEQ	Translate the equation into a polynomial identity in the matrix or a commutator relation.	PT-LA-SPEC; TR-LA-01; TR-LA-05; TR-RA-05
2002-D2-P6	TPL-INEQ	TPL-GEO; TPL-FUNC; TPL-SHARP; TPL-EXIST	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-FE; TR-LA-07; TR-RA-01; TR-RA-05
2003-D1-P1	TPL-LIMIT	TPL-EXIST; TPL-CONSTRUCT; TPL-GF; TPL-INTEGRAL	Identify the scale and normalize the expression.	PT-LIM; TR-RA-02; TR-RA-04; TR-RA-06
2003-D1-P2	TPL-MOD	TPL-EXIST; TPL-LIMIT; TPL-CONSTRUCT; TPL-SHARP	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-04; TR-RA-06; TR-PC-05
2003-D1-P3	TPL-SPEC	TPL-POLY; TPL-LIMIT; TPL-DET; TPL-INTEGRAL	Translate the equation into a polynomial identity in the matrix or a commutator relation.	PT-LA-SPEC; TR-LA-01; TR-LA-03; TR-LA-04
2003-D1-P4	TPL-COUNT	TPL-MOD; TPL-DIR; TPL-EQUIV; TPL-SHARP	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-LA-01; TR-LA-05; TR-RA-07
2003-D1-P5	TPL-LIMIT	TPL-EXIST; TPL-CONSTRUCT; TPL-DERIV; TPL-COUNT	Identify the scale and normalize the expression.	PT-LIM; TR-LA-01; TR-LA-05; TR-RA-02
2003-D1-P6	TPL-POLY	TPL-GF; TPL-IND; TPL-SHARP; TPL-SPEC	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-LA-05; TR-RA-06; TR-RA-07
2003-D2-P1	TPL-DET	TPL-SPEC; TPL-EQUIV; TPL-DIR	Choose a basis, row/column order, or block decomposition adapted to the hypotheses.	PT-LA-DET; TR-LA-01; TR-LA-02; TR-LA-05
2003-D2-P2	TPL-LIMIT	TPL-POLY; TPL-DERIV; TPL-FUNC; TPL-INEQ	Identify the scale and normalize the expression.	PT-LIM; TR-RA-04; TR-RA-07; TR-PC-04
2003-D2-P3	TPL-EXIST	TPL-COUNT; TPL-SHARP; TPL-CONSTRUCT; TPL-GEO	Identify the ambient compact/finite set or create one by truncation/localization.	PT-COMB; TR-RA-01; TR-RA-07; TR-RA-10
2003-D2-P4	TPL-COUNT	TPL-MOD; TPL-CLASS; TPL-GROUP; TPL-CASE	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-RA-07; TR-ANT-01; TR-ANT-03
2003-D2-P5	TPL-COUNT	TPL-LIMIT; TPL-FUNC; TPL-EXIST; TPL-CONSTRUCT	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-FE; TR-RA-02; TR-RA-04; TR-RA-06
2003-D2-P6	TPL-LIMIT	TPL-GF; TPL-INTEGRAL; TPL-IND; TPL-SHARP	Identify the scale and normalize the expression.	PT-LIM; TR-RA-02; TR-RA-04; TR-RA-06
2004-D1-P1	TPL-COUNT	TPL-CONTR; TPL-SHARP; TPL-GF; TPL-EXIST	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-LA-05; TR-RA-07; TR-ANT-04
2004-D1-P2	TPL-COUNT	TPL-INEQ; TPL-SHARP; TPL-INTEGRAL; TPL-IND	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-INEQ; TR-RA-02; TR-RA-04; TR-RA-06
2004-D1-P3	TPL-LIMIT	TPL-SHARP; TPL-COUNT; TPL-DET; TPL-TOPO	Identify the scale and normalize the expression.	PT-LIM; TR-LA-01; TR-RA-01; TR-RA-02
2004-D1-P4	TPL-GEO	TPL-COUNT; TPL-IND; TPL-EQUIV; TPL-INEQ	Normalize by translation, rotation, scaling, affine transformation, or coordinate choice.	PT-GEO; TR-RA-04; TR-RA-06; TR-RA-07
2004-D1-P5	TPL-EXIST	TPL-SHARP; TPL-INEQ; TPL-CONSTRUCT ; TPL-DERIV	Identify the ambient compact/finite set or create one by truncation/localization.	PT-INEQ; TR-LA-05; TR-RA-02; TR-RA-04
2004-D1-P6	TPL-INTEGRAL	TPL-COMPLEX; TPL-POLY; TPL-LIMIT; TPL-DERIV	Look for a weight or kernel that converts the hypothesis into the desired integral.	PT-INT; TR-LA-05; TR-RA-04; TR-RA-06

Problem	Primary template	Secondary templates	First proof move	Type / triggers
2004-D2-P1	TPL-DET	TPL-SPEC; TPL-EQUIV; TPL-DIR	Choose a basis, row/column order, or block decomposition adapted to the hypotheses.	PT-LA-DET; TR-LA-01; TR-LA-02; TR-LA-05
2004-D2-P2	TPL-FUNC	TPL-INEQ; TPL-COUNT; TPL-DERIV; TPL-SHARP	Substitute special values and freeze parameters to obtain monotonicity, injectivity, or invariance.	PT-FE; TR-RA-02; TR-RA-05; TR-RA-07
2004-D2-P3	TPL-GEO	TPL-LIMIT; TPL-EXIST; TPL-CONSTRUCT; TPL-INEQ	Normalize by translation, rotation, scaling, affine transformation, or coordinate choice.	PT-GEO; TR-RA-04; TR-RA-06; TR-RA-07
2004-D2-P4	TPL-SPEC	TPL-LIMIT; TPL-DET; TPL-SHARP; TPL-DIR	Translate the equation into a polynomial identity in the matrix or a commutator relation.	PT-LA-SPEC; TR-LA-01; TR-LA-05; TR-RA-04
2004-D2-P5	TPL-INEQ	TPL-SHARP; TPL-DERIV; TPL-DIR; TPL-INTEGRAL	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-INEQ; TR-RA-02; TR-RA-04; TR-RA-07
2004-D2-P6	TPL-DET	TPL-INEQ; TPL-COUNT; TPL-LIMIT; TPL-GF	Choose a basis, row/column order, or block decomposition adapted to the hypotheses.	PT-LA-DET; TR-LA-01; TR-LA-02; TR-LA-05
2005-D1-P1	TPL-DET	TPL-SPEC; TPL-SHARP; TPL-DIR; TPL-EQUIV	Choose a basis, row/column order, or block decomposition adapted to the hypotheses.	PT-LA-DET; TR-LA-01; TR-LA-02; TR-LA-03
2005-D1-P2	TPL-COUNT	TPL-MOD; TPL-IND; TPL-GF; TPL-SHARP	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-RA-07; TR-PC-05; TR-ANT-01
2005-D1-P3	TPL-INEQ	TPL-SHARP; TPL-FUNC; TPL-DERIV; TPL-DIR	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-FE; TR-RA-02; TR-RA-04; TR-RA-07
2005-D1-P4	TPL-POLY	TPL-CLASS; TPL-MOD; TPL-DERIV; TPL-DIR	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-RA-06; TR-RA-07; TR-ANT-02
2005-D1-P5	TPL-LIMIT	TPL-DERIV; TPL-DIR; TPL-FUNC; TPL-INEQ	Identify the scale and normalize the expression.	PT-LIM; TR-RA-02; TR-RA-04; TR-RA-06
2005-D1-P6	TPL-GROUP	TPL-SPEC; TPL-MOD; TPL-IND; TPL-SHARP	Translate the statement into generators, relations, subgroups, quotients, or commutators.	PT-GRP; TR-LA-03; TR-LA-04; TR-RA-07
2005-D2-P1	TPL-INTEGRAL	TPL-INEQ; TPL-SHARP; TPL-LIMIT; TPL-EXIST	Look for a weight or kernel that converts the hypothesis into the desired integral.	PT-INT; TR-RA-01; TR-RA-02; TR-RA-04
2005-D2-P2	TPL-POLY	TPL-MOD; TPL-FUNC; TPL-COMPLEX; TPL-DIR	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-RA-06; TR-ANT-02
2005-D2-P3	TPL-SPEC	TPL-INEQ; TPL-LIMIT; TPL-DET; TPL-SHARP	Translate the equation into a polynomial identity in the matrix or a commutator relation.	PT-LA-SPEC; TR-LA-01; TR-LA-02; TR-LA-03
2005-D2-P4	TPL-EXIST	TPL-FUNC; TPL-DERIV; TPL-CONSTRUCT; TPL-COUNT	Identify the ambient compact/finite set or create one by truncation/localization.	PT-FE; TR-RA-02; TR-RA-05; TR-ANT-04
2005-D2-P5	TPL-CLASS	TPL-FUNC; TPL-SHARP; TPL-DERIV; TPL-INEQ	Start with an arbitrary solution and derive necessary restrictions.	PT-FE; TR-RA-01; TR-RA-02; TR-RA-05
2005-D2-P6	TPL-MOD	TPL-DET; TPL-EXIST; TPL-CONSTRUCT; TPL-POLY	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-LA-01; TR-LA-02; TR-LA-03
2006-D1-P1	TPL-FUNC	TPL-LIMIT; TPL-EXIST; TPL-CONSTRUCT; TPL-CONTR	Substitute special values and freeze parameters to obtain monotonicity, injectivity, or invariance.	PT-FE; TR-RA-02; TR-RA-04; TR-RA-06
2006-D1-P2	TPL-COUNT	TPL-MOD; TPL-GF; TPL-POLY; TPL-INEQ	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-FINITE; TR-RA-07; TR-ANT-01; TR-ANT-02
2006-D1-P3	TPL-MOD	TPL-DET; TPL-LIMIT; TPL-IND; TPL-SPEC	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-LA-01; TR-LA-02; TR-LA-03
2006-D1-P4	TPL-POLY	TPL-MOD; TPL-LIMIT; TPL-COMPLEX; TPL-FUNC	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-RA-04; TR-RA-06; TR-PC-02
2006-D1-P5	TPL-INEQ	TPL-DERIV; TPL-SHARP; TPL-DIR; TPL-EQUIV	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-INEQ; TR-RA-02; TR-RA-04; TR-RA-07
2006-D1-P6	TPL-CLASS	TPL-POLY; TPL-FUNC; TPL-DERIV; TPL-COUNT	Start with an arbitrary solution and derive necessary restrictions.	PT-FE; TR-LA-01; TR-RA-04; TR-RA-06
2006-D2-P1	TPL-COUNT	TPL-MOD; TPL-SHARP; TPL-IND; TPL-INEQ	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-FINITE; TR-RA-06; TR-RA-07; TR-ANT-01

Problem	Primary template	Secondary templates	First proof move	Type / triggers
2006-D2-P2	TPL-FUNC	TPL-CLASS; TPL-DET; TPL-SHARP; TPL-TOPO	Substitute special values and freeze parameters to obtain monotonicity, injectivity, or invariance.	PT-FE; TR-LA-01; TR-LA-05; TR-RA-01
2006-D2-P3	TPL-INEQ	TPL-SHARP; TPL-DET; TPL-DIR; TPL-EQUIV	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-INEQ; TR-RA-02; TR-RA-04; TR-RA-07
2006-D2-P4	TPL-MOD	TPL-GEO; TPL-INEQ; TPL-DET; TPL-SHARP	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-LA-01; TR-LA-07; TR-RA-06
2006-D2-P5	TPL-POLY	TPL-MOD; TPL-EXIST; TPL-CONSTRUCT; TPL-COUNT	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-RA-06; TR-ANT-01; TR-ANT-02
2006-D2-P6	TPL-SPEC	TPL-POLY; TPL-DET; TPL-INEQ; TPL-LIMIT	Translate the equation into a polynomial identity in the matrix or a commutator relation.	PT-LA-SPEC; TR-LA-01; TR-LA-02; TR-LA-03
2007-D1-P1	TPL-POLY	TPL-MOD; TPL-DERIV; TPL-DIR; TPL-FUNC	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-PC-01; TR-PC-02; TR-ANT-01
2007-D1-P2	TPL-MOD	TPL-DET; TPL-INEQ; TPL-SHARP; TPL-COUNT	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-LA-01; TR-LA-02; TR-LA-03
2007-D1-P3	TPL-DET	TPL-POLY; TPL-CLASS; TPL-SPEC; TPL-FUNC	Choose a basis, row/column order, or block decomposition adapted to the hypotheses.	PT-LA-DET; TR-LA-01; TR-LA-02; TR-LA-03
2007-D1-P4	TPL-COUNT	TPL-GROUP; TPL-DERIV; TPL-DIR; TPL-MOD	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-GRP; TR-PC-05; TR-ANT-01; TR-ANT-03
2007-D1-P5	TPL-POLY	TPL-MOD; TPL-COMPLEX; TPL-SHARP; TPL-DET	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-LA-01; TR-RA-06; TR-RA-07
2007-D1-P6	TPL-POLY	TPL-COUNT; TPL-COMPLEX; TPL-INEQ; TPL-LIMIT	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-RA-05; TR-RA-06; TR-RA-07
2007-D2-P1	TPL-COUNT	TPL-FUNC; TPL-SHARP; TPL-DERIV; TPL-CASE	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-FE; TR-RA-02; TR-RA-04; TR-ANT-04
2007-D2-P2	TPL-MOD	TPL-INTEGRAL; TPL-POLY; TPL-IND; TPL-EQUIV	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-06; TR-ANT-01; TR-ANT-02
2007-D2-P3	TPL-FUNC	TPL-EXIST; TPL-CONSTRUCT; TPL-DERIV; TPL-SHARP	Substitute special values and freeze parameters to obtain monotonicity, injectivity, or invariance.	PT-FE; TR-RA-01; TR-RA-02; TR-RA-04
2007-D2-P4	TPL-MOD	TPL-DET; TPL-SPEC; TPL-CASE; TPL-EQUIV	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-LA-01; TR-LA-02; TR-LA-03
2007-D2-P5	TPL-MOD	TPL-DET; TPL-SPEC; TPL-SHARP; TPL-CASE	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-LA-01; TR-RA-07; TR-ANT-01
2007-D2-P6	TPL-POLY	TPL-EXIST; TPL-CONSTRUCT; TPL-DERIV; TPL-FUNC	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-LA-05; TR-RA-01; TR-RA-03
2008-D1-P6	TPL-COUNT	TPL-POLY; TPL-INEQ; TPL-DET; TPL-GEO	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-LA-01; TR-LA-02; TR-LA-03
2008-D1-P1	TPL-FUNC	TPL-CLASS; TPL-DERIV; TPL-SHARP; TPL-TOPO	Substitute special values and freeze parameters to obtain monotonicity, injectivity, or invariance.	PT-FE; TR-RA-01; TR-RA-02; TR-RA-05
2008-D1-P2	TPL-DET	TPL-POLY; TPL-SPEC; TPL-IND; TPL-SHARP	Choose a basis, row/column order, or block decomposition adapted to the hypotheses.	PT-LA-DET; TR-LA-01; TR-LA-05; TR-RA-06
2008-D1-P3	TPL-POLY	TPL-EXIST; TPL-CONSTRUCT; TPL-MOD; TPL-SHARP	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-LA-05; TR-RA-06; TR-PC-01
2008-D1-P4	TPL-COUNT	TPL-CLASS; TPL-INTEGRAL; TPL-DIR; TPL-EQUIV	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-RA-06; TR-RA-07; TR-ANT-04
2008-D1-P5	TPL-MOD	TPL-GROUP; TPL-EXIST; TPL-CONSTRUCT; TPL-CONTR	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-GRP; TR-LA-01; TR-LA-07; TR-RA-02
2008-D2-P1	TPL-POLY	TPL-MOD; TPL-INEQ; TPL-COMPLEX; TPL-DERIV	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-INEQ; TR-RA-04; TR-PC-01; TR-PC-04
2008-D2-P2	TPL-POLY	TPL-GEO; TPL-DERIV; TPL-INEQ; TPL-DIR	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-GEO; TR-RA-07; TR-CMB-06; TR-GEO-01

Problem	Primary template	Secondary templates	First proof move	Type / triggers
2008-D2-P3	TPL-INEQ	TPL-MOD; TPL-GF; TPL-POLY; TPL-SHARP	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-POLY; TR-RA-07; TR-PC-05; TR-ANT-01
2008-D2-P4	TPL-POLY	TPL-MOD; TPL-COUNT; TPL-GROUP; TPL-SHARP	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-GRP; TR-PC-01; TR-PC-02; TR-ANT-01
2008-D2-P5	TPL-DET	TPL-MOD; TPL-POLY; TPL-SPEC; TPL-INEQ	Choose a basis, row/column order, or block decomposition adapted to the hypotheses.	PT-FINITE; TR-LA-01; TR-LA-02; TR-LA-03
2008-D2-P6	TPL-COUNT	TPL-DET; TPL-POLY; TPL-INEQ; TPL-SHARP	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-LA-DET; TR-LA-01; TR-LA-05; TR-LA-07
2009-D1-P1	TPL-COUNT	TPL-LIMIT; TPL-INEQ; TPL-FUNC; TPL-DERIV	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-INEQ; TR-RA-02; TR-RA-04; TR-RA-06
2009-D1-P2	TPL-DET	TPL-SPEC; TPL-EQUIV; TPL-DIR; TPL-IND	Choose a basis, row/column order, or block decomposition adapted to the hypotheses.	PT-LA-DET; TR-LA-01; TR-LA-05; TR-CMB-02
2009-D1-P3	TPL-COUNT	TPL-CLASS; TPL-POLY; TPL-DET; TPL-TOPO	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-LA-01; TR-RA-07; TR-CMB-01
2009-D1-P4	TPL-INEQ	TPL-POLY; TPL-SHARP; TPL-DET; TPL-APPROX	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-INEQ; TR-LA-01; TR-LA-05; TR-RA-04
2009-D1-P5	TPL-DET	TPL-MOD; TPL-COUNT; TPL-GEO; TPL-IND	Choose a basis, row/column order, or block decomposition adapted to the hypotheses.	PT-FINITE; TR-LA-01; TR-LA-02; TR-LA-03
2009-D2-P1	TPL-GEO	TPL-POLY; TPL-INEQ; TPL-SHARP; TPL-DIR	Normalize by translation, rotation, scaling, affine transformation, or coordinate choice.	PT-GEO; TR-RA-07; TR-ANT-04; TR-CMB-03
2009-D2-P2	TPL-DERIV	TPL-FUNC; TPL-INEQ; TPL-SHARP; TPL-DIR	Construct an auxiliary function whose derivative is exactly the target expression or has a fixed sign.	PT-FE; TR-RA-02; TR-RA-03; TR-RA-05
2009-D2-P3	TPL-SPEC	TPL-MOD; TPL-POLY; TPL-EXIST; TPL-CONSTRUCT	Translate the equation into a polynomial identity in the matrix or a commutator relation.	PT-FINITE; TR-LA-01; TR-LA-02; TR-LA-03
2009-D2-P4	TPL-POLY	TPL-COUNT; TPL-MOD; TPL-GROUP	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-ANT-01; TR-ANT-02; TR-ANT-03
2009-D2-P5	TPL-DET	TPL-INEQ; TPL-POLY; TPL-MOD; TPL-SHARP	Choose a basis, row/column order, or block decomposition adapted to the hypotheses.	PT-FINITE; TR-LA-01; TR-LA-02; TR-LA-03
2010-D1-P1	TPL-INEQ	TPL-FUNC; TPL-DERIV; TPL-SHARP; TPL-DIR	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-FE; TR-RA-02; TR-RA-05; TR-RA-07
2010-D1-P2	TPL-LIMIT	TPL-INTEGRAL; TPL-EQUIV; TPL-DIR; TPL-SHARP	Identify the scale and normalize the expression.	PT-INT; TR-RA-02; TR-RA-04; TR-RA-06
2010-D1-P3	TPL-LIMIT	TPL-FUNC; TPL-IND; TPL-SHARP; TPL-DIR	Identify the scale and normalize the expression.	PT-FE; TR-RA-04; TR-RA-05; TR-RA-07
2010-D1-P4	TPL-MOD	TPL-COUNT; TPL-SHARP; TPL-CASE; TPL-INEQ	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-06; TR-RA-07; TR-ANT-01
2010-D1-P5	TPL-INEQ	TPL-MOD; TPL-DET; TPL-DERIV; TPL-SHARP	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-FINITE; TR-LA-01; TR-LA-02; TR-LA-03
2010-D2-P1	TPL-LIMIT	TPL-INEQ; TPL-FUNC; TPL-SHARP; TPL-INTEGRAL	Identify the scale and normalize the expression.	PT-INEQ; TR-LA-05; TR-RA-02; TR-RA-04
2010-D2-P2	TPL-INEQ	TPL-SHARP; TPL-INTEGRAL; TPL-IND; TPL-GF	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-INEQ; TR-RA-02; TR-RA-04; TR-RA-06
2010-D2-P3	TPL-GROUP	TPL-EXIST; TPL-CONSTRUCT; TPL-DERIV; TPL-DIR	Translate the statement into generators, relations, subgroups, quotients, or commutators.	PT-GRP; TR-RA-06; TR-RA-07; TR-ANT-01
2010-D2-P4	TPL-MOD	TPL-DET; TPL-IND; TPL-SPEC; TPL-INEQ	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-LA-01; TR-LA-02; TR-LA-07
2010-D2-P5	TPL-POLY	TPL-MOD; TPL-LIMIT; TPL-SPEC; TPL-FUNC	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-LA-05; TR-RA-04; TR-PC-04
2011-D1-P1	TPL-EXIST	TPL-FUNC; TPL-LIMIT; TPL-CONSTRUCT; TPL-SHARP	Identify the ambient compact/finite set or create one by truncation/localization.	PT-FE; TR-RA-01; TR-RA-02; TR-RA-04

Problem	Primary template	Secondary templates	First proof move	Type / triggers
2011-D1-P2	TPL-SPEC	TPL-POLY; TPL-EXIST; TPL-CONSTRUCT; TPL-CONTR	Translate the equation into a polynomial identity in the matrix or a commutator relation.	PT-LA-SPEC; TR-LA-01; TR-LA-02; TR-LA-03
2011-D1-P3	TPL-POLY	TPL-MOD; TPL-INEQ; TPL-SHARP; TPL-FUNC	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-RA-07; TR-PC-02; TR-ANT-01
2011-D1-P4	TPL-COUNT	TPL-FUNC; TPL-DERIV; TPL-SHARP; TPL-INEQ	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-RA-07; TR-ANT-04; TR-CMB-01
2011-D1-P5	TPL-MOD	TPL-POLY; TPL-EXIST; TPL-CONSTRUCT; TPL-DET	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-LA-01; TR-RA-06; TR-RA-07
2011-D2-P1	TPL-LIMIT	TPL-FUNC; TPL-INTEGRAL; TPL-DERIV; TPL-IND	Identify the scale and normalize the expression.	PT-LIM; TR-RA-02; TR-RA-04; TR-RA-06
2011-D2-P2	TPL-INEQ	TPL-COUNT; TPL-MOD; TPL-POLY; TPL-DET	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-FINITE; TR-LA-02; TR-LA-05; TR-RA-06
2011-D2-P3	TPL-FUNC	TPL-DET; TPL-DIR; TPL-EQUIV; TPL-SHARP	Substitute special values and freeze parameters to obtain monotonicity, injectivity, or invariance.	PT-FE; TR-LA-01; TR-RA-05; TR-RA-06
2011-D2-P4	TPL-POLY	TPL-DET; TPL-DERIV; TPL-IND; TPL-DIR	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-LA-01; TR-RA-06; TR-RA-07
2011-D2-P5	TPL-GEO	TPL-SHARP; TPL-COUNT; TPL-DET; TPL-CASE	Normalize by translation, rotation, scaling, affine transformation, or coordinate choice.	PT-GEO; TR-LA-01; TR-RA-06; TR-RA-07
2012-D1-P1	TPL-COUNT	TPL-MOD; TPL-LIMIT; TPL-INTEGRAL; TPL-INEQ	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-FINITE; TR-RA-04; TR-RA-06; TR-RA-07
2012-D1-P2	TPL-MOD	TPL-DET; TPL-SPEC; TPL-DIR; TPL-EQUIV	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-LA-01; TR-LA-02; TR-LA-03
2012-D1-P3	TPL-GROUP	TPL-MOD; TPL-COUNT; TPL-DIR; TPL-EQUIV	Translate the statement into generators, relations, subgroups, quotients, or commutators.	PT-GRP; TR-RA-07; TR-ANT-01; TR-ANT-03
2012-D1-P4	TPL-INEQ	TPL-SHARP; TPL-FUNC; TPL-DERIV; TPL-DIR	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-FE; TR-RA-02; TR-RA-04; TR-RA-07
2012-D1-P5	TPL-POLY	TPL-MOD; TPL-GROUP; TPL-COMPLEX; TPL-INEQ	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-ANT-01; TR-ANT-03; TR-ANT-04
2012-D2-P1	TPL-POLY	TPL-MOD; TPL-FUNC; TPL-COMPLEX; TPL-DIR	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-ANT-01; TR-ANT-02
2012-D2-P2	TPL-INTEGRAL	TPL-LIMIT; TPL-GF; TPL-IND; TPL-DERIV	Look for a weight or kernel that converts the hypothesis into the desired integral.	PT-INT; TR-LA-01; TR-RA-02; TR-RA-04
2012-D2-P3	TPL-POLY	TPL-MOD; TPL-GROUP; TPL-DIR; TPL-EQUIV	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-ANT-01; TR-ANT-03; TR-ANT-04
2012-D2-P4	TPL-MOD	TPL-CLASS; TPL-GROUP; TPL-DIR; TPL-EQUIV	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-07; TR-ANT-01; TR-ANT-03
2012-D2-P5	TPL-GROUP	TPL-INEQ; TPL-MOD; TPL-DERIV; TPL-DIR	Translate the statement into generators, relations, subgroups, quotients, or commutators.	PT-GRP; TR-RA-07; TR-ANT-01; TR-ANT-03
2013-D1-P1	TPL-SPEC	TPL-DET; TPL-IND; TPL-EQUIV	Translate the equation into a polynomial identity in the matrix or a commutator relation.	PT-LA-SPEC; TR-LA-01; TR-LA-05; TR-GEO-01
2013-D1-P2	TPL-EXIST	TPL-FUNC; TPL-CONSTRUCT; TPL-DERIV; TPL-SHARP	Identify the ambient compact/finite set or create one by truncation/localization.	PT-FE; TR-RA-02; TR-RA-04
2013-D1-P3	TPL-COUNT	TPL-SHARP; TPL-EXIST; TPL-GF; TPL-INEQ	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-RA-07; TR-CMB-01; TR-CMB-06
2013-D1-P4	TPL-INEQ	TPL-SHARP; TPL-DET; TPL-DIR; TPL-EQUIV	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-INEQ; TR-RA-02; TR-RA-04; TR-RA-07
2013-D1-P5	TPL-MOD	TPL-CONTR; TPL-EXIST; TPL-CONSTRUCT; TPL-POLY	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-06; TR-PC-04; TR-ANT-01
2013-D2-P1	TPL-POLY	TPL-COMPLEX; TPL-EQUIV; TPL-DIR; TPL-FUNC	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-PC-04

Problem	Primary template	Secondary templates	First proof move	Type / triggers
2013-D2-P2	TPL-MOD	TPL-POLY; TPL-IND; TPL-EQUIV; TPL-INEQ	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-ANT-01; TR-ANT-02; TR-ANT-03
2013-D2-P3	TPL-EXIST	TPL-SHARP; TPL-INEQ; TPL-CONSTRUCT; TPL-GEO	Identify the ambient compact/finite set or create one by truncation/localization.	PT-INEQ; TR-LA-07; TR-RA-02; TR-RA-04
2013-D2-P4	TPL-MOD	TPL-CONTR; TPL-LIMIT; TPL-EXIST; TPL-CONSTRUCT	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-04; TR-RA-06; TR-ANT-01
2013-D2-P5	TPL-COUNT	TPL-GF; TPL-SHARP; TPL-INEQ; TPL-GEO	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-RA-07; TR-CMB-01; TR-CMB-06
2014-D1-P1	TPL-SPEC	TPL-DET; TPL-CLASS; TPL-DIR; TPL-INEQ	Translate the equation into a polynomial identity in the matrix or a commutator relation.	PT-LA-SPEC; TR-LA-01; TR-LA-02; TR-LA-03
2014-D1-P2	TPL-LIMIT	TPL-CLASS; TPL-DIR; TPL-EQUIV; TPL-DERIV	Identify the scale and normalize the expression.	PT-LIM; TR-RA-02; TR-RA-04; TR-UNI-03
2014-D1-P3	TPL-POLY	TPL-DIR; TPL-EQUIV; TPL-COMPLEX; TPL-FUNC	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-PC-01
2014-D1-P4	TPL-MOD	TPL-INTEGRAL; TPL-POLY; TPL-DIR; TPL-EQUIV	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-04; TR-RA-06; TR-RA-07
2014-D1-P5	TPL-GEO	TPL-DIR; TPL-EQUIV; TPL-SHARP; TPL-INEQ	Normalize by translation, rotation, scaling, affine transformation, or coordinate choice.	PT-GEO; TR-RA-07; TR-GEO-01; TR-GEO-02
2014-D2-P1	TPL-MOD	TPL-POLY; TPL-DIR; TPL-EQUIV; TPL-GROUP	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-ANT-01; TR-ANT-03; TR-ANT-04
2014-D2-P2	TPL-CLASS	TPL-SPEC; TPL-INEQ; TPL-SHARP; TPL-DET	Start with an arbitrary solution and derive necessary restrictions.	PT-INEQ; TR-LA-01; TR-LA-05; TR-RA-02
2014-D2-P3	TPL-INEQ	TPL-DERIV; TPL-DET; TPL-DIR; TPL-EQUIV	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-INEQ; TR-RA-02; TR-RA-04
2014-D2-P4	TPL-GEO	TPL-INEQ; TPL-COUNT; TPL-SHARP; TPL-DIR	Normalize by translation, rotation, scaling, affine transformation, or coordinate choice.	PT-GEO; TR-RA-07; TR-ANT-04; TR-CMB-03
2014-D2-P5	TPL-INEQ	TPL-SHARP; TPL-GEO; TPL-DET; TPL-DIR	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-INEQ; TR-RA-02; TR-RA-04; TR-RA-07
2015-D1-P1	TPL-DET	TPL-SPEC; TPL-MOD; TPL-POLY; TPL-INEQ	Choose a basis, row/column order, or block decomposition adapted to the hypotheses.	PT-LA-DET; TR-LA-01; TR-LA-02; TR-LA-03
2015-D1-P2	TPL-MOD	TPL-INEQ; TPL-SHARP; TPL-GROUP; TPL-DET	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-07; TR-ANT-01; TR-ANT-03
2015-D1-P3	TPL-FUNC	TPL-DET; TPL-DIR; TPL-EQUIV; TPL-SHARP	Substitute special values and freeze parameters to obtain monotonicity, injectivity, or invariance.	PT-FE; TR-LA-01; TR-RA-02; TR-RA-05
2015-D1-P4	TPL-MOD	TPL-DIR; TPL-EQUIV; TPL-SHARP; TPL-GROUP	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-LA-01; TR-RA-07; TR-ANT-01
2015-D1-P5	TPL-INEQ	TPL-GEO; TPL-DET; TPL-GF; TPL-SHARP	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-INEQ; TR-LA-01; TR-RA-02; TR-RA-04
2015-D2-P6	TPL-INEQ	TPL-DERIV; TPL-FUNC; TPL-EQUIV; TPL-DIR	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-INEQ; TR-RA-02; TR-RA-04
2015-D2-P7	TPL-LIMIT	TPL-COUNT; TPL-DIR; TPL-EQUIV; TPL-DERIV	Identify the scale and normalize the expression.	PT-LIM; TR-RA-02; TR-RA-04; TR-ANT-05
2015-D2-P8	TPL-COUNT	TPL-LIMIT; TPL-INTEGRAL; TPL-GF; TPL-INEQ	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-RA-04; TR-RA-06; TR-CMB-01
2015-D2-P9	TPL-INEQ	TPL-DET; TPL-SPEC; TPL-SHARP; TPL-GEO	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-LA-DET; TR-LA-01; TR-LA-05; TR-RA-07
2015-D2-P10	TPL-POLY	TPL-INEQ; TPL-MOD; TPL-SHARP; TPL-FUNC	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-RA-07; TR-PC-01; TR-ANT-01

Problem	Primary template	Secondary templates	First proof move	Type / triggers
2016-D1-P1	TPL-FUNC	TPL-DERIV; TPL-SHARP; TPL-CASE; TPL-DIR	Substitute special values and freeze parameters to obtain monotonicity, injectivity, or invariance.	PT-FE; TR-RA-02; TR-RA-05; TR-UNI-02
2016-D1-P2	TPL-MOD	TPL-INEQ; TPL-DET; TPL-SHARP; TPL-GROUP	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-07; TR-ANT-01; TR-ANT-03
2016-D1-P3	TPL-MOD	TPL-INEQ; TPL-SHARP; TPL-DIR; TPL-EQUIV	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-07; TR-ANT-01; TR-ANT-03
2016-D1-P4	TPL-COUNT	TPL-MOD; TPL-EQUIV; TPL-DIR; TPL-IND	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-RA-07; TR-ANT-01; TR-ANT-03
2016-D1-P5	TPL-COUNT	TPL-MOD; TPL-GROUP; TPL-FUNC; TPL-INEQ	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-RA-07; TR-ANT-01; TR-ANT-02
2016-D2-P6	TPL-INEQ	TPL-SHARP; TPL-DIR; TPL-EQUIV; TPL-DERIV	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-INEQ; TR-RA-02; TR-RA-04; TR-RA-07
2016-D2-P7	TPL-FUNC	TPL-SHARP; TPL-EXIST; TPL-DERIV; TPL-DIR	Substitute special values and freeze parameters to obtain monotonicity, injectivity, or invariance.	PT-FE; TR-RA-01; TR-RA-02; TR-RA-04
2016-D2-P8	TPL-GROUP	TPL-MOD; TPL-EXIST; TPL-CONSTRUCT; TPL-LIMIT	Translate the statement into generators, relations, subgroups, quotients, or commutators.	PT-GRP; TR-ANT-01; TR-ANT-02; TR-ANT-03
2016-D2-P9	TPL-MOD	TPL-COUNT; TPL-INEQ; TPL-SHARP; TPL-GROUP	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-07; TR-ANT-01; TR-ANT-03
2016-D2-P10	TPL-SPEC	TPL-INEQ; TPL-SHARP; TPL-DET; TPL-POLY	Translate the equation into a polynomial identity in the matrix or a commutator relation.	PT-LA-SPEC; TR-LA-01; TR-LA-05; TR-RA-07
2017-D1-P1	TPL-SPEC	TPL-CLASS; TPL-MOD; TPL-POLY; TPL-DET	Translate the equation into a polynomial identity in the matrix or a commutator relation.	PT-LA-SPEC; TR-LA-01; TR-LA-04; TR-LA-05
2017-D1-P2	TPL-EXIST	TPL-SHARP; TPL-FUNC; TPL-CONSTRUCT; TPL-DIR	Identify the ambient compact/finite set or create one by truncation/localization.	PT-FE; TR-RA-02; TR-RA-04; TR-RA-07
2017-D1-P3	TPL-MOD	TPL-EXIST; TPL-CONSTRUCT; TPL-SHARP; TPL-DIR	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-LA-01; TR-RA-07; TR-ANT-01
2017-D1-P4	TPL-COUNT	TPL-GROUP; TPL-MOD; TPL-GF	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-GRP; TR-ANT-01; TR-ANT-03; TR-ANT-04
2017-D1-P5	TPL-POLY	TPL-MOD; TPL-DERIV; TPL-COMPLEX; TPL-FUNC	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-RA-07; TR-PC-04; TR-ANT-01
2017-D2-P6	TPL-LIMIT	TPL-DERIV; TPL-DIR; TPL-EQUIV; TPL-SHARP	Identify the scale and normalize the expression.	PT-LIM; TR-RA-02; TR-RA-04; TR-ANT-05
2017-D2-P7	TPL-POLY	TPL-MOD; TPL-COMPLEX; TPL-FUNC; TPL-IND	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-RA-07; TR-ANT-01
2017-D2-P8	TPL-MOD	TPL-SPEC; TPL-GF; TPL-IND; TPL-EQUIV	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-LA-01; TR-LA-05; TR-PC-05
2017-D2-P9	TPL-LIMIT	TPL-DERIV; TPL-FUNC; TPL-DIR; TPL-EQUIV	Identify the scale and normalize the expression.	PT-LIM; TR-LA-01; TR-RA-02; TR-RA-04
2017-D2-P10	TPL-MOD	TPL-EXIST; TPL-CONSTRUCT; TPL-SHARP; TPL-GEO	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-07; TR-ANT-01; TR-ANT-03
2018-D1-P1	TPL-INEQ	TPL-DERIV; TPL-INTEGRAL; TPL-EQUIV; TPL-DIR	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-INEQ; TR-RA-02; TR-RA-04; TR-RA-06
2018-D1-P2	TPL-GROUP	TPL-CONTR; TPL-EXIST; TPL-MOD; TPL-CONSTRUCT	Translate the statement into generators, relations, subgroups, quotients, or commutators.	PT-GRP; TR-ANT-01; TR-ANT-03; TR-ANT-04
2018-D1-P3	TPL-DET	TPL-CLASS; TPL-SPEC; TPL-DIR; TPL-EQUIV	Choose a basis, row/column order, or block decomposition adapted to the hypotheses.	PT-LA-DET; TR-LA-01; TR-LA-05; TR-GEO-01
2018-D1-P4	TPL-CLASS	TPL-DET; TPL-FUNC; TPL-DIR; TPL-EQUIV	Start with an arbitrary solution and derive necessary restrictions.	PT-FE; TR-RA-02; TR-RA-04; TR-UNI-03

Problem	Primary template	Secondary templates	First proof move	Type / triggers
2018-D1-P5	TPL-MOD	TPL-INEQ; TPL-GEO; TPL-SHARP; TPL-POLY	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-07; TR-ANT-01; TR-ANT-02
2018-D2-P6	TPL-MOD	TPL-GEO; TPL-DIR; TPL-EQUIV; TPL-SHARP	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-LA-07; TR-RA-07; TR-ANT-01
2018-D2-P7	TPL-LIMIT	TPL-GF; TPL-INTEGRAL; TPL-EQUIV; TPL-DIR	Identify the scale and normalize the expression.	PT-INT; TR-RA-02; TR-RA-04; TR-RA-06
2018-D2-P8	TPL-COUNT	TPL-MOD; TPL-SHARP; TPL-GF; TPL-DET	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-FINITE; TR-LA-01; TR-RA-07; TR-ANT-01
2018-D2-P9	TPL-POLY	TPL-CLASS; TPL-COMPLEX; TPL-FUNC; TPL-DIR	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-LA-01; TR-PC-01; TR-PC-04
2018-D2-P10	TPL-LIMIT	TPL-DIR; TPL-EQUIV; TPL-DERIV; TPL-SHARP	Identify the scale and normalize the expression.	PT-LIM; TR-RA-02; TR-RA-04; TR-ANT-05
2019-D1-P1	TPL-INEQ	TPL-DERIV; TPL-FUNC; TPL-EQUIV; TPL-DIR	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-INEQ; TR-RA-02; TR-RA-04
2019-D1-P2	TPL-MOD	TPL-CLASS; TPL-POLY; TPL-DIR; TPL-EQUIV	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-ANT-01; TR-ANT-03; TR-ANT-04
2019-D1-P3	TPL-INEQ	TPL-SHARP; TPL-FUNC; TPL-DET; TPL-DIR	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-FE; TR-RA-02; TR-RA-04; TR-RA-07
2019-D1-P4	TPL-MOD	TPL-GF; TPL-EQUIV; TPL-DIR; TPL-SHARP	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-07; TR-PC-05; TR-ANT-01
2019-D1-P5	TPL-MOD	TPL-DET; TPL-SPEC; TPL-DIR; TPL-EQUIV	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-LA-01; TR-LA-02; TR-LA-03
2019-D2-P6	TPL-EXIST	TPL-FUNC; TPL-CONSTRUCT; TPL-DERIV; TPL-INTEGRAL	Identify the ambient compact/finite set or create one by truncation/localization.	PT-FE; TR-RA-02; TR-RA-04; TR-RA-05
2019-D2-P7	TPL-MOD	TPL-LIMIT; TPL-DIR; TPL-EQUIV; TPL-INTEGRAL	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-LA-01; TR-RA-04; TR-RA-06
2019-D2-P8	TPL-COUNT	TPL-CONTR; TPL-GF; TPL-INEQ; TPL-LIMIT	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-RA-04; TR-RA-06; TR-ANT-04
2019-D2-P9	TPL-MOD	TPL-CLASS; TPL-DET; TPL-SPEC; TPL-DIR	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-LA-01; TR-LA-02; TR-ANT-01
2019-D2-P10	TPL-COUNT	TPL-GEO; TPL-INTEGRAL; TPL-SHARP; TPL-INEQ	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-RA-02; TR-RA-07; TR-ANT-04
2020-D1-P1	TPL-COUNT	TPL-MOD; TPL-CASE; TPL-SHARP; TPL-EQUIV	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-RA-07; TR-ANT-01; TR-ANT-03
2020-D1-P2	TPL-DET	TPL-SPEC; TPL-INEQ; TPL-COUNT; TPL-SHARP	Choose a basis, row/column order, or block decomposition adapted to the hypotheses.	PT-LA-SPEC; TR-LA-01; TR-LA-02; TR-LA-03
2020-D1-P3	TPL-MOD	TPL-COUNT; TPL-GEO; TPL-EXIST; TPL-CONSTRUCT	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-07; TR-ANT-01; TR-ANT-03
2020-D1-P4	TPL-INEQ	TPL-POLY; TPL-SHARP; TPL-DERIV; TPL-FUNC	Identify the source of the inequality: convexity, Cauchy-Schwarz, Jensen, tangent/chord, rearrangement, AM-GM, monotonicity, or positivity of a square.	PT-INEQ; TR-RA-04; TR-RA-07
2020-D2-P5	TPL-CLASS	TPL-FUNC; TPL-DERIV; TPL-SHARP; TPL-DIR	Start with an arbitrary solution and derive necessary restrictions.	PT-FE; TR-RA-02; TR-RA-03; TR-RA-04
2020-D2-P6	TPL-MOD	TPL-CLASS; TPL-POLY; TPL-DIR; TPL-CASE	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-ANT-01; TR-ANT-02; TR-ANT-03
2020-D2-P7	TPL-GROUP	TPL-EXIST; TPL-COUNT; TPL-MOD; TPL-CONSTRUCT	Translate the statement into generators, relations, subgroups, quotients, or commutators.	PT-GRP; TR-RA-07; TR-ANT-01; TR-ANT-03

Problem	Primary template	Secondary templates	First proof move	Type / triggers
2020-D2-P8	TPL-LIMIT	TPL-DIR; TPL-EQUIV; TPL-DERIV; TPL-SHARP	Identify the scale and normalize the expression.	PT-LIM; TR-RA-02; TR-RA-04; TR-ANT-05
2021-D1-P1	TPL-DET	TPL-SPEC; TPL-EQUIV; TPL-DIR; TPL-IND	Choose a basis, row/column order, or block decomposition adapted to the hypotheses.	PT-LA-DET; TR-LA-01; TR-LA-04; TR-LA-05
2021-D1-P2	TPL-COUNT	TPL-MOD; TPL-GF; TPL-INEQ; TPL-INTEGRAL	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-ANT-01; TR-ANT-03; TR-ANT-04
2021-D1-P3	TPL-EXIST	TPL-GF; TPL-INEQ; TPL-CONSTRUCT; TPL-SHARP	Identify the ambient compact/finite set or create one by truncation/localization.	PT-INEQ; TR-RA-02; TR-RA-04; TR-PC-05
2021-D1-P4	TPL-LIMIT	TPL-EXIST; TPL-FUNC; TPL-CONSTRUCT ; TPL-SHARP	Identify the scale and normalize the expression.	PT-FE; TR-RA-02; TR-RA-04
2021-D2-P5	TPL-MOD	TPL-EXIST; TPL-CONSTRUCT; TPL-SHARP; TPL-SPEC	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-LA-01; TR-RA-07; TR-ANT-01
2021-D2-P6	TPL-GROUP	TPL-MOD; TPL-DERIV; TPL-DIR; TPL-DET	Translate the statement into generators, relations, subgroups, quotients, or commutators.	PT-GRP; TR-ANT-01; TR-ANT-02; TR-ANT-03
2021-D2-P7	TPL-POLY	TPL-INEQ; TPL-COMPLEX; TPL-TOPO; TPL-SHARP	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-RA-07; TR-PC-04; TR-PC-06
2021-D2-P8	TPL-COUNT	TPL-MOD; TPL-GEO; TPL-INEQ; TPL-DIR	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-FINITE; TR-LA-07; TR-RA-07; TR-ANT-01
2022-D1-P1	TPL-COUNT	TPL-INEQ; TPL-SHARP; TPL-INTEGRAL; TPL-DIR	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-RA-02; TR-RA-07; TR-CMB-05
2022-D1-P2	TPL-MOD	TPL-CLASS; TPL-SPEC; TPL-DET; TPL-SHARP	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-LA-01; TR-LA-05; TR-RA-07
2022-D1-P3	TPL-COUNT	TPL-MOD; TPL-SHARP; TPL-CASE; TPL-FUNC	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-FINITE; TR-ANT-01; TR-ANT-02; TR-ANT-03
2022-D1-P4	TPL-COUNT	TPL-MOD; TPL-INEQ; TPL-SHARP; TPL-GROUP	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-FINITE; TR-RA-07; TR-ANT-01; TR-ANT-03
2022-D2-P5	TPL-COUNT	TPL-GEO; TPL-SPEC; TPL-INEQ; TPL-GROUP	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-RA-02; TR-RA-07; TR-ANT-04
2022-D2-P6	TPL-MOD	TPL-COUNT; TPL-GROUP; TPL-IND; TPL-EQUIV	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-COMB; TR-ANT-01; TR-ANT-02; TR-ANT-03
2022-D2-P7	TPL-SPEC	TPL-DET; TPL-INEQ; TPL-SHARP; TPL-POLY	Translate the equation into a polynomial identity in the matrix or a commutator relation.	PT-LA-SPEC; TR-LA-01; TR-LA-02; TR-LA-03
2022-D2-P8	TPL-COUNT	TPL-MOD; TPL-SHARP; TPL-INEQ; TPL-INTEGRAL	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-FINITE; TR-RA-07; TR-ANT-01; TR-ANT-03
2023-D1-P1	TPL-CLASS	TPL-FUNC; TPL-DERIV; TPL-DET; TPL-DIR	Start with an arbitrary solution and derive necessary restrictions.	PT-FE; TR-RA-02; TR-RA-04; TR-UNI-03
2023-D1-P2	TPL-DET	TPL-DERIV; TPL-SPEC; TPL-POLY; TPL-EQUIV	Choose a basis, row/column order, or block decomposition adapted to the hypotheses.	PT-LA-DET; TR-LA-01; TR-LA-04; TR-LA-05
2023-D1-P3	TPL-POLY	TPL-CLASS; TPL-COMPLEX; TPL-FUNC; TPL-DIR	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-UNI-03
2023-D1-P4	TPL-MOD	TPL-POLY; TPL-DIR; TPL-EQUIV; TPL-INEQ	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-ANT-01; TR-ANT-02; TR-ANT-03
2023-D1-P5	TPL-COUNT	TPL-MOD; TPL-EQUIV; TPL-DIR; TPL-GROUP	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-RA-07; TR-ANT-01; TR-ANT-03
2023-D2-P6	TPL-DET	TPL-SPEC; TPL-DIR; TPL-EQUIV; TPL-INEQ	Choose a basis, row/column order, or block decomposition adapted to the hypotheses.	PT-LA-DET; TR-LA-01; TR-LA-02; TR-LA-05
2023-D2-P7	TPL-CLASS	TPL-FUNC; TPL-DERIV; TPL-COUNT; TPL-DIR	Start with an arbitrary solution and derive necessary restrictions.	PT-FE; TR-LA-01; TR-RA-02; TR-RA-04
2023-D2-P8	TPL-COUNT	TPL-INEQ; TPL-GEO; TPL-TOPO; TPL-SHARP	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-COMB; TR-RA-01; TR-RA-04; TR-RA-06
2023-D2-P9	TPL-GEO	TPL-COUNT; TPL-DET; TPL-SHARP; TPL-TOPO	Normalize by translation, rotation, scaling, affine transformation, or coordinate choice.	PT-GEO; TR-LA-01; TR-RA-07; TR-ANT-04

Problem	Primary template	Secondary templates	First proof move	Type / triggers
2023-D2-P10	TPL-MOD	TPL-EXIST; TPL-CONSTRUCT; TPL-SHARP; TPL-DIR	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-LA-01; TR-RA-07; TR-ANT-01
2024-D1-P1	TPL-POLY	TPL-CLASS; TPL-DIR; TPL-EQUIV; TPL-FUNC	Move between root factorization and coefficient identities using Vieta or logarithmic derivatives.	PT-POLY; TR-LA-01; TR-UNI-03
2024-D1-P2	TPL-LIMIT	TPL-SHARP; TPL-DIR; TPL-EQUIV; TPL-DERIV	Identify the scale and normalize the expression.	PT-LIM; TR-RA-02; TR-RA-04; TR-RA-07
2024-D1-P3	TPL-MOD	TPL-CONTR; TPL-EXIST; TPL-CONSTRUCT; TPL-DET	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-ANT-01; TR-ANT-03; TR-ANT-04
2024-D1-P4	TPL-GROUP	TPL-MOD; TPL-DERIV; TPL-DIR; TPL-FUNC	Translate the statement into generators, relations, subgroups, quotients, or commutators.	PT-GRP; TR-RA-05; TR-RA-07; TR-ANT-01
2024-D1-P5	TPL-MOD	TPL-INEQ; TPL-COUNT; TPL-SHARP; TPL-GEO	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-04; TR-RA-07; TR-ANT-01
2024-D2-P6	TPL-FUNC	TPL-INEQ; TPL-SHARP; TPL-DIR; TPL-EQUIV	Substitute special values and freeze parameters to obtain monotonicity, injectivity, or invariance.	PT-FE; TR-RA-05; TR-RA-07
2024-D2-P7	TPL-DET	TPL-CLASS; TPL-SPEC; TPL-MOD; TPL-INEQ	Choose a basis, row/column order, or block decomposition adapted to the hypotheses.	PT-LA-DET; TR-LA-01; TR-LA-02; TR-LA-03
2024-D2-P8	TPL-LIMIT	TPL-INEQ; TPL-GF; TPL-DIR; TPL-EQUIV	Identify the scale and normalize the expression.	PT-LIM; TR-RA-02; TR-RA-04; TR-RA-07
2024-D2-P9	TPL-COUNT	TPL-DET; TPL-MOD; TPL-SHARP; TPL-FUNC	Choose the right encoding: graph vertices/edges, incidence pairs, words, states, functions, or subsets.	PT-FINITE; TR-LA-02; TR-RA-07; TR-ANT-01
2024-D2-P10	TPL-MOD	TPL-POLY; TPL-SHARP; TPL-IND; TPL-EQUIV	Decide the correct modulus: prime field for inverses/root-counts, prime powers for valuations, or CRT moduli for construction.	PT-FINITE; TR-RA-04; TR-RA-07; TR-ANT-01

34 Reproducible Proof-Review Protocol

1. **Template prediction.** Before solving, write one primary template and at most two backup templates. This prevents uncontrolled wandering among methods.
2. **Skeleton first.** Write only the skeleton: object, invariant/auxiliary construction, closure theorem, and edge cases. Do not start algebra before this skeleton exists.
3. **Proof execution.** Fill the skeleton line by line, explicitly naming nontrivial theorems and hypotheses.
4. **Rigor audit.** Mark every step as reversible, one-way implication, estimate, construction, or theorem application. Check zero denominators, compactness, endpoints, equality cases, and convergence justifications.
5. **Post-solution extraction.** Record whether failure was theory, template recognition, method execution, calculation, or proof writing. Update the problem row in the structured CSV accordingly.

34.1 Minimal written-solution standard

A competition solution should contain: (i) a precise target statement; (ii) definitions of introduced objects; (iii) a justified reduction or normalization; (iv) the main proof template move; (v) the closure theorem or final calculation; (vi) edge/equality/construction checks; and (vii) a final sentence returning to the original statement. Any solution missing one of these seven elements should be treated as incomplete during practice, even if the main idea is correct.

Chapter 3

Problems and Solutions

3.1 1994

Day 1

1994-D1-P1 — Statement

Day 1, P1 | 13 points

a) Let A be a $n \times n$, $n \geq 2$, symmetric, invertible matrix with real positive elements. Show that $z_n \leq n^2 - 2n$, where z_n is the number of zero elements in A^{-1}.

b) How many zero elements are there in the inverse of the $n \times n$ matrix

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & \dots & 1 \\ 1 & 2 & 2 & 2 & \dots & 2 \\ 1 & 2 & 1 & 1 & \dots & 1 \\ 1 & 2 & 1 & 2 & \dots & 2 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 1 & 2 & 1 & 2 & \dots & \dots \end{pmatrix} ?$$

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Theory and methods for 1994-D1-P1

Theory Determinants, block identities and inverses [T-LA-DETERMINANT]; Rank, dimension and exact linear constraints [T-LA-RANK]
Methods Read the entries before choosing a theorem [M-LA-ENTRY]
Strategy Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

1994-D1-P1 — Solution

For this record: Theory: T-LA-DETERMINANT; T-LA-RANK. Methods: M-LA-ENTRY. Full learning route

Denote by a_{ij} and b_{ij} the elements of A and A^{-1}, respectively. Then for $k \neq m$ we have $\sum_{i=1}^{n} a_{ki} b_{im} = 0$ and from the positivity of a_{ij} we conclude that at least one of $\{b_{im} : i = 1, 2, \dots, n\}$ is positive and at least one is negative. Hence we have at least two non-zero elements in every column of A^{-1}. This proves part a). For part b) all b_{ij} are zero except $b_{1,1} = 2$, $b_{n,n} = (-1)^n$, $b_{i,i+1} = b_{i+1,i} = (-1)^i$ for $i = 1, 2, \dots, n-1$.

1994-D1-P2 — Statement

Day 1, P2 | 13 points

Let $f \in C^1(a, b)$, $\lim_{x \to a+} f(x) = +\infty$, $\lim_{x \to b-} f(x) = -\infty$ and $f'(x) + f^2(x) \geq -1$ for $x \in (a, b)$. Prove that $b - a \geq \pi$ and give an example where $b - a = \pi$.

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Theory and methods for 1994-D1-P2

Theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Differential inequalities and integrating factors [T-AN-ODE]
Methods Design an auxiliary derivative [M-AN-AUXILIARY]
Strategy Control the limiting operation before taking the limit [S-ANALYTIC]; Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

1994-D1-P2 — Solution

For this record: Theory: T-AN-DERIVATIVE; T-AN-ODE. Methods: M-AN-AUXILIARY. Full learning route

From the inequality we get

$$\frac{d}{dx}\left(\operatorname{arctg} f(x)+x\right)=\frac{f'(x)}{1+f^2(x)}+1\geq 0$$

for $x\in(a,b)$. Thus $\operatorname{arctg} f(x)+x$ is non-decreasing in the interval and using the limits we get $\frac{\pi}{2}+a\leq -\frac{\pi}{2}+b$. Hence $b-a\geq\pi$. One has equality for $f(x)=\operatorname{cotg} x$, $a=0$, $b=\pi$.

1994-D1-P3 — Statement

Day 1, P3 | 13 points

Given a set S of $2n-1$, $n\in\mathbb{N}$, different irrational numbers. Prove that there are n different elements $x_1,x_2,\dots,x_n\in S$ such that for all non-negative rational numbers $a_1,a_2,\dots,a_n$ with $a_1+a_2+\dots+a_n>0$ we have that $a_1x_1+a_2x_2+\dots+a_nx_n$ is an irrational number.

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Theory and methods for 1994-D1-P3

Theory Rational linear relations and discrete-value rigidity [T-AL-RATIONALITY]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Methods Eliminate rational relations, then exploit discreteness [M-AL-RATIONAL]
Strategy Reduce size while preserving the defining conditions [S-INDUCTION]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

1994-D1-P3 — Solution

For this record: Theory: T-AL-RATIONALITY; T-DS-INDUCTION. Methods: M-AL-RATIONAL. Full learning route

Let I be the set of irrational numbers, $\mathbb{Q}$ – the set of rational numbers, $\mathbb{Q}^+=\mathbb{Q}\cap[0,\infty)$. We work by induction. For $n=1$ the statement is trivial. Let it be true for $n-1$. We start to prove it for n. From the induction argument there are $n-1$ different elements $x_1,x_2,\dots,x_{n-1}\in S$ such that

$$a_1x_1+a_2x_2+\dots+a_{n-1}x_{n-1}\in I \tag{1}$$

for all $a_1,a_2,\dots,a_n\in\mathbb{Q}^+$ with $a_1+a_2+\dots+a_{n-1}>0$.

Denote the other elements of S by $x_n,x_{n+1},\dots,x_{2n-1}$. Assume the statement is not true for n. Then for $k=0,1,\dots,n-1$ there are $r_k\in\mathbb{Q}$ such that

$$\sum_{i=1}^{n-1}b_{ik}x_i+c_kx_{n+k}=r_k\quad\text{for some } b_{ik},c_k\in\mathbb{Q}^+,\ \sum_{i=1}^{n-1}b_{ik}+c_k>0. \tag{2}$$

Also

$$\sum_{k=0}^{n-1}d_kx_{n+k}=R\quad\text{for some } d_k\in\mathbb{Q}^+,\ \sum_{k=0}^{n-1}d_k>0,\ R\in\mathbb{Q}. \tag{3}$$

1994-D1-P3 — Solution (continued)

If in (2) $c_k = 0$ then (2) contradicts (1). Thus $c_k \neq 0$ and without loss of generality one may take $c_k = 1$. In (2) also $\sum_{i=1}^{n-1} b_{ik} > 0$ in view of $x_{n+k} \in I$. Replacing (2) in (3) we get

$$\sum_{k=0}^{n-1} d_k \left(-\sum_{i=1}^{n-1} b_{ik} x_i + r_k \right) = R \quad \text{or} \quad \sum_{i=1}^{n-1} \left(\sum_{k=0}^{n-1} d_k b_{ik} \right) x_i \in \mathbb{Q},$$

which contradicts (1) because of the conditions on b's and d's.

1994-D1-P4 — Statement

Day 1, P4 | 18 points

Let $\alpha \in \mathbb{R} \setminus \{0\}$ and suppose that F and G are linear maps (operators) from $\mathbb{R}^n$ into $\mathbb{R}^n$ satisfying $F \circ G - G \circ F = \alpha F$.
a) Show that for all $k \in \mathbb{N}$ one has $F^k \circ G - G \circ F^k = \alpha k F^k$.

b) Show that there exists $k \geq 1$ such that $F^k = 0$.

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Theory and methods for 1994-D1-P4

Theory Trace pairing and commutator calculus [T-LA-TRACE]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Nilpotents, projections and involutions [T-LA-NILPOTENT]
Methods Propagate a commutator by telescoping [M-LA-DERIVATION]; Treat the unknown matrix as a vector [M-LA-MATRIXOP]
Strategy Reduce size while preserving the defining conditions [S-INDUCTION]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

1994-D1-P4 — Solution

For this record: Theory: T-LA-TRACE; T-LA-SPECTRAL; T-LA-NILPOTENT. Methods: M-LA-DERIVATION; M-LA-MATRIXOP. Full learning route

For a) using the assumptions we have

$$\begin{aligned} F^k \circ G - G \circ F^k &= \sum_{i=1}^{k} \left(F^{k-i+1} \circ G \circ F^{i-1} - F^{k-i} \circ G \circ F^i \right) = \\ &= \sum_{i=1}^{k} F^{k-i} \circ (F \circ G - G \circ F) \circ F^{i-1} = \\ &= \sum_{i=1}^{k} F^{k-i} \circ \alpha F \circ F^{i-1} = \alpha k F^k. \end{aligned}$$

b) Consider the linear operator $L(F) = F \circ G - G \circ F$ acting over all $n \times n$ matrices F. It may have at most n^2 different eigenvalues. Assuming that $F^k \neq 0$ for every k we get that L has infinitely many different eigenvalues αk in view of a) – a contradiction.

1994-D1-P5 — Statement

Day 1, P5 | 18 points

a) Let $f \in C[0, b]$, $g \in C(\mathbb{R})$ and let g be periodic with period b. Prove that $\int_0^b f(x) g(nx)\, dx$ has a limit as $n \to \infty$ and

$$\lim_{n \to \infty} \int_0^b f(x) g(nx)\, dx = \frac{1}{b} \int_0^b f(x)\, dx \cdot \int_0^b g(x)\, dx.$$

1994-D1-P5 — Statement (continued)

b) Find

$$\lim_{n\to\infty}\int_0^{\pi}\frac{\sin x}{1+3\cos^2 nx}\,dx.$$

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Theory and methods for 1994-D1-P5

Theory Continuity, compactness, and order [T-AN-CONTINUITY]; Riemann sums, periodic averaging, and tails [T-AN-RIEMANN]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Trigonometric identities and safe branches [T-AN-TRIG]
Methods Expose the mesh and control its tails [M-AN-RIEMANN]; Pair an integral through symmetry [M-AN-PAIR]
Strategy Control the limiting operation before taking the limit [S-ANALYTIC]; Pass between local constraints and a global conclusion [S-LOCALGLOBAL]

Statement Recorded solutions Catalog

1994-D1-P5 — Solution

For this record: Theory: T-AN-CONTINUITY; T-AN-RIEMANN; T-AN-INTEGRAL; T-AN-TRIG. Methods: M-AN-RIEMANN; M-AN-PAIR. Full learning route

Set $\|g\|_1=\int_0^b|g(x)|\,dx$ and

$$\omega(f,t)=\sup\{|f(x)-f(y)|:x,y\in[0,b],\ |x-y|\le t\}.$$

In view of the uniform continuity of f we have $\omega(f,t)\to 0$ as $t\to 0$. Using the periodicity of g we get

$$\begin{aligned}
\int_0^b f(x)g(nx)\,dx&=\sum_{k=1}^n\int_{b(k-1)/n}^{bk/n}f(x)g(nx)\,dx\\
&=\sum_{k=1}^n f(bk/n)\int_{b(k-1)/n}^{bk/n}g(nx)\,dx+\sum_{k=1}^n\int_{b(k-1)/n}^{bk/n}\{f(x)-f(bk/n)\}g(nx)\,dx\\
&=\frac{1}{n}\sum_{k=1}^n f(bk/n)\int_0^b g(x)\,dx+O(\omega(f,b/n)\|g\|_1)\\
&=\frac{1}{b}\sum_{k=1}^n\int_{b(k-1)/n}^{bk/n}f(x)\,dx\int_0^b g(x)\,dx\\
&\quad+\frac{1}{b}\sum_{k=1}^n\left(\frac{b}{n}f(bk/n)-\int_{b(k-1)/n}^{bk/n}f(x)\,dx\right)\int_0^b g(x)\,dx+O(\omega(f,b/n)\|g\|_1)\\
&=\frac{1}{b}\int_0^b f(x)\,dx\int_0^b g(x)\,dx+O(\omega(f,b/n)\|g\|_1).
\end{aligned}$$

This proves a). For b) we set $b=\pi$, $f(x)=\sin x$, $g(x)=(1+3\cos^2 x)^{-1}$. From a) and

$$\int_0^{\pi}\sin x\,dx=2,\qquad\int_0^{\pi}(1+3\cos^2 x)^{-1}\,dx=\frac{\pi}{2}$$

we get

$$\lim_{n\to\infty}\int_0^{\pi}\frac{\sin x}{1+3\cos^2 nx}\,dx=1.$$

1994-D1-P6 — Statement

Day 1, P6 | 25 points

Let $f \in C^2[0,N]$ and $|f'(x)| < 1$, $f''(x) > 0$ for every $x \in [0,N]$. Let $0 \le m_0 < m_1 < \cdots < m_k \le N$ be integers such that $n_i = f(m_i)$ are also integers for $i = 0, 1, \ldots, k$. Denote $b_i = n_i - n_{i-1}$ and $a_i = m_i - m_{i-1}$ for $i = 1, 2, \ldots, k$.

a) Prove that

$$-1 < \frac{b_1}{a_1} < \frac{b_2}{a_2} < \cdots < \frac{b_k}{a_k} < 1.$$

b) Prove that for every choice of $A > 1$ there are no more than N/A indices j such that $a_j > A$.

c) Prove that $k \le 3N^{2/3}$ (i.e. there are no more than $3N^{2/3}$ integer points on the curve $y = f(x)$, $x \in [0,N]$).

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Theory and methods for 1994-D1-P6

Theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Localize an extremum or an accumulation point [M-AN-COMPACT]; Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

1994-D1-P6 — Solution

For this record: Theory: T-AN-DERIVATIVE; T-AN-CONVEXITY; T-DS-COUNTING. Methods: M-AN-COMPACT; M-AN-WEIGHTS. Full learning route

a) For $i = 1, 2, \ldots, k$ we have

$$b_i = f(m_i) - f(m_{i-1}) = (m_i - m_{i-1}) f'(x_i)$$

for some $x_i \in (m_{i-1}, m_i)$. Hence $\frac{b_i}{a_i} = f'(x_i)$ and so $-1 < \frac{b_i}{a_i} < 1$. From the convexity of f we have that f' is increasing and $\frac{b_i}{a_i} = f'(x_i) < f'(x_{i+1}) = \frac{b_{i+1}}{a_{i+1}}$ because of $x_i < m_i < x_{i+1}$.

b) Set $S_A = \{j \in \{0, 1, \ldots, k\} : a_j > A\}$. Then

$$N \ge m_k - m_0 = \sum_{i=1}^{k} a_i \ge \sum_{j \in S_A} a_j > A|S_A|$$

and hence $|S_A| < N/A$.

c) All different fractions in $(-1,1)$ with denominators less or equal A are no more $2A^2$. Using b) we get $k < N/A + 2A^2$. Put $A = N^{1/3}$ in the above estimate and get $k < 3N^{2/3}$.

Day 2

1994-D2-P1 — Statement

Day 2, P7 | 14 points

Let $f \in C^1[a,b]$, $f(a) = 0$ and suppose that $\lambda \in \mathbb{R}$, $\lambda > 0$, is such that

$$|f'(x)| \le \lambda |f(x)|$$

for all $x \in [a,b]$. Is it true that $f(x) = 0$ for all $x \in [a,b]$?

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Theory and methods for 1994-D2-P1

Theory Continuity, compactness, and order [T-AN-CONTINUITY]; Differential inequalities and integrating factors [T-AN-ODE]
Methods Design an auxiliary derivative [M-AN-AUXILIARY]; Use a first crossing or a forced-growth contradiction [M-AN-BARRIER]
Strategy Make the contradiction quantitative or structural [S-CONTRADICTION]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

1994-D2-P1 — Solution

For this record: Theory: T-AN-CONTINUITY; T-AN-ODE. Methods: M-AN-AUXILIARY; M-AN-BARRIER. Full learning route

Assume that there is $y \in (a,b]$ such that $f(y) \neq 0$. Without loss of generality we have $f(y) > 0$. In view of the continuity of f there exists $c \in [a,y)$ such that $f(c) = 0$ and $f(x) > 0$ for $x \in (c,y]$. For $x \in (c,y]$ we have $|f'(x)| \leq \lambda f(x)$. This implies that the function $g(x) = \ln f(x) - \lambda x$ is not increasing in $(c,y]$ because of $g'(x) = \frac{f'(x)}{f(x)} - \lambda \leq 0$. Thus $\ln f(x) - \lambda x \geq \ln f(y) - \lambda y$ and $f(x) \geq e^{\lambda x - \lambda y} f(y)$ for $x \in (c,y]$. Thus

$$0 = f(c) = f(c+0) \geq e^{\lambda c - \lambda y} f(y) > 0$$

— a contradiction. Hence one has $f(x) = 0$ for all $x \in [a,b]$.

1994-D2-P2 — Statement

Day 2, P8 | 14 points

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be given by $f(x,y) = (x^2 - y^2)e^{-x^2-y^2}$.
a) Prove that f attains its minimum and its maximum.

b) Determine all points (x,y) such that $\frac{\partial f}{\partial x}(x,y) = \frac{\partial f}{\partial y}(x,y) = 0$ and determine for which of them f has global or local minimum or maximum.

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Theory and methods for 1994-D2-P2

Theory Continuity, compactness, and order [T-AN-CONTINUITY]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Methods Localize an extremum or an accumulation point [M-AN-COMPACT]
Strategy Choose an extremal object and force rigidity [S-EXTREMAL]; Prove necessity and sufficiency separately [S-CLASSIFICATION]

Statement Recorded solutions Catalog

1994-D2-P2 — Solution

For this record: Theory: T-AN-CONTINUITY; T-AN-DERIVATIVE. Methods: M-AN-COMPACT. Full learning route

We have $f(1,0) = e^{-1}$, $f(0,1) = -e^{-1}$ and $te^{-t} \leq 2e^{-2}$ for $t \geq 2$. Therefore $|f(x,y)| \leq (x^2+y^2)e^{-x^2-y^2} \leq 2e^{-2} < e^{-1}$ for $(x,y) \notin M = \{(u,v) : u^2 + v^2 \leq 2\}$ and f cannot attain its minimum and its maximum outside M. Part a) follows from the compactness of M and the continuity of f. Let (x,y) be a point from part b). From $\frac{\partial f}{\partial x}(x,y) = 2x(1 - x^2 + y^2)e^{-x^2-y^2}$ we get

$$x(1 - x^2 + y^2) = 0. \tag{1}$$

Similarly

$$y(1 + x^2 - y^2) = 0. \tag{2}$$

All solutions (x,y) of the system (1), (2) are $(0,0)$, $(0,1)$, $(0,-1)$, $(1,0)$ and $(-1,0)$. One has $f(1,0) = f(-1,0) = e^{-1}$ and f has global maximum at the points $(1,0)$ and $(-1,0)$. One has $f(0,1) = f(0,-1) = -e^{-1}$ and f has global minimum at the points $(0,1)$ and $(0,-1)$. The point $(0,0)$ is not an extrema point because of $f(x,0) = x^2e^{-x^2} > 0$ if $x \neq 0$ and $f(y,0) = -y^2e^{-y^2} < 0$ if $y \neq 0$.

1994-D2-P3 — Statement

Day 2, P9 | 14 points

Let f be a real-valued function with $n+1$ derivatives at each point of $\mathbb{R}$. Show that for each pair of real numbers a, b, $a < b$, such that

$$\ln\left(\frac{f(b)+f'(b)+\cdots+f^{(n)}(b)}{f(a)+f'(a)+\cdots+f^{(n)}(a)}\right)=b-a$$

there is a number c in the open interval (a,b) for which

$$f^{(n+1)}(c)=f(c).$$

Note that ln denotes the natural logarithm.

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Theory and methods for 1994-D2-P3

Theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Differential inequalities and integrating factors [T-AN-ODE]
Methods Count zeros and subtract an interpolant [M-AN-ROLLE]; Design an auxiliary derivative [M-AN-AUXILIARY]
Strategy Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

1994-D2-P3 — Solution

For this record: Theory: T-AN-DERIVATIVE; T-AN-ODE. Methods: M-AN-ROLLE; M-AN-AUXILIARY. Full learning route

Set $g(x)=\left(f(x)+f'(x)+\cdots+f^{(n)}(x)\right)e^{-x}$. From the assumption one get $g(a)=g(b)$. Then there exists $c\in(a,b)$ such that $g'(c)=0$. Replacing in the last equality $g'(x)=\left(f^{(n+1)}(x)-f(x)\right)e^{-x}$ we finish the proof.

1994-D2-P4 — Statement

Day 2, P10 | 18 points

Let A be a $n\times n$ diagonal matrix with characteristic polynomial

$$(x-c_1)^{d_1}(x-c_2)^{d_2}\ldots(x-c_k)^{d_k},$$

where $c_1,c_2,\ldots,c_k$ are distinct (which means that c_1 appears d_1 times on the diagonal, c_2 appears d_2 times on the diagonal, etc. and $d_1+d_2+\cdots+d_k=n$). Let V be the space of all $n\times n$ matrices B such that $AB=BA$. Prove that the dimension of V is

$$d_1^2+d_2^2+\cdots+d_k^2.$$

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Theory and methods for 1994-D2-P4

Theory Rank, dimension and exact linear constraints [T-LA-RANK]; Adapted bases, centralizers and common eigenspaces [T-LA-INVARIANT]
Methods Read the entries before choosing a theorem [M-LA-ENTRY]; Choose a basis adapted to the structure [M-LA-BASIS]
Strategy Expose the structure before calculating [S-STRUCTURE]; Choose the objects and count them without ambiguity [S-COUNTING]

Statement Recorded solutions Catalog

1994-D2-P4 — Solution

For this record: Theory: T-LA-RANK; T-LA-INVARIANT. Methods: M-LA-ENTRY; M-LA-BASIS. Full learning route

Set $A=(a_{ij})_{i,j=1}^n$, $B=(b_{ij})_{i,j=1}^n$, $AB=(x_{ij})_{i,j=1}^n$ and $BA=(y_{ij})_{i,j=1}^n$. Then $x_{ij}=a_{ii}b_{ij}$ and $y_{ij}=a_{jj}b_{ij}$. Thus $AB=BA$ is equivalent to $(a_{ii}-a_{jj})b_{ij}=0$ for $i,j=1,2,\dots,n$. Therefore $b_{ij}=0$ if $a_{ii}\neq a_{jj}$ and b_{ij} may be arbitrary if $a_{ii}=a_{jj}$. The number of indices (i,j) for which $a_{ii}=a_{jj}=c_m$ for some $m=1,2,\dots,k$ is d_m^2. This gives the desired result.

1994-D2-P5 — Statement

Day 2, P11 | 18 points

Let $x_1,x_2,\dots,x_k$ be vectors of m-dimensional Euclidian

space, such that $x_1+x_2+\cdots+x_k=0$. Show that there exists a permutation π of the integers $\{1,2,\dots,k\}$ such that

$$\left\|\sum_{i=1}^n x_{\pi(i)}\right\| \leq \left(\sum_{i=1}^k \|x_i\|^2\right)^{1/2} \quad \text{for each } n=1,2,\dots,k.$$

Note that $\|\cdot\|$ denotes the Euclidian norm.

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Theory and methods for 1994-D2-P5

Theory Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Inner products, orthogonality and polarization [T-LA-INNER]
Methods Strengthen the statement before the induction step [M-DS-INDUCT]
Strategy Build a witness and verify every constraint [S-CONSTRUCTION]; Reduce size while preserving the defining conditions [S-INDUCTION]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

1994-D2-P5 — Solution

For this record: Theory: T-DS-INDUCTION; T-LA-INNER. Methods: M-DS-INDUCT. Full learning route

We define π inductively. Set $\pi(1)=1$. Assume π is defined for $i=1,2,\dots,n$ and also

$$\left\|\sum_{i=1}^n x_{\pi(i)}\right\|^2 \leq \sum_{i=1}^n \|x_{\pi(i)}\|^2. \tag{1}$$

Note (1) is true for $n=1$. We choose $\pi(n+1)$ in a way that (1) is fulfilled with $n+1$ instead of n. Set $y=\sum_{i=1}^n x_{\pi(i)}$ and $A=\{1,2,\dots,k\}\setminus\{\pi(i):i=1,2,\dots,n\}$. Assume that $(y,x_r)>0$ for all $r\in A$. Then $\left(y,\sum_{r\in A}x_r\right)>0$ and in view of $y+\sum_{r\in A}x_r=0$ one gets $-(y,y)>0$, which is impossible. Therefore there is $r\in A$ such that

$$(y,x_r)\leq 0. \tag{2}$$

Put $\pi(n+1)=r$. Then using (2) and (1) we have

$$\left\|\sum_{i=1}^{n+1} x_{\pi(i)}\right\|^2 = \|y+x_r\|^2 = \|y\|^2+2(y,x_r)+\|x_r\|^2 \leq \|y\|^2+\|x_r\|^2 \leq \sum_{i=1}^n \|x_{\pi(i)}\|^2+\|x_r\|^2 = \sum_{i=1}^{n+1}\|x_{\pi(i)}\|^2,$$

which verifies (1) for $n+1$. Thus we define π for every $n=1,2,\dots,k$. Finally from (1) we get

$$\left\|\sum_{i=1}^n x_{\pi(i)}\right\|^2 \leq \sum_{i=1}^n \|x_{\pi(i)}\|^2 \leq \sum_{i=1}^k \|x_i\|^2.$$

1994-D2-P6 — Statement

Day 2, P12 | 22 points

Find

$$\lim_{N\to\infty} \frac{\ln^2 N}{N} \sum_{k=2}^{N-2} \frac{1}{\ln k \cdot \ln(N-k)}.$$

Note that ln denotes the natural logarithm.

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Theory and methods for 1994-D2-P6

Theory Scale separation and limit comparisons [T-AN-ASYMPTOTICS]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Methods Separate a main range from exceptional scales [M-AN-SPLIT]
Strategy Build a chain of bounds and track equality [S-BOUND]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

1994-D2-P6 — Solution

For this record: Theory: T-AN-ASYMPTOTICS; T-AN-CONVERGENCE. Methods: M-AN-SPLIT. Full learning route

Obviously

$$A_N = \frac{\ln^2 N}{N} \sum_{k=2}^{N-2} \frac{1}{\ln k \cdot \ln(N-k)} \geq \frac{\ln^2 N}{N} \cdot \frac{N-3}{\ln^2 N} = 1 - \frac{3}{N}. \tag{1}$$

Take M, $2 \leq M < N/2$. Then using that $\dfrac{1}{\ln k \cdot \ln(N-k)}$ is decreasing in $[2, N/2]$ and the symmetry with respect to $N/2$ one get

$$\begin{aligned} A_N &= \frac{\ln^2 N}{N} \left\{ \sum_{k=2}^{M} + \sum_{k=M+1}^{N-M-1} + \sum_{k=N-M}^{N-2} \right\} \frac{1}{\ln k \cdot \ln(N-k)} \leq \\ &\leq \frac{\ln^2 N}{N} \left\{ 2\frac{M-1}{\ln 2 \cdot \ln(N-2)} + \frac{N-2M-1}{\ln M \cdot \ln(N-M)} \right\} \leq \\ &\leq \frac{2}{\ln 2} \cdot \frac{M \ln N}{N} + \left(1 - \frac{2M}{N}\right) \frac{\ln N}{\ln M} + O\left(\frac{1}{\ln N}\right). \end{aligned}$$

Choose $M = \left[\dfrac{N}{\ln^2 N}\right] + 1$ to get

$$A_N \leq \left(1 - \frac{2}{N \ln^2 N}\right) \frac{\ln N}{\ln N - 2\ln\ln N} + O\left(\frac{1}{\ln N}\right) \leq 1 + O\left(\frac{\ln\ln N}{\ln N}\right). \tag{2}$$

Estimates (1) and (2) give

$$\lim_{N\to\infty} \frac{\ln^2 N}{N} \sum_{k=2}^{N-2} \frac{1}{\ln k \cdot \ln(N-k)} = 1.$$

3.2 1995

Day 1

1995-D1-P1 — Statement

Day 1, P1 | 10 points

Let X be a nonsingular matrix with columns $X_1, X_2, \dots, X_n$. Let Y be a matrix with columns $X_2, X_3, \dots, X_n, 0$. Show that the matrices $A = YX^{-1}$ and $B = X^{-1}Y$ have rank $n-1$ and have only 0's for eigenvalues.

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Theory and methods for 1995-D1-P1

Theory Rank, dimension and exact linear constraints [T-LA-RANK]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Nilpotents, projections and involutions [T-LA-NILPOTENT]
Methods Choose a basis adapted to the structure [M-LA-BASIS]
Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

1995-D1-P1 — Solution

For this record: Theory: T-LA-RANK; T-LA-SPECTRAL; T-LA-NILPOTENT. Methods: M-LA-BASIS. Full learning route

Let $J = (a_{ij})$ be the $n \times n$ matrix where $a_{ij} = 1$ if $i = j+1$ and $a_{ij} = 0$ otherwise. The rank of J is $n-1$ and its only eigenvalues are 0's. Moreover $Y = XJ$ and $A = YX^{-1} = XJX^{-1}$, $B = X^{-1}Y = J$. It follows that both A and B have rank $n-1$ with only 0's for eigenvalues.

1995-D1-P2 — Statement

Day 1, P2 | 15 points

Let f be a continuous function on $[0,1]$ such that for every $x \in [0,1]$ we have $\int_x^1 f(t)\,dt \geq \frac{1-x^2}{2}$. Show that $\int_0^1 f^2(t)\,dt \geq \frac{1}{3}$.

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Theory and methods for 1995-D1-P2

Theory Nonnegative sums, Fubini, and integral kernels [T-AN-FUBINI]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]
Methods Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]; Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]
Strategy Expose the structure before calculating [S-STRUCTURE]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

1995-D1-P2 — Solution

For this record: Theory: T-AN-FUBINI; T-AN-INTEGRAL; T-AN-INEQUALITIES. Methods: M-AN-PRIMITIVE; M-AN-WEIGHTS. Full learning route

From the inequality

$$0 \leq \int_0^1 (f(x)-x)^2 dx = \int_0^1 f^2(x)\,dx - 2\int_0^1 x f(x)\,dx + \int_0^1 x^2 dx$$

1995-D1-P2 — Solution (continued)

we get

$$\int_0^1 f^2(x)\,dx \geq 2\int_0^1 xf(x)\,dx - \int_0^1 x^2 dx = 2\int_0^1 xf(x)\,dx - \frac{1}{3}.$$

From the hypotheses we have $\int_0^1 \int_x^1 f(t)\,dt\,dx \geq \int_0^1 \frac{1-x^2}{2}\,dx$ or $\int_0^1 tf(t)\,dt \geq \frac{1}{3}$. This completes the proof.

1995-D1-P3 — Statement

Day 1, P3 | 15 points

Let f be twice continuously differentiable on $(0,+\infty)$ such that $\lim_{x\to 0+} f'(x) = -\infty$ and $\lim_{x\to 0+} f''(x) = +\infty$. Show that

$$\lim_{x\to 0+} \frac{f(x)}{f'(x)} = 0.$$

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Theory and methods for 1995-D1-P3

Theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Methods Expand near equality, bound away from it [M-AN-LOCAL]; Compare growth through derivatives [M-AN-RATIO]
Strategy Control the limiting operation before taking the limit [S-ANALYTIC]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

1995-D1-P3 — Solution

For this record: Theory: T-AN-DERIVATIVE; T-AN-CONVEXITY; T-AN-CONVERGENCE. Methods: M-AN-LOCAL; M-AN-RATIO. Full learning route

Since f' tends to $-\infty$ and f'' tends to $+\infty$ as x tends to $0+$, there exists an interval $(0,r)$ such that $f'(x) < 0$ and $f''(x) > 0$ for all $x \in (0,r)$. Hence f is decreasing and f' is increasing on $(0,r)$. By the mean value theorem for every $0 < x < x_0 < r$ we obtain

$$f(x) - f(x_0) = f'(\xi)(x - x_0) > 0,$$

for some $\xi \in (x, x_0)$. Taking into account that f' is increasing, $f'(x) < f'(\xi) < 0$, we get

$$x - x_0 < \frac{f'(\xi)}{f'(x)}(x - x_0) = \frac{f(x) - f(x_0)}{f'(x)} < 0.$$

Taking limits as x tends to $0+$ we obtain

$$-x_0 \leq \liminf_{x\to 0+} \frac{f(x)}{f'(x)} \leq \limsup_{x\to 0+} \frac{f(x)}{f'(x)} \leq 0.$$

Since this happens for all $x_0 \in (0,r)$ we deduce that $\lim_{x\to 0+} \frac{f(x)}{f'(x)}$ exists and

$$\lim_{x\to 0+} \frac{f(x)}{f'(x)} = 0.$$

1995-D1-P4 — Statement

Day 1, P4 | 15 points

Let $F : (1, \infty) \to \mathbb{R}$ be the function defined by

$$F(x) := \int_x^{x^2} \frac{dt}{\ln t}.$$

Show that F is one-to-one (i.e. injective) and find the range (i.e. set of values) of F.

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Theory and methods for 1995-D1-P4

Theory Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Continuity, compactness, and order [T-AN-CONTINUITY]
Methods Design an auxiliary derivative [M-AN-AUXILIARY]; Pair an integral through symmetry [M-AN-PAIR]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

1995-D1-P4 — Solution

For this record: Theory: T-AN-INTEGRAL; T-AN-DERIVATIVE; T-AN-CONTINUITY. Methods: M-AN-AUXILIARY; M-AN-PAIR. Full learning route

From the definition we have

$$F'(x) = \frac{x-1}{\ln x}, \quad x > 1.$$

Therefore $F'(x) > 0$ for $x \in (1, \infty)$. Thus F is strictly increasing and hence one-to-one. Since

$$F(x) \geq (x^2 - x) \min\left\{\frac{1}{\ln t} : x \leq t \leq x^2\right\} = \frac{x^2 - x}{\ln x^2} \to \infty$$

as $x \to \infty$, it follows that the range of F is $(F(1+), \infty)$. In order to determine $F(1+)$ we substitute $t = e^v$ in the definition of F and we get

$$F(x) = \int_{\ln x}^{2\ln x} \frac{e^v}{v}\, dv.$$

Hence

$$F(x) < e^{2\ln x} \int_{\ln x}^{2\ln x} \frac{1}{v}\, dv = x^2 \ln 2$$

and similarly $F(x) > x \ln 2$. Thus $F(1+) = \ln 2$.

1995-D1-P5 — Statement

Day 1, P5 | 20 points

Let A and B be real $n \times n$ matrices. Assume that there exist $n+1$ different real numbers $t_1, t_2, \ldots, t_{n+1}$ such that the matrices

$$C_i = A + t_i B, \quad i = 1, 2, \ldots, n+1,$$

are nilpotent (i.e. $C_i^n = 0$).

Show that both A and B are nilpotent.

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Theory and methods for 1995-D1-P5

Theory Nilpotents, projections and involutions [T-LA-NILPOTENT]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Compare coefficients in a matrix polynomial [M-LA-PARAMETER]
Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

1995-D1-P5 — Solution

For this record: Theory: T-LA-NILPOTENT; T-AL-POLYNOMIAL. Methods: M-LA-PARAMETER. Full learning route

We have that

$$(A+tB)^n = A^n + tP_1 + t^2P_2 + \cdots + t^{n-1}P_{n-1} + t^nB^n$$

for some matrices $P_1, P_2, \ldots, P_{n-1}$ not depending on t.

Assume that $a, p_1, p_2, \ldots, p_{n-1}, b$ are the (i,j)-th entries of the corresponding matrices $A^n, P_1, P_2, \ldots, P_{n-1}, B^n$. Then the polynomial

$$bt^n + p_{n-1}t^{n-1} + \cdots + p_2t^2 + p_1t + a$$

has at least $n+1$ roots $t_1, t_2, \ldots, t_{n+1}$. Hence all its coefficients vanish. Therefore $A^n = 0$, $B^n = 0$, $P_i = 0$; and A and B are nilpotent.

1995-D1-P6 — Statement

Day 1, P6 | 25 points

Let $p > 1$. Show that there exists a constant $K_p > 0$ such that for every $x, y \in \mathbb{R}$ satisfying $|x|^p + |y|^p = 2$, we have

$$(x-y)^2 \le K_p\left(4 - (x+y)^2\right).$$

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Theory and methods for 1995-D1-P6

Theory Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Continuity, compactness, and order [T-AN-CONTINUITY]; Taylor remainders and local scale [T-AN-TAYLOR]
Methods Expand near equality, bound away from it [M-AN-LOCAL]; Choose the correct convexity comparison [M-AN-CONVEX]
Strategy Pass between local constraints and a global conclusion [S-LOCALGLOBAL]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

1995-D1-P6 — Solution

For this record: Theory: T-AN-CONVEXITY; T-AN-CONTINUITY; T-AN-TAYLOR. Methods: M-AN-LOCAL; M-AN-CONVEX. Full learning route

Let $0 < \delta < 1$. First we show that there exists $K_{p,\delta} > 0$ such that

$$f(x,y) = \frac{(x-y)^2}{4-(x+y)^2} \le K_{p,\delta}$$

for every $(x,y) \in D_\delta = \{(x,y) : |x-y| \ge \delta,\ |x|^p + |y|^p = 2\}$.
Since D_δ is compact it is enough to show that f is continuous on D_δ. For this we show that the denominator of f is different from zero. Assume the contrary. Then $|x+y| = 2$, and $\left|\frac{x+y}{2}\right|^p = 1$. Since $p > 1$, the function $g(t) = |t|^p$ is strictly convex, in other words $\left|\frac{x+y}{2}\right|^p < \frac{|x|^p+|y|^p}{2}$ whenever $x \ne y$. So for some $(x,y) \in D_\delta$ we have $\left|\frac{x+y}{2}\right|^p < \frac{|x|^p+|y|^p}{2} = 1 = \left|\frac{x+y}{2}\right|^p$. We get a contradiction.

1995-D1-P6 — Solution (continued)

If x and y have different signs then $(x,y) \in D_\delta$ for all $0 < \delta < 1$ because then $|x-y| \geq \max\{|x|,|y|\} \geq 1 > \delta$. So we may further assume without loss of generality that $x > 0$, $y > 0$ and $x^p + y^p = 2$. Set $x = 1 + t$. Then

$$\begin{aligned} y &= (2 - x^p)^{1/p} = (2 - (1+t)^p)^{1/p} = \left(2 - \left(1 + pt + \frac{p(p-1)}{2}t^2 + o(t^2)\right)\right)^{1/p} \\ &= \left(1 - pt - \frac{p(p-1)}{2}t^2 + o(t^2)\right)^{1/p} \\ &= 1 + \frac{1}{p}\left(-pt - \frac{p(p-1)}{2}t^2 + o(t^2)\right) + \frac{1}{2p}\left(\frac{1}{p} - 1\right)(-pt + o(t))^2 + o(t^2) \\ &= 1 - t - \frac{p-1}{2}t^2 + o(t^2) - \frac{p-1}{2}t^2 + o(t^2) = 1 - t - (p-1)t^2 + o(t^2). \end{aligned}$$

We have

$$(x-y)^2 = (2t + o(t))^2 = 4t^2 + o(t^2)$$

and

$$4 - (x+y)^2 = 4 - (2 - (p-1)t^2 + o(t^2))^2 = 4 - 4 + 4(p-1)t^2 + o(t^2) = 4(p-1)t^2 + o(t^2).$$

So there exists $\delta_p > 0$ such that if $|t| < \delta_p$ we have $(x-y)^2 < 5t^2$, $4 - (x+y)^2 > 3(p-1)t^2$. Then

$$(x-y)^2 < 5t^2 = \frac{5}{3(p-1)} \cdot 3(p-1)t^2 < \frac{5}{3(p-1)}\left(4 - (x+y)^2\right) \qquad (*)$$

if $|x-1| < \delta_p$. From the symmetry we have that $(*)$ also holds when $|y-1| < \delta_p$.

To finish the proof it is enough to show that $|x-y| \geq 2\delta_p$ whenever $|x-1| \geq \delta_p$, $|y-1| \geq \delta_p$ and $x^p + y^p = 2$. Indeed, since $x^p + y^p = 2$ we have that $\max\{x,y\} \geq 1$. So let $x - 1 \geq \delta_p$. Since $\left(\frac{x+y}{2}\right)^p \leq \frac{x^p+y^p}{2} = 1$ we get $x + y \leq 2$. Then $x - y \geq 2(x-1) \geq 2\delta_p$.

Day 2

1995-D2-P1 — Statement

Day 2, P7 | 10 points

Let A be 3×3 real matrix such that the vectors Au and u are orthogonal for each column vector $u \in \mathbb{R}^3$. Prove that:
a) $A^\top = -A$, where $A^\top$ denotes the transpose of the matrix A;

b) there exists a vector $v \in \mathbb{R}^3$ such that $Au = v \times u$ for every $u \in \mathbb{R}^3$, where $v \times u$ denotes the vector product in $\mathbb{R}^3$.

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Theory and methods for 1995-D2-P1

Theory Inner products, orthogonality and polarization [T-LA-INNER]
Methods Read the entries before choosing a theorem [M-LA-ENTRY]
Strategy Expose the structure before calculating [S-STRUCTURE]; Translate geometry into a controlled representation [S-GEOMETRIC]

Statement Recorded solutions Catalog

1995-D2-P1 — Solution

For this record: Theory: T-LA-INNER. Methods: M-LA-ENTRY. Full learning route

a) Set $A = (a_{ij})$, $u = (u_1, u_2, u_3)^\top$. If we use the orthogonality condition

$$(Au, u) = 0 \tag{1}$$

with $u_i = \delta_{ik}$ we get $a_{kk} = 0$. If we use (1) with $u_i = \delta_{ik} + \delta_{im}$ we get

$$a_{kk} + a_{km} + a_{mk} + a_{mm} = 0$$

and hence $a_{km} = -a_{mk}$.

b) Set $v_1 = -a_{23}$, $v_2 = a_{13}$, $v_3 = -a_{12}$. Then

$$Au = (v_2u_3 - v_3u_2,\ v_3u_1 - v_1u_3,\ v_1u_2 - v_2u_1)^\top = v \times u.$$

1995-D2-P2 — Statement

Day 2, P8 | 15 points

Let $\{b_n\}_{n=0}^{\infty}$ be a sequence of positive real numbers such that $b_0 = 1$, $b_n = 2 + \sqrt{b_{n-1}} - 2\sqrt{1 + \sqrt{b_{n-1}}}$. Calculate

$$\sum_{n=1}^{\infty} b_n 2^n.$$

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Theory and methods for 1995-D2-P2

Theory Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Series, finite cancellation, and summation by parts [T-AN-SERIES]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Methods Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]; Find a finite telescoping identity [M-AN-TELESCOPE]
Strategy Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

1995-D2-P2 — Solution

For this record: Theory: T-AN-CONVERGENCE; T-AN-SERIES; T-AN-DERIVATIVE. Methods: M-AN-RECURRENCE; M-AN-TELESCOPE. Full learning route

Put $a_n = 1 + \sqrt{b_n}$ for $n \geq 0$. Then $a_n > 1$, $a_0 = 2$ and

$$a_n = 1 + \sqrt{1 + a_{n-1} - 2\sqrt{a_{n-1}}} = \sqrt{a_{n-1}},$$

so $a_n = 2^{2^{-n}}$. Then

$$\begin{aligned}\sum_{n=1}^{N} b_n 2^n &= \sum_{n=1}^{N} (a_n - 1)^2 2^n = \sum_{n=1}^{N} \left[a_n^2 2^n - a_n 2^{n+1} + 2^n\right] \\ &= \sum_{n=1}^{N} \left[(a_{n-1} - 1)2^n - (a_n - 1)2^{n+1}\right] \\ &= (a_0 - 1)2^1 - (a_N - 1)2^{N+1} = 2 - 2\,\frac{2^{2^{-N}} - 1}{2^{-N}}.\end{aligned}$$

Put $x = 2^{-N}$. Then $x \to 0$ as $N \to \infty$ and so

$$\sum_{n=1}^{\infty} b_n 2^n = \lim_{N\to\infty} \left(2 - 2\,\frac{2^{2^{-N}} - 1}{2^{-N}}\right) = \lim_{x\to 0} \left(2 - 2\,\frac{2^x - 1}{x}\right) = 2 - 2\ln 2.$$

1995-D2-P3 — Statement

Day 2, P9 | 15 points

Let all roots of an n-th degree polynomial $P(z)$ with complex coefficients lie on the unit circle in the complex plane. Prove that all roots of the polynomial

$$2zP'(z) - nP(z)$$

lie on the same circle.

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Theory and methods for 1995-D2-P3

Theory Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]; Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]
Methods Use a logarithmic derivative to expose root geometry [M-AL-LOGDERIVATIVE]
Strategy Expose the structure before calculating [S-STRUCTURE]; Translate geometry into a controlled representation [S-GEOMETRIC]

Statement Recorded solutions Catalog

1995-D2-P3 — Solution

For this record: Theory: T-AL-POLYNOMIAL; T-AL-ROOTS; T-AL-COMPLEX. Methods: M-AL-LOGDERIVATIVE. Full learning route

It is enough to consider only polynomials with leading coefficient 1. Let $P(z) = (z-\alpha_1)(z-\alpha_2)\dots(z-\alpha_n)$ with $|\alpha_j| = 1$, where the complex numbers $\alpha_1, \alpha_2, \dots, \alpha_n$ may coincide.

We have

$$\widetilde{P}(z) \equiv 2zP'(z) - nP(z) = (z+\alpha_1)(z-\alpha_2)\dots(z-\alpha_n) + (z-\alpha_1)(z+\alpha_2)\dots(z-\alpha_n) + \dots + (z-\alpha_1)(z-\alpha_2)\dots(z+\alpha_n).$$

Hence, $\dfrac{\widetilde{P}(z)}{P(z)} = \sum\limits_{k=1}^{n} \dfrac{z+\alpha_k}{z-\alpha_k}$. Since $\operatorname{Re} \dfrac{z+\alpha}{z-\alpha} = \dfrac{|z|^2 - |\alpha|^2}{|z-\alpha|^2}$ for all complex z, α, $z \neq \alpha$, we deduce that in our case $\operatorname{Re} \dfrac{\widetilde{P}(z)}{P(z)} = \sum\limits_{k=1}^{n} \dfrac{|z|^2 - 1}{|z-\alpha_k|^2}$. From $|z| \neq 1$ it follows that $\operatorname{Re} \dfrac{\widetilde{P}(z)}{P(z)} \neq 0$. Hence $\widetilde{P}(z) = 0$ implies $|z| = 1$.

1995-D2-P4 — Statement

Day 2, P10 | 15 points

a) Prove that for every $\varepsilon > 0$ there is a positive integer n and real numbers $\lambda_1, \dots, \lambda_n$ such that

$$\max_{x \in [-1,1]} \left| x - \sum_{k=1}^{n} \lambda_k x^{2k+1} \right| < \varepsilon.$$

b) Prove that for every odd continuous function f on $[-1,1]$ and for every $\varepsilon > 0$ there is a positive integer n and real numbers $\mu_1, \dots, \mu_n$ such that

$$\max_{x \in [-1,1]} \left| f(x) - \sum_{k=1}^{n} \mu_k x^{2k+1} \right| < \varepsilon.$$

Recall that f is odd means that $f(x) = -f(-x)$ for all $x \in [-1,1]$.

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Theory and methods for 1995-D2-P4

Theory Uniform approximation and positive kernels [T-AN-APPROXIMATION]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Preserve structure while approximating [M-AN-APPROXIMATE]; Separate a main range from exceptional scales [M-AN-SPLIT]
Strategy Build a witness and verify every constraint [S-CONSTRUCTION]; Pass between local constraints and a global conclusion [S-LOCALGLOBAL]

Statement Recorded solutions Catalog

1995-D2-P4 — Solution

For this record: Theory: T-AN-APPROXIMATION; T-AL-POLYNOMIAL. Methods: M-AN-APPROXIMATE; M-AN-SPLIT. Full learning route

a) Let n be such that $(1-\varepsilon^2)^n \le \varepsilon$. Then $|x(1-x^2)^n| < \varepsilon$ for every $x \in [-1,1]$. Thus one can set $\lambda_k = (-1)^{k+1}\binom{n}{k}$ because then

$$x - \sum_{k=1}^{n} \lambda_k x^{2k+1} = \sum_{k=0}^{n} (-1)^k \binom{n}{k} x^{2k+1} = x(1-x^2)^n.$$

b) From the Weierstrass theorem there is a polynomial, say $p \in \Pi_m$, such that

$$\max_{x\in[-1,1]} |f(x) - p(x)| < \frac{\varepsilon}{2}.$$

Set $q(x) = \frac{1}{2}\{p(x) - p(-x)\}$. Then

$$f(x) - q(x) = \frac{1}{2}\{f(x) - p(x)\} - \frac{1}{2}\{f(-x) - p(-x)\}$$

and

$$\max_{|x|\le 1} |f(x) - q(x)| \le \frac{1}{2}\max_{|x|\le 1} |f(x) - p(x)| + \frac{1}{2}\max_{|x|\le 1} |f(-x) - p(-x)| < \frac{\varepsilon}{2}. \tag{1}$$

But q is an odd polynomial in Π_m and it can be written as

$$q(x) = \sum_{k=0}^{m} b_k x^{2k+1} = b_0 x + \sum_{k=1}^{m} b_k x^{2k+1}.$$

If $b_0 = 0$ then (1) proves b). If $b_0 \neq 0$ then one applies a) with $\dfrac{\varepsilon}{2|b_0|}$ instead of ε to get

$$\max_{|x|\le 1} \left| b_0 x - b_0 \sum_{k=1}^{n} \lambda_k x^{2k+1} \right| < \frac{\varepsilon}{2} \tag{2}$$

for appropriate n and $\lambda_1, \lambda_2, \ldots, \lambda_n$. Now b) follows from (1) and (2) with $\max\{n, m\}$ instead of n.

1995-D2-P5 — Statement

Day 2, P11 | 10+15 points

a) Prove that every function of the form

$$f(x) = \frac{a_0}{2} + \cos x + \sum_{n=2}^{N} a_n \cos(nx)$$

1995-D2-P5 — Statement (continued)

with $|a_0| < 1$, has positive as well as negative values in the period $[0, 2\pi)$.

b) Prove that the function

$$F(x) = \sum_{n=1}^{100} \cos(n^{\frac{3}{2}} x)$$

has at least 40 zeros in the interval $(0, 1000)$.

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Theory and methods for 1995-D2-P5

Theory Trigonometric identities and safe branches [T-AN-TRIG]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Continuity, compactness, and order [T-AN-CONTINUITY]; Series, finite cancellation, and summation by parts [T-AN-SERIES]
Methods Exploit identities before estimating oscillations [M-AN-TRIG]; Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]
Strategy Make the contradiction quantitative or structural [S-CONTRADICTION]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

1995-D2-P5 — Solution

For this record: Theory: T-AN-TRIG; T-AN-INTEGRAL; T-AN-CONTINUITY; T-AN-SERIES. Methods: M-AN-TRIG; M-AN-WEIGHTS. Full learning route

a) Let us consider the integral

$$\int_0^{2\pi} f(x)(1 \pm \cos x)\, dx = \pi(a_0 \pm 1).$$

The assumption that $f(x) \geq 0$ implies $a_0 \geq 1$. Similarly, if $f(x) \leq 0$ then $a_0 \leq -1$. In both cases we have a contradiction with the hypothesis of the problem.
b) We shall prove that for each integer N and for each real number $h \geq 24$ and each real number y the function

$$F_N(x) = \sum_{n=1}^{N} \cos(x\, n^{\frac{3}{2}})$$

changes sign in the interval $(y, y+h)$. The assertion will follow immediately from here.

Consider the integrals

$$I_1 = \int_y^{y+h} F_N(x)\, dx, \qquad I_2 = \int_y^{y+h} F_N(x) \cos x\, dx.$$

If $F_N(x)$ does not change sign in $(y, y+h)$ then we have

$$|I_2| \leq \int_y^{y+h} |F_N(x)|\, dx = \left| \int_y^{y+h} F_N(x)\, dx \right| = |I_1|.$$

Hence, it is enough to prove that

$$|I_2| > |I_1|.$$

Obviously, for each $\alpha \neq 0$ we have

$$\left| \int_y^{y+h} \cos(\alpha x)\, dx \right| \leq \frac{2}{|\alpha|}.$$

Hence

$$|I_1| = \left| \sum_{n=1}^{N} \int_y^{y+h} \cos(x\, n^{\frac{3}{2}})\, dx \right| \leq 2 \sum_{n=1}^{N} \frac{1}{n^{\frac{3}{2}}} < 2\left(1 + \int_1^{\infty} \frac{dt}{t^{\frac{3}{2}}}\right) = 6. \tag{1}$$

1995-D2-P5 — Solution (continued)

On the other hand we have

$$I_2 = \sum_{n=1}^{N} \int_y^{y+h} \cos x \cos(x\, n^{\frac{3}{2}})\, dx = \frac{1}{2} \int_y^{y+h} (1 + \cos(2x))\, dx$$
$$+ \frac{1}{2} \sum_{n=2}^{N} \int_y^{y+h} \left(\cos\left(x(n^{\frac{3}{2}} - 1)\right) + \cos\left(x(n^{\frac{3}{2}} + 1)\right) \right) dx = \frac{1}{2} h + \Delta,$$

where

$$|\Delta| \le \frac{1}{2} \left(1 + 2 \sum_{n=2}^{N} \left(\frac{1}{n^{\frac{3}{2}} - 1} + \frac{1}{n^{\frac{3}{2}} + 1} \right) \right) \le \frac{1}{2} + 2 \sum_{n=2}^{N} \frac{1}{n^{\frac{3}{2}} - 1}.$$

We use that $n^{\frac{3}{2}} - 1 \ge \frac{2}{3} n^{\frac{3}{2}}$ for $n \ge 3$ and we get

$$|\Delta| \le \frac{1}{2} + \frac{2}{2^{\frac{3}{2}} - 1} + 3 \sum_{n=3}^{N} \frac{1}{n^{\frac{3}{2}}} < \frac{1}{2} + \frac{2}{2\sqrt{2} - 1} + 3 \int_2^{\infty} \frac{dt}{t^{\frac{3}{2}}} < 6.$$

Hence

$$|I_2| > \frac{1}{2} h - 6. \tag{2}$$

We use that $h \ge 24$ and inequalities (1), (2) and we obtain $|I_2| > |I_1|$. The proof is completed.

1995-D2-P6 — Statement

Day 2, P12 | 20 points

Suppose that $\{f_n\}_{n=1}^{\infty}$ is a sequence of continuous functions on the interval $[0, 1]$ such that

$$\int_0^1 f_m(x) f_n(x)\, dx = \begin{cases} 1 & \text{if } n = m \\ 0 & \text{if } n \ne m \end{cases}$$

and

$$\sup\{|f_n(x)| : x \in [0, 1] \text{ and } n = 1, 2, \dots\} < +\infty.$$

Show that there exists no subsequence $\{f_{n_k}\}$ of $\{f_n\}$ such that $\lim_{k \to \infty} f_{n_k}(x)$ exists for all $x \in [0, 1]$.

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Theory and methods for 1995-D2-P6

Theory Dominated convergence and orthonormal separation [T-AN-MEASURE]; Inner products, orthogonality and polarization [T-LA-INNER]
Methods Pass a limit through an integral safely [M-AN-DOMINATE]
Strategy Make the contradiction quantitative or structural [S-CONTRADICTION]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

1995-D2-P6 — Solution

For this record: Theory: T-AN-MEASURE; T-LA-INNER. Methods: M-AN-DOMINATE. Full learning route

Let $M < \infty$ satisfy $|f_n(x)| \le M$ for every n and $x \in [0, 1]$. Suppose that a subsequence f_{n_k} converges pointwise. Then $|f_{n_{k+1}}(x) - f_{n_k}(x)|^2 \to 0$ pointwise and these functions are bounded by $4M^2$. The dominated convergence theorem gives

$$\int_0^1 |f_{n_{k+1}}(x) - f_{n_k}(x)|^2\, dx \longrightarrow 0.$$

1995-D2-P6 — Solution (continued)

But orthonormality and $n_{k+1} \neq n_k$ give, for every k,

$$\int_0^1 |f_{n_{k+1}} - f_{n_k}|^2 = \int_0^1 f_{n_{k+1}}^2 + \int_0^1 f_{n_k}^2 - 2\int_0^1 f_{n_{k+1}} f_{n_k} = 2,$$

a contradiction.

3.3 1996

Day 1

1996-D1-P1 — Statement

Day 1, P1 | 10 points

Let for $j = 0, \ldots, n$, $a_j = a_0 + jd$, where a_0, d are fixed real numbers. Put

$$A = \begin{pmatrix} a_0 & a_1 & a_2 & \ldots & a_n \\ a_1 & a_0 & a_1 & \ldots & a_{n-1} \\ a_2 & a_1 & a_0 & \ldots & a_{n-2} \\ \ldots & \ldots & \ldots & \ldots & \ldots \\ a_n & a_{n-1} & a_{n-2} & \ldots & a_0 \end{pmatrix}.$$

Calculate $\det(A)$, where $\det(A)$ denotes the determinant of A.

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Theory and methods for 1996-D1-P1

Theory Determinants, block identities and inverses [T-LA-DETERMINANT]
Methods Eliminate systematically and track the invariant [M-LA-ELIM]
Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

1996-D1-P1 — Solution

For this record: Theory: T-LA-DETERMINANT. Methods: M-LA-ELIM. Full learning route

Adding the first column of A to the last column we get that

$$\det(A) = (a_0 + a_n) \det \begin{pmatrix} a_0 & a_1 & a_2 & \ldots & 1 \\ a_1 & a_0 & a_1 & \ldots & 1 \\ a_2 & a_1 & a_0 & \ldots & 1 \\ \ldots & \ldots & \ldots & \ldots & \ldots \\ a_n & a_{n-1} & a_{n-2} & \ldots & 1 \end{pmatrix}.$$

Subtracting the n-th row of the above matrix from the $(n+1)$-st one, $(n-1)$-st from n-th, ..., first from second we obtain that

$$\det(A) = (a_0 + a_n) \det \begin{pmatrix} a_0 & a_1 & a_2 & \ldots & 1 \\ d & -d & -d & \ldots & 0 \\ d & d & -d & \ldots & 0 \\ \ldots & \ldots & \ldots & \ldots & \ldots \\ d & d & d & \ldots & 0 \end{pmatrix}.$$

1996-D1-P1 — Solution (continued)

Hence,

$$\det(A) = (-1)^n(a_0 + a_n)\det\begin{pmatrix} d & -d & -d & \dots & -d \\ d & d & -d & \dots & -d \\ d & d & d & \dots & -d \\ \dots & \dots & \dots & \dots & \dots \\ d & d & d & \dots & d \end{pmatrix}.$$

Adding the last row of the above matrix to the other rows we have

$$\det(A) = (-1)^n(a_0 + a_n)\det\begin{pmatrix} 2d & 0 & 0 & \dots & 0 \\ 2d & 2d & 0 & \dots & 0 \\ 2d & 2d & 2d & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ d & d & d & \dots & d \end{pmatrix} = (-1)^n(a_0 + a_n)2^{n-1}d^n.$$

1996-D1-P2 — Statement

Day 1, P2 | 10 points

Evaluate the definite integral

$$\int_{-\pi}^{\pi} \frac{\sin nx}{(1+2^x)\sin x}\,dx,$$

where n is a natural number.

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Theory and methods for 1996-D1-P2

Theory Trigonometric identities and safe branches [T-AN-TRIG]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Methods Pair an integral through symmetry [M-AN-PAIR]; Exploit identities before estimating oscillations [M-AN-TRIG]; Find a finite telescoping identity [M-AN-TELESCOPE]
Strategy Expose the structure before calculating [S-STRUCTURE]; Reduce size while preserving the defining conditions [S-INDUCTION]

Statement Recorded solutions Catalog

1996-D1-P2 — Solution

For this record: Theory: T-AN-TRIG; T-AN-INTEGRAL. Methods: M-AN-PAIR; M-AN-TRIG; M-AN-TELESCOPE. Full learning route

We have

$$\begin{aligned} I_n &= \int_{-\pi}^{\pi} \frac{\sin nx}{(1+2^x)\sin x}\,dx \\ &= \int_0^{\pi} \frac{\sin nx}{(1+2^x)\sin x}\,dx + \int_{-\pi}^{0} \frac{\sin nx}{(1+2^x)\sin x}\,dx. \end{aligned}$$

In the second integral we make the change of variable $x = -x$ and obtain

$$\begin{aligned} I_n &= \int_0^{\pi} \frac{\sin nx}{(1+2^x)\sin x}\,dx + \int_0^{\pi} \frac{\sin nx}{(1+2^{-x})\sin x}\,dx \\ &= \int_0^{\pi} \frac{(1+2^x)\sin nx}{(1+2^x)\sin x}\,dx \\ &= \int_0^{\pi} \frac{\sin nx}{\sin x}\,dx. \end{aligned}$$

1996-D1-P2 — Solution (continued)

For $n \geq 2$ we have

$$I_n - I_{n-2} = \int_0^\pi \frac{\sin nx - \sin(n-2)x}{\sin x}\,dx$$
$$= 2\int_0^\pi \cos(n-1)x\,dx = 0.$$

The answer

$$I_n = \begin{cases} 0 & \text{if } n \text{ is even,} \\ \pi & \text{if } n \text{ is odd} \end{cases}$$

follows from the above formula and $I_0 = 0$, $I_1 = \pi$.

1996-D1-P3 — Statement

Day 1, P3 | 15 points

The linear operator A on the vector space V is called an involution if $A^2 = E$ where E is the identity operator on V. Let $\dim V = n < \infty$.
(i) Prove that for every involution A on V there exists a basis of V consisting of eigenvectors of A.

(ii) Find the maximal number of distinct pairwise commuting involutions on V.

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Theory and methods for 1996-D1-P3

Theory Nilpotents, projections and involutions [T-LA-NILPOTENT]; Adapted bases, centralizers and common eigenspaces [T-LA-INVARIANT]
Methods Choose a basis adapted to the structure [M-LA-BASIS]; Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Reduce size while preserving the defining conditions [S-INDUCTION]

Statement Recorded solutions Catalog

1996-D1-P3 — Solution

For this record: Theory: T-LA-NILPOTENT; T-LA-INVARIANT. Methods: M-LA-BASIS; M-LA-SPECTRAL. Full learning route

(i) Let $B = \frac{1}{2}(A + E)$. Then

$$B^2 = \frac{1}{4}(A^2 + 2AE + E) = \frac{1}{4}(2AE + 2E) = \frac{1}{2}(A + E) = B.$$

Hence B is a projection. Thus there exists a basis of eigenvectors for B, and the matrix of B in this basis is of the form $\operatorname{diag}(1, \dots, 1, 0, \dots, 0)$.
Since $A = 2B - E$ the eigenvalues of A are ± 1 only.
(ii) Let $\{A_i : i \in I\}$ be a set of commuting diagonalizable operators on V, and let A_1 be one of these operators. Choose an eigenvalue λ of A_1 and denote $V_\lambda = \{v \in V : A_1 v = \lambda v\}$. Then V_λ is a subspace of V, and since $A_1 A_i = A_i A_1$ for each $i \in I$ we obtain that V_λ is invariant under each A_i. If $V_\lambda = V$ then A_1 is either E or $-E$, and we can start with another operator A_i. If $V_\lambda \neq V$ we proceed by induction on $\dim V$ in order to find a common eigenvector for all A_i. Therefore $\{A_i : i \in I\}$ are simultaneously diagonalizable.

If they are involutions then $|I| \leq 2^n$ since the diagonal entries may equal 1 or -1 only.

1996-D1-P4 — Statement

Day 1, P4 | 15 points

Let $a_1 = 1$, $a_n = \dfrac{1}{n}\sum_{k=1}^{n-1} a_k a_{n-k}$ for $n \geq 2$. Show that

(i) $\limsup_{n\to\infty} |a_n|^{1/n} < 2^{-1/2}$;

(ii) $\limsup_{n\to\infty} |a_n|^{1/n} \geq 2/3$.

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Theory and methods for 1996-D1-P4

Theory Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Methods Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]
Strategy Reduce size while preserving the defining conditions [S-INDUCTION]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

1996-D1-P4 — Solution

For this record: Theory: T-AN-CONVERGENCE; T-DS-INDUCTION. Methods: M-AN-RECURRENCE. Full learning route

(i) We show by induction that

$$a_n \leq q^n \quad \text{for } n \geq 3, \tag{$*$}$$

where $q = 0.7$ and use that $0.7 < 2^{-1/2}$. One has $a_1 = 1$, $a_2 = \frac{1}{2}$, $a_3 = \frac{1}{3}$, $a_4 = \frac{11}{48}$. Therefore $(*)$ is true for $n = 3$ and $n = 4$. Assume $(*)$ is true for $n \leq N-1$ for some $N \geq 5$. Then

$$a_N = \frac{2}{N}a_{N-1} + \frac{1}{N}a_{N-2} + \frac{1}{N}\sum_{k=3}^{N-3} a_k a_{N-k} \leq \frac{2}{N}q^{N-1} + \frac{1}{N}q^{N-2} + \frac{N-5}{N}q^N \leq q^N$$

because $\dfrac{2}{q} + \dfrac{1}{q^2} \leq 5$.

(ii) We show by induction that

$$a_n \geq q^n \quad \text{for } n \geq 2,$$

where $q = \dfrac{2}{3}$. One has $a_2 = \dfrac{1}{2} > \left(\dfrac{2}{3}\right)^2 = q^2$. Going by induction we have for $N \geq 3$

$$a_N = \frac{2}{N}a_{N-1} + \frac{1}{N}\sum_{k=2}^{N-2} a_k a_{N-k} \geq \frac{2}{N}q^{N-1} + \frac{N-3}{N}q^N = q^N$$

because $\dfrac{2}{q} = 3$.

1996-D1-P5 — Statement

Day 1, P5 | 25 points

(i) Let a, b be real numbers such that $b \leq 0$ and $1 + ax + bx^2 \geq 0$ for every x in $[0, 1]$. Prove that

$$\lim_{n \to +\infty} n \int_0^1 (1 + ax + bx^2)^n dx = \begin{cases} -\dfrac{1}{a} & \text{if } a < 0, \\ +\infty & \text{if } a \geq 0. \end{cases}$$

(ii) Let $f : [0, 1] \to [0, \infty)$ be a function with a continuous second derivative and let $f''(x) \leq 0$ for every x in $[0, 1]$. Suppose that $L = \lim_{n \to \infty} n \int_0^1 (f(x))^n dx$ exists and $0 < L < +\infty$. Prove that f' has a constant sign and $\min_{x \in [0,1]} |f'(x)| = L^{-1}$.

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Theory and methods for 1996-D1-P5

Theory Scale separation and limit comparisons [T-AN-ASYMPTOTICS]; Taylor remainders and local scale [T-AN-TAYLOR]; Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Methods Separate a main range from exceptional scales [M-AN-SPLIT]; Expand near equality, bound away from it [M-AN-LOCAL]; Choose the correct convexity comparison [M-AN-CONVEX]
Strategy Control the limiting operation before taking the limit [S-ANALYTIC]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

1996-D1-P5 — Solution

For this record: Theory: T-AN-ASYMPTOTICS; T-AN-TAYLOR; T-AN-CONVEXITY; T-AN-INTEGRAL. Methods: M-AN-SPLIT; M-AN-LOCAL; M-AN-CONVEX. Full learning route

(i) With a linear change of the variable (i) is equivalent to:
(i′) Let a, b, A be real numbers such that $b \leq 0$, $A > 0$ and $1 + ax + bx^2 > 0$ for every x in $[0, A]$. Denote $I_n = n \int_0^A (1 + ax + bx^2)^n dx$. Prove that $\lim_{n \to +\infty} I_n = -\dfrac{1}{a}$ when $a < 0$ and $\lim_{n \to +\infty} I_n = +\infty$ when $a \geq 0$.
Let $a < 0$. Set $f(x) = e^{ax} - (1 + ax + bx^2)$. Using that $f(0) = f'(0) = 0$ and $f''(x) = a^2 e^{ax} - 2b$ we get for $x > 0$ that

$$0 < e^{ax} - (1 + ax + bx^2) < cx^2$$

where $c = \dfrac{a^2}{2} - b$. Using the mean value theorem we get

$$0 < e^{anx} - (1 + ax + bx^2)^n < cx^2 n e^{a(n-1)x}.$$

Therefore

$$0 < n \int_0^A e^{anx} dx - n \int_0^A (1 + ax + bx^2)^n dx < cn^2 \int_0^A x^2 e^{a(n-1)x} dx.$$

Using that

$$n \int_0^A e^{anx} dx = \frac{e^{anA} - 1}{a} \xrightarrow[n \to \infty]{} -\frac{1}{a}$$

and

$$\int_0^A x^2 e^{a(n-1)x} dx < \frac{1}{|a|^3 (n-1)^3} \int_0^\infty t^2 e^{-t} dt$$

we get (i′) in the case $a < 0$.

1996-D1-P5 — Solution (continued)

Let $a \geq 0$. Then for $n > \max\{A^{-2}, -b\} - 1$ we have

$$\begin{aligned} n\int_0^A (1+ax+bx^2)^n dx &> n\int_0^{\frac{1}{\sqrt{n+1}}} (1+bx^2)^n dx \\ &> n \cdot \frac{1}{\sqrt{n+1}} \cdot \left(1+\frac{b}{n+1}\right)^n \\ &> \frac{n}{\sqrt{n+1}}\, e^b \xrightarrow[n\to\infty]{} \infty. \end{aligned}$$

(i) is proved.

(ii) Denote $I_n = n\int\limits_0^1 (f(x))^n dx$ and $M = \max\limits_{x\in[0,1]} f(x)$.

For $M < 1$ we have $I_n \leq nM^n \xrightarrow[n\to\infty]{} 0$, a contradiction.

If $M > 1$ since f is continuous there exists an interval $I \subset [0,1]$ with $|I| > 0$ such that $f(x) > 1$ for every $x \in I$. Then $I_n \geq n|I| \xrightarrow[n\to\infty]{} +\infty$, a contradiction. Hence $M = 1$. Now we prove that f' has a constant sign. Assume the opposite. Then $f'(x_0) = 0$ for some $x_0 \in (0,1)$. Then $f(x_0) = M = 1$ because $f'' \leq 0$. For $x_0 + h$ in $[0,1]$, $f(x_0+h) = 1 + \frac{h^2}{2} f''(\xi)$, $\xi \in (x_0, x_0+h)$. Let $m = \min\limits_{x\in[0,1]} f''(x)$. So, $f(x_0+h) \geq 1 + \frac{h^2}{2} m$.

Let $\delta > 0$ be such that $1 + \frac{\delta^2}{2} m > 0$ and $x_0 + \delta < 1$. Then

$$I_n \geq n\int_{x_0}^{x_0+\delta} (f(x))^n dx \geq n\int_0^{\delta} \left(1+\frac{m}{2}h^2\right)^n dh \xrightarrow[n\to\infty]{} \infty$$

in view of (i) – a contradiction. Hence f' is monotone and $M = f(0)$ or $M = f(1)$.

Let $M = f(0) = 1$. For h in $[0,1]$

$$1 + hf'(0) \geq f(h) \geq 1 + hf'(0) + \frac{m}{2}h^2,$$

where $f'(0) \neq 0$, because otherwise we get a contradiction as above. Since $f(0) = M$ the function f is decreasing and hence $f'(0) < 0$. Let $0 < A < 1$ be such that $1 + Af'(0) + \frac{m}{2}A^2 > 0$. Then

$$n\int_0^A (1+hf'(0))^n dh \geq n\int_0^A (f(x))^n dx \geq n\int_0^A \left(1+hf'(0)+\frac{m}{2}h^2\right)^n dh.$$

From (i) the first and the third integral tend to $-\frac{1}{f'(0)}$ as $n \to \infty$, hence so does the second.

Also $n\int\limits_A^1 (f(x))^n dx \leq n(f(A))^n \xrightarrow[n\to\infty]{} 0$ ($f(A) < 1$). We get $L = -\frac{1}{f'(0)}$ in this case.

If $M = f(1)$ we get in a similar way $L = \frac{1}{f'(1)}$.

1996-D1-P6 — Statement

Day 1, P6 | 25 points

Upper content of a subset E of the plane $\mathbb{R}^2$ is defined as

$$\mathcal{C}(E) = \inf\left\{\sum_{i=1}^n \operatorname{diam}(E_i)\right\}$$

where inf is taken over all finite families of sets $E_1, \ldots, E_n$, $n \in \mathbb{N}$, in $\mathbb{R}^2$ such that $E \subset \bigcup\limits_{i=1}^n E_i$.

1996-D1-P6 — Statement (continued)

Lower content of E is defined as

$$\mathcal{K}(E) = \sup\{\text{lenght}(L) : L \text{ is a closed line segment onto which } E \text{ can be contracted}\}.$$

Show that
(a) $\mathcal{C}(L) = \text{lenght}(L)$ if L is a closed line segment;
(b) $\mathcal{C}(E) \geq \mathcal{K}(E)$;
(c) the equality in (b) needs not hold even if E is compact.
Hint. If $E = T \cup T'$ where T is the triangle with vertices $(-2,2)$, $(2,2)$ and $(0,4)$, and T' is its reflexion about the x-axis, then $\mathcal{C}(E) = 8 > \mathcal{K}(E)$.

Remarks: All distances used in this problem are Euclidian. Diameter of a set E is $\text{diam}(E) = \sup\{\text{dist}(x,y) : x,y \in E\}$. Contraction of a set E to a set F is a mapping $f : E \mapsto F$ such that $\text{dist}(f(x), f(y)) \leq \text{dist}(x,y)$ for all $x, y \in E$. A set E can be contracted onto a set F if there is a contraction f of E to F which is onto, i.e., such that $f(E) = F$. Triangle is defined as the union of the three segments joining its vertices, i.e., it does not contain the interior.

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Theory and methods for 1996-D1-P6

Theory Metric contractions, covers, and nearest points [T-DS-METRIC]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Continuity, compactness, and order [T-AN-CONTINUITY]
Methods Push covers forward or move toward a nearest point [M-DS-METRIC]; Strengthen the statement before the induction step [M-DS-INDUCT]; Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Strategy Translate geometry into a controlled representation [S-GEOMETRIC]; Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

1996-D1-P6 — Solution

For this record: Theory: T-DS-METRIC; T-DS-INDUCTION; T-AN-CONTINUITY. Methods: M-DS-METRIC; M-DS-INDUCT; M-DS-GEOMETRIC. Full learning route

(a) The choice $E_1 = L$ gives $\mathcal{C}(L) \leq \text{lenght}(L)$. If $E \subset \bigcup_{i=1}^{n} E_i$ then $\sum_{i=1}^{n} \text{diam}(E_i) \geq \text{lenght}(L)$: By induction, $n = 1$ obvious, and assuming that E_{n+1} contains the end point a of L, define the segment $L_\varepsilon = \{x \in L : \text{dist}(x,a) \geq \text{diam}(E_{n+1}) + \varepsilon\}$ and use induction assumption to get $\sum_{i=1}^{n+1} \text{diam}(E_i) \geq \text{lenght}(L_\varepsilon) + \text{diam}(E_{n+1}) \geq \text{lenght}(L) - \varepsilon$; but $\varepsilon > 0$ is arbitrary.
(b) If f is a contraction of E onto L and $E \subset \bigcup_{n=1}^{n} E_i$, then $L \subset \bigcup_{i=1}^{n} f(E_i)$ and $\text{lenght}(L) \leq \sum_{i=1}^{n} \text{diam}(f(E_i)) \leq \sum_{i=1}^{n} \text{diam}(E_i)$.
(c1) Let $E = T \cup T'$ where T is the triangle with vertices $(-2,2)$, $(2,2)$ and $(0,4)$, and T' is its reflexion about the x-axis. Suppose $E \subset \bigcup_{i=1}^{n} E_i$. If no set among E_i meets both T and T', then E_i may be partitioned into covers of segments $[(-2,2),(2,2)]$ and $[(-2,-2),(2,-2)]$, both of length 4, so $\sum_{i=1}^{n} \text{diam}(E_i) \geq 8$. If at least one set among E_i, say E_k, meets both T and T', choose $a \in E_k \cap T$ and $b \in E_k \cap T'$ and note that the sets $E_i' = E_i$ for $i \neq k$, $E_k' = E_k \cup [a,b]$ cover $T \cup T' \cup [a,b]$, which is a set of upper content at least 8, since its orthogonal projection onto y-axis is a segment of length 8. Since $\text{diam}(E_j) = \text{diam}(E_j')$, we get $\sum_{i=1}^{n} \text{diam}(E_i) \geq 8$.

(c2) Let f be a contraction of E onto $L = [a', b']$. Choose $a = (a_1, a_2)$, $b = (b_1, b_2) \in E$ such that $f(a) = a'$ and $f(b) = b'$. Since $\text{lenght}(L) = \text{dist}(a', b') \leq \text{dist}(a, b)$ and since the triangles have diameter only 4, we may assume that $a \in T$ and $b \in T'$. Observe that if $a_2 \leq 3$ then a lies on one of the segments joining some of the points $(-2,2)$, $(2,2)$, $(-1,3)$, $(1,3)$; since all these points have distances

1996-D1-P6 — Solution (continued)

from vertices, and so from points, of T' at most $\sqrt{50}$, we get that lenght$(L) \leq$ dist$(a,b) \leq \sqrt{50}$. Similarly if $b_2 \geq -3$. Finally, if $a_2 > 3$ and $b_2 < -3$, we note that every vertex, and so every point of T' is in the distance at most $\sqrt{10}$ for a and every vertex, and so every point, of T is in the distance at most $\sqrt{10}$ of b. Since f is a contraction, the image of T' lies in a segment containing a' of length at most $\sqrt{10}$ and the image of T lies in a segment containing b' of length at most $\sqrt{10}$. Since the union of these two images is L, we get lenght$(L) \leq 2\sqrt{10} \leq \sqrt{50}$. Thus $\mathcal{K}(E) \leq \sqrt{50} < 8$.

Day 2

1996-D2-P1 — Statement

Day 2, P7 | 10 points

Prove that if $f : [0,1] \to [0,1]$ is a continuous function, then the sequence of iterates $x_{n+1} = f(x_n)$ converges if and only if

$$\lim_{n\to\infty} (x_{n+1} - x_n) = 0.$$

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Theory and methods for 1996-D2-P1

Theory Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Continuity, compactness, and order [T-AN-CONTINUITY]
Methods Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]; Use a first crossing or a forced-growth contradiction [M-AN-BARRIER]
Strategy Make the contradiction quantitative or structural [S-CONTRADICTION]; Pass between local constraints and a global conclusion [S-LOCALGLOBAL]

Statement Recorded solutions Catalog

1996-D2-P1 — Solution

For this record: Theory: T-AN-CONVERGENCE; T-AN-CONTINUITY. Methods: M-AN-RECURRENCE; M-AN-BARRIER. Full learning route

The "only if" part is obvious. Now suppose that $\lim_{n\to\infty} (x_{n+1} - x_n) = 0$ and the sequence $\{x_n\}$ does not converge. Then there are two cluster points $K < L$. There must be points from the interval (K, L) in the sequence. There is an $x \in (K, L)$ such that $f(x) \neq x$. Put $\varepsilon = \dfrac{|f(x) - x|}{2} > 0$. Then from the continuity of the function f we get that for some $\delta > 0$ for all $y \in (x - \delta, x + \delta)$ it is $|f(y) - y| > \varepsilon$. On the other hand for n large enough it is $|x_{n+1} - x_n| < 2\delta$ and $|f(x_n) - x_n| = |x_{n+1} - x_n| < \varepsilon$. So the sequence cannot come into the interval $(x - \delta, x + \delta)$, but also cannot jump over this interval. Then all cluster points have to be at most $x - \delta$ (a contradiction with L being a cluster point), or at least $x + \delta$ (a contradiction with K being a cluster point).

1996-D2-P2 — Statement

Day 2, P8 | 10 points

Let θ be a positive real number and let $\cosh t = \dfrac{e^t + e^{-t}}{2}$ denote the hyperbolic cosine. Show that if $k \in \mathbb{N}$ and both $\cosh k\theta$ and $\cosh(k+1)\theta$ are rational, then so is $\cosh \theta$.

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Theory and methods for 1996-D2-P2

Theory Linear recurrences and generating functions [T-AL-RECURRENCE]; Rational linear relations and discrete-value rigidity [T-AL-RATIONALITY]
Methods Find a recurrence that reveals the invariant [M-AL-RECURRENCE]; Eliminate rational relations, then exploit discreteness [M-AL-RATIONAL]
Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

1996-D2-P2 — Solution

For this record: Theory: T-AL-RECURRENCE; T-AL-RATIONALITY. Methods: M-AL-RECURRENCE; M-AL-RATIONAL. Full learning route

First we show that
(1) If $\cosh t$ is rational and $m \in \mathbb{N}$, then $\cosh mt$ is rational.

Since $\cosh 0 \cdot t = \cosh 0 = 1 \in \mathbb{Q}$ and $\cosh 1 \cdot t = \cosh t \in \mathbb{Q}$, (1) follows inductively from

$$\cosh(m+1)t = 2\cosh t \cdot \cosh mt - \cosh(m-1)t.$$

The statement of the problem is obvious for $k = 1$, so we consider $k \geq 2$. For any m we have

$$\begin{aligned} \cosh\theta &= \cosh((m+1)\theta - m\theta) = \\ &= \cosh(m+1)\theta \cdot \cosh m\theta - \sinh(m+1)\theta \cdot \sinh m\theta \\ &= \cosh(m+1)\theta \cdot \cosh m\theta - \sqrt{\cosh^2(m+1)\theta - 1} \cdot \sqrt{\cosh^2 m\theta - 1} \end{aligned} \tag{2}$$

Set $\cosh k\theta = a$, $\cosh(k+1)\theta = b$, $a, b \in \mathbb{Q}$. Then (2) with $m = k$ gives

$$\cosh\theta = ab - \sqrt{a^2-1}\sqrt{b^2-1}$$

and then

$$\begin{aligned} (a^2-1)(b^2-1) &= (ab - \cosh\theta)^2 \\ &= a^2b^2 - 2ab\cosh\theta + \cosh^2\theta. \end{aligned} \tag{3}$$

Set $\cosh(k^2-1)\theta = A$, $\cosh k^2\theta = B$. From (1) with $m = k-1$ and $t = (k+1)\theta$ we have $A \in \mathbb{Q}$. From (1) with $m = k$ and $t = k\theta$ we have $B \in \mathbb{Q}$. Moreover $k^2 - 1 > k$ implies $A > a$ and $B > b$. Thus $AB > ab$. From (2) with $m = k^2 - 1$ we have

$$\begin{aligned} (A^2-1)(B^2-1) &= (AB - \cosh\theta)^2 \\ &= A^2B^2 - 2AB\cosh\theta + \cosh^2\theta. \end{aligned} \tag{4}$$

So after we cancel the $\cosh^2\theta$ from (3) and (4) we have a non-trivial linear equation in $\cosh\theta$ with rational coefficients.

1996-D2-P3 — Statement

Day 2, P9 | 15 points

Let G be the subgroup of $GL_2(\mathbb{R})$, generated by A and B, where

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Let H consist of those matrices $\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ in G for which $a_{11} = a_{22} = 1$.
(a) Show that H is an abelian subgroup of G.
(b) Show that H is not finitely generated.

Remarks. $GL_2(\mathbb{R})$ denotes, as usual, the group (under matrix multiplication) of all 2×2 invertible matrices with real entries (elements). Abelian means commutative. A group is finitely generated

1996-D2-P3 — Statement (continued)

if there are a finite number of elements of the group such that every other element of the group can be obtained from these elements using the group operation.

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Theory and methods for 1996-D2-P3

Theory Groups, orders, generators, and quotients [T-AL-GROUP]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Methods Prove generation or obstruct it by an image [M-AL-GENERATORS]
Strategy Expose the structure before calculating [S-STRUCTURE]; Make the contradiction quantitative or structural [S-CONTRADICTION]

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1996-D2-P3 — Solution

For this record: Theory: T-AL-GROUP; T-AL-CONGRUENCE. Methods: M-AL-GENERATORS. Full learning route

(a) All of the matrices in G are of the form

$$\begin{bmatrix} * & * \\ 0 & * \end{bmatrix}.$$

So all of the matrices in H are of the form

$$M(x) = \begin{bmatrix} 1 & x \\ 0 & 1 \end{bmatrix},$$

so they commute. Since $M(x)^{-1} = M(-x)$, H is a subgroup of G.

(b) A generator of H can only be of the form $M(x)$, where x is a binary rational, i.e., $x = \frac{p}{2^n}$ with integer p and non-negative integer n. In H it holds

$$M(x)M(y) = M(x+y)$$
$$M(x)M(y)^{-1} = M(x-y).$$

The matrices of the form $M\left(\frac{1}{2^n}\right)$ are in H for all $n \in \mathbb{N}$. With only finite number of generators all of them cannot be achieved.

1996-D2-P4 — Statement

Day 2, P10 | 20 points

Let B be a bounded closed convex symmetric (with respect to the origin) set in $\mathbb{R}^2$ with boundary the curve Γ. Let B have the property that the ellipse of maximal area contained in B is the disc D of radius 1 centered at the origin with boundary the circle C. Prove that $A \cap \Gamma \neq \emptyset$ for any arc A of C of length $l(A) \geq \frac{\pi}{2}$.

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Theory and methods for 1996-D2-P4

Theory Convex bodies, supporting functionals, and packing [T-DS-CONVEX]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Continuity, compactness, and order [T-AN-CONTINUITY]
Methods Contradict an extremum by a controlled perturbation [M-DS-PERTURB]; Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Strategy Choose an extremal object and force rigidity [S-EXTREMAL]; Make the contradiction quantitative or structural [S-CONTRADICTION]; Translate geometry into a controlled representation [S-GEOMETRIC]

Statement Recorded solutions Catalog

1996-D2-P4 — Solution

For this record: Theory: T-DS-CONVEX; T-DS-GEOMETRY; T-AN-CONTINUITY. Methods: M-DS-PERTURB; M-DS-GEOMETRIC. Full learning route

Assume the contrary – there is an arc $A \subset C$ with length $l(A) = \dfrac{\pi}{2}$ such that $A \subset B \setminus \Gamma$. Without loss of generality we may assume that the ends of A are $M = (1/\sqrt{2}, 1/\sqrt{2})$, $N = (1/\sqrt{2}, -1/\sqrt{2})$. A is compact and Γ is closed. From $A \cap \Gamma = \emptyset$ we get $\delta > 0$ such that $\operatorname{dist}(x, y) > \delta$ for every $x \in A$, $y \in \Gamma$. Given $\varepsilon > 0$ with E_ε we denote the ellipse with boundary:

$$\frac{x^2}{(1+\varepsilon)^2} + \frac{y^2}{b^2} = 1,$$

such that $M, N \in E_\varepsilon$. Since $M \in E_\varepsilon$ we get

$$b^2 = \frac{(1+\varepsilon)^2}{2(1+\varepsilon)^2 - 1}.$$

Then we have

$$\operatorname{area} E_\varepsilon = \pi \frac{(1+\varepsilon)^2}{\sqrt{2(1+\varepsilon)^2 - 1}} > \pi = \operatorname{area} D.$$

In view of the hypotheses, $E_\varepsilon \setminus B \neq \emptyset$ for every $\varepsilon > 0$. Let $S = \{(x, y) \in \mathbb{R}^2 : |x| > |y|\}$. From $E_\varepsilon \setminus S \subset D \subset B$ it follows that $E_\varepsilon \setminus B \subset S$. Taking $\varepsilon < \delta$ we get that

$$\emptyset \neq E_\varepsilon \setminus B \subset E_\varepsilon \cap S \subset D_{1+\varepsilon} \cap S \subset B$$

– a contradiction (we use the notation $D_t = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq t^2\}$).

Remark. The ellipse with maximal area is well known as John's ellipse. Any coincidence with the President of the Jury is accidental.

1996-D2-P5 — Statement

Day 2, P11 | 20 points

(i) Prove that

$$\lim_{x \to +\infty} \sum_{n=1}^{\infty} \frac{nx}{(n^2 + x)^2} = \frac{1}{2}.$$

(ii) Prove that there is a positive constant c such that for every $x \in [1, \infty)$ we have

$$\left| \sum_{n=1}^{\infty} \frac{nx}{(n^2 + x)^2} - \frac{1}{2} \right| \leq \frac{c}{x}.$$

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Theory and methods for 1996-D2-P5

Theory Riemann sums, periodic averaging, and tails [T-AN-RIEMANN]; Taylor remainders and local scale [T-AN-TAYLOR]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Methods Expose the mesh and control its tails [M-AN-RIEMANN]
Strategy Control the limiting operation before taking the limit [S-ANALYTIC]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

1996-D2-P5 — Solution

For this record: Theory: T-AN-RIEMANN; T-AN-TAYLOR; T-AN-INTEGRAL. Methods: M-AN-RIEMANN. Full learning route

(i) Set $f(t) = \dfrac{t}{(1+t^2)^2}$, $h = \dfrac{1}{\sqrt{x}}$. Then

$$\sum_{n=1}^{\infty} \frac{nx}{(n^2+x)^2} = h \sum_{n=1}^{\infty} f(nh) \xrightarrow[h\to 0]{} \int_0^{\infty} f(t)\,dt = \frac{1}{2}.$$

The convergence holds since $h \sum\limits_{n=1}^{\infty} f(nh)$ is a Riemann sum of the integral $\int\limits_0^{\infty} f(t)\,dt$. There are no problems with the infinite domain because f is integrable and $f \downarrow 0$ for $t \to \infty$ (thus $h \sum\limits_{n=N}^{\infty} f(nh) \geq \int\limits_{Nh}^{\infty} f(t)\,dt \geq h \sum\limits_{n=N+1}^{\infty} f(nh)$).

(ii) We have

$$\begin{aligned}
&\left| \sum_{n=1}^{\infty} \frac{nx}{(n^2+x)^2} - \frac{1}{2} \right| \\
&= \left| \sum_{n=1}^{\infty} \left(hf(nh) - \int_{nh-\frac{h}{2}}^{nh+\frac{h}{2}} f(t)\,dt \right) - \int_0^{\frac{h}{2}} f(t)\,dt \right| \qquad (1) \\
&\leq \sum_{n=1}^{\infty} \left| hf(nh) - \int_{nh-\frac{h}{2}}^{nh+\frac{h}{2}} f(t)\,dt \right| + \int_0^{\frac{h}{2}} f(t)\,dt
\end{aligned}$$

Using twice integration by parts one has

$$2b\,g(a) - \int_{a-b}^{a+b} g(t)\,dt = -\frac{1}{2} \int_0^b (b-t)^2 (g''(a+t) + g''(a-t))\,dt \qquad (2)$$

for every $g \in C^2[a-b, a+b]$. Using $f(0) = 0$, $f \in C^2[0, h/2]$ one gets

$$\int_0^{h/2} f(t)\,dt = O(h^2). \qquad (3)$$

From (1), (2) and (3) we get

$$\begin{aligned}
&\left| \sum_{n=1}^{\infty} \frac{nx}{(n^2+x)^2} - \frac{1}{2} \right| \\
&\leq \sum_{n=1}^{\infty} h^2 \int_{nh-\frac{h}{2}}^{nh+\frac{h}{2}} |f''(t)|\,dt + O(h^2) \\
&= h^2 \int_{\frac{h}{2}}^{\infty} |f''(t)|\,dt + O(h^2) \\
&= O(h^2) \\
&= O(x^{-1}).
\end{aligned}$$

For completeness, the bound is uniform for $x \geq 1$: then $0 < h \leq 1$, $f(t) \leq t$ for $t \geq 0$, and

$$f''(t) = \frac{12t(t^2-1)}{(1+t^2)^4}, \qquad \int_0^{\infty} |f''(t)|\,dt = \frac{3}{2}.$$

Indeed, $f'(0) = 1$, $f'(1) = -1/4$, and $f'(t) \to 0$ as $t \to \infty$; f'' changes sign only at 1 on $(0, \infty)$. The preceding error bound therefore gives

$$\left| \sum_{n=1}^{\infty} \frac{nx}{(n^2+x)^2} - \frac{1}{2} \right| \leq h^2 \frac{3}{2} + \int_0^{h/2} t\,dt = \frac{13}{8} h^2 = \frac{13}{8x}.$$

Thus $c = 13/8$ is one admissible constant in (ii).

1996-D2-P6 — Statement

Day 2, P12 | 25 points

(Carleman's inequality)

(i) Prove that for every sequence $\{a_n\}_{n=1}^{\infty}$, such that $a_n > 0$, $n = 1, 2, \ldots$ and $\sum_{n=1}^{\infty} a_n < \infty$, we have

$$\sum_{n=1}^{\infty} (a_1 a_2 \cdots a_n)^{1/n} < e \sum_{n=1}^{\infty} a_n,$$

where e is the natural log base.

(ii) Prove that for every $\varepsilon > 0$ there exists a sequence $\{a_n\}_{n=1}^{\infty}$, such that $a_n > 0$, $n = 1, 2, \ldots$, $\sum_{n=1}^{\infty} a_n < \infty$ and

$$\sum_{n=1}^{\infty} (a_1 a_2 \cdots a_n)^{1/n} > (e - \varepsilon) \sum_{n=1}^{\infty} a_n.$$

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Theory and methods for 1996-D2-P6

Theory Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Nonnegative sums, Fubini, and integral kernels [T-AN-FUBINI]; Series, finite cancellation, and summation by parts [T-AN-SERIES]
Methods Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]; Find a finite telescoping identity [M-AN-TELESCOPE]
Strategy Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

1996-D2-P6 — Solution

For this record: Theory: T-AN-INEQUALITIES; T-AN-FUBINI; T-AN-SERIES. Methods: M-AN-WEIGHTS; M-AN-TELESCOPE. Full learning route

(i) Put for $n \in \mathbb{N}$

$$c_n = (n+1)^n / n^{n-1}. \tag{1}$$

Observe that $c_1 c_2 \cdots c_n = (n+1)^n$. Hence, for $n \in \mathbb{N}$,

$$(a_1 a_2 \cdots a_n)^{1/n} = (a_1 c_1 a_2 c_2 \cdots a_n c_n)^{1/n} / (n+1) \le (a_1 c_1 + \cdots + a_n c_n)/n(n+1).$$

Consequently,

$$\sum_{n=1}^{\infty} (a_1 a_2 \cdots a_n)^{1/n} \le \sum_{n=1}^{\infty} a_n c_n \left(\sum_{m=n}^{\infty} (m(m+1))^{-1} \right). \tag{2}$$

Since

$$\sum_{m=n}^{\infty} (m(m+1))^{-1} = \sum_{m=n}^{\infty} \left(\frac{1}{m} - \frac{1}{m+1} \right) = 1/n$$

we have

$$\sum_{n=1}^{\infty} a_n c_n \left(\sum_{m=n}^{\infty} (m(m+1))^{-1} \right) = \sum_{n=1}^{\infty} a_n c_n / n = \sum_{n=1}^{\infty} a_n ((n+1)/n)^n < e \sum_{n=1}^{\infty} a_n$$

(by (1)). Combining the last inequality with (2) we get the result.

(ii) Set $a_n = n^{n-1}(n+1)^{-n}$ for $n = 1, 2, \ldots, N$ and $a_n = 2^{-n}$ for $n > N$, where N will be chosen later. Then

$$(a_1 \cdots a_n)^{1/n} = \frac{1}{n+1} \tag{3}$$

for $n \le N$. Let $K = K(\varepsilon)$ be such that

$$\left(\frac{n+1}{n} \right)^n > e - \frac{\varepsilon}{2} \quad \text{for } n > K. \tag{4}$$

1996-D2-P6 — Solution (continued)

Choose N from the condition

$$\sum_{n=1}^{K} a_n + \sum_{n=N+1}^{\infty} 2^{-n} \le \frac{\varepsilon}{(2e-\varepsilon)(e-\varepsilon)} \sum_{n=K+1}^{N} \frac{1}{n}, \tag{5}$$

which is always possible because the harmonic series diverges. Using (3), (4) and (5) we have

$$\begin{aligned}\sum_{n=1}^{\infty} a_n &= \sum_{n=1}^{K} a_n + \sum_{n=N+1}^{\infty} 2^{-n} + \sum_{n=K+1}^{N} \frac{1}{n}\left(\frac{n}{n+1}\right)^n < \\ &< \frac{\varepsilon}{(2e-\varepsilon)(e-\varepsilon)} \sum_{n=K+1}^{N} \frac{1}{n} + \left(e - \frac{\varepsilon}{2}\right)^{-1} \sum_{n=K+1}^{N} \frac{1}{n} = \\ &= \frac{1}{e-\varepsilon} \sum_{n=K+1}^{N} \frac{1}{n} \le \frac{1}{e-\varepsilon} \sum_{n=1}^{\infty} (a_1 \cdots a_n)^{1/n}.\end{aligned}$$

3.4 1997

Day 1

1997-D1-P1 — Statement

Day 1, P1

Let $\{\varepsilon_n\}_{n=1}^{\infty}$ be a sequence of positive real numbers, such that $\lim_{n\to\infty} \varepsilon_n = 0$. Find

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \ln\left(\frac{k}{n} + \varepsilon_n\right),$$

where ln denotes the natural logarithm.

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Theory and methods for 1997-D1-P1

Theory Riemann sums, periodic averaging, and tails [T-AN-RIEMANN]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Methods Expose the mesh and control its tails [M-AN-RIEMANN]; Separate a main range from exceptional scales [M-AN-SPLIT]
Strategy Build a chain of bounds and track equality [S-BOUND]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

1997-D1-P1 — Solution

For this record: Theory: T-AN-RIEMANN; T-AN-CONVERGENCE; T-AN-INTEGRAL. Methods: M-AN-RIEMANN; M-AN-SPLIT. Full learning route

It is well known that

$$-1 = \int_0^1 \ln x \, dx = \lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \ln\left(\frac{k}{n}\right)$$

(Riemman's sums). Then

$$\frac{1}{n} \sum_{k=1}^{n} \ln\left(\frac{k}{n} + \varepsilon_n\right) \ge \frac{1}{n} \sum_{k=1}^{n} \ln\left(\frac{k}{n}\right) \xrightarrow[n\to\infty]{} -1.$$

1997-D1-P1 — Solution (continued)

Given $\varepsilon > 0$ there exist n_0 such that $0 < \varepsilon_n \le \varepsilon$ for all $n \ge n_0$. Then

$$\frac{1}{n}\sum_{k=1}^{n} \ln\left(\frac{k}{n} + \varepsilon_n\right) \le \frac{1}{n}\sum_{k=1}^{n} \ln\left(\frac{k}{n} + \varepsilon\right).$$

Since

$$\lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} \ln\left(\frac{k}{n} + \varepsilon\right) = \int_0^1 \ln(x+\varepsilon)\,dx = \int_\varepsilon^{1+\varepsilon} \ln x\,dx$$

we obtain the result when ε goes to 0 and so

$$\lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} \ln\left(\frac{k}{n} + \varepsilon_n\right) = -1.$$

1997-D1-P2 — Statement

Day 1, P2

Suppose $\sum_{n=1}^{\infty} a_n$ converges. Do the following sums have to converge as well?

a) $a_1 + a_2 + a_4 + a_3 + a_8 + a_7 + a_6 + a_5 + a_{16} + a_{15} + \cdots + a_9 + a_{32} + \cdots$

b) $a_1 + a_2 + a_3 + a_4 + a_5 + a_7 + a_6 + a_8 + a_9 + a_{11} + a_{13} + a_{15} + a_{10} + a_{12} + a_{14} + a_{16} + a_{17} + a_{19} + \cdots$

Justify your answers.

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Theory and methods for 1997-D1-P2

Theory Series, finite cancellation, and summation by parts [T-AN-SERIES]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]

Methods Test convergence through blocks and differences [M-AN-SERIES]

Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

1997-D1-P2 — Solution

For this record: Theory: T-AN-SERIES; T-AN-CONVERGENCE. Methods: M-AN-SERIES. Full learning route

a) Yes. Let $S = \sum_{n=1}^{\infty} a_n$, $S_n = \sum_{k=1}^{n} a_k$. Fix $\varepsilon > 0$ and a number n_0 such that $|S_n - S| < \varepsilon$ for $n > n_0$. The partial sums of the permuted series have the form $L_{2^{n-1}+k} = S_{2^{n-1}} + S_{2^n} - S_{2^n-k}$, $0 \le k < 2^{n-1}$ and for $2^{n-1} > n_0$ we have $|L_{2^{n-1}+k} - S| < 3\varepsilon$, i.e. the permuted series converges.

b) No. Take $a_n = \dfrac{(-1)^{n+1}}{\sqrt{n}}$. Then $L_{3\cdot 2^{n-2}} = S_{2^{n-1}} + \sum_{k=2^{n-2}}^{2^{n-1}-1} \dfrac{1}{\sqrt{2k+1}}$ and $L_{3\cdot 2^{n-2}} - S_{2^{n-1}} \ge \dfrac{2^{n-2}}{\sqrt{2^n}} \xrightarrow[n\to\infty]{} \infty$, so $L_{3\cdot 2^{n-2}} \xrightarrow[n\to\infty]{} \infty$.

1997-D1-P3 — Statement

Day 1, P3

Let A and B be real $n \times n$ matrices such that $A^2 + B^2 = AB$. Prove that if $BA - AB$ is an invertible matrix then n is divisible by 3.

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Theory and methods for 1997-D1-P3

Theory Determinants, block identities and inverses [T-LA-DETERMINANT]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Factor while preserving matrix order [M-LA-FACTOR]
Strategy Expose the structure before calculating [S-STRUCTURE]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

1997-D1-P3 — Solution

For this record: Theory: T-LA-DETERMINANT; T-AL-POLYNOMIAL. Methods: M-LA-FACTOR. Full learning route

Set $S = A + \omega B$, where $\omega = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$. We have

$$\begin{aligned} S\overline{S} &= (A+\omega B)(A+\overline{\omega}B) = A^2 + \omega BA + \overline{\omega}AB + B^2 \\ &= AB + \omega BA + \overline{\omega}AB = \omega(BA - AB), \end{aligned}$$

because $\overline{\omega}+1 = -\omega$. Since $\det(S\overline{S}) = \det S \cdot \det \overline{S}$ is a real number and $\det \omega(BA-AB) = \omega^n \det(BA-AB)$ and $\det(BA-AB) \neq 0$, then ω^n is a real number. This is possible only when n is divisible by 3.

1997-D1-P4 — Statement

Day 1, P4

Let α be a real number, $1 < \alpha < 2$.
a) Show that α has a unique representation as an infinite product

$$\alpha = \left(1 + \frac{1}{n_1}\right)\left(1 + \frac{1}{n_2}\right)\ldots$$

where each n_i is a positive integer satisfying

$$n_i^2 \leq n_{i+1}.$$

b) Show that α is rational if and only if its infinite product has the following property:

For some m and all $k \geq m$,

$$n_{k+1} = n_k^2.$$

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Theory and methods for 1997-D1-P4

Theory Greedy products, tail bounds, and integer descent [T-AL-GREEDY]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Methods Choose the next factor and control the entire tail [M-AL-GREEDY]
Strategy Build a witness and verify every constraint [S-CONSTRUCTION]; Choose an extremal object and force rigidity [S-EXTREMAL]

Statement Recorded solutions Catalog

1997-D1-P4 — Solution

For this record: Theory: T-AL-GREEDY; T-AN-CONVERGENCE. Methods: M-AL-GREEDY. Full learning route

a) We construct inductively the sequence $\{n_i\}$ and the ratios

$$\theta_k = \frac{\alpha}{\prod_1^k (1 + \frac{1}{n_i})}$$

1997-D1-P4 — Solution (continued)

so that

$$\theta_k > 1 \quad \text{for all } k.$$

Choose n_k to be the least n for which

$$1 + \frac{1}{n} < \theta_{k-1}$$

($\theta_0 = \alpha$) so that for each k,

$$1 + \frac{1}{n_k} < \theta_{k-1} \leq 1 + \frac{1}{n_k - 1}. \tag{1}$$

Since

$$\theta_{k-1} \leq 1 + \frac{1}{n_k - 1}$$

we have

$$1 + \frac{1}{n_{k+1}} < \theta_k = \frac{\theta_{k-1}}{1 + \frac{1}{n_k}} \leq \frac{1 + \frac{1}{n_k - 1}}{1 + \frac{1}{n_k}} = 1 + \frac{1}{n_k^2 - 1}.$$

Hence, for each k, $n_{k+1} \geq n_k^2$.
Since $n_1 \geq 2$, $n_k \to \infty$ so that $\theta_k \to 1$. Hence

$$\alpha = \prod_1^\infty \left(1 + \frac{1}{n_k}\right).$$

The uniquness of the infinite product will follow from the fact that on every step n_k has to be determine by (1).
Indeed, if for some k we have

$$1 + \frac{1}{n_k} \geq \theta_{k-1}$$

then $\theta_k \leq 1$, $\theta_{k+1} < 1$ and hence $\{\theta_k\}$ does not converge to 1.
Now observe that for $M > 1$,

$$\left(1 + \frac{1}{M}\right)\left(1 + \frac{1}{M^2}\right)\left(1 + \frac{1}{M^4}\right)\cdots = 1 + \frac{1}{M} + \frac{1}{M^2} + \frac{1}{M^3} + \cdots = 1 + \frac{1}{M-1}. \tag{2}$$

Assume that for some k we have

$$1 + \frac{1}{n_k - 1} < \theta_{k-1}.$$

Then we get

$$\begin{aligned} \frac{\alpha}{(1+\frac{1}{n_1})(1+\frac{1}{n_2})\ldots} &= \frac{\theta_{k-1}}{(1+\frac{1}{n_k})(1+\frac{1}{n_{k+1}})\ldots} \\ &\geq \frac{\theta_{k-1}}{(1+\frac{1}{n_k})(1+\frac{1}{n_k^2})\ldots} = \frac{\theta_{k-1}}{1+\frac{1}{n_k-1}} > 1 \end{aligned}$$

– a contradiction.
b) From (2) α is rational if its product ends in the stated way.

Conversely, suppose α is the rational number $\frac{p}{q}$. Our aim is to show that for some m,

$$\theta_{m-1} = \frac{n_m}{n_m - 1}.$$

Suppose this is not the case, so that for every m,

$$\theta_{m-1} < \frac{n_m}{n_m - 1}. \tag{3}$$

For each k we write

$$\theta_k = \frac{p_k}{q_k}$$

1997-D1-P4 — Solution (continued)

as a fraction (not necessarily in lowest terms) where

$$p_0 = p, \quad q_0 = q$$

and in general

$$p_k = p_{k-1}n_k, \quad q_k = q_{k-1}(n_k + 1).$$

The numbers $p_k - q_k$ are positive integers: to obtain a contradiction it suffices to show that this sequence is strictly decreasing. Now,

$$\begin{aligned} p_k - q_k - (p_{k-1} - q_{k-1}) &= n_k p_{k-1} - (n_k + 1)q_{k-1} - p_{k-1} + q_{k-1} \\ &= (n_k - 1)p_{k-1} - n_k q_{k-1} \end{aligned}$$

and this is negative because $\dfrac{p_{k-1}}{q_{k-1}} = \theta_{k-1} < \dfrac{n_k}{n_k - 1}$ by inequality (3).

1997-D1-P5 — Statement

Day 1, P5

For a natural n consider the hyperplane

$$\mathbb{R}_0^n = \left\{ x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n : \sum_{i=1}^{n} x_i = 0 \right\}$$

and the lattice $\mathbb{Z}_0^n = \{y \in \mathbb{R}_0^n : \text{all } y_i \text{ are integers}\}$. Define the (quasi–)norm in $\mathbb{R}^n$ by $\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}$ if $0 < p < \infty$, and $\|x\|_\infty = \max_i |x_i|$.

a) Let $x \in \mathbb{R}_0^n$ be such that

$$\max_i x_i - \min_i x_i \leq 1.$$

For every $p \in [1, \infty]$ and for every $y \in \mathbb{Z}_0^n$ prove that

$$\|x\|_p \leq \|x + y\|_p.$$

b) For every $p \in (0, 1)$, show that there is an n and an $x \in \mathbb{R}_0^n$ with $\max_i x_i - \min_i x_i \leq 1$ and an $y \in \mathbb{Z}_0^n$ such that

$$\|x\|_p > \|x + y\|_p.$$

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Theory and methods for 1997-D1-P5

Theory Metric contractions, covers, and nearest points [T-DS-METRIC]; Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Methods Split a family by traces, pieces, or labels [M-DS-PARTITION]; Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Strategy Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]; Prove necessity and sufficiency separately [S-CLASSIFICATION]

Statement Recorded solutions Catalog

1997-D1-P5 — Solution

For this record: Theory: T-DS-METRIC; T-AN-CONVEXITY; T-AN-CONVERGENCE. Methods: M-DS-PARTITION; M-DS-GEOMETRIC. Full learning route

a) For $x = 0$ the statement is trivial. Let $x \neq 0$. Then $\max_i x_i > 0$ and $\min_i x_i < 0$. Hence $\|x\|_\infty < 1$. From the hypothesis on x it follows that:

1997-D1-P5 — Solution (continued)

i) If $x_j \le 0$ then $\max_i x_i \le x_j + 1$.
ii) If $x_j \ge 0$ then $\min_i x_i \ge x_j - 1$.
Consider $y \in \mathbb{Z}_0^n$, $y \neq 0$. We split the indices $\{1, 2, \dots, n\}$ into five sets:

$$I(0) = \{i : y_i = 0\},$$

$$I(+,+) = \{i : y_i > 0, x_i \ge 0\}, \quad I(+,-) = \{i : y_i > 0, x_i < 0\},$$
$$I(-,+) = \{i : y_i < 0, x_i > 0\}, \quad I(-,-) = \{i : y_i < 0, x_i \le 0\}.$$

As least one of the last four index sets is not empty. If $I(+,+) \neq \emptyset$ or $I(-,-) \neq \emptyset$ then $\|x+y\|_\infty \ge 1 > \|x\|_\infty$. If $I(+,+) = I(-,-) = \emptyset$ then $\sum y_i = 0$ implies $I(+,-) \neq \emptyset$ and $I(-,+) \neq \emptyset$. Therefore i) and ii) give $\|x+y\|_\infty \ge \|x\|_\infty$ which completes the case $p = \infty$.
Now let $1 \le p < \infty$. Then using i) for every $j \in I(+,-)$ we get $|x_j+y_j| = y_j - 1 + x_j + 1 \ge |y_j| - 1 + \max_i x_i$. Hence

$$|x_j + y_j|^p \ge |y_j| - 1 + |x_k|^p \quad \text{for every } k \in I(-,+) \text{ and } j \in I(+,-).$$

Similarly

$$|x_j + y_j|^p \ge |y_j| - 1 + |x_k|^p \quad \text{for every } k \in I(+,-) \text{ and } j \in I(-,+);$$
$$|x_j + y_j|^p \ge |y_j| + |x_j|^p \quad \text{for every } j \in I(+,+) \cup I(-,-).$$

Assume that $\sum_{j \in I(+,-)} 1 \ge \sum_{j \in I(-,+)} 1$. Then

$$\begin{aligned}
\|x+y\|_p^p - \|x\|_p^p &= \sum_{j \in I(+,+) \cup I(-,-)} (|x_j + y_j|^p - |x_j|^p) + \left(\sum_{j \in I(+,-)} |x_j + y_j|^p - \sum_{k \in I(-,+)} |x_k|^p \right) \\
&\quad + \left(\sum_{j \in I(-,+)} |x_j + y_j|^p - \sum_{k \in I(+,-)} |x_k|^p \right) \\
&\ge \sum_{j \in I(+,+) \cup I(-,-)} |y_j| + \sum_{j \in I(+,-)} (|y_j| - 1) + \sum_{j \in I(-,+)} (|y_j| - 1) - \sum_{j \in I(+,-)} 1 + \sum_{j \in I(-,+)} 1 \\
&= \sum_{i=1}^n |y_i| - 2 \sum_{j \in I(+,-)} 1 = 2 \sum_{j \in I(+,-)} (y_j - 1) + 2 \sum_{j \in I(+,+)} y_j \ge 0.
\end{aligned}$$

The case $\sum_{j \in I(+,-)} 1 \le \sum_{j \in I(-,+)} 1$ is similar. This proves the statement.

b) Fix $p \in (0,1)$ and a rational $t \in (\frac{1}{2}, 1)$. Choose a pair of positive integers m and l such that $mt = l(1-t)$ and set $n = m + l$. Let

$$x_i = t, \quad i = 1, 2, \dots, m; \qquad x_i = t - 1, \quad i = m+1, m+2, \dots, n;$$

$$y_i = -1, \quad i = 1, 2, \dots, m; \qquad y_{m+1} = m; \qquad y_i = 0, \quad i = m+2, \dots, n.$$

Then $x \in \mathbb{R}_0^n$, $\max_i x_i - \min_i x_i = 1$, $y \in \mathbb{Z}_0^n$ and

$$\|x\|_p^p - \|x+y\|_p^p = m(t^p - (1-t)^p) + (1-t)^p - (m - 1 + t)^p,$$

which is possitive for m big enough.

1997-D1-P6 — Statement

Day 1, P6

Suppose that F is a family of finite subsets of $\mathbb{N}$ and for any two sets $A, B \in F$ we have $A \cap B \neq \emptyset$.
a) Is it true that there is a finite subset Y of $\mathbb{N}$ such that for any $A, B \in F$ we have $A \cap B \cap Y \neq \emptyset$?

1997-D1-P6 — Statement (continued)

b) Is the statement a) true if we suppose in addition that all of the members of F have the same size?

Justify your answers.

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Theory and methods for 1997-D1-P6

Theory Finite set systems and antichains [T-DS-SETS]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Split a family by traces, pieces, or labels [M-DS-PARTITION]; Strengthen the statement before the induction step [M-DS-INDUCT]
Strategy Reduce size while preserving the defining conditions [S-INDUCTION]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

1997-D1-P6 — Solution

For this record: Theory: T-DS-SETS; T-DS-INDUCTION; T-DS-COUNTING. Methods: M-DS-PARTITION; M-DS-INDUCT.
Full learning route

a) No. Consider $F = \{A_1, B_1, \dots, A_n, B_n, \dots\}$, where $A_n = \{1, 3, 5, \dots, 2n-1, 2n\}$, $B_n = \{2, 4, 6, \dots, 2n, 2n+1\}$.
b) Yes. We will prove inductively a stronger statement:
Suppose F, G are two families of finite subsets of $\mathbb{N}$ such that:
1) For every $A \in F$ and $B \in G$ we have $A \cap B \neq \emptyset$;
2) All the elements of F have the same size r, and elements of G – size s. (we shall write $\#(F) = r$, $\#(G) = s$).
Then there is a finite set Y such that $A \cap B \cap Y \neq \emptyset$ for every $A \in F$ and $B \in G$.
The problem b) follows if we take $F = G$.
Proof of the statement: Empty families are trivial. If $r = 1$, fix $B_0 \in G$ and take $Y = B_0$: every singleton $A \in F$ meets B_0 and every $B \in G$, so $A \cap B \cap Y \neq \emptyset$. The case $s = 1$ is symmetric. Fix the numbers r, s and suppose the statement is proved for all pairs F', G' with $\#(F') < r$, $\#(G') < s$. Fix $A_0 \in F$, $B_0 \in G$. For any subset $C \subset A_0 \cup B_0$, denote

$$F(C) = \{A \in F : A \cap (A_0 \cup B_0) = C\}.$$

Then $F = \bigcup_{\emptyset \neq C \subset A_0 \cup B_0} F(C)$. It is enough to prove that for any pair of non-empty sets $C, D \subset A_0 \cup B_0$ the families $F(C)$ and $G(D)$ satisfy the statement.
Indeed, if we denote by $Y_{C,D}$ the corresponding finite set, then the finite set $\bigcup_{C,D \subset A_0 \cup B_0} Y_{C,D}$ will satisfy the statement for F and G. The proof for $F(C)$ and $G(D)$.
If $C \cap D \neq \emptyset$, it is trivial.

If $C \cap D = \emptyset$, then any two sets $A \in F(C)$, $B \in G(D)$ must meet outside $A_0 \cup B_0$. Then if we denote $\widetilde{F}(C) = \{A \setminus C : A \in F(C)\}$, $\widetilde{G}(D) = \{B \setminus D : B \in G(D)\}$, then $\widetilde{F}(C)$ and $\widetilde{G}(D)$ satisfy the conditions 1) and 2) above, with $\#(\widetilde{F}(C)) = \#(F) - \#C < r$, $\#(\widetilde{G}(D)) = \#(G) - \#D < s$, and the inductive assumption works.

Day 2

1997-D2-P1 — Statement

Day 2, P7

Let f be a $C^3(\mathbb{R})$ non-negative function, $f(0) = f'(0) = 0$, $0 < f''(0)$. Let

$$g(x) = \left(\frac{\sqrt{f(x)}}{f'(x)} \right)'$$

for $x \neq 0$ and $g(0) = 0$. Show that g is bounded in some neighbourhood of 0. Does the theorem hold for $f \in C^2(\mathbb{R})$?

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Theory and methods for 1997-D2-P1

Theory Taylor remainders and local scale [T-AN-TAYLOR]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Methods Expand near equality, bound away from it [M-AN-LOCAL]
Strategy Pass between local constraints and a global conclusion [S-LOCALGLOBAL]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

1997-D2-P1 — Solution

For this record: Theory: T-AN-TAYLOR; T-AN-DERIVATIVE. Methods: M-AN-LOCAL. Full learning route

Let $c = \frac{1}{2} f''(0)$. We have

$$g = \frac{(f')^2 - 2ff''}{2(f')^2\sqrt{f}},$$

where

$$f(x) = cx^2 + O(x^3), \quad f'(x) = 2cx + O(x^2), \quad f''(x) = 2c + O(x).$$

Therefore $(f'(x))^2 = 4c^2x^2 + O(x^3)$,

$$2f(x)f''(x) = 4c^2x^2 + O(x^3)$$

and

$$2(f'(x))^2\sqrt{f(x)} = 2(4c^2x^2 + O(x^3))|x|\sqrt{c + O(x)}.$$

g is bounded because

$$\frac{2(f'(x))^2\sqrt{f(x)}}{|x|^3} \xrightarrow[x\to 0]{} 8c^{5/2} \neq 0$$

and $f'(x)^2 - 2f(x)f''(x) = O(x^3)$.
The theorem does not hold for some C^2-functions.

Let $f(x) = (x + |x|^{3/2})^2 = x^2 + 2x^2\sqrt{|x|} + |x|^3$, so f is C^2. For $x > 0$,

$$g(x) = \frac{1}{2}\left(\frac{1}{1 + \frac{3}{2}\sqrt{x}} \right)' = -\frac{1}{2} \cdot \frac{1}{(1 + \frac{3}{2}\sqrt{x})^2} \cdot \frac{3}{4} \cdot \frac{1}{\sqrt{x}} \xrightarrow[x\to 0]{} -\infty.$$

1997-D2-P2 — Statement

Day 2, P8

Let M be an invertible matrix of dimension $2n \times 2n$, represented in block form as

$$M = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \quad \text{and} \quad M^{-1} = \begin{bmatrix} E & F \\ G & H \end{bmatrix}.$$

Show that $\det M \cdot \det H = \det A$.

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Theory and methods for 1997-D2-P2

Theory Determinants, block identities and inverses [T-LA-DETERMINANT]
Methods Exploit a compatible block decomposition [M-LA-BLOCK]
Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

1997-D2-P2 — Solution

For this record: Theory: T-LA-DETERMINANT. Methods: M-LA-BLOCK. Full learning route

Let I denote the identity $n \times n$ matrix. Then

$$\det M \cdot \det H = \det \begin{bmatrix} A & B \\ C & D \end{bmatrix} \cdot \det \begin{bmatrix} I & F \\ 0 & H \end{bmatrix} = \det \begin{bmatrix} A & 0 \\ C & I \end{bmatrix} = \det A.$$

1997-D2-P3 — Statement

Day 2, P9

Show that $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} \sin(\log n)}{n^{\alpha}}$ converges if and only if $\alpha > 0$.

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Theory and methods for 1997-D2-P3

Theory Series, finite cancellation, and summation by parts [T-AN-SERIES]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Trigonometric identities and safe branches [T-AN-TRIG]
Methods Test convergence through blocks and differences [M-AN-SERIES]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

1997-D2-P3 — Solution

For this record: Theory: T-AN-SERIES; T-AN-DERIVATIVE; T-AN-TRIG. Methods: M-AN-SERIES. Full learning route

Set $f(t) = \frac{\sin(\log t)}{t^{\alpha}}$. We have

$$f'(t) = -\frac{\alpha}{t^{\alpha+1}} \sin(\log t) + \frac{\cos(\log t)}{t^{\alpha+1}}.$$

So $|f'(t)| \leq \frac{1+\alpha}{t^{\alpha+1}}$ for $\alpha > 0$. Then from Mean value theorem for some $\theta \in (0,1)$ we get $|f(n+1) - f(n)| = |f'(n+\theta)| \leq \frac{1+\alpha}{n^{\alpha+1}}$. Since $\sum \frac{1+\alpha}{n^{\alpha+1}} < +\infty$ for $\alpha > 0$ and $f(n) \xrightarrow[n\to\infty]{} 0$ we get that $\sum_{n=1}^{\infty} (-1)^{n-1} f(n) = \sum_{n=1}^{\infty} (f(2n-1) - f(2n))$ converges.

1997-D2-P3 — Solution (continued)

Now we have to prove that $\dfrac{\sin(\log n)}{n^\alpha}$ does not converge to 0 for $\alpha \le 0$. It suffices to consider $\alpha = 0$.

We show that $a_n = \sin(\log n)$ does not tend to zero. Assume the contrary. There exist $k_n \in \mathbb{N}$ and $\lambda_n \in [-\frac{1}{2}, \frac{1}{2}]$ for $n > e^2$ such that $\dfrac{\log n}{\pi} = k_n + \lambda_n$. Then $|a_n| = \sin \pi|\lambda_n|$. Since $a_n \to 0$ we get $\lambda_n \to 0$. We have

$$\begin{aligned} k_{n+1} - k_n &= \frac{\log(n+1) - \log n}{\pi} - (\lambda_{n+1} - \lambda_n) \\ &= \frac{1}{\pi} \log\left(1 + \frac{1}{n}\right) - (\lambda_{n+1} - \lambda_n). \end{aligned}$$

Then $|k_{n+1} - k_n| < 1$ for all n big enough. Hence there exists n_0 so that $k_n = k_{n_0}$ for $n > n_0$. So $\dfrac{\log n}{\pi} = k_{n_0} + \lambda_n$ for $n > n_0$. Since $\lambda_n \to 0$ we get contradiction with $\log n \to \infty$.

1997-D2-P4 — Statement

Day 2, P10

a) Let the mapping $f : M_n \to \mathbb{R}$ from the space $M_n = \mathbb{R}^{n^2}$ of $n \times n$ matrices with real entries to reals be linear, i.e.:

$$f(A + B) = f(A) + f(B), \quad f(cA) = cf(A) \tag{1}$$

for any $A, B \in M_n$, $c \in \mathbb{R}$. Prove that there exists a unique matrix $C \in M_n$ such that $f(A) = \operatorname{tr}(AC)$ for any $A \in M_n$. (If $A = \{a_{ij}\}_{i,j=1}^n$ then $\operatorname{tr}(A) = \sum_{i=1}^n a_{ii}$).

b) Suppose in addition to (1) that

$$f(A \cdot B) = f(B \cdot A) \tag{2}$$

for any $A, B \in M_n$. Prove that there exists $\lambda \in \mathbb{R}$ such that $f(A) = \lambda \cdot \operatorname{tr}(A)$.

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Theory and methods for 1997-D2-P4

Theory Dual bases, kernels and separating functionals [T-LA-DUALITY]; Trace pairing and commutator calculus [T-LA-TRACE]; Rank, dimension and exact linear constraints [T-LA-RANK]
Methods Use trace to extract a scalar invariant [M-LA-TRACE]; Prove a dimension obstruction and attain it [M-LA-DIMENSION]
Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

1997-D2-P4 — Solution

For this record: Theory: T-LA-DUALITY; T-LA-TRACE; T-LA-RANK. Methods: M-LA-TRACE; M-LA-DIMENSION. Full learning route

a) If we denote by E_{ij} the standard basis of M_n consisting of elementary matrix (with entry 1 at the place (i, j) and zero elsewhere), then the entries c_{ij} of C can be defined by $c_{ij} = f(E_{ji})$.

b) Denote by L the $n^2 - 1$-dimensional linear subspace of M_n consisting of all matrices with zero trace. The elements E_{ij} with $i \ne j$ and the elements $E_{ii} - E_{nn}$, $i = 1, \dots, n-1$ form a linear basis for L. Since

$$\begin{aligned} E_{ij} &= E_{ij} \cdot E_{jj} - E_{jj} \cdot E_{ij}, \quad i \ne j \\ E_{ii} - E_{nn} &= E_{in} \cdot E_{ni} - E_{ni} \cdot E_{in}, \quad i = 1, \dots, n-1, \end{aligned}$$

then the property (2) shows that f is vanishing identically on L. Now, for any $A \in M_n$ we have $A - \frac{1}{n}\operatorname{tr}(A) \cdot E \in L$, where E is the identity matrix, and therefore $f(A) = \frac{1}{n} f(E) \cdot \operatorname{tr}(A)$.

1997-D2-P5 — Statement

Day 2, P11

Let X be an arbitrary set, let f be an one-to-one function mapping X onto itself. Prove that there exist mappings $g_1, g_2 : X \to X$ such that $f = g_1 \circ g_2$ and $g_1 \circ g_1 = id = g_2 \circ g_2$, where id denotes the identity mapping on X.

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Theory and methods for 1997-D2-P5

Theory Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]; Groups, orders, generators, and quotients [T-AL-GROUP]
Methods Work orbit by orbit and track parity [M-AL-ORBIT]
Strategy Expose the structure before calculating [S-STRUCTURE]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

1997-D2-P5 — Solution

For this record: Theory: T-AL-PERMUTATION; T-AL-GROUP. Methods: M-AL-ORBIT. Full learning route

Let $f^n = \underbrace{f \circ f \circ \ldots \circ f}_{n \text{ times}}$, $f^0 = id$, $f^{-n} = (f^{-1})^n$ for every natural number n. Let $T(x) = \{f^n(x) : n \in \mathbb{Z}\}$ for every $x \in X$. The sets $T(x)$ for different x's either coinside or do not intersect. Each of them is mapped by f onto itself. It is enough to prove the theorem for every such set. Let $A = T(x)$. If A is finite, then we can think that A is the set of all vertices of a regular n polygon and that f is rotation by $\frac{2\pi}{n}$. Such rotation can be obtained as a composition of 2 symmetries mapping the n polygon onto itself (if n is even then there are axes of symmetry making $\frac{\pi}{n}$ angle; if $n = 2k+1$ then there are axes making $k\frac{2\pi}{n}$ angle). If A is infinite then we can think that $A = \mathbb{Z}$ and $f(m) = m + 1$ for every $m \in \mathbb{Z}$. In this case we define g_1 as a symmetry relative to $\frac{1}{2}$, g_2 as a symmetry relative to 0.

1997-D2-P6 — Statement

Day 2, P12

Let $f : [0, 1] \to \mathbb{R}$ be a continuous function. Say that f "crosses the axis" at x if $f(x) = 0$ but in any neighbourhood of x there are y, z with $f(y) < 0$ and $f(z) > 0$.
a) Give an example of a continuous function that "crosses the axis" infiniteley often.
b) Can a continuous function "cross the axis" uncountably often?

Justify your answer.

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Theory and methods for 1997-D2-P6

Theory Countability, accumulation, and Cantor constructions [T-AN-SETS]; Continuity, compactness, and order [T-AN-CONTINUITY]; Uniform approximation and positive kernels [T-AN-APPROXIMATION]
Methods Use a dense set, an enumeration, or a category argument [M-AN-TOPOLOGY]
Strategy Build a witness and verify every constraint [S-CONSTRUCTION]; Pass between local constraints and a global conclusion [S-LOCALGLOBAL]

Statement Recorded solutions Catalog

1997-D2-P6 — Solution

For this record: Theory: T-AN-SETS; T-AN-CONTINUITY; T-AN-APPROXIMATION. Methods: M-AN-TOPOLOGY. Full learning route

a) $f(x) = x \sin \frac{1}{x}$.

b) Yes. The Cantor set is given by

$$C = \left\{ x \in [0,1) : x = \sum_{j=1}^{\infty} b_j 3^{-j},\ b_j \in \{0,2\} \right\}.$$

There is an one-to-one mapping $f : [0,1) \to C$. Indeed, for $x = \sum\limits_{j=1}^{\infty} a_j 2^{-j}$, $a_j \in \{0,1\}$ we set $f(x) = \sum\limits_{j=1}^{\infty} (2a_j) 3^{-j}$. Hence C is uncountable.

For $k = 1, 2, \ldots$ and $i = 0, 1, 2, \ldots, 2^{k-1} - 1$ we set

$$a_{k,i} = 3^{-k} \left(6 \sum_{j=0}^{k-2} a_j 3^j + 1 \right), \qquad b_{k,i} = 3^{-k} \left(6 \sum_{j=0}^{k-2} a_j 3^j + 2 \right),$$

where $i = \sum\limits_{j=0}^{k-2} a_j 2^j$, $a_j \in \{0,1\}$. Then

$$[0,1) \setminus C = \bigcup_{k=1}^{\infty} \bigcup_{i=0}^{2^{k-1}-1} (a_{k,i}, b_{k,i}),$$

i.e. the Cantor set consists of all points which have a trinary representation with 0 and 2 as digits and the points of its compliment

have some 1's in their trinary representation. Thus, $\bigcup\limits_{i=0}^{2^{k-1}-1} (a_{k,i}, b_{k,i})$ are all points (exept $a_{k,i}$) which have 1 on k-th place and 0 or 2 on the j-th ($j < k$) places.

Noticing that the points with at least one digit equals to 1 are everywhere dence in $[0,1]$ we set

$$f(x) = \sum_{k=1}^{\infty} (-1)^k g_k(x).$$

where g_k is a piece-wise linear continuous functions with values at the knots

$$g_k \left(\frac{a_{k,i} + b_{k,i}}{2} \right) = 2^{-k}, \qquad g_k(0) = g_k(1) = g_k(a_{k,i}) = g_k(b_{k,i}) = 0,$$

$i = 0, 1, \ldots, 2^{k-1} - 1$.

Then f is continuous and f "crosses the axis" at every point of the Cantor set.

3.5 1998

Day 1

1998-D1-P1 — Statement

Day 1, P1 | 20 points

Let V be a 10-dimensional real vector space and U_1 and U_2 two linear subspaces such that $U_1 \subseteq U_2$, $\dim_{\mathbb{R}} U_1 = 3$ and $\dim_{\mathbb{R}} U_2 = 6$. Let $\mathcal{E}$ be the set of all linear maps $T : V \longrightarrow V$ which have U_1 and U_2 as invariant subspaces (i.e., $T(U_1) \subseteq U_1$ and $T(U_2) \subseteq U_2$). Calculate the dimension of $\mathcal{E}$ as a real vector space.

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Theory and methods for 1998-D1-P1

Theory Adapted bases, centralizers and common eigenspaces [T-LA-INVARIANT]; Rank, dimension and exact linear constraints [T-LA-RANK]
Methods Choose a basis adapted to the structure [M-LA-BASIS]
Strategy Expose the structure before calculating [S-STRUCTURE]; Choose the objects and count them without ambiguity [S-COUNTING]

Statement Recorded solutions Catalog

1998-D1-P1 — Solution

For this record: Theory: T-LA-INVARIANT; T-LA-RANK. Methods: M-LA-BASIS. Full learning route

First choose a basis $\{v_1, v_2, v_3\}$ of U_1. It is possible to extend this basis with vectors v_4, v_5 and v_6 to get a basis of U_2. In the same way we can extend a basis of U_2 with vectors $v_7, \dots, v_{10}$ to get as basis of V.

Let $T \in \mathcal{E}$ be an endomorphism which has U_1 and U_2 as invariant subspaces. Then its matrix, relative to the basis $\{v_1, \dots, v_{10}\}$ is of the form

$$\begin{pmatrix} * & * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & * & * & * & * & * & * & * \\ 0 & 0 & 0 & * & * & * & * & * & * & * \\ 0 & 0 & 0 & * & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & * & * & * & * \end{pmatrix}.$$

So $\dim_{\mathbb{R}} \mathcal{E} = 9 + 18 + 40 = 67$.

1998-D1-P2 — Statement

Day 1, P2

Prove that the following proposition holds for $n = 3$ (5 points) and $n = 5$ (7 points), and does not hold for $n = 4$ (8 points).

"For any permutation π_1 of $\{1, 2, \dots, n\}$ different from the identity there is a permutation π_2 such that any permutation π can be obtained from π_1 and π_2 using only compositions (for example, $\pi = \pi_1 \circ \pi_1 \circ \pi_2 \circ \pi_1$)."

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Theory and methods for 1998-D1-P2

Theory Groups, orders, generators, and quotients [T-AL-GROUP]; Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]
Methods Prove generation or obstruct it by an image [M-AL-GENERATORS]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

1998-D1-P2 — Solution

For this record: Theory: T-AL-GROUP; T-AL-PERMUTATION. Methods: M-AL-GENERATORS. Full learning route

Let S_n be the group of permutations of $\{1, 2, \dots, n\}$.
1) When $n = 3$ the proposition is obvious: if $x = (12)$ we choose $y = (123)$; if $x = (123)$ we choose $y = (12)$.
2) $n = 4$. Let $x = (12)(34)$. Assume that there exists $y \in S_n$, such that $S_4 = \langle x, y\rangle$. Denote by K the invariant subgroup

$$K = \{id,\ (12)(34), (13)(24), (14)(23)\}.$$

By the fact that x and y generate the whole group S_4, it follows that the factor group S_4/K contains only powers of $\bar{y} = yK$, i.e., S_4/K is cyclic. It is easy to see that this factor-group is not comutative (something more this group is not isomorphic to S_3).
3) $n = 5$
a) If $x = (12)$, then for y we can take $y = (12345)$.
b) If $x = (123)$, we set $y = (124)(35)$. Then $y^3xy^3 = (125)$ and $y^4 = (124)$. Therefore $(123), (124), (125) \in \langle x, y\rangle$ – the subgroup generated by x and y. From the fact that $(123), (124), (125)$ generate the alternating subgroup A_5, it follows that $A_5 \subset \langle x, y\rangle$. Moreover y is an odd permutation, hence $\langle x, y\rangle = S_5$.
c) If $x = (123)(45)$, then as in b) we see that for y we can take the element (124).
d) If $x = (1234)$, we set $y = (12345)$. Then $(yx)^3 = (24) \in \langle x, y\rangle$, $x^2(24) = (13) \in \langle x, y\rangle$ and $y^2 = (13524) \in \langle x, y\rangle$. By the fact $(13) \in \langle x, y\rangle$ and $(13524) \in \langle x, y\rangle$, it follows that $\langle x, y\rangle = S_5$.
e) If $x = (12)(34)$, then for y we can take $y = (1354)$. Then $y^2x = (125)$, $y^3x = (124)(53)$ and by c) $S_5 = \langle x, y\rangle$.

f) If $x = (12345)$, then it is clear that for y we can take the element $y = (12)$.

1998-D1-P3 — Statement

Day 1, P3

Let $f(x) = 2x(1 - x)$, $x \in \mathbb{R}$. Define

$$f_n = \underbrace{f \circ \dots \circ f}_{n}.$$

a) (10 points) Find $\lim\limits_{n\to\infty} \int_0^1 f_n(x)\,dx$.

b) (10 points) Compute $\int_0^1 f_n(x)\,dx$ for $n = 1, 2, \dots$.

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Theory and methods for 1998-D1-P3

Theory Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Dominated convergence and orthonormal separation [T-AN-MEASURE]
Methods Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]; Strengthen the statement before the induction step [M-DS-INDUCT]
Strategy Control the limiting operation before taking the limit [S-ANALYTIC]; Reduce size while preserving the defining conditions [S-INDUCTION]

Statement Recorded solutions Catalog

1998-D1-P3 — Solution

For this record: Theory: T-DS-DYNAMICS; T-DS-INDUCTION; T-AN-CONVERGENCE; T-AN-INTEGRAL; T-AN-MEASURE. Methods: M-DS-ITERATE; M-DS-INDUCT. Full learning route

a) Fix $x = x_0 \in (0,1)$. If we denote $x_n = f_n(x_0)$, $n = 1, 2, \ldots$ it is easy to see that $x_1 \in (0, 1/2]$, $x_1 \le f(x_1) \le 1/2$ and $x_n \le f(x_n) \le 1/2$ (by induction). Then $(x_n)_n$ is a bounded non-decreasing sequence and, since $x_{n+1} = 2x_n(1-x_n)$, the limit $l = \lim_{n\to\infty} x_n$ satisfies $l = 2l(1-l)$, which implies $l = 1/2$. Now the monotone convergence theorem implies that

$$\lim_{n\to\infty} \int_0^1 f_n(x)\,dx = 1/2.$$

b) We prove by induction that

$$f_n(x) = \frac{1}{2} - 2^{2^n-1}\left(x - \frac{1}{2}\right)^{2^n} \tag{1}$$

holds for $n = 1, 2, \ldots$. For $n = 1$ this is true, since $f(x) = 2x(1-x) = \frac{1}{2} - 2(x-\frac{1}{2})^2$. If (1) holds for some $n = k$, then we have

$$\begin{aligned} f_{k+1}(x) = f_k(f(x)) &= \frac{1}{2} - 2^{2^k-1}\left(\left(\tfrac{1}{2} - 2(x-\tfrac{1}{2})^2\right) - \tfrac{1}{2}\right)^{2^k} \\ &= \frac{1}{2} - 2^{2^k-1}\left(-2(x-\tfrac{1}{2})^2\right)^{2^k} = \frac{1}{2} - 2^{2^{k+1}-1}\left(x-\tfrac{1}{2}\right)^{2^{k+1}} \end{aligned}$$

which is (2) for $n = k+1$.

Using (1) we can compute the integral,

$$\int_0^1 f_n(x)\,dx = \left[\frac{x}{2} - \frac{2^{2^n-1}}{2^n+1}\left(x-\frac{1}{2}\right)^{2^n+1}\right]_{x=0}^{1} = \frac{1}{2} - \frac{1}{2(2^n+1)}.$$

1998-D1-P4 — Statement

Day 1, P4 | 20 points

The function $f : \mathbb{R} \to \mathbb{R}$ is twice differentiable and satisfies $f(0) = 2$, $f'(0) = -2$ and $f(1) = 1$. Prove that there exists a real number $\xi \in (0,1)$ for which

$$f(\xi)\cdot f'(\xi) + f''(\xi) = 0.$$

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Theory and methods for 1998-D1-P4

Theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Continuity, compactness, and order [T-AN-CONTINUITY]; Differential inequalities and integrating factors [T-AN-ODE]

Methods Design an auxiliary derivative [M-AN-AUXILIARY]; Count zeros and subtract an interpolant [M-AN-ROLLE]

Strategy Control the limiting operation before taking the limit [S-ANALYTIC]; Prove necessity and sufficiency separately [S-CLASSIFICATION]

Statement Recorded solutions Catalog

1998-D1-P4 — Solution

For this record: Theory: T-AN-DERIVATIVE; T-AN-CONTINUITY; T-AN-ODE. Methods: M-AN-AUXILIARY; M-AN-ROLLE. Full learning route

Define the function

$$g(x) = \frac{1}{2}f^2(x) + f'(x).$$

1998-D1-P4 — Solution (continued)

Because $g(0) = 0$ and

$$f(x) \cdot f'(x) + f''(x) = g'(x),$$

it is enough to prove that there exists a real number $0 < \eta \leq 1$ for which $g(\eta) = 0$.
a) If f is never zero, let

$$h(x) = \frac{x}{2} - \frac{1}{f(x)}.$$

Because $h(0) = h(1) = -\frac{1}{2}$, there exists a real number $0 < \eta < 1$ for which $h'(\eta) = 0$. But $g = f^2 \cdot h'$, and we are done.
b) If f has at least one zero, let z_1 be the first one and z_2 be the last one. (The set of the zeros is closed.) By the conditions, $0 < z_1 \leq z_2 < 1$.
The function f is positive on the intervals $[0, z_1)$ and $(z_2, 1]$; this implies that $f'(z_1) \leq 0$ and $f'(z_2) \geq 0$. Then $g(z_1) = f'(z_1) \leq 0$ and $g(z_2) = f'(z_2) \geq 0$, and there exists a real number $\eta \in [z_1, z_2]$ for which $g(\eta) = 0$.

Remark. For the function $f(x) = \dfrac{2}{x+1}$ the conditions hold and $f \cdot f' + f''$ is constantly 0.

1998-D1-P5 — Statement

Day 1, P5

Let P be an algebraic polynomial of degree n having only real zeros and real coefficients.
a) (15 points) Prove that for every real x the following inequality holds:

$$(n-1)(P'(x))^2 \geq nP(x)P''(x). \tag{2}$$

b) (5 points) Examine the cases of equality.

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Theory and methods for 1998-D1-P5

Theory Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]
Methods Use a logarithmic derivative to expose root geometry [M-AL-LOGDERIVATIVE]
Strategy Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

1998-D1-P5 — Solution

For this record: Theory: T-AL-ROOTS; T-AN-INEQUALITIES. Methods: M-AL-LOGDERIVATIVE. Full learning route

Observe that both sides of (2) are identically equal to zero if $n = 1$. Suppose that $n > 1$. Let $x_1, \ldots, x_n$ be the zeros of P. Clearly (2) is true when $x = x_i$, $i \in \{1, \ldots, n\}$, and equality is possible only if $P'(x_i) = 0$, i.e., if x_i is a multiple zero of P. Now suppose that x is not a zero of P. Using the identities

$$\frac{P'(x)}{P(x)} = \sum_{i=1}^{n} \frac{1}{x - x_i}, \qquad \frac{P''(x)}{P(x)} = \sum_{1 \leq i < j \leq n} \frac{2}{(x - x_i)(x - x_j)},$$

we find

$$(n-1)\left(\frac{P'(x)}{P(x)}\right)^2 - n\,\frac{P''(x)}{P(x)} = \sum_{i=1}^{n} \frac{n-1}{(x - x_i)^2} - \sum_{1 \leq i < j \leq n} \frac{2}{(x - x_i)(x - x_j)}.$$

But this last expression is simply

$$\sum_{1 \leq i < j \leq n} \left(\frac{1}{x - x_i} - \frac{1}{x - x_j}\right)^2,$$

1998-D1-P5 — Solution (continued)

and therefore is positive. The inequality is proved. In order that (2) holds with equality sign for every real x it is necessary that $x_1 = x_2 = \cdots = x_n$. A direct verification shows that indeed, if $P(x) = c(x - x_1)^n$, then (2) becomes an identity.

1998-D1-P6 — Statement

Day 1, P6

Let $f : [0, 1] \to \mathbb{R}$ be a continuous function with the property that for any x and y in the interval,

$$xf(y) + yf(x) \leq 1.$$

a) (15 points) Show that

$$\int_0^1 f(x)\,dx \leq \frac{\pi}{4}.$$

b) (5 points) Find a function, satisfying the condition, for which there is equality.

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Theory and methods for 1998-D1-P6

Theory Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Trigonometric identities and safe branches [T-AN-TRIG]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]
Methods Pair an integral through symmetry [M-AN-PAIR]
Strategy Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

1998-D1-P6 — Solution

For this record: Theory: T-AN-INTEGRAL; T-AN-TRIG; T-AN-INEQUALITIES. Methods: M-AN-PAIR. Full learning route

Observe that the integral is equal to

$$\int_0^{\frac{\pi}{2}} f(\sin\theta)\cos\theta\,d\theta$$

and to

$$\int_0^{\frac{\pi}{2}} f(\cos\theta)\sin\theta\,d\theta$$

So, twice the integral is at most

$$\int_0^{\frac{\pi}{2}} 1\,d\theta = \frac{\pi}{2}.$$

Now let $f(x) = \sqrt{1 - x^2}$. If $x = \sin\theta$ and $y = \sin\varphi$ then

$$xf(y) + yf(x) = \sin\theta\cos\varphi + \sin\varphi\cos\theta = \sin(\theta + \varphi) \leq 1.$$

Day 2

1998-D2-P1 — Statement

Day 2, P7 | 20 points

Let V be a real vector space, and let $f, f_1, f_2, \dots, f_k$ be linear maps from V to $\mathbb{R}$. Suppose that $f(x) = 0$ whenever $f_1(x) = f_2(x) = \cdots = f_k(x) = 0$. Prove that f is a linear combination of $f_1, f_2, \dots, f_k$.

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Theory and methods for 1998-D2-P1

Theory Dual bases, kernels and separating functionals [T-LA-DUALITY]; Rank, dimension and exact linear constraints [T-LA-RANK]
Methods Prove a dimension obstruction and attain it [M-LA-DIMENSION]
Strategy Reduce size while preserving the defining conditions [S-INDUCTION]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

1998-D2-P1 — Solution

For this record: Theory: T-LA-DUALITY; T-LA-RANK. Methods: M-LA-DIMENSION. Full learning route

We use induction on k. By passing to a subset, we may assume that $f_1, \dots, f_k$ are linearly independent.

Since f_k is independent of $f_1, \dots, f_{k-1}$, by induction there exists a vector $a_k \in V$ such that $f_1(a_k) = \cdots = f_{k-1}(a_k) = 0$ and $f_k(a_k) \neq 0$. After normalising, we may assume that $f_k(a_k) = 1$. The vectors $a_1, \dots, a_{k-1}$ are defined similarly to get

$$f_i(a_j) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j. \end{cases}$$

For an arbitrary $x \in V$ and $1 \leq i \leq k$, $f_i(x - f_1(x)a_1 - f_2(x)a_2 - \cdots - f_k(x)a_k) = f_i(x) - \sum_{j=1}^{k} f_j(x) f_i(a_j) = f_i(x) - f_i(x) f_i(a_i) = 0$, thus $f(x - f_1(x)a_1 - \cdots - f_k(x)a_k) = 0$. By the linearity of f this implies $f(x) = f_1(x)f(a_1) + \cdots + f_k(x)f(a_k)$, which gives $f(x)$ as a linear combination of $f_1(x), \dots, f_k(x)$.

1998-D2-P2 — Statement

Day 2, P8 | 20 points

Let

$$P = \left\{ f : f(x) = \sum_{k=0}^{3} a_k x^k,\ a_k \in \mathbb{R},\ |f(\pm 1)| \leq 1,\ \left|f\left(\pm \tfrac{1}{2}\right)\right| \leq 1 \right\}.$$

Evaluate

$$\sup_{f \in P} \max_{-1 \leq x \leq 1} |f''(x)|$$

and find all polynomials $f \in P$ for which the above "sup" is attained.

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Theory and methods for 1998-D2-P2

Theory Interpolation, divided differences, and extremal signs [T-AL-INTERPOLATION]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Methods Turn nodal bounds into a sharp linear functional [M-AL-INTERPOLATE]

Theory and methods for 1998-D2-P2 (continued)

Strategy Choose an extremal object and force rigidity [S-EXTREMAL]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

1998-D2-P2 — Solution

For this record: Theory: T-AL-INTERPOLATION; T-AN-DERIVATIVE. Methods: M-AL-INTERPOLATE. Full learning route

Denote $x_0 = 1$, $x_1 = -1$, $x_2 = \frac{1}{2}$, $x_3 = -\frac{1}{2}$,

$$w(x) = \prod_{i=0}^{3}(x - x_i),$$

$$w_k(x) = \frac{w(x)}{x - x_k}, \quad k = 0, \dots, 3,$$

$$l_k(x) = \frac{w_k(x)}{w_k(x_k)}.$$

Then for every $f \in P$

$$f''(x) = \sum_{k=0}^{3} l_k''(x) f(x_k),$$

$$|f''(x)| \le \sum_{k=0}^{3} |l_k''(x)|.$$

Since f'' is a linear function $\max\limits_{-1\le x\le 1} |f''(x)|$ is attained either at $x = -1$ or at $x = 1$. Without loss of generality let the maximum point is $x = 1$. Then

$$\sup_{f\in P} \max_{-1\le x\le 1} |f''(x)| = \sum_{k=0}^{3} |l_k''(1)|.$$

In order to have equality for the extremal polynomial f_* there must hold

$$f_*(x_k) = \operatorname{sign} l_k''(1), \quad k = 0, 1, 2, 3.$$

It is easy to see that $\{l_k''(1)\}_{k=0}^{3}$ alternate in sign, so $f_*(x_k) = (-1)^{k-1}$, $k = 0, \dots, 3$. Hence $f_*(x) = T_3(x) = 4x^3 - 3x$, the Chebyshev polynomial of the first kind, and $f_*''(1) = 24$. The other extremal polynomial, corresponding to $x = -1$, is $-T_3$.

1998-D2-P3 — Statement

Day 2, P9 | 20 points

Let $0 < c < 1$ and

$$f(x) = \begin{cases} \dfrac{x}{c} & \text{for } x \in [0, c], \\ \dfrac{1 - x}{1 - c} & \text{for } x \in [c, 1]. \end{cases}$$

We say that p is an n-periodic point if

$$\underbrace{f(f(\dots f}_{n}(p))) = p$$

and n is the smallest number with this property. Prove that for every $n \ge 1$ the set of n-periodic points is non-empty and finite.

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Theory and methods for 1998-D2-P3

Theory Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]; Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Methods Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]; Strengthen the statement before the induction step [M-DS-INDUCT]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Reduce size while preserving the defining conditions [S-INDUCTION]

Statement Recorded solutions Catalog

1998-D2-P3 — Solution

For this record: Theory: T-DS-DYNAMICS; T-DS-COUNTING; T-DS-INDUCTION. Methods: M-DS-ITERATE; M-DS-INDUCT. Full learning route

Let $f_n(x) = \underbrace{f(f(\dots f}_{n}(x)))$. It is easy to see that $f_n(x)$ is a picewise monotone function and its graph contains 2^n linear segments; one endpoint is always on $\{(x,y) : 0 \le x \le 1,\ y = 0\}$, the other is on $\{(x,y) : 0 \le x \le 1,\ y = 1\}$. Thus the graph of the identity function intersects each segment once, so the number of points for which $f_n(x) = x$ is 2^n.
Since for each n-periodic points we have $f_n(x) = x$, the number of n-periodic points is finite.

A point x is n-periodic if $f_n(x) = x$ but $f_k(x) \neq x$ for $k = 1, \dots, n-1$. But as we saw before $f_k(x) = x$ holds only at 2^k points, so there are at most $2^1 + 2^2 + \dots + 2^{n-1} = 2^n - 2$ points x for which $f_k(x) = x$ for at least one $k \in \{1, 2, \dots, n-1\}$. Therefore at least two of the 2^n points for which $f_n(x) = x$ are n-periodic points.

1998-D2-P4 — Statement

Day 2, P10 | 20 points

Let $A_n = \{1, 2, \dots, n\}$, where $n \ge 3$. Let F be the family of all non-constant functions $f : A_n \to A_n$ satisfying the following conditions:
(1) $f(k) \le f(k+1)$ for $k = 1, 2, \dots, n-1$,
(2) $f(k) = f(f(k+1))$ for $k = 1, 2, \dots, n-1$.

Find the number of functions in F.

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Theory and methods for 1998-D2-P4

Theory Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]; Finite counting and incidence identities [T-DS-COUNTING]; Extremal choice, exchange arguments, and invariants [T-DS-EXTREMAL]
Methods Force structure by admissible substitutions [M-DS-FUNCTIONAL]; Choose information that makes an encoding injective [M-DS-ENCODE]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Choose the objects and count them without ambiguity [S-COUNTING]

Statement Recorded solutions Catalog

1998-D2-P4 — Solution

For this record: Theory: T-DS-FUNCTIONAL; T-DS-COUNTING; T-DS-EXTREMAL. Methods: M-DS-FUNCTIONAL; M-DS-ENCODE. Full learning route

It is clear that $id : A_n \longrightarrow A_n$, given by $id(x) = x$, does not verify condition (2). Since id is the only increasing injection on A_n, F does not contain injections. Let us take any $f \in F$ and suppose that $\#(f^{-1}(k)) \ge 2$. Since f is increasing, there exists $i \in A_n$ such that $f(i) = f(i+1) = k$. In view of (2), $f(k) = f(f(i+1)) = f(i) = k$. If $\{i < k : f(i) < k\} = \emptyset$, then taking $j = \max\{i < k : f(i) < k\}$ we get $f(j) < f(j+1) = k = f(f(j+1))$, a contradiction. Hence $f(i) = k$ for $i \le k$. If $\#(f^{-1}(\{l\})) \ge 2$ for some $l \ge k$, then the similar consideration shows that $f(i) = l = k$ for $i \le k$. Hence $\#(f^{-1}\{i\}) = 0$ or 1 for every $i > k$. Therefore $f(i) \le i$ for $i > k$. If $f(l) = l$, then taking $j = \max\{i < l : f(i) < l\}$ we get

1998-D2-P4 — Solution (continued)

$f(j) < f(j+1) = l = f(f(j+1))$, a contradiction. Thus, $f(i) \le i-1$ for $i > k$. Let $m = \max\{i : f(i) = k\}$. Since f is non-constant $m \le n-1$. Since $k = f(m) = f(f(m+1))$, $f(m+1) \in [k+1, m]$. If $f(l) > l-1$ for some $l > m+1$, then $l-1$ and $f(l)$ belong to $f^{-1}(f(l))$ and this contradicts the facts above. Hence $f(i) = i-1$ for $i > m+1$. Thus we show that every function f in F is defined by natural numbers k, l, m, where $1 \le k < l = f(m+1) \le m \le n-1$.

$$f(i) = \begin{cases} k & \text{if } i \le m \\ l & \text{if } i = m+1 \\ i-1 & \text{if } i > m+1. \end{cases}$$

Then

$$\#(F) = \binom{n}{3}.$$

1998-D2-P5 — Statement

Day 2, P11 | 20 points

Suppose that S is a family of spheres (i.e., surfaces of balls of positive radius) in $\mathbb{R}^n$, $n \ge 2$, such that the intersection of any two contains at most one point. Prove that the set M of those points that belong to at least two different spheres from S is countable.

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Theory and methods for 1998-D2-P5

Theory Metric contractions, covers, and nearest points [T-DS-METRIC]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Countability, accumulation, and Cantor constructions [T-AN-SETS]; Continuity, compactness, and order [T-AN-CONTINUITY]
Methods Choose information that makes an encoding injective [M-DS-ENCODE]; Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Strategy Translate geometry into a controlled representation [S-GEOMETRIC]; Choose the objects and count them without ambiguity [S-COUNTING]

Statement Recorded solutions Catalog

1998-D2-P5 — Solution

For this record: Theory: T-DS-METRIC; T-DS-GEOMETRY; T-AN-SETS; T-AN-CONTINUITY. Methods: M-DS-ENCODE; M-DS-GEOMETRIC. Full learning route

For every $x \in M$ choose spheres $S, T \in S$ such that $S \ne T$ and $x \in S \cap T$; denote by U, V, W the three components of $\mathbb{R}^n \setminus (S \cup T)$, where the notation is such that $\partial U = S$, $\partial V = T$ and x is the only point of $\overline{U} \cap \overline{V}$, and choose points with rational coordinates $u \in U$, $v \in V$, and $w \in W$. We claim that x is uniquely determined by the triple $\langle u, v, w\rangle$; since the set of such triples is countable, this will finish the proof.

To prove the claim, suppose, that from some $x' \in M$ we arrived to the same $\langle u, v, w\rangle$ using spheres $S', T' \in S$ and components U', V', W' of $\mathbb{R}^n \setminus (S' \cup T')$. Since $S \cap S'$ contains at most one point and since $U \cap U' \ne \emptyset$, we have that $U \subset U'$ or $U' \subset U$; similarly for V's and W's. Exchanging the role of x and x' and/or of U's and V's if necessary, there are only two cases to consider: (a) $U \supset U'$ and $V \supset V'$ and (b) $U \subset U'$, $V \supset V'$ and $W \subset W'$. In case (a) we recall that $\overline{U} \cap \overline{V}$ contains only x and that $x' \in \overline{U'} \cap \overline{V'}$, so $x = x'$. In case (b) we get from $W \subset W'$ that $U' \subset \overline{U} \cup \overline{V}$; so since U' is open and connected, and $\overline{U} \cap \overline{V}$ is just one point, we infer that $U' = U$ and we are back in the already proved case (a).

1998-D2-P6 — Statement

Day 2, P12 | 20 points

Let $f : (0,1) \to [0,\infty)$ be a function that is zero except at the distinct points $a_1, a_2, \ldots$. Let $b_n = f(a_n)$.

(a) Prove that if $\sum_{n=1}^{\infty} b_n < \infty$, then f is differentiable at at least one point $x \in (0,1)$.

(b) Prove that for any sequence of non-negative real numbers $(b_n)_{n=1}^{\infty}$, with $\sum_{n=1}^{\infty} b_n = \infty$, there exists a sequence $(a_n)_{n=1}^{\infty}$ such that the function f defined as above is nowhere differentiable.

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Theory and methods for 1998-D2-P6

Theory Interval coverings and level-set integration [T-AN-COVERING]; Countability, accumulation, and Cantor constructions [T-AN-SETS]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Series, finite cancellation, and summation by parts [T-AN-SERIES]
Methods Turn local bad intervals into a global measure bound [M-AN-COVER]; Use a dense set, an enumeration, or a category argument [M-AN-TOPOLOGY]
Strategy Build a witness and verify every constraint [S-CONSTRUCTION]; Pass between local constraints and a global conclusion [S-LOCALGLOBAL]

Statement Recorded solutions Catalog

1998-D2-P6 — Solution

For this record: Theory: T-AN-COVERING; T-AN-SETS; T-AN-DERIVATIVE; T-AN-SERIES. Methods: M-AN-COVER; M-AN-TOPOLOGY. Full learning route

a) We first construct a sequence c_n of positive numbers such that $c_n \to \infty$ and $\sum_{n=1}^{\infty} c_n b_n < \frac{1}{2}$. Let $B = \sum_{n=1}^{\infty} b_n$, and for each $k = 0, 1, \ldots$ denote by N_k the first positive integer for which

$$\sum_{n=N_k}^{\infty} b_n \le \frac{B}{4^k}.$$

Now set $c_n = \dfrac{2^k}{5B}$ for each n, $N_k \le n < N_{k+1}$. Then we have $c_n \to \infty$ and

$$\sum_{n=1}^{\infty} c_n b_n = \sum_{k=0}^{\infty} \sum_{N_k \le n < N_{k+1}} c_n b_n \le \sum_{k=0}^{\infty} \frac{2^k}{5B} \sum_{n=N_k}^{\infty} b_n \le \sum_{k=0}^{\infty} \frac{2^k}{5B} \cdot \frac{B}{4^k} = \frac{2}{5}.$$

Consider the intervals $I_n = (a_n - c_n b_n,\ a_n + c_n b_n)$. The sum of their lengths is $2\sum c_n b_n < 1$, thus there exists a point $x_0 \in (0,1)$ which is not contained in any I_n. We show that f is differentiable at x_0, and $f'(x_0) = 0$. Since x_0 is outside of the intervals I_n, $x_0 \ne a_n$ for any n and $f(x_0) = 0$. For arbitrary $x \in (0,1) \setminus \{x_0\}$, if $x = a_n$ for some n, then

$$\left|\frac{f(x) - f(x_0)}{x - x_0}\right| = \frac{f(a_n) - 0}{|a_n - x_0|} \le \frac{b_n}{c_n b_n} = \frac{1}{c_n},$$

otherwise $\left|\frac{f(x)-f(x_0)}{x-x_0}\right| = 0$. Since $c_n \to \infty$, this implies that for arbitrary $\varepsilon > 0$ there are only finitely many $x \in (0,1) \setminus \{x_0\}$ for which

$$\left|\frac{f(x) - f(x_0)}{x - x_0}\right| < \varepsilon$$

does not hold, and we are done.
Remark. The variation of f is finite, which implies that f is differentiable almost everywhere.
b) We remove the zero elements from sequence b_n. Since $f(x) = 0$ except for a countable subset of $(0,1)$, if f is differentiable at some point x_0, then $f(x_0)$ and $f'(x_0)$ must be 0.

1998-D2-P6 — Solution (continued)

It is easy to construct a sequence β_n satisfying $0 < \beta_n \le b_n$, $\beta_n \to 0$ and $\sum_{n=1}^{\infty} \beta_n = \infty$.

Choose the numbers $a_1, a_2, \dots$ such that the intervals $I_n = (a_n - \beta_n,\ a_n + \beta_n)$ $(n = 1, 2, \dots)$ cover each point of $(0,1)$ infinitely many times (it is possible since the sum of lengths is $2\sum \beta_n = \infty$). Then for arbitrary $x_0 \in (0,1)$, $f(x_0) = 0$ and $\varepsilon > 0$ there is an n for which $\beta_n < \varepsilon$ and $x_0 \in I_n$ which implies

$$\frac{|f(a_n) - f(x_0)|}{|a_n - x_0|} > \frac{b_n}{\beta_n} \ge 1.$$

3.6 1999

Day 1

1999-D1-P1 — Statement

Day 1, P1 | 6+14 points

a) Show that for any $m \in \mathbb{N}$ there exists a real $m \times m$ matrix A such that $A^3 = A + I$, where I is the $m \times m$ identity matrix. (6 points)

b) Show that $\det A > 0$ for every real $m \times m$ matrix satisfying $A^3 = A + I$. (14 points)

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Theory and methods for 1999-D1-P1

Theory Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Determinants, block identities and inverses [T-LA-DETERMINANT]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Continuity, compactness, and order [T-AN-CONTINUITY]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Methods Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

1999-D1-P1 — Solution

For this record: Theory: T-LA-POLYNOMIAL; T-LA-SPECTRAL; T-LA-DETERMINANT; T-AL-POLYNOMIAL; T-AN-CONTINUITY; T-AN-DERIVATIVE. Methods: M-LA-SPECTRAL. Full learning route

a) The diagonal matrix

$$A = \lambda I = \begin{pmatrix} \lambda & & 0 \\ & \ddots & \\ 0 & & \lambda \end{pmatrix}$$

is a solution for equation $A^3 = A + I$ if and only if $\lambda^3 = \lambda + 1$, because $A^3 - A - I = (\lambda^3 - \lambda - 1)I$. This equation, being cubic, has real solution.
b) It is easy to check that the polynomial $p(x) = x^3 - x - 1$ has a positive real root λ_1 (because $p(0) < 0$) and two conjugated complex roots λ_2 and λ_3 (one can check the discriminant of the polynomial, which is $\left(\frac{-1}{3}\right)^3 + \left(\frac{-1}{2}\right)^2 = \frac{23}{108} > 0$, or the local minimum and maximum of the polynomial).

If a matrix A satisfies equation $A^3 = A + I$, then its eigenvalues can be only λ_1, λ_2 and λ_3. The multiplicity of λ_2 and λ_3 must be the same, because A is a real matrix and its characteristic polynomial has only real coefficients. Denoting the multiplicity of λ_1 by α and the common multiplicity of λ_2 and λ_3 by β,

$$\det A = \lambda_1^{\alpha} \lambda_2^{\beta} \lambda_3^{\beta} = \lambda_1^{\alpha} \cdot (\lambda_2 \lambda_3)^{\beta}.$$

Because λ_1 and $\lambda_2 \lambda_3 = |\lambda_2|^2$ are positive, the product on the right side has only positive factors.

1999-D1-P2 — Statement

Day 1, P2 | 20 points

Does there exist a bijective map $\pi : \mathbb{N} \to \mathbb{N}$ such that

$$\sum_{n=1}^{\infty} \frac{\pi(n)}{n^2} < \infty?$$

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Theory and methods for 1999-D1-P2

Theory Series, finite cancellation, and summation by parts [T-AN-SERIES]; Finite counting and incidence identities [T-DS-COUNTING]; Nonnegative sums, Fubini, and integral kernels [T-AN-FUBINI]
Methods Test convergence through blocks and differences [M-AN-SERIES]; Find a finite telescoping identity [M-AN-TELESCOPE]; Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

1999-D1-P2 — Solution — Record 1 of 2

For this record: Theory: T-AN-SERIES; T-DS-COUNTING. Methods: M-AN-SERIES. Full learning route

No. For, let π be a permutation of $\mathbb{N}$ and let $N \in \mathbb{N}$. We shall argue that

$$\sum_{n=N+1}^{3N} \frac{\pi(n)}{n^2} > \frac{1}{9}.$$

In fact, of the $2N$ numbers $\pi(N+1), \dots, \pi(3N)$ only N can be $\le N$ so that at least N of them are $> N$. Hence

$$\sum_{n=N+1}^{3N} \frac{\pi(n)}{n^2} \ge \frac{1}{(3N)^2} \sum_{n=N+1}^{3N} \pi(n) > \frac{1}{9N^2} \cdot N \cdot N = \frac{1}{9}.$$

1999-D1-P2 — Solution — Record 2 of 2

For this record: Theory: T-AN-SERIES; T-AN-FUBINI; T-DS-COUNTING. Methods: M-AN-TELESCOPE; M-AN-WEIGHTS. Full learning route

Let π be a permutation of $\mathbb{N}$. For any $n \in \mathbb{N}$, the numbers $\pi(1), \dots, \pi(n)$ are distinct positive integers, thus $\pi(1) + \cdots + \pi(n) \ge 1 + \cdots + n = \frac{n(n+1)}{2}$. By this inequality,

$$\sum_{n=1}^{\infty} \frac{\pi(n)}{n^2} = \sum_{n=1}^{\infty} (\pi(1) + \cdots + \pi(n)) \left(\frac{1}{n^2} - \frac{1}{(n+1)^2} \right) \ge$$
$$\ge \sum_{n=1}^{\infty} \frac{n(n+1)}{2} \cdot \frac{2n+1}{n^2(n+1)^2} = \sum_{n=1}^{\infty} \frac{2n+1}{2n(n+1)} \ge \sum_{n=1}^{\infty} \frac{1}{n+1} = \infty.$$

1999-D1-P3 — Statement

Day 1, P3 | 20 points

Suppose that a function $f : \mathbb{R} \to \mathbb{R}$ satisfies the inequality

$$\left| \sum_{k=1}^{n} 3^k (f(x+ky) - f(x-ky)) \right| \le 1 \tag{1}$$

for every positive integer n and for all $x, y \in \mathbb{R}$. Prove that f is a constant function.

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Theory and methods for 1999-D1-P3

Theory Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Methods Force structure by admissible substitutions [M-DS-FUNCTIONAL]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

1999-D1-P3 — Solution

For this record: Theory: T-DS-FUNCTIONAL; T-AN-CONVERGENCE. Methods: M-DS-FUNCTIONAL. Full learning route

Writing (1) with $n-1$ instead of n,

$$\left|\sum_{k=1}^{n-1} 3^k\big(f(x+ky)-f(x-ky)\big)\right| \le 1. \tag{2}$$

From the difference of (1) and (2),

$$\left|3^n\big(f(x+ny)-f(x-ny)\big)\right| \le 2;$$

which means

$$\left|f(x+ny)-f(x-ny)\right| \le \frac{2}{3^n}. \tag{3}$$

For arbitrary $u, v \in \mathbb{R}$ and $n \in \mathbb{N}$ one can choose x and y such that $x-ny=u$ and $x+ny=v$, namely $x=\frac{u+v}{2}$ and $y=\frac{v-u}{2n}$. Thus, (3) yields

$$\left|f(u)-f(v)\right| \le \frac{2}{3^n}$$

for arbitrary positive integer n. Because $\frac{2}{3^n}$ can be arbitrary small, this implies $f(u)=f(v)$.

1999-D1-P4 — Statement

Day 1, P4 | 20 points

Find all strictly monotonic functions $f:(0,+\infty)\to(0,+\infty)$ such that $f\left(\dfrac{x^2}{f(x)}\right)\equiv x$.

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Theory and methods for 1999-D1-P4

Theory Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]; Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Methods Force structure by admissible substitutions [M-DS-FUNCTIONAL]; Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

1999-D1-P4 — Solution

For this record: Theory: T-DS-FUNCTIONAL; T-DS-DYNAMICS; T-DS-INDUCTION; T-AN-CONVERGENCE. Methods: M-DS-FUNCTIONAL; M-DS-ITERATE. Full learning route

Let $g(x) = \dfrac{f(x)}{x}$. We have $g\left(\dfrac{x}{g(x)}\right) = g(x)$. By induction it follows that $g\left(\dfrac{x}{g^n(x)}\right) = g(x)$, i.e.

$$f\left(\frac{x}{g^n(x)}\right) = \frac{x}{g^{n-1}(x)}, \quad n \in \mathbb{N}. \tag{1}$$

On the other hand, let substitute x by $f(x)$ in $f\left(\dfrac{x^2}{f(x)}\right) = x$. From the injectivity of f we get $\dfrac{f^2(x)}{f(f(x))} = x$, i.e. $g(xg(x)) = g(x)$. Again by induction we deduce that $g(xg^n(x)) = g(x)$ which can be written in the form

$$f(xg^n(x)) = xg^{n-1}(x), \quad n \in \mathbb{N}. \tag{2}$$

Set $f^{(m)} = \underbrace{f \circ f \circ \ldots \circ f}_{m \text{ times}}$. It follows from (1) and (2) that

$$f^{(m)}(xg^n(x)) = xg^{n-m}(x), \quad m, n \in \mathbb{N}. \tag{3}$$

Now, we shall prove that g is a constant. Assume $g(x_1) < g(x_2)$. Then we may find $n \in \mathbb{N}$ such that $x_1 g^n(x_1) \le x_2 g^n(x_2)$. On the other hand, if m is even then $f^{(m)}$ is strictly increasing and from (3) it follows that $x_1^m g^{n-m}(x_1) \le x_2^m g^{n-m}(x_2)$. But when n is fixed the opposite inequality holds $\forall m \gg 1$. This contradiction shows that g is a constant, i.e. $f(x) = Cx$, $C > 0$.

Conversely, it is easy to check that the functions of this type verify the conditions of the problem.

1999-D1-P5 — Statement

Day 1, P5 | 20 points

Suppose that $2n$ points of an $n \times n$ grid are marked. Show that for some $k > 1$ one can select $2k$ distinct marked points, say $a_1, \ldots, a_{2k}$, such that a_1 and a_2 are in the same row, a_2 and a_3 are in the same column, ..., a_{2k-1} and a_{2k} are in the same row, and a_{2k} and a_1 are in the same column.

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Theory and methods for 1999-D1-P5

Theory Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]; Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Methods Encode a relation as a graph and extract a cycle [M-DS-GRAPH]; Strengthen the statement before the induction step [M-DS-INDUCT]
Strategy Expose the structure before calculating [S-STRUCTURE]; Reduce size while preserving the defining conditions [S-INDUCTION]

Statement Recorded solutions Catalog

1999-D1-P5 — Solution — Record 1 of 2

For this record: Theory: T-DS-GRAPH; T-DS-INDUCTION; T-DS-COUNTING. Methods: M-DS-GRAPH; M-DS-INDUCT. Full learning route

We prove the more general statement that if at least $n + k$ points are marked in an $n \times k$ grid, then the required sequence of marked points can be selected.
If a row or a column contains at most one marked point, delete it. This decreases $n + k$ by 1 and the number of the marked points by at most 1, so the condition remains true. Repeat this step until each row and column contains at least two marked points. Note that the condition implies that there are at least two marked points, so the whole set of marked points cannot be deleted.

1999-D1-P5 — Solution — Record 1 of 2 (continued)

We define a sequence $b_1, b_2, \ldots$ of marked points. Let b_1 be an arbitrary marked point. For any positive integer n, let b_{2n} be an other marked point in the row of b_{2n-1} and b_{2n+1} be an other marked point in the column of b_{2n}.
Let m be the first index for which b_m is the same as one of the earlier points, say $b_m = b_l$, $l < m$.
If $m-l$ is even, the line segments $b_l b_{l+1}$, $b_{l+1}b_{l+2}$, ..., $b_{m-1}b_l = b_{m-1}b_m$ are alternating horizontal and vertical. So one can choose $2k = m-l$, and $(a_1, \ldots, a_{2k}) = (b_l, \ldots, b_{m-1})$ or $(a_1, \ldots, a_{2k}) = (b_{l+1}, \ldots, b_m)$ if l is odd or even, respectively.

If $m-l$ is odd, then the points $b_l = b_m$, b_{l+1} and b_{m-1} are in the same row/column. In this case chose $2k = m-l-1$. Again, the line segments $b_{l+1}b_{l+2}$, $b_{l+2}b_{l+3}$, ..., $b_{m-1}b_{l+1}$ are alternating horizontal and vertical and one can choose $(a_1, \ldots, a_{2k}) = (b_{l+1}, \ldots, b_{m-1})$ or $(a_1, \ldots, a_{2k}) = (b_{l+2}, \ldots, b_{m-1}, b_{l+1})$ if l is even or odd, respectively.

1999-D1-P5 — Solution — Record 2 of 2

For this record: Theory: T-DS-GRAPH; T-DS-COUNTING. Methods: M-DS-GRAPH. Full learning route

Define the graph G in the following way: Let the vertices of G be the rows and the columns of the grid. Connect a row r and a column c with an edge if the intersection point of r and c is marked.

The graph G has $2n$ vertices and $2n$ edges. As is well known, if a graph of N vertices contains no circle, it can have at most $N-1$ edges. Thus G does contain a circle. A circle is an alternating sequence of rows and columns, and the intersection of each neighbouring row and column is a marked point. The required sequence consists of these intersection points.

1999-D1-P6 — Statement

Day 1, P6 | 10+10 points

a) For each $1 < p < \infty$ find a constant $c_p < \infty$ for which the following statement holds: If $f : [-1,1] \to \mathbb{R}$ is a continuously differentiable function satisfying $f(1) > f(-1)$ and $|f'(y)| \le 1$ for all $y \in [-1,1]$, then there is an $x \in [-1,1]$ such that $f'(x) > 0$ and $|f(y) - f(x)| \le c_p (f'(x))^{1/p} |y - x|$ for all $y \in [-1,1]$. (10 points)

b) Does such a constant also exist for $p = 1$? (10 points)

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Theory and methods for 1999-D1-P6

Theory Interval coverings and level-set integration [T-AN-COVERING]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Methods Turn local bad intervals into a global measure bound [M-AN-COVER]; Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]
Strategy Build a chain of bounds and track equality [S-BOUND]; Make the contradiction quantitative or structural [S-CONTRADICTION]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

1999-D1-P6 — Solution

For this record: Theory: T-AN-COVERING; T-AN-INTEGRAL; T-AN-DERIVATIVE. Methods: M-AN-COVER; M-AN-PRIMITIVE. Full learning route

Editorially corrected proof.
(a) Put $g = (f')_+ = \max(0, f')$, $v = |f'|$, and $A = \int_{-1}^{1} g$. Since $f(1) > f(-1)$,

$$A > 0, \qquad \int_{-1}^{1} v < 2A, \qquad 0 \le g \le 1.$$

1999-D1-P6 — Solution (continued)

We use the following elementary interval estimate: if $v \geq 0$ is integrable and E is the set of points belonging to an interval I for which $\int_I v > \lambda |I|$, then

$$|E| \leq \frac{3}{\lambda} \int_{-1}^{1} v.$$

Indeed, strictness allows each witnessing interval to be enlarged slightly within $[-1,1]$ while retaining the inequality. Cover a compact subset of E by finitely many such intervals, and greedily select disjoint intervals in decreasing order of length. Every rejected interval lies in the concentric triple of an intersecting selected interval. Thus the compact subset has measure at most three times the sum of the selected lengths, which is at most $3\lambda^{-1} \int v$. Inner regularity gives the claim.
Take $c_p = 6p/(p-1)$. Suppose that no required x exists. For each x with $g(x) > 0$, some $y \neq x$ then satisfies

$$|f(y) - f(x)| > c_p g(x)^{1/p} |y - x|.$$

The interval with endpoints x, y has $\int_I v \geq |f(y) - f(x)|$. Consequently, for every $0 < t < 1$, the interval estimate gives

$$|\{g > t\}| \leq \frac{3}{c_p} t^{-1/p} \int_{-1}^{1} v.$$

Integrating with respect to t and using $g \leq 1$ yields

$$\begin{aligned} A &= \int_0^1 |\{g > t\}| \, dt \\ &\leq \frac{3p}{c_p(p-1)} \int_{-1}^{1} v \\ &< \frac{6p}{c_p(p-1)} A = A, \end{aligned}$$

a contradiction.

(b) No. It suffices to disprove every constant $c > 1$, since failure for such a constant also implies failure for every smaller one. Fix $c > 1$, put $\alpha = 1/c$, and choose $0 < \varepsilon < 1/2$ such that

$$\left(\frac{1+\varepsilon}{2\varepsilon}\right)^{-\alpha} < \frac{1}{4}.$$

For $\varepsilon \leq t \leq 1$, define

$$h(t) = \alpha \left(\frac{t+\varepsilon}{2\varepsilon}\right)^{-\alpha-1}.$$

Then $0 < h \leq \alpha < 1$ and

$$\int_\varepsilon^1 h(t)\, dt = 2\varepsilon \left[1 - \left(\frac{1+\varepsilon}{2\varepsilon}\right)^{-\alpha}\right] > \frac{3}{2}\varepsilon.$$

Choose $0 < \delta < \varepsilon/32$. Let ϕ be continuous on $[\varepsilon, 1]$, equal to zero at ε, increasing linearly to one at $\varepsilon + \delta$, and equal to one thereafter. Define a continuous even function g by

$$g(t) = \begin{cases} -1, & |t| \leq \varepsilon, \\ -(1 - \phi(|t|)) + \dfrac{3}{4}\phi(|t|)h(|t|), & |t| > \varepsilon. \end{cases}$$

Thus $-1 \leq g \leq 1$ and $g(t) \leq \frac{3}{4}h(t) < h(t)$ for $t > \varepsilon$. The difference between g and $\frac{3}{4}h$ is supported in an interval of length δ and has absolute value less than 2. Hence

$$\int_\varepsilon^1 g(t)\, dt \geq \frac{3}{4}\int_\varepsilon^1 h(t)\, dt - 2\delta > \frac{9}{8}\varepsilon - \frac{1}{16}\varepsilon > \varepsilon.$$

Set $f(x) = \int_0^x g(t)\, dt$. Then $f \in C^1[-1,1]$, $|f'| \leq 1$, and

$$f(1) - f(-1) = 2\left(-\varepsilon + \int_\varepsilon^1 g(t)\, dt\right) > 0.$$

1999-D1-P6 — Solution (continued)

If $f'(x) = g(x) > 0$ and $x > 0$, then $x > \varepsilon$. Choose $y = -\varepsilon$. We have

$$\begin{aligned} |f(x) - f(y)| &\geq 2\varepsilon - \int_{\varepsilon}^{x} g(t)\,dt \\ &\geq 2\varepsilon - \int_{\varepsilon}^{x} h(t)\,dt \\ &= 2\varepsilon \left(\frac{x+\varepsilon}{2\varepsilon}\right)^{-\alpha} \\ &= c\,h(x)(x+\varepsilon) \\ &> c\,g(x)|x-y|. \end{aligned}$$

For $x < 0$, the same conclusion follows by evenness of g (and oddness of f), taking $y = \varepsilon$. Thus every point with $f'(x) > 0$ violates the required inequality for some y, completing the counterexample.

Day 2

1999-D2-P1 — Statement

Day 2, P7 | 20 points

Suppose that in a not necessarily commutative ring R the square of any element is 0. Prove that $abc + abc = 0$ for any three elements a, b, c.

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Theory and methods for 1999-D2-P1

Theory Noncommutative rings, polarization, and torsion [T-AL-RING]
Methods Polarize identities and isolate commutators [M-AL-COMMUTATOR]
Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

1999-D2-P1 — Solution

For this record: Theory: T-AL-RING. Methods: M-AL-COMMUTATOR. Full learning route

From $0 = (a+b)^2 = a^2 + b^2 + ab + ba = ab + ba$, we have $ab = -(ba)$ for arbitrary a, b, which implies

$$abc = a(bc) = -((bc))a = -(b(ca)) = (ca)b = c(ab) = -((ab)c) = -abc.$$

1999-D2-P2 — Statement

Day 2, P8 | 20 points

We throw a dice (which selects one of the numbers $1, 2, \ldots, 6$ with equal probability) n times. What is the probability that the sum of the values is divisible by 5?

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Theory and methods for 1999-D2-P2

Theory Finite probability, conditioning, and expectations [T-AN-PROBABILITY]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Finite counting and incidence identities [T-DS-COUNTING]; Roots-of-unity filters and positive trigonometric kernels [T-AL-FOURIER]
Methods Count outcomes or condition on a useful structure [M-AN-PROBABILITY]; Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]; Encode convolution or denominators by a power series [M-AN-GENERATING]

Theory and methods for 1999-D2-P2 (continued)

Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

1999-D2-P2 — Solution — Record 1 of 3

For this record: Theory: T-AN-PROBABILITY; T-AL-CONGRUENCE; T-DS-INDUCTION. Methods: M-AN-PROBABILITY; M-AN-RECURRENCE. Full learning route

For all nonnegative integers n and modulo 5 residue class r, denote by $p_n^{(r)}$ the probability that after n throwing the sum of values is congruent to r modulo n. It is obvious that $p_0^{(0)} = 1$ and $p_0^{(1)} = p_0^{(2)} = p_0^{(3)} = p_0^{(4)} = 0$.

Moreover, for any $n > 0$ we have

$$p_n^{(r)} = \sum_{i=1}^{6} \frac{1}{6} p_{n-1}^{(r-i)}. \tag{1}$$

From this recursion we can compute the probabilities for small values of n and can conjecture that $p_n^{(r)} = \frac{1}{5} + \frac{4}{5 \cdot 6^n}$ if $n \equiv r \pmod 5$ and $p_n^{(r)} = \frac{1}{5} - \frac{1}{5 \cdot 6^n}$ otherwise. From (1), this conjecture can be proved by induction.

1999-D2-P2 — Solution — Record 2 of 3

For this record: Theory: T-AN-PROBABILITY; T-DS-COUNTING; T-AL-CONGRUENCE. Methods: M-AN-PROBABILITY. Full learning route

Let S be the set of all sequences consisting of digits $1, \dots, 6$ of length n. We create collections of these sequences.
Let a collection contain sequences of the form

$$\underbrace{66 \dots 6}_{k} X Y_1 \dots Y_{n-k-1},$$

where $X \in \{1, 2, 3, 4, 5\}$ and k and the digits $Y_1, \dots, Y_{n-k-1}$ are fixed. Then each collection consists of 5 sequences, and the sums of the digits of sequences give a whole residue system mod 5.
Except for the sequence $66 \dots 6$, each sequence is the element of one collection. This means that the number of the sequences, which have a sum of digits divisible by 5, is $\frac{1}{5}(6^n - 1) + 1$ if n is divisible by 5, otherwise $\frac{1}{5}(6^n - 1)$.

Thus, the probability is $\frac{1}{5} + \frac{4}{5 \cdot 6^n}$ if n is divisible by 5, otherwise it is $\frac{1}{5} - \frac{1}{5 \cdot 6^n}$.

1999-D2-P2 — Solution — Record 3 of 3

For this record: Theory: T-AN-PROBABILITY; T-AL-FOURIER. Methods: M-AN-PROBABILITY; M-AN-GENERATING. Full learning route

For arbitrary positive integer k denote by p_k the probability that the sum of values is k. Define the generating function

$$f(x) = \sum_{k=1}^{\infty} p_k x^k = \left(\frac{x + x^2 + x^3 + x^4 + x^5 + x^6}{6} \right)^n.$$

(The last equality can be easily proved by induction.)

Our goal is to compute the sum $\sum_{k=1}^{\infty} p_{5k}$. Let $\varepsilon = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$ be the first 5th root of unity. Then

$$\sum_{k=1}^{\infty} p_{5k} = \frac{f(1) + f(\varepsilon) + f(\varepsilon^2) + f(\varepsilon^3) + f(\varepsilon^4)}{5}.$$

1999-D2-P2 — Solution — Record 3 of 3 (continued)

Obviously $f(1)=1$, and $f(\varepsilon^j)=\frac{\varepsilon^{jn}}{6^n}$ for $j=1,2,3,4$. This implies that $f(\varepsilon)+f(\varepsilon^2)+f(\varepsilon^3)+f(\varepsilon^4)$ is $\frac{4}{6^n}$ if n is divisible by 5, otherwise it is $\frac{-1}{6^n}$. Thus, $\sum_{k=1}^{\infty} p_{5k}$ is $\frac{1}{5}+\frac{4}{5\cdot 6^n}$ if n is divisible by 5, otherwise it is $\frac{1}{5}-\frac{1}{5\cdot 6^n}$.

1999-D2-P3 — Statement

Day 2, P9 | 20 points

Assume that $x_1,\dots,x_n\geq -1$ and $\sum_{i=1}^{n} x_i^3=0$. Prove that $\sum_{i=1}^{n} x_i\leq \frac{n}{3}$.

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Theory and methods for 1999-D2-P3

Theory Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]
Strategy Build a chain of bounds and track equality [S-BOUND]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

1999-D2-P3 — Solution

For this record: Theory: T-AN-INEQUALITIES; T-AL-POLYNOMIAL. Methods: M-AN-WEIGHTS. Full learning route

The inequality

$$0\leq x^3-\frac{3}{4}x+\frac{1}{4}=(x+1)\left(x-\frac{1}{2}\right)^2$$

holds for $x\geq -1$.
Substituting $x_1,\dots,x_n$, we obtain

$$0\leq \sum_{i=1}^{n}\left(x_i^3-\frac{3}{4}x_i+\frac{1}{4}\right)=\sum_{i=1}^{n}x_i^3-\frac{3}{4}\sum_{i=1}^{n}x_i+\frac{n}{4}=0-\frac{3}{4}\sum_{i=1}^{n}x_i+\frac{n}{4},$$

so $\sum_{i=1}^{n} x_i\leq \frac{n}{3}$.

Remark. Equailty holds only in the case when $n=9k$, k of the $x_1,\dots,x_n$ are -1, and $8k$ of them are $\frac{1}{2}$.

1999-D2-P4 — Statement

Day 2, P10 | 20 points

Prove that there exists no function $f:(0,+\infty)\to(0,+\infty)$ such that $f^2(x)\geq f(x+y)(f(x)+y)$ for any $x,y>0$.

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Theory and methods for 1999-D2-P4

Theory Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Methods Force structure by admissible substitutions [M-DS-FUNCTIONAL]; Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]

Theory and methods for 1999-D2-P4 (continued)

Strategy Make the contradiction quantitative or structural [S-CONTRADICTION]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

1999-D2-P4 — Solution

For this record: Theory: T-DS-FUNCTIONAL; T-DS-INDUCTION. Methods: M-DS-FUNCTIONAL; M-DS-ITERATE. Full learning route

Assume that such a function exists. The initial inequality can be written in the form $f(x)-f(x+y) \geq f(x)-\dfrac{f^2(x)}{f(x)+y}=\dfrac{f(x)y}{f(x)+y}$. Obviously, f is a decreasing function. Fix $x>0$ and choose $n \in \mathbb{N}$ such that $nf(x+1) \geq 1$. For $k=0,1,\ldots,n-1$ we have

$$f\left(x+\frac{k}{n}\right)-f\left(x+\frac{k+1}{n}\right) \geq \frac{f\left(x+\frac{k}{n}\right)}{nf\left(x+\frac{k}{n}\right)+1} \geq \frac{1}{2n}.$$

The additon of these inequalities gives $f(x+1) \leq f(x)-\frac{1}{2}$. From this it follows that $f(x+2m) \leq f(x)-m$ for all $m \in \mathbb{N}$. Taking $m \geq f(x)$, we get a contradiction with the conditon $f(x)>0$.

1999-D2-P5 — Statement

Day 2, P11 | 20 points

Let S be the set of all words consisting of the letters x, y, z, and consider an equivalence relation $\sim$ on S satisfying the following conditions: for arbitrary words $u, v, w \in S$
(i) $uu \sim u$;
(ii) if $v \sim w$, then $uv \sim uw$ and $vu \sim wu$.

Show that every word in S is equivalent to a word of length at most 8.

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Theory and methods for 1999-D2-P5

Theory Extremal choice, exchange arguments, and invariants [T-DS-EXTREMAL]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Methods Reduce equivalent words with a congruence [M-DS-REWRITE]; Strengthen the statement before the induction step [M-DS-INDUCT]
Strategy Expose the structure before calculating [S-STRUCTURE]; Reduce size while preserving the defining conditions [S-INDUCTION]

Statement Recorded solutions Catalog

1999-D2-P5 — Solution

For this record: Theory: T-DS-EXTREMAL; T-DS-INDUCTION. Methods: M-DS-REWRITE; M-DS-INDUCT. Full learning route

First we prove the following lemma: If a word $u \in S$ contains at least one of each letter, and $v \in S$ is an arbitrary word, then there exists a word $w \in S$ such that $uvw \sim u$.
If v contains a single letter, say x, write u in the form $u=u_1xu_2$, and choose $w=u_2$. Then $uvw=(u_1xu_2)xu_2=u_1((xu_2)(xu_2)) \sim u_1(xu_2)=u$.
In the general case, let the letters of v be $a_1,\ldots,a_k$. Then one can choose some words $w_1,\ldots,w_k$ such that $(ua_1)w_1 \sim u$, $(ua_1a_2)w_2 \sim ua_1$, ..., $(ua_1\ldots a_k)w_k \sim ua_1\ldots a_{k-1}$. Then $u \sim ua_1w_1 \sim ua_1a_2w_2w_1 \sim \cdots \sim ua_1\ldots a_kw_k\ldots w_1=uv(w_k\ldots w_1)$, so $w=w_k\ldots w_1$ is a good choice.
Consider now an arbitrary word a, which contains more than 8 digits. We shall prove that there is a shorter word which is equivalent to a. If a can be written in the form $uvvw$, its length can be reduced by $uvvw \sim uvw$. So we can assume that a does not have this form.
Write a in the form $a=bcd$, where b and d are the first and last four letters of a, respectively. We prove that $a \sim bd$.

1999-D2-P5 — Solution (continued)

It is easy to check that b and d contain all the three letters x, y and z, otherwise their length could be reduced. By the lemma there is a word e such that $b(cd)e \sim b$, and there is a word f such that $def \sim d$. Then we can write

$$\begin{aligned} & a \\ &= bcd \\ &\sim bc(def) \\ &\sim bc(dedef) \\ &= (bcde)(def) \\ &\sim bd. \end{aligned}$$

Remark. Of course, it is enough to give for every word of length 9 a shorter word. Assuming that the first letter is x and the second is y, it is easy (but a little long) to check that there are 18 words of length 9 which cannot be written in the form $uvvw$.

Some of these words admit short reductions, for example

$$\begin{aligned} & xyxzyzxzy \\ &\sim xyxzyzxzyzy \\ &\sim xyxzyzy \\ &\sim xyxzy. \end{aligned}$$

Other cases can require longer reductions. The general algorithm given by the Solution works for these cases as well, but needs also very long words. For example, to reduce the length of the word $a = xyzyxzxyz$, we have set $b = xyzy$, $c = x$, $d = zxyz$, $e = xyxzxzyxyzy$, $f = zyxyxzyxzxzxzxyxyzxyz$. The longest word in the algorithm was

$$bcdedef = xyzyxzxyzxyxzxzyxyzyzxyzxyxzxzyxyzyzyxyxzyxzxzxzxyxyzxyz,$$

which is of length 56. This is not the shortest way: reducing the length of word a can be done for example by the following steps:

$$\begin{aligned} & xyzyxzxyz \\ &\sim xyzyxzxyzyz \\ &\sim xyzyxzxyzyxyzyz \\ &\sim xyzyxzxyzyxzyxyzyz \\ &\sim xyzyxzyxyz \\ &\sim xyzyxyz. \end{aligned}$$

(The last example is due to Nayden Kambouchev from Sofia University.)

1999-D2-P6 — Statement

Day 2, P12 | 20 points

Let A be a subset of $\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z}$ containing at most $\frac{1}{100}\ln n$ elements. Define the rth Fourier coefficient of A for $r \in \mathbb{Z}_n$ by

$$f(r) = \sum_{s \in A} \exp\left(\frac{2\pi i}{n} sr\right).$$

Prove that there exists an $r \neq 0$, such that $\left|f(r)\right| \geq \frac{|A|}{2}$.

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Theory and methods for 1999-D2-P6

Theory Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Evaluate at roots of unity or average over them [M-AL-UNITY]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Translate geometry into a controlled representation [S-GEOMETRIC]

Statement Recorded solutions Catalog

1999-D2-P6 — Solution

For this record: Theory: T-AL-COMPLEX; T-DS-COUNTING. Methods: M-AL-UNITY. Full learning route

Let $A = \{a_1, \dots, a_k\}$. Consider the k-tuples

$$\left(\exp \frac{2\pi i a_1 t}{n}, \dots, \exp \frac{2\pi i a_k t}{n}\right) \in \mathbb{C}^k, \quad t = 0, 1, \dots, n-1.$$

Each component is in the unit circle $|z| = 1$. Split the circle into 6 equal arcs. This induces a decomposition of the k-tuples into 6^k classes. By the condition $k \le \frac{1}{100} \ln n$ we have $n > 6^k$, so there are two k-tuples in the same class say for $t_1 < t_2$. Set $r = t_2 - t_1$. Then

$$\operatorname{Re} \exp \frac{2\pi i a_j r}{n} = \cos\left(\frac{2\pi a_j t_2}{n} - \frac{2\pi a_j t_1}{n}\right) \ge \cos \frac{\pi}{3} = \frac{1}{2}$$

for all j, so

$$|f(r)| \ge \operatorname{Re} f(r) \ge \frac{k}{2}.$$

3.7 2000

Day 1

2000-D1-P1 — Statement

Day 1, P1

Is it true that if $f : [0,1] \to [0,1]$ is
a) monotone increasing
b) monotone decreasing

then there exists an $x \in [0,1]$ for which $f(x) = x$?

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Theory and methods for 2000-D1-P1

Theory Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]; Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]
Methods Force structure by admissible substitutions [M-DS-FUNCTIONAL]; Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2000-D1-P1 — Solution

For this record: Theory: T-DS-DYNAMICS; T-DS-FUNCTIONAL. Methods: M-DS-FUNCTIONAL; M-DS-ITERATE. Full learning route

a) Yes.

2000-D1-P1 — Solution (continued)

Proof: Let $A = \{x \in [0,1] : f(x) > x\}$. If $f(0) = 0$ we are done, if not then A is non-empty (0 is in A) bounded, so it has supremum, say a. Let $b = f(a)$.
I. case: $a < b$. Then, using that f is monotone and a was the sup, we get $b = f(a) \leq f((a+b)/2) \leq (a+b)/2$, which contradicts $a < b$.
II. case: $a > b$. Then we get $b = f(a) \geq f((a+b)/2) > (a+b)/2$ contradiction.
Therefore we must have $a = b$.

b) No. Let, for example,

$$f(x) = 1 - x/2 \quad \text{if } x \leq 1/2$$

and

$$f(x) = 1/2 - x/2 \quad \text{if } x > 1/2$$

This is clearly a good counter-example.

2000-D1-P2 — Statement

Day 1, P2

Let $p(x) = x^5 + x$ and $q(x) = x^5 + x^2$. Find all pairs (w, z) of complex numbers with $w \neq z$ for which $p(w) = p(z)$ and $q(w) = q(z)$.

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Theory and methods for 2000-D1-P2

Theory Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]; Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]
Methods Compare coefficients after a structural substitution [M-AL-COEFFICIENT]; Translate root restrictions into symmetric coefficients [M-AL-VIETA]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]

Statement Recorded solutions Catalog

2000-D1-P2 — Solution

For this record: Theory: T-AL-POLYNOMIAL; T-AL-ROOTS; T-AL-COMPLEX. Methods: M-AL-COEFFICIENT; M-AL-VIETA.
Full learning route

Let

$$P(x,y) = \frac{p(x) - p(y)}{x - y} = x^4 + x^3y + x^2y^2 + xy^3 + y^4 + 1$$

and

$$Q(x,y) = \frac{q(x) - q(y)}{x - y} = x^4 + x^3y + x^2y^2 + xy^3 + y^4 + x + y.$$

We need those pairs (w, z) which satisfy $P(w, z) = Q(w, z) = 0$.

From $P - Q = 0$ we have $w + z = 1$. Let $c = wz$. After a short calculation we obtain $c^2 - 3c + 2 = 0$, which has the solutions $c = 1$ and $c = 2$. From the system $w + z = 1$, $wz = c$ we obtain the following pairs:

$$\left(\frac{1 \pm \sqrt{3}i}{2}, \frac{1 \mp \sqrt{3}i}{2}\right) \quad \text{and} \quad \left(\frac{1 \pm \sqrt{7}i}{2}, \frac{1 \mp \sqrt{7}i}{2}\right).$$

2000-D1-P3 — Statement

Day 1, P3

A and B are square complex matrices of the same size and

$$\operatorname{rank}(AB - BA) = 1.$$

Show that $(AB - BA)^2 = 0$.

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Theory and methods for 2000-D1-P3

Theory Rank, dimension and exact linear constraints [T-LA-RANK]; Trace pairing and commutator calculus [T-LA-TRACE]; Nilpotents, projections and involutions [T-LA-NILPOTENT]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]
Methods Use trace to extract a scalar invariant [M-LA-TRACE]; Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]
Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2000-D1-P3 — Solution

For this record: Theory: T-LA-RANK; T-LA-TRACE; T-LA-NILPOTENT; T-LA-SPECTRAL. Methods: M-LA-TRACE; M-LA-SPECTRAL. Full learning route

Let $C = AB - BA$. Since $\operatorname{rank} C = 1$, at most one eigenvalue of C is different from 0. Also $\operatorname{tr} C = 0$, so all the eigevalues are zero. In the Jordan canonical form there can only be one 2×2 cage and thus $C^2 = 0$.

2000-D1-P4 — Statement

Day 1, P4

a) Show that if (x_i) is a decreasing sequence of positive numbers then

$$\left(\sum_{i=1}^{n} x_i^2\right)^{1/2} \leq \sum_{i=1}^{n} \frac{x_i}{\sqrt{i}}.$$

b) Show that there is a constant C so that if (x_i) is a decreasing sequence of positive numbers then

$$\sum_{m=1}^{\infty} \frac{1}{\sqrt{m}} \left(\sum_{i=m}^{\infty} x_i^2\right)^{1/2} \leq C \sum_{i=1}^{\infty} x_i.$$

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Theory and methods for 2000-D1-P4

Theory Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Nonnegative sums, Fubini, and integral kernels [T-AN-FUBINI]; Series, finite cancellation, and summation by parts [T-AN-SERIES]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Methods Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]; Find a finite telescoping identity [M-AN-TELESCOPE]
Strategy Build a chain of bounds and track equality [S-BOUND]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2000-D1-P4 — Solution

For this record: Theory: T-AN-INEQUALITIES; T-AN-FUBINI; T-AN-SERIES; T-AN-INTEGRAL. Methods: M-AN-WEIGHTS; M-AN-TELESCOPE. Full learning route

a)

$$\left(\sum_{i=1}^{n}\frac{x_i}{\sqrt{i}}\right)^2=\sum_{i,j}\frac{x_ix_j}{\sqrt{i}\sqrt{j}}\geq\sum_{i=1}^{n}\frac{x_i}{\sqrt{i}}\sum_{j=1}^{i}\frac{x_j}{\sqrt{j}}\geq\sum_{i=1}^{n}\frac{x_i}{\sqrt{i}}\,i\,\frac{x_i}{\sqrt{i}}=\sum_{i=1}^{n}x_i^2$$

b)

$$\sum_{m=1}^{\infty}\frac{1}{\sqrt{m}}\left(\sum_{i=m}^{\infty}x_i^2\right)^{1/2}\leq\sum_{m=1}^{\infty}\frac{1}{\sqrt{m}}\sum_{i=m}^{\infty}\frac{x_i}{\sqrt{i-m+1}}$$

by a)

$$=\sum_{i=1}^{\infty}x_i\sum_{m=1}^{i}\frac{1}{\sqrt{m}\sqrt{i-m+1}}$$

You can get a sharp bound on

$$\sup_i\sum_{m=1}^{i}\frac{1}{\sqrt{m}\sqrt{i-m+1}}$$

by checking that it is at most

$$\int_0^{i+1}\frac{1}{\sqrt{x}\sqrt{i+1-x}}\,dx=\pi$$

Alternatively you can observe that

$$\sum_{m=1}^{i}\frac{1}{\sqrt{m}\sqrt{i+1-m}}=2\sum_{m=1}^{i/2}\frac{1}{\sqrt{m}\sqrt{i+1-m}}\leq$$

$$\leq 2\frac{1}{\sqrt{i/2}}\sum_{m=1}^{i/2}\frac{1}{\sqrt{m}}\leq 2\frac{1}{\sqrt{i/2}}\cdot 2\sqrt{i/2}=4$$

2000-D1-P5 — Statement

Day 1, P5

Let R be a ring of characteristic zero (not necessarily commutative). Let e, f and g be idempotent elements of R satisfying $e+f+g=0$. Show that $e=f=g=0$.

(R is of characteristic zero means that, if $a\in R$ and n is a positive integer, then $na\neq 0$ unless $a=0$. An idempotent x is an element satisfying $x=x^2$.)

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Theory and methods for 2000-D1-P5

Theory Noncommutative rings, polarization, and torsion [T-AL-RING]
Methods Polarize identities and isolate commutators [M-AL-COMMUTATOR]
Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2000-D1-P5 — Solution

For this record: Theory: T-AL-RING. Methods: M-AL-COMMUTATOR. Full learning route

Suppose that $e+f+g=0$ for given idempotents $e,f,g\in R$. Then

$$g=g^2=\left(-(e+f)\right)^2=e+(ef+fe)+f=(ef+fe)-g,$$

i.e. $ef+fe=2g$, whence the additive commutator

$$[e,f]=ef-fe=[e,ef+fe]=2[e,g]=2[e,-e-f]=-2[e,f],$$

2000-D1-P5 — Solution (continued)

i.e. $ef = fe$ (since R has zero characteristic). Thus $ef+fe = 2g$ becomes $ef = g$, so that $e+f+ef = 0$. On multiplying by e, this yields $e + 2ef = 0$, and similarly $f + 2ef = 0$, so that $f = -2ef = e$, hence $e = f = g$ by symmetry. Hence, finaly, $3e = e + f + g = 0$, i.e. $e = f = g = 0$.

For part (i) just omit some of this.

2000-D1-P6 — Statement

Day 1, P6

Let $f : \mathbb{R} \to (0, \infty)$ be an increasing differentiable function for which $\lim_{x\to\infty} f(x) = \infty$ and f' is bounded.

Let $F(x) = \int_0^x f$. Define the sequence (a_n) inductively by

$$a_0 = 1, \qquad a_{n+1} = a_n + \frac{1}{f(a_n)}, \tag{1}$$

and the sequence (b_n) simply by $b_n = F^{-1}(n)$. Prove that $\lim_{n\to\infty}(a_n - b_n) = 0$.

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Theory and methods for 2000-D1-P6

Theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Methods Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]; Compare growth through derivatives [M-AN-RATIO]; Separate a main range from exceptional scales [M-AN-SPLIT]
Strategy Control the limiting operation before taking the limit [S-ANALYTIC]; Pass between local constraints and a global conclusion [S-LOCALGLOBAL]

Statement Recorded solutions Catalog

2000-D1-P6 — Solution

For this record: Theory: T-AN-DERIVATIVE; T-AN-INTEGRAL; T-AN-CONVERGENCE. Methods: M-AN-RECURRENCE; M-AN-RATIO; M-AN-SPLIT. Full learning route

From the conditions it is obvious that F is increasing and $\lim_{n\to\infty} b_n = \infty$.

By Lagrange's theorem and the recursion in (1), for all $k \geq 0$ integers there exists a real number $\xi \in (a_k, a_{k+1})$ such that

$$F(a_{k+1}) - F(a_k) = f(\xi)(a_{k+1} - a_k) = \frac{f(\xi)}{f(a_k)}. \tag{2}$$

By the monotonity, $f(a_k) \leq f(\xi) \leq f(a_{k+1})$, thus

$$1 \leq F(a_{k+1}) - F(a_k) \leq \frac{f(a_{k+1})}{f(a_k)} = 1 + \frac{f(a_{k+1}) - f(a_k)}{f(a_k)}. \tag{3}$$

Summing (3) for $k = 0, \ldots, n-1$ and substituting $F(b_n) = n$, we have

$$F(b_n) < n + F(a_0) \leq F(a_n) \leq F(b_n) + F(a_0) + \sum_{k=0}^{n-1} \frac{f(a_{k+1}) - f(a_k)}{f(a_k)}. \tag{4}$$

From the first two inequalities we already have $a_n > b_n$ and $\lim_{n\to\infty} a_n = \infty$.

2000-D1-P6 — Solution (continued)

Let ε be an arbitrary positive number. Choose an integer K_ε such that $f(a_{K_\varepsilon}) > \frac{2}{\varepsilon}$. If n is sufficiently large, then

$$\begin{aligned} F(a_0) + \sum_{k=0}^{n-1} \frac{f(a_{k+1}) - f(a_k)}{f(a_k)} = \\ = F(a_0) + \left(\sum_{k=0}^{K_\varepsilon - 1} \frac{f(a_{k+1}) - f(a_k)}{f(a_k)} \right) + \sum_{k=K_\varepsilon}^{n-1} \frac{f(a_{k+1}) - f(a_k)}{f(a_k)} < \\ < O_\varepsilon(1) + \frac{1}{f(a_{K_\varepsilon})} \sum_{k=K_\varepsilon}^{n-1} (f(a_{k+1}) - f(a_k)) < \\ < O_\varepsilon(1) + \frac{\varepsilon}{2}(f(a_n) - f(a_{K_\varepsilon})) < \varepsilon f(a_n). \end{aligned} \tag{5}$$

Inequalities (4) and (5) together say that for any positive ε, if n is sufficiently large,

$$F(a_n) - F(b_n) < \varepsilon f(a_n).$$

Again, by Lagrange's theorem, there is a real number $\zeta \in (b_n, a_n)$ such that

$$F(a_n) - F(b_n) = f(\zeta)(a_n - b_n) > f(b_n)(a_n - b_n), \tag{6}$$

thus

$$f(b_n)(a_n - b_n) < \varepsilon f(a_n). \tag{7}$$

Let B be an upper bound for f'. Apply $f(a_n) < f(b_n) + B(a_n - b_n)$ in (7):

$$f(b_n)(a_n - b_n) < \varepsilon(f(b_n) + B(a_n - b_n)),$$

$$(f(b_n) - \varepsilon B)(a_n - b_n) < \varepsilon f(b_n). \tag{8}$$

Due to $\lim\limits_{n\to\infty} f(b_n) = \infty$, the first factor is positive, and we have

$$a_n - b_n < \varepsilon \frac{f(b_n)}{f(b_n) - \varepsilon B} < 2\varepsilon \tag{9}$$

for sufficiently large n.

Thus, for arbitrary positive ε we proved that $0 < a_n - b_n < 2\varepsilon$ if n is sufficiently large.

Day 2

2000-D2-P1 — Statement

Day 2, P7

a) Show that the unit square can be partitioned into n smaller squares if n is large enough.

b) Let $d \geq 2$. Show that there is a constant $N(d)$ such that, whenever $n \geq N(d)$, a d-dimensional unit cube can be partitioned into n smaller cubes.

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Theory and methods for 2000-D2-P1

Theory Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]
Methods Split a family by traces, pieces, or labels [M-DS-PARTITION]; Strengthen the statement before the induction step [M-DS-INDUCT]
Strategy Build a witness and verify every constraint [S-CONSTRUCTION]; Reduce size while preserving the defining conditions [S-INDUCTION]

Statement Recorded solutions Catalog

2000-D2-P1 — Solution

For this record: Theory: T-DS-INDUCTION; T-AL-CONGRUENCE; T-DS-GEOMETRY. Methods: M-DS-PARTITION; M-DS-INDUCT. Full learning route

We start with the following lemma: If a and b be coprime positive integers then every sufficiently large positive integer m can be expressed in the form $ax + by$ with x, y non-negative integers.
Proof of the lemma. The numbers $0, a, 2a, \ldots, (b-1)a$ give a complete residue system modulo b. Consequently, for any m there exists a $0 \leq x \leq b-1$ so that $ax \equiv m \pmod{b}$. If $m \geq (b-1)a$, then $y = (m - ax)/b$, for which $x + by = m$, is a non-negative integer, too.

Now observe that any dissection of a cube into n smaller cubes may be refined to give a dissection into $n + (a^d - 1)$ cubes, for any $a \geq 1$. This refinement is achieved by picking an arbitrary cube in the dissection, and cutting it into a^d smaller cubes. To prove the required result, then, it suffices to exhibit two relatively prime integers of form $a^d - 1$. In the 2-dimensional case, $a_1 = 2$ and $a_2 = 3$ give the coprime numbers $2^2 - 1 = 3$ and $3^2 - 1 = 8$. In the general case, two such integers are $2^d - 1$ and $(2^d - 1)^d - 1$, as is easy to check.

2000-D2-P2 — Statement

Day 2, P8

Let f be continuous and nowhere monotone on $[0, 1]$. Show that the set of points on which f attains local minima is dense in $[0, 1]$.

(A function is nowhere monotone if there exists no interval where the function is monotone. A set is dense if each non-empty open interval contains at least one element of the set.)

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Theory and methods for 2000-D2-P2

Theory Continuity, compactness, and order [T-AN-CONTINUITY]
Methods Localize an extremum or an accumulation point [M-AN-COMPACT]
Strategy Choose an extremal object and force rigidity [S-EXTREMAL]; Pass between local constraints and a global conclusion [S-LOCALGLOBAL]

Statement Recorded solutions Catalog

2000-D2-P2 — Solution

For this record: Theory: T-AN-CONTINUITY. Methods: M-AN-COMPACT. Full learning route

Let $(x - \alpha, x + \alpha) \subset [0, 1]$ be an arbitrary non-empty open interval. The function f is not monoton in the intervals $[x - \alpha, x]$ and $[x, x + \alpha]$, thus there exist some real numbers $x - \alpha \leq p < q \leq x$, $x \leq r < s \leq x + \alpha$ so that $f(p) > f(q)$ and $f(r) < f(s)$.

By Weierstrass' theorem, f has a global minimum in the interval $[p, s]$. The values $f(p)$ and $f(s)$ are not the minimum, because they are greater than $f(q)$ and $f(s)$, respectively. Thus the minimum is in the interior of the interval, it is a local minimum. So each non-empty interval $(x-\alpha, x+\alpha) \subset [0, 1]$ contains at least one local minimum.

2000-D2-P3 — Statement

Day 2, P9

Let $p(z)$ be a polynomial of degree n with complex coefficients. Prove that there exist at least $n+1$ complex numbers z for which $p(z)$ is 0 or 1.

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Theory and methods for 2000-D2-P3

Theory Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Methods Replace many equal values by too many roots [M-AL-ROOTCOUNT]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]

Statement Recorded solutions Catalog

2000-D2-P3 — Solution

For this record: Theory: T-AL-POLYNOMIAL; T-AN-DERIVATIVE. Methods: M-AL-ROOTCOUNT. Full learning route

The statement is not true if p is a constant polynomial. We prove it only in the case if n is positive. For an arbitrary polynomial $q(z)$ and complex number c, denote by $\mu(q,c)$ the largest exponent α for which $q(z)$ is divisible by $(z-c)^\alpha$. (With other words, if c is a root of q, then $\mu(q,c)$ is the root's multiplicity. Otherwise 0.)

Denote by S_0 and S_1 the sets of complex numbers z for which $p(z)$ is 0 or 1, respectively. These sets contain all roots of the polynomials $p(z)$ and $p(z)-1$, thus

$$\sum_{c\in S_0}\mu(p,c)=\sum_{c\in S_1}\mu(p-1,c)=n. \tag{1}$$

The polynomial p' has at most $n-1$ roots ($n>0$ is used here). This implies that

$$\sum_{c\in S_0\cup S_1}\mu(p',c)\le n-1. \tag{2}$$

If $p(c)=0$ or $p(c)-1=0$, then

$$\mu(p,c)-\mu(p',c)=1 \quad \text{or} \quad \mu(p-1,c)-\mu(p',c)=1, \tag{3}$$

respectively. Putting (1), (2) and (3) together we obtain

$$\begin{aligned}|S_0|+|S_1| &= \sum_{c\in S_0}(\mu(p,c)-\mu(p',c))+\sum_{c\in S_1}(\mu(p-1,c)-\mu(p',c))= \\ &= \sum_{c\in S_0}\mu(p,c)+\sum_{c\in S_1}\mu(p-1,c)-\sum_{c\in S_0\cup S_1}\mu(p',c)\ge n+n-(n-1)=n+1.\end{aligned}$$

2000-D2-P4 — Statement

Day 2, P10

Suppose the graph of a polynomial of degree 6 is tangent to a straight line at 3 points A_1, A_2, A_3, where A_2 lies between A_1 and A_3.
a) Prove that if the lengths of the segments A_1A_2 and A_2A_3 are equal, then the areas of the figures bounded by these segments and the graph of the polynomial are equal as well.

b) Let $k=\dfrac{A_2A_3}{A_1A_2}$, and let K be the ratio of the areas of the appropriate figures. Prove that

$$\frac{2}{7}k^5<K<\frac{7}{2}k^5.$$

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Theory and methods for 2000-D2-P4

Theory Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Methods Factor first and compare irreducible exponents [M-AL-FACTOR]; Compare coefficients after a structural substitution [M-AL-COEFFICIENT]
Strategy Translate geometry into a controlled representation [S-GEOMETRIC]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2000-D2-P4 — Solution — Part a

For this record: Theory: T-AL-POLYNOMIAL; T-AN-INTEGRAL. Methods: M-AL-FACTOR. Full learning route

a) Without loss of generality, we can assume that the point A_2 is the origin of system of coordinates. Then the polynomial can be presented in the form

$$y = (a_0x^4 + a_1x^3 + a_2x^2 + a_3x + a_4)x^2 + a_5x,$$

where the equation $y = a_5x$ determines the straight line A_1A_3. The abscissas of the points A_1 and A_3 are $-a$ and a, $a > 0$, respectively. Since $-a$ and a are points of tangency, the numbers $-a$ and a must be double roots of the polynomial $a_0x^4 + a_1x^3 + a_2x^2 + a_3x + a_4$. It follows that the polynomial is of the form

$$y = a_0x^2(x^2 - a^2)^2 + a_5x.$$

The equality follows from the equality of the integrals

$$\int_{-a}^{0} a_0(x^2 - a^2)^2x^2\,dx = \int_{0}^{a} a_0(x^2 - a^2)^2x^2\,dx$$

due to the fact that the function $y = a_0x^2(x^2 - a^2)^2$ is even.

2000-D2-P4 — Solution — Part b

For this record: Theory: T-AL-POLYNOMIAL; T-AN-INTEGRAL; T-AN-DERIVATIVE. Methods: M-AL-COEFFICIENT. Full learning route

b) Without loss of generality, we can assume that $a_0 = 1$. Then the function is of the form

$$y = (x + a)^2(x - b)^2x^2 + a_5x,$$

where a and b are positive numbers and $b = ka$, $0 < k < \infty$. The areas of the figures at the segments A_1A_2 and A_2A_3 are equal respectively to

$$\int_{-a}^{0} (x + a)^2(x - b)^2x^2\,dx = \frac{a^7}{210}(7k^2 + 7k + 2)$$

and

$$\int_{0}^{b} (x + a)^2(x - b)^2x^2\,dx = \frac{a^7k^5}{210}(2k^2 + 7k + 7)$$

Then

$$K = k^5\,\frac{2k^2 + 7k + 7}{7k^2 + 7k + 2}.$$

The derivative of the function $f(k) = \dfrac{2k^2 + 7k + 7}{7k^2 + 7k + 2}$ is negative for $0 < k < \infty$. Therefore $f(k)$ decreases from $\frac{7}{2}$ to $\frac{2}{7}$ when k increases from 0 to ∞. Inequalities $\frac{2}{7} < \frac{2k^2+7k+7}{7k^2+7k+2} < \frac{7}{2}$ imply the desired inequalities.

2000-D2-P5 — Statement

Day 2, P11

Let $\mathbb{R}^+$ be the set of positive real numbers. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that for all $x, y \in \mathbb{R}^+$

$$f(x)f(yf(x)) = f(x+y).$$

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Theory and methods for 2000-D2-P5

Theory Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Dominated convergence and orthonormal separation [T-AN-MEASURE]
Methods Force structure by admissible substitutions [M-DS-FUNCTIONAL]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2000-D2-P5 — Solution — Record 1 of 2

For this record: Theory: T-DS-FUNCTIONAL. Methods: M-DS-FUNCTIONAL. Full learning route

First, if we assume that $f(x) > 1$ for some $x \in \mathbb{R}^+$, setting $y = \dfrac{x}{f(x)-1}$ gives the contradiction $f(x) = 1$. Hence $f(x) \le 1$ for each $x \in \mathbb{R}^+$, which implies that f is a decreasing function.
If $f(x) = 1$ for some $x \in \mathbb{R}^+$, then $f(x+y) = f(y)$ for each $y \in \mathbb{R}^+$, and by the monotonicity of f it follows that $f \equiv 1$.
Let now $f(x) < 1$ for each $x \in \mathbb{R}^+$. Then f is strictly decreasing function, in particular injective. By the equalities

$$\begin{aligned} f(x)f(yf(x)) = f(x+y) = \\ = f(yf(x)+x+y(1-f(x))) = f(yf(x))f\Big((x+y(1-f(x)))f(yf(x))\Big) \end{aligned}$$

we obtain that $x = (x+y(1-f(x)))f(yf(x))$. Setting $x = 1$, $z = xf(1)$ and $a = \dfrac{1-f(1)}{f(1)}$, we get $f(z) = \dfrac{1}{1+az}$.

Combining the two cases, we conclude that $f(x) = \dfrac{1}{1+ax}$ for each $x \in \mathbb{R}^+$, where $a \ge 0$. Conversely, a direct verification shows that the functions of this form satisfy the initial equality.

2000-D2-P5 — Solution — Record 2 of 2

For this record: Theory: T-DS-FUNCTIONAL; T-AN-MEASURE; T-AN-DERIVATIVE. Methods: M-DS-FUNCTIONAL. Full learning route

As in the first solution we get that f is a decreasing function, in particular differentiable almost everywhere. Write the initial equality in the form

$$\frac{f(x+y)-f(x)}{y} = f^2(x)\,\frac{f(yf(x))-1}{yf(x)}.$$

It follows that if f is differentiable at the point $x \in \mathbb{R}^+$, then there exists the limit $\lim\limits_{z\to 0+} \dfrac{f(z)-1}{z} =: -a$.
Therefore $f'(x) = -af^2(x)$ for each $x \in \mathbb{R}^+$, i.e. $\left(\dfrac{1}{f(x)}\right)' = a$, which means that $f(x) = \dfrac{1}{ax+b}$.
Substituting in the initial relaton, we find that $b = 1$ and $a \ge 0$.

2000-D2-P6 — Statement

Day 2, P12

For an $m \times m$ real matrix A, e^A is defined as $\sum_{n=0}^{\infty} \frac{1}{n!} A^n$. (The sum is convergent for all matrices.)

Prove or disprove, that for all real polynomials p and $m \times m$ real matrices A and B, $p(e^{AB})$ is nilpotent if and only if $p(e^{BA})$ is nilpotent. (A matrix A is nilpotent if $A^k = 0$ for some positive integer k.)

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Theory and methods for 2000-D2-P6

Theory Matrix power series and transfer between AB and BA [T-LA-SERIES]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Nilpotents, projections and involutions [T-LA-NILPOTENT]
Methods Transfer spectral information between products [M-LA-TRANSFER]
Strategy Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2000-D2-P6 — Solution

For this record: Theory: T-LA-SERIES; T-LA-POLYNOMIAL; T-LA-NILPOTENT. Methods: M-LA-TRANSFER. Full learning route

First we prove that for any polynomial q and $m \times m$ matrices A and B, the characteristic polinomials of $q(e^{AB})$ and $q(e^{BA})$ are the same. It is easy to check that for any matrix X, $q(e^X) = \sum_{n=0}^{\infty} c_n X^n$ with some real numbers c_n which depend on q. Let

$$C = \sum_{n=1}^{\infty} c_n \cdot (BA)^{n-1} B = \sum_{n=1}^{\infty} c_n \cdot B(AB)^{n-1}.$$

Then $q(e^{AB}) = c_0 I + AC$ and $q(e^{BA}) = c_0 I + CA$. It is well-known that the characteristic polynomials of AC and CA are the same; denote this polynomial by $f(x)$. Then the characteristic polynomials of matrices $q(e^{AB})$ and $q(e^{BA})$ are both $f(x - c_0)$.

Now assume that the matrix $p(e^{AB})$ is nilpotent, i.e. $\left(p(e^{AB})\right)^k = 0$ for some positive integer k. Chose $q = p^k$. The characteristic polynomial of the matrix $q(e^{AB}) = 0$ is x^m, so the same holds for the matrix $q(e^{BA})$. By the theorem of Cayley and Hamilton, this implies that $\left(q(e^{BA})\right)^m = \left(p(e^{BA})\right)^{km} = 0$. Thus the matrix $q(e^{BA})$ is nilpotent, too.

3.8 2001

Day 1

2001-D1-P1 — Statement

Day 1, P1

Let n be a positive integer. Consider an $n \times n$ matrix with entries $1, 2, \ldots, n^2$ written in order starting top left and moving along each row in turn left–to–right. We choose n entries of the matrix such that exactly one entry is chosen in each row and each column. What are the possible values of the sum of the selected entries?

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Theory and methods for 2001-D1-P1

Theory Finite counting and incidence identities [T-DS-COUNTING]; Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]
Methods Choose information that makes an encoding injective [M-DS-ENCODE]
Strategy Expose the structure before calculating [S-STRUCTURE]; Choose the objects and count them without ambiguity [S-COUNTING]

Statement Recorded solutions Catalog

2001-D1-P1 — Solution

For this record: Theory: T-DS-COUNTING; T-AL-PERMUTATION. Methods: M-DS-ENCODE. Full learning route

Since there are exactly n rows and n columns, the choice is of the form

$$\{(j, \sigma(j)) : j = 1, \dots, n\}$$

where $\sigma \in S_n$ is a permutation. Thus the corresponding sum is equal to

$$\begin{aligned}
&\sum_{j=1}^{n} (n(j-1) + \sigma(j)) \\
&= \sum_{j=1}^{n} nj - \sum_{j=1}^{n} n + \sum_{j=1}^{n} \sigma(j) \\
&= n\sum_{j=1}^{n} j - \sum_{j=1}^{n} n + \sum_{j=1}^{n} j \\
&= (n+1)\frac{n(n+1)}{2} - n^2 \\
&= \frac{n(n^2+1)}{2},
\end{aligned}$$

which shows that the sum is independent of σ.

2001-D1-P2 — Statement

Day 1, P2

Let r, s, t be positive integers which are pairwise relatively prime. If a and b are elements of a commutative multiplicative group with unity element e, and $a^r = b^s = (ab)^t = e$, prove that $a = b = e$.

Does the same conclusion hold if a and b are elements of an arbitrary non-commutative group?

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Theory and methods for 2001-D1-P2

Theory Groups, orders, generators, and quotients [T-AL-GROUP]; Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Methods Prove generation or obstruct it by an image [M-AL-GENERATORS]
Strategy Expose the structure before calculating [S-STRUCTURE]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2001-D1-P2 — Solution

For this record: Theory: T-AL-GROUP; T-AL-PERMUTATION; T-AL-CONGRUENCE. Methods: M-AL-GENERATORS. Full learning route

1. There exist integers u and v such that $us + vt = 1$. Since $ab = ba$, we obtain

$$ab = (ab)^{us+vt} = (ab)^{us}\left((ab)^t\right)^v = (ab)^{us}e = (ab)^{us} = a^{us}(b^s)^u = a^{us}e = a^{us}.$$

2001-D1-P2 — Solution (continued)

Therefore, $b^r = eb^r = a^r b^r = (ab)^r = a^{usr} = (a^r)^{us} = e$. Since $xr + ys = 1$ for suitable integers x and y,

$$b = b^{xr+ys} = (b^r)^x (b^s)^y = e.$$

It follows similarly that $a = e$ as well.

2. This is not true. Let $a = (123)$ and $b = (34567)$ be cycles of the permutation group S_7 of order 7. Then $ab = (1234567)$ and $a^3 = b^5 = (ab)^7 = e$.

2001-D1-P3 — Statement

Day 1, P3

Find $\lim_{t \nearrow 1} (1-t) \sum_{n=1}^{\infty} \frac{t^n}{1+t^n}$, where $t \nearrow 1$ means that t approaches 1 from below.

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Theory and methods for 2001-D1-P3

Theory Riemann sums, periodic averaging, and tails [T-AN-RIEMANN]; Scale separation and limit comparisons [T-AN-ASYMPTOTICS]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Methods Expose the mesh and control its tails [M-AN-RIEMANN]
Strategy Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2001-D1-P3 — Solution

For this record: Theory: T-AN-RIEMANN; T-AN-ASYMPTOTICS; T-AN-INTEGRAL. Methods: M-AN-RIEMANN. Full learning route

$$\lim_{t\to 1-0} (1-t) \sum_{n=1}^{\infty} \frac{t^n}{1+t^n} = \lim_{t\to 1-0} \frac{1-t}{-\ln t} \cdot (-\ln t) \sum_{n=1}^{\infty} \frac{t^n}{1+t^n} =$$

$$= \lim_{t\to 1-0} (-\ln t) \sum_{n=1}^{\infty} \frac{1}{1+e^{-n\ln t}} = \lim_{h\to +0} h \sum_{n=1}^{\infty} \frac{1}{1+e^{nh}} = \int_0^{\infty} \frac{dx}{1+e^x} = \ln 2.$$

2001-D1-P4 — Statement

Day 1, P4

Let k be a positive integer. Let $p(x)$ be a polynomial of degree n each of whose coefficients is -1, 1 or 0, and which is divisible by $(x-1)^k$. Let q be a prime such that $\frac{q}{\ln q} < \frac{k}{\ln(n+1)}$. Prove that the complex qth roots of unity are roots of the polynomial $p(x)$.

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Theory and methods for 2001-D1-P4

Theory Roots of unity, cyclotomic polynomials, and algebraic norms [T-AL-CYCLOTOMIC]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]
Methods Evaluate at roots of unity or average over them [M-AL-UNITY]
Strategy Build a chain of bounds and track equality [S-BOUND]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

2001-D1-P4 — Solution

For this record: Theory: T-AL-CYCLOTOMIC; T-AL-POLYNOMIAL; T-AL-COMPLEX. Methods: M-AL-UNITY. Full learning route

Let $p(x) = (x-1)^k \cdot r(x)$ and $\varepsilon_j = e^{2\pi i \cdot j/q}$ $(j = 1, 2, \dots, q-1)$. As is well-known, the polynomial $x^{q-1} + x^{q-2} + \cdots + x + 1 = (x - \varepsilon_1) \dots (x - \varepsilon_{q-1})$ is irreducible, thus all $\varepsilon_1, \dots, \varepsilon_{q-1}$ are roots of $r(x)$, or none of them.

Suppose that none of $\varepsilon_1, \dots, \varepsilon_{q-1}$ is a root of $r(x)$. Then $\prod_{j=1}^{q-1} r(\varepsilon_j)$ is a rational integer, which is not 0 and

$$(n+1)^{q-1} \geq \left| \prod_{j=1}^{q-1} p(\varepsilon_j) \right| = \left| \prod_{j=1}^{q-1} (1-\varepsilon_j)^k \right| \cdot \left| \prod_{j=1}^{q-1} r(\varepsilon_j) \right| \geq$$

$$\geq \left| \prod_{j=1}^{q-1} (1-\varepsilon_j) \right|^k = (1^{q-1} + 1^{q-2} + \cdots + 1^1 + 1)^k = q^k.$$

This contradicts the condition $\dfrac{q}{\ln q} < \dfrac{k}{\ln(n+1)}$.

2001-D1-P5 — Statement

Day 1, P5

Let A be an $n \times n$ complex matrix such that $A \neq \lambda I$ for all $\lambda \in \mathbb{C}$. Prove that A is similar to a matrix having at most one non-zero entry on the main diagonal.

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Theory and methods for 2001-D1-P5

Theory Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Adapted bases, centralizers and common eigenspaces [T-LA-INVARIANT]; Trace pairing and commutator calculus [T-LA-TRACE]
Methods Choose a basis adapted to the structure [M-LA-BASIS]
Strategy Reduce size while preserving the defining conditions [S-INDUCTION]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2001-D1-P5 — Solution

For this record: Theory: T-LA-SPECTRAL; T-LA-INVARIANT; T-LA-TRACE. Methods: M-LA-BASIS. Full learning route

The statement will be proved by induction on n. For $n = 1$, there is nothing to do. In the case $n = 2$, write $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $b \neq 0$, and $c \neq 0$ or $b = c = 0$ then A is similar to

$$\begin{bmatrix} 1 & 0 \\ a/b & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -a/b & 1 \end{bmatrix} = \begin{bmatrix} 0 & b \\ c - ad/b & a+d \end{bmatrix}$$

or

$$\begin{bmatrix} 1 & -a/c \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & a/c \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & b - ad/c \\ c & a+d \end{bmatrix},$$

respectively. If $b = c = 0$ and $a \neq d$, then A is similar to

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & d-a \\ 0 & d \end{bmatrix},$$

and we can perform the step seen in the case $b \neq 0$ again.

2001-D1-P5 — Solution (continued)

Assume now that $n > 3$ and the problem has been solved for all $n' < n$. Let $A = \begin{bmatrix} A' & * \\ * & \beta \end{bmatrix}$, where A' is $(n-1) \times (n-1)$ matrix. Clearly we may assume that $A' \neq \lambda' I$, so the induction provides a P with, say, $P^{-1}A'P = \begin{bmatrix} 0 & * \\ * & \alpha \end{bmatrix}$. But then the matrix

$$B = \begin{bmatrix} P^{-1} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} A' & * \\ * & \beta \end{bmatrix} \begin{bmatrix} P & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} P^{-1}A'P & * \\ * & \beta \end{bmatrix}$$

is similar to A and its diagonal is $(0, 0, \dots, 0, \alpha, \beta)$. On the other hand, we may also view B as $\begin{bmatrix} 0 & * \\ * & C \end{bmatrix}$, where C is an $(n-1) \times (n-1)$ matrix with diagonal $(0, \dots, 0, \alpha, \beta)$. If the inductive hypothesis is applicable to C, we would have $Q^{-1}CQ = D$, with $D = \begin{bmatrix} 0 & * \\ * & \gamma \end{bmatrix}$ so that finally the matrix

$$E = \begin{bmatrix} 1 & 0 \\ 0 & Q^{-1} \end{bmatrix} \cdot B \cdot \begin{bmatrix} 1 & 0 \\ 0 & Q \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & Q^{-1} \end{bmatrix} \begin{bmatrix} 0 & * \\ * & C \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & Q \end{bmatrix} = \begin{bmatrix} 0 & * \\ * & D \end{bmatrix}$$

is similar to A and its diagonal is $(0, 0, \dots, 0, \gamma)$, as required.

The inductive argument can fail only when $n - 1 = 2$ and the resulting matrix applying P has the form

$$P^{-1}AP = \begin{bmatrix} 0 & a & b \\ c & d & 0 \\ e & 0 & d \end{bmatrix}$$

where $d \neq 0$. The numbers a, b, c, e cannot be 0 at the same time. If, say, $b \neq 0$, A is similar to

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & a & b \\ c & d & 0 \\ e & 0 & d \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -b & a & b \\ c & d & 0 \\ e-b-d & a & b+d \end{bmatrix}.$$

Performing half of the induction step again, the diagonal of the resulting matrix will be $(0, d-b, d+b)$ (the trace is the same) and the induction step can be finished. The cases $a \neq 0$, $c \neq 0$ and $e \neq 0$ are similar.

2001-D1-P6 — Statement

Day 1, P6

Suppose that the differentiable functions $a, b, f, g : \mathbb{R} \to \mathbb{R}$ satisfy

$$f(x) \geq 0,\ f'(x) \geq 0,\ g(x) > 0,\ g'(x) > 0 \quad \text{for all } x \in \mathbb{R},$$

$$\lim_{x\to\infty} a(x) = A > 0, \quad \lim_{x\to\infty} b(x) = B > 0, \quad \lim_{x\to\infty} f(x) = \lim_{x\to\infty} g(x) = \infty,$$

and

$$\frac{f'(x)}{g'(x)} + a(x)\frac{f(x)}{g(x)} = b(x).$$

Prove that

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \frac{B}{A+1}.$$

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Theory and methods for 2001-D1-P6

Theory Differential inequalities and integrating factors [T-AN-ODE]; Scale separation and limit comparisons [T-AN-ASYMPTOTICS]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Methods Design an auxiliary derivative [M-AN-AUXILIARY]; Compare growth through derivatives [M-AN-RATIO]
Strategy Build a chain of bounds and track equality [S-BOUND]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2001-D1-P6 — Solution

For this record: Theory: T-AN-ODE; T-AN-ASYMPTOTICS; T-AN-DERIVATIVE. Methods: M-AN-AUXILIARY; M-AN-RATIO.
Full learning route

Let $0 < \varepsilon < A$ be an arbitrary real number. If x is sufficiently large then $f(x) > 0$, $g(x) > 0$, $|a(x) - A| < \varepsilon$, $|b(x) - B| < \varepsilon$ and

$$B - \varepsilon < b(x) = \frac{f'(x)}{g'(x)} + a(x)\frac{f(x)}{g(x)} < \frac{f'(x)}{g'(x)} + (A+\varepsilon)\frac{f(x)}{g(x)} < \tag{1}$$
$$< \frac{(A+\varepsilon)(A+1)}{A} \cdot \frac{f'(x)(g(x))^A + A \cdot f(x) \cdot (g(x))^{A-1} \cdot g'(x)}{(A+1) \cdot (g(x))^A \cdot g'(x)} =$$
$$= \frac{(A+\varepsilon)(A+1)}{A} \cdot \frac{\left(f(x) \cdot (g(x))^A\right)'}{\left((g(x))^{A+1}\right)'},$$

thus

$$\frac{\left(f(x) \cdot (g(x))^A\right)'}{\left((g(x))^{A+1}\right)'} > \frac{A(B-\varepsilon)}{(A+\varepsilon)(A+1)}. \tag{2}$$

It can be similarly obtained that, for sufficiently large x,

$$\frac{\left(f(x) \cdot (g(x))^A\right)'}{\left((g(x))^{A+1}\right)'} < \frac{A(B+\varepsilon)}{(A-\varepsilon)(A+1)}. \tag{3}$$

From $\varepsilon \to 0$, we have

$$\lim_{x\to\infty} \frac{\left(f(x) \cdot (g(x))^A\right)'}{\left((g(x))^{A+1}\right)'} = \frac{B}{A+1}.$$

By l'Hospital's rule this implies

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \lim_{x\to\infty} \frac{f(x) \cdot (g(x))^A}{(g(x))^{A+1}} = \frac{B}{A+1}.$$

Day 2

2001-D2-P1 — Statement

Day 2, P7

Let $r, s \geq 1$ be integers and $a_0, a_1, \ldots, a_{r-1}, b_0, b_1, \ldots, b_{s-1}$ be real non-negative numbers such that

$$(a_0 + a_1 x + a_2 x^2 + \cdots + a_{r-1} x^{r-1} + x^r)(b_0 + b_1 x + b_2 x^2 + \cdots + b_{s-1} x^{s-1} + x^s) = \\ = 1 + x + x^2 + \cdots + x^{r+s-1} + x^{r+s}.$$

Prove that each a_i and each b_j equals either 0 or 1.

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Theory and methods for 2001-D2-P1

Theory Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]
Methods Compare coefficients after a structural substitution [M-AL-COEFFICIENT]
Strategy Reduce size while preserving the defining conditions [S-INDUCTION]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2001-D2-P1 — Solution

For this record: Theory: T-AL-POLYNOMIAL; T-AL-ROOTS. Methods: M-AL-COEFFICIENT. Full learning route

Multiply the left hand side polynomials. We obtain the following equalities:

$$a_0 b_0 = 1, \quad a_0 b_1 + a_1 b_0 = 1, \quad \ldots$$

Among them one can find equations

$$a_0 + a_1 b_{s-1} + a_2 b_{s-2} + \cdots = 1$$

and

$$b_0 + b_1 a_{r-1} + b_2 a_{r-2} + \cdots = 1.$$

From these equations it follows that $a_0, b_0 \leq 1$. Taking into account that $a_0 b_0 = 1$ we can see that $a_0 = b_0 = 1$.

Now looking at the following equations we notice that all a's must be less than or equal to 1. The same statement holds for the b's. It follows from $a_0 b_1 + a_1 b_0 = 1$ that one of the numbers a_1, b_1 equals 0 while the other one must be 1. Follow by induction.

2001-D2-P2 — Statement

Day 2, P8

Let $a_0 = \sqrt{2}$, $b_0 = 2$, $a_{n+1} = \sqrt{2 - \sqrt{4 - a_n^2}}$, $b_{n+1} = \dfrac{2b_n}{2 + \sqrt{4 + b_n^2}}$.

a) Prove that the sequences (a_n), (b_n) are decreasing and converge to 0.
b) Prove that the sequence $(2^n a_n)$ is increasing, the sequence $(2^n b_n)$ is decreasing and that these two sequences converge to the same limit.

c) Prove that there is a positive constant C such that for all n the following inequality holds: $0 < b_n - a_n < \dfrac{C}{8^n}$.

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Theory and methods for 2001-D2-P2

Theory Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Taylor remainders and local scale [T-AN-TAYLOR]; Trigonometric identities and safe branches [T-AN-TRIG]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Methods Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]; Expand near equality, bound away from it [M-AN-LOCAL]
Strategy Reduce size while preserving the defining conditions [S-INDUCTION]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2001-D2-P2 — Solution

For this record: Theory: T-AN-CONVERGENCE; T-AN-TAYLOR; T-AN-TRIG; T-DS-INDUCTION. Methods: M-AN-RECURRENCE; M-AN-LOCAL. Full learning route

Obviously $a_2 = \sqrt{2-\sqrt{2}} < \sqrt{2}$.
Since the function $f(x) = \sqrt{2-\sqrt{4-x^2}}$ is increasing on the interval $[0,2]$ the inequality $a_1 > a_2$ implies that $a_2 > a_3$. Simple induction ends the proof of monotonicity of (a_n). In the same way we prove that (b_n) decreases (just notice that $g(x) = \dfrac{2x}{2+\sqrt{4+x^2}} = 2\Big/\big(2/x + \sqrt{1+4/x^2}\big)$). It is a matter of simple manipulation to prove that $2f(x) > x$ for all $x \in (0,2)$, this implies that the sequence $(2^n a_n)$ is strictly increasing. The inequality $2g(x) < x$ for $x \in (0,2)$ implies that the sequence $(2^n b_n)$ strictly decreases. By an easy induction one can show that $a_n^2 = \dfrac{4b_n^2}{4+b_n^2}$ for positive integers n. Since the limit of the decreasing sequence $(2^n b_n)$ of positive numbers is finite we have

$$\lim 4^n a_n^2 = \lim \frac{4\cdot 4^n b_n^2}{4+b_n^2} = \lim 4^n b_n^2.$$

We know already that the limits $\lim 2^n a_n$ and $\lim 2^n b_n$ are equal. The first of the two is positive because the sequence $(2^n a_n)$ is strictly increasing. The existence of a number C follows easily from the equalities

$$2^n b_n - 2^n a_n = \left(4^n b_n^2 - \frac{4^{n+1} b_n^2}{4+b_n^2}\right) \Big/ (2^n b_n + 2^n a_n) = \frac{(2^n b_n)^4}{4+b_n^2}\cdot\frac{1}{4^n}\cdot\frac{1}{2^n(b_n+a_n)}$$

and from the existence of positive limits $\lim 2^n b_n$ and $\lim 2^n a_n$.

Remark. The last problem may be solved in a much simpler way by someone who is able to make use of sine and cosine. It is enough to notice that

$$a_n = 2\sin\frac{\pi}{2^{n+1}} \quad \text{and} \quad b_n = 2\tan\frac{\pi}{2^{n+1}}.$$

2001-D2-P3 — Statement

Day 2, P9

Find the maximum number of points on a sphere of radius 1 in $\mathbb{R}^n$ such that the distance between any two of these points is strictly greater than $\sqrt{2}$.

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Theory and methods for 2001-D2-P3

Theory Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Inner products, orthogonality and polarization [T-LA-INNER]
Methods Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]; Strengthen the statement before the induction step [M-DS-INDUCT]

Theory and methods for 2001-D2-P3 (continued)

Strategy Translate geometry into a controlled representation [S-GEOMETRIC]; Reduce size while preserving the defining conditions [S-INDUCTION]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2001-D2-P3 — Solution

For this record: Theory: T-DS-GEOMETRY; T-DS-INDUCTION; T-LA-INNER. Methods: M-DS-GEOMETRIC; M-DS-INDUCT.
Full learning route

The unit sphere in $\mathbb{R}^n$ is defined by

$$S_{n-1}=\left\{(x_1,\dots,x_n)\in\mathbb{R}^n \;\middle|\; \sum_{k=1}^{n}x_k^2=1\right\}.$$

The distance between the points $X=(x_1,\dots,x_n)$ and $Y=(y_1,\dots,y_n)$ is:

$$d^2(X,Y)=\sum_{k=1}^{n}(x_k-y_k)^2.$$

We have

$$\begin{aligned} d(X,Y)>\sqrt{2} &\Leftrightarrow d^2(X,Y)>2\\ &\Leftrightarrow \sum_{k=1}^{n}x_k^2+\sum_{k=1}^{n}y_k^2+2\sum_{k=1}^{n}x_ky_k>2\\ &\Leftrightarrow \sum_{k=1}^{n}x_ky_k<0 \end{aligned}$$

Taking account of the symmetry of the sphere, we can suppose that

$$A_1=(-1,0,\dots,0).$$

For $X=A_1$, $\sum\limits_{k=1}^{n}x_ky_k<0$ implies $y_1>0$, $\forall Y\in M_n$.
Let $X=(x_1,\overline{X})$, $Y=(y_1,\overline{Y})\in M_n\setminus\{A_1\}$, $\overline{X},\overline{Y}\in\mathbb{R}^{n-1}$.
We have

$$\sum_{k=1}^{n}x_ky_k<0\Rightarrow x_1y_1+\sum_{k=1}^{n-1}\overline{x}_k\overline{y}_k<0\Leftrightarrow\sum_{k=1}^{n-1}x'_ky'_k<0,$$

where

$$x'_k=\frac{\overline{x}_k}{\sqrt{\sum\overline{x}_k^2}},\qquad y'_k=\frac{\overline{y}_k}{\sqrt{\sum\overline{y}_k^2}}.$$

therefore

$$(x'_1,\dots,x'_{n-1}),(y'_1,\dots,y'_{n-1})\in S_{n-2}$$

and verifies $\sum\limits_{k=1}^{n}x'_ky'_k<0$.
If a_n is the search number of points in $\mathbb{R}^n$ we obtain $a_n\le 1+a_{n-1}$ and $a_1=2$ implies that $a_n\le n+1$.
We show that $a_n=n+1$, giving an example of a set M_n with $(n+1)$ elements satisfying the conditions of the problem.

$$\begin{aligned} A_1&=(-1,0,0,0,\dots,0,0)\\ A_2&=\left(\tfrac{1}{n},-c_1,0,0,\dots,0,0\right)\\ A_3&=\left(\tfrac{1}{n},\tfrac{1}{n-1}\cdot c_1,-c_2,0,\dots,0,0\right)\\ A_4&=\left(\tfrac{1}{n},\tfrac{1}{n-1}\cdot c_1,\tfrac{1}{n-2}\cdot c_2,-c_3,\dots,0,0\right)\\ &\vdots\\ A_{n-1}&=\left(\tfrac{1}{n},\tfrac{1}{n-1}\cdot c_1,\tfrac{1}{n-2}\cdot c_2,\tfrac{1}{n-3}\cdot c_3,\dots,-c_{n-2},0\right)\\ A_n&=\left(\tfrac{1}{n},\tfrac{1}{n-1}\cdot c_1,\tfrac{1}{n-2}\cdot c_2,\tfrac{1}{n-3}\cdot c_3,\dots,\tfrac{1}{2}\cdot c_{n-2},-c_{n-1}\right)\\ A_{n+1}&=\left(\tfrac{1}{n},\tfrac{1}{n-1}\cdot c_1,\tfrac{1}{n-2}\cdot c_2,\tfrac{1}{n-3}\cdot c_3,\dots,\tfrac{1}{2}\cdot c_{n-2},c_{n-1}\right) \end{aligned}$$

2001-D2-P3 — Solution (continued)

where

$$c_k = \sqrt{\left(1+\frac{1}{n}\right)\left(1-\frac{1}{n-k+1}\right)}, \quad k = \overline{1, n-1}.$$

We have $\sum_{k=1}^{n} x_k y_k = -\frac{1}{n} < 0$ and $\sum_{k=1}^{n} x_k^2 = 1$, $\forall X, Y \in \{A_1, \dots, A_{n+1}\}$.
These points are on the unit sphere in $\mathbb{R}^n$ and the distance between any two points is equal to

$$d = \sqrt{2}\sqrt{1+\frac{1}{n}} > \sqrt{2}.$$

Remark. For $n = 2$ the points form an equilateral triangle in the unit circle; for $n = 3$ the four points from a regular tetrahedron and in $\mathbb{R}^n$ the points from an n dimensional regular simplex.

2001-D2-P4 — Statement

Day 2, P10

Let $A = (a_{k,\ell})_{k,\ell=1,\dots,n}$ be an $n \times n$ complex matrix such that for each $m \in \{1, \dots, n\}$ and $1 \le j_1 < \cdots < j_m \le n$ the determinant of the matrix $(a_{j_k,j_\ell})_{k,\ell=1,\dots,m}$ is zero. Prove that $A^n = 0$ and that there exists a permutation $\sigma \in S_n$ such that the matrix

$$(a_{\sigma(k),\sigma(\ell)})_{k,\ell=1,\dots,n}$$

has all of its nonzero elements above the diagonal.

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Theory and methods for 2001-D2-P4

Theory Determinants, block identities and inverses [T-LA-DETERMINANT]; Nilpotents, projections and involutions [T-LA-NILPOTENT]; Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]
Methods Exploit a compatible block decomposition [M-LA-BLOCK]; Interpret matrix multiplication as a count [M-LA-WALKS]
Strategy Choose an extremal object and force rigidity [S-EXTREMAL]; Make the contradiction quantitative or structural [S-CONTRADICTION]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2001-D2-P4 — Solution

For this record: Theory: T-LA-DETERMINANT; T-LA-NILPOTENT; T-DS-GRAPH. Methods: M-LA-BLOCK; M-LA-WALKS.
Full learning route

We will only prove (2), since it implies (1). Consider a directed graph G with n vertices $V_1, \dots, V_n$ and a directed edge from V_k to V_ℓ when $a_{k,\ell} \neq 0$. We shall prove that it is acyclic.
Assume that there exists a cycle and take one of minimum length m. Let $j_1 < \cdots < j_m$ be the vertices the cycle goes through and let $\sigma_0 \in S_n$ be a permutation such that $a_{j_k,j_{\sigma_0(k)}} \neq 0$ for $k = 1, \dots, m$. Observe that for any other $\sigma \in S_n$ we have $a_{j_k,j_{\sigma(k)}} = 0$ for some $k \in \{1, \dots, m\}$, otherwise we would obtain a different cycle through the same set of vertices and, consequently, a shorter cycle. Finally

$$0 = \det(a_{j_k,j_\ell})_{k,\ell=1,\dots,m} = (-1)^{\operatorname{sign}\sigma_0} \prod_{k=1}^{m} a_{j_k,j_{\sigma_0(k)}} + \sum_{\sigma \neq \sigma_0} (-1)^{\operatorname{sign}\sigma} \prod_{k=1}^{m} a_{j_k,j_{\sigma(k)}} \neq 0,$$

which is a contradiction.

Since G is acyclic there exists a topological ordering i.e. a permutation $\sigma \in S_n$ such that $k < \ell$ whenever there is an edge from $V_{\sigma(k)}$ to $V_{\sigma(\ell)}$. It is easy to see that this permutation solves the problem.

2001-D2-P5 — Statement

Day 2, P11

Let $\mathbb{R}$ be the set of real numbers. Prove that there is no function $f : \mathbb{R} \to \mathbb{R}$ with $f(0) > 0$, and such that

$$f(x+y) \geq f(x) + yf(f(x)) \quad \text{for all } x, y \in \mathbb{R}.$$

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Theory and methods for 2001-D2-P5

Theory Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Methods Force structure by admissible substitutions [M-DS-FUNCTIONAL]; Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]
Strategy Make the contradiction quantitative or structural [S-CONTRADICTION]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2001-D2-P5 — Solution

For this record: Theory: T-DS-FUNCTIONAL; T-AN-CONVERGENCE. Methods: M-DS-FUNCTIONAL; M-DS-ITERATE. Full learning route

Suppose that there exists a function satisfying the inequality. If $f(f(x)) \leq 0$ for all x, then f is a decreasing function in view of the inequalities $f(x+y) \geq f(x)+yf(f(x)) \geq f(x)$ for any $y \leq 0$. Since $f(0) > 0 \geq f(f(x))$, it implies $f(x) > 0$ for all x, which is a contradiction. Hence there is a z such that $f(f(z)) > 0$. Then the inequality $f(z+x) \geq f(z)+xf(f(z))$ shows that $\lim\limits_{x\to\infty} f(x) = +\infty$ and therefore $\lim\limits_{x\to\infty} f(f(x)) = +\infty$. In particular, there exist $x, y > 0$ such that $f(x) \geq 0$, $f(f(x)) > 1$, $y \geq \dfrac{x+1}{f(f(x))-1}$ and $f(f(x+y+1)) \geq 0$. Then $f(x+y) \geq f(x) + yf(f(x)) \geq x+y+1$ and hence

$$\begin{aligned} f(f(x+y)) &\geq f(x+y+1) + (f(x+y)-(x+y+1))f(f(x+y+1)) \geq \\ &\geq f(x+y+1) \geq f(x+y) + f(f(x+y)) \geq \\ &\geq f(x) + yf(f(x)) + f(f(x+y)) > f(f(x+y)). \end{aligned}$$

This contradiction completes the solution of the problem.

2001-D2-P6 — Statement

Day 2, P12

For each positive integer n, let $f_n(\vartheta) = \sin\vartheta \cdot \sin(2\vartheta) \cdot \sin(4\vartheta) \cdots \sin(2^n\vartheta)$. For all real ϑ and all n, prove that

$$|f_n(\vartheta)| \leq \frac{2}{\sqrt{3}}|f_n(\pi/3)|.$$

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Theory and methods for 2001-D2-P6

Theory Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Trigonometric identities and safe branches [T-AN-TRIG]
Methods Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]; Exploit identities before estimating oscillations [M-AN-TRIG]
Strategy Build a chain of bounds and track equality [S-BOUND]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2001-D2-P6 — Solution

For this record: Theory: T-AN-INEQUALITIES; T-AN-TRIG. Methods: M-AN-WEIGHTS; M-AN-TRIG. Full learning route

We prove that $g(\vartheta) = |\sin\vartheta||\sin(2\vartheta)|^{1/2}$ attains its maximum value $(\sqrt{3}/2)^{3/2}$ at points $2k\pi/3$ (where k is a positive integer). This can be seen by using derivatives or a classical bound like

$$g(\vartheta) = |\sin\vartheta||\sin(2\vartheta)|^{1/2} = \frac{\sqrt{2}}{\sqrt[4]{3}}\left(\sqrt[4]{|\sin\vartheta|\cdot|\sin\vartheta|\cdot|\sin\vartheta|\cdot|\sqrt{3}\cos\vartheta|}\right)^2$$
$$\le \frac{\sqrt{2}}{\sqrt[4]{3}}\cdot\frac{3\sin^2\vartheta + 3\cos^2\vartheta}{4} = \left(\frac{\sqrt{3}}{2}\right)^{3/2}.$$

Hence

$$\left|\frac{f_n(\vartheta)}{f_n(\pi/3)}\right| = \left|\frac{g(\vartheta)\cdot g(2\vartheta)^{1/2}\cdot g(4\vartheta)^{3/4}\cdots g(2^{n-1}\vartheta)^E}{g(\pi/3)\cdot g(2\pi/3)^{1/2}\cdot g(4\pi/3)^{3/4}\cdots g(2^{n-1}\pi/3)^E}\right|\cdot\left|\frac{\sin(2^n\vartheta)}{\sin(2^n\pi/3)}\right|^{1-E/2}$$
$$\le \left|\frac{\sin(2^n\vartheta)}{\sin(2^n\pi/3)}\right|^{1-E/2} \le \left(\frac{1}{\sqrt{3}/2}\right)^{1-E/2} \le \frac{2}{\sqrt{3}}.$$

where $E = \frac{2}{3}(1-(-1/2)^n)$. This is exactly the bound we had to prove.

3.9 2002

Day 1

2002-D1-P1 — Statement

Day 1, P1

A standard parabola is the graph of a quadratic polynomial $y = x^2 + ax + b$ with leading coefficient 1. Three standard parabolas with vertices V_1, V_2, V_3 intersect pairwise at points A_1, A_2, A_3. Let $A \mapsto s(A)$ be the reflection of the plane with respect to the x axis.

Prove that standard parabolas with vertices $s(A_1)$, $s(A_2)$, $s(A_3)$ intersect pairwise at the points $s(V_1)$, $s(V_2)$, $s(V_3)$.

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Theory and methods for 2002-D1-P1

Theory Conics, reflections, and inversion [T-DS-TRANSFORMS]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]
Methods Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Strategy Translate geometry into a controlled representation [S-GEOMETRIC]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2002-D1-P1 — Solution

For this record: Theory: T-DS-TRANSFORMS; T-DS-GEOMETRY. Methods: M-DS-GEOMETRIC. Full learning route

First we show that the standard parabola with vertex V contains point A if and only if the standard parabola with vertex $s(A)$ contains point $s(V)$.
Let $A = (a,b)$ and $V = (v,w)$. The equation of the standard parabola with vertex $V = (v,w)$ is $y = (x-v)^2 + w$, so it contains point A if and only if $b = (a-v)^2 + w$. Similarly, the equation of the

2002-D1-P1 — Solution (continued)

parabola with vertex $s(A) = (a, -b)$ is $y = (x-a)^2 - b$; it contains point $s(V) = (v, -w)$ if and only if $-w = (v-a)^2 - b$. The two conditions are equivalent.

Now assume that the standard parabolas with vertices V_1 and V_2, V_1 and V_3, V_2 and V_3 intersect each other at points A_3, A_2, A_1, respectively. Then, by the statement above, the standard parabolas with vertices $s(A_1)$ and $s(A_2)$, $s(A_1)$ and $s(A_3)$, $s(A_2)$ and $s(A_3)$ intersect each other at points V_3, V_2, V_1, respectively, because they contain these points.

2002-D1-P2 — Statement

Day 1, P2

Does there exist a continuously differentiable function $f : \mathbb{R} \to \mathbb{R}$ such that for every $x \in \mathbb{R}$ we have $f(x) > 0$ and $f'(x) = f(f(x))$?

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Theory and methods for 2002-D1-P2

Theory Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Methods Force structure by admissible substitutions [M-DS-FUNCTIONAL]
Strategy Control the limiting operation before taking the limit [S-ANALYTIC]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

2002-D1-P2 — Solution

For this record: Theory: T-DS-FUNCTIONAL; T-AN-DERIVATIVE. Methods: M-DS-FUNCTIONAL. Full learning route

Assume that there exists such a function. Since $f'(x) = f(f(x)) > 0$, the function is strictly monotone increasing.
By the monotonity, $f(x) > 0$ implies $f(f(x)) > f(0)$ for all x. Thus, $f(0)$ is a lower bound for $f'(x)$, and for all $x < 0$ we have $f(x) < f(0) + x \cdot f(0) = (1+x)f(0)$. Hence, if $x \le -1$ then $f(x) \le 0$, contradicting the property $f(x) > 0$.

So such function does not exist.

2002-D1-P3 — Statement

Day 1, P3

Let n be a positive integer and let

$$a_k = \frac{1}{\binom{n}{k}}, \qquad b_k = 2^{k-n}, \qquad \text{for } k = 1, 2, \ldots, n.$$

Show that

$$\frac{a_1 - b_1}{1} + \frac{a_2 - b_2}{2} + \cdots + \frac{a_n - b_n}{n} = 0. \tag{1}$$

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Theory and methods for 2002-D1-P3

Theory Linear recurrences and generating functions [T-AL-RECURRENCE]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Find a recurrence that reveals the invariant [M-AL-RECURRENCE]
Strategy Reduce size while preserving the defining conditions [S-INDUCTION]

Statement Recorded solutions Catalog

2002-D1-P3 — Solution

For this record: Theory: T-AL-RECURRENCE; T-DS-INDUCTION; T-DS-COUNTING. Methods: M-AL-RECURRENCE. Full learning route

Since $k\binom{n}{k} = n\binom{n-1}{k-1}$ for all $k \geq 1$, (1) is equivalent to

$$\frac{2^n}{n}\left[\frac{1}{\binom{n-1}{0}} + \frac{1}{\binom{n-1}{1}} + \cdots + \frac{1}{\binom{n-1}{n-1}}\right] = \frac{2^1}{1} + \frac{2^2}{2} + \cdots + \frac{2^n}{n}. \tag{2}$$

We prove (2) by induction. For $n = 1$, both sides are equal to 2.

Assume that (2) holds for some n. Let

$$x_n = \frac{2^n}{n}\left[\frac{1}{\binom{n-1}{0}} + \frac{1}{\binom{n-1}{1}} + \cdots + \frac{1}{\binom{n-1}{n-1}}\right];$$

then

$$x_{n+1} = \frac{2^{n+1}}{n+1}\sum_{k=0}^{n}\frac{1}{\binom{n}{k}} = \frac{2^n}{n+1}\left(1 + \sum_{k=0}^{n-1}\left(\frac{1}{\binom{n}{k}} + \frac{1}{\binom{n}{k+1}}\right) + 1\right) =$$
$$= \frac{2^n}{n+1}\sum_{k=0}^{n-1}\frac{\frac{n-k}{n} + \frac{k+1}{n}}{\binom{n-1}{k}} + \frac{2^{n+1}}{n+1} = \frac{2^n}{n}\sum_{k=0}^{n-1}\frac{1}{\binom{n-1}{k}} + \frac{2^{n+1}}{n+1} = x_n + \frac{2^{n+1}}{n+1}.$$

This implies (2) for $n+1$.

2002-D1-P4 — Statement

Day 1, P4

Let $f : [a,b] \to [a,b]$ be a continuous function and let $p \in [a,b]$. Define $p_0 = p$ and $p_{n+1} = f(p_n)$ for $n = 0, 1, 2, \ldots$. Suppose that the set $T_p = \{p_n : n = 0, 1, 2, \ldots\}$ is closed, i.e., if $x \notin T_p$ then there is a $\delta > 0$ such that for all $x' \in T_p$ we have $|x' - x| \geq \delta$. Show that T_p has finitely many elements.

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Theory and methods for 2002-D1-P4

Theory Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]; Continuity, compactness, and order [T-AN-CONTINUITY]; Countability, accumulation, and Cantor constructions [T-AN-SETS]; Baire class one and controlled discontinuities [T-AN-BAIRE]
Methods Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]
Strategy Control the limiting operation before taking the limit [S-ANALYTIC]; Make the contradiction quantitative or structural [S-CONTRADICTION]

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2002-D1-P4 — Solution

For this record: Theory: T-DS-DYNAMICS; T-AN-CONTINUITY; T-AN-SETS; T-AN-BAIRE. Methods: M-DS-ITERATE. Full learning route

If for some $n > m$ the equality $p_m = p_n$ holds then T_p is a finite set. Thus we can assume that all points $p_0, p_1, \ldots$ are distinct. There is a convergent subsequence p_{n_k} and its limit q is in T_p. Since f is continuous $p_{n_k+1} = f(p_{n_k}) \to f(q)$, so all, except for finitely many, points p_n are accumulation points of T_p. Hence we may assume that all of them are accumulation points of T_p. Let $d = \sup\{|p_m - p_n| : m, n \geq 0\}$. Let δ_n be positive numbers such that $\sum_{n=0}^{\infty}\delta_n < \frac{d}{2}$. Let I_n be an interval of length less than δ_n centered at p_n such that there are there are infinitely many k's such that $p_k \notin \bigcup_{j=0}^{n} I_j$, this can be done by induction. Let $n_0 = 0$ and n_{m+1} be the smallest integer $k > n_m$ such that $p_k \notin \bigcup_{j=0}^{n_m} I_j$.

2002-D1-P4 — Solution (continued)

Since T_p is closed the limit of the subsequence (p_{n_m}) must be in T_p but it is impossible because of the definition of I_n's, of course if the sequence (p_{n_m}) is not convergent we may replace it with its convergent subsequence. The proof is finished.

Remark. If $T_p = \{p_1, p_2, \dots\}$ and each p_n is an accumulation point of T_p, then T_p is the countable union of nowhere dense sets (i.e. the single-element sets $\{p_n\}$). If T_p is closed then this contradicts the Baire Category Theorem.

2002-D1-P5 — Statement

Day 1, P5

Prove or disprove the following statements:
(a) There exists a monotone function $f : [0,1] \to [0,1]$ such that for each $y \in [0,1]$ the equation $f(x) = y$ has uncountably many solutions x.

(b) There exists a continuously differentiable function $f : [0,1] \to [0,1]$ such that for each $y \in [0,1]$ the equation $f(x) = y$ has uncountably many solutions x.

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Theory and methods for 2002-D1-P5

Theory Countability, accumulation, and Cantor constructions [T-AN-SETS]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Interval coverings and level-set integration [T-AN-COVERING]
Methods Use a dense set, an enumeration, or a category argument [M-AN-TOPOLOGY]; Turn local bad intervals into a global measure bound [M-AN-COVER]
Strategy Make the contradiction quantitative or structural [S-CONTRADICTION]; Choose the objects and count them without ambiguity [S-COUNTING]

Statement Recorded solutions Catalog

2002-D1-P5 — Solution

For this record: Theory: T-AN-SETS; T-AN-DERIVATIVE; T-AN-COVERING. Methods: M-AN-TOPOLOGY; M-AN-COVER.
Full learning route

a. It does not exist. For each y the set $\{x : y = f(x)\}$ is either empty or consists of 1 point or is an interval. These sets are pairwise disjoint, so there are at most countably many of the third type.
b. Let f be such a map. Then for each value y of this map there is an x_0 such that $y = f(x)$ and $f'(x) = 0$, because an uncountable set $\{x : y = f(x)\}$ contains an accumulation point x_0 and clearly $f'(x_0) = 0$. For every $\varepsilon > 0$ and every x_0 such that $f'(x_0) = 0$ there exists an open interval I_{x_0} such that if $x \in I_{x_0}$ then $|f'(x)| < \varepsilon$. The union of all these intervals I_{x_0} may be written as a union of pairwise disjoint open intervals J_n. The image of each J_n is an interval (or a point) of length $< \varepsilon \cdot \text{length}(J_n)$ due to Lagrange Mean Value Theorem. Thus the image of the interval $[0,1]$ may be covered with the intervals such that the sum of their lengths is $\varepsilon \cdot 1 = \varepsilon$. This is not possible for $\varepsilon < 1$.
Remarks. 1. The proof of part b is essentially the proof of the easy part of A. Sard's theorem about measure of the set of critical values of a smooth map.

2. If only continuity is required, there exists such a function, e.g. the first co-ordinate of the very well known Peano curve which is a continuous map from an interval onto a square.

2002-D1-P6 — Statement

Day 1, P6

For an $n \times n$ matrix M with real entries let $\|M\| = \sup_{x \in \mathbb{R}^n \setminus \{0\}} \frac{\|Mx\|_2}{\|x\|_2}$, where $\|\cdot\|_2$ denotes the Euclidean norm on $\mathbb{R}^n$. Assume that an $n \times n$ matrix A with real entries satisfies $\|A^k - A^{k-1}\| \leq \frac{1}{2002k}$ for all positive integers k. Prove that $\|A^k\| \leq 2002$ for all positive integers k.

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Theory and methods for 2002-D1-P6

Theory Operator norms and convergence of matrix powers [T-LA-NORM]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Methods Combine norm bounds with algebraic decay [M-LA-NORM]
Strategy Control the limiting operation before taking the limit [S-ANALYTIC]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2002-D1-P6 — Solution

For this record: Theory: T-LA-NORM; T-AN-CONVERGENCE. Methods: M-LA-NORM. Full learning route

Lemma 1. Let $(a_n)_{n\geq 0}$ be a sequence of non-negative numbers such that $a_{2k} - a_{2k+1} \leq a_k^2$, $a_{2k+1} - a_{2k+2} \leq a_k a_{k+1}$ for any $k \geq 0$ and $\limsup na_n < 1/4$. Then $\limsup \sqrt[n]{a_n} < 1$.
Proof. Let $c_l = \sup_{n \geq 2^l}(n+1)a_n$ for $l \geq 0$. We will show that $c_{l+1} \leq 4c_l^2$. Indeed, for any integer $n \geq 2^{l+1}$ there exists an integer $k \geq 2^l$ such that $n = 2k$ or $n = 2k+1$. In the first case there is $a_{2k} - a_{2k+1} \leq a_k^2 \leq \frac{c_l^2}{(k+1)^2} \leq \frac{4c_l^2}{(2k+1)(2k+2)}$, whereas in the second case there is $a_{2k+1} - a_{2k+2} \leq a_k a_{k+1} \leq \frac{c_l^2}{(k+1)(k+2)} \leq \frac{4c_l^2}{(2k+2)(2k+3)}$.
Hence a sequence $\left(a_n - \frac{4c_l^2}{n+1}\right)_{n \geq 2^{l+1}}$ is non-decreasing and its terms are non-positive since it converges to zero. Therefore $a_n \leq \frac{4c_l^2}{n+1}$ for $n \geq 2^{l+1}$, meaning that $c_{l+1} \leq 4c_l^2$. This implies that a sequence $((4c_l)^{2^{-l}})_{l \geq 0}$ is non-increasing and therefore, for all sufficiently large l, bounded from above by some number $q \in (0,1)$ since all its terms except finitely many are less than 1. Hence $c_l \leq q^{2^l}$ for l large enough. For any n between 2^l and 2^{l+1} there is $a_n \leq \frac{c_l}{n+1} \leq q^{2^l} \leq (\sqrt{q})^n$ yielding $\limsup \sqrt[n]{a_n} \leq \sqrt{q} < 1$, yielding $\limsup \sqrt[n]{a_n} \leq \sqrt{q} < 1$,
which ends the proof.
Lemma 2. Let T be a linear map from $\mathbb{R}^n$ into itself. Assume that $\limsup n\|T^{n+1} - T^n\| < 1/4$. Then $\limsup \|T^{n+1} - T^n\|^{1/n} < 1$. In particular T^n converges in the operator norm and T is power bounded.
Proof. Put $a_n = \|T^{n+1} - T^n\|$. Observe that

$$T^{k+m+1} - T^{k+m} = (T^{k+m+2} - T^{k+m+1}) - (T^{k+1} - T^k)(T^{m+1} - T^m)$$

implying that $a_{k+m} \leq a_{k+m+1} + a_k a_m$. Therefore the sequence $(a_m)_{m \geq 0}$ satisfies assumptions of Lemma 1 and the assertion of Lemma 2 follows.
To recover the numerical bound in the problem, put $c = 1/2002$ and $a_j = \|A^{j+1} - A^j\|$ for $j \geq 0$. The hypothesis gives $a_j \leq c/(j+1)$. For the quantities c_l in Lemma 1, $c_0 \leq c$ and $c_{l+1} \leq 4c_l^2$, hence $c_l \leq (4c)^{2^l}/4$. Therefore

$$\sum_{j=1}^{\infty} a_j \leq \sum_{l=0}^{\infty} c_l \leq \frac{1}{4}\sum_{l=0}^{\infty}(4c)^{l+1} = \frac{c}{1-4c} = \frac{1}{1998}.$$

Here each dyadic block $2^l \leq j < 2^{l+1}$ contributes at most c_l. For every $k \geq 1$, telescoping from $A^0 = I$ now gives

$$\|A^k\| \leq 1 + \sum_{j=0}^{k-1} a_j \leq 1 + \frac{1}{2002} + \frac{1}{1998} < 2 < 2002.$$

2002-D1-P6 — Solution (continued)

Remarks. 1. The theorem proved above holds in the case of a bounded linear operator T on a Banach space X; X need not be finite dimensional.
2. The constant $1/4$ in Lemma 1 cannot be replaced by any greater number since a sequence $a_n = \frac{1}{4n}$ satisfies the inequality $a_{k+m} - a_{k+m+1} \le a_k a_m$ for any positive integers k and m whereas it does not have exponential decay.
3. The constant $1/4$ in Lemma 2 cannot be replaced by any number greater that $1/e$. Consider an operator $(Tf)(x) = xf(x)$ on $L_2([0,1])$. One can easily check that $\lim_{n\to\infty} n\|T^{n+1} - T^n\| = 1/e$, whereas T^n does not converge in the operator norm. The original 2002 solution described the question whether, in general, $\limsup n\|T^{n+1} - T^n\| < \infty$ implies power boundedness as open; no claim about its present status is made here.

Remark The problem was incorrectly stated during the competition: instead of the inequality $\|A^k - A^{k-1}\| \le \frac{1}{2002k}$, the inequality $\|A^k - A^{k-1}\| \le \frac{1}{2002n}$ was assumed. If $A = \begin{pmatrix} 1 & \varepsilon \\ 0 & 1 \end{pmatrix}$ then $A^k = \begin{pmatrix} 1 & k\varepsilon \\ 0 & 1 \end{pmatrix}$. Therefore $A^k - A^{k-1} = \begin{pmatrix} 0 & \varepsilon \\ 0 & 0 \end{pmatrix}$, so for sufficiently small ε the condition is satisfied although the sequence $(\|A^k\|)$ is clearly unbounded.

Day 2

2002-D2-P1 — Statement

Day 2, P7

Compute the determinant of the $n \times n$ matrix $A = [a_{ij}]$,

$$a_{ij} = \begin{cases} (-1)^{|i-j|}, & \text{if } i \neq j, \\ 2, & \text{if } i = j. \end{cases}$$

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Theory and methods for 2002-D2-P1

Theory Determinants, block identities and inverses [T-LA-DETERMINANT]
Methods Eliminate systematically and track the invariant [M-LA-ELIM]
Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2002-D2-P1 — Solution

For this record: Theory: T-LA-DETERMINANT. Methods: M-LA-ELIM. Full learning route

Adding the second row to the first one, then adding the third row to the second one, ..., adding the nth row to the $(n-1)$th, the determinant does not change and we have

$$\det(A) = \begin{vmatrix} 2 & -1 & +1 & \dots & \pm 1 & \mp 1 \\ -1 & 2 & -1 & \dots & \mp 1 & \pm 1 \\ +1 & -1 & 2 & \dots & \pm 1 & \mp 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \mp 1 & \pm 1 & \mp 1 & \dots & 2 & -1 \\ \pm 1 & \mp 1 & \pm 1 & \dots & -1 & 2 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 0 & 0 & \dots & 0 & 0 \\ 0 & 1 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & 1 & 1 \\ \pm 1 & \mp 1 & \pm 1 & \mp 1 & \dots & -1 & 2 \end{vmatrix}.$$

2002-D2-P1 — Solution (continued)

Now subtract the first column from the second, then subtract the resulting second column from the third, ..., and at last, subtract the $(n-1)$th column from the nth column. This way we have

$$\det(A) = \begin{vmatrix} 1 & 0 & 0 & \dots & 0 & 0 \\ 0 & 1 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & 0 \\ 0 & 0 & 0 & \dots & 0 & n+1 \end{vmatrix} = n+1.$$

2002-D2-P2 — Statement

Day 2, P8

Two hundred students participated in a mathematical contest. They had 6 problems to solve. It is known that each problem was correctly solved by at least 120 participants. Prove that there must be two participants such that every problem was solved by at least one of these two students.

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Theory and methods for 2002-D2-P2

Theory Finite counting and incidence identities [T-DS-COUNTING]
Methods Count incidences with their exact multiplicity [M-DS-DOUBLECOUNT]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2002-D2-P2 — Solution

For this record: Theory: T-DS-COUNTING. Methods: M-DS-DOUBLECOUNT. Full learning route

For each pair of students, consider the set of those problems which was not solved by them. There exist $\binom{200}{2} = 19900$ sets; we have to prove that at least one set is empty.
For each problem, there are at most 80 students who did not solve it. From these students at most $\binom{80}{2} = 3160$ pairs can be selected, so the problem can belong to at most 3160 sets. The 6 problems together can belong to at most $6 \cdot 3160 = 18960$ sets.

Hence, at least $19900 - 18960 = 940$ sets must be empty.

2002-D2-P3 — Statement

Day 2, P9

For each $n \geq 1$ let

$$a_n = \sum_{k=0}^{\infty} \frac{k^n}{k!}, \qquad b_n = \sum_{k=0}^{\infty} (-1)^k \frac{k^n}{k!}.$$

Show that $a_n \cdot b_n$ is an integer.

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Theory and methods for 2002-D2-P3

Theory Linear recurrences and generating functions [T-AL-RECURRENCE]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Methods Find a recurrence that reveals the invariant [M-AL-RECURRENCE]
Strategy Reduce size while preserving the defining conditions [S-INDUCTION]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2002-D2-P3 — Solution

For this record: Theory: T-AL-RECURRENCE; T-AN-CONVERGENCE; T-DS-INDUCTION. Methods: M-AL-RECURRENCE. Full learning route

We prove by induction on n that a_n/e and $b_n e$ are integers, we prove this for $n=0$ as well. (For $n=0$, the term 0^0 in the definition of the sequences must be replaced by 1.)
From the power series of e^x, $a_0 = e^1 = e$ and $b_0 = e^{-1} = 1/e$.

Suppose that for some $n \geq 0$, $a_0, a_1, \ldots, a_n$ and $b_0, b_1, \ldots, b_n$ are all multipliers of e and $1/e$, respectively. Then, by the binomial theorem,

$$a_{n+1} = \sum_{k=0}^{\infty} \frac{(k+1)^{n+1}}{(k+1)!} = \sum_{k=0}^{\infty} \frac{(k+1)^n}{k!} = \sum_{k=0}^{\infty} \sum_{m=0}^{n} \binom{n}{m} \frac{k^m}{k!} = \sum_{m=0}^{n} \binom{n}{m} \sum_{k=0}^{\infty} \frac{k^m}{k!} = \sum_{m=0}^{n} \binom{n}{m} a_m$$

and similarly

$$b_{n+1} = \sum_{k=0}^{\infty} (-1)^{k+1} \frac{(k+1)^{n+1}}{(k+1)!} = -\sum_{k=0}^{\infty} (-1)^k \frac{(k+1)^n}{k!} =$$
$$= -\sum_{k=0}^{\infty} (-1)^k \sum_{m=0}^{n} \binom{n}{m} \frac{k^m}{k!} = -\sum_{m=0}^{n} \binom{n}{m} \sum_{k=0}^{\infty} (-1)^k \frac{k^m}{k!} = -\sum_{m=0}^{n} \binom{n}{m} b_m.$$

The numbers a_{n+1} and b_{n+1} are expressed as linear combinations of the previous elements with integer coefficients which finishes the proof.

2002-D2-P4 — Statement

Day 2, P10

In the tetrahedron $OABC$, let $\angle BOC = \alpha$, $\angle COA = \beta$ and $\angle AOB = \gamma$. Let σ be the angle between the faces OAB and OAC, and let τ be the angle between the faces OBA and OBC. Prove that

$$\gamma > \beta \cdot \cos\sigma + \alpha \cdot \cos\tau.$$

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Theory and methods for 2002-D2-P4

Theory Spherical distance and signed dihedral derivatives [T-DS-SPHERICAL]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Inner products, orthogonality and polarization [T-LA-INNER]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Methods Turn a spherical comparison into an endpoint integral [M-DS-SPHERICAL]; Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Strategy Translate geometry into a controlled representation [S-GEOMETRIC]; Control the limiting operation before taking the limit [S-ANALYTIC]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2002-D2-P4 — Solution

For this record: Theory: T-DS-SPHERICAL; T-DS-GEOMETRY; T-LA-INNER; T-AN-DERIVATIVE; T-AN-INTEGRAL. Methods: M-DS-SPHERICAL; M-DS-GEOMETRIC. Full learning route

Editorially corrected proof. Normalize $OA = OB = OC = 1$ and let $X(t)$, $0 \leq t \leq \gamma$, be the unit-speed minor great-circle arc from A to B. Define $f(t) = \arccos\langle C, X(t)\rangle$, the spherical distance from C to $X(t)$. Since the tetrahedron is nondegenerate, C does not lie in the plane OAB. Consequently $0 < f(t) < \pi$ and $|f'(t)| < 1$ throughout the arc. The endpoint values and derivatives are

$$f(0) = \beta, \qquad f(\gamma) = \alpha, \qquad f'(0) = -\cos\sigma, \qquad f'(\gamma) = \cos\tau.$$

2002-D2-P4 — Solution (continued)

For completeness, the derivative identities follow by taking the scalar product of the unit tangent to AB with the unit tangents to AC and BC at the corresponding endpoints. The strict bound $|f'| < 1$ follows because equality would place C in the great-circle plane OAB.
Since $X''(t) = -X(t)$, twice differentiating $\cos f(t) = \langle C, X(t)\rangle$ gives

$$f''(t) = \cot f(t)(1 - f'(t)^2).$$

For $0 < u < \pi$ we have $u \cot u < 1$: indeed, $\sin u - u\cos u > 0$, since its derivative is $u \sin u > 0$ and its limit at 0 is 0. Therefore

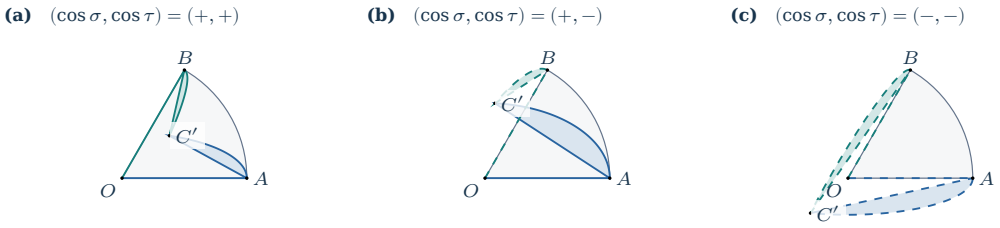
$$(ff')' = f'^2 + f \cot f\,(1 - f'^2) < 1.$$

Integrating from 0 to γ now yields

$$\alpha\cos\tau + \beta\cos\sigma = f(\gamma)f'(\gamma) - f(0)f'(0) < \gamma,$$

which proves the required strict inequality.

Geometric orientation. The signed projection cases below retain the geometric interpretation of the source argument. They illustrate signs; the rigorous inequality is established by the calculation above.



Blue: projection of AOC, signed area $\frac{1}{2}\beta\cos\sigma$. Teal: projection of COB, signed area $\frac{1}{2}\alpha\cos\tau$.
Solid/dashed outlines indicate positive/negative signed area. Representative projections; the $(-,+)$ case is symmetric.

2002-D2-P5 — Statement

Day 2, P11

Let A be an $n \times n$ matrix with complex entries and suppose that $n > 1$. Prove that

$$A\overline{A} = I_n \iff \exists S \in GL_n(\mathbb{C}) \text{ such that } A = S\overline{S}^{-1}.$$

(If $A = [a_{ij}]$ then $\overline{A} = [\overline{a_{ij}}]$, where $\overline{a_{ij}}$ is the complex conjugate of a_{ij}; $GL_n(\mathbb{C})$ denotes the set of all $n \times n$ invertible matrices with complex entries, and I_n is the identity matrix.)

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Theory and methods for 2002-D2-P5

Theory Determinants, block identities and inverses [T-LA-DETERMINANT]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]
Methods Compare coefficients in a matrix polynomial [M-LA-PARAMETER]; Complete the equation to a product equal to identity [M-LA-INVERSE]
Strategy Build a witness and verify every constraint [S-CONSTRUCTION]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2002-D2-P5 — Solution

For this record: Theory: T-LA-DETERMINANT; T-LA-SPECTRAL. Methods: M-LA-PARAMETER; M-LA-INVERSE. Full learning route

The direction $\Leftarrow$ is trivial, since if $A = S\overline{S}^{-1}$, then

$$A\overline{A} = S\overline{S}^{-1} \cdot \overline{S}S^{-1} = I_n.$$

For the direction $\Rightarrow$, we must prove that there exists an invertible matrix S such that $A\overline{S} = S$.

Let w be an arbitrary complex number which is not 0. Choosing $S = wA + \overline{w}I_n$, we have $A\overline{S} = A(\overline{w}\,\overline{A} + wI_n) = \overline{w}I_n + wA = S$. If S is singular, then $\frac{1}{w}S = A - (-\overline{w}/w)\,I_n$ is singular as well, so $-\overline{w}/w$ is an eigenvalue of A. Since A has finitely many eigenvalues and $-\overline{w}/w$ can be any complex number on the unit circle, there exist such w that S is invertible.

2002-D2-P6 — Statement

Day 2, P12

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function whose gradient $\nabla f = \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}\right)$ exists at every point of $\mathbb{R}^n$ and satisfies the condition

$$\exists L > 0\ \forall x_1, x_2 \in \mathbb{R}^n \quad \|\nabla f(x_1) - \nabla f(x_2)\| \le L\|x_1 - x_2\|.$$

Prove that

$$\forall x_1, x_2 \in \mathbb{R}^n \quad \|\nabla f(x_1) - \nabla f(x_2)\|^2 \le L\langle \nabla f(x_1) - \nabla f(x_2),\ x_1 - x_2\rangle. \tag{1}$$

In this formula $\langle a, b\rangle$ denotes the scalar product of the vectors a and b.

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Theory and methods for 2002-D2-P6

Theory Lipschitz derivatives and convex smoothness [T-AN-SMOOTH]; Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Inner products, orthogonality and polarization [T-LA-INNER]
Methods Choose the correct convexity comparison [M-AN-CONVEX]
Strategy Expose the structure before calculating [S-STRUCTURE]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2002-D2-P6 — Solution

For this record: Theory: T-AN-SMOOTH; T-AN-CONVEXITY; T-AN-INTEGRAL; T-LA-INNER. Methods: M-AN-CONVEX. Full learning route

Let $g(x) = f(x) - f(x_1) - \langle \nabla f(x_1), x - x_1\rangle$. It is obvious that g has the same properties. Moreover, $g(x_1) = \nabla g(x_1) = 0$ and, due to convexity, g has 0 as the absolute minimum at x_1. Next we prove that

$$g(x_2) \ge \frac{1}{2L}\|\nabla g(x_2)\|^2. \tag{2}$$

2002-D2-P6 — Solution (continued)

Let $y_0 = x_2 - \frac{1}{L}\nabla g(x_2)$

and $y(t) = y_0 + t(x_2 - y_0)$. Then

$$g(x_2) = g(y_0) + \int_0^1 \langle \nabla g(y(t)),\ x_2 - y_0\rangle\, dt =$$

$$= g(y_0) + \langle \nabla g(x_2),\ x_2 - y_0\rangle - \int_0^1 \langle \nabla g(x_2) - \nabla g(y(t)),\ x_2 - y_0\rangle\, dt \geq$$

$$\geq 0 + \frac{1}{L}\|\nabla g(x_2)\|^2 - \int_0^1 \|\nabla g(x_2) - \nabla g(y(t))\| \cdot \|x_2 - y_0\|\, dt \geq$$

$$\geq \frac{1}{L}\|\nabla g(x_2)\|^2 - \|x_2 - y_0\| \int_0^1 L\|x_2 - y(t)\|\, dt =$$

$$= \frac{1}{L}\|\nabla g(x_2)\|^2 - L\|x_2 - y_0\|^2 \int_0^1 (1-t)\, dt = \frac{1}{2L}\|\nabla g(x_2)\|^2.$$

Substituting the definition of g into (2), we obtain

$$f(x_2) - f(x_1) - \langle \nabla f(x_1),\ x_2 - x_1\rangle \geq \frac{1}{2L}\|\nabla f(x_2) - \nabla f(x_1)\|^2,$$

$$\|\nabla f(x_2) - \nabla f(x_1)\|^2 \leq 2L\langle \nabla f(x_1),\ x_1 - x_2\rangle + 2L(f(x_2) - f(x_1)). \tag{3}$$

Exchanging variables x_1 and x_2, we have

$$\|\nabla f(x_2) - \nabla f(x_1)\|^2 \leq 2L\langle \nabla f(x_2),\ x_2 - x_1\rangle + 2L(f(x_1) - f(x_2)). \tag{4}$$

The statement (1) is the average of (3) and (4).

3.10 2003

Day 1

2003-D1-P1 — Statement

Day 1, P1 | 10+10 points

(a) Let $a_1, a_2, \ldots$ be a sequence of real numbers such that $a_1 = 1$ and $a_{n+1} > \frac{3}{2}a_n$ for all n. Prove that the sequence

$$\frac{a_n}{\left(\frac{3}{2}\right)^{n-1}}$$

has a finite limit or tends to infinity. (10 points)

(b) Prove that for all $\alpha > 1$ there exists a sequence $a_1, a_2, \ldots$ with the same properties such that

$$\lim \frac{a_n}{\left(\frac{3}{2}\right)^{n-1}} = \alpha.$$

(10 points)

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Theory and methods for 2003-D1-P1

Theory Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Methods Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]
Strategy Build a witness and verify every constraint [S-CONSTRUCTION]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2003-D1-P1 — Solution

For this record: Theory: T-AN-CONVERGENCE. Methods: M-AN-RECURRENCE. Full learning route

(a) Let $b_n = \dfrac{a_n}{\left(\frac{3}{2}\right)^{n-1}}$. Then $a_{n+1} > \frac{3}{2}a_n$ is equivalent to $b_{n+1} > b_n$, thus the sequence (b_n) is strictly increasing. Each increasing sequence has a finite limit or tends to infinity.

(b) For all $\alpha > 1$ there exists a sequence $1 = b_1 < b_2 < \ldots$ which converges to α. Choosing $a_n = \left(\frac{3}{2}\right)^{n-1} b_n$, we obtain the required sequence (a_n).

2003-D1-P2 — Statement

Day 1, P2 | 20 points

Let $a_1, a_2 \ldots, a_{51}$ be non-zero elements of a field. We simultaneously replace each element with the sum of the 50 remaining ones. In this way we get a sequence $b_1 \ldots, b_{51}$. If this new sequence is a permutation of the original one, what can be the characteristic of the field? (The characteristic of a field is p, if p is the smallest positive integer such that $\underbrace{x + x + \cdots + x}_{p} = 0$ for any element x of the field. If there exists no such p, the characteristic is 0.)

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Theory and methods for 2003-D1-P2

Theory Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]
Methods Compare coefficients after a structural substitution [M-AL-COEFFICIENT]; Work orbit by orbit and track parity [M-AL-ORBIT]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]

Statement Recorded solutions Catalog

2003-D1-P2 — Solution

For this record: Theory: T-AL-FINITEFIELD; T-AL-PERMUTATION. Methods: M-AL-COEFFICIENT; M-AL-ORBIT. Full learning route

Let $S = a_1 + a_2 + \cdots + a_{51}$. Then $b_1 + b_2 + \cdots + b_{51} = 50S$. Since $b_1, b_2, \ldots, b_{51}$ is a permutation of $a_1, a_2, \ldots, a_{51}$, we get $50S = S$, so $49S = 0$. Assume that the characteristic of the field is not equal to 7. Then $49S = 0$ implies that $S = 0$. Therefore $b_i = -a_i$ for $i = 1, 2, \ldots, 51$. On the other hand, $b_i = a_{\varphi(i)}$, where $\varphi \in S_{51}$. Therefore, if the characteristic is not 2, the sequence $a_1, a_2, \ldots, a_{51}$ can be partitioned into pairs $\{a_i, a_{\varphi(i)}\}$ of additive inverses. But this is impossible, since 51 is an odd number. It follows that the characteristic of the field is 7 or 2.

The characteristic can be either 2 or 7. For the case of 7, $x_1 = \cdots = x_{51} = 1$ is a possible choice. For the case of 2, any elements can be chosen such that $S = 0$, since then $b_i = -a_i = a_i$.

2003-D1-P3 — Statement

Day 1, P3 | 20 points

Let A be an $n \times n$ real matrix such that $3A^3 = A^2 + A + I$ (I is the identity matrix). Show that the sequence A^k converges to an idempotent matrix. (A matrix B is called idempotent if $B^2 = B$.)

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Theory and methods for 2003-D1-P3

Theory Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Operator norms and convergence of matrix powers [T-LA-NORM]; Nilpotents, projections and involutions [T-LA-NILPOTENT]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Methods Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]
Strategy Control the limiting operation before taking the limit [S-ANALYTIC]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2003-D1-P3 — Solution

For this record: Theory: T-LA-POLYNOMIAL; T-LA-NORM; T-LA-NILPOTENT; T-AN-CONVERGENCE. Methods: M-LA-SPECTRAL. Full learning route

The minimal polynomial of A is a divisor of $3x^3-x^2-x-1$. This polynomial has three different roots. This implies that A is diagonalizable: $A = C^{-1}DC$ where D is a diagonal matrix. The eigenvalues of the matrices A and D are all roots of polynomial $3x^3 - x^2 - x - 1$. One of the three roots is 1, the remaining two roots have smaller absolute value than 1. Hence, the diagonal elements of D^k, which are the kth powers of the eigenvalues, tend to either 0 or 1 and the limit $M = \lim D^k$ is idempotent. Then $\lim A^k = C^{-1}MC$ is idempotent as well.

2003-D1-P4 — Statement

Day 1, P4 | 20 points

Determine the set of all pairs (a, b) of positive integers for which the set of positive integers can be decomposed into two sets A and B such that $a \cdot A = b \cdot B$.

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Theory and methods for 2003-D1-P4

Theory Prime valuations, divisors, and factorials [T-AL-VALUATION]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Methods Track one prime or one extreme divisor [M-AL-VALUATION]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2003-D1-P4 — Solution

For this record: Theory: T-AL-VALUATION; T-AL-CONGRUENCE. Methods: M-AL-VALUATION. Full learning route

Clearly a and b must be different since A and B are disjoint.
Let $\{a, b\}$ be a solution and consider the sets A, B such that $a \cdot A = b \cdot B$. Denoting $d = (a, b)$ the greatest common divisor of a and b, we have $a = d \cdot a_1$, $b = d \cdot b_1$, $(a_1, b_1) = 1$ and $a_1 \cdot A = b_1 \cdot B$. Thus $\{a_1, b_1\}$ is a solution and it is enough to determine the solutions $\{a, b\}$ with $(a, b) = 1$.
If $1 \in A$ then $a \in a \cdot A = b \cdot B$, thus b must be a divisor of a. Similarly, if $1 \in B$, then a is a divisor of b. Therefore, in all solutions, one of numbers a, b is a divisor of the other one.

Now we prove that if $n \geq 2$, then $(1, n)$ is a solution. For each positive integer k, let $f(k)$ be the largest non-negative integer for which $n^{f(k)} \mid k$. Then let $A = \{k : f(k) \text{ is odd}\}$ and $B = \{k : f(k) \text{ is even}\}$. This is a decomposition of all positive integers such that $A = n \cdot B$.

2003-D1-P5 — Statement

Day 1, P5 | 20 points

Let $g : [0,1] \to \mathbb{R}$ be a continuous function and let $f_n : [0,1] \to \mathbb{R}$ be a sequence of functions defined by $f_0(x) = g(x)$ and

$$f_{n+1}(x) = \frac{1}{x}\int_0^x f_n(t)\,dt \quad (x \in (0,1],\ n = 0,1,2,\dots).$$

Determine $\lim_{n\to\infty} f_n(x)$ for every $x \in (0,1]$.

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Theory and methods for 2003-D1-P5

Theory Uniform approximation and positive kernels [T-AN-APPROXIMATION]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Nonnegative sums, Fubini, and integral kernels [T-AN-FUBINI]
Methods Preserve structure while approximating [M-AN-APPROXIMATE]; Separate a main range from exceptional scales [M-AN-SPLIT]
Strategy Build a chain of bounds and track equality [S-BOUND]; Pass between local constraints and a global conclusion [S-LOCALGLOBAL]

Statement Recorded solutions Catalog

2003-D1-P5 — Solution

For this record: Theory: T-AN-APPROXIMATION; T-AN-CONVERGENCE; T-AN-INTEGRAL; T-AN-FUBINI. Methods: M-AN-APPROXIMATE; M-AN-SPLIT. Full learning route

B. We shall prove in two different ways that $\lim_{n\to\infty} f_n(x) = g(0)$ for every $x \in (0,1]$. (The second one is more lengthy but it tells us how to calculate f_n directly from g.)
Proof I. First we prove our claim for non-decreasing g. In this case, by induction, one can easily see that
1. each f_n is non-decrasing as well, and
2. $g(x) = f_0(x) \geq f_1(x) \geq f_2(x) \geq \dots \geq g(0)$ $(x \in (0,1])$.
Then (2) implies that there exists

$$h(x) = \lim_{n\to\infty} f_n(x) \quad (x \in (0,1]).$$

Clearly h is non-decreasing and $g(0) \leq h(x) \leq f_n(x)$ for any $x \in (0,1]$, $n = 0,1,2,\dots$. Therefore to show that $h(x) = g(0)$ for any $x \in (0,1]$, it is enough to prove that $h(1)$ cannot be greater than $g(0)$. Suppose that $h(1) > g(0)$. Then there exists a $0 < \delta < 1$ such that $h(1) > g(\delta)$. Using the definition, (2) and (1) we get

$$f_{n+1}(1) = \int_0^1 f_n(t)\,dt \leq \int_0^\delta g(t)\,dt + \int_\delta^1 f_n(t)\,dt \leq \delta g(\delta) + (1-\delta) f_n(1).$$

Hence

$$f_n(1) - f_{n+1}(1) \geq \delta(f_n(1) - g(\delta)) \geq \delta(h(1) - g(\delta)) > 0,$$

so $f_n(1) \to -\infty$, which is a contradiction.
Similarly, we can prove our claim for non-increasing continuous functions as well.
Now suppose that g is an arbitrary continuous function on $[0,1]$. Let

$$M(x) = \sup_{t\in[0,x]} g(t), \qquad m(x) = \inf_{t\in[0,x]} g(t) \qquad (x \in [0,1])$$

Then on $[0,1]$ m is non-increasing, M is non-decreasing, both are continuous, $m(x) \leq g(x) \leq M(x)$ and $M(0) = m(0) = g(0)$. Define the sequences of functions $M_n(x)$ and $m_n(x)$ in the same way as f_n is defined but starting with $M_0 = M$ and $m_0 = m$.
Then one can easily see by induction that $m_n(x) \leq f_n(x) \leq M_n(x)$. By the first part of the proof, $\lim_n m_n(x) = m(0) = g(0) = M(0) = \lim_n M_n(x)$ for any $x \in (0,1]$. Therefore we must have $\lim_n f_n(x) = g(0)$.

2003-D1-P5 — Solution (continued)

Proof II. To make the notation clearer we shall denote the variable of f_j by x_j. By definition (and Fubini theorem) we get that

$$\begin{aligned} f_{n+1}(x_{n+1}) &= \frac{1}{x_{n+1}} \int_0^{x_{n+1}} \frac{1}{x_n} \int_0^{x_n} \frac{1}{x_{n-1}} \int_0^{x_{n-1}} \dots \int_0^{x_2} \frac{1}{x_1} \int_0^{x_1} g(x_0)\, dx_0\, dx_1 \dots dx_n \\ &= \frac{1}{x_{n+1}} \iint\limits_{0 \le x_0 \le x_1 \le \dots \le x_n \le x_{n+1}} g(x_0)\, \frac{dx_0\, dx_1 \dots dx_n}{x_1 \dots x_n} \\ &= \frac{1}{x_{n+1}} \int_0^{x_{n+1}} g(x_0) \left(\iint\limits_{x_0 \le x_1 \le \dots \le x_n \le x_{n+1}} \frac{dx_1 \dots dx_n}{x_1 \dots x_n} \right) dx_0. \end{aligned}$$

Therefore with the notation

$$h_n(a,b) = \iint\limits_{a \le x_1 \le \dots \le x_n \le b} \frac{dx_1 \dots dx_n}{x_1 \dots x_n}$$

and $x = x_{n+1}$, $t = x_0$ we have

$$f_{n+1}(x) = \frac{1}{x} \int_0^x g(t)\, h_n(t,x)\, dt.$$

Using that $h_n(a,b)$ is the same for any permutation of $x_1, \dots, x_n$ and the fact that the integral is 0 on any hyperplanes ($x_i = x_j$) we get that

$$n!\, h_n(a,b) = \iint\limits_{a \le x_1, \dots, x_n \le b} \frac{dx_1 \dots dx_n}{x_1 \dots x_n} = \int_a^b \dots \int_a^b \frac{dx_1 \dots dx_n}{x_1 \dots x_n} = \left(\int_a^b \frac{dx}{x} \right)^n = (\log(b/a))^n.$$

Therefore

$$f_{n+1}(x) = \frac{1}{x} \int_0^x g(t)\, \frac{(\log(x/t))^n}{n!}\, dt.$$

Note that if g is constant then the definition gives $f_n = g$. This implies on one hand that we must have

$$\frac{1}{x} \int_0^x \frac{(\log(x/t))^n}{n!}\, dt = 1$$

and on the other hand that, by replacing g by $g - g(0)$, we can suppose that $g(0) = 0$.

Let $x \in (0,1]$ and $\varepsilon > 0$ be fixed. By continuity there exists a $0 < \delta < x$ and an M such that $|g(t)| < \varepsilon$ on $[0,\delta]$ and $|g(t)| \le M$ on $[0,1]$. Since

$$\lim_{n \to \infty} \frac{(\log(x/\delta))^n}{n!} = 0$$

there exists an n_0 sucht that $(\log(x/\delta))^n/n! < \varepsilon$ whenever $n \ge n_0$. Then, for any $n \ge n_0$, we have

$$\begin{aligned} |f_{n+1}(x)| &\le \frac{1}{x} \int_0^x |g(t)|\, \frac{(\log(x/t))^n}{n!}\, dt \\ &\le \frac{1}{x} \int_0^\delta \varepsilon\, \frac{(\log(x/t))^n}{n!}\, dt + \frac{1}{x} \int_\delta^x |g(t)|\, \frac{(\log(x/\delta))^n}{n!}\, dt \\ &\le \varepsilon\, \frac{1}{x} \int_0^x \frac{(\log(x/t))^n}{n!}\, dt + \frac{1}{x} \int_\delta^x M\varepsilon\, dt \\ &\le \varepsilon + M\varepsilon. \end{aligned}$$

Therefore $\lim_n f_n(x) = 0 = g(0)$.

2003-D1-P6 — Statement

Day 1, P6 | 20 points

Let $f(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$ be a polynomial with real coefficients. Prove that if all roots of f lie in the left half-plane $\{z \in \mathbb{C} : \operatorname{Re} z < 0\}$ then

$$a_k a_{k+3} < a_{k+1} a_{k+2}$$

holds for every $k = 0, 1, \ldots, n-3$.

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Theory and methods for 2003-D1-P6

Theory Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Methods Compare coefficients after a structural substitution [M-AL-COEFFICIENT]
Strategy Reduce size while preserving the defining conditions [S-INDUCTION]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2003-D1-P6 — Solution

For this record: Theory: T-AL-ROOTS; T-AL-POLYNOMIAL; T-DS-INDUCTION. Methods: M-AL-COEFFICIENT. Full learning route

The polynomial f is a product of linear and quadratic factors, $f(z) = \prod_i (k_i z + l_i) \cdot \prod_j (p_j z^2 + q_j z + r_j)$, with $k_i, l_i, p_j, q_j, r_j \in \mathbb{R}$. Since all roots are in the left half-plane, for each i, k_i and l_i are of the same sign, and for each j, p_j, q_j, r_j are of the same sign, too. Hence, multiplying f by -1 if necessary, the roots of f don't change and f becomes the polynomial with all positive coefficients.
For the simplicity, we extend the sequence of coefficients by $a_{n+1} = a_{n+2} = \cdots = 0$ and $a_{-1} = a_{-2} = \cdots = 0$ and prove the same statement for $-1 \le k \le n-2$ by induction.
For $n \le 2$ the statement is obvious: a_{k+1} and a_{k+2} are positive and at least one of a_{k-1} and a_{k+3} is 0; hence,
$a_{k+1} a_{k+2} > a_k a_{k+3} = 0$.
Now assume that $n \ge 3$ and the statement is true for all smaller values of n. Take a divisor of $f(z)$ which has the form $z^2 + pz + q$ where p and q are positive real numbers. (Such a divisor can be obtained from a conjugate pair of roots or two real roots.) Then we can write

$$f(z) = (z^2 + pz + q)(b_{n-2} z^{n-2} + \cdots + b_1 z + b_0) = (z^2 + pz + q) g(x). \tag{1}$$

The roots polynomial $g(z)$ are in the left half-plane, so we have $b_{k+1} b_{k+2} < b_k b_{k+3}$
for all $-1 \le k \le n-4$. Defining $b_{n-1} = b_n = \cdots = 0$ and $b_{-1} = b_{-2} = \cdots = 0$ as well, we also have $b_{k+1} b_{k+2} \le b_k b_{k+3}$
for all integer k.
Now we prove $a_{k+1} a_{k+2} > a_k a_{k+3}$. If $k = -1$ or $k = n-2$ then this is obvious since $a_{k+1} a_{k+2}$ is positive and $a_k a_{k+3} = 0$. Thus, assume $0 \le k \le n-3$. By an easy computation,

$$\begin{aligned} a_{k+1} a_{k+2} - a_k a_{k+3} = \\ &= (qb_{k+1} + pb_k + b_{k-1})(qb_{k+2} + pb_{k+1} + b_k) - (qb_k + pb_{k-1} + b_{k-2})(qb_{k+3} + pb_{k+2} + b_{k+1}) = \\ &= (b_{k-1} b_k - b_{k-2} b_{k+1}) + p(b_k^2 - b_{k-2} b_{k+2}) + q(b_{k-1} b_{k+2} - b_{k-2} b_{k+3}) + \\ &\quad + p^2 (b_k b_{k+1} - b_{k-1} b_{k+2}) + q^2 (b_{k+1} b_{k+2} - b_k b_{k+3}) + pq(b_{k+1}^2 - b_{k-1} b_{k+3}). \end{aligned}$$

We prove that all the six terms are non-negative and at least one is positive. Term $p^2(b_k b_{k+1} - b_{k-1} b_{k+2})$ is positive since $0 \le k \le n-3$. Also terms $b_{k-1} b_k - b_{k-2} b_{k+1}$ and $q^2(b_{k+1} b_{k+2} - b_k b_{k+3})$ are non-negative by the induction hypothesis.
To check the sign of $p(b_k^2 - b_{k-2} b_{k+2})$ consider

$$b_{k-1}(b_k^2 - b_{k-2} b_{k+2}) = b_{k-2}(b_k b_{k+1} - b_{k-1} b_{k+2}) + b_k (b_{k-1} b_k - b_{k-2} b_{k+1}) \ge 0.$$

2003-D1-P6 — Solution (continued)

If $b_{k-1} > 0$ we can divide by it to obtain $b_k^2 - b_{k-2}b_{k+2} \geq 0$. Otherwise, if $b_{k-1} = 0$, either $b_{k-2} = 0$ or $b_{k+2} = 0$ and thus $b_k^2 - b_{k-2}b_{k+2} = b_k^2 \geq 0$. Therefore, $p(b_k^2 - b_{k-2}b_{k+2}) \geq 0$ for all k. Similarly, $pq(b_{k+1}^2 - b_{k-1}b_{k+3}) \geq 0$.
The sign of $q(b_{k-1}b_{k+2} - b_{k-2}b_{k+3})$ can be checked in a similar way. Consider

$$b_{k+1}(b_{k-1}b_{k+2} - b_{k-2}b_{k+3}) = b_{k-1}(b_{k+1}b_{k+2} - b_k b_{k+3}) + b_{k+3}(b_{k-1}b_k - b_{k-2}b_{k+1}) \geq 0.$$

If $b_{k+1} > 0$, we can divide by it. Otherwise either $b_{k-2} = 0$ or $b_{k+3} = 0$. In all cases, we obtain $b_{k-1}b_{k+2} - b_{k-2}b_{k+3} \geq 0$.

Now the signs of all terms are checked and the proof is complete.

Day 2

2003-D2-P1 — Statement

Day 2, P7

Let A and B be $n \times n$ real matrices such that $AB + A + B = 0$. Prove that $AB = BA$.

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Theory and methods for 2003-D2-P1

Theory Determinants, block identities and inverses [T-LA-DETERMINANT]
Methods Complete the equation to a product equal to identity [M-LA-INVERSE]
Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2003-D2-P1 — Solution

For this record: Theory: T-LA-DETERMINANT. Methods: M-LA-INVERSE. Full learning route

Since $(A + I)(B + I) = AB + A + B + I = I$ (I is the identity matrix), matrices $A + I$ and $B + I$ are inverses of each other. Then $(A + I)(B + I) = (B + I)(A + I)$ and $AB = BA$.

2003-D2-P2 — Statement

Day 2, P8

Evaluate the limit

$$\lim_{x \to 0+} \int_x^{2x} \frac{\sin^m t}{t^n}\, dt \quad (m, n \in \mathbb{N}).$$

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Theory and methods for 2003-D2-P2

Theory Trigonometric identities and safe branches [T-AN-TRIG]; Scale separation and limit comparisons [T-AN-ASYMPTOTICS]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Methods Separate a main range from exceptional scales [M-AN-SPLIT]; Pair an integral through symmetry [M-AN-PAIR]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2003-D2-P2 — Solution

For this record: Theory: T-AN-TRIG; T-AN-ASYMPTOTICS; T-AN-INTEGRAL. Methods: M-AN-SPLIT; M-AN-PAIR. Full learning route

We use the fact that $\dfrac{\sin t}{t}$ is decreasing in the interval $(0,\pi)$ and $\lim\limits_{t\to 0+0}\dfrac{\sin t}{t}=1$. For all $x\in\left(0,\frac{\pi}{2}\right)$ and $t\in[x,2x]$ we have $\dfrac{\sin 2x}{2x}<\dfrac{\sin t}{t}<1$, thus

$$\left(\frac{\sin 2x}{2x}\right)^m\int_x^{2x}\frac{t^m}{t^n}\,dt<\int_x^{2x}\frac{\sin^m t}{t^n}\,dt<\int_x^{2x}\frac{t^m}{t^n}\,dt,$$

$$\int_x^{2x}\frac{t^m}{t^n}\,dt=x^{m-n+1}\int_1^2 u^{m-n}\,du.$$

The factor $\left(\dfrac{\sin 2x}{2x}\right)^m$ tends to 1. If $m-n+1<0$, the limit of x^{m-n+1} is infinity; if $m-n+1>0$ then 0. If $m-n+1=0$ then $x^{m-n+1}\int_1^2 u^{m-n}\,du=\ln 2$. Hence,

$$\lim_{x\to 0+0}\int_x^{2x}\frac{\sin^m t}{t^n}\,dt=\begin{cases}0, & m\geq n\\ \ln 2, & n-m=1\\ +\infty, & n-m>1.\end{cases}$$

2003-D2-P3 — Statement

Day 2, P9

Let A be a closed subset of $\mathbb{R}^n$ and let B be the set of all those points $b\in\mathbb{R}^n$ for which there exists exactly one point $a_0\in A$ such that

$$|a_0-b|=\inf_{a\in A}|a-b|.$$

Prove that B is dense in $\mathbb{R}^n$; that is, the closure of B is $\mathbb{R}^n$.

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Theory and methods for 2003-D2-P3

Theory Metric contractions, covers, and nearest points [T-DS-METRIC]; Continuity, compactness, and order [T-AN-CONTINUITY]; Inner products, orthogonality and polarization [T-LA-INNER]
Methods Push covers forward or move toward a nearest point [M-DS-METRIC]
Strategy Translate geometry into a controlled representation [S-GEOMETRIC]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2003-D2-P3 — Solution

For this record: Theory: T-DS-METRIC; T-AN-CONTINUITY; T-LA-INNER. Methods: M-DS-METRIC. Full learning route

Let $b_0\notin A$ (otherwise $b_0\in A\subset B$), $\varrho=\inf\limits_{a\in A}|a-b_0|$. The intersection of the ball of radius $\varrho+1$ with centre b_0 with set A is compact and there exists $a_0\in A$: $|a_0-b_0|=\varrho$.
Denote by $B_r(a)=\{x\in\mathbb{R}^n:|x-a|\leq r\}$ and $\partial B_r(a)=\{x\in\mathbb{R}^n:|x-a|=r\}$ the ball and the sphere of center a and radius r, respectively.

If a_0 is not the unique nearest point then for any point a on the open line segment (a_0,b_0) we have $B_{|a-a_0|}(a)\subset B_\varrho(b_0)$ and $\partial B_{|a-a_0|}(a)\cap\partial B_\varrho(b_0)=\{a_0\}$, therefore $(a_0,b_0)\subset B$ and b_0 is an accumulation point of set B.

2003-D2-P4 — Statement

Day 2, P10

Find all positive integers n for which there exists a family F of three-element subsets of $S = \{1, 2, \dots, n\}$ satisfying the following two conditions:
(i) for any two different elements $a, b \in S$, there exists exactly one $A \in F$ containing both a, b;
(ii) if a, b, c, x, y, z are elements of S such that if

$\{a, b, x\}, \{a, c, y\}, \{b, c, z\} \in F$, then $\{x, y, z\} \in F$.

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Theory and methods for 2003-D2-P4

Theory Finite set systems and antichains [T-DS-SETS]; Groups, orders, generators, and quotients [T-AL-GROUP]; Rank, dimension and exact linear constraints [T-LA-RANK]
Methods Choose information that makes an encoding injective [M-DS-ENCODE]
Strategy Expose the structure before calculating [S-STRUCTURE]; Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2003-D2-P4 — Solution

For this record: Theory: T-DS-SETS; T-AL-GROUP; T-LA-RANK. Methods: M-DS-ENCODE. Full learning route

The condition (i) of the problem allows us to define a (well-defined) operation $*$ on the set S given by

$$a * b = c \quad \text{if and only if} \quad \{a, b, c\} \in F, \text{ where } a \neq b.$$

We note that this operation is still not defined completely (we need to define $a*a$), but nevertheless let us investigate its features. At first, due to (i), for $a \neq b$ the operation obviously satisfies the following three conditions:
(a) $a \neq a * b \neq b$;
(b) $a * b = b * a$;
(c) $a * (a * b) = b$.
What does the condition (ii) give? It claims that
(e') $x * (a * c) = x * y = z = b * c = (x * a) * c$
for any three different x, a, c, i.e. that the operation is associative if the arguments are different. Now we can complete the definition of $*$. In order to save associativity for non-different arguments, i.e. to make $b = a * (a * b) = (a * a) * b$ hold, we will add to S an extra element, call it 0, and define
(d) $a * a = 0$ and $a * 0 = 0 * a = a$.
Now it is easy to check that, for any $a, b, c \in S \cup \{0\}$, (a), (b), (c) and (d), still hold, and
(e) $a * b * c := (a * b) * c = a * (b * c)$.
We have thus obtained that $(S \cup \{0\}, *)$ has the structure of a finite Abelian group, whose elements are all of order two. Since the order of every such group is a power of 2, we conclude that $|S \cup \{0\}| = n + 1 = 2^m$ and $n = 2^m - 1$ for some integer $m \geq 1$.
Given $n = 2^m - 1$, according to what we have proven till now, we will construct a family of three-element subsets of S satisfying (i) and (ii). Let us define the operation $*$ in the following manner:

> if $a = a_0 + 2a_1 + \cdots + 2^{m-1}a_{m-1}$ and $b = b_0 + 2b_1 + \cdots + 2^{m-1}b_{m-1}$, where a_i, b_i are either 0 or 1, we put $a * b = |a_0 - b_0| + 2|a_1 - b_1| + \cdots + 2^{m-1}|a_{m-1} - b_{m-1}|$.

It is simple to check that this $*$ satisfies (a), (b), (c) and (e'). Therefore, if we include in F all possible triples $a, b, a * b$, the condition (i) follows from (a), (b) and (c), whereas the condition (ii) follows from (e')

The answer is: $n = 2^m - 1$.

2003-D2-P5 — Statement

Day 2, P11

(a) Show that for each function $f : \mathbb{Q} \times \mathbb{Q} \to \mathbb{R}$ there exists a function $g : \mathbb{Q} \to \mathbb{R}$ such that $f(x, y) \leq g(x) + g(y)$ for all $x, y \in \mathbb{Q}$.

(b) Find a function $f : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ for which there is no function $g : \mathbb{R} \to \mathbb{R}$ such that $f(x, y) \leq g(x)+g(y)$ for all $x, y \in \mathbb{R}$.

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Theory and methods for 2003-D2-P5

Theory Countability, accumulation, and Cantor constructions [T-AN-SETS]; Continuity, compactness, and order [T-AN-CONTINUITY]
Methods Use a dense set, an enumeration, or a category argument [M-AN-TOPOLOGY]
Strategy Build a witness and verify every constraint [S-CONSTRUCTION]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

2003-D2-P5 — Solution — Record 1 of 2

For this record: Theory: T-AN-SETS; T-AN-CONTINUITY. Methods: M-AN-TOPOLOGY. Full learning route

a) Let $\varphi : \mathbb{Q} \to \mathbb{N}$ be a bijection. Define $g(x) = \max\{|f(s, t)| : s, t \in \mathbb{Q},\ \varphi(s) \leq \varphi(x),\ \varphi(t) \leq \varphi(x)\}$. We have $f(x, y) \leq \max\{g(x), g(y)\} \leq g(x) + g(y)$.

b) We shall show that the function defined by $f(x, y) = \dfrac{1}{|x - y|}$ for $x \neq y$ and $f(x, x) = 0$ satisfies the problem. If, by contradiction there exists a function g as above, it results, that $g(y) \geq \dfrac{1}{|x - y|} - f(x)$ for $x, y \in \mathbb{R}$, $x \neq y$; one obtains that for each $x \in \mathbb{R}$, $\lim\limits_{y \to x} g(y) = \infty$. We show, that there exists no function g having an infinite limit at each point of a bounded and closed interval $[a, b]$.
For each $k \in \mathbb{N}_+$ denote $A_k = \{x \in [a, b] : |g(x)| \leq k\}$.

We have obviously $[a, b] = \bigcup_{k=1}^{\infty} A_k$. The set $[a, b]$ is uncountable, so at least one of the sets A_k is infinite (in fact uncountable). This set A_k being infinite, there exists a sequence in A_k having distinct terms. This sequence will contain a convergent subsequence $(x_n)_{n \in \mathbb{N}}$ convergent to a point $x \in [a, b]$. But $\lim\limits_{y \to x} g(y) = \infty$ implies that $g(x_n) \to \infty$, a contradiction because $|g(x_n)| \leq k$, $\forall n \in \mathbb{N}$.

2003-D2-P5 — Solution — Record 2 of 2

For this record: Theory: T-AN-SETS. Methods: M-AN-TOPOLOGY. Full learning route

Second solution for part (b). Let S be the set of all sequences of real numbers. The cardinality of S is $|S| = |\mathbb{R}|^{\aleph_0} = 2^{\aleph_0^2} = 2^{\aleph_0} = |\mathbb{R}|$. Thus, there exists a bijection $h : \mathbb{R} \to S$.
Now define the function f in the following way. For any real x and positive integer n, let $f(x, n)$ be the nth element of sequence $h(x)$. If y is not a positive integer then let $f(x, y) = 0$. We prove that this function has the required property.

Let g be an arbitrary $\mathbb{R} \to \mathbb{R}$ function. We show that there exist real numbers x, y such that $f(x, y) > g(x) + g(y)$. Consider the sequence $(n + g(n))_{n=1}^{\infty}$. This sequence is an element of S, thus $(n + g(n))_{n=1}^{\infty} = h(x)$ for a certain real x. Then for an arbitrary positive integer n, $f(x, n)$ is the nth element, $f(x, n) = n + g(n)$. Choosing n such that $n > g(x)$, we obtain $f(x, n) = n + g(n) > g(x) + g(n)$.

2003-D2-P6 — Statement

Day 2, P12

Let $(a_n)_{n \in \mathbb{N}}$ be the sequence defined by

$$a_0 = 1, \qquad a_{n+1} = \frac{1}{n+1} \sum_{k=0}^{n} \frac{a_k}{n - k + 2}.$$

2003-D2-P6 — Statement (continued)

Find the limit

$$\lim_{n\to\infty}\sum_{k=0}^{n}\frac{a_k}{2^k},$$

if it exists.

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Theory and methods for 2003-D2-P6

Theory Power series and generating functions [T-AN-POWER]; Differential inequalities and integrating factors [T-AN-ODE]; Series, finite cancellation, and summation by parts [T-AN-SERIES]
Methods Encode convolution or denominators by a power series [M-AN-GENERATING]
Strategy Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2003-D2-P6 — Solution

For this record: Theory: T-AN-POWER; T-AN-ODE; T-AN-SERIES. Methods: M-AN-GENERATING. Full learning route

Consider the generating function $f(x)=\sum\limits_{n=0}^{\infty}a_nx^n$. By induction $0<a_n\le 1$, thus this series is absolutely convergent for $|x|<1$, $f(0)=1$ and the function is positive in the interval $[0,1)$. The goal is to compute $f\left(\frac{1}{2}\right)$.

By the recurrence formula,

$$f'(x)=\sum_{n=0}^{\infty}(n+1)a_{n+1}x^n=\sum_{n=0}^{\infty}\sum_{k=0}^{n}\frac{a_k}{n-k+2}\,x^n=\sum_{k=0}^{\infty}a_kx^k\sum_{n=k}^{\infty}\frac{x^{n-k}}{n-k+2}=f(x)\sum_{m=0}^{\infty}\frac{x^m}{m+2}.$$

Then

$$\ln f(x)=\ln f(x)-\ln f(0)=\int_0^x\frac{f'}{f}=\sum_{m=0}^{\infty}\frac{x^{m+1}}{(m+1)(m+2)}=$$

$$=\sum_{m=0}^{\infty}\left(\frac{x^{m+1}}{(m+1)}-\frac{x^{m+1}}{(m+2)}\right)=1+\left(1-\frac{1}{x}\right)\sum_{m=0}^{\infty}\frac{x^{m+1}}{(m+1)}=1+\left(1-\frac{1}{x}\right)\ln\frac{1}{1-x},$$

$$\ln f\left(\frac{1}{2}\right)=1-\ln 2,$$

and thus $f\left(\frac{1}{2}\right)=\frac{e}{2}$.

3.11 2004

Day 1

2004-D1-P1 — Statement

Day 1, P1 | 20 points

Let S be an infinite set of real numbers such that $|s_1 + s_2 + \cdots + s_k| < 1$ for every finite subset $\{s_1, s_2, \ldots, s_k\} \subset S$. Show that S is countable.

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Theory and methods for 2004-D1-P1

Theory Countability, accumulation, and Cantor constructions [T-AN-SETS]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Split a family by traces, pieces, or labels [M-DS-PARTITION]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2004-D1-P1 — Solution

For this record: Theory: T-AN-SETS; T-DS-COUNTING. Methods: M-DS-PARTITION. Full learning route

Let $S_n = S \cap \left(\frac{1}{n}, \infty\right)$ for any integer $n > 0$. It follows from the inequality that $|S_n| < n$. Similarly, if we define $S_{-n} = S \cap \left(-\infty, -\frac{1}{n}\right)$, then $|S_{-n}| < n$. Any nonzero $x \in S$ is an element of some S_n or S_{-n}, because there exists an n such that $x > \frac{1}{n}$, or $x < -\frac{1}{n}$. Then $S \subset \{0\} \cup \bigcup_{n \in \mathbb{N}} (S_n \cup S_{-n})$, S is a countable union of finite sets, and hence countable.

2004-D1-P2 — Statement

Day 1, P2 | 20 points

Let $P(x) = x^2 - 1$. How many distinct real solutions does the following equation have:

$$\underbrace{P(P(\ldots(P}_{2004}(x))\ldots)) = 0\ ?$$

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Theory and methods for 2004-D1-P2

Theory Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Finite counting and incidence identities [T-DS-COUNTING]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]; Strengthen the statement before the induction step [M-DS-INDUCT]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Reduce size while preserving the defining conditions [S-INDUCTION]

Statement Recorded solutions Catalog

2004-D1-P2 — Solution

For this record: Theory: T-DS-DYNAMICS; T-DS-INDUCTION; T-DS-COUNTING; T-AL-POLYNOMIAL. Methods: M-DS-ITERATE; M-DS-INDUCT. Full learning route

Put $P_n(x) = \underbrace{P(P(\dots(P}_{n}(x))\dots))$. As $P_1(x) \geq -1$, for each $x \in \mathbb{R}$, it must be that $P_{n+1}(x) = P_1(P_n(x)) \geq -1$, for each $n \in \mathbb{N}$ and each $x \in \mathbb{R}$. Therefore the equation $P_n(x) = a$, where $a < -1$ has no real solutions.

Let us prove that the equation $P_n(x) = a$, where $a > 0$, has exactly two distinct real solutions. To this end we use mathematical induction by n. If $n = 1$ the assertion follows directly. Assuming that the assertion holds for a $n \in \mathbb{N}$ we prove that it must also hold for $n+1$. Since $P_{n+1}(x) = a$ is equivalent to $P_1(P_n(x)) = a$, we conclude that $P_n(x) = \sqrt{a+1}$ or $P_n(x) = -\sqrt{a+1}$. The equation $P_n(x) = \sqrt{a+1}$, as $\sqrt{a+1} > 1$, has exactly two distinct real solutions by the inductive hypothesis, while the equation $P_n(x) = -\sqrt{a+1}$ has no real solutions (because $-\sqrt{a+1} < -1$). Hence the equation $P_{n+1}(x) = a$, has exactly two distinct real solutions.

Let us prove now that the equation $P_n(x) = 0$ has exactly $n+1$ distinct real solutions. Again we use mathematical induction. If $n = 1$ the solutions are $x = \pm 1$, and if $n = 2$ the solutions are $x = 0$ and $x = \pm\sqrt{2}$, so in both cases the number of solutions is equal to $n+1$. Suppose that the assertion holds for some $n \in \mathbb{N}$. Note that $P_{n+2}(x) = P_2(P_n(x)) = P_n^2(x)(P_n^2(x) - 2)$, so the set of all real solutions of the equation $P_{n+2} = 0$ is exactly the union of the sets of all real solutions of the equations $P_n(x) = 0$, $P_n(x) = \sqrt{2}$ and $P_n(x) = -\sqrt{2}$. By the inductive hypothesis the equation $P_n(x) = 0$ has exactly $n+1$ distinct real solutions, while the equations $P_n(x) = \sqrt{2}$ and $P_n(x) = -\sqrt{2}$ have two and no distinct real solutions, respectively. Hence, the sets above being pairwise disjoint, the equation $P_{n+2}(x) = 0$ has exactly $n+3$ distinct real solutions. Thus we have proved that, for each $n \in \mathbb{N}$, the equation $P_n(x) = 0$ has exactly $n+1$ distinct real solutions, so the answer to the question posed in this problem is 2005.

2004-D1-P3 — Statement

Day 1, P3 | 10+10 points

Let S_n be the set of all sums $\sum_{k=1}^{n} x_k$, where $n \geq 2$, $0 \leq x_1, x_2, \dots, x_n \leq \frac{\pi}{2}$ and

$$\sum_{k=1}^{n} \sin x_k = 1.$$

a) Show that S_n is an interval. [10 points]

b) Let l_n be the length of S_n. Find $\lim_{n\to\infty} l_n$. [10 points]

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Theory and methods for 2004-D1-P3

Theory Continuity, compactness, and order [T-AN-CONTINUITY]; Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Trigonometric identities and safe branches [T-AN-TRIG]; Scale separation and limit comparisons [T-AN-ASYMPTOTICS]

Methods Choose the correct convexity comparison [M-AN-CONVEX]; Localize an extremum or an accumulation point [M-AN-COMPACT]

Strategy Choose an extremal object and force rigidity [S-EXTREMAL]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2004-D1-P3 — Solution

For this record: Theory: T-AN-CONTINUITY; T-AN-CONVEXITY; T-AN-TRIG; T-AN-ASYMPTOTICS. Methods: M-AN-CONVEX; M-AN-COMPACT. Full learning route

(a) Equivalently, we consider the set

$$Y = \{y = (y_1, y_2, \dots, y_n) \mid 0 \le y_1, y_2, \dots, y_n \le 1,\ y_1 + y_2 + \cdots + y_n = 1\} \subset \mathbb{R}^n$$

and the image $f(Y)$ of Y under

$$f(y) = \arcsin y_1 + \arcsin y_2 + \cdots + \arcsin y_n.$$

Note that $f(Y) = S_n$. Since Y is a connected subspace of $\mathbb{R}^n$ and f is a continuous function, the image $f(Y)$ is also connected, and we know that the only connected subspaces of $\mathbb{R}$ are intervals. Thus S_n is an interval.

(b) We prove that

$$n \arcsin \frac{1}{n} \le x_1 + x_2 + \cdots + x_n \le \frac{\pi}{2}.$$

Since the graph of $\sin x$ is concave down for $x \in [0, \frac{\pi}{2}]$, the chord joining the points $(0,0)$ and $(\frac{\pi}{2}, 1)$ lies below the graph. Hence

$$\frac{2x}{\pi} \le \sin x \quad \text{for all } x \in \left[0, \frac{\pi}{2}\right]$$

and we can deduce the right-hand side of the claim:

$$\frac{2}{\pi}(x_1 + x_2 + \cdots + x_n) \le \sin x_1 + \sin x_2 + \cdots + \sin x_n = 1.$$

The value 1 can be reached choosing $x_1 = \frac{\pi}{2}$ and $x_2 = \cdots = x_n = 0$.

The left-hand side follows immediately from Jensen's inequality, since $\sin x$ is concave down for $x \in [0, \frac{\pi}{2}]$ and $0 \le \frac{x_1 + x_2 + \cdots + x_n}{n} < \frac{\pi}{2}$

$$\frac{1}{n} = \frac{\sin x_1 + \sin x_2 + \cdots + \sin x_n}{n} \le \sin \frac{x_1 + x_2 + \cdots + x_n}{n}.$$

Equality holds if $x_1 = \cdots = x_n = \arcsin \frac{1}{n}$.

Now we have computed the minimum and maximum of interval S_n; we can conclude that $S_n = [n \arcsin \frac{1}{n}, \frac{\pi}{2}]$. Thus $l_n = \frac{\pi}{2} - n \arcsin \frac{1}{n}$ and

$$\lim_{n\to\infty} l_n = \frac{\pi}{2} - \lim_{n\to\infty} \frac{\arcsin(1/n)}{1/n} = \frac{\pi}{2} - 1.$$

2004-D1-P4 — Statement

Day 1, P4 | 20 points

Suppose $n \ge 4$ and let M be a finite set of n points in $\mathbb{R}^3$, no four of which lie in a plane. Assume that the points can be coloured black or white so that any sphere which intersects M in at least four points has the property that exactly half of the points in the intersection of M and the sphere are white. Prove that all of the points in M lie on one sphere.

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Theory and methods for 2004-D1-P4

Theory Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Finite counting and incidence identities [T-DS-COUNTING]; Rank, dimension and exact linear constraints [T-LA-RANK]

Methods Count incidences with their exact multiplicity [M-DS-DOUBLECOUNT]; Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]

Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Make the contradiction quantitative or structural [S-CONTRADICTION]; Translate geometry into a controlled representation [S-GEOMETRIC]

Statement Recorded solutions Catalog

2004-D1-P4 — Solution

For this record: Theory: T-DS-GEOMETRY; T-DS-COUNTING; T-LA-RANK. Methods: M-DS-DOUBLECOUNT; M-DS-GEOMETRIC. Full learning route

Define $f : M \to \{-1, 1\}$, $f(X) = \begin{cases} -1, & \text{if } X \text{ is white} \\ 1, & \text{if } X \text{ is black} \end{cases}$. The given condition becomes $\sum_{X \in S} f(X) = 0$ for any sphere S which passes through at least 4 points of M. For any 3 given points A, B, C in M, denote by $S(A, B, C)$ the set of all spheres which pass through A, B, C and at least one other point of M and by $|S(A, B, C)|$ the number of these spheres. Also, denote by $\sum$ the sum $\sum_{X \in M} f(X)$.

We have

$$0 = \sum_{S \in S(A,B,C)} \sum_{X \in S} f(X) = (|S(A, B, C)| - 1)(f(A) + f(B) + f(C)) + \sum \tag{1}$$

since the values of A, B, C appear $|S(A, B, C)|$ times each and the other values appear only once.
If there are 3 points A, B, C such that $|S(A, B, C)| = 1$, the proof is finished.
If $|S(A, B, C)| > 1$ for any distinct points A, B, C in M, we will prove at first that $\sum = 0$.
Assume that $\sum > 0$. From (1) it follows that $f(A) + f(B) + f(C) < 0$ and summing by all $\binom{n}{3}$ possible choices of (A, B, C) we obtain that $\binom{n}{3} \sum < 0$, which means $\sum < 0$ (contradicts the starting assumption). The same reasoning is applied when assuming $\sum < 0$.

Now, from $\sum = 0$ and (1), it follows that $f(A) + f(B) + f(C) = 0$ for any distinct points A, B, C in M. Taking another point $D \in M$, the following equalities take place

$$\begin{aligned} f(A) + f(B) + f(C) &= 0 \\ f(A) + f(B) + f(D) &= 0 \\ f(A) + f(C) + f(D) &= 0 \\ f(B) + f(C) + f(D) &= 0 \end{aligned}$$

which easily leads to $f(A) = f(B) = f(C) = f(D) = 0$, which contradicts the definition of f.

2004-D1-P5 — Statement

Day 1, P5 | 20 points

Let X be a set of $\binom{2k-4}{k-2} + 1$ real numbers, $k \geq 2$. Prove that there exists a monotone sequence $\{x_i\}_{i=1}^{k} \subseteq X$ such that

$$|x_{i+1} - x_1| \geq 2|x_i - x_1|$$

for all $i = 2, \ldots, k-1$.

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Theory and methods for 2004-D1-P5

Theory Order, colouring, and binomial threshold induction [T-DS-RAMSEY]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Split a family by traces, pieces, or labels [M-DS-PARTITION]; Strengthen the statement before the induction step [M-DS-INDUCT]
Strategy Reduce size while preserving the defining conditions [S-INDUCTION]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2004-D1-P5 — Solution

For this record: Theory: T-DS-RAMSEY; T-DS-INDUCTION; T-DS-COUNTING. Methods: M-DS-PARTITION; M-DS-INDUCT. Full learning route

We prove a more general statement:

2004-D1-P5 — Solution (continued)

Lemma. Let $k, l \geq 2$, let X be a set of $\binom{k+l-4}{k-2}+1$ real numbers. Then either X contains an increasing sequence $\{x_i\}_{i=1}^{k} \subseteq X$ of length k and

$$|x_{i+1} - x_1| \geq 2|x_i - x_1| \quad \forall i = 2, \dots, k-1,$$

or X contains a decreasing sequence $\{x_i\}_{i=1}^{l} \subseteq X$ of length l and

$$|x_{i+1} - x_1| \geq 2|x_i - x_1| \quad \forall i = 2, \dots, l-1.$$

Proof of the lemma. We use induction on $k+l$. In case $k=2$ or $l=2$ the lemma is obviously true. Now let us make the induction step. Let m be the minimal element of X, M be its maximal element. Let

$$X_m = \left\{x \in X : x \leq \frac{m+M}{2}\right\}, \qquad X_M = \left\{x \in X : x > \frac{m+M}{2}\right\}.$$

Since $\binom{k+l-4}{k-2} = \binom{k+(l-1)-4}{k-2} + \binom{(k-1)+l-4}{(k-1)-2}$, we can see that either

$$|X_m| \geq \binom{(k-1)+l-4}{(k-1)-2} + 1, \quad \text{or} \quad |X_M| \geq \binom{k+(l-1)-4}{k-2} + 1.$$

In the first case we apply the inductive assumption to X_m and either obtain a decreasing sequence of length l with the required properties (in this case the inductive step is made), or obtain an increasing sequence $\{x_i\}_{i=1}^{k-1} \subseteq X_m$ of length $k-1$. Then we note that the sequence $\{x_1, x_2, \dots, x_{k-1}, M\} \subseteq X$ has length k and all the required properties.
In the case $|X_M| \geq \binom{k+(l-1)-4}{k-2} + 1$ the inductive step is made in a similar way. Thus the lemma is proved.

The reader may check that the number $\binom{k+l-4}{k-2}+1$ cannot be smaller in the lemma.

2004-D1-P6 — Statement

Day 1, P6 | 10+10 points

For every complex number $z \notin \{0,1\}$ define

$$f(z) := \sum (\log z)^{-4},$$

where the sum is over all branches of the complex logarithm.
a) Show that there are two polynomials P and Q such that $f(z) = P(z)/Q(z)$ for all $z \in \mathbb{C} \setminus \{0,1\}$. [10 points]

b) Show that for all $z \in \mathbb{C} \setminus \{0,1\}$

$$f(z) = z\,\frac{z^2+4z+1}{6(z-1)^4}.$$

[10 points]

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Theory and methods for 2004-D1-P6

Theory Holomorphic bounds, poles, and logarithmic-branch sums [T-AL-COMPLEXANALYSIS]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Build a holomorphic function with a useful boundary [M-AL-ANALYTIC]
Strategy Control the limiting operation before taking the limit [S-ANALYTIC]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2004-D1-P6 — Solution — Record 1 of 2

For this record: Theory: T-AL-COMPLEXANALYSIS; T-AN-CONVERGENCE; T-AL-POLYNOMIAL. Methods: M-AL-ANALYTIC. Full learning route

It is clear that the left hand side is well defined and independent of the order of summation, because we have a sum of the type $\sum n^{-4}$, and the branches of the logarithms do not matter because all branches are taken. It is easy to check that the convergence is locally uniform on $\mathbb{C}\setminus\{0,1\}$; therefore, f is a holomorphic function on the complex plane, except possibly for isolated singularities at 0 and 1. (We omit the detailed estimates here.)
The function log has its only (simple) zero at $z=1$, so f has a quadruple pole at $z=1$.
Now consider $|z|\geq e$. Write $c=\log|z|\geq 1$ and choose $\theta\in[-\pi,\pi]$ with $z=|z|e^{i\theta}$. All logarithmic values are $c+i(\theta+2\pi k)$, $k\in\mathbb{Z}$. For $k\neq 0$, $|\theta+2\pi k|\geq\pi|k|$, so

$$\begin{aligned}|f(z)| &\leq \sum_{k\in\mathbb{Z}} \frac{1}{\left(c^2+(\theta+2\pi k)^2\right)^2}\\ &\leq c^{-4}+2\sum_{k=1}^{\infty}(c^2+\pi^2k^2)^{-2}\\ &\leq c^{-4}+2\int_0^{\infty}(c^2+\pi^2t^2)^{-2}\,dt = c^{-4}+\frac{1}{2c^3}\leq\frac{3}{2c^3}.\end{aligned}$$

Therefore, the infinite sum tends to 0 as $|z|\to\infty$. In particular, the isolated singularity at ∞ is not essential, but rather has (at least a single) zero at ∞.
The remaining singularity is at $z=0$. It is readily verified that $f(1/z)=f(z)$ (because $\log(1/z)=-\log(z)$); this implies that f has a zero at $z=0$.
We conclude that the infinite sum is holomorphic on $\mathbb{C}$ with at most one pole and without an essential singularity at ∞, so it is a rational function, i.e. we can write $f(z)=P(z)/Q(z)$ for some polynomials P and Q which we may as well assume coprime. This solves the first part.
Since f has a quadruple pole at $z=1$ and no other poles, we have $Q(z)=(z-1)^4$ up to a constant factor which we can as well set equal to 1, and this determines P uniquely. Since $f(z)\to 0$ as $z\to\infty$, the degree of P is at most 3, and since $P(0)=0$, it follows that $P(z)=z(az^2+bz+c)$ for yet undetermined complex constants a,b,c.
There are a number of ways to compute the coefficients a,b,c, which turn out to be $a=c=1/6$, $b=2/3$. Since $f(z)=f(1/z)$, it follows easily that $a=c$. Moreover, the fact $\lim_{z\to1}(z-1)^4f(z)=1$ implies $a+b+c=1$ (this fact follows from the observation that at $z=1$, the sum over nonprincipal branches stays bounded near $z=1$, whereas the principal branch contributes the quadruple pole).
Finally, we can calculate

$$f(-1)=\pi^{-4}\sum_{n\text{ odd}}n^{-4}=2\pi^{-4}\sum_{n\geq1\text{ odd}}n^{-4}=2\pi^{-4}\left(\sum_{n\geq1}n^{-4}-\sum_{n\geq1\text{ even}}n^{-4}\right)=\frac{1}{48}.$$

This implies $a-b+c=-1/3$. These three equations easily yield a,b,c.
Moreover, the function f satisfies $f(z)+f(-z)=16f(z^2)$: this follows because the branches of $\log(z^2)=\log((-z)^2)$ are the numbers $2\log(z)$ and $2\log(-z)$. This observation supplies the two equations $b=4a$ and $a=c$, which can be used instead of some of the considerations above.
Another way is to compute $g(z)=\sum\frac{1}{(\log z)^2}$ first. In the same way, $g(z)=\frac{dz}{(z-1)^2}$. The unknown coefficient d can be computed from $\lim_{z\to1}(z-1)^2g(z)=1$; it is $d=1$. Then the exponent 2 in the denominator can be increased by taking derivatives (see Solution 2). Similarly, one can start with exponent 3 directly.

A more straightforward, though tedious way to find the constants is computing the first four terms of the Laurent series of f around $z=1$. For that branch of the logarithm which vanishes at 1, for all $|w|<1$ we have

$$\log(1+w)=w-\frac{w^2}{2}+\frac{w^3}{3}-\frac{w^4}{4}+O(|w|^5);$$

after some computation, one can obtain

$$\frac{1}{\log(1+w)^4}=w^{-4}+2w^{-3}+\frac{7}{6}w^{-2}+\frac{1}{6}w^{-1}+O(1).$$

2004-D1-P6 — Solution — Record 1 of 2 (continued)

The remaining branches of logarithm give a bounded function. So

$$f(1+w) = w^{-4} + 2w^{-3} + \frac{7}{6}w^{-2} + \frac{1}{6}w^{-1}$$

(the remainder vanishes) and

$$f(z) = \frac{1 + 2(z-1) + \frac{7}{6}(z-1)^2 + \frac{1}{6}(z-1)^3}{(z-1)^4} = \frac{z(z^2+4z+1)}{6(z-1)^4}.$$

2004-D1-P6 — Solution — Record 2 of 2

For this record: Theory: T-AL-COMPLEXANALYSIS; T-AN-CONVERGENCE. Methods: M-AL-ANALYTIC. Full learning route

From the well-known series for the cotangent function,

$$\lim_{N\to\infty} \sum_{k=-N}^{N} \frac{1}{w + 2\pi i \cdot k} = \frac{i}{2} \cot \frac{iw}{2}$$

and

$$\lim_{N\to\infty} \sum_{k=-N}^{N} \frac{1}{\log z + 2\pi i \cdot k} = \frac{i}{2} \cot \frac{i \log z}{2} = \frac{i}{2} \cdot i \frac{e^{2i \cdot \frac{i \log z}{2}} + 1}{e^{2i \cdot \frac{i \log z}{2}} - 1} = \frac{1}{2} + \frac{1}{z-1}.$$

Taking derivatives we obtain

$$\sum \frac{1}{(\log z)^2} = -z \cdot \left(\frac{1}{2} + \frac{1}{z-1}\right)' = \frac{z}{(z-1)^2},$$

$$\sum \frac{1}{(\log z)^3} = -\frac{z}{2} \cdot \left(\frac{z}{(z-1)^2}\right)' = \frac{z(z+1)}{2(z-1)^3}$$

and

$$\sum \frac{1}{(\log z)^4} = -\frac{z}{3} \cdot \left(\frac{z(z+1)}{2(z-1)^3}\right)' = \frac{z(z^2+4z+1)}{6(z-1)^4}.$$

Day 2

2004-D2-P1 — Statement

Day 2, P7 | 20 points

Let A be a real 4×2 matrix and B be a real 2×4 matrix such that

$$AB = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}.$$

Find BA.

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Theory and methods for 2004-D2-P1

Theory Determinants, block identities and inverses [T-LA-DETERMINANT]; Rank, dimension and exact linear constraints [T-LA-RANK]
Methods Exploit a compatible block decomposition [M-LA-BLOCK]; Complete the equation to a product equal to identity [M-LA-INVERSE]

Theory and methods for 2004-D2-P1 (continued)

Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2004-D2-P1 — Solution

For this record: Theory: T-LA-DETERMINANT; T-LA-RANK. Methods: M-LA-BLOCK; M-LA-INVERSE. Full learning route

Let $A = \begin{pmatrix} A_1 \\ A_2 \end{pmatrix}$ and $B = \begin{pmatrix} B_1 & B_2 \end{pmatrix}$ where A_1, A_2, B_1, B_2 are 2×2 matrices. Then

$$\begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} A_1 \\ A_2 \end{pmatrix} \begin{pmatrix} B_1 & B_2 \end{pmatrix} = \begin{pmatrix} A_1B_1 & A_1B_2 \\ A_2B_1 & A_2B_2 \end{pmatrix}$$

therefore, $A_1B_1 = A_2B_2 = I_2$ and $A_1B_2 = A_2B_1 = -I_2$. Then $B_1 = A_1^{-1}$, $B_2 = -A_1^{-1}$ and $A_2 = B_2^{-1} = -A_1$. Finally,

$$BA = \begin{pmatrix} B_1 & B_2 \end{pmatrix} \begin{pmatrix} A_1 \\ A_2 \end{pmatrix} = B_1A_1 + B_2A_2 = 2I_2 = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

2004-D2-P2 — Statement

Day 2, P8 | 20 points

Let $f, g : [a, b] \to [0, \infty)$ be continuous and non-decreasing functions such that for each $x \in [a, b]$ we have

$$\int_a^x \sqrt{f(t)}\, dt \leq \int_a^x \sqrt{g(t)}\, dt$$

and $\int_a^b \sqrt{f(t)}\, dt = \int_a^b \sqrt{g(t)}\, dt$.

Prove that $\int_a^b \sqrt{1 + f(t)}\, dt \geq \int_a^b \sqrt{1 + g(t)}\, dt$.

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Theory and methods for 2004-D2-P2

Theory Convex graph length and ordered primitives [T-AN-ARCLENGTH]; Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]

Methods Choose the correct convexity comparison [M-AN-CONVEX]; Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]

Strategy Translate geometry into a controlled representation [S-GEOMETRIC]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2004-D2-P2 — Solution

For this record: Theory: T-AN-ARCLENGTH; T-AN-CONVEXITY; T-AN-INTEGRAL; T-DS-GEOMETRY. Methods: M-AN-CONVEX; M-AN-PRIMITIVE. Full learning route

Let $F(x) = \int_a^x \sqrt{f(t)}\, dt$ and $G(x) = \int_a^x \sqrt{g(t)}\, dt$. The functions F, G are convex, $F(a) = 0 = G(a)$ and $F(b) = G(b)$ by the hypothesis. We are supposed to show that

$$\int_a^b \sqrt{1 + (F'(t))^2}\, dt \geq \int_a^b \sqrt{1 + (G'(t))^2}\, dt$$

i.e. The length ot the graph of F is $\geq$ the length of the graph of G. This is clear since both functions are convex, their graphs have common ends and the graph of F is below the graph of G — the length of the graph of F is the least upper bound of the lengths of the graphs of piecewise linear

2004-D2-P2 — Solution (continued)

functions whose values at the points of non-differentiability coincide with the values of F, if a convex polygon P_1 is contained in a polygon P_2 then the perimeter of P_1 is $\leq$ the perimeter of P_2.

2004-D2-P3 — Statement

Day 2, P9 | 20 points

Let D be the closed unit disk in the plane, and let $p_1, p_2, \ldots, p_n$ be fixed points in D. Show that there exists a point p in D such that the sum of the distances of p to each of $p_1, p_2, \ldots, p_n$ is greater than or equal to 1.

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Theory and methods for 2004-D2-P3

Theory Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Inner products, orthogonality and polarization [T-LA-INNER]
Methods Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Strategy Translate geometry into a controlled representation [S-GEOMETRIC]; Build a witness and verify every constraint [S-CONSTRUCTION]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2004-D2-P3 — Solution

For this record: Theory: T-DS-GEOMETRY; T-LA-INNER. Methods: M-DS-GEOMETRIC. Full learning route

considering as vectors, thoose p to be the unit vector which points into the opposite direction as $\sum_{i=1}^{n} p_i$. Then, by the triangle inequality,

$$\sum_{i=1}^{n} |p - p_i| \geq \left| np - \sum_{i=1}^{n} p_i \right| = n + \left| \sum_{i=1}^{n} p_i \right| \geq n..$$

2004-D2-P4 — Statement

Day 2, P10 | 20 points

For $n \geq 1$ let M be an $n \times n$ complex matrix with distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_k$, with multiplicities $m_1, m_2, \ldots, m_k$, respectively. Consider the linear operator L_M defined by $L_M(X) = MX + XM^T$, for any complex $n \times n$ matrix X. Find its eigenvalues and their multiplicities. (M^T denotes the transpose of M; that is, if $M = (m_{k,l})$, then $M^T = (m_{l,k})$.)

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Theory and methods for 2004-D2-P4

Theory Tensor products and operators on matrix spaces [T-LA-TENSOR]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Continuity, compactness, and order [T-AN-CONTINUITY]
Methods Treat the unknown matrix as a vector [M-LA-MATRIXOP]; Lift products of inner products to new vectors [M-LA-TENSOR]
Strategy Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2004-D2-P4 — Solution

For this record: Theory: T-LA-TENSOR; T-LA-SPECTRAL; T-LA-POLYNOMIAL; T-AN-CONTINUITY. Methods: M-LA-MATRIXOP; M-LA-TENSOR. Full learning route

We first solve the problem for the special case when the eigenvalues of M are distinct and all sums $\lambda_r + \lambda_s$ are different. Let λ_r and λ_s be two eigenvalues of M and $\vec{v}_r$, $\vec{v}_s$ eigenvectors associated to them, i.e. $M\vec{v}_j = \lambda\vec{v}_j$ for $j = r, s$. We have $M\vec{v}_r(\vec{v}_s)^T + \vec{v}_r(\vec{v}_s)^T M^T = (M\vec{v}_r)(\vec{v}_s)^T + \vec{v}_r(M\vec{v}_s)^T = \lambda_r\vec{v}_r(\vec{v}_s)^T + \lambda_s\vec{v}_r(\vec{v}_s)^T$, so $\vec{v}_r(\vec{v}_s)$
is an eigenmatrix of L_M with the eigenvalue $\lambda_r + \lambda_s$.
Notice that if $\lambda_r \neq \lambda_s$ then vectors $\vec{u}, \vec{w}$ are linearly independent and matrices $\vec{u}(\vec{w})^T$ and $\vec{w}(\vec{u})^T$ are linearly independent, too. This implies that the eigenvalue $\lambda_r + \lambda_s$ is double if $r \neq s$.
The map L_M maps n^2–dimensional linear space into itself, so it has at most n^2 eigenvalues. We already found n^2 eigenvalues, so there exists no more and the problem is solved for the special case.

In the general case, matrix M is a limit of matrices $M_1, M_2, \ldots$ such that each of them belongs to the special case above. By the continuity of the eigenvalues we obtain that the eigenvalues of L_M are

- $2\lambda_r$ with multiplicity m_r^2 $(r = 1, \ldots, k)$;
- $\lambda_r + \lambda_s$ with multiplicity $2m_r m_s$ $(1 \leq r < s \leq k)$.

(It can happen that the sums $\lambda_r + \lambda_s$ are not pairwise different; for those multiple values the multiplicities should be summed up.)

2004-D2-P5 — Statement

Day 2, P11 | 20 points

Prove that

$$\int_0^1 \int_0^1 \frac{dx\,dy}{x^{-1} + |\ln y| - 1} \leq 1.$$

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Theory and methods for 2004-D2-P5

Theory Nonnegative sums, Fubini, and integral kernels [T-AN-FUBINI]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]
Methods Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]; Pair an integral through symmetry [M-AN-PAIR]; Choose the correct convexity comparison [M-AN-CONVEX]
Strategy Build a chain of bounds and track equality [S-BOUND]; Control the limiting operation before taking the limit [S-ANALYTIC]

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2004-D2-P5 — Solution — Record 1 of 2

For this record: Theory: T-AN-FUBINI; T-AN-INTEGRAL; T-AN-DERIVATIVE. Methods: M-AN-PRIMITIVE; M-AN-PAIR. Full learning route

First we use the inequality

$$x^{-1} - 1 \geq |\ln x|, \quad x \in (0, 1],$$

which follows from

$$(x^{-1} - 1)\big|_{x=1} = |\ln x|\big|_{x=1} = 0,$$

$$(x^{-1} - 1)' = -\frac{1}{x^2} \leq -\frac{1}{x} = |\ln x|', \quad x \in (0, 1].$$

Therefore

$$\int_0^1 \int_0^1 \frac{dx\,dy}{x^{-1} + |\ln y| - 1} \leq \int_0^1 \int_0^1 \frac{dx\,dy}{|\ln x| + |\ln y|} = \int_0^1 \int_0^1 \frac{dx\,dy}{|\ln(x \cdot y)|}.$$

2004-D2-P5 — Solution — Record 1 of 2 (continued)

Substituting $y = u/x$, we obtain

$$\int_0^1 \int_0^1 \frac{dx\,dy}{|\ln(x\cdot y)|} = \int_0^1 \left(\int_u^1 \frac{dx}{x}\right) \frac{du}{|\ln u|} = \int_0^1 |\ln u| \cdot \frac{du}{|\ln u|} = 1.$$

2004-D2-P5 — Solution — Record 2 of 2

For this record: Theory: T-AN-FUBINI; T-AN-INTEGRAL; T-AN-CONVEXITY. Methods: M-AN-PAIR; M-AN-CONVEX. Full learning route

Substituting $s = x^{-1} - 1$ and $u = s - \ln y$,

$$\int_0^1 \int_0^1 \frac{dx\,dy}{x^{-1} + |\ln y| - 1} = \int_0^\infty \int_s^\infty \frac{e^{s-u}}{(s+1)^2 u}\,du\,ds = \int_0^\infty \left(\int_0^u \frac{e^s}{(s+1)^2}\,ds\right) \frac{e^{-u}}{u}\,du.$$

Since the function $\dfrac{e^s}{(s+1)^2}$ is convex,

$$\int_0^u \frac{e^s}{(s+1)^2}\,ds \le \frac{u}{2}\left(\frac{e^u}{(u+1)^2} + 1\right)$$

so

$$\int_0^1 \int_0^1 \frac{dx\,dy}{x^{-1} + |\ln y| - 1} \le \int_0^\infty \frac{u}{2}\left(\frac{e^u}{(u+1)^2} + 1\right) \frac{e^{-u}}{u}\,du = \frac{1}{2}\left(\int_0^\infty \frac{du}{(u+1)^2} + \int_0^\infty e^{-u}\,du\right) = 1.$$

2004-D2-P6 — Statement

Day 2, P12 | 20 points

For $n \ge 0$ define matrices A_n and B_n as follows: $A_0 = B_0 = (1)$ and for every $n > 0$

$$A_n = \begin{pmatrix} A_{n-1} & A_{n-1} \\ A_{n-1} & B_{n-1} \end{pmatrix} \quad \text{and} \quad B_n = \begin{pmatrix} A_{n-1} & A_{n-1} \\ A_{n-1} & 0 \end{pmatrix}.$$

Denote the sum of all elements of a matrix M by $S(M)$. Prove that $S(A_n^{k-1}) = S(A_k^{n-1})$ for every $n, k \ge 1$.

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Theory and methods for 2004-D2-P6

Theory Finite counting and incidence identities [T-DS-COUNTING]; Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]; Rank, dimension and exact linear constraints [T-LA-RANK]
Methods Interpret matrix multiplication as a count [M-LA-WALKS]; Exploit a compatible block decomposition [M-LA-BLOCK]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2004-D2-P6 — Solution

For this record: Theory: T-DS-COUNTING; T-DS-GRAPH; T-LA-RANK. Methods: M-LA-WALKS; M-LA-BLOCK. Full learning route

The quantity $S(A_n^{k-1})$ has a special combinatorical meaning. Consider an $n \times k$ table filled with 0's and 1's such that no 2×2 contains only 1's. Denote the number of such fillings by F_{nk}. The filling of each row of the table corresponds to some integer ranging from 0 to $2^n - 1$ written in base 2. F_{nk} equals to the number of k-tuples of integers such that every two consecutive integers correspond to the filling of $n \times 2$ table without 2×2 squares filled with 1's.

2004-D2-P6 — Solution (continued)

Consider binary expansions of integers i and j: $i_n i_{n-1} \dots i_1$ and $j_n j_{n-1} \dots j_1$. There are two cases:
1. If $i_n j_n = 0$ then i and j can be consecutive iff $i_{n-1} \dots i_1$ and $j_{n-1} \dots j_1$ can be consequtive.
2. If $i_n = j_n = 1$ then i and j can be consecutive iff $i_{n-1} j_{n-1} = 0$ and $i_{n-2} \dots i_1$ and $j_{n-2} \dots j_1$ can be consecutive.
Hence i and j can be consecutive iff $(i{+}1, j{+}1)$-th entry of A_n is 1. Denoting this entry by $a_{i,j}$, the sum $S(A_n^{k-1}) = \sum_{i_1=0}^{2^n-1} \dots \sum_{i_k=0}^{2^n-1} a_{i_1 i_2} a_{i_2 i_3} \dots a_{i_{k-1} i_k}$ counts the possible fillings. Therefore $F_{nk} = S(A_n^{k-1})$.

The the obvious statement $F_{nk} = F_{kn}$ completes the proof.

3.12 2005

Day 1

2005-D1-P1 — Statement

Day 1, P1

Let A be the $n \times n$ matrix, whose (i,j)th entry is $i + j$ for all $i, j = 1, 2, \dots, n$. What is the rank of A?

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Theory and methods for 2005-D1-P1

Theory Rank, dimension and exact linear constraints [T-LA-RANK]; Determinants, block identities and inverses [T-LA-DETERMINANT]
Methods Prove a dimension obstruction and attain it [M-LA-DIMENSION]; Eliminate systematically and track the invariant [M-LA-ELIM]
Strategy Build a chain of bounds and track equality [S-BOUND]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2005-D1-P1 — Solution — Record 1 of 2

For this record: Theory: T-LA-RANK; T-LA-DETERMINANT. Methods: M-LA-DIMENSION. Full learning route

For $n = 1$ the rank is 1. Now assume $n \geq 2$. Since $A = (i)_{i,j=1}^{n} + (j)_{i,j=1}^{n}$, matrix A is the sum of two matrixes of rank 1. Therefore, the rank of A is at most 2. The determinant of the top-left 2×2 minor is -1, so the rank is exactly 2.

Therefore, the rank of A is 1 for $n = 1$ and 2 for $n \geq 2$.

2005-D1-P1 — Solution — Record 2 of 2

For this record: Theory: T-LA-RANK. Methods: M-LA-ELIM. Full learning route

Consider the case $n \geq 2$. For $i = n, n-1, \dots, 2$, subtract the $(i-1)$th row from the nth row. Then subtract the second row from all lower rows.

$$\operatorname{rank} \begin{pmatrix} 2 & 3 & \dots & n+1 \\ 3 & 4 & \dots & n+2 \\ \vdots & & \ddots & \vdots \\ n+1 & n+2 & \dots & 2n \end{pmatrix} = \operatorname{rank} \begin{pmatrix} 2 & 3 & \dots & n+1 \\ 1 & 1 & \dots & 1 \\ \vdots & & \ddots & \vdots \\ 1 & 1 & \dots & 1 \end{pmatrix} = \operatorname{rank} \begin{pmatrix} 1 & 2 & \dots & n \\ 1 & 1 & \dots & 1 \\ 0 & 0 & \dots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix} = 2.$$

2005-D1-P2 — Statement

Day 1, P2

For an integer $n \geq 3$ consider the sets

$$S_n = \{(x_1, x_2, \dots, x_n) : \forall i \; x_i \in \{0, 1, 2\}\}$$

$$A_n = \{(x_1, x_2, \dots, x_n) \in S_n : \forall i \leq n-2 \; |\{x_i, x_{i+1}, x_{i+2}\}| \neq 1\}$$

and

$$B_n = \{(x_1, x_2, \dots, x_n) \in S_n : \forall i \leq n-1 \; (x_i = x_{i+1} \Rightarrow x_i \neq 0)\}.$$

Prove that $|A_{n+1}| = 3 \cdot |B_n|$.

($|A|$ denotes the number of elements of the set A.)

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Theory and methods for 2005-D1-P2

Theory Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]

Methods Partition into states and propagate a recurrence [M-DS-STATE]; Encode, decode, and verify a bijection [M-DS-BIJECTION]; Strengthen the statement before the induction step [M-DS-INDUCT]

Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2005-D1-P2 — Solution — Record 1 of 2

For this record: Theory: T-DS-INDUCTION; T-DS-COUNTING. Methods: M-DS-STATE; M-DS-INDUCT. Full learning route

Extend the definitions also for $n = 1, 2$. Consider the following sets

$$A'_n = \{(x_1, x_2, \dots, x_n) \in A_n : x_n = x_{n-1}\}, \quad A''_n = A_n \setminus A'_n,$$

$$B'_n = \{(x_1, x_2, \dots, x_n) \in B_n : x_n = 0\}, \quad B''_n = B_n \setminus B'_n$$

and denote $a_n = |A_n|$, $a'_n = |A'_n|$, $a''_n = |A''_n|$, $b_n = |B_n|$, $b'_n = |B'_n|$, $b''_n = |B''_n|$.
It is easy to observe the following relations between the a–sequences

$$\begin{cases} a_n = a'_n + a''_n \\ a'_{n+1} = a''_n \\ a''_{n+1} = 2a'_n + 2a''_n \end{cases},$$

which lead to $a_{n+1} = 2a_n + 2a_{n-1}$.
For the b–sequences we have the same relations

$$\begin{cases} b_n = b'_n + b''_n \\ b'_{n+1} = b''_n \\ b''_{n+1} = 2b'_n + 2b''_n \end{cases},$$

therefore $b_{n+1} = 2b_n + 2b_{n-1}$.

By computing the first values of (a_n) and (b_n) we obtain

$$\begin{cases} a_1 = 3, \; a_2 = 9, \; a_3 = 24 \\ b_1 = 3, \; b_2 = 8 \end{cases}$$

which leads to

$$\begin{cases} a_2 = 3b_1 \\ a_3 = 3b_2 \end{cases}$$

Now, reasoning by induction, it is easy to prove that $a_{n+1} = 3b_n$ for every $n \geq 1$.

2005-D1-P2 — Solution — Record 2 of 2

For this record: Theory: T-DS-COUNTING; T-AL-CONGRUENCE. Methods: M-DS-BIJECTION. Full learning route

Regarding x_i to be elements of $\mathbb{Z}_3$ and working "modulo 3", we have that

$$(x_1, x_2, \dots, x_n) \in A_n \Rightarrow (x_1+1, x_2+1, \dots, x_n+1) \in A_n,\ (x_1+2, x_2+2, \dots, x_n+2) \in A_n$$

which means that $1/3$ of the elements of A_n start with 0. We establish a bijection between the subset of all the vectors in A_{n+1} which start with 0 and the set B_n by

$$(0, x_1, x_2, \dots, x_n) \in A_{n+1} \longmapsto (y_1, y_2, \dots, y_n) \in B_n$$

$$y_1 = x_1,\ y_2 = x_2 - x_1,\ y_3 = x_3 - x_2,\ \dots,\ y_n = x_n - x_{n-1}$$

(if $y_k = y_{k+1} = 0$ then $x_k - x_{k-1} = x_{k+1} - x_k = 0$ (where $x_0 = 0$), which gives $x_{k-1} = x_k = x_{k+1}$, which is not possible because of the definition of the sets A_p; therefore, the definition of the above function is correct).

The inverse is defined by

$$(y_1, y_2, \dots, y_n) \in B_n \longmapsto (0, x_1, x_2, \dots, x_n) \in A_{n+1}$$

$$x_1 = y_1,\ x_2 = y_1 + y_2,\ \dots,\ x_n = y_1 + y_2 + \cdots + y_n$$

2005-D1-P3 — Statement

Day 1, P3

Let $f : \mathbb{R} \to [0, \infty)$ be a continuously differentiable function. Prove that

$$\left| \int_0^1 f^3(x)\,dx - f^2(0) \int_0^1 f(x)\,dx \right| \leq \max_{0 \leq x \leq 1} |f'(x)| \left(\int_0^1 f(x)\,dx \right)^2.$$

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Theory and methods for 2005-D1-P3

Theory Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Methods Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]
Strategy Build a chain of bounds and track equality [S-BOUND]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2005-D1-P3 — Solution — Record 1 of 2

For this record: Theory: T-AN-INTEGRAL; T-AN-DERIVATIVE. Methods: M-AN-PRIMITIVE. Full learning route

Let $M = \max\limits_{0 \leq x \leq 1} |f'(x)|$. By the inequality $-M \leq f'(x) \leq M$, $x \in [0,1]$ it follows:

$$-Mf(x) \leq f(x)f'(x) \leq Mf(x), \quad x \in [0,1].$$

By integration

$$-M \int_0^x f(t)\,dt \leq \frac{1}{2}f^2(x) - \frac{1}{2}f^2(0) \leq M \int_0^x f(t)\,dt, \quad x \in [0,1]$$

$$-2Mf(x) \int_0^x f(t)\,dt \leq f^3(x) - f^2(0)f(x) \leq 2Mf(x) \int_0^x f(t)\,dt, \quad x \in [0,1].$$

2005-D1-P3 — Solution — Record 1 of 2 (continued)

Integrating the last inequality on $[0,1]$ it follows that

$$-M\left(\int_0^1 f(x)\,dx\right)^2 \le \int_0^1 f^3(x)\,dx - f^2(0)\int_0^1 f(x)\,dx,$$

$$\int_0^1 f^3(x)\,dx - f^2(0)\int_0^1 f(x)\,dx \le M\left(\int_0^1 f(x)\,dx\right)^2.$$

Equivalently,

$$\left|\int_0^1 f^3(x)\,dx - f^2(0)\int_0^1 f(x)\,dx\right| \le M\left(\int_0^1 f(x)\,dx\right)^2.$$

2005-D1-P3 — Solution — Record 2 of 2

For this record: Theory: T-AN-INTEGRAL; T-AN-DERIVATIVE. Methods: M-AN-PRIMITIVE. Full learning route

Let $M = \max\limits_{0\le x\le 1}|f'(x)|$ and $F(x) = -\int_x^1 f$; then $F' = f$, $F(0) = -\int_0^1 f$ and $F(1) = 0$. Integrating by parts,

$$\begin{aligned}
&\int_0^1 f^3 \\
&= \int_0^1 f^2\cdot F' \\
&= [f^2F]_0^1 - \int_0^1 (f^2)'F \\
&= f^2(1)F(1) - f^2(0)F(0) - \int_0^1 2Fff' \\
&= f^2(0)\int_0^1 f - \int_0^1 2Fff'.
\end{aligned}$$

Since $f\ge 0$ and $F\le 0$, we obtain

$$\begin{aligned}
&\left|\int_0^1 f^3(x)\,dx - f^2(0)\int_0^1 f(x)\,dx\right| \\
&= \left|\int_0^1 2Fff'\right| \\
&\le \int_0^1 2|F|f|f'| \\
&\le M\int_0^1 (-2Ff) \\
&= -M\cdot[F^2]_0^1 \\
&= M\left(\int_0^1 f\right)^2.
\end{aligned}$$

2005-D1-P4 — Statement

Day 1, P4

Find all polynomials $P(x) = a_nx^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$ ($a_n \ne 0$) satisfying the following two conditions:
(i) $(a_0, a_1, \ldots, a_n)$ is a permutation of the numbers $(0, 1, \ldots, n)$

2005-D1-P4 — Statement (continued)

and

(ii) all roots of $P(x)$ are rational numbers.

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Theory and methods for 2005-D1-P4

Theory Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]; Polynomial factorization and coprime divisibility [T-AL-FACTORIZATION]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Prime valuations, divisors, and factorials [T-AL-VALUATION]

Methods Translate root restrictions into symmetric coefficients [M-AL-VIETA]; Factor first and compare irreducible exponents [M-AL-FACTOR]

Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2005-D1-P4 — Solution — Record 1 of 2

For this record: Theory: T-AL-ROOTS; T-AL-FACTORIZATION; T-AN-INEQUALITIES. Methods: M-AL-VIETA. Full learning route

Note that $P(x)$ does not have any positive root because $P(x) > 0$ for every $x > 0$. Thus, we can represent them in the form $-\alpha_i$, $i = 1, 2, \dots, n$, where $\alpha_i \geq 0$. If $a_0 \neq 0$ then there is a $k \in \mathbb{N}$, $1 \leq k \leq n-1$, with $a_k = 0$, so using Viete's formulae we get

$$\alpha_1\alpha_2 \dots \alpha_{n-k-1}\alpha_{n-k} + \alpha_1\alpha_2 \dots \alpha_{n-k-1}\alpha_{n-k+1} + \dots + \alpha_{k+1}\alpha_{k+2} \dots \alpha_{n-1}\alpha_n = \frac{a_k}{a_n} = 0,$$

which is impossible because the left side of the equality is positive. Therefore $a_0 = 0$ and one of the roots of the polynomial, say α_n, must be equal to zero. Consider the polynomial $Q(x) = a_n x^{n-1} + a_{n-1}x^{n-2} + \dots + a_1$. It has zeros $-\alpha_i$, $i = 1, 2, \dots, n-1$. Again, Viete's formulae, for $n \geq 3$, yield:

$$\alpha_1\alpha_2 \dots \alpha_{n-1} = \frac{a_1}{a_n} \tag{1}$$

$$\alpha_1\alpha_2 \dots \alpha_{n-2} + \alpha_1\alpha_2 \dots \alpha_{n-3}\alpha_{n-1} + \dots + \alpha_2\alpha_3 \dots \alpha_{n-1} = \frac{a_2}{a_n} \tag{2}$$

$$\alpha_1 + \alpha_2 + \dots + \alpha_{n-1} = \frac{a_{n-1}}{a_n}. \tag{3}$$

Dividing (2) by (1) we get

$$\frac{1}{\alpha_1} + \frac{1}{\alpha_2} + \dots + \frac{1}{\alpha_{n-1}} = \frac{a_2}{a_1}. \tag{4}$$

From (3) and (4), applying the AM-HM inequality we obtain

$$\frac{a_{n-1}}{(n-1)a_n} = \frac{\alpha_1 + \alpha_2 + \dots + \alpha_{n-1}}{n-1} \geq \frac{n-1}{\frac{1}{\alpha_1} + \frac{1}{\alpha_2} + \dots + \frac{1}{\alpha_{n-1}}} = \frac{(n-1)a_1}{a_2},$$

therefore $\frac{a_2 a_{n-1}}{a_1 a_n} \geq (n-1)^2$. Hence $\frac{n^2}{2} \geq \frac{a_2 a_{n-1}}{a_1 a_n} \geq (n-1)^2$, implying $n \leq 3$. So, the only polynomials possibly satisfying (i) and (ii) are those of degree at most three. These polynomials can easily be found and they are $P(x) = x$, $P(x) = x^2 + 2x$, $P(x) = 2x^2 + x$, $P(x) = x^3 + 3x^2 + 2x$ and $P(x) = 2x^3 + 3x^2 + x$.

2005-D1-P4 — Solution — Record 2 of 2

For this record: Theory: T-AL-FACTORIZATION; T-AL-POLYNOMIAL; T-AL-VALUATION. Methods: M-AL-FACTOR. Full learning route

Consider the prime factorization of P in the ring $\mathbb{Z}[x]$. Since all roots of P are rational, P can be written as a product of n linear polynomials with rational coefficients. Therefore, all prime factor of P are linear and P can be written as

$$P(x) = \prod_{k=1}^{n} (b_k x + c_k)$$

where the coefficients b_k, c_k are integers. Since the leading coefficient of P is positive, we can assume $b_k > 0$ for all k. The coefficients of P are nonnegative, so P cannot have a positive root. This implies $c_k \geq 0$. It is not possible that $c_k = 0$ for two different values of k, because it would imply $a_0 = a_1 = 0$. So $c_k > 0$ in at least $n-1$ cases.
Now substitute $x = 1$.

$$P(1) = a_n + \cdots + a_0 = 0 + 1 + \cdots + n = \frac{n(n+1)}{2} = \prod_{k=1}^{n} (b_k + c_k) \geq 2^{n-1};$$

therefore it is necessary that $2^{n-1} \leq \frac{n(n+1)}{2}$, therefore $n \leq 4$. Moreover, the number $\frac{n(n+1)}{2}$ can be written as a product of $n-1$ integers greater than 1.
If $n = 1$, the only solution is $P(x) = 1x + 0$.
If $n = 2$, we have $P(1) = 3 = 1 \cdot 3$, so one factor must be x, the other one is $x + 2$ or $2x + 1$. Both $x(x+2) = 1x^2 + 2x + 0$ and $x(2x+1) = 2x^2 + 1x + 0$ are solutions.
If $n = 3$, then $P(1) = 6 = 1 \cdot 2 \cdot 3$, so one factor must be x, another one is $x+1$, the third one is again $x + 2$ or $2x + 1$. The two polynomials are $x(x+1)(x+2) = 1x^3 + 3x^2 + 2x + 0$ and $x(x+1)(2x+1) = 2x^3 + 3x^2 + 1x + 0$, both have the proper set of coefficients.
In the case $n = 4$, there is no solution because $\frac{n(n+1)}{2} = 10$ cannot be written as a product of 3 integers greater than 1.

Altogether we found 5 solutions: $1x+0$, $1x^2+2x+0$, $2x^2+1x+0$, $1x^3+3x^2+2x+0$ and $2x^3+3x^2+1x+0$.

2005-D1-P5 — Statement

Day 1, P5

Let $f : (0, \infty) \to \mathbb{R}$ be a twice continuously differentiable function such that

$$|f''(x) + 2xf'(x) + (x^2 + 1)f(x)| \leq 1$$

for all x. Prove that $\lim_{x \to \infty} f(x) = 0$.

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Theory and methods for 2005-D1-P5

Theory Differential inequalities and integrating factors [T-AN-ODE]; Scale separation and limit comparisons [T-AN-ASYMPTOTICS]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Methods Design an auxiliary derivative [M-AN-AUXILIARY]; Compare growth through derivatives [M-AN-RATIO]
Strategy Control the limiting operation before taking the limit [S-ANALYTIC]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2005-D1-P5 — Solution — Record 1 of 2

For this record: Theory: T-AN-ODE; T-AN-ASYMPTOTICS; T-AN-INTEGRAL. Methods: M-AN-AUXILIARY; M-AN-RATIO. Full learning route

Let $g(x) = f'(x) + xf(x)$; then $f''(x) + 2xf'(x) + (x^2 + 1)f(x) = g'(x) + xg(x)$.

2005-D1-P5 — Solution — Record 1 of 2 (continued)

We prove that if h is a continuously differentiable function such that $h'(x)+xh(x)$ is bounded then $\lim_{\infty} h = 0$. Applying this lemma for $h = g$ then for $h = f$, the statement follows.

Let M be an upper bound for $|h'(x)+xh(x)|$ and let $p(x) = h(x)e^{x^2/2}$. (The function $e^{-x^2/2}$ is a solution of the differential equation $u'(x) + xu(x) = 0$.) Then

$$|p'(x)| = |h'(x) + xh(x)|e^{x^2/2} \leq Me^{x^2/2}$$

and

$$|h(x)| = \left|\frac{p(x)}{e^{x^2/2}}\right| = \left|\frac{p(0) + \int_0^x p'}{e^{x^2/2}}\right| \leq \frac{|p(0)| + M\int_0^x e^{x^2/2}\,dx}{e^{x^2/2}}.$$

Since $\lim_{x\to\infty} e^{x^2/2} = \infty$ and $\lim_{x\to\infty} \frac{\int_0^x e^{x^2/2}\,dx}{e^{x^2/2}} = 0$ (by L'Hospital's rule), this implies $\lim_{x\to\infty} h(x) = 0$.

2005-D1-P5 — Solution — Record 2 of 2

For this record: Theory: T-AN-ODE; T-AN-ASYMPTOTICS; T-AN-DERIVATIVE. Methods: M-AN-RATIO; M-AN-AUXILIARY. Full learning route

Apply L'Hospital rule twice on the fraction $\frac{f(x)e^{x^2/2}}{e^{x^2/2}}$. (Note that L'Hospital rule is valid if the denominator converges to infinity, without any assumption on the numerator.)

$$\begin{aligned}
&\lim_{x\to\infty} f(x) \\
&= \lim_{x\to\infty} \frac{f(x)e^{x^2/2}}{e^{x^2/2}} \\
&= \lim_{x\to\infty} \frac{(f'(x) + xf(x))e^{x^2/2}}{xe^{x^2/2}} \\
&= \lim_{x\to\infty} \frac{(f''(x) + 2xf'(x) + (x^2+1)f(x))e^{x^2/2}}{(x^2+1)e^{x^2/2}} \\
&= \lim_{x\to\infty} \frac{f''(x) + 2xf'(x) + (x^2+1)f(x)}{x^2+1} = 0.
\end{aligned}$$

2005-D1-P6 — Statement

Day 1, P6

Given a group G, denote by $G(m)$ the subgroup generated by the mth powers of elements of G. If $G(m)$ and $G(n)$ are commutative, prove that $G(\gcd(m,n))$ is also commutative. ($\gcd(m,n)$ denotes the greatest common divisor of m and n.)

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Theory and methods for 2005-D1-P6

Theory Groups, orders, generators, and quotients [T-AL-GROUP]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Methods Polarize identities and isolate commutators [M-AL-COMMUTATOR]; Prove generation or obstruct it by an image [M-AL-GENERATORS]
Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2005-D1-P6 — Solution

For this record: Theory: T-AL-GROUP; T-AL-CONGRUENCE. Methods: M-AL-COMMUTATOR; M-AL-GENERATORS. Full learning route

Write $d = \gcd(m,n)$. It is easy to see that $\langle G(m), G(n)\rangle = G(d)$; hence, it will suffice to check commutativity for any two elements in $G(m) \cup G(n)$, and so for any two generators a^m and b^n. Consider their commutator $z = a^{-m}b^{-n}a^m b^n$; then the relations

$$z = (a^{-m}ba^m)^{-n}b^n = a^{-m}(b^{-n}ab^n)^m$$

show that $z \in G(m) \cap G(n)$. But then z is in the center of $G(d)$. Now, from the relation $a^m b^n = b^n a^m z$, it easily follows by induction that

$$a^{ml}b^{nl} = b^{nl}a^{ml}z^{l^2}.$$

Setting $l = m/d$ and $l = n/d$ we obtain $z^{(m/d)^2} = z^{(n/d)^2} = e$, but this implies that $z = e$ as well.

Day 2

2005-D2-P1 — Statement

Day 2, P7

Let $f(x) = x^2 + bx + c$, where b and c are real numbers, and let

$$M = \{x \in \mathbb{R} : |f(x)| < 1\}.$$

Clearly the set M is either empty or consists of disjoint open intervals. Denote the sum of their lengths by $|M|$. Prove that

$$|M| \le 2\sqrt{2}.$$

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Theory and methods for 2005-D2-P1

Theory Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]
Methods Compare coefficients after a structural substitution [M-AL-COEFFICIENT]
Strategy Build a chain of bounds and track equality [S-BOUND]; Prove necessity and sufficiency separately [S-CLASSIFICATION]

Statement Recorded solutions Catalog

2005-D2-P1 — Solution

For this record: Theory: T-AL-POLYNOMIAL; T-AN-INEQUALITIES. Methods: M-AL-COEFFICIENT. Full learning route

Write $f(x) = \left(x + \frac{b}{2}\right)^2 + d$ where $d = c - \frac{b^2}{4}$. The absolute minimum of f is d.
If $d \ge 1$ then $f(x) \ge 1$ for all x, $M = \emptyset$ and $|M| = 0$.

If $-1 < d < 1$ then $f(x) > -1$ for all x,

$$-1 < \left(x + \frac{b}{2}\right)^2 + d < 1 \iff \left|x + \frac{b}{2}\right| < \sqrt{1-d}$$

so

$$M = \left(-\frac{b}{2} - \sqrt{1-d},\ -\frac{b}{2} + \sqrt{1-d}\right)$$

and

$$|M| = 2\sqrt{1-d} < 2\sqrt{2}.$$

If $d \le -1$ then

$$-1 < \left(x + \frac{b}{2}\right)^2 + d < 1 \iff \sqrt{|d|-1} < \left|x + \frac{b}{2}\right| < \sqrt{|d|+1}$$

2005-D2-P1 — Solution (continued)

so

$$M = \left(-\sqrt{|d|+1}, -\sqrt{|d|-1}\right) \cup \left(\sqrt{|d|-1}, \sqrt{|d|+1}\right)$$

and

$$|M| = 2\left(\sqrt{|d|+1} - \sqrt{|d|-1}\right) = 2\,\frac{(|d|+1)-(|d|-1)}{\sqrt{|d|+1}+\sqrt{|d|-1}} \le 2\,\frac{2}{\sqrt{1+1}+\sqrt{1-0}} = 2\sqrt{2}.$$

2005-D2-P2 — Statement

Day 2, P8

Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $(f(x))^n$ is a polynomial for every $n = 2, 3, \dots$. Does it follow that f is a polynomial?

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Theory and methods for 2005-D2-P2

Theory Polynomial factorization and coprime divisibility [T-AL-FACTORIZATION]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Factor first and compare irreducible exponents [M-AL-FACTOR]
Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2005-D2-P2 — Solution — Record 1 of 2

For this record: Theory: T-AL-FACTORIZATION; T-AL-POLYNOMIAL. Methods: M-AL-FACTOR. Full learning route

Yes, it is even enough to assume that f^2 and f^3 are polynomials.
Let $p = f^2$ and $q = f^3$. Write these polynomials in the form of

$$p = a \cdot p_1^{a_1} \cdot \dots \cdot p_k^{a_k}, \qquad q = b \cdot q_1^{b_1} \cdot \dots \cdot q_l^{b_l},$$

where $a, b \in \mathbb{R}$, $a_1, \dots, a_k, b_1, \dots b_l$ are positive integers and $p_1, \dots, p_k, q_1, \dots, q_l$ are irreducible polynomials with leading coefficients 1. For $p^3 = q^2$ and the factorisation of $p^3 = q^2$ is unique we get that $a^3 = b^2$, $k = l$ and for some $(i_1, \dots, i_k)$ permutation of $(1, \dots, k)$ we have $p_1 = q_{i_1}, \dots, p_k = q_{i_k}$ and $3a_1 = 2b_{i_1}, \dots, 3a_k = 2b_{i_k}$. Hence $b_1, \dots, b_l$ are divisible by 3 let

$r = b^{1/3} \cdot q_1^{b_1/3} \cdot \dots \cdot q_l^{b_l/3}$ be a polynomial. Since $r^3 = q = f^3$ we have $f = r$.

2005-D2-P2 — Solution — Record 2 of 2

For this record: Theory: T-AL-FACTORIZATION; T-AL-POLYNOMIAL. Methods: M-AL-FACTOR. Full learning route

Let $\frac{p}{q}$ be the simplest form of the rational function $\frac{f^3}{f^2}$. Then the simplest form of its square is $\frac{p^2}{q^2}$. On the other hand $\frac{p^2}{q^2} = \left(\frac{f^3}{f^2}\right)^2 = f^2$ is a polynomial therefore q must be a constant and so $f = \frac{f^3}{f^2} = \frac{p}{q}$ is a polynomial.

2005-D2-P3 — Statement

Day 2, P9

In the linear space of all real $n \times n$ matrices, find the maximum possible dimension of a linear subspace V such that

$$\forall X, Y \in V \quad \operatorname{trace}(XY) = 0.$$

(The trace of a matrix is the sum of the diagonal entries.)

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Theory and methods for 2005-D2-P3

Theory Trace pairing and commutator calculus [T-LA-TRACE]; Rank, dimension and exact linear constraints [T-LA-RANK]; Inner products, orthogonality and polarization [T-LA-INNER]
Methods Prove a dimension obstruction and attain it [M-LA-DIMENSION]; Use trace to extract a scalar invariant [M-LA-TRACE]
Strategy Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2005-D2-P3 — Solution

For this record: Theory: T-LA-TRACE; T-LA-RANK; T-LA-INNER. Methods: M-LA-DIMENSION; M-LA-TRACE. Full learning route

If A is a nonzero symmetric matrix, then $\operatorname{trace}(A^2) = \operatorname{trace}(A^tA)$ is the sum of the squared entries of A which is positive. So V cannot contain any symmetric matrix but 0.
Denote by S the linear space of all real $n \times n$ symmetric matrices; $\dim V = \frac{n(n+1)}{2}$.
Since $V \cap S = \{0\}$, we have $\dim V + \dim S \le n^2$ and thus $\dim V \le n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}$.
The space of strictly upper triangular matrices has dimension $\frac{n(n-1)}{2}$ and satisfies the condition of the problem.

Therefore the maximum dimension of V is $\frac{n(n-1)}{2}$.

2005-D2-P4 — Statement

Day 2, P10

Prove that if $f : \mathbb{R} \to \mathbb{R}$ is three times differentiable, then there exists a real number $\xi \in (-1, 1)$ such that

$$\frac{f'''(\xi)}{6} = \frac{f(1) - f(-1)}{2} - f'(0).$$

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Theory and methods for 2005-D2-P4

Theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Count zeros and subtract an interpolant [M-AN-ROLLE]
Strategy Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2005-D2-P4 — Solution — Record 1 of 2

For this record: Theory: T-AN-DERIVATIVE; T-AL-POLYNOMIAL. Methods: M-AN-ROLLE. Full learning route

Let

$$g(x) = -\frac{f(-1)}{2}\,x^2(x-1) - f(0)(x^2-1) + \frac{f(1)}{2}\,x^2(x+1) - f'(0)\,x(x-1)(x+1).$$

It is easy to check that $g(\pm 1) = f(\pm 1)$, $g(0) = f(0)$ and $g'(0) = f'(0)$.

Apply Rolle's theorem for the function $h(x) = f(x) - g(x)$ and its derivatives. Since $h(-1) = h(0) = h(1) = 0$, there exist $\eta \in (-1, 0)$ and $\vartheta \in (0, 1)$ such that $h'(\eta) = h'(\vartheta) = 0$. We also have $h'(0) = 0$, so there exist $\varrho \in (\eta, 0)$ and $\sigma \in (0, \vartheta)$ such that $h''(\varrho) = h''(\sigma) = 0$. Finally, there exists a $\xi \in (\varrho, \sigma) \subset (-1, 1)$ where $h'''(\xi) = 0$. Then

$$f'''(\xi) = g'''(\xi) = -\frac{f(-1)}{2} \cdot 6 - f(0) \cdot 0 + \frac{f(1)}{2} \cdot 6 - f'(0) \cdot 6 = \frac{f(1) - f(-1)}{2} - f'(0).$$

2005-D2-P4 — Solution — Record 2 of 2

For this record: Theory: T-AN-DERIVATIVE; T-AL-POLYNOMIAL. Methods: M-AN-ROLLE. Full learning route

The expression $\dfrac{f(1)-f(-1)}{2}-f'(0)$ is the divided difference $f[-1,0,0,1]$ and there exists a number $\xi\in(-1,1)$ such that $f[-1,0,0,1]=\dfrac{f'''(\xi)}{3!}$.

2005-D2-P5 — Statement

Day 2, P11

Find all $r>0$ such that whenever $f:\mathbb{R}^2\to\mathbb{R}$ is a differentiable function such that $|\operatorname{grad} f(0,0)|=1$ and $|\operatorname{grad} f(u)-\operatorname{grad} f(v)|\le|u-v|$ for all $u,v\in\mathbb{R}^2$, then the maximum of f on the disk $\{u\in\mathbb{R}^2:|u|\le r\}$ is attained at exactly one point. ($\operatorname{grad} f(u)=(\partial_1 f(u),\partial_2 f(u))$ is the gradient vector of f at the point u. For a vector $u=(a,b)$, $|u|=\sqrt{a^2+b^2}$.)

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Theory and methods for 2005-D2-P5

Theory Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Metric contractions, covers, and nearest points [T-DS-METRIC]; Lipschitz derivatives and convex smoothness [T-AN-SMOOTH]; Continuity, compactness, and order [T-AN-CONTINUITY]
Methods Contradict an extremum by a controlled perturbation [M-DS-PERTURB]; Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Strategy Translate geometry into a controlled representation [S-GEOMETRIC]; Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2005-D2-P5 — Solution

For this record: Theory: T-DS-GEOMETRY; T-DS-METRIC; T-AN-SMOOTH; T-AN-CONTINUITY. Methods: M-DS-PERTURB; M-DS-GEOMETRIC. Full learning route

To get an upper bound for r, set $f(x,y)=x-\dfrac{x^2}{2}+\dfrac{y^2}{2}$. This function satisfies the conditions, since $\operatorname{grad} f(x,y)=(1-x,y)$, $\operatorname{grad} f(0,0)=(1,0)$ and $|\operatorname{grad} f(x_1,y_1)-\operatorname{grad} f(x_2,y_2)|=|(x_2-x_1,y_1-y_2)|=|(x_1,y_1)-(x_2,y_2)|$.
In the disk $D_r=\{(x,y):x^2+y^2\le r^2\}$

$$f(x,y)=\frac{x^2+y^2}{2}-\left(x-\frac{1}{2}\right)^2+\frac{1}{4}\le\frac{r^2}{2}+\frac{1}{4}.$$

If $r>\frac{1}{2}$ then the absolute maximum is $\frac{r^2}{2}+\frac{1}{4}$, attained at the points $\left(\frac{1}{2},\pm\sqrt{r^2-\frac{1}{4}}\right)$. Therefore, it is necessary that $r\le\frac{1}{2}$ because if $r>\frac{1}{2}$ then the maximum is attained twice.

Suppose now that $r\le 1/2$ and that f attains its maximum on D_r at u,v, $u\ne v$. Since $|\operatorname{grad} f(z)-\operatorname{grad} f(0)|\le r$, $|\operatorname{grad} f(z)|\ge 1-r>0$ for all $z\in D_r$. Hence f may attain its maximum only at the boundary of D_r, so we must have $|u|=|v|=r$ and $\operatorname{grad} f(u)=au$ and $\operatorname{grad} f(v)=bv$, where $a,b\ge 0$. Since $au=\operatorname{grad} f(u)$ and $bv=\operatorname{grad} f(v)$ belong to the disk D with centre $\operatorname{grad} f(0)$ and radius r, they do not belong to the interior of D_r. Hence $|\operatorname{grad} f(u)-\operatorname{grad} f(v)|=|au-bv|\ge|u-v|$ and this inequality is strict since $D\cap D_r$ contains no more than one point. But this contradicts the assumption that $|\operatorname{grad} f(u)-\operatorname{grad} f(v)|\le|u-v|$. So all $r\le\frac{1}{2}$ satisfies the condition.

2005-D2-P6 — Statement

Day 2, P12

Prove that if p and q are rational numbers and $r = p + q\sqrt{7}$, then there exists a matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \neq \pm \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ with integer entries and with $ad - bc = 1$ such that

$$\frac{ar + b}{cr + d} = r.$$

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Theory and methods for 2005-D2-P6

Theory Quadratic norms and Pell-type constructions [T-AL-PELL]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Determinants, block identities and inverses [T-LA-DETERMINANT]
Methods Construct roots or units before the coefficients [M-AL-PARAMETRIZE]; Find a recurrence that reveals the invariant [M-AL-RECURRENCE]
Strategy Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2005-D2-P6 — Solution

For this record: Theory: T-AL-PELL; T-AL-CONGRUENCE; T-AL-POLYNOMIAL; T-LA-DETERMINANT. Methods: M-AL-PARAMETRIZE; M-AL-RECURRENCE. Full learning route

First consider the case when $q = 0$ and r is rational. Choose a positive integer t such that $r^2 t$ is an integer and set

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 + rt & -r^2 t \\ t & 1 - rt \end{pmatrix}.$$

Then

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = 1 \quad \text{and} \quad \frac{ar + b}{cr + d} = \frac{(1 + rt)r - r^2 t}{tr + (1 - rt)} = r.$$

Now assume $q \neq 0$. Let the minimal polynomial of r in $\mathbb{Z}[x]$ be $ux^2 + vx + w$. The other root of this polynomial is $\bar{r} = p - q\sqrt{7}$, so $v = -u(r + \bar{r}) = -2up$ and $w = ur\bar{r} = u(p^2 - 7q^2)$. The discriminant is $v^2 - 4uw = 7 \cdot (2uq)^2$. The left-hand side is an integer, implying that also $\Delta = 2uq$ is an integer.
The equation $\frac{ar+b}{cr+d} = r$ is equivalent to $cr^2 + (d - a)r - b = 0$. This must be a multiple of the minimal polynomial, so we need

$$c = ut, \quad d - a = vt, \quad -b = wt$$

for some integer $t \neq 0$. Putting together these equalities with $ad - bc = 1$ we obtain that

$$(a + d)^2 = (a - d)^2 + 4ad = 4 + (v^2 - 4uw)t^2 = 4 + 7\Delta^2 t^2.$$

Therefore $4 + 7\Delta^2 t^2$ must be a perfect square. Introducing $s = a + d$, we need an integer solution (s, t) for the Diophantine equation

$$s^2 - 7\Delta^2 t^2 = 4 \tag{1}$$

such that $t \neq 0$.
The numbers s and t will be even. Then $a + d = s$ and $d - a = vt$ will be even as well and a and d will be really integers.
Let $(8 \pm 3\sqrt{7})^n = k_n \pm l_n\sqrt{7}$ for each integer n. Then $k_n^2 - 7l_n^2 = (k_n + l_n\sqrt{7})(k_n - l_n\sqrt{7}) = ((8 + 3\sqrt{7})(8 - 3\sqrt{7}))^n = 1$ and the sequence (l_n) also satisfies the linear recurrence $l_{n+1} = 16l_n - l_{n-1}$. Consider the residue of l_n modulo Δ. There are Δ^2 possible residue pairs for (l_n, l_{n+1}) so some are the same. Starting from such two positions, the recurrence shows that the sequence of residues is periodic in both directions. Then there are infinitely many indices such that $l_n \equiv l_0 = 0 \pmod{\Delta}$.
Taking such an index n, we can set $s = 2k_n$ and $t = 2l_n/\Delta$.
Remarks. 1. It is well-known that if $D > 0$ is not a perfect square then the Pell-like Diophantine equation

$$x^2 - Dy^2 = 1$$

2005-D2-P6 — Solution (continued)

has infinitely many solutions. Using this fact the solution can be generalized to all quadratic algebraic numbers.

2. It is also known that the continued fraction of a real number r is periodic from a certain point if and only if r is a root of a quadratic equation. This fact can lead to another solution.

3.13 2006

Day 1

2006-D1-P1 — Statement

Day 1, P1 | 20 points

Let $f : \mathbb{R} \to \mathbb{R}$ be a real function. Prove or disprove each of the following statements.
(a) If f is continuous and range$(f) = \mathbb{R}$ then f is monotonic.
(b) If f is monotonic and range$(f) = \mathbb{R}$ then f is continuous.

(c) If f is monotonic and f is continuous then range$(f) = \mathbb{R}$.

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Theory and methods for 2006-D1-P1

Theory Continuity, compactness, and order [T-AN-CONTINUITY]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Methods Use a dense set, an enumeration, or a category argument [M-AN-TOPOLOGY]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2006-D1-P1 — Solution

For this record: Theory: T-AN-CONTINUITY; T-AN-DERIVATIVE. Methods: M-AN-TOPOLOGY. Full learning route

(a) False. Consider function $f(x) = x^3 - x$. It is continuous, range$(f) = \mathbb{R}$ but, for example, $f(0) = 0$, $f\left(\frac{1}{2}\right) = -\frac{3}{8}$ and $f(1) = 0$, therefore $f(0) > f\left(\frac{1}{2}\right)$, $f\left(\frac{1}{2}\right) < f(1)$ and f is not monotonic.
(b) True. Assume first that f is non-decreasing. For an arbitrary number a, the limits $\lim\limits_{a-} f$ and $\lim\limits_{a+} f$ exist and $\lim\limits_{a-} f \le \lim\limits_{a+} f$. If the two limits are equal, the function is continuous at a. Otherwise, if $\lim\limits_{a-} f = b < \lim\limits_{a+} f = c$, we have $f(x) \le b$ for all $x < a$ and $f(x) \ge c$ for all $x > a$; therefore range$(f) \subset (-\infty, b) \cup (c, \infty) \cup \{f(a)\}$ cannot be the complete $\mathbb{R}$.
For non-increasing f the same can be applied writing reverse relations or $g(x) = -f(x)$.

(c) False. The function $g(x) = \arctan x$ is monotonic and continuous, but range$(g) = (-\pi/2, \pi/2) \neq \mathbb{R}$.

2006-D1-P2 — Statement

Day 1, P2 | 20 points

Find the number of positive integers x satisfying the following two conditions:
1. $x < 10^{2006}$;

2. $x^2 - x$ is divisible by 10^{2006}.

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Theory and methods for 2006-D1-P2

Theory Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Prime valuations, divisors, and factorials [T-AL-VALUATION]
Methods Choose a modulus that exposes the obstruction [M-AL-MODULAR]; Construct and count compatible local residues [M-AL-CRT]
Strategy Pass between local constraints and a global conclusion [S-LOCALGLOBAL]; Choose the objects and count them without ambiguity [S-COUNTING]

Statement Recorded solutions Catalog

2006-D1-P2 — Solution — Record 1 of 2

For this record: Theory: T-AL-CONGRUENCE; T-AL-VALUATION. Methods: M-AL-MODULAR. Full learning route

Let $S_k = \{0 < x < 10^k \mid x^2 - x \text{ is divisible by } 10^k\}$ and $s(k) = |S_k|$, $k \geq 1$. Let $x = a_{k+1}a_k \ldots a_1$ be the decimal writing of an integer $x \in S_{k+1}$, $k \geq 1$. Then obviously $y = a_k \ldots a_1 \in S_k$. Now, let $y = a_k \ldots a_1 \in S_k$ be fixed. Considering a_{k+1} as a variable digit, we have $x^2 - x = (a_{k+1}10^k + y)^2 - (a_{k+1}10^k + y) = (y^2 - y) + a_{k+1}10^k(2y - 1) + a_{k+1}^2 10^{2k}$. Since $y^2 - y = 10^k z$ for an integer z, it follows that $x^2 - x$ is divisible by 10^{k+1} if and only if $z + a_{k+1}(2y - 1) \equiv 0 \pmod{10}$. Since $y \equiv 3 \pmod{10}$ is obviously impossible, the congruence has exactly one solution. Hence we obtain a one-to-one correspondence between the sets S_{k+1} and S_k for every $k \geq 1$. Therefore $s(2006) = s(1) = 3$, because $S_1 = \{1, 5, 6\}$.

2006-D1-P2 — Solution — Record 2 of 2

For this record: Theory: T-AL-CONGRUENCE; T-AL-VALUATION. Methods: M-AL-CRT. Full learning route

Since $x^2 - x = x(x - 1)$ and the numbers x and $x - 1$ are relatively prime, one of them must be divisible by 2^{2006} and one of them (may be the same) must be divisible by 5^{2006}. Therefore, x must satisfy the following two conditions:

$$x \equiv 0 \text{ or } 1 \pmod{2^{2006}};$$

$$x \equiv 0 \text{ or } 1 \pmod{5^{2006}}.$$

Altogether we have 4 cases. The Chinese remainder theorem yields that in each case there is a unique solution among the numbers $0, 1, \ldots, 10^{2006} - 1$. These four numbers are different because each two gives different residues modulo 2^{2006} or 5^{2006}. Moreover, one of the numbers is 0 which is not allowed.

Therefore there exist 3 solutions.

2006-D1-P3 — Statement

Day 1, P3 | 20 points

Let A be an $n \times n$-matrix with integer entries and $b_1, \ldots, b_k$ be integers satisfying $\det A = b_1 \cdot \cdots \cdot b_k$. Prove that there exist $n \times n$-matrices $B_1, \ldots, B_k$ with integer entries such that $A = B_1 \cdot \cdots \cdot B_k$ and $\det B_i = b_i$ for all $i = 1, \ldots, k$.

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Theory and methods for 2006-D1-P3

Theory Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Prime valuations, divisors, and factorials [T-AL-VALUATION]; Determinants, block identities and inverses [T-LA-DETERMINANT]
Methods Track one prime or one extreme divisor [M-AL-VALUATION]
Strategy Pass between local constraints and a global conclusion [S-LOCALGLOBAL]; Reduce size while preserving the defining conditions [S-INDUCTION]

Statement Recorded solutions Catalog

2006-D1-P3 — Solution

For this record: Theory: T-AL-CONGRUENCE; T-AL-VALUATION; T-LA-DETERMINANT. Methods: M-AL-VALUATION. Full learning route

By induction, it is enough to consider the case $m = 2$.
Furthermore, we can multiply A with any integral matrix with determinant 1 from the right or from the left, without changing the problem. Hence we can assume A to be upper triangular.
Lemma. Let A be an integral upper triangular matrix, and let b, c be integers satisfying $\det A = bc$. Then there exist integral upper triangular matrices B, C such that $\det B = b$, $\det C = c$, $A = BC$.
Proof. The proof is done by induction on n, the case $n = 1$ being obvious. Assume the statement is true for $n-1$. Let A, b, c as in the statement of the lemma. Define B_{nn} to be the greatest common divisor of b and A_{nn}, and put $C_{nn} = \frac{A_{nn}}{B_{nn}}$. Since A_{nn} divides bc, C_{nn} divides $\frac{b}{B_{nn}}c$, which divides c. Hence C_{nn} divides c. Therefore, $b' = \frac{b}{B_{nn}}$ and $c' = \frac{c}{C_{nn}}$ are integers. Define A' to be the upper-left $(n-1) \times (n-1)$-submatrix of A; then $\det A' = b'c'$. By induction we can find the upper-left $(n-1) \times (n-1)$-part of B and C in such a way that $\det B = b$, $\det C = c$ and $A = BC$ holds on the upper-left $(n-1) \times (n-1)$-submatrix of A. It remains to define $B_{i,n}$ and $C_{i,n}$ such that $A = BC$ also holds for the (i, n)-th entry for all $i < n$.

First we check that B_{ii} and C_{nn} are relatively prime for all $i < n$. Since B_{ii} divides b', it is certainly enough to prove that b' and C_{nn} are relatively prime, i.e.

$$\gcd\left(\frac{b}{\gcd(b, A_{nn})}, \frac{A_{nn}}{\gcd(b, A_{nn})}\right) = 1,$$

which is obvious. Now we define $B_{j,n}$ and $C_{j,n}$ inductively: Suppose we have defined $B_{i,n}$ and $C_{i,n}$ for all $i = j+1, j+2, \dots, n-1$. Then $B_{j,n}$ and $C_{j,n}$ have to satisfy

$$A_{j,n} = B_{j,j}C_{j,n} + B_{j,j+1}C_{j+1,n} + \cdots + B_{j,n}C_{n,n}$$

Since $B_{j,j}$ and $C_{n,n}$ are relatively prime, we can choose integers $C_{j,n}$ and $B_{j,n}$ such that this equation is satisfied. Doing this step by step for all $j = n-1, n-2, \dots, 1$, we finally get B and C such that $A = BC$.

2006-D1-P4 — Statement

Day 1, P4 | 20 points

Let f be a rational function (i.e. the quotient of two real polynomials) and suppose that $f(n)$ is an integer for infinitely many integers n. Prove that f is a polynomial.

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Theory and methods for 2006-D1-P4

Theory Rational linear relations and discrete-value rigidity [T-AL-RATIONALITY]; Polynomial factorization and coprime divisibility [T-AL-FACTORIZATION]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Rank, dimension and exact linear constraints [T-LA-RANK]
Methods Eliminate rational relations, then exploit discreteness [M-AL-RATIONAL]
Strategy Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2006-D1-P4 — Solution

For this record: Theory: T-AL-RATIONALITY; T-AL-FACTORIZATION; T-AL-POLYNOMIAL; T-LA-RANK. Methods: M-AL-RATIONAL. Full learning route

Let S be an infinite set of integers such that rational function $f(x)$ is integral for all $x \in S$. Suppose that $f(x) = p(x)/q(x)$ where p is a polynomial of degree k and q is a polynomial of degree n. Then p, q are solutions to the simultaneous equations $p(x) = q(x)f(x)$ for all $x \in S$ that are not roots of q. These are linear simultaneous equations in the coefficients of p, q with rational coefficients. Since they have a solution, they have a rational solution.

2006-D1-P4 — Solution (continued)

Thus there are polynomials p', q' with rational coefficients such that $p'(x) = q'(x)f(x)$ for all $x \in S$ that are not roots of q. Multiplying this with the previous equation, we see that $p'(x)q(x)f(x) = p(x)q'(x)f(x)$ for all $x \in S$ that are not roots of q. If x is not a root of p or q, then $f(x) \neq 0$, and hence $p'(x)q(x) = p(x)q'(x)$ for all $x \in S$ except for finitely many roots of p and q. Thus the two polynomials $p'q$ and pq' are equal for infinitely many choices of value. Thus $p'(x)q(x) = p(x)q'(x)$. Dividing by $q(x)q'(x)$, we see that $p'(x)/q'(x) = p(x)/q(x) = f(x)$. Thus $f(x)$ can be written as the quotient of two polynomials with rational coefficients. Multiplying up by some integer, it can be written as the quotient of two polynomials with integer coefficients.

Suppose $f(x) = p''(x)/q''(x)$ where p'' and q'' both have integer coefficients. Then by Euler's division algorithm for polynomials, there exist polynomials s and r, both of which have rational coefficients such that $p''(x) = q''(x)s(x)+r(x)$ and the degree of r is less than the degree of q''. Dividing by $q''(x)$, we get that $f(x) = s(x) + r(x)/q''(x)$. Now there exists an integer N such that $Ns(x)$ has integral coefficients. Then $Nf(x) - Ns(x)$ is an integer for all $x \in S$. However, this is equal to the rational function Nr/q'', which has a higher degree denominator than numerator, so tends to 0 as x tends to ∞. Thus for all sufficiently large $x \in S$, $Nf(x) - Ns(x) = 0$ and hence $r(x) = 0$. Thus r has infinitely many roots, and is 0. Thus $f(x) = s(x)$, so f is a polynomial.

2006-D1-P5 — Statement

Day 1, P5 | 20 points

Let $a, b, c, d, e > 0$ be real numbers such that $a^2 + b^2 + c^2 = d^2 + e^2$ and $a^4 + b^4 + c^4 = d^4 + e^4$. Compare the numbers $a^3 + b^3 + c^3$ and $d^3 + e^3$.

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Theory and methods for 2006-D1-P5

Theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Methods Count zeros and subtract an interpolant [M-AN-ROLLE]; Compare growth through derivatives [M-AN-RATIO]
Strategy Make the contradiction quantitative or structural [S-CONTRADICTION]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2006-D1-P5 — Solution

For this record: Theory: T-AN-DERIVATIVE; T-AN-CONVERGENCE. Methods: M-AN-ROLLE; M-AN-RATIO. Full learning route

Without loss of generality $a \geq b \geq c$, $d \geq e$. Let $c^2 = e^2 + \Delta$, $\Delta \in \mathbb{R}$. Then $d^2 = a^2 + b^2 + \Delta$ and the second equation implies

$$a^4 + b^4 + (e^2 + \Delta)^2 = (a^2 + b^2 + \Delta)^2 + e^4, \qquad \Delta = -\frac{a^2b^2}{a^2 + b^2 - e^2}.$$

(Here $a^2 + b^2 - e^2 \geq \frac{2}{3}(a^2 + b^2 + c^2) - \frac{1}{2}(d^2 + e^2) = \frac{1}{6}(d^2 + e^2) > 0$.)
Since $c^2 = e^2 - \frac{a^2b^2}{a^2+b^2-e^2} = \frac{(a^2-e^2)(e^2-b^2)}{a^2+b^2-e^2} > 0$ then $a > e > b$.
Therefore $d^2 = a^2 + b^2 - \frac{a^2b^2}{a^2+b^2-e^2} < a^2$ and $a > d \geq e > b \geq c$.

Consider a function $f(x) = a^x + b^x + c^x - d^x - e^x$, $x \in \mathbb{R}$. We shall prove that $f(x)$ has only two zeroes $x = 2$ and $x = 4$ and changes the sign at these points. Suppose the contrary. Then Rolle's theorem implies that $f'(x)$ has at least two distinct zeroes. Without loss of generality $a = 1$. Then $f'(x) = \ln b \cdot b^x + \ln c \cdot c^x - \ln d \cdot d^x - \ln e \cdot e^x$, $x \in \mathbb{R}$. If $f'(x_1) = f'(x_2) = 0$, $x_1 < x_2$, then

$$\ln b \cdot b^{x_i} + \ln c \cdot c^{x_i} = \ln d \cdot d^{x_i} + \ln e \cdot e^{x_i}, \quad i = 1, 2,$$

but since $1 > d \geq e > b \geq c$ we have

$$\frac{(-\ln b) \cdot b^{x_2} + (-\ln c) \cdot c^{x_2}}{(-\ln b) \cdot b^{x_1} + (-\ln c) \cdot c^{x_1}} \leq b^{x_2-x_1} < e^{x_2-x_1} \leq \frac{(-\ln d) \cdot d^{x_2} + (-\ln e) \cdot e^{x_2}}{(-\ln d) \cdot d^{x_1} + (-\ln e) \cdot e^{x_1}},$$

2006-D1-P5 — Solution (continued)

a contradiction. Therefore $f(x)$ has a constant sign at each of the intervals $(-\infty, 2)$, $(2, 4)$ and $(4, \infty)$. Since $f(0) = 1$ then $f(x) > 0$, $x \in (-\infty, 2) \cup (4, \infty)$ and $f(x) < 0$, $x \in (2, 4)$. In particular, $f(3) = a^3 + b^3 + c^3 - d^3 - e^3 < 0$.

2006-D1-P6 — Statement

Day 1, P6 | 20 points

Find all sequences $a_0, a_1, \ldots, a_n$ of real numbers where $n \geq 1$ and $a_n \neq 0$, for which the following statement is true:

If $f : \mathbb{R} \to \mathbb{R}$ is an n times differentiable function and $x_0 < x_1 < \cdots < x_n$ are real numbers such that $f(x_0) = f(x_1) = \cdots = f(x_n) = 0$ then there exists an $h \in (x_0, x_n)$ for which

$$a_0 f(h) + a_1 f'(h) + \cdots + a_n f^{(n)}(h) = 0.$$

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Theory and methods for 2006-D1-P6

Theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Differential inequalities and integrating factors [T-AN-ODE]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Count zeros and subtract an interpolant [M-AN-ROLLE]; Use a first crossing or a forced-growth contradiction [M-AN-BARRIER]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2006-D1-P6 — Solution

For this record: Theory: T-AN-DERIVATIVE; T-AN-ODE; T-AL-POLYNOMIAL. Methods: M-AN-ROLLE; M-AN-BARRIER. Full learning route

Let $A(x) = a_0 + a_1 x + \cdots + a_n x^n$. We prove that sequence $a_0, \ldots, a_n$ satisfies the required property if and only if all zeros of polynomial $A(x)$ are real.
(a) Assume that all roots of $A(x)$ are real. Let us use the following notations. Let I be the identity operator on $\mathbb{R} \to \mathbb{R}$ functions and D be differentiation operator. For an arbitrary polynomial $P(x) = p_0 + p_1 x + \cdots + p_n x^n$, write $P(D) = p_0 I + p_1 D + p_2 D^2 + \cdots + p_n D^n$. Then the statement can written as $(A(D)f)(\xi) = 0$.
First prove the statement for $n = 1$. Consider the function

$$g(x) = e^{\frac{a_0}{a_1} x} f(x).$$

Since $g(x_0) = g(x_1) = 0$, by Rolle's theorem there exists a $\xi \in (x_0, x_1)$ for which

$$g'(\xi) = e^{\frac{a_0}{a_1}\xi} \frac{a_0}{a_1} f(\xi) + e^{\frac{a_0}{a_1}\xi} f'(\xi) = \frac{e^{\frac{a_0}{a_1}\xi}}{a_1} (a_0 f(\xi) + a_1 f'(\xi)) = 0.$$

Now assume that $n > 1$ and the statement holds for $n - 1$. Let $A(x) = (x - c)B(x)$ where c is a real root of polynomial A. By the $n = 1$ case, there exist $y_0 \in (x_0, x_1)$, $y_1 \in (x_1, x_2)$, ..., $y_{n-1} \in (x_{n-1}, x_n)$ such that $f'(y_j) - c f(y_j) = 0$ for all $j = 0, 1, \ldots, n - 1$. Now apply the induction hypothesis for polynomial $B(x)$, function $g = f' - cf$ and points $y_0, \ldots, y_{n-1}$. The hypothesis says that there exists a $\xi \in (y_0, y_{n-1}) \subset (x_0, x_n)$ such that

$$(B(D)g)(\xi) = (B(D)(D - cI)f)(\xi) = (A(D)f)(\xi) = 0.$$

(b) Assume that $u + vi$ is a complex root of polynomial $A(x)$ such that $v \neq 0$. Consider the linear differential equation $a_n g^{(n)} + \cdots + a_1 g' + g = 0$.
A solution of this equation is $g_1(x) = e^{ux} \sin vx$ which has infinitely many zeros.

2006-D1-P6 — Solution (continued)

Let k be the smallest index for which $a_k \neq 0$. Choose a small $\varepsilon > 0$ and set $f(x) = g_1(x) + \varepsilon x^k$. If ε is sufficiently small then g has the required number of roots but $a_0 f + a_1 f' + \cdots + a_n f^{(n)} = a_k \varepsilon \neq 0$ everywhere.

Day 2

2006-D2-P1 — Statement

Day 2, P7 | 20 points

Let V be a convex polygon with n vertices.
(a) Prove that if n is divisible by 3 then V can be triangulated (i.e. dissected into non-overlapping triangles whose vertices are vertices of V) so that each vertex of V is the vertex of an odd number of triangles.

(b) Prove that if n is not divisible by 3 then V can be triangulated so that there are exactly two vertices that are the vertices of an even number of the triangles.

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Theory and methods for 2006-D2-P1

Theory Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Methods Strengthen the statement before the induction step [M-DS-INDUCT]; Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Strategy Reduce size while preserving the defining conditions [S-INDUCTION]; Build a witness and verify every constraint [S-CONSTRUCTION]

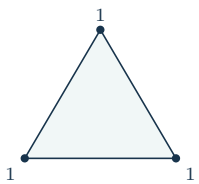
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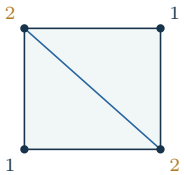
For this record: Theory: T-DS-INDUCTION; T-DS-GEOMETRY; T-AL-CONGRUENCE. Methods: M-DS-INDUCT; M-DS-GEOMETRIC. Full learning route

Apply induction on n. For the initial cases $n = 3, 4, 5$, choose the triangulations shown below to prove the statement.

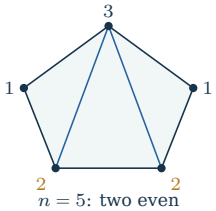
Initial triangulations: numbers of incident triangles



$n = 3$: all odd

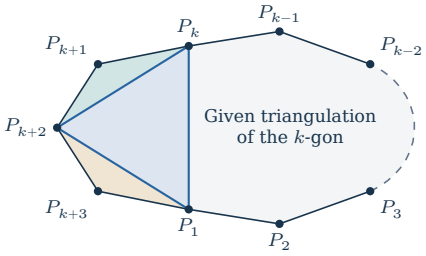


$n = 4$: two even



$n = 5$: two even

Induction step: attach three triangles



P_1, P_k: two additional triangles each.
New vertices: $1, 3, 1$ triangles.
All existing parities are preserved.

2006-D2-P1 — Solution (continued)

Now assume that the statement is true for some $n = k$ and consider the case $n = k + 3$. Denote the vertices of V by $P_1, \dots, P_{k+3}$. Apply the induction hypothesis on the polygon $P_1P_2 \dots P_k$; in this triangulation each of vertices $P_1, \dots, P_k$ belong to an odd number of triangles, except two vertices if n is not divisible by 3. Now add triangles $P_1P_kP_{k+2}$, $P_kP_{k+1}P_{k+2}$ and $P_1P_{k+2}P_{k+3}$. This way we introduce two new triangles at vertices P_1 and P_k so parity is preserved. The vertices P_{k+1}, P_{k+2} and P_{k+3} share an odd number of triangles. Therefore, the number of vertices shared by even number of triangles remains the same as in polygon $P_1P_2 \dots P_k$.

2006-D2-P2 — Statement

Day 2, P8 | 20 points

Find all functions $f : \mathbb{R} \longrightarrow \mathbb{R}$ such that for any real numbers $a < b$, the image $f([a, b])$ is a closed interval of length $b - a$.

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Theory and methods for 2006-D2-P2

Theory Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]; Metric contractions, covers, and nearest points [T-DS-METRIC]; Continuity, compactness, and order [T-AN-CONTINUITY]
Methods Force structure by admissible substitutions [M-DS-FUNCTIONAL]; Push covers forward or move toward a nearest point [M-DS-METRIC]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2006-D2-P2 — Solution

For this record: Theory: T-DS-FUNCTIONAL; T-DS-METRIC; T-AN-CONTINUITY. Methods: M-DS-FUNCTIONAL; M-DS-METRIC. Full learning route

The functions $f(x) = x + c$ and $f(x) = -x + c$ with some constant c obviously satisfy the condition of the problem. We will prove now that these are the only functions with the desired property.

Let f be such a function. Then f clearly satisfies $|f(x) - f(y)| \le |x - y|$ for all x, y; therefore, f is continuous. Given x, y with $x < y$, let $a, b \in [x, y]$ be such that $f(a)$ is the maximum and $f(b)$ is the minimum of f on $[x, y]$. Then $f([x, y]) = [f(b), f(a)]$; hence

$$y - x = f(a) - f(b) \le |a - b| \le y - x$$

This implies $\{a, b\} = \{x, y\}$, and therefore f is a monotone function. Suppose f is increasing. Then $f(x) - f(y) = x - y$ implies $f(x) - x = f(y) - y$, which says that $f(x) = x + c$ for some constant c. Similarly, the case of a decreasing function f leads to $f(x) = -x + c$ for some constant c.

2006-D2-P3 — Statement

Day 2, P9 | 20 points

Compare $\tan(\sin x)$ and $\sin(\tan x)$ for all $x \in (0, \frac{\pi}{2})$.

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Theory and methods for 2006-D2-P3

Theory Trigonometric identities and safe branches [T-AN-TRIG]; Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Methods Exploit identities before estimating oscillations [M-AN-TRIG]; Choose the correct convexity comparison [M-AN-CONVEX]

Theory and methods for 2006-D2-P3 (continued)

Strategy Build a chain of bounds and track equality [S-BOUND]; Prove necessity and sufficiency separately [S-CLASSIFICATION]

Statement Recorded solutions Catalog

2006-D2-P3 — Solution

For this record: Theory: T-AN-TRIG; T-AN-CONVEXITY; T-AN-INEQUALITIES; T-AN-DERIVATIVE. Methods: M-AN-TRIG; M-AN-CONVEX. Full learning route

Let $f(x) = \tan(\sin x) - \sin(\tan x)$. Then

$$f'(x) = \frac{\cos x}{\cos^2(\sin x)} - \frac{\cos(\tan x)}{\cos^2 x} = \frac{\cos^3 x - \cos(\tan x)\cdot\cos^2(\sin x)}{\cos^2 x\cdot\cos^2(\tan x)}$$

Let $0 < x < \arctan\frac{\pi}{2}$. It follows from the concavity of cosine on $(0, \frac{\pi}{2})$ that

$$\sqrt[3]{\cos(\tan x)\cdot\cos^2(\sin x)} < \frac{1}{3}\left[\cos(\tan x) + 2\cos(\sin x)\right] \le \cos\left[\frac{\tan x + 2\sin x}{3}\right] < \cos x\ ,$$

the last inequality follows from

$$\left(\frac{\tan x + 2\sin x}{3}\right)' = \frac{1}{3}\left[\frac{1}{\cos^2 x} + 2\cos x\right] \ge \sqrt[3]{\frac{1}{\cos^2 x}\cdot\cos x\cdot\cos x} = 1.$$

This proves that $\cos^3 x - \cos(\tan x)\cdot\cos^2(\sin x) > 0$, so $f'(x) > 0$, so f increases on the interval $[0, \arctan\frac{\pi}{2}]$.

To end the proof it is enough to notice that (recall that $4 + \pi^2 < 16$)

$$\tan\left[\sin\left(\arctan\frac{\pi}{2}\right)\right] = \tan\frac{\pi/2}{\sqrt{1+\pi^2/4}} > \tan\frac{\pi}{4} = 1\ .$$

This implies that if $x \in [\arctan\frac{\pi}{2}, \frac{\pi}{2}]$ then $\tan(\sin x) > 1$ and therefore $f(x) > 0$.

2006-D2-P4 — Statement

Day 2, P10 | 20 points

Let v_0 be the zero vector in $\mathbb{R}^n$ and let $v_1, v_2, \ldots, v_{n+1} \in \mathbb{R}^n$ be such that the Euclidean norm $|v_i - v_j|$ is rational for every $0 \le i, j \le n+1$. Prove that $v_1, \ldots, v_{n+1}$ are linearly dependent over the rationals.

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Theory and methods for 2006-D2-P4

Theory Gram matrices and positive semidefinite forms [T-LA-GRAM]; Inner products, orthogonality and polarization [T-LA-INNER]; Determinants, block identities and inverses [T-LA-DETERMINANT]; Rank, dimension and exact linear constraints [T-LA-RANK]
Methods Encode distances and orthogonality by a Gram matrix [M-LA-GRAM]
Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2006-D2-P4 — Solution

For this record: Theory: T-LA-GRAM; T-LA-INNER; T-LA-DETERMINANT; T-LA-RANK. Methods: M-LA-GRAM. Full learning route

By passing to a subspace we can assume that $v_1, \dots, v_n$ are linearly independent over the reals. Then there exist $\lambda_1, \dots, \lambda_n \in \mathbb{R}$ satisfying

$$v_{n+1} = \sum_{j=1}^{n} \lambda_j v_j$$

We shall prove that λ_j is rational for all j. From

$$-2\langle v_i, v_j \rangle = |v_i - v_j|^2 - |v_i|^2 - |v_j|^2$$

we get that $\langle v_i, v_j \rangle$ is rational for all i, j. Define A to be the rational $n \times n$-matrix $A_{ij} = \langle v_i, v_j \rangle$, $w \in \mathbb{Q}^n$ to be the vector $w_i = \langle v_i, v_{n+1} \rangle$, and $\lambda \in \mathbb{R}^n$ to be the vector $(\lambda_i)_i$. Then,

$$\langle v_i, v_{n+1} \rangle = \sum_{j=1}^{n} \lambda_j \langle v_i, v_j \rangle$$

gives $A\lambda = w$. Since $v_1, \dots, v_n$ are linearly independent, A is invertible. The entries of A^{-1} are rationals, therefore $\lambda = A^{-1}w \in \mathbb{Q}^n$, and we are done.

2006-D2-P5 — Statement

Day 2, P11 | 20 points

Prove that there exists an infinite number of relatively prime pairs (m, n) of positive integers such that the equation

$$(x + m)^3 = nx$$

has three distinct integer roots.

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Theory and methods for 2006-D2-P5

Theory Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]; Prime valuations, divisors, and factorials [T-AL-VALUATION]
Methods Construct roots or units before the coefficients [M-AL-PARAMETRIZE]; Translate root restrictions into symmetric coefficients [M-AL-VIETA]
Strategy Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2006-D2-P5 — Solution

For this record: Theory: T-AL-ROOTS; T-AL-VALUATION. Methods: M-AL-PARAMETRIZE; M-AL-VIETA. Full learning route

Substituting $y = x + m$, we can replace the equation by

$$y^3 - ny + mn = 0.$$

Let two roots be u and v; the third one must be $w = -(u + v)$ since the sum is 0. The roots must also satisfy

$$uv + uw + vw = -(u^2 + uv + v^2) = -n, \quad \text{i.e.} \quad u^2 + uv + v^2 = n$$

and

$$uvw = -uv(u + v) = mn.$$

So we need some integer pairs (u, v) such that $uv(u + v)$ is divisible by $u^2 + uv + v^2$. Look for such pairs in the form $u = kp$, $v = kq$. Then

$$u^2 + uv + v^2 = k^2(p^2 + pq + q^2),$$

2006-D2-P5 — Solution (continued)

and

$$uv(u+v) = k^3pq(p+q).$$

Chosing p, q such that they are coprime then setting $k = p^2+pq+q^2$ we have $\dfrac{uv(u+v)}{u^2+uv+v^2} = p^2+pq+q^2$.

Substituting back to the original quantites, we obtain the family of cases

$$n = (p^2+pq+q^2)^3, \qquad m = p^2q+pq^2,$$

and the three roots are

$$x_1 = p^3, \qquad x_2 = q^3, \qquad x_3 = -(p+q)^3.$$

2006-D2-P6 — Statement

Day 2, P12 | 20 points

Let A_i, B_i, S_i $(i = 1, 2, 3)$ be invertible real 2×2 matrices such that
(1) not all A_i have a common real eigenvector;
(2) $A_i = S_i^{-1}B_iS_i$ for all $i = 1, 2, 3$;
(3) $A_1A_2A_3 = B_1B_2B_3 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

Prove that there is an invertible real 2×2 matrix S such that $A_i = S^{-1}B_iS$ for all $i = 1, 2, 3$.

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Theory and methods for 2006-D2-P6

Theory Adapted bases, centralizers and common eigenspaces [T-LA-INVARIANT]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Trace pairing and commutator calculus [T-LA-TRACE]; Determinants, block identities and inverses [T-LA-DETERMINANT]
Methods Choose a basis adapted to the structure [M-LA-BASIS]; Exploit a compatible block decomposition [M-LA-BLOCK]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2006-D2-P6 — Solution

For this record: Theory: T-LA-INVARIANT; T-LA-SPECTRAL; T-LA-TRACE; T-LA-DETERMINANT. Methods: M-LA-BASIS; M-LA-BLOCK. Full learning route

We note that the problem is trivial if $A_j = \lambda I$ for some j, so suppose this is not the case. Consider then first the situation where some A_j, say A_3, has two distinct real eigenvalues. We may assume that $A_3 = B_3 = \begin{pmatrix} \lambda & \\ & \mu \end{pmatrix}$ by conjugating both sides. Let $A_2 = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $B_2 = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$. Then

$$\begin{aligned} a + d &= \operatorname{Tr} A_2 = \operatorname{Tr} B_2 = a' + d' \\ a\lambda + d\mu &= \operatorname{Tr}(A_2A_3) = \operatorname{Tr} A_1^{-1} = \operatorname{Tr} B_1^{-1} = \operatorname{Tr}(B_2B_3) = a'\lambda + d'\mu. \end{aligned}$$

Hence $a = a'$ and $d = d'$ and so also $bc = b'c'$. Now we cannot have $c = 0$ or $b = 0$, for then $(1, 0)^\top$ or $(0, 1)^\top$ would be a common eigenvector of all A_j. The matrix $S = \begin{pmatrix} \frac{c'}{c} & \\ & 1 \end{pmatrix}$ conjugates $A_2 = S^{-1}B_2S$, and as S commutes with $A_3 = B_3$, it follows that $A_j = S^{-1}B_jS$ for all j.
If the distinct eigenvalues of $A_3 = B_3$ are not real, we know from above that $A_j = S^{-1}B_jS$ for some $S \in GL_2\mathbb{C}$ unless all A_j have a common eigenvector over $\mathbb{C}$. Even if they do, say $A_jv = \lambda_jv$, by taking the conjugate square root it follows that A_j's can be simultaneously diagonalized. If $A_2 = \begin{pmatrix} a & \\ & d \end{pmatrix}$ and $B_2 = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$, it follows as above that $a = a'$, $d = d'$ and so $b'c' = 0$. Now B_2 and B_3 (and

2006-D2-P6 — Solution (continued)

hence B_1 too) have a common eigenvector over $\mathbb{C}$ so they too can be simultaneously diagonalized. And so $SA_j = B_jS$ for some $S \in GL_2\mathbb{C}$ in either case. Let $S_0 = \operatorname{Re} S$ and $S_1 = \operatorname{Im} S$. By separating the real and imaginary components, we are done if either S_0 or S_1 is invertible. If not, S_0 may be conjugated to some $T^{-1}S_0T = \begin{pmatrix} x & 0 \\ y & 0 \end{pmatrix}$, with $(x, y)^\top \neq (0, 0)^\top$, and it follows that all A_j have a common eigenvector $T(0, 1)^\top$, a contradiction.
We are left with the case when no A_j has distinct eigenvalues; then these eigenvalues by necessity are real. By conjugation and division by scalars we may assume that $A_3 = \begin{pmatrix} 1 & b \\ & 1 \end{pmatrix}$ and $b \neq 0$. By further conjugation by upper-triangular matrices (which preserves the shape of A_3 up to the value of b) we can also assume that $A_2 = \begin{pmatrix} 0 & u \\ 1 & v \end{pmatrix}$. Here $v^2 = \operatorname{Tr}^2 A_2 = 4 \det A_2 = -4u$. Now $A_1 = A_3^{-1}A_2^{-1} = \begin{pmatrix} -(b+v)/u & 1 \\ 1/u & 0 \end{pmatrix}$, and hence $(b+v)^2/u^2 = \operatorname{Tr}^2 A_1 = 4 \det A_1 = -4/u$. Comparing these two it follows that $b = -2v$.

What we have done is simultaneously reduced all A_j to matrices whose all entries depend on u and v ($= -\det A_2$ and $\operatorname{Tr} A_2$, respectively) only, but these themselves are invariant under similarity. So B_j's can be simultaneously reduced to the very same matrices.

3.14 2007

Day 1

2007-D1-P1 — Statement

Day 1, P1

Let f be a polynomial of degree 2 with integer coefficients. Suppose that $f(k)$ is divisible by 5 for every integer k. Prove that all coefficients of f are divisible by 5.

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Theory and methods for 2007-D1-P1

Theory Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]
Methods Choose a modulus that exposes the obstruction [M-AL-MODULAR]; Replace many equal values by too many roots [M-AL-ROOTCOUNT]
Strategy Pass between local constraints and a global conclusion [S-LOCALGLOBAL]

Statement Recorded solutions Catalog

2007-D1-P1 — Solution — Record 1 of 2

For this record: Theory: T-AL-CONGRUENCE; T-AL-POLYNOMIAL. Methods: M-AL-MODULAR. Full learning route

Let $f(x) = ax^2 + bx + c$. Substituting $x = 0$, $x = 1$ and $x = -1$, we obtain that $5 \mid f(0) = c$, $5 \mid f(1) = (a+b+c)$ and $5 \mid f(-1) = (a-b+c)$. Then $5 \mid f(1)+f(-1)-2f(0) = 2a$ and $5 \mid f(1)-f(-1) = 2b$. Therefore 5 divides $2a$, $2b$ and c and the statement follows.

2007-D1-P1 — Solution — Record 2 of 2

For this record: Theory: T-AL-FINITEFIELD; T-AL-POLYNOMIAL. Methods: M-AL-MODULAR; M-AL-ROOTCOUNT. Full learning route

Consider $f(x)$ as a polynomial over the 5-element field (i.e. modulo 5). The polynomial has 5 roots while its degree is at most 2. Therefore $f \equiv 0 \pmod 5$ and all of its coefficients are divisible by 5.

2007-D1-P2 — Statement

Day 1, P2

Let $n \geq 2$ be an integer. What is the minimal and maximal possible rank of an $n \times n$ matrix whose n^2 entries are precisely the numbers $1, 2, \ldots, n^2$?

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Theory and methods for 2007-D1-P2

Theory Rank, dimension and exact linear constraints [T-LA-RANK]; Determinants, block identities and inverses [T-LA-DETERMINANT]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Methods Prove a dimension obstruction and attain it [M-LA-DIMENSION]; Reduce integer matrices modulo an integer [M-LA-RESIDUE]
Strategy Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2007-D1-P2 — Solution

For this record: Theory: T-LA-RANK; T-LA-DETERMINANT; T-AL-CONGRUENCE. Methods: M-LA-DIMENSION; M-LA-RESIDUE. Full learning route

The minimal rank is 2 and the maximal rank is n. To prove this, we have to show that the rank can be 2 and n but it cannot be 1.
(i) The rank is at least 2. Consider an arbitrary matrix $A = [a_{ij}]$ with entries $1, 2, \ldots, n^2$ in some order. Since permuting rows or columns of a matrix does not change its rank, we can assume that $1 = a_{11} < a_{21} < \cdots < a_{n1}$ and $a_{11} < a_{12} < \cdots < a_{1n}$. Hence $a_{n1} \geq n$ and $a_{1n} \geq n$ and at least one of these inequalities is strict. Then

$$\det \begin{bmatrix} a_{11} & a_{1n} \\ a_{n1} & a_{nn} \end{bmatrix} < 1 \cdot n^2 - n \cdot n = 0$$

so $\operatorname{rk}(A) \geq \operatorname{rk} \begin{bmatrix} a_{11} & a_{1n} \\ a_{n1} & a_{nn} \end{bmatrix} \geq 2$.
(ii) The rank can be 2. Let

$$T = \begin{pmatrix} 1 & 2 & \ldots & n \\ n+1 & n+2 & \ldots & 2n \\ \vdots & \vdots & \ddots & \vdots \\ n^2-n+1 & n^2-n+2 & \ldots & n^2 \end{pmatrix}$$

The ith row is $(1, 2, \ldots, n) + n(i-1) \cdot (1, 1, \ldots, 1)$ so each row is in the two-dimensional subspace generated by the vectors $(1, 2, \ldots, n)$ and $(1, 1, \ldots, 1)$. We already proved that the rank is at least 2, so $\operatorname{rk}(T) = 2$.

(iii) The rank can be n, i.e. the matrix can be nonsingular. Put odd numbers into the diagonal, only even numbers above the diagonal and arrange the entries under the diagonal arbitrarily. Then the determinant of the matrix is odd, so the rank is complete.

2007-D1-P3 — Statement

Day 1, P3

Call a polynomial $P(x_1, \ldots, x_k)$ good if there exist 2×2 real matrices $A_1, \ldots, A_k$ such that

$$P(x_1, \ldots, x_k) = \det \left(\sum_{i=1}^{k} x_i A_i \right).$$

Find all values of k for which all homogeneous polynomials with k variables of degree 2 are good. (A polynomial is homogeneous if each term has the same total degree.)

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Theory and methods for 2007-D1-P3

Theory Rank, dimension and exact linear constraints [T-LA-RANK]; Determinants, block identities and inverses [T-LA-DETERMINANT]
Methods Prove a dimension obstruction and attain it [M-LA-DIMENSION]; Exploit a compatible block decomposition [M-LA-BLOCK]
Strategy Build a witness and verify every constraint [S-CONSTRUCTION]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

2007-D1-P3 — Solution

For this record: Theory: T-LA-RANK; T-LA-DETERMINANT. Methods: M-LA-DIMENSION; M-LA-BLOCK. Full learning route

The possible values for k are 1 and 2.
If $k = 1$ then $P(x) = \alpha x^2$ and we can choose $A_1 = \left(\begin{smallmatrix} 1 & 0 \\ 0 & \alpha \end{smallmatrix}\right)$.
If $k = 2$ then $P(x, y) = \alpha x^2 + \beta y^2 + \gamma xy$ and we can choose matrices $A_1 = \left(\begin{smallmatrix} 1 & 0 \\ 0 & \alpha \end{smallmatrix}\right)$ and $A_2 = \left(\begin{smallmatrix} 0 & \beta \\ -1 & \gamma \end{smallmatrix}\right)$.

Now let $k \geq 3$. We show that the polynomial $P(x_1, \dots, x_k) = \sum_{i=0}^{k} x_i^2$

is not good. Suppose that $P(x_1, \dots, x_k) = \det\left(\sum_{i=0}^{k} x_i A_i\right)$. Since the first columns of $A_1, \dots, A_k$ are linearly dependent, the first column of some non-trivial linear combination $y_1 A_1 + \cdots + y_k A_k$ is zero. Then $\det(y_1 A_1 + \cdots + y_k A_k) = 0$ but $P(y_1, \dots, y_k) \neq 0$, a contradiction.

2007-D1-P4 — Statement

Day 1, P4

Let G be a finite group. For arbitrary sets $U, V, W \subset G$, denote by N_{UVW} the number of triples $(x, y, z) \in U \times V \times W$ for which xyz is the unity.

Suppose that G is partitioned into three sets A, B and C (i.e. sets A, B, C are pairwise disjoint and $G = A \cup B \cup C$). Prove that $N_{ABC} = N_{CBA}$.

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Theory and methods for 2007-D1-P4

Theory Groups, orders, generators, and quotients [T-AL-GROUP]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Convert group data into orbits or cosets [M-AL-ACTION]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]

Statement Recorded solutions Catalog

2007-D1-P4 — Solution

For this record: Theory: T-AL-GROUP; T-DS-COUNTING. Methods: M-AL-ACTION. Full learning route

We start with three preliminary observations.

Let U, V be two arbitrary subsets of G. For each $x \in U$ and $y \in V$ there is a unique $z \in G$ for which $xyz = e$. Therefore,

$$N_{UVG} = |U \times V| = |U| \cdot |V|. \tag{1}$$

2007-D1-P4 — Solution (continued)

Second, the equation $xyz = e$ is equivalent to $yzx = e$ and $zxy = e$. For arbitrary sets $U, V, W \subset G$, this implies

$$\begin{aligned}&\{(x,y,z) \in U \times V \times W : xyz \\ &= e\} \\ &= \{(x,y,z) \in U \times V \times W : yzx \\ &= e\} \\ &= \{(x,y,z) \in U \times V \times W : zxy \\ &= e\}\end{aligned}$$

and therefore

$$N_{UVW} = N_{VWU} = N_{WUV}. \tag{2}$$

Third, if $U, V \subset G$ and W_1, W_2, W_3 are disjoint sets and $W = W_1 \cup W_2 \cup W_3$ then, for arbitrary $U, V \subset G$,

$$\begin{aligned}&\{(x,y,z) \in U \times V \times W : xyz = e\} \\ &= \{(x,y,z) \in U \times V \times W_1 : xyz = e\} \\ &\quad \cup \{(x,y,z) \in U \times V \times W_2 : xyz = e\} \\ &\quad \cup \{(x,y,z) \in U \times V \times W_3 : xyz = e\}\end{aligned}$$

so

$$N_{UVW} = N_{UVW_1} + N_{UVW_2} + N_{UVW_3}. \tag{3}$$

Applying these observations, the statement follows as

$$\begin{aligned}N_{ABC} &= N_{ABG} - N_{ABA} - N_{ABB} \\ &= |A| \cdot |B| - N_{BAA} - N_{BAB} \\ &= N_{BAG} - N_{BAA} - N_{BAB} = N_{BAC} = N_{CBA}.\end{aligned}$$

2007-D1-P5 — Statement

Day 1, P5

Let n be a positive integer and $a_1, \dots, a_n$ be arbitrary integers. Suppose that a function $f : \mathbb{Z} \to \mathbb{R}$ satisfies $\sum_{i=1}^{n} f(k + a_i \ell) = 0$ whenever k and ℓ are integers and $\ell \neq 0$. Prove that $f = 0$.

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Theory and methods for 2007-D1-P5

Theory Finite differences and shift-operator ideals [T-AL-DIFFERENCE]; Polynomial factorization and coprime divisibility [T-AL-FACTORIZATION]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Take the gcd of annihilating shift polynomials [M-AL-SHIFT]
Strategy Expose the structure before calculating [S-STRUCTURE]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

2007-D1-P5 — Solution

For this record: Theory: T-AL-DIFFERENCE; T-AL-FACTORIZATION; T-AL-POLYNOMIAL. Methods: M-AL-SHIFT. Full learning route

Let us define a subset I of the polynomial ring $\mathbb{R}[X]$ as follows:

$$I = \left\{ P(X) = \sum_{j=0}^{m} b_j X^j : \sum_{j=0}^{m} b_j f(k + j\ell) = 0 \text{ for all } k, \ell \in \mathbb{Z},\ \ell \neq 0 \right\}.$$

2007-D1-P5 — Solution (continued)

This is a subspace of the real vector space $\mathbb{R}[X]$. Furthermore, $P(X) \in I$ implies $X \cdot P(X) \in I$. Hence, I is an ideal, and it is non-zero, because the polynomial $R(X) = \sum_{i=1}^{n} X^{a_i}$ belongs to I. Thus, I is generated (as an ideal) by some non-zero polynomial Q.

If Q is constant then the definition of I implies $f = 0$, so we can assume that Q has a complex zero c. Again, by the definition of I, the polynomial $Q(X^m)$ belongs to I for every natural number $m \geq 1$; hence $Q(X)$ divides $Q(X^m)$. This shows that all the complex numbers

$$c, c^2, c^3, c^4, \ldots$$

are roots of Q. Since Q can have only finitely many roots, we must have $c^N = 1$ for some $N \geq 1$; in particular, $Q(1) = 0$, which implies $P(1) = 0$ for all $P \in I$. This contradicts the fact that $R(X) = \sum_{i=1}^{n} X^{a_i} \in I$, and we are done.

2007-D1-P6 — Statement

Day 1, P6

How many nonzero coefficients can a polynomial $P(z)$ have if its coefficients are integers and $|P(z)| \leq 2$ for any complex number z of unit length?

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Theory and methods for 2007-D1-P6

Theory Roots-of-unity filters and positive trigonometric kernels [T-AL-FOURIER]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]
Methods Evaluate at roots of unity or average over them [M-AL-UNITY]; Replace many equal values by too many roots [M-AL-ROOTCOUNT]
Strategy Build a chain of bounds and track equality [S-BOUND]; Choose an extremal object and force rigidity [S-EXTREMAL]

Statement Recorded solutions Catalog

2007-D1-P6 — Solution

For this record: Theory: T-AL-FOURIER; T-AL-POLYNOMIAL; T-AL-COMPLEX. Methods: M-AL-UNITY; M-AL-ROOTCOUNT.
Full learning route

We show that the number of nonzero coefficients can be 0, 1 and 2. These values are possible, for example the polynomials $P_0(z) = 0$, $P_1(z) = 1$ and $P_2(z) = 1 + z$ satisfy the conditions and they have 0, 1 and 2 nonzero terms, respectively.
Now consider an arbitrary polynomial $P(z) = a_0 + a_1 z + \cdots + a_n z^n$ satisfying the conditions and assume that it has at least two nonzero coefficients. Dividing the polynomial by a power of z and optionally replacing $p(z)$ by $-p(z)$, we can achieve $a_0 > 0$ such that conditions are not changed and the number of nonzero terms is preserved. So, without loss of generality, we can assume that $a_0 > 0$.
Let $Q(z) = a_1 z + \cdots + a_{n-1} z^{n-1}$. Our goal is to show that $Q(z) = 0$.
Consider those complex numbers $w_0, w_1, \ldots, w_{n-1}$ on the unit circle for which $a_n w_k^n = |a_n|$; namely, let

$$w_k = \begin{cases} e^{2k\pi i/n} & \text{if } a_n > 0 \\ e^{(2k+1)\pi i/n} & \text{if } a_n < 0 \end{cases} \quad (k = 0, 1, \ldots, n).$$

Notice that

$$\sum_{k=0}^{n-1} Q(w_k) = \sum_{k=0}^{n-1} Q(w_0 e^{2k\pi i/n}) = \sum_{j=1}^{n-1} a_j w_0^j \sum_{k=0}^{n-1} (e^{2j\pi i/n})^k = 0.$$

Taking the average of polynomial $P(z)$ at the points w_k, we obtain

$$\frac{1}{n} \sum_{k=0}^{n-1} P(w_k) = \frac{1}{n} \sum_{k=0}^{n-1} (a_0 + Q(w_k) + a_n w_k^n) = a_0 + |a_n|$$

2007-D1-P6 — Solution (continued)

and

$$2 \geq \frac{1}{n}\sum_{k=0}^{n-1}|P(w_k)| \geq \frac{1}{n}\left|\sum_{k=0}^{n-1}P(w_k)\right| = a_0 + |a_n| \geq 2.$$

This obviously implies $a_0 = |a_n| = 1$ and $P(w_k) = 2 + Q(w_k) = 2$ for all k. Therefore, all values of $Q(w_k)$ must lie on the circle $|2+z| = 2$, while their sum is 0. This is possible only if $Q(w_k) = 0$ for all k. Then polynomial $Q(z)$ has at least n distinct roots while its degree is at most $n-1$. So $Q(z) = 0$ and $P(z) = a_0 + a_n z^n$ has only two nonzero coefficients.

Remark. From Parseval's formula (i.e. integrating $|P(z)|^2 = P(z)\overline{P(z)}$ on the unit circle) it can be obtained that

$$|a_0|^2 + \cdots + |a_n|^2 = \frac{1}{2\pi}\int_0^{2\pi}\left|P(e^{it})\right|^2 dt \leq \frac{1}{2\pi}\int_0^{2\pi} 4\,dt = 4. \tag{4}$$

Hence, there cannot be more than four nonzero coefficients, and if there are more than one nonzero term, then their coefficients are ± 1.

It is also easy to see that equality in (4) cannot hold two or more nonzero coefficients, so it is sufficient to consider only polynomials of the form $1 \pm x^m \pm x^n$. However, we do not know (yet :-)) any simpler argument for these cases than the proof above.

Day 2

2007-D2-P1 — Statement

Day 2, P7

Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function. Suppose that for any $c > 0$, the graph of f can be moved to the graph of cf using only a translation or a rotation. Does this imply that $f(x) = ax + b$ for some real numbers a and b?

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Theory and methods for 2007-D2-P1

Theory Continuity, compactness, and order [T-AN-CONTINUITY]; Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]
Methods Design an auxiliary derivative [M-AN-AUXILIARY]
Strategy Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2007-D2-P1 — Solution

For this record: Theory: T-AN-CONTINUITY; T-DS-FUNCTIONAL. Methods: M-AN-AUXILIARY. Full learning route

No. The function $f(x) = e^x$ also has this property since $ce^x = e^{x+\log c}$.

2007-D2-P2 — Statement

Day 2, P8

Let x, y, and z be integers such that $S = x^4 + y^4 + z^4$ is divisible by 29. Show that S is divisible by 29^4.

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Theory and methods for 2007-D2-P2

Theory Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Prime valuations, divisors, and factorials [T-AL-VALUATION]
Methods Choose a modulus that exposes the obstruction [M-AL-MODULAR]
Strategy Pass between local constraints and a global conclusion [S-LOCALGLOBAL]

Statement Recorded solutions Catalog

2007-D2-P2 — Solution

For this record: Theory: T-AL-FINITEFIELD; T-AL-VALUATION. Methods: M-AL-MODULAR. Full learning route

We claim that $29 \mid x, y, z$. Then, $x^4 + y^4 + z^4$ is clearly divisible by 29^4.
Assume, to the contrary, that 29 does not divide all of the numbers x, y, z. Without loss of generality, we can suppose that $29 \nmid x$. Since the residue classes modulo 29 form a field, there is some $w \in \mathbb{Z}$ such that $xw \equiv 1 \pmod{29}$. Then, $(xw)^4 + (yw)^4 + (zw)^4$ is also divisible by 29. So we can assume that $x \equiv 1 \pmod{29}$.

Thus, we need to show that $y^4 + z^4 \equiv -1 \pmod{29}$, i.e. $y^4 \equiv -1 - z^4 \pmod{29}$, is impossible. There are only eight fourth powers modulo 29,

$$\begin{aligned}
0 &\equiv 0^4, \\
1 &\equiv 1^4 \equiv 12^4 \equiv 17^4 \equiv 28^4 \pmod{29}, \\
7 &\equiv 8^4 \equiv 9^4 \equiv 20^4 \equiv 21^4 \pmod{29}, \\
16 &\equiv 2^4 \equiv 5^4 \equiv 24^4 \equiv 27^4 \pmod{29}, \\
20 &\equiv 6^4 \equiv 14^4 \equiv 15^4 \equiv 23^4 \pmod{29}, \\
23 &\equiv 3^4 \equiv 7^4 \equiv 22^4 \equiv 26^4 \pmod{29}, \\
24 &\equiv 4^4 \equiv 10^4 \equiv 19^4 \equiv 25^4 \pmod{29}, \\
25 &\equiv 11^4 \equiv 13^4 \equiv 16^4 \equiv 18^4 \pmod{29}.
\end{aligned}$$

The differences $-1 - z^4$ are congruent to 28, 27, 21, 12, 8, 5, 4, and 3. None of these residue classes is listed among the fourth powers.

2007-D2-P3 — Statement

Day 2, P9

Let C be a nonempty closed bounded subset of the real line and $f : C \to C$ be a nondecreasing continuous function. Show that there exists a point $p \in C$ such that $f(p) = p$.

(A set is closed if its complement is a union of open intervals. A function g is nondecreasing if $g(x) \leq g(y)$ for all $x \leq y$.)

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Theory and methods for 2007-D2-P3

Theory Continuity, compactness, and order [T-AN-CONTINUITY]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Methods Localize an extremum or an accumulation point [M-AN-COMPACT]; Use a first crossing or a forced-growth contradiction [M-AN-BARRIER]
Strategy Choose an extremal object and force rigidity [S-EXTREMAL]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

2007-D2-P3 — Solution

For this record: Theory: T-AN-CONTINUITY; T-AN-CONVERGENCE. Methods: M-AN-COMPACT; M-AN-BARRIER. Full learning route

Suppose $f(x) \neq x$ for all $x \in C$. Let $[a, b]$ be the smallest closed interval that contains C. Since C is closed, $a, b \in C$. By our hypothesis $f(a) > a$ and $f(b) < b$. Let $p = \sup\{x \in C : f(x) > x\}$. Since C is closed and f is continuous, $f(p) \geq p$, so $f(p) > p$. For all $x > p$, $x \in C$ we have $f(x) < x$. Therefore $f(f(p)) < f(p)$ contrary to the fact that f is non-decreasing.

2007-D2-P4 — Statement

Day 2, P10

Let $n > 1$ be an odd positive integer and $A = (a_{ij})_{i,j=1\ldots n}$ be the $n \times n$ matrix with

$$a_{ij} = \begin{cases} 2 & \text{if } i = j \\ 1 & \text{if } i - j \equiv \pm 2 \pmod{n} \\ 0 & \text{otherwise.} \end{cases}$$

Find $\det A$.

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Theory and methods for 2007-D2-P4

Theory Determinants, block identities and inverses [T-LA-DETERMINANT]; Rank, dimension and exact linear constraints [T-LA-RANK]
Methods Eliminate systematically and track the invariant [M-LA-ELIM]; Exploit a compatible block decomposition [M-LA-BLOCK]
Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2007-D2-P4 — Solution

For this record: Theory: T-LA-DETERMINANT; T-LA-RANK. Methods: M-LA-ELIM; M-LA-BLOCK. Full learning route

Notice that $A = B^2$, with $b_{ij} = \begin{cases} 1 & \text{if } i - j \equiv \pm 1 \pmod{n} \\ 0 & \text{otherwise} \end{cases}$. So it is sufficient to find $\det B$.

To find $\det B$, expand the determinant with respect to the first row, and then expad both terms with respect to the first column.

$$\det B = \begin{vmatrix} 0 & 1 & & & 1 \\ 1 & 0 & 1 & & \\ & 1 & 0 & \ddots & \\ & & \ddots & \ddots & 1 \\ 1 & & & 1 & 0 \end{vmatrix}$$

$$= -\begin{vmatrix} 1 & 1 & & & \\ & 0 & \ddots & & \\ & \ddots & \ddots & 1 & \\ 1 & & 1 & 0 & \end{vmatrix} + \begin{vmatrix} 1 & 0 & 1 & & \\ & 1 & \ddots & & \\ & & \ddots & 1 & \\ 1 & & 1 & 0 & \end{vmatrix}$$

$$= -\left(\begin{vmatrix} 0 & 1 & & & \\ & \ddots & \ddots & & \\ & & \ddots & 1 & \\ 1 & 0 & 1 & & \end{vmatrix} - 1 \cdot \begin{vmatrix} 1 & & & \\ & \ddots & & \\ & & \ddots & \\ 1 & 0 & 1 & \ddots \end{vmatrix} \right)$$

$$+ \left(\begin{vmatrix} 1 & 0 & 1 & & \\ & \ddots & \ddots & & \\ 1 & 0 & \ddots & 1 & \\ & 1 & & 0 & \end{vmatrix} - 1 \cdot \begin{vmatrix} 0 & 1 & & \\ 1 & 0 & \ddots & \\ & \ddots & \ddots & 1 \\ & & 1 & 0 \end{vmatrix} \right)$$

$$= -(0 - 1) + (1 - 0) = 2,$$

2007-D2-P4 — Solution (continued)

since the second and the third matrices are lower/upper triangular, while in the first and the fourth matrices we have $\text{row}_1 - \text{row}_3 + \text{row}_5 - \cdots \pm \text{row}_{n-2} = \bar{0}$.

So $\det B = 2$ and thus $\det A = 4$.

2007-D2-P5 — Statement

Day 2, P11

For each positive integer k, find the smallest number n_k for which there exist real $n_k \times n_k$ matrices $A_1, A_2, \ldots, A_k$ such that all of the following conditions hold:
(1) $A_1^2 = A_2^2 = \cdots = A_k^2 = 0$,
(2) $A_i A_j = A_j A_i$ for all $1 \leq i, j \leq k$, and
(3) $A_1 A_2 \ldots A_k \neq 0$.

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Theory and methods for 2007-D2-P5

Theory Nilpotents, projections and involutions [T-LA-NILPOTENT]; Rank, dimension and exact linear constraints [T-LA-RANK]; Dual bases, kernels and separating functionals [T-LA-DUALITY]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Prove a dimension obstruction and attain it [M-LA-DIMENSION]; Choose a basis adapted to the structure [M-LA-BASIS]
Strategy Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2007-D2-P5 — Solution

For this record: Theory: T-LA-NILPOTENT; T-LA-RANK; T-LA-DUALITY; T-DS-COUNTING. Methods: M-LA-DIMENSION; M-LA-BASIS. Full learning route

The anwser is $n_k = 2^k$. In that case, the matrices can be constructed as follows: Let V be the n-dimensional real vector space with basis elements $[S]$, where S runs through all $n = 2^k$ subsets of $\{1, 2, \ldots, k\}$. Define A_i as an endomorphism of V by

$$A_i[S] = \begin{cases} 0 & \text{if } i \in S \\ [S \cup \{i\}] & \text{if } i \notin S \end{cases}$$

for all $i = 1, 2, \ldots, k$ and $S \subset \{1, 2, \ldots, k\}$. Then $A_i^2 = 0$ and $A_i A_j = A_j A_i$. Furthermore,

$$A_1 A_2 \ldots A_k[\emptyset] = [\{1, 2, \ldots, k\}],$$

and hence $A_1 A_2 \ldots A_k \neq 0$.
Now let $A_1, A_2, \ldots, A_k$ be $n \times n$ matrices satisfying the conditions of the problem; we prove that $n \geq 2^k$. Let v be a real vector satisfying $A_1 A_2 \ldots A_k v \neq 0$. Denote by $\mathcal{P}$ the set of all subsets of $\{1, 2, \ldots, k\}$. Choose a complete ordering $\prec$ on $\mathcal{P}$ with the property

$$X \prec Y \Rightarrow |X| \leq |Y| \quad \text{for all } X, Y \in \mathcal{P}.$$

For every element $X = \{x_1, x_2, \ldots, x_r\} \in \mathcal{P}$, define $A_X = A_{x_1} A_{x_2} \ldots A_{x_r}$ and $v_X = A_X v$. Finally, write $\bar{X} = \{1, 2, \ldots, k\} \setminus X$ for the complement of X.

Now take $X, Y \in \mathcal{P}$ with $X \prec Y$. Then $A_{\bar{X}}$ annihilates v_Y, because $X \prec Y$ implies the existence of some $y \in Y \setminus X = Y \cap \bar{X}$, and

$$A_{\bar{X}} v_Y = A_{\bar{X} \setminus \{y\}} A_y A_y v_{Y \setminus \{y\}} = 0,$$

since $A_y^2 = 0$. So, $A_{\bar{X}}$ annihilates the span of all the v_Y with $X \prec Y$. This implies that v_X does not lie in this span, because $A_{\bar{X}} v_X = v_{\{1,2,\ldots,k\}} \neq 0$. Therefore, the vectors v_X (with $X \in \mathcal{P}$) are linearly independent; hence $n \geq |\mathcal{P}| = 2^k$.

2007-D2-P6 — Statement

Day 2, P12

Let $f \neq 0$ be a polynomial with real coefficients. Define the sequence $f_0, f_1, f_2, \ldots$ of polynomials by $f_0 = f$ and $f_{n+1} = f_n + f_n'$ for every $n \geq 0$. Prove that there exists a number N such that for every $n \geq N$, all roots of f_n are real.

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Theory and methods for 2007-D2-P6

Theory Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Methods Use a logarithmic derivative to expose root geometry [M-AL-LOGDERIVATIVE]
Strategy Reduce size while preserving the defining conditions [S-INDUCTION]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2007-D2-P6 — Solution

For this record: Theory: T-AL-ROOTS; T-AN-DERIVATIVE; T-AN-INTEGRAL; T-DS-INDUCTION. Methods: M-AL-LOGDERIVATIVE. Full learning route

For the proof, we need the following
Lemma 1. For any polynomial g, denote by $d(g)$ the minimum distance of any two of its real zeros ($d(g) = \infty$ if g has at most one real zero). Assume that g and $g + g'$ both are of degree $k \geq 2$ and have k distinct real zeros. Then $d(g + g') \geq d(g)$.
Proof of Lemma 1: Let $x_1 < x_2 < \cdots < x_k$ be the roots of g. Suppose a, b are roots of $g + g'$ satisfying $0 < b - a < d(g)$. Then, a, b cannot be roots of g, and

$$\frac{g'(a)}{g(a)} = \frac{g'(b)}{g(b)} = -1. \tag{1}$$

Since $\frac{g'}{g}$ is strictly decreasing between consecutive zeros of g, we must have $a < x_j < b$ for some j. For all $i = 1, 2, \ldots, k-1$ we have $x_{i+1} - x_i > b - a$, hence $a - x_i > b - x_{i+1}$. If $i < j$, both sides of this inequality are negative; if $i \geq j$, both sides are positive. In any case, $\frac{1}{a - x_i} < \frac{1}{b - x_{i+1}}$, and hence

$$\frac{g'(a)}{g(a)} = \sum_{i=1}^{k-1} \frac{1}{a - x_i} + \underbrace{\frac{1}{a - x_k}}_{<0} < \sum_{i=1}^{k-1} \frac{1}{b - x_{i+1}} + \underbrace{\frac{1}{b - x_1}}_{>0} = \frac{g'(b)}{g(b)}$$

This contradicts (1).
Now we turn to the proof of the stated problem. Denote by m the degree of f. We will prove by induction on m that f_n has m distinct real zeros for sufficiently large n. The cases $m = 0, 1$ are trivial; so we assume $m \geq 2$. Without loss of generality we can assume that f is monic. By induction, the result holds for f', and by ignoring the first few terms we can assume that f_n' has $m - 1$ distinct real zeros for all n. Let us denote these zeros by $x_1^{(n)} > x_2^{(n)} > \cdots > x_{m-1}^{(n)}$. Then f_n has minima in $x_1^{(n)}, x_3^{(n)}, x_5^{(n)}, \ldots$, and maxima in $x_2^{(n)}, x_4^{(n)}, x_6^{(n)}, \ldots$. Note that in the interval $(x_{i+1}^{(n)}, x_i^{(n)})$, the function $f_{n+1}' = f_n' + f_n''$ must have a zero (this follows by applying Rolle's theorem to the function $e^x f_n'(x)$); the same is true for the interval $(-\infty, x_{m-1}^{(n)})$. Hence, in each of these $m - 1$ intervals, f_{n+1}' has exactly one zero. This shows that

$$x_1^{(n)} > x_1^{(n+1)} > x_2^{(n)} > x_2^{(n+1)} > x_3^{(n)} > x_3^{(n+1)} > \ldots \tag{2}$$

Lemma 2. We have $\lim_{n \to \infty} f_n(x_j^{(n)}) = -\infty$ if j is odd, and $\lim_{n \to \infty} f_n(x_j^{(n)}) = +\infty$ if j is even.
Lemma 2 immediately implies the result: For sufficiently large n, the values of all maxima of f_n are positive, and the values of all minima of f_n are negative; this implies that f_n has m distinct zeros.
Proof of Lemma 2: Let $d = \min\{d(f'), 1\}$; then by Lemma 1, $d(f_n') \geq d$ for all n. Define $\varepsilon = \frac{(m-1)d^{m-1}}{m^{m-1}}$; we will show that

$$f_{n+1}(x_j^{(n+1)}) \geq f_n(x_j^{(n)}) + \varepsilon \quad \text{for } j \text{ even}. \tag{3}$$

2007-D2-P6 — Solution (continued)

(The corresponding result for odd j can be shown similarly.) Do to so, write $f = f_n$, $b = x_j^{(n)}$, and choose a satisfying $d \le b - a \le 1$ such that f' has no zero inside (a, b). Define ξ by the relation $b - \xi = \frac{1}{m}(b - a)$; then $\xi \in (a, b)$. We show that $f(\xi) + f'(\xi) \ge f(b) + \varepsilon$.

Notice, that

$$\begin{aligned}\frac{f''(\xi)}{f'(\xi)} &= \sum_{i=1}^{m-1} \frac{1}{\xi - x_i^{(n)}} \\ &= \underbrace{\sum_{i<j} \frac{1}{\xi - x_i^{(n)}}}_{<\frac{1}{\xi-a}} + \frac{1}{\xi - b} + \underbrace{\sum_{i>j} \frac{1}{\xi - x_i^{(n)}}}_{<0} \\ &< (m-1)\frac{1}{\xi - a} + \frac{1}{\xi - b} = 0.\end{aligned}$$

The last equality holds by definition of ξ. Since f' is positive and $\frac{f''}{f'}$ is decreasing in (a, b), we have that f'' is negative on (ξ, b). Therefore,

$$f(b) - f(\xi) = \int_\xi^b f'(t)\,dt \le \int_\xi^b f'(\xi)\,dt = (b - \xi)f'(\xi)$$

Hence,

$$\begin{aligned}f(\xi) + f'(\xi) &\ge f(b) - (b - \xi)f'(\xi) + f'(\xi) \\ &= f(b) + (1 - (\xi - b))f'(\xi) \\ &= f(b) + \left(1 - \tfrac{1}{m}(b - a)\right) f'(\xi) \\ &\ge f(b) + \left(1 - \tfrac{1}{m}\right) f'(\xi).\end{aligned}$$

Together with

$$f'(\xi) = |f'(\xi)| = m \prod_{i=1}^{m-1} \underbrace{\left|\xi - x_i^{(n)}\right|}_{\ge |\xi - b|} \ge m|\xi - b|^{m-1} \ge \frac{d^{m-1}}{m^{m-2}}$$

we get

$$f(\xi) + f'(\xi) \ge f(b) + \varepsilon.$$

Together with (2) this shows (3). This finishes the proof of Lemma 2.

3.15 2008

Day 1

2008-D1-P1 — Statement

Day 1, P1

Find all continuous functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) - f(y)$ is rational for all reals x and y such that $x - y$ is rational.

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Theory and methods for 2008-D1-P1

Theory Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]; Continuity, compactness, and order [T-AN-CONTINUITY]; Countability, accumulation, and Cantor constructions [T-AN-SETS]
Methods Force structure by admissible substitutions [M-DS-FUNCTIONAL]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2008-D1-P1 — Solution

For this record: Theory: T-DS-FUNCTIONAL; T-AN-CONTINUITY; T-AN-SETS. Methods: M-DS-FUNCTIONAL. Full learning route

We prove that $f(x) = ax + b$ where $a \in \mathbb{Q}$ and $b \in \mathbb{R}$. These functions obviously satify the conditions. Suppose that a function $f(x)$ fulfills the required properties. For an arbitrary rational q, consider the function $g_q(x) = f(x+q) - f(x)$. This is a continuous function which attains only rational values, therefore g_q is constant.
Set $a = f(1) - f(0)$ and $b = f(0)$. Let n be an arbitrary positive integer and let $r = f(1/n) - f(0)$. Since $f(x + 1/n) - f(x) = f(1/n) - f(0) = r$ for all x, we have

$$f(k/n) - f(0) = (f(1/n) - f(0)) + (f(2/n) - f(1/n)) + \cdots + (f(k/n) - f((k-1)/n) = kr$$

and

$$f(-k/n) - f(0) = -(f(0) - f(-1/n)) - (f(-1/n) - f(-2/n)) - \cdots - (f(-(k-1)/n) - f(-k/n) = -kr$$

for $k \geq 1$. In the case $k = n$ we get $a = f(1) - f(0) = nr$, so $r = a/n$. Hence, $f(k/n) - f(0) = kr = ak/n$ and then $f(k/n) = a \cdot k/n + b$ for all integers k and $n > 0$.

So, we have $f(x) = ax + b$ for all rational x. Since the function f is continous and the rational numbers form a dense subset of $\mathbb{R}$, the same holds for all real x.

2008-D1-P2 — Statement

Day 1, P2

Denote by V the real vector space of all real polynomials in one variable, and let $P : V \to \mathbb{R}$ be a linear map. Suppose that for all $f, g \in V$ with $P(fg) = 0$ we have $P(f) = 0$ or $P(g) = 0$. Prove that there exist real numbers x_0, c such that $P(f) = cf(x_0)$ for all $f \in V$.

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Theory and methods for 2008-D1-P2

Theory Dual bases, kernels and separating functionals [T-LA-DUALITY]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Test a linear map on a basis and normalize its values [M-LA-DUAL]
Strategy Reduce size while preserving the defining conditions [S-INDUCTION]; Prove necessity and sufficiency separately [S-CLASSIFICATION]

Statement Recorded solutions Catalog

2008-D1-P2 — Solution

For this record: Theory: T-LA-DUALITY; T-AL-POLYNOMIAL. Methods: M-LA-DUAL. Full learning route

We can assume that $P \neq 0$.
Let $f \in V$ be such that $P(f) \neq 0$. Then $P(f^2) \neq 0$, and therefore $P(f^2) = aP(f)$ for some non-zero real a. Then $0 = P(f^2 - af) = P(f(f-a))$ implies $P(f-a) = 0$, so we get $P(a) \neq 0$. By rescaling, we can assume that $P(1) = 1$. Now $P(X + b) = 0$ for $b = -P(X)$. Replacing P by $\hat{P}$ given as

$$\hat{P}(f(X)) = P(f(X + b))$$

2008-D1-P2 — Solution (continued)

we can assume that $P(X) = 0$.

Now we are going to prove that $P(X^k) = 0$ for all $k \geq 1$. Suppose this is true for all $k < n$. We know that $P(X^n + e) = 0$ for $e = -P(X^n)$. From the induction hypothesis we get

$$P((X + e)(X + 1)^{n-1}) = P(X^n + e) = 0,$$

and therefore $P(X + e) = 0$ (since $P(X + 1) = 1 \neq 0$). Hence $e = 0$ and $P(X^n) = 0$, which completes the inductive step. From $P(1) = 1$ and $P(X^k) = 0$ for $k \geq 1$ we immediately get $P(f) = f(0)$ for all $f \in V$.

2008-D1-P3 — Statement

Day 1, P3

Let p be a polynomial with integer coefficients and let $a_1 < a_2 < \cdots < a_k$ be integers.
a) Prove that there exists $a \in \mathbb{Z}$ such that $p(a_i)$ divides $p(a)$ for all $i = 1, 2, \ldots, k$.

b) Does there exist an $a \in \mathbb{Z}$ such that the product $p(a_1) \cdot p(a_2) \cdot \cdots \cdot p(a_k)$ divides $p(a)$?

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Theory and methods for 2008-D1-P3

Theory Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Prime valuations, divisors, and factorials [T-AL-VALUATION]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Construct and count compatible local residues [M-AL-CRT]
Strategy Build a witness and verify every constraint [S-CONSTRUCTION]; Pass between local constraints and a global conclusion [S-LOCALGLOBAL]

Statement Recorded solutions Catalog

2008-D1-P3 — Solution

For this record: Theory: T-AL-CONGRUENCE; T-AL-VALUATION; T-AL-POLYNOMIAL. Methods: M-AL-CRT. Full learning route

The theorem is obvious if $p(a_i) = 0$ for some i, so assume that all $p(a_i)$ are nonzero and pairwise different.
There exist numbers s, t such that $s \mid p(a_1)$, $t \mid p(a_2)$, $st = \operatorname{lcm}(p(a_1), p(a_2))$ and $\gcd(s, t) = 1$.
As s, t are relatively prime numbers, there exist $m, n \in \mathbb{Z}$ such that $a_1 + sn = a_2 + tm =: b_2$. Obviously $s \mid p(a_1 + sn) - p(a_1)$ and $t \mid p(a_2 + tm) - p(a_2)$, so $st \mid p(b_2)$.
Similarly one obtains b_3 such that $p(a_3) \mid p(b_3)$ and $p(b_2) \mid p(b_3)$ thus also $p(a_1) \mid p(b_3)$ and $p(a_2) \mid p(b_3)$.
Reasoning inductively we obtain the existence of $a = b_k$ as required.
The polynomial $p(x) = 2x^2 + 2$ shows that the second part of the problem is not true, as $p(0) = 2$, $p(1) = 4$ but no value of $p(a)$ is divisible by 8 for integer a.

Remark. One can assume that the $p(a_i)$ are nonzero and ask for a such that $p(a)$ is a nonzero multiple of all $p(a_i)$. In the solution above, it can happen that $p(a) = 0$. But every number $p(a + np(a_1)p(a_2)\ldots p(a_k))$ is also divisible by every $p(a_i)$, since the polynomial is nonzero, there exists n such that $p(a + np(a_1)p(a_2)\ldots p(a_k))$ satisfies the modified thesis.

2008-D1-P4 — Statement

Day 1, P4

We say a triple (a_1, a_2, a_3) of nonnegative reals is better than another triple (b_1, b_2, b_3) if two out of the three following inequalities $a_1 > b_1$, $a_2 > b_2$, $a_3 > b_3$ are satisfied. We call a triple (x, y, z) special if x, y, z are nonnegative and $x + y + z = 1$. Find all natural numbers n for which there is a set S of n special triples such that for any given special triple we can find at least one better triple in S.

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Theory and methods for 2008-D1-P4

Theory Convex bodies, supporting functionals, and packing [T-DS-CONVEX]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Extremal choice, exchange arguments, and invariants [T-DS-EXTREMAL]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]; Repair an extremal configuration by a local exchange [M-DS-EXTREMAL]; Split a family by traces, pieces, or labels [M-DS-PARTITION]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2008-D1-P4 — Solution

For this record: Theory: T-DS-CONVEX; T-DS-GEOMETRY; T-DS-EXTREMAL; T-DS-COUNTING. Methods: M-DS-GEOMETRIC; M-DS-EXTREMAL; M-DS-PARTITION. Full learning route

The answer is $n \geq 4$. Adding distinct special triples preserves the required property, and any set with fewer than three elements can be enlarged to three. Thus it suffices to construct a four-element set and rule out every three-element set.
Consider the following set of special triples:

$$\left(0, \frac{8}{15}, \frac{7}{15}\right), \quad \left(\frac{2}{5}, 0, \frac{3}{5}\right), \quad \left(\frac{3}{5}, \frac{2}{5}, 0\right), \quad \left(\frac{2}{15}, \frac{11}{15}, \frac{2}{15}\right).$$

We will prove that any special triple (x, y, z) is worse than one of these (triple a is worse than triple b if triple b is better than triple a). We suppose that some special triple (x, y, z) is actually not worse than the first three of the triples from the given set, derive some conditions on x, y, z and prove that, under these conditions, (x, y, z) is worse than the fourth triple from the set.
If (x, y, z) is not beaten by any of the first three triples, then

$$(y \geq 8/15 \text{ or } z \geq 7/15), \qquad (x \geq 2/5 \text{ or } z \geq 3/5), \qquad (x \geq 3/5 \text{ or } y \geq 2/5).$$

The three inequalities $x \geq 2/5$, $y \geq 2/5$, $z \geq 7/15$ cannot all hold, since their lower bounds sum to more than one. If $x < 2/5$, the last two alternatives imply $y \geq 2/5$ and $z \geq 3/5$. Thus $(x, y, z) = (0, 2/5, 3/5)$, which the fourth triple beats in the first two coordinates. If $y < 2/5$, the first and third alternatives give $x \geq 3/5$ and $z \geq 7/15$, contradicting $x + y + z = 1$. Finally, if $z < 7/15$, the first two alternatives give $y \geq 8/15$ and $x \geq 2/5$. Consequently $z \leq 1/15$ and $y \leq 3/5$; the fourth triple beats (x, y, z) in its second and third coordinates. These cases prove that the four triples suffice.
We will prove that for any given set of three special triples one can find a special triple which is not worse than any triple from the set. Suppose we have a set S of three special triples

$$(x_1, y_1, z_1), \quad (x_2, y_2, z_2), \quad (x_3, y_3, z_3).$$

Denote $a(S) = \min(x_1, x_2, x_3)$, $b(S) = \min(y_1, y_2, y_3)$, $c(S) = \min(z_1, z_2, z_3)$. It is easy to check that S_1:

$$\left(\frac{x_1 - a}{1 - a - b - c}, \frac{y_1 - b}{1 - a - b - c}, \frac{z_1 - c}{1 - a - b - c}\right)$$

$$\left(\frac{x_2 - a}{1 - a - b - c}, \frac{y_2 - b}{1 - a - b - c}, \frac{z_2 - c}{1 - a - b - c}\right)$$

$$\left(\frac{x_3 - a}{1 - a - b - c}, \frac{y_3 - b}{1 - a - b - c}, \frac{z_3 - c}{1 - a - b - c}\right)$$

is a set of three special triples also (we may suppose that $a + b + c < 1$, because otherwise all three triples are equal and our statement is trivial).
If there is a special triple (x, y, z) which is not worse than any triple from S_1, then the triple

$$((1 - a - b - c)x + a, (1 - a - b - c)y + b, (1 - a - b - c)z + c)$$

is special and not worse than any triple from S. We also have $a(S_1) = b(S_1) = c(S_1) = 0$, so we may suppose that the same holds for our starting set S.
Suppose that one element of S has two entries equal to 0.

2008-D1-P4 — Solution (continued)

Note that one of the two remaining triples from S is not worse than the other. This triple is also not worse than all triples from S because any special triple is not worse than itself and the triple with two zeroes.

So we have $a = b = c = 0$ but we may suppose that all triples from S contain at most one zero. By transposing triples and elements in triples (elements in all triples must be transposed simultaneously) we may achieve the following situation $x_1 = y_2 = z_3 = 0$ and $x_2 \geq x_3$. If $z_2 \geq z_1$, then the second triple $(x_2, 0, z_2)$ is not worse than the other two triples from S. So we may assume that $z_1 > z_2$. If $y_1 \geq y_3$, then the first triple is not worse than the second and the third and we assume $y_3 > y_1$. Consider the three pairs of numbers x_2, y_1; z_1, x_3; y_3, z_2. The sum of all these numbers is three and consequently the sum of the numbers in one of the pairs is less than or equal to one. If it is the first pair then the triple $(x_2, 1 - x_2, 0)$ is not worse than all triples from S, for the second we may take $(1 - z_1, 0, z_1)$ and for the third — $(0, y_3, 1 - y_3)$. So we found a desirable special triple for any given S.

2008-D1-P5 — Statement

Day 1, P5

Does there exist a finite group G with a normal subgroup H such that $|\operatorname{Aut} H| > |\operatorname{Aut} G|$?

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Theory and methods for 2008-D1-P5

Theory Groups, orders, generators, and quotients [T-AL-GROUP]; Group actions, stabilizers, and semidirect products [T-AL-ACTION]; Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Rank, dimension and exact linear constraints [T-LA-RANK]
Methods Convert group data into orbits or cosets [M-AL-ACTION]
Strategy Build a witness and verify every constraint [S-CONSTRUCTION]; Choose the objects and count them without ambiguity [S-COUNTING]

Statement Recorded solutions Catalog

2008-D1-P5 — Solution

For this record: Theory: T-AL-GROUP; T-AL-ACTION; T-AL-FINITEFIELD; T-LA-RANK. Methods: M-AL-ACTION. Full learning route

Yes. Let H be the commutative group $H = \mathbb{F}_2^3$, where $\mathbb{F}_2 \cong \mathbb{Z}/2\mathbb{Z}$ is the field with two elements. The group of automorphisms of H is the general linear group $GL_3\mathbb{F}_2$; it has

$$(8-1)\cdot(8-2)\cdot(8-4) = 7\cdot 6\cdot 4 = 168$$

elements. One of them is the shift operator $\varphi : (x_1, x_2, x_3) \mapsto (x_2, x_3, x_1)$.
Now let $T = \{a^0, a^1, a^2\}$ be a group of order 3 (written multiplicatively); it acts on H by $\tau(a) = \varphi$. Let G be the semidirect product $G = H \rtimes_\tau T$. In other words, G is the group of 24 elements

$$G = \{ba^i : b \in H,\ i \in (\mathbb{Z}/3\mathbb{Z})\}, \qquad ab = \varphi(b)a.$$

G has one element e of order 1 and seven elements b, $b \in H$, $b \neq e$ of order 2.
If $g = ba$, we find that $g^2 = baba = b\varphi(b)a^2 \neq e$, and that

$$g^3 = b\varphi(b)a^2ba = b\varphi(b)a\varphi(b)a^2 = b\varphi(b)\varphi^2(b)a^3 = \psi(b),$$

where the homomorphism $\psi : H \to H$ is defined as $\psi : (x_1, x_2, x_3) \mapsto (x_1 + x_2 + x_3)(1, 1, 1)$. It is clear that $g^3 = \psi(b) = e$ for 4 elements $b \in H$, while $g^6 = \psi^2(b) = e$ for all $b \in H$.
We see that G has 8 elements of order 3, namely ba and ba^2 with $b \in \operatorname{Ker}\psi$, and 8 elements of order 6, namely ba and ba^2 with $b \notin \operatorname{Ker}\psi$. That accounts for orders of all elements of G.
Let $b_0 \in H \setminus \operatorname{Ker}\psi$ be arbitrary; it is easy to see that G is generated by b_0 and a. As every automorphism of G is fully determined by its action on b_0 and a, it follows that G has no more than

$$7 \cdot 8 = 56$$

2008-D1-P5 — Solution (continued)

automorphisms.

Remark. G and H can be equivalently presented as subgroups of S_6, namely as $H = \langle (12), (34), (56) \rangle$ and $G = \langle (135)(246), (12) \rangle$.

2008-D1-P6 — Statement

Day 1, P6

For a permutation $\sigma = (i_1, i_2, \dots, i_n)$ of $(1, 2, \dots, n)$ define $D(\sigma) = \sum_{k=1}^{n} |i_k - k|$. Let $Q(n, d)$ be the number of permutations σ of $(1, 2, \dots, n)$ with $d = D(\sigma)$. Prove that $Q(n, d)$ is even for $d \geq 2n$.

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Theory and methods for 2008-D1-P6

Theory Generating functions, coefficient filters, and finite differences [T-DS-GENERATING]; Finite counting and incidence identities [T-DS-COUNTING]; Determinants, block identities and inverses [T-LA-DETERMINANT]; Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Methods Encode the statistic and extract the required coefficient [M-DS-GENERATING]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2008-D1-P6 — Solution

For this record: Theory: T-DS-GENERATING; T-DS-COUNTING; T-LA-DETERMINANT; T-AL-PERMUTATION; T-AL-CONGRUENCE. Methods: M-DS-GENERATING. Full learning route

Consider the $n \times n$ determinant

$$\Delta(x) = \begin{vmatrix} 1 & x & \dots & x^{n-1} \\ x & 1 & \dots & x^{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ x^{n-1} & x^{n-2} & \dots & 1 \end{vmatrix}$$

where the ij-th entry is $x^{|i-j|}$. From the definition of the determinant we get

$$\Delta(x) = \sum_{(i_1,\dots,i_n) \in S_n} (-1)^{\mathrm{inv}(i_1,\dots,i_n)} x^{D(i_1,\dots,i_n)}$$

2008-D1-P6 — Solution (continued)

where S_n is the set of all permutations of $(1, 2, \dots, n)$ and $\text{inv}(i_1, \dots, i_n)$ denotes the number of inversions in the sequence $(i_1, \dots, i_n)$. So $Q(n, d)$ has the same parity as the coefficient of x^d in $\Delta(x)$.

It remains to evaluate $\Delta(x)$. In order to eliminate the entries below the diagonal, subtract the $(n-1)$-th row, multiplied by x, from the n-th row. Then subtract the $(n-2)$-th row, multiplied by x, from the $(n-1)$-th and so on. Finally, subtract the first row, multiplied by x, from the second row.

$$\begin{aligned}
&\Delta(x)\\
&= \begin{vmatrix} 1 & x & \dots & x^{n-2} & x^{n-1} \\ x & 1 & \dots & x^{n-3} & x^{n-2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x^{n-2} & x^{n-3} & \dots & 1 & x \\ x^{n-1} & x^{n-2} & \dots & x & 1 \end{vmatrix}\\
&= \dots\\
&= \begin{vmatrix} 1 & x & \dots & x^{n-2} & x^{n-1} \\ 0 & 1-x^2 & \dots & x^{n-3}-x^{n-1} & x^{n-2}-x^n \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1-x^2 & x-x^3 \\ 0 & 0 & \dots & 0 & 1-x^2 \end{vmatrix}\\
&= (1-x^2)^{n-1}.
\end{aligned}$$

For $d \geq 2n$, the coefficient of x^d is 0 so $Q(n, d)$ is even.

Day 2

2008-D2-P1 — Statement

Day 2, P7

Let n, k be positive integers and suppose that the polynomial $x^{2k} - x^k + 1$ divides $x^{2n} + x^n + 1$. Prove that $x^{2k} + x^k + 1$ divides $x^{2n} + x^n + 1$.

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Theory and methods for 2008-D2-P1

Theory Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]; Roots of unity, cyclotomic polynomials, and algebraic norms [T-AL-CYCLOTOMIC]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Evaluate at roots of unity or average over them [M-AL-UNITY]; Replace many equal values by too many roots [M-AL-ROOTCOUNT]
Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2008-D2-P1 — Solution

For this record: Theory: T-AL-COMPLEX; T-AL-CYCLOTOMIC; T-AL-POLYNOMIAL. Methods: M-AL-UNITY; M-AL-ROOTCOUNT. Full learning route

Let $f(x) = x^{2n} + x^n + 1$, $g(x) = x^{2k} - x^k + 1$, $h(x) = x^{2k} + x^k + 1$. The complex number $x_1 = \cos\left(\frac{\pi}{3k}\right) + i \sin\left(\frac{\pi}{3k}\right)$ is a root of $g(x)$.
Let $\alpha = \frac{\pi n}{3k}$. Since $g(x)$ divides $f(x)$, $f(x_1) = g(x_1) = 0$. So,

$$\begin{aligned}
&0\\
&= x_1^{2n} + x_1^n + 1\\
&= (\cos(2\alpha) + i\sin(2\alpha)) + (\cos\alpha + i\sin\alpha) + 1\\
&= 0,
\end{aligned}$$

2008-D2-P1 — Solution (continued)

and $(2\cos\alpha+1)(\cos\alpha+i\sin\alpha)=0$. Hence $2\cos\alpha+1=0$, i.e. $\alpha=\pm\frac{2\pi}{3}+2\pi c$, where $c\in\mathbb{Z}$.

Let x_2 be a root of the polynomial $h(x)$. Since $h(x)=\frac{x^{3k}-1}{x^k-1}$, the roots of the polynomial $h(x)$ are distinct and they are $x_2=\cos\left(\frac{2\pi s}{3k}\right)+i\sin\left(\frac{2\pi s}{3k}\right)$, where $s=3a\pm1$, $a\in\mathbb{Z}$. It is enough to prove that $f(x_2)=0$. We have

$$\begin{aligned}&f(x_2)\\&=x_2^{2n}+x_2^n+1\\&=(\cos(4s\alpha)+i\sin(4s\alpha))+(\cos(2s\alpha)+i\sin(2s\alpha))+1\\&=(2\cos(2s\alpha)+1)(\cos(2s\alpha)+i\sin(2s\alpha))\\&=0\end{aligned}$$

(since $2\cos\left(\frac{4\pi}{3}s\right)+1=2\cos\left(\frac{4\pi}{3}(3a\pm1)\right)+1=0$).

2008-D2-P2 — Statement

Day 2, P8

Two different ellipses are given. One focus of the first ellipse coincides with one focus of the second ellipse. Prove that the ellipses have at most two points in common.

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Theory and methods for 2008-D2-P2

Theory Conics, reflections, and inversion [T-DS-TRANSFORMS]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Strategy Translate geometry into a controlled representation [S-GEOMETRIC]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2008-D2-P2 — Solution

For this record: Theory: T-DS-TRANSFORMS; T-DS-GEOMETRY; T-AL-POLYNOMIAL. Methods: M-DS-GEOMETRIC. Full learning route

It is well known that an ellipse might be defined by a focus (a point) and a directrix (a straight line), as a locus of points such that the distance to the focus divided by the distance to the directrix is equal to a given number $e<1$. So, if a point X belongs to both ellipses with the same focus F and directrices l_1, l_2, then $e_1\cdot l_1X=FX=e_2\cdot l_2X$ (here we denote by l_1X, l_2X distances between the corresponding line and the point X). The equation $e_1\cdot l_1X=e_2\cdot l_2X$ defines two lines, whose equations are linear combinations with coefficients $e_1,\pm e_2$ of the normalized equations of lines l_1,l_2 but of those two only one is relevant, since X and F should lie on the same side of each directrix. So, we have that all possible points lie on one line. The intersection of a line and an ellipse consists of at most two points.

2008-D2-P3 — Statement

Day 2, P9

Let n be a positive integer. Prove that 2^{n-1} divides

$$\sum_{0\le k<n/2}\binom{n}{2k+1}5^k.$$

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Theory and methods for 2008-D2-P3

Theory Linear recurrences and generating functions [T-AL-RECURRENCE]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Find a recurrence that reveals the invariant [M-AL-RECURRENCE]
Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2008-D2-P3 — Solution

For this record: Theory: T-AL-RECURRENCE; T-AL-POLYNOMIAL. Methods: M-AL-RECURRENCE. Full learning route

As is known, the Fibonacci numbers F_n can be expressed as $F_n = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right)$. Expanding this expression, we obtain that $F_n = \frac{1}{2^{n-1}}\left(\binom{n}{1} + \binom{n}{3}5 + \cdots + \binom{n}{l}5^{\frac{l-1}{2}}\right)$, where l is the greatest odd number such that $l \leq n$ and $s = \frac{l-1}{2} \leq n/2$.

So, $F_n = \frac{1}{2^{n-1}}\sum\limits_{k=0}^{s}\binom{n}{2k+1}5^k$, which implies that 2^{n-1} divides $\sum\limits_{0\leq k<n/2}\binom{n}{2k+1}5^k$.

2008-D2-P4 — Statement

Day 2, P10

Let $\mathbb{Z}[x]$ be the ring of polynomials with integer coefficients, and let $f(x), g(x) \in \mathbb{Z}[x]$ be nonconstant polynomials such that $g(x)$ divides $f(x)$ in $\mathbb{Z}[x]$. Prove that if the polynomial $f(x) - 2008$ has at least 81 distinct integer roots, then the degree of $g(x)$ is greater than 5.

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Theory and methods for 2008-D2-P4

Theory Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Prime valuations, divisors, and factorials [T-AL-VALUATION]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Replace many equal values by too many roots [M-AL-ROOTCOUNT]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]

Statement Recorded solutions Catalog

2008-D2-P4 — Solution

For this record: Theory: T-AL-POLYNOMIAL; T-AL-VALUATION; T-DS-COUNTING. Methods: M-AL-ROOTCOUNT. Full learning route

Let $f(x) = g(x)h(x)$ where $h(x)$ is a polynomial with integer coefficients.
Let $a_1, \ldots, a_{81}$ be distinct integer roots of the polynomial $f(x) - 2008$. Then $f(a_i) = g(a_i)h(a_i) = 2008$ for $i = 1, \ldots, 81$, Hence, $g(a_1), \ldots, g(a_{81})$ are integer divisors of 2008.

Since $2008 = 2^3 \cdot 251$ (2, 251 are primes) then 2008 has exactly 16 distinct integer divisors (including the negative divisors as well). By the pigeonhole principle, there are at least 6 equal numbers among $g(a_1), \ldots, g(a_{81})$ (because $81 > 16 \cdot 5$). For example, $g(a_1) = g(a_2) = \cdots = g(a_6) = c$. So $g(x) - c$ is a nonconstant polynomial which has at least 6 distinct roots (namely $a_1, \ldots, a_6$). Then the degree of the polynomial $g(x) - c$ is at least 6.

2008-D2-P5 — Statement

Day 2, P11

Let n be a positive integer, and consider the matrix $A = (a_{ij})_{1 \le i,j \le n}$, where

$$a_{ij} = \begin{cases} 1 & \text{if } i+j \text{ is a prime number,} \\ 0 & \text{otherwise.} \end{cases}$$

Prove that $|\det A| = k^2$ for some integer k.

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Theory and methods for 2008-D2-P5

Theory Determinants, block identities and inverses [T-LA-DETERMINANT]; Rank, dimension and exact linear constraints [T-LA-RANK]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Methods Exploit a compatible block decomposition [M-LA-BLOCK]
Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2008-D2-P5 — Solution

For this record: Theory: T-LA-DETERMINANT; T-LA-RANK; T-AL-CONGRUENCE. Methods: M-LA-BLOCK. Full learning route

Call a square matrix of type (B), if it is of the form

$$\begin{pmatrix} 0 & b_{12} & 0 & \dots & b_{1,2k-2} & 0 \\ b_{21} & 0 & b_{23} & \dots & 0 & b_{2,2k-1} \\ 0 & b_{32} & 0 & \dots & b_{3,2k-2} & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ b_{2k-2,1} & 0 & b_{2k-2,3} & \dots & 0 & b_{2k-2,2k-1} \\ 0 & b_{2k-1,2} & 0 & \dots & b_{2k-1,2k-2} & 0 \end{pmatrix}.$$

Note that every matrix of this form has determinant zero, because it has k columns spanning a vector space of dimension at most $k-1$.
Call a square matrix of type (C), if it is of the form

$$C' = \begin{pmatrix} 0 & c_{11} & 0 & c_{12} & \dots & 0 & c_{1,k} \\ c_{11} & 0 & c_{12} & 0 & \dots & c_{1,k} & 0 \\ 0 & c_{21} & 0 & c_{22} & \dots & 0 & c_{2,k} \\ c_{21} & 0 & c_{22} & 0 & \dots & c_{2,k} & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & c_{k,1} & 0 & c_{k,2} & \dots & 0 & c_{k,k} \\ c_{k,1} & 0 & c_{k,2} & 0 & \dots & c_{k,k} & 0 \end{pmatrix}.$$

By permutations of rows and columns, we see that

$$|\det C'| = \left|\det \begin{pmatrix} 0 & C \\ C & 0 \end{pmatrix}\right| = |\det C|^2,$$

where C denotes the $k \times k$-matrix with coefficients $c_{i,j}$. Therefore, the determinant of any matrix of type (C) is a perfect square (up to a sign).
Now let X' be the matrix obtained from A by replacing the first row by $(1 \quad 0 \quad 0 \quad \dots \quad 0)$, and let Y be the matrix obtained from A by replacing the entry a_{11} by 0. By multi-linearity of the determinant, $\det(A) = \det(X') + \det(Y)$. Note that X' can be written as

$$X' = \begin{pmatrix} 1 & 0 \\ v & X \end{pmatrix}$$

for some $(n-1) \times (n-1)$-matrix X and some column vector v. Then $\det(A) = \det(X) + \det(Y)$. Now consider two cases. If n is odd, then X is of type (C), and Y is of type (B). Therefore, $|\det(A)| =$

2008-D2-P5 — Solution (continued)

$|\det(X)|$ is a perfect square. If n is even, then X is of type (B), and Y is of type (C); hence $|\det(A)| = |\det(Y)|$ is a perfect square.

The set of primes can be replaced by any subset of $\{2\} \cup \{3, 5, 7, 9, 11, \dots\}$.

2008-D2-P6 — Statement

Day 2, P12

Let $\mathcal{H}$ be an infinite-dimensional real Hilbert space, let $d > 0$, and suppose that S is a set of points (not necessarily countable) in $\mathcal{H}$ such that the distance between any two distinct points in S is equal to d. Show that there is a point $y \in \mathcal{H}$ such that

$$\left\{ \frac{\sqrt{2}}{d}(x - y) : x \in S \right\}$$

is an orthonormal system of vectors in $\mathcal{H}$.

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Theory and methods for 2008-D2-P6

Theory Completeness, barycentres and equidistant sets [T-LA-HILBERT]; Inner products, orthogonality and polarization [T-LA-INNER]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Methods Encode distances and orthogonality by a Gram matrix [M-LA-GRAM]
Strategy Translate geometry into a controlled representation [S-GEOMETRIC]; Control the limiting operation before taking the limit [S-ANALYTIC]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2008-D2-P6 — Solution

For this record: Theory: T-LA-HILBERT; T-LA-INNER; T-AN-CONVERGENCE. Methods: M-LA-GRAM. Full learning route

If S is empty, the assertion is immediate. If $S = \{x\}$, choose a unit vector $e \in \mathcal{H}$ and set $y = x - (d/\sqrt{2})e$. Henceforth assume $|S| \geq 2$.

It is clear that, if B is an orthonormal system in a Hilbert space $\mathcal{H}$, then $\{(d/\sqrt{2})e : e \in B\}$ is a set of points in $\mathcal{H}$, any two of which are at distance d apart. We need to show that every set S of equidistant points is a translate of such a set.

We begin with identities that hold whenever the points occurring in the respective identity are distinct:

$$\begin{aligned}
&\langle x_2 - x_1, x_2 - x_1 \rangle = d^2, \\
&\langle x_2 - x_1, x_3 - x_1 \rangle \\
&= \tfrac{1}{2}(\|x_2 - x_1\|^2 + \|x_3 - x_1\|^2 - \|x_2 - x_3\|^2) = \tfrac{d^2}{2}, \\
&\langle x_2 - x_1, x_4 - x_3 \rangle \\
&= \langle x_2 - x_1, x_4 - x_1 \rangle - \langle x_2 - x_1, x_3 - x_1 \rangle \\
&= \tfrac{d^2}{2} - \tfrac{d^2}{2} = 0.
\end{aligned}$$

More explicitly, for any distinct $x_1, \dots, x_n \in S$ and real coefficients a_i with $\sum_i a_i = 0$, expansion of squared distances gives

$$\begin{aligned}
\left\| \sum_{i=1}^{n} a_i x_i \right\|^2 &= -\frac{1}{2} \sum_{i,j=1}^{n} a_i a_j \|x_i - x_j\|^2 \\
&= \frac{d^2}{2} \sum_{i=1}^{n} a_i^2.
\end{aligned}$$

Polarization gives the corresponding inner-product identity for two coefficient lists with sum zero. Hence all these norms and scalar products can be evaluated in the canonical example $x_i = (d/\sqrt{2})e_i$ in $\mathbb{R}^n$.

2008-D2-P6 — Solution (continued)

If $S = \{x_1, x_2, \dots, x_n\}$ is a finite set, we form $x = \frac{1}{n}(x_1 + x_2 + \cdots + x_n)$, pick a non-zero vector $z \in [\text{span}(x_1 - x, x_2 - x, \dots, x_n - x)]^\perp$ and seek y in the form $y = x + \lambda z$ for a suitable $\lambda \in \mathbb{R}$. We find that

$$\langle x_1 - y, x_2 - y\rangle = \langle x_1 - x - \lambda z, x_2 - x - \lambda z\rangle = \langle x_1 - x, x_2 - x\rangle + \lambda^2\|z\|^2.$$

$\langle x_1 - x, x_2 - x\rangle$ may be computed by our remark above as

$$\begin{aligned}
&\langle x_1 - x, x_2 - x\rangle \\
&= \frac{d^2}{2}\left\langle \begin{pmatrix} \frac{n-1}{n} \\ -\frac{1}{n} \\ -\frac{1}{n} \\ \vdots \\ -\frac{1}{n} \end{pmatrix}, \begin{pmatrix} -\frac{1}{n} \\ \frac{n-1}{n} \\ -\frac{1}{n} \\ \vdots \\ -\frac{1}{n} \end{pmatrix} \right\rangle_{\mathbb{R}^n} \\
&= \frac{d^2}{2}\left(-\frac{2(n-1)}{n^2} + \frac{n-2}{n^2}\right) \\
&= -\frac{d^2}{2n}.
\end{aligned}$$

So the choice $\lambda = \dfrac{d}{\sqrt{2n}\,\|z\|}$ will make all vectors $\frac{\sqrt{2}}{d}(x_i - y)$ orthogonal to each other; it is easily checked as above that they will also be of length one.
Let now S be an infinite set. Pick an infinite sequence $T = \{x_1, x_2, \dots, x_n, \dots\}$ of distinct points in S. We claim that the sequence

$$y_n = \frac{1}{n}(x_1 + x_2 + \cdots + x_n)$$

is a Cauchy sequence in $\mathcal{H}$. (This is the crucial observation.) Indeed, for $m > n$, the norm $\|y_m - y_n\|$ may be computed by the above remark as

$$\begin{aligned}
&\|y_m - y_n\|^2 \\
&= \frac{d^2}{2}\left\|\left(\tfrac{1}{m} - \tfrac{1}{n}, \dots, \tfrac{1}{m} - \tfrac{1}{n}, \tfrac{1}{m}, \dots, \tfrac{1}{m}\right)^\top\right\|_{\mathbb{R}^m}^2 \\
&= \frac{d^2}{2}\left(\frac{n(m-n)^2}{m^2n^2} + \frac{m-n}{m^2}\right) \\
&= \frac{d^2}{2}\cdot\frac{m-n}{mn} \\
&= \frac{d^2}{2}\left(\frac{1}{n} - \frac{1}{m}\right) \\
&\to 0, \quad m, n \to \infty.
\end{aligned}$$

By completeness of $\mathcal{H}$, it follows that there exists a limit

$$y = \lim_{n\to\infty} y_n \in \mathcal{H}.$$

We claim that y satisfies all conditions of the problem. For fixed distinct positive integers n, p and $m > \max\{n, p\}$, we compute

$$\begin{aligned}
&\|x_n - y_m\|^2 \\
&= \frac{d^2}{2}\left\|\left(-\tfrac{1}{m}, \dots, -\tfrac{1}{m}, 1 - \tfrac{1}{m}, -\tfrac{1}{m}, \dots, -\tfrac{1}{m}\right)^\top\right\|_{\mathbb{R}^m}^2 \\
&= \frac{d^2}{2}\left(\frac{m-1}{m^2} + \frac{(m-1)^2}{m^2}\right) \\
&= \frac{d^2}{2}\cdot\frac{m-1}{m} \\
&\to \frac{d^2}{2}, \quad m \to \infty,
\end{aligned}$$

2008-D2-P6 — Solution (continued)

showing that $\|x_n - y\| = d/\sqrt{2}$, as well as

$$\langle x_n - y_m, x_p - y_m \rangle = \frac{d^2}{2}\left(\frac{m-2}{m^2} - \frac{2}{m}\left(1 - \frac{1}{m}\right)\right) = -\frac{d^2}{2m} \to 0, \quad m \to \infty,$$

showing that $\langle x_n - y, x_p - y \rangle = 0$, so that

$$\left\{\frac{\sqrt{2}}{d}(x_n - y) : n \in \mathbb{N}\right\}$$

is indeed an orthonormal system of vectors.

This proves the claim on T. To cover all of S, let F be any subset of $S \setminus T$ with one or two elements, and write $k = |F|$. Apply the preceding construction to the sequence obtained by placing the points of F before $x_1, x_2, \ldots$. Its averages have the same limit because

$$\frac{\sum_{z \in F} z + x_1 + \cdots + x_n}{n+k} = \frac{n}{n+k} y_n + \frac{1}{n+k} \sum_{z \in F} z \longrightarrow y.$$

Thus all vectors in

$$\left\{\frac{\sqrt{2}}{d}(x - y) : x \in T \cup F\right\}$$

have norm one and are pairwise orthogonal. Taking every possible one- or two-point set F proves these properties for the entire system

$$\left\{\frac{\sqrt{2}}{d}(x - y) : x \in S\right\},$$

as required.

3.16 2009

Day 1

2009-D1-P1 — Statement

Day 1, P1

Suppose that f and g are real-valued functions on the real line and $f(r) \leq g(r)$ for every rational r. Does this imply that $f(x) \leq g(x)$ for every real x if
a) f and g are non-decreasing?

b) f and g are continuous?

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Theory and methods for 2009-D1-P1

Theory Continuity, compactness, and order [T-AN-CONTINUITY]; Countability, accumulation, and Cantor constructions [T-AN-SETS]
Methods Use a dense set, an enumeration, or a category argument [M-AN-TOPOLOGY]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2009-D1-P1 — Solution

For this record: Theory: T-AN-CONTINUITY; T-AN-SETS. Methods: M-AN-TOPOLOGY. Full learning route

a) No. Counter-example: f and g can be chosen as the characteristic functions of $[\sqrt{3}, \infty)$ and $(\sqrt{3}, \infty)$, respectively.

b) Yes. By the assumptions $g - f$ is continuous on the whole real line and nonnegative on the rationals. Since any real number can be obtained as a limit of rational numbers we get that $g - f$ is nonnegative on the whole real line.

2009-D1-P2 — Statement

Day 1, P2

Let A, B and C be real square matrices of the same size, and suppose that A is invertible. Prove that if $(A - B)C = BA^{-1}$, then $C(A - B) = A^{-1}B$.

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Theory and methods for 2009-D1-P2

Theory Determinants, block identities and inverses [T-LA-DETERMINANT]
Methods Complete the equation to a product equal to identity [M-LA-INVERSE]
Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2009-D1-P2 — Solution

For this record: Theory: T-LA-DETERMINANT. Methods: M-LA-INVERSE. Full learning route

A straightforward calculation shows that $(A - B)C = BA^{-1}$ is equivalent to $AC - BC - BA^{-1} + AA^{-1} = I$, where I denotes the identity matrix. This is equivalent to $(A - B)(C + A^{-1}) = I$. Hence, $(A - B)^{-1} = C + A^{-1}$, meaning that $(C + A^{-1})(A - B) = I$ also holds. Expansion yields the desired result.

2009-D1-P3 — Statement

Day 1, P3

In a town every two residents who are not friends have a friend in common, and no one is a friend of everyone else. Let us number the residents from 1 to n and let a_i be the number of friends of the i-th resident. Suppose that $\sum_{i=1}^{n} a_i^2 = n^2 - n$. Let k be the smallest number of residents (at least three) who can be seated at a round table in such a way that any two neighbors are friends. Determine all possible values of k.

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Theory and methods for 2009-D1-P3

Theory Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Count incidences with their exact multiplicity [M-DS-DOUBLECOUNT]; Encode a relation as a graph and extract a cycle [M-DS-GRAPH]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2009-D1-P3 — Solution

For this record: Theory: T-DS-GRAPH; T-DS-COUNTING. Methods: M-DS-DOUBLECOUNT; M-DS-GRAPH. Full learning route

Let us define the simple, undirected graph G so that the vertices of G are the town's residents and the edges of G are the friendships between the residents. Let $V(G) = \{v_1, v_2, \dots, v_n\}$ denote the vertices of G; a_i is degree of v_i for every i. Let $E(G)$ denote the edges of G. In this terminology, the problem asks us to describe the length k of the shortest cycle in G.

Let us count the walks of length 2 in G, that is, the ordered triples (v_i, v_j, v_l) of vertices with $v_iv_j, v_jv_l \in E(G)$ ($i = l$ being allowed). For a given j the number is obviously a_j^2, therefore the total number is $\sum_{i=1}^{n} a_i^2 = n^2 - n$.

Now we show that there is an injection f from the set of ordered pairs of distinct vertices to the set of these walks. For $v_iv_j \notin E(G)$, let $f(v_i, v_j) = (v_i, v_l, v_j)$ with arbitrary l such that $v_iv_l, v_lv_j \in E(G)$. For $v_iv_j \in E(G)$, let $f(v_i, v_j) = (v_i, v_j, v_i)$. f is an injection since for $i = l$, (v_i, v_j, v_l) can only be the image of (v_i, v_j), and for $i \neq l$, it can only be the image of (v_i, v_l).

Since the number of ordered pairs of distinct vertices is $n^2 - n$, $\sum_{i=1}^{n} a_i^2 \geq n^2 - n$. Equality holds iff f is surjective, that is, iff there is exactly one l with $v_iv_l, v_lv_j \in E(G)$ for every i, j with $v_iv_j \notin E(G)$ and there is no such l for any i, j with $v_iv_j \in E(G)$. In other words, iff G contains neither C_3 nor C_4 (cycles of length 3 or 4), that is, G is either a forest (a cycle-free graph) or the length of its shortest cycle is at least 5.

It is easy to check that if every two vertices of a forest are connected by a path of length at most 2, then the forest is a star (one vertex is connected to all others by an edge). But G has n vertices, and none of them has degree $n-1$. Hence G is not forest, so it has cycles. On the other hand, if the length of a cycle C of G is at least 6 then it has two vertices such that both arcs of C connecting them are longer than 2. Hence there is a path connecting them that is shorter than both arcs. Replacing one of the arcs by this path, we have a closed walk shorter than C. Therefore length of the shortest cycle is 5.

Finally, we must note that there is at least one G with the prescribed properties – e.g. the cycle C_5 itself satisfies the conditions. Thus 5 is the sole possible value of k.

2009-D1-P4 — Statement

Day 1, P4

Let $p(z) = a_0 + a_1z + a_2z^2 + \cdots + a_nz^n$ be a complex polynomial. Suppose that $1 = c_0 \geq c_1 \geq \dots \geq c_n \geq 0$ is a sequence of real numbers which is convex (i.e. $2c_k \leq c_{k-1} + c_{k+1}$ for every $k = 1, 2, \dots, n-1$), and consider the polynomial

$$q(z) = c_0a_0 + c_1a_1z + c_2a_2z^2 + \cdots + c_na_nz^n.$$

Prove that

$$\max_{|z|\leq 1} |q(z)| \leq \max_{|z|\leq 1} |p(z)|.$$

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Theory and methods for 2009-D1-P4

Theory Roots-of-unity filters and positive trigonometric kernels [T-AL-FOURIER]; Holomorphic bounds, poles, and logarithmic-branch sums [T-AL-COMPLEXANALYSIS]; Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]

Methods Evaluate at roots of unity or average over them [M-AL-UNITY]; Build a holomorphic function with a useful boundary [M-AL-ANALYTIC]

Strategy Build a chain of bounds and track equality [S-BOUND]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2009-D1-P4 — Solution

For this record: Theory: T-AL-FOURIER; T-AL-COMPLEXANALYSIS; T-AN-CONVEXITY. Methods: M-AL-UNITY; M-AL-ANALYTIC. Full learning route

The polynomials p and q are regular on the complex plane, so by the Maximum Principle, $\max_{|z|\le 1}|q(z)| = \max_{|z|=1}|q(z)|$, and similarly for p. Let us denote $M_f = \max_{|z|=1}|f(z)|$ for any regular function f. Thus it suffices to prove that $M_q \le M_p$.

First, note that we can assume $c_n = 0$. Indeed, for $c_n = 1$, we get $p = q$ and the statement is trivial; otherwise, $q(z) = c_n p(z) + (1-c_n)r(z)$, where $r(z) = \sum_{j=0}^{n} \frac{c_j - c_n}{1-c_n} a_j z^j$. The sequence $c'_j = \frac{c_j - c_n}{1-c_n}$ also satisfies the prescribed conditions (it is a positive linear transform of the sequence c_j with $c'_0 = 1$), but $c'_n = 0$ too, so we get $M_r \le M_p$. This is enough: $M_q = |q(z_0)| \le c_n|p(z_0)| + (1-c_n)|r(z_0)| \le c_n M_p + (1-c_n)M_r \le M_p$.

Using the Cauchy formulas, we can express the coefficients a_j of p from its values taken over the positively oriented circle $S = \{|z| = 1\}$:

$$a_j = \frac{1}{2\pi i}\int_S \frac{p(z)}{z^{j+1}}\,dz = \frac{1}{2\pi}\int_S \frac{p(z)}{z^j}\,|dz|$$

for $0 \le j \le n$, otherwise

$$\int_S \frac{p(z)}{z^j}\,|dz| = 0.$$

Let us use these identities to get a new formula for q, using only the values of p over S:

$$2\pi \cdot q(w) = \sum_{j=0}^{n} c_j \left(\int_S p(z) z^{-j}\,|dz|\right) w^j.$$

We can exchange the order of the summation and the integration (sufficient conditions to do this obviously apply):

$$2\pi \cdot q(w) = \int_S \left(\sum_{j=0}^{n} c_j (w/z)^j\right) p(z)\,|dz|.$$

It would be nice if the integration kernel (the sum between the brackets) was real. But this is easily arranged — for $-n \le j \le -1$, we can add the conjugate expressions, because by the above remarks, they are zero anyway:

$$2\pi \cdot q(w) = \sum_{j=0}^{n} c_j \left(\int_S p(z) z^{-j}\,|dz|\right) w^j = \sum_{j=-n}^{n} c_{|j|} \left(\int_S p(z) z^{-j}\,|dz|\right) w^j,$$

$$2\pi \cdot q(w) = \int_S \left(\sum_{j=-n}^{n} c_{|j|} (w/z)^j\right) p(z)\,|dz| = \int_S K(w/z)\,p(z)\,|dz|,$$

where

$$K(u) = \sum_{j=-n}^{n} c_{|j|} u^j = c_0 + 2\sum_{j=1}^{n} c_j\,\Re(u^j)$$

for $u \in S$.

Let us examine $K(u)$. It is a real-valued function. Again from the Cauchy formulas, $\int_S K(u)\,|du| = 2\pi c_0 = 2\pi$.

If $\int_S |K(u)|\,|du| = 2\pi$ still holds (taking the absolute value does not increase the integral), then for every w:

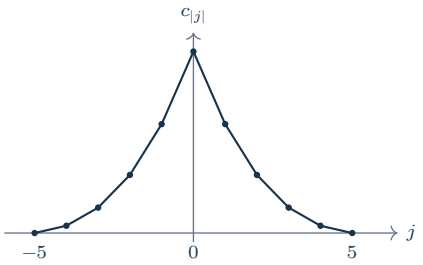
$$2\pi|q(w)| = \left|\int_S K(w/z)\,p(z)\,|dz|\right| \le \int_S |K(w/z)|\cdot|p(z)|\,|dz| \le M_p \int_S |K(u)|\,|du| = 2\pi M_p;$$

this would conclude the proof. So it suffices to prove that $\int_S |K(u)|\,|du| = \int_S K(u)\,|du|$, which is to say, K is non-negative.

Now let us decompose K into a sum using the given conditions for the numbers c_j (including $c_n = 0$). Let $d_k = c_{k-1} - 2c_k + c_{k+1}$ for $k = 1, \dots, n$ (setting $c_{n+1} = 0$); we know that $d_k \ge 0$. Let $F_k(u) = \sum_{j=-k+1}^{k-1} (k - |j|)u^j$. Then $K(u) = \sum_{k=1}^{n} d_k F_k(u)$ by easy induction (or see the diagram below for a graphical illustration).

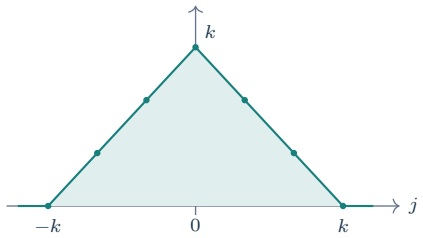
2009-D1-P4 — Solution (continued)

Symmetric extension of the coefficients



Example: $c = (1, .60, .32, .14, .04, 0)$.

One tent of coefficients



F_k: coefficients $(k - |j|)_+$ (shown for $k = 3$).

$$c_{|j|} = \sum_{k=|j|+1}^{n} d_k(k - |j|), \quad d_k \geq 0, \quad |j| \leq n \ (c_n = 0).$$

The points are discrete coefficients; connecting segments guide the eye.

So it suffices to prove that $F_k(u) \geq 0$ for $u \in S$. This is reasonably well-known (as F_k is the Fejér kernel), and also very easy:

$$F_k(u) = (1 + u + u^2 + \cdots + u^{k-1})(1 + u^{-1} + u^{-2} + \cdots + u^{-(k-1)}) =$$
$$= (1 + u + u^2 + \cdots + u^{k-1})\overline{(1 + u + u^2 + \cdots + u^{k-1})} = \left|1 + u + u^2 + \cdots + u^{k-1}\right|^2 \geq 0$$

This completes the proof.

2009-D1-P5 — Statement

Day 1, P5

Let n be a positive integer. An n-simplex in $\mathbb{R}^n$ is given by $n + 1$ points $P_0, P_1, \ldots, P_n$, called its vertices, which do not all belong to the same hyperplane. For every n-simplex S we denote by $v(S)$ the volume of S, and we write $C(S)$ for the center of the unique sphere containing all the vertices of S.

Suppose that P is a point inside an n-simplex S. Let S_i be the n-simplex obtained from S by replacing its i-th vertex by P. Prove that

$$v(S_0)C(S_0) + v(S_1)C(S_1) + \cdots + v(S_n)C(S_n) = v(S)C(S).$$

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Theory and methods for 2009-D1-P5

Theory Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Inner products, orthogonality and polarization [T-LA-INNER]; Determinants, block identities and inverses [T-LA-DETERMINANT]; Rank, dimension and exact linear constraints [T-LA-RANK]

Methods Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]; Strengthen the statement before the induction step [M-DS-INDUCT]

Strategy Translate geometry into a controlled representation [S-GEOMETRIC]; Reduce size while preserving the defining conditions [S-INDUCTION]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2009-D1-P5 — Solution — Record 1 of 2

For this record: Theory: T-DS-GEOMETRY; T-DS-INDUCTION; T-LA-INNER. Methods: M-DS-GEOMETRIC; M-DS-INDUCT. Full learning route

We will prove this by induction on n, starting with $n = 1$. In that case we are given an interval $[a, b]$ with a point $p \in (a, b)$, and we have to verify

$$(b-p)\frac{b+p}{2} + (p-a)\frac{p+a}{2} = (b-a)\frac{b+a}{2},$$

which is true.
Now let assume the result is true for $n-1$ and prove it for n. We have to show that the point

$$X = \sum_{j=0}^{n} \frac{v(S_j)}{v(S)} O(S_j)$$

has the same distance to all the points $P_0, P_1, \dots, P_n$. Let $i \in \{0, 1, 2, \dots, n\}$ and define the sets $M_i = \{P_0, P_1, \dots, P_{i-1}, P_{i+1}, \dots, P_n\}$. The set of all points having the same distance to all points in M_i is a line h_i orthogonal to the hyperplane E_i determined by the points in M_i. We are going to show that X lies on every h_i. To do so, fix some index i and notice that

$$X = \frac{v(S_i)}{v(S)} O(S_i) + \frac{v(S) - v(S_i)}{v(S)} \cdot \underbrace{\sum_{j \neq i} \frac{v(S_j)}{v(S) - v(S_i)} O(S_j)}_{Y}$$

and $O(S_i)$ lies on h_i, so that it is enough to show that Y lies on h_i.
A map $f : \mathbb{R}_{>0} \to \mathbb{R}^n$ will be called affine if there are points $A, B \in \mathbb{R}^n$ such that $f(\lambda) = \lambda A + (1-\lambda)B$. Consider the ray g starting in P_i and passing through P. For $\lambda > 0$ let $P_\lambda = (1-\lambda)P_i + \lambda P$, so that P_λ is an affine function describing the points of g. For every such λ let S_j^λ be the n-simplex obtained from S by replacing the j-th vertex by P_λ. The point $O(S_j^\lambda)$ is the intersection of the fixed line h_j with the hyperplane orthogonal to g and passing through the midpoint of the segment $P_i P_\lambda$ which is given by an affine function. This implies that also $O(S_j^\lambda)$ is an affine function. We write $\varphi_j = \frac{v(S_j)}{v(S) - s(S_i)}$, and then

$$Y_\lambda = \sum_{j \neq i} \varphi_j O(S_j^\lambda)$$

is an affine function. We want to show that $Y_\lambda \in h_i$ for all λ (then specializing to $\lambda = 1$ gives the desired result). It is enough to do this for two different values of λ.

Let g intersect the sphere containing the vertices of S in a point Z; then $Z = P_\lambda$ for a suitable $\lambda > 0$, and we have $O(S_j^\lambda) = O(S)$ for all j, so that $Y_\lambda = O(S) \in h_i$. Now let g intersect the hyperplane E_i in a point Q; then $Q = P_\lambda$ for some $\lambda > 0$, and Q is different from Z. Define T to be the $(n-1)$-simplex with vertex set M_i, and let T_j be the $(n-1)$-simplex obtained from T by replacing the vertex P_j by Q. If we write v' for the volume of $(n-1)$-simplices in the hyperplane E_i, then

$$\frac{v'(T_j)}{v'(T)} = \frac{v(S_j^\lambda)}{v(S)} = \frac{v(S_j^\lambda)}{\sum_{k \neq i} v(S_k^\lambda)} = \frac{\lambda v(S_j)}{\lambda \sum_{k \neq i} v(S_k)} = \frac{v(S_j)}{v(S) - v(S_i)} = \varphi_j.$$

If p denotes the orthogonal projection onto E_i then $p(O(S_j^\lambda)) = O(T_j)$, so that $p(Y_\lambda) = \sum_{j \neq i} \varphi_j O(T_j)$ equals $O(T)$ by induction hypothesis, which implies $Y_\lambda \in p^{-1}(O(T)) = h_i$, and we are done.

2009-D1-P5 — Solution — Record 2 of 2

For this record: Theory: T-DS-GEOMETRY; T-LA-INNER; T-LA-DETERMINANT; T-LA-RANK. Methods: M-DS-GEOMETRIC. Full learning route

For $n = 1$, the statement is checked easily.

2009-D1-P5 — Solution — Record 2 of 2 (continued)

Assume $n \geq 2$. Denote $O(S_j) - O(S)$ by q_j and $P_j - P$ by p_j. For all distinct j and k in the range $0, \dots, n$ the point $O(S_j)$ lies on a hyperplane orthogonal to p_k and P_j lies on a hyperplane orthogonal to q_k. So we have

$$\begin{cases} \langle p_i, q_j - q_k \rangle = 0 \\ \langle q_i, p_j - p_k \rangle = 0 \end{cases}$$

for all $j \neq i \neq k$. This means that the value $\langle p_i, q_j \rangle$ is independent of j as long as $j \neq i$, denote this value by λ_i. Similarly, $\langle q_i, p_j \rangle = \mu_i$ for some μ_i. Since $n \geq 2$, these equalities imply that all the λ_i and μ_i values are equal, in particular, $\langle p_i, q_j \rangle = \langle p_j, q_i \rangle$ for any i and j.
We claim that for such p_i and q_i, the volumes

$$V_j = |\det(p_0, \dots, p_{j-1}, p_{j+1}, \dots, p_n)|$$

and

$$W_j = |\det(q_0, \dots, q_{j-1}, q_{j+1}, \dots, q_n)|$$

are proportional. Indeed, first assume that $p_0, \dots, p_{n-1}$ and $q_0, \dots, q_{n-1}$ are bases of $\mathbb{R}^n$, then we have

$$\begin{aligned} V_j &= \frac{1}{|\det(q_0, \dots, q_{n-1})|} \left| \det\Big((\langle p_k, q_l \rangle)_{\substack{k \neq j \\ l < n}} \Big) \right| = \\ &= \frac{1}{|\det(q_0, \dots, q_{n-1})|} \left| \det\Big((\langle p_k, q_l \rangle)_{\substack{l \neq j \\ k < n}} \Big) \right| = \frac{|\det(p_0, \dots, p_{n-1})|}{|\det(q_0, \dots, q_{n-1})|} W_j. \end{aligned}$$

If our assumption did not hold after any reindexing of the vectors p_i and q_i, then both p_i and q_i span a subspace of dimension at most $n-1$ and all the volumes are 0.

Finally, it is clear that $\sum_j q_j W_j / \det(q_0, \dots, q_n) = 0$: the weight of p_j is the height of 0 over the hyperplane spanned by the rest of the vectors q_k relative to the height of p_j over the same hyperplane, so the sum is parallel to all the faces of the simplex spanned by $q_0, \dots, q_n$. By the argument above, we can change the weights to the proportional set of weights $V_j / \det(p_0, \dots, p_n)$ and the sum will still be 0. That is,

$$0 = \sum \frac{V_j}{\det(p_0, \dots, p_n)} q_j = \sum \frac{v(S_j)}{v(S)} (O(S_j) - O(S)) =$$

$$= \frac{1}{v(S)} \left(\sum O(S_j) v(S_j) - O(S) \sum v(S_j) \right) = \frac{1}{v(S)} \left(\sum O(S_j) v(S_j) - O(S) v(S) \right),$$

q.e.d.

Day 2

2009-D2-P1 — Statement

Day 2, P6

Let ℓ be a line and P a point in $\mathbb{R}^3$. Let S be the set of points X such that the distance from X to ℓ is greater than or equal to two times the distance between X and P. If the distance from P to ℓ is $d > 0$, find the volume of S.

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Theory and methods for 2009-D2-P1

Theory Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Conics, reflections, and inversion [T-DS-TRANSFORMS]; Determinants, block identities and inverses [T-LA-DETERMINANT]
Methods Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Strategy Translate geometry into a controlled representation [S-GEOMETRIC]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2009-D2-P1 — Solution

For this record: Theory: T-DS-GEOMETRY; T-DS-TRANSFORMS; T-LA-DETERMINANT. Methods: M-DS-GEOMETRIC. Full learning route

We can choose a coordinate system of the space such that the line ℓ is the z-axis and the point P is $(d,0,0)$. The distance from the point (x,y,z) to ℓ is $\sqrt{x^2+y^2}$, while the distance from P to X is $|PX|=\sqrt{(x-d)^2+y^2+z^2}$. Square everything to get rid of the square roots. The condition can be reformulated as follows: the square of the distance from ℓ to X is at least $4|PX|^2$.

$$x^2+y^2 \geq 4((x-d)^2+y^2+z^2)$$

$$0 \geq 3x^2-8dx+4d^2+3y^2+4z^2$$

$$\frac{4}{3}d^2 \geq 3\left(x-\frac{4}{3}d\right)^2+3y^2+4z^2$$

A translation by $\frac{4}{3}d$ in the x-direction does not change the volume, so we get

$$\frac{4}{3}d^2 \geq 3x_1^2+3y^2+4z^2$$

$$1 \geq \left(\frac{3x_1}{2d}\right)^2+\left(\frac{3y}{2d}\right)^2+\left(\frac{\sqrt{3}z}{d}\right)^2,$$

where $x_1=x-\frac{4}{3}d$. This equation defines a solid ellipsoid in canonical form. To compute its volume, perform a linear transformation: we divide x_1 and y by $\frac{2d}{3}$ and z by $\frac{d}{\sqrt{3}}$. This changes the volume by the factor $\left(\frac{2d}{3}\right)^2\frac{d}{\sqrt{3}}=\frac{4d^3}{9\sqrt{3}}$ and turns the ellipsoid into the unit ball of volume $\frac{4\pi}{3}$. So before the transformation the volume was $\frac{4d^3}{9\sqrt{3}}\cdot\frac{4\pi}{3}=\frac{16\pi d^3}{27\sqrt{3}}$.

2009-D2-P2 — Statement

Day 2, P7

Suppose $f:\mathbb{R}\to\mathbb{R}$ is a two times differentiable function satisfying $f(0)=1$, $f'(0)=0$, and for all $x\in[0,\infty)$,

$$f''(x)-5f'(x)+6f(x)\geq 0.$$

Prove that for all $x\in[0,\infty)$,

$$f(x)\geq 3e^{2x}-2e^{3x}.$$

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Theory and methods for 2009-D2-P2

Theory Differential inequalities and integrating factors [T-AN-ODE]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Methods Design an auxiliary derivative [M-AN-AUXILIARY]
Strategy Expose the structure before calculating [S-STRUCTURE]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2009-D2-P2 — Solution

For this record: Theory: T-AN-ODE; T-AN-DERIVATIVE; T-AN-INTEGRAL. Methods: M-AN-AUXILIARY. Full learning route

We have $f''(x)-2f'(x)-3(f'(x)-2f(x))\geq 0$, $x\in[0,\infty)$.

Let $g(x)=f'(x)-2f(x)$, $x\in[0,\infty)$. It follows that

$$g'(x)-3g(x)\geq 0,\quad x\in[0,\infty),$$

2009-D2-P2 — Solution (continued)

hence

$$(g(x)e^{-3x})' \geq 0, \quad x \in [0, \infty),$$

therefore

$$g(x)e^{-3x} \geq g(0) = -2, \quad x \in [0, \infty) \quad \text{or equivalently}$$
$$f'(x) - 2f(x) \geq -2e^{3x}, \quad x \in [0, \infty).$$

Analogously we get

$$(f(x)e^{-2x})' \geq -2e^{x}, \quad x \in [0, \infty) \quad \text{or equivalently}$$
$$(f(x)e^{-2x} + 2e^{x})' \geq 0, \quad x \in [0, \infty).$$

It follows that

$$f(x)e^{-2x} + 2e^{x} \geq f(0) + 2 = 3, \quad x \in [0, \infty) \quad \text{or equivalently}$$
$$f(x) \geq 3e^{2x} - 2e^{3x}, \quad x \in [0, \infty).$$

2009-D2-P3 — Statement

Day 2, P8

Let $A, B \in M_n(\mathbb{C})$ be two $n \times n$ matrices such that

$$A^2B + BA^2 = 2ABA.$$

Prove that there exists a positive integer k such that $(AB - BA)^k = 0$.

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Theory and methods for 2009-D2-P3

Theory Trace pairing and commutator calculus [T-LA-TRACE]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Nilpotents, projections and involutions [T-LA-NILPOTENT]
Methods Propagate a commutator by telescoping [M-LA-DERIVATION]; Use trace to extract a scalar invariant [M-LA-TRACE]
Strategy Expose the structure before calculating [S-STRUCTURE]; Reduce size while preserving the defining conditions [S-INDUCTION]

Statement Recorded solutions Catalog

2009-D2-P3 — Solution — Record 1 of 2

For this record: Theory: T-LA-TRACE; T-LA-POLYNOMIAL; T-LA-NILPOTENT. Methods: M-LA-DERIVATION. Full learning route

Let us fix the matrix $A \in M_n(\mathbb{C})$. For every matrix $X \in M_n(\mathbb{C})$, let $\Delta X := AX - XA$. We need to prove that the matrix ΔB is nilpotent.
Observe that the condition $A^2B + BA^2 = 2ABA$ is equivalent to

$$\Delta^2 B = \Delta(\Delta B) = 0. \tag{1}$$

Δ is linear; moreover, it is a derivation, i.e. it satisfies the Leibniz rule:

$$\Delta(XY) = (\Delta X)Y + X(\Delta Y), \quad \forall X, Y \in M_n(\mathbb{C}).$$

Using induction, one can easily generalize the above formula to k factors:

$$\Delta(X_1 \cdots X_k) = (\Delta X_1)X_2 \cdots X_k + \cdots + X_1 \cdots X_{j-1}(\Delta X_j)X_{j+1} \cdots X_k + \cdots + X_1 \cdots X_{k-1}\Delta X_k, \tag{2}$$

for any matrices $X_1, X_2, \ldots, X_k \in M_n(\mathbb{C})$. Using the identities (1) and (2) we obtain the equation for $\Delta^k(B^k)$:

$$\Delta^k(B^k) = k!(\Delta B)^k, \quad \forall k \in \mathbb{N}. \tag{3}$$

2009-D2-P3 — Solution — Record 1 of 2 (continued)

By the last equation it is enough to show that $\Delta^n(B^n) = 0$.

To prove this, first we observe that equation (3) together with the fact that $\Delta^2 B = 0$ implies that $\Delta^{k+1} B^k = 0$, for every $k \in \mathbb{N}$. Hence, we have

$$\Delta^k(B^j) = 0, \quad \forall k, j \in \mathbb{N},\ j < k. \tag{4}$$

By the Cayley–Hamilton Theorem, there are scalars $\alpha_0, \alpha_1, \ldots, \alpha_{n-1} \in \mathbb{C}$ such that

$$B^n = \alpha_0 I + \alpha_1 B + \cdots + \alpha_{n-1} B^{n-1},$$

which together with (4) implies that $\Delta^n B^n = 0$.

2009-D2-P3 — Solution — Record 2 of 2

For this record: Theory: T-LA-TRACE; T-LA-POLYNOMIAL; T-LA-NILPOTENT. Methods: M-LA-TRACE. Full learning route

Set $X = AB - BA$. The matrix X commutes with A because

$$AX - XA = (A^2B - ABA) - (ABA - BA^2) = A^2B + BA^2 - 2ABA = 0.$$

Hence for any $m \geq 0$ we have

$$X^{m+1} = X^m(AB - BA) = AX^mB - X^mBA.$$

Take the trace of both sides:

$$\operatorname{tr} X^{m+1} = \operatorname{tr} A(X^mB) - \operatorname{tr}(X^mB)A = 0$$

(since for any matrices U and V, we have $\operatorname{tr} UV = \operatorname{tr} VU$). As $\operatorname{tr} X^{m+1}$ is the sum of the $m+1$-st powers of the eigenvalues of X, the values of $\operatorname{tr} X$, ..., $\operatorname{tr} X^n$ determine the eigenvalues of X uniquely, therefore all of these eigenvalues have to be 0. This implies that X is nilpotent.

2009-D2-P4 — Statement

Day 2, P9

Let p be a prime number and $\mathbb{F}_p$ be the field of residues modulo p. Let W be the smallest set of polynomials with coefficients in $\mathbb{F}_p$ such that

- the polynomials $x + 1$ and $x^{p-2} + x^{p-3} + \cdots + x^2 + 2x + 1$ are in W, and
- for any polynomials $h_1(x)$ and $h_2(x)$ in W the polynomial $r(x)$, which is the remainder of $h_1(h_2(x))$ modulo $x^p - x$, is also in W.

How many polynomials are there in W?

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Theory and methods for 2009-D2-P4

Theory Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Choose a modulus that exposes the obstruction [M-AL-MODULAR]; Prove generation or obstruct it by an image [M-AL-GENERATORS]
Strategy Expose the structure before calculating [S-STRUCTURE]; Choose the objects and count them without ambiguity [S-COUNTING]

Statement Recorded solutions Catalog

2009-D2-P4 — Solution

For this record: Theory: T-AL-FINITEFIELD; T-AL-PERMUTATION; T-AL-POLYNOMIAL. Methods: M-AL-MODULAR; M-AL-GENERATORS. Full learning route

Note that both of our polynomials are bijective functions on $\mathbb{F}_p$: $f_1(x) = x + 1$ is the cycle $0 \to 1 \to 2 \to \ldots \to (p-1) \to 0$ and $f_2(x) = x^{p-2} + x^{p-3} + \cdots + x^2 + 2x + 1$ is the transposition $0 \leftrightarrow 1$ (this follows from the formula $f_2(x) = \frac{x^{p-1}-1}{x-1} + x$ and Fermat's little theorem). So any composition formed from them is also a bijection, and reduction modulo $x^p - x$ does not change the evaluation in $\mathbb{F}_p$. Also note that the transposition and the cycle generate the symmetric group ($f_1^k \circ f_2 \circ f_1^{p-k}$ is the transposition $k \leftrightarrow (k+1)$, and transpositions of consecutive elements clearly generate S_p), so we get all $p!$ permutations of the elements of $\mathbb{F}_p$.

The set W only contains polynomials of degree at most $p-1$. This means that two distinct elements of W cannot represent the same permutation. So W must contain those polynomials of degree at most $p-1$ which permute the elements of $\mathbb{F}_p$. By minimality, W has exactly these $p!$ elements.

2009-D2-P5 — Statement

Day 2, P10 | 8+2 points

Let M be the vector space of $m \times p$ real matrices. For a vector subspace $S \subseteq M$, denote by $\delta(S)$ the dimension of the vector space generated by all columns of all matrices in S.
Say that a vector subspace $T \subseteq M$ is a covering matrix space if

$$\bigcup_{A \in T, A \neq 0} \ker A = \mathbb{R}^p.$$

Such a T is minimal if it does not contain a proper vector subspace $S \subset T$ which is also a covering matrix space.
(a) (8 points) Let T be a minimal covering matrix space and let $n = \dim T$. Prove that

$$\delta(T) \leq \binom{n}{2}.$$

(b) (2 points) Prove that for every positive integer n we can find m and p, and a minimal covering matrix space T as above such that $\dim T = n$ and $\delta(T) = \binom{n}{2}$.

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Theory and methods for 2009-D2-P5

Theory Rank, dimension and exact linear constraints [T-LA-RANK]; Evaluation maps and rank under perturbation [T-LA-PERTURBATION]; Dual bases, kernels and separating functionals [T-LA-DUALITY]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Methods Use a generic parameter to force an image inclusion [M-LA-PERTURB]; Prove a dimension obstruction and attain it [M-LA-DIMENSION]
Strategy Choose an extremal object and force rigidity [S-EXTREMAL]; Reduce size while preserving the defining conditions [S-INDUCTION]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2009-D2-P5 — Solution — Record 1 of 2

For this record: Theory: T-LA-RANK; T-LA-PERTURBATION; T-LA-DUALITY; T-AL-POLYNOMIAL. Methods: M-LA-PERTURB; M-LA-DIMENSION. Full learning route

(a) We will prove the claim by constructing a suitable decomposition $T = Z_0 \oplus Z_1 \oplus \cdots$ and a corresponding decomposition of the space spanned by all columns of T as $W_0 \oplus W_1 \oplus \cdots$, such that $\dim W_0 \leqslant n-1$, $\dim W_1 \leqslant n-2$, etc., from which the bound follows.
We first claim that, in every covering matrix space S, we can find an $A \in S$ with $\operatorname{rk} A \leqslant \dim S - 1$. Indeed, let $S_0 \subseteq S$ be some minimal covering matrix space. Let $s = \dim S_0$ and fix some subspace $S' \subset S_0$ of dimension $s-1$. S' is not covering by minimality of S_0, so that we can find an $u \in \mathbb{R}^p$

2009-D2-P5 — Solution — Record 1 of 2 (continued)

with $u \notin \bigcup_{B\in S', B\neq 0} \operatorname{Ker} B$. Let $V = S'(u)$; by the rank-nullity theorem, $\dim V = s-1$. On the other hand, as S_0 is covering, we have that $Au = 0$ for some $A \in S_0 \setminus S'$. We claim that $\operatorname{Im} A \subset V$ (and therefore $\operatorname{rk}(A) \leqslant s-1$).

For suppose that $Av \notin V$ for some $v \in \mathbb{R}^p$. For every $\alpha \in \mathbb{R}$, consider the map $f_\alpha : S_0 \to \mathbb{R}^m$ defined by $f_\alpha : (\tau+\beta A) \mapsto \tau(u+\alpha v)+\beta Av$, $\tau \in S'$, $\beta \in \mathbb{R}$. Note that f_0 is of rank $s = \dim S_0$ by our assumption, so that some $s \times s$ minor of the matrix of f_0 is non-zero. The corresponding minor of f_α is thus a nonzero polynomial of α, so that it follows that $\operatorname{rk} f_\alpha = s$ for all but finitely many α. For such an $\alpha \neq 0$, we have that $\operatorname{Ker} f_\alpha = \{0\}$ and thus

$$0 \neq \tau(u+\alpha v) + \beta Av = (\tau + \alpha^{-1}\beta A)(u+\alpha v)$$

for all $\tau \in S'$, $\beta \in \mathbb{R}$ not both zero, so that $B(u+\alpha v) \neq 0$ for all nonzero $B \in S_0$, a contradiction.

Let now T be a minimal covering matrix space, and write $\dim T = n$. We have shown that we can find an $A \in T$ such that $W_0 = \operatorname{Im} A$ satisfies $w_0 = \dim W_0 \leqslant n-1$. Denote $Z_0 = \{B \in T : \operatorname{Im} B \subset W_0\}$; we know that $t_0 = \dim Z_0 \geqslant 1$. If $T = Z_0$, then $\delta(T) \leqslant n-1$ and we are done. Else, write $T = Z_0 \oplus T_1$, also write $\mathbb{R}^m = W_0 \oplus V_1$ and let $\pi_1 : \mathbb{R}^m \to \mathbb{R}^m$ be the projection onto the V_1-component. We claim that

$$T_1^\sharp = \{\pi_1\tau_1 : \tau_1 \in T_1\}$$

is also a covering matrix space. Note here that $\pi_1^\sharp : T_1 \to T_1^\sharp$, $\tau_1 \mapsto (\pi_1\tau_1)$ is an isomorphism. In particular we note that $\delta(T) = w_0 + \delta(T_1^\sharp)$.

Suppose that $T_1^\sharp$ is not a covering matrix space, so we can find a $v_1 \in \mathbb{R}^p$ with $v_1 \notin \bigcup_{\tau_1\in T_1, \tau_1\neq 0} \operatorname{Ker}(\pi_1\tau_1)$. On the other hand, by minimality of T we can find a $u_1 \in \mathbb{R}^p$ with $u_1 \notin \bigcup_{\tau_0\in Z_0, \tau_0\neq 0} \operatorname{Ker}\tau_0$. The maps $g_\alpha : Z_0 \to V_0$, $\tau_0 \mapsto \tau_0(u_1+\alpha v_1)$ and $h_\beta : T_1 \to V_1$, $\tau_1 \mapsto \pi_1(\tau_1(v_1+\beta u_1))$ have $\operatorname{rk} g_0 = t_0$ and $\operatorname{rk} h_0 = n - t_0$ and thus both $\operatorname{rk} g_\alpha = t_0$ and $\operatorname{rk} h_{\alpha^{-1}} = n - t_0$ for all but finitely many $\alpha \neq 0$ by the same argument as above. Pick such an α and suppose that

$$(\tau_0 + \tau_1)(u_1 + \alpha v_1) = 0$$

for some $\tau_0 \in Z_0$, $\tau_1 \in T_1$. Applying π_1 to both sides we see that we can only have $\tau_1 = 0$, and then $\tau_0 = 0$ as well, a contradiction given that T is a covering matrix space.

In fact, the exact same proof shows that, in general, if T is a minimal covering matrix space, $\mathbb{R}^m = V_0 \oplus V_1$, $T_0 = \{\tau \in T : \operatorname{Im}\tau \subset V_0\}$, $T = T_0 \oplus T_1$, $\pi_1 : \mathbb{R}^m \to \mathbb{R}^m$ is the projection onto the V_1-component, and $T_1^\sharp = \{\pi_1\tau_1 : \tau_1 \in T_1\}$, then $T_1^\sharp$ is a covering matrix space.

We can now repeat the process. We choose a $\pi_1 A_1 \in T_1^\sharp$ such that $W_1 = (\pi_1 A_1)(\mathbb{R}^p)$ has $w_1 = \dim W_1 \leqslant n - t_0 - 1 \leqslant n-2$. We write $Z_1 = \{\tau_1 \in T_1 : \operatorname{Im}(\pi_1\tau_1) \subset W_1\}$, $T_1 = Z_1 \oplus T_2$ (and so $T = (Z_0 \oplus Z_1) \oplus T_2$), $t_1 = \dim Z_1 \geqslant 1$, $V_1 = W_1 \oplus V_2$ (and so $\mathbb{R}^m = (W_0 \oplus W_1) \oplus V_2$), $\pi_2 : \mathbb{R}^m \to \mathbb{R}^m$ is the projection onto the V_2-component, and $T_2^\sharp = \{\pi_2\tau_2 : \tau_2 \in T_2\}$, so that $T_2^\sharp$ is also a covering matrix space, etc.

We conclude that

$$\delta(T) = w_0 + \delta(T_1^\sharp) = w_0 + w_1 + \delta(T_2^\sharp) = \cdots \leqslant (n-1) + (n-2) + \cdots \leqslant \binom{n}{2}.$$

(b) We consider $\binom{n}{2} \times n$ matrices whose rows are indexed by $\binom{n}{2}$ pairs (i,j) of integers $1 \leqslant i < j \leqslant n$. For every $u = (u_1, u_2, \ldots, u_n) \in \mathbb{R}^n$, consider the matrix $A(u)$ whose entries $(A(u))_{(i,j),k}$ with $1 \leqslant i < j \leqslant n$ and $1 \leqslant k \leqslant n$ are given by

$$(A(u))_{(i,j),k} = \begin{cases} u_i, & k = j, \\ -u_j, & k = i, \\ 0, & \text{otherwise.} \end{cases}$$

It is immediate that $\operatorname{Ker} A(u) = \mathbb{R}\cdot u$ for every $u \neq 0$, so that $S = \{A(u) : u \in \mathbb{R}^n\}$ is a covering matrix space, and in fact a minimal one.

On the other hand, for any $1 \leqslant i < j \leqslant n$, we have that $A(e_i)_{(i,j),j}$ is the (i,j)-th vector in the standard basis of $\mathbb{R}^{\binom{n}{2}}$, where e_i denotes the i-th vector in the standard basis of $\mathbb{R}^n$. This means that $\delta(S) = \binom{n}{2}$, as required.

2009-D2-P5 — Solution — Record 2 of 2

For this record: Theory: T-LA-RANK; T-LA-PERTURBATION; T-AN-DERIVATIVE. Methods: M-LA-PERTURB; M-LA-DIMENSION. Full learning route

(for part a)
Let us denote $X = \mathbb{R}^p$, $Y = \mathbb{R}^m$. For each $x \in X$, denote by $\mu_x : T \to Y$ the evaluation map $\tau \mapsto \tau(x)$. As T is a covering matrix space, $\ker \mu_x > 0$ for every $x \in X$. Let $U = \{x \in X : \dim \ker \mu_x = 1\}$.
Let T_1 be the span of the family of subspaces $\{\ker \mu_x : x \in U\}$. We claim that $T_1 = T$. For suppose the contrary, and let $T' \subset T$ be a subspace of T of dimension $n-1$ such that $T_1 \subseteq T'$. This implies that T' is a covering matrix space. Indeed, for $x \in U$, $(\ker \mu_x) \cap T' = \ker \mu_x \neq 0$, while for $x \notin U$ we have $\dim \mu_x \geq 2$, so that $(\ker \mu_x) \cap T' \neq 0$ by computing dimensions. However, this is a contradiction as T is minimal.
Now we may choose $x_1, x_2, \ldots, x_n \in U$ and $\tau_1, \tau_2, \ldots, \tau_n \in T$ in such a way that $\ker \mu_{x_i} = \mathbb{R}\tau_i$ and τ_i form a basis of T. Let us complete $x_1, \ldots, x_n$ to a sequence $x_1, \ldots, x_d$ which spans X. Put $y_{ij} = \tau_i(x_j)$. It is clear that y_{ij} span the vector space generated by the columns of all matrices in T. We claim that the subset $\{y_{ij} : i > j\}$ is enough to span this space, which clearly implies that $\delta(T) \leqslant \binom{n}{2}$.
We have $y_{ii} = 0$. So it is enough to show that every y_{ij} with $i < j$ can be expressed as a linear combination of y_{ki}, $k = 1, \ldots, n$. This follows from the following lemma:
Lemma. For every $x_0 \in U$, $0 \neq \tau_0 \in \ker \mu_{x_0}$ and $x \in X$, there exists a $\tau \in T$ such that $\tau_0(x) = \tau(x_0)$.
Proof. The operator μ_{x_0} has rank $n-1$, which implies that for small ε the operator $\mu_{x_0+\varepsilon x}$ also has rank $n-1$. Therefore one can produce a rational function $\tau(\varepsilon)$ with values in T such that $m_{x_0+\varepsilon x}(\tau(\varepsilon)) = 0$.
Taking the derivative at $\varepsilon = 0$ gives $\mu_{x_0}(\tau_0) + \mu_x(\tau'(0)) = 0$.
Therefore $\tau = -\tau'(0)$ satisfies the desired property.
Remark. Lemma in solution 2 is the same as the claim $\operatorname{Im} A \subset V$ at the beginning of solution 1, but the proof given here is different. It can be shown that all minimal covering spaces T with $\dim T = \binom{n}{2}$

are essentially the ones described in our example.

3.17 2010

Day 1

2010-D1-P1 — Statement

Day 1, P1 | 10 points

Let $0 < a < b$. Prove that

$$\int_a^b (x^2+1)e^{-x^2}dx \geq e^{-a^2} - e^{-b^2}.$$

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Theory and methods for 2010-D1-P1

Theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]
Methods Count zeros and subtract an interpolant [M-AN-ROLLE]; Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]; Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]
Strategy Build a chain of bounds and track equality [S-BOUND]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2010-D1-P1 — Solution — Record 1 of 2

For this record: Theory: T-AN-DERIVATIVE; T-AN-INTEGRAL; T-AN-INEQUALITIES. Methods: M-AN-ROLLE; M-AN-WEIGHTS. Full learning route

Let $f(x) = \int_0^x (t^2+1)e^{-t^2}dt$ and let $g(x) = -e^{-x^2}$; both functions are increasing.

By Cauchy's Mean Value Theorem, there exists a real number $x \in (a,b)$ such that

$$\frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(x)}{g'(x)} = \frac{(x^2+1)e^{-x^2}}{2xe^{-x^2}} = \frac{1}{2}\left(x+\frac{1}{x}\right) \geq \sqrt{x\cdot\frac{1}{x}} = 1.$$

Then

$$\int_a^b (x^2+1)e^{-x^2}dx = f(b)-f(a) \geq g(b)-g(a) = e^{-a^2}-e^{-b^2}.$$

2010-D1-P1 — Solution — Record 2 of 2

For this record: Theory: T-AN-INTEGRAL; T-AN-INEQUALITIES. Methods: M-AN-WEIGHTS; M-AN-PRIMITIVE. Full learning route

$$\int_a^b (x^2+1)e^{-x^2}dx \geq \int_a^b 2xe^{-x^2}dx = \left[-e^{-x^2}\right]_a^b = e^{-a^2}-e^{-b^2}.$$

2010-D1-P2 — Statement

Day 1, P2 | 10 points

Compute the sum of the series

$$\sum_{k=0}^{\infty} \frac{1}{(4k+1)(4k+2)(4k+3)(4k+4)} = \frac{1}{1\cdot 2\cdot 3\cdot 4} + \frac{1}{5\cdot 6\cdot 7\cdot 8} + \cdots.$$

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Theory and methods for 2010-D1-P2

Theory Power series and generating functions [T-AN-POWER]; Nonnegative sums, Fubini, and integral kernels [T-AN-FUBINI]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Series, finite cancellation, and summation by parts [T-AN-SERIES]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]

Methods Encode convolution or denominators by a power series [M-AN-GENERATING]; Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]; Find a finite telescoping identity [M-AN-TELESCOPE]; Test convergence through blocks and differences [M-AN-SERIES]

Strategy Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2010-D1-P2 — Solution — Record 1 of 2

For this record: Theory: T-AN-POWER; T-AN-FUBINI; T-AN-INTEGRAL. Methods: M-AN-GENERATING; M-AN-PRIMITIVE. Full learning route

Let

$$F(x) = \sum_{k=0}^{\infty} \frac{x^{4k+4}}{(4k+1)(4k+2)(4k+3)(4k+4)}.$$

This power series converges for $|x| \leq 1$ and our goal is to compute $F(1)$.

Differentiating 4 times, we get

$$F^{(IV)}(x) = \sum_{k=0}^{\infty} x^{4k} = \frac{1}{1-x^4}.$$

2010-D1-P2 — Solution — Record 1 of 2 (continued)

Since $F(0) = F'(0) = F''(0) = F'''(0) = 0$ and F is continuous at $1-0$ by Abel's continuity theorem, integrating 4 times (with the formulas interpreted for $0 \le x, y, z, t < 1$ and endpoint values obtained by limits) we get

$$\begin{aligned} F'''(y) &= F'''(0) + \int_0^y F^{(IV)}(x)\,dx = \int_0^y \frac{dx}{1-x^4} \\ &= \frac{1}{2}\arctan y + \frac{1}{4}\log(1+y) - \frac{1}{4}\log(1-y). \end{aligned}$$

$$\begin{aligned} F''(z) &= F''(0) + \int_0^z F'''(y)\,dy \\ &= \int_0^z \left(\frac{1}{2}\arctan y + \frac{1}{4}\log(1+y) - \frac{1}{4}\log(1-y)\right) dy \\ &= \frac{1}{2}\left(z\arctan z - \int_0^z \frac{y}{1+y^2}\,dy\right) \\ &\quad + \frac{1}{4}\left((1+z)\log(1+z) - \int_0^z dy\right) + \frac{1}{4}\left((1-z)\log(1-z) + \int_0^z dy\right) \\ &= \frac{1}{2}z\arctan z - \frac{1}{4}\log(1+z^2) \\ &\quad + \frac{1}{4}(1+z)\log(1+z) + \frac{1}{4}(1-z)\log(1-z). \end{aligned}$$

$$\begin{aligned} F'(t) &= \int_0^t \Big(\frac{1}{2}z\arctan z - \frac{1}{4}\log(1+z^2) \\ &\qquad\qquad + \frac{1}{4}(1+z)\log(1+z) + \frac{1}{4}(1-z)\log(1-z)\Big)dz \\ &= \frac{1}{4}\left((1+t^2)\arctan t - t\right) - \frac{1}{4}\left(t\log(1+t^2) - 2t + 2\arctan t\right) \\ &\quad + \frac{1}{8}\left((1+t)^2\log(1+t) - t - \frac{1}{2}t^2\right) \\ &\quad - \frac{1}{8}\left((1-t)^2\log(1-t) + t - \frac{1}{2}t^2\right) \\ &= \frac{1}{4}(-1+t^2)\arctan t - \frac{1}{4}t\log(1+t^2) \\ &\quad + \frac{1}{8}(1+t)^2\log(1+t) - \frac{1}{8}(1-t)^2\log(1-t). \end{aligned}$$

$$\begin{aligned} F(1) &= \int_0^1 \Big(\frac{1}{4}(-1+t^2)\arctan t - \frac{1}{4}t\log(1+t^2) \\ &\qquad\qquad + \frac{1}{8}(1+t)^2\log(1+t) - \frac{1}{8}(1-t)^2\log(1-t)\Big)dt \\ &= \left[\frac{-3t+t^3}{12}\arctan t + \frac{1-3t^2}{24}\log(1+t^2)\right. \\ &\qquad \left. + \frac{(1+t)^3}{24}\log(1+t) + \frac{(1-t)^3}{24}\log(1-t)\right]_0^1 \\ &= \frac{\ln 2}{4} - \frac{\pi}{24}. \end{aligned}$$

2010-D1-P2 — Solution — Record 1 of 2 (continued)

Remark. The computation can be shorter if we change the order of integrations. The integrand is nonnegative, so Tonelli's theorem justifies this step.

$$\begin{aligned} F(1) &= \int_{t=0}^{1}\int_{z=0}^{t}\int_{y=0}^{z}\int_{x=0}^{y}\frac{1}{1-x^4}\,dx\,dy\,dz\,dt \\ &= \int_{x=0}^{1}\frac{1}{1-x^4}\left(\int_{y=x}^{1}\int_{z=y}^{1}\int_{t=z}^{1}dt\,dz\,dy\right)dx \\ &= \int_{x=0}^{1}\frac{1}{1-x^4}\cdot\frac{(1-x)^3}{6}\,dx \\ &= \left[-\frac{1}{6}\arctan x-\frac{1}{12}\log(1+x^2)+\frac{1}{3}\log(1+x)\right]_0^1 \\ &= \frac{\ln 2}{4}-\frac{\pi}{24}. \end{aligned}$$

2010-D1-P2 — Solution — Record 2 of 2

For this record: Theory: T-AN-SERIES; T-AN-INTEGRAL; T-AL-POLYNOMIAL. Methods: M-AN-TELESCOPE; M-AN-SERIES. Full learning route

Let

$$\begin{aligned} A_m &= \sum_{k=0}^{m}\frac{1}{(4k+1)(4k+2)(4k+3)(4k+4)} \\ &= \sum_{k=0}^{m}\left(\frac{1}{6}\cdot\frac{1}{4k+1}-\frac{1}{2}\cdot\frac{1}{4k+2}+\frac{1}{2}\cdot\frac{1}{4k+3}-\frac{1}{6}\cdot\frac{1}{4k+4}\right). \end{aligned}$$

$$B_m = \sum_{k=0}^{m}\left(\frac{1}{4k+1}-\frac{1}{4k+3}\right),$$

$$C_m = \sum_{k=0}^{m}\left(\frac{1}{4k+1}-\frac{1}{4k+2}+\frac{1}{4k+3}-\frac{1}{4k+4}\right) \quad \text{and}$$

$$D_m = \sum_{k=0}^{m}\left(\frac{1}{4k+2}-\frac{1}{4k+4}\right).$$

It is easy check that

$$A_m = \frac{1}{3}C_m - \frac{1}{6}B_m - \frac{1}{6}D_m.$$

Therefore,

$$\lim A_m = \lim\frac{2C_m - B_m - D_m}{6} = \frac{2\ln 2 - \frac{\pi}{4} - \frac{1}{2}\ln 2}{6} = \frac{\ln 2}{4} - \frac{\pi}{24}.$$

2010-D1-P3 — Statement

Day 1, P3 | 10 points

Define the sequence $x_1, x_2, \ldots$ inductively by $x_1 = \sqrt{5}$ and $x_{n+1} = x_n^2 - 2$ for each $n \geq 1$. Compute

$$\lim_{n\to\infty}\frac{x_1 \cdot x_2 \cdot x_3 \cdots x_n}{x_{n+1}}.$$

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Theory and methods for 2010-D1-P3

Theory Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Methods Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]; Strengthen the statement before the induction step [M-DS-INDUCT]
Strategy Control the limiting operation before taking the limit [S-ANALYTIC]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2010-D1-P3 — Solution

For this record: Theory: T-DS-DYNAMICS; T-DS-INDUCTION; T-AN-CONVERGENCE. Methods: M-DS-ITERATE; M-DS-INDUCT. Full learning route

Let $y_n = x_n^2$. Then $y_{n+1} = (y_n - 2)^2$ and $y_{n+1} - 4 = y_n(y_n - 4)$. Since $y_2 = 9 > 5$, we have $y_3 = (y_2 - 2)^2 > 5$ and inductively $y_n > 5$, $n \geq 2$. Hence, $y_{n+1} - y_n = y_n^2 - 5y_n + 4 > 4$ for all $n \geq 2$, so $y_n \to \infty$.

By $y_{n+1} - 4 = y_n(y_n - 4)$,

$$\left(\frac{x_1 \cdot x_2 \cdot x_3 \cdots x_n}{x_{n+1}}\right)^2 = \frac{y_1 \cdot y_2 \cdot y_3 \cdots y_n}{y_{n+1}}$$
$$= \frac{y_{n+1} - 4}{y_{n+1}} \cdot \frac{y_1 \cdot y_2 \cdot y_3 \cdots y_n}{y_n - 4} = \frac{y_{n+1} - 4}{y_{n+1}} \cdot \frac{y_1 \cdot y_2 \cdot y_3 \cdots y_{n-1}}{y_n - 4} \cdot y_n = \cdots =$$
$$= \frac{y_{n+1} - 4}{y_{n+1}} \cdot \frac{1}{y_1 - 4} = \frac{y_{n+1} - 4}{y_{n+1}} \to 1.$$

Therefore,

$$\lim_{n \to \infty} \frac{x_1 \cdot x_2 \cdot x_3 \cdots x_n}{x_{n+1}} = 1.$$

2010-D1-P4 — Statement

Day 1, P4 | 10 points

Let a, b be two integers and suppose that n is a positive integer for which the set

$$\mathbb{Z} \setminus \{ax^n + by^n \mid x, y \in \mathbb{Z}\}$$

is finite. Prove that $n = 1$.

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Theory and methods for 2010-D1-P4

Theory Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Prime valuations, divisors, and factorials [T-AL-VALUATION]
Methods Choose a modulus that exposes the obstruction [M-AL-MODULAR]
Strategy Pass between local constraints and a global conclusion [S-LOCALGLOBAL]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

2010-D1-P4 — Solution

For this record: Theory: T-AL-CONGRUENCE; T-AL-FINITEFIELD; T-AL-VALUATION. Methods: M-AL-MODULAR. Full learning route

Assume that $n > 1$. Notice that n may be replaced by any prime divisor p of n. Moreover, a and b should be coprime, otherwise the numbers not divisible by the greatest common divisor of a, b cannot be represented as $ax^n + by^n$.
If $p = 2$, then the number of the form $ax^2 + by^2$ takes not all possible remainders modulo 8. If, say, b is even, then ax^2 takes at most three different remainders modulo 8, by^2 takes at most two, hence

2010-D1-P4 — Solution (continued)

ax^2+by^2 takes at most $3\times 2=6$ different remainders. If both a and b are odd, then $ax^2+by^2\equiv x^2\pm y^2 \pmod 4$; the expression x^2+y^2 does not take the remainder 3 modulo 4 and x^2-y^2 does not take the remainder 2 modulo 4.

Consider the case when $p\geq 3$. The pth powers take exactly p different remainders modulo p^2. Indeed, $(x+kp)^p$ and x^p have the same remainder modulo p^2, and all numbers $0^p, 1^p, \dots, (p-1)^p$ are different even modulo p. So, ax^p+by^p take at most p^2 different remainders modulo p^2. If it takes strictly less then p^2 different remainders modulo p^2, we get infinitely many non-representable numbers. If it takes exactly p^2 remainders, then ax^p+by^p is divisible by p^2 only if both x and y are divisible by p. Hence if ax^p+by^p is divisible by p^2, it is also divisible by p^p. Again we get infinitely many non-representable numbers, for example the numbers congruent to p^2 modulo p^3 are non-representable.

2010-D1-P5 — Statement

Day 1, P5 | 10 points

Suppose that a, b, c are real numbers in the interval $[-1, 1]$ such that

$$1+2abc\geq a^2+b^2+c^2.$$

Prove that

$$1+2(abc)^n\geq a^{2n}+b^{2n}+c^{2n}$$

for all positive integers n.

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Theory and methods for 2010-D1-P5

Theory Gram matrices and positive semidefinite forms [T-LA-GRAM]; Tensor products and operators on matrix spaces [T-LA-TENSOR]; Determinants, block identities and inverses [T-LA-DETERMINANT]; Inner products, orthogonality and polarization [T-LA-INNER]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Encode distances and orthogonality by a Gram matrix [M-LA-GRAM]; Lift products of inner products to new vectors [M-LA-TENSOR]; Read the entries before choosing a theorem [M-LA-ENTRY]
Strategy Expose the structure before calculating [S-STRUCTURE]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2010-D1-P5 — Solution — Record 1 of 2

For this record: Theory: T-LA-GRAM; T-LA-TENSOR; T-LA-DETERMINANT. Methods: M-LA-GRAM; M-LA-TENSOR. Full learning route

Consider the symmetric matrix

$$A=\begin{pmatrix}1 & a & b\\ a & 1 & c\\ b & c & 1\end{pmatrix}.$$

By the constraint we have $\det A\geq 0$ and $\det\begin{pmatrix}1 & a\\ a & 1\end{pmatrix}, \det\begin{pmatrix}1 & b\\ b & 1\end{pmatrix}, \det\begin{pmatrix}1 & c\\ c & 1\end{pmatrix}\geq 0$. Hence A is positive semidefinite, and $A=B^2$ for some symmetric real matrix B.

Let the rows of B be x, y, z. Then $|x|=|y|=|z|=1$, $a=x\cdot y$, $b=y\cdot z$ and $c=z\cdot x$, where $|x|$ and $x\cdot y$ denote the Euclidean norm and scalar product. Denote by $X=\otimes^n x$, $Y=\otimes^n y$, $Z=\otimes^n z$ the nth tensor powers, which belong to $\mathbb{R}^{3^n}$. Then $|X|=|Y|=|Z|=1$, $X\cdot Y=a^n$, $Y\cdot Z=b^n$ and $Z\cdot X=c^n$. So, the matrix

$$\begin{pmatrix}1 & a^n & b^n\\ a^n & 1 & c^n\\ b^n & c^n & 1\end{pmatrix},$$

2010-D1-P5 — Solution — Record 1 of 2 (continued)

being the Gram matrix of three vectors in $\mathbb{R}^{3^n}$, is positive semidefinite, and its determinant, $1+2(abc)^n-a^{2n}-b^{2n}-c^{2n}$ is non-negative.

2010-D1-P5 — Solution — Record 2 of 2

For this record: Theory: T-LA-INNER; T-AN-INEQUALITIES; T-AL-POLYNOMIAL. Methods: M-LA-ENTRY. Full learning route

The constraint can be written as

$$(a-bc)^2 \le (1-b^2)(1-c^2). \tag{1}$$

By the Cauchy-Schwarz inequality,

$$\begin{aligned}\left(a^{n-1}+a^{n-2}bc+\cdots+b^{n-1}c^{n-1}\right)^2 &\le \left(|a|^{n-1}+|a|^{n-2}|bc|+\cdots+|bc|^{n-1}\right)^2 \le \\ &\le \left(1+|bc|+\cdots+|bc|^{n-1}\right)^2 \le \left(1+|b|^2+\cdots+|b|^{2(n-1)}\right)\left(1+|c|^2+\cdots+|c|^{2(n-1)}\right)\end{aligned}$$

Multiplying by (1), we get

$$(a-bc)^2\left(a^{n-1}+a^{n-2}bc+\cdots+b^{n-1}c^{n-1}\right)^2 \le (1-b^2)\left(1+|b|^2+\cdots+|b|^{2(n-1)}\right)(1-c^2)\left(1+|c|^2+\cdots+|c|^{2(n-1)}\right),$$

$$(a^n-b^nc^n)^2 \le (1-b^n)(1-c^n),$$

$$1+2(abc)^n \ge a^{2n}+b^{2n}+b^{2n}.$$

Day 2

2010-D2-P1 — Statement

Day 2, P6 | 5+5 points

(a) A sequence $x_1, x_2, \ldots$ of real numbers satisfies

$$x_{n+1} = x_n \cos x_n \quad \text{for all } n \ge 1.$$

Does it follow that this sequence converges for all initial values x_1? (5 points)

(b) A sequence $y_1, y_2, \ldots$ of real numbers satisfies

$$y_{n+1} = y_n \sin y_n \quad \text{for all } n \ge 1.$$

Does it follow that this sequence converges for all initial values y_1? (5 points)

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Theory and methods for 2010-D2-P1

Theory Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Trigonometric identities and safe branches [T-AN-TRIG]; Continuity, compactness, and order [T-AN-CONTINUITY]
Methods Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2010-D2-P1 — Solution — Record 1 of 2

For this record: Theory: T-AN-CONVERGENCE; T-AN-TRIG; T-AN-CONTINUITY. Methods: M-AN-RECURRENCE. Full learning route

(a) NO. For example, for $x_1 = \pi$ we have $x_n = (-1)^{n-1}\pi$, and the sequence is divergent.

2010-D2-P1 — Solution — Record 1 of 2 (continued)

(b) YES. Notice that $|y_n|$ is nonincreasing and hence converges to some number $a \geq 0$.
If $a = 0$, then $\lim y_n = 0$ and we are done. If $a > 0$, then $a = \lim |y_{n+1}| = \lim |y_n \sin y_n| = a \cdot |\sin a|$, so $\sin a = \pm 1$ and $a = (k + \frac{1}{2})\pi$ for some nonnegative integer k.
Since the sequence $|y_n|$ is nonincreasing, there exists an index n_0 such that $(k+\frac{1}{2})\pi \leq |y_n| < (k+1)\pi$ for all $n > n_0$. Then all the numbers $y_{n_0+1}, y_{n_0+2}, \ldots$ lie in the union of the intervals $\left[(k + \frac{1}{2})\pi, (k + 1)\pi\right)$ and $\left(-(k + 1)\pi, -(k + \frac{1}{2})\pi\right]$.
Depending on the parity of k, in one of the intervals $\left[(k + \frac{1}{2})\pi, (k + 1)\pi\right)$ and $\left(-(k + 1)\pi, -(k + \frac{1}{2})\pi\right]$ the values of the sine function is positive; denote this interval by I_+. In the other interval the sine function is negative; denote this interval by I_-. If $y_n \in I_-$ for some $n > n_0$ then y_n and $y_{n+1} = y_n \sin y_n$ have opposite signs, so $y_{n+1} \in I_+$. On the other hand, if If $y_n \in I_+$ for some $n > n_0$ then y_n and y_{n+1} have the same sign, so $y_{n+1} \in I_+$. In both cases, $y_{n+1} \in I_+$.

We obtained that the numbers $y_{n_0+2}, y_{n_0+3}, \ldots$ lie in I_+, so they have the same sign. Since $|y_n|$ is convergent, this implies that the sequence (y_n) is convergent as well.

2010-D2-P1 — Solution — Record 2 of 2

For this record: Theory: T-AN-CONVERGENCE; T-AN-TRIG; T-AN-CONTINUITY. Methods: M-AN-RECURRENCE. Full learning route

(for part (b)) Similarly to the first solution, $|y_n| \to a$ for some real number a.

Notice that $t \cdot \sin t = (-t)\sin(-t) = |t| \sin |t|$ for all real t, hence $y_{n+1} = |y_n| \sin |y_n|$ for all $n \geq 2$. Since the function $t \mapsto t \sin t$ is continuous, $y_{n+1} = |y_n| \sin |y_n| \to |a| \sin |a| = a$.

2010-D2-P2 — Statement

Day 2, P7 | 10 points

Let $a_0, a_1, \ldots, a_n$ be positive real numbers such that $a_{k+1} - a_k \geq 1$ for all $k = 0, 1, \ldots, n-1$. Prove that

$$1 + \frac{1}{a_0}\left(1 + \frac{1}{a_1 - a_0}\right) \cdots \left(1 + \frac{1}{a_n - a_0}\right) \leq \left(1 + \frac{1}{a_0}\right)\left(1 + \frac{1}{a_1}\right) \cdots \left(1 + \frac{1}{a_n}\right).$$

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Theory and methods for 2010-D2-P2

Theory Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Methods Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]; Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]
Strategy Reduce size while preserving the defining conditions [S-INDUCTION]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2010-D2-P2 — Solution

For this record: Theory: T-AN-INEQUALITIES; T-DS-INDUCTION. Methods: M-AN-WEIGHTS; M-AN-RECURRENCE. Full learning route

Apply induction on n. Considering the empty product as 1, we have equality for $n = 0$. Now assume that the statement is true for some n and prove it for $n+1$. For $n+1$, the statement can be written as the sum of the inequalities

$$1 + \frac{1}{a_0}\left(1 + \frac{1}{a_1 - a_0}\right) \cdots \left(1 + \frac{1}{a_n - a_0}\right) \leq \left(1 + \frac{1}{a_0}\right) \cdots \left(1 + \frac{1}{a_n}\right)$$

2010-D2-P2 — Solution (continued)

(which is the induction hypothesis) and

$$\frac{1}{a_0}\left(1+\frac{1}{a_1-a_0}\right)\cdots\left(1+\frac{1}{a_n-a_0}\right)\cdot\frac{1}{a_{n+1}-a_0}\le\left(1+\frac{1}{a_0}\right)\cdots\left(1+\frac{1}{a_n}\right)\cdot\frac{1}{a_{n+1}}. \tag{1}$$

Hence, to complete the solution it is sufficient to prove (1).
To prove (1), apply a second induction. For $n=0$, we have to verify

$$\frac{1}{a_0}\cdot\frac{1}{a_1-a_0}\le\left(1+\frac{1}{a_0}\right)\frac{1}{a_1}.$$

Multiplying by $a_0a_1(a_1-a_0)$, this is equivalent with

$$\begin{aligned} a_1 &\le (a_0+1)(a_1-a_0)\\ a_0 &\le a_0a_1-a_0^2\\ 1 &\le a_1-a_0. \end{aligned}$$

For the induction step it is sufficient that

$$\left(1+\frac{1}{a_{n+1}-a_0}\right)\cdot\frac{a_{n+1}-a_0}{a_{n+2}-a_0}\le\left(1+\frac{1}{a_{n+1}}\right)\cdot\frac{a_{n+1}}{a_{n+2}}.$$

Multiplying by $(a_{n+2}-a_0)a_{n+2}$,

$$\begin{aligned} (a_{n+1}-a_0+1)a_{n+2} &\le (a_{n+1}+1)(a_{n+2}-a_0)\\ a_0 &\le a_0a_{n+2}-a_0a_{n+1}\\ 1 &\le a_{n+2}-a_{n+1}. \end{aligned}$$

Remark 1. It is easy to check from the solution that equality holds if and only if $a_{k+1}-a_k=1$ for all k.

Remark 2. The statement of the problem is a direct corollary of the identity

$$1+\sum_{i=0}^{n}\frac{1}{x_i}\prod_{j\ne i}\left(1+\frac{1}{x_j-x_i}\right)=\prod_{i=0}^{n}\left(1+\frac{1}{x_i}\right).$$

2010-D2-P3 — Statement

Day 2, P8 | 10 points

Denote by S_n the group of permutations of the sequence $(1,2,\dots,n)$. Suppose that G is a subgroup of S_n, such that for every $\pi\in G\setminus\{e\}$ there exists a unique $k\in\{1,2,\dots,n\}$ for which $\pi(k)=k$. (Here e is the unit element in the group S_n.) Show that this k is the same for all $\pi\in G\setminus\{e\}$.

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Theory and methods for 2010-D2-P3

Theory Group actions, stabilizers, and semidirect products [T-AL-ACTION]; Groups, orders, generators, and quotients [T-AL-GROUP]
Methods Convert group data into orbits or cosets [M-AL-ACTION]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2010-D2-P3 — Solution

For this record: Theory: T-AL-ACTION; T-AL-GROUP. Methods: M-AL-ACTION. Full learning route

Let us consider the action of G on the set $X = \{1, \dots, n\}$. Let

$$G_x = \{g \in G : g(x) = x\} \quad \text{and} \quad Gx = \{g(x) : g \in G\}$$

be the stabilizer and the orbit of $x \in X$ under this action, respectively. The condition of the problem states that

$$G = \bigcup_{x \in X} G_x \tag{1}$$

and

$$G_x \cap G_y = \{e\} \quad \text{for all } x \neq y. \tag{2}$$

We need to prove that $G = G_x$ for some $x \in X$.

Let $Gx_1, \dots, Gx_k$ be the distinct orbits of the action of G. Then one can write (1) as

$$G = \bigcup_{i=1}^{k} \bigcup_{y \in Gx_i} G_y. \tag{3}$$

It is well known that

$$|Gx| = \frac{|G|}{|G_x|}. \tag{4}$$

Also note that if $y \in Gx$ then $Gy = Gx$ and thus $|Gy| = |Gx|$. Therefore,

$$|G_x| = \frac{|G|}{|Gx|} = \frac{|G|}{|Gy|} = |G_y| \quad \text{for all } y \in Gx. \tag{5}$$

Combining (3), (2), (4) and (5) we get

$$|G| - 1 = |G \setminus \{e\}| = \left| \bigcup_{i=1}^{k} \bigcup_{y \in Gx_i} G_y \setminus \{e\} \right| = \sum_{i=1}^{k} \frac{|G|}{|G_{x_i}|}(|G_{x_i}| - 1),$$

hence

$$1 - \frac{1}{|G|} = \sum_{i=1}^{k} \left(1 - \frac{1}{|G_{x_i}|}\right). \tag{6}$$

If for some $i, j \in \{1, \dots, k\}$ $|G_{x_i}|, |G_{x_j}| \geq 2$ then

$$\sum_{i=1}^{k} \left(1 - \frac{1}{|G_{x_i}|}\right) \geq \left(1 - \frac{1}{2}\right) + \left(1 - \frac{1}{2}\right) = 1 > 1 - \frac{1}{|G|}$$

which contradicts with (6), thus we can assume that

$$|G_{x_1}| = \cdots = |G_{x_{k-1}}| = 1.$$

Then from (6) we get $|G_{x_k}| = |G|$, hence $G_{x_k} = G$.

2010-D2-P4 — Statement

Day 2, P9 | 10 points

Let A be a symmetric $m \times m$ matrix over the two-element field all of whose diagonal entries are zero. Prove that for every positive integer n each column of the matrix A^n has a zero entry.

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Theory and methods for 2010-D2-P4

Theory Linear algebra in characteristic two [T-LA-FINITEFIELD]
Methods Exploit parity with linear algebra over F2 [M-LA-FINITEFIELD]
Strategy Reduce size while preserving the defining conditions [S-INDUCTION]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

2010-D2-P4 — Solution

For this record: Theory: T-LA-FINITEFIELD. Methods: M-LA-FINITEFIELD. Full learning route

Denote by e_k ($1 \le k \le m$) the m-dimensional vector over $\mathbb{F}_2$, whose k-th entry is 1 and all the other elements are 0. Furthermore, let u be the vector whose all entries are 1. The k-th column of A^n is $A^n e_k$. So the statement can be written as $A^n e_k \neq u$ for all $1 \le k \le m$ and all $n \ge 1$.
For every pair of vectors $x = (x_1, \dots, x_m)$ and $y = (y_1, \dots, y_m)$, define the bilinear form $(x, y) = x^T y = x_1 y_1 + \cdots + x_m y_m$. The product (x, y) has all basic properties of scalar products (except the property that $(x, x) = 0$ implies $x = 0$). Moreover, we have $(x, x) = (x, u)$ for every vector $x \in \mathbb{F}_2^m$. It is also easy to check that $(w, Aw) = w^T A w = 0$ for all vectors w, since A is symmetric and its diagonal elements are 0.
Lemma. Suppose that $v \in \mathbb{F}_2^m$ a vector such that $A^n v = u$ for some $n \ge 1$. Then $(v, v) = 0$.
Proof. Apply induction on n. For odd values of n we prove the lemma directly. Let $n = 2k + 1$ and $w = A^k v$. Then

$$(v, v) = (v, u) = (v, A^n v) = v^T A^n v = v^T A^{2k+1} v = (A^k v, A^{k+1} v) = (w, Aw) = 0.$$

Now suppose that n is even, $n = 2k$, and the lemma is true for all smaller values of n. Let $w = A^k v$; then $A^k w = A^n v = u$ and thus we have $(w, w) = 0$ by the induction hypothesis. Hence,

$$(v, v) = (v, u) = v^T A^n v = v^T A^{2k} v = (A^k v)^T (A^k v) = (A^k v, A^k v) = (w, w) = 0.$$

The lemma is proved.

Now suppose that $A^n e_k = u$ for some $1 \le k \le m$ and positive integer n. By the Lemma, we should have $(e_k, e_k) = 0$. But this is impossible because $(e_k, e_k) = 1 \neq 0$.

2010-D2-P5 — Statement

Day 2, P10 | 10 points

Suppose that for a function $f : \mathbb{R} \to \mathbb{R}$ and real numbers $a < b$ one has $f(x) = 0$ for all $x \in (a, b)$. Prove that $f(x) = 0$ for all $x \in \mathbb{R}$ if

$$\sum_{k=0}^{p-1} f\left(y + \frac{k}{p}\right) = 0$$

for every prime number p and every real number y.

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Theory and methods for 2010-D2-P5

Theory Finite differences and shift-operator ideals [T-AL-DIFFERENCE]; Roots of unity, cyclotomic polynomials, and algebraic norms [T-AL-CYCLOTOMIC]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Methods Take the gcd of annihilating shift polynomials [M-AL-SHIFT]
Strategy Pass between local constraints and a global conclusion [S-LOCALGLOBAL]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2010-D2-P5 — Solution

For this record: Theory: T-AL-DIFFERENCE; T-AL-CYCLOTOMIC; T-AL-CONGRUENCE. Methods: M-AL-SHIFT. Full learning route

Let $N > 1$ be some integer to be defined later, and consider set of real polynomials

$$J_N = \left\{ c_0 + c_1 x + \cdots + c_n x^n \in \mathbb{R}[x] \;\middle|\; \forall x \in \mathbb{R} \ \sum_{k=0}^{n} c_k f\left(x + \frac{k}{N}\right) = 0 \right\}.$$

Notice that $0 \in J_N$, any linear combinations of any elements in J_N is in J_N, and for every $P(x) \in J_N$ we have $xP(x) \in J_N$. Hence, J_N is an ideal of the ring $\mathbb{R}[x]$.

By the problem's conditions, for every prime divisors p of N we have $\dfrac{x^N - 1}{x^{N/p} - 1} \in J_N$. Since $\mathbb{R}[x]$ is a principal ideal domain (due to the Euclidean algorithm), the greatest common divisor of these polynomials is an element of J_N. The complex roots of the polynomial $\dfrac{x^N - 1}{x^{N/p} - 1}$ are those Nth roots of unity whose order does not divide N/p. The roots of the greatest common divisor is the intersection of such sets; it can be seen that the intersection consist of the primitive Nth roots of unity. Therefore,

$$\gcd\left\{ \frac{x^N - 1}{x^{N/p} - 1} \;\middle|\; p \mid N \right\} = \Phi_N(x)$$

is the Nth cyclotomic polynomial. So $\Phi_N \in J_N$, which polynomial has degree $\varphi(N)$.

Now choose N in such a way that $\dfrac{\varphi(N)}{N} < b - a$. It is well-known that $\liminf\limits_{N\to\infty} \dfrac{\varphi(N)}{N} = 0$, so there exists such a value for N. Let $\Phi_N(x) = a_0 + a_1 x + \cdots + a_{\varphi(N)} x^{\varphi(N)}$ where $a_{\varphi(N)} = 1$ and $|a_0| = 1$. Then, by the definition of J_N, we have $\sum\limits_{k=0}^{\varphi(N)} a_k f\left(x + \frac{k}{N}\right) = 0$ for all $x \in \mathbb{R}$.

If $x \in \left[b, b + \frac{1}{N}\right)$, then

$$f(x) = -\sum_{k=0}^{\varphi(N)-1} a_k f\left(x - \frac{\varphi(N) - k}{N}\right).$$

On the right-hand side, all numbers $x - \frac{\varphi(N)-k}{N}$ lie in (a, b). Therefore the right-hand side is zero, and $f(x) = 0$ for all $x \in \left[b, b + \frac{1}{N}\right)$. It can be obtained similarly that $f(x) = 0$ for all $x \in \left(a - \frac{1}{N}, a\right]$ as well. Hence, $f = 0$ in the interval $\left(a - \frac{1}{N}, b + \frac{1}{N}\right)$. Continuing in this fashion we see that f must vanish everywhere.

3.18 2011

Day 1

2011-D1-P1 — Statement

Day 1, P1 | 10 points

Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function. A point x is called a shadow point if there exists a point $y \in \mathbb{R}$ with $y > x$ such that $f(y) > f(x)$. Let $a < b$ be real numbers and suppose that

- all the points of the open interval $I = (a, b)$ are shadow points;
- a and b are not shadow points.

Prove that
a) $f(x) \le f(b)$ for all $a < x < b$;
b) $f(a) = f(b)$.

(José Luis Díaz-Barrero, Barcelona)

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Theory and methods for 2011-D1-P1

Theory Continuity, compactness, and order [T-AN-CONTINUITY]
Methods Localize an extremum or an accumulation point [M-AN-COMPACT]
Strategy Choose an extremal object and force rigidity [S-EXTREMAL]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

2011-D1-P1 — Solution

For this record: Theory: T-AN-CONTINUITY. Methods: M-AN-COMPACT. Full learning route

(a) We prove by contradiction. Suppose that exists a point $c \in (a,b)$ such that $f(c) > f(b)$.
By Weierstrass' theorem, f has a maximal value m on $[c,b]$; this value is attained at some point $d \in [c,b]$. Since $f(d) = \max\limits_{[c,b]} f \geq f(c) > f(b)$, we have $d \neq b$, so $d \in [c,b) \subset (a,b)$. The point d, lying in (a,b), is a shadow point, therefore $f(y) > f(d)$ for some $y > d$. From combining our inequalities we get $f(y) > f(d) > f(b)$.
Case 1: $y > b$. Then $f(y) > f(b)$ contradicts the assumption that b is not a shadow point.
Case 2: $y \leq b$. Then $y \in (d,b] \subset [c,b]$, therefore $f(y) > f(d) = m = \max\limits_{[c,b]} f \geq f(y)$, contradiction again.

(b) Since $a < b$ and a is not a shadow point, we have $f(a) \geq f(b)$.

By part (a), we already have $f(x) \leq f(b)$ for all $x \in (a,b)$. By the continuity at a we have

$$f(a) = \lim_{x \to a+0} f(x) \leq \lim_{x \to a+0} f(b) = f(b)$$

Hence we have both $f(a) \geq f(b)$ and $f(a) \leq f(b)$, so $f(a) = f(b)$.

2011-D1-P2 — Statement

Day 1, P2 | 10 points

Does there exist a real 3×3 matrix A such that $\mathrm{tr}(A) = 0$ and $A^2 + A^t = I$? ($\mathrm{tr}(A)$ denotes the trace of A, A^t is the transpose of A, and I is the identity matrix.)

(Moubinool Omarjee, Paris)

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Theory and methods for 2011-D1-P2

Theory Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Trace pairing and commutator calculus [T-LA-TRACE]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]
Methods Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]; Use trace to extract a scalar invariant [M-LA-TRACE]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

2011-D1-P2 — Solution

For this record: Theory: T-LA-POLYNOMIAL; T-LA-TRACE; T-LA-SPECTRAL. Methods: M-LA-SPECTRAL; M-LA-TRACE. Full learning route

The answer is NO.
Suppose that $\mathrm{tr}(A) = 0$ and $A^2 + A^t = I$. Taking the transpose, we have

$$A = I - (A^2)^t = I - (A^t)^2 = I - (I - A^2)^2 = 2A^2 - A^4,$$

$$A^4 - 2A^2 + A = 0.$$

The roots of the polynomial $x^4 - 2x^2 + x = x(x-1)(x^2+x-1)$ are $0, 1, \frac{-1 \pm \sqrt{5}}{2}$ so these numbers can be the eigenvalues of A; the eigenvalues of A^2 can be $0, 1, \frac{1 \pm \sqrt{5}}{2}$.

2011-D1-P2 — Solution (continued)

By $\mathrm{tr}(A) = 0$, the sum of the eigenvalues is 0, and by $\mathrm{tr}(A^2) = \mathrm{tr}(I - A^t) = 3$ the sum of squares of the eigenvalues is 3. It is easy to check that this two

conditions cannot be satisfied simultaneously.

2011-D1-P3 — Statement

Day 1, P3 | 10 points

Let p be a prime number. Call a positive integer n interesting if

$$x^n - 1 = (x^p - x + 1)f(x) + pg(x)$$

for some polynomials f and g with integer coefficients.
a) Prove that the number $p^p - 1$ is interesting.
b) For which p is $p^p - 1$ the minimal interesting number?

(Eugene Goryachko and Fedor Petrov, St. Petersburg)

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Theory and methods for 2011-D1-P3

Theory Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Groups, orders, generators, and quotients [T-AL-GROUP]
Methods Choose a modulus that exposes the obstruction [M-AL-MODULAR]
Strategy Expose the structure before calculating [S-STRUCTURE]; Prove necessity and sufficiency separately [S-CLASSIFICATION]

Statement Recorded solutions Catalog

2011-D1-P3 — Solution

For this record: Theory: T-AL-FINITEFIELD; T-AL-POLYNOMIAL; T-AL-GROUP. Methods: M-AL-MODULAR. Full learning route

(a) Let's reformulate the property of being interesting: n is interesting if $x^n - 1$ is divisible by $x^p - x + 1$ in the ring of polynomials over $\mathbb{F}_p$ (the field of residues modulo p). All further congruences are modulo $x^p - x + 1$ in this ring. We have $x^p \equiv x - 1$, then $x^{p^2} = (x^p)^p \equiv (x-1)^p \equiv x^p - 1 \equiv x - 2$, $x^{p^3} = (x^{p^2})^p \equiv (x-2)^p \equiv x^p - 2^p \equiv x - 2^p - 1 \equiv x - 3$ and so on by Fermat's little theorem, finally

$$x^{p^p} \equiv x - p \equiv x,$$

$$x\left(x^{p^p - 1} - 1\right) \equiv 0.$$

Since the polynomials $x^p - x + 1$ and x are coprime, this implies $x^{p^p - 1} - 1 \equiv 0$.
(b) We write

$$x^{1+p+p^2+\cdots+p^{p-1}} = x \cdot x^p \cdot x^{p^2} \cdot \ldots \cdot x^{p^{p-1}} \equiv x(x-1)(x-2)\ldots(x-(p-1)) = x^p - x \equiv -1,$$

hence $x^{2(1+p+p^2+\cdots+p^{p-1})} \equiv 1$ and $a = 2(1 + p + p^2 + \cdots + p^{p-1})$ is an interesting number.
If $p > 3$, then $a = \frac{2}{p-1}(p^p - 1) < p^p - 1$, so we have an interesting number less than $p^p - 1$. On the other hand, we show that $p = 2$ and $p = 3$ do satisfy the condition. First notice that by $\gcd(x^m - 1, x^k - 1) = x^{\gcd(m,k)} - 1$, for every fixed p the greatest common divisors of interesting numbers is also an interesting number. Therefore the minimal interesting number divides all interesting numbers. In particular, the minimal interesting number is a divisor of $p^p - 1$.
For $p = 2$ we have $p^p - 1 = 3$, so the minimal interesting number is 1 or 3. But $x^2 - x + 1$ does not divide $x - 1$, so 1 is not interesting. Then the minimal interesting number is 3.
For $p = 3$ we have $p^p - 1 = 26$ whose divisors are $1, 2, 13, 26$. The numbers 1 and 2 are too small and $x^{13} \equiv -1 \neq +1$ as shown above, so none of 1, 2 and 13 is interesting. So 26 is the minimal interesting number.

Hence, $p^p - 1$ is the minimal interesting number if and only if $p = 2$ or $p = 3$.

2011-D1-P4 — Statement

Day 1, P4 | 10 points

Let $A_1, A_2, \ldots, A_n$ be finite, nonempty sets. Define the function

$$f(t) = \sum_{k=1}^{n} \sum_{1 \le i_1 < i_2 < \cdots < i_k \le n} (-1)^{k-1} t^{|A_{i_1} \cup A_{i_2} \cup \cdots \cup A_{i_k}|}.$$

Prove that f is nondecreasing on $[0, 1]$.
($|A|$ denotes the number of elements in A.)

(Levon Nurbekyan and Vardan Voskanyan, Yerevan)

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Theory and methods for 2011-D1-P4

Theory Finite counting and incidence identities [T-DS-COUNTING]; Finite probability, conditioning, and expectations [T-AN-PROBABILITY]
Methods Realize all inclusion probabilities on one probability space [M-DS-COUPLING]; Count incidences with their exact multiplicity [M-DS-DOUBLECOUNT]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2011-D1-P4 — Solution

For this record: Theory: T-DS-COUNTING; T-AN-PROBABILITY. Methods: M-DS-COUPLING; M-DS-DOUBLECOUNT. Full learning route

Let $\Omega = \bigcup_{i=1}^{n} A_i$. Consider a random subset X of Ω which chosen in the following way: for each $x \in \Omega$, choose the element x for the set X with probability t, independently from the other elements.

Then for any set $C \subset \Omega$, we have

$$P(C \subset X) = t^{|C|}.$$

By the inclusion-exclusion principle,

$$P((A_1 \subset X) \text{ or } (A_2 \subset X) \text{ or } \ldots \text{ or } (A_n \subset X)) =$$
$$= \sum_{k=1}^{n} \sum_{1 \le i_1 < i_2 < \cdots < i_k \le n} (-1)^{k-1} P(A_{i_1} \cup A_{i_2} \cup \cdots \cup A_{i_k} \subset X) =$$
$$= \sum_{k=1}^{n} \sum_{1 \le i_1 < i_2 < \cdots < i_k \le n} (-1)^{k-1} t^{|A_{i_1} \cup A_{i_2} \cup \cdots \cup A_{i_k}|}.$$

The probability $P((A_1 \subset X) \text{ or } \ldots \text{ or } (A_n \subset X))$ is a nondecreasing function of the probability t.

2011-D1-P5 — Statement

Day 1, P5 | 10 points

Let n be a positive integer and let V be a $(2n-1)$-dimensional vector space over the two-element field. Prove that for arbitrary vectors $v_1, \ldots, v_{4n-1} \in V$, there exists a sequence $1 \le i_1 < \cdots < i_{2n} \le 4n-1$ of indices such that $v_{i_1} + \cdots + v_{i_{2n}} = 0$.

(Ilya Bogdanov, Moscow and Géza Kós, Budapest)

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Theory and methods for 2011-D1-P5

Theory Linear algebra in characteristic two [T-LA-FINITEFIELD]; Rank, dimension and exact linear constraints [T-LA-RANK]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Exploit parity with linear algebra over F2 [M-LA-FINITEFIELD]; Prove a dimension obstruction and attain it [M-LA-DIMENSION]
Strategy Reduce size while preserving the defining conditions [S-INDUCTION]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2011-D1-P5 — Solution

For this record: Theory: T-LA-FINITEFIELD; T-LA-RANK; T-DS-COUNTING. Methods: M-LA-FINITEFIELD; M-LA-DIMENSION. Full learning route

Let $V = \operatorname{aff}\{v_1, \dots, v_{4n-1}\}$. The statement $v_{i_1} + \dots + v_{i_{2n}} = 0$ is translation-invariant (i.e. replacing the vectors by $v_1 - a, \dots, v_{4n-1} - a$), so we may assume that $0 \in V$. Let $d = \dim V$.
Lemma. The vectors can be permuted in such a way that $v_1 + v_2, v_3 + v_4, \dots, v_{2d-1} + v_{2d}$ form a basis of V.
Proof. We prove by induction on d. If $d = 0$ or $d = 1$ then the statement is trivial.
First choose the vector v_1 such a way that $\operatorname{aff}(v_1, v_2, v_3, \dots, v_{4n-1}) = V$; this is possible since V is generated by some $d+1$ vectors and we have $d+1 \le 2n < 4n-1$. Next, choose v_2 such that $v_2 \neq v_1$. (By $d > 0$, not all vectors are the same.)
Now let $\ell = \{0, v_1 + v_2\}$ and let $V' = V/\ell$. For any $w \in V$, let $\tilde{w} = \ell + w = \{w, w + v_1 + v_2\}$ be the class of the factor space V' containing w. Apply the induction hypothesis to the vectors $\tilde{v}_3, \dots, \tilde{v}_{4n-1}$. Since $\dim V' = d - 1$, the vectors can permuted in such a way that $\tilde{v}_3 + \tilde{v}_4$, ..., $\tilde{v}_{2d-1} + \tilde{v}_{2d}$ is a basis of V'. Then $v_1 + v_2, v_3 + v_4, \dots, v_{2d-1} + v_{2d}$ is a basis of V.

Now we can assume that $v_1 + v_2, v_3 + v_4, \dots, v_{2d-1} + v_{2d}$ is a basis of V. The vector $w = (v_1 + v_3 + \dots + v_{2d-1}) + (v_{2d+1} + v_{2d+2} + \dots + v_{2n+d})$ is the sum of $2n$ vectors, so $w \in V$. Hence, $w + \varepsilon_1(v_1 + v_2) + \dots + \varepsilon_d(v_{2d-1} + v_{2d}) = 0$ with some $\varepsilon_1, \dots, \varepsilon_d \in \mathbb{F}_2$, therefore

$$\sum_{i=1}^{d}((1-\varepsilon_i)v_{2i-1} + \varepsilon_i v_{2i}) + \sum_{i=2d+1}^{2n+d} v_i = 0.$$

The left-hand side is the sum of $2n$ vectors.

Day 2

2011-D2-P1 — Statement

Day 2, P6 | 10 points

Let $(a_n)_{n=0}^{\infty}$ be a sequence with $\frac{1}{2} < a_n < 1$ for all $n \ge 0$. Define the sequence $(x_n)_{n=0}^{\infty}$ by

$$x_0 = a_0, \qquad x_{n+1} = \frac{a_{n+1} + x_n}{1 + a_{n+1}x_n} \quad (n \ge 0).$$

What are the possible values of $\lim_{n\to\infty} x_n$? Can such a sequence diverge?

(Johnson Olaleru, Lagos)

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Theory and methods for 2011-D2-P1

Theory Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Trigonometric identities and safe branches [T-AN-TRIG]
Methods Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]; Exploit identities before estimating oscillations [M-AN-TRIG]
Strategy Control the limiting operation before taking the limit [S-ANALYTIC]; Expose the structure before calculating [S-STRUCTURE]

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2011-D2-P1 — Solution — Record 1 of 2

For this record: Theory: T-AN-CONVERGENCE; T-DS-INDUCTION. Methods: M-AN-RECURRENCE. Full learning route

We prove by induction that

$$0 < 1 - x_n < \frac{1}{2^{n+1}}.$$

Then we will have $(1 - x_n) \to 0$ and therefore $x_n \to 1$.
The case $n = 0$ is true since $\frac{1}{2} < x_0 = a_0 < 1$.

Supposing that the induction hypothesis holds for n, from the recurrence relation we get

$$1 - x_{n+1} = 1 - \frac{a_{n+1} + x_n}{1 + a_{n+1}x_n} = \frac{1 - a_{n+1}}{1 + a_{n+1}x_n}(1 - x_n).$$

By

$$0 < \frac{1 - a_{n+1}}{1 + a_{n+1}x_n} < \frac{1 - \frac{1}{2}}{1 + 0} = \frac{1}{2}$$

we obtain

$$0 < 1 - x_{n+1} < \frac{1}{2}(1 - x_n) < \frac{1}{2} \cdot \frac{1}{2^{n+1}} = \frac{1}{2^{n+2}}.$$

Hence, the sequence converges in all cases and $x_n \to 1$.

2011-D2-P1 — Solution — Record 2 of 2

For this record: Theory: T-AN-CONVERGENCE; T-AN-TRIG. Methods: M-AN-TRIG; M-AN-RECURRENCE. Full learning route

As is well-known,

$$\tanh(u + v) = \frac{\tanh u + \tanh v}{1 + \tanh u \tanh v}$$

for all real numbers u and v.
Setting $u_n = \operatorname{ar}\tanh a_n$ we have $x_n = \tanh(u_0 + u_1 + \cdots + u_n)$. Then $u_0 + u_1 + \cdots + u_n > (n+1)\operatorname{ar}\tanh \frac{1}{2}$ and $\lim_{n\to\infty} x_n = \lim_{u\to\infty} \tanh u = 1$.

Remark. If the condition $a_n \in (\frac{1}{2}, 1)$ is replaced by $a_n \in (0, 1)$ then the sequence remains increasing and bounded, but the limit can be less than 1.

2011-D2-P2 — Statement

Day 2, P7 | 10 points

An alien race has three genders: male, female, and emale. A married triple consists of three persons, one from each gender, who all like each other. Any person is allowed to belong to at most one married triple. A special feature of this race is that feelings are always mutual — if x likes y, then y likes x.
The race is sending an expedition to colonize a planet. The expedition has n males, n females, and n emales. It is known that every expedition member likes at least k persons of each of the two other genders. The problem is to create as many married triples as possible to produce healthy offspring so the colony could grow and prosper.
a) Show that if n is even and $k = \frac{n}{2}$, then it might be impossible to create even one married triple.

2011-D2-P2 — Statement (continued)

b) Show that if $k \geq \frac{3n}{4}$, then it is always possible to create n disjoint married triples, thus marrying all of the expedition members.

(Fedor Duzhin and Nick Gravin, Singapore)

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Theory and methods for 2011-D2-P2

Theory Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]; Finite counting and incidence identities [T-DS-COUNTING]; Extremal choice, exchange arguments, and invariants [T-DS-EXTREMAL]; Hall matching and augmenting paths [T-DS-MATCHING]
Methods Repair an extremal configuration by a local exchange [M-DS-EXTREMAL]; Match classes in stages and verify Hall each time [M-DS-MATCHING]; Split a family by traces, pieces, or labels [M-DS-PARTITION]
Strategy Build a witness and verify every constraint [S-CONSTRUCTION]; Choose an extremal object and force rigidity [S-EXTREMAL]; Build a chain of bounds and track equality [S-BOUND]

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2011-D2-P2 — Solution — Record 1 of 2

For this record: Theory: T-DS-GRAPH; T-DS-COUNTING; T-DS-EXTREMAL. Methods: M-DS-EXTREMAL; M-DS-PARTITION. Full learning route

(a) Let M be the set of males, F the set of females, and E the set of emales. Consider the (tripartite) graph G with vertices $M \cup F \cup E$ and edges for likes. A 3-cycle is then a possible family. We'll call G the graph of likes.
First, let $k = \frac{n}{2}$. Then n has to be even and we need to construct a graph of likes with no 3-cycles. We'll do the following: divide each of the sets M, F, and E into two equal parts and draw all edges between two parts as shown below:
Clearly, there is no 3-cycle.
(b) First divide the the expedition into male-emale-female triples arbitrarily. Let the unhappiness of such a subdivision be the number of pairs of aliens that belong to the same triple but don't like each other. We shall show that if unhappiness is positive, then the unhappiness can be decreased by a simple operation. It will follow that after several steps the unhappiness will be reduced to zero, which will lead to the happy marriage of everybody.
Assume that we have an emale which doesn't like at least one member of its triple (the other cases are similar). We perform the following operation: we swap this emale with another emale, so that each of these two emales will like the members of their new triples. Thus the unhappiness related to this emales will decrease, and the other pairs that contribute to the unhappiness remain unchanged, therefore the unhappiness will be decreased.
So, it remains to prove that such an operation is always possible. Enumerate the triples with $1, 2, \ldots, n$ and denote by E_i, F_i, M_i the emale, female, and male members of the ith triple, respectively. Without loss of generality we may assume that E_1 doesn't like either F_1 or M_1 or both. We have to find an index $i > 1$ such that E_1 likes the couple F_i, M_i and E_i likes the couple F_1, M_1; then we can swap E_1 and E_i.

There are at most $n/4$ indices i for which E_1 dislikes F_i and at most $n/4$ indices for which E_1 dislikes M_i, so there are no more than $n/2$ indices i for which E_1 dislikes someone from the couple M_i, F_i, and the set of these undesirable indexes includes 1. Similarly, there are no more than $n/2$ indices such that either M_1 or F_1 dislikes E_i. Since both undesirable sets of indices have at most $n/2$ elements and both contain 1, their union doesn't cover all indices, so we have some i which satisfies all conditions. Therefore we can always perform the operation that decreases unhappiness.

2011-D2-P2 — Solution — Record 2 of 2

For this record: Theory: T-DS-GRAPH; T-DS-MATCHING; T-DS-COUNTING. Methods: M-DS-MATCHING. Full learning route

(for part b) Suppose that $k \geq \frac{3n}{4}$ and let's show that it's possible to marry all of the colonists. First, we'll prove that there exists a perfect matching between M and F. We need to check the

2011-D2-P2 — Solution — Record 2 of 2 (continued)

condition of Hall's marriage theorem. In other words, for $A \subset M$, let $B \subset F$ be the set of all vertices of F adjacent to at least one vertex of A. Then we need to show that $|A| \leq |B|$. Let us assume the contrary, that is $|A| > |B|$. Clearly, $|B| \geq k$ if A is not empty. Let's consider any $f \in F \setminus B$. Then f is not adjacent to any vertex in A, therefore, f has degree in M not more than $n - |A| < n - |B| \leq n - k \leq \frac{n}{4}$, a contradiction.
Let's now construct a new bipartite graph, say H. The set of its vertices is $P \cup E$, where P is the set of pairs male–female from the perfect matching we just found. We will have an edge from $(m, f) = p \in P$ to $e \in E$ for each 3-cycle (m, f, e) of the graph G, where $(m, f) \in P$ and $e \in E$. Notice that the degree of each vertex of P in H is then at least $2k - n$.
What remains is to show that H satisfies the condition of Hall's marriage theorem and hence has a perfect matching. Assume, on the contrary, that the following happens. There is $A \subset P$ and $B \subset E$ such that $|A| = l$, $|B| < l$, and B is the set of all vertices of E adjacent to at least one vertex of A. Since the degree of each vertex of P is at least $2k - n$, we have $2k - n \leq |B| < l$. On the other hand, let $e \in E \setminus B$. Then for each pair $(m, f) = p \in P$, at most one of the pairs (e, m) and (e, f) is joined by an edge and hence the degree of e in G is at most $|M \setminus A| + |F \setminus A| + |A| = 2(n - l) + l = 2n - l$. But the degree of any vertex of G is $2k$
and thus we get $2k \leq 2n - l$, that is, $l \leq 2n - 2k$.

Finally, $2k - n < l \leq 2n - 2k$ implies that $k < \frac{3n}{4}$. This contradiction concludes the solution.

2011-D2-P3 — Statement

Day 2, P8 | 10 points

Determine the value of

$$\sum_{n=1}^{\infty} \ln\left(1 + \frac{1}{n}\right) \cdot \ln\left(1 + \frac{1}{2n}\right) \cdot \ln\left(1 + \frac{1}{2n+1}\right).$$

(Gerhard Woeginger, Utrecht)

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Theory and methods for 2011-D2-P3

Theory Series, finite cancellation, and summation by parts [T-AN-SERIES]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Find a finite telescoping identity [M-AN-TELESCOPE]
Strategy Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]

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2011-D2-P3 — Solution

For this record: Theory: T-AN-SERIES; T-AN-INEQUALITIES; T-AL-POLYNOMIAL. Methods: M-AN-TELESCOPE. Full learning route

Define $f(n) = \ln \frac{n+1}{n}$ for $n \geq 1$, and observe that $f(2n) + f(2n+1) = f(n)$. The well-known inequality $\ln(1 + x) \leq x$ implies $f(n) \leq 1/n$. Furthermore introduce

$$g(n) = \sum_{k=n}^{2n-1} f^3(k) < n f^3(n) \leq 1/n^2.$$

Then

$$\begin{aligned} g(n) - g(n+1) &= f^3(n) - f^3(2n) - f^3(2n+1) \\ &= (f(2n) + f(2n+1))^3 - f^3(2n) - f^3(2n+1) \\ &= 3(f(2n) + f(2n+1)) f(2n) f(2n+1) \\ &= 3 f(n) f(2n) f(2n+1), \end{aligned}$$

2011-D2-P3 — Solution (continued)

therefore

$$\sum_{n=1}^{N} f(n)f(2n)f(2n+1) = \frac{1}{3}\sum_{n=1}^{N} g(n) - g(n+1) = \frac{1}{3}(g(1) - g(N+1)).$$

Since $g(N+1) \to 0$ as $N \to \infty$, the value of the considered sum hence is

$$\sum_{n=1}^{\infty} f(n)f(2n)f(2n+1) = \frac{1}{3}g(1) = \frac{1}{3}\ln^3(2).$$

2011-D2-P4 — Statement

Day 2, P9 | 10 points

Let $f(x)$ be a polynomial with real coefficients of degree n. Suppose that $\dfrac{f(k) - f(m)}{k - m}$ is an integer for all integers $0 \le k < m \le n$. Prove that $a - b$ divides $f(a) - f(b)$ for all pairs of distinct integers a and b.

(Fedor Petrov, St. Petersburg)

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Theory and methods for 2011-D2-P4

Theory Interpolation, divided differences, and extremal signs [T-AL-INTERPOLATION]; Finite differences and shift-operator ideals [T-AL-DIFFERENCE]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Methods Turn nodal bounds into a sharp linear functional [M-AL-INTERPOLATE]; Compare coefficients after a structural substitution [M-AL-COEFFICIENT]
Strategy Reduce size while preserving the defining conditions [S-INDUCTION]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2011-D2-P4 — Solution — Record 1 of 2

For this record: Theory: T-AL-INTERPOLATION; T-AL-DIFFERENCE; T-AL-CONGRUENCE. Methods: M-AL-INTERPOLATE.
Full learning route

We need the following

Lemma. Denote the least common multiple of $1, 2, \dots, k$ by $L(k)$, and define

$$h_k(x) = L(k) \cdot \binom{x}{k} \quad (k = 1, 2, \dots).$$

Then the polynomial $h_k(x)$ satisfies the condition, i.e. $a - b$ divides $h_k(a) - h_k(b)$ for all pairs of distinct integers a, b.

Proof. It is known that

$$\binom{a}{k} = \sum_{j=0}^{k} \binom{a-b}{j}\binom{b}{k-j}.$$

(This formula can be proved by comparing the coefficient of x^k in $(1+x)^a$ and $(1+x)^{a-b}(1+x)^b$.)
From here we get

$$\begin{aligned} h_k(a) - h_k(b) &= L(k)\left(\binom{a}{k} - \binom{b}{k}\right) \\ &= L(k)\sum_{j=1}^{k} \binom{a-b}{j}\binom{b}{k-j} \\ &= (a-b)\sum_{j=1}^{k} \frac{L(k)}{j}\binom{a-b-1}{j-1}\binom{b}{k-j}. \end{aligned}$$

2011-D2-P4 — Solution — Record 1 of 2 (continued)

On the right-hand side all fractions $\frac{L(k)}{j}$ are integers, so the right-hand side is a multiple of $a-b$. The lemma is proved.
Expand the polynomial f in the basis $1, \binom{x}{1}, \binom{x}{2}, \ldots$ as

$$f(x) = A_0 + A_1\binom{x}{1} + A_2\binom{x}{2} + \cdots + A_n\binom{x}{n}. \tag{1}$$

We prove by induction on j that A_j is a multiple of $L(j)$ for $1 \le j \le n$. (In particular, A_j is an integer for $j \ge 1$.) Assume that $L(j)$ divides A_j for $1 \le j \le m-1$. Substituting m and some $k \in \{0, 1, \ldots, m-1\}$ in (1),

$$\frac{f(m)-f(k)}{m-k} = \sum_{j=1}^{m-1} \frac{A_j}{L(j)} \cdot \frac{h_j(m)-h_j(k)}{m-k} + \frac{A_m}{m-k}.$$

Since all other terms are integers, the last term $\frac{A_m}{m-k}$ is also an integer. This holds for all $0 \le k < m$, so A_m is an integer that is divisible by $L(m)$.

Hence, A_j is a multiple of $L(j)$ for every $1 \le j \le n$. By the lemma this implies the problem statement.

2011-D2-P4 — Solution — Record 2 of 2

For this record: Theory: T-AL-DIFFERENCE; T-AL-POLYNOMIAL; T-DS-INDUCTION. Methods: M-AL-COEFFICIENT. Full learning route

The statement of the problem follows immediately from the following claim, applied to the polynomial $g(x,y) = \dfrac{f(x)-f(y)}{x-y}$.

Claim. Let $g(x,y)$ be a real polynomial of two variables with total degree less than n. Suppose that $g(k,m)$ is an integer whenever $0 \le k < m \le n$ are integers. Then $g(k,m)$ is a integer for every pair k, m of integers.
Proof. Apply induction on n. If $n = 1$ then g is a constant. This constant can be read from $g(0,1)$ which is an integer, so the claim is true.
Now suppose that $n \ge 2$ and the claim holds for $n-1$. Consider the polynomials

$$g_1(x,y) = g(x+1,y+1) - g(x,y+1) \quad \text{and} \quad g_2(x,y) = g(x,y+1) - g(x,y). \tag{1}$$

For every pair $0 \le k < m \le n-1$ of integers, the numbers $g(k,m)$, $g(k,m+1)$ and $g(k+1,m+1)$ are all integers, so $g_1(k,m)$ and $g_2(k,m)$ are integers, too. Moreover, in (1) the maximal degree terms of g cancel out, so $\deg g_1, \deg g_2 < \deg g$. Hence, we can apply the induction hypothesis to the polynomials g_1 and g_2 and we thus have $g_1(k,m), g_2(k,m) \in \mathbb{Z}$ for all $k, m \in \mathbb{Z}$.
In view of (1), for all $k, m \in \mathbb{Z}$, we have that
(a) $g(0,1) \in \mathbb{Z}$;
(b) $g(k,m) \in \mathbb{Z}$ if and only if $g(k+1,m+1) \in \mathbb{Z}$;
(c) $g(k,m) \in \mathbb{Z}$ if and only if $g(k,m+1) \in \mathbb{Z}$.

For arbitrary integers k, m, apply (b) $|k|$ times then apply (c) $|m-k-1|$ times as

$$g(k,m) \in \mathbb{Z} \Leftrightarrow \cdots \Leftrightarrow g(0,m-k) \in \mathbb{Z} \Leftrightarrow \cdots \Leftrightarrow g(0,1) \in \mathbb{Z}.$$

Hence, $g(k,m) \in \mathbb{Z}$. The claim has been proved.

2011-D2-P5 — Statement

Day 2, P10 | 10 points

Let $F = A_0A_1 \ldots A_n$ be a convex polygon in the plane. Define for all $1 \le k \le n-1$ the operation f_k which replaces F with a new polygon

$$f_k(F) = A_0 \ldots A_{k-1}A_k'A_{k+1} \ldots A_n,$$

where A_k' is the point symmetric to A_k with respect to the perpendicular bisector of $A_{k-1}A_{k+1}$. Prove that $(f_1 \circ f_2 \circ \ldots \circ f_{n-1})^n(F) = F$. We suppose that all operations are well-defined on the polygons, to

2011-D2-P5 — Statement (continued)

which they are applied, i.e. results are convex polygons again. (A_0, $A_1, \ldots, A_n$ are the vertices of F in consecutive order.)

(Mikhail Khristoforov, St. Petersburg)

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Theory and methods for 2011-D2-P5

Theory Conics, reflections, and inversion [T-DS-TRANSFORMS]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Adjacent-transposition relations and finite presentations [T-AL-COXETER]; Groups, orders, generators, and quotients [T-AL-GROUP]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]; Choose information that makes an encoding injective [M-DS-ENCODE]
Strategy Expose the structure before calculating [S-STRUCTURE]; Translate geometry into a controlled representation [S-GEOMETRIC]

Statement Recorded solutions Catalog

2011-D2-P5 — Solution

For this record: Theory: T-DS-TRANSFORMS; T-DS-GEOMETRY; T-AL-COXETER; T-AL-GROUP; T-AL-POLYNOMIAL. Methods: M-DS-GEOMETRIC; M-DS-ENCODE. Full learning route

The operations f_i are rational maps on the $2(n-1)$-dimensional phase space of coordinates of the vertices $A_1, \ldots, A_{n-1}$. To show that $(f_1 \circ f_2 \circ \ldots \circ f_{n-1})^n$ is the identity, it is sufficient to verify this on some open set. For example, we can choose a neighborhood of the regular polygon, then all intermediate polygons in the proof will be convex.
Consider the operations f_i. Notice that (i) $f_i \circ f_i = id$ and (ii) $f_i \circ f_j = f_j \circ f_i$ for $|i-j| \geq 2$. We also show that (iii) $(f_i \circ f_{i+1})^3 = id$ for $1 \leq i \leq n-1$.
The operations f_i and f_{i+1} change the order of side lengths by interchanging two consecutive sides; after performing $(f_i \circ f_{i+1})^3$, the side lengths are in the original order. Moreover, the sums of opposite angles in the convex quadrilateral $A_{i-1}A_iA_{i+1}A_{i+2}$ are preserved in all operations. These quantities uniquely determine the quadrilateral, because with fixed sides, both angles $\angle A_1A_2A_3$ and $\angle A_1A_4A_3$ decrease when A_1A_3 increases. Hence, property (iii) is proved.

In the symmetric group S_n, the transpositions $\sigma_i = (i, i+1)$, which from a generator system, satisfy the same properties (i–iii). It is well-known that S_n is the maximal group with $n-1$ generators, satisfying (i–iii). In S_n we have $(\sigma_1 \circ \sigma_2 \circ \ldots \circ \sigma_{n-1})^n = (1, 2, 3, \ldots, n)^n = id$, so this implies $(f_1 \circ f_2 \circ \ldots \circ f_{n-1})^n = id$.

3.19 2012

Day 1

2012-D1-P1 — Statement

Day 1, P1 | 10 points

For every positive integer n, let $p(n)$ denote the number of ways to express n as a sum of positive integers. For instance, $p(4) = 5$ because

$$4 = 3+1 = 2+2 = 2+1+1 = 1+1+1+1.$$

Also define $p(0) = 1$.
Prove that $p(n) - p(n-1)$ is the number of ways to express n as a sum of integers each of which is strictly greater than 1.

(Proposed by Fedor Duzhin, Nanyang Technological University)

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Theory and methods for 2012-D1-P1

Theory Finite counting and incidence identities [T-DS-COUNTING]; Generating functions, coefficient filters, and finite differences [T-DS-GENERATING]
Methods Encode, decode, and verify a bijection [M-DS-BIJECTION]; Split a family by traces, pieces, or labels [M-DS-PARTITION]; Encode the statistic and extract the required coefficient [M-DS-GENERATING]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2012-D1-P1 — Solution — Record 1 of 2

For this record: Theory: T-DS-COUNTING. Methods: M-DS-BIJECTION; M-DS-PARTITION. Full learning route

The statement is true for $n = 1$, because $p(0) = p(1) = 1$ and the only partition of 1 contains the term 1. In the rest of the solution we assume $n \geq 2$.
Let $P_n = \{(a_1, \dots, a_k) : k \in \mathbb{N},\ a_1 \geq \dots \geq a_k,\ a_1 + \dots + a_k = n\}$ be the set of partitions of n, and let $Q_n = \{(a_1, \dots, a_k) \in P_n : a_k = 1\}$ the set of those partitions of n that contain the term 1. The set of those partitions of n that do not contain 1 as a term, is $P_n \setminus Q_n$. We have to prove that $|P_n \setminus Q_n| = |P_n| - |P_{n-1}|$.

Define the map $\varphi : P_{n-1} \to Q_n$ as

$$\varphi(a_1, \dots, a_k) = (a_1, \dots, a_k, 1).$$

This is a partition of n containing 1 as a term (so indeed $\varphi(a_1, \dots, a_k) \in Q_n$). Moreover, each partition $(a_1, \dots, a_k, 1) \in Q_n$ uniquely determines $(a_1, \dots, a_k)$. Therefore the map φ is a bijection between the sets P_{n-1} and Q_n. Then $|P_{n-1}| = |Q_n|$. Since $Q_n \subset P_n$,

$$|P_n \setminus Q_n| = |P_n| - |Q_n| = |P_n| - |P_{n-1}| = p(n) - p(n-1).$$

2012-D1-P1 — Solution — Record 2 of 2

For this record: Theory: T-DS-GENERATING; T-DS-COUNTING. Methods: M-DS-GENERATING. Full learning route

(outline) Denote by $q(n)$ the number of partitions of n not containing 1 as term ($q(0) = 1$ as the only partition of 0 is the empty sum), and define the generating functions

$$F(x) = \sum_{n=0}^{\infty} p(n)x^n \quad \text{and} \quad G(x) = \sum_{n=0}^{\infty} q(n)x^n.$$

Since $q(n) \leq p(n) < 2^n$, these series converge in some interval, say for $|x| < \frac{1}{2}$, and the values uniquely determine the coefficients.

According to Euler's argument, we have

$$F(x) = \sum_{n=0}^{\infty} p(n)x^n = \prod_{k=1}^{\infty}(1 + x^k + x^{2k} + \dots) = \prod_{k=1}^{\infty} \frac{1}{1 - x^k}$$

and

$$G(x) = \sum_{n=0}^{\infty} q(n)x^n = \prod_{k=2}^{\infty}(1 + x^k + x^{2k} + \dots) = \prod_{k=2}^{\infty} \frac{1}{1 - x^k}.$$

Then $G(x) = (1-x)F(x)$. Comparing the coefficient of x^n in this identity we get $q(n) = p(n) - p(n-1)$.

2012-D1-P2 — Statement

Day 1, P2 | 10 points

Let n be a fixed positive integer. Determine the smallest possible rank of an $n \times n$ matrix that has zeros along the main diagonal and strictly positive real numbers off the main diagonal.

(Proposed by Ilya Bogdanov and Grigoriy Chelnokov, MIPT, Moscow)

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Theory and methods for 2012-D1-P2

Theory Rank, dimension and exact linear constraints [T-LA-RANK]
Methods Read the entries before choosing a theorem [M-LA-ENTRY]; Prove a dimension obstruction and attain it [M-LA-DIMENSION]
Strategy Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2012-D1-P2 — Solution

For this record: Theory: T-LA-RANK. Methods: M-LA-ENTRY; M-LA-DIMENSION. Full learning route

For $n = 1$ the only matrix is (0) with rank 0. For $n = 2$ the determinant of such a matrix is negative, so the rank is 2. We show that for all $n \geq 3$ the minimal rank is 3.
Notice that the first three rows are linearly independent. Suppose that some linear combination of them, with coefficients c_1, c_2, c_3, vanishes. Observe that from the first column one deduces that c_2 and c_3 either have opposite signs or both zero. The same applies to the pairs (c_1, c_2) and (c_1, c_3). Hence they all must be zero.

It remains to give an example of a matrix of rank (at most) 3. For example, the matrix

$$\begin{pmatrix} 0^2 & 1^2 & 2^2 & \dots & (n-1)^2 \\ (-1)^2 & 0^2 & 1^2 & \dots & (n-2)^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ (-n+1)^2 & (-n+2)^2 & (-n+3)^2 & \dots & 0^2 \end{pmatrix} = \left((i-j)^2\right)_{i,j=1}^{n} =$$

$$= \begin{pmatrix} 1^2 \\ 2^2 \\ \vdots \\ n^2 \end{pmatrix} (1, 1, \dots, 1) - 2 \begin{pmatrix} 1 \\ 2 \\ \vdots \\ n \end{pmatrix} (1, 2, \dots, n) + \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} (1^2, 2^2, \dots, n^2)$$

is the sum of three matrices of rank 1, so its rank cannot exceed 3.

2012-D1-P3 — Statement

Day 1, P3 | 10 points

Given an integer $n > 1$, let S_n be the group of permutations of the numbers $1, 2, \dots, n$. Two players, A and B, play the following game. Taking turns, they select elements (one element at a time) from the group S_n. It is forbidden to select an element that has already been selected. The game ends when the selected elements generate the whole group S_n. The player who made the last move loses the game. The first move is made by A. Which player has a winning strategy?

(Proposed by Fedor Petrov, St. Petersburg State University)

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Theory and methods for 2012-D1-P3

Theory Groups, orders, generators, and quotients [T-AL-GROUP]; Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]
Methods Prove generation or obstruct it by an image [M-AL-GENERATORS]; Work orbit by orbit and track parity [M-AL-ORBIT]
Strategy Expose the structure before calculating [S-STRUCTURE]; Choose an extremal object and force rigidity [S-EXTREMAL]

Statement Recorded solutions Catalog

2012-D1-P3 — Solution

For this record: Theory: T-AL-GROUP; T-AL-PERMUTATION. Methods: M-AL-GENERATORS; M-AL-ORBIT. Full learning route

Player A can win for $n = 2$ (by selecting the identity) and for $n = 3$ (selecting a 3-cycle).
We prove that B has a winning strategy for $n \geq 4$. Consider the moment when all permitted moves lose immediately, and let H be the subgroup generated by the elements selected by the players. Choosing another element from H would not lose immediately, so all elements of H must have been selected. Since H and any other element generate S_n, H must be a maximal subgroup in S_n.
If $|H|$ is even, then the next player is A, so B wins. Denote by n_i the order of the subgroup generated by the first i selected elements; then $n_1 \mid n_2 \mid n_3 \mid \ldots$. We show that B can achieve that n_2 is even and $n_2 < n!$; then $|H|$ will be even and A will be forced to make the final – losing – move.
Denote by g the element chosen by A on his first move. If the order n_1 of g is even, then B may choose the identical permutation id and he will have $n_2 = n_1$ even and $n_2 = n_1 < n!$.
If n_1 is odd, then g is a product of disjoint odd cycles, so it is an even permutation. Then B can chose the permutation $h = (1, 2)(3, 4)$ which is another even permutation. Since g and h are elements of the alternating group A_n, they cannot generate the whole S_n. Since the order of h is 2, B achieves $2 \mid n_2$.

Remark. If $n \geq 4$, all subgrups of odd order are subgroups of A_n which has even order. Hence, all maximal subgroups have even order and B is never forced to lose.

2012-D1-P4 — Statement

Day 1, P4 | 10 points

Let $f : \mathbb{R} \to \mathbb{R}$ be a continuously differentiable function that satisfies $f'(t) > f(f(t))$ for all $t \in \mathbb{R}$. Prove that $f(f(f(t))) \leq 0$ for all $t \geq 0$.

(Proposed by Tomáš Bárta, Charles University, Prague)

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Theory and methods for 2012-D1-P4

Theory Differential inequalities and integrating factors [T-AN-ODE]; Continuity, compactness, and order [T-AN-CONTINUITY]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Methods Use a first crossing or a forced-growth contradiction [M-AN-BARRIER]
Strategy Make the contradiction quantitative or structural [S-CONTRADICTION]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2012-D1-P4 — Solution

For this record: Theory: T-AN-ODE; T-AN-CONTINUITY; T-AN-DERIVATIVE. Methods: M-AN-BARRIER. Full learning route

Lemma 1. Either $\lim_{t \to +\infty} f(t)$ does not exist or $\lim_{t \to +\infty} f(t) \neq +\infty$.
Proof. Assume that the limit is $+\infty$. Then there exists $T_1 > 0$ such that for all $t > T_1$ we have $f(t) > 2$. There exists $T_2 > 0$ such that $f(t) > T_1$ for all $t > T_2$. Hence, $f'(t) > f(f(t)) > 2$ for $t > T_2$. Hence, there exists T_3 such that $f(t) > t$ for $t > T_3$. Then $f'(t) > f(f(t)) > f(t)$, $f'(t)/f(t) > 1$, after

2012-D1-P4 — Solution (continued)

integration $\ln f(t) - \ln T_3 > t - T_3$, i.e. $f(t) > T_3 e^{t-T_3}$ for all $t > T_3$. Then $f'(t) > f(f(t)) > T_3 e^{f(t)-T_3}$ and $f'(t)e^{-f(t)} > T_3 e^{-T_3}$. Integrating from T_3 to t yields $e^{-f(T_3)} - e^{-f(t)} > (t-T_3)T_3 e^{-T_3}$. The right-hand side tends to infinity, but the left-hand side is bounded from above, a contradiction.

Lemma 2. For all $t > 0$ we have $f(t) < t$.

Proof. By Lemma 1, there are some positive real numbers t with $f(t) < t$. Hence, if the statement is false then there is some $t_0 > 0$ with $f(t_0) = t_0$.

Case I: There exist some value $t \geq t_0$ with $f(t) < t_0$. Let $T = \inf\{t \geq t_0 : f(t) < t_0\}$. By the continuity of f, $f(T) = t_0$. Then $f'(T) > f(f(T)) = f(t_0) = t_0 > 0$. This implies $f > f(T) = t_0$ in a right neighbourhood, contradicting the definition of T.

Case II: $f(t) \geq t_0$ for all $t \geq t_0$. Now we have $f'(t) > f(f(t)) \geq t_0 > 0$. So, f' has a positive lower bound over (t_0, ∞), which contradicts Lemma 1.

Lemma 3. (a) If $f(s_1) > 0$ and $f(s_2) \geq s_1$, then $f(s) > s_1$ for all $s > s_2$.

(b) In particular, if $s_1 \leq 0$ and $f(s_1) > 0$, then $f(s) > s_1$ for all $s > s_1$.

Proof. Suppose that there are values $s > s_2$ with $f(s) \leq s_1$ and let $S = \inf\{s > s_2 : f(s) \leq s_1\}$. By the continuity we have $f(S) = s_1$. Similarly to Lemma 2, we have $f'(S) > f(f(S)) = f(s_1) > 0$. If $S > s_2$ then in a left neighbourhood of S we have $f < s_1$, contradicting the definition of S. Otherwise, if $S = s_2$ then we have $f > s_1$ in a right neighbourhood of s_2, contradiction again.

Part (b) follows if we take $s_2 = s_1$.

With the help of these lemmas the proof goes as follows. Assume for contradiction that there exists some $t_0 > 0$ with $f(f(f(t_0))) > 0$. Let $t_1 = f(t_0)$, $t_2 = f(t_1)$ and $t_3 = f(t_2) > 0$. We show that $0 < t_3 < t_2 < t_1 < t_0$. By lemma 2 it is sufficient to prove that t_1 and t_2 are positive. If $t_1 < 0$, then $f(t_1) \leq 0$ (if $f(t_1) > 0$ then taking $s_1 = t_1$ in Lemma 3(b) yields $f(t_0) > t_1$, contradiction). If $t_1 = 0$ then $f(t_1) \leq 0$ by lemma 2 and the continuity of f. Hence, if $t_1 \leq 0$, then also $t_2 \leq 0$. If $t_2 = 0$ then $f(t_2) \leq 0$ by lemma 2 and the continuity of f (contradiction, $f(t_2) = t_3 > 0$). If $t_2 < 0$, then by lemma 3(b), $f(t_0) > t_2$, so $t_1 > t_2$. Applying lemma 3(a) we obtain $f(t_1) > t_2$, contradiction. We have proved $0 < t_3 < t_2 < t_1 < t_0$.

By lemma 3(a) ($f(t_1) > 0$, $f(t_0) \geq t_1$) we have $f(t) > t_1$ for all $t > t_0$ and similarly $f(t) > t_2$ for all $t > t_1$. It follows that for $t > t_0$ we have $f'(t) > f(f(t)) > t_2 > 0$. Hence, $\lim\limits_{t \to +\infty} f(t) = +\infty$, which is a contradiction. This contradiction proves that $f(f(f(t))) \leq 0$ for all $t > 0$. For $t = 0$ the inequality follows from the continuity of f.

2012-D1-P5 — Statement

Day 1, P5 | 10 points

Let a be a rational number and let n be a positive integer. Prove that the polynomial $X^{2^n}(X+a)^{2^n}+1$ is irreducible in the ring $\mathbb{Q}[X]$ of polynomials with rational coefficients.

(Proposed by Vincent Jugé, École Polytechnique, Paris)

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Theory and methods for 2012-D1-P5

Theory Polynomial factorization and coprime divisibility [T-AL-FACTORIZATION]; Roots of unity, cyclotomic polynomials, and algebraic norms [T-AL-CYCLOTOMIC]; Prime valuations, divisors, and factorials [T-AL-VALUATION]

Methods Factor first and compare irreducible exponents [M-AL-FACTOR]; Replace a minimal algebraic counterexample by a smaller one [M-AL-DESCENT]

Strategy Make the contradiction quantitative or structural [S-CONTRADICTION]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2012-D1-P5 — Solution

For this record: Theory: T-AL-FACTORIZATION; T-AL-CYCLOTOMIC; T-AL-VALUATION. Methods: M-AL-FACTOR; M-AL-DESCENT. Full learning route

First let us consider the case $a = 0$. The roots of $X^{2^{n+1}} + 1$ are exactly all primitive roots of unity of order 2^{n+2}, namely $e^{2\pi i \frac{k}{2^{n+2}}}$ for odd $k = 1, 3, 5, \dots, 2^{n+2} - 1$. It is a cyclotomic polynomial, hence irreducible in $\mathbb{Q}[X]$.

Let now $a \neq 0$ and suppose that the polynomial in the question is reducible. Substituting $X = Y - \frac{a}{2}$ we get a polynomial $\left(Y - \frac{a}{2}\right)^{2^n} \left(Y + \frac{a}{2}\right)^{2^n} + 1 = \left(Y^2 - \frac{a^2}{4}\right)^{2^n} + 1$. It is again a cyclotomic polynomial in the variable $Z = Y^2 - \frac{a^2}{4}$, and therefore it is not divisible by any polynomial in Y^2 with rational coefficients. Let us write this polynomial as the product of irreducible monic polynomials in Y with appropriate multiplicities, i.e.

$$\left(Y^2 - \frac{a^2}{4}\right)^{2^n} + 1 = \prod_{i=1}^{r} f_i(Y)^{m_i} \quad f_i \text{ monic, irreducible, all different.}$$

Since the left-hand side is a polynomial in Y^2 we must have $\prod_i f_i(Y)^{m_i} = \prod_i f_i(-Y)^{m_i}$. By the above argument non of the f_i is a polynomial in Y^2, i.e. $f_i(-Y) \neq f_i(Y)$. Therefore for every i there is $i' \neq i$ such that $f_i(-Y) = \pm f_{i'}(Y)$. In particular r is even and irreducible factors f_i split into pairs. Let us renumber them so that $f_1, \dots, f_{\frac{r}{2}}$ belong to different pairs and we have $f_{i+\frac{r}{2}}(-Y) = \pm f_i(Y)$. Consider the polynomial $f(Y) = \prod_{i=1}^{r/2} f_i(Y)^{m_i}$. This polynomial is monic of degree 2^n and $\left(Y^2 - \frac{a^2}{4}\right)^{2^n} + 1 = f(Y)f(-Y)$. Let us write $f(Y) = Y^{2^n} + \dots + b$ where $b \in \mathbb{Q}$ is the constant term, i.e. $b = f(0)$.

Comparing constant terms we then get $\left(\frac{a}{2}\right)^{2^{n+1}} + 1 = b^2$. Denote $c = \left(\frac{a}{2}\right)^{2^{n-1}}$. This is a nonzero rational number and we have $c^4 + 1 = b^2$.

It remains to show that there are no rational solutions $c, b \in \mathbb{Q}$ to the equation $c^4 + 1 = b^2$ with $c \neq 0$ which will contradict our assumption that the polynomial under consideration is reducible. Suppose there is a solution. Without loss of generality we can assume that $c, b > 0$. Write $c = \frac{u}{v}$ with u and v coprime positive integers. Then $u^4 + v^4 = (bv^2)^2$. Let us denote $w = bv^2$, this must be a positive integer too since u, v are positive integers. Let us show that the set $T = \{(u, v, w) \in \mathbb{N}^3 \mid u^4 + v^4 = w^2 \text{ and } u, v, w \geq 1\}$ is empty. Suppose the contrary and consider some triple $(u, v, w) \in T$ such that w is minimal. Without loss of generality, we may assume that u is odd. (u^2, v^2, w) is a primitive Pythagorean triple and thus there exist relatively prime integers $d > e \geq 1$ such that $u^2 = d^2 - e^2$, $v^2 = 2de$ and $w = d^2 + e^2$. In particular, considering the equation $u^2 = d^2 - e^2$ in $\mathbb{Z}/4\mathbb{Z}$ proves that d is odd and e is even. Therefore, we can write $d = f^2$ and $e = 2g^2$. Moreover, since $u^2 + e^2 = d^2$, (u, e, d) is also a primitive Pythagorean triple: there exist relatively prime integers $h > i \geq 1$ such that $u = h^2 - i^2$, $e = 2hi = 2g^2$ and $d = h^2 + i^2$. Once again, we can write $h = k^2$ and $i = l^2$, so that we obtain the relation $f^2 = d = h^2 + i^2 = k^4 + l^4$ and $(k, l, f) \in T$. Then, the inequality $w > d^2 = f^4 \geq f$ contradicts the minimality of w.

Remark 1. One can also use Galois theory arguments in order to solve this question. Let us denote the polynomial in the question by $P(X) = X^{2^n}(X + a)^{2^n} + 1$ and we will also need the cyclotomic polynomial $T(X) = X^{2^n} + 1$. As we already said, when $a = 0$ then $P(X)$ is itself cyclotomic and hence irreducible. Let now $a \neq 0$ and x be any complex root of $P(x) = 0$. Then $\zeta = x(x + a)$ satisfies $T(\zeta) = 0$, hence it is a primitive root of unity of order 2^{n+1}. The field $\mathbb{Q}[x]$ is then an extension of $\mathbb{Q}[\zeta]$. The latter field is cyclotomic and its degree over $\mathbb{Q}$ is $\dim_{\mathbb{Q}}(\mathbb{Q}[\zeta]) = 2^n$. Since the polynomial in the question has degree 2^{n+1} we see that it is reducible if and only if the above mentioned extension is trivial, i.e. $\mathbb{Q}[x] = \mathbb{Q}[\zeta]$. For the sake of contradiction we will now assume that this is indeed the case. Let $S(X)$ be the minimal polynomial of x over $\mathbb{Q}$. The degree of S is then 2^n and we can number its roots by odd numbers in the set $I = \{1, 3, \dots, 2^{n+1} - 1\}$ so that $S(X) = \prod_{k \in I}(X - x_k)$ and $x_k(x_k + a) = \zeta^k$ because Galois automorphisms of $\mathbb{Q}[\zeta]$ map ζ to ζ^k, $k \in I$. Then one has

$$S(X)S(-a - X) = \prod_{k \in I}(X - x_k)(-a - X - x_k) = (-1)^{|I|} \prod_{k \in I}\Big(X(X + a) - \zeta^k\Big) = T(X(X + a)) = P(X).$$

2012-D1-P5 — Solution (continued)

In particular $P\left(-\frac{a}{2}\right) = S\left(-\frac{a}{2}\right)^2$, i.e. $\left(\frac{a}{2}\right)^{2^{n+1}} + 1 = \left(\left(\frac{a}{2}\right)^{2^n} + 1\right)^2$. Therefore the rational numbers $c = \left(\frac{a}{2}\right)^{2^{n-1}} \neq 0$ and $b = \left(\frac{a}{2}\right)^{2^n} + 1$ satisfy $c^4 + 1 = b^2$ which is a contradiction as it was shown in the first proof.

Remark 2. It is well-known that the Diophantine equation $x^4 + y^4 = z^2$ has only trivial solutions (i.e. with $x = 0$ or $y = 0$). This implies immediately that $c^4 + 1 = b^2$ has no rational solution with nonzero c.

Day 2

2012-D2-P1 — Statement

Day 2, P6 | 10 points

Consider a polynomial

$$f(x) = x^{2012} + a_{2011}x^{2011} + \cdots + a_1 x + a_0.$$

Albert Einstein and Homer Simpson are playing the following game. In turn, they choose one of the coefficients $a_0, \ldots, a_{2011}$ and assign a real value to it. Albert has the first move. Once a value is assigned to a coefficient, it cannot be changed any more. The game ends after all the coefficients have been assigned values.
Homer's goal is to make $f(x)$ divisible by a fixed polynomial $m(x)$ and Albert's goal is to prevent this.
(a) Which of the players has a winning strategy if $m(x) = x - 2012$?
(b) Which of the players has a winning strategy if $m(x) = x^2 + 1$?

(Proposed by Fedor Duzhin, Nanyang Technological University)

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Theory and methods for 2012-D2-P1

Theory Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Rank, dimension and exact linear constraints [T-LA-RANK]
Methods Compare coefficients after a structural substitution [M-AL-COEFFICIENT]
Strategy Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2012-D2-P1 — Solution

For this record: Theory: T-AL-POLYNOMIAL; T-LA-RANK. Methods: M-AL-COEFFICIENT. Full learning route

We show that Homer has a winning strategy in both part (a) and part (b).
(a) Notice that the last move is Homer's, and only the last move matters. Homer wins if and only if $f(2012) = 0$, i.e.

$$2012^{2012} + a_{2011}2012^{2011} + \cdots + a_k 2012^k + \cdots + a_1 2012 + a_0 = 0. \tag{1}$$

Suppose that all of the coefficients except for a_k have been assigned values. Then Homer's goal is to establish (1) which is a linear equation on a_k. Clearly, it has a solution and hence Homer can win.
(b) Define the polynomials

$$g(y) = a_0 + a_2 y + a_4 y^2 + \cdots + a_{2010}y^{1005} + y^{1006} \quad \text{and} \quad h(y) = a_1 + a_3 y + a_5 y^2 + \cdots + a_{2011}y^{1005},$$

so $f(x) = g(x^2) + h(x^2) \cdot x$. Homer wins if he can achieve that $g(y)$ and $h(y)$ are divisible by $y + 1$, i.e. $g(-1) = h(-1) = 0$.

Notice that both $g(y)$ and $h(y)$ have an even number of undetermined coefficients in the beginning of the game. A possible strategy for Homer is to follow Albert: whenever Albert assigns a value

2012-D2-P1 — Solution (continued)

to a coefficient in g or h, in the next move Homer chooses the value for a coefficient in the same polynomial. This way Homer defines the last coefficient in g and he also chooses the last coefficient in h. Similarly to part (a), Homer can choose these two last coefficients in such a way that both $g(-1) = 0$ and $h(-1) = 0$ hold.

2012-D2-P2 — Statement

Day 2, P7 | 10 points

Define the sequence $a_0, a_1, \ldots$ inductively by $a_0 = 1$, $a_1 = \frac{1}{2}$ and

$$a_{n+1} = \frac{na_n^2}{1 + (n+1)a_n} \quad \text{for } n \geq 1.$$

Show that the series $\sum_{k=0}^{\infty} \frac{a_{k+1}}{a_k}$ converges and determine its value.

(Proposed by Christophe Debry, KU Leuven, Belgium)

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Theory and methods for 2012-D2-P2

Theory Series, finite cancellation, and summation by parts [T-AN-SERIES]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]

Methods Find a finite telescoping identity [M-AN-TELESCOPE]; Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]

Strategy Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2012-D2-P2 — Solution

For this record: Theory: T-AN-SERIES; T-AN-CONVERGENCE. Methods: M-AN-TELESCOPE; M-AN-RECURRENCE. Full learning route

Observe that

$$ka_k = \frac{(1 + (k+1)a_k)a_{k+1}}{a_k} = \frac{a_{k+1}}{a_k} + (k+1)a_{k+1} \quad \text{for all } k \geq 1,$$

and hence

$$\sum_{k=0}^{n} \frac{a_{k+1}}{a_k} = \frac{a_1}{a_0} + \sum_{k=1}^{n} (ka_k - (k+1)a_{k+1}) = \frac{1}{2} + 1 \cdot a_1 - (n+1)a_{n+1} = 1 - (n+1)a_{n+1} \tag{1}$$

for all $n \geq 0$.

By (1) we have $\sum_{k=0}^{n} \frac{a_{k+1}}{a_k} < 1$. Since all terms are positive, this implies that the series $\sum_{k=0}^{\infty} \frac{a_{k+1}}{a_k}$ is convergent. The sequence of terms, $\frac{a_{k+1}}{a_k}$ must converge to zero. In particular, there is an index n_0 such that $\frac{a_{k+1}}{a_k} < \frac{1}{2}$ for $n \geq n_0$. Then, by induction on n, we have $a_n < \frac{C}{2^n}$ with some positive constant C. From $na_n < \frac{Cn}{2^n}$ we get $na_n \to 0$, and therefore

$$\sum_{k=0}^{\infty} \frac{a_{k+1}}{a_k} = \lim_{n\to\infty} \sum_{k=0}^{n} \frac{a_{k+1}}{a_k} = \lim_{n\to\infty} (1 - (n+1)a_{n+1}) = 1.$$

Remark. The inequality $a_n \leq \frac{1}{2^n}$ can be proved by a direct induction as well.

2012-D2-P3 — Statement

Day 2, P8 | 10 points

Is the set of positive integers n such that $n!+1$ divides $(2012n)!$ finite or infinite?

(Proposed by Fedor Petrov, St. Petersburg State University)

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Theory and methods for 2012-D2-P3

Theory Prime valuations, divisors, and factorials [T-AL-VALUATION]; Factorial growth and the prime-number input [T-AL-PRIMEGROWTH]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Track one prime or one extreme divisor [M-AL-VALUATION]
Strategy Build a chain of bounds and track equality [S-BOUND]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2012-D2-P3 — Solution — Record 1 of 2

For this record: Theory: T-AL-VALUATION; T-AL-PRIMEGROWTH; T-DS-COUNTING. Methods: M-AL-VALUATION. Full learning route

Consider a positive integer n with $n!+1 \mid (2012n)!$. It is well-known that for arbitrary nonnegative integers $a_1, \dots, a_k$, the number $(a_1+\dots+a_k)!$ is divisible by $a_1!\cdot\dots\cdot a_k!$. (The number of sequences consisting of a_1 digits 1, ..., a_k digits k, is $\frac{(a_1+\dots+a_k)!}{a_1!\dots a_k!}$.) In particular, $(n!)^{2012}$ divides $(2012n)!$.
Since $n!+1$ is co-prime with $(n!)^{2012}$, their product $(n!+1)(n!)^{2012}$ also divides $(2012n)!$, and therefore

$$(n!+1)\cdot(n!)^{2012} \le (2012n)!.$$

By the known inequalities $\left(\frac{n+1}{e}\right)^n < n! \le n^n$, we get

$$\left(\frac{n}{e}\right)^{2013n} < (n!)^{2013} < (n!+1)\cdot(n!)^{2012} \le (2012n)! < (2012n)^{2012n}$$

$$n < 2012^{2012}e^{2013}.$$

Therefore, there are only finitely many such integers n.

Remark. Instead of the estimate $\left(\frac{n+1}{e}\right)^n < n!$, we may apply the Multinomial theorem:

$$(x_1+\dots+x_\ell)^N = \sum_{k_1+\dots+k_\ell=N} \frac{N!}{k_1!\cdot\dots\cdot k_\ell!} x_1^{k_1}\dots x_\ell^{k_\ell}.$$

Applying this to $N=2012n$, $\ell=2012$ and $x_1=\dots=x_\ell=1$,

$$\frac{(2012n)!}{(n!)^{2012}} < (\underbrace{1+1+\dots+1}_{2012})^{2012n} = 2012^{2012n},$$

$$n! < n!+1 \le \frac{(2012n)!}{(n!)^{2012}} < 2012^{2012n}.$$

On the right-hand side we have a geometric progression which increases slower than the factorial on the left-hand side, so this is true only for finitely many n.

2012-D2-P3 — Solution — Record 2 of 2

For this record: Theory: T-AL-VALUATION; T-AL-PRIMEGROWTH. Methods: M-AL-VALUATION. Full learning route

Assume that $n > 2012$ is an integer with $n! + 1 \mid (2012n)!$. Notice that all prime divisors of $n! + 1$ are greater than n, and all prime divisors of $(2012n)!$ are smaller than $2012n$.

Consider a prime p with $n < p < 2012n$. Among $1, 2, \ldots, 2012n$ there are $\left\lfloor \frac{2012n}{p} \right\rfloor < 2012$ numbers divisible by p; by $p^2 > n^2 > 2012n$, none of them is divisible by p^2. Therefore, the exponent of p in the prime factorization of $(2012n)!$ is at most 2011. Hence,

$$n! + 1 = \gcd(n! + 1, (2012n)!) < \prod_{n<p<2012p} p^{2011}.$$

Applying the inequality $\prod_{p \le X} p < 4^X$,

$$n! < \prod_{n<p<2012p} p^{2011} < \left(\prod_{p<2012n} p \right)^{2011} < \left(4^{2012n}\right)^{2011} = \left(4^{2012 \cdot 2011}\right)^n. \tag{2}$$

Again, we have a factorial on the left-and side and a geometric progression on the right-hand side.

2012-D2-P4 — Statement

Day 2, P9 | 10 points

Let $n \ge 2$ be an integer. Find all real numbers a such that there exist real numbers x_1, ..., x_n satisfying

$$x_1(1 - x_2) = x_2(1 - x_3) = \cdots = x_{n-1}(1 - x_n) = x_n(1 - x_1) = a. \tag{1}$$

(Proposed by Walther Janous and Gerhard Kirchner, Innsbruck)

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Theory and methods for 2012-D2-P4

Theory Fractional-linear recurrences and matrix powers [T-LA-PROJECTIVE]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Linearize a fractional recurrence in dimension two [M-LA-PROJECTIVE]; Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2012-D2-P4 — Solution

For this record: Theory: T-LA-PROJECTIVE; T-LA-POLYNOMIAL; T-AL-POLYNOMIAL. Methods: M-LA-PROJECTIVE; M-LA-SPECTRAL. Full learning route

Throughout the solution we will use the notation $x_{n+1} = x_1$.
We prove that the set of possible values of a is

$$\left(-\infty, \frac{1}{4}\right] \cup \left\{ \frac{1}{4\cos^2 \frac{k\pi}{n}} \; ; \; k \in \mathbb{N}, \; 1 \le k < \frac{n}{2} \right\}.$$

In the case $a \le \frac{1}{4}$ we can choose x_1 such that $x_1(1 - x_1) = a$ and set $x_1 = x_2 = \cdots = x_n$. Hence we will now suppose that $a > \frac{1}{4}$.
The system (1) gives the recurrence formula

$$x_{i+1} = \varphi(x_i) = 1 - \frac{a}{x_i} = \frac{x_i - a}{x_i}, \quad i = 1, \ldots, n.$$

The fractional linear transform φ can be interpreted as a projective transform of the real projective line $\mathbb{R} \cup \{\infty\}$; the map φ is an element of the group $PGL_2(\mathbb{R})$, represented by the linear transform

2012-D2-P4 — Solution (continued)

$M = \begin{pmatrix} 1 & -a \\ 1 & 0 \end{pmatrix}$. (Note that $\det M \neq 0$ since $a \neq 0$.) The transform φ^n can be represented by M^n. A point $[u, v]$ (written in homogenous coordinates) is a fixed point of this transform if and only if $(u, v)^T$ is an eigenvector of M^n. Since the entries of M^n and the coordinates u, v are real, the corresponding eigenvalue is real, too. The characteristic polynomial of M is $x^2 - x + a$, which has no real root for $a > \frac{1}{4}$. So M has two conjugate complex eigenvalues $\lambda_{1,2} = \frac{1}{2}\left(1 \pm \sqrt{4a-1}\, i\right)$. The eigenvalues of M^n are $\lambda_{1,2}^n$, they are real if and only if $\arg \lambda_{1,2} = \pm\frac{k\pi}{n}$ with some integer k; this is equivalent with

$$\pm\sqrt{4a-1} = \tan\frac{k\pi}{n},$$

$$a = \frac{1}{4}\left(1 + \tan^2\frac{k\pi}{n}\right) = \frac{1}{4\cos^2\frac{k\pi}{n}}.$$

If $\arg\lambda_1 = \frac{k\pi}{n}$ then $\lambda_1^n = \overline{\lambda_2^n}$, so the eigenvalues of M^n are equal. The eigenvalues of M are distinct, so M and M^n have two linearly independent eigenvectors. Hence, M^n is a multiple of the identity. This means that the projective transform φ^n is the identity; starting from an arbitrary point $x_1 \in \mathbb{R} \cup \{\infty\}$, the cycle $x_1, x_2, \ldots, x_n$ closes at $x_{n+1} = x_1$. There are only finitely many cycles $x_1, x_2, \ldots, x_n$ containing the point ∞; all other cycles are solutions for (1).

Remark. If we write $x_j = P + Q\tan t_j$ where P, Q and $t_1, \ldots, t_n$ are real numbers, the recurrence relation re-writes as

$$(P + Q\tan t_j)(1 - P - Q\tan t_{j+1}) = a$$

$$(1-P)Q\tan t_j - PQ\tan t_{j+1} = a + P(P-1) + Q^2\tan t_j \tan t_{j+1} \quad (j = 1, 2, \ldots, n).$$

In view of the identity $\tan(\alpha - \beta) = \frac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta}$, it is reasonable to choose $P = \frac{1}{2}$, and $Q = \sqrt{a - \frac{1}{4}}$. Then the recurrence leads to

$$t_j - t_{j+1} \equiv -\arctan\sqrt{4a-1} \pmod{\pi}.$$

2012-D2-P5 — Statement

Day 2, P10 | 10 points

Let $c \geq 1$ be a real number. Let G be an abelian group and let $A \subset G$ be a finite set satisfying $|A + A| \leq c|A|$, where $X + Y := \{x + y \mid x \in X,\ y \in Y\}$ and $|Z|$ denotes the cardinality of Z. Prove that

$$|\underbrace{A + A + \cdots + A}_{k \text{ times}}| \leq c^k|A|$$

for every positive integer k. (Plünnecke's inequality)

(Proposed by Przemyslaw Mazur, Jagiellonian University)

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Theory and methods for 2012-D2-P5

Theory Sumsets and the minimal-growth lemma [T-DS-ADDITIVE]; Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Groups, orders, generators, and quotients [T-AL-GROUP]
Methods Repair an extremal configuration by a local exchange [M-DS-EXTREMAL]; Strengthen the statement before the induction step [M-DS-INDUCT]; Count incidences with their exact multiplicity [M-DS-DOUBLECOUNT]
Strategy Choose an extremal object and force rigidity [S-EXTREMAL]; Reduce size while preserving the defining conditions [S-INDUCTION]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2012-D2-P5 — Solution

For this record: Theory: T-DS-ADDITIVE; T-DS-COUNTING; T-DS-INDUCTION; T-AL-GROUP. Methods: M-DS-EXTREMAL; M-DS-INDUCT; M-DS-DOUBLECOUNT. Full learning route

If $A=\emptyset$, every sumset in the conclusion is empty and the claim is immediate. Assume $A\neq\emptyset$. Let B be a nonempty subset of A for which the value of the expression $c_1=\frac{|A+B|}{|B|}$ is the least possible. Note that $c_1\leq c$ since A is one of the possible choices of B.
Lemma 1. For any finite set $D\subset G$ we have $|A+B+D|\leq c_1|B+D|$.
Proof. If $D=\emptyset$, both sides are zero. Otherwise apply induction on the cardinality of D. For $|D|=1$ the Lemma is true by the definition of c_1. Suppose it is true for some D and let $x\notin D$. Let $B_1=\{y\in B\mid x+y\in B+D\}$. Then $B+(D\cup\{x\})$ decomposes into the union of two disjoint sets:

$$B+(D\cup\{x\})=(B+D)\cup((B\setminus B_1)+\{x\})$$

and therefore $|B+(D\cup\{x\})|=|B+D|+|B|-|B_1|$. Now we need to deal with the cardinality of the set $A+B+(D\cup\{x\})$. Writing $A+B+(D\cup\{x\})=(A+B+D)\cup(A+B+\{x\})$ we count some of the elements twice: for example if $y\in B_1$, then $A+\{y\}+\{x\}\subset(A+B+D)\cap(A+B+\{x\})$. Therefore all the elements of the set $A+B_1+\{x\}$ are counted twice and thus

$$\begin{aligned}|A+B+(D\cup\{x\})|&\leq|A+B+D|+|A+B+\{x\}|-|A+B_1+\{x\}|\\&=|A+B+D|+|A+B|-|A+B_1|\\&\leq c_1(|B+D|+|B|-|B_1|)\\&=c_1|B+(D\cup\{x\})|.\end{aligned}$$

where the last inequality follows from the inductive hypothesis and the observation that $\frac{|A+B|}{|B|}=c_1\leq\frac{|A+B_1|}{|B_1|}$ (or B_1 is the empty set).
Lemma 2. For every $k\geq1$ we have $|\underbrace{A+\cdots+A}_{k\text{ times}}+B|\leq c_1^k|B|$.
Proof. Induction on k. For $k=1$ the statement is true by definition of c_1. For greater k set $D=\underbrace{A+\cdots+A}_{k-1\text{ times}}$ in the previous lemma: $|\underbrace{A+\cdots+A}_{k\text{ times}}+B|\leq c_1|\underbrace{A+\cdots+A}_{k-1\text{ times}}+B|\leq c_1^k|B|$.
Choose any $b\in B$. Translation by b injects each k-fold sumset of A into that sumset plus B. Thus

$$|\underbrace{A+\cdots+A}_{k\text{ times}}|\leq|\underbrace{A+\cdots+A}_{k\text{ times}}+B|\leq c_1^k|B|\leq c^k|A|.$$

Remark. The proof above due to Giorgios Petridis and can be found at the source note for this problem

3.20 2013

Day 1

2013-D1-P1 — Statement

Day 1, P1 | 10 points

Let A and B be real symmetric matrices with all eigenvalues strictly greater than 1. Let λ be a real eigenvalue of matrix AB. Prove that $|\lambda|>1$.

(Proposed by Pavel Kozhevnikov, MIPT, Moscow)

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Theory and methods for 2013-D1-P1

Theory Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Operator norms and convergence of matrix powers [T-LA-NORM]
Methods Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]; Combine norm bounds with algebraic decay [M-LA-NORM]
Strategy Build a chain of bounds and track equality [S-BOUND]; Translate geometry into a controlled representation [S-GEOMETRIC]

Statement Recorded solutions Catalog

2013-D1-P1 — Solution

For this record: Theory: T-LA-SPECTRAL; T-LA-NORM. Methods: M-LA-SPECTRAL; M-LA-NORM. Full learning route

The transforms given by A and B strictly increase the length of every nonzero vector, this can be seen easily in a basis where the matrix is diagonal with entries greater than 1 in the diagonal. Hence their product AB also strictly increases the length of any nonzero vector, and therefore its real eigenvalues are all greater than 1 or less than -1.

2013-D1-P2 — Statement

Day 1, P2 | 10 points

Let $f : \mathbb{R} \to \mathbb{R}$ be a twice differentiable function. Suppose $f(0) = 0$. Prove that there exists $\xi \in (-\pi/2, \pi/2)$ such that

$$f''(\xi) = f(\xi)(1 + 2\tan^2 \xi).$$

(Proposed by Karen Keryan, Yerevan State University, Yerevan, Armenia)

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Theory and methods for 2013-D1-P2

Theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Trigonometric identities and safe branches [T-AN-TRIG]
Methods Count zeros and subtract an interpolant [M-AN-ROLLE]; Design an auxiliary derivative [M-AN-AUXILIARY]
Strategy Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2013-D1-P2 — Solution

For this record: Theory: T-AN-DERIVATIVE; T-AN-TRIG. Methods: M-AN-ROLLE; M-AN-AUXILIARY. Full learning route

Let $g(x) = f(x)\cos x$. Since $g(-\pi/2) = g(0) = g(\pi/2) = 0$, by Rolle's theorem there exist some $\xi_1 \in (-\pi/2, 0)$ and $\xi_2 \in (0, \pi/2)$ such that

$$g'(\xi_1) = g'(\xi_2) = 0.$$

Now consider the function

$$h(x) = \frac{g'(x)}{\cos^2 x} = \frac{f'(x)\cos x - f(x)\sin x}{\cos^2 x}.$$

2013-D1-P2 — Solution (continued)

We have $h(\xi_1) = h(\xi_2) = 0$, so by Rolle's theorem there exist $\xi \in (\xi_1, \xi_2)$ for which

$$0 = h'(\xi) = \frac{g''(\xi)\cos^2\xi + 2\cos\xi\sin\xi\, g'(\xi)}{\cos^4\xi} =$$
$$= \frac{(f''(\xi)\cos\xi - 2f'(\xi)\sin\xi - f(\xi)\cos\xi)\cos\xi + 2\sin\xi(f'(\xi)\cos\xi - f(\xi)\sin\xi)}{\cos^3\xi} =$$
$$= \frac{f''(\xi)\cos^2\xi - f(\xi)(\cos^2\xi + 2\sin^2\xi)}{\cos^3\xi} = \frac{1}{\cos\xi}(f''(\xi) - f(\xi)(1 + 2\tan^2\xi)).$$

The last yields the desired equality.

2013-D1-P3 — Statement

Day 1, P3 | 10 points

There are $2n$ students in a school ($n \in \mathbb{N}$, $n \geq 2$). Each week n students go on a trip. After several trips the following condition was fulfilled: every two students were together on at least one trip. What is the minimum number of trips needed for this to happen?

(Proposed by Oleksandr Rybak, Kiev, Ukraine)

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Theory and methods for 2013-D1-P3

Theory Finite counting and incidence identities [T-DS-COUNTING]
Methods Count incidences with their exact multiplicity [M-DS-DOUBLECOUNT]; Split a family by traces, pieces, or labels [M-DS-PARTITION]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Build a witness and verify every constraint [S-CONSTRUCTION]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2013-D1-P3 — Solution

For this record: Theory: T-DS-COUNTING. Methods: M-DS-DOUBLECOUNT; M-DS-PARTITION. Full learning route

We prove that for any $n \geq 2$ the answer is 6.
First we show that less than 6 trips is not sufficient. In that case the total quantity of students in all trips would not exceed $5n$. A student meets $n-1$ other students in each trip, so he or she takes part on at least 3 excursions to meet all of his or her $2n-1$ schoolmates. Hence the total quantity of students during the trips is not less then $6n$ which is impossible.
Now let's build an example for 6 trips.
If n is even, we may divide $2n$ students into equal groups A, B, C, D. Then we may organize the trips with groups (A, B), (C, D), (A, C), (B, D), (A, D) and (B, C), respectively.
If n is odd and divisible by 3, we may divide all students into equal groups E, F, G, H, I, J. Then the members of trips may be the following: (E, F, G), (E, F, H), (G, H, I), (G, H, J), (E, I, J), (F, I, J).

In the remaining cases let $n = 2x+3y$ be, where x and y are natural numbers. Let's form the groups A, B, C, D of x students each, and E, F, G, H, I, J of y students each. Then we apply the previous cases and organize the following trips: (A, B, E, F, G), (C, D, E, F, H), (A, C, G, H, I), (B, D, G, H, J), (A, D, E, I, J), (B, C, F, I, J).

2013-D1-P4 — Statement

Day 1, P4 | 10 points

Let $n \geq 3$ and let $x_1, \dots, x_n$ be nonnegative real numbers. Define $A = \sum_{i=1}^{n} x_i$, $B = \sum_{i=1}^{n} x_i^2$ and $C = \sum_{i=1}^{n} x_i^3$. Prove that

$$(n+1)A^2B + (n-2)B^2 \geq A^4 + (2n-2)AC.$$

(Proposed by Géza Kós, Eötvös University, Budapest)

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Theory and methods for 2013-D1-P4

Theory Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]
Methods Translate root restrictions into symmetric coefficients [M-AL-VIETA]
Strategy Build a chain of bounds and track equality [S-BOUND]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2013-D1-P4 — Solution

For this record: Theory: T-AL-ROOTS; T-AN-DERIVATIVE; T-AN-INEQUALITIES. Methods: M-AL-VIETA. Full learning route

Let

$$\begin{aligned} p(X) &= \prod_{i=1}^{n}(X - x_i) \\ &= X^n - AX^{n-1} + \frac{A^2 - B}{2}X^{n-2} - \frac{A^3 - 3AB + 2C}{6}X^{n-3} + \dots . \end{aligned}$$

By repeated Rolle's theorem, the $(n-3)$th derivative of p has three nonnegative real roots $0 \leq u \leq v \leq w$. Hence,

$$\begin{aligned} \frac{6}{n!}\,p^{(n-3)}(X) &= X^3 - \frac{3A}{n}X^2 + \frac{3(A^2 - B)}{n(n-1)}X - \frac{A^3 - 3AB + 2C}{n(n-1)(n-2)} \\ &= (X-u)(X-v)(X-w). \end{aligned}$$

so

$$u + v + w = \frac{3A}{n}, \qquad uv + vw + wu = \frac{3(A^2 - B)}{n(n-1)},$$
$$\text{and} \qquad uvw = \frac{A^3 - 3AB + 2C}{n(n-1)(n-2)}.$$

From these we can see that

$$\begin{aligned} \frac{9}{n^2(n-1)^2(n-2)}&\left((n+1)A^2B + (n-2)B^2 - A^4 - (2n-2)AC\right) \\ &= (uv + vw + wu)^2 - 3uvw(u + v + w) \\ &= u^2v^2 + v^2w^2 + w^2u^2 - uvw(u + v + w) \\ &= uv(u-w)(v-w) + vw(v-u)(w-u) + wu(w-v)(u-v) \\ &\geq 0 + uw(v-u)(w-v) + wu(w-v)(u-v) = 0. \end{aligned}$$

2013-D1-P5 — Statement

Day 1, P5 | 10 points

Does there exist a sequence (a_n) of complex numbers such that for every positive integer p we have that $\sum_{n=1}^{\infty} a_n^p$ converges if and only if p is not a prime?

(Proposed by Tomáš Bárta, Charles University, Prague)

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Theory and methods for 2013-D1-P5

Theory Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Methods Translate root restrictions into symmetric coefficients [M-AL-VIETA]
Strategy Build a witness and verify every constraint [S-CONSTRUCTION]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2013-D1-P5 — Solution

For this record: Theory: T-AL-ROOTS; T-AL-POLYNOMIAL; T-AN-CONVERGENCE. Methods: M-AL-VIETA. Full learning route

The answer is YES. We prove a more general statement; suppose that $\mathbb{N} = C \cup D$ is an arbitrary decomposition of $\mathbb{N}$ into two disjoint sets. Then there exists a sequence $(a_n)_{n=1}^{\infty}$ such that $\sum_{n=1}^{\infty} a_n^p$ is convergent for $p \in C$ and divergent for $p \in D$.

Define $C_k = C \cap [1,k]$ and $D_k = D \cap [1,k]$.

Lemma. For every positive integer k there exists a positive integer N_k and a sequence $X_k = (x_{k,1}, \dots, x_{k,N_k})$ of complex numbers with the following properties:

(a) For $p \in D_k$, we have $\left|\sum_{j=1}^{N_k} x_{k,j}^p\right| \geq 1$.

(b) For $p \in C_k$, we have $\sum_{j=1}^{N_k} x_{k,j}^p = 0$; moreover, $\left|\sum_{j=1}^{m} x_{k,j}^p\right| \leq \frac{1}{k}$ holds for $1 \leq m \leq N_k$.

Proof. First we find some complex numbers $z_1, \dots, z_k$ with

$$\sum_{j=1}^{k} z_j^p = \begin{cases} 0 & p \in C_k \\ 1 & p \in D_k \end{cases} \tag{1}$$

As is well-known, this system of equations is equivalent to another system $\sigma_\nu(z_1, \dots, z_k) = w_\nu$ ($\nu = 1, 2, \dots, k$) where σ_ν is the νth elementary symmetric polynomial, and the constants w_ν are uniquely determined by the Newton–Waring–Girard formulas. Then the numbers $z_1, \dots, z_k$ are the roots of the polynomial $z^k - w_1 z^{k-1} + - \cdots + (-1)^k w_k$ in some order.

Now let

$$M = \left\lceil \max_{1 \leq m \leq k,\ p \in C_k} \left|\sum_{j=1}^{m} z_j^p\right| \right\rceil$$

and let $N_k = k \cdot (kM)^k$. We define the numbers $x_{k,1}, \dots, x_{k,N_k}$ by repeating the sequence $(\frac{z_1}{kM}, \frac{z_2}{kM}, \dots, \frac{z_k}{kM})$ $(kM)^k$ times, i.e. $x_{k,\ell} = \frac{z_j}{kM}$ if $\ell \equiv j \pmod k$. Then we have

$$\sum_{j=1}^{N_k} x_{k,j}^p = (kM)^k \sum_{j=1}^{k} \left(\frac{z_j}{kM}\right)^p = (kM)^{k-p} \sum_{j=1}^{k} z_j^p;$$

then from (1) the properties (a) and the first part of (b) follows immediately. For the second part of (b), suppose that $p \in C_k$ and $1 \leq m \leq N_k$; then $m = kr + s$ with some integers r and $1 \leq s \leq k$ and hence

$$\left|\sum_{j=1}^{m} x_{k,j}^p\right| = \left|\sum_{j=1}^{kr} + \sum_{j=kr+1}^{kr+s}\right| = \left|\sum_{j=1}^{s} \left(\frac{z_j}{kM}\right)^p\right| \leq \frac{M}{(kM)^p} \leq \frac{1}{k}.$$

2013-D1-P5 — Solution (continued)

The lemma is proved.
Now let $S_k = N_1 + \cdots + N_k$ (we also define $S_0 = 0$). Define the sequence (a_n) by simply concatenating the sequences $X_1, X_2, \dots$:

$$(a_1, a_2, \dots) = (x_{1,1}, \dots, x_{1,N_1}, x_{2,1}, \dots, x_{2,N_2}, \dots, x_{k,1}, \dots, x_{k,N_k}, \dots);$$

$$a_{S_k+j} = x_{k+1,j} \quad (1 \le j \le N_{k+1}).$$

If $p \in D$ and $k \ge p$ then

$$\left| \sum_{j=S_k+1}^{S_{k+1}} a_j^p \right| = \left| \sum_{j=1}^{N_{k+1}} x_{k+1,j}^p \right| \ge 1;$$

By Cauchy's convergence criterion it follows that $\sum a_n^p$ is divergent.

If $p \in C$ and $S_{u-1} < n \le S_u$ with some $u \ge p$ then

$$\left| \sum_{n=S_p+1}^{n} a_n^p \right| = \left| \sum_{k=p+1}^{u-1} \sum_{j=1}^{N_k} x_{k,j}^p + \sum_{j=1}^{n-S_{u-1}} x_{u,j}^p \right| = \left| \sum_{j=1}^{n-S_{u-1}} x_{u,j}^p \right| \le \frac{1}{u}.$$

Then it follows that $\sum\limits_{n=S_p+1}^{\infty} a_n^p = 0$, and thus $\sum\limits_{n=1}^{\infty} a_n^p = 0$ is convergent.

Day 2

2013-D2-P1 — Statement

Day 2, P6 | 10 points

Let z be a complex number with $|z+1| > 2$. Prove that $|z^3+1| > 1$.

(Proposed by Walther Janous and Gerhard Kirchner, Innsbruck)

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Theory and methods for 2013-D2-P1

Theory Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]
Methods Compare coefficients after a structural substitution [M-AL-COEFFICIENT]
Strategy Build a chain of bounds and track equality [S-BOUND]; Translate geometry into a controlled representation [S-GEOMETRIC]

Statement Recorded solutions Catalog

2013-D2-P1 — Solution

For this record: Theory: T-AL-COMPLEX; T-AL-POLYNOMIAL; T-AN-INEQUALITIES. Methods: M-AL-COEFFICIENT. Full learning route

Since $z^3+1 = (z+1)(z^2-z+1)$, it suffices to prove that $|z^2-z+1| \ge \frac{1}{2}$.

Assume that $z+1 = re^{\varphi i}$, where $r = |z+1| > 2$, and $\varphi = \arg(z+1)$ is some real number. Then

$$z^2 - z + 1 = (re^{\varphi i} - 1)^2 - (re^{\varphi i} - 1) + 1 = r^2 e^{2\varphi i} - 3re^{\varphi i} + 3,$$

2013-D2-P1 — Solution (continued)

and

$$\begin{aligned}|z^2-z+1|^2 &= (r^2e^{2\varphi i}-3re^{\varphi i}+3)(r^2e^{-2\varphi i}-3re^{-\varphi i}+3) = \\ &= r^4+9r^2+9-(6r^3+18r)\cos\varphi+6r^2\cos 2\varphi = \\ &= r^4+9r^2+9-(6r^3+18r)\cos\varphi+6r^2(2\cos^2\varphi-1) = \\ &= 12\left(r\cos\varphi-\frac{r^2+3}{4}\right)^2+\frac{1}{4}(r^2-3)^2>0+\frac{1}{4}=\frac{1}{4}.\end{aligned}$$

This finishes the proof.

2013-D2-P2 — Statement

Day 2, P7 | 10 points

Let p and q be relatively prime positive integers. Prove that

$$\sum_{k=0}^{pq-1}(-1)^{\lfloor\frac{k}{p}\rfloor+\lfloor\frac{k}{q}\rfloor}=\begin{cases}0 & \text{if } pq \text{ is even,}\\ 1 & \text{if } pq \text{ is odd.}\end{cases} \qquad (*)$$

(Here $\lfloor x\rfloor$ denotes the integer part of x.)

(Proposed by Alexander Bolbot, State University, Novosibirsk)

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Theory and methods for 2013-D2-P2

Theory Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Construct and count compatible local residues [M-AL-CRT]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2013-D2-P2 — Solution

For this record: Theory: T-AL-CONGRUENCE; T-DS-COUNTING. Methods: M-AL-CRT. Full learning route

Suppose first that pq is even (which implies that p and q have opposite parities), and let $a_k=(-1)^{\lfloor\frac{k}{p}\rfloor+\lfloor\frac{k}{q}\rfloor}$. We show that $a_k+a_{pq-1-k}=0$, so the terms on the left-and side of $(*)$ cancel out in pairs.
For every positive integer k we have $\left\{\frac{k}{p}\right\}+\left\{\frac{p-1-k}{p}\right\}=\frac{p-1}{p}$, hence

$$\left\lfloor\frac{k}{p}\right\rfloor+\left\lfloor\frac{pq-1-k}{p}\right\rfloor=\frac{k}{p}-\left\{\frac{k}{p}\right\}+\frac{pq-1-k}{p}-\left\{\frac{pq-1-k}{p}\right\}=\frac{pq-1}{p}-\frac{p-1}{p}=q-1$$

and similarly

$$\left\lfloor\frac{pq-1-k}{q}\right\rfloor+\left\lfloor\frac{k}{q}\right\rfloor=p-1.$$

Since p and q have opposite parities, it follows that $\lfloor\frac{k}{p}\rfloor+\lfloor\frac{k}{q}\rfloor$ and $\lfloor\frac{pq-1-k}{p}\rfloor+\lfloor\frac{pq-1-k}{q}\rfloor$ have opposite parities and therefore $a_{pq-1-k}=-a_k$.

Now suppose that pq is odd. For every index k, denote by p_k and q_k the remainders of k modulo p and q, respectively. (I.e., $0\le p_k<p$ and $0\le q_k<q$ such that $k\equiv p_k \pmod p$ and $k\equiv q_k \pmod q$.) Notice that

$$\left\lfloor\frac{k}{p}\right\rfloor+\left\lfloor\frac{k}{q}\right\rfloor\equiv p\left\lfloor\frac{k}{p}\right\rfloor+q\left\lfloor\frac{k}{q}\right\rfloor=(k-p_k)+(k-q_k)\equiv p_k+q_k \pmod 2.$$

2013-D2-P2 — Solution (continued)

Since p and q are co-prime, by the Chinese remainder theorem the map $k \mapsto (p_k, q_k)$ is a bijection between the sets $\{0, 1, \dots, pq-1\}$ and $\{0, 1, \dots, p-1\} \times \{0, 1, \dots, q-1\}$. Hence

$$\sum_{k=0}^{pq-1} (-1)^{\lfloor \frac{k}{p} \rfloor + \lfloor \frac{k}{q} \rfloor} = \sum_{k=0}^{pq-1} (-1)^{p_k+q_k} = \sum_{i=0}^{p-1} \sum_{j=0}^{q-1} (-1)^{i+j} = \left(\sum_{i=0}^{p-1} (-1)^i \right) \cdot \left(\sum_{j=0}^{q-1} (-1)^j \right) = 1.$$

2013-D2-P3 — Statement

Day 2, P8 | 10 points

Suppose that $v_1, \dots, v_d$ are unit vectors in $\mathbb{R}^d$. Prove that there exists a unit vector u such that

$$|u \cdot v_i| \leq 1/\sqrt{d}$$

for $i = 1, 2, \dots, d$.
(Here $\cdot$ denotes the usual scalar product on $\mathbb{R}^d$.)

(Proposed by Tomasz Tkocz, University of Warwick)

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Theory and methods for 2013-D2-P3

Theory Dual bases, kernels and separating functionals [T-LA-DUALITY]; Inner products, orthogonality and polarization [T-LA-INNER]; Rank, dimension and exact linear constraints [T-LA-RANK]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Prove a dimension obstruction and attain it [M-LA-DIMENSION]; Test a linear map on a basis and normalize its values [M-LA-DUAL]
Strategy Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2013-D2-P3 — Solution

For this record: Theory: T-LA-DUALITY; T-LA-INNER; T-LA-RANK; T-DS-COUNTING. Methods: M-LA-DIMENSION; M-LA-DUAL. Full learning route

If $v_1, \dots, v_d$ are linearly dependent then we can simply take a unit vector u perpendicular to $\operatorname{span}(v_1, \dots, v_d)$. So assume that $v_1, \dots, v_d$ are linearly independent. Let $w_1, \dots, w_d$ be the dual basis of $(v_1, \dots, v_d)$, i.e.

$$w_i \cdot v_j = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad \text{for } 1 \leq i, j \leq d.$$

From $w_i \cdot v_i = 1$ we have $|w_i| \geq 1$.

For every sequence $\varepsilon = (\varepsilon_1, \dots, \varepsilon_d) \in \{+1, -1\}^d$ of signs define $u_\varepsilon = \sum_{i=1}^d \varepsilon_i w_i$. Then we have

$$|u_\varepsilon \cdot v_k| = \left| \sum_{i=1}^d \varepsilon_i (w_i \cdot v_k) \right| = \left| \sum_{i=1}^d \varepsilon_i \delta_{ik} \right| = |\varepsilon_k| = 1 \quad \text{for } k = 1, \dots, d.$$

Now estimate the average of $|u_\varepsilon|^2$.

$$\frac{1}{2^d} \sum_{\varepsilon \in \{+1,-1\}^n} |u_\varepsilon|^2 = \frac{1}{2^d} \sum_{\varepsilon \in \{+1,-1\}^n} \left(\sum_{i=1}^d \varepsilon_i w_i \right) \cdot \left(\sum_{j=1}^d \varepsilon_j w_j \right) =$$

$$= \sum_{i=1}^d \sum_{j=1}^d (w_i \cdot w_j) \left(\frac{1}{2^d} \sum_{\varepsilon \in \{+1,-1\}^n} \varepsilon_i \varepsilon_j \right) = \sum_{i=1}^d \sum_{j=1}^d (w_i \cdot w_j) \delta_{ij} = \sum_{i=1}^d |w_i|^2 \geq d.$$

It follows that there is a ε such that $|u_\varepsilon|^2 \geq d$. For that ε the vector $u = \frac{u_\varepsilon}{|u_\varepsilon|}$ satisfies the conditions.

2013-D2-P4 — Statement

Day 2, P9 | 10 points

Does there exist an infinite set M consisting of positive integers such that for any $a, b \in M$, with $a < b$, the sum $a + b$ is square-free?
(A positive integer is called square-free if no perfect square greater than 1 divides it.)

(Proposed by Fedor Petrov, St. Petersburg State University)

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Theory and methods for 2013-D2-P4

Theory Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Prime valuations, divisors, and factorials [T-AL-VALUATION]; Finite counting and incidence identities [T-DS-COUNTING]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Methods Construct by excluding a sparse set of bad choices [M-AL-SIEVE]
Strategy Build a witness and verify every constraint [S-CONSTRUCTION]; Reduce size while preserving the defining conditions [S-INDUCTION]

Statement Recorded solutions Catalog

2013-D2-P4 — Solution

For this record: Theory: T-AL-CONGRUENCE; T-AL-VALUATION; T-DS-COUNTING; T-AN-CONVERGENCE. Methods: M-AL-SIEVE. Full learning route

The answer is yes. We construct an infinite sequence $1 = n_1 < 2 = n_2 < n_3 < \ldots$ so that $n_i + n_j$ is square-free for all $i < j$. Suppose that we already have some numbers $n_1 < \cdots < n_k$ ($k \geq 2$), which satisfy this condition and find a suitable number n_{k+1} to be the next element of the sequence.
We will choose n_{k+1} of the form $n_{k+1} = 1 + Mx$, with $M = ((n_1 + \cdots + n_k + 2k)!)^2$ and some positive integer x. For $i = 1, 2, \ldots, k$ we have $n_i + n_{k+1} = 1 + Mx + n_i = (1 + n_i)m_i$, where m_i and M are co-prime, so any perfect square dividing $1 + Mx + n_i$ is co-prime with M.
In order to find a suitable x, take a large N and consider the values $x = 1, 2, \ldots, N$. If a value $1 \leq x \leq N$ is not suitable, this means that there is an index $1 \leq i \leq k$ and some prime p such that $p^2 \mid 1 + Mx + n_i$. For $p \leq 2k$ this is impossible because $p \mid M$. Moreover, we also have $p^2 \leq 1 + Mx + n_i < M(N + 1)$, so $2k < p < \sqrt{M(N+1)}$.
For any fixed i and p, the values for x for which $p^2 \mid 1 + Mx + n_i$ form an arithmetic progression with difference p^2. Therefore, there are at most $\frac{N}{p^2} + 1$ such values. In total, the number of unsuitable values x is less than

$$\sum_{i=1}^{k} \sum_{2k<p<\sqrt{M(N+1)}} \left(\frac{N}{p^2} + 1\right) < k\left(N \cdot \sum_{p>2k} \frac{1}{p^2} + \sum_{p<\sqrt{M(N+1)}} 1\right) <$$

$$< kN \sum_{p>2k} \frac{1}{(p-1)p} + k\sqrt{M(N+1)} < \frac{N}{2} + k\sqrt{M(N+1)}.$$

If N is big enough then this is less than N, and there exist

a suitable choice for x.

2013-D2-P5 — Statement

Day 2, P10 | 10 points

Consider a circular necklace with 2013 beads. Each bead can be painted either white or green. A painting of the necklace is called good, if among any 21 successive beads there is at least one green bead. Prove that the number of good paintings of the necklace is odd.
(Two paintings that differ on some beads, but can be obtained from each other by rotating or flipping the necklace, are counted as different paintings.)

(Proposed by Vsevolod Bykov and Oleksandr Rybak, Kiev)

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Theory and methods for 2013-D2-P5

Theory Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Finite counting and incidence identities [T-DS-COUNTING]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Methods Strengthen the statement before the induction step [M-DS-INDUCT]; Split a family by traces, pieces, or labels [M-DS-PARTITION]; Pair objects canonically to prove parity [M-DS-INVOLUTION]; Partition into states and propagate a recurrence [M-DS-STATE]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2013-D2-P5 — Solution — Record 1 of 2

For this record: Theory: T-DS-INDUCTION; T-DS-COUNTING; T-AL-CONGRUENCE. Methods: M-DS-INDUCT; M-DS-PARTITION; M-DS-STATE. Full learning route

For $k = 0, 1, \ldots$ let N_k denote the number of good open laces, consisting of k (white and green) beads in a row, such that among any 21 successive beads there is at least one green bead. For $k \le 20$ all laces have this property, so $N_k = 2^k$ for $0 \le k \le 20$; in particular, N_0 is odd, and $N_1, \ldots, N_{20}$ are even.
For $k \ge 21$, there must be a green bead among the last 21 ones. Suppose that the last green bead is at the ℓth position; then $\ell \ge k-20$. The previous $\ell - 1$ beads have $N_{\ell-1}$ good colorings, and every such good coloring provides a good lace of length k. Hence,

$$N_k = N_{k-1} + N_{k-2} + \cdots + N_{k-21} \quad \text{for } k \ge 21. \tag{1}$$

From (1) we can see that $N_{21} = N_0 + \cdots + N_{20}$ is odd, and $N_{22} = N_1 + \cdots + N_{21}$ is also odd.
For $k \ge 22$, applying (1) again to the term N_{k-1},

$$\begin{aligned} N_k &= N_{k-1} + \cdots + N_{k-21} \\ &= \left(N_{k-2} + \cdots + N_{k-22}\right) + N_{k-2} + \cdots + N_{k-21} \\ &\equiv N_{k-22} \pmod 2. \end{aligned}$$

so the sequence of parities in (N_k) is periodic with period 22. We conclude that

- N_k is odd if $k \equiv 0 \pmod{22}$ or $k \equiv 21 \pmod{22}$;
- N_k is even otherwise.

Now consider the good circular necklaces of 2013 beads. At a fixed point between two beads cut each. The resulting open lace may have some consecutive white beads at the two ends, altogether at most 20. Suppose that there are x white beads at the beginning and y white beads at the end; then we have $x, y \ge 0$ and $x + y \le 20$, and we have a good open lace in the middle, between the first and the last green beads. That middle lace consist of $2011 - x - y$ beads. So, for any fixed values of x and y the number of such cases is $N_{2011-x-y}$.

It is easy to see that from such a good open lace we can reconstruct the original circular lace. Therefore, the number of good circular necklaces is

$$\begin{aligned} \sum_{x+y \le 20} N_{2011-x-y} &= N_{2011} + 2N_{2010} + 3N_{2009} + \cdots + 21N_{1991} \\ &\equiv N_{2011} + N_{2009} + N_{2007} + \cdots + N_{1991} \pmod 2. \end{aligned}$$

By $91 \cdot 22 - 1 = 2001$ the term N_{2001} is odd, the other terms are all even, so the number of the good circular necklaces is odd.

2013-D2-P5 — Solution — Record 2 of 2

For this record: Theory: T-DS-COUNTING. Methods: M-DS-INVOLUTION; M-DS-PARTITION. Full learning route

(by Yoav Krauz, Israel) There is just one good monochromatic necklace. Let us count the parity of good necklaces having both colors.

2013-D2-P5 — Solution — Record 2 of 2 (continued)

For each necklace, we define an adjusted necklace, so that at position 0 we have a white bead and at position 1 we have a green bead. If the necklace is satisfying the condition, it corresponds to itself; if both beads 0 and 1 are white we rotate it (so that the bead 1 goes to place 0) until bead 1 becomes green; if bead 1 is green, we rotate it in the opposite direction until the bead 0 will be white. This procedure is called adjusting, and the place between the white and green bead which are rotated into places 0 and 1 will be called distinguished place. The interval consisting of the subsequent green beads after the distinguished place and subsequent white beads before it will be called distinguished interval.
For each adjusted necklace we have several necklaces corresponding to it, and the number of them is equal to the length of distinguished interval (the total number of beads in it) minus 1. Since we count only the parity, we can disregard the adjusted necklaces with even distinguished intervals and count once each adjusted necklace with odd distinguished interval.
Now we shall prove that the number of necklaces with odd distinguished intervals is even by grouping them in pairs. The pairing is the following. If the number of white beads with in the distinguished interval is even, we turn the last white bead (at the distinguished place) into green. The white interval remains, since a positive even number minus 1 is still positive. If the number of white beads in the distinguished interval is odd, we turn the green bead next to the distinguished place into white. The green interval remains since it was even; the white interval was odd and at most 19 so it will become even and at most 20, so we still get a good necklace.

This pairing on good necklaces with distinguished intervals of odd length shows, that the number of such necklaces is even; hence the total number of all good necklaces using both colors is even. Therefore, together with monochromatic green necklace, the number of good necklaces is odd.

3.21 2014

Day 1

2014-D1-P1 — Statement

Day 1, P1 | 10 points

Determine all pairs (a, b) of real numbers for which there exists a unique symmetric 2×2 matrix M with real entries satisfying $\text{trace}(M) = a$ and $\det(M) = b$.

(Proposed by Stephan Wagner, Stellenbosch University)

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Theory and methods for 2014-D1-P1

Theory Determinants, block identities and inverses [T-LA-DETERMINANT]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]
Methods Read the entries before choosing a theorem [M-LA-ENTRY]; Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]

Statement Recorded solutions Catalog

2014-D1-P1 — Solution — Record 1 of 2

For this record: Theory: T-LA-DETERMINANT. Methods: M-LA-ENTRY. Full learning route

Let the matrix be

$$M = \begin{bmatrix} x & z \\ z & y \end{bmatrix}.$$

The two conditions give us $x + y = a$ and $xy - z^2 = b$. Since this is symmetric in x and y, the matrix can only be unique if $x = y$. Hence $2x = a$ and $x^2 - z^2 = b$. Moreover, if (x, y, z) solves the system of

2014-D1-P1 — Solution — Record 1 of 2 (continued)

equations, so does $(x, y, -z)$. So M can only be unique if $z = 0$. This means that $2x = a$ and $x^2 = b$, so $a^2 = 4b$.

If this is the case, then M is indeed unique: if $x + y = a$ and $xy - z^2 = b$, then

$$(x - y)^2 + 4z^2 = (x + y)^2 + 4z^2 - 4xy = a^2 - 4b = 0,$$

so we must have $x = y$ and $z = 0$, meaning that

$$M = \begin{bmatrix} a/2 & 0 \\ 0 & a/2 \end{bmatrix}$$

is the only solution.

2014-D1-P1 — Solution — Record 2 of 2

For this record: Theory: T-LA-SPECTRAL; T-LA-POLYNOMIAL; T-LA-DETERMINANT. Methods: M-LA-SPECTRAL. Full learning route

Note that $\operatorname{trace}(M) = a$ and $\det(M) = b$ if and only if the two eigenvalues λ_1 and λ_2 of M are solutions of $x^2 - ax + b = 0$. If $\lambda_1 \neq \lambda_2$, then

$$M_1 = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \quad \text{and} \quad M_2 = \begin{bmatrix} \lambda_2 & 0 \\ 0 & \lambda_1 \end{bmatrix}$$

are two distinct solutions, contradicting uniqueness. Thus $\lambda_1 = \lambda_2 = \lambda = a/2$, which implies $a^2 = 4b$ once again. In this case, we use the fact that M has to be diagonalisable as it is assumed to be symmetric. Thus there exists a matrix T such that

$$M = T^{-1} \cdot \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \cdot T,$$

however this reduces to $M = \lambda(T^{-1} \cdot I \cdot T) = \lambda I$, which shows again that M is unique.

2014-D1-P2 — Statement

Day 1, P2 | 10 points

Consider the following sequence

$$(a_n)_{n=1}^{\infty} = (1, 1, 2, 1, 2, 3, 1, 2, 3, 4, 1, 2, 3, 4, 5, 1, \dots).$$

Find all pairs (α, β) of positive real numbers such that $\lim_{n\to\infty} \dfrac{\sum_{k=1}^{n} a_k}{n^{\alpha}} = \beta$.

(Proposed by Tomas Barta, Charles University, Prague)

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Theory and methods for 2014-D1-P2

Theory Scale separation and limit comparisons [T-AN-ASYMPTOTICS]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Normalize, trap, and identify a recurrence [M-AN-RECURRENCE]; Separate a main range from exceptional scales [M-AN-SPLIT]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2014-D1-P2 — Solution

For this record: Theory: T-AN-ASYMPTOTICS; T-AN-CONVERGENCE; T-DS-COUNTING. Methods: M-AN-RECURRENCE; M-AN-SPLIT. Full learning route

Let $N_n = \binom{n+1}{2}$ (then a_{N_n} is the first appearance of number n in the sequence) and consider limit of the subsequence

$$b_{N_n} := \frac{\sum_{k=1}^{N_n} a_k}{N_n^\alpha} = \frac{\sum_{k=1}^{n}(1+\cdots+k)}{\binom{n+1}{2}^\alpha} = \frac{\sum_{k=1}^{n}\binom{k+1}{2}}{\binom{n+1}{2}^\alpha} = \frac{\binom{n+2}{3}}{\binom{n+1}{2}^\alpha} = \frac{\frac{1}{6}n^3(1+2/n)(1+1/n)}{(1/2)^\alpha n^{2\alpha}(1+1/n)^\alpha}.$$

We can see that $\lim_{n\to\infty} b_{N_n}$ is positive and finite if and only if $\alpha = 3/2$. In this case the limit is equal to $\beta = \frac{\sqrt{2}}{3}$. So, this pair $(\alpha,\beta) = (\frac{3}{2}, \frac{\sqrt{2}}{3})$ is the only candidate for solution.

We will show convergence of the original sequence for these values of α and β. Let N be a positive integer in $[N_n+1, N_{n+1}]$, i.e., $N = N_n + m$ for some $1 \le m \le n+1$. Then we have

$$b_N = \frac{\binom{n+2}{3} + \binom{m+1}{2}}{\left(\binom{n+1}{2} + m\right)^{3/2}}$$

which can be estimated by

$$\frac{\binom{n+2}{3}}{\left(\binom{n+1}{2} + n\right)^{3/2}} \le b_N \le \frac{\binom{n+2}{3} + \binom{n+1}{2}}{\binom{n+1}{2}^{3/2}}.$$

Since both bounds converge to $\frac{\sqrt{2}}{3}$, the sequence b_N has the same limit and we are done.

2014-D1-P3 — Statement

Day 1, P3 | 10 points

Let n be a positive integer. Show that there are positive real numbers $a_0, a_1, \ldots, a_n$ such that for each choice of signs the polynomial

$$\pm a_n x^n \pm a_{n-1}x^{n-1} \pm \cdots \pm a_1 x \pm a_0$$

has n distinct real roots.

(Proposed by Stephan Neupert, TUM, München)

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Theory and methods for 2014-D1-P3

Theory Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Continuity, compactness, and order [T-AN-CONTINUITY]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Methods Compare coefficients after a structural substitution [M-AL-COEFFICIENT]
Strategy Build a witness and verify every constraint [S-CONSTRUCTION]; Reduce size while preserving the defining conditions [S-INDUCTION]

Statement Recorded solutions Catalog

2014-D1-P3 — Solution

For this record: Theory: T-AL-POLYNOMIAL; T-AN-CONTINUITY; T-AN-DERIVATIVE; T-DS-INDUCTION. Methods: M-AL-COEFFICIENT. Full learning route

We proceed by induction on n. The statement is trivial for $n = 1$. Thus assume that we have some $a_0, \ldots, a_n$ which satisfy the conditions for some n. Consider now the polynomials

$$\tilde{P}(x) = \pm a_n x^{n+1} \pm a_{n-1}x^n \pm \cdots \pm a_1 x^2 \pm a_0 x$$

By induction hypothesis and $a_0 \neq 0$, each of these polynomials has $n+1$ distinct zeros, including the nonzero roots of $\pm a_n x^n \pm a_{n-1}x^{n-1} \pm \cdots \pm a_1 x \pm a_0$ and 0. In particular none of the polynomials

2014-D1-P3 — Solution (continued)

has a root which is a local extremum. Hence we can choose some $\varepsilon > 0$ such that for each such polynomial $\tilde{P}(x)$ and each s of its local extrema we have $|\tilde{P}(s)| > \varepsilon$. We claim that then each of the polynomials

$$P(x) = \pm a_n x^{n+1} \pm a_{n-1}x^n \pm \cdots \pm a_1 x^2 \pm a_0 x \pm \varepsilon$$

has exactly $n+1$ distinct zeros as well. As $\tilde{P}(x)$ has $n+1$ distinct zeros, it admits a local extremum at n points. Call these local extrema $-\infty = s_0 < s_1 < s_2 < \cdots < s_n < s_{n+1} = \infty$. Then for each $i \in \{0, 1, \ldots, n\}$ the values $\tilde{P}(s_i)$ and $\tilde{P}(s_{i+1})$ have opposite signs (with the obvious convention at infinity). By choice of ε the same holds true for $P(s_i)$ and $P(s_{i+1})$. Hence there is at least one real zero of $P(x)$ in each interval (s_i, s_{i+1}), i.e. $P(x)$ has at least (and therefore exactly) $n+1$ zeros. This shows that we have found a set of positive reals $a'_{n+1} = a_n, a'_n = a_{n-1}, \ldots, a'_1 = a_0, a'_0 = \varepsilon$ with the desired properties.

2014-D1-P4 — Statement

Day 1, P4 | 10 points

Let $n > 6$ be a perfect number, and let $n = p_1^{e_1} \cdots p_k^{e_k}$ be its prime factorisation with $1 < p_1 < \cdots < p_k$. Prove that e_1 is an even number.
A number n is perfect if $s(n) = 2n$, where $s(n)$ is the sum of the divisors of n.

(Proposed by Javier Rodrigo, Universidad Pontificia Comillas)

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Theory and methods for 2014-D1-P4

Theory Prime valuations, divisors, and factorials [T-AL-VALUATION]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Methods Track one prime or one extreme divisor [M-AL-VALUATION]
Strategy Choose an extremal object and force rigidity [S-EXTREMAL]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

2014-D1-P4 — Solution

For this record: Theory: T-AL-VALUATION; T-AL-CONGRUENCE. Methods: M-AL-VALUATION. Full learning route

Suppose that e_1 is odd, contrary to the statement.

We know that $s(n) = \prod_{i=1}^{k}(1 + p_i + p_i^2 + \cdots + p_i^{e_i}) = 2n = 2p_1^{e_1} \cdots p_k^{e_k}$. Since e_1 is an odd number, $p_1 + 1$ divides the first factor $1 + p_1 + p_1^2 + \cdots + p_1^{e_1}$, so $p_1 + 1$ divides $2n$. Due to $p_1 + 1 > 2$, at least one of the primes $p_1, \ldots, p_k$ divides $p_1 + 1$. The primes $p_3, \ldots, p_k$ are greater than $p_1 + 1$ and p_1 cannot divide $p_1 + 1$, so p_2 must divide $p_1 + 1$. Since $p_1 + 1 < 2p_2$, this possible only if $p_2 = p_1 + 1$, therefore $p_1 = 2$ and $p_2 = 3$. Hence, $6|n$.
Now n, $\frac{n}{2}$, $\frac{n}{3}$, $\frac{n}{6}$ and 1 are distinct divisors of n, so

$$s(n) \geq n + \frac{n}{2} + \frac{n}{3} + \frac{n}{6} + 1 = 2n + 1 > 2n,$$

contradiction.

Remark. It is well-known that all even perfect numbers have the form $n = 2^{p-1}(2^p - 1)$ such that p and $2^p - 1$ are primes. So if e_1 is odd then $k = 2$, $p_1 = 2$, $p_2 = 2^p - 1$, $e_1 = p - 1$, and $e_2 = 1$. If $n > 6$ then $p > 2$ so p is odd and $e_1 = p - 1$ should be even.

2014-D1-P5 — Statement

Day 1, P5 | 10 points

Let $A_1A_2\ldots A_{3n}$ be a closed broken line consisting of $3n$ line segments in the Euclidean plane. Suppose that no three of its vertices are collinear, and for each index $i = 1, 2, \ldots, 3n$, the triangle $A_iA_{i+1}A_{i+2}$ has counterclockwise orientation and $\angle A_iA_{i+1}A_{i+2} = 60^\circ$, using the notation $A_{3n+1} = A_1$ and $A_{3n+2} = A_2$. Prove that the number of self-intersections of the broken line is at most $\frac{3}{2}n^2 - 2n + 1$.

(Proposed by Martin Langer)

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Theory and methods for 2014-D1-P5

Theory Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Finite counting and incidence identities [T-DS-COUNTING]; Extremal choice, exchange arguments, and invariants [T-DS-EXTREMAL]
Methods Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]; Choose information that makes an encoding injective [M-DS-ENCODE]; Repair an extremal configuration by a local exchange [M-DS-EXTREMAL]
Strategy Translate geometry into a controlled representation [S-GEOMETRIC]; Choose an extremal object and force rigidity [S-EXTREMAL]; Choose the objects and count them without ambiguity [S-COUNTING]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

2014-D1-P5 — Solution

For this record: Theory: T-DS-GEOMETRY; T-DS-COUNTING; T-DS-EXTREMAL. Methods: M-DS-GEOMETRIC; M-DS-ENCODE; M-DS-EXTREMAL. Full learning route

Place the broken line inside an equilateral triangle T such that their sides are parallel to the segments of the broken line. For every $i = 1, 2, \ldots, 3n$, denote by x_i the distance between the segment A_iA_{i+1} and that side of T which is parallel to A_iA_{i+1}. We will use indices modulo $3n$ everywhere.

Choose T so that its counterclockwise side directions agree with the successive segment directions. For distinct $i \equiv j \pmod 3$, we have $x_i \neq x_j$: coincident supporting lines would contain at least three vertices. In the affine coordinates given by distances to the two relevant sides of T, the polyline $A_iA_{i+1}A_{i+2}$ lies on the two positive coordinate rays from (x_i, x_{i+1}). Two such polylines intersect at most once, and only when their corner coordinates have opposite orders. Every crossing of original segments belongs to one such comparison, since their nonparallel directions determine the starting residue class. Consequently, the number of self-intersections is at most the number of unordered pairs $\{i, j\}$ satisfying

$$(*) \qquad i \equiv j \pmod 3, \text{ and } (x_i < x_j \text{ and } x_{i+1} > x_{j+1}) \text{ or } (x_i > x_j \text{ and } x_{i+1} < x_{j+1}).$$

Grouping the indices $1, 2, \ldots, 3n$ by remainders modulo 3, we have n indices in each residue class. Altogether there are $3\binom{n}{2}$ index pairs (i, j) with $i \equiv j \pmod 3$. We will show that for every integer k with $1 \le k < \frac{n}{2}$, there is some index i such that the pair $(i, i + 6k)$ does not satisfy $(*)$. We first explain why this supplies at least $\lfloor (n-1)/2 \rfloor$ distinct missing unordered pairs.

For odd n, the displacements $6k$ give distinct classes up to sign modulo $3n$. For even n, the only duplication pairs k with $n/2 - k$. Fix $d = 6k$ and put $s_i = \operatorname{sgn}(x_i - x_{i+d}) \in \{-1, 1\}$. The pair $\{i, i + d\}$ fails $(*)$ exactly when $s_i = s_{i+1}$. A cyclic sequence of $3n$ signs has an even number of sign changes; when n is even, its number of nonchanges is therefore also even. The existence of one failure, proved below, thus gives at least two. If $2d \not\equiv 0 \pmod{3n}$, different failing indices give different unordered pairs, so these two failures account for both duplicated values of k. If $2d \equiv 0 \pmod{3n}$, then $k = n/4$ is its own partner and only one missing pair is needed. Distinct displacement classes cannot share an unordered pair. This proves the required count, and hence there are at most

$$3\binom{n}{2} - \left[\frac{n-1}{2}\right] \le \frac{3}{2}n^2 - 2n + 1$$

self-intersections.

2014-D1-P5 — Solution (continued)

Without loss of generality we may assume that $x_{3n} = x_0$ is the smallest among $x_1, \dots, x_{3n}$. Suppose that all of the pairs

$$(**) \qquad (-6k, 0),\ (-6k+1, 1),\ (-6k+2, 2),\ \dots,\ (-1, 6k-1),\ (0, 6k)$$

satisfy $(*)$. Since x_0 is minimal, we have $x_{-6k} > x_0$. The pair $(-6k, 0)$ satisfies $(*)$, so $x_{-6k+1} < x_1$. Then we can see that $x_{-6k+2} > x_2$, and so on; finally we get $x_0 > x_{6k}$. But this contradicts the minimality of x_0. Therefore, there is a pair in $(**)$ that does not satisfy $(*)$.

Remark. The bound $3\binom{n}{2} - \left[\frac{n-1}{2}\right] = \left[\frac{3n^2}{2} - 2n + 1\right]$ is sharp.

Day 2

2014-D2-P1 — Statement

Day 2, P6 | 10 points

For a positive integer x, denote its nth decimal digit by $d_n(x)$, i.e. $d_n(x) \in \{0, 1, \dots, 9\}$ and $x = \sum_{n=1}^{\infty} d_n(x) 10^{n-1}$. Suppose that for some sequence $(a_n)_{n=1}^{\infty}$, there are only finitely many zeros in the sequence $(d_n(a_n))_{n=1}^{\infty}$. Prove that there are infinitely many positive integers that do not occur in the sequence $(a_n)_{n=1}^{\infty}$.

(Proposed by Alexander Bolbot, State University, Novosibirsk)

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Theory and methods for 2014-D2-P1

Theory Finite counting and incidence identities [T-DS-COUNTING]; Countability, accumulation, and Cantor constructions [T-AN-SETS]
Methods Choose information that makes an encoding injective [M-DS-ENCODE]; Encode, decode, and verify a bijection [M-DS-BIJECTION]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Build a witness and verify every constraint [S-CONSTRUCTION]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

2014-D2-P1 — Solution — Record 1 of 2

For this record: Theory: T-DS-COUNTING. Methods: M-DS-ENCODE. Full learning route

By the assumption there is some index n_0 such that $d_n(a_n) \neq 0$ for $n \geq n_0$. We show that

$$a_{n+1}, a_{n+2}, \dots \geq 10^n \quad \text{for } n \geq n_0. \tag{1}$$

Notice that in the sum $a_n = \sum_{k=1}^{\infty} d_k(a_n) 10^{k-1}$ we have the term $d_n(a_n) 10^{n-1}$ with $d_n(a_n) \geq 1$. Therefore, $a_n \geq 10^{n-1}$. Then for $m > n$ we have $a_m \geq 10^{m-1} \geq 10^n$. This proves (1).

From (1) we know that only the first n elements, $a_1, a_2, \dots, a_n$ may lie in the interval $[1, 10^n - 1]$. Hence, at least $10^n - 1 - n$ integers in this interval do not occur in the sequence at all. As $\lim(10^n - 1 - n) = \infty$, this shows that there are infinitely many numbers that do not appear among $a_1, a_2, \dots$.

2014-D2-P1 — Solution — Record 2 of 2

For this record: Theory: T-AN-SETS; T-DS-COUNTING. Methods: M-DS-ENCODE; M-DS-BIJECTION. Full learning route

We will use Cantor's diagonal method to construct infinitely many positive integers that do not occur in the sequence (a_n).

2014-D2-P1 — Solution — Record 2 of 2 (continued)

Assume that $d_n(a_n) \neq 0$ for $n > n_0$. Define the sequence of digits

$$g_n = \begin{cases} 2 & d_n(a_n) = 1 \\ 1 & d_n(a_n) \neq 1. \end{cases}$$

Hence $g_n \neq d_n(a_n)$ for every positive integer n. Let

$$x_k = \sum_{n=1}^{k} g_n \cdot 10^{n-1} \quad \text{for } k = 1, 2, \ldots .$$

As $x_{k+1} \geq 10^k > x_k$, the sequence (x_k) is increasing and so it contains infinitely many distinct positive integers. We show that the numbers $x_{n_0}, x_{n_0+1}, x_{n_0+2}, \ldots$ no not occur in the sequence (a_n); in other words, $x_k \neq a_n$ for every pair $n \geq 1$ and $k \geq n_0$ of integers.
Indeed, if $k \geq n$ then $d_n(x_k) = g_n \neq d_n(a_n)$, so $x_k \neq a_n$.

If $n > k \geq n_0$ then $d_n(x_k) = 0 \neq d_n(a_n)$, so $x_k \neq a_n$.

2014-D2-P2 — Statement

Day 2, P7 | 10 points

Let $A = (a_{ij})_{i,j=1}^{n}$ be a symmetric $n \times n$ matrix with real entries, and let $\lambda_1, \lambda_2, \ldots, \lambda_n$ denote its eigenvalues. Show that

$$\sum_{1 \leq i < j \leq n} a_{ii} a_{jj} \geq \sum_{1 \leq i < j \leq n} \lambda_i \lambda_j,$$

and determine all matrices for which equality holds.

(Proposed by Martin Niepel, Comenius University, Bratislava)

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Theory and methods for 2014-D2-P2

Theory Trace pairing and commutator calculus [T-LA-TRACE]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Inner products, orthogonality and polarization [T-LA-INNER]
Methods Use trace to extract a scalar invariant [M-LA-TRACE]
Strategy Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2014-D2-P2 — Solution

For this record: Theory: T-LA-TRACE; T-LA-SPECTRAL; T-LA-INNER. Methods: M-LA-TRACE. Full learning route

Eigenvalues of a real symmetric matrix are real, hence the inequality makes sense. Similarly, for Hermitian matrices diagonal entries as well as eigenvalues have to be real.
Since the trace of a matrix is the sum of its eigenvalues, for A we have

$$\sum_{i=1}^{n} a_{ii} = \sum_{i=1}^{n} \lambda_i,$$

and consequently

$$\sum_{i=1}^{n} a_{ii}^2 + 2 \sum_{i<j} a_{ii} a_{jj} = \sum_{i=1}^{n} \lambda_i^2 + 2 \sum_{i<j} \lambda_i \lambda_j.$$

Therefore our inequality is equivalent to

$$\sum_{i=1}^{n} a_{ii}^2 \leq \sum_{i=1}^{n} \lambda_i^2.$$

2014-D2-P2 — Solution (continued)

Matrix A^2, which is equal to A^TA (or A^*A in Hermitian case), has eigenvalues $\lambda_1^2, \lambda_2^2, \dots, \lambda_n^2$. On the other hand, the trace of A^TA gives the square of the Frobenius norm of A, so we have

$$\sum_{i=1}^{n} a_{ii}^2 \le \sum_{i,j=1}^{n} |a_{ij}|^2 = \operatorname{tr}(A^TA) = \operatorname{tr}(A^2) = \sum_{i=1}^{n} \lambda_i^2.$$

The inequality follows, and it is clear that the equality holds for diagonal matrices only.

Remark. Same statement is true for Hermitian matrices.

2014-D2-P3 — Statement

Day 2, P8 | 10 points

Let $f(x) = \frac{\sin x}{x}$, for $x > 0$, and let n be a positive integer. Prove that $\left|f^{(n)}(x)\right| < \frac{1}{n+1}$, where $f^{(n)}$ denotes the nth derivative of f.

(Proposed by Alexander Bolbot, State University, Novosibirsk)

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Theory and methods for 2014-D2-P3

Theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Trigonometric identities and safe branches [T-AN-TRIG]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Methods Design an auxiliary derivative [M-AN-AUXILIARY]; Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]
Strategy Build a chain of bounds and track equality [S-BOUND]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2014-D2-P3 — Solution — Record 1 of 2

For this record: Theory: T-AN-DERIVATIVE; T-AN-TRIG; T-AN-INTEGRAL. Methods: M-AN-AUXILIARY. Full learning route

Putting $f(0) = 1$ we can assume that the function is analytic in $\mathbb{R}$. Let $g(x) = x^{n+1}\left(f^{(n)}(x) - \frac{1}{n+1}\right)$. Then $g(0) = 0$ and

$$g'(x) = (n+1)x^n\left(f^{(n)}(x) - \frac{1}{n+1}\right) + x^{n+1}f^{(n+1)}(x) =$$
$$= x^n\Big((n+1)f^{(n)}(x) + xf^{(n+1)}(x) - 1\Big) = x^n\big((xf(x))^{(n+1)} - 1\big) = x^n\big(\sin^{(n+1)}(x) - 1\big) \le 0.$$

Hence $g(x) \le 0$ for $x > 0$. Taking into account that $g'(x) < 0$ for $0 < x < \frac{\pi}{2}$ we obtain the strict upper bound for $x > 0$. To obtain the lower bound, set $h(x) = x^{n+1}\big(f^{(n)}(x) + 1/(n+1)\big)$. Then $h(0) = 0$ and

$$h'(x) = x^n\big(\sin^{(n+1)}(x) + 1\big) \ge 0,$$

with strict inequality for $0 < x < \pi/2$. Thus $h(x) > 0$ for $x > 0$, so $f^{(n)}(x) > -1/(n+1)$. Combining the two bounds gives the claimed absolute-value inequality.

2014-D2-P3 — Solution — Record 2 of 2

For this record: Theory: T-AN-INTEGRAL; T-AN-TRIG. Methods: M-AN-PRIMITIVE. Full learning route

$$\left(\frac{\sin x}{x}\right)^{(n)} = \frac{d^n}{dx^n}\int_0^1 \cos(xt)\,dt = \int_0^1 \frac{\partial^n}{\partial x^n}\cos(xt)\,dt = \int_0^1 t^n g_n(xt)\,dt$$

2014-D2-P3 — Solution — Record 2 of 2 (continued)

where the function $g_n(u)$ can be $\pm\sin u$ or $\pm\cos u$, depending on n. We only need that $|g_n| \le 1$ and equality occurs at finitely many points. So,

$$\left|\left(\frac{\sin x}{x}\right)^{(n)}\right| \le \int_0^1 t^n\,|g_n(xt)|\,dt < \int_0^1 t^n\,dt = \frac{1}{n+1}.$$

2014-D2-P4 — Statement

Day 2, P9 | 10 points

We say that a subset of $\mathbb{R}^n$ is k-almost contained by a hyperplane if there are less than k points in that set which do not belong to the hyperplane. We call a finite set of points k-generic if there is no hyperplane that k-almost contains the set. For each pair of positive integers k and n, find the minimal number $d(k,n)$ such that every finite k-generic set in $\mathbb{R}^n$ contains a k-generic subset with at most $d(k,n)$ elements.

(Proposed by Shachar Carmeli, Weizmann Inst. and Lev Radzivilovsky, Tel Aviv Univ.)

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Theory and methods for 2014-D2-P4

Theory Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Extremal choice, exchange arguments, and invariants [T-DS-EXTREMAL]; Rank, dimension and exact linear constraints [T-LA-RANK]
Methods Repair an extremal configuration by a local exchange [M-DS-EXTREMAL]; Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Strategy Choose an extremal object and force rigidity [S-EXTREMAL]; Translate geometry into a controlled representation [S-GEOMETRIC]; Build a witness and verify every constraint [S-CONSTRUCTION]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2014-D2-P4 — Solution

For this record: Theory: T-DS-GEOMETRY; T-DS-EXTREMAL; T-LA-RANK. Methods: M-DS-EXTREMAL; M-DS-GEOMETRIC. Full learning route

The answer is:

$$d(k,n) = \begin{cases} k\cdot n & k,n>1 \\ k+n & \text{otherwise} \end{cases}$$

Throughout the solution, we shall often say that a hyperplanes skips a point to signify that the plane does not contain that point.
For $n=1$ the claim is obvious.
For $k=1$ we have an arbitrary finite set of points in $\mathbb{R}^n$ such that neither hyperplane contains it entirely. We can build a subset of $n+1$ points step by step: on each step we add a point, not contained in the minimal plane spanned by the previous points. Thus any 1-generic set contains a non-degenerate simplex of $n+1$ points, and obviously a non-degenerate simplex of $n+1$ points cannot be reduced without loosing 1-generality.
In the case $k,n>1$ we shall give an example of $k\cdot n$ points. On each of the Cartesian axes choose k distinct points, different from the origin. Let's show that this set is k-generic. There are two types of planes: containing the origin and skipping it. If a plane contains the origin, it either contains all the chose points of a axis or skips all of them. Since no plane contains all axes, it skips the chosen k points on one of the axes. If a plane skips the origin, it it contains at most one point of each axis. Therefore it skips at least $n(k-1)$ points. It remains to verify a simple inequality $n(k-1)\ge k$ which is equivalent to $(n-1)(k-1)\ge 1$ which holds for $n,k>1$.
The example we have shown is minimal by inclusion: if any point is removed, say a point from axis i, then the hyperplane $x_i=0$ skips only $k-1$ points, and our set stops being k-generic. Hence $d(k,n)\ge kn$.
It remains to prove that $d(k,n)\le kn$ for $k,n>1$,

2014-D2-P4 — Solution (continued)

meaning: for each k-generic finite set of points, it is possible to choose a k-generic subset of at most kn points. Let us call a subset of points minimal if by taking out any point, we loose k-generality. It suffices to prove that any minimal k-generic subset in $\mathbb{R}^n$ has at most kn points. A hyperplane will be called ample if it skips precisely k points. A point cannot be removed from a k-generic set, if and only if it is skipped by an ample hyperplane. Thus, in a minimal set each point is skipped by an ample hyperplane.

Organise the following process: on each step we choose an ample hyperplane, and paint blue all the points which are skipped by it. Each time we choose an ample hyperplane, which skips one of the unpainted points. The unpainted points at each step (after the beginning) is the intersection of all chosen hyperplanes. The intersection set of chosen hyperplanes is reduced with each step (since at least one point is being painted on each step).

Notice, that on each step we paint at most k points. So if we start with a minimal set of more then nk points, we can choose n planes and still have at least one unpainted points. The intersection of the chosen planes is a point (since on each step the dimension of the intersection plane was reduced), so there are at most $nk+1$ points in the set. The last unpainted point will be denoted by O. The last unpainted line (which was formed on the step before the last) will be denoted by ℓ_1. This line is an intersection of all the chosen hyperplanes except the last one. If we have more than nk points, then ℓ_1 contains exactly $k+1$ points from the set, one of which is O.

We could have executed the same process with choosing the same hyperplanes, but in different order. Anyway, at each step we would paint at most k points, and after n steps only O would remain unpainted; so it was precisely k points on each step. On step before the last, we might get a different line, which is intersection of all planes except the last one. The lines obtained in this way will be denoted $\ell_1, \ell_2, \dots, \ell_n$, and each contains exactly k points except O. Since we have O and k points on n lines, that is the entire set. Notice that the vectors spanning these lines are linearly independent (since for each line we have a hyperplane containing all the other lines except that line). So by removing O we obtain the example that we've described already, which is k-generic.

Remark. From the proof we see, that the example is unique.

2014-D2-P5 — Statement

Day 2, P10 | 10 points

For every positive integer n, denote by D_n the number of permutations $(x_1, \dots, x_n)$ of $(1, 2, \dots, n)$ such that $x_j \neq j$ for every $1 \leq j \leq n$. For $1 \leq k \leq \frac{n}{2}$, denote by $\Delta(n, k)$ the number of permutations of $(1, 2, \dots, n)$ such that $x_i = k + i$ for every $1 \leq i \leq k$ and $x_j \neq j$ for every $1 \leq j \leq n$. Prove that

$$\Delta(n,k) = \sum_{i=0}^{k-1} \binom{k-1}{i} \frac{D_{(n+1)-(k+i)}}{n-(k+i)}.$$

(Proposed by Combinatorics; Ferdowsi University of Mashhad, Iran; Mirzavaziri)

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Theory and methods for 2014-D2-P5

Theory Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]

Methods Encode, decode, and verify a bijection [M-DS-BIJECTION]; Strengthen the statement before the induction step [M-DS-INDUCT]

Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Reduce size while preserving the defining conditions [S-INDUCTION]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2014-D2-P5 — Solution

For this record: Theory: T-DS-COUNTING; T-DS-INDUCTION; T-AL-PERMUTATION. Methods: M-DS-BIJECTION; M-DS-INDUCT. Full learning route

Write $[n] = \{1, \dots, n\}$ and set $D_0 = 1$. Choose distinct $i_1, \dots, i_k$ and distinct $a_1, \dots, a_k$ in $[n]$ such that $\{i_1, \dots, i_k\} \cap \{a_1, \dots, a_k\} = \emptyset$. Let $\Delta(n, k, 0)$ count the derangements x of $[n]$ satisfying $x_{i_j} = a_j$ for all j. Relabelling shows that this count depends only on n and k; choosing $i_j = j$ and $a_j = k + j$ gives precisely $\Delta(n, k)$ from the problem. Only this disjoint case is needed. We evaluate $\Delta(n, k, 0)$, where $2k \le n$. For $k = 0$, we obviously have $\Delta(n, 0, 0) = D_n$. For $k > 1$, we claim that

$$\Delta(n, k, 0) = \Delta(n-1, k-1, 0) + \Delta(n-2, k-1, 0).$$

For a derangement $x = (x_1, \dots, x_n)$ satisfying $x_{i_j} = a_j$ there are two cases: $x_{a_1} = i_1$ or $x_{a_1} \neq i_1$.
If the first case occurs then we have to evaluate the number of derangements of the set $[n] \setminus \{i_1, a_1\}$ for the given sets $\{i_2, \dots, i_k\}$ and $\{a_2, \dots, a_k\}$ with 0 elements in their intersections. The number is equal to $\Delta(n-2, k-1, 0)$.
If the second case occurs, delete a_1 from its permutation cycle, setting $x'_{i_1} = x_{a_1}$ and leaving the other arrows unchanged. Since $x_{a_1} \neq i_1$, no fixed point is created. Conversely, insert a_1 immediately after i_1 to recover x. Thus we count the derangements of the set $[n] \setminus \{a_1\}$ for the given sets $\{i_2, \dots, i_k\}$ and $\{a_2, \dots, a_k\}$ with 0 elements in their intersections. The number is equal to $\Delta(n-1, k-1, 0)$.
We now use induction on k to show that

$$\Delta(n, k, 0) = \sum_{i=0}^{k-1} \binom{k-1}{i} \frac{D_{(n+1)-(k+i)}}{n-(k+i)}, \qquad 2 \le 2k \le n.$$

For $k = 1$ we have

$$\Delta(n, 1, 0) = \Delta(n-1, 0, 0) + \Delta(n-2, 0, 0) = D_{n-1} + D_{n-2} = \frac{D_n}{n-1}.$$

Now let the result be true for $k-1$. We can write

$$\begin{aligned}
\Delta(n, k, 0) &= \Delta(n-1, k-1, 0) + \Delta(n-2, k-1, 0) \\
&= \sum_{i=0}^{k-2} \binom{k-2}{i} \frac{D_{n-(k-1+i)}}{(n-1)-(k-1+i)} + \sum_{i=0}^{k-2} \binom{k-2}{i} \frac{D_{(n-1)-(k-1+i)}}{(n-2)-(k-1+i)} \\
&= \sum_{i=0}^{k-2} \binom{k-2}{i} \frac{D_{(n+1)-(k+i)}}{n-(k+i)} + \sum_{i=1}^{k-1} \binom{k-2}{i-1} \frac{D_{n-(k+i-1)}}{(n-1)-(k+i-1)} \\
&= \frac{D_{(n+1)-k}}{n-k} + \sum_{i=1}^{k-2} \binom{k-2}{i} \frac{D_{(n+1)-(k+i)}}{n-(k+i)} \\
&\quad + \frac{D_{(n+1)-(2k-1)}}{n-(2k-1)} + \sum_{i=1}^{k-2} \binom{k-2}{i-1} \frac{D_{(n+1)-(k+i)}}{n-(k+i)} \\
&= \frac{D_{(n+1)-k}}{n-k} + \sum_{i=1}^{k-2} \left[\binom{k-2}{i} + \binom{k-2}{i-1} \right] \frac{D_{(n+1)-(k+i)}}{n-(k+i)} + \frac{D_{(n+1)-(2k-1)}}{n-(2k-1)} \\
&= \frac{D_{(n+1)-k}}{n-k} + \sum_{i=1}^{k-2} \binom{k-1}{i} \frac{D_{(n+1)-(k+i)}}{n-(k+i)} + \frac{D_{(n+1)-(2k-1)}}{n-(2k-1)} \\
&= \sum_{i=0}^{k-1} \binom{k-1}{i} \frac{D_{(n+1)-(k+i)}}{n-(k+i)}.
\end{aligned}$$

Remark. As a corollary of the above problem, we can solve the first problem. Let $n = 2k$, and $i_j = j$, $a_j = k + j$ for $j = 1, \dots, k$. Then a derangement $x = (x_1, \dots, x_n)$ satisfies the condition $x_{i_j} = a_j$ if and only if $x' = (x_{k+1}, \dots, x_n)$ is a permutation of $[k]$. The number of such permutations x' is $k!$. Thus $\sum_{i=0}^{k-1} \binom{k-1}{i} \frac{D_{k+1-i}}{k-i} = k!$.

3.22 2015

Day 1

2015-D1-P1 — Statement

Day 1, P1 | 10 points

For any integer $n \geq 2$ and two $n \times n$ matrices with real entries A, B that satisfy the equation

$$A^{-1} + B^{-1} = (A + B)^{-1}$$

prove that $\det(A) = \det(B)$.
Does the same conclusion follow for matrices with complex entries?

(Proposed by Zbigniew Skoczylas, Wrocław University of Technology)

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Theory and methods for 2015-D1-P1

Theory Determinants, block identities and inverses [T-LA-DETERMINANT]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Complete the equation to a product equal to identity [M-LA-INVERSE]; Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]
Strategy Expose the structure before calculating [S-STRUCTURE]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2015-D1-P1 — Solution

For this record: Theory: T-LA-DETERMINANT; T-LA-POLYNOMIAL; T-AL-POLYNOMIAL. Methods: M-LA-INVERSE; M-LA-SPECTRAL. Full learning route

Multiplying the equation by $(A + B)$ we get

$$I = (A + B)(A + B)^{-1} = (A + B)(A^{-1} + B^{-1}) =$$
$$= AA^{-1} + AB^{-1} + BA^{-1} + BB^{-1} = I + AB^{-1} + BA^{-1} + I$$
$$AB^{-1} + BA^{-1} + I = 0.$$

Let $X = AB^{-1}$; then $A = XB$ and $BA^{-1} = X^{-1}$, so we have $X + X^{-1} + I = 0$; multiplying by $(X - I)X$,

$$0 = (X - I)X \cdot (X + X^{-1} + I) = (X - I) \cdot (X^2 + X + I) = X^3 - I.$$

Hence,

$$X^3 = I$$
$$(\det X)^3 = \det(X^3) = \det I = 1$$
$$\det X = 1$$
$$\det A = \det(XB) = \det X \cdot \det B = \det B.$$

In case of complex matrices the statement is false. Let $\omega = \frac{1}{2}(-1 + i\sqrt{3})$. Obviously $\omega \notin \mathbb{R}$ and $\omega^3 = 1$, so $0 = 1 + \omega + \omega^2 = 1 + \omega + \bar{\omega}$.

Let $A = I$ and let B be a diagonal matrix with all entries along the diagonal equal to either ω or $\bar{\omega} = \omega^2$ such a way that $\det(B) \neq 1$ (if n is not divisible by 3 then one may set $B = \omega I$). Then $A^{-1} = I$, $B^{-1} = \bar{B}$. Obviously $I + B + \bar{B} = 0$ and

$$(A + B)^{-1} = (-\bar{B})^{-1} = -B = I + \bar{B} = A^{-1} + B^{-1}.$$

By the choice of A and B, $\det A = 1 \neq \det B$.

2015-D1-P2 — Statement

Day 1, P2 | 10 points

For a positive integer n, let $f(n)$ be the number obtained by writing n in binary and replacing every 0 with 1 and vice versa. For example, $n = 23$ is 10111 in binary, so $f(n)$ is 1000 in binary, therefore $f(23) = 8$. Prove that

$$\sum_{k=1}^{n} f(k) \leq \frac{n^2}{4}.$$

When does equality hold?

(Proposed by Stephan Wagner, Stellenbosch University)

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Theory and methods for 2015-D1-P2

Theory Linear recurrences and generating functions [T-AL-RECURRENCE]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Compare coefficients after a structural substitution [M-AL-COEFFICIENT]
Strategy Build a chain of bounds and track equality [S-BOUND]; Prove necessity and sufficiency separately [S-CLASSIFICATION]

Statement Recorded solutions Catalog

2015-D1-P2 — Solution

For this record: Theory: T-AL-RECURRENCE; T-AL-CONGRUENCE; T-AL-POLYNOMIAL. Methods: M-AL-COEFFICIENT. Full learning route

If r and k are positive integers with $2^{r-1} \leq k < 2^r$ then k has r binary digits, so $k + f(k) = \underbrace{11\ldots1}_{r}{}_{(2)} = 2^r - 1$.

Assume that $2^{s-1} - 1 \leq n \leq 2^s - 1$. Then

$$\begin{aligned}
\frac{n(n+1)}{2} + \sum_{k=1}^{n} f(k) &= \sum_{k=1}^{n} (k + f(k)) = \\
&= \sum_{r=1}^{s-1} \sum_{2^{r-1} \leq k < 2^r} (k + f(k)) + \sum_{2^{s-1} \leq k \leq n} (k + f(k)) = \\
&= \sum_{r=1}^{s-1} 2^{r-1} \cdot (2^r - 1) + (n - 2^{s-1} + 1) \cdot (2^s - 1) = \\
&= \sum_{r=1}^{s-1} 2^{2r-1} - \sum_{r=1}^{s-1} 2^{r-1} + (n - 2^{s-1} + 1)(2^s - 1) = \\
&= \frac{2}{3}(4^{s-1} - 1) - (2^{s-1} - 1) + (2^s - 1)n - 2^{2s-1} + 3 \cdot 2^{s-1} - 1 = \\
&= (2^s - 1)n - \frac{1}{3}4^s + 2^s - \frac{2}{3}
\end{aligned}$$

and therefore

$$\begin{aligned}
\frac{n^2}{4} - \sum_{k=1}^{n} f(k) &= \frac{n^2}{4} - \left((2^s - 1)n - \frac{1}{3}4^s + 2^s - \frac{2}{3} - \frac{n(n+1)}{2}\right) = \\
&= \frac{3}{4}n^2 - \left(2^s - \frac{3}{2}\right)n + \frac{1}{3}4^s - 2^s + \frac{2}{3} = \\
&= \frac{3}{4}\left(n - \frac{2^{s+1} - 2}{3}\right)\left(n - \frac{2^{s+1} - 4}{3}\right).
\end{aligned}$$

Notice that the difference of the last two factors is less than 1, and one of them must be an integer: $\frac{2^{s+1}-2}{3}$ is integer if s is even, and $\frac{2^{s+1}-4}{3}$ is integer if s is odd. Therefore, either one of them is 0, resulting a zero product, or both factors have the same sign, so the product is strictly positive.

2015-D1-P2 — Solution (continued)

This solves the problem and shows that equality occurs if $n = \frac{2^{s+1}-2}{3}$ (s is even) or $n = \frac{2^{s+1}-4}{3}$ (s is odd).

2015-D1-P3 — Statement

Day 1, P3 | 10 points

Let $F(0) = 0$, $F(1) = \frac{3}{2}$, and $F(n) = \frac{5}{2}F(n-1) - F(n-2)$ for $n \geq 2$.

Determine whether or not $\displaystyle\sum_{n=0}^{\infty} \frac{1}{F(2^n)}$ is a rational number.

(Proposed by Gerhard Woeginger, Eindhoven University of Technology)

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Theory and methods for 2015-D1-P3

Theory Linear recurrences and generating functions [T-AL-RECURRENCE]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Prime valuations, divisors, and factorials [T-AL-VALUATION]
Methods Find a recurrence that reveals the invariant [M-AL-RECURRENCE]
Strategy Control the limiting operation before taking the limit [S-ANALYTIC]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2015-D1-P3 — Solution — Record 1 of 2

For this record: Theory: T-AL-RECURRENCE; T-AN-CONVERGENCE. Methods: M-AL-RECURRENCE. Full learning route

The characteristic equation of our linear recurrence is $x^2 - \frac{5}{2}x + 1 = 0$, with roots $x_1 = 2$ and $x_2 = \frac{1}{2}$. So $F(n) = a \cdot 2^n + b \cdot \left(\frac{1}{2}\right)^n$ with some constants a, b. By $F(0) = 0$ and $F(1) = \frac{3}{2}$, these constants satisfy $a + b = 0$ and $2a + \frac{b}{2} = \frac{3}{2}$. So $a = 1$ and $b = -1$, and therefore

$$F(n) = 2^n - 2^{-n}.$$

Observe that

$$\frac{1}{F(2^n)} = \frac{2^{2^n}}{(2^{2^n})^2 - 1} = \frac{1}{2^{2^n} - 1} - \frac{1}{(2^{2^n})^2 - 1} = \frac{1}{2^{2^n} - 1} - \frac{1}{2^{2^{n+1}} - 1},$$

so

$$\sum_{n=0}^{\infty} \frac{1}{F(2^n)} = \sum_{n=0}^{\infty} \left(\frac{1}{2^{2^n} - 1} - \frac{1}{2^{2^{n+1}} - 1} \right) = \frac{1}{2^{2^0} - 1} = 1.$$

Hence the sum takes the value 1, which is rational.

2015-D1-P3 — Solution — Record 2 of 2

For this record: Theory: T-AL-RECURRENCE; T-AL-VALUATION; T-AN-CONVERGENCE. Methods: M-AL-RECURRENCE. Full learning route

As in the first solution we find that $F(n) = 2^n - 2^{-n}$. Then

$$\begin{aligned}
\sum_{n=0}^{\infty} \frac{1}{F(2^n)} &= \sum_{n=0}^{\infty} \frac{1}{2^{2^n} - 2^{-2^n}} = \sum_{n=0}^{\infty} \frac{\left(\frac{1}{2}\right)^{2^n}}{1 - \left(\frac{1}{2}\right)^{2^{n+1}}} \\
&= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^{2^n} \sum_{k=0}^{\infty} \left(\left(\frac{1}{2}\right)^{2^{n+1}} \right)^k = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^{2^n} \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^{2k \cdot 2^n} \\
&= \sum_{n=0}^{\infty} \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^{2^n(2k+1)} = \sum_{m=1}^{\infty} \left(\frac{1}{2}\right)^m = 1.
\end{aligned}$$

2015-D1-P3 — Solution — Record 2 of 2 (continued)

(Here we used the fact that every positive integer m has a unique representation $m = 2^n(2k+1)$ with non-negative integers n and k.)

This shows that the series converges to 1.

2015-D1-P4 — Statement

Day 1, P4 | 10 points

Determine whether or not there exist 15 integers $m_1, \dots, m_{15}$ such that

$$\sum_{k=1}^{15} m_k \cdot \arctan(k) = \arctan(16). \tag{1}$$

(Proposed by Gerhard Woeginger, Eindhoven University of Technology)

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Theory and methods for 2015-D1-P4

Theory Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]; Prime valuations, divisors, and factorials [T-AL-VALUATION]
Methods Track one prime or one extreme divisor [M-AL-VALUATION]
Strategy Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

2015-D1-P4 — Solution

For this record: Theory: T-AL-COMPLEX; T-AL-VALUATION. Methods: M-AL-VALUATION. Full learning route

We show that such integers $m_1, \dots, m_{15}$ do not exist.
Suppose that (1) is satisfied by some integers $m_1, \dots, m_{15}$. Then the argument of the complex number $z_1 = 1 + 16i$ coincides with the argument of the complex number

$$z_2 = (1+i)^{m_1}(1+2i)^{m_2}(1+3i)^{m_3} \cdots \cdots (1+15i)^{m_{15}}.$$

Therefore the ratio $R = z_2/z_1$ is real (and not zero). As $\operatorname{Re} z_1 = 1$ and $\operatorname{Re} z_2$ is an integer, R is a nonzero integer.

By considering the squares of the absolute values of z_1 and z_2, we get

$$(1+16^2)R^2 = \prod_{k=1}^{15} (1+k^2)^{m_k}.$$

Notice that $p = 1+16^2 = 257$ is a prime (the fourth Fermat prime), which yields an easy contradiction through p-adic valuations: all prime factors in the right hand side are strictly below p (as $k < 16$ implies $1+k^2 < p$). On the other hand, in the left hand side the prime p occurs with an odd exponent.

2015-D1-P5 — Statement

Day 1, P5 | 10 points

Let $n \geq 2$, let $A_1, A_2, \dots, A_{n+1}$ be $n+1$ points in the n-dimensional Euclidean space, not lying on the same hyperplane, and let B be a point strictly inside the convex hull of $A_1, A_2, \dots, A_{n+1}$. Prove that $\angle A_i B A_j > 90^\circ$ holds for at least n pairs (i, j) with $1 \leq i < j \leq n+1$.

(Proposed by Géza Kós, Eötvös University, Budapest)

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Theory and methods for 2015-D1-P5

Theory Convex bodies, supporting functionals, and packing [T-DS-CONVEX]; Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]; Inner products, orthogonality and polarization [T-LA-INNER]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]
Methods Encode a relation as a graph and extract a cycle [M-DS-GRAPH]; Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Strategy Translate geometry into a controlled representation [S-GEOMETRIC]; Expose the structure before calculating [S-STRUCTURE]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

2015-D1-P5 — Solution

For this record: Theory: T-DS-CONVEX; T-DS-GRAPH; T-LA-INNER; T-DS-GEOMETRY. Methods: M-DS-GRAPH; M-DS-GEOMETRIC. Full learning route

Let $v_i = \overrightarrow{BA_i}$. The condition $\angle A_iBA_j > 90^\circ$ is equivalent with $v_i \cdot v_j < 0$. Since B is an interior point of the simplex, there are some weights $w_1, \dots, w_{n+1} > 0$ with $\sum_{i=1}^{n+1} w_i v_i = 0$.

Let us build a graph on the vertices $1, \dots, n+1$. Let the vertices i and j be connected by an edge if $v_i \cdot v_j < 0$. We show that this graph is connected. Since every connected graph on $n+1$ vertices has at least n edges, this will prove the problem statement.

Suppose the contrary that the graph is not connected; then the vertices can be split in two disjoint nonempty sets, say V and W such that $V \cup W = \{1, 2, \dots, n+1\}$. Since there is no edge between the two vertex sets, we have $v_i \cdot v_j \geq 0$ for all $i \in V$ and $j \in W$. Consider

$$\begin{aligned} 0 &= \left(\sum_{i \in V \cup W} w_i v_i \right)^2 \\ &= \left(\sum_{i \in V} w_i v_i \right)^2 + \left(\sum_{i \in W} w_i v_i \right)^2 \\ &\quad + 2 \sum_{i \in V} \sum_{j \in W} w_i w_j (v_i \cdot v_j). \end{aligned}$$

Notice that all terms are nonnegative on the right-hand side. Moreover, $\sum_{i \in V} w_i v_i \neq 0$ and $\sum_{i \in W} w_i v_i \neq 0$: otherwise, after normalizing the corresponding positive weights, B would lie in the convex hull of a proper subset of the vertices, which is a proper face of the simplex. This contradicts strict interiority. Thus there are at least two strictly nonzero terms, contradiction.

Remark 1. The number n in the statement is sharp; if $v_{n+1} = (1, 1, \dots, 1)$ and $v_i = (\underbrace{0, \dots, 0}_{i-1}, -1, \underbrace{0, \dots, 0}_{n-i})$ for $i = 1, \dots, n$ then $v_i \cdot v_j < 0$ holds only when $i = n+1$ or $j = n+1$.

Remark 2. The origin of the problem is here: the source note for this problem

Day 2

2015-D2-P6 — Statement

Day 2, P6 | 10 points

Prove that

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}(n+1)} < 2.$$

(Proposed by Ivan Krijan, University of Zagreb)

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Theory and methods for 2015-D2-P6

Theory Series, finite cancellation, and summation by parts [T-AN-SERIES]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]
Methods Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]; Find a finite telescoping identity [M-AN-TELESCOPE]
Strategy Build a chain of bounds and track equality [S-BOUND]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2015-D2-P6 — Solution

For this record: Theory: T-AN-SERIES; T-AN-INEQUALITIES. Methods: M-AN-WEIGHTS; M-AN-TELESCOPE. Full learning route

We prove that

$$\frac{1}{\sqrt{n}(n+1)} < \frac{2}{\sqrt{n}} - \frac{2}{\sqrt{n+1}}. \tag{1}$$

Multiplying by $\sqrt{n}(n+1)$, the inequality (1) is equivalent with

$$1 < 2(n+1) - 2\sqrt{n(n+1)}$$
$$2\sqrt{n(n+1)} < n + (n+1)$$

which is true by the AM-GM inequality.

Applying (1) to the terms in the left-hand side,

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}(n+1)} < \sum_{n=1}^{\infty} \left(\frac{2}{\sqrt{n}} - \frac{2}{\sqrt{n+1}} \right) = 2.$$

2015-D2-P7 — Statement

Day 2, P7 | 10 points

Compute

$$\lim_{A \to +\infty} \frac{1}{A} \int_1^A A^{\frac{1}{x}} \, dx.$$

(Proposed by Jan Šustek, University of Ostrava)

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Theory and methods for 2015-D2-P7

Theory Scale separation and limit comparisons [T-AN-ASYMPTOTICS]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Methods Separate a main range from exceptional scales [M-AN-SPLIT]; Compare growth through derivatives [M-AN-RATIO]; Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]
Strategy Control the limiting operation before taking the limit [S-ANALYTIC]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2015-D2-P7 — Solution — Record 1 of 2

For this record: Theory: T-AN-ASYMPTOTICS; T-AN-INTEGRAL. Methods: M-AN-SPLIT. Full learning route

We prove that

$$\lim_{A \to +\infty} \frac{1}{A} \int_1^A A^{\frac{1}{x}} \, dx = 1.$$

2015-D2-P7 — Solution — Record 1 of 2 (continued)

For $A > 1$ the integrand is greater than 1, so

$$\frac{1}{A}\int_1^A A^{\frac{1}{x}}\,dx > \frac{1}{A}\int_1^A 1\,dx = \frac{1}{A}(A-1) = 1 - \frac{1}{A}.$$

In order to find a tight upper bound, fix two real numbers, $\delta > 0$ and $K > 0$, and split the interval into three parts at the points $1+\delta$ and $K\log A$. Notice that for sufficiently large A (i.e., for $A > A_0(\delta, K)$ with some $A_0(\delta, K) > 1$) we have $1+\delta < K\log A < A$.) For $A > 1$ the integrand is decreasing, so we can estimate it by its value at the starting points of the intervals:

$$\begin{aligned}\frac{1}{A}\int_1^A A^{\frac{1}{x}}\,dx &= \frac{1}{A}\left(\int_1^{1+\delta} + \int_{1+\delta}^{K\log A} + \int_{K\log A}^{A}\right) < \\ &= \frac{1}{A}\left(\delta\cdot A + (K\log A - 1 - \delta)A^{\frac{1}{1+\delta}} + (A - K\log A)A^{\frac{1}{K\log A}}\right) < \\ &< \frac{1}{A}\left(\delta A + KA^{\frac{1}{1+\delta}}\log A + A\cdot A^{\frac{1}{K\log A}}\right) = \delta + KA^{-\frac{\delta}{1+\delta}}\log A + e^{\frac{1}{K}}.\end{aligned}$$

Hence, for $A > A_0(\delta, K)$ we have

$$1 - \frac{1}{A} < \frac{1}{A}\int_1^A A^{\frac{1}{x}}\,dx < \delta + KA^{-\frac{\delta}{1+\delta}}\log A + e^{\frac{1}{K}}.$$

Taking the limit $A \to \infty$ we obtain

$$1 \le \liminf_{A\to\infty}\frac{1}{A}\int_1^A A^{\frac{1}{x}}\,dx \le \limsup_{A\to\infty}\frac{1}{A}\int_1^A A^{\frac{1}{x}}\,dx \le \delta + e^{\frac{1}{K}}.$$

Now from $\delta \to +0$ and $K \to \infty$ we get

$$1 \le \liminf_{A\to\infty}\frac{1}{A}\int_1^A A^{\frac{1}{x}}\,dx \le \limsup_{A\to\infty}\frac{1}{A}\int_1^A A^{\frac{1}{x}}\,dx \le 1,$$

so $\liminf_{A\to\infty}\frac{1}{A}\int_1^A A^{\frac{1}{x}}\,dx = \limsup_{A\to\infty}\frac{1}{A}\int_1^A A^{\frac{1}{x}}\,dx = 1$ and therefore

$$\lim_{A\to+\infty}\frac{1}{A}\int_1^A A^{\frac{1}{x}}\,dx = 1.$$

2015-D2-P7 — Solution — Record 2 of 2

For this record: Theory: T-AN-INTEGRAL; T-AN-ASYMPTOTICS. Methods: M-AN-RATIO; M-AN-PRIMITIVE. Full learning route

We will employ l'Hospital's rule.

Let $f(A,x) = A^{\frac{1}{x}}$, $g(A,x) = \frac{1}{x}A^{\frac{1}{x}}$, $F(A) = \int_1^A f(A,x)\,dx$ and $G(A) = \int_1^A g(A,x)\,dx$. Since $\frac{\partial}{\partial A}f$ and $\frac{\partial}{\partial A}g$ are continuous, the parametric integrals $F(A)$ and $G(A)$ are differentiable with respect to A, and

$$F'(A) = f(A,A) + \int_1^A \frac{\partial}{\partial A}f(A,x)\,dx = A^{\frac{1}{A}} + \int_1^A \frac{1}{x}A^{\frac{1}{x}-1}\,dx = A^{\frac{1}{A}} + \frac{G(A)}{A},$$

and

$$\begin{aligned}G'(A) &= g(A,A) + \int_1^A \frac{\partial}{\partial A}g(A,x)\,dx = \frac{A^{\frac{1}{A}}}{A} + \int_1^A \frac{1}{x^2}A^{\frac{1}{x}-1}\,dx = \\ &= \frac{A^{\frac{1}{A}}}{A} + \left[\frac{-1}{\log A}A^{\frac{1}{x}-1}\right]_1^A = \frac{A^{\frac{1}{A}}}{A} - \frac{A^{\frac{1}{A}}}{A\log A} + \frac{1}{\log A}.\end{aligned}$$

2015-D2-P7 — Solution — Record 2 of 2 (continued)

Since $\lim\limits_{A\to\infty} A^{\frac{1}{A}} = 1$, we can see that $\lim\limits_{A\to\infty} G'(A) = 0$. Aplying l'Hospital's rule to $\lim\limits_{A\to\infty} \frac{G(A)}{A}$ we get

$$\lim_{A\to\infty} \frac{G(A)}{A} = \lim_{A\to\infty} \frac{G'(A)}{1} = 0,$$

so

$$\lim_{A\to\infty} F'(A) = \lim_{A\to\infty} \left(A^{\frac{1}{A}} + \frac{G(A)}{A} \right) = 1 + 0 = 1.$$

Now applying l'Hospital's rule to $\lim\limits_{A\to\infty} \frac{F(A)}{A}$ we get

$$\lim_{A\to+\infty} \frac{1}{A} \int_1^A A^{\frac{1}{x}}\, dx = \lim_{A\to\infty} \frac{F(A)}{A} = \lim_{A\to\infty} \frac{F'(A)}{1} = 1.$$

2015-D2-P8 — Statement

Day 2, P8 | 10 points

Consider all 26^{26} words of length 26 in the Latin alphabet. Define the weight of a word as $1/(k+1)$, where k is the number of letters not used in this word. Prove that the sum of the weights of all words is 3^{75}.

(Proposed by Fedor Petrov, St. Petersburg State University)

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Theory and methods for 2015-D2-P8

Theory Generating functions, coefficient filters, and finite differences [T-DS-GENERATING]; Finite counting and incidence identities [T-DS-COUNTING]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Encode the statistic and extract the required coefficient [M-DS-GENERATING]; Count incidences with their exact multiplicity [M-DS-DOUBLECOUNT]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2015-D2-P8 — Solution

For this record: Theory: T-DS-GENERATING; T-DS-COUNTING; T-AL-POLYNOMIAL. Methods: M-DS-GENERATING; M-DS-DOUBLECOUNT. Full learning route

Let $n = 26$, then $3^{75} = (n+1)^{n-1}$. We use the following well-known
Lemma. If $f(x)$ is a polynomial of degree at most n, then its $(n+1)$-st finite difference vanishes: $\Delta^{n+1} f(x) := \sum\limits_{i=0}^{n+1} (-1)^{n+1-i} \binom{n+1}{i} f(x+i) \equiv 0$.
Proof. If Δ is the operator which maps $f(x)$ to $f(x+1) - f(x)$, then Δ^{n+1} is indeed $(n+1)$-st power of Δ and the claim follows from the observation that Δ decreases the power of a polynomial.

In other words, $f(x) = \sum\limits_{i=1}^{n+1} (-1)^{i+1} \binom{n+1}{i} f(x+i)$. Applying this for $f(x) = (n-x)^n$, substituting $x = -1$ and denoting $i = j+1$ we get

$$(n+1)^n = \sum_{j=0}^{n} (-1)^j \binom{n+1}{j+1} (n-j)^n = (n+1) \sum_{j=0}^{n} \binom{n}{j} \cdot \frac{(-1)^j}{j+1} \cdot (n-j)^n.$$

The j-th summand $\binom{n}{j} \cdot \frac{(-1)^j}{j+1} \cdot (n-j)^n$ may be interpreted as follows: choose j letters, consider all $(n-j)^n$ words without those letters and sum up $\frac{(-1)^j}{j+1}$ over all those words. Now we change the order of summation, counting at first by words. For any fixed word W with k absent letters we

2015-D2-P8 — Solution (continued)

get $\sum_{j=0}^{k}\binom{k}{j}\cdot\frac{(-1)^j}{j+1}=\frac{1}{k+1}\cdot\sum_{j=0}^{k}(-1)^j\cdot\binom{k+1}{j+1}=\frac{1}{k+1}$, since the alternating sum of binomial coefficients $\sum_{j=-1}^{k}(-1)^j\cdot\binom{k+1}{j+1}$ vanishes. That is, after changing order of summation we get exactly initial sum, and it equals $(n+1)^{n-1}$.

2015-D2-P9 — Statement

Day 2, P9 | 10 points

An $n\times n$ complex matrix A is called t-normal if $AA^t=A^tA$ where A^t is the transpose of A. For each n, determine the maximum dimension of a linear space of complex $n\times n$ matrices consisting of t-normal matrices.

(Proposed by Shachar Carmeli, Weizmann Institute of Science)

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Theory and methods for 2015-D2-P9

Theory Rank, dimension and exact linear constraints [T-LA-RANK]; Trace pairing and commutator calculus [T-LA-TRACE]; Dual bases, kernels and separating functionals [T-LA-DUALITY]; Adapted bases, centralizers and common eigenspaces [T-LA-INVARIANT]

Methods Prove a dimension obstruction and attain it [M-LA-DIMENSION]; Use trace to extract a scalar invariant [M-LA-TRACE]

Strategy Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2015-D2-P9 — Solution

For this record: Theory: T-LA-RANK; T-LA-TRACE; T-LA-DUALITY; T-LA-INVARIANT. Methods: M-LA-DIMENSION; M-LA-TRACE. Full learning route

Answer: The maximum dimension of such a space is $\frac{n(n+1)}{2}$.

The number $\frac{n(n+1)}{2}$ can be achieved, for example the symmetric matrices are obviously t-normal and they form a linear space with dimension $\frac{n(n+1)}{2}$. We shall show that this is the maximal possible dimension.

Let M_n denote the space of $n\times n$ complex matrices, let $S_n\subset M_n$ be the subspace of all symmetric matrices and let $A_n\subset M_n$ be the subspace of all anti-symmetric matrices, i.e. matrices A for which $A^t=-A$.

Let $V\subset M_n$ be a linear subspace consisting of t-normal matrices. We have to show that $\dim(V)\leq\dim(S_n)$. Let $\pi: V\to S_n$ denote the linear map $\pi(A)=A+A^t$. We have

$$\dim(V)=\dim(\mathrm{Ker}(\pi))+\dim(\mathrm{Im}(\pi))$$

so we have to prove that $\dim(\mathrm{Ker}(\pi))+\dim(\mathrm{Im}(\pi))\leq\dim(S_n)$. Notice that $\mathrm{Ker}(\pi)\subseteq A_n$.

We claim that for every $A\in\mathrm{Ker}(\pi)$ and $B\in V$, $A\pi(B)=\pi(B)A$. In other words, $\mathrm{Ker}(\pi)$ and $\mathrm{Im}(\pi)$ commute. Indeed, if $A,B\in V$ and $A=-A^t$ then

$$\begin{gathered}(A+B)(A+B)^t=(A+B)^t(A+B)\Leftrightarrow\\ \Leftrightarrow AA^t+AB^t+BA^t+BB^t=A^tA+A^tB+B^tA+B^tB\Leftrightarrow\\ \Leftrightarrow AB^t-BA=-AB+B^tA\Leftrightarrow A(B+B^t)=(B+B^t)A\Leftrightarrow\\ \Leftrightarrow A\pi(B)=\pi(B)A.\end{gathered}$$

Our bound on the dimension on V follows from the following lemma:

Lemma. Let $X\subseteq S_n$ and $Y\subseteq A_n$ be linear subspaces such that every element of X commutes with every element of Y. Then

$$\dim(X)+\dim(Y)\leq\dim(S_n)$$

2015-D2-P9 — Solution (continued)

Proof. Without loss of generality we may assume $X = Z_{S_n}(Y) := \{x \in S_n : xy = yx \ \forall y \in Y\}$. Define the bilinear map $B : S_n \times A_n \to \mathbb{C}$ by $B(x,y) = \mathrm{tr}(d[x,y])$ where $[x,y] = xy - yx$ and $d = \mathrm{diag}(1,\dots,n)$ is the matrix with diagonal elements $1,\dots,n$ and zeros off the diagonal. Clearly $B(X,Y) = \{0\}$. Furthermore, if $y \in Y$ satisfies that $B(x,y) = 0$ for all $x \in S_n$ then $\mathrm{tr}(d[x,y]) = -\mathrm{tr}([d,x],y]) = 0$ for every $x \in S_n$.

We claim that $\{[d,x] : x \in S_n\} = A_n$. Let E_i^j denote the matrix with 1 in the entry (i,j) and 0 in all other entries. Then a direct computation shows that $[d, E_i^j] = (j-i)E_i^j$ and therefore $[d, E_i^j + E_j^i] = (j-i)(E_i^j - E_j^i)$ and the collection $\{(j-i)(E_i^j - E_j^i)\}_{1 \le i < j \le n}$ span A_n for $i \neq j$.

It follows that if $B(x,y) = 0$ for all $x \in S_n$ then $\mathrm{tr}(yz) = 0$ for every $z \in A_n$. But then, taking $z = \bar{y}$, where $\bar{y}$ is the entry-wise complex conjugate of y, we get $0 = \mathrm{tr}(y\bar{y}) = -\mathrm{tr}(y\bar{y}^t)$ which is the sum of squares of all the entries of y. This means that $y = 0$.

It follows that if $y_1,\dots,y_k \in Y$ are linearly independent then the equations

$$B(x,y_1) = 0, \ \dots, \ B(x,y_k) = 0$$

are linearly independent as linear equations in x, otherwise there are $a_1,\dots,a_k$ such that $B(x, a_1y_1 + \cdots + a_ky_k) = 0$ for every $x \in S_n$, a contradiction to the observation above. Since the solution of k linearly independent linear equations is of codimension k,

$$\begin{aligned} &\dim(\{x \in S_n : [x,y_i] = 0, \text{ for } i = 1,..,k\}) \le \\ &\le \dim(x \in S_n : B(x,y_i) = 0 \text{ for } i = 1,\dots,k) = \dim(S_n) - k. \end{aligned}$$

The lemma follows by taking $y_1,\dots,y_k$ to be a basis of Y.

Since $\mathrm{Ker}(\pi)$ and $\mathrm{Im}(\pi)$ commute, by the lemma we deduce that

$$\dim(V) = \dim(\mathrm{Ker}(\pi)) + \dim(\mathrm{Im}(\pi)) \le \dim(S_n) = \frac{n(n+1)}{2}.$$

2015-D2-P10 — Statement

Day 2, P10 | 10 points

Let n be a positive integer, and let $p(x)$ be a polynomial of degree n with integer coefficients. Prove that

$$\max_{0 \le x \le 1} |p(x)| > \frac{1}{e^n}.$$

(Proposed by Géza Kós, Eötvös University, Budapest)

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Theory and methods for 2015-D2-P10

Theory Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Factorial growth and the prime-number input [T-AL-PRIMEGROWTH]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]

Methods Build a holomorphic function with a useful boundary [M-AL-ANALYTIC]

Strategy Build a chain of bounds and track equality [S-BOUND]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2015-D2-P10 — Solution

For this record: Theory: T-AL-POLYNOMIAL; T-AL-PRIMEGROWTH; T-AN-INTEGRAL; T-AN-CONVERGENCE. Methods: M-AL-ANALYTIC. Full learning route

Let

$$M = \max_{0 \le x \le 1} |p(x)|.$$

2015-D2-P10 — Solution (continued)

For every positive integer k, let

$$J_k = \int_0^1 \big(p(x)\big)^{2k} dx.$$

Obviously $0 < J_k < M^{2k}$ is a rational number. If $(p(x))^{2k} = \sum_{i=0}^{2kn} a_{k,i}x^i$ then $J_k = \sum_{i=0}^{2kn} \frac{a_{k,i}}{i+1}$. Taking the least common denominator, we can see that $J_k \geq \dfrac{1}{\operatorname{lcm}(1,2,\dots,2kn+1)}$.

An equivalent form of the prime number theorem is that $\log \operatorname{lcm}(1,2,\dots,N) \sim N$ if $N \to \infty$. Therefore, for every $\varepsilon > 0$ and sufficiently large k we have

$$\operatorname{lcm}(1,2,\dots,2kn+1) < e^{(1+\varepsilon)(2kn+1)}$$

and therefore

$$M^{2k} > J_k \geq \frac{1}{\operatorname{lcm}(1,2,\dots,2kn+1)} > \frac{1}{e^{(1+\varepsilon)(2kn+1)}},$$

$$M > \frac{1}{e^{(1+\varepsilon)(n+\frac{1}{2k})}}.$$

Taking $k \to \infty$ and then $\varepsilon \to +0$ we get

$$M \geq \frac{1}{e^n}.$$

Since e is transcendent, equality is impossible.

Remark. The constant $\frac{1}{e} \approx 0.3679$ is not sharp. It is known that the best constant is between 0.4213 and 0.4232. (See I. E. Pritsker, The Gelfond–Schnirelman method in prime number theory, Canad. J. Math. 57 (2005), 1080–1101.)

3.23 2016

Day 1

2016-D1-P1 — Statement

Day 1, P1 | 10 points

Let $f : [a,b] \to \mathbb{R}$ be continuous on $[a,b]$ and differentiable on (a,b). Suppose that f has infinitely many zeros, but there is no $x \in (a,b)$ with $f(x) = f'(x) = 0$.
(a) Prove that $f(a)f(b) = 0$.
(b) Give an example of such a function on $[0,1]$.

(Proposed by Alexandr Bolbot, Novosibirsk State University)

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Theory and methods for 2016-D1-P1

Theory Continuity, compactness, and order [T-AN-CONTINUITY]; Countability, accumulation, and Cantor constructions [T-AN-SETS]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Trigonometric identities and safe branches [T-AN-TRIG]
Methods Localize an extremum or an accumulation point [M-AN-COMPACT]; Use a dense set, an enumeration, or a category argument [M-AN-TOPOLOGY]
Strategy Make the contradiction quantitative or structural [S-CONTRADICTION]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2016-D1-P1 — Solution

For this record: Theory: T-AN-CONTINUITY; T-AN-SETS; T-AN-DERIVATIVE; T-AN-TRIG. Methods: M-AN-COMPACT; M-AN-TOPOLOGY. Full learning route

(a) Choose a convergent sequence z_n of zeros and let $c = \lim z_n \in [a, b]$. By the continuity of f we obtain $f(c) = 0$. We want to show that either $c = a$ or $c = b$, so $f(a) = 0$ or $f(b) = 0$; then the statement follows.

If c was an interior point then we would have $f(c) = 0$ and $f'(c) = \lim\limits_{n\to\infty} \dfrac{f(z_n) - f(c)}{z_n - c} = \lim\limits_{n\to\infty} \dfrac{0-0}{z_n - c} = 0$ simultaneously, contradicting the conditions. Hence, $c = a$ or $c = b$.

(b) Let

$$f(x) = \begin{cases} x \sin \dfrac{1}{x} & \text{if } 0 < x \le 1 \\ 0 & \text{if } x = 0. \end{cases}$$

This function has zeros at the points $\dfrac{1}{k\pi}$ for $k = 1, 2, \dots$, and it is continuous at 0 as well.

In $(0, 1)$ we have

$$f'(x) = \sin \frac{1}{x} - \frac{1}{x} \cos \frac{1}{x}.$$

Since $\sin \frac{1}{x}$ and $\cos \frac{1}{x}$ cannot vanish at the same point, we have either $f(x) \neq 0$ or $f'(x) \neq 0$ everywhere in $(0, 1)$.

2016-D1-P2 — Statement

Day 1, P2 | 10 points

Let k and n be positive integers. A sequence $(A_1, \dots, A_k)$ of $n \times n$ real matrices is preferred by Ivan the Confessor if $A_i^2 \neq 0$ for $1 \le i \le k$, but $A_i A_j = 0$ for $1 \le i, j \le k$ with $i \neq j$. Show that $k \le n$ in all preferred sequences, and give an example of a preferred sequence with $k = n$ for each n.

(Proposed by Fedor Petrov, St. Petersburg State University)

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Theory and methods for 2016-D1-P2

Theory Rank, dimension and exact linear constraints [T-LA-RANK]; Dual bases, kernels and separating functionals [T-LA-DUALITY]
Methods Prove a dimension obstruction and attain it [M-LA-DIMENSION]
Strategy Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2016-D1-P2 — Solution — Record 1 of 2

For this record: Theory: T-LA-RANK; T-LA-DUALITY. Methods: M-LA-DIMENSION. Full learning route

For every $i = 1, \dots, n$, since $A_i \cdot A_i \neq 0$, there is a column $v_i \in \mathbb{R}^n$ in A_i such that $A_i v_i \neq 0$. We will show that the vectors $v_1, \dots, v_k$ are linearly independent; this immediately proves $k \le n$.

Suppose that a linear combination of $v_1, \dots, v_k$ vanishes:

$$c_1 v_1 + \dots + c_k v_k = 0, \qquad c_1, \dots, c_k \in \mathbb{R}.$$

For $i \neq j$ we have $A_i A_j = 0$; in particular, $A_i v_j = 0$. Now, for each $i = 1, \dots, n$, from

$$0 = A_i(c_1 v_1 + \dots + c_k v_k) = \sum_{j=1}^{k} c_j (A_i v_j) = c_i (A_i v_i)$$

we can see that $c_i = 0$. Hence, $c_1 = \dots = c_k = 0$.

2016-D1-P2 — Solution — Record 1 of 2 (continued)

The case $k = n$ is possible: if A_i has a single 1 in the main diagonal at the ith position and its other entries are zero then $A_i^2 = A_i$ and $A_i A_j = 0$ for $i \neq j$.

Remark. The solution above can be re-formulated using block matrices in the following way. Consider

$$\begin{pmatrix} A_1 \\ A_2 \\ \vdots \\ A_k \end{pmatrix} (A_1 \quad A_2 \quad \ldots \quad A_k) = \begin{pmatrix} A_1^2 & 0 & \ldots & 0 \\ 0 & A_2^2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & 0 & \ldots & A_k^2 \end{pmatrix}.$$

It is easy to see that the rank of the left-hand side is at most n; the rank of the right-hand side is at least k.

2016-D1-P2 — Solution — Record 2 of 2

For this record: Theory: T-LA-RANK. Methods: M-LA-DIMENSION. Full learning route

Let U_i and K_i be the image and the kernel of the matrix A_i (considered as a linear operator on $\mathbb{R}^n$), respectively. For every pair i, j of indices, we have $U_j \subset K_i$ if and only if $i \neq j$.
Let $X_0 = \mathbb{R}^n$ and let $X_i = K_1 \cap K_2 \cap \cdots \cap K_i$ for $i = 1, \ldots, k$, so $X_0 \supset X_1 \supset \cdots \supset X_k$. Notice also that $U_i \subset X_{i-1}$ because $U_i \subset K_j$ for every $j < i$, and $U_i \not\subset X_i$ because $U_i \not\subset K_i$. Hence, $X_i \neq X_{i-1}$; X_i is a proper subspace of X_{i-1}.

Now, from

$$n = \dim X_0 > \dim X_1 > \cdots > \dim X_k \geq 0$$

we get $k \geq n$.

2016-D1-P3 — Statement

Day 1, P3 | 10 points

Let n be a positive integer. Also let $a_1, a_2, \ldots, a_n$ and $b_1, b_2, \ldots, b_n$ be real numbers such that $a_i + b_i > 0$ for $i = 1, 2, \ldots, n$. Prove that

$$\sum_{i=1}^{n} \frac{a_i b_i - b_i^2}{a_i + b_i} \leq \frac{\sum_{i=1}^{n} a_i \cdot \sum_{i=1}^{n} b_i - \left(\sum_{i=1}^{n} b_i\right)^2}{\sum_{i=1}^{n} (a_i + b_i)}.$$

(Proposed by Daniel Strzelecki, Nicolaus Copernicus University in Toruń, Poland)

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Theory and methods for 2016-D1-P3

Theory Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Inner products, orthogonality and polarization [T-LA-INNER]
Methods Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]
Strategy Expose the structure before calculating [S-STRUCTURE]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2016-D1-P3 — Solution

For this record: Theory: T-AN-INEQUALITIES; T-LA-INNER. Methods: M-AN-WEIGHTS. Full learning route

By applying the identity

$$\frac{XY - Y^2}{X + Y} = Y - \frac{2Y^2}{X + Y}$$

2016-D1-P3 — Solution (continued)

with $X = a_i$ and $Y = b_i$ to the terms in the LHS and $X = \sum_{i=1}^{n} a_i$ and $Y = \sum_{i=1}^{n} b_i$ to the RHS,

$$\text{LHS} = \sum_{i=1}^{n} \frac{a_i b_i - b_i^2}{a_i + b_i} = \sum_{i=1}^{n} \left(b_i - \frac{2b_i^2}{a_i + b_i} \right) = \sum_{i=1}^{n} b_i - 2 \sum_{i=1}^{n} \frac{b_i^2}{a_i + b_i},$$

$$\text{RHS} = \frac{\sum_{i=1}^{n} a_i \cdot \sum_{i=1}^{n} b_i - \left(\sum_{i=1}^{n} b_i \right)^2}{\sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i} = \sum_{i=1}^{n} b_i - 2 \frac{\left(\sum_{i=1}^{n} b_i \right)^2}{\sum_{i=1}^{n} (a_i + b_i)}.$$

Therefore, the statement is equivalent with

$$\sum_{i=1}^{n} \frac{b_i^2}{a_i + b_i} \geq \frac{\left(\sum_{i=1}^{n} b_i \right)^2}{\sum_{i=1}^{n} (a_i + b_i)},$$

which is the same as the well-known variant of the Cauchy-Schwarz inequality,

$$\sum_{i=1}^{n} \frac{X_i^2}{Y_i} \geq \frac{(X_1 + \cdots + X_n)^2}{Y_1 + \cdots + Y_n} \qquad (Y_1, \ldots, Y_n > 0)$$

with $X_i = b_i$ and $Y_i = a_i + b_i$.

2016-D1-P4 — Statement

Day 1, P4 | 10 points

Let $n \geq k$ be positive integers, and let $\mathcal{F}$ be a family of finite sets with the following properties:
(i) $\mathcal{F}$ contains at least $\binom{n}{k} + 1$ distinct sets containing exactly k elements;
(ii) for any two sets $A, B \in \mathcal{F}$, their union $A \cup B$ also belongs to $\mathcal{F}$.
Prove that $\mathcal{F}$ contains at least three sets with at least n elements.

(Proposed by Fedor Petrov, St. Petersburg State University)

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Theory and methods for 2016-D1-P4

Theory Finite set systems and antichains [T-DS-SETS]; Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Extremal choice, exchange arguments, and invariants [T-DS-EXTREMAL]
Methods Count incidences with their exact multiplicity [M-DS-DOUBLECOUNT]; Split a family by traces, pieces, or labels [M-DS-PARTITION]; Repair an extremal configuration by a local exchange [M-DS-EXTREMAL]; Strengthen the statement before the induction step [M-DS-INDUCT]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Reduce size while preserving the defining conditions [S-INDUCTION]; Choose an extremal object and force rigidity [S-EXTREMAL]

Statement Recorded solutions Catalog

2016-D1-P4 — Solution — Record 1 of 2

For this record: Theory: T-DS-SETS; T-DS-COUNTING. Methods: M-DS-DOUBLECOUNT; M-DS-PARTITION. Full learning route

If $n = k$ then we have at least two distinct sets in the family with exactly n elements and their union, so the statement is true. From now on we assume that $n > k$.
Fix $\binom{n}{k} + 1$ sets of size k in $\mathcal{F}$, call them 'generators'. Let $V \in \mathcal{F}$ be the union of the generators. Since V has at least $\binom{n}{k} + 1$ subsets of size k, we have $|V| > n$.

2016-D1-P4 — Solution — Record 1 of 2 (continued)

Call an element $v \in V$ 'appropriate' if v belongs to at most $\binom{n-1}{k-1}$ generators. Then there exist at least $\binom{n}{k}+1-\binom{n-1}{k-1} = \binom{n-1}{k}+1$ generators not containing v. Their union contains at least n elements, and the union does not contain v.
Now we claim that among any n elements $x_1, \dots, x_n$ of V, there exists an appropriate element. Consider all pairs (G, x_i) such that G is a generator and $x_i \in G$. Every generator has exactly k elements, so the number of such pairs is at most $\left(\binom{n}{k}+1\right) \cdot k$. If some x_i is not appropriate then x_i is contained in at least $\binom{n-1}{k-1}+1$ generators; if none of $x_1, \dots, x_n$ was appropriate then we wold have at least $n \cdot \left(\binom{n-1}{k-1}+1\right)$ pairs. But $n \cdot \left(\binom{n-1}{k-1}+1\right) > \left(\binom{n}{k}+1\right) \cdot k$,
so this is not possible; at least one of $x_1, \dots, x_n$ must be appropriate.

Since $|V| > n$, the set V contains some appropriate element v_1. Let $U_1 \in \mathcal{F}$ be the union of all generators not containing v_1. Then $|U_1| \geq n$ and $v_1 \notin U_1$. Now take an appropriate element v_2 from U_1 and let $U_2 \in \mathcal{F}$ be the union of all generators not containing v_2. Then $|U_2| \geq n$, so we have three sets, V, U_1 and U_2 in $\in \mathcal{F}$ with at least n elements: $V \neq U_1$ because $v_1 \in V$ and $v_1 \notin U_1$, and U_2 is different from V and U_1 because $v_2 \in V, U_1$ but $v_2 \notin U_2$.

2016-D1-P4 — Solution — Record 2 of 2

For this record: Theory: T-DS-SETS; T-DS-INDUCTION; T-DS-EXTREMAL. Methods: M-DS-EXTREMAL; M-DS-INDUCT.
Full learning route

We proceed by induction on k, so we can assume that the statement of the problem is known for smaller values of k. By contradiction, assume that $\mathcal{F}$ has less than 3 sets with at least n elements, that is the number of such sets is 0, 1 or 2. We can assume without loss of generality that $\mathcal{F}$ consists of exactly $N := \binom{n}{k}+1$ distinct sets of size k and all their possible unions. Denote the sets of size k by $S_1, S_2, \dots$.
Consider a maximal set $I \subset \{1, \dots, N\}$ such that $A := \bigcup_{i \in I} S_i$ has size less than n, $|A| < n$. This means that adding any S_j for $j \notin I$ makes the size at least n, $|S_j \cup A| \geq n$. First, let's prove that such j exist. Otherwise, all the sets S_i are contained in A. But there are only $\binom{|A|}{k} \leq \binom{n-1}{k} < N$ distinct k-element subsets of A, this is a contradiction. So there is at least one j such that $|S_j \cup A| \geq n$.
Consider all possible sets that can be obtained as $S_j \cup A$ for $j \notin I$. Their size is at least n, so their number can be 1 or 2. If there are two of them, say B and C then $B \subset C$ or $C \subset B$, for otherwise the union of B and C would be different from both B and C, so we would have three sets B, C and $B \cup C$ of size at least n. We see that in any case there must exist $x \notin A$ such that $x \in S_j$ for all $j \notin I$.
Consider sets $S'_j = S_j \setminus \{x\}$ for $j \notin I$. Their sizes are equal to $k-1$. Their number is at least

$$N - \binom{n-1}{k} = \binom{n-1}{k-1} + 1.$$

By the induction hypothesis, we can form 3 sets of size at least $n-1$ by taking unions of the sets S'_j for $j \notin I$. Adding x back we see that the corresponding unions of the sets S_j will have sizes at least n, so we are done proving the induction step.

The above argument allows us to decrease k all the way to $k = 0$, so it remains to check the statement for $k = 0$. The assumption says that we have at least $\binom{n}{0} + 1 = 2$ sets of size 0. This is impossible, because there is only one empty set. Thus the statement trivially holds for $k = 0$.

2016-D1-P5 — Statement

Day 1, P5 | 10 points

Let S_n denote the set of permutations of the sequence $(1, 2, \dots, n)$. For every permutation $\pi = (\pi_1, \dots, \pi_n) \in S_n$, let $\text{inv}(\pi)$ be the number of pairs $1 \leq i < j \leq n$ with $\pi_i > \pi_j$; i.e. the number of inversions in π. Denote by $f(n)$ the number of permutations $\pi \in S_n$ for which $\text{inv}(\pi)$ is divisible by $n+1$.

2016-D1-P5 — Statement (continued)

Prove that there exist infinitely many primes p such that $f(p-1) > \frac{(p-1)!}{p}$, and infinitely many primes p such that $f(p-1) < \frac{(p-1)!}{p}$.

(Proposed by Fedor Petrov, St. Petersburg State University)

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Theory and methods for 2016-D1-P5

Theory Generating functions, coefficient filters, and finite differences [T-DS-GENERATING]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]; Roots-of-unity filters and positive trigonometric kernels [T-AL-FOURIER]

Methods Encode the statistic and extract the required coefficient [M-DS-GENERATING]; Choose information that makes an encoding injective [M-DS-ENCODE]

Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Control the limiting operation before taking the limit [S-ANALYTIC]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2016-D1-P5 — Solution

For this record: Theory: T-DS-GENERATING; T-AL-POLYNOMIAL; T-AL-CONGRUENCE; T-AN-CONVERGENCE; T-AL-PERMUTATION; T-AL-FOURIER. Methods: M-DS-GENERATING; M-DS-ENCODE. Full learning route

We will use the well-known formula

$$\sum_{\pi\in S_n} x^{\mathrm{inv}(\pi)} = 1\cdot(1+x)\cdot(1+x+x^2)\dots(1+x+\dots+x^{n-1}).$$

(This formula can be proved by induction on n. The cases $n=1,2$ are obvious. From each permutation of $(1,2,\dots,n-1)$, we can get a permutation of $(1,2,\dots,n)$ such that we insert the element n at one of the n possible positions before, between or after the numbers $1,2,\dots,n-1$; the number of inversions increases by $n-1,n-2,\dots,1$ or 0, respectively.)
Now let

$$G_n(x) = \sum_{\pi\in S_n} x^{\mathrm{inv}(\pi)}.$$

and let $\varepsilon = e^{\frac{2\pi i}{n+1}}$. The sum of coefficients of the powers divisible by $n+1$ may be expressed as a trigonometric sum as

$$f(n) = \frac{1}{n+1}\sum_{k=0}^{n} G_n(\varepsilon^k) = \frac{n!}{n+1} + \frac{1}{n+1}\sum_{k=1}^{n} G_n(\varepsilon^k).$$

Hence, we are interested in the sign of

$$f(n) - \frac{n!}{n+1} = \frac{1}{n+1}\sum_{k=1}^{n} G_n(\varepsilon^k)$$

with $n = p-1$ where p is a (large, odd) prime.

For every fixed $1 \le k \le p-1$ we have

$$G_{p-1}(\varepsilon^k) = \prod_{j=1}^{p-1}(1+\varepsilon^k+\varepsilon^{2k}+\dots+\varepsilon^{(j-1)k}) = \prod_{j=1}^{p-1}\frac{1-\varepsilon^{jk}}{1-\varepsilon^k} = \frac{(1-\varepsilon^k)(1-\varepsilon^{2k})\cdots(1-\varepsilon^{(p-1)k})}{(1-\varepsilon^k)^{p-1}}.$$

Notice that the factors in the numerator are $(1-\varepsilon)$, $(1-\varepsilon^2)$, ..., $(1-\varepsilon^{p-1})$; only their order is different. So, by the identity $(z-\varepsilon)(z-\varepsilon^2)\dots(z-\varepsilon^{p-1}) = 1+z+\dots+z^{p-1}$,

$$G_{p-1}(\varepsilon^k) = \frac{p}{(1-\varepsilon^k)^{p-1}} = \frac{p}{(1-e^{\frac{2k\pi i}{p}})^{p-1}}.$$

2016-D1-P5 — Solution (continued)

Hence, $f(p-1)-\frac{(p-1)!}{p}$ has the same sign as

$$\sum_{k=1}^{p-1}\left(1-e^{\frac{2k\pi i}{p}}\right)^{1-p}=\sum_{k=1}^{p-1}e^{\frac{k(1-p)\pi i}{p}}\left(-2i\sin\frac{\pi k}{p}\right)^{1-p}=$$

$$=2\cdot 2^{1-p}(-1)^{\frac{p-1}{2}}\sum_{k=1}^{\frac{p-1}{2}}\cos\frac{\pi k(p-1)}{p}\left(\sin\frac{\pi k}{p}\right)^{1-p}.$$

Put $\theta=\pi/p$ and $m=(p-1)/2$. For $p\geq 5$, write the last real sum as $\sum_{k=1}^{m}b_k$, where $b_k=\cos((p-1)k\theta)\sin(k\theta)^{1-p}$. Its first term is $b_1=-\cos\theta\,\sin(\theta)^{1-p}<0$. Since $\sin(k\theta)\geq\sin(2\theta)$ for $2\leq k\leq m$,

$$\frac{\sum_{k=2}^{m}|b_k|}{|b_1|}\leq\frac{p-3}{2\cos\theta}\left(\frac{1}{2\cos\theta}\right)^{p-1}\longrightarrow 0.$$

Thus the first term determines the sign for all sufficiently large primes. The factor $(-1)^m$ makes the discrepancy positive when m is odd and negative when m is even. Consequently, $f(p-1)-(p-1)!/p$ is positive for $p\equiv 3\pmod 4$ and negative for $p\equiv 1\pmod 4$, once p is sufficiently large. There are infinitely many primes in each of these two residue classes, so both required strict inequalities occur infinitely often.

Day 2

2016-D2-P6 — Statement

Day 2, P6 | 10 points

Let $(x_1,x_2,\dots)$ be a sequence of positive real numbers satisfying $\sum_{n=1}^{\infty}\frac{x_n}{2n-1}=1$. Prove that

$$\sum_{k=1}^{\infty}\sum_{n=1}^{k}\frac{x_n}{k^2}\leq 2.$$

(Proposed by Gerhard J. Woeginger, The Netherlands)

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Theory and methods for 2016-D2-P6

Theory Nonnegative sums, Fubini, and integral kernels [T-AN-FUBINI]; Series, finite cancellation, and summation by parts [T-AN-SERIES]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]
Methods Find a finite telescoping identity [M-AN-TELESCOPE]; Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]
Strategy Build a chain of bounds and track equality [S-BOUND]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2016-D2-P6 — Solution

For this record: Theory: T-AN-FUBINI; T-AN-SERIES; T-AN-INEQUALITIES. Methods: M-AN-TELESCOPE; M-AN-WEIGHTS. Full learning route

By interchanging the sums,

$$\sum_{k=1}^{\infty}\sum_{n=1}^{k}\frac{x_n}{k^2}=\sum_{1\leq n\leq k}\frac{x_n}{k^2}=\sum_{n=1}^{\infty}x_n\left(\sum_{k=n}^{\infty}\frac{1}{k^2}\right).$$

2016-D2-P6 — Solution (continued)

Then we use the upper bound

$$\sum_{k=n}^{\infty}\frac{1}{k^2}\le\sum_{k=n}^{\infty}\frac{1}{k^2-\frac14}=\sum_{k=n}^{\infty}\left(\frac{1}{k-\frac12}-\frac{1}{k+\frac12}\right)=\frac{1}{n-\frac12}$$

and get

$$\sum_{k=1}^{\infty}\sum_{n=1}^{k}\frac{x_n}{k^2}=\sum_{n=1}^{\infty}x_n\left(\sum_{k=n}^{\infty}\frac{1}{k^2}\right)<\sum_{n=1}^{\infty}x_n\cdot\frac{1}{n-\frac12}=2\sum_{n=1}^{\infty}\frac{x_n}{2n-1}=2.$$

2016-D2-P7 — Statement

Day 2, P7 | 10 points

Today, Ivan the Confessor prefers continuous functions $f:[0,1]\to\mathbb{R}$ satisfying $f(x)+f(y)\ge|x-y|$ for all pairs $x,y\in[0,1]$. Find the minimum of $\int_0^1 f$ over all preferred functions.

(Proposed by Fedor Petrov, St. Petersburg State University)

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Theory and methods for 2016-D2-P7

Theory Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]
Methods Pair an integral through symmetry [M-AN-PAIR]
Strategy Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2016-D2-P7 — Solution

For this record: Theory: T-AN-INTEGRAL; T-AN-INEQUALITIES. Methods: M-AN-PAIR. Full learning route

The minimum of $\int_0^1 f$ is $\frac14$.

Applying the condition with $0\le x\le\frac12$, $y=x+\frac12$ we get

$$f(x)+f(x+\tfrac12)\ge\tfrac12.$$

By integrating,

$$\int_0^1 f(x)\,dx=\int_0^{1/2}(f(x)+f(x+\tfrac12))\,dx\ge\int_0^{1/2}\tfrac12\,dx=\tfrac14.$$

On the other hand, the function $f(x)=\left|x-\frac12\right|$ satisfies the conditions because

$$|x-y|=\left|(x-\tfrac12)+(\tfrac12-y)\right|\le\left|x-\tfrac12\right|+\left|\tfrac12-y\right|=f(x)+f(y),$$

and establishes

$$\int_0^1 f(x)\,dx=\int_0^{1/2}(\tfrac12-x)\,dx+\int_{1/2}^1(x-\tfrac12)\,dx=\frac18+\frac18=\frac14.$$

2016-D2-P8 — Statement

Day 2, P8 | 10 points

Let n be a positive integer, and denote by $\mathbb{Z}_n$ the ring of integers modulo n. Suppose that there exists a function $f:\mathbb{Z}_n\to\mathbb{Z}_n$ satisfying the following three properties:
(i) $f(x)\ne x$,
(ii) $f(f(x))=x$,

2016-D2-P8 — Statement (continued)

(iii) $f(f(f(x+1)+1)+1) = x$ for all $x \in \mathbb{Z}_n$.
Prove that $n \equiv 2 \pmod 4$.

(Proposed by Ander Lamaison Vidarte, Berlin Mathematical School, Germany)

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Theory and methods for 2016-D2-P8

Theory Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]; Groups, orders, generators, and quotients [T-AL-GROUP]
Methods Work orbit by orbit and track parity [M-AL-ORBIT]
Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2016-D2-P8 — Solution

For this record: Theory: T-AL-PERMUTATION; T-AL-GROUP. Methods: M-AL-ORBIT. Full learning route

From property (ii) we can see that f is surjective, so f is a permutation of the elements in $\mathbb{Z}_n$, and its order is at most 2. Therefore, the permutation f is the product of disjoint transpositions of the form $(x, f(x))$. Property (i) yields that this permutation has no fixed point, so n is be even, and the number of transpositions is precisely $n/2$.
Consider the permutation $g(x) = f(x+1)$. If g was odd then $g \circ g \circ g$ also would be odd. But property (iii) constraints that $g \circ g \circ g$ is the identity which is even. So g cannot be odd; g must be even. The cyclic permutation $h(x) = x - 1$ has order n, an even number, so h is odd. Then $f(x) = g \circ h$ is odd. Since f is the product of $n/2$ transpositions, this shows that $n/2$ must be odd, so $n \equiv 2 \pmod 4$.
Remark. There exists a function with properties (i–iii) for every $n \equiv 2 \pmod 4$. For $n = 2$ take $f(1) = 2$, $f(2) = 1$. Here we outline a possible construction for $n \geq 6$.
Let $n = 4k + 2$, take a regular polygon with $k + 2$ sides, and divide it into k triangles with $k - 1$ diagonals. Draw a circle that intersects each side and each diagonal twice; altogether we have $4k + 2$ intersections. Label the intersection points clockwise around the circle. On every side and diagonal we have two intersections; let f send them to each other.

This function f obviously satisfies properties (i) and (ii). For every x we either have $f(x + 1) = x$ or the effect of adding 1 and taking f three times is going around the three sides of a triangle, so this function satisfies property (iii).

2016-D2-P9 — Statement

Day 2, P9 | 10 points

Let k be a positive integer. For each nonnegative integer n, let $f(n)$ be the number of solutions $(x_1, \ldots, x_k) \in \mathbb{Z}^k$ of the inequality $|x_1| + \cdots + |x_k| \leq n$. Prove that for every $n \geq 1$, we have $f(n-1)f(n+1) \leq f(n)^2$.

(Proposed by Esteban Arreaga, Renan Finder and José Madrid, IMPA, Rio de Janeiro)

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Theory and methods for 2016-D2-P9

Theory Log-concave sequences and coefficient smoothing [T-DS-LOGCONCAVITY]; Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Generating functions, coefficient filters, and finite differences [T-DS-GENERATING]
Methods Strengthen the statement before the induction step [M-DS-INDUCT]; Split a family by traces, pieces, or labels [M-DS-PARTITION]; Encode the statistic and extract the required coefficient [M-DS-GENERATING]
Strategy Reduce size while preserving the defining conditions [S-INDUCTION]; Build a chain of bounds and track equality [S-BOUND]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2016-D2-P9 — Solution — Record 1 of 2

For this record: Theory: T-DS-LOGCONCAVITY; T-DS-COUNTING; T-DS-INDUCTION. Methods: M-DS-INDUCT; M-DS-PARTITION. Full learning route

We prove by induction on k. If $k = 1$ then we have $f(n) = 2n + 1$ and the statement immediately follows from the AM-GM inequality.

Assume that $k \geq 2$ and the statement is true for $k-1$. Let $g(m)$ be the number of integer solutions of $|x_1| + \cdots + |x_{k-1}| \leq m$; by the induction hypothesis $g(m-1)g(m+1) \leq g(m)^2$ holds; this can be transformed to

$$\frac{g(0)}{g(1)} \leq \frac{g(1)}{g(2)} \leq \frac{g(2)}{g(3)} \leq \dots .$$

For any integer constant c, the inequality $|x_1| + \cdots + |x_{k-1}| + |c| \leq n$ has $g(n - |c|)$ integer solutions. Therefore, we have the recurrence relation

$$f(n) = \sum_{c=-n}^{n} g(n - |c|) = g(n) + 2g(n-1) + \cdots + 2g(0).$$

It follows that

$$\begin{aligned}\frac{f(n-1)}{f(n)} &= \frac{g(n-1) + 2g(n-2) + \cdots + 2g(0)}{g(n) + 2g(n-1) + \cdots + 2g(1) + 2g(0)} \leq \\ &\leq \frac{g(n) + g(n-1) + (g(n-1) + \cdots + 2g(0) + 2 \cdot 0)}{g(n+1) + g(n) + (g(n) + \cdots + 2g(1) + 2g(0))} = \frac{f(n)}{f(n+1)}\end{aligned}$$

as required.

2016-D2-P9 — Solution — Record 2 of 2

For this record: Theory: T-DS-LOGCONCAVITY; T-DS-GENERATING. Methods: M-DS-GENERATING; M-DS-INDUCT. Full learning route

We first compute the generating function for $f(n)$:

$$\sum_{n=0}^{\infty} f(n)q^n = \sum_{(x_1,x_2,\dots,x_k)\in\mathbb{Z}^k} \sum_{c=0}^{\infty} q^{|x_1|+|x_2|+\cdots+|x_k|+c} = \left(\sum_{x\in\mathbb{Z}} q^{|x|}\right)^k \frac{1}{1-q} = \frac{(1+q)^k}{(1-q)^{k+1}}.$$

For each $a = 0, 1, \dots$ denote by $g_a(n)$ ($n = 0, 1, 2, \dots$) the coefficients in the following expansion:

$$\frac{(1+q)^a}{(1-q)^{k+1}} = \sum_{n=0}^{\infty} g_a(n)q^n.$$

So it is clear that $g_{a+1}(n) = g_a(n) + g_a(n-1)$ ($n \geq 1$), $g_a(0) = 1$. Call a sequence of positive numbers $g(0)$, $g(1)$, $g(2)$, ... good if $\frac{g(n-1)}{g(n)}$ ($n = 1, 2, \dots$) is an increasing sequence. It is straightforward to check that g_0 is good:

$$g_0(n) = \binom{k+n}{k}, \qquad \frac{g_0(n-1)}{g_0(n)} = \frac{n}{k+n}.$$

If g is a good sequence then a new sequence g' defined by $g'(0) = g(0)$, $g'(n) = g(n) + g(n-1)$ ($n \geq 1$) is also good:

$$\frac{g'(n-1)}{g'(n)} = \frac{g(n-1) + g(n-2)}{g(n) + g(n-1)} = \frac{1 + \frac{g(n-2)}{g(n-1)}}{1 + \frac{g(n)}{g(n-1)}},$$

where define $g(-1) = 0$. Thus we see that each of the sequences g_1, g_2, ..., $g_k = f$ are good. So the desired inequality holds.

2016-D2-P10 — Statement

Day 2, P10 | 10 points

Let A be a $n \times n$ complex matrix whose eigenvalues have absolute value at most 1. Prove that

$$\|A^n\| \leq \frac{n}{\ln 2}\|A\|^{n-1}.$$

(Here $\|B\| = \sup_{\|x\|\leq 1} \|Bx\|$ for every $n \times n$ matrix B and $\|x\| = \sqrt{\sum_{i=1}^{n} |x_i|^2}$ for every complex vector $x \in \mathbb{C}^n$.)

(Proposed by Ian Morris and Fedor Petrov, St. Petersburg State University)

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Theory and methods for 2016-D2-P10

Theory Operator norms and convergence of matrix powers [T-LA-NORM]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]

Methods Combine norm bounds with algebraic decay [M-LA-NORM]; Exploit a compatible block decomposition [M-LA-BLOCK]

Strategy Build a chain of bounds and track equality [S-BOUND]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2016-D2-P10 — Solution — Record 1 of 2

For this record: Theory: T-LA-NORM; T-LA-POLYNOMIAL; T-AL-POLYNOMIAL; T-AN-INEQUALITIES. Methods: M-LA-NORM. Full learning route

Let $r = \|A\|$. We have to prove $\|A^n\| \leq \frac{n}{\ln 2} r^{n-1}$.
As is well-known, the matrix norm satisfies $\|XY\| \leq \|X\| \cdot \|Y\|$ for any matrices X, Y, and as a simple consequence, $\|A^k\| \leq \|A\|^k = r^k$ for every positive integer k.

Let $\chi(t) = (t-\lambda_1)(t-\lambda_2)\dots(t-\lambda_n) = t^n + c_1 t^{n-1} + \dots + c_n$ be the characteristic polynomial of A. From Vieta's formulas we get

$$|c_k| = \Big|\sum_{1\leq i_1<\dots<i_k\leq n} \lambda_{i_1}\cdots\lambda_{i_k}\Big| \leq \sum_{1\leq i_1<\dots<i_k\leq n} |\lambda_{i_1}\cdots\lambda_{i_k}| \leq \binom{n}{k} \qquad (k = 1, 2, \dots, n)$$

By the Cayley–Hamilton theorem we have $\chi(A) = 0$, so

$$\|A^n\| = \|c_1 A^{n-1} + \dots + c_n\| \leq \sum_{k=1}^{n} \binom{n}{k} \|A^k\| \leq \sum_{k=1}^{n} \binom{n}{k} r^k = (1+r)^n - r^n.$$

Combining this with the trivial estimate $\|A^n\| \leq r^n$, we have

$$\|A^n\| \leq \min(r^n, (1+r)^n - r^n)).$$

Let $r_0 = \frac{1}{\sqrt[n]{2}-1}$; it is easy to check that the two bounds are equal if $r = r_0$, moreover

$$r_0 = \frac{1}{e^{\ln 2/n} - 1} < \frac{n}{\ln 2}.$$

For $r \leq r_0$ apply the trivial bound:

$$\|A^n\| \leq r^n \leq r_0 \cdot r^{n-1} < \frac{n}{\ln 2} r^{n-1}.$$

For $r > r_0$ we have

$$\|A^n\| \leq (1+r)^n - r^n = r^{n-1} \cdot \frac{(1+r)^n - r^n}{r^{n-1}}.$$

2016-D2-P10 — Solution — Record 1 of 2 (continued)

Notice that the function $f(r) = \frac{(1+r)^n - r^n}{r^{n-1}}$ is decreasing because the numerator has degree $n-1$ and all coefficients are positive, so

$$\frac{(1+r)^n - r^n}{r^{n-1}} < \frac{(1+r_0)^n - r_0^n}{r_0^{n-1}} = r_0((1+1/r_0)^n - 1) = r_0 < \frac{n}{\ln 2},$$

so $\|A^n\| < \frac{n}{\ln 2} r^{n-1}$.

2016-D2-P10 — Solution — Record 2 of 2

For this record: Theory: T-LA-NORM; T-LA-SPECTRAL. Methods: M-LA-NORM; M-LA-BLOCK. Full learning route

We will use the following facts which are easy to prove:

- For any square matrix A there exists a unitary matrix U such that UAU^{-1} is upper-triangular.
- For any matrices A, B we have $\|A\| \le \|(A|B)\|$ and $\|B\| \le \|(A|B)\|$ where $(A|B)$ is the matrix whose columns are the columns of A and the columns of B.
- For any matrices A, B we have $\|A\| \le \left\|\binom{A}{B}\right\|$ and $\|B\| \le \left\|\binom{A}{B}\right\|$ where $\binom{A}{B}$ is the matrix whose rows are the rows of A and the rows of B.
- Adding a zero row or a zero column to a matrix does not change its norm.

We will prove a stronger inequality

$$\|A^n\| \le n\|A\|^{n-1}$$

for any $n \times n$ matrix A whose eigenvalues have absolute value at most 1. We proceed by induction on n. The case $n = 1$ is trivial. Without loss of generality we can assume that the matrix A is upper-triangular. So we have

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix}.$$

Note that the eigenvalues of A are precisely the diagonal entries. We split A as the sum of 3 matrices, $A = X + Y + Z$ as follows:

$$X = \begin{pmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \cdots & \cdots & \cdots & \\ 0 & 0 & \cdots & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & a_{12} & \cdots & a_{1n} \\ 0 & 0 & \cdots & 0 \\ \cdots & \cdots & \cdots & \\ 0 & 0 & \cdots & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix}.$$

Denote by A' the matrix obtained from A by removing the first row and the first column:

$$A' = \begin{pmatrix} a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \\ 0 & \cdots & a_{nn} \end{pmatrix}.$$

We have $\|X\| \le 1$ because $|a_{11}| \le 1$. We also have

$$\|A'\| = \|Z\| \le \|Y + Z\| \le \|A\|.$$

Now we decompose A^n as follows:

$$A^n = XA^{n-1} + (Y + Z)A^{n-1}.$$

We substitute $A = X + Y + Z$ in the second term and expand the parentheses. Because of the following identities:

$$Y^2 = 0, \quad YX = 0, \quad ZY = 0, \quad ZX = 0$$

only the terms YZ^{n-1} and Z^n survive. So we have

$$A^n = XA^{n-1} + (Y + Z)Z^{n-1}.$$

By the induction hypothesis we have $\|A'^{n-1}\| \le (n-1)\|A'\|^{n-2}$, hence $\|Z^{n-1}\| \le (n-1)\|Z\|^{n-2} \le (n-1)\|A\|^{n-2}$. Therefore

$$\|A^n\| \le \|XA^{n-1}\| + \|(Y + Z)Z^{n-1}\| \le \|A\|^{n-1} + (n-1)\|Y + Z\|\|A\|^{n-2} \le n\|A\|^{n-1}.$$

3.24 2017

Day 1

2017-D1-P1 — Statement

Day 1, P1 | 10 points

Determine all complex numbers λ for which there exist a positive integer n and a real $n \times n$ matrix A such that $A^2 = A^T$ and λ is an eigenvalue of A.

(Proposed by Alexandr Bolbot, Novosibirsk State University)

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Theory and methods for 2017-D1-P1

Theory Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]
Methods Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2017-D1-P1 — Solution

For this record: Theory: T-LA-POLYNOMIAL; T-LA-SPECTRAL. Methods: M-LA-SPECTRAL. Full learning route

By taking squares,

$$A^4 = (A^2)^2 = (A^T)^2 = (A^2)^T = (A^T)^T = A,$$

so

$$A^4 - A = 0;$$

it follows that all eigenvalues of A are roots of the polynomial $X^4 - X$.
The roots of $X^4 - X = X(X^3 - 1)$ are 0, 1 and $\frac{-1 \pm \sqrt{3}i}{2}$. In order to verify that these values are possible, consider the matrices

$$A_0 = (0), \quad A_1 = (1), \quad A_2 = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}, \quad A_4 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ 0 & 0 & -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}.$$

The numbers 0 and 1 are the eigenvalues of the 1×1 matrices A_0 and A_1, respectively. The numbers $\frac{-1 \pm \sqrt{3}i}{2}$ are the eigenvalues of A_2; it is easy to check that

$$A_2^2 = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} = A_2^T.$$

The matrix A_4 establishes all the four possible eigenvalues in a single matrix.

Remark. The matrix A_2 represents a rotation by $2\pi/3$.

2017-D1-P2 — Statement

Day 1, P2 | 10 points

Let $f : \mathbb{R} \to (0, \infty)$ be a differentiable function, and suppose that there exists a constant $L > 0$ such that

$$\left|f'(x) - f'(y)\right| \leq L\left|x - y\right|$$

2017-D1-P2 — Statement (continued)

for all x, y. Prove that

$$\big(f'(x)\big)^2 < 2Lf(x)$$

holds for all x.

(Proposed by Jan Šustek, University of Ostrava)

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Theory and methods for 2017-D1-P2

Theory Lipschitz derivatives and convex smoothness [T-AN-SMOOTH]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Methods Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]; Choose the correct convexity comparison [M-AN-CONVEX]
Strategy Build a chain of bounds and track equality [S-BOUND]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2017-D1-P2 — Solution

For this record: Theory: T-AN-SMOOTH; T-AN-INTEGRAL; T-AN-DERIVATIVE. Methods: M-AN-PRIMITIVE; M-AN-CONVEX.
Full learning route

Notice that f' satisfies the Lipschitz-property, so f' is continuous and therefore locally integrable. Consider an arbitrary $x \in \mathbb{R}$ and let $d = f'(x)$. We need to prove $f(x) > \frac{d^2}{2L}$.
If $d = 0$ then the statement is trivial.

If $d > 0$ then the condition provides $f'(x-t) \geq d - Lt$; this estimate is positive for $0 \leq t < \frac{d}{L}$. By integrating over that interval,

$$f(x) > f(x) - f\left(x - \tfrac{d}{L}\right) = \int_0^{d/L} f'(x-t)\,dt \geq \int_0^{d/L} (d - Lt)\,dt = \frac{d^2}{2L}.$$

If $d < 0$ then apply $f'(x+t) \leq d + Lt = -|d| + Lt$ and repeat the same argument as

$$f(x) > f(x) - f\left(x + \tfrac{|d|}{L}\right) = \int_0^{|d|/L} \left(-f'(x+t)\right)dt \geq \int_0^{|d|/L} (|d| - Lt)\,dt = \frac{d^2}{2L}.$$

2017-D1-P3 — Statement

Day 1, P3 | 10 points

For any positive integer m, denote by $P(m)$ the product of positive divisors of m (e.g. $P(6) = 36$). For every positive integer n define the sequence

$$a_1(n) = n, \qquad a_{k+1}(n) = P(a_k(n)) \quad (k = 1, 2, \ldots, 2016).$$

Determine whether for every set $S \subseteq \{1, 2, \ldots, 2017\}$, there exists a positive integer n such that the following condition is satisfied:
For every k with $1 \leq k \leq 2017$, the number $a_k(n)$ is a perfect square if and only if $k \in S$.

(Proposed by Matko Ljulj, University of Zagreb)

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Theory and methods for 2017-D1-P3

Theory Prime valuations, divisors, and factorials [T-AL-VALUATION]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Methods Choose a modulus that exposes the obstruction [M-AL-MODULAR]; Find a recurrence that reveals the invariant [M-AL-RECURRENCE]
Strategy Build a witness and verify every constraint [S-CONSTRUCTION]; Reduce size while preserving the defining conditions [S-INDUCTION]

Statement Recorded solutions Catalog

2017-D1-P3 — Solution

For this record: Theory: T-AL-VALUATION; T-AL-CONGRUENCE; T-DS-INDUCTION. Methods: M-AL-MODULAR; M-AL-RECURRENCE. Full learning route

We prove that the answer is yes; for every $S \subset \{1, 2, \dots, 2017\}$ there exists a suitable n. Specially, n can be a power of 2: $n = 2^{w_1}$ with some nonnegative integer w_1. Write $a_k(n) = 2^{w_k}$; then

$$2^{w_{k+1}} = a_{k+1}(n) = P(a_k(n)) = P(2^{w_k}) = 1 \cdot 2 \cdot 4 \cdots 2^{w_k} = 2^{\frac{w_k(w_k+1)}{2}},$$

so

$$w_{k+1} = \frac{w_k(w_k+1)}{2}.$$

The proof will be completed if we prove that for each choice of S there exists an initial value w_1 such that w_k is even if and only if $k \in S$.
Lemma. Suppose that the sequences $(b_1, b_2, \dots)$ and $(c_1, c_2, \dots)$ of nonnegative integers satisfy $b_{k+1} = \frac{b_k(b_k+1)}{2}$ and $c_{k+1} = \frac{c_k(c_k+1)}{2}$ for $k \geq 1$, and $c_1 = b_1 + 2^m$ for an integer $m \geq 0$. Then for each $k = 1, \dots, m+1$ we have $c_k \equiv b_k + 2^{m-k+1} \pmod{2^{m-k+2}}$.
As an immediate corollary, we have $b_k \equiv c_k \pmod 2$ for $1 \leq k \leq m$ and $b_{m+1} \equiv c_{m+1} + 1 \pmod 2$.
Proof. We prove the claim by induction. For $k = 1$ we have $c_1 = b_1 + 2^m$ so the statement holds. Suppose the statement is true for some $k \leq m$, then for $k+1$ we have

$$\begin{aligned} c_{k+1} &= \frac{c_k(c_k+1)}{2} \\ &\equiv \frac{(b_k + 2^{m-k+1})(b_k + 2^{m-k+1} + 1)}{2} \\ &= \frac{b_k^2 + 2^{m-k+2}b_k + 2^{2m-2k+2} + b_k + 2^{m-k+1}}{2} \\ &= \frac{b_k(b_k+1)}{2} + 2^{m-k} + 2^{m-k+1}b_k + 2^{2m-2k+1} \\ &\equiv \frac{b_k(b_k+1)}{2} + 2^{m-k} \pmod{2^{m-k+1}}. \end{aligned}$$

therefore $c_{k+1} \equiv b_{k+1} + 2^{m-(k+1)+1} \pmod{2^{m-(k+1)+2}}$.
Going back to the solution of the problem, for every $1 \leq m \leq 2017$ we construct inductively a sequence $(v_1, v_2, \dots)$ such that $v_{k+1} = \frac{v_k(v_k+1)}{2}$, and for every $1 \leq k \leq m$, v_k is even if and only if $k \in S$. For $m = 1$ we can choose $v_1 = 0$ if $1 \in S$ or $v_1 = 1$ if $1 \notin S$. If we already have such a sequence $(v_1, v_2, \dots)$ for a positive integer m, we can choose either the same sequence or choose $v_1' = v_1 + 2^m$ and apply the same recurrence $v_{k+1}' = \frac{v_k'(v_k'+1)}{2}$. By the Lemma, we have $v_k \equiv v_k' \pmod 2$ for $k \leq m$, but v_{m+1} and v_{m+1}' have opposite parities; hence, either the sequence (v_k) or the sequence (v_k') satisfies the condition for $m+1$.

Repeating this process for $m = 1, 2, \dots, 2016$, we obtain a suitable sequence (w_k).

2017-D1-P4 — Statement

Day 1, P4 | 10 points

There are n people in a city, and each of them has exactly 1000 friends (friendship is always symmetric). Prove that it is possible to select a group S of people such that at least $n/2017$ persons in S have exactly two friends in S.

(Proposed by Rooholah Majdodin and Fedor Petrov, St. Petersburg State University)

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Theory and methods for 2017-D1-P4

Theory Finite probability, conditioning, and expectations [T-AN-PROBABILITY]; Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Random selection and the expected number of successes [M-DS-EXPECTATION]
Strategy Build a witness and verify every constraint [S-CONSTRUCTION]; Choose the objects and count them without ambiguity [S-COUNTING]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2017-D1-P4 — Solution

For this record: Theory: T-AN-PROBABILITY; T-DS-GRAPH; T-DS-COUNTING. Methods: M-DS-EXPECTATION. Full learning route

Let $d = 1000$ and let $0 < p < 1$. Choose the set S randomly such that each people is selected with probability p, independently from the others.

The probability that a certain person is selected for S and knows exactly two members of S is

$$q = \binom{d}{2} p^3 (1-p)^{d-2}.$$

Choose $p = 3/(d+1)$ (this is the value of p for which q is maximal); then

$$q = \binom{d}{2}\left(\frac{3}{d+1}\right)^3\left(\frac{d-2}{d+1}\right)^{d-2} =$$
$$= \frac{27d(d-1)}{2(d+1)^3}\left(1+\frac{3}{d-2}\right)^{-(d-2)} > \frac{27d(d-1)}{2(d+1)^3}\cdot e^{-3} > \frac{1}{2017}.$$

Let X be the number of persons in S having exactly two friends in S. Summing their indicator variables and using linearity of expectation gives

$$\mathsf{E}X = nq > \frac{n}{2017}.$$

Therefore some choice of S has $X > n/2017$, as required.

2017-D1-P5 — Statement

Day 1, P5 | 10 points

Let k and n be positive integers with $n \geq k^2 - 3k + 4$, and let

$$f(z) = z^{n-1} + c_{n-2}z^{n-2} + \cdots + c_0$$

be a polynomial with complex coefficients such that

$$c_0 c_{n-2} = c_1 c_{n-3} = \cdots = c_{n-2}c_0 = 0.$$

Prove that $f(z)$ and $z^n - 1$ have at most $n - k$ common roots.

(Proposed by Vsevolod Lev and Fedor Petrov, St. Petersburg State University)

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Theory and methods for 2017-D1-P5

Theory Roots-of-unity filters and positive trigonometric kernels [T-AL-FOURIER]; Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Evaluate at roots of unity or average over them [M-AL-UNITY]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2017-D1-P5 — Solution

For this record: Theory: T-AL-FOURIER; T-AL-COMPLEX; T-DS-COUNTING. Methods: M-AL-UNITY. Full learning route

Let $M=\{z : z^n=1\}$, $A=\{z\in M : f(z)\neq 0\}$ and $A^{-1}=\{z^{-1} : z\in A\}$. We have to prove $|A|\geq k$.
Claim.

$$A\cdot A^{-1}=M.$$

That is, for any $\eta\in M$, there exist some elements $a,b\in A$ such that $ab^{-1}=\eta$.
Proof. As is well-known, for every integer m,

$$\sum_{z\in M} z^m=\begin{cases} n & \text{if } n|m \\ 0 & \text{otherwise.}\end{cases}$$

Define $c_{n-1}=1$ and consider

$$\sum_{z\in M} z^2 f(z)f(\eta z)=\sum_{z\in M} z^2\sum_{j=0}^{n-1} c_j z^j\sum_{\ell=0}^{n-1} c_\ell(\eta z)^\ell=\sum_{j=0}^{n-1}\sum_{\ell=0}^{n-1} c_j c_\ell\eta^\ell\sum_{z\in M} z^{j+\ell+2}=$$
$$=\sum_{j=0}^{n-1}\sum_{\ell=0}^{n-1} c_j c_\ell\eta^\ell\left\{\begin{matrix} n & \text{if } n|j+\ell+2 \\ 0 & \text{otherwise}\end{matrix}\right\}=c_{n-1}^2 n+\sum_{j=0}^{n-2} c_j c_{n-2-j}\eta^{n-2-j}n=n\neq 0.$$

Therefore there exists some $b\in M$ such that $f(b)\neq 0$ and $f(\eta b)\neq 0$, i.e. $b\in A$, and $a=\eta b\in A$, satisfying $ab^{-1}=\eta$.

By double-counting the elements of M, from the Claim we conclude

$$|A|(|A|-1)\geq \big|M\setminus\{1\}\big|=n-1\geq k^2-3k+3>(k-1)(k-2)$$

which shows $|A|>k-1$.

Day 2

2017-D2-P6 — Statement

Day 2, P6 | 10 points

Let $f:[0;+\infty)\to\mathbb{R}$ be a continuous function such that $\lim\limits_{x\to+\infty} f(x)=L$ exists (it may be finite or infinite). Prove that

$$\lim_{n\to\infty}\int_0^1 f(nx)\,dx=L.$$

(Proposed by Alexandr Bolbot, Novosibirsk State University)

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Theory and methods for 2017-D2-P6

Theory Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Scale separation and limit comparisons [T-AN-ASYMPTOTICS]
Methods Separate a main range from exceptional scales [M-AN-SPLIT]; Compare growth through derivatives [M-AN-RATIO]
Strategy Control the limiting operation before taking the limit [S-ANALYTIC]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2017-D2-P6 — Solution — Record 1 of 2

For this record: Theory: T-AN-CONVERGENCE; T-AN-INTEGRAL. Methods: M-AN-SPLIT. Full learning route

Case 1: L is finite. Take an arbitrary $\varepsilon > 0$. We construct a number $K \geq 0$ such that $\left|\int_0^1 f(nx)\,dx - L\right| < \varepsilon$.
Since $\lim_{x\to+\infty} f(x) = L$, there exists a $K_1 \geq 1$ such that $\left|f(x) - L\right| < \frac{\varepsilon}{2}$ for every $x \geq K_1$. Hence, for $n > K_1$ we have

$$\begin{aligned}
\left|\int_0^1 f(nx)\,dx - L\right| &= \left|\frac{1}{n}\int_0^n f(x)\,dx - L\right| \\
&= \frac{1}{n}\left|\int_0^n (f - L)\right| \\
&\leq \frac{1}{n}\int_0^n |f - L| \\
&= \frac{1}{n}\left(\int_0^{K_1} |f - L| + \int_{K_1}^n |f - L|\right) \\
&< \frac{1}{n}\left(\int_0^{K_1} |f - L| + \int_{K_1}^n \frac{\varepsilon}{2}\right) \\
&= \frac{1}{n}\int_0^{K_1} |f - L| + \frac{n - K_1}{n}\cdot\frac{\varepsilon}{2} \\
&< \frac{1}{n}\int_0^{K_1} |f - L| + \frac{\varepsilon}{2}.
\end{aligned}$$

If $n > K_2 = \frac{2}{\varepsilon}\int_0^{K_1} |f - L|$ then the first term is less than $\frac{\varepsilon}{2}$. Then for $n > K := \max(K_1, K_2)$ we have

$$\left|\int_0^1 f(nx)\,dx - L\right| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Case 2: $L = +\infty$. Take an arbitrary real M; we need a $K \geq 0$ such that $\int_0^1 f(nx)\,dx > M$ for every $n > K$.
Since $\lim_{x\to+\infty} f(x) = \infty$, there exists a $K_1 \geq 1$ such that $f(x) > M + 1$ for every $x \geq K_1$. Hence, for $n > K_1$ we have

$$\begin{aligned}
\int_0^1 f(nx)\,dx = \frac{1}{n}\int_0^n f(x)\,dx &= \frac{1}{n}\int_0^n f \\
&= \frac{1}{n}\left(\int_0^{K_1} f + \int_{K_1}^n f\right) \\
&> \frac{1}{n}\left(\int_0^{K_1} f + \int_{K_1}^n (M+1)\right) \\
&= \frac{1}{n}\left(\int_0^{K_1} f - K_1(M+1)\right) + M + 1.
\end{aligned}$$

2017-D2-P6 — Solution — Record 1 of 2 (continued)

If $n \geq K_2 := \left|\int_0^{K_1} f - K_1(M+1)\right|$ then the first term is at least -1. For $n > K := \max(K_1, K_2)$ we have $\int_0^1 f(nx)\,dx > M$.

Case 3: $L = -\infty$. We can repeat the steps in Case 2 for the function $-f$.

2017-D2-P6 — Solution — Record 2 of 2

For this record: Theory: T-AN-ASYMPTOTICS; T-AN-INTEGRAL. Methods: M-AN-RATIO. Full learning route

Let $F(x) = \int_0^x f$. For $t > 0$ we have

$$\int_0^1 f(tx)\,dx = \frac{F(t)}{t}.$$

Since $\lim_{t\to\infty} t = \infty$ in the denominator and $\lim_{t\to\infty} F'(t) = \lim_{t\to\infty} f(t) = L$, L'Hospital's rule proves $\lim_{t\to\infty} \frac{F(t)}{t} = \lim_{t\to\infty} \frac{F'(t)}{1} = \lim_{t\to\infty} \frac{f(t)}{1} = L$. Then it follows that $\lim_{n\to\infty} \frac{F(n)}{n} = L$.

2017-D2-P7 — Statement

Day 2, P7 | 10 points

Let $p(x)$ be a nonconstant polynomial with real coefficients. For every positive integer n, let

$$q_n(x) = (x+1)^n p(x) + x^n p(x+1).$$

Prove that there are only finitely many numbers n such that all roots of $q_n(x)$ are real.

(Proposed by Alexandr Bolbot, Novosibirsk State University)

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Theory and methods for 2017-D2-P7

Theory Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Translate root restrictions into symmetric coefficients [M-AL-VIETA]
Strategy Build a chain of bounds and track equality [S-BOUND]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

2017-D2-P7 — Solution

For this record: Theory: T-AL-ROOTS; T-AL-POLYNOMIAL. Methods: M-AL-VIETA. Full learning route

Lemma. If $f(x) = a_m x^m + \cdots + a_1 x + a_0$ is a polynomial with $a_m \neq 0$, and all roots of f are real, then

$$a_{m-1}^2 - 2a_m a_{m-2} \geq 0.$$

Proof. Let the roots of f be $w_1, \ldots, w_n$. By the Viète-formulas,

$$\sum_{i=1}^m w_i = -\frac{a_{m-1}}{a_m}, \qquad \sum_{i<j} w_i w_j = \frac{a_{m-2}}{a_m},$$

$$0 \leq \sum_{i=1}^m w_i^2 = \left(\sum_{i=1}^m w_i\right)^2 - 2\sum_{i<j} w_i w_j = \left(\frac{a_{m-1}}{a_m}\right)^2 - 2\frac{a_{m-2}}{a_m} = \frac{a_{m-1}^2 - 2a_m a_{m-2}}{a_m^2}.$$

2017-D2-P7 — Solution (continued)

In view of the Lemma we focus on the asymptotic behavior of the three terms in $q_n(x)$ with the highest degrees. Let $p(x) = ax^k + bx^{k-1} + cx^{k-2} + \dots$ and $q_n(x) = A_n x^{n+k} + B_n x^{n+k-1} + C_n x^{n+k-2} + \dots$; then

$$\begin{aligned} q_n(x) &= (x+1)^n p(x) + x^n p(x+1) = \\ &= \left(x^n + nx^{n-1} + \frac{n(n-1)}{2}x^{n-2} + \dots\right)(ax^k + bx^{k-1} + cx^{k-2} + \dots) \\ &\quad + x^n\left(a\left(x^k + kx^{k-1} + \frac{k(k-1)}{2}x^{k-2} + \dots\right)\right. \\ &\quad \left. + b(x^{k-1} + (k-1)x^{k-2} + \dots) + c(x^{k-2}\dots) + \dots\right) \\ &= 2a \cdot x^{n+k} + ((n+k)a + 2b)x^{n+k-1} \\ &\quad + \left(\frac{n(n-1)+k(k-1)}{2}a + (n+k-1)b + 2c\right)x^{n+k-2} + \dots, \end{aligned}$$

so

$$A_n = 2a, \quad B_n = (n+k)a + 2b = C_n = \frac{n(n-1)+k(k-1)}{2}a + (n+k-1)b + 2c.$$

If $n \to \infty$ then

$$B_n^2 - 2A_nC_n = (na + O(1))^2 - 2 \cdot 2a\left(\frac{n^2 a}{2} + O(n)\right) = -an^2 + O(n) \to -\infty,$$

so $B_n^2 - 2A_nC_n$ is eventually negative, indicating that q_n cannot have only real roots.

2017-D2-P8 — Statement

Day 2, P8 | 10 points

Define the sequence $A_1, A_2, \dots$ of matrices by the following recurrence:

$$A_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad A_{n+1} = \begin{pmatrix} A_n & I_{2^n} \\ I_{2^n} & A_n \end{pmatrix} \quad (n = 1, 2, \dots)$$

where I_m is the $m \times m$ identity matrix.
Prove that A_n has $n+1$ distinct integer eigenvalues $\lambda_0 < \lambda_1 < \cdots < \lambda_n$ with multiplicities $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$, respectively.

(Proposed by Snježana Majstorović, University of J. J. Strossmayer in Osijek, Croatia)

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Theory and methods for 2017-D2-P8

Theory Tensor products and operators on matrix spaces [T-LA-TENSOR]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Lift products of inner products to new vectors [M-LA-TENSOR]
Strategy Reduce size while preserving the defining conditions [S-INDUCTION]; Choose the objects and count them without ambiguity [S-COUNTING]

Statement Recorded solutions Catalog

2017-D2-P8 — Solution

For this record: Theory: T-LA-TENSOR; T-LA-SPECTRAL; T-DS-COUNTING. Methods: M-LA-TENSOR. Full learning route

For each $n \in \mathbb{N}$, matrix A_n is symmetric $2^n \times 2^n$ matrix with elements from the set $\{0, 1\}$, so that all elements on the main diagonal are equal to zero. We can write

$$A_n = I_{2^{n-1}} \otimes A_1 + A_{n-1} \otimes I_2, \tag{1}$$

2017-D2-P8 — Solution (continued)

where $\otimes$ is binary operation over the space of matrices, defined for arbitrary $B \in \mathbb{R}^{n\times p}$ and $C \in \mathbb{R}^{m\times s}$ as

$$B \otimes C := \begin{pmatrix} b_{11}C & b_{12}C & \dots & b_{1p}C \\ b_{21}C & b_{22}C & \dots & b_{2p}C \\ \vdots & & & \\ b_{n1}C & b_{12}C & \dots & b_{np}C \end{pmatrix}_{nm\times ps}.$$

Lemma 1. If $B \in \mathbb{R}^{n\times n}$ has eigenvalues λ_i, $i = 1, \dots, n$ and $C \in \mathbb{R}^{m\times m}$ has eigenvalues μ_j, $j = 1, \dots, m$, then $B \otimes C$ has eigenvalues $\lambda_i\mu_j$, $i = 1, \dots, n$, $j = 1, \dots, m$. If B and C are diagonalizable, then $A \otimes B$ has eigenvectors $y_i \otimes z_j$, with (λ_i, y_i) and (μ_j, z_j) being eigenpairs of B and C, respectively.
Proof 1. Let (λ, y) be an eigenpair of B and (μ, z) an eigenpar of C. Then

$$(B \otimes C)(y \otimes z) = By \otimes Cz = \lambda y \otimes \mu z = \lambda\mu(y \otimes z).$$

If we take (λ, y) to be an eigenpair of A_1 and (μ, z) to be an eigenpair of A_{n-1}, then from (1) and Lemma 1 we get

$$\begin{aligned} A_n(z \otimes y) &= (I_{2^{n-1}} \otimes A_1 + A_{n-1} \otimes I_2)(z \otimes y) \\ &= (I_{2^{n-1}} \otimes A_1)(z \otimes y) + (A_{n-1} \otimes I_2)(z \otimes y) \\ &= (\lambda + \mu)(z \otimes y). \end{aligned}$$

So the entire spectrum of A_n can be obtained from eigenvalues of A_{n-1} and A_1: just sum up each eigenvalue of A_{n-1} with each eigenvalue of A_1. Since the spectrum of A_1 is $\sigma(A_1) = \{-1, 1\}$, we get

$$\begin{aligned} \sigma(A_2) &= \{-1 + (-1), -1 + 1, 1 + (-1), 1 + 1\} = \{-2, 0^{(2)}, 2\} \\ \sigma(A_3) &= \{-1 + (-2), -1 + 0, -1 + 0, -1 + 2, 1 + (-2), 1 + 0, 1 + 0, 1 + 2\} = \{-3, (-1)^{(3)}, 1^{(3)}, 3\} \\ \sigma(A_4) &= \{-1 + (-3), -1 + (-1^{(3)}), -1 + 1^{(3)}, -1 + 3, 1 + (-3), 1 + (-1^{(3)}), 1 + 1^{(3)}, 1 + 3\} \\ &= \{-4, (-2)^{(4)}, 0^{(3)}, 2^{(4)}, 4\}. \end{aligned}$$

Inductively, A_n has $n + 1$ distinct integer eigenvalues $-n, -n + 2, -n + 4, \dots, n - 4, n - 2, n$ with multiplicities $\binom{n}{0}, \binom{n}{1}, \binom{n}{2}, \dots, \binom{n}{n}$, respectively.

2017-D2-P9 — Statement

Day 2, P9 | 10 points

Define the sequence $f_1, f_2, \dots : [0, 1) \to \mathbb{R}$ of continuously differentiable functions by the following recurrence:

$$f_1 = 1; \qquad f'_{n+1} = f_n f_{n+1} \text{ on } (0, 1), \text{ and } f_{n+1}(0) = 1.$$

Show that $\lim_{n\to\infty} f_n(x)$ exists for every $x \in [0, 1)$ and determine the limit function.

(Proposed by Tomáš Bárta, Charles University, Prague)

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Theory and methods for 2017-D2-P9

Theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]; Uniform approximation and positive kernels [T-AN-APPROXIMATION]
Methods Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]; Pass a monotone functional iteration through its integral operator [M-DS-VOLTERRA]
Strategy Control the limiting operation before taking the limit [S-ANALYTIC]; Reduce size while preserving the defining conditions [S-INDUCTION]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2017-D2-P9 — Solution

For this record: Theory: T-AN-DERIVATIVE; T-AN-INTEGRAL; T-AN-CONVERGENCE; T-AN-CONVEXITY; T-DS-DYNAMICS; T-AN-APPROXIMATION. Methods: M-DS-ITERATE; M-DS-VOLTERRA. Full learning route

First of all, the sequence f_n is well defined and it holds that

$$f_{n+1}(x) = e^{\int_0^x f_n(t)\,dt}. \tag{2}$$

The mapping $\Phi : C([0,1)) \to C([0,1))$ given by

$$\Phi(g)(x) = e^{\int_0^x g(t)\,dt}$$

is monotone, i.e. if $f < g$ on $(0,1)$ then

$$\Phi(f)(x) = e^{\int_0^x f(t)\,dt} < e^{\int_0^x g(t)\,dt} = \Phi(g)(x)$$

on $(0,1)$. Since $f_2(x) = e^{\int_0^x 1\,\mathrm{d}t} = e^x > 1 = f_1(x)$ on $(0,1)$, we have by induction $f_{n+1}(x) > f_n(x)$ for all $x \in (0,1)$, $n \in \mathbb{N}$. Moreover, function $f(x) = \frac{1}{1-x}$ is the unique solution to $f' = f^2$, $f(0) = 1$, i.e. it is the unique fixed point of Φ in $\{\varphi \in C([0,1)) : \varphi(0) = 1\}$. Since $f_1 < f$ on $(0,1)$, by induction we have $f_{n+1} = \Phi(f_n) < \Phi(f) = f$ for all $n \in \mathbb{N}$. Hence, for every $x \in (0,1)$ the sequence $f_n(x)$ is increasing and bounded, so a finite limit exists.

Let us denote the limit $g(x)$. We show that $g(x) = f(x) = \frac{1}{1-x}$. Obviously, $g(0) = \lim f_n(0) = 1$. By $f_1 \equiv 1$ and (2), we have $f_n > 0$ on $[0,1)$ for each $n \in \mathbb{N}$, and therefore (by (2) again) the function f_{n+1} is increasing. Since f_n, f_{n+1} are positive and increasing also f'_{n+1} is increasing (due to $f'_{n+1} = f_n f_{n+1}$), hence f_{n+1} is convex. A pointwise limit of a sequence of convex functions is convex, since we pass to a limit $n \to \infty$ in

$$f_n(\lambda x + (1-\lambda)y) \le \lambda f_n(x) + (1-\lambda) f_n(y)$$

and obtain

$$g(\lambda x + (1-\lambda)y) \le \lambda g(x) + (1-\lambda) g(y)$$

for any fixed $x, y \in [0,1)$ and $\lambda \in (0,1)$. Hence, g is convex, and therefore continuous on $(0,1)$. Moreover, g is continuous in 0, since $1 \equiv f_1 \le g \le f$ and $\lim_{x\to 0+} f(x) = 1$. By Dini's Theorem, convergence $f_n \to g$ is uniform on $[0, 1-\varepsilon]$ for each $\varepsilon \in (0,1)$ (a monotone sequence converging to a continuous function on a compact interval). We show that Φ is continuous and therefore f_n have to converge to a fixed point of Φ.

In fact, let us work on the space $C([0,1-\varepsilon])$ with any fixed $\varepsilon \in (0,1)$, $\|\cdot\|$ being the supremum norm on $[0,1-\varepsilon]$. Then for a fixed function h and $\|\varphi - h\| < \delta$ we have

$$\sup_{x\in[0,1-\varepsilon]} |\Phi(h)(x) - \Phi(\varphi)(x)| = \sup_{x\in[0,1-\varepsilon]} e^{\int_0^x h(t)\,dt} \left|1 - e^{\int_0^x \varphi(t)-h(t)\,dt}\right| \le C(e^\delta - 1) < 2C\delta$$

for $\delta > 0$ small enough. Hence, Φ is continuous on $C([0,1-\varepsilon])$. Let us assume for contradiction that $\Phi(g) \ne g$. Hence, there exists $\eta > 0$ and $x_0 \in [0,1-\varepsilon]$ such that $|\Phi(g)(x_0) - g(x_0)| > \eta$. There exists $\delta > 0$ such that $\|\Phi(\varphi) - \Phi(g)\| < \frac{1}{3}\eta$ whenever $\|\varphi - g\| < \delta$. Take n_0 so large that $\|f_n - g\| < \min\{\delta, \frac{1}{3}\eta\}$ for all $n \ge n_0$. Hence, $\|f_{n+1} - \Phi(g)\| = \|\Phi(f_n) - \Phi(g)\| < \frac{1}{3}\eta$. On the other hand, we have $|f_{n+1}(x_0) - \Phi(g)(x_0)| > |\Phi(g)(x_0) - g(x_0)| - |g(x_0) - f_{n+1}(x_0)| > \eta - \frac{1}{3}\eta = \frac{2}{3}\eta$, contradiction. So, $\Phi(g) = g$.

Since f is the only fixed point of Φ in $\{\varphi \in C([0,1-\varepsilon]) : \varphi(0) = 1\}$, we have $g = f$ on $[0,1-\varepsilon]$. Since $\varepsilon \in (0,1)$ was arbitrary, we have $\lim_{n\to\infty} f_n(x) = \frac{1}{1-x}$ for all $x \in [0,1)$.

2017-D2-P10 — Statement

Day 2, P10 | 10 points

Let K be an equilateral triangle in the plane. Prove that for every $p > 0$ there exists an $\varepsilon > 0$ with the following property: If n is a positive integer, and $T_1, \ldots, T_n$ are non-overlapping triangles inside K such that each of them is homothetic to K with a negative ratio, and

$$\sum_{\ell=1}^{n} \mathrm{area}(T_\ell) > \mathrm{area}(K) - \varepsilon,$$

2017-D2-P10 — Statement (continued)

then

$$\sum_{\ell=1}^{n} \text{perimeter}(T_\ell) > p.$$

(Proposed by Fedor Malyshev, Steklov Math. Inst. and Ilya Bogdanov, MIPT, Moscow)

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Theory and methods for 2017-D2-P10

Theory Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Methods Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]; Split a family by traces, pieces, or labels [M-DS-PARTITION]
Strategy Translate geometry into a controlled representation [S-GEOMETRIC]; Control the limiting operation before taking the limit [S-ANALYTIC]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2017-D2-P10 — Solution

For this record: Theory: T-DS-GEOMETRY; T-AN-INTEGRAL; T-AN-DERIVATIVE; T-AN-CONVERGENCE. Methods: M-DS-GEOMETRIC; M-DS-PARTITION. Full learning route

For an arbitrary $\varepsilon > 0$ we will establish a lower bound for the sum of perimeters that would tend to $+\infty$ as $\varepsilon \to +0$; this solves the problem.
Rotate and scale the picture so that one of the sides of K is the segment from $(0,0)$ to $(0,1)$, and stretch the picture horizontally in such a way that the projection of K to the x axis is $[0,1]$. Evidently, we may work with the lengths of the projections to the x or y axis instead of the perimeters and consider their sum, that is why we may make any affine transformation.
Let $f_i(a)$ be the length of intersection of the straight line $\{x = a\}$ with T_i and put $f(a) = \sum_i f_i(a)$. Then f is piece-wise increasing with possible downward gaps, $f(a) \le 1 - a$, and

$$\int_0^1 f(x)\,dx \ge \frac{1}{2} - \varepsilon.$$

Let $d_1, \dots, d_N$ be the values of the gaps of f. Every gap is a sum of side-lengths of some of T_i and every T_i contributes to one of d_j, we therefore estimate the sum of the gaps of f.
In the points of differentiability of f we have $f'(a) \ge f(a)/a$; this follows from $f_i'(a) \ge f_i(a)/a$ after summation. Indeed, if f_i is zero this inequality holds trivially, and if not then $f_i' = 1$ and the inequality reads $f_i(a) \le a$, which is clear from the definition.

Choose an integer $m = \lfloor 1/(8\varepsilon) \rfloor$ (considering ε sufficiently small). Then for all $k = 0, 1, \dots, [(m-1)/2]$ in the section of K by the strip $k/m \le x \le (k+1)/m$ the area, covered by the small triangles T_i is no smaller than $1/(2m) - \varepsilon \ge 1/(4m)$. Thus

$$\int_{k/m}^{(k+1)/m} f'(x)\,dx \ge \int_{k/m}^{(k+1)/m} \frac{f(x)\,dx}{x} \ge \frac{m}{k+1} \int_{k/m}^{(k+1)/m} f(x)\,dx \ge \frac{m}{k+1} \cdot \frac{1}{4m} = \frac{1}{4(k+1)}.$$

Hence,

$$\int_0^{1/2} f'(x)\,dx \ge \frac{1}{4}\left(\frac{1}{1} + \dots + \frac{1}{[(m-1)/2]}\right).$$

The right hand side tends to infinity as $\varepsilon \to +0$. On the other hand, the left hand side equals

$$f(1/2) + \sum_{x_i < 1/2} d_i;$$

hence $\sum_i d_i$ also tends to infinity.

3.25 2018

Day 1

2018-D1-P1 — Statement

Day 1, P1 | 10 points

Let $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ be two sequences of positive numbers. Show that the following statements are equivalent:

(1) There is a sequence $(c_n)_{n=1}^{\infty}$ of positive numbers such that $\sum_{n=1}^{\infty} \frac{a_n}{c_n}$ and $\sum_{n=1}^{\infty} \frac{c_n}{b_n}$ both converge;

(2) $\sum_{n=1}^{\infty} \sqrt{\frac{a_n}{b_n}}$ converges.

(Proposed by Tomáš Bárta, Charles University, Prague)

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Theory and methods for 2018-D1-P1

Theory Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Series, finite cancellation, and summation by parts [T-AN-SERIES]
Methods Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2018-D1-P1 — Solution

For this record: Theory: T-AN-INEQUALITIES; T-AN-SERIES. Methods: M-AN-WEIGHTS. Full learning route

Note that the sum of a series with positive terms can be either finite or $+\infty$, so for such a series, "converges" is equivalent to "is finite".
Proof for (1) $\Longrightarrow$ (2): By the AM-GM inequality,

$$\sqrt{\frac{a_n}{b_n}} = \sqrt{\frac{a_n}{c_n} \cdot \frac{c_n}{b_n}} \leq \frac{1}{2}\left(\frac{a_n}{c_n} + \frac{c_n}{b_n}\right),$$

so

$$\sum_{n=1}^{\infty} \sqrt{\frac{a_n}{b_n}} \leq \frac{1}{2}\sum_{n=1}^{\infty} \frac{a_n}{c_n} + \frac{1}{2}\sum_{n=1}^{\infty} \frac{c_n}{b_n} < +\infty.$$

Hence, $\sum_{n=1}^{\infty} \sqrt{\frac{a_n}{b_n}}$ is finite and therefore convergent.

Proof for (2) $\Longrightarrow$ (1): Choose $c_n = \sqrt{a_n b_n}$. Then

$$\frac{a_n}{c_n} = \frac{c_n}{b_n} = \sqrt{\frac{a_n}{b_n}}.$$

By the condition $\sum_{n=1}^{\infty} \sqrt{\frac{a_n}{b_n}}$ converges, therefore $\sum_{n=1}^{\infty} \frac{a_n}{c_n}$ and $\sum_{n=1}^{\infty} \frac{c_n}{b_n}$ converge, too.

2018-D1-P2 — Statement

Day 1, P2 | 10 points

Does there exist a field such that its multiplicative group is isomorphic to its additive group?

(Proposed by Alexandre Chapovalov, New York University, Abu Dhabi)

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Theory and methods for 2018-D1-P2

Theory Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Groups, orders, generators, and quotients [T-AL-GROUP]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Prove generation or obstruct it by an image [M-AL-GENERATORS]; Replace many equal values by too many roots [M-AL-ROOTCOUNT]
Strategy Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

2018-D1-P2 — Solution

For this record: Theory: T-AL-FINITEFIELD; T-AL-GROUP; T-AL-POLYNOMIAL. Methods: M-AL-GENERATORS; M-AL-ROOTCOUNT. Full learning route

There exist no such field.
Suppose that F is such a field and $g : F^* \to F^+$ is a group isomorphism. Then $g(1) = 0$.
Let $a = g(-1)$. Then $2a = 2 \cdot g(-1) = g((-1)^2) = g(1) = 0$; so either $a = 0$ or $\operatorname{char} F = 2$. If $a = 0$ then $-1 = g^{-1}(a) = g^{-1}(0) = 1$; we have $\operatorname{char} F = 2$ in any case.

For every $x \in F$, we have $g(x^2) = 2g(x) = 0 = g(1)$, so $x^2 = 1$. But this equation has only one or two solutions. Hence F is the 2-element field; but its additive and multiplicative groups have different numbers of elements and are not isomorphic.

2018-D1-P3 — Statement

Day 1, P3 | 10 points

Determine all rational numbers a for which the matrix

$$\begin{pmatrix} a & -a & -1 & 0 \\ a & -a & 0 & -1 \\ 1 & 0 & a & -a \\ 0 & 1 & a & -a \end{pmatrix}$$

is the square of a matrix with all rational entries.

(Proposed by Daniël Kroes, University of California, San Diego)

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Theory and methods for 2018-D1-P3

Theory Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Determinants, block identities and inverses [T-LA-DETERMINANT]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]
Strategy Make the contradiction quantitative or structural [S-CONTRADICTION]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2018-D1-P3 — Solution

For this record: Theory: T-LA-POLYNOMIAL; T-LA-DETERMINANT; T-AL-POLYNOMIAL. Methods: M-LA-SPECTRAL. Full learning route

We will show that the only such number is $a = 0$.

Let $A = \begin{pmatrix} a & -a & -1 & 0 \\ a & -a & 0 & -1 \\ 1 & 0 & a & -a \\ 0 & 1 & a & -a \end{pmatrix}$ and suppose that $A = B^2$. It is easy to compute the characteristic polynomial of A, which is

$$p_A(x) = \det(A - xI) = (x^2 + 1)^2.$$

By the Cayley-Hamilton theorem we have $p_A(B^2) = p_A(A) = 0$.

2018-D1-P3 — Solution (continued)

Let $\mu_B(x)$ be the minimal polynomial of B. The minimal polynomial divides all polynomials that vanish at B; in particular $\mu_B(x)$ must be a divisor of the polynomial $p_A(x^2) = (x^4+1)^2$. The polynomial $\mu_B(x)$ has rational coefficients and degree at most 4. On the other hand, the polynomial x^4+1, being the 8th cyclotomic polynomial, is irreducible in $\mathbb{Q}[x]$. Hence the only possibility for μ_B is $\mu_B(x) = x^4+1$. Therefore,

$$A^2 + I = \mu_B(B) = 0. \tag{1}$$

Since we have

$$A^2 + I = \begin{pmatrix} 0 & 0 & -2a & 2a \\ 0 & 0 & -2a & 2a \\ 2a & -2a & 0 & 0 \\ 2a & -2a & 0 & 0 \end{pmatrix},$$

the relation (1) forces $a = 0$.

In case $a = 0$ we have

$$A = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}^2,$$

hence $a = 0$ satisfies the condition.

2018-D1-P4 — Statement

Day 1, P4 | 10 points

Find all differentiable functions $f : (0,\infty) \to \mathbb{R}$ such that

$$f(b) - f(a) = (b-a) f'(\sqrt{ab}) \quad \text{for all } a, b > 0. \tag{2}$$

(Proposed by Orif Ibrogimov, National University of Uzbekistan)

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Theory and methods for 2018-D1-P4

Theory Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Methods Force structure by admissible substitutions [M-DS-FUNCTIONAL]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2018-D1-P4 — Solution

For this record: Theory: T-DS-FUNCTIONAL; T-AN-DERIVATIVE. Methods: M-DS-FUNCTIONAL. Full learning route

First we show that f is infinitely many times differentiable. By substituting $a = \frac{1}{2}t$ and $b = 2t$ in (2),

$$f'(t) = \frac{f(2t) - f(\frac{1}{2}t)}{\frac{3}{2}t}. \tag{3}$$

Inductively, if f is k times differentiable then the right-hand side of (3) is k times differentiable, so the $f'(t)$ on the left-hand-side is k times differentiable as well; hence f is $k+1$ times differentiable.

2018-D1-P4 — Solution (continued)

Now substitute $b = e^h t$ and $a = e^{-h} t$ in (2), differentiate three times with respect to h then take limits with $h \to 0$:

$$f(e^h t) - f(e^{-h} t) - (e^h t - e^{-h} t) f'(t) = 0$$

$$\left(\frac{\partial}{\partial h}\right)^3 \left(f(e^h t) - f(e^{-h} t) - (e^h t - e^{-h} t) f'(t)\right) = 0$$

$$e^{3h} t^3 f'''(e^h t) + 3e^{2h} t^2 f''(e^h t) + e^h t f'(e^h t) + e^{-3h} t^3 f'''(e^{-h} t) + 3e^{-2h} t^2 f''(e^{-h} t) + e^{-h} t f'(e^{-h} t) -$$
$$- (e^h t + e^{-h} t) f'(t) = 0$$

$$2t^3 f'''(t) + 6t^2 f''(t) = 0$$

$$t f'''(t) + 3f''(t) = 0$$

$$(t f(t))''' = 0.$$

Consequently, $tf(t)$ is an at most quadratic polynomial of t, and therefore

$$f(t) = C_1 t + \frac{C_2}{t} + C_3 \tag{4}$$

with some constants C_1, C_2 and C_3.

It is easy to verify that all functions of the form (4) satisfy the equation (2).

2018-D1-P5 — Statement

Day 1, P5 | 10 points

Let p and q be prime numbers with $p < q$. Suppose that in a convex polygon $P_1 P_2 \ldots P_{pq}$ all angles are equal and the side lengths are distinct positive integers. Prove that

$$P_1 P_2 + P_2 P_3 + \cdots + P_k P_{k+1} \geq \frac{k^3 + k}{2}$$

holds for every integer k with $1 \leq k \leq p$.

(Proposed by Ander Lamaison Vidarte, Berlin Mathematical School, Berlin)

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Theory and methods for 2018-D1-P5

Theory Roots of unity, cyclotomic polynomials, and algebraic norms [T-AL-CYCLOTOMIC]; Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Evaluate at roots of unity or average over them [M-AL-UNITY]; Construct and count compatible local residues [M-AL-CRT]
Strategy Translate geometry into a controlled representation [S-GEOMETRIC]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2018-D1-P5 — Solution

For this record: Theory: T-AL-CYCLOTOMIC; T-AL-COMPLEX; T-AL-CONGRUENCE; T-DS-COUNTING. Methods: M-AL-UNITY; M-AL-CRT. Full learning route

Place the polygon in the complex plane counterclockwise, so that $P_2 - P_1$ is a positive real number. Let $a_i = |P_{i+2} - P_{i+1}|$, which is an integer, and define the polynomial $f(x) = a_{pq-1} x^{pq-1} + \cdots + a_1 x + a_0$. Let $\omega = e^{\frac{2\pi i}{pq}}$; then $P_{i+1} - P_i = a_{i-1} \omega^{i-1}$, so $f(\omega) = 0$.

The minimal polynomial of ω over $\mathbb{Q}[x]$ is the cyclotomic polynomial $\Phi_{pq}(x) = \frac{(x^{pq}-1)(x-1)}{(x^p-1)(x^q-1)}$, so $\Phi_{pq}(x)$ divides $f(x)$. At the same time, $\Phi_{pq}(x)$ is the greatest common divisor of $s(x) = \frac{x^{pq}-1}{x^p-1} = \Phi_q(x^p)$ and $t(x) = \frac{x^{pq}-1}{x^q-1} = \Phi_p(x^q)$, so by Bézout's identity (for real polynomials), we can write $f(x) =$

2018-D1-P5 — Solution (continued)

$s(x)u(x)+t(x)v(x)$, with some polynomials $u(x), v(x)$. These polynomials can be replaced by $u^*(x) = u(x) + w(x)\frac{x^p-1}{x-1}$ and $v^*(x) = v(x) - w(x)\frac{x^q-1}{x-1}$, so without loss of generality we may assume that $\deg u \le p-1$. Since $\deg a = pq-1$, this forces $\deg v \le q-1$.

Let $u(x) = u_{p-1}x^{p-1} + \cdots + u_1x + u_0$ and $v(x) = v_{q-1}x^{q-1} + \cdots + v_1x + v_0$. Denote by (i,j) the unique integer $n \in \{0,1,\dots,pq-1\}$ with $n \equiv i \pmod p$ and $n \equiv j \pmod q$. By the choice of s and t, we have $a_{(i,j)} = u_i + v_j$. Then

$$P_1P_2 + \cdots + P_kP_{k+1} = \sum_{i=0}^{k-1} a_{(i,i)} = \sum_{i=0}^{k-1}(u_i+v_i) = \frac{1}{k}\sum_{i=0}^{k-1}\sum_{j=0}^{k-1}(u_i+v_j)$$
$$= \frac{1}{k}\sum_{i=0}^{k-1}\sum_{j=0}^{k-1} a_{(i,j)} \overset{(*)}{\ge} \frac{1}{k}(1+2+\cdots+k^2) = \frac{k^3+k}{2}$$

where $(*)$ uses the fact that the numbers (i,j) are pairwise different.

Day 2

2018-D2-P6 — Statement

Day 2, P6 | 10 points

Let k be a positive integer. Find the smallest positive integer n for which there exist k nonzero vectors $v_1,\dots,v_k$ in $\mathbb{R}^n$ such that for every pair i,j of indices with $|i-j|>1$ the vectors v_i and v_j are orthogonal.

(Proposed by Alexey Balitskiy, Moscow Institute of Physics and Technology and M.I.T.)

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Theory and methods for 2018-D2-P6

Theory Inner products, orthogonality and polarization [T-LA-INNER]; Rank, dimension and exact linear constraints [T-LA-RANK]
Methods Prove a dimension obstruction and attain it [M-LA-DIMENSION]
Strategy Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2018-D2-P6 — Solution

For this record: Theory: T-LA-INNER; T-LA-RANK. Methods: M-LA-DIMENSION. Full learning route

First we prove that if $2n+1 \le k$ then no sequence $v_1,\dots,v_k$ of vectors can satisfy the condition. Suppose to the contrary that $v_1,\dots,v_k$ are vectors with the required property and consider the vectors

$$v_1, v_3, v_5, \dots, v_{2n+1}.$$

By the condition these $n+1$ vectors should be pairwise orthogonal, but this is not possible in $\mathbb{R}^n$. Next we show a possible construction for every pair k,n of positive integers with $2n \ge k$. Take an orthogonal basis $(e_1,\dots,e_n)$ of $\mathbb{R}^n$ and consider the vectors

$$v_1 = v_2 = e_1, \quad v_3 = v_4 = e_2, \quad \dots, \quad v_{2n-1} = v_{2n} = e_n.$$

For every pair (i,j) of indices with $1 \le i,j \le 2n$ and $|i-j|>1$ the vectors v_i and v_j are distinct basis vectors, so they are orthogonal. Evidently the subsequence $v_1, v_2, \dots, v_k$ also satisfies the same property.

Hence, such a sequence of vectors exists if and only if $2n \ge k$; that is, for a fixed k, the smallest suitable n is $\left\lceil \frac{k}{2} \right\rceil$.

2018-D2-P7 — Statement

Day 2, P7 | 10 points

Let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers such that $a_0 = 0$ and

$$a_{n+1}^3 = a_n^2 - 8 \quad \text{for } n = 0, 1, 2, \ldots$$

Prove that the following series is convergent:

$$\sum_{n=0}^{\infty} |a_{n+1} - a_n|. \tag{1}$$

(Proposed by Orif Ibrogimov, National University of Uzbekistan)

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Theory and methods for 2018-D2-P7

Theory Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]; Strengthen the statement before the induction step [M-DS-INDUCT]
Strategy Build a chain of bounds and track equality [S-BOUND]; Reduce size while preserving the defining conditions [S-INDUCTION]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2018-D2-P7 — Solution

For this record: Theory: T-DS-DYNAMICS; T-AN-CONVERGENCE; T-AL-POLYNOMIAL. Methods: M-DS-ITERATE; M-DS-INDUCT. Full learning route

We will estimate the ratio between the terms $|a_{n+2} - a_{n+1}|$ and $|a_{n+1} - a_n|$.
Before doing that, we localize the numbers a_n; we prove that

$$-2 \le a_n \le -\sqrt[3]{4} \quad \text{for } n \ge 1. \tag{2}$$

The lower bound simply follows from the recurrence: $a_n = \sqrt[3]{a_{n-1}^2 - 8} \ge \sqrt[3]{-8} = -2$. The proof of the upper bound can be done by induction: we have $a_1 = -2 < -\sqrt[3]{4}$, and whenever $-2 \le a_n < 0$, it follows that $a_{n+1} = \sqrt[3]{a_n^2 - 8} \le \sqrt[3]{2^2 - 8} = -\sqrt[3]{4}$.

Now compare $|a_{n+2} - a_{n+1}|$ with $|a_{n+1} - a_n|$. By applying $x^3 - y^3 = (x-y)(x^2+xy+y^2)$, $x^2 - y^2 = (x-y)(x+y)$ and the recurrence,

$$\begin{aligned}(a_{n+2}^2 + a_{n+2}a_{n+1} + a_{n+1}^2) \cdot |a_{n+2} - a_{n+1}| = \\ = |a_{n+2}^3 - a_{n+1}^3| = |(a_{n+1}^2 - 8) - (a_n^2 - 8)| = \\ = |a_{n+1} + a_n| \cdot |a_{n+1} - a_n|.\end{aligned}$$

On the left-hand side we have

$$a_{n+2}^2 + a_{n+2}a_{n+1} + a_{n+1}^2 \ge 3 \cdot 4^{2/3};$$

on the right-hand side

$$|a_{n+1} + a_n| \le 4.$$

Hence,

$$|a_{n+2} - a_{n+1}| \le \frac{4}{3 \cdot 4^{2/3}} |a_{n+1} - a_n| = \frac{\sqrt[3]{4}}{3} |a_{n+1} - a_n|.$$

By a trivial induction it follows that

$$|a_{n+1} - a_n| \le \left(\frac{\sqrt[3]{4}}{3}\right)^{n-1} |a_2 - a_1|.$$

Hence the series $\sum_{n=0}^{\infty} |a_{n+1} - a_n|$ can be majorized by a geometric series with quotient $\frac{\sqrt[3]{4}}{3} < 1$; that proves that the series converges.

2018-D2-P8 — Statement

Day 2, P8 | 10 points

Let $\Omega = \{(x, y, z) \in \mathbb{Z}^3 : y + 1 \geq x \geq y \geq z \geq 0\}$. A frog moves along the points of Ω by jumps of length 1. For every positive integer n, determine the number of paths the frog can take to reach (n, n, n) starting from $(0, 0, 0)$ in exactly $3n$ jumps.

(Proposed by Fedor Petrov and Anatoly Vershik, St. Petersburg State University)

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Theory and methods for 2018-D2-P8

Theory Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Methods Encode, decode, and verify a bijection [M-DS-BIJECTION]; Strengthen the statement before the induction step [M-DS-INDUCT]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]; Reduce size while preserving the defining conditions [S-INDUCTION]

Statement Recorded solutions Catalog

2018-D2-P8 — Solution

For this record: Theory: T-DS-COUNTING; T-DS-INDUCTION. Methods: M-DS-BIJECTION; M-DS-INDUCT. Full learning route

Let $\Psi = \{(u, v) \in \mathbb{Z}^2 : v \geq 0, u \geq 2v\}$.
Notice that the map $\pi : \Omega \to \Psi$, $\pi(x, y, z) = (x + y, z)$ is a bijection between the two sets; moreover π projects all allowed paths of the frogs to paths inside the set Ψ, using only unit jump vectors. Hence, we are interested in the number of paths from $\pi(0, 0, 0) = (0, 0)$ to $\pi(n, n, n) = (2n, n)$ in the set Ψ, using only jumps $(1, 0)$ and $(0, 1)$.
For every lattice point $(u, v) \in \Psi$, let $f(u, v)$ be the number of paths from $(0, 0)$ to (u, v) in Ψ with $u + v$ jumps. Evidently we have $f(0, 0) = 1$. Extend this definition to the points with $v = -1$ and $2v = u + 1$ by setting

$$f(u, -1) = 0 \ (u \geq 1), \qquad f(2v - 1, v) = 0 \ (v \geq 1). \tag{3}$$

To any point (u, v) of Ψ other than the origin, the path can come either from $(u - 1, v)$ or from $(u, v - 1)$, so

$$f(u, v) = f(u - 1, v) + f(u, v - 1) \quad \text{for } (u, v) \in \Psi \setminus \{(0, 0)\}. \tag{4}$$

If we ignore the boundary condition (3), there is a wide family of functions that satisfy (4); namely, for every integer c, $(u, v) \mapsto \binom{u+v}{v+c}$ is such a function, with defining this binomial coefficient to be 0 if $v + c$ is negative or greater than $u + v$.
Along the line $2v = u + 1$ we have $\binom{u+v}{v} = \binom{3v-1}{v} = 2\binom{3v-1}{v-1} = 2\binom{u+v}{v-1}$. Hence, the function

$$f^*(u, v) = \binom{u + v}{v} - 2\binom{u + v}{v - 1}$$

satisfies (3), (4) and $f(0, 0) = 1$. These properties uniquely define the function f, so $f = f^*$.
In particular, the number of paths of the frog from $(0, 0, 0)$ to (n, n, n) is

$$f(\pi(n, n, n)) = f(2n, n) = \binom{3n}{n} - 2\binom{3n}{n - 1} = \frac{\binom{3n}{n}}{2n + 1}.$$

Remark. There exist direct proofs for the formula $\binom{3n}{n}/(2n + 1)$. For instance, we can replicate the well-known proof of the formula for the Catalan numbers using the Cycle Lemma of Dvoretzky and Motzkin (related to the petrol station replenishment problem). See the source note for this problem

2018-D2-P9 — Statement

Day 2, P9 | 10 points

Determine all pairs $P(x)$, $Q(x)$ of complex polynomials with leading coefficient 1 such that $P(x)$ divides $Q(x)^2+1$ and $Q(x)$ divides $P(x)^2+1$.

(Proposed by Rodrigo Angelo, Princeton University and Matheus Secco, PUC, Rio de Janeiro)

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Theory and methods for 2018-D2-P9

Theory Polynomial factorization and coprime divisibility [T-AL-FACTORIZATION]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]
Methods Replace a minimal algebraic counterexample by a smaller one [M-AL-DESCENT]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Choose an extremal object and force rigidity [S-EXTREMAL]

Statement Recorded solutions Catalog

2018-D2-P9 — Solution

For this record: Theory: T-AL-FACTORIZATION; T-AL-POLYNOMIAL; T-AL-ROOTS. Methods: M-AL-DESCENT. Full learning route

The answer is all pairs $(1,1)$ and $(P, P+i)$, $(P, P-i)$, where P is a non-constant monic polynomial in $\mathbb{C}[x]$ and i is the imaginary unit.
Notice that if $P|Q^2+1$ and $Q|P^2+1$ then P and Q are coprime and the condition is equivalent with $PQ|P^2+Q^2+1$.
Lemma. If $P, Q \in \mathbb{C}[x]$ are monic polynomials such that P^2+Q^2+1 is divisible by PQ, then $\deg P = \deg Q$.
Proof. Assume for the sake of contradiction that there is a pair (P,Q) with $\deg P \neq \deg Q$. Among all these pairs, take the one with smallest sum $\deg P + \deg Q$ and let (P,Q) be such pair. Without loss of generality, suppose that $\deg P > \deg Q$. Let S be the polynomial such that

$$\frac{P^2+Q^2+1}{PQ} = S.$$

Notice that P a solution of the polynomial equation $X^2 - QSX + Q^2 + 1 = 0$, in variable X. By Vieta's formulas, the other solution is $R = QS - P = \dfrac{Q^2+1}{P}$. By $R = QS - P$, the R is indeed a polynomial, and because P, Q are monic, $R = \dfrac{Q^2+1}{P}$ is also monic. Therefore the pair (R,Q) satisfies the conditions of the Lemma. Notice that $\deg R = 2\deg Q - \deg P < \deg P$, which contradicts the minimality of $\deg P + \deg Q$. This contradiction establishes the Lemma.

By the Lemma, we have that $\deg(PQ) = \deg(P^2+Q^2+1)$ and therefore $\dfrac{P^2+Q^2+1}{PQ}$ is a constant polynomial. If P and Q are constant polynomials, we have $P = Q = 1$. Assuming that $\deg P = \deg Q \geq 1$, as P and Q are monic, the leading coefficient of P^2+Q^2+1 is 2 and the leading coefficient of PQ is 1, which give us $\dfrac{P^2+Q^2+1}{PQ} = 2$. Finally we have that $P^2+Q^2+1 = 2PQ$ and therefore $(P-Q)^2 = -1$, i.e $Q = P+i$ or $Q = P-i$. It's easy to check that these pairs are indeed solutions of the problem.

2018-D2-P10 — Statement

Day 2, P10 | 10 points

For $R > 1$ let $D_R = \{(a, b) \in \mathbb{Z}^2 : 0 < a^2 + b^2 < R\}$. Compute

$$\lim_{R\to\infty} \sum_{(a,b)\in D_R} \frac{(-1)^{a+b}}{a^2+b^2}.$$

(Proposed by Rodrigo Angelo, Princeton University and Matheus Secco, PUC, Rio de Janeiro)

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Theory and methods for 2018-D2-P10

Theory Riemann sums, periodic averaging, and tails [T-AN-RIEMANN]; Scale separation and limit comparisons [T-AN-ASYMPTOTICS]; Series, finite cancellation, and summation by parts [T-AN-SERIES]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]
Methods Separate a main range from exceptional scales [M-AN-SPLIT]; Expose the mesh and control its tails [M-AN-RIEMANN]
Strategy Expose the structure before calculating [S-STRUCTURE]; Translate geometry into a controlled representation [S-GEOMETRIC]

Statement Recorded solutions Catalog

2018-D2-P10 — Solution

For this record: Theory: T-AN-RIEMANN; T-AN-ASYMPTOTICS; T-AN-SERIES; T-DS-GEOMETRY. Methods: M-AN-SPLIT; M-AN-RIEMANN. Full learning route

Define $E_R = \{(a, b) \in \mathbb{Z}^2 \setminus \{(0, 0)\} : a^2 + b^2 < R \text{ and } a + b \text{ is even}\}$. Then

$$\sum_{(a,b)\in D_R} \frac{(-1)^{a+b}}{a^2+b^2} = 2\sum_{(a,b)\in E_R} \frac{1}{a^2+b^2} - \sum_{(a,b)\in D_R} \frac{1}{a^2+b^2}. \tag{5}$$

But $a+b$ is even if and only if one can write $(a, b) = (m-n, m+n)$, and such m, n are unique. Notice also that $a^2 + b^2 = (m - n)^2 + (m + n)^2 = 2m^2 + 2n^2$, hence $a^2 + b^2 < R$ if and only if $m^2 + n^2 < R/2$. With that we get:

$$2\sum_{(a,b)\in E_R} \frac{1}{a^2+b^2} = 2\sum_{(m,n)\in D_{R/2}} \frac{1}{(m-n)^2+(m+n)^2} = \sum_{(m,n)\in D_{R/2}} \frac{1}{m^2+n^2}. \tag{6}$$

Replacing (6) in (5), we obtain

$$\sum_{(a,b)\in D_R} \frac{(-1)^{a+b}}{a^2+b^2} = -\sum_{R/2\le a^2+b^2<R} \frac{1}{a^2+b^2},$$

where the second sum is evaluated for a and b integers.

Denote by $N(r)$ the number of lattice points in the open disk $x^2 + y^2 < r^2$. Along the circle with radius r with $\sqrt{R/2} \le r < \sqrt{R}$, there are $N(r + 0) - N(r - 0)$ lattice points; each of them contribute $\frac{1}{r^2}$ in the sum (7). So we can re-write the sum as a Stieltjes integral:

$$\sum_{R/2\le a^2+b^2<R} \frac{1}{a^2+b^2} = \int_{\sqrt{R/2}}^{\sqrt{R}} \frac{1}{r^2}\, dN(r).$$

2018-D2-P10 — Solution (continued)

It is well-known that $N(r) = \pi r^2 + O(r)$. (Putting a unit square around each lattice point, these squares cover the disk with radius $r-1$ and lie inside the disk with radius $r+1$, so there their total area is between $\pi(r-1)^2$ and $\pi(r+1)^2$). By integrating by parts,

$$\begin{aligned}\int_{\sqrt{R/2}}^{\sqrt{R}} \frac{1}{r^2}\,dN(r) &= \left[N(r)\frac{1}{r^2}\right]_{\sqrt{R/2}}^{\sqrt{R}} + \int_{\sqrt{R/2}}^{\sqrt{R}} \frac{2}{r^3}N(r)\,dr \\ &= \left[\frac{\pi r^2 + O(r)}{r^2}\right]_{\sqrt{R/2}}^{\sqrt{R}} + 2\int_{\sqrt{R/2}}^{\sqrt{R}} \frac{\pi r^2 + O(r)}{r^3}\,dr \\ &= 2\pi \int_{\sqrt{R/2}}^{\sqrt{R}} \frac{dr}{r} + O(1/\sqrt{R}) = \pi \log 2 + O(1/\sqrt{R}).\end{aligned}$$

Therefore,

$$\lim_{R\to\infty} \sum_{(a,b)\in D_R} \frac{(-1)^{a+b}}{a^2+b^2} = -\lim_{R\to\infty} \sum_{R/2\le a^2+b^2<R} \frac{1}{a^2+b^2} = -\lim_{R\to\infty} \int_{\sqrt{R/2}}^{\sqrt{R}} \frac{1}{r^2}\,dN(r) = -\pi\log 2.$$

3.26 2019

Day 1

2019-D1-P1 — Statement

Day 1, P1 | 10 points

Evaluate the product

$$\prod_{n=3}^{\infty} \frac{(n^3+3n)^2}{n^6-64}.$$

Proposed by Orif Ibrogimov, ETH Zurich and National University of Uzbekistan and Karen Keryan, Yerevan State University and American University of Armenia, Yerevan

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Theory and methods for 2019-D1-P1

Theory Series, finite cancellation, and summation by parts [T-AN-SERIES]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Find a finite telescoping identity [M-AN-TELESCOPE]
Strategy Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2019-D1-P1 — Solution

For this record: Theory: T-AN-SERIES; T-AL-POLYNOMIAL. Methods: M-AN-TELESCOPE. Full learning route

Hint: Telescoping product.

Let

$$a_n = \frac{(n^3+3n)^2}{n^6-64}.$$

2019-D1-P1 — Solution (continued)

Notice that

$$a_n = \frac{(n^3+3n)^2}{(n^3-8)(n^3+8)} = \frac{n^2(n^2+3)^2}{(n-2)(n^2+2n+4)\cdot(n+2)(n^2-2n+4)}$$
$$= \frac{n}{n-2}\cdot\frac{n}{n+2}\cdot\frac{n^2+3}{(n-1)^2+3}\cdot\frac{n^2+3}{(n+1)^2+3}.$$

Hence, for $N \geq 3$ we have

$$\prod_{n=3}^{N} a_n = \left(\prod_{n=3}^{N}\frac{n}{n-2}\right)\left(\prod_{n=3}^{N}\frac{n}{n+2}\right)\left(\prod_{n=3}^{N}\frac{n^2+3}{(n-1)^2+3}\right)\left(\prod_{n=3}^{N}\frac{n^2+3}{(n+1)^2+3}\right)$$
$$= \frac{N(N-1)}{1\cdot 2}\cdot\frac{3\cdot 4}{(N+1)(N+2)}\cdot\frac{N^2+3}{2^2+3}\cdot\frac{3^2+3}{(N+1)^2+3}$$
$$= \frac{72}{7}\cdot\frac{N(N-1)(N^2+3)}{(N+1)(N+2)((N+1)^2+3)}$$
$$= \frac{72}{7}\cdot\frac{(1-\frac{1}{N})(1+\frac{3}{N^2})}{(1+\frac{1}{N})(1+\frac{2}{N})((1+\frac{1}{N})^2+\frac{3}{N^2})},$$

so

$$\prod_{n=3}^{\infty} a_n = \lim_{N\to\infty}\prod_{n=3}^{N} a_n = \lim_{N\to\infty}\left(\frac{72}{7}\cdot\frac{(1-\frac{1}{N})(1+\frac{3}{N^2})}{(1+\frac{1}{N})(1+\frac{2}{N})((1+\frac{1}{N})^2+\frac{3}{N^2})}\right) = \frac{72}{7}.$$

2019-D1-P2 — Statement

Day 1, P2 | 10 points

A four-digit number $YEAR$ is called very good if the system

$$\begin{aligned} Yx + Ey + Az + Rw &= Y \\ Rx + Yy + Ez + Aw &= E \\ Ax + Ry + Yz + Ew &= A \\ Ex + Ay + Rz + Yw &= R \end{aligned}$$

of linear equations in the variables x, y, z and w has at least two solutions. Find all very good $YEAR$s in the 21st century.
(The 21st century starts in 2001 and ends in 2100.)

Proposed by Tomáš Bárta, Charles University, Prague

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Theory and methods for 2019-D1-P2

Theory Rank, dimension and exact linear constraints [T-LA-RANK]; Determinants, block identities and inverses [T-LA-DETERMINANT]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Inner products, orthogonality and polarization [T-LA-INNER]
Methods Eliminate systematically and track the invariant [M-LA-ELIM]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]

Statement Recorded solutions Catalog

2019-D1-P2 — Solution

For this record: Theory: T-LA-RANK; T-LA-DETERMINANT; T-LA-SPECTRAL; T-LA-INNER. Methods: M-LA-ELIM. Full learning route

Editorially corrected proof.

2019-D1-P2 — Solution (continued)

Hint: If the solution of the system is not unique then $\det \begin{pmatrix} Y & E & A & R \\ R & Y & E & A \\ A & R & Y & E \\ E & A & R & Y \end{pmatrix} = 0$.

Let us apply row transformations to the augmented matrix of the system to find its rank. First we add the second, third and fourth row to the first one and divide by $Y+E+A+R$ to get

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ R & Y & E & A & E \\ A & R & Y & E & A \\ E & A & R & Y & R \end{array}\right)$$

$$\sim \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & Y-R & E-R & A-R & E-R \\ 0 & R-A & Y-A & E-A & 0 \\ 0 & A-E & R-E & Y-E & R-E \end{array}\right)$$

$$\sim \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & Y-R & E-R & A-R & E-R \\ 0 & R-A & Y-A & E-A & 0 \\ 0 & A-E+Y-R & 0 & Y-E+A-R & 0 \end{array}\right)$$

To justify consistency, let M be the original 4×4 coefficient matrix. It is real circulant, hence $MM^T = M^TM$. Therefore

$$\|Mz\|^2 = \|M^Tz\|^2 \quad \text{for every } z\in\mathbb{R}^4,$$

so $\ker M = \ker M^T$ and consequently $\operatorname{im} M = \operatorname{im} M^T$. The right-hand side is M^Te_1, where $e_1 = (1,0,0,0)^T$. It therefore belongs to $\operatorname{im} M$, and the system is always consistent. It follows that the system has more than one solution if and only if the 4×4 matrix (with the last column omitted) is singular. Let us first assume that $E\neq R$. We apply one more transform to get

$$\sim \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & Y-R & E-R & A-R \\ 0 & \begin{matrix}(R-A)(E-R)\\ -(Y-R)(Y-A)\end{matrix} & 0 & \begin{matrix}(E-A)(E-R)\\ -(A-R)(Y-A)\end{matrix} \\ 0 & A-E+Y-R & 0 & Y-E+A-R \end{pmatrix}$$

Obviously, this matrix is singular if and only if $A-E+Y-R=0$ or the two expressions in the third row are equal, i.e.

$$\begin{aligned} &RE-R^2-AE+AR-Y^2+RY+AY-AR \\ &= E^2-AE-ER+AR-AY+RY+A^2-AR \\ &0 = (E-R)^2+(A-Y)^2, \end{aligned}$$

but this is impossible if $E\neq R$. If $E=R$, we have

$$\sim \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & Y-R & 0 & A-R \\ 0 & R-A & Y-A & R-A \\ 0 & A+Y-2R & 0 & Y+A-2R \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & Y-R & 0 & A-R \\ 0 & R-A & Y-A & R-A \\ 0 & A-R & 0 & Y-R \end{pmatrix}.$$

If $A=Y$, this matrix is singular. If $A\neq Y$, the matrix is regular if and only if $(Y-R)^2\neq(A-R)^2$ and since $Y\neq A$, it means that $Y-R\neq -(A-R)$, i.e. $Y+A\neq 2R$. We conclude that $YEAR$ is very good if and only if

1. $E\neq R$ and $A+Y=E+R$, or
2. $E=R$ and $Y=A$, or
3. $E=R$, $A\neq Y$ and $Y+A=2R$.

We can see that if $Y=2$, $E=0$, then the very good years satisfying 1 are $A+2=R\neq 0$, i.e. 2002, 2013, 2024, 2035, 2046, 2057, 2068, 2079, condition 2 is satisfied for 2020 and condition 3 is never satisfied. The remaining century endpoint, 2100, satisfies none of the three conditions. Thus the complete list is $2002, 2013, 2020, 2024, 2035, 2046, 2057, 2068, 2079$.

2019-D1-P3 — Statement

Day 1, P3 | 10 points

Let $f : (-1, 1) \to \mathbb{R}$ be a twice differentiable function such that

$$2f'(x) + xf''(x) \geq 1 \quad \text{for } x \in (-1, 1).$$

Prove that

$$\int_{-1}^{1} xf(x)\,dx \geq \frac{1}{3}.$$

Proposed by Orif Ibrogimov, ETH Zurich and National University of Uzbekistan and Karim Rakhimov, Scuola Normale Superiore and National University of Uzbekistan

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Theory and methods for 2019-D1-P3

Theory Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Methods Design an auxiliary derivative [M-AN-AUXILIARY]; Choose the correct convexity comparison [M-AN-CONVEX]
Strategy Build a chain of bounds and track equality [S-BOUND]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2019-D1-P3 — Solution

For this record: Theory: T-AN-CONVEXITY; T-AN-DERIVATIVE; T-AN-INTEGRAL. Methods: M-AN-AUXILIARY; M-AN-CONVEX. Full learning route

Hint: $2f'(x) + xf''(x)$ is the second derivative of a certain function.

Let

$$g(x) = xf(x) - \frac{x^2}{2}.$$

Notice that

$$g''(x) = 2f'(x) + xf''(x) - 1 \geq 0,$$

so g is convex. Estimate g by its tangent at 0: let $g'(0) = a$, then

$$g(x) = g(0) + g'(0)x = ax$$

and therefore

$$\int_{-1}^{1} xf(x)\,dx = \int_{-1}^{1} \left(g(x) + \frac{x^2}{2}\right) dx \geq \int_{-1}^{1} \left(ax + \frac{x^2}{2}\right) dx = \frac{1}{3}.$$

2019-D1-P4 — Statement

Day 1, P4 | 10 points

Define the sequence $a_0, a_1, \ldots$ of numbers by the following recurrence:

$$a_0 = 1, \quad a_1 = 2, \quad (n+3)a_{n+2} = (6n+9)a_{n+1} - na_n \quad \text{for } n \geq 0.$$

Prove that all terms of this sequence are integers.

Proposed by Khakimboy Egamberganov, ICTP, Italy

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Theory and methods for 2019-D1-P4

Theory Linear recurrences and generating functions [T-AL-RECURRENCE]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Methods Find a recurrence that reveals the invariant [M-AL-RECURRENCE]
Strategy Expose the structure before calculating [S-STRUCTURE]; Reduce size while preserving the defining conditions [S-INDUCTION]

Statement Recorded solutions Catalog

2019-D1-P4 — Solution

For this record: Theory: T-AL-RECURRENCE; T-AN-DERIVATIVE; T-AN-CONVERGENCE. Methods: M-AL-RECURRENCE.
Full learning route

Hint: Determine the generating function $\sum a_n x^n$.
Take the generating function of this sequence

$$f(x) = \sum_{n=0}^{\infty} a_n x^n.$$

It is easy to see that the sequence is increasing and

$$\frac{a_{n+1}}{a_n} = \frac{(6n+3)a_n - (n-1)a_{n-1}}{(n+2)a_n} < \frac{6n+3}{n+2} \quad \Rightarrow \quad \limsup_{n\to\infty} \frac{a_{n+1}}{a_n} \le 6.$$

So the generating function converges in some neighbourhood of 0. Then, we have

$$f(x) = 1 + 2x + \sum_{n=2}^{\infty} a_n x^n = 1 + 2x + \sum_{n=0}^{\infty} a_{n+2} x^{n+2} = 1 + 2x + \sum_{n=0}^{\infty} \frac{6n+9}{n+3} a_{n+1} x^{n+2} - \sum_{n=0}^{\infty} \frac{n}{n+3} a_n x^{n+2}.$$

Let $f_1(x) = \sum\limits_{n=0}^{\infty} \frac{6n+9}{n+3} a_{n+1} x^{n+2}$ and $f_2(x) = \sum\limits_{n=0}^{\infty} \frac{n}{n+3} a_n x^{n+2}$. Then

$$(xf_1(x))' = \sum_{n=0}^{\infty} (6n+9) a_{n+1} x^{n+2} = 6x^2 \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n + 3x \sum_{n=0}^{\infty} a_{n+1} x^{n+1} = 6x^2 f'(x) + 3x(f(x) - 1)$$

and

$$(xf_2(x))' = \sum_{n=0}^{\infty} n a_n x^{n+2} = x^2 \sum_{n=0}^{\infty} (n+1) a_n x^n - x^2 \sum_{n=0}^{\infty} a_n x^n = x^2 (xf(x))' - x^2 f(x) = x^3 f'(x).$$

Using this relations, we arrive at the following differential equation for f:

$$(xf(x))' = 1 + 4x + (xf_1(x))' - (xf_2(x))' = 1 + x + (6x^2 - x^3) f'(x) + 3xf(x)$$

or, equivalently,

$$(x^3 - 6x^2 + x) f'(x) + (1 - 3x) f(x) - 1 - x = 0.$$

So, we need solve this differential equation in some sufficiently smaller neighbourhood of 0. We know that $f(0) = 1$ and we need a neighbourhood of 0 such that $x^2 - 6x + 1 > 0$. Then

$$f'(x) + \frac{1 - 3x}{x(x^2 - 6x + 1)} f(x) = \frac{1 + x}{x(x^2 - 6x + 1)}$$

for $x \neq 0$. So the integral multiplier is $\mu(x) = \dfrac{x}{\sqrt{x^2 - 6x + 1}}$ and

$$(f(x)\mu(x))' = \frac{x + x^2}{(x^2 - 6x + 1)^{\frac{3}{2}}},$$

so

$$f(x) = \left(\frac{1 - x}{2\sqrt{x^2 - 6x + 1}} - \frac{1}{2} \right) \frac{\sqrt{x^2 - 6x + 1}}{x} = \frac{1 - x - \sqrt{x^2 - 6x + 1}}{2x}.$$

2019-D1-P4 — Solution (continued)

We found the generating function of (a_n) in some neighbourhood of 0, which $x^2 - 6x + 1 > 0$. So our series uniformly converges to $f(x) = \dfrac{1 - x - \sqrt{x^2 - 6x + 1}}{2x}$ in $|x| < 3 - 2\sqrt{2}$.

Instead of computing the coefficients of the Taylor series of $f(x)$ directly, we will find another recurrence relation for (a_n). It is easy to see that $f(x)$ satisfies the quadratic equation $xt^2 - (1 - x)t + 1 = 0$. So

$$xf(x)^2 - (1 - x)f(x) + 1 = 0.$$

Then

$$x\left(\sum_{n=0}^{\infty} a_n x^n\right)^2 + 1 = \sum_{n=0}^{\infty} a_n x^n - \sum_{n=0}^{\infty} a_n x^{n+1} \Rightarrow \sum_{n=0}^{\infty}\left(\sum_{k=0}^{n} a_k a_{n-k}\right) x^{n+1} = \sum_{n=0}^{\infty}(a_{n+1} - a_n)x^{n+1}$$

and from here, we get

$$a_{n+1} = a_n + \sum_{k=0}^{n} a_k a_{n-k}.$$

If $a_0, a_1, \ldots, a_n$ be integers, then a_{n+1} is also integer. We know that $a_0 = 1, a_1 = 2$ are integer numbers, so all terms of the sequence (a_n) are integers by induction.

2019-D1-P5 — Statement

Day 1, P5 | 10 points

Determine whether there exist an odd positive integer n and $n \times n$ matrices A and B with integer entries, that satisfy the following conditions:
(1) $\det(B) = 1$;
(2) $AB = BA$;
(3) $A^4 + 4A^2B^2 + 16B^4 = 2019I$.
(Here I denotes the $n \times n$ identity matrix.)

Proposed by Orif Ibrogimov, ETH Zurich and National University of Uzbekistan

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Theory and methods for 2019-D1-P5

Theory Determinants, block identities and inverses [T-LA-DETERMINANT]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Methods Factor while preserving matrix order [M-LA-FACTOR]; Reduce integer matrices modulo an integer [M-LA-RESIDUE]
Strategy Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

2019-D1-P5 — Solution — Record 1 of 2

For this record: Theory: T-LA-DETERMINANT; T-AL-CONGRUENCE. Methods: M-LA-FACTOR; M-LA-RESIDUE. Full learning route

Hint: Consider the determinants modulo 4.
Remark. The proposed solution was more complicated and involved; during the contest it turned out that a signficantly simplified solution exists — which we now provide below.
We show that there are no such matrices.

Notice that $A^4 + 4A^2B^2 + 16B^4$ can factorized as

$$A^4 + 4A^2B^2 + 16B^4 = (A^2 + 2AB + 4B^2)(A^2 - 2AB + 4B^2).$$

Let $C = A^2 + 2AB + 4B^2$ and $D = A^2 - 2AB + 4B^2$ be the two factors above. Then

$$\det C \cdot \det D = \det(CD) = \det(A^4 + 4A^2B^2 + 16B^4) = \det(2019I) = 2019^n.$$

2019-D1-P5 — Solution — Record 1 of 2 (continued)

The matrices C, D have integer entries, so their determinants are integers. Moreover, from $C \equiv D \pmod 4$ we can see that

$$\det C \equiv \det D \pmod 4.$$

This implies that $\det C \cdot \det D \equiv (\det C)^2 \pmod 4$, but this is a contradiction because $2019^n \equiv 3 \pmod 4$ is a quadratic nonresidue modulo 4.

2019-D1-P5 — Solution — Record 2 of 2

For this record: Theory: T-LA-DETERMINANT; T-AL-CONGRUENCE. Methods: M-LA-RESIDUE. Full learning route

Notice that

$$A^4 \equiv A^4 + 4A^2B^2 + 16B^4 = 2019I \mod 4$$

so

$$(\det A)^4 = \det A^4 \equiv \det(2109I) = 2019^n \pmod 4.$$

But $2019^n \equiv 3$ is a quadratic nonresidue modulo 4, contradiction.

Day 2

2019-D2-P6 — Statement

Day 2, P6 | 10 points

Let $f, g : \mathbb{R} \longrightarrow \mathbb{R}$ be continuous functions such that g is differentiable. Assume that $(f(0) - g'(0))(g'(1) - f(1)) > 0$. Show that there exists a point $c \in (0, 1)$ such that $f(c) = g'(c)$.

Proposed by Fereshteh Malek, K. N. Toosi University of Technology

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Theory and methods for 2019-D2-P6

Theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Methods Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]; Count zeros and subtract an interpolant [M-AN-ROLLE]
Strategy Control the limiting operation before taking the limit [S-ANALYTIC]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2019-D2-P6 — Solution

For this record: Theory: T-AN-DERIVATIVE; T-AN-INTEGRAL. Methods: M-AN-PRIMITIVE; M-AN-ROLLE. Full learning route

Define $F(x) = \int_0^x f(t)\,dt$ and let $h(x) = F(x) - g(x)$. By the continuouity of f we have $F' = f$, so $h' = f - g'$.

The assumption can be re-written as $h'(0)(-h'(1)) > 0$, so $h'(0)$ and $h'(1)$ have opposite signs. Then, by the Mean Value Theorem For Derivatives (Darboux property of derivatives) it follows that there is a point c between 0 and 1 where $h'(c) = 0$, so $f(c) = g'(c)$.

2019-D2-P7 — Statement

Day 2, P7 | 10 points

Let $C = \{4, 6, 8, 9, 10, \dots\}$ be the set of composite positive integers. For each $n \in C$ let a_n be the smallest positive integer k such that $k!$ is divisible by n. Determine whether the following series converges:

$$\sum_{n \in C} \left(\frac{a_n}{n}\right)^n. \tag{1}$$

Proposed by Orif Ibrogimov, ETH Zurich and National University of Uzbekistan

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Theory and methods for 2019-D2-P7

Theory Prime valuations, divisors, and factorials [T-AL-VALUATION]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Methods Track one prime or one extreme divisor [M-AL-VALUATION]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2019-D2-P7 — Solution

For this record: Theory: T-AL-VALUATION; T-AN-CONVERGENCE. Methods: M-AL-VALUATION. Full learning route

The series (1) converges. We will show that $\frac{a_n}{n} \le \frac{2}{3}$ for $n > 4$; then the geometric series $\sum \left(\frac{2}{3}\right)^n$ majorizes (1).

Case 1: n has at least two distinct prime divisors. Then n can be factored as $n = qr$ with some co-prime positive integers $q, r \ge 2$; without loss of generality we can assume $q > r$. Notice that $q \mid q!$ and $r \mid r! \mid q!$, so $n = qr \mid q!$; this shows $a_n \le q$ and therefore

$$\frac{a_n}{n} \le \frac{q}{n} = \frac{1}{r} \le \frac{1}{2}.$$

Case 2: n is the square of a prime, $n = p^2$ with some prime $p \ge 3$. From $p^2 \mid p \cdot 2p \mid (2p)!$ we obtain $a_n = 2p$, so

$$\frac{a_n}{n} = \frac{2p}{p^2} = \frac{2}{p} \le \frac{2}{3}.$$

Case 3: n is a prime power, $n = p^k$ with some prime p and $k \ge 3$. Notice that $n = p^k \mid p \cdot p^2 \cdots p^{k-1}$, so $a_n \le p^{k-1}$ and therefore

$$\frac{a_n}{n} \le \frac{p^{k-1}}{p^k} = \frac{1}{p} \le \frac{1}{2}.$$

2019-D2-P8 — Statement

Day 2, P8 | 10 points

Let $x_1, \dots, x_n$ be real numbers. For any set $I \subset \{1, 2, \dots, n\}$ let $s(I) = \sum_{i \in I} x_i$. Assume that the function $I \mapsto s(I)$ takes on at least 1.8^n values where I runs over all 2^n subsets of $\{1, 2, \dots, n\}$. Prove that the number of sets $I \subset \{1, 2, \dots, n\}$ for which $s(I) = 2019$ does not exceed 1.7^n.

Proposed by Fedor Part and Fedor Petrov, St. Petersburg State University

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Theory and methods for 2019-D2-P8

Theory Finite counting and incidence identities [T-DS-COUNTING]; Inner products, orthogonality and polarization [T-LA-INNER]
Methods Choose information that makes an encoding injective [M-DS-ENCODE]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

2019-D2-P8 — Solution

For this record: Theory: T-DS-COUNTING; T-LA-INNER. Methods: M-DS-ENCODE. Full learning route

Choose sets $I_1, \dots, I_A \subset \{1, 2, \dots, n\}$ whose subset sums $s(I_1), \dots, s(I_A)$ are pairwise distinct, where $A \geq 1.8^n$, and let $J_1, \dots, J_B \subset \{1, 2, \dots, n\}$ be all sets so that $s(J_i) = 2019$; for the sake of contradiction, assume that $B \geq 1.7^n$.
Every set $I \subset \{1, 2, \dots, n\}$ can be identified with a $0-1$ vector of length n: the kth coordinate in the vector is 1 if $k \in I$. Then $s(I) = \langle I, X \rangle$, where $X = (x_1, \dots, x_n)$ and $\langle \cdot, \cdot \rangle$ stands for the usual scalar product.

For all ordered pairs $(a, b) \in \{1, \dots, A\} \times \{1, \dots, B\}$ consider the vector $I_a - J_b \in \{-1, 0, 1\}^n$. By the pigeonhole principle, since $AB \geq (1.8 \cdot 1.7)^n > 3^n$, there are two pairs (a, b) and (c, d) such that $I_a - J_b = I_c - J_d$. Multiplying this by X we get $s(I_a) - 2019 = s(I_c) - 2019$; that implies $a = c$. But then $J_b = J_d$, that is, $b = d$, and our pairs coincide. Contradiction.

2019-D2-P9 — Statement

Day 2, P9 | 10 points

Determine all positive integers n for which there exist $n \times n$ real invertible matrices A and B that satisfy $AB - BA = B^2A$.

Proposed by Karen Keryan, Yerevan State University & American University of Armenia, Yerevan

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Theory and methods for 2019-D2-P9

Theory Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Determinants, block identities and inverses [T-LA-DETERMINANT]
Methods Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]; Exploit a compatible block decomposition [M-LA-BLOCK]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2019-D2-P9 — Solution

For this record: Theory: T-LA-SPECTRAL; T-LA-POLYNOMIAL; T-LA-DETERMINANT. Methods: M-LA-SPECTRAL; M-LA-BLOCK. Full learning route

We prove that there exist such matrices A and B inf and only if n is even.
I. Assume that n is odd and some invertible $n \times n$ matrices A, B satisfy $AB - BA = B^2A$. Hence $B = A^{-1}(B^2+B)A$, so the matrices B and B^2+B are similar and therefore have the same eigenvalues. Since n is odd, the matrix B has a real eigenvalue, denote it by λ_1. Therefore $\lambda_2 := \lambda_1^2 + \lambda_1$ is an eigenvalue of $B^2 + B$, hence an eigenvalue of B. Similarly, $\lambda_3 := \lambda_2^2 + \lambda_2$ is an eigenvalue of $B^2 + B$,

2019-D2-P9 — Solution (continued)

hence an eigenvalue of B. Repeating this process and taking into account that the number of eigenvalues of B is finite we will get there exist numbers $k \le l$ so that $\lambda_{l+1} = \lambda_k$. Hence

$$\begin{aligned}\lambda_{k+1} &= \lambda_k^2 + \lambda_k \\ \lambda_{k+2} &= \lambda_{k+1}^2 + \lambda_{k+1} \\ &\dots\dots\dots \\ \lambda_l &= \lambda_{l-1}^2 + \lambda_{l-1} \\ \lambda_k &= \lambda_l^2 + \lambda_l.\end{aligned}$$

Adding this equations we get $\lambda_k^2 + \lambda_{k+1}^2 + \dots + \lambda_l^2 = 0$. Taking into account that all λ_i's are real (as λ_1 is real), we have $\lambda_k = \dots = \lambda_l = 0$, which implies that B is not invertible, contradiction.

II. Now we construct such matrices A, B for even n. Let $A_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $B_2 = \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix}$. It is easy to check that the matrices A_2, B_2 are invertible and satisfy the condition. For $n = 2k$ the $n \times n$ block matrices

$$A = \begin{bmatrix} A_2 & 0 & \dots & 0 \\ 0 & A_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & A_2 \end{bmatrix}, \qquad B = \begin{bmatrix} B_2 & 0 & \dots & 0 \\ 0 & B_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & B_2 \end{bmatrix}$$

are also invertible and satisfy the condition.

2019-D2-P10 — Statement

Day 2, P10 | 10 points

2019 points are chosen at random, independently, and distributed uniformly in the unit disc $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \le 1\}$. Let C be the convex hull of the chosen points. Which probability is larger: that C is a polygon with three vertices, or a polygon with four vertices?

Proposed by Fedor Petrov, St. Petersburg State University

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Theory and methods for 2019-D2-P10

Theory Uniform random points and convex-hull events [T-AN-GEOMETRICPROB]; Finite probability, conditioning, and expectations [T-AN-PROBABILITY]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]
Methods Condition and average a convex-hull event [M-AN-GEOMETRIC]; Count outcomes or condition on a useful structure [M-AN-PROBABILITY]
Strategy Translate geometry into a controlled representation [S-GEOMETRIC]; Choose the objects and count them without ambiguity [S-COUNTING]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2019-D2-P10 — Solution

For this record: Theory: T-AN-GEOMETRICPROB; T-AN-PROBABILITY; T-DS-GEOMETRY. Methods: M-AN-GEOMETRIC; M-AN-PROBABILITY. Full learning route

We will show that the quadrilateral has larger probability.
Let $D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \le 1\}$. Denote the random points by $X_1, \dots, X_{2019}$ and let

$$\begin{aligned}p &= \mathsf{P}(C \text{ is a triangle with vertices } X_1, X_2, X_3), \\ q &= \mathsf{P}(C \text{ is a convex quadrilateral with vertices } X_1, X_2, X_3, X_4).\end{aligned}$$

By symmetry we have $\mathsf{P}(C \text{ is a triangle}) = \binom{2019}{3}p$, $\mathsf{P}(C \text{ is a quadrilateral}) = \binom{2019}{4}q$ and we need to prove that $\binom{2019}{4}q > \binom{2019}{3}p$, or equivalently $p < \frac{2016}{4}q = 504q$.
Note that p is the average over X_1, X_2, X_3 of the following expression:

$$u(X_1, X_2, X_3) = \mathsf{P}(X_4 \in \triangle X_1X_2X_3) \cdot \mathsf{P}(X_5, X_6, \dots, X_{2019} \in \triangle X_1X_2X_3),$$

2019-D2-P10 — Solution (continued)

and q is not less than the average over X_1, X_2, X_3 of

$$v(X_1, X_2, X_3) = \mathsf{P}(X_1, X_2, X_3, X_4 \text{ form a convex quad.}) \cdot \mathsf{P}(X_5, X_6, \dots, X_{2019} \in \triangle X_1X_2X_3).$$

Thus it suffices to prove that $u(X_1, X_2, X_3) \le 500\, v(X_1, X_2, X_3)$ for all X_1, X_2, X_3. It reads as $\operatorname{area}(\triangle X_1X_2X_3) \le 500 \operatorname{area}(\Omega)$, where $\Omega = \{Y : X_1, X_2, X_3, Y \text{ form a convex quadrilateral}\}$. Assume the contrary, i.e., $\operatorname{area}(\triangle X_1X_2X_3) > 500 \operatorname{area}(\Omega)$.
Let the lines X_1X_2, X_1X_3, X_2X_3 meet the boundary of D at $A_1, A_2, A_3, B_1, B_2, B_3$; these lines divide D into 7 regions as shown in the picture; $\Omega = D_4 \cup D_5 \cup D_6$.
By our indirect assumption,

$$\operatorname{area}(D_4) + \operatorname{area}(D_5) + \operatorname{area}(D_6) = \operatorname{area}(\Omega) < \frac{1}{500} \operatorname{area}(D_0) < \frac{1}{500} \operatorname{area}(D) = \frac{\pi}{500}.$$

From $\triangle X_1X_3B_3 \subset \Omega$ we get $X_3B_3/X_3X_2 = \operatorname{area}(\triangle X_1X_3B_3)/\operatorname{area}(\triangle X_1X_2X_3) < 1/500$, so $X_3B_3 < \frac{1}{500} X_2X_3 < \frac{1}{250}$. Similarly, the lengths segments $A_1X_1, B_1X_1, A_2X_2, B_2X_2, A_3X_2$ are less than $\frac{1}{250}$.
The regions D_1, D_2, D_3 can be covered by disks with radius $\frac{1}{250}$, so

$$\operatorname{area}(D_1) + \operatorname{area}(D_2) + \operatorname{area}(D_3) < 3 \cdot \frac{\pi}{250^2}.$$

Finally, it is well-known that the area of any triangle inside the unit disk is at most $\frac{3\sqrt{3}}{4}$, so $\operatorname{area}(D_0) \le \frac{3\sqrt{3}}{4}$.

But then

$$\sum_{i=0}^{6} \operatorname{area}(D_i) < \frac{3\sqrt{3}}{4} + 3 \cdot \frac{\pi}{250^2} + \frac{\pi}{500} < \operatorname{area}(D),$$

contradiction.

3.27 2020

Day 1

2020-D1-P1 — Statement

Day 1, P1

Let n be a positive integer. Compute the number of words w (finite sequences of letters) that satisfy all the following three properties:
(1) w consists of n letters, all of them are from the alphabet $\{a, b, c, d\}$;
(2) w contains an even number of letters a;
(3) w contains an even number of letters b.
(For example, for $n = 2$ there are 6 such words: aa, bb, cc, dd, cd and dc.)

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Theory and methods for 2020-D1-P1

Theory Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Generating functions, coefficient filters, and finite differences [T-DS-GENERATING]
Methods Split a family by traces, pieces, or labels [M-DS-PARTITION]; Encode, decode, and verify a bijection [M-DS-BIJECTION]; Strengthen the statement before the induction step [M-DS-INDUCT]; Encode the statistic and extract the required coefficient [M-DS-GENERATING]; Partition into states and propagate a recurrence [M-DS-STATE]

Theory and methods for 2020-D1-P1 (continued)

Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2020-D1-P1 — Solution — Record 1 of 3

For this record: Theory: T-DS-COUNTING. Methods: M-DS-PARTITION; M-DS-BIJECTION. Full learning route

Let $N = \{1, 2, \dots, n\}$. Consider a word w that satisfies the conditions and let $A, B, C, D \subset N$ be the sets of positions of letters a, b, c and d in w, respectively. By the definition of the words we have $A \sqcup B \sqcup C \sqcup D = N$. The sets A and B are constrained to have even sizes.
In order to construct all suitable words w, choose the set $S = A \cup B$ first; by the conditions, $|S| = |A| + |B|$ must be even. It is well-known that an n-element set (with $n \geq 1$) has 2^{n-1} even subsets, so there are 2^{n-1} possibilities for S.
If $S = \emptyset$ then we can choose $C \subset N$ arbitrarily, and then the set $D = N \setminus C$ is determined uniquely. Since N has 2^n subsets, we have 2^n options for set C and therefore 2^n suitable words w with $S = \emptyset$. Otherwise, if $k = |S| > 0$, we have to choose an arbitrary subset C of $N \setminus S$ and an even subset A of S; then $D = (N \setminus S) \setminus C$ and $B = S \setminus A$ are determined and $|B| = |S| - |A|$ will automatically be even. We have 2^{n-k} choices for C and 2^{k-1} independent choices for A; so for each nonempty even S we have $2^{n-k} \cdot 2^{k-1} = 2^{n-1}$ suitable words.

The number of nonempty even sets S is $2^{n-1} - 1$, so in total, the number of words satisfying the conditions is

$$1 \cdot 2^n + (2^{n-1} - 1) \cdot 2^{n-1} = 4^{n-1} + 2^{n-1}.$$

2020-D1-P1 — Solution — Record 2 of 3

For this record: Theory: T-DS-INDUCTION; T-DS-COUNTING. Methods: M-DS-INDUCT; M-DS-PARTITION; M-DS-STATE. Full learning route

Let a_n denote the number of words of length n over $\mathcal{A} = \{a, b, c, d\}$ such that a and b appear even number of times. Further, we define the following sequences for the number of words of length n, all over $\mathcal{A}$.

- b_n - the number of words with an odd number of a's and even number of b's
- c_n - the number of words with even number of a's and an odd number of b's
- d_n - the number of words with an odd number of a's and an odd number of b's

We will call them A-words, B-words, C-words and D-words, respectively.
It is clear that $a_1 = 2$ and that

$$a_n + b_n + c_n + d_n = 4^n.$$

First, we find a recurrence relation for a_n. If an A-word of length n begins with c or d, it can be followed by any A-word of length $n-1$, contributing with $2a_{n-1}$. If an A-word of length n begins with a, it can be followed by any word of length $n-1$ that contains an odd number of a's and even number of b's, thus contributing with b_{n-1}. If an A-word of length n begins with b, it can be followed by any word of length $n-1$ that contains even number of a's and an odd number of b's, thus contributing with c_{n-1}. Therefore we have the following recurrence relation:

$$a_n = 2a_{n-1} + b_{n-1} + c_{n-1}. \tag{1}$$

Next, we find a recurrence relation for b_n.

If a B-word of length n begins with c or d, it can be followed by any B-word of length $n-1$, contributing with $2b_{n-1}$. If a B-word of length n begins with a, it can be followed by any word of length $n-1$ that contains even number of a's and even number of b's, contributing with a_{n-1}. If a B-word of length n begins with b, it can be followed by any word of length $n-1$ that contains an odd number of a's and an odd number of b's, contributing with $d_{n-1} = 4^{n-1} - a_{n-1} - b_{n-1} - c_{n-1}$. Therefore we have the following recurrence relation:

$$b_n = b_{n-1} + 4^{n-1} - c_{n-1}. \tag{2}$$

2020-D1-P1 — Solution — Record 2 of 3 (continued)

Now observe that $b_k = c_k$ for all k, since simultaneously replacing a's to b's and vice versa we get a C-word from a B-word. Therefore (2) yields $b_n = 4^{n-1}$. Now (1) yields

$$a_n = 2 \cdot a_{n-1} + 2 \cdot 4^{n-2}.$$

Solving the last recurrence relation (for example, diving by 2^n we get $x_n := a_n 2^{-n}$ satisfies $x_n - x_{n-1} = 2^{n-3}$, and it remains to sum up consecutive powers of 2) we get

$$a_n = 2^{n-1} + 4^{n-1}.$$

2020-D1-P1 — Solution — Record 3 of 3

For this record: Theory: T-DS-GENERATING; T-DS-COUNTING. Methods: M-DS-GENERATING. Full learning route

Consider the sum

$$\frac{(a+b+c+d)^n + (-a-b+c+d)^n + (-a+b+c+d)^n + (a-b+c+d)^n}{4}. \qquad (*)$$

Expanding the parentheses as

$$(a+b+c+d)^n = (a+b+c+d)(a+b+c+d)\dots(a+b+c+d),$$

we get a sum of products $x_1 \dots x_n$, $x_i \in \{a,b,c,d\}$, naturally corresponding to the words of length n over the alphabet $\{a,b,c,d\}$. Consider the other terms in the numerator similarly.

If a word $x_1 \dots x_n$ contains A, B, C, D letters a, b, c and d respectively, we get $a^A b^B c^C d^D$ with the coefficient

$$\frac{1+(-1)^{A+B}+(-1)^A+(-1)^B}{4} = \frac{(1+(-1)^A)(1+(-1)^B)}{4} = \begin{cases} 1, & \text{if } A \text{ and } B \text{ are even} \\ 0, & \text{otherwise.} \end{cases}$$

Hence, by substituting $a = b = c = d = 1$ in $(*)$ we get the answer $(4^n + 2^{n+1})/4 = 4^{n-1} + 2^{n-1}$.

2020-D1-P2 — Statement

Day 1, P2

Let A and B be $n \times n$ real matrices such that

$$\operatorname{rk}(AB - BA + I) = 1$$

where I is the $n \times n$ identity matrix.
Prove that

$$\operatorname{trace}(ABAB) - \operatorname{trace}(A^2B^2) = \frac{1}{2}n(n-1).$$

($\operatorname{rk}(M)$ denotes the rank of matrix M, i.e., the maximum number of linearly independent columns in M. $\operatorname{trace}(M)$ denotes the trace of M, that is the sum of diagonal elements in M.)

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Theory and methods for 2020-D1-P2

Theory Trace pairing and commutator calculus [T-LA-TRACE]; Rank, dimension and exact linear constraints [T-LA-RANK]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]
Methods Use trace to extract a scalar invariant [M-LA-TRACE]
Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2020-D1-P2 — Solution

For this record: Theory: T-LA-TRACE; T-LA-RANK; T-LA-POLYNOMIAL. Methods: M-LA-TRACE. Full learning route

Let $X = AB - BA$. The first important observation is that

$$\operatorname{trace}(X^2) = \operatorname{trace}(ABAB - ABBA - BAAB + BABA) = 2\operatorname{trace}(ABAB) - 2\operatorname{trace}(A^2B^2)$$

using that the trace is cyclic. So we need to prove that $\operatorname{trace}(X^2) = n(n-1)$.

By assumption, $X + I$ has rank one, so we can write $X + I = v^t w$ for two vectors v, w. So

$$X^2 = (v^t w - I)^2 = I - 2v^t w + v^t w v^t w = I + (wv^t - 2)v^t w.$$

Now by definition of X we have $\operatorname{trace}(X) = 0$ and hence $wv^t = \operatorname{trace}(wv^t) = \operatorname{trace}(v^t w) = n$ so that indeed

$$\operatorname{trace}(X^2) = n + (n-2)n = n(n-1).$$

An alternative way to use the rank one condition is via eigenvalues: Since $X + I$ has rank one, it has eigenvalue 0 with multiplicity $n - 1$. So X has eigenvalue -1 with multiplicity $n - 1$. Since $\operatorname{trace}(X) = 0$ the remaining eigenvalue of X must be $n - 1$. Hence

$$\operatorname{trace}(X^2) = (n-1)^2 + (n-1) \cdot 1^2 = n(n-1).$$

2020-D1-P3 — Statement

Day 1, P3

Let $d \geq 2$ be an integer. Prove that there exists a constant $C(d)$ such that the following holds: For any convex polytope $K \subset \mathbb{R}^d$, which is symmetric about the origin, and any $\varepsilon \in (0, 1)$, there exists a convex polytope $L \subset \mathbb{R}^d$ with at most $C(d)\varepsilon^{1-d}$ vertices such that

$$(1 - \varepsilon)K \subseteq L \subseteq K.$$

(For a real α, a set $T \subset \mathbb{R}^d$ with nonempty interior is a convex polytope with at most α vertices, if T is a convex hull of a set $X \subset \mathbb{R}^d$ of at most α points, i.e., $T = \{\sum_{x \in X} t_x x \mid t_x \geq 0, \sum_{x \in X} t_x = 1\}$. For a real λ, put $\lambda K = \{\lambda x \mid x \in K\}$. A set $T \subset \mathbb{R}^d$ is symmetric about the origin if $(-1)T = T$.)

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Theory and methods for 2020-D1-P3

Theory Convex bodies, supporting functionals, and packing [T-DS-CONVEX]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Extremal choice, exchange arguments, and invariants [T-DS-EXTREMAL]; Metric contractions, covers, and nearest points [T-DS-METRIC]
Methods Repair an extremal configuration by a local exchange [M-DS-EXTREMAL]; Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]
Strategy Choose an extremal object and force rigidity [S-EXTREMAL]; Translate geometry into a controlled representation [S-GEOMETRIC]; Build a chain of bounds and track equality [S-BOUND]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

2020-D1-P3 — Solution — Record 1 of 2

For this record: Theory: T-DS-CONVEX; T-DS-GEOMETRY; T-DS-EXTREMAL. Methods: M-DS-EXTREMAL; M-DS-GEOMETRIC. Full learning route

[in elementary terms] Let $\{p_1, \dots, p_m\}$ be an inclusion-maximal collection of points on the boundary ∂K of K such that the homothetic copies $K_i := p_i + \frac{\varepsilon}{2}K$ have disjoint interiors. We claim that the convex hull $L := \operatorname{conv}\{p_1, \dots, p_m\}$ satisfies all the conditions.

2020-D1-P3 — Solution — Record 1 of 2 (continued)

First, note that by convexity of K we have $aK+bK=(a+b)K$ for $a,b>0$. It follows that $K_i\subset(1+\frac{\varepsilon}{2})K$. On the other hand, if $k\in K$, $a>0$ and and $ak\in K_i$, then

$$p_i\in ak-\frac{\varepsilon}{2}K=ak+\frac{\varepsilon}{2}K\subset\left(a+\frac{\varepsilon}{2}\right)K,$$

and since p_i is a boundary point of K, we get $a+\frac{\varepsilon}{2}\geqslant 1$, $a\geqslant 1-\frac{\varepsilon}{2}$. It means that all K_i lie between $(1-\frac{\varepsilon}{2})K$ and $(1+\frac{\varepsilon}{2})K$. Since their interiors are disjoint, by the volume counting we obtain

$$m\left(\frac{\varepsilon}{2}\right)^d\leqslant\left(1+\frac{\varepsilon}{2}\right)^d-\left(1-\frac{\varepsilon}{2}\right)^d\leqslant(3/2)^d\varepsilon$$

(since $F(\varepsilon)=(1+\frac{\varepsilon}{2})^d-(1-\frac{\varepsilon}{2})^d$ is a polynomial in ε without constant term with non-negative coefficients which sum up to $(3/2)^d-(1/2)^d$), therefore $m\leqslant 3^d\varepsilon^{1-d}$.

It is clear that $L\subseteq K$, so it remains to prove that $(1-\varepsilon)K\subseteq L$. Assume the contrary: there exists a point $p\in(1-\varepsilon)K\setminus L$. Separate p from L by a hyperplane: Choose a linear functional ℓ such that $\ell(p)>\max\limits_{x\in L}\ell(x)=\max_i\ell(p_i)$. Choose $x\in K$ such that $\ell(x)=:a$ is maximal possible. Note that by our construction $x+\frac{\varepsilon}{2}K$ has a common point with some K_i: there exists a point $z\in(x+\frac{\varepsilon}{2}K)\cap(p_i+\frac{\varepsilon}{2}K)$. We have

$$\ell(p_i)+\frac{\varepsilon}{2}a\geqslant\ell(z)\geqslant\ell(x)-\frac{\varepsilon}{2}a,$$

and therefore $\ell(p_i)\geqslant a(1-\varepsilon)$. Since $p\in(1-\varepsilon)K$, we obtain $\ell(p)\leqslant a(1-\varepsilon)$. A contradiction.

2020-D1-P3 — Solution — Record 2 of 2

For this record: Theory: T-DS-CONVEX; T-DS-METRIC; T-DS-EXTREMAL. Methods: M-DS-EXTREMAL; M-DS-GEOMETRIC. Full learning route

[in the language of Banach spaces] Equip $\mathbb{R}^d$ with the norm $\|\cdot\|$, whose unit ball is K, call this Banach space V. Choose an inclusion maximal set $X\subset\partial K$ whose pairwise distances are $\geq\varepsilon$. Put $L=\operatorname{conv}X$.
The inclusion $L\subseteq K$ follows from the convexity of K. If the inclusion $(1-\varepsilon)K\subseteq L$ fails then the Hahn–Banach theorem provides a unit linear functional $\lambda\in V^*$ such that $\max\{\lambda(L)\}=\max\{\lambda X\}\leq 1-\varepsilon$. Then the point $x\in K$, where the maximum $\max\{\lambda(K)\}=1$ is attained (thanks to the finite dimension and compactness) is in ∂K and, as λ witnesses, at distance $\geq\varepsilon$ from all other points of L and X, contradicting the inclusion-maximality of X.

The upper bound for the cardinality $|X|$ is obtained by noting that the $\varepsilon/2$ balls centered at the points of X are pairwise disjoint and lie in the difference of balls $(1+\varepsilon/2)K\setminus(1-\varepsilon/2)K$, whose volume is $((1+\varepsilon/2)^d-(1-\varepsilon/2)^d)\operatorname{vol}K$, the volume of each of the small balls being $\varepsilon^d/2^d\operatorname{vol}K$. Hence

$$|X|\leq\frac{(2+\varepsilon)^d-(2-\varepsilon)^d}{\varepsilon^d}=O(\varepsilon^{1-d}).$$

2020-D1-P4 — Statement

Day 1, P4

A polynomial p with real coefficients satisfies the equation $p(x+1)-p(x)=x^{100}$ for all $x\in\mathbb{R}$. Prove that $p(1-t)\geqslant p(t)$ for $0\leqslant t\leqslant 1/2$.

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Theory and methods for 2020-D1-P4

Theory Finite differences and shift-operator ideals [T-AL-DIFFERENCE]; Holomorphic bounds, poles, and logarithmic-branch sums [T-AL-COMPLEXANALYSIS]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]
Methods Build a holomorphic function with a useful boundary [M-AL-ANALYTIC]; Compare coefficients after a structural substitution [M-AL-COEFFICIENT]
Strategy Control the limiting operation before taking the limit [S-ANALYTIC]; Reduce size while preserving the defining conditions [S-INDUCTION]

Statement Recorded solutions Catalog

2020-D1-P4 — Solution — Record 1 of 2

For this record: Theory: T-AL-DIFFERENCE; T-AL-COMPLEXANALYSIS; T-AL-POLYNOMIAL. Methods: M-AL-ANALYTIC. Full learning route

Denote $h(z) = p(1-\bar{z}) - p(z)$ for complex z. For $t \in \mathbb{R}$ we have $h(it) = p(1+it) - p(it) = t^{100}$, $h(1/2+it) = 0$.
If $p(z) = c_n z^n + \cdots + c_0$, $c_n \neq 0$, we have

$$h(a+it) = p((1-a)+it) - p(a+it) = (1-2a)(nc_n i^{n-1} t^{n-1} + Q(t,a))$$

for some polynomial Q having degree at most $n-2$ with respect to the variable t. Substituting $a = 0$ we get $n = 101$, $c_n = 1/101$.
Next, for large $|t|$ we see that $\Re(h(a+it)) > 0$ for $0 \leqslant a < 1/2$.

Therefore by Maximum Principle for the harmonic function $\Re h$ and the rectangle $[0,1/2] \times [-N,N]$ for large enough N we conclude that $\Re h$ is non-negative in this rectangle, in particular on $[0,1/2]$, as we need.

2020-D1-P4 — Solution — Record 2 of 2

For this record: Theory: T-AL-DIFFERENCE; T-AN-DERIVATIVE; T-AN-CONVEXITY; T-DS-INDUCTION. Methods: M-AL-COEFFICIENT. Full learning route

Let $p(x) = \sum_{j=0}^{m} a_j x^j$. Then

$$p(x+1) - p(x) = \sum_{j=0}^{m} a_j((x+1)^j - x^j) = a_1 + a_2(2x+1) + \cdots + a_m \left(m x^{m-1} + \binom{m}{2} x^{m-2} + \cdots + 1 \right).$$

This implies that $m = 101$, $m a_m = 1$ so $a_{101} = \frac{1}{101}$, $(m-1)a_{m-1} + a_m \binom{m}{2} = 0$ so $a_{100} = -\frac{1}{2}$ etc. For $j \geqslant 1$ a_j is uniquely defined, a_0 may be chosen arbitrarily.

The equality $p_{2n}(\frac{1}{2}) = 0$ holds because $0 = p_{2n}(\frac{1}{2}) + p_{2n}(1-\frac{1}{2}) = 2p_{2n}(\frac{1}{2})$. Let $n \geqslant 1$ be an integer and let p_n be a polynomial such that $p_n(x+1) - p_n(x) = x^n$ for all x and $p_n(0) = 0 = p_n(1)$. The above considerations prove the uniqueness of p_n. We have $p_1(x) = \frac{1}{2}x^2 - \frac{1}{2}x$. Also $p_n'(x+1) - p_n'(x) = nx^{n-1} = n(p_{n-1}(x+1) - p_{n-1}(x))$. Therefore $p_n'(x) = np_{n-1}(x) + c_{n-1}$ for a properly chosen constant c_{n-1}. We shall prove that

$$p_{2n-1}(x) - p_{2n-1}(1-x) = 0, \quad p_{2n}(x) + p_{2n}(1-x) = 0, \quad c_{2n} = 0, \quad p_{2n}''(x) = 2n(2n-1)p_{2n-2}(x) \qquad (1)$$

for $n = 1, 2, \ldots$ and for all x. Simple computation shows that $p_1(x) - p_1(1-x) = 0$. We have $(p_2(x) + p_2(1-x))' = 2p_1(x) + c_1 - (2p_1(1-x) + c_1) = 0$ so the map $x \mapsto p_2(x) + p_2(1-x)$ is constant thus $p_2(x) + p_2(1-x) = p_2(0) + p_2(1-0) = 0$. If the first two equalities hold for some n then

$$(p_{2n+1}(x) - p_{2n+1}(1-x))' = (2n+1)p_{2n}(x) + c_{2n} + (p_{2n}(1-x) + c_{2n}) = 2c_{2n}$$

so there exists $b \in \mathbb{R}$ such that $p_{2n+1}(x) - p_{2n+1}(1-x) = 2c_{2n}x + b$ for all x. $p_{2n+1}(0) - p_{2n+1}(1-0) = 0$ and $p_{2n+1}(1) - p_{2n+1}(1-1) = 0$ so $2c_{2n} = 0 = b$. This proves that $p_{2n+1}(x) - p_{2n+1}(1-x) = 0$ for all x. In a similar way we shall prove the second equality:

$$(p_{2n+2}(x) + p_{2n+2}(1-x))' = (2n+2)p_{2n+1}'(x) + c_{2n+1} - (2n+2)(p_{2n+1}(1-x) + c_{2n+1}) = 0$$

2020-D1-P4 — Solution — Record 2 of 2 (continued)

so the map $x \mapsto p_{2n+2}(x)+p_{2n+2}(1-x)$ is constant hence $p_{2n+2}(x)+p_{2n+2}(1-x) = p_{2n+2}(0)+p_{2n+2}(1-0) = 0$ for all x. Now $p''_{2n+2}(x) = ((2n+2)p_{2n+1}(x)+c_{2n+1})' = (2n+2)p'_{2n+1}(x) = (2n+2)((2n+1)p_{2n}(x)+c_{2n}) = (2n+2)(2n+1)p_{2n}(x)$. Since $p'_2(x) = 2p_1(x)+c_1 = x^2-x+c_1$ we obtain $p''_2(x) = 2x-1 < 0$ for $x < \frac{1}{2}$. The function p_2 is strictly concave on $[0,\frac{1}{2}]$ and $p_2(0) = 0 = p_2(\frac{1}{2})$. Therefore $p_2(x) > 0$ for $x \in (0,\frac{1}{2})$. This together with the equality $p''_4(x) = 12p_2(x)$ implies that p_4 is strictly convex on $[0,\frac{1}{2}]$ so in view of $p_4(0) = 0 = p_4(\frac{1}{2})$ we conclude that $p_4(x) < 0$ for $x \in (0,\frac{1}{2})$. Easy induction shows that for $x \in (0,\frac{1}{2})$ one has $p_{2n}(x) > 0$ for an odd n and $p_{2n}(x) < 0$ for an even n. If $t \in (0,\frac{1}{2})$ then by (1) we get $p_{100}(1-t)-p_{100}(t) = -2p_{100}(t) > 0$ as required.

Day 2

2020-D2-P5 — Statement

Day 2, P5

Find all twice continuously differentiable functions $f : \mathbb{R} \to (0,+\infty)$ satisfying

$$f''(x)f(x) \geq 2(f'(x))^2$$

for all $x \in \mathbb{R}$.

Karen Keryan, Yerevan State University & American University of Armenia, Yerevan

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Theory and methods for 2020-D2-P5

Theory Convexity, Jensen, and supporting lines [T-AN-CONVEXITY]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]
Methods Design an auxiliary derivative [M-AN-AUXILIARY]; Choose the correct convexity comparison [M-AN-CONVEX]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2020-D2-P5 — Solution

For this record: Theory: T-AN-CONVEXITY; T-AN-DERIVATIVE. Methods: M-AN-AUXILIARY; M-AN-CONVEX. Full learning route

We shall show that only positive constant functions satisfy the condition.
Let $g(x) = \dfrac{1}{f(x)}$. Notice that

$$g'' = \left(\frac{1}{f}\right)'' = \left(\frac{-f'}{f^2}\right)' = \left(\frac{2(f')^2 - f''f}{f^3}\right) \leq 0,$$

so the positive function $g(x)$ is concave. We show that g must be constant.
Take two arbitrary real numbers $a < b$. By the concavity of g, for all $u < a$ and $v > b$ we have

$$\frac{g(a)-g(u)}{a-u} \geq \frac{g(b)-g(a)}{b-a} \geq \frac{g(v)-g(b)}{v-b}.$$

Combining this with $g(u), g(v) > 0$ we get

$$\frac{g(a)}{a-u} > \frac{g(b)-g(a)}{b-a} > \frac{-g(b)}{v-b}$$

Now by taking limits $u \to -\infty$ and $v \to \infty$ we obtain

$$0 \geq \frac{g(b)-g(a)}{b-a} \geq 0,$$

2020-D2-P5 — Solution (continued)

so $g(a) = g(b)$. This holds for any pair (a, b), so $g(x)$ is constant and $f(x) = 1/g(x)$ also is constant. If f is constant then $f' = f'' = 0$, so the condition is satisfied.

Remark. Instead of the function $1/f(x)$, the same idea works with $\arctan f(x)$:

$$(\arctan f(x))'' = \frac{f''(1+f^2) - 2(f')^2}{(1+f^2)^2} = \frac{f''(1+f^2) - 2(f')^2(1+f^2)}{(1+f^2)^2} = \frac{f'' - 2(f')^2}{1+f^2} \geq 0.$$

As can be seen, $\arctan f(x)$ is a bounded convex function, therefore it must be constant.

2020-D2-P6 — Statement

Day 2, P6

Find all prime numbers p for which there exists a unique $a \in \{1, 2, \ldots, p\}$ such that $a^3 - 3a + 1$ is divisible by p.

Géza Kós, Loránd Eötvös University, Budapest

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Theory and methods for 2020-D2-P6

Theory Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Root identities, symmetric sums, and logarithmic derivatives [T-AL-ROOTS]; Polynomial factorization and coprime divisibility [T-AL-FACTORIZATION]
Methods Choose a modulus that exposes the obstruction [M-AL-MODULAR]; Factor first and compare irreducible exponents [M-AL-FACTOR]; Translate root restrictions into symmetric coefficients [M-AL-VIETA]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2020-D2-P6 — Solution — Record 1 of 3

For this record: Theory: T-AL-FINITEFIELD; T-AL-POLYNOMIAL; T-AL-ROOTS. Methods: M-AL-MODULAR; M-AL-FACTOR. Full learning route

We show that $p = 3$ the only prime that satsfies the condition.
Let $f(x) = x^3 - 3x + 1$. As preparation, let's compute the roots of $f(x)$. By Cardano's formula, it can be seen that the roots are

$$2\,\mathrm{Re}\sqrt[3]{\frac{-1}{2} + \sqrt{\left(\frac{-1}{2}\right)^2 + \left(\frac{-3}{3}\right)^3}} = 2\,\mathrm{Re}\sqrt[3]{\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}} = \left\{2\cos\frac{2\pi}{9}, 2\cos\frac{4\pi}{9}, 2\cos\frac{8\pi}{9}\right\}$$

where all three values of the complex cubic root were taken.
Notice that, by the trigonometric identity $2\cos 2t = (2\cos t)^2 - 2$, the map $\varphi(x) = x^2 - 2$ cyclically permutes the three roots. We will use this map to find another root of f, when it is considered over $\mathbb{F}_p$.
Suppose that $f(a) = 0$ for some $a \in \mathbb{F}_p$ and consider

$$g(x) = \frac{f(x)}{x-a} = \frac{f(x) - f(a)}{x-a} = x^2 + ax + (a^2 - 3).$$

We claim that $b = a^2 - 2$ is a root of $g(x)$. Indeed,

$$g(b) = (a^2-2)^2 + a(a^2-2) + (a^2-3) = (a+1)\cdot f(a) = 0.$$

By Vieta's formulas, the other root of $g(x)$ is $c = -a - b = -a^2 - a + 2$.
If f has a single root then the three roots must coincide, so

$$a = a^2 - 2 = -a^2 - a + 2.$$

2020-D2-P6 — Solution — Record 1 of 3 (continued)

Here the quadratic equation $a = a^2 - 2$, or equivalently $(a+1)(a-2) = 0$, has two solutions, $a = -1$ and $a = 2$. By $f(-1) = f(2) = 3$, in both cases we have $0 = f(a) = 3$, so the only choice is $p = 3$.

Finally, for $p = 3$ we have $f(1) = -1$, $f(2) = 3$ and $f(3) = 19$, from these values only $f(2)$ is divisible by 3, so $p = 3$ satisfies the condition.

2020-D2-P6 — Solution — Record 2 of 3

For this record: Theory: T-AL-FINITEFIELD; T-AL-POLYNOMIAL; T-AL-ROOTS. Methods: M-AL-MODULAR; M-AL-FACTOR. Full learning route

(outline) Define $f(x)$ and $g(x)$ like in Solution 1. The discriminant of $g(x)$ is

$$\Delta_g = a^2 - 4(a^2 - 3) = 12 - 3a^2.$$

We show that Δ_g has a square root in $\mathbb{F}_p$.
Take two integers k, m (to be determinated later) and consider

$$\Delta_g = \Delta_g + (ka + m)f(a) = ka^4 + ma^3 - (3k + 1)a^2 + (k - 3m)a + (m + 12).$$

Our goal is to choose k, m in such a way that the last expression is a complete square. Either by direct calculations or guessing, we can find that $k = m = 4$ works:

$$\Delta_g = \Delta_g + (4a + 4)f(a) = 4a^4 + 4a^3 - 15a^2 - 8a + 16 = (2a^2 + a - 4)^2.$$

If $p \neq 2$ then we can conclude that $f(x)$ has either no or three roots, therefore p is suitable if and only is $f(x)$ is a complete cube: $x^3 - 3x + 1 = (x - a)^3$. From Vieta's formulas $a^3 = 1$, so $a \neq 0$ and $3a = 0$, which is possible if $p = 3$.
For $p = 3$ we have $f(x) = (x + 1)^3$, so $p = 3$ is suitable.

The case $p = 2$ must be checked separately because the quadratic formula contains a division by 2. $f(1) = -1$ and $f(2) = 3$, so $p = 2$ is not suitable.

2020-D2-P6 — Solution — Record 3 of 3

For this record: Theory: T-AL-FINITEFIELD; T-AL-ROOTS; T-AL-FACTORIZATION. Methods: M-AL-MODULAR; M-AL-VIETA. Full learning route

(outline) Assume $p > 3$; the cases $p = 2$ and $p = 3$ will be checked separately.
Let $f(x) = x^3 - 3x + 1$ and suppose that $a \in \mathbb{F}_p$ is a root of $f(x)$, and let $b, c \in \mathbb{F}_{p^2}$ be the other two roots. The discriminant Δ_f of $f(x)$ can be expressed by the elementary symmetric polynomials of a, b, c; it can be calculated that

$$\Delta_f = (b - c)^2(a - b)^2(a - c)^2 = 81 = 9^2,$$

so

$$(b - c)(a - b)(a - c) = \pm 9 \in \mathbb{F}_p.$$

Notice that $\Delta_f \neq 0$, so the three roots are distinct.
Either $b, c \in \mathbb{F}_p$ or b, c are conjugate elements in $\mathbb{F}_{p^2} \setminus \mathbb{F}_p$, we have $(a - b)(a - c) \in \mathbb{F}_p$, so $b - c = \frac{(b-c)(a-b)(a-c)}{(a-b)(a-c)} \in \mathbb{F}_p$. From Vieta's formulas we have $b + c \in \mathbb{F}_p$ as well; since $p \neq 2$, it follows that $b, c \in \mathbb{F}_p$. Now $f(x)$ has three distinct roots in $\mathbb{F}_p$, so p cannot be suitable.

$p = 2$ does not satisfies the condition because both $f(1) = -1$ and $f(2) = 3$ are odd. $p = 3$ is suitable, because $f(2) = 3$ is divisible by 3 while $f(1) = -1$ and $f(3) = 19$ are not.

2020-D2-P7 — Statement

Day 2, P7

Let G be a group and $n \geq 2$ be an integer. Let H_1 and H_2 be two subgroups of G that satisfy

$$[G : H_1] = [G : H_2] = n \quad \text{and} \quad [G : (H_1 \cap H_2)] = n(n-1).$$

Prove that H_1 and H_2 are conjugate in G.
(Here $[G : H]$ denotes the index of the subgroup H, i.e. the number of distinct left cosets xH of H in G. The subgroups H_1 and H_2 are conjugate if there exists an element $g \in G$ such that $g^{-1}H_1g = H_2$.)

Ilya Bogdanov and Alexander Matushkin, Moscow Institute of Physics and Technology

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Theory and methods for 2020-D2-P7

Theory Groups, orders, generators, and quotients [T-AL-GROUP]; Group actions, stabilizers, and semidirect products [T-AL-ACTION]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Convert group data into orbits or cosets [M-AL-ACTION]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2020-D2-P7 — Solution — Record 1 of 2

For this record: Theory: T-AL-GROUP; T-AL-ACTION. Methods: M-AL-ACTION. Full learning route

Denote $K = H_1 \cap H_2$. Since

$$n(n-1) = [G : K] = [G : H_1][H_1 : K] = n[H_1 : K],$$

we obtain that $[H_1 : K] = n-1$. Thus, the subgroup H_1 is partitioned into $n-1$ left cosets of K, say $H_1 = \bigsqcup_{i=1}^{n-1} h_iK$. Therefore, the set $H_1H_2 = \{ab : a \in H_1, b \in H_2\}$ is partitioned as

$$H_1H_2 = \left(\bigsqcup_{i=1}^{n-1} h_iK\right) H_2 = \bigsqcup_{i=1}^{n-1} h_iKH_2 = \bigsqcup_{i=1}^{n-1} h_iH_2.$$

The last equality holds because $K \subseteq H_2$, so $KH_2 = H_2$. The last expression is a disjoint union since

$$h_iH_2 \cap h_jH_2 \neq \emptyset \iff h_i^{-1}h_j \in H_2 \iff h_i^{-1}h_j \in K \iff h_i = h_j.$$

Thus, H_1H_2 is a disjoint union of $n-1$ left cosets with respect to H_2; hence $L = G \setminus (H_1H_2)$ is the remaining such left coset. Similarly, L is a right coset with respect to H_1. Therefore, for each $g \in L$ we have $L = gH_2 = H_1g$, which yields $H_2 = g^{-1}H_1g$.

2020-D2-P7 — Solution — Record 2 of 2

For this record: Theory: T-AL-GROUP; T-AL-ACTION; T-DS-COUNTING. Methods: M-AL-ACTION. Full learning route

Put $G/H_1 = X$ and $G/H_2 = Y$, those are n-element sets acted on by G from the left. Let G act on $X \times Y$ from the left coordinate-wise, consider this product as a table, with rows being copies of X and columns being copies of Y.
The stabilizer of a point (x, y) in $X \times Y$ is $H_1 \cap H_2$. By the orbit-stabilizer theorem, we obtain that the orbit Z of (x, y) has size $[G : H_1 \cap H_2] = n(n-1)$.
If Z contains a whole column then there is a subgroup G_1 of G that stabilizes x and acts transitively on Y. If we conjugate G_1 to a group G_1', then its action remains transitive on Y, so by conjugation we obtain columns of the table. Since G acts transitively on X, we cover all the columns. It follows that $Z = X \times Y$, so

$$n(n-1) = |Z| = |X \times Y| = n^2,$$

which is a contradiction.

2020-D2-P7 — Solution — Record 2 of 2 (continued)

Hence every column of $X \times Y$ has an element not from Z. The same holds for the rows of $X \times Y$. There are n elements not from Z in total and they induce a bijection between X and Y which allows us to identify $X = Y$.
After this identification, every element (x, x) from the diagonal of $X \times X$ (i.e. from $(X \times X) \setminus Z$) is moved to a diagonal element by any $g \in G$, because $gx = gx$. In this formula the action of g in the left hand side and the action of g in the right hand side are the actions of g on X and Y respectively. Therefore our bijection between X and Y is an isomorphism of sets with a left action of G. Since H_1 and H_2 are stabilizers of the points in the same transitive action of G, we conclude that they are conjugate.

Remark. The situation in the problem is possible for every $n \geq 2$: let $G = S_n$ and let H_1 an H_2 be the stabilizer subgroups of two elements.

2020-D2-P8 — Statement

Day 2, P8

Compute

$$\lim_{n\to\infty} \frac{1}{\log\log n} \sum_{k=1}^{n} (-1)^k \binom{n}{k} \log k.$$

(Here log denotes the natural logarithm.)

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Theory and methods for 2020-D2-P8

Theory Nonnegative sums, Fubini, and integral kernels [T-AN-FUBINI]; Scale separation and limit comparisons [T-AN-ASYMPTOTICS]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Finite counting and incidence identities [T-DS-COUNTING]; Series, finite cancellation, and summation by parts [T-AN-SERIES]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Encode convolution or denominators by a power series [M-AN-GENERATING]; Separate a main range from exceptional scales [M-AN-SPLIT]
Strategy Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2020-D2-P8 — Solution — Record 1 of 2

For this record: Theory: T-AN-FUBINI; T-AN-ASYMPTOTICS; T-AN-INTEGRAL; T-DS-COUNTING. Methods: M-AN-GENERATING; M-AN-SPLIT. Full learning route

Answer: 1.
The idea is that if $f(k) = \int g^k$, then

$$\sum (-1)^k \binom{n}{k} f(k) = \int (1-g)^n.$$

To relate this to logarithm, we may use the Frullani integrals

$$\begin{aligned}
\int_0^\infty \frac{e^{-x} - e^{-kx}}{x}\,dx &= \lim_{c\to+0} \left(\int_c^\infty \frac{e^{-x}}{x}\,dx - \int_c^\infty \frac{e^{-kx}}{x}\,dx \right) \\
&= \lim_{c\to+0} \left(\int_c^\infty \frac{e^{-x}}{x}\,dx - \int_{kc}^\infty \frac{e^{-x}}{x}\,dx \right) \\
&= \lim_{c\to+0} \int_c^{kc} \frac{e^{-x}}{x}\,dx \\
&= \log k + \lim_{c\to+0} \int_c^{kc} \frac{e^{-x}-1}{x}\,dx = \log k.
\end{aligned}$$

2020-D2-P8 — Solution — Record 1 of 2 (continued)

This gives the integral representation of our sum:

$$A := \sum_{k=1}^{n} (-1)^k \binom{n}{k} \log k = \int_0^\infty \frac{-e^{-x} + 1 - (1-e^{-x})^n}{x}\, dx.$$

Now the problem is reduced to a rather standard integral asymptotics.

We have $(1-e^{-x})^n \geqslant 1 - ne^{-x}$ by Bernoulli inequality, thus $0 \leqslant -e^{-x} + 1 - (1-e^{-x})^n \leqslant ne^{-x}$, and we get

$$\begin{aligned} 0 &\leqslant \int_M^\infty \frac{-e^{-x} + 1 - (1-e^{-x})^n}{x}\, dx \\ &\leqslant n \int_M^\infty \frac{e^{-x}}{x}\, dx \leqslant nM^{-1} \int_M^\infty e^{-x}\, dx = nM^{-1}e^{-M}. \end{aligned}$$

So choosing M such that $Me^M = n$ (such M exists and goes to ∞ with n) we get

$$A = O(1) + \int_0^M \frac{-e^{-x} + 1 - (1-e^{-x})^n}{x}\, dx.$$

Note that for $0 \leqslant x \leqslant M$ we have $e^{-x} \geqslant e^{-M} = M/n$, and $(1-e^{-x})^{n-1} \leqslant e^{-e^{-x}(n-1)} \leqslant e^{-M(n-1)/n}$ tends to 0 uniformly in x. Therefore

$$\int_0^M \frac{(1-e^{-x})(1-(1-e^{-x})^{n-1})}{x}\, dx = (1+o(1)) \int_0^M \frac{1-e^{-x}}{x}\, dx.$$

Finally

$$\begin{aligned} \int_0^M \frac{1-e^{-x}}{x}\, dx &= \int_0^1 \frac{1-e^{-x}}{x}\, dx + \int_1^M \frac{-e^{-x}}{x}\, dx + \int_1^M \frac{dx}{x} \\ &= \log M + O(1) = \log(M + \log M) + O(1) \\ &= \log\log n + O(1). \end{aligned}$$

and we get $A = (1+o(1)) \log\log n$.

2020-D2-P8 — Solution — Record 2 of 2

For this record: Theory: T-AN-ASYMPTOTICS; T-AN-INTEGRAL; T-AN-SERIES; T-AL-POLYNOMIAL; T-DS-COUNTING. Methods: M-AN-GENERATING; M-AN-SPLIT. Full learning route

Editorially corrected proof.
The limit is 1. Put

$$S_n = \sum_{k=1}^{n} (-1)^k \binom{n}{k} \log k.$$

We first obtain an integral representation. Partial fractions give, for $m \geq 0$ and $x > 0$,

$$\sum_{k=0}^{m} (-1)^k \binom{m}{k} \frac{1}{x+k} = \frac{m!}{x(x+1)\cdots(x+m)}.$$

Indeed, the coefficient of $1/(x+k)$ on the right is $(-1)^k \binom{m}{k}$. Pascal's identity also gives

$$\sum_{k=j+1}^{n} (-1)^k \binom{n}{k} = (-1)^{j+1} \binom{n-1}{j}, \qquad 0 \leq j < n.$$

Since

$$\log k = \sum_{j=1}^{k-1} \int_0^1 \frac{dx}{x+j},$$

2020-D2-P8 — Solution — Record 2 of 2 (continued)

interchanging finite sums and integrals yields

$$\begin{aligned} S_n &= \int_0^1 \sum_{j=1}^{n-1} \frac{(-1)^{j+1}}{x+j}\binom{n-1}{j}\,dx \\ &= \int_0^1 \left\{ \frac{1}{x} - \frac{(n-1)!}{x(x+1)\cdots(x+n-1)} \right\} dx \\ &= \int_0^1 \frac{1-R_n(x)}{x}\,dx, \end{aligned}$$

where

$$R_n(x) = \prod_{j=1}^{n-1} \left(1+\frac{x}{j}\right)^{-1}.$$

The apparent singularity at $x=0$ is removable, as is clear from the finite sum in the first integral.

Write $H = H_{n-1} = \sum_{j=1}^{n-1} 1/j$. The elementary inequalities

$$y - y^2 \le \log(1+y) \le y, \qquad y \ge 0,$$

give, for $0 \le x \le 1$,

$$e^{-Hx} \le R_n(x) \le \exp\left(-Hx + x^2 \sum_{j=1}^{n-1} \frac{1}{j^2} \right) \le e^{-(H-2)x}.$$

Here we used $\sum_{j=1}^{\infty} j^{-2} \le 2$ and $x^2 \le x$. In particular, when $H > 2$,

$$J(H-2) \le S_n \le J(H), \qquad J(a) := \int_0^1 \frac{1-e^{-ax}}{x}\,dx \quad (a>0).$$

The parameters $H-2$ and H are constants with respect to the integration variable; the squeeze therefore permits a valid change of variables in each bounding integral. For $a \ge 1$,

$$\begin{aligned} J(a) &= \int_0^a \frac{1-e^{-t}}{t}\,dt \\ &= \log a + \int_0^1 \frac{1-e^{-t}}{t}\,dt - \int_1^a \frac{e^{-t}}{t}\,dt \\ &= \log a + O(1). \end{aligned}$$

The last two integrals are bounded uniformly in $a \ge 1$. Finally, integral comparison gives

$$\log n \le H_{n-1} \le 1 + \log(n-1),$$

so $H = \log n + O(1)$. Consequently both bounds satisfy

$$J(H-2) = \log\log n + O(1), \qquad J(H) = \log\log n + O(1).$$

Thus $S_n = \log\log n + O(1)$, and division by $\log\log n \to \infty$ proves the claimed limit.

3.28 2021

Day 1

2021-D1-P1 — Statement

Day 1, P1 | 10 points

Let A be a real $n \times n$ matrix such that $A^3 = 0$.
(a) Prove that there is a unique real $n \times n$ matrix X that satisfies the equation

$$X + AX + XA^2 = A.$$

(b) Express X in terms of A.

(proposed by Bekhzod Kurbonboev, Institute of Mathematics, Tashkent)

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Theory and methods for 2021-D1-P1

Theory Nilpotents, projections and involutions [T-LA-NILPOTENT]; Determinants, block identities and inverses [T-LA-DETERMINANT]
Methods Treat the unknown matrix as a vector [M-LA-MATRIXOP]; Complete the equation to a product equal to identity [M-LA-INVERSE]
Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2021-D1-P1 — Solution — Record 1 of 3

For this record: Theory: T-LA-NILPOTENT. Methods: M-LA-MATRIXOP. Full learning route

Hint: (a) Multiply the equation by some power of A from left and another power of A from right. (b) Substitute repeatedly $X = A - AX - XA^2$.
First suppose that some matrix X satisfies the equation. We can obtain new equations if we multiply the given equation by some power of A from left and another power of A from right. For example,

$$A^2(X + AX + XA^2)A^2 = A^2XA^2 + A^3 \cdot XA^2 + A^2XA \cdot A^3 = A^2XA^2.$$

The right-hand side is $A^2 \cdot A \cdot A^2 = A^3 \cdot A^2 = 0$, so

$$A^2XA^2 = A^2(X + AX + XA^2)A^2 = A^5 = 0. \quad \text{Similarly,}$$

$$\begin{aligned} A^2X &= A^2(X + AX + XA^2) = A^3 = 0 \\ AXA &= A(X + AX + XA^2)A = A^3 = 0 \\ XA^2 &= (X + AX + XA^2)A^2 = A^3 = 0 \\ AX &= A(X + AX + XA^2) = A^2. \quad \text{Finally} \\ X &= A - AX - XA^2 = A - A^2. \end{aligned}$$

Hence, no matrix other than $A - A^2$ can satisfy the equation.
Note that the argument above does not prove that the matrix $X = A - A^2$ satisfies the equation, because the steps cannot be done in reverse order. That must be verified separately. Indeed,

$$X + AX + XA^2 = (A - A^2) + A(A - A^2) + (A - A^2)A^2 = A - A^4 = A.$$

Hence, $X = A - A^2$ is the unique solution of the equation.

Remark. By multiplying the equation by A^n from left and by A^k from right we can get 9 different equations:

$$\begin{gathered} X + AX + XA^2 = A \qquad XA + AXA = A^2 \qquad XA^2 + AXA^2 = 0 \\ AX + A^2X + AXA^2 = A^2 \qquad AXA + A^2XA = 0 \qquad AXA^2 + A^2XA^2 = 0 \\ A^2X + A^2XA^2 = 0 \qquad A^2XA = 0 \qquad A^2XA^2 = 0 \end{gathered}$$

These formulas provide a system of linear equations for the nine matrices X, AX, A^2X, XA, AXA, A^2XA, XA^2, AXA^2 and A^2XA^2.

2021-D1-P1 — Solution — Record 2 of 3

For this record: Theory: T-LA-NILPOTENT. Methods: M-LA-MATRIXOP. Full learning route

We use a different approach to express X in terms of A. If some matrix X satisfies the equation then

$$X = A - AX - XA^2.$$

Let us substitute this identity in the right-hand side repeatedly until X cancels out everywhere.

Notice that by the condition $A^3 = 0$ we have $A^3 = A^4 = A^5 = A^3X = XA^4 = AXA^4 = A^3XA^2 = 0$, so

$$\begin{aligned}
X &= A - AX - XA^2 \\
&= A - A(A - AX - XA^2) - (A - AX - XA^2)A^2 \\
&= A - (A^2 - A^2X - AXA^2) - (A^3 - AXA^2 - XA^4) \\
&= A - A^2 + A^2X + 2AXA^2 \\
&= A - A^2 + A^2(A - AX - XA^2) + 2A(A - AX - XA^2)A^2 \\
&= A - A^2 + (A^3 - A^3X - A^2XA^2) + 2(A^4 - A^2XA^2 - AXA^4) \\
&= A - A^2 - 3A^2XA^2 \\
&= A - A^2 - 3A^2(A - AX - XA^2)A^2 \\
&= A - A^2 - 3(A^5 - A^3XA^2 - A^2XA^4) \\
&= A - A^2.
\end{aligned}$$

To complete the solution, we have to verify that $X = A - A^2$ is indeed a solution. This step is the same as in Solution 1.

2021-D1-P1 — Solution — Record 3 of 3

For this record: Theory: T-LA-NILPOTENT; T-LA-DETERMINANT. Methods: M-LA-INVERSE. Full learning route

Let $B = I - A + A^2$ so that B is the inverse of $I + A$. Multiplying by B from the left, the equation is equivalent to

$$X + BXA^2 = BA. \tag{1}$$

Now assume X satisfies the equation. Multiplying by A^2 from the right and using $A^3 = 0$ we get $XA^2 = 0$. Hence the equation simplifies to $X = BA = A - A^2$.

On the other hand, $X = BA$ obviously satisfies (1).

2021-D1-P2 — Statement

Day 1, P2 | 10 points

Let n and k be fixed positive integers, and let a be an arbitrary non-negative integer. Choose a random k-element subset X of $\{1, 2, \ldots, k+a\}$ uniformly (i.e., all k-element subsets are chosen with the same probability) and, independently of X, choose a random n-element subset Y of $\{1, \ldots, k+n+a\}$ uniformly.
Prove that the probability

$$\mathsf{P}(\min(Y) > \max(X))$$

does not depend on a.

(proposed by Fedor Petrov, St. Petersburg State University)

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Theory and methods for 2021-D1-P2

Theory Finite probability, conditioning, and expectations [T-AN-PROBABILITY]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Count outcomes or condition on a useful structure [M-AN-PROBABILITY]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2021-D1-P2 — Solution — Record 1 of 2

For this record: Theory: T-AN-PROBABILITY; T-DS-COUNTING. Methods: M-AN-PROBABILITY. Full learning route

Hint: The sets X and Y with $\min(Y) > \max(X)$ are uniquely determined by $X \cup Y$.
The number of choices for (X, Y) is $\binom{k+a}{k} \cdot \binom{n+k+a}{n}$.
The number of such choices with $\min(Y) > \max(X)$ is equal to $\binom{n+k+a}{n+k}$ since this is the number of choices for the $(n+k)$-element set $X \cup Y$ and this union together with the condition $\min(Y) > \max(X)$ determines X and Y uniquely (note in particular that no elements of X will be larger than $k+a$).
Hence the probability is

$$\frac{\binom{n+k+a}{n+k}}{\binom{k+a}{k} \cdot \binom{n+k+a}{n}} = \frac{1}{\binom{n+k}{k}}$$

where the identity can be seen by expanding the binomial coefficients on both sides into factorials and canceling.

Since the right hand side is independent of a, the claim follows.

2021-D1-P2 — Solution — Record 2 of 2

For this record: Theory: T-AN-PROBABILITY; T-DS-COUNTING. Methods: M-AN-PROBABILITY. Full learning route

Let f be the increasing bijection from $\{1, 2, \dots, k+a\}$ to $\{1, \dots, k+a+n\} \setminus Y$. Note that $\min(Y) > \max(X)$ if and only if $\min(Y) > \max(f(X))$.
Note that

$$\{Z_n := Y,\ Z_k := f(X),\ Z_a := f(\{1, 2, \dots, k+a\} \setminus X)\}$$

is a random partition of

$$\{1, \dots, n+k+a\} = Z_n \sqcup Z_k \sqcup Z_a$$

into an n-subset, k-subset, and a-subset.
If an a-subset Z_a is fixed, the conditional probability that $\min(Z_k) > \max(Z_n)$

always equals $1/\binom{n+k}{k}$. Therefore the total probability also equals $1/\binom{n+k}{k}$.

2021-D1-P3 — Statement

Day 1, P3 | 10 points

We say that a positive real number d is good if there exists an infinite sequence $a_1, a_2, a_3, \dots \in (0, d)$ such that for each n, the points $a_1, \dots, a_n$ partition the interval $[0, d]$ into segments of length at most $1/n$ each. Find

$$\sup\{d \mid d \text{ is good}\}.$$

(proposed by Josef Tkadlec)

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Theory and methods for 2021-D1-P3

Theory Finite counting and incidence identities [T-DS-COUNTING]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Methods Split a family by traces, pieces, or labels [M-DS-PARTITION]; Strengthen the statement before the induction step [M-DS-INDUCT]
Strategy Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2021-D1-P3 — Solution

For this record: Theory: T-DS-COUNTING; T-DS-INDUCTION; T-AN-INTEGRAL; T-AN-CONVERGENCE. Methods: M-DS-PARTITION; M-DS-INDUCT. Full learning route

Hint: To get an upper bound, use that some of the gaps after n steps are still intact some steps later.
Let $d^\star = \sup\{d \mid d \text{ is good}\}$. We will show that $d^\star = \ln(2) \doteq 0.693$.
1. $d^\star \le \ln 2$:
Assume that some d is good and let $a_1, a_2, \dots$ be the witness sequence.
Fix an integer n. By assumption, the prefix $a_1, \dots, a_n$ of the sequence splits the interval $[0, d]$ into $n+1$ parts, each of length at most $1/n$.
Let $0 \le \ell_1 \le \ell_2 \le \dots \le \ell_{n+1}$ be the lengths of these parts. Now for each $k = 1, \dots, n$ after placing the next k terms $a_{n+1}, \dots, a_{n+k}$, at least $n+1-k$ of these initial parts remain intact. Hence $\ell_{n+1-k} \le \frac{1}{n+k}$.
Hence

$$d = \ell_1 + \dots + \ell_{n+1} \le \frac{1}{n} + \frac{1}{n+1} + \dots + \frac{1}{2n}. \tag{2}$$

As $n \to \infty$, the RHS tends to $\ln(2)$ showing that $d \le \ln(2)$.
Hence $d^\star \le \ln 2$ as desired.
2. $d^\star \ge \ln 2$:
Observe that

$$\ln 2 = \ln 2n - \ln n = \sum_{i=1}^{n} (\ln(n+i) - \ln(n+i-1)) = \sum_{i=1}^{n} \ln\left(1 + \frac{1}{n+i-1}\right).$$

Interpreting the summands as lengths, we think of the sum as the lengths of a partition of the segment $[0, \ln 2]$ in n parts. Moreover, the maximal length of the parts is $\ln(1+1/n) < 1/n$.
Changing n to $n+1$ in the sum keeps the values of the sum, removes the summand $\ln(1+1/n)$, and adds two summands

$$\ln\left(1 + \frac{1}{2n}\right) + \ln\left(1 + \frac{1}{2n+1}\right) = \ln\left(1 + \frac{1}{n}\right).$$

This transformation may be realized by adding one partition point in the segment of length $\ln(1 + 1/n)$.
In total, we obtain a scheme to add partition points one by one, all the time keeping the assumption that once we have $n-1$ partition points and n partition segments, all the partition segments are smaller than $1/n$.
The first terms of the constructed sequence will be $a_1 = \ln \frac{3}{2}$, $a_2 = \ln \frac{5}{4}$, $a_3 = \ln \frac{7}{4}$, $a_4 = \ln \frac{9}{8}$, ...
Remark. This remark describes in fact the same solution from a different view and some ideas behind it. It could be erased after marking is finished.

Estimate (2) is quite natural. To prove that RHS tends to $\ln 2$ we use some integral estimates by

$$\int_n^{2n+1} \frac{1}{x}\,dx = \ln(2n+1) - \ln n.$$

Here we can observe that

$$\int_n^{2n} \frac{1}{x}\,dx = \ln 2$$

is independent of n. This can help us with the construction since the above equality means

$$I_1 = \int_n^{n+1} \frac{1}{x}\,dx = \int_{2n}^{2n+1} \frac{1}{x}\,dx + \int_{2n+1}^{2n+2} \frac{1}{x}\,dx = I_2 + I_3,$$

2021-D1-P3 — Solution (continued)

so, interval of length I_1 can be splitted into two intervals of lengths I_2 and I_3. In fact, after placing the point a_n in the construction for $d = \ln 2$, the lengths of the $n+1$ intervals are

$$\int_{n+1}^{n+2} \frac{1}{x}, \quad \int_{n+2}^{n+3} \frac{1}{x}, \quad \ldots, \quad \int_{2n+1}^{2n+2} \frac{1}{x}$$

with total length

$$d = \int_{n+1}^{2n+2} \frac{1}{x} = \ln 2.$$

2021-D1-P4 — Statement

Day 1, P4 | 10 points

Let $f : \mathbb{R} \to \mathbb{R}$ be a function. Suppose that for every $\varepsilon > 0$, there exists a function $g : \mathbb{R} \to (0, \infty)$ such that for every pair (x, y) of real numbers,

$$\text{if } |x - y| < \min\{g(x), g(y)\}, \text{ then } |f(x) - f(y)| < \varepsilon.$$

Prove that f is the pointwise limit of a sequence of continuous $\mathbb{R} \to \mathbb{R}$ functions, i.e., there is a sequence $h_1, h_2, \ldots$ of continuous $\mathbb{R} \to \mathbb{R}$ functions such that $\lim_{n\to\infty} h_n(x) = f(x)$ for every $x \in \mathbb{R}$.

(proposed by Camille Mau, Nanyang Technological University, Singapore)

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Theory and methods for 2021-D1-P4

Theory Baire class one and controlled discontinuities [T-AN-BAIRE]; Continuity, compactness, and order [T-AN-CONTINUITY]; Uniform approximation and positive kernels [T-AN-APPROXIMATION]; Countability, accumulation, and Cantor constructions [T-AN-SETS]
Methods Use a dense set, an enumeration, or a category argument [M-AN-TOPOLOGY]; Preserve structure while approximating [M-AN-APPROXIMATE]
Strategy Build a witness and verify every constraint [S-CONSTRUCTION]; Pass between local constraints and a global conclusion [S-LOCALGLOBAL]

Statement Recorded solutions Catalog

2021-D1-P4 — Solution — Record 1 of 3

For this record: Theory: T-AN-BAIRE; T-AN-CONTINUITY; T-AN-APPROXIMATION. Methods: M-AN-TOPOLOGY; M-AN-APPROXIMATE. Full learning route

Hint: Start from a segment in place of $\mathbb{R}$ and use its compactness. Or recall the cool things called "the Lebesgue characterization theorem" and "the Baire characterization theorem".
Since g depends also on ε, let us use the notation $g(x, \varepsilon)$. Considering only $\varepsilon = 1/n$ for positive integer n will suffice to reach our conclusions, hence we may use $\min\{g(x, 1/m) \mid m \le n\}$ in place of $g(x, 1/n)$ and thus assume $g(x, \varepsilon)$ decreasing in ε.
For any $x \in \mathbb{R}$, choose $\delta_n(x) = \min\{1/n, g(x, 1/n)\}$. Of the $\delta_n(x)$-neighborhoods of all x select (using local compactness of the reals) an inclusion-minimal locally finite covering $\{U_i\}$. From its inclusion-minimality it follows that we may enumerate U_i with $i \in \mathbb{Z}$ so that $U_i \cap U_j \neq \emptyset$ only when $|i - j| \le 1$ and the enumeration goes from left to right on the real line. For an assumed n, let x_i be the center of U_i and $\delta_i = \delta_n(x_i)$, so that $U_i = (x_i - \delta_i, x_i + \delta_i)$ and $\delta_i < 1/n$ for all i.
Now define a continuous $f_n : \mathbb{R} \to \mathbb{R}$ so that it equals $f(x_i)$ in $U_i \setminus (U_{i-1} \cup U_{i+1})$, and so that f_n changes continuously between $f(x_{i-1})$ and $f(x_i)$ in the intersection $U_{i-1} \cap U_i$.
Now we show that $f_n \to f$ pointwise. Fix a point x and $\varepsilon = 1/m > 0$, and choose $n > \max\{1/g(x, \varepsilon), m\}$.

Examine the construction of f_n for any such n. Observe that $g(x, \varepsilon) > 1/n > \delta_i$ and $1/n < 1/m$. There are two cases:

2021-D1-P4 — Solution — Record 1 of 3 (continued)

- x belongs to the unique U_i. Then using the monotonicity of $g(x,\varepsilon)$ in ε we have

$$|x_i - x| < \delta_i \leq \min\Big\{g\Big(x_i, \frac{1}{n}\Big), g(x,\varepsilon)\Big\} \leq \min\{g(x_i,\varepsilon), g(x,\varepsilon)\}.$$

  Hence

$$|f(x) - f_n(x)| = |f(x) - f(x_i)| < \varepsilon.$$

- x belongs to $U_{i-1} \cap U_i$. Similar to the previous case,

$$|f(x) - f(x_{i-1})|,\ |f(x) - f(x_i)| < \varepsilon.$$

  Since $f_n(x)$ is between $f_n(x_{i-1}) = f(x_{i-1})$ and $f_n(x_i) = f(x_i)$ by construction, we have

$$|f(x) - f_n(x)| < \varepsilon.$$

We have that $|f(x) - f_n(x)| < \varepsilon$ holds for sufficiently large n, which proves the pointwise convergence.

2021-D1-P4 — Solution — Record 2 of 3

For this record: Theory: T-AN-BAIRE; T-AN-SETS. Methods: M-AN-TOPOLOGY. Full learning route

This solution uses the Baire characterization theorem: A function $f : \mathbb{R} \to \mathbb{R}$ is a pointwise limit of continuous functions if and only if its restriction to every non-empty closed subset of $\mathbb{R}$ has a point of continuity.
Assume the contrary in view of the above theorem: $A \subseteq \mathbb{R}$ is a non-empty closed set and f has no point of continuity in A. Let's think that f is defined only on A.
Then for all $x \in A$ there exist rationals $p < q$ for which $\limsup_x f > q$, $\liminf_x f < p$. Apply the Baire category theorem: If a complete metric space A is a countable union of sets then some of the sets is dense in a positive radius metric ball of A. It follows that there exist p and q, which serve for a subset $B \subset A$ which is dense on a certain ball (in the induced metric of the real line) $A_1 \subset A$. It yields that both sets $Q = f^{-1}(q,\infty)$ and $P = f^{-1}(-\infty, p)$ are dense in A_1.
Choose $\varepsilon = (q-p)/10$ and find k for which the set $S = \{x : g(x) > 1/k\}$ is also dense on a certain ball $A_2 \subset A_1$. Partition S into subsets where $f(x) > (p+q)/2$ and $f(x) \leqslant (p+q)/2$, one of them is again dense somewhere in A_3, say the latter.

Now take any point $y \in A_3 \cap Q$ and a very close (within distance $\min(1/k, g(y))$) to y point x with $g(x) > 1/k$ but $f(x) \leqslant (p+q)/2$. This pair x, y contradicts the property of f from the problem statement.

2021-D1-P4 — Solution — Record 3 of 3

For this record: Theory: T-AN-BAIRE; T-AN-SETS. Methods: M-AN-TOPOLOGY. Full learning route

This solution uses the Lebesgue characterization theorem: If $f : \mathbb{R} \to \mathbb{R}$ is a function and, for all real c, the sublevel and superlevel sets $\{x \mid f(x) \geqslant c\}$, $\{x \mid f(x) \leqslant c\}$ are countable intersections of open sets then f is a pointwise limit of continuous functions.
Now the solution follows from the formula with a countable intersection of the unions of intervals:

$$\{x \mid f(x) \geqslant c\} = \bigcap_{n,k=1}^{\infty} \bigcup_{\substack{y\in\mathbb{R} \\ f(y)\geqslant c}} \Big(y - \min\Big\{\frac{1}{k}, g\Big(y, \frac{1}{n}\Big)\Big\}, y + \min\Big\{\frac{1}{k}, g\Big(y, \frac{1}{n}\Big)\Big\}\Big) \qquad (*)$$

and the similar formula for $\{x : f(x) \leqslant c\}$. It remains to prove $(*)$.
The left hand side is obviously contained in the right hand side, just put $y = x$.

To prove the opposite inclusion assume the contrary, that $f(x) < c$, but x is contained in the right hand side. Choose a positive integer n such that $f(x) < c - 1/n$ and k such that $g(x, 1/n) > 1/k$. Then, since x belongs to the right hand side, we see that there exists y such that $f(y) \geqslant c$ and

$$|x - y| < \min\Big\{g\Big(y, \frac{1}{n}\Big), \frac{1}{k}\Big\} \leqslant \min\Big\{g\Big(y, \frac{1}{n}\Big), g\Big(x, \frac{1}{n}\Big)\Big\},$$

2021-D1-P4 — Solution — Record 3 of 3 (continued)

which yields $f(x) \geqslant f(y) - 1/n \geqslant c - 1/n$, a contradiction.

Day 2

2021-D2-P5 — Statement

Day 2, P5 | 10 points

Let A be a real $n \times n$ matrix and suppose that for every positive integer m there exists a real symmetric matrix B such that

$$2021B = A^m + B^2.$$

Prove that $|\det A| \leq 1$.

(proposed by Rafael Filipe dos Santos, Instituto Militar de Engenharia, Rio de Janeiro)

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Theory and methods for 2021-D2-P5

Theory Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Determinants, block identities and inverses [T-LA-DETERMINANT]
Methods Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]
Strategy Build a chain of bounds and track equality [S-BOUND]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

2021-D2-P5 — Solution — Record 1 of 2

For this record: Theory: T-LA-SPECTRAL; T-LA-POLYNOMIAL; T-LA-DETERMINANT. Methods: M-LA-SPECTRAL. Full learning route

Hint: The determinant is the product of the eigenvalues.
Let B_m be the corresponding matrix B depending on m:

$$2021B_m = A^m + B_m^2.$$

For $m = 1$, we obtain $A = 2021B_1 - B_1^2$. Since B_1 is real and symmetric, so is A. Thus A is diagonalizable and all eigenvalues of A are real.
Now fix a positive integer m and let λ be any real eigenvalue of A. Since $A^m = 2021B_m - B_m^2$ and B_m is real symmetric, the spectral mapping theorem gives a real eigenvalue μ of B_m such that

$$2021\mu = \lambda^m + \mu^2 \quad \Rightarrow \quad \mu^2 - 2021\mu + \lambda^m = 0.$$

The last equation is a second degree equation with a real root. Therefore, the discriminant is non-negative:

$$2021^2 - 4\lambda^m \geq 0 \quad \Rightarrow \quad \lambda^m \leq \frac{2021^2}{4}.$$

If $|\lambda| > 1$, letting m even sufficiently large we reach a contradiction. Thus $|\lambda| \leq 1$.

Finally, since $\det A$ is the product of the eigenvalues of A and each of them has absolute value less then or equal to 1, we get $|\det A| \leq 1$ as desired.

2021-D2-P5 — Solution — Archival source pointer

For this record: Archival source pointer; use the complete proof above. Full learning route

Archival source note. The official sheet contains the following pointer, which supplies no additional proof in this volume: Different solution can be found in paper s2002
Use the complete first record above. The unidentified paper is not treated as a verified reference.

2021-D2-P6 — Statement

Day 2, P6 | 10 points

For a prime number p, let $\mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z})$ be the group of invertible 2×2 matrices of residues modulo p, and let S_p be the symmetric group (the group of all permutations) on p elements. Show that there is no injective group homomorphism $\varphi : \mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z}) \to S_p$.

(proposed by Thiago Landim, Sorbonne University, Paris)

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Theory and methods for 2021-D2-P6

Theory Groups, orders, generators, and quotients [T-AL-GROUP]; Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]; Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]
Methods Prove generation or obstruct it by an image [M-AL-GENERATORS]
Strategy Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

2021-D2-P6 — Solution

For this record: Theory: T-AL-GROUP; T-AL-PERMUTATION; T-AL-FINITEFIELD. Methods: M-AL-GENERATORS. Full learning route

Hint: First find what the monomorphism must do with elements of order p.
For $p = 2$, just note that $\mathrm{GL}_2(\mathbb{Z}/2\mathbb{Z})$ has more than $2 = |S_2|$ elements.
From now on, let p be an odd prime and suppose that there exists such a homomorphism.

The matrix

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

has order p and commutes with the matrix

$$B = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

of order 2, hence AB has order $2p$. But there is no permutation in S_p of order $2p$ since only p-cycles have order divisible by p, and their order is exactly p.

2021-D2-P7 — Statement

Day 2, P7 | 10 points

Let $D \subseteq \mathbb{C}$ be an open set containing the closed unit disk $\{z : |z| \le 1\}$. Let $f : D \to \mathbb{C}$ be a holomorphic function, and let $p(z)$ be a monic polynomial. Prove that

$$\big|f(0)\big| \le \max_{|z|=1} \big|f(z)p(z)\big|.$$

(proposed by Lars Hörmander)

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Theory and methods for 2021-D2-P7

Theory Holomorphic bounds, poles, and logarithmic-branch sums [T-AL-COMPLEXANALYSIS]; Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]
Methods Build a holomorphic function with a useful boundary [M-AL-ANALYTIC]
Strategy Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2021-D2-P7 — Solution

For this record: Theory: T-AL-COMPLEXANALYSIS; T-AL-COMPLEX. Methods: M-AL-ANALYTIC. Full learning route

Hint: Apply the maximum principle or the Cauchy formula to a suitable function $f(z)q(z)$.

Let $q(z) = z^n \cdot p(1/z)$, or more explicitly, if

$$p(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0,$$

let

$$q(z) = 1 + a_{n-1}z + \cdots + a_0 z^n.$$

Note that for $|z| = 1$ we have $1/z = \bar{z}$ and hence $|q(z)| = |p(z)|$. Hence by the maximum principle or the Cauchy formula for the product of f and q, it follows that

$$\big|f(0)\big| = \big|f(0)q(0)\big| \le \max_{|z|=1}\big|f(z)q(z)\big| = \max_{|z|=1}\big|f(z)p(z)\big|.$$

2021-D2-P8 — Statement

Day 2, P8 | 10 points

Let n be a positive integer. At most how many distinct unit vectors can be selected in $\mathbb{R}^n$ such that from any three of them, at least two are orthogonal?

(proposed by Alexander Polyanskii, Moscow Institute of Physics and Technology; based on results of Paul Erdős and Moshe Rosenfeld)

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Theory and methods for 2021-D2-P8

Theory Gram matrices and positive semidefinite forms [T-LA-GRAM]; Trace pairing and commutator calculus [T-LA-TRACE]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]
Methods Encode distances and orthogonality by a Gram matrix [M-LA-GRAM]; Use trace to extract a scalar invariant [M-LA-TRACE]
Strategy Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2021-D2-P8 — Solution — Record 1 of 2

For this record: Theory: T-LA-GRAM; T-LA-TRACE; T-LA-SPECTRAL; T-AN-INEQUALITIES. Methods: M-LA-GRAM; M-LA-TRACE. Full learning route

Hint: Play with the Gram matrix of these vectors.
$2n$ is the maximal number.
An example of $2n$ vectors in the set is given by a basis and its opposite vectors. In the rest of the text we prove that it is impossible to have $2n + 1$ vectors in the set.

2021-D2-P8 — Solution — Record 1 of 2 (continued)

Consider the Gram matrix A with entries $a_{ij} = e_i \cdot e_j$. Its rank is at most n, its eigenvalues are real and non-negative. Put $B = A - I_{2n+1}$, this is the same matrix, but with zeros on the diagonal. The eigenvalues of B are real, greater or equal to -1, and the multiplicity of -1 is at least $n+1$.

The matrix $C = B^3$ has the following diagonal entries

$$c_{ii} = \sum_{i \neq j \neq k \neq i} a_{ij} a_{jk} a_{ki}.$$

The problem statement implies that in every summand of this expression at least one factor is zero. Hence $\operatorname{tr} C = 0$. Let $x_1, \dots, x_m$ be the positive eigenvalues of B, their number is $m \leq n$ as noted above. From $\operatorname{tr} B = \operatorname{tr} C$ we deduce (taking into account that the eigenvalues between -1 and 0 satisfy $\lambda^3 \geq \lambda$):

$$x_1 + \dots + x_m \geq x_1^3 + \dots + x_m^3$$

Applying $\operatorname{tr} C = 0$ once again and noting that C has eigenvalue -1 of multiplicity at least $n+1$, we obtain

$$x_1^3 + \dots + x_m^3 \geq n + 1.$$

It also follows that

$$(x_1 + \dots + x_m)^3 \geq (x_1^3 + \dots + x_m^3)(n+1)^2.$$

By Hölder's inequality, we obtain

$$(x_1^3 + \dots + x_m^3) m^2 \geq (x_1 + \dots + x_m)^3,$$

which is a contradiction with $m \leq n$.

2021-D2-P8 — Solution — Record 2 of 2

For this record: Theory: T-LA-GRAM; T-LA-TRACE; T-LA-POLYNOMIAL. Methods: M-LA-GRAM; M-LA-TRACE. Full learning route

Let P_i denote the projection onto i-th vector, $i = 1, \dots, N$. Then our relation reads as $\operatorname{tr}(P_i P_j P_k) = 0$ for distinct i, j, k. Consider the operator $Q = \sum_{i=1}^{N} P_i$, it is non-negative definite, let $t_1, \dots, t_n$ be its eigenvalues, $\sum t_i = \operatorname{tr} Q = N$. We get

$$\sum t_i^3 = \operatorname{tr} Q^3 = N + 6 \sum_{i<j} \operatorname{tr} P_i P_j = N + 3(\operatorname{tr} Q^2 - N) = 3 \sum t_i^2 - 2N$$

(we used the obvious identities like $\operatorname{tr} P_i P_j P_i = \operatorname{tr} P_i^2 P_j = \operatorname{tr} P_i P_j$). But $(t_i - 2)^2 (t_i + 1) = t_i^3 - 3t_i^2 + 4 \geqslant 0$, thus $-2N = \sum t_i^3 - 3t_i^2 \geqslant -4n$ and $N \leqslant 2n$.

3.29 2022

Day 1

2022-D1-P1 — Statement

Day 1, P1 | 10 points

Let $f : [0,1] \to (0,\infty)$ be an integrable function such that $f(x) \cdot f(1-x) = 1$ for all $x \in [0,1]$. Prove that

$$\int_0^1 f(x)\,dx \geq 1.$$

(proposed by Mike Daas, Universiteit Leiden)

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Theory and methods for 2022-D1-P1

Theory Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]; Inner products, orthogonality and polarization [T-LA-INNER]
Methods Pair an integral through symmetry [M-AN-PAIR]; Choose a square, factor, or AM–GM weight [M-AN-WEIGHTS]
Strategy Build a chain of bounds and track equality [S-BOUND]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2022-D1-P1 — Solution — Record 1 of 2

For this record: Theory: T-AN-INTEGRAL; T-AN-INEQUALITIES. Methods: M-AN-PAIR; M-AN-WEIGHTS. Full learning route

Hint: Apply the AM–GM inequality.

By the AM–GM inequlity we have

$$f(x) + f(1-x) \geq 2\sqrt{f(x)f(1-x)} = 2.$$

By integrating in the interval $[0, \frac{1}{2}]$ we get

$$\int_0^1 f(x)\,dx = \int_0^{\frac{1}{2}} f(x)\,dx + \int_0^{\frac{1}{2}} f(1-x)\,dx = \int_0^{\frac{1}{2}} (f(x) + f(1-x))\,dx \geq \int_0^{\frac{1}{2}} 2\,dx = 1.$$

2022-D1-P1 — Solution — Record 2 of 2

For this record: Theory: T-AN-INTEGRAL; T-AN-INEQUALITIES; T-LA-INNER. Methods: M-AN-PAIR; M-AN-WEIGHTS. Full learning route

From the condition, we have

$$\int_0^1 f(x)\,dx = \int_0^1 f(1-x)\,dx = \int_0^1 \frac{1}{f(x)}\,dx$$

and hence, using the positivity of f, the claim follows since

$$\left(\int_0^1 f(x)\,dx\right)^2 = \int_0^1 f(x)\,dx \cdot \int_0^1 \frac{1}{f(x)}\,dx \geq \left(\int_0^1 1\,dx\right)^2 \geq 1$$

by the Cauchy-Schwarz inequality.

2022-D1-P2 — Statement

Day 1, P2 | 10 points

Let n be a positive integer. Find all $n \times n$ real matrices A with only real eigenvalues satisfying

$$A + A^k = A^T$$

for some integer $k \geq n$.
(A^T denotes the transpose of A.)

(proposed by Camille Mau, Nanyang Technological University)

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Theory and methods for 2022-D1-P2

Theory Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Nilpotents, projections and involutions [T-LA-NILPOTENT]; Eigenvalues, similarity and orthogonal diagonalization [T-LA-SPECTRAL]; Trace pairing and commutator calculus [T-LA-TRACE]
Methods Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]; Use trace to extract a scalar invariant [M-LA-TRACE]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]

Statement Recorded solutions Catalog

2022-D1-P2 — Solution — Record 1 of 2

For this record: Theory: T-LA-POLYNOMIAL; T-LA-NILPOTENT; T-LA-SPECTRAL. Methods: M-LA-SPECTRAL. Full learning route

Hint: Consider the eigenvalues of A.
Taking the transpose of the matrix equation and substituting we have

$$A^T + (A^T)^k = A \implies A + A^k + (A + A^k)^k = A \implies A^k(I + (I + A^{k-1})^k) = 0.$$

Hence $p(x) = x^k(1 + (1 + x^{k-1})^k)$ is an annihilating polynomial for A. It follows that all eigenvalues of A must occur as roots of p (possibly with different multiplicities). Note that for all $x \in \mathbb{R}$ (this can be seen by considering even/odd cases on k),

$$(1 + x^{k-1})^k \geq 0,$$

and we conclude that the only eigenvalue of A is 0 with multiplicity n.
Thus A is nilpotent, and since A is $n \times n$, $A^n = 0$. It follows $A^k = 0$, and $A = A^T$. Hence A can only be the zero matrix: A is real symmetric and so is orthogonally diagonalizable, and all its eigenvalues are 0.

Remark. It's fairly easy to prove that eigenvalues must occur as roots of any annihilating polynomial. If λ is an eigenvalue and v an associated eigenvector, then $f(A)v = f(\lambda)v$. If f annihilates A, then $f(\lambda)v = 0$, and since $v \neq 0$, $f(\lambda) = 0$.

2022-D1-P2 — Solution — Record 2 of 2

For this record: Theory: T-LA-POLYNOMIAL; T-LA-TRACE; T-LA-NILPOTENT; T-LA-SPECTRAL. Methods: M-LA-SPECTRAL; M-LA-TRACE. Full learning route

If λ is an eigenvalue of A, then $\lambda + \lambda^k$ is an eigenvalue of $A^T = A + A^k$, thus of A too. Now, if k is odd, then taking λ with maximal absolute value we get a contradiction unless all eigenvalues are 0. If k is even, the same contradiction is obtained by comparing the traces of A^T and $A + A^k$.

Hence all eigenvalues are zero and A is nilpotent. The hypothesis that $k \geq n$ ensures $A = A^T$. A nilpotent self-adjoint operator is diagonalizable and is necessarily zero.

2022-D1-P3 — Statement

Day 1, P3 | 10 points

Let p be a prime number. A flea is staying at point 0 of the real line. At each minute, the flea has three possibilities: to stay at its position, or to move by 1 to the left or to the right. After $p-1$ minutes, it wants to be at 0 again. Denote by $f(p)$ the number of its strategies to do this (for example, $f(3)=3$: it may either stay at 0 for the entire time, or go to the left and then to the right, or go to the right and then to the left). Find $f(p)$ modulo p.

(proposed by Fedor Petrov, St. Petersburg)

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Theory and methods for 2022-D1-P3

Theory Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Permutation cycles, parity, and orbit decomposition [T-AL-PERMUTATION]; Linear recurrences and generating functions [T-AL-RECURRENCE]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Choose a modulus that exposes the obstruction [M-AL-MODULAR]; Find a recurrence that reveals the invariant [M-AL-RECURRENCE]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2022-D1-P3 — Solution — Record 1 of 2

For this record: Theory: T-AL-FINITEFIELD; T-AL-PERMUTATION; T-AL-RECURRENCE; T-DS-COUNTING. Methods: M-AL-MODULAR; M-AL-RECURRENCE. Full learning route

Hint: Find a recurrence for $f(p)$ or use generating functions.
The answer is $f(p)\equiv 0 \mod 3$
for $p=3$, $f(p)\equiv 1 \mod 3$ for $p=3k+1$, and $f(p)\equiv -1 \mod 3$ for $p=3k-1$.

The case $p=3$ is already considered, let further $p\neq 3$. For a residue i modulo p denote by $a_i(k)$ the number of Flea strategies for which she is at position i modulo p after k minutes. Then $f(p)=a_0(p-1)$. The natural recurrence is $a_i(k+1)=a_{i-1}(k)+a_i(k)+a_{i+1}(k)$, where the indices are taken modulo p. The idea is that modulo p we have $a_0(p)\equiv 3$ and $a_i(p)\equiv 0$. Indeed, for all strategies for p minutes for which not all p actions are the same, we may cyclically shift the actions, and so we partition such strategies onto groups by p strategies which result with the same i. Remaining three strategies correspond to $i=0$. Thus, if we denote $x_i=a_i(p-1)$, we get a system of equations $x_{-1}+x_0+x_1=3$, $x_{i-1}+x_i+x_{i+1}=0$ for all $i=1,\dots,p-1$. It is not hard to solve this system (using the 3-periodicity, for example). For $p=3k+1$ we get $(x_0,x_1,\dots,x_{p-1})=(1,1,-2,1,1,-2,\dots,1)$, and $(x_0,x_1,\dots,x_{p-1})=(-1,2,-1,-1,2,\dots,2)$ for $p=3k+2$.

2022-D1-P3 — Solution — Record 2 of 2

For this record: Theory: T-AL-FINITEFIELD; T-AL-RECURRENCE. Methods: M-AL-MODULAR; M-AL-RECURRENCE. Full learning route

Note that $f(p)$ is the constant term of the Laurent polynomial $(x+1+1/x)^{p-1}$ (the moves to right, to left and staying are in natural correspondence with x, $1/x$ and 1.) Thus, working with power series over $\mathbb{F}_p$ we get (using the notation $[x^k]P(x)$ for the coefficient of x^k in P)

$$\begin{aligned} f(p) &= [x^{p-1}](1+x+x^2)^{p-1} \\ &= [x^{p-1}](1-x^3)^{p-1}(1-x)^{1-p} \\ &= [x^{p-1}](1-x^3)^{p}(1-x)^{-p}(1-x^3)^{-1}(1-x) \\ &= [x^{p-1}](1-x^{3p})(1-x^p)^{-1}(1-x^3)^{-1}(1-x) \\ &= [x^{p-1}](1-x^3)^{-1}(1-x). \end{aligned}$$

2022-D1-P3 — Solution — Record 2 of 2 (continued)

and expanding $(1-x^3)^{-1} = \sum_{k\geq 0} x^{3k}$ gives

$$f(p) \equiv \begin{cases} 0 & \text{if } p = 3, \\ 1 & \text{if } p \equiv 1 \pmod 3, \\ -1 & \text{if } p \equiv 2 \pmod 3 \end{cases} \pmod p.$$

2022-D1-P4 — Statement

Day 1, P4 | 10 points

Let $n > 3$ be an integer. Let Ω be the set of all triples of distinct elements of $\{1, 2, \dots, n\}$. Let m denote the minimal number of colours which suffice to colour Ω so that whenever $1 \leq a < b < c < d \leq n$, the triples $\{a, b, c\}$ and $\{b, c, d\}$ have different colours. Prove that

$$\frac{1}{100} \log\log n \leqslant m \leqslant 100 \log\log n.$$

(proposed by Danila Cherkashin, St. Petersburg)

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Theory and methods for 2022-D1-P4

Theory Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Encode a relation as a graph and extract a cycle [M-DS-GRAPH]; Choose information that makes an encoding injective [M-DS-ENCODE]
Strategy Expose the structure before calculating [S-STRUCTURE]; Choose the objects and count them without ambiguity [S-COUNTING]; Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2022-D1-P4 — Solution

For this record: Theory: T-DS-GRAPH; T-DS-COUNTING. Methods: M-DS-GRAPH; M-DS-ENCODE. Full learning route

Hint: Define two graphs, one on Ω and another graph on pairs (2-element sets).
For $k = 1, 2, \dots, n$ denote by Ω_k the set of all $\binom{n}{k}$ k-subsets of $[n]$. For each $k = 1, 2, \dots, n-1$ define a directed graph G_k whose vertices are elements of Ω_k, and edges correspond to elements of Ω_{k+1} as follows: if $1 \leqslant a_1 < a_2 < \dots < a_{k+1} \leqslant n$, then the edge of G_k corresponding to $(a_1, \dots, a_{k+1})$ goes from $(a_1, \dots, a_k)$ to $(a_2, \dots, a_{k+1})$.
For a directed graph $G = (V, E)$ we call a subset $E_1 \subset E$ admissible, if E_1 does not contain a directed path $a - b - c$ of length 2. Define b-index $b(G)$ of the G as the minimal number of admissible sets which cover E. As usual, a subset $V_1 \subset V$ is called independent, if there are no edges with both endpoints in V_1; a chromatic number of G is defined as the minimal number of independent sets which cover V.
A straightforward but crucial observation is the following
Lemma. For all $k = 2, 3, \dots, n$ a subset $A \subset \Omega_k$ is independent in G_k if and only if it is admissible as a set of edges of G_{k-1}.
Corollary. $\chi(G_k) = b(G_{k-1})$ for all $k = 2, 3, \dots, n$.
Now the bounds for numbers $\chi(G_k)$ follow by induction using the following general
Lemma. For a directed graph $G = (V, E)$ we have

$$\log_2 \chi(G) \leqslant b(G) \leqslant 2\lceil \log_2 \chi(G) \rceil.$$

Proof. 1) Denote $b(G) = m$ and prove that $\log_2 \chi(G) \leqslant m$. For this we take a covering of E by m admissible subsets $E_1, \dots, E_m$ and define a color $c(v)$ of a vertex $v \in V$ as the following subset of $[m]$: $c(v) := \{i \in [m] : \exists vw \in E_i\}$. Note that for any edge $vw \in E$ there exists i such that $vw \in E_i$ which yields $i \in c(v)$ and $i \notin c(w)$, therefore $c(v) \neq c(w)$. So, each color class is an independent set and we get $\chi(G) \leqslant 2^m$ as needed.

2022-D1-P4 — Solution (continued)

2) Denote $\chi(G) = k$ and prove that $b(G) \leqslant 2\lceil \log_2 k \rceil$. Take a proper coloring $\tau : V \to \{0, 1, \dots, k-1\}$ (that means that $\tau(u) \neq \tau(v)$ for all edges $vu \in E$). For an integer $x \in \{0, 1, \dots, k-1\}$ take a binary representation $x = \sum_{i=0}^{r-1} \varepsilon_i(x) 2^i$, $\varepsilon_i(x) \in \{0, 1\}$, where $r = \lceil \log_2 k \rceil$. Consider the following $2r$ subsets of E, two subsets $E_{i,+}$ and $E_{i,-}$ for each $i \in \{0, 1, \dots, r-1\}$:

$$E_{i,+} = \{vu \in E : \varepsilon_i(\tau(v)) = 0, \varepsilon_i(\tau(u)) = 1\},$$
$$E_{i,-} = \{vu \in E : \varepsilon_i(\tau(v)) = 1, \varepsilon_i(\tau(u)) = 0\}.$$

Each of them is admissible, and they cover E, thus $b(G) \leqslant 2r$.
Note that $\chi(G_1) = n$, thus $b(G_1) \geqslant \log_2 n$. Actually we have $b(G_1) = \lceil \log_2 n \rceil$: indeed, if we define $\tau(v) = v - 1$ for all $v \in [n] = \Omega_1$, then the above sets $E_{i,+}$ cover all edges of G_1.
The Lemma above now yields for our number $m = \chi(G_3) = b(G_2)$ the following bounds, which are better than required:

$$b(G_2) \geqslant \log_2 \chi(G_2) = \log_2 b(G_1) = \log_2 \lceil \log_2 n \rceil$$
$$b(G_2) \leqslant 2\lceil \log_2 \chi(G_2) \rceil = 2\lceil \log_2 b(G_1) \rceil = 2\lceil \log_2 \lceil \log_2 n \rceil \rceil.$$

Remark. Actually the upper bound in the Lemma may be improved to $(1+o(1)) \log_2 \chi(G)$ that yields $m = (1 + o(1)) \log_2 \log_2 n$.

Day 2

2022-D2-P5 — Statement

Day 2, P5 | 10 points

We colour all the sides and diagonals of a regular polygon P with 43 vertices either red or blue in such a way that every vertex is an endpoint of 20 red segments and 22 blue segments. A triangle formed by vertices of P is called monochromatic if all of its sides have the same colour. Suppose that there are 2022 blue monochromatic triangles. How many red monochromatic triangles are there?

(proposed by Mike Daas, Universiteit Leiden)

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Theory and methods for 2022-D2-P5

Theory Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Count incidences with their exact multiplicity [M-DS-DOUBLECOUNT]; Encode a relation as a graph and extract a cycle [M-DS-GRAPH]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Pass between local constraints and a global conclusion [S-LOCALGLOBAL]

Statement Recorded solutions Catalog

2022-D2-P5 — Solution

For this record: Theory: T-DS-GRAPH; T-DS-COUNTING. Methods: M-DS-DOUBLECOUNT; M-DS-GRAPH. Full learning route

Hint: Call two connecting edges a cherry. Double-count cherries.

Define a cherry to be a set of two distinct edges from K_{43} that have a vertex in common. We observe that a monochromatic triangle always contains three monochromatic cherries, and that a polychromatic triangle always contains one monochromatic cherry and two polychromatic cherries. Therefore we study the quantity $2M-P$, where M is the number of monochromatic cherries and P is the number of polychromatic cherries. By observing that every cherry is part of a unique triangle, we can split this quantity up into all the distinct triangles in K_{43}. By construction the contribution of a polychromatic triangle will vanish, whereas a monochromatic triangle will contribute 6. We conclude that

$$2M-P=6\cdot\{\text{number of monochromatic triangles}\}.$$

Consider any vertex v. Let M_v be the number of monochromatic cherries with central vertex v and P_v the number such polychromatic cherries. It then follows that

$$M_v=\frac{20\cdot 19}{2}+\frac{22\cdot 21}{2}=421 \quad \text{and} \quad P_v=20\cdot 22=440.$$

In other words, for any vertex v it holds that $2M_v-P_v=402$. Adding up all these contributions, we find that

$$2M-P=43\cdot 402.$$

We conclude that there are $43\cdot 402/6=43\cdot 67=2881$ monochromatic triangles in total. Since 2022 of these were blue, 859 must be red.

2022-D2-P6 — Statement

Day 2, P6 | 10 points

Let $p > 2$ be a prime number. Prove that there is a permutation $(x_1, x_2, \dots, x_{p-1})$ of the numbers $(1, 2, \dots, p-1)$ such that

$$x_1x_2 + x_2x_3 + \cdots + x_{p-2}x_{p-1} \equiv 2 \pmod{p}.$$

(proposed by Giorgi Arabidze, Tbilisi Free University, Georgia)

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Theory and methods for 2022-D2-P6

Theory Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Choose a modulus that exposes the obstruction [M-AL-MODULAR]; Construct roots or units before the coefficients [M-AL-PARAMETRIZE]; Compare coefficients after a structural substitution [M-AL-COEFFICIENT]
Strategy Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2022-D2-P6 — Solution — Record 1 of 2

For this record: Theory: T-AL-FINITEFIELD; T-AL-CONGRUENCE. Methods: M-AL-MODULAR; M-AL-PARAMETRIZE. Full learning route

Hint:
We show such a permutation.

Let $x_i \equiv i^{-1} \pmod{p}$ for $i = 1, 2, \cdots, p-1$. Then

$$\sum_{i=1}^{p-2} x_i x_{i+1} \equiv \sum_{i=1}^{p-2} \frac{1}{i} \cdot \frac{1}{i+1} \equiv \sum_{i=1}^{p-2} \left(\frac{1}{i} - \frac{1}{i+1} \right) \equiv 1 - \frac{1}{p-1} \equiv \frac{p-2}{p-1} \equiv 2 \pmod{p}$$

2022-D2-P6 — Solution — Record 2 of 2

For this record: Theory: T-AL-CONGRUENCE; T-AL-POLYNOMIAL. Methods: M-AL-COEFFICIENT. Full learning route

We begin by noting that the identity permutation yields the value

$$1 \cdot 2 + 2 \cdot 3 + \cdots + (p-2)(p-1) = 2 \cdot \binom{p}{3} \equiv 0 \pmod{p}$$

as soon as $p > 3$. The idea now is to perturb that permutation to obtain the desired value 2.
One thing we can do is to replace $(i, i+1, i+2, i+3)$ by $(i, i+2, i+1, i+3)$. Indeed, this will decrease the sum by 3. So if $p \equiv 2 \pmod 3$, we can just take the permutation $(1, 3, 2, 4, 6, 5, 7, \dots, p-4, p-2, p-3, p-1)$ i.e. exchanging $3k-1$ and $3k$ whenever $k = 1, 2, \dots, \frac{p-2}{3}$. This means we decrease the sum $\frac{p-2}{3}$ times by 3, leading to a remaining sum of $-(p-2) \equiv 2 \pmod p$.
If $p \equiv 1 \pmod 3$, this strategy does not work immediately. Instead, we can change $(1, 2, 3, 4, 5)$ to $(1, 4, 3, 2, 5)$ resulting in a decrement of the sum by 8. If we then exchange $3k$ and $3k+1$ for $k = 2, 3, \dots, \frac{p-7}{3}$ as before, we get another $\frac{p-10}{3}$ times a decrement by 3, leading to a remaining sum of $-8 - \frac{p-10}{3} \cdot 3 \equiv 2 \pmod p$.

Of course this only works if $p \geq 13$. It thus remains to consider the cases $p = 3$ and $p = 7$ by hand. For $p = 3$, we just take $(1, 2)$ and for $p = 7$ we can take $(1, 4, 5, 2, 3, 6)$.

2022-D2-P7 — Statement

Day 2, P7 | 10 points

Let $A_1, A_2, \ldots, A_k$ be $n \times n$ idempotent complex matrices such that

$$A_i A_j = -A_j A_i \quad \text{for all } i \neq j.$$

Prove that at least one of the given matrices has rank $\leq \frac{n}{k}$.

(A matrix A is called idempotent if $A^2 = A$.)

(proposed by Danila Belousov, Novosibirsk)

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Theory and methods for 2022-D2-P7

Theory Nilpotents, projections and involutions [T-LA-NILPOTENT]; Trace pairing and commutator calculus [T-LA-TRACE]; Rank, dimension and exact linear constraints [T-LA-RANK]; Dual bases, kernels and separating functionals [T-LA-DUALITY]

Methods Use trace to extract a scalar invariant [M-LA-TRACE]; Prove a dimension obstruction and attain it [M-LA-DIMENSION]

Strategy Build a chain of bounds and track equality [S-BOUND]

Statement Recorded solutions Catalog

2022-D2-P7 — Solution — Record 1 of 2

For this record: Theory: T-LA-NILPOTENT; T-LA-TRACE; T-LA-RANK. Methods: M-LA-TRACE. Full learning route

Hint: Consider the trace and the rank of A.

Lemma. For any idempotent matrix B

$$\operatorname{tr}(B) = \operatorname{rank}(B)$$

Proof. Observe that an idempotent matrix satisfies the equation $\lambda(1-\lambda) = 0$. Hence the minimal polynomial is a product of linear factors and the matrix is diagonalizable. Therefore, the rank of the matrix equals the number of non-zero eigenvalues. Since the matrix has eigenvalues 0 or 1, this provides that the trace is equal to the number of unity eigenvalues, or non-zero eigenvalues.

It can be shown that $\sum_{i=1}^{k} A_i$ is also an idempotent. Indeed,

$$\left(\sum_{i=1}^{k} A_i\right)^2 = \sum_{i=1}^{k} A_i^2 + \sum_{i \neq j} (A_i A_j + A_j A_i) = \sum_{i=1}^{k} A_i$$

Applying the lemma one can obtain

$$\sum_{i=1}^{k} \operatorname{rank}(A_i) = \sum_{i=1}^{k} \operatorname{tr}(A_i) = \operatorname{tr}\left(\sum_{i=1}^{k} A_i\right) = \operatorname{rank}\left(\sum_{i=1}^{k} A_i\right) \leqslant n$$

The required inequality follows.

2022-D2-P7 — Solution — Record 2 of 2

For this record: Theory: T-LA-NILPOTENT; T-LA-RANK; T-LA-DUALITY. Methods: M-LA-DIMENSION. Full learning route

We first prove that for idempotents A, B with $AB = -BA$ we already must have $AB = BA = 0$. Indeed, it is clear that $ABx = BAx = 0$ for $x \in \ker(A)$ so it suffices to prove the same for $x \in \operatorname{im}(A)$, i.e. when $Ax = x$. But then writing $Bx = y$ we have $Ay = -y$ i.e. $y = -Ay = -A^2 y = Ay = -y$ and hence $y = 0$ so that again $ABx = BAx = 0$.

Henceforth, we can assume the stronger condition $A_i A_j = 0$ for all $i \neq j$. We next claim that all the image spaces V_i of A_i are linearly independent. This will imply the claim, since then the sum

2022-D2-P7 — Solution — Record 2 of 2 (continued)

of their dimensions can be at most n, and so one of them has to be $\leq \frac{n}{k}$. Now, for the sake of contradiction, suppose that $\sum v_i = 0$ with $v_i \in V_i$ and w.l.o.g. $v_1 \neq 0$. But then

$$0 = A_1(v_1 + \cdots + v_k) = v_1 + A_1v_2 + \cdots + A_1v_k = v_1 + A_1A_2v_2 + \cdots + A_1A_kv_k = v_1$$

since $A_1A_i = 0$ for all i.

Remark. Here is a different argument for $AB = BA = 0$, without eigenvectors: multiplying by A and using its idempotence and the super-commutativity, we have

$$-BA = AB = A^2B = AAB = -ABA = BAA = BA^2 = BA$$

thus $BA = 0$.

2022-D2-P8 — Statement

Day 2, P8 | 10 points

Let $n, k \geq 3$ be integers, and let S be a circle. Let n blue points and k red points be chosen uniformly and independently at random on the circle S. Denote by F the intersection of the convex hull of the red points and the convex hull of the blue points. Let m be the number of vertices of the convex polygon F (in particular, $m = 0$ when F is empty). Find the expected value of m.

(proposed by Fedor Petrov, St. Petersburg)

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Theory and methods for 2022-D2-P8

Theory Uniform random points and convex-hull events [T-AN-GEOMETRICPROB]; Finite probability, conditioning, and expectations [T-AN-PROBABILITY]; Finite counting and incidence identities [T-DS-COUNTING]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Methods Condition and average a convex-hull event [M-AN-GEOMETRIC]; Count outcomes or condition on a useful structure [M-AN-PROBABILITY]; Turn a cumulative constraint into a weighted identity [M-AN-PRIMITIVE]
Strategy Choose the objects and count them without ambiguity [S-COUNTING]; Translate geometry into a controlled representation [S-GEOMETRIC]

Statement Recorded solutions Catalog

2022-D2-P8 — Solution — Record 1 of 3

For this record: Theory: T-AN-GEOMETRICPROB; T-AN-PROBABILITY; T-DS-COUNTING; T-DS-GEOMETRY. Methods: M-AN-GEOMETRIC; M-AN-PROBABILITY. Full learning route

Hint:

We prove that

$$\mathsf{E}(m) = \frac{2kn}{n+k-1} - 2\frac{k!\,n!}{(k+n-1)!}.$$

Let $A_1, \ldots, A_n$ be blue points. Fix $i \in \{1, \ldots, n\}$. Enumerate our $n + k$ points starting from a blue point A_i counterclockwise as $A_i, X_{1,i}, X_{2,i}, \ldots, X_{(n+k-1),i}$. Denote the minimal index j for which the point $X_{j,i}$ is blue as $m(i)$. So, $A_iX_{m(i),i}$ is a side of the convex hull of blue points. Denote by b_i the following random variable:

$$b_i = \begin{cases} 1, & \text{if the chord } A_iX_{m(i),i} \text{ contains a side of } F \\ 0, & \text{otherwise.} \end{cases}$$

Define analogously k random variables $r_1, \ldots, r_k$ for the red points. Clearly,

$$m = b_1 + \cdots + b_n + r_1 + \cdots + r_k. \qquad (\heartsuit)$$

We proceed with computing the expectation of each b_i and r_j. Note that $b_i = 0$ if and only if all red points lie on the side of the line $A_iX_{m(i),i}$. This happens either if $m(i) = 1$, i.e., the point $X_{i,1}$

2022-D2-P8 — Solution — Record 1 of 3 (continued)

is blue (which happens with probability $\frac{n-1}{k+n-1}$), or if $i = k+1$, points $X_{1,i}, \dots, X_{k,i}$ are red, and points $X_{k+1,i}, \dots, X_{k+n-1,i}$ are blue (which happens with probability $1/\binom{k+n-1}{k}$, since all subsets of size k of $\{1, 2, \dots, n+k-1\}$ have equal probabilities to correspond to the indices of red points between $X_{1,i}, \dots, X_{n+k-1,i}$). Thus the expectation of b_i equals $1 - \frac{n-1}{k+n-1} - 1/\binom{k+n-1}{k} = \frac{k}{n+k-1} - \frac{k!(n-1)!}{(k+n-1)!}$. Analogously, the expectation of r_j equals $\frac{n}{n+k-1} - \frac{n!(k-1)!}{(k+n-1)!}$. It remains to use ($\heartsuit$) and linearity of expectation.

2022-D2-P8 — Solution — Record 2 of 3

For this record: Theory: T-AN-GEOMETRICPROB; T-AN-PROBABILITY; T-DS-COUNTING. Methods: M-AN-PROBABILITY; M-AN-GEOMETRIC. Full learning route

Let $C_1, \dots, C_{n+k}$ be the colours of the points, scanned counterclockwise from a fixed point on the circle. We consider the sequence as cyclic (so C_{n+k} is also adjacent to C_1). There are two cases: Either (i) all red points appear contiguously, followed by all blue points contiguously, or (ii) the red and blue points alternate at least twice. It can be seen that in the second case, m is exactly equal to the number of colour changes in the C_i sequence: For example, if C_i is red and C_{i+1} is blue, then the intersection of the red chord from C_i to the next red point with the blue chord from C_{i+1} to the previous blue point is a vertex of F, and every vertex is of this form. Case (i) is exceptional, as we have two colour changes, but $m = 0$, so it is 2 less than the number of changes in that case.
Now observe that the distribution of C_i is purely combinatorial: Each of the $\binom{n+k}{n,k}$ distributions of colours is equally likely (for example, because we can generate the distribution by first choosing all $n+k$ points on the circle, and then assigning colours uniformly). In particular the probability that C_iC_{i+1} is a colour change is exactly $\frac{2nk}{(n+k)(n+k-1)}$, and by lineraity of expectation, the total expected number of color changes (including $i = n+k$) is $n+k$ times this, i.e. $\frac{2nk}{n+k-1}$.

To get the expected value of m, we must subtract from the above 2 times the probability of case (i). Exactly $n+k$ of the $\binom{n+k}{n,k}$ distributions belong to case (i), so we must subtract $2(n+k)\binom{n+k}{n,k}^{-1} = 2\frac{n!k!}{(n+k-1)!}$, as claimed.

2022-D2-P8 — Solution — Record 3 of 3

For this record: Theory: T-AN-GEOMETRICPROB; T-AN-PROBABILITY; T-AN-INTEGRAL. Methods: M-AN-GEOMETRIC; M-AN-PRIMITIVE. Full learning route

Let $A_1, \dots, A_n$ be the blue points and $B_1, \dots, B_k$ be the red points. For every pair of blue points A_i, A_j, $1 \leqslant i < j \leqslant n$, we evaluate the probability p that A_iA_j contains a side of F (it obviously does not depend on the choice of i and j). By q denote the analogous probability for the red points. Then by linearity of expectation we have $\mathsf{E}m = \binom{n}{2}p + \binom{k}{2}q$.
We proceed with finding p. Without loss of generality $i = 1$, $j = 2$. Let the length of the circle be 1, and the length of arc A_1A_2 (counterclockwise from A_1 to A_2) be x. Then x is uniformly distributed on $[0, 1]$. Then A_1A_2 contains a side of F if
(i) all blue points are on the same side of A_1A_2, but
(ii) the red points are not on the same side of A_1A_2.
The probability of (i) is $x^{n-2} + (1-x)^{n-2}$. The probability of (ii) is $1 - (x^k + (1-x)^{n-k})$.

Thus, using Beta function value $B(a,b) = \int_0^1 x^{a-1}(1-x)^{b-1}\,dx = B(a,b) = \frac{(a-1)!(b-1)!}{(a+b-1)!}$ for positive integers a, b

$$p = \int_0^1 (x^{n-2} + (1-x)^{n-2})(1 - (x^k + (1-x)^{n-k}))\,dx = \frac{2}{n-1} - \frac{2}{n+k-1} - 2B(n-1, k+1)$$
$$= \frac{2}{n-1} - \frac{2}{n+k-1} - 2\frac{(n-2)!\,k!}{(n+k-1)!}.$$

Next,

$$\binom{n}{2}p = n - \frac{n(n-1)}{n+k-1} - \frac{n!\,k!}{(n+k-1)!} = \frac{nk}{n+k-1} - \frac{n!\,k!}{(n+k-1)!},$$

2022-D2-P8 — Solution — Record 3 of 3 (continued)

and by symmetry $\binom{k}{2}q$ takes the same value (that is in agreement with the observation that red and blue sides of F alternate).

3.30 2023

Day 1

2023-D1-P1 — Statement

Day 1, P1 | 10 points

Find all functions $f : \mathbb{R} \to \mathbb{R}$ that have a continuous second derivative and for which the equality $f(7x+1) = 49f(x)$ holds for all $x \in \mathbb{R}$.

(proposed by Alex Avdiushenko, Neapolis University Paphos, Cyprus)

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Theory and methods for 2023-D1-P1

Theory Functional equations: domain, injectivity, and regularity [T-DS-FUNCTIONAL]; Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]; Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Continuity, compactness, and order [T-AN-CONTINUITY]
Methods Force structure by admissible substitutions [M-DS-FUNCTIONAL]; Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2023-D1-P1 — Solution

For this record: Theory: T-DS-FUNCTIONAL; T-DS-DYNAMICS; T-AN-DERIVATIVE; T-AN-CONTINUITY. Methods: M-DS-FUNCTIONAL; M-DS-ITERATE. Full learning route

Hint:

- The fixed point of $7x+1$ is $-1/6$.
- Differentiating twice cancels out the coefficient 49.

Differentiating the equation twice, we get

$$f''(7x+1) = f''(x) \quad \text{or} \quad f''(x) = f''\left(\frac{x-1}{7}\right). \tag{1}$$

Take an arbitrary $x \in \mathbb{R}$, and construct a sequence by the recurrence

$$x_0 = x, \qquad x_{k+1} = \frac{x_k - 1}{7}.$$

By (1), the values of f'' at all points of this sequence are equal. The limit of this sequence is $-\frac{1}{6}$, since $\left|x_{k+1} + \frac{1}{6}\right| = \frac{1}{7}\left|x_k + \frac{1}{6}\right|$.
Due to the continuity of f'', the values of f'' at all points of this sequence are equal to $f''(-\frac{1}{6})$, which means that $f''(x)$ is a constant.

Then f is an at most quadratic polynomial, $f(x) = ax^2 + bx + c$. Substituting this expression into the original equation, we get a system of equations, from which we find $a = 36c$, $b = 12c$, and hence

$$f(x) = c(6x+1)^2.$$

2023-D1-P2 — Statement

Day 1, P2 | 10 points

Let A, B and C be $n \times n$ matrices with complex entries satisfying

$$A^2 = B^2 = C^2 \quad \text{and} \quad B^3 = ABC + 2I.$$

Prove that $A^6 = I$.

(proposed by Mike Daas, Universiteit Leiden)

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Theory and methods for 2023-D1-P2

Theory Determinants, block identities and inverses [T-LA-DETERMINANT]; Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]
Methods Complete the equation to a product equal to identity [M-LA-INVERSE]
Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2023-D1-P2 — Solution

For this record: Theory: T-LA-DETERMINANT; T-LA-POLYNOMIAL. Methods: M-LA-INVERSE. Full learning route

Hint: Factorize $B^3 - ABC$.

Note that $B^3 = A^2B$, from which it follows that

$$A^2B - ABC = 2I \implies A(AB - BC) = 2I.$$

Similarly, using that $B^3 = BC^2$, we find that

$$BC^2 - ABC = 2I \implies (BC - AB)C = 2I.$$

It follows that A is a left-inverse of $(AB - BC)/2$, whereas $-C$ is a right inverse. Hence $A = -C$ and as such, it must hold that $ABA = 2I - B^3$. It follows that ABA must commute with B, and so it follows that $(AB)^2 = (BA)^2$. Now we compute that

$$(AB - BA)(AB + BA) = (AB)^2 + AB^2A - BA^2B - (BA)^2 = (AB)^2 + A^4 - B^4 - (AB)^2 = 0.$$

However, we noted before that the matrix $AB - BC = AB + BA$ must be invertible. As such, it must follow that $AB = BA$. We conclude that $ABA = A^2B = B^3$ and so it readily follows that $B^3 = I$. Finally, $A^6 = B^6 = (B^3)^2 = I^2 = I$, completing the proof.

2023-D1-P3 — Statement

Day 1, P3 | 10 points

Find all polynomials P in two variables with real coefficients satisfying the identity

$$P(x, y)P(z, t) = P(xz - yt, xt + yz).$$

(proposed by Giorgi Arabidze, Free University of Tbilisi, Georgia)

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Theory and methods for 2023-D1-P3

Theory Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Polynomial factorization and coprime divisibility [T-AL-FACTORIZATION]; Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]
Methods Compare coefficients after a structural substitution [M-AL-COEFFICIENT]; Factor first and compare irreducible exponents [M-AL-FACTOR]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]

Statement Recorded solutions Catalog

2023-D1-P3 — Solution

For this record: Theory: T-AL-POLYNOMIAL; T-AL-FACTORIZATION; T-AL-COMPLEX. Methods: M-AL-COEFFICIENT; M-AL-FACTOR. Full learning route

Hint: The polynomials $(x+iy)^n$ and $(x-iy)^m$ are trivial complex solutions. Suppose that $P(x,y) = (x+iy)^n(x-iy)^m Q(x,y)$, where $Q(x,y)$ is divisible neither by $x+iy$ nor $x=iy$ and consider $Q(x,y)$.
First we find all polynomials $P(x,y)$ with complex coefficients which satisfies the condition of the problem statement. The identically zero polynomial clearly satisfies the condition. Let consider other polynomials.
Let $i^2=-1$ and $P(x,y) = (x+iy)^n(x-iy)^m Q(x,y)$, where n and m are non-negative integers and $Q(x,y)$ is a polynomial with complex coefficients such that it is not divisible neither by $x+iy$ nor by $x-iy$. By the problem statement we have $Q(x,y)Q(z,t) = Q(xz-yt, xt+yz)$. Note that $z=t=0$ gives $Q(x,y)Q(0,0) = Q(0,0)$. If $Q(0,0)\neq 0$, then $Q(x,y)=1$ for all x and y. Thus $P(x,y) = (x+iy)^n(x-iy)^m$.
Now consider the case when $Q(0,0)=0$.
Let $x=iy$ and $z=-it$. We have $Q(iy,y)Q(-it,t) = Q(0,0) = 0$ for all y and t. Since $Q(x,y)$ is not divisible by $x-iy$, $Q(iy,y)$ is not identically zero and since $Q(x,y)$ is not divisible by $x+iy$, $Q(-it,t)$ is not identically zero. Thus there exist y and t such that $Q(iy,y)\neq 0$ and $Q(-it,t)\neq 0$ which is impossible because $Q(iy,y)Q(-it,t)=0$ for all y and t.
Finally, $P(x,y)$ polynomials with complex coefficients which satisfies the condition of the problem statement are $P(x,y)=0$ and $P(x,y) = (x+iy)^n(x-iy)^m$. It is clear that if $n\neq m$, then $P(x,y) = (x+iy)^n(x-iy)^m$ cannot be polynomial with real coefficients. So we need to require $n=m$, and for this case $P(x,y) = (x+iy)^n(x-iy)^n = (x^2+y^2)^n$.

So, the answer of the problem is $P(x,y)=0$ and $P(x,y)=(x^2+y^2)^n$ where n is any non-negative integer.

2023-D1-P4 — Statement

Day 1, P4 | 10 points

Let p be a prime number and let k be a positive integer. Suppose that the numbers $a_i = i^k + i$ for $i = 0, 1, \ldots, p-1$ form a complete residue system modulo p. What is the set of possible remainders of a_2 upon division by p?

(proposed by Tigran Hakobyan, Yerevan State University, Armenia)

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Theory and methods for 2023-D1-P4

Theory Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Groups, orders, generators, and quotients [T-AL-GROUP]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Choose a modulus that exposes the obstruction [M-AL-MODULAR]; Evaluate at roots of unity or average over them [M-AL-UNITY]
Strategy Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2023-D1-P4 — Solution

For this record: Theory: T-AL-FINITEFIELD; T-AL-GROUP; T-AL-POLYNOMIAL. Methods: M-AL-MODULAR; M-AL-UNITY.
Full learning route

Hint: Consider $\prod_{i=0}^{p-1}(i^k+i)$.

First observe that $p=2$ does not satisfy the condtion, so p must be an odd prime.

Lemma. If $p>2$ is a prime and $\mathbb{F}_p$ is the field containing p elements, then for any integer $1\le n<p$ one has the following equality in the field $\mathbb{F}_p$

$$\prod_{\alpha\in\mathbb{F}_p^*}(1+\alpha^n)=\begin{cases}0, & \text{if } \frac{p-1}{\gcd(p-1,n)} \text{ is even}\\ 2^n, & \text{otherwise}\end{cases}$$

Proof. We may safely assume that $n|p-1$ since it can be easily proved that the set of n-th powers of the elements of $\mathbb{F}_p^*$ coincides with the set of $\gcd(p-1,n)$-th powers of the same elements. Assume that $t_1,t_2,\dots,t_n$ are the roots of the polynomial $t^n+1\in\mathbb{F}_p[x]$ in some extension of the field $\mathbb{F}_p$. It follows that

$$\prod_{\alpha\in\mathbb{F}_p^*}(1+\alpha^n)=\prod_{\alpha\in\mathbb{F}_p^*}\prod_{i=1}^{n}(\alpha-t_i)=\prod_{i=1}^{n}\prod_{\alpha\in\mathbb{F}_p^*}(\alpha-t_i)=\prod_{i=1}^{n}\prod_{\alpha\in\mathbb{F}_p^*}(t_i-\alpha)=\prod_{i=1}^{n}\Phi(t_i),$$

where we define $\Phi(t)=\prod_{\alpha\in\mathbb{F}_p^*}(t-\alpha)=t^{p-1}-1$. Therefore

$$\prod_{\alpha\in\mathbb{F}_p^*}(1+\alpha^n)=\prod_{i=1}^{n}(t_i^{p-1}-1)=\prod_{i=1}^{n}((t_i^n)^{\frac{p-1}{n}}-1)=\prod_{i=1}^{n}((-1)^{\frac{p-1}{n}}-1)=\begin{cases}0, & \text{if } \frac{p-1}{n} \text{ is even}\\ 2^n, & \text{otherwise}\end{cases}$$

Let us now get back to our problem. Suppose the numbers i^k+i, $0\le i\le p-1$ form a complete residue system modulo p. It follows that

$$\prod_{\alpha\in\mathbb{F}_p^*}(\alpha^k+\alpha)=\prod_{\alpha\in\mathbb{F}_p^*}\alpha$$

so that $\prod_{\alpha\in\mathbb{F}_p^*}(\alpha^{k-1}+1)=1$ in $\mathbb{F}_p$. According to the Lemma, this means that $2^{k-1}=1$ in $\mathbb{F}_p$, or equivalently, that $2^{k-1}\equiv 1\pmod p$. Therefore $a_2=2^k+2\equiv 4\pmod p$ so that the remainder of a_2 upon division by p is either 4 when $p>3$ or is 1, when $p=3$.

2023-D1-P5 — Statement

Day 1, P5 | 10 points

Fix positive integers n and k such that $2\le k\le n$ and a set M consisting of n fruits. A permutation is a sequence $x=(x_1,x_2,\dots,x_n)$ such that $\{x_1,\dots,x_n\}=M$. Ivan prefers some (at least one) of these permutations. He realized that for every preferred permutation x, there exist k indices $i_1<i_2<\dots<i_k$ with the following property: for every $1\le j<k$, if he swaps x_{i_j} and $x_{i_{j+1}}$, he obtains another preferred permutation.
Prove that he prefers at least $k!$ permutations.

(proposed by Ivan Mitrofanov, École Normale Superieur Paris)

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Theory and methods for 2023-D1-P5

Theory Extremal choice, exchange arguments, and invariants [T-DS-EXTREMAL]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Repair an extremal configuration by a local exchange [M-DS-EXTREMAL]; Count incidences with their exact multiplicity [M-DS-DOUBLECOUNT]

Theory and methods for 2023-D1-P5 (continued)

Strategy Choose an extremal object and force rigidity [S-EXTREMAL]; Choose the objects and count them without ambiguity [S-COUNTING]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2023-D1-P5 — Solution

For this record: Theory: T-DS-EXTREMAL; T-DS-COUNTING. Methods: M-DS-EXTREMAL; M-DS-DOUBLECOUNT. Full learning route

Hint: For every permutation z of M, choose a preferred permutation x such that $\sum_{m\in M} x^{-1}(m)z^{-1}(m)$ is maximal.

Let S be the set of all $n!$ permutations of M, and let P be the set of preferred permutations. For every permutation $x \in S$ and $m \in M$, let $x^{-1}(m)$ denote the unique number $i \in \{1, 2, \dots, n\}$ with $x_i = m$.

For every $x \in P$, define

$$A(x) = \left\{ z \in S : \forall y \in P \ \sum_{m\in M} x^{-1}(m)z^{-1}(m) \geq \sum_{m\in M} y^{-1}(m)z^{-1}(m) \right\}.$$

For every permutation $z \in S$, we can choose a permutation $x \in P$ for which $\sum_{m\in M} x^{-1}(m)z^{-1}(m)$ is maximal, and then we have $z \in A(x)$; hence, all $z \in S$ is contained in at least one set $A(x)$.

So, it suffices to prove that $|A(x)| \leq \frac{n!}{k!}$ for every preferred permutation x. Fix $x \in P$, and consider an arbitrary $z \in A(x)$. Let the indices $i_1 < \dots < i_k$ be as in the statement of the problem, and let $m_j = x_{i_j}$ for $j = 1, 2, \dots, k$.

For $s = 1, 2, \dots, k-1$ consider the permutation y obtained from x by swapping m_s and m_{s+1}. Since $y \in P$, the definition of $A(x)$ provides

$$i_s z^{-1}(m_s) + i_{s+1} z^{-1}(m_{s+1}) \geq i_{s+1} z^{-1}(m_s) + i_s z^{-1}(m_{s+1}),$$
$$z^{-1}(m_{s+1}) \geq z^{-1}(m_s).$$

Therefore, the elements $m_1, m_2, \dots, m_k$ appear in z in this order. There are exactly $n!/k!$ permutations with this property, so $|A(x)| \leq \frac{n!}{k!}$.

Day 2

2023-D2-P6 — Statement

Day 2, P6 | 10 points

Ivan writes the matrix $\begin{pmatrix} 2 & 3 \\ 2 & 4 \end{pmatrix}$ on the board. Then he performs the following operation on the matrix several times:

- he chooses a row or a column of the matrix, and
- he multiplies or divides the chosen row or column entry-wise by the other row or column, respectively.

Can Ivan end up with the matrix $\begin{pmatrix} 2 & 4 \\ 2 & 3 \end{pmatrix}$ after finitely many steps?

(proposed by Alex Avdiushenko, Neapolis University Paphos, Cyprus)

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Theory and methods for 2023-D2-P6

Theory Determinants, block identities and inverses [T-LA-DETERMINANT]
Methods Eliminate systematically and track the invariant [M-LA-ELIM]
Strategy Expose the structure before calculating [S-STRUCTURE]; Make the contradiction quantitative or structural [S-CONTRADICTION]

Statement Recorded solutions Catalog

2023-D2-P6 — Solution

For this record: Theory: T-LA-DETERMINANT. Methods: M-LA-ELIM. Full learning route

Hint: Construct an invariant quantity that does not change during Ivan's procedure.

We show that starting from $A = \begin{pmatrix} 2 & 3 \\ 2 & 4 \end{pmatrix}$, Ivan cannot reach the matrix $B = \begin{pmatrix} 2 & 4 \\ 2 & 3 \end{pmatrix}$.

Notice first that the allowed operations preserve the positivity of entries; all matrices Ivan can reach have only positive entries.

For every matrix $X = \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix}$ with positive entries, let $L(X) = \begin{pmatrix} \log_2 x_{11} & \log_2 x_{12} \\ \log_2 x_{21} & \log_2 x_{22} \end{pmatrix}$. By taking logarithms of the entries, the steps in Ivan's game will be replaced by adding or subtracting a row or column to the other row. Such standard row and column operations preserve the determinant. Hence, if the matrices in the game are $A = X_0, X_1, X_2, \dots$, then we have $\det L(A) = \det L(X_1) = \det L(X_2) = \dots$, and it suffices to verify that $\det L(A) \neq \det L(B)$.

Indeed,

$$\det L(A) = \log_2 2 \cdot \log_2 4 - \log_2 2 \cdot \log_2 3 = \log_2(4/3) > 0$$

and similarly $\det L(B) < 0$, so $\det L(A) \neq \det L(B)$.

2023-D2-P7 — Statement

Day 2, P7 | 10 points

Let V be the set of all continuous functions $f : [0,1] \to \mathbb{R}$, differentiable on $(0,1)$, with the property that $f(0) = 0$ and $f(1) = 1$. Determine all $\alpha \in \mathbb{R}$ such that for every $f \in V$, there exists some $\xi \in (0,1)$ such that

$$f(\xi) + \alpha = f'(\xi).$$

(proposed by Mike Daas, Leiden University)

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Theory and methods for 2023-D2-P7

Theory Rolle, mean values, and the Darboux property [T-AN-DERIVATIVE]; Differential inequalities and integrating factors [T-AN-ODE]
Methods Count zeros and subtract an interpolant [M-AN-ROLLE]; Design an auxiliary derivative [M-AN-AUXILIARY]; Compare growth through derivatives [M-AN-RATIO]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2023-D2-P7 — Solution — Record 1 of 2

For this record: Theory: T-AN-DERIVATIVE; T-AN-ODE. Methods: M-AN-ROLLE; M-AN-AUXILIARY. Full learning route

Hint: Find a function $h \in V$ such that $h' - h$ is constant, then apply Rolle's theorem to $f - h$. Alternatively, you can apply Cauchys's mean value theorem with some auxiliary functions.

First consider the function

$$h(x) = \frac{e^x - 1}{e - 1}, \quad \text{which has the property that} \quad h'(x) = \frac{e^x}{e - 1}.$$

2023-D2-P7 — Solution — Record 1 of 2 (continued)

Note that $h \in V$ and that $h'(x) - h(x) = 1/(e-1)$ is constant. As such, $\alpha = 1/(e-1)$ is the only possible value that could possibly satisfy the condition from the problem. For $f \in V$ arbitrary, let

$$g(x) = f(x)e^{-x} + h(-x), \quad \text{with } g(0) = 0 \text{ and also } g(1) = e^{-1} + \frac{e^{-1}-1}{e-1} = 0.$$

We compute that

$$g'(x) = f'(x)e^{-x} - f(x)e^{-x} - h'(-x).$$

Now apply Rolle's Theorem to g on the interval $[0,1]$; it yields some $\xi \in (0,1)$ with the property that

$$g'(\xi) = 0 \implies f'(\xi)e^{-\xi} - f(\xi)e^{-\xi} - \frac{e^{-\xi}}{e-1} = 0 \implies f'(\xi) = f(\xi) + \frac{1}{e-1},$$

showing that $\alpha = 1/(e-1)$ indeed satisfies the condition from the problem.

2023-D2-P7 — Solution — Record 2 of 2

For this record: Theory: T-AN-DERIVATIVE; T-AN-ODE. Methods: M-AN-ROLLE; M-AN-RATIO. Full learning route

Notice that the expression $f'(x) - f(x)$ appears in the derivative of the function $F(x) = f(x) \cdot e^{-x}$: $F'(x) = (f'(x) - f(x))e^{-x}$.
Apply Cauchy's mean value theorem to $F(x)$ and the function $G(x) = -e^{-x}$. By the theorem, there is some $\xi \in (0,1)$ such that

$$\frac{F'(\xi)}{G'(\xi)} = \frac{F(1) - F(0)}{G(1) - G(0)}$$
$$f'(\xi) - f(\xi) = \frac{e^{-1} - 0}{-e^{-1} + 1} = \frac{1}{e-1}.$$

This proves the required property for $a = \dfrac{1}{e-1}$.

Now we show that no other α is possible. Choose f and F in such a way that $\frac{F'(x)}{G'(x)} = f'(x) - f(x) = \frac{1}{e-1}$ is constant. That means

$$F'(x) = \frac{G'(x)}{e-1} = \frac{e^{-x}}{e-1},$$
$$F(x) = \frac{1 - e^{-x}}{e-1},$$
$$f(x) = F(x) \cdot e^{x} = \frac{e^{x} - 1}{e-1}.$$

With this choice we have $f(0) = 0$ and $f(1) = 1$, so $f' \in V$, and $f'(x) - f(x) \equiv \dfrac{1}{e-1}$ for all x, so for this function the only possible value for α is $\dfrac{1}{e-1}$.

2023-D2-P8 — Statement

Day 2, P8 | 10 points

Let T be a tree with n vertices; that is, a connected simple graph on n vertices that contains no cycle. For every pair u, v of vertices, let $d(u,v)$ denote the distance between u and v, that is, the number of edges in the shortest path in T that connects u with v.

Consider the sums

$$W(T) = \sum_{\substack{\{u,v\} \subseteq V(T) \\ u \neq v}} d(u,v) \quad \text{and} \quad H(T) = \sum_{\substack{\{u,v\} \subseteq V(T) \\ u \neq v}} \frac{1}{d(u,v)}.$$

2023-D2-P8 — Statement (continued)

Prove that

$$W(T)\cdot H(T)\geq \frac{(n-1)^3(n+2)}{4}.$$

(proposed by Slobodan Filipovski, University of Primorska, Koper)

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Theory and methods for 2023-D2-P8

Theory Trees, unique paths, and distance sums [T-DS-TREES]; Cauchy–Schwarz, AM–GM, and sharpness [T-AN-INEQUALITIES]
Methods Split a family by traces, pieces, or labels [M-DS-PARTITION]
Strategy Build a chain of bounds and track equality [S-BOUND]; Translate geometry into a controlled representation [S-GEOMETRIC]

Statement Recorded solutions Catalog

2023-D2-P8 — Solution

For this record: Theory: T-DS-TREES; T-AN-INEQUALITIES. Methods: M-DS-PARTITION. Full learning route

Hint: There are $n-1$ pairs u,v with $d(u,v)=1$; in all other cases $d(u,v)\geq 2$.
Let $k=\binom{n}{2}$ and let $x_1\leq x_2\leq ...\leq x_k$ be the distances between the pairs of vertices in the tree T. Thus

$$W(T)\cdot H(T)=(x_1+x_2+\cdots+x_k)\cdot\left(\frac{1}{x_1}+\frac{1}{x_2}+\cdots+\frac{1}{x_k}\right).$$

Since the tree has exactly $n-1$ edges, there are exactly $n-1$ pairs of vertices at distance one, that is, $x_1=x_2=\cdots=x_{n-1}=1$. Thus

$$\begin{aligned}W(T)\cdot H(T)&=(n-1+x_n+x_{n+1}+\cdots+x_k)\cdot\left(n-1+\frac{1}{x_n}+\frac{1}{x_{n+1}}+\cdots+\frac{1}{x_k}\right)=\\&=(n-1)^2+(n-1)\left(\left(x_n+\frac{1}{x_n}\right)+\cdots+\left(x_k+\frac{1}{x_k}\right)\right)+\\&\quad+(x_n+\cdots+x_k)\left(\frac{1}{x_n}+\cdots+\frac{1}{x_k}\right).\end{aligned}$$

From Cauchy inequality we have

$$(x_n+\cdots+x_k)\left(\frac{1}{x_n}+\cdots+\frac{1}{x_k}\right)\geq(1+1+\cdots+1)^2=(k-n+1)^2=\frac{(n-1)^2(n-2)^2}{4}.$$

The equality holds if and only if $x_n=x_{n+1}=\cdots=x_k$.
Now we minimize the expression $\left(x_n+\frac{1}{x_n}\right)+\cdots+\left(x_k+\frac{1}{x_k}\right)$, where $x_i\in[2,n-1]$.

It is clear that the minimal value is achieved for $x_n=x_{n+1}=\cdots=x_k=2$. Therefore we get

$$W(T)\cdot H(T)\geq(n-1)^2+(n-1)\left(2+\frac{1}{2}\right)(k-n+1)+\frac{(n-1)^2(n-2)^2}{4}=\frac{(n-1)^3(n+2)}{4}.$$

The equality holds for $x_1=\cdots=x_{n-1}=1$ and $x_n=x_{n+1}=\cdots=x_k=2$, that is, the smallest value is achieved for the tree where $n-1$ pairs are at distance one, and the remaining $k-(n-1)=\frac{(n-1)(n-2)}{2}$ pairs are at distance two. The unique tree which satisfies these conditions is the star graph S_n. In this case it holds

$$W(S_n)\cdot H(S_n)=(n-1)^2\cdot\frac{(n-1)(n+2)}{4}=\frac{(n-1)^3(n+2)}{4}.$$

2023-D2-P9 — Statement

Day 2, P9 | 10 points

We say that a real number V is good if there exist two closed convex subsets X, Y of the unit cube in $\mathbb{R}^3$, with volume V each, such that for each of the three coordinate planes (that is, the planes spanned by any two of the three coordinate axes), the projections of X and Y onto that plane are disjoint.
Find $\sup\{V \mid V \text{ is good}\}$.

(proposed by Josef Tkadlec and Arseniy Akopyan)

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Theory and methods for 2023-D2-P9

Theory Convex bodies, supporting functionals, and packing [T-DS-CONVEX]; Euclidean geometry, simplices, and barycentric weights [T-DS-GEOMETRY]; Finite counting and incidence identities [T-DS-COUNTING]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]
Methods Normalize coordinates and linearize squared distances [M-DS-GEOMETRIC]; Split a family by traces, pieces, or labels [M-DS-PARTITION]
Strategy Translate geometry into a controlled representation [S-GEOMETRIC]; Choose the objects and count them without ambiguity [S-COUNTING]; Build a chain of bounds and track equality [S-BOUND]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2023-D2-P9 — Solution

For this record: Theory: T-DS-CONVEX; T-DS-GEOMETRY; T-DS-COUNTING; T-AN-INTEGRAL. Methods: M-DS-GEOMETRIC; M-DS-PARTITION. Full learning route

Hint: The two bodies can be replaced by a pair symmetric to the midpoint of the cube.
We prove that $\sup\{V \mid V \text{ is good}\} = 1/4$.
We will use the unit cube $U = [-1/2, 1/2]^3$.
For $\varepsilon \to 0$, the axis-parallel boxes $X = [-1/2, -\varepsilon] \times [-1/2, -\varepsilon] \times [-1/2, 1/2]$ and $Y = [\varepsilon, 1/2] \times [\varepsilon, 1/2] \times [-1/2, 1/2]$ show that $\sup\{V\} \geq 1/4$.
To prove the other bound, consider two admissible convex bodies X, Y. For any point $P = [x, y, z] \in U$ with $xyz \neq 0$, let $\mathcal{P} = \{[\pm x, \pm y, \pm z]\}$ be the set consisting of 8 points (the original P and its 7 "symmetric" points).
If for each such P we have $|\mathcal{P} \cap (X \cup Y)| \leq 4$, then the conclusion follows by integrating. Suppose otherwise and let P be a point with $|\mathcal{P} \cap (X \cup Y)| \geq 5$. Below we will complete the proof by arguing that:
(1) we can replace one of the two bodies (the "thick" one) with the reflection of the other body about the origin, and
(2) for such symmetric pairs of bodies we in fact have $|\mathcal{P} \cap (X \cup Y)| \leq 4$, for all P.
To prove Claim (1), we say that a convex body is thick if each of its three projections contains the origin. We claim that one of the two bodies X, Y is thick. This is a short casework on the 8 points of $\mathcal{P}$. Since $|\mathcal{P} \cap (X \cup Y)| \geq 5$, by pigeonhole principle, we find a pair of points in $\mathcal{P} \cap (X \cup Y)$ symmetric about the origin. If both points belong to one body (say to X), then by convexity of X the origin belongs to X, thus X is thick. Otherwise, label $\mathcal{P}$ as $ABCDA'B'C'D'$. Wlog $A \in X$, $C' \in Y$ is the pair of points in $\mathcal{P}$ symmetric about the origin. Wlog at least 3 points of $\mathcal{P}$ belong to X. Since X, Y have disjoint projections, we have $C, B', D' \notin X$, so wlog $B, D \in X$. Then Y can contain no other point of $\mathcal{P}$ (apart from C'), so X must contain at least 4 points of $\mathcal{P}$ and thus $A' \in X$. But then each projection of X contains the origin, so X is indeed thick.
Note that if X is thick then none of the three projections of Y contains the origin. Consider the reflection $Y' = -Y$ of Y about the origin. Then (Y, Y') is an admissible pair with the same volume as (X, Y): the two bodies Y and Y' clearly have equal volumes V and they have disjoint projections (by convexity, since the projections of Y miss the origin). This proves Claim (1).
Claim (2) follows from a similar small casework on the 8-tuple $\mathcal{P}$: For contradiction, suppose $|\mathcal{P} \cap Y'| = |\mathcal{P} \cap Y| \geq 3$. Wlog $A \in Y'$. Then $C' \in Y$, so $C, B', D' \notin Y'$, so wlog $B, D \in Y'$. Then $B', D' \in Y$, a contradiction with (Y, Y') being admissible.

2023-D2-P9 — Solution (continued)

Remark. There are more examples with $V \to 1/4$, e.g. X a union of two triangular pyramids with base ACD' – one with apex D, one with apex at the origin (and Y symmetric with X about the origin).

Remark. The word "convex" matters. E.g., in a $3 \times 3 \times 3$ cube, one can set X to be a $2 \times 2 \times 2$ sub-cube, and Y to be the (non-convex) 3D L-shape consisting of 7 unit cubes. This shows that without convexity we have $V \geq 7/27 > 1/4$.

2023-D2-P10 — Statement

Day 2, P10 | 10 points

For every positive integer n, let $f(n)$, $g(n)$ be the minimal positive integers such that

$$1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!} = \frac{f(n)}{g(n)}.$$

Determine whether there exists a positive integer n for which $g(n) > n^{0.999n}$.

(proposed by Fedor Petrov, St. Petersburg State University)

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Theory and methods for 2023-D2-P10

Theory Prime valuations, divisors, and factorials [T-AL-VALUATION]; Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]; Factorial growth and the prime-number input [T-AL-PRIMEGROWTH]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Track one prime or one extreme divisor [M-AL-VALUATION]; Replace many equal values by too many roots [M-AL-ROOTCOUNT]
Strategy Build a chain of bounds and track equality [S-BOUND]; Choose the objects and count them without ambiguity [S-COUNTING]

Statement Recorded solutions Catalog

2023-D2-P10 — Solution

For this record: Theory: T-AL-VALUATION; T-AL-FINITEFIELD; T-AL-POLYNOMIAL; T-AL-PRIMEGROWTH; T-DS-COUNTING. Methods: M-AL-VALUATION; M-AL-ROOTCOUNT. Full learning route

We show that there does exist such a number n.
Let $\varepsilon = 10^{-10}$. Call a prime p special, if for certain $k \in \{1, 2, \ldots, p-1\}$ there exist at least $\varepsilon \cdot k$ positive integers $j \leq k$ for which p divides $f(j)$.
Lemma. There exist only finitely many special primes.
Proof. Let p be a special prime number, and p divides $f(j)$ for at least $\varepsilon \cdot k$ values of $j \in \{1, 2, \ldots, k\}$. Note that if p divides $f(j)$ and $f(j+r)$, then p divides

$$(j+r)! \left(\frac{f(j+r)}{g(j+r)} - \frac{f(j)}{g(j)} \right) = 1 + (j+r) + (j+r)(j+r-1) + \cdots + (j+r)\ldots(j+2)$$

that is a polynomial of degree $r-1$ with respect to j. Thus, for fixed j it equals to 0 modulo p for at most $r-1$ values of j. Look at our $\geq \varepsilon \cdot k$ values of $j \in \{1, 2, \ldots, k\}$ and consider the gaps between consecutive j's. The number of such gaps which are greater than $2/\varepsilon$ does not exceed $\varepsilon \cdot k/2$ (since the total sum of gaps is less than k). Therefore, at least $\varepsilon \cdot k/2 - 1$ gaps are at most $2/\varepsilon$. But the number of such small gaps is bounded from above by a constant (not depending on k) by the above observation. Therefore, k is bounded, and, since p divides $f(1)f(2)\ldots f(k)$, p is bounded too.
Now we want to bound the product $g(1)g(2)\ldots g(n)$ (for a large integer n) from below. Let $p \leq n$ be a non-special prime. Our nearest goal is to prove that

$$\nu_p(g(1)g(2)\ldots g(n)) \geq (1-\varepsilon)\nu_p(1! \cdot 2! \cdot \cdots \cdot n!) \tag{1}$$

Partition the numbers $p, p+1, \ldots, n$ onto the intervals of length p (except possibly the last interval which may be shorter): $\{p, p+1, \ldots, 2p-1\}$, ..., $\{p\lfloor n/p \rfloor, \ldots, n\}$. Note that in every interval $\Delta =$

2023-D2-P10 — Solution (continued)

$[a\cdot p, a\cdot p+k]$, all factorials $x!$ with $x\in\Delta$ have the same p-adic valuation, denote it $T=\nu_p((ap)!)$. We claim that at least $(1-\varepsilon)(k+1)$ valuations of $g(x)$, $x\in\Delta$, are equal to the same number T. Indeed, if $j=0$ or $1\le j\le k$ and $f(j)$ is not divisible by p, then

$$\frac{1}{(ap)!}+\frac{1}{(ap+1)!}+\cdots+\frac{1}{(ap+j)!}=\frac{1}{(ap)!}\cdot\frac{A}{B}$$

where $A\equiv f(j)\pmod p$, $B\equiv g(j)\pmod p$, so, this sum has the same p-adic valuation as $1/(ap)!$, which is strictly less than that of the sum $\sum_{i=0}^{ap-1}1/i!$, that yields $\nu_p(g(ap+j))=\nu_p((ap)!)$. Using this for every segment Δ, we get (1).
Now, using (1) for all non-special primes, we get

$$A\cdot g(1)g(2)\dots g(n)\ge(1!\cdot 2!\cdot\dots\cdot n!)^{1-\varepsilon},$$

where $A=\prod_{p,k}p^{\nu_p(g(k))}$, p runs over non-special

primes, k from 1 to n. Since $\nu_p(g(k))\le\nu_p(k!)=\sum_{i=1}^{\infty}\lfloor k/p^i\rfloor\le k$, we get

$$A\le\left(\prod_p p\right)^{1+2+\dots+n}\le C^{n^2}$$

for some constant C. But if we had $g(n)\le n^{0.999n}\le e^n n!^{0.999}$ for all n, then

$$\log(A\cdot g(1)g(2)\dots g(n))\le O(n^2)+0.999\log(1!\cdot 2!\cdot\dots\cdot n!)<(1-\varepsilon)\log(1!\cdot 2!\cdot\dots\cdot n!)$$

for large n, a contradiction.

3.31 2024

Day 1

2024-D1-P1 — Statement

Day 1, P1 | 10 points

Determine all pairs $(a,b)\in\mathbb{C}\times\mathbb{C}$ satisfying

$$|a|=|b|=1\quad\text{and}\quad a+b+a\bar{b}\in\mathbb{R}.$$

(proposed by Mike Daas, Universiteit Leiden)

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Theory and methods for 2024-D1-P1

Theory Complex coordinates, conjugation, and unit-circle geometry [T-AL-COMPLEX]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Compare coefficients after a structural substitution [M-AL-COEFFICIENT]; Factor first and compare irreducible exponents [M-AL-FACTOR]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Translate geometry into a controlled representation [S-GEOMETRIC]

Statement Recorded solutions Catalog

2024-D1-P1 — Solution — Record 1 of 2

For this record: Theory: T-AL-COMPLEX; T-AL-POLYNOMIAL. Methods: M-AL-COEFFICIENT. Full learning route

Hint: Write $a=e^{ix}$ and $b=e^{iy}$, and transform the RHS to a product.

2024-D1-P1 — Solution — Record 1 of 2 (continued)

Write $a = e^{ix}$ and $b = e^{iy}$ for some $x, y \in [0, 2\pi)$. Using Euler's formula, and the well-known identities

$$\sin x + \sin y = 2 \sin \frac{x+y}{2} \cos \frac{x-y}{2} \quad \text{and} \quad \sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2},$$

we get a product form of the left-hand side as

$$\begin{aligned} \operatorname{Im}(a + b + a\bar{b}) &= \sin x + \sin y + \sin(x - y) \\ &= 2 \sin \frac{x+y}{2} \cos \frac{x-y}{2} + 2 \sin \frac{x-y}{2} \cos \frac{x-y}{2} \\ &= 2 \left(\sin \frac{x+y}{2} + \sin \frac{x-y}{2} \right) \cos \frac{x-y}{2} \\ &= 4 \sin \frac{x}{2} \cdot \cos \frac{y}{2} \cdot \cos \frac{x-y}{2}. \end{aligned}$$

Hence, $a + b + a\bar{b}$ is real if and only if either $\sin \frac{x}{2} = 0$, $\cos \frac{y}{2} = 0$ or $\cos \frac{x-y}{2} = 0$, which respectivelly correspond to $x = 2k\pi$, $y = (2k+1)\pi$ and $x = y + (2k+1)\pi$.

Therefore, the solutions are

$$(1, b), \quad (a, -1) \quad \text{and} \quad (a, -a) \quad \text{with } |a| = 1,\ |b| = 1.$$

2024-D1-P1 — Solution — Record 2 of 2

For this record: Theory: T-AL-COMPLEX; T-AL-POLYNOMIAL. Methods: M-AL-FACTOR. Full learning route

Notice that

$$a + b + a\bar{b} \in \mathbb{R} \iff 1 + a + b + a\bar{b} \in \mathbb{R}.$$

Let $c \in \mathbb{C}$ be such that $a = c^2$. Now observe that

$$\begin{aligned} \bar{c}(1 + a + b + a\bar{b}) &= \bar{c} + \bar{c}c^2 + \bar{c}b + \bar{c}c^2\bar{b} \\ &= \bar{c} + c + \bar{c}b + c\bar{b} \in \mathbb{R}, \end{aligned}$$

where we used that $c\bar{c} = 1$ and $z + \bar{z} \in \mathbb{R}$ for any $z \in \mathbb{C}$. We conclude that either $c \in \mathbb{R}$, or $1 + a + b + a\bar{b} = 0$. In the first case, $c = \pm 1$ and so $a = 1$. In the second case, we factor the equation as

$$(a + b)(1 + \bar{b}) = 1 + a + b + a\bar{b} = 0, \quad \text{and as such, } a = -b \text{ or } b = -1.$$

We find precisely three families of pairs (a, b): the pairs $(1, b)$ for b on the unit circle; the pairs $(a, -1)$ for a on the unit circle; and the pairs $(a, -a)$ for a on the unit circle.

2024-D1-P2 — Statement

Day 1, P2 | 10 points

For $n = 1, 2, \ldots$ let

$$S_n = \log\left(\sqrt[n^2]{1^1 \cdot 2^2 \cdot \cdots \cdot n^n} \right) - \log(\sqrt{n}),$$

where log denotes the natural logarithm. Find $\lim_{n \to \infty} S_n$.

(proposed by Sergey Chernov, Belarusian State University, Minsk)

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Theory and methods for 2024-D1-P2

Theory Riemann sums, periodic averaging, and tails [T-AN-RIEMANN]; Primitives, substitution, and integration by parts [T-AN-INTEGRAL]; Scale separation and limit comparisons [T-AN-ASYMPTOTICS]
Methods Expose the mesh and control its tails [M-AN-RIEMANN]
Strategy Expose the structure before calculating [S-STRUCTURE]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2024-D1-P2 — Solution

For this record: Theory: T-AN-RIEMANN; T-AN-INTEGRAL; T-AN-ASYMPTOTICS. Methods: M-AN-RIEMANN. Full learning route

Hint: S_n is (close to) a Riemann sum of a certain integral.

Transform S_n as

$$\begin{aligned} S_n &= \frac{1}{n^2}\sum_{k=1}^{n} k\log k - \frac{1}{2}\log n \\ &= \frac{1}{n}\sum_{k=1}^{n}\frac{k}{n}\left(\log\frac{k}{n}+\log n\right) - \frac{1}{2}\log n \\ &= \frac{1}{n}\sum_{k=1}^{n}\frac{k}{n}\log\frac{k}{n} + \frac{\log n}{n^2}\sum_{k=1}^{n} k - \frac{1}{2}\log n \\ &= \frac{1}{n}\sum_{k=1}^{n}\frac{k}{n}\log\frac{k}{n} + \frac{\log n}{2n}. \end{aligned}$$

Here the last term $\dfrac{\log n}{2n}$ converges to 0. The sum $\dfrac{1}{n}\sum\limits_{k=1}^{n}\dfrac{k}{n}\log\dfrac{k}{n}$ is a Riemann sum for the integrable function $f(x) = x\log x$ on the segment $[0,1]$ with the uniform grid $\left\{\dfrac{1}{n}, \dfrac{2}{n}, \ldots, \dfrac{n-1}{n}, 1\right\}$. Therefore

$$\lim\frac{1}{n}\sum_{k=1}^{n}\frac{k}{n}\log\frac{k}{n} = \lim\frac{1}{n}\sum_{k=1}^{n} f\left(\frac{k}{n}\right) = \int_0^1 x\log x\,dx = \left[\frac{x^2}{2}\log x - \frac{x^2}{4}\right]_0^1 = -\frac{1}{4}.$$

Hence, $\lim S_n$ exists, and $\lim S_n = -\dfrac{1}{4}$.

2024-D1-P3 — Statement

Day 1, P3 | 10 points

For which positive integers n does there exist an $n\times n$ matrix A whose entries are all in $\{0,1\}$, such that A^2 is the matrix of all ones?

(proposed by Alex Avdiushenko, Neapolis University Paphos, Cyprus)

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Theory and methods for 2024-D1-P3

Theory Rank, dimension and exact linear constraints [T-LA-RANK]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Read the entries before choosing a theorem [M-LA-ENTRY]; Exploit a compatible block decomposition [M-LA-BLOCK]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2024-D1-P3 — Solution

For this record: Theory: T-LA-RANK; T-DS-COUNTING. Methods: M-LA-ENTRY; M-LA-BLOCK. Full learning route

Hint: Let J be the $n \times n$ matrix with all ones. Consider $A^3 = AJ = JA$.
Answer: Such a matrix A exists if and only if n is a complete square.
Let J_n be the $n \times n$ matrix with all ones, so $A^2 = J_n$. Consider the equality

$$A^3 = AJ_n = J_nA.$$

In the matrix AJ_n, all columns are equal to the sum of colums in A, that is, the (i,j)th entry in AJ_n is the number of ones in the ith row of A. Similarly, the (i,j)th entry in J_nA is the number of ones in the jth column of A. These numbers must be equal, so A contains the same number of ones in every row and every column. Let this common number be k; then $AJ_n = J_nA = kJ_n$.
Now from

$$nJ_n = J_n^2 = (A^2)^2 = A(AJ_n) = A(kJ_n) = k(AJ_n) = k^2J_n$$

we can read $n = k^2$, so n must be a complete square.

It remains to show an example for a matrix A of order $n = k^2$. For $l = 0, 1, \dots, k-1$, let B_l be the $k \times k$ matrix whose (i,j)th entry is 1 if $j - i \equiv l \pmod k$ and 0 otherwise, i.e., B_l can be obtained from the identity matrix by cyclically shifting the colums l times, and let

$$A = \begin{pmatrix} B_0 & B_1 & B_2 & \dots & B_{k-1} \\ B_0 & B_1 & B_2 & \dots & B_{k-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ B_0 & B_1 & B_2 & \dots & B_{k-1} \end{pmatrix};$$

The (i,j)th block in A^2 is

$$\begin{pmatrix} B_0 & B_1 & \dots & B_{k-1} \end{pmatrix} \begin{pmatrix} B_{j-1} \\ \vdots \\ B_{j-1} \end{pmatrix} = (B_0 + B_1 + \dots + B_{k-1})B_{j-1} = J_kB_{j-1} = J_k,$$

so this matrix indeed satisfies $A^2 = J_{k^2}$.

2024-D1-P4 — Statement

Day 1, P4 | 10 points

Let g and h be two distinct elements of a group G, and let n be a positive integer. Consider a sequence $w = (w_1, w_2, \dots)$ which is not eventually periodic and where each w_i is either g or h. Denote by H the subgroup of G generated by all elements of the form $w_kw_{k+1}\dots w_{k+n-1}$ with $k \geq 1$. Prove that H does not depend on the choice of the sequence w (but may depend on n).

(proposed by Ivan Mitrofanov, Saarland University)

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Theory and methods for 2024-D1-P4

Theory Groups, orders, generators, and quotients [T-AL-GROUP]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Prove generation or obstruct it by an image [M-AL-GENERATORS]
Strategy Reduce size while preserving the defining conditions [S-INDUCTION]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2024-D1-P4 — Solution

For this record: Theory: T-AL-GROUP; T-DS-INDUCTION; T-DS-COUNTING. Methods: M-AL-GENERATORS. Full learning route

Let X_m denote the subset of G of products of the form $g_1 \dots g_m$, where each g_i is either g or h.

2024-D1-P4 — Solution (continued)

Lemma. For all $j = 1, 2, \dots, n$ and for all $a, b \in X_j$ the ratio $a^{-1}b$ is contained in H.
Proof. Induction in j.
We start with the base case $j = 1$. By the pigeonhole principle, there exist $k < \ell$ for which the sequences $(w_{k+1}, \dots, w_{k+n-1})$ and $(w_{\ell+1}, \dots, w_{\ell+n-1})$ coincide. If $w_{k+m} = w_{\ell+m}$ for all positive integer m, then the sequence w is eventually periodic with period $\ell - k$. Thus, there exists $m > 0$ for which $w_{k+m} \neq w_{\ell+m}$. We have $m \geqslant n$, so $w_{k+m-i} = w_{\ell+m-i}$ for $i = 1, 2, \dots, n-1$. Therefore, since the products $x = w_{k+m-n+1} \dots w_{k+m}$ and $y = w_{\ell+m-n+1} \dots w_{\ell+m}$ both are elements of H, the subgroup H contains their ratios $x^{-1}y$ and $y^{-1}x$. These ratios are equal to $g^{-1}h$ and $h^{-1}g$ (in some order), that finishes the proof for $j = 1$.
Induction step from $j-1$ to j, $2 \leqslant j \leqslant n$. We say that an element $a \in X_j$ is a g-element, correspondingly an h-element, if it can be represented as $a = ga_1$, correspondingly $a = ha_1$, where $a_1 \in X_{j-1}$. The ratio of two g-elements, or of two h-elements, is a ratio of two elements of X_{j-1}, thus, it is in H by the induction hypothesis. Since the property $a^{-1}b \in H$ is an equivalence relation on pairs (a, b), it suffices to find a g-element and h-element whose ratio is in H.
Define k, ℓ, m, as in the base case. The subgroup H contains the products

$$v = w_{k+m-n+j} \dots w_{k+m} w_{k+m+1} \dots w_{k+m+j-1},$$
$$u = w_{\ell+m-n+j} \dots w_{\ell+m} w_{\ell+m+1} \dots w_{\ell+m+j-1}.$$

Their ratio $u^{-1}v$ is a ratio of g-element and an h-element in X_j, since $\{w_{k+m}, w_{\ell+m}\} = \{g, h\}$ and $w_{k+m-i} = w_{\ell+m-i}$ for all $i = 1, 2, \dots, n-j$.

The Lemma for $j = n$ yields that H is the subgroup of G generated by X_n, and this description does not depend on w.

2024-D1-P5 — Statement

Day 1, P5 | 10 points

Let $n > d$ be positive integers. Choose n independent, uniformly distributed random points $x_1, \dots, x_n$ in the unit ball $B \subset \mathbb{R}^d$ centered at the origin. For a point $p \in B$ denote by $f(p)$ the probability that the convex hull of $x_1, \dots, x_n$ contains p. Prove that if $p, q \in B$ and the distance of p from the origin is smaller than the distance of q from the origin, then $f(p) \geq f(q)$.

(proposed by Fedor Petrov, St Petersburg State University)

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Theory and methods for 2024-D1-P5

Theory Uniform random points and convex-hull events [T-AN-GEOMETRICPROB]; Finite probability, conditioning, and expectations [T-AN-PROBABILITY]; Facets and signed simplex decompositions [T-DS-POLYTOPE]
Methods Condition and average a convex-hull event [M-AN-GEOMETRIC]
Strategy Translate geometry into a controlled representation [S-GEOMETRIC]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2024-D1-P5 — Solution

For this record: Theory: T-AN-GEOMETRICPROB; T-AN-PROBABILITY; T-DS-POLYTOPE. Methods: M-AN-GEOMETRIC. Full learning route

By radial symmetry of the distribution, $f(p)$ depends only on $|op|$ (the distance between o and p), so, we may assume that p lies on the segment between o and q. For points $x_1, \dots, x_n$ and $x \in B$ denote by $f_x(x_1, \dots, x_n)$ the indicator function of the event "x is in the convex hull of $x_1, \dots, x_n$". The claim follows from the following deterministic inequality

$$\sum f_p(\pm x_1, \dots, \pm x_n) \geqslant \sum f_q(\pm x_1, \dots, \pm x_n), \tag{1}$$

where $x_1, \dots, x_n \in B$ are arbitrary points in general position and the summations are over all 2^n choices of signs (here o is identified with the origin, that is, x and $-x$ are symmetric with re-

2024-D1-P5 — Solution (continued)

spect to o). Indeed, taking the expectation in (1) over independent random uniform $x_1, \dots, x_n$, we get $2^n f(p) \geqslant 2^n f(q)$. (To be specific, here "general position" means that for any point set $A \subset \{\pm x_1, \dots, \pm x_n, p, q\}$, which does not contain simultaneosuly x_i and $-x_i$, is not contained in an (affine) $(|A|-2)$-dimensional plane. This holds with probability 1.)
To prove (1), we use the following formula for the characteristic function χ_P of the convex polyhedron $P \subset \mathbb{R}^d$: if $P_1, \dots, P_k$ are all facets of P, and Q_i is the convex hull of o and P_i, then $\chi_P = \sum \pm \chi_{Q_i}$, where the sign is plus if o and P are on the same side of P_i, and minus otherwise. Indeed, for every point p in general position look how the ray op intersects the boundary of P and realize that for at most two summands the contribution of the RHS at point p is non-zero, and the total contribution equals 1 when p is inside P and 0 (possibly as $0 = 1 - 1$) otherwise. Use this formula for every polyhedron P with n vertices $y_1, \dots, y_n$, where each y_i is $\pm x_i$. These polyhedrons are simplicial (all facets are simplices) because of the general position condition. Sum up over all 2^n such P, we get the expression of $\sum_P \chi_P$ as a linear combination of χ_S, where S are simplices formed by o and some d points in $\{\pm x_1, \dots, \pm x_n\}$ (not containing x_i and $-x_i$ simultaneously).

For proving (1), it suffices to verify that all coefficients of χ_S in this linear combination are positive (since two sides of (1) are the values of the sum $\sum_P \chi_P$ at p and q). Let's find a coefficient of χ_S, where, say, S is a simplex with vertices $o, x_1, \dots, x_d$. The plane α through $x_1, \dots, x_d$ partitions $\mathbb{R}^d$ onto two parts H^+ (containing o) and H^- (not containing o). For every pair $\{x_i, -x_i\}$ with $i > d$, either both points belong to H^+, or one belongs to H^- and another to H^+. χ_S goes with the plus sign for P with vertices $x_1, \dots, x_d$ and other vertices from H^+, and with the minus sign for P with vertices $x_1, \dots, x_d$ and other vertices from H^-. It is immediate that there are at least as many pluses as minuses.

Day 2

2024-D2-P6 — Statement

Day 2, P6 | 10 points

Prove that for any function $f : \mathbb{Q} \to \mathbb{Z}$, there exist $a, b, c \in \mathbb{Q}$ such that $a < b < c$, $f(b) \geq f(a)$, and $f(b) \geq f(c)$.

(proposed by Mehdi Golafshan & Markus A. Whiteland, University of Liège, Liège)

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Theory and methods for 2024-D2-P6

Theory Countability, accumulation, and Cantor constructions [T-AN-SETS]; Finite counting and incidence identities [T-DS-COUNTING]; Extremal choice, exchange arguments, and invariants [T-DS-EXTREMAL]
Methods Split a family by traces, pieces, or labels [M-DS-PARTITION]; Repair an extremal configuration by a local exchange [M-DS-EXTREMAL]
Strategy Make the contradiction quantitative or structural [S-CONTRADICTION]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2024-D2-P6 — Solution — Record 1 of 2

For this record: Theory: T-AN-SETS; T-DS-COUNTING. Methods: M-DS-PARTITION. Full learning route

We can replace $f(x)$ by the function $g(x) = f(1-x)$, so without loss of generality we can assume $f(0) \leq f(1)$.
If $f(1) \geq f(2)$ then we can choose $(a, b, c) = (0, 1, 2)$. Otherwise we have $f(0) \leq f(1) < f(2)$. If there is some $x \in (1, 2)$ such that $f(x) \geq f(2)$ then we can chose $(a, b, c) = (1, x, 2)$; similarly, if there is

2024-D2-P6 — Solution — Record 1 of 2 (continued)

some $x \in (1,2)$ with $f(x) \le f(1)$ then choose $(a,b,c) = (0,1,x)$. Hence, in the remaining cases we have $f(1) \le f(x) \le f(2)$ for all $x \in (1,2)$.

Now f is bonded on the interval $[1,2]$, so it has only finitely many values on this interval. Since there are infintely many rational numbers in $[1,2]$, there is a value y that is attained infinitely many times. The we can choose $1 \le a < b < c \le 2$ such that $f(a) = f(b) = f(c) = y$.

2024-D2-P6 — Solution — Record 2 of 2

For this record: Theory: T-AN-SETS; T-DS-EXTREMAL. Methods: M-DS-EXTREMAL. Full learning route

Assume towards a contradiction that there is a function f which does not satisfy the claim: for all rationals a, b, c with $a < b < c$ we have $f(b) < f(a)$ or $f(b) < f(c)$.
Let x and y be arbitrary rationals with $x < y$. Let $I(x,y) = [x,y] \cap \mathbb{Q}$. We first observe that $\inf f(I(x,y)) = -\infty$. Indeed, if the infimum was finite, then, as the set $f(I(x,y))$ is bounded ($\sup f(I(x,y)) = \max\{f(x), f(y)\}$) and thus finite, there are three points having the same value under f, which leads to a contradiction regarding our assumption on f.

So, going back to the question at hand, let x, b, y be arbitrary rationals with $x < b < y$. Applying the above observation to the set $I(x,b)$, there exists a point $a \in I(x,b)$ such that $f(a) < f(b)$. Similarly, there exists a point $c \in I(b,y)$ such that $f(c) < f(b)$. Hence we have the points a, b, c with $a < b < c$ and $f(b) > \max\{f(a), f(c)\}$, which contradicts our assumption on f.

2024-D2-P7 — Statement

Day 2, P7 | 10 points

Let n be a positive integer. Suppose that A and B are invertible $n \times n$ matrices with complex entries such that $A + B = I$ (where I is the identity matrix) and

$$(A^2 + B^2)(A^4 + B^4) = A^5 + B^5.$$

Find all possible values of $\det(AB)$ for the given n.

(proposed by Sergey Bondarev, Sergey Chernov, Belarusian State University, Minsk)

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Theory and methods for 2024-D2-P7

Theory Annihilating polynomials and spectral mapping [T-LA-POLYNOMIAL]; Determinants, block identities and inverses [T-LA-DETERMINANT]; Polynomial identities, degree, and multiplicity [T-AL-POLYNOMIAL]
Methods Factor while preserving matrix order [M-LA-FACTOR]; Reduce an operator identity to its scalar roots [M-LA-SPECTRAL]
Strategy Prove necessity and sufficiency separately [S-CLASSIFICATION]; Build a witness and verify every constraint [S-CONSTRUCTION]

Statement Recorded solutions Catalog

2024-D2-P7 — Solution

For this record: Theory: T-LA-POLYNOMIAL; T-LA-DETERMINANT; T-AL-POLYNOMIAL. Methods: M-LA-FACTOR; M-LA-SPECTRAL. Full learning route

Hint: Find a polynomial $p(x)$ such that $p(AB) = 0$.
Notice first that $AB = A(I-A) = A - A^2 = (I-A)A = BA$, so A and B commute. Let $C = AB = BA$; then

$$\begin{aligned}
A^2 + B^2 &= (A+B)^2 - 2AB = I - 2C, \\
A^4 + B^4 &= (A+B)^4 - 4AB(A+B)^2 + 2A^2B^2 = I - 4C + 2C^2, \\
A^5 + B^5 &= (A+B)^5 - 5AB(A+B)^3 + 5A^2B^2(A+B) = I - 5C + 5C^2,
\end{aligned}$$

2024-D2-P7 — Solution (continued)

so

$$0 = (A^5 + B^5) - (A^2 + B^2)(A^4 + B^4) = (I - 5C + 5C^2) - (I - 2C)(I - 4C + 2C^2)$$
$$= 4C^3 - 5C^2 + C = 4C(C - I)(C - \tfrac{1}{4}I);$$

since C is invertible, we have

$$(C - I)(C - \tfrac{1}{4}I) = 0.$$

Hence, the polynomial $p(x) = (x - 1)(x - \frac{1}{4})$ annihilates the matrix $C = AB$ and therefore all eigenvalues of C are roots of $p(x)$, so the possible eigenvalues are 1 and $\frac{1}{4}$. The determinant is the product of the n eigenvalues, so

$$\det(AB) = \det C \in \left\{1, \frac{1}{4}, \frac{1}{4^2}, \dots, \frac{1}{4^n}\right\}.$$

Now show that these values are indeed possible.

If

$$A = \operatorname{diag}(\underbrace{\tfrac{1}{2}, \dots, \tfrac{1}{2}}_{k}, \underbrace{e^{i\pi/3}, \dots, e^{i\pi/3}}_{n-k}) \quad \text{and} \quad B = \operatorname{diag}(\underbrace{\tfrac{1}{2}, \dots, \tfrac{1}{2}}_{k}, \underbrace{e^{-i\pi/3}, \dots, e^{-i\pi/3}}_{n-k}),$$

then $A + B = I$, $AB = \operatorname{diag}(\underbrace{\frac{1}{4}, \dots, \frac{1}{4}}_{k}, \underbrace{1, \dots, 1}_{n-k})$ and $\det(AB) = \dfrac{1}{4^k}$.

2024-D2-P8 — Statement

Day 2, P8 | 10 points

Define the sequence $x_1, x_2, \dots$ by the initial terms $x_1 = 2$, $x_2 = 4$, and the recurrence relation

$$x_{n+2} = 3x_{n+1} - 2x_n + \frac{2^n}{x_n} \quad \text{for } n \geq 1.$$

Prove that $\lim\limits_{n\to\infty} \dfrac{x_n}{2^n}$ exists and satisfies

$$\frac{1 + \sqrt{3}}{2} \leq \lim_{n\to\infty} \frac{x_n}{2^n} \leq \frac{3}{2}.$$

(proposed by Karen Keryan, Yerevan State University & American University of Armenia, Armenia)

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Theory and methods for 2024-D2-P8

Theory Iteration, fixed points, and periodic orbits [T-DS-DYNAMICS]; Strong induction, state recurrences, and constructive invariants [T-DS-INDUCTION]; Sequences, subsequences, and recurrence limits [T-AN-CONVERGENCE]
Methods Trap an orbit and telescope its transformed recurrence [M-DS-ITERATE]; Strengthen the statement before the induction step [M-DS-INDUCT]
Strategy Reduce size while preserving the defining conditions [S-INDUCTION]; Build a chain of bounds and track equality [S-BOUND]; Control the limiting operation before taking the limit [S-ANALYTIC]

Statement Recorded solutions Catalog

2024-D2-P8 — Solution

For this record: Theory: T-DS-DYNAMICS; T-DS-INDUCTION; T-AN-CONVERGENCE. Methods: M-DS-ITERATE; M-DS-INDUCT. Full learning route

Hint: Prove that $2x_n \leq x_{n+1} \leq 2x_n + n$.

2024-D2-P8 — Solution (continued)

Let's prove by induction that $x_{n+1} \geq 2x_n$. It holds for $n = 1$. Assume it holds for n. Then by the induction hypothesis we have that $x_n \geq 2x_{n-1} \geq \ldots \geq 2^{n-1}x_1 > 0$ and

$$x_{n+2} = 2x_{n+1} + (x_{n+1} - 2x_n) + \frac{2^n}{x_n} > 2x_{n+1}.$$

Similarly we prove that $x_{n+1} \leq 2x_n + n$. Again it holds for $n = 1$. Assume that the inequality holds for n. Then using that $x_n \geq 2^n$ and the induction hypothesis we obtain

$$x_{n+2} \leq 3x_{n+1} - 2x_n + 1 \leq 2x_{n+1} + (2x_n + n) - 2x_n + 1 = 2x_{n+1} + n + 1.$$

Using the previous inequalities we obtain that the sequence $y_n = \frac{x_n}{2^n}$ is increasing and $y_{n+1} \leq y_n + \frac{n}{2^n} \leq \ldots \leq y_1 + \sum_{k=1}^{n} \frac{k}{2^k} < \infty$, thus $\lim_{n\to\infty} y_n = c$ exists.

The recurrence relation has the following form for y_n:

$$4y_{n+2} - 2y_{n+1} = 4y_{n+1} - 2y_n + \frac{1}{2^n \cdot y_n}.$$

By summing up the above equality for $n = 1, \ldots, m$ we obtain

$$4y_{m+2} - 2y_{m+1} = 4y_2 - 2y_1 + \sum_{n=1}^{m} \frac{1}{2^n \cdot y_n} = 2 + \sum_{n=1}^{m} \frac{1}{2^n \cdot y_n}. \tag{1}$$

Now using the facts that $y_1 = 1$, y_n increases and $\lim_{n\to\infty} y_n = c$ we obtain $1 \leq y_n \leq c$. Hence

$$\frac{1}{c} \leq \sum_{n=1}^{\infty} \frac{1}{2^n \cdot y_n} \leq 1.$$

Thus we get from (1)

$$2c = \lim_{m\to\infty} (4y_{m+2} - 2y_{m+1}) = 2 + \sum_{n=1}^{\infty} \frac{1}{2^n \cdot y_n} \in \left[2 + \frac{1}{c}, 3\right].$$

So we have $2c^2 \geq 2c + 1$ and $2c \leq 3$. Recall that $c \geq 1$. Therefore $1 + \sqrt{3} \leq 2c \leq 3$, which finishes the proof.

2024-D2-P9 — Statement

Day 2, P9 | 10 points

A matrix $A = (a_{ij})$ is called nice, if it has the following properties:
(i) the set of all entries of A is $\{1, 2, \ldots, 2t\}$ for some integer t;
(ii) the entries are non-decreasing in every row and in every column: $a_{i,j} \leq a_{i,j+1}$ and $a_{i,j} \leq a_{i+1,j}$;
(iii) equal entries can appear only in the same row or the same column: if $a_{i,j} = a_{k,\ell}$, then either $i = k$ or $j = \ell$;
(iv) for each $s = 1, 2, \ldots, 2t - 1$, there exist $i \neq k$ and $j \neq \ell$ such that $a_{i,j} = s$ and $a_{k,\ell} = s + 1$.
Prove that for any positive integers m and n, the number of nice $m \times n$ matrices is even.
For example, the only two nice 2×3 matrices are $\begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \end{pmatrix}$ and $\begin{pmatrix} 1 & 1 & 3 \\ 2 & 4 & 4 \end{pmatrix}$.

(proposed by Fedor Petrov, St Petersburg State University)

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Theory and methods for 2024-D2-P9

Theory Graphs, walks, cycles, and local degree counts [T-DS-GRAPH]; Finite counting and incidence identities [T-DS-COUNTING]
Methods Encode a relation as a graph and extract a cycle [M-DS-GRAPH]; Encode, decode, and verify a bijection [M-DS-BIJECTION]
Strategy Expose the structure before calculating [S-STRUCTURE]; Choose the objects and count them without ambiguity [S-COUNTING]

Statement Recorded solutions Catalog

2024-D2-P9 — Solution

For this record: Theory: T-DS-GRAPH; T-DS-COUNTING. Methods: M-DS-GRAPH; M-DS-BIJECTION. Full learning route

Define a standard Young tableaux of shape $m \times n$ as an $m \times n$ matrix with the set of entries $\{1, 2, \dots, mn\}$, increasing in every row and in every column as in (ii).
Call two standard Young tableaux Y_1, Y_2 friends, if they differ by a switch of two consecutive numbers $x, x+1$ (the places of x and $x+1$ must be not neighbouring, for such a switch preserving the monotonicity in rows and columns).
For a nice $m \times n$ matrix A we construct a standard Young tableaux Y_A of shape $m \times n$ as follows: if A has n_i entries equal to i ($i = 1, 2, \dots 2t$), we replace them by the numbers from $n_1 + \cdots + n_{i-1} + 1$ to $n_1 + \cdots + n_i$ preserving monotonicity.

Note that our Y_A has exactly $2t - 1$ friends, where $2t$ is the number of distinct entries in A, and moreover, every standard Young tableaux with odd number of friends corresponds to a unique nice matrix. It remains to apply the handshaking lemma (i.e., the sum of the degrees equals twice the number of edges in this graph).

2024-D2-P10 — Statement

Day 2, P10 | 10 points

We say that a square-free positive integer n is almost prime if

$$n \mid x^{d_1} + x^{d_2} + \cdots + x^{d_k} - kx$$

for all integers x, where $1 = d_1 < d_2 < \cdots < d_k = n$ are all the positive divisors of n. Suppose that r is a Fermat prime (i.e. it is a prime of the form $2^{2^m} + 1$ for an integer $m \geq 0$), p is a prime divisor of an almost prime integer n, and $p \equiv 1 \pmod{r}$. Show that, with the above notation, $d_i \equiv 1 \pmod{r}$ for all $1 \leq i \leq k$.
(An integer n is called square-free if it is not divisible by d^2 for any integer $d > 1$.)

(proposed by Tigran Hakobyan, Yerevan State University, Vanadzor, Armenia)

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Theory and methods for 2024-D2-P10

Theory Finite fields, Frobenius, and polynomial functions [T-AL-FINITEFIELD]; Groups, orders, generators, and quotients [T-AL-GROUP]; Prime valuations, divisors, and factorials [T-AL-VALUATION]; Congruences, Bézout, and the Chinese remainder theorem [T-AL-CONGRUENCE]
Methods Choose a modulus that exposes the obstruction [M-AL-MODULAR]
Strategy Pass between local constraints and a global conclusion [S-LOCALGLOBAL]; Expose the structure before calculating [S-STRUCTURE]

Statement Recorded solutions Catalog

2024-D2-P10 — Solution

For this record: Theory: T-AL-FINITEFIELD; T-AL-GROUP; T-AL-VALUATION; T-AL-CONGRUENCE. Methods: M-AL-MODULAR. Full learning route

We first prove the following claims.

2024-D2-P10 — Solution (continued)

Lemma 1. If n is almost prime then $\gcd(n, \varphi(n)) = 1$.
Proof. Assume to the contrary that $\gcd(n, \varphi(n)) > 1$ so that there are primes p and q dividing n such that $p \equiv 1 \pmod{q}$. For $0 \le i \le p-2$ let h_i be the number of positive divisors of n congruent to i modulo $p-1$ and similarly for $0 \le j \le q-1$ let ν_j denote the number of positive divisors of n congruent to j modulo q. Observe that the polynomial $F_n(x) = x^{d_1} + x^{d_2} + \cdots + x^{d_k} - kx$ defines the zero function on $\mathbb{F}_p$ due to the condition of the problem. On the other hand, $F_n(x) = (h_1 - k)x + \sum_{i \ne 1} h_i x^i$ in $\mathbb{F}_p[x]$, so that $p | h_i$ for all $0 \le i \le p-2$, $i \ne 1$. It follows that $2^{\omega(n)-1} = \nu_0 = h_0 + h_q + h_{2q} + \cdots \equiv 0 \pmod{p}$ which is a contradiction (here $\omega(n)$ means the number of distinct prime divisors of n). Therefore our assumption was wrong and the lemma is proved. □
Lemma 2. Let q be a prime number and let h be a positive integer coprime to $q-1$. If l is the order of h modulo $q-1$, then there exists $a \in \mathbb{F}_q$ such that $a^{h^l} = a$ and

$$a - a^h + a^{h^2} - \cdots + (-1)^{l-1} a^{h^{l-1}} \ne 0$$

Proof. Observe that $a^{h^l} = a$ for any $a \in \mathbb{F}_q$ since $q - 1 \mid h^l - 1$. On the other hand, the numbers $h^0, h^1, \ldots, h^{l-1}$ leave different remainders upon division by $q-1$ and therefore the polynomial

$$f(x) = x - x^h + x^{h^2} - \cdots + (-1)^{l-1} x^{h^{l-1}}$$

defines a function on $\mathbb{F}_q$, which is not identically zero. Hence the existence of an element with the required properties is proved. □
Lemma 3. If n is almost prime then for any primes p and q dividing n, the order of p modulo $q-1$ is an odd number.
Proof. Observe that due to Lemma 1 the order l of p modulo $q-1$ is well defined and assume to the contrary that l is an even number. According to Lemma 2 there exists $a \in \mathbb{F}_q$ such that $a^{p^l} = a$ and $f(a) \ne 0$, where $f(x) = x - x^p + x^{p^2} - \cdots + (-1)^{l-1} x^{p^{l-1}}$. Let us consider the sequence $(a_i)_{i=0}^{l} \subset \mathbb{F}_q$ defined by $a_0 = a$ and $a_{i+1} = -a_i^p$ for $0 \le i \le l-1$. Notice that since l is even by the assumption, we have $a_l = a_0^{p^l} = a_0$. It follows that

$$\sum_{i=0}^{l-1} \sum_{d | n} a_i^d = \sum_{i=0}^{l-1} \left(\sum_{d | \frac{n}{p}} a_i^d + \sum_{d | \frac{n}{p}} a_i^{pd} \right) = \sum_{i=0}^{l-1} \sum_{d | \frac{n}{p}} (a_{i+1}^d + a_i^{pd}) = 0,$$

since d is always odd being a divisor of n (Recall that $\gcd(n, \varphi(n)) = 1$ due to Lemma 1, so that n is odd, except the trivial case $n = 2$), and $a_{i+1} = -a_i^p$ for all $0 \le i \le l-1$. On the other hand, according to the condition of the problem, $\sum_{d|n} a_i^d = k a_i$ in $\mathbb{F}_q$ for all i, which shows that

$$k f(a) = k \sum_{i=0}^{l-1} a_i = \sum_{i=0}^{l-1} k a_i = \sum_{i=0}^{l-1} \sum_{d | n} a_i^d = 0$$

in $\mathbb{F}_q$ which is impossible, since $f(a) \ne 0$ by construction and $k = 2^{\omega(n)-1}$
is coprime to q. The attained contradiction shows that our assumption was wrong and concludes the proof of the lemma. □

Let us get back to the problem. Suppose that $p | n$ is prime and $r = 2^{2^m} + 1$ is a Fermat's prime such that $p \equiv 1 \pmod{r}$. If q is any prime divisor of n, then by Lemma 3 we have that $q^l \equiv 1 \pmod{p-1}$ for some odd l, so that $q^l \equiv 1 \pmod{r}$ and therefore $q = q^{\gcd(l, r-1)} \equiv 1 \pmod{r}$. Hence $d \equiv 1 \pmod{r}$ for any divisor d of n. □

Chapter 4

Correction Register

This register gives the canonical replacements for 31 identified theory defects. The stable correction IDs connect the retained teaching pages to their controlling statements. Cross-references below point to pages in this edition, rather than to the obsolete physical page numbers of earlier PDFs.

Reading rule. Where a retained passage conflicts with a numbered replacement, use the replacement. The register corrects specific known defects; it is not a claim that every proof or every alternative route has received independent mathematical review. The problem corpus is the historical 1994–2024 collection of 334 problems.

4.1 Canonical mathematical replacements

MATH-001 – Lagrange interpolation variable

Retained passage: pp. 447, 505

Let F be a field, let $n \geq 0$ be an integer, and let $x_0, \dots, x_n \in F$ be distinct. Set

$$\ell_k(x) = \prod_{\substack{0 \leq j \leq n \\ j \neq k}} \frac{x - x_j}{x_k - x_j}.$$

For every polynomial $p \in F[x]$ with $\deg p \leq n$,

$$p(x) = \sum_{k=0}^{n} p(x_k)\ell_k(x), \qquad \ell_k(x_j) = \delta_{kj}.$$

The legacy factor $\ell_k(x_k)$ is incorrect.

MATH-002 – Bézout identity

Retained passage: pp. 345, 390, 453

Let $a, b \in \mathbb{Z}$ be not both zero and put $g = \gcd(a, b) > 0$. Then

$$\{ua + vb : u, v \in \mathbb{Z}\} = g\mathbb{Z}.$$

Equivalently, $d = \gcd(a, b)$ if and only if $d > 0$, $d \mid a$, $d \mid b$, and $d = ua + vb$ for some $u, v \in \mathbb{Z}$. Mere representability as $ua + vb$ does not characterize the greatest common divisor.

MATH-003 – Characteristic zero and additive torsion

Retained passage: pp. 347, 456

For a unital ring, $\operatorname{char} R = 0$ means $n1_R \neq 0$ for every $n \geq 1$; this alone does not imply $na = 0 \Rightarrow a = 0$ for all $a \in R$. If the additive group $(R, +)$ is torsion-free, then

$$na = 0, \quad n \in \mathbb{Z} \setminus \{0\} \quad \Longrightarrow \quad a = 0.$$

This holds, for example, in a characteristic-zero integral domain or an algebra over a characteristic-zero field.

MATH-004 – Rank and trace of an idempotent

Retained passage: p. 423

Let F be a field and let $n \geq 1$ be an integer. If $P \in M_n(F)$ and $P^2 = P$, then $F^n = \operatorname{im} P \oplus \ker P$ and P is similar to $\operatorname{diag}(I_r, 0)$, where $r = \operatorname{rank} P$. Hence

$$\operatorname{tr} P = r\, 1_F.$$

Over $\mathbb{R}$ or $\mathbb{C}$ this is the usual equality $\operatorname{tr} P = r$; in positive characteristic the trace records the image of r in F.

MATH-005 – Endpoint Laplace principle

Retained passage: p. 499

Let $A > 0$ and let $f : [0, A] \to \mathbb{R}$ be continuous, $f(0) = 1$, $|f(x)| < 1$ for $0 < x \leq A$, and

$$f(x) = 1 - \lambda x + O(x^2) \qquad (x \downarrow 0), \qquad \lambda > 0.$$

Then

$$n \int_0^A f(x)^n \, dx \longrightarrow \frac{1}{\lambda}.$$

For a nonnegative-only version, replace $|f(x)| < 1$ by $0 \leq f(x) < 1$.

MATH-006 – Normalized convex-coefficient kernel

Retained passage: p. 447

Let $n \geq 0$ be an integer, let $p(z) = \sum_{k=0}^{n} a_k z^k \in \mathbb{C}[z]$, and let $1 = c_0 \geq c_1 \geq \cdots \geq c_n \geq 0$ satisfy $2c_k \leq c_{k-1} + c_{k+1}$ for $1 \leq k < n$. Then

$$q(z) = \sum_{k=0}^{n} c_k a_k z^k, \qquad \|q\|_{\infty,\mathbb{D}} \leq \|p\|_{\infty,\mathbb{D}}.$$

Here $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ and $T_c p = q$. If $c_n < 1$, put $d_k = (c_k - c_n)/(1 - c_n)$, so $d_0 = 1$, $d_n = 0$, and $T_c = c_n I + (1 - c_n) T_d$, where $T_d p = \sum_{k=0}^{n} d_k a_k z^k$; the case $c_n = 1$ is trivial. With $d_{n+1} = 0$,

$$K(t) = d_0 + 2\sum_{k=1}^{n} d_k \cos(kt) = \sum_{j=0}^{n-1} (d_j - 2d_{j+1} + d_{j+2}) \left| \sum_{r=0}^{j} e^{irt} \right|^2 \geq 0, \qquad \frac{1}{2\pi} \int_{-\pi}^{\pi} K(t)\, dt = 1.$$

Convolution gives $\|T_d p\|_\infty \leq \|p\|_\infty$, and the endpoint reduction gives the stated result. Convexity and monotonicity without normalization do not imply a contraction.

MATH-007 – Equidistant sets in Hilbert space

Retained passage: pp. 274, 473

In a real Hilbert space, let $m \geq 2$ and let $x_1, \ldots, x_m$ be pairwise at distance $d > 0$. Put $\bar{x} = m^{-1} \sum_i x_i$. Then

$$\langle x_i - \bar{x}, x_j - \bar{x} \rangle = \frac{d^2}{2} \left(\delta_{ij} - \frac{1}{m} \right).$$

Thus the finite set is a regular $(m-1)$-simplex, not generally a translate of m orthogonal vectors. For a countably infinite equidistant family in a real Hilbert space, the finite barycentres converge to a point y and $\langle x_i - y, x_j - y \rangle = (d^2/2)\delta_{ij}$; hence $\sqrt{2}(x_i - y)/d$ is orthonormal. The finite and infinite statements must not be conflated. In a complex Hilbert space, these claims apply to the underlying real inner product $\operatorname{Re}\langle \cdot, \cdot \rangle$; distances alone do not imply complex orthogonality.

MATH-008 – Block determinant and Schur complement

Retained passage: p. 253

Let F be a field and let $r, s \geq 1$ be integers. For $A \in M_r(F)$, $B \in F^{r\times s}$, $C \in F^{s\times r}$, and $D \in M_s(F)$, with A invertible,

$$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det(A)\det(D - CA^{-1}B).$$

If D is invertible, the analogous formula is $\det(D)\det(A - BD^{-1}C)$. An unidentified formula $\det(M)\det(H) = \det(A)$ is not a theorem.

MATH-009 – Rank under scalar extension

Retained passage: p. 254

For a matrix A with real entries,

$$\operatorname{rank}_{\mathbb{R}} A = \operatorname{rank}_{\mathbb{C}} A.$$

This follows from the nonzero-minor characterization. Scalar extension can change eigenvalue and Jordan descriptions, but not rank.

MATH-010 – Trace pairing for symmetric matrices

Retained passage: p. 256

Over $\mathbb{R}$, if $S = S^{\mathsf{T}} \neq 0$, then

$$\operatorname{tr}(S^2) = \|S\|_F^2 > 0.$$

Hence a subspace $W \leq M_n(\mathbb{R})$ satisfying $\operatorname{tr}(XY) = 0$ for all $X, Y \in W$ contains no nonzero real symmetric matrix. The corresponding claim is false for complex symmetric matrices without additional structure.

MATH-011 – Meaning of $q(e^X)$

Retained passage: p. 256

Let q be a polynomial and X a square complex matrix. Matrix functional calculus defines $q(e^X)$. If

$$q(e^z) = \sum_{r\geq 0} c_r z^r,$$

then $q(e^X) = \sum_{r\geq 0} c_r X^r$, with convergence for every X. Any broader function class must be named together with its domain and functional-calculus convention.

MATH-012 – Rational coordinates from Gram data

Retained passage: p. 256

In a real inner-product space, let $u_1, \ldots, u_m$ be linearly independent, $G = (\langle u_i, u_j\rangle) \in M_m(\mathbb{Q})$, and $v \in \operatorname{span}\{u_i\}$. If every $\langle u_i, v\rangle$ is rational, then the coordinate vector c of v in this basis is rational, because

$$Gc = (\langle u_i, v\rangle)_i.$$

A rational Gram matrix alone does not make the right-hand side rational.

MATH-013 – Even and odd symmetrization

Retained passage: p. 261

On a symmetric compact interval, if p approximates an even target f, then

$$p_e(x) = \frac{p(x) + p(-x)}{2}, \qquad \|p_e - f\|_\infty \leq \|p - f\|_\infty.$$

For an odd target use $p_o(x) = [p(x) - p(-x)]/2$. The target symmetry and the chosen sign must both be stated.

MATH-014 – Orthonormal sequences and pointwise convergence

Retained passage: p. 261
Let (X, Σ, μ) be a measure space with $\mu(X) < \infty$. An orthonormal sequence $(f_n) \subset L^2(X, \mu)$ satisfying $\sup_n \|f_n\|_{L^\infty} < \infty$ has no pointwise-almost-everywhere convergent subsequence. Dominated convergence would make such a subsequence L^2-Cauchy, whereas $\|f_n - f_m\|_2 = \sqrt{2}$ for $n \neq m$.

MATH-015 – Uniqueness of a maximizer

Retained passage: p. 262
Lipschitz continuity of ∇f gives smoothness estimates, not uniqueness. On a convex feasible set, strict concavity (in particular, strong concavity) gives at most one maximizer. Existence requires a separate hypothesis, such as compactness of the feasible set and continuity of f.

MATH-016 – Cayley-transform identity

Retained passage: p. 263
For $\alpha, z \in \mathbb{C}$ with $|\alpha| = 1$ and $z \neq -\alpha$,

$$\operatorname{Re}\left(\frac{z-\alpha}{z+\alpha}\right) = \frac{|z|^2 - 1}{|z+\alpha|^2}.$$

Thus the real part is negative, zero, or positive according as $|z| < 1$, $|z| = 1$, or $|z| > 1$. This replaces the corrupted formula containing the literal word “alpha.”

MATH-017 – Root-of-unity filter

Retained passage: p. 264
Let $m \geq 1$ be an integer, $\omega = e^{2\pi i/m}$, and $N \in \mathbb{Z}$. Then

$$\frac{1}{m}\sum_{j=0}^{m-1} \omega^{jN} = \begin{cases} 1, & m \mid N, \\ 0, & m \nmid N. \end{cases}$$

MATH-018 – Maximum-modulus principle on a bounded domain

Retained passage: p. 264
Let $\Omega \subset \mathbb{C}$ be a bounded domain and $f \in C(\overline{\Omega}) \cap \mathcal{O}(\Omega)$. Then

$$\max_{\overline{\Omega}} |f| = \max_{\partial\Omega} |f|.$$

The underlying maximum-modulus principle says that a nonconstant holomorphic function cannot attain its maximum modulus in Ω.

MATH-019 – Normalized Fejér kernel

Retained passage: p. 264
For an integer $k \geq 1$, define

$$F_k(t) = \frac{1}{k}\left|\sum_{j=0}^{k-1} e^{ijt}\right|^2 = \sum_{|r|<k} \left(1 - \frac{|r|}{k}\right) e^{irt} \geq 0.$$

Moreover, $(2\pi)^{-1}\int_{-\pi}^{\pi} F_k(t)\,dt = 1$. If the unnormalized square is retained elsewhere, label it kF_k and state that its integral is $2\pi k$.

MATH-020 – Telescoping infinite product

Retained passage: p. 269

For real $M > 1$,

$$\prod_{j=0}^{\infty}\left(1 + M^{-2^j}\right) = \frac{M}{M-1}.$$

Indeed, with $x = M^{-1}$, $\prod_{j=0}^{n}(1 + x^{2^j}) = (1 - x^{2^{n+1}})/(1 - x)$, and the result follows as $n \to \infty$.

MATH-021 – Equation of a vertical parabola

Retained passage: p. 273

In the mapped 2002 problem, “standard parabola” is a defined normalization: $y = (x - v)^2 + w$. A general vertical parabola has equation

$$y = a(x - v)^2 + w, \qquad a \neq 0.$$

MATH-022 – Existence of a nearest point

Retained passage: p. 274

For every nonempty closed $A \subset \mathbb{R}^n$ and $x \in \mathbb{R}^n$, $d(x, A)$ is attained: a minimizing sequence can be restricted to a compact ball. In an arbitrary Hilbert space, the projection theorem instead requires A to be nonempty, closed, and convex; the nearest point is then unique.

MATH-023 – Closed forward orbit on an interval

Retained passage: pp. 276, 360

Let I be a compact interval, let $f : I \to I$ be continuous, fix $x \in I$, and define

$$O^+(x) = \{f^n(x) : n \geq 0\}.$$

If $O^+(x)$ itself is closed in I, then it is finite; equivalently, x is eventually periodic. Here $f^0 = \mathrm{id}$, and “orbit” means $O^+(x)$, not its closure.

MATH-024 – Graph intersections and fixed points

Retained passage: p. 276

Let I be an interval, let $f : I \to I$, and let $n \geq 1$ be an integer. Fixed points of f^n are exactly the intersections of its graph with the diagonal. If I is the union of finitely many intervals and $f^n(x) - x$ is strictly monotone on each interval, then there is at most one zero on each interval, so the fixed-point set is finite. Monotonicity of f^n alone is insufficient; in particular, a nondegenerate interval on which $f^n = \mathrm{id}$ contains infinitely many fixed points.

MATH-025 – Geometric probability

Retained passage: p. 277

If X is uniformly distributed on a measurable set $D \subset \mathbb{R}^n$ with $0 < \mathrm{vol}(D) < \infty$, then for every measurable $A \subset D$,

$$\mathbb{P}(X \in A) = \frac{\mathrm{vol}(A)}{\mathrm{vol}(D)}.$$

The volume ratio is not valid without the uniform-distribution assumption.

MATH-026 – Differentiability of bounded-variation functions

Retained passage: p. 278

A real-valued function of bounded variation on an interval is differentiable almost everywhere with respect to Lebesgue measure.

MATH-027 – Cauchy’s mean value theorem

Retained passage: p. 431
Let $a < b$ and let $f, g \in C[a,b] \cap C^1(a,b)$. Then some $c \in (a,b)$ satisfies

$$[f(b) - f(a)]g'(c) = [g(b) - g(a)]f'(c).$$

Use a quotient form only when its denominators are nonzero.

MATH-028 – Dominated convergence

Retained passage: p. 438
On a measure space (X, Σ, μ), let f and each f_n be measurable real- or complex-valued functions, suppose $f_n \to f$ almost everywhere, and assume $|f_n| \le g$ almost everywhere for some $g \in L^1(X,\mu)$. Then $f \in L^1(X,\mu)$ and

$$\int f_n \longrightarrow \int f.$$

A positive constant bound is an integrable dominator only when the underlying measure is finite.

MATH-029 – Fixed points of a Möbius map

Retained passage: p. 456
Let $a, b, c, d \in \mathbb{C}$ and put $T(z) = (az+b)/(cz+d)$ where $cz + d \neq 0$; for a genuine Möbius map require $ad - bc \neq 0$. A finite r is fixed if and only if $cr + d \neq 0$ and

$$q(r) = 0, \qquad q(x) = cx^2 + (d-a)x - b.$$

If $a, b, c, d \in \mathbb{Q}$, r is quadratic irrational, and $q \neq 0$, then the minimal polynomial of r divides q; when $\deg q = 2$, q is a nonzero scalar multiple of that minimal polynomial.

MATH-030 – Intersections of conics

Retained passage: p. 471
Two projective plane conics over $\mathbb{C}$ with no common component have total intersection multiplicity four by Bézout’s theorem. Hence they have at most four distinct intersection points, and in particular at most four distinct real affine intersections. No finite bound holds when a component is shared.

MATH-031 – Spike construction and limsup coverage

Retained passage: p. 502
Let $b_n > 0$ and suppose $\sum_{n\ge1} b_n = \infty$. There exist distinct $a_n \in (0,1)$ such that the function $h : [0,1] \to \mathbb{R}$ defined by $h(a_n) = b_n$ and $h(x) = 0$ otherwise is nowhere differentiable on $(0,1)$.
For the construction, choose consecutive finite blocks B_k inductively so that

$$\sum_{n \in B_k} \min(b_n, 1/k) > 2, \qquad \beta_n = \min(b_n, 1/k) \quad (n \in B_k).$$

Such a block always exists: for fixed k, the sum of $\min(b_n, 1/k)$ over any infinite tail diverges. Thus $0 < \beta_n \le b_n$, $\beta_n \to 0$, and each block has enough total interval length to cover $[0,1]$. Place the intervals of radii β_n with centres in $(0,1)$ to cover $[0,1]$ in every block. Small perturbations preserve the finite open cover and allow all centres to be distinct, including from earlier blocks.
Every $x \in [0,1]$ belongs to one of these intervals in each block. For $x \notin \{a_n\}$ this supplies distinct centres tending to x for which

$$\left| \frac{h(a_n) - h(x)}{a_n - x} \right| = \frac{b_n}{|a_n - x|} \ge 1.$$

The complement of the countable support is dense, so a different sequence of zero values forces any derivative to equal zero. Hence no derivative exists at such $x \in (0,1)$. At every support point, h is discontinuous because $b_n > 0$ and the zero set is dense.

4.2 Data and classification corrections

DATA-001 – Legacy annotation counts

The 22 broad-theory counts printed in the July edition total 996. The earlier register's equality $996 = 3 \times 334$ was an arithmetic error: $3 \times 334 = 1002$. Consequently that total does not establish an exact quota of three labels per problem. A recovered aggregate file reports 997 broad links, which differs from the printed table; this discrepancy must be reconciled before republishing a broad-theory census.

The 107 detailed-node counts retained on pp. 246–249 total $2338 = 7 \times 334$. This total is consistent with seven assignments per problem; the total alone does not prove a per-record quota or validate any mathematical assignment. These legacy tables are exploratory navigation metadata, not estimates of actual theorem use or future competition prevalence.

Distinguish required prerequisites, decisive methods, verified alternative routes, and contextual associations. Publish the coding rule, per-problem records, denominators, reconciliation process, version, and checksum when reporting a validated census. A problem can require several methods; multi-label method counts need not sum to 334. Pace labels are editorial training guidance.

CLASS-001 – Known contradictory classifications

The following legacy assignments are withdrawn. Any retained occurrence list must be read subject to these corrections until the relevant mappings are reconciled against the solutions.

- **1994-D1-P1**: inverse-matrix sign/count problem; the broad convexity/Jensen tag is unsupported by the recorded solution route.
- **1994-D1-P3**: irrational-number selection; withdraw integrals/series, graph-colouring, and rational-roots/Vieta labels, including the mapping on p. 279.
- **2007-D2-P1**: "graph" means a function graph, not a combinatorial graph or cycle.
- **2007-D2-P2**: modular fourth powers; use congruence/residue methods instead of the unsupported integrals/series labels.
- **2019-D1-P1**: infinite product/real analysis; withdraw the Schur-complement main-theory assignment on p. 316.
- **2020-D1-P3**: symmetric convex body/polytope approximation; withdraw the primary algebra/number-theory assignment.
- **2024-D2-P9**: matrix enumeration; withdraw the generic prime/GCD trap unless a specific solution justifies it.
- **2024-D2-P10**: modular/CRT/valuation structure; withdraw geometry, calculus, convexity, and integral/limit assignments, including p. 741.

CLASS-002 – Method, trigger, and template fields

The determinant-trigger table on p. 751, trigger contradictions on pp. 792–805, and primary-template totals on pp. 836–837 are not validated measures of proof architecture. Direct proof, contradiction, induction, extremal argument, invariant argument, construction, and sharpness require proof-grounded coding distinct from topical solution families. Essential, alternative, and incidental relations must not be aggregated.

4.3 Theorem-writing and audit standard

Every canonical theorem record should provide a quantified statement; all necessary domain, field, regularity, finiteness, measurability, and nondegeneracy hypotheses; a conclusion supported by the proof; a proof or precise source reference; and one stable source ID used by its rendered views. Where omitting a hypothesis invites a false inference, include a boundary example or counterexample.

Scope of the retained coverage audit

The audit on p. 490 targets selected theory gaps in the 1994–2024 annotated corpus. Residual omissions, statement errors, and alternative-solution prerequisites may remain. Independent mathematical review and a versioned errata record are required to establish broader correctness.

4.4 Publication review status

The correction register supplies specific mathematical and editorial repairs. The accompanying build report records the compilation and PDF checks actually performed. These production checks do not establish publisher approval or replace independent scholarly review.

Review area	Completion evidence
OPEN Mathematics	Independent review of all 334 problems, all included solution variants, and all theorem occurrences, with a reconciled review record. The 31-item register addresses identified errors rather than certifying all mathematics.
OPEN Data	Complete solution-grounded reconciliation of retained theory, method, trigger, and template relations; publish the codebook, per-problem records, denominators, and versions. Resolve the broad-count discrepancy noted in DATA-001.
OPEN Rights	Establish the applicable licence or permission and required credit for reproduced third-party statements, solutions, diagrams, and other materials; obtain the publisher's required clearance before commercial release.
OPEN Accessibility	Verify semantic headings, lists, tables, links, mathematics descriptions, reading order, and the publisher's required screen and print profiles. Searchable vector content alone is not proof of accessible reading order or PDF/UA conformance.

Sources and Provenance

The catalog and solution chapters use internal links throughout. A Source note opens the corresponding entry below; its year link leads to the official-source bibliography in this section. Only links explicitly marked web or website open a website. You can return to a problem's statement, solutions, or catalog card directly from its entry.

Reading the source record

Official contest materials. The International Mathematics Competition archive is the primary historical source for the contest statements and published solutions. The annual references below identify the relevant official pages.

Archived transcription. The source project contains 334 structured transcription records retrieved from Olympstat on 12 September 2026, together with a retrieval manifest and SHA-256 checksums. Each filename in the register identifies an included file under `data/corpus/`; it is an archival identifier, not a web link. The record includes its statement and solution variants. A matching filename and checksum establish the reconstruction input; they do not independently certify the mathematics.

Stable problem identities. A guide ID preserves the navigation used in earlier editions. The archived filename preserves the transcription source's year, day, and problem numbering. These labels may differ: the source register records their exact correspondence rather than silently renumbering the guide.

Editorial lineage. The July 2026 guide supplies the preparation architecture. The integrated August edition supplies the retained theory and method atlas and the catalog reference. The reconstructed chapters use the archived transcriptions with the documented corrections and equation-layout repairs. The source project retains the unmodified records, the normalized mathematics, and the correction logs.

Problem source register

The year column opens an internal bibliography entry. The three return links open the statement (S), solutions (L), and catalog card (C). A dagger identifies a historical citation preserved in the notes after this table.

Guide ID	Archived transcription record	Year	Return
1994-D1-P1	`imc-1994-d1-p1.json`	1994	S L C
1994-D1-P2	`imc-1994-d1-p2.json`	1994	S L C
1994-D1-P3	`imc-1994-d1-p3.json`	1994	S L C
1994-D1-P4	`imc-1994-d1-p4.json`	1994	S L C
1994-D1-P5	`imc-1994-d1-p5.json`	1994	S L C
1994-D1-P6	`imc-1994-d1-p6.json`	1994	S L C
1994-D2-P1	`imc-1994-d2-p1.json`	1994	S L C
1994-D2-P2	`imc-1994-d2-p2.json`	1994	S L C
1994-D2-P3	`imc-1994-d2-p3.json`	1994	S L C
1994-D2-P4	`imc-1994-d2-p4.json`	1994	S L C
1994-D2-P5	`imc-1994-d2-p5.json`	1994	S L C
1994-D2-P6	`imc-1994-d2-p6.json`	1994	S L C
1995-D1-P1	`imc-1995-d1-p1.json`	1995	S L C
1995-D1-P2	`imc-1995-d1-p2.json`	1995	S L C

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Problem source register (continued)

Guide ID	Archived transcription record	Year	Return
1995-D1-P3	imc-1995-d1-p3.json	1995	S L C
1995-D1-P4	imc-1995-d1-p4.json	1995	S L C
1995-D1-P5	imc-1995-d1-p5.json	1995	S L C
1995-D1-P6	imc-1995-d1-p6.json	1995	S L C
1995-D2-P1	imc-1995-d2-p1.json	1995	S L C
1995-D2-P2	imc-1995-d2-p2.json	1995	S L C
1995-D2-P3	imc-1995-d2-p3.json	1995	S L C
1995-D2-P4	imc-1995-d2-p4.json	1995	S L C
1995-D2-P5	imc-1995-d2-p5.json	1995	S L C
1995-D2-P6	imc-1995-d2-p6.json	1995	S L C
1996-D1-P1	imc-1996-d1-p1.json	1996	S L C
1996-D1-P2	imc-1996-d1-p2.json	1996	S L C
1996-D1-P3	imc-1996-d1-p3.json	1996	S L C
1996-D1-P4	imc-1996-d1-p4.json	1996	S L C
1996-D1-P5	imc-1996-d1-p5.json	1996	S L C
1996-D1-P6	imc-1996-d1-p6.json	1996	S L C
1996-D2-P1	imc-1996-d2-p1.json	1996	S L C
1996-D2-P2	imc-1996-d2-p2.json	1996	S L C
1996-D2-P3	imc-1996-d2-p3.json	1996	S L C
1996-D2-P4	imc-1996-d2-p4.json	1996	S L C
1996-D2-P5	imc-1996-d2-p5.json	1996	S L C
1996-D2-P6	imc-1996-d2-p6.json	1996	S L C
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1997-D1-P4	imc-1997-d1-p4.json	1997	S L C
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1997-D1-P6	imc-1997-d1-p6.json	1997	S L C
1997-D2-P1	imc-1997-d2-p1.json	1997	S L C
1997-D2-P2	imc-1997-d2-p2.json	1997	S L C
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1998-D1-P4	imc-1998-d1-p4.json	1998	S L C
1998-D1-P5	imc-1998-d1-p5.json	1998	S L C
1998-D1-P6	imc-1998-d1-p6.json	1998	S L C
1998-D2-P1	imc-1998-d2-p1.json	1998	S L C
1998-D2-P2	imc-1998-d2-p2.json	1998	S L C
1998-D2-P3	imc-1998-d2-p3.json	1998	S L C
1998-D2-P4	imc-1998-d2-p4.json	1998	S L C
1998-D2-P5	imc-1998-d2-p5.json	1998	S L C
1998-D2-P6	imc-1998-d2-p6.json	1998	S L C
1999-D1-P1	imc-1999-d1-p1.json	1999	S L C
1999-D1-P2	imc-1999-d1-p2.json	1999	S L C
1999-D1-P3	imc-1999-d1-p3.json	1999	S L C
1999-D1-P4	imc-1999-d1-p4.json	1999	S L C

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Problem source register (continued)

Guide ID	Archived transcription record	Year	Return
1999-D1-P5	imc-1999-d1-p5.json	1999	S L C
1999-D1-P6	imc-1999-d1-p6.json	1999	S L C
1999-D2-P1	imc-1999-d2-p1.json	1999	S L C
1999-D2-P2	imc-1999-d2-p2.json	1999	S L C
1999-D2-P3	imc-1999-d2-p3.json	1999	S L C
1999-D2-P4	imc-1999-d2-p4.json	1999	S L C
1999-D2-P5	imc-1999-d2-p5.json	1999	S L C
1999-D2-P6	imc-1999-d2-p6.json	1999	S L C
2000-D1-P1	imc-2000-d1-p1.json	2000	S L C
2000-D1-P2	imc-2000-d1-p2.json	2000	S L C
2000-D1-P3	imc-2000-d1-p3.json	2000	S L C
2000-D1-P4	imc-2000-d1-p4.json	2000	S L C
2000-D1-P5	imc-2000-d1-p5.json	2000	S L C
2000-D1-P6	imc-2000-d1-p6.json	2000	S L C
2000-D2-P1	imc-2000-d2-p1.json	2000	S L C
2000-D2-P2	imc-2000-d2-p2.json	2000	S L C
2000-D2-P3	imc-2000-d2-p3.json	2000	S L C
2000-D2-P4	imc-2000-d2-p4.json	2000	S L C
2000-D2-P5	imc-2000-d2-p5.json	2000	S L C
2000-D2-P6	imc-2000-d2-p6.json	2000	S L C
2001-D1-P1	imc-2001-d1-p1.json	2001	S L C
2001-D1-P2	imc-2001-d1-p2.json	2001	S L C
2001-D1-P3	imc-2001-d1-p3.json	2001	S L C
2001-D1-P4	imc-2001-d1-p4.json	2001	S L C
2001-D1-P5	imc-2001-d1-p5.json	2001	S L C
2001-D1-P6	imc-2001-d1-p6.json	2001	S L C
2001-D2-P1	imc-2001-d2-p1.json	2001	S L C
2001-D2-P2	imc-2001-d2-p2.json	2001	S L C
2001-D2-P3	imc-2001-d2-p3.json	2001	S L C
2001-D2-P4	imc-2001-d2-p4.json	2001	S L C
2001-D2-P5	imc-2001-d2-p5.json	2001	S L C
2001-D2-P6	imc-2001-d2-p6.json	2001	S L C
2002-D1-P1	imc-2002-d1-p1.json	2002	S L C
2002-D1-P2	imc-2002-d1-p2.json	2002	S L C
2002-D1-P3	imc-2002-d1-p3.json	2002	S L C
2002-D1-P4	imc-2002-d1-p4.json	2002	S L C
2002-D1-P5	imc-2002-d1-p5.json	2002	S L C
2002-D1-P6	imc-2002-d1-p6.json	2002	S L C
2002-D2-P1	imc-2002-d2-p1.json	2002	S L C
2002-D2-P2	imc-2002-d2-p2.json	2002	S L C
2002-D2-P3	imc-2002-d2-p3.json	2002	S L C
2002-D2-P4	imc-2002-d2-p4.json	2002	S L C
2002-D2-P5	imc-2002-d2-p5.json	2002	S L C
2002-D2-P6	imc-2002-d2-p6.json	2002	S L C
2003-D1-P1	imc-2003-d1-p1.json	2003	S L C
2003-D1-P2	imc-2003-d1-p2.json	2003	S L C
2003-D1-P3	imc-2003-d1-p3.json	2003	S L C
2003-D1-P4	imc-2003-d1-p4.json	2003	S L C
2003-D1-P5	imc-2003-d1-p5.json	2003	S L C
2003-D1-P6	imc-2003-d1-p6.json	2003	S L C

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Problem source register (continued)

Guide ID	Archived transcription record	Year	Return
2003-D2-P1	imc-2003-d2-p1.json	2003	S L C
2003-D2-P2	imc-2003-d2-p2.json	2003	S L C
2003-D2-P3	imc-2003-d2-p3.json	2003	S L C
2003-D2-P4	imc-2003-d2-p4.json	2003	S L C
2003-D2-P5	imc-2003-d2-p5.json	2003	S L C
2003-D2-P6	imc-2003-d2-p6.json	2003	S L C
2004-D1-P1	imc-2004-d1-p1.json	2004	S L C
2004-D1-P2	imc-2004-d1-p2.json	2004	S L C
2004-D1-P3	imc-2004-d1-p3.json	2004	S L C
2004-D1-P4	imc-2004-d1-p4.json	2004	S L C
2004-D1-P5	imc-2004-d1-p5.json	2004	S L C
2004-D1-P6	imc-2004-d1-p6.json	2004	S L C
2004-D2-P1	imc-2004-d2-p1.json	2004	S L C
2004-D2-P2	imc-2004-d2-p2.json	2004	S L C
2004-D2-P3	imc-2004-d2-p3.json	2004	S L C
2004-D2-P4	imc-2004-d2-p4.json	2004	S L C
2004-D2-P5	imc-2004-d2-p5.json	2004	S L C
2004-D2-P6	imc-2004-d2-p6.json	2004	S L C
2005-D1-P1	imc-2005-d1-p1.json	2005	S L C
2005-D1-P2	imc-2005-d1-p2.json	2005	S L C
2005-D1-P3	imc-2005-d1-p3.json	2005	S L C
2005-D1-P4	imc-2005-d1-p4.json	2005	S L C
2005-D1-P5	imc-2005-d1-p5.json	2005	S L C
2005-D1-P6	imc-2005-d1-p6.json	2005	S L C
2005-D2-P1	imc-2005-d2-p1.json	2005	S L C
2005-D2-P2	imc-2005-d2-p2.json	2005	S L C
2005-D2-P3	imc-2005-d2-p3.json	2005	S L C
2005-D2-P4	imc-2005-d2-p4.json	2005	S L C
2005-D2-P5	imc-2005-d2-p5.json	2005	S L C
2005-D2-P6	imc-2005-d2-p6.json	2005	S L C
2006-D1-P1	imc-2006-d1-p1.json	2006	S L C
2006-D1-P2	imc-2006-d1-p2.json	2006	S L C
2006-D1-P3	imc-2006-d1-p3.json	2006	S L C
2006-D1-P4	imc-2006-d1-p4.json	2006	S L C
2006-D1-P5	imc-2006-d1-p5.json	2006	S L C
2006-D1-P6	imc-2006-d1-p6.json	2006	S L C
2006-D2-P1	imc-2006-d2-p1.json	2006	S L C
2006-D2-P2	imc-2006-d2-p2.json	2006	S L C
2006-D2-P3	imc-2006-d2-p3.json	2006	S L C
2006-D2-P4	imc-2006-d2-p4.json	2006	S L C
2006-D2-P5	imc-2006-d2-p5.json	2006	S L C
2006-D2-P6	imc-2006-d2-p6.json	2006	S L C
2007-D1-P1	imc-2007-d1-p1.json	2007	S L C
2007-D1-P2	imc-2007-d1-p2.json	2007	S L C
2007-D1-P3	imc-2007-d1-p3.json	2007	S L C
2007-D1-P4	imc-2007-d1-p4.json	2007	S L C
2007-D1-P5	imc-2007-d1-p5.json	2007	S L C
2007-D1-P6	imc-2007-d1-p6.json	2007	S L C
2007-D2-P1	imc-2007-d2-p1.json	2007	S L C
2007-D2-P2	imc-2007-d2-p2.json	2007	S L C

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Problem source register (continued)

Guide ID	Archived transcription record	Year	Return
2007-D2-P3	imc-2007-d2-p3.json	2007	S L C
2007-D2-P4	imc-2007-d2-p4.json	2007	S L C
2007-D2-P5	imc-2007-d2-p5.json	2007	S L C
2007-D2-P6	imc-2007-d2-p6.json	2007	S L C
2008-D1-P1	imc-2008-d1-p1.json	2008	S L C
2008-D1-P2	imc-2008-d1-p2.json	2008	S L C
2008-D1-P3	imc-2008-d1-p3.json	2008	S L C
2008-D1-P4	imc-2008-d1-p4.json	2008	S L C
2008-D1-P5	imc-2008-d1-p5.json	2008	S L C
2008-D1-P6	imc-2008-d1-p6.json	2008	S L C
2008-D2-P1	imc-2008-d2-p1.json	2008	S L C
2008-D2-P2	imc-2008-d2-p2.json	2008	S L C
2008-D2-P3	imc-2008-d2-p3.json	2008	S L C
2008-D2-P4	imc-2008-d2-p4.json	2008	S L C
2008-D2-P5	imc-2008-d2-p5.json	2008	S L C
2008-D2-P6	imc-2008-d2-p6.json	2008	S L C
2009-D1-P1	imc-2009-d1-p1.json	2009	S L C
2009-D1-P2	imc-2009-d1-p2.json	2009	S L C
2009-D1-P3	imc-2009-d1-p3.json	2009	S L C
2009-D1-P4	imc-2009-d1-p4.json	2009	S L C
2009-D1-P5	imc-2009-d1-p5.json	2009	S L C
2009-D2-P1	imc-2009-d2-p1.json	2009	S L C
2009-D2-P2	imc-2009-d2-p2.json	2009	S L C
2009-D2-P3	imc-2009-d2-p3.json	2009	S L C
2009-D2-P4	imc-2009-d2-p4.json	2009	S L C
2009-D2-P5	imc-2009-d2-p5.json	2009	S L C
2010-D1-P1	imc-2010-d1-p1.json	2010	S L C
2010-D1-P2	imc-2010-d1-p2.json	2010	S L C
2010-D1-P3	imc-2010-d1-p3.json	2010	S L C
2010-D1-P4	imc-2010-d1-p4.json	2010	S L C
2010-D1-P5	imc-2010-d1-p5.json	2010	S L C
2010-D2-P1	imc-2010-d2-p1.json	2010	S L C
2010-D2-P2	imc-2010-d2-p2.json	2010	S L C
2010-D2-P3	imc-2010-d2-p3.json	2010	S L C
2010-D2-P4	imc-2010-d2-p4.json	2010	S L C
2010-D2-P5	imc-2010-d2-p5.json	2010	S L C
2011-D1-P1	imc-2011-d1-p1.json	2011	S L C
2011-D1-P2	imc-2011-d1-p2.json	2011	S L C
2011-D1-P3	imc-2011-d1-p3.json	2011	S L C
2011-D1-P4	imc-2011-d1-p4.json	2011	S L C
2011-D1-P5	imc-2011-d1-p5.json	2011	S L C
2011-D2-P1	imc-2011-d2-p1.json	2011	S L C
2011-D2-P2	imc-2011-d2-p2.json	2011	S L C
2011-D2-P3	imc-2011-d2-p3.json	2011	S L C
2011-D2-P4	imc-2011-d2-p4.json	2011	S L C
2011-D2-P5	imc-2011-d2-p5.json	2011	S L C
2012-D1-P1	imc-2012-d1-p1.json	2012	S L C
2012-D1-P2	imc-2012-d1-p2.json	2012	S L C
2012-D1-P3	imc-2012-d1-p3.json	2012	S L C
2012-D1-P4	imc-2012-d1-p4.json	2012	S L C

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Problem source register (continued)

Guide ID	Archived transcription record	Year	Return
2012-D1-P5	imc-2012-d1-p5.json	2012	S L C
2012-D2-P1	imc-2012-d2-p1.json	2012	S L C
2012-D2-P2	imc-2012-d2-p2.json	2012	S L C
2012-D2-P3	imc-2012-d2-p3.json	2012	S L C
2012-D2-P4	imc-2012-d2-p4.json	2012	S L C
2012-D2-P5†	imc-2012-d2-p5.json	2012	S L C
2013-D1-P1	imc-2013-d1-p1.json	2013	S L C
2013-D1-P2	imc-2013-d1-p2.json	2013	S L C
2013-D1-P3	imc-2013-d1-p3.json	2013	S L C
2013-D1-P4	imc-2013-d1-p4.json	2013	S L C
2013-D1-P5	imc-2013-d1-p5.json	2013	S L C
2013-D2-P1	imc-2013-d2-p1.json	2013	S L C
2013-D2-P2	imc-2013-d2-p2.json	2013	S L C
2013-D2-P3	imc-2013-d2-p3.json	2013	S L C
2013-D2-P4	imc-2013-d2-p4.json	2013	S L C
2013-D2-P5	imc-2013-d2-p5.json	2013	S L C
2014-D1-P1	imc-2014-d1-p1.json	2014	S L C
2014-D1-P2	imc-2014-d1-p2.json	2014	S L C
2014-D1-P3	imc-2014-d1-p3.json	2014	S L C
2014-D1-P4	imc-2014-d1-p4.json	2014	S L C
2014-D1-P5	imc-2014-d1-p5.json	2014	S L C
2014-D2-P1	imc-2014-d2-p1.json	2014	S L C
2014-D2-P2	imc-2014-d2-p2.json	2014	S L C
2014-D2-P3	imc-2014-d2-p3.json	2014	S L C
2014-D2-P4	imc-2014-d2-p4.json	2014	S L C
2014-D2-P5	imc-2014-d2-p5.json	2014	S L C
2015-D1-P1	imc-2015-d1-p1.json	2015	S L C
2015-D1-P2	imc-2015-d1-p2.json	2015	S L C
2015-D1-P3	imc-2015-d1-p3.json	2015	S L C
2015-D1-P4	imc-2015-d1-p4.json	2015	S L C
2015-D1-P5†	imc-2015-d1-p5.json	2015	S L C
2015-D2-P6	imc-2015-d2-p1.json	2015	S L C
2015-D2-P7	imc-2015-d2-p2.json	2015	S L C
2015-D2-P8	imc-2015-d2-p3.json	2015	S L C
2015-D2-P9	imc-2015-d2-p4.json	2015	S L C
2015-D2-P10	imc-2015-d2-p5.json	2015	S L C
2016-D1-P1	imc-2016-d1-p1.json	2016	S L C
2016-D1-P2	imc-2016-d1-p2.json	2016	S L C
2016-D1-P3	imc-2016-d1-p3.json	2016	S L C
2016-D1-P4	imc-2016-d1-p4.json	2016	S L C
2016-D1-P5	imc-2016-d1-p5.json	2016	S L C
2016-D2-P6	imc-2016-d2-p1.json	2016	S L C
2016-D2-P7	imc-2016-d2-p2.json	2016	S L C
2016-D2-P8	imc-2016-d2-p3.json	2016	S L C
2016-D2-P9	imc-2016-d2-p4.json	2016	S L C
2016-D2-P10	imc-2016-d2-p5.json	2016	S L C
2017-D1-P1	imc-2017-d1-p1.json	2017	S L C
2017-D1-P2	imc-2017-d1-p2.json	2017	S L C
2017-D1-P3	imc-2017-d1-p3.json	2017	S L C
2017-D1-P4	imc-2017-d1-p4.json	2017	S L C

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Problem source register (continued)

Guide ID	Archived transcription record	Year	Return
2017-D1-P5	imc-2017-d1-p5.json	2017	S L C
2017-D2-P6	imc-2017-d2-p1.json	2017	S L C
2017-D2-P7	imc-2017-d2-p2.json	2017	S L C
2017-D2-P8	imc-2017-d2-p3.json	2017	S L C
2017-D2-P9	imc-2017-d2-p4.json	2017	S L C
2017-D2-P10	imc-2017-d2-p5.json	2017	S L C
2018-D1-P1	imc-2018-d1-p1.json	2018	S L C
2018-D1-P2	imc-2018-d1-p2.json	2018	S L C
2018-D1-P3	imc-2018-d1-p3.json	2018	S L C
2018-D1-P4	imc-2018-d1-p4.json	2018	S L C
2018-D1-P5	imc-2018-d1-p5.json	2018	S L C
2018-D2-P6	imc-2018-d2-p1.json	2018	S L C
2018-D2-P7	imc-2018-d2-p2.json	2018	S L C
2018-D2-P8[†]	imc-2018-d2-p3.json	2018	S L C
2018-D2-P9	imc-2018-d2-p4.json	2018	S L C
2018-D2-P10	imc-2018-d2-p5.json	2018	S L C
2019-D1-P1	imc-2019-d1-p1.json	2019	S L C
2019-D1-P2	imc-2019-d1-p2.json	2019	S L C
2019-D1-P3	imc-2019-d1-p3.json	2019	S L C
2019-D1-P4	imc-2019-d1-p4.json	2019	S L C
2019-D1-P5	imc-2019-d1-p5.json	2019	S L C
2019-D2-P6	imc-2019-d2-p1.json	2019	S L C
2019-D2-P7	imc-2019-d2-p2.json	2019	S L C
2019-D2-P8	imc-2019-d2-p3.json	2019	S L C
2019-D2-P9	imc-2019-d2-p4.json	2019	S L C
2019-D2-P10	imc-2019-d2-p5.json	2019	S L C
2020-D1-P1	imc-2020-d1-p1.json	2020	S L C
2020-D1-P2	imc-2020-d1-p2.json	2020	S L C
2020-D1-P3	imc-2020-d1-p3.json	2020	S L C
2020-D1-P4	imc-2020-d1-p4.json	2020	S L C
2020-D2-P5	imc-2020-d2-p1.json	2020	S L C
2020-D2-P6	imc-2020-d2-p2.json	2020	S L C
2020-D2-P7	imc-2020-d2-p3.json	2020	S L C
2020-D2-P8	imc-2020-d2-p4.json	2020	S L C
2021-D1-P1	imc-2021-d1-p1.json	2021	S L C
2021-D1-P2	imc-2021-d1-p2.json	2021	S L C
2021-D1-P3	imc-2021-d1-p3.json	2021	S L C
2021-D1-P4	imc-2021-d1-p4.json	2021	S L C
2021-D2-P5	imc-2021-d2-p1.json	2021	S L C
2021-D2-P6	imc-2021-d2-p2.json	2021	S L C
2021-D2-P7	imc-2021-d2-p3.json	2021	S L C
2021-D2-P8	imc-2021-d2-p4.json	2021	S L C
2022-D1-P1	imc-2022-d1-p1.json	2022	S L C
2022-D1-P2	imc-2022-d1-p2.json	2022	S L C
2022-D1-P3	imc-2022-d1-p3.json	2022	S L C
2022-D1-P4	imc-2022-d1-p4.json	2022	S L C
2022-D2-P5	imc-2022-d2-p1.json	2022	S L C
2022-D2-P6	imc-2022-d2-p2.json	2022	S L C
2022-D2-P7	imc-2022-d2-p3.json	2022	S L C
2022-D2-P8	imc-2022-d2-p4.json	2022	S L C

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Problem source register (continued)

Guide ID	Archived transcription record	Year	Return
2023-D1-P1	`imc-2023-d1-p1.json`	2023	S L C
2023-D1-P2	`imc-2023-d1-p2.json`	2023	S L C
2023-D1-P3	`imc-2023-d1-p3.json`	2023	S L C
2023-D1-P4	`imc-2023-d1-p4.json`	2023	S L C
2023-D1-P5	`imc-2023-d1-p5.json`	2023	S L C
2023-D2-P6	`imc-2023-d2-p1.json`	2023	S L C
2023-D2-P7	`imc-2023-d2-p2.json`	2023	S L C
2023-D2-P8	`imc-2023-d2-p3.json`	2023	S L C
2023-D2-P9	`imc-2023-d2-p4.json`	2023	S L C
2023-D2-P10	`imc-2023-d2-p5.json`	2023	S L C
2024-D1-P1	`imc-2024-d1-p1.json`	2024	S L C
2024-D1-P2	`imc-2024-d1-p2.json`	2024	S L C
2024-D1-P3	`imc-2024-d1-p3.json`	2024	S L C
2024-D1-P4	`imc-2024-d1-p4.json`	2024	S L C
2024-D1-P5	`imc-2024-d1-p5.json`	2024	S L C
2024-D2-P6	`imc-2024-d2-p1.json`	2024	S L C
2024-D2-P7	`imc-2024-d2-p2.json`	2024	S L C
2024-D2-P8	`imc-2024-d2-p3.json`	2024	S L C
2024-D2-P9	`imc-2024-d2-p4.json`	2024	S L C
2024-D2-P10	`imc-2024-d2-p5.json`	2024	S L C

Historical citation notes

These three pointers were present in the archived solution remarks. Their original addresses are retained as plain, non-clickable provenance; they are not required for navigation or for reading any proof. They are recorded as historical citations, without a claim that the third-party pages remain available or unchanged.

2012-D2-P5. The source solution attributes its sumset argument to Petridis and cites a discussion on Timothy Gowers’s weblog. Return to solution.

`http://gowers.wordpress.com/2011/02/10/a-new-way-of-proving-sumset-estimates/`

2015-D1-P5. The source solution identifies a Mathematics Stack Exchange discussion as the origin of the simplex problem. Return to solution.

`http://math.stackexchange.com/questions/476640/n-simplex-in-an-intersection-of-n-balls/789390`

2018-D2-P8. The source solution points to the sixth proof in the Wikipedia article on Catalan numbers for a related cycle-lemma argument. Return to solution.

`https://en.wikipedia.org/wiki/Catalan_number#Sixth_proof`

Official-source bibliography

Website links. The entries in this bibliography open external sites. They provide access to the official annual archive and the transcription provider; the book's statements, solutions, and internal navigation do not depend on these sites being available.

International Mathematics Competition: problems and solutions archive (web).

Olympstat: structured transcription provider (web).

Year	Official contest source
1994	IMC 1994: official problems and solutions (web)
1995	IMC 1995: official problems and solutions (web)
1996	IMC 1996: official problems and solutions (web)
1997	IMC 1997: official problems and solutions (web)
1998	IMC 1998: official problems and solutions (web)
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2013	IMC 2013: official problems and solutions (web)
2014	IMC 2014: official problems and solutions (web)
2015	IMC 2015: official problems and solutions (web)
2016	IMC 2016: official problems and solutions (web)
2017	IMC 2017: official problems and solutions (web)
2018	IMC 2018: official problems and solutions (web)
2019	IMC 2019: official problems and solutions (web)
2020	IMC 2020: official problems and solutions (web)
2021	IMC 2021: official problems and solutions (web)
2022	IMC 2022: official problems and solutions (web)
2023	IMC 2023: official problems and solutions (web)
2024	IMC 2024: official problems and solutions (web)

Mathematical reference notes

The new reference entries give compact statements, proof cores and examples. The following primary or author-hosted sources support the indicated advanced material. Exercise and theory links elsewhere in the book stay inside this PDF.

[1] Matthew B. Day and Trevor Nakamura, A 3-skeleton for a classifying space for the symmetric group (2024) (website).

[2] Giorgis Petridis, New Proofs of Plünnecke-type Estimates for Product Sets in Groups, Combinatorica 32 (2012), 721–733 (website).

[3] NIST Digital Library of Mathematical Functions, Section 4.22: Infinite Products and Partial Fractions (website).

[4] Sheldon Axler, Linear Algebra Done Right, fourth edition (Springer, 2024) (website).

[5] Keith Conrad, Cyclicity of the multiplicative group modulo a prime (website).

[6] Jeremy Orloff, MIT 18.04: Complex Variables with Applications, lecture notes (2018) (website).

[7] Andrew V. Sutherland, MIT 18.785 Lecture 16: Riemann's Zeta Function and the Prime Number Theorem (2021) (website).

[8] MIT 6.042J, Lecture 11: Graphs—Handshaking, Connectivity and Trees (2005) (website).

[9] MIT 6.1200J, Lecture 13: Connectivity and Trees (2024) (website).

[10] Terence Tao, 245A Notes 5: Differentiation theorems (2010) (website).

[11] A. Freire, Notes on Baire's theorem (website).

[12] IMC 2017, Day 1, Problem 4: official solution (website).

[13] IMC 2016, Day 1, Problem 5: official solutions (website).

[14] IMC 2015, Day 2, Problem 8: official solutions (website).

[15] IMC 2016, Day 2, Problem 9: official solutions (website).

[16] IMC 2008: official problem and solution archive (website).

[17] IMC 2009: official problem and solution archive (website).

[18] IMC 2012: official problems and solutions (website).

[19] IMC 2023, Day 2, Problem 8: official solution (website).

[20] IMC 2020, Day 1, Problem 3: official solutions (website).

[21] IMC 2021: official first-day solutions (website).

[22] IMC 2024, Day 1, Problem 5: official solution (website).

[23] J. S. Milne, Fields and Galois Theory, author-hosted course notes (website).

[24] John K. Hunter, Measure Theory, University of California, Davis (website).

Corrections identified during the learning-route review

The theory and method mapping prompted checks of the formulas used in the recorded solutions. The source project preserves the original transcription and an exact correction ledger. The following exercise entries include verified repairs; their Source notes identify the official contest material. These local checks do not claim independent refereeing of every archived proof.

- 1995-D2-P6: corrected solution record(s); source note.
- 1997-D1-P6: corrected solution record(s); source note.
- 2000-D2-P4: corrected solution record(s); source note.
- 2002-D1-P6: corrected solution record(s); source note.
- 2002-D2-P6: corrected solution record(s); source note.
- 2003-D2-P1: corrected solution record(s); source note.
- 2004-D1-P6: corrected solution record(s); source note.
- 2005-D1-P4: corrected solution record(s); source note.
- 2008-D1-P4: corrected solution record(s); source note.
- 2014-D1-P5: corrected solution record(s); source note.
- 2014-D2-P1: corrected solution record(s); source note.
- 2014-D2-P3: corrected solution record(s); source note.
- 2014-D2-P4: corrected solution record(s); source note.
- 2014-D2-P5: corrected solution record(s); source note.
- 2015-D2-P6: corrected solution record(s); source note.
- 2015-D2-P8: corrected solution record(s); source note.
- 2016-D1-P4: corrected solution record(s); source note.
- 2016-D1-P5: corrected solution record(s); source note.
- 2017-D1-P4: corrected solution record(s); source note.
- 2018-D1-P4: corrected solution record(s); source note.
- 2018-D2-P7: corrected solution record(s); source note.
- 2018-D2-P8: corrected solution record(s); source note.
- 2019-D2-P8: corrected solution record(s); source note.
- 2020-D1-P1: corrected solution record(s); source note.
- 2021-D2-P5: corrected solution record(s); source note.
- 2022-D1-P4: corrected solution record(s); source note.
- 2024-D2-P6: corrected solution record(s); source note.

The 401 archived solution records include alternative proofs, complementary parts, and one explicitly labelled source pointer (2021-D2-P5, second record). They are not described as 401 independent complete proofs. The first record for that problem supplies its complete proof. Theory reference Method reference Strategy reference

Design and Reading References

The visual revision uses descriptive headings and consistent labels as structural cues [1], keeps elaborate artwork outside instructional passages [2], and uses a restrained, legible composition informed by research on perceptual fluency and first impressions [3–5]. Contrast checks and non-color labels follow selected W3C criteria [6].

These are editorial applications of research conducted in other settings. The cited studies do not validate this particular book, establish an optimal color palette for mathematics, or demonstrate improved competition results. Exact type sizes, colors and cover geometry are design choices. The accompanying design-evidence note records the findings, applications and limitations separately.

[1] P. D. Mautone and R. E. Mayer (2001). Signaling as a cognitive guide in multimedia learning. Journal of Educational Psychology, 93(2), 377–389. DOI / publisher record (website).

[2] S. F. Harp and R. E. Mayer (1998). How seductive details do their damage: A theory of cognitive interest in science learning. Journal of Educational Psychology, 90(3), 414–434. DOI / publisher record (website).

[3] R. Reber, P. Winkielman and N. Schwarz (1998). Effects of perceptual fluency on affective judgments. Psychological Science, 9(1), 45–48. DOI / publisher record (website).

[4] A. Y. Lee and A. A. Labroo (2004). The effect of conceptual and perceptual fluency on brand evaluation. Journal of Marketing Research, 41(2), 151–165. DOI / publisher record (website).

[5] A. N. Tuch, E. E. Presslaber, M. Stöcklin, K. Opwis and J. A. Bargas-Avila (2012). The role of visual complexity and prototypicality regarding first impression of websites: Working towards understanding aesthetic judgments. International Journal of Human-Computer Studies, 70(11), 794–811. DOI / publisher record (website).

[6] World Wide Web Consortium. Web Content Accessibility Guidelines (WCAG) 2.2. Contrast, use of color, headings, link purpose and consistent identification criteria. Official recommendation (website).

Edition and Design Notes

Editorial lineage. The July 2026 guide supplies the preparation architecture. The August integrated edition supplies the retained theory and method atlas and the catalog reference. The September reconstruction supplies the editable problem and solution chapters. The present pass adds solution-grounded theory, method and strategy links, compact reference entries and explicit exercise returns. The established visual design is retained.

Mathematical cover. The cover construction uses the seventeen points

$$\zeta_k = \cos\frac{2\pi k}{17} + i\sin\frac{2\pi k}{17}, \qquad k = 0, \dots, 16.$$

They lie on the unit circle. Successive points define a regular polygon; additional chords join indices differing by four, and one triangle is highlighted. The illustration is a reproducible vector construction. It is a visual motif, not a claim about the distribution of contest topics.

Visual system. Navy marks major structure, dark teal marks instructional emphasis, and blue identifies internal navigation. Words and stable IDs accompany these colors. The retained atlas also uses distinct violet, blue and ochre labels for its established categories. Gold is reserved for decorative rules and cover geometry. The design references distinguish relevant research from editorial judgment; no claim of improved examination performance is inferred from the appearance of this book.

Learning reference. Each new theory, method and strategy entry has a canonical destination. Problem-specific links describe the recorded proofs, including distinct requirements of different solution records. They do not assert that a listed technique is indispensable to every alternative proof. The closed source register documents exact corrections found while checking these dependencies.

Source project. The front and back matter, catalog, problem and solution chapters, and correction register are editable LaTeX. The 581-page theory and method atlas remains a searchable vector PDF component. Its original semantic LaTeX was not present in the recovered source. Native LaTeX replacements repair 69 exposed mathematical fragments in 51 atlas paragraphs. Fonts, data, assets, generators and verification tools accompany the build project.

Review status. This independent manuscript is prepared for author and editorial review. Independent mathematical refereeing, full review of retained classification judgments, third-party permissions, and semantic-accessibility production remain separate publication work. The design and automated checks do not constitute publisher endorsement or PDF/UA certification.

Corrections and correspondence. Contact Leonidas Askianakis at `askianakis.leonidas@gmail.com`. Please include the edition date, problem or theorem ID, and the precise passage concerned.

Contents | Sources and provenance | Correction register

IMC FULL PREPARATION GUIDE

From a first idea to a complete proof

A preparation atlas organized around 334 competition problems from 1994–2024. Theory, reusable methods, recognition cues, proof templates, and 401 recorded solution records are connected by stable identifiers and internal navigation.

PREPARE Connect definitions and theorem hypotheses to a concrete problem.

PRACTICE Attempt the statement, identify the obstruction, and consult a focused method cue.

RECONSTRUCT Compare solution records, check exceptional cases, and write the proof independently.

Return to contents | **Reading routes**

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